



AutoSys Workload Automation 12.0

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Release Notes

Welcome to the AutoSys Workload Automation Release Notes.

This section contains information on the AutoSys Workload Automation and AutoSys Web UI release.

AutoSys Version 12.0

Product Name Change

In AutoSys Version 12.0,

- CA Workload Automation AE is now known as AutoSys Workload Automation (AutoSys)
- CA Workload Control Center (WCC) is now known as AutoSys Web UI (Web UI)

This section includes information about the AutoSys Workload Automation and Web UI Version 12.0 new features, changes to existing features, and fixed issues.

- [New Features - 12](#)
- [Changes to Existing Features - 12](#)
- [Fixed Issues - 12](#)
- [Third-Party Software Acknowledgements - 12](#)

New Features - 12.0

This article describes new features added to AutoSys Workload Automation and Web UI in Version 12.0

New Simplified Installer

- Simplifies install and upgrade experience.
- Installs AutoSys, Web UI, EEM, Agents, and other components from a single DVD.
- A single DVD per platform that is self-contained for AutoSys, EEM, Web UI, and Agent installations. You no longer have to deal with multiple DVDs for AutoSys, Web UI, Common Component, Agent, and so on.
- AutoSys and Web UI are installed in the same directory.
- Bundled with AdoptOpenJDK. However, if you have the license, you can use your own Oracle JRE or any other JRE.
- Analyses prerequisites and validates user inputs, upfront.
- Eliminates the need to manually install 32-bit RPMs as compared to the previous installer.
- Creates databases, users, and tablespaces for the Web UI. For Oracle database, the operational user need not create or modify the tables, and index privileges.

Native 64-bit Support

AutoSys (Version 12.0) has native 64-bit support and is backward compatible with supported 32-bit clients, agents, and AutoSys servers.

Web Services Orchestration (REST and SOAP)

Web Service Orchestration provides you the ability to run one web service with the response data value(s) from another associated web service.

For example, if you have a web service that performs authentication and another web service that returns the security token, then the security token has to be passed on to all the subsequent web services associated with that session.

The Web Service Document/Literal (WSDOC) job lets you call a WSDL operation with given inputs to the operation parameters in the form of SOAP envelope. AutoSys Web UI provides "WSDL Wizard" helper dialog to auto-generate SOAP envelopes with user given inputs. SOAP jobtype also allows users to copy-paste or specify SOAP envelopes generated from external tools in the 'Envelope XML' text area from Web UI.

Improved Web UI Collector Design for Performance

A Web server provides the GUI functionality using Tomcat and serves the web application requests. Also, this communicates with the AutoSys Application Servers to perform operations over the AutoSys artifacts. A Server component "Collector" collects the AutoSys artifacts like Job definitions, Job runs, Alarms, and Machines from AutoSys and populates the Web UI database. The collected information is used to monitor and manage the workload. A Web UI Collector component is now dedicated for collection but not for user workload. When Web UI is in High Availability (HA), collection is distributed between the HA nodes. Additionally, you can install collectors only on separate machines.

Password Vault Support - AutoSys Integration with Symantec Privileged Access Manager (PAM)

As an AutoSys user, you will be using many applications in your daily work and they may require credentials for their functionality. Managing the credentials is made easy by integrating AutoSys with the Symantec Privileged Access Manager (PAM) vaulting system. Your credentials of various applications that you use are protected and managed by PAM against your alias.

Centralized Agents Upgrade

You can perform the following options:

- Upload and stage agent installers from the Web UI Server and upgrade the Agent. Control and upgrade policy access to perform these actions are required.
- Use the JVM bundled with the product or specify the JVM path in your computer against the operating system.
- Perform the upgrade whenever you want or select the date and time at which you want to schedule the upgrade.

NOTE

Agent Inventory supports only the agents that are in AutoSys compatible mode.

Rename a Job

The `rename_job` subcommand renames an existing job, and all dependencies reflect the changed name for that specific job in local instance and remote cross instances which are at Version 12.0 or higher. Cross instance dependencies of only type 'a' are changed. For all other instances, you must manually change the condition.

You can use the Rename job option under the Actions drop-down in Web UI.

RESTful Web Service JIL Interface

- The Manage JIL (AEWS) method is added to Web Services. You can now import and export JIL objects definitions using RESTful APIs as follows:
 - The POST `AEWS/jil` method imports jil definitions given in the request body.
 - The GET `AEWS/jil/{type}` method exports jil definitions of the given type based on the query parameters.
- The Manage Calendars (AEWS) method is added to Web Services. You can now import and export calendar definitions using RESTful APIs as follows:
 - The POST `AEWS/calendar` method imports calendar definitions given in the request body and supports standard, extended, and cycle calendars.
 - The GET `AEWS/calendar/{type}` method exports calendar definitions of the given type based on the query parameters.

Custom Schema Owner for Oracle and Custom Database User for All Databases

You can now install multiple AutoSys instances on the same Oracle SID.

You can provide custom schema owner for Oracle database. In the following example the custom schema owner is dbadmin.

Example: Schema owner is now stored in the config file EventServer_1=AEORA02:dbadmin,1521,ae-oracle-1.ibn.broadcom.net

You can provide custom database users for all the databases. In the following example, the username can be customized.

Example: UNIX -- Edit the following entry in the configuration file:

```
DBAccess=username/password
```

You can also specify the schema owner for Oracle database and database user for all databases in Web UI.

Platforms and Databases Certifications

AutoSys Workload Automation supports the following:

- MSSQL Server Support for Linux
- Red Hat Enterprise Linux 8
- Oracle Enterprise Linux 8
- SUSE Linux Enterprise Server 15
- MSSQL Server 2019
- Oracle 19c
- Google Cloud Platform

Changes to Existing Features - 12.0

This article details the changes to existing features in AutoSys Workload Automation and Web UI Version 12.0

Release Numbering Convention

The numbering standard for AutoSys released to customers through any medium shall have version numbers in the form of "aa.b.cc.dd-eeee" where

- "aa" indicates the "Version" (major) level
- "b" indicates the "Release" (minor) level
- "cc" indicates the "Service Pack" (maintenance) level
- "dd" indicates the "Increment" number
- "eeee" indicates the "Build" number

Changed Features

The following features have changed in this release:

- We now support Certificate-based EEM authentication for AutoSys REST web services. Therefore, any change in the EIAMAdmin password will not affect AutoSys components/configuration.
- The extended calendar date conditions field limit is increased to 1024 characters.
- The j2ee_parameter and web_parameter fields limit is now 4000 characters.
- Command Sponsor component is discontinued from this release. Support for the Web UI Enterprise Command Line and Forecast applications is retained using RESTful Web server.

NOTE

In releases before Version 12.0, the ECLI and Forecast tabs of Web UI continue to use Command Sponsor.

- In Web UI, an AutoSys Server can be configured with multiple AutoSys REST endpoints now; if the first AutoSys REST endpoint is not available/reachable, it tries with the next url.
In addition, there is one-to-one mapping between an AutoSys instance and a web server. Therefore, multiple AutoSys instances should have multiple web servers.
- You can now cancel a POJO job.
- To change the database used by the Web UI component, use the installer to create the schema and then use the wcc_config utility to switch the database.
- You can enable the Web UI Reporting option to see the Reporting configuration. If the Reporting option is disabled, you cannot see the Reporting tab and also the "Connection status" wizard of the Web UI Dashboard does not show the Reporting database entry. Reporting collection and cleanup will not start.
- When upgrading to Web UI Version 12.0, if the existing connection.properties file has incorrect port number or instance name for the MSSQL database instances, then the installer may report a connection error. You must provide the correct port number or instance name in the connection.properties file.
- Agent Inventory now has two more options: Quiesce Agents and Unquiesce Agents.
- When viewing job details, you can now click the links in the job description field.
- You can add duration column to existing / new views, which display the duration of running jobs.

Discontinued Support - 12.0

Support for the following products and features have been discontinued.

Discontinued Support

The following products are not supported in this release:

- CA Spectrum Automation Manager
- Unicenter Notification Services

NOTE

While notification_id will be deprecated, the notification_msg part of Unicenter Notification will continue to work as the notificaiton_msg attribute is used by the email notification feature.

- Command Sponsor
- Event Management (formerly known as CA Unicenter NSM Event Management) currently shipped as part of CA Common Components. The OneAutomation Event Engine introduced with 11.3.6 SP7 is intended to provide similar automation needs using a more modern infrastructure.

Operating Systems

- **Solaris** - withdrawn for Server, Client, and SDK. We continue to support Solaris for Agents.
- **HP PA-RISC** - withdrawn for all components.
- HP Itanium server support is dropped.

Features

clean_files command is discontinued.

Plans for Upcoming Release

The Web Service RPC/Encoded Job type will not be supported from AutoSys Workload Automation Release 12.1

Fixed Issues - 12.0

This article details the issues fixed in AutoSys Workload Automation and Web UI Version 12.0.

AutoSys Fixed Issues

The following issues were fixed in AutoSys in this release:

NOTE

AutoSys Workload Automation Version 12.0 also includes the issues fixed in each of the prior releases of 11.3.6.

SCHEDULER DOES NOT SEND AGENT USER PASSWORDS FOR WSDOC JOBS

Problem Number: 4771

When you define and schedule a Web Service Document/Literal (WSDOC) job that specifies values for any two of the ws_conn_user, ws_proxy_user, or the wss_user attributes after using the autosys_secure utility to insert passwords for those users, the scheduler generated decryption errors and did not propagate the password for the users.

JOB DOES NOT START BASED ON STARTING CONDITIONS

Problem Number: 4778

Jobs enter the PEND_MACH status when a machine is already offline or when the machine cannot be reached. In the latter case, the job first goes into the RESTART status before being placed in the PEND_MACH status. If this job specifies date conditions and is then placed in the ON_HOLD status, the job does not start as per the date conditions when it is taken off hold.

WEBSERVER LOG CONTAINS WSQUERYNOTFOUNDEXCEPTION EVERY MIN

Problem Number: 4781

The CA Workload Automation AE web service that is used by the agent inventory module of CA Workload Control Center continuously wrote WsQueryNotFoundException into the \$AUTOUSER/out/waae_webservices.log file. The agent inventory module makes a request for delta changes of machine definitions from the instance. When the query retrieving the results does not return any rows, then the web service writes WsQueryNotFoundException along with the complete stack although this is a desired result few times.

REINDEX DB NO SUCH FILE OR DIR

Problem Number: 4782

The reindex script "reindexDB.pl" throws the following error intermittently when executed simultaneously against different databases.

Error message:

Cannot open /tmp/index.out: No such file or directory.

JOB-RUN-INFO SUMMARY SHOWS NTRY=1 NOT LATEST

Problem Number: 4787

When a child job is force started the run_num will remain same but the ntry gets incremented. The job-run-info web service always returns the summary from the ntry=1 instead of showing the latest ntry similar to autorep output.

JOB STATUS FROM Z/OS DELAYED AND QUERY FAILURES

Problem Number: 4789

Job status from z/OS agents are delayed due to SQL failures from ujo_comm_update stored procedure causing scheduling delays.

SCH DOESNT PROCESS JOB EVENTS AFTER DB ERROR ON STARTING EVT**Problem Number: 4790**

When the scheduler experiences a database error after scheduling the job and processing the STARTING event for a job, it did not process any other events for that job. The same problem can also occur when the scheduler experiences a database error while processing the RUNNING event for a box job.

SCHEDULER START WITH WRONG DATABASE VERSION IN SECOND SERVER**Problem Number: 4795**

CA Workload Automation services start even when the second database is not upgraded to the latest level causing unwarranted errors.

AGENT NOTIFICATION ON SCHEDULER START ON DIFFERENT SERVER**Problem Number: 4796**

CA Workload Automation AE Scheduler does not send notifications to the active agents when the scheduler is started in a different machine as primary scheduler.

OPTIMIZE QUERY TO DELETE AGENT ALIAS**Problem Number: 4797**

When a machine is deleted, the 'JIL' utility takes a long time to complete.

EXECUTE USER SCRIPT WHEN DB_PROBLEM ALARM IS RAISED**Problem Number: 4799**

AutoSys can be configured to execute scripts when problems with database connectivity are encountered, such as DB_PROBLEM, DB_ROLLOVER. In the current design, DB_PROBLEM alarm generated manually when running 'dbspace' command does support execution of user defined scripts.

SCHEDULER EVALUATES DUPLICATE JOB STARTS AFTER RESTARTING**Problem Number: 4803**

When you issue multiple STARTJOB events for the same job and the scheduler must restart the job after evaluating the first STARTJOB event, the scheduler evaluated the job and restarted it a second time while processing the next STARTJOB.

SCHEMA CREATION FAILS ON ORACLE**Problem Number: 4805**

CA Workload Automation installer fails to create the database on Oracle with failure to create ujo_cleanup_jobruns stored procedure due to insufficient privileges to run the DBMS_UTILITY command.

SCP JOB FAILS WHEN USER WITH KEYFILE IS USED**Problem Number: 4819**

SCP Job fails when the owner of the job is created with a key credential.

The scheduler generates the following error:

CAUAJM_E_10540 Owner User/Password error due to: USER ERROR: The username and /or password specified for "user" on machine "mach1" is invalid. Run autosys_secure to enter the correct password.

CREATEAEDB.PL ON ORACLE TAKES LONG**Problem Number: 4821**

CA WAAE r11.3.6 SP08 upgrade from earlier version takes more than two hours during database upgrade from earlier version.

NO STARTJOB WHEN CALENDAR AND EXCLUDE_CALENDAR HAVE SAME DAY**Problem Number: 4822**

When a job has run_calendar and an exclude_calendar and tomorrow's date appears in both the calendars, the JIL commands trying to create the job will fail to add STARTJOB to the job when both start_mins and start_times attributes are not defined for the job.

AUTOREP -M -P DUMPS CORE FILE**Problem Number: 4823**

Autosys server is running on 11.3.6 SP02 and client is 11.3.6 SP08. When running the command autorep -M <Machine Name> -p, it terminates with a sig 11 and dumps a core file.

EXTENDED CALENDAR DESCRIPTION ATTRIBUTE BLANK ISSUE**Problem Number: 4826**

When we import an extended calendar with empty description, the date condition attribute value is blank.

AUTOCAL_ASC CALENDAR PROBLEM**Problem Number: 4828**

- User did not get an error when trying to export a standard calendar by passing the extended calendar name as input.
- autocal_asc did not get an error when they selected sub menu option: List dates for a Calendar under menu option: Administer Standard Calendar by passing input as extended calendar name.

WRONG MESSAGE REMOVED JOB FROM RUNNING BOX**Problem Number: 4829**

When a user deletes a job from a box that never ran, Scheduler log displayed an ALERT event as follows:
CAUAJM_I_40245 EVENT: ALERT JOB: Test
<Removed job 'Test' from running box.>

SCHEDULER CRASH WITH ONE AUTOMATION INTEGRATION**Problem Number: 4830**

CA Workload Automation AE Scheduler crashes when the Automic server is unreachable over the network for posting the log messages to Automic server in One-Automation integration environment.

GLOBAL VARIABLE SUBSTITUTION FOR SOME ATTRIBUTES**Problem Number: 4831**

The following JIL attributes did not support global variables substitution:

- ps_run_cntrl_id
- ps_output_dest
- oracle_args

SAP - PASSWORD MISSING IN KILLJOB

Problem Number: 4832

When a user attempts to kill a SAP job, the agent returns the following error without attempting to kill the job:
Not Connected - Exception Persistent connection rejected. Missing Parameters: Password.

SAP KILLJOB FAIL DUE TO PASSWORD

Problem Number: 4833

When a user attempts to kill a SAP job, the scheduler tries to terminate the job, but agent fails to kill the job with the following error:
Missing Parameters: Password.

INCONSISTENT BEHAVIOR WHEN AUTO_HOLD ATTRIBUTE IS USED

Problem Number: 4834

When a job is created with auto_hold=1 within multiple levels of boxes, and the job is placed in ON_HOLD status multiple times in the following scenario:

- Top level job starts and places the said job in ON_HOLD status.
- Issue a JOB_OFF_HOLD to the job.
- Before the job's immediate box_job starts, the scheduler is restarted or scheduler will failover from high-available status.

IMPROVE THE PERFORMANCE OF AUTOREP DETAILED JOB REPORTS

Problem Number: 4835

When you accumulate hundreds of thousands of job runs, the application server delays the creation of detailed job reports by tens of minutes.

AUDIT INFO ARCHIVE CAUSES DEADLOCK ON SYBASE

Problem Number: 4836

Archive Events from audit tables causes deadlock on scheduler processing as the archive_events utility takes a long time to complete.

CHASE DOES NOT IDENTIFY RUNNING JOBS ON DELETED MACHINE

Problem Number: 4837

CA AutoSys JM CA does not identify jobs in RUNNING state when the machine where job has been running is deleted.

JIL PERFORMANCE AFFECTED DUE TO AUDIT INFORMATION

Problem Number: 4838

Client utility 'JIL' takes a longer time to insert/update a large batch of jobs when autotrack level is set to 2.

RENEW AGENT SOCKET CONNECTION AFTER CAUAJM_W_00060 TIMEOUT**Problem Number: 4840**

When the network experiences a failure after the scheduler opens a socket connection with the agent and sends it a request, the request can get lost and not arrive at the agent end of the socket connection. In this case, after the scheduler times out while waiting for an acknowledgement, it continues to reuse the socket connection to resend the request instead of re-creating the socket connection.

IMPROVE THE PERFORMANCE OF AUTOREP DETAILED JOB REPORTS**Problem Number: 4842**

When a fatal error occurs during a database operation, the application server process goes into a spin.

SCHEDULER DOES NOT ROLLOVER MSSQL DATABASE DUE TO DISK SPACE**Problem Number: 4843**

When the Microsoft SQL Server database server fails due to insufficient disk space, the scheduler continues executing queries against the database resulting in more query failures.

Web UI Fixed Issues

The following issues were fixed in Web UI in this release:

WCC SHOWS TRANSFER TYPE BINARY WHEN BACKEND IS SET TO ASCII**Problem Number: 454**

Transfer Type in WCC shows as binary when the job at the backend is set to ASCII.

AVOID UNNECESSARY JOBS DURING DELTA COLLECTION**Problem Number: 455**

After applying patches T6FU003 and T5RF018, WCC Monitoring Views is not refreshing in a timely manner. It takes a long time (around 1 minute) to display the updated job status.

SCP JOB TYPE DOES NOT HAVE OPTION TO SEE SPOOL FILE IN QV**Problem Number: 456**

SCP job type does not have an option to Retrieve spool files under the Job Runs section in Quick View.

HIGH LEVEL VULNERABILITIES IN WCC 11.4.SP7**Problem Number: 457**

Vulnerabilities reported on jackson-databind, spring-services, commons-fileupload, and xstream api.

ERRORS IN RESOURCES FIELD IF LAST JOB RUN MACHINE IS DELETED**Problem Number: 458**

If the last run machine of a job having a resource dependency is deleted, then we see errors in the Resources field while displaying job details in Quick View.

ALERT STATUS SUMMARY BOX TURNS RED WITHOUT ANY REAL ALERTS

Problem Number: 459

Alert summary box turns red in the Monitoring tab even when no new alerts are generated for a particular view.

SESSION TIMEOUT ISSUE IN REPORTING TAB

Problem Number: 460

When Auto Refresh checkbox is selected in the Reporting tab, the user session never gets timed out.

QUICK VIEW SHOW JIL DOES NOT DISPLAY ALARM_IF_TERMINATED

Problem Number: 461

Show JIL in Quick View does not display the alarm_if_terminated attribute when it is not set.

WCC - New High Vulnerabilities

Problem Number: 462

Vulnerabilities reported on jackson-databind-2.9.8.jar.

Third-Party Software Acknowledgements - 12.0

The AutoSys Workload Automation Platform links to [acknowledgements.txt](#) which contains the copyright and license agreements for third-party software that this AutoSys Workload Automation release uses.

AutoSys Known Issues

NOTE

- For information about agent limitations and known issues, see the Readme or [Release Information](#) for your agent.
- For AutoSys Workload Automation patch information, see [Broadcom support](#).

Contents

AutoSys Known Issues

This section contains information on the known issues in AutoSys.

Duplicate warning message

In the console mode, the AutoSys installer gives duplicate warning messages with an empty title. This is because of the InstallAnywhere issue. It does not impact the AutoSys functionality.

autorep -p

autorep, when used with the *-p* option displays incorrect Windows Operating System version. This behavior is due to the AdoptOpenJDK 8 shipped with the Workload Automation Agents. There is no issue when the agent uses Oracle's JDK or AdoptOpenJDK 11.

Using Microsoft Edge browser to upload agent installers

When using Microsoft Edge browser to upload agent installers, the Upload button is not enabled in Web UI Agent Inventory.

Application Server Stops Responding (UNIX Only)

If you installed AutoSys Workload Automation on a UNIX computer, the application server may stop responding if the `/etc/hosts` file contains the `::1` entry (IPv6 loop back entry) that includes the host name. To resolve this issue, remove the host name from the `::1` entry in the `/etc/hosts` file.

Defined Aliases (UNIX Only)

Defined aliases that change the native behavior of operating system commands can cause installation problems. Remove all defined aliases before installing AutoSys Workload Automation.

CAPKI in FIPS Mode (Linux Only)

If you enable Security-Enhanced Linux (SELinux) feature on your computer, CAPKI cannot run in FIPS mode. To enable CAPKI to run in FIPS mode, you must disable the SELinux feature.

Datamover Fails to Validate Oracle Real Application Cluster (RAC) Database (Oracle Only)

When you migrate data to an Oracle RAC database, the datamover utility fails to validate the database. This happens because you cannot connect to the Oracle RAC database using the JDBC thin client drivers.

As a workaround, you must run the datamover utility on individual nodes of the Oracle RAC database.

archive_jobs Command Does Not Delete All Logically Deleted Jobs

The `archive_jobs` command does not remove all jobs marked for deletion that meet the obsolete criteria as expected. In addition, the optional audit report produced using the `-A` option in the command may contain incomplete information or the message “CAUAJM_E_10257 Potential corruption of job data for joid <nnn>”. The database is not corrupted; the CAUAJM_E_10257 message is a symptom of the jobs that were not completely deleted.

Application Server is Not Able to Connect to the Second CA EEM Host

If you configure AutoSys Workload Automation to work with CA EEM that is running in high availability mode, the client utilities like `forecast`, `autoaggr`, or `DBMaint` fail if the `itechpoz-router` service on the primary CA EEM host is down and an error message similar to the following is displayed:

```
log4cxx: No appender could be found for logger (PozFactory).
log4cxx: Please initialize the log4cxx system properly.
CAUAJM_E_10436 Security server unreachable or invalid authentication certificate file.
CAUAJM_E_10434 Error initiating security session.
CAUAJM_E_10437 Detailed Error Information:
[EE_NOTALLOWED Operation not allowed]
[attach to application instance not allowed]
CAUAJM_E_10433 Could not obtain security server from the database.
CAUAJM_E_10434 Error initiating security session.
```

This happens because CA EEM does not fail over to the second CA EEM host and the application server is not able to connect to the second CA EEM host.

As a workaround, you can do *one* of the following:

- Allow CA EEM failover to take place by either shutting down the primary CA EEM host (that is, the iGateway service is shut down) or taking the primary CA EEM host completely offline.
- Restart the primary CA EEM host.

Client Utilities Time Out When External Security is Enabled

When external security is enabled, client utilities (such as autorep) can time out. These timeouts happen when there are a large number of jobs to process and restrictive policies are specified in the as-job resource class.

Owner Attribute and user host Value Must Match

The owner attribute and the user ID (user@host) defined in the database using autosys_secure are case-sensitive. When specifying the owner attribute in a job definition, the owner value must match the user@host value exactly, including case. Otherwise, the password associated with the owner cannot be retrieved.

Running SQL Transaction Cannot be Cancelled

If the transaction submitted by an SQL job is already running on the database server, the sendevent -E KILLJOB command does not cancel the transaction. Workload Automation Agent for UNIX, Linux, or Windows sends a confirmation to AutoSys Workload Automation that the cancel request was submitted to the database server, but the transaction continues to run. The job goes to a SUCCESS state when the transaction is complete.

sap_job_count Attribute

If you run a job without specifying the sap_job_count attribute on an r11.3 SAP agent plug-in, the job fails. This happens because the sap_job_count attribute is required when you run a SAP Job Copy job on an agent with an r11.3 SAP agent plug-in.

Search in Web UI Fails and Returns Database Errors

If you search non-object attributes (for example, job description) in Web UI using the exclamation point character (!), the search fails and the AutoSys Workload Automation API returns database errors.

Update Job Fails with Multiple Machines

If you specify multiple machines in a job definition by separating each machine with a comma (,), the jil command fails to update the job with resource dependencies.

(Linux Only) Oracle 12c Client Memory Leak

Oracle 12c client libraries have a known memory leak when the glibc version on the machine is below 2.13. The root cause of the memory leak is due to a bug in GNU glibc that is related to a multi-thread program that uses dlopen() to load the libraries.

To resolve this issue, apply glibc 2.14 or higher. Contact your Linux vendor to get the GNU glibc compatible patch for bug [12650](#) based on your Linux operating environment.

Machine Status Displayed Incorrectly

When a critical disk threshold breach occurs, the agent shuts itself down. If you try to restart the agent without freeing up enough disk space, the machine status may not match with the actual status of the agent.

autorep Machine Report Displays the Current Load Incorrectly

When you define multiple virtual machines that specify the same machine and their own max load values and then run jobs with loads on those machines, the `autorep -m ALL -d` command displays the current load on the machine incorrectly. The detailed machine report might show a load value that exceeds the maximum load that is defined on the machine and this gives a false impression that the scheduler is not honoring the load balancing attributes of jobs and machines.

False KILLJOBFAIL Alarms on UNIX

When you send a comma separated list of kill signals to a job, the agent successfully sends the first signal and this terminates the process. As the process no longer exists, the second signal fails and the agent notifies the scheduler of the failure. As a result, the scheduler generates the KILLJOBFAIL alarm. The scheduler might receive a failure notification for each additional signal that follows. Shortly after that, the agent informs the scheduler that the job was successfully terminated and so the previous failure notifications were false. The false KILLJOBFAIL alarms might be confusing as the job process was successfully killed.

Web Services Jobs Fail

If Workload Automation Agent for Web Services is installed on your computer and you are upgrading Workload Automation Agent for UNIX, Linux, or Windows to 11.3 SP5 or 11.3 SP6 or CA Workload Automation AE to Release 11.3.6 SP5, the web services jobs fail. For the web services jobs to run, rerun the Workload Automation Agent for Web Services installer immediately after the upgrade.

Web UI Known Issues

This section contains information on the known issues in Web UI.

Staging Agent Fails on Windows

Staging the agent fails on Windows when AutoSys, Web UI, CA EEM, and WLA Agent are on the same computer or the Windows environment variables size exceeds 16KB.

The agent transmitter.log displays the following error:

**State SUBERROR Failed SetEnd Status("The environment is incorrect.") Cmpc(10) User(AGENT_01)
Host(Ivntest015869)**

As a workaround, before staging the agent reduce the size of the Windows environment variables and after the agent is staged restore the environment variables.

HTTP Job UI

In the HTTP Job UI, if the job type is POST or PUT and "Payload", "Request parameters" are specified, the request parameters are ignored while executing the REST API. As a workaround, specify the request parameters as part of the `provider_url` in the `name=value` pair format. For example, `http://host:port/abc/xyz?name1=value1&name2=value2`.

Renaming Job in Web UI

When a job is renamed in Web UI, all the dependent jobs in the current instance are displayed. If dependent jobs also exist in other instances, the other instance name along with only the first dependent job is displayed, whereas we should be displaying only instance names to be consistent with the CLI.

AutoSys Web UI Startup

During the AutoSys Web UI startup, the logs will contain a few generic warnings. And a few warnings similar to the ones mentioned below. You can ignore such warnings as they have no impact on the functionality.

WARNING: INSERT INTO CFG_GLOBAL_PREF (PREFKEY, PREFVALUE) VALUES (?, ?)

In a (Remote) Web UI Collector Environment, wcc_config Utility Fails to Execute

In an AutoSys Web UI environment, where only the **(Remote)** Web UI Collector(s) is/are present but **not** the Web UI component and the wcc_config script is executed, it fails with following error in the console:

E150012 - Unexpected error occurred during processing the command.

Additionally, the following error message is logged in the **<CA_WCC_INSTALL_LOCATION>/log/application/cli/wcc_config.log** file:

root Could not load properties; nested exception is java.io.FileNotFoundException: /opt/CA/WorkloadAutomationAE/wcc/data/config/application/config/resources/internal-componentURLs.properties (No such file or directory)

When a user tries to execute commands **other than** the following through the wcc_config script on a (Remote) Web UI Collector, then the script tries to execute the command by **pointing to a Web UI configuration location which does not exist**.

- Specify a Different Collector Database
- Specify a Different Collector Database for Reporting
- Change the Port for the Collector

Application Editor Performance Overhead When Importing from Scheduler or Committing to Scheduler

In the Application Editor, when you use either the Import from Scheduler or Commit to Scheduler option under the Select an action, it can result in performance overhead.

Agent Inventory Issues

- The Internet Explorer browser displays the following message when you log in to the application and click the Agent Inventory tab:
The application could not be loaded inside Web UI Tab.
Please *click here* to open in new window.
Click the link and Agent Inventory is available for use.
- In the Internet Explorer browser, the Show drop-down at the bottom of the page is empty and does not display the default for the Selected agents count.

Messages are displayed during Console mode installation in InstallAnywhere 2017

When installing Web UI in console mode using InstallAnywhere 2017, 'Please wait...' messages may be displayed. You can ignore these messages.

Browser Issues

- Graphical representation of reports do not display on Microsoft Edge browser version 38.14393
Note: You can view the graphical representation of reports on previous versions of Microsoft Edge browser.
- In Informatica job type, the Object Hierarchy button does not display in the Internet Explorer browser.

Right-clicking to Open Features

CA WCC Release 11.4 does not support the browser feature that lets you right-click an object or tab and open it in another window.

CA EEM Performance Issue on Linux

There can be a CA EEM performance issue when the kernel threading model version used by CA EEM and the Linux kernel version differ. The default kernel threading model used by CA EEM is based on the 2.4 kernel. The Linux kernel version can be either 2.4 or 2.6. To display the kernel version, run the following command:

```
uname -r
```

The first two digits in the returned text indicate the Linux kernel version. Examples are as follows:

- 2.4.19-4GB -- indicates Linux kernel version 2.4.
- 2.6.5-4GB -- indicates Linux kernel version 2.6.

To select the appropriate kernel threading model

1. Stop iGateway.
2. Do one of the following:
 - Red Hat**
Open /etc/init.d/igatewayd.
 - SuSE**
Open /etc/init.d/S99igateway.
3. Do one of the following:
 - If the Linux kernel version is 2.4, verify that the file contains the line LD_ASSUME_KERNEL=2.4.1 and then close the file.
 - If the Linux kernel version is 2.6, comment out the line LD_ASSUME_KERNEL=2.4.1 and then save and close the file.
4. Open \$IGW_LOC/S99igateway, and do one of the following:
 - If the Linux kernel version is 2.4, verify that the file contains the line LD_ASSUME_KERNEL=2.4.1 and then close the file.
 - If the Linux kernel version is 2.6, comment out the line LD_ASSUME_KERNEL=2.4.1, and then save and close the file.
5. Restart iGateway.

Providing Incorrect Password For EiamAdmin User in CA EEM Registration Panel of Web UI Installation

If you provide an incorrect password for the EiamAdmin user in the CA EEM registration panel during the Web UI installation, and multiple CA EEM servers are configured in fail-over mode, the EiamAdmin user is locked automatically. Correct the password and wait five minutes. After five minutes, the EiamAdmin user is automatically unlocked, and you can continue the Web UI installation.

Reports Do Not Include Job Events Until After The First Run

Events from the AutoSys Workload Automation server are not counted in Job Statistics and Job Run Event Summary reports until a job has run at least one time. For example, if a job that has not run at least one time is issued a force start, the Job Statistics Report indicates 0 in the Force Start column. Similarly, if a job that has not run at least one time is issued Job On Ice event, the event is not counted in the Total Events column of the Job Run Event Summary report.

Session Data May Be Lost When a Web UI Instance in an HA Group Fails

Failure of a Web UI instance in a High Availability group may result in data loss for users. When a Web UI instance fails, users' sessions end and users are notified. Users must then log back into Web UI using the URL for Apache HTTPD. Because users will be assigned a different Web UI instance, unsaved data from the previous session may be lost.

Configuration Migration Update Error in the Validation Thread Pool Minimum Size

When migrating from CA WCC r11.3 to CA WCC r11.3.5 (and later service packs), the Validation Thread Pool Minimum Size field fails to update. To work around this error, change the property name in the file that contains the exported data from CA WCC r11.3:

```
config_env_validation_thread_pool_core_size
to
config_env_cred_validation_thread_pool_core_size
```

The import works properly.

Web UI Fails to Respond if Oracle Database is Restarted

When an Oracle database used with CA WCC is restarted while CA WCC is running, CA WCC will no longer respond. To resolve this issue, restart CA WCC services. For information about restarting services, see [Implementation](#).

Encryption Performance Tuning of Query Parameters

When installing Web UI on UNIX or UNIX virtualized machines, you may observe slower performance of Quick Edit, Quick View, Resource, and Monitoring Flow applications. The performance issue is a result of the low entropy needed for random number generation used for internal Web UI encryption procedures. The workaround for this is to uncomment the line:

```
#wrapper.java.additional.101=-Djava.security.egd=file:/dev/./urandom
```

In both the WCC_Installation_Root files found at:

```
WCC_INSTALLATION_ROOT/tomcat/conf/wrapper.conf
```

and

```
WCC_INSTALLATION_ROOT/tomcat_32/conf/wrapper.conf
```

NOTE

The workaround improves performance but it leads to a generation of less quality random numbers that are used as an input to cryptography algorithms performed in Web UI.

Multiple \$ Characters in Text Fields

The installer interprets the text between dollar sign characters (\$) as an incorrect path. When this situation occurs, installer operations fail. Ensure that all installer fields, EEM passwords, and database passwords contain a maximum of one dollar sign character (\$).

NOTE

If an existing password for (EEM or database) contains two or more \$ characters, a password change is required prior to Web UI installation.

% Character in Install Path

The installer of AutoSys SDK fails if install patch is containing "%" character

Compatibility View Mode in Internet Explorer

Some Web UI features may not display correctly or work properly in Internet Explorer (IE) browsers. Ensure that the Internet Explorer compatibility view mode is not activated for any Web UI pages.

Defining AutoSys instance with different than default (9000) port when Web UI is running in HA

(Only when Web UI is running in HA environment). When AutoSys instance is being added or modified in Web UI Configuration and the port of AutoSys instance is different than default (9000), the manual step of running csamconfigedit

command on the other nodes is required. For details, see Server Properties and Application Preferences topic in Installing - GUI Implementation.

Web UI Upgrade Does Not Detect Correct Installation Directory

When upgrading Web UI from a WCC Release 11.4.X that is installed without any other AutoSys component in a custom path, the upgrade does not correctly detect the install base directory. It takes the default path as the installation location. To resolve this issue, see the [Knowledge Base article 212314](#).

Other Known Issues

This section contains information on the known issues other than AutoSys and Web UI.

XML Compliant Audit Logs

If you are using a CA EEM version older than CA EEM Release 12.51.2.11 CR02, the Service fails sometimes when importing data to the database. Previous versions of CA EEM may record data in audit logs in a way that causes the import to fail when reading the data. If bad data is detected, the import process is stopped and the log file is updated with the failure information.

NOTE

This does not cause the Service to stop; other operations continue as normal.

AutoSys Workload Automation Operating System and Database Support

Broadcom supports the following databases and operating systems during their life cycle. This is as determined by the manufacturer of the operating system, or until Broadcom announces that we are ending support. The supported operating systems are subject to change. For more information about compatibility and to see the latest operating systems that are supported, see the [Broadcom Support website](#).

NOTE

The UNIX instructions in this article also apply to Linux systems unless otherwise noted. The term Windows refers to any Microsoft Windows operating system supported by AutoSys Workload Automation unless otherwise noted.

Support on Cloud

CA Workload Automation AE Release 11.3.6 and later and CA WCC Release 11.4 and later (including the agents) are supported on the following cloud platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

NOTE

The following database limitation applies to CA Workload Automation AE Release 11.3.6 to 11.3.6 SP6 and CA WCC Release 11.4 to 11.4 SP5:

- Amazon RDS and Microsoft Azure PaaS databases are not supported.

Consider the following points when hosting AutoSys Workload Automation or Web UI on the cloud:

- The AutoSys Workload Automation database should be hosted within the same environment as the AutoSys Workload Automation server for optimal performance. For example, hosting the database and the AutoSys Workload Automation application servers on premise and the AutoSys Workload Automation scheduler on the cloud will most likely significantly increase scheduler latency and general response times.
- Enable the [ports](#) that are related to AutoSys Workload Automation as well as traffic in the respective network.
- In Amazon Web Services (AWS) environment, ensure that the IP address is resolvable across the respective domains to enable proper communication between the Broadcom products. For example, you can resolve the IP address by updating the host file, providing the DNS server details, or by navigating to EC2 configService settings > EC2 Service properties and selecting the Set Computer Name option.
- We recommend that you host both AutoSys Workload Automation and Web UI on the cloud.

AutoSys Operating System and Database Support

Database Support

AutoSys Workload Automation supports the following databases:

Database	Supported Versions	Architecture	Notes
Microsoft SQL Server	2012	64-bit	Bundled Event Manager is supported on Microsoft SQL Server 2012 Enterprise and Standard Editions. Microsoft SQL Server “Always On” feature is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2014	64-bit	Bundled Event Manager is not supported. Microsoft SQL Server “Always On” feature is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2016	64-bit	Bundled Event Manager is not supported. Microsoft SQL Server “Always On” feature is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2019	64-bit	Microsoft SQL Server is supported on Windows and Linux operating systems. Bundled Event Manager is not supported. Microsoft SQL Server “Always On” feature is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Azure	—	—	Microsoft SQL Server 2012 or higher client is supported.

Oracle	12c	64-bit RAC	Oracle RAC is supported. Oracle Container database is supported. The AutoSys Workload Automation schema must be created only on a pluggable database. Do not use CDB to create the schema and the same is recommended by Oracle. Setting this up is outside the scope of Broadcom Support. For further information, consult Oracle documentation or Oracle Support. (Linux only) Oracle 12c client libraries have a known AutoSys Known Issues when the glibc version on the machine is below 2.14. The root cause of the memory leak is due to a bug in GNU glibc that is related to a multi-thread program that uses dlopen() to load the libraries. To resolve this issue, apply glibc 2.14 or higher. Contact your Linux vendor to get the GNU glibc compatible patch for bug 12650 based on your Linux operating environment. Amazon Relational Database Service (Amazon RDS) is supported.
Oracle	18c	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Oracle	19c	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Sybase ASE	15.7	64-bit	The minimum page size must be at least 4 KB. If you are running CA WA Advanced Integration for Hadoop jobs, the minimum page size must be at least 8 KB. Up to 137 free Sybase user connections are required, depending on which AutoSys Workload Automation components you install. Sybase jConnect 7 is required. For more information about installing Sybase jConnect 7, contact your database vendor.
Sybase	16.0	64-bit	Sybase 16.0 or higher is supported. The minimum page size must be at least 4 KB. If you are running CA WA Advanced Integration for Hadoop jobs, the minimum page size must be at least 8 KB. Up to 137 free Sybase user connections are required, depending on which AutoSys Workload Automation components you install. Sybase jConnect 7 is required. For more information about installing Sybase jConnect 7, contact your database vendor.

Server Supported Versions

UNIX

The AutoSys Workload Automation server component supports the following UNIX versions:

Operating Environment	Supported Versions	Architecture	Notes
AIX	7.1	64-bit	In AIX, to see what level your system is at, run: xlC.aix61.rte 13.1.3.1 COMMITTED IBM XL C++ Runtime for AIX 6.1 or later The operating system must be at TL5.
AIX	7.2	64-bit	The AIX C++ Runtime Environment must be at level 11.1.0.0 or higher. To see what level your system is at, run: lspp -l xlC.aix61.rte

Linux

The AutoSys Workload Automation server component supports the following Linux versions:

Operating Environment	Supported Versions	Architecture	Notes
Oracle Enterprise Linux	6	64bit	
Oracle Enterprise Linux	7	64-bit	
Oracle Enterprise Linux	8	64-bit	
Red Hat Enterprise Linux	6	64-bit	
Red Hat Enterprise Linux	7	64-bit	
Red Hat Enterprise Linux	8	64-bit	
SuSE Linux Enterprise Server	12	64-bit	
SuSE Linux Enterprise Server	15	64-bit	
Amazon Linux	2017 and 2018	64-bit	
Amazon Linux	2	64-bit	
Red Hat Cluster Suite			

NOTE

- SUSE Linux Enterprise Server 12 and SP1 are only supported.
- For Oracle Enterprise Linux and Red Hat Enterprise Linux, both UEK and RedHat Kernels are supported.

Windows

The AutoSys Workload Automation server component supports the following Windows versions:

Version	Architecture	Notes
Microsoft Windows 2012	64-bit	
Microsoft Windows 2016	64-bit	
Microsoft Windows 2019	64-bit	
Microsoft Cluster Server		

NOTE

Windows R2 and SQL server R2 releases are supported unless noted otherwise.

Client and SDK Supported Versions**UNIX**

The AutoSys Workload Automation client and SDK components support the following UNIX versions:

Operating Environment	Supported Versions	Architecture	Notes
AIX	7.1	64-bit	The AIX C++ Runtime Environment must be at level 11.1.0.0 or higher. To see what level your system is at, run: lspp -l xIC.aix61.rte
AIX	7.2	64-bit	The AIX C++ Runtime Environment must be at level 11.1.0.0 or higher. To see what level your system is at, run: lspp -l xIC.aix61.rte
HP-UX	11i v3 (11.31)	IA 64-bit (Itanium)	

Linux

The AutoSys Workload Automation client and SDK components support the following Linux versions:

Operating Environment	Supported Versions	Architecture	Notes
Oracle Enterprise Linux	6	64bit	
Oracle Enterprise Linux	7	64-bit	
Oracle Enterprise Linux	8	64-bit	
Red Hat Enterprise Linux	6	64-bit	
Red Hat Enterprise Linux	7	64-bit	
Red Hat Enterprise Linux	8	64-bit	
SuSE Linux Enterprise Server	12	64-bit	
SuSE Linux Enterprise Server	15	64-bit	
Amazon Linux	2017 and 2018	64-bit	
Amazon Linux	2	64-bit	

NOTE

- SUSE Linux Enterprise Server 12 and SP1 are only supported.
- For Oracle Enterprise Linux and Red Hat Enterprise Linux, both UEK and RedHat Kernels are supported.

Windows

The AutoSys Workload Automation client and SDK components support the following Windows versions:

Version	Architecture	Notes
Microsoft Windows 10	64-bit	
Microsoft Windows 2012	64-bit	
Microsoft Windows 2016	64-bit	

Microsoft Windows 2019	64-bit	
------------------------	--------	--

NOTE

Windows R2 and SQL server R2 releases are supported unless noted otherwise.

Agent Supported Versions

AutoSys Workload Automation installs Workload Automation Agent for UNIX, Linux, or Windows. For information about the platforms that are supported by the agent, see [Workload Automation System Agent Platform Support Matrix](#).

For more information on the agent or configuring the agent to behave like the legacy agent, see the [Workload Automation Agents](#) documentation.

Patch 979306 -- Time Zone Update for Windows

For AutoSys Workload Automation to work correctly with daylight saving time (DST), install patch 979306 on the AutoSys Workload Automation client computers.

You can download the patch from <http://support.microsoft.com/kb/955839>.

For more information about configuring DST for Windows, see <http://support.microsoft.com/kb/914387>.

NOTE

The patch number is 979306 at the time of General Availability (GA). This patch may be superseded post GA. Ensure that you install the latest patch. For current information about the time zone patches, check the Microsoft Support web site.

Web UI Operating System and Database Support**Operating Systems**

This section contains the supported operating systems.

An unzip application must be installed on the Web UI server.

Linux

Web UI server supports the following Linux versions:

Operating Environment	Supported Versions	Architecture	Notes
Red Hat Enterprise Linux	5	64-bit	
Red Hat Enterprise Linux	6	64-bit	
Red Hat Enterprise Linux	7.2	64-bit	
Red Hat Enterprise Linux	8	64-bit	
SuSE Linux Enterprise Server	11	64-bit	
SuSE Linux Enterprise Server	12	64-bit	
SuSE Linux Enterprise Server	15	64-bit	
Amazon Linux	2017 and 2018	64-bit	
Amazon Linux	2	64-bit	

NOTE

- For Red Hat, the required libraries can be installed by electing to install the Legacy Software Development Package using the Add/Remove Packages application. For SuSE, the required libraries can be installed as needed from their respective RPM Package Managers using YaST.
- RHEL Versions 7.2 and higher are supported. RHEL 7.0 and 7.1 are not supported.

For information about changes required to kernel parameters on Linux, see the Common Components section in the Installing documentation.

Windows

Web UI server supports the following Windows versions:

Version	Architecture	Notes
Microsoft Windows 2012	x86 64-bit	
Microsoft Windows 2012 R2	x86 64-bit	
Microsoft Windows 2016	x86 64-bit	
Microsoft Windows 2019	x86 64-bit	

NOTE

- You must install Web UI on a supported 64-bit server. If you attempt to install Web UI on a 32-bit server, the following error message appears:
Windows error 216 occurred while loading the Java VM
- The cumulative time zone update for Microsoft Windows operating systems may be required. For information, go to <http://microsoft.com>.

Database Support

Web UI supports the following databases:

Database	Supported Versions	Architecture	Notes
Microsoft SQL Server	2008	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2008 R2	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2012	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2014	64-bit	Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2016	64-bit	MS SQL Server <i>Always On</i> is supported. MS SQL Server <i>Encryption</i> is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Microsoft SQL Server	2019	64-bit	Microsoft SQL Server is supported on Windows and Linux operating systems.
Microsoft SQL Azure	-	-	Microsoft SQL Server 2012 or higher client is supported.
Oracle	11g Release 2 (11.2.0.3.0) and later versions	64-bit	RAC is required for Web UI High Availability. Amazon Relational Database Service (Amazon RDS) is supported.

Oracle	12c (12.1.0.1.0) and later versions	64-bit	RAC is required for Web UI High Availability. Container Database is supported. Amazon Relational Database Service (Amazon RDS) is supported.
Oracle	18c		Amazon Relational Database Service (Amazon RDS) is supported.
Oracle	19c		Amazon Relational Database Service (Amazon RDS) is supported.
Derby	(10.14.2.0)		Installed with Web UI.

Supported Operating Systems for Agents in Agent Inventory

The following are the operating systems that are supported for downloading agent logs and agent parameters. You can also update agent parameters, stop and restart agents.

- AIX
- HP PA RISC
- HP IA
- Linux
- Microsoft Windows
- Solaris SPARC
- Solaris x86

NOTE

You can view agent details in the Agent List table which displays all the agents running on agent supported platforms.

The following operating systems are not supported and an error message displays if you perform any action on agents installed on these operating systems :

- HP Non-Stop (Tandem)
- OpenVMS
- z/OS

NOTE

For more information about the operating systems support, see the [Workload Automation System Agent Platform Support Matrix](#) documentation.

The following requirement is for the support of the Agent Inventory actions for the Windows operating system:

- When the agent is installed for AutoSys Workload Automation manager, the property in agentparm.txt file, oscomponent.lookupcommand, is set to true by default. When it is set to false, the Workload Automation Agent for Windows is unable to locate the file that must be executed, and all agent inventory actions will fail.

When the requirements are not supported, the following error messages are issued:

- ERROR: E140179 Unexpected error occurred while performing the agent action. File is not available
- ERROR: E140178 Unexpected error occurred while executing agent command. Communication attempt with the CA WAAE agent has failed!
- ERROR: E140179 Unexpected error occurred while downloading agent file. Refused by agent security

AutoSys Workload Automation System Requirements

This article documents the system requirements for AutoSys Workload Automation.

AutoSys System Requirements

The following are the system requirements for AutoSys:

System Requirements for UNIX and Linux

NOTE

As the number of users and the workload increases, you may see a performance benefit from adding additional memory and/or additional processing power. Additional processing power can be added either by using a faster CPU or by adding additional CPUs.

To install and run the AutoSys Workload Automation server in a UNIX or Linux environment, the minimum recommended requirements are as follows:

- Processor: 1 GHz, 2CPU
- Physical memory: 4 GB RAM
- Swap space: 2 GB
- Hard disk free space: 5 GB
- fsize: The file size ulimit must be set to unlimited. To determine the current setting, issue the following command:

```
ulimit -f
```

If the value is not unlimited, change this ulimit value by issuing the following command:

```
ulimit -f unlimited
```

- Disk space requirements for AutoSys Workload Automation:
 - Full product: 700 MB
 - Scheduler, Client, Agent: 430 MB
 - Application Server, Web Server: 360 MB
 - Application Server only: 300 MB
 - Agent only: 420 MB
 - Client only: 300 MB
 - SDK only: 250 MB
 - Documentation only: 250 MB

NOTE

- The system temporary directory (/tmp or \$TMPDIR) requires a minimum of 400 MB disk space for the installation.
- The installer creates a directory to store control files. By default, the /opt/CA/installer directory is created. The file system that the installer directory belongs to requires a minimum of 10 MB disk space. For more information about the /opt/CA/installer directory, see [\(UNIX\) Changes to System Files and Directories](#).
- SDK runtime environment: 250 MB
- Disk space for the database (to support default installation values):
 - Oracle: Data—800 MB, Index—80 MB
 - Sybase: Data—800 MB, Log—100 MB, tempdb—500 MB, tempdb log—50 MB

NOTE

For more information about the database requirements, see the appropriate database documentation or consult your database administrator. For information about tuning the database, see [Maintain the Event Server](#).

System Requirements for Windows

NOTE

As the number of users and the workload increases, you may see a performance benefit from adding additional memory and/or additional processing power. Additional processing power can be added either by using a faster CPU or by adding additional CPUs.

To install and run the AutoSys Workload Automation server in a Windows environment, the minimum recommended requirements are as follows:

- Processor: 2 GHz, 2CPU
- Physical memory: 4GB RAM
- Swap file: 2GB
- Hard disk free space: 5 GB
- Disk space requirements for AutoSys Workload Automation:
 - Full product: 630 MB
 - Scheduler, Client, Agent: 310 MB
 - Application Server, Web Server: 350 MB
 - Application Server only: 180 MB
 - Agent only: 290 MB
 - Client only: 180 MB
 - SDK only: 40 MB
 - Documentation only: 40 MB
- SDK runtime environment: 50 MB
- Disk space requirements for the database (to support default installation values):
 - Microsoft SQL Server: 40 MB plus 1 MB for log file

NOTE

SQL Server adjusts the sizes incrementally over time based on usage.

- Oracle: Data—800 MB, Index—80 MB
- Sybase: Data—800 MB, Log—100 MB, tempdb—500 MB, tempdb log—50 MB

NOTE

For more information about the database requirements, see the appropriate database documentation or consult your database administrator. For information about tuning the database, see [Maintain the Event Server](#).

Java Runtime Environment (JRE) and Adopt OpenJDK Supported Versions

This section contains JRE support information for AutoSys Workload Automation that includes Web UI.

The AdoptOpenJDK 8.x is distributed with the installer. The other supported JREs include the OpenJDK based JRE (such as Zulu) and licensed Oracle JRE (Java 8 64-bit JRE).

The minimum supported JRE versions are as follows:

Operating Environment	Supported JRE Version
Linux	1.8.0_252-b09
Windows	1.8.0_252-b09
AIX	1.8.0_600
HP-UX	1.8.0_19

NOTE

- All minor versions of the above specific major JRE versions are supported.
- The supported JRE for all AutoSys Workload Automation components is 64-bit.
- On HP-UX Itanium (IA-64), use the java from IA64W that is inside the /opt/CA/WorkloadAutomationAE/JRE64_WA/bin/IA64W/java directory.
- Java 8 does not support X.509 certificates signed with MD5withRSA algorithm and are no longer acceptable by default. If you are using MD5withRSA algorithm to create the certificates, you cannot use the AutoSys Workload Automation web services and Compliance Application with Oracle PKI certificates. For information about how to resolve this problem, see the Java documentation.

TCP/IP Port Usage

AutoSys Workload Automation uses SSA for communication, which uses IANA assigned port 7163. SSA is automatically installed with AutoSys Workload Automation.

By default, the AutoSys Workload Automation communication with the agent is set up to use plain socket communication. The agent's default port number is 7520.

su Command on UNIX

On UNIX, the AutoSys Workload Automation installer must be allowed to run the following command with no user response or interaction:

```
su - root - c "command"
```

The installer embeds other installation packages that are executed by running the su command.

Graphical Mode Requirements for UNIX and Linux

Before you install AutoSys Workload Automation on UNIX or Linux, verify the following requirements are met so that installation dialogs can appear in graphical mode:

- Java 1.3 or higher is installed, the PATH environment variable is set to include the Java executable, and the JAVA_HOME environment variable is set.
- The monitor is graphics-enabled.

Base Application Development Math Library Package for AIX

Before you install AutoSys Workload Automation on AIX, verify the system has the Base Application Development Math Library package (bos.adt.libm). The AutoSys Workload Automation application server and the scheduler installations require this package.

User Resource Limits for AIX 64-bit

To install and run AutoSys Workload Automation on AIX, verify the following variable=value pairs exist in /etc/security/limits for a 64-bit installation:

```
default:
fsize = -1
core = 2097151
cpu = -1
data = -1
rss = -1
stack = 262144
```

```
nfiles = 4096
```

You must restart the computer for the changes to take effect and before starting the installation.

Perl

On UNIX or Linux installations, Perl, version 5.8 or greater must be installed. Perl must also be included in the PATH environment variable to run the AutoSys Workload Automation installation.

Web UI System Requirements

The following are the system requirements for Web UI:

System requirements are provided for the servers in the Web UI environment.

NOTE

- For AutoSys Workload Automation system requirements, see AutoSys [System Requirements](#).
- To check the version of the components and other environmental information after Web UI is installed, click About, System Information.

Web UI Server

The main Web UI product is installed on the Web UI server.

As the number of users and the number of jobs monitored by Web UI increases, you may see a performance benefit from adding additional memory and/or additional processing power. Additional processing power can be added either by using a faster CPU or by adding additional CPUs.

Hardware Requirements

Depending on the operating system, specific Web UI server hardware requirements must be met or exceeded for Web UI to install and run correctly.

WARNING

The following requirements apply to installing Web UI on the Web UI server only. Installing CA EEM on the Web UI server is not recommended; however, if you plan to install other components such as CA EEM or AutoSys Workload Automation on the Web UI server, the RAM and hard disk space requirements for those components must be added to the RAM and hard disk space requirements listed for Web UI. Additionally, the fastest processor recommended for any single component should be used.

NOTE

The requirements that follow are for a typical enterprise with up to two AutoSys Workload Automation instances and up to 15,000 jobs in the database of each AutoSys Workload Automation instance.

The requirements for a Windows or Linux x86 computer are as follows:

- 2 GHz or higher processor recommended.
- Two CPUs minimum, four CPUs recommended for large enterprises.
- At least 6 GB of RAM; 8 GB recommended.
- 2 GB of hard disk space for the installation, including Web UI and the required common components.

NOTE

On UNIX and Linux, the /tmp directory should have a minimum of 500 MB of space available. On Windows, %TEMP% and %TMP% must point to a directory with a minimum of 500 MB of space available. Installer has a pre-check for temp space, but the minimum requirement is 500 MB; depending on the selected components, it may be increased.

Software Requirements

The following software requirements must be met for Web UI to run correctly:

- BSAFE Crypto J is replaced with Bouncy Castle from WCC Release 11.4 SP07.
Before you install or upgrade Web UI, you must modify the CA EEM server certificate as follows:
 - Upgrade CA EEM.
 - On the CA EEM server, set the KeyLength to 2048.

NOTE

For more information about the KeyLength, see the following links:

To generate certificate with the Keylength as 2048 - [How to Generate the Certificates](#)

To generate certificate with the Keylength as 2048 in case of CA EEM Failover - [Generate Certificates with Custom Key Length for CA EEM Servers in Failover or Cluster Environment](#)

- Perform the Web UI installation or upgrade.
- Adobe Flash Player version 8 or higher is required for Reporting. From Version 12.0, Monitoring summary does not use Adobe Flash Player.

NOTE

- Web UI can communicate with CA EEM running in FIPS or non-FIPS mode.
- CA WCC 11.4 SP6 supports all common components that are installed using the CA Common Components DVD that is distributed with the CA Workload Automation AE Release 11.3.6 SP7 media. For information about the common components, see [Common Components Release Notes](#).
- On Linux with kernel version 2.4, CA EEM and other common components cannot be upgraded. In this case, CA EEM and other common components must be installed on a clean computer with Linux kernel version 2.6 and above. CA EEM policies can then be migrated to the new CA EEM installation.

The following required software is automatically installed with Web UI:

- Apache™ Tomcat 9.0.33

Database Requirements

The Configuration, Forecast, and Monitoring applications require a database, known as the Web UI database. The Reporting application also requires a database, known as the Reporting database. An internal Derby database is provided that can be used by all Web UI applications. Otherwise, you can use a supported database that Web UI supports.

Web UI Client

You can access Web UI remotely.

NOTE

Access to Web UI is available only to users created in or referenced by CA EEM.

Hardware Requirements

The following client hardware requirements must be met or exceeded for Web UI to run correctly on a Windows x86 computer:

- 1 GB or more recommended.
- 1 GHz processor; 2 GHz or more recommended.
- Minimum screen resolution of 1024 x 768.

Software Requirements

Web UI is certified on the following web browsers and versions:

- Google Chrome 53, 57, 61, and 64
- Mozilla Firefox 49, 53, and 58
- Safari 10.0 10.1, and 11
- Microsoft Edge 38.14393.1066.0
- Microsoft Internet Explorer 11.0

NOTE

We recommend Google Chrome, Mozilla Firefox, and Safari browsers for better results.

In Safari, you must turn off the pop-up blocking function in Safari > Preferences > Security > Web content - uncheck "Block pop-up window" For more information on Browser Issues, see [Known Issues](#).

NOTE

TLS 1.2 secure connection protocol needs to be enabled in the web browser.

AutoSys Workload Automation Servers

Web UI Version 12.0 supports CA Workload Automation AE Release 11.3.5, Release 11.3.6, Release 11.3.6 SP1, Release 11.3.6 SP2, Release 11.3.6 SP3, Release 11.3.6 SP4, Release 11.3.6 SP5, Release 11.3.6 SP6, Release 11.3.6 SP7, Release 11.3.6 SP8, and AutoSys Workload Automation Version 12.0 servers.

NOTE

Certain job types are supported only by CA Workload Automation AE Release 11.3.5, Release 11.3.6, Release 11.3.6 SP2, Release 11.3.6 SP3, Release 11.3.6 SP4, Release 11.3.6 SP5, Release 11.3.6 SP6, Release 11.3.6 SP7, and AutoSys Workload Automation Version 12.0.

Hardware Requirements

For AutoSys Workload Automation hardware requirements, see the [AutoSys Workload Automation Release Notes](#).

Software Requirements

The following are Web UI requirements for AutoSys Workload Automation servers:

- If CA EEM without failover is implemented with AutoSys Workload Automation, the AutoSys Workload Automation servers and the Web UI servers must reference the same instance of CA EEM. If CA EEM with failover is implemented with AutoSys Workload Automation, the AutoSys Workload Automation servers and the Web UI servers do not need to reference the same instance of CA EEM; however, the CA EEM instances referenced must be in the same failover setup.

NOTE

For information about configuring CA EEM for failover, see the *CA Embedded Entitlements Manager Getting Started*, available at the [Documentation](#) website.

For information on compatibility with CA EEM, see [AutoSys Workload Automation, Web UI, and CA EEM Compatibility](#).

AutoSys Compatibility Matrix

This article includes the support matrix for AutoSys Workload Automation. Support is limited only to the platforms, databases, Broadcom products, CA EEM server, and SDK-client combinations that are listed in this article. Any platforms, Broadcom products, CA EEM server, and SDK combinations or versions that are not listed in this document are not supported.

For information about the platforms supported by:

- Workload Automation Agent, see the [Workload Automation System Agent Platform Support Matrix](#)
- Web UI, see the [Web UI Certification Information](#)
- Workload Automation iDash, see the [Workload Automation iDash Compatibility Matrix](#)
- Workload Automation iXp, see the [Workload Automation iXp Certification Information](#)

For information about Workload Automation Solutions Certification/Supported Platforms, see the Workload Automation Solutions Certification/Supported Platform Information.

For information about the CA Embedded Entitlements Manager Compatibility, see the CA Embedded Entitlements Manager Compatibility Matrix.

NOTE

We recommend that you upgrade to the latest release to bring the product to the latest maintenance level.

DISCLAIMER:

Broadcom supports third party product versions only when they are supported by the original vendor. If a vendor discontinues support for a product, Broadcom no longer supports it.

You can engage Broadcom Support for Workload Automation products that:

- Have not reached an End of Support date or End of Life date. For more information, see [Workload Automation Release and Support Lifecycle Dates](#).
- Run on third party product versions that are currently supported by the original vendor.

Support on Cloud

CA Workload Automation AE Release 11.3.6 and later and CA WCC Release 11.4 and later (including the agents) are supported on the following cloud platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

NOTE

The following database limitation applies to CA Workload Automation AE Release 11.3.6 to 11.3.6 SP6 and CA WCC Release 11.4 to 11.4 SP5:

- Amazon RDS and Microsoft Azure PaaS databases are not supported.

Consider the following information when hosting AutoSys Workload Automation or Web UI on the cloud:

- The AutoSys Workload Automation database should be hosted within the same environment as the AutoSys Workload Automation server for optimal performance. For example, hosting the database and the AutoSys Workload Automation application servers on premise and the AutoSys Workload Automation scheduler on the cloud will most likely significantly increase scheduler latency and general response times.
- Enable the [ports](#) that are related to AutoSys Workload Automation as well as traffic in the respective network.
- In Amazon Web Services (AWS) environment, ensure that the IP address is resolvable across the respective domains to enable proper communication between the Broadcom products. For example, you can resolve the IP address by updating the host file, providing the DNS server details, or by navigating to EC2 configService settings > EC2 Service properties and selecting the Set Computer Name option.
- We recommend that you host both AutoSys Workload Automation and Web UI on the cloud.

Server Database Support Matrix

Database	AutoSys Workload Automation Server Versions							Version 12.0
	R11.3.5	R11.3.6	R11.3.6SP1 to SP4	R11.3.6SP5	R11.3.6SP6	R11.3.6 SP7	R11.3.6 SP8	
Microsoft SQL Server 2012	YES	YES	YES	YES	YES	YES	YES	YES
Microsoft SQL Server 2014	NO	NO	YES	YES	YES	YES	YES	YES
Microsoft SQL Server 2016	NO	NO	NO	YES	YES	YES	YES	YES
Microsoft SQL Server 2017	NO	NO	NO	NO	NO	NO	YES	YES
Microsoft SQL Server 2019	NO	NO	NO	NO	NO	NO	NO	YES
Microsoft SQL Azure	NO	NO	NO	NO	NO	YES	YES	YES
Amazon Relational Database Service	NO	NO	NO	NO	NO	YES	YES	YES
Oracle 11gr2/ RAC	YES	YES	YES	YES	YES	YES	YES	NO
Oracle 12c/ RAC 2,3,4,5	YES	NO	YES	YES	YES	YES	YES	YES
Oracle 18c	NO	NO	NO	NO	NO	NO	YES	YES
Oracle 19c	NO	NO	NO	NO	NO	NO	YES	YES
Sybase 15.7 or higher	NO	NO	NO	YES	YES	YES	YES	YES
Sybase 16 or higher	NO	NO	NO	YES	YES	YES	YES	YES

NOTE**Specific Notes:**

1. For Sybase, the minimum page size must be at least 4 KB. If you are running Workload Automation Agents Advanced Integration for Hadoop jobs, the minimum page size must be at least 8 KB. Up to 137 free Sybase user connections are required, depending on the AutoSys Workload Automation components that are being installed. Sybase jConnect 7 or higher is required. For more information about installing Sybase jConnect 7 or higher, contact your database vendor.
2. Oracle RAC is supported.
3. Oracle TAF is supported from Release 11.3.6 SP07.
4. Oracle Container database is supported.

- The AutoSys Workload Automation schema must be created only on a pluggable database. Do not use CDB to create the schema and the same is recommended by Oracle. Setting this up is outside the scope of Broadcom Support. For further information, consult Oracle documentation or Oracle Support.
- 5. (Linux only) Oracle 12c client libraries have a known memory leak when the glibc version on the machine is below 2.14. The root cause of the memory leak is due to a bug in GNU glibc that is related to a multi-thread program that uses dlopen() to load the libraries. To resolve this issue, apply glibc 2.14 or higher. Contact your Linux vendor to get the GNU glibc compatible patch for bug [12650](#) based on your Linux operating environment.
- 6. CA Workload Automation AE 11.3.5 INCR03 or above is required; Rest Web services requires ojdbc6.jar, which is shipped with CA Workload Automation AE 11.3.5 INCR03 onwards.

Server Platform Support Matrix – UNIX

Platforms	AutoSys Workload Automation Server Versions				Version 12.0
	R11.3.5	R11.3.6	R11.3.6 SP1 to SP5	R11.3.6 SP6 to SP8	
IBM-AIX					
AIX 7.1 (64bit)	YES	YES	YES	YES	YES
AIX 7.2 (64bit)	NO	NO	NO	YES	YES
HP					
HP-UX 11i v3 (Itanium)	YES	YES	YES	YES	NO

NOTE

General Notes:

- On AIX, to see what level your system is at run: xlc.aix61.rte 13.1.3.1 COMMITTED IBM XL C++ Runtime for AIX 6.1 or later .
- On AIX 7.1, TL5 or later is required.

Server Platform Support Matrix – Linux

Platforms	AutoSys Workload Automation Server Versions					Version 12.0
	R11.3.5	R11.3.6	R11.3.6SP1 to SP4	R11.3.6SP5 to SP6	R11.3.6SP7 to SP8	
Linux						
Oracle Enterprise Linux 6 (64bit)	YES	YES	YES	YES	YES	YES
Oracle Enterprise Linux 7 (64bit)	NO	NO	NO	YES	YES	YES
Oracle Enterprise Linux 8 (64bit)	NO	NO	NO	NO	NO	YES
Red Hat Enterprise Linux 6 (64bit)	YES	YES	YES	YES	YES	YES
Red Hat Enterprise Linux 7.2 (64bit)	NO	NO	NO	YES	YES	YES

Red Hat Enterprise Linux 8 (64bit)	NO	NO	NO	NO	NO	YES
SUSE Linux Enterprise Server 11 (64bit)	YES	YES	YES	YES	YES	NO
SUSE Linux Enterprise Server 12 (64bit)	NO	NO	NO	YES	YES	YES
SUSE Linux Enterprise Server 15 (64bit)	NO	NO	NO	NO	NO	YES
Amazon Linux 2017 and 2018 (64-bit)	NO	NO	NO	NO	YES (SP8 Only)	YES
Amazon Linux 2	NO	NO	NO	NO	YES (SP8 Only)	YES
Clusters						
Unix/Linux Cluster Server	YES	YES	YES	YES	YES	YES

NOTE**Specific Notes:**

1. For OEL 6 and RHEL 6 support for some releases, see RI38228.
2. CCI is not supported on RHEL8.
3. RHEL Versions 7.2 and higher are supported.
4. Version 12.0 supports only 64-bit.
5. If you are using Amazon Linux 2, follow these steps:
 - a. Set the SKIP_PRE_CHECKS to 1.

```
export SKIP_PRE_CHECKS=1
```
 - b. Run the Installer.

```
./install.bin
```

NOTE**General Notes:**

- For information about the required library packages for AutoSys Workload Automation on Linux (64), see [AutoSys Operating System and Database Support](#).
- AutoSys Workload Automation Web Server component is not supported on 32-bit versions of Linux.

Server Platform Support Matrix – Windows

Platforms	AutoSys Workload Automation Server Versions							Version 12.0
	R11.3.5	R11.3.6	R11.3.6SP1 to SP2	R11.3.6 SP3 to SP5	R11.3.6 SP6	R11.3.6 SP7	R11.3.6 SP8	
Windows 2012 (64bit)	YES	YES	YES	YES	YES	YES	YES	YES
Windows 2016 (64bit)	NO	NO	NO	NO	NO	YES	YES	YES

Windows 2019 (64bit)	NO	NO	NO	NO	NO	NO	YES	YES
(SDK-Client only) Windows 7 (x86 or x64)	YES	YES	YES	YES	YES	YES	YES	YES
(SDK-Client only) Windows 8 (x86 or x64)	NO	YES	YES	YES	YES	YES	YES	YES
(SDK-Client only) Windows 10 (x86 or x64)	NO	NO	NO	NO	YES	YES	YES	YES
Clusters								
Microsoft Cluster Server (MSCS)	YES	YES	YES	YES	YES	YES	YES	YES

NOTE**Specific Notes:**

- In Release 11.3.5, Windows 2012 is supported only by the AutoSys Workload Automation SDK-Client.
- CCI is supported on Windows 2016 and 2019.

NOTE**General Notes:**

- The AutoSys Workload Automation web server component is not supported on 32-bit version of Windows.
- Windows R2 and SQL server R2 releases are supported unless noted otherwise.

Client Server Compatibility Matrix

Client Version	AutoSys Workload Automation Server Versions			
	Version 12.0	R11.3.6SP1 to SP8	R11.3.6	R11.3.5
R11.3.5	YES	YES	YES	YES
R11.3.6	YES	YES	YES	YES
R11.3.6 SP1 to SP8	YES	YES	YES	YES
Version 12.0	YES	YES	YES	YES

NOTE

- Client versions prior to R11.3.6 do not authenticate with the external security server and are automatically denied access to all secured objects by the R11.3.6 or higher application server when the instance is running under external security and STRICT or EXTERNAL unauthenticated user mode is set. Refer to the R11.3.6 and higher supported platforms section for more details.
- R11.3.6 or higher clients are denied access to all secured objects by the R11.3.5 application server when the respective instance is running under external security and STRICT unauthenticated user mode is set. Refer

to the R11.3.6 and higher supported platforms section for more details. Post 11.3 release, client enhanced functionalities are not supported. For example: autoaggr

- If the client and the server versions are not at the same level, the features implemented in the latest version are not supported.

Agent Compatibility Matrix

	AutoSys Workload Automation Server Versions				
	R11.3.5	R11.3.6	R11.3.6SP1 to SP6	R11.3.6SP7 & SP8	Version 12.0
R11SP5	YES	YES	YES	YES	NO
R11.3	YES	YES	YES	YES	YES
R11.3 SP1	YES	YES	YES	YES	YES
R11.3 SP2	YES	YES	YES	YES	YES
R11.3 SP3	YES	YES	YES	YES	YES
R11.3 SP4	YES	YES	YES	YES	YES
R11.3 SP5	YES	YES	YES	YES	YES
R11.3 SP6	YES	YES	YES	YES	YES
R11.3 SP7	YES	YES	YES	YES	YES
R11.4	YES	YES	YES	YES	YES
R11.4 SP1	YES	YES	YES	YES	YES
R11.5	YES	YES	YES	YES	YES
Version 12.0	YES	YES	YES	YES	YES

Server Upgrade Compatibility Matrix

Server Version	CA Workload Automation AE Server versions										
	R11.3.5	R11.3.6	R11.3.6 SP1	R11.3.6SP2	R11.3.6SP3	R11.3.6SP4	R11.3.6SP5	R11.3.6SP6	R11.3.6SP7	R11.3.6 SP8	Version 12.0
R11.3.5	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES
R11.3.6	NO	NO	YES	YES	YES	YES	YES	YES	YES	YES	YES
R11.3.6 SP1	NO	NO	NO	YES	YES	YES	YES	YES	YES	YES	YES
R11.3.6 SP2	NO	NO	NO	NO	YES	YES	YES	YES	YES	YES	YES
R11.3.6 SP3	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES	YES
R11.3.6 SP4	NO	NO	NO	NO	NO	NO	YES	YES	YES	YES	YES
R11.3.6 SP5	NO	NO	NO	NO	NO	NO	NO	YES	YES	YES	YES
R11.3.6 SP6	NO	NO	NO	NO	NO	NO	NO	NO	YES	YES	YES
R11.3.6 SP7	NO	NO	NO	NO	NO	NO	NO	NO	NO	YES	YES

R11.3.6 SP8	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	YES
Version 12.0	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO

Web UI/AutoSys Workload Automation Server Compatibility Matrix

Web UI Server Version	AutoSys Workload Automation Server Versions										
	R11.3.5	R11.3.6	R11.3.6 SP1	R11.3.6 SP2	R11.3.6SP3	R11.3.6SP4	R11.3.6SP5	R11.3.6SP6	R11.3.6SP7	R11.3.6 SP8	Version 12.0
R11.3.5	YES	NO	NO	NO	NO	NO	NO	NO	NO	NO	NO
R11.3.6	YES	YES	YES	NO	NO	NO	NO	NO	NO	NO	NO
R11.4 SP1	YES	YES	YES	YES	NO	NO	NO	NO	NO	NO	NO
R11.4 SP2	YES	YES	YES	YES	YES	NO	NO	NO	NO	NO	NO
R11.4 SP3	YES	YES	YES	YES	YES	YES	NO	NO	NO	NO	NO
R11.4 SP4	YES	YES	YES	YES	YES	YES	YES	NO	NO	NO	NO
R11.4 SP5	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO	NO
R11.4 SP6	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO	NO
R11.4 SP7	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	NO
Version 12.0	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES	YES

NOTE

For Web UI release 11.3.6 compatibility with AutoSys 11.3.6 SP1, see RO77628 and RO80613.

Embedded Broadcom Products Shipped/Supported

Broadcom products	AutoSys Workload Automation Server Versions										
	R11.3.5	R11.3.6	R11.3.6 SP1	R11.3.6 SP2	R11.3.6SP3	R11.3.6SP4	R11.3.6SP5	R11.3.6SP6	R11.3.6SP7	R11.3.6SP8	Version 12.0
CA Secure Socket Adapter	2.2.8	2.2.8	2.3.0	2.3.2	2.4.0	2.4.0	2.4.0	2.4.0	2.4.0	2.4.1	2.5.0
CAPKI/ETPKI	4.2.9	4.3.0	4.3.6	4.3.8	5.0.2	5.1.0	5.2.0	5.2.4	5.2.5	5.2.6	5.2.6
CA NSM Event Manager	r11.2 SP2	r11.2 SP2	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	NO

CAICCI	r11.2 SP2	r11.2 SP2	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1	r11.2 SP2 CUM1
CA EEM	r12 CR07 and higher	r12.51 and higher	r12.51 CR02 and higher	r12.51 CR03 and higher	r12.51 CR04 and higher	r12.51 CR04 and higher	r12.51 CR05 and higher	r12.6 and higher	r12.6 and higher	r12.6 and higher

Other Broadcom Products Supported

Broadcom Products	AutoSys Workload Automation Server Versions						
	R11.3.5	R11.3.6	R11.3.6 SP1 to SP5	R11.3.6SP6	R11.3.6SP7	R11.3.6 SP8	Version 12.0
CA Service Desk	r11 through r12.6	r11 through r12.6	r12.9, 14.1	r12.9, 14.1	r12.9, 14.1	r12.9, 14.1	17.1
CA UJMA	r3.1 for AS400	r3.1 for AS400	r3.1 for AS400	r3.1 for AS400	r3.1 for AS400	r3.1 for AS400	r3.1 for AS400
CA Workload Automation iDash	12.0 or higher	12.0 or higher	12.0 or higher	12.0 or higher	12.0.05.00 or higher	12.1	12.1.01.00
CA One Automation Platform	No	No	No	No	12.1 or higher	12.1 or higher	12.1 or higher

NOTE

For information on Automic® Automation Intelligence (AAI) compatibility with AutoSys, see the AAI documentation.

IPv6 Support, FIPS 140-2 and Section 508 Compliance

Compliance	AutoSys Workload Automation Server Versions			
	Version 12.0	R11.3.6 SP1 to SP8	R11.3.6	R11.3.5
IPv6 Support	YES	YES	YES	YES
FIPS 140-2	YES	YES	YES	YES
Section 508 Compliance	NO	NO	NO	NO

Web UI Compatibility Matrix

The Compatibility Matrix provides information about Web UI supported operating systems and databases, platform, browser compatibility, and support for other Broadcom products.

For information about the platforms supported by:

- Workload Automation Agent, see the [Workload Automation System Agent Platform Support Matrix](#)
- AutoSys Workload Automation, see the [AutoSys Compatibility Matrix](#)

For information about legacy WCC releases, see the [Legacy CA Workload Control Center Certification Information](#)

For information about the CA Embedded Entitlements Manager Compatibility, see the CA Embedded Entitlements Manager Compatibility Matrix

NOTE

We recommend that you upgrade to the latest release to bring the product to the latest maintenance level.

DISCLAIMER:

Broadcom supports third party product versions only when they are supported by the original vendor. If a vendor discontinues support for a product, Broadcom no longer supports it.

You can engage Broadcom Support for Workload Automation products that:

- Have not reached an End of Support date or End of Life date. For more information, see [Workload Automation Release and Support Lifecycle Dates](#).
- Run on third party product versions that are currently supported by the original vendor.

Support on Cloud

CA Workload Automation AE Release 11.3.6 and later and CA WCC Release 11.4 and later (including the agents) are supported on the following cloud platforms:

- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)

NOTE

The following database limitation applies to CA Workload Automation AE Release 11.3.6 to 11.3.6 SP6 and CA WCC Release 11.4 to 11.4 SP5:

- Amazon RDS and Microsoft Azure PaaS databases are not supported.

Consider the following information when hosting AutoSys Workload Automation or Web UI on the cloud:

- The AutoSys Workload Automation database should be hosted within the same environment as the AutoSys Workload Automation server for optimal performance. For example, hosting the database and the AutoSys Workload Automation application servers on premise and the AutoSys Workload Automation scheduler on the cloud will most likely significantly increase scheduler latency and general response times.
The Web UI database must be hosted within the same environment as the Web UI server. For example, do not host the Web UI database on premise and the Web UI on the cloud.
- Enable the [ports](#) that are related to AutoSys Workload Automation & Web UI as well as traffic in the respective network.
- In Amazon Web Services (AWS) environment, ensure that the IP address is resolvable across the respective domains to enable proper communication between the Broadcom products. For example, you can resolve the IP address by updating the host file, providing the DNS server details, or by navigating to EC2 configService settings > EC2 Service properties and selecting the Set Computer Name option.
- We recommend that you host both AutoSys Workload Automation and Web UI on the cloud.

Server Database Support Matrix

Databases	Web UI Versions						
	Version 12.0	11.4.7	11.4.4 to 11.4.6	11.4.3	11.4.2	11.4.1	11.3.5 to 11.4
Microsoft SQL Server 2019	Yes						
Microsoft SQL Server 2016	Yes	Yes	Yes				

Microsoft SQL Server 2014	Yes	Yes	Yes	Yes	Yes	Yes	
Microsoft SQL Server 2012 R2	Yes	Yes	Yes	Yes			
Microsoft SQL Server 2012	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Microsoft SQL Server 2008 R2		Yes	Yes	Yes	Yes	Yes	Yes
Microsoft SQL Azure	Yes						
Microsoft Azure PaaS	Yes	Yes	Yes (11.4.6 only)				
Amazon RDS	Yes	Yes	Yes (11.4.6 only)				
Oracle 19c	Yes	Yes					
Oracle 18c	Yes	Yes					
Oracle 12c (12.1.0.2)	Yes	Yes	Yes	Yes	Yes	Yes	
Oracle 11G R2 (11.2.0.3.0)		Yes	Yes	Yes	Yes	Yes	Yes
Derby (10.14.2.0)	Yes						
Derby (10.12.1.1)		Yes	Yes				
Derby (10.8.22)				Yes	Yes	Yes	Yes

Note:**Microsoft SQL:**

Microsoft SQL Server 2016 - Microsoft Cluster Server is supported.

Microsoft SQL Server 2016 - MS SQL Server Always On is supported.

Oracle:**Oracle 12c (12.1.0.2)**

Oracle Real Application Cluster (RAC) is supported.

PKI certificate authentication is supported.

Container Database is supported.

Derby:

Installed along with Web UI.

Server Platform Support Matrix - Linux

Linux	Web UI Versions				
Operating System and Version	12.0	11.4.4 to 11.4.7	11.4.3 to 11.4	11.3.6	11.3.5

Oracle Enterprise Linux 8 (64 bit)	Yes				
Red Hat Enterprise Linux 8 (64bit)	Yes				
Red Hat Enterprise Linux 7.2 (64bit)	Yes	Yes			
Red Hat Enterprise Linux 6 (64bit)	Yes	Yes	Yes	Yes	Yes
SUSE Linux Enterprise Server 15 (64bit)	Yes				
SUSE Linux Enterprise Server 12 (64bit)	Yes	Yes			
SUSE Linux Enterprise Server 11 (64bit)		Yes	Yes	Yes	Yes
Amazon Linux 2017 and 2018 (64-bit)	Yes	Yes (SP7 only)			
Amazon Linux 2 (64-bit)	Yes	Yes (SP7 only)			

NOTE

- For more information about Linux RPM packages and patches requirements, see the Release Notes section.
- RHEL Versions 7.2 and higher are supported.
- For Release 11.4 SP1:
EEM 8.3 and BO 2.0 do not support SLES 10. Please use EEM 8.4 SR01 and install BO on a separate machine.

Server Platform Support Matrix - Windows

Windows	Web UI Versions			
Operating System and Version	12.0	11.4.6 to 11.4.7	11.4.5 to 11.4.1	11.4 to 11.3.5
Microsoft Windows 2019 (64-bit)	Yes			
Microsoft Windows 2016 (64-bit)	Yes	Yes		
Microsoft Windows 2012 R2 (64bit)	Yes	Yes	Yes	Yes
Microsoft Windows 2012 (64bit)	Yes	Yes	Yes	Yes
Microsoft Windows 2008 R2 (64bit)		Yes	Yes	Yes

Third Party Product/s and Other Information

Third Party Product/s and Other Information	Web UI Versions									
	12.0	11.4.7	11.4.6	11.4.5	11.4 .4	11.4.2 and 11.4.3	11.4.1	11.4	11.3.6	11.3.5
Apache HTTPD Server 2.2.x	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Jaws-3.5.x								Yes	Yes	Yes
Apache Tomcat 7.0.22										Yes
Apache Tomcat 7.0.54							Yes	Yes	Yes	
Apache Tomcat 8.0.24						Yes				
Apache Tomcat 8.0.36					Yes					
Apache Tomcat 8.5.11				Yes						
Apache Tomcat 8.5.23			Yes							
Apache Tomcat 8.5.24			Yes							
Apache Tomcat 9.0.33	Yes									
AdoptOpenJDK 8	Yes									
JRE 1.7 update 17								Yes	Yes	
JRE 1.7 update 79						Yes	Yes			
JRE 1.7 update 101					Yes					
JRE 1.7 update 131				Yes						

JRE 1.7 update 161			Yes							
JRE 1.8 update 31							Yes			
JRE 1.8 update 65, 60						Yes				
JRE 1.8 update 91					Yes					
JRE 1.8 update 121				Yes						
JRE 1.8 update 152			Yes							
Adobe Flash Player version 8 or above			Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
CA SiteMinder 12.5 CR2								Yes		

Note:

- Apache HTTPD Server 2.2.x is used for load balancing of High Availability Web UI Servers.
- Apache Tomcat is installed during Web UI default installation.
- JRE 1.8.0 updates are installed along with Web UI on Windows and Linux platforms.
- To launch Jaws from Web UI Client machine, "JRE 1.5.0_09" or later is required.
- For 11.4.2, Apache Tomcat 5.5.33, fix RI33636 is required.

Client Browser Compatibility Matrix**NOTE**

- The following table lists the minimum supported browser versions in the second column. The certified browser is listed with the browser name and version.
- Web UI is supported by various browsers, listed in the following table, that are supported on Desktop and Server operating systems.

Product Version	Supported Browsers	Notes
Web UI 12.0	• Google Chrome 80 • Mozilla Firefox 76 • Safari 10.0 • Microsoft Internet Explorer 11.0 • Microsoft Edge 38.14393.1066.0	
CA Workload Control Center Client r11.4.7	• Google Chrome 53, 57, 61, 64, 70, and 72 • Mozilla Firefox 49, 53, 58, and 65 • Safari 10.0 10.1, 11, and 12 • Microsoft Internet Explorer 11.0 • Microsoft Edge 42.17134	
CA Workload Control Center Client r11.4.6	• Google Chrome 53, 57, 61, and 64 • Mozilla Firefox 49, 53, and 58 • Safari 10.0 10.1, and 11 • Microsoft Internet Explorer 11.0 • Microsoft Edge 38.14393.1066.0	
CA Workload Control Center Client r11.4.5	• Google Chrome 53, 57 • Mozilla Firefox 49, 53 • Safari 10.0, 10.1 • Microsoft Internet Explorer 11.0 • Microsoft Edge 38.14393.1066.0	See the product for any known issues.
CA Workload Control Center Client r11.4.4	• Microsoft Internet Explorer 11.0 • Mozilla Firefox 12,17,22,33,38,42,44,49 • Google Chrome 17,22,27,38,42,46,48,53 • Safari 7.1, 8.0.4, 9.0.2, 10.0	
CA Workload Control Center Client r11.4.3	• Microsoft Internet Explorer 11.0 • Mozilla Firefox 12, 17, 22, 33, 38, 42, 44 • Google Chrome 17, 22, 27, 38, 42, 46, 48 • Safari 7.1, 8.0.4, 9.0.2	
CA Workload Control Center Client r11.4.2	• Microsoft Internet Explorer 10.0, 11.0 • Mozilla Firefox 12, 17, 22, 33, 38, 42 • Google Chrome 17, 22, 27, 38, 42, 46 • Safari 7.1, 8.0.4, 9.0.1	
CA Workload Control Center Client r11.4.1	• Microsoft Internet Explorer 9.0, 10.0, 11.0 • Mozilla Firefox 12, 17, 22, 33, 38 • Google Chrome 17, 22, 27, 38, 42 • Safari 7.1, 8.0.4	

Product Version	Supported Browsers	Notes
CA Workload Control Center Client r11.4	- Microsoft Internet Explorer 9.0, 10.0, 11.0 - Mozilla Firefox 12, 33 - Google Chrome 22,27,38	
CA Workload Control Center Client r11.3.6	- Microsoft Internet Explorer 8.0, 9.0, 10.0 - Microsoft Internet Explorer 11.0 in Compatibility mode - Mozilla Firefox 10,17,22 - Google Chrome 17,22,27	
CA Workload Control Center Client r11.3.5	- Microsoft Internet Explorer 8.0, 9.0, 10.0 - Microsoft Internet Explorer 11.0 in Compatibility mode - Mozilla Firefox 10 - Mozilla Firefox 17 - Google Chrome 17	RO57256 is required for Mozilla Firefox 17

Web UI/AutoSys Workload Automation Server Compatibility Matrix

Web UI Server Version	AutoSys Workload Automation Server Versions										
	R11.3.5	R11.3.6	R11.3.6 SP1	R11.3.6 SP2	R11.3.6 SP3	R11.3.6 SP4	R11.3.6 SP5	R11.3.6 SP6	R11.3.6 SP7	R11.3.6 SP8	12.0
R11.3.5	Yes										
R11.3.6	Yes	Yes	RO77628 RO80613								
R11.4 SP1	Yes	Yes	Yes	Yes							
R11.4 SP2	Yes	Yes	Yes	Yes	Yes						
R11.4 SP3	Yes	Yes	Yes	Yes	Yes	Yes					
R11.4 SP4	Yes	Yes	Yes	Yes	Yes	Yes	Yes				
R11.4 SP5	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes			
R11.4 SP6	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes		
R11.4 SP7	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	
12.0	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

CA Embedded Entitlements Manager (EEM) - Supported Releases

Web UI versions	12.0	11.4.7	11.4.6	11.4.5	11.4.4	11.4 to 11.4.3	11.3.6	11.3.5
Supported CA Embedded Entitlements Manager (EEM) versions	r12.6 and higher	r12.6 and higher	r12.6 and higher	r12.51 CR05 and higher	r12.51 CR04 and higher	r12.51 CR02 and higher	r12.51 and higher	r12 CR07 and higher

Release History (AE)

The following table lists the features that were implemented from CA Workload Automation AE r11.3 to the latest release:

Features	Release Implemented
IPAddressPreference —New Parameter in the Configuration File Specifies the criteria that CA Workload Automation AE uses to select an IP address from the list of IP addresses that are returned by the operating system after translating the host name.	11.3.6 SP8
Telemetry CA Workload Automation AE supports CA Remote Engineer r4.0. The CA Remote Engineer tool collects configuration and diagnostic data that helps you to troubleshoot problems with the system or software. In this release, you can integrate CA Remote Engineer with CA Telemetry Service and this integration helps to capture customer environment and topology information.	11.3.6 SP7
Change the Job Status Using Web Services You can use web services to change the status of a job.	11.3.6 SP7
ComponentRegister Utility The ComponentRegister utility registers the REST web server that is installed with CA Workload Automation AE and updates the database.	11.3.6 SP7
New Features - 11.3.6 SP7 - The alarm_if_fail attribute is updated to specify whether to post an alarm only when the job completes with a FAILURE status. In previous releases, the alarm_if_fail attribute specified whether to post an alarm when a job completes with FAILURE or TERMINATED status. - The alarm_if_terminated attribute is added to specify whether to post the JOBTERMINATED alarm when the job completes with a TERMINATED status. - The JOBTERMINATED (552) alarm is added that indicates that a job is terminated. - The trapJobTerminated (552) SNMP trap is added that indicates that the job terminated.	11.3.6 SP7
emailtemplate Blob You can create an emailtemplate blob using the insert_glob subcommand (specify the blob_type attribute value as emailtemplate) and use it in your job definition or in the notification template that is used to send the email notification.	11.3.6 SP7

Global Variable Substitution You can now reference a global variable as part of the syntax of the following attributes: <code>hadoop_hdfs_destination</code>, <code>hadoop_hdfs_path</code>, <code>hadoop_hdfs_source</code>, <code>informatica_folder_name</code>, <code>informatica_param_file</code>, <code>informatica_repository_name</code>, <code>informatica_security_domain</code>, <code>informatica_target_name</code>, <code>informatica_task_name</code>, <code>informatica_workflow_inst_name</code>, <code>informatica_workflow_name</code>, <code>sqlagent_jobname</code>, <code>sqlagent_server_name</code>, <code>sqlagent_step_name</code>, <code>sqlagent_target_db</code>, and <code>ws_parameter</code>.	11.3.6 SP7
Increase autosys_secure Password Limit The limit for the password that you specify when you create an user using the <code>autosys_secure</code> command is increased from 20 characters to 127 characters. The database user password limit is 20 characters.	11.3.6 SP7
Installation Pre-checks The installer performs some pre-checks to verify the <code>aedbadmin</code> privileges, database privileges, ports, operating system versions, and database versions and libraries and displays appropriate warning or error messages. This helps prevent any post-installation failures.	11.3.6 SP7
Integrate CA Workload Automation AE with CA Automatic One Automation Platform In previous releases, you integrated CA Workload Automation AE with Event Management to automate manual problem resolution tasks, filter and consolidate multiple events, monitor for unusual conditions, and take proper corrective action. In the current release, you can integrate CA Workload Automation AE with CA Automatic One Automation Platform or Event Management to automate manual problem resolution tasks, filter and consolidate multiple events, monitor for unusual conditions, and take proper corrective action.	11.3.6 SP7
Machine Status Changes In the previous releases, when you insert a machine, the machine status is set to online. From this release onwards, when you insert a machine, the scheduler calibrates to the actual machine status and sets the machine status accordingly. Also, when you update the port, <code>node_name</code>, <code>encryption_type</code>, <code>agent_name</code>, or <code>key_to_agent</code> machine attribute, the scheduler calibrates to the actual machine status and sets the machine status accordingly. In the previous releases, when you issue the <code>MACH_ONLINE</code> event, the machine status is set to online. From this release onwards, the <code>MACH_ONLINE</code> event places the machine in the online status after the scheduler calibrates that the machine is actually available and accepts jobs to run. If the machine is not available, the machine status is set to missing or blocked.	11.3.6 SP7
Machine Methods to Achieve Load Balancing You can use the new <code>machine_method</code> attribute in your job definition or the <code>MachineMethod</code> parameter in the configuration file to specify the machine methods to achieve load balancing. The <code>MachineMethod</code> parameter is updated to include the following two new machine methods: <code>roundrobin</code> and <code>roundrobin_jobload</code>.	11.3.6 SP7
New Attribute for Informatica Jobs: <code>informatica_security_domain</code>.	11.3.6 SP7

New <code>update_blob</code> and <code>update_glob</code> Subcommands • <code>update_glob</code>—Updates the content of an existing glob. • <code>update_blob</code>—Updates the content of an existing blob.	11.3.6 SP7
Retrieve Jobs with jobtype Filter Using Web Services You can use web services to retrieve jobs with the jobtype filter.	11.3.6 SP7
Search is Case Insensitive for the Description Field In web services, you can do a case insensitive search for the description field while searching for jobs, connection profiles, or machines. In SDK, you can do a case insensitive search for the description field while searching for jobs, connection profiles, machines, calendars, or virtual resources.	11.3.6 SP7
Send Email Notifications You can send an email notification using the email notification template that you specified in the <code>notification_template</code> attribute when a job fails, terminates, or completes successfully or when an alarm is raised. To send an email notification using an email notification template, set the <code>send_notification</code> attribute value to <code>y</code> or <code>f</code> and specify the <code>notification_template</code> and the <code>notification_emailaddress</code>, <code>notification_emailaddress_on_success</code>, <code>notification_emailaddress_on_failure</code>, <code>notification_emailaddress_on_terminated</code>, or <code>notification_emailaddress_on_alarm</code> attributes in your job definition based on the job status or alarm you want to send the email notification for. The following attributes are added to send email notifications using template: • <code>notification_template</code>—Specifies the email notification template that is used to send the email notification. • <code>notification_emailaddress_on_success</code> — Specifies the email addresses of the recipients of the email notification when the job completes successfully. • <code>notification_emailaddress_on_failure</code> — Specifies the email addresses of the recipients of the email notification when the job fails. • <code>notification_emailaddress_on_terminated</code> — Specifies the email addresses of the recipients of the email notification when the job terminates. • <code>notification_emailaddress_on_alarm</code> — Specifies the email addresses of the recipients of the email notification when an alarm is raised. You can specify the alarm for which you want to send an email notification using the <code>notification_alarm_types</code> attribute. In a job run, for each alarm type, you can now use the <code>RepeatAlarmNotification</code> parameter to configure whether you want to send an email notification for all alarms, just the first alarm, or only the last alarm.	11.3.6 SP7
Specify the Agent Status Poll Interval The <code>AgentStatusPollInterval</code> parameter in the configuration file specifies the time interval (in minutes) between status polls that the scheduler waits before polling all the agents.	11.3.6 SP7
TAF Certification CA Workload Automation AE can now run in Transparent Application Failover (TAF) mode.	11.3.6 SP7

Updated as-owner Resource Class The as-owner resource class is updated to include the SENDEVENT_JOBEXECUTE access mode that controls whether you can execute the sendevent command.	11.3.6 SP7
Updated autotrack Command The following options are added to the autotrack command: • -B group—To define the group about which to report. • -I application—To define the application about which to report. • -R resource—To define the virtual resource about which to report.	11.3.6 SP7
Updated blob_type Attribute You can specify the blob_type as generic or emailtemplate.	11.3.6 SP7
Updated Limits for the command Attribute The limits for the command attribute is increased from 512 to 1000 characters.	11.3.6 SP7
Update Security Policies After Migration You can use the -m and -v options of the as_safetool command to update the security policies after migration.	11.3.6 SP7
Global Variable Substitution Lets you reference a global variable as part of the syntax of the following attributes: destination_name, ftp_server_name, initial_context_factory, j2ee_parameter, notification_id, notification_msg, notification_emailaddress, provider_url, sap_recipients (only recipient parameter), sap_step_parms (only variant and step_user parameters), sap_target_sys, and sp_arg (only name and value parameters).	11.3.6 SP6
Informatica Job Type Lets you run Informatica jobs to start a predefined workflow on Informatica PowerCenter.	11.3.6 SP6
New Attributes for JMS Publish and JMS Subscribe Jobs: correlation_id , destination_user , and (JMS Publish jobs only) reply_to .	11.3.6 SP6
New Attributes for SCP and FTP Jobs: • run_external and external_os_user • (SCP jobs only) scp_transfer_mode, scp_delete_sourcefile, scp_delete_sourcedir, scp_rename_sourcefile, and scp_create_targetdir.	11.3.6 SP6
Specify Whether to Use the MachineMethod Parameter When a Job Is Force Started The UseMachineMethodOnForceStartJob parameter in the configuration file lets you specify whether the scheduler ignores or uses the MachineMethod parameter value to select the machine where the job must run when you force start a job and multiple non-zero factor online machines are available.	11.3.6 SP6
SQL Server Agent Job Type Lets you execute predefined jobs in the SQL Server Agent to perform administrative tasks.	11.3.6 SP6
(Sybase only) Support Kerberos Authentication Lets you configure CA Workload Automation AE to use Kerberos authentication to access the database.	11.3.6 SP6

Updated autosys_secure Command The autosys_secure command is updated to let you define a user with both password and key credentials.	11.3.6 SP6
Updated dbtype Attribute The dbtype attribute is updated to monitor the POSTGRESQL or TERADATA database.	11.3.6 SP6
Updated LogMaxEndLines Parameter The default value of the LogMaxEndLines parameter in the configuration file is updated from 0 to 2000. If you have set the LogMaxEndLines parameter to a non-zero value, after you upgrade this value is set to the default value, that is, 2000.	11.3.6 SP6
Change the Status of the Job that is in the ON_NOEXEC Status Lets you issue the CHANGE_STATUS, JOB_ON_NOEXEC, or JOB_OFF_NOEXEC event to change the status of a job that is in the ON_NOEXEC status.	11.3.6 SP5
KILLJOB Send Signal Behavior On UNIX agents (r11.3, Release 11.3.5, or Release 11.3.6), you can send multiple signals by entering a comma-delimited list as the value for the signal_num variable. The agent sends the first signal, sleeps for a second, sends the next signal, and so on. In the previous releases, only the legacy agents supported this behavior.	11.3.6 SP5
New as-sendevent Resource Class Lets you control access to execute the sendevent command.	11.3.6 SP5
New Attributes for Oracle E-Business Suite Single Request Jobs: oracle_oprunit and oracle_template_name.	11.3.6 SP5
Priority is Honored for Jobs with Load Balancing and Resources Attributes If you defined jobs with load balancing and resources attributes, jobs with highest priority run ahead of jobs with lower priority. If higher priority jobs are queued in the QUE_WAIT or RESWAIT state, then lower priority jobs are queued behind the higher priority jobs and remain queued until all higher priority jobs leave their queued states. You can send a CHANGE_PRIORITY event to change the priority of RESWAIT jobs.	11.3.6 SP5
STATE_CHANGE Events Display in the Detailed Report of the Previous Run When Jobs are Queued in the RESWAIT State The STATE_CHANGE events that are associated with the job entering and leaving the RESWAIT state are displayed in the detailed autorep report of the previous run of the job. These STATE_CHANGE events are no longer displayed in the detailed autorep report of the current run of the job.	11.3.6 SP5
Support Global Resources for Box Job Lets you specify only global resources on a box job. However, the resource specified for a box applies only to the box and not to the jobs within the box.	11.3.6 SP5
Updated autorep Command The -Q option is added to the autorep command to generate a report on jobs that are queued due to insufficient load balancing units, machine resources, or global resources. The ST column heading in the autorep report is updated to ST/Ex. Note: These updates are available only when you use the command line interface and are not available when you use the CA Workload Automation AE SDK or web services.	11.3.6 SP5

Updated autostatus Command If the job is in the ON_NOEXEC status, the autostatus command displays the current status and the execution mode of the job.	11.3.6 SP5
Updated job_depends Command The Status column in the job_depends report is updated to Status/Exec. Mode.	11.3.6 SP5
watch_no_change Attribute -- Specify the Time the File Must Remain Unchanged The watch_no_change attribute is updated to specify the value in seconds, minutes, hours, or days.	11.3.6 SP5
(Oracle only) New Features - 11.3.6 SP4Lets you configure CA Workload Automation AE to use single sign-on wallets with certificates that are present on the file system to access the database.	11.3.6 SP4
CA WA Advanced Integration for Hadoop Job Types: HDFS (Hadoop Distributed file System), Oozie, Hive, Sqoop, and Define a Pig JobLets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. These job types implementation includes: • Connection Profiles—Set of attributes that are required for connecting to environments where the job runs. • as-connectionprofile Resource Class—Controls access to a connection profile. • StatusIndicates that the Hadoop Oozie job is suspended. • RESUMEJOB (149) Event—Resumes a Hadoop Oozie job and places it in the RUNNING state. • SUSPENDJOB (148) Event—Suspends the Hadoop Oozie job and places it in the SUSPENDED state. • RESUMEJOBFAIL(551) Alarm—Indicates that the RESUMEJOB event failed; typically when you send the event with the Hadoop Oozie job not in the SUSPENDED state. • SUSPENDJOBFAIL (550) Alarm—Indicates that the SUSPENDJOB event failed; typically when you send the event with the Hadoop Oozie job not in the RUNNING state. • Authenticate a User Using Key Credentials —Lets you authenticate users that run Hadoop jobs using key credentials instead of a password.	11.3.6 SP3
Performance Improvement with Multi-threaded Listener From Release 11.3.6 SP3The scheduler runs in multi-threaded listener mode where connections from multiple agents are processed simultaneously as opposed to the serial processing of statuses from agents earlier; thus providing increased throughput of jobs and decrease in event latency. Also, the number of autoping timeouts and false chase alarms are reduced.	11.3.6 SP3
Updated autosys_secure Command The autosys_secure command is updated to let you add a regular user with key credentials, change the key credentials of the specified user, display only the list of users with key credentials. or display only the list of users with password.	11.3.6 SP3
Agent Version Information Listed in the autorep OutputThe autorep -p option generates a report that displays all the active agent plug-ins and advanced integrations on the machine and the agent release and build information.	11.3.6 SP2

Add or Remove Jobs In Boxes Lets you add or remove jobs in a box.	11.3.6 SP2
Calendar Name and Description Lets you use the autocal_asc command to add or modify a description for a calendar. The calendar name can now contain up to 64 characters.	11.3.6 SP2
Retrieve Information of a Machine Using Web Services Lets you use web services to retrieve information of a specific machine or multiple machines.	11.3.6 SP2
Sending More Detailed Events to External Instances When a local scheduler sends events to external instances, it populates the Event text field to indicate the action in the local instance that resulted in the external event.	11.3.6 SP2
Updates to the ON_NOEXEC Feature The following changes are made for the ON_NOEXEC feature: • JOB_ON_NOEXEC event—The scheduler processes the JOB_ON_NOEXEC event similar to that of the CHANGE_STATUS event to INACTIVE. For example, if you send the JOB_ON_NOEXEC event to a job in a box, the effect is the same as sending the CHANGE_STATUS event to INACTIVE for a job in a box. The scheduler sets the job to the ON_NOEXEC status and also sets the internal secondary status to INACTIVE, so that the scheduler evaluates the overall box status as if the job entered the INACTIVE status. To take the job off the ON_NOEXEC status, you must send the JOB_OFF_NOEXEC event. Sending manual CHANGE_STATUS event does not change the status of the job. To complete the NOEXEC job immediately so that the scheduler evaluates the job as if the job entered the SUCCESS status, send the FORCE_STARTJOB event. • JOB_OFF_NOEXEC event—When the scheduler processes the JOB_OFF_NOEXEC event for a job, it places the job in the INACTIVE status. If the job is in a box, the scheduler evaluates the overall box status as the job is entering the INACTIVE status. If you send the JOB_OFF_NOEXEC event to a box, all jobs in the box (including all jobs contained in lower level boxes within the box) are reset to the INACTIVE status. • Evaluation of downstream dependent jobs—When you send the JOB_ON_NOEXEC event to a job, the effect is the same as if the job enters the INACTIVE status. The scheduler does not immediately schedule jobs that have a dependency on the NOEXEC job nor does it evaluate their success conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the INACTIVE status. When the job's starting conditions are met and the scheduler sends the BYPASS event, the effect is the same as if the job enters the SUCCESS status with an exit code of 0. The scheduler schedules jobs that have a dependency on the NOEXEC job and evaluates their success conditions to success.	11.3.6 SP2
Aggregate Statistics The default value of the AggregateStatistics parameter in the configuration file is set to 1, that is, CA Workload Automation AE aggregates the job, alarm, and scheduler statistics automatically by default.	11.3.6 SP1

Email Notifications If you specify an application or group attribute in your job definition, the application name or the group name is included in the email notification.	11.3.6 SP1
(UNIX) Enabling Core File Generation Core file creation is enabled for the server processes (scheduler and application server) to enhance supportability.	11.3.6 SP1
Integrate with CA Service Desk Lets you configure CA Workload Automation AE to work with CA Service Desk using SSL communication.	11.3.6 SP1
Job Name Supports Colon When you insert, update, override, or delete a job, you can include colons in the job name or the box job name. Enclose the job name that includes a colon with quotation marks (" ") or precede it with a backslash (\).	11.3.6 SP1
How Machine Status Changes When Disk Threshold is Breached vAE_12 Lets the agent to monitor the amount of available disk space and send notifications to warn you when the space is too low. The MACHINE_DISKTHRESHOLD (549) alarm is introduced to indicate that a disk threshold has been breached on the agent.	11.3.6 SP1
(Oracle only) Configure CA Workload Automation AE to Run in Dual Event Server Mode Lets you synchronize the databases quickly; thereby improving performance.	11.3.6 SP1
Alarms Indicates that the DBMaint command failed during automated database maintenance.	11.3.6 SP1
New Machine Status: Blocked Indicates that the agent blocks communication.	11.3.6 SP1
status Attribute—Set an Initial Status for a Job During Insertion Lets you set an initial status for a job during insertion.	11.3.6 SP1
SNMP Traps In addition to the alarmName, alarmJobName, alarmText, alarmCode, alarmExitCode, trapDate, and trapMessge values, the SNMP trap passes the following values: alarmMachineName, alarmBoxName, alarmApplicationName, alarmGroupName, alarmRunNum, alarmNtry, and alarmInstanceName.	11.3.6 SP1
STATE_CHANGE Event is Issued When a Job Enters the QUE_WAIT, PEND_MACH, or RESWAIT State When a job is placed in the QUE_WAIT, PEND_MACH, or RESWAIT state, the scheduler issues a STATE_CHANGE event with an informative message. The event and the associated message are written to the scheduler log and are also displayed in the detailed autorep report that you can generate using the autorep -J <i>jobname</i> -d command.	11.3.6 SP1
Updated as-owner Resource Class The as-owner resource class is updated to control whether a user has permissions to the existing job owner value to update, override, or delete any attribute of a job.	11.3.6 SP1

Updated EvaluateQueuedJobStarts Parameter The EvaluateQueuedJobStarts parameter in the configuration file is updated to support the following three options for scheduling jobs leaving a queued state: 0, 1, or 2. On Windows, in the Administrator utility, you can set this parameter to any one of the following options: Off, Skip same day check, or Include same day check.	11.3.6 SP1
Updated LOGROLLOVER Parameter The LOGROLLOVER parameter in the configuration file is updated to include a new value named PURGE(x) that purges all log files that are older than the specified number of days at midnight. The default value of the LOGROLLOVER parameter is set to MIDNIGHT,PURGE(7).	11.3.6 SP1
Authenticating Command-Line Utilities with External Security The addition of the -usr command-line option lets you to authenticate certain command-line utilities with external security. Using this option improves security by ensuring that the utility runs as an authenticated external security user.	11.3.6
Cannot Issue the STARTJOB Event to Start a Job With Virtual Resource Dependency You cannot issue the STARTJOB event to start a job that has a virtual resource dependency with free=Y or free=N and has already held the resource. To start such a job, take one of the following actions: - Manually release the held resource by issuing the RELEASE_RESOURCE event. - Force start the job in FAILURE or TERMINATED status by issuing the FORCE_STARTJOB event. The virtual resource is released if the job has the virtual resource dependency with free=Y and completes successfully.	11.3.6
Cross-platform Job Requests When the scheduler receives the cross-platform start job requests, it now requests the application server to authenticate and authorize that request. For the job to execute successfully, you must have EXECUTE access for the following resource classes: as-appl, as-group, as-machine, and as-job.	11.3.6
Force Start a Job in FAILURE or TERMINATED Status Lets you force start a job in FAILURE or TERMINATED status that has a virtual resource dependency with free=Y or free=N and has not released the virtual resources. The FORCE_STARTJOB event verifies if the current status of the job is FAILURE or TERMINATED and schedules the job using the already held virtual resources. Note: Before force starting the job, the scheduler does not reevaluate other resource dependencies.	11.3.6
Manage the Unauthenticated User Mode Lets you authenticate client utilities using external authentication protocols when customized authentication libraries are installed on all client and server machines in the instance.	11.3.6
Set the Maximum Number of Lines to Retrieve from a Log File The LogMaxEndLines parameter in the configuration file lets you specify the maximum number of lines to retrieve from a log file.	11.3.6

Updating the resources Attribute in an Existing Job Definition You cannot update the resources attribute in the existing job definition if the job has a resource dependency and has held the resource. To release the held resource, take one of the following actions: - Manually release the held resource by issuing the <code>RELEASE_RESOURCE</code> event. - Force start the job in <code>FAILURE</code> or <code>TERMINATED</code> status by issuing the <code>FORCE_STARTJOB</code> event. The virtual resource is released if the job has the virtual resource dependency with <code>free=Y</code> and completes successfully.	11.3.6
Aggregate the Job, Alarm, and Scheduler Statistics into the Database Tables The AggregateStatistics parameter in the configuration file lets you specify whether CA Workload Automation AE aggregates the job, alarm, and scheduler statistics automatically at a specified interval.	11.3.5
Alarm Removed The <code>APP_SERVER_COMM</code> alarm is removed from CA Workload Automation AE.	11.3.5
Allow the Shadow Scheduler to Failback to the Primary Scheduler The PrimaryFailbackMode parameter in the configuration file lets you define the mode that the primary scheduler enters when you restart CA Workload Automation AE after a failover.	11.3.5
AUTOPIDRelease 11.3.5 does not support setting AUTOPID as an environment variable during job execution. Note: The AUTOPID environment variable is still valid for jobs that run on the legacy agent.	11.3.5
Bypass a Job to Run Downstream Dependent Jobs Lets you bypass a job to run downstream dependent jobs. To bypass a job, use the sendevent <code>-E JOB_ON_NOEXEC</code> command. This command sets the job status to <code>ON_NOEXEC</code> and instructs the scheduler to bypass the execution of the job. CA Workload Automation AE evaluates <code>ON_NOEXEC</code> jobs as successfully completed even though CA Workload Automation AE does not execute the commands for these jobs. CA Workload Automation AE reports the machine field for jobs in <code>ON_NOEXEC</code> status as <code>***NOEXEC***</code> to reflect that these jobs were not executed on any computer. Downstream dependent jobs still run and boxes with <code>ON_NOEXEC</code> jobs evaluate as if the <code>ON_NOEXEC</code> jobs or boxes had run to success. CA Workload Automation AE continues to bypass jobs in <code>ON_NOEXEC</code> status until it receives and processes the <code>JOB_OFF_NOEXEC</code> event.	11.3.5
Compliance ApplicationLets you audit the system utilization. It installs automatically with CA Workload Automation AE and runs as part of the <code>DBMaint</code> command.	11.3.5
Configuring CA Workload Automation AELets you configure CA Workload Automation AE by editing the parameters in the configuration file. In previous releases, only the CA Workload Automation AE Administrator was used to configure CA Workload Automation AE on Windows.	11.3.5

Configure the Agent to Behave Like the Legacy Agent If you select the Configure Agent with Legacy Remote Agent Compatibility check box on the Agent Attributes (on UNIX) or Agent Properties (on Windows) panel during the installation, the installer automatically configures the agent to behave like the legacy agent. That is, the job processing behavior of the agent closely matches the job processing behavior of the legacy agent. If you want to configure the agent to behave like the legacy agent, we recommend that you select the Configure Agent with Legacy Remote Agent Compatibility check box. If you did not select the Configure Agent with Legacy Remote Agent Compatibility check box during the installation, but later decide to configure the agent to behave like the legacy, you must manually add or edit the compatibility parameters in the agent's agentparm.txt file.	11.3.5
Control the Starting of Jobs in PEND_MACH Status The GlobalPendMachInterval parameter in the configuration file lets you control the starting of jobs in PEND_MACH status to reduce the load on the scheduler and agent computers.	11.3.5
Control the Status of Jobs Scheduled on an Offline Machine The GlobalPendMachStatus parameter in the configuration file lets you define the completion status that the scheduler assigns to jobs that are scheduled on an offline machine. The jobs temporarily remain in the PEND_MACH status before the scheduler assigns the status that is specified in the GlobalPendMachStatus parameter.	11.3.5
Creating a Forecast Report for Multiple Days You can use the -F and -T options of the forecast command to specify any time frame when creating a forecast report. There are no limits on the number of days you can specify; however, reports with longer time frames consume more memory and time. In the previous release, you could specify a time frame up to 24 hours.	11.3.5
Disable IP Address Caching The EnableIPCaching parameter in the configuration file lets you specify whether you want to enable IP address caching.	11.3.5
Eligibility of Machines with a Factor Value of Zero The scheduler does not disqualify machines with a factor value of zero (0). When all available machines have a factor value of zero (0), the agent selects one of these machines at random and runs the job on that machine.	11.3.5
Enable FIPS Mode The EnableFIPSMODE parameter in the configuration file lets you specify whether you want CA Workload Automation AE to operate in FIPS mode.	11.3.5
Exit Code is Returned When a jil Command is Issued When you issue a jil command, an exit code is returned to indicate the status of the command. A zero (0) exit code indicates success, while a non-zero exit code indicates an error.	11.3.5

Global Variables vAE_12 Lets you reference a global variable as part of the syntax of the following attributes: command, connect_string, destination_file, ftp_local_name, ftp_remote_name, i5_library_list, i5_name, i5_params, monitor_cond, scp_local_name, scp_remote_dir, scp_remote_name, sp_name, sql_command, std_err_file, , std_out_file, success_criteria, text_file_name, trigger_cond, and watch_file.	11.3.5
Increase the CA EEM Server List Input in autosys_secure When you enable external security using the autosys_secure command, you are prompted to enter the CA EEM back end server host name. In r11, you could specify only one CA EEM back end server host name. In r11.3, you could specify a comma-separated list of CA EEM servers if you configured multiple CA EEM servers for failover. The entire list of CA EEM servers could consist of at most 64 characters. In Release 11.3.5 you can specify a comma-separated list of CA EEM servers. The entire list of CA EEM servers can consist of at most 255 characters.	11.3.5
Micro Focus (MICROFOCUS) Job Type Lets you define jobs to run your mainframe JCL and COBOL programs in a Windows environment.	11.3.5
CA EEM Resource Classes for CA Workload Automation AE Lets you control whether you can aggregate CA Workload Automation AE statistics.	11.3.5
CA EEM Resource Classes for CA Workload Automation AE Lets you support the authorization of jobs based on their job types.	11.3.5
New CA EEM Global User Group for CA Workload Automation AE Named WorkloadAutomationAEWebService A predefined CA EEM global user group with users who are authorized to access the CA Workload Automation AE web service.	11.3.5
Install CA Workload Automation AE with Oracle Lets you create the Oracle database tablespaces, users, and roles before you install CA Workload Automation AE by running the waae_oracle.sql script. The installer does not prompt you to specify the Oracle administrator user and password information.	11.3.5
Process Automation Process Execution (PAPROC) Job Type Lets you define jobs to execute a CA Process Automation process.	11.3.5
Process Automation Start Request Form (PAREQ) Job Type Lets you define jobs to submit a Start Request Form to execute a CA Process Automation process.	11.3.5
Remote Execution (PROXY) Job Type Lets you define jobs to run commands to a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Telnet or Secure Shell (SSH2).	11.3.5
Removed EXECUTE_CATALOG_ROLE Privilege In the previous release, the Oracle aedbadm user was granted the EXECUTE_CATALOG_ROLE privilege. In Release 11.3.5, the EXECUTE_CATALOG_ROLE privilege is not granted to the Oracle aedbadm user because the aedbadm user no longer requires it.	11.3.5

Renamed Web Service (WBSVC) to Web Service RPC/Encoded (WBSVC) The Web Service (WBSVC) job types is renamed to Web Service RPC/Encoded (WBSVC) in the documentation.	11.3.5
Repair an Existing CA Workload Automation AE Installation The installer does the following when you repair an existing installation: • Uses the RefreshAEDB script to repair the CA Workload Automation AE database. • Regenerates the agentparm.txt file and the instance profile files.	11.3.5
Send Email Notifications Lets you configure CA Workload Automation AE to send email notifications to operators or administrators by setting the NotifyMethod, NotifySMTPHost, UseSMTPAuthentication, NotifySMTPUser, and NotifySMTPFromAddress parameters in the configuration file.	11.3.5
Set Job Attribute Environment VariablesThe SetJobAttributeEnvironmentals parameter in the configuration file lets you specify whether CA Workload Automation AE automatically sets the supported job definition JIL attributes as environment variables. The legacy agent supports the setting of job definition JIL attributes as environment variables, which are sourced as part of running a job. The agent supports the setting of only the following environment variables based on the job definition JIL attribute values: __job_name, __box_name, __machine, __run_machine, and __max_exit_success. You can set these job definitions JIL attributes as environment variables only for command and file watcher jobs.	11.3.5
Simple Network Management Protocol Value Get (SNMPGET) Job Type Lets you define jobs to retrieve the value of an SNMP variable.	11.3.5
Simple Network Management Protocol Value Set (SNMPSET) Job Type Lets you define jobs to set the value of an SNMP variable.	11.3.5
Skip Starting Condition Evaluation for Queued Jobs The EvaluateQueuedJobStarts parameter in the configuration file lets you specify whether CA Workload Automation AE skips starting condition evaluation for jobs leaving a queued state. When CA Workload Automation AE skips starting condition evaluation, queued jobs start immediately upon leaving a queued state.	11.3.5
SNMP Support on Windows Lets you configure CA Workload Automation AE to send SNMP traps to SNMP managers on Windows. In the previous releases, this feature was supported only on UNIX.	11.3.5

Status Changes for Jobs with Cross-Instance Dependencies The cross-instance interface design supports reporting status changes to the remote instance for jobs with cross-instance dependencies when those changes result from one of the following: • The scheduler changes the status of the job when unavailable machine load units, resources or agents prevent a job from running. • The user changes the status of the job by issuing a sendevent command for one of the following events: JOB_ON_HOLD, JOB_OFF_HOLD, JOB_ON_ICE, JOB_OFF_ICE, JOB_ON_NOEXEC, JOB_OFF_NOEXEC.	11.3.5
Oracle E-Business Suite Jobs Workload Automation Agent for Oracle E-Business Suite r11.3.1 includes the following new features: • Quote resolved expressions in default values • Give a higher priority to request set defaults over concurrent program defaults • Specify value set expressions to resolve profile and flexfield expressions in SQL statements • Specify the output format, template language, and template territory settings for the layout template • Specify a list of Oracle users to notify when the job completes • Identify request sets, single request programs, and Oracle users by display name or short name	11.3.5
Updated Application and Group Level Security In previous releases, if you defined the application or group attributes in your job definition, security checks were performed to validate execute access when the job started. However, these security checks were not performed against jobs when they were submitted through the cross-platform interface. In Release 11.3.5, CA Workload Automation AE also performs application and group level security checks against jobs when they are submitted through the cross-platform interface.	11.3.5

autoaggr Command -- Report Aggregated StatisticsIn r11 and r11.3, the autoaggr command was used to aggregate statistics into the hourly, daily, weekly, and monthly tables that other programs use to generate reports. In Release 11.3.5, the autoaggr command generates reports based on the aggregated job, alarm, and scheduler statistics retrieved from the database. You can aggregate the job, alarm, and scheduler statistics into the database tables either manually by using the sendevent command or automatically by configuring the AggregateStatistics parameter in the configuration file. Notes: • In r11 and r11.3, the total_latency statistics were collected. In Release 11.3.5, the total_latency statistics are not collected. Instead, the MAX LATENCY, AVERAGE LATENCY, MAX LAG TIME, and AVERAGE LAG TIME statistics are collected. • In r11 and r11.3, the total_events and total_latency statistics were stored in the jc_jrestart_n and jc_quewait_n database columns respectively. When you upgrade from r11 or r11.3 to a later release, the values in the jc_jrestart_n database column are copied to the new tc_events_n database column and are then zeroed out. The aggregation process does not generate statistics into the jc_quewait_n, jc_jedit_n, js_activated_n, and js_activated_p database columns and so these database columns are dropped in the Release 11.3.5.	11.3.5
Updated as_info Command The -C option is added to the as_info command to return the configuration information for all CA Workload Automation AE instances installed on a computer.	11.3.5
Updated autoflags Command The -f option is added to the autoflags command to write the fully qualified host name to standard output.	11.3.5
Updated autosys_secure command The autosys_secure command is updated to let you enable and disable external security in batch mode. The addition of the -secadmu option lets you to authenticate autosys_secure with external security as a user with administrative privileges. The addition of the -e parameter lets you to enable external security and generate a CA EEM certificate. The addition of the -n parameter lets you to disable external security and revert to native security.	11.3.5
Updated autosyslog Command The autosyslog command is updated to display the following data: • The spool file for a job. Use the S argument with the -T option. • The available z/OS job log types and dataset names. Use the -j parameter with the -z option. • The z/OS job log dataset. Use the -d parameter.	11.3.5

Updated CA EEM Security Policy Authorizations for Jobs In Release 11.3.5, CA Workload Automation AE performs more security policy authorizations for jobs than in previous releases. So, you must create additional security policies to authorize users to do the following tasks: • To define, update, override, or delete a job in a box: – If you are defining a job, create a policy in the as-base-jobtype resource class authorizing EXECUTE access to the job_type attribute value that represents a predefined CA Workload Automation AE job type. – If you are updating a job, create a policy in the as-job resource class authorizing EXECUTE access to the existing unmodified box_name attribute value to modify the contents of the box with that name. – Create a policy in the as-job resource class authorizing EXECUTE access to the box_name attribute value to modify the contents of the box with that name. – Create a policy in the as-base-jobtype resource class authorizing EXECUTE access to the predefined CA Workload Automation AE BOX job type to modify the box with the name represented by the box_name attribute. • To update, override, or delete a job belonging to an application or a group: – If you are updating a job, create a policy in the as-appl resource class authorizing WRITE access to the existing unmodified application attribute value. – Create a policy in the as-appl resource class authorizing WRITE access to the application attribute value. – If you are updating a job, create a policy in the as-group resource class authorizing WRITE access to the existing unmodified group attribute value. – Create a policy in the as-group resource class authorizing WRITE access to the group attribute value.	11.3.5
Updated EP_SHUTDOWN Alarm The EP_SHUTDOWN alarm is raised when the active scheduler (either the primary scheduler or the shadow scheduler after it takes over processing events while running in high availability mode) is shutting down because of a normal shutdown process or an error condition.	11.3.5
Updated Highly Available Cluster Environment Options Release 11.3.5 can cooperate with cluster management software to form a clustered server, a clustered database, or a clustered agent. A clustered server is an alternative to high-availability mode and provides a more stable failover solution. A clustered database is an alternative to dual event server mode and is recommended if your database vendor is cluster-aware. A clustered agent complements the load-balancing capabilities of the scheduler and is recommended when you are running jobs that execute client utilities, use shared resources or have high CPU consumption. We recommend setting up a clustered server before you set up a clustered agent. You cannot set up the clustered agent on the same cluster as the clustered server. In Release 11.3.5, you can also specify a manager host alias for the clustered server.	11.3.5
Wake on LAN (WOL) Job Type Lets you define jobs to send a signal to a server to turn it on.	11.3.5

Web Service Document/Literal (WSDOC) Job Type Lets you define jobs that call an operation within a web service and pass parameters to the operation using document/literal style binding.	11.3.5
AE Architecture Lets you host web services. The web server component is installed and configured as part of the CA Workload Automation AE installation.	11.3.5
Append Event Message Text in the Scheduler Log File The AppendEventMessageText parameter in the configuration file lets you specify whether the text associated with an event is appended to the corresponding event message or printed as a standalone message in the scheduler log file.	11.3
Change in as_owner Policy Validation CA Workload Automation AE validates the as_owner policy using either the owner specified by the owner attribute in the job definition or the default owner of the job.	11.3
Changes to Cross-Instance Job Dependencies Cross-instance job dependencies have changed as follows: - To improve efficiency, external events are stored in a new database table named ujo_ext_event. - To define cross-instance job dependencies between r11.3 and 4.5, a new lightweight application server that supports 4.5 is installed on the r11.3 instance. - You can define cross-instance job dependencies between CA Workload Automation AE instances that have different encryption settings. You specify the external instance's encryption key using the xcrypt_type attribute when you define the instance to CA Workload Automation AE.	11.3
Command (CMD) Job Type Lets you define jobs to run workload on UNIX and Windows client computers.	11.3
Command Substitutions Not Allowed in the watch_file Attribute In 4.0 and r4.5, you could use back ticks or the grave accent (` ) to specify command substitutions in the watch_file attribute. In r11 and r11.3, you cannot use back ticks when you specify the path in the watch_file attribute. For example, the watch_file attribute cannot resolve the date if you specify the path as follows: watch_file: `tmp`date`	11.3
CPU Monitoring (OMCPU) Job Type Lets you define jobs to monitor the CPU usage of the computer where the specified agent is installed.	11.3
Cross-Instance Job Dependencies with CA Workload Automation EE Lets you define and monitor cross-instance (external) job dependencies between CA Workload Automation AE and CA Workload Automation EE. These job dependencies let you create job flows between distributed and mainframe systems.	11.3
Database Monitor (DBMON) Job Type Lets you define jobs to monitor a database table for an increase or decrease in the number of rows.	11.3
Database Stored Procedure (DBPROC) Job Type Lets you define jobs to invoke a procedure stored in a database.	11.3

Database Trigger (DBTRIG) Job Type Lets you define jobs to monitor a database table for added, deleted, or updated rows.	11.3
Disk Monitoring (OMD) Job Type Lets you define jobs to monitor the available or used space on a disk or logical partition.	11.3
Encryption and FIPS 140-2 Compliance In r11, CA Workload Automation AE used SSL encryption. In r11.3, CA Workload Automation AE uses Advanced Encryption Standard (AES) encryption to comply with the U.S. Government encryption standard FIPS 140-2. This standard requires a FIPS-certified library and FIPS-certified cipher algorithm, such as AES. Note: For r11.3, CA Workload Automation AE supports AES encryption only. While the product still supports running SSL encryption over SSA, the preferred data encryption method is AES (SSL encryption under SSA is disabled by default when SSA is installed).	11.3
Entity Bean (ENTYBEAN) Job Type Lets you define jobs to create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database.	11.3
File Transfer Protocol (FTP) Job Type Lets you define jobs to automate FTP transfers.	11.3
Define a File Trigger Job Lets you define jobs to monitor file activity.	11.3
Hypertext Transfer Protocol (HTTP) Job Type Lets you define jobs to invoke a program over HTTP.	11.3
i5/OS (I5) Job Type Lets you define jobs to schedule workload to run on an i5/OS system.	11.3
Improve Log Maintenance The LOGROLLOVER parameter in the configuration file lets you specify when the scheduler or the application server logs roll over.	11.3
IP Monitoring (OMIP) Job Type Lets you define jobs to monitor an IP address or a port at an IP address.	11.3
IPv6 Support r11.3 supports IPv6 between CA Workload Automation AE r11.3, CA 7, CA Workload Automation EE, and agents. JIL accepts IPv6 addresses in addition to hostnames and IPv4 addresses. For example, when you use the jil command to define a new machine, you can specify an IPv6 address in the machine attribute.	11.3
Java Remote Method Invocation (JAVARMI) Job Type Lets you define jobs to call a method on a remote server and store the output of the method.	11.3
JMS Publish (JMSPUB) Job Type Lets you define jobs to send a message to a queue or publish a message to a topic.	11.3
JMS Subscribe (JMSSUB) Job Type Lets you define jobs to consume messages from a queue or topic.	11.3

JMX-MBean Attribute Get (JMXMAG) Job Type Lets you define jobs to query a JMX server for the value of an MBean attribute.	11.3
JMX-MBean Attribute Set (JMXMAS) Job Type Lets you define jobs to change the value of an MBean attribute on a JMX server.	11.3
JMX-MBean Create Instance (JMXMC) Job Type Lets you define jobs to create an MBean on a JMX server.	11.3
JMX-MBean Operation (JMXMOP) Job Type Lets you define jobs to invoke an operation on an MBean.	11.3
JMX-MBean Remove Instance (JMXMREM) Job Type Lets you define jobs to remove an MBean from a JMX server.	11.3
JMX-MBean Subscribe (JMXSUB) Job Type Lets you define jobs to monitor an MBean for a single notification or monitor continuously for notifications.	11.3
KILLJOB and SEND_SIGNAL Behavior The following new alarms are generated when the scheduler experiences a problem communicating with the agent while killing a job or sending a signal: • KILLJOBFAIL – generated when the attempt to kill a job fails • SENDSIGFAIL – generated when the attempt to send a signal fails Although Windows agents do not support sending signals to jobs, you can signal a named Windows semaphore. To signal a named Windows semaphore, set the value of the opsys attribute for a type-a machine to the Windows operating system ('windows').	11.3
Legacy Agent Replaced by the Workload Automation Agent Workload Automation Agent for UNIX, Linux, or Windows is replaced by the legacy remote agent that was available for r4.5 and r11. By default, all new machine definitions are set to type "a" . Type "a" machines represent the Workload Automation Agent and require new machine attributes representing the agent name, type of data encryption, encryption key, and so on. For information about the differences between the legacy agent and the agent, see Differences Between the Legacy Agent and the Agent .	11.3
Log a Job's State Changes and the New STATE_CHANGE event Some of the job types go through different state changes when they run. For example, a z/OS Regular job can go through state changes for each step that runs. The scheduler log file records the job's state changes using the STATE_CHANGE event. These events are also displayed when you create a report using the autorep -J -d command. The report includes the events that are generated during the most recent job runs.	11.3

Manual Intervention for i5/OS Job Types and the New REPLY_RESPONSE Event You can define an i5/OS job to schedule workload to run on an i5/OS system. The job can run a program or an i5/OS command. You can run i5/OS jobs in the root file system, open systems file system (QOpenSys), and library file system (QSYS). A program run on an i5/OS system might require additional feedback from the end user before it can continue execution. The Workload Automation Agent for i5/OS notifies the scheduler when a manual response is required. In this case, the scheduler raises a WAIT_REPLY_ALARM and places the job in a new state: WAIT_REPLY. The text of the WAIT_REPLY_ALARM contains the query of the i5/OS program and might show the expected responses. You must send a REPLY_RESPONSE event with a valid response in order for the job to proceed. When the scheduler experiences a problem communicating with the Workload Automation Agent for i5/OS to send the REPLY_RESPONSE event, the REPLY_RESPONSE_FAIL alarm is raised. The Workload Automation Agent for i5/OS resumes sending job status updates to the scheduler upon receipt of an accepted response.	11.3
Monitor a Condition Continuously and the New ALERT event Lets you define the following job types to monitor a condition continuously: CPU Monitoring (OMCPU), Database Monitor (DBMON), Database Trigger (DBTRIG), Disk Monitoring (OMD), File Trigger (FT), Text File Reading and Monitoring (OMTF), Windows Event Log Monitoring (OMEL), and Windows Services Monitoring (OMS). Each time the specified condition occurs, an ALERT event is written to the scheduler log file. These events are also displayed when you create a report using the autorep -J -d command. The report includes the events that are generated during the most recent job runs. To stop a continuous monitor, you must complete the job manually by issuing the sendevent -E KILLJOB command.	11.3
Must Start Times and Must Complete Times Lets you specify the must start times and must complete times so you get notifications when a job has not started or completed on time. The must_start_times attribute lets you define the time or a list of times that a job must start by. The must_complete_times attribute lets you define the time or a list of times that a job must complete by.	11.3
New as_test Command A test utility that can run for a specified amount of time, write a message to either stdout or stderr or to both stdout and stderr, and exit with a specific exit code.	11.3
New archive_jobs Command Deletes obsolete job versions from the database.	11.3
(Windows only) New autoprofm Command Converts job profiles to the new format.	11.3
forecast Command -- Report Job Flow Reports job flows based on a date or date range.	11.3
New reindex.pl Script for Rebuilding Database Table Indexes lets you rebuild the table indexes of a specified CA Workload Automation AE database.	11.3

New Scheduler Startup Parameters In the previous release, the eventor command lets you specify whether the scheduler starts in Global AutoHold mode (eventor -G option) and whether to run the chase command at startup (eventor -n option). These two options have been removed from the eventor command. Instead, you can now control the startup behavior of the scheduler by using the new GlobalAutoHold and ChaseOnStartup parameters in the configuration file.	11.3
Oracle E-Business Suite Copy Single Request (OACOPY) Job Type Lets you define jobs to copy an existing single request that is defined on Oracle E-Business Suite and run it under the agent.	11.3
Oracle E-Business Suite Request Set (OASET) Job Type Lets you define jobs to run a request set program.	11.3
Oracle E-Business Suite Single Request (OASG) Job Type Lets you define jobs to run a single request program.	11.3
PeopleSoft (PS) Job Type Lets you define jobs to schedule workload to run in PeopleSoft.	11.3
Plain Old Java Object (POJO) Job Type Lets you define jobs to create a Java object instance with no arguments, invoke a method on the object instance, and store the output of the method.	11.3
Poll for Resource Availability The ResourceWaitPollInterval parameter in the configuration file lets you define how frequently the scheduler polls for resource availability when jobs are waiting on resources.	11.3
Pound Sign (#) Allowed in Object Names The pound sign or hash character (#) is allowed in all object names (for example, job and calendar names).	11.3
Process Monitoring (OMP) Job Type Lets you define jobs to monitor the status of a process on the computer where the agent is installed.	11.3
Removed Commands The following commands have been removed from r11.3: • autodwp • autosys_report—The autosysreport.exe application is now part of CA Workload Automation AE Administrator (Windows only). • autosys_wv • job_delete—This command is replaced by the archive_jobs command. • job_profiles—This application is now part of CA Workload Automation AE Administrator (Windows only). • ntgetdate • xql • zql	11.3
SAP Batch Input Session (SAPBDC) Job Type Lets you define jobs to import large amounts of data from external systems to the SAP system.	11.3
SAP BW InfoPackage (SAPBWIP) Job Type Lets you define jobs to transfer data from any data source into an SAP Business Warehouse system.	11.3

SAP BW Process Chain (SAPBWPC) Job Type Lets you define jobs to run a sequence of background processes on the SAP system.	11.3
SAP Data Archiving (SAPDA) Job Type Lets you define jobs to store information that is described in an SAP Archiving Object into an SAP data archive.	11.3
SAP Event Monitor (SAPEVT) Job Type Lets you define jobs to schedule workload based on the activity of an SAP event or trigger an SAP event at the appropriate time in your schedule.	11.3
SAP Job Copy (SAPJC) Job Type Lets you define jobs to copy an existing SAP R/3 job.	11.3
SAP Process Monitor (SAPPM) Job Type Lets you define jobs to monitor for a specific SAP process status and end after detecting a process.	11.3
SAP R/3 (SAP) Job Type Lets you define jobs to schedule an SAP R/3 job on your SAP system.	11.3
Secure Copy (SCP) Job Type Lets you define jobs to transfer binary files using the Secure Copy Protocol.	11.3
Session Bean (SESSBEAN) Job Type Lets you define jobs to access a stateless or stateful session bean, invoke a method on the bean, and return the results.	11.3
Specify an Instance-Wide Encryption Key The UseEncryption parameter in the configuration file lets you define the instance-specific encryption key for all communication between the CA Workload Automation AE components of the same instance.	11.3
Specify the Localhost In r11.3, the localhost machine name is a reserved name. You can no longer define a machine for localhost by creating an insert_machine: localhost definition. By default, the localhost value is resolved to the name of the machine where the CA Workload Automation AE scheduler was started. You can override the reserved localhost value to the name of another real machine by using the LocalMachineDefinition parameter in the configuration file.	11.3
Structured Query Language (SQL) Job Type Lets you define jobs to run an SQL query against an Oracle, SQL Server, Sybase, or DB2 database.	11.3

Supported Agents/Agent-plug-ins Supports new agents and agent plug-ins that let you automate, monitor, and manage workload on all major platforms, applications, and databases. To run workload on a particular system, you must install an agent on that system and add a machine definition to CA Workload Automation AE. You can install multiple agents on the same machine. Each agent on that machine must have a unique name and port number. The following agents are supported: • Workload Automation Agent for i5/OS • Workload Automation Agent for Linux • Workload Automation Agent for UNIX • Workload Automation Agent for Windows • Workload Automation Agent for z/OS You can extend the functionality of the agent by installing one or more agent plug-ins into the agent installation directory. If you have a relational database such as Oracle, for example, you can install a database agent plug-in to query and monitor the database. The following agent plug-ins are supported: • Workload Automation Agent for Application Services • Workload Automation Agent for Databases • Workload Automation Agent for PeopleSoft • Workload Automation Agent for Oracle E-Business Suite • Workload Automation Agent for SAP • Workload Automation Agent for Web Services Note: The agent plug-ins are only available for UNIX, Linux, and Windows operating environments.	11.3
Support for Virtual Resources Lets you define virtual resources to CA Workload Automation AE and specify those resources as job dependencies.	11.3
Text File Reading and Monitoring (OMTF) Job Type Lets you define jobs to search a text file on a Windows, UNIX, or i5/OS computer for a text string.	11.3
Updated autoping Command The -S option is added to the autoping command to test the connectivity between the application server and the new Workload Automation Agent. If you issue autoping -M <i>machine</i> -S against the legacy agent, the command reverts to its previous behavior and tests the database connectivity between CA Workload Automation AE and the agent.	11.3
Updated clean_files Command The clean_files command now applies to legacy agent log files only.	11.3
Web Service (WBSVC) Job Type Lets you define jobs that call an operation within a web service.	11.3
Windows Event Log Monitoring Jobs (OMEL) Job Type Lets you define jobs to monitor a Windows event log in the computer where the agent is running.	11.3
Define a Windows Service Monitoring Job Lets you define jobs to monitor a service on a Windows computer where the agent is running.	11.3
z/OS Data Set Trigger (ZOSDST) Job Type Lets you define jobs to create dependencies on data set activities.	11.3

z/OS Manual (ZOSM) Job Type Lets you define jobs to create dependencies on z/OS jobs that are submitted outside the scheduling manager, such as a job that a user submits manually .	11.3
z/OS Regular (ZOS) Job Type Lets you define jobs to schedule a z/OS job.	11.3

Release History (WCC)

The following table lists the features that were implemented from CA Workload Automation WCC r11.4 SP1 to the latest release:

Features	Release Implemented
Updated Key Length for EEM Server Certificate CA WCC now requires root certificates with a key length of 2048 or more.	11.4 SP7
Agent Inventory The Agent Inventory display lets you see and export information about each separately installable component (system agent, plug-ins, and advanced integrations). Agent information is retrieved periodically from CA Workload Automation AE and displayed in Agent Inventory.	11.4 SP6
Sending Email Notifications Using Template You can now send an email notification using the email notification template when a job fails, terminates, completes successfully, or when an alarm is raised.	11.4 SP6
Machine Method to Achieve Load Balancing With Virtual Machines You can use the Machine Method field in the Virtual Machine properties section in Quick Edit to achieve load balancing.	11.4 SP6
Distinguish Alarms Based on the Failure and Terminated Job Status Alarms on the FAILURE and TERMINATED job statuses can now be individually specified.	11.4 SP6
Overwrite option for Calendars in Export/Import You can now select the Overwrite option when importing a calendar from a text file. This option helps you to overwrite an existing calendar when you import a calendar.	11.4 SP6
CA Remote Engineer CA Workload Automation WCC supports CA Remote Engineer r4.0. The CA Remote Engineer tool collects configuration and diagnostic data that helps you to troubleshoot problems with the system or software. In this release, you can integrate CA Remote Engineer with CA Telemetry Service and this integration helps to capture customer environment and topology information. For more information about CA Remote Engineer and CA Telemetry Service integration, see the CA Remote Engineer documentation.	11.4 SP6
Microsoft SQL Server Encryption Support WCC can be installed with MS SQL Server 2016 supporting encryption.	11.4 SP6

LDAP Authentication for Informatica Jobs The Informatica domain property in the job definition authenticates the user against the specified LDAP security domain that is configured with the Informatica Server.	11.4 SP6
Updated Limits for the Command Property The limits for the Command property is now increased from 512 to 1000 characters. This limit change is applicable if the CA Workload Automation AE server that is configured in CA WCC is at Release 11.3.6 SP7 and later.	11.4 SP6
Application Editor - Changes and Enhancements The changes and enhancements to the Application Editor application in this release are available here .	11.4 SP6
Informatica Job Type Informatica jobs let you start predefined workflows on the Informatica PowerCenter. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.	11.4 SP5
SQL Server Agent Job Type SQL Server Agent jobs let you execute predefined jobs in the SQL Server Agent to perform administrative tasks.	11.4 SP5
Database Trigger Job Type PostgreSQL and Teradata databases are supported.	11.4 SP5
Support SQL Server Always On CA Workload Automation WCC now supports Microsoft SQL Server Always On feature.	11.4 SP5
CA WCC Upgrade From Older Versions CA WCC supports upgrade from 11.3.5, 11.3.6, 11.4, 11.4 SP1, 11.4 SP2, 11.4 SP3, or 11.4 SP4 to current version 11.4 SP5.	11.4 SP5
New Fields for JMS Publish and JMS Subscribe Jobs The following fields are added for the JMS Publish and JMS Subscribe jobs: - correlation_id (JMS Publish jobs only) Reply to - Destination user name	11.4 SP5
New Fields for FTP, SCP, and SFTP Jobs The following fields are added for the FTP, SCP, and SFTP jobs: - FTP - Run as user - SCP - Transfer type and Run as user - SFTP - Transfer type, File operation, and Run as user	11.4 SP5
Global Variables Global variables substitution is now supported by more properties, simplifying job administration.	11.4 SP5
New as-sendevent Resource Class The as-sendevent resource class and its default CA EEM policy are added to control access to execute the sendevent command: CA WCC functionality allows for hiding the send event actions on the UI that is based on the CA Workload Automation AE Resource Class as-sendevent.	11.4 SP4
Multiple Dashboard Tab Creation A user can create and remove Dashboard tabs, alter a Dashboard tab title, and select a choice of a primary Dashboard.	11.4 SP4

Creation of Multiple Jobs Widget Instances A user can create various jobs widgets in the dashboard that allow the user to simultaneously monitor jobs from different CA Workload Automation AE instances, monitoring views, and grid filters.	11.4 SP4
CA WCC Monitoring Enhancement of Jobs Filter Preferences • A Monitoring preference added into Configuration allows an empty Jobs Widget filter in the Servers section. • By default it is checked. If the checkbox is unchecked it will force the users to use the Jobs filter in all Jobs Widgets for Servers, otherwise no data will be shown.	11.4 SP4
New Attributes for Oracle E-Business Suite Request Set and Oracle E-Business Suite Single Request Jobs The following attributes are added for the Oracle E-Business Suite Request Set and Oracle E-Business Suite Single Request jobs: • oracle_oprunit • oracle_template_name	11.4 SP4
CA WCC Enhanced Security Using New Cryptography To enhance security of CA WCC, the CA WCC server is now using a X-XSS-Protection header with value "1;mode=block" for HTTP responses. In addition, CA WCC is now using Elliptic-curve cryptography for https connections as a preferred option. For normal cryptography operation, DH group of 2048-bit is now being used.	11.4 SP4
CA WCC Login and Logout Audit CA WCC administrators have a log file containing user login and logout events.	11.4 SP4
CA WCC Upgrade From Older Versions CA WCC supports upgrade from 11.3.5, 11.3.6, 11.4, 11.4 SP1, 11.4 SP2, or 11.4 SP3 to current version 11.4 SP4.	11.4 SP4
CA WCC Support of CA Workload Automation AE Version 11.3.6 SP5 CA Workload Automation AE 11.3.6 SP5 is supported by CA WCC 11.4 SP4.	11.4 SP4
Job Search By Command Attribute in Quick Edit A new command field is provided in Quick Edit to enhance search queries. The command jobs are filtered by its command attribute.	11.4 SP4
Job Search By Multiple Job Names in Quick View Enhancement of the Quick View job name search field that accepts multiple job names. The job names are separated by a comma, new line separator, or a space. Each job name supports wildcards.	11.4 SP4
Enhanced Scheduler Log Functionality An additional search mode has been created to find a pattern and display a subsequent number of lines as specified. Enhanced navigation allows you to move to a next or previous option and highlights the matching pattern.	11.4 SP4
Interval Specification for File Trigger Job Type For File Trigger job type it is possible to specify the watch_no_change attribute in seconds, minutes, hours, or days.	11.4 SP4

Support Global Resources for Box Job You can now specify global resources on a box job. However, the resource specified for a box applies only to the box and not to the jobs within the box.	11.4 SP4
Monitoring Expand/Collapse Flows Synced Expand/Collapse All functionality, in a particular view, is supported through the flow toolbar buttons and is synced to the job tree when these flow actions are triggered.	11.4 SP4
CA WCC Performance Improvements for Bigger Loads CA WCC startup load time of a flow model is faster as collection speed is improved when many views are defined. Collection time is decreased when adding a server with pre-existing view definitions.	11.4 SP3
CA WCC Security Improvements CA WCC allows the usage of single sign-on wallets with certificates that are present on the file system to access the database. The database access mode can be set to password or SSL certificate either during installation or later.	11.4 SP3
Oracle PKI Authentication Connection to Oracle database now is supported using Public Key Infrastructure (PKI) for authentication.	11.4 SP3
Update to Tree Navigation in Monitoring Update to the UI and tree structure allow for a more efficient customer experience while navigating CA WCC Monitoring. Three tabs concerning Views, Servers, and Alerts have been added to the left navigation pane. The tree navigation provides the same context menu and action functionality as previously found in the table view.	11.4 SP3
Forecast Reports Export The Table Views of Forecast Reports can be exported into .csv format.	11.4 SP2
Embedded Security Against Hijacking of Clicks CA WCC 11.4 SP2 prevents this version from being embedded into frames for increased security against Clickjacking. Clickjacking (UI Redressing) is a malicious technique that utilizes hidden layers of website frames to trick you into clicking on the disguised buttons.	11.4 SP2
Hadoop Job Types CA WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define new job types using CA WA Advanced Integration for Hadoop.	11.4 SP2
Connection Profiles A connection profile is a set of attributes that are required for connecting to environments where the job runs. It also lets you save the connection information and you can use the connection profile in the job definition.	11.4 SP2
CA WCC Upgrade From Older Versions CA WCC supports upgrade from 11.3.x and 11.4 or 11.4 SP1 to current version 11.4 SP2.	11.4 SP2
CA WCC Support of AE Version 11.3.6 SP3 AE 11.3.6 SP3 is supported by CA WCC 11.4 SP2.	11.4 SP2

Using Urandom as Default RNG Source of Entropy On Linux and Unix systems, CA WCC installer and CA WCC application are using /dev/urandom as a default source of entropy for its non blocking pool behaviour.	11.4 SP2
Forbidden Characters in Install Path The forbidden characters "%", "(", and ")" are not to be used. Use of these characters will not be permitted in the installer.	11.4 SP2
Enterprise Command Line Import and Export Supports Jobs, Calendars, Machines, Resources, and Connection Profiles The Enterprise Command Line application lets you export one or more jobs, a single or all calendars, machines, resources, or connection profiles residing on a CA Workload Automation AE server to a file. You can modify the calendar, job, machine, resource, or connection profile definitions using a text editor, and then import the modified file to any CA Workload Automation AE server you specify.	11.4 SP2
Addition of Machines Widget to Dashboard A Machines Widget is added to the Dashboard functionality.	11.4 SP1
Auto-Refresh Persistent Functionality The auto-refresh check box in the Dashboard and Monitoring widgets remains persistent between user sessions.	11.4 SP1
Widget Filter Persistent Functionality The visibility of the widget filters in Dashboard and Monitoring are persistent between user sessions and is turn ON by default.	11.4 SP1
Alert Defined in Alert Policy Dialog You have the option to select view names from the View pane to assign an alert that is defined to one or more existing views.	11.4 SP1
Default time zone for CA WCC Servers You have the option to select time zone which will be used as default for CA WCC users. Users are still able to customize the time zone using My Profile dialog.	11.4 SP1
Description Field for Calendar Calendar, extended calendar or cycle has a new Description field with limit of 1024 characters, if you are using only CA AE 11.3.6 SP2 (and any later versions).	11.4 SP1
Servers widget displays detail of connection failure Servers widget on CA WCC Dashboard displays detailed error information in case the connection to CA AE instance fails. It is necessary to hover the mouse cursor over the status icon which will trigger the tooltip with detailed error message.	11.4 SP1
Secure Cookies for HTTPS For HTTPS connections, all cookies are flagged as secure.	11.4 SP1
CA WCC Upgrade from 11.3.x and 11.4 to 11.4 SP1 CA WCC supports upgrade to 11.4 SP1.	11.4 SP1
Colon Support in Job Names Colon (:) support is introduced for New Job Names and Default Job Names.	11.4 SP1
Help File Consolidation The CA WCC Helps are consolidated, that enables you to search through all Help content. Also if Internet access is enabled link to the full on-line doc on the wiki DocOps platform is available from top of each Help page.	11.4 SP1

Monitoring Monitoring has the following enhancements for this release: • A command is displayed in the Command Job tooltip located in the flow diagram. • The Search job field in Monitoring Flow is containing tooltip with example. • The visibility of the widget filter is persistent between user sessions and is turn ON by default. • The number of rows in the widget filter is persistent between user sessions. • The auto-refresh option in widgets remain persistent between user sessions.	11.4 SP1
Quick View Quick View has the following enhancements for this release: • The Maximum Flow Height limits of the Job Flow section are increased to 1500 • A command is displayed in the Command Job tooltip located in the flow diagram.	11.4 SP1
Auto Refresh No Longer Effects Session Timeout Operations in CA WCC Auto refresh is no longer a user action when you are idle in the application and remain logged in.	11.4 SP1
Application Editor Re-Inclusion The Application Editor is available for Release 11.4 SP1.	11.4 SP1
HTTPS Enabled for HA Configuration It is possible to install a new HA node using HTTPS. Only supported configuration for an HA cluster is to have all the nodes in either HTTP or HTTPS.	11.4 SP1
Simplification of EEM Settings in the Installer When using EEM, in failover mode, the comma separated list of the EEM servers are no longer required. Instead, a single EEM server (possibly the primary server) is required. The other servers in the failover mode are automatically retrieved and displayed. The installer will verify the network access and the state of all EEM servers. In case there is an issue, a dialog with a summary will be displayed.	11.4 SP1
Change in Calendar Name Length Limits The character length for the name of a calendar, extended calendar or cycle has a limit of 64 characters, if you are using only CA AE 11.3.6 SP2 (and any later versions).	11.4 SP1
CA Workload Automation AE Server Version Auto Discovery During validation, there is an automatic discovery of the CA Workload Automation AE server version. Instead of an error message, the CA Workload Automation AE version number is updated.	11.4 SP1
Enterprise Command Line and Forecast commands Used by CA WCC Logged On Users Context CA WCC Enterprise Command Line and Forecast run under user context of the currently logged in user to CA WCC, not under the user defined in Credentials. Credentials user is used only to access AE machine, but the remotely launched commands from Enterprise Command Line and Forecast are run under CA WCC user.	11.4 SP1

Status of Cross Instance Dependencies CA WCC displays dependencies on jobs running on External Instances. The status of the Cross Instance Job is visible in the Job Dependencies Table in QuickView and the flows in Monitoring and QuickView.	11.4 SP1
Fix for Forecast DST issue WCC Forecast displays correct data for periods involving DST shift periods. The prerequisite is to set time zone in CA AE instance definition in CA WCC Configuration.	11.4 SP1
Controlling EEM Caching Behavior When controlling EEM SDK caching behavior, you can switch on/off the full cache updates and set the interval between the cache updates. The caching behavior can be controlled by the connection.properties file located under the CA WCC root directory.	11.4 SP1

AutoSys Support for Other Broadcom Products

This article documents other Broadcom products supported by AutoSys Workload Automation.

CA EEM Support

AutoSys Workload Automation supports CA EEM Release 12.6. EEM includes the following features:

- Scalability
- Event Management and Persistence
- Reliable Event Delivery
- Authentication

You can install CA EEM using the CA Common Components DVD.

NOTE

- On Windows, if you are upgrading from a 32-bit EEM to a 64-bit EEM or migrating from CA EEM Release 8.x to 12.6, follow the instructions in the [CA Common Components Upgrade](#) page.
- If the iGateway built on Linux 2.4 kernel version is already installed, EEM, iGateway, and all other iGateway plug-ins cannot be installed or upgraded. If \$IGW_LOC/igw.linux_k26 file is present, then the iGateway built on Linux 2.6 kernel version is installed. In the case where the iGateway built on Linux 2.4 kernel version is already installed, uninstall CA EEM, iGateway, and all other iGateway plug-ins, and then install these products using the CA Common Components DVD. EEM policies can then be migrated to the new EEM installation. For information about migrating CA EEM policies, see [Security Policy Customization](#). For information about upgrading EEM, see [EEM documentation](#).
- For more information about installing EEM, see [CA Common Components Installation](#).

Secure Socket Adapter Support

AutoSys Workload Automation supports CA Secure Socket Adapter (SSA) 2.4.1. SSA is an application that lets Broadcom components use a single multiplexed communication port to ease firewall administration and minimize conflicts with other applications. SSA is installed automatically during the AutoSys Workload Automation installation.

NOTE

For more information about configuring port numbers and settings, see [Configure AutoSys Workload Automation to Work with Other Broadcom Products](#).

CA Service Desk Support

AutoSys Workload Automation supports CA Service Desk r12.9 and 14.1. AutoSys Workload Automation lets you open a service desk ticket (request or incident) when a job fails.

NOTE

For more information about configuring AutoSys Workload Automation to work with CA Service Desk, see [Configure AutoSys Workload Automation to Work with Other Broadcom Products](#).

CA NSM and CA UJMA Support

AutoSys Workload Automation supports CA Network and Systems Management Job Management Option (CA NSM JMO) and CA Universal Job Management Agent (CA UJMA). CA UJMA lets AutoSys Workload Automation schedule jobs to and receive job submissions from CA NSM JMO. AutoSys Workload Automation can also submit jobs to CA NSM JMO.

WARNING

When you install AutoSys Workload Automation components on a server that has any component of CA NSM r3.1 installed, you must follow these rules:

- If you do not want to perform cross-platform scheduling using AutoSys Workload Automation, you can install AutoSys Workload Automation on a server that has CA NSM r3.1 installed.
- After you install AutoSys Workload Automation on a server, you can no longer install any CA NSM r3.1 components on that server. The CA NSM r3.1 components must be installed first.
- The AutoSys Workload Automation scheduler (excluding the client and agent) only supports CA NSM r11.2 SP2 CUM1 components, including the Job Management Option. The r11.2 SP2 CUM1 components are supported because CAICCI r11.x is required to perform cross-platform scheduling.

NOTE

For more information about configuring cross-platform scheduling, see [Configuring Cross-Platform Scheduling](#).

CAPKI

AutoSys Workload Automation supports CAPKI 5.2.5. CAPKI is required to implement Information Security services in products.

AutoSys Workload Automation, Web UI, and CA EEM Compatibility

The following table displays the releases of AutoSys Workload Automation, Web UI, and CA EEM that are compatible:

Web UI Release	AutoSys Workload Automation Release	CA EEM Release	Notes
12.0	12.0	r12.6 and higher	These products must run either in FIPS mode or non-FIPS mode.
11.4. SP7	11.3.6 SP8	12.6.05	These products must run either in FIPS mode or non-FIPS mode.
11.4 SP6	11.3.6 SP7	12.6	These products must run either in FIPS mode or non-FIPS mode.
11.4 SP5	11.3.6 SP6	12.51 CR05	These products must run either in FIPS mode or non-FIPS mode.

11.4 SP4	11.3.5 to 11.3.6 SP5	12.51 CR04	These products must run either in FIPS mode or non-FIPS mode.
11.4 SP1	11.3 SP1 or later	12.51 CR03	AutoSys Workload Automation r11.3 SP1 does not support FIPS mode. So, these products must run in non-FIPS mode.

Note:

- In this topic, AutoSys Workload Automation refers to any or all components that can be installed using the AutoSys Workload Automation DVD, AutoSys Workload Automation Standalone DVDs (UNIX), or the Workload Automation Agent DVD.
- To install these products on the same or separate computers, install CA EEM first, then AutoSys Workload Automation, and then Web UI. If these products are installed on the same computer, ensure that AutoSys Workload Automation and Web UI releases are at the same level (see table below). We recommend that you install CA EEM, AutoSys Workload Automation components, Web UI, and additional agents on separate computers as follows:

Figure 1: AutoSys Workload Automation, Web UI, and EEM compatability**Non High Availability Implementation of AutoSys Workload Automation**

- To upgrade these products installed on separate computers, upgrade CA EEM first, then Web UI, and then AutoSys Workload Automation.
- To upgrade these products installed on the same computer, update EEM to use 2048 certificates and perform the upgrade.
- CA EEM Release 12.6.2.0 supports RHEL Version 7 or higher. You can upgrade to AutoSys Version 12.0 with CA EEM Release 12.6.0.5 on RHEL 6, but cannot log in to CA EEM.
- The following table displays the AutoSys Workload Automation and Web UI releases that are at the same level:

Web UI Release	AutoSys Workload Automation Release
11.1 SP3	11.0 SP5
11.1 SP4	11.0 SP5
11.3 SP1	11.3 SP1
11.3.5	11.3.5
11.3.6	11.3.6
11.3.6 INCR3 (RO77628 + RO80613)	11.3.6 SP1
11.4 SP1	11.3.6 SP2
11.4 SP2	11.3.6 SP3
11.4 SP3	11.3.6 SP4
11.4 SP4	11.3.6 SP5
11.4 SP5	11.3.6 SP6

11.4 SP6	11.3.6 SP7
11.4 SP7	11.3.6 SP8
12.0	12.0

AutoSys - Previous Release Notes

This section contains the Release Notes of all the earlier releases of AE, WCC, and CA Common Components.

AE Release Notes

This section contains the Release Information for the following AutoSys Workload Automation releases and service packs:

AE Release 11.3.6 SP8

This section includes information about the CA Workload Automation AE Release 11.3.6 SP8 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 SP8.

New Features - 11.3.6 SP8

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP8.

Telemetry

[Telemetry](#) is a capability that is integrated into CA Workload Automation AE to send product usage and system configuration data to CA Technologies, a Broadcom Company (CA).

NOTE

- Telemetry does not collect Personally Identifiable Information (PII) or sensitive information.
- For information about configuring CA Workload Automation AE to collect and send telemetry data to CA, see [Configure CA Workload Automation AE to Collect and Send Telemetry Data to CA](#).
- Telemetry is not supported on HP-UX Itanium and HP-UX PA-RISC operating environments.
- You can also extract the telemetry data offline by using the telemetry.bat (Windows) or telemetry.sh (UNIX) script. For more information about the telemetry script, see [telemetry Script—Extracts Telemetry Data Offline](#).

IPAddressPreference—New Parameter in the Configuration File

The [IPAddressPreference](#) parameter is added to the configuration file. This parameter specifies the criteria that CA Workload Automation AE uses to select an IP address from the list of IP addresses that are returned by the operating system after translating the host name.

Changes to Existing Features - 11.3.6 SP8

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP8.

Web Service Self-signed Certificate

In this release, a new CA Workload Automation AE web service self-signed certificate based on Bouncy Castle FIPS is created and stored in the %AUTOUSER%\webserver\conf\keystore (Windows) or \$AUTOUSER/webserver/conf/.keystore (UNIX) file.

When you upgrade from any previous release to Release 11.3.6 SP8 or later, if an error occurs when the installer is migrating the JKS keystore file to the BCFKS keystore file, the existing JKS keystore file is backed up as follows:

- (Windows) %AUTOUSER%\webserver\conf\keystore is backed up as %AUTOUSER%\webserver\conf\keystore_backup.
- (UNIX) \$AUTOUSER/webserver/conf/.keystore is backed up as \$AUTOUSER/webserver/conf/.keystore_backup.

CA Remote Engineer is Removed from CA Workload Automation AE

Beginning with Release 11.3.6 SP8, CA Remote Engineer is removed from CA Workload Automation AE.

Fixed Issues - 11.3.6 SP8

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP8.

NOTE

CA Workload Automation AE Release 11.3.6 SP8 also includes the issues fixed in each of the prior releases of 11.3.6.

CHK_AUTO_UP AND DBSPACE FAIL IN SYBASE KERBEROS

Problem Number: 4620

When you use Kerberos authentication for Sybase database and you specify a Service Principal name that differs from the server name in the configuration file, the chk_auto_up and dspace commands failed to connect to the database.

AUTOSYSLOG RETURNS LOG FILE FOR JOB NOT FOUND

Problem Number: 4712

If a job has std_out_file file as part of a long running job that started several weeks ago and is still in the running status, then when you run the DBMaint script, you cannot retrieve the std_out_file file using the autosyslog -J job -tO command.

SILENT INSTALL DOES NOT WORK - PREDEFINED USERS AND TS

Problem Number: 4741

CA Workload Automation AE Release 11.3.6 SP7 installer on UNIX failed when you try to install the product in silent mode and select not to create the users and tablespaces as part of the installation. The installer tries to connect to the database with an invalid user and password combination in the pre-installation phase and exits with an error.

ROUNDROBIN FEATURE NOT WORKING CORRECTLY

Problem Number: 4744

When MachineMethod is set to 'roundrobin' or 'roundrobin_jobload', the scheduler functionality is affected:

1. If real machines in a virtual machine are offline, then those machines are likely to be selected repeatedly instead of in a round robin manner.
2. Jobs using global resources go into RE status instead of RW status when resources are unavailable.
3. Jobs using resources and running on real machines run out of turn when resources are unavailable.
4. When a job that is running on a virtual machine is force started and all the machines are offline, the job goes into RE status due to unavailability of machine instead of selecting one of the offline machine to run the job on.

WEBSERVICE: JOB-RUN-INFO DOES NOT GET ALL ROWS FOR COUNT**Problem Number: 4746**

The CA Workload Automation AE web services component did not get all records for the job-run-info request when the count option is used. The following request starts retrieving lesser records after the job is updated:

```
AEWS/job-run-info/<job_name>?runcount=<number>
```

ARCHIVE_EVENTS WRITING DUPLICATE INFO TO ARCHIVE FILE**Problem Number: 4748**

The DBMaint command failed to archive job runs data.

JOB INSERT FAILS WHEN ATTR USES \$IBS IN ENVVAR AFTER PR4664**Problem Number: 4750**

When you define a job with two consecutive job attribute values that specify environment variables and the environment variable value of the first attribute begins with the "\$IBS" or "\$IBI" text, the application server failed to insert the job with a database error.

QUE_WAIT JOB STUCK IN STARTING AFTER LOAD BALANCE JOB ENDS**Problem Number: 4751**

When you define load balancing jobs that execute on the same virtual machine and you manually change the status of one of the jobs from the RUNNING status to the INACTIVE status and then to the SUCCESS status after other jobs are in the QUE_WAIT status, the scheduler placed a queued job in the STARTING status but did not execute it.

ABLE TO CREATE JOB ON NON-EXISTING MACHINE**Problem Number: 4752**

After you delete a machine using the delete_machine command, you are still able to insert jobs to run on the deleted machine.

JIL SYNTAX CHECK RETURN FALSE PASS**Problem Number: 4753**

Custom job validation failed but JIL syntax checker returns syntax check passed with warnings.

SENDEVENT CHANGE_PRIORITY CHECKING FOR WRITE ACCESS**Problem Number: 4755**

The sendevent -E CHANGE_PRIORITY -j *jobname* command checked the as-job write access instead of checking for the as-job execute access.

APPSERVER HANGS WHEN RESPONDING TO CLIENTS DURING SHUTDOWN**Problem Number: 4756**

When you shut down the application server while it is sending responses to clients, the application server process hung indefinitely.

UNIQUE CONSTRAINT AEDBADMIN.XPKUJO_RESWAIT_QUE AFTER PR4479**Problem Number: 4757**

When multiple predecessor jobs that trigger the same successor job having virtual resources complete simultaneously and virtual resources are not available, the scheduler tried to place the job into RESWAIT status twice resulting in the following database errors:

```
CAUAJM_E_18416 Event Server: <CAAP> Failed Query: <INSERT INTO ujo_reswait_que(joid,
priority, stamp, run_num, ntry, wf_joid) VALUES (xxxx, 0, nnnnn, mmm, 1, 1)>
CAUAJM_E_18402 ORA-00001: unique constraint (AEDBADMIN.XPKUJO_RESWAIT_QUE) violated
CAUAJM_I_18403 Processing OCI function OEXEC, OEXN(3)
CAUAJM_E_18400 An error has occurred while interfacing with ORACLE.
CAUAJM_E_18401 Function <doExecute> invoked from <execute> failed <862>
```

UNPROCESSED EVENTS IN DUAL SERVER MODE**Problem Number: 4758**

When the CA Workload Automation AE scheduler loses communication with both the event servers for a few minutes, then the scheduler might leave some events in the ujo_event table unprocessed. These events are not picked for processing until the scheduler is restarted.

SCHEDULER CRASH ON INVALID AFM**Problem Number: 4759**

Scheduler cores dump while processing an invalid or corrupted AFM that is coming from an agent. The AFM starts with few junk characters and does not commit to the AFM protocol.

JIL SYNTAX CHECK FALSELY REPORTS EXTENDED CALENDAR DOES NOT EXIST**Problem Number: 4760**

The JIL syntax checker falsely reported that the extended calendar does not exist, even though it existed.

SERVICES CRASH ON SUSE LINUX 12 SP2,SP3**Problem Number: 4761**

The CA Workload Automation AE scheduler and application server crash while running on Suse Linux 12 SP2 or Suse Linux 12 SP3 with the following error:

```
event_demon: ../sysdeps/unix/sysv/linux/createthread.c:120: create_thread:
Assertion `stopped_start' failed.
```

SCHED DELAYS AGENT COMM FOR SECURE SOCKETS ADAPTER CONFIG**Problem Number: 4762**

The scheduler delayed communication with agents to start jobs as it interrogates the Secure Sockets Adapter (SSA) port configuration.

DBSTATISTICS NOT REFRESHING SYBASE INDEX DATA**Problem Number: 4764**

The dbstatistics command did not update the Sybase index statistics. This resulted in job latencies up to 2 minutes.

SCHEDULER RAN A JOB TWICE AFTER TAKING IT OFF HOLD**Problem Number: 4765**

When the scheduler gets delayed by starting many load balancing jobs at one time and you place a quick running job on hold and then off hold it, the scheduler started the job multiple times.

UPGRADE DATABASE FOR AGGREGATE RUNS FOREVER**Problem Number: 4766**

When you upgrade your CA Workload Automation AE instance to Release 11.3.6 SP6 and the tables that are associated with the aggregation of statistics have many rows, the installer completes after many minutes.

BINS ON IPV4 NETWORK DO NOT CHOOSE IPV4 ADDRESS TO CONNECT**Problem Number: 4768**

Suppose that the application server resides in an IPv6 network and its hostname resolves to both an IPv4 and an IPv6 address. Clients that reside in an IPv4 only network always resolved the application server hostname to its IPv6 address and failed to connect to it.

Also, suppose that the client resides in an IPv6 network and its hostname resolves to both an IPv4 and an IPv6 address. Application servers that reside in an IPv4 only network always resolved the client hostname to its IPv6 address and failed to connect to it.

ARCHIVE_EVENTS RUNS FOREVER**Problem Number: 4769**

The CA Workload Automation AE archive_events utility runs too long.

OPTIMIZE COMMUNICATIONS WITH WA SYSTEM AGENT RELEASE 11.4**Problem Number: 4770**

When the scheduler or application server communicates with the CA Workload Automation System Agent Release 11.4, it took longer to retrieve the network data from the socket than when it communicates with the CA Workload Automation System Agent Release 11.3.

SCH EXPERIENCES DUAL EVENT SYNC DEADLOCK AFTER KILLING JOBS**Problem Number: 4772**

When you configure a CA Workload Automation AE instance to run in dual event server mode and you define many jobs in boxes that specify the job_terminator attribute, as the scheduler terminates boxes and jobs, it detected a dual event synchronization deadlock condition and shuts itself down gracefully.

JOB WITH RESOURCE RUNS ON OFFLINE MACHINE**Problem Number: 4774**

The CA Workload Automation AE scheduler did not check the offline status of a real machine when a previous run of the same job ran on a machine that is placed in the offline state.

The scheduler log contains the following contradicting lines:

```
CAUAJM_I_10960 Resource dependency evaluations for job <test_ren_res_dep>:
CAUAJM_I_10964 Global virtual resource <ren_res> is already reserved from a previous run
on machine <machine-01>.
CAUAJM_I_10961 The resource manager successfully satisfied all resource dependencies for
job <test_ren_res_dep>.
CAUAJM_I_10208 Machine usages obtained for job <test_ren_res_dep>:
CAUAJM_E_10199 Machine <machine-01> is unavailable because it is not online
CAUAJM_I_10549 Machine <machine-01> selected by Resource Manager.
CAUAJM_I_10556 Machine <machine-02>: Method <CPU monitor>. Usage ignored because this is
the only available machine.
CAUAJM_I_10209 Machine <machine-01> selected for job <test_ren_res_dep>.
```

SILENT AGENT INSTALL FROM SERVER DVD FAILS W/ EXIT CODE 119

Problem Number: 4775

Silent installation of the agent from the CA Workload Automation AE Release 11.3.6 SP7 Server DVD failed with exit code 119.

SHADOW FAILOVER WHILE PRIMARY EVALS JOB STARTS ON SHUTDOWN

Problem Number: 4776

When you configure the CA Workload Automation AE instance to run in high availability mode and you shut down the primary scheduler while it is processing the completion event of a job that has many downstream jobs, the shadow scheduler failed over and ran while the primary scheduler is still running.

SHADOW SCH DOES NOT SEND QUEUED EVENTS TO EXTERNAL INSTANCE

Problem Number: 4777

When you configure the CA Workload Automation AE instance to run in high availability mode and the external instance is not available during the run of the primary scheduler, after the shadow scheduler failed over, it did not send events to external instances that were queued by the primary scheduler.

Third-Party Software Acknowledgements - 11.3.6 SP8

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

AE Release 11.3.6 SP7

This section includes information about the CA Workload Automation AE Release 11.3.6 SP7 new features, changes to existing features, and fixed issues.

WARNING

IMPORTANT!

On UNIX, if you are upgrading from a 32-bit CA EEM to a 64-bit CA EEM and CA EEM and CA Workload Automation AE (with the command sponsor component) are installed on the same computer, you must upgrade CA EEM first using the CA Common Components DVD only and then immediately upgrade CA Workload

Automation AE. If CA EEM and CA Workload Automation AE are installed on separate computers, the limitation does not apply and so you can upgrade CA EEM using the CA Common Components DVD or the Standalone CA EEM DVD.

On Windows, if you are upgrading from a 32-bit CA EEM to a 64-bit CA EEM or migrating from CA EEM Release 8.x to 12.6, follow the instructions in the [CA Common Components Upgrade](#) page.

New Features - 11.3.6 SP7

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP7.

TIP

<https://communities.ca.com/videos/6698> and <https://communities.ca.com/videos/6699>: These videos provide detailed information about the new features and changes to existing features in CA Workload Automation AE Release 11.3.6 SP7 and CA WCC Release 11.4 SP6.

Machine Methods to Achieve Load Balancing

You can now use the [machine_method](#) attribute in your job definition to specify any of the following machine methods to achieve load balancing: `cpu_mon`, `vmstat`, `rstatd`, `job_load`, `roundrobin`, or `roundrobin_jobload`.

The [MachineMethod](#) parameter in the configuration file is updated to include the following two new machine methods:

- `roundrobin`—uses the round robin method to determine the real machine where the job runs from the list of available real machines belonging to a virtual machine.
- `roundrobin_jobload`—uses the `job_load` method first and then the `roundrobin` method is used on the resulting real machines to determine the real machine where the job runs. However, if none of the machines have sufficient load units, CA Workload Automation AE puts the job in `QUE_WAIT` status. The job stays in `QUE_WAIT` status until one of the machines has sufficient load units available. When sufficient load units are available, the job runs on that machine. In this situation, the `roundrobin` method is not considered.

NOTE

If you set the `MachineMethod` parameter value and also specify the `machine_method` attribute in your job definition, the `machine_method` attribute value overrides the value you specified for the `MachineMethod` parameter.

Sending Email Notifications Using Template

You can now [send an email notification](#) using the email notification template that you specified in the [notification_template](#) attribute when a job fails, terminates, or completes successfully or when an alarm is raised.

To send an email notification using an email notification template, set the `send_notification` attribute value to **y** or **f** and specify the `notification_template` and the `notification_emailaddress`, `notification_emailaddress_on_success`, `notification_emailaddress_on_failure`, `notification_emailaddress_on_terminated`, or `notification_emailaddress_on_alarm` attributes in your job definition based on the job status or alarm you want to send the email notification for.

The following attributes are added to send email notifications using template:

- [notification_template](#)—Specifies the email notification template that is used to send the email notification.
- [notification_emailaddress_on_success](#)—Specifies the email addresses of the recipients of the email notification when the job completes successfully.
- [notification_emailaddress_on_failure](#)—Specifies the email addresses of the recipients of the email notification when the job fails.
- [notification_emailaddress_on_terminated](#)—Specifies the email addresses of the recipients of the email notification when the job terminates.
- [notification_emailaddress_on_alarm](#)—Specifies the email addresses of the recipients of the email notification when an alarm is raised. You can specify the alarm for which you want to send an email notification using the [notification_alarm_types](#) attribute. In a job run, for each alarm type, you can now use the [RepeatAlarmNotification](#) parameter to configure whether you want to send an email notification for all alarms, just the first alarm, or only the last alarm.

emailtemplate Blob

You can create an emailtemplate blob using the [insert_glob subcommand](#) (specify the [blob_type](#) attribute value as emailtemplate) and use it in your job definition or in the [notification template](#) that is used to send the email notification.

Distinguish Alarms Based on the Failure and Terminated Job Status

The following changes are made to distinguish alarms based on the FAILURE and TERMINATED job statuses:

- Updated [alarm_if_fail](#) attribute: Specifies whether to post an alarm only when the job completes with a FAILURE status. In previous releases, the [alarm_if_fail](#) attribute specified whether to post an alarm when a job completes with FAILURE or TERMINATED status.
- New [alarm_if_terminated](#) attribute: Specifies whether to post the JOBTERMINATED alarm when the job completes with a TERMINATED status.

NOTE

- When you insert a job, the [alarm_if_terminated](#) attribute is set to y. That is, the JOBTERMINATED alarm is posted to the scheduler when a job is terminated.
- For existing jobs, when you upgrade, if the [alarm_if_fail](#) attribute is set to n, then the [alarm_if_terminated](#) attribute is also set to n. However, if the [alarm_if_fail](#) attribute is set to y, then the [alarm_if_terminated](#) attribute is also set to y. So, any user actions that you might have set for the JOBFAILURE alarm might have to be revisited to accommodate the JOBTERMINATED alarm.
- New JOBTERMINATED (552) alarm: Indicates that a job is terminated.
- New [trapJobTerminated](#) (552) SNMP trap: Indicates that the job terminated.

Specify the Agent Status Poll Interval

You can specify the time interval (in minutes) between status polls that the scheduler waits before polling all the agents. On UNIX, you can configure this setting using the [AgentStatusPollInterval](#) parameter in the configuration file. On Windows, you can configure this setting using the Agent Status Poll Interval field on the Scheduler - CA Workload Automation AE Administrator window of the Administrator utility.

NOTE

When you define machines, if the [agent_name](#) and [node_name](#) attributes are the same and the ports are different, the machine goes to a Missing or Unqualified state. We recommend that you look for machines that are accessible from autoping but are placed in the Missing or Unqualified state by the scheduler and correct the machine definitions to specify the correct port.

Integrate CA Workload Automation AE with CA Automatic One Automation Platform

You can now integrate CA Workload Automation AE with CA Automatic One Automation Platform to automate manual problem resolution tasks, filter and consolidate multiple events, monitor for unusual conditions, and take proper corrective action.

NOTE

- For information about integrating CA Workload Automation AE with CA Automatic One Automation Platform, see [Integrate with CA Automatic One Automation Platform](#).
- In previous releases, you integrated CA Workload Automation AE with Event Management to automate manual problem resolution tasks, filter and consolidate multiple events, monitor for unusual conditions, and take proper corrective action. In the current release, you can integrate CA Workload Automation AE with CA Automatic One Automation Platform or Event Management to automate manual problem resolution tasks, filter and consolidate multiple events, monitor for unusual conditions, and take proper corrective action.

New Attribute for Informatica Jobs

The following attribute is added for Informatica jobs:

- [informatica_security_domain](#)—Specifies the name of the LDAP security domain.

ComponentRegister Utility

The [ComponentRegister](#) utility registers the REST web server that is installed with CA Workload Automation AE and updates the database.

New update_blob and update_glob SubCommands

The following subcommands are added to update the content of a blob or glob:

- [update_glob](#)—Updates the content of an existing glob.
- [update_blob](#)—Updates the content of an existing blob.

TAF Certification

CA Workload Automation AE can now run in Transparent Application Failover (TAF) mode. When the database fails over, CA Workload Automation AE resumes working with the failed over database.

NOTE

- For information about configuring the tnsnames.ora file for TAF enablement, see [Install CA Workload Automation AE with Oracle](#).
- When the Oracle client environment is configured for TAF, the [DBEventReconnect](#) parameter defines the number of times the Oracle client rechecks the failover status.

Update Security Policies After Migration

You can now use the [as_safetool](#) command to update the security policies after migration.

The following options are added to the [as_safetool](#) command:

- [-m](#)—Updates the security policies after migration.
- [-v version](#)—Specifies the CA Workload Automation AE release level from which the security policies are imported.

NOTE

For information about updating the migrated security policies, see [Migrate CA Workload Automation AE Security Policies from Release 11.3](#).

CA Workload Automation AE Certifications

CA Workload Automation AE now supports the following databases and operating systems:

- Microsoft SQL Server 2016
- Microsoft Windows 10 (client only)
- Windows 2016 (x86 64bit)
- RHEL on Amazon Cloud with RDS
- Amazon Linux 09
- Microsoft SQL Azure

Changes to Existing Features - 11.3.6 SP7

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP7.

Updated send_notification Attribute

You can now use the [send_notification](#) attribute to send an email notification when an alarm is raised. In the previous releases, an email notification is sent when the job completes successfully, fails, or terminates. To send an email notification using the email notification template when an alarm is raised, set the `send_notification` attribute value to **y** or **f** and specify the `notification_template`, `notification_alarm_types`, and the `notification_emailaddress_on_alarm` or `notification_emailaddress` attributes in your job definition.

To send an email notification using the email notification template when a job completes successfully or terminates, set the `send_notification` attribute value to **y** and specify the `notification_template` and the `notification_emailaddress`, `notification_emailaddress_on_success`, or `notification_emailaddress_on_terminated` attributes in your job definition.

To send an email notification using the email notification template when a job fails, set the `send_notification` attribute value to **f** and specify the `notification_template` and the `notification_emailaddress` or `notification_emailaddress_on_failure` attributes in your job definition.

Updated notification_emailaddress Attribute

You can now use the `notification_emailaddress` attribute to send an email notification when an alarm is raised. You can specify the alarm for which you want to send an email notification using the [notification_alarm_types](#) attribute.

Updated blob_type Attribute

You can now specify the [blob_type](#) as generic or emailtemplate.

Increase autosys_secure Password Limit

The limit for the password that you specify when you create a user using the `autosys_secure` command is increased from 20 characters to 127 characters. The database user password limit is 20 characters.

Updated Limits for the command Attribute

The limits for the [command](#) attribute is now increased from 512 to 1000 characters.

Updated as-owner Resource Class

The [as-owner resource class](#) is updated to include the `SENDEVENT_JOBEXECUTE` access mode that controls whether you can execute the `sendevent` command.

Search is Case Insensitive for the Description Field

- In web services, you can now do a case insensitive search for the description field while searching for jobs, connection profiles, or machines.
- In SDK, you can now do a case insensitive search for the description field while searching for jobs, connection profiles, machines, calendars, or virtual resources.

Updated autotrack Command

The following options are added to the [autotrack](#) command:

- *-B group*—To define the group about which to report.
- *-I application*—To define the application about which to report.
- *-R resource*—To define the virtual resource about which to report.

CA Remote Engineer

CA Workload Automation AE supports [CA Remote Engineer](#) r4.0. The CA Remote Engineer tool collects configuration and diagnostic data that helps you to troubleshoot problems with the system or software. In this release, you can integrate CA Remote Engineer with CA Telemetry Service and this integration helps to capture customer environment and topology information.

NOTE

For more information about CA Remote Engineer and CA Telemetry Service integration, see the [CA Remote Engineer](#) documentation.

Global Variable Substitution

You can now reference a [global variable](#) as part of the syntax of the following attributes:

- [hadoop_hdfs_destination](#)
- [hadoop_hdfs_path](#)
- [hadoop_hdfs_source](#)
- [informatica_folder_name](#)
- [informatica_param_file](#)
- [informatica_repository_name](#)
- [informatica_security_domain](#)
- [informatica_target_name](#)
- [informatica_task_name](#)
- [informatica_workflow_inst_name](#)
- [informatica_workflow_name](#)
- [sqlagent_jobname](#)
- [sqlagent_server_name](#)
- [sqlagent_step_name](#)
- [sqlagent_target_db](#)
- [ws_parameter](#)

Retrieve Jobs with jobtype Filter Using Web Services

You can now use web services to retrieve jobs with the jobtype filter. For more information about retrieving jobs with the jobtype filter using web services, see [Manage Jobs](#).

Change the Job Status Using Web Services

You can now use web services to change the status of a job. For more information about changing the status of a job using web services, see [Manage Jobs](#).

Installation Pre-checks

The installer performs some pre-checks to verify the aedbadmin privileges, database privileges, ports, operating system versions, and database versions and libraries and displays appropriate warning or error messages. This helps prevent any post-installation failures.

On UNIX, the installer also verifies if you created an instance by manually copying an existing instance environment and then modified the copied environment to assign it a new instance name and displays an error message. For information about converting a manually copied instance environment into an environment that can be upgraded successfully, see [\(UNIX\) Convert a Manually Copied Instance Environment into an Environment that can be Upgraded Successfully](#).

Machine Status Changes

- In the previous releases, when you insert a machine, the machine status is set to online. From this release onwards, when you insert a machine, the scheduler calibrates to the actual machine status and sets the machine status accordingly. Also, when you update the port, node_name, encryption_type, agent_name, or key_to_agent machine attribute, the scheduler calibrates to the actual machine status and sets the machine status accordingly.
- In the previous releases, when you issue the MACH_ONLINE event, the machine status is set to online. From this release onwards, the MACH_ONLINE event places the machine in the online status after the scheduler calibrates that the machine is actually available and accepts jobs to run. If the machine is not available, the machine status is set to missing or blocked.

Fixed Issues - 11.3.6 SP7

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP7.

TIP

CA Workload Automation AE Release 11.3.6 SP7 also includes the issues fixed in each of the prior releases of 11.3.6.

NOTE

IMPORTANT! The following issues that are fixed in Release 11.3.6 SP7 help reduce the time and resources it takes for the database server to execute the same set of repeating SQL queries that differ only in input values:

- [SCHEDULER ORACLE QUERY FOR THE EXISTENCE OF EVENTS TIMES OUT](#): Improves the set of SQL queries that the scheduler uses to check for the existence of events that belong to a job run to prevent it from starting a concurrent run.
- [SQL INJECTION VULNERABILITY](#): Improves the set of SQL queries that the application server uses to send an event. The fix also ensures that the application server does not accidentally execute malicious SQL queries that are embedded in the input values of the sendevent command that is issued by hackers.
- [Problem Number 4688 - AUTOREP PERFORMANCE POOR WITH DEFAULT CURSOR_SHARING](#) : Improves the set of SQL queries that the application server uses to generate reports and export definitions.

CHASE DOES NOT REPORT CROSS-PLATFORM JOBS IN STARTING STATE

Problem Number: 4648

The chase utility did not report cross-platform jobs that are in the STARTING status for more than 120 seconds.

SCHEDULER GENERATES CORRUPTED EVENT ID STRINGS ON SHUTDOWN**Problem Number: 4650**

When the scheduler is shut down while it is communicating with agents, the scheduler process generated invalid event identification values for job events. Also, when the scheduler processes more than 1,296 incoming job updates from an agent that executes a monitor-type job, the scheduler generated invalid event identification values for job events.

AUA GENERATING WRONG DATES**Problem Number: 4652**

AutoSys Upgrade Assistant(AUA) using 12-hour format, while migrating calendars to destination database, due to 12 hour format standard calendars migrated with default time as 12:00, extender calendar generating dates incorrectly, when standard holiday calendar has time as 12:00.

DBSTATISTICS RUNS ON ALL SERVERS DURING INSTALLATION**Problem Number: 4653**

Installer runs DBStatistics on all three (Primary, Shadow and Tie-breaker) servers.

USER DEFINED TIMEZONE INFO LOST DURING INCREMENTALS**Problem Number: 4654**

User defined timezones and alias are lost during upgrade to any incremental or patches. Even running RefreshAEDB.pl manually to update the procedures removes all the timezone information.

UJO_POST_VIRT_RESOURCES SP ERROR IN DUAL SERVER BLOCKS JOBS**Problem Number: 4655**

Suppose you define a job having a dependency on a virtual resource. Also, suppose you define a job having a dependency on two virtual resources where you specify the same virtual resource as the first job and list it as the second of the two virtual resource dependencies. When the CA Workload Automation instance is configured to run in dual event server mode and the scheduler evaluates the starting conditions of both jobs simultaneously, the scheduler experiences database errors while trying to reserve the common virtual resource units.

SCHEDULER CRASHES TRYING TO RESOLVE IP ADDRESS**Problem Number: 4656**

The scheduler stopped responding when the IP address resolution of an agent failed.

MACHINE NAME ATTRIBUTE MISSING FROM UJO_JOBST**Problem Number: 4657**

ujo_jobst view created by the installer does not include machine attribute used as part of the job definition. You find it hard to query database directly with new schema and metadata used by CA Workload Automation AE.

INSTALL_INCR.SH FAILED ON CLIENT SDK INSTALLATION**Problem Number: 4659**

The Install_INCR.sh script failed to apply Release 11.3.6 SP6 CUM1 on the client and SDK installation.

SCHEDULER HANGS AFTER RECONNECTING WITH THE DB AFTER PR4600**Problem Number: 4660**

After the scheduler's database monitoring system successfully reconnects to the database after detecting the temporary loss of the database, the scheduler experienced an internal deadlock condition and stopped all workflow activity.

PASSWORD FOR PROXY USER IS NOT SENT FOR HTTP JOB TYPE**Problem Number: 4661**

When an HTTP job is defined with the j2ee_proxy_user attribute, the job failed with the <Invalid format,null> message.

SCHEDULER DOES NOT CONSISTENTLY RUN QUICK SELF-DEPENDENT BOX**Problem Number: 4662**

When you define a small box that is dependent on itself containing quick running jobs one of which is dependent on another job in the box, the scheduler did not execute the job with the dependency after one or more box runs during a time of heavy job activity.

SCHEDULER FAILS TO START WITH ORA-24374**Problem Number: 4663**

CA Workload Automation AE Scheduler fails to start after upgrading from pre-SP5 to latest version with following error messages in the log:

```
CAUAJM_E_18416 Event Server: <ASYDA7MD> Failed Query: <SELECT queue_id, queue_type, sequence, stamp, afm,
s.afm_queue_id, s.afm_num, events, source FROM ujo_afm a JOIN ujo_afm_strings s ON a.afm_num=s.afm_num AND
a.afm_queue_id=s.afm_queue_id ORDER BY s.afm_queue_id ASC NULLS FIRST , s.afm_num ASC NULLS FIRST ,
s.piece ASC NULLS FIRST >.
```

SCHEDULER ORACLE QUERY FOR THE EXISTENCE OF EVENTS TIMES OUT**Problem Number: 4664**

When you configure the CURSOR_SHARING=FORCE Oracle database server parameter to improve the processing of similar queries, the scheduler delayed scheduling jobs in boxes that specify a condition attribute. This happened because the queries for CHANGE_STATUS events in the ujo_proc_event or ujo_event table timed out.

JOBS W/ RESRC STUCK ORA-24347 NULL COLUMN IN AGGREGATE FUNC**Problem Number: 4665**

When you define jobs with resource dependencies, the scheduler experienced a query failure while evaluating the job starting conditions and did not execute the job.

INSTALL CRASHES WHEN PROCESSING SYBASE INTERFACES FILE**Problem Number: 4666**

On UNIX, the CA Workload Automation AE installation failed while processing the Sybase database interface file that listed thousands of servers.

WEBSERVICE JOB RUN INFO - NEXT PAGE ERROR**Problem Number: 4667**

CA Workload Automation AE Web Service for Job-Run-Info generates HTTP 500 error when the link generated for "Next".

DB_PROBLEM ALARM FOR QUERY ID #92 IN DUAL EVENT SRV INSTANCE**Problem Number: 4668**

When you define multiple machines that specify different node_name attribute values that refer to the same network address and then try to execute the autoping or chase commands, the application server or scheduler raised the DB_PROBLEM alarm for each command invocation.

UPGRADE TO SP6 FAILED TO COMPLETE AND LOOP IN REFRESHAEDB.PL**Problem Number: 4669**

Upgrade to CA Workload Automation AE Release 11.3.6 SP6 from an earlier version leads to an infinite loop in the database upgrade phase. Log confirms that the RefreshAEDB.pl script loops for ever.

REINDEXDB.PL INCOORECT USAGE OUTPUT ON INVALID ARGUMENTS**Problem Number: 4670**

The reindexDB.pl script displayed incorrect usage information when you pass invalid arguments.

SENDEVENT WITH -M OPTION NOT WORKING AS EXPECTED**Problem Number: 4671**

When you specify sendevent -M option with no value followed by -T option, the event processes immediately instead of displaying sendevent usage.

JOB_DEPENDS MAN PAGE CORRECTIONS**Problem Number: 4672**

job_depends man page has no information related to -e option.

PRE-1136SP2 DB UPDATE FAILS DUE TO UJO_COMM_SEND_ALIAS QUERY**Problem Number: 4673**

Suppose that the version of your CA Workload Automation AE Release 11.3.6 instance is less than CA Workload Automation AE Release 11.3.6 SP2 and you upgrade your instance to the current version of CA Workload Automation AE Release 11.3.6. When you run the upgrade installer, the upgrade failed because it was unable to update the ujo_comm_send_alias table.

TIMEOUT OF ORACLE QUERY TO REQUE EVENTS PREVENTS SCH STARTUP**Problem Number: 4674**

When you restart the scheduler, it was not able to startup and shut itself down immediately. This happened because the query that the scheduler made to retrieve previously fetched and unprocessed events from the ujo_event table experienced timeout errors.

APPLICATION SERVER CRASHED WHILE PROCESSING A BOX JOB**Problem Number: 4675**

When responding to a request to fetch a job definition, application server crashed while processing a box job with the box_success attribute exceeding 256 characters.

SQL INJECTION VULNERABILITY**Problem Number: 4676**

CA Workload Automation AE SQL is vulnerable to SQL injection when you send event operations from CA WCC webservices. The following URLs are affected:

```
/jsc-rest/sendevent/FORCESTART (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/KILL (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/SEND_SIGNAL (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/OFFHOLD (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/OFFICE (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/OFF_NOEXEC (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/RELEASE_RESOURCE (jobs[0][ name ] JSON parameter)
/jsc-rest/sendevent/START (jobs[0][ name ] JSON parameter)
```

SCH LONG DELAYS FETCHING DB EVENTS DUE TO EVENTS FROM AGENTS**Problem Number: 4678**

When you define a job with date and time conditions and the scheduler is processing many jobs, it delayed in starting the job on time when many agents sent events to the scheduler.

CREATEAEDB.PL FAILS WHEN INDEX WITH SAME COLUMNS EXIST**Problem Number: 4679**

The CreateAEDB.pl script failed with ORA-01408 when an index being created is already present in the schema.

11.3.6 SP6 WINDOWS INSTALL FAILS AGAINST ORACLE 12C**Problem Number: 4680**

CA Workload Automation AE windows installation fails with the below error against Oracle 12c which did not have container feature that is enabled. The specified Oracle service ID is a container database which cannot be used for the CA Workload Automation AE installation. Provide Pluggable Database to proceed further.

LEGACY AUTOCAL_ASC DOES NOT DISPLAY DATES IN APPENDED MODE**Problem Number: 4681**

When you try to append the dates using legacy procedure, the dates were not displayed.

AUTOREP ESCAPES DOUBLE QUOTES FOR I5_PARAMS ATTRIBUTE**Problem Number: 4682**

AutoRep output to display the job definition for i5 job_type escaped the i5_param attribute when the value is enclosed in double quotes, but JIL does not remove the escaping and returns a failure while importing the job definition generated by autorep command.

I5_OTHERS ATTRIBUTE DOES NOT ALLOW SPACES**Problem Number: 4683**

JIL does not allow creation of an i5 job with space in i5_others attribute. i5_others attribute allows wrapping the value in single quote but AFM request retains the single quote which causes problems at agent.

AUTOTRACK TRACK LEVEL TRUNS TO ZERO WHEN INPUT IS INVALID**Problem Number: 4684**

Autotrack set track level to zero when using invalid input parameter as input for the option u.

UPDATE ISDBGACTIV=SEC TO MEASURE PERFORMANCE**Problem Number: 4685**

When the CA Workload Automation AE instance is configured to run under external security and you enable ISDBGACTIV=SEC traces, the application server log did not display any useful information.

REDUCE SCHEDULER MEMORY UTILIZATION WHEN RUNNING MANY BOXES**Problem Number: 4686**

When you define and schedule many boxes with jobs, the scheduler consumed large quantities of system memory.

SCHED DEADLOCKS WHEN RUNNING MULTILEVEL BOXES AFTER PR4606**Problem Number: 4687**

When you define and schedule boxes with multiple inner boxes, the scheduler deadlocked.

AUTOREP PERFORMANCE POOR WITH DEFAULT CURSOR_SHARING**Problem Number: 4688**

On the Oracle server, when you change the cursor_sharing parameter from FORCE to EXACT, the autorep command took more time to fetch the job information. This problem is fixed and now the time taken by the autorep -j ALL - q command to fetch the job information is as follows:

EXACT WITHOUT FIX	EXACT WITH FIX
42 minutes	13 minutes

NOTE

When cursor_sharing parameter is set to FORCE, the CA Workload Automation AE performance is degraded in the long run and also FORCE increases the load on the Oracle server. Hence, we do not recommend you to use FORCE.

The autorep performance testing was performed on the Red Hat Enterprise Linux Server Release 6.7 with Oracle 12c database.

NOTE

CA Workload Automation AE and database server are on different machines.

The following artifacts were used for testing the autorep performance:

Following machines were used for testing the autorep performance:

- 1K machines
- 100 virtual machines (each VM has 10 machines)
- 1K machines with load
- 100 virtual machines with load (each VM has 10 machines with load)
- 100 renewable resources
- 100 machine renewable resources (each resource is inserted with 10 machines)

About 50K jobs were used to run the performance test that was defined to generate a report. The following list states the distribution of 50K jobs that were used to run the performance test:

- 5K jobs on real machines
- 5K jobs on real machines with condition attribute
- 5K jobs on virtual machines
- 5K jobs on real machines with load
- 5K jobs on virtual machines with load
- 5K jobs with resources
- 5K jobs with machine resources
- 5.5K of all job types (each job type has 100 jobs)
- 10K jobs in a box (number of levels: 100, each level has 100 jobs and a box)

USER CANNOT DELETE A JOB EVEN WHEN PERMISSIONS ARE PRESENT

Problem Number: 4689

You failed to delete a job even with proper permissions. The following error message is returned when an attempt is made to delete a

job:

CAUAJM_W_10225 Only the CA WAAE EDIT superuser can set the owner.

JOBS STUCK IN QUE_WAIT INDEFINITELY

Problem Number: 4690

Intermittently jobs having job_load and priority that is stuck in QUE_WAIT state indefinitely even though job_load and priority are met later.

DORMANT SCHED CORES ROLLING DB OVER AFTER ACTIVE SCHED

Problem Number: 4691

When you configure your CA Workload Automation AE instance to run in dual event server mode, and the primary scheduler rolls over to Event Server B after losing connectivity with Event Server A, the primary scheduler notifies all other schedulers and application server to also roll over to Event Server B. When the shadow scheduler attempted to roll over to Event

Server B, it cored. Similarly, when you configure the PrimaryFailbackMode configuration file variable to Dormant, and the primary scheduler functions in dormant mode after the shadow scheduler fails over, the same issue occurs with the scheduler roles reversed. When the primary scheduler attempted to roll over to Event Server B after the shadow rolled over to Event Server B, it cored.

INFINITE LOOP IN APP SERVER AFTER ORACLE UNRECOVERABLE ERROR

Problem Number: 4692

The application server reports a potential problem after Oracle fails to run a query and returns ORA-01652. In such cases, the application server should exit and administrators must address the issue, but the application server continuously displayed the same error and filled the disk space within minutes that caused more damage to the system resources.

The following error is repeated in applicaiton server log:

CAUAJM_E_10656 The database has encountered a critical error. Please contact the database administrator.

FORWARD_AFM_MESSAGE PUBLIC API SUPPORTS NON-SYSTEM AGENTS**Problem Number: 4693**

CA Workload Automation AE application server tried to forward an AFM passed through public API FORWARD_AFM_MESSAGE. The application server should reject the AFM forwarding for a non-system agent machine.

OVERRIDE FROM WCC INSERTS JOBROW WITH NULL BOX_JOID**Problem Number: 4694**

CA Workload Automation AE scheduler complains about invalid job definition when the job inside box a gets an override from CA Workload Control Center interface. Job's box_joid becomes null and an EVENT_HDLR_ERROR is logged into the scheduler log with the following error message:

CAUAJM_E_10333 Error retrieving active override for joid <6,039>. Possible job definition corruption.

CAUAJM_E_40111 Unable to fetch job details using internal job identifier <6,039>. Event processing for HR1z10007231 aborted.

UPGRADE-NOT ALL TABLES UPDATED PROPERLY**Problem Number: 4695**

CreateAEDB.pl fails to alter tables with proper column length when an old column of smaller size is increased to a larger value to accommodate longer values.

UJO_COMM_SEND_ALIAS UNIQUE CONSTRAINT VIOLATIONS OPCODE:0**Problem Number: 4696**

When both the scheduler and application server contact the same agent for the first time after starting up, either the scheduler or application server experienced a database error after the agent begins sending data.

Fixed Issues - 11.3.6 SP7 CUM1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP7 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 SP7 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

NOTE

IMPORTANT! The following issues that are fixed in Release 11.3.6 SP7 CUM1 help reduce the time and resources it takes for the database server to execute the same set of repeating SQL queries that differ only in input values:

- **ORACLE DATABASE QUERY PERFORMANCE DEGRADES AFTER PR4688:** Improves the set of SQL queries that the scheduler uses to process events for jobs that specify the must_start, must_complete, max_run_alarm, and term_run_time attributes.
- **Problem Number 4734 - QUERY FOR GET_JOB_RUNS_WITH_FILTER TIMED OUT:** Improves the set of SQL queries that the application server uses to process SDK client requests to retrieve job run information.

APPSERVER/SCH STUCK IN SQL QUERY DURING STARTUP AFTER PR4308**Problem Number: 4697**

When you start up the application server or scheduler after communicating with many agents, the process experienced long delays while executing a SQL query that validates the rows in the communication tables.

FORECAST REPORT GENERATION FAILS FROM WCC**Problem Number: 4698**

CA Workload Control Center fails to generate the CA Workload Automation AE forecast report when a job depends on a job that runs more than a day every time it runs. CA Workload Control Center validates the date and time for every job and reports an error when the job duration is more than 24 hours.

IMPROVE ACCURACY OF IP ADDRESSES SHOWN IN ISDBGACTIV TRACES**Problem Number: 4701**

When you run the CA Workload Automation AE product binaries on a machine that supports both IPv4 and IPv6 address resolution and you enable ISDBGACTIV=COMM,LIGHT traces, the traces were ambiguous about the display of the IPv4 or IPv6 address.

CROSS INSTANCE DEPENDENCY BETWEEN 11.3.6 and 4.5 FAILS**Problem Number: 4705**

When a predecessor job of type BOX is defined in CA Workload Automation AE Release 11.3.6 and the successor job resides in 4.5, the successor job did not start when the predecessor job that is in the NO_EXEC status is started.

The following message is displayed in the Release 11.3.6 scheduler log:

CAUAJM_W_40284 Postponing send of event <ACE00G318g00> due to offline external instance <R45>.

SETTING TIMEOUT TO EMAIL NOTIFICATION**Problem Number: 4707**

Email notification feature sometimes gets stuck in sending emails for almost five (5) mins without sending other emails. This fix will allow user to specify a timeout in cases where SMTP server becomes unresponsive and try the email notification for next job. Use [ASNOTIFY_SMTP_TIMEOUT](#) environment variable to specify the timeout in seconds.

If ASNOTIFY_SMTP_TIMEOUT=20 then email notification will return timeout after 20 seconds when the SMPT server becomes unresponsive.

JOB STUCK IN STARTING AFTER FIRST COMMUNICATION WITH AGENT**Problem Number: 4708**

When the scheduler schedules a job to a remote machine and contacts the agent for the first time, it initiates an exchange to introduce itself to the agent. After the exchange completes, the scheduler placed the job in the STARTING status but the agent did not execute the job.

AUTOREP -J -Q DISPLAY BLANK OR GARBAGE DATA**Problem Number: 4709**

autorep -J -q ALL prints blank or garbage data when jobs are inserted and deleted while running autorep.

SAP PLUGIN RECEIVER OBJECT COULD NOT BE CREATED**Problem Number: 4711**

When the sap_recipients attribute of the SAP job_type contains a space in the recipient part of the value, the agent returns the following message to the scheduler:

```
<JobName> function failed. R/3 Code: Receiver object could not be created.
```

EXT CAL DATE_CONDITION #E AND #O OPTIONS**Problem Number: 4713**

You must add every EVEN or ODD week of the year to an extended calendar date_condition field, however field length is limited to 255 characters.

DBMAINT FAILS WHEN DBMAINTCMD VALUE INCLUDES AN ENVIRONMENT VARIABLE WITH SPACES**Problem Number: 4714**

CA Workload Automation Scheduler fails to run the database maintenance activity on Windows machine when the product is installed on a drive that does not support 8dot3name file naming convention and the DBMaintCmd variable has an environment variable with spaces in the value of the environment variable.

DATABASE ERROR IN 'UJO_GET_PREDS_AND_SUCCS' STORED PROCEDURE**Problem Number: 4715**

When you try to edit a job using the Quick Edit tab in the CA WCC, the job details are not populated, and the page remains blank. This problem occurs when the CA Workload Automation AE is running on MSSQL or SYBASE databases.

UNABLE TO CONNECT TO SQL SERVER DATABASE HAVING TLS ENABLED**Problem Number: 4717**

The CA Workload Automation AE scheduler and application server on Windows failed to connect to the SQL Server database having TLS enabled with the following error message:
SQLDriverConnect failed with -1(SQL_ERROR) and State: IM008.

To resolve this problem, follow these steps:

NOTE

Follow these steps only when you want the CA Workload Automation AE server to connect to the SQL Server that is TLS enabled.

1. Add the MSSQLDriverConnString variable in the configuration file (%AUTOUSER%/config.%AUTOSERV%) and set its value (within curly brackets) to the driver that are available on the CA Workload Automation AE server computer.

NOTE

- By default, the value is set to {SQL Server}.
- You can download the latest drivers from the SQL Server support site.
- To view the list of drivers that are installed on the CA Workload Automation AE server, follow these steps:
 - a. Navigate to the Windows Control Panel.
 - b. Double-click Administrative Tools.
 - c. Double-click ODBC (32-bit). The ODBC Data Source Administrator (32-bit) panel is displayed.
 - d. Click the Drivers tab.

2. Save the configuration file.

Once the CA Workload Automation AE services are restarted, CA Workload Automation AE connects to the SQL Server that is TLS enabled.

For example, if the Microsoft ODBC Driver 11.0 is available on the CA Workload Automation AE server computer, set the value of the MSSQLDriverConnString variable as follows:

```
MSSQLDriverConnString={ODBC Driver 11 for SQL Server}
```

INFORMATICA JOB INSERT FAILS WHEN YOU DEFINE SECURITY DOMAIN**Problem Number: 4718**

When you define an INFORMATICA-type job and specify a value for the `informatica_security_domain` JIL attribute, the application server fails the job insert operation.

HANDLING ESCAPED SINGLE/DOUBLE QUOTES OF STRING**Problem Number: 4719**

When the POJO job uses string with escaped double quotes as part of the input for `j2ee_parameter`, those string values are truncated in parsing, which results in incorrect AFM formation.

FAILED TO UPDATE CONFIG DURING DATABASE ROLLOVER**Problem Number: 4720**

In dual server mode, the scheduler fails to rename the configuration file `$AUTOUSER/config.$AUTOSERV`, while attempting to rollover to single server mode. The following messages are displayed:

```
CAUAJM_W_10648 The database: DATABASE1 is unavailable. Attempting to rollover to single
server mode.
CAUAJM_E_10646 Failed to update configuration information regarding rollover.
```

SHADOW IN BAD STATE AFTER RESTARTING WHILE AGENTPOLLER EXECS**Problem Number: 4723**

When the primary scheduler fails back after the shadow scheduler took over processing jobs, the shadow scheduler stops processing jobs and restarts in dormant mode without exiting the process. When the shadow scheduler restart takes place while the 11.3.6 SP7 agent poller feature is in progress and the shadow fails over a second time following the loss of the primary scheduler, the shadow sets the status of some of the machines to the unqualified status. When the primary fails back again, the shadow scheduler deadlocks while trying to restart without exiting the process.

SCHEDULER DEADLOCKS WITH ISDBGACTIV WHEN GETADDRINFO DELAYS**Problem Number: 4726**

When you have enabled `ISDBGACTIV=LIGHT` traces for the UNIX scheduler and the scheduler's attempts to resolve the agent's node name into an IP address delay, the scheduler process deadlocks and hangs indefinitely.

JOBS STUCK WHEN PRIMARY AND SHADOW SCHEDS RUN AT SAME TIME**Problem Number: 4727**

When you configure CA Workload Automation AE instance to run in high availability mode and you manually restart the primary scheduler following an ungraceful exit, while the shadow scheduler begins to take over, both schedulers run simultaneously for a moment. During this time, both schedulers compete for job events stealing events from the other resulting in jobs getting stuck in the wrong status.

UPGRADE OF AGGREGATOR TABLES TO 11.3.6 SP6 TAKES A LONG TIME**Problem Number: 4728**

When you upgrade your CA Workload Automation AE instance to Release 11.3.6 SP6 and the tables that are associated with the aggregation of statistics have many rows, the installer completes after many minutes.

UNDER NATIVE SECURITY NON OWNER COULD NOT EXECUTE SENDEVENT**Problem Number: 4729**

When you are not a job owner and have job execute permissions, you cannot execute sendevent command in native security mode.

SCHED HANGS WHEN RUNNING LOAD BALANCING JOBS AFTER PR4606**Problem Number: 4730**

When you define a box that contains many load balancing jobs, and the jobs in the box specify a dependency on one another and the scheduler schedules most of the jobs for execution, the scheduler process hung while it processed the incoming completion events for the jobs.

CCI COMMUNICATION DELAY WHEN MANAGERHOSTALIAS IS FQDN**Problem Number: 4731**

When ManagerHostAlias is defined with fully qualified domain name, then the CA Workload Automation AE scheduler sends all messages over cross-platform using fully qualified domain name and mainframe takes almost 40 seconds more to resolve the hostname when compared to node name without fqdn.

ORACLE DATABASE QUERY PERFORMANCE DEGRADES AFTER PR4688**Problem Number: 4732**

This problem addresses multiple issues that are related to the performance of the database queries:

- When you define many jobs that specify the must_start, must_complete, max_run_alarm, and term_run_time attributes, and the scheduler schedules the jobs, it delayed while it processed the incoming completion events for the jobs.
- When you create filters to view jobs in CA Workload Control Center (CA WCC) and the application server executes a query on behalf of CA WCC, the Oracle database took too long to parse the query.

SCHEDULER DID NOT RETRY UPON UJO_BATCH_GET_EVENT DB ERRORS**Problem Number: 4733**

The scheduler executes database stored procedure ujo_batch_pkg.ujo_batch_get_event twice every second to get the events from the database. When there is a failover from Oracle dataguard, execution of this stored procedure is considered as success even though there is a database error.

QUERY FOR GET_JOB_RUNS_WITH_FILTER TIMED OUT**Problem Number: 4734**

When you create filters to view jobs in the CA Workload Command Center (CA WCC) and the application server executes a query on behalf of CA WCC, the query gets timed out because oracle server takes more time in parsing.

ENABLE DEBUG FOR SIMPLE EMAIL NOTIFICATION**Problem Number: 4736**

When VERBOSE option is enabled for simple email feature in CA Workload Automation AE, the debug output is written to the end of the file and after the scheduler is stopped gracefully. Debug information is useful when the verbose information is written to the log file immediately after the event occurs.

AVOID SENDING EMAILS WHEN AN EMAIL IS WRONG**Problem Number: 4737**

In CA Workload Automation AE, the scheduler does not send emails to all the addresses when one of the address that is specified as part of the job definition is wrong. Some SMTP servers are strict in checking the email addresses and will block the email sending attempts.

FILE TRIGGER DELETE/EXIST/NOTEXIST OPERATIONS NOT WORKING**Problem Number: 4739**

When a file trigger job is created (job_type: FT) for watch_file_type: DELETE or EXIST or NOTEXIST, then "watch_no_change: 1" is added by default and the job fails. The following messages are displayed:

```
Invalid format, "NoChange does not make sense here"
```

or

```
File does not Exist
```

SCHED HANGS WHEN RUNNING LOAD BALANCING JOBS AFTER PR4730**Problem Number: 4740**

When you define many load balancing jobs that specify multiple dependencies on one another and the scheduler schedules most of the jobs for execution while queuing other jobs, the scheduler process hung while it processed the incoming completion events for the jobs.

'AUTOREP -J %' COMMAND TAKES LONGER TIME TO EXECUTE**Problem Number: 4742**

The 'autorep -j ALL' command fetches all the job runs and displays the results faster than the 'autorep -j %' command.

DUAL DATABASE EVENT SYNCHRONIZATION DEADLOCK AFTER PR4603**Problem Number: 4743**

When you configure the CA Workload Automation AE instance to run in dual event server mode and the scheduler is busy processing hundreds of events at one time, the scheduler was unable to synchronize events between both databases and shut itself down.

RESOURCES ARE NOT HONORED WHEN BOX JOBS ARE FORCE STARTED**Problem Number: 4745**

When multiple box jobs are force started and are waiting on resources, the scheduler does not honor resources.

AUTOREP -J -Q RETURNS J2EE_PARAMETER VALUES IN REVERSE ORDER**Problem Number: 4747**

autorep -j -q returns j2ee_parameter values in reverse order because of Standard Template Library (STL) sort.

IMPROVE RATE OF LOAD BALANCE JOB STARTS DURING HIGH ACTIVITY**Problem Number: 4749**

When the scheduler must start many load balancing jobs at once, it reduced the number of job start evaluations to one or two jobs starts per second.

SCHEDULER FAILED TO CREATE SERVICEDESK TICKETS

Problem Number: 4754

The scheduler failed to create servicedesk tickets intermittently.

Third-Party Software Acknowledgements - 11.3.6 SP7

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

AE Release 11.3.6 SP6

This section includes information about the CA Workload Automation AE Release 11.3.6 SP6 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 SP6 and Release 11.3.6 SP6 CUM1.

New Features - 11.3.6 SP6

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP6.

Informatica Job Type

[Informatica jobs](#) let you start predefined workflows on the Informatica PowerCenter. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

NOTE

To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.

You can cancel a running Informatica workflow using the KILLJOB event.

You can retrieve the step log files for an Informatica job. For information about how to retrieve the step log files, see [autosyslog Command—Display Log Files](#).

You can restart an Informatica job. For more information about restarting an Informatica job, see [sendevent Command -- Start Jobs](#).

SQL Server Agent Job Type

[SQL Server Agent jobs](#) let you execute predefined jobs in the SQL Server Agent to perform administrative tasks. The jobs can be Transact-SQL scripts, command prompt applications, Microsoft ActiveX scripts, Integration Services packages, Analysis Services commands and queries, or Replication tasks.

NOTE

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

You can cancel a SQL Server Agent job using the KILLJOB event.

You can retrieve the step log files or session log files for a SQL Server Agent job. For information about how to retrieve these log files, see [autosyslog Command—Display Log Files](#).

AgentCLI_To_JobConverter Utility

The [AgentCLI_To_JobConverter](#) utility lets you convert:

- Workload Automation Agent for Informatica CLI command jobs to CA Workload Automation AE Informatica jobs—This lets you to use the inbuilt capabilities of the Informatica job type.
- CA WA Agent for Microsoft SQL Server CLI command jobs to CA Workload Automation AE SQL Server Agent jobs—This lets you to use the inbuilt capabilities of the SQL Server Agent job type.

(Sybase Only) Support Kerberos Authentication

You can now configure CA Workload Automation AE to use Kerberos authentication to access the database. During installation, you can set the database access mode to either password or Kerberos authentication.

The [DBAccessMode](#) parameter in the configuration file or the Access Mode field on the Event Server - CA Workload Automation AE Administrator window of the Administrator utility (Windows) displays the access mode that CA Workload Automation AE is using to establish connections to the database.

After you installed CA Workload Automation AE, you can use the [switchDBAccessMode.pl](#) script to [switch the database access mode](#) from password to Kerberos authentication or vice versa.

NOTE

For information about configuring MIT Kerberos on CA Workload Automation AE, see [Configure MIT Kerberos on CA Workload Automation AE](#).

Support AIX 7.2

CA Workload Automation AE now supports AIX 7.2.

Support Microsoft SQL Server "Always On" Feature

CA Workload Automation AE now supports Microsoft SQL Server "Always On" feature.

New Attributes for JMS Publish and JMS Subscribe Jobs

The following attributes are added for the JMS Publish and JMS Subscribe jobs:

- [correlation_id Attribute](#)—Specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue.
- [destination_user Attribute](#)—Specifies the user name that is required to connect to the topic or queue in a JMS job.
- (JMS Publish jobs only) [reply_to Attribute](#)—Specifies the JNDI name of the reply to a queue.

New Attributes for SCP and FTP Jobs

The following attributes are added for the SCP and FTP jobs:

- [run_external run_external Attribute -- Specify Whether to Define a User ID](#).
- [external_os_user](#)—specifies the user ID of the user with the authority to download the file from the remote FTP server or upload the file to the remote FTP server.

The following attributes are added for the SCP jobs:

- [scp_transfer_mode scp_transfer_mode Attribute -- Specify the Type of Data to Transfer](#).
- [scp_delete_sourcefile](#)—Specifies whether you want to delete the local source files (if uploading) after they are uploaded from the agent computer or the remote source files (if downloading) after they are downloaded from the remote SFTP server. Specify this attribute only if you are using SFTP protocol.
- [scp_delete_sourcedir](#)—Specifies whether to delete the directory that contains the local source files (if uploading) or the remote source files (if downloading) after the files are uploaded or downloaded. Specify this attribute only if you are using SFTP protocol.
- [scp_rename_sourcefile](#)—Specifies the name that the source file must be renamed to after the file is uploaded or downloaded. Specify this attribute only if you are using SFTP protocol.
- [scp_create_targetdir](#)—Specifies whether to create a target directory before you upload or download a file. Specify this attribute only if you are using SFTP protocol.

Specify Whether to Use the MachineMethod Parameter When a Job Is Force Started

You can specify whether the scheduler ignores or uses the MachineMethod parameter value to select the machine where the job must run when you force start a job and multiple non-zero factor online machines are available. On UNIX, you can configure this setting using the UseMachineMethodOnForceStartJob parameter in the configuration file. On Windows, you can configure this setting using the Machine Method On Force Starts check box on the Scheduler - CA Workload Automation AE Administrator window of the Administrator utility.

NOTE

For more information about the UseMachineMethodOnForceStartJob parameter, see [UseMachineMethodOnForceStartJob](#).

Changes to Existing Features - 11.3.6 SP6

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP6.

Updated dbtype Attribute

You can now use the [dbtype](#) attribute to monitor the POSTGRESQL or TERADATA database.

Updated autosys_secure Command

You can now use autosys_secure (from the command line or using the AutoSys Security Utility interactive menu) to define a user with both password and key credentials.

Global Variable Substitution

You can now reference a [global variable](#) as part of the syntax of the following attributes:

- destination_name
- ftp_server_name
- initial_context_factory
- j2ee_parameter
- notification_id
- notification_msg
- notification_emailaddress
- provider_url
- sap_recipients (only recipient parameter)
- sap_step_parms (only variant and step_user parameters)
- sap_target_sys
- sp_arg (only name and value parameters)

Updated LogMaxEndLines Parameter

The default value of the [LogMaxEndLines](#) parameter is now updated from 0 to 2000. If you have set the LogMaxEndLines parameter to a non-zero value, after you upgrade this value is set to the default value, that is, 2000.

Command Sponsor is Not Supported on Microsoft Windows (32-bit) Operating Systems

Common Sponsor is not supported on Microsoft Windows 32-bit operating systems. If you want to use the 32-bit Command Sponsor, install it using the CA Workload Automation AE Release 11.3.6 SP5 or a previous release DVD .

Oracle 12c Container Database Support

Oracle 12c container database is supported. For more information, see [CA Workload Automation AE Operating System and Database Support](#).

Fixed Issues - 11.3.6 SP6

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP6.

TIP

CA Workload Automation AE Release 11.3.6 SP6 also includes the issues fixed in each of the prior releases of 11.3.6.

STANDALONE AGENT AND CLIENT INSTALL MISSING INSTANCE PROFILE

Problem Number: 4461

When the client and agent are installed using the Standalone CA Workload Automation AE DVD, the instance specific profile file was missing in the agent profile directory. The `oscomponent.environment.variable_manager_instance_name_SCH` parameter was missing in the agent parameter file. This happens when the agent is installed before the client or the client is installed before the agent.

UPGRADE DB USER VALIDATION FAILURE DOES NOT SPECIFY USER

Problem Number: 4512

(UNIX Installer) During upgrade, the database user validation failed without specifying the user for which the validation failed.

CA ERU FAILED TO CONNECT TO ORACLE RAC DATABASE

Problem Number: 4523

CA EEM Reporting Utility (CA ERU) failed to connect to the Oracle RAC database. It failed with the following error:

```
ORA-12505 Connection refused, the specified SID (TESTDB) was not recognized by the
Oracle server.
```

MACHINE REPORT DOESNT CALCULATE THE CORRECT CURRENT LOAD

Problem Number: 4525

When you define multiple virtual machines that specify the same machine and their own max load values and then run jobs with loads on those machines, the autorep command machine report displayed incorrect total load on the machine.

DATABASE COLUMN ORDER DIFFERENCE AFTER UPGRADES

Problem Number: 4526

On Sybase, the autobcpSYB.pl script failed to complete when the source database is upgraded from older versions of CA Workload Automation AE and the target database is newly created using the CreateAEDB script as the column order is different in some tables.

SAP JOB COPY JOB DOES NOT HONOR KILLJOB COMMAND

Problem Number: 4527

The scheduler failed to terminate a SAP Job Copy job(SAPJC) when you issue the KILLJOB event.

AUTOREP -P DISPLAYS DUPLICATE MACHINE INFORMATION

Problem Number: 4529

The autorep -p request returned two sections for each response when the application is running in the high availability mode.

AUTOSYSADMIN PROFILE SCREEN SHOWS VARIABLES IN SORTED ORDER

Problem Number: 4531

The Job Profiles - CA Workload Automation AE Administrator window of the CA Workload Automation AE Administrator utility shows the environment variables in a sorted order even when they are created in a different order. The profile file stored in the agent's profile folder lists the variables in the order of their creation.

DST/ST SWITCH DATE CONDITION AND WINDOW CALCULATION UPDATES

Problem Number: 4532

This problem record contains the fixes to the following four issues that are related to the scheduling of jobs with date and time conditions:

1. When you define a job with a run window that opens on the current day and ends the next day, and the scheduler schedules a job to execute inside the run window that ends on the current day, the scheduler generated the CHK_RUN_WINDOW event for the following day leaving the current day without a CHK_RUN_WINDOW event.
2. When the scheduler completes the final run of a job with a run window on the day before the Standard Time change, it skipped a day when calculating the next start time of the job.

3. When you define a job that runs every hour with a run window that ends in the missing hour on the day of the Daylight Savings Time change, the scheduler did not schedule the next job run in the hour that the change occurred.
4. When you define a job that specifies the `must_start` or `must_complete` attribute as an absolute time and the time falls either in the repeated Standard hour or in the missing Daylight Savings hour, the scheduler generated the `CHK_START` or `CHK_COMPLETE` event for the wrong time.

GLOBAL VARIABLE - ESCAPED SINGLE QUOTE NOT REMOVED

Problem Number: 4533

The global variable value with a single quote gets escaped with two single quotes during global variable creation. While sending the global variable value to the agent, the scheduler should replace two quotes with a single quote, but the scheduler always sends two quotes causing job failures.

DATABASE ERR WHEN SCH PUTS JOB W/ MULTIPLE MACHS IN RESWAIT

Problem Number: 4534

When you start or force start a job with a resource dependency and no load-balancing attributes that specifies a comma-separated list of machines or a virtual machine with multiple machine components and no resources are available, the scheduler generated database errors while placing the job in the RESWAIT status.

AUTOSYS_SECURE OPTION 3 CORRUPTS DATABASE PASSWORD

Problem Number: 4535

On Solaris, when you select option 3 from the `autosys_secure` interactive menu to change the password of the `autosys` user, the `autosys_secure` command generated errors and stored a corrupted password in the `ujo_alamode` database table.

SCH IGNORES AMOUNT WHEN IT FORCESTARTS JOBS SHARING RESOURCE

Problem Number: 4536

When you force start multiple jobs that specify the same virtual resource, the scheduler started the jobs even when no more resources are available.

SENDEVENT CORES WHEN SET_GLOBAL VALUES SPECIFY SINGLE QUOTE

Problem Number: 4537

When you execute the `sendevent` command to set a global variable and you specify a value that contains more characters than the global variable name with more than two single quote characters, the `sendevent` process aborted ungracefully.

CYCLE EXISTENCE IN THE EXTENDED CALENDAR

Problem Number: 4538

When you use the `autocal_asc` command from the CLI to create an extended calendar with a non-existing cycle calendar, the following warning is displayed:

```
CAUAJM_W_50414 *** WARNING - The Cycle calendar: cycle_name does not exist.
```

However, when you perform the same task using the SDK or CA WCC, the warning is not displayed.

AS_SERVER SLOWER THAN 11.3 SP1 IN DOING JIL OPERATIONS**Problem Number: 4539**

The job or machine insert, update, or delete operation was taking 40 to 50 percent more time when compared with the time it takes with the r11.3 latest maintenance.

AUTOSYSLOG DOES NOT FETCH LOG WHEN STD_OUT_FILE IS INVALID**Problem Number: 4540**

When you set the value of the std_out_file attribute to a non-existent path and try to run the job, a non-zero exit code is generated by the agent. This prevents a row to be created in the ujo_job_run table. As a result, the autosyslog command is unable to display the agent log. You cannot fetch the previous run information using the autorep -J <Job_name> -r <number> command or the job run details using CA WCC.

SCH PROCESSES DUPLICATE STATUSES AFTER PROBLEM 4455**Problem Number: 4541**

When the scheduler is running with a multi-threaded agent listener and the agent resends job status messages for which it did not receive an acknowledgement, the scheduler did not detect that the message is a duplicate and it generated a repeating job event.

N_RETRY AND CONDITION LOOKBACK TOGETHER DOESN'T RESTART JOB**Problem Number: 4543**

When a job is defined with both the condition attribute with lookback and the n_retrys attribute, a RESTART event is generated due to an application failure. However, the job fails to start and the following message is displayed in the scheduler log:

```
CAUAJM_I_40162 The starting conditions for job <Job1> have not been met. Cannot be started.
```

```
CAUAJM_I_40149 Cancelling restart for job <Job1> and setting to INACTIVE.
```

The job must restart as the job's condition succeeded in the first time.

AUTOBCPORADP DOES NOT EXCLUDE STATISTICS**Problem Number: 4544**

The autobcpORAdp.pl script failed to synchronize data with ORA-30926 and ORA-39126 on KUPW \$WORKER.STATS_LOAD.

TIMEZONE UPDATES 2017**Problem Number: 4545**

AUTOTIMEZONE UPDATES 2016-2017.

COMMAND LINE UTILITIES BREAK WITH MISSING OPTION**Problem Number: 4546**

When you specify only the dash (-) character as an argument to the command line utilities like jil, sendevent, autorep, and so on, that command line utility process aborted ungracefully.

JIL CRASHED WHEN KEYWORD AND VALUE IN DIFFERENT LINES

Problem Number: 4547

The jil command crashed intermittently when the job attribute and its value are in different lines. For example, the below job definition read from the jil input file can crash jil:

```
insert_job: testjob1
command: echo hello
machine: localhost
owner
:user1
```

11.3.6 SP5 SCHEDULER WRONGLY PLACES UPDATED JOB IN RESWAIT

Problem Number: 4549

When you define two virtual resources and you define a job that initially uses one virtual resource and then update the job to use the other virtual resource, the scheduler placed the updated job in the RESWAIT state when higher priority jobs are queued in the RESWAIT state due to the first resource.

11.3.6 SP5 FIX SCHEDULER QUEUING W/ RESOURCE JOB AND NO LOAD

Problem Number: 4550

This problem addresses two issues that occur when you define jobs with resource dependencies, a non-zero priority attribute value, and a job_load attribute value of zero:

1. When you specify a lower priority job with a job_load attribute and no resource attributes having the same machine attribute value as the resource job, and the resource job is in the RESWAIT state, the scheduler placed the job in the QUE_WAIT state.
2. When you specify a lower priority job with a different resource name and a job_load attribute value of zero having the same machine attribute value as the resource job, and the resource job is in the RESWAIT state, the scheduler placed the lower priority job in the RESWAIT state.

11.3.6 SP5 FIX SCHEDULER LOAD BALANCING W/ VIRTUAL RESOURCES

Problem Number: 4551

When you define a job with virtual resource dependencies and specify a virtual machine as the machine attribute value, the scheduler submitted jobs at random when two or more machines satisfied the resource requirements even when a machine with better load or factor values exists.

SCHED DEADLOCKS IN GLOBALAUTOHOLD WHEN JOBS LEAVE QUE_WAIT

Problem Number: 4552

When you configure the scheduler to start in the GlobalAutoHold mode, the scheduler experienced a deadlock as it placed a load balancing job in the ON_HOLD status from the QUE_WAIT status.

SCHED REMOVES OVERRIDE ON 1ST FAIL INSTEAD OF AFTER N_RETRYS**Problem Number: 4553**

When you insert a job definition that specifies a `n_retrys` job attribute value greater than 0 and you insert a job override, the scheduler removed the override after the first failure instead of removing the override after completing all application restarts.

BOX RUNNING EVENT PROCESSED AFTER JOB IN THE BOX EXECUTES**Problem Number: 4554**

When you configure the CA Workload Automation AE instance to run in the dual event server mode and start a box during a period when the job starting activities are high, the scheduler executed a job in a box that is triggered by its predecessor job before it sets the box itself to the `RUNNING` status.

11.3.6 SP5 UJO_SETUP_NEXT DELETES CHK_RUN_WINDOW ON SAME DAY**Problem Number: 4555**

When you define a job with date and time conditions that specifies multiple times in a day for the `start_times`, `must_start_times`, and `must_complete_times` attributes, and you update the attributes while the job executes, the scheduler deleted the `CHK_COMPLETE` event of the run that was still executing. Also, when you specify the `run_window` attribute for a job that runs multiple times in a day, the scheduler deleted the `CHK_RUN_WINDOW` event after the first execution of the job.

11.3.6 SP2 SCHED RUNS JOBS W/ NOTRUNNING COND AT SAME TIME**Problem Number: 4556**

When you define a set of jobs with `notrunning` dependencies on each other, all the jobs start at once, as the scheduler did not always honor the job condition.

SCHED IN GLOBALAUTOHOLD MODE DOESNT HOLD JOBS FROM PEND_MACH**Problem Number: 4557**

When you restart the scheduler in global auto hold mode after it sets jobs to the `PEND_MACH` status due to an offline agent and when the agent comes online, the scheduler executed the job instead of placing it in the `ON_HOLD` status.

AUTOREP -V ERROR WITH CUSTOM AUTHENTICATION LIBRARY**Problem Number: 4558**

When you use custom authentication library for authentication, the `autorep -V` command returned a message that the authentication failed.

UPGRADE DATABASE LOG OVERRIDDEN IN MULTIPLE INSTANCE UPGRADE**Problem Number: 4559**

When you perform an upgrade or repair on a machine where multiple instances are installed, then the `AEDB DATABASE` update or refresh log is overridden.

SERVICEDESK TICKET IGNORES CATEGORY ATTRIBUTE**Problem Number: 4560**

The scheduler did not pass the non-default attributes that are used as part of the `svcdesk_attr` attribute value in your job definition. This causes inconvenience to advance users.

UPGRADE STUCK AT EEM SERVER INTERVIEW DIALOG

Problem Number: 4561

The product was installed without CA EEM security. When you try to upgrade to Release 11.3.6 SP5, it prompts for the CA EEM Server name and did not proceed further with the upgrade process.

Solution: The installer uses the `autosec_test` STAT command to determine whether CA Workload Automation AE runs in CA EEM mode or native security mode. To execute the `autosec_test` command, it sources the `$AUTOUSER.$SHELL.hostname` file, where the `hostname` is obtained from the `autoflags -n` command. If the `hostname` that is returned by the `autoflags -n` command does not match the `hostname` that is part of the `$AUTOUSER.$SHELL.hostname` file, then `autosec_test` command fails to source the file and wrongly prompts for the CA EEM server name during the upgrade. So, ensure the `hostname` (machine where the upgrade is run) that is returned by the `autoflags -n` command exactly matches with the `hostname` that is part of the `$AUTOUSER.$SHELL.hostname` file.

DATABASE PWD CHANGE FAILED IN PASSWORD_VERIFY_FUNCTION MODE

Problem Number: 4562

When the ORACLE database server is configured in the `PASSWORD_VERIFY_FUNCTION` mode, the `autosys_secure` command failed when you change the `autosys` database user password.

JOB IN QUE_WAIT DID NOT RUN UPON CHANGING THE QUEUE PRIORITY

Problem Number: 4563

When the scheduler is running with `MachineMethod=job_load` and the job has a priority of non-zero and a `job_load` of non-zero and it went to the `QUE_WAIT` status. The job did not run after changing its queue priority to zero using the event: `CHANGE_PRIORITY`.

WEBSERVICE - DOES NOT SHOW JOB INFO AFTER ARCHIVE EVENTS

Problem Number: 4564

Job-run-info web service did not show the job status and other information when the `ujo_job_runs` table is archived. This is different from `autorep` that shows all jobs summary even when `JobRuns` is archived.

UNDO CROSS-QUEUE PRI CHK BTWN RESWAIT/QUE_WAIT AFTER PR4466

Problem Number: 4566

Suppose that you define two jobs:

1. A job with a priority attribute value that has a `job_load` value and runs on a virtual machine.
2. A load balancing job with virtual resource dependencies that runs on the same virtual machine as the first job and has a lower priority attribute value than the first job.

When the scheduler evaluated the first job and the second job is queued in the `RESWAIT` status, the scheduler immediately queued the first job in the `QUE_WAIT` status.

Suppose that you define two jobs:

1. A job with a priority attribute value that has a resource dependency.
2. A load balancing job with a dependency on the same virtual resource dependency as the first job and has a lower priority attribute value than the first job.

When the scheduler evaluated the first job and the second job is queued in the QUE_WAIT status, the scheduler immediately queued the first job in the RESWAIT status.

AUTOREP -Q XTRA ROWS FOR JOB W/ GLOBAL RSRC AND VIRTUAL MACH

Problem Number: 4567

Suppose that you define a job that specifies a priority attribute value, a job_load attribute value of zero, a dependency on a global virtual resource and it runs on a virtual machine. When the scheduler placed this job in the RESWAIT status, the autorep utility generated one entry for each machine component that is part of the virtual machine in the queue report (using autorep -Q).

JOB GETTING STARTED EVEN CURRENT STATUS IS ??????

Problem Number: 4568

1. The job_depends -c command displayed the default value as T.
2. When a job has multiple cross-instance predecessors, out of which one predecessor's current status is ?????? (undefined) and the other predecessor's condition is evaluated to FALSE, it lead to the successor job getting started.

WINDOWS INSTALL RESPONSE FILE SHOWS PASSWORD

Problem Number: 4569

The agent password is stored in the plain text format in the Windows response file. Other passwords are stored and read in the encrypted format.

SCH PROCESSES AS TRUE FAILURE ON 1ST FAIL NOT AFTER N_RETRYS

Problem Number: 4570

When you insert a job definition that specifies a n_retrys job attribute value that is greater than zero (0), the scheduler performed the following actions after the first failure instead of processing it after completing all the application restarts:

1. Processed the CHK_COMPLETE event when the job specifies the must_complete attribute.
2. Generated extra DELETEJOB events when the job specifies the auto_delete attribute.
3. Deleted the job override when you insert a job override.
4. Evaluated the start of jobs in the QUE_WAIT state when the job specifies the priority and job_load attributes.
5. Delayed the processing of the FAILURE event to evaluate downstream dependent jobs.

SCHED IGNORES TERM_RUN_TIME WHEN OTHER ATTRIBUTES OVERRIDEN

Problem Number: 4571

When you insert a job definition that specifies a term_run_time job attribute value that is greater than zero (0) and you insert a job override for a different attribute, the scheduler ignored the original term_run_time value when it executed the job.

SCHED DOES NOT PROPAGATE GROUP OR APPLICATION IN SNMP TRAP

Problem Number: 4572

When you insert a job definition that specifies a group or application value and the scheduler runs a job that fails to start, sometimes the scheduler prepared a SNMP trap for the STARTJOBFAIL alarm without specifying the job's group or application value.

AGENT FAILS TO START IN MULTI AGENT ENVIRONMENT**Problem Number: 4573**

When multiple agents are installed on a machine and only one agent is up and running and you try to start another agent, it did not start as it assumes that it is already running.

SCHED FOREVER RUNS JOB WITH VIRTUAL RESOURCES AND N_RETRYs**Problem Number: 4574**

When you insert a job definition with virtual resource dependencies and a `n_retrys` job attribute value that is greater than zero (0), the scheduler continually created a new run every time the job restarted and then entered and exited the RESWAIT status.

PROFILE FILE IGNORED FOR USER-DEFINED JOB TYPES**Problem Number: 4575**

User defined jobs that include the profile attribute did not run with the profile file that is referenced in the job.

AUTOCAL_ASC BATCH UPDATE IN LEGACY MODE**Problem Number: 4576**

When you use the `autocal_asc` import through legacy option (redirection from file), it added dates to the database one by one; refreshing jobs every time it added a date. This caused the calendar updates to take hours to complete and miss job runs during this update window.

RESTARTING MICROFOCUS JOB AT STEP FAILS**Problem Number: 4577**

When you define a Microfocus job that consists of multiple named execution steps and you manually send the RESTARTJOB event with the execution step name to execute the job, the scheduler failed to restart the job.

DETAILED AUTOREP JOB REPORT DOES NOT SHOW CHK_START EVENT**Problem Number: 4578**

When you define a job that specifies the `must_start` attribute and the scheduler is able to schedule the job, communicate the job details to the agent, and place the job in the RUNNING status in the same second, the detailed job reports did not show the CHK_START event for the job run.

MUST_START/COMPLETE DOES NOT WORK WITH JOB OVERRIDES**Problem Number: 4579**

This problem record contains the fixes to the following issues that are related to the overriding of jobs that specify a `must_start_times` or `must_complete_times` attribute value:

1. When you insert a job override for a different attribute, the scheduler did not process the CHK_START and CHK_COMPLETE event immediately after it executed or completed the job respectively.
2. When you insert a job override and specify the `start_times` time values after the values in the `must_start_times` or `must_complete_times` attribute, CA Workload Automation AE did not reject the override.

JOB_DEPENDS DOES NOT SHOW X-INSTANCE DEPENDENCIES**Problem Number: 4580**

The job_depends command did not show the successor cross instance dependency and so you do not know if there is any dependency on the given job.

GENERIC SERVICE DESK REQUEST PRINTS JUNK CHARS

Problem Number: 4581

When you try to create a Service Desk ticket using the auto_svcdesk command with the -G option, junk characters were generated in place of the job name.

AGENT INSTALLER CREATES UNWANTED FOLDERS

Problem Number: 4582

The agent installer created unwanted folders like dbobj, JRE64_WA, JRE_WA, and perl.

SOAP RPC WEB SERVICES SETJOB AND SETCALENDAR

Problem Number: 4583

The Simple Object Access Protocol (SOAP) Remote Procedure Call (RPC) web services that were provided with CA Workload Control Center (CA WCC) r11.0 were dropped with CA WCC r11.3. A subset of those are added back to CA Workload Automation AE (Autosys Edition) Release 11.3.5. Following were still missed:

- setJob
- setCalendar

WEB SERVICE STATUS INCORRECT WHEN USERID TOO LONG

Problem Number: 4585

CA Workload Automation AE web server status is displayed as not running even when the web server is running on the Red Hat Linux server with user id longer than 8 characters.

IMPROVE APP SERVER PERFORMANCE WHEN SERVICING R11 AGENTS

Problem Number: 4586

When you define a job to execute on a machine where the CA Unicenter AutoSys JM Release r11 agent is installed, the application server delayed in servicing repeating agent requests due to database table unique constraint violations error.

DATABASE UPDATE REDUNDANT ON SHADOW AND TIE-BREAKER

Problem Number: 4588

While running in the upgrade mode, the CreateAEDB script did not check if the database was already upgraded and tries to update the schema multiple times.

AUA DID NOT MIGRATE GLOBAL VARIABLES

Problem Number: 4589

AutoSys Upgrade Assistant (AUA) did not migrate global variables that are referenced in the job command and attributes such as std_in_file, std_out_file, std_err_file, watch_file and so on. But, it migrated global variables that are referenced in the job condition attribute.

RECONNECTION DOES NOT HAPPEN FOR ORA-02399 AND ORA-01012

Problem Number: 4590

CA Workload Automation AE did not try to reconnect to the Oracle server when the Oracle server logs off the idle connection after CONNECT_TIMEOUT with ORA-02399 and ORA-01012.

LOGMAXENDLINES DEFAULT VALUE 0

Problem Number: 4591

In the configuration file, the default value for the LogMaxEndLine parameter is set to 0. When the LogMaxEndLine parameter is set to the default, the application server retrieves the entire job output log file. Sometimes, the application server crashed or quit without responding as huge memory is consumed while retrieving a large job log file or when multiple requests are made for retrieving large job log files.

SENDEVENT CHANGE_STATUS RESWAIT PEND_MECH MISSING

Problem Number: 4592

The SENDEVENT command displayed invalid status on changing the job status to RESWAIT or PEND_MACH.

DUPLICATE EVENT WHILE SENDING INACTIVE STATUS TO X-INSTANCE

Problem Number: 4593

When a box contains multiple jobs and some of them have external dependent successor jobs and they go into the inactive status due to box job failure, the scheduler generated duplicate eoids while sending the event to an external instance and this lead to duplicate sendevent status.

WCC CANNOT DETECT 11.3.6 SP5 JOB STATUS CHANGE TO ON_NOEXEC

Problem Number: 4594

When you manually send the JOB_ON_NOEXEC event, the scheduler updated the status in the database but did not update the time of the status change. Since CA WCC relies on the updated time stamp to detect status changes, it did not reflect the ON_NOEXEC status change of the job.

NOTIFY COMMUNICATION INFO TO ONLY ACTIVE AGENTS

Problem Number: 4595

During the failover process, the shadow scheduler tries to notify all the online agents. In this process, the scheduler creates two to three threads for each agent it tries to communicate and eventually runs out of memory or hits the limits and then starts complaining about resource unavailability.

SENDEVENT FOR GROUP OR APPLICATION INCOMPLETE

Problem Number: 4596

When the sendevent command is used to change the status of all jobs in a group or application to ON_HOLD, the status change is done only until a job that is picked in the process hits a problem. The rest of the jobs are ignored.

AUTOAGGR TO DATE/TIME HIGHER THEN CURRENT TIME

Problem Number: 4597

In the autoaggr command, if the time you specified in the TO DATE/TIME option is higher than the current time, the following output was displayed:

```
No statistics found for the specified time range.
```

AUTOTRACK LIST USER WITH DIGITS 64 AND 80**Problem Number: 4598**

The autotrack -U command returned the user details with 64 and 80 digits.

JOB NOT RESCHEDULED AFTER UJO_SETUP_NEXT SP DEADLOCK ERROR**Problem Number: 4599**

When you define a job with date conditions and the scheduler experiences a database deadlock error while executing the ujo_setup_next database stored procedure to reschedule the job after it completes, the scheduler did not execute the stored procedure again.

IMPROVE DETECTION OF TWO ACTIVE SCHEDS AFTER DB MONITOR EVAL**Problem Number: 4600**

This problem addresses the following issues that occur when the database monitoring system detects a potential problem with the database:

1. When any scheduler loses connection to any event server and rolls over into the single event server mode or reestablishes connection with the lost event server, the scheduler did not check if another scheduler is running in its place.
2. When the primary scheduler loses connection to any event server during a period of much activity, it did not update the ujo_ha_process table causing the shadow scheduler to fail over and it continued processing job events.
3. When the active scheduler runs in dual event server mode and it experiences errors creating process threads after losing connection to any event server, it did not update the ujo_ha_process table causing the dormant scheduler to fail over as the new active scheduler.

SCHEDULER DOES NOT HONOR RESTRICT_FORCE_STARTJOB**Problem Number: 4601**

The scheduler did not honor the RESTRICT_FORCE_STARTJOB environment variable when multiple FORCE_STARTJOB events are processed in the same batch due to a race condition.

SCH PROCESSES RUNNING AFTER SUCCESS IN DUAL EVENTSERVER MODE**Problem Number: 4603**

When the scheduler is running in the dual event server mode and it detects that a job's RUNNING event is missing in one of the event servers, it wrote a new RUNNING event and fetched the event after the job's corresponding SUCCESS event. This caused the scheduler to process the SUCCESS event before the RUNNING event.

JOB STUCK AFTER SCH REQUEUES STARTING EVT FROM EXTERNAL INST**Problem Number: 4605**

When you define a job that has a dependency on a job in an external instance and the scheduler process in the local instance ungracefully exits as the scheduler in the external instance schedules a job, the local scheduler ignored events for an unrelated local job every time it starts up.

NON EDIT SUPER USER IS WRONGLY ABLE TO MODIFY JOB OWNER**Problem Number: 4607**

In native security mode, the current job owner who is not an EDIT Superuser was able to set a new job owner.

JOB STUCK IN STARTING AFTER SCHEDULER SHUTDOWN AFTER PR.4437

Problem Number: 4609

When you shut down the scheduler at the time when it contacts an agent to start a job, the scheduler processed the job's STARTING event and did not resume contacting the agent after it was restarted.

JOB STUCK WITH EVENT_HDLR_ERROR AND NO ORACLE ERROR

Problem Number: 4631

As the scheduler processes many jobs and receives inbound job statuses from agents, it raised an EVENT_HDLR_ERROR alarm without specifying an Oracle database error code in the scheduler log file.

Fixed Issues - 11.3.6 SP6 CUM1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP6 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 SP6 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

POJO JOB STRING ARRAY VALUES EQUAL SIGN PARSING ERROR

Problem Number: 4548

When you specify a string array as the value for the j2ee_parameter attribute in a POJO job, the value got truncated in parsing and this resulted in incorrect AFM formation. For example,

```
insert_job: test_POJO_String_arrays
job_type: POJO
class_name: com.ca.waae.Testing
method_name: setArrays
j2ee_parameter: String[5]="This\=That, ,is, ,Testing"
```

EXT CAL WITH HOL CAL CAN SOMETIMES GEN INVALID RUNDATES

Problem Number: 4584

The following three issues are corrected in this fix:

1. When you create an extended calendar that has a holiday calendar, invalid run dates were created if the holiday calendar extends to more than 337 days in the future.
2. If you create an extended calendar that has both a holiday calendar and a cycle, a maximum of 368 days of the holiday calendar was used to determine run dates, even if the holiday calendar was defined for many years in advance.
3. When you insert a standard calendar that has more than 16 dates and the application server is in HEAVY debug mode, the application server stopped responding.

AUTOBCP DATA PUMP FAILED WITH ORA-31684 ERRORS

Problem Number: 4602

The autobcpORAdp.pl script failed with the following Oracle error messages:

```
Processing object type SCHEMA_EXPORT/TYPE/TYPE_SPEC
ORA-31684: Object type TYPE:"AEDBADMIN"."UJO_PREDS_SUCCS_ROW" already exists
ORA-31684: Object type TYPE:"AEDBADMIN"."UJO_PREDS_SUCCS_TYPE" already exists
```

SCH DELAYS BTWN FETCH OF BOX RUNNING EVENT AND STARTING JOBS

Problem Number: 4606

When you define several box jobs with many jobs and the scheduler schedules the box jobs at nearly the same time, the scheduler delayed for several minutes from the moment it fetched the RUNNING event of the box job to the moment it scheduled the first job in the box.

LOGLIST OF DELETEJOB EVENTS DUE TO AUTO_DELETE

Problem Number: 4608

When you define a job with the auto_delete attribute, the DELETEJOB event is generated after each completion of the job. When you execute the autorep -J *job_name* -d command, a long list of DELETEJOB events are displayed; the first delete event is enough and the subsequent ones are just causing confusion.

MONTHLY AGGREGATION MISSING AFTER DST CHANGE

Problem Number: 4610

The scheduler failed to generate monthly aggregation information immediately after the DST change in the month of March.

SCHED FOREVER RUNS JOB WITH SYSTEM FAILURE ON FORCE_STARTJOB

Problem Number: 4611

The scheduler runs the job forever with the new run number every time the job is restarted due to system failures like the agent is down, user is not available, and so on, upon multiple FORCE_STARTJOB. This leads to generation of huge number of events in the database and database size grows.

APP SERVER CORES WHEN VERSION R11 AGENT CONNECTS TO IT

Problem Number: 4612

When you run jobs to a Unicenter AutoSys JM r11 agent, the application server process aborted ungracefully when the r11 agent connects to it.

RESTRICT_FORCE_STARTJOB DOES NOT ALLOW RESTART

Problem Number: 4613

When the CA Workload Automation AE scheduler is running with configuration RESTRICT_FORCE_STARTJOB, job once goes into RESTART due to environmental problems will stay in RESTART status for ever.

AUTHORIZATION CHECK FAILS FOR ZOS JOBS

Problem Number: 4614

AUTHSTR on CA Workload Automation AE manual job did not work with the zOS agent because the wrong AFM string is passed from the CA Workload Automation AE scheduler.

DB ERRORS ENCOUNTERED WHILE SENDING STATUS TO CA7**Problem Number: 4615**

When the description attribute value of the job exceeded 255 characters, the scheduler attempted to send the status of jobs to CA7. The scheduler encountered the following errors:

```
CAUAJM_E_10283 Exhausted maximum number of retries for scheduler
operation[get_legacy_job<73901>]
```

```
CAUAJM_E_50178 Could not get row for joid = 73,901
```

SCHED MANGLES ENVIRONMENT VARIABLE VALUES FOR MULTIPLE JOBS**Problem Number: 4616**

When you define several jobs that specified the envvars job attribute with a long value and executed the jobs at the same time, the scheduler corrupted the environment variable value of one or more jobs.

CROSS-PLATFORM JOBS RECEIVED ERRORS ON UJO_CHKPNT_RSTART**Problem Number: 4617**

When you issue FORCE_STARTJOB to a box job whose last run was triggered from cross-platform manager (CA7 or UJMO), following errors are noticed in the scheduler log while scheduler was attempting to send the status to the cross-platform manager (CA7 or UJMO):

```
CAUAJM_E_18402 ORA-12899: value too large for column
"AEDBADMIN"."UJO_CHKPNT_RSTART"."UBC_JOBNUMB"(actual: 12, maximum: 5)
CAUAJM_E_40321 Unable to process checkpoint restart event.
```

JOB FAILS TO START WITH QUEUEDJOB_STARTFAIL**Problem Number: 4618**

The scheduler failed to start a job and generated the QUEUEDJOB_STARTFAIL alarm when a job comes out of the QUE_WAIT, RES_WAIT, or PEND_MACH status and the job has a dependency on the global variable that matches the job's name.

EDIT SUPER USER CANNOT INSERT JOB INTO A BOX**Problem Number: 4619**

When CA Workload Automation AE is running in native mode, it did not allow job insert when an EDIT Superuser tries to insert a job into an existing box job. The JIL returns a failure stating that the user does not have execute permissions on the box. But, the EDIT Superuser by definition must be able to perform CRUD on any job in the instance.

CHK_AUTO_UP AND DBSPACE FAIL IN SYBASE KERBEROS**Problem Number: 4620**

When you use Kerberos authentication for Sybase database and you specify a Service Principal name that differs from the server name in the configuration file, the chk_auto_up and dspace commands failed to connect to the database.

MSSQL VIOLATION OF PRIMARY KEY CONSTRAINT XPKUJO_AFM TABLE**Problem Number: 4621**

When the scheduler executed many jobs at once in a Microsoft SQL Server instance, the scheduler occasionally experienced database errors while executing the ujo_sendafm stored procedure.

NULL VALUE ELIMINATED BY AGGREGATE MSGS AFTER REFRESHAEDB.PL**Problem Number: 4622**

When the scheduler started after executing the RefreshAEDB.pl script, the scheduler experienced repeated database errors while executing the ujo_get_jobstart_depends stored procedure.

AUTOPING -M ALL DOES NOT VALIDATE ALL MACHINES**Problem Number: 4623**

The autoping -M ALL command was not validating machines of type 'r' and 'n'.

SYS PASSWORD VISIBLE IN CREATEAEDB.PL**Problem Number: 4624**

The CreateAEDB.pl script generated logs that contained Oracle sys password, which is a security threat.

DB_ROLLOVER DOES NOT HAPPEN**Problem Number: 4625**

In HADS mode, the primary scheduler raised a DB_ROLLOVER alarm. The other components like the shadow scheduler, application servers did not rollover which caused interruptions in service and the instance became unusable.

ERRORS DISPLAYED WHEN UPDATING A JOB THROUGH JAVA SDK**Problem Number: 4626**

When InsertUpdateJobReq interface is used to update a job, the following error message was displayed when the same job is updated the second time:

```
CAUAJM_E_10281 ERROR for Job: pseudo_xiddm_ViewToAKill < Field 'send_notification = -1'
is not valid. Only '0', 'N/n', '1', 'Y/y', or 'F/f' are valid >
```

SLOW AUTOREP REPORTING MACHINES UNDER AS-LIST EEM SECURITY**Problem Number: 4627**

When you define the CA Embedded Entitlements Manager (CA EEM) security policies and enable the CA Workload Automation AE instance to use CA EEM as the external security service, and you then execute the autorep command to generate a report for hundreds of machines, the application server delayed evaluating policies although you are granted access to the as-list/AUTOREP CA EEM security policy resource.

WINDOWS INSTALL PATCH SCRIPT DOES NOT ALLOW (!) IN PWD**Problem Number: 4628**

CA Workload Automation AE Release 11.3.6 SP5 CUM1 patch installer script did not allow the special character (!) in the password. The installer script failed to authenticate with the SQL server when the password had (!) in the string.

MEMORY LEAK IN WAAE SDK**Problem Number: 4630**

When you run CA Workload Automation AE operations using the Java SDK, native memory is being exhausted and this resulted in the crash of the Java process.

THREAD CREATION PROBLEM AT EP_FAILOVER**Problem Number: 4632**

CA Workload Automation AE shadow scheduler complained about thread creation problems as shown below at the time of EP_FAILOVER. The shadow scheduler tried to send communication information to all the active agents, but failed to complete the notification process due to resource unavailability.

```
CAUAJM_E_00051 Unable to create thread! [31850240][0]:Resource temporarily unavailable
```

The shadow scheduler never completes failover causing scheduling problems.

MONITOR ASSOCIATES X-INSTANCE EVENTS TO DELETED JOBS**Problem Number: 4633**

CA Workload Automation AE monitor associates events of a x-instance job to the deleted job of the home instance and reported the deleted job in the report.

UJO_JOB_TREE TABLE NOT POLULATED DUE TO INVALID DATA**Problem Number: 4634**

Upgrades failed to populate the ujo_job_tree table due to invalid active rows in the ujo_job table.

FILE TRIGGER JOB FAILS WHEN WATCH_FILE_WIN_USER DEFINED**Problem Number: 4635**

The CA Workload Automation AE scheduler did not pass the correct password to the agent when the watch_file_win_user attribute is defined in a File Trigger job. The agent rejects the password and fails to start the job.

AUTOREP DOES NOT DISPLAY JOB RUN INFORMATION**Problem Number: 4636**

The autorep utility did not report job run information when a job's prior run number is 0 (zero). A job runs with a run_num of 0 when it is defined inside a box and is force started before the box has ever run after its creation.

CREATEAEDB FOR UPGRADE REQUIRES SA CREDENTIALS - SYB/MSQ**Problem Number: 4637**

The CreateAEDB.pl script for upgrade did not allow running the script with a non-sa user with dbowner privilege, which is enough to update the schema.

NOT ABLE TO UPDATE BOX_NAME USING IINSERTUPDATEJOBREQ**Problem Number: 4639**

When you try to update the box_name of a job using the IinsertUpdateJobReq API, the box_name was not updated.

SQL DEADLOCKS WHEN AUTO DELETING BOX JOBS**Problem Number: 4640**

When the auto_delete attribute is defined for box jobs and multiple DELETEJOB events for box jobs are processed simultaneously, SQL deadlocks were encountered with Microsoft SQL and Sybase databases.

BOX WITH AUTO_HOLD DOES NOT EXECUTE WHEN TAKEN OFF HOLD**Problem Number: 4641**

Suppose that you define a box with the auto_hold job attribute that is contained in an outer box. Also, suppose that you define jobs inside the inner box. When you send the JOB_OFF_HOLD event after the scheduler runs the outer box and sets the inner box to on hold, the scheduler did not schedule jobs contained in the inner box.

SCHEDULER DOES NOT PROCESS DUAL SERVER EVENTS AFTER PR4603**Problem Number: 4643**

This problem occurs when an CA Workload Automation AE instance is configured with dual event servers and the scheduler process executes on a powerful and fast machine. When the scheduler is busy running many jobs such that it accumulates hundreds of unprocessed job events in the ujo_event table, the scheduler fetched but did not process several events until it is restarted.

FALSE CAUAJM_W_10575 OUT OF SEQUENCE REPEATED MESSAGE**Problem Number: 4644**

When the instance is configured in high availability mode and the scheduler is configured in immediate primary fail back mode and resumes processing job events after the shadow scheduler fails over, the primary scheduler accidentally considered job statuses from the agent as out of sequence and discarded them.

CAUAJM_E_00051 THREAD CREATE ERRORS WHILE RUNNING MANY JOBS**Problem Number: 4645**

When the scheduler starts many jobs, it was unable to process incoming job statuses from the agents because it experienced operating system errors while trying to create an internal process thread.

SCHEDULER IS UNABLE TO AUTHENTICATE CROSS PLATFORM JOBS**Problem Number: 4646**

When you trigger multiple CA Workload Automation AE jobs simultaneously from a remote manager like CA-7, the scheduler sporadically failed to start one or more of the jobs due to authentication failures.

SCH/APPSRV SKIPS DAY FOR MULTI-DAY RUN WINDOW AFTER PR4532**Problem Number: 4647**

When you define a job that specifies a run window that spans across two days and other date time attributes that restrict the number of days the job can execute and the scheduler or application server evaluates the next start time at a time that is outside the run window, it incorrectly scheduled the job to a future date instead of on the current day.

SDK/CLIENT TOOLS TAKE LONG TIME TO LOAD JOB DETAILS**Problem Number: 4649**

CA WCC and autorep utility takes a long time to load job details when all the jobs in the database are retrieved at a time.

Third-Party Software Acknowledgements - 11.3.6 SP6

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

AE Release 11.3.6 SP5

This section includes information about the CA Workload Automation AE Release 11.3.6 SP5 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 SP5 and Release 11.3.6 SP5 CUM1.

WARNING

- On UNIX, if you are upgrading from a 32-bit CA EEM to a 64-bit CA EEM and CA EEM and CA Workload Automation AE (with the command sponsor component) are installed on the same computer, you must upgrade CA EEM first using the CA Common Components DVD only and then immediately upgrade CA Workload Automation AE. If CA EEM and CA Workload Automation AE are installed on separate computers, the limitation does not apply and so you can upgrade CA EEM using the CA Common Components DVD or the Standalone CA EEM DVD.
- The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6 SP5

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP5.

Support RHEL 7, OEL 7, and SUSE 12

CA Workload Automation AE supports the following Linux versions:

- Oracle Enterprise Linux 7
- Red Hat Enterprise Linux 7
- SuSE Linux Enterprise Server 12

New Attributes for Oracle E-Business Suite Single Request Jobs

The following attributes are added for the Oracle E-Business Suite Single Request jobs:

- [oracle_oprunit](#)—Specifies the operating unit name for a single request.
- [oracle_template_name](#)—Specifies the template name for a single request.

New as-sendevent Resource Class

The [as-sendevent resource class](#) and its default CA EEM policy with the following actions is added to control access to execute the sendevent command:

- ALARM
- CHANGE_PRIORITY
- CHANGE_STATUS
- COMMENT
- DELETEJOB
- FORCE_STARTJOB
- JOB_OFF_HOLD
- JOB_OFF_ICE
- JOB_OFF_NOEXEC
- JOB_ON_HOLD
- JOB_ON_ICE
- JOB_ON_NOEXEC
- KILLJOB
- MACH_OFFLINE
- MACH_ONLINE
- RELEASE_RESOURCE
- REPLY_RESPONSE
- RESTARTJOB
- RESUMEJOB
- SEND_SIGNAL
- SET_GLOBAL
- STARTJOB
- SUSPENDJOB
- UNSEND_EVENT

Changes to Existing Features - 11.3.6 SP5

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP5.

Support Global Resources for Box Job

You can now specify only global resources on a box job. However, the resource specified for a box applies only to the box and not to the jobs within the box.

Updated watch_no_change Attribute

You can now specify the watch_no_change attribute value in seconds, minutes, hours, or days. For more information about the watch_no_change attribute, see [watch_no_change Attribute -- Specify the Time the File Must Remain Unchanged](#).

Change the Status of the Job that is in the ON_NOEXEC Status

You can now issue the following events to change the status of a job that is in the ON_NOEXEC status:

- The CHANGE_STATUS event to change the status of a job that is in the ON_NOEXEC status to either the INACTIVE or SUCCESS status while remaining in the non-execution mode. When you change the status of a non-execution job to SUCCESS, the scheduler completes the job's run and schedules the downstream dependent jobs. Changing the

status of a non-execution box job to INACTIVE also changes the status of all the jobs in that box job to INACTIVE in non-execution mode. Changing the status of a non-execution box job to the RUNNING status starts the box job.

- THE JOB_ON_NOEXEC event to an ACTIVATED job that is within a running box. The job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job is met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.
- The JOB_OFF_NOEXEC event for a job. It places the job in the INACTIVE, ACTIVATED, or SUCCESS status based on the job's existing status while in non-execution mode. If the job is in a box, the scheduler does not evaluate the overall box status. If you send the JOB_OFF_NOEXEC event to a box, all jobs in the box (including all jobs contained in lower level boxes within the box) are reset to the INACTIVE, ACTIVATED, or SUCCESS status.

KILLJOB Send Signal Behavior

On UNIX agents (r11.3, Release 11.3.5, or Release 11.3.6), you can now send multiple signals by entering a comma-delimited list as the value for the signal_num variable. The agent sends the first signal, sleeps for a second, sends the next signal, and so on.

In the previous releases, only the legacy agents supported this behavior.

Priority is Honored for Jobs with Load Balancing and Resources Attributes

If you defined jobs with load balancing and resources attributes, jobs with highest priority run ahead of jobs with lower priority. If higher priority jobs are queued in the QUE_WAIT or RESWAIT state, then lower priority jobs are queued behind the higher priority jobs and remain queued until all higher priority jobs leave their queued states.

CA Workload Automation AE follows these rules to run jobs with priority, load balancing attributes, and resource attributes:

- Jobs with the highest priority run ahead of jobs with lower priority.
- Jobs with lower priority and load balancing attributes enter a queued state when higher priority jobs for the same queue are in the RESWAIT or QUE_WAIT state.
- Jobs with lower priority and resource attributes enter a queued state when higher priority jobs for the same queue are in the RESWAIT or QUE_WAIT state.
- Jobs with equal or higher priority and either load balancing or resource attributes or both run if there is sufficient load on the machine and sufficient resource units to execute.

You can now send a CHANGE_PRIORITY event to change the priority of RESWAIT jobs.

STATE_CHANGE Events Display in the Detailed Report of the Previous Run When Jobs are Queued in the RESWAIT State

The STATE_CHANGE events that are associated with the job entering and leaving the RESWAIT state are displayed in the detailed autorep report of the previous run of the job. These STATE_CHANGE events are no longer displayed in the detailed autorep report of the current run of the job.

Updated autorep Command

The -Q option is added to the autorep command to generate a report on jobs that are queued due to insufficient load balancing units, machine resources, or global resources. To generate a report on all the jobs waiting in the queue, specify -Q ALL. To generate a report for all jobs waiting on the same queues as a specific queued job, specify -Q *job_name*.

The ST column heading in the autorep report is now updated to ST/Ex. ST reports the completion status of the most recent run of the specified job or if the job is still running it reports the current status. If the job is in the ON_NOEXEC status, Ex reports the execution mode of the job as NE. Depending on the current status of the job, the ST/Ex column reports the status as IN/NE, AC/NE, RU/NE (for box jobs only), or SU/NE.

NOTE

The updates to the autorep command (the -Q option and the ST column updated to ST/Ex) are available only when you use the command line interface. These updates are not available if you use the CA Workload Automation AE SDK or web services.

Updated job_depends Command

The Status column in the job_depends report is now updated to Status/Exec. Mode. Status reports the completion status of the most recent run of the specified job or if the job is still running it reports the current status. If the job is in the ON_NOEXEC status, Exec. Mode reports the execution mode of the job as ON_NOEXEC.

Updated autostatus Command

If the job is in the ON_NOEXEC status, the autostatus command displays the current status and the execution mode of the job.

Fixed Issues - 11.3.6 SP5

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP5.

TIP

CA Workload Automation AE Release 11.3.6 SP5 also includes the issues fixed in each of the prior releases of 11.3.6.

GLOBAL VARIABLE TAKES LONG TIME TO STARTJOB**Problem Number: 4372**

When many (excess of 20000) jobs depend on a global variable, the scheduler took a long time to determine the list of jobs that depend on the global variable. You notice a long pause between the global variable setting and the first job starting due to the global variable setting.

SCHED LEAVES JOBS IN QUE_WAIT**Problem Number: 4416**

The scheduler performs machine load calculations to evaluate jobs that enter or leave the QUE_WAIT status. Depending on the timing of when a load-balancing job starts and another job completes, the part of the scheduler that evaluates a job to place it in the QUE_WAIT state competes with the part of the scheduler that evaluates queued jobs to take them out of the QUE_WAIT state. When a race condition occurred, the job remained in the QUE_WAIT status instead of executing.

SCHED UNFAIRLY DELAYS JOB START HANDSHAKES ACROSS NODE NAMES**Problem Number: 4427**

When the scheduler has to start a steady stream of jobs across many different node_name machine attribute values, the scheduler might remain focused on starting jobs on a small subset of the node names. This results in delay of starting jobs on the rest of the node names.

AUTOREP -T -D DISPLAYS WRONG TIME**Problem Number: 4434**

The autorep -j -t -d command displayed a wrong time zone when the server is on a UNIX or Linux platform and the client is on a Windows platform and the time zone is specified in the Olson format.

TIMEZONE IN DB IS WRONG**Problem Number: 4436**

The time zone mentioned in the database was incorrect for few time zones.

SENDEVENT WASTES ONE EVENT ID FOR EACH EVENT**Problem Number: 4443**

When you invoke the sendevent command to send a single event, the application server reserves two event ID values but only assigns one to the event.

EEM REPORTING UTILITY FAILS ON 32 BIT EEM**Problem Number: 4452**

The CA EEM Reporting utility failed to install if the 32-bit CA EEM is installed on a Windows machine.

UNIX INSTALLER SHOWS SUMMARY SCREEN WITH WRONG DATA**Problem Number: 4459**

On UNIX, when you select the Modify option to add a non-existing feature, the installer displayed wrong information about the database, ORACLE_HOME, and so on.

SCHED DOES NOT RUN JOB WITH 24 HOUR RUN_WINDOW**Problem Number: 4463**

When you define a job to have a run_window attribute that spans 24 hours (for example, "00:00-23:59"), the scheduler only executed the job on the first day, but not on subsequent days.

SCHEDULER DOES NOT HONOR PRIORITY FOR QUE_WAIT/RESWAIT JOBS**Problem Number: 4466**

When the scheduler evaluates jobs and determines that there is insufficient load or virtual resources to execute the job, it places the job in either the QUE_WAIT or RESWAIT status. While the scheduler honors the priority of QUE_WAIT jobs when evaluating other load balancing jobs, it allows load balancing jobs with a higher priority job attribute value to be scheduled ahead of jobs that have a lower priority attribute value and are in RESWAIT. Likewise, while the scheduler honors the priority of RESWAIT jobs when evaluating other jobs having virtual resource dependencies, it allows jobs having virtual resource dependencies with a higher priority job attribute value to be scheduled ahead of jobs that have a lower priority attribute value and are in the QUE_WAIT status.

AUDIT TABLE AND APPSERVER LOG EXPOSE PASSWORDS IN PLAIN TEXT**Problem Number: 4468**

CA Workload Automation AE tracking feature stores the plain-text passwords and displayed the same when the external security credentials are used with the sendevent command.

JOBS LOOP INFINITELY IN RESTART/STARTJOB**Problem Number: 4469**

Jobs in the RESTART or STARTJOB state loop indefinitely when the GlobalPendMachStatus parameter is set to FAILURE and the job has the n_retry attribute set to a valid value and machine is not available for running the job to terminal status.

AUTOREP DETAIL COMMAND TAKES A LONG TIME**Problem Number: 4470**

'autorep -j <job> -d' command took a long time to display the job details. This problem occurs when the ujo_proc_event table has large number of rows in it.

SCHEDULER DOES NOT RETRY RESOURCE QUERY WHEN NEXTROW FAILS**Problem Number: 4473**

The scheduler performs the following actions to execute a database query: parse the SQL, bind the input and output parameters to memory, execute the SQL, and fetch each row from the result set in a loop. When the scheduler performs a query related to virtual resources and the execution operation fails due to a recoverable error, the scheduler retries the query. However, if the scheduler experienced a recoverable database error during the row fetching phase after the virtual resource-based query executed successfully, the scheduler did not retry the query.

SCHED SENT DB_PROBLEM ALARM AFTER BOX AND JOB TAKEN OFF HOLD**Problem Number: 4478**

Suppose you define a box with jobs inside an outer box job and you place the inner box and a job inside the inner box on hold. When you run the instance in dual event server mode then take the inner box and the job in the box off hold at nearly the same time, the scheduler generated a DB_PROBLEM alarm.

UNIQUE CONSTRAINT AEDBADMIN.XPKUJO_RESWAIT_QUE VIOLATED**Problem Number: 4479**

When a job using resources and having multiple successors is run, the scheduler log showed the following database errors intermittently:

```
CAUAJM_E_18416 Event Server: <CAAP> Failed Query: <INSERT INTO ujo_reswait_que(joid,
  priority, stamp, run_num, ntry, wf_joid) VALUES (xxxx, 0, nnnnn, mmm, 1, 1)>
CAUAJM_E_18402 ORA-00001: unique constraint (AEDBADMIN.XPKUJO_RESWAIT_QUE) violated
CAUAJM_I_18403 Processing OCI function OEXEC, OEXN(3)
CAUAJM_E_18400 An error has occurred while interfacing with ORACLE.
CAUAJM_E_18401 Function <doExecute> invoked from <execute> failed <862>
```

JIL ACCEPTING WRONG VALUE FOR START_TIMES**Problem Number: 4480**

If the start_times attribute value is more than 9 characters in length (for example: 12:00000000023), the job was scheduled to run in the wrong time (for example: 10:39).

JIL MISMATCHED DOUBLE QUOTES CAN CAUSE WRONG ERROR MSG**Problem Number: 4481**

When there is no closing double-quote in a jil insert, sometimes jil indicated an error message for the wrong field.

```
insert_job: test_unbalanced_qte
job_type: cmd
```

```

command: "echo hello wall
machine: local_machine
owner: autosys
permission:
std_out_file: /tmp/$AUTO_JOB_NAME.out
std_err_file: /tmp/$AUTO_JOB_NAME.err

```

```
jil < test.jil
```

```

CAUAJM_I_50323 Inserting/Updating job: test_unbalanced_qte
CAUAJM_E_50442 Invalid attribute value for <permission>: Unexpected end of character
string: "echo hello wall
CAUAJM_E_10302 Database Change WAS NOT successful.

CAUAJM_E_50198 Exit Code = 1

```

SCHED TAKES LONG TO START JOBS DUE TO UJO_COMM_UPDATE SP

Problem Number: 4483

The scheduler delayed job starts for up to 30 to 45 minutes when the ujo_comm_update stored procedure executed for a number of seconds as it updated the outbound message sequence number in the ujo_comm_send_seq database table.

JIL SYNTAX CHECKER DOES NOT DETECT BOXES

Problem Number: 4486

The jil syntax checker did not detect boxes while validating the syntax of child jobs and it failed with the following error even if the box existed:

```
CAUAJM_E_18947 Invalid box name box1 specified, it does not exist.
```

SCHEDULER DOES NOT RUN DOWNSTREAM JOB AFTER CONDITION UPDATE

Problem Number: 4488

When you update a job definition to specify a new dependency on another job in a condition that contains a notrunning dependency and the predecessor job completes, the scheduler did not start the updated job.

CHILD JOBS STAY IN ACTIVATED STATE AFTER RUNNING ONCE IN BOX

Problem Number: 4490

The child jobs inside a nested box were wrongly placed in the ACTIVATED state after they run to completion status. This happened because a FORCE_STARTJOB event is issued to its immediate parent job.

SCHED HANGS ON SHUTDOWN AFTER ACCEPT THREAD CREATE ERRORS

Problem Number: 4491

When the scheduler fails to create a thread to process incoming job status messages from an agent during its run time, and you initiate a shut down, the scheduler process remained indefinitely in a deadlocked state.

GETTING RESOURCE DEFINITION THROWS DATABASE ERRORS

Problem Number: 4492

The description attribute that describes a virtual resource can contain up to 256 characters. Insertion of a 256 character description is successful. But, when you try to get the resource definition report either using CA WCC or the autorep -V name -q command , it threw database errors. For example, in case of Oracle, the following errors were displayed:

```
CAUAJM_E_40200 GetResourcesWithFilter failed.
GetResourcesWithFilter info: CAUAJM_E_18416 Event Server: <INCR3> Failed Query:
<select distinct L.res_name, V.mach_name, V.description, V.type,V.amount_defined from
  ujo_virt_resource_lookup L join ujo_virt_resource Von L.roid=V.roid where (L.res_name
    like '%' or L.res_name is null) order by1 ASC NULLS FIRST ,2 ASC NULLS FIRST >
CAUAJM_E_18402 ORA-01406: fetched column value was truncated
CAUAJM_I_18403 Processing OCI function not used(4)
CAUAJM_E_18400 An error has occurred while interfacing with ORACLE.
CAUAJM_E_18401 Function <doExecute> invoked from <execute> failed <882>
```

CAUAJM_W_19102 ADDED JOB INTO AN ALREADY RUNNING BOX

Problem Number: 4493

When you insert a job into an already running box, the following sequence of messages are displayed. These messages can confuse you into thinking that the job is starting immediately.

```
CAUAJM_I_50323 Inserting/Updating job: boxA_newjob1
CAUAJM_I_10122 Job 'boxA_newjob1' scheduled: 05/17/2016 23:00:00
CAUAJM_W_19102 Added job boxA_newjob1 into an already running box. Sent the STARTJOB
  event to schedule the job immediately.
CAUAJM_I_50205 Database Change WAS Successful!
```

The message sequence is now updated as follows:

```
CAUAJM_I_50323 Inserting/Updating job: boxA_newjob1
CAUAJM_I_10122 Job 'boxA_newjob1' scheduled: 05/18/2016 23:00:00
CAUAJM_W_19102 Added job boxA_newjob1 into an already running box. Sent the STARTJOB
  event to schedule the job immediately, if necessary, or on its appointed time.
CAUAJM_I_50205 Database Change WAS Successful!
```

JIL SYNTAX VERIFICATION PERFORMANCE POOR FOR BOX JOBS

Problem Number: 4494

Jil syntax verification of a job in a box takes minutes when there are thousands of jobs in the database. For example update_job of a job in a box takes 2 to 3 minutes when there are 100K jobs in the database. Note that the syntax verification of jobs that are not in the boxes is not the problem here.

JOB GOES TO PE INSTEAD OF OH WHEN AGENT IS OFFLINE

Problem Number: 4495

Job is put in the PEND_MACH state instead of the ON_HOLD state when an agent is offline and the scheduler is running in the global auto hold mode.

AUTOREMOTEDIR CONFIG FILE VALUE OVERWRITES AUTOSERVER VALUE

Problem Number: 4496

When you specify the AutoRemoteDir parameter value in the configuration file containing more than 64 characters, the scheduler overwrote the AutoServer parameter value in the configuration file.

DATABASE CONNECTION POOL USAGE OPTIMIZATIONS AFTER 4428

Problem Number: 4497

After applying the solution to problem 4428 (SCH SKIPS HA HEARTBEATS WHEN A DATABASE IS TEMPORARILY DOWN), the scheduler indefinitely lost access to one of the database connections in the database connection pool.

APP SERVER HANGS OR LEAKS RESOURCES ON THREAD CREATION ERROR

Problem Number: 4498

When the application server process fails to create threads to service SDK client requests, it increased in system memory, lost access to database connections, or remained indefinitely in a deadlocked state.

SCHED RESETS JOBS IN BOX TO INACTIVE WHEN BOX GOES OFF HOLD

Problem Number: 4499

When you send the JOB_OFF_HOLD event for a box that is in the ON_HOLD status, the scheduler reset all jobs contained in the box to INACTIVE. Similarly, when you send the JOB_OFF_ICE event for a box that is in the ON_ICE status, the scheduler reset all jobs contained in the box to INACTIVE.

HOLIDAY CALENDAR EXISTANCE IN EXTENDED CALENDARS

Problem Number: 4501

When an extended calendar is created with a non-existing holiday calendar using the autocal_asc command from the CLI, it threw the following warning. However, when you perform the same task using the SDK or CA WCC, it did not display the following warning:

```
CAUAJM_W_50413 *** WARNING - The Holiday calendar: cal_name does not exist.
```

UNIQUE CONSTRAINT VIOLATION IN UJO_REP_% TABLES

Problem Number: 4502

Unique constraint violation in the ujo_rep_% tables when the first date falls on the DST failover or rollback day, week, or month. The scheduler reported the following error:

```
ORA-00001: unique constraint (AEDBADMIN.XPKUJO_REP_WEEKLY) violated
```

SERVICEDESK REQUEST FAILS TO OVERRIDE ATTRIBUTE

Problem Number: 4503

The scheduler did not override attributes that are defined as part of the svcdesk_attr attribute value in your job definition and complained about duplicate service desk keyword.

DBMAINT THROWN ERRORS WHEN IT RUNS ARCHIVE_JOBS

Problem Number: 4504

The DBMaint command threw errors when it runs the archive_jobs command. This problem occurs when you defined a job to send an email notification using the send_notification, notification_emailaddress, and notification_msg attribute values and then try to archive the job information using the DBMaint command. The DBMaint command internally calls the archive_jobs command, which threw errors similar to the following:

```
Invalid column name 'notification_emailaddressNOT'
```

ARCHIVING CHECKPOINTS THROWS DATABASE ERRORS

Problem Number: 4505

The archive_events command with the -r option failed and returned an error saying that the fetched column value was truncated. This happens when you insert maximum number of elements into the database column and try to fetch column data using the archive_events command with the -A and -r options.

FALSE MUST_COMPLETE_ALARM AFTER JOB UPDATE ON RUNNING JOB

Problem Number: 4508

A child job with must_start_times or must_complete_times in the RUNNING state raises false MUST_START_ALARM and MUST_COMPLETE_ALARM alarms when the job is updated or the calendar associated with the job is updated.

REDUCE TIME TAKEN TO COMMIT IN UJO_BUMP_OID

Problem Number: 4509

The scheduler and application server locked the ujo_next_oid table for a longer time than required; causing others threads to wait for the lock on the ujo_next_oid table.

SEND_SIGNAL KILLED UNIX SYSTEM PROCESSES

Problem Number: 4511

When you issue the SEND_SIGNAL event, it terminated the UNIX system processes.

ARCHIVE_EVENTS DELETES INVALID DATA FROM EXT_RUNINFO

Problem Number: 4517

If you update a job while it is in the RUNNING state, the archive_events utility deleted that job information from the ujo_job_runs and ujo_extended_jobrun_info tables.

ARCHIVE_EVENTS DID NOT ARCHIVE OLD EVENTS IF ON_HOLD ON_ICE**Problem Number: 4519**

The archive_events command did not archive the latest run events for jobs even when they are older than the days specified in the archive_events utility. In other words, the most current run events for a job are kept in the database if the status of the job is ON_HOLD or ON_ICE.

BLOB JOB RETURNS UNABLE TO STORE STDOUT TO APP SERVER**Problem Number: 4520**

The autorep -d command displayed an error message 'unable to store STDOUT to the Application Server'. This problem occurs when you define a job to capture the output and store the output as a textual formatted output blob associated with the current job.

DUPLICATE MACHINE INFORMATION IN WEB SERVICE**Problem Number: 4521**

The web service request generated duplicate machine definitions when the application is running in the High Availability mode. In few cases, the machine definition does not get listed.

QUERY NOT RETRIED AFTER ORA-08103**Problem Number: 4522**

When Oracle returns ORA-08103, the failed query is not retried. This results in the termination of the transaction.

Fixed Issues - 11.3.6 SP5 CUM1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP5 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 SP5 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

MACHINE REPORT DOESNT CALCULATE THE CORRECT CURRENT LOAD**Problem Number: 4525**

When you define multiple virtual machines that specify the same machine and their own max load values and then run jobs with loads on those machines, the autorep command machine report displayed incorrect total load on the machine.

SAP JOB COPY JOB DOES NOT HONOR KILLJOB COMMAND**Problem Number: 4527**

The scheduler failed to terminate a SAP Job Copy job(SAPJC) when you issue the KILLJOB event.

GLOBAL VARIABLE - ESCAPED SINGLE QUOTE NOT REMOVED**Problem Number: 4533**

The global variable value with a single quote gets escaped with two single quotes during global variable creation. While sending the global variable value to the agent, the scheduler should replace two quotes with a single quote, but the scheduler always sends two quotes causing job failures.

DATABASE ERR WHEN SCH PUTS JOB W/ MULTIPLE MACHS IN RESWAIT**Problem Number: 4534**

When you start or force start a job with a resource dependency and no load-balancing attributes that specify a comma-separated list of machines or a virtual machine with multiple machine components and no resources are available, the scheduler generated database errors while placing the job in the RESWAIT status.

AUTOSYS_SECURE OPTION 3 CORRUPTS DATABASE PASSWORD**Problem Number: 4535**

On Solaris, when you select option 3 from the autosys_secure interactive menu to change the password of the autosys user, the autosys_secure command generated errors and stored a corrupted password in the ujo_alamode database table.

SCH IGNORES AMOUNT WHEN IT FORCESTARTS JOBS SHARING RESOURCE**Problem Number: 4536**

When you force start multiple jobs that specify the same virtual resource, the scheduler started the jobs even when no more resources are available.

AS_SERVER SLOWER THAN 11.3 SP1 IN DOING JIL OPERATIONS**Problem Number: 4539**

The job or machine insert, update, or delete operation was taking 40 to 50 percent more time when compared with the time it takes with the r11.3 latest maintenance.

AUTOSYSLOG DOES NOT FETCH LOG WHEN STD_OUT_FILE IS INVALID**Problem Number: 4540**

When you set the value of the std_out_file attribute to a non-existent path and try to run the job, a non-zero exit code is generated by the agent. This prevents a row to be created in the ujo_job_run table. As a result, the autosyslog command is unable to display the agent log. You cannot fetch the previous run information using the autorep -J *job_name* -r *number* command or the job run details using CA WCC.

SCH PROCESSES DUPLICATE STATUSES AFTER PROBLEM 4455**Problem Number: 4541**

When the scheduler is running with a multi-threaded agent listener and the agent resends job status messages for which it did not receive an acknowledgment, the scheduler did not detect that the message is a duplicate and it generated a repeating job event.

N_RETRYS AND CONDITION LOOKBACK TOGETHER DOESN'T RESTART JOB**Problem Number: 4543**

When a job is defined with both the condition attribute with lookback and the n_retrys attribute, a RESTART event is generated due to an application failure. However, the job fails to start and the following message is displayed in the scheduler log:

```
CAUAJM_I_40162 The starting conditions for job <Job1> have not been met.
Cannot be started.
```

CAUAJM_I_40149 Cancelling restart for job <Job1> and setting to INACTIVE.

The job must restart as the job's condition succeeded in the first time.

COMMAND LINE UTILITIES BREAK WITH MISSING OPTION

Problem Number: 4546

When you specify only the dash (-) character as an argument to the command line utilities like jil, sendevent, autorep, and so on, that command line utility process aborted ungracefully.

11.3.6 SP5 SCHEDULER WRONGLY PLACES UPDATED JOB IN RESWAIT

Problem Number: 4549

When you define two virtual resources and you define a job that initially uses one virtual resource and then update the job to use the other virtual resource, the scheduler placed the updated job in the RESWAIT state when higher priority jobs are queued in the RESWAIT state due to the first resource.

11.3.6 SP5 FIX SCHEDULER QUEUING W/ RESOURCE JOB AND NO LOAD

Problem Number: 4550

This problem addresses two issues that occur when you define jobs with resource dependencies, a non-zero priority attribute value, and a job_load attribute value of zero:

1. When you specify a lower priority job with a job_load attribute and no resource attributes having the same machine attribute value as the resource job, and the resource job is in the RESWAIT state, the scheduler placed the job in the QUE_WAIT state.
2. When you specify a lower priority job with a different resource name and a job_load attribute value of zero having the same machine attribute value as the resource job, and the resource job is in the RESWAIT state, the scheduler placed the lower priority job in the RESWAIT state.

11.3.6 SP5 FIX SCHEDULER LOAD BALANCING W/ VIRTUAL RESOURCES

Problem Number: 4551

When you define a job with virtual resource dependencies and specify a virtual machine as the machine attribute value, the scheduler submitted jobs at random when two or more machines satisfied the resource requirements even when a machine with better load or factor values exists.

SCHED DEADLOCKS IN GLOBALAUTOHOLD WHEN JOBS LEAVE QUE_WAIT

Problem Number: 4552

When you configure the scheduler to start in the GlobalAutoHold mode, the scheduler experienced a deadlock as it placed a load balancing job in the ON_HOLD status from the QUE_WAIT status.

SCHED REMOVES OVERRIDE ON 1ST FAIL INSTEAD OF AFTER N_RETRYS

Problem Number: 4553

When you insert a job definition that specifies a n_retrys job attribute value greater than 0 and you insert a job override, the scheduler removed the override after the first failure instead of removing the override after completing all application restarts.

BOX RUNNING EVENT PROCESSED AFTER JOB IN THE BOX EXECUTES**Problem Number: 4554**

When you configure the CA Workload Automation AE instance to run in the dual event server mode and start a box during a period when the job starting activities are high, the scheduler executed a job in a box that is triggered by its predecessor job before it sets the box itself to the RUNNING status.

11.3.6 SP5 UJO_SETUP_NEXT DELETES CHK_RUN_WINDOW ON SAME DAY**Problem Number: 4555**

When you define a job with date and time conditions that specifies multiple times in a day for the start_times, must_start_times, and must_complete_times attributes, and you update the attributes while the job executes, the scheduler deleted the CHK_COMPLETE event of the run that was still executing.

Also, when you specify the run_window attribute for a job that runs multiple times in a day, the scheduler deleted the CHK_RUN_WINDOW event after the first execution of the job.

11.3.6 SP2 SCHED RUNS JOBS W/ NOTRUNNING COND AT SAME TIME**Problem Number: 4556**

When you define a set of jobs with notrunning dependencies on each other, all the jobs started at once as the scheduler did not always honor the job condition.

SCHED IN GLOBALAUTOHOLD MODE DOESNT HOLD JOBS FROM PEND_MACH**Problem Number: 4557**

When you restart the scheduler in the global auto-hold mode after it sets jobs to the PEND_MACH status due to an offline agent and when the agent comes online, the scheduler executed the job instead of placing it in the ON_HOLD status.

SERVICEDESK TICKET IGNORES CATEGORY ATTRIBUTE**Problem Number: 4560**

The scheduler did not pass the non-default attributes that are used as part of the svcdesk_attr attribute value in your job definition; this causes inconvenience to advance users.

JOB IN QUE_WAIT DID NOT RUN UPON CHANGING THE QUEUE PRIORITY**Problem Number: 4563**

When the scheduler is running with MachineMethod=job_load and the job has a priority of non-zero and a job_load of nonzero and it went to the QUE_WAIT status. The job did not run after changing its queue priority to zero using the event: CHANGE_PRIORITY.

UNDO CROSS-QUEUE PRI CHK BTWN RESWAIT/QUE_WAIT AFTER PR4466**Problem Number: 4566**

Suppose that you define two jobs:

1. A job with a priority attribute value that has a job_load value and runs on a virtual machine.
2. A load balancing job with virtual resource dependencies that runs on the same virtual machine as the first job and has a lower priority attribute value than the first job.

When the scheduler evaluated the first job and the second job is queued in the RESWAIT status, the scheduler immediately queued the first job in the QUE_WAIT status.

Suppose that you define two jobs:

1. A job with a priority attribute value that has a resource dependency.
2. A load balancing job with a dependency on the same virtual resource dependency as the first job and has a lower priority attribute value than the first job.

When the scheduler evaluated the first job and the second job is queued in the QUE_WAIT status, the scheduler immediately queued the first job in the RESWAIT status.

AUTOREP -Q XTRA ROWS FOR JOB W/ GLOBAL RSRC AND VIRTUAL MACH

Problem Number: 4567

Suppose that you define a job that specifies a priority attribute value, a job_load attribute value of zero, a dependency on a global virtual resource and it runs on a virtual machine. When the scheduler placed this job in the RESWAIT status, the autorep utility generated one entry for each machine component that is part of the virtual machine in the queue report (using autorep -Q).

SCHED FOREVER RUNS JOB WITH VIRTUAL RESOURCES AND N_RETRYS

Problem Number: 4574

When you insert a job definition with virtual resource dependencies and a n_retrys job attribute value that is greater than zero (0), the scheduler continually created a new run every time the job restarted and then entered and exited the RESWAIT status.

RESTARTING MICROFOCUS JOB AT STEP FAILS

Problem Number: 4577

When you define a Microfocus job that consists of multiple named execution steps and you manually send the RESTARTJOB event with the execution step name to execute the job, the scheduler failed to restart the job.

Changes to Existing Features - 11.3.6 SP5 CUM1

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP5 CUM1.

Priority is Honored for Jobs with Load Balancing and Resources Attributes

If you defined jobs with load balancing and resources attributes, jobs with highest priority run ahead of jobs with lower priority. If higher priority jobs are queued in the QUE_WAIT or RESWAIT state, then lower priority jobs are queued behind the higher priority jobs and remain queued until all higher priority jobs leave their queued states.

CA Workload Automation AE follows these rules to run jobs with priority, load balancing attributes, and resource attributes:

- Jobs with the highest priority and load balancing attributes run ahead of jobs with lower priority for the same machine-based queue.
- Jobs with the highest priority and resource attributes run ahead of jobs with lower priority for the same virtual resource-based queue.
- Jobs with lower priority and load balancing attributes enter the QUE_WAIT state when higher priority jobs for the same machine-based queue are in the QUE_WAIT state.
- Jobs with lower priority and resource attributes enter the RESWAIT state when higher priority jobs for the same virtual resource-based queue are in the RESWAIT state.
- Jobs with equal or higher priority and load balancing attributes run if there is sufficient load on the machine to execute.
- Jobs with equal or higher priority and resource attributes run if there are sufficient resource units to execute.
- Jobs with equal or higher priority and both load balancing and resource attributes run if there is sufficient load on the machine and sufficient resource units to execute.

You can send a CHANGE_PRIORITY event to change the priority of RESWAIT jobs.

AE Release 11.3.6 SP4

This section includes information about the CA Workload Automation AE Release 11.3.6 SP4 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 SP4 and Release 11.3.6 SP4 CUM1.

WARNING

The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6 SP4

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP4.

(Oracle only) Support Single Sign-On Wallets with Certificates for Database Access

You can now configure CA Workload Automation AE to use single sign-on wallets with certificates that are present on the file system to access the database. During installation, you can set the database access mode to either password or SSL certificate mode.

The [DBAccessMode](#) parameter in the configuration file or the Access Mode field on the Event Server - CA Workload Automation AE Administrator window of the Administrator utility (Windows) displays the access mode that CA Workload Automation AE is using to establish connections to the database.

After you installed CA Workload Automation AE, you can use the switchDBAccessMode.pl script to [switch the database access mode](#) from password to SSL certificate mode or vice versa.

Fixed Issues - 11.3.6 SP4

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP4.

TIP

CA Workload Automation AE Release 11.3.6 SP4 also includes the issues fixed in each of the prior releases of 11.3.6.

UPDATE_MACHINE LOOKS EXEC SUPERUSER UNDER NATIVE SECURITY

Problem Number: 4149

When CA Workload Automation AE is running in native security mode, you could not update a machine using the update_machine subcommand if you are not an EXEC superuser. You can insert and delete a machine, but cannot update it.

JOBS STUCK IN STARTING/RUNNING AFTER FAIL-OVER/FAIL-BACK

Problem Number: 4390

Some jobs were stuck in the STARTING or RUNNING state after the scheduler fails over or fails back. This happened because the scheduler did not contact all active agents to send updated scheduler communication attributes. This happened for agents whose port is changed after the last successful communication.

SCHED HANGS WHEN PROCESSING MACH_ONLINE EVENT

Problem Number: 4393

When the scheduler processes the MACH_ONLINE event, it expects to evaluate all jobs in the PEND_MACH state. When this expectation is not met and the scheduler skips a job start evaluation, the scheduler did not evaluate the start of other PEND_MACH jobs. Also, during shutdown, the scheduler blocked indefinitely.

SCHED LEAVES EVENTS BEHIND WHEN SP FAILS DUE TO BIND ERRORS

Problem Number: 4415

The scheduler performs the following actions to execute a database query: parse the SQL, bind the input and output parameters to memory, execute the SQL, and fetch each row from the result set in a loop. Normally, when the execute operation fails due to a recoverable error, the scheduler retries the query. The exception is the ujo_batch_get_event stored procedure as it is executed at regular intervals. If, during one interval, the ujo_batch_get_event stored procedure fails to return a batch of events, those events are marked so that they remain behind out of the reach of the scheduler. For this reason, the scheduler does not retry the query so that it can recover and fetch the events as part of the next event polling iteration.

However, when the database operation to fetch events failed because of errors specific to the bind action, the scheduler retried the query and left a batch of events in the ujo_event table.

SCHED DOES NOT UPDATE JOB STATUS AFTER FAILOVER

Problem Number: 4417

When the primary scheduler detects that the shadow scheduler has taken over, it shuts itself down. Due to a timing issue, the primary scheduler may not send events for a job that has already completed as it shuts down. The shadow scheduler is not aware of the primary scheduler's final job activities and ignored the job. The job remained stuck in the RUNNING state until the primary scheduler is restarted.

COMM TABLE RECONCILE SCRIPT DOESNT WORK DUE TO SQL ERROR

Problem Number: 4419

Prior to upgrading the product binaries, it is recommended that you execute a SQL script that repairs table data that is used to store information about all agents. You may be required to execute the SQL script several times especially if your networked environment uses multiple levels of virtual node name associations. For example, you may be running in a networked environment where a virtual node name resolves to one of the many virtual IP addresses, each of which resolves to numerous physical IP addresses. In this case, the output of the SQL script may prompt you to execute one or more additional times to repair each level of virtual node name association that is detected in the table data. However, in

some cases, the script may experience a database error and leave the table data unrepaired regardless of the number of executions.

COMM TABLE RECONCILE SCRIPT COMPLETES AFTER HALF AN HOUR

Problem Number: 4420

Prior to upgrading the product binaries, it is recommended that you execute a SQL script that repairs table data that is used to store information about all agents. You may be required to execute the SQL script several times especially if your networked environment uses multiple levels of virtual node name associations. For example, you may be running in a networked environment where a virtual node name resolves to one of the many virtual IP addresses, each of which resolves to numerous physical IP addresses. In this case, the output of the SQL script may prompt you to execute one or more additional times to repair each level of virtual node name association that is detected in the table data. However, in some cases, the script may take 30 minutes or more to complete.

UNWANTED WORLD WRITE PERMISSIONS TO FILES

Problem Number: 4422

The files created by the installer had write permissions to everyone. Security audits complained about world write permissions.

SENDEVENT FROM WCC DOES NOT COMPLAIN FOR INACTIVE FOR ON_ICE

Problem Number: 4423

CA WCC did not complain when a job in the ON_ICE state is placed in the INACTIVE state. However, the scheduler rejected the sendevent command silently and displayed the following message that lead to the assumption that the sendevent command executed without any errors:

```
CAUAJM_E_40127 Cannot change the status of job <test> to INACTIVE because it is in the
ON_ICE status. Please send the JOB_OFF_ICE event to take the job off the ON_ICE status.
```

SCH SKIPS HA HEARTBEATS WHEN A DATABASE IS TEMPORARILY DOWN

Problem Number: 4428

When the instance is running in high availability mode with dual event servers and one of the event servers is temporarily down or unreachable, the primary scheduler may not update the ujo_ha_process table of the available event server if the loss occurs during a period of heavy database activity. Since the primary scheduler could not obtain a database connection to the live event server in order to notify the shadow of its presence, the shadow prematurely failed over.

WATCH_FILE_MIN_SIZE ALWAYS DISPLAYED AS 0 FROM SDK

Problem Number: 4429

When the watch_file_min_size attribute is retrieved from the Java SDK, its value was always displayed as zero even when you assigned a proper value to the attribute.

JOBS IN OFF HOLD BOXES RUN WITH WRONG NUMBER AFTER PR#4408

Problem Number: 4430

When you run a multi-level box after placing one of the inner level boxes on hold, the scheduler executed all jobs except the box that is in the ON_HOLD status. When you take the box off hold, the scheduler executed the box, but the jobs in the box retained the previous run number instead of advancing it.

AGGREGATOR STATISTICS NOT GENERATED AFTER SP3 UPGRADE

Problem Number: 4431

After you upgrade to Release 11.3.6 SP3, the scheduler failed to generate aggregator statistics.

SCH/APPSRV HANG IN FINDCLIENT AFTER AGENT COMMUNICATIONS

Problem Number: 4433

As the scheduler and application server communicate with many agents, the internal communications engine may develop an internal deadlock condition that causes the scheduler or application server to stop responding in the communications. Also, depending on the timing of the shutdown, the scheduler may deadlock or shutdown.

AUTOCAL_ASC IMPORT SUPPORT DATE FORMAT

Problem Number: 4435

The calendar utility (autocal_asc) did not support the instance date format in the import operation.

EVENT_HDLR_ERROR DUE TO UJO_POST_EVENT SP

Problem Number: 4438

The scheduler encountered the EVENT_HDLR_ERROR error when the ujo_post_event event fails to execute. The failure is not accompanied by any database error and the query is not retried resulting in the EVENT_HDLR_ERROR error.

APP SERVER SHUTS DOWN ON A QUERY FAILURE

Problem Number: 4442

The application server shuts down on a query failure in the database monitor thread; incorrectly assuming that the database connectivity is lost.

CAPKI5 - UPGRADE PROBLEM - WINDOWS ONLY

Problem Number: 4444

CA Workload Automation AE Release 11.3.6 SP3 installer failed to upgrade a multi-instance machine when all the instances in the machine are running in the CA EEM security mode.

JOB DOES NOT RUN WITH MULTIPLE GV CHANGES

Problem Number: 4447

The scheduler ignored jobs with global variable condition when two or more sendevents are issued to the same global variable with the same value.

AUTOSYS_SECURE CANNOT DELETE/UPDATE USER

Problem Number: 4449

The CA Workload Automation AE security utility (autosys_secure) failed to update or delete a user when the application server is running on an older version and the following message is displayed:

CAUAJM_I_60405 Use the Manage users with key credentials menu.

Fixed Issues - 11.3.6 SP4 CUM1

This article details the issues that are fixed in CA Workload Automation AE Release 11.3.6 SP4 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 SP4 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

JOBS IN OFF HOLD BOXES RUN WITH WRONG NUMBER AFTER PR#4408

Problem Number: 4430

When you run a multi-level box after placing one of the inner level boxes on hold, the scheduler executes all jobs except the box that is in the ON_HOLD status. When you take the box off hold, the scheduler executed the box but the jobs in the box retained the previous run number instead of receiving a new one.

SCHEDULER CORE ON SHUTDOWN IN LIBMGRCOMM PINGING DOWN AGENTS

Problem Number: 4437

As the scheduler shuts down, it might try to initiate a ping to an agent that is in the Missing state. Through an internal race condition, the scheduler might try to access an invalid memory address resulting in a core.

SCHED PROCESSES EVENTS OUT OF ORDER WITH DUAL EVENT SERVERS

Problem Number: 4441

In dual event server mode, when you send events manually and specify the highest event priority during a period of time when the scheduler has many job events in the ujo_event table, the scheduler fetched job events out of order.

DECOUPLE SCHED_SCALE WITH NUMBER OF AGENT LISTENER THREAD

Problem Number: 4500

Starting in CA Workload Automation AE Release 11.3.6 SP3, the scheduler's agent listener thread is multi-threaded. The number of agent listener threads were controlled by the SCHED_SCALE environment variable. When SCHED_SCALE is set to 64 or to a non-default value, it impacted the horizontal latency (average processing time required for all the events).

To address this issue, a new environment variable named AGENTLISTENER_THREAD_CEILING=*n* is introduced.

NOTE

For more information about the AGENTLISTENER_THREAD_CEILING environment variable, see [Environment Variables -- Network Communication](#).

WITH NOLOCK - IGNORE WHEN JOB_NAME CONTAINS TABLENAMES

Problem Number: 4454

On Windows, if any one of the following table names: ujo_event, ujo_proc_event, ujo_job_status, ujo_job_runs, is part of the job name, the query optimization logic added the With NOLOCK clause in the wrong position in the SELECT queries that are used in the product.

SCHED CREATES DUPLICATE EVENTS WHEN AGENT COMM TIMES OUT**Problem Number: 4455**

When the scheduler is running with a multi-threaded agent listener and the agent resends job status messages for which it did not receive an acknowledgement, the scheduler did not detect that the message is a duplicate and it generated a repeating job event.

JOBS RUN WITH THE OLD CONNECTION PROFILE**Problem Number: 4464**

When the job definition is updated to use a new connection profile (than the original connection profile that you specified while defining the job initially), the job still continued to use the original connection profile. This problem is applicable to the CA WA Advanced Integration for Hadoop job types (HDFS, OOZIE, HIVE, SQOOP, and PIG).

IMPROVED SCHEDULER PERFORMANCE FOR FETCHING EVENTS**Problem Number: 4465**

The scheduler always sleeps for a minimum of half a second for each event fetching cycle to avoid executing an intensive database stored procedure too many times in the same second. During periods of high workflow activity where many events accumulate in the database, the scheduler may spend an extended period of time on a single cycle because the scheduler worker threads are busy processing events from previous cycles. The scheduler is not optimized to skip the half second sleep during the extended cycles.

AUTOSYS_SECURE INPUT DOES NOT ALLOW SPACES IN PASSPHRASE**Problem Number: 4471**

When you specify a passphrase having space characters for a key credentialed user at the interactive menu prompt of the CA Workload Automation AE security utility, the utility rejected the passphrase.

SHADOW CORES WHEN SHUTDOWN INTERRUPTS NOTIFICATION OF AGENTS**Problem Number: 4472**

When the shadow scheduler fails over, it initiates contact with all agents that have run jobs to notify them to send job statuses back to the shadow scheduler. When you set the PrimaryFailbackMode parameter to Immediate, and the shadow scheduler detects the return of the primary scheduler, the shadow scheduler restarts. When the shut down of the shadow scheduler occurs while active agent communications are in progress, the shadow scheduler process exited ungracefully instead of automatically starting back up.

INFINITE PTHREAD_ATTR_SETDETACHSTATE INVALID ATTRIBUTE ERROR**Problem Number: 4475**

When the operating system does not allow the scheduler to start a thread, the scheduler repeatedly logged errors due to a failure in the pthread_attr_setdetachstate operating system library call.

SCHED HANGS ON SHUTDOWN WHEN COMMUNICATIONS ARE INTERRUPTED**Problem Number: 4484**

When the scheduler is shut down while it is communicating with agents, the scheduler process remained indefinitely in a deadlocked state.

SHADOW HANGS WHEN SHUTDOWN INTERRUPTS AGENT COMM AFTER 4472**Problem Number: 4487**

When you configure the instance for high-availability, the shadow scheduler takes over from the primary scheduler when it detects that the primary scheduler is missing. Before it resumes processing events, the shadow scheduler communicates with all agents to update the manager communication attributes so the agent knows to send job statuses to the shadow scheduler. When you shut down the shadow scheduler while it is communicating with agents to update its manager communication attributes, the shadow scheduler process remained indefinitely in a deadlocked state.

SCHED HANGS ON SHUTDOWN AFTER ACCEPT THREAD CREATE ERRORS**Problem Number: 4491**

When the scheduler fails to create a thread to process incoming job status messages from an agent during its run time, and you initiate a shut down, the scheduler process remained indefinitely in a deadlocked state.

DATABASE CONNECTION POOL USAGE OPTIMIZATIONS AFTER PROBLEM NUMBER: 4428**Problem Number: 4497**

After applying the solution to problem 4428 (SCH SKIPS HA HEARTBEATS WHEN A DATABASE IS TEMPORARILY DOWN), the scheduler indefinitely lost access to one of the database connections in the database connection pool.

ORACLE INDEXES GOT CREATED IN THE DATA TABLESPACE**Problem Number: 4518**

Oracle database indexes were created in the Oracle data tablespace rather than getting created in the index tablespace. This problem occurs when you use the Repair option in the product installation media or when you use the incremental/cumulative patch install script to apply product maintenance. This problem applies to Oracle database only.

Changes to Existing Features - 11.3.6 SP4 CUM1

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP4.CUM1.

Updated IJobRunsFilterDate API

The FLT_EVENT_QUE_STAMP attribute is added to the IJobRunsFilterDate API interface. The FLT_EVENT_QUE_STAMP attribute lets you filter events based on the event processed time (que_status_stamp) for CA Workload Automation AE job runs.

AE Release 11.3.6 SP3

This section includes information about the CA Workload Automation AE Release 11.3.6 SP3 new features, changes to existing features, and fixed issues.

WARNING

The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6 SP3

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP3.

New Job Types

CA WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define the following new job types using CA WA Advanced Integration for Hadoop:

- [HDFS](#) (Hadoop Distributed file System)
- [Oozie](#)
- [Hive](#)
- [Sqoop](#)
- [Pig](#)

Connection Profiles

A *connection profile* is a set of attributes that are required for connecting to environments where the job runs. For example, to run a job in the Hadoop environment, you can create a connection profile with the appropriate parameters to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database.

A connection profile lets you save the connection information and you can use the connection profile in the job definition. For example, when you define a Hadoop HDFS job, you can assign a connection profile to it.

NOTE

For information about the JIL subcommands and attributes that you can use to define, update, and delete connection profiles, see [JIL Connection Profile Definitions](#).

A connection profile applies to the following job types:

- HDFS
- Hive
- Oozie
- Sqoop
- Pig

You can use web services to retrieve information of a specific connection profile or multiple connection profiles. For more information, see [Managing Connection Profiles](#).

New as-connectionprofile Resource Class

The as-connectionprofile resource class and its default CA EEM policy with READ, CREATE, DELETE, EXECUTE, and WRITE access modes is added to control access to a connection profile.

Authenticate a User Using Key Credentials

For robust security, you can authenticate users that run Hadoop (HDFS, Hive, Oozie, Sqoop, or Pig) jobs using key credentials instead of a password.

You can use the autosys_secure command to specify a password or key credentials for a user.

WARNING

IMPORTANT! You can authenticate only those users that run Hadoop jobs using key credentials.

New Job State

The SUSPENDED (17) job state is added in this release. This job state indicates that the Hadoop Oozie job is suspended.

You can use the *suspended* monitor or report attribute to specify whether to track the job status event generated when a job changes to the SUSPENDED status. For more information about the suspended attribute, see [suspended Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to SUSPENDED](#).

New Events

The following events are added in this release:

- RESUMEJOB (149)—Resumes a Hadoop Oozie job and places it in the RUNNING state. This event is manually generated.
- SUSPENDJOB (148)—Suspends the Hadoop Oozie job and places it in the SUSPENDED state. This event is manually generated.

Use the sendevent command to suspend a Hadoop Oozie job that is in the RUNNING state or to resume a Hadoop Oozie job that is in the SUSPENDED state.

Using web services, you can now issue RESUMEJOB and SUSPENDJOB events. For more information, see [Managing Jobs](#).

New Alarms

The following alarms are added in this release:

- RESUMEJOBFAIL(551)—Indicates that the RESUMEJOB event failed; typically when you send the event with the Hadoop Oozie job not in the SUSPENDED state.
- SUSPENDJOBFAIL (550)—Indicates that the SUSPENDJOB event failed; typically when you send the event with the Hadoop Oozie job not in the RUNNING state.

Changes to Existing Features - 11.3.6 SP3

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP3.

Updated autosys_secure Command

You can now use autosys_secure (from the command line or using the AutoSys Security Utility interactive menu) to perform the following tasks:

- Add a regular user with key credentials.
- Change the key credentials of the specified user.
- Display only the list of users with key credentials.
- Display only the list of users with password.

NOTE

For more information about the autosys_secure command, see [Manage Security](#).

Performance Improvement With Multi-threaded Listener

The scheduler runs in [multi-threaded listener mode](#) where connections from multiple agents are processed simultaneously as opposed to the serial processing of statuses from agents earlier; thus providing increased throughput of jobs and decrease in event latency. Also, the number of autoping timeouts and false chase alarms are reduced.

Fixed Issues - 11.3.6 SP3

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP3.

TIP

CA Workload Automation AE Release 11.3.6 SP3 also includes the issues fixed in each of the prior releases of 11.3.6.

Contents**FREQUENT DEADLOCKS IN SCHEDULER****Problem Number: 4272**

The scheduler displayed frequent deadlock messages while running on MSSQL; thus leading to EVENT_HDLR_ERROR alarms and causing scheduling problems. This issue was fixed in Release 11.3.6 SP2.

The application server displayed frequent deadlock messages while running on MSSQL; thus leading to EVENT_HDLR_ERROR alarms. This issue is fixed in Release 11.3.6 SP3.

ASM DISKGROUP VERIFICATION CHECK FAILS DURING INSTALL**Problem Number: 4282**

During installation, if you select the ASM (Automatic Storage Management) option, the Data Disk Group not found error occurred and prevented from continuing with the installation.

SCHEDULER TRUNCATES ISDBGACTIV TRACES**Problem Number: 4294**

When the UNIX scheduler runs under ISDBGACTIV=JOB traces and it evaluates the start of jobs across different time zones, the scheduler begins to output just the file and line number but no other debug information.

DECOUPLE CONFIGURATION OF AGENT LISTENER FROM RECV_TIMEOUT**Problem Number: 4315**

The RECV_TIMEOUT environment variable is used primarily to configure the CA Workload Automation AE clients to wait longer for the application server to process client commands and respond. Over time, the RECV_TIMEOUT environment has been associated with other time outs that are not related to request/response interactions. One example is the scheduler agent listener, which also uses RECV_TIMEOUT to reduce the time to wait for incoming data to arrive from an agent that has established a connection over the network. To avoid conflicting uses of RECV_TIMEOUT, a new environment variable named MSGSVC_LISTEN_TIMEOUT is introduced.

NOTE

If CA Technical Support previously instructed you to set the RECV_TIMEOUT variable to a low value to time out poor socket connections sooner due to frequent network delays and you upgraded from Release 11.3.6 SP2 or lower, unset the RECV_TIMEOUT variable and set the MSGSVC_LISTEN_TIMEOUT variable to the same low value instead. Restart the scheduler or application server for the changes to take effect. For more information about the related network communication environment variables, see [Configuring CA Workload Automation AE](#).

There is no relationship between RECV_TIMEOUT and MSGSVC_LISTEN_TIMEOUT. After upgrading to Release 11.3.6 SP3 or higher, you may set both without consequence.

CHASE/AGENT INTEGRATION FOR DELAYED STATUS**Problem Number: 4323**

Part of the chase utilities task is to contact each agent where a job is running and confirm with the agent that the job is still actually in the RUNNING state. However, in times of intense load (lots of events to process, from many agents),

the scheduler can sometimes briefly fall behind processing incoming events from the agents (that is, SU/FA) and subsequently the agent cannot report the jobs completion status in a timely manner. At that moment, if chase is run, it raised an alarm indicating that the job should be running on the agent but it is in fact not running.

As part of this fix, the CA Workload Automation AE application server now requests the patched agent for the list of jobs that have completed execution but whose statuses have not yet been communicated to the scheduler. This fix requires Workload Automation Agent for UNIX, Linux, Windows or i5/OS 11.3 SP4 Incremental 2. For all agents earlier than the specified version, the chase utility continues to operate as it has in the past.

I5 JOB_PARMS AND OTHERS COLUMN SIZE

Problem Number: 4325

On Sybase installations, the database field size for ujo_i5_job columns *others* and *job_parms* is SMALLINT. However, those fields may need to store values up to 2 31, which is the size of ujo_next_oid.oid. Over time, the value of oid can increase beyond the ability for *others/job_parms* columns to hold the key values.

SCHED ACCUMULATES LARGE AMOUNTS OF MEMORY PINGING AGENTS

Problem Number: 4328

When multiple agents compete to communicate with a scheduler, machines may enter the offline status. As the scheduler pings these agents, it may end up accumulating memory as it waits for responses that do not arrive.

FORECAST CRASHES WHEN PROCESSING JOBS WITH GROUP DEFINED

Problem Number: 4339

The forecast command stopped responding when a wildcard is used for the job name and one of more of those jobs have group or application attribute defined and CA EEM security is enabled.

AGENT INSTALL DOES NOT CREATE /ETC/AUTO.PROFILE

Problem Number: 4348

If you install the agent after the standalone client installation, the standalone agent installer created an incorrect /etc/auto.profile file.

IMPROVE SCHED RECOVERY FROM DB FAILURES WHEN RUNNING BOXES

Problem Number: 4358

1. When the scheduler executes the Oracle ujo_send_event or ujo_sendafm stored procedures, it did not handle Oracle exceptions when inserting into the ujo_event or ujo_afm tables.
2. As the scheduler evaluates the starting conditions of a large number of jobs in a box and it experiences a database error with one job, it moves the box RUNNING event to the ujo_proc_event table and raises an EVENT_HDLR_ERROR. However, it continued to evaluate the rest of the jobs in the box. A momentary recoverable database error with one job should not cause the scheduler to prematurely move the containing box's RUNNING event into the ujo_proc_event table.

CROSS-INSTANCE JOBS OF 4.X INSTANCE DID NOT RUN BY 11.X

Problem Number: 4359

When there are multiple jobs in the 4.x instance depending on the 11.x instance and if one of the job from the 4.x instance is deleted, the 11.x instance deleted the dependency for all jobs of 4.x. So, other dependent jobs in the 4.x instance did not run although the predecessor in the 11.3.x instance ran.

AUTOSEC_TEST CORES ON EXIT AFTER UPGRADING SSA TO 2.3.2**Problem Number: 4360**

After you upgrade SSA to 2.3.2, when you execute the `autosec_test` command, it experiences a core when it exits.

SCHEDULER DUPLICATES RUNNING EVENT AFTER RUN_NUM OVERFLOWS**Problem Number: 4361**

After the scheduler completes the job with run number 2147483647, it sent extra `RUNNING` events for every subsequent job run. Also, `autorep` no longer reports job runs after run number 2147483647.

ARCHIVE_EVENTS DOES NOT DELETE JOB_RUNS DATA**Problem Number: 4362**

Job runs data does not get archived even when the job runs information in the table is eligible for archiving causing job runs table containing stale information.

JOB RUNS ALTERNATE MINS USING CAL**Problem Number: 4364**

If a job is using a run calendar with no `start_mins` or `start_times` attribute and every consecutive minute of an hour is defined in the calendar itself, it failed to run every minute. Instead, the job got scheduled for every alternate minute.

UPDATE STATEMENT IN REFRESHAEDB.PL GOES INTO INFINITE LOOP**Problem Number: 4365**

The `RefreshAEDB` or `CreateAEDB` script went into an infinite loop when the number of rows in the `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_weekly`, or `ujo_rep_monthly` table is more than 50000 rows. This happens only if you are using aggregator and any one of the four tables has more than 50000 rows. One row is inserted every hour in the `ujo_rep_hourly` table and at this speed it takes 5.6 years to get more than 50000 rows and so the probability of this occurring is low.

If you are maintaining old data and made any manipulations with data (in dev/test), then the tables may have more than 50000 rows.

NOTE

The `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_weekly`, and `ujo_rep_monthly` tables were introduced in CA Workload Automation AE Release 11.3.5.

SENDEVENT MAN PAGE CORRECTIONS**Problem Number: 4366**

In the `sendevent` man page, the `STARTJOB` section included invalid information.

TIMEZONE INCORRECT FOR AMERICA/SANTIAGO**Problem Number: 4367**

The timezone for 'America/Santiago' is currently listed as `America/Santiago Z CST4CDT,M10.3.0/00:00:00,M3.2.0/00:00:00`. However, from 2015, Santiago does not observe DST and hence must be corrected.

SCHED/APPSERVER HANGS ON SHUTDOWN DURING DATABASE MONITORING**Problem Number: 4369**

When the scheduler or application server experiences a database error that requires the database connection pool to be refreshed, the database monitoring system is invoked. If the product is shutdown while the database monitoring runs at just the wrong time, the product deadlocked internally and the process stopped responding indefinitely.

MID-NIGHT LOG ROLLOVER HAPPENS AT 23:00**Problem Number: 4370**

On Windows, the midnight log rollover happens one hour early in the non-US time zones that follow DST. Both the scheduler and application server logs rollover early. This occurred between the period of US timezone DST(second Sunday of March) and non-US timezone DST(Last Sunday of March).

APPSERVER LOG DISPLAYS UNKNOWN EEM USER UNAUTHENTICATED MODE**Problem Number: 4371**

When you enable CA EEM security for an instance using the default setting for the unauthenticated user mode, the application server log file displayed the mode as ****UNKNOWN****.

AUTOREP PERFORMANCE DUE TO UJO_JOBROW VIEW**Problem Number: 4373**

The autorep command took long time to retrieve the job definitions from the database due to inefficient query.

DEPENDENT JOB RUNS UPON COMPLETION OF ONE CROSS-INSTANCE JOB**Problem Number: 4376**

A job is dependent on a local instance job and a cross-instance job with the same name. When the cross-instance job is completed, the dependent job ran ignoring the status of the local instance job.

In this example, the job *dep_job* runs when *job1* of the external instance (RMT) is completed. The status of the local instance job *job1* is ignored.

```
insert_job: dep_job
condition: s(job1) and s(job1@RMT)
```

CHANGE_STATUS OF BOX TO INACTIVE UNDOES JOB ON NOEXEC**Problem Number: 4377**

When the scheduler processes a **CHANGE_STATUS** or **INACTIVE** event on behalf of a box job, it overwrites the statuses of jobs that are set to **ON_NOEXEC**. This caused the jobs that are supposed to be in **ON_NOEXEC** state to execute on the next box run.

AUTOCAL_ASC -F OPTION**Problem Number: 4378**

The CA Workload Automation AE calendar utility(*autocal_asc*) overwrites the calendar in import function without prompting the user about calendar replacement.

CA DIRECTORY INSTALLS IN WRONG DRIVE (WINDOWS)**Problem Number: 4379**

CA Directory gets installed in the wrong drive even when the CA Common Components installation folder is selected as non-system drive (C:).

ASM DISKGRP VALIDATION FOR EXISTING TABLESPACES**Problem Number: 4380**

The CA Workload Automation AE installer failed to verify the Oracle disk group when you select ASM (Automatic Storage Management) for schema and also clear the Create the tablespaces, Database Users, and Roles check box in the Event Server Properties page during installation.

UJO_COMM_UPDATE STORED PROCEDURE ERRORS AFTER 4334**Problem Number: 4381**

In environments with many agents, the scheduler or application server may occasionally experience database deadlock errors while executing the ujo_comm_update stored procedure during periods of high activity. Also, when multiple agents are installed on the same nodes in a clustered agent environment, the scheduler or application server may generate repeated database errors when first contacting one of the agents on the node after successfully communicating with the other agent on the same node.

TIMEZONE CHANGE FOR CAIRO**Problem Number: 4382**

The timezone for 'Africa/Cairo' is currently aliased to 'Egypt' timezone as Egypt Z EET-2EETDST,J121/01:00:00,J274/03:00:00 . However, from 2015, Egypt does not observe DST and hence must be corrected.

UPGRADE TO SP2 FAILS WITH WRONG TABLESPACE NAMES**Problem Number: 4384**

The upgrade to the latest release failed when the installer passes the default tablespace names to the database that is using non-default tablespaces.

SCHED HANGS ON SHUTDOWN WHEN EVALUATING BOX AFTER DB LOSS**Problem Number: 4385**

When the scheduler experiences the failure and loss of the database connections, it automatically initiates a shut down. If the scheduler happens to be evaluating box jobs and the timing is just right, then database failures occurring during the evaluation of jobs in a box resulted in the indefinite blocking of a thread that prevented the shut down from completing.

SCHED HANGS ON SHUTDOWN AS BROKER SENDS TO REMOTE MANAGERS**Problem Number: 4386**

When the scheduler starts up with cross platform scheduling enabled, it initiates a handshake with remote managers to obtain job information that was pending from the previous run of the scheduler. Also, if cross platform job dependencies are established, the scheduler communicates with the remote application to send its job status. If the scheduler is shutdown while this communication takes place, the scheduler process stopped responding indefinitely.

NEXTROW FAILURE WHEN FETCHING EVENT ID CAUSES SCHED SHUTDOWN**Problem Number: 4387**

When the scheduler experiences a database error while fetching the result set following the successful execution of the Oracle ujo_get_eoid stored procedure, it automatically shuts down instead of retrying the query.

BATCH EVENTS NOT REQUEUED ON FAILURE OF NEXTROW()**Problem Number: 4388**

When timeout occurs in the nextRow() call to the database while fetching batch events, the events were not re-queued to be fetched again and were not retrieved from the ujo_event table unless the scheduler is recycled.

LATENCY IN SCHEDULER DUE TO SERIAL PROCESSING OF MESSAGES**Problem Number: 4389**

Latency issues were seen in the scheduler as it processes incoming messages serially; throttling flow of communication from agents to the scheduler.

As part of this fix, the scheduler now runs in multi-threaded listener mode where connections from multiple agents are processed simultaneously as opposed to the serial processing of statuses from agents earlier. By default, the number of listener threads is three if the SCHED_SCALE parameter is set to the default value of five. The formula used to calculate the number of threads is $(\text{SCHED_SCALE} + 1)/2$. To increase the number of threads, set the SCHED_SCALE value accordingly.

If you want the scheduler to run in single threaded listener mode, set the USE_SINGLETHREADED_AGENTLISTENER variable to any value.

AUTOSTATUS RETURN ZERO ON INVALID GLOBAL VARIABLE**Problem Number: 4391**

The autostatus -G command always returned a zero exit code if the global variable does not exist.

UPGRADE TO 11.3.6 FAILS W/ COMMENTED EVENTSERVER_2**Problem Number: 4392**

On Windows, during the upgrade from Release 11.3.5 to 11.3.6, the installer prompted for the secondary event server although the EventServer_2 parameter was commented out in the config.%AUTOSERV% file. So, the upgrade process did not complete without providing the secondary event server details. Actually, Release 11.3.5 was installed with dual event servers and later moved to single server mode by commenting the EventServer_2 line in the configuration file.

SCHED HANGS WHEN PROCESSING MACH_ONLINE EVENT**Problem Number: 4393**

When the scheduler processes the MACH_ONLINE event and a job that is in the PEND_MACH state is not evaluated to start; this resulted in the indefinite blocking of a thread that prevents the shut down from completing.

SCHED CORES WHEN IOS AGENT HAS LONG REPLY INPUT MESSAGE**Problem Number: 4394**

When the scheduler runs an iOS job that requires manual intervention, the iOS agent sends a message containing the response choices. If this message is longer than 255 characters, the scheduler cored while creating the WAIT_REPLY_ALARM alarm.

REPLACE NEW OJDBC6.JAR IN BUILDS**Problem Number: 4395**

The compliance application failed to open connection with the Oracle server when TNS alias contains a dot(.) in its name. The following errors were displayed:

```
CAUAJM_E_85100 Unable to open database connection.  
* IO Error: Invalid connection string format, a valid format is: "host:port:sid"  
CAUAJM_E_85002 Compliance Application terminated with error [100]
```

EXIT CODE IS PASSED IN AS UNSIGNED INTEGER IN SNMP TRAP**Problem Number: 4397**

Negative exit codes in JOBFAILURE alarms were sent in as large numbers in the SNMP trap.

MIDNIGHT LOG ROLL-OVER FILE HAS CURRENT DATE**Problem Number: 4398**

On Windows, the midnight log rollover files have current date instead of previous day. This occurred between the period of US timezone DST(second Sunday of March) and non-US timezone DST(Last Sunday of March).

SCHEDULER CANNOT PARSE INCOMING JOB AFM CONTAINING QUOTES**Problem Number: 4399**

When the agent executes a SQL job that fails, it may prepare an extended status message containing quotes and include this message in the job status data that it sends to the scheduler. The scheduler may experience errors parsing the extended status resulting in its inability to uniquely identify an agent through its host name. When the scheduler attempts to update its communication tables using the ujo_comm_update stored procedure, the stored procedure failed because the host name parameter is missing.

JIL INCORRECTLY POSTS AN ERROR WHEN AUTOTRACK IS ENABLED**Problem Number: 4400**

When a job with the resources attribute is updated, JIL posts an error although the job has been successfully updated in the database. This happened only when autotrack is enabled.

END TIME OF COMPLETED JOB GETS UPDATED VIA THE MANUAL EVENTS**Problem Number: 4401**

Job end time value gets replaced with the current time when you perform manual CHANGE_STATUS events against jobs that were already completed. This created a false positive record with a skewed job run duration.

SCHEDULER GENERATES BUT DOES NOT ASSIGN EVENT IDS TO EVENTS**Problem Number: 4402**

On occasions, the scheduler generates event IDs but never assigned them to any events.

SCHEDULER DOES NOT RETRY QUERIES THAT FAIL DUE TO ORA-00604**Problem Number: 4403**

The scheduler did not retry queries that fail due to Oracle error ORA-00604. Oracle raises this error when the database server is in the process of shutting down. The scheduler must retry the query in the event the database is restarted.

SCHEDULER SKIPS THE EVALUATION OF JOBS IN QUE_WAIT**Problem Number: 4404**

If the scheduler restarts following an ungraceful shutdown and processes a completion event for a load balancing job, it skipped the evaluation of jobs that are in the QUE_WAIT state.

SCHEDULER DOES NOT RETRY QUERIES DUE TO NEXTROW FAILURES**Problem Number: 4405**

The scheduler performs the following actions to execute a database query: parse the SQL, bind the input and output parameters to memory, execute the SQL, and fetch each row from the result set in a loop.

When the execute operation fails due to a recoverable error, the scheduler retries the query. However, if the scheduler experiences a recoverable database error during the row fetching phase after the query executes successfully, the scheduler does not retry the query. When the database operation that fails is a SELECT query, the scheduler gave up after one try resulting in an unfinished job processing.

NOTRUNNING CONDITION IS NOT CONSIDERING WAIT_REPLY STATE**Problem Number: 4406**

When a job *jobA* is dependent on not running of another job *jobB* and *jobB* is started and goes to WAIT_REPLY state, then *jobA* evaluation considered that *jobB* is not running and *jobA* can start if other conditions are met.

JOB ON WR CAN BE FORCE STARTED WHEN RESTRICT FS IS SET**Problem Number: 4407**

A job in the WAIT_REPLY state was force started even when the RESTRICT_FORCE_STARTJOB variable is set.

JOBS IN BOXES TAKEN OFF HOLD DO NOT INHERIT RUN NUMBERS**Problem Number: 4408**

When you run a multi-level box after placing one of the inner level boxes on hold, the scheduler will execute all jobs except the box that is in the ON_HOLD status. When you take the box off hold, the scheduler executes the box but the jobs in the box retained the previous run number instead of advancing it.

SCH SKIPS MUST COMPLETE CRITERIA WHEN JOB RUNS OUT OF WINDOW**Problem Number: 4409**

When the scheduler tries to restart a job due to an application failure and the next start is outside of the jobs window, it places the job in the INACTIVE state. However, in the process of doing so, it automatically moved the job's rescheduled CHK_COMPLETE event to the ujo_proc_event table.

JOB_DEPENDS DOES NOT SHOW WAIT_REPLY JOB STATUS**Problem Number: 4410**

The job status of a job in the WAIT_REPLY state was displayed as ?? in the job_depends output.

RESTRICT DELETE OF JOB IN WAIT_REPLY STATE

Problem Number: 4411

A job in the WAIT_REPLY status was deleted when the RESTRICT_DELETE_JOB environment variable is set.

SCHED HANG WHEN RUNNING BOXES UNDER ISDBGACTIV=LIGHT,DBQUERY

Problem Number: 4413

When you enable ISDBGACTIV=LIGHT, DBQUERY tracing in the scheduler, the scheduler may experience a deadlock condition when it executes a box with large number of jobs in it.

i5OS JOB DOES NOT GO TO RUNNING AFTER SENDING REPLY_RESPONSE

Problem Number: 4418

When the scheduler runs an i5OS job that requires user intervention, it places the job in the WAIT_REPLY status. When you send the REPLY_RESPONSE event and the agent resumes executing the job, the job remained in the WAIT_REPLY status and did not go to the RUNNING status.

AE Release 11.3.6 SP2

This section includes information about the CA Workload Automation AE Release 11.3.6 SP2 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 SP2 and Release 11.3.6 SP2 CUM1.

WARNING

The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6 SP2

No new features were added to CA Workload Automation AE in Release 11.3.6 SP2.

Changes to Existing Features - Release 11.3.6 SP2

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP2.

Documentation in the Product Image

The product documentation is no longer included in the product image. You can access the documentation online using this DocOps platform.

Retrieve Information of a Machine Using Web Services

Using web services, you can now retrieve information of a specific machine or multiple machines. For more information, see [Manage Machines](#).

Calendar Name and Description

You can now use the [autocal_asc](#) utility to add or modify a description for a calendar. The description can be up to 1024 characters.

The calendar name can now contain up to 64 characters.

Updates to the ON_NOEXEC Feature

In the current release, the following changes are made for the ON_NOEXEC feature:

- **JOB_ON_NOEXEC event**
The scheduler processes the JOB_ON_NOEXEC event similar to that of the CHANGE_STATUS event to INACTIVE. For example, if you send the JOB_ON_NOEXEC event to a job in a box, the effect is the same as sending the CHANGE_STATUS event to INACTIVE for a job in a box. The scheduler sets the job to the ON_NOEXEC status and also sets the internal secondary status to INACTIVE, so that the scheduler evaluates the overall box status as if the job entered the INACTIVE status.
To take the job off the ON_NOEXEC status, you must send the JOB_OFF_NOEXEC event. Sending manual CHANGE_STATUS event does not change the status of the job. To complete the NOEXEC job immediately so that the scheduler evaluates the job as if the job entered the SUCCESS status, send the FORCE_STARTJOB event.
- **JOB_OFF_NOEXEC event**
When the scheduler processes the JOB_OFF_NOEXEC event for a job, it places the job in the INACTIVE status. If the job is in a box, the scheduler evaluates the overall box status as the job is entering the INACTIVE status. If you send the JOB_OFF_NOEXEC event to a box, all jobs in the box (including all jobs contained in lower level boxes within the box) are reset to the INACTIVE status.
- **Evaluation of downstream dependent jobs**
When you send the JOB_ON_NOEXEC event to a job, the effect is the same as if the job enters the INACTIVE status. The scheduler does not immediately schedule jobs that have a dependency on the NOEXEC job nor does it evaluate their success conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the INACTIVE status.
When the job's starting conditions are met and the scheduler sends the BYPASS event, the effect is the same as if the job enters the SUCCESS status with an exit code of 0. The scheduler schedules jobs that have a dependency on the NOEXEC job and evaluates their success conditions to success.

Sending More Detailed Events to External Instances

When a local scheduler sends events to external instances, it populates the Event text field to indicate the action in the local instance that resulted in the external event. The event text can be as follows:

- Job *job name* entered *status*
- Job *job name* was placed ON_HOLD|ON_ICE
- Job *job name* was bypassed and placed ON_NOEXEC
- Job *job name* was not scheduled and was reset to ON_NOEXEC
- Job *job name* was taken off ON_HOLD|ON_ICE|ON_NOEXEC
- Job *job name* was placed in RESTART due to application failure

Adding or Removing Jobs In Boxes

You can add a job into a box by inserting a new job and specifying the box name in the box name attribute value of the inserted job. You can also add a job into a box by modifying the box name in the box name attribute value of an existing job. Similarly, you can remove a job out of a box by deleting the job or by removing the box name in the box name attribute value of the job. When you add or remove jobs in a box, CA Workload Automation AE performs the following actions:

- Modify the job status based on the status of the box
- Send a STARTJOB event to immediately schedule a job that is added to a box in the RUNNING state.
- Send an ALERT event to notify you that the content of the box job is changed. CA Workload Automation AE sends the ALERT event only when the jobs are removed or added in the box that is in the RUNNING status.

NOTE

For more information about the actions that CA Workload Automation AE performs on jobs that are added into or removed out of a box, see [Basic Box Job Concepts](#).

Agent Version Information Listed in the autorep Output

The autorep -p option generates a report that displays all the active agent plug-ins and advanced integrations on the machine and the agent release and build information.

Fixed Issues - 11.3.6 SP2

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP2.

TIP

CA Workload Automation AE Release 11.3.6 SP2 also includes the issues fixed in each of the prior releases of 11.3.6.

Contents**Global Variable Modification Time Displayed Incorrectly****Problem Number = 4291**

Global value modification date was displayed incorrectly using the Java SDK. This impacted CA WCC as it did not match the modified date or times returned by the autorep – G command.

SYBASE DEADLOCK ON UJO_HA_PROCESS FROM UJO_SETNGET_STATUS**Problem Number: 4318**

The Sybase server reported numerous deadlock errors on a select query on the ujo_ha_process table that is executed from the CA Workload Automation AE database stored procedure named ujo_setnget_status. This applies to MS-SQL also, but not Oracle.

R11 JOBS STUCK IN STARTING AFTER 2ND TIME SHADOW TOOK OVER**Problem Number: 4317**

Jobs that are supposed to start on r11 agents were stuck in the STARTING state after the shadow scheduler took over the second time. This problem did not occur when the shadow scheduler took over for the first time. It occurred only when the primary scheduler was restarted with PrimaryFailbackMode=2 after the shadow scheduler took over for the first time and then the primary scheduler went missing again.

AS_SERVER STUCK IN SQL QUERY DURING STARTUP**Problem Number: 4308**

The application server was stuck in a SQL query during startup. The SQL query deleted the invalid records from the ujo_comm_send_seq table.

NO SPECIFIC EXITCODE WHEN ATLEAST ONE AS_SERVER IS DOWN**Problem Number: 4326**

When there are multiple application servers, chk_auto_up did not report a specific exit code to differentiate if at least one application server is down versus all application servers are up. It used to work in r11.0. In r11.0, if any one of the

application server is down, `chk_auto_up` returns an exit code of 0 whereas in r11.3 and above, it returns an exit code of 100 if all the application servers are up.

SEND_NOTIFICATION NOT SHOWN IN THE JOB DEFINITION REPORT

Problem Number: 4332

When you define a job, if you specify the `notification_emailaddress` attribute and the `send_notification` attribute is set to 0, the job definition report (for example, `autorep -q`) did not display `send_notification=0`.

MUST START/MUST COMPLETE ALARMS NOT GENERATED UPON OVERRIDE

Problem Number: 4300

When the `must_start_times` or `must_complete_times` attribute is overridden, the `CHK_START` and `CHK_COMPLETE` events were not generated. This caused the job not to generate the `MUST_START_ALARM` or `MUST_COMPLETE_ALARM` alarm.

NOTE

If you are overriding the `start_times` attribute of a job and your job definition includes the `must_start_times` or `must_complete_times` attribute, ensure that the `start_times` value is earlier than the `must_start_times` or `must_complete_times` value to avoid a false `MUST_START_ALARM` alarm; until overrides are supported for the `must_start_times` and `must_complete_times` attributes.

GAPS IN DEBUG SCHEDULER OR AS_SERVER LOGS

Problem Number: 4281

When the scheduler or application server logs roll over, there were hours of gaps in the logs. So, the debug logs have partial log contents. This issue occurs when the `ISDBGACTIV` environment variable is set to `HEAVY`.

FREQUENT DEADLOCKS IN SCHEDULER

Problem Number: 4272

The scheduler displayed frequent deadlock messages while running on MSSQL; thus leading to `EVENT_HDLR_ERROR` alarms and causing scheduling problems.

JOB WITH FAILED MACHINE CHOOSER SCRIPT REMAIN STUCK IN RESTART

Problem Number: 4147

When you define a job with a machine chooser script as the machine attribute and the scheduler experiences an error while executing the script, the scheduler placed the job in the `RESTART` state. If the job had completed a prior run and the process ID of the job that is carried over from the prior run began with the number four, the scheduler mistakenly treated the process ID as an error condition leaving the job stuck in the `RUNNING` state.

LEGACY __TIMEZONE ENVIRONMENT VARIABLE NOT EXPOSED TO JOBS

Problem Number: 4301

The legacy agent supports the setting of job definition JIL attributes as environment variables, which are sourced as part of running a job.

The r11.3 agent supports `job_name`, `box_name`, `machine`, `run_machine`, and `max_exit_success` attributes. But, the `timezone` attribute was not exported. So, jobs referring the `__timezone` environment variable failed. It used to work with the r11 and r4.x agents.

JIL RETURNS INVALID KEYWORD VALUE ERROR FOR i5_LIBRARY_LIST**Problem Number: 4304**

The JIL utility returned an invalid keyword value error for the i5_library_list attribute when the list contains a space; although space is a delimiter for the i5_library_list attribute.

ARCHIVE_JOBS FAILS IN NATIVE SECURITY MODE**Problem Number: 4327**

On Windows, when CA Workload Automation AE is running in native security mode, the archive_jobs command failed to run when the scheduler triggered the DBMaint command as the Local System user.

SOLARIS-X86 AGENT DOES NOT START AFTER REBOOT**Problem Number: 4331**

The agent installed on a Solaris x86 machine did not start after a reboot.

AUTOBCPORADP.PL TRIES TO MIGRATE AEDBADMIN USER AND FAILS**Problem Number: 4276**

On some Oracle versions, the autobcpORAdp.pl script tried to import all the objects including users (like autosys and aedbadmin), but these users are already present on the target database. This resulted in failures as shown below:

```
ORA-31684: Object type USER:"AEDBADMIN" already exists
```

ARCHIVE_EVENTS FAILS TO WRITE TO LOG WHEN IT GROWS TO 2GB**Problem Number: 4280**

The archive_events command failed to produce files over 2GB.

CREATEAEDB FAILS TO CREATE USER IN UPGRADE MODE**Problem Number: 4305**

The upgrade from earlier versions to CA Workload Automation AE Release 11.3.6 failed on MSSQL database as the CreateAEDB script failed to create an already existing user.

REINDEXDB.PL FAILS AGAINST ORACLE**Problem Number: 4284**

On Oracle, the reindexDB.pl script failed intermittently and displayed the following errors:

```
ERROR at line 1:
```

```
ORA-00054: resource busy and acquire with NOWAIT specified or timeout expired
```

AUTOCAL_ASC VALIDATION CHECK FOR STD_CAL DELETE**Problem Number: 4290**

The autocal_asc command did not validate the existence of a standard calendar when the -D command line option is used with an existing extended calendar.

SAP JOB DOES NOT HONOR KILLJOB COMMAND

Problem Number: 4307

When you issue the KILLJOB event on the SAP job, it did not kill the job. It continued to run and went to SUCCESS after the job completed.

ARCHIVE_EVENTS ARCHIVES LESS THAN BATCH SIZE

Problem Number: 4283

The archive_events command archived less than the input batch size.

INCLUDE REINDEXDB AS PART OF CREATEAEDB.PL

Problem Number: 4296

On Sybase, if the application is running for a long time and the indexes are not re-organized, there is a possibility that the indexes on table spawn too many data pages causing Sybase to lock huge number of pages for updating the database. Running the reindexDB script re-organizes the index and running fix_data_syb.sql script is easy. Now, the reindexDB and fix_data_syb.sql scripts are run as part of the CreateAEDB.pl script and this avoids manual intervention in schema updates.

BOX STATUS WRONG - FORCE_START FAILED CHILD TO SUCCESS

Problem Number: 4314

A child job with resources attribute went into the FAILURE state due to a wrong command. If the same job is restarted with the correct command, it entered the RES_WAIT state and eventually went into STARTING, RUNNING, and SUCCESS. In this case, the box where the job resides must go to a SUCCESS state, but it went into a FAILURE state.

GLOBAL VARIABLE EVALUATION WRONG WHEN VALUE IS NUMERIC

Problem Number: 4316

The scheduler evaluated strings as numeric value and treated every string as zero(0). If the value of the global variable is set to zero, then the condition evaluation always became true leading to incorrect scheduling.

WCC REPORTS JOB_RUN WITH \$\$AGENT\$\$ AS MACHINE NAME

Problem Number: 4279

CA WCC Monitoring window displayed jobs running on \$AGENT\$\$; causing unnecessary confusion to operators.

HAPOLLINTERVAL DEFAULT VALUE WRONG IN NON-HA MODE

Problem Number: 4278

In non-HA mode, if you comment the HAPollInterval parameter, it did not work as documented in the configuration file or the product documentation. The application server and scheduler spinned infinitely calling the SetNGet stored procedure immediately without waiting for 15 seconds.

INBOUND CROSS-PLATFORM JOBS AUTH FAILURE IGNORED

Problem Number: 4170

Although the cross-platform jobs failed authentication on the agent, the scheduler still started the jobs.

JIL FAILS WHEN OWNER STARTS WITH #

Problem Number: 4277

JIL failed when the owner attribute starts with a # sign.

AUTOCAL_ASC IMPORT FAILS WITH DUPLICATE DATES

Problem Number: 4287

When a standard calendar is updated with dates from an import file and there is a duplicate date in the import file, the job(s) using that calendar were not updated with the next valid date.

CONFIGURABLE CHASE STARTING INTERVAL

Problem Number: 4298

When you run the chase utility, it checks for any jobs in the STARTING status for more than 2 minutes (120 seconds). During times of intense load (lots of events to process, from many agents), the scheduler can sometimes fall behind and not process a job's RUNNING event before the 120 second limit. When that happened, the chase utility raised an alarm.

The CHASE_STARTING_WAIT_INTERVAL environment variable is added that lets you reconfigure the chase STARTING timeout. For example, CHASE_STARTING_WAIT_INTERVAL=5 sets the timeout to 5 minutes (300 seconds). Set this environment variable on the computer where you are running the chase command.

SCHEDULER RUNS QUICK LOAD BALANCING BOX JOB MULTIPLE TIMES SCHED FS JOB ON 0 FACTOR MACH WHEN NON-0 FACTOR MACH EXISTS

Problem Number: 4312

The Reference Guide states the following about machines with a factor of 0: Setting the factor value to zero (0) results in a calculated usage of zero (0) but does not disqualify the machine. The scheduler selects an agent machine with a factor value of zero (0) only when all other available machines specified in the job definition also have a factor value of zero (0)." When you force start a job, however, it may run jobs on machines with a factor value of zero when available machines exist with a factor value that is non-zero.

UJO_COMM_UPDATE ERRORS WITH MULTIPLE AGENTS ON SAME HOSTS

Problem Number: 4321

When you install multiple agents on the same host (configured on different ports) in a clustered agent or as part of an environment where the virtual node names are used, the scheduler or application server may experience database errors communicating with one of the agents on the host.

SCHEDULER DEADLOCKS WHEN LOAD BALANCING BOXES

Problem Number: 4335

Due to a rare race condition, the scheduler may experience a deadlock condition while executing boxes that contain jobs defined with load balancing attributes (job_load, priority).

UJO_COMM_UPDATE STORED PROCEDURE DEADLOCK ERRORS

Problem Number: 4334

In environments with many agents, the scheduler may occasionally experience database deadlock errors while executing the ujo_comm_update stored procedure during periods of high activity.

SCHEDULER CORES IMMEDIATELY AFTER YOU FORCE START A BOX**Problem Number: 4303**

When you force-start a box inside another box, the scheduler evaluates the start of jobs contained in the inner box. Shortly afterwards, the scheduler terminated ungracefully.

SCHEDULER RUNS QUICK LOAD BALANCING BOX JOB MULTIPLE TIMES**Problem Number: 4309**

When a load balancing job (specifies both the priority and job load attributes) in a box is dependent on a large number of upstream jobs and the upstream jobs complete at nearly the same time, the scheduler may execute the job multiple times over the course of the same box run. The larger the condition and the quicker the job, the more times it may iterate.

UNIX SCHEDULER CORES WHEN QUEUING A LOAD BALANCING JOB**Problem Number: 4310**

When you define a load-balancing job on a virtual machine and there is no available load, the UNIX scheduler places the job in the QUE_WAIT state. As it does so, the scheduler may core due to a timing issue when machines happen to go offline at around the same time. The implementation of the Windows scheduler differs such that it does not have the same timing issue.

SCHEDULER CORES ON SHUTDOWN AS IT COMMUNICATES WITH AGENT**Problem Number: 4311**

Depending on the timing, on rare occasions, the scheduler process may terminate ungracefully as it processes the final inbound data from an agent while shutting down.

CLIENT UTILITIES CORE UNDER DBG WITH NO APP SERVER**Problem Number: 4285**

The client utilities terminated ungracefully after failing to connect to an application server and outputting a debug trace while the ISDBGACTIV environment variable is set to HEAVY.

ON_NOEXEC JOBS RESET TO INACTIVE AND DOWNSTREAM DEP ISSUES**Problem Number: 4247**

The scheduler erroneously resets a job that is in the ON_NOEXEC status to INACTIVE through the following activities:

1. The job's starting condition is met outside its run_window
2. The job's starting condition is met on a day in the exclude_calendar
3. The activated job is part of a box with a box_success or box_failure and the conditions are met to complete the box
4. The job is part of a box that is ON_HOLD or ON_ICE and the box is taken off ON_ICE or ON_HOLD

Also, the scheduler should only evaluate the condition of a downstream dependent job with a success dependency on the ON_NOEXEC job to true after the starting conditions of the ON_NOEXEC job are met and the scheduler bypasses its execution.

WRONG USER IN SDK AUTH ATTEMPT**Problem Number: 4322**

CA WCC intermittently specified the wrong user to perform an action in CA Workload Automation AE that is controlled by CA EEM. The result is that the user may be granted or denied access to an action based on the privileges of the wrong

user. For example, when executing the sendevent command using CA WCC, CA WCC may specify the Monitoring user to the application server instead of specifying the user who logged in to CA WCC and is executing the command.

EXTRA CHK_COMPLETE EVENT IS GENERATED

Problem Number: 4273

If a job is defined with must_complete_times and run_calendar attributes and the calendar is updated using the autocal_asc command or CA WCC when the job is in the ACTIVATED state, an extra CHK_COMPLETE event was generated in the ujo_event table, which resulted in a false alarm.

EXTRA SINGLE QUOTES IS ADDED IN OVERRIDE JOBS

Problem Number: 4288

When a job is overridden and single quotes are used in the command attribute, extra quotes were added to the command. For example, if a job is defined as follows:

```
override_job: over  
  
command: echo 'test'
```

After the job is overridden, the command becomes:

```
command: echo ''test''
```

SCHEDULER EVALUATES NON-RUNNING BOX STATUS WITH JOB_ON_ICE

Problem Number = 4337

When you send a JOB_ON_ICE event to place a job that is in a box on ice, the scheduler is supposed to only evaluate the completion status of the box when the box is running. Instead, the scheduler always evaluated the box status regardless of whether the box is running or not.

WRONG EEM OPTIONS DURING UNINSTALL

Problem Number: 4340

On UNIX, the CA EEM options were displayed improperly when you select the uninstallation of an older release using the latest version of the CA Workload Automation AE installer.

DATAMOVER THROWS NO DMOVER ERROR ON HPIA IN VERFIY MODE

Problem Number: 4341

On HP-IA, the Datamover utility failed with NO DMOVER error in Verify mode.

WINDOWS UPGRADE STOPS WITH AN ERROR OF UNABLE TO DELETE FOLD

Problem Number: 4343

On Windows, when you upgrade CA Workload Automation AE, the installer displayed the following error: Unable to delete the JRE folder; and prevented from continuing with the installation.

D-40201 SSA SSL COMMUNICATION FAILS IN ITS JNI**Problem Number: 4344**

On Linux, if you configure SSA communication by setting EnableSSL=True and EnablePMUX=True, the communication between the scheduler and the agent or the application server and the agent or the application server and a Java application that uses the CA Workload Automation AE SDK failed with errors.

LINUX STANDALONE SDK INSTALLER EXITS INTERMITTENTLY**Problem Number: 4342**

On Linux, when you upgrade the CA Workload Automation AE standalone SDK, the installer exited intermittently and generated a core.

Fixed Issues - 11.3.6 SP2 CUM1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP2 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 SP2 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

Contents**QUE_WAIT JOB RUNS ON OFFLINE MACHINE****Problem Number: 3813**

Jobs in the QUE_WAIT (QU) state ignore the OFFLINE machine status, hence the jobs ran although the status of the machine is OFFLINE.

SCHEDULER DOES NOT REACTIVATE JOBS IN A FAILED BOX ON RESTART**Problem Number: 4338**

When you define a box within another box and specify the n_retrys attribute value and when the scheduler executes the jobs in the inner box and the inner box runs to FAILURE, the inner box was re-run but the jobs inside the box were not re-activated.

SCHEDULER CORES ON STARTUP WHEN REQUEUEING AGENT STATUSES**Problem Number: 4349**

When the scheduler starts up to process job status messages that were saved from a previous run of the scheduler, the scheduler exited ungracefully.

In this issue, the scheduler might have been shut down while it was processing many incoming statuses from the agent. Or, the scheduler process might have been abnormally terminated during the previous run leaving unprocessed agent statuses in the database.

When the issue occurs, the scheduler ends immediately on startup. The scheduler log file may contain the following trace message:

```
Assertion failed: NULL !=
g_ssf._setNget_comm_update,
file /<path>/SystemAgentComm.cpp, line 2619
```

UJO_COMM_SEND_SEQ TABLE CONTAINS UN-USED COLUMNS**Problem Number: 4350**

The UJO_COMM_SEND_SEQ table contains two unused columns (destination and dest_host_alias) that occupy unnecessary space in the database. These two columns were introduced in one of the previous incremental, but were not used in the later versions. Hence, they do not have any significance in the later versions.

UPDATE STATEMENTS IN DB UPGRADE TO USE BATCH MODE**Problem Number: 4351**

Applying incremental failed for fix_data.sql when the update statement defined in the fix_data.sql is going to update huge number of rows. The RefreshAEDB.pl script complains that the transaction log is full or there is not enough memory or no space while running fix_data.sql. fix_data.sql runs partially and the required updates do not happen resulting in invalid data in the database.

BOX ENDS INCORRECTLY WHEN FINAL ACTIVATED JOB GOES ON_ICE**Problem Number: 4352**

The scheduler always evaluates a box to SUCCESS after you manually place the last remaining ACTIVATED job on ice even if the box contains jobs that completed in a FAILURE status. Also, when you manually place the last remaining ACTIVATED job on ice, the scheduler did not honor the box's box_success or box_failure conditions of the box.

The box can evaluate to SUCCESS after you manually place the last remaining ACTIVATED job on ice even when jobs in the box failed or the box_failure condition is satisfied.

SCHEDULER CORES WHILE CONTACTING THE AGENT FOR CPU USAGE**Problem Number: 4353**

When you define a workload consisting of load-balancing jobs (defined with the priority and job_load attributes) that runs on a virtual machine defined with two or more machine components, the scheduler contacts the agent to obtain the CPU load. As the scheduler contacts the agent to calculate the CPU load, the agent may delay in sending a response. If the response arrives in exactly 20 seconds, the scheduler enters a race condition which may result in ending ungracefully.

The scheduler log ended abruptly while starting a load-balancing job to a virtual machine containing two or more machine components.

FORECAST FAILS WITH CAL LEN > 30 CHARS**Problem Number: 4355**

The forecast utility failed to generate a report on jobs when any of the jobs is associated with a calendar whose name length is greater than 30 characters.

You cannot forecast jobs from both CA Workload Automation AE and CA WCC.

JOB EDIT FAILS DESPITE GROUP EDIT PERMS**Problem Number: 4356**

Job update on a non-owner field failed even when the job has group/world edit permissions. So, you cannot update the non-owner attribute of the job.

INSERT_XINST GENERATES MSSQL DB ERRORS WITH LARGE NODE NAME

Problem Number: 4357

When you specify a node name with greater than 256 characters as the xmachine attribute value, JIL failed with database errors. This issue manifests only when you run CA Workload Automation AE using a Microsoft SQL Server database that was installed after specifying a case-insensitive collation type for strings.

AE Release 11.3.6 SP1

This section includes information about the CA Workload Automation AE Release 11.3.6 SP1 new features, changes to existing features, and fixed issues.

WARNING

The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6 SP1

This article describes new features added to CA Workload Automation AE in Release 11.3.6 SP1.

(UNIX) Enabling Core File Creation

On UNIX, core file creation is enabled for the server processes (scheduler and application server) to enhance supportability. Core file creation is enabled only when the server processes are started using the unisvrctr command or by a root user.

Enabling SSL Communication Between CA Workload Automation AE and CA Service Desk

You can enable SSL communication between CA Workload Automation AE and CA Service Desk.

NOTE

On HP-UX, you cannot enable SSL communication between CA Workload Automation AE and CA Service Desk.

To configure CA Workload Automation AE to work with CA Service Desk using SSL communication, perform the following tasks:

1. Enable SSL communication between CA Workload Automation AE and CA Service Desk.

NOTE

For information about enabling SSL communication between CA Workload Automation AE and CA Service Desk, see [Configure CA Workload Automation AE to Work with Other Broadcom products](#).

2. Specify a HTTPS URL in the ServiceDeskURL parameter in the configuration file (UNIX) or the URL Location field in the Service Desk pane on the Integration - CA Workload Automation AE Administrator window of the Administrator utility (Windows).

New DBMAINT_FAILURE (548) Alarm

The DBMAINT_FAILURE (548) alarm indicates that the DBMaint command failed during automated database maintenance.

New autobcpORAdp.pl Script

On Oracle, you can now use the autobcpORAdp.pl script to synchronize the databases quickly; thereby improving performance.

New status Attribute

The status attribute sets an initial status for a job during insertion. This can prevent jobs from running during the insertion process.

Monitoring Available Disk Space

The agent is now configured to monitor the amount of available disk space and send notifications to warn you when the space is too low. The agent has three disk space warning thresholds:

- Notice—The agent logs a warning notice when the available disk space is less than the size specified in the agent.resourcemon.threshold.disk.warning.notice parameter in the agentparm.txt file. The agent continues to run; it accepts new and eligible pending jobs requests.
- Severe—The agent logs a severe warning when the available disk space is less than the size specified in the agent.resourcemon.threshold.disk.warning.severe parameter in the agentparm.txt file. The agent stops accepting new job requests. All jobs that are scheduled to start on this agent are put in a PEND_MACH status.
- Critical—The agent logs a critical warning and shuts down when the available disk space is less than the size specified in the agent.resourcemon.threshold.disk.critical parameter in the agentparm.txt file.

NOTE

For information about configuring the agent to monitor the available disk space, see [Workload Automation Agents](#).

New MACHINE_DISKTHRESHOLD Alarm

The MACHINE_DISKTHRESHOLD alarm indicates that a disk threshold has been breached on the agent. This alarm is raised when the notice, severe, or critical threshold is breached.

New Machine Status

The Blocked machine status indicates that the agent blocks communication. All jobs that are scheduled to start on the agent are put in a PEND_MACH status.

Changes to Existing Features - 11.3.6 SP1

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6 SP1.

Aggregate Statistics

In the current release, CA Workload Automation AE aggregates the job, alarm, and scheduler statistics automatically by default. The default value of the AggregateStatistics parameter in the configuration file is now set to 1. Also, the Aggregate Statistics check box in the Scheduler – CA Workload Automation AE window of the Administrator utility is now selected by default.

NOTE

If you upgrade from Release 11.3.5 or Release 11.3.6 to the current release, the value of the AggregateStatistics parameter in the configuration file is reset to 1.

STATE_CHANGE Event is Issued When a Job Enters the QUE_WAIT, PEND_MACH, or RESWAIT State

When a job is placed in the QUE_WAIT, PEND_MACH, or RESWAIT state, the scheduler now issues a STATE_CHANGE event with an informative message. The event and the associated message are written to the scheduler log and are also displayed in the detailed autorep report that you can generate using the autorep -J jobname -d command.

Updated LOGROLLOVER Parameter

You can now set the **LOGROLLOVER** parameter to PURGE(x) to purge all log files that are older than the specified number of days at midnight. The default value of the LOGROLLOVER parameter is now set to MIDNIGHT,PURGE(7). All log files that are older than seven days are purged by default.

NOTE

If you upgrade to the current release, the log files are not purged by default. Use the LOGROLLOVER parameter to specify that all log files that are older than the specified number of days are purged at midnight. For example, to purge log files that are older than 10 days, set the LOGROLLOVER parameter as follows:

```
LOGROLLOVER=PURGE (10)
```

Email Notifications

If you specify an application or group attribute in your job definition, the application name or the group name is now included in the email notification.

NOTE

For better readability, view the email notification as plain text. This may require a change to the font used by your email editor.

SNMP Traps

In addition to the alarmName, alarmJobName, alarmText, alarmCode, alarmExitCode, trapDate, and trapMessge values, the SNMP trap now passes the following values:

- alarmMachineName—The name of the machine where the job runs.
- alarmBoxName—The name of the box that includes the job.
- alarmApplicationName—The name of the application that is associated with the job.
- alarmGroupName—The name of the group that is associated with the job.
- alarmRunNum—The run number of the job.
- alarmNtry—The number of times the job was restarted.
- alarmInstanceName—The name of the instance.

Updated as-owner Resource Class

The as-owner resource class controls whether a user has permissions to include the owner attribute in the job definition. In addition, it now also controls whether a user has permissions to the existing job owner value to update, override, or delete any attribute of a job.

Job Name Supports Colon

When you insert, update, override, or delete a job, you can include colons in the job name or the box job name.

NOTE

Enclose the job name that includes a colon with quotation marks (" ") or precede it with a backslash (\).

Updated EvaluateQueuedJobStarts Parameter

The [EvaluateQueuedJobStarts](#) parameter is now updated to support three options for scheduling jobs leaving a queued state.

On UNIX, you can set the EvaluateQueuedJobStarts parameter in the configuration file to 0, 1, or 2.

On Windows, you can select any one of the following options in the Evaluate Queued Job Starts pane under the Options tab on the Scheduler – CA Workload Automation AE Administrator window of the Administrator utility:

- Off
- Skip same day check
- Include same day check

If you set the EvaluateQueuedJobStarts parameter to 0 (UNIX) or Off (Windows), the scheduler immediately starts jobs leaving a queued state.

If you set the EvaluateQueuedJobStarts parameter to 1 (UNIX) or Skip same day check (Windows), the scheduler does not reevaluate date and time conditions. Jobs that meet their original date and time conditions before entering a queued state start immediately after they leave the queued state unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are rescheduled to their next start time.

If you set the EvaluateQueuedJobStarts parameter to 2 (UNIX) or Include same day check (Windows), the scheduler reevaluates date conditions but not time conditions. Jobs that meet their date conditions after leaving a queued state start unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are rescheduled to their next start time.

Fixed Issues - 11.3.6 SP1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 SP1.

TIP

CA Workload Automation AE Release 11.3.6 SP1 also includes the issues fixed in each of the prior releases of 11.3.6.

Contents**NOTIFY.\$AUTOSERV NOT PASSING \$2****Problem Number=4229**

When the scheduler is shut down, it calls the program specified in the EP_SHUTDOWN entry in \$AUTOUSER/notify. \$AUTOSERV and passes the following two parameters: an alarm number and a message associated with the alarm. However, only the alarm number was passed but not the message associated with the alarm.

LOGS FOR A DAY MISSING**Problem Number=4036**

The scheduler log for a part of or an entire day was lost.

ARCHIVE_EVENTS DOES NOT DELETE MANUAL CHANGE_STATUS DETAILS**Problem Number=4243**

When a manual CHANGE_STATUS completion (SUCCESS/FAILURE/TERMINATED) event is issued to a job, the following problems were noticed:

1. The archive_events command did not delete the job run information that corresponds to the manual completion events.
2. The historical reports generated using the autorep -r command reported the job status as RUNNING, ignoring the manual completion events.

SCHEDULER WAITS FOR DBMAINT TO COMPLETE ON A NORMAL SHUTDOWN**Problem Number=4237**

When DBMaint was run as part of the scheduler, a graceful shutdown command such as unisrvctr stop waae_sched.XXX could cause the scheduler to wait indefinitely until DBMaint completes.

APPLICATION SERVER STUCK DURING SHUTDOWN**Problem Number=4238**

The application server got stuck while shutting down and continued to receive client requests such as autorep and autoping. Hence, the client requests were timed out and the timed out requests were not reissued to any other application server.

JIL SYNTAX CHECK FAILS ON BOX_NAME > 62 CHARS**Problem Number=4255**

The JIL syntax check failed on a box job when the number of characters of the box_name is between 61 and 64. The syntax check passes when the number of characters is less than or equal to 60.

REFRESHAEDB DOES NOT CONTAIN CORRECT INDEX**Problem Number=4253**

RefreshAEDB ignored index creation when the index is introduced in lower CA Workload Automation AE version but the same index is applicable for all versions of the product causing unnecessary database errors and sometimes affecting the performance.

Sometimes, the scheduler or application server logs contained unique constraint errors as shown below:

```
CAUAJM_E_18402 ORA-00001: unique constraint (AEDBADMIN.XPKUJO_REQ_JOB) violated
```

AUTOTRACKING INFORMATION FOR EXT-CALENDARS**Problem Number=4241**

Modifications to extended calendars using autocal_asc did not update auto-tracking information.

SAP_JOB_COUNT DID NOT ACCEPT ALPHANUMERIC CHARACTERS**Problem Number=4214**

The sap_job_count attribute accepted only numeric characters. Hence, when you specified alphabetic characters to the sap_job_count attribute (1133150M), job insert or update failed with the following error:

```
CAUAJM_E_18907 Invalid integer specified "1133150M" for JIL keyword "sap_job_count".
```

With SAP Note 942804, the job count generation process has been enhanced. The `sap_job_count` attribute now accepts alphanumeric characters from this Release 11.3.6 SP1.

LOOKBACK ON EXTERNAL JOB CAUSES SQL FAILURES

Problem Number=4260

Scheduler dumps SQL errors when a job's condition was evaluated with an external job containing a lookback validation. This affected jobs with external jobs and lookback condition to stay in current status and never start even when the job's conditions are met.

IMPROVED IMPLEMENTATION OF EVALUATEQUEUEDJOBSTARTS

Problem Number=4257

The `EvaluateQueuedJobStarts=1` did not allow the scheduler to run queued jobs that no longer meets its date condition when the job is taken out of the queued state except when a `run_window` is specified.

For example, when a job specifies the `start_times` attribute value without specifying a `run_window` attribute and the scheduler re-evaluates a job that was in the `PEND_MACH` state for one minute after the `start_times` value, the scheduler does not run the job that day. The scheduler does not consider date and time conditions as previously met and so the job goes to `INACTIVE`.

The `EvaluateQueuedJobStarts` parameter is now updated to support three options for scheduling jobs leaving a queued state.

On UNIX, you can set the `EvaluateQueuedJobStarts` parameter in the configuration file to 0, 1, or 2.

On Windows, you can select any one of the following options in the Evaluate Queued Job Starts pane under the Options tab on the Scheduler - CA Workload Automation AE Administrator window of the Administrator utility:

- Off
- Skip same day check
- Include same day check

NOTE

- If you set the `EvaluateQueuedJobStarts` parameter to 0 (UNIX) or Off (Windows), the scheduler immediately starts jobs leaving a queued state.
- If you set the `EvaluateQueuedJobStarts` parameter to 1 (UNIX) or Skip same day check (Windows), the scheduler does not re-evaluate date and time conditions. Jobs that meet their original date and time conditions before entering a queued state start immediately after they leave the queued state unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- If you set the `EvaluateQueuedJobStarts` parameter to 2 (UNIX) or Include same day check (Windows), the scheduler re-evaluates date conditions but not time conditions. Jobs that meet their date conditions after leaving a queued state start unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.

BLOCK CHANGE_STATUS EVENT TO INACTIVE WHEN JOB IS IN THE ON_ICE STATUS

Problem Number=4262

This is a follow up to problem 4252. Although the scheduler ignored `CHANGE_STATUS` events to `INACTIVE` when a job is in the `ON_ICE` status, you are still allowed to manually send the `CHANGE_STATUS` event.

AUTOTRACKING INFORMATION FOR CYCLE CALENDARS**Problem Number=4263**

Modifications to cycle calendars using autocal_asc did not update auto-tracking information.

LOAD BALANCING PROBLEM WITH VIRTUAL RESOURCES**Problem Number=4180**

A job with virtual resources defined with a virtual machine was always submitted to the first machine satisfying the resource requirements even when there are other machines with more available load.

FORECAST REPORT PROBLEM**Problem Number=4246**

Forecast report did not list a job when its parent job has a dependency on another job from a different job flow with its own date and time conditions.

AUTOTRACK CONDITIONS TAGGED IMPROPERLY**Problem Number=4256**

When autotrack is at level 2, adding or modifying a job using jil or CA WCC with either "condition:", "box_success:" or "box_failure" conditions caused the following auditing problems:

1. An invalid field "condition: 1", "box_success: 1" or "box_failure: 1" was added to the autotrack audit info tables. This is not a valid field.
2. The valid "condition:", "box_success:" or "box_failure:" field was marked as modified, even though it did not actually change.

FORECAST LISTS JOBS DEFINED WITH ONLY EXCLUDE_CALENDAR**Problem Number=4245**

Job with only exclude_calendar and no run_calendar were reported in the forecast report. For example, job with the following definition was listed in the forecast report:

```
insert_job: test_forecast
job_type: CMD
command: echo "Wrong forecast"
machine: localhost
date_conditions: 1
exclude_calendar: test_exc_cal
```

Issue the following command and observe that the job gets listed in the forecast report:

```
forecast -J test_forecast -F "01/01/2015"
```

BOX CONDITIONED WITH A CHILD JOB NOEXEC STATE DOES NOT COMPLETE**Problem Number=4266**

A box job that was conditioned with box_success or box_failure on a child job in a NOEXEC state never goes to the terminal status and remains in the running state.

JIL ACCEPTS UNSUPPORTED GLOBAL VARIABLE CONDITION**Problem Number=4259**

JIL accepted a cross-instance global variable condition that is not supported, for example, condition: v(VAR~ABC) = "1"

JMS JOB_TYPE DOES NOT SUPPORT GVAR IN CONNECTION_FACTORY ATTRIBUTE**Problem Number=4244**

Most attributes that support strings did not allow global variable substitution. Jobs that use fields such as connection strings should support global variable substitution to let users modify the variable once and reflect the change in all jobs without modifying the job definition.

CROSS-INSTANCE STATUS CANNOT BE SENT INTERMITTENTLY**Problem Number=4221**

Intermittently, the scheduler was unable to send the INACTIVE status to the cross-instance and the following error message appeared:

```
[09/11/2014 10:06:46.1735] 15335636 9354 CAUAJM_I_40256 [Sending CHANGE_STATUS to
  INACTIVE for Job: PVSLMUPD01D_P to AUTOSERV: ASP]
[09/11/2014 10:06:46.2048] 15335636 9354 CAUAJM_W_40283 Error from Instance
<<ASP>: "CAUAJM_E_10248 send_event SP detected a duplicate event.
```

ALL QUEUED UP CROSS-INSTANCE STATUSES ARE NOT SENT**Problem Number=4231**

When a cross-instance application server is offline, the scheduler in the native instance queued up all its cross-instance statuses. When the cross-instance app server is online, only 128 queued up events were sent to the cross-instance application server and the rest were not sent.

AUTOAGGR CREATES AN EMPTY CON FILE**Problem Number=4224**

The autoaggr command created an empty file named "CON" when it was used with the -o option.

DATABASE CREATION FAILS WHEN USING EXISTING DATABASE**Problem Number=4267**

The CreateAEDB.pl script failed to place the CA Workload Automation AE schema when it is instructed to use the existing Sybase or MS-SQL database.

WEB SERVICES INSTALLATION FAILS ON AN 8DOT3-DISABLED WINDOWS MACHINE**Problem Number=4269**

The Web Services installation failed on a Windows machine where 8dot3 naming is disabled due to spaces in the AUTOSYS path and JAVA_HOME variables.

UJO_COMM_UPDATE SP ERRORS WITH VIRTUAL NODE NAMES**Problem Number=4213**

The scheduler or application server experienced errors while communicating with an agent when the machine definition uses a virtual node name that point to an agent in a load balancing cluster. Hence, temporary send failures occurred, causing the scheduler to restart a job. Additionally, autopings to agents failed consistently and machines might remain stuck in the Missing status.

The scheduler or application server log file showed one of the following messages:

- The log file shows errors while contacting an agent due to Oracle error "ORA-01422: exact fetch returns more than requested number of rows" while executing the ujo_comm_update stored procedure. The scheduler or application server creates conflicting entries in the communications tables resulting in the Oracle error.
- The log file shows errors while contacting an agent due to Oracle error "ORA-20870: ujo_comm_update update comm_send_seq - could not find row for unqualified agent" while executing the ujo_comm_update stored procedure. The scheduler or application server orphans entries in the communications tables resulting in the Oracle error.

HEARTBEAT SUPPORT FOR SAP ADAPTER JOBS

Problem Number=4120

The SAP Adapter jobs could not send HEARTBEAT events to the 11.3 scheduler due to the new system agent.

EXTENDED CALENDAR CREATES WRONG DATES

Problem Number=4270

The extended calendars added unwanted dates to the calendar and ignored valid dates by adding the holiday or cycle calendar that is used by a different extended calendar. This causes issues in job scheduling.

JOBS NOT EXECUTED UPON MACH_ONLINE

Problem Number = 4274

When the scheduler receives a MACH_ONLINE event, jobs in PEND_MACH status were set to INACTIVE and a QUEUEDJOB_STARTFAIL alarm was generated. This issue occurred when EvaluateQueuedJobStarts=1.

DEPENDENT JOB DID NOT RUN WHEN RUN_WINDOW HAS MISSING HOUR

Problem Number = 4275

When the run window of jobs falls during the DST missing hour, jobs were set to INACTIVE stating that the run window is missed.

CA WCC Upgrade Failure

Problem Number = 4346

On Linux, if CA Workload Automation AE and CA WCC are on the same computer and you try to upgrade CA WCC to Release 11.3.6 after you upgraded CA Workload Automation AE, the upgrade failed with the following error:

```
install of ca-waae-common-11.3.6-185.i386 conflicts with file from package ca-waae-common-cs2
Script or command "rpm -U "<PATH>/ca-waae-common-11.3.6-185.i386.rpm"" failed with exit code 2.
package ca-waae-common-11.3.6-185 is not installed
Terminating install. Reverting changes.
Job executed with error: 7
```

This error occurs if you upgraded the installed CA Workload Automation AE components to Release 11.3.6 using a different type of DVD than what you used for installation.

AE Release 11.3.6

This section includes information about the CA Workload Automation AE Release 11.3.6 new features and changes to existing features. It also lists the issues fixed in CA Workload Automation AE Release 11.3.6 INCR1 and Release 11.3.6 INCR1 CUM1.

WARNING

The acknowledgements for third-party components used by CA Workload Automation AE are in the acknowledgements.txt file that is located in the root directory of the installed product.

New Features - 11.3.6

This article describes new features added to CA Workload Automation AE in Release 11.3.6.

New Unauthenticated User Mode Setting

The EXTERNAL unauthenticated user mode setting enables you to authenticate client utilities using external authentication protocols when customized authentication libraries are installed on all client and server machines in the instance.

NOTE

For information about changing the unauthenticated user mode setting to EXTERNAL and about creating and installing customized authentication libraries, see [Customize Security Policy and Settings](#).

Setting the Maximum Number of Lines to Retrieve from a Log File

You can set the maximum number of lines to retrieve from a log file. On UNIX, you can configure this setting using the [LogMaxEndLines](#) parameter in the configuration file. On Windows, you can configure this setting using the Log Max End Lines field on the Application Server window of CA Workload Automation AE Administrator (autosysadmin).

Changes to Existing Features - 11.3.6

This article details the changes to existing features in CA Workload Automation AE Release 11.3.6.

Authenticating Command-Line Utilities with External Security

The addition of the -usr command-line option enables you to authenticate certain command-line utilities with external security. Using this option improves security by ensuring that the utility runs as an authenticated external security user.

The authentication of the external security user is successful only when the user password is accurately specified using the -pw or -pwx parameter. When the authentication fails, the utility does not run and exits with an error.

Following the authentication, the external security system assigns a security policy identity to the utility. The security policy determines which protected CA Workload Automation AE objects are accessible based on the assigned identity and grants the utility access to those objects.

NOTE

- This option is required when the CA Workload Automation AE instance is configured to run in external security mode and the unauthenticated user mode is set to STRICT; otherwise, it is optional.
- The utility ignores this option when the CA Workload Automation AE instance is operating in native security mode.
- For more information about which utilities support this option and how to use it, see [AE Commands](#).

Cross-platform Job Requests

When the scheduler receives the cross-platform start job requests, it now requests the application server to authenticate and authorize that request. For the job to execute successfully, you must have EXECUTE access for the following resource classes:

- as-appl
- as-group
- as-machine
- as-job

FORCE_STARTJOB (108)

In the current release, you can force start a job in FAILURE or TERMINATED status that has a virtual resource dependency with free=Y or free=N and has not released the virtual resources. The FORCE_STARTJOB event verifies if the current status of the job is FAILURE or TERMINATED and schedules the job using the already held virtual resources.

NOTE

Before force starting the job, the scheduler does not reevaluate other resource dependencies.

STARTJOB (107)

In the current release, you cannot issue the STARTJOB event to start a job that has a virtual resource dependency with free=Y or free=N and has already held the resource.

To start such a job, take *one* of the following actions:

- Manually release the held resource by issuing the RELEASE_RESOURCE event.
- Force start the job in FAILURE or TERMINATED status by issuing the FORCE_STARTJOB event. The virtual resource is released if the job has the virtual resource dependency with free=Y and completes successfully.

Updating the resources Attribute in an Existing Job Definition

You cannot update the resources attribute in the existing job definition if the job has a resource dependency and has held the resource.

To release the held resource, take *one* of the following actions:

- Manually release the held resource by issuing the RELEASE_RESOURCE event.
- Force start the job in FAILURE or TERMINATED status by issuing the FORCE_STARTJOB event. The virtual resource is released if the job has the virtual resource dependency with free=Y and completes successfully.

Fixed Issues - 11.3.6 INCR1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 INCR1.

Contents

AUCKLAND-Z POSIX VALUE WRONG IN UJO_TIMEZONE TABLE

Problem Number = 3907

The Pacific/Auckland Posix TZ value did not reflect the correct DST change for Australia/Pacific. The jobs running with Auckland-Z time zone were scheduled incorrectly.

SAP JOB INTRODUCES IN PS_ARGS ATTRIBUTE**Problem Number = 3918**

The JIL utility escaped the colon (:) character in the ps_args attribute with backslash (\). This resulted in miss interpretation of the attribute. For example, if the ps_args value contains a URL, the link did not work in the browser.

UJO_COMM_SEND_SEQ TABLE HAS INVALID RECORDS**Problem Number = 4060**

The application server and scheduler intermittently failed to start due to the unique constraint error from the ujo_comm_send_seq table as the invalid data was not deleted at application server startup.

LOGMAXENDLINES VALUE NOT DISPLAYED ON STARTUP**Problem Number = 4019**

The application server log did not display the LogMaxEndLines configuration parameter value at startup.

SUCCESS_CODES/FAIL_CODES ATTRIBUTES NOT DISPLAYED**Problem Number = 4034**

If you set the success_codes or fail_codes attribute value to -1 and used the autorep command with the -q option to list the job details, the report did not display the success_codes or fail_codes fields.

JOB TERMINATION EVENTS NOT SENT TO CA7**Problem Number = 4064**

If you defined job dependencies between CA7 and CA Workload Automation AE, only the JOBINITU and RUNNING events were sent successfully. The JOBTERMU, SUCCESS, and FAILURE events were not sent to CA7.

SERVER LOGS TIMESTAMP DELAYED BY ONE HOUR**Problem Number = 4089**

The scheduler and application server printed timestamps with a delay of one hour; especially after applying the Microsoft August-2013 patch KB2863058 (Windows) or Time zone data 2013 (UNIX) patch for timezones such as Israel, Taipei, and Seoul.

JOB DOES NOT HONOR LOOKBACK '0'**Problem Number = 4113**

When you define a job with a look-back dependency of 0 (zero), the scheduler examines the last end time of the condition job and if the condition job has not run since the last run of the job for which the condition is set for, the job should not run. However, the scheduler ignored the look-back dependency of 0 and scheduled the job as if the look-back dependency is not defined when the condition job and scheduled job went into terminal status (such as SUCCESS or FAILURE) in same second.

DEFAULT TIMEZONE FOR ISRAEL WRONG**Problem Number = 3594**

The CA Workload Automation AE timezones table did not reflect the current DST of the Israel time zone.

USER WITH AS-OWNER DENY ACCESS INCORRECTLY ALLOWED TO ALTER JOB**Problem Number = 4144**

A user without the as-owner access on a job was allowed to modify the job definition.

JOB_DEPENDS DISSATISFIED FOR JOB WITH HELD RENEWABLE RESOURCE**Problem Number = 4138**

The job_depends report displayed renewable resource availability as not satisfied when the job held resource usage from the prior run.

WEB SERVICES AUTHENTICATION EXCEPTION WITH ORACLE 12C**Problem Number = 4145**

When you ran web services with Oracle 12C, it failed with the following error:

```
ORA-28040: No matching authentication protocol
```

ORIGINAL CREATE USER/TIMESTAMP LOST AFTER UPDATING JOB**Problem Number = 4100**

When you updated a job, the original values of the create user and create stamp fields were replaced with the values of the current user (performing the update) and the current date (the update is performed on). So, the original values of the create user and create stamp fields were lost permanently.

APP SERVER CORES WHEN EMPTY CALENDAR IS PASSED TO AUTOCAL_ASC**Problem Number = 4117**

The application server core dumped when an empty calendar name was passed as input to the autocal_asc command.

JOBS STUCK IN RESWAIT STATE ENCOUNTER UNIQUE CONSTRAINT ERROR**Problem Number = 4137**

Jobs did not run when the resources are available and the unique constraint errors are observed in the scheduler log. The same behavior was observed with jobs in RESWAIT state.

MAXRESTARTTRYs IGNORED DURING AGENT COMMUNICATION PROBLEMS**Problem Number = 4123**

The MaxRestartTrys parameter value in the configuration file was not honoured when there is a communication problem with an agent during job restart causing the restart attempts to end prematurely like this:

```
<Have EXCEEDED the Max # of times (1) to attempt a restart.
```

```
Something is wrong and needs to be investigated manually.>
```

As a result, the job was not tried to restart the number of times specified in the MaxRestartTrys parameter in case of communication problems with the agent causing the job to stay in the FAILURE state indefinitely.

CA WCC -203 ERROR VIEWING JOB RUN DETAIL

Problem Number = 4071

Job retrieval using CA WCC GetJobsWithFilterReq failed in Sybase DMI layer with a return code of -203.

GET JOB LOG FAILS WHEN NO INITIATORS

Problem Number = 4079

When the agent is busy and processing many CONTROL AFMs (such as those received from explicit or implicit autoping or job log file retrievals), the application server crashed and displayed memory errors.

AUTOREP BROKEN N_RETRY REPORT

Problem Number = 3863

Historical reports for previous runs of box jobs displayed incorrect run details for jobs in the box that were not scheduled for the corresponding box run.

The autorep command with the -r option was not working when a job ran to failure due to application failure.

AUTOPING -S FAILS

Problem Number = 4055

When communicating in an environment where DNS returns a different name than the hostname, the SDK client must provide an alias that matches the DNS name so that the return messages are routed correctly.

This communication problem was seen with "autoping -m mach.dom.com -S" when the network environment has fully qualified machine names only in the DNS, but the local hostname returns the short (not fully qualified) name. The "-S" option requests the agent to use the SDK to establish a connection with the application server. This connection failed unless the application server recognized the name (one that matches with the DNS name returned on the application server).

JAVA SDK MEMORY LEAKS

Problem Number = 4039

When CA WCC JSC ran, the memory usage of the Java process grew and eventually exceeded the maximum user process memory, causing JSC to stop responding.

CHASE_SLEEP BEFORE STATUS CHANGE CHECK

Problem Number = 4066

The CHASE_SLEEP environment variable intermittently reported completed jobs within a box as failure before the correct status of jobs got updated; this resulted in the failure of the box.

DEPRECATED ATTRIBUTES OF SDK THROWS EXCEPTION

Problem Number = 3940

The deprecated attribute ATTR_TOTAL_EVENT_LATENCY threw BadAttributes exception.

I5_PARAMS JOB VALIDATION FAILS

Problem Number = 4114

The i5_params attribute that is valid only for RUN_PROGRAM or COMMAND values of the i5_action attribute was not validated properly by JIL.

JOB RUN TWICE WHEN FORCE_START

Problem Number = 4049

If a job is in the QUE_WAIT state on a machine and you increased the max_load attribute value of the machine and force started it, it resulted in the job running twice.

The job ran once for the normal STARTJOB event and once for the FORCE_STARTJOB event.

START TIMES FOR JOBS RECALCULATED WRONGLY

Problem Number = 4108

When you added new dates to an existing calendar, the next start time of job was recalculated and displayed even if the updated calendar is linked only to an inactive version of the job.

INCONSISTENT JIL BEHAVIOR WITH BOX_TERMINATOR FIELD

Problem Number = 4095

If you set the box_terminator attribute to a box job that is not inside another box, the JIL syntax check (-v) failed with an error. If you set the box_terminator attribute to a non-box job that is not inside another box, the JIL syntax check (-v) passed successfully.

DATE CONDITIONS EVALUATED IN JOB RESTART

Problem Number = 4086

The scheduler did not restart jobs that are defined with date and time conditions and the n_retry attribute when the original starting conditions are no longer met. The scheduler log file displayed log traces similar to the following:

```
CAUAJM_I_40159 The date/  
time conditions for job <job1> have not been met. Cannot start job.
```

```
CAUAJM_I_40149 Cancelling restart for job <job1> and setting to INACTIVE.
```

MAX_LOAD UPDATE IS NOT EFFECTIVE WHEN BRING JOBS OUT OF QU

Problem Number = 4059

When the max_load of a machine was updated while the jobs are in the QU status, the new value was not loaded when the jobs come out of the QU state.

GET AEWS/JOB WITH FILTER RETURNS 204 IF NO MATCHES FOUND

Problem Number = 4072

Job retrieval with filter using web services returned wrongly 204 (NO_CONTENT) error when none of the jobs satisfied the filter criteria. . The return code must be 404 (NOT_FOUND) in accordance with the other exposed web services.

EVENT_TIME_GMT IN LOG

Problem Number = 4016

The scheduler log format for event display is modified to show the time the event is marked to be picked up. A new ISDBGACTIV setting named UTC is added to allow the scheduler to display the time information against the event.

MISSING JOBS IN AUTOREP OUTPUT

Problem Number = 4047

The autorep command did not return a consistent count of jobs every time it is run. Two out of five runs of the same command was found to return a different count of jobs.

AS_SERVER CRASHES IN JIL SYNTAX

Problem Number = 4088

The job information that is exchanged between JIL and the application server was getting corrupted. As a result of this corrupt data, the application server terminated abnormally.

FORECAST FAILS ON GROUP ATTRIBUTE OF A JOB

Problem Number = 4148

When a job with the group attribute was forecasted, CA EEM error messages similar to the following were displayed and the report generation failed:

```
CAUAJM_W_10509 Group Read Access Denied!
```

```
CAUAJM_W_10439 No policies granting access to resource.
```

```
CAUAJM_W_10440 Class: as-group
```

```
Resource: User: autosys Access: read
```

```
CAUAJM_W_10442 Time: 1391784486
```

```
Delegator: None
```

JOB IS NOT SCHEDULED IN REPEAT HOUR

Problem Number = 4143

When the DST to Standard time transition occurred at 23:59, a job scheduled to run every few minutes was not scheduled in the repeated hour.

SCHEDULER/APPLICATION SERVER SHUTS DOWN WHEN ORA-12541 ERROR IS

Problem Number = 3989

The shadow or primary scheduler or application server shut down when an ORA-12541 error was encountered during the failover process.

JOB WITH RESOURCES GET STUCK IN ACTIVATED STATE

Problem Number = 4098

When a job that held resource consumption from prior run was scheduled again as part of the date and time conditions, it was stuck in the activated state. On issuing the `RELEASE_RESOURCE` event, it freed up the held resources but the job continued to be stuck in the activated state.

MIXED JOB TZ FORMATS CONFUSE SCHEDULER

Problem Number = 3810

When jobs with date and time conditions were defined with timezone attribute values in the Olson format (America/New_York) mixed along with jobs having timezone attribute values in the POSIX format (EST5EDT), the scheduler may have calculated the wrong scheduled start time for any job with date and time conditions irrespective of whether the timezone attribute was defined for the job or not.

RESTRICT_FORCE_STARTJOB NOT WORKING

Problem Number = 4093

When a `FORCE_STARTJOB` event was issued against a `RUNNING` job, the scheduler reran the job although the `RESTRICT_FORCESTARTJOB` environmental variable is set to restrict it.

ON_NOEXEC ALLOWED OVER ON_ICE

Problem Number = 4097

Jobs that are placed in the `OFF_HOLD` or `NO_EXEC` status did not generate new start time; resulting in the job not being scheduled the next time.

AUTOTRACK FAILS TO REPORT USAGE ON SPECIFIC MACHINE

Problem Number = 4069

When autotrack was enabled, the report for a specific machine did not contain any output. However, % mask for a given machine produced the expected output.

SCHED DEADLOCKS WHILE DEQUEUEING LOAD BALANCING JOBS

Problem Number = 4116

The scheduler deadlocked while running job flows containing load balancing jobs.

CLIENT AUTHENTICATION TO STRICT MODE SERVER FAILS

Problem Number = 4126

The CA Workload Automation AE Release 11.3.6 client was not able to detect that the CA Workload Automation AE Release 11.3.5 server is in `STRICT` mode and failed authenticating with it.

SCHEDULER CORES WHILE CALCULATING NEXT DAY OF BAD CALENDAR

Problem Number = 4131

On Windows, the scheduler terminated abnormally while trying to calculate a job's next start time after the job completed its run and there are no more valid dates in the job's `run_calendar`.

SCHEDULER RUNS JOBS TO OFFLINE MACHINE IN VIRTUAL MACHINE

Problem Number = 4135

When you defined a machine with a factor value of 0 (zero), the scheduler did not contact the agent to obtain the usage, which resulted in the scheduler to run jobs on offline machines at random.

ARCHIVE_EVENTS DOES NOT REMOVE

Problem Number = 3986

The archive_events command did not delete the data from ujo_extended_jobrun_info table. In fact there were some rows left in the ujo_extended_jobrun_info table forever and the corresponding ujo_job_runs entries were not available in ujo_job_runs table. So over a period of time it accumulated millions of rows causing unnecessary maintenance. This did also impact the performance of SQL queries on ujo_extended_jobrun_info table.

If you ran the archive_events command with the -j option, it did not delete the rows from the ujo_job_runs table for inactive jobs. As a result, the archive_jobs command did not delete the inactive job rows from the ujo_job table for jobs having endtime=0 in ujo_job_runs table.

EVENTS MISSED FROM GETMONITOREVENT API

Problem Number = 3498

The GetMonitorEvent API provides direct programmatic access to all events in the system. Calling this API in a loop returns CA Workload Automation AE events as they occur.

When the GetMonitorEvent API was used, the events were missed intermittently.

The monbro command stopped responding intermittently and the events were missing in the monbro output.

The sample get_auto_event.cpp file that is distributed with CA Workload Automation AE did not compile as it has references to non-existing header files.

JOB_TERMINATOR BROKEN FOR ACTIVATED BOXES

Problem Number = 4068

If one job (for which you set the box_terminator attribute) in a box failed, the other jobs in the box remained in the activated state.

BOX STATUS WRONG

Problem Number = 4070

The scheduler set wrong status to a box job after a job goes to failure when the agent is shut down and the chase command is run.

After the agent came back online, the job completed and displayed the status as SUCCESS, however, the box job status was set to FAILURE.

NO RESCHEDULE OF DEQUEUED JOBS WITH DATETIME CONDITION NOT MET

Problem Number = 4011

When the EvaluateQueuedJobStarts configuration parameter is set to 1, the scheduler did not reschedule jobs coming out of the PEND_MACH, RESWAIT, or QUE_WAIT state if the jobs had a date or time condition and the starting conditions were not met.

EVENT PROCESSING DELAYED

Problem Number = 4076

On Windows, the scheduler experienced severe lag times in changing the job status from STARTING to RUNNING or from RUNNING to a completion status when the network performance was poor. The scheduler begins to accumulate many expired TCP/IP connections within the internal operating system socket listening queue. As the scheduler falls behind in responding to the agent, the agent experienced frequent read time outs due to unacknowledged messages causing it to start new TCP/IP connections. The scheduler was also unable to detect that the socket connection has expired to the degree that it can frequently timeout after 20 seconds causing it to fall further behind.

NEXT START TIME NOT CALCULATED

Problem Number = 3836

The next start time was not calculated when a job was first put in the ON_ICE state and then in the ON_HOLD state and subsequently released from the ON_HOLD state.

UNIQUE CONSTRAINT ERRORS THROWN

Problem Number = 4056

The application server log was filled with Oracle ujo_audit_msg_log unique constraint errors intermittently when the stored procedure failed initially and attempted to retry.

JOB STATUS OUT OF SEQUENCE

Problem Number = 3866

When CA Workload Automation AE is running in dual event server mode, the scheduler copies the missing events from one database to other database. If a RUNNING event was copied from the secondary database to the primary database, the scheduler may have processed the RUNNING event after the SUCCESS event, which will lead to a job staying in the RUNNING status forever.

REINDEX INDEXES SINGLE INDEX

Problem Number = 4043

When the reindexDB perl script was run to reindex the database indexes, the script terminated after reindexing a single database index.

SCHEDULER TERMINATED ABNORMALLY ON FORCE_STARTJOB

Problem Number = 4096

When the FORCE_STARTJOB event was issued against a job that uses the machine chooser script and if the machine chooser script did not exist on the machine or if it returned blank machine, the scheduler was terminated abnormally.

AUTOREP -D RESPONSE TIME BAD

Problem Number = 3472

In a heavily loaded instance, when you issued the autorep command with the -d option, it stopped responding and timed out. In r11.0, there is an index on the ujo_proc_event table that begins with the joid column, was missing in Release 11.3.6.

APPLICATION SERVER HANGS

Problem Number = 3912

On Windows, the application server ran for a certain duration of time and then stopped responding. So, the incoming API requests were not processed. The client utilities and CA WCC that depend on the application server did not function.

ARCHIVE_EVENTS NOT REMOVE SOME ROWS UJO_EXTENDED**Problem Number = 3986**

When the archive_events command was issued against job runs, it left some entries in the ujo_extended_jobrun_info table. Some of these entries cannot be deleted and so the number of rows kept increasing and this affected the performance of the component that accesses the ujo_extended_jobrun_info table.

SCHED SETS BOXJOB TO ACTIVATED AND RUNS TWICE**Problem Number = 4026**

In a single box run, the scheduler executed the jobs in boxes twice in the following situations:

- a) If the processing of a job start for a job in a box experiences a sufficiently long delay before evaluating the job's condition and the upstream job completes resulting in competing job start evaluations, the child job can run twice.
- b) If there is a shared dependency between the ancestor job and the descendant job or if the dependency for both are satisfied at the same time (for example, if the ancestor job is dependent on jobA and the descendant job is dependent on jobB and both jobA and jobB complete at the same time), the competing job start evaluations can result in a strange race condition that could allow a descendant job to be executed before the parent box is executed.

If the parent box is executed after the child job has completed, the child job is reset to ACTIVATED; this potentially causes the child job to run twice or remain in the ACTIVATED state indefinitely.

- c) If there is a miscommunication between the scheduler and the agent, such that the agent is able to run the job but the scheduler fails the job start, the job can end up in the PEND_MACH state. When the scheduler re-evaluates the job following a MACH_ONLINE event, the job may have already run successfully but the second evaluation may determine that the starting conditions are not met resulting in the job being placed indefinitely in the ACTIVATED state.

WIN32 PATCH INSTALL FAILS**Problem Number = 3955**

When you defined CA Workload Automation AE without quotes, the following error was displayed:

```
'Files' is not recognized as an internal or external command, operable program or batch file.
```

When you defined CA Workload Automation AE with quotes, the following error was displayed:

```
'Files' was unexpected at this time.
```

AUTOSYSLOG SPOOL FROM I5/OS**Problem Number = 4041**

When you issued the "autosyslog -J i5jobname -tS" command, it did not retrieve the job spool file from the Workload Automation Agent for i5/OS.

COMM ENGINE DEADLOCKS CMADDR NAK**Problem Number = 4015**

The agent communication failure occurred when you set the communication.manager.dynamic.modifications.lock parameter in the agentparam.txt file to true. However, these communication failures can occur anytime the agent sends a negative acknowledgment. Essentially, the scheduler prepares a registration request message when communicating

with an agent for the first time after starting up. When the agent receives the registration request, it adds an entry for that manager in the agentparm.txt to allow the agent to dynamically recognize the manager for future communications and sends a response to the manager.

When you set the communication.manager.dynamic.modifications.lock parameter, the agent acknowledges the receipt of the registration request but does not send a response back to the manager.

The scheduler internally keeps track of the acknowledged requests that are not fulfilled and assumes that successful acknowledgment means that the agent fulfills the request and awaits for the response. In the meantime, the scheduler proceeds to send job start requests to the agent. The agent rejects the requests because it does not recognize the manager. This causes the scheduler to resend a registration request but the request and all subsequent communications to that node are blocked internally because the scheduler is still awaiting the response of the previously acknowledged registration request. This blocked state carries over when the scheduler must be shut down effectively resulting in the scheduler not responding indefinitely.

UJO_JOB_TREE LINEAGE COL OVERFLOWS SIZE 2048 ON JOB UPDATES

Problem Number = 4028

The lineage column was calculated incorrectly when a box (boxA) with children (inside a box) was moved into another box (boxH) with children (inside a box). After the update or move, the lineage for boxA was calculated correctly. However, the lineage for children inside boxA was calculated incorrectly where it still reflected the old value and the new value concatenated together. The lineage for children should be the parent box lineage (boxA lineage pointing to boxH) plus the parent box (boxA).

When boxA was updated to move multiple times between boxes, it caused the lineage value for the children to grow off the maximum 2048 limit. This caused the updates using JIL to fail with an error similar to the following:

```
CAUAJM_E_18402 ORA-12899: value too large for column "AEDBADMIN"."UJO_JOB_TREE"."LINEAGE" (actu
```

The same happened with SDK. CA WCC collector was not able to store the lineage into the database which resulted in stalled data in the UI.

SQL DEADLOCK ERRORS

Problem Number = 4208

The SQL deadlock errors were displayed during the run of the scheduler.

EXTERNAL AUTHENTICATION IN SDK

Problem Number = 4152

When the Unauthenticated User Mode is set to EXTERNAL, the CLIs authenticate using user-written callouts. The equivalent authentication was not added to the SDKs.

EXTENDED CALENDAR ISSUE

Problem Number = 4061

When a large number of jobs were defined with the same extended calendar and the run of the jobs is on the last day defined in the calendar, the next start time of the job resulted in the regeneration of the days and this action when performed by more than one job resulted in the unique constraint error for one job which caused the job to not get scheduled.

CALENDAR CHANGE DOES NOT REFLECT WCC RESULTS**Problem Number = 4199**

The changes made in a calendar did not get reflected in the job's next_start attribute in CA WCC.

JOB RAN OFF THE SCHEDULE DUE TO TIMEZONE**Problem Number = 3810**

On Windows, a job running on an agent ran from Monday to Saturday at 7:00. On 12/11/12, the job failed at 7:00:16. The job was rescheduled at 2:00 (7 hours later) instead of the next day at 7:00.

SCHED RUNS JOB TWICE AFTER NO ACK FOR JOB RUN AFM**Problem Number = 4164**

The agent executed the same job twice. This happened when the scheduler did not receive an acknowledgment of the first job start from the agent and it restarted the job.

WIN32 AUTOSYSADMIN BLANK MUL INSTANC**Problem Number = 4087**

In a multi-instance environment on Windows, if you launched autosysadmin from the first instance, autosysadmin failed to load the second instance information.

WEBSERVICE DENIES ACCESS**Problem Number = 4092**

Web services failed to connect to CA EEM Release 12.51.

UPGRADE FAILS TO 11.3.6**Problem Number = 4099**

If the ujo_web_services table contains data, the CA Workload Automation AE installer failed to upgrade from Release 11.3.x to Release 11.3.6.

EVT_NUM AT MAX INTEGER**Problem Number = 4030**

When evt_num was incremented beyond the maximum of a signed integer (2147483647), Oracle threw an error when you retrieved it into a signed integer variable.

Error is "integer overflow".

JOB STATUS CHANGES NE->AC**Problem Number = 4101**

When a job inside a box was in RW state and then sent to NE state, the status changed to AC upon FORCE_STARTING the box.

MAXBINARYTRANSFERSIZE LIMIT HIT**Problem Number = 4168**

When you performed a syntax check for a large set of jobs, the blob size limit (1MB) which is the transfer data size between the application server and clients was hit and the syntax check failed.

AUTOREP -MALL REPORTS NEGATIVE VALUES FOR LOAD AND PRIORITY

Problem Number = 4161

The autorep -MALL -dn report incorrectly reported boxes as part of the "Current Jobs". For the jobs listed, the machine, load, and priority had incorrect or negative values.

SSL ENABLEMENT AUTOSYS - SERVICE DESK MANAGER

Problem Number = 4050

The Service Desk Manager (SDM) integration supported only HTTP communication. This fix adds the necessary support for connecting to a SDM web service that is configured to use HTTPS/SSL.

UJO_WAIT_QUE HAS DELETED JOBS

Problem Number = 4094

The ujo_wait_que table had references of deleted jobs that are in the queue wait state. So, when a new job was defined on the same machines, the scheduler picked up the deleted jobs in ujo_wait_que and so the original jobs which should trigger were going to the queue wait state.

POSTPONED CROSS-INSTANCE EVENTS ARE NEVER SENT AGAIN

Problem Number = 4197

When a communication problem occurred with the cross-instance application server, the scheduler postponed sending the cross-instance events, but at the same time, it incorrectly flagged those events as "successful sends". This resulted in those events never being sent again to the cross-instance.

SIGNAL MASK PASSED FROM AGENT TO JOB PROCESSES

Problem Number = 4150

The job processes started by the agent may have inherited a signal mask from the agent, which rendered them unable to receive signals such as SIGHUP, SIGTERM, and so on from the operating system.

SHADOW DUPLICATES JOB EVENTS AFTER FAILING OVER

Problem Number = 4212

The scheduler generated duplicate job events after the shadow fails over.

APP SERVER DB MONITOR DOES NOT DETECT CLUSTER FAILOVER

Problem Number = 4210

The application server did not process the sendevent commands and must be restarted after the Oracle RAC server internally fails over following the loss of a cluster node.

UJO_COMM_UPDATE SP ERRORS WITH VIRTUAL NODE NAMES

Problem Number = 4213

In an environment where the virtual node names were defined so that it can be configured to point to one agent node in a load-balancing cluster, the scheduler and application server failed to receive data from the agent after the virtual node name was configured to point to a different agent.

AEDB UPGRADE FAILS WHEN CREATEAEDB.PL IS INTERRUPTED

Problem Number = 4211

The CreateAEDB.pl script failed when either the user terminated the script or the script got terminated while running the upgrades. This left the database in intermediate state and caused database failures while trying to add a column that was already added in an earlier unsuccessful iteration.

JOB INSIDE A BOX WITH OVERRIDE DOES NOT START

Problem Number = 4209

A child job with an active override did not start even when the job's conditions were met. As a result, the jobs depending on the child job were never scheduled.

TRAILING SPACE IN ISDBGACTIV VALUE IMPACTS THE SETTING

Problem Number = 4200

When the ISDBGACTIV value had trailing spaces at the end of the value, the last trace setting was ignored.

CANNOT DOWNLOAD EVENT_DEMON LOG FROM WCC IF IT HAS >7K LINES

Problem Number = 4204

If the event_demon log file included more than 7000 lines, you could not download the log file from CA WCC.

ON_HOLD CREATES BAD BOX RUN

Problem Number = 4196

Jobs in a box did not execute with the proper run number when they were taken OFF_HOLD.

JAVA SDK DUP WHERE CLAUSE

Problem Number = 4191

Duplicate, superfluous WHERE clauses can be generated when using the Java SDK and making calls to the GetAlarmsWithFilterReq API (or any of the GetXXXWithFilterReq APIs) when the filter has either an AND or OR filter inside another AND or OR filter. The lower filter has its AND or OR clause duplicated in the resultant WHERE clause on the application server.

This can also be manifested by CA WCC monitoring views that have complex filters applied to them.

AS_SERVER CRASH ON JOB DELETE

Problem Number = 4187

The application server stopped responding when a job with more than 40 jobs with external dependencies in the condition statement were inserted or deleted.

JOB WITH A LEADING SPACE IN COMMAND FAILS WITH EXITCODE 3005

Problem Number = 4188

A command job with a leading space in the command attribute failed to run with an exit code of 3005.

INTRODUCE AUTOBCPORADP.PL

Problem Number = 4175

The autobcpORA.pl script slowed down on huge database as it dumped data from the source database to the target database table-by-table and consumed more time; it never used the data-pump capability available in Oracle 10g and later versions.

JOB HAVING LOOK-BACK CONDITION DOES NOT START

Problem Number = 4190

A successor job having a look-back dependency was not triggered when its predecessor was overridden before the start of the successor.

QUICK RUNNING BOXES STUCK IN RUNNING

Problem Number = 4194

The scheduler scheduled and ran quick running boxes that complete before the STARTJOB event is processed.

EP LOG TRUNCATED

Problem Number = 4167

The log files that the scheduler generates were truncated.

PERFORMANCE PROBLEM - GLOBAL VARIABLE

Problem Number = 4181

Global variable setting did not change the value of a global variable from the current value to a new value until all the jobs depending on the global variable were evaluated based on the current value. This caused delay in evaluating the jobs based on the new value.

JOB DEFINITION IN AN UNKNOWN STATE

Problem Number = 4192

The job definition changed to an unknown state when you tried to update a job with date_conditions such that it could not calculate the next start time of the job. When you tried to update a job, messages similar to the following were displayed:

```
CAUAJM_E_19028 Job <newjob1> could not be modified properly due to a database failure. The job
CAUAJM_E_10302 Database Change WAS NOT successful.
```

GETJOB SOAP FAILURE

Problem Number = 4163

If the start_mins attribute value has a space (for example, "start_mins: 00, 15"), the GetJob Soap web service failed with the JobManagerException error.

FTP JOB - QUOTES PROBLEM**Problem Number = 4195**

An FTP job failed when the local_file_name or remote_file_name attribute contained a slash and the scheduler tried to add an extra slash as shown below:

"\"C:\\test\\out\\NumJobsRun.csv\""" gets converted to \"C:\\test\\out\\NumJobsRun.csv\". It should be C:\\test\\out\\NumJobsRun.csv

SCHED DOES NOT PROCESS BOXES OF JOBS W/VIRT MACHS AND RESRC**Problem Number = 4162**

The scheduler stopped processing boxes when multiple boxes contained jobs that run on virtual machines and have resource conditions.

JIL DOES NOT ALLOW FOR COMMAND**Problem Number = 4151**

JIL did not allow # as the first character in the attribute value of a command. JIL treated the complete command as a comment if the first character in an attribute value is #. The command should not treat # as a comment.

AUTOCAL_ASC -I CYCLE WILL FAIL**Problem Number = 4160**

The autocal_asc command with the -I cycle failed randomly once in thousand attempts displaying the following errors and so the cycle insertion failed:

```
CAUAJM_I_50439 Insert Successful for Cycle: QAcycle_cal2
```

```
CAUAJM_I_50440 Insert Failed for Cycle: QAcycle_cal2
```

```
CAUAJM_E_50313 Invalid Cycle Period. Periods must be between 1 and 366 days
```

```
Return Code=1
```

WEBSERVICES AUTHENTICATION EXCEPTION WITH ORACLE 12C**Problem Number = 4145**

Webservices fail with ORA-28040: No matching authentication protocol when running with Oracle 12C.

UNABLE TO DELETE JOB**Problem Number = 4156**

If you set the RESTRICT_DELETE_DEPENDENT_JOB and RESTRICT_DELETE_JOB variables, a predecessor job could not be deleted even after the dependent jobs were deleted.

UPDATE_MACHINE LOOKS EXEC SUPERUSER UNDER NATIVE SECURITY**Problem Number = 4149**

If CA Workload Automation AE is running in native security mode, you could not update a machine using the `update_machine` subcommand if you are not a EXEC Superuser. However, you could insert or delete a machine.

CANNOT KILL LONG RUNNING JOB

Problem Number = 4185

You could not kill a long running job using the KILLJOB event.

JOBS W/R11 MACH CHOOSER SCRIPT GET STUCK IN RE RANDOMLY

Problem Number = 4147

Job were stuck in the RESTART state randomly instead of going to the FAILURE state after many tries. As a result, the parent box was stuck in the RUNNING state forever.

DBSPACE FAILS LOGON WHEN PASSWORD HAS SPECIAL CHARS. SYB

Problem Number = 4132

The dbspace command failed while logging into the database with an error message similar to the following:

```
CAUAJM_I_60020 dbspace: Running dbspace at : Thu Feb 13 11:03:44 2014
```

```
sh: sys: No such file or directory
```

```
CAUAJM_E_10046 Could not open file: /opt/CA/WorkloadAutomationAE/autouser.xxx/out/  
spaceused.txt
```

JOB OVERRIDE - UPDATE NOT SYNC IN DUAL SERVER MODE

Problem Number = 4178

In dual event server mode, when you ran an overridden job, the override information was removed from the primary database but not from the secondary database.

PS JOB ARGS ATTRIBUTE IN AFM WRONG

Problem Number = 4155

The automated framework messages sent from CA Workload Automation AE Release 11.3.5 INCR2 for PeopleSoft job with the `ps_args` attribute contained wrong format.

BRASILIA AND BUENOSAIRES TIMEZONE CFG WRONG

Problem Number = 4124

The DST rules for the BRASILIA and BUENOSAIRES time zones were incorrect and so the jobs were scheduled incorrectly.

AS CRASHES IN JIL SYNTAX

Problem Number = 4215

The application server stopped responding during the syntax check of jobs (other than box and command in a delete operation).

LEADING SPACE IN STD_OUT_FILE FAILS JOB

Problem Number = 4216

When you defined a job with leading spaces in std_out_file and std_err_file, the job failed with an exit code of 2005.

JOB LOGS NOT RETRIEVED

Problem Number = 4217

When you ran a job with machine length more than 64 characters, the associated job log could not be retrieved.

Fixed Issues - 11.3.6 INCR1 CUM1

This article details the issues fixed in CA Workload Automation AE Release 11.3.6 INCR1 CUM1.

TIP

CA Workload Automation AE Release 11.3.6 INCR1 CUM1 also includes the issues fixed in each of the prior releases of 11.3.6.

Contents

AUTOREP DISPLAYS BACKSLASH IN COMMAND

Problem Number=3771

When you issue the autorep command with the -q option, the command attribute reported backslashes although the command value is enclosed in double quotes.

INACTIVE ALLOWED OVER ON_ICE JOBS

Problem Number=4252

You could manually set the status of a job in the ON_ICE state to INACTIVE. This caused intermittent job scheduling issues.

ON_HOLD CREATES BAD BOX RUN

Problem Number=4196

Jobs in a box did not execute properly when they were put in an OFF_HOLD state.

A detailed report shows that a job ran under the previous run number with an incremented ntry value.

NOTE

This issue was fixed in Release 11.3.6 INCR1.

B. A box with n_retrys an OFF_HOLD state was resuming ntry = 0 instead of incrementing by 1 when it was put in ON_HOLD state.

NOTE

This issue is fixed in Release 11.3.6 INCR1 CUM1.

CALENDAR CHANGE DOES NOT REFLECT WCC RESULTS

Problem Number=4199

A change to a calendar did not get reflected to the job's next_start attribute in CA WCC. CA WCC must reflect the new next_start attribute in following cases:

A. Calendar is updated to include new dates.

NOTE

This issue is fixed in Release 11.3.6 INCR1.

B. Calendar is updated to remove old dates affecting the job's next_start.

NOTE

This issue is fixed in Release 11.3.6 INCR1.

C. Calendar is deleted from the database.

Note: This issue is fixed in Release 11.3.6 INCR1 CUM1.

SCHEDULER DEADLOCK CONTROL MGRADDR AFM

Problem Number=4226

The scheduler stopped processing events due to a deadlock in the agent communication module. This resulted in the scheduler not processing any events causing a production down situation.

UNIQUE CONSTRAINT ERROR ON DBMAINT OR DBSTATISTICS

Problem Number=4227

The DBMaint or DBStatistics command failed with a unique constraint violated error on the XPKUJO_AVG_JOB_RUNS key.

SCHEDULER HANG WHILE COMMUNICATING WITH R11 AGENTS

Problem Number=4228

The scheduler stopped responding intermittently while communicating with the r11 agents.

CAUAJM_E_50033 ERROR INITIAL - APPSERVER PERFORMANCE

Problem Number=4234

The clients utilities (CA WCC, sendevent, jil, and so on) occasionally received an error while trying to connect to a busy application server. This happened because the application server checks SSA parameters in an inefficient manner. This fix improved the application server performance and response rates.

EXCESSIVE EXTERNAL_DEPENDENCY EVENTS

Problem Number=4235

After migrating jobs from 4.5 to Release 11.3.6, when a job with a dependency was deleted, the following EXTERNAL_DEPENDENCY events were continuously displayed in the scheduler log, although the job had a dependency on a local job.

```
CAUAJM_I_40245 EVENT: EXTERNAL_DEPENDENCY JOB: test_job1 **Deleted**
```

AFTER MIGRATION FROM 4.5 TO 11.3 VIRTUAL

Problem Number=4242

After migrating data from 4.5 to r11.3, the default factor for a virtual machine which is 1.0 in 4.5 is set to 0 in r11.3. Jobs submitted to the virtual machine did not use load balancing correctly as all the available load is multiplied by a factor of 0 leading to choosing a random machine in the list to run the job every time, instead of choosing a machine based on load.

WCC Release Notes

This release notes article explains the key features and details for the following WCC versions, releases, and service packs:

CA WCC Release - 11.4 SP7

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP7 changes to existing features.

Installation and Upgrade Prerequisite - Modify the CA EEM Server Certificate

From Release 11.4 SP7, CA WCC, requires root certificates with a key length of 2048 or more. By default, CA EEM generates root certificates with a key length of 1024. CA Workload Automation AE uses 1024 key length certificates.

NOTE

For more information about the changes to CA EEM, see the [CA EEM - Installation and Upgrade Prerequisite](#) section.

You can generate root certificates with various key lengths (2048 or more) and use them on a EEM Server.

Also, in some instances, previous and current releases of CA WCC using root certificates with key lengths of 1024 and 2048 may point to the same CA EEM server. Before you install or upgrade WCC, you must modify the CA EEM server certificate.

For CA EEM to work with key lengths of 1024 and 2048 (or more) follow these steps:

1. Stop the AE and WCC servers.
2. Upgrade CA EEM.
3. Configure the CA EEM server to work with both 1024 and 2048 key length certificates.

NOTE

For more information about different key lengths, see the [Configure CA EEM for Root Certificates Generated with Different Key Lengths](#) section.

4. Restart the AE and WCC servers.
5. CA WCC is now ready for installation or upgrade.

Changes and Enhancements To Existing Features - 11.4 SP7

This article details the changes to existing features in CA Workload Automation WCC Release 11.4 SP7.

Installation and Upgrade Prerequisite - Modify the CA EEM Server Certificate

From Release 11.4 SP7, CA WCC, requires root certificates with a key length of 2048 or more. By default, CA EEM generates root certificates with a key length of 1024. CA Workload Automation AE uses 1024 key length certificates.

NOTE

For more information about the changes to CA EEM, see the [CA EEM - Installation and Upgrade Prerequisite](#) section.

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NOTE

For more information about different key lengths, [Configure CA EEM for Root Certificates Generated with Different Key Lengths](#)

4. Restart the AE and WCC servers.
5. CA WCC is now ready for installation or upgrade.

Bouncy Castle (BC FIPS) replaces RSA BSAFE Crypto J

Bouncy Castle replaces RSA BSAFE Crypto J in this release, to enable a secure environment to run CA Workload Automation AE and WCC.

Compatibility Matrix

The CA Workload Control Center Certification Information (support compatibility matrix) is available.

You can also view the compatibility matrix in the documentation at [CA WCC Compatibility Matrix](#).

Discontinued Support for CA Workload Automation AE Release 11.3 Server

Support for CA Workload Automation AE Release 11.3 Server is discontinued from this release.

CA Remote Engineer is Removed from CA Workload Automation WCC

Beginning with Release 11.4 SP7, CA Remote Engineer is removed from CA Workload Automation WCC.

CA EEM - Installation and Upgrade Prerequisite

You can upgrade CA Workload Automation AE and WCC, after implementing the required changes to CA EEM.

If the EEM servers are configured in a failover or multi-write group, then the following process can be adopted to minimize the impact to CA Workload Automation AE and WCC. The EEM Certificate change must be applied upon all servers within the EEM failover group.

Before you begin:

- If a single EEM system is being used, stop all the applications that are using EEM.
- Ensure that CA Workload Automation AE and WCC are connected to your EEM systems.

We recommend that no changes are made to users and policies, within any of the EEM systems, until all nodes in the failover or multi-write group have been updated and synchronization has been verified.

Follow these steps:

1. Apply the EEM Certificate change to the Primary Server in the EEM failover or multi-write configuration.

NOTE

You can also refer the following topics, for more information on the process:

[Configure CA EEM for Root Certificates Generated with Different Key Lengths](#)

[How to Generate the Certificates](#)

1.
 - a. Backup the existing rootcert.cer and rootcert.key files to a folder on the EEM server, for example, EEMprimary_1024_certs.
 - b. Copy the folder created in the step (1a) to your local computer.
 - c. Stop applications using EEM (if not using an EEM failover or multi-write group).
 - d. Generate the new CA EEM Server Certificate with key length 2048.
 - e. Again add the previously backed up CA EEM Server Certificate with key length 1024.

The following changes are observed in an EEM failover or multi-write environment:

 - AE and WCC connections to EEM node 1 automatically switch to another node in the failover group. This is caused by the restart of igateway during the certificate creation process.
 - EEM synchronization stops. For example, any change that is now made within the EEM node 1 UI does not get reflected within the EEM node 2 or EEM node 3 systems.
2. Apply the EEM Certificate change to all nodes in the EEM failover or multi-write group. On each server, complete the following actions:
 - a. Backup the existing rootcert.cer and rootcert.key files to a folder on the EEM server, for example, <server_name>_1024_certs.
 - b. Copy the folder created in the step (2a) to your local computer.
 - c. Generate the new CA EEM Server Certificate with key length 2048.
 - d. Again add the previously backed up CA EEM Server Certificate with key length 1024.
 - e. Run the [EEM Cluster Setup Utility to perform a 'sync' \(delta only\)](#) to resume synchronization with the Primary EEM Server.
3. Verify that full EEM synchronization has resumed.
 Apply a change within each EEM node and verify that the change is reflected upon the other nodes.
 For example:
 - a. Add or amend a description within a policy on EEM node 1.
 - Check that the change is reflected upon EEM nodes 2 and 3.
 - b. Add or amend a description within a policy on EEM node 2.
 - Check that the change is reflected on EEM nodes 1 and 3.
 - c. Add or amend a description within a policy on EEM node 3.
 - Check that the change is reflected upon EEM nodes 1 and 2.
4. Upgrade CA Workload Control Center to Release 11.4 SP7.
5. Upgrade CA Workload Automation AE (AutoSys Edition) to Release 11.3.6 SP8.

Fixed Issues - 11.4 SP7

This article details the issues fixed in CA Workload Automation WCC Release 11.4 SP7.

CLOSE/ACKNOWLEDGE ALERT GREYS OUT ALERT ACTIONS**Problem Number: 418**

Alerts in the dashboard Alerts widget are not getting closed and the other options like acknowledge are grayed out after trying to close alerts.

WCC REPORTING SHOWS DELETED INSTANCE

Problem Number: 419

When generating reports for job runs, the Reporting tab displays deleted AE instances.

AUTOSYS_SECURE FROM ECLI FAILS WHEN PASSWORD HAS ***Problem Number: 422**

In WCC ECLI, users can change their password using the autosys_secure command (if they know the existing password). Because WCC interchanges * with %, users were not able to use * as part of their passwords as WCC was interchanging * in their passwords with %.

WCC "STATUS SUMMARY" BOX NEVER CHANGES IN MONITORING**Problem Number: 425**

In the status summary of a Monitoring View, the status color for a view remains unchanged even after the alert is generated in WCC.

WCC R11.4.SP6 - ERROR IN MONITORING, VIEW TREE REFRESH**Problem Number: 426**

Refresh is not working in the Monitoring View/Server Tree section. Status of jobs and time of refresh are not getting updated on refresh when the box jobs, under View/Server tree, are in an expanded state.

QUICK EDIT SCREEN COMES UP BLANK AFTER UPGRADE TO WCC-11.4.6**Problem Number: 428**

When opening a job with many successors in Quick Edit, it takes a longer time to load and sometimes results in a blank screen.

WCC IMPORT FUNCTION NOT WORKING PROPERLY**Problem Number: 431**

When users try to import jobs in WCC through ECLI, then it fails to import jobs, without any message.

WCC SHOWS AN ECMASCRIPT6 ERROR MESSAGE**Problem Number: 432**

When using Internet Explorer 11 or earlier to open a job in Monitoring or Quick View, an error message displays.

REPORTS NOT GENERATED CORRECTLY IN REPORTING TAB**Problem Number: 434**

When generated report data contains more than one page, the last row of a page is sometimes duplicated in the subsequent page.

NOT ABLE TO COMMIT JOB OVERRIDE**Problem Number: 439**

After opening create override page for a job that is defined with 'Alarm Type' property in the Notification section, none of the buttons in WCC Quick Edit tab respond.

WCC - ALARMS DOESN'T APPEARS IN DASHBOARD, AND ALERT'S VIEW**Problem Number: 440**

When the alert policy is created with the % symbol in the job name filter, Alerts do not appear in the dashboard and monitoring.

REPORTING CLEANUP FAILS

Problem Number: 441

When there is a lot of data in reporting tables, reporting tables cleanup fails. Reporting cleanup gets stuck on a delete query which takes a lot of time.

WCC AGENT INVENTORY TAB GREYED OUT

Problem Number: 442

When user logs in with single sign-on, the agent inventory tab is greyed out and user cannot perform any action.

DASHBOARD: ASCENDING AZ ORDER FOR STATUSES IN JOBS WIDGET

Problem Number: 443

User is not able to sort statuses in the Jobs Widget table in the Monitoring tab and Dashboard.

QUICK VIEW SHOWS 404 ERROR

Problem Number: 444

Quick View page displays 404 error and job details are not displayed.

UNABLE TO CREATE OR UPDATE SECURE COPY JOBS

Problem Number: 445

User is not able to create or update secure copy (SCP) jobs from quick edit when AE server 11.3.6 SP6 or earlier is used.

FULL COLLECTION HAPPENS AFTER 11.4SP6 UPGRADE

Problem Number: 446

When WCC is upgraded to 11.4 SP6 and the collector background process starts, the very first jobs-collection request does a full jobs collection, instead of delta updates from the last successful time.

VALIDATE ACTION CLEARS JOBS COLLECTION

Problem Number: 447

When 'Validate' action is performed against an upgraded AE server, the version gets refreshed with the new value but internally the entire jobs collection of the respective AE server gets cleared and the full collection is requested subsequently from the AE server.

JOBS COLLECTION GET TIMEDOUT DURING FULL COLLECTION

Problem Number: 448

When full jobs collection is requested from an AE server, the request can get timed out at WCC, if the AE server takes more than the default 30 minutes.

TOMCAT32 PROCESS HEAP USAGE INCREASES OVER TIME

Problem Number: 449

The CA-wcc-services (for example, Tomcat32bit) process heap memory usage keeps growing in size over a period of time and reaches the limit which causes the WCC application to crash. Usually requires WCC services to be restarted for WCC GUI to function normal.

NOT ABLE TO ADD EXTERNAL APPLICATION

Problem Number: 450

When adding an external application in the "Add external application dialog", if all values are not filled, on clicking ok, nothing happens.

NOT ABLE TO FILTER ALERTS BASED ON DATE

Problem Number: 451

When trying to filter alerts using "Start Time" or "End Time", nothing is displayed in the Dashboard Alerts widget.

ECMA SCRIPT ERROR WHEN VIEWING TOOLTIP OF A NODE IN IE11

Problem Number: 452

When viewing the tooltip of a node in the flow section of the Monitoring tab in Internet Explorer version 11, Ecma script error keeps coming, eventually fills the screen, and the user has to repeatedly refresh the page.

REPORTING CLEANUP NOT STARTING ON TIME

Problem Number: 453

If reporting cleanup is set to start once a day, it is actually starting after a few days. This is causing reporting tables to grow in size.

CA WCC Release - 11.4 SP6

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP6 new features and changes to existing features.

New Features

This article describes the new features added to CA WCC Release 11.4 SP6.

Agent Inventory

The Agent Inventory display lets you see and export information about each separately installable component (system agent, plug-ins, and advanced integrations). Agent information is retrieved periodically from CA Workload Automation AE and displayed in Agent Inventory.

The Agent Inventory tab is available to CA WCC users with permission for Agent Inventory functions.

You must specify the Web Services URL in the Application Server category under the CA Workload Automation AE Server Properties for use by Agent Inventory to collect data. Agents are displayed only for CA Workload Automation AE instances that have the CA Workload Automation AE Web Services component enabled.

Action Status in Agent Inventory

Action Status is not updated for agents with Java Runtime Environment (JRE) version 1.6. For more information, see the [Troubleshooting](#) section.

Sending Email Notifications Using Template

You can now send an email notification using the email notification template when a job fails, terminates, completes successfully, or when an alarm is raised.

Machine Method to Achieve Load Balancing With Virtual Machines

You can use the Machine Method field in the Virtual Machine properties section in Quick Edit to specify any one of the following [Specify the Method to Achieve Load Balancing](#):

- vmstat
- rstatd
- job_load
- cpu_mon
- roundrobin
- roundrobin_jobload

Distinguish Alarms Based on the Failure and Terminated Job Status

The following changes are made to capture alarms based on the FAILURE and TERMINATED job statuses:

- – The new JOBTERMINATED alarm can be specified in the Alert policy to capture an alarm when a job ends with a terminated status.
- In previous releases, the JOBFAILURE alarm was specified in the Alert policy to capture an alarm when a job ended with failure or terminated status. The JOBFAILURE alarm now captures only when the job ends with a failure status.

NOTE

If you want to capture both job failure and job terminated alarms, you must now specify both JOBFAILURE and JOBTERMINATED alarms in the Alert policy.

If you created a Alert policy for JOBFAILURE alarm in previous releases, and upgrade to the current release, JOBTERMINATED alarm will also be added to the Alert policy.

Overwrite Option for Calendars in Export/Import

You can now select the Overwrite option when importing a calendar from a text file. This option helps you to overwrite an existing calendar when you import a calendar.

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Changes and Enhancements To Existing Features - 11.4 SP6

This article details the changes and enhancements to existing features in CA WCC Release 11.4 SP6.

CA Remote Engineer

CA Workload Automation WCC supports CA Remote Engineer r4.0. The CA Remote Engineer tool collects configuration and diagnostic data that helps you to troubleshoot problems with the system or software. In this release, you can integrate CA Remote Engineer with CA Telemetry Service and this integration helps to capture customer environment and topology information.

NOTE

For more information about CA Remote Engineer and CA Telemetry Service integration, see the [CA Remote Engineer](#) documentation.

Microsoft SQL Server Encryption Support

WCC can be installed with MS SQL Server 2016 supporting *encryption*.

NOTE

For more information about how to use the MS SQL Server encryption, see the section [Use Microsoft SQL Server Database with Encryption](#).

LDAP Authentication for Informatica Jobs

The Informatica domain property in the job definition authenticates the user against the specified LDAP security domain that is configured with the Informatica Server.

Updated Limits for the Command Property

The limits for the Command property is now increased from 512 to 1000 characters. This limit change is applicable if the CA Workload Automation AE server that is configured in CA WCC is at Release 11.3.6 SP7 and later.

WCC Collectors Performance

The following improvements have been made to the WCC Collectors:

- The WCC collectors processing logic is done by 64-bit Tomcat.
- You can now seamlessly run more jobs and servers.

Application Editor - Changes and Enhancements

The following are the changes and enhancements in the **Application Editor** application:

New Look and Feel for Objects in the Application Editor Graph

The Application Editor graph now has a new look and feel. Box jobs are now displayed in the graph similar to the Monitoring application.

NOTE

After migration from ILOG to yFiles in Release 11.4 SP6, job flows saved in releases before 11.4 SP5 are not displayed.

Support Enter Key to Edit Objects

You can now select an object and press the Enter key in your keyboard to edit the object.

- You can change the name of a global variable displayed in the Graph and also edit the dependency link between objects to modify the condition that must be met before the next step in the job flow is performed.
- You can modify date and time conditions, change dependencies or triggers, manage permissions, or change notification settings for a specific job.

Simplified Creation of Links Between Objects

You can now directly create links between objects. Hover the cursor on the job object or the link object to view the corresponding tooltip.

Download Job Flow as PDF

You can download the job flow as a PDF document or PNG image and save it to a file.

Expand or Collapse Box Jobs

Box jobs in the graph can be expanded or collapsed either by double-click or using the flow toolbar buttons.

Enhanced Job Search

The flow toolbar now lets you search the job flow by job name to locate specific jobs.

The application now prompts you with a list of jobs as soon as you start entering a search value.

Delete Global Variable, Logical Operator, and Job Objects Using the Delete Key

You can use the keyboard to delete one or more jobs, logical operators, global variable, or dependencies from the job flow in the Graph.

CA WCC Release - 11.4 SP5

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP5 new features, changes to existing features, and fixed issues.

New Features

Informatica Job Type

Informatica jobs let you start predefined workflows on the Informatica PowerCenter. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

NOTE

To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.

SQL Server Agent Job Type

SQL Server Agent jobs let you execute predefined jobs in the SQL Server Agent to perform administrative tasks.

NOTE

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

Database Trigger Job Type

PostgreSQL and Teradata databases are supported.

Support SQL Server Always On

CA Workload Automation WCC now supports Microsoft SQL Server *Always On* feature.

CA WCC Upgrade From Older Versions

CA WCC supports upgrade from 11.3.5, 11.3.6, 11.4, 11.4 SP1, 11.4 SP2, 11.4 SP3, or 11.4 SP4 to current version 11.4 SP5.

New Features - 11.4 SP5

This article describes new features added to CA WCC in Release 11.4 SP5.

Informatica Job Type

Informatica jobs let you start predefined workflows on the Informatica PowerCenter. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

NOTE

To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.

SQL Server Agent Job Type

SQL Server Agent jobs let you execute predefined jobs in the SQL Server Agent to perform administrative tasks.

NOTE

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

Database Trigger Job Type

PostgreSQL and Teradata databases are supported.

Support SQL Server Always On

CA Workload Automation WCC now supports Microsoft SQL Server *Always On* feature.

CA WCC Upgrade From Older Versions

CA WCC supports upgrade from 11.3.5, 11.3.6, 11.4, 11.4 SP1, 11.4 SP2, 11.4 SP3, or 11.4 SP4 to current version 11.4 SP5.

Changes and Enhancements To Existing Features - 11.4 SP5

This article details the changes and enhancements to existing features in CA WCC Release 11.4 SP5.

New Fields for JMS Publish and JMS Subscribe Jobs

The following fields are added for the JMS Publish and JMS Subscribe jobs:

- `correlation_id` — Specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue.
- (JMS Publish jobs only) `Reply to`—Specifies the JNDI name of the reply to a queue.
- `Destination user name`—Specifies the user name that is required to connect to the topic or queue in a JMS job.

New Fields for FTP, SCP, and SFTP Jobs

The following fields are added for the FTP, SCP, and SFTP jobs:

- FTP
 - Run as user - Specifies the user ID that runs the job on behalf of the agent user.
- SCP
 - Transfer type - Specifies the type of data involved in an SCP transfer: ASCII, Binary, or /tmp.
 - Run as user - Specifies the user ID that runs the job on behalf of the agent user.
- SFTP
 - Transfer type - Specifies the type of data involved in an SFTP transfer
 - File operation - Specifies if you want to rename, delete or not delete files.
 - Do not delete
 - Delete source directory
 - Delete source file(s)
 - Rename source file
 - Run as user - Specifies the user ID that runs the job on behalf of the agent user.

Global Variables

Global variables substitution is now supported by more properties, simplifying job administration. **Note:** For more information, see the section [Global Variables vAE_12](#).

Fixed Issues - 11.4 SP5

This article details the issues fixed in CA Workload Automation WCC Release 11.4 SP5.

SCHEDULER LOG NOT LOADING IN INTERNET EXPLORER

JavaScript exception resulted in the incomplete loading of the scheduler log in Internet Explorer.

TREE VIEW CONTENT NOT DISPLAYED IF MONITORING COMPONENT IS THE FIRST PAGE

When the Dashboard access was disabled for the user (and Monitoring component was the first page) then the content of the tree view was not displayed until the user clicked on a different tab. The job tree did not initialize.

CROSS-INSTANCE EXIT CODE CONDITIONS SHOW ERROR MESSAGE IN QUICK VIEW

The Quick View displayed error message when loading a job with cross-instance exit code condition defined, due to a bug in the code when only status conditions were expected.

JOB WERE NOT FILTERED IN MONITORING VIEWS BY AUTOSYS INSTANCE

Monitoring view configured with filter (or filters) for some AutoSys instance also displayed jobs from different AutoSys instance, if several AutoSys instances were set up in the configuration.

SEARCHING FOR CALENDARS IN QUICK EDIT DISPLAYED ERROR MESSAGE

When only standard or extended calendars were accessible by the user, searching for calendars in Quick Edit displayed error message instead of the calendar list.

CREDENTIALS ARE NOT VALID AFTER UPDATING THEM FROM COMMAND LINE

After updating the credentials from the command line, they did not validate anymore due to the double encryption of the password.

MULTIPLE SEND EVENTS TO JOB FROM QUICK VIEW

Multiple send events to job were sent if the user clicked several times (for example, double clicks) on the “Yes” button in the Quick View confirmation dialog.

CONDITION STATUS NOT DISPLAYED IN QUICK VIEW

When a job name was used more than once, the condition status displayed as *UNDEFINED* in the list of job conditions in Quick View.

JOB NEXT START TIME WAS NOT UPDATED

When assigned calendar was modified Job next start time was not updated in the Monitoring. This was because detection of the support of this feature on AutoSys side did not work with the latest AutoSys releases.

LENGTHY JOB NAMES WERE WRAPPED IN QUICK EDIT JOB LIST

The long job names in QE job list were wrapped on multiple lines. Browsers usually wrapped long text on underscore (“_”) or dash (“-”) boundaries to fit content into the browser window.

FETCHING LOGS FROM Z/OS AGENTS FAILED

Attempts to fetch logs from z/OS agents in Quick View always failed with an error message due to a bug in the code.

MONITORING COLLECTORS CRASHED DURING STARTUP

When collection (or cleaner) run interval was set to high values, the monitoring collectors and alert cleaner crashed during startup.

AUTO_DELETE JOB ATTRIBUTE DID NOT DISPLAY IN QUICK VIEW

When the value of the auto_delete job attribute was set to zero (0), the attribute did not display in Quick View.

REPORTING COLLECTION FAILED WITH “TOO MANY CURSORS OPEN” ERROR

When Oracle platform was used as reporting database, the reporting collections failed after some time with “too many cursors open” error. This was because database objects were not properly closed in the code when some exception was thrown.

REPORTING CLEANUP DID NOT WORK

When huge amount of reporting data was deleted after configured retention period, the cleanup failed and data piled up in the database.

WIDGETS DISAPPEARED IN MONITORING OR ON DASHBOARD

When several users use the same account and they maximize different widgets in concurrent sessions, the widgets disappeared after login from the Dashboard or from Monitoring page.

ON/OFF NO EXEC OPTIONS DID NOT WORK FROM JOB FLOW

When wrong values handled JavaScript code, nothing happened when “ON/OFF NO EXEC” options were selected from the job flow context menu in Monitoring.

WSDL WIZARD CRASHED OR PROVIDED WRONG PARAMETERS

Wrong formatting of the parameter names caused the WSDL wizard to crash with some web services and most of the time did not provide correct parameter names.

CANNOT DELETE VIEW WITH “,” OR “:” IN ITS NAME

Due to missing quotes around the view name, the monitoring views with comma (“,”) or colon (“:”) characters in their names cannot be deleted (without manual intervention). Also, after exporting such views they cannot be imported.

QUICK VIEW CRASHES IF SAP ATTRIBUTE CONTAINS “{“ OR “}”

Wrong parsing of SAP job attribute containing “{“ or “}” in Quick View displayed error message.

XSS VULNERABILITY IN ECLI

With missing input validation in the code, the cross-site scripting vulnerability exists in Enterprise Command Line.

MONITORING COLLECTORS CAN BE DISABLED

Monitoring collectors could not be disabled (if not needed) – only run time interval could be set to some high value (but even this did not stop the first collection).

INITIAL STATE OF JOB FLOW CAN BE SELECTED

Job flows in Quick View were always displayed as expanded after new login.

CA WCC Release - 11.4 SP4

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP4 new features, changes to existing features, and fixed issues.

New Features - 11.4 SP4

This article describes new features added to CA WCC in Release 11.4 SP4.

Support RHEL 7 and SUSE 12

CA Workload Automation AE and CA WCC support the following Linux versions:

- Red Hat Enterprise Linux 7
- SuSE Linux Enterprise Server 12

New as-sendevent Resource Class

The [as-sendevent resource class](#) and its default CA EEM policy are added to control access to execute the sendevent command: CA WCC functionality allows for hiding the send event actions on the UI that is based on the CA Workload Automation AE Resource Class as-sendevent.

Multiple Dashboard Tab Creation

A user can create and remove Dashboard tabs, alter a Dashboard tab title, and select a choice of a primary Dashboard.

NOTE

Note: By using the EEM *ConfigurationControl* resource class policy, the Security Administrator can define the user roles that are allowed to create multiple dashboards.

Creation of Multiple Jobs Widget Instances

A user can create various jobs widgets in the dashboard that allow the user to simultaneously monitor jobs from different CA Workload Automation AE instances, monitoring views, and grid filters.

CA WCC Monitoring Enhancement of Jobs Filter Preferences

A Monitoring preference added into Configuration allows an empty Jobs Widget filter in the Servers section.

By default it is checked. If the checkbox is unchecked it will force the users to use the Jobs filter in all Jobs Widgets for Servers, otherwise no data will be shown.

NOTE

Add the new preference to the CA WCC Configuration Application under the Preferences Section in Monitoring:

Allow empty Jobs Widget filter in Servers section

Specifies whether filter usage is required in all Jobs Widgets for Server.

Default: Selected

New Attributes for Oracle E-Business Suite Request Set and Oracle E-Business Suite Single Request Jobs

The following attributes are added for the Oracle E-Business Suite Request Set and Oracle E-Business Suite Single Request jobs:

- [oracle_oprunit](#)—Specifies the operating unit name for a single request or request set.
- [oracle_template_name](#)—Specifies the template name for a single request or request set.

CA WCC Enhanced Security Using New Cryptography

To enhance security of CA WCC, the CA WCC server is now using a X-XSS-Protection header with value "1;mode=block" for HTTP responses. In addition, CA WCC is now using Elliptic-curve cryptography for https connections as a preferred option. For normal cryptography operation, DH group of 2048-bit is now being used.

CA WCC Login and Logout Audit

CA WCC administrators have a log file containing user login and logout events.

NOTE

The log file is located in the \$CA_WCC_INSTALL_LOCATION/log/audit

CA WCC Upgrade From Older Versions

CA WCC supports upgrade from 11.3.5, 11.3.6, 11.4, 11.4 SP1, 11.4 SP2, or 11.4 SP3 to current version 11.4 SP4.

CA WCC Support of CA Workload Automation AE Version 11.3.6 SP5

CA Workload Automation AE 11.3.6 SP5 is supported by CA WCC 11.4 SP4.

Job Search By Command Attribute in Quick Edit

A new command field is provided in Quick Edit to enhance search queries. The command jobs are filtered by its command attribute.

Job Search By Multiple Job Names in Quick View

Enhancement of the Quick View job name search field that accepts multiple job names. The job names are separated by a comma, new line separator, or a space. Each job name supports wildcards.

NOTE

New line separators in job name are supported by Chrome, Firefox, and Safari.

Changes To Existing Features - 11.4 SP4

This article details the changes to existing features in CA WCC Release 11.4 SP4.

Enhanced Scheduler Log Functionality

An additional search mode has been created to find a pattern and display a subsequent number of lines as specified. Enhanced navigation allows you to move to a next or previous option and highlights the matching pattern.

Interval Specification for File Trigger Job Type

For File Trigger job type it is possible to specify the watch_no_change attribute in seconds, minutes, hours, or days.

NOTE

Applies only for AE 11.3.6 SP5 and later.

Support Global Resources for Box Job

You can now specify global resources on a box job. However, the resource specified for a box applies only to the box and not to the jobs within the box.

NOTE

Global resource specification on a box job is only available for AE 11.3.6 SP5 and later.

Internal Database Upgrade

Derby database upgraded to version 10.12.1.1.

Monitoring Expand/Collapse Flows Synced

Expand/Collapse All functionality, in a particular view, is supported through the flow toolbar buttons and is synced to the job tree when these flow actions are triggered.

Fixed Issues - 11.4 SP4

DATE PICKER DOES NOT EXPAND

When using Internet Explorer in Quick Edit, expanded information of the date picker fails.

CROSS-INSTANCE DEPENDENCIES MISSING

Flows and dependent jobs have missing cross-instance dependencies.

NO VERIFICATION OF BROKEN HOLIDAY CALENDAR REFERENCES IN EXTENDED CALENDAR

When an extended calendar is created with a non-existing holiday calendar from CA Workload Automation AE, it displays a warning saying "CAUAJM_W_50413 *** WARNING - The Holiday calendar: testhol does not exist.", However, the calendar is created with the non-existing holiday calendar. Through CA WCC, the extended calendar is created with a non-existing holiday calendar and does not display any warning.

In addition, a holiday calendar that is used by an extended calendar can be deleted with no warning messages displayed, both from CA WCC and CA Workload Automation AE.

CANNOT EXPORT OR PRINT AppEditor FLOW

When CA WCC is running on Windows, you are unable to export or print a flow in AppEditor.

UPGRADE TO CA WCCSP3/SP4 COULD FAIL DUE TO OPENED FILES

There is a workaround for Release 11.4 SP3 that allows you to upgrade from CA WCC SP3 to CA WCC SP4. This will avoid failures to open files.

CA-wcc-services ARE CHECKING ALL DELETED JOBS DURING STARTUP

This check was unnecessary and CA WCC 11.4 SP4 no longer performs this check. This enhancement could speed up the start up of CA-wcc-services service.

DOES NOT LOAD JOBS IN MONITORING JOBS WIDGET WHEN NO FILTER IS SET

This feature should be configured in Monitoring preferences and is turned off by default.

CORRECT BRAZILIAN TIMEZONE SUPPORT

Problem Number: 404

Several issues were fixed in the Quick Edit component when working with calendars whereas the CA WCC was installed on a system operating in Brazilian time zone.

MONITORING DB UPDATES FAILED

Due to cached entities from JPA entity manager, sometimes the instances are out of sync which lead to insertion of already inserted data.

ECLI MIGRATION TOOL CANNOT BE LAUNCHED

Migration tool for ECLI commands refused to start.

GLOBAL VARIABLES DELETED ON CA WORKLOAD AUTOMATION AE ARE STILL VISIBLE IN QV**Problem Number: 405**

Global variables deleted on CA Workload Automation AE are still visible in QV until the next services restart. The fix is available in Release 11.4 SP3 branch.

MISSING SERVER AND COMMAND ATTRIBUTES IN CommandExecute AUTHORIZATION CHECKS

CommandExecute resource policy authorization checks that are triggered by the Enterprise command line application, do not contain values for server and command attributes.

NullPointerException ERROR MESSAGE DURING SAVING OF PREFERENCES IN CONFIGURATION

In CA WCC Configuration Preferences section, there is an error message displayed during the save operation when the component is selected from the dropdown and not confirmed using the Go button.

INSTALLER INSTALLATION ERROR - RETURN CODE 1 AFTER INSTALL IN HA

CA WCC installation has no error (return code 1) after installing in HA mode.

JOB TREE - REDUCE HIGH LOAD WHEN LOCATING JOB ON TOP LEVEL

When there are large numbers of nodes listed in the job tree, selecting a job in the jobs widget may take a longer time to load or may not complete the request.

ADDING A BOX JOB IN QUICK EDIT FAILS WHILE USING IE11

When the customer selects an option to create a box job in Quick Edit, a pop up window does not appear to "Create Job in this Box". This functionality works in Chrome and Firefox but failed while using IE11.

NOTE

The option to "Create a Box Job" is not functional in all versions of Internet Explorer.

GA DEFECT - HA INSTALLATION FAIL FOR FRESH INSTALL AND SHARED DATABASE

HA installation fails for fresh install and shared database. Add another node to HA installation so that the EEM hostname field is disabled and not filled automatically. It cannot be set.

ADDING NO WIDGETS ARE VISIBLE ON MONITORING TAB

Jobs widgets on All Views and All Servers use the same ID for storing user preferences - in some cases all the widgets can disappear.

CA WCC Release - 11.4 SP3

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP3 new features, changes to existing features, and fixed issues.

New Features - 11.4 SP3

This article describes new features added to CA WCC in Release 11.4 SP3.

CA WCC Performance Improvements for Bigger Loads

CA WCC startup load time of a flow model is faster as collection speed is improved when many views are defined. Collection time is decreased when adding a server with pre-existing view definitions.

CA WCC Security Improvements

CA WCC allows the usage of single sign-on wallets with certificates that are present on the file system to access the database. The database access mode can be set to password or SSL certificate either during installation or later.

- WCC-ASID session cookie is using "/wcc" path instead of "/".
- Client initiated SSL renegotiation is disabled.

Oracle PKI Authentication

Connection to Oracle database now is supported using Public Key Infrastructure (PKI) for authentication.

Changes to Existing Features - 11.4 SP3

This article details the changes to existing features in CA WCC Release 11.4 SP3.

Update to Tree Navigation in Monitoring

Update to the UI and tree structure allow for a more efficient customer experience while navigating CA WCC Monitoring.

Three tabs concerning Views, Servers, and Alerts have been added to the left navigation pane. The tree navigation provides the same context menu and action functionality as previously found in the table view.

The structure of the monitoring navigation provides a tree hierarchy of boxes and jobs. This enables supervision of the job status and allows for quick access to the job flow. When you select a job or a box job in the flow, it will be reflected in the tree and vice versa. When you select a job or a box job in the Jobs table, it will be reflected in the tree.

NOTE

Multi-selection functionality is not available.

Fixed Issues - 11.4 SP3

This article details the issues fixed in CA Workload Control Center Release 11.4 SP3.

Values in job runs table in Quick View are not wrapped to two lines

Problem Number: 398

Due to changes in design long job names are truncated in job flows. This patch allows to display full job names - if the name is too long it'll be wrapped on multiple lines.

Export limitations in monitoring views

Problem Number: 399

It isn't possible to export monitoring views from WCC if more than 1000 views exist. This patch will allow to export monitoring views in such situation.

Slow fetching of CLOB columns in monitoring table

Problem Number: 400

In some environments the startup of WCC can take around 50 minutes with many jobs stored in monitoring database and Quick View won't be available after startup. This is caused by slow fetching of CLOB columns in monitoring table. This patch will change table definition and avoid using CLOB columns whenever possible.

Importing Alert Policies Could Fail

Problem Number: 401

Import of the views and alert policies could fail if some of the view names contain one or more spaces. After patching the view names will be enclosed in quotes to prevent import failure. It means that you need to export views and alert policies again after patching.

wcc_monitor can use "-d ALL" option

WCC_monitor command line tool can use -d ALL option.

WCC can use characters with diacritics

Certain languages such as German or French that use diacritics, created problems when entered as values to certain CA WCC forms.

WCC Credentials could use password longer than 20 characters

Credentials defined the WCC credentials application could use passwords longer than 20 characters. The new limit is 80 characters.

CA WCC Release - 11.4 SP2

This section includes information about the CA Workload Automation WCC Release 11.4 SP2 new features and changes to existing features.

New Features - 11.4 SP2

This article describes new features added to CA WCC in Release 11.4 SP2.

Contents

CA WCC Upgrade From Older Versions

CA WCC supports upgrade from 11.3.x and 11.4 or 11.4 SP1 to current version 11.4 SP2.

CA WCC Support of AE Version 11.3.6 SP3

AE 11.3.6 SP3 is supported by CA WCC 11.4 SP2.

Forecast Reports Export

The Table Views of Forecast Reports can be exported into .csv format.

Embedded Security Against Hijacking of Clicks

CA WCC 11.4 SP2 prevents this version from being embedded into frames for increased security against Clickjacking. Clickjacking (UI Redressing) is a malicious technique that utilizes hidden layers of website frames to trick you into clicking on the disguised buttons.

NOTE

If you are intentionally embedding CA WCC in your environment, refer to further information to disable security features in [Embedding CA WCC into an HTML Frame](#).

Hadoop Job Types

CA WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define the following new job types using CA WA Advanced Integration for Hadoop:

- HDFS (Hadoop Distributed file System)
- OOZIE
- HIVE
- SQOOP
- PIG

Connection Profiles

A *connection profile* is a set of attributes that are required for connecting to environments where the job runs. For example, to run a job in the Hadoop environment, you can create a connection profile with the appropriate parameters to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database.

A connection profile lets you save the connection information and you can use the connection profile in the job definition. For example, when you define a Hadoop (HDFS, Hive, Oozie, Sqoop, or Pig) job, you can assign a connection profile to it.

A connection profile applies to the following Hadoop job types:

- HDFS
- Hive
- Oozie
- Sqoop
- Pig

Changes to Existing Features - 11.4 SP2

This article details the changes to existing features in CA WCC Release 11.4 SP2.

Contents

Apache Tomcat upgrade to Version 8

WCC now operates on Apache Tomcat version 8.

Using Urandom as Default RNG Source of Entropy

On Linux and Unix systems, CA WCC installer and CA WCC application are using /dev/urandom as a default source of entropy for its non blocking pool behaviour.

Forbidden Characters in Install Path

The forbidden characters "%", "(", and ")" are not to be used. Use of these characters will not be permitted in the installer.

Enterprise Command Line Import and Export Supports Jobs, Calendars, Machines, Resources, and Connection Profiles

The Enterprise Command Line application lets you export one or more jobs, a single or all calendars, machines, resources, or connection profiles residing on a CA Workload Automation AE server to a file. You can modify the calendar, job, machine, resource, or connection profile definitions using a text editor, and then import the modified file to any CA Workload Automation AE server you specify.

CA WCC Release - 11.4 SP1

Welcome to CA WCC. This section includes information about the CA WCC Release 11.4 SP1 new features and changes to existing features.

New Features - 11.4 SP1

This article describes new features added to CA WCC in Release 11.4 SP1.

Contents

Secure Cookies for HTTPS

For HTTPS connections, all cookies are flagged as secure.

CA WCC Upgrade from 11.3.x and 11.4 to 11.4 SP1

CA WCC supports upgrade to 11.4 SP1.

Addition of Machines Widget to Dashboard

A Machines Widget is added to the Dashboard functionality.

NOTE

You can open CA WLA Agent monitor from the Machine widget, if you configured the functionality in CA WCC Configuration.

Auto-Refresh Persistent Functionality

The auto-refresh check box in the Dashboard and Monitoring widgets remains persistent between user sessions.

Widget Filter Persistent Functionality

The visibility of the widget filters in Dashboard and Monitoring are persistent between user sessions and is turn ON by default.

Alert Defined in Alert Policy Dialog

You have the option to select view names from the View pane to assign an alert that is defined to one or more existing views.

Safari browser support

CA WCC 11.4 SP1 is supporting Safari web browser from version 7.1.6 and newer.

NOTE

In order to work properly, you must turn off the pop-up blocking function in Safari > Preferences > Security > Web content - uncheck "Block pop-up window"

Default time zone for CA WCC Servers

You have the option to select time zone which will be used as default for CA WCC users. Users are still able to customize the time zone using My Profile dialog.

Description Field for Calendar

Calendar, extended calendar or cycle has a new Description field with limit of 1024 characters, if you are using only CA AE 11.3.6 SP2 (and any later versions).

Servers widget displays detail of connection failure

Servers widget on CA WCC Dashboard displays detailed error information in case the connection to CA AE instance fails. It is necessary to hover the mouse cursor over the status icon which will trigger the tooltip with detailed error message.

Changes to Existing Features - 11.4 SP1

This article details the changes to existing features in CA WCC Release 11.4 SP1.

Contents

Colon Support in Job Names

Colon (:) support is introduced for New Job Names and Default Job Names.

Help File Consolidation

The CA WCC Helps are consolidated, that enables you to search through all Help content. Also if Internet access is enabled link to the full on-line doc on the wiki DocOps platform is available from top of each Help page.

Monitoring

Monitoring has the following enhancements for this release:

- A command is displayed in the Command Job tooltip located in the flow diagram.
- The Search job field in Monitoring Flow is containing tooltip with example.
- The visibility of the widget filter is persistent between user sessions and is turn ON by default.
- The number of rows in the widget filter is persistent between user sessions.
- The auto-refresh option in widgets remain persistent between user sessions.

Quick View

Quick View has the following enhancements for this release:

- The Maximum Flow Height limits of the Job Flow section are increased to 1500
- A command is displayed in the Command Job tooltip located in the flow diagram.

Auto Refresh No Longer Effects Session Timeout Operations in CA WCC

Auto refresh is no longer a user action when you are idle in the application and remain logged in.

NOTE

If you want to keep the original auto refresh functionality, prolonging session timeout, set in Configuration preferences - Configuration - Security section "Auto Refresh prevents session timeout" to ON.

Application Editor Re-Inclusion

The Application Editor is available for Release 11.4 SP1.

HTTPS Enabled for HA Configuration

It is possible to install a new HA node using HTTPS. Only supported configuration for an HA cluster is to have all the nodes in either HTTP or HTTPS.

NOTE

- If all HA nodes in a cluster are in the same mode, it is not possible to modify the protocol to HTTP or HTTPS.
- You can continue installing HTTP or HTTPS in mixed mode. In this case, the list of HA nodes and the protocol value are displayed. It is only possible to change the protocol using the batch interface.

Simplification of EEM Settings in the Installer

When using EEM, in failover mode, the comma separated list of the EEM servers are no longer required. Instead, a single EEM server (possibly the primary server) is required. The other servers in the failover mode are automatically retrieved and displayed. The installer will verify the network access and the state of all EEM servers. In case there is an issue, a dialog with a summary will be displayed.

Support for Oracle 12c

Oracle 12c is available as a CA WCC database component. Container Database of Oracle 12c is not supported.

Support for MS SQL Server 2014

MS SQL Server 2014 is available as a CA WCC database component.

Change in Calendar Name Length Limits

The character length for the name of a calendar, extended calendar or cycle has a limit of 64 characters, if you are using only CA AE 11.3.6 SP2 (and any later versions).

CA Workload Automation AE Server Version Auto Discovery

During validation, there is an automatic discovery of the CA Workload Automation AE server version. Instead of an error message, the CA Workload Automation AE version number is updated.

Enterprise Command Line and Forecast commands Used by CA WCC Logged On Users Context

CA WCC Enterprise Command Line and Forecast run under user context of the currently logged in user to CA WCC, not under the user defined in Credentials. Credentials user is used only to access AE machine, but the remotely launched commands from Enterprise Command Line and Forecast are run under CA WCC user.

NOTE

- This functionality only works with CA AE 11.3.6 and later versions
- CA AE and CA WCC need to operate on the same EEM

Status of Cross Instance Dependencies

CA WCC displays dependencies on jobs running on External Instances. The status of the Cross Instance Job is visible in the Job Dependencies Table in QuickView and the flows in Monitoring and QuickView.

Fix for Forecast DST issue

WCC Forecast displays correct data for periods involving DST shift periods. The prerequisite is to set time zone in CA AE instance definition in CA WCC Configuration.

Controlling EEM Caching Behavior

When controlling EEM SDK caching behavior, you can switch on/off the full cache updates and set the interval between the cache updates. The caching behavior can be controlled by the connection.properties file located under the CA WCC root directory. See Performance tuning topic for more details.

NOTE

The connection with the EEM server is established during CA WCC startup. CA WCC services must be restarted before the new settings take effect.

Common Components Release Notes

Use the Common Components DVD to install the following common components:

- Embedded Entitlements Manager (EEM) Release 12.6

NOTE

- On Windows, if you are upgrading from a 32-bit EEM to a 64-bit EEM or migrating from CA EEM Release 8.x to 12.6, follow the instructions in the [Common Components Upgrade](#) page.
- For the latest EEM support information, see the CA Embedded Entitlements Manager Compatibility Matrix on the [Broadcom Support website](#). This document lists the following:
 - Supported operating systems
 - IPv6 support
 - FIPS 140-2 compliance
 - Section 508 compliance
 - Third-party products that EEM supports
 - Embedded Broadcom products that are shipped with EEM
- Secure Socket Adapter (SSA) 2.4.1
- Event Management r11.2 SP2 CUM1
- CA, Inc. Common Communications Interface (CAICCI) r11.2 SP2 CUM1
- Management Command Center r11.2 SP2 CUM1

NOTE

On Windows, you cannot install Management Command Center using the Common Components DVD.

- CAPKI 5.2.5

CA EEM Reporting Utility-An Overview

The CA EEM Reporting Utility (CA ERU) audits CA EEM log files. It imports event data from the audit log files to a database. Optionally, it archives the data to an auxiliary data set and deletes audit data from the database after a specified period. You can use this utility to generate customized reports in PDF or .CSV format.

The CA EEM Reporting Utility is designed for use with CA EEM Release 12.0, Release 12.51, Release 12.51 CR02, Release 12.51 CR03, Release 12.51 CR04, or Release 12.6. To install the CA EEM Reporting utility, use the CA EEMReporting_1.0.1.18_unix.tar (UNIX) or CA EEMReporting_1.0.1.18_win.zip (Windows) file that is available in the following location in the CA Common Components DVD:

- UNIX—Mount the DVD and go to the following folder:

`CA EEM-Reporting-Utility`

- Windows

`DVD_drive\Disk1\InstData\VM\CA EEM\CA EEM-Reporting-Utility`

DVD_drive

Identifies the CA Common Components installation media drive.

NOTE

For more information about installing and configuring the CA EEM Reporting Utility, see [CA EEM Reporting Utility](#).

Common Components Operating System Support

Broadcom supports the subsequent operating systems for the duration of their life cycle (as determined by the operating system's manufacturer or until Broadcom announces that we are dropping support). The supporting operating systems are subject to change. For more information about compatibility and to see the latest operating systems supported, visit the [Broadcom Support website](#).

NOTE

- For information about CA EEM system requirements, installation considerations, and known issues, see the [CA EEM documentation](#).
- For information about Event Management, Management Command Center, and CA, Inc. Common Communications Interface (CAICCI) system requirements and installation considerations, see the [CA NSM Release Notes](#).
- For information about the supported operating systems, system requirements, and known issues of the Broadcom product that uses the common components, see [Release Notes](#).

If you want to install a common component on an operating system that is not listed in this article, do the following:

1. Verify if the common component supports that operating system.

NOTE

- For information about CA EEM supported operating systems, see the [CA EEM documentation](#), or visit the [Broadcom Support website](#).
- For information about the Event Management, Management Command Center, and CA, Inc. Common Communications Interface (CAICCI) supported operating systems, see the [CA NSM Release Notes](#) or visit the [Broadcom Support website](#).

2. Install the common component using the common component installation media.

For example, if you want to install Event Management on an operating system that is not listed in this topic, see the [CA NSM Release Notes](#) to verify if Event Management supports that operating system and then install Event Management using the CA NSM media.

Supported UNIX Versions

The CA Common Components support the following UNIX versions:

- Solaris 10 (SPARC 64-bit)
- HP-UX 11iv3 (Itanium 64-bit)
- AIX 5.3 (POWER 32-bit and 64-bit)

NOTE

Event Management requires Maintenance Level 3.

- AIX 6.1 (POWER 64-bit)
- AIX 7.1 (POWER 64-bit)
- AIX 7.2 (POWER 64-bit)

NOTE

On AIX 6.1, 7.1, or 7.2 Workload Partitions (WPARs) computers, install the Event Manager using the CA NSM media. The Event Manager that is installed using the CA Common Components DVD is not supported on AIX WPARs computers. However, the Event Agent is supported on AIX 6.1, 7.1, or 7.2 WPARs computers and so you can install it using the CA Common Components DVD.

NOTE

For information about AutoSys Workload Automation supported operating systems, see the [AutoSys Workload Automation Release Information](#).

Supported Linux Versions

The CA Common Components support the following Linux versions:

- Red Hat Enterprise Linux Server 5 (x86 32-bit and 64-bit)
- Red Hat Enterprise Linux Server 6 (x86 32-bit and 64-bit)
- Red Hat Enterprise Linux Server 7 (64-bit)
- Oracle Enterprise Linux 6 (x86 32-bit and 64-bit)
- Oracle Enterprise Linux 7 (64-bit)
- SuSE Linux Enterprise Server 11 (x86 32-bit and 64-bit) (Event Management Agent only)
- SuSE Linux Enterprise Server 12 (64-bit)

NOTE

- The gtk+-1.2.10 rpm package is required to install Management Command Center using the CA Common Components DVD.
- For information about AutoSys Workload Automation supported operating systems, see the [AutoSys Workload Automation Release Information](#).

Supported Windows Versions

The CA Common Components support the following Windows versions:

- Microsoft Windows Server 2003 (x86 32-bit and 64-bit) with any later maintenance release
- Microsoft Windows Server 2003 R2 (x86 32-bit and 64-bit)

NOTE

In 64-bit environment (agents only)-refer to PIB QI86232 for restrictions.

- Microsoft Windows 7 (x86 32-bit and 64-bit)
- Microsoft Windows 2008 (Notification client r11.2 SP2 only)
- Microsoft Windows 2008 R2
- Microsoft Windows Server 2008 (x64)
- Microsoft Windows Server 2008 (x86 32-bit and 64-bit)
- Microsoft Windows Server 2008 R2 (x86 64-bit)
- Microsoft Windows Server 2008 R2 (x64)
- Microsoft Windows 2008 R2 Cluster
- Microsoft Windows Server 2008 R2 SP1
- Microsoft Windows 2008 SP2
- Microsoft Windows XP Professional (32-bit version only), with any later maintenance
- Microsoft Windows Vista (x86 64-bit) for CA NSM remote admin clients
- Microsoft Windows Server 2012
- Microsoft Windows Server 2012 R2

Unicenter Managers are not supported on Microsoft client environments (Windows XP, Windows Vista, and Windows 7). Only the Enterprise Management Event Agent client component is supported on those environments.

NOTE

For information about AutoSys Workload Automation supported operating systems, see the [AutoSys Workload Automation Release Information](#).

CA Common Components IPv6 and Browser Support

This article documents IPv6 and browser support information for the following common components:

- SSA
- Event Management
- CAICCI
- Management Command Center

NOTE

For information about CA EEM browser support, see the [CA EEM documentation](#).

IPv6 Support

CA Common Components Release 11.3.6 SP8 supports Internet Protocol version 6 (IPv6), as well as the legacy IPv4 only operation, and dual IPv4 and IPv6 stacks.

Web Browser Support

The following web browsers are supported:

- Microsoft Internet Explorer 6 SP1 through 8

Note: Internet Explorer 11 is supported in compatibility mode only.

- Mozilla Firefox 1.6 and higher

Web Servers and Servlet Environment Support

For the common components, the web applications run as Java Servlets and conform to the Java Servlet 2.3 and JSP 1.2 specifications. CA Common Components Release 11.3.6 SP7 is shipped with Apache Tomcat, so a web server is not required to install or run the web applications. JavaScript must be enabled on all supported web browsers for the common components to function correctly.

Java Browser Support

The common components web applications use Java applications in their web pages and require the following browser support for Java:

- Built-in Java support in a web browser, if present. If the built-in support for Java is not present, the dashboards require that you install the Java Runtime Environment (JRE) from the Sun Microsystems web site.
- Install the JRE from the Sun Microsystems web site for the web browser.

If JRE is not detected on your computer when you use the common components web applications, a link appears so you can download JRE from the Sun Microsystems web site. When you install JRE for Windows, ensure that you select the Offline Installation option instead of the Install from the Web option. The common components web applications support JRE 1.5.10 or later.

The graphical displays on dashboards and reports do not appear unless built-in support or the JRE is present.

CA Common Components System Requirements

This article documents the system requirements for the following common components:

- SSA
- Event Management
- CAICCI
- Management Command Center

NOTE

For information about CA EEM system requirements, see the [CA EEM documentation](#).

Contents

Hardware Requirements

If the computer does not have the minimum resources necessary to support the installed components, you will see messages stating that one or more CA services have failed to start. In some cases, even computers that meet or exceed the necessary hardware requirements may experience this behavior if the combination of software products installed cause a temporary shortage of resources during startup. The services may be operational shortly after the errors appear.

If this occurs, it is necessary to delay the startup of CA services to allow other critical services time to completely initialize. In cases where the resources available are insufficient, CA services and other services may fail causing other cascading failures. The overall stability of your system is affected if you install the common components on systems with insufficient resources.

Depending on your operating environment, the following hardware requirements must be met or exceeded for the common components to install and run correctly:

AIX

- RS 6000 1.0 GHz
- Physical Memory: 1 GB (minimum) 4 GB (recommended)
- Swap Space: 2 GB
- Disk Space: At least 1 GB of free space

HP-UX

- Physical Memory: 1 GB (minimum) 4 GB (recommended)
- Swap Space: 4 GB
- Disk Space: At least 1 GB of free space

Linux

- Intel P4 1.6 GHz
- Physical Memory: 1 GB (minimum) 4 GB (recommended)
- Swap Space: 2 GB
- Disk Space: At least 1 GB of free space

Solaris SPARC

- Processor: SPARC 1.0 GHz
- Physical Memory: 1 GB (minimum) 4 GB (recommended)
- Swap Space: 2 GB
- Disk Space: At least 1 GB of free space

Windows

- Processor: 1 GHz
- Physical Memory: 2 GB
- Disk Space: At least 2 GB of free space

Software Requirements

Depending on your operating environment, the following prerequisite software must be installed for the common components to install and run correctly:

AIX

- xlc.rte.5.0, version 8.0.0.9 through 9.0.0.8
- GIMP Toolkit (GTK+) 1.2

HP-UX

- TOUR packages for IPv6 support
- GTK+ 1.2

Linux

- compat-libstdc++-296 or higher for Red Hat Enterprise Linux 4.0
- compat-libstdc++-33 or higher for Red Hat Enterprise Linux 5.0 or above. The 32-bit version is required on the 64-bit version of Linux.
- libstdc++.so.6 or higher is required.
- glibc 2.4 (32-bit) or higher is required.
- compat for SuSE Linux
- ncurses (32-bit) version 5 or higher on Linux (64-bit)
- GTK+ 1.2

Solaris SPARC

- libgcc-3.4.6
- GTK+ 1.2
- If you install CA EEM on Solaris 9, you must apply Patch ID 114129-02 or later.

NOTE

You can download the patch from <http://www.sunsolve.sun.com>.

Solaris Intel

- libgcc-3.4.6
- GTK+ 1.2

GTK+ Requirements for the Management Command Center

The Event Manager includes the Management Command Center, which requires libraries from the GNU Image Manipulation Program (GIMP) Toolkit. The GIMP Toolkit (GTK+) libraries are not included in the CA Common Components package. If you install the Event Manager or the Management Command Center, you must ensure that specific packages have been installed on the target systems as follows:

AIX

You must apply the following required packages:

- glib-1.2.10-2
- gtk+-1.2.10-4

NOTE

You can download the packages from <http://www.ibm.com>. The files are installed in /opt/freeware/lib.

Linux

You must apply GTK+-1.2.10 patch and its dependencies.

The packages are available on your installation media and installed with the currently supported operating systems. The only exception is Red Hat Enterprise Linux Server 5, where the packages must be downloaded externally and then installed.

Solaris 10 SPARC and Solaris 10 x86

You must apply the following required packages:

- SMCglib
- SMCgtk

NOTE

The packages are available on your installation media and installed in /usr/sfw/lib. You can also download the packages from <http://www.sunfreeware.com>, but these files are installed in /usr/local/lib.

You must also add /usr/sfw/lib and /usr/local/lib to LD_LIBRARY_PATH as follows:

```
LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/sfw/lib:/usr/local/lib
export LD_LIBRARY_PATH
```

Database Requirements - UNIX

CA EEM uses DxGrid as its data store. The DxGrid data store is installed when you install CA EEM. The installer performs a pre-installation check to determine if an Ingres database exists. If an Ingres database exists, the data in the Ingres database is migrated to the DxGrid data store.

NOTE

eTrust IAM r8.3 use an Ingres database.

Event Management requires PostgreSQL 9.5.2. PostgreSQL is automatically installed if you select the Event Manager component during installation. The installer performs a pre-installation check to determine if an Ingres database with the MDB or any CA database exists. If an Ingres database with the MDB or any CA database exists, the installer displays the option to migrate the database data to the PostgreSQL database. If you select the option to migrate the data, the post installation process performs the database data migration.

NOTE

The migration may take a considerable amount of time depending on the data to be migrated. If you do not select the option to migrate the data, you can migrate the database data manually at a later time. For more information, see the [AE Installation](#).

Database Requirements - Windows

CA EEM uses DxGrid as its data store. The DxGrid data store is installed when you install CA EEM. The installer performs a pre-installation check to determine if an Ingres database exists. If an Ingres database exists, the data in the Ingres database is migrated to the DxGrid data store.

NOTE

eTrust IAM r8.3 use an Ingres database.

Event Management requires Microsoft SQL Server, which must be installed before you install Event Management. The following database versions are supported:

- Microsoft SQL Server 2005 (64-bit)
- Microsoft SQL Server 2005 (32-bit) with SP1 or any later maintenance release
- Microsoft SQL Server 2000 (32-bit) with a minimum of SP4 or any later maintenance release
- Microsoft SQL Server 2008 (x86 64-bit)
- Microsoft SQL Server 2012

Ensure that you adhere to the following database requirements when setting up Microsoft SQL Server:

- Mixed-mode authentication is required.
- A dictionary sort order of case-sensitive or case-insensitive is required.

NOTE

When installing Microsoft SQL Server 2008, select any of the following collation settings that CA NSM supports:

- SQL_Latin1_General_CP1_CS_AS: Dictionary order, case-sensitive for use with 1252 Character Set
- SQL_Latin1_General_CP1_CI_AS: Dictionary order, case-insensitive for use with 1252 Character Set
- The MDB uses JDBC, which requires TCP/IP to be enabled.
- The MDB password is not limited to six characters.
- You can use the English version of Microsoft SQL Server with a code page/character set of 1252 on any of the Windows language versions that Broadcom has certified for use with the common components.
- You can use localized versions of Microsoft SQL Server other than English on any of their respective and matching Windows language versions that Broadcom has certified for use with the common components, with the default code page/character set, or with a code page/character set of 1252 if you are not using specific localized character data for that region, even though you are installing on a localized system of that region. You must use the default code page/character set if you plan to use specific localized characters.
- You cannot use a Microsoft SQL Server alias that matches the local computer name but which really points to a remote Microsoft SQL Server (whether in another resource group on the same physical node of a cluster or actually on a different computer).

NOTE

MDB creation uses Windows Authentication and must be run by an operating system user who belongs to the System Administrators server role.

su Command on UNIX

On UNIX, the CA Common Components Event Management installation process must be allowed to run the following command with no user response or interaction:

```
su - root - c "command"
```

The installer embeds other installation packages that are executed by running the su command.

Adobe Acrobat Reader Support

WRS and Systems Performance reports let you export reports in PDF format. To view reports in PDF format, the computer running the web browser or the Management Command Center must have Adobe Acrobat Reader version 7.0 or any later release (for Windows) or Adobe Acrobat Reader version 5.0.9 or any later release installed.

To download the appropriate version of Adobe Acrobat Reader, visit <http://www.adobe.com>.

NOTE

To help ensure that you can correctly open reports using Adobe Acrobat Reader, open Adobe Acrobat Reader and clear the Internet category option "Display PDF in browser" under Edit, Preferences.

Kernel Parameter Requirements - HP-UX Installation

To install and run the common components on HP-UX, the following kernel parameter requirements must be met or exceeded:

- The msgmax and msgmnb upper limits must be set to the following values to ensure that the common components function properly:

```
msgmax = 32768
msgmnb = 65535
```

After the upper limits are set and the parameters are updated during the installation, issue the following command to check whether a restart is required:

```
$ /usr/sbin/kmtune -u
```

Restart your computer for the changes to take effect if you encounter a message similar to the following statements:

```
WARNING: maxfiles cannot be set dynamically.
WARNING: maxusers cannot be set dynamically.
```

- Many kernel parameter values must be set at required minimum levels. If you change a kernel parameter, you must rebuild the kernel.

To update the kernel parameter values and rebuild the kernel

- Enter **sam** at the command prompt.
The sam utility starts running.
- Click kernel configuration, configurable parameters.
A list of configurable parameters appear.
- Select the following kernel parameters and update the values, if the values in your environment are less than the following required minimum values:

```
max_thread_proc = 512
maxdsiz = 1073741824
maxfiles = 576
maxssiz = 67108864
maxtsiz = 536870912
```

```

maxusers = 64 #obsolete for HP-UX 11.2
maxuprc = 200
msgmap = 258
msgmni = 128
msgseg = 8192
msgssz = 32
msgtql = 256
nfile = 2048
ninode = 2048
nkthread = 1500
nproc = 512
npty = 720
sema = 1
semmap = 258 #obsolete for HP-UX 11.2
semmni = 256
semmns = 512
semmnu = 80
semume = 40
semvmx = 32767
shmmax = 268435456
shmmni = 512

```

4. Select File, Exit, Creating New Kernel, and follow the steps to create the new kernel.
The sam utility restarts your computer and the updated kernel parameters take effect.

Kernel Parameter Requirements - Linux Installation

To install and run the common components on Linux, the following kernel parameter requirements must be met or exceeded:

```

msgmax = 32768
msgmnb = 65535
msgmni = 128
msgseg = 8192
semmni = 256
semmns = 512
semmsl = 100
shmmni = 512
shmmax = 1073741824
file-max = 8192
threads-max = 65262

```

To update the kernel parameter values

1. Issue the following commands depending on the kernel parameter values you want to see:

Parameter	Command
msgmax, msgmnb, msgmni	/sbin/sysctl -a grep msg
semmni, semmns, semmsl	/sbin/sysctl -a grep sem The results are similar to the following: kernel.sem = <semmsl> <semmns> <semopm> <semmni>
shmmax, shmmni	/sbin/sysctl -a grep shm
file-max	/sbin/sysctl -a grep file-max

threads-max	/sbin/sysctl -a grep threads-max
-------------	------------------------------------

The current kernel parameter values are displayed.

2. Edit the /etc/sysctl.conf file by adding or updating the following kernel parameters, if the values in your environment are less than the following recommended values:

```
fs.file-max = 8192
kernel.msgmnb = 65535
kernel.msgmni = 128
kernel.msgmax = 32768
kernel.sem = 100 512 100 256
kernel.shmmni = 512
kernel.shmmax = 1073741824
kernel.threads-max = 65262
```

3. Issue the following command:

```
/sbin/sysctl -p /etc/sysctl.conf
```

The sysctl settings are loaded from /etc/sysctl.conf file.

4. Restart your computer.

The updated kernel parameters take effect.

Kernel Parameter and Resource Controls Requirements - Solaris Installation

To install and run the common components on Solaris 9, the following kernel parameter requirements must be met or exceeded:

```
forceload: sys/msgsys
forceload: sys/semsys
forceload: sys/shmsys
set msgsys:msginfo_msgmax = 32768
set msgsys:msginfo_msgmnb = 65535
set msgsys:msginfo_msgmni = 128
set msgsys:msginfo_msgseg = 8192
set msgsys:msginfo_msgtql = 256
set semsys:seminfo_semmap = 258
set semsys:seminfo_semmni = 256
set semsys:seminfo_semmns = 512
set semsys:seminfo_semmnl = 100
set semsys:seminfo_semmnu = 80
set semsys:seminfo_semopm = 20
set semsys:seminfo_semume = 40
set semsys:seminfo_semusz = 96
set semsys:seminfo_sevmx = 32767
set semsys:seminfo_semaem = 16384
set shmsys:shminfo_shmmax = 1073741824
set shmsys:shminfo_shmmni = 512
set shmsys:shminfo_shmseg = 50
set rlim_fd_cur = 576
```

To update the kernel parameter values in Solaris 9

1. Issue the sysdef command.

The kernel parameters of the Solaris computer are displayed.

2. Add the appropriate set and forceload commands (previously mentioned) to the end of the /etc/system file.
3. Restart your computer.
The updated kernel parameters take effect.

To install and run the common components on Solaris 10, the following resource controls requirements must be met or exceeded:

```
process.max-msg-qbytes = 65535
project.max-msg-ids = 128
project.max-sem-ids = 256
project.max-shm-ids = 512
project.max-shm-memory = 1073741824
process.max-file-descriptor = 1024
```

To update the resource controls values in Solaris 10, use the Resource Controls Facility.

NOTE

For more information about this facility, see the *Solaris 10 System Administration Guide: Solaris Containers-Resource Management and Solaris Zones*.

tr Command in Unicode Environment on UNIX

On UNIX, the common components installer runs the following command:

```
tr '[:upper:]' '[:lower:]'
```

In a Unicode environment, the tr command must be Unicode compliant. The Unicode compliant version is /usr/xpg4/bin/tr. If your environment is Unicode, check the version of tr by running the following command:

```
which tr
```

If the output is /usr/bin/tr, you must change the PATH variable to use the Unicode compliant version of tr. To change the PATH environment variable, run the following command:

```
PATH=/usr/xpg4/bin:$PATH
Export PATH
```

CA Common Components General Considerations and Published Fixes

This article provides general information you should know when using the common components.

HAS Considerations - UNIX

Consider the following guidelines for HAS on UNIX:

- CA Common Components Release 11.3.6 SP8 includes High Availability services. These services enable HA-ready components in clustered configurations. HA-ready components provided by this deliverable are consistent with those enabled in CCS r11 or CA NSM r11.2 SP2 CUM1. Any components or configurations that have not been certified as HA-ready in CCS r11 or CA NSM r11.2 SP2 CUM1 will not be considered HA-ready in CA Common Components Release 11.3.6 SP8.
- HA readiness is limited to Linux platforms only.
- On Linux, Event Management r11 was not certified on a cluster and is not planned to be HA-ready in CA Common Components Release 11.3.6 SP8.

HAS Considerations - Windows

You cannot install Event Manager HAS using the CA Common Components DVD. For high availability services, install the Event Manager using the CA NSM media.

CA EEM Service Name Changes

The eTrust Directory - iTechPoz-*hostname* service in CA EEM r8.3 or r8.4 is named as CA Directory - iTechPoz in CA EEM Release 12.6.

NOTE

The eTrust Directory - iTechPoz-*hostname*-Router, eTrust Directory Administration Daemon - master, and eTrust Directory SSL daemon - iTechPoz-Server services in CA EEM r8.3 or r8.4 are deprecated.

CA Common Components Published Fixes

The complete list of published bug fixes for this product can be found through Published Solutions on CA Support Online.

Third-Party Software Acknowledgements - Release 11.3.6 SP6

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

WARNING

The acknowledgements for third-party components used by CA Common Components are in the acknowledgements.txt file that is located in the root directory of the installed product.

Third-Party Software Acknowledgements Release - 11.3.6 SP7

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

WARNING

The acknowledgements for third-party components used by CA Common Components are in the acknowledgements.txt file that is located in the root directory of the installed product.

Third-Party Software Acknowledgements Release - 11.3.6 SP8

This product uses third-party components. [Download](#) the document locally so you can review the license agreements.

WARNING

The acknowledgements for third-party components used by CA Common Components are in the acknowledgements.txt file that is located in the root directory of the installed product.

Product Accessibility Features

Broadcom is committed to making sure that all customers, regardless of ability, can successfully use its products and supporting documentation to accomplish vital business tasks. This section outlines the accessibility features that are part of *AutoSys Workload Automation and Web UI*.

Product Enhancements

AutoSys Workload Automation offers accessibility enhancements in the following areas:

- Display
- Keyboard
- Mouse
- Icons
- Timeout

Note: The following information applies to Windows-based and Macintosh-based applications. Java applications run on many host operating systems, some of which already have assistive technologies available to them. For these existing assistive technologies to provide access to programs written in JPL, they need a bridge between themselves in their native environments and the Java Accessibility support that is available from within the Java virtual machine (or Java VM). This bridge has one end in the Java VM and the other on the native platform, so it will be slightly different for each platform it bridges to. Sun is currently developing both the JPL and the Win32 sides of this bridge.

Display

To increase visibility on your computer display, you can adjust the following options:

- **Font style, color, and size of items**

Defines font color, size, and other visual combinations.

- **Screen resolution**

Defines the pixel count to enlarge objects on the screen.

- **Cursor width and blink rate**

Defines the cursor width or blink rate, which makes the cursor easier to find or minimize its blinking.

- **Icon size**

Defines the size of icons. You can make icons larger for visibility or smaller for increased screen space.

- **High contrast schemes**

Defines color combinations. You can select colors that are easier to see.

Keyboard

You can make the following keyboard adjustments:

- **Repeat Rate**

Defines how quickly a character repeats when a key is struck.

- **Filter Keys**

Disables the multiple displaying of one key, if it is press and hold.

- **Sticky Keys**

Defines the modifier key, such as Shift, Ctrl, Alt, or the Windows Logo key, for shortcut key combinations. Sticky keys remain active until another key is pressed.

Keyboard Shortcuts

The following table lists the keyboard shortcuts that AutoSys Workload Automation supports:

Keyboard	Description
Alt+N	Next
Alt+B	Back

Alt+C	Cancel
Alt+O	OK
Alt+T	Third Party License

Mouse

You can use the following options to make your mouse faster and easier to use:

- **Click Speed**

Defines how fast to click the mouse button to make a selection.

- **Click Lock**

Sets the mouse to highlight or drag without holding down the mouse button.

- **Reverse Action**

Sets the reverse function controlled by the left and right mouse keys.

- **Blink Rate**

Defines how fast the cursor blinks or if it blinks at all.

- **Pointer Options**

Let you do the following:

- - Hide the pointer while typing
 - Show the location of the pointer
 - Set the speed that the pointer moves on the screen
 - Choose the pointer's size and color for increased visibility
 - Move the pointer to a default location in a dialog box

Consistent Use of Icons

All icons in the product are unique. The bitmap images means the same at all of the locations.

Timeout Warning

User is alerted, when the session timeout occurs and can continue working.

Before You Begin

Before you begin with the AutoSys installation, it is recommended that you go through the following topics carefully.

Plan your AutoSys Environment

The following table lists the tasks that you must perform to set up AutoSys Workload Automation and Web UI in a typical environment.

NOTE

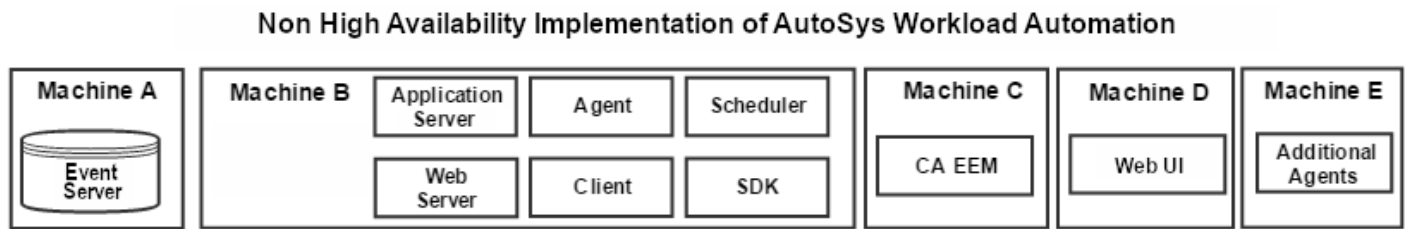
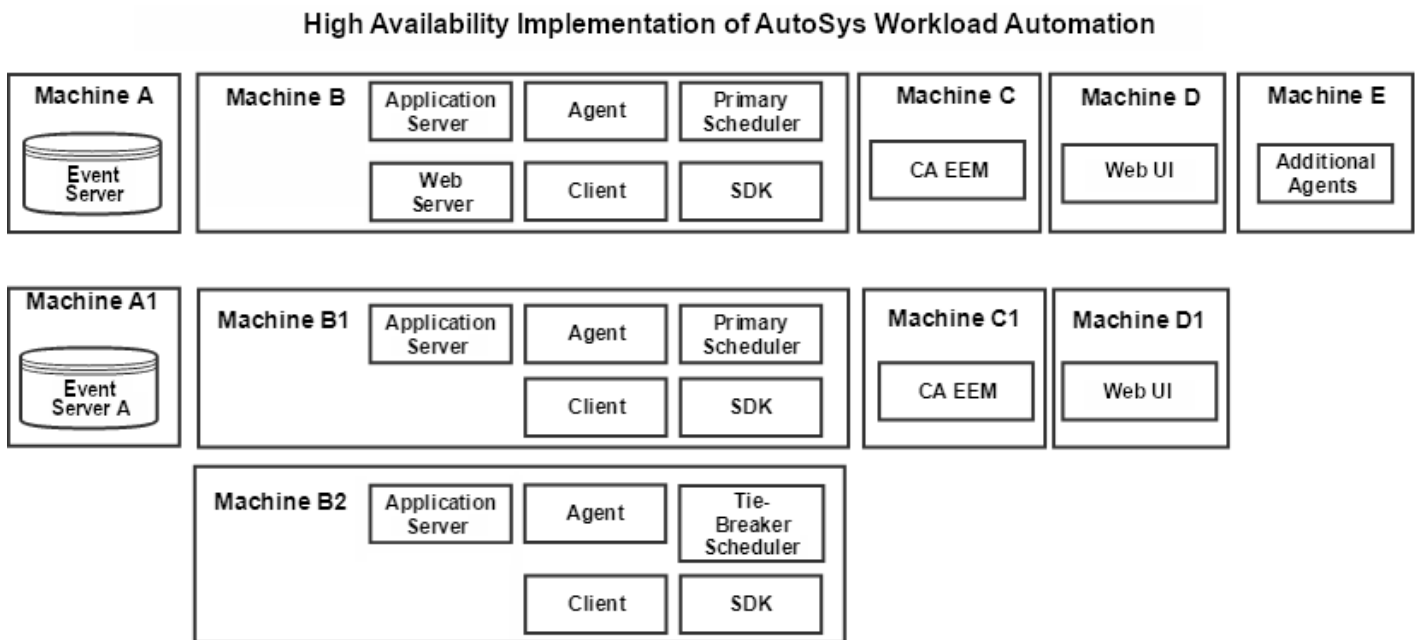
- You can perform the tasks that are listed in the table if you are installing AutoSys Workload Automation for the first time or installing AutoSys Workload Automation in a test environment before upgrading and migrating your data to the current release.
- We recommend that you install the Broadcom products in the order listed in the table.
- For installing AutoSys Workload Automation, see [Installing the Server](#), [Installing the Client](#), [Installing the Agent](#).
- For installing Workload Automation Agents and Agents Plug-ins, see [Workload Automation Agents](#).
- For installing CA EEM, see [CA Common Components Installation](#).

Installation Phase	Tasks to Perform
Pre-Installation	Plan your environment Set up the database for AutoSys Workload Automation
Installation	Install and verify the AutoSys Workload Automation server Before you install AutoSys Workload Automation, you must ensure that the system requirements are met on the server computer. For information about the AutoSys Workload Automation installation considerations, see the following topics: Installation Considerations Verify Web UI Install and verify additional agents on other computers Install and verify agent plug-ins
Post-Installation	Set up custom CA EEM security policies

Plan Your Environment

Before you install the products complete the following tasks:

- Ensure that you install environment prerequisites. For more information on prerequisites, see [AutoSys Operating System and Database Support](#).
- [Identify the computers](#) on which you want to install the products, and decide which product to install on each computer. We recommend that you install the AutoSys Workload Automation components, CA EEM, Web UI, and more agents on separate computers as follows:

Figure 2: Non-HA implementation of AutoSys Workload Automation and Web UI**Figure 3: HA implementation of AutoSys Workload Automation and Web UI****Set Up the Database for AutoSys Workload Automation**

Ask your Database Administrator to set up the database (event server) for AutoSys Workload Automation. Record the Database Administrator password. You need this information during the AutoSys Workload Automation installation.

NOTE

For more information about the considerations, before setting up the AutoSys Workload Automation database, see the following topics:

- [Host Machine Checklist](#)
- [Installing into an Existing MDB \(Oracle Only\)](#)
- [Install AutoSys Workload Automation with Sybase](#)
- [Install AutoSys Workload Automation with Oracle](#)

Install CA EEM

CA EEM is optional for AutoSys Workload Automation; however, it is required for Web UI. We recommend that you configure AutoSys Workload Automation to use CA EEM for enhanced security.

CA EEM must be installed and running before you install Web UI or AutoSys Workload Automation (if you are using CA EEM security). Record the CA EEM password. You need this information during the AutoSys Workload Automation and Web UI installation.

On UNIX, if you install CA EEM on a computer where AutoSys Workload Automation is already installed and active, the CA EEM installation process shuts down the AutoSys Workload Automation active processes. Start the AutoSys Workload Automation processes manually after the CA EEM installation is complete.

Install and Verify the AutoSys Workload Automation Server

If you perform a custom installation, we recommend that you select all the components.

NOTE

- On UNIX, if you want to install AutoSys Workload Automation on a computer where Web UI Release is already installed, you must stop the Web UI services before you install AutoSys Workload Automation Release.
- During the AutoSys Workload Automation installation, select the option to create the CA EEM security policies for the AutoSys Workload Automation instance.

Verify Web UI

You can verify that Web UI works with AutoSys Workload Automation by creating a job using the Web UI Quick Edit application. You can monitor the job using the Web UI Job Status Console application.

Install and Verify Additional Agents on Other Computers

You can install additional agents on other computers.

NOTE

(Windows) You can also use the silent installation files to install more agents. Before you install the agent using the silent installation files, you must modify the silent installation files to match the required configuration.

Install and Verify Agent Plug-ins

You can use the agent plug-in installation files on the Workload Automation Agent for UNIX, Linux, or Windows ISO to install the agent plug-ins.

NOTE

(Windows) You cannot install the agent plug-ins using CA SDM.

Set Up Custom CA EEM Security Policies

After you install and verify CA EEM, AutoSys Workload Automation, and Web UI, you can [customize the default security policies](#) or create security policies and grant access modes based on your requirements.

Download AutoSys

You can install the server using the AutoSys Workload Automation ISO file that you can download from the download center using the following procedure.

To download AutoSys:

1. Go to <https://support.broadcom.com>.
2. Go to **Enterprise software > Product Downloads > Download Management**.
3. In the search box, enter AutoSys Workload Automation.

4. Click **AutoSys Workload Automation**. The product gets downloaded as an ISO file. You can mount this file on your system to install AutoSys.

Prerequisites

The following are prerequisites when installing AutoSys Workload Automation.

- For installation on Windows, log in as a user with Windows Administrators group privileges. And ensure that Windows has all the latest updates installed.
- To perform an AutoSys installation on Unix, log in as sudo or root user.
- On Windows, AutoSys cannot be installed from a network drive that is not mapped to a letter drive (for example, "X"). To install from a network drive, map the shared network drive (for example, map \\ server\ drive to X:\) and then install from that drive. Note that using a mapped drive instead of local media may extend the amount of time required for the installation.
- On UNIX, consider the following information:
The DISPLAY environment variable is required only for GUI mode of installation.
 - To install AutoSys in Console mode which is the default mode, unset the DISPLAY environment variable as it is not required for Console mode of installation.
 - To install AutoSys in GUI mode, the X-Windows(X11) packages for the respective operating systems are the prerequisites. These are the required dependencies from the Installer Technology (InstallAnywhere).

Example:

On Red Hat Enterprise Linux (RHEL), to install AutoSys in GUI mode, the following are the required packages:

- libXext.x86_64
- libXrender.x86_64
- libXtst.x86_64
- AppServices and WebServices are required to be installed on the Agent for the HTTP and SOAP features to work. You must start the installation program using one of the following commands changing to the Agent install directory:

Example:

- **UNIX:** `cd opt/CA/WA_Agent`
- **Windows:** `cd C:\Program Files\CA\WA Agent`

You install these agent plug-ins using an interactive program that prompts you for the required information.

- AppServices
 - **UNIX:** `./PluginInstaller addons/appservices.cfg <agent directory>`
 - **Windows:** `PluginInstaller addons/appservices.cfg <agent directory>`
- WebServices
 - **UNIX:** `./PluginInstaller addons/webservices.cfg <agent directory>`
 - **Windows:** `PluginInstaller addons/webservices.cfg <agent directory>`

For more information on how to download and install the plug-ins, refer to the Workload Automation Agents Documentation.

- To install Web UI, the recommended machine memory size is 8GB RAM.
- On Linux, before starting AutoSys installation, ensure Oracle vendor file: `tnsnames.ora` has read permissions (`chmod 644`).
- On Linux and AIX, the `set-user-ID` set on `AutoSysCommandWrapper` is not honored if the underlying file system is mounted with `nosuid`.
To resolve this issue, remount the file system with `suid` using the following command:


```
mount -o remount,suid <filesystem>
```

- AutoSys uses 1024 key length certificates. By default, CA EEM generates root certificates with a key length of 1024. You can generate root certificates with various key lengths (2048 or more) and use them on an EEM Server. For more information, refer to [Configure CA EEM](#).
- If you want to install AutoSys with the MSSQL server on Linux, you must install the SQL Server client tools (RPM package name: mssql-tools) and ODBC software (RPM package name: unixodbc-dev). For more information, refer to SQL Server Command Line Tools.
- On Windows, we recommend that you use a display resolution of 1920 X 1080 or below. Otherwise, the installation wizard may appear distorted.
- If there is no sufficient space available in the \$TMPDIR or %TEMP% directory, you can set the IATEMPDIR environment variable to configure the temp location to a path in which there is sufficient space.

During an upgrade, if you are using the Sybase Kerberos environment, the JDBC connectivity issue occurs. This happens because CA Workload Automation AE 11.3.6 SP6's JASS configuration file has a duplicate login entry.

To resolve the JDBC connectivity issue:

1. Navigate to %AUTOSYS%\config\cmplapp on Windows, or \$AUTOSYS/config/cmplapp on UNIX.
2. Locate the JAAS.conf file and open it.
3. In the file, replace the following content

Content to be replaced:

```
initiate {
    initiate {
        com.sun.security.auth.module.Krb5LoginModule
        requireduseTicketCache=true;
    };
};
```

Replace with:

```
initiate {
    com.sun.security.auth.module.Krb5LoginModule required
    useTicketCache=true;    };
```

- For Linux Rhel 8, the libnsl-2.28-101.el8.x86_64 package is required for the Scheduler, Application Server, and CA EEM Server.

Installation Preparation

Review the following information before you install AutoSys Workload Automation:

Components to Install

You can use the setup program to install different combinations of components on various computers, and also to install these components for each instance that you want to run. A custom installation provides greater flexibility in component selection, while a typical installation automatically selects all applicable product components. Run the setup program on each computer on which you install a AutoSys Workload Automation component.

The following components are available for installation:

- **Scheduler**
Interprets AutoSys Workload Automation events and, based on job definitions, initiates actions through the agent.
- **Application Server**

Enables programs to securely access the database without installing a database client.

- **Agent**
Performs tasks such as running jobs and reporting their status.
- **Client**
Lets you define, run, and maintain all AutoSys Workload Automation instances and jobs.
- **SDK**
Provides the necessary tools to build your own applications to manipulate product data.
- **SSA**
Enables components use in a single multiplexed communication port.
- **RESTful Web Server**
Provides a web service environment to host the AutoSys Workload Automation web services.
- **SOAP**
Hosts web services using Simple Object Access Protocol (SOAP) service-oriented architecture.
- **Web UI**
Provides a graphical user interface to define, schedule, monitor, or generate report on workload objects.
- **Web UI Collector**
The Web UI Collector collects the AutoSys artifacts such as Job definitions, Alarms, and populates the Web UI database, which helps in monitoring and managing the workload.
- **CA EEM**
Secures AutoSys Workload Automation objects and manages role-based access to users in the enterprise.

Identify Computers

Before you install AutoSys Workload Automation, identify the computers on which you want to install the required components, and decide which components to install on each computer.

Server Computers

The *server* is a computer on which the database, the scheduler, or both reside.

Identify at least one computer on which to install the database. To ensure high availability of the database, you can install dual event servers; in that case, you need two computers on which to install databases.

NOTE

The terms *event server* and *database* are often used interchangeably.

You can also install a shadow scheduler to ensure high availability of the scheduler. This requires two extra computers: a shadow computer and a tie-breaker computer. The primary, shadow, and tie-breaker computers must all be of the same type, either Windows or UNIX. All three computers must be defined in the same instance.

Host Machine Checklist

Use the following table to collect information about your host machine:

Information	Your Selection or Value
Platform	
Operating system/version	
Host name	
(UNIX only) Password of root	
Minimum requirements — Available free memory (2 GB)	
Minimum requirements — Available free disk space (2 GB)	

Instance name (AUTOSERV)	
Name of the data server that contains the database	
The location of the Oracle tnsnames.ora file	
The location of the Sybase interfaces file	
Password for the database user who has been granted DBA role	

Client and Agent Computers

Identify one or more computers on which to install the client, the agent, or both. You can define an agent computer to run jobs only, or a client computer to run both the jobs and the client utilities that let you define and monitor jobs.

Database Considerations

General Database Considerations

These considerations are applicable to all AutoSys Workload Automation components.

Microsoft SQL Database Considerations

The following are important considerations when you specify a Microsoft SQL database:

- An external Microsoft SQL database instance specified during Web UI installation must be configured for standard user authentication. Windows user authentication is not supported by Web UI.
- For an external Microsoft SQL database named instance, the host name and instance name should be specified as follows: *host_name\instance_name*
- A named Microsoft SQL instance cannot run on a dynamic port with Web UI; however you can use named instance with a static port. You must specify the correct static port number. Web UI does not validate the port number field of the named instance.
- The instance name is applicable only if SQL Browser is running.
- For the default instance of Microsoft SQL Server, it is not required to specify the instance name.
- In the Microsoft SQL database, the AutoSys server must be configured with Binary or Case Sensitive collation to take advantage of case sensitivity. You can create AutoSys objects on a database with Case Sensitive collation, multiple objects with different cases but same value. If your database is configured with Case Insensitive collation, you may not be able to take advantage of additional benefits of SQL server.

Example:

- If your database collation is SQL_Latin1_General_CP1_CS_AS you can insert jobs with specific cases; a job with the job names such as test, Test_, TEST_ are supported.
- If your database collation is SQL_Latin1_General_CP1_CI_AS and you already have a job called test_, then trying to add a job name Test_, the following error messages are displayed:
CAUAJM_I_50323 Inserting/Updating job: Test_
CAUAJM_E_10288 *** CANNOT INSERT Job: Test_, because it ALREADY EXISTS ***
CAUAJM_E_10302 Database Change WAS NOT successful.

We recommend, if the database is configured with Case Insensitive collation, your Administrator must create the database with Case Sensitive collation.

Oracle Database Considerations

The following are important considerations when you specify an Oracle database:

- If you specify Oracle TNS and the Oracle database is on a remote computer, copy the tnsnames.ora file to the computer on which Web UI will be installed prior to installation. Do not change the name of the tnsnames.ora file.
- The value specified for the tnsnames.ora path field must include the file name. For example, c:\oracle\tnsnames.ora.
- After Web UI is installed, do not delete the tnsnames.ora file or move the file to a different location.
- If Oracle TNS is selected in conjunction with UNIX Run As User, that user must have read permission for the tnsnames.ora file and the path to the tnsnames.ora file.
- The AutoSys installer creates the database and users for you.
- Web UI allows the service-name based connectivity.

AutoSys-Specific Database Configuration Considerations

AutoSys Workload Automation uses database-specific environment variables and configuration settings to locate and connect to the database (event server).

AutoSys Workload Automation uses the TNS alias name to look up Oracle database information. To ensure that the compliance application and the RESTful web services can connect to the server, configure the Oracle client to store the TNS name in the tnsnames.ora file or change the default settings on the JDBC driver.

Microsoft SQL Server

For Microsoft SQL Server, you must install the Microsoft SQL Server client utilities on each scheduler and application server computer. Then, ensure that there is database connectivity between the event server, scheduler, and application server computers. When you install AutoSys Workload Automation, you must indicate the appropriate event server information, including the event server name (Microsoft SQL Server name) and database name.

In addition, you must meet Microsoft SQL Server security requirements for the AutoSys Workload Automation database users.

NOTE

For AutoSys Workload Automation to support Microsoft SQL Server, mixed mode authentication must be enabled.

Microsoft SQL Server Always On

When you configure AutoSys Workload Automation to connect with Microsoft SQL Server Always On, ensure that you create a database user on the secondary replicas for AutoSys Workload Automation to work properly when a fail over occurs.

Check the database user SID on the primary server and then create a user with the same SID on the secondary replicas. Run the following command to create the database user SID on the primary server; for example, database user SID is 'autosys'.

```
select * from sys.server_principals where name='autosys'
```

Use this SID to create a user with the same SID on the secondary replicas, as follows:

```
CREATE LOGIN autosys
WITH PASSWORD='interOP@1234',
SID=0x6A1A04178F98BD4C9C2F7988FD075071
```

NOTE

For information about connecting to Microsoft SQL Server Always On, see the Microsoft SQL Server Always On documentation.

Microsoft SQL Azure

Consider the following information before using Microsoft SQL Azure:

- The password policy is governed by the security policies in the Windows Operating System.
- A connection is associated with a single database. When you connect to Azure, specify the connection. If you are creating any additional users to access the AutoSys Workload Automation database, ensure that you mention the default database for the users.

NOTE

For more information about Microsoft SQL Azure, see the Microsoft SQL Azure documentation.

Oracle

AutoSys Workload Automation uses the Transparent Network Substrate (TNS) alias name to locate and connect to Oracle databases. Depending on how the Oracle client is configured to use the TNS names file (tnsnames.ora) or Lightweight Directory Access Protocol (LDAP) to access the TNS alias name, it uses either the tnsnames.ora file or the sqlnet.ora and ldap.ora files to look up the alias name. These files typically reside in the administrative directories. When configuring the TNS alias name and Oracle drivers, consider the following points:

- The thin Oracle Java Database Connectivity (JDBC) driver supports the lookup of the TNS alias name in the tnsnames.ora file only. To ensure that the AutoSys Workload Automation compliance application and RESTful web service can connect to the database, configure the Oracle client to store the alias name in that file or change the default configuration of the JDBC driver.
- The Oracle thin driver does not accept TNS alias names in LDAP. When you configure an alias name in this format, ensure that you change the configuration of the JDBC driver to use Oracle Client Interface (OCI) instead of Oracle thin and store the alias name in the LDAP file (ldap.ora) instead of the TNS names file.

NOTE

- For more information about configuring the JDBC Oracle driver or the Oracle client, see the Oracle documentation.
- If you edit the tnsnames.ora file, follow the best practices that are recommended by Oracle to ensure that you do not disturb the formatting. Ensure that no extra spaces or unnecessary characters are added in the tnsnames.ora file as they might result in errors while accessing the database.

Sybase

The compliance application, web services, and datamover access the AutoSys Workload Automation database using Java Database Connectivity (JDBC). To connect to the Sybase database, you must install Sybase jConnect 7 from the Sybase installation media or download jConnect 7 from Sybase. AutoSys Workload Automation is configured to use the following default location:

- (UNIX) \$SYBASE/jConnect-7_0/classes/jconn4.jar
- (Windows) %SYBASE%\jConnect-7_0\classes\jconn4.jar

NOTE

For information about installing the JDBC driver for Sybase databases, see the Sybase documentation.

The Sybase configuration file defines database information using the following environment variables:

- **DSQUERY**
Defines the name of the Sybase data server.
- **SYBASE**
Identifies the complete path to the Sybase software directory.

To ensure that the AutoSys Workload Automation installer sets the appropriate default values for these variables, set the SYBASE environment variable before running the setup script.

Installation Considerations

Consider the following information when installing AutoSys Workload Automation.

EEM Install Considerations

We recommend that you review the following important considerations before you install CA EEM on UNIX:

- We recommend that you install CA EEM on a dedicated server.
- If your environment uses CA EEM with AutoSys Workload Automation, all AutoSys Workload Automation and Web UI servers must reference the same instance of CA EEM. Also, all instances of Web UI in a High Availability group must reference the same instance of CA EEM.
- If you install CA EEM on a computer where AutoSys Workload Automation is installed and active, the CA EEM installation process shuts down the AutoSys Workload Automation active processes. You must start the AutoSys Workload Automation processes manually after the CA EEM installation is complete.
- On Linux, CA EEM supports kernel version 2.6 and later.
- For information about installing CA EEM, see the [CA EEM](#) documentation.

NOTE

A CA EEM administrator user (EiamAdmin) is created when you install CA EEM. You must remember the password that you provide for this administrator user. You need this information when you install the Broadcom product that uses CA EEM or to manage and configure CA EEM.

Install AutoSys Workload Automation with Sybase

When installing AutoSys Workload Automation with a Sybase database, consider the following points:

Database User Privileges

Ensure that the database user has the sa_role and sso_role privileges.

(UNIX) Locale

The LANG variable specifies the locale of a computer. To enable the Sybase computer and the AutoSys Workload Automation computer to communicate, ensure that the two computers support the same locale.

To display the LANG environment variable, run the following command from both the AutoSys Workload Automation session and the session from which Sybase was started:

```
echo $LANG
```

If they are different, change the LANG value to match the value on the Sybase session by running the following command from the AutoSys Workload Automation session:

```
LANG=value; export LANG
```

Remote Connections

Verify that the computer where you plan to install the AutoSys Workload Automation server can connect to the remote database using one of the database client utilities before installing the server. Contact your database administrator about any problems connecting to the remote server.

User Connections

Before you install AutoSys Workload Automation, set the Sybase available user connections as appropriate. AutoSys Workload Automation requires up to 115 free Sybase user connections, depending on which components you install.

The following list states the number of Sybase user connections required for each component of AutoSys Workload Automation:

- Scheduler: 16+4
- Application Server: 35
- High Availability (two Schedulers and two Application Servers): 110
- Tie-breaker Scheduler: 5

For example, a typical AutoSys Workload Automation server installation with High Availability requires 115 Sybase user connections. When determining the minimum number of user connections that are required to support your configuration, account for the user connections used by Sybase itself, typically about 15 for each machine. The number may vary depending upon the version of Sybase. Thus, if AutoSys Workload Automation is the only Sybase application in a typical High Availability environment, the minimum number of user connections that are needed would be calculated as follows: (2 Schedulers x 20) + (2 Application Servers x 35) + (1 Tie-breaker Scheduler x 5) + (3 machines x 15) = 150.

Run the following SQL command to determine the number of configured user connections:

```
1>sp_configure 'user connections'
2>go
```

Run the following SQL command to determine the number of user connections currently in use:

```
1>sp_who
2>go
```

The number of rows returned from the aforementioned command represent the number of user connections currently in use. This number can be subtracted from the configured amount to determine the number of free user connections.

Run the following SQL command to set the number of user connections to 64:

```
1>sp_configure 'user connections', 64
2>go
```

Page Size

The default Sybase installation comes with a page size of 2 KB. However, AutoSys Workload Automation requires the minimum page size to be at least 4 KB. If you are running CA WA Advanced Integration for Hadoop jobs, the minimum page size must be at least 8 KB.

Enter the following command to check the current page size:

```
select @@maxpagesize
```

NOTE

For information about changing the page size, see the Sybase documentation.

Runtime Configuration

AutoSys Workload Automation uses the Open Client/Open Server interface with Sybase. If your Sybase environment uses the runtime configuration file (ocs.cfg) and Sybase Open Client/Open Server products are used in your Sybase environment, you must update ocs.cfg with the corresponding application sections. Add the following information to ocs.cfg, at \$SYBASE/\$SYBASE_OCS/config (UNIX) or %SYBASE%\%SYBASE_OCS%\ini (Windows), before starting the AutoSys Workload Automation processes:

```
[CA WAAE Application Server]
CS_SEC_ENCRYPTION = CS_TRUE
[CA WAAE Scheduler]
CS_SEC_ENCRYPTION = CS_TRUE
```

Install AutoSys Workload Automation with Oracle

When installing AutoSys Workload Automation with an Oracle database, consider the following points:

Remote Connections

Verify that the computer where you plan to install the AutoSys Workload Automation server can connect to the remote database using one of the database client utilities before installing the server. Contact your database administrator about any problems connecting to the remote server.

Oracle Memory Management

Oracle Memory management involves maintaining optimal sizes for the Oracle database instance memory structures (System Global Area (SGA) and the instance Program Global Area (PGA)) as demands on the database change. Oracle database supports various memory management methods, which are chosen by the initialization parameter settings. Oracle recommends Automatic Memory Management (AMM) for databases when the total size of the SGA and PGA memory is less than or equal to 4 GB and Automatic Shared Memory Management (ASMM) for databases when the total size of the SGA and PGA memory is 4 GB or larger.

For large enterprises, we recommend that Oracle ASMM is enabled with the SGA and PGA size set as 4832 MB and 1608 MB respectively. When cursor_sharing value is equal to SIMILAR, the SGA and PGA memory requirements are less when compared to that when cursor_sharing value is equal to EXACT. For better resource management, we recommend the Oracle DBA to monitor the Oracle memory usage as suggested by Oracle and tune the SGA and PGA size to suit their workload needs. For more information about Oracle Memory Management, see the [Oracle Memory Management documentation](#).

Configure the tnsnames.ora File for TAF Enablement

The following configuration changes on the Oracle client do not impede AutoSys Workload Automation from supporting nodes failing over on the Oracle Server. It only prevents the Oracle failover mechanism from controlling the failover on the Oracle Server for the database connections that are opened by AutoSys Workload Automation.

If the Oracle client is configured for the Oracle Server side failover and not Transparent Application Failover (TAF), the entries in the tnsnames.ora file are similar to the following:

```
RAC-NOTAF =
  (DESCRIPTION =
    (ADDRESS = (PROTOCOL = TCP) (HOST = xxx) (PORT = 1521))
    (ADDRESS = (PROTOCOL = TCP) (HOST = xxxx) (PORT = 1521))
    (LOAD_BALANCE = yes)
    (FAILOVER = false)
    (ENABLE = broken
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACDB)
    )
  )
)
```

TAF mode prevents AutoSys Workload Automation from accessing its database schema on the active database node following a failover on the Oracle Server. So, for AutoSys Workload Automation to work with the Oracle Server that is running in the TAF mode, ensure that the Oracle client on the AutoSys Workload Automation server is configured for the Oracle RAC failover (FAILOVER = ON). No configuration changes are required on the Oracle server.

If the entries in the tnsnames.ora file are similar to the following, Oracle client is configured for TAF:


```

RACTAF =
  (DESCRIPTION =
    (FAILOVER = ON)
    (LOAD_BALANCE = ON)
    (ADDRESS_LIST =
      (ADDRESS = (PROTOCOL = TCP) (HOST = XXX ) (PORT = 1521))
      (ADDRESS = (PROTOCOL = TCP) (HOST = XXXX ) (PORT = 1521))
    )
    (CONNECT_DATA =
      (SERVER = DEDICATED)
      (SERVICE_NAME = RACTAF)
      (failover_mode=
        (type=SELECT)
        (method=PRECONNECT)
      )
    )
  )
)

```

When the Oracle client environment is configured for TAF, the Oracle client library proactively notifies the scheduler or application server when it detects a loss of connection to the database. When notified, the scheduler or application server sleeps for five seconds and instructs the Oracle client library to either resume checking the status of the database connection or end all further status checks. The [DBEventReconnect](#) parameter controls the number of times the scheduler or application server instructs the Oracle client library to resume checking the status of the database connection before shutting down.

NOTE

- For more information about running the Oracle server in the TAF mode, see the Oracle documentation.
- When the nodes on the Oracle server are frequently going down while the jobs are running continuously, you might encounter an intermittent issue with jobs stuck in the intermediate state.

Define an Oracle User to Install AutoSys Workload Automation Oracle Objects

The installer connects to the Oracle AEDB database to create database objects if you select the Create the Tablespaces, Database Users, and Roles check box during installation and define an Oracle user with sufficient permissions. To define an Oracle user with authority to install AutoSys Workload Automation, use the following SQL statements:

```

CREATE USER <user> IDENTIFIED BY <password>;
GRANT DBA TO <user>;
GRANT CREATE SESSION TO <user>;
GRANT SELECT ON "SYS"."DBA_TABLESPACES" TO <user> WITH GRANT OPTION;
GRANT SELECT ON "SYS"."V_$PARAMETER" TO <user>;

```

However, if you want to create the database objects manually, you must grant the following privileges to the database schema owner, database ujoadmin role, and the database user before you install AutoSys Workload Automation. These grants are also required to upgrade the product. Do not revoke these grants after you install or upgrade the product.

NOTE

The schema owner is aedbadmin and the database user is autosys in the following SQL statements.

schema owner

```

GRANT CREATE VIEW TO aedbadmin;
GRANT 'CONNECT' TO aedbadmin;

```

```
GRANT 'RESOURCE' TO aedbadmin;
GRANT 'SELECT_CATALOG_ROLE' TO aedbadmin;
GRANT CREATE PUBLIC SYNONYM TO aedbadmin
```

ujoadmin role

```
GRANT ANALYZE ANY TO ujoadmin;
GRANT SELECT ON DBA_TABLESPACES TO ujoadmin;
GRANT ujoadmin TO aedbadmin;
```

database user

```
GRANT 'CONNECT' TO autosys;
GRANT ujoadmin TO autosys;
```

NOTE

If you created the database objects manually, verify that you clear the Create the Tablespaces, Database Users, and Roles check box on the Primary Event Server Properties page during the installation.

Use Single Sign-On Wallets with Certificates for Database Access

On Oracle, you can use password or single sign-on wallets with certificates that are present on the file system to access the database. AutoSys Workload Automation uses the sqlnet.ora and tnsnames.ora files from the location that you specified in the TNS_ADMIN environment variable and the cwallet.sso file from the configured Oracle wallet location to connect to the database.

NOTE

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

You can set the database access mode that you want to use either during installation or later by using the switchDBAccessMode.pl script. For more information about switching the database access mode after installation, see [Switch the Database Access Mode](#).

Before switching AutoSys Workload Automation to use an SSL certificate to access the database, perform the following tasks:

- Generate the SSL certificates for the Oracle system administrator user. Execute the following SQL statements to create an Oracle system administrator user that is mapped to the SSL certificate:


```
create user <user> identified externally as '<Certificate Distinguished Name>';
GRANT DBA TO <user>;
GRANT CREATE SESSION TO <user>;
GRANT SELECT ON "SYS"."DBA_TABLESPACES" TO <user> WITH GRANT OPTION;
GRANT SELECT ON "SYS"."V_$PARAMETER" TO <user>;
```
- Generate the SSL certificates for the AutoSys Workload Automation database schema owner (for example, aedbadmin) and the AutoSys Workload Automation database user (for example, autosys).
- Gather the following data as you need this information during the installation process:

- TNS_ADMIN value of the Oracle system administrator user for connecting to the primary and secondary event servers.
- TNS_ADMIN value of the database user for connecting to the primary and secondary event servers.
- TNS_ADMIN value of the schema owner for connecting to the primary and secondary event servers.
- Distinguished name of the SSL certificate that you want to associate with the schema owner on the primary and secondary event servers.
- Distinguished name of the SSL certificate that you want to associate with the database user on the primary event server. The same distinguished name is used for the secondary event server.
- Absolute path of the Oracle wallet file (cwallet.sso) in the file system.
- Directory that contains the oraclepki.jar, osdt_core.jar, and osdt_cert.jar files.

NOTE

On UNIX, the database user must have read privileges on the Oracle wallet that is specified using the TNS_ADMIN environment variable. On Windows, the local system account must have read privileges on the Oracle wallet that is specified using the TNS_ADMIN environment variable.

Create the Oracle Tablespaces, Users, and Roles

An Oracle database user with administrator privileges (such as SYS) is required to create the AutoSys Workload Automation tablespaces, database users, and roles when you install server components. To bypass this requirement, run the waae_oracle.sql script before you install a server component. The waae_oracle.sql script creates the tablespaces, database users, and roles.

AutoSys Workload Automation uses the TNS alias name to locate and connect to Oracle databases. Depending on how the Oracle client is configured, it uses either the tnsnames.ora file or the sqlnet.ora and ldap.ora files to look up the alias name. These files typically reside in the administrative directories.

NOTE

- When you run the waae_oracle.sql script, connect as an Oracle database user with administrative privileges and run the waae_oracle.sql script from the Oracle command prompt.
- When you run the waae_oracle.sql script, verify that you clear the Create the Tablespaces, Database Users, and Roles check box on the Primary Event Server Properties page during the installation. The installer does not prompt you to specify the Oracle administrator user and password information.

waae_oracle.sql script—Creates Oracle Tablespaces, Users, and Roles

The waae_oracle.sql script creates the Oracle tablespaces, users, and roles.

```

Rem
Rem  Parameters to this SQL are:
Rem
Rem  1 - Data tablespace name. For example: AEDB_DATA
Rem  2 - Data tablespace size in MB. For example: 800
Rem  3 - Data tablespace directory file. For example: /home/oracle/oradata/orcl/AEDB_DATA.DBF
Rem  4 - Index tablespace name. For example: AEDB_INDEX
Rem  5 - Index tablespace size in MB. For example: 80
Rem  6 - Index tablespace directory file. For example: /home/oracle/oradata/orcl/AEDB_INDEX.DBF
Rem  7 - 'aedbadmin' database user password OR certificate distinguished name.
Rem  8 - 'autosys' database user password OR certificate distinguished name.
Rem

Rem  Create the Oracle tablespaces:
Rem
Rem  Uncomment if using OMF:

```

```

Rem CREATE TABLESPACE &1 LOGGING;

Rem  Uncomment if using ASM:
Rem CREATE TABLESPACE &1 DATAFILE '&3' SIZE &2.M AUTOEXTEND

Rem  If not using Oracle-managed space:
Rem  Create the data tablespace:
CREATE TABLESPACE &1 DATAFILE '&3' SIZE &2.M REUSE AUTOEXTEND ON NEXT 300K MAXSIZE UNLIMITED LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;

Rem  Create the index tablespace:
CREATE TABLESPACE &4 DATAFILE '&6' SIZE &5.M REUSE AUTOEXTEND ON NEXT 300K MAXSIZE UNLIMITED LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;

Rem  Create 'aedbadmin' user:
Rem  Comment if using certificate based authentication.
CREATE USER aedbadmin PROFILE DEFAULT IDENTIFIED BY "&7" DEFAULT TABLESPACE &1 ACCOUNT UNLOCK;

Rem  Uncomment if using certificate based authentication.
Rem  CREATE USER aedbadmin PROFILE DEFAULT IDENTIFIED EXTERNALLY AS '&7' DEFAULT TABLESPACE &1 ACCOUNT
UNLOCK;

SET VERIFY ON;
GRANT CREATE VIEW TO aedbadmin;
GRANT 'CONNECT' TO aedbadmin;
Rem GRANT 'EXECUTE_CATALOG_ROLE' TO aedbadmin;
GRANT 'RESOURCE' TO aedbadmin;
GRANT 'SELECT_CATALOG_ROLE' TO aedbadmin;
GRANT CREATE PUBLIC SYNONYM TO aedbadmin;

Rem  Create 'autosys' user:
Rem  Comment if using certificate based authentication.
CREATE USER autosys PROFILE DEFAULT IDENTIFIED BY "&8" DEFAULT TABLESPACE &1 ACCOUNT UNLOCK;

Rem  Uncomment if using certificate based authentication.
Rem  CREATE USER autosys PROFILE DEFAULT IDENTIFIED EXTERNALLY AS '&8' DEFAULT TABLESPACE &1 ACCOUNT UNLOCK;

Rem  Create 'ujoadmin' role:
CREATE ROLE ujoadmin NOT IDENTIFIED;
GRANT ANALYZE ANY TO ujoadmin;
GRANT SELECT ON DBA_TABLESPACES TO ujoadmin;
GRANT ujoadmin TO aedbadmin;
GRANT 'CONNECT' TO autosys;
GRANT ujoadmin TO autosys;

Rem  Alter 'aedbadmin' user:
ALTER USER aedbadmin QUOTA UNLIMITED ON &1;

Rem  Comment if using OMF:
ALTER USER aedbadmin QUOTA UNLIMITED ON &4;

```

On UNIX, the `waee_oracle.sql` script is located in the `ORA/DB` directory in the AutoSys Workload Automation Release 12.0 installer.

On Windows, you can download the `waae_oracle.sql` script by selecting the SQL script to create Oracle database tablespaces, users, and roles option under Database Scripts on the AutoSys Workload Automation Release 12.0 Product Explorer window.

NOTE

Follow the instructions in the `waae_oracle.sql` script file if you are using Oracle Managed Files (OMF), Automatic Storage Management (ASM), or SSL certificates to access the database.

The `waae_oracle.sql` script has the following format:

```
waae_oracle.sql ADB_TS_DATANM ADB_TS_DATASZ ADB_TS_DATADIR ADB_TS_IXNM ADB_TS_IXSZ
ADB_TS_IXDIR aedbadmin_password|certificate_DN autosys_password|certificate_DN
```

- **ADB_TS_DATANM**
Specifies the Oracle data tablespace name.
- **ADB_TS_DATASZ**
Specifies the Oracle data tablespace size (in MB). The minimum value is 800.
- **ADB_TS_DATADIR**
Specifies the Oracle data tablespace directory path, including the database file (.dbf). This directory must exist on the Oracle server.
- **ADB_TS_IXNM**
Specifies the Oracle index tablespace name.
- **ADB_TS_IXSZ**
Specifies the Oracle index tablespace size (in MB). The minimum value is 80.
- **ADB_TS_IXDIR**
Specifies the Oracle index tablespace directory path, including the database file (.dbf). This directory must exist on the Oracle server.
- **aedbadmin_password | certificate_DN**
Specifies the password of the database schema owner (for example, aedbadmin) or the distinguished name of the SSL certificate that is associated with the database schema owner (for example, aedbadmin).
- **autosys_password | certificate_DN**
Specifies the password of the database user (autosys) or the distinguished name of the SSL certificate that is associated with the database user (autosys).

Example: Create the Oracle Tablespaces, Users, and Roles on UNIX

This example creates the Oracle tablespaces, users, and roles when you are using password to connect to the Oracle database. The `waae_oracle.sql` script is copied to the `/tmp/waae` directory. The Oracle data tablespace that is named `AEDB_DATA` stores up to 800 MB of data and is located in the `/home/oracle/product/10.2.0/oradata/orcl` directory. The database file is named `AEDB_DATA.DBF`. The Oracle index tablespace that is named `AEDB_INDEX` stores up to 80 MB of data and is located in the `/home/oracle/product/10.2.0/oradata/orcl` directory. The database file is named `AEDB_INDEX.DBF`. The schema owner password is `AedbAdm1` and the database user password is `AutoSys1`.

The database user example is 'autosys' and the schema owner example is 'aedbadmin'.

```
> sqlplus /nolog
SQL*Plus: Release 10.2.0.1.0 - Production on Tue Feb 15 14:39:35 2011
Copyright (c) 1982, 2005, Oracle. All rights reserved.
SQL> connect SYS/***** as SYSDBA
Connected.
SQL> @/tmp/waae/waae_oracle.sql AEDB_DATA 800 /home/oracle/oracle/product/10.2.0/
oradata/orcl/AEDB_DATA.DBF AEDB_INDEX 80 /home/oracle/oracle/product/10.2.0/oradata/
orcl/AEDB_INDEX.DBF AedbAdm1 AutoSys1
```

The output is displayed as follows:

```
old 1: CREATE TABLESPACE &1 DATAFILE '&3' SIZE &2.M REUSE AUTOEXTEND ON NEXT 300K
MAXSIZE UNLIMITED LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO
new 1: CREATE TABLESPACE AEDB_DATA DATAFILE '/home/oracle/oracle/product/10.2.0/
oradata/orcl/AEDB_DATA.DBF' SIZE 800M REUSE AUTOEXTEND ON NEXT 300K MAXSIZE UNLIMITED
LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO
Tablespace created.
old 1: CREATE TABLESPACE &4 DATAFILE '&6' SIZE &5.M REUSE AUTOEXTEND ON NEXT 300K
MAXSIZE UNLIMITED LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO
```

Create the Oracle Tablespaces, Users, and Grants Before Installing AutoSys

Before installing AutoSys, you can create tablespaces, users, and grants as follows:

Tablespaces

```
CREATE SMALLFILE TABLESPACE <TABLESPACENAME>_DATA DATAFILE '<TABLESPACE_PATH>' SIZE 1000M LOGGING EXTENT
MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
CREATE SMALLFILE TABLESPACE <TABLESPACENAME>_INDEX DATAFILE '<TABLESPACE_PATH>' SIZE 80M REUSE AUTOEXTEND ON
NEXT 300K MAXSIZE UNLIMITED LOGGING EXTENT MANAGEMENT LOCAL SEGMENT SPACE MANAGEMENT AUTO;
CREATE USER <schema_user> PROFILE "DEFAULT" IDENTIFIED BY autosys DEFAULT TABLESPACE <TABLESPACENAME>_DATA
TEMPORARY TABLESPACE TEMP QUOTA UNLIMITED ON <TABLESPACENAME>_DATA ACCOUNT UNLOCK;
CREATE USER <application_user> PROFILE "DEFAULT" IDENTIFIED BY autosys DEFAULT TABLESPACE
<TABLESPACENAME>_DATA TEMPORARY TABLESPACE TEMP QUOTA UNLIMITED ON <TABLESPACENAME>_DATA ACCOUNT UNLOCK;
```

Schema User

```
GRANT CREATE SEQUENCE TO <schema_user>;
GRANT CREATE TABLE TO <schema_user>;
GRANT CONNECT TO <schema_user>;
grant alter any table to <schema_user>;
grant create any index to <schema_user>;
grant drop any index to <schema_user>;
grant SELECT_CATALOG_ROLE to <schema_user>;
grant unlimited tablespace to <schema_user>;
```

In addition to the above, you must grant the following Schema user and Application DB user permissions for Autosys:

Schema User

```
GRANT CREATE PUBLIC SYNONYM to <schema_user>;
CREATE ROLE ujoadmin NOT IDENTIFIED;
GRANT ANALYZE ANY TO ujoadmin;
GRANT SELECT ON DBA_TABLESPACES TO ujoadmin;
GRANT ujoadmin TO <schema_user>;
GRANT 'CONNECT' TO <application_user>;
```

Application DB User

```
GRANT CREATE SESSION TO <application_user>;
```

(UNIX) Installing AutoSys Workload Automation with Oracle

Consider the following points when installing AutoSys Workload Automation with Oracle on UNIX or Linux:

Oracle Directory Permissions

AutoSys Workload Automation requires access to the Oracle shared libraries. By default, Oracle does not allow this access. Perform one of the following procedures for AutoSys Workload Automation to function properly:

- Before you run the AutoSys Workload Automation installation, create the autosys OS user and specify the OSDBA group of Oracle as the autosys primary group, or add the OSDBA group of Oracle to the autosys supplemental group list.
- Allow the AutoSys Workload Automation installation to automatically create the autosys OS user. When you are prompted to specify the autosys owner and group, specify the OSDBA group of Oracle as the autosys group.

If one of these two procedures is not followed, the AutoSys Workload Automation startup fails with the following message:

```
CAUAJM_E_90013 Unable to load Oracle client libraries </opt/CA/CALib/libclntsh.so:
cannot open shared object file: No such file or directory>
CAUAJM_E_10368 Failed to connect to Oracle server: <SID>
Cannot find key="" in the resource bundle
CAUAJM_E_10649 Server orcl was not available during connection operation.
```

Determine the OSDBA group

To determine the OSDBA group, issue the following command:

```
ls -dl $ORACLE_HOME | awk '{print $4}'
```

Modify an Existing autosys OS User

To add the owner of an existing autosys OS user to the OSDBA group, issue the following command. For this example, the OSDBA group is dba and the autosys owner is autosys.

HP-UX, Linux, Solaris:

```
usermod -G dba autosys # To add to the supplemental group list
usermod -g dba autosys # To set as the primary group
```

AIX:

```
chuser groups=dba autosys # To add to the supplemental group list
chuser pgrp=dba autosys # To set as the primary group
```

Create the autosys OS User

To create the autosys OS user and set its primary or supplemental group as the OSDBA group, issue the following command:

HP-UX, Linux, Solaris:

```
useradd -G dba autosys # To add to the supplemental group list
useradd -g dba autosys # To set as the primary group
```

AIX:

```
mkuser groups=dba autosys # To add to the supplemental group list
mkuser pgrp=dba autosys # To set as the primary group
```

Use Oracle Instant Client

If an Oracle user decides to use the Oracle Instant Client instead of a native Oracle client, the following modifications must be made to the base Oracle Instant Client installation:

- Oracle Instant Client users must set up a directory structure that looks exactly like a native Oracle client installation.
- The shared libraries must reside in the ORACLE_HOME/lib directory and the sqlplus binary must reside in the ORACLE_HOME/bin directory. In addition, the user must create a soft link for libclntsh in the ORACLE_HOME/lib directory.

NOTE

If you install AutoSys Workload Automation (that uses Oracle Instant Client to connect to the database) as a non-root user having root privileges, you must set the Oracle Instant Client library path in the `/etc/profile` file before you install AutoSys Workload Automation. For example, on Linux, set the Oracle Instant Client library path as follows:

```
export LD_LIBRARY_PATH=oracle_instant_client_path:$LD_LIBRARY_PATH
```

oracle_instant_client_path Specifies the path where the Oracle Instant Client is installed.

Example: `/home/instantclient_11_2`

Install AutoSys with MSSQL

The following are important considerations when you specify a Microsoft SQL database:

- For an external Microsoft SQL database named instance, the host name and instance name should be specified as follows: `host_name\instance_name`
- A named Microsoft SQL instance cannot run on a dynamic port with Web UI; however you can use named instance with a static port. You must specify the correct static port number.
- The instance name is applicable only if SQL Browser is running.
- For the default instance of Microsoft SQL Server, it is not required to specify the instance name.
- **Microsoft SQL Server Always On**

When you configure Web UI to connect with Microsoft SQL Server Always On, follow these steps:

- Create a Web UI db in the MS SQL server and then create a Web UI user.
- Create a Web UI user on the secondary replicas for Web UI to work properly when a fail over occurs.
- Check the Web UI user SID on the primary server and then create a user with the same SID on the secondary replicas.

NOTE

For information about connecting to Microsoft SQL Server Always On, see the Microsoft SQL Server Always On documentation.

Install Dual Event Servers

Install and configure the two databases before you can use them, and then set the appropriate configuration parameters.

When installing and configuring dual event servers, consider the following points:

- The two event servers must reside on two different database servers, running on different computers, to avoid a single point of failure.
- The two event servers must have unique names.
- Both databases must be of the same type; for example, Oracle.
- The scheduler does not start unless it can connect to both databases.
- The scheduler and application server installations for an instance must have the same event server settings in the `$AUTOUSER/config.$AUTOSERV` file (UNIX) or the AutoSys Workload Automation Administrator utility (Windows).
- The two event servers system time is synchronized.

Install Shadow and Tie-Breaker Schedulers

Consider the following points when installing shadow and tie-breaker schedulers:

- Running AutoSys Workload Automation in high availability mode requires both primary and shadow schedulers.
- Running AutoSys Workload Automation in high availability mode with dual event servers requires primary, shadow, and tie-breaker schedulers.
- The scheduler computers must have application server and agents installed.
- The primary, shadow, and tie-breaker schedulers are at the same release level.
- The primary, shadow, and tie-breaker schedulers must all have the same instance name.
- The primary, shadow, and tie-breaker schedulers must use the same type of database and must connect to the same primary and secondary event servers.
- The high availability poll interval configuration parameter must be identical for all schedulers to help ensure that the primary, shadow, and tie-breaker schedulers properly evaluate the high availability state of the system.
- Install the software on a local drive on the primary, shadow, and tie-breaker scheduler computers, *not* on a network drive.

iGateway Configuration Information

The following information applies to iGateway configuration:

- Invalid values are not allowed for JVM memory options. For example, iGateway does not start when JVM minimum memory is set to 10 bytes.
- All configuration files are case-sensitive.
- Performance of iGateway can be affected if the number of concurrent connections exceeds 1024. This limitation comes from the select() API.
- For all UNIX platforms, if:::1 is available then:::1 must be resolved to localhost.
- On UNIX platforms, if the IPV6 protocol is enabled, the loopback address (:::1) must be available; otherwise SafetyNet can fail.
- If you are using safetynet in iGateway 4.5 and upgrade to iGateway 4.7.2.3, your existing safetynet group files are not upgraded. The safetynet group files are required to run the safetynet process. Therefore, your safetynet process does not start automatically when iGateway server 4.7.2.3 starts.
To resolve this issue, update your safetyent group files according to sponsors.group.template file that is shipped with the iGateway installer. CertificateManager tag has been introduced from iTech 4.6.0 which is a mandatory tag in both gateway.conf and safetynet group files. The tag must be present for the corresponding process to start.

AutoSys Workload Automation and VMWare vMotion

If you are using VMWare, we recommend that you set up and configure your AutoSys Workload Automation virtual machines on physical ESXi hosts with plenty of compute resources and storage available on both the virtual machines and ESXi hosts. Do not use vMotion to move the virtual machines to other ESXi hosts.

- Set up and configure the following components on physical ESXi hosts:
 - Scheduler
 - Application server (as_server)
 - Event server
 - Web UI server
 - Web UI database
 - CA EEM server
 - Workload Automation Agent for UNIX, Linux, or Windows – If you are using vMotion or IPs change frequently, [IP caching must be disabled](#) on the application servers.
- AutoSys Workload Automation establishes hundreds of network connections and sends thousands of communications per second. If you move the virtual machine using vMotion, the communications are interrupted and in most cases the connections must be re-established as they become stale or broken; thus, requiring the application process to be recycled.

- By default, IP caching is utilized for Workload Automation Agent for UNIX, Linux, or Windows and application servers and so IPs must remain the same when the instances are up and running. IP caching is used to quickly establish the connection and send the appropriate data, rather than calling upon DNS for each transaction that might cause some latency. If you must perform an IP change, in most cases the application server process needs a SIGHUP and in some cases the entire instance must be recycled (scheduler and application server processes).

If you cannot dedicate these virtual machines to specific ESXi hosts and must run in a cluster, we recommend that you:

- Either disable DRS, set it to Manual or utilize Partially Automated, as this only manages what host it gets initialized to at startup, though every other move is manual.
- Before allowing a move, shut down the AutoSys Workload Automation processes as you typically would when performing any maintenance. This way processing ceases until the processes are brought back up, and then new connections are re-established.

NOTE

For more information about how to move or migrate a virtual machine, see the VMWare and Oracle documentation.

Environment and Database Connection

Environment Variables

AutoSys Workload Automation consults the following environment variables to run properly and to determine which instance to connect to:

- **AUTOSYS**
Identifies the full path to the AutoSys Workload Automation installation directory.
- **AUTOUSER**
Identifies the directory containing instance-wide configuration files, scheduler or application server output files, web server files, encryption files, archive output files generated during database maintenance, and sound files (for operating environments supporting audio functionality).
- **AUTOSERV**
Identifies the unique, uppercase three-letter name of a AutoSys Workload Automation instance.

To communicate with the Oracle, Sybase, or Microsoft SQL Server database, AutoSys Workload Automation relies on the environment variables.

For Windows, these variables are set during installation, and you can view the settings on the System - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.

On UNIX, if only one instance is running, the environment variables can be set during login by including their definitions in either the `.profile` or `.cshrc` file for each user accessing AutoSys Workload Automation.

The installation script generates files that are sourced by a user wanting to access AutoSys Workload Automation. The following files can be sourced from a `.profile` or `.cshrc` file and are found in the `$AUTOUSER` directory:

- `autosys.sh.hostname`—for Bourne shell users
- `autosys.csh.hostname`—for C shell users
- `autosys.ksh.hostname`—for Korn shell users
- `autosys.bash.hostname`—for Bash shell users

Optionally, frequently used variations for the `sendevent`, `autorep`, and `eventor` commands can be aliased in the files to be sourced.

Configuration Parameters

You can define the AutoSys Workload Automation environment using configuration parameters. The configuration parameters include information about the agents, event servers, scheduler, application servers, web servers, and many tunable parameters that control the behavior of AutoSys Workload Automation.

Some configuration parameters are defined when you install AutoSys Workload Automation and the rest have default settings. If the installation specifications are acceptable, do not modify these settings.

NOTE

Properly setting the required environment variables in the environment of every user and configuring AutoSys Workload Automation correctly helps to prevent potential problems. The most common problems are that AutoSys Workload Automation cannot determine which event server to connect to and it cannot locate various executables or files.

(UNIX) `/etc/auto.profile` File

The `/etc/auto.profile` file is one of the several objects that source the environment for a job. The `/etc/auto.profile` file is automatically created during UNIX installation and contains variable definitions such as `AUTOUSER`. The file is located in the computer where AutoSys Workload Automation is installed.

System environment variables are automatically set in the environment for a job. When a job is submitted, the agent processes the following additional information to source the environment, in the following order:

1. `/etc/auto.profile`
2. Environment variables defined using the `envvars` attribute in the job definition (if specified)
3. The job profile defined using the `profile` attribute (if specified)

(UNIX) `/etc/auto.inst` File

The `/etc/auto.inst` file (where *inst* specifies the AutoSys Workload Automation instance name) is automatically created during installation and is located on the computer where AutoSys Workload Automation is installed. This file is also created when you add an instance.

This file is used:

- By the application server, during startup, to defend the accidental restarts on the same machine.
- By the installer to identify the already installed instances on the machine. This file along with the `/etc/auto.profile` file is used to verify the instance information. The installer gets the configuration details of the already installed instances from the `/etc/auto.$AUTOSERV` and `$AUTOSYS/install/vars$AUTOSERV` files.
- By the `as_info` utility to get the instance information.

Database Connection

All information is stored in a Relational Database Management System (RDBMS) named event server, which is configured for AutoSys Workload Automation. Access to AutoSys Workload Automation requires a connection to this database. You must connect to the database to add, modify, control, report on, or monitor jobs, and to change certain configuration settings.

AutoSys Workload Automation stores database connection information in the `AUTOSERV` file:

- (UNIX) `$AUTOUSER/config.$AUTOSERV`
- (Windows) `%AUTOUSER%\config.%AUTOSERV%`

Configuration parameters and database environment variables tell the software which databases to connect to for an instance.

WARNING

We recommend that you do not use the same database across different AutoSys Workload Automation instances.

If you try to connect to an existing database that is in use with another AutoSys Workload Automation instance, the connection fails with the following error message:

```
# autosys_secure
CAUAJM_W_10446 Security Access to Resource Denied!
CAUAJM_E_10448 FATAL SECURITY ERROR: Tampering or corruption of the security-related
database keys detected!
CAUAJM_E_10449 Please contact CA WAAE technical support!
CAUAJM_E_60203 Program aborting due to an invalid security environment.
```

How AutoSys Workload Automation Connects to a Microsoft SQL Server Database

The following steps describe how AutoSys Workload Automation connects to a Microsoft SQL Server database:

1. Reads the AutoSys Workload Automation Administrator settings in the %AUTOUSER%\config.%AUTOSERV% file to get the connection information.
2. Uses an internally built open database connectivity (ODBC) connection to connect to the database. It requires to know the Microsoft SQL Server name or servername/instance and the Microsoft SQL Server database name.

How AutoSys Workload Automation Connects to an Oracle Database

The following steps describe how AutoSys Workload Automation connects to an Oracle database:

1. Reads the following file to get the connection information:
 - (Windows) AutoSys Workload Automation Administrator settings in the %AUTOUSER%\config.%AUTOSERV% file
 - (UNIX) \$AUTOUSER/config.\$AUTOSERV file
2. Invokes the OCI client libraries to open the connection. The OCI client libraries read the tnsnames.ora file to identify the network information to the Oracle database.

How AutoSys Workload Automation Connects to a Sybase Database

The following steps describe how AutoSys Workload Automation connects to a Sybase database:

1. Reads the following file to get the connection information:
 - (Windows) AutoSys Workload Automation Administrator settings in the %AUTOUSER%\config.%AUTOSERV% file
 - (UNIX) \$AUTOUSER/config.\$AUTOSERV file
2. Invokes the Sybase client libraries to open the connection. The Sybase client libraries read the interfaces file to identify the network information to the Sybase database.

Configure the Java Database Connectivity Driver to Use Oracle Call Interface

The compliance application and RESTful web services can connect to an Oracle database only when a JDBC driver is installed. The JDBC driver is included in the installation software for these components.

NOTE

SSL_SERVER_DN_MATCH will be read from the connection.properties first. If not found, then it would be read from TNS_ADMIN/sqlnet.ora

oracle.net.ssl_server_dn_match=false is added to connection.properties during installation.

If oracle wallet is chosen, then the installer will update the values as entered by the user during the interview phase.

By default the Oracle JDBC driver is configured to use thin mode, which does not support storing TNS alias names in LDAP. To store a TNS alias name in LDAP, configure the JDBC driver to use Oracle Call Interface (OCI).

Follow these steps:

1. Stop the AutoSys Workload Automation web server.
2. Configure the application to use the OCI JDBC driver as follows:
 - a. Open the following file:
 (UNIX) : \$AUTOSYS/config/cmplapp/connection.properties (Windows) : %AUTOSYS%\config\cmplapp\connection.properties
 - b. Locate the following parameter:
`oracle.tns.connection.string=jdbc:oracle:thin:@${dbname}`
 - c. Edit the parameter as follows:
`oracle.tns.connection.string=jdbc:oracle:oci:@${dbname}`
 - d. Save and close the file.
3. (RESTful web services) Configure the application to use the 64-bit Oracle client libraries as follows:
 - a. Open the following file:
 (UNIX) \$AUTOUSER/webserver/conf/wrapper.conf
 (Windows) %AUTOUSER%\webserver\conf\wrapper.conf
 - b. Locate the following parameter:
`wrapper.java.library.path.1=../lib`
 - c. Edit the file to include an extra parameter as follows:
`wrapper.java.library.path.1=../lib`
`wrapper.java.library.path.2=Oracle 64-bit client library path`

WARNING

Ensure that 64-bit Oracle client is installed and that the client and the JDBC driver are the same version.

- d. Save and close the file.
4. Add the LD_LIBRARY_PATH variable to install_directory/webserver/bin/waae_webserver.\$AUTOSERV file as shown below.
`LD_LIBRARY_PATH=<Oracle 64-bit client library path>:$LD_LIBRARY_PATH`
`export LD_LIBRARY_PATH`
Note: install_directory specifies the root directory where AutoSys Workload Automation is installed.
5. Oracle 11g and above) Replace the JDBC driver file as follows:
 (UNIX) - "\$AUTOSYS/lib and \$AUTOUSER/webserver/webapps/AEWS/WEB-INF/lib" with the JDBC driver file from "\$ORACLE_HOME/jdbc/lib".
 (Windows) - "%AUTOSYS%\lib and %AUTOUSER%\webserver\webapps\AEWS\WEB-INF\lib" with the JDBC driver file from "%ORACLE_HOME%\jdbc\lib".

WARNING

The ODBC jar is located in the Oracle home directory only if you use the full Oracle installer to install the database. If you use the instant installer, the file is saved to the location that you specify when you unzip the Oracle installation files..

6. Start the AutoSys Workload Automation web server.
 The JDBC driver is configured to use OCI and the Java components can connect to the database using TNS names that are stored in LDAP.

Change the Sybase JDBC JAR File Default Location for Web Server

The Sybase JDBC JAR files are not included with this release of AutoSys Workload Automation. To use web services, install Sybase jConnect 7 from the Sybase installation media or download jConnect 7 from Sybase. AutoSys Workload Automation is configured to use the following default location:

- (UNIX) \$SYBASE/jConnect-7_0/classes/jconn4.jar
- (Windows) %SYBASE%\jConnect-7_0\classes\jconn4.jar

If you do not want to use the default location of the JAR files, change it during the installation.

NOTE

Use this procedure to change the Sybase JDBC JAR file default location after AutoSys Workload Automation has been installed.

Follow these steps:

1. Locate the wrapper.conf file from the AutoSys Workload Automation sourced shell window or the instance command prompt window:
(UNIX) \$AUTOUSER/webserver/conf
(Windows) %AUTOUSER%\webserver\conf
2. Open the file using a text editor and locate the following line:
(UNIX) set.NATIVEJDBCJARPATH=\$SYBASE/jConnect-7_0/classes/jconn4.jar
(Windows) set.NATIVEJDBCJARPATH=%SYBASE%\jConnect-7_0\classes\jconn4.jar
3. Replace the path after the = with the appropriate path to the Sybase jconn4.jar file.
4. Save the file and restart the AutoSys Workload Automation web server.

(Sybase Only) Switch the Database Access Mode

After you installed AutoSys Workload Automation, you can use the switchDBAccessMode.pl script to switch the database access mode that AutoSys Workload Automation is using to connect to the database. You can switch the database access mode from password to Kerberos authentication and also do the reverse.

NOTE

For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#).

Switch to Use Kerberos Authentication to Access the Database

Before you switch to use Kerberos authentication to access the database, perform the following tasks:

- Ensure that you created a ticket granting ticket (TGT).
- Gather the JAAS configuration file details.

Follow these steps:

1. Perform any one of the following tasks based on your operating system:
(UNIX) Assuming that AutoSys Workload Automation is installed in /opt/CA/WorkloadAutomationAE and the AutoSys Workload Automation instance ID is ACE, source the AutoSys Workload Automation environment as follows:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.shell.host
cd $AUTOSYS/dbobj/switchDBAccessMode
```

– **shell**

Specifies the name of the shell that is used to execute the script or command file.

– **host**

Specifies the name of the computer where AutoSys Workload Automation is installed.

(Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*) to open the instance command prompt and enter the following command:

```
cd %AUTOSYS%\dbobj\switchDBAccessMode
```

The switchDBAccessMode directory appears.

2. Edit the following parameters in the *paramfile.sybase.EXTAUTH* parameter file as appropriate and save the file:

NOTE

The *paramfile.sybase.EXTAUTH* parameter file is available in the \$AUTOSYS/dbobj/switchDBAccessMode (UNIX) or %AUTOSYS%\dbobj\switchDBAccessMode (Windows) directory.

- **DB_ACCESS_MODE=EXTAUTH** Specifies that AutoSys Workload Automation uses Kerberos authentication to connect to the database.
 - **JAASCONFIGDETAILS** Specifies the JAAS configuration file details. Specify the details between JAASCONFIGDETAILS_START and JAASCONFIGDETAILS_END.
 - **SERVICE_PRINCIPAL_NAME_1**
(Optional) Specifies the service principal name of the primary event server.
 - **VERIFYONLY** Specifies whether you want to validate the environment and parameter file and proceed with switching the database access mode. Valid values are:
 - **YES**
Validates whether the environment and the parameter file are set up properly, but does not proceed to switch the database access mode.
 - **NO**
Validates whether the environment and the parameter file are set up properly and proceeds with switching the database access mode only if the validation is successful.
 - **RESTART_CAWAAE** Specifies whether you want to restart the AutoSys Workload Automation services after successfully updating the configurations to use Kerberos authentication to access the database. Valid values are:
 - **YES**
Restarts the AutoSys Workload Automation services after successfully updating the configurations to use Kerberos authentication to access the database.
 - **NO**
Does not restart the AutoSys Workload Automation services.
 - **SERVICE_PRINCIPAL_NAME_2**(Optional) Specifies the service principal name of the secondary event server.
3. Run the switchDBAccessMode.pl script using the following command:

```
perl switchDBAccessMode.pl paramfile
```

The script completes successfully with a return code of zero and AutoSys Workload Automation starts using Kerberos authentication to access the database. If the script fails, a non-zero return code is displayed.

The AutoSys Workload Automation services must be restarted for the settings to take effect. So, if the RESTART_CAWAAE parameter is set to NO, restart the AutoSys Workload Automation services manually.

NOTE

You can also use the switchDBAccessMode.pl script to refresh the database access mode values. Edit the values in the parameter file and run the switchDBAccessMode.pl script for the new values to take effect.

Switch to Use Password to Access the Database

Before invoking the `switchDBAccessMode.pl` script to switch the database access mode from Kerberos authentication to password mode, gather the following data as you need this information to complete the parameter file:

- Sybase system administrator user ID and password for connecting to the primary and secondary event servers
- database user password

NOTE

Perform this procedure on all computers where the scheduler, application server, or both of a same instance are installed.

Follow these steps:

1. Perform any one of the following tasks based on your operating system:(UNIX) Assuming that AutoSys Workload Automation is installed in `/opt/CA/WorkloadAutomationAE` and the AutoSys Workload Automation instance ID is `ACE`, source the AutoSys Workload Automation environment as follows:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.shell.host
cd $AUTOSYS/dbobj/switchDBAccessMode
```

– **shell**

Specifies the name of the shell that is used to execute the script or command file.

– **host**

Specifies the name of the computer where AutoSys Workload Automation is installed.

(Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*) to open the instance command prompt and enter the following command:

```
cd %AUTOSYS%\dbobj\switchDBAccessMode
```

The `switchDBAccessMode` directory appears.

2. Edit the following parameters in the *paramfile.sybase.PASSMODE* parameter file as appropriate and save the file:

NOTE

The *paramfile.sybase.PASSMODE* parameter file is available in the `$AUTOSYS/dbobj/switchDBAccessMode` (UNIX) or `%AUTOSYS%\dbobj\switchDBAccessMode` (Windows) directory.

- **DB_ACCESS_MODE=PASSWORD** Specifies that AutoSys Workload Automation uses password to connect to the database.
- **ALTER_AEDB_USER_LOGIN** Specifies whether you want to alter the database user (autosys) to log in using passwords. Valid values are:
 - **YES**
Alters the database user (autosys) to log in using passwords.
 - **NO**
Does not connect to the database to alter the database user (autosys).
- **SYSTEM_USER_NAME**
Defines the Sybase system administrator user ID to connect to the primary event server
- **SYSTEM_USER_PASSWORD**
Defines the password that is associated with the Sybase system administrator user ID (**SYSTEM_USER_NAME**) to connect to the primary event server.
- **AUTOSYS_USER_PASSWORD** Defines the password that you want to set for the database user on the primary and secondary event servers.
- **VERIFYONLY** Specifies whether you want to validate the environment and parameter file and proceed with switching the database access mode. Valid values are:
 - **YES**

Validates whether the environment and the parameter file are set up properly, but does not proceed to switch the database access mode.

- **NO**

Validates whether the environment and the parameter file are set up properly and proceeds with switching the database access mode only if the validation is successful.

- **RESTART_CAWAAE** Specifies whether you want to restart the AutoSys Workload Automation services after successfully updating the configurations to use password to access the database. Valid values are:

- **YES**

Restarts the AutoSys Workload Automation services after successfully updating the configurations to use password to access the database.

- **NO**

Does not restart the AutoSys Workload Automation services.

- **SYSTEM_USER_NAME2**

Defines the Sybase system administrator user ID to connect to the secondary event server.

- **SYSTEM_USER_PASSWORD2**

Defines the password that is associated with the Sybase system administrator user ID (SYSTEM_USER_NAME2) to connect to the secondary event server.

3. Run the switchDBAccessMode.pl script using the following command:

```
perl switchDBAccessMode.pl paramfile
```

The script completes successfully with a return code of zero and AutoSys Workload Automation starts using password to access the database. If the script fails, a non-zero return code is displayed.

The AutoSys Workload Automation services must be restarted for the settings to take effect. So, if the RESTART_CAWAAE parameter is set to NO, restart the AutoSys Workload Automation services manually.

NOTE

You can also use the switchDBAccessMode.pl script to refresh the database access mode values. Edit the values in the parameter file and run the switchDBAccessMode.pl script for the new values to take effect.

Sample Parameter Files

The following sample parameter files are located in the \$AUTOSYS/dbobj/switchDBAccessMode (UNIX) or %AUTOSYS%\dbobj\switchDBAccessMode (Windows) directory:

- paramfile.sybase.EXTAUTH—Sample file to switch the database access mode from password to Kerberos authentication.
- paramfile.sybase.PASSMODE—Sample file to switch the database access mode from Kerberos authentication to password.

An example of a parameter file for switching the database access mode from password to Kerberos authentication is as follows:

```
DB_ACCESS_MODE=EXTAUTH
JAASCONFIGDETAILS_START
com.sun.security.auth.module.Krb5LoginModule required
useTicketCache=true;
JAASCONFIGDETAILS_END
SERVICE_PRINCIPAL_NAME_1=KSPN
VERIFYONLY=NO
RESTART_CAWAAE=YES
```

An example of a parameter file for switching the database access mode from Kerberos authentication to password is as follows:

```
DB_ACCESS_MODE=PASSWORD
ALTER_AEDB_USER_LOGIN=YES
SYSTEM_USER_NAME=sa
SYSTEM_USER_PASSWORD=sa_password
AUTOSYS_USER_PASSWORD=autosys_password
VERIFYONLY=NO
RESTART_CAWAAE=YES
SYSTEM_USER_NAME2=sa
SYSTEM_USER_PASSWORD2=sa_password
```

Configure MIT Kerberos on AutoSys Workload Automation

On Sybase, you can configure AutoSys Workload Automation to use Kerberos authentication. Kerberos is a computer network authentication protocol that works on tickets to let nodes communicate over a non-secure network to prove their identity to one another in a secure manner.

You can configure AutoSys Workload Automation to use Kerberos authentication using different methods. This article describes how to configure MIT Kerberos on AutoSys Workload Automation supported operating systems.

NOTE

- The Kerberos client must be installed on the machine where AutoSys Workload Automation is installed.
- The Kerberos Key Distribution Center (KDC), Kerberos database, and AutoSys Workload Automation must use the same date and time zone.
- When you install AutoSys Workload Automation or perform database operations, the logged in OS user must have a preauthenticated session with Kerberos that is mapped with a Sybase user with sa_role and sso_role privileges. The database user must also have a preauthenticated session with Kerberos.

(Windows) Configure MIT Kerberos on AutoSys Workload Automation

To configure MIT Kerberos on AutoSys Workload Automation, follow these steps:

1. Download the 64-bit MIT Kerberos client for Windows and install it.

NOTE

For information about installing the 64-bit MIT Kerberos client for Windows and creating the Kerberos configuration file, see the MIT Kerberos documentation.

2. Create the Kerberos configuration file (for example, krb5.ini file).
3. Set the KRB5_CONFIG and KRB5CCNAME environment variables in the Windows environment: **KRB5_CONFIG** Specifies the path to the Kerberos configuration file.
Example: C:\winnt\krb5.ini **KRB5CCNAME** Specifies the path to the credentials cache file. Specify the entire path.**Example:** C:\temp\krb5cache
4. Copy the Kerberos configuration file in the location that you specified in the KRB5_CONFIG environment variable and the C:\Windows directories.

NOTE

The Java utilities check for the krb5.ini file in the C:\Windows directory.

At a minimum, the Kerberos configuration file must contain the following entries:

NOTE

The default encryption must be rc4-hmac.

[logging]

```
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log
```

[libdefaults]

```
dns_lookup_realm = false
ticket_lifetime = 24h
renew_lifetime = 7d
forwardable = true
rdns = false
default_ccache_name = KEYRING:persistent:%{uid}
default_tkt_enctypes = rc4-hmac
default_tgs_enctypes = rc4-
hmac
default_realm = WA.COM
dns_lookup_realm = false
dns_lookup_kdc = false
ticket_lifetime = 124h
renew_lifetime = 7d
forwardable = true
```

[realms]

```
WA.COM = {
kdc = wa-domain
admin_server = wa-domain
default_domain = wa.com
}
```

[domain_realm]

```
wa.com = WA.COM
.wa.com = WA.COM
.ae.com = WA.COM
ae.com = WA.COM
```

(Linux) Configure MIT Kerberos on AutoSys Workload Automation

To configure MIT Kerberos on AutoSys Workload Automation, follow these steps:

1. Download 64-bit MIT Kerberos client for Linux and install it using yum.

NOTE

For information about installing the 64-bit MIT Kerberos client for Linux and creating the Kerberos configuration file, see the MIT Kerberos documentation.

2. Create (if it does not already exist) or modify the Kerberos configuration file (for example, krb5.conf file).

NOTE

By default, the Kerberos configuration file is located at /etc/krb5.conf.

3. Edit the Kerberos configuration file to ensure that the realms and domain_realm entries are set properly. At a minimum, the Kerberos configuration file must contain the following entries:

[logging]

```
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log
```

[libdefaults]

```
dns_lookup_realm = false
dns_lookup_kdc = false
ticket_lifetime = 3d
renew_lifetime = 7d
forwardable = true
default_realm = WA.COM
```

[realms]

```
WA.COM = {
kdc = wa-domain
admin_server = wa-domain
default_domain = wa.com
}
```

[domain_realm]

```
wa.com = WA.COM
.wa.com = WA.COM
```

Configure MIT Kerberos on AutoSys Workload Automation on AIX

To configure MIT Kerberos on AutoSys Workload Automation, follow these steps:

1. Download 64-bit MIT Kerberos client for AIX and install it.

NOTE

For information about installing the 64-bit MIT Kerberos client for AIX and creating the Kerberos configuration file, see the MIT Kerberos documentation.

2. Set the PATH and LIBPATH variables as follows:

```
PATH=path/krb5/bin:$PATH ; export PATH
LIBPATH=lib:$LIBPATH ; export LIBPATH
```

path

Specifies the path to the location where Kerberos is installed.

3. Create the Kerberos configuration file (if it does not already exist) and edit it.

NOTE

By default, the Kerberos configuration file is located in the /usr/local/etc/krb5.conf and /etc/krb5/krb5.conf directories.

At a minimum, the Kerberos configuration file must contain the following entries:

[logging]

```
default = FILE:/var/log/krb5libs.log
kdc = FILE:/var/log/krb5kdc.log
admin_server = FILE:/var/log/kadmind.log
```

[libdefaults]

```
dns_lookup_realm = false
dns_lookup_kdc = false
ticket_lifetime = 124h
renew_lifetime = 7d
forwardable = true
default_realm = WA.COM
default_tkt_enctypes = rc4-hmac
default_tgs_enctypes = rc4-hmac
```

[realms]

```
WA.COM = {
kdc = wa-domain
admin_server = wa-domain
default_domain = wa.com
}
```

[domain_realm]

```
wa.com = WA.COM
.wa.com = WA.COM
ae.com = WA.COM
.ae.com = WA.COM
```

NOTE

Use the kinit utility to create pre-authenticated tickets for a user as follows:

```
kinit username
```

Copy the created ticket to the /tmp directory, for example, klist

Ticket Management

A ticket is a temporary set of electronic credentials that verify the identity of a client for a particular service. You can create, list, or destroy tickets.

To create a ticket, run the following command:

```
kinit username
```

- ***username***

Specifies the user name for which you want to create the ticket. You must also specify the password that is associated with this user.

To list the tickets, run the following command:

```
klist
```

To destroy a ticket, run the following command:

```
kdestroy -c path
```

- **path**
Specifies the path where the ticket is located.

NOTE

For more information about ticket management, see the Kerberos documentation.

Configure the Sybase Client to Support Kerberos Authentication

The libtcl.cfg file is located in the following directory:

- Windows: %SYBASE%\%SYBASE_OCS%\config\libtcl.cfg
- Linux, AIX: \$SYBASE/\$SYBASE_OCS/config/libtcl.cfg

To configure the Sybase client to support Kerberos authentication, configure the libtcl.cfg file to:

- Include the krb5 libraries and realm name.
- Pick the correct security driver.
- Uncomment the CSFKRB5 parameter and ensure that proper values are set.

The following are examples of the libtcl.cfg file where the CSFKRB5 parameter is set with a properly configured driver:

- Windows: CSFKRB5=LIBSYBSKRB secbase=@WA.COM libgss=gssapi64.dll
- Linux: csfkrb5=libsybskrb.so secbase=@WA.COM libgss=libgssapi_krb5.so.2.2
- Solaris: csfkrb5=libsybskrb.so secbase=@WA.COM libgss=/usr/lib/libgss.so
- AIX: csfkrb5=libsybskrb.so secbase=@WA.COM libgss=/AIX/lib/libgssapi_krb5.so
- HP-UX: csfkrb5=libsybskrb.so secbase=@WA.COM libgss=/usr/lib/hpux64/libgssapi_krb5.so

Configure JAAS Configuration File

Java applications require the presence of a JAAS configuration file for connecting to the database successfully. This configuration file contains the login module details and other cache related information.

The AutoSys Workload Automation Compliance Application looks for a JAAS configuration file that is named *JAAS.conf* and a section that is named *initiate* within this file that contains the login module details. The JAAS.conf file is located in the (UNIX) \$AUTOSYS/config/cmplapp or (Windows) %AUTOSYS%\config\cmplapp directory.

For web services, the logon.conf file is located in the (UNIX) \$AUTOUSER/webserver/config/logonconf or (Windows) %AUTOUSER%\webserver\config\logonconf directory.

For Java login modules to connect to the Sybase, the JAAS configuration file must include the following entries:

Windows:

```
initiate {
com.sun.security.auth.module.Krb5LoginModule required
useTicketCache=true
debug=true;
};
```

AIX:

```
initiate {
com.ibm.security.auth.module.Krb5LoginModule required
useCcache="FILE:/var/krb5/security/creds/krb5cc_213";
};
```

(Oracle Only) Switch the Database Access Mode

After you installed AutoSys Workload Automation, you can use the `switchDBAccessMode.pl` script to switch the database access mode that AutoSys Workload Automation is using to connect to the database. You can switch the database access mode from password to SSL certificate and also do the reverse.

NOTE

- If you are using the database user or schema owner accounts to connect CA Workload Automation iDash or CA Workload Automation iXp to the AutoSys Workload Automation database, you must reconfigure the CA Workload Automation iDash or CA Workload Automation iXp connections to use different user/password accounts before switching to use SSL certificates to access the database.
- For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

Switch to Use SSL Certificate to Access the Database

Before you switch to use SSL certificate to access the database, perform the following tasks:

- Identify the Oracle system administrator user who can log in to the database using an Oracle wallet with SSL certificate and can alter the AutoSys Workload Automation database users. Execute the following SQL statements:

```
SELECT DISTINCT count(*) FROM ROLE_SYS_PRIVS WHERE PRIVILEGE='ALTER USER';
```

If the output displays a non-zero number, it confirms that the Oracle system administrator user can log in to the database using an Oracle wallet with SSL certificate and the system administrator has the required permissions to alter user.
- Generate the Oracle wallets with SSL certificates for the AutoSys Workload Automation Database schema owner (for example, aedbadmin) and the AutoSys Workload Automation database user (for example, autosys).
- Gather the following data before invoking the `switchDBAccessMode.pl` script as you need this information to complete the parameter file:
 - TNS_ADMIN value of the Oracle system administrator user for connecting to the primary and secondary event servers.
 - TNS_ADMIN value of the database user for connecting to the primary and secondary event servers.
 - Distinguished name of the SSL certificate that you want to associate with the schema owner on the primary and secondary event servers.
 - Distinguished name of the SSL certificate that you want to associate with the database user on the primary and secondary event servers.
 - Absolute path of the Oracle wallet file (`cwallet.sso`) in the file system.
 - Directory that contains the `oraclepki.jar`, `osdt_core.jar`, and `osdt_cert.jar` files.

NOTE

- On UNIX, the database user must have read privileges on the Oracle wallet file in the file system.
- On Windows, the local system account must have read privileges on the Oracle wallet file in the file system.
- Perform this procedure on all computers where the scheduler, application server, or both of a same instance are installed.

Follow these steps:

1. Perform any one of the following tasks based on your operating system:
 (UNIX) Assuming that AutoSys Workload Automation is installed in `/opt/CA/WorkloadAutomationAE` and the AutoSys Workload Automation instance ID is ACE, source the AutoSys Workload Automation environment as follows:


```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.shell.host
cd $AUTOSYS/dbobj/switchDBAccessMode
```

– **shell**

Specifies the name of the shell that is used to execute the script or command file.

– **host**

Specifies the name of the computer where AutoSys Workload Automation is installed.

(Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*) to open the instance command prompt and enter the following command:

```
cd %AUTOSYS%\dbobj\switchDBAccessMode
```

The switchDBAccessMode directory appears.

2. Edit the following parameters in the *paramfile.oracle.CERTMODE* parameter file as appropriate and save the file:

NOTE

The *paramfile.oracle.CERTMODE* parameter file is available in the \$AUTOSYS/dbobj/switchDBAccessMode (UNIX) or %AUTOSYS%\dbobj\switchDBAccessMode (Windows) directory.

- **DB_ACCESS_MODE=CERTIFICATE** Specifies that AutoSys Workload Automation uses SSL certificate to connect to the database.
- **ALTER_AEDB_USER_LOGIN**
Specifies whether you want to alter the database users (schema owner and database) to log in using SSL certificates. Valid values are:
 - **YES**
Alters the database users (schema owner and database) to log in using SSL certificates.
 - **NO**
Does not connect to the database to alter the database users (schema owner and database).
- **SYSTEM_USER_TNS_ADMIN**
Defines the TNS_ADMIN value of the Oracle system administrator user for connecting to the primary event server.
- **AEDBADMIN_USER_DN**
Defines the distinguished name of the SSL certificate that you want to associate with the schema owner on the primary event server.
- **AUTOSYS_USER_TNS_ADMIN**
Defines the TNS_ADMIN value of the database user for connecting to the primary and secondary event servers.
- **AUTOSYS_USER_DN**
Defines the distinguished name of the SSL certificate that you want to associate with the database user on the primary and secondary event servers.
- **AUTOSYS_USER_WALLET_LOCATION**
Defines the absolute path of the Oracle wallet file (cwallet.sso) in the file system. This Oracle wallet file must include the SSL certificate that you want to associate with the database user on the primary and secondary event servers.
- **ADDITIONAL_JARS_PATH**
Defines the location of the directory that contains the oraclepki.jar, osdt_core.jar, and osdt_cert.jar files.
- **SSL_SERVER_DN_MATCH**
- **VERIFYONLY** Specifies whether you want to validate the environment and parameter file and proceed with switching the database access mode. Valid values are:
 - **YES**
Validates whether the environment and the parameter file are set up properly, but does not proceed to switch the database access mode.
 - **NO**
Validates whether the environment and the parameter file are set up properly and proceeds with switching the database access mode only if the validation is successful.
- **RESTART_CAWAAE**
Specifies whether you want to restart the AutoSys Workload Automation services after successfully updating the configurations to use an SSL certificate to access the database. Valid values are:
 - **YES**

Restarts the AutoSys Workload Automation services after successfully updating the configurations to use an SSL certificate to access the database.

- **NO**

Does not restart the AutoSys Workload Automation services.

- **SYSTEM_USER_TNS_ADMIN2**

Defines the TNS_ADMIN value of the Oracle system administrator user for connecting to the secondary event server.

- **AEDBADMIN_USER_DN2**

Defines the distinguished name of the SSL certificate that you want to associate with the schema owner on the secondary event server.

3. Run the switchDBAccessMode.pl script using the following command:

```
perl switchDBAccessMode.pl paramfile
```

The script completes successfully with a return code of zero and AutoSys Workload Automation starts using an SSL certificate to access the database. If the script fails, a non-zero return code is displayed.

The AutoSys Workload Automation services must be restarted for the settings to take effect. So, if the RESTART_CAWAAE parameter is set to NO, restart the AutoSys Workload Automation services manually.

NOTE

You can also use the switchDBAccessMode.pl script to refresh the database access mode values. Edit the values in the parameter file and run the switchDBAccessMode.pl script for the new values to take effect.

Switch to Use Password to Access the Database

Before invoking the switchDBAccessMode.pl script to switch the database access mode from SSL certificate to password mode, gather the following data as you need this information to complete the parameter file:

- Oracle system administrator user ID and password for connecting to the primary and secondary event servers
- schema owner password for connecting to the primary and secondary event servers
- database user password
- Primary and secondary event server names
- Machine names where the primary and secondary event servers are running
- Port that is used to connect to the primary and secondary event servers

NOTE

Perform this procedure on all computers where the scheduler, application server, or both of a same instance are installed.

Follow these steps:

1. Perform any one of the following tasks based on your operating system:(UNIX) Assuming that AutoSys Workload Automation is installed in /opt/CA/WorkloadAutomationAE and the AutoSys Workload Automation instance ID is ACE, source the AutoSys Workload Automation environment as follows:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.shell.host
cd $AUTOSYS/dbobj/switchDBAccessMode
```

- **shell**

Specifies the name of the shell that is used to execute the script or command file.

- **host**

Specifies the name of the computer where AutoSys Workload Automation is installed.

(Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*) to open the instance command prompt and enter the following command:

```
cd %AUTOSYS%\dbobj\switchDBAccessMode
```

The switchDBAccessMode directory appears.

2. Edit the following parameters in the *paramfile.oracle.PASSMODE* parameter file as appropriate and save the file:

NOTE

The *paramfile.oracle.PASSMODE* parameter file is available in the \$AUTOSYS/dbobj/switchDBAccessMode (UNIX) or %AUTOSYS%\dbobj\switchDBAccessMode (Windows) directory.

- **DB_ACCESS_MODE=PASSWORD** Specifies that AutoSys Workload Automation uses password to connect to the database.
- **ALTER_AEDB_USER_LOGIN**
Specifies whether you want to alter the database users (schema owner and database) to log in using passwords. Valid values are:
 - **YES**
Alters the database users (schema owner and database) to log in using passwords.
 - **NO**
Does not connect to the database to alter the database users (schema owner and database).
- **SYSTEM_USER_NAME**
Defines the Oracle system administrator user ID to connect to the primary event server (DB_NAME).
- **SYSTEM_USER_PASSWORD**
Defines the password that is associated with the Oracle system administrator user ID (SYSTEM_USER_NAME) to connect to the primary event server (DB_NAME).
- **AEDBADMIN_USER_PASSWORD**
Defines the password that you want to set for the schema owner on the primary event server (DB_NAME).
- **AUTOSYS_USER_PASSWORD** Defines the password that you want to set for the database user on the primary and secondary event servers.
- **DB_NAME**
Defines the primary event server name where the database tables are already created.
- **DB_MACHINE**
Defines the name of the machine where the primary event server (DB_NAME) is running.
- **DB_PORT**
Defines the TCP port that is used to connect to the primary event server (DB_NAME) using username and password.
- **VERIFYONLY** Specifies whether you want to validate the environment and parameter file and proceed with switching the database access mode. Valid values are:
 - **YES**
Validates whether the environment and the parameter file are set up properly, but does not proceed to switch the database access mode.
 - **NO**
Validates whether the environment and the parameter file are set up properly and proceeds with switching the database access mode only if the validation is successful.
- **RESTART_CAWAAE** Specifies whether you want to restart the AutoSys Workload Automation services after successfully updating the configurations to use password to access the database. Valid values are:
 - **YES**
Restarts the AutoSys Workload Automation services after successfully updating the configurations to use password to access the database.
 - **NO**
Does not restart the AutoSys Workload Automation services.
- **SYSTEM_USER_NAME2**
Defines the Oracle system administrator user ID to connect to the secondary event server (DB_NAME2).
- **SYSTEM_USER_PASSWORD2**

Defines the password that is associated with the Oracle system administrator user ID (SYSTEM_USER_NAME2) to connect to the secondary event server (DB_NAME2).

– **AEDBADMIN_USER_PASSWORD2**

Defines the password that you want to set for the schema owner on the secondary event server (DB_NAME2).

– **DB_NAME2**

Defines the secondary event server name where the database tables are already created.

– **DB_MACHINE2**

Defines the name of the machine where the secondary event server (DB_NAME2) is running.

– **DB_PORT2**

Defines the TCP port that is used to connect to the secondary event server (DB_NAME2) using username and password.

3. Run the switchDBAccessMode.pl script using the following command:

```
perl switchDBAccessMode.pl paramfile
```

The script completes successfully with a return code of zero and AutoSys Workload Automation starts using password to access the database. If the script fails, a non-zero return code is displayed.

The AutoSys Workload Automation services must be restarted for the settings to take effect. So, if the RESTART_CAWAAE parameter is set to NO, restart the AutoSys Workload Automation services manually.

NOTE

You can also use the switchDBAccessMode.pl script to refresh the database access mode values. Edit the values in the parameter file and run the switchDBAccessMode.pl script for the new values to take effect.

Sample Parameter Files

The following sample parameter files are located in the \$AUTOSYS/dbobj/switchDBAccessMode (UNIX) or %AUTOSYS%\dbobj\switchDBAccessMode (Windows) directory:

- paramfile.oracle.CERTMODE—Sample file to switch the database access mode from password to SSL certificate.
- paramfile.oracle.PASSMODE—Sample file to switch the database access mode from SSL certificate to password.

An example of a UNIX parameter file for switching the database access mode from password to SSL certificate is as follows:

```
DB_ACCESS_MODE=CERTIFICATE
ALTER_AEDB_USER_LOGIN=YES
SYSTEM_USER_TNS_ADMIN=/home/client/system_user_tns_admin
AEDBADMIN_USER_DN='CN=AEDBADMIN,OU=ITC,O=CA Technologies,C=IN'
AUTOSYS_USER_TNS_ADMIN=/home/client/autosys_user_tns_admin
AUTOSYS_USER_DN='CN=AUTOSYS,OU=ITC,O=CA Technologies,C=IN'
AUTOSYS_USER_WALLET_LOCATION=/home/client/autosys_wallet/cwallet.sso
ADDITIONAL_JARS_PATH=/opt/oracle/app/oracle/product/12.1.0/dbhome_1/jlib
VERIFYONLY=NO
RESTART_CAWAAE=YES
SYSTEM_USER_TNS_ADMIN2=/home/client/system_user_tns_admin
AEDBADMIN_USER_DN2='CN=AEDBADMIN,OU=ITC,O=CA Technologies,C=IN'
```

The following is an example of a Windows parameter file for switching the database access mode from password to SSL certificate

```
DB_ACCESS_MODE=CERTIFICATE
ALTER_AEDB_USER_LOGIN=YES
SYSTEM_USER_TNS_ADMIN=C:\app\client\system_user_tns_admin
```

```

AEDBADMIN_USER_DN='CN=AEDBADMIN,OU=ITC,O=CA Technologies,C=IN'
AUTOSYS_USER_TNS_ADMIN=C:\app\client\autosys_user_tns_admin
AUTOSYS_USER_DN='CN=AUTOSYS,OU=ITC,O=CA Technologies,C=IN'
AUTOSYS_USER_WALLET_LOCATION=C:\autosys_wallet\cwallet.sso
ADDITIONAL_JARS_PATH=C:\app\oracle\product\12.1.0\dbhome_1\jlib
VERIFYONLY=NO
RESTART_CAWAAE=YES
SYSTEM_USER_TNS_ADMIN2=C:\app\client\system_user_tns_admin
AEDBADMIN_USER_DN2='CN=AEDBADMIN,OU=ITC,O=CA Technologies,C=IN'

```

The following is an example of a parameter file for switching the database access mode from SSL certificate to password on Windows and UNIX:

```

DB_ACCESS_MODE=PASSWORD
ALTER_AEDB_USER_LOGIN=YES
SYSTEM_USER_NAME=system
SYSTEM_USER_PASSWORD=system_password
AEDBADMIN_USER_PASSWORD=aedbadmin_password
AUTOSYS_USER_PASSWORD=autosys_password
DB_NAME=AEDB
DB_MACHINE=dbmachine.ca.com
DB_PORT=1521
VERIFYONLY=NO
RESTART_CAWAAE=YES
SYSTEM_USER_NAME2=system
SYSTEM_USER_PASSWORD2=system_password
AEDBADMIN_USER_PASSWORD2=aedbadmin_password
DB_NAME2=AEDB2
DB_MACHINE2=dbmachine2.ca.com
DB_PORT2=1521

```

Create Oracle Wallets with Certificates

You can create an Oracle wallet with certificates and configure the Oracle server and client to use these certificates to access the database.

NOTE

The example procedures in this article describe one of the ways to create a single sign-on Oracle wallet with SSL certificates.

NOTE

For more information about creating Oracle wallets, see the following Oracle documentation:

- Using Oracle Wallet Manager section from the Oracle Database Advanced Security Administrator's Guide.
- orapki Utility section from the Oracle Database Advanced Security Administrator's Guide.

Create an Oracle Wallet with Trusted Certificate

This procedure is an example on how to create an Oracle wallet with trusted certificate on Linux.

Follow these steps:

1. Log in to the machine where the Oracle database is installed.
2. Execute the following commands to create a new directory where you want to create the wallets:

```
mkdir /wallet_directory_path
cd /wallet_directory_path
```

3. Execute the following command to create an Oracle wallet with default trusted certificates:

```
orapki wallet create -wallet ./trustedWallet -pwd password
```

NOTE

To remove all of the existing default trusted certificates in the wallet, execute the following command:

```
orapki wallet remove -wallet ./trustedWallet -trusted_cert_all -pwd password
```

4. Execute the following command to add a self-signed certificate in the wallet with the distinguished name as CN=CA TrustedCertificate,OU=Certificate OU,O=Certificate Org,C=IN:

```
orapki wallet add -wallet ./trustedWallet -dn "CN=CA
TrustedCertificate,OU=Certificate OU,O=Certificate Org,C=IN" -keysize 2048 -
self_signed -validity 3650 -pwd password -sign_alg sha512
```

TIP

To view the contents of the Oracle wallet, execute the following command:

```
orapki wallet display -wallet ./trustedWallet -pwd password
```

5. Execute the following command to export the self-signed certificate from the Oracle wallet to use it to sign other certificates:

```
orapki wallet export -wallet ./trustedWallet -dn "CN=CA
TrustedCertificate,OU=Certificate OU,O=Certificate Org,C=IN" -cert ./trustedWallet/
b64certificate.txt -pwd password
```

An Oracle wallet with trusted certificate is created in the /wallet_directory_path/trustedWallet directory.

Create an Oracle Wallet with User Certificate

This procedure is an example on how to create an Oracle wallet with a user certificate that is signed by a trusted certificate (created using the above procedure) on Linux.

Follow these steps:

1. Log in to the machine where the Oracle database is installed.
2. Execute the following commands to create a new directory where you want to create the wallets:

```
mkdir /wallet_directory_path
cd /wallet_directory_path
```

3. Execute the following command to create an Oracle wallet with auto login enabled:

```
orapki wallet create -wallet ./userWallet -auto_login -pwd password
```

The /wallet_directory_path/userWallet/cwallet.sso file is created.

NOTE

To remove all of the existing default trusted certificates in the wallet, execute the following command:

```
orapki wallet remove -wallet ./userWallet -trusted_cert_all -pwd password
```

4. Execute the following command to add a user certificate in the wallet with a distinguished name as CN=USERNAME,OU=ITC,O=Certificate Org,C=IN:

```
orapki wallet add -wallet ./userWallet -dn "CN=USERNAME,OU=ITC,O=Certificate
Org,C=IN" -keysize 2048 -pwd password -sign_alg sha512
```

5. Execute the following command to export the user certificate request to a file:

```
orapki wallet export -wallet ./userWallet -dn "CN=USERNAME,OU=ITC,O=Certificate
Org,C=IN" -request ./userWallet/creq.txt
```

6. Execute the following command to sign the user certificate request using the trusted certificate that you created using the above procedure.

```
orapki cert create -wallet ./trustedWallet -request ./userWallet/creq.txt -cert ./
userWallet/cert.txt -validity 3650 -pwd password
```

TIP

To view the signed certificate, execute the following command:

```
orapki cert display -cert ./userWallet/cert.txt -complete
```

7. Execute the following command to add the trusted certificate that you created using the above procedure to the wallet:

```
orapki wallet add -wallet ./userWallet -trusted_cert -cert ./trustedWallet/
b64certificate.txt -pwd password -sign_alg sha512
```

8. Execute the following command to add the signed user certificate to the Oracle wallet:

```
orapki wallet add -wallet ./userWallet -user_cert -cert ./userWallet/cert.txt -pwd
password -sign_alg sha512
```

TIP

To view the contents of the Oracle wallet, execute the following command:

```
orapki wallet display -wallet ./userWallet -pwd password
```

A single sign-on Oracle wallet with a user certificate that is signed by a trusted certificate is created in the / wallet_directory_path/userWallet directory.

NOTE

- Use this procedure to create Oracle wallets for the following users:
 - Oracle system administrator user (to use during AutoSys Workload Automation installation to create the database).
To create a user in the database associating with a Oracle wallet, see [Verify the Database Access Using Oracle Wallet](#).
 - AutoSys Workload Automation database schema owner (for example, aedbadmin)
 - AutoSys Workload Automation database user (for example, autosys)
- Ensure that only a desired user who is logged in to the machine has appropriate access privileges to the respective single sign-on Oracle wallet (cwallet.sso).

Configuring the Oracle Server and Client to Use Certificates

Configure the listener.ora, sqlnet.ora, and tnsnames.ora files using the Oracle Net Manager tool so that the listener opens the TCP/IP with SSL port and the communication between the Oracle server and client can happen through the same using single sign-on Oracle Wallets using SSL certificates.

NOTE

For information about configuring the listener.ora file on the Oracle server and the sqlnet.ora and tnsnames.ora files on the Oracle server and client, see the following Oracle documentation:

- Configuring Secure Sockets Layer Authentication section from the Database Advanced Security Administrator's Guide
- Configuring and Administering Oracle Net Listener section from the Database Net Services Administrator's Guide
- Configuring Naming Methods section from the Database Net Services Administrator's Guide

A properly configured listener.ora file on the Oracle server looks similar to the following:

```
SSL_CLIENT_AUTHENTICATION = FALSE

WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = /wallet_directory_path/userWallet)
    )
  )

LISTENER =
  (DESCRIPTION_LIST =
    (DESCRIPTION =
      (ADDRESS = (PROTOCOL = IPC) (KEY = EXTPROC1521))
      (ADDRESS = (PROTOCOL = TCP) (HOST = hostname) (PORT = 1521))
      (ADDRESS = (PROTOCOL = TCPS) (HOST = hostname) (PORT = 2484))
    )
  )
```

A properly configured sqlnet.ora file on the Oracle server and client looks similar to the following:

```
SQLNET.AUTHENTICATION_SERVICES= (BEQ, TCPS)

SSL_VERSION = 0

NAMES.DIRECTORY_PATH= (TNSNAMES, EZCONNECT)

SSL_CLIENT_AUTHENTICATION = TRUE

WALLET_LOCATION =
  (SOURCE =
    (METHOD = FILE)
    (METHOD_DATA =
      (DIRECTORY = /wallet_directory_path/userWallet)
    )
  )

SSL_CIPHER_SUITES= (<user chosen values>)
```

A properly configured tnsnames.ora file on the Oracle server and client looks similar to the following:

```
ORACLE_SID_ALIAS =
  (DESCRIPTION =
```

```

(ADDRESS_LIST =
  (ADDRESS = (PROTOCOL = TCPS) (HOST = hostname) (PORT = 2484))
)
(CONNECT_DATA =
  (SERVER = DEDICATED)
  (SERVICE_NAME = ORACLE_SID)
)
)

```

NOTE

- Ensure that the TNS_ADMIN directory includes a properly configured sqlnet.ora and tnsnames.ora files on the Oracle server and client.
- After you configure the Oracle server to use Oracle wallets with SSL certificates, restart the Oracle listener.

Verify the Database Access Using Oracle Wallet

You must verify if you are able to access the database using Oracle wallets.

Follow these steps:

1. Login in as sysdba, create a user associating with the respective Oracle wallet, and grant create session privileges:


```

create user username identified externally as 'CN=USERNAME,OU=ITC,O=Certificate
Org,C=IN';
grant create session to username;

```
2. Set the TNS_ADMIN variable to the absolute path of the directory that contains the properly configured tnsnames.ora and sqlnet.ora files:


```

TNS_ADMIN=tnsadmin_directory ; export TNS_ADMIN

```
3. Execute the following command to connect to the database using the Oracle wallet specific configurations set in the tnsnames.ora and sqlnet.ora files in the TNS_ADMIN directory:


```

sqlplus /@ORACLE_SID_ALIAS

```

The user logs in to the database using Oracle wallet and the ORACLE_SID_ALIAS that is configured in the sqlnet.ora and tnsnames.ora files in the TNS_ADMIN directory.

NOTE

The database login might fail due to any of the following reasons:

- The files in the TNS_ADMIN directory of the Oracle server and client are improperly configured.
 - The Oracle wallet that is configured in the files in the TNS_ADMIN directory was not associated with any of the database users who can login.
 - The user who logged in to the machine does not have appropriate permissions to access the Oracle wallet that is configured in the files in the TNS_ADMIN directory.
4. Execute the following command to verify the user logged in to the database:


```

show user;

```

The Oracle user associated with the Oracle wallet that is configured in the tnsnames.ora and sqlnet.ora files in the TNS_ADMIN directory is displayed.

AutoSys Workload Automation Deployment Best Practices

This section describes the best practices for deploying your Workload Automation solution using AutoSys Workload Automation. The Broadcom solution presents an enterprise and business-wide view of the business process as it is

executing, offers tailored views based on the role of the solution user, and provides granular role-based security. The solution also matches work to do with the resources available to do the work according to the business process policies that define the service levels.

Whether you are responsible for a line of business, production control, security, or are the person who deploys and administers the products, this section has information to help you execute your role. This section is designed for the technical staff who is charged with deploying and administering the products, and staff responsible for a line of business, production control, and security. This section provides an explanation of capabilities that you can deploy today to automate the scheduling and deployment of workload across mainframe and distributed systems.

The articles in this section explain the product architecture and the solution itself. The articles also explain best practices to set up the solution and to integrate with other products, such as CA Workload Automation ESP Edition and CA Service Desk Manager.

Deployment Architecture

This article explains deployment architectures for planning purposes. The scheduling environment has changed over the past couple of years, becoming more complicated, with ever-increasing demands to complete more work in shorter times. In addition to the traditional date and time-triggered processing, we also see an increasing need to treat scheduling as a service available for improvised requests or run from a predefined series of processes based on different types of events that occur on the system, such as an arriving file from an external client or an update to a database. Most companies also have a mixture of hardware and platforms that need to be able to communicate and synchronize the complex processes running across them. These complex processes may have interdependencies that did not exist in previous years.

Synchronizing certain processes can be a difficult challenge. For example, you may have an application such as SAP with a UNIX-based front end that uses a database on the mainframe. At times, you may need to apply maintenance to the database and stop the submission of any jobs that communicate with SAP, or you may want to stop processing while backups are performed. Without tight control of when and how these events occur, the chances of making your batch window and meeting your SLAs (Service-Level Agreements) are lessened.

Architecture Choices

With consolidation, changes to existing platforms, and new platforms being created, the number of options for a scheduling architecture is increasing. What is the best one for your company and how do you decide?

- **Mainframe-centric architecture**
In a mainframe-centric architecture, the mainframe is the focal point for administering and managing the workload; the jobs and processes themselves can run on multiple platforms. A mainframe-centric approach lets you use a familiar and centralized tool to manage and monitor the enterprise workload.
- **Distributed-centric architecture**
A distributed-centric architecture runs the same way as the mainframe-centric architecture except the manager is on a UNIX, Linux, or Windows platform. The distributed-centric architecture has the same advantage as the mainframe-centric architecture.
- **Multiple managers**
The third choice is to have more than one manager. Each manager controls sections of the workload, but each should communicate with the other managers as required. Multiple managers offer the additional advantage of having independent managers that can operate their critical work even if there was a failed router, for example, between the mainframe and a Linux manager.

Planning Considerations

Before you decide on architecture, answer some basic questions, considering the long-term needs and requirements for your business. This approach helps you avoid retooling down the road. Consider the following questions:

1. **What is the main reason for the exercise and the desired outcome?**

- Is it a consolidation based on reducing complexity, where your aim is to reduce the number of engines that are required to run the workload? It could also be to put workloads using an OS native scheduler, like Unix or Linux "cron", into an enterprise-wide scheduling engine.
 - Is it a consolidation based on cost savings, where your aim is to reduce the number of computers that are required to run the current workload? Advances in hardware have increased the capacity and virtualization is also on the rise.
 - Is it because of an acquisition? New workload impacts your current system and needs to be investigated before trying to decide on how to integrate any additional workload. If your system is at or near maximum capacity, then more systems may be required or an upgrade to the current one.
 - Is it part of a regular review on process? Complete a regular review of your system to see how changes made because the last review have impacted the performance of the scheduling process.
- 2. Where is the new workload coming from?**
- Is this new workload from an external acquisition or a new, in-house-developed application? If the new workload is external, does the acquired entity use the same hardware and management infrastructure as your company does and will that be managed separately or integrated? Ideally, you want to reduce the complexity as much as is practical to maximize your current investments, both software and staffing requirements.
- 3. What is the expertise of your existing staff? Can you hire new staff?**
- Having a staff that can work on multiple platforms makes it much easier to deploy work and troubleshoot any problems that may occur. Suppose that you have a mainframe-based staff and you inherit some Cron-scheduled UNIX servers that transfer files across systems to be used as input for another task. Using Cron, this scenario depends on everything working correctly and with enough extra time padded on, in slow networks. Because there is no real control on when the second job should run, if there is a problem, there is no built-in way to abort the process.
- Two options can help you improve this situation. One, use a distributed manager to handle the UNIX work, which is a viable option if you have UNIX expertise or if you have the time and budget to learn and install the new system. Another option would be to maintain your mainframe as the central manager and deploy an agent on UNIX instead. There would still be a learning curve on how to navigate UNIX and to debug problems, but it should reduce the curve by using a familiar tool and should make the integration more transparent. Using a familiar interface also improves the monitoring of the workload.
- 4. Do you have or expect to have cross-platform dependencies?**
- Cross-platform dependencies require a careful search through the current schedules. Some sites use multiple scheduling engines and have workload that does require synchronization across engines, using dummy ftp files to signal each other. While this works to a degree, it makes tracking the application difficult. To coordinate work, it may be more practical to run two scheduling engines that can communicate to each other instead of centralizing.
- 5. What happens if one of the platforms is unavailable?**
- Most sites have a scheduled maintenance window. If the windows are such that they do not pose a threat to the successful completion of the workload, then keeping a centric view is still possible. If, however, the cycles conflict so that there will be too much down time between the central manager and the agents, a distributed view may be a better choice.
- 6. What is the geographic location of the workload and the dependency on the network?**
- If you have work that is scheduled in various states or between countries, verify how reliable the network has been. If network drops are common, distributing the work helps mitigate problems when that occurs. Depending on your business, you may need to plan for natural disasters. A home supply company needs to be open to help people in case of a hurricane. In cases where workloads have dependencies across these borders, cross-instance communication is possible.
- 7. What is the future direction of platform choice?**
- If the direction of the company is to migrate an application away from one platform in favor of another, then it only makes sense to think about moving the manager away from that platform and then use an agent instead. Agents allow a phased approach, where the staff will learn the new platform when an application moves, and will make the migration of the manager easier to deal with in a later phase.
- 8. What is the alerting mechanism being used?**

- If you use a tool to track problem tickets and have automation in place, think about how the new workload impacts this and if any changes need to be made to accommodate it. For example, you may use AutoSys Workload Automation to send SNMP traps to a help desk or escalation type tool. If you decide to start using or want to integrate the mainframe engine into this, you need a way to get those traps sent, which CA OPS/MVS could easily handle with the addition of a new message rule.

9. What features of the product are being used?

- If one of the engines has a unique feature that makes migration awkward, a workaround or compromise will need to be made if consolidation is the goal. For example, if one product can open a help desk ticket when a critical job fails, losing that ability may impact your monitoring and possibly your reporting abilities.

10. What are the pain points that are currently being experienced?

- Part of data gathering should be to find what parts of the current system work well and what parts could be improved.

After you work out the technical requirements, consider the cultural ones. In many shops, the mainframe group is separate from the UNIX and Windows groups. Sometimes even having different management structures inhibits communications or impede building a cohesive team. If the mainframe group is separated from the distributed group and there is little interaction between them, then using a centralized scheduling solution is bound to run into organizational challenges. To manage a centric view effectively, the communication and teamwork issues need to be worked out before an optimum setup can be configured. In addition, when you maintain or troubleshoot problems in the future, that communication is crucial. Another consideration is determining ownership; who will be monitoring the workload and who is responsible for correcting any problems that may arise.

From an operations standpoint, it is helpful to track workload that crosses platforms and possibly different engines. A common GUI speeds the time to resolve issues and also gives a better idea of the impact of failures or jobs running later than usual.

Architecture Summary

In summary, there is no one size fits all answer to how best to deploy your solution. Multiple ways are available to accomplish the same goal, each with different challenges and rewards. The only way to decide is to conduct an analysis of your current and future environment (including timeframes and budget considerations), determine your criteria for success, and then work to overcome any obstacles, technical or otherwise.

Deploy the Solution

This solution deployment is based on our Workload Automation strategy. The AutoSys Workload Automation solutions include support for z/OS, UNIX, Linux, and Windows and integrate with leading applications from Oracle, PeopleSoft, and SAP to provide end-to-end business process automation. The solutions can be configured to suit any skill level and let users manage and monitor business processes at any appropriate level of granularity. The AutoSys Workload Automation High Availability option provides extra stability for distributed jobs.

CA Service Desk Manager is available to manage incidents relating to workload processing and incident management.

The deployment solution is designed to reduce administrative overhead by using best of breed scheduling engines for the mainframe and distributed environments and to provide a unified scheduling service for distributed environments. The service lets users log in to a web-based command center and create, administer, and monitor jobs. CA EEM security is used to personalize the experience and let users focus on their specific tasks. Web UI also abstracts the complexities specific to various operating systems and job scheduling platforms, empowering users to work with minimal skills and experience.

Workloads (Job Flows) within the organization are created primarily by two groups, System Administrators and Application Developers.

Lower labor costs are achieved by having operators monitor and administer the jobs while also providing job scheduling access to applications for system administrators and application developers. Implementing the solution requires changes

to the organization. These changes include a consolidation of scheduling administrators for the mainframe and distributed environments to create a single scheduling team. The team is responsible for managing the scheduling services in the development and production environments. The team also implements development jobs in the production service once they have gone through a review and test process in the test environment.

User Roles

This table summarizes the actors that typically play a role in deploying the solution. This table may introduce new actors (for example, new administrative roles). Where appropriate, this table may also include systems that can initiate or otherwise participate in a use case as actors.

Actor	Description	Current Use of the System	Expected Use of the New Solution
IT Operations Manager Monitor	Responsible for monitoring the status of jobs and reacting to Job Management Alerts.	System Administrators monitor the job output periodically. No real-time monitoring or notification is in place.	Monitoring by exception is implemented. Failed jobs will automatically page job owners. Role-based views are set up to let users see only jobs of interest.
Development Manager	Responsible for creating jobs to automate transactions or tasks directly related to a business transaction.	Application Development Managers create job specifications and then pass them off to system administrators for implementation. When things do not work correctly, administrators need to disable the job stream and send it back to the creators. Sometimes poorly written jobs affect other users by locking up the scheduling system.	Application Development Managers are able to test and implement their own jobs using the development scheduling services. They have read-only access to production job information.
IT Operations Manager System Administrators	Responsible for creating jobs from a system administration perspective. Typically, users use the command interface for all job creation, execution, and monitoring.	IT Operations Manager System Administrators create jobs for automating administrative tasks. All users have access to all scheduling components. Mistakes or malicious acts can bring the system down. There is no auditing or control of system changes.	Users are given access to scheduling objects on an as needed basis. Changes made are logged.
Job Scheduler	Responsible for testing and implementing jobs in production.	Schedulers are not part of the system.	Schedulers implement the jobs in their enterprise scheduling environment.
Scheduling Service	Responsible for executing jobs and tracking job status on distributed all systems. The solution provides a development, test, and production system.	Many different schedulers with different interfaces on different systems.	Single central scheduler that looks and feels the same on every platform.

Solution Components

- **AutoSys Workload Automation**

AutoSys Workload Automation is an industry-leading job management system that provides unparalleled reliability to support your enterprise-wide, mission-critical applications. Designed for distributed environments, AutoSys Workload Automation delivers event-driven scheduling, centralized real-time monitoring, and programmable error recovery—providing reliability and scalability to your production environment.

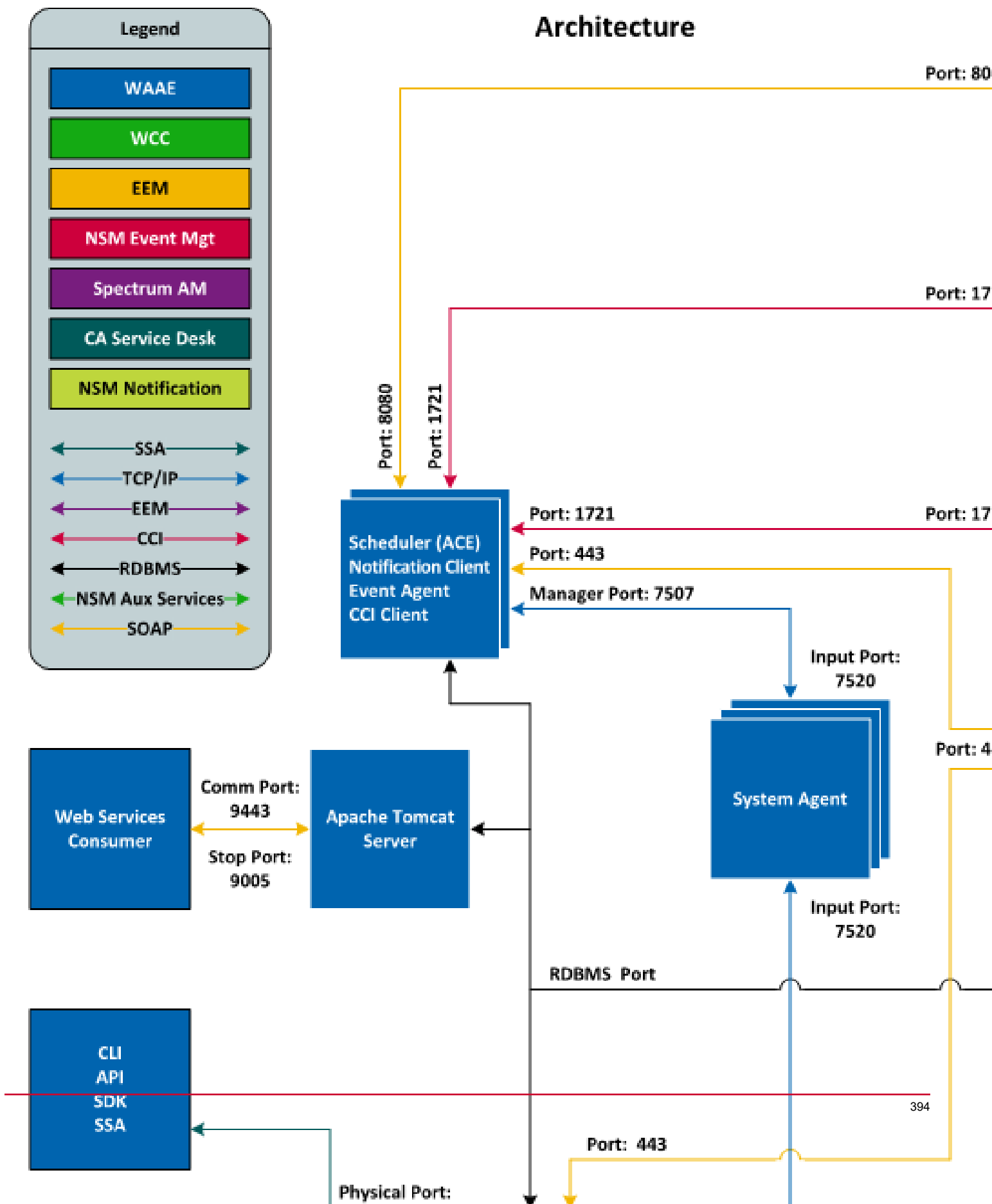
- **Web UI**

A key component of the AutoSys Workload Automation solution, Web UI provides a seamless interface for Workload Automation engines on multiple platforms and helps visualize business processes from end-to-end through a consolidated console. Web UI lets you manage, administer, monitor, and visualize your jobs in distributed environments from a business relevance perspective. As an innovative environment for the AutoSys Workload Automation solution, it provides a consolidated and secure interface to administer and monitor workloads. Web UI lets companies customize deployment of scheduling engines on their platforms of choice, and view and manage them from a central point.

CA Common Communications Interface (CA CCI)

This service enables communication among CA management products. CA has also developed it to view multiple data formats as object-oriented systems, thus any format of management data can be delivered to any CA product without requiring it to translate that data on its own. In this solution, we are using CA CCI to communicate between AutoSys Workload Automation on the distributed platform and the CA WA CA 7 Edition instance on the mainframe. The following diagram depicts the relationship of the Job Scheduling components, including the Enterprise Job Manager and components requiring CA CCI:

CA Workload Automation AE Architecture



Deploy a Distributed-Centric Solution

AutoSys Workload Automation provides a complete set of articles to help you take advantage of its full functionality and unique features. You may also find these documents helpful:

- **Release Notes** Provides important information about the current release including hardware and software requirements, new features, and changes to existing features. Read this section before proceeding with product installation.
- **Getting Started**
Provides an overview about AutoSys Workload Automation architecture, jobs, security, and key features.
- **Installing**
Describes how to install and configure AutoSys Workload Automation on UNIX and Windows.
- **Upgrading**
Describes how to upgrade AutoSys Workload Automation, Web UI, and CA Common Components.
- **Securing**
Detailed information about setting up security for AutoSys Workload Automation and Web UI, including native security and CA EEM security.
- **Administrating**
Describes how to configure, maintain, and troubleshoot AutoSys Workload Automation components
- **Scheduling**
Describes how to use AutoSys Workload Automation to define, run, manage, monitor, and report on jobs.
- **Monitoring and Reporting**
Describes how to identify problems with the workflow, use any combination of the following monitoring and reporting tools
- **Reference**
Describes the products RESTful web services, their syntax, and examples for their use. Lists product commands and jobs, machine, definition parameters. It also describes system states and database tables and views.
- **Messages**
Lists the messages that you see while working with AutoSys Workload Automation.
- **Troubleshooting**
Provides troubleshooting procedures for AutoSys Workload Automation and Web UI.
- **Web UI Help Systems**
Provides task-based help about using the various tabs on the Web UI Graphical User Interface.

System Components

The main AutoSys Workload Automation system components are as follows:

- **Event Server**
The Event Server, or database, is the data repository for all events and system information. It also serves as a repository for all job, monitor, and report definitions. Occasionally, the database is called a data server, which actually describes a server instance. That is, it is either a UNIX process or Windows service, and associated data space (or raw disk storage), which can include multiple databases or tablespaces. You can configure AutoSys Workload Automation to run using two databases, or Dual Event Servers. This feature provides complete redundancy. Therefore, if you lose one Event Server due to hardware, software, or network problems, operations can continue on the second Event Server without loss of information or functionality.
- **Application Server**
The Application Server is a UNIX process or Windows service that communicates between the AutoSys Workload Automation Agent or client utilities, and the Event Server. After you start the Application Server, the communication between the Event Server, Agent, or client utilities begins.
- **Scheduler**

The Scheduler (formerly called the Event Processor) is the heart of AutoSys Workload Automation. The Scheduler is the program, running either as a UNIX process or a Windows service, that actually runs AutoSys Workload Automation. It interprets and processes all the events that it reads from the AutoSys Workload Automation database. An active Scheduler continually scans the database for events to be processed. When one is found, the Scheduler verifies whether the event satisfies the starting conditions for any job in the database. Based on this information, the Scheduler first determines what actions to take, and instructs the appropriate Agent process to perform the actions. These actions may include starting or stopping of jobs, checking for resources, monitoring existing jobs, or initiating corrective procedures.

You can set up a second Scheduler, called the Shadow Scheduler. If the Primary Scheduler fails for some reason, the Shadow Scheduler takes over the responsibility of interpreting and processing events.

A client is any executable that interfaces with the application server. AutoSys Workload Automation provides utilities to help you define, run, and maintain instances and jobs. The utilities include the command-line interface (CLI) applications such as Job Information Language (JIL) and autorep. The included utilities are operating environment specific; however, all operating environments include the client utilities. The client utilities let you define, manage, monitor, and report on jobs.

- **Agents and Agent Plug-ins**

The agent (formerly called the remote agent) is a UNIX process or Windows service that is directed by the Scheduler to perform specific tasks. Agents are the key integration components of AutoSys Workload Automation products. Agents let you automate, monitor, and manage workload on all major platforms, applications, and databases.

To run workload on a particular system, you install an agent on that system. You can extend the functionality of agents by installing one or more agent plug-ins in the agent installation directory. If you have a relational database such as Oracle, for example, you can install a database agent plug-in to query and monitor the database. Other agent plug-ins are also available.

Instance

An AutoSys Workload Automation instance is a licensed version of AutoSys Workload Automation software running on a server with one or more clients or agents. Clients and agents can run on a single computer or on multiple computers. An instance uses its own scheduler, application server, and event server and operates independently of other instances.

An AutoSys Workload Automation instance is defined by the following components:

- At least one Event Server
- At least one Application Server
- At least one Scheduler
- At least one Agent

An instance ID is an uppercase, three-character alphanumeric name defined by the AUTOSERV environment variable. You must set the instance ID during installation, and it cannot be changed. The default instance ID is ACE.

You can install multiple AutoSys Workload Automation instances. For example, you may want to have one instance for production and another for development. Each instance operates independently of other AutoSys Workload Automation instances.

Computers Used

From a hardware perspective, the AutoSys Workload Automation architecture comprises the following types of computers that are attached to a network:

- **Server computer**
The server is the computer on which the Scheduler and the Application Server reside.
- **Client computer**

The client is the computer on which the Client software resides.

- **Agent computer**
- The agent is the computer on which the Agent software resides. An agent must be installed on the computer with the Scheduler, and it can also be installed on separate physical computers.

Configure Cross-Platform Scheduling for AutoSys

A Broadcom scheduling solution can request work to be run on its behalf by another Broadcom scheduling solution on a different platform. For example, AutoSys Workload Automation on a UNIX platform can request that WA CA 7 Edition schedule z/OS jobs on its behalf. The following Broadcom scheduling solutions can participate in this type of cross-platform scheduling:

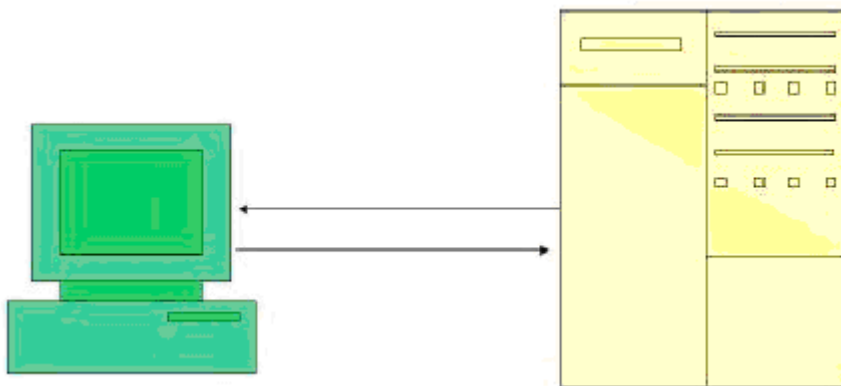
- WA CA 7 Edition
- CA Workload Automation AE Release 11.3.6
- WA ESP Edition
- Workload Automation Agent for z/OS Release 11.4

Configure Cross-Platform Scheduling

When submitting a job, the AutoSys Workload Automation scheduler processes the start job request and it transmits it to the mainframe using a cross-platform-enabled scheduling system. The scheduling system can be any of the Broadcom-developed mainframe schedulers or the AutoSys Workload Automation Connect Option product. After the product receives the request on the mainframe, it submits and tracks the mainframe job either directly by the XPS-enabled mainframe scheduling product or through the AutoSys Workload Automation Connect Option.

AutoSys Workload Automation not only permits job submission to the mainframe, but it can also accept a job that is submitted from any of the Broadcom-developed mainframe scheduling products, including WA CA 7 Edition, and report its status back. For more information about submitting the job, see the appropriate product documentation.

This section provides a brief tutorial to guide you through the process of enabling cross-directional scheduling in AutoSys Workload Automation. The tutorial's scenario involves the following systems:



- **AutoSys Workload Automation Machine**
 Hostname—local_host Operating System—UNIX Software Installed—CA CCI, AutoSys Workload Automation
 Parameter Configuration: AUTOTREE/AUTOUSER AUTOSERV - ACE
- **Mainframe**
 Hostname—remote_host Operating System—z/OS Software Installed—CA Common Services Interface (CAIRIM, CA CCI, CAIENF)

Configure with WA ESP Edition

The configuration setup must be correct on each instance of AutoSys Workload Automation and WA ESP Edition. Before you start the configuration, determine whether you want to encrypt the data that is transferred between the two systems. If encryption is not required, the steps are simplified and the task is easy. If encryption is required, decide which encryption keys to use. AutoSys Workload Automation can only support AES 128-bit encryption or no encryption, so both sides must be defined the same way.

Configure AutoSys Workload Automation

To enable communication the auxiliary ports for the scheduler and application server must be defined. By default, the auxiliary ports are enabled during installation, but they can be disabled. On UNIX or Linux, look for the SchedAuxiliaryListeningPort=xxxx and AppSrvAuxiliaryListeningPort=xxxx settings in the \$AUTOUSER/config.\$AUTOSERV file. On Windows, you can use the Administrator GUI to enable the ports.

Decide if you want to encrypt the data between the two systems. AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with WA ESP Edition. If you are using AES encryption, generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory. The cryptkey_alias.txt file stores the communication alias encryption key. AutoSys Workload Automation expects WA ESP Edition to encrypt the data using the key that is specified in the cryptkey_alias.txt file.

To use AES encryption, do the following tasks:

- On UNIX, set the UseCommAliasEncryption parameter to 2 in the configuration file.
- On Windows, select the Use AES 128-bit encryption when communicating with z/OS managers check box (located under the z/OS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator).

After you set the communication alias encryption setting, generate the cryptkey_alias.txt file. When you generate the cryptkey_alias.txt file, you specify a key to secure and store in that file. On WA ESP Edition, issue the CRYPTKEY command using that same key. The CRYPTKEY command secures and stores the key using a key name on WA ESP Edition.

Generate an Instance-Wide Communication Alias Encryption File

AutoSys Workload Automation uses either no encryption or AES 128-bit encryption to exchange data with WA ESP Edition and Workload Automation Agent for z/OS. If you are not using encryption, you do not need to create the alias encryption file. If you are using AES encryption, generate the instance-wide communication alias encryption file (cryptkey_alias.txt).

An AutoSys Workload Automation instance can have only one cryptkey_alias.txt file. Before you perform this procedure, check whether the file already exists. If the file exists, skip this procedure. You must provide the key that is associated with that file to the WA ESP Edition or agent administrator. The administrator needs the key to configure the AGENTDEF data set.

You can enter the following command at the operating system prompt to generate a key:

```
as_config - a key
```

- *key*
Specifies the communication alias encryption key. You must prefix the hexadecimal identifier 0x to this value.
Limits: Must contain 32-hex characters; valid characters are 0-9 and A-F.
This key must match the key that is stored in the ENCRYPT KEYNAME(keyname) parameter in the AGENTDEF data set of WA ESP Edition or Workload Automation Agent on z/OS.

The communication alias encryption file (cryptkey_alias.txt) is generated with the encryption key. AES 128-bit encryption is used. This setup creates the cryptkey_alias.txt file in your \$AUTOUSER (UNIX) or %AUTOUSER% (Windows) directory.

Create an External Instance Definition

To use external job dependencies, the scheduling manager or remote machine must be defined as an external instance in the AutoSys Workload Automation database. To define WA ESP Edition as an external instance, specify xtype: e JIL attribute in the definition. This attribute indicates that the external scheduling manager is WA ESP Edition. Ensure that you define each instance of WA ESP Edition to use external job dependencies.

Example: Define an External Instance with No Encryption

This example defines an external AutoSys Workload Automation instance named WAE. The instance's machine name is WAEHOST. The instance communicates using port 12345 and no encryption.

```
insert_xinst: WAE
xtype: e
xmanager: WAEEMGR
xmachine: WAEHOST
xport: 12345
xcrypt_type: NONE
xcomm_alias: ACE_SCH_ALIAS
```

Example: Define an External Instance with AES Encryption

This example defines an external AutoSys Workload Automation instance named WAE. The instance's machine name is WAEHOST. The instance communicates using port 12345. The data is encrypted using AES 128-bit encryption.

```
insert_xinst: WAE
xtype: e
xmanager: WAEEMGR
xmachine: WAEHOST
xcomm_alias: ACE_SCH_ALIAS
xport: 12345
xkey_to_manager: 0x012345678901234567890ABCDEFABCDEF
xcrypt_type: AES
```

Configure WA ESP Edition and Workload Automation Agent on z/OS

If you are using no encryption, AutoSys Workload Automation expects the data that it receives from WA ESP Edition to be unencrypted. You must specify the same AutoSys Workload Automation encryption setting in the WA ESP Edition AGENTDEF data set.

To configure the AGENTDEF data set, add the following two entries:

```
COMMCHAN alias_name ADDRESS(sch_ip_address) PORT(sch_aux_port) platform +
ASCII TCPIP PREF(2) {NOENCRYPT|ENCRYPT KEYNAME(keyname)}

EXTSCHED alias_name SAFUSER(mfuserid) EVENTPREFIX(ZOSEVENT) -
```

```
APPLPREFIX(ZOS) ID(SYSZAG)
```

Example: Using No Encryption When Receiving Data from WA ESP Edition

Suppose that you do not want the data that are received from WA ESP Edition to be encrypted. To use no encryption, do the following tasks:

- On UNIX, set the UseCommAliasEncryption parameter to 0 in the configuration file.
- On Windows, clear the Use AES encryption when communicating with z/OS managers check box (located under the z/OS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator).

When you define your AutoSys Workload Automation instance to WA ESP Edition, specify the NOENCRYPT operand, as follows:

```
COMMCHAN ACE_SCH_ALIAS_AGT ADDRESS(address) PORT(sched_aux_port) UNIX
ASCII TCPIP PREF(2) NOENCRYPT
```

Example: Using AES 128-Bit Encryption When Receiving Data from WA ESP Edition

Suppose that you want WA ESP Edition to encrypt data with AES encryption. To use AES encryption, do the following tasks:

- On UNIX, set the UseCommAliasEncryption parameter to 2 in the configuration file.
- On Windows, select the Use AES 128-bit encryption when communicating with z/OS managers check box (located under the z/OS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator).

After you set the communication alias encryption setting, generate the cryptkey_alias.txt file. When you generate the cryptkey_alias.txt file, you specify a key to secure and store in that file. On WA ESP Edition, issue the CRYPTKEY command using that same key. The CRYPTKEY command secures and stores the key using a key name on WA ESP Edition.

When you define your AutoSys Workload Automation instance to WA ESP Edition, specify the key name using the ENCRYPT operand, as follows:

```
COMMCHAN ACE_SCH_ALIAS_AGT ADDRESS(address) PORT(sched_aux_port) UNIX
ASCII TCPIP PREF(2) ENCRYPT KEYNAME(keyname)
```

Example: Using No Encryption to Send Data to the Workload Automation Agent on z/OS

This example displays a mainframe job definition to run an AE job daily.

```
APPL WAAEAPPL
CCCHK RC(5:4095) FAIL
  EXTSCHEDULER ACE_SCH_ALIAS
  EXTSCH WAAEDEP
  EXTJOB waaejob
  RUN DAILY
ENDJOB
```

After you have defined the external instance and determined the encryption, you can create external dependencies. The following example defines an external dependency on a z/OS job that is named WAEEJOB for an external instance named WAE:

```
insert_job: waeedep
...
condition: s(WAEEJOB^WAE)
```

Configure with Workload Automation Agent for z/OS

The configuration setup must be correct on each side of AutoSys Workload Automation and Workload Automation Agent for z/OS. Before you start the configuration, determine whether you want to encrypt the data transferred between the two systems. If encryption is not required, the steps are simplified and the task is easy. If encryption is required, you must decide which encryption keys to use. AutoSys Workload Automation can only use AES 128-bit encryption or no encryption, so both sides must be defined the same way.

In the AGENTDEF dataset, define the agent communication properties.

The name of the agent used in the machine definition comes from the AGENTDEF dataset from the NAME attribute. For example, set the agent name to PRODTSO you would code:

```
ZOSAGENT NAME(PRODTSO) TCPIP
```

The next sections in the AGENTDEF dataset contain the external schedulers and communication chain. Assuming the AE instance name is ACE two entries would be added:

```
COMMCHAN ACE_SWATRH5X642 -
  ADDRESS(10.130.66.145) -
  PORT(7500) -
  UNIX ASCII TCPIP NOENCRYPT PREF(2)

EXTSCHED ACE_SWATRH5X642 -
  SAFUSER(MFUSER01) -
  EVENTPREFIX(ZOSEVENT) -
  APPLPREFIX(ZOS) -
  ID(PRODTSO)
```

The COMMCHAN and EXTSCHED entries are set in the AutoServerAliasId entry in the config file for AE. This entry is optional, and it must be updated as a post installation step. In this case, set it to AutoServerAliasId=ACE_SWATRH5X64-2.

Suppose that you want to enable encryption between AE and the Workload Automation Agent for z/OS with AES 128-bit encryption. To use AES encryption, do the following tasks:

- On UNIX, set the UseCommAliasEncryption parameter to 2 in the configuration file.
- On Windows, select the Use AES 128-bit encryption when communicating with z/OS managers check box (located under the z/OS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator).

After you set the communication alias encryption setting, generate the cryptkey_alias.txt file. When you generate the cryptkey_alias.txt file, you specify a key to secure and store in that file. On the Workload Automation Agent for z/OS, issue

the CRYPTKEY command using that same key. The CRYPTKEY command secures and stores the key using a key name on the Workload Automation Agent for z/OS.

When you define your AutoSys Workload Automation instance to CA WA ESP Edition, you must specify the key name using the ENCRYPT operand, as follows:

```
COMMCHAN ACE_SCH_ALIAS_AGT ADDRESS(address) PORT(sched_aux_port) UNIX
ASCII TCPIP PREF(2) ENCRYPT KEYNAME(keyname)
```

The following example shows the CRYPTKEY named COMKEY:

```
CRYPTKEY DEFINE KEYNAME(COMKEY) KEY(X'0123456789ABCDEF0123456789ABCDEF') AES
```

Example: Create Machine Definition for a Workload Automation Agent on z/OS

When you submit work to the Workload Automation Agent on z/OS, you must define the machine. The keywords opsys: zos and character_code: EBCDIC is used. All the other keywords are used in the standard way.

This example creates a machine definition for Workload Automation Agent on z/OS:

```
insert_machine: ZOS
type: a
opsys: zos
port: 9680
node_name: PRODTSO
agent_name: PRODTSO
encryption_type: NONE
character_code: EBCIDIC
```

Example: WA AE Job Definition for z/OS Manual Job Type

This example posts a z/OS Manual job as complete when the manually submitted job ABC runs under user zosuser. The zosagent agent monitors job ABC.

```
insert_job: ABC_job
job_type: ZOSM
machine: zosagent
zos_jobname: ABC
owner: zosuser
auth_string: zosuser
```

Example: Define a z/OS Regular Job

This example submits the JCL in member CYBDL01A in the CYBDL01.JCLLIB library.

```
insert_job: zos_job
job_type: ZOS
owner: zosuser
machine: zosagent
```

```
jcl_library: CYBDL01.JCLLIB  
jcl_member: CYBDL01A
```

Configure with All Other Schedulers

For cross-directional scheduling to occur, enable AutoSys Workload Automation cross-platform scheduling, configure and start CA CCI, and define and implement the job.

Also, for AutoSys Workload Automation to submit a job to the mainframe, the job to be executed (specified by the command line argument in the AutoSys Workload Automation definition) must be defined as a valid job in the mainframe scheduling system.

Conversely, for the mainframe to submit a job to AutoSys Workload Automation, the job to be executed (specified by the SUBFILE parameter of the mainframe job) must be defined as a valid job in the AutoSys Workload Automation system.

Enable Cross-Platform Scheduling on Windows

To enable cross-platform scheduling support on Windows, set a parameter using the AutoSys Workload Automation Administrator GUI.

Follow these steps:

1. Open the AutoSys Workload Automation Scheduler screen in the AutoSys Workload Automation Administrator GUI.
2. Select 2 from the Cross Platform Scheduling drop-down list.
The outbound and inbound cross-platform scheduling (manager and agent) is enabled.

Enable Cross-Platform Scheduling on UNIX

To enable cross-platform scheduling support on UNIX, set the Cross Platform Scheduling parameter in the instance configuration file to 2, as follows:

```
$AUTOUSER/config.$AUTOSERV: CrossPlatformScheduling=2
```

Cross-Platform Dependencies

Cross-platform dependencies are jobs that execute on an external instance, but were not initiated on behalf of the local AutoSys Workload Automation instance. When the cross-platform dependent job completes, status information is forwarded back to the local AutoSys Workload Automation instance, and recorded in the database. The AutoSys Workload Automation Connect Option supports cross-platform dependencies. Currently, this support cannot be obtained directly with a mainframe scheduler.

Configure the CA Common Communications Interface (CA CCI)

CA Common Communications Interface (CA CCI) provides the underlying communication infrastructure that connects applications running on OpenVMS, z/OS, AS/400, Windows, various versions of UNIX, and many other operating systems. CA CCI forms the basis for the cross-platform workload automation agent that you can manage from AutoSys Workload Automation.

You do not need to configure the ccirmt file on a Windows to Windows platform environment. However, you must configure the ccirmt file on any cross-platform environment, such as Windows to UNIX, UNIX to mainframe, mainframe to Windows.

Configure CA CCI on Windows

To configure CA CCI on Windows, set the local and remote parameters in the ccirmttd.rc file.

Follow these steps:

1. Log in to the Windows computer AutoSys Workload Automation is installed on with an account having administrative privileges.
2. Locate and open the ccirmttd.rc file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:

```
LOCAL = local_host local_host 32768 startup  
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CA CCI is configured.

NOTE

: local_host cannot be longer than eight characters. If local_host exceeds eight characters and the first eight characters are not unique for the computers that are CCI nodes, use the ALIAS parameter. For more information, see the *Windows section under Installing*.

Start CA CCI on Windows

You must start or restart CA CCI to enable the configuration settings. To start CA CCI on Windows, use the following command:

```
ccicntrl start
```

Configure CA CCI on UNIX

To configure CA CCI on UNIX, set the local and remote parameters in the ccirmttd.prf file.

Follow these steps:

1. Log in as root to the AutoSys Workload Automation instance.
2. Open the \$CAIGLBL0000/cci/config/local_host/ccirmttd.prf file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:

```
LOCAL = local_host local_host 32768 startup  
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CA CCI is configured.

NOTE

By default the local_host name is used as the CCI sysid. If it is longer than eight characters, it is truncated to the first eight characters. If local_host exceeds eight characters and the first eight characters are not unique for the computers that are CCI nodes, use the ALIAS parameter. For more information, see the *UNIX section under Installing*.

Start CA CCI on UNIX

You must start or restart CA CCI to enable the configuration settings. To start CA CCI on UNIX, use the following command:

```
unicntrl start cci
```


Define the z/OS Computer

For AutoSys Workload Automation to submit a job directly to a CA mainframe scheduler, the z/OS computer must be defined in the AutoSys Workload Automation database, identifying the computer name and type.

If AutoSys Workload Automation submits a job directly to a CA mainframe scheduler, enter the following command at the operating system command prompt:

```
jil
insert_machine: remote_host
type: u
```

Alternatively, if AutoSys Workload Automation submits a job to a CA mainframe scheduler and it supports cross-platform dependencies through the AutoSys Workload Automation Connect Option on the mainframe, enter the following command at the operating system command prompt:

```
jil
insert_machine: remote_host
type: c
```

Use Jobs

As with most control systems, there are many ways to define and implement jobs correctly. It is likely that the way you use AutoSys Workload Automation to address your distributed computing needs will evolve over time. AutoSys Workload Automation is an automated Workload Automation system for scheduling, monitoring, and reporting on any AutoSys Workload Automation configured computer that is attached to a network. As you become more familiar with the features and the characteristics of your jobs, refine your use of AutoSys Workload Automation.

Define External Instances

You can define an external instance in multiple scenarios, as described in the following examples.

Example: Define an External Instance for AutoSys Workload Automation

Use the following JIL to define an external instance for AutoSys Workload Automation instance ACE on a computer named remote using port 9000:

```
insert_xinst: ACE
xtype: a
xmachine: remote\:9000
```

Example: Define an External Instance for CA WA ESP Edition

Use the following JIL to define an external instance for CA WA ESP Edition instance EE1 on a computer named ZOS1:

```
insert_xinst: EE1
xtype: e
xmachine: ZOS1
```

Example: Define an External Instance for CA WA CA 7 Edition

Use the following JIL to define an external instance for CA WA CA 7 Edition instance CA7 on a computer named remote3:

```
insert_xinst: CA7
xtype: u
xmachine: remote3
```

CA CCI must be configured to use the CA WA CA 7 Edition computer. When running multiple instances of CA WA CA 7 Edition, submit the proper monitor name. You can also submit the monitor name by using monitor=.

Submit Jobs

You can define a command job to depend on certain starting conditions to tell AutoSys Workload Automation when to run the job. However, if the job does not have any starting conditions, you must start it manually. Multiple scenarios can occur, as described in the following examples.

Example: Submit a Job Directly to CA WA CA 7 Edition

Use the following JIL to submit a job directly to CA WA CA 7 Edition, where the CA WA CA 7 Edition-defined job to run on the mainframe is CA7JOBNM and connect to XPS Monitor CA72XXX using userid MIKEW:

```
insert_job: ca7_job1
owner: MIKEW@CA72XXX
command: CA7JOBNM
machine: remote_host *
date_conditions: 1
days_of_week: all
start_mins: 25
```

The computer definition of remote_host must have type that is coded as *t* for CA NSM.

Example: Submit a Job to CA WA CA 7 Edition through the AutoSys Workload Automation Connect Option

Use the following JIL to submit a job to CA WA CA 7 Edition through the AutoSys Workload Automation Connect Option, where the CA WA CA 7 Edition-defined job to run on the mainframe is CA7JOBNM:

```
insert_job: ca7_job1
command: auto_cnct -a remote_host -j CA7JOBNM -s CA7 -c RUN -d
machine: remote_host *
date_conditions: 1
days_of_week: all
start_mins: 25
```

The computer definition of remote_host must have type that is coded as *c* for AutoSys Workload Automation Connect Option.

Example: Define Two Jobs with External Dependencies

Use the following JIL to define two jobs, each having a dependency on two z/OS cross-platform dependent jobs named JB5MINS and JB5HRS respectively:

```
insert_job: test_dep1
command: sleep 100
condition: success(JB5MINS^RMT)
machine: local_host
insert_job: test_dep2
command: sleep 100
condition: success(JB5HRS^RMT)
machine: local_host
```

This job definition requires that you use JIL to insert an entry for an external instance named RMT. The computer definition of remote_host must have type that is coded as c for AutoSys Workload Automation Connect Option.

Example: Submit a Job that Executes a CA WA CA 7 Edition-Defined Job after a Cross-platform Dependent Job Completes

You can also define jobs as a combination of both command jobs and external dependencies.

Use the following JIL to submit a job that executes a CA WA CA 7 Edition-defined job that is named ASYS7002 through AutoSys Workload Automation Connect Option after the cross-platform dependent job JB5HRS completes:

```
insert_job: CA7job4
command: auto_cnct -a remote_host -j ASYS7002 -s CA7 -c RUN -d
machine: remote_host
condition: success(JB5HRS^RMT)
```

Configure the AutoSys Workload Automation Connect Option

AutoSys Workload Automation Connect Option provides cross-platform scheduling capabilities to mainframe users by letting them schedule jobs into a mainframe-based scheduling system.

AutoSys Workload Automation on UNIX and Windows, as the manager, initiates a job that starts a job that is defined in a mainframe-based scheduling system using the AutoSys Workload Automation Connect Option. By using CA Common Services for z/OS (CCS), AutoSys Workload Automation Connect Option tracks the mainframe-based job and reports its status back to AutoSys Workload Automation on the distributed platform.

This product was developed using the foundation that is provided by CCS on UNIX and Windows, and on other distributed platforms. AutoSys Workload Automation Connect Option is a stand-alone product that can be installed on any z/OS system that needs to interface existing third-party mainframe scheduling systems with AutoSys Workload Automation.

Requirements

The hardware that is required to install and operate AutoSys Workload Automation Connect Option varies depending on whether you install all components on a single server or across multiple servers. Consequently, it is impossible to make a definitive statement of your hardware requirements.

Operating System Requirements

This product runs as a started task. It adds one active task to the system. The MAXUSER parameter of IEASYSxx PARMLIB member specifies the maximum number of tasks. If the system runs near the maximum number of tasks, you might need to increase this value.

Note: If you use AutoSys Workload Automation Connect Option to interface with OPC and the FILTER=OPC option is specified at startup, the AutoSys Workload Automation Connect Option load library must be placed in the APF authorization list. The FILTER=OPC option requires APF authorization.

This section provides an overview of the AutoSys Workload Automation Connect Option during its usual operation.

Message Processing

AutoSys Workload Automation Connect Option functionality is driven by message requests from remote AutoSys Workload Automation nodes. AutoSys Workload Automation Connect Option receives various message types, which let it do the following tasks:

- STARTJOB (SJ)
- ADD AGENT (4A)
- ADD DEPENDENCY (AD)
- PING (0A)

The hexadecimal codes following the message types are the internal one-byte codes that are used when the ASBROKERII transmits a request to AutoSys Workload Automation Connect Option. These codes are listed for reference as they are commonly used throughout debug messages that are produced by enabling the DEBUG control option.

STARTJOB

The STARTJOB message request is transmitted to AutoSys Workload Automation Connect Option following successful completion of a started job. The start job request on the distributed platform occurs against a job that is defined to execute on a AutoSys Workload Automation Connect Option-based system.

The following sample shows a AutoSys Workload Automation job definition:

```
insert_job: JOBB
owner: master
machine: MAINF
type: c
condition: success(JOBA)
command: auto_cnct -a MAINF -c RUN -j JOBC -s CA7 -d
```

The job definition adds a job that is named JOBB to AutoSys Workload Automation when inserted into the database. The job has a dependency on JOBA, also defined in AutoSys Workload Automation, which must complete before JOBB can start. If JOBA completes successfully, AutoSys Workload Automation starts JOBB. The scheduler on the distributed platform interrogates the start job request, determines the job is not a request to be executed on a AutoSys Workload Automation Remote agent, and forwards it the scheduler component. The Scheduler component packages up the request and transmits it to the mainframe-based AutoSys Workload Automation Connect Option system (based on the computer definition) using the CA Common Communications Interface (CA CCI).

When AutoSys Workload Automation Connect Option receives the request on the mainframe, it performs a two-step process to submit and track the mainframe-based job.

The first step involves parsing the command string that is passed to it by the scheduler component on the distributed platform. AutoSys Workload Automation Connect Option must determine which mainframe-based scheduling system

the job runs on by scanning the command string for the “-s” parameter. Using our preceding sample job definition, the scheduling system this job would be executed on is CA WA CA 7 Edition. To execute or initiate this request in CA WA CA 7 Edition, AutoSys Workload Automation Connect Option must first locate the scheduling system's batch interface job control it has loaded in memory (read from its parmlib dataset during startup). AutoSys Workload Automation Connect Option uses the three-character scheduling name that is passed to it in the command string and appends it to prefix “ATCJ”. Using our preceding sample job definition, AutoSys Workload Automation Connect Option would attempt to locate the job control for ATCJCA7.

The batch interface job control has now been successfully located. Next, AutoSys Workload Automation Connect Option extracts the “-j” and “-c” parameters to let it build the input command string that the scheduler's batch interface program executes. Using our preceding sample job definition, AutoSys Workload Automation Connect Option would insert a command string (replace \$CMDTEXT located within ATCJCA7) with “RUN,JOB=JOBC”. The scheduler's batch interface job control is now ready to be submitted on the internal reader for execution.

This completes the first step.

Before submitting the batch interface job to the internal reader for execution, AutoSys Workload Automation Connect Option adds two dependencies to its in-core dependency table. The first dependency to be added is an auto-added dependency for the batch interface job control (ATCJCA7). The second dependency to be added is a start job dependency for the job (JOBC) to be executed within the mainframe-based scheduling system. The start job dependency lets AutoSys Workload Automation Connect Option not only track the job originally requested to be run but to also track the batch interface job control should it result in a failure, which can be transmitted back to CA Workload Automation A. You can view AutoSys Workload Automation Connect Option dependencies at the operator console by executing the DEPLIST operator command.

After the dependencies are successfully added, the batch interface job control is submitted. Following successful execution of this job, the mainframe-based scheduler should start the requested job.

AutoSys Workload Automation Connect Option can track execution of both these jobs through its integration with the CAIENF component. The AutoSys Workload Automation Connect Option application subscribes to ENF for the following events:

- JOBINIT
- JOBTTERM
- STEPTERM
- JOBFAIL
- MEMTERM
- ENFTERM

If any of the preceding events occur on the mainframe system, the CAIENF component notifies AutoSys Workload Automation Connect Option of the event. AutoSys Workload Automation Connect Option interrogates the event data and determines based on jobname using its dependency array if the event matches a dependency it is tracking. If matched, the event is then further evaluated and eventually transmitted back to AutoSys Workload Automation (ASBROKERII). The appropriate status (RUNNING, SUCCESS, or FAILURE) is set on the original job that had been submitted to the mainframe for execution.

After the mainframe job completes and the JOBTTERM event has been transmitted back to the AutoSys Workload Automation system, the auto-added (ATCJCA7) and start job (JOBC) dependencies are deleted from the dependency array.

This completes the job life cycle (two-step process) within the AutoSys Workload Automation Connect Option system. For additional information about implementing AutoSys Workload Automation cross-platform scheduling with AutoSys Workload Automation Connect Option, see the "Initiating Jobs on a Mainframe" appendix in the AutoSys Workload Automation Connect Option Getting Started *documentation*.

ADD AGENT

When AutoSys Workload Automation starts up on the distributed platform, the scheduler process transmits an ADD AGENT message request to the AutoSys Workload Automation Connect Option based systems it will communicate with. This request lets AutoSys Workload Automation Connect Option register this remote AutoSys Workload Automation node as a node that uses its services.

You can view AutoSys Workload Automation Connect Option remote agent nodes at the operator console by executing the AGTLIST operator command.

ADD DEPENDENCY

The ADD DEPENDENCY request is received when a AutoSys Workload Automation job definition contains a cross-instance (external) dependency reference. For example, the following AutoSys Workload Automation job definition initiates an ADD DEPENDENCY message request from the scheduler process.

```
insert_job: JOBD
owner: master
machine: autosys
type: c
condition: success(JOBC^ACE)
command: sleep 10
```

The ADD DEPENDENCY message request differs from the STARTJOB message request in that AutoSys Workload Automation Connect Option monitors for the start of job JOBC on the mainframe system. It does not initiate the start of JOBC manually or within a mainframe-based scheduling system. When this job is started and completes, event status is reported back to the AutoSys Workload Automation system similar to the way STARTJOB dependencies are processed.

Cross-instance (external) dependencies are not deleted following success or failure of the job on the mainframe. If the same job is submitted again for execution, AutoSys Workload Automation Connect Option treats this submission as a new execution of the job, and it tracks it and sends event status to AutoSys Workload Automation.

You can view AutoSys Workload Automation Connect Option dependencies at the operator console by executing the DEPLIST operator command.

PING

The AutoSys Workload Automation Connect Option system receives at two-minute intervals an internal PING message request from registered AutoSys Workload Automation remote nodes. The ASBROKERII process initiates and transmits this internal ping by CA CCI to AutoSys Workload Automation Connect Option-based systems. This internal ping lets the scheduler process keep statuses of all AutoSys Workload Automation Connect Option-based systems that it communicates with. AutoSys Workload Automation Connect Option internally registers the request, and no further processing takes place.

AutoSys Workload Automation Connect Option does not deploy an internal ping subtask to monitor its remote connections, but it does track status through SUCCESS or FAILURE of send event buffers or through execution the ASPING operator command.

You can view AutoSys Workload Automation Connect Option remote agent nodes at the operator console by executing the AGTLIST operator command.

For more installation and implementation details, see the AutoSys Workload Automation Connect Option Installation and Administration *documentation* and the Getting Started *documentation*. These manuals provide a tutorial that steps you through initiating jobs on a mainframe from AutoSys Workload Automation.

Security Best Practices with CA EEM

This section provides general best practices on setting up and implementing security using the CA EEM.

Create Operational Roles

Most departments or subgroups have a requirement to access the same types of resources. Therefore, it makes sense to use that information to create separate roles for them. In general terms, the best practice is to define profiles or groups based on role usage as opposed to assigning permissions based on user ID. This setup lets you quickly get new employees configured; it also shortens the time that it takes to transition existing employees to a new role and remove any obsolete permissions that may pose audit problems. One of the challenges with security implementations is that most administrators do not go back and remove policies that are no longer needed. It is a lot like your garage, where you keep storing items and eventually they pile up. A better approach is to do some spring cleaning and go back regularly to remove items that are no longer required. There are always exceptions to this rule, but, as long as they are exceptions and not general practice, removing items should reduce overhead. If a particular user requires more access based on being a manager or supervisor, consider creating a role instead of giving that user individual access.

Enforce Naming Standards

A good naming standard not only makes it easier to create security policies, it also speeds identifying objects and their impact on the business. If all the Human Resources PeopleSoft jobs start with HRPS, it is easy to figure out which application HRPS_XXX belongs to. Many shops have a naming convention that they would like the developers to follow. Using security that you can enforce these rules. This setup frees resources that check for policy infractions to do other things. In addition to the naming standard, add a description to the policy that can be used to describe the original intent of the policy. Using this feature makes future decisions about that policy more informed and more likely to be effective without disrupting intended access.

Functional IDs

In the past, many sites would create a functional or service user account that would be used as a job owner and also a user that member of an application team would use to perform job administration. This setup would prevent having to create multiple super editor super exec users and creating an even larger security concern.

Because the introduction of CA EEM, a growing number of AutoSys clients have embraced CA EEM to let users perform all actions under their own user accounts. This setup removes the use of super exec, super edit, and the common practice of having to switch to a service or functional account to perform job administration. This setup provides more accountability and a clearer audit trail of who did what and when it occurred. Autotrack reports contain the actual userid that is used and not a generic userid. CA EEM audit reports also contain more useful information.

Use Root to Run Jobs

Using root as a job owner is a potential security risk. Best practice is to restrict the use of root completely or only to a selected group to create and execute root jobs if required. The ability to run a job as root always has the danger of somebody doing something intentionally or unintentionally that can disrupt the business. An alternative is to use sudo or equivalent to define what commands a non-root account to run a command that requires root access. This provides a clear audit trail of what user account issued the command, instead of having to find who issued the command.

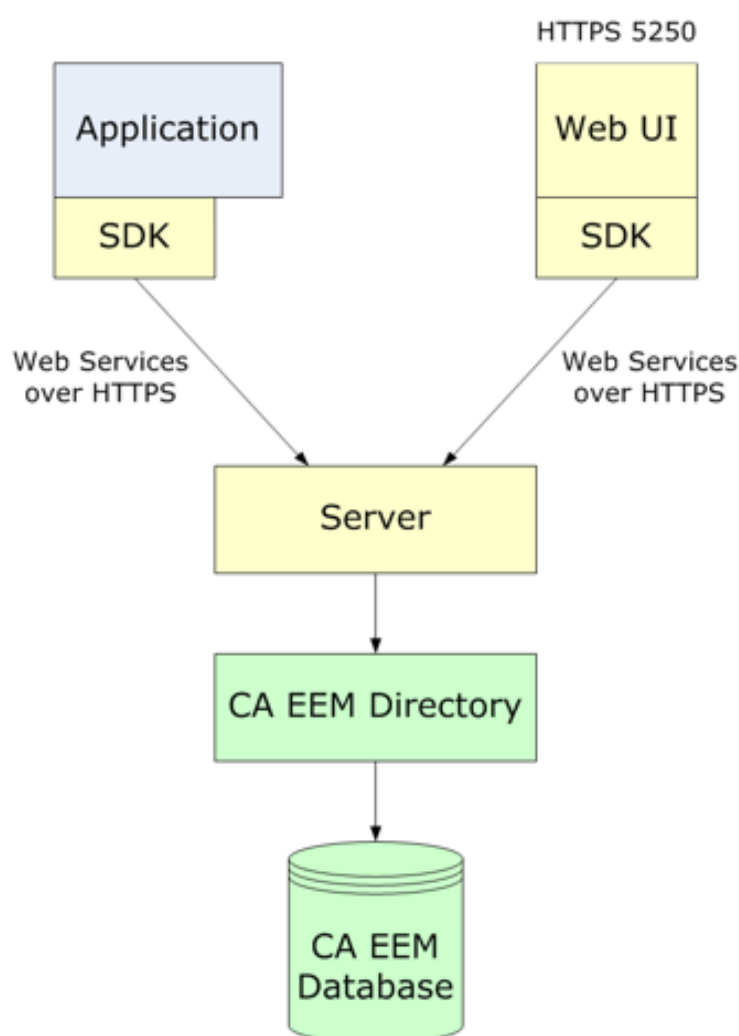
To implement this best practice, create a deny policy for the as-owner resource class with a resource name of root and root@* to cover the use of root with a hostname attached and root by itself.

Create a CA EEM Test Environment

Having a test environment lets you confirm new initiatives without impacting production. CA EEM provides a calendar function that lets you define the policies before they become active and also a 'pre-deployment' checkbox that prevents them from being used immediately. This setup would let you use the permission check function to verify the policies, but not to run them in an actual environment.

CA EEM Architecture

CA EEM is an embedded tool that is available to CA products, so that they can implement a common interface and management tool for securing applications. It was designed specifically for applications as opposed to a full system security system like CA Access Control. A single CA EEM instance can support multiple applications (products), with each having a separate area to store its application resources and policies. The client piece is embedded into the application, so there is no client-side installation for the application. Instead the product installs whatever part of the CA EEM SDK that it requires as part of the standard installation.



CA EEM provides several choices of which user stores are available to use. CA EEM has its own database that you can use to add and store user IDs, or you can point CA EEM to an external directory such as Microsoft Active Directory, an LDAP, or CA SiteMinder. When using an external directory, it is helpful to have someone from your security group involved in mapping the schema to ensure that all required users are found. For example: By default, active directory uses port

389 is used; for better performance, we recommend that you use port 3268, which is the default global catalog port. The current R12 releases of CA EEM also allow for multiple domain support.

Active Directory authentication can impact CA EEM and overall application performance when the user must be authenticated from a large enterprise directory. We recommend that you use filters to retrieve target groups of users rather than the entire Active Directory.

Terminology

Before you continue, note the following terms:

- **Identity**An *identity* is either the user ID or group.
- **Resource Class**A *resource class* is the type of object to be protected, for example, a directory or a file.
- **Resource**
A *resource* is the name of the object being checked against the resource class. For example, "c:\temp" is the resource name and the resource class is directory.

Best Match Algorithm

CA EEM has an option of using what is called the "Best matching algorithm". This option helps determine if access is allowed or denied to a resource. When it is not flagged, any policy that grants access is used to allow access. If it is flagged, then the most specific policy takes precedence. A simple example would be if you wanted to protect the jobname PAYROLL123. Suppose that you write two policies, one which has a resource name PAY* that allows access and another named PAYROLL* that denies access. With Best matching algorithm turned off, access is granted to the user because the PAY* resource grants access. With Best matching algorithm turned on, access is denied because the PAYROLL* resource is more specific to the actual resource name PAYROLL123.

Workload Automation AE uses this setup by default. It is considered best practice.

Policy Types

You can create three types of policies. Each policy type applies better to a different situation.

Access List

An Access List policy grants a list of users the same type of access to a group of resources. This policy type is useful when you have one or more resources that require the same access for a group of users. If all the actions are left blank, it is the same as turning them all on. A NONE access is not necessary or available; to achieve this access, you would simply omit any identity that should not have access.

Filter

An Access list policy may contain an optional filter is used to do additional, finer-grained checks for which the identity list and resource list are not sufficient. For example, you may need to check which department the identity is part of, which cannot be done using a group or user ID. Alternatively, you may want to give slightly different access to someone with a manager title within a group that is used to define who should be considered for access within this policy.

Access Control List

An Access Control List policy lets you give a group of users different access to each resource used in the policy. This type of policy is useful when a common set of identities require different access to a list of resources.

Identity Control List

An Identity Control List policy lets you define a default access and give each identity listed a separate access list to all the resources entered. This type of policy is useful because it has a default row that applies to everyone that is not explicitly defined as part of the policy. For example, if you wanted to deny access to everyone except a few users or groups, then you would leave the actions blank, providing no access.

Set Up and Test CA EEM

This section explains how to set up CA EEM for testing purposes.

Enable AutoSys Workload Automation to Use CA EEM Security

When AutoSys Workload Automation first starts up, it is enabled to use CA EEM if selected during the installation. Otherwise, it defaults to using the native security model. To check what security mode is in use issue the command: `autosec_test STAT`

Follow these steps:

1. Activate CA EEM using the following command:

```
autosys_secure
```

2. Respond to the prompts as follows:

NOTE

eIAM_server is the computer CA EEM is installed on.

-Please select from the following options:

- [1] Activate EIAM instance security.
- [2] Manage EDIT/EXEC superusers.
- [3] Change AutoSys database password.
- [4] Change AutoSys remote authentication method.
- [5] Manage AutoSys User@Host users.
- [6] Get Encrypted Password for Adapters.
- [0] Exit AutoSys Security Utility.

```
> 1
```

Are you sure you wish to activate EIAM security? [1(yes)/0(no)]: 1

Input the EIAM backend server name (or hit enter to cancel): eIAM_server

CAUAJM_I_60201 EIAM instance security successfully set.

Please select from the following options:

- [1] Revert to NATIVE instance security.
- [2] Manage CA EEM security settings.
- [3] Change database password.
- [4] Change remote authentication method.
- [5] Manage user@host or user@domain users.
- [6] Get encrypted password.
- [0] Exit AutoSys Security Utility. The CA EEM security model is now active.

Log in to the CA EEM User Interface

CA EEM is now the active security model for AutoSys Workload Automation. To go back to the Native security model, execute `autosys_secure` again and select option 1. CA EEM comes pre-configured with several default security policies. To modify any of those policies or create policies, log in to CA EEM.

Follow these steps:

1. Go to URL `https://EEM_Hostname:5250/spin/eiam`.
The login screen appears.
2. Select the Application Instance that you would like to manage security for, in this case `CA WorkloadAutomationAE`.
The User Name is filled in by default.
Enter the Password, which was given during the installation process.
3. Select Log in.
The UI opens. To manage the security policies, select the Manage Access Policies tab.

Modify the Default as-job Security Policy

Our first task is to secure jobs using the as-job policy. With modifications to this policy, we prevent jobs from being read by `autorep` and from being deleted by the `JIL` command. In general, the as-job policy controls access to jobs, but other checks are made for the owner, group, application, and machine. Selecting the as-job policy in the left pane Access Policies list, and then selecting the Default Job Policy for a specified server in the Policy Table opens the default policy.

CA Embedded Entitlements Manager

Home | Manage Identities | **Manage Access Policies** | Manage Reports | Configure

▼ Policies | ► Calendars | ► Permission Check

Search Policies

Explicit Grants | **Explicit Denies**

☒ Show policies matching name
Name:

☐ Show policies matching identity

☐ Show policies matching resource

Go

Access Policies

- as-appl
- as-calendar
- as-control
- as-cycle
- as-group
- as-gvar
- as-job
- as-joblog
- as-jobtype
- as-list
- as-machine
- as-owner
- as-resource
- Delegation Policies
- Dynamic User Group Policies
- Event Policies
- Obligation Policies

Type: ☐ Access Policy
☐ Access Control List
☒ Identity Access Control List

Identity Access Control List Configuration

Enter / Search Identities

Type: **Search Identities**

Identity:

Selected Identities

Identities	read	write	create	delete	execute
[Default]	☒	☒	☒	☒	☒

Resources

This illustration shows the default settings for the as-job for instance NY5: Default Policy. This policy states that all resources (jobs in this case) in the AutoSys Workload Automation NY5 instance have all actions that are enabled for the user. In other words, users can read, write, create, delete, and execute jobs in the AutoSys Workload Automation NY5 instance. When the user unchecks an action and saves the policy, that action is disabled.

The resource name in a CA EEM policy consists of the name of the AutoSys Workload Automation instance, a period, and the name of the corresponding asset. You can use wildcards to create policies that apply to multiple assets across different instances. CA EEM also supports the use of regular expressions to define an asset in a policy.

If a user unchecks all actions for as-job or any policy, the policy actually enables all actions. The way to disable all actions is to check Explicit Deny in the policy or ensure that the grant policy does not give permission to the undesired actions. This point is important to remember, especially with resource classes which have only one action. The first impulse is to uncheck that one action but the proper way to disable that action is to check Explicit Deny.

Example: Read a Job

Suppose that we want to disable read access for all jobs in the AutoSys Workload Automation NY5 instance (remember NY5.* specifies all assets in the AutoSys Workload Automation NY5 instance). We do this step by unchecking the read action and saving the policy.

To verify this, read a job using the autorep command.

Follow these steps:

1. From the command prompt, execute the following command:

```
autorep -J as-job1 -q
```

Assuming that as-job1 exists, you get the following security warnings:

```
CAUAJM_I_50200 Attempting to Read Job: as-job1
CAUAJM_W_10421 Job Read Access Denied!
CAUAJM_W_10439 No Policies Granting Access To Resource
CAUAJM_W_10440 Class: as-job Resource: NY5.as-job1 User:
BOCOTNG2\Administrator
Access: Read
CAUAJM_W_10442 Time: 1,110,470,876 Delegator: None
CAUAJM_E_50201 Read was not performed!
```

```
CAUAJM_E_50198 Exit Code = 1
```

2. To protect the job from being deleted, uncheck the delete action and save the policy.
By unchecking delete, you tell CA EEM to disable the delete action for the jobs that are listed in the resource box.
3. To confirm this test case, delete the job with JIL as follows:

```
jil
jil>>1>delete_job: as-job1
jil>>2>exit
```

```
CAUAJM_I_50200 Attempting to Delete Job: as-job1
CAUAJM_W_10421 Job Delete Access Denied!
CAUAJM_W_10439 No Policies Granting Access To Resource
CAUAJM_W_10440 Class: as-job Resource: NY5.as-job1 User:
BOCOTNG2\Administrator
Access: delete
CAUAJM_W_10442 Time: 1,110,470,876 Delegator: None
CAUAJM_E_50201 Delete was not performed!
```

```
CAUAJM_E_50198 Exit Code = 1
```

These actions work the same for all policies. Do not uncheck All, as each action is still enabled.

Modify the as-calendar Policy

The as-calendar policy controls access to calendar objects.

In this example, we modify the as-calendar default policy by turning it into an Explicit Deny policy, meaning we explicitly deny any access to calendar objects.

Follow these steps:

1. Create a standard AutoSys Workload Automation calendar as follows:

```
autocal_asc
```

2. Respond to the prompts as follows:

```
AutoSys Calendar Utility
```

```
Please select from the following options:
```

```
[1] Administer Standard Calendar.
[2] Administer Extended Calendar.
[3] Administer Cycle Calendar.
[4] List all Calendars.
[0] Exit AutoSys Calendar Utility.
> 1
```

```
Administer AutoSys Standard Calendar
```

```
Please select from the following options:
```

```
[1] Create a Calendar.
[2] Modify a Calendar.
[3] Delete a Calendar date.
[4] Delete an entire Calendar.
[5] Print dates for a Calendar.
[6] List all Standard Calendars.
[9] Exit from "Administer AutoSys Standard Calendar" menu.
[0] Exit AutoSys Calendar Utility.
> 1
```

```
Calendar Name: cal_1
```

```
Date (MM/dd/yyyy [HH:MM]): 05/02/2005
```

```
Date (MM/dd/yyyy [HH:MM]): 05/10/2005
```

```
Date (MM/dd/yyyy [HH:MM]):
```

```
CAUAJM_I_50400 ...Successful
```

3. Create a job that uses the newly created calendar using the following commands:

```
jil
jil>>1>insert_job: as_call
```

```
jil>>2>command: sleep 3
jil>>3>machine: localhost
jil>>4>run_calendar: cal_1
jil>>5>exit
```

```
CAUAJM_I_50204 Insert/Updating Job: as_cal1
CAUAJM_I_50205 Database Change WAS Successful!
```

4. Log in to CA EEM.
5. Go to Manage Access Policies, and click the default as-calendar policy to open the policy table. Select NY5: Calendar Default Policy from the table.
6. In the upper-right corner, select the check box that is labeled Explicit Deny, and save the policy.

The screenshot displays the CA Embedded Entitlements Manager web interface. The top navigation bar includes links for Home, Manage Identities, Manage Access Policies (which is the active tab), Manage Reports, and Configure. Below this, a breadcrumb trail shows Policies > Calendars > Permission Check.

On the left side, there is a 'Search Policies' section with two tabs: 'Explicit Grants' and 'Explicit Denies'. The 'Explicit Grants' tab is selected. It contains checkboxes for 'Show policies matching name', 'Show policies matching identity', and 'Show policies matching resource'. A 'Name:' text box and a 'Go' button are also present.

Below the search section is a tree view of policy categories. The 'Access Policies' category is expanded, showing a list of sub-policies: as-appl, as-calendar, as-control, as-cycle, as-group, as-gvar, as-job, as-joblog, as-jobtype, as-list, as-machine, as-owner, and as-resource.

The main content area is titled 'Policy' and contains a 'General' section. This section includes fields for 'Folder' (set to /WorkloadAutomationAll_NYS), 'Name' (set to NY5: Default Calendar Policy), and 'Description'. There are also dropdown menus for 'Calendar' and 'Resource Class Name' (set to as-calendar). The 'Type' section has three radio buttons: 'Access Policy', 'Access Control List', and 'Identity Access Control List' (which is selected).

Below the 'General' section is the 'Identity Access Control List Configuration' section. It features an 'Enter / Search Identities' header, a 'Type' dropdown (set to User), and a 'Search Identities' button. An 'Identity:' text box is also present. Below this is a 'Selected Identities' section with a table showing identities and their actions.

Identities	Actions
	read
	write

7. Return to the AutoSys Workload Automation command prompt and try to use JIL to load in a different job, which uses the standard calendar we created, as follows:

```
C:\Program Files\CA\UnicenterAutoSysJM\autosys\bin>jil
jil>>1>insert_job: as_cal2
jil>>2>command: sleep 3
jil>>3>machine: localhost
jil>>4>run_calendar: cal_1
jil>>5>exit
```

```

CAUAJM_I_50204 Insert/Updating Job: as_cal2
CAUAJM_W_10403 Calendar Execute Access Denied!
CAUAJM_W_10438 Explicit Deny Policy: "NY5: Default Calendar Policy"
CAUAJM_W_10440 Class: as-calendar Resource: NY5.cal_1 User:
MTSHERMAN2\Administrator Access: execute
CAUAJM_W_10442 Time: 1,114,796,178 Delegator: None
CAUAJM_E_10302 Database Change WAS NOT successful.

```

```

CAUAJM_E_50198 Exit Code = 1

```

Create Custom as-job Policies

You can also create custom as-job policies that represent the job name. This process lets some jobs be seen and other jobs be hidden, displaying a partial list of jobs.

The following steps assume that the AutoSys Workload Automation jobs cal_job and as_job1 exist in the NY5 instance.

Follow these steps:

1. Open the as-job default policy and add the resource name NY5.cal_job.
2. Select Save As to save the policy under a different name.
3. Enter NY5.cal_job in the Save As dialog box.
4. Delete the default resource name NY5.* in the new policy.
5. Select Explicit Deny and save the policy.
6. Open the as-job default policy and add the resource name NY5.as_job1.
7. Select Save As to save the policy under a different name.
8. Enter NY5.as_job1 in the Save As dialog box.
9. Delete the default resource name NY5.* in the new policy.
10. Save the policy.

Ensure the as-job default policy does not have explicit deny checked, so that cal_job is explicitly denied and as-job1 is not.

When you use the autorep command, the following results should appear:

```

autorep -J ALL
Job Name                Last Start                Last End                ST    Run Pri/Xit
-----
*****
as_job1                  -----                  -----                  IN    0/0
CAUAJM_W_10269 Access rights could not be obtained to view all jobs for Instance
GWH!
CAUAJM_W_50094 A partial list is displayed.
CAUAJM_W_10422 List Read Access Denied!
CAUAJM_W_10438 Explicit Deny Policy: "AUTOREP Policy"
CAUAJM_W_10440 Class: as-list Resource: NY5.AUTOREP User: MTSHERMAN2\Administrator
Access: read

```

CAUAJM_W_10442 Time: 1,114,809,778 Delegator: None

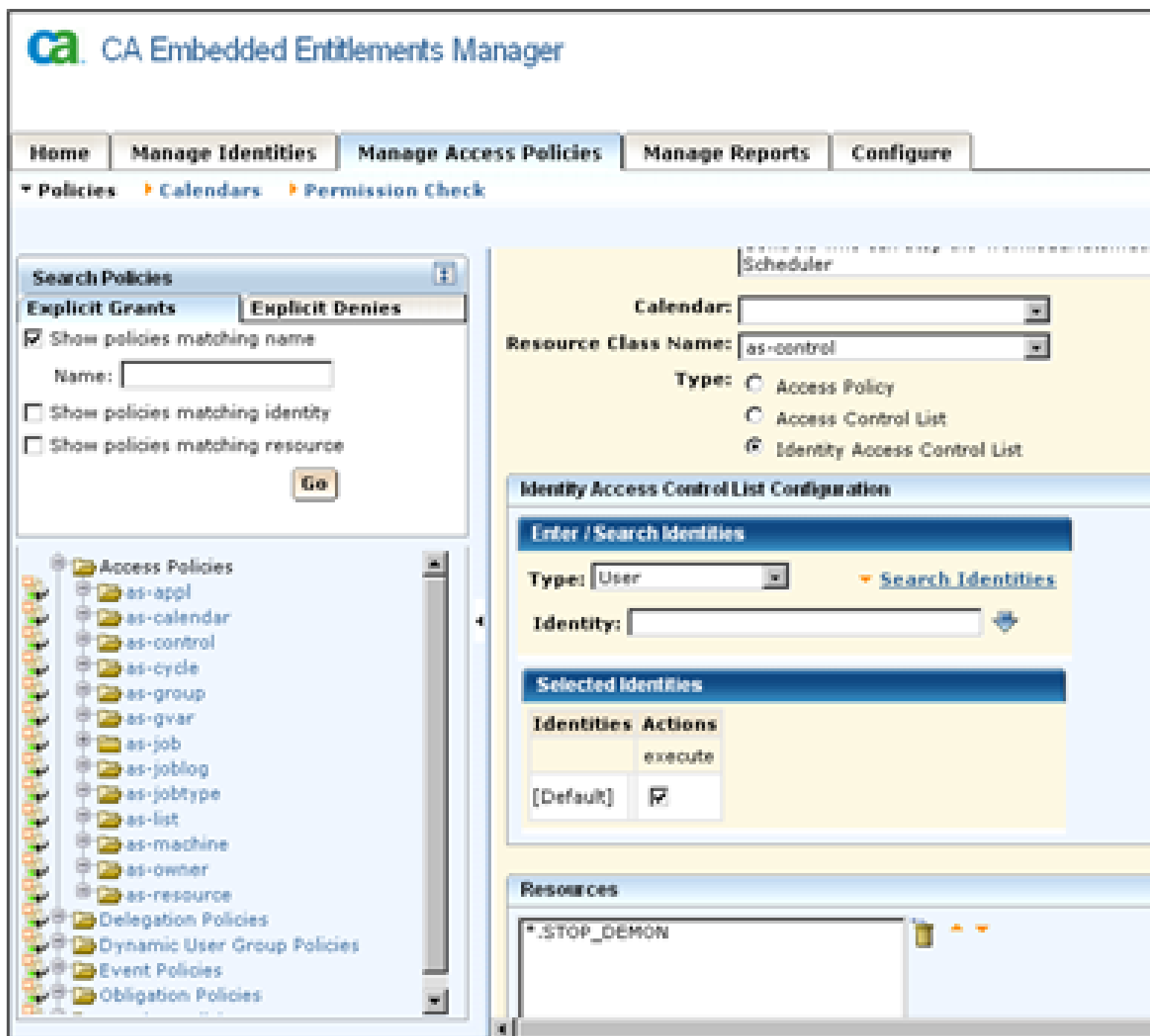
These actions work the same for all policies. Remember NOT to uncheck All, as each action remains enabled.

Example of the as-control/STOP_DEMON Policy

The as-control class controls access to critical AutoSys Workload Automation services. The only action is EXECUTE, which controls critical resources through the sendevent -e STOP_DEMON command and autosys_secure binary.

In this example, we use the as-control/STOP_DEMON policy to prevent the Scheduler from being shut down.

Selecting as-control in the Access Policies list, and selecting the STOP_DEMON Policy from the Policy Table opens the default policy.



With this action, the only option is Explicit Deny because, when you uncheck the action "execute", the action remains enabled. The only way to deny this action is to make this an Explicit Deny policy.

To prevent the Scheduler from being shut down, complete these steps:

1. From the illustration, scroll up, check Explicit Deny, and save the policy.
2. From a command prompt, enter the following command:

```
sendevent -E STOP_DEMON
```

You should get a security warning and this command is disabled, as follows:

```
CAUAJM_E_60033 SEND_EVENT API failed.
CAUAJM_W_10406 Control Execute Access Denied!
CAUAJM_W_10438 Explicit Deny Policy: "STOP_DEMON Policy"
CAUAJM_W_10440 Class: as-control Resource: NY5.STOP_DEMON User:
BOCOTNG2\Administrator Access: execute
CAUAJM_W_10442 Time: 1,110,471,641 Delegator: None
```

Protection of Binaries by Resource Classes

The following table shows which binaries and their actions are controlled and protected by which resource classes:

Binary	Class Protection	Access	Security Checkpoints
autocal_asc	as-calendar	R W C D	PRINT ADD (existing) ADD (new) DELETE
autorep	as-list\AUTOREP	R	On - Do NOT use security; Denied - Use security
	as-job	R	-J job
	as-machine	R	-M machine
	as-gvar	R	-G global variable
autostatad	as-list\AUTOSTAT	R	On - Do NOT use security; Denied - Use security
	as-job	R	-J job
autostatus	as-job	R	-J job
	as-gvar	R	-g variable
autosys_secure	as-control\SECADM	X	Controls option to disable external security
jil	as-calendar	X	run_calendar, exclude_calendar
	as-job	W C D	insert_job update_job delete_job
	as-machine	C D X	insert_machine delete_machine machine
	as-owner	X	owner
	as-group	R W X	group
	as-appl	R W X	application
job_depends	as-list\JOBDEP	R	On - Do NOT use security; Denied - Use security
	as-job	R	-J job

Binary	Class Protection	Access	Security Checkpoints
monbro	as-list\MONBRO	R	On - Do NOT use security; Denied - Use security
	as-job	R	jobs
sendevent	as-control	X	-e STOP_DEMON
	as-job	X W D	-e STARTJOB, -e KILLJOB, -e FORCE_STARTJOB, -e JOB_ON_ICE, -e JOB_OFF_ICE, -e JOB_ON_HOLD, -e JOB_OFF_HOLD, -e COMMENT (not global) -e SEND_SIGNAL -e CHANGE_STATUS -e CHANGE_PRIORITY -e DELETEJOB
	as-machine	X	-e STARTJOB, -e KILLJOB, -e FORCE_STARTJOB, -e JOB_ON_ICE, -e JOB_OFF_ICE, -e JOB_ON_HOLD, -e JOB_OFF_HOLD, -e COMMENT (not global)
	as-gvar	X W C D	-e SET_GLOBAL: -g options -g (existing variable) -g (new variable) -g variable=DELETE

NOTE

For a complete list of the resource classes and actions, see [Securing](#).

Create More Useful Policies

To this point, we have discussed only simple policies. Often, this type of approach would require many policies and maintaining them could take much time.

After you complete the installation, you will have one default policy for most resource classes. If you have multiple instances that have the same requirements, you can combine the resources to include several instances together, for example, two test instances. The default policy only contains the one instance. You could add the other instance to the resources and could cut your number of policies in half. To reduce the administrative costs, some other approaches can be used to reduce the number of policies required usually. This approach takes advantage of the ability of CA EEM to use an Identities attribute as variables with strong naming conventions.

For example, assume that your naming convention is to use the application code as the first part of the job name. By creating or using an existing group definition, you can either use the group name directly if it matches or map it to a Dynamic User Group. Use the group name in the policy as part of the filter.

The following policy allows full access to the TST instance for all users that belong to a Dynamic User Group (DUG) and have a jobname that starts with the DUG name.

 CA Embedded Entitlements Manager

Home Manage Identities **Manage Access Policies** Manage Reports Configure

▾ Policies ▸ Calendars ▸ Permission Check

☐ Assign Labels

Calendar:

Resource Class Name:

Type: ☒ Access Policy
☐ Access Control List
☐ Identity Access Control List

Identities

Enter / Search Identities

Type:

Identity:

Selected Identities

[All Identities]

Access Policy Configuration

Resources	Actions
	read
	write

CA Embedded Entitlements Manager

Home | Manage Identities | **Manage Access Policies** | Manage Reports | Configure

▾ Policies ▸ Calendars ▸ Permission Check

Add resource:

TST.*

☐ Treat resource names as regular expressions

Filters

Logic	(	Left type/value	Operator	Right type/value	)	Actions
None ▾	▾	request ▾ resource ▾	STRING ▾ STARTSWITH --" ▾	calculation ▾ STRING("TST.*" + dupName) Show Calculation Editor	▾	

Add Filter

This setup could also be extended to other resources such as calendar names and global variables by reusing the same policy, but with a different resource class. To do so, do a 'Save As' and in the new policy, switch the resource class to the one desired. This step only works when the resource class has the same actions and naming conventions. Which is most of the classes. For more details, see the Securing section.

Most organizations have an operations group that needs to be able to execute jobs, but maybe not to create or maintain production jobs. Most likely, they have another administration type group that can create and maintain, but not execute jobs in production. The example can be expanded to include additional filters to include these types of functions without having to resort to creating a set of policies.

CA Embedded Entitlements Manager

Backend: swatrh5x64-1 Application: WorkloadAutomationAE Welcome: eiamadmin (Log Out)

Home Manage Identities **Manage Access Policies** Configure

▼ Policies ▶ Calendars ▶ Permission Check

Resources

Add resource:



Actions

read
write
create
delete
execute
[All Actions]

NY5.*



☐ Treat resource names as regular expressions

Filters

Logic	(	Left type/value	Operator	Right type/value	)	Actions
NONE		request	REGEX	calculation		
		resource	MATCH (E)	STRING("NY5." + STRING(gug:PrincipalName) + "_")		
				Show Calculation Editor		
OR		global user group	STRING	value		
		Principal Name	EQUAL ==	autosys_admin_group		
OR	(	global user group	STRING	value		
		Principal Name	EQUAL ==	autosys_read_group		
AND		request	STRING	value	)	
		action	EQUAL ==	read		
OR	(	dynamic user group	STRING	value		
		Group Name	EQUAL ==	autosys_operations		
AND		request	STRING	value	)	
		action	EQUAL ==	execute,read		

Add Filter

In the example, if the resource name, for example, NY5.ABC_testjob, then a member of dynamic group ABC has full access to the job. A member of the autosys_admin_group also has full access to the job. A member of the autosys_read_group only has access to read the job. This step is accomplished by using an 'and' condition in parentheses to identify the group membership and the action requested. The same method is employed to limit what the autosys_operations group to only read and execute access.

Assuming you have the naming convention in place, you only need to ensure that the users are part of the correct DUGs. You can do this step by using LDAP groups and then creating the DUG policy based on that group membership or, in the worst case, by adding users to the DUG individually. Even in the worst case, you are still maintaining fewer policies while limiting the number of actual resource class policies as in the example.

Sometimes, you require more granular access to certain applications. If so, you can have specific policies for those applications by using the resource name to include the specific cases with filters that meet those needs. For other cases, a more generic policy applies because the Best Match algorithm is used.

Use Deny Policies

The use of deny policies can make it easier to implement certain things. For example, you want to deny all users to use root as the owner for any job. This setup is easier than creating an Identity control access list grant policy that uses the default access with no actions specified. When the best match algorithm is selected, care must be taken. When an authorization request is made and best match is enabled, CA EEM gathers all the policies that have the same number of relevant characters. This step includes both deny and grant policies. Therefore, if you have a deny policy that has xxx as the resource name and no corresponding grant policy that has xxx as a resource name, no access will be granted, even if there is a grant policy that allows access to resource '*'. To deny some cases of resource xxx and allow others, you need two policies with the same resource name. Depending on the use case, using an Identity Access Control List may be an easier way to implement this.

Implementation Best Practices

Before you start implementing AutoSys Workload Automation, consider the following best practices:

Instance Wide Encryption

Instance-wide encryption defines what if any AES key should be used to communicate to the application server. There are three choices to select from: NONE|DEFAULT|AES, the default being DEFAULT. The setting is defined in the \$AUTOUSER/config.\$AUTOSERV file under the UseEncryption entry. Any client utility must have the same key set to communicate to the application server.

- **NONE**
No encryption is used. This setting is required when Version 11.0 agents are being used in the environment. The Version 11.0 agent is designed to send status updates to the application server directly and cannot encrypt messages. The Version 11.0 agent also uses the CSAM or Secure Sockets Adaptor communication protocol which can be configured to use its own layer of encryption if desired.
- **DEFAULT**
uses a hard coded 128-bit AES encryption key. This key is the same for any installation of WA AE. This setup would be similar to using a default password and is not recommended for production environments.
- **AES**
Uses a customized 128-bit AES encryption key. This setting complies with the FIPS-140-2 standards.

We recommend that you use the AES setting and then create a custom key for the most secure setting. For more security, a different key for each tier level or even per instance could be used.

Naming Conventions

Thoughtful naming conventions offer the following benefits:

- Reduces downtime and allows for easier monitoring and reporting by job, machine, or group.
- Easier migration to CA EEM available with AutoSys Workload Automation to enforce by namespace.
- Web UI filtering and view creation by name.
- Simplifies the task of mass updates.
- A number at the end specifying job criticality can be used in notification mechanism or disaster recovery selection.

Naming should be a component of the current application architecture and consistently enforced during the implementation.

- Jobs
- Calendars
- Global variables
- Virtual machines
- Web UI views
- Tied to groups, applications, geography, criticality, machines, and job profiles

Example naming convention: Job

HR_aaaaa.bc_ddd_eeeeeeee

- HR (business process name)
- aaaaa – machine
- b – p or t (production or test) for those without test environments
- c - command (c), filewatcher (f), box (b)
- ddd - application group (ora, psft, sap)
- eeeeeeee – description of process

Example: Calendar

HR_1st_busday; AR_3rd_wed

Local File Locations

AutoSys Workload Automation job logs (standard out, standard error) should write to the same locations. Note the following best practices:

- Use jobname in logfile name. For example, /opt/CA/WorkloadAutomationAE/jobs/logs/jobname.out/err.\$TAG could be used regardless of the machine the job is set up to run on.
- jobname.out: stores general program logging information.
- jobname.err: stores any AutoSys Workload Automation-related errors and command errors.
- jobname.out.date: both logs can have date-stamped extensions, but they should be in the format YYMMDDHHMM (for example, 1409251224) for easy sorting.

You can view job log files in Web UI or using the autosyslog -j command.

- autosyslog -j job1 -t O to retrieve stdout
- autosyslog -j job1 -t E to retrieve stderr
- autosyslog -j job1 to retrieve the agent job log.

Standardized Batch Errors and Return Codes

Exit codes exist in all programs that are executed through AutoSys Workload Automation. AutoSys Workload Automation interprets these exit codes to determine whether the program has succeeded or failed. If these exit codes are not in place, AutoSys Workload Automation and the scheduling group cannot guarantee the results (failure of job when program fails).

The best guideline is to have an exit code check done after every major area in the program. If there are many tasks that are performed in one program, consider splitting the program up into separate processes to be run through AutoSys Workload Automation to allow for easier support and exit coding.

Example: Batch File

```
call c:\program files\scripts\vb\upload.exe
RC=%ERRORLEVEL%
```

Example: Exit %RCShell Program

```
/opt/apps/programs/scripts/upload.exe
RC=$?
```

Example: Exit \$RCVB Program

```
#Call program
upload.exe parm1 parm2
RC=$?
if ( $RC != 0)
then
print "FATAL ERROR: Exiting program"
exit $RC
fi
# continue on
```

System Agents

The agent that is installed with Release 11.3 and higher is called the System Agent. This agent replaced the older auto remote agent and was designed to be a common agent between several Broadcom Workload Automation products. To use a system agent, it must be defined to the instance or scheduler. The definition provides the details on how to communicate to the agent, such as the port number, the name, and hostname of the agent, which encryption if any is to be used. While AutoSys still communicates with the Release 11.0 SP5 version of the agent, this section is devoted to the system agent only.

Agent Names

The agent name is defined in the agentparm.txt file when the agent is installed and must match what is used in the machine definition in AE. The default is to use WA_AGENT or AGENT depending on which media is used to install the agent. While using the same name is supported in AE, it does create a limitation in terms of the encryption key. To use different keys or to change the keys for one or more agents, use unique agent names as the agent name is tied to only one key. Using unique names also provides an easier way to look at certain messages and differentiates which machine

the message came from. If the wrong agent name is entered, the agent rejects the message and the request is not fulfilled.

If clusters are used, the agent can still use a unique name, but a common alias would be required to enable a job to run on any node in the cluster.

Agent Encryption

Each agent installation is asked to specify the encryption type and Key. WA AE currently supports only AES encryption to comply with the FIPS compliance rules. This key is separate from the Instance Wide encryption which is specifically for communication to the application server. The agent key defines how to communicate to that specific agent and is part of the machine definition.

The same options are available per agent: NONE|DEFAULT|AES with the default being DEFAULT.

- **NONE**
No encryption is used. No encryption is not recommended for a production environment.
- **DEFAULT**
Uses a hard coded 128-bit AES encryption key. This key is the same for any installation of WA AE. This setup would be similar to using a default password and is not recommended for production environments.
- **AES**
Uses a customized 128-bit AES encryption key. This value complies with the FIPS-140-2 standards.

While in theory each agent could have a unique encryption key, maintaining them could become a chore. The recommended best practice is to have a unique key per tier level at a minimum. This should prevent an accidental submission to a production agent from a non-production scheduler.

Clusters

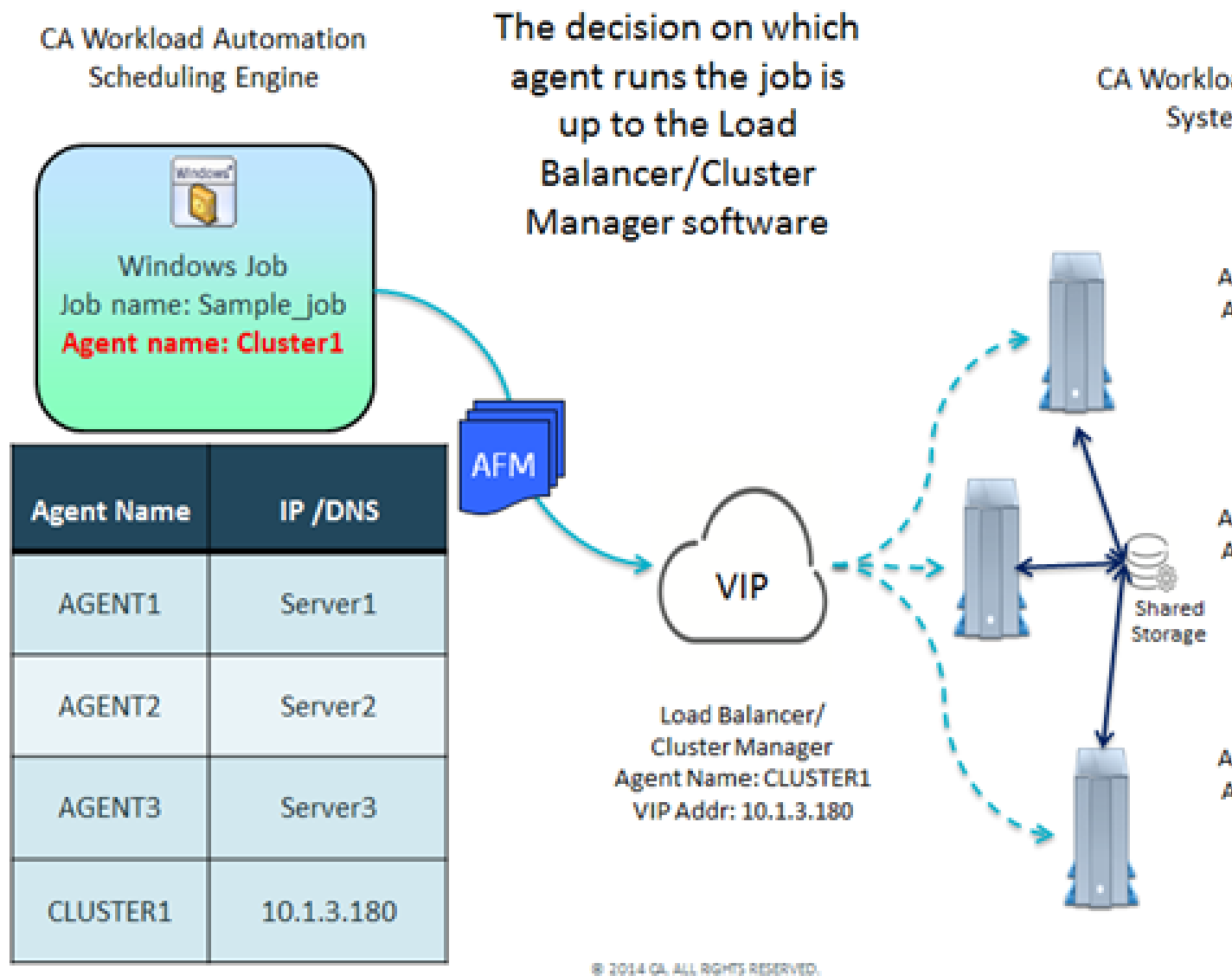
The use of clusters is common. Clusters often raise questions about how to configure them within AutoSys. The simple rules are as follows:

- Ensure each node has an agent installed with the same port.
- Ensure each node has an agent installed with the same encryption key.
- We recommend that they also share the spool directory. This setup helps give a complete picture in the spool file if the machine changes between restarts.
- Each node must be defined in AutoSys as a real machine.
- The cluster name must also be defined as a real machine.
- The cluster machine name is what should be used in any job that is destined for the cluster.

Consider the the following example. Three servers make up a cluster. The three hostnames are Server1, Server2, and Server3. The cluster name is Cluster1.

- Server1: Agent name Agent1, nodename: Server1
- Server2: Agent name Agent2, nodename: Server2
- Server3: Agent name Agent3, nodename: Server3
- Cluster1: Agent name Cluster1Agent1, nodename: Cluster1

Assuming that unique agent names are used, the jobs that run on this cluster uses "machine: Cluster1".



The alias name Cluster1 is defined in each of the agents agentparm.txt file with the configuration setting:

communication.alias_N=alias_name

- **N**
Represents the alias sequence number starting with 1.

If a job should only run on Server1, then you would use “machine: Server1” in the job definition.

Lock Down the Agent

Sometimes, you need to prevent unwanted schedulers or application servers from communicating to an agent. This step can be done to prevent a non-prod agent talking to a production agent or to prevent any other scheduler or application servers from using this agent. Besides the firewall approach, locking down the agents to communicate only to known managers is another way to help accomplish this task.

To do so, the manager must be defined to the agentparm.txt file before the agent starts. The agentparm.txt configuration setting is:

```
communication.manager.dynamic.modifications.lock=true
```

This configuration is applicable to both the Scheduler and Application server.

When this value is set to true, define only instances that are used to communicate to this agent.

Define the scheduler and application server manager entries in the agentparam.txt, if log files, autoping, and some other functions are expected to work.

An example scheduler manager setting:

```
communication.manageraddress_1=scheduler_hostname
communication.managerid_1=ACE_SCH
communication.managerport_1=7507
communication.socket_1=plain
```

An example application server setting:

```
communication.manageraddress_1=appServer_hostname
communication.managerid_1=ACE_APP_appServer_hostname
communication.managerport_1=7500
communication.socket_1=plain
```

If you have multiple application servers installed, you need an entry per application server. This setup is because the application managerID contains the hostname to allow for multiple application servers.

For HA configurations, only one SCH manager entry is required as only one scheduler is active at a time. When a failover occurs, the shadow sends a message to all agents that it has taken over and all messages should be forwarded to it. The same type of message occurs when the primary is actively processing events again.

Virtual Machines

We recommend virtual machines as a best practice for load balancing across clients. Using the System Agent, the only method that is used to obtain CPU usage is using the CPU_MON method. Even if another method is configured, the System Agent already has a CPU monitor in place and uses that instead of rstatd or vstat. This setup provides better performance. On average, using the other methods can take 6 or more seconds per machine in the virtual as opposed to less than a second using the new method for all machines.

Some clients use a virtual machine for all machines. When a single real machine exists in a virtual machine, the priority should be set to 0. This setup ensures that the load test is not run for every job.

Virtual machines can also be used to simplify moving machines between instances. For example, by having the same virtual machine that is defined in both environments, the job definition does not need to change as each one has the appropriate real machines defined. If only one machine is defined, then the CPU usage check is not made and the job is run on the machine.

Virtual Resources to Control Job Flow

Virtual resources provide another way to control the flow of jobs to prevent overloading machines, better regulation of resources such as software licenses, restricting how many jobs access an application such as a database and other practical uses.

Because they are maintained by the scheduler, they offer a better alternative than using global variables or the not running starting condition, which can run into race conditions and not provide the desired results. Resources are defined and added to a job definition using the jil syntax, thus making them easy to set up and use.

Configure Email Notifications

AutoSys Workload Automation can send an email when a job terminates or when a job fails.

Follow these steps:

1. Configure WA AE to use an SMTP server.
2. Update the job definition to have the following attribute to have a job send an email:
 - **send_notification: y|n|f**
 - **notification_emailaddress:**
 Specifies the email address of either a user or distribution list. You can specify multiple email addresses or can use multiple entries of notification_emailaddress.
 For example,
 notification_emailaddress: user1@ca.com,dlist1@ca.com
 is equivalent to
 notification_emailaddress: user1@ca.com
 notification_emailaddress: dlist1@ca.com
 - **notification_msg:**
 Specifies an option message to append to the end of the email. Enclose the message in double quotes and escape any embedded double quotes with a backslash.
 notification_msg: "The job has failed, contact the on call person"

The job is configured to send emails.

Backup and Recovery

This article includes basic backup and recovery data points:

- DBA only: Total database dump and load
- Utility based: See [Back Up and Restore Definitions](#)

autobcp is the only supported mechanism available for synchronizing the dual server databases. Although autobcp is recommended, it is not the fastest way to synchronize the AutoSys Workload Automation databases. Database dump and load are faster, but should only be done by a DBA.

Job Profile Usage

Note the following job profile usage best practices:

- We recommend that you use a profile management policy.
- Job profiles should be strictly enforced with file permissions.
- Users should source profiles in their batch scripts.
- Profile names should be tied to the naming conventions.

Job Architecture

To reduce job stream overhead, consider the following job architecture tips:

- Jobs with the same date and time conditions produce many unprocessed STARTJOBS for the same time; stagger start times when possible.
- Schedule jobs based on when they need to FINISH rather than when they need to start.
- Nested boxes: use "flat" boxes to reduce overhead of recursively evaluating starting conditions for child jobs.
- Boxes containing jobs with no starting conditions increase the time to process the box's RUNNING event; serialize jobs within a box whenever possible.
- The Scheduler that processes the Box STARTJOB event starts all the jobs in the box. For example, 10 boxes of 100 are more efficient than 1 box of 1000. Job starts are processed simultaneously.

AutoSys Database User Password

The database user for AutoSys Workload Automation must always be available to use. Many clients have password policies that force them to change regularly. If not coordinated with AutoSys, you run the risk of stopping your workload from processing. To prevent this from occurring, you could request an exemption from the policy, and set the password to never expire. If this setup is not possible, use the `autosys_secure` command when the password needs to be changed. This command updates the DB password and also updates the configuration file with the new encrypted password.

Schedule and Performance Tuning

Before you start an implementation of AutoSys Workload Automation, consider the following best practices:

Set IP Caching

You can cache the IP address for each machine the first time it is contacted by the scheduler or application server.

The default is to enable this feature to improve performance by reducing the number of system calls to retrieve the IP address of a machine node name. If your network is designed such that you do not use hostnames that switch to another IP address while leaving the old address still active, this feature should help improve your performance. WA AE automatically recovers from a dynamic IP address when the old IP address becomes invalid. WA AE does not function as desired if the old IP address is still active with an active agent still running, such as not using a VIP address for a cluster.

The setting is found in the configuration file under the `EnableIPCaching` entry. The default is 1, enable IP caching.

To disable this feature, reset to 0. This setting can be dynamically changed by issuing a `kill -HUP scheduler_pid application_server_pid`

Database Sizing

This topic includes AutoSys Workload Automation and Web UI database sizing best practices.

AutoSys Database Sizing

The following requirements must be met or exceeded for AutoSys Workload Automation to install and run correctly in a Windows environment:

- Processor: 1-GHz processor
- Physical memory:
 - MSSQL: 1 GB
 - Oracle: 1 GB
 - Sybase: 1 GB

Note: On Linux, check your memory configuration if you have hugepages enabled and enough memory allocated. This configuration improves the performance of the DBMaint processes.

- Swap file:
 - MSSQL: 2 GB (twice the size of RAM)
 - Oracle: 2 GB (twice the size of RAM)
 - Sybase: 2 GB (twice the size of RAM)
- Disk space:
 - Full product: 320 MB
 - Client only: 120 MB
 - Agent only: 105 MB
 - CA EEM: add 95 MB
- Database:
 - MSSQL: 40 MB plus an additional 1 MB for the log file. MSSQL adjusts the sizes incrementally over time based on usage.
 - Oracle: data - 400 MB; index - 100 MB
 - Sybase: 200 MB

The following requirements must be met or exceeded for AutoSys Workload Automation to install and run correctly in a UNIX or Linux environment:

- Processor: 1-GHz processor
- Physical memory: 1 GB
- Swap space: 2 GB available swap space
- fsize: The file size ulimit must be set to unlimited. To determine the current setting, issue the following command:

```
ulimit -f
```

If the value is not "unlimited", change the ulimit value by issuing the following command:

```
ulimit -f unlimited
```

- Disk space:
 - Full product: 1 GB
 - Client only: 45 MB
 - Agent only: 20 MB
 - CA EEM: add 120 MB
- Database:
 - Oracle: data - 400 MB; index - 100 MB
 - Sybase: 200 MB

Database Sizing Recommendations

The following list is based on Release 11.3.x. There is no precise calculation due to numerous variables, including items such as lengths of string values (object names, text data), optional job definition attributes, job types, and job run behavior. Job definitions alone range on average from approximately 3,500 bytes (box) to 14,000 bytes (wsdoc) with an average of about 6,400 bytes without optional attributes like conditions and resources. Although variables exist, the algorithm that is provided is based on typical implementations and should help you to determine the appropriate next steps for your environment.

Assumptions

- Object name (job, machine, calendar) average lengths are half the maximum
- 80 percent of jobs defined are CMD job types
- 25 percent of defined jobs have date conditions
- 50 percent of defined jobs have one condition
- 10 percent of defined jobs use virtual resources
- Defined calendars include one day/week
- 3 percent of jobs run generate alarms
- Less than 500 calendar
- Blobs are not stored in the database
- Autotrack level is 0

Estimation

- 10-MB static data and overhead (including less significant objects such as calendars, user credentials)
- 2.7 MB per year of aggregated events
- 140 jobs / MB
- 750 job runs / MB
- 750 calendars / MB
- 900 cycle calendars / MB
- 1800 machines / MB
- 2500 virtual resources / MB
- 5500 global variables / MB

1. Overhead 10

2. Jobs defined (J) _____ / 140 = _____

3. Job Runs

a. Job runs per day _____

b. # of days retained _____

c. Total job runs (multiply a by b) _____ / 750 = _____

4. Machines _____ / 1800 = _____

5. Virtual resources _____ / 2500 = _____

6. Global variables _____ / 5500 = _____

7. Aggregate events (years) _____ * 2.7 _____

Total Data Size (MB) =====

Index Size (MB) .67 * Data Size = _____

Example

Stored procedure cache: 40 percent

WCC Database Sizing

When using an external database, create a plan for configuration/monitoring and reporting. You can use a single database or two databases. If you plan to keep a long history or many job runs, using two databases gives better performance.

Monitoring

The sizing for the monitoring and configuration database is influenced by a couple of factors. The main ones are the number of views and the number of jobs in those views, the number of alerts that are stored. In general, it should only require 500M for up to 100k jobs and 1000 alerts.

Reporting

Database	Jobs	Job Runs	Size
MS-SQL	50,000	150,000	650M
MS-SQL	50,000	1,000,000	3.2G
MS-SQL	150,000	10,000,000	15G
Oracle	50,000	150,000	2G
Oracle	50,000	10,000,000	12G
Oracle	150,000	30,000,000	31G

Tuning and Connectivity

This article explains tuning and connectivity best practices.

Scheduler Tuning

- **DB_CONNECTIONS:**

Is an environment variable that identifies the number of persistent connections that are used for job scheduling and for making SDK (CLI and Web UI) requests. Each DB connection does require memory, so using too high a setting can have an adverse effect. For most large sites a maximum setting of 64 is then adequate. The default settings are designed for shops with 10,000 jobs defined and 50,000 job runs per day.

- **SCHED_SCALE**

Is an environment variable that identifies the number of “pipes” uses to process events. The default setting is 5 and increasing it can help improve the throughput of the scheduler. The limits are 1-64, with 5 the default. Setting it too high can have adverse effects by overwhelming agents with too many requests. For most large sites, a maximum setting of 25 is adequate.

Both of these variables can be used with each other to provide the best performance for your scheduler. The DB_CONNECTIONS is used by both the scheduler and the application server, so the total number of database connections that are required are the sum of each application server times the setting plus the number of schedulers times the setting plus 10 to allow for some other connections for DBMaint, forecast commands, and some internal scheduler processing for HA requirements. After you change the settings, monitoring the memory and CPU consumption is recommended and checking the lag time using the autoaggr command to see what impact is being made regarding the lag and latency in processing events.

Database Connectivity

- **DBLibWaitTime:**

Lower the value to time out more quickly.

Default: 90 seconds.

- **EventProcSleepTime:**
Applicable for low number of unprocessed events in the event table. Set to 0.5 seconds.
- **XInstanceDBDropTime:**
Leave set unless cross-instance dependencies are rare.
- **ShadowPingDelay:**
Lower the value to failover more quickly (but not too quickly); reduce in decrements of 50 percent.
- **DBEventReconnect:**
Lower the value to failover more quickly (but not too quickly); reduce in decrements of 50 percent.

Database Tuning

- **Database bottlenecks:**
 - Poor memory usage
 - Database device file sharing
 - Fragmented indexing
 - Poor disk hardware
- **Memory**
 - Allocate enough real memory for the database engine.
 - Database total memory should fit within real memory. No paging!
- **Sybase memory (sp_configure)**
 - Total memory (physical memory in Sybase 15.x)
 - Additional network memory
 - Procedure cache percentage: 40 percent

Note: On Sybase 15.x, procedure cache is an absolute value and total memory is dynamic.
- **Oracle memory (query the v\$parameter table)**
 - db_block_buffers: area used for processing data (db_block_buffers x db_block_size = total memory)
 - shared_pool_size: area used for stored procedure library and data dictionary (40 percent of memory)
 - sort_area_size: amount of real memory set aside for sorting data. Otherwise, temp segment on disk is used.
 - On Linux, use of huge pages provides better performance than standard-size paging.
- **Indexes**
 - Update statistics frequently when job definitions change frequently.
 - Drop and recreate indexes. You can use the reindexDB.pl Perl script to determine which indexes are available. It is located in the \$AUTOSYS/dbobj directory.
 - Over 10000 updates after 500 definition changes. Statistics update after 5 percent of job definitions change.
- **Database I/O**
 - Create the AutoSys Workload Automation database before installing AutoSys Workload Automation on separate devices.
- **DBA:**
 - Split data and log in to their own devices.
 - If you are not using RAID, data and log devices should reside on separate disks. Use RAW disk partitions whenever possible. RAW disk partitions have faster I/O than device files.

Web UI Best Practices

Web UI is the user interface for AutoSys Workload Automation. Web UI lets you administer and monitor workload objects such as jobs, machines, global variables, calendars, and virtual resources.

The architecture has changed over the past couple of versions to its present form. This architecture consists of two services that are divided into two functions. The first is the main Web UI component that makes up the UI itself and is contained into a single 64-bit Tomcat server. The second is the collector instance which runs as a separate service.

Installation

Web UI can be installed on Windows or Linux platforms. For more information on supported version, see [Release Notes](#) and [AutoSys Workload Automation Operating System and Database Support](#). The two most important decisions to make are the user that is used assigned for Web UI to run as, and which database to use for the reporting database and also the configuration/monitoring database. They can be the same database or two different databases.

When installing on Linux, the user used to run Web UI is root by default. In many sites, the use of root is restricted and either not desired to run applications, or making it more difficult for the AutoSys admin team to control Web UI or even gather logs if required. If these conditions apply to your site, then using another userid, such as the same one that is used to run AutoSys, to run Web UI will make it easier to administer Web UI.

The database choice can have an impact on performance and your users' experience. The default Derby database is not meant for mid-size to large sites. If you have any of the following conditions, we recommend using an external database (Microsoft SQL Server or Oracle):

- Over 50k defined jobs
- Over 50k job runs
- 70 or more concurrent users
- Over 200 views
- Over 500 alerts defined
- Over 5 AutoSys servers defined

For the reporting database, if there are over 1 million job runs, we recommend using a separate external database for reporting.

Design Your Web UI Environment

Define Roles

Web UI requires the use of EEM to allow only the desired components of the UI to be displayed. This setup suits a role-based design. The first step is to define the roles, and what components or tabs should they have access to. A common set of roles that are found in many sites are:

- **Workload Administrator**
Normally has access to all aspects of Workload, but may have restricted access to production.
- **Operations**
Primarily concerned with monitoring production environments only. They typically have start and stop powers, but not the ability to edit jobs.
- **Scheduler**
Could be called change control or other department or titles. Involved with migrating jobs from lower environments to production, but may not have execute permission on jobs.
- **Application Developer**
Develops the application and job streams that make up a particular business application. Works mostly in Dev and UAT, but may only have limited, if any, access to production.

Your site may have more or different sets of roles and responsibilities, but the main point is that you first define these roles and what types of functions they perform.

Define Components to Display

The functionality that is made available to a Web UI user is controlled mainly with the `ApplicationAccess` resource class in EEM. Each action determines whether a functional tab is displayed. Most users that log in to Web UI do not have to and should not be able to log in to EEM. The Security Configuration action controls whether the EEM link is displayed when a user logs in to Web UI. If access is not granted to the action, the link does not appear. The same applies to the configuration action. In most sites, only the Administrators can add or remove AutoSys instances or change the base configuration for all users.

Define Content

After you have defined the roles and primary responsibilities, implement the design with EEM policy. The first step is typically configuring the `ApplicationAccess` resource class so that the basic screen for each role is set. The next step is to determine what those screens contain. If you have defined multiple AutoSys servers, you can also limit what servers are displayed in each Web UI component. For example, if you have full access in development, and read-only access in production, it is not necessary to display the Quick edit tab. Use the `ServerAccess` resource class to limit what is displayed in each of the tabs.

Another area where you can enhance your users' experience is by reducing the number of job types. Not all users will need all types. For example, AutoSys supports three different ERP products, but most sites have one or at most two of those products, so you can limit the different job type menus to include only job types that your site needs or has agents for. The Web UI batch utilities have a `filter_jobtypes` script that can be called to remove job types that are not desired.

Design Monitoring Views

Monitoring views provide an easy way for your end users to determine the status of jobs that are of concern to them. Depending on their function, they are most likely only concerned with a subset of the total jobs that based on certain criteria. For example, operations may be most concerned with jobs that are in a certain status like failure or some other abnormal ending state, while an application group user may only be concerned with jobs that belong to this application group, no matter the status.

The first step is to develop a list of the different groups and by which criteria to differentiate the jobs they are concerned with. Typically, a status or job name are the most common criteria, but there are other fields that can be used, such as group, application, machine, owner, job type, or time zone. When you create the list, consider the naming convention that is used for the views. As with the AutoSys resource classes, a good naming convention makes it easier to secure. In this case, security can be used to limit what is displayed to make it easier to find what you are looking for without having to scroll through a long list of views.

When creating the view, consider the following points. Depending on the type of criteria, displaying the flow view or graphical view may not make sense. For example, if the views criteria is based on the failed status of jobs, they most likely are not part of the same stream. Turning off the default rendering of a flow would help speed up the display of the view. This step can be done when creating the view in the VMC mode or if you are automating the onboarding of new applications.

The number of jobs in a flow can also affect the performance and behavior of the view. By default, any view over 2000 elements does not have the flow display. Any view with over 5000 elements by default does not display as a graphical view. These defaults can be overridden in the configuration tab for the monitoring preferences, but exercise caution when you are planning to increase them. Large numbers of elements makes the view harder to understand, and it has an impact on performance to generate a graph to display the relationships between the jobs.

Backup and Recovery

In case of disk failure or database corruption situations, we recommend that you back up your Web UI configurations. Web UI provides a set of batch utilities that can be used to export and import most of the settings. The two sets of data to save are the monitoring views and the configuration tab settings. To export all of your existing configuration data, issue the following command:

```
wcc_config -u ejmcommander -p ejmcommander -x config.txt
```

Next, export all your monitoring views by issuing the command:

```
wcc_monitor -u ejmcommander -p ejmcommander -x views.txt
```

To import your configurations, issue the following command:

```
wcc_config -u ejmcommander -p ejmcommander -i config.txt
```

To import your monitoring views, issue the following command:

```
wcc_monitor -u ejmcommander -p ejmcommander -i views.txt
```

Integrate Service Desk Manager

Service Desk Manager is an enterprise-level service desk solution that lets you automate IT processes and provide audit trails for regulatory compliance. Service Desk Manager lets you improve efficiency, while fostering customer satisfaction and improved productivity. It is easily configured to support the ITIL Service Desk model; you can leverage our collection of years of service and support best practices, or implement your own processes. Service Desk Manager provides a self-service interface that helps your customers resolve their own issues. From this web interface, they can submit tickets, can check status, and can browse the knowledge base.

When AutoSys Workload Automation is integrated with Service Desk Manager, default data files added to the Service Desk Manager database during configuration are used to create requests through the web services. In addition to the data files, Service Desk Manager also provides the default templates for AutoSys Workload Automation.

Configure Service Desk Manager for AutoSys Workload Automation

To configure Service Desk Manager for AutoSys Workload Automation, you must load the default data files, templates, and contacts to the Service Desk Manager database.

Only the Service Desk Manager Administrator is authorized to modify the default templates that are provided during the configuration process.

Follow these steps:

1. Open the Service Desk Manager application and verify that it operates properly.
2. Use a Windows command prompt to access the following location:
C:\Program Files\CA\Service Desk\data\integrations
3. Run one of the following commands to complete the installation:
4. • For an ITIL configured Service Desk Manager installation:
pdm_load -f integAutoSys.dat
pdm_load -f itil_integAutoSys.dat
- For a "default" or non-ITIL configured Service Desk Manager installation:
pdm_load -f integAutoSys.dat

Configure AutoSys Workload Automation to Activate Its Service Desk Manager Interface

After you install and configure the basic software, configure the AutoSys Workload Automation server to activate its Service Desk Manager interface. To initiate a service desk ticket to Service Desk Manager, AutoSys Workload Automation requires the Web Service Desk URL, login identifier, and password. You can substitute the Web Service Desk Customer for the Web Service Desk login identifier and password.

Service Desk Manager is installed as a stand-alone product. AutoSys Workload Automation requires at least Service Desk Manager r11. Contact your Service Desk Manager administrator if you need assistance with setting the parameters in this section.

Follow these steps:

1. Log in to a AutoSys Workload Automation-configured computer as the EXEC Superuser and issue the following command at an instance command prompt:

```
sendevent -E STOP_DEMON
```

The Scheduler completes any processing that it is currently performing and then stops.

2. Open the Integration screen of the AutoSys Workload Automation Administrator and update the Web Service Login ID, Web Service Login Password, and URL Location fields.

If you use a Web Service Customer identifier to access Service Desk Manager, update the Web Service Customer field instead of the Web Service Login ID and Web Service Login Password fields.

3. Set the ServiceDeskURL parameter in the \$AUTOUSER/config.\$AUTOSERV configuration file to the URL of your Service Desk Manager Web server (for example, http://localhost:8080/axis/services/USD_R11_WebService?wsdl).
4. Set the ServiceDeskUser parameter in the \$AUTOUSER/config.\$AUTOSERV configuration file to your Service Desk Manager user ID and password.

If you use a Service Desk Customer identifier to access Service Desk Manager instead of a user ID and password, update the ServiceDeskCust parameter instead of the ServiceDeskUser field.

5. Issue the following command at an instance command prompt:

```
eventor
```

The Scheduler starts.

Initiate a Service Desk Ticket with AutoSys Workload Automation

The integration of Service Desk Manager into AutoSys Workload Automation lets you open a service desk ticket (request or incident) when a job fails. The AutoSys Workload Automation Scheduler opens a service desk ticket during terminal status processing based on the job's service_desk attribute. If the job's service_desk attribute is set, the Scheduler prepares and sends the service desk ticket using the job's svcdesk_desc, svcdesk_pri, svcdesk_imp, svcdesk_sev, and svcdesk_attr attributes. Messages are written to the Scheduler log indicating whether the service desk ticket was sent and processed successfully.

Only the service_desk attribute is mandatory. If the job's svcdesk_desc, svcdesk_pri, svcdesk_imp, svcdesk_sev, and svcdesk_attr attributes are not set, they use the AutoSys Workload Automation service desk template values set in Service Desk Manager. This template is included in the Service Desk Manager installation. Before you initiate a service desk ticket, ensure that Web Services for CA Service Desk Manager are active.

To use the svcdesk_attr field, find the identifier for the type of attribute you want to use. For example, for a specific category, an entry like "category/pcat:5108".

Example: Initiate a Service Desk Incident

This example shows the JIL statements that are used in a job definition to initiate a service desk incident with a priority of 1 for a job named service_desk_on_failure_1:

```
insert_job: service_desk_on_failure_1
machine: localhost
command: false
owner: user@localhost
service_desk: y
svcdesk_pri: 1
svcdesk_desc: "service_desk_on_failure_1 has failed."
EXIT
```

Example: Initiate a Service Desk Request

This example shows the JIL statements that are used in a job definition to initiate a service desk request with an impact of 3 and a severity of 4 for a job named `service_desk_on_failure_2`:

```
insert_job: service_desk_on_failure_2
machine: localhost
command: false
owner: user@localhost
service_desk: y
svcdesk_imp: 3
svcdesk_sev: 4
svcdesk_desc: "service_desk_on_failure_2 has failed."
EXIT
```

Getting Started

AutoSys Workload Automation is an automated job control system that lets you manage workload from a command-line interface (CLI) or graphical user interface (GUI). Web UI is the GUI of AutoSys Workload Automation.

Integration with the supported Common Components expands its product capabilities.

This section contains the following topics:

AutoSys Overview

AutoSys Workload Automation provides end-to-end visibility of application workloads and dependencies across platforms. Event-based triggering, real-time alerting and dynamic critical path management, ensure timely availability of business applications. It readily scales to enterprise demands by intelligently and efficiently allocating workloads based on real-time information about available computing resources and service level agreements. It also enables organizations to quickly provision and process workloads across physical, virtualized and cloud environments.

AutoSys Workload Automation is an automated job control system that lets you manage workload from a command-line interface (CLI) or graphical user interface (GUI). [Web UI](#) is the GUI for AutoSys Workload Automation.

AutoSys Workload Automation lets you:

- Schedule, monitor, and report on jobs. You can schedule jobs across platforms.
- Monitor each job and provide reports when the job enters various phases of the production schedule. For example, if a job fails, you are automatically notified of the problem. You can diagnose the problem and start the job again.
- Generate reports that improve control and facilitate regulatory compliance.
- Forecast to identify and understand the business impact of a job within a job stream. You can send notifications for overdue jobs and can take other types of actions, such as bypassing less critical jobs.
- Manage resources. It evaluates the available resources and schedules jobs to meet a defined SLA; thereby improving resource utilization.

All activity controlled by AutoSys Workload Automation is based on jobs. A *job* is any single command, executable, UNIX shell script, or a Windows batch file. You define jobs to AutoSys Workload Automation by creating [job definitions](#). Each job definition contains attributes that specify the job's properties and behavior. For example, you can specify conditions that determine when and where a job runs.

You can define jobs using the following methods:

- Job Information Language (JIL)—A scripting language that lets you define and modify jobs, global variables, machines, user-defined job types, external instances, and blobs.

NOTE

For more information about JIL, see [Manage JIL](#). For information about how to define or modify jobs, see [Define Jobs](#).

- Web UI: A GUI that lets you interactively set the attributes that describe when, where, and how a job should run. The fields in the Web UI GUI correspond to the JIL subcommands and attributes. In addition, from the Web UI GUI, you can define calendars, global variables, and monitor and manage jobs.

A [job type](#) defines the type of work that you want to schedule based on your business requirements. AutoSys Workload Automation supports numerous job. While creating a job definition, you must specify the job type and its required and optional attributes. For example, you can create a command job to run a Windows command, an FTP job to download a file from a server, and so on. You can also define box jobs, which are containers that hold other jobs or box jobs.

NOTE

For information about each of these job types, see the articles under [AutoSys Scheduling](#).

Communications

Network data between the AutoSys Workload Automation Software Development Kit (SDK) client and the application server is prepared using the proprietary AutoSys Workload Automation Request Response Protocol (RRP). The SDK clients are as follows:

- The AutoSys Workload Automation CLI utilities
- Web UI
- The scheduler when transmitting external instance information to the application server of another instance
- Any product that links with the AutoSys Workload Automation SDK libraries.

Network data between the scheduler and the agent, the application server and the agent, the application server and the scheduler, or between the scheduler and the CA Workload Automation EE manager is prepared using the proprietary Automation Framework Message (AFM) protocol.

NOTE

For more information about the CA Workload Automation ESP edition, see the [CA Workload Automation ESP Edition](#) documentation.

SSA is an application that lets the CA components use a single multiplexed communication port to ease firewall administration and minimize conflicts with other applications. SSA provides port multiplexing and SSL encryption.

Security

AutoSys Workload Automation lets you secure objects such as jobs, calendars, cycles, global variables, machines, and resources. You can delegate administrative privileges to these objects to specific users or user groups.

AutoSys Workload Automation provides security in the following ways:

- System-level security
- Native security
- External security

Integration with Other Broadcom products

You can integrate AutoSys Workload Automation with other Broadcom products to expand its product capabilities.

NOTE

For information about the products that you can integrate with AutoSys Workload Automation, see [Support for Other Broadcom Products](#) and [Configure AutoSys Workload Automation to Work with Other Broadcom Products](#).

AutoSys Architecture

From a hardware perspective, the AutoSys Workload Automation architecture comprises of the following types of computers that are attached to a network:

- Server
The *server* is the computer where the server is installed. The server is the core of the AutoSys Workload Automation system and is installed with a scheduler, application server, agent, web server, SDK, and client.
- Client

The *client* is the computer where the client is installed. A client is any executable that interfaces with the application server.

- Agent

The *agent* is the computer where the agent is installed. You must install the agent on every computer that you use to run jobs.

Instance

An AutoSys Workload Automation *instance* is a licensed version of AutoSys Workload Automation software running as a server with one or more clients or agents. Clients and agents can run on a single computer or on multiple computers. An instance uses its own scheduler, application server, and event server, and web server and operates independently of other instances.

The instance ID (an uppercase, three-character alphanumeric name) identifies an AutoSys Workload Automation server installation on a particular computer. The AUTOSERV environment variable references it. The default instance ID is ACE. However, you can specify a different instance ID only during installation.

Multiple instances can run on the same computer, but they must have different instance IDs. For example, one instance for production and another for development. Multiple instances can run on the same computer using a single copy of the binaries. You can schedule jobs on the same computers without interfering or affecting other instances.

Schedulers are associated and identified with specific instance names. They have a unique ID, named as _SCH, which is suffixed to the instance name. This naming convention ensures the uniqueness and trace-ability of a scheduler across multiple instances.

For example, for the instance name ACE, the scheduler name is ACE_SCH.

- In non-HA mode, two running schedulers in the enterprise cannot share the same instance name
- In HA mode, two primary schedulers cannot share the same instance name

If two schedulers share the same instance name and try communicating with an agent, the agent cannot determine the scheduler with which it must communicate.

NOTE

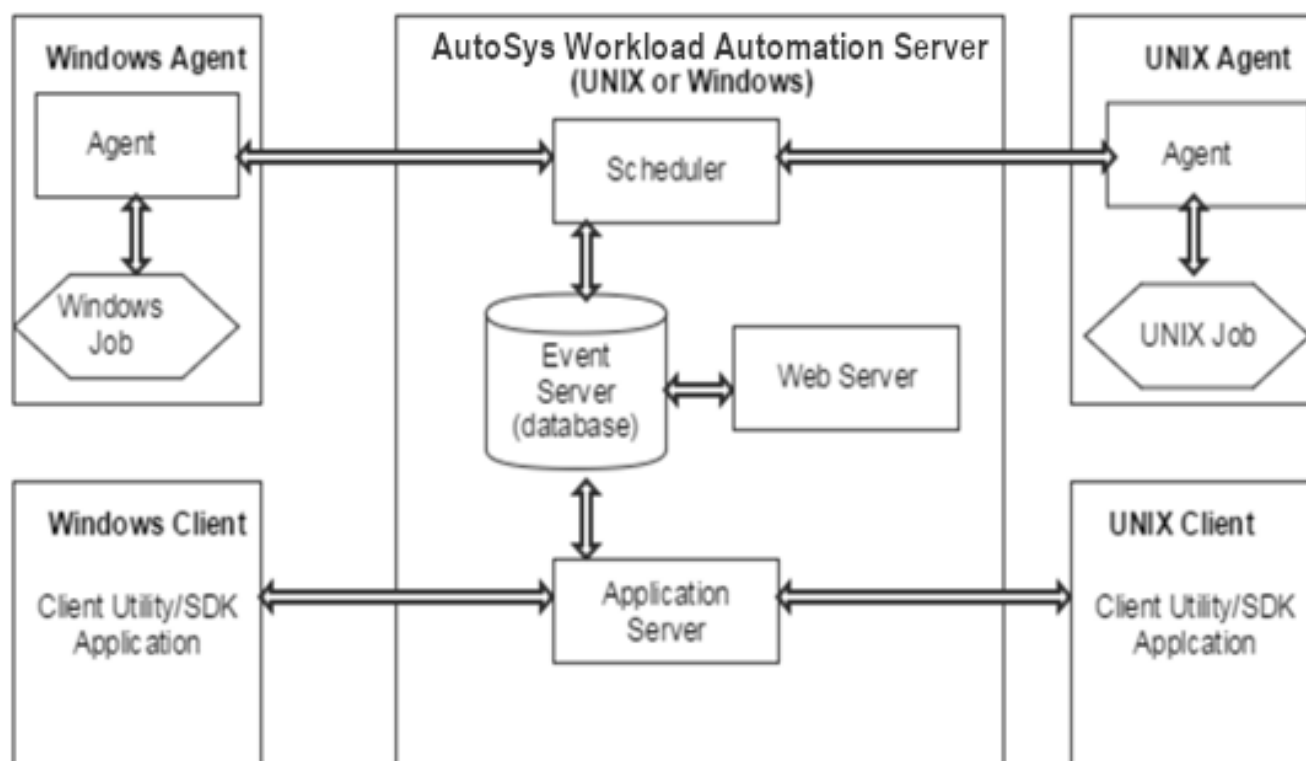
- Additional instances can be added later by either running the wa_setup.sh script (UNIX) or using the installation wizard (Windows).
- (Windows) Multiple instances of AutoSys Workload Automation are supported and these instances will share the %AUTOSYS% directory. Only a new %AUTOUSER% directory is created.

AutoSys Workload Automation Components

The main AutoSys Workload Automation components are as follows:

- Event server (database)
- Application server
- Web server
- Scheduler
- Web UI
- Web UI Collector
- Agent
- Client

The following illustration shows the components in a basic configuration, and displays the communication paths between them:



Event Server

The *event server* (database) stores all the objects that AutoSys Workload Automation uses. The job, machine, and calendar object definitions comprise a subset of the data that is contained in the event server as do job events. The application server manages the creation, update, and deletion of the AutoSys Workload Automation objects in the event server. The scheduler polls the event server for job events and fetches the corresponding object definitions that the event references when necessary.

Occasionally, the database is referred to as a data server, which actually describes a server instance. That is, it is a UNIX process or a Windows service and associated data space (or raw disk storage), which can include multiple databases or tablespaces.

AutoSys Workload Automation supports various databases including Oracle, Sybase, and Microsoft SQL Server. The scheduler and the application server processes interface directly with the database. Therefore, these processes require a vendor database client installation to access the database. All other AutoSys Workload Automation processes interface with the application server and do not require database client installations. The scheduler and the application server interact with the database using vendor-specific native code libraries.

NOTE

While AutoSys Workload Automation uses the database solely as a SQL engine, it does use Sybase Open Client C Library communications protocol, Oracle Common Interface, or Microsoft SQL Server Multiprotocol Net-Library to communicate with the vendor database server installation. For more information, see the vendor documentation.

Dual Event Servers

You can configure an AutoSys Workload Automation instance to run using two event servers (databases), and this configuration is named *dual event server mode*. The dual event server mode provides high availability by running two

event servers that are synchronized to maintain identical data, including object definitions and events. AutoSys Workload Automation reads from one event server and writes to both the event servers simultaneously. If you lose one event server due to hardware, software, or network problems, operations can continue on the second event server without losing data or functionality. This feature is independent of any database replication or redundancy.

For various reasons, database users often run multiple instances of servers that are unaware of the other servers on the network. When implementing AutoSys Workload Automation, the database can run for AutoSys Workload Automation only, or it can be shared with other applications.

Application Server

The *application server* acts as the communication interface between the event server and the client utilities. The application server receives requests from the client utilities, queries the event server, and returns the responses to the client utilities.

Scheduler

The *scheduler* is the program, running either as a UNIX daemon process or a Windows service, that runs AutoSys Workload Automation. The scheduler processes all the events that it reads from the event server.

When you start the scheduler, it continually scans the database for events to process. For example, when the scheduler finds a STARTJOB event, it verifies whether the event satisfies the starting conditions for that job in the database. Based on this information, the scheduler determines the actions to take and instructs the appropriate agent to perform the actions. These actions include starting or stopping jobs, checking for resources, monitoring existing jobs, or initiating corrective procedures.

High Availability

To detect and recover from a failure, you can configure AutoSys Workload Automation with a second scheduler, named the *shadow scheduler*. This shadow scheduler must run on a separate computer, and it takes over if the primary scheduler fails. This configuration is named *high availability*.

If AutoSys Workload Automation is running in high availability and dual event server mode, a third scheduler that is named the *tie-breaker scheduler* is required. The tie-breaker scheduler is a scheduler process that runs on a third computer. The tie-breaker scheduler remains permanently idle and periodically updates its heartbeat in the event servers to indicate its presence. The tie-breaker scheduler resolves contentions and eliminates situations where one scheduler takes over because of network problems.

NOTE

Shadow and tie-breaker schedulers and dual event servers are independent features. If you configure AutoSys Workload Automation to run in high availability mode, these components run together. For more information about shadow and tie-breaker schedulers, installing dual event servers, and configuring high availability, see the [\(UNIX\) How to Improve Workload Performance in Highly-Available Environments](#) or [\(Windows\) How to Improve Workload Performance in Highly-Available Environments](#).

Agent and Agent Plug-ins

The *agent* (named as Workload Automation Agent for UNIX, Linux, or Windows) is the key integration component of AutoSys Workload Automation that lets you automate, monitor, and manage workload on different operating environments, applications, and databases. To run jobs on a particular system, your administrator must install an agent on that system. For example, if your workload must run on a UNIX computer, your administrator must install and configure the Workload Automation Agent for UNIX, which lets you create jobs to run UNIX scripts, execute UNIX commands, transfer files using FTP, monitor file activity on the agent computer, and perform many other tasks.

You can extend the core functionality of the agent by installing one or more agent plug-ins. For example, if you have a relational database such as Oracle, you can install the Database Agent plug-in with the agent to query and monitor the

database. Other agent plug-ins, such as Application Services, Oracle, PeopleSoft, SAP, and Web Services, are available. You can perform all actions for the agent plug-ins, such as starting and stopping them, on the agent.

The agent lets you perform the following tasks:

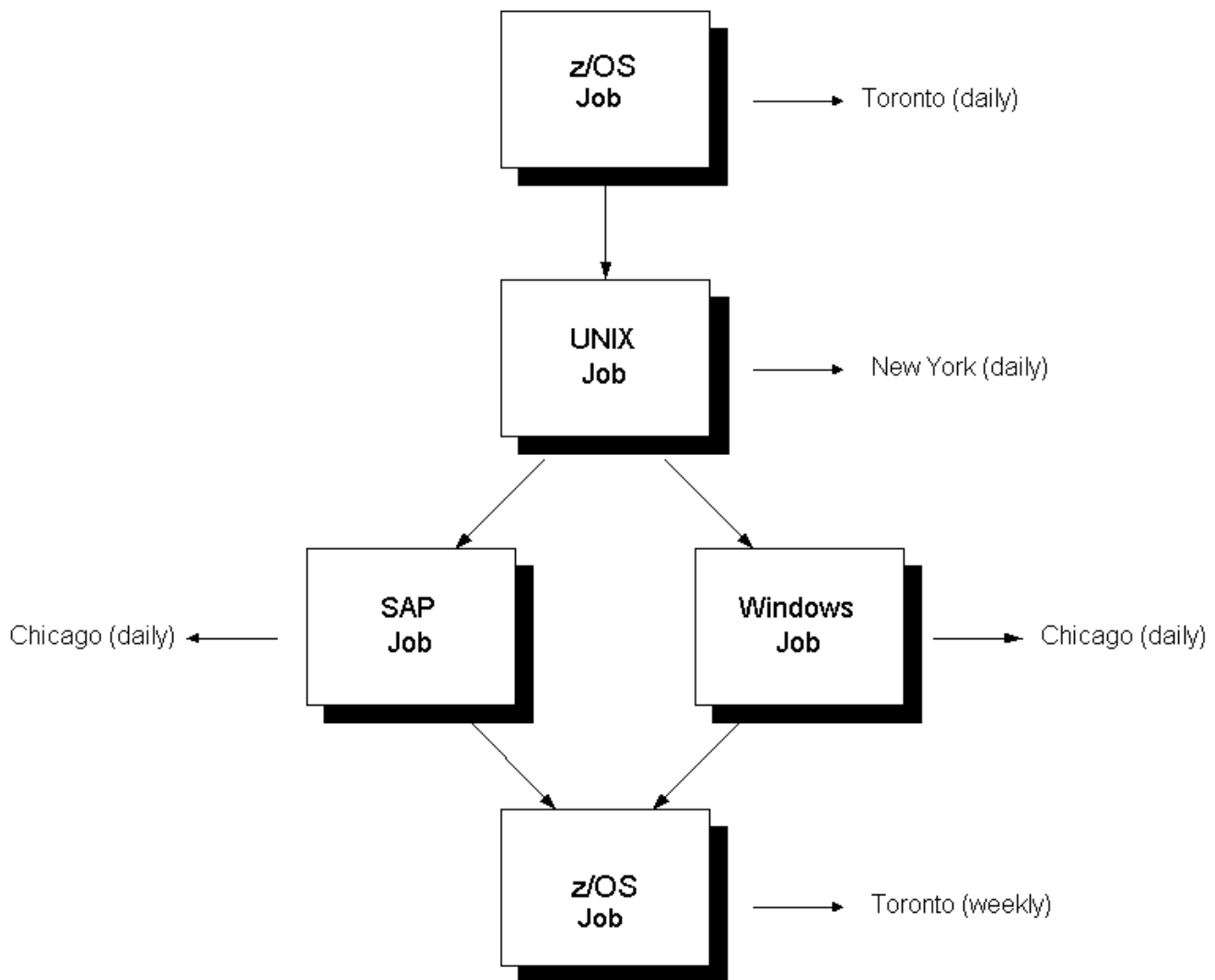
- Run the Windows command files and UNIX scripts.
- Execute the UNIX commands.
- Monitor file activity and release jobs that are based on that activity.
- Transfer files using FTP.
- Monitor the agent computer for CPU usage, disk space, IP address, process execution, and text files.
- Monitor the Windows agent computer for Windows event logs and the status of Windows services.
- Retrieve or set the value of an SNMP variable.
- Subscribe for the SNMP trap information or publish.

NOTE

- AutoSys Workload Automation also works with agents that run on different operating environments such as i5/OS. The agent plug-ins only work with the agent for Windows and UNIX operating environments.
- For more information about agents and agent plug-ins, see the *Workload Automation Agent for UNIX, Linux, or Windows documentation*.

Example: Workload with Different Types of Jobs

The following workload contains z/OS jobs, a UNIX job, an SAP job, and a Windows job, running on different computers, in different locations, and at different times:

Figure 4: Agent and Agent-plugins**Legacy Agent Replaced by Workload Automation Agent**

The Workload Automation Agent for UNIX, Linux, or Windows replaces the Remote Agent (auto_remote) that was provided with Unicenter AutoSys JM 4.5.1 and r11. The current release documentation refers to auto_remote as the *legacy agent*.

The new agent provides more job types, including monitoring and FTP jobs. The agent is automatically installed on the computer where AutoSys Workload Automation is installed. You can also install the agent on remote computers to run jobs on those computers.

NOTE

- For information about the differences between the legacy agent and the new agent, see the Differences Between the Legacy Agent and the Agent.
- If you have not yet decommissioned or upgraded the legacy agents (r11 agents) to Release 11.3 or later:

If you are running CA Workload Automation AE Release 11.3.6 SP5, you must apply the following fixes, or you will experience problems 4586 and 4612 that relate to the legacy agents. For more information, see the documentation for previous releases:

- Linux patch RO94612 is available in [Release 11.3.6 SP5](#).
- The solution for Problem Number 4568 is available in [Release 11.3.6 SP6](#).
- The solution for Problem Number 4612 is available in [Release 11.3.6 SP6 CUM1](#).

Client

A *client* is any executable that interfaces with the application server. This includes AutoSys Workload Automation CLI applications such as JIL and autorep. It also includes the Web UI services, which are clients of the application server and service the Web UI components, and any user-defined binaries that link to the AutoSys Workload Automation SDK.

The client applications work by calling Application Programming Interfaces (APIs) that are available in the application server. A client can run anywhere in the enterprise provided it can reach the computer where the application server is running. It does not require the installation of a database vendor client. Clients are the means by which you can control the scheduling environment by creating and monitoring the scheduling resources.

Web Server

Apache Tomcat is the designated web server that is used to host web services. This web server is installed and configured as part of the AutoSys Workload Automation installation. Apache Tomcat uses the AutoSys Workload Automation configuration parameters for database access and security. By default, web services use port 9443 to communicate with the Apache Tomcat server. The Apache Tomcat server uses port 5250 to communicate with CA EEM.

AutoSys Workload Automation SDK

The AutoSys Workload Automation Software Delivery Kit (SDK) is a ready-to-use programmatic interface that includes the Java and C++ APIs. Developers can use one or more of the SDK APIs to develop applications.

The AutoSys Workload Automation SDK is a client application that can be run from any AutoSys Workload Automation client installation.

The AutoSys Workload Automation SDK is available as follows:

- The runtime environment—Is installed on the Web UI server when you install Web UI. It enables communication between Web UI and the AutoSys Workload Automation application server.

View SDK API Documentation

NOTE

If you upgraded to CA Workload Automation AE Release 11.3.6 SP2 or a later release, recompile the Java SDK applications for the functionality to work correctly.

The following two JAR files contain the SDK API documentation:

- `asapidoc.jar` -- Includes Java API documentation in HTML format.
- `asapicppdoc.jar` -- Includes C++ API documentation in HTML format.

The `asapidoc` and `asapicppdoc` JAR files are located in the Documentation folder of your installation directory. The default locations are as follows:

UNIX: `install_directory/Documentation`

Windows: `install_directory\Documentation`

- ***install_directory***
Specifies the root directory where AutoSys Workload Automation is installed.

To view the SDK API documentation

1. Install SDK on the client.
2. Enter any one the following commands based on whether you want to view the Java API documentation or the C++ API documentation:

```
jar -xf asapidoc.jar
```

Or

```
jar -xf asapicppdoc.jar
```

The JAR file is expanded.

3. Open the index.html file in your web browser.
You can now view the SDK API documentation.

Interface Components

You can use the client utilities (commands) or Web UI to define, monitor, and report on jobs.

NOTE

For more information about the commands, see [AutoSys Commands](#).

On Windows, you can use the Administrator utility to view or modify the configuration parameters of all the AutoSys Workload Automation instances that you have installed. You can also define the job profiles that contain the environment variables that must be set for a job to run.

How the Event Server, Scheduler, and Agent Interact

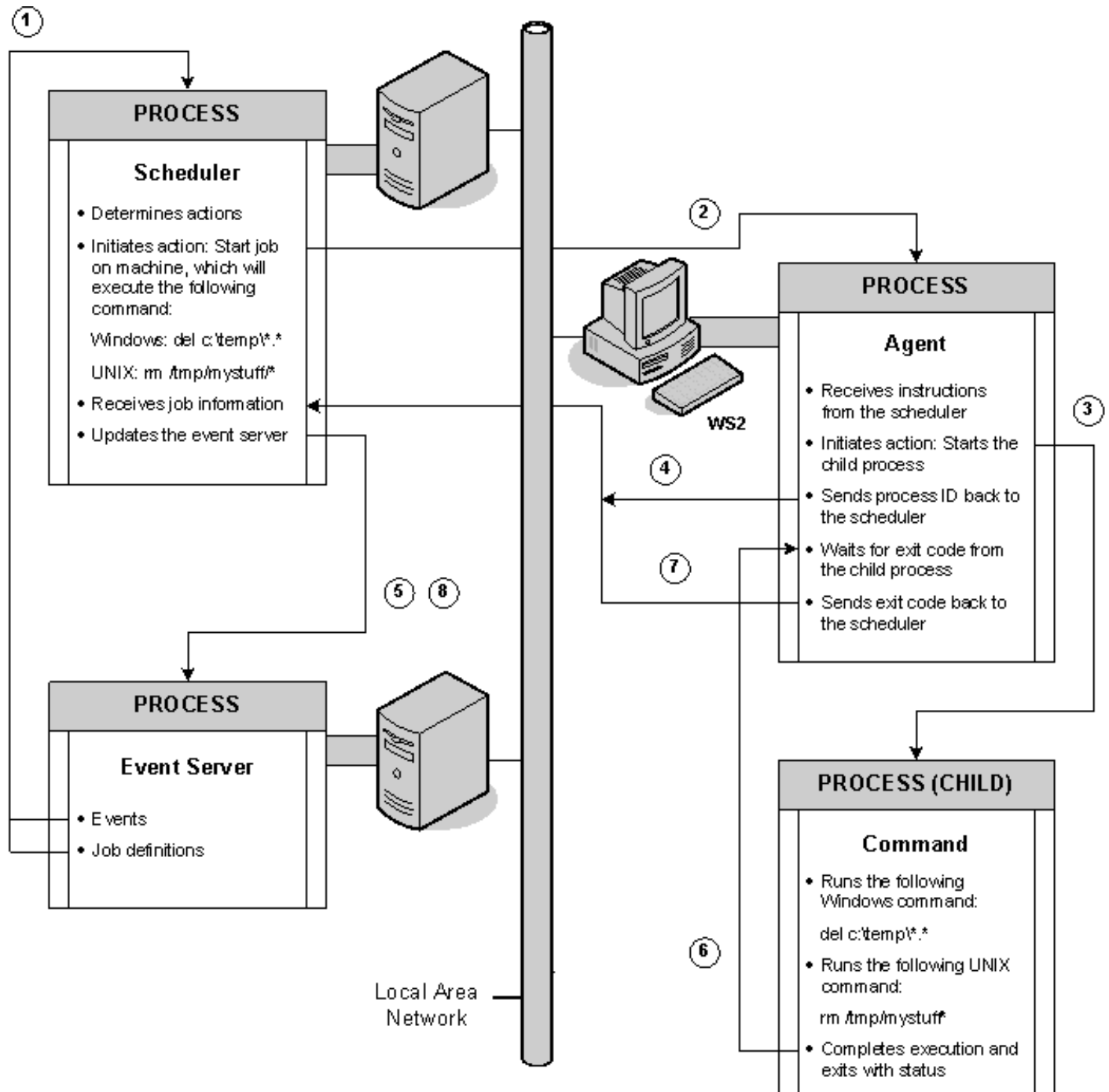
The following steps explain the interactions between the event server, scheduler, and agent with an example:

1. From the event server, the scheduler reads a new event, which is a STARTJOB event with a start time condition that has been met. Then, the scheduler reads the appropriate job definition from the database and, based on that definition, determines what action to take.
2. The scheduler communicates with the agent to run the job. The agent receives the instructions to run the job.
3. The agent performs resource checks and creates a process that actually runs the specified command.
4. The agent communicates the job execution information (such as the process ID, agent log file name, job output log file name) to the scheduler.
5. The scheduler converts the job execution information into a job event and updates the event server with the event information.
6. The command completes and exits, and the agent captures the exit code of the command.
7. The agent communicates the job completion information (such as exit code, status) to the scheduler.
8. The scheduler converts the job completion information into a job event and updates the event server with the event information.

The scheduler and the event server must be running to make AutoSys Workload Automation fully operational.

Example: Interaction Between the Event Server, Scheduler, and Agent

This example illustrates the event server, scheduler, and agent running on different computers. At a start date and time specified in the job definition, suppose that you run the command that is shown in the illustration on WorkStation_2 (WS2):

Figure 5: Interaction Between the Event Server, Scheduler, and Agent**NOTE**

- The application server communicates with the agent only when client utilities like chase and autoping are run or when jobs contain globs or blobs as input or output.
- Typically, the scheduler and the event server run on the same computer.

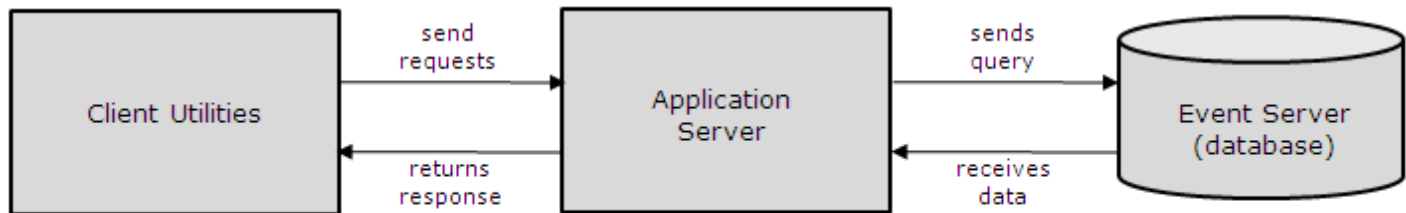
How the Event Server, Application Server, and Client Utilities Interact

The following steps explain the interactions between the event server, application server, and client utilities:

1. The client utilities send requests to the application server.
2. The application server executes the request by contacting the event server. The information is either being inserted, updated, retrieved, or removed from the event server. The responses are returned to the client as the operation executes or after the operation completes.

The following illustration shows how the event server, application server, and client utilities interact.

Figure 6: How the Event Server, Application Server, and Client Utilities Interact



NOTE

The application server communicates with the agent only when client utilities like chase and autoping are run or when jobs contain globs or blobs as input or output.

Example: Interaction Between the Event Server, Application Server, and Client Utilities

Suppose that you issue the autorep command at an UNIX operating system prompt or the Windows instance command prompt, the event server, application server, and the client utilities interact with each other as follows:

1. The autorep client sends a request to the application server.
2. The application server queries the database, receives the data from the event server, prepares one or more responses, and sends all the responses to the autorep client.
3. The autorep client receives all the responses and displays the report.

How the Event Server, Web Server, and Web Service Consumer Interact

The following steps explain the interactions between the event server, web server, and web service consumer:

1. The web service consumer sends requests to the web server.
2. The web server executes the request by contacting the event server. The information is being either inserted, updated, retrieved, or removed from the event server. The responses are returned to the web service consumer as the operation executes or after the operation completes.

The following illustration shows how the event server, web server, and web service consumer interact:

Figure 7: How the Event Server, Web Server, and Web Service Consumer Interact



Example: Interaction Between the Event Server, Web Server, and Web Service Consumer

Suppose that your application program invokes the web service to get information about the jobs defined, the event server, web server, and the web service consumer interact with each other as follows:

1. Your application program sends a request to the web server.
2. The web server queries the database, receives the data from the event server, prepares the response, and sends it back to your application program.
3. Your application program receives the response and processes it.

AutoSys Web UI

AutoSys Web UI is the GUI for AutoSys Workload Automation that lets you define, manage, schedule, monitor, or report on jobs. You can monitor and control workload on different instances of AutoSys Workload Automation, providing all the information you need about your jobs from one centralized location.

AutoSys Workload Automation lets you view and manage all relevant jobs and their interdependencies from a single point and presents the various aspects of workload automation through a sophisticated visualization user interface, Web UI.

Web UI lets you interactively set the attributes that describe when, where, and how a job should run. You can create and manage jobs, box jobs, and job flows using Web UI without needing extensive knowledge of JIL. The fields in the Web UI GUI correspond to the JIL subcommands and attributes. From within Web UI, you can also enter job-related and other common JIL commands so that, for example, you can quickly enter large quantities of similar jobs or run autorep without having to move back and forth between the GUI and the CLI.

In addition, from the Web UI GUI, you can define calendars, global variables, and monitor and manage jobs. Both AutoSys Workload Automation and Web UI set the same attributes and the job definition is always stored in the database. You can also update and delete existing jobs.

Web UI displays the Dashboard upon log in and provides status information at a summary and detail level through its View, Alerts, and Job Status Console, and Monitoring applications. From the summary information, you can access applications of Web UI such as Quick View and Quick Edit in context to get detailed information with which to analyze warnings and failure points and take appropriate corrective action. There are additional applications that enable administrators and other users to perform certain system configuration tasks, including running commands on back-end servers, and changing credentials.

Web UI Applications

Web UI applications are accessed using the following tabs in the user interface. When you move from one application to another, the displayed information is retained and redisplayed when you return.

- **Dashboard**—Contains widgets that are useful to schedulers, console operators, and executives in your enterprise.
- **Monitoring**—Lets you add, edit, delete, export, and import views, monitor jobs and box jobs within the views, and view any associated alerts. You can view the jobs and box jobs by server (workload automation scheduling manager instances) or by user-defined views.
- **Quick View**—Provides detailed AutoSys Workload Automation job information in a single page view. Quick View lets you view a job definition, access Quick Edit, view a job flow, send an event to one or more jobs, view a prior job run

report, and view logs. Quick View can be used as an alternative to search through a command line report for the job status information that you want.

- **Enterprise Command Line**—Lets you run a command line interface command such as `jil` or `autorep` on the selected AutoSys Workload Automation server. Enterprise Command Line also lets you save commands, modify and run saved commands, and import and export AutoSys Workload Automation jobs and calendars.
- **Reporting**—Lets you run predefined reports for multiple servers. These reports display server statistical information and job run information in a table and graphical view.
- **Forecast**—Lets you create, modify, and delete report definitions that specify the time period of the forecast window. Forecast report output includes jobs that are forecasted to start during the forecast window period. You can view forecast report output in table and Gantt chart formats.
- **Quick Edit**— Lets you manage all the properties of your AutoSys Workload Automation workload objects, such as jobs, cycles, calendars, global variables, and machines from a single page view.
- **Application Editor**—Provides a graphical representation of all the relationships between AutoSys Workload Automation jobs and their dependencies. It displays the conditions of the relationships that must be met before a job can run, and the successor jobs triggered when a job finishes running.
- **Resources**—Lets you add, edit, or delete virtual resources, monitor resources, and view resource dependencies for a job or collection of jobs in a graphical view. You can specify box job or job predecessors that must complete successfully before a box job or job can run. Similarly, you can specify virtual and real resource requirements that must be met before a box job or job can run.
- **Credentials**—Provides a summary of the servers to which the logged-in user has access. Credential also lets you add or delete credentials for a selected server, update the password for a selected user and server, and validate credentials for the active users against a back-end server.
- **Configuration**—Lets you perform all system setup for Web UI. For example, you can configure Quick Edit, Application Editor, and Monitoring run-time parameters, and connectivity to back end servers.
- **Agent Inventory**—Lets you see and export information about each separately installable component (system agent, plug-ins, and advanced integrations).

NOTE

Consider the following points when using Web UI:

- When two users edit or delete the same object, the last change that is entered prevails. This behavior is consistent with what occurs on the back-end workload automation server.
- The Web UI features are available based on the user roles and their associated access rights.

Web UI Architecture

The Web UI architecture is modular, providing maximum flexibility. When planning your workload automation environment, there are multiple factors to consider in determining the appropriate server architecture. The number of servers required depends on which features you want to install, the amount of data you foresee being stored in the applicable databases, and the number of security policies you plan to implement.

The Web UI architecture has the following three tiers:

Web Client Tier

From the web client tier, you use the Web UI user interface to access the primary Web UI functions.

Server Tier

Most of the work is performed in the server tier. The primary function of the server tier is to provide an interface to the Web UI application to let you configure and monitor jobs. This level includes the interfaces, servlets, servlet engines, and adapters. Each functional category has its own servlet deployed in a private servlet engine. Each servlet acts as a central point of communication and management between the web application and the back-end management server.

The ideal Web UI environment must be set up with the following dedicated servers:

- The Web UI server hosts Web UI, one or more Web UI Collectors, and the AutoSys Workload Automation SDK (including SSA). Additionally, one Web UI server in a High Availability group can host Apache HTTPD Server.
- The AutoSys Workload Automation server hosts AutoSys Workload Automation and the system agent.
- The CA EEM server hosts CA EEM, which provides security through role-based login access to Web UI applications and objects.

Back-end System Tier

The back-end management server or system tier includes AutoSys Workload Automation instances.

Common Components Overview

AutoSys Workload Automation and Web UI use CA Common Components. You can install the following common components using the AutoSys ISO.

- **CA Embedded Entitlements Manager (CA EEM)**
Enables you to secure objects such as jobs, calendars, cycles, global variables, machines, and resources, and delegate role-based access to the users in your enterprise.

NOTE

For more information about CA EEM, see [CA EEM documentation](#).

- **CA Secure Socket Adapter (SSA)**
Lets CA components use a single multiplexed communication port to ease firewall administration and minimize conflicts with other applications. SSA uses port 7163.
- **Management Command Center**
Integrates all enterprise and network monitoring functionality into a single console. Using the Management Command Center, you can do the following:
 - Create customized views of data
 - Apply filters to data
 - Create class-based queries
 - View alerts and event consoles
 - Create reports

NOTE

On Windows, you cannot install Management Command Center using the AutoSys ISO.

- **CA, Inc. Common Communications Interface (CAICCI)**
Enables communication with mainframe products.
- **CAPKI**
Implements Information Security services in products.

The Broadcom products that use the common components are AutoSys Workload Automation and Web UI. During installation, you select the components that you want to install depending on whether each is required or optional for the Broadcom product that you use.

NOTE

We recommend that you install the common components before you install the Broadcom product.

The following table lists the required and optional common components for AutoSys Workload Automation and Web UI:

Common Component	AutoSys Workload Automation	Web UI
CA EEM	Optional	Required
SSA	Required SSA is automatically installed when you install AutoSys Workload Automation.	Required SSA is automatically installed when you install Web UI.

CAICCI	Required if cross-platform scheduling is enabled or AutoSys Workload Automation communicates with any of the following products or agent: CA Workload Automation SE, CA Jobtrac JM, or CA Scheduler JM on the mainframe, Notification Services component of CA NSM, or CA UJMA agent; optional otherwise	Optional
Management Command Center	Optional	Optional

NOTE

SSA installs a default certificate that can be used to install and run SSL without any additional configuration. This certificate is the same on all installations and does not provide any level of authentication. To authenticate this certificate, you must replace it with a valid certificate that is issued and signed by your own Certificate Authority (CA). For information about configuring SSA to use valid certificates, see [Configure SSA to Use a Valid Certificate](#).

Compliance Application

The compliance application is a console application that audits system utilization. This application installs automatically with AutoSys Workload Automation.

The compliance application runs queries to collect the following statistics:

- The number of defined jobs, based on the job type
- The number of jobs run in a particular time window
- The number of defined machines
- The number of machines used in a particular time window

The compliance application collects statistics on jobs that physically run on the agent. The application does not collect statistics on the following jobs:

- Box jobs -- A box job acts as a container for other jobs but performs no action itself, so the box job does not physically run on the agent.
- Bypassed jobs -- The scheduler starts bypassed jobs but does not invoke the agent to execute job commands. These jobs evaluate as successfully completed but do not physically run on the agent. To bypass a job, place the job in ON_NOEXEC status using the sendevent command.

The compliance application accesses the AutoSys Workload Automation database using Java Database Connectivity (JDBC). The application software includes the required drivers for Oracle and SQL server databases. To connect to a Sybase database, you must install Sybase jConnect 7 from the Sybase installation media or download jConnect 7 from Sybase. AutoSys Workload Automation is configured to use the following default location:

- (UNIX) \$SYBASE/jConnect-7_0/classes/jconn4.jar
- (Windows) %SYBASE%\jConnect-7_0\classes\jconn4.jar

To aggregate statistics on all job runs, including box jobs and bypassed jobs, enable the Aggregate Statistics parameter in the UNIX configuration file or select the Aggregate Statistics checkbox in the Scheduler--AutoSys Workload Automation Administrator window of the Windows AutoSys Workload Automation Administrator.

The compliance application runs as part of the DBMaint command. The application stores each query result in a separate comma-separated (.csv) file in the \$AUTOUSER/out (UNIX) or %AUTOUSER%\out (Windows) directory.

The compliance application appends the output files with new data each time it runs. AutoSys Workload Automation does not manage the size of the output file. To prevent the log file from becoming too large, back-up older files in another location and manually delete those files.

You can load the query results in Microsoft Office Excel or any other compatible programs that support .csv files.

WARNING

The compliance application shares a connection properties file with the RESTful web services. Ensure that you take appropriate precautions when updating URLs in this file.

NOTE

For information about installing the JDBC driver for Sybase databases, see the Sybase documentation.

CA EEM Reporting Utility

The CA EEM Reporting Utility is composed of two programs:

- **Service**—A process that resides and runs in the background on a CA EEM system. It reads CA EEM audit log files and imports event data to a database that you provide. Optionally, the Service also deletes audit data from the database after a specified period and archives the data to an auxiliary data set. It supports all CA EEM supported operating systems.
- **Client**—The reporting program that lets local or remote users query the database to produce customized PDF reports and/or CSV files for detailed reporting or analysis. The Client can generate queries on any field that is provided in the CA EEM audit log files. The client runs on any operating system with a Java SE Runtime Environment 6 or higher, provided that the system has connectivity to the database.

Setup scripts are provided to install required components and manage configuration files for both the Service and Client programs.

Install the CA EEM Reporting Utility

You can install the CA EEM Reporting Utility using the CA EEM Reporting Utility software that is distributed in two files:

- **eem_reporting_service**—Contains the Service that loads audit data and manages the database. The Service must be installed on any system that runs CA EEM where audit data should be included in reports.
- **eem_reporting_client**—Contains the Client to generate reports. The Client can be installed on any system from which audit reports are produced.

NOTE

Before installing the CA EEM Reporting Utility, verify that the CA EEM components are installed and operational. For more information about installing CA EEM components, see the [CA EEM documentation](#).

To install the CA EEM Reporting Utility, follow these steps:

1. [Configure the Database for the Service](#)
2. [Install the Service](#)
3. [Configure the Database for Clients](#)
4. [Install the Client](#)

Configure the Database for the Service

A database administrator must set up a database to store the audit data that is extracted from the CA EEM audit log files. The database can be installed on any system that has network connectivity to the CA EEM server and to any Client systems requesting reports.

NOTE

- Optionally, an internal database (H2) can be used to avoid additional database resources. For more information about internal database, see <http://www.h2database.com/html/quickstart.html>.
- If you use H2 as an internal database, you can skip this procedure and start installing the Service.

The available size in the database depends on the number of events generated and the time duration the event data is stored in the database. Normally, a database of 1 GB size can support approximately 150,000 audit records.

CA EEM Reporting Utility supports the following databases:

- **Oracle:**
 - Oracle 12c R1 (12.1)
 - Oracle 11g R2, R1 (11.2, 11.1)
- **Microsoft SQL Server:**
 - Microsoft SQL Server 2014
 - Microsoft SQL Server 2012
 - Microsoft SQL Server 2008 R1, R2
 - Microsoft SQL Server 2005
- **Sybase:**
 - Sybase Adaptive Server Enterprise 15.0, 15.5, and 15.7

To configure the Oracle database for the Service, follow these steps:

1. Create a user named *eemserveruser* with CONNECT and RESOURCE privileges.
2. Verify that the permissions are defined correctly by running the following commands:


```
connect eemserveruser/password@schema;
select * from USER_ROLE_PRIVS;
```
3. Use the *eemserveruser* user for setup with the service.

To configure the Microsoft SQL Server database for the Service, follow these steps:

1. Create a user named *eemserveruser* with the db_owner role in the database.
The access privileges of db_owner are limited to connect, execute, create table, insert, update, and delete.
2. Use the *eemserveruser* user for setup with the service.

To configure the Sybase database for the Service, follow these steps:

1. Create a user named *eemserveruser* with the sa_role permission:


```
sp_role "grant", sa_role, eemserveruser
```
2. Use the *eemserveruser* user for setup with the service.

Install the Service

Install the Service to read and store the CA EEM audit log data and manage the database. Install the Service on the same system that runs CA EEM.

NOTE

- The installation prerequisites and system requirements of CA EEM apply to the Service.
- On Sybase, if you are using Kerberos authentication, the [AutoSys Workload Automation System Requirements](#) of AutoSys Workload Automation are applicable to the Service. If you are using H2 database, the supported JRE version is 1.6 or 1.7.
- The *.conf/JAAS.conf* file contains the default login module settings. Before you run the CA EEM Reporting utility, you can configure the login module settings as appropriate to ensure that it connects to the Sybase database successfully.

Follow these steps:

1. On the CA EEM system, unpack the *eem_reporting_service* file.

NOTE

On UNIX, you must have proper read, write, and execute permissions:

```
chmod 755 eem_reporting_service
```

2. Change to the installation directory at the command prompt.
3. Run one of the following setup scripts:
 - Windows:**
setup.bat
 - UNIX or Linux:**
setup.sh
4. Specify the values for the following options when prompted. Some options are preset to default values that appear enclosed in brackets. If the script is rerun, the value that you entered previously is enclosed in brackets. To use the default or previous value, press Enter.
 - a. Database engine (h2/oracle/sqlserver/sybase/sybasekerb)
 - b. Jconn path

NOTE

The installer prompts you for the jconn4.jar file path only if you entered sybasekerb as the Database engine. Specify the full path. The supported jconn4.jar version is 7.0.7. Do not use the 7.0 or a lower version as it includes a known issue with Sybase.

- c. Java path

NOTE

The installer prompts you for the Java path only if you entered sybasekerb as the Database engine. Specify the full Java path. For example, /java6/bin/java.

- d. Database server hostname
- e. Database port
- f. Database name/SID
- g. Service Principal Name

NOTE

The installer prompts for the service principal name only if you entered sybasekerb as the Database engine.

- h. Database username
- i. Database password
- j. Verify password

NOTE

- If you entered sybasekerb as the Database engine, the installer does not prompt you for the Database username, Database password, and Verify password.
- If the setup is unable to connect to the database, repeat Steps a to i.

- k. Delete processed logs (true/false)
- l. Delete expired data during cleanup (true/false)
- m. Archive expired data during cleanup (true/false)

NOTE

If you set this value to true, an archive directory is required.

- n. Archive directory

NOTE

If you set the *Delete expired data during cleanup* and/or *Archive expired data during cleanup* options to true, you must specify the *Cleanup interval* and *Retention period for expired data* values.

- o. Cleanup interval (Hours)
- p. Retention period for expired data (Days)

The setup creates the `./conf/service.xml` configuration file, updates the `./conf/wrapper.conf` file with the CA EEM JRE path, configures tables within the specified database, and launches the CA EEM Reproting Utility Service. After the Service is installed, the process automatically starts upon system startup.

NOTE

- To review or change the configuration settings, run `setup.bat` or `setup.sh` again. Any changes made to an existing configuration forces a running service to stop.
- Actions performed by the Service are logged in the logs directory. These log files can be used for troubleshooting problems with the Service.
- The Service can be automatically started after the setup completes. However, if you want to stop or restart the Service at any time, or if the Service fails to start after a system reboot, perform the following actions:
 - To install and start the Service, execute `install.bat` (Windows) or `install.sh` (UNIX or Linux).
 - To stop and remove the Service, execute `uninstall.bat` (Windows) or `uninstall.sh` (UNIX or Linux).

Configure the Database for Clients

You can configure the database to provide report requests.

NOTE

If you use H2 as an internal database, you can skip this procedure and start installing the Client.

To configure the Oracle database for the Client, follow these steps:

1. Create a user named `eemreportuser` with CONNECT privileges.
2. Grant SELECT ON permissions for the following tables to `eemreportuser`:

```
eemserveruser.safe_admin_event
eemserveruser.safe_authentication_event
eemserveruser.safe_authorization_event
```

3. Create public synonyms for all three tables by issuing the following commands:

```
Create PUBLIC SYNONYM safe_admin_event for eemserveruser.safe_admin_event;
Create PUBLIC SYNONYM safe_authentication_event
for eemserveruser.safe_authentication_event;
Create PUBLIC SYNONYM safe_authorization_event
for eemserveruser.safe_authorization_event;
```

4. Grant privileges to the following tablespaces:

```
ALTER USER eemserveruser quota unlimited on safe_admin_event;

ALTER USER eemserveruser quota unlimited on safe_authentication_event;

ALTER USER eemserveruser quota unlimited on safe_authorization_event;

ALTER USER eemserveruser QUOTA 100M ON USERS;
```

5. Use the `eemreportuser` user for generating reports with a Client.

To configure the Microsoft SQL Server database for the Client, follow these steps:

1. Create a user named *eemreportuser* with the db_datareader role.
The access privileges to the db_datareader role are limited to connect and select.
2. Use the *eemreportuser* user for generating reports with a Client.

To configure the Sybase database for the Client, follow these steps:

1. Create a user named *eemreportuser* with CONNECT privileges.
2. Grant SELECT ON permissions for the following tables to *eemreportuser*:

```
SAFE_ADMIN_EVENT
SAFE_AUTHENTICATION_EVENT
SAFE_AUTHORIZATION_EVENT
```
3. Test the permissions by connecting to the database with *eemreportuser* and running the following SELECT queries on the event tables:

```
SELECT * FROM SAFE_ADMIN_EVENT
SELECT * FROM SAFE_AUTHENTICATION_EVENT
SELECT * FROM SAFE_AUTHORIZATION_EVENT
```
4. Use the *eemreportuser* user for generating reports with a Client.

Install the Client

You can install the Client on any system where the reports are generated. The system must have network connectivity to the database and requires Java SE Runtime Environment 6 or higher.

NOTE

- On Sybase, if you are using Kerberos authentication, the [AutoSys Workload Automation System Requirements](#) of AutoSys Workload Automation are applicable to the Client. If you are using H2 database, the supported JRE version is 1.6 or 1.7.
- The `./conf/JAAS.conf` file contains the default login module settings. Before you run the CA EEM Reporting utility, you can configure the login module settings as appropriate to ensure that it connects to the Sybase database successfully.

Follow these steps:

1. Unpack the `eem_reporting_client` file.

NOTE

On UNIX, you must have proper read, write, and execute permissions:

```
chmod 755 eem_reporting_client -R
```

2. Change to the installation directory at the command prompt.
3. Run one of the following setup scripts:
Windows:
`setup.bat`
UNIX or Linux:
`setup.sh`
4. Enter the values for the following options when prompted. If the script is rerun, the value that you entered previously appear enclosed in brackets. To use the previously entered value, press Enter.
 - a. Database engine (h2/oracle/sqlserver/sybase/sybasekerb)
 - b. Jconn path

NOTE

The installer prompts you for the jconn4.jar file path only if you entered sybasekerb as the Database engine. Specify the full path. The supported jconn4.jar version is 7.0.7. Do not use the 7.0 or a lower version as it includes a known issue with Sybase.

c. Java path

NOTE

The installer prompts you for the Java path only if you entered sybasekerb as the Database engine. Specify the full Java path. For example, /java6/bin/java.

d. Database server hostname

e. Database port

f. Database name/SID

g. Service Principal Name

NOTE

The installer prompts for the service principal name only if you entered sybasekerb as the Database engine.

h. Database username

i. Database password

j. Verify password

NOTE

- If you entered sybasekerb as the Database engine, the installer does not prompt you for the Database username, Database password, and Verify password.
- If the setup is unable to connect to the database, repeat Steps a to h.

The setup script creates the ./conf/reporting.xml configuration file. To review or change the configuration settings, run the setup.bat or setup.sh script again.

Generate Reports

You can create reports in different ways. To generate a report, run the EEMReporting.jar file followed by a list of report parameters. This procedure describes how to create a batch file on Windows that executes the jar file and generates a report.

Follow these steps:

1. Create a new batch file.
2. Open the batch file and add the following:

```
java -jar EEMReporting.jar parms...
```

EEMReporting.jar

Specifies the path to the EEMReporting.jar file. **parms**

Specifies a list of options/parameters that are referenced in the following table.

NOTE

- The parameters that are indicated with asterisk (*) must be enclosed in quotes if spaces are present within the parameter.
- To create a report, list the required option with each option followed by one or more parameters associated with that option. The options can be specified in any order.

3. Save the batch file and exit.
4. Run the batch file.

The report is generated.

Reports are requested with a set of parameters followed by specific options that are listed in the table:

Option	Description	Required
-n	Name of the report*	Yes
-d	Destination (file path) for report*	Yes
-e	Type of report ("pdf" and/or "csv")	Yes
-t	Database table to include in the report (see Audit Events)	Yes
-c	Database column to include in the report (see Audit Events)	Yes
-f	Filters to apply to report* (see Audit Events)	No
-o	Database columns used to sort the report (see Audit Events)	No
-p	Specifies a file that contains a predefined set of parameters*	No

Example 1

```
-n "New Report" -d "C:\Program Files\reports" -e pdf -t admin -c method datetime status
source resource identity -f status=success -o datetime ASC
```

In this example, the name of the report is New Report and the report file is written to C:\Program Files\reports. A PDF report is created. It contains information from the admin table in the database and includes the method, datetime, status, source, resource, and identity columns. The report only includes entries whose status value is success. Results for this report are listed by the datetime column in ascending order.

Example 2

```
-n "New Report" -p "C:\Authentication" -d "C:\Program Files\reports" -e pdf csv
Contents of "C:\Authentication":
-t authentication -c method datetime status source resource identity
```

In this example, a predefined report is created. The name of the report is New Report and the report files are written to C:\Program Files\reports. Two types of reports are created: PDF and CSV. The -p file is read to obtain additional report parameters. The Authentication file contains parameters that include the method, datetime, status, source, resource, and identity columns from the authentication table in the report.

Example 3

```
-n "New Report" -d "C:\Program Files\reports" -e pdf -c status source identity -t admin
-o datetime ASC -f datetime=2014-06-05
```

In this example, a PDF titled New Report is written to C:\Program Files\reports. The report contains the status, source, and identity columns for all events that are generated on June 5, 2014 from the admin table. Results are listed in sequential order.

Audit Events

Events from CA EEM audit logs are grouped into one of the following types:

- Admin

- Authentication
- Authorization

The CA EEM Reporting Utility stores data in the database tables that correspond to each event type. The columns of data available for reports are as follows:

Admin

Column Name	Description
guid	Unique key
identity	The identity for which the authorization check is performed
resourceclass	Resource class the operation is performed on
resource	Resource for which the authorization check is performed
method	Authentication method
severity	Event severity
status	Status of the operation
source	Source application instance
error	Result of the operation
errorcode	Result of the operation as a string. Example: EE_SUCCESS, EE_NOTALLOWED
serverhostname	CA EEM server host name
datetime	Date and time at which the event is generated (see Specifying the Date and Time)
serveros	Operating system the CA EEM server machine is running on
clienthostname	Host name of the CA EEM client reported this event

Authentication

Column Name	Description
guid	Unique key
identity	The identity for which the authorization check is performed
resourceclass	Resource class the operation is performed on
resource	Resource for which the authorization check is performed
method	Authentication method
severity	Event severity
status	Status of the operation
error	Result of the operation
errorcode	Result of the operation as a string. Example: EE_SUCCESS, EE_NOTALLOWED
serverhostname	CA EEM server host name
datetime	Date and time at which the event is generated (see Specifying the Date and Time)
serveros	Operating system the CA EEM server machine is running on
clienthostname	Host name of the CA EEM client reported this event

Authorization

Column Name	Description
guid	Unique key
identity	The identity for which the authorization check is performed
resource	Resource for which the authorization check is performed
method	Authentication method
severity	Event severity
status	Status of the operation
source	Source application instance
error	Result of the operation
errorcode	Result of the operation as a string. Example: EE_SUCCESS, EE_NOTALLOWED
serverhostname	CA EEM server host name
datetime	Date and time at which the event is generated (see Specifying the Date and Time)
serveros	Operating system the CA EEM server machine is running on
clienthostname	Host name of the CA EEM client reported this event
delegator	Delegator
Policy	Policy name

Specifying the Date and Time

Relative

A relative filter specifies an amount of time before the current time. It can be specified as follows:

```
nnn[Y] [M] [D] [h] [m] [s]
```

nnn Specifies the time value.

Y

Specifies the time value in years.

M

Specifies the time value in months.

D

Specifies the time value in days.

h

Specifies the time value in hours.

m

Specifies the time value in minutes.

s

Specifies the time value in seconds.

NOTE

These parameters are case-sensitive.

Examples:

Example	Description
<code>datetime>2Y</code>	includes events that occurred within the past two years
<code>datetime=5M</code>	includes events that occurred 5 months ago (for example, if the current month is February, the report includes events from September)
<code>datetime>1h</code>	includes events that occurred within the last hour
<code>datetime<7D</code>	includes events that occurred more than one week ago
<code>datetime<30m datetime>60m</code>	includes events that occurred between 30 and 60 minutes ago
<code>datetime>30m datetime<60m</code>	error – logically impossible

Absolute

An absolute filter specifies a specific date and time. It can be specified as follows:

yyyy-mm-dd hh:mm:ss

yyyy

Specifies the year.

mm

Specifies the month.

dd

Specifies the day.

hh

Specifies the hour.

mm

Specifies the minute.

ss

Specifies the second.

Partial absolute datetime filters can be specified. When a partial filter is specified, only the part of the date and time specified are matched.

Examples:

Example	Description
<code>datetime=2013</code>	includes events that occurred during the year 2013
<code>datetime=2014-05</code>	includes events that occurred in May 2014
<code>datetime=2014-06-01 13</code>	includes events that occurred on June 1 2014 from 13:00:00:000-13:59:59:999
<code>datetime>2014-04-19 datetime<2014-04-21</code>	includes events that occurred from April 19-20
<code>datetime=2013-05-14 13:25:10</code>	includes events that occurred on May 14, 2013 between 13:25:10:000 and 13:25:10:999
<code>datetime<2013-12-05 02:30</code>	includes events that occurred before December 5, 2013 at 2:30

Configuration Files

The parameters in the Service and Client configuration files are listed in the following table. Most of these parameters are initially set and can be modified using the setup scripts.

Service

Parameter	Description	Default Value
Size	In memory catch size	1000
Eiamhome	CA EEM home directory	\$EIAM_HOME
Eventdir	Path to the CA EEM log folder	\$EIAM_HOME/logs/
Deleteprocessed	Delete the log files after processing the data (true/false)	True
Auditlogfilename	Name of the log file prefix	audit.log
Driver	Database engine (oracle/sqlserver)	
url	Database URL that contains the host, port, and database name	
User	Database username (with connect, create table, insert, and delete permissions)	
Password	Database password—Can be stored either in plain text or MUNGE2 format	
Batchsize	Audit events are inserted into the database batch by batch based on this size	4000
Datadirectext	Datadirect extension class	DO NOT CHANGE
Datadirectooempassword	Datadirect unlock password	DO NOT CHANGE
Classpath	The path to a preexisting jdbc driver based on the chosen database driver	
Import	Schedule import (true/false)	True
Importinterval	The time interval (minutes) to import data	1
Archive	Archive expired data (true/false)	True
Archivedir	Path to the archive directory	./archive/
Cleanup	Delete expired data (true/false)	True
Interval	The time interval (hours) to archive and/or cleanup expired data in the database	24
Retention	Data retention (days)	60

Client

Parameter	Description	Default Value
Driver	Database engine (oracle/sqlserver)	
url	Database URL that contains the host, port, and database name	
User	Database username (with connect, create table, insert, and delete permissions)	
Password	Database password – Encrypted in AES 128-bit	

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Installing

This section contains information about how to install and upgrade the AutoSys Workload Automation components, Web UI, and CA EEM.

This content is meant for system administrators who are responsible for installing, configuring, and upgrading these products. Procedures and best practices assume familiarity with the operating system and with the database server you use.

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This section includes the following topics:

- **Installation**
Install the server, agent and client. This section also includes configuration and migration procedures, with procedures specific to UNIX and Windows platforms.
- **Common Components Installation**
Install common components on UNIX and Windows. This section also includes upgrade, removal, and troubleshooting procedures.
- **Web UI Implementation**
Install the Web UI on UNIX and Windows. This section also includes migration and uninstallation procedures

After you log in, use the navigation tree on the left or search for Installing content:

Installing

Install AutoSys Workload Automation

The following topics describe the different ways that you can install AutoSys.

AutoSys Installation Options

You can install AutoSys using:

- [The installation wizard](#) (GUI).
- [Console](#)
- [Silent install](#)

AutoSys Installation Modes

When you install AutoSys, select one of the following installation modes:

NOTE

The instructions in this section follow the installation wizard steps that appear in the UI.

Custom Mode

Custom mode lets you install AutoSys in a customized way where you can specify your requirements and values. It also offers you the following features.

- Dual event server
- High availability
- External authentication using PKI (Oracle) and Kerberos (Sybase)
- Encryption: Instance-level data encryption, Agent-level data encryption, and FIPS compliance
- Ability to configure SOAP and ports
- Ability to choose a custom installation path

NOTE

Click **Next** as you proceed with the installation.

Follow these steps:

1. Double-click on the AutoSys installer. The installation wizard appears.
2. Select **I accept the terms of the License Agreement**.
3. In the Installation Options window, select **Install the product**.
4. Select **Custom**. Follow the instructions available on the panels to proceed. You will find detailed lists of panels and their fields in later sections of this guide.

Express Mode

Express mode lets you install AutoSys with minimal inputs. In Express mode, default values are automatically entered for the components that you install in order to expedite your AutoSys installation process. If you want to customize values, select the Custom mode instead.

NOTE

Click **Next** as you proceed with the installation.

Follow these steps:

1. Double-click on the AutoSys installer. The installation wizard appears.
2. Select **I accept the terms of the License Agreement**.
3. In the Installation Options window, select **Install the product**.
4. Select **Express**. Follow the instructions available on the panels to proceed. You will find detailed lists of panels and their fields in later sections of this guide.

Fresh Install AutoSys Using GUI

This section describes how to install AutoSys using the GUI. Select your operating system:

Windows

These instructions describe how to perform a fresh install of AutoSys on Windows, using the GUI.

NOTE

Click **Next** to proceed with the installation.

Follow these steps:

1. Double-click the install.exe file to open the AutoSys Workload Automation installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In Installation Options, select **Install the product** or **Create or refresh the database schema**.
4. Click **Custom** or **Express** based on your requirements. For more information, see [AutoSys Installation Modes](#).
5. Proceed with the installation panels, following the instructions in the installation wizard. Each panel and its fields are covered in detail in later sections of this guide.

NOTE

Refer to the following sections for details on installing the components:

- [AutoSys Server](#)
- [Agent](#)
- [Client](#)

Unix

These instructions describe how to perform a fresh install of AutoSys on UNIX.

Execute the following command:

```
./install.bin -i gui
```

The installation wizard will appear, and the procedure will be similar to the process described for [Windows](#).

Fresh Install AutoSys Using Console

This section describes how to install AutoSys using the console. Select your operating system:

Windows

These instructions describe how to perform a fresh install of AutoSys on Windows, using the console.

Follow these steps.

1. In the command prompt, go to the directory where you have stored the AutoSys installer.
2. Run the following command.

```
install.exe -i console
```

The installer opens in console mode. Follow the instructions in the console, and refer to the descriptions provided there.

Unix

Run the following command to perform a fresh install of AutoSys on UNIX using the console.

```
install.bin
```

The installer opens in console mode. Follow the instructions in the console, and refer to the descriptions provided there.

Fresh Install AutoSys Using Silent Install

This section describes how to perform a silent install/upgrade of Autosys, without user interaction.

NOTE

- The steps mentioned here are valid for both fresh install/upgrade of AutoSys.
- The steps to record response files are the same for Windows and Linux.
- The command line options and arguments are the same for Windows and Linux.
- You can record a response file and use it to execute an unattended AutoSys silent install.
- On Windows, login with Windows Administrators group privileges.
- On Unix, login as the root user.
- Once the silent installation is successfully completed, the exit code will be zero. Otherwise, refer to the log files for more information.

Pre-Silent-Install Validation (Optional)

Before proceeding with the silent install, you can perform a pre-silent-install validation for AutoSys by setting the `IA_AE_SKIP_INSTALL` variable in the silent response file to 1.

The actual AutoSys installation is skipped, but all the required validations as part of the pre-installation are performed until an error occurs. If an error occurs, the details are recorded in the log file to help you identify any shortcomings before performing the actual install.

The "Skipping installation as per the user choice" message appears in the log when all the pre-install validations are successful. The exit code on Windows is "19999", and on Unix it is "31".

In addition, if you are using the GUI or Console option for AutoSys installation, you can set the `IA_AE_SKIP_INSTALL=1` variable as an environment variable. This will help you generate the response file without installing AutoSys.

On Unix, you can use the `export IA_AE_SKIP_INSTALL=1` command to skip the actual installation.

NOTE

- On Unix, the "31" code is not specific to pre-silent-install validation. Therefore, please refer to the logs.
- On Windows, the logs can be found at, for example, `<install_dir>:Program Files\CA\WorkloadAutomationAE\Logs`.

Silent Install AutoSys

Follow these steps:

1. Record the response file in a sandbox machine.
2. Execute silent install with the recorded response file in your target machine.

Record the Response File

1. In your sandbox machine, mount the AutoSys Workload Automation media.
2. In your target machine, go to the directory where you stored the mounted file system, and run the following command:
`cd <dir name>`
3. In a sandbox machine, use the following command to record the response file.

Note: If you do not want to install AutoSys and just test the installation process, in the response file set `IA_AE_SKIP_INSTALL=1`. Please refer to the Pre-Silent-Install Validation section above.

Syntax:

Windows: `<Install dir>:install.exe -r <response_file_name.txt>`

Unix: `<Install dir>/install.bin -r <response_file_name.txt>`

Example:

Windows: `D:\install.exe -r c:\response.txt`

Unix: `install.bin -r /root/r12_install.properties`

Where (for Windows):

- `D:` is the source directory where the AutoSys installer is present.
- `install.exe` is the command to run the AutoSys setup.
- `-r` records the user inputs that you enter in the installation wizard.
- `C:\response.txt` is the filename and location where the recorded file containing key-value pairs is saved. This file is also called the response file.

In the command prompt, when you execute the command above, the AutoSys installation wizard appears. Run through the desired installation mode using the UI. At each step, the variable that you enter and the modes that you select are recorded in the `response.txt` file.

NOTE

If you want to use different modes of installation to perform a silent install, you will have to use different sandbox configurations. For example, to perform a silent install to upgrade AutoSys, you will need a sandbox machine that already has an older version of AutoSys installed.

Execute Silent Install with a Response File

You can modify the variables present in the generated response file in your sandbox machine to represent your target machine. Copy the response file to your target machine at a specific location, modify the key-value pairs, and execute the following command.

Note: To install AutoSys, ensure that in the response file `IA_AE_SKIP_INSTALL=0`.

Syntax:

```
Windows:<install-dir>: install.exe -i silent -f <absolute path of the response-file>
Unix:<install-dir>
\install.bin -i silent -f <absolute path of the response-file>
```

Example:

```
Windows:D:\install.exe -i silent -f C:\installer\response.txt
Unix:install.bin -i silent -f /root/r12_install.properties
```

Where (for Windows):

- `D:` is the source directory where you have the AutoSys installer present.
- `install.exe` is the command to run the AutoSys setup.
- `-i` silently performs the AutoSys installation on your target machine in the backend.
- `-f` is the absolute path of the response file.
- `C:\installer\response.txt` is the response file name and location on your target machine. This file is the modified response file that you generated in your sandbox.

NOTE

You can deploy AutoSys to machines in your enterprise through the CA SDM. For more information about setting up AutoSys to be delivered through CA SDM, see the Software Distribution Manager documentation.

AutoSys: (UNIX) Apply a Patch

This article provides information about how to apply a patch on UNIX, for example how to apply a cumulative on UNIX.

NOTE

- Database updates must be run only on server-based machines.
- If multiple instances are installed on a machine, the patch upgrades all the instances to the patch level, except for the database changes. It upgrades only one database server that is supplied in the command line.

For upgrading the database of the rest of the instances, use the `-db` (applies only the database portion of the patch) option. Do not use the `-r` option as it starts all the services and might cause database conflicts for other instances.

- If Web UI is installed on the same machine where the AutoSys Workload Automation server or client is installed, execute the patch installer only once. Common files that are shared between the two installations are updated as part of patching the AutoSys Workload Automation server or client installation.
- Before applying the patch, ensure that you:
 - Close all the AutoSys Workload Automation client commands or user interfaces that are currently opened.
 - Stop all the AutoSys Workload Automation services. If the patch installer finds any active services, it shuts them down.
 - Back up the AEDB databases on the machines where the patch is being installed.
 - Shut down all the Web UI services if Web UI is installed on the same machine where the AutoSys Workload Automation server is installed as the common files that are shared between AutoSys Workload Automation and Web UI might require updating. The patch installer does not shut down the Web UI services. So, you must shut down all the active Web UI services manually to avoid file copy failures due to active processes.
- Before applying a patch to an SDK-only environment, set the `$AUTOSYS` and library path environment variables prior to launching the patch installer. For example, if AutoSys Workload Automation is installed in the default location, set the `$AUTOSYS` environment variables as follows:.

```
$ AUTOSYS=/opt/CA/WorkloadAutomationAE/autosys
$ export AUTOSYS
```

Set the library path based on your operating system as follows:

For AIX:

```
$ LIBPATH=$AUTOSYS/lib:$LIBPATH
$ export LIBPATH
```

For HP-UX:

```
$ SHLIB_PATH=$AUTOSYS/lib:$SHLIB_PATH
$ export SHLIB_PATH
```

For Linux and Solaris:

```
$ LD_LIBRARY_PATH=$AUTOSYS/lib:$LD_LIBRARY_PATH
$ export LD_LIBRARY_PATH
```

- Run the following command to verify that the installed AutoSys Workload Automation SDK version is at the pre-patch level:


```
> autoflags -x
```

Follow these steps:

1. Retrieve the patch tar file (*.tar.Z) from the CA Technologies FTP site.
2. Move the tar file to a temporary directory on the machine where AutoSys Workload Automation is installed..
3. Log in as the owner of the AutoSys Workload Automation executables and run the following commands from the operating system prompt to stop the AutoSys Workload Automation services:

```
$ $CASHCOMP/bin/unisrvcntr stop waae_agent-$WA_AGENT
$ $CASHCOMP/bin/unisrvcntr stop waae_sched.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr stop waae_server.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr stop waae_webserver.$AUTOSERV
```

NOTE

- Run these commands for all instances.
- `WA_AGENT` specifies the name of the agent.

4. Back up the WorkloadAutomationAE directory by running the following commands:

```
$ cd /opt/CA
$ cp -R WorkloadAutomationAE WorkloadAutomationAE.prepatch
```

NOTE

- Run these commands for all instances.
- WorkloadAutomationAE.prepatch is the backup of the WorkloadAutomationAE directory.

5. Assuming the tar file was downloaded to the /tmp/patch directory, uncompress and extract the file contents by running the following commands:

```
$ cd /tmp/patch
$ gunzip patch_file_name.tar.Z;tar -xvf patch_file_name.tar
```

NOTE

- You can change the file permissions as required.
- *patch_file_name* specifies the name of the patch file.

6. Change the file permission of the Install_INCR.sh script by running the following command:

```
$ chmod +x Install_INCR.sh
```

7. Apply the patch by running the Install_INCR.sh script from the extracted directory.

8. Verify that the patch is applied correctly by:

- Running the following command:

```
#autoflags -a
```

The command returns the following output: *build_number* OS_TYPE DB_TYPE *patch_version* hostid MACH_NAME FQDN

- ***build_number*** Specifies the build number of the patch.
- ***patch_version*** Specifies the version of the patch.

- Executing the following SQL query:

```
select str_val from ujo_alamode where type='AEDB_VERSION'
```

The query must return the patch version. In dual event server mode, run the query on both the databases.

9. Start the AutoSys Workload Automation services by running the following commands:

```
$ $CASHCOMP/bin/unisrvcntr start waae_agent-WA_AGENT
$ $CASHCOMP/bin/unisrvcntr start waae_sched.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr start waae_server.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr start waae_webserver.$AUTOSERV
```

NOTE

Run these commands for all instances.

Apply the SDK Patch on the Web UI Server

Follow these steps:

1. Log in as the root user and stop all the Web UI services manually.
2. Set the \$AUTOSYS and library path environment variables prior to launching the patch installer. For example, if AutoSys Workload Automation SDK is installed in the default location, set the \$AUTOSYS environment variables as follows:.

```
$ AUTOSYS=/opt/CA/WorkloadAutomationAE/autosys
$ export AUTOSYS
```

Set the library path based on your operating system as follows:

For AIX:

```
$ LIBPATH=$AUTOSYS/lib:$LIBPATH
$ export LIBPATH
```

For HP-UX:

```
$ SHLIB_PATH=$AUTOSYS/lib:$SHLIB_PATH
$ export SHLIB_PATH
```

For Linux and Solaris:

```
$ LD_LIBRARY_PATH=$AUTOSYS/lib:$LD_LIBRARY_PATH
$ export LD_LIBRARY_PATH
```

3. Apply the SDK patch to the Web UI server by executing the following script:

```
./Install_INCR.sh -t K -f patch_file_name
```

4. Start all the Web UI services manually.

NOTE

The patch updates the log file named *update_history.log* that is located in the \$AUTOSYS directory.

5. Verify that the patch was applied correctly by running the following command:

```
$ autoflags -a
```

The command returns the following output:

```
build_number OS_TYPE DB_TYPE patch_version hostid MACH_NAME FQDN
```

- **build_number**
Specifies the build number of the patch.
- **patch_version**
Specifies the version of the patch.

Install_INCR.sh Script—Applies the Patch

The Install_INCR.sh script applies the patch. The script verifies that the pre-patch version of AutoSys Workload Automation is currently installed on the machine and stops all the AutoSys Workload Automation processes. It then saves the existing files and replaces the original files with the patched files. If database updates are required, the script connects to the database and makes the changes. If the restart option is specified, the AutoSys Workload Automation processes are started.

This script has the following format:

```
Install_INCR.sh -t A|C|K|S|W -f fix_number [-d db_type] [-db]
[-j java_path] [-n db_name] [-n2 db_name]
[-p db_password] [-p2 db_password] [-r]
[-s db_server] [-s2 db_server]
[-u db_user] [-u2 db_user]
[-i SPN] [-i2 SPN2]
[-tns TNS_ADMIN of aedbadmin]
[-tns2 TNS_ADMIN of aedbadmin]
[-m] [-v] [-x] [-h|?]
```

The following two options are required if -v or -h are not specified:

- **-t A|C|K|S|W**
Specifies the type of the original installation. The installation options are as follows:

- A—AutoSys Workload Automation agent installation.
- C—AutoSys Workload Automation client installation.
- K—AutoSys Workload Automation SDK installation.
- S—AutoSys Workload Automation server, shadow, or tie-breaker installation.
- W—Web UI installation.
- **-f fix_number**
(Optional) Specifies the number that is associated with the patch. This number is reflected in the tar file name. For example, RO82896.

The following options are required if you specify the -t S option:

- **-d db_type**
(Optional) Specifies the database type. Valid values are ORA (for Oracle) and SYB (for Sybase). This option is required if you specify the -t S option on a server type installation. It is invalid on a shadow or tie-breaker installation.
- **-db**
Applies only the database portion of the patch.
- **-j java_path**
(Optional) Specifies the home directory of Java. If this option is not specified, the following path is used: \$AUTOSYS/./JRE_WA/java_version. For example, \$AUTOSYS/./JRE_WA/1.6.0_33
- **-n db_name**
(Optional) Specifies the database name that is used to identify the Sybase database during the AutoSys Workload Automation installation. This option is required if you specify the -t S option on a server type installation and the database type is Sybase.
- **-n2 db_name**
(Optional) Specifies the database name of the secondary event server. This option is required if you specify the -t S option on a dual event server server type installation and the database type is Sybase.

NOTE

The -n and -n2 options are invalid when the database type is Oracle.

- **-p db_password**
(Optional) For Sybase, specifies the database administrator (sa) user password for the primary event server. For Oracle, specifies the schema owner database user password for the primary event server. If -p is not specified, you are prompted to type the password.
- **-p2 db_password**
(Optional) For Sybase, specifies the database administrator (sa) user password for the secondary event server. For Oracle, specifies the schema owner and database user password for the secondary event server. If -p2 is not specified, you are prompted to type the password.
- **-r**
(Optional) Restarts (or starts) the AutoSys Workload Automation processes after the patch is applied. By default, the processes are not started.
- **-s db_server**
(Optional) For Sybase, specifies the database server name as identified in the interfaces file. For Oracle, specifies the Oracle SID as defined in the tnsnames.ora file. This option is required if you specify the -t S option on a server type installation. It is invalid on a shadow or tie-breaker installation.
- **-s2 db_server**
(Optional) For Sybase, specifies the database server name of the secondary event server. For Oracle, specifies the Oracle SID of the secondary event server. This option is required if you specify the -t S option on a dual event server server type installation. It is invalid on a shadow or tie-breaker installation.
- **-u db_user**

(Optional) For Sybase, specifies the database administrator (sa) user ID of the primary event server that is used to update the database. For Oracle, this is an optional parameter and specifies 'schema owner' as the database user ID. This option is required if you specify the -t S option on a server type installation.

- **-u2 db_user**
(Optional) For Sybase, specifies the database administrator (sa) user ID of the secondary event server in a dual event server installation that is used to update the database. For Oracle, this is an optional parameter and specifies 'schema owner' as the database user ID. This option is required if you specify the -t S option on a dual event server type installation.
- **-i SPN** (Optional) Valid for Sybase only. Specifies the service principal name to be used for Kerberos authentication for the primary event server. It is mutually exclusive to the user -u or -u2 and password -p or -p2 parameters.
- **-i2 SPN2** (Optional) Valid for Sybase only. Specifies the service principal name to be used for Kerberos authentication for the secondary event server. It is mutually exclusive to the user -u or -u2 and password -p or -p2 parameters.
- **-tns TNS_ADMIN of schema owner**
(Optional) Valid for Oracle only. Specifies the TNS_ADMIN value for the Oracle schema owner of the primary event server. It is used for SSL certificate based authentication to establish connection to the primary event server. It is mutually exclusive to the user (-u) and password (-p) parameters.
- **-tns2 TNS_ADMIN of schema owner**
(Optional) Valid for Oracle only. Specifies the TNS_ADMIN value for the Oracle schema owner of the secondary event server. It is used for SSL certificate based authentication to establish connection to the secondary event server. It is mutually exclusive to the user (-u2) and password (-p2) parameters.
- **-m**
(Optional) Runs (or starts) the AutoSys Workload Automation DBMaint processes after the patch is applied. By default, DBMaint is not started. This option is applicable only for server type installations.
- **-h | ?**
(Optional) Displays help for this command.
- **-v**
(Optional) Ignores all other options and indicates whether the database updates are included in this patch. This is helpful to determine options you want to specify. For example, if database updates are not included in the patch, you need not specify the database related options when applying the patch.
- **-x**
(Optional) Turns on debugging when the database is updated.

Examples

Example 1: To verify that the database changes are included in the patch, execute the following script :

```
./Install_INCR.sh -v
```

Example 2: To apply the patch to a client installation, execute the following script:

```
./Install_INCR.sh -t C -f patch_file_name
```

Example 3: To apply the patch that has database updates to a server product installation on a Sybase database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d SYB
-n AEDB -u sa -p sapswd -s WAAEPROD
```

The Sybase database name is AEDB. The Sybase server is WAAEPROD. Use the default java path.

Example 4: To apply the patch that has database updates to a server product installation running in dual event server mode on Sybase databases, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d SYB
```

```
-n AEDB -u sa -s WAAEPROD -n2 AEDB2
-u2 sa -s2 WAAEPROD2
```

The Sybase database names are AEDB and AEDB2. The Sybase servers are WAAEPROD and WAAEPROD2. Use the default java path. You are prompted for the password.

Example 5: To apply the patch that has database updates to a server product installation on an Oracle database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
```

The Oracle SID is WAAETEST. Use the default java path.

Example 6: To apply the patch that has database updates to a server product installation running in dual event server mode on an Oracle database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
-u2 aedbadmin -p2 aedb2passwd -s2 WAAETEST2
```

The Oracle SIDs are WAAETEST and WAAETEST2. Use the default java path.

Example 7: To apply only the database portion of the patch, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST -db
```

The Oracle SID is WAAETEST. Use the default java path.

Example 8: To apply the patch if both the Web UI and AutoSys Workload Automation servers are installed on the same machine, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
```

The AutoSys Workload Automation server is running on Oracle in single event server mode using WAAETEST as SID. Use the default java path.

NOTE

Before you apply the patch, shut down all Web UI services.

Example 9: To apply the patch on the application servers or shadow or tie-breaker machines, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name
```

Example 10: To apply the patch that has database updates to a server installation that is configured with Kerberos based authentication and running in single event server mode on a Sybase database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d SYB -s WAAEPROD -n AEDB -i SPN
```

Example 11: To apply the patch that has database updates to a server installation that is configured with Kerberos based authentication and running in dual event server mode on a Sybase database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d SYB -s WAAEPROD -n AEDB -i SPN -s2
WAAEPROD2 -n2 AEDB2 -i2 SPN2
```

Example 12: To apply the patch that has database updates to a server installation that is configured with Oracle wallets and running in single event server mode on an Oracle database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA -s WAAETEST -tns /opt/oracle/tnsaedbadmin
```

NOTE

- The Oracle SID is WAAETEST.
- Use the default Java path.
- The TNS_ADMIN value of the schema owner and database user for the primary database is /opt/oracle/tnsaedbadmin1

Example 13: To apply the patch that has database updates to a server installation that is configured with Oracle wallets and running in dual event server mode on an Oracle database, execute the following script:

```
./Install_INCR.sh -t S -f patch_file_name -d ORA -s WAAETEST -tns /opt/oracle/tnsaedbadmin -s2 WAAETEST2 -tns2 /opt/oracle/tnsaedbadmin2
```

NOTE

- The Oracle SIDs are WAAETEST and WAAETEST2.
- Use the default Java path.
- The TNS_ADMIN value of the schema owner and database user for the primary database is /opt/oracle/tnsaedbadmin1
- The TNS_ADMIN value of the schema owner database user for the secondary database is /opt/oracle/tnsaedbadmin2

AutoSys: (Windows) Apply a Patch

This article provides information about how to apply a patch on Windows, for example how to apply a cumulative on Windows.

NOTE

- Database updates must be run only on server-based machines.
- If multiple instances are installed on a machine, the patch upgrades all the instances to the patch level, except for the database changes. It upgrades only one database server that is supplied in the command line. For upgrading the database of the rest of the instances, use the -db (applies only the database portion of the patch) option. Do not use the -r option as it starts all the services and might cause database conflicts for other instances.
- If Web UI is installed on the same machine where the AutoSys Workload Automation server or client is installed, execute the patch installer only once. Common files that are shared between the two installations are updated as part of patching the AutoSys Workload Automation server or client installation.
- Before applying the patch, ensure that you:
 - Close all the AutoSys Workload Automation client commands or user interfaces that are currently opened.
 - Stop all the AutoSys Workload Automation services. If the patch installer finds any active services, it shuts them down.
 - Back up the AEDB databases on the machines where the patch is being installed.
 - Back up the WorkloadAutomationAE directory as WorkloadAutomationAE.prepatch.
 - Shut down all the Web UI services if Web UI is installed on the same machine where the AutoSys Workload Automation server is installed as the common files that are shared between AutoSys Workload Automation and Web UI might require updating. The patch installer does not shut down the Web UI

services. So, you must shut down all the active Web UI services manually to avoid file copy failures due to active processes.

- Before applying the patch to an SDK-only environment, set the AUTOSYS and PATH environment variables prior to launching the patch installer. For example, if AutoSys Workload Automation is installed in the default location, set the %AUTOSYS% and PATH environment variables as follows:.

```
> set AUTOSYS=C:\Program Files (x86)\CA\Workload Automation AE\autosys
> set PATH=%AUTOSYS%\bin;%PATH%
```

Run the following command to verify that the installed AutoSys Workload Automation SDK version is at the pre-patch level:

```
> autoflags -x
```

Follow these steps:

1. Retrieve the patch zip from the CA Technologies FTP site.
2. Move the zip file to a temporary directory on the machine where AutoSys Workload Automation is installed.
3. Extract the files from the patch to a drive where the product is installed using any *one* of the following methods:
 - If you are using WinZip, assuming that the patch zip file was downloaded to C:\temp\patch, unzip and extract the files from the zip file as follows:

```
C:\AutoSys> cd C:\temp\patch
C:\temp\patch> winzip
```

NOTE

This step assumes that the WinZip utility is in the system path.

If the WinZip utility is not in the system path, open WinZip from the Start menu, select open, choose the zip file, and select extract all.

NOTE

The zip file is the WinZip compatible compressed folder that contains the patch. You can also use Windows Explorer to open the compressed folder.

- Extract the patch with cazipxp.exe. Assuming that the CAZ file was downloaded to C:\temp\patch, run the cazipxp.exe command to extract the files, as follows:
- ```
C:\AutoSys> cd C:\temp\patch
C:\temp\patch> cazipxp.exe -u patch_file_name.caz
```
4. Apply the patch by executing the Install\_INCR\_NT.bat script from the AutoSys Workload Automation instance command prompt from the extracted directory.

#### NOTE

For the server and client installations, run the Install\_INCR\_NT.bat script from the AutoSys Workload Automation instance command prompt. Prior to running the script, set the AutoSys Workload Automation environment variables. For server type installation, either set JAVA\_HOME at the AutoSys Workload Automation instance command prompt or use the -j option with the appropriate JAVA\_HOME path. For example, for the default JAVA\_HOME, execute the following command from the AutoSys Workload Automation instance command prompt prior to executing the Install\_INCR\_NT.bat script:

```
set JAVA_HOME="C:\Program Files (x86)\CA\Workload Automation AE\JRE_WA"
```

5. Verify that the patch is applied correctly by:
  - Running the following command:

```
C:\> autoflags.exe -a
```

The command returns the following output: *build\_number* OS\_TYPE DB\_TYPE *patch\_version* hostid MACH\_NAME FQDN

- **build\_number** Specifies the build number of the patch.
- **patch\_version** Specifies the version of the patch.
- Executing the following SQL query:
 

```
select str_val from ujo_alamode where type='AEDB_VERSION'
```

The query must return the patch version. In dual event server mode, run the query on both the databases.

6. Start the AutoSys Workload Automation services as follows:
  - a. Click Start, Programs, AutoSys, Administrator
  - b. Select an instance from the Instance drop-down list in the Settings pane and click Apply.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears displaying a list of services installed on the selected instance.
  - d. Right-click the scheduler service and select the Start option.  
The scheduler starts.
  - e. Right-click the application server and select the Start option.  
The application server starts.
  - f. Right-click the agent service and select the Start option.  
The agent starts.
  - g. Right-click the web server service and select the Start option.  
The web server starts.

#### NOTE

Repeat this step for all AutoSys Workload Automation instances.

### Apply the SDK Patch on the Web UI Server

#### Follow these steps:

1. Stop all the Web UI services manually.
2. Apply the SDK patch to the Web UI server by using the Install\_INCR\_NT.bat script as follows:
 

```
Install_INCR_NT.bat -t W
```
3. Start all the Web UI services manually.

#### NOTE

The patch updates the log file named *WAAE\_Update\_Install.log* that is located in the %AUTOSYS% directory.

4. Verify that the patch was applied correctly by running the following command:

```
C:\> autoflags.exe -a
```

The command returns the following output:

```
build_number OS_TYPE DB_TYPE patch_version hostid MACH_NAME FQDN
```

1. – **build\_number**  
Specifies the build number of the patch.
- **patch\_version**  
Specifies the version of the patch.

### Install\_INCR\_NT.bat Script—Applies the Patch

The Install\_INCR\_NT.bat script applies the patch. The script verifies that the pre-patch version of AutoSys Workload Automation is currently installed on the machine and stops all the AutoSys Workload Automation processes. It then saves the existing files and replaces the original files with the patched files. If database updates are required, the script connects



to the database and makes the changes. If the restart option is specified, the AutoSys Workload Automation processes are started.

This script has the following format:

```
Install_INCR_NT.bat -t A|C|K|S|W -f fix_number [-d db_type] [-db]
[-j java_path] [-n db_name] [-n2 db_name]
[-p db_password] [-p2 db_password] [-r] [-e] [-e2]
[-s db_server] [-s2 db_server]
[-u db_user] [-u2 db_user]
[-i SPN] [-i2 SPN2]
[-tns TNS_ADMIN of aedbadmin]
[-tns2 TNS_ADMIN of aedbadmin]
[-m] [-v] [-x] [-h|?]
```

The following options are required if -h is not specified:

- **-t A|C|K|S|W**  
Specifies the type of the original installation. The installation options are as follows:
  - A—AutoSys Workload Automation agent installation.
  - C—AutoSys Workload Automation client installation.
  - K—AutoSys Workload Automation SDK installation.
  - S—AutoSys Workload Automation server, shadow, or tie-breaker installation.
  - W—Web UI installation.
- **-f *fix\_number***  
Specifies the number that is associated with the patch. This number is reflected in the patch file name. For example, RO82896.

The following options are required if you specify the -t S option:

- **-d *db\_type***  
(Optional) Specifies the database type. Valid values are ORA (for Oracle), SYB (for Sybase), and MSQ (for MSSQL). This option is required if you specify the -t S option on a server type installation. It is invalid on a shadow or tie-breaker installation.
- **-db**  
Applies only the database portion of the patch.
- **-j *java\_path***  
Specifies the home directory of Java. If this option is not specified, JAVA\_HOME path is used.
- **-n *db\_name***  
(Optional) Specifies the database name that is used to identify the Sybase or MSSQL database during the AutoSys Workload Automation installation. This option is required if you specify the -t S option on a server type installation and the database type is Sybase or MSSQL.
- **-n2 *db\_name***  
(Optional) Specifies the database name of the secondary event server. This option is required if you specify the -t S option on a dual event server server type installation and the database type is Sybase or MSSQL.

#### NOTE

The -n and -n2 options are invalid when the database type is Oracle.

- **-p *db\_password***  
(Optional) For Sybase and MSSQL, specifies the database administrator (sa) user password for the primary event server. For Oracle, specifies the schema owner and database user password for the primary event server.
- **-p2 *db\_password***

(Optional) For Sybase and MSSQL, specifies the database administrator (sa) user password for the secondary event server. For Oracle, specifies the schema owner and database user password for the secondary event server.

- **-r**  
(Optional) Restarts (or starts) the AutoSys Workload Automation processes after the patch is applied. By default, the processes are not started.
- **-e**  
Specifies to use a trusted connection (windows authentication mode) to connect to the primary event server.
- **-e2**  
Specifies to use a trusted connection (windows authentication mode) to connect to the secondary event server.

#### NOTE

The -e and -e2 options are valid only for MSSQL.

- **-s db\_server**  
(Optional) For Sybase and MSSQL, specifies the database server name. For Oracle, specifies the Oracle SID. This option is required if you specify the -t S option on a server type installation. It is invalid on a shadow or tie-breaker installation.
- **-s2 db\_server**  
(Optional) For Sybase and MSSQL, specifies the database server name of the secondary event server. For Oracle, specifies the Oracle SID of the secondary event server. This option is required if you specify the -t S option on a dual event server server type installation. It is invalid on a shadow or tie-breaker installation.
- **-u db\_user**  
(Optional) For Sybase and MSSQL, specifies the database administrator (sa) user ID of the primary event server that is used to update the database. For Oracle, this is an optional parameter and specifies 'schema owner' as the database user ID..
- **-u2 db\_user**  
(Optional) For Sybase and MSSQL, specifies the database administrator (sa) user ID of the secondary event server in a dual event server server installation that is used to update the database. For Oracle, this is an optional parameter and specifies 'schema owner' as the database user ID.
- **-i SPN** (Optional) Valid for Sybase only. Specifies the service principal name to be used for Kerberos authentication for the primary event server. It is mutually exclusive to the user -u or -u2 and password -p or -p2 parameters.
- **-i2 SPN2** (Optional) Valid for Sybase only. Specifies the service principal name to be used for Kerberos authentication for the secondary event server. It is mutually exclusive to the user -u or -u2 and password -p or -p2 parameters.
- **-tns TNS\_ADMIN of schema owner**  
(Optional) Valid for Oracle only. Specifies the TNS\_ADMIN value for the Oracle schema owner of the primary event server. It is used for SSL certificate based authentication to establish connection to the primary event server. It is mutually exclusive to the user (-u) and password (-p) parameters.
- **-tns2 TNS\_ADMIN of schema owner**  
(Optional) Valid for Oracle only. Specifies the TNS\_ADMIN value for the Oracle schema owner of the secondary event server. It is used for SSL certificate based authentication to establish connection to the secondary event server. It is mutually exclusive to the user (-u2) and password (-p2) parameters.
- **-m**  
(Optional) Runs (or starts) the AutoSys Workload Automation DBMaint processes after the patch is applied. By default, DBMaint is not started. This option is applicable only for server type installations.
- **-h | ?**  
(Optional) Displays help for this command.
- **-v**  
(Optional) Ignores all other options and indicates whether the database updates are included in the patch. This is helpful to determine options you want to specify. For example, if the database updates are not included in the patch, you need not specify the database related options when applying the patch.
- **-x**  
(Optional) Turns on debugging when the database is updated.

## Examples

**Example 1:** To apply the patch to a client product installation, execute the following script:

```
Install_INCR_NT.bat -t C
```

**Example 2:** To apply the patch that has database updates to a server product installation on a Sybase database, execute the following script:

```
Install_INCR_NT.bat -t S -d SYB
-n AEDB -u sa -p sapswd -s WAAEPROD
```

The Sybase database name is AEDB. The Sybase server is WAAEPROD. Use the default Java path.

**Example 3:** To apply the patch that has database updates to a server product installation on a MSSQL database, execute the following script :

```
Install_INCR_NT.bat -t S -d MSQ
-n AEDB -u sa -p sapswd -s WAAEPROD
```

The MS-SQL database name is AEDB. The MS-SQL server is WAAEPROD. Use the default Java path.

**Example 4:** To apply the patch that has database updates to a server product installation running in dual event server mode on MSSQL databases, execute the following script:

```
Install_INCR_NT.bat -t S -d MSQ
-n AEDB1 -u sa -p sapswd -s WAAEPROD1
-n2 AEDB2 -u2 sa -p2 sapswd -s2 WAAEPROD2
```

The MSSQL primary database name is AEDB1. The MSSQL primary server is WAAEPROD1. The MSSQL secondary database name is AEDB2. The MSSQL secondary server is WAAEPROD2. Use the default Java path.

**Example 5:** To apply the patch that has database updates to a server product installation running in dual event server mode on Sybase databases, execute the following script:

```
Install_INCR_NT.bat -t S -d SYB -n AEDB
-u sa -s WAAEPROD -p sapasswd -n2 AEDB2
-u2 sa -s2 WAAEPROD2 -p2 sapasswd
```

The Sybase database names are AEDB and AEDB2. The Sybase servers are WAAEPROD and WAAEPROD2. Use the default Java path.

**Example 6:** To apply the patch that has database updates to a server product installation on an Oracle database, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
```

The Oracle SID is WAAETEST. Use the default Java path.

**Example 7:** To apply the patch that has database updates to a server product installation running in dual event server mode on Oracle databases, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
-u2 aedbadmin -p2 aedb2passwd -s2 WAAETEST2
```

The Oracle SIDs are WAAETEST and WAAETEST2. Use the default Java path.

**Example 8:** To apply the patch that has database updates to a server installation that is configured with DBAccessMode=1, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-s WAAETEST -tns C:\oracle\tnsaedbadmin
```

The Oracle SID is WAAETEST. Use the default Java path. The TNS\_ADMIN value for the schema owner and database user is C:\oracle\tnsaedbadmin

**Example 9:** To apply the patch that has database updates to a server installation that is configured with DBAccessMode=1 and running in dual event server mode on Oracle databases, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-s WAAETEST -tns C:\oracle\tnsaedbadmin
-s2 WAAETEST -tns2 C:\oracle\tnsaedbadmin2
```

The Oracle SIDs are WAAETEST and WAAETEST2. Use the default Java path. The TNS\_ADMIN value for the schema owner database user of the primary event server is C:\oracle\tnsaedbadmin. The TNS\_ADMIN value for the schema owner and database user of the secondary event server is C:\oracle\tnsaedbadmin2.

**Example 10:** To apply only the database portion of the patch if an earlier attempt to apply it failed during the database update stage, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST -db
```

**Example 11:** To apply the patch to a shadow scheduler machine or a tie-breaker scheduler machine, execute the following script:

```
Install_INCR_NT.bat -t S
```

**Example 12:** To apply a patch where both Web UI and AutoSys Workload Automation servers are installed on the same machine, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA
-u aedbadmin -p aedbpasswd -s WAAETEST
```

The AutoSys Workload Automation server is running on a single event server using WAAETEST SID. The product uses the default Java path. Prior to applying the patch, shut down all the Web UI services.

**Example 13:** To apply the patch to a server installation that is running in single event server mode on a MSSQL database; where the MSSQL database uses Windows authentication, execute the following script:

```
Install_INCR_NT.bat -t S -d MSQ
-n AEDB -s WAAEPROD -e
```

**Example 14:** To apply the patch on a machine where the client, SDK, and Web UI are installed, execute the following script:

```
Install_INCR_NT.bat -t C
```

#### NOTE

The -t C option also updates the SDK.

**Example 15:** To apply the patch that has database updates to a server installation that is configured with Kerberos based authentication and running in single event server mode on a Sybase database, execute the following script:

```
Install_INCR_NT.bat -t S -d SYB -s WAAEPROD -n AEDB -i SPN
```

**Example 16:** To apply the patch that has database updates to a server installation that is configured with Kerberos based authentication and running in dual event server mode on a Sybase database, execute the following script:

```
Install_INCR_NT.bat -t S -d SYB -s WAAEPROD -n AEDB -i SPN -s2 WAAEPROD2 -n2 AEDB2 -
i2 SPN2
```

**Example 17** To apply the patch that has database updates to a server installation that is configured with Oracle wallets and running in single event server mode on an Oracle database, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA -s WAAETEST -tns /opt/oracle/tnsaedbadmin
```

#### NOTE

- The Oracle SID is WAAETEST.
- Use the default Java path.
- The TNS\_ADMIN value of the schema owner and database user for the primary database is /opt/oracle/tnsaedbadmin1

**Example 18:** To apply the patch that has database updates to a server installation that is configured with Oracle wallets and running in dual event server mode on an Oracle database, execute the following script:

```
Install_INCR_NT.bat -t S -d ORA -s WAAETEST -tns /opt/oracle/tnsaedbadmin -s2 WAAETEST2
-tns2 /opt/oracle/tnsaedbadmin2
```

#### NOTE

- The Oracle SIDs are WAAETEST and WAAETEST2.
- Use the default Java path.
- The TNS\_ADMIN value of the schema owner and database user for the primary database is /opt/oracle/tnsaedbadmin1
- The TNS\_ADMIN value of the schema owner and database user for the secondary database is /opt/oracle/tnsaedbadmin2

## AutoSys: Back Out a Patch

You can back out from applying a patch and move the system back to a state that it was in before applying the patch.

#### NOTE

This procedure assumes that you backed up the WorkloadAutomationAE directory before applying the patch and the backup of the WorkloadAutomationAE directory is WorkloadAutomationAE.prerelease.

#### (UNIX) Follow these steps:

1. Stop the AutoSys Workload Automation services by logging in as the root user and running the following commands at the operating system prompt for all AutoSys Workload Automation instances:

```
$ $CASHCOMP/bin/unisrvctr stop waae_sched.$AUTOSERV
$ $CASHCOMP/bin/unisrvctr stop waae_server.$AUTOSERV
$ $CASHCOMP/bin/unisrvctr stop waae_webserver.$AUTOSERV
$ $CASHCOMP/bin/unisrvctr stop waae_agent-WA_AGENT
```

#### – WA\_AGENT

Specifies the name of the agent.

2. Back up the database.
3. Move the current WorkloadAutomationAE directory by running the following commands:

```
$ cd /opt/CA
$ mv WorkloadAutomationAE WorkloadAutomationAE.current
```

The WorkloadAutomationAE directory is renamed to WorkloadAutomationAE.current.

4. Restore the backed up WorkloadAutomationAE directory by running the following commands:

```
$ cd /opt/CA
$ mv WorkloadAutomationAE.prerelease WorkloadAutomationAE
```

The WorkloadAutomationAE.prerelease directory is renamed to WorkloadAutomationAE.

#### NOTE

If the above command prompts for overwrite, then select Y.

5. Refresh the databases by executing the [RefreshAEDB.pl script](#) as follows and providing appropriate input to it:

```
$ cd $AUTOSYS/dbobj/DB_TYPE
$ perl RefreshAEDB.pl
```

#### – DB\_TYPE

Specifies the database provider; SYB for Sybase or ORA for Oracle.

#### NOTE

- This step is required only for server installation.
- Repeat this step for each event server in an instance.
- Repeat this step for all AutoSys Workload Automation instances.

The RefreshAEDB.pl script refreshes only the stored procedures. You must restore the database schema to the pre-patch level. To restore the database schema to the pre-patch level, restore the database that you backed up before applying the patch.

6. Start the AutoSys Workload Automation services by running the following commands at the operating system prompt for all AutoSys Workload Automation instances:

```
$ $CASHCOMP/bin/unisrvcntr start waae_sched.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr start waae_server.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr start waae_webserver.$AUTOSERV
$ $CASHCOMP/bin/unisrvcntr start waae_agent-WA_AGENT
```

The AutoSys Workload Automation services start and the patch is backed out to a state that it was in before you applied a patch.

7. Verify that the patch is backed out by running the following commands at the operating system prompt:

```
autoflags -i
select str_val from ujo_alamode where type='AEDB_VERSION'
```

The autoflags -i command output must not display the build number of the current patch. The output of the SQL command must not display the release number of the current patch.

#### (Windows) Follow these steps:

1. Log in as a user with Windows Administrators group privileges and stop the AutoSys Workload Automation services as follows:
  - a. Click Start, Programs, AutoSys, Administrator.
  - b. Select an instance from the Instance drop-down list in the Settings pane and click Apply.
  - c. Click the Services icon on the toolbar.
  - d. Right-click the scheduler service and select the Stop option.
  - e. Right-click the application server service and select the Stop option.
  - f. Right-click the web server service and select the Stop option.
  - g. Right-click the agent service and select the Stop option.

#### NOTE

Repeat these steps for all AutoSys Workload Automation instances.

2. Back up the database.
3. Rename the current WorkloadAutomationAE directory to WorkloadAutomationAE.current.
4. Rename the backed up WorkloadAutomationAE.prerelease directory to WorkloadAutomationAE.
5. Refresh the databases by executing the RefreshAEDB.pl script as follows and providing appropriate input to it:

```
cd %AUTOSYS%\dbobj\DB_TYPE
perl RefreshAEDB.pl
```

– **DB\_TYPE**

Specifies the database provider: SYB for Sybase, ORA for Oracle, or MSQ for Microsoft SQL Server.

**NOTE**

- This step is required only for server installation.
- Repeat this step for each event server in an instance.
- Repeat this step for all AutoSys Workload Automation instances.

The RefreshAEDB.pl script refreshes only the stored procedures. You must restore the database schema to the pre-patch level. To restore the database schema to the pre-patch level, restore the database that you backed up before applying the patch.

6. Start the AutoSys Workload Automation services as follows:
  - a. Click Start, Programs, AutoSys, Administrator.
  - b. Select an instance from the Instance drop-down list in the Settings pane and click Apply.
  - c. Click the Services icon on the toolbar.
  - d. Right-click the scheduler service and select the Start option.
  - e. Right-click the application server service and select the Start option.
  - f. Right-click the web server service and select the Start option.
  - g. Right-click the agent service and select the Start option.

**NOTE**

Repeat these steps for all AutoSys Workload Automation instances.

7. Verify that the patch is backed out by running the following commands at the operating system prompt:

```
autoflags -i
select str_val from ujo_alamode where type='AEDB_VERSION'
```

The autoflags -i command output must not display the build number of the current patch. The output of the SQL command must not display the release number of the current patch.

## JRE Update

To update the JRE, use any JRE or the JRE provided by OpenJDK in AutoSys Workload Automation and perform the steps mentioned below.

### Update JRE

In the case of any other JRE, follow these steps:

1. Download 64-bit JRE from the OpenJDK website in /tmp location on your AE server machine.
2. Create folder JRE in the /tmp location
3. Extract the JRE in /tmp/jre folder.
4. Stop the AutoSys Workload Automation web services., Java utilities, AutoSys Rest web server, SOAP web server, Web UI, Web UI collector and, agent services.
5. Follow the steps mentioned in Complete the JRE Update.

In the case of bundled JRE, follow these steps:

1. Go to the AUTOROOT directory (/opt/CA/WorkloadAutomationAE), and backup the JRE folders if it is the bundled JRE.
2. Move the /tmp/jre folder to the AUTOROOT directory.
3. Follow the steps mentioned in Complete the JRE Update.

### **Complete the JRE Update**

1. Unlink the folders pointing to JRE\_WA and JRE64\_WA.

**Linux:**

```
rm -rf /opt/CA/WorkloadAutomationAE/JRE_WA
rm -rf /opt/CA/WorkloadAutomationAE/JRE64_WA
```

**Windows:** Delete the links WorkloadAutomationAE/JRE\_WA and WorkloadAutomationAE/JRE64\_WA

2. Create the below soft links pointing to the new extracted JRE.

**Linux:**

```
ln -s /opt/CA/WorkloadAutomationAE/jre JRE_WA
ln -s /opt/CA/WorkloadAutomationAE/jre JRE64_WA
```

**Windows:**

```
MKLINK /D JRE_WA "C:\Program Files\CA\WorkloadAutomationAE\jre"
MKLINK /D JRE64_WA "C:\Program Files\CA\WorkloadAutomationAE\jre"
```

3. For all agents, stop the Agent Service.
4. For all agents create the soft link pointing to the new extracted JRE.

**Linux:** `ln -s /opt/CA/WorkloadAutomationAE/jre/ /opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT/jre`

**Windows:** `Cd "C:\Program Files\CA\WorkloadAutomationAE\SystemAgent\WA_AGENT"`

`MKLINK /D jre "C:\Program Files\CA\WorkloadAutomationAE\jre"`

5. Stop the Web UI (WCC) service.
6. Ensure the link is proper or else create the link for the Web UI.

**Linux:** `ln -s /opt/CA/WorkloadAutomationAE/JRE_WA/ /opt/CA/WorkloadAutomationAE/wcc/jre`

**Windows:** `Cd "C:\Program Files\CA\WorkloadAutomationAE\wcc"`

`MKLINK /D jre "C:\Program Files\CA\WorkloadAutomationAE\jre"`

### **Regenerate Certificates for AutoSys REST Web Server and Web UI Server**

When using custom JRE with "security.useSystemPropertiesFile=true" in the java.security file and if the system is configured not to accept certificates of 1024 key length, then the certificates must be re-generated for the required key length for the AutoSys REST Web Server and Web UI Server.

You can generate certificates of 2048 key length for the AutoSys REST Web Server.

**NOTE**

Replace the <hostname\_fqn> with the fully qualified host name.

#### **Follow these steps:**

1. Stop the AutoSys REST Web Server.
2. Back up the keystore file for the AutoSys REST Web Server as follows:

```
mkdir -p /backupDir
mv $AUTOUSER/webserver/conf/.keystore /backupDir/.keystoreRestWebServer
```

3. Generate certificates of 2048 key length as follows:



```
$AUTOSYS_INSTALL_LOCATION/JRE_WA/bin/keytool -genkey -alias tomcat -keyalg RSA -keystore $AUTOUSER/
webserver/conf/.keystore -provider org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider -providerpath
$AUTOSYS/lib/bc-fips.jar -storepass changeit -keypass changeit -dname "CN=<hostname_fqn>, OU=WAAE, O=CA"
-validity 14600 -keysize 2048 -storetype BCFKS
```

#### 4. Start the AE Web Server.

You can generate certificates of 2048 key length for the Web UI Server.

#### Follow these steps:

1. Stop WCC services.
2. Back up the existing Web UI keystore as follows:

```
mkdir -p /backupDir
mv $AUTOSYS_INSTALL_LOCATION/wcc/data/config/.keystore /backupDir/.keystoreUITomcat
```

3. Generate certificates of 2048 key length for Web UI Tomcat as follows:

```
$AUTOSYS_INSTALL_LOCATION/JRE_WA/bin/keytool -genkey -keyalg RSA -keysize 2048 -alias tomcat -keystore
$AUTOSYS_INSTALL_LOCATION/wcc/data/config/.keystore -storepass changeit -keypass changeit -validity 3652
-dname "CN=<hostname_fqn>, OU=WCC, O=CA"
```

4. Start WCC services.

## Upgrade Tomcat Version for AutoSys

You can upgrade the Tomcat version for AutoSys and Web UI.

### NOTE

The procedure may vary based on the versions of AutoSys and Tomcat because of changes to both AutoSys and Tomcat.

Download the required Tomcat version from the Apache website.

#### Follow these steps:

1. Perform the following action:
  - (Linux) Run the shell that is sourced to use AutoSys Workload Automation.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (instance\_name).
2. Stop all the web services and Web UI.
3. Backup the \$AUTOSYS\_INSTALL\_LOCATION/webserver folder as follows:

#### Linux

```
mv $AUTOSYS_INSTALL_LOCATION/webserver $AUTOSYS_INSTALL_LOCATION/webserver.bkp
```

#### Windows

```
cd %AUTOSYS_INSTALL_LOCATION%
mkdir webserver.bkp
copy webserver* webserver.bkp
```

4. Use the following command to extract the required Tomcat version:

#### Linux

Assuming you downloaded the required Tomcat version into your /tmp folder.

```
mv /tmp/apache-tomcat.tar.gz $AUTOSYS_INSTALL_LOCATION
cd $AUTOSYS_INSTALL_LOCATION
gunzip apache-tomcat.tar.gz
tar -xvf apache-tomcat.tar
mv apache-tomcat webserver
```

#### Windows

- a. Delete the %AUTOSYS\_INSTALL\_LOCATION%\webserver folder.

- b. Extract the contents of the required Apache version into the %AUTOSYS\_INSTALL\_LOCATION%\webserver folder.
5. (Linux only) Update the owner-group and set the permissions.
 

```
chown -R <install_owner>:<install_group> $AUTOSYS_INSTALL_LOCATION/webserver
chmod -R 755 $AUTOSYS_INSTALL_LOCATION/webserver
```
6. Start all the webservices and Web UI.  
The Tomcat version upgrade is now complete.

## Install AutoSys Server

The following topics describe how to install the AutoSys server, and verify the installation.

### Install Considerations

Before you install the server, review the following sections:

#### Installation Considerations

When installing the server, consider the following points:

- Completing the server installation checklist provides you with information that you need during the installation process. We recommend that you complete the checklist before you start the installation process.
- The considerations on this list include recommendations and requirements that are specific to certain situations. We recommend that you review all these considerations and take actions that apply to your situation.
- Ensure that the relational database management system (RDBMS) is installed before you begin the server installation process.
- The default installation locations are as follows:
  - CA Workload Automation AE
    - (UNIX) /opt/CA/WorkloadAutomationAE
    - (Windows) C:\Program Files\CA\WorkloadAutomationAE
  - CA common components
    - (UNIX) /opt/CA/SharedComponents
    - (Windows) C:\Program Files\CA\SC or C:\Program Files\CA\SharedComponents
  - (Windows) Shared Files—C:\Program Files\CA\CA\_APPSW or C:\CA\_APPSW

#### **NOTE**

- You cannot specify another target installation directory if a common component or CA Workload Automation AE r11.3 component is already installed on your computer.
- Previously installed Unicenter AutoSys JM 4.5.1 or r11 remain usable only if you specify a different directory name when you install the server.
- The installer automatically installs the following components:

- CAPKI
- Safex
- If you select a component for installation, the installer automatically accepts dependent components. For example, the installer automatically selects the client when you select the Scheduler.
- 32-bit computers do not support web server installations. If you select to install the web server on a 64-bit computer, the 64-bit JRE is installed along with the web server.
- To enable FIPS 140-2 compliancy, perform a custom installation and ensure that you also enable external security mode. AutoSys Workload Automation operates in FIPS 140-2 compliancy only when all instances and any associated external security servers are also configured to run in FIPS 140-2 compliancy.
- Before you begin the installation process verify whether the IP address that the hostname command returns, matches the IP address that the nslookup command returns. You cannot change host names after the installation is complete.
- If you install the scheduler component during installation, the post-installation process creates the default fully qualified host name machine definition. The scheduler uses this machine definition when it starts running jobs on the localhost for the locally installed agent.
- (UNIX) For all the Unicenter NSM Job Management Option (JMO) components work as expected when AutoSys Workload Automation and JMO 3.1 are installed on the same computer, configure the environment as follows:
  - a. Open the operating system prompt and issue the following command:
 

```
echo $TERM
env -i bash
```

The terminal code value is obtained and the environment is cleared.
  - b. Export the TERM environment variable to the terminal code value that the echo command returned.
  - c. Source the /etc/profile.CA (for CA Workload Automation AE) and /etc/profile.uni (for Unicenter JMO) scripts.
- Ensure that the database and scheduler are configured to run in the same time zone.

### Agent Installed on the Scheduler Computer

When you install the scheduler, the agent is installed on the server computer. You can use this agent to run jobs on the server computer.

#### NOTE

The agent is installed on the server computer whether you perform a typical or custom scheduler installation.

## Install AutoSys Server

This topic describes how to install the AutoSys server.

#### NOTE

Click **Next** to proceed with the installation.

#### Follow these steps:

1. Double-click the install.exe file to open the AutoSys Workload Automation installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In [Installation Options](#), select **Install the product**.
4. Click **Custom** or **Express** based on your requirements.

#### NOTE

In this topic, we are proceeding with a custom installation procedure. For more information about Custom and Express mode, see [AutoSys Installation Modes](#).

5. Select the [Java Virtual Machine](#).

#### NOTE

By default, all the components required for a server are selected in the Components to install window.

6. In the installation path, proceed with the default installation paths or select your desired installation paths. For more information, see [Installation Path](#).
7. In the [Instance information window](#), enter a name for the AutoSys instance that you are currently installing.
8. In the Instance-level Data Encryption window, select **Use the instance-level encryption values as agent encryption values** if you intend to use the instance key for agent.
9. In the FIPS encryption window, you can select the **FIPS 140-2 compliance** checkbox. However, for the purposes of these instructions, we are not selecting this checkbox. For more information, see [FIPS Encryption](#).
10. In the Database Type window, you can select the desired database type. However, for the purposes of these instructions, we are using Microsoft SQL Server. For more information, see [Database Type](#).
11. In the Primary Event Server Properties window, enter the required details. For more information, see [Primary Event Server Properties](#).

#### NOTE

This window will depend upon the database type you select.

12. In the Primary Admin Details window, enter the required details. For more information, see [Primary Admin Details](#).
13. In the AutoSys Database User Details window, enter the required details. For more information, see [AutoSys Database User Details](#).
14. In the Instance Level CA EEM Security window, enter the required details. For more information, see [Instance Level CA EEM Security](#).
15. In the Scheduler Properties window, enter the **Auxiliary port**. By default, the value is set to 7507.
16. In the Application Server Information window, enter the required details. For more information, see [Application Server Information](#).
17. In the Web Services Properties window, enter the REST Communication port. By default, the value is 9443.
18. In the Agent properties window, enter the required details. For more information, see [Agent Properties](#) and [SNMP Information](#).
19. In the Agent Application Server Port window, enter the comma-separated port numbers for the agent to communicate with the application server. By default, the value is set to 9000.
20. In the Services Startup Options window, select the desired services. For more information, see [Services Startup Options](#).
21. In the Activate Your Product window, enter the required details. For more information, see [Activate your product](#).
22. The Pre-Installation Summary window displays summary of what is going to be installed on your machine.
23. Click **Install**. This will begin the installation.
24. The Install Complete window displays a summary of what has been installed. Click **Done**.

#### NOTE

The **Whether you want to restart your computer to complete the installation** option is displayed if the CAPKI installer or other components require a reboot or the installation fails to change any SSA-related binaries or libraries. This happens if SSA or any other application using SSA is active during the installation.

## Define the Agent on the Server

To enable communication between AutoSys Workload Automation and the agent, [define the agent](#) that is installed with the server to the database.

#### NOTE

Ensure that the parameters you specify when you define an agent on AutoSys Workload Automation match the corresponding parameters in the agentparm.txt file of the agent.

## (UNIX) Verify the Server Installation

Before continuing with the post-installation procedures that may include creating more EDIT and EXEC superusers and installing clients, you must test the product to make sure that it is installed properly.

To verify the server installation, perform the following steps:

### **Set the Time Zone**

Before you start the scheduler or application server, ensure that the TZ environment variable is set. The scheduler or application server references this setting to determine the default time zone. After you upgrade your database, you must start the scheduler to insert a time zone offset value (calculated from the value of the TZ environment variable) into the database. Do this before executing jil or autorep.

#### **WARNING**

Ensure that the database is running in the same time zone that the scheduler starts up with.

Jobs with time-based starting conditions that do not specify a time zone have their start event scheduled based on the time zone under which the database runs. This time zone is also used to report event times, using the autorep command.

### **Set Up the Environment**

To set up the environment, log in to the server computer as the owner of AutoSys Workload Automation (for example, *autosys*) and source the proper file in AUTOUSER.

#### **NOTE**

Because AUTOUSER is not defined yet, you must enter the full path.

In the following examples, *fiji* is the name of the server computer, */opt/CA/WorkloadAutomationAE* represents the path to the directory where you installed CA Workload Automation AE, and *ACE* is the name of the AutoSys Workload Automation instance.

If you are running the C shell (csh), enter the following command:

```
source /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.csh.fiji
```

If you are running the Bourne shell (sh), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.sh.fiji
```

If you are running the Korn shell (ksh), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.ksh.fiji
```

If you are running the Bash shell (bash), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.bash.fiji
```

### **Start the Scheduler**

To start the scheduler, enter the following command:

```
eventor
```

When eventor is run, you lose control of the open command prompt window. To run any other commands while the eventor is running, you must open a new command window. This lets you view the eventor output while running commands through the second command prompt.

To let the eventor run silently and return a command prompt, enter the following command:

```
eventor - q
```

This lets you manually scroll through the eventor output without losing the command prompt.

#### NOTE

The eventor script is designed to make sure that the environment is in the right state before starting the scheduler; specifically, it ensures that another scheduler is not already running.

### **Start the Application Server**

Before you start the application server, make sure that the TZ environment variable is set. The application server references the TZ environment variable to determine the default time zone.

To start the application server, enter the following command at the operating system prompt:

```
as_server -A $AUTOSERV
```

### **Verify the Agent and Database Accessibility**

You can use the autoping command to do the following:

- Verify that the agent on the server computer is functioning.
- Check the database connection of the computer. If you are running dual event servers, the autoping command checks both the databases.

### **Verify Agent Accessibility**

You can verify that an agent is functional and can set up properly by running the autoping command to ping the agent from the server or client computer.

#### NOTE

After you install the agent, you must [define that agent on AutoSys Workload Automation](#) to enable communication between AutoSys Workload Automation and the agent. So, you must ensure that the agent is defined on AutoSys Workload Automation before you verify the agent accessibility.

To verify that the agent is functional on the server computer, enter the following command:

```
autoping -m servername
```

- **servername**  
Specifies the name of the server computer.

The following message appears:

```
AutoPinging Machine [servername]
AutoPing WAS SUCCESSFUL!
```

If you do not get this message, the agent is not configured properly and, as a result, AutoSys Workload Automation cannot start jobs on that computer (even if it is the same computer as the server).

### **Verify the Database Connection**

You can check the running status of the database. If you are running dual event servers, the autoping command checks both the databases.

To check the database connection on the computer, enter the following command:

```
autoping -m servername -S
```

- **servername**

Specifies the name of the server computer.

The following message appears:

```
AutoPinging Machine [servername]
AutoPing WAS SUCCESSFUL!
```

If you do not get this message, the database is not accessible.

### **Run a Test Job**

You can test your configured installation and can verify your AutoSys Workload Automation environment by running a test job.

If CA EEM controls the instance, before executing any command-line interface, first verify that the AutoSys Workload Automation subscriber security word is set by running the AutoSys Workload Automation Secure Utility (autosys\_secure).

Job definitions are specified using Job Information Language (JIL). The jil command is a language processor that parses the language and updates the database.

A test job named test\_install is included with the product. Its job definition is in the file named \$AUTOSYS/test/jil/test\_install. If you view this job definition in a text editor, you see the following:

```
JIL file to test the installation
It will write a line to the Output File
insert_job: test_install
machine: localhost
command: /bin/echo "AUTOSYS install test"
std_out_file: /tmp/test_install.out
std_err_file: /tmp/test_install.err
```

Use the test\_install job as a template to verify the installation.

#### **NOTE**

If your computer is not aliased to localhost, you must modify the test\_install job definition to specify your computer's actual name.

### **Specify a Computer Name in the test\_install Job**

If your computer is not aliased to localhost and you intend to use the test job that is provided with the product, you must modify the test\_install job to specify the actual name of your computer.

#### **Follow these steps:**

1. Open \$AUTOSYS/test/jil/test\_install to modify the line that reads as follows:

```
machine: localhost
```

2. Replace localhost with the host name of your computer. For example, you could use the following code:

```
machine: myhost
```

**NOTE**

This name must match the value that you entered in the `insert_machine` subcommand when you defined the agent to the database.

3. Save and close the `test_install` file.

**Add the Test Job to the Database**

After you have modified the `test_install` job, you must insert it into the database.

**Follow these steps:**

1. Run the shell that is sourced to use CA Workload Automation AE.  
The operating system command prompt appears.
2. Enter the following command:

```
jil < $AUTOSYS/test/jil/test_install
```

A message appears indicating whether the installation of the test job was successful.

**NOTE**

The test job runs only when you start it manually. To start jobs manually, use the `sendevent` command to issue a `STARTJOB` event.

**Run the Test Job**

To send an event to start the `test_install` job, enter the following command:

```
sj test_install
```

This command starts the `test_install` job by using the `sj` alias that is defined in the environment file. The `sj` alias represents the full command line as follows (which could also be used to start the job, if you do not have the aliases defined):

```
sendevent -E STARTJOB -J test_install
```

The event to start the job is now in the database, but the job itself does not start until the scheduler is up and running.

**Verify the Test Job**

To verify that the job started and ran successfully, monitor the scheduler output log with the following command:

```
autosyslog -e
```

If the job ran successfully, the following message is written to the `/tmp/test_install.out` file:

```
AUTOSYS install test
```

This indicates that the basic AutoSys Workload Automation environment is set up properly.

If the job did not run successfully, an error message appears indicating the problem in the `/tmp/test_install.err` file.

To close the scheduler output log, press `Ctrl+C` on your keyboard.



## **Test the Environment Setup**

After a test job is completed, you must test the navigation through the data server. A successful test confirms that the environment variables needed for AutoSys Workload Automation are set up properly.

To test the data server navigation, enter the following command:

```
autorep -J ALL
```

The following message appears:

| Job Name | Last Start          | Last End            | ST Run/Ntry | Pri/Xit |
|----------|---------------------|---------------------|-------------|---------|
| -----    | -----               | -----               | -----       | -----   |
| Jobname1 | 06/24/2009 01:01:01 | 06/24/2009 01:01:05 | SU 3/1      |         |

If the environment is not set up correctly, diagnostic messages are displayed to inform you of what is incorrect. For example:

```
CAUAJM_E_10029 Communication attempt with the AutoSys Workload Automation Application
Server has failed! [machine1:9,000]
CAUAJM_E_10221 Exhausted list of application servers. Failing request.
CAUAJM_E_50033 Error initializing tx subsystem: CAUAJM_E_10062 Failed to get initial
configuration from AutoSys Workload Automation Application Server(s)
```

## **(Windows) Verify the Server Installation**

Before continuing with the post-installation procedures that may include creating additional EDIT and EXEC superusers, you must test the product to make sure it is installed properly.

To verify the server installation, do the following:

### **Set the Time Zone**

Before you start the scheduler, ensure that the correct time zone is specified in the Control Panel, Date/Time dialog. The scheduler references this setting to determine the default time zone. After you upgrade your database, you must start the scheduler to insert a time zone offset value (calculated from the value specified in the Control Panel, Date/Time dialog) into the database. Do this before executing jil or autorep.

#### **WARNING**

Ensure that the database is running in the same time zone that the scheduler starts up with.

Jobs with time-based starting conditions that do not specify a time zone have their start event scheduled based on the time zone under which the database runs. This time zone is also used to report event times, using the autorep command.

### **Stop the Scheduler**

To stop the scheduler, enter the following command:

```
sendevent -E STOP_DEMON
```

This command enables the scheduler to complete any processing before it shuts down.

**NOTE**

- You must be an EXEC superuser to issue commands and stop the scheduler.
- After you stop the scheduler, you must verify its status using the Services - AutoSys Workload Automation Administrator window of AutoSys Workload Automation administrator, or from the Services dialog in the Control Panel. For more information about verifying the status of a service using the AutoSys Workload Automation Administrator, see [Control the Services](#).

**Start the Scheduler**

You must start the scheduler manually using the AutoSys Workload Automation Administrator, or from the Services dialog in the Control Panel.

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list.
3. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the scheduler service, and click Start.  
The scheduler starts. The Status column indicates the status.

**NOTE**

You can also use the autosyslog -e command at a AutoSys Workload Automation instance command prompt to monitor messages from the scheduler. To stop the log command, press Ctrl+C on your keyboard.

**Start the Application Server**

You must start the application server manually using the Services - AutoSys Workload Automation Administrator window, or from the Services dialog in the Control Panel.

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list.
3. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the application server service, and click Start.  
The application server starts. The Status column indicates the status.

**Verifying the Agent and Database Accessibility**

You can test the product installation by doing the following:

- Verify the agent accessibility.
- Verify that the database is running.
- Verify the environment and configuration.

**Verify Agent Accessibility**

You can verify that an agent is functional and set up properly by running the autoping command to ping the agent from the server or client computer.

**NOTE**

After you install the agent, you must [define that agent on AutoSys Workload Automation](#) to enable communication between AutoSys Workload Automation and the agent. So, you must ensure that the agent is defined on AutoSys Workload Automation before you verify the agent accessibility.

**Follow these steps:**

1. Open the instance command prompt window from the program group.
2. Run the autoping command. For example, if the computer's name is jupiter, enter the following:

```
autoping -m jupiter
```

You must get the following message:

```
CAUAJM_I_50023 AutoPinging Machine [jupiter]
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL!
```

If you do not get the success message in the instance command prompt window, the agent is not configured properly, or it is not started. As a result, AutoSys Workload Automation cannot start jobs on that computer (even if it is the same computer as the event server).

**NOTE**

You must run commands from an instance command prompt (located in the AutoSys Workload Automation program group) because it sets several environment variables required to run the commands.

**Verify the Database is Running**

You can verify that a database (event server) is running by following the procedure for your database type.

**Verify a Microsoft SQL Server Database is Running**

You can check the running status of a Microsoft SQL Server database.

**Follow these steps:**

1. Open the IS graphical query interface (located in your Microsoft SQL Server program group). If you are using Microsoft SQL 2005, you can use sqlcmd or OSQL to accomplish the same task.
2. Log in to the server using the database user and password. The default password is autosys, but you can change it using the autosys\_secure command.
3. If you cannot log on to the server using the ISQL/w interface, check the following:
  - a. Log in to a configured computer with the event server and use the Microsoft SQL Service Manager interface to verify that the service is running or to start the service if it is not running.
  - b. On the client computers, use the AutoSys Workload Automation Administrator to make sure that the parameters on the Event Server - AutoSys Workload Automation Administrator window are the same as those entered for the Microsoft SQL Server.

**NOTE**

If the problem still exists, contact your database administrator.

**Verify an Oracle Database is Running**

You can check the running status of an Oracle database.

**Follow these steps:**

1. Enter the following command at a AutoSys Workload Automation instance command prompt:

```
sqlplus User_Name/Password@TNS_Alias
```

– **User\_Name**

Defines your user name.

– **Password**

Defines your password.

– **TNS\_Alias**

Defines your TNS database name.

The following prompt is displayed:

```
sql>
```

2. If you do not get the SQL prompt in the instance command prompt window, follow these steps:
  - a. Log in to the configured computer with the event server and verify that SQL\*Plus works from there. If the event server is running and accessible on that computer, AutoSys Workload Automation was not installed properly on the client computer. In particular, one of the following parameters was probably entered incorrectly: event server name, port number, database name, or event server host.
  - b. On the client computer, use the AutoSys Workload Automation Administrator to make sure that everything was entered correctly. You can locate the parameter information on the Event Server - AutoSys Workload Automation Administrator window.

**NOTE**

If the problem still exists, contact your database administrator.

3. Enter exit.  
You exit SQL\*Plus.

### **Verify a Sybase Database is Running**

You can check the running status of a Sybase database.

**Follow these steps:**

1. Enter the following command at a AutoSys Workload Automation instance command prompt:

```
isql - Uautosys - Pautosys -SServer_Name -DDatabase_Name
```

– **-Uautosys**

Defines your user name.

– **-Pautosys**

Defines your user password.

– **-SServer\_Name**

Defines your server name.

– **-DDatabase\_Name**

Defines your database name.

The following prompt is displayed:

```
1>
```

2. If you do not get the SQL prompt, follow these steps:
  - a. Log in to a configured computer with the event server and verify that the ISQL utility works from there. If the event server is running and accessible on that computer, AutoSys Workload Automation was not installed properly on the client computer. In particular, one of the following parameters was probably entered incorrectly: event server name, port number, database name, or event server host.
  - b. Use the AutoSys Workload Automation Administrator on the client computer to make sure that all information was entered correctly. You can locate the parameter information on the Event Server - AutoSys Workload Automation Administrator window.

**NOTE**

For assistance on additional [debugging techniques](#), contact Technical Support at [Broadcom support](#).

3. Enter exit.  
You exit ISQL.

## Verify the Environment and Configuration

You must verify the connection to the event server, which is based on the setting of the environment variables (at this point, the database should be running).

To verify the connection to the event server from a AutoSys Workload Automation instance command prompt, enter the following command:

```
chk_auto_up
```

You must see a message similar to this, which indicates success:

```
CAUAJM_I_50054 Attempting (1) to Connect with Database: machine:AEDB
CAUAJM_I_50055 *** Have Connected successfully with Database: machine:AEDB. ***
```

```
CAUAJM_I_50128 Connected with Event Server: machine:AEDB
```

```
CAUAJM_I_50038 Checking CA WAAE Scheduler on Machine: machine
CAUAJM_I_50044 Primary Scheduler is RUNNING on machine: machine
```

The `chk_auto_up` command also lets you check whether the event servers and the schedulers are running.

You must run commands from the instance command prompt (located in the AutoSys Workload Automation program group) because it sets several environment variables required to run the commands. If you used this command prompt and did not get the previous message, some diagnostic messages that contain information about the problem are displayed.

## Run a Test Job

You can test your configured installation and verify your AutoSys Workload Automation environment by running a test job.

If the instance is being controlled by CA EEM, before executing any command line interface or GUI programs, first verify that AutoSys Workload Automation has been enabled using the AutoSys Secure Utility (`autosys_secure`).

Job definitions are specified using Job Information Language (JIL). The `jil` command is a language processor that parses the language and updates the database. You can also define jobs using the Job Editor, which you can open from your AutoSys Workload Automation program group.

A test job named `test_install` is included with the product. Its job definition is in the file named `%AUTOSYS%\test\jil\test_install`. Use the `test_install` job as a template to verify the installation.

## Specify a Computer Name in the test\_install Job

If you intend to use the test job provided with the product, you must modify the `test_install` job to specify your computer's actual name.

### Follow these steps:

1. Log in as the user whose user ID and password you entered following the steps in [Adding the Superusers and the Windows User IDs and Passwords](#).

**NOTE**

The user who defines the job and is the owner of the job must be a valid user on the computer on which you want to run the job.

2. Edit the test\_install script by using the following command:

```
notepad %AUTOSYS%\test\jil\test_install
```

3. Replace NT\_HOSTNAME in the computer: NT\_HOSTNAME line with the host name of the Windows computer you are currently logged on to. For example, if the host name is called sales, you must enter the following:

```
insert_machine: sales
```

**NOTE**

The NT\_HOSTNAME computer must be defined using jil, otherwise the job is not inserted.

4. Save the test\_install script and enter the following command at the instance command prompt:

```
jil < %AUTOSYS%\test\jil\test_install
```

The machine (sales) is inserted into the database. The output is similar to the following:

```
Insert/Updating Machine: "sales"
```

```
Database Change was Successful!
```

5. Delete the following line from the %AUTOSYS%\test\jil\test\_install script, and save and close the file:

```
insert_machine: sales
```

**Add the Test Job to the Database**

After you have modified the test\_install job, you must insert it into the database.

To insert the test\_install job into the database, enter the following command:

```
jil < %AUTOSYS%\test\jil\test_install
```

The following message appears:

```
Insert/Updating Job: test_install
```

```
Database Change WAS Successful!
```

**NOTE**

If c:\tmp directory does not exist, create it so that the job can write the files to this directory.

**Run the Test Job**

To send an event to start the test\_install job, enter the following command at the instance command prompt:

```
sendevent -E STARTJOB -J test_install
```

The event to start the job is now in the database, but the job itself does not start until the scheduler is up and running.

**Verify the Test Job**

To verify that the job started and ran successfully, monitor the scheduler output log with the following command:

```
autosyslog -e
```

If the job ran successfully, the following message is written to the /tmp/test\_install.out file:

```
AUTOSYS install test
```

This indicates that the basic AutoSys Workload Automation environment is set up properly.

If the job did not run successfully, you should see an error message indicating the problem in the /tmp/test\_install.err file.

If you see an error message in the scheduler output log that indicates it was unable to log you on as user@domain, you either entered an incorrect password or entered a password for a user other than the job owner. You can identify the job owner by running the following command at the instance command prompt:

```
autorep -J test_install -q
```

You can verify that you entered this specific user ID and password with the autosys\_secure command for the job owner. If it is necessary, you can use the autosys\_secure command to enter the correct user logon ID and password.

## Install Agent

The following topics describe how to install the agent, and verify the installation.

### Install Scenarios and Considerations

Before you install the agent, review the following sections:

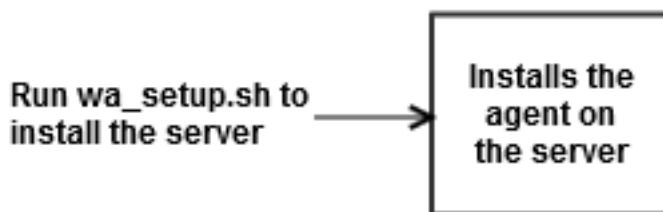
#### NOTE

- For more information about installing the agent on i5/OS, agent limitations, and agent known issues, see [Workload Automation Agents](#).
- For more information about installing the agent on z/OS, see [Workload Automation Agent for z/OS](#).

#### (UNIX) Installation Scenarios

You must install the agent on every computer that you use to run jobs.

When you install the server, the agent is installed by default on the server computer.



Depending on your requirements, you can install additional agents on other computers using the Add additional agent option in AutoSys maintenance mode.

- Running install.bin on a physically accessible computer or distributed network lets you install the agent on any supported computer.

#### NOTE

- If you are using the Workload Automation Agent DVD to install the agent, we recommend that you select to install the agent in the AutoSys Workload Automation compatibility mode as the installer configures the agent to work with AutoSys Workload Automation. If you do not select to install the agent in the AutoSys Workload Automation compatibility mode, take the following actions as the installer does not configure the agent to use all the available features of AutoSys Workload Automation:
  - a. Back up the agentparm.txt file.
  - b. Uninstall the agent.
  - c. Use the AutoSys Workload Automation ISO to install the agent.

- d. Compare the backed up agentparm.txt file with the new agentparm.txt file and merge the appropriate parameters into the new agentparm.txt file.
- If you are installing the agent for the first time, use the AutoSys Workload Automation ISO. If you use this method, the agent is automatically configured to work with AutoSys Workload Automation.

### **(Windows) Installation Scenarios**

You must install the agent on every computer that you use to run jobs.

You can install the agent in the following ways:

- When you install the server, the agent is installed by default on the server computer.
- When you install the client, the agent is installed by default on the client computer.
- Depending on your requirements, you can install additional agents on other computers as follows:
  - By selecting the Additional Workload Automation Agent option on the AutoSys Workload Automation Product Explorer window. This option lets you install the agent on remote computers or to install multiple agents on a single computer.
  - By selecting the Register packages to Unicenter SD for Windows option on the AutoSys Workload Automation Product Explorer window. This option lets you register the agent with CA SDM on your computer so that you can use CA SDM to install additional agents.

#### **NOTE**

The itemproc.dat procedures file located at SDO\agent\reginfo is used to register procedures in CA SDM. Currently, this procedures file registers AutoSys Workload Automation in CA SDM. So, to install only the additional agents, you must modify the itemproc.dat procedures file to call the command as Setup.exe /s /f1"agent.iss". For example, you can modify the following keywords in the itemproc.dat procedures file as follows:

```
Parameters = "/s /f1\"agent.iss\""
```

```
ProcToRun = setup.exe
```

#### **NOTE**

- If you are using the Workload Automation Agent DVD to install the agent, we recommend that you select to install the agent in the AutoSys Workload Automation compatibility mode as the installer configures the agent to work with AutoSys Workload Automation. If you do not select to install the agent in the AutoSys Workload Automation compatibility mode, take the following actions as the installer does not configure the agent to use all the available features of AutoSys Workload Automation:
  - a. Back up the agentparm.txt file.
  - b. Uninstall the agent.
  - c. Use the AutoSys Workload Automation ISO to install the Workload Automation Agent.
  - d. Compare the backed up agentparm.txt file with the new agentparm.txt file and merge the appropriate parameters into the new agentparm.txt file.
- If you are installing the agent for the first time, use the AutoSys Workload Automation ISO. If you use this method, the agent is automatically configured to work with AutoSys Workload Automation.

### **(UNIX) User Account Considerations**

We recommend that you use the root account to install and start the agent on UNIX. Using the root account lets you run jobs under different user accounts.

If you start the agent with an account other than root, you cannot run jobs under different user accounts. The agent cannot switch users. If you plan to run the agent under a specific user account instead of root, consider the following notes:



- Verify that the user account has the permissions to access the required directories and run the commands and scripts on the agent computer.
- You can run the agent under the user account when the agent is installed under root. However, you can only run jobs that belong to the user account. We recommend that you install the agent using the specific user account to avoid permission problems.

## Install the Agent

Use the AutoSys installation wizard to install the agent.

### NOTE

Click **Next** to proceed with the installation.

### Follow these steps:

1. Double-click the install.exe file to open the AutoSys Workload Automation installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In Installation Options, select **Install the product**.
4. Click **Custom** or **Express** based on your requirements.

### NOTE

For the purposes of these instructions, we are proceeding in Custom mode.

5. Select the [Java Virtual Machine](#).
6. In the Components to install window, select Agent and deselect the rest. By default, all the dependent components are selected automatically.
7. In the installation path, proceed with the default installation paths or select your desired installation paths. For more information, refer to the [Installation Path](#).
8. In the Agent properties window, enter the required details. For more information, see [Agent Properties](#) and [SNMP Information](#).
9. In the [Agent And Application Server Port](#) window, enter the comma-separated port numbers for the agent to communicate with the application server. By default, the value is set to 9000.
10. In the [Agent Level Data Encryption](#) window, select **Encrypt data using a user-specified key** if required. However, for the purposes of these instructions, we are not selecting it.
11. In the Services Startup Options window, select the desired services. For more information, see [Services Startup Options](#).
12. The Pre-Installation Summary window displays a summary of what is going to be installed on your machine.
13. Click **Install**.

## (UNIX) Install the Agent Using wa\_setup.sh

You can install the agent using the wa\_setup.sh script from the AutoSys Workload Automation ISO. The agent installation must be performed on every computer that you use to run jobs.

### NOTE

- You can install the agent using the AutoSys Workload Automation ISO by downloading the ISO file by logging in to *Download Center, Products* in the Broadcom Support Online website ([Broadcom support](#)).
- If you have installed the AutoSys Workload Automation server or client on this computer, the agent is already installed.
- We recommend that you use wa\_setup.sh to install the agent only if the computer where you want to install the agent is physically accessible. The wa\_setup.sh script is large in size and is not as efficient for distributing across the network.

### Follow these steps:

**NOTE**

Click **Next** to proceed through the installation steps described below.

1. Log in as root.
2. Mount the AutoSys Workload Automation media.
3. Change directories to the mounted file system and run the following command:

```
./wa_setup.sh
```

The Welcome page appears.

4. On the License Agreement page, read the license text, and select **I Accept the Terms of the License Agreement**.
5. On the Installation Option page, select **New**.
6. On the Installation Type page, select **Typical** or **Custom** based on the type of installation you want to perform. If you select Typical mode, the Installation Definition page appears. If you select Custom mode, the Components page appears.

**NOTE**

For the purposes of these instructions, we are proceeding in Custom mode.

7. On the Components page, select **Agent**.
8. Continue with the installation by entering the required information on each wizard page.
9. After you complete the last data entry page of the wizard and click **Next**, the Review Settings page appears, listing the information you entered. Review the information and, if it is correct, click **Install**.

**NOTE**

If you need to change an entry, click **Back** as many times as necessary to locate that entry, make the appropriate change, and proceed as before. Verify the change on the Review Settings page.

10. When the installation completes, the Installation Complete page appears. Click **Finish**.

The agent installation is complete.

## Define the Agent on AutoSys Workload Automation

To enable communication between AutoSys Workload Automation and an agent that you install, define the agent on AutoSys Workload Automation.

**NOTE**

When you install more agents on computers that are not server or client computers, define these agents on the computer where the AutoSys Workload Automation server or client is installed. Ensure that the parameters you specify when you define an agent on AutoSys Workload Automation match the corresponding parameters in the agentparm.txt file.

### Follow these steps:

1. Perform the following action:
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter **jil** at the Windows instance command prompt or the UNIX operating system prompt.  
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Enter the following commands:

```
insert_machine: machine_name
type: a
node_name: address
agent_name: agent_name
port: port_number
encryption_type: NONE | DEFAULT | AES
```

`key_to_agent: key`

– **machine\_name**

Defines a unique name for the agent. When defining jobs, specify this name in the machine attribute.

– **a**

Specifies that the machine is a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS.

– **address**

(Optional) Defines the IP address or DNS name of the computer where the agent is installed.

**Default:** The value specified in the `insert_machine: machine_name` command.

**NOTE**

If you do not specify the `node_name` attribute, `insert_machine: machine_name` (the default) must be the DNS name of the agent machine. Otherwise, AutoSys Workload Automation cannot connect to the agent.

– **agent\_name**

(Optional) Specifies the name of an agent.

**Default:** `WA_AGENT`

**NOTE**

- This name must match the `agentname` parameter specified in the `agentparm.txt` file.
- You can specify the alias name for the agent in the `agent_name` parameter to configure the alias on AutoSys Workload Automation. For more information about creating an alias for the agent plug in, see the appropriate agent plug-in implementation details in the [Workload Automation Agents](#) documentation.

– **port\_number**

(Optional) Specifies the port that the agent uses to listen for traffic.

**Default:** 7520

**NOTE**

This port number must match the `communication.inputport` parameter in the `agentparm.txt` file.

– **NONE | DEFAULT | AES**

(Optional) Specifies the type of encryption to be used by the agent. You can set the encryption type to one of the following:

- **NONE**  
Specifies that the agent uses no encryption.
- **DEFAULT**  
Specifies that the agent uses the default encryption key and type. This is the default.
- **AES**  
Specifies that the agent uses AES 128-bit encryption.

**NOTE**

Specify a key using the `key_to_agent` attribute.

– **key**

(Optional) Specifies the key that is used to encrypt data from AutoSys Workload Automation to the agent. This value must match the `security.cryptkey` parameter in the `agentparm.txt` file of the agent, without the prefix 0x. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. Specify one of the following:

- A 32-digit hexadecimal key
- A passphrase with up to 16 characters

4. (Optional) Specify optional machine attributes:

- character\_code
- description
- opsys
- max\_load
- factor
- heartbeat\_attempts
- heartbeat\_freq

5. Enter exit.

The data is loaded into the database. The agent is defined on AutoSys Workload Automation.

**NOTE**

For more information about the parameters in the agentparm.txt file, see [Workload Automation Agents](#).

**Example: Define an Agent Using the Default Values**

This example defines an agent using the default values for the following attributes: type: a, node\_name: sysagt, port: 7520, and agent\_name: WA\_AGENT.

```
insert_machine: sysagt
```

**Example: Define an Agent**

This example defines a machine that is named eagle where the agent WA\_AGENT runs on the node myagenthostname and uses 49154 as its main input port.

```
insert_machine: eagle
type: a
agent_name: WA_AGENT
node_name: myagenthostname
port: 49154
max_load: 100
factor: 1.0
```

## Verify Agent Installation

You can verify the agent was installed successfully and that the agent can communicate with AutoSys Workload Automation by defining, running, and monitoring a test job.

To verify the agent installation, follow these steps:

### (UNIX) Set Up the Environment

Before verifying the agent installation, you must set up your environment as follows:

- If you are running the C shell (csh), enter the following command:  

```
source /opt/CA/WorkloadAutomationAE/autosys.csh
```
- If you are running the Bourne shell (sh), enter the following command:  

```
./opt/CA/WorkloadAutomationAE/autosys.sh
```
- If you are running the Korn shell (ksh), enter the following command:  

```
./opt/CA/WorkloadAutomationAE/autosys.ksh
```
- If you are running the Bash shell (bash), enter the following command:  

```
./opt/CA/WorkloadAutomationAE/autosys.bash
```

## Test Communication Between AutoSys Workload Automation and the Agent

You can verify that AutoSys Workload Automation communicates with the agent by issuing the autoping command to ping the server computer.

### Follow these steps:

1. Perform the following action:
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (instance\_name).
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the Windows instance command prompt or the UNIX operating system prompt:
 

```
autoping -m machine_name
```

#### – **machine\_name**

Specifies the name of the machine where the agent runs.

The following messages appear, which indicates that autoping was successful:

```
CAUAJM_I_50023 AutoPinging Machine [machine_name]
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL!
```

### NOTE

- If you do not get this message, the agent is not configured properly and, as a result, AutoSys Workload Automation cannot start jobs on that computer (even if it is the same computer as the server).
- The agent on z/OS does not support connectivity to the application server through the SDK. Therefore, if you issue the autoping command with the -S option for the agent on z/OS, the command skips the connectivity test with the application server.

## Define a Test Job

You can define a job, such as a Command job that runs a Windows or UNIX script, to test communication between AutoSys Workload Automation and the agent.

### Follow these steps:

1. Perform the following action:
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (instance\_name).
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter **jil** at the Windows instance command prompt or the UNIX operating system prompt. The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Enter the following commands:

#### – Windows:

```
insert_job: job_name
machine: machine_name
command: "C:\sleep.exe" 1
owner: user@host
```

#### – UNIX:

```
insert_job: job_name
machine: machine_name
command: "/usr/bin/sleep" 1
owner: user@host
```

A test job is defined. The following message appears:

```
CAUAJM_I_50323 Inserting/Updating job: job_name
CAUAJM_I_50205 Database Change WAS Successful!
```

4. Enter exit.  
The data is loaded into the database.

### Example: Define an i5/OS Job

This example runs the command named CALC on the i5agent computer.

```
insert_job: i5job_runcmd
job_type: I5
machine: i5agent
i5_name: CALC
```

### Example: Define an UNIX Job

This example runs the /usr/bin/sleep command on the unixagent computer.

```
insert_job: UNIXjob_runcmd
job_type: CMD
machine: unixagent
command: "/usr/bin/sleep" 1
owner: autosys@unixagent
```

### Example: Define a Windows Job

This example runs the C:\sleep.exe command on the winagent computer. The command is enclosed in quotation marks because the path contains a colon.

```
insert_job: WINjob_runcmd
job_type: CMD
machine: winagent
command: "C:\sleep.exe" 1
owner: autosys@winagent
```

### Run the Test Job

You can verify whether jobs will run on the agent using the test job.

#### Follow these steps:

1. Perform the following action:
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (instance\_name).
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the Windows instance command prompt or the UNIX operating system prompt:

```
sendevent -e STARTJOB -J job_name
```

The sendevent command sends an event to start the test job. The event to start the job is now in the database, but the job itself does not start until the scheduler is up and running.

### Monitor the Test Job

You can use the scheduler log to monitor the test job and verify that it started and ran successfully.

#### Follow these steps:

1. Perform the following action:
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (instance\_name).
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following command at the Windows instance command prompt or the UNIX operating system prompt:

```
autosyslog -e
```

The scheduler log displays the following messages, which indicates that communication between AutoSys Workload Automation and the agent is successful:

```
CAUAJM_I_40245 EVENT: CHANGE_STATUS STATUS: STARTING JOB: job_name
MACHINE: machine_name
CAUAJM_I_40245 EVENT: CHANGE_STATUS STATUS: RUNNING JOB: job_name
MACHINE: machine_name
CAUAJM_I_40245 EVENT: CHANGE_STATUS STATUS: SUCCESS JOB: job_name
MACHINE: machine_name
```

## How the Installer Configures the Agent to Work with AutoSys Workload Automation

### (UNIX) How the Installer Configures the Agent to Work with AutoSys Workload Automation

You can install the agent using the AutoSys Workload Automation ISO. If you install the agent using either of these ISOs, the installer automatically configures the agent to work with AutoSys Workload Automation.

#### NOTE

- If you are using the Workload Automation Agent ISO to install the agent, we recommend that you select to install the agent in the AutoSys Workload Automation compatibility mode as the installer configures the agent to work with AutoSys Workload Automation. If you do not select to install the agent in the AutoSys Workload Automation compatibility mode, take the following actions as the installer does not configure the agent to use all the available features of AutoSys Workload Automation:
  - a. Back up the agentparm.txt file.
  - b. Uninstall the agent.
  - c. Use the AutoSys Workload Automation ISO to install the agent.
  - d. Compare the backed up agentparm.txt file with the new agentparm.txt file and merge the appropriate parameters into the new agentparm.txt file.

If you are installing the agent for the first time, use the AutoSys Workload Automation ISO or the AutoSys Workload Automation Standalone Agent ISO.
- For more information about configuring the agent to behave like the legacy agent, see [Workload Automation Agents](#).

To configure the agent to work with AutoSys Workload Automation, the installer performs the following tasks:

- Creates the WAAE.txt file in the \$AUTOROOT/SystemAgent/*agent\_name*/profiles directory.

#### – **agent\_name**

Defines the name of the agent.

The oscomponent.environment.variable parameter in the agent's agentparm.txt is set to the location of the WAAE.txt file. The WAAE.txt file defines the environment settings for jobs started on behalf of all managers for all instances of AutoSys Workload Automation.

- Appends the CA\_FIPS1402\_ENABLE=1 parameter and its value to the WAAE.txt file name if you select the Enable FIPS 140-2 Compliance check box on the Agent Level Data Encryption page during the installation.
- Creates the *instance\_name*.txt file in the \$AUTOROOT/SystemAgent/*agent\_name*/profiles directory. The oscomponent.environment.variable\_manager\_*instance\_name*\_SCH parameter in the agent's agentparm.txt file is set to the location of the *instance\_name*.txt file. The *instance\_name*.txt file contains a set of environment variables

for each AutoSys Workload Automation instance. This file includes the path to the `auto.profile` file, which is one of the several objects that source the environment for a job.

- Creates the `cryptkey.txt` file in the `$AUTOROOT/SystemAgent/agent_name` directory. The `security.cryptkey` parameter in the agent's `agentparm.txt` file is set to the location of the `cryptkey.txt` file. The `cryptkey.txt` file stores the encryption key for the agent. The `as_config` command is used to define the encryption key and store it in the `cryptkey.txt` file.
- Sets the following parameters in the agent's `agentparm.txt` file that is located in the `$AUTOROOT/SystemAgent/agent_name` directory:
 

```
agent.resourcemon.enable=true
agent.resourcemon.threshold.disk.warning.notice=21M
agent.resourcemon.threshold.disk.warning.severe=20M
agent.resourcemon.threshold.disk.critical=10M
core.health.monitor.enable=true
oscomponent.auth.pam.svc=sshd
agent.spool.success.autocleanup=true
oscomponent.joblog=true
oscomponent.joblog.success.autocleanup=true
runnerplugin.spool.clean.enable=true
runnerplugin.spool.expire=7d
security.cryptkey=cryptkey.txt
oscomponent.lookupcommand=true
```
- Sets the following parameters in the agent's `agentparm.txt` file if you enable SNMP capabilities on the Agent Attributes panel during the installation:
 

```
management.snmp.host=manager_host
snmp.trap.listener.port=listener_port
management.snmp.agent.trapsink.host=host
management.snmp.agent.port=port
```

  - **manager\_host**  
Identifies the SNMP Manager IP address or DNS name. Your SNMP administrator can provide the host name.
  - **listener\_port**  
Specifies the agent port listening for trap information.
  - **host**  
Specifies the host name or the IP address of the SNMP listener that receives trap information. The management connector uses this host to send the trap.
  - **port**  
Specifies the agent port that receives the trap information.
- Sets the following compatibility parameters in the agent's `agentparm.txt` file if you select the Configure Agent with Legacy Remote Agent Compatibility check box on the Agent Attributes panel during the installation:
 

```
filemon.firstscan.skip=true
oscomponent.noexitcode=256
oscomponent.cmdprefix.force=true
oscomponent.noforceprofile=true
oscomponent.initialworkingdirectory=USER_CONT_NOT_AVAIL
oscomponent.profiles.src.delay=true
oscomponent.profiles.src.order.global.first=true
oscomponent.profiles.src.verify=true
```



**NOTE**

- When the installer sets these compatibility parameters, the job processing behavior of the agent closely matches the job processing behavior of the legacy agent. We recommend that you select the Configure Agent with Legacy Remote Agent Compatibility check box if you want to configure the agent to behave like the legacy agent.
- If you did not select the Configure Agent with Legacy Remote Agent Compatibility check box during the installation, but later decide to configure the agent to behave like the legacy agent, manually add or edit the compatibility parameters in the agent's agentparm.txt file.
- Stores the following information in the \$AUTOROOT/SystemAgent/agent\_name/cybspawn and \$AUTOROOT/SystemAgent/agent\_name/ESPMgr directories:  

```
EWA_SHARED=sharedcomponents_directory
export EWA_SHARED
CSAM_SOCKADAPTER=$EWA_SHARED/Csam/SockAdapter
export CSAM_SOCKADAPTER
LIBPATH_VAR_NAME=agent_install_directory
:$CSAM_SOCKADAPTER/lib:install_directory
/autosys/lib:$LIBPATH_VAR_NAME"
export LIBPATH_VAR_NAME
```

  - **sharedcomponents\_directory**  
Specifies the path to the directory where the shared components are installed.
  - **LIBPATH\_VAR\_NAME**  
Specifies the path to the directory where the LIBPATH environment variable is installed. Based on your operating environment, the LIBPATH environment variable is named as follows:
    - On Linux—LD\_LIBRARY\_PATH
    - On SunOS—LD\_LIBRARY\_PATH
    - On AIX—LIBPATH
    - On HP-UX—SHLIB\_PATH
  - **agent\_install\_directory**  
Specifies the path to the directory where the agent is installed.
  - **install\_directory**  
Specifies the path to the root directory where AutoSys Workload Automation is installed.

**NOTE**

For more information about the parameters in the agentparm.txt file, see [Workload Automation Agents](#).

## (Windows) Install Additional Agents Using CA IT Client Manager

You can also install additional agents by registering the agent with the Software Delivery Manager component of CA IT Client Manager on your computer. After the agent is registered with the Software Delivery Manager component of CA IT Client Manager, you can use the Software Delivery Manager component of CA IT Client Manager to install additional agents.

**NOTE**

- The itemproc.dat procedures file located at SDO\agent\reginfo is used to register procedures in the Software Delivery Manager component of CA IT Client Manager. Currently, this procedures file registers the agent in the Software Delivery Manager component of CA IT Client Manager. So, to install only the additional agents, you must modify the itemproc.dat procedures file to call the command as Setup.exe /s /f1"agent.iss". For example, you can modify the following keywords in the itemproc.dat procedures file as follows:

```
Parameters = "/s /f1"C:\EWA\AutoSys\agent.iss""
ProcToRun = setup.exe
```

Set the Parameters keyword value to the path where the agent.iss file is located on the computer where the Software Delivery Manager component of CA IT Client Manager is installed.

- For more information about setting up the agent to be delivered through the Software Delivery Manager component of CA IT Client Manager, see the Software Distribution Manager documentation.

#### Follow these steps:

##### NOTE

Click **Next** to proceed with the installation.

1. Log in as a user with Windows Administrators group privileges.
2. Insert the installation media into the drive and mount it.

##### NOTE

If autorun is enabled, the installation starts automatically.

3. Run setup.exe.  
The Product Explorer appears.
4. Select **Register packages to Unicenter SD for Windows**, and click **Install**.  
The Choose Products to Register page appears. By default, the Workload Automation Agent check box is selected.
5. Proceed to the Unicenter Software Delivery User Details page, and complete the fields as required.
6. Click **Complete** on the Registering Products page when the registration is complete.

The registration is complete. You can now install additional agents using the Software Delivery Manager component of CA IT Client Manager.

## Install Client

The following topics describe how to install the client, and verify the installation.

### Install Considerations

Before you install the client, review the following considerations:

- Before you install the client, ensure that the server was installed successfully on a machine and that you have completed your client installation checklist. You can use the information from your checklist during the installation.
- If you are also using the server computer as a client (running jobs and utilities on the server computer), you do not need to install the client software on the server. The client software is installed during the server installation.

### Install the Client

Use the installation wizard to install the client. The client installation must be performed on every computer that you use to run, monitor, or define jobs.

##### NOTE

You can install the client using the AutoSys Workload Automation ISO file. It can be downloaded from the Download Center.

#### Follow these steps:

##### NOTE

Click **Next** to proceed through the steps of the installation wizard.

1. Double-click the install.exe file to open the AutoSys Workload Automation installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In Installation Options, select **Install the product**.
4. Click **Custom** or **Express** based on your requirements.

**NOTE**

For the purposes of these instructions, we are proceeding in Custom mode.

5. Select the [Java Virtual Machine](#).
6. In the Components to install window, select the client and deselect the rest. By default, all the dependent components are selected.
7. In the installation path, proceed with the default installation paths or select your desired installation paths. For more information, see [Installation Path](#).
8. In the Instance information window, enter a name for the AutoSys instance that you are currently installing.
9. In the Instance-level Data Encryption window, you can select a key format for encryption. However, for the purposes of these instructions, we are not selecting any encryption. For more information, see [Instance level data encryption](#).
10. In the FIPS encryption window, you can select the **FIPS 140-2 compliance** checkbox. However, for the purposes of these instructions, we are not selecting this checkbox. For more information, see [FIPS Encryption](#).
11. In the Application Server Information window, enter the required details. For more information, see [Application Server Information](#).
12. The Pre-Installation Summary window displays a summary of what will to be installed on your machine.
13. Click **Install**.

The agent is installed on the computer.

## Verify the Client Installation

Verify the client installation after installation is completed.

### Follow these steps:

1. (UNIX) [Set up the client environment](#).
2. Verify the client.

### (UNIX) Set Up the Environment

To set up the environment, log in to the client computer as the owner of AutoSys Workload Automation (for example, autosys) and source the proper file in AUTOUSER.

**NOTE**

Because AUTOUSER is not defined yet, you must enter the full path.

In the following examples, fiji is the name of the server computer, /opt/CA/WorkloadAutomationAE represents the path to the directory where you installed AutoSys Workload Automation, and ACE is the name of the AutoSys Workload Automation instance.

If you are running the C shell (csh), enter the following command:

```
source /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.csh.fiji
```

If you are running the Bourne shell (sh), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.sh.fiji
```

If you are running the Korn shell (ksh), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.ksh.fiji
```

If you are running the Bash shell (bash), enter the following command:

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.bash.fiji
```

## Back Out Upgrade

We recommend that prior to performing an upgrade, you take a backup of the AutoSys related components file system including the Shared components. You must take a backup of the registry in case of Windows, and /etc/init.d or /etc/rc.c/init.d files in the case of Unix.

The following procedure describes how to restore/rollback an Autosys Workload Automation 12.0 installation upgrade failure. If the upgrade fails during an installation, it is possible to manually revert the Autosys upgrade to the original release. For this, you must have a backup of the file system and the database.

### NOTE

For more information, see the [Back Out Upgrade Troubleshooting](#) section.

### Follow these steps:

1. Log in to either a UNIX or Windows system.
2. If the AUTOSYS\_INSTALL\_LOCATION variable is set then unset it.
3. There will be the original folders.
4. Rename the original folder \_FAILURE.
5. Rename the backup folders to the original name.
6. Replace the older database that was backed up.
7. Run the upgrade again.

## Install Web UI

This section describes installation and configuration tasks you must perform to enable Web UI to function in your enterprise. It is intended for system administrators.

The AutoSys installer upgrades all components installed on the system. Therefore, if Web UI is installed along with AutoSys, both will be upgraded. You do not need to run the upgrade twice.

### NOTE

- Remember to install all operating system patches before installing any of the product components.
- For AutoSys Workload Automation upgrades or install, older versions of Workload Automation Agent for UNIX, Linux, or Windows are supported.
- An agent is configured with the FTP plug-in and as an FTP server to enable transfers of large amounts of data to Web UI applications. For more information see the *Workload Automation Agent for UNIX, Linux, or Windows Implementation documentation*.
- For patch information, see [Release Notes](#). The patches are available from [Broadcom support](#).
- The AutoSys installer lets you create the database and database users. For more information, see [Web UI Database Type](#) and the sections under it.
- REST web server, which is installed by default when you install AutoSys Workload Automation, is required to enable Enterprise Command Line and Forecast functionality with Web UI.
- You can install one or more components on the same system in a single step.
- CA EEM, a common component, is required for Web UI and must be installed before you install Web UI. We recommend that you install CA EEM on a dedicated server.
- BSAFE Crypto J is replaced with Bouncy Castle.  
Before you install Web UI, you must modify the CA EEM server certificate.

For more information about modifying the CA EEM server certificate, see the Web UI Server, Software Requirements in the [System Requirements](#) section.

- Install and configure Apache HTTPD Server to manage load balancing in a High Availability environment.
- CA EEM must be running when you install Web UI so that registration can occur; otherwise, you cannot install Web UI.
- Before you begin the installation, stop performance-monitoring, antivirus, and antispyware software.
- If Web UI is already installed on the server, you cannot reinstall Web UI. Uninstall Web UI first, then install Web UI.
- If the database you specify for the Web UI database is used with another Web UI instance, the option to install Web UI in High Availability mode appears.
- Verify that the Web UI servers have started before you use or perform operations with Web UI. You can display Web UI server startup status by running the following command:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

This section contains the following topics:

## Web UI Install Considerations

### WARNING

If you are running performance monitoring, antivirus, or antispyware software, you may experience a problem completing the installation. If you receive a message indicating that a .DLL is in use or that a file is locked, stop the performance monitoring, antivirus, or antispyware software, and select the option to retry the installation.

### General Installation Considerations

Note the following general installation considerations:

- Review the "System Requirements" section of the *Release Notes* before beginning the installation.
- When installing Web UI on Windows servers that are Terminal Services server machines running in Application mode, you must install and uninstall using the Windows Control Panel Add/Remove Programs feature.
- Web UI cannot be installed from a network drive that is not mapped to a letter drive (for example, "X"). To install from a network drive, map the shared network drive (for example, map \\server\drive to X:\) and then install from that drive. Note that using a mapped drive instead of local media may extend the amount of time required for the installation.
- If an instance of CA EEM resides on a server other than the Web UI server, Web UI can reference that instance if it was installed using the current Common Components DVD.
- If you do not want to reference an existing instance of CA EEM and you have more than one Broadcom product in your environment, we recommend that you install CA EEM on a dedicated server, and that you configure all Broadcom products to reference the same instance of CA EEM.
- For CA EEM to work with key lengths of 2048 (or more), follow these steps:
  - a. Stop the AE and Web UI servers.
  - b. Upgrade CA EEM.
  - c. Configure the CA EEM server to work with 2048 key length certificates.

### NOTE

For more information about different key lengths, see [Configure CA EEM for Root Certificates Generated with Different Key Lengths](#).

- d. Restart the AE and Web UI servers.
- e. Web UI is now ready for installation or upgrade.

## **Web UI Installation Considerations**

The following are important considerations when you install Web UI:

- An instance of CA EEM must be installed and accessible; otherwise, you cannot install Web UI.
- The database schema should be set to case-sensitive prior to installation.
- The installation lets you do the following:
  - Specify the installation directory path. A default path is provided. The forbidden character in the path is: "%".
  - Specify the CA EEM server host name and the password for the user EiamAdmin.
  - Specify the ports for the Tomcat application servers that Web UI installs.
  - Specify the Event Bus port. The Event Bus routes event information to the various applications in Web UI.
  - Specify the Web UI database properties. If you select Internal, a database is installed. Alternatively, you can select an existing supported database.
  - Specify the Reporting database properties. You can select the Web UI database or an existing supported database.

The following are considerations for installing Web UI in a High Availability group:

- To install Web UI in High Availability mode, specify an external database for the Web UI DB that another Web UI instance is using. You can change the SSL mode using the Configuration Batch Interface. For more information, see *Configuration Batch Interface Help*.
- The internal database included with Web UI is not compatible with High Availability.
- When installing Web UI in High Availability mode, all Web UI instances in the group must be of the same version. You cannot continue the installation if a Web UI instance of a different version is detected.
- Only supported configuration for an HA cluster is to have all the nodes in either HTTP or HTTPS.
- If all HA nodes in a cluster are in the same mode, it is not possible to modify the protocol to HTTP or HTTPS.
- You can continue installing HTTP or HTTPS in mixed mode. In this case, the list of HA nodes and the protocol value are displayed. It is only possible to change the protocol using the batch interface.

### **NOTE**

- Default port values are provided. If the value specified for a port is in use or is not in the defined range, you are prompted to specify a different value.
- The password for the user EiamAdmin is the same password entered during the installation of CA EEM, a common component.
- If your environment includes AutoSys Workload Automation servers that are registered with CA EEM and CA EEM is not configured for failover, specify the same CA EEM instance when you install Web UI to enable pass-through security. If CA EEM is configured for failover, only the primary server is required. The other servers are automatically retrieved.

The Web UI installer validates the encrypted link to CA EEM. Random numbers provided by the operating system are used to create an encrypted link. On Linux, random number generation is affected by the entropy value, which in turn is affected by random events. On virtual machines, the entropy value can be very low (below 200) which causes random numbers to be generated slowly. As a result, it can take as long as 30 minutes for Web UI to generate an encrypted connection with CA EEM. The entropy value can be found in the following location: `cat /proc/sys/kernel/random/entropy_avail`. One way to increase the entropy value when needed is to use `rng-tools`.

## **Common Components Installation Considerations**

The following are important considerations when you install the common components:

- On Linux, verify that the appropriate RPM package and operating system patches for your Linux platform have been installed. For more information, see the "Operating System and Database Support" article of the *Web UI Release Notes*.
- If an instance of CA EEM resides on a server other than the Web UI server, Web UI can reference that instance.
- If you do not want to reference an existing instance of CA EEM and you have more than one Broadcom product in your environment, we recommend that you install CA EEM on a dedicated server, and that you configure all Broadcom products to reference the same instance of CA EEM.
- For CA EEM to work with key lengths of 1024 and 2048 (or more), follow these steps:
  - a. Stop the AutoSys and WCC servers.
  - b. Upgrade CA EEM.
  - c. Configure the CA EEM server to work with both 1024 and 2048 key length certificates.

#### NOTE

For more information about different key lengths, see the CA EEM documentation section [Configure CA EEM for Root Certificates Generated with Different Key Lengths](#).

- d. Restart the AutoSys and WCC servers.
- e. Web UI is now ready for installation or upgrade.

#### NOTE

For information about installing or upgrading common components, see the CA Common Components Implementation *documentation*.

## Web UI Database and Dependencies

This section describes the dependencies of the Web UI and its components.

### Web UI Dependencies

The Web UI environment has the following dependencies:

- From Web UI Version 12.0, ECLI and Forecast tabs require the REST web server to be configured for the AutoSys servers.
- Install the Scheduler, Client and REST Webserver on your computer, using the AutoSys Workload Automation ISO. Configure the REST Webserver on Web UI to enable Web UI Enterprise Command Line and Forecast functionality.

### Web UI Component Dependencies

The CA Workload Automation AE SDK (SDK) is installed on the Web UI server when you install Web UI. The SDK enables Web UI to communicate with the CA Workload Automation AE database. The SDK installer also installs CA Secure Socket Adapter (SSA), which lets CA Technologies components use a single communications multiplexing port to ease firewall administration and minimize conflicts with other applications.

### Required Databases

The Configuration, Forecast, and Monitoring applications require a database, which is known as the WCC database. The Reporting application also requires a database, which is known as the Reporting database. An internal Derby database is provided that can be used by all Web UI applications. Otherwise, you can specify an existing supported database to which Web UI has access. If you specify an existing database, the Derby database is not installed.

#### NOTE

Using the Internal Derby database may not be suitable for large workload operations. Performance issues may occur. Broadcom recommends using a commercial-grade database to handle high volume transactions.

The AutoSys Workload Automation installer creates the Web UI database and schema similar to the AE installers of previous releases. Web UI users do not need to manually create the databases and tablespaces. The installer uses the database admin or schema owner credentials to create or update the database and schema. Web UI users also do need to create or modify the tables, and index privileges. However, they will have access and privileges to modify the data as required.

## Web UI Deployment Scenarios

The installation for Web UI is modular, providing maximum flexibility. When planning your workload automation environment, there are multiple factors to consider in determining the appropriate server architecture. The number of servers required depends on which features you want to install, the amount of data you foresee being stored in the applicable databases, and the number of security policies you plan to implement.

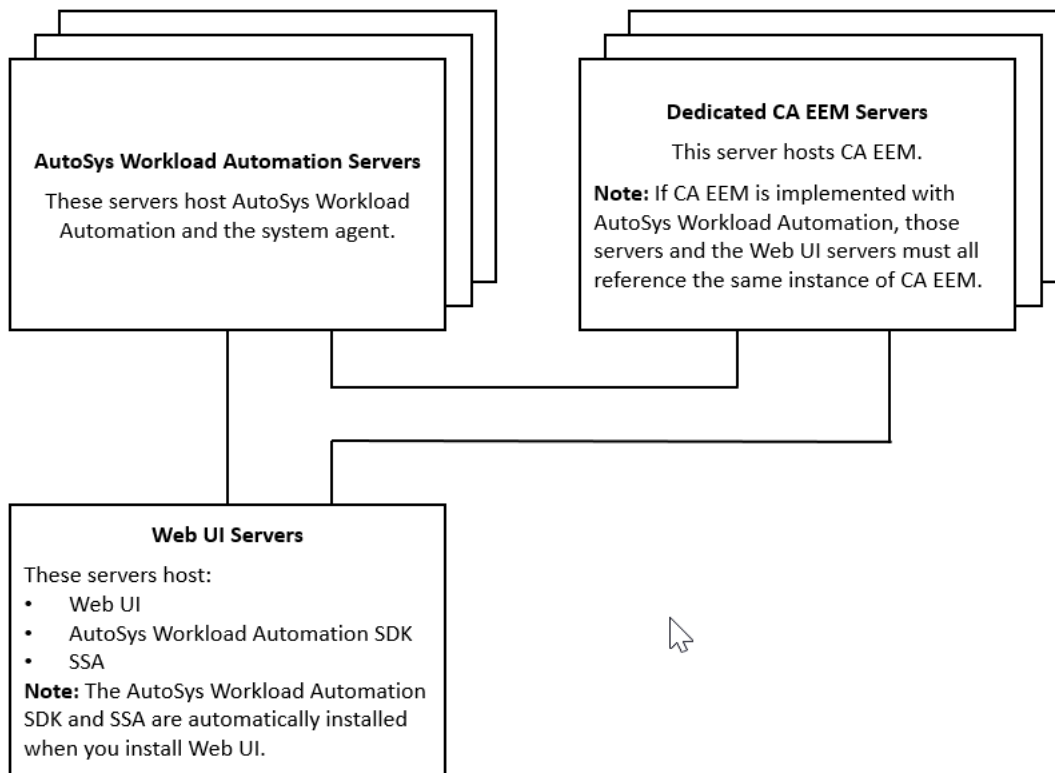
If the resources are available, set up your workload automation environment with the following dedicated servers:

- Web UI server: This server hosts Web UI and the AutoSys Workload Automation SDK (including SSA). Additionally, one Web UI server in a High Availability group can host Apache HTTPD Server.
- AutoSys Workload Automation server: This server hosts AutoSys Workload Automation and the system agent. This requires a REST web server to enable Web UI Enterprise Command Line and Forecast functionality.
- (Optional) Server for Apache HTTPD Server: This server hosts Apache HTTPD Server for a High Availability group, when it is not installed on a Web UI server.

### **Deployment Example Using Dedicated Servers**

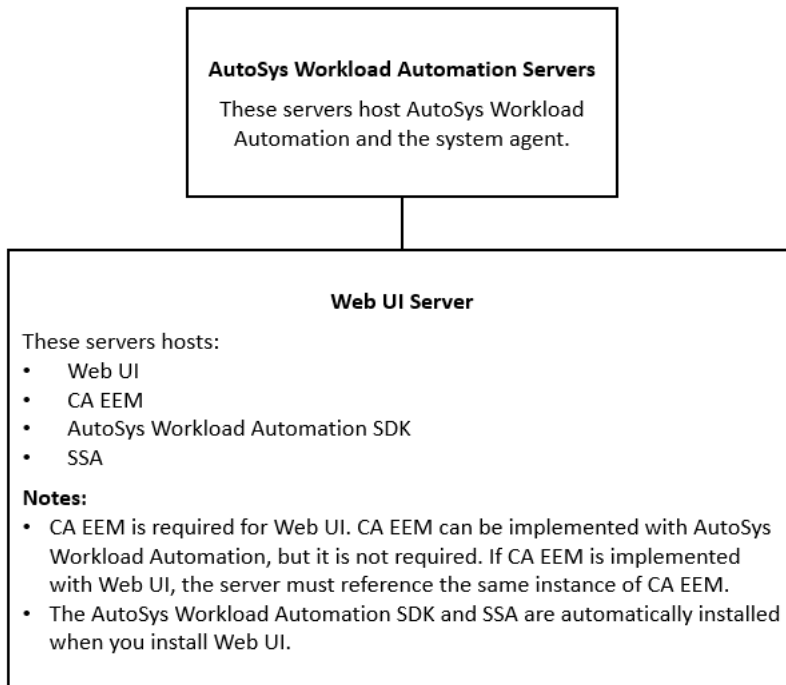
For a large-scale enterprise with multiple scheduling manager servers and extensive workload management requirements, we recommend using dedicated servers for each function. The following diagram shows the recommended deployment of Web UI, the common components, and the supported scheduling managers in an environment with four or more servers available:



**Figure 8: Typical Deployment**

### **Deployment Example Using Two Servers**

For a small-scale enterprise with light workload management requirements, deployment can be implemented using two servers. The following diagram shows the recommended deployment of Web UI, the common components, and the supported scheduling managers in an environment with only two servers available:

**Figure 9: Typical Deployment Using Two Servers**

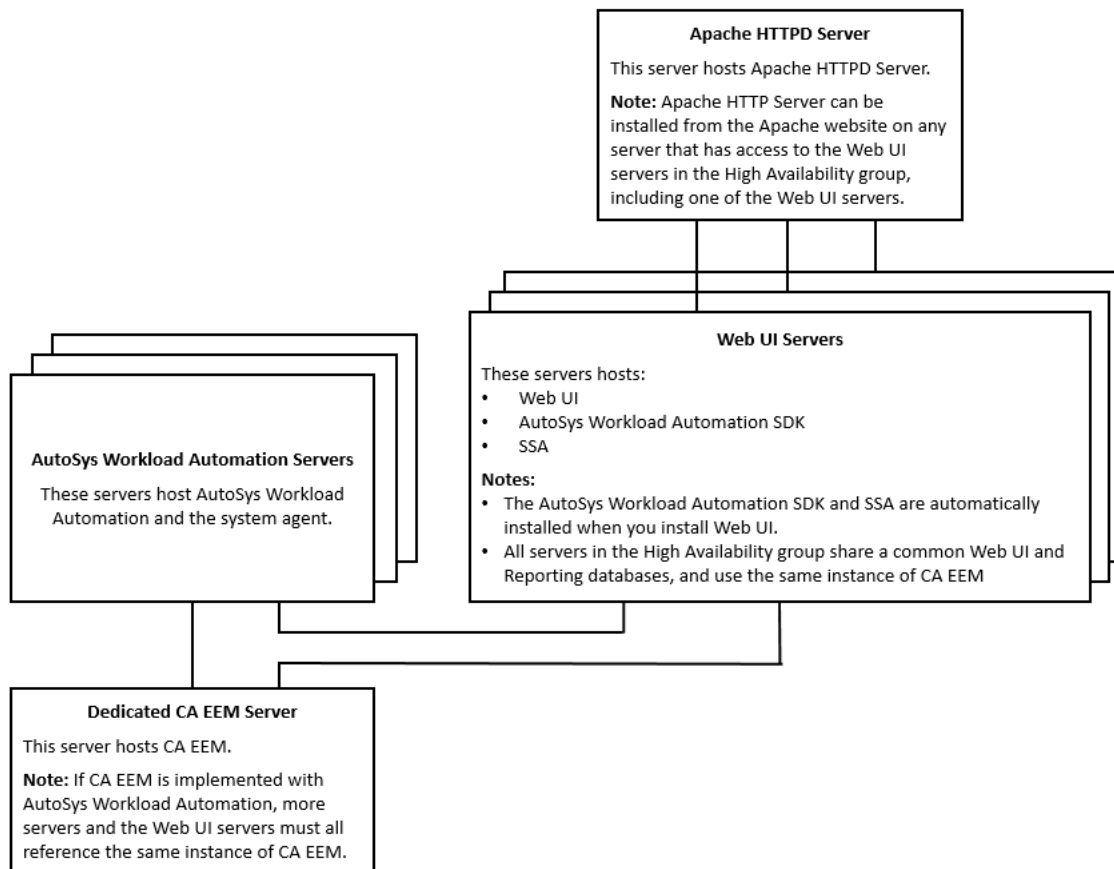
### **High Availability**

Web UI can be installed in High Availability mode. High Availability includes load balancing, which is implemented using Apache HTTPD Server. Apache HTTPD Server can be installed on any server that has access to the Web UI servers in a High Availability group, including a server on which Web UI is installed.

Web UI instances in a High Availability group share the Web UI and Reporting databases, and use a common instance of CA EEM. Web UI configurations are stored in the common database, and changes to the configurations are available to all Web UI instances in the High Availability group.

### **Deployment Example with High Availability**

The following diagram shows the deployment of Web UI in a High Availability environment. Users access Web UI using the URL for Apache HTTPD Server.

**Figure 10: Typical Deployment in an HA group**

## Install Web UI

This overview lists the steps required to install the Web UI.

1. Double-click the install.exe file to open the AutoSys Workload Automation installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In [Install Options](#), select **Install the product**.
4. Click **Custom** or **Express** based on your requirements.

### NOTE

For the purposes of these instructions, we are proceeding in Custom mode.

5. Select the [Java Virtual Machine](#).
6. In [Components to Install](#), select **Workload Control Center (Web UI)**. The dependent components get selected automatically.
7. In the installation path, proceed with the default installation paths or select your desired installation paths. For more information, see [Installation Path](#).
8. In the FIPS encryption window, you can select the **FIPS 140-2 compliance** checkbox. However, for the purposes of these instructions, we are not selecting this checkbox. For more information, refer to [FIPS Encryption](#).
9. In [Web UI Database Type](#), select the desired Database Type and appropriate options. For this procedure, we are using the MS SQL server.
10. In [Web UI Database Properties](#), enter the required details.
11. In [Web UI Admin Details](#), enter the required details.
12. In [Web UI Database User Details](#), enter the required details.

13. In [CA EEM Security for Web UI](#), enter the required details.
14. In [Web UI Port Configuration](#), enter the required details.
15. In [Services Startup Options](#), select the desired options.
16. The [Pre-Installation Summary](#) window displays a summary of what will be installed on your machine.
17. Click **Install**. The installation process begins.
18. The [Install Complete](#) window displays a summary of what has been installed. Click **Done**.

## Install Web UI Miscellaneous Topics

### Install and Configure Apache HTTPD Server for Load Balancing

Apache HTTPD Server is required for load balancing of the Web UI servers in a High Availability group. After installation, configure Apache HTTPD Server for load balancing.

#### Follow these steps:

1. Download and install the latest release of Apache HTTPD Server to the load balancing server from the following URL:  
<http://httpd.apache.org>
2. Open the httpd.conf file in a text editor.
3. Delete the pound (#) character appearing at the beginning of the following lines:
  - (only 2.4.x) #LoadModule lbmethod\_byrequests\_module modules/mod\_lbmethod\_byrequests.so
  - #LoadModule proxy\_module modules/mod\_proxy.so
  - #LoadModule proxy\_balancer\_module modules/mod\_proxy\_balancer.so
  - #LoadModule proxy\_http\_module modules/mod\_proxy\_http.so
  - (only 2.4.x) #LoadModule slotmem\_shm\_module modules/mod\_slotmem\_shm.so
  - #LoadModule status\_module modules/mod\_status.so

#### NOTE

- On some platforms those lines do not contain the (#) character and therefore should not be modified.
- Tested with version 2.4.10

The modules that are related to the load balancer are enabled.

4. Go to the following URL: [http://CA\\_WCC\\_instance:port/wcc/rest/configuration/httpd-config](http://CA_WCC_instance:port/wcc/rest/configuration/httpd-config)
  - **CA\_WCC\_instance**  
Specifies the Web UI instance. You can specify any Web UI instance in the High Availability group.
  - **port**  
Specifies the Application Port of the Web UI instance.

You are redirected to a Web UI login page.
5. Log in as a user with access to Web UI Configuration.  
Text that is related to the load balancer appears in the browser.

#### Example:

```
<Location /balancer-manager>

 SetHandler balancer-manager

</Location>

ProxyPass /balancer-manager !
```

# failonstatus status is available with Apache HTTP Server 2.2.17 and later. Remove the parameter if necessary.

```
failonstatus=503

ProxyPass /wcc balancer://wccbalancer stickysession=WCC-ASID

ProxyPassReverse /wcc balancer://wccbalancer

<Proxy balancer://wccbalancer>

BalanceMember http://
 CA_WCC_instance_A:
 port/wcc route=
 CA_WCC_instance_A

BalanceMember http://
 CA_WCC_instance_B:
 port/wcc route=
 CA_WCC_instance_B

</Proxy>
```

#### NOTE

In the example, *CA\_WCC\_instance\_A* and *CA\_WCC\_instance\_B* represent two Web UI instances. In your environment, the actual names of the Web UI instances are displayed, and lines for one or more instances can be displayed.

6. Copy the load balancer text that appears in the browser and paste it in the httpd.conf file after the line that contains the last instance of `</IfModule>`. For example:

```
#Various default settings

#Include conf/extra/httpd-default.conf

#Secure (SSL/TLS) connections

#Include conf/extra/httpd-ssl.conf

<IfModule ssl_module>

SSLRandomSeed startup builtin

SSLRandomSeed connect builtin

</IfModule>
```

```
<Location /balancer-manager>

.

.

</Proxy>
```

7. Save and close the httpd.conf file.

8. Restart Apache HTTPD Server.

#### NOTE

- Users must access Web UI from the following URL: `http://apache_httpd_server/wcc`  
**apache\_httpd\_server**  
 Specifies the name of the server on which Apache HTTPD Server is installed.  
 If Apache HTTPD Server is configured for SSL, specify https in the URL.
- You can test the load balancer setup and view the statuses of the Web UI instances from the following URL:  
`http://apache_httpd_server/balancer-manager`

## Verify the Installation with Secure Socket Layer Not Enabled

After you install Web UI, you can test the installation by starting a Web UI session. Use this login URL if you selected the Do not enable SSL option during the installation of Web UI.

#### Follow these steps:

1. Open a browser window and navigate to the following URL: `http://host_name:tomcat_port/wcc`
  - **host\_name**  
Specifies the host name for the Web UI server.
  - **tomcat\_port**  
Specifies the Tomcat non-SSL port.

The login page appears.

#### NOTE

- The default Tomcat non-SSL port is 8080. If you specified a different port during installation, use that port number in the URL.
  - If you specified the list of servers in a CA EEM failover setup when you installed Web UI, display of the login page can be delayed if the first server in the list is not available.
2. Enter the appropriate credentials and click Log In.  
 The CA Workload Automation window opens to the default page for your user ID, displaying the tabs and applications of Web UI that you are authorized to access.

#### NOTE

If a database connection error message appears when you log in, verify that the Web UI database is running, log out, and then log back in to Web UI.

## Verify the Installation with Secure Socket Layer Enabled

After you install Web UI, you can test the installation by starting a Web UI session. Use this login URL if you selected the Enable SSL option during the installation of Web UI.

**Follow these steps:**

1. Open a browser window and navigate to the following URL: `https://host_name:tomcat_ssl_port/wcc`

- **host\_name**  
Specifies the host name for the Web UI server.
- **tomcat\_ssl\_port**  
Specifies the Tomcat SSL port.

The login page appears.

**NOTE**

- The default Tomcat SSL port is 8443. If you specified a different port during installation, use that port number in the URL.
- If you specified the list of servers in a CA EEM failover setup when you installed Web UI, display of the login page can be delayed if the first server in the list is not available.

2. Enter the appropriate credentials and click Log In.

The CA Workload Automation window opens to the default page for your user ID, displaying the tabs and applications of Web UI that you are authorized to access.

**NOTE**

If a database connection error message appears when you log in, verify that the Web UI database is running, log out, and then log back in to Web UI.

## Verify the Installation In a High Availability Environment

After you install Web UI, you can test the installation by starting a Web UI session. Use this login URL if you installed Web UI in a High Availability environment and have installed Apache HTTPD Server and configured it for load balancing.

**Follow these steps:**

1. Open a browser window and navigate to the following URL for Web UI: `http://apache_httpd_server/wcc`

`apache_httpd_server`  
Specifies the name of the server on which Apache HTTPD Server is installed.

**NOTE**

If Apache HTTPD Server is configured for SSL, specify https in the URL.

The login page appears.

**NOTE**

If you specified the list of servers in a CA EEM failover setup when you installed Web UI, display of the login page can be delayed if the first server in the list is not available.

2. Enter the appropriate credentials and click Log In.

The CA Workload Automation window opens to the default page for your user ID, displaying the tabs and applications of Web UI that you are authorized to access.

**NOTE**

If a database connection error message appears when you log in, verify that the Web UI database is running, log out, and then log back in to Web UI.

## Check the Web UI Version

You can use the uejmver.bat file, located in the root installation directory of the Web UI server, to display the version information for the Web UI instance.

**NOTE**

The version information for the Web UI instance is also available by clicking About at the bottom of the Web UI page.

**Follow these steps:**

1. Open a command prompt.
2. Navigate to the Web UI installation directory. By default, this is c:\Program Files\CA\Workload Control Center.
3. Type uejmver and press Enter.  
The version details are displayed.

## Add a Web UI Instance to the Load Balance Manager

When you add a Web UI instance to your High Availability group, configure the Load Balance Manager to include that instance.

**NOTE**

Before adding a Web UI instance, verify that the server can communicate with the other Web UI instances in the High Availability group. For example, use a ping command to test communication.

**Follow these steps:**

1. Open the httpd.conf file in a text editor on the Web UI server on which Apache HTTPD Server was downloaded. The file is typically located in the following directory: \$APACHE\_INSTALL\_DIR/conf
2. Go to the section that begins with <Location /balancer-manager> and add a BalanceMember line for the new Web UI instance.

**Example:**

```
<Location /balancer-manager>

 SetHandler balancer-manager

</Location>

ProxyPass /balancer-manager !

 # failonstatus status is available with Apache HTTP Server
 2.2.17 and later. Remove the parameter if necessary.

ProxyPass /wcc balancer://wccbalancer stickysession=WCC-ASID
failonstatus=503

ProxyPassReverse /wcc

<Proxy balancer://wccbalancer>

 BalanceMember http://
 CA_WCC_instance_A:
 port/wcc route=
 CA_WCC_instance_A
```



```

BalanceMember http://
 CA_WCC_instance_B:
 port/wcc route=
 CA_WCC_instance_B

BalanceMember http://
 CA_WCC_new_instance:
 port/wcc route=
 CA_WCC_new_instance

</Proxy>

```

3. Save and close the httpd.conf file.
4. Restart Apache HTTPD Server.

#### NOTE

- Users must access Web UI from the following URL: `http://apache_httpd_server/wcc`
  - **apache\_httpd\_server**  
Specifies the name of the server on which Apache HTTPD Server is installed.  
If Apache HTTPD Server is configured for SSL, specify https in the URL.
- You can display all registered Web UI instances from the following URL: `http://CA_WCC_instance:port/wcc/rest/configuration/httpd-config`
  - **CA\_WCC\_instance**  
Specifies the Web UI instance. You can specify any Web UI instance in the High Availability group.
  - **port**  
Specifies the Application Port of the Web UI instance.

## Install CCI

CA Common Communications Interface (CA CCI) provides the underlying communication infrastructure that connects applications running on OpenVMS, z/OS, AS/400, Windows, various versions of UNIX, and many other operating systems. CA CCI forms the basis for the cross-platform workload automation agent that you can manage from AutoSys Workload Automation.

The following sections describe how you can install CCI on Windows and Unix.

#### Prerequisites

- Install AutoSys Workload Automation with the Scheduler component.
- The compressed file setup for CCI is present at the following location.
  - **For Windows:** %AUTOSYS%\install\cci\cci-12.0.00.00-Windows.zip
  - **For Linux:** \$AUTOSYS/autosys/install/CCI/cci-12.0.00.00-Linux.tar
  - **For AIX:** \$AUTOSYS/autosys/install/CCI/cci-12.0.00.00-AIX.tar

## Windows

To install CCI on Windows.

1. Extract the zip files to `C:\program files (x86)\CA\SC`.
2. Set `CASHCOMP(x86)=C:\Program Files (x86)\CA\SC`.
3. Run `CCI_SETUP_WIN.bat`.  
This creates services and registry entries, updates the PATH variable to include `%CASHCOMP(x86)%\CA_APPSW`, and updates `CCIRMTD.RC`.

### Notes:

- In the procedure below "<OS>" can be replaced with "AIX" or "LINUX".
- When you extract the zip files, `CA_APPSW` and `CCS` directories will appear.

## Unix

To install CCI on Unix.

1. Install 32-bit CAPKI 2.5.6 from `ISO/modules/CAPKI/32`.  

```
cd ISO/modules//CAPKI/32
./setup install caller=WAAE instdir=$CASHCOMP verbose env=all
```
2. Extract `$AUTOSYS/install/CCI/cci-12.0.00.00-<OS>.tar` using below command:  

```
cd $AUTOSYS/install/CCI
tar -xvf cci-12.0.00.00-<OS>.tar
```
3. Provide execute permissions to `$AUTOSYS/install/CCI/CCI_AIX/CCI_SETUP_<OS>.sh`  

```
cd $AUTOSYS/install/CCI/CCI_<OS>
chmod +x $AUTOSYS/install/CCI/CCI_AIX/CCI_SETUP_<OS>.sh
```
4. Execute `$AUTOSYS/install/CCI/CCI_AIX/CCI_SETUP_AIX.sh`  

```
./CCI_SETUP_<OS>.sh
```

## Post-Installation Procedures for the Server

You can perform the following tasks to customize AutoSys Workload Automation after it has been installed.

### (UNIX) Startup Scripts

AutoSys Workload Automation creates the following scripts to start the agent, application server, scheduler, and web server at system startup:

```
waee_agent-WA_AGENT
waee_sched.$AUTOSERV
waee_server.$AUTOSERV
waee_webserver.$AUTOSERV
```

- **WA\_AGENT**

Specifies the name of the agent assigned during installation. The default is `WA_AGENT`.

`$AUTOSERV` is the instance name. These scripts are active if you select the corresponding options in the setup wizard.

The scripts are located in the following directories:

- /etc/init.d (on Linux)
- /sbin/init.d (on HP-UX)
- /etc/rc.d/init.d (on AIX)

Each script accepts the conventional start or stop argument. The root user can run these scripts to start or stop the AutoSys Workload Automation services manually. For example, the following script starts the *WA\_AGENT* agent and instance ACE's application server on a Linux computer:

```
/etc/init.d/waee_agent-WA_AGENT start
/etc/init.d/waee_server.ACE start
```

The database must be available for the application server and scheduler to start. If AutoSys Workload Automation uses an Oracle or Sybase database on the same computer, you can start the application server and scheduler at startup only if you start the database first.

## (UNIX) Define Additional EDIT and EXEC Superusers

The *EDIT superuser* is the only user who can change the database password and remote authentication method, change the owner of a job, or edit any job regardless of who owns it.

The *EXEC superuser* is the only user who has permissions to stop the scheduler. The EXEC superuser can also start and stop all jobs, regardless of their ownership or permissions.

By default, the EDIT and EXEC superusers are assigned to one of the following user accounts:

- The owner account of the AutoSys Workload Automation product files (if the account exists or you selected the Create the owner and group account option during the installation)
- root (if the owner account does not exist or was not created during installation)

After installation, you can define additional EDIT and EXEC superusers.

### Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation and enter the following command:  

```
autosys_secure
```

The AutoSys Security Utility submenu appears.
2. Enter the following command at the instance command prompt:  

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.
3. Enter option [2] Manage EDIT/EXEC superusers.  

The manage EDIT/EXEC superuser submenu appears.
4. Enter option [1] Create an EDIT/EXEC superuser.  

The Utility prompts you to create an EDIT superuser or and EXEC superuser.
5. Enter the EDIT or EXEC superuser name and domain.  

A message appears to indicate whether the user was successfully added.
6. If error messages appear, navigate the interactive menu options to make the required changes. Repeat this action until the message indicates that the user was successfully added.
7. Enter option [0] Exit AutoSys Security Utility.  

The EDIT or EXEC superuser data for the new user is stored in the database.

### NOTE

The Manage EDIT/EXEC superusers menu is only available when `autosys_secure` is first run and option 2 is selected. After the superusers are modified for the first time, only the EDIT superuser can access this menu.

When this menu is accessed again, the current settings are displayed. The EDIT superuser can accept the same users by pressing Enter, or change the users by entering a new specification.

## (Windows) Add the Superusers and the Windows User IDs and Passwords

When an agent runs a job, the agent logs on to the remote computer as the owner of the job. To do this, the agent uses the encrypted passwords that were passed to it with the job request by the scheduler. The scheduler gets these passwords from the event server (database). Therefore, after the installation is complete and before you can run jobs, you must enter Windows user IDs and passwords for users who define and run jobs.

Before you can enter the Windows user IDs and passwords in the database, you must first establish the EDIT and EXEC superusers.

The *EDIT superuser* is the only user who can change the database password and remote authentication method, change the owner of a job, or edit any job regardless of who owns it.

The *EXEC superuser* is the only user who has permissions to stop the scheduler. The EXEC superuser can also start and stop all jobs, regardless of their ownership or permissions.

### Define Additional EDIT and EXEC Superusers

After installation, you can define additional EDIT and EXEC superusers.

#### **Follow these steps:**

1. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt opens.
2. Enter the following command at the instance command prompt:

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

3. Enter option [2] Manage EDIT/EXEC superusers.  
The manage EDIT/EXEC superuser submenu appears.
4. Enter option [1] Create an EDIT/EXEC superuser.  
The Utility prompts you to create an EDIT superuser or and EXEC superuser.
5. Enter the EDIT or EXEC superuser name and domain.  
A message appears to indicate whether the user was successfully added.
6. If error messages appear, navigate the interactive menu options to make the required changes. Repeat this action until the message indicates that the user was successfully added.
7. Enter option [0] Exit AutoSys Security Utility.  
The EDIT or EXEC superuser data for the new user is stored in the database.

### Add the Windows User IDs and Passwords

Before you can run jobs on Windows, you must enter the Windows user IDs and passwords for all users on all domains who define and run jobs. We recommend that you add only the user and password needed to run a test job before adding other users. After you have successfully created and run the test job, you can add additional Windows user IDs and passwords.

#### **Follow these steps:**

1. Log in to Windows as the EDIT superuser.
2. Open a AutoSys Workload Automation instance command prompt window from your program group.
3. Enter the following command at the instance command prompt:

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

4. Enter option [5] Manage users.  
The Manage users submenu appears.
5. Enter option [1] Manage users with password.  
The Manage users with password menu appears.
6. Enter option [1] Create a user.  
The utility prompts you to enter the user name, host or domain name, and password.
7. Enter the user name, host or domain name, new password, and password confirmation.

#### NOTE

Windows user IDs must not exceed 20 characters, and they can include any characters except the following:

" / ; : < > | = + \*

Windows passwords are case-sensitive and must not exceed 127 characters, and they can contain any character except a space.

If the user is created successfully, the user information is entered into the database with the encrypted password.

## Change the Web Server Java Heap Size

After you install AutoSys Workload Automation, you can change the web server Java heap size. We recommend that you increase the web service Java heap size value when you see CA EEM exceptions similar to the following while using web services:

```
[com.ca.eiam.SafeCache] ThreadLoop - exception in cache update
java.lang.OutOfMemoryError: Java heap space
```

#### Follow these steps:

1. Open the following file:  
(UNIX) \$AUTOUSER/webserver/conf/wrapper.conf  
(Windows) %AUTOUSER%\webserver\conf\wrapper.conf
2. Locate the following parameter and set its value to 256:

```
wrapper.java.maxmemory
```

#### NOTE

The default value is 128.

3. Save and close the file.
4. Start the web server as follows:  
UNIX:  
Enter the following command at the operating system prompt:

```
unisrvctr restart waee_webserver.$AUTOSERV
```

Windows:

- a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
- b. Select an instance from the Instance drop-down list in the Settings pane.

- c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
- d. Right-click the web server and select Start.  
The web server starts and the changes take effect.

## Change the Web Server Shutdown Port Number

### **(UNIX) Change the Web Server Shutdown Port Number**

You can change the web server shutdown port number to resolve a port conflict after you install AutoSys Workload Automation.

#### **Follow these steps:**

1. Open the server.xml file in a text editor. The file is in the following location:  
`$AUTOUSER/webserver/conf/server.xml`
2. Locate the following text:  

```
<!-- Note: A "Server" is not itself a "Container", so you may not
 define subcomponents such as "Valves" at this level.
 Documentation at /docs/config/server.html
-->
<Server port="-1" shutdown="SHUTDOWN">
```
3. Change `Server port="-1"` to the appropriate port number.

#### **NOTE**

The web server shutdown port is disabled by default when AutoSys Workload Automation is installed. The value in your instance can be different.

4. Save and close the server.xml file.
5. Enter the following command at the operating system prompt:  
`unisrvctr restart waae_webserver.$AUTOSERV`  
The web server restarts and the changes take effect.

### **(Windows) Change the Web Server Shutdown Port Number**

You can change the web server shutdown port number to resolve a port conflict after you install AutoSys Workload Automation.

#### **Follow these steps:**

1. Stop the web server as follows:
  - a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
  - d. Right-click the web server and select Stop.  
The web server stops.
2. Open the server.xml file in a text editor. The file is in the following location:  
`%AUTOUSER%\webserver\conf\server.xml`

## 3. Locate the following text:

```
<!-- Note: A "Server" is not itself a "Container", so you may not
 define subcomponents such as "Valves" at this level.
 Documentation at /docs/config/server.html
-->
<Server port="-1" shutdown="SHUTDOWN">
```

## 4. Change Server port="-1" to the appropriate port number.

**NOTE**

The web server shutdown port is disabled by default when AutoSys Workload Automation is installed. The value in your instance can be different.

## 5. Save and close server.xml.

## 6. Start the web server as follows:

- a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
- b. Select an instance from the Instance drop-down list in the Settings pane.
- c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
- d. Right-click the web server and select Start.  
The web server starts and the changes take effect.

## Change the Web Server SSL HTTP/1.1 Port Number

### (UNIX) Change the Web Server SSL HTTP/1.1 Port Number

You can change the web server SSL HTTP/1.1 port number to resolve a port conflict after you install AutoSys Workload Automation.

**Follow these steps:**

## 1. Open the server.xml file in a text editor. The file is in the following location:

```
$AUTOUSER/webserver/conf/server.xml
```

## 2. Locate the following text:

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443
 This connector uses the JSSE configuration, when using APR, the
 connector should be using the OpenSSL style configuration
 described in the APR documentation -->
<!-- -->
<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"
 ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
 TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
 SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"
 keystoreFile="/opt/CA/WorkloadAutomationAE/autouser.ACE/webserver/
 conf/.keystore"
 keystorePass="changeit"
 maxThreads="400" scheme="https" secure="true"
 clientAuth="false" sslProtocol="TLS" />
```

3. Change Connector port="9443" to the appropriate port number.

#### NOTE

The connector port is set to 9443 by default when AutoSys Workload Automation is installed. The value in your instance can be different.

4. Save and close the server.xml file.
5. Enter the following command at the operating system prompt:

```
unisrvctr restart waae_webserver.$AUTOSERV
```

The web server restarts and the changes take effect.

### **(Windows) Change the Web Server SSL HTTP/1.1 Port Number**

You can change the web server SSL HTTP/1.1 port number to resolve a port conflict after you install AutoSys Workload Automation.

#### **Follow these steps:**

1. Stop the web server as follows:
  - a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
  - d. Right-click the web server and select Stop.  
The web server stops.

2. Open the server.xml file in a text editor. The file is in the following location:

```
%AUTOUSER%\webserver\conf\server.xml
```

3. Locate the following text:

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443
This connector uses the JSSE configuration, when using APR, the
connector should be using the OpenSSL style configuration
described in the APR documentation -->
<!-- -->
<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"
maxThreads="400" scheme="https" secure="true"
clientAuth="false" sslProtocol="TLS"
ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"
keystoreFile="C:\Program Files (x86)\CA\
Workload Automation AE\autouser.SQL\webserver\conf\.keystore"
keystorePass="changeit" />
```

4. Change Connector port="9443" to the appropriate port number.

#### NOTE

The connector port is set to 9443 by default when AutoSys Workload Automation is installed. The value in your instance can be different.

5. Save and close the server.xml file.



6. Start the web server as follows:
  - a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
  - d. Right-click the web server and select Start.  
The web server starts and the changes take effect.

## Change the Web Server SSL Protocol to TLSv1.2

### (UNIX) Change the Web Server SSL Protocol to TLSv1.2

You can change the web server SSL Protocol to TLSv1.2 to adhere to the standards.

#### **Follow these steps:**

1. Open the server.xml file in a text editor. The file is in the following location:

```
$AUTOUSER/webserver/conf/server.xml
```

2. Locate the following text:

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443

This connector uses the JSSE configuration, when using APR, the
connector should be using the OpenSSL style configuration
described in the APR documentation -->

<!-- -->

<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"

ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"

keystoreFile="/opt/CA/WorkloadAutomationAE/autouser.ACE/webserver/
conf/.keystore"

keystorePass="changeit"

maxThreads="400" scheme="https" secure="true"

clientAuth="false" sslProtocol="TLS" />
```

3. Ensure that the sslProtocol is set to TLS and then add the sslEnabledProtocols="TLSv1.2" parameter.

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443

This connector uses the JSSE configuration, when using APR, the
connector should be using the OpenSSL style configuration
described in the APR documentation -->

<!-- -->

<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"

 ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
 TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
 TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
 SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"

 keystoreFile="/opt/CA/WorkloadAutomationAE/autouser.ACE/webserver/
 conf/.keystore"

 keystorePass="changeit"

 maxThreads="400" scheme="https" secure="true"

 clientAuth="false" sslProtocol="TLS" sslEnabledProtocols="TLSv1.2"/>
```

4. Save and close the server.xml file.
5. Enter the following command at the operating system prompt:

```
unisrvctr restart waee_webserver.$AUTOSERV
```

The web server restarts and the changes take effect.

### **(Windows) Change the Web Server SSL Protocol to TLSv1.2**

You can change the web server SSL Protocol to TLSv1.2 to adhere to the standards.

#### **Follow these steps:**

1. Stop the web server as follows:
  - a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
  - d. Right-click the web server and select Stop.  
The web server stops.
2. Open the server.xml file in a text editor. The file is in the following location:

```
%AUTOUSER%\webserver\conf\server.xml
```

### 3. Locate the following text:

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443

This connector uses the JSSE configuration, when using APR, the
connector should be using the OpenSSL style configuration
described in the APR documentation -->

<!-- -->

<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"

maxThreads="400" scheme="https" secure="true"

clientAuth="false" sslProtocol="TLS"

ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"

keystoreFile="C:\Program Files (x86)\CA\
Workload Automation AE\autouser.SQL\webserver\conf\.keystore"

keystorePass="changeit" />
```

### 4. Ensure that the sslProtocol is set to TLS and then add the sslEnabledProtocols="TLSv1.2" parameter.

```
<!-- Define a SSL HTTP/1.1 Connector on port 9443

This connector uses the JSSE configuration, when using APR, the
connector should be using the OpenSSL style configuration
described in the APR documentation -->

<!-- -->

<Connector port="9443" protocol="HTTP/1.1" SSLEnabled="true" acceptCount="100"

maxThreads="400" scheme="https" secure="true"

clientAuth="false" sslProtocol="TLS" sslEnabledProtocols="TLSv1.2"
```

```

ciphers="SSL_RSA_WITH_RC4_128_MD5, SSL_RSA_WITH_RC4_128_SHA,
TLS_RSA_WITH_AES_128_CBC_SHA, TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_DSS_WITH_AES_128_CBC_SHA, SSL_RSA_WITH_3DES_EDE_CBC_SHA,
SSL_DHE_RSA_WITH_3DES_EDE_CBC_SHA, SSL_DHE_DSS_WITH_3DES_EDE_CBC_SHA"

keystoreFile="C:\Program Files (x86)\CA\
Workload Automation AE\autouser.SQL\webserver\conf\.keystore"

keystorePass="changeit" />

```

5. Save and close the server.xml file.
6. Start the web server as follows:
  - a. Click Start, Programs, CA, AutoSys Workload Automation, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
  - d. Right-click the web server and select Start.  
The web server starts and the changes take effect.

## Configure the Firewall

Firewalls may block the ports AutoSys Workload Automation uses to communicate with client utilities and agents that are located outside of the firewall. To prevent communication problems, configure the firewall to allow communication with client utilities, agents, and CA common components.

### NOTE

Unless otherwise noted, the UNIX parameters listed in the following table are defined in the configuration file, and the Windows fields are located in the AutoSys Workload Automation Administrator. For more information about the UNIX parameters or the Windows fields, see [AutoSys Workload Automation Administration](#).

To configure the firewall, add the following ports as exceptions:

Ports	Details
Database access ports	The configuration location and communication specifications for database access ports depend on the database vendor. For more information, see the vendor documentation.
CA iTechnology iGateway IANA-defined port	<b>Communication:</b> The iGateway port enables communication between the CA EEM client (the application server) and the CA EEM security server. The client uses this port to connect to the security server. To prevent communication problems, configure the client firewall to allow outgoing communication and the security server firewall to allow incoming communication.   <b>Configuration location:</b> The "port" XML tag in the \$IGW_LOC/igateway.conf file (UNIX) or the %IGW_LOC%\igateway.conf file (Windows).   <b>Default:</b> 5250; this is a static port and must not be changed.

common components interface (CCI) IANA-defined port   <b>Note:</b> This port includes integration with the Notification Services component of CA NSM.	<b>Communication:</b> The CCI port enables communication between the CCI client (the scheduler) and the CCI server. The client uses this port to connect to the CCI server. To prevent communication problems, configure the client firewall to allow outgoing communication and the CCI server firewall to allow incoming communication.   <b>Configuration location:</b> The "port" tag in the \$CASHCOMP/ccs/cci/config/&lt;hostname&gt;/ccirmtd.prf file (UNIX) or the %CASHCOMP%\CA_APPSW\CCIRMTD.RC file (Windows).   <b>Defaults:</b> 7001 (Mainframe, not IANA-defined), 1721 (Distributed)
Integration with CA Service Desk through HTTP	<b>Communication:</b> The CA Service Desk integration ports enable communication between the HTTP client (the scheduler) and the web service. The client uses these ports to connect to the web service. To prevent communication problems, configure the client firewall to allow outgoing communication and the web service firewall to allow incoming communication.   <b>Configuration locations:</b> The "ServiceDeskURL" parameter (UNIX) or the "URL Location" field in the Service Desk tab on the Integration - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (Windows).   <b>Default:</b> 8080   <b>Note:</b> If the web service is installed on the same computer where the CA Service Desk is installed, ensure that the CA Service Desk integration port number is not the same as the web server communication port number to avoid communication problems.
Integration with SNMP IANA-defined ports	<b>Communication:</b> The SNMP integration port enables communication between the SNMP agent (the scheduler) and the SNMP manager. The agent uses this port to connect to the manager. To prevent communication problems, configure the agent firewall to allow outgoing communication and the manager firewall to allow incoming communication.   <b>Configuration location:</b> The "snmp" tag in the /etc/services file (UNIX) or the %SystemRoot%\System32\Drivers\Etc\SERVICES file (Windows).   <b>Defaults:</b> 161 and 10161 (Agent), 162 and 10162 (Manager)
SSA IANA-defined port	<b>Communication:</b> The SSA-port enables communication between all SSA clients (all agents configured to run on SSA ports, the scheduler, the application server, and all application server clients) and the SSA broker. The clients use this port to connect to the broker. To prevent communication problems, configure the client firewalls to allow outgoing communication and the broker firewall to allow incoming communication.   <b>Configuration location:</b> The "PmuxServerPort" parameter displayed by the "csamconfigedit display" command.   <b>Default:</b> 7163; this is a static port and must not be changed.   <b>Note:</b> If the agent is configured to use the SSA ports, then the 7163 port must be open bidirectionally. For more information, see <a href="#">Configure the Agent to Communicate Using SSA Ports</a>.

Scheduler auxiliary listening port (for non-SSA brokered communication)	<b>Communication:</b> The scheduler listening port enables the scheduler to receive data from the application server and the agent. The application server and agent use this port to connect to the scheduler. To prevent communication problems, configure the application server and agent firewalls to allow outgoing communication and the scheduler firewall to allow incoming communication.   <b>Configuration location:</b> The "SchedAuxiliaryListeningPort" parameter (UNIX) or the "Auxiliary Listening Port" field in the Communications Ports tab on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (Windows).   <b>Default:</b> 7507
Application server auxiliary listening port (for non-SSA brokered communication)	<b>Communication:</b> The application server listening port enables the application to receive data from the scheduler and the agent. The scheduler and agent use this port to connect to the application server. To prevent communication problems, configure the scheduler and agent firewalls to allow outgoing communication and the application server firewall to allow incoming communication.   <b>Configuration location:</b> The "AppSrvAuxiliaryListeningPort" parameter (UNIX) or the "Auxiliary Listening Port" field on the Application Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (Windows).   <b>Default:</b> 7500
Web server ports	<b>Communication:</b> The web server communication port enables the web server to listen to HTTPS requests. The stop port number is used to send the shutdown request to the web server. To prevent communication problems, configure the firewalls to allow the outgoing and incoming communication to the web server listening port.   <b>Configuration location:</b> The "Communication Port Number" and "Stop Port Number" fields on the Web Server Properties installation page.   <b>Default:</b> 9443 (communication port number)
Workload Automation Agent listening ports (for non-SSA brokered communication)	<b>Communication:</b> The agent listening ports enables the agent to receive data from the application server and the scheduler. The application server and scheduler use this port to connect to the agent. To prevent communication problems, configure the application server and scheduler firewalls to allow outgoing communication and the agent firewall to allow incoming communication.   <b>Configuration location:</b> The "port attribute" value in the machine definition.   <b>Default:</b> 7520

## Database Tracking

You can run the autotrack command to set your appropriate database tracking level. The autotrack command tracks changes to the database (for example, job definition changes, sendevent calls, and job overrides) and writes this information to the database. Changes to job definitions made through the command utilities can be tracked. Changes made directly to the database through SQL statements cannot be tracked.

When you query for this information, the autotrack command prints a report to the screen, or you can redirect the output to a file.

Automatic tracking is useful for the following:

- Sites that require monitoring of the job definition environment.
- Sites where multiple users have permission to edit job definitions or send events.

### **Set Up the Database Tracking Level**

You can run the autotrack command to set your appropriate database tracking level.

To set up the database tracking level, issue the following command:

```
autotrack -u 0|1|2
```

- **0**  
Does not track changes to the database. This is the default.
- **1**  
Tracks changes to the database and condenses each tracked event to a one-line summary.
- **2**  
Tracks the same information as level 1, but also writes the entire job definition for overrides and job definition changes.

#### **NOTE**

This level is database-intensive and significantly impairs JIL performance.

The database tracking level is set.

## **Installation Panels**

During fresh installation, upgrade, and maintenance procedures, you need to enter various values. The following topics describe the values and properties that you define on different panels during those procedures.

### **MSSQL Database Panels**

This section provides reference information for the values and properties that you define for Microsoft SQL Server during fresh installation, upgrade, or maintenance procedures.

#### **Database Type**

The following information is required on the Database Type panel when you select Microsoft SQL Server during the installation procedure.

#### **NOTE**

This database type is supported only on Linux and Windows.

Field name	Silent install variable	Description
Database type	IA_AE_DATABASE_TYPE=MSSQL	Select the Microsoft SQL Server.
Dual event server	IA_AE_DUAL_DATABASE	Select this checkbox if you want to implement a dual event server using MS SQL. For more information, see <a href="#">Dual Event Server Mode</a> .

Create the database schema	IA_AE_CREATE_DATABASE	Select this checkbox if you want to create the database schema.
----------------------------	-----------------------	-----------------------------------------------------------------

## Primary Event Server Properties

The following information is required on the Primary Event Server Properties panel when you select Microsoft SQL Server during the installation procedure.

### NOTE

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields mentioned in this topic also apply to the Secondary Event Server Properties window.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SERVERNAME, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SERVERNAME.
- An MSSQL client should be already installed on the machine. In the case of Linux, the location is at `/opt/mssql-tools`

Field name	Silent install variable	Description
Server name	IA_AE_PRIMARY_DB_SERVERNAME	The server name on which your MSSQL database is installed. It is the logical name of the database server. The name of the host computer where the server is installed is the default. For named instance, specify the server name as "hostname/instance name".
Database name	IA_AE_PRIMARY_DB_NAME	The name of the database that is to be created in MSSQL for the AutoSys instance that you are creating. The default database name is AEDB. The entire value of the Database Name and Server Name fields can be up to 64 characters.
Database TCP port	IA_AE_PRIMARY_DB_PORT	The database TCP/IP listener port. Specify the port using which AutoSys should establish a connection with MSSQL. By default, the port no is 1433. The database TCP/IP listener port is retrieved from the Microsoft SQL Server Windows Registry.

## Primary Admin Details

The following information is required on the Primary Admin Details panel when you select Microsoft SQL Server during the installation procedure.



**NOTE**

- If you are using a dual server configuration, a Secondary Admin Details window will appear. The fields mentioned here also apply to the Secondary Admin Details window.
- The silent variable for Secondary Admin Details will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ADMIN\_USER, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ADMIN\_USER.

Field name	Silent install variable	Description
Authentication type	IA_AE_PRIMARY_DB_MSSQL_AUTH_TYPE	Integrated security or Password-based. This field is not applicable for Linux.
Username	IA_AE_PRIMARY_DB_ADMIN_USER	The username of your MSSQL server for authentication.
Password	IA_AE_PRIMARY_DB_ADMIN_PWD	The password of your MS SQL server for authentication.

**AutoSys Database user Details**

The following information is required on the AutoSys Admin User Details panel when you select Microsoft SQL Server during the installation procedure.

Field name	Silent install variable	Description
Database username	IA_AE_DB_USER	The username for the AutoSys database.
Password	IA_AE_DB_USER_PWD	The password of AutoSys database user.
Verify password	NA	Enter the same password as entered above.

**Oracle Database Panels**

This section provides reference information for the values and properties that you define for the Oracle database during fresh installation, upgrade, or maintenance procedures.

**NOTE**

A 64-bit Oracle client should be present on the machine.

**Database Type**

The following information is required on the Database Type panel when you select Oracle during the installation procedure.

Field name	Silent install variable	Description
Database Type	IA_AE_DATABASE_TYPE=ORACLE	Select Oracle.
Dual event server	IA_AE_DUAL_DATABASE	Select this checkbox if you want to implement a dual event server. For more information, see <a href="#">Dual Event Server Mode</a> .
Use Oracle public key infrastructure (PKI) external authentication	IA_AE_DB_USE_EXTERNAL_AUTH	Select this checkbox if you want to enable the public key infrastructure authentication.
Create the database schema	IA_AE_CREATE_DATABASE	Select this checkbox if you want to create a database schema.

Storage management	IA_AE_DB_ORA_STORAGE_TYPE	Select the desired storage management.
Create the tablespace, database user, and role	IA_AE_DB_ORA_CREATE_TBL_SPACE	Select this checkbox if you want to create tablespace, database, and roles. This check box is selected by default. Clear this check box if you plan to create the tablespaces, database users, and roles manually by running the waae_oracle.sql script before you install AutoSys Workload Automation.

## Primary Event Server Properties

The following information is required on the Primary Event Server Properties panel when you select Oracle during the installation procedure.

### NOTE

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields mentioned here apply to the Secondary Event Server Properties window as well.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ORA\_TNSALIAS, then for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ORA\_TNSALIAS.

Field name	Silent install variable	Description
TNS alias name	IA_AE_PRIMARY_DB_ORA_TNSALIAS	The TNS alias name to connect to. The Oracle Net TNS alias name (Oracle service ID) for the event server that contains the database.
Database host	IA_AE_PRIMARY_DB_HOSTNAME	The server location where the database is hosted.
Database TCP port	IA_AE_PRIMARY_DB_PORT	The database TCP port that is to be used for communication between AutoSys and the database. In case of PKI, the default port used for authentication will be 2484.
Oracle home directory	IA_AE_DB_ORA_HOME	The directory location of the Oracle home directory.
TNS admin directory	IA_AE_DB_ORA_TNS_ADMIN	The directory location of the Oracle TNS admin directory.

## Primary Admin Details

The following information is required on the Primary Admin Details panel when you select Oracle during the installation procedure.

**NOTE**

- If you are using a dual server configuration, a Secondary Admin Details window will appear. The fields mentioned here also apply to the Secondary Admin Details window.
- The silent variable for Secondary Admin Details will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ADMIN\_USER, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ADMIN\_USER.

Field name	Silent install variable	Description
User name	IA_AE_PRIMARY_DB_ADMIN_USER	The Oracle username that has DBA ROLE, DBA TABLESPACE SELECT, V \$PARAMETER, and CREATE SESSION authority.
Password	IA_AE_PRIMARY_DB_ADMIN_PWD	The password for the username entered above in case you are using password authentication.
TNS admin directory	IA_AE_PRIMARY_DB_ORA_ADMIN_USE R_TNS_ADMIN	The TNS admin directory path.
Oracle wallet SSO file path (In case of PKI authentication).	IA_AE_PRIMARY_DB_ORA_ADMIN_USE R_WALLET_LOC	The Oracle wallet SSO file path (in case of PKI authentication)

**Primary Schema Admin Details**

The following information is required on the Primary Schema Admin Details panel when you select Oracle during the installation procedure.

**NOTE**

- If you are using a dual server configuration, a Secondary Schema Admin Details window will appear. The fields mentioned here also apply to the Secondary Schema Admin Details window.
- The silent variable for Secondary Schema Admin Details will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ORA\_SCHEMA\_ADMIN\_USER, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ORA\_SCHEMA\_ADMIN\_USER.

Field name	Silent install variable	Description
Schema owner name	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_USER	Enter a name to create the Oracle database schema admin.
Password	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_PWD	In case of password authentication, enter the password for the owner's name entered above.
Verify password	NA	Re-enter the password entered above.
Certificate distinguished name	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_DN_NAME	In case of PKI authentication, enter the certificate distinguished name.
TNS admin directory	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_TNS_ADMIN	In case of PKI authentication, enter the TNS admin directory path.

Oracle wallet SSO file path	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_WALLET_LOC	The Oracle wallet SSO file path. In case of PKI authentication, ensure that the wallet is unique. If the schema owner already exists, ensure that the wallet points to the user with the provided schema owner name.
Use same user credentials for secondary database	IA_AE_USE_SAME_SCHEMA_CRED_FOR_SECONDARY	Select this checkbox if you want to use the credentials mentioned above for the secondary database as well.

## Primary Data And Index Tablespace Information

The following information is required on the Primary Data And Index Tablespace Information panel when you select Oracle during the installation procedure.

### NOTE

- If you are using a dual server configuration, a Secondary data & Index Tablespace Information window will appear. The fields mentioned here also apply to the Secondary data & Index Tablespace Information window.
- The silent variable for Secondary Schema Admin Details will have the word **SECONDARY** instead of **PRIMARY**. For example, for your primary server, if the silent variable is `IA_AE_PRIMARY_DB_ORA_DATA_TBSLSPACE_NAME`, for the secondary server, it will be `IA_AE_SECONDARY_DB_ORA_DATA_TBSLSPACE_NAME`.

Field name	Silent install variable	Description
Data tablespace name	IA_AE_PRIMARY_DB_ORA_DATA_TBSLSPACE_NAME	The data tablespace name. By default, the value is <code>AEDB_DATA</code> . A tablespace file is created with this name in the data tablespace directory.
Data tablespace size (in MB)	IA_AE_PRIMARY_DB_ORA_DATA_TBSLSPACE_SIZE	The tablespace file size in MB. By default, the file size is 800 MB.
Data tablespace directory	IA_AE_PRIMARY_DB_ORA_DATA_TBSLSPACE_PATH	The data tablespace directory where the file is created or stored.
Data Disk Group	IA_AE_PRIMARY_DB_ORA_DATA_DISKGROUP_NAME	If you are using Automatic Storage Management (ASM), the data disk group to be used.
Index tablespace name	IA_AE_PRIMARY_DB_ORA_INDEX_TBSLSPACE_NAME	The index tablespace name. By default, the value is <code>AEDB_INDEX</code> . An index tablespace file is created with this name in the index tablespace directory.
Index tablespace size (in MB)	IA_AE_PRIMARY_DB_ORA_INDEX_TBSLSPACE_SIZE	The index tablespace file size in MB. By default, the file size is 80 MB.
Index tablespace directory	IA_AE_PRIMARY_DB_ORA_INDEX_TBSLSPACE_PATH	The index tablespace directory where the file is created or stored.

Index Disk Group	IA_AE_PRIMARY_DB_ORA_INDEX_DISKGROUP_NAME	If you are using Automatic Storage Management (ASM), the index disk group to be used.
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## AutoSys Database User Details

The following information is required on the AutoSys Database User Details panel when you select Oracle during the installation procedure.

Field name	Silent install variable	Description
Database username	IA_AE_DB_USER	The username for the AutoSys database.
Password	IA_AE_DB_USER_PWD	In case of password authentication, the password of the AutoSys database user.
Verify password	NA	Enter the same password as entered above.
Certificate distinguished name	IA_AE_DB_ORA_USER_DN_NAME	In the case of PKI authentication, the certificate distinguished name.
TNS admin directory	IA_AE_DB_ORA_USER_TNS_ADMIN	The TNS admin directory path. In the case of PKI authentication, ensure that this is unique for the user. If the user is already defined, ensure that the wallet is mapped to the user name provided.
Oracle wallet SSO file path	IA_AE_DB_ORA_USER_WALLET_LOC	In the case of Oracle PKI, the Oracle wallet SSO file path.

## Sybase Database Panels

This section provides reference information for the values and properties that you define for the Sybase database during fresh installation, upgrade, or maintenance procedures.

### NOTE

- A 64-bit Sybase client should be pre-installed on the machine.
- In the case of Kerberos the ticket cache should be properly configured.

## Database Type

The following information is required on the Database Type panel when you select Sybase during the installation procedure.

Field name	Silent install variable	Description
Database Type	IA_AE_DATABASE_TYPE=SYBASE	Select Sybase.
Dual event server	IA_AE_DUAL_DATABASE	Select this checkbox if you want to implement a dual event server using Sybase. For more information, see <a href="#">Dual Event Server Mode</a> .
Use Kerberos external authentication to access database	IA_AE_DB_USE_EXTERNAL_AUTH	Select this checkbox if you want to implement external authentication using Kerberos.

Create the database schema	IA_AE_CREATE_DATABASE	Select this checkbox if you want to create the database schema.
Create the database devices	IA_AE_DB_SYB_CREATE_DEVICE	Select this checkbox if you want to create database devices.

## Primary Event Server Properties

The following information is required on the Primary Event Server Properties panel when you select Sybase during the installation procedure.

### NOTE

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields mentioned here apply to the Secondary Event Server Properties window as well.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SERVERNAME, then for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SERVERNAME.

Field name	Silent install variable	Description
Server name	IA_AE_PRIMARY_DB_SERVERNAME	The name of the Sybase server where you want to install the database for the primary event server.
Database name	IA_AE_PRIMARY_DB_NAME	The Sybase database name to connect to.
Database host	IA_AE_PRIMARY_DB_HOSTNAME	The server location where the database is hosted.
Database TCP port	IA_AE_PRIMARY_DB_PORT	The database port that can be used to communicate with AutoSys. By default, the value is 5000.
Sybase directory	IA_AE_DB_SYB_HOME	The directory location where Sybase is installed.
JDBC JAR path	IA_AE_DB_SYB_JCONN_PATH	The jconn JAR file location in the Sybase installation directory. The path should include the file name as well.   <b>Example:</b> opt/jconn16/jconn4.jar   If you specify the path in the Sybase JDBC Jar File Path field, the NATIVEJDBCJARPAT environment variable is added to the autosys.sh.hostname, autosys.csh.hostname, autosys.ksh.hostname, and autosys.bash.hostname files in the \$AUTOUSER directory and its value is set to the value specified in the Sybase JDBC Jar File Path field.

## Primary Admin Details

The following information is required on the Primary Admin Details panel when you select Sybase during the installation procedure.

**NOTE**

- If you are using a dual server configuration, a Secondary Admin Details window will appear. The fields mentioned here also apply to the Secondary Admin Details window.
- The silent variable for Secondary Admin Details will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ADMIN\_USER, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ADMIN\_USER.
- In the case of Sybase Kerberos, the ticket available for the session is used to log in and that should map to the user that has the admin privileges.

Field name	Silent install variable	Description
Username	IA_AE_PRIMARY_DB_ADMIN_USER	The username of the Sybase server for authentication.
Password	IA_AE_PRIMARY_DB_ADMIN_PWD	The password of the Sybase server for authentication.

**Primary Data Device Information**

The following information is required on the Primary Data Device Information panel when you select Sybase during the installation procedure.

**NOTE**

- If you are using a dual server configuration, a Secondary Data Device Information window will appear. The fields mentioned here also apply to the Secondary Data Device Information window.
- The silent variable for Secondary Data Device Information will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SYB\_DATA\_DEVICE\_NAME, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SYB\_DATA\_DEVICE\_NAME.

Field name	Silent install variable	Description
Data device name	IA_AE_PRIMARY_DB_SYB_DATA_DEVICE_NAME	Enter a name for the data device. By default, the name is AEDB_DATA.
Data device size (in MB)	IA_AE_PRIMARY_DB_SYB_DATA_DEVICE_SIZE	The size of the data device in MB. By default, the size is 800 MB.
Data device directory	IA_AE_PRIMARY_DB_SYB_DATA_DEVICE_PATH	The directory location to store the data device.
Create a different device for Log details	IA_AE_PRIMARY_DB_SYB_USE_LOG	Select this checkbox if you want to create a different device for log details.

**Primary Log Device Information**

The following information is required on the Primary Log Device Information panel when you select Sybase during the installation procedure.

**Notes:**

- If you are using a dual server configuration, a Secondary Log Device Information window will appear. The fields mentioned here also apply to the Secondary Log Device Information window.
- The silent variable for Secondary Log Device Information will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SYB\_LOG\_DEVICE\_NAME, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SYB\_LOG\_DEVICE\_NAME.

Field name	Silent install variable	Description
Log device name	IA_AE_PRIMARY_DB_SYB_LOG_DEVICE_NAME	Enter a name for the log device. By default, the name is AEDB_LOG.
Log device size (in MB)	IA_AE_PRIMARY_DB_SYB_LOG_DEVICE_SIZE	The size of the data device in MB. By default, the size is 100 MB.
Log device directory	IA_AE_PRIMARY_DB_SYB_LOG_DEVICE_PATH	The directory location to store the log device.

**AutoSys Database User Details**

The following information is required on the AutoSys Database User Details panel when you select Sybase during the installation procedure.

**NOTE**

In the case of the application user, the ticket available for the user of the services will be used.

Field name	Silent install variable	Description
Database username	IA_AE_DB_USER	The username for the AutoSys database.
Password	IA_AE_DB_USER_PWD	The password of AutoSys database user.
Verify password	NA	Enter the same password as entered above.

**Component Panels**

This section provides reference information for the values and properties that you define for components such as CA EEM and the agent during fresh installation, upgrade, or maintenance procedures. This section contains the following topics:

**Instance Level Security**

The following information is required on the Instance Level Security panel during the installation procedure.



**NOTE**

This panel appears when you are installing the Application Server and web services are not selected.

Field name	Silent install variable	Description
Native security	IA_AE_INSTANCE_SECURITY_MODE=NATIVE	Select this checkbox if you want to use the native security mode for AutoSys installation. For more information, see <a href="#">Native Security Mode</a> .
CA EEM security	IA_AE_INSTANCE_SECURITY_MODE=EEM	Select this checkbox if you want to use CA EEM security. For more information, refer to <a href="#">CA EEM Security</a> .
Use local CA EEM server	IA_AE_USE_LOCAL_EEM_AS_INSTANCE_EEM	Select this checkbox if you want to use an already configured local CA EEM server. This field appears only when you select EEM installation along with the Application Server.

## Instance Level CA EEM Security

The following information is required on the Instance Level CA EEM Security panel during the installation procedure.

Field name	Silent install variable	Description
CA EEM server(s)	IA_AE_INSTANCE_EEM_SERVER_NAME	Specify the comma-separated hostnames of the EEM servers installed.
CA EEM administrator	NA	The EEM administrator username EiamAdmin appears as the label.
Password	IA_AE_INSTANCE_EEM_PASSWORD	Specify the password to access your EEM server.
Use the same CA EEM server(s) for Web UI	IA_AE_INSTANCE_EEM_FOR_WCC	Select this checkbox if you want to use the above mentioned EEM server for Web UI. This checkbox is shown only when the Web UI component is selected.

## CA EEM Configuration

The following information is required on the CA EEM Configuration panel during the installation procedure.

Field name	Silent install variable	Description
CA EEM administrator	NA	The EEM administrator username EiamAdmin appears as the label.
EEM host FQDN name	IA_AE_EEM_FQDN_NAME	The fully qualified domain name of the EEM server or host.
Password	IA_AE_EEM_PASSWORD	Set the password for the EEM administrator.
Verify password	NA	Enter the same password that you entered above.
Data DSA port	IA_AE_DSA_PORT	Enter the DSA port number. By default, the port number is 509.
DB size	IA_AE_EEM_DB_SIZE	Enter the database size in MB for EEM. By default, the database size is 256 MB.

Use the same CA EEM server for instance	IA_AE_EEM_USE_SAME_FOR_INST	Select this checkbox if you want to use the above mentioned EEM server for this instance. This field is shown when the Rest web server is selected.
Use the same CA EEM server for Web UI	IA_AE_EEM_USE_SAME_FOR_WCC	Select this checkbox if you want to use the above mentioned EEM server for Web UI as well. This field is shown when Web UI is selected.

## Scheduler Properties

The following information is required on the Scheduler Properties panel during the installation procedure. For more information, see [Configure a Scheduler](#).

Field name	Silent install variable	Description
Auxiliary port	IA_AE_SCHEDULER_AUX_PORT	The auxiliary port that is to be used for communication between AutoSys and Scheduler. By default, the port is set to 7507.
Configure high availability	IA_AE_CONFIGURE_HA=<0/1>	Select to configure the scheduler for High Availability.
Scheduler role	IA_AE_SCHED_ROLE = PRIMARY/ SHADOW/TIEBREAKER	TIEBREAKER - applicable only when installing in dual database mode.

## Application Server Information

The following information is required on the Application Server Information panel during the installation procedure. For more information, see [Configure an Application Server](#).

Field name	Silent install variable	Description
Application server host(s)	IA_AE_APP_SERVER_HOSTNAME	The server location where Application Server is installed.
Port	IA_AE_APP_SERVER_PORT	The Application Server that is to be used for communication between AutoSys and Application server. By default, the port is set to 9000.
Auxiliary port	IA_AE_APP_SERVER_AUX_PORT	The auxiliary port to be used for communication between AutoSys and Application Server. By default, the port is set to 7500.

## Web Services Properties

The following information is required on the Web Services Properties panel during the installation procedure. For more information, see [Web Services](#).

**NOTE**

By default, the stop ports are disabled. If you want to configure them, see [Change the Web Server Shutdown Port Number](#).

Field name	Silent install variable	Description
REST communication port	IA_AE_RESTWS_COMM_PORT	The port that can be used for communication between AutoSys and Web server using REST. By default, the port is set to 9443.
SOAP Communication Port	IA_AE_SOAPWS_COMM_PORT	The port that can be used for communication between AutoSys and Web server using the SOAP protocol. By default, the port is set to 7443.

## Agent Properties

The following information is required on the Agent Properties panel during the installation procedure. For more information, see [Install the Agent](#).

**NOTE**

A temporary directory that is required for the agent installation process. Ensure that you specify the name of an existing directory with at least 500 MB of free space. Custom 588.

Field name	Silent install variable	Description
Agent name	IA_AE_AGENT_NAME	The name of the Agent. By default, the value is set to WA_AGENT.      - This name must match the agentname parameter specified in the agentparm.txt file.    - You can specify the alias name for the agent in the agent_name parameter to configure the alias on AutoSys. For more information about creating an alias for the agent plugin, see the appropriate agent plug-in implementation details in Workload Automation Agents.       <b>Limits:</b>      - Up to 16 characters    - The first character must be a letter    - The name can contain any alphanumeric characters and the special characters @, \$, and _
Enable port-multiplexing	IA_AE_ENABLE_PORT_MUX	Select this checkbox if you want to enable port multiplexing.

Port	IA_AE_AGENT_PORT	The port that can be used for AutoSys and Agent communication. By default, the value is set to 7520.      - If you enable multiplexing without setting a non-default port number, the default changes to 41952.    - This port number must match the communication.inputport parameter in the agentparm.txt file
Enable SNMP capabilities	NA	Select this checkbox if you want to enable SNMP capabilities.   For more information, see <a href="#">SNMP Jobs Overview</a>.

## SNMP Information

The following information is required on the SNMP Information panel during the installation procedure.

Field name	Silent install variable	Description
Control the agent using SNMP	IA_AE_CONTROL_BY_SNMP	Select this checkbox if you want to control the agent using an SNMP. For more information, see <a href="#">SNMP Job Attributes with Default Values</a> .
SNMP hostname	IA_AE_SNMP_HOSTNAME	Enter the hostname or IP address of the network device. For more information, see <a href="#">snmp_host Attribute -- Specify an SNMP Host Name or IP Address</a> .
SNMP control port	IA_AE_SNMP_CONTROL_PORT	Enter the port of the network device. For more information, see <a href="#">port Attribute -- Specify an SNMP Port</a>.   The SNMP control port. The default value is 161.
Enable support for the SNMP job type	IA_AE_ENABLE_SNMPJOB_TYPE	Select this checkbox if you want to enable support for the SNMP job type. For more information, see <a href="#">SNMP Jobs Overview</a> .
SNMP trap listener port	IA_AE_SNMP_TRAP_LISTENER_PORT	The SNMP trap listener port. For more information, see <a href="#">SnmpManagerHosts and SnmpCommunity</a>.   The SNMP trap listener port. The default value is 162.

## Agent And Application Server Port

The following information is required on the Agent and Application Server Port panel during the installation procedure. For more information, see [Configure an Application Server](#).

Field name	Silent install variable	Description
Port(s)	IA_AE_AGENT_APP_SERVER_PORT	The Application Server that is to be used for communication between AutoSys and Application server. By default, the port is set to 9000.

## CA EEM Security for Web UI

The following information is required on the CA EEM Security for Web UI panel during the installation procedure. For more information, see [CA EEM and Security](#).

### NOTE

This screen appears only when installing the Web UI alone.

Field name	Silent install variable	Description
CA EEM server	IA_AE_WCC_EEM_SERVER_NAME	The EEM server location.
CA EEM administrator	NA	The EEM administrator username.
Administrator password	IA_AE_WCC_EEM_PASSWORD	The EEM administrator password.

## Web UI Panels

This section provides reference information for the values and properties that you define for the Web UI during fresh installation, upgrade, or maintenance procedures.

### Web UI Database Type

The following information is required on the Web UI Database Properties panel during the installation procedure.

### NOTE

- If you specify a database that is used by another Web UI instance, the option to install Web UI in High Availability mode appears.
- By default, the internal database is selected for the Web UI database. Alternatively, you can select an existing supported database.
- The Host Name, Database Name, User, tnsnames.ora.path, and Oracle Service ID fields apply to an external database only.
- The Confirm Password field applies to the internal database only.
- For an external database, the database and user must exist, and the user must have Connect, Read/Select, Write, and Insert permissions.
- The user must have Grant, Commit, and Execute permissions for the Web UI database and its objects.

Field name	Silent install variable	Description
Database type	IA_AE_WCC_DATABASE_TYPE	Select the database type. We currently support MS SQL, Oracle, and Derby.
Use the reporting module for Web UI	IA_AE_WCC_USE_REPORT_DATABASE	Select this checkbox if you want to use the same database for the reporting module and Web UI.
Create or refresh the database schema	IA_AE_WCC_CREATE_DATABASE	Select this checkbox if you want to create a new database schema or if you want to refresh an existing one.
Transparent Network Substrate	IA_AE_WCC_DB_ORA_USE_TNS	Select this checkbox if you want to use the TNS. For more information, click <a href="#">here</a> . This field is available only for the Oracle database.

Use Oracle public key infrastructure (PKI) external authentication	IA_AE_WCC_DB_USE_EXTERNAL_AUTH	Select this checkbox if you want to enable the public key infrastructure authentication. This field is available only for the Oracle database.
Storage management	IA_AE_WCC_DB_ORA_STORAGE_TYPE	Select the desired storage management. This field is available only for the Oracle database.
Create the tablespace, database user, and roles	IA_AE_WCC_DB_ORA_CREATE_TBL_SPACE	Select this checkbox if you want to create a tablespace, database, and roles in Oracle. This field is available only for the Oracle database.

## Web UI Database Properties

The following information is required on the Web UI Database Properties panel during the installation procedure.

### NOTE

- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_DB\_SERVERNAME, then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_DB\_SERVERNAME.
- For the Oracle database, the Web UI database owner account needs to be assigned the SELECT\_CATALOG\_ROLE role.

Field name	Silent install variable	Description
Server name	IA_AE_WCC_DB_SERVERNAME	The server name where the selected database type is present. This field is available only for the MSSQL database.
Database name	IA_AE_WCC_DB_NAME	The database name where you want to store the Web UI related details.
Service name	IA_AE_WCC_DB_ORA_SERVICENAME	The Oracle service name that you want to connect to.     - This field is available only for the Oracle database and if you are not using TNS.    - This field is non-editable during an upgrade.
TNS alias name	IA_AE_WCC_DB_ORA_TNSALIAS	The TNS alias name.     - This field is available only when you are using TNS.    - This field is non-editable during an upgrade.
Database host	IA_AE_WCC_DB_HOSTNAME	The database host location where you want to store the Web UI related details. This field is available only for the Oracle database.

Database TCP port	IA_AE_WCC_DB_PORT	The TCP port that is to be used for communication with the Web UI server. <b>Default values:</b>     - MSSQL server: 1433    - Oracle: 1521    - Derby: 1527
Use same database as reporting database	IA_AE_WCC_USE_SAME_DB_FOR_REPORTING	Select this checkbox if you want to use this database configuration (Web UI database) for the reporting database as well.
Use same server for reporting database	IA_AE_WCC_USE_SAME_DB_FOR_REPORTING	Select this checkbox if you want to use this configuration for the reporting module of Web UI.
Oracle wallet Support JAR directory	IA_AE_DB_ORA_WALLET_JAR_PATH	The directory path where you have all the Oracle wallet JAR files. This field appears only when you use Oracle as the database.

## Web UI Admin Details

The following information is required on the Web UI Admin Details panel during the installation procedure.

### NOTE

The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_DB\_ADMIN\_USER, then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_DB\_ADMIN\_USER.

Field name	Silent install variable	Description
Username	IA_AE_WCC_DB_ADMIN_USER	Enter the username of the database server.
Password	IA_AE_WCC_DB_ADMIN_PWD	Enter the password of the database server.
Use same user credentials for reporting database	IA_AE_WCC_USE_SAME_CRED_FOR_REPORTING	Select this checkbox if you want to use the same credentials for the reporting database and the Web UI database.
TNS admin directory	IA_AE_WCC_DB_ORA_ADMIN_USER_TNS_ADMIN	The TNS admin directory path. This field will appear only for Oracle.
Oracle wallet SSO file path	IA_AE_WCC_DB_ORA_ADMIN_USER_WALLET_LOC	The Oracle wallet SSO file path. This field will appear only for Oracle.
DN matching	IA_AE_DB_ORA_USER_DN_MATCH	Select this checkbox if you want the distinguished name to match the service name. If the name matches, the database connection succeeds. Otherwise, the connection fails.

## Web UI Schema Admin Details

The following information is required on the Web UI Schema Admin Details panel when you select Oracle during the installation procedure.

**NOTE**

- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_DB\_ORA\_SCHEMA\_ADMIN\_USER , then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_DB\_ORA\_SCHEMA\_ADMIN\_USER .
- This panel is available only for the Oracle database.

Field name	Silent install variable	Description
Schema owner name	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_USER	Enter a name to create a Web UI database schema admin.
Password	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_PWD	Enter the password for the owner's name entered above.
Verify password	NA	Re-enter the password entered above.

**Web UI Data And Index Tablespace Information**

The following information is required on the Web UI Data And Index Tablespace Information panel during the installation procedure.

**NOTE**

- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_DB\_ORA\_DATA\_TBLSPACE\_NAME, then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_DB\_ORA\_DATA\_TBLSPACE\_NAME.
- This panel is available only for the Oracle database.

Field name	Silent install variable	Description
Data tablespace name	IA_AE_WCC_DB_ORA_DATA_TBLSPACE_NAME	The data tablespace file name.
Data tablespace size (in MB)	IA_AE_WCC_DB_ORA_DATA_TBLSPACE_SIZE	The size of the data tablespace file name in MB. By default, the value is set to 800 MB.
Data tablespace directory	IA_AE_WCC_DB_ORA_DATA_TBLSPACE_PATH	The directory location where the data tablespace file is to be stored.
Index tablespace name	IA_AE_WCC_DB_ORA_INDEX_TBLSPACE_NAME	The index tablespace file name.
Index tablespace size (in MB)	IA_AE_WCC_DB_ORA_INDEX_TBLSPACE_SIZE	The size of the index tablespace file name in MB. By default, the value is set to 80 MB.



Index tablespace directory	IA_AE_WCC_DB_ORA_INDEX_TBSLSPACE_PATH	The directory location where the index tablespace file is to be stored.
Index diskgroup	IA_AE_WCC_DB_ORA_INDEX_DISKGROUP_NAME	The index diskgroup.

## Web UI Database User Details

The following information is required on the Web UI Database User Details panel during the installation procedure.

### NOTE

The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_DB\_USER, then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_DB\_USER.

Field name	Silent install variable	Description
Database username	IA_AE_WCC_DB_USER	The username that is to be created for the Web UI database.
Password	IA_AE_WCC_DB_USER_PWD	The password for the username you entered above.
Verify password	NA	Re-enter the password you entered above.

## Web UI Reporting Database Properties

The following information is required on the Web UI Reporting Database Properties panel during the installation procedure.

### NOTE

- You can select the Web UI database or an existing supported database for Reporting.
- Reporting database properties cannot be modified when you select Web UI for the Reporting database or if you opt to install Web UI in High Availability mode.
- When installing Web UI in High Availability mode and the Reporting database is Web UI, Microsoft SQL, and Oracle, database properties cannot be modified. However, if the Reporting database is Oracle TNS, you are prompted to specify the tnsnames.ora file name and path, and the Oracle Service ID.
- The Host Name, Database Name, User, tnsnames.ora.path, and Oracle Service ID fields apply to an external database only.
- The Confirm Password field applies to the internal database only.
- For an external database, the database and user must exist, and the user must have Connect, Read/Select, Write, and Insert permissions.
- The user must have Grant, Commit, and Execute permissions for the Web UI database and its objects.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_WCC\_REPORT\_DB\_SERVERNAME, then for the secondary server, it will be IA\_AE\_SECONDARY\_WCC\_REPORT\_DB\_SERVERNAME.

Field name	Silent Install Variable	Description
Server name	IA_AE_WCC_REPORT_DB_SERVERNAME	The server where the Web UI reporting database is present.

Service name	IA_AE_WCC_REPORT_DB_ORA_SERVICENAME	The Oracle service name that you want to connect to. This field will appear for Oracle and if TNS is not being used.
TNS alias name	IA_AE_WCC_REPORT_DB_ORA_TNSALIAS	The TNS alias name. This field will appear in Oracle and if you are using the TNS.
Database name	IA_AE_WCC_REPORT_DB_NAME	The database name that is to be created in the database server.
Database TCP port	IA_AE_WCC_REPORT_DB_PORT	The TCP port that the AUTOSYS can use to communicate with the Web UI reporting database.
Database host	IA_AE_WCC_REPORT_DB_HOSTNAME	The server where the Web UI reporting database is present. This field is available only for the Oracle database.

## Web UI Reporting Admin User Details

The following information is required on the Web UI Reporting Admin User Details panel during the installation procedure.

### NOTE

The silent variable for Secondary Event Server will have the word **SECONDARY** instead of **PRIMARY**. For example, for your primary server, if the silent variable is **IA\_AE\_PRIMARY\_WCC\_REPORT\_DB\_ADMIN\_USER**, then for the secondary server, it will be **IA\_AE\_SECONDARY\_WCC\_REPORT\_DB\_ADMIN\_USER**.

Field name	Silent install variable	Description
Username	IA_AE_WCC_REPORT_DB_ADMIN_USER	Enter the username of the database server. For Oracle, enter the username that has DBA ROLE, DBA TABLESPACE SELECT, V\$PARAMETER SELECT and CREATE SESSION authority.
Password	IA_AE_WCC_REPORT_DB_ADMIN_PWD	Enter the password of the database server.

## Web UI Reporting Schema Admin Details

The following information is required on the Web UI Reporting Schema Admin Details panel during the installation procedure.

### NOTE

- The silent variable for Secondary Event Server will have the word **SECONDARY** instead of **PRIMARY**. For example, for your primary server, if the silent variable

is `IA_AE_PRIMARY_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_USER`, then for the secondary server, it will be `IA_AE_SECONDARY_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_USER`.

- This panel is available only for the Oracle database.

Field name	Silent install variable	Description
Schema owner name	<code>IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_USER</code>	Enter a name to create a Web UI reporting database schema admin.
Password	<code>IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_PWD</code>	Enter a password for the owner's name entered above.
Verify password	NA	Re-enter the password entered above.

## Reporting Data And Index Tablespace Information

The following information is required on the Reporting Data And Index Tablespace Information panel during the installation procedure.

### NOTE

- The silent variable for Secondary Event Server will have the word `SECONDARY` instead of `PRIMARY`. For example, for your primary server, if the silent variable is `IA_AE_PRIMARY_WCC_REPORT_DB_ORA_DATA_TBLSPACE_NAME`, then for the secondary server, it will be `IA_AE_SECONDARY_WCC_REPORT_DB_ORA_DATA_TBLSPACE_NAME`.
- This panel is available only for the Oracle database.

Field name	Silent install variable	Description
Data tablespace name	<code>IA_AE_WCC_REPORT_DB_ORA_DATA_TBLSPACE_NAME</code>	The data tablespace name for reporting. By default, the value is <code>WCC_DATA</code> . A tablespace file is created with this name in the data tablespace directory.
Data tablespace size (in MB)	<code>IA_AE_WCC_REPORT_DB_ORA_DATA_TBLSPACE_SIZE</code>	The tablespace file size in MB. By default, the file size is 800 MB.
Data tablespace directory	<code>IA_AE_WCC_REPORT_DB_ORA_DATA_TBLSPACE_PATH</code>	The data tablespace directory where the file is created or stored.
Index tablespace name	<code>IA_AE_WCC_REPORT_DB_ORA_INDEX_TBLSPACE_NAME</code>	The index tablespace name for reporting. By default, the value is <code>WCC_INDEX</code> . An index tablespace file is created with this name in the index tablespace directory.
Index tablespace size (in MB)	<code>IA_AE_WCC_REPORT_DB_ORA_INDEX_TBLSPACE_SIZE</code>	The index tablespace file size in MB. By default, the file size is 80 MB.
Index tablespace directory	<code>IA_AE_WCC_REPORT_DB_ORA_INDEX_TBLSPACE_PATH</code>	The index tablespace directory where the file is created or stored.

## CA EEM Security for Web UI

The following information is required on the CA EEM Security for Web UI panel during the installation procedure. For more information, see [CA EEM and Security](#).

**NOTE**

- If CA WCC is being installed in High Availability mode, you cannot specify a different instance of CA EEM; however, you must specify the password for the EiamAdmin user.
- If CA EEM is configured for failover, you should only specify the primary server. The installer will automatically retrieve all other secondary servers and configure Web UI accordingly.
- Default port numbers are provided. If the default value for a port is already assigned to another function in your environment, you are prompted to specify a different port value.
- When you change a port value, the value you enter becomes the new default value.
- The Use SSL option toggles the Tomcat port value between the SSL value (with Use SSL selected) and the non-SSL value (with Use SSL not selected).
- SSL is not available for the Services Tomcat.

Field name	Silent install variable	Description
CA EEM server	IA_AE_WCC_EEM_SERVER_NAME	The EEM server location.
CA EEM administrator	NA	The EEM administrator username.
Administrator password	IA_AE_WCC_EEM_PASSWORD	The EEM administrator password.

**Web UI Port Configuration**

The following information is required on the Web UI Port Configuration panel during the installation procedure.

**NOTE**

The fields will vary depending on the component selected.

Field name	Silent install variable	Description
Web UI collector name	IA_AE_WCC_COLLECTOR_NAME	Enter a name for the Web UI Collector. For more information, refer to <a href="#">Web UI Collector</a>.       • This field will be editable if you are performing a fresh install of AutoSys or are maintaining AutoSys.     • The collector name specified here is appended to the service name. For example, if you specify the name as "MyCollector", its service name will be "CA-wcc-services.MyCollector". However, this does not stand true for the default collector that gets installed during the Web UI installation. The default collector name is "CA-wcc-services".
Communication port	IA_AE_WCC_COMM_PORT	The communication port to be used between AutoSys and Web UI.

Collector event bus port	IA_AE_WCC_COLLECTOR_EVENTBUS_PORT	The event bus port to be used between AutoSys and Web UI Collector. By default, it is set to 7004.      - During upgrade, the port that is already configured for CA-wcc-services is released. And the port that was configured was event bus (default 7004) will be used as the collector event bus port.    - This field will appear as a label during AutoSys upgrade.
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## Web UI Database Schema Options

The following information is required.

Field name	Silent install variable	Description
Web UI application database schema	IA_AE_WCC_APPLICATION_DB_SCHEMA_CREATION	Select this option if you want to create a database schema for the Web UI application.
Reporting database schema	IA_AE_WCC_REPORTING_DB_SCHEMA_CREATION	Select this option if you want to create a database schema for reporting.

## AutoSys Install Miscellaneous Panels

This section provides reference information for values and properties that you define on various panels during fresh installation, upgrade, or maintenance procedures that are not covered in other sections under [Installation Panels](#).

### Choose Java Virtual Machine

The following information is required on the Choose Java Virtual Machine panel during the installation procedure.

#### NOTE

Adopt OpenJDK is distributed with the AutoSys installer. However, you can use Oracle JRE if you have the license.

Field name	Silent Install variable	Description
Use the Java VM installed with the application	IA_AE_USE_BUNDLED_JVM	Select this option if you want to use the Java VM that is distributed with the AutoSys installer.
Choose a Java VM already installed on this system	IA_AE_JAVA_DOT	Select this option if you want to use an already installed Java VM.
Choose Java Executable	IA_AE_JAVA_EXE	Click this button if you want to select a Java executable file.

## Components to Install

You can select one or more components to install. The dependent components, if any, are selected automatically.

Field name	Silent Variable	Description
Scheduler	IA_AE_SELECTED_SCHEDULER	Select this checkbox if you want to install the Scheduler on a machine.
Application Server	IA_AE_SELECTED_APPLICATIONSERVER	Select this checkbox if you want to install the Application Server on a machine.
Client	IA_AE_SELECTED_CLIENT	Select this checkbox if you want to install the Client on a machine.
Agent	IA_AE_SELECTED_AGENT	Select this checkbox if you want to install the Agent on a machine.
REST Web Service	IA_AE_SELECTED_RESTWS	Select this checkbox if you want to configure the REST web Service on a machine.
SOAP Web Service	IA_AE_SELECTED_SOAPWS	Select this checkbox if you want to configure the SOAP web service on a machine.
Workload Control Center (Web UI)	IA_AE_SELECTED_WCC	Select this checkbox if you want to install the Web UI on a machine.
Software Development Kit (SDK)	IA_AE_SELECTED_SDK	Select this checkbox if you want to install the Software Development Kit on a machine.
CA Embedded Entitlements Manager (CA EEM)	IA_AE_SELECTED_EEM	Select this checkbox if you want to install the CA Embedded Entitlements Manager on a machine.
Secure Socket Adapter (SSA)	IA_AE_SELECTED_SSA	Select this checkbox if you want to install the Secure Socket Adapter on a machine.

## Installation Options

The following options are available on the Installation Options panel during the installation procedure.

Field name	Silent install variable	Description
Install the product		Select this option if you want to install AutoSys or any of its components.
Create or refresh the database schema		Select this option if you want to create or refresh the database schema.

## Instance Information

The following information is required on the Instance Information panel during the installation procedure.

Field name	Silent Install Variable	Description
Instance name	IA_AE_INSTANCE_NAME	The AutoSys instance ID, which is an uppercase, three-character alphanumeric name that identifies a specific installation of AutoSys. The default instance ID is ACE. The first character is represented by an alpha character (A-Z). The remaining characters can be alphanumeric (A-Z,0-9).

## Installation Path

The following information is required on the Installation Path panel during the installation procedure.

Field name	Silent Install Variable	Description
Installation path	USER_INSTALL_DIR	The installation path for AutoSys Workload Automation. The default installation path is C:\Program Files\CA\WorkloadAutomationAE. This field will appear as a label if Version 12.0 agents are already present.
Shared components installation path	SC_INSTALL_DIR	The installation path for shared components. The default installation path is C:\Program Files\CA\SC.     - CA EEM, SSA, Command Sponsor, and CAPKI are the shared components that are installed in the shared components installation path.    - This field will appear as a label if shared components are already present.

## Instance Level Data Encryption

The following information is required on the Instance Level Data Encryption panel during the installation procedure.

Field name	Silent Install Variable	Description
Encrypt data using a user-specified key	IA_AE_ENCRYPTION_KEY_INSTANCE	Select this checkbox if you want to encrypt data using a user-specified key. If you select to encrypt the data using a user-specified key, you must also specify the key format and key contents. A passphrase format specifies 1-16 alphanumeric characters. A hexadecimal format specifies exactly 32 alphanumeric characters, consisting of 0-9 or A-F.     - In the case of silent install, if this variable value is empty, it implies use the default encryption.    - If the <b>Encrypt Data Using a User specified Key</b> checkbox is not selected, the default key is used for data encryption.      For more information, see <a href="#">FIPS Encryption</a> .
Use the instance-level encryption values as agent encryption values.	NA	Select this checkbox if you want to use the instance-level data encryption configuration for agent-level data encryption as well. This field is displayed when the Agent is selected.

## Agent Level Data Encryption

The following information is required on the Agent Level Data Encryption panel during the installation procedure.

Field name	Silent Install Variable	Description
Encrypt data using a user-specified key	IA_AE_ENCRYPTION_KEY_AGENT	Select this checkbox if you want to encrypt data using a user-specified key.   If you select to encrypt the data using a user-specified key, you must also specify the key format and key contents.   A passphrase format specifies 1-16 alphanumeric characters.   A hexadecimal format specifies exactly 32 alphanumeric characters, consisting of 0-9 or A-F.       • If the <b>Encrypt Data Using a User specified Key</b> checkbox is not selected, the default key is used for data encryption.     • In the case of silent install, if this variable value is empty it implies, use the default encryption.       For more information, see <a href="#">FIPS Encryption</a>.

## FIPS Encryption

The following information is required on the FIPS Encryption panel during the installation procedure.

Field name	Silent Install Variable	Description
FIPS 140-2 compliance	IA_AE_IS_FIPS_ENABLED	Select this checkbox to use FIPS 140-2 certified cryptographic module for encryption. For more information, see <a href="#">FIPS Mode</a>.

## Services Startup Options

You can use the Services Startup Options window during the installation procedure to select the services that you want to start.

Select checkboxes under the **Start post-installation** column to start the service immediately after the AutoSys installation is complete. You can also select checkboxes under the **Start on system startup** column to start the AutoSys services after your system is restarted. By default, services under the **Start post-installation** column are unchecked and the services under the **Start on system startup** column are checked.

Field name	Silent Variable	Description
Services	NA	Lists all the services that be started post-installation or on system startup.



Start post-installation	<b>IA_AE_SERVICE_START_ONINSTALL</b> // Set this variable to YES to start the services post-installation. Otherwise, set to NO. In addition, you can also set it to CUSTOM. // When CUSTOM is specified, the value specified for <COMPONENT> is used to start the services post-installation. <b>IA_AE_START_SERVICE_&lt;INSTANCE&gt;_&lt;COMPONENT&gt;</b> // Set this variable to 1 to start the services post-installation. Otherwise, set to 0.	Select this checkbox to start the selected services post-installation.
Start on system startup	<b>IA_AE_SERVICE_START_ON_SYSTEM_START</b> // Set this variable to YES to start the services post-installation. Otherwise, set to NO. In addition, you can also set it to CUSTOM. // When CUSTOM is specified, the value specified for <COMPONENT> is used to start the services post-installation. <b>IA_AE_AUTOSTART_SERVICE_&lt;INSTANCE&gt;_&lt;COMPONENT&gt;</b> // Set this variable to 1 to start the services post-installation. Otherwise, set to 0.	Select this checkbox to start the selected services on the system startup.

The following are a few examples of how the **IA\_AE\_START\_SERVICE\_<INSTANCE>\_<COMPONENT>** variable will look if your AutoSys instance name is ACE and if you may want to select specific services that you want to start post-installation.

Components	<b>IA_AE_START_SERVICE_&lt;INSTANCE&gt;_&lt;COMPONENT&gt;</b>
Scheduler	<b>IA_AE_START_SERVICE_ACE_SCHEDULER</b>
Application Server	<b>IA_AE_START_SERVICE_ACE_APPLICATIONSERVER</b>
REST Web Service	<b>IA_AE_START_SERVICE_ACE_RESTWS</b>
Agent	<b>IA_AE_START_SERVICE_WA_AGENT_AGENT</b>
WCC	<b>IA_AE_START_SERVICE_WCC</b>
Collector	<b>IA_AE_START_SERVICE_WCC_COLLECTOR_WCC_COLLECTOR</b>
SOAP Web Service	<b>IA_AE_START_SERVICE_SOAPWS</b>

Similarly, you modify the **IA\_AE\_AUTOSTART\_SERVICE\_<INSTANCE>\_<COMPONENT>** variable.

## Activate your product

You can activate your product during the installation procedure by entering the company domain, enterprise site ID, and internal identifier, which is optional. Select if you are installing AutoSys as per the Portfolio License Agreement.

Field name	Silent Install Variable	Description
Instance name		The instance name where the product will be activated. This field appears only when you are upgrading AutoSys.
Is this instance install or upgrade related to new or additional planned usage under a Portfolio License Agreement?	<b>IA_AE_TELE_PLA_APPLICABLE_&lt;INSTANCE&gt;</b>	Details of whether the AutoSys installation or upgrade is a new or additional planned usage under the portfolio license agreement (PLA) with Broadcom. Select Yes or No based on the PLA agreement.

Company Domain	IA_AE_PLA_COMPANY_DOMAIN_<INSTANCE>	Your company's domain name (the last part of your company's email address). For example, Broadcom.com.
Enterprise Site ID	IA_AE_PLA_ENTERPRISE_ID_<INSTANCE>	Your company's site ID that is listed on the Broadcom's Support portal.
Internal Identifier (optional)	IA_AE_PLA_INTERNAL_ID_<INSTANCE>	Your cost-center or department's non-personally identifiable information (non-PII) identifier that you use for your company's tracking needs.

## Pre-installation Summary

The Pre-installation Summary panel gives a detailed list of entities to be installed and their paths based on your selection during the installation, upgrade, or maintenance process.

## Deregister CA EEM Policies

The Deregister CA EEM Policies panel appears when you are about to uninstall AutoSys Workload Automation, an instance of AutoSys, or Web UI.

### NOTE

- You cannot record the uninstallation procedure.
- For instances that are in native mode, this panel will not appear.
- For instances and Web UI for which the policies are already up to date, this panel will appear only during an upgrade.

Field name	Silent Uninstall Variable	Description
Instance (checkbox)	IA_AE_INSTANCE_EEM_DELETE_POLICIES_FOR_<INSTANCE_NAME> // Set the variable to one to select the instance. Otherwise, set to zero. // In the silent way, if uninstalling, you will have to populate this variable for every AutoSys instance. // <b>Note:</b> If you are using Console to uninstall AutoSys, for multiple AutoSys instances that are using the same EEM server, the password field will appear only once.	Select the instance that you want to deregister.
EiamAdmin Password (Instance)	IA_AE_EEM_USER_PASSWORD_FOR_<INSTANCE_NAME> = <Password_encrypted_AES_fomat> // The specified password should be AES encrypted. // You will have to populate this variable for every AutoSys instance.	The EiamAdmin password for the instance you selected above.
Web UI (checkbox)	IA_AE_WCC_EEM_DELETE_POLICIES // Set the variable to one to select the Web UI. Otherwise, set to zero.	Select the instance that you want to deregister.
EiamAdmin Password (Web UI)	IA_AE_WCC_EEM_PASSWORD= <Password_encrypted_AES_fomat> // The specified password should be AES encrypted.	The EiamAdmin password for the Web UI you selected above.

## Database Schema Options

The following information is required.

Field name	Silent install variable	Description
Instance database schema	IA_AE_SCHEMA_TYPE	Select this option if you want to create a database for the AutoSys instance.
Web UI database schema		Select this option if you want to create a database for the Web UI application and the reporting database.

## Cluster Resource Group

The Cluster Resource Group panel only appears when you are installing a Scheduler, Application Server, or both.

### Windows

The following information is required on Windows.

Field name	Silent install variable	Description
Configure the instance for use in clustered environment	IA_AE_USE_CLUSTER =< 0/1> // Set to "1" to configure the instance in a cluster environment. Otherwise, set to one.	Select this option if you want to use a cluster environment for this instance.
Resource groups	IA_AE_CLUSTER_GROUP_NAME	Select a resource group that is to be used to configure this instance in a cluster environment.

### Unix

The following information is required on Unix.

Field name	Silent install variable	Description
Owner (Default: autosys)	IA_AE_OWNER	Specify the owner name to be assigned to AutoSys Workload Automation files.
Group (Default: autosys)	IA_AE_GROUP	Specify the group name to be assigned to AutoSys Workload Automation files.

## Installation Complete

The Installation Complete panel appears when the installation has finished. It displays an overview of what has been installed as part of the installation. You must click **Done** on this panel to complete the installation.

Field name	Silent install variable	Description
Yes, restart my system	NA	Select this option if you want to restart your system immediately.
No, I will restart my system myself	NA	Select this option if you want to restart your system later.

## Miscellaneous Topics

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### Set Up the Database

The AutoSys Workload Automation server requires a database. By default, the AutoSys Workload Automation installer creates the tablespaces (for Oracle) or database devices (for Microsoft SQL Server or Sybase) and schema objects. However, by using the AutoSys Workload Automation silent installer, you can bypass the database setup during the AutoSys Workload Automation server installation and then create a database manually. You run the CreateAEDB.pl Perl script to create the tablespaces or database devices and schema objects that form the Event server for AutoSys Workload Automation. You can run the script in the interactive or console mode. Or, you can use the installer option "Create the database schema" to create the database.

### Create Instance Database or Web UI Database

#### Create Instance Database Schema

To create an instance database:

1. Double-click the install.exe file to open the installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In [Installation Options](#), select **Create the database schema**.
4. In [Database Schema Options](#), select **Instance database schema**.
5. From **Database type**, select Microsoft SQL Server.  
**Note:** You can select any database type that you want. However, for this procedure, we are using MS SQL server. You can fill up the required details in the panels that appear based on your selected database type
6. In [Primary Event Server Properties](#), enter the required details.
7. In [Primary Admin Details](#), select **Microsoft SQL server authentication** and enter the required details.
8. In [AutoSys Database User Details](#), enter the required details.
9. Click **Install**.
10. The [Installation Complete](#) panel displays a summary of what all has been installed.
11. Click **Done**.

#### Create Web UI Database Schema

1. Double-click the install.exe file to open the installation wizard.
2. Click **I accept the terms of the License Agreement**.
3. In [Installation Options](#), select **Create the database schema**.
4. In [Database Schema Options](#), select **Web UI database schema**.
5. In [Web UI Database Schema Options](#), select the desired options.
6. From **Database type**, select Microsoft SQL Server.  
**Note:** You can select any database type that you want. However, for this procedure, we are using MS SQL server. You can fill up the required details in the panels that appear based on your selected database type
7. In [Web UI Database Properties](#), enter the required details.
8. In [Web UI Admin Details](#), select **Microsoft SQL server authentication** and enter the required details.
9. In [Web UI Database User Details](#), enter the required details.
10. Click **Install**.

11. The [Installation Complete](#) panel displays a summary of what all has been installed.
12. Click **Done**.

## CreateAEDB Script-Create a Database

The CreateAEDB script creates the database required by AutoSys Workload Automation. The script creates the tablespaces (for Oracle) and devices (for Microsoft SQL Server or Sybase) and all the schema objects.

A CreateAEDB script is included for each database vendor in the following directories:

On Windows:

- %AUTOSYS%\dbobj\MSQ\CreateAEDB.pl (for Microsoft SQL Server)
- %AUTOSYS%\dbobj\ORA\CreateAEDB.pl (for Oracle)
- %AUTOSYS%\dbobj\SYB\CreateAEDB.pl (for Sybase)

On UNIX:

- \$AUTOSYS/dbobj/ORA/CreateAEDB.pl (for Oracle)
- \$AUTOSYS/dbobj/SYB/CreateAEDB.pl (for Sybase)

### NOTE

You can enter the CreateAEDB script with no options. You are prompted for the required information line by line.

You can also run the CreateAEDB script in console mode. In this case, the script has the following format:

### For Microsoft SQL Server

```
perl CreateAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER"
"ADB_SA_PSWD" "ADB_WA_DB_USER" "ADB_WA_DB_PSWD" "JAVA_HOME" "ADB_OUTDIR" "CREATE_DB"
"ADB_DEBUG"
```

### For Oracle; if using password to access the database

- Windows

```
perl CreateAEDB.pl "ADB_SID" "ADB_SA_USER" "ADB_SA_PSWD" "ADB_ADMIN_USER"
"ADB_ADMIN_PSWD" "ADB_WA_DB_USER" "ADB_WA_DB_PSWD" "JAVA_HOME" "ADB_OUTDIR"
"CREATE_TBLSPC" "ADB_TS_DATANM" "ADB_TS_DATADIR" "ADB_TS_DATASZ" "ADB_TS_IXNM"
"ADB_TS_IXDIR" "ADB_TS_IXSZ" "ADB_DEBUG"
```

- UNIX

```
perl ./CreateAEDB.pl "ADB_SID" "ADB_SA_USER" "ADB_SA_PSWD" "ADB_ADMIN_USER"
"ADB_ADMIN_PSWD" "ADB_WA_DB_USER" "ADB_WA_DB_PSWD" "JAVA_HOME" "ADB_OUTDIR"
"CREATE_TBLSPC" "ADB_TS_DATANM" "ADB_TS_DATADIR" "ADB_TS_DATASZ" "ADB_TS_IXNM"
"ADB_TS_IXDIR" "ADB_TS_IXSZ" "ADB_DEBUG"
```

### For Oracle; if using an SSL certificate to access the database

- Windows

```
perl CreateAEDB.pl -CERTMODE "ADB_SID" "ADB_SYS_TNSADMIN" "ASI_ADBADMIN_USER"
"ADB_AEDBADMIN_TNSADMIN" "ADB_ADMIN_USER" "JAVA_HOME" "ADB_OUTDIR" "CREATE_TBLSPC"
"ADB_TS_DATANM" "ADB_TS_DATADIR" "ADB_TS_DATASZ" "ADB_TS_IXNM" "ADB_TS_IXDIR"
"ADB_TS_IXSZ" "ADB_DEBUG" "ADB_AEDBADMIN_DN" "ADB_AUTOSYS_DN"
```

- UNIX

```
perl ./CreateAEDB.pl -CERTMODE "ADB_SID" "ADB_SYS_TNSADMIN" "ASI_ADBADMIN_USER"
"ADB_AEDBADMIN_TNSADMIN" "ADB_ADMIN_USER" "JAVA_HOME" "ADB_OUTDIR" "CREATE_TBLSPC"
"ADB_TS_DATANM" "ADB_TS_DATADIR" "ADB_TS_DATASZ" "ADB_TS_IXNM" "ADB_TS_IXDIR"
"ADB_TS_IXSZ" "ADB_DEBUG" "ADB_AEDBADMIN_DN" "ADB_AUTOSYS_DN"
```

#### For Sybase, if using password to access the database

- Windows

```
perl CreateAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER"
"ADB_SA_PSWD" "ADB_WA_DB_USER" "ADB_WA_DB_PSWD" "JAVA_HOME" "ADB_OUTDIR" "CREATE_DB"
"ADB_DV_DATADEV" "ADB_DV_DATADIR" "ADB_DV_DATASZ" "ADB_DV_LOGDEV" "ADB_DV_LOGDIR"
"ADB_DV_LOGSZ" "ADB_DEBUG"
```

- UNIX

```
perl ./CreateAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER"
"ADB_SA_PSWD" "ADB_WA_DB_USER" "ADB_WA_DB_PSWD" "JAVA_HOME" "ADB_OUTDIR" "CREATE_DB"
"ADB_DV_DATADEV" "ADB_DV_DATADIR" "ADB_DV_DATASZ" "ADB_DV_LOGDEV" "ADB_DV_LOGDIR"
"ADB_DV_LOGSZ" "ADB_DEBUG"
```

#### For Sybase, if using an Kerberos authentication to access the database

##### NOTE

To use Kerberos authentication:

- Ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which has sa\_role and sso\_role.
- Ensure that the database user is created in the database. Sybase does not let you create a login without the password. So, the createaedb script assigns a default password *autosys* for the database user and modifies the login to enable only Kerberos authentication.

- Windows

```
perl CreateAEDB.pl -EXTAUTH "ADB_DATASERVER" "ADB_DATABASE"
"ADB_SPN" "ADB_WA_DB_USER" "JAVA_HOME" "ADB_OUTDIR" "CREATE_DB" "ADB_DV_DATADEV"
"ADB_DV_DATADIR" "ADB_DV_DATASZ" "ADB_DV_LOGDEV" "ADB_DV_LOGDIR" "ADB_DV_LOGSZ"
"ADB_DEBUG"
```

- UNIX

```
perl ./CreateAEDB.pl -EXTAUTH "ADB_DATASERVER" "ADB_DATABASE"
"ADB_SPN" "ADB_WA_DB_USER" "JAVA_HOME" "ADB_OUTDIR" "CREATE_DB" "ADB_DV_DATADEV"
"ADB_DV_DATADIR" "ADB_DV_DATASZ" "ADB_DV_LOGDEV" "ADB_DV_LOGDIR" "ADB_DV_LOGSZ"
"ADB_DEBUG"
```

- **ADB\_ADMIN\_PSWD** Specifies the Oracle 'schema owner' password. If the schema owner is already defined in Oracle, you must specify the valid password. If the schema owner is not defined in Oracle, the Installer creates the user with the specified password.

##### NOTE

On UNIX, if you specify a password that contains a dollar sign (\$) using the command line, escape the dollar sign and enclose the password in single quotes. For example, 'aedb\\$admin'

- **EXTAUTH** Specifies that Kerberos authentication is used to access the database.

**NOTE**

For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)

- **ADB\_DATABASE** Specifies the Microsoft SQL Server or Sybase database name.
- **ADB\_DATASERVER**
  - Microsoft SQL Server—Specifies the machine name where the Microsoft SQL Server is running.
  - Sybase—Specifies the Sybase server name, as defined in the %SYBASE%\ini\sql.ini file (Windows) or the \$SYBASE/interfaces file (UNIX).
- **ADB\_DEBUG** Indicates whether to set debugging on during the installation, as follows:
  - Y—Sets debugging on
  - N—Disables debugging
- **ADB\_DV\_DATADEV** Specifies the Sybase data device name.
- **ADB\_DV\_DATADIR** Specifies the Sybase data device path.
- **ADB\_DV\_DATASZ** Specifies the Sybase data device size (in MB). The minimum value is 400.
- **ADB\_DV\_LOGDEV** Specifies the Sybase log device name.
- **ADB\_DV\_LOGDIR** Specifies the Sybase log device path.
- **ADB\_DV\_LOGSZ** Specifies the Sybase log device size (in MB). The minimum value is 80.
- **ADB\_OUTDIR** Specifies the directory where you want to store the output from the CreateAEDB script. This directory must already exist and be empty prior to running the CreateAEDB script.
- **ADB\_SA\_PSWD** Specifies the Microsoft SQL Server, Oracle, or Sybase system administrator user password.

**NOTE**

On UNIX, if you specify a password that contains a dollar sign (\$) using the command line, escape the dollar sign and enclose the password in single quotes. For example, 'sy\\$tem'.

- **ADB\_SA\_USER** Specifies the Microsoft SQL Server, Oracle, or Sybase system administrator user ID.
- **CERTMODE** Specifies that an SSL certificate is used to access the database.

**NOTE**

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- **ADB\_SID** Specifies the Oracle SID name.
- **ADB\_SYS\_TNSADMIN** Specifies the TNS\_ADMIN value for the Oracle user with system administrator privileges.
- **ADB\_AEDBADMIN\_TNSADMIN** Specifies the TNS\_ADMIN value for the schema owner.
- **ADB\_TS\_DATADIR** Specifies the Oracle data tablespace directory path. This directory must already exist on the Oracle server.
- **ADB\_TS\_DATANM** Specifies the Oracle data tablespace name.
- **ADB\_TS\_DATASZ** Specifies the Oracle data tablespace size (in MB). The minimum value is 800.
- **ADB\_TS\_IDXDIR** Specifies the Oracle index tablespace directory path. This directory must already exist on the Oracle server.
- **ADB\_TS\_IXNM** Specifies the Oracle index tablespace name.
- **ADB\_TS\_IXSZ** Specifies the Oracle index tablespace size (in MB). The minimum value is 80.
- **ADB\_WA\_DB\_PSWD** Specifies the Microsoft SQL Server, Oracle, or Sybase 'database' user password. If the database user is already defined in Oracle or Sybase, you must specify the valid password. If the database user is not defined in Oracle or Sybase, the installer creates the user with the specified password.

**NOTE**

On UNIX, if you specify a password that contains a dollar sign (\$) using the command line, escape the dollar sign and enclose the password in single quotes. For example, 'auto\\$ys'.

- **CREATE\_DB** Indicates whether to create a database for Sybase, as follows:

- Y—Creates a database
- N—Overwrites the existing database

**NOTE**

If you enter N when you execute the CreateAEDB.pl script on the AutoSys Workload Automation r11.3.5 database, the script upgrades the database to the current release.

- **CREATE\_TBLSPC** Indicates whether to create tablespaces for Oracle, as follows:
  - Y—Creates the data and index tablespaces
  - N—Overwrites the existing tablespaces
- **ADB\_SPN** (Optional) Specifies the service principal name to be used for Kerberos authentication. If you do not want to specify the service principal name, include an empty space for this parameter, for example, " ".
- **JAVA\_HOME** Specifies the JAVA\_HOME path.
- **ADB\_AEDBADMIN\_DN**  
Specifies the distinguished name of the SSL certificate that is associated with the database schema owner (for example, aedbadmin).
- **ADB\_AUTOSYS\_DN**  
Specifies the distinguished name of the SSL certificate that is associated with the database user (autosys).

**NOTE**

If you try to connect to an existing database that is in use with another AutoSys Workload Automation instance, the connection fails. So, we recommend that you do not use the same database across different instances.

**Example: Create a Database on Microsoft SQL Server**

This example creates a Microsoft SQL Server database named AEDB on server LAM04.

```
perl CreateAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "dbuser" "dbuserpw" "C:\Program
Files\CA\Workload Automation AE\JRE_WA" "C:\tmp\adbldog" "Y" "N"
```

**Example: Create Oracle Tablespaces**

This example creates Oracle tablespaces in the Oracle orcl instance. The output of the script is stored in the \tmp\adbldog directory (Windows) or the /tmp/adbldog directory (UNIX). The script creates two Oracle tablespaces; one named AEDB\_DATA that stores up to 800 MB of data, and another named AEDB\_INDEX that stores up to 80 MB of data. The script runs without debugging.

Windows:

```
perl CreateAEDB.pl "orcl" "SYS" "syspassword" "adbuser" "adbpassword" "dbuser"
"dbuserpassword" "C:\Program Files\CA\Workload Automation AE\JRE_WA" "\tmp\adbldog" "Y"
"AEDB_DATA" "\home\oracle\oradata" "800" "AEDB_INDEX" "C:\oracle\oradata\SERVER_NAME"
"80" "N"
```

UNIX:

```
perl ./CreateAEDB.pl "orcl" "SYS" "syspassword" "adbuser" "adbpassword" "dbuser"
"dbuserpassword" "/opt/CA/WorkloadAutomationAE/JRE_WA" "/tmp/adbldog" "Y" "AEDB_DATA" "/"
home/oracle/oradata" "800" "AEDB_INDEX" "/home/oracle/oradata" "80" "N"
```

**Example: Create a Database on Sybase**

This example creates a Sybase database named AEDB on Sybase server LAM04. The script creates two Sybase devices; one for the data and one for the log. The data device is named AEDB\_DATA and stores up to 800 MB of data. The log device is named AEDB\_LOG and stores up to 80 MB of data.

Windows:



```
perl CreateAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "dbuser" "dbuserpw" "C:\Program
Files\CA\Workload Automation AE\JRE_WA" "\tmp\adblog" "Y" "AEDB_DATA" "\opt\sybase
\data" "800" "AEDB_LOG" "%SYBASE%\data" "80" "N"
```

#### UNIX:

```
perl ./CreateAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "dbuser" "dbuserpw" "/opt/CA/
WorkloadAutomationAE/JRE_WA" "/tmp/adblog" "Y" "AEDB_DATA" "/opt/sybase/data" "800"
"AEDB_LOG" "/opt/sybase/data" "80" "N"
```

#### Example: Update Oracle Tablespaces on UNIX

This example updates the existing Oracle tablespaces in the Oracle orcl instance. The schema owner and database user passwords contain a dollar sign (\$). In each password, the dollar sign is escaped and the password is enclosed in single quotes. The output of the script is stored in the /tmp/adblog directory. The script updates two Oracle tablespaces named AEDB\_DATA and AEDB\_INDEX. The script runs with debugging.

```
perl ./CreateAEDB.pl "orcl" "SYS" "" 'aedbadmin' 'aedb\$admin' 'dbuser' 'auto\$ys' /
opt/CA/WorkloadAutomationAE/JRE_WA" "/tmp/adblog N AEDB_DATA "" "0" "AEDB_INDEX" "" "0"
"Y"
```

## Run the CreateAEDB Script for Microsoft SQL Server in Interactive Mode

To create a CA Workload Automation AE Microsoft SQL Server database, you can run the CreateAEDB script in interactive mode. The script prompts you for the required information.

#### Follow these steps:

1. Make sure that the MSSQL bin path is in System PATH, and then issue the following commands:

```
cd %AUTOSYS%\dbobj\MSQ
perl CreateAEDB.pl
```

2. Verify the required information for each of the following prompts:

- **Server name?**  
Enter the Microsoft SQL Server server name.  
**Default:** HOSTNAME
- **Database name?**  
Enter the name that you want the Microsoft SQL Server database to be called.  
**Default:** AEDB
- **User name with system admin privileges?**  
Enter the Microsoft SQL Server system administrator user ID.  
**Default:** sa
- **sa user's password?**  
Enter the Microsoft SQL Server system administrator user password.  
**Default:** sa

#### NOTE

When you press Enter to accept the password, CreateAEDB verifies that it can connect to Microsoft SQL Server. If it cannot connect, CreateAEDB displays the following message and exits:

```
The host, userid, and password combination for the administrator is incorrect.
```

- **database user's name?**  
Enter the Microsoft SQL Server 'database' user name.  
**Default:** autosys
- **database user's password?**

Enter the Microsoft SQL Server 'database' user password.

**Default:** autosys

#### NOTE

If the database user is defined in Microsoft SQL Server, you must specify the valid password. If the database user is not defined in Microsoft SQL Server, the installer creates the user with the specified password.

#### – JRE Directory?

Enter the JRE\_HOME path.

**Default:** *install\_directory\JRE\_WA*

##### • *install\_directory*

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE

##### • *version\_number*

Specifies the JRE version that is installed with CA Workload Automation AE.

3. Enter Y or N when prompted to create the Microsoft SQL Server database, as follows:

#### – **Create a new DB? (Y|N)**

If you have previously defined a Microsoft SQL database for CA Workload Automation AE, enter N. Otherwise, enter Y.

**Default:** N

4. Enter Y or N when prompted to run the script, as follows:

#### – **Continue and create the database tables (Y|N)**

Enter Y to execute the script.

**Default:** N

#### NOTE

If CA Workload Automation AE r11.3 database was previously installed on your computer, the script displays the following prompt:

#### – **Continue and upgrade the database tables where required? (Y|N)** Enter Y to execute the script.

**Default:** N

The CA Workload Automation AE database is created or upgraded in Microsoft SQL Server if you chose to create or upgrade the database. Otherwise, the database is refreshed.

## Run the CreateAEDB Script for Oracle in Interactive Mode

To create the CA Workload Automation AE Oracle tablespaces or load existing tablespaces, you can run the CreateAEDB script in interactive mode. The script prompts you for the required information.

### Follow these steps:

1. Issue the following commands:

#### – Windows:

```
cd %AUTOSYS%\dbobj\ORA
perl CreateAEDB.pl
```

#### – UNIX:

```
cd $AUTOSYS/dbobj/ORA
perl ./CreateAEDB.pl
```

2. Enter the required information for each of the following prompts:

#### – **Database authentication mode? (Certificate:1 | [Password:0])** Enter 1 to use an SSL certificate to access to database. Enter 0 to use password to access the database.

**Default:** 0

**NOTE**

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

– **Do you want to create the Oracle tablespaces, users, and roles? (Y|N)**

If you have previously defined the Oracle tablespaces, users, and roles using the waae\_oracle.sql script or SQL, enter N. Otherwise, enter Y

**Default:** N

If you enter N, the following message is displayed:

Tablespaces and database users will not be created.

– **Service Identifier?**

Enter the Oracle SID name.

**Default:** AEDB

3. Enter the information for each of the following prompts if you entered 1 when prompted for database authentication mode and Y when prompted to create the database and continue with Step 5 :

– **TNS\_ADMIN value for the Oracle user with system administrator privileges?**

Enter the TNS\_ADMIN value for the Oracle system administrator user ID.

– **TNS\_ADMIN value for the schema owner?**

Enter the TNS\_ADMIN value for the schema owner.

– **Distinguished Name for the schema owner?** Enter the distinguished name for the schema owner.

– **Distinguished Name for the database user?**

Enter the distinguished name for the database user.

**NOTE**

If you entered 1 when prompted for database authentication mode and N when prompted to create the database, enter the TNS\_ADMIN value for the schema owner and continue with Step 6.

4. Enter the information for each of the following prompts if you entered 0 when prompted for database authentication mode and Y when prompted to create the database and continue with Step 5:

– **User name with system administrator privileges?**

Enter the Oracle system administrator user ID.

**Default:** sys

– **SYS user's password?**

Enter the Oracle system administrator user password.

**Default:** sys

**NOTE**

- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, sy\\$tem.
- When you press Enter to accept the password, CreateAEDB verifies that it can connect to Oracle. If it cannot connect to Oracle, CreateAEDB displays the following message and exits:

The userid and password combination for the administrator is incorrect.

– **Schema owner's password?**

Enter the Oracle schema owner password.

**Default:** aedbadmin

**NOTE**

- If the schema owner is defined in Oracle, you must specify the valid password. If the schema owner is not defined in Oracle, the installer creates the user with the specified password.
- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, aedb\\$admin.

– **database user's password?**

Enter the Oracle database user password.

**Default:** autosys

**NOTE**

- If the database user is defined in Oracle, you must specify the valid password. If the database user is not defined in Oracle, the installer creates the user with the specified password.
- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, auto\\$ys.

**NOTE**

If you entered 0 when prompted for database authentication mode and N when prompted to create the database, enter the schema owner's password and the database user's password and continue with Step 6.

5. Enter the information for each of the following prompts and continue with Step 7:

- **JRE directory?** Enter the JRE\_HOME path.

**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA*.(UNIX).

- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

- **Data tablespace name?**

Specify the name to create the data tablespace with.

**Default:** AEDB\_DATA

- **Data tablespace device path ?**

Specify the full path to create the data tablespace device in.

- **Data tablespace size MB ?**

Specify the data tablespace size, in MB.

**Default:** 800

- **Index tablespace name?**

Specify the name to create the index tablespace with.

**Default:** AEDB\_INDEX

- **Index tablespace device path?**

Specify the full path to create the index tablespace device in.

- **Index tablespace size MB?**

Specify the index tablespace size, in MB.

**Default:** 80

6. Enter the information for each of the following prompts as follows:

- **JRE directory?** Enter the JRE\_HOME path.

**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA*.(UNIX).

- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

- **Data Tablespace name?**

Enter the name of the defined data tablespace.

**Default:** AEDB\_DATA

- **Index Tablespace name?**

Enter the name of the defined index tablespace.

**Default:** AEDB\_INDEX

An information summary appears.

7. Enter Y or N when prompted to run the script, as follows:

- **Continue and create the database tables (Y|N)**

Enter Y to execute the script.

**Default:** N

#### NOTE

If CA Workload Automation AE r11.3 database was previously installed on your computer, the script displays the following prompt:

- **Continue and upgrade the database tables where required? (Y|N)**

Enter Y to execute the script.

**Default:** N

The CA Workload Automation AE tablespaces are created or upgraded in Oracle if you chose to create or upgrade the Oracle tablespaces. Otherwise, the tablespaces are refreshed

## Run the CreateAEDB Script for Sybase in Interactive Mode

To create the CA Workload Automation AE Sybase database and devices, you can run the CreateAEDB script in interactive mode. The script prompts you for the required information.

### Follow these steps:

1. Ensure that %SYBASE% (Windows) or \$SYBASE (UNIX) is set, and then issue the following commands:

- Windows

```
cd %AUTOSYS%\dbobj\SYB
perl CreateAEDB.pl
```

- UNIX:

```
cd $AUTOSYS/dbobj/SYB
perl ./CreateAEDB.pl
```

2. Enter the information for each of the following prompts:

- **Database authentication mode? (Kerberos:1 | [Password:0])**

Enter 1 to use Kerberos authentication to access the database. Enter 0 to use password to access the database.

#### NOTE

- To use Kerberos authentication:
  - Ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login with sa\_role and sso\_role privileges.
  - Ensure that the database user is created in the database. Sybase does not let you create a login without the password. So, the createaedb script assigns a default password *autosys* for the database user and modifies the login to enable only Kerberos authentication.
- For information about configuring MIT Kerberos on CA Workload Automation AE, see [Configure MIT Kerberos on CA Workload Automation AE](#)

**Default:** 0

- **Server name?**

Enter the Sybase server name.

**Default:** DEFAULT\_SERVER

- **Database name?**

Enter the name of the Sybase database.

**Default:** AEDB

- **Service Principal Name**(Optional) Specifies the service principal name to be used for Kerberos authentication.

- **JRE Directory?**

Enter the JRE\_HOME path.

**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA* (UNIX).

- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

3. Enter the information for each of the following prompts if you entered 0 when prompted for database authentication mode and continue with Step 4:

- **User name with system admin privileges?**

Enter the Sybase system administrator user ID.

**Default:** sa

- **sa user's password?**

Enter the Sybase system administrator user password.

**Default:** sa

**NOTE**

- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, sau\\$er.
- When you press Enter to accept the password, CreateAEDB verifies that it can connect to Sybase. If it cannot connect, CreateAEDB displays the following message and exits:

```
The host, userid, and password combination for the administrator is
incorrect.
```

- **database user's password?**

Enter the Sybase 'database' user password.

**Default:** autosys

**NOTE**

- If the database user is defined in Sybase, you must specify the valid password. If the database user is not defined in Sybase, the installer creates the user with the specified password.
- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, auto\\$ys.

4. Enter y or n when prompted to create the Sybase database, as follows:

- **Do you want to create a new database (y|[n])**

If you have previously defined a Sybase database for CA Workload Automation AE, enter n. Otherwise, enter y.

**Default:** n

- If you enter n, the script refreshes the database when it runs. Continue with Step 4.
- If you enter y, you are prompted for the following information:

- **Target data device name?**

Enter the Sybase data device name to be created.

**Default:** AEDB\_DATA

- **Target data device path?**

Enter the Sybase data device path, where the device is created.

**Default:** \opt\sybase\data\ (Windows) or /opt/sybase/data/ (UNIX)

- **Target data device size MB?**

Enter the Sybase data device size in MB.

**Default:** 800

Next, you are prompted for the following information related to creating the database device:

- **Do you want to create the database device (y|[n])**

If you have previously defined a Sybase device for CA Workload Automation AE, enter n. Otherwise, enter y.

Next, you are prompted for the following information related to creating a log device:

- • **Do you want to create a log device? (y|[n])**

If you have previously defined a Sybase device to be used for logs by CA Workload Automation AE, enter n. Otherwise, enter y.

- If you enter n, continue with Step 4.
- If you enter y, you are prompted for the following information:
  - **Target log device name?**  
Enter the Sybase log device name to be created.  
**Default:** AEDB\_LOG
  - **Target log device path?**  
Enter the Sybase log device path, where the device is created.  
**Default:** %SYBASE%\data (Windows) or /opt/sybase/data/ (UNIX)
  - **Target log size MB?**  
Enter the Sybase log device size in MB.  
**Default:** 100

An information summary appears.

5. Enter y or n when prompted to run the script, as follows:

- Continue and create the database tables? **(y|[n])**  
Enter y to execute the script.  
**Default:** n

**NOTE**

If CA Workload Automation AE r11.3 database was previously installed on your computer, the script displays the following prompt:

- **Continue and upgrade the database tables where required? (Y|N)**  
Enter Y to execute the script.  
**Default:** N

The CA Workload Automation AE database is created or upgraded in Sybase if you chose to create or upgrade the database. Otherwise, the database is refreshed.

## Refresh a CA Workload Automation AE Database

The RefreshAEDB.pl Perl script is used to refresh, update, or add read-only data in the CA Workload Automation AE database, for example, metadata, real-time resources, and stored procedures. RefreshAEDB should only be run when directed by technical support. You can run the script in interactive or console mode.

**NOTE**

The CA Workload Automation AE database must have already been installed before you can use the RefreshAEDB script.

### RefreshAEDB Script—Refresh a Database

The RefreshAEDB script updates the database used by CA Workload Automation AE. The script updates the tablespaces (for Oracle), devices (for Microsoft SQL Server or Sybase), and all the schema objects.

A RefreshAEDB script is included for each database vendor in the following directories:

Windows:

- %AUTOSYS%\dbobj\MSQ\RefreshAEDB.pl (for Microsoft SQL Server)
- %AUTOSYS%\dbobj\ORA\RefreshAEDB.pl (for Oracle)
- %AUTOSYS%\dbobj\SYB\RefreshAEDB.pl (for Sybase)

UNIX:

- \$AUTOSYS/dbobj/ORA/RefreshAEDB.pl (for Oracle)
- \$AUTOSYS/dbobj/SYB/RefreshAEDB.pl (for Sybase)

**NOTE**

You can enter the RefreshAEDB script with no options. You are prompted for the required information line by line.

You can also run the RefreshAEDB script in console mode. In this case, the script has the following format:

**For Microsoft SQL Server**

```
perl RefreshAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER" "ADB_SA_PSWD"
"JAVA_HOME" "ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

**For Oracle; if using password to access the database**

- Windows

```
perl RefreshAEDB.pl "ADB_SID" "SCHEMA_OWNER_USER" "SCHEMA_OWNER_PSWD" "JAVA_HOME"
"ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

- UNIX

```
perl ./RefreshAEDB.pl "ADB_SID" "SCHEMA_OWNER_USER" "SCHEMA_OWNER_PSWD" "JAVA_HOME"
"ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

**For Oracle; if using an SSL certificate to access the database**

- Windows

```
perl RefreshAEDB.pl -CERTMODE "ADB_SID" "ADB_SYS_TNSADMIN" "JAVA_HOME" "ADB_OUTDIR"
"ADB_DEBUG_VERIFY"
```

- UNIX

```
perl ./RefreshAEDB.pl -CERTMODE "ADB_SID" "ADB_SYS_TNSADMIN" "JAVA_HOME" "ADB_OUTDIR"
"ADB_DEBUG_VERIFY"
```

**For Sybase; if using password to access the database**

- Windows

```
perl RefreshAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER" "ADB_SA_PSWD"
"JAVA_HOME" "ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

- UNIX

```
perl ./RefreshAEDB.pl "ADB_DATASERVER" "ADB_DATABASE" "ADB_SA_USER" "ADB_SA_PSWD"
"JAVA_HOME" "ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

**For Sybase; if using Kerberos authentication to access the database**

- Windows

```
perl RefreshAEDB.pl -EXTAUTH "ADB_DATASERVER" "ADB_DATABASE" "ADB_SPN" "JAVA_HOME"
"ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

- UNIX

```
perl ./RefreshAEDB.pl -EXTAUTH "ADB_DATASERVER" "ADB_DATABASE" "ADB_SPN" "JAVA_HOME"
"ADB_OUTDIR" "ADB_DEBUG_VERIFY"
```

**NOTE**

To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which has sa\_role.

- **EXTAUTH** Specifies that Kerberos authentication is used to access the database.



**NOTE**

For information about configuring MIT Kerberos on CA Workload Automation AE, see [Configure MIT Kerberos on CA Workload Automation AE](#)

- **ADB\_DATABASE** Specifies the Microsoft SQL Server or Sybase database name.
- **ADB\_DATASERVER** Specifies the Microsoft SQL Server or Sybase server name.
- **ADB\_DEBUG\_VERIFY** Indicates whether to set debugging on during the update, as follows:
  - Y—Sets debugging on
  - V—Verifies only the connection parameters (no refresh occurs)
- **ADB\_OUTDIR** Specifies the directory where you want to store the output from the RefreshAEDB script. This directory must already exist and be empty prior to running the RefreshAEDB script.
- **ADB\_SA\_PSWD** Specifies the Microsoft SQL Server, Oracle, or Sybase system administrator user password.

**NOTE**

On UNIX, if you specify a password that contains a dollar sign (\$) using the command line, escape the dollar sign and enclose the password in single quotes. For example, 'sy\$tem'.

- **ADB\_SA\_USER** Specifies the Microsoft SQL Server, Oracle, or Sybase system administrator user ID.
- **CERTMODE** Specifies that an SSL certificate is used to access the database.

**NOTE**

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- **ADB\_SID** Specifies the Oracle SID name.
- **ADB\_SYS\_TNSADMIN** Specifies the TNS\_ADMIN value for the Oracle user with system administrator privileges.
- **SCHEMA\_OWNER\_USER** Specifies the Oracle 'schema owner' name.
- **SCHEMA\_OWNER\_PSWD** Specifies the Oracle 'schema owner' password.

**NOTE**

On UNIX, if you specify a password that contains a dollar sign (\$) using the command line, escape the dollar sign and enclose the password in single quotes. For example, 'aedb\$admin'.

- **ADB\_SPN**(Optional) Specifies the service principal name to be used for Kerberos authentication. If you do not want to specify the service principal name, include an empty space for this parameter, for example, " ".
- **JAVA\_HOME** Specifies the JAVA\_HOME path.

**Example: Refresh a Database on Microsoft SQL Server**

This example updates a Microsoft SQL Server database named AEDB on server LAM04.

```
perl RefreshAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "C:\Program Files\CA\Workload
Automation AE\JRE_WA" "C:\tmp\adblog" "Y" "N"
```

**Example: Refresh Oracle Tablespaces**

This example updates Oracle tablespaces in the Oracle orcl instance. The output of the script is stored in the \tmp\adblog directory (Windows) or /tmp/adblog directory (UNIX). The script runs without debugging.

Windows:

```
perl RefreshAEDB.pl "orcl" "adminuser" "aspassword" "C:\Program Files\CA\Workload
Automation AE\JRE_WA" "C:\tmp\adblog" "N"
```

UNIX:

```
perl ./RefreshAEDB.pl "orcl" "aspassword" "/opt/CA/WorkloadAutomationAE/JRE_WA" "/tmp/adbldlog" "N"
```

### Example: Refresh a Database on Sybase

This example updates a Sybase database named AEDB on Sybase server LAM04. The script runs without debugging.

Windows:

```
perl RefreshAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "C:\Program Files\CA\WorkloadAutomation AE\JRE_WA" "C:\tmp\adbldlog" "N"
```

UNIX:

```
perl ./RefreshAEDB.pl "LAM04" "AEDB" "sa" "sapassword" "/opt/CA/WorkloadAutomationAE/JRE_WA" "/tmp/adbldlog" "N"
```

## Run the RefreshAEDB Script for Microsoft SQL Server in Interactive Mode

To update a CA Workload Automation AE MSSQL database, you can run the RefreshAEDB script in interactive mode. The script prompts you for the required information.

### Follow these steps:

1. Ensure MSSQL osql is in PATH, and then issue the following commands:

```
cd %AUTOSYS%\dbobj\MSQ
```

```
perl RefreshAEDB.pl
```

2. Verify the required information for each of the following prompts:

- **Server name?**  
Enter the MSSQL server name.  
**Default:** Hostname
- **Database name?**  
Enter the MSSQL database to be called.  
**Default:** AEDB
- **User name with system admin privileges?**  
Enter the MSSQL system administrator user ID.  
**Default:** sa
- **sa user's password?**  
Enter the MSSQL system administrator user password.  
**Default:** sa

### NOTE

When you press Enter to accept the password, RefreshAEDB verifies that it can connect to MSSQL. If it cannot connect, RefreshAEDB displays the following message and exits:

```
The host,/userid/password combination is valid.
```

- **JRE Directory?**  
Enter the JRE\_HOME path.  
**Default:** *install\_directory*\JRE\_WA

- **install\_directory**  
Specifies the root directory where CA Workload Automation AE is installed.  
**Default:** C:\Program Files\CA\Workload Automation AE
  - **version\_number**  
Specifies the JRE version that is installed with CA Workload Automation AE.
3. Enter Y or N when prompted to run the script, as follows:
- **Are you sure? (Y|N)**  
Enter Y to execute the script.  
**Default:** N
- The CA Workload Automation AE database is updated in MSSQL.

## Run the RefreshAEDB Script for Oracle in Interactive Mode

To update the CA Workload Automation AE Oracle tablespaces, you can run the RefreshAEDB script in interactive mode. The script prompts you for the required information.

### Follow these steps:

1. Issue the following commands:  
Windows:  

```
cd %AUTOSYS%\dbobj\ORA
perl RefreshAEDB.pl
```

UNIX:  

```
cd $AUTOSYS/dbobj/ORA
perl ./RefreshAEDB.pl
```
2. Enter the required information for each of the following prompts:
  - **Database authentication mode? (Certificate:1 | [Password:0])** Enter 1 to use an SSL certificate to access to database. Enter 0 to use password to access the database.  
**Default:** 0

### NOTE

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- **Service Identifier?**  
Enter the Oracle SID name.  
**Default:** AEDB
3. Enter the information for each of the following prompts if you entered 0 when prompted for database authentication mode and continue with Step 5:
    - **Schema owner name?**  
Enter the valid Oracle 'schema owner' name.  
**Default:** aedbadmin
    - **Schema owners password?**  
Enter the valid Oracle 'schema owner' password.  
**Default:** aedbadmin

### NOTE

On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, aedb\\$admin.

- **JRE Directory?**  
Enter the JRE\_HOME path.  
**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA* (UNIX).
- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

4. Enter the information for each of the following prompts if you entered 1 when prompted for database authentication mode and continue with Step 5:

- **TNS\_ADMIN value for the schema owner?**

Enter the TNS\_ADMIN value for the schema owner.

- **JRE Directory?**

Enter the JRE\_HOME path.

**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA* (UNIX).

- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

5. Enter Y or N when prompted to run the script, as follows:

- **Are you sure? (Y|N)**

Enter Y to execute the script.

**Default:** N

The CA Workload Automation AE tablespaces are updated.

## Run the RefreshAEDB Script for Sybase in Interactive Mode

To update a CA Workload Automation AE Sybase database, run the RefreshAEDB script in interactive mode. The script prompts you for the required information.

### Follow these steps:

1. Ensure that %SYBASE% (Windows) or \$SYBASE (UNIX) is set, and then issue the following commands:

Windows:

```
cd %AUTOSYS%\dbobj\SYB
```

```
perl RefreshAEDB.pl
```

UNIX:

```
cd $AUTOSYS/dbobj/SYB
```

```
perl ./RefreshAEDB.pl
```

2. Enter the information for each of the following prompts:

- **Database authentication mode? (Kerberos:1 | [Password:0])**

Enter 1 to use Kerberos authentication to access the database. Enter 0 to use password to access the database.

### NOTE

- To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which has sa\_role.
- For information about configuring MIT Kerberos on CA Workload Automation AE, see [Configure MIT Kerberos on CA Workload Automation AE](#)

**Default:** 0

– **Server name?**

Enter the Sybase server name.

**Default:** DEFAULT\_SERVER

– **Database name?**

Enter the Sybase database to be called.

**Default:** AEDB

– **Service Principal Name**(Optional) Specifies the service principal name to be used for Kerberos authentication.

– **JRE Directory?**

Enter the JRE\_HOME path.

**Default:** Based on the operating system, the default path is *install\_directory\JRE\_WA* (Windows) or *install\_directory/JRE\_WA* (UNIX).

- **install\_directory**

Specifies the root directory where CA Workload Automation AE is installed.

**Default:** C:\Program Files\CA\Workload Automation AE (Windows) or /opt/CA/WorkloadAutomationAE (UNIX)

- **version\_number**

Specifies the JRE version that is installed with CA Workload Automation AE.

3. Enter the information for each of the following prompts if you entered 0 when prompted for database authentication mode and continue with Step 4:

– **User name with system admin privileges?**

Enter the Sybase system administrator user ID.

**Default:** sa

– **sa user's password?**

Enter the Sybase system administrator user password.

**Default:** sa

**NOTE**

- On UNIX, if you enter a password that contains a dollar sign (\$), escape it. For example, sau\\$.er.
- When you press Enter to accept the password, RefreshAEDB verifies that it can connect to Sybase. If it cannot connect, RefreshAEDB displays the following message and exits:

```
The host, userid, and password combination for the administrator is incorrect. Ex
```

4. Enter y or n when prompted to run the script, as follows:

– **Are you sure? (y|n)**

Enter y to execute the script.

**Default:** n

The CA Workload Automation AE database is updated in Sybase.

## (UNIX) Convert a Manually Copied Instance of 11.3.x version into an Environment that can be Upgraded Successfully

The CA Workload Automation AE installation includes the Add Instance option to create an instance environment on a machine that already has another installed instance. We recommend that you use this option to install all the necessary

files that are required for a new instance to run. However, some system administrators create an instance by manually copying an existing instance environment and then modifying the copied environment to assign it a new instance name.

### **WARNING**

**IMPORTANT!** CA Technologies does not support the manual creation of an CA Workload Automation AE instance by copying an existing instance environment. While this manual method can result in a functioning instance, it makes the new instance difficult to upgrade using the CA Workload Automation AE installer. However, if you have already created an instance using this manual method earlier, follow the below procedures to convert a manually copied instance environment into an environment that can be upgraded successfully.

To convert an instance environment that was manually copied from a previously installed instance into an environment that can be upgraded, follow these steps:

1. Determine if your instance environment was created using the manual process
2. Repair the Instance Created Using the Manual Process

### **Creating the Instance Environment Using the Manual Process**

Follow these steps to determine if you have an instance environment that was created using the manual process instead of using the Add Instance option of the CA Workload Automation AE installer.

1. Obtain the list of all instance environments on your machine. Execute the following commands to get the list of all directories beginning with autouser:

#### **NOTE**

Every instance contains a folder that begins with “autouser” and ends with the instance name.

```
$ cd /opt/CA/WorkloadAutomationAE
$ ls | grep ^autouser
autouser.ACE
autouser.BCE
```

In the above sample output, observe that two autouser directories are listed, indicating that two instances are present. However, the directory listing alone does not indicate how the instances are installed.

2. Determine if the manual process was used to install one or more instances by executing the following commands:
  - a. Verify that each instance has a file in the \$AUTOSYS/install directory that begins with vars and ends with the instance name:

```
$ cd $AUTOSYS/install
$ ls | grep ^vars
varsACE
```

Observe from the above sample output that the \$AUTOSYS/install directory contains a vars file for the instance ACE alone. This indicates that the BCE instance was most likely created using the manual process. If you had used the CA Workload Automation AE installer to add the BCE instance, then the output of the above command would have also listed a file named varsBCE.

- b. Verify if the installation folder contains the corresponding installer log files, as follows:

```
$ cd /opt/CA/SharedComponents/installer/log
$ ls | grep CAWorkload
CAWorkloadAutomationAE.install.log
CAWorkloadAutomationAE.log
```

Observe from the above sample output that the \$AUTOSYS/install directory does not contain a log file beginning with CAWorkloadAutomationAE.AddInstance for the BCE instance. This means that you most likely did not invoke the Add Instance option of the installer to add the BCE instance. If you had invoked the Add Instance option of the

CA Workload Automation AE installer to add the BCE instance, then the output of the command mentioned above would have also listed the file named CAWorkloadAutomationAE.AddInstance.BCE.log.

### **Repair the Instance Created Using the Manual Process**

Follow these steps to repair the instance that you created using the manual process:

1. Log in as the root user, open the command prompt, and execute the `unisrvctr` command to stop all the CA Workload Automation AE processes for all instances:

```
$ unisrvctr stop waee_agent-WA_AGENT waee_server.ACE waee_server.BCE waee_sched.ACE
waee_sched.BCE
```

#### **NOTE**

- The `unisrvctr` command is located in the directory `$CASHCOMP/bin`. The default value of the `CASHCOMP` environment variable is `opt/CA/SharedComponents`.
  - Execute the UNIX `ps` command to verify that no CA Workload Automation AE processes are still running. If the `unisrvctr` command did not successfully stop a process, manually execute the UNIX command “kill -9” to forcefully end the process.
2. Execute the following command to rename the `autouser` folder for the instance so that it no longer contains the phrase `autouser`:

```
$ mv autouser.BCE autobackup.BCE
```

3. Check if a file named `/etc/auto.BCE` exists. If it exists, execute the following command to rename it. For example, rename it to `/etc/autobackup.BCE`.

```
$ mv /etc/auto.BCE /etc/autobackup.BCE
```

4. Execute the `autoflags` command to get the base version of CA Workload Automation AE that is installed on your machine:

```
$ autoflags -a
546 LINUX ORA 11.3.5 INCR2 830a193b supermachine supermachine.domain.com
```

In the above sample output, the base version of CA Workload Automation AE that is installed on the machine is 11.3.5.

5. Create a backup of the binaries and libraries by executing the following commands:

```
$ cp -fpr $AUTOSYS/lib $AUTOSYS/backup_lib
$ cp -fpr $AUTOSYS/bin $AUTOSYS/backup_bin
$ cp -fpr $AUTOSYS/messages $AUTOSYS/backup_messages
$ cp -fpr $AUTOSYS/dbobj $AUTOSYS/backup_dbobj
```

6. Load the CA Workload Automation AE Release 11.3.5 DVD that is used to install the first instance on your machine and run the installer.  
The Welcome page appears.
7. Click Next.  
The CA Workload Automation AE page appears.
8. Select Modify and click Next.  
The Modify Installation Function page appears.
9. Select Add Instance and click Next.  
The Add Instance page appears.
10. Specify the name of the instance you want to repair and click Next.  
The Instance Type page appears.
11. Select the components that you want to add to the instance and click Next.
12. Proceed to the Primary Event Server Properties page, specify the database details, and click Next.

**NOTE**

Ensure that you clear both the Create the Tablespaces, Database Users, and Roles and Create or Refresh the Database check boxes. This ensures that the installer does not make the database changes.

13. Proceed to the Review Settings page and review the information. If the information is correct, click Add Instance.  
The installer adds the instance and the Instance Modification Results page appears.
14. Click OK.  
The Modify Installation Function page appears.
15. Repeat Steps 8 through 13 for each instance you must repair. After you have completed this procedure for each instance that must be repaired, click Cancel.  
The Exit Setup page appears.
16. Click Yes.
17. Return to the directory where CA Workload Automation AE is installed. Use the UNIX diff command to manually compare the files in the newly created autouser directory with the files in the autouser directory you backed up. Make any necessary changes to the configuration file in the new autouser directory to preserve your previous settings.
18. Verify that the installer created the instance properly by executing the steps described in Determine If Your Instance Environment Was Created Using the Manual Process procedure.
19. Restore your previously backed up binaries by executing the following commands:

**NOTE**

```
$ rm -rf $AUTOSYS/lib $AUTOSYS/bin $AUTOSYS/messages
$ mv $AUTOSYS/backup_lib $AUTOSYS/lib
$ mv $AUTOSYS/backup_bin $AUTOSYS/bin
$ mv $AUTOSYS/backup_messages $AUTOSYS/messages

$ mv $AUTOSYS/backup_dbobj $AUTOSYS/dbobj
```

**WARNING**

Do not start the CA Workload Automation AE processes until you have completed the repair process for all instances and have restored your previously backed up binaries.

You have completed the reparation of your instance environments. Now, you can run a future release of the CA Workload Automation AE installer to upgrade your instances successfully without experiencing the issues caused by the manual process of adding instances.



---

# Uninstall

---

This section describes how to uninstall AutoSys and its components.

## Uninstall AutoSys

You can uninstall AutoSys using the following options.

### WARNING

After uninstalling AutoSys, some AutoSys files and folders may remain present. You must rename or delete them if you try to install AutoSys again.

## Uninstall AutoSys Using GUI

The following sections describe how you can uninstall AutoSys using the GUI.

### Windows

To uninstall AutoSys on Windows, follow these steps.

### NOTE

Click **Next** to proceed through the step described below.

1. Go to the AutoSys install folder. For example, C:\Program Files\CA\WorkloadAutomationAE\uninstaller.
2. Double-click the UninstallAutoSysWorkloadAutomation application file. The Maintenance Mode window appears and **Uninstall Product** is already selected.  
**Note:** For Linux, ./UninstallAutoSysWorkloadAutomation.
3. In [Deregister CA EEM Policies](#), select the instances from which you want to deregister the CA EEM policies.
4. In [Pre-Installation Summary](#), review the summary.
5. Click **Uninstall**.

### NOTE

The uninstall can also be performed using the **Add or remove programs** option of **System settings**.

### Unix

Follow these steps to uninstall AutoSys on Unix.

1. Go to /opt/CA/WorkloadAutomationAE/uninstaller.
2. Run the following command to launch the uninstaller in the GUI mode.  

```
./UninstallAutoSysWorkloadAutomation -i gui
```

## Uninstall AutoSys Using Console

The following sections describe how to uninstall AutoSys using the console.

### Windows

1. In the command prompt, go to <install-dir>:\Program Files\CA\WorkloadAutomationAE\uninstaller.
2. Execute the UninstallAutoSysWorkloadAutomation.exe -i console command.

## Unix

1. In the console, go to `/opt/CA/WorkloadAutomationAE/uninstaller`.
2. Execute the following binary: `./UninstallAutoSysWorkloadAutomation`.

## Uninstall AutoSys using Silent Uninstall

A silent uninstallation is supported. However, for uninstallation, you cannot record a response file in the silent mode.

To uninstall AutoSys run the following command.

```
UninstallAutoSysWorkloadAutomation -i silent -f /respfile_name.txt
```

If you want to delete policies for an AutoSys instance, the below variable must be set to 1.

`IA_AE_INSTANCE_EEM_DELETE_POLICIES_FOR_<INSTANCE>=1`

or example, to delete "ACE" instance policies, in the response file, set

`IA_AE_INSTANCE_EEM_DELETE_POLICIES_FOR_ACE=1`.

If you want to delete the Web UI policies, in the response file, set `IA_AE_WCC_EEM_DELETE_POLICIES=1`.

Password should be in AES encrypted form for variables `IA_AE_EEM_USER_PASSWORD_FOR_<INSTNACE_NAME>` and `IA_AE_EEM_USER_PASSWORD_FOR_WCC`.

## Uninstall Miscellaneous Topics

### (UNIX) Post-Uninstall, Uninstall Failure

If necessary, you can uninstall AutoSys Workload Automation using the installation wizard. After you uninstall AutoSys Workload Automation, remove the database tables and delete the `auto.profile` file. If a catastrophic uninstallation failure occurs, clean up the server manually.

To uninstall AutoSys Workload Automation, follow these steps:

1. [Remove the database tables](#)
2. [Delete the auto.profile file](#)
3. [Clean up after a catastrophic uninstallation failure](#)

### Remove the Database Tables

After AutoSys Workload Automation is uninstalled, remove the database tables and data.

#### Follow these steps:

1. Remove the database tables.

**Note:** The following commands can be used for Web UI and Secondary database by giving appropriate values.

**Examples:**

Database Type	Sample names in database	Commands to run to run database tables
Oracle	Tablespace names are: AEDB_DATA and AEDB_INDEX Schema owner name is aedbadmin Database user name is autosys	To drop the tablespaces run the following commands: drop tablespace AEDB_DATA including contents and datafiles cascade constraints; drop tablespace AEDB_INDEX including contents and datafiles cascade constraints; To drop user and roles, run the following commands: drop user aedbadmin cascade; drop user autosys cascade;
Sybase	Database name is AEDB Database user name is autosys	To drop the devices, run the following commands: drop database AEDB go exec sp_dropdevice AEDB_DATA go exec sp_dropdevice AEDB_LOG go To drop the login, run the following command: sp_droplogin autosys go <b>Note:</b> Remove the Sybase device files from the directory that is specified in the Sybase Data and Log Directories. For example, if the device files are in /opt/sybase/data, run the following commands: cd /opt/sybase/data rm -f AEDB_DATA.DAT rm -f AEDB_LOG.DAT
MSSQL	Database name is AEDB Database user name is autosys	To drop the database, run the following command: drop database AEDB go To drop the database user, run the following command: drop user autosys go

- Remove the directory where AutoSys Workload Automation was installed (if it is the only AutoSys Workload Automation instance). For example, if AutoSys Workload Automation was installed in /opt/CA/WorkloadAutomationAE, run the following commands:

```
cd /opt/CA
rm -rf WorkloadAutomationAE
```

**Delete the auto.profile File**

After AutoSys Workload Automation is uninstalled, delete the auto.profile file from the /etc directory manually.

**Clean Up After a Catastrophic Uninstallation Failure**

Cleaning up the server is necessary after a catastrophic failure of the AutoSys Workload Automation uninstallation process.

**Follow these steps:**

**NOTE**

Enter all commands at the operating system prompt.

1. Enter the following command:

```
unisrvctr stop CA-WAAE
```

All AutoSys Workload Automation processes stop.

2. Enter the following command for each *process\_name*, and note the processes that are running:

```
ps -ef | grep process_name
```

– **process\_name**

Specifies the name of the process searched for in the ps output. Process names to use with the command are as follows:

- as\_server
- event\_demon
- cybAgent
- csampmxf
- lic98fds

An active process displays its process ID. Use this process ID in the next step.

3. Enter the following command using each process ID obtained in the previous step:

```
kill -9 process_ID
```

The process is stopped.

4. Deregister the services.

```
uniregister -D <service_name>
```

5. Enter the following command:

```
rm -rf install_directory
```

- **install\_directory** Specifies the root directory where AutoSys Workload Automation is installed. By default, /opt/CA/WorkloadAutomationAE is the installation directory.

The installation directory is removed.

6. Enter the following commands:

```
rm -f /etc/auto.profile
```

```
rm -f /etc/auto.instance_name
```

– **instance\_name**

Specifies the AutoSys Workload Automation instance name.

The profile files are removed.

7. Enter the following commands:

– On AIX:

```
rm -f /etc/rc.d/waae_*
```

```
rm -f /etc/rc.d/CA-WAAE
```

```
rm -f /etc/rc.d/init.d/waae_*
```

```
rm -f /etc/rc.d/init.d/CA-WAAE
```

– On HP-UX:

```
rm -f /sbin/init.d/waae_*
```

```
rm -f /sbin/init.d/CA-WAAE
```

– On Linux:

```
rm -f /etc/init.d/waae_*
```

```
rm -f /etc/init.d/CA-WAAE
```

The waae\_\* and CA-WAAE system startup files are removed.

**NOTE**

The server cleanup is complete if you do not need to perform the optional steps that follow.

8. (Optional) Enter the following commands only if other Broadcom products are not installed on the server:

```
rm -f /etc/profile.CA
rm -f /etc/csh_login.CA
```

The profile files are removed.

## (Windows) Post-Uninstall, Uninstall Failure

If necessary, you can uninstall AutoSys Workload Automation using the installation wizard. If a catastrophic uninstallation failure occurs, clean up the server manually.

To uninstall AutoSys Workload Automation, follow these steps:

1. [Uninstall AutoSys Workload Automation](#)
2. [Clean up after a catastrophic uninstallation failure](#)

### Uninstall AutoSys Workload Automation

You can uninstall AutoSys Workload Automation using the installation wizard.

**WARNING**

Before uninstalling AutoSys Workload Automation, stop all AutoSys Workload Automation jobs or processes that are running on that computer. Otherwise, the uninstallation process may halt because the jobs or processes are still running. In that situation, you can wait until the jobs or processes complete, and click the Retry option on the Uninstall Complete page to continue the uninstallation process. Alternatively, you can try to stop the processes manually by terminating the su.exe processes in the Windows Task Manager or by using a diagnostic tool.

If a common component is being used by a Broadcom product, the common component is not uninstalled. It cannot be uninstalled until all the Broadcom products have deregistered from it.

### Clean Up After a Catastrophic Uninstallation Failure

Cleaning up the server is necessary after a catastrophic failure of the AutoSys Workload Automation uninstallation process.

#### **Follow these steps:**

1. Click Start, Run.  
The Run dialog opens.
2. Enter the following command and click OK.  
`services.msc`  
The Services window opens.
3. Stop all services with the AutoSys Workload Automation prefix.
4. Open Task Manager and stop the following processes if they are running:
  - as\_server.exe
  - event\_demon.exe
  - eventsysd.exe
  - cybAgent.exe
  - wrapper.exe
  - java.exe (related to the Tomcat services)
  - CSAMPmuxf.exe

**NOTE**

The CSAMPmuxf.exe process cannot be stopped while another Broadcom product is using it.

5. Open Add or Remove Programs from the Windows Control Panel.
6. Remove the Workload Automation Agent for UNIX, Linux, or Windows and the Command Sponsor components as follows:
  - a. Select the component and select uninstall.  
The Product Removal Selection page appears.
  - b. Click Next and follow the prompts to complete the uninstallation procedure.  
The component is removed.
7. Enter the following command at the AutoSys Workload Automation instance command prompt for each application server:
 

```
sc delete "CA Workload Automation AE Application Server(instance_name)"
```

  - **instance\_name**  
Defines the name of the AutoSys Workload Automation instance.  
The application server service is removed.
8. Enter the following command at the AutoSys Workload Automation instance command prompt for each scheduler:
 

```
sc delete "CA Workload Automation AE Scheduler(instance_name)"
```

The scheduler service is removed.
9. Enter the following command at the AutoSys Workload Automation instance command prompt for each agent:
 

```
sc delete "Workload Automation Agent (agent_name)"
```

  - **agent\_name**  
Defines the name of the Workload Automation Agent for Windows.  
The agent service is removed.
10. Enter the following command at the AutoSys Workload Automation instance command prompt for each web server:
 

```
sc delete "CA Workload Automation AE Web Server(instance_name)"
```

The web server service is removed.
11. Enter the following command at the CA Workload Automation AE instance command prompt:
 

```
rmdir "install_directory"
```

  - **install\_directory**  
Specifies the root directory where AutoSys Workload Automation is installed.

**NOTE**

If a file in the root directory is in use, this command may fail. In such cases, run this command after you restart the server.

The AutoSys Workload Automation installation directory is removed.

12. Restart the server.  
The server cleanup is complete.

## Uninstall Common Components

This section describes how you can uninstall common components, such as CCI, Event Manager, and CommandSponsor (including iGateway). The scripts available for uninstalling common components identify the components on your server, and perform the following actions.

- Stop services
- Delete services
- Delete folders
- Delete registry entries
- Delete unwanted environment variables

## **Windows**

1. As an Administrator, go to the command prompt.
2. Run the `RemoveWAC3Components.bat` file.

### **NOTE**

You may need to reboot the system for the environment variables and registry entries to be effective.

## **Unix**

As a root user or sudo user, run the `RemoveWAC3Components.sh` script located in `<ISO>\modules\C3UninstallScript\`

The `RemoveWAC3Components.bat` script is designed to remove components that are not used by AutoSys Workload Automation r12. This script depends on environment variables and user inputs. It deletes services, folders, and registry entries from the server.

### **NOTE**

- If you have EEM installed on the server, the iGateway service and its related folders and registry entries will not be removed. You will receive the following message: "Found EEM and Directory, iGateway service will not be deleted and iTechnology folder will not be deleted."
- If you have CCI installed on the server, you will be prompted with the following message: "Do you want to remove CCI service and its path? [Y/N]". The removal of the CCI will depend on your response.

## Upgrading

This section contains information about how to upgrade AutoSys Workload Automation, AutoSys Web UI, and Common Components.

This content is meant for system administrators who are responsible for upgrading these products.

### NOTE

Procedures and best practices assume familiarity with the operating system and with the database server you use.

## Upgrade AutoSys

Follow these steps to upgrade AutoSys:

1. Double-click the AutoSys installer.
2. Select the desired upgrade option. For more information, see [Upgrade Options](#).
3. Follow the instructions that appear in the upgrade panels.

For details on the values and properties you need to define in the upgrade process, see the following topics:

## Silent Upgrade AutoSys

The steps to perform a silent upgrade of AutoSys and fresh install are the same for both Windows and Linux.

For more information, refer to [Fresh Install AutoSys Using Silent Install](#).

## Upgrade Options

When you [upgrade AutoSys](#), you must select the type of upgrade to perform.

Field name	Silent Install Variable	Description
Upgrade the product	IA_AE_UPGRADE_OPTIONS= UPGRADE_PRODUCT	Select this option if you already have AutoSys version 11.3.6 or below.
Create or upgrade the database schema	IA_AE_UPGRADE_OPTIONS= CREATEDB	Select this option if you want to create a new database schema or update an existing one. The panels that come up after you select this option are similar to the ones for a fresh installation.

## Running Services

During upgrade procedures, the Running Services panel displays a list of services that will be automatically stopped during an upgrade or maintenance of AutoSys. Click **Cancel** if you want to stop the listed services manually.

### NOTE

As a best practice, with the exception of the Derby database service, it is recommended that you stop all the AutoSys services manually. This will ensure a clean closure of the running services and will shorten the installation time.



Define the following options on the Running Services panel.

Field name	Silent Install Variable	Description
Start all the above services immediately after installation	IA_AE_SERVICE_START_AFTER_INSTALL //Set this variable to 1 to start all the above services after installation. Otherwise, set to 0.	Select this checkbox if you want to start all the mentioned services after you finish the upgrade. By default, after the upgrade, all the mentioned services will be automatically stopped.

## Upgrade: Database Panels

When you [upgrade AutoSys](#), there are various values and options that pertain to the database. The following sections describe the fields in you need to define depending on your database:

### MSSQL: Upgrade: Primary Event Server Properties

When you [upgrade AutoSys](#) with Microsoft SQL Server, you need to define the primary event server properties. The following information is required.

#### NOTE

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields mentioned here are applicable to the Secondary Event Server Properties window as well.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_ADMIN\_USER\_FOR\_<INSTANCE>, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_ADMIN\_USER\_FOR\_<INSTANCE>.
- For Web UI and Reporting panels, the required silent variables are similar to that of the fresh install.
- This window will repeat for each AutoSys instance.

Field name	Silent Variable	Description
Instance name	NA	The AutoSys instance name for which the upgrade has to be performed. This will appear as a label.
Database type	NA	It will be Microsoft SQL Server. This will appear as a label.
Server name	NA	The server name on which your MSSQL database is installed. This will appear as a label.
Database name	NA	The MSSQL database name. This will appear as a label.
Windows authentication	IA_AE_PRIMARY_DB_MSSQL_AUTH_TYPE_FOR_<INSTANCE>	Select this option if you want to use Windows authentication.
Microsoft SQL server authentication	IA_AE_PRIMARY_DB_MSSQL_AUTH_TYPE_FOR_<INSTANCE>	Select this option if you want to use MSSQL server authentication.
Username	IA_AE_PRIMARY_DB_ADMIN_USER_FOR_<INSTANCE>	The username for Microsoft SQL server authentication.
Password	IA_AE_PRIMARY_DB_ADMIN_PWD_FOR_<INSTANCE>	The password for Microsoft SQL server authentication.

## Upgrade: Oracle Properties

When you [upgrade AutoSys](#) with Oracle, you need to define the following properties.

Field name	Silent Install Variable	Description
Oracle home directory	IA_AE_DB_ORA_HOME	The Oracle home directory path.
Oracle wallet support JAR directory	IA_AE_DB_ORA_WALLET_JAR_PATH	The Oracle wallet support JAR directory path.

## Oracle: Upgrade: Primary Event Server Properties

When you [upgrade AutoSys](#) with Oracle, you need to define the primary event server properties. The following information is required.

### NOTE

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields described in this topic are also applicable to the Secondary Event Server Properties window.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SCHEMA\_ADMIN\_PWD\_FOR\_<INSTANCE>, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SCHEMA\_ADMIN\_PWD\_FOR\_<INSTANCE>.
- For the Web UI and Reporting panels, the required silent variables are similar to those of the fresh install.
- This window will repeat for each AutoSys instance.

Field name	Silent Install Variable	Description
Instance name	NA	The AutoSys instance name for which the upgrade is to be performed. This will appear as a label.
Database type	NA	The type will be Oracle. This will appear as a label.
TNS alias name	NA	The TNS alias name that appears as a label to connect to your database.
Schema owner name	NA For Web UI, when upgrading from the 11.4 version, this field is editable. The corresponding silent install variable is IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_USER for the Web UI application database. For reporting, it is IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_USE.	The Oracle database schema owner name. This will appear as a label.
Password	IA_AE_PRIMARY_DB_SCHEMA_ADMIN_PWD_FOR_<INSTANCE>	The password to connect to the Oracle database mentioned above.
TNS admin directory	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_USER_TNS_ADMIN_FOR_<INSTANCE>	The Oracle TNS admin directory.

Oracle wallet SSO file path	IA_AE_PRIMARY_DB_ORA_SCHEMA_ADMIN_USER_WALLET_LOC_FOR_<INSTANCE>	The Oracle wallet SSO file path.
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## Upgrade: Sybase Properties

When you [upgrade AutoSys](#) with Sybase, you need to define the following properties.

Field name	Silent Install Variable	Description
Sybase directory	IA_AE_DB_SYB_HOME	The Sybase directory path.
JDBC JAR path	IA_AE_DB_SYB_JCONN_PATH	The JDBC jconn JAR file path. The path should include the jconn filename as well. <b>Example:</b> opt/SAP/jconn16/jconn4.jar

## Sybase: Upgrade: Primary Event Server Properties

When you [upgrade AutoSys](#) with Sybase, you need to define the primary event server properties. The following information is required.

### Notes:

- If you are using a dual server configuration, a Secondary Event Server Properties window will appear. The fields mentioned here are also applicable to the Secondary Event Server Properties window.
- The silent variable for Secondary Event Server will have the word SECONDARY instead of PRIMARY. For example, for your primary server, if the silent variable is IA\_AE\_PRIMARY\_DB\_SCHEMA\_ADMIN\_PWD\_FOR\_<INSTANCE>, for the secondary server, it will be IA\_AE\_SECONDARY\_DB\_SCHEMA\_ADMIN\_PWD\_FOR\_<INSTANCE>.
- For Web UI and Reporting panels, the required silent variables are similar to those of the fresh install.
- This window will repeat for each AutoSys instance.

Field name	Silent Install Variable	Description
Instance name		The AutoSys instance name for which the upgrade is to be performed. This will appear as a label.
Database type		The type will be Sybase. This will appear as a label.
Database host		The server location where the Sybase database is hosted. This will appear as a label.
Server name		The sever name where the Sybase database is installed. This will appear as a label.
Database name		The Sybase database to be used during the upgrade. This will appear as a label.
Username		The username to access the Sybase database.
Password		The password to access the Sybase database.

## Upgrade: Web UI Panels

When you [upgrade AutoSys](#), there are various values and options that pertain to the Web UI. The following sections describe the fields in you need to define, depending on your database type:

### MSSQL: Upgrade: Web UI Database Properties

When you [upgrade AutoSys](#) with Microsoft SQL Server, you need to define the following WEB UI database properties:

Field name	Silent Install Variable	Description
Windows authentication	IA_AE_WCC_DB_MSSQL_AUTH_TYPE=0	Select this option if you want to use Windows authentication.
Microsoft SQL server authentication	IA_AE_WCC_DB_MSSQL_AUTH_TYPE=1	Select this option if you want to use MSSQL server authentication
Username	IA_AE_WCC_DB_ADMIN_USER	The username for Microsoft SQL server authentication.
Password	IA_AE_WCC_DB_ADMIN_PWD	The password for Microsoft SQL server authentication.

### MSSQL: Upgrade: Web UI Reporting Database Properties

When you [upgrade AutoSys](#) with Microsoft SQL Server, you need to define the following WEB UI Reporting database properties:

Field name	Silent Install Variable	Description
Windows authentication	IA_AE_WCC_REPORT_DB_MSSQL_AUTH_TYPE=0	Select this option if you want to use Windows authentication.
Microsoft SQL server authentication	IA_AE_WCC_REPORT_DB_MSSQL_AUTH_TYPE=1	Select this option if you want to use MSSQL server authentication.
Username	IA_AE_WCC_REPORT_DB_ADMIN_USER	The username for Microsoft SQL server authentication.
Password	IA_AE_WCC_REPORT_DB_ADMIN_PWD	The password for Microsoft SQL server authentication.

### Oracle: Upgrade: Web UI Database Properties

When you [upgrade AutoSys](#) with Oracle, you need to define the following WEB UI database properties:

Field name	Silent Install Variable	Description
Schema owner name	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_USER	The schema owner name.
Password	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_PWD	The schema owner password.
TNS admin directory	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_TNS_ADMIN	The Oracle TNS admin directory.
Oracle wallet SSO file path	IA_AE_WCC_DB_ORA_SCHEMA_ADMIN_WALLET_LOC	The Oracle wallet SSO file path.

## Oracle: Upgrade: Web UI Reporting Database Properties

When you [upgrade AutoSys](#) with Oracle, you need to define the following WEB UI Reporting database properties:

Field name	Silent Install Variable	Description
Schema owner name	IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_USER	The schema owner name.
Password	IA_AE_WCC_REOPRT_DB_ORA_SCHEMA_ADMIN_PWD	The schema owner password.
TNS admin directory	IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_TNS_ADMIN	The Oracle TNS admin directory.
Oracle wallet SSO file path	IA_AE_WCC_REPORT_DB_ORA_SCHEMA_ADMIN_WALLET_LOC	The Oracle wallet SSO file path.

## Upgrade: EEM Panels

When you [upgrade AutoSys](#), the EEM panels appear if the security interaction components, such as the Application Server configured in EEM security mode, REST web server, and WCC, are installed. The following sections describe the values you need to define in the EEM panels:

### NOTE

- If your 64-bit EEM was not installed using the AutoSys installer and if your system has other AutoSys components, the panel will prompt for an upgrade of EEM during the AutoSys upgrade. If no other AE components are available on your system, the installation will be treated as a fresh install. Later, EEM can be upgraded using the EEM installer available in the AutoSys ISO.
- The instances that are in native mode will not appear in this panel.
- This panel will not appear if the the policies for the instances and Web UI are already up to date.

## CA EEM Component Upgrade

When you [upgrade AutoSys](#), the CA EEM Component Upgrade panel appears if your 64-bit EEM was not installed using the AutoSys installer.

Select **Yes** to update EEM. If you have just installed CA EEM, a fresh install will be triggered on the system.

## Upgrade CA EEM Policies

When you [upgrade AutoSys](#), the Upgrade CA EEM Policies panel appears only if the EEM policies are at a lower version.

The panel lists all the EEM servers where the EEM policies need to be upgraded.

## CA EEM Properties

When you [upgrade AutoSys](#), the CA EEM Properties panel appears for each unique EEM server available for your AutoSys instance.

Define the following values in this panel:

Field name	Silent Install Variable	Description
CA EEM Server(s)	For AutoSys instance: IA_AE_INSTANCE_EEM_SERVER_NAME For Web UI: IA_AE_WCC_EEM_SERVER_NAME	The server name where EEM is configured. This appears as a label.
CA EEM administrator		The EEM administrator username EiamAdmin appears as the label.
Administrator password	For AutoSys instance: IA_AE_INSTANCE_EEM_PASSWORD For Web UI: IA_AE_WCC_EEM_PASSWORD	Enter the password to connect to the EEM server. In the response file, the password must be AES encrypted. Outside the installer, this can be generated using the autosys as_config utility.
Deregister CA EEM policies	IA_AE_INSTANCE_EEM_DELETE_POLICIES_FOR_<INSTANCE>	Select this check box if you want to deregister the CA EEM policies. This field appears only when you are using the AutoSys installer for maintenance while deleting an instance.

## Migrate Command Sponsor

When you [upgrade AutoSys](#), the Migrate Command Sponsor panel lists instances where the existing Command Sponsor can be migrated to AutoSys 12.0.

### NOTE

- This panel will only appear if you have the Command Sponsor component installed in CA Workload Automation AE version 11.3.6.
- CA Workload Automation AE version 11.3.6 instances configured with the native security mode cannot be migrated.

To migrate, you must define the Webserver start port details:

Field Name	Silent Install Variable	Description
Instances	NA	Lets you see the AutoSys instance from which the Command Sponsor has to be migrated. This field appears as a label.
Webserver start port	IA_AE_RESTWS_COMM_PORT_FOR_<INSTANCE>	Specify the web server start port to use for migrating the Command Sponsor as a REST web service.
Start post-installation	IA_AE_START_SERVICE_<INSTANCE>_RESTWS //Set this variable to 1 to start the web service post installation. Otherwise, set to 0.	Select this checkbox if you want to start the web service post-installation.

Start on system startup	IA_AE_AUTOSTART_SERVICE_<INSTANCE>_RESTWS //Set this variable to 1 to start the web service on system startup. Otherwise, set to 0.	Select this checkbox if you want to start the web service on system startup.
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## Miscellaneous Topics

This section covers the following topics:

### Upgrade Considerations

- If you are running in high availability mode, upgrade the primary scheduler first, followed by the shadow and tie-breaker schedulers. If required, you can upgrade the shadow and tie-breaker schedulers simultaneously. We recommend that you complete the database upgrade on the primary scheduler first and then proceed with running the upgrade on the shadow and tie-breaker schedulers.
- After the upgrade is complete, the CA Workload Automation AE services are automatically restarted if you selected the options to restart them during the initial installation. Else, you must restart the services manually.
- The database schema owner must have the following privileges. Do not revoke these privileges after you upgrade the product.

```
GRANT CREATE VIEW TO aedbadmin;
GRANT 'CONNECT' TO aedbadmin;
GRANT 'RESOURCE' TO aedbadmin;
GRANT 'SELECT_CATALOG_ROLE' TO aedbadmin;
GRANT CREATE PUBLIC SYNONYM TO aedbadmin;
```

- The database ujo\_admin role must have the following privileges. Do not revoke these privileges after you upgrade the product.

```
GRANT ANALYZE ANY TO ujoadmin;
GRANT SELECT ON DBA_TABLESPACES TO ujoadmin;
GRANT ujoadmin TO aedbadmin;
```

- The database user (for example, autosys) must have the following privileges. Do not revoke these privileges after you upgrade the product.

```
GRANT 'CONNECT' TO autosys;
GRANT ujoadmin TO autosys;
```

- Java 8 does not support X.509 certificates signed with MD5withRSA algorithm and are no longer acceptable by default. If you are using MD5withRSA algorithm to create the certificates, you cannot use the CA Workload Automation AE web services and Compliance Application with Oracle PKI certificates. For information about how to resolve this problem, see the Java documentation.

The following upgrade considerations apply for UNIX:

- If iGateway 32-bit is being used by a Broadcom product, the installer does not upgrade the Command Sponsor. To install or upgrade the Command Sponsor, issue the following commands and follow the instructions on the installation wizard:

```
cd CA Workload Automation AE_Install_Directory/CommandSponsor
chmod +x WCC_CommandSponsor.bin
./WCC_CommandSponsor.bin
```

- If the iGateway built on Linux 2.6 kernel version is already installed, you can install or upgrade the Command Sponsor. To verify if iGateway built on Linux 2.6 kernel version is installed, check for the existence of the \$IGW\_LOC/igw.linux\_k26 file.
- If the iGateway built on Linux 2.4 kernel version is already installed, you cannot install or upgrade the Command Sponsor. So, before you upgrade to the current release, we recommend that you uninstall the Command Sponsor,

iGateway, and all other iGateway plug-ins. However, if you upgrade without uninstalling these components, the upgrade detects an unsupported version of iGateway and the following message is displayed:

The CA Workload Command Center (WCC) Command Sponsor bundles the CA iGateway component. The r11.3 CA WCC Command Sponsor is installed with the 2.4 kernel version of the iGateway component. The r11.3.6 CA WCC Command Sponsor bundles the CA iGateway component with a 2.6 kernel version. The two iGateway versions are incompatible. Therefore, the r11.3 CA WCC Command Sponsor cannot be upgraded. Click OK to continue and upgrade CA Workload Automation AE and skip the r11.3 CA WCC Command Sponsor upgrade. After the CA Workload Automation AE is upgraded, you will need to manually uninstall the r11.3 CA WCC Command Sponsor and then install the r11.3.6 CA WCC Command Sponsor. Please see the CA Workload Automation AE Installing section for more information, and then install the Command Sponsor.

Click OK to continue with the upgrade. After you upgrade to the current release, uninstall and install the Command Sponsor.

#### NOTE

- To uninstall the Command Sponsor, issue the following commands and then log out and log in to reset the environment:

```
unisrvctr stop CA-WAAE
$IGW_LOC/S99gateway stop
cd $IGW_LOC/"Uninstall_CA Workload Control Center Command Sponsor"
./Uninstall CA Workload Control Center Command Sponsor -i silent
cd $IGW_LOC
./gw_remove.sh
```

If iGateway is being used by a Broadcom product, the `./gw_remove.sh` command does not uninstall iGateway. To uninstall iGateway, you must either uninstall all the dependent Broadcom products on the same computer or remove the dependencies forcefully by deleting the `$IGW_LOC/*.conf` and `$IGW_LOC/*.jar` files and then renaming the `igateway.conf.backup` file to `igateway.conf` in the `$IGW_LOC` directory.

- To install the Command Sponsor, issue the following commands and follow the instructions on the installation wizard:

```
cd CA Workload Automation AE_Install_Directory/CommandSponsor
chmod +x WCC_CommandSponsor.bin
./WCC_CommandSponsor.bin
```

- If the CA Workload Automation AE r11.3 instance is set up to integrate with CA EEM r8.4, we recommend that you upgrade CA EEM r8.4 to Release 12.6 before you upgrade the CA Workload Automation AE r11.3 instance to the current release. If your operating system does not support upgrading CA EEM, take the following actions before you upgrade CA Workload Automation AE r11.3 to the current release:
  - Install CA EEM Release 12.6 on another computer.
  - Migrate your security policies manually to CA EEM Release 12.6.
  - Set up CA Workload Automation AE r11.3 instance to integrate with CA EEM Release 12.6.

#### NOTE

- If the iGateway built on Linux 2.6 kernel version is already installed, you can upgrade CA EEM r8.4 to Release 12.6. To verify if iGateway built on Linux 2.6 kernel version is installed, check for the existence of the `$IGW_LOC/igw.linux_k26` file.
- If the iGateway built on Linux 2.4 kernel version is already installed, you cannot upgrade CA EEM r8.4 to Release 12.6. So, before you import your migrated policies, take the following actions:
  - Export your security policies.



- b. Uninstall CA EEM r8.4, iGateway, and all other iGateway plug-ins.
  - c. Install CA EEM Release 12.6.
- For more information about migrating your security policies from Unicenter AutoSys JM 4.5.1 or r11 to the current release, see [Migrate Security Policies from Unicenter AutoSys JM 4.5.1 to Current Release](#) and [Migrate CA Workload Automation AE Security Policies from Unicenter AutoSys JM Release 11](#).
- Before you upgrade, verify that the following parameter or variable values are set to the primary event server name that you specified during the installation:

The following upgrade considerations apply for Windows:

- Before you upgrade to the current release, if the tnsalias name contains a dot (.), set the TNS\_ADMIN environment variable to the path where the tnsnames.ora file is located.
- (If you are upgrading from a previous release to Release 11.3.6 SP7 or above) If iGateway 32-bit is being used by a Broadcom product, the installer does not upgrade the Command Sponsor. To upgrade to a 64-bit iGateway version, follows these steps:
  - Verify that iGateway 32-bit is not being used by any other Broadcom product, as follows:
    - Check the conf files in the Command Sponsor installation directory. The value of this directory is stored in the %IGW\_LOC% environment variable.
    - If only AutoSysCommandSponsor.conf file is found, it indicates that only CA Workload Automation AE is using iGateway.
    - If other conf files that contain ISponsor tags (<ISponsor>) are found, it indicates that other Broadcom products are using iGateway.
  - Uninstall the iGateway 32-bit version, as follows:
    - Open the Windows Control Panel's Add or Remove Programs feature.
    - Select 'CA iTechnology iGateway' from the Currently installed programs list.
    - Click Remove.
- To avoid database deadlocks, we recommend that you set the READ COMMITTED isolation level with the READ\_COMMITTED\_SNAPSHOT option to ON, as follows:
  - a. Stop all applications that are connected to the AEDB database.
  - b. Log in as the sa user and run the following command:

```
alter database AEDB set read_committed_snapshot on
```

You can perform this task either before or after the upgrade.

## Define the localhost Machine After Database Migration

To define a localhost machine after the database migration.

### Follow these steps:

1. Run the scheduler, open the scheduler log, and search for the CAUAJM\_W\_10109 message. The message indicates which *scheduler\_machine\_name* the scheduler is trying to resolve as the localhost.
2. Enter **jil** at the UNIX operating system prompt or the Windows instance command prompt. The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Delete all the legacy localhost, hostname, and fully qualified domain name machines where the scheduler runs from the database by specifying the following definition:

```
delete_machine: machine_name
force: y
```

4. Specify the following definition:

```
insert_machine: scheduler_machine_name
node_name: address
```

```
type: type
```

5. Specify additional optional attributes as required:

- agent\_name
- character\_code
- description
- encryption\_type
- factor
- heartbeat\_attempts (Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS only)
- heartbeat\_freq (Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS only)
- key\_to\_agent
- max\_load
- opsys
- port

6. Enter exit.

The localhost machine is defined in the database. The scheduler can now resolve the localhost machine in a job definition.

7. Refresh the scheduler log and look for the CAUAJM\_I\_10116 message.

The message confirms that the scheduler resolved the localhost value to the machine defined in Step 4 and that it can run jobs on this machine.

Alternatively, instead of defining a new machine, you can set the machine that the scheduler resolves as the localhost to any existing machine definition. The job definitions that include the **machine: localhost** attribute run on that existing machine. To change the machine that the scheduler resolves as the localhost on UNIX, modify the LocalMachineDefinition parameter in the configuration file. On Windows, modify the Local Machine Definition field on the Scheduler window of AutoSys Workload Automation Administrator (autosysadmin).

### Example: Define the localhost Machine After a Database Migration

Suppose that the following job definition was migrated from a previous release of AutoSys Workload Automation:

```
insert_job: h1
job_type: C
command: ls
machine: localhost
owner: root@localhost
```

After you run the scheduler, the scheduler log contains a message similar to the following:

```
CAUAJM_W_10109 Please define a single machine with the name 'prod' or specify an
existing machine as the LocalMachineDefinition configuration variable.
```

This message indicates that the scheduler tried to resolve the localhost to the prod machine. To run the job in the current release, define the prod machine in the database. Assuming that the prod machine has a Workload Automation Agent installed, the machine definition can be similar to the following:

```
insert_machine: prod
type: a
```

After you define the prod machine, the scheduler log contains a message similar to the following:

CAUAJM\_I\_10116 Localhost machine definition has been successfully set to prod.

The job can now run on the prod machine.

## (UNIX) Migrate a Unicenter AutoSys JM 4.5.1 or r11 Database Manually

To migrate the Unicenter AutoSys JM 4.5.1 or r11 database manually, follow these steps:

1. [Reviewing the Migration Considerations](#)
2. [Archive Information on the Unicenter AutoSys JM4.5.1 or r11 Database](#)
3. [Locating the TCP/IP Database ListenerPortNumber](#)
4. [Determine the Native JDBC jar and Additional jars Path](#)
5. [Migrate a Unicenter AutoSys JM4.5.1 or r11 Database](#)
6. [RebuildTableIndexesforaAutoSys Workload AutomationDatabase](#)

After you complete these steps, the database is migrated manually.

### Reviewing the Migration Considerations

As an administrator, before starting the migration process, consider the following information:

- Migrating large amounts of data takes a considerable amount of time. We recommend scheduling the migration accordingly.
- We recommend archiving old information from the database using `archive_events` on the old instance before invoking the migration utility. Archiving reduces the amount of data for migration and is more time-efficient.
- If the source database is Unicenter AutoSys JM 4.5.1, the migration utility runs prerequisite checks to ensure that the data will be migrated successfully.

#### **NOTE**

If the prerequisite checks fail, informational messages are displayed. Contact your database administrator and correct the database entries in the source database.

- Superuser information is migrated from Unicenter AutoSys JM 4.5.1 only if that instance is using native security. Unicenter AutoSys JM r11 (including service packs) superuser information is migrated if the instance is using native security or CA Embedded Entitlements Manager (CA EEM) for security. However, you can still use the `autosys_secure` utility in the current release to set up superuser information. After migrating from an instance that is using CA EEM for security, it is necessary to regenerate the security certificate for AutoSys Workload Automation by running the `autosys_secure` utility.
- We recommend creating a new parameter file using the one supplied with the installation to verify that you have the correct information for your migration. When the migration is complete, delete the parameter file or change the AutoSys Workload Automation user password in the parameter file to avoid a security exposure on the AutoSys Workload Automation user password, which is stored in ASCII format.
- The source database remains intact during the migration process with the exception of Oracle. If the source and target instances are using the same Oracle installation, several global synonyms are replaced, causing the replacing of previous definitions. The list of objects with global synonyms is as follows: `send_event`, `alamode`, `event`, `intcodes`, `proc_event`, and `timezones`.
- If the `TNS_ADMIN` environment variable is set, the source and target databases information must be included in the `$TNS_ADMIN/tnsnames.ora` file.
- We recommend reviewing the schema-related information in the [Migration Utility](#) and also gathering the following data before invoking the migration utility. You need this information to complete the migration utility parameter file.

- Source and target database computers
- Source and target database names
- TCP/IP listener port number for the source and target database
- Source and target database users
- Source and target database users passwords
- Native JDBC JAR path
- Database JAR file
- (Oracle only, if using SSL certificate to access the database) Location of the directory that contains the oraclepki.jar, osdt\_core.jar, and osdt\_cert.jar files.
- (Oracle only, if using SSL certificate to access the database) Absolute path of the Oracle wallet file (cwallet.sso) in the file system.

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- On AIX 7.1, when the operating system is not at TL1 SP2 or later, the migration utility displays the following error messages:

```
Error: Port Library failed to initialize: -125
Error: Could not create the Java Virtual Machine
```

## Migration Utility

Unlike in previous releases, the Unicenter AutoSys JM r11 (including service packs) database schema was uniquely identified in a large MDB that included the schemas of many CA products. All AutoSys Workload Automation tables were prefixed with ujo\_. For example, the job\_cond table in Unicenter AutoSys JM 4.5.1 was renamed ujo\_job\_cond in Unicenter AutoSys JM r11 (including service packs). Additionally, the Unicenter AutoSys JM r11 (including service packs) schema included new tables and new columns in existing tables, and was updated to delete unneeded tables. AutoSys 12.0 expanded on those changes and added tables for the newly supported job types. In CA Workload Automation AE r11.3, the schema was consolidated into a form that is more normalized. As a result of these changes, upgrading data from previous releases to the current release requires a migration process that retrofits the old data into the corresponding new tables and columns.

The migration utility, which comprises a single JAR file and dynamic library on the AutoSys Workload Automation installation media, is written in Java using the Java Database Connectivity (JDBC) API. This utility migrates data in the following instances:

From 4.5.1 or r11 (including service packs) Schema	To the Current Schema
<i>Sybase 11.9.2, 11.9.3, 12.0, 12.5, or 15.0.x</i>	Sybase 15.0.2 and higher
Oracle 11g	Oracle 11g or higher

### NOTE

For more information about supported database versions, see [AutoSys Compatibility Matrix](#).

You can invoke the AutoSys Workload Automation migration utility by a Perl script that takes a file name as a parameter. The input file contains the list of parameters that specify the credentials of the source database (the Unicenter AutoSys JM 4.5.1 or r11 (including service packs) instance) and the target database (the current AutoSys Workload Automation instance). The utility requires access to both the source and target databases. Because the utility is based on JDBC, it does not mandate the existence of database-specific client or server software on the computers involved.

### NOTE

The migration utility uses some of the AutoSys Workload Automation libraries. Therefore, ensure that you set the LD\_LIBRARY\_PATH environment variable to the *install\_directory*/autosys/lib directory, where *install\_directory* specifies the root directory where AutoSys Workload Automation is installed.

## **Sample UNIX Parameter File**

Sample parameter files are located in the \$AUTOSYS/dbobj/DataMover directory.

The following is an example of a UNIX parameter file migrating data to an Oracle database when you are using password to access the database:

```
SRCDBTYPE=Oracle
SRCDBMACHINE=machine1
SRCDBNAME=4.5.1_oracle_sid
SRCDBPORT=1521
SRCDBUSER=autosys
SRCDBPWD=srcdbuser_password
TGTDDBTYPE=oracle
TGTDDBMACHINE=machine2
TGTDDBNAME=11.3.6_oracle_sid
TGTDDBPORT=1521
TGTDDBUSER=aedbadmin
TGTDDBPWD=tgtdbuser_password
NATIVEJDBCJARPATH=/opt/oracle/11.2.0/jdbc/lib/ojdbc6.jar
JREPATH=/opt/CA/WorkloadAutomationAE/JRE_WA
VERIFY=no
MAXMEMORYLIMIT=1024
```

The following is an example of a UNIX parameter file migrating data to an Oracle database when you are using an SSL certificate to access the database:

```
SRCDBTYPE=Oracle
SRCDBMACHINE=machine1
SRCDBNAME=4.5.1_oracle_sid
SRCDBPORT=1521
SRCDBUSER=autosys
SRCDBPWD=srcdbuser_password
TGTDDBTYPE=oracle
TGTDDBNAME=11.3.6_oracle_sid
TGTAEDBADMINWALLETPATH=/home/aedbadmin_wallet/cwallet.sso
NATIVEJDBCJARPATH=/opt/oracle/11.2.0/jdbc/lib/ojdbc6.jar
JREPATH=/opt/CA/WorkloadAutomationAE/JRE_WA
ADDITIONALJARSPATH=/usr/lib/oracle/product/11.1.0/db_1/jlib/oraclepki.jar:/usr/lib/oracle/product/11.1.0/db_1/jlib/osdt_core.jar:/usr/lib/oracle/product/11.1.0/db_1/jlib/osdt_cert.jar
JREPATH=/opt/CA/WorkloadAutomationAE/JRE_WA
VERIFY=no
MAXMEMORYLIMIT=1024
```

## **Archive Information on the Unicenter AutoSys JM 4.5.1 or r11 Database**

Archiving reduces the amount of data for migration. To archive information from the Unicenter AutoSys JM 4.5.1 or r11 database, run the DBMaint command on the old instance before invoking the migration utility.

The DBMaint command runs the dbstatistics, archive\_events, and archive\_jobs commands to perform maintenance on the database.

The DBMaint command has the following format:

```
DBMaint [-x] [-?]
```

- **-x**  
(Optional) Returns version information.
- **-?**  
(Optional) Displays help for the command.

The DBMaint command performs the following tasks:

- Runs the archive\_events command to remove old information from the database tables.
- Runs the archive\_jobs command to remove obsolete jobs from the database tables.
- Runs the dbstatistics command to update the statistics for all database tables except the ujo\_meta\_enumerations, ujo\_meta\_properties, ujo\_meta\_rules, ujo\_meta\_types, and ujo\_real\_resource tables. The dbstatistics command also calculates and updates the average job run statistics in the ujo\_avg\_job\_run table.
- Runs the dbspace command to determine the available space in the database.

### **Locating the TCP/IP Database Listener Port Number**

The TCP/IP database listener port numbers are required during the migration process to complete the migration utility parameter file, based on your installation. The port numbers facilitate communications across database instances.

Depending on your type of database, do *one* of the following:

- [Locate the TCP/IP Port for Oracle](#)
- [Locate the TCP/IP Port for Sybase](#)

#### **Locate the TCP/IP Port for Oracle**

To find the TCP/IP port for Oracle, open the tnsnames.ora file typically located in \$ORACLE\_HOME/network/admin . Locate the following parameter:

```
(ADDRESS = (PROTOCOL = TCP) (HOST = host.domain.com) (PORT = 1521))
```

#### **NOTE**

The default port number is 1521.

#### **Locate the TCP/IP Port for Sybase**

To find the TCP/IP port for Sybase, open the interfaces file for Adaptive Server Enterprise. The default location is \$SYBASE/interfaces. Locate the following database entry:

```
AUTOSYSDB
master tcp ether machine1 5000
query tcp ether machine1 5000
```

#### **NOTE**

The default port number is 5000.

### **Determine the Native JDBC jar and Additional jars Path**

Use the NATIVEJDBCJARPATH parameter in the parameter file to specify the native JDBC jar file path for the Oracle and Sybase database types. The following tables list native JDBC jar files and their example sample paths:

**Oracle: Oracle Database 11g or 12c JDBC Drivers**

JDBC JAR	Example Path
ojdbc6.jar	\$ORACLE_HOME/jdbc/lib/ojdbc6.jar

**Oracle: Oracle Database 18c or 19c JDBC Drivers**

JDBC JAR	Example Path
ojdbc7.jar	\$ORACLE_HOME/jdbc/lib/ojdbc7.jar

**Sybase: Sybase jConnect 7 Drivers**

JDBC JAR	Example Path
jconn4.jar	\$SYBASE/jConnect-7_0/classes/jconn4.jar

Use the ADDITIONALJARSPATH parameter in the parameter file to specify the additional jar files path if you are using SSL certificate to access the Oracle database. The following tables list additional jar files and their example sample paths:

**Oracle: Additional Jars**

JDBC JAR	Example Path
oraclepki.jar	\$ORACLE_HOME/jlib/oraclepki.jar
osdt_core.jar	\$ORACLE_HOME/jlib/osdt_core.jar
osdt_cert.jar	\$ORACLE_HOME/jlib/osdt_cert.jar

**NOTE**

- The paths in the example are for illustrative purposes and are not necessarily the same on every installation.
- The migration utility requires access to these JAR files to perform the migration. Verify that these sets of drivers are installed and accessible on the installation before proceeding with the migration.
- If you are using the Datamover utility to migrate data to r11.3 with Oracle 12c, ensure that you perform the following tasks before you migrate the data:
  - Install Java 1.6 on the computer where you are performing the migration and set the JREPATH parameter in the parameter file to the path where Java 1.6 is located.
  - Set the NATIVEJDBCJARPATH parameter in the parameter file to the path where ojdbc6.jar or ojdbc7.jar file is located.
- If you do not have the database vendor JAR file installed, you can download the respective JAR files for each database vendor. Follow the installation instructions as specified by each vendor. Contact the specific vendor for any licensing requirements.

**Migrate a Unicenter AutoSys JM 4.5.1 or r11 Database****WARNING**

Before using AutoSys Workload Automation, verify that the data migration is completed successfully. Any data in the target database is removed before the actual migration starts with the exception of AutoSys Workload Automation users, which are retained.

**Follow these steps:**

1. Assuming that AutoSys Workload Automation is installed in /opt/CA/WorkloadAutomationAE and the AutoSys Workload Automation instance ID is ACE, source the AutoSys Workload Automation environment as follows:
 

```
. /opt/CA/WorkloadAutomationAE/autouser.ACE/autosys.shell.host
```

```
cd $AUTOSYS/dbobj/DataMover
```

- **shell**  
Specifies the name of the shell that is used to execute the script or command file.
- **host**  
Specifies the name of the computer where AutoSys Workload Automation is installed.

2. Enter the following commands:

```
unisrvctr stop waae_sched.$AUTOSERV
unisrvctr stop waae_server.$AUTOSERV
unisrvctr stop waae_webserver.$AUTOSERV
```

The scheduler, application server, and the web server stop.

3. Edit the following parameters in the parameter file, using values based on your installation:

- **SRCDBTYPE** Defines the source database type. Valid values are *oracle* and *sybase*.
- **SRCDBMACHINE**  
Defines the source database computer.
- **SRCDBNAME**  
Defines the source database name. For Oracle, specify the Oracle System ID.
- **SRCDBPORT**  
Defines the source database port (the TCP/IP listener port for the database).
- **SRCDBUSER**  
Defines the source database user. For Oracle, the source database user is *autosys* (if you are migrating from 4.5.1 to the current release) or *mdbadmin* (if you are migrating from r11 to the current release).
- **SRCDBPWD**  
Defines the password that is associated with the source database user.
- **TGTDBTYPE**  
Defines the target database type. Valid values are *oracle* and *sybase*.
- **TGTDBMACHINE**  
Defines the target database computer.
- **TGTDBNAME**  
Defines the target database name. For Oracle, specify the Oracle System ID.
- **TGTDBPORT**  
Defines the target database port (the TCP/IP listener port for the database).
- **TGTDBUSER**  
Defines the target database user. For Oracle, this must be *schema owner*.
- **TGTDBPWD**  
Defines the password that is associated with the target database user.

**NOTE**

Before the actual migration starts, any data in the target database is removed and the deleted rows are logged. However, if you use the system administrator user ID and password for the target database, the tables are truncated, which removes the data without logging it and the migration is faster.

- **(Oracle only) TGTAEADBADMINWALLETPATH**

Defines the absolute path of the Oracle wallet file (*cwallet.sso*) in the file system. This Oracle wallet file must include the SSL certificate that is associated with the schema owner on the target database.

**NOTE**

Specify this parameter only if you are using SSL certificate to access the database. If you specify **TGTAEADBADMINWALLETPATH** parameter, you must specify a value for **ADDITIONALJARSPATH** parameter and the values specified for the **TGTDBMACHINE**, **TGTDBPORT**, **TGTDBUSER**, and **TGTDBPWD** parameters are ignored.

- **NATIVEJDBCJARPATH**



Defines an Oracle or Sybase native JDBC JAR path.

– **JREPATH**

Defines the JRE root path.

**NOTE**

The migration utility requires 32-bit JRE, which is installed by default with AutoSys Workload Automation.

- **(Oracle only) ADDITIONALJARSPATH** Defines the location of the directory that contains the oraclepki.jar, osdt\_core.jar, and osdt\_cert.jar files. Separate each jar file path with a colon.

**NOTE**

Specify this parameter only if you are using SSL certificate to access the database.

– **VERIFY**

**NOTE**

We recommend that you first set the VERIFY parameter to YES. This ensures that all the connection information is correct before actually migrating any data.

Specifies whether to verify the source and target credentials. Valid values are:

- **yes**  
Tests the source and target credentials and prints a message if the verification test is successful, but does not proceed with the migration. This verification lets you verify that the input parameters for the source and target databases are correct and perform the data migration later.
- **no**  
Continues the migration process without verifying the source and target credentials.
- **source**  
Tests the source credentials and prints a message if the verification test is successful.

– **MAXMEMORYLIMIT**

Defines the maximum heap memory with which the migration utility runs.

**Default:** 1024 MB

4. Run the uajmdatamover.pl utility using the following command:

```
perl uajmdatamover.pl paramfile
```

A message appears if the verification test is successful.

5. Repeat Step 2, setting the VERIFY parameter to NO.

6. Repeat Step 3.

The migration utility completes the migration. When the migration is successful, the command prompt returns with a timestamp message. A log of the actions taken is written to DataMover.log in the dbobj directory.

**NOTE**

- To stop the migration utility during the migration process, press Ctrl+C while the utility is running. The following message appears:  

```
User Interrupt Encountered. AutoSys Database Not Migrated.
Please rerun the utility. Exiting...
```
- If the migration process is interrupted, reinvoke the migration utility to start the process again.

7. Enter the following commands:

```
unisrvctr start waee_sched.$AUTOSERV
unisrvctr start waee_server.$AUTOSERV
unisrvctr stop waee_webserver.$AUTOSERV
```

The scheduler, application server, and the web server start.

8. Run autosys\_secure to set the superuser information, and verify that the migration was successful.

**NOTE**

Superuser information is migrated from prior releases (if they are using native security) or from Unicenter AutoSys JM r11 (including service packs). CA EEM deemphasized the need for superusers.

**WARNING**

The scheduler adjusts AutoSys Workload Automation information in the migrated database. Run the scheduler at least once before you run any other AutoSys Workload Automation processes.

**Rebuild Table Indexes for a AutoSys Workload Automation Database**

Over time, the database table indexes can become inefficient while you run jobs and update them. You can rebuild the table indexes of a specified AutoSys Workload Automation database to renew the efficiency.

After the migration is complete, recreate the table indexes. If you do not recreate the table indexes, it may impact the performance of client utilities in retrieving jobs from the database.

**NOTE**

- The `reindexDB` script is located in the `$AUTOSYS/dbobj` directory.
- We recommend that you run the `reindexDB` script when the system activity is minimal. Otherwise, AutoSys Workload Automation may experience a slow down or time-out condition while performing database transactions.

**Follow these steps:**

1. Open the operating system prompt.
2. Run the following script:

```
reindexDB.pl database_type server_name server_userid server_password database_name
```

On Sybase, if you are using Kerberos authentication to access the database, run the following script:

```
reindexDB.pl -EXTAUTH SYB server_name database_name SPN
```

**NOTE**

To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which has `sa_role`.

On Oracle, if you are using SSL certificate to access the database, run the following script:

```
reindexDB.pl -EXTAUTH ORA server_name aedbadmin_tnsadmin
```

**– database\_type**

*Specifies the database type. This value can be one of the following:*

- **ORA**  
Identifies Oracle as the database.
- **SYB**  
Identifies Sybase as the database.

**– EXTAUTH**

*Specifies that an SSL certificate (Oracle) or Kerberos authentication (Sybase) is used to access the database.*

**NOTE**

- On Oracle, you can also use the CERTMODE parameter (instead of the EXTAUTH parameter) to specify that SSL certificate is used to access the database.
- For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).
- For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)

– **server\_name**

Defines the name of the Oracle System ID or Sybase server.

– **aedbadmin\_tnsadmin**

Specifies the TNS\_ADMIN value for the schema owner.

– **server\_userid**

Defines the user ID that is used to connect to the Oracle or Sybase server.

**Default:** sa (Sybase)

**NOTE**

For Oracle, you must use schema owner as the server user ID.

– **server\_password**

Defines the password that corresponds to the user ID that is used to connect to the Oracle or Sybase server.

**Default:** autosys

– **database\_name**

Defines the name of the Sybase database.

**NOTE**

The database\_name parameter does not apply to Oracle.

- **SPN** (Optional) Specifies the service principal name to be used for Kerberos authentication. The CA Workload Automation AE table indexes are rebuild.

**Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Sybase**

This example rebuilds table indexes for a AutoSys Workload Automation database on Sybase where SYBASESRV is the name of the Sybase server.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl SYB SYBASESRV sa password
AEDB
```

**Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Oracle**

This example rebuilds table indexes for a AutoSys Workload Automation database on Oracle where ORACLESRV is the Oracle System ID.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl ORA ORACLESRV aedbadmin
password
```

You have successfully migrated the database manually. You can now connect to the database and perform any database-related operation or run any AutoSys Workload Automation service or job.

**(Windows) Migrate a Unicenter AutoSys JM 4.5.1 or r11 Database Manually**

To migrate the Unicenter AutoSys JM 4.5.1 or r11 database manually, follow these steps:

1. [Reviewing the Migration Considerations](#)
2. [Archive Information on the Unicenter AutoSys JM 4.5.1 or r11 Database](#)
3. [Locating the TCP/IP Database Listener Port Number](#)

4. [Determine the Native JDBC jar and Additional jars Path](#)
5. [Migrate a Unicenter AutoSys JM 4.5.1 or r11 Database](#)
6. [Rebuild Table Indexes for a AutoSys Workload Automation Database](#)

After you complete these steps, the database is migrated manually.

### **Reviewing the Migration Considerations**

As an administrator, before starting the migration process, consider the following information:

- Migrating large amounts of data takes a considerable amount of time. We recommend scheduling the migration accordingly.
- We recommend archiving old information from the database using `archive_events` on the old instance before invoking the migration utility. Archiving reduces the amount of data for migration and is more time-efficient.
- If the source database is Unicenter AutoSys JM 4.5.1, the migration utility runs prerequisite checks to ensure that the data will be migrated successfully.

#### **NOTE**

If the prerequisite checks fail, informational messages are displayed. Contact your database administrator and correct the database entries in the source database.

- Superuser information is migrated from Unicenter AutoSys JM 4.5.1 only if that instance is using native security. Unicenter AutoSys JM r11 (including service packs) superuser information is migrated if the instance is using native security or CA Embedded Entitlements Manager (CA EEM) for security. However, you can still use the `autosys_secure` utility in the current release to set up superuser information. After migrating from an instance that is using CA EEM for security, it is necessary to regenerate the security certificate for AutoSys Workload Automation by running the `autosys_secure` utility.
- We recommend creating a new parameter file using the one supplied with the installation to verify that you have the correct information for your migration. When the migration is complete, delete the parameter file or change the AutoSys Workload Automation user password in the parameter file to avoid a security exposure on the AutoSys Workload Automation user password, which is stored in ASCII format.
- The source database remains intact during the migration process with the exception of Oracle. If the source and target instances are using the same Oracle installation, several global synonyms are replaced, causing the replacing of previous definitions. The list of objects with global synonyms is as follows: `send_event`, `alamode`, `event`, `intcodes`, `proc_event`, and `timezones`.
- If the `TNS_ADMIN` environment variable is set, the source and target databases information must be included in the `%TNS_ADMIN%\tnsnames.ora` file.
- We recommend reviewing the schema-related information in the [migration utility](#) and also gathering the following data before invoking the migration utility. You need this information to complete the migration utility parameter file.
  - Source and target database computers
  - Source and target database names
  - TCP/IP listener port number for the source and target database
  - Source and target database users
  - Source and target database users passwords
  - Native JDBC JAR path
  - Database JAR file
  - (Oracle only, if using SSL certificate to access the database) Location of the directory that contains the `oraclepi.jar`, `osdt_core.jar`, and `osdt_cert.jar` files.
  - (Oracle only, if using SSL certificate to access the database) Absolute path of the Oracle wallet file (`cwallet.sso`) in the file system.

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

## Migration Utility

Unlike in previous releases, the Unicenter AutoSys JM r11 (including service packs) database schema was uniquely identified in a large MDB that included the schemas of many CA products. All AutoSys Workload Automation tables were prefixed with ujo\_. For example, the job\_cond table in Unicenter AutoSys JM 4.5.1 was renamed ujo\_job\_cond in Unicenter AutoSys JM r11 (including service packs). Additionally, the Unicenter AutoSys JM r11 (including service packs) schema included new tables and new columns in existing tables, and was updated to delete unneeded tables. AutoSys Workload Automation expanded on those changes and added tables for the newly supported job types. In CA Workload Automation AE r11.3, the schema was consolidated into a form that is more normalized. As a result of these changes, upgrading data from previous releases to the current release requires a migration process that retrofits the old data into the corresponding new tables and columns.

The migration utility, which comprises a single JAR file and dynamic library on the AutoSys Workload Automation installation media, is written in Java using the Java Database Connectivity (JDBC) API. This utility migrates data in the following instances:

From 4.5.1 or r11 (including service packs) Schema	To the Current Schema
Microsoft SQL Server 7, 2000, or 2005	Microsoft SQL Server 2008, 2008 R2, or 2012
Oracle 11g	Oracle 11g and higher
Sybase 11.9.2, 11.9.3, 12.0, 12.5, or 15.0.x	Sybase 15.0.2 and higher

### NOTE

For more information about supported database versions, see [AutoSys Compatibility Matrix](#).

You can invoke the AutoSys Workload Automation migration utility by a Perl script that takes a file name as a parameter. The input file contains the list of parameters that specify the credentials of the source database (the Unicenter AutoSys JM 4.5.1 or r11 (including service packs) instance) and the target database (the current AutoSys Workload Automation instance). The utility requires access to both the source and target databases. Because the utility is based on JDBC, it does not mandate the existence of database-specific client or server software on the computers involved.

### NOTE

The migration utility uses some of the AutoSys Workload Automation libraries. Therefore, ensure that you set the PATH environment variable to the *install\_directory\autosys\lib* directory, where *install\_directory* specifies the root directory where AutoSys Workload Automation is installed.

## Sample Windows Parameter File

Sample parameter files are located in the %AUTOSYS%\dbobj\DataMover directory.

The following is an example of a Windows parameter file migrating data to an Oracle database when you are using password to access the database:

```

SRCDBTYPE=Oracle
SRCDBMACHINE=machine1
SRCDBNAME=4.5.1_oracle_sid
SRCDBPORT=1521
SRCDBUSER=autosys
SRCDBPWD=srcdbuser_password
TGTDDBTYPE=oracle
TGTDDBMACHINE=machine2
TGTDDBNAME=11.3.6 SP1_oracle_sid
TGTDDBPORT=1521
TGTDDBUSER=aedbadmin

```

```

TGTDBPWD=tgtdbuser_password
NATIVEJDBCJARPATH=C:\oracle\product\11.2.0\Db_1\jdbc\lib\ojdbc6.jar
JREPATH=C:\Program Files\CA\Workload Automation AE\JRE_WA
VERIFY=no
MAXMEMORYLIMIT=1024

```

The following is an example of a Windows parameter file migrating data to an Oracle database when you are using an SSL certificate to access the database:

```

SRCDBTYPE=Oracle
SRCDBMACHINE=machine1
SRCDBNAME=4.5.1_oracle_sid
SRCDBPORT=1521
SRCDBUSER=autosys
SRCDBPWD=srcdbuser_password
TGTDDBTYPE=oracle
TGTDDBNAME=11.3.6 SP1_oracle_sid
TGTAEDBADADMINWALLETPATH=C:\aedbadmin_wallet\cwallet.sso
NATIVEJDBCJARPATH=C:\oracle\product\11.2.0\Db_1\jdbc\lib\ojdbc6.jar
JREPATH=C:\Program Files\CA\Workload Automation AE\JRE_WA
ADDITIONALJARSPATH=C:\oracle\product\11.2.0\Db_1\jlib\oraclepki.jar;C:\oracle\product\11.2.0\Db_1\jlib\osdt_core.jar;C:\oracle\product\11.2.0\Db_1\jlib\osdt_cert.jar
VERIFY=no
MAXMEMORYLIMIT=1024

```

### **Archive Information on the Unicenter AutoSys JM 4.5.1 or r11 Database**

Archiving reduces the amount of data for migration. To archive information from the Unicenter AutoSys JM 4.5.1 or r11 database, run the DBMaint command on the old instance before invoking the migration utility.

The DBMaint command runs the dbstatistics, archive\_events, and archive\_jobs commands to perform maintenance on the database.

The DBMaint command has the following format:

```
DBMaint [-x] [-?]
```

- **-x**  
(Optional) Returns version information.
- **-?**  
(Optional) Displays help for the command.

The DBMaint command performs the following tasks:

- Runs the archive\_events command to remove old information from the database tables.
- Runs the archive\_jobs command to remove obsolete jobs from the database tables.
- Runs the dbstatistics command to update the statistics for all database tables except the ujo\_meta\_enumerations, ujo\_meta\_properties, ujo\_meta\_rules, ujo\_meta\_types, and ujo\_real\_resource tables. The dbstatistics command also calculates and updates the average job run statistics in the ujo\_avg\_job\_run table.
- Runs the dbspace command to determine the available space in the database.

## **Locating the TCP/IP Database Listener Port Number**

The TCP/IP database listener port numbers are required during the migration process to complete the migration utility parameter file, based on your installation. The port numbers facilitate communications across database instances.

Depending on your type of database, do *one* of the following:

- [Locate the TCP/IP Port for Microsoft SQL Server](#)
- [Locate the TCP/IP Port for Oracle](#)
- [Locate the TCP/IP Port for Sybase](#)

### **Locate the TCP/IP Port for Microsoft SQL Server**

#### **To find the TCP/IP port for Microsoft SQL Server**

1. Start SQL Query Analyzer, and connect to the instance of Microsoft SQL Server.

#### **NOTE**

If you have installed AutoSys Workload Automation on Microsoft SQL Server 2008, you must use SQL Server Management Studio to query the database.

2. Run the following Transact-SQL statement in SQL Query Analyzer:

```
Use master
Go
Xp_readerrorlog
```

The Results pane appears.

3. Locate the following text:  
SQL server listening on X.X.X.X: Y
  - **X.X.X.X**  
Indicates the IP address of the instance of Microsoft SQL Server.
  - **Y**  
Indicates the TCP/IP port where Microsoft SQL Server is listening.

#### **NOTE**

The default port number is 1433.

#### **Example: Microsoft SQL Server TCP/IP port**

Suppose the text reads 10.150.158.246: 1433. 10.150.158.246 indicates the IP address of the Microsoft SQL Server, and 1433 indicates the TCP/IP port where the instance of Microsoft SQL Server is listening.

### **Locate the TCP/IP Port for Oracle**

To find the TCP/IP port for Oracle, open the tnsnames.ora file typically located in %ORACLE\_HOME%\network\admin. Locate the following parameter:

```
(ADDRESS = (PROTOCOL = TCP) (HOST = host.domain.com) (PORT = 1521))
```

#### **NOTE**

The default port number is 1521.

### **Locate the TCP/IP Port for Sybase**

To find the TCP/IP port for Sybase, open the sql.ini file for Adaptive Server Enterprise. The default location is %SYBASE%\ini\sql.ini. Locate the following database entry:

```
[AUTOSYSDB]
MASTER=NLWNSCK,machine1,5000
```

QUERY=NLWNSCK,machine1,5000

#### NOTE

The default port number is 5000.

### Determine the Native JDBC jar and Additional jars Path

Use the NATIVEJDBCJARPATH parameter in the parameter file to specify the native JDBC JAR file path for the Microsoft SQL Server, Oracle, and Sybase database types. The following tables list native JDBC JAR files and their example sample paths:

#### SQL Server: Microsoft SQL Server 2008 JDBC Drivers

JDBC JAR	Example Path
sqljdbc.jar	%MSSQL_HOME%\sqljdbc_1.2\enu\sqljdbc4.jar

#### Oracle: Oracle Database 11g or 12c JDBC Drivers

JDBC JAR	Example Path
ojdbc6.jar	%ORACLE_HOME%\jdbc\lib\ojdbc6.jar

#### Oracle: Oracle Database 18c or 19c JDBC Drivers

JDBC JAR	Example Path
ojdbc7.jar	%ORACLE_HOME%\jdbc\lib\ojdbc7.jar

#### Sybase: Sybase jConnect 7 Drivers

JDBC JAR	Example Path
jconn4.jar	%SYBASE%\jConnect-7_0\classes\jconn4.jar

Use the ADDITIONALJARSPATH parameter in the parameter file to specify the additional jar files path if you are using SSL certificate to access the Oracle database. The following tables list additional jar files and their example sample paths:

#### Oracle: Additional Jars

JDBC JAR	Example Path
oraclepki.jar	%ORACLE_HOME%\jlib\oraclepki.jar
osdt_core.jar	%ORACLE_HOME%\jlib\osdt_core.jar
osdt_cert.jar	%ORACLE_HOME%\jlib\osdt_cert.jar

#### NOTE

- The paths in the example are for illustrative purposes and are not necessarily the same on every installation.
- The migration utility requires access to these JAR files to perform the migration. Verify that these sets of drivers are installed and accessible on the installation before proceeding with the migration.
- If you are using the Datamover utility to migrate data to r11.3 with Oracle 12c, ensure that you perform the following tasks before you migrate the data:



- Install Java 1.6 on the computer where you are performing the migration and set the JREPATH parameter in the parameter file to the path where Java 1.6 is located.
- Set the NATIVEJDBCJARPATH parameter in the parameter file to the path where ojdbc6.jar or ojdbc7.jar file is located.
- If you do not have the database vendor JAR file installed, you can download the respective JAR files for each database vendor. Follow the installation instructions as specified by each vendor. Contact the specific vendor for any licensing requirements.

### **Migrate a Unicenter AutoSys JM 4.5.1 or r11 Database**

If you bypassed the automatic database migration when you upgraded to the current release or encountered errors during migration, run the migration utility before you can use your new AutoSys Workload Automation instance.

#### **WARNING**

Before using AutoSys Workload Automation, verify that the data migration is completed successfully. Any data in the target database is removed before the actual migration starts with the exception of AutoSys Workload Automation users, which are retained.

#### **Follow these steps:**

1. Stop the scheduler and application server as follows:
  - a. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
  - c. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - d. Right-click the scheduler and select Stop. Repeat this action for the application server and web server.  
The scheduler, application server, and web server stop.
2. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The AutoSys Workload Automation instance command prompt opens.
3. Enter the following command:
 

```
cd %AUTOSYS%\dbobj\DataMover
```

 The DataMover directory appears.
4. Edit the following parameters in the parameter file, using the values that are based on your installation:
  - **SRCDBTYPE**  
Defines the source database type. Valid values are *oracle*, *mssql*, and *sybase*.
  - **SRCDBMACHINE**  
Defines the source database computer.
  - **SRCDBNAME**  
Defines the source database name. For Oracle, specify the Oracle System ID.
  - **SRCDBPORT**  
Defines the source database port (the TCP/IP listener port for the database).
  - **SRCDBUSER**  
Defines the source database user. For Oracle, the source database user is *autosys* (if you are migrating from 4.5.1 to the current release) or *mbadmin* (if you are migrating from r11 to the current release).
  - **SRCDBPWD**  
Defines the password that is associated with the source database user.
  - **TGTDBTYPE**  
Defines the target database type. Valid values are *mssql*, *oracle*, and *sybase*.
  - **TGTDBMACHINE**

Defines the target database computer.

– **TGTDBNAME**

Defines the target database name. For Oracle, specify the Oracle System ID.

– **TGTDBPORT**

Defines the target database port (the TCP/IP listener port for the database).

– **TGTDBUSER**

Defines the target database user. For Oracle, this must be *schema owner*.

– **TGTDBPWD**

Defines the password that is associated with the target database user.

**NOTE**

Before the actual migration starts, any data in the target database is removed and the deleted rows are logged. However, if you use the system administrator user ID and password for the target database, the tables are truncated, which removes the data without logging it and the migration is faster.

- (Oracle only) **TGTAEDBADMINWALLETPATH** Defines the absolute path of the Oracle wallet file (cwallet.sso) in the file system. This Oracle wallet file must include the SSL certificate that is associated with the schema owner on the target database.

**NOTE**

Specify this parameter only if you are using SSL certificate to access the database. If you specify TGTAEDBADMINWALLETPATH parameter, you must specify a value for ADDITIONALJARSPATH parameter and the values specified for the TGTDBMACHINE, TGTDBPORT, TGTDBUSER, and TGTDBPWD parameters are ignored.

– **NATIVEJDBCJARPATH**

Defines a Microsoft SQL Server, Oracle, or Sybase native JDBC JAR path.

– **JREPATH**

Defines the JRE root path.

**NOTE**

The migration utility requires JRE 1.7.0\_72, which is installed by default with AutoSys Workload Automation.

**Example:** C:\Program Files\CA\Workload Automation AE\JRE\_WA

– (Oracle only) **ADDITIONALJARSPATH**

Defines the location of the directory that contains the oraclepki.jar, osdt\_core.jar, and osdt\_cert.jar files. Separate each jar file path with a semicolon.

**NOTE**

Specify this parameter only if you are using SSL certificate to access the database.

– **VERIFY**

**NOTE**

We recommend that you first set the VERIFY parameter to YES. This ensures that all the connection information is correct before actually migrating any data.

Specifies whether to verify the source and target credentials. Valid values are:

- **yes**  
Tests the source and target credentials and prints a message if the verification test is successful, but does not proceed with the migration. This verification lets you verify that the input parameters for the source and target databases are correct and perform the data migration later.
- **no**

Continues the migration process without verifying the source and target credentials.

- **source**

Tests the source credentials and prints a message if the verification test is successful.

- **MAXMEMORYLIMIT**

Defines the maximum heap memory with which the migration utility runs.

**Default:** 1024 MB

5. Run the `uajmdatamover.pl` utility using the following command:

```
perl uajmdatamover.pl paramfile
```

A message appears if the verification test is successful.

6. Repeat Step 4, setting the `VERIFY` parameter to `NO`.

7. Repeat Step 5.

The migration utility completes the migration. When the migration is successful, the command prompt returns with a timestamp message. A log of the actions taken is written to `DataMover.log` in the `dbobj` directory.

#### NOTE

- To stop the migration utility during the migration process, press `Ctrl+C` while the utility is running. The following message appears:

```
User Interrupt Encountered. AutoSys Database Not Migrated.
Please rerun the utility. Exiting...
```

- If the migration process is interrupted, reinvoke the migration utility to start the process again.

8. Start the scheduler and application server as follows:

- a. Click **Start, Programs, AutoSys, Administrator**.

The **Instance - AutoSys Workload Automation Administrator** window opens.

- b. Select your AutoSys Workload Automation instance from the Instance drop-down list.

- c. Click the **Services** icon on the toolbar.

The **Services - AutoSys Workload Automation Administrator** window appears, displaying a list of services installed on the selected instance.

- d. Right-click the scheduler and select **Start**. Repeat this action for the application server and web server.

The scheduler, application server, and web server start.

9. Run `autosys_secure` to set the superuser information, and verify that the migration was successful.

#### NOTE

Superuser information is migrated from prior releases (if they are using native security) or from Unicenter AutoSys JM r11 (including service packs). CA EEM deemphasized the need for superusers.

#### WARNING

The scheduler adjusts AutoSys Workload Automation information in the migrated database. Run the scheduler at least once before you run any other AutoSys Workload Automation processes.

### **Rebuild Table Indexes for a AutoSys Workload Automation Database**

Over time, the database table indexes can become inefficient while you run jobs and update them. You can rebuild the table indexes of a specified AutoSys Workload Automation database to renew the efficiency.

After the migration is complete, recreate the table indexes. If you do not recreate the table indexes, it may impact the performance of client utilities in retrieving jobs from the database.

#### **Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Sybase**

This example rebuilds table indexes for a AutoSys Workload Automation database on Sybase where `SYBASESRV` is the name of the Sybase server.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl SYB SYBASESRV sa password
AEDB
```

### Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Oracle

This example rebuilds table indexes for a AutoSys Workload Automation database on Oracle where ORACLESRV is the Oracle System ID.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl ORA ORACLESRV aedbadmin
password
```

#### NOTE

- The reindexDB script is located in the %AUTOSYS%\dbobj directory.
- We recommend that you run the reindexDB script when the system activity is minimal. Otherwise, AutoSys Workload Automation may experience a slow down or time-out condition while performing database transactions

#### Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The AutoSys Workload Automation instance command prompt opens.
2. Run the following script:

```
reindexDB.pl database_type server_name server_userid server_password database_name
```

On Sybase, if you are using Kerberos authentication to access the database, run the following script:

```
reindexDB.pl -EXTAUTH SYB server_name database_name SPN
```

#### NOTE

To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which has sa\_role.

On Oracle, if you are using SSL certificate to access the database, run the following script:

```
reindexDB.pl -EXTAUTH ORA server_name aedbadmin_tnsadmin
```

#### – **database\_type**

*Specifies the database type. This value can be one of the following:*

- **ORA**  
Identifies Oracle as the database.
- **SYB**  
Identifies Sybase as the database.
- **MSQ**  
Identifies Microsoft SQL Server as the database.

#### – **EXTAUTH**

*Specifies that an SSL certificate (Oracle) or Kerberos authentication (Sybase) is used to access the database.*

#### NOTE

- On Oracle, you can also use the CERTMODE parameter (instead of the EXTAUTH parameter) to specify that SSL certificate is used to access the database.
- For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).
- For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)

#### – **server\_name**

Defines the name of the Oracle System ID, Sybase server, or Microsoft SQL Server server.

– ***aedbadmin\_tnsadmin***

Specifies the TNS\_ADMIN value for the schema owner.

– ***server\_userid***

Defines the user ID that is used to connect to the Oracle, Sybase, or Microsoft SQL Server server.

**Default:** sa (Sybase, Microsoft SQL Server)

**NOTE**

For Oracle, you must use schema owner as the server user ID.

– ***server\_password***

Defines the password that corresponds to the user ID that is used to connect to the Oracle, Sybase, or Microsoft SQL Server server.

**Default:** autosys

– ***database\_name***

Defines the name of the Sybase or Microsoft SQL Server database

**NOTE**

The database\_name parameter does not apply to Oracle.

– ***SPN*** (Optional) Specifies the service principal name to be used for Kerberos authentication.

The AutoSys Workload Automation table indexes are rebuild.

### Example: Rebuild Table Indexes for a AutoSys Workload Automation database on Microsoft SQL Server

This example rebuilds table indexes for a AutoSys Workload Automation database on Microsoft SQL Server where MSQSRV is the name of the Microsoft SQL Server.

```
c:\Program Files (x86)\CA\Workload Automation AE\autosys\dbobj>perl reindexDB.pl MSQ
MSQSRV sa password AEDB
```

### Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Sybase

This example rebuilds table indexes for a AutoSys Workload Automation database on Sybase where SYBASESRV is the name of the Sybase server.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl SYB SYBASESRV sa password
AEDB
```

### Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Oracle

This example rebuilds table indexes for a AutoSys Workload Automation database on Oracle where ORACLESRV is the Oracle System ID.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl ORA ORACLESRV aedbadmin
password
```

You have successfully migrated the database manually. You can now connect to the database and perform any database-related operation or run any AutoSys Workload Automation service or job.

## Configure SockAdapter to Change the Socket Type of the Agent

You can change, the socket type used for communication by the Agent and CA Workload Automation AE, from Plain to Dylan mode.

This is applicable only to the agents upgraded from CA Workload Automation AE Releases 11.3.6 SP5 and SP6.

You must configure the following files located under \$AUTOROOT/SystemAgent/<Agent Name> to run the Agent in the port multiplexer (Dylan Socket Adapter) mode:

- cybAgent
- cybspawn
- ESPMgr

#### NOTE

For information on how to run csamconfigedit and to configure the ports to run in Dylan mode, see [Configure the Agent to Communicate Using SSA Ports](#) and [Configure the Agent to Communicate with CA Workload Automation AE](#).

The variables specific to various operating systems are listed as follows:

#### NOTE

Review the paths and change as required, if you had provided custom directories during installation.

### HP-IA

```
CAPKIHOM=$EWA_SHARED/CAPKI
export CAPKIHOM
CSAM_SOCKADAPTER=$EWA_SHARED/Csam/SockAdapter
export CSAM_SOCKADAPTER
```

```
SHLIB_PATH="/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT:/opt/CA/
WorkloadAutomationAE/SystemAgent/WA_AGENT/jre/lib/IA64W/se
rver:/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT/jre/lib/IA64W:$ETPKIHOM/lib:
$CSAM_SOCKADAPTER/lib:$CSAM_SOCKADAPTER/lib6
4:/opt/CA/WorkloadAutomationAE/autosys/lib:$SHLIB_PATH"
export SHLIB_PATH
```

```
LD_LIBRARY_PATH=$SHLIB_PATH:$CSAM_SOCKADAPTER/lib_ia64
export LD_LIBRARY_PATH
```

### LINUX

```
EWA_SHARED=/opt/CA/SharedComponents
export EWA_SHARED
ETPKIHOM=$EWA_SHARED/ETPKI
export ETPKIHOM
CAPKIHOM=$EWA_SHARED/CAPKI
export CAPKIHOM
CSAM_SOCKADAPTER=$EWA_SHARED/Csam/SockAdapter
export CSAM_SOCKADAPTER
```

```
LD_LIBRARY_PATH="/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT:/opt/CA/
WorkloadAutomationAE/SystemAgent/WA_AGENT/jre/lib/amd64/server:
/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT/jre/lib/amd64:$ETPKIHOM/lib:
$CSAM_SOCKADAPTER/lib:
$CSAM_SOCKADAPTER/lib64:/opt/CA/WorkloadAutomationAE/autosys/lib:$LD_LIBRARY_PATH"
export LD_LIBRARY_PATH
```

**AIX**

```
EWA_SHARED=/opt/CA/SharedComponents
export EWA_SHARED
ETPKIHOME=$EWA_SHARED/ETPKI
export ETPKIHOME
CAPKIHOME=$EWA_SHARED/CAPKI
export CAPKIHOME
CSAM_SOCKADAPTER=$EWA_SHARED/Csam/SockAdapter
export CSAM_SOCKADAPTER
```

```
LIBPATH="/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT:/opt/CA/WorkloadAutomationAE/
SystemAgent/WA_AGENT/jre/lib/ppc64/classic:
/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT/jre/lib/ppc64:$ETPKIHOME/lib:
$CSAM_SOCKADAPTER/lib:$CSAM_SOCKADAPTER/lib64:
/opt/CA/WorkloadAutomationAE/autosys/lib:$LIBPATH"
export LIBPATH
```

## Web UI Upgrade

This describes how to upgrade Workload Control Center (WCC) Release 11.3.x to Web UI Release 12.0.

### **How the Web UI Server is Upgraded**

The Web UI server is the main server in a typical Web UI environment. This topic provides an overview of upgrading a WCC Release 11.3.x server to Web UI Release 12.0.

#### **Follow these steps:**

1. Verify that you have access to the supported version of CA EEM.
2. Upgrade the instance of WCC Release 11.3.x using the Web UI Release 12.0 DVD.

#### **NOTE**

Data migration between WCC Release 11.3.x and Web UI Release 12.0 is not required.

3. Verify the upgrade by logging in to Web UI.

### **Web UI Upgrade Considerations**

Review the following considerations before you upgrade WCC Release 11.3.5 or higher to Web UI Release 12.0:

- When upgrading from a WCC Release 11.4.X that is installed without any other AutoSys component in a custom path, the upgrade does not correctly detect the install base directory. For more information, review the procedure in the [Knowledge Base article 212314](#).
- WCC Release 11.3.5 or higher must be installed before you perform the upgrade.
- A minimum of 2 GB free disk space is required to upgrade.
- An instance of CA EEM must be accessible to Web UI.
- We recommend that you set the database schema to be case-sensitive prior to upgrading.
- Before you upgrade, you can create a backup of your views and alerts, job templates, and configuration settings using the export options of the associated batch interfaces. For more information about exporting your data, see the *Monitoring Batch Interface Help*, *Job Templates Batch Interface Help*, and the *Configuration Batch Interface Help*.
- We recommend that you clear the cache for your web browser before using Web UI.
- On Windows, you must run the upgrade as an Administrator user or as a user from the Administrative group
- On UNIX, to run the installation in graphical mode, use a graphics-enabled monitor and define the DISPLAY environment variable correctly.
- Web UI cannot be upgraded from a network drive that is not mapped to a letter drive (for example, "X"). To upgrade from a network drive, map the shared network drive (for example, map \\server\drive to X:\) and then run the upgrade from that drive. Using a mapped drive instead of local media may extend the amount of time that is required for the upgrading the server.
- We strongly recommend that you mount the ISO image instead of using an image "buster" tool. The use of an image "buster" tool produces unexpected results.
- The upgrade procedure does not allow changes to any of the following:
  - The AutoSys SDK install location
  - CA EEM server host name and password
  - The assigned Tomcat application server ports
  - Event Bus port.
  - Web UI database properties
  - Reporting database properties
- Web UI Release 12.0 installs the Jobs widget during the upgrade procedure. However, this widget is not enabled by default. To enable access, you must select the MonitoringJobs action in the WidgetAccess policy in CA EEM.
- Web UI Release 12.0 installs the ActiveUsers widget during the upgrade procedure. However, this widget is not enabled by default. To enable access, you must select the UserStatus action in the WidgetAccess policy in CA EEM.
- Before you upgrade WCC to Web UI, you must modify the CA EEM server certificate as follows:
  - Upgrade CA EEM.
  - On the CA EEM server, set the KeyLength to 2048 and set the Algorithm to SHA1.

#### NOTE

For more information about the KeyLength, see the following links in the CA EEM documentation:

- To generate a certificate with the Keylength as 2048: [How to Generate the Certificates](#)
- To generate a certificate with the Keylength as 2048 in case of CA EEM Failover: [Generate Certificates with Custom Key Length for CA EEM Servers in Failover or Cluster Environment](#)
- Perform the WCC upgrade.
- To enable access to Monitoring and Reporting for a CA EEM Executive user, you must select the appropriate check boxes in the ApplicationAccess default policy after upgrading from Web UI Release 11.3.5.
- Prior to WCC Release 11.3.6, the job status order range, set in the Monitoring preferences in Configuration, was 100-300. The current job status order range is 80-127. We recommend that you set your parameters to a value within the current range prior to upgrading.
- Changes made in the previous version of Web UI to properties in the wrapper.conf and \*log4j.properties files, located in the install folder, are over-written by default values after upgrading to Web UI Release 12.0. To restore the values, you can set the values manually.



**WARNING**

If your Web UI servers are configured in a High Availability group, you can only upgrade a server if the other servers in the group are not running. If you attempt to upgrade a server while another server in the group is running, an error message is displayed. Also, after one server in the group has been upgraded, attempting to start any server in the group that has not been upgraded will result in an error.

**Upgrade Web UI on Windows**

This procedure describes how to upgrade Web UI on a Windows system.

**NOTE**

- CA EEM must be running.
- Before you begin the installation, stop performance monitoring, antivirus, and antispyware software.

**NOTE**

- Installation log files are in the following location:  
`<AUTOSYS_INSTALL_LOCATION>\wcc\log`
- Verify that the Web UI servers have started before you use or perform operations with Web UI. You can display Web UI server startup status by running the following command:  
`findstr /C:"startup in" "AUTOSYS_INSTALL_LOCATION\wcc\log\*.log"`
- (Optional) Log in to CA EEM and enable the MonitoringJobs action in the WidgetAccess policy to enable the Jobs widget option on the Dashboard tab.
- (Optional) Log in to CA EEM and enable the UsersStatus action in the WidgetAccess policy to enable the ActiveUsers widget option on the Dashboard tab.

**Upgrade Web UI on UNIX in Graphical Mode**

This procedure describes how to upgrade on a Unix server using graphical mode.

**NOTE**

- CA EEM must be running.
- Before you begin the installation, stop performance monitoring, antivirus, and antispyware software.
- Use graphical mode when you are upgrading locally.
- To use graphical mode, the monitor must be graphics enabled, and the DISPLAY environment variable must be correctly defined.

**NOTE**

- Log out and log back in to the computer and verify that the Web UI servers have started before you use or perform operations with Web UI. You can display Web UI server startup status by running the following command:

```
grep "startup in"$AUTOSYS_INSTALL_LOCATION\wcc\log/log/*.log
```

- Installation log files are in the following location:

**NOTE**

`AUTOSYS_INSTALL_LOCATION\Logs`

- (Optional) Log in to CA EEM and enable the MonitoringJobs action in the WidgetAccess policy to enable the Jobs widget option on the Dashboard tab.
- (Optional) Log in to CA EEM and enable the UsersStatus action in the WidgetAccess policy to enable the ActiveUsers widget option on the Dashboard tab.

## Upgrade Web UI on UNIX Using Console Mode

This procedure describes how to upgrade to Web UI Release 11.4 on the UNIX server using console mode.

### NOTE

- CA EEM must be running when you install Web UI.
- Before you begin the upgrade, stop performance monitoring, antivirus, and antispyware software.
- Use console mode when you are upgrading remotely.

### NOTE

- Log out and log back in to the computer and verify that the Web UI servers have started before you use or perform operations with Web UI. You can display Web UI server startup status by running the following command:  

```
grep "startup in" $AUTOSYS_INSTALL_LOCATION/wcc/log/*.log
```
- Installation log files are in the following location:  
*AUTOSYS\_INSTALL\_LOCATION/Logs*
- (Optional) Log in to CA EEM and enable the MonitoringJobs action in the WidgetAccess policy to enable the Jobs widget option on the Dashboard tab.
- (Optional) Log in to CA EEM and enable the UsersStatus action in the WidgetAccess policy to enable the ActiveUsers widget option on the Dashboard tab.

## Verify the Upgrade

After you upgrade Web UI, you can test the upgrade by starting a Web UI session.

### Follow these steps:

1. Open a browser window, navigate to the correct URL, and log in using the appropriate credentials. The AutoSys Workload Automation window opens to the default page for your user ID, displaying the tabs and applications of Web UI that you are authorized to access.

### NOTE

If a database connection error message appears when you log in, verify that the Web UI database is running, log out, and then log back in to Web UI.

2. Verify the following items:
  - AutoSys Workload Automation servers are configured and accessible.
  - Modifications to any Configuration preferences, if any, are still implemented.
  - Views are displayed in the Widgets on the Dashboard tab and on the Monitoring tab.
  - ECLI commands are present.
  - Forecast reports created in the previous Web UI release are available.
  - SAP filters, job filters, and job templates, if any, are available.

## AutoSys Common Components Upgrade

This article explains how to upgrade common components on UNIX and Windows.

### Upgrade Policy and Sample Scenarios

The common components are highly integrated; therefore, all installed common components must be upgraded at one time and must be at the same release level. You cannot upgrade individual common components because the components that were not upgraded may not function properly. This policy affects the installation or upgrade of any Broadcom product that uses the common components from the AutoSys Workload Automation ISO. Because of

this policy, the process to upgrade the common components depends on which components are installed and which Broadcom products are interfacing with them.

#### **NOTE**

An upgrade is allowed only if all the common components that are installed on your computer can be upgraded from the AutoSys ISO. If this requirement cannot be met, you must first upgrade the Broadcom product that originally installed one or more common components, and then upgrade the remaining common components using the AutoSys ISO.

The following examples explain how the common components can be upgraded in different scenarios:

#### **Example: Web UI is Installed on a Computer**

In this scenario, all the common components can be upgraded..

#### **Example: CA EEM is Installed on a Computer**

In this scenario, CA EEM can be upgraded using the AutoSys ISO because it does not use any other common components.

### **Common Components Upgrade Overview**

You can upgrade previously installed common components CA EEM and SSA using the Autosys ISO.

### **UNIX Upgrade Considerations**

We recommend that you review the following important considerations before upgrading the common components on UNIX:

- Before you upgrade the common components, stop all its active processes. The only exception is CA EEM, which must be active to be upgraded.
- If the iGateway built on Linux 2.4 kernel version is already installed, CA EEM, iGateway, and all other iGateway plug-ins cannot be installed or upgraded. If \$IGW\_LOC/igw.linux\_k26 file is present, then the iGateway built on Linux 2.6 kernel version is installed. In the case where the iGateway built on Linux 2.4 kernel version is already installed, uninstall CA EEM, iGateway, and all other iGateway plug-ins, and then install these products using the AutoSys ISO. CA EEM policies can then be migrated to the new CA EEM installation. For information about migrating CA EEM policies, see [Security Policy Customization](#). For information about upgrading CA EEM, see the [CA EEM documentation](#).

### **Windows Upgrade Considerations**

We recommend that you review the following important considerations before upgrading the common components on Windows:

- During the common components upgrade, if SSA or any other application using SSA is active, the upgrade fails to change any SSA-related binaries or libraries and you may have to restart your computer to complete the common components upgrade. If you must restart your computer, the installer displays the option to restart your computer during the common components upgrade. You can select to restart your computer immediately or later.
- Stop CA EEM while upgrading to the current release.
- If you are upgrading from a 32-bit CA EEM to a 64-bit CA EEM or migrating from CA EEM Release 8.x to 12.6, follow these steps:
  - a. Stop Web UI, AutoSys Workload Automation, and CA EEM.
  - b. Record your LDAP configuration (if used).
  - c. Stop CA EEM by executing the following commands:
 

```
dxserver stop all
net stop igateway
```

- d. Export CA EEM database to a LDIF file by executing the following command:  

```
dxdumpdb -f itechpoz_old.ldif itechpoz
```
- e. Uninstall command sponsor.
- f. Uninstall CA EEM 32-bit, all other iGateway plug-ins, and iGateway manually, as follows:  

```
"%EIAM_HOME%\uninstall\eamuninstall.exe" -DFORCE_UNINSTALL=true
```
- g. Restart the server (if prompted).
- h. Install the latest release of CA EEM using the AutoSys ISO.
- i. Stop CA EEM by executing the following commands:  

```
dxserver stop all
net stop igateway
```
- j. Export the new CA EEM database to a LDIF file by executing the following command:  

```
dxdumpdb -f newly_installed.ldif itechpoz
```
- k. Open the new LDIF file named *newly\_installed.ldif* and do the following:
  - a. Find the following entry:  

```
- dn: cn=PozAdmin,cn=Admins,cn=Entities,cn=iTechPoz
```
  - b. Copy the following user password line:  

```
- userPassword: {SSHA512}xxxxx
```
- l. Open the old LDIF file named *itechpoz\_old.ldif* and do the following:
  - a. Find the following entry:  

```
- dn: cn=PozAdmin,cn=Admins,cn=Entities,cn=iTechPoz
```
  - b. Replace the user password line with the one you copied from the *newly\_installed.ldif*:  

```
file - userPassword: {SSHA512}xxxxx
```
- m. Import the LDIF file by executing the following command:  

```
dxloaddb itechpoz itechpoz_old.ldif
```
- n. Start CA EEM by executing the following commands:  

```
dxserver start all
net start igateway
```
- o. Verify CA EEM by opening a browser and navigating to the following URL:  
<https://hostname:5250/spin/eiam/eiam.csp>
- p. Login to WCC0004 and verify that the policy is in place.
- q. Login to WorkloadAutomationAE and verify that the policy is in place.
- r. Reapply your LDAP configuration.
- s. Restart Web UI.
- t. Restart AutoSys Workload Automation.

**NOTE**

If you are using a different EiamAdmin or PozAdmin password for the old and new installations, you must replace the passwords in the LDIF file. For more information about performing each of these steps, see [CA EEM documentation](#).

## Maintain AutoSys

When AutoSys is already installed and you double-click the AutoSys installer with the same version, the installation wizard launches in the maintenance mode. The following sections describe the panels that appear in the maintenance mode.

### Modify Installation

The following options can be used to manage multiple AutoSys instances. You can select the appropriate option based on your requirement.

Field Name	Silent Install Variable	Description
Add Component	IA_AE_MODIFY_OPTIONS = ADDCOMPONENT	Select this option if you want to add a component that you have missed or not added to the AutoSys instance during the fresh install.
Add Instance	IA_AE_MODIFY_OPTIONS =ADDINSTANCE	Select this option if you want to add another AutoSys instance.
Modify Instance	IA_AE_MODIFY_OPTIONS =MODIFYINSTANCE	Select this option if you want to modify an existing AutoSys instance.
Delete Instance	IA_AE_MODIFY_OPTIONS =DELETEINSTANCE	Select this option if you want to delete an existing AutoSys instance.
Add Additional Agent	IA_AE_MODIFY_OPTIONS =ADDAGENT	Select this option if you want to add an addition AutoSys agent.
Add Web UI Collector	IA_AE_MODIFY_OPTIONS =ADDWEBUICOLLECTOR	Select this option if you want to add a Web UI collector.
Delete Web UI Collector	IA_AE_MODIFY_OPTIONS =DELETEWEBUICOLLECTOR	Select this option if you want to delete an existing Web UI collector.

### Add Components

During the fresh install, if you have missed or have not added any components, you can use this procedure to add the missing component.

To add a missing AutoSys component, follow these steps.

#### NOTE

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Component**.
2. In [Components to Install](#), select the component to be installed.
3. Follow the instructions on the panels that follow.
4. Click **Install**.

## **Add Instance**

In addition to an existing AutoSys instance, if you want to use additional AutoSys instances, you can use this procedure.

This can be because of different requirements. For example, you might be experiencing an increased workload or activity on your current AutoSys instance.

This procedure helps you create an additional AutoSys instance to distribute your workload efficiently.

To add an AutoSys instance, follow these steps.

### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Instance**.
2. In Instance Information, enter the desired instance name.
3. In [Enable Components](#), select the desired component.
4. (Optional) In [Instance-level Data Encryption](#), enter the required details.
5. In Database Type, select and enter the required details.

### **NOTE**

We will use MSSQL as a database for this procedure. You can choose the desired database and refer to the respective panel documentation in this document.

6. Provide the details in the panel.
7. Click **Install**.

## **Modify Instance**

You can modify any existing AutoSys instance to add a Scheduler and Client, Application Server, or REST web service.

If the above-mentioned components are already installed for the selected instance, you cannot proceed any further.

To modify an AutoSys instance, follow these steps:

### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Modify Instance**.
2. In [Select Instance](#), select the AutoSys instance that you want to modify.
3. In [Running Services](#), do the necessary selection.
4. In [Enable Components](#), select the components that you want to install.
5. Review the [Pre-installation Summary](#).
6. Click **Install**.

## **Delete Instance**

If you have multiple AutoSys instances running, you can use this procedure to delete a specific AutoSys instance.

### **NOTE**

You cannot use this option if you have only one AutoSys instance.

To delete an AutoSys instance, follow these steps.

### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Delete Instance**.
2. In [Select Instance](#), select the AutoSys instance that you want to modify.
3. In [CA EEM Properties](#), enter the appropriate details.

4. Review the [Pre-installation Summary](#).
5. Click **Install**.

### **Add Additional Agent**

You can install additional agents to run job types from your system. Adding additional agents can help you to structure or group your job types in one place.

To add an additional AutoSys agent, follow these steps.

#### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Additional Agent**.
2. In [Agent Properties](#), enter the required details.
3. In [Agent and Application Server Port](#), enter the required details.
4. In [Agent Level Data Encryption](#), enter the required details.
5. In [Services Startup Options](#), select the appropriate options.
6. Review the [Pre-installation Summary](#).
7. Click **Install**.

### **Add Web UI Collector**

You can add additional Web UI collector to distribute the data collection load coming from various application servers. Eventually, increasing data collection performance.

To add a Web UI Collector, follow these steps.

#### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Web UI Collector**.
2. In [Web UI Database Type](#), enter the required details.

#### **NOTE**

We will use MSSQL as a database for this procedure. You can choose the desired database and refer to the respective panel documentation in this document.

3. In [Web UI Database User Details](#), enter the required database user credentials.
4. In [Web UI Database Properties](#), enter the required database details.
5. In [Web UI Port Configuration](#), enter the required port configuration.
6. In [Services Startup Options](#), select the appropriate options.
7. Review the [Pre-installation Summary](#).
8. Click **Install**.

### **Delete Web UI Collector**

To delete a Web UI Collector:

#### **NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select Add Web UI Collector.
2. In Delete Web UI Collector, from **Web UI collector name**, select the Web UI collector to be deleted.
3. Review the [Pre-installation Summary](#).
4. Click **Install**.



## **Miscellaneous Panels**

The following are the other miscellaneous panels that appear when you are maintaining your AutoSys instance.

### **Enable Components**

The following information is required.

Field Name	Silent Install Variable	Description
Scheduler	IA_AE_SELECTED_SCHEDULER	Select this checkbox if you want to install the scheduler in the selected AutoSys instance, or in the instance that you are about to create. Client will be automatically selected.
Application Server	IA_AE_SELECTED_APPLICATIONSERVER	Select this checkbox if you want to install the application server in the selected AutoSys instance, or in the instance that you are about to create.
Client	IA_AE_SELECTED_CLIENT	Select this checkbox if you want to install the client in the selected AutoSys instance, or in the instance that you are about to create.
REST web service	IA_AE_SELECTED_RESTWS	Select this checkbox if you want to install the REST web service in the selected AutoSys instance, or in the instance that you are about to create.

### **Select Instance**

The following information is required. This panel will appear when you have more than one AutoSys instance installed.

Field Name	Silent Install Variable	Description
Instance Name	IA_AE_INSTANCE_NAME	Select the AutoSys instance in which you want to make the required modification.

### **Machine Definition - Local Agent**

You will encounter this panel while adding a scheduler with an already installed agent. If you have multiple agents, you will get a drop-down box to select the desired agent. The following information is required.

Field name	Silent Install Variable	Description
Agent name	IA_AE_LOCAL_MAC_DEF_AGENT_NAME	The agent name that you want to select.

Encrypt data using a user-specified key	IA_AE_ENCRYPTION_KEY_AGENT   // An empty value indicates that the default key will be used.   // A value with 0x prefix indicates that the value is Hexadecimal. It should be AES encrypted.   // Any other indicates that the value is a Passphrase. It should be AES encrypted.	Select this checkbox if you want to encrypt data using a user-specified key.   If you select to encrypt the data using a user-specified key, you must also specify the key format and key contents.   A passphrase format specifies 1-16 alphanumeric characters.   A hexadecimal format specifies exactly 32 alphanumeric characters, consisting of 0-9 or A-F.   If the <b>Encrypt Data Using a User specified Key</b> checkbox is not selected, the default key is used for data encryption.   <b>See Also:</b> <a href="#">FIPS Encryption</a>
Key	IA_AE_ENCRYPTION_KEY_AGENT	Enter the key for encryption.

## Modify Existing AutoSys Installation

The following topics describe how to modify components in an existing instance of AutoSys.

### Modify Installation

The following options can be used to manage multiple AutoSys instances. You can select the appropriate option based on your requirement.

Field Name	Silent Install Variable	Description
Add Component	IA_AE_MODIFY_OPTIONS = ADDCOMPONENT	Select this option if you want to add a component that you have missed or not added to the AutoSys instance during the fresh install.
Add Instance	IA_AE_MODIFY_OPTIONS =ADDINSTANCE	Select this option if you want to add another AutoSys instance.
Modify Instance	IA_AE_MODIFY_OPTIONS =MODIFYINSTANCE	Select this option if you want to modify an existing AutoSys instance.

Delete Instance	IA_AE_MODIFY_OPTIONS =DELETEINSTANCE	Select this option if you want to delete an existing AutoSys instance.
Add Additional Agent	IA_AE_MODIFY_OPTIONS =ADDAGENT	Select this option if you want to add an addition AutoSys agent.
Add Web UI Collector	IA_AE_MODIFY_OPTIONS =ADDWEBUICollector	Select this option if you want to add a Web UI collector.
Delete Web UI Collector	IA_AE_MODIFY_OPTIONS =DELETEWEBUICollector	Select this option if you want to delete an existing Web UI collector.

## Add Components

During the fresh install, if you have missed or have not added any components, you can use this procedure to add the missing component.

To add a missing AutoSys component, follow these steps.

### NOTE

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Component**.
2. In [Components to Install](#), select the component to be installed.
3. Follow the instructions on the panels that follow.
4. Click **Install**.

## Add Instance

In addition to an existing AutoSys instance, if you want to use additional AutoSys instances, you can use this procedure.

This can be because of different requirements. For example, you might be experiencing an increased workload or activity on your current AutoSys instance.

This procedure helps you create an additional AutoSys instance to distribute your workload efficiently.

To add an AutoSys instance, follow these steps.

### NOTE

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Instance**.
2. In Instance Information, enter the desired instance name.
3. In [Enable Components](#), select the desired component.
4. (Optional) In [Instance-level Data Encryption](#), enter the required details.
5. In Database Type, select and enter the required details.  
**Note:** We will use MSSQL as a database for this procedure. You can choose the desired database and refer to the respective panel documentation in this document.
6. Provide the details in the panel.
7. Click **Install**.

## Modify Instance

You can modify any existing AutoSys instance to add a Scheduler and Client, Application Server, or REST web service.

If the above-mentioned components are already installed for the selected instance, you cannot proceed any further.

To modify an AutoSys instance, follow these steps:

**NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Modify Instance**.
2. In [Select Instance](#), select the AutoSys instance that you want to modify.
3. In [Running Services](#), do the necessary selection.
4. In [Enable Components](#), select the components that you want to install.
5. Review the [Pre-installation Summary](#).
6. Click **Install**.

## Delete Instance

If you have multiple AutoSys instances running, you can use this procedure to delete a specific AutoSys instance.

**NOTE**

You cannot use this option if you have only one AutoSys instance.

To delete an AutoSys instance, follow these steps.

**NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Delete Instance**.
2. In [Select Instance](#), select the AutoSys instance that you want to modify.
3. In [CA EEM Properties](#), enter the appropriate details.
4. Review the [Pre-installation Summary](#).
5. Click **Install**.

## Add Additional Agent

You can install additional agents to run job types from your system. Adding additional agents can help you to structure or group your job types in one place.

To add an additional AutoSys agent, follow these steps.

**NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Additional Agent**.
2. In [Agent Properties](#), enter the required details.
3. In [Agent and Application Server Port](#), enter the required details.
4. In [Agent Level Data Encryption](#), enter the required details.
5. In [Services Startup Options](#), select the appropriate options.
6. Review the [Pre-installation Summary](#).
7. Click **Install**.

## Add Web UI Collector

You can add additional Web UI collector to distribute the data collection load coming from various application servers. Eventually, increasing data collection performance.

**NOTE**

The database details should be the same as that of the existing Web UI for which an additional collector is being added.

To add a Web UI Collector, follow these steps.

**NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Add Web UI Collector**.
2. In [Web UI Database Type](#), enter the required details.

**NOTE**

We will use MSSQL as a database for this procedure. You can choose the desired database and refer to the respective panel documentation in this document.

3. In [Web UI Database User Details](#), enter the required database user credentials.
4. In [Web UI Database Properties](#), enter the required database details.
5. In [Web UI Port Configuration](#), enter the required port configuration.
6. In [Services Startup Options](#), select the appropriate options.
7. Review the [Pre-installation Summary](#).
8. Click **Install**.

## Delete Web UI Collector

To delete a Web UI Collector:

**NOTE**

Click **Next** to proceed through the steps described below.

1. In Modify Installation, select **Delete Web UI Collector**.
2. In Delete Web UI Collector, from **Web UI collector name**, select the Web UI collector to be deleted.
3. Review the [Pre-installation Summary](#).
4. Click **Install**.

## AutoSys Maintenance Miscellaneous Panels

The following are the other miscellaneous panels that appear when you are maintaining your AutoSys instance.

### Installation Maintenance

The following information is required.

Field name	Silent install variable	Description
Modify the product		Select this option if you want to modify the existing AutoSys instance.

Create the database schema	//Variable to be used when the installer is launched for AutoSys maintenance. IA_AE_MAINT_OPTIONS //Variable to be used when the installer is launched for AutoSys upgrade. IA_AE_UPGRADE_OPTIONS //Variable to be used when the installer is launched for a fresh AutoSys install. IA_AE_FRESH_OPTIONS	Select this option if you want to create a database schema for the AutoSys instance, Web UI application, and reporting. Note: This field appears while performing a fresh install, upgrade, or maintenance of an AutoSys instance.
----------------------------	------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------	---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------

## Enable Components

The following information is required.

Field Name	Silent Install Variable	Description
Scheduler	IA_AE_SELECTED_SCHEDULER	Select this checkbox if you want to install the scheduler in the selected AutoSys instance, or in the instance that you are about to create. Client will be automatically selected.
Application Server	IA_AE_SELECTED_APPLICATIONSERVER	Select this checkbox if you want to install the application server in the selected AutoSys instance, or in the instance that you are about to create.
Client	IA_AE_SELECTED_CLIENT	Select this checkbox if you want to install the client in the selected AutoSys instance, or in the instance that you are about to create.
REST web service	IA_AE_SELECTED_RESTWS	Select this checkbox if you want to install the REST web service in the selected AutoSys instance, or in the instance that you are about to create.

## Select Instance

The following information is required. This panel will appear when you have more than one AutoSys instance installed.

Field Name	Silent Install Variable	Description
Instance Name	IA_AE_INSTANCE_NAME	Select the AutoSys instance in which you want to make the required modification.

## Machine Definition - Local Agent

You will encounter this panel while adding a scheduler with an already installed agent. If you have multiple agents, you will get a drop-down box to select the desired agent. The following information is required.

Field name	Silent Install Variable	Description
Agent name	IA_AE_LOCAL_MAC_DEF_AGENT_NAME	The agent name that you want to select.
Encrypt data using a user-specified key	IA_AE_ENCRYPTION_KEY_AGENT // An empty value indicates that the default key will be used. // A value with 0x prefix indicates that the value is Hexadecimal. It should be AES encrypted. // Any other indicates that the value is a Passphrase. It should be AES encrypted.	Select this checkbox if you want to encrypt data using a user-specified key. If you select to encrypt the data using a user-specified key, you must also specify the key format and key contents. A passphrase format specifies 1-16 alphanumeric characters. A hexadecimal format specifies exactly 32 alphanumeric characters, consisting of 0-9 or A-F. If the <b>Encrypt Data Using a User specified Key</b> checkbox is not selected, the default key is used for data encryption. <b>See Also:</b> <a href="#">FIPS Encryption</a>
Key	IA_AE_ENCRYPTION_KEY_AGENT	Enter the key for encryption.

# Securing

---

AutoSys Workload Automation provides various options to meet your security needs, which depend on your interface.

## **WARNING**

This section contains confidential subsections that are not viewable to anonymous users. To view all subsections, select Sign In in the upper-right corner. Enter your CA Support Online credentials. If you are logged in, ignore this message.

## **AutoSys Workload Automation Security**

The AutoSys Workload Automation CLI lets you operate an instance in one of the two security modes and provides system-level protections.

Native security mode lets you control the security of your system as follows:

- Grant job-specific permissions.
- Delegate administrative privileges with superuser assignments.

## **NOTE**

For more information about Job Ownership, Permission Types, and Superusers, see [Native Security Mode](#).

External security mode lets users with administrative privileges customize security policy and settings.

## **NOTE**

For more information, see [Customize Security Policy and Settings](#).

System-level security is in effect in all instances regardless of the active security mode and provides the following safeguards:

- **Password protection**

User name and password credentials protect workload objects as follows:

- The administrator defines a database password during installation to ensure that only the database administrator can access the database.
- To execute a job, you must specify the login credentials of the job owner, so that the agent can log in as the owner.

## **NOTE**

For more information, see [Job Ownership](#).

- The system requires every user to specify a password at login and grants or denies permissions-based user type.
- The system uses login credentials to determine user permissions and then grants or denies access to the product accordingly.

## **Web UI Security**

The Web UI uses CA EEM to control access to workload objects based on user role. CA EEM lets you customize security policy.

## **NOTE**

For more information, see [Customize your Web UI Policies](#) and [CA EEM and Security](#).



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## **Navigate Securing Content**

After you log in, use the navigation tree on the left or search for Securing content:

Securing

## **Security Best Practices**

Use these best practices when setting up security. To secure workload automation objects, you can use native security on AutoSys Workload Automation with CA EEM policies on Web UI. Conversely, you can define CA EEM policies on AutoSys Workload Automation and Web UI. We strongly recommend that you implement CA EEM security on both AutoSys Workload Automation and Web UI.

### **AutoSys Workload Automation Security Mode**

By default, AutoSys Workload Automation uses external security. After you install AutoSys Workload Automation, you can change the security mode using the `autosys_secure` command. The protection level that native security mode provides is limited compared to that of external security (integration with CA EEM).

#### **NOTE**

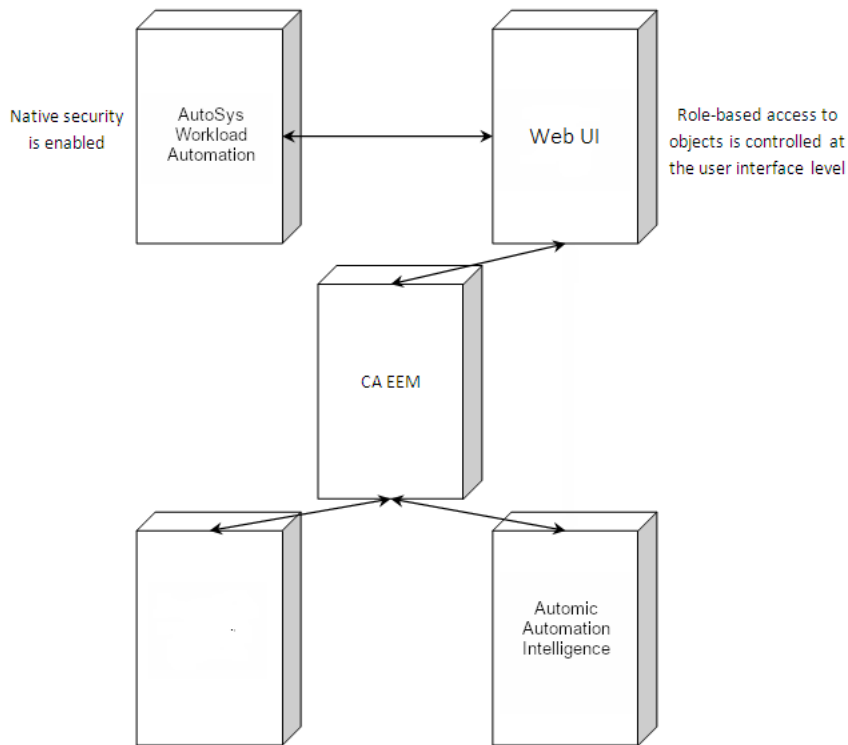
We recommend that you use external security. Only external security lets you control role-based access to objects (such as jobs, calendars, cycles, global variables, machines, and resources) at a granular level.

However, if you must use native security mode on AutoSys Workload Automation, we recommend the following tactics:

- Remove icons and options from Web UI that users do not need. By removing options from the user interface, the users in your environment cannot try to access features that they do not need.
- Specify the owner and permission attributes in the job definitions that are created using the `jil` command on AutoSys Workload Automation. CA EEM policies defined on Web UI do not check security for objects that are defined using the `jil` command, so you can use the owner and permission attributes to grant the appropriate permissions to users. Define appropriate operating system user IDs and passwords in the database that restrict who can log in to and run jobs on client computers.
- Create CA EEM policies on Web UI that control what objects users can see and modify when they use Web UI. These policies control access to some objects that are stored on AutoSys Workload Automation that you want to secure. This behavior is because granular, role-based security is only enforced at the user interface level in this environment.

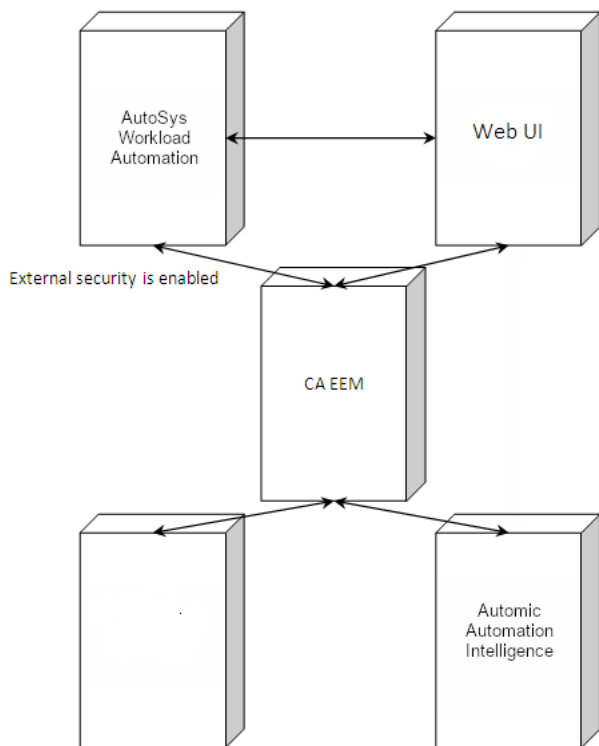
This diagram shows the configuration where AutoSys Workload Automation uses native security mode and Web UI uses CA EEM on a dedicated server:

**Figure 11: AutoSys Workload Automation uses native security mode and Web UI uses CA EEM on a dedicated server**



### **Recommended Security Configuration**

To secure your environment, we recommend that you configure AutoSys Workload Automation and Web UI to use CA EEM as shown in this diagram:

**Figure 12: AutoSys Workload Automation and Web UI to use CA EEM**

In this configuration, CA EEM is a common component used by multiple products, including AutoSys Workload Automation and Web UI.

#### NOTE

- This recommended configuration shows AutoSys Workload Automation and Web UI using the same CA EEM instance that is installed on a dedicated server. Alternatively, you can configure AutoSys Workload Automation to work with an existing CA EEM instance that is installed on the Web UI server. If CA EEM and Web UI are installed on the server, ensure that the server has sufficient memory to handle the processing for both components. We also recommend that you use a multi-CPU or a high performance CPU.
- To avoid a single point of failure and to avoid potential performance issues, we do *not* recommend installing CA EEM and AutoSys Workload Automation on the same server. Most security policies are typically created and checked on the AutoSys Workload Automation server.
- If CA EEM is not configured for failover and your environment utilizes CA EEM for AutoSys Workload Automation security, the AutoSys Workload Automation and Web UI servers must reference the same instance of CA EEM.
- If CA EEM is configured for failover, the AutoSys Workload Automation and Web UI servers do not need to reference the same instance of CA EEM; however, the CA EEM instances referenced must be in the same failover setup.
- If Web UI and a AutoSys Workload Automation server that is defined in Web UI reference different CA EEM instances, EEM Enabled should not be selected in the definition for that server in Web UI Configuration. AutoSys Workload Automation native security is used.

## Policy Differences

The CA EEM policies for AutoSys Workload Automation are categorized by object type (for example, jobs, calendars, and resources). AutoSys Workload Automation provides a resource class for each object type (for example, as-job, as-calendar, and as-resource). You can define a policy in the appropriate resource class that controls all access modes (read, create, modify, delete, execute) for a particular object or group of objects.

In contrast, CA EEM policies for Web UI are classified according to their functions. For example, the ObjectAccess resource class controls which objects a user can view in Web UI (read access). The ObjectControl resource class controls the actions that can be performed on a class of objects (create, modify, and delete access).

### NOTE

- Individual objects are not controlled in Web UI. For example, a user must be authorized to create a job.
- Users and user groups are associated with these policies to restrict the objects that the users have access to.

### NOTE

We recommend the best practices in this section so that you can coordinate the policy designs between AutoSys Workload Automation and Web UI to achieve the best possible performance.

## Use a Naming Convention for Objects and Policies

When defining objects (such as jobs, calendars, cycles, global variables, machines, and resources), we recommend that you use a naming convention that groups objects by categories. For example, you can prefix the names of all payroll jobs with **payroll**, and you can prefix the names of all renewable resources with **renewable**.

By using a naming convention for objects, you can coordinate the security configurations in AutoSys Workload Automation, Web UI, and CA EEM:

- You can define CA EEM policies that apply to groups of related objects that share similar security restrictions.
- You can ensure that the correct objects are secured as determined by the best match policy evaluation that CA EEM uses.

Examples of how the naming conventions are used are described in the following topics.

## Using a Naming Convention for the Resources Specified in Policies

When you create a policy in CA EEM, the following conventions apply when naming the resources (objects) that you want to secure:

Broadcom product	Resource Class	Naming Convention for the Resources (Objects) Specified in the Policy
AutoSys Workload Automation	All resource classes except as-owner and as_sendevent.	<i>INS.objectName</i>, where INS is the 3-digit AutoSys Workload Automation instance name.   <b>Example:</b> ACE.payroll_employeeID5701   The as-owner resource class does not require the instance name. For example, suppose that you want to update the job owner field of a job to parrot@jungle in the ACE instance. AutoSys Workload Automation queries CA EEM as follows: parrot@jungle   The as_sendevent resource class requires only the instance name.

Web UI	ObjectControl	server/serverName/objectType <b>Example:</b> server/AEserver/GlobalVariable server/serverName/Job/jobType <b>Example:</b> server/AEserver/Job/JMX-MBeanSubscribe
Web UI	ObjectAccess	server/serverName/objectType/objectName <b>Example:</b> server/SvrACE/Forecast/myReport

In all these conventions, *objectName* specifies the name of the object that you want to secure.

If you use a standard naming convention for your object names, you can use wildcards in the *objectName* value to create CA EEM policies that restrict access to groups of objects.

Also, CA EEM uses a best match policy evaluation to determine whether to grant access to objects. CA EEM only considers policies whose resource names most closely match the requested object name. Therefore, by following a standard object naming convention, you can ensure that the correct objects are matched.

#### NOTE

If you have not used a standard naming convention for your existing objects, we recommend that you start using a naming convention for new objects.

### Use a Naming Convention for Policy Names

To coordinate policies between Web UI and AutoSys Workload Automation, we recommend that you use a standard naming convention for policy names. By using a naming convention that describes the intent of the policies, the policies are easier to understand and manage.

For example, the following Web UI and AutoSys Workload Automation policies use the *product-roles-accessType-objectType* naming convention:

```
WCC-SCHEDOPER-CONTROL-ALLJOBS
WCC-SCHEDOPER-ACCESS-ALLJOBS
AE-SCHED-ACCESSCONTROL-PAYROLLJOBS
AE-OPER-ACCESSCONTROL-PAYROLLJOBS
```

The names indicate that a Web UI ObjectControl policy and a Web UI ObjectAccess policy specify the object permissions for the scheduler and operator roles for all jobs. On AutoSys Workload Automation, an as-job policy specifies the permissions for the scheduler role for payroll jobs. Another as-job policy specifies the permissions for the operator role for payroll jobs.

### Example: Define Policies for Payroll Jobs

This example shows policies that follow the security best practices. Suppose that your enterprise runs jobs that are associated with payroll in the AutoSys Workload Automation instance ACE. CA EEM policies that govern who can access these jobs are created using the following guidelines:

- Avoid redundant policies.  
The Web UI policies are general and are categorized by user roles. The AutoSys Workload Automation policies are specific and define the access modes for the objects.
- Use a standard naming policy for objects.  
The names of all the payroll jobs are prefixed with **payroll**. This naming convention lets you define CA EEM policies that protect these specific objects. Note the following sample job names:

- payroll\_employeeID5701
- payroll\_employeeID5702
- payroll\_employeeID5703
- payroll\_employeeID6002
- payroll\_employeeID6003
- Use a standard naming convention for the resources specified in the policies.  
The resources (job objects) specified in the Web UI policies use the following naming standard:
  - server/serverName/objectType (Object Control)  
**Example:** server/AEserver/Calendar
  - server/serverName/Job/jobType (Object Control)  
**Example:** server/AEserver/Job/Command
  - server/serverName/objectType/objectName (Object Access)  
**Example:** server/AEserver/AlertPolicy/FailedJobs
 The resources that are specified in the AutoSys Workload Automation policies use the following naming standard:
  - *INS.objectName INS* Specifies the 3-digit AutoSys Workload Automation instance name.  
**Example:** ACE.payroll\_employeeID5701
- Use a standard naming convention for policy names.  
To coordinate the policies among AutoSys Workload Automation, Web UI, and CA EEM, the following naming convention is used for the policy names:
  - *product-roles-accessType-objectType*  
**Examples:** WCC-SCHEDOPER-CONTROL-ALLJOBS, WCC-SCHEDOPER-ACCESS-ALLJOBS, AE-SCHED-ACCESSCONTROL-PAYROLLJOBS, AE-OPER-ACCESSCONTROL-PAYROLLJOBS.

The following application groups (user groups) are defined in CA EEM:

- **SCHEDGRP**  
The users in this group need full control (read, create, delete, execute, and write) to all payroll jobs in the AutoSys Workload Automation instance ACE. This user group includes the default Web UI ejmscheduler, ejmexec, and ejmsupervisor users.
- **OPERGRP**  
The users in this group are primarily responsible for creating monitors. They only need read access to payroll jobs whose names start with **payroll\_employeeID6** in the AutoSys Workload Automation instance ACE. This user group includes the default Web UI ejmoperator user.

## AutoSys Workload Automation Data Encryption

This article explains how to manage AutoSys Workload Automation agent, data, and instance-wide encryption.

### Data Encryption

AutoSys Workload Automation supports the encryption of data and messages that are shared between the command-line utilities, agent, scheduler, and the application server. AutoSys Workload Automation uses the Advanced Encryption Standard (AES) algorithm to encrypt and decrypt data. This algorithm requires an encryption key to encrypt data.

Data is encrypted in the following communication scenarios:

- **Between the application server and client utilities**  
The exchanged data is encrypted using an instance-wide encryption key. This key is specific to an instance. The key must be the same on all computers where the server and clients are installed. During the AutoSys Workload Automation installation, a default instance-wide encryption key is created. This key is stored in the \$AUTOUSER/cryptkey.txt (on UNIX) or %AUTOUSER%/cryptkey.txt (on Windows) file. However, you can define a user-specific encryption key using the as\_config command or using AutoSys Workload Automation Administrator on Windows.

- **Between the application server and agent or the scheduler and agent**

The exchanged data is encrypted based on the encryption type and the encryption key that is specified in the machine definition and the agent encryption setting that is specified in the agentparm.txt file. On AutoSys Workload Automation, you can set the encryption type and encryption key to be used for each agent using the encryption\_type and key\_to\_agent JIL attributes. The encryption key that is specified on AutoSys Workload Automation must match the encryption key that is specified in the agentparm.txt file.

**NOTE**

For more information about setting up encryption on the agent, see the [Workload Automation Agents](#) documentation.

## **Instance-Wide Encryption**

The data that is exchanged between the command-line utilities and the application server is encrypted using an instance-wide encryption key. This key is specific to an instance. This key must be the same on all computers where the server and clients are installed. During the AutoSys Workload Automation installation, a default instance-wide encryption key is created. This key is stored in the \$AUTOUSER/cryptkey.txt (on UNIX) or %AUTOUSER%\cryptkey.txt (on Windows) file. However, you can define a user-specific encryption key using the as\_config command or using AutoSys Workload Automation Administrator on Windows.

**NOTE**

- The agent encryption and instance-wide encryption are mutually exclusive. For example, you can disable instance-wide encryption and can set the agent encryption type to either the default or AES. All combinations of encryption are supported.
- Install eTrust Public Key Infrastructure (ETPKI). AutoSys Workload Automation installs ETPKI when any of the following components are selected during the AutoSys Workload Automation installation:
  - Server
  - Agent
  - Client

## **Set Instance-Wide Encryption on UNIX**

You can set an instance-wide encryption key for all communication between the AutoSys Workload Automation components of the same instance. For AutoSys Workload Automation to work properly, the encryption key must be the same on all client computers. This encryption key is stored in the \$AUTOUSER/cryptkey.txt file.

**NOTE**

- During the AutoSys Workload Automation installation, a default encryption key is created. However, if you perform a custom installation, you can specify the encryption key during the installation.
- You can modify the encryption key after the AutoSys Workload Automation installation.

### **Follow these steps:**

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt:

```
unisrvctr stop waee_sched.$AUTOSERV
unisrvctr stop waee_server.$AUTOSERV
```

The scheduler and the application server stop.

3. Edit the following parameter in the configuration file, and save the file:

```
UseEncryption=0|1|2
```

- **0**  
Specifies that no encryption is used.
- **1**

Specifies that the default encryption key is used to encrypt data.

– 2

Specifies that a user-specified encryption key is used to encrypt data.

#### NOTE

If you set the UseEncryption parameter to 2, specify the encryption key using the as\_config command.

4. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. The instance-wide encryption is set.

#### NOTE

On Windows, you can select the equivalent value using the Use Instance Wide AES 128-bit Data Encryption check box on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.

## Agent Encryption

The data that is exchanged between the application server and the agent or the scheduler and the agent is encrypted based on the encryption type and the encryption key specified in the machine definition and the agent encryption setting specified in the agentparm.txt file. On AutoSys Workload Automation, you can set the encryption type and the encryption key to be used for each agent using the encryption\_type and key\_to\_agent JIL attributes. On the agent, the encryption key is stored in the cryptkey.txt file located in the agent installation directory. The key that is specified on AutoSys Workload Automation must match the key specified on the agent.

#### NOTE

- The agent encryption and instance-wide encryption are mutually exclusive. For example, you can disable instance-wide encryption and can set the agent encryption type to either the default or AES. All combinations of encryption are supported.
- An instance can communicate with agents using different encryption settings.
- You can copy the cryptkey.txt file that is generated on one computer to another, provided the encryption type and the encryption key are the same.
- The encryption type and the encryption key must be the same for all agents defined with the same name. If you want to define the agents with different encryption types or encryption keys, define the agents with different names. You can define the agent name using the agentname parameter in the agentparm.txt file.

## CA Workload Automation EE External Instance and Workload Automation Agent for z/OS Encryption

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, generate an instance-wide communication alias encryption file (cryptkey\_alias.txt) in the \$AUTOUSER.instance\_name (on UNIX) or %AUTOUSER%.instance\_name (on Windows) directory. The cryptkey\_alias.txt file stores the communication alias encryption key.

#### NOTE

For more information about generating the instance-wide communication alias encryption file, see [Configure Cross-Instance Dependencies with CA Workload Automation EE](#).

#### WARNING

Set AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.



**NOTE**

The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, disable AES 128-bit encryption and use no encryption for z/OS communication. For more information about setting encryption for z/OS communication, see [Configure an Instance](#).

**Encrypt a Password**

Use the security utility to generate an encrypted password when you need to update passwords in the configuration file or when you want to use encrypted authentication to run client utilities.

**NOTE**

You can also use the `as_config` command to manage encrypted passwords. Doing so is useful when you change passwords regularly or need to use encrypted authentication when running commands in batch. This method presents a security risk because the command does not obscure your plain text password. We recommend using the security utility to encrypt passwords unless you need to automate password management.

**Follow these steps:**

1. Open the operating system or instance command prompt as follows:  
**UNIX:**  
 Run the shell that is sourced to use AutoSys Workload Automation.  
 The operating system command prompt appears.  
**Windows:**  
 Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
 The instance command prompt appears.
2. Enter the following command:  

```
autosys_secure
```

 The interactive menu appears.
3. Type 6 and press the Enter key.  
 The Get encrypted password submenu appears.
4. Type 1 and press the Enter key.  
 The utility prompts you to specify your password.
5. Type your password in plain text and press the Enter key. Confirm your password and press the Enter key.  
 The utility displays the encrypted password.

**Automate Encrypted Password Management**

Environment variables can help you automate password management. Doing so is useful if you frequently update passwords that are stored in the configuration file or if you use encrypted authentication when running commands in batch.

**NOTE**

The `as_config` utility does not obscure your plain text password. If password security is a concern, manage encrypted passwords manually using the security utility.

**Follow these steps:**

1. Open the operating system or instance command prompt as follows:  
**UNIX:**  
 Run the shell that is sourced to use AutoSys Workload Automation.  
 The operating system command prompt appears.  
**Windows:**  
 Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
 The instance command prompt appears.

2. Set an environment variable for the password as follows:

– (UNIX)

```
pw_var=password; export pw_var
```

– (Windows)

```
set pw_var=password
```

The *pw\_var* variable is set to the specified value.

3. Set an environment variable for the encrypted password as follows:

– (UNIX)

```
pw_var=`as_config - p - n pw_var`; export pw_var
```

– (Windows)

```
for /f %i in ('as_config - p - n %pw_var%') do set pw_var=%i
```

The configuration utility encrypts the password and sets the *pw\_var* variable to the encrypted value. You can use this variable in the configuration file or to specify the encrypted password in scripts that run commands in batch.

## AutoSys Workload Automation System-Level Security

This article explains how to manage AutoSys Workload Automation system-level security.

### User and Database Administrator Passwords

When you install AutoSys Workload Automation and you configure the database, for example, a database user named *autosys* is added, and a database password is defined. The database user is granted rights to the AutoSys Workload Automation objects and can make changes to specific information in the database. AutoSys Workload Automation uses the database user and password specified in the DBAccess parameter in the configuration file (on UNIX) or the User and Password fields on the Event Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (on Windows).

Provide the user IDs and passwords to run jobs successfully. For example, user IDs and passwords are required when running Command, File Watcher, or File Trigger jobs to a Windows agent or when running jobs of other job types.

#### NOTE

- Only the EDIT superuser can use the *autosys\_secure* command to change the *autosys* database user password.
- Every event server in an instance must have the same database password. If you are running in dual event server mode, the *autosys\_secure* command changes the password on both event servers. If AutoSys Workload Automation has rolled over to single event server mode, do not change the password until you have re-established dual event server mode.
- Use the *autosys\_secure* command to generate the encrypted password that is specified in the DBAccess parameter in the UNIX configuration file.
- Set the value of the *opsys* attribute of the machine definition to *windows* for the Windows agent to support Windows Active Domain user accounts properly.

### Add a User ID and Password to Run Jobs

To run jobs on an agent computer, you must define user IDs and passwords that the jobs run under. The user IDs are specified in job definitions using the owner attribute or a user-specific attribute as required by the job type.

#### NOTE

The user ID you enter in the owner attribute for Command, File Watcher, or File Trigger jobs does not require a password if the job is scheduled to run on a UNIX agent computer.

**Follow these steps:**

1. Log in to AutoSys Workload Automation as the EDIT superuser. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:  

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.
2. Enter option [5] Manage users.  

The Manage users submenu appears.
3. Enter option [1] Manage users with password.  

The Manage users with password menu appears.
4. Enter option [1] Create a user.  

The utility prompts you to enter the user name, host or domain name, and password.
5. Enter the user name, user host or domain, and password information when prompted.  

The utility issues a message to indicate that the user was successfully added.
6. If the utility issues an error message, navigate the menu to make the appropriate changes. Repeat this action until the utility issues a message that the user is successfully added.
7. Enter option [0] Exit AutoSys Security Utility.  

The utility closes and the user information is saved in the database.
8. Open the operating system or instance command prompt, and enter the following command:  

```
jil
```

The JIL command prompt opens.
9. Enter the following jil command and attributes:  

```
insert_job: job_name
job_type: c
owner: user@host_or_domain
machine: machine_name
command: command argument
```

**NOTE**

The owner must have an account on the target agent computer. Specify the owner of the job definition as *user@host\_or\_domain*. The entry in the AutoSys Workload Automation database must match the owner name exactly for the job to run on a Windows agent. The new user is defined as the owner of the specified job.

**Example: Specify a User ID in a Job Definition**

This example runs a command on a Windows agent computer named winagenthost. The command runs under the sched user. This user is the job owner and is specified using the owner attribute.

```
insert_job: aswinjob
job_type: CMD
owner: sched@winagenthost
machine: winagenthost
command: my_command
```

**Integrate with Symantec Privileged Access Manager (PAM) Vaulting System**

As an Autosys user, you will be using many applications in your daily work and they may require credentials for their functionality. Managing the credentials is made easy by integrating AutoSys with the Symantec Privileged Access Manager (PAM) vaulting system. Your credentials of various applications that you use are protected and managed by PAM against your alias.

AutoSys can use the credentials stored in PAM to run the applications through alias which you can define using AutoSys Secure. Because your alias is used to access your credentials, whenever you must change the credentials for an application, it is enough if you change the credentials stored in PAM.

#### Prerequisites:

- The cspm client must be available in the machine.
- On Linux and AIX operating systems, the environment variable CSPM\_CLIENT\_HOME must be available in the source files so that the application server and scheduler can resolve them.

### **Add and Manage Password Vaults**

As a workload user, you can use the password vaulting solution so that your credentials are more secure.

#### Follow these steps:

1. Log in to AutoSys as the EDIT superuser. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

2. Enter option [7] Manage password vault.  
The Manage password vault submenu appears.
3. Enter option [1] Add password vault.  
The utility prompts you to enter the Password Vault Name and Password Vault Type.
4. Enter the Password Vault Name and Password Vault Type when prompted.  
The utility displays a message to indicate that the password vault was successfully added.
5. Enter option [2] Delete password vault.  
The utility prompts you to enter the password vault name and vault type.
6. Enter the Password Vault Name and Password Vault Type when prompted.  
The utility displays a message to indicate that the selected password vault is deleted.

#### **NOTE**

The utility displays an error message if there are references to the password vault through user with an alias.

7. Enter option [3] Show password vault.  
The utility displays the Password vault name and Password vault type.
8. Enter option [0] Exit AutoSys Security Utility.  
The utility closes and the user information is saved in the database.

### **Manage Users with Password Vaults**

As a workload user, you can use the password vaulting solution so that your credentials are more secure.

#### **NOTE**

A user defined with password cannot coexist with the password vault user. Only one of the users can exist.

#### Follow these steps:

1. Log in to AutoSys as the EDIT superuser. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

2. Enter option [5] Manage users.  
The Manage users submenu appears.
3. Enter option [3] Manage users with password vault.  
The Manage users with password vault menu appears.

4. Enter option [1] Create a user.  
The utility prompts you to enter the user name, host or domain name, and the alias as defined in the password vault.
  - a. Enter the user name, user host or domain, and the alias as defined in the password vault when prompted.  
The utility displays a message to indicate that the user was successfully added.
  - b. If the utility displays an error message, navigate the menu to make the appropriate changes. Repeat this action until the utility displays a message that the user is successfully added.
5. Enter option [2] Change a user.  
The system prompts you to enter the name for the existing user.
  - a. Enter the name for the existing user.  
A message appears to indicate whether the change was successful.
  - b. If the message indicates that there was an error changing the user, navigate the menu to make the required changes. Repeat this action until the utility displays a message that the change is successful.
6. Enter option [3] Delete a user.
7. Enter option [4] Show all users.  
Displays all the password vault users along with the vault name these are associated with.
8. Exit from the Manage users with password Vault menu.
9. Enter option [0] Exit AutoSys Security Utility.  
The utility closes and the user information is saved in the database.

### **File System Access Restriction**

You can restrict unauthorized use of AutoSys Workload Automation at the file system level as follows:

- Ensure that only authorized users can change permissions on the files and directories in the directory structure.
- Check the level of security you must use. For example, you can restrict access as follows:
  - Only authorized users can use AutoSys Workload Automation.
  - Any user can view reports about jobs. For example, use the autorep command to view the status of a job, but only authorized users can create jobs and calendars or can change them.

If you want only authorized users to access AutoSys Workload Automation, ensure that only those users have execute permissions for the files in the \$AUTOSYS/bin directory. If you want all users to view reports about jobs, but only authorized users to create and edit jobs and calendars, ensure that only the authorized users have execute permission for the following files in the \$AUTOSYS/bin directory:

- autocal\_asc
- archive\_events
- autotimezone
- DBMaint
- dbspace
- dbstatistics
- jil
- sendevent

#### **NOTE**

By default all users are granted execute permissions for the autorep file in the \$AUTOSYS/bin directory.

Restricting execute permission for these files prevents unauthorized users from making configuration changes. Protect the files in the \$AUTOUSER directory from modification by ensuring that only users authorized to make configuration changes have write permission for the files. Read permission is necessary to source the environment files.

## Native Security Mode

Native security mode provides a limited level of security protections. This mode simplifies your operations, but may expose your enterprise to security risks.

Native security mode grants permissions and controls access to AutoSys Workload Automation objects based on pre-defined user types. Native security settings are not customizable. To customize security settings, operate in external security mode.

### Job Ownership

Every job runs under a specified user (the owner of the job). By default, the owner is the user who defines that job on a particular computer. The owner is defined by the owner job attribute and its value is specified as *user@machine*.

The default owner is automatically assigned when you define a job. However, if you have EDIT superuser privileges, you can override the default owner by specifying the owner attribute in the job definition.

#### **UNIX:**

The owner of the job is the user ID retrieved from the UNIX environment. The owner value is attached to the job definition in the form of *user@machine*. By default, only the owner can edit and execute the job.

The *user@machine* combination must have execute permission for any command that is specified in a job on the computer where the job is to run. The job owner must also have permission to access any device, resource, and so on, that the command must access. Therefore, the job owner must have the appropriate system permissions.

The owner's umask write permission is used as the default edit permission for the job, and the umask execute permission is used as the default execute permission for the job.

#### **Windows:**

The owner of the job (the Windows user ID that a job runs under and its corresponding password) must be defined in the database. The user ID and password are defined in the database by the EDIT superuser.

### User Types

To provide security, AutoSys Workload Automation associates the following four user types with each job:

- **Owner**

Creates the job.

#### **NOTE**

The owner of a job can let other users edit and execute the job by setting the permissions in the job definition.

- **Group**

Resides in the same primary group as the owner.

#### **NOTE**

This type is valid on UNIX only.

- **World**

Specifies all users.

- **Machine**

Specifies the users that are defined on a machine.

On UNIX, AutoSys Workload Automation uses the user ID (uid) and group ID (gid) of the owner of a job to control the following permissions:

- Who can edit, override, or delete a job definition.
- Who can execute the command specified in a job.

## **Superusers**

Under the native security model, users with administrative privileges are named as *superusers*. AutoSys Workload Automation lets you define two levels of superusers: EDIT and EXEC. You can use the `autosys_secure` command to define these superusers.

### **NOTE**

For more information about defining the superusers, see [Post-Installation Procedures for the Server](#).

## **EDIT Superusers**

The EDIT superuser is the only user with permission to do the following tasks:

- Edit or delete any job regardless of its owner or its permissions.
- Enter valid operating system user IDs and passwords in the database.  
These user IDs and passwords are required to log on to and run jobs on client computers. When an agent runs a job on a computer, it logs on as the user defined in the owner attribute for the job. To do this task, the scheduler retrieves encrypted versions of the IDs and passwords for the `user@host_or_domain` and the `user@machine` from the event server and passes them to the agent.

### **NOTE**

Users who do not have EDIT superuser permission cannot add user IDs and passwords in the database. However, any user who knows an existing user ID and password can change that password or delete that user and password.

- Change the priority of a job.
- Change the owner attribute of a job.  
The EDIT superuser can override user authentication (if enabled) on a job by job basis by changing the owner of the job from the `user@machine` form to the `user` form. When the owner of the job is specified in the `user` form, user authentication of the job at run time is not performed on the client computer.  
The purpose of the `user@machine` form is to prevent users from running jobs on machines where they do not have the appropriate permission. For example, `root@machine` prevents `root` on any machine from running jobs on all machines.
- Use the `autosys_secure` command to do the following tasks:
  - Enable external security.
  - Add or change Windows user IDs and passwords.
  - Change the database password.
  - Change the remote authentication method.

## **EXEC Superusers**

The EXEC superuser is the only user with permission to do the following tasks:

- Use the `sendevent` command or Web UI to issue commands that affect the running or the state of a job.  
Use the `sendevent` command to stop the scheduler as follows:

```
sendevent -E STOP_DEMON
```

### **NOTE**

EXEC superuser privileges are typically granted to the night operator.

## **Permission Types**

AutoSys Workload Automation associates different permission types with each job. Every job has the following permission types:

- **Edit**

Lets users edit, override, or delete a job definition.

- **Execute**

Lets users use the sendevent command or Web UI to send an execute event that affects the running or the state of a job.

- **Machine**

Lets users who are logged on to a computer, other than the one where a job was created, edit, or execute the job.

**NOTE**

For a job to run on a computer other than the one where it was defined, the owner of that job must have an account on that computer.

## **Grant Permissions**

In the native security mode, individuals or groups of users are provided with edit and execute permissions on a job-by-job basis as follows:

- On UNIX, AutoSys Workload Automation supports Owner, Group, Machine, and World Edit and Execute permissions.
- On Windows, AutoSys Workload Automation supports Owner, Machine, and World Edit and Execute permissions.

The owner of a job cannot override his or her ownership designation. By default, only the owner has edit and execute permissions for a job. All edit and execute permissions are valid only on the computer where the job is defined. Only the EDIT superuser has the permission to change the owner attribute for a job. However, the owner can use JIL to set the permission attribute in the job definition to grant other users edit and execute permissions for a job.

The following table shows the permissions that you can set using JIL:

JIL	Description
permission: gx	Valid on UNIX only. Grants execute permissions to all users in the job owner's primary group. Users in the group can execute the job if the user is logged on to the computer where the job was created (the computer that is specified in the owner attribute; that is, <i>user@machine</i> ).  Group execute permissions are ignored in Windows job definitions. On Windows, the user executing the job must be the owner of the job, or world execute (wx) permissions must be specified for the job.
permission: ge	Valid on UNIX only. Grants edit permissions to all users in the job owner's primary group. Users in the group can edit the job if the user is logged on to the computer where the job was created (the computer that is specified in the owner attribute; that is, <i>user@machine</i> ).  Group edit permissions are ignored in Windows job definitions. On Windows, the user editing the job must be the owner of the job, or world edit (we) permissions must be specified for the job.
permission: mx	Grants execute permissions to any authorized user, regardless of the computer they are logged on to.  If this permission is not granted, the user must be logged on to the computer specified in the owner attribute ( <i>user@machine</i> ) to execute the job.
permission: me	Grants edit permissions to any authorized user, regardless of the computer they are logged on to.  If this permission is not granted, the user must be logged on to the computer specified in the owner attribute ( <i>user@machine</i> ) to edit the job.



permission: wx	Grants execute permissions to all users (world execute permissions). Any user can execute the job if the user is logged on to the computer where the job was created (the computer that is specified in the owner attribute; that is, <i>user@machine</i> ).
permission: we	Grants edit permissions to all users (world edit permissions). Any user can edit the job if the user is logged on to the computer where the job was created (the computer that is specified in the owner attribute; that is, <i>user@machine</i> ).

**NOTE**

A job and the command it runs always runs under the user specified in the owner attribute of the job definition. Execute permissions determine who can execute events against the job, but not the user that the job runs under. Even if world execute permissions (permission: we) are granted, the job still runs under the user defined in the owner attribute.

**Security on Events Sent by Users (sendevent)**

If you have the appropriate permissions, use the sendevent command to send the following execute events that affect the running of a job or the state of a job:

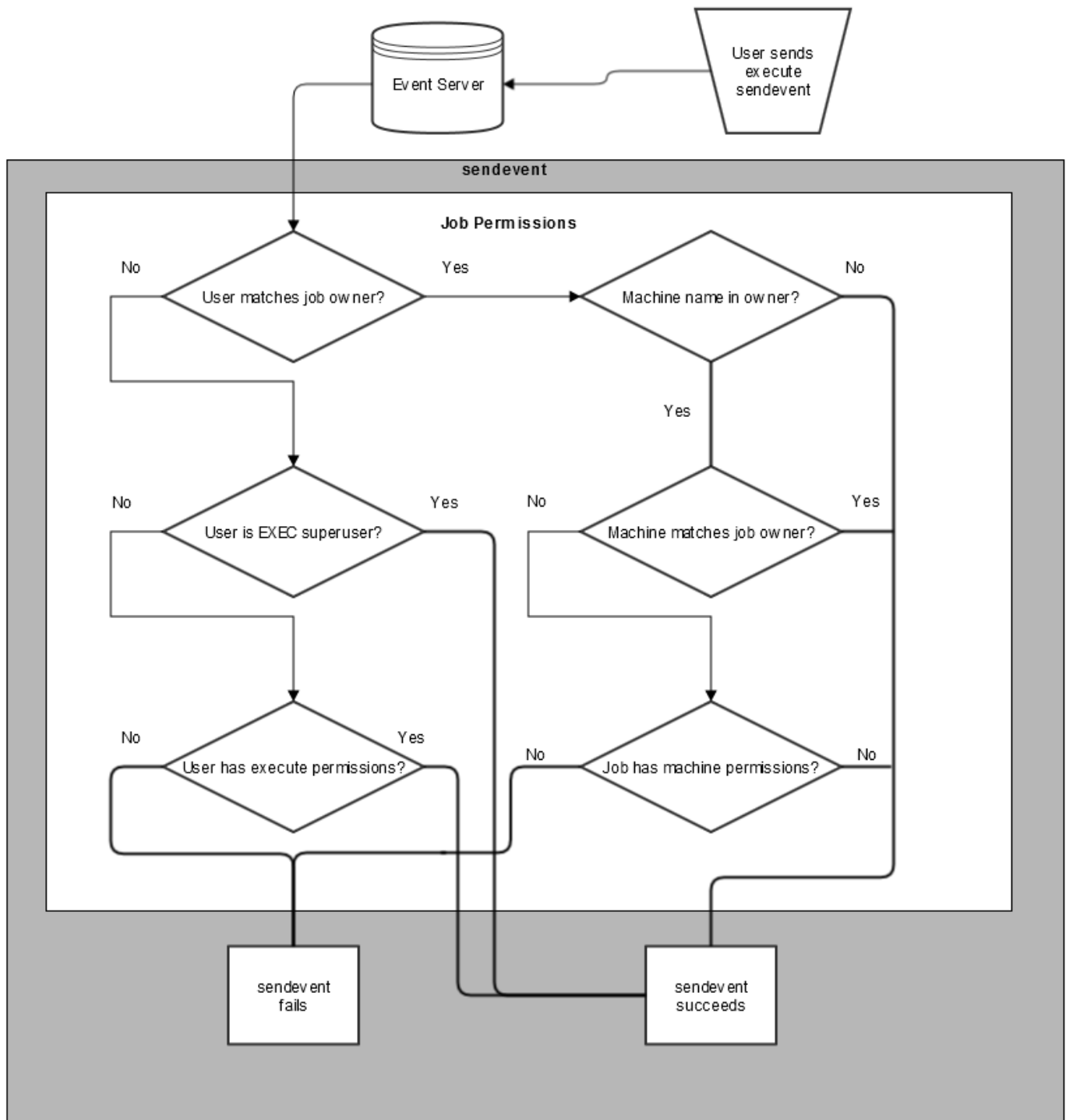
- CHANGE\_PRIORITY
- CHANGE\_STATUS
- DELETEJOB
- FORCE\_STARTJOB
- JOB\_OFF\_HOLD
- JOB\_OFF\_ICE
- JOB\_ON\_HOLD
- JOB\_ON\_ICE
- KILLJOB
- SEND\_SIGNAL
- STARTJOB
- JOB\_ON\_NOEXEC
- JOB\_OFF\_NOEXEC
- RELEASE\_RESOURCE

**Verify Job Permissions**

AutoSys Workload Automation verifies the following points when a user sends an event to start a job:

- Does the user match the owner as indicated in the job definition?
- Is the user the EXEC superuser as defined with autosys\_secure?
- Does the user have job execute permissions as indicated in the job definition?
- Is there a machine name in the owner value of the job definition?
- Does the machine portion of the user login credentials match the machine portion of the job owner definition?
- Does the job have machine permission as indicated by the job definition?

When you start a job by sending an event, the job permissions are verified as shown in the following illustration:

**Figure 13: Sendevent flow**

## Revert to Native Security

This article explains how to revert to native security.

As an administrator, your responsibilities include ensuring that your enterprise runs as efficiently as possible. External security mode requires that you register the AutoSys Workload Automation instance with an external security server and may add other administrative requirements depending on your customized settings. Reverting to native security mode simplifies your operations, but is recommended only when the limited protections offered by native security mode meet your specific security requirements.

**Follow these steps:**

1. [Change the security mode](#)
2. [Unregister the external security server](#)

### **Change the Security Mode**

External security provides more comprehensive access control than native security, but you can enable external security only when you have an external security server installed.

You can activate external security during installation or you can convert from native security mode to external security mode using the interactive security utility. You can also use the interactive utility to disable external security and revert to native security mode.

To improve access control, enable external security. To uninstall the external security server for maintenance or troubleshooting purposes, revert to native security.

#### **NOTE**

- AutoSys Workload Automation requires a valid external security authentication certificate to enable external security. If the certificate is missing or invalid, AutoSys Workload Automation requires the credentials of an external security user with administrative privileges to generate a new one. If you generate a new certificate and AutoSys Workload Automation cannot store the certificate, contact the database administrator.
- Only superusers with edit privileges can enable external security in an instance that is operating in native security mode.
- Users can revert to native security only if existing security policy authorizes them to do so.
- This procedure provides instructions for using the interactive security utility to change the security mode. You can also use `autosys_secure` command line options to change the security mode.
- For more information about activating external security when you install AutoSys Workload Automation, see [Install AutoSys Server](#).

**Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:

- (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
- (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).a  
The instance command prompt appears.

2. Execute the following command:

```
autosys_secure
```

The interactive security utility opens. The available options depend on your configuration settings.

3. (As applicable) Authenticate by typing 1, pressing the Enter key, specifying requested information, and pressing the Enter key again. Continue specifying requested information and pressing the Enter key until the utility stops issuing prompts.

#### **WARNING**

When the menu includes an option to authenticate, you can continue with the procedure only after successful authentication. In this case, the menu does not display options to change the security mode, set the unauthenticated user mode, or add trusted Windows domains. If the menu displays these options, the option to authenticate is not available and this step is not applicable.

The utility validates your credentials and redirects to the full interactive menu.

4. Change the security mode as follows:

- a. Type 1 and press the Enter key.  
The utility requests confirmation.
- b. Type 1, press the Enter key, specify any requested information and press the Enter key. Continue specifying information and pressing the Enter key until the utility stops issuing prompts.

**WARNING**

If you registered with a group of external security servers that is configured to be highly available, use a comma separate list to specify the individual server host names. Ensure that less than 80 characters and that the entire list is less than 255 characters.

**NOTE**

The utility prompts you to specify external security server information when you enable external security in an instance that is operating in native security mode. The utility does not prompt you to specify any information when you revert to native security.

The utility displays a message indicating that the instance security mode is set.

5. Type 0 (zero) and press the Enter key.

The utility closes and the AutoSys Workload Automation instance is operating in the selected security mode.

### **Remove the Default Security Policies**

When a AutoSys Workload Automation instance is operating in external security mode, the instance is registered with an external security server. After you change the security mode from external to native, unregister the instance with the external security server.

**Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears.
2. Enter the following command:
 

```
as_safetool
```

The interactive menu appears.
3. Type 1 and select Enter.  
The Manage Embedded Entitlements Manager server location submenu appears.
4. Type 1 and select Enter.  
The system prompts you to enter the host name of the external security server.
5. Type the server host name and select Enter.  
The utility prompts you to specify the application instance.
6. Type 0 (zero) and select Enter.  
The utility prompts you to provide your credentials.
7. Type the name of a user with administrative privileges to the specified external security host server and press the Enter key, then type the corresponding password and select Enter again.

**NOTE**

If the utility prompts you to re-enter credentials, ensure that you are using an accurate and valid username and password combination and attempt to authenticate again. Repeat this action until authentication is successful.

The utility redirects to Manage Embedded Entitlements Manager server location submenu.

8. Type 9 and select Enter.  
The main menu appears.
9. Type 3 and select Enter.  
The Manage Instance Security submenu appears.
10. Type 2 and select Enter.  
The utility prompts you to enter the instance name.
11. Enter the name of the AutoSys Workload Automation instance for which you want to remove default security policies and select Enter, then type 0 (zero) and select Enter again.  
The utility closes, the instance is unregistered with CA EEM, and the default security policy is removed.

You have reverted to native security mode.

## Security Policy Customization

CA Embedded Entitlements Manager (CA EEM) lets you customize security in all GUI instances and in CLI instances that are operating in external security mode:

### Active Directory Authentication

You can configure CA EEM to reference more users from an external directory source, such as Active Directory. An active directory is a directory structure that is used on Microsoft Windows-based systems to store information. This structure includes resources such as users and user groups.

#### NOTE

- Default Web UI users cannot log in to Web UI unless you add them to the external directory source.
- The default port for Active Directory is 389. For improved performance, use port 3268.

### Configure CA EEM to Reference Active Directory Global Users

Use this procedure to configure CA EEM to reference the Active Directory global users. Doing so lets the applications using CA EEM, such as AutoSys Workload Automation and Web UI, share the user information from the Active Directory.

#### NOTE

You can configure filters so that users do not access all entries in the global Active Directory.

#### Follow these steps:

1. Access the CA EEM Home page.
2. Click the Configure tab.  
The Configure page opens.
3. Click the User Store link.  
The User Store page appears.
4. Click User Store.  
The User Store Configuration page appears in the right pane.
5. Select the Reference from an external LDAP directory option.  
The Directory Information section appears.
6. Leave the default configuration type as Basic LDAP Directory and click Add external LDAP directory.  
Additional fields relating to the external directory appear.
7. Leave the default attribute map as Microsoft Active Directory, and complete the following fields:
  - a. Enter a name for the configuration.
  - b. Enter the host computer on which Active Directory is running.

- c. Enter the port on which Active Directory is listening and click



default port to enter is 389.

**NOTE**

Enter a port number other than 389 only if the Active Directory Server administrator has reconfigured the default Active Directory Server port.

The host and port appear in the Selected Hostnames section.

- d. (Optional) Add other host computers and ports to the Selected Hostnames section as needed.

**NOTE**

Other host computers can be added only if Active Directory failover is configured.

- e. (Optional) Select the appropriate protocol from the Protocol drop-down list. The default is LDAP.

**NOTE**

If TLS or SSL is enabled in the Active Directory, you can select a different protocol.

- f. Enter the Base DN. Only global users and groups that are discovered under this DN are mapped to CA EEM.  
 g. Enter the User DN that is used to attach to the Active Directory server.  
 h. Enter the password for the user specified in the User DN field. This value is required.  
 i. Re-enter the password in the Confirm Password field.  
 j. (Optional) Click Show Advanced Configuration and enter the values in the fields based on the enterprise configuration.
8. Click Save.  
 The values that you have specified are saved, and CA EEM is configured to the Active Directory.

**NOTE**

- You can configure CA EEM to reference users from any external LDAP directory. To reference users from another external directory, edit the attribute map.
- For more information about referencing users from an external LDAP directory, see the [CA EEM](#) documentation.

### **Configure Group Cache and Resolution**

You can use the Group Configuration page to configure cache and resolution properties for the global and application groups.

**Follow these steps:**

1. Click the Configure tab.
2. The Configure page opens.
3. Click the User Store link.  
The User Store page appears.
4. Click Group Configuration.  
The Group Configuration page appears in the right pane.
5. Specify the following items in the Global Group Configuration section:
  - Group Resolution Level
  - Enable Group Cache (select or clear the check box)
  - Group Cache Size
  - Cache Expiry (in seconds)
6. Specify the following items in the Application Group Configuration section:

- The Group Resolution level
  - Enable Group Cache (select or clear the check box)
  - Group Cache Size
  - Cache Expiry (in seconds)
7. Click Save.  
The changes are saved.

**NOTE**

For more information about configuring cache and resolution properties, see the [CA EEM](#) documentation.

**Retrieve Groups Using a Pattern in Group Names**

You can configure a filter to retrieve groups using a pattern in their group names.

**Follow these steps:**

1. Click the Configure tab.  
The Configure page opens.
2. Click the User Store link.  
The User Store page appears.
3. Click Attribute Maps.  
The LDAP Mapping page appears in the right pane.
4. Edit the Group Search Filter field to include the groups to be included.  
For example, to include a single group pattern, the string may resemble the following example:  
`(&(objectClass=group)(sAMAccountName=T*))`

To include multiple group patterns, the string may resemble the following example:

```
(&(objectClass=group)(|(sAMAccountName=t*)(sAMAccountName=a*)))
```

5. Click Save As, enter a custom directory map name, and click OK.  
The changes are saved.

**NOTE**

The custom directory map name can be selected from the Mapping Name drop-down list in the LDAP Mapping page.

6. Select the Manage Identities tab and then click the Groups link.  
The Search Groups section is displayed.
7. Search for groups and verify that you can view the groups as defined in the filter.  
The groups that match the pattern that is entered in the Group Search Filter field are retrieved.

**Retrieve Users from a Group**

As a CA EEM default behavior, you can authenticate and retrieve all users who are defined in the configuration. To limit users due to the directory size or to restrict unnecessary users from gaining access, apply a filter.

LDAP filters use the following syntax rules:

- The segments of the expressions must be enclosed in parentheses.
- The following relational operators can be applied: <=, =, =substring\*, =\*, ~=, and >=.
- Expressions are formed using the prefix operators & and |.

For example: `(&(&(objectClass=user)(!(objectClass=computer)))(telephoneNumber>=0))`

**Follow these steps:**

1. Click the Configure tab.  
The Configure page opens.
2. Click the User Store link.  
The User Store page appears.
3. Click Attribute Maps.  
The LDAP Mapping page appears in the right pane.
4. Leave Microsoft Active Directory selected or select the appropriate directory from the Mapping Name drop-down list.
5. Edit the User Search Filter field under User Lookup as follows:
  - a. Enter (& at the beginning of the existing filter.
  - b. Enter ) at the end of the 'objectClass=computer))' string.
  - c. Enter (memberOf=groupName)).

**NOTE**

The 'memberOf' string must contain the fully qualified name for the group.

- d. Change other values as required.
6. Click Save As, enter a custom directory map name, and click OK.  
The changes are saved.

**NOTE**

The custom directory map name can be selected from the Mapping Name drop-down list in the LDAP Mapping page.

7. Select the Manage Identities tab and then click the Users link.  
The Search Users section is displayed.
8. Search for users and verify that you can view the users as defined in the filter.  
The users from the specified group are retrieved.

## Customize Security Policy and Settings

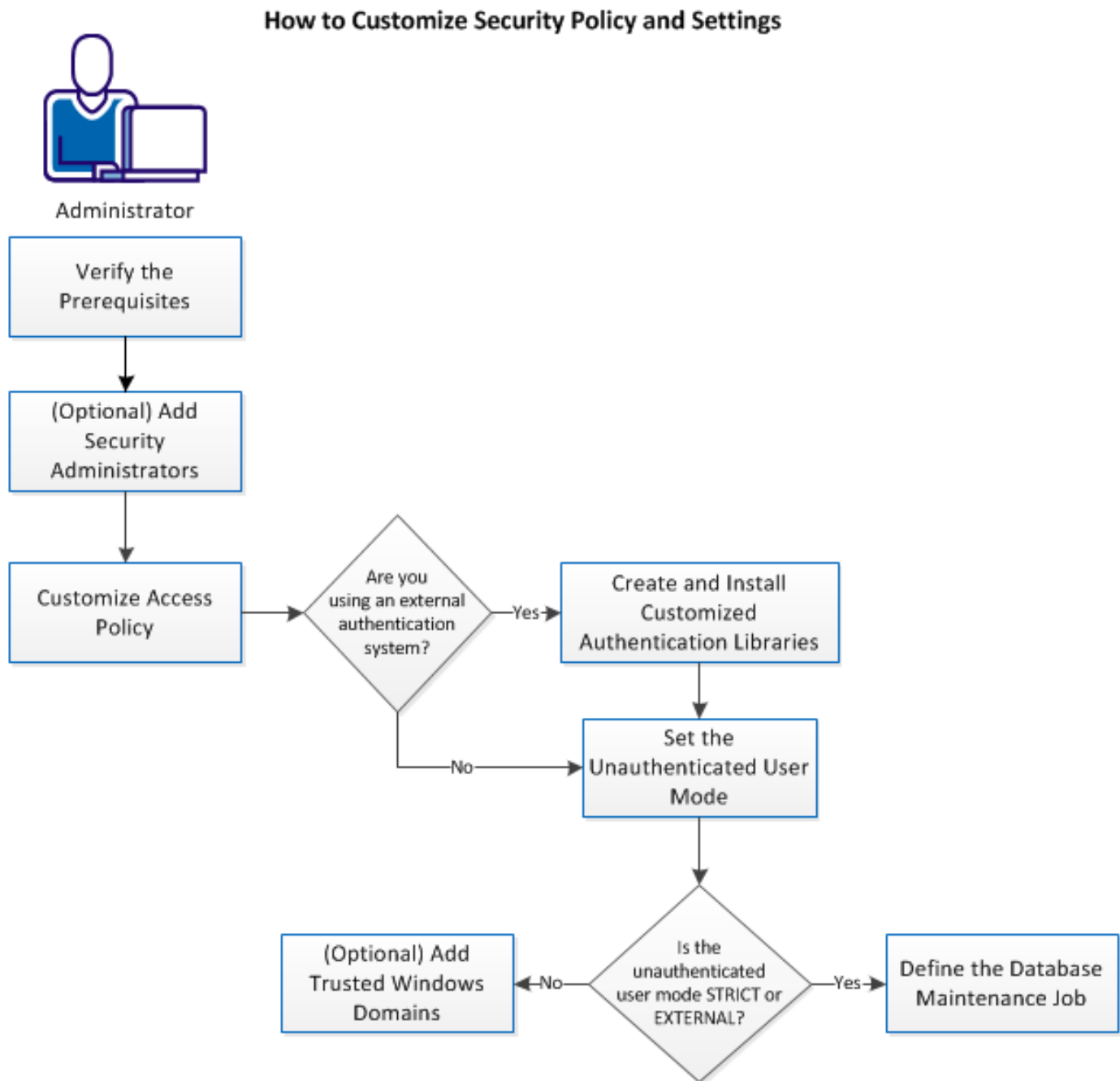
As an administrator, you manage the security of your enterprise. External security mode lets you customize security policy and settings. This customization increases the control you have over the security of your environment.

Default external security policy provides no more protection than native security mode. We recommend that you customize access policy to protect resources and set the unauthenticated user mode setting to prevent potentially malicious from impersonating authenticated users. By default, only the EiamAdmin can change security policy, but you can add security administrators.

This scenario explains the process of customizing security policy and settings:



Figure 14: How to Customize Security Policy and Settings

**Follow these steps:**

1. [Verify the prerequisites.](#)
2. (Optional) [Add Security Administrators.](#)  
Use this option the first time that you customize policy, or when you want to add administrators.
3. [Customize access policy.](#)
4. (As required) [Create and install customized authentication libraries.](#)  
This procedure is required when you are using an external authentication system; otherwise, it is not applicable.

5. (Recommended) [Set the unauthenticated user mode](#).  
This procedure is required when you are using an external authentication system or when the default setting does not meet your security requirements. This procedure is also useful in situations where you have sufficient control of your enterprise and you want to simplify operations.
6. (As required) [Define the database maintenance job](#).  
This procedure is required when you configure AutoSys Workload Automation to require all users to authenticate.
7. (Optional) [Add trusted Windows domains](#).  
Use this option to enable users who are managed by the specified domains to execute utilities without authenticating through the external security server. The availability of this option depends on the unauthenticated user mode setting.

## Install the Default External Security Policy

You can customize security policy and settings only when default security policy is installed. If you followed the recommendation to activate external security during installation, default policy is installed. If you did not activate external security during installation, install default security policy.

### Register with the External Security Server

We recommend that you activate external security during installation, but the product provides the option to enable it post-installation. You can change the security mode from native to external only in instances that are registered with the external security server.

#### Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears.
2. Enter the following command:  
`as_safetool`  
The interactive menu appears.
3. Type 1 and press Enter. The Manage Embedded Entitlements Manager server location submenu appears.
4. Type 1 and press the Enter key. The system prompts you to enter the host names of the servers to register with.
5. Specify the servers as follows:  
`host_name [, host_name2...]`

#### **WARNING**

When you specify multiple servers, ensure that each server is part of the group of servers that are configured to be highly available.

#### **NOTE**

For information about how to configure CA Embedded Entitlements Manager servers to be highly available, see [CA EEM documentation](#).

6. Press the Enter key, type 0 (zero) at the prompt, and then select the Enter key. The utility prompts you to provide your credentials.
7. Type the name of an external security user with administrative privileges and press the Enter key, then type the corresponding password and press the Enter key again.

#### **NOTE**

The utility redirects to Manage Embedded Entitlements Manager server location submenu.

8. Type 9 and press the Enter key. The main menu appears.
9. Type 3 and press the Enter key. The Manage instance security submenu appears.

10. Type 1 and press the Enter key. The utility prompts you to specify the name of a AutoSys Workload Automation instance.
11. Type the name of the instance for which you want to install default security policy and select the Enter key. The instance is registered with the external security server, default external security policy is installed, and the utility redirects to the Manage instance security submenu.
12. Type 0 and press the Enter key. The utility closes.

### **Change the Security Mode**

External security provides more comprehensive access control than native security, but you can enable external security only when you have an external security server installed.

You can activate external security during installation or you can convert from native security mode to external security mode using the interactive security utility. You can also use the interactive utility to disable external security and revert to native security mode.

To improve access control, enable external security. To uninstall the external security server for maintenance or troubleshooting purposes, revert to native security.

#### **NOTE**

- AutoSys Workload Automation requires a valid external security authentication certificate to enable external security. If the certificate is missing or invalid, AutoSys Workload Automation requires the credentials of an external security user with administrative privileges to generate a new one. If you generate a new certificate and AutoSys Workload Automation cannot store the certificate, contact the database administrator.
- Only superusers with edit privileges can enable external security in an instance that is operating in native security mode.
- Users can revert to native security only if existing security policy authorizes them to do so.
- This procedure provides instructions for using the interactive security utility to change the security mode. You can also use `autosys_secure` command line options to change the security mode.
- For more information about activating external security when you install AutoSys Workload Automation, see [Installing the Server](#).

#### **Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).a  
The instance command prompt appears.
2. Execute the following command:
 

```
autosys_secure
```

The interactive security utility opens. The available options depend on your configuration settings.
3. (As applicable) Authenticate by typing 1, pressing the Enter key, specifying requested information, and pressing the Enter key again. Continue specifying requested information and pressing the Enter key until the utility stops issuing prompts.

#### **WARNING**

When the menu includes an option to authenticate, you can continue with the procedure only after successful authentication. In this case, the menu does not display options to change the security mode, set the unauthenticated user mode, or add trusted Windows domains. If the menu displays these options, the option to authenticate is not available and this step is not applicable.

The utility validates your credentials and redirects to the full interactive menu.

4. Change the security mode as follows:

- a. Type 1 and press the Enter key.  
The utility requests confirmation.
- b. Type 1, press the Enter key, specify any requested information and press the Enter key. Continue specifying information and pressing the Enter key until the utility stops issuing prompts.

**WARNING**

If you registered with a group of external security servers that is configured to be highly available, use a comma separate list to specify the individual server host names. Ensure that less than 80 characters and that the entire list is less than 255 characters.

**NOTE**

The utility prompts you to specify external security server information when you enable external security in an instance that is operating in native security mode. The utility does not prompt you to specify any information when you revert to native security.

The utility displays a message indicating that the instance security mode is set.

5. Type 0 (zero) and press the Enter key.  
The utility closes and the AutoSys Workload Automation instance is operating in the selected security mode.

## Add Security Administrators

This article explains how to add security administrators. Administrative external security privileges let you activate external security and customize policy for AutoSys Workload Automation instances that are registered with an external security server. When you register an instance with the external security server, default policy applies to the instance. To enable users to activate external security or customize policy, delegate administrative privileges.

### Follow these steps:

1. Open a browser and go to the following web site:  
`http://eemserverhost:5250/spin/eiam`  
– **eemserverhost**  
Specifies the IP address or host name of the computer where CA EEM is installed.  
If the specified server is available, the CA EEM login page appears.  
Select WorkloadAutomationAE from the Application drop-down list, type the name of a user with administrative privileges in the User Name field, type the corresponding password in the Password field, and click Log In.

**NOTE**

Only existing administrators can delegate administrative privileges. When default policy is installed, EiamAdmin is the only external security administrator. If you are modifying a customized policy that grants administrative privileges to other users, log in as an administrator.

The CA EEM web interface opens.

2. Select the Manage Identities tab.  
The User Search and Users panes appear.
3. (Optional) Create a user as follows:
  - a. Click the new user icon (insert icon).  
The New User form appears.
  - b. Complete the User name, New password, and Confirm password fields and, optionally, the General Information section. Click Save.

**NOTE**

When CA Embedded Entitlements Manager (EEM) is configured to reference users from an external user source, ensure that you create this user in that external source. For more information about configuring EEM to reference users from an external user source, see the *CA Embedded Entitlements Manager Online Help*.

A message appears confirming that the new user was created.

4. Click Go in the User Search pane.  
The Users folder expands displaying a list of external security users.
5. Select a user to whom you want to delegate administrative privileges, and click the Add User Application Details button.  
User groups to which the user already belongs are displayed in the Selected User Groups box. User groups to which you can add the user are displayed in the Available user groups box.
6. Select the WorkloadAutomationAEAdmin group from available groups and click the right arrow and click Save.  
The group moves from the Available box to the Selected box and the user is added to the group.
7. (Optional) Continue selecting or creating users and adding them to the WorkloadAutomationAEAdmin user group as needed.  
Administrative privileges are delegated to all users that you added to the WorkloadAutomationAEAdmin user group. These users can create or update access policy and change the security.

## Customize Access Policy

*Access policy* is the collective term for the set of policies that protects access to AutoSys Workload Automation resources. Users who are authorized to create, modify, view, execute, or delete resources may have access to sensitive information or permission to control AutoSys Workload Automation administrative tasks.

Default external security policy grants all users full access to all resources. To ensure that only trusted users can access sensitive information or control critical services, customize access policy.

### NOTE

- When security policy grants the user who executes a utility the requested access to an object, the utility runs normally. When security policy specifies that the user does not have the requested access, the actions that the utility performs are limited.
- For more information about the events, commands, or jil attributes that are mentioned in this procedure, see [Reference](#).

### Follow these steps:

1. Open a browser and go to the following web site:  
`http://eemserverhost:5250/spin/eiam`  
– **eemserverhost**  
Specifies the IP address or host name of the computer where CA EEM is installed.  
If the specified server is available, the CA EEM login page appears.  
Select WorkloadAutomationAE from the Application drop-down list, type the name of a user with administrative privileges in the User Name field, type the corresponding password in the Password field, and click Log In.

### NOTE

Only existing administrators can delegate administrative privileges. When default policy is installed, EiamAdmin is the only external security administrator. If you are modifying a customized policy that grants administrative privileges to other users, log in as any administrator.

The CA EEM web interface opens.

2. Select the Manage Access Policies tab.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name check box is selected by default.
3. Expand the Access Policies folder in the left pane.  
The resource class folders appear.
4. Open a resource class folder based on the type of resource you want to control access to.

5. Open a policy in the folder, click Save As, specify a name for the policy you are adding, and click OK. Ensure that the policy you open is associated with the AutoSys Workload Automation instance that you are adding policies to.
6. (Optional) Describe the policy.
7. (Optional) Select an external security calendar to associate the policy with. Use this option to enforce a policy only on the dates or times specified in the calendar.

**WARNING**

The calendars in the drop-down list are external security objects, not AutoSys Workload Automation objects. To ensure that a policy is effective only on particular dates, define a calendar in the WAAE application instance on the external security server and then select that calendar when you create the policy.

8. (Optional) Select the Explicit Deny check box to specify that the policy denies the specified users access to the specified resources.

**WARNING**

A policy that denies a group of users access to resources does not grant other users access to the same resources. To grant other users access to these resources, create a separate policy.

9. (Optional) Temporarily disable the policy. Use this option to troubleshoot a policy.
10. (Optional) Prevent the external security server from deploying the policy. Use this option to test a new policy within the CA EEM web interface before you deploy it to AutoSys Workload Automation. When the Pre-deployment check box is not selected, the external security server deploys the policy when you save it.
11. (Optional) Associate the policy with other pre-deployment policies as follows:
  - a. Select the Assign Labels check box.
  - b. Enter at least label that is also assigned to the policies you want to associate this policy with.

**NOTE**

Labels do not affect policies that are already deployed to AutoSys Workload Automation.

The policy is grouped with all policies that share at least one of the assigned labels.

12. Select the identities that the policy applies to. Identities represent multiple users, including users who are not authenticated by the external security server.

**WARNING**

Ensure that you include and exclude all appropriate users. We recommend that you verify selected users based on identity format. If you are unsure about the identity that is associated with a user that you want to include or exclude, verify the identity of the user. For information about identity verification, see [Verify Security Policy](#).

13. Select the Actions to the permissions you want to control.

**NOTE**

The available actions depend on the resource class.

14. Specify the resources that policy applies to and then click Save.  
Pre-deployment policies are saved on the external security server. All other policies are saved, deployed to the specified AutoSys Workload Automation instance.
15. Repeat steps 5 through 15 until all resources are adequately protected.
16. Verify pre-deployment policies, then make any necessary modifications and repeat the verification. Use labels to verify groups of pre-deployment policies simultaneously. Repeat this action as needed.

**NOTE**

For information about pre-deployment policy verification, see the [CA EEM documentation](#).

17. Open each verified pre-deployment policy, clear the pre-deployment check box, and click Save.  
The policies are saved and deployed. AutoSys Workload Automation polls for security policy updates at the interval that is specified in the configuration file (30 seconds by default) and enforces newly deployed policies when it detects them.

## CA EEM Policies for AutoSys Workload Automation

During the AutoSys Workload Automation installation, when AutoSys Workload Automation is registered with CA EEM, the default security policies are created. These default security policies are included in the corresponding resource classes within the AutoSysWorkloadAutomation application instance.

### NOTE

The default security policies grant all access modes to all users. We recommend that you use the CA EEM web interface to customize the default policies. Conversely, you can create policies and grant access modes based on your requirements.

Before performing an action on a specified object, AutoSys Workload Automation issues a security call to the appropriate resource class in the repository. For example, for jobs, AutoSys Workload Automation queries policies in the as-job resource class. For global variables, AutoSys Workload Automation queries policies in the as-gvar resource class.

## CA EEM Resource Classes for AutoSys Workload Automation

The CA EEM AutoSysWorkloadAutomation application instance supports the following resource classes:

Each resource class uses at least one of the following access modes:

- READ
- CREATE
- DELETE
- EXECUTE
- WRITE
- SENDEVENT\_JOBEXECUTE

### as-appl Resource Class

The actions in the as-appl resource class control access to job sets as follows:

- **READ**  
View jobs that belong to an application group.

### NOTE

If the as-list resource class policies do not enable the utilities to bypass security checks, reports will include only information about resources that the utility has read access to. When a utility runs individual security checks, reports processing time increases.

- **EXECUTE**  
Send events to an application group. Authorize the cross-platform start job requests.
- **WRITE**  
Edit jobs that belong to an application group.

### as-base-jobtype Resource Class

The actions in the as-base-jobtype resource class control access pre-defined job types as follows:

- **EXECUTE**
  - Send events that affect jobs that belong to the specified job types.
  - Create, edit or delete jobs that belong to the specified job type.
  - Edit a box job as follows:

- Add a job to it.
- Remove a job from it.
- Edit a job in it.

**WARNING**

This resource class controls access to pre-defined job types only. To control access to user-defined job types, use the `as-jobtype` class.

Specify the pre-defined job types that a policy applies to as follows:

`INSTANCE_NAME.BASE_JOB_TYPE`

**WARNING**

Unlike JIL definitions, these values are case sensitive. Ensure that you use capital letters.

- **INSTANCE\_NAME**  
Specifies the name of the AutoSys Workload Automation instance.
- **BASE\_JOB\_TYPE**  
Specifies the base job type.

**Example:** To apply a policy to box jobs in the instance named ACE, type ACE.BOX in the Type field.

**NOTE**

For a list of pre-defined job types, see [AE Scheduling](#).

**as-calendar Resource Class**

The actions in the `as-calendar` resource class control access to standard and extended calendars as follows:

- **READ**  
View calendars.

**NOTE**

If the `as-list` resource class policies do not enable the utilities to bypass security checks, reports will include only information about resources that the utility has read access to. When a utility runs individual security checks, reports processing time increases.

- **CREATE**  
Add calendars.
- **DELETE**  
Delete calendars.
- **EXECUTE**  
Use calendars to schedule jobs.
- **WRITE**  
Edit calendars.

**WARNING**

An extended calendar can reference a cycle and this class does not control access to cycles. Ensure that a complimentary policy in the `as-cycle` class is created.

**as-connectionprofile Resource Class**

The action in the `as_connectionprofile` resource class controls access to connection profiles as follows:

- **READ**  
View connection profiles to generate reports with the `autorep` command.
- **CREATE**



Create connection profiles.

- **DELETE**  
Delete connection profiles.
- **EXECUTE**  
Insert a connection profile in the job.
- **WRITE**  
Edit connection profiles.

### **as-control Resource Class**

The actions in the as-control resource class control access to administrative to the following resources:

- **<instance>.AGGREGATE**  
Aggregate statistics.
- **<instance>.ARCHIVE\_JOBS**  
Archive jobs.
- **<instance>.AUTOTRACK**  
Alter the tracking level.
- **<instance>.EPLOG**  
Retrieve scheduler log files.
- **<instance>.SECADM**  
Disable external security.
- **<instance>.SENDEVENT\_GRPAPP**  
Send events to job sets.
- **<instance>.STOP\_DEMON**  
Send an event to stop server processes.

The following actions are available:

- **EXECUTE**  
Execute the specified tasks.

### **as-cycle Resource Class**

A *cycle* is a list of date ranges (periods) that you can apply criteria to when you define an extended calendar. Ensure that the policies in this resource class complement the policies in the as-calendar resource class.

The actions in the as-cycle resource class control access to cycles as follows:

- **READ**  
View cycles.

#### **NOTE**

If the as-list resource class policies do not enable the utilities to bypass security checks, reports will include only information about resources that the utility has read access to. When a utility runs individual security checks, reports processing time increases.

- **CREATE**  
Add cycles.
- **DELETE**  
Delete cycles.
- **WRITE**  
Edit cycles.

#### **WARNING**

This action is valid only in combination with the READ action.

### **as-group Resource Class**

The actions in the as-group resource class control access to job groups as follows:

- **READ**

View jobs that belong to the specified groups.

**NOTE**

If the as-list resource class policies do not enable the utilities to bypass security checks, reports will include only information about resources that the utility has read access to. When a utility runs individual security checks, reports processing time increases.

- **EXECUTE**

Send events to the specified groups or to jobs in those groups.

Authorize the cross-platform start job requests.

- **WRITE**

- Add jobs to the specified groups.

- Edit jobs that belong to the specified groups.

- Remove jobs from the specified groups.

### **as-gvar Resource Class**

The actions in the as-gvar class control access to global variables as follows:

- **READ**

View variables.

- **CREATE**

Add variables.

- **DELETE**

Delete variables.

- **EXECUTE**

Send the SET\_GLOBAL event.

The SET\_GLOBAL event controls global variables as follows:

- **sendevent - e SET\_GLOBAL *new\_var=new\_value***

Creates the variable.

- **sendevent - e SET\_GLOBAL *existing\_var=DELETE***

Deletes the variable.

- **sendevent - e SET\_GLOBAL *existing\_var=new\_value***

Updates the value of the variable.

- **WRITE**

Edit variables.

**WARNING**

Ensure that you select EXECUTE when you select CREATE, DELETE, or WRITE.

**NOTE**

For more information about the SET\_GLOBAL event, see [Events](#) .

### **as-job Resource Class**

The actions in the as-job resource class control access to jobs and job blobs as follows:

- **READ**

View jobs and job blobs to generate reports with the autorep, autostatad, autostatus, job\_depends, or monbro command.

**WARNING**

To limit read access to jobs, ensure that you also modify the default policies in the as-list resource class. These policies override as-job policies and grant all users read access to all jobs. Users can execute the previously mentioned commands successfully only when as-job or as-list policies grant them read access to the jobs specified in the command parameters.

- **CREATE**  
Create jobs.
- **DELETE**  
Delete jobs.
- **EXECUTE**
  - Send events that affect jobs.
  - Create, edit, or delete jobs that are contained in box jobs.
  - Authorize the cross-platform start job requests.
- **WRITE**
  - Edit jobs, including jobs contained in box jobs.
  - Add jobs to or remove jobs from box jobs.
  - Change the priority of jobs.

**NOTE**

To apply policies from this resource class to Web UI Monitoring, select the Use AutoSysWorkloadAutomation Policies in CA EEM check box in Monitoring Preferences in Configuration. This configuration can negatively impact Web UI performance.

**as-joblog Resource Class**

The actions in the as-joblog resource class control access to job log files as follows:

- **READ**  
Retrieve job log files from the application server.  
Job log files include files that contain normal or error output from a job run, output from the job command, or output from the job profile files.

**as-jobtype Resource Class**

The actions in the as-jobtype resource class control access to user-defined job types as follows:

- **READ**  
Generate autorep and job\_depends reports for the specified types of jobs.

**NOTE**

If the as-list resource class policies do not enable the utilities to bypass security checks, reports will include only information about resources that the utility has read access to. When a utility runs individual security checks, reports processing time increases.

- **CREATE**  
Add job types.
- **DELETE**  
Delete job types.
- **EXECUTE**  
Create jobs that belong to the specified job types.
- **WRITE**  
Edit job types.

### **as-list Resource Class**

The actions in the as-list resource control whether reporting utilities bypass individual security checks as follows:

- **READ**

View listed resources

An as-list policy can specify any combination of the following resources:

- **<instance>.AUTOCAL**  
Specifies the calendar utility (autocal\_asc command)
- **<instance>.AUTOREP**  
Specifies the report utility (autorep command)
- **<instance>.AUTOSTAT**  
Specifies the adapter status utility (autostatad command)
- **<instance>.FORECAST**  
Specifies the forecast utility (forecast command)
- **<instance>.JOBDEP**  
Specifies the job dependencies utility (job\_depends command)
- **<instance>.MONBRO**  
Specifies the monitor browser utility (monbro command)

#### **NOTE**

- When a reporting utility does not have as-list access to the requested resources, the utility conducts individual security checks. The utility generates a report that includes only information about resources to which the utility has read access. Individual security checks increase report processing time.
- For more information about the specified reporting utilities, see [AutoSys Commands](#).

### **as-machine Resource Class**

The actions in the as-machine resource class control access to machines as follows:

- **READ**

View machines with the autorep utility.

- **CREATE**

Create machines.

- **DELETE**

Delete machines.

- **EXECUTE**

Send events that affect jobs that run on the specified machines or specify the machine names in job definitions. Authorize the cross-platform start job requests.

- **WRITE**

Edit machine definitions.

### **as-owner Resource Class**

The as-owner resource class controls the following user permissions:

- Whether a user has permissions to include the owner attribute in the job definition. When the user defines a job, AutoSys Workload Automation evaluates the policies in the as-owner resource class to find a policy that grants the user access to the owner value.
- Whether a user has permissions to update, override, or delete any attribute of a job. When the user updates, overrides, or deletes a job, AutoSys Workload Automation evaluates the policies in the as-owner resource class to find a policy that grants the user access to the job existing owner value.
- Whether a user has permissions to execute the job.

The actions in the as-owner resource class controls user permissions as follows:

- **EXECUTE** Controls whether you can specify a different user as the owner of the job. Also, controls whether you can update, override, or delete any attribute of a job.
- **SENDEVENT\_JOBEXECUTE** Controls whether you can execute the sendevent command.

### **as-resource Resource Class**

The actions in the as-resource class control access to virtual resources as follows:

- **READ**  
View a resource to generate autorep or job\_depends reports.
- **CREATE**  
Create resources.
- **DELETE**  
Delete resources.
- **EXECUTE**  
Specify resources in job definitions.
- **WRITE**  
Edit resources.

### **as-sendevent Resource Class**

The actions in the as-sendevent resource class control access to execute the sendevent command as follows:

- **ALARM**  
Indicates the occurrence of one of the specific alarms. For more information about alarms, see [Alarms](#).
- **CHANGE\_PRIORITY**  
Modifies the job queue priority of the specified job to the priority specified in the -q parameter of the sendevent command.
- **CHANGE\_STATUS**  
Changes the status of the specified job to the status defined in the -s parameter of the sendevent command.
- **COMMENT**  
Generates a comment in the scheduler log.
- **DELETEJOB**  
Deletes the specified job from the database.
- **FORCE\_STARTJOB**  
Starts the specified job immediately on the machine specified in the job definition.
- **JOB\_OFF\_HOLD**  
Removes the specified job from the ON\_HOLD status.
- **JOB\_OFF\_ICE**  
Returns the specified job to the job stream. The job is removed from the ON\_ICE status.
- **JOB\_OFF\_NOEXEC**  
Removes the specified job from the ON\_NOEXEC status.
- **JOB\_ON\_HOLD**  
Places the specified job in the ON\_HOLD status.
- **JOB\_ON\_ICE**  
Suspends scheduled runs of the specified job by temporarily removing it from the job stream.
- **JOB\_ON\_NOEXEC**  
Places the specified job in the ON\_NOEXEC status and bypasses execution of commands that are associated with the job.
- **KILLJOB**

Notifies the agent to stop execution of the specified job and place it in the TERMINATED status.

- **MACH\_OFFLINE** Takes the machine that the -N parameter of the sendevent command specifies offline.
- **MACH\_ONLINE** Places the machine specified in the -N parameter of the sendevent command specifies online.
- **RELEASE\_RESOURCE**  
Releases a held resource back to the pool of usable resources.
- **REPLY\_RESPONSE**  
Issues a response to a WAIT\_REPLY\_ALARM.
- **RESTARTJOB**  
Restarts a job that ends in failure.
- **RESUMEJOB**  
Resumes a Hadoop Oozie workflow and places it in the RUNNING state.
- **SEND\_SIGNAL**  
Sends a signal to a running job.
- **SET\_GLOBAL**  
Sets a global variable.
- **STARTJOB** Starts the specified job when its starting conditions are satisfied.
- **SUSPENDJOB**  
Suspends the Hadoop Oozie workflow and places it in the SUSPENDED state.
- **UNSEND\_EVENT**  
Cancels unprocessed events.

#### NOTE

For more information about the events, see [Events](#).

## CA EEM Identities for AutoSys Workload Automation

CA EEM employs *identities* to represent authenticated users (or their roles) and groups of users. You can write security policies against CA EEM users to grant or deny access to AutoSys Workload Automation objects.

When you install CA EEM, the installer creates the default administrative CA EEM user named *EiamAdmin* and the global CA EEM application instance. The global CA EEM application instance is the central repository for administrative policies that are used to control CA EEM. The instance defines the global CA EEM users that are visible by all Broadcom applications that use CA EEM.

When you activate external security during the AutoSys Workload Automation installation or register AutoSys Workload Automation with CA EEM using the `as_safetool` command, the CA EEM application instance for AutoSys Workload Automation named *AutoSys* is created. This CA EEM application instance is the central repository for all policies that are used by AutoSys Workload Automation. The instance includes following pre-defined CA EEM user groups:

- WorkloadAutomationAEAdmin
- WorkloadAutomationAEWebService

#### NOTE

By default the AutoSysWorkloadAutomationAdmin and AutoSysWorkloadAutomationWebService CA EEM user groups have no members. You can add CA EEM users to the CA EEM user groups.

Sometimes you must add or delete CA EEM users or grant or restrict their permissions:

- When an employee transfers departments
- When a new employee joins the company
- When an employee leaves the company
- When the duties of a user or group of users change

Only EiamAdmin and CA EEM users with administrative privileges to the global CA EEM application instance can add, delete, or modify global CA EEM users. CA EEM users with administrative privileges to the AutoSys CA EEM application instance can add, delete, or modify CA EEM users that are scoped to the CA EEM application instance. EiamAdmin and members of the Admin CA EEM user group are authorized administrators to the AutoSys CA EEM application instance. Only members of the AutoSysWorkloadAutomationWebService CA EEM user group are authorized to access the AutoSys Workload Automation web service.

By default, the AutoSysWorkloadAutomationAdmin and AutoSysWorkloadAutomationWebService CA EEM user groups have no members. You can add CA EEM users that you create in either the global CA EEM application instance or the AutoSys CA EEM application instance to these CA EEM user groups.

To add, delete, or modify CA EEM users, log in as EiamAdmin or as an authorized CA EEM administrator that you define. To define a CA EEM user as an authorized administrator, take *one* of the following actions:

- Add a CA EEM administrative scoping policy that grants the CA EEM user read and write access to the AutoSys CA EEM application instance.
- Add the CA EEM user to the AutoSysWorkloadAutomationAdmin CA EEM user group.

When you add a CA EEM user to the AutoSysWorkloadAutomationAdmin CA EEM user group, the following administrative privileges are granted to this user:

- Permission to log in to the AutoSys CA EEM application instance.
- Permission to add or remove CA EEM users from CA EEM user groups.
- Permission to modify the privileges of CA EEM users scoped to the AutoSys CA EEM application instance.
- Permission to create or delete CA EEM users scoped to the AutoSys CA EEM application instance.

#### NOTE

- Use a CA EEM administrative scoping policy to grant a CA EEM user the permission to modify security policies and manage CA EEM users. For more information about scoping policies and creating them using the CA EEM web interface, see the [CA EEM documentation](#).
- When you log in to the AutoSys CA EEM application instance and create a CA EEM user, the CA EEM user is only valid on that application instance. Only the EiamAdmin or CA EEM users granted administrative privileges to the global CA EEM application instance can create CA EEM global users.

When you add a CA EEM user to the AutoSysWorkloadAutomationWebService CA EEM user group, the CA EEM user is granted the permission to access the AutoSys Workload Automation web service.

To create or delete a CA EEM user or user group, use the Manage Identities tab of the CA EEM web interface.

To restrict permissions for a CA EEM user, edit the CA EEM user in the Manage Identities tab of the CA EEM web interface. Delete the CA EEM user from any CA EEM user group that grants those permissions to members.

To grant permissions to a CA EEM user, edit the CA EEM user in the Manage Identities tab of the CA EEM web interface. Add the CA EEM user to a CA EEM user group that grants the required permissions.

To modify permissions for a CA EEM user group, edit the user group definition in the Manage Access Policies tab of the CA EEM web interface.

#### NOTE

- You can configure CA EEM to reference users from the CA EEM repository, from the Active Directory, or from any external LDAP directory using the Configure tab of the CA EEM web interface.
- When you configure CA EEM to reference users from an external LDAP directory, you cannot log in to CA EEM as a CA EEM user that is stored in the CA EEM repository unless you add that CA EEM user as a user on the external directory.

## Representative Identities

An *identity* is a value that the external security server uses to represent a AutoSys Workload Automation user, role, or group of users. You can specify a AutoSys Workload Automation user when you customize access policy only if the user has an assigned identity.

The external security server identifies authenticated users by AutoSys Workload Automation username, group name, or role.

The security subsystem can assign identities to users who do not validate their identities through the external security server, depending on the unauthenticated user mode.

The subsystem sometimes uses an identity format that distinguishes unauthenticated users from authenticated users. This distinction prevents the users from impersonating authenticated users. When the subsystem is configured to trust types of unauthenticated users, the system assigns those users an identity in the form of the real username of an authenticated user.

The format that the subsystem uses to assign an identity to a user depends on the unauthenticated user mode and the user type.

Unauthenticated User Mode Setting	User Type	Representative Identity Format
OFF	Non-local Windows users	<i>WIN_Domain\user</i>
OFF	Local Windows users Local UNIX users Non-local UNIX users	<i>authenticated_user</i>
STRICT	All users	N/A You cannot grant access to users who do not authenticate through the external security server.
QUALIFY	All users	<i>user@fully_qualified_host</i>
TRUST	Local users	<i>user@fully_qualified_host</i>
TRUST	Non-local users	<i>authenticated_user</i>
TRUSTALL	All users	<i>authenticated_user</i>
EXTERNAL	Unauthenticated users	N/A You cannot grant access control to users who do not authenticate through the customized authentication library.

## Verify Security Policy

This article explains how to verify security policy and test the access level of a secured resource.

### Verify Your Security Policy

The `autosec_test` enables you to verify security policy. Run this command to verify *any* of the following information:

- The product is running in external security mode.
- Customized access policy grants appropriate access permissions.
- The external security identity and credentials of an unauthenticated AutoSys Workload Automation user.

#### Follow these steps:

1. Open the operating system or instance command prompt as follows:



- (UNIX) Run the shell that is sourced to use AutoSys Workload Automation  
The operating command prompt appears.
- (Windows) Click Start, Programs, AutoSys Workload Automation, Command Prompt (*instance\_name*).  
The instance command prompt appears.

## 2. Enter the following command:

- `autosec_test [{CALENDAR|GLOBAL_VAR|JOB|JOBTYP|ACHINE|RESOURCES|CYCLE|APPL|GROUP|BASE_JOBTYP|CONTROL|OWNER|LIST|JOBLOG} resource {C|D|R|W|X} [testuser]] [STAT] [{-usr username {-pw|pwx} password|-saml auth_artifact}]`
- `autosec_test [{CALENDAR|GLOBAL_VAR|JOB|JOBTYP|ACHINE|RESOURCES|CYCLE|APPL|GROUP|BASE_JOBTYP|CONTROL|OWNER|LIST|JOBLOG} resource {C|D|R|W|X} [testuser]]`

(Optional) Issues a test to verify that the specified user (or the logged-in user when no test user is specified) has the specified level of access to the specified resource.

### WARNING

Ensure that you specify an access level that applies to the specified resource class. To verify which access levels apply, display command help.

- `autosec_test[STAT]`

(Optional) Issues a test to determine the security mode that the product is operating in, the identity of the current (logged-in) user, and the credentials that the current user can specify to authenticate client utilities.

### WARNING

This parameter enables you to determine information about the current (logged-in) user only. If you are an external security server and you want to obtain information about another AutoSys Workload Automation user, login to AutoSys Workload Automation as that user and then issue the command.

- `[-usr username {-pw|-pwx} password|-saml auth_artifact}]`

(Optional) Specifies credentials to use to authenticate the utility with external security. Use this option if your settings require authentication and cached credentials are missing or expired.

The utility runs the specified tests and displays relevant messages.

## Example: Verify the Access Level of a Secured Object

This example verifies whether the current user has access permission to start the job named `good_test` manually:

```
autosec_test JOB good_test X
```

This example verifies whether the user named `sched_3` has access permission to start the job named `good_test` manually:

```
autosec_test JOB good_test X sched_3
```

## Test the Access Level of a Secured Resource

You can test the behavior of a AutoSys Workload Automation secured application. You can also verify the enforcement of security policies for AutoSys Workload Automation assets (jobs, calendars, machines).

When you customize external security policy, you specify an action (execute, write, read, create, or delete) to grant users a particular level of access to particular resources. If AutoSys Workload Automation is running in external security mode, the `autosec_test` command returns a message indicating whether a security access check for a given resource to an action passes or fails as governed by the security policy. If no username value is specified, the application issues a security check as the currently logged-on user.

### NOTE

For more information about customized security policy, see [Security Policy Customization](#).

To test the security policies, run the `autosec_test` command.

**NOTE**

For information about the `autosec_test` command syntax, see [AE Commands](#).

**Example: Determine Whether the Current User May Issue an Event for a Job**

This example simulates a request by the AutoSys Workload Automation secured application to the security subsystem to determine whether the current user has access to issue a `STARTJOB` event for the job `good_test`:

```
autosec_test JOB good_test X
```

If the user has execute access to the job, the following message is displayed:

```
Security check passed!
```

If the user does not have execute access to the job, the following message is displayed:

```
Security check failed!
```

**Example: Determine Whether a Specified User May Issue an Event for a Job**

This example simulates a request by the AutoSys Workload Automation secured application to the security subsystem to determine whether the user `admin@example.com` can issue a `STARTJOB` event for the job `good_test`:

```
autosec_test JOB good_test X admin@example.com
```

If `admin@example.com` has execute access to the job, the following message appears:

```
Security check passed!
```

If `admin@example.com` does not have execute access to the job, the following message appears:

```
Security check failed!
```

## Create and Install Customized Authentication Libraries

Customized authentication libraries specify how the security subsystem authenticates client utilities using external authentication information:

- The customized authentication library that you install on client computers (client authentication library) specifies how client utilities obtain the external authentication information that they submit to the security subsystem.
- The customized authentication library that you install on server computers (server authentication library) specifies how the security subsystem validates authentication information and passes an identity to the security subsystem.
- The client utility that executes the client library is assigned the identity that the server library passes to the security subsystem.

After you write the library code for a customized authentication library, build the library using a compiler and linker.

AutoSys Workload Automation supports the following compilers:

- (Windows) Visual Studio 2010, 10.0.40219.1 SP1Rel
- (Linux) GCC version 4.1.2
- (HP-UX) HP aC++/HP C A.06.29
- (AIX) IBM XL C/C++ for AIX, V11.1 (5724-X13)

## Create and Install Customized Authentication Libraries

You can create and install customized authentication libraries only when the AutoSys Workload Automation SDK runtime environment and a supported compiler are installed on the computer that you use to build the libraries.

### NOTE

- The AutoSys Workload Automation codes directory provides sample code for customized authentication libraries.
- For information about installing the AutoSys Workload Automation SDK runtime environment, see [Installing the SDK Runtime Environment](#). For information about installing a compiler, see the product documentation for a supported compiler that is compatible with your operating system.

### Follow these steps:

1. Create a .c or .cpp file for the client authentication library code.
2. Write the client authentication library code in the .c or .cpp file using the C or C++ programming language as follows:

- a. Include the following preprocessor directives:

```
#include "autoextauth_defs.h"
#include "autoextauth_cli.h"
```

- b. Define a function named `ext_GetIdentityToken` that accepts the following parameters as is declared in the `autoextauth_cli.h` file by including the following code:

```
int ext_GetIdentityToken(const char * hostname,
const unsigned int hostname_len,
const char ** app_servers,
const unsigned int app_servers_len,
const char * instance,
unsigned char * token_buf,
unsigned int * token_buf_len,
char * error_message,
unsigned int * error_message_len,
int * error_code)
```

- **hostname**  
(Input) Specifies the application server host to which the utility is connected.
- **hostname\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `hostname` parameter.
- **app\_servers**  
(Input) Specifies a list of application server hosts that are configured for the instance.
- **app\_servers\_len**  
(Input) Specifies the total number of application server hosts in the list represented by the `app_servers` parameter.
- **instance**  
(Input) Specifies the AutoSys Workload Automation instance.
- **token\_buf**  
(Output) Stores the client authentication information.
- **token\_buf\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `token_buf` parameter.  
(Output) Stores the total number of bytes of the client authentication information.
- **error\_message**  
(Output) Stores the error message to explain the reason for the failure, if any.
- **error\_message\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `error_message` parameter.

(Output) Stores the total number of bytes of the error message.

- **error\_code**

(Output) Stores the internal failure code associated with the reason for the failure, if any.

- c. Define the circumstances under which the `ext_GetIdentityToken` function obtains the authentication information to pass to the client utility. Ensure that the definition specifies the return code that the function passes to the client utility.

#### NOTE

AutoSys Workload Automation accepts only the return codes that are declared in the `autoextauth_defs.h` file.

- d. (Optional) Include code that instructs client utilities to perform actions that help you troubleshoot problems with the library. For example, you can write the library code to ensure that the library creates a log file and writes informational messages to that file whenever a client utility executes the library.
3. Create a `.c` or `.cpp` file for the server authentication library code.
  4. Write the server authentication library code in the `.c` or `.cpp` file using the C or C++ programming language as follows:
    - a. Include the following preprocessor directives:
 

```
#include "autoextauth_defs.h"
#include "autoextauth_srv.h"
```
    - b. Define a function named `ext_AuthenticateIdentityToken` that accepts the following parameters as is declared in the `autoextauth_srv.h` file by including the following code:
 

```
int ext_AuthenticateIdentityToken(const char * instance,
const unsigned char * token_buf,
const unsigned int * token_buf_len,
unsigned char * identity,
unsigned int * identity_len,
char * error_message,
unsigned int * error_message_len,
int * error_code)
```

      - **instance**  
(Input) Specifies the AutoSys Workload Automation instance.
      - **token\_buf**  
(Input) Specifies the client authentication information.
      - **token\_buf\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `token_buf` parameter.
      - **identity**  
(Output) Stores the assigned external security user identity.
      - **identity\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `identity` parameter.  
(Output) Stores the total number of bytes of the assigned external security user identity.
      - **error\_message**  
(Output) Stores the error message to explain the reason for the failure, if any.
      - **error\_message\_len**  
(Input) Specifies the total number of bytes of the buffer represented by the `error_message` parameter.  
(Output) Stores the total number of bytes of the error message.
      - **error\_code**  
(Output) Stores the internal failure code associated with the reason for the failure, if any.
    - c. Define how the `ext_AuthenticateIdentityToken` function validates the authentication information and passes the identity to the security subsystem. Ensure that the definition specifies the return code that the function passes to the security subsystem.

**WARNING**

Ensure that you specify a valid return code. For more information, see Customized Authentication Library Return Code Values.

5. Build the authentication libraries as follows:

a. Execute the compiler and specify the following options:

- (Windows)
 

```
/Oy- /D "WIN32" /D "_WINDOWS" /D "_USRDLL" /D "_CRT_SECURE_NO_DEPRECATED" /
D "_USE_32BIT_TIME_T" /D "_WINDLL" /D "_MBCS" /GF /Gm- /EHsc /MD /GS /Gy /
fp:precise /Zc:wchar_t /Gd
```
- (Linux)
 

```
-m32 -fpic -D_REENTRANT
```
- (HP-UX)
 

```
-DHP_UX -D_REENTRANT -D_FILE_OFFSET_BITS=64 -Ae +DD64 -mt +z +Z
```
- (AIX)
 

```
-DAIX433 -DAIX -D__AIX -Dss_family=__ss_family
```

b. Execute the linker and specify the following options:

- (Windows)
 

```
/DLL /DYNAMICBASE /FIXED:NO /NXCOMPAT /MACHINE:X86
```
- (Linux)
 

```
-m32 -shared
```
- (AIX)
 

```
-bh:5 -bmaxdata:0x80000000/dsa -qmkshrojb -brtl
```

c. Name the client authentication library as follows:

- (Windows)
 

```
libasxauthc.dll
```
- (Linux)
 

```
libasxauthc.so
```
- (HP-UX)
 

```
libasxauthc.so
```
- (AIX)
 

```
libasxauthc.so
```

d. Name the server authentication library as follows:

- (Windows)
 

```
libasxauths.dll
```
- (Linux)
 

```
libasxauthc.so
```
- (HP-UX)
 

```
libasxauthc.so
```
- (AIX)
 

```
libasxauthc.so
```

Customized client and server authentication libraries are built.

6. Save a copy of the client library to every AutoSys Workload Automation client computer within the instance. Save the library to the following location:

- (Windows)

```
%AUTOSYS%\bin
```

– (UNIX)

```
$AUTOSYS/lib
```

The client library is installed.

7. (Optional) Configure the instance to resolve library conflicts between Oracle and Kerberos as follows:

- a. Log in to a server computer in the instance and add the following environment variable definition:

```
AUTOPRELOADKRB5=1
```

- b. Repeat this action on every server computer within the instance.

**WARNING**

The system cannot resolve library conflicts between Oracle and Kerberos when this environment variable definition is missing or when the value of the variable is not set to 1. If you write library code to execute Kerberos functions, ensure that you define this variable with a value of 1.

**NOTE**

You can add environment variable definitions using the operating system console (UNIX) or the Systems page of the Administrator Utility (Windows). For more information about defining environment variables, see [Configuring AutoSys Workload Automation](#).

8. Save a copy of the server library to every AutoSys Workload Automation server computer within the instance. Save the library to the following location:

– (Windows)

```
%AUTOSYS%\bin
```

– (UNIX)

```
$AUTOSYS/lib
```

The server library is installed. You can configure the instance to authenticate client utilities by changing the unauthenticated user mode setting to EXTERNAL and starting the application server.

### **Customized Authentication Library Return Code Values**

The `autoextauth_defs.h` file specifies the return codes that you can use when you write customized authentication library code:

- **0**  
(Client libraries) Specifies that the client authentication routine successfully obtained the client authentication information and stored the information in the output buffer. The routine must store the authentication information in the `token_buf` output parameter and the total number of bytes of the information in the `token_buf_len` output parameter.  
(Server libraries) Specifies that the server authentication routine successfully authenticated the user using the client authentication information and stored the assigned external security user identity in the output buffer. The routine must store the assigned identity in the `identity` output parameter and the total number of bytes of the identity in the `identity_len` output parameter.
- **1**  
(Client libraries) Specifies that the client authentication routine successfully obtained the client authentication information but the output buffer is too small to store the information. The routine must store the total number of bytes of the information in the `token_buf_len` output parameter.  
(Server libraries) Specifies that the server authentication routine successfully authenticated the user using the client authentication information but the output buffer is too small to store the assigned external security user identity. The routine must store the total number of bytes of the identity in the `identity_len` output parameter.
- **2**  
(Client libraries) Specifies that the client authentication routine failed to obtain the client authentication information but the error buffer is too small to store the reason for the failure. The routine must store the total number of bytes of the failure message in the `error_message_len` output parameter.

(Server libraries) Specifies that the server authentication routine failed to authenticate the user using the client authentication information but the error buffer is too small to store the reason for the failure. The routine must store the total number of bytes of the failure message in the `error_message_len` output parameter.

- **-1**  
 (Client libraries) Specifies that the client authentication routine failed to obtain the client authentication information and stored the reason for the failure in the error buffer. The routine must store the failure message in the `error_message` output parameter and the total number of bytes of the failure message in the `error_message_len` output parameter.  
 (Server libraries) Specifies that the server authentication routine failed to authenticate the user using the client authentication information and stored the reason for the failure in the error buffer. The routine must store the failure message in the `error_message` output parameter and the total number of bytes of the failure message in the `error_message_len` output parameter.

## Manage the Unauthenticated User Mode

This article explains how to set unauthenticated user mode and generate cached credentials. When AutoSys Workload Automation operates in external security mode and you do not authenticate with external security, the security subsystem considers you an unauthenticated user. You can control whether users are required to authenticate with external security when they execute utilities by setting the unauthenticated user mode.

### Set the Unauthenticated User Mode

The external security server assigns identities to users when they authenticate. The security subsystem assigns identities to unauthenticated users only when you enable users to execute utilities without authenticating.

The security subsystem prevents potentially malicious users from impersonating authenticated users by assigning identities to these users in a different format than the external security server uses to assign identities to users.

The subsystem assigns identities to trusted unauthenticated users in a format that does not distinguish them from authenticated external security users. Several options exist to help you mitigate risks when you enable users to execute utilities without authenticating.

#### NOTE

Users can authenticate through the external security server by generating cached credentials or using command-line options. These options let you specify credentials using a user name and password combination or a SAML artifact. You can view, generate, or delete cached credentials using the interactive security utility or the `autosys_secure` command. For more information about the `autosys_secure` command or about using command-line options to authenticate, see [AutoSys Commands](#).

#### Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (`instance_name`).  
The instance command prompt appears.
2. Enter the following command:
 

```
autosys_secure
```

The interactive menu appears. The available options depend on your configuration settings.
3. (As applicable) Authenticate by typing 1, pressing the Enter key, specifying requested information, and pressing the Enter key again. Continue specifying requested information and pressing the Enter key until the utility stops issuing prompts.

**WARNING**

When the menu includes an option to authenticate, you can continue with the procedure only after successful authentication. In this case, the menu does not display options to change the security mode, set the unauthenticated user mode, or add trusted Windows domains. If the menu displays these options, the option to authenticate is not available and this step is not applicable.

The utility validates your credentials and redirects to the full menu.

4. Type 2 and press the Enter key.

The Manage CA EEM security setting submenu appears.

5. Type 1 and press the Enter key.

The Manage CA EEM server setting submenu appears.

6. Type 3 and press the enter key.

The unauthenticated user mode settings appear as follows:

1=OFF, 2=STRICT, 3=QUALIFY, 4=TRUST, 5=TRUSTALL, 6=EXTERNAL

- **1=OFF**

(Default) Specifies that the security subsystem does not trust non-local Windows users; enables users to execute utilities without authenticating.

This setting enables unauthenticated local Windows users and all unauthenticated UNIX users to impersonate authenticated external security users. Use this option only when you can ensure that these users do not present a security threat.

- **2=STRICT**

(Recommended) Specifies that the security subsystem does not trust unauthenticated users and does not assign identities to them. This setting requires all users to authenticate when they execute client utilities. Use this option to ensure the complete security of your enterprise.

**WARNING**

This setting prevents the automatic database maintenance tool from operating properly and requires that you define a job to handle maintenance.

- **3=QUALIFY**

Specifies that the security subsystem does not trust unauthenticated users; enables users to execute utilities without authenticating. Use this option to eliminate security risks without requiring users to authenticate.

- **4=TRUST**

Specifies that the security subsystem does not trust unauthenticated local users; enables users to execute utilities without authenticating. Use this option to reduce the risk of external security user impersonations by local users while treating non-local users who do not present a security risk as authenticated external security users.

- **5=TRUSTALL**

Specifies that the security subsystem treats all users as trusted users; enables users to execute utilities without authenticating. Use this option to reduce policy administration requirements that are associated with qualifying unauthenticated users.

- **6=EXTERNAL**

Specifies that the security subsystem trusts unauthenticated users who validate their identities using an external authentication system such as the Kerberos network authentication protocol. Use this option to bypass the external security server when you are using an external authentication system.

**WARNING**

Ensure that you create and install customized authentication libraries before selecting this setting. This setting prevents the automatic database maintenance tool from operating properly and requires that you define a job to handle maintenance.

**NOTE**

This setting prevents you from using a user name and password combination to authenticate through the external security server, but you can still use a SAML artifact to authenticate.



7. Type the number that corresponds with the setting that you want to select and press the Enter key.
8. Type 0 (zero) and press the Enter key.  
The utility closes. The security subsystem assigns security policy identities according to the selected configuration setting.

### **Generate Cached Credentials**

When AutoSys Workload Automation is operating in external security mode, the unauthenticated user mode can specify that certain users can access secured objects only when they authenticate with external security. The recommended unauthenticated user mode requires all users to authenticate, and most other modes require some users to authenticate. If you are required to authenticate based on this configuration setting and do not, client utilities that you execute cannot access secured objects and can fail.

Cached credentials enable you to authenticate without using command-line options to specify your credentials every time that you execute a utility.

#### **NOTE**

- If you do not generate cached credentials and you do not specify them using command-line options, the utility prompts you to specify them.
- Cached credentials expire 24 hours after you generate them unless you specify a different validity period. If you are relying on cached credentials to authenticate client utilities and receive an authentication error message, cached credentials can be missing or expired. You can view cached credentials using the security utility interactive menu or `autosys_secure` command-line options.
- You can delete cached credentials using the security utility interactive menu or `autosys_secure` command-line options. You can accomplish this task by using the interactive menu, enter the appropriate option in the Manage Cached Credentials submenu and then exit the utility. For information about using `autosys_secure` command-line options to delete cached credentials, see [AutoSys Commands](#).
- This procedure provides instructions for using the security utility interactive menu to generate cached credentials. We recommend this method, but you can also generate cached credentials using `autosys_secure` command-line options.

#### **Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears.
2. Enter the following command:  
`autosys_secure`  
The interactive menu appears. The options in this menu depend on your user privileges.
3. Navigate to the Manage Cached Credentials submenu as follows:
  - (External Security Administrators) Type 2 and press the Enter key, and then type 2 and press the Enter key again.
  - (Users without administrative privileges) Type 2 and press the Enter key.
4. (Optional) Type 2 and press the Enter key.  
The utility displays the current cached credentials, including the expiration date.
5. Type 1 and press the enter key.  
The utility prompts you to specify your user name.
6. Type your user name and press the Enter key.  
The utility prompts you to specify password and then to confirm your password.
7. Type your password, press the Enter key, confirm your password, and press the Enter key again.  
The utility prompts you to enter an expiration date and time.

8. (Optional) Enter the following value:

`{-1 | DDHH}`

– **-1**

Specifies that the credentials never expire.

– **DDHH**

Defines a specific validity period for the credentials in days and hours.

#### NOTE

- To specify that the credentials are valid for less than one day, enter a value of 00 for days and then specify a value between 01 and 24 for hours.
- If you specify that a non-zero value for days and a 00 value for hours, the credentials expire after the specified number of days at the same time of day that you generated them.
- If you specify 0000, the utility returns an error.
- The utility sets the validity period to 24 hours by default when no expiration date and time is specified.

9. Press the Enter key.

#### NOTE

Press the enter key without specifying a validity period, the utility sets the default validity period.

10. Type 0 and press the Enter key.

#### NOTE

If the credential information that the utility displays is not accurate, do not exit the utility. In this case, type 2 and re-generate credentials.

The utility closes.

## Define the Database Maintenance Job

To ensure that the database is properly maintained, disable automated database maintenance and define a job to maintain the database. The STRICT and EXTERNAL unauthenticated user mode settings interfere with the automatic database tool.

### Follow these steps:

1. Open the operating system or Instance command prompt:

– (UNIX) Run the shell that is sourced to use AutoSys Workload Automation

The operating system command prompt appears.

– (Windows) Click Start, Programs, CA, AutoSys Workload Automation, Command Prompt (instance\_name).

The AutoSys Workload Automation instance command prompt opens. The command prompt presets all the environment variables for the instance.

2. Enter the following command:

```
sendevent -E STOP_DEMON [-v ROLE=ALL]
```

– **ALL**

(Optional) Stops all server processes. Use this option in high availability mode so that you can perform maintenance on all server machines without shutting down each server process individually.

All schedulers and application servers running on the instance stop.

3. Disable automated maintenance on each scheduler machine as follows:

– (UNIX) Edit the database maintenance start time in the configuration file as follows:

a. Open the configuration file and locate the following parameter:

```
DBMaintTime=hh:mm[:ss]
```

- b. Update the parameter as follows:

```
DBMaintTime=
```

- c. Save and close the configuration file.

- (Windows) Delete the automated database maintenance start time using the AutoSys Workload Automation Administrator as follows:
  - a. Click Start, All Programs, CA, Enterprise Workload Automation, Administrator.  
The AutoSys Workload Automation Enterprise Administrator Window opens.
  - b. Click the WAAE Scheduler icon, select the Database Maintenance tab, delete the value from the StartTime field, and click Apply.  
Automated database maintenance is disabled.
- 4. Start the schedulers and any other components that are not running as follows:
  - (UNIX) run the shell that is sourced to use AutoSys Workload Automation, wait for the operating system command prompt to appear, and enter the following command:
 

```
/etc/init.d/CA-WAAE start all
```
  - (Windows) Click Start, All Programs, CA, Workload Automation, Administrator, select the instance from the instance drop-down list and click Apply, select Services from the WAAE menu, select the schedulers and any other components that you want to start from the services list, and click the start icon.
- 5. Enter the following command:
 

```
jil
```

The JIL command prompt appears.

- 6. Enter the following JIL subcommand and attribute-value combinations:

```
insert_job: DBMaint
command: DBMaint
machine: localhost
date_conditions: y
days_of_week: all
start_times: "HH:mm [:ss]"[, "HH:mm [:ss]"...]
exit
```

- **machine: localhost**

Specifies that the job runs on the AutoSys Workload Automation machine named localhost.

#### NOTE

Specifying the localhost instead of the scheduler ensures that the job continues running even when the shadow scheduler is handling workload. The shadow scheduler takes over workload when AutoSys Workload Automation is operating in high availability mode and the primary scheduler fails. For more information about highly available environments, see (UNIX) [Improve Workload Performance: Highly Available Environments](#) or (Windows) [Improve Workload Performance: Highly Available Environments](#).

- **start\_times: "HH:mm [:ss]"[, "HH:mm [:ss]"...**

Specifies a starting time for the job other than the default time. To specify multiple times, use a comma-separated list.

#### WARNING

Specify times when other jobs do not run; otherwise, this job may interfere with other scheduled workflow.

The database maintenance job is defined and the JIL utility closes.

You have customized Security Settings. The instance is operating according to the specified security policy and unauthenticated user mode.

## Add Trusted Windows Domains

This article explains how to add trusted Windows domains. In enterprises where the external security server is linked to Windows Active Directory, external security users can authenticate through Active Directory. The security subsystem does not automatically treat these users as authenticated external security users unless the unauthenticated user mode setting specifies that non-local Windows users are trusted users.

A *trusted user* is a user that the security subsystem treats as an authenticated external security user. Trusting unauthenticated users presents a security risk.

When no domains are trusted, Active Directory users are treated as unauthenticated users. They are unauthenticated unless the unauthenticated user mode is set to trust them. These settings present a significant security risk.

When a domain is listed as a trusted Windows domain, the security subsystem treats users who are managed by that domain as trusted when they log in through Active Directory but does not trust other unauthenticated users. This option mitigates security risks that are associated with trusting unauthenticated users, but it does not eliminate them.

Add trusted Windows domains only when you can ensure that the users who are managed by those domains do not present a security risk.

### NOTE

- This option is not supported when the unauthenticated user mode setting is OFF, TRUST, TRUSTALL, or EXTERNAL.
- To eliminate all security risks, set the unauthenticated user mode to 2: "STRICT." Do not list any domains as trusted Windows domains. In this case, generate cached credentials or use command-line options to authenticate client utilities even if you logged in through Active Directory.

### Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears.
2. Enter the following command:
 

```
autosys_secure
```

The interactive menu appears. The available options depend on your configuration settings.
3. (As applicable) Authenticate by typing 1, pressing the Enter key, specifying requested information, and pressing the Enter key again. Continue specifying requested information and pressing the Enter key until the utility stops issuing prompts.

### WARNING

When the menu includes an option to authenticate, you can continue with the procedure only after successful authentication. In this case, the menu does not display options to change the security mode, set the unauthenticated user mode, or add trusted Windows domains. If the menu displays these options, the option to authenticate is not available and this step is not applicable.

The utility validates your credentials and redirects to the full menu.

4. Type 2 and press the Enter key.  
The Manage CA EEM security setting submenu appears.
5. Type 1 and press the Enter key.  
The Manage CA EEM server setting submenu appears.
6. Type 1 and press the Enter key.  
The current server settings appear.
7. Verify that the unauthenticated user mode setting is 2: "STRICT", 3: "QUALIFY".

**NOTE**

The menu option to add trusted Windows domains is disabled when the unauthenticated user mode setting is OFF, TRUST, TRUSTALL, or EXTERNAL.

8. Type 4 and press the Enter key.  
The Manage trusted domains submenu appears.
9. Type 1 and press the Enter key.  
The system prompts you to enter the name of the domain that you want to add to the list of trusted domains.
10. Enter the domain name.  
The Windows domain is added to the list of trusted domains, and the utility redirects to the Manage trusted domains submenu.
11. (Optional) Type 1, press the Enter key, and enter the name of another domain at the prompt.  
Repeat this action until you have added all the domains that you want AutoSys Workload Automation to treat as trusted domains.
12. Type 0 (zero) and press the Enter key.  
The utility closes. The security subsystem treats users who are managed by the specified domains as authenticated external security users.

**NOTE**

You can also remove or list trusted Windows domains using `autosys_secure` command-line options. For more information about the `autosys_secure` command, see [AutoSys Commands](#).

## Customize Web UI Policies

This article explains how to customize Web UI policies. Web UI installs a default policy for each resource class in the CA EEM application for Web UI. The rights that are granted by these policies are designed to reflect typical user roles.

CA EEM provides highly granular and flexible capabilities for creating customized policies to reflect the requirements of your enterprise. This section describes the default policy for each resource class. This section also provides sample procedures to create another policy for the resource class, enabling you to customize your enterprise security. This section concludes with permission checks information. We recommend that you review these procedures before proceeding with any customizations. In doing so, you have all materials and information that you need to ensure a successful CA EEM security setup.

Before you use the sample policy procedures, perform the following steps:

1. Create views in Monitoring.

**NOTE**

For more information about creating views, see the *Monitoring Help*.

2. Log in to CA EEM using the Web UI application.
3. Create a CA EEM Application group. The sample policies use this Application Group.
4. Create one or more CA EEM users that are members of the application group created in Step 3.
5. Create the necessary policies, and perform a permission check after each.
6. Verify, in Web UI, that your policy setup works properly.

### Log in to CA EEM for Web UI

Before you complete the remaining steps of the policy creation process, you must be logged in to the CA EEM component of Web UI.

#### Follow these steps:

1. Log in to Web UI as a user who has security administrator rights.  
Web UI opens to the user's default view.

2. Click the EEM link.  
A CA EEM login screen is displayed with EiamAdmin displayed in the User Name field.
3. Select the Web UI application from the Application drop-down list, enter the appropriate password in the Password field, and click Log In.  
CA EEM opens with the Home tab displayed by default.

### **Create CA EEM User Groups**

To give multiple users the same access rights, you can create user groups. We recommend that you create multiple user groups by user role. For example, you could create one user group for schedulers and another for administrators. This procedure describes creating application groups; however, you can also create global groups, dynamic groups, and individual users.

#### **Follow these steps:**

1. Select the Manage Identities tab in the CA EEM application.  
The Users and Groups links appear, with the Users link selected by default.
2. (Optional) Click Groups, leave the Show application groups check box selected, and click Go.  
All the available application groups are listed under Application Groups in the User Groups section.
3. Click the New Application Group icon in the left pane.  
The New Application User Group page appears in the right pane.
4. Enter a name for the new application group, and click Save.  
A confirmation message appears.
5. Repeat Steps 3 and 4 to create more application groups as necessary.

### **Create CA EEM Users**

To give users access to Web UI, first create CA EEM users. All CA EEM users are global users. This procedure describes how to add CA EEM users manually. You can also add users by referencing an Active Directory.

You can add users to global groups from any application. To add a user to an application group or a dynamic group, you must be logged in to the specific CA EEM application that contains the application group or dynamic group. This procedure shows you how to add a user to an application group.

#### **Follow these steps:**

1. Select the Manage Identities tab in the CA EEM application.  
The Users and Groups links appear, with the Users link selected by default.
2. Select the Application User Details option button.
3. (Optional) Leave User Name selected in the Attribute drop-down list, leave LIKE selected in the Operation drop-down list, leave the Value field blank, and click Go to view all the users currently available.  
All application instance-specific user details are listed under Users in the Users pane.
4. Click the New User icon in the left pane.  
The New User page appears in the right pane.
5. Enter a name for the new user in the User Name field.
6. Click Add Application User Details.
7. Select the appropriate application groups from the Available User Groups box in the Application Group Membership section, and click



The application groups are added to the Selected User Groups box.

8. Enter the password for the user in the New Password and Confirm Password fields in the Authentication section. Click Save.  
A confirmation message appears.
9. Repeat Steps 4 through 8 to create more users as necessary.

## Create an AlertAction Policy

To enable a user to access specified alert actions, you can create an AlertAction policy. For this policy, a type of Access Policy is recommended. This example is written based on that selection. This example assumes that you have a Monitoring view, and describes giving access to acknowledge and close alerts for that view and the AEServer server.

The default policy, AlertActionDefault, lets users in the Console Operator, Supervisor, and Commander (superuser) groups acknowledge, close, and re-open alerts and alarms.

### Follow these steps:

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click AlertAction in the left pane.  
All the AlertAction policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to AlertAction in the left pane.  
The New Access Policy tab appears in the right pane.
4. Enter a unique policy name in the General pane, leave AlertAction selected in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of application groups is displayed.
7. Select the application group that you created, and click  
  
The application group appears in the Selected Identities box to the right.
8. Enter view/viewName (where viewName indicates the name of the applicable Monitoring view) in the Add Resource field in the Access Policy Configuration section, and click  
  
The resource is added to the resource list, providing access to performing actions on alerts/alarms in that Monitoring view.
9. Enter server/AEServer in the Add Resource field, and click  
  
The resource is added to the resource list, providing access to performing actions on alerts/alarms from the AEServer server.
10. Select the Acknowledge and Close checkboxes under Actions for your group.
11. Click Save at the top or bottom right of the page.  
The policy is created.
12. Perform a permission check as appropriate.

## Create an ApplicationAccess Policy

To enable a user to access specified applications reflecting that user role, you can create an ApplicationAccess policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection.

The default policy, ApplicationAccessDefault, establishes different application access levels for each CA EEM group (role) installed with the Web UI application. For example, only the Administrator and Commander (superuser) groups have access to the Configuration application; whereas, the Commander, Console Operator, Scheduler, and Supervisor groups have access to the Monitoring application.

**Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ApplicationAccess in the left pane.  
All the ApplicationAccess policies are listed in the right pane.
3. Click the New Access Policy icon  
 next to ApplicationAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave ApplicationAccess selected in the Resource Class Name drop-down list.
5. Select the Identity Access Control List option button in the Type field.  
A confirmation prompt appears.
6. Click OK.  
The Identity Access Control List Configuration section appears.
7. Select Application Group from the Type drop-down list in the Identity Access Control List Configuration section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
8. Click Search.  
A list of application groups is displayed.
9. Select the application group that you created, and click  
  
The application group appears in the Selected Identities list.
10. Select the following checkboxes under Actions for your group:
  - Reporting
  - Monitoring
  - QuickView
  - Credentials
11. Click Save at the top or bottom right of the page.  
Your new policy appears in the list.
12. Perform a permission check as appropriate.

**AgentAccess Policy**

The AgentAccess policy provides permissions to the users and groups to perform various tasks in the Agent List tab against the machine name, such as Download Logs, Status Details, Stop or Restart Agents.

**NOTE**

For more information on AgentAccess Policy, see [Administer Agent Inventory](#).

**Create a CommandExecute Policy**

To enable a user to execute commands in the Enterprise Command-Line application, create a CommandExecute policy. For this policy, we recommend an Access Policy type. This example is written based on that selection. This example assumes that the users in the group have rights to execute commands on all servers to which they have access.

The default policy, CommandExecuteDefault, lets users in the Administrator, Console Operator, Scheduler, Supervisor, and Commander (superuser) groups perform commands on the servers identified in the resource class.



**Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click CommandExecute in the left pane.  
All the CommandExecute policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to CommandExecute in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, leave CommandExecute in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of application groups is displayed.
7. Select the application group that you created, and click  
  
The application group appears in the Selected Identities box to the right.
8. Enter server/\* in the Add Resource field in the Resources section, and click  
  
The resource is added to the resource list.
9. Select the Execute check box under Actions in the Access Policy Configuration section.
10. Click Save at the top or bottom right of the page.  
The policy is created.
11. Perform a permission check as appropriate.

**Create a CommandSetup Policy**

To enable a user to create and save commands in the Enterprise Command-Line application, you can create a CommandSetup policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection. This example assumes that the users in the group have rights to view, create, and modify commands in all tabs, but cannot delete commands.

The default policy, CommandSetupDefault, lets users in the Supervisor and Commander (superuser) groups view, create, modify, and delete commands, and lets users in the Administrator, Console Operator, and Scheduler groups view commands but not create or delete them.

**Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click CommandSetup in the left pane.  
All the CommandSetup policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to CommandSetup in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave CommandSetup in the Resource Class Name drop-down list.
5. Select the Identity Access Control List option button in the Type field.

- A confirmation prompt appears.
6. Click OK.  
The Identity Access Control List Configuration section appears.
  7. Select Application Group from the Type drop-down list in the Identity Access Control List Configuration section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
  8. Click Search.  
A list of application groups is displayed.
  9. Select the application group that you created, and click .  
The application group appears in the Selected Identities list.
  10. Enter local/\* in the Add Resource field in the Resources section, and click .  
The resource is added to the resource list. Doing so provides access to the local command list (My Commands tab) in the Enterprise Command-Line application.
  11. Enter global/\* in the Add Resource field, and click .  
The resource is added to the resource list. Doing so provides access to the global command list (Global Commands tab) in the Enterprise Command-Line application.
  12. Select the Create, Modify, and View checkboxes under Actions for your group.
  13. Click Save at the top or bottom right of the page.  
The policy is created.
  14. Perform a permission check as appropriate.

### **Create a ConfigurationControl Policy**

The ConfigurationControl policy can be used to control the configuration and operation of several disparate resources. Currently, you can create policies for configuring the Credentials, Monitoring, and external applications. For this policy, a type of Access Policy is recommended. This example is written based on that selection. In this example, we create a policy allowing members of the Administrator group to change the credentials for members of the Operator user group.

#### **NOTE**

We recommended that you create a separate policy for each type of application to be configured.

The following default policies establish different access levels for the Credentials, Quick View, Resources, Enterprise Command Line, Job Status Console, Monitoring, external applications, and widgets respectively:

- ConfigurationControlCredentialsAdminDefault
- ConfigurationControlCredentialsUserDefault
- ConfigurationControlEventDefault
- ConfigurationControlMonitoringAdminDefault
- ConfigurationControlMonitoringUserDefault
- ConfigurationControlExternalAppsAdminDefault
- ConfigurationControlWidgetsDefault

For example, only the Commander (superuser), Console Operator, Scheduler, and Supervisor groups can configure access to the Quick View application. However, only the Commander and Administrator groups can configure access to the Credentials application (although all users can modify their own credentials).

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.

The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.

2. (Optional) Click ConfigurationControl in the left pane.  
All the ConfigurationControl policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to ConfigurationControl in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, leave ConfigurationControl in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of application groups is displayed.
7. Select the application group that you created, and click  
  
The group appears in the Selected Identities box to the right.
8. Enter Credentials/server/AEserver/admin/password/operator in the Add resource field in the Access Policy Configuration section, and click  
  
The resource is added to the resource list, allowing members of the Administrator group to change the credentials for members of the Operator user group.
9. Select the Delete check box under Actions for this resource.  
Members of the Administrator group can now delete the password of any Operator group member.
10. Click Save at the top or bottom right of the page.  
The policy is created.
11. Perform a permission check as appropriate.

### **Create an ExternalApplicationAccess Policy**

To enable a user to access configured external applications reflecting that user role, you can create an ExternalApplicationAccess policy. For this policy, a type of Access Policy is recommended. This example is written based on that selection.

The default policy, ExternalApplicationAccessDefault, grants access to all external applications for all users and groups.

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ExternalApplicationAccess in the left pane.  
All the ExternalApplicationAccess policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to ExternalApplicationAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave ExternalApplicationAccess selected in the Resource Class Name drop-down list.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.

6. Click Search.

A list of application groups is displayed.

7. Select the application group that you created, and click



The application group appears in the Selected Identities box to the right.

8. Enter externalApplication/casupport in the Add Resource field, and click



The resource is added to the resource list, providing access to the application assigned the EEM Resource ID casupport in Web UI.

#### NOTE

For more information about assigning EEM resource IDs, see *Dashboard Help*.

9. Enter externalApplication/ca\* in the Add Resource field, and click



#### NOTE

Creating a policy with this entry gives you access to all applications assigned Web UI EEM Resource IDs that start with the characters ca.

10. Select the View check box under Actions in the Access Policy Configuration section.
11. Click Save at the top or bottom right of the page.  
The policy is created.
12. Perform a permission check as appropriate.

### Create a LogAccess Policy

To enable a user to access specified logs, you can create a LogAccess policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection.

The default policy, LogAccessDefault, lets users in the Console Operator, Scheduler, Supervisor, and Commander (superuser) groups view all logs.

#### Follow these steps:

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies left selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click LogAccess in the left pane.  
All the LogAccess policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to LogAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave LogAccess selected in the Resource Class Name drop-down list.
5. Select the Identity Access Control List Type option button in the Type field.  
A confirmation prompt appears.
6. Click OK.  
The Identity Access Control List Type Configuration section appears.
7. Select Application Group from the Type drop-down list in the Identity Access Control List Configuration section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
8. Click Search.

A list of application groups is displayed.

9. Select the application group that you created, and click



The application group appears in the Selected Identities list.

10. Enter server/AEServer in the Add Resource field in the Resources section, and click



The resource is added to the resource list, providing access to the AEServer server.

11. Select the JobLog check box under Actions for your group.
12. Click Save at the top or bottom right of the page.  
The policy is created.
13. Perform a permission check as appropriate.

### **Create a MonitorViewControl Policy**

To enable a user to access specified job flows, you can create a MonitorViewControl policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection. This procedure assumes that you have a Monitoring view and that the user has view-only access.

The default policy, MonitorViewControlDefault, lets users in the Console Operator, Scheduler, Supervisor, and Commander (superuser) groups control all objects. Members of the Executive group have view-only access to job flows.

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear across the top, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click MonitorViewControl in the left pane.  
All the MonitorViewControl policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to MonitorViewControl in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave MonitorViewControl in the Resource Class Name drop-down list.
5. Select the Identity Access Control List option button in the Type field.  
A confirmation prompt appears.
6. Click OK.  
The Identity Access Control List Configuration section appears.
7. Select Application Group from the Type drop-down list in the Identity Access Control List Configuration section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
8. Click Search.  
A list of application groups is displayed.
9. Select the application group that you created, and click  
  
The application group appears in the Selected Identities list.
10. Enter view/viewName/JobFlow (where *viewName* indicates the name of the applicable job flow) in the Add Resource field in the Resources section, and click  
  
The resource is added to the resource list, providing access to that job flow.
11. Select the View check box under Actions for your group.

12. Click Save at the top or bottom right of the page.  
The policy is created.
13. Perform a permission check as appropriate.

### **Create an ObjectAccess Policy**

To assign a user-specified view access to alert policies or forecast reports, you can create an ObjectAccess policy. For this policy type, we recommend the Access Policy or Access Control List type. To permit the same access to all the resource types assigned in the policy, use the Access Policy type. This example is written based on that selection. To permit different types of access to the resource types assigned in the policy, use the Access Control List type instead.

The default policy, ObjectAccessDefault, lets all users access all objects.

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ObjectAccess in the left pane.  
All the ObjectAccess policies are listed in the right pane.
3. Click New Access Policy icon  
 next to ObjectAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, leave ObjectAccess in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of application groups is displayed.
7. Select the application group that you created, and click  
  
The application group appears in the Selected Identities box to the right.
8. Enter AlertPolicy/\* in the Add Resource field in the Access Policy Configuration section, and click  
  
The resource is added to the resource list.

#### **NOTE**

To provide view access to a subset of the objects available in each of these categories, use the final asterisk (\*) as a wildcard, and insert an identifier. For example, to view only the alert policies whose names begin with ACC\* (the ACC application alert policies), enter AlertPolicy/ACC\*.

9. Select the Access check box under Actions.
10. Click Save at the top or bottom right of the page.  
The policy is created.
11. Perform a permission check as appropriate.

### **Create an ObjectControl Policy**

To assign a user specified levels of access to the various objects available, you can create an ObjectControl policy. For this policy, a type of either Access Policy or Access Control List is recommended. To permit the same access to all the resource types assigned in the policy, use the Access Policy type. This example is written based on that selection. To permit different types of access to the resource types assigned in the policy, use the Access Control List type instead.

The default policy, ObjectControlDefault, lets users in the Scheduler, Supervisor, and Commander (superuser) groups control all objects.

**Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ObjectControl in the left pane.  
All the ObjectControl policies are listed in the right pane.
3. Click New Access Policy icon  
  
next to ObjectControl in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, leave ObjectControl in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of application groups is displayed.
7. Select the application group that you created, and click  
  
The application group appears in the Selected Identities box to the right.
8. Enter \*/Forecast/\* in the Add Resource field in the Access Policy Configuration section, and click  
  
The resource is added to the resource list.
9. Enter \*/Job\* in the Add Resource field, and click  
  
The resource is added to the resource list.
10. Select the Create and Modify check boxes under Actions for your group.
11. Click Save at the top or bottom right of the page.  
The policy is created.
12. Perform a permission check as appropriate.

**Create a ReportTemplateAccess Policy**

To enable a user to access specified reporting templates based on a user role, you can create a ReportTemplateAccess policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection.

The default policy, ReportTemplateAccessDefault, allows access to all templates for all CA EEM user groups.

**Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ReportTemplateAccess in the left pane.  
All the ReportTemplateAccess policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to ReportTemplateAccess in the left pane.  
The New Access Policy page appears in the right pane.

4. Enter a unique policy name in the Name field, and leave ReportTemplateAccess selected in the Resource Class Name drop-down list.
5. Select the Identity Access Control List option button in the Type field.  
A confirmation prompt appears.
6. Click OK.  
The Identity Access Control List Configuration section appears.
7. Select Application Group from the Type drop-down list in the Identity Access Control List Configuration section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
8. Click Search.  
A list of application groups is displayed.
9. Select the application group that you created, and click .  
The application group appears in the Selected Identities list.
10. Select the following checkboxes under Actions for your group:
  - JobRunSummary
  - JobRunDetails
  - JobRunThroughput
  - JobRunEventStatus
11. Click Save at the top or bottom right of the page.  
Your new policy appears in the list.
12. Perform a permission check as appropriate.

### **Create a ServerAccess Policy**

To enable a user to access specified servers reflecting that user role, you can create a ServerAccess policy. You define each of the servers as a resource in the server/*serverName* format. For this policy, we recommend an Access Policy type. This example is written based on that selection.

#### **NOTE**

You can create policies to reflect the naming convention that you use when you define the server name for your servers in Web UI Configuration. For example, you can create a policy that gives a user access to only the servers with the identifier AE\*, as shown in the sample policy.

The default policy, ServerAccessDefault, lets all users access all servers.

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click ServerAccess in the left pane.  
All the ServerAccess policies are listed in the right pane.
3. Click the New Access Policy icon  next to ServerAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, leave ServerAccess in the Resource Class Name drop-down list, and leave the Access Policy option button selected in the Type field.
5. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
6. Click Search.



A list of application groups is displayed.

7. Select the application group that you created, and click



The application group appears in the Selected Identities box to the right.

8. Enter server/AEServer in the Add Resource field, and click



The resource is added to the resource list, providing access to the AEServer server.

9. Enter server/AESrvr\* in the Add Resource field, and click



#### NOTE

Creating a policy with this entry gives you access to all servers that start with the characters AESrvr. If you use a naming convention for your servers (for example, AESrvrACE, AESrvrAC2, AESrvrAC3), you can use a wildcard in the resource.

10. Select the Access check box under Actions in the Access Policy Configuration section.
11. Click Save at the top or bottom right of the page.  
The policy is created.
12. Perform a permission check as appropriate.

### Create a WidgetAccess Policy

To enable a user to access widgets reflecting that user role, you can create a WidgetAccess policy. For this policy, we recommend an Identity Access Control List type. This example is written based on that selection.

The default policy, WidgetAccessDefault, grants access to all widgets for all users and groups.

#### Follow these steps:

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
2. (Optional) Click WidgetAccess in the left pane.  
All the WidgetAccess policies are listed in the right pane.
3. Click the New Access Policy icon  
  
next to WidgetAccess in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter a unique policy name in the Name field, and leave WidgetAccess selected in the Resource Class Name drop-down list.
5. Select the Identity Access Control List option button in the Type field.  
A confirmation prompt appears.
6. Click OK.
7. The Identity Access Control List Configuration section appears.
8. Select Application Group from the Type drop-down list in the Identities section, and click Search Identities.  
The Attribute, Operator, and Value fields appear.
9. Click Search.  
A list of application groups is displayed.
10. Select the application group that you created, and click



The application group appears in the Selected Identities box to the right.

11. Select the following checkboxes under Actions for your group:

- MonitoringAlerts
  - MonitoringViews
12. Click Save at the top or bottom right of the page.  
The policy is created.
  13. Perform a permission check as appropriate.

### **Perform a Permission Check**

To check whether access permission has been properly assigned, you can perform a permission check.

#### **Follow these steps:**

1. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default.
2. Click the Permission Check link.  
The Permission Check page opens.
3. Click Synchronize Cache.  
A confirmation message appears.
4. Select the appropriate policy from the Resource Class drop-down list in the Permission Check Parameters section.
5. Select the action for the respective policy from the Action drop-down list.
6. Enter a value for the respective policy in the Resource field.
7. Enter the user ID you created in the Identity field.
8. Click Run Permission Check.  
The results of the permission check appear in the Permission Check Results section. If correct, this check returns a result of Allow.

<b>Action Drop-Down List Options</b>	
<b>Policy</b>	<b>Action</b>
AlertAction	Acknowledge
ApplicationAccess	Reporting
CommandExecute	Execute
CommandSetup	View
ConfigurationControl	Delete
ExternalApplicationAccess	View
LogAccess	JobLog
MonitorViewControl	View
ObjectAccess	Access
ObjectControl	Create
ObjectControl	Modify
ReportTemplateAccess	JobRunSummary
ServerAccess	Access
WidgetAccess	MonitoringViews

<b>Resource Field Options</b>	
<b>Policy</b>	<b>Resource</b>
AlertAction	view/viewName
ApplicationAccess	*

CommandExecute	server/*
CommandSetup	global/*
ConfigurationControl	Credentials/server/AEserver/admin/password/operator
ExternalApplicationAccess	externalApplication/ <i>eemResourceId</i>
LogAccess	server/SrvrAutoSys
MonitorViewControl	view/ <i>viewName</i> /JobFlow
ObjectAccess	AlertPolicy/*
ObjectControl	*/Forecast/*
ObjectControl	*/Override
ReportTemplateAccess	*
ServerAccess	server/SrvrAutoSys
WidgetAccess	*

### **Verify the Access Set by Your Policies**

After you have completed the policy creation and permission checks, log in to Web UI as a user in your application group. Verify that all policies that you added are working as designed.

## **CA EEM Identities for Web UI**

Web UI installs multiple default users. You can use the default users when you configure your Web UI environment initially.

CA EEM can be configured to reference more users from an external directory source, such as Active Directory, or you can add users to the native CA EEM repository. However, if you configure CA EEM to reference users from an external directory source, the default users cannot log in to Web UI unless you add them to that external directory source.

You can manage your identities using the Manage Identities tab of CA EEM.

### **Users**

The following CA EEM users are created by default during the Web UI installation:

- **EJADMIN**  
Provides system configuration access for Web UI. Typically, these users are system administrators or security administrators.  
**User name:** ejmadmin  
**Password:** ejmadmin
- **EJMEEXEC**  
Provides access to summary information on job status for Web UI. Typically, these users are executives and managers from other disciplines.  
**User name:** ejmexec  
**Password:** ejmexec
- **EJMOPERATOR**  
Provides a basic level of access for Web UI. Typically, these users are console operators.  
**User name:** ejmoperator  
**Password:** ejmoperator
- **EJMSCHEDULER**  
Provides a level of access that includes the ability to create and modify jobs and job objects. Typically, these users are job administrators.

**User name:** ejmscheduler

**Password:** ejmscheduler

- **EJMSUPERVISOR**

Provides a higher level of access than that granted to the scheduler. Typically, these users are scheduling or monitoring supervisors and are responsible for creating views.

**User name:** ejmsupervisor

**Password:** ejmsupervisor

- **EJMCOMMANDER**

Provides the highest level of access. These users have access to all in Web UI features. Typically, these users have a combination of the rights of the system administrator and the scheduling supervisor.

**User name:** ejmcommander

**Password:** ejmcommander

These users are set up in CA EEM, each with a default display in Web UI, letting them access the applications to which they have rights.

Typically, you would use the system administrator (ejmadmin) or the superuser (ejmcommander) to perform system configuration.

## **User Roles**

Several preconfigured user roles that are provided with Web UI are used in the default CA EEM policy definitions. These roles are defined as follows:

- **Console Operator**

The *console operator* is responsible for monitoring job streams, correcting minor errors that are resolved through forcing job status or through one-time overrides of data. The console operator also reports significant application errors to other operators and to the application owners. To complete these tasks, console operators use Monitoring to monitor the jobs for which they are responsible. Optionally, they may use Reporting to generate server statistics and job run activity reports. Console operators may also use Quick View to view the detailed properties for a particular job in the flow and Forecast to create reports that show jobs that started or that are scheduled to start during a specific time period.

- **Scheduler**

The *scheduler* is responsible for the creation and maintenance of the job definitions. The schedulers use Quick Edit and Application Editor to create, modify, and delete machines, jobs, calendars, and other job-related information such as constraints. Schedulers may also monitor job streams to make sure that they are running properly. Schedulers receive reports about problems with jobs that have already run, and they might need to review historical information to analyze and correct a job stream.

- **Supervisor**

The *supervisor* role performs all the tasks of a scheduler, and an additional level of tasks to help oversee and assist all the schedulers. The supervisor also performs other tasks such as defining job streams and monitoring views.

- **Administrator**

The *administrator* is responsible for configuring and maintaining the Web UI environment. This includes defining the scheduling manager servers, configuring software-specific properties, and defining host connection links for mainframe systems (if applicable). System administrators may also be responsible for security, and for creating users and user groups assigned to specific roles. Also, the administrator can create tabs in Web UI for external applications and instances.

- **Executive**

The *executive*, or *manager*, has a different needs compared to the other roles. The executive tasks include monitoring, at the highest level, the overall status of the scheduling managers operating in their area. They can use Reporting to generate and view reports. The executive also uses the Views widget to track the status of their views.

- **Security Administrator**

The *security administrator* is responsible for configuring and maintaining security, and creating users and user groups assigned to specific roles. The security administrator also creates and maintains policies for CA EEM.

- **Commander**

The *commander* is a superuser that is created as a convenience, and is a member of all the application groups.

The following table shows the application groups and default users for each role:

Role	Application Group	Default User
Console Operator	Console Operator	ejmoperator
Supervisor	Supervisor	ejmsupervisor
Scheduler	Scheduler	ejmscheduler
Administrator	Administrator	ejmadmin
Executive	Executive	ejmexec
Security Administrator	Security Administrator	ejmadmin and ejmcommander
Commander	Commander	ejmcommander

## Web UI Policies

Web UI policies define the access rights of a user or user group to a resource. They associate identities with resources. A default policy exists for each resource class. The default policy for each class is delivered with the Web UI application in CA EEM.

### Application Security Table

By default, all user roles are granted access to the Main page and the applications available from the Main page (as tabs or from links).

The ApplicationAccess policy controls access to each Web UI application, including the Main page. Each tab on the Main page can be hidden or displayed based on the authorizations you set up in this policy.

The ExternalApplicationAccess policy controls access to each external application. Each configured external application tab on the Main page can be hidden or displayed based on the authorizations you set up in this policy.

The ServerAccess policy controls access to the back-end server instances. Using this policy, you can restrict user roles to specific instances of the scheduling manager. By default, members of the Administrator, Commander, ConsoleOperator, Executive, Scheduler, SecurityAdministrator, and Supervisor roles have access to all configured servers.

This table lists all available Web UI applications and the CA EEM policies associated with each one:

Web UI Application	CA EEM Policy	Usage
<b>Agent Inventory</b> Used to view and export information about each separately installable component (system agent, plug-ins, and advanced integrations).	ApplicationAccess	Lets you limit access to users that require access to Agent Inventory. By default, members of the Administrator and Commander roles are granted access to Agent Inventory.
<b>Application Editor</b> Used to manage job flows and dependencies using a graphical, drag-and-drop user interface.	ApplicationAccess	Lets you limit access to users that require access to Application Editor. By default, members of the Commander, Scheduler, and Supervisor roles are granted access to Application Editor.

	ObjectControl	Specifies the server and class of objects to control. Individual objects are not controlled. You can specify jobType (Command, J2EE) or objectType (global variable).   By default, members of the Commander, Supervisor, and Scheduler roles have create, modify, and delete control for all Application Editor object classes.
<b>Configuration</b> Used to configure Web UI and to perform administrative tasks related to job management and scheduling.	ApplicationAccess	Lets you limit access to users that are responsible for configuring your system.   By default, members of the Administrator and Commander roles are granted access to Configuration.
<b>Credentials</b> Used to let users add, modify, and delete their credentials for back-end servers in real time.	ApplicationAccess	Lets you limit access to users that are responsible for setting credentials for back-end servers.   By default, members of the Administrator, Commander, ConsoleOperator, Executive, Scheduler, and Supervisor roles are granted access to Credentials.
	ConfigurationControl - Credentials (Admin)	Lets you create, modify, and delete credentials for global users.   By default, members of the Commander or Administrator roles are granted access.
	ConfigurationControl - Credentials (User)	Lets you set access to allow users to update their own credentials.   By default, members of the Administrator, Commander, ConsoleOperator, Executive, Scheduler, and Supervisor roles are granted access to create, delete, and save their personal credentials.
<b>ECL (Enterprise Command Line)</b> Used to run JIL commands from the Windows command line or UNIX console.	ApplicationAccess	Lets you limit access to users that require access to the Enterprise Command Line.   By default, members of the Administrator, Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted access to the Enterprise Command Line.
	CommandExecute	Lets you restrict the commands that can be executed and who can perform those commands on a specific AutoSys Workload Automation server. Using this policy, you could grant a user group access only to canceling jobs on the server.   By default, members of the Administrator, Commander, ConsoleOperator, Scheduler, and Supervisor roles have execute access to all configured servers.

	CommandSetup	Specifies whether the user can view, create, modify, or delete commands based on command names. Using this policy, you could limit users to defining and viewing commands in their own My Commands list, or could let them view and execute commands in the global list, but not modify them.   By default, members of the Commander and Supervisor roles have access to create, modify, delete, and view commands; members of the Administrator, ConsoleOperator, and Scheduler roles have view access only.
<b>Launcher</b>	ApplicationAccess	Specifies which users or user roles can log in to Web UI. By default, all user roles can log in to Web UI.
<b>Monitoring</b> Used to create, manage, and monitor job flow views and alerts.	ApplicationAccess	Lets you limit access to users responsible for monitoring jobs and job flows.   <b>Notes:</b>   You must have access to Monitoring Views for it to be displayed on the Main page.   You must have access to Quick View or Quick Edit to view the job details or to edit the properties of a job.   By default, members of the Commander, ConsoleOperator, Scheduler, Executive, and Supervisor roles are granted access to Monitoring.
	AlertAction	Specifies the actions that can be performed on an alarm or alert. Using this policy, you can grant rights to acknowledge and close alerts for a specified view and server.   By default, members of the Commander, ConsoleOperator, and Supervisor roles are granted acknowledge, close, and open access to all views and configured servers.
	LogAccess	Specifies the logs that are accessible, by server. Using this policy, you could grant access to the scheduler log (SchedulerLog) on a particular server.   By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted access to the SchedulerLog and the JobLog on all servers.

	MonitorViewControl	Controls the actions that can be performed on a view and the views to which users have access. Using this policy, you could grant specific users create, modify, delete, and view access to all views with names that begin with a certain naming convention, such as Payroll.   By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted create, modify, delete, and view actions for all monitoring views.   <b>Note:</b> Because members of the Executive role are not granted ApplicationAccess to Monitoring, only view access is available.
	ObjectAccess	Specifies the object types the user can access. Using this policy, you could give a user access to specific alert policies by name.   By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles can access all alert policies and views.
	ObjectControl	Controls create, delete, or modify access to a class of objects. Individual objects are not controlled. You can specify AlertPolicy.   By default, members of the Commander, Supervisor, and Scheduler roles have create, modify, and delete control for all objects.
	ConfigurationControl - Event	Specifies whether you have access to execute the events and job actions that can be issued for a job on a back-end server. Using this policy, you could assign one user group access to the rights to send and cancel events on a specified server, while assigning another group access to send events on a different server.   By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles have access to all job actions and events for all servers.
	ConfigurationControl - Monitoring (Admin)	Controls user permission to delete the monitoring cache for a server.   By default, members of the Commander and Administrator roles are granted access.
	ConfigurationControl - Monitoring (User)	Controls whether a user has access to change the layouts in the Flow section in Monitoring.   By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor are granted access.



	WidgetAccess	Specifies the widgets that are visible on the Dashboard tab to each user. Default access settings depend on the access settings for widget parent applications.
<b>Quick Edit</b> Used to create and manage all of your workload objects, including jobs and calendars.	ApplicationAccess	Lets you limit access to users that are responsible for creating, modifying, or deleting workload objects. By default, members of the Commander, Scheduler, and Supervisor roles are granted access to Quick Edit.
	ObjectControl	Specifies the server and class of objects to control. Individual objects are not controlled. You can specify jobType (Command, J2EE) or objectType (global variable, calendar, extended calendar, machine, or cycle). By default, members of the Commander, Supervisor, and Scheduler roles have create, modify, and delete control for all Quick Edit object classes.
	ObjectControlOverride	Controls create and delete access to job overrides based on server name. If the override actions are not allowed, the icons and buttons to create or delete overrides are not rendered. By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles have create and delete override access on all back-end servers.
<b>Quick View</b> Used to access consolidated job detail information for AutoSys Workload Automation servers.	ApplicationAccess	Lets you limit access to users that are responsible for monitoring jobs. <b>Note:</b> You must have access to Quick Edit to edit the properties of a job. By default, members of the Commander, ConsoleOperator, Executive, Scheduler, and Supervisor roles are granted access to Quick View.
	ConfigurationControl - Event	Specifies whether you have access to execute the events and job actions that can be issued for a job on a back-end server. Using this policy, you could assign one user group access to the rights to send and cancel events on a specified server, while assigning another group access to send events on a different server. By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles have access to all job actions and events for all servers.

	LogAccess	Specifies the logs that are accessible, by server. Using this policy, you could grant access to the job run log (JobLog) on a particular server. By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted access to the JobLog on all servers.
<b>Forecast</b> Used to forecast job completion on a AutoSys Workload Automation server and to generate reports.	ApplicationAccess	Lets you limit access to users that require access to Forecast. By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted access to Forecast.
	ObjectAccess	Lets you limit access to users that require access to Forecast reports. By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles are granted access to Forecast reports.
	ObjectControl	Specifies whether the user can create, modify, or delete Forecast reports. By default, members of the Commander, Supervisor, and Scheduler roles have create, modify, and delete control for all Forecast reports.
<b>Reporting</b> Used to generate reports on demand on server statistics and job run activity for AutoSys Workload Automation servers.	ApplicationAccess	Lets you limit access to users that are responsible for running and viewing job run and server statistical reports. By default, members of the Commander, ConsoleOperator, Scheduler, Executive, and Supervisor roles are granted access to Reporting.
	ReportTemplateAccess	Lets you limit access to specific Reporting templates to only those users with access to the Reporting application. By default, members of all user roles are granted access to all report templates.
<b>Resources</b> Used to create and manage AutoSys Workload Automation virtual resources, run resource reports, and view job resource usage.	ApplicationAccess	Lets you limit access to users that are responsible for monitoring resources. You must have access to Quick View or Quick Edit to view the job details or to edit the properties of a job. By default, members of the Commander, Scheduler, and Supervisor roles are granted access to Resources.

	ConfigurationControl - Event	Specifies whether you have access to execute the events and job actions that can be issued for a job on a back-end server. Using this policy, you could assign one user group access to the rights to send and cancel events on a specified server, while assigning another group access to send events on a different server.  By default, members of the Commander, ConsoleOperator, Scheduler, and Supervisor roles have access to all job actions and events for all servers.
<b>External Applications</b>	ExternalApplicationAccess	Lets you limit access to users that require access to each external application  By default, members of all roles have access to all configured external applications.
	ConfigurationControl	Specifies whether the user can add, modify, or delete external applications.  By default, only members of the Administrator roles have create, modify, and delete control for all external applications.
<b>Dashboard</b>	ApplicationAccess	Lets you limit access to users that require access to the Dashboard.  Default access settings depend on the access settings for widget parent applications.
	WidgetAccess	Specifies which widgets are visible on the dashboard to each user.  Default access settings depend on the access settings for widget parent applications
	ConfigurationControl	Specifies whether the user can add or remove Dashboard widgets and Dashboard tabs.  By default, members of all roles can add or remove Dashboard widgets and Dashboard tabs.

## Web UI Resource Classes

Resource classes are used in CA EEM to classify logical or physical entities. A resource class contains the following information:

- Name
- Actions (List)
- Named Attributes (List)

The following resource classes are available for the Web UI application. The classes can be categorized into the following types:

- **Access Resource Classes:** The ApplicationAccess and ExternalApplicationAccess classes control access to applications. The WidgetAccess class controls access to widgets. The ServerAccess class controls access to servers.

The ConfigurationControl class lets you configure additional levels of access to certain application features, such as those in Credentials.

- **Action or Log Resource Classes:** The AlertAction, LogAccess, CommandSetup, and CommandExecute classes control access to alert actions, job and scheduler logs, and Enterprise Command Line commands respectively.
- **Object Access Resource Classes:** The MonitorViewControl, ReportTemplateAccess, ObjectAccess, and ObjectControl classes are used to control specific types of access to a variety of objects that are displayed and maintained by Web UI. Each of the actions that can be performed on these objects is enumerated in the resource actions. The name of the object is specified for the resource name. There are typically attributes specified for the server type and server name. These resource classes are only implemented in server-side code.

### **Alert or Alarm Actions**

The AlertAction resource identifies the actions that can be performed on an alarm or an alert in Monitoring. Alarms and alerts are considered equivalent with respect to actions.

**Resource Name Value:** Specifies a view or a server. The resource name value is structured as one of the following: view/viewName or server/serverName

- viewName -- the name of the view as defined in Monitoring
- serverName -- the name of the server as defined in Web UI Configuration

#### **Defaults:**

view/\*  
server/\*

#### **Examples:**

server/autosys1  
view/myView

**Actions:** The following actions are available:

- Acknowledge
- Close
- Open

#### **Attributes:**

```
component -- the name of the component issuing the authorization check
nameType -- View or Server
name -- viewName or serverName
```

- viewName -- the name of the view as defined in Monitoring
- serverName -- the name of the server as defined in Web UI Configuration

### **Application Access**

The ApplicationAccess resource controls access to the applications that operate in the Web UI framework. Most of these applications are available as tabs from the main page of Web UI; however, CA EEM is opened in a separate browser window when you click the EEM link in the top right corner of the main page. If the user does not have access to an application, that application or the link to that application is not displayed. Each application is represented by an action in the resource.

#### **NOTE**

- In order to view and/or edit job properties from Monitoring and Resources, you must also have application access to Quick View and Quick Edit respectively.
- To edit job properties from Quick View, you must also have application access to Quick Edit.

**Resource Name Value:** Specifies an application category and, if there are or can be multiple applications in that category, may additionally indicate a particular application. The resource name value is structured as follows: category/name

category -- the category for a set of applications  
name -- a specific application within the application category

**Example:** Configuration/Security

**Actions:** The following actions are available:

**Table 1: Application Access Actions**

Action Code	Default Resource Name Mapping
Configuration	Configuration
SecurityConfiguration	Configuration/Security
QuickView	QuickView
CommandLine	CommandLine
QuickEdit	QuickEdit
ApplicationEditor	ApplicationEditor
Resources	Resources
Monitoring	Monitoring
Forecast	Forecast
Credentials	Credentials
Reporting	Reporting
Dashboard	Dashboard
Launcher	Launcher
AgentInventory	AgentInventory

### Attributes

component -- the name of the component issuing the authorization check  
category -- the category for a set of applications  
name -- a specific application within the application category

### Command Execution

The CommandExecute resource identifies the command to be executed on a given back-end server. The Enterprise Command Line View performs an authorization check prior to executing the requested command. A policy can be written to control who can perform a specified command on a specified server.

**Resource Name Value:** Specifies the command to execute. The resource name value is structured as follows: server/serverName/command

serverName -- the name of the server as defined in CA WCC Configuration  
command -- command to be executed on the given server

**Default:** server/\*

**Example:** server/AEserver/cmd-spec

**Actions:** The following action is available:

Execute

### Attributes

**component** -- the name of the component issuing the authorization check  
**serverName** -- the name of the server as defined in CA WCC Configuration  
**command** -- command to be executed on the given server

#### NOTE

The command value is the actual command with all the parameters. It might be somewhat difficult to parse the command and its parameters using CA EEM expression processing; however, it is usually easy to isolate the first word of the command, so it is possible to identify the name of the command being executed.

### AgentAccess

The AgentAccess resource controls access to the various resource classes in Agent Inventory.

#### NOTE

User must have **execute** permission for the **as-owner** resource class and the following specific permissions for the respective actions.

Actions	Permissions
Download Logs	<b>read</b> permission for the <b>as-machine</b> resource class
Download AgentParms	<b>read</b> permission for the <b>as-machine</b> resource class
View Details	<b>read</b> permission for the <b>as-machine</b> resource class
Stop	<b>execute</b> permission for the <b>as-machine</b> resource class <b>mach-offline</b> permission for the <b>as-sendevent</b> resource class
Restart	<b>execute</b> permission for the <b>as-machine</b> resource class <b>mach-offline</b> permission for the <b>as-sendevent</b> resource class <b>mach-online</b> permission for the <b>as-sendevent</b> resource class

### Command Setup

The CommandSetup resource identifies the actions that can be performed on an entry in the specified command lists of the Enterprise Command Line. It controls what the Enterprise Command Line allows during the specification, saving, and listing of commands.

This resource is used together with the CommandExecute resource to control what a user can do with the Enterprise Command Line. The CommandExecute resource makes a call to CA EEM for each command being executed. In addition, the Command Line back-end agent checks that command against the list of commands allowed to be run.

**Resource Name Value** Specifies the command to use. The resource name value is structured as follows: mode/cmdName

**mode** -- the mode to use. (Local, Global)

**cmdName** -- the name of the command specification in the local or global command list

#### Defaults:

local/\*  
 global/\*

#### Examples:

local/ListMyJobs  
 global/CancelJob

**Actions:** The following actions are available: Create, Modify, Delete, View

#### Attributes:

**component** -- the name of the component issuing the authorization check  
**mode** -- local or global

`cmdName` -- the name of the command specification in the local or global command list

#### Notes:

`local/*` limits users to defining and viewing commands in their own My Commands list.

`global/*` + Action of View lets users view and execute commands in the global list, but not modify them.

### Configuration Control

The ConfigurationControl resource controls the configuration of various applications and functionalities within Web UI, including Credentials, Monitoring, and Event. This security resource class is different than other resource classes because it addresses various disparate resources. This is done to minimize the number of classes that must be defined to protect various Web UI resources. For that reason, the description of the class is separated into sections based on usage.

**Resource Name Value:** Specifies a particular application to be configured. The resource name value is structured as follows: `name/options`.

`name` -- one of the applications that makes use of this resource class; currently, this is "Credentials", "Event", or "Monitoring" to match the names specified for these applications in the action codes for the ApplicationAccess policy, "ExternalApplication" for external applications, or "Widget" for widgets

`options` -- application-specific configuration options

**Actions:** The actions available vary depending on the specified application. For more information about actions, see the actions list for the appropriate application.

#### Attributes

`component` -- the name of the component issuing the authorization check

`name` -- the ApplicationAccess action code for a specific application, "ExternalApplication" for external applications, or "Widget" for widgets

`options` -- everything after the name section in the resource name, which is unique for each application utilizing the ConfigurationControl resource class

**Credentials:** For the Credentials application, this class controls the setting of credentials, as follows.

Use credential - An authorization request is made for each use of a credential.

Edit credential - An authorization request is made for each create/modify/delete of a credential.

**Resource Name Value:** Specifies the Credentials application and the options to configure for it. The resource name value is structured as follows: `Credentials/server/serverName/userid/credentialtype/creduserid`

`serverName` -- the server name in a Credentials policy definition

`userid` -- the user ID of the user requesting the credential use or modification

`credentialtype` -- the credential type in a Credentials policy definition; currently, only "password" is supported but additional types are planned for future releases

`creduserid` -- the credential user ID in a Credentials policy definition

**Default:** `Credentials/server/*`

**Examples:** `Credentials/server/ServerACE/brifr/password/brifr` `Credentials/server/AEserver/admin/password/opuser`

**Actions:** The actions available vary depending on the specified option. The following actions are available.

Create -- specified when a credential is created; this lets you control who can create a

credential.

Save -- specified when a credential has been created or modified and a new value is being saved for it; this lets you control who can save a credential

Delete -- specifi

**Monitoring:** For the Monitoring application, this class controls Monitoring settings, as follows.

Delete cache - controls who can drop all jobs in the database cache and rebuild the cache for the specified server

Note: This action is accessible from the Monitoring Batch Interface only.

Use flow layout - controls who can access the flow layouts

View job dependencies - controls who can view job dependencies for a specified view

**Resource Name Value:** Specifies the Monitoring application and the options to configure for it. The resource name value is structured as follows.

Monitoring/server/serverName/cache

Monitoring/view/viewName/JobDeps

serverName -- the name of the server as defined in CA WCC Configuration

viewName -- the name of the view as defined in CA WCC Monitoring

**Defaults:**

Monitoring/server/./cache

Monitoring/view/./JobDeps

Monitoring/flow/layout

**Examples:**

Monitoring/server/SrvrACE/cache

Monitoring/server/AEserver/cache

Monitoring/view/MyView/JobDeps

**Actions:** The actions available vary depending on the specified option. The following actions are available:

Delete -- specified when the job database cache is being deleted in the Monitoring Batch Interface; this lets you control who can delete the cache for a specified server

Access -- specified when a job dependency or a flow layout is accessed; this lets you control who can view job dependencies change the flow layout

**Event:** For the Monitoring, Quick View, and Resources applications, this class controls access to sending events.

**Resource Name Value:** Specifies the application and the options to configure for it. The resource name value is structured as follows: applicationName/Event

applicationName -- the name of the application

**Default:** \*/Event

**Examples>:**

Monitoring/Event QuickView/Event

**Actions:** The actions available vary depending on the specified option. The following option is available:

Access -- specified when a job action, send event, or send server event is accessed; this lets you control who can send or cancel events for a job or server

**Widget:** For widgets, this class controls who can add and remove widgets from the Dashboard tab.

**Resource Name Value:** Specifies the widget and the options to configure for it. The resource name value is structured as follows: Widget/widgetName



widgetName -- the name of the widget

**Default:** Widget/\*

**Examples:**

```
Widget/MonitoringViews
Widget/MonitoringStatus
Widget/ConnectionStatus
Widget/MonitoringAlerts
Widget/MonitoringJobs
Widget/MonitoringMachines
Widget/UsersStatus
```

**NOTE**

The name of the widget is the action code name in the WidgetAccess resource class.

**Actions:** The actions available vary depending on the specified option. The following options are available:

```
Create -- lets you control who can add widgets to the Dashboard
Delete -- lets you control who can remove widgets from the Dashboard
```

**Dashboard**

For Dashboard, this class controls who has access to create and manage multiple dashboard tabs in Web UI.

- **Resource Name Value**  
Specifies the general resource available for this feature. The only value is:  
DashboardManagement/\*
- **Actions**  
The actions available vary depending on the specified option. The following option is available:
  - Access — lets you control who has access to create and manage multiple dashboard tabs

**External Application Access**

The ExternalApplicationAccess resource controls access to the external applications and instances added as tabs to Web UI.

**Resource Name Value:** Specifies a particular application. The resource name value is structured as follows: name/eemResourceId

```
name -- type of application. Currently one type is supported: externalApplication
eemResourceId -- the unique EEM resource ID, assigned to the application in CA WCC
when the tab is created
```

**NOTE**

For information about assigning EEM resource IDs, see Dashboard Help.

**Examples:** externalApplication/casupport

**Actions:** The following action is available: View.

**Attributes:**

```
component -- the name of the component issuing the authorization check
type -- the external application type
name -- a specific application
```

**Log Access**

The LogAccess resource identifies the logs that are accessible, by server, in Monitoring and Quick View.

- **Resource Name Value**

Specifies a server. The resource name value is structured as follows:

`server/serverName`

– *serverName* -- the name of the server as defined in Web UI Configuration

**Default:** `server/*`

**Examples:**

`server/serverAE`

`server/SrvrACE`

- **Actions**

The following action is available for Monitoring:

– SchedulerLog

The following action is available for Quick View:

– JobLog

- **Attributes**

`component` -- the name of the component issuing the authorization check

`serverName` -- the name of the server as defined in Web UI Configuration

### **Monitor View Control**

The MonitorViewControl resource controls the actions that can be performed on a particular monitoring view in Monitoring.

- **Resource Name Value**

Specifies the view to control. The resource name value is structured as follows:

`view/viewName/JobFlow`

– *viewName* -- the name of the view as defined in Web UI Monitoring

**Default:** `*/JobFlow`

**Examples:**

`view/myview/JobFlow`

`view/FailedJobs/JobFlow`

- **Actions**

The following actions are available:

– Create

– Modify

– Delete

– View

- **Attributes**

`component` -- the name of the component issuing the authorization check

`viewName` -- the name of the view as defined in Monitoring

### **Object Access**

The ObjectAccess resource controls access to a particular object. Policies that are written against this class can control which objects a user can view in Forecast and Monitoring.

- **Resource Name Value**

Specifies the server and object to control. The resource name value is structured as follows:

`server/serverName/objectType/objectName`

– *serverName* -- the name of the server as defined in Web UI Configuration

– *objectType* -- the type of object to control:

- AlertPolicy

- Forecast

– *objectName* -- the name of the object

**Defaults:**

The following resource is used for Monitoring:

- AlertPolicy/\*

The following resource is used for Forecast:

- \*/Forecast/\*

**Examples:**

AlertPolicy/FailedJobs

server/SrvrACE/Forecast/myReport

- **Actions**

The following action is available for Forecast and Monitoring:

- Access

- **Attributes**

component -- the name of the component issuing the authorization check

serverName -- the name of the server as defined in Web UI Configuration

objectType -- AlertPolicy or Forecast

objectName -- the name of the object

## **Object Control**

The ObjectControl resource controls the actions that can be performed on a particular class of objects. Individual objects are not controlled. For example, a user must be authorized to create a job. The ObjectControl policy is used by Application Editor, Monitoring, Quick Edit, and Forecast.

### **NOTE**

The ObjectControlOverrideDefault policy controls whether you have authorization to create or delete job overrides for a job on a particular server. The ObjectControlOverrideDefault policy is applicable to Quick Edit only.

- **Resource Name Value**

Specifies the server and the class of objects to control. The resource name value is structured as one of the following:

*objectType*

server/*serverName/objectType*

server/*serverName/Job/jobType*

server/*serverName/Forecast/reportNameserver/serverName/ConnectionProfile/connectionType*

- *serverName* -- the name of the server as defined in Web UI Configuration

- *objectType* -- the type of object to control:

- AlertPolicy
- Calendar
- Cycle
- ExtendedCalendar
- GlobalVariable
- Job
- Machine
- Forecast
- ConnectionProfile
- Override

- *jobType* -- the job type for a Job object:

- 
- Box
  - Command
  - CPUMonitoring
  - DatabaseMonitor
  - DatabaseStoredProcedure
  - DatabaseTrigger
  - DiskMonitoring
  - EntityBean
  - FileTrigger
  - FileWatcher
  - FTP
  - HDFS
  - Hive
  - HTTP
  - iOS
  - Informatica
  - IPMonitoring
  - JMSPublish
  - JMSSubscribe
  - JMX-MBeanAttributeGet
  - JMX-MBeanAttributeSet
  - JMX-MBeanCreateInstance
  - JMX-MBeanOperation
  - JMX-MBeanRemoveInstance
  - JMX-MBeanSubscribe
  - MicroFocus
  - MSSQLServer
  - Oozie
  - OracleEBusinessSuiteCopy
  - OracleEBusinessSuiteRequestSet
  - OracleEBusinessSuiteSingleRequest
  - PeopleSoft
  - Pig
  - POJO
  - ProcessAutomationExecution
  - ProcessAutomationStartRequestForm
  - ProcessMonitoring
  - Proxy
  - RMI
  - SAPBatchInputSession
  - SAPBWInfoPackage
  - SAPBWProcessChain
  - SAPDataArchiving
  - SAPEventMonitor
  - SAPJobCopy
  - SAPProcessMonitor
  - SAPR3
  - SecureCopy
  - SessionBean
  - SNMPValueGet
  - SNMPValueSet
  - SQL
  - Sqoop
  - TextFileReadingAndMonitoring
-

**NOTE**

Hadoop is the only available connection type.

**Defaults:**

The following resources are used for Quick Edit:

- \*/GlobalVariable
- \*/Job\*
- \*/Calendar
- \*/ExtendedCalendar
- \*/Machine
- \*/Cycle
- \*/ConnectionProfile/Hadoop
- \*/Override

The following resource is used for Application Editor:

- \*/Job\*

The following resource is used for Monitoring:

- AlertPolicy

The following resource is used for Forecast:

- \*/Forecast/\*

**Examples:**

```
server/AEserver/GlobalVariable
server/AEserver/Calendar
server/AEserver/Job/Command
server/AEserver/Job/JMX-MBeanSubscribe
server/AEserver/Job/UserDefined2
server/SrvrACE/Forecast/Week1
server/AEserver/ConnectionProfile/Hadoop
server/AEserver/Override
server/AEserver/Machine
```

- **Actions**

The following actions are available for Quick Edit, Application Editor, Monitoring, and Forecast:

- Create
- Delete
- Modify

**NOTE**

The modify action cannot be used for the Override resource in Quick Edit; only Create and Delete apply.

- **Attributes**

component -- the name of the component issuing the authorization check  
 serverName -- the name of the server as defined in Web UI Configuration  
 reportName -- the name of the Forecast report  
 connectionType -- the type of connection profile  
 objectType -- AlertPolicy, Calendar, Cycle, ExtendedCalendar, GlobalVariable, Forecast, ConnectionProfile, Machine, or Job  
 jobType -- Box, Command, FileWatcher, FileTrigger, FTP, iOS, SecureCopy, zOSRegular, zOSDataSetTrigger, zOSManual, CPUMonitoring, DiskMonitoring, IPMonitoring, ProcessMonitoring, TextFileReadingAndMonitoring, WindowsEventLogMonitoring, WindowsServiceMonitoring, OracleEBusinessSuiteCopy, OracleEBusinessSuiteSingleRequest, OracleEBusinessSuiteRequestSet, PeopleSoft, SAPBatchInputSession, SAPBWInfoPackage, SAPBWProcessChain, SAPDataArchiving, SAPEventMonitor, SAPJobCopy, SAPProcessMonitor, SAPR3, DatabaseMonitor, DatabaseStoredProcedure, DatabaseTrigger, SQL, EntityBean, HTTP, JMSPublish, JMSSubscribe, JMX-MBeanAttributeGet, JMX-MBeanAttributeSet, JMX-MBeanCreateInstance, JMX-

MBeanOperation, JMX-MBeanRemoveInstance, JMX-MBeanSubscribe, POJO, RMI, SessionBean, WebService, WebServicesDocumentLiteral, SNMPValueGet, SNMPValueSet, MicroFocus, ProcessAutomationExecution, ProcessAutomationStartRequestForm, Proxy, WakeOnLAN, HDFS, Oozie, Hive, Sqoop, Pig, UserDefined0, UserDefined1, UserDefined2, UserDefined3, UserDefined4, UserDefined5, UserDefined6, UserDefined7, UserDefined8, UserDefined9

## **Report Template Access**

The ReportTemplateAccess resource controls user access to the reports that can be generated using the Reporting application. If the user does not have access to a specific report template, then that template is not displayed. Each template is represented by an action in the resource.

### **NOTE**

The ReportTemplateAccess resource class has no defined resources or attributes associated with it. Only the resource class and its actions are defined.

- **Actions**

The following actions are available:

- JobRunSummary
- JobRunDetails
- JobRunThroughput
- JobRunEventSummary
- JobRunEventStatus
- JobStatusSuccessFailure
- JobStatusStartingRunning
- JobStatusInactiveTerminated
- JobRunFailureTotal
- JobRunForceStartRestart
- ServerStatisticsByDay
- JobStatistics
- JobRunEventStatistics

## **Server Access**

The ServerAccess resource identifies the scheduling manager servers that a user can access. Each Web UI component will only allow access to servers, if applicable, that the current user is permitted to access.

### **NOTE**

A view can be edited only by a user who has access to all servers defined in the view.

- **Resource Name Value**

Corresponds to the name specified for the server in Web UI Configuration. The resource name value is structured as follows:

*server/serverName*

- *serverName* -- the name of the server as defined in Web UI Configuration

**Default:** *server/\**

**Examples:**

*server/myAEServer*

*server/AEServer*

- **Actions**

The following action is available:

- Access

- **Attributes**

component -- the name of the component issuing the authorization check  
 type -- AutoSys Workload Automation  
 serverName -- the name of the server as defined in Web UI Configuration

## Widget Access

The WidgetAccess resource controls access to the Web UI application widgets. The widgets are displayed on the Dashboard tab of Web UI. If the user does not have access to a widget, that widget is not displayed. Each widget is represented by an action in the resource.

- **Resource Name Value**

Specifies a widget category and, if there are or can be multiple widgets in that category, may additionally indicate a particular widget. The resource name value is structured as follows:

*category/name*

- *category* -- the category for a set of widgets
- *name* -- a specific widget within the widget category

**Examples:**

Monitoring/Views  
 Monitoring/Servers

- **Actions**

The following actions are available:

	Action Code	Default Resource Name Mapping
	ConnectionStatus	ConnectionStatus
	MonitoringAlerts	Monitoring/Alerts
	MonitoringServers	Monitoring/Servers
	MonitoringViews	Monitoring/Views
	MonitoringJobs	Monitoring/Jobs
	MonitoringMachines	Monitoring/Machines
	UsersStatus	UsersStatus

- **Attributes**

component -- the name of the component issuing the authorization check  
 category -- the category for a set of widgets  
 name -- a specific widget within the widget category

## FIPS Mode

The Federal Information Processing Standards (FIPS) publications set guidelines for best practices for software and hardware computer security products. In FIPS mode, AutoSys Workload Automation complies with the standards in FIPS 140-2: Security Requirements for Cryptographic Modules.

DES encryption and unencrypted communication violate the FIPS 140-2 guidelines. To ensure compliance with FIPS standards, AutoSys Workload Automation instances running in FIPS mode block the following communication:

- Messages sent to 4.5.1 of type 'l','L','r', or 'n'.
- Messages sent to or received from external AutoSys Workload Automation (xtype: a) and CA Workload Automation ESP Edition (xtype: e) instances with no encryption key

**NOTE**

The xcrypt\_type attribute in the instance definition specifies whether the instance has an encryption key.

- Non-encrypted messages from clients that are sent to the application server

- Messages sent to Release 11.3, Release 11.3.5, or Release 11.3.6 agents of type 'a' with no encryption key.

#### NOTE

The encryption\_type attribute in the machine definition specifies whether the agent has an encryption key.

## Disable FIPS Mode for Web Services

To disable FIPS mode for web services and switch to non-FIPS mode, modify the eiam.ws.config file.

#### Follow these steps:

1. Stop the AutoSys Workload Automation web service.
2. Modify and save the \$AUTOUSER/eiam.ws.config file (UNIX) or %AUTOUSER%\eiam.ws.config file (Windows) as follows:

```
<SDK type="Java">
<ITechSDK>
 <!-- possible values are OFF/ON. This tag is used to override the default
initialization. By default it is set to OFF -->
 <FIPSMODE>OFF</FIPSMODE>
```

3. Save the eiam.ws.config file.
4. Start the AutoSys Workload Automation web service.

## Enable FIPS Mode for an Agent Only Installation on UNIX

AutoSys Workload Automation uses FIPS-approved and non-FIPS approved security algorithms unless you enable FIPS mode. When AutoSys Workload Automation operates in FIPS mode, the system only uses security algorithms that comply with the standards in the FIPS 140-2 publication.

#### NOTE

- This procedure applies primarily to U.S. government agencies required to comply with FIPS guidelines.
- This procedure applies to Agent-only installations.
- This procedure enables FIPS mode for a particular AutoSys Workload Automation instance and the common components associated with that instance (CA Secure Sockets Adapter (SSA) Broker). Running all AutoSys Workload Automation instances, CA common components, and other Broadcom products that are installed on your system in FIPS mode ensures that your system is fully compliant with the guidelines in the FIPS 140-2 publication.

#### Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvctr stop waae_agent-WA_AGENT
```

The agent stops.

3. Enter the following command at the operating system prompt:

```
$CSAM_SOCKETADAPTER/bin/csampmux stop
```

The CA SSA Broker service stops.

4. Open the file specified by the oscomponent.environment.variable parameter in the agentparm.txt file and append the following environment variable and value to the file name:

```
CA_FIPS1402_ENABLE=1
```



**NOTE**

To disable FIPS mode, reset the value of this parameter to default:

```
CA_FIPS1402_ENABLE=0
```

**NOTE**

You define the file that is specified by the `oscomponent.environment.variable` parameter when you install AutoSys Workload Automation. For more information about the `oscomponent.environment.variable` parameter, see [Workload Automation Agents](#).

5. Open the operating system command prompt that is sourced to use AutoSys Workload Automation.
6. Enter the following commands at the operating system prompt:

```
unisrvctr start waee_agent-WA_AGENT
```

The agent starts. The agent automatically restarts the CA SSA Broker service. AutoSys Workload Automation runs in FIPS mode.

## Enable FIPS Mode for an Agent Only Installation on Windows

AutoSys Workload Automation uses FIPS-approved and non-FIPS approved security algorithms unless you enable FIPS mode. When AutoSys Workload Automation operates in FIPS mode, the system only uses security algorithms that comply with the standards in the FIPS 140-2 publication.

**NOTE**

- This procedure applies primarily to U.S. government agencies required to comply with FIPS guidelines.
- This procedure applies to agent-only installations.
- This procedure enables FIPS mode for a particular AutoSys Workload Automation instance and the common components associated with that instance (CA Secure Sockets Adapter (SSA) Broker). Running all AutoSys Workload Automation instances, CA common components, and other Broadcom products that are installed on your system in FIPS mode ensures that your system is fully compliant with the guidelines in the FIPS 140-2 publication.
- The Windows Service Control Manager displays the name of the CA SSA Broker service as "CA Connection Broker."

**Follow these steps:**

1. Open the Windows Service Control Manager and stop the following services:
  - Workload Automation Agent (*WA\_AGENT*)
  - CA Connection Broker
 The agent and the CA SSA Broker service stop.
2. Open the file specified by the `oscomponent.environment.variable` parameter in the `agentparm.txt` file and append the following environment variable and value to the file name:

```
CA_FIPS1402_ENABLE=1
```

**NOTE**

To disable FIPS mode, reset the value of this parameter to default:

```
CA_FIPS1402_ENABLE=0
```

**NOTE**

You define the file that is specified by the `oscomponent.environment.variable` parameter when you install AutoSys Workload Automation. For more information about the `oscomponent.environment.variable` parameter, see [Workload Automation Agents](#).

3. Open the Windows Service Control Manager and start the Workload Automation Agent (*WA\_AGENT*) service. The agent service starts. The agent automatically restarts the CA SSA Broker service. AutoSys Workload Automation runs in FIPS mode.

## Enable FIPS Mode for a Server or Client Installation on UNIX

AutoSys Workload Automation uses FIPS-approved and non-FIPS approved security algorithms unless you enable FIPS mode. Use this procedure to enable FIPS mode with UNIX. When AutoSys Workload Automation operates in FIPS mode, the system only uses security algorithms that comply with the standards in the FIPS 140-2 publication.

### NOTE

- This procedure applies primarily to U.S. government agencies required to comply with FIPS guidelines.
- This procedure applies only to server or client installations.
- This procedure enables FIPS mode for a particular AutoSys Workload Automation instance and the common components that are associated with that instance (the CA iTechnology iGateway service, the CA EEM server, and the CA Secure Sockets Adapter (SSA) Broker). Running all AutoSys Workload Automation instances, common components, and other Broadcom products that are installed on your system in FIPS mode ensures that your system is fully compliant with the guidelines in the FIPS 140-2 publication.

### Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
unisrvcntr stop waae_agent-WA_AGENT
```

The scheduler, application server, and the agent stop.

3. Enter the following command:

```
unisrvcntr stop all
```

### WARNING

These services do not comply with the FIPS 140-2 guidelines. If you restart these services, your system does not comply with the FIPS 140-2 guidelines even if all other Broadcom products, common components, and AutoSys Workload Automation instances are running in FIPS mode.

CAICCI and Management Command Center stop running.

4. Enter the following command:

```
$CSAM_SOCKADAPTER/bin/csampmux stop
```

The CA SSA Broker stops.

5. Enter the following command:

```
ps -ef | grep pmux
```

The system displays active SSA processes.

6. (As applicable) Kill the csampmuxf process.

### NOTE

This step is necessary only if the system displays csampmuxf as an active SSA process.

7. Stop CA EEM and active common components as follows:

- a. Log in to the CA EEM server that the AutoSys Workload Automation instance is registered with.
- b. Open operating system command prompt, and issue the following commands:

```
su - dsa -c "dxserver stop all"
$IGW_LOC/S99gateway stop
```

8. Enable FIPS mode in the iTechnology iGateway configuration file on the CA EEM server as follows:
  - a. Open file that is named `$IGW_LOC\igateway.conf` and locate the following parameter:
 

```
<FIPSMODE>off</FIPSMODE>
```
  - b. Edit the parameter to read as follows:
 

```
<FIPSMODE>on</FIPSMODE>
```
  - c. Save and close the file.
9. Open the operating system command prompt on the CA EEM server and issue the following commands:
 

```
$IGW_LOC/S99igateway start
su - dsa -c "dxserver start all"
```
10. Edit the AutoSys Workload Automation configuration file as follows:
  - a. Log in to AutoSys Workload Automation, open the file named `$AUTOUSER/config.$AUTOSERV`, and locate the following parameters:
 

```
EnableFIPSMODE=0
UseEncryption=0|1|2
UseCommAliasEncryption=0|1|2
```
  - b. Edit the parameters to read as follows:
 

```
EnableFIPSMODE=1
UseEncryption=1|2
UseCommAliasEncryption=2
```
  - • **EnableFIPSMODE=1**  
Specifies that the instance uses only FIPS-142 approved security algorithms.
  - **UseEncryption=1**  
Specifies that the instance uses the default encryption key
  - **UseEncryption=2**  
Specifies that the instance uses a user-specified encryption key.
  - **UseCommAliasEncryption=2**  
Specifies that the instance uses a user-specified communication alias encryption key.
11. Open the operating system or instance command prompt and enter the following commands:
 

```
as_config -a hex
```

  - **-d**  
Sets the encryption key to the default value.
  - **-g passphrase**  
Sets the specified 16-character string as the encryption key.
  - **-g hex**  
Sets the specified hexadecimal string as the encryption key.
12. Open the agent profile and add the following variable:
 

```
CA_FIPS1402_ENABLE=1
```

## NOTE

The agent parameter file specifies the filename for the agent profile as follows:

```
oscomponent.environment.variable=filename
```

13. Open the operating system or instance command prompt and enter the following commands:
 

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
unisrvcntr start waae_agent-WA_AGENT
```

The scheduler, application server, and agent start in FIPS mode. The scheduler automatically restarts the CA SSA Broker in FIPS mode. The AutoSys Workload Automation instance runs in FIPS mode.

**NOTE**

For more information about the agentparm.txt file, see [Configuring AutoSys Workload Automation](#) or [Workload Automation Agents](#).

## Enable FIPS Mode for a Server or Client Installation on Windows

AutoSys Workload Automation uses FIPS-approved and non-FIPS approved security algorithms unless you enable FIPS mode. When AutoSys Workload Automation operates in FIPS mode, the system only uses security algorithms that comply with the standards in the FIPS 140-2 publication.

**NOTE**

- This procedure applies primarily to U.S. government agencies required to comply with FIPS guidelines.
- This procedure applies only to server or client installations.
- This procedure enables FIPS mode for a particular AutoSys Workload Automation instance and the common components that are associated with that instance (the CA iTechnology iGateway service, the CA EEM server, and the CA Secure Sockets Adapter (SSA) Broker). Running all AutoSys Workload Automation instances, common components, and other Broadcom products that are installed on your system in FIPS mode ensures that your system is fully compliant with the guidelines in the FIPS 140-2 publication.
- The Windows Service Control Manager displays the name of the CA SSA Broker service as "CA Connection Broker."

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select the instance that you want to enable FIPS mode for from the instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.  
Right-click the scheduler and select Stop. Repeat this action for the application server and agent.  
The scheduler, application server, and the agent stop.
4. Open the Windows Service Control Manager and stop the following services:
  - CA Connection Broker
  - CA-Unicenter
  - CA-Unicenter (NR-Server)
  - CA-Unicenter (Remote)
  - CA-Unicenter (Transport)
 The CA SSA Broker and CAICCI services stop.

**WARNING**

CAICCI does not comply with the FIPS 140-2 guidelines. If you restart these services, your system does not comply with the FIPS 140-2 guidelines even if all other Broadcom products, common components, and AutoSys Workload Automation instances are running in FIPS mode.

5. Perform the following tasks on AutoSys Workload Automation and on the CA EEM server:
  - a. Open the Windows Service Control Manager and stop the following services:
    - CA iTechnology iGateway
    - CA Directory iTechPoz
 The CA iTechnology iGateway and CA Directory iTechPoz services stop. All active common components are stopped.
  - b. Edit the following parameter in the iTechnology iGateway configuration file:
 

```
<FIPSMODE>on</FIPSMODE>
```

**NOTE**

- To disable FIPS mode, reset the value of this parameter to default: `<FIPSMODE>off</FIPSMODE>`
- The CA iTechnology iGateway configuration file is named `%IGW_LOC%\gateway.conf`.

c. Open the Windows Service Control Manager and start the following services:

- CA iTechnology iGateway
- CA Directory iTechPoz

The CA iTechnology iGateway and CA Directory iTechPoz services start.

The CA EEM server runs in FIPS mode.

**NOTE**

Stop and start the CA Directory iTechPoz service only on the CA EEM server as it is applicable only to CA EEM.

6. Click Start, Programs, AutoSys, Administrator.

The Instance - AutoSys Workload Automation Administrator window opens.

7. Click the Instance Wide Encryption tab, select the Use AES 128-bit encryption check box, and select the appropriate encryption option.

8. Click the zOS Encryption tab, select the Use AES 128-bit encryption when communicating with zOS managers check box, and enter the encryption key.

9. Click the FIPS Mode tab, select the Enable FIPS Mode check box, and click Apply.

10. Open the file specified by the `oscomponent.environment.variable` parameter in the `agentparm.txt` file and append the following environment variable and value to the file name:

```
CA_FIPS1402_ENABLE=1
```

**NOTE**

To disable FIPS mode, reset the value of this parameter to default:

```
CA_FIPS1402_ENABLE=0
```

11. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation window appears, displaying a list of services that are installed on the selected instance.

Right-click the scheduler and select Start. Repeat this action for the application server and agent.

The scheduler, application server, and the agent start and the AutoSys Workload Automation Administrator closes.

The scheduler automatically restarts the CA SSA Broker in FIPS mode, and the AutoSys Workload Automation instance runs in FIPS mode.

## Enable FIPS Mode for Web Services Using IBM JRE

To enable FIPS mode for web services using the IBM JRE, the IBMJCEFIPS must be configured in the CA EEM SDK and the JRE. Web services use a CA EEM JAAS login module and CA EEM does not add IBMJCEFIPS to the JCE providers list. Modify the JRE to add IBMJCEFIPS to the JCE providers list and add value ON for FIPSMODE.

### Follow these steps:

1. Stop the AutoSys Workload Automation web service.
2. Modify and save the `eam.ws.config` file as follows:

```
<SDK type="Java">
 <iTechSDK>
 <!-- possible values are OFF/ON. This tag is used to override the default
initialization. By default it is set to OFF -->
 <FIPSMODE>ON</FIPSMODE>
```

```

 <JCEProvider>IBMJCEFIPS</JCEProvider>
 <Security>
 <!-- possible values are MD5/SHA1/SHA256/SHA384/SHA512 -->
 <digestAlgorithm>SHA1</digestAlgorithm>
 </Security>
 <Debug>
 <!-- possible values are severe/warning/info/config/fine/finer/finest/all/
off -->
 <logLevel></logLevel>
 <!-- possible values are true/false -->
 <logToFile></logToFile>
 <!-- log file name -->
 <logFile></logFile>
 <!-- log file size in MB (positive integer) -->
 <maxLogSize></maxLogSize>
 </Debug>
</iTechSDK>
</SDK>

```

3. Locate and edit the \$JAVA\_HOME/lib/security/java.security file and add the following line to the security provider list:  
security.provider.1=com.ibm.crypto.fips.provider.IBMJCEFIPS
4. Locate and edit the wrapper.conf file and add the following lines:  
wrapper.java.additional.9=-D com.ibm.security.useFIPS=true  
wrapper.java.additional.10=-Dcom.ibm.jsse2.JSSEFIPS=false
5. Start the AutoSys Workload Automation web service.  
The IBM JRE file is configured to run in FIPS mode.

## Enable FIPS Mode for Web Services Using Oracle JRE

To enable FIPS mode for web services using the JRE, Bouncy Castle must be configured in the CA EEM SDK and the JRE. Web services use a CA EEM JAAS login module and CA EEM does not add Bouncy Castle to the JCE providers list. Modify the JRE to add Bouncy Castle to the JCE providers list and add value ON for FIPSMODE.

### Follow these steps:

1. Stop the AutoSys Workload Automation web service.
2. Modify and save the \$AUTOUSER/eiam.ws.config (UNIX) or %AUTOUSER%\eiam.ws.config (Windows) file as follows:

#### From Release 11.3.6 SP8 to the latest release:

```

<SDK type="Java">
 <iTechSDK>
 <!-- possible values are OFF/ON. This tag is used to override the default
initialization. By default it is set to OFF -->
 <FIPSMODE>ON</FIPSMODE>
 <JCEProvider>BCFIPS</JCEProvider>
 <Security>
 <!-- possible values are MD5/SHA1/SHA256/SHA384/SHA512 -->
 <digestAlgorithm>SHA1</digestAlgorithm> <keyLength>2048</keyLength>
 </Security>
 </iTechSDK>
</SDK>

```

```

 <Debug>
 <!-- possible values are severe/warning/info/config/fine/finer/finest/all/
off -->
 <logLevel></logLevel>
 <!-- possible values are true/false -->
 <logToFile></logToFile>
 <!-- log file name -->
 <logFile></logFile>
 <!-- log file size in MB (positive integer) -->
 <maxLogSize></maxLogSize>
 </Debug>
 </iTechSDK>
</SDK>

```

### From Release 11.3.6 to 11.3.6 SP7:

```

<SDK type="Java">
 <iTechSDK>
 <!-- possible values are OFF/ON. This tag is used to override the default
initialization. By default it is set to OFF -->
 <FIPSMODE>ON</FIPSMODE>
 <JCEProvider>JsafeJCE</JCEProvider>
 <Security>
 <!-- possible values are MD5/SHA1/SHA256/SHA384/SHA512 -->
 <digestAlgorithm>SHA1</digestAlgorithm>
 </Security>
 <Debug>
 <!-- possible values are severe/warning/info/config/fine/finer/finest/all/
off -->
 <logLevel></logLevel>
 <!-- possible values are true/false -->
 <logToFile></logToFile>
 <!-- log file name -->
 <logFile></logFile>
 <!-- log file size in MB (positive integer) -->
 <maxLogSize></maxLogSize>
 </Debug>
 </iTechSDK>
</SDK>

```

3. Locate and edit the \$AUTOSYS/./JRE64\_WA/lib/security/java.security (UNIX) or %AUTOSYS%\./JRE64\_WA\lib\security\java.security (Windows) file, add the following line to the security provider list.

### From Release 11.3.6 SP8 to the latest release:

```
security.provider.11=org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider
```

### From Release 11.3.6 to 11.3.6 SP7:

```
security.provider.9=com.rsa.jsafe.provider.JsafeJCE
```

and add the following lines at the end of the java.security file:

```
com.rsa.cryptoj.jce.no.verify.jar=true
com.rsa.cryptoj.jce.fips140initialmode=NON_FIPS140_MODE
com.rsa.cryptoj.jce.kat.strategy=on.load
```

4. Copy the jar files as follows:

**From Release 11.3.6 to 11.3.6 SP4:**

- On UNIX, copy the cryptojFIPS.jar file that is located in the \$AUTOUSER/webserver/webapps/AEWS/WEB-INF/lib directory to the \$AUTOSYS/..JRE64\_WA/lib/ext directory.
- On Windows, copy the cryptojFIPS.jar file that is located in the %AUTOUSER%\webserver\webapps\AEWS\WEB-INF\lib directory to the %AUTOSYS%\..JRE64\_WA\lib\ext directory.

**From Release 11.3.6 SP5 to 11.3.6 SP7:**

- On UNIX, copy the cryptojce.jar, cryptojcommon.jar, and jcmFIPS.jar files that are located in the \$AUTOUSER/webserver/webapps/AEWS/WEB-INF/lib directory to the \$AUTOSYS/..JRE64\_WA/lib/ext directory.
- On Windows, copy the cryptojce.jar, cryptojcommon.jar, and jcmFIPS.jar files that are located in the %AUTOUSER%\webserver\webapps\AEWS\WEB-INF\lib directory to the %AUTOSYS%\..JRE64\_WA\lib\ext directory.

5. Start the AutoSys Workload Automation web service.  
The JRE file is configured to run in FIPS mode.

## Enable Support for FIPS Mode Communication on Web UI

If you have configured your AutoSys Workload Automation servers to use FIPS mode, enable support for FIPS mode communication on your Web UI servers.

**Follow these steps:**

1. Run the following command in a Command Prompt window (on Windows) or in a shell window (on UNIX) on one Web UI server in your environment:  
**Windows**  
Set CA\_FIPS1402\_ENABLE=1  
**UNIX**  
Export CA\_FIPS1402\_ENABLE=1
2. Restart the CA-wcc-services service.  
The CA\_FIPS1402 variable is added and enabled.
3. Repeat Steps 1 and 2 for each Web UI server in your environment.  
The Web UI servers are configured to run in FIPS mode.

## CA EEM and Security

Embedded Entitlements Manager (CA EEM) lets applications share common access policy management, authentication, and authorization services. CA EEM provides a set of security services. Use the articles in this section to manage security with AutoSys Workload Automation and Autosys Web UI (Web UI).

**CA EEM server:** This server hosts CA EEM, which provides security through role-based login access to Web UI applications and objects. If more than one Broadcom product in your environment uses CA EEM for security, we recommend that you use a dedicated server for CA EEM and that all Broadcom products reference the same instance of CA EEM. If an existing instance of CA EEM previously installed by another Broadcom product on a Windows or UNIX server is available, Web UI can reference that instance.



**NOTE**

- If CA EEM is not configured for failover and your environment utilizes CA EEM for CA Workload Automation AE security, the AutoSys Workload Automation and Web UI servers must reference the same instance of CA EEM.
- If CA EEM is configured for failover, the AutoSys Workload Automation and Web UI servers do not need to reference the same instance of CA EEM; however, the CA EEM instances referenced are in the same failover setup.
- For information about installing CA EEM, see [CA Common Components Installation](#).

## Prerequisites for Configuring NTLM Authentication

Complete the following tasks before you configure the NTLM authentication:

- Verify that CA Embedded Entitlements Manager (CA EEM) is installed on a server with a Microsoft Windows operating system.
- Verify that CA EEM uses Microsoft Active Directory (AD) as the external user store.
- Verify that CA EEM is not configured for Multiple Active Directories or for an Active Directory forest.
- Verify that the users browse to AutoSys Web UI from a Windows computer.
- Verify that the CA EEM Server and the computer from which users browse to Web UI are in the same network domain.
- Verify that the CA EEM Server and the computer where you start Web UI belong to domains with an established trust relationship if they are in nested domains.
- Verify that the domain users are added to the User Groups on the computer where you start the application.

## Import and Manage Identities in CA EEM

This article explains how to import and manage identities in CA Embedded Entitlements Manager (CA EEM). As an administrator, you want to enable access to specific objects with specific permissions. To do so, you manage identities in CA EEM. This product is the default security protocol for AutoSys Workload Automation and AutoSys Web UI (Web UI). CA EEM consists of these three *applications*:

- The Global application lets you import *identities* (users and groups) from an external store and manage identities that are shared between AutoSys Workload Automation and Web UI.
- The AutoSysWorkloadAutomation application lets you manage identities and security policies for AutoSys Workload Automation only.
- The WCC0004 application lets you manage identities and security policies for Web UI only. You can select the appropriate application when you log in to CA EEM. CA EEM contains global identities, AutoSys Workload Automation-specific identities, and Web UI-specific identities.

To create users and groups in CA EEM, we recommend that you reference an external user store such as Microsoft Active Directory. We also recommend that you do all user and group management within that store instead of within CA EEM. To simplify user and group management, you can opt to create dynamic groups in CA EEM to contain selected groups and users. One or more dynamic groups can be assigned to the policies you create, enabling you to update only those dynamic groups when membership or rights change. Otherwise, it would be necessary to update your policies directly.

As a prerequisite, your external user store should already contain all users and groups that you intend to use during security setup.

To import and manage identities in CA EEM, follow these steps:

1. [Configure CA EEM to reference users and groups from an external LDAP store](#)
2. (Optional) [Create a custom directory map](#)
3. (Optional) [Create a dynamic group](#)

**NOTE**

You can create additional users and groups directly in CA EEM. They can also be used in your dynamic groups.

## Example: Configure CA EEM to Reference Users and Groups in Your Active Directory and Create a Dynamic Group to Use in Web UI Policies

Suppose that these global groups exist in your Active Directory. They are organized by the Web UI actions that you want group members to perform:

- **MonitorSupers**  
Supervisors who can access job monitoring and editing applications, and can take corrective action when needed. These users can view, create, modify, and delete all Monitoring views. Also, they can view and send events against all jobs.
- **AllOps**  
Console operators who can access job monitoring applications, and can take corrective action when needed. These users can view all Monitoring views but cannot modify them. Also, they can view and send events against all jobs.
- **AcctOps, PayrollOps**  
Console operators who can access job monitoring applications, and can take corrective action when needed. These users can view Monitoring views only for their department, and cannot modify them. Also, they can view and send events against jobs only for their department.
- **AcctReadOnly, PayrollReadOnly**  
Console operators who can access job monitoring applications but cannot take any corrective action. These users can view Monitoring views only for their department, and cannot modify them. Also, they can view jobs only for their department, and cannot send events against them.

To import these users and groups, begin by configuring CA EEM to reference your Active Directory. Next, create a custom directory map in CA EEM so that only Monitoring users and groups are visible for the policies you plan to create. We will not create any additional application groups for this example, but we will create a dynamic group that includes all of the visible global groups except MonitorSupers. This dynamic group can be used in the policies that control access to Web UI applications and dashboard widgets.

## Configure CA EEM to Reference Users and Groups from an External LDAP Store

Configuring CA EEM to reference users and groups from an external LDAP store enables all applications using CA EEM, such as AutoSys Workload Automation and Web UI, to share the user information from it. These users and groups are therefore considered global in CA EEM.

### NOTE

- Use the Global application in CA EEM to perform this procedure.
- An understanding of how to access your external user store is necessary to perform this procedure.

### Follow these steps:

1. Verify that you are logged in to the Global application in CA EEM.
2. Click the Configure tab.  
The Configure page appears.
3. Click the User Store link.  
The User Store page appears.
4. Click User Store in the User Store section.  
The User Store Configuration section appears in the right pane.
5. Select the Reference from an external LDAP directory option.  
The Directory Information section appears.
6. Leave the default configuration type as Basic LDAP Directory and click Add external LDAP directory.  
The LDAP Directory Configuration section appears.
7. Select the appropriate attribute map and complete the fields as required.

**NOTE**

You can configure CA EEM to reference users from any external LDAP directory. To reference users from another external directory, edit the attribute map.

## 8. Click Save.

The specified values are saved, the User Store Configuration section appears, and CA EEM is configured to reference your external store.

**NOTE**

- You can verify the import by clicking Groups, and then Go on the Manage Identities tab. All the groups from your Active Directory appear in the User Groups section.
- You can use Multiple Microsoft Active Directory Domains or Microsoft Active Directory Forest to configure multiple LDAP directories. For more information about the LDAP directories configurations that CA EEM supports, see the [CA EEM documentation](#).

**Example: Configure CA EEM to Reference Active Directory Users and Groups**

Suppose that your external store is Active Directory. You can configure CA EEM to reference Active Directory, and then reference the users and groups in CA EEM.

**NOTE**

We recommend that you have an understanding of the properties that are required to access your external user store is required.

**To configure CA EEM to reference Active Directory users and groups**

1. Click the Configure tab.  
The Configure page appears.
2. Click the User Store link.  
The User Store page appears.
3. Click User Store in the User Store section.  
The User Store Configuration section appears in the right pane.
4. Select the Reference from an external LDAP directory option.  
The Directory Information section appears.
5. Leave the default configuration type as Basic LDAP Directory and click Add external LDAP directory.  
The LDAP Directory Configuration section appears.
6. Leave the default attribute map as Microsoft Active Directory, and complete the following fields:
  - a. Enter a name for the configuration.
  - b. Enter the host computer on which Active Directory is running.
  - c. Enter the port on which Active Directory is listening and click the right arrow icon

**NOTE**

Usually, enter 389 for the port value. Enter a port number other than 389 only if the Active Directory Server administrator has reconfigured the default Active Directory Server port.

The host and port appear in the Selected Hostnames section.

- d. (Optional) Add other host computers and ports to the Selected Hostnames section as needed.

**NOTE**

Other host computers can be added only if Active Directory failover is configured.

- e. (Optional) Select the appropriate protocol from the Protocol drop-down list. The default is LDAP.

**NOTE**

You can select a different protocol when TLS or SSL is enabled in the Active Directory.

- f. Enter the Base DN. Only global users and groups that are discovered under this DN are mapped to CA EEM.

- g. Enter the User DN that is used to attach to the Active Directory server.
  - h. Enter the password for the specified user in the User DN field. This value is required.
  - i. Enter the password in the Confirm Password field.
  - j. (Optional) Click Show Advanced Configuration and enter the values in the fields that are based on the enterprise configuration.
7. Click Save.  
The specified values are saved, the User Store Configuration section appears, and CA EEM is configured to reference your external store.

### Create a Custom Directory Map

After you have linked your external store and CA EEM, you can create a custom directory map. The custom directory map includes a filter so that only the appropriate global users and groups are visible in CA EEM.

#### NOTE

Use the Global application in CA EEM to perform this procedure.

#### Follow these steps:

1. Click the Configure tab.  
The Configure page appears.
2. Click the User Store link.  
The User Store section appears.
3. Click Attribute Maps in the User Store section.  
The LDAP Mapping page appears in the right pane.
4. Enter the appropriate string in the Group Search Filter field and click Save As.  
The Explorer User Prompt dialog appears.
5. Enter a name for your custom directory map and click OK.  
The changes are saved and a confirmation message is displayed.
6. Click User Store in the User Store section.

#### NOTE

The Global Users/Global Groups section appears in the right pane.

7. Click the link in the Name column for the user store configuration you created previously.  
The LDAP Directory Configuration section appears in the right pane.
8. Select your custom directory map from the Attribute Map drop-down list and click Save.  
The custom directory map is associated with the user store and a confirmation message is displayed.

#### NOTE

You can verify that the custom map is applied by clicking the Groups link on the Manage Identities tab. The Search Groups section displays the groups that match the Group Search Filter string.

### Example: Filter out Monitoring Users and Groups

Suppose that you want only these global users and groups to be available when you create policies:

Global Group	Users in the Global Group
MonitorSupers	MonitorSupers_user1, MonitorSupers_user2
AcctOps	AcctOps_user1, AcctOps_user2
AcctRead	AcctRead_user1, AcctRead_user2
PayrollOps	PayrollOps_user1, PayrollOps_user2
PayrollRead	PayrollRead_user1, PayrollRead_user2
AllOps	AllOps_user1, AllOps_user2

You can create a custom directory map with those requirements. The custom directory map that you create in this example allows only the Monitoring users and groups to be visible.

### To filter out Monitoring users and groups

1. Click the Configure tab.  
The Configure page appears.
2. Click the User Store link.  
The User Store section appears.
3. Click Attribute Maps in the User Store section.  
The LDAP Mapping page appears in the right pane.
4. Enter this string in the Group Search Filter field and click Save As:  
((&(objectClass=group)(!(sAMAccountName=Acct\*)(sAMAccountName=Payroll\*)(sAMAccountName=All\*)  
(sAMAccountName=Monitor\*))))  
The Explorer User Prompt dialog appears.
5. Enter a name for your custom directory map in the dialog that appears, and click OK.  
The changes are saved and a confirmation message is displayed.
6. Click User Store in the User Store section.

#### NOTE

The Global Users/Global Groups section appears in the right pane.

7. Click the link in the Name column for the user store configuration you created previously.  
The LDAP Directory Configuration section appears in the right pane.
8. Select your custom directory map from the Attribute Map drop-down list and click Save.  
The custom directory map is associated with the user store and a confirmation message is displayed.

### Create a Dynamic Group

You can create dynamic groups to control the overall level of access for other users and groups in CA EEM. A dynamic group is created in a CA EEM application as a policy.

#### NOTE

- Use the WCC0004 application in CA EEM to create dynamic groups.
- Creating more than 100 dynamic groups can result in slow performance.

### Follow these steps:

1. Verify that you are logged in to the WCC0004 application in CA EEM.
2. Click the Manage Access Policies tab and verify that Policies is selected.
3. Click the New Dynamic Group Policy icon  
 next to Dynamic User Group Policies in the left pane.  
The New Dynamic Group pane is displayed on the right side of the page. SafeDynamicUserGroup is selected by default in the Resource Class Name drop-down list, and Access Policy is selected by default for Type. Do not change these selections.
4. Enter the appropriate dynamic group name in the Name field.
5. Select User or the type group you want to include in this dynamic group from the Type drop-down list in the Enter/Search Identities pane, and click the Search Identities link.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of users or groups is displayed.
7. Select the appropriate users or groups, and click the right arrow icon



The users or groups are added to the Selected Identities list.

8. (Optional) Repeat Steps 4 through 7 as necessary to add users or groups of another type.
9. Enter the name for your dynamic group in the Add resource field and click the plus icon



The resource is added to the resource list.

10. Select the belong check box under Actions and click Save.  
The dynamic group is saved.

### Example: Create a Dynamic Group to Contain Multiple Monitoring Global Groups

Suppose that you want a single group to contain all global groups that have access to job monitoring applications but not job editing applications. A single group simplifies your work with your Web UI application and widget-related policies. You can create a dynamic group that contains the appropriate users.

#### To create a dynamic group to contain multiple Monitoring global groups

1. Click the Manage Access Policies tab and verify that Policies is selected.
2. Click the New Dynamic Group Policy icon



next to Dynamic User Group Policies in the left pane.

The New Dynamic Group Policy pane is displayed on the right side of the page. SafeDynamicUserGroup is selected by default in the Resource Class Name drop-down list, and Access Policy is selected by default for Type. Do not change these selections.

3. Enter MonitorOperators in the Name field.
4. Select Global Group from the Type drop-down list in the Enter/Search Identities pane and click the Search Identities link.  
The Attribute, Operator, and Value fields appear.
5. Click Search.  
A list of global groups is displayed.
6. Select AcctOps, AcctRead, AllOps, PayrollOps, and PayrollRead, and click the right arrow icon



The global groups are added to the Selected Identities list.

7. Enter MonitorOperators in the Add resource field and click the plus icon



The resource is added to the resource list.

8. Select the belong check box under Actions and click Save.  
The MonitorOperators dynamic group is saved.

You have now completed a comprehensive example of setting up CA EEM to reference an external store, creating a filter to display only the appropriate users, and creating a dynamic group that simplifies the creation and maintenance of security policies. You can now create security policies for your users.

## Set Up and Verify CA EEM Security Policies

As an administrator, you want to set up security policies in CA Embedded Entitlements Manager (CA EEM). You do so to control access to objects. A *policy* is a set of rules that are associated with users or groups to define access to an object.

A *resource class* defines a type of object to which the CA EEM application controls access. For AutoSys Workload Automation, objects include jobs, calendars, machines, and so on. For Web UI, objects include views, logs, user interface components, and so on.

AutoSys Workload Automation creates a default policy for each resource class in the AutoSysWorkloadAutomation application in CA EEM. Web UI creates a default policy for each resource class in the WCC0004 application in CA EEM. The rights that are granted by these policies are designed to reflect typical user roles.

As a prerequisite for setting up and verifying that security, sample users, groups, views, and jobs are created. Sample users and groups that exist in your user store can be imported into CA EEM. For information about referencing an external store in CA EEM, see [Import and Manage Identities in CA EEM](#). For information about creating views and using Monitoring, see *Monitoring Help*. For information about creating jobs and using Quick View, see *Quick Edit Help*. For information about AgentAccess, see the Manage Access Policies in CA Embedded Entitlements Manager section in the [Administer Agent Inventory](#) page.

To set up and verify security, follow these steps:

1. [Create policies](#)
2. [Perform a policy permission check](#)
3. [Check access in the product](#)

### Example: Set Up Security Policies and Verify Access for Monitoring Users

Suppose that you want security policies that provide various levels of access for Monitoring users. Assume that you have the following users and groups in CA EEM:

Group	Group Type	Users contained in the Group
MonitorSupers	Global	MonitorSupers_user1, MonitorSupers_user2
AcctOps	Global	AcctOps_user1, AcctOps_user2
AcctRead	Global	AcctRead_user1, AcctRead_user2
PayrollOps	Global	PayrollOps_user1, PayrollOps_user2
PayrollRead	Global	PayrollRead_user1, PayrollRead_user2
AllOps	Global	AllOps_user1, AllOps_user2
MonitorOperators	Dynamic	Users in the following global groups: AcctOps, AcctRead, PayrollOps, PayrollRead, AllOps

To accomplish this task, follow these steps:

1. Create an ApplicationAccess policy for users in the MonitorOperators dynamic group and MonitorSupers global group. This policy lets users access the specific Web UI applications that their group allows.
2. Create a WidgetAccess policy for users in the MonitorOperators dynamic group and the MonitorSupers global group. This policy lets all users access the Monitoring Views, Servers, and Alerts widgets.
3. Create a MonitorViewControl read-only policy for users in the MonitorOperators dynamic group. This policy lets users view, but not modify, Monitoring views for their department.
4. Create a MonitorViewControl full access policy for users in the MonitorSupers global group. This policy lets users view, create, modify, and delete all Monitoring views.
5. Create an as-job read-only policy that lets department-specific global users view jobs for their department only.
6. Create an as-job policy that lets department-specific global users view and run jobs for their department only. Also, this policy lets users in the AllOps and MonitorSupers global groups view and run all jobs.
7. Perform permission checks to verify that the policies are valid.

After you create Monitoring views and jobs, you can log in to Web UI with your user credentials. You can then verify that the users have the correct access to those objects.



## Create Policies

Security at your site can be set up in many ways. The method for setting up security depends on:

- The objects you want to secure
- The criteria for securing them
- The levels of access that you provide to them.

The way a policy is created differs depending on this criteria:

- The resource class on which the policy is based
- The type of policy selected
- The users and groups assigned to the policy
- The security that the policy should reflect

We recommend that you create your security policies based on the default policies that come with AutoSys Workload Automation and Web UI.

### Example: Create an ApplicationAccess Policy for Monitoring Users

The ApplicationAccess policy controls user access to Web UI applications. Suppose that you want your Monitoring users to be able to use Monitoring-related applications. The MonitorSupers global group should be able to access Quick Edit. The MonitorOperators dynamic group, which contains all the other global groups, should not. You can create an ApplicationAccess policy that provides the appropriate application access for the users in those two groups.

#### NOTE

- Use the WCC0004 application in CA EEM to create ApplicationAccess policies.
- For this policy, the Identity Access Control List type is recommended.

#### Follow these steps:

1. Verify that you are logged in to the WCC0004 application.
2. Click the Manage Access Policies tab and verify that Policies is selected.
3. Click the New Access Policy icon  next to ApplicationAccess in the left pane.  
The New Access Policy pane is displayed on the right side of the page.
4. Enter ApplicationAccessMonitor in the Name field.  
ApplicationAccess is selected by default in the Resource Class Name drop-down list. Do not change this selection.
5. Select the Identity Access Control List option and click OK in response to the prompt that appears.  
The page refreshes.
6. Select Global Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
7. Click Search.  
A list of global groups is displayed.
8. Select MonitorSupers and then click the down arrow icon   
The global group is added to the Selected Identities list.
9. Select Dynamic Group from the Type drop-down list and click Search.  
A list of dynamic groups is displayed.
10. Select MonitorOperators and then click the down arrow icon   
The dynamic group is added to the Selected Identities list.



11. Scroll down to view the Selected Identities pane and select the specified application checkboxes for the following identities:
  - MonitorSupers -- QuickView, QuickEdit, Monitoring, Forecast, Dashboard, and Launcher.
  - MonitorOperators -- QuickView, Monitoring, Dashboard, and Launcher.
12. Click Save at the bottom of the page.  
The ApplicationAccessMonitor policy is saved.

### Example: Create a WidgetAccess Policy for Monitoring Users

The WidgetAccess policy controls user access to the Web UI application widgets. The widgets that a user can access are displayed on the Dashboard tab of Web UI. Suppose that you want all your Monitoring users to be able to use the Monitoring-related widgets only. You have a MonitorSupers global group and a MonitorOperators dynamic group, which together contain of all the Monitoring users. You can create a WidgetAccess policy that provides the appropriate widget access for the users in those two groups. The policy that you create in this example leverages a dynamic group and a global group so that you do not have to assign all the actions to each user.

#### NOTE

- Use the WCC0004 application in CA EEM to create WidgetAccess policies.
- For this policy, we recommend a type of Identity Access Control List.

### To create a WidgetAccess policy for Monitoring users

1. Click the Manage Access Policies tab in CA EEM and verify that Policies is selected.
2. Click the New Access Policy icon  next to WidgetAccess in the left pane.  
The New Access Policy pane is displayed on the right side of the page.
3. Enter WidgetAccessMonitor in the Name field.  
WidgetAccess is selected by default in the Resource Class Name drop-down list. Do not change this selection.
4. Select the Identity Access Control List option button and click OK in response to the prompt that appears.  
The page refreshes.
5. Select Global Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of global groups is displayed.
7. Select MonitorSupers and then click the down arrow icon .  
The global group is added to the Selected Identities list.
8. Select Dynamic Group from the Type drop-down list and click Search.  
A list of dynamic groups is displayed.
9. Select MonitorOperators and then click the down arrow icon .  
The dynamic group is added to the Selected Identities list.
10. Scroll down to view the Selected Identities pane and select the following Actions checkboxes for MonitorOperators and MonitorSupers:
  - MonitoringViews
  - MonitoringServers
  - MonitoringAlerts
  - MonitoringJobs
11. Click Save at the bottom of the page.  
The WidgetAccessMonitor policy is saved.

### Example: Create a Read-Only MonitorViewControl policy

The MonitorViewControl resource class controls the actions that can be performed on a view in Monitoring. Suppose that you want users in the MonitorOperators dynamic group to be able to view Monitoring views created previously. You can create a MonitorViewControl policy that lets those users view Monitoring views only.

#### NOTE

- Use the WCC0004 application in CA EEM to create MonitorViewControl policies.
- This example includes a best practices approach, which uses a view (job flow) naming convention that embeds the application group name as the first element in the view name. For example, all views for the Accounting application group have names that begin with Acct. In this example, you create a policy to leverage the naming convention. We recommend that you use a strong naming convention for view names and application group names. Doing so reduces the number of policies required.

#### To create a read-only MonitorViewControl policy

1. Click the Manage Access Policies tab and verify that Policies is selected.
2. Click the New Access Policy icon  next to MonitorViewControl in the left pane.  
The New Access Policy pane is displayed on the right side of the page.
3. Enter MonitorViewsReadOnly in the Names field.  
MonitorViewControl is selected by default in the Resource Class Name drop-down list and Access Policy is selected by default for Type. Do not change these selections.
4. Select Dynamic Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
5. Click Search.  
A list of dynamic groups is displayed.
6. Select the MonitorOperators dynamic group and click the right arrow icon .  
The dynamic group is added to the Selected Identities list.
7. Enter view/.+/.JobFlow in the Add resource field and click the plus icon .  
The resource is added to the resource list.

#### NOTE

Specifying .+ in the path indicates that the name must contain at least one character.

8. Select the View check box under Actions.
9. Select the Treat resource names as the regular expressions check box.
10. Scroll down the page and click Add Filter.  
The filter definition fields appear.
11. Enter or select the values in the following table, clicking Add Filter to add filter rows as needed:

Logic	(	Left type/value	Operator	Right type/value	)
NONE	(	named attribute	STRING	value	
		viewName	STARTSWITH --*	Acct	
AND	(	global user group	STRING	value	
		Principal Name	EQUAL ==	AcctRead	

OR		global user group	STRING	value	))
		Principal Name	EQUAL ==	AcctOps	
OR	(	named attribute	STRING	value	
		viewName	STARTSWITH --*	Payroll	
AND	(	global user group	STRING	value	
		Principal Name	EQUAL ==	PayrollRead	
OR		global user group	STRING	value	))
		Principal Name	EQUAL ==	PayrollOps	
OR		global user group	STRING	value	
		Principal Name	EQUAL ==	AllOps	

12. Click Save.

The MonitorViewsReadOnly policy is saved.

### Example: Create a Full Access MonitorViewControl Policy

The MonitorViewControl resource class controls the actions that can be performed on a view in Monitoring. Suppose that you want users in the MonitorSupers global group to create Monitoring views, and modify, delete, and view Monitoring views created previously. You can create a MonitorViewControl policy that lets those users have full access to Monitoring views.

#### NOTE

- Use the WCC0004 application in CA EEM to create MonitorViewControl policies.
- This example includes a best practices approach, which uses a view (job flow) naming convention that embeds the application group name as the first element in the view name. For example, all views for the Accounting application group have names that begin with Acct. In this example, you create a policy to leverage the naming convention. We recommend that you use a strong naming convention for view names and application group names. Doing so reduces the number of policies required.

### To create a full access MonitorViewControl policy

1. Click the Manage Access Policies tab and verify that Policy is selected.
2. Click the New Access Policy icon  next to MonitorViewControl in the left pane.  
The New Access Policy pane is displayed on the right side of the page.
3. In the Name field, enter MonitorViewsFullAccess.  
MonitorViewControl is selected by default in the Resource Class Name drop-down list and Access Policy is selected by default for Type. Do not change these selections.
4. Select Global Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
5. Click Search.  
A list of global groups is displayed.
6. Select the MonitorSupers global group and click the right arrow icon .  
The global group is added to the Selected Identities list.
7. Enter view./+JobFlow in the Add resource field and click the plus icon .  
The resource is added to the resource list.

**NOTE**

Specifying .+ in the path indicates that the name must contain at least one character.

8. Select the Create, Modify, Delete, and View checkboxes under Actions.
9. Select the Treat resource names as regular expressions check box.
10. Click Save.

The MonitorViewsFullAccess policy is saved.

**Example: Create a Read-Only as-job policy**

To enable user access to jobs reflecting that user role, you can create an as-job policy. Suppose that you want users in the AcctRead and PayrollRead global groups to be able to only view jobs that are created for their respective departments. You can create an as-job policy that lets those users view only the appropriate jobs.

**NOTE**

- Use the WorkloadAutomationAE application in CA EEM to create as-job policies.
- An as-job policy is not applied in Monitoring views unless the Use AutoSys Workload Automation Security option is enabled in the Monitoring Properties in Configuration.
- An as-job policy is always applied in Quick Edit.

**To create a read-only as-job policy**

1. Log out of CA EEM.  
You are logged out of the WCC0004 application and the login page for CA EEM appears.
2. Select WorkloadAutomationAE for the application, enter your credentials, and click Log In.  
CA EEM opens with the Home tab selected.
3. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
4. Click the New Access Policy icon  
  
next to the as-job policy in the left pane.  
The New Access Policy page appears in the right pane.
5. Enter MonitorJobsRead in the Name field, and leave as-job selected in the Resource Class Name drop-down list.
6. Select Global Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
7. Click Search.  
A list of global groups is displayed.
8. Select the AcctRead and PayrollRead global groups and click the right arrow icon  
  
The global groups are added to the Selected Identities list.
9. Enter PUB.\* in the Add resource field and click the plus icon  
  
The resource is added to the resource list.

**NOTE**

PUB is the name of the AutoSys Workload Automation server instance in this example.

10. Select the read check box under Actions.
11. Click Add Filter.  
The filter definition fields appear.

12. Enter or select the values in the following table, clicking Add Filter to add filter rows as needed:

Logic	(	Left type/value	Operator	Right type/value	)
NONE	(	request	STRING	value	
		resource	STARTSWITH --*	PUB.Acct	
AND		global user group	STRING	value	)
		Principal Name	EQUAL ==	AcctRead	
OR	(	request	STRING	value	
		resource	STARTSWITH --*	PUB.Payroll	
AND		global user group	STRING	value	)
		Principal Name	EQUAL ==	PayrollRead	

13. Click Save.

The MonitorJobsRead policy is saved.

### Example: Create a Read and Execute as-job Policy

To enable user access to jobs reflecting the role of the user, you can create an as-job policy. Suppose that you want users in the AcctOps and PayrollOps global groups to be able to view and run jobs that are created for their respective departments. Also, you want users in the AllOps and MonitorSupers global groups to be able to view and run all jobs. You can create an as-job policy that lets those users view and run the appropriate jobs.

#### NOTE

Use the WorkloadAutomationAE application in CA EEM to create as-job policies.

### To create a read and execute as-job policy

1. Verify that you are logged in to the AutoSysWorkloadAutomation application in CA EEM.
2. Select the Manage Access Policies tab in the CA EEM application.  
The Policies, Calendars, and Permission Check links appear, with the Policies link selected by default. In the left pane, the Show policies matching name option is selected by default.
3. Click the New Access Policy icon  
  
next to the as-job policy in the left pane.  
The New Access Policy page appears in the right pane.
4. Enter MonitorJobsExecute in the Name field, and leave as-job selected in the Resource Class Name drop-down list.
5. Select Global Group from the Type drop-down list and click the Search Identities link in the Enter/Search Identities section.  
The Attribute, Operator, and Value fields appear.
6. Click Search.  
A list of global groups is displayed.
7. Select the AcctOps, AllOps, MonitorSupers, and PayrollOps global groups and click the right arrow icon  
  
The global groups are added to the Selected Identities list.
8. Enter PUB.\* in the Add resource field and click the plus icon  
  
The resource is added to the resource list.
9. Select the read and execute checkboxes under Actions.
10. Click Add Filter.  
The filter definition fields appear.

11. Enter or select the values in the following table, clicking Add Filter to add filter rows as needed:

Logic	(	Left type/value	Operator	Right type/value	)
NONE	(	request	STRING	value	
		resource	STARTSWITH --*	PUB.Acct	
AND		global user group	STRING	value	)
		Principal Name	EQUAL ==	AcctOps	
OR	(	request	STRING	value	
		resource	STARTSWITH --*	PUB.Payroll	
AND		global user group	STRING	value	)
		Principal Name	EQUAL ==	PayrollOps	
OR		global user group	STRING	value	
		Principal Name	EQUAL ==	AllOps	
OR		global user group	STRING	value	
		Principal Name	EQUAL ==	MonitorSupers	

12. Click Save.

The MonitorJobsExecute policy is saved.

### **Perform a Policy Permission Check**

You can perform a permission check to verify whether permission has been properly assigned in a policy that you have created.

#### **Follow these steps:**

1. Click Permission Check on the Manage Access Policies tab.  
The Permission Check page opens.
2. Click Synchronize Cache.  
The web user interface is synchronized with the latest sets of policies and user attributes.
3. Enter or select the appropriate values in the Permission Check Parameters fields.
4. Click Run Permission Check.  
The results of the permission check are displayed in the Permission Check Results pane.

#### **NOTE**

A permission check can return a result of Allow or Deny. If the value is Deny, select the Display debug information check box to view the details of the policy checks. The information can assist you in determining the problem.

### **Example: Perform Permission Checks on Your Web UI Policies**

Suppose that you want to verify whether permission has been properly assigned in the Web UI policies that you have created. To do so, you can perform a permission check.

#### **NOTE**

Use the WCC0004 application in CA EEM to check Web UI policies.

The following table contains the values to use in the procedure to test the policies you have created:

Resource Class	Action	Resource	Identity	Attribute	Value
ApplicationAccess	Monitoring	*	AcctRead_user1	(None)	(None)

ApplicationAccess	QuickView	*	PayrollOps_user1	(None)	(None)
ApplicationAccess	Dashboard	*	AllOps_user1	(None)	(None)
ApplicationAccess	Forecast	*	MonitorSupers_user1	(None)	(None)
ApplicationAccess	QuickEdit	*	MonitorSupers_user1	(None)	(None)
MonitorViewControl	View	view/./+JobFlow	AcctRead_user1	viewName	Acct
MonitorViewControl	View	view/./+JobFlow	PayrollOps_user1	viewName	Payroll
MonitorViewControl	View	view/./+JobFlow	AllOps_user1	viewName	Sales
MonitorViewControl	Create	view/./+JobFlow	MonitorSupers_user1	viewName	Sales
WidgetAccess	MonitoringViews	*	AcctRead_user1	(None)	(None)
WidgetAccess	MonitoringServers	*	MonitorSupers_user1	(None)	(None)

### To perform permission checks on your Web UI policies

1. Click Permission Check on the Manage Access Policies tab.  
The Permission Check page opens.
2. Click Synchronize Cache.  
The web user interface is synchronized with the latest sets of policies and user attributes.
3. Enter or select the values from the first row of the previous table in the Permission Check Parameters fields.
4. Click Run Permission Check.  
In the Permission Check Results pane, the results of the permission check are displayed.

#### NOTE

A permission check can return a result of Allow or Deny. If the value is Deny, select the Display debug information check box to view the details of the policy checks. The information can assist you in determining the problem.

5. Repeat Steps 3 and 4. Enter or select the values from the second through tenth rows of the previous table in the Permission Check Parameters fields.
6. (Optional) Perform more permission checks to verify the appropriate policy for each user.

### Example: Perform Permission Checks on Your AutoSys Workload Automation Policies

Suppose that you want to verify whether permission has been properly assigned in the as-job policy that you have created. To do so, you can perform a permission check.

#### NOTE

Use the AutoSys Workload Automation application in CA EEM to check AutoSys Workload Automation policies.

The following table contains the values to use in the procedure to test the policies you have created:

Resource Class	Action	Resource	Identity
as-job	read	PUB.Acct*	AcctRead_user1
as-job	read	PUB.Payroll*	PayrollRead_user1
as-job	execute	PUB.Acct*	AcctOps_user1
as-job	execute	PUB.Payroll*	PayrollOps_user1
as-job	execute	PUB.Sales*	AllOps_user1
as-job	execute	PUB.Sales*	MonitorSupers_user1

## To perform a permission check on your AutoSys Workload Automation policies

1. Click Permission Check on the Manage Access Policies tab.  
The Permission Check page opens.
2. Click Synchronize Cache.  
The web user interface is synchronized with the latest sets of policies and user attributes.
3. Enter or select the values from the first row of the previous table in the Permission Check Parameters fields.
4. Click Run Permission Check.  
The results of the permission check are displayed in the Permission Check Results pane.

### NOTE

A permission check can return a result of Allow or Deny. If the value is Deny, select the Display debug information check box to view the details of the policy checks. The information can assist you in determining the problem.

5. Repeat Steps 3 and 4. Enter or select the values from the second through sixth rows of the previous table in the Permission Check Parameters fields.
6. (Optional) Perform more permission checks to verify the appropriate policy for each user.

## Check Access in the Product

To ensure that your security policies are working as designed, you can log in to the AutoSys Workload Automation and Web UI products directly once those policies are set up. You can log in as any user you have defined in CA EEM. You can then verify that access is granted and denied for the appropriate objects at the appropriate levels (from read-only all the way to full access).

### Example: Check Access in Web UI

For example, you can verify policies that you have created are valid by logging in to Web UI as the user for whom the policy was created.

### NOTE

- To test the sample policies, you would need to create sample views in the Web UI Monitoring application with names that begin with Acct and Payroll. Also, you would need to create sample jobs that begin with Acct and Payroll in the Quick Edit application.
- An as-job policy is not applied in Monitoring views unless the Use AutoSys Workload Automation Security option is enabled in the Monitoring Properties in Configuration.
- An as-job policy is always applied in Quick Edit.

### Follow these steps:

1. Go to the URL for Web UI, enter the credentials for one of the users in the following table, and click Log In:

User Name	Available Applications	Actions User Can Perform
AcctRead_user1	Dashboard	View Alerts, Servers, and Views widgets
	Monitoring	View Accounting views
	Quick View	View Accounting jobs
PayrollOps_user1	Dashboard	View Alerts, Servers, and Views widgets
	Monitoring	View Payroll views
	Quick View	View, run Payroll jobs
AllOps_user1	Dashboard	View Alerts, Servers, and Views widgets
	Monitoring	View any Monitoring view
	Quick View	View, run any job



MonitorSupers_user1	Dashboard	View Alerts, Servers, and Views widgets
	Monitoring	View, create, modify, delete any Monitoring view
	Quick View	View, run any job
	Quick Edit	View job overrides
	Forecast	Run forecast reports to troubleshoot job flow execution issues

Web UI opens with the Dashboard tab selected.

- Verify that the applications are available and actions can be performed as shown in the previous table.
- (Optional) Log in to Web UI with each user you have created. Verify that the appropriate applications are available and actions can be performed.

#### NOTE

If a tab is not visible or an action cannot be performed, review your security policy definitions.

You have now completed a comprehensive example of setting up CA EEM security policies for a particular user role. The users that you have created can now log in to Web UI. They can access the applications and objects to which they have permission.

## Update the EiamAdmin Password

You can use the following procedure to update the EiamAdmin password for Web UI.

### Update the Password for Web UI

The following procedure helps you to update the EiamAdmin password in Web UI, by using the command syntax:

```
wcc_config -u user_name -p password --eemhostname eem_host_name --
eemadmin eem_administrator --eempassword eem_password --eemcertname eem_certificate_name
--eemcertkey eem_certificate_key
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--eemhostname eem\_host\_name[,eem\_host\_name2,eem\_host\_name3...]**  
Specifies the host names of the new CA EEM servers that you want the Web UI instance to reference. The user name and password are authenticated against this host.

#### NOTE

If CA EEM is configured for failover, you can optionally specify a list of the servers in the failover setup. Server names should be separated by a comma only and listed in the order of reliability.

- **--eemadmin eem\_administrator**  
Specifies the administrator name for the new instance of CA EEM.
- **--eempassword eem\_password**  
Specifies the password for the administrator of the new instance of CA EEM.
- **--eemcertname eem\_certificate\_name** Specifies the CA EEM certificate file name. This is required only if you are using the CA EEM certificate. This file is in the following location:

```
$CA_WCC_INSTALL_LOCATION/data/config
```

**--eemcertkey eem\_certificate\_key** Specifies the CA EEM certificate key file name. This is required only if you are using the CA EEM certificate. This file is in the following location:

```
$CA_WCC_INSTALL_LOCATION/data/config
```

### Example: Update the EiamAdmin password

The following command updates the EiamAdmin password in Web UI when you use the optional CA EEM certificate name and keyfile:

```
wcc_config -u ejmcommander -p ejmcommander --eemhostname EEM_01 --eemadmin EiamAdmin --eempassword admin --eemcertname wcc.pem --eemcertkey wcc.key
```

#### NOTE

After running the command, restart the services CA-wcc and CA-wcc-services on Windows, and the service CA-wcc-services on Unix.

#### NOTE

For more information, see the [Administer the Configuration Batch Interface](#) section.

## Migrate Security Policies from Unicenter AutoSys JM 4.5.1 to Current Release

This article describes how to migrate your security policies from the previous security protocols used by Unicenter AutoSys JM 4.5.1 to CA Workload Automation AE Release 11.3.6. Unicenter AutoSys JM 4.5.1 used CA Access Control (CA AC) to store the user and group definitions and the access level definitions for your security policies. Later releases of AutoSys Workload Automation offer CA Embedded Entitlements Manager (CA EEM) security in addition to native security.

Typically, security policy in an enterprise consists of these entities:

- User and group definitions
- Policy definitions (access-level definitions of resources) for the users and groups

Unicenter AutoSys JM 4.5.1 uses CA Access Control (CA AC) to store the user and group definitions and the access-level definitions. To migrate security policies from Unicenter AutoSys JM 4.5.1 to CA Workload Automation AE Release 11.3.6, complete the following tasks:

1. [Migrate users and group definitions from CA AC to CA EEM.](#)
2. [Migrate policy definitions from CA AC to CA EEM.](#)
3. [Change CA EEM backup XML file values.](#)
4. [Register the product in the target CA EEM instance using as\\_safetool.](#)
5. [Import the CA EEM XML file using safex.](#)
6. [Verify imported security policies.](#)

#### NOTE

- The migration of all users and groups must occur independently of the individual policy migrations.
- When the process is complete, remove the selang file and all the intermediate XML files created during the process.

### Migration Requirements

The CA EEM policy migration requires these files:

- antl.jar
- se2xml.jar
- AutoSys.xsl
- PostRegex.xsl
- selang2eem.xsl
- serializer.jar
- xalan.jar

**NOTE**

These files are included with the AutoSys Workload Automation installation. They are located in the \$AUTOSYS/EEMmigrate (on UNIX) or %AUTOSYS%\EEMmigrate (on Windows) directory after AutoSys Workload Automation is installed. On UNIX, the \$AUTOSYS/EEMmigrate directory is created when you install the AutoSys Workload Automation scheduler.

The following tools are used to migrate your existing CA AC policy to CA EEM:

- **AutoSys Workload Automation as\_safetool Utility** Installs the new and updated resource classes and default policies for all instances that require migration. AutoSys Workload Automation installs the as\_safetool utility.
- **CA EEM safex Utility** Imports the final generated XML file containing the migrated policies to the CA EEM back-end server.

**NOTE**

The safex utility is only available on a computer where the CA EEM back-end server is installed. This utility is located in the bin subdirectory of the CA EEM installation path

The safex utility directs all output to stderr. We recommend that you capture this output and save it to a file so you can examine errors.

- **Xalan and Serializer JAR files** Parses the XML files.

**NOTE**

The Xalan and Serializer Java jar files are located in the \$AUTOSYS/EEMmigrate (on UNIX) or %AUTOSYS%\EEMmigrate (on Windows) directory after AutoSys Workload Automation is installed. The EEMmigrate directory is created when you install the AutoSys Workload Automation scheduler.

- **Java Runtime Environment (JRE)** Runs the Java commands that are part of the migration policy.

**NOTE**

The JRE is located in the SharedComponents directory. Ensure that the PATH environment variable is updated to include the location of the java binary.

**Security Policy Changes from Unicenter AutoSys JM 4.5.1**

This topic describes the changes to the security policy implementation in CA Workload Automation AE Release 11.3.6. You cannot import the Unicenter AutoSys JM 4.5.1 security policies into CA EEM servers that are linked with CA Workload Automation AE Release 11.3.6. First complete the migration process to apply these changes and other changes associated with previous releases.

This list details the security policy changes from Unicenter AutoSys JM 4.5.1 to CA Workload Automation AE Release 11.3.6:

- Every CA AC resource class contains a default resource (with the name \_default) defining the security policy for objects that do not have a matching policy. CA EEM does not automatically define a default policy for resource classes for resources with no matching policy. The migration process converts the CA AC default resource policies to equivalent CA EEM policies for resources with no matching policy.
- CA Workload Automation AE Release 11.3.6 does not use the as-view resources class and certain as-control and as-list resources. Deprecated as-control resources include those ending with \_ON or \_OFF and those beginning with

WEBADM. Deprecated as-list resources include those beginning with AUTOCONS, JOBDEF, or XPERT. The migration process does not import the deprecated as-control and as-list resources, or the as-view resource class.

- The naming convention for CA EEM resources is *instance.object* instead of *object.instance*. The migration process converts CA AC resource names following the Unicenter AutoSys JM 4.5.1 convention to match the new naming convention. Resource names in CA AC that use asterisks (\*) to apply to multiple instances change to the following CA EEM resource name equivalents:
  - object.\* becomes \*.object
  - object\* becomes \*.object\*
- CA EEM interprets resource names containing asterisks (\*) differently than CA AC. The migration process changes CA AC resource names containing asterisks (\*) to conform to the regular expression language used by CA EEM. Resource names in CA AC containing asterisks (\*) change to the following CA EEM resource name equivalents:
  - \*[object] remains unchanged
  - [object]\* remains unchanged
  - \*[object]\* remains unchanged
  - [object]\*[object] becomes [object].\*[object]
  - \*[object]\*[object] becomes .\*[object].\*[object]
  - [object]\*[object]\* becomes [object].\*[object].\*
  - \*[object]\*[object]\* becomes .\*[object].\*[object].\*

#### NOTE

The migration process relies on the old naming convention to detect the instance. Do not change the resource names before initiating the migration process.

- The addition of the as-appl resource class supports the authorization of jobs based on their association with a AutoSys Workload Automation application.
- The addition of the as-base-jobtype resource class with EXECUTE access mode supports the authorization of jobs based on their job types. This new resource class includes a default policy. This policy grants EXECUTE access for all jobs in the AutoSys Workload Automation instance.
- The addition of the as-cycle class supports the authorization of AutoSys Workload Automation cycle calendars.
- The addition of the as-group resource class supports the authorization of jobs based on their association with a AutoSys Workload Automation group.
- The addition of the as-joblog resource class supports the authorization of the retrieval of job logs.
- The addition of the as-jobtype resource class supports the authorization of jobs based on their user-defined job type.
- The addition of the as-resource resource class supports the authorization of AutoSys Workload Automation resources.
- Updates to the as-control resource class support the new policy resources named *AGGREGATE* and *SENDEVENT\_GRPAPP* for the sendevent command.
- Updates to the as-control resource class support the new policy resource named *ARCHIVE\_JOBS* for the archive\_jobs command.
- Updates to the as-list resource class support a new policy resource named *FORECAST* for the forecast command.

### **Migrate Users and Groups from CA AC to CA EEM**

Users and groups are defined differently in CA AC and CA EEM. In CA AC, users and groups are represented by a script file containing specific commands written in selang, the CA AC command language. In CA EEM, users and groups are represented by an XML file using specific CA EEM XML tags. Also, in CA EEM, *application* groups work for AutoSys Workload Automation policies only (they are application-specific), while *global users and groups* can be shared across all application instances registered with CA EEM.

When you migrate users from CA AC to CA EEM, you must define all users to CA EEM as global users. When you migrate user groups from CA AC to CA EEM, you must define the user groups to CA EEM as global user groups, application-specific user groups, or any mix of the two. To migrate users and user groups from CA AC to CA EEM, create one or more CA EEM XML files manually.

**NOTE**

For this process to be successful, confirm that the PATH environment variable is updated to include the location of the Java binary.

To migrate users and groups from CA AC to CA EEM, complete the following tasks:

1. [Export CA AC users and groups to a selang file.](#)
2. [Convert the selang file to a selang XML file.](#)
3. Do one or both of the following tasks:
  - a. (For global users and groups) [Manually create a CA EEM XML file for global users and groups from the selang XML file.](#)
  - b. (For application groups) [Manually create a CA EEM XML file to add global users to application user groups from the selang XML file.](#)

**Export CA AC Users and Groups to a selang File**

CA AC provides the dbmgr utility to export the necessary users and groups into a script file containing the selang commands that are required to duplicate the database.

**NOTE**

For more information about how to use the dbmgr utility to export users and groups into a selang file, see the *CA Access Control Reference documentation*.

**Convert the selang File to a selang XML File**

The JRE is required to convert the selang file to a selang XML file. You must identify the selang commands from the exported file and generate an equivalent XML file containing the selang commands as XML tags. After the script file is converted to an XML file, you can use an XML parser to translate the CA AC XML tags to the equivalent CA EEM XML tags.

To convert the selang file to a selang XML file, go to the \$AUTOSYS/EEMmigrate (on UNIX) or %AUTOSYS%\EEMmigrate (on Windows) directory and enter the following command:

```
java -jar se2xml.jar exported_selang_file_name
```

- ***exported\_selang\_file\_name***  
Identifies the name of the selang file to convert.

An XML file with the name *exported\_selang\_file\_name.xml* is generated.

**Create a CA EEM XML File for Global Users and Groups from the selang XML File Manually**

To migrate users and groups from CA AC to CA EEM as global users and groups, generate one or more XML files containing the users and user groups defined in the selang XML tags.

**Follow these steps:**

1. Create a CA EEM XML file.
2. Add the following lines to the file:
 

```
<Safex>
<Attach />
<Add>
```
3. To define a global user group, add the following lines:
 

```
<GlobalUserGroup folder="root_path_to_CA_EEM_server" name="global_group_name">
```

```

<global_group_attribute>global_group_attribute_value</global_group_attribute>
<global_group_attribute>global_group_attribute_value</global_group_attribute>
<global_group_attribute>global_group_attribute_value</global_group_attribute>
...
</GlobalUserGroup>

```

- **global\_group\_name** Specifies the name of the global group.
  - **global\_group\_attribute** Specifies an attribute for the global group, as follows:
    - GroupMembership
    - Description
  - **global\_group\_attribute\_value** Specifies a value for the attribute of the global group.
4. Repeat Step 3 for each global user group you want to define.
5. To define a global user, add the following lines:

```

<GlobalUser folder="root_path_to_CA_EEM_server" name="global_user_name">
<global_user_attribute>global_user_attribute_value</global_user_attribute>
<global_user_attribute>global_user_attribute_value</global_user_attribute>
<global_user_attribute>global_user_attribute_value</global_user_attribute>
...
</GlobalUser>

```

- **global\_user\_name** Specifies the name of the global user.
- **global\_user\_attribute** Specifies an attribute for the user, as follows:

- UserName
- GroupMembership
- FirstName
- MiddleName
- LastName
- EmailAddress
- Alias
- Department
- DisplayName
- HomePhoneNumber
- WorkPhoneNumber
- MobilePhoneNumber
- FaxPhoneNumber
- Address
- City
- State
- PostalCode
- Country
- Office
- Company
- PasswordDigest
- IncorrectLoginCount
- SuspendDate
- DisableDate
- EnableDate
- Description
- Comments
- JobTitle
- MailStop

– ***global\_user\_attribute\_value*** Specifies a value for the attribute of the global user.

6. Repeat Step 5 for each additional global user you want to define.

7. Add the following lines to the end of the file:

```
</Add>
</Safex>
```

8. Save the XML file. The CA EEM XML file is created.

### Example: Create an XML File for a Global User

This example defines a user XML file for global user “johndoe”. This file contains all user attributes that you can use.

```
<Safex>
<Attach />
<Add>
<GlobalUser folder="/" name="johndoe">
<UserName>doejo33</UserName>
<GroupMembership>Administrators</GroupMembership>
<FirstName>john</FirstName>
<MiddleName>dennis</MiddleName>
```

```

<LastName>doe</LastName>
<EmailAddress>jdoe@example.com</EmailAddress>
<Alias>jdoe</Alias>
<Department>accounting</Department>
<DisplayName>John D Doe</DisplayName>
<HomePhoneNumber>718-264-8966</HomePhoneNumber>
<WorkPhoneNumber>508-628-7076</WorkPhoneNumber>
<MobilePhoneNumber>508-593-0963</MobilePhoneNumber>
<FaxPhoneNumber>508-628-2319</FaxPhoneNumber>
<Address>331 Main St</Address>
<Address>Jones Building</Address>
<Address>Suite 3200</Address>
<Address>Acme Corp.</Address>
<City>Smallville</City>
<State>Quebec</State>
<PostalCode>H4M2X4</PostalCode>
<Country>Canada</Country>
<Office>C-42</Office>
<Company>Acme</Company>
<PasswordDigest>xxxxxxxxxxxxxx</PasswordDigest>
<IncorrectLoginCount>0</IncorrectLoginCount>
<SuspendDate>0</SuspendDate>
<DisableDate>0</DisableDate>
<EnableDate>0</EnableDate>
<Description>Working in Finance</Description>
<Comments>12 month temp</Comments>
<JobTitle>Billing Manager</JobTitle>
<MailStop>C-42-2-12</MailStop>
</GlobalUser>
</Add>
</Safex>

```

### Example: Create an XML File for Global User Groups

This example defines a user group XML file for global user groups “Staff” and “Administrator”. This file contains all user group attributes that you can use.

```

<Safex>
<Attach />
<Add>
<GlobalUserGroup folder="/" name="Staff">
<Description>Staff group description</Description>
</GlobalUserGroup>
<GlobalUserGroup folder="/" name="Administrators">
<GroupMembership>Staff</GroupMembership>
<Description>Administrator group description</Description>
</GlobalUserGroup>

```



```
</Add>
</Safex>
```

### **Manually Create a CA EEM XML File to Add Global Users to Application Groups from the selang XML File**

To migrate user groups from CA AC to CA EEM as application user groups and to add global users to application groups, you must generate one or more XML files containing the users and user groups defined in the selang XML tags.

#### **NOTE**

After you complete these steps, register the current release of the product with CA EEM before you can import it to the CA EEM back-end server. This requirement is because the application instance you import to does not exist in CA EEM.

#### **Follow these steps:**

1. Create a CA EEM XML file.

2. Add the following lines to the file:

```
<Safex>
<Attach label="WorkloadAutomationAE"/>
<Add>
```

3. To define an application user group, add the following lines:

```
<UserGroup folder="/UserGroups name=">
<Description>description of user group</Description>
</UserGroup>
```

4. Repeat Step 3 for each application user group you want to define.

5. To add a global user to one or more application groups, add the following lines:

```
<User folder="/Users" name="global_user_name">
<GroupMembership>application_user_group_name</GroupMembership>
<GroupMembership>application_user_group_name</GroupMembership>
<GroupMembership>application_user_group_name</GroupMembership>
...
</User>
```

– **global\_user\_name** Specifies the name of an existing global user.

– **application\_user\_group\_name** Specifies the name of an existing application user group.

6. Repeat Step 5 for each additional application user you want to define.

7. Add the following lines to the end of the file:

```
</Add>
</Safex>
```

8. Save the XML file. The CA EEM XML file is created.

#### **Example: Create an XML File for an Application Group**

This example defines a user group XML file for application user group “operator”. This file contains all user attributes that you can use.

```
<Safex>
<Attach label="WorkloadAutomationAE"/>
```

```

<Add>
<UserGroup folder="/UserGroups" name="operator">
<Description>The group of operators</Description>
</UserGroup>
</Add>
</Safex>

```

### Example: Create an XML File to Add a Global User to an Application Group

This example defines a user XML file to add global user “johndoe” to application user groups “operator” and “reporter”. This file contains all the user attributes that you can use.

```

<Safex>
<Attach />
<Add>
<User folder="/Users" name="johndoe">
<GroupMembership>operator</GroupMembership>
<GroupMembership>reporter</GroupMembership>
</User>
</Add>
</Safex>

```

### Migrate Policies from CA AC to CA EEM

In CA AC, policies are represented by a script file containing specific commands written in selang, the CA AC command language. In CA EEM, policies are represented by an XML file using specific CA EEM XML tags. The migration process involves obtaining the subset of CA AC security policies used by Unicenter AutoSys JM 4.5.1 and translating them to an XML file containing equivalent policies for Release 11. These steps are necessary because of the differences in the policy evaluation of the two security engines and changes in the resource naming convention between Unicenter AutoSys JM 4.5.1 and Release 11.

#### NOTE

For this process to be successful, confirm that the PATH environment variable is updated to include the Java binary location.

To migrate policies from CA AC to CA EEM, complete the following tasks:

1. [Export CA AC policies to a selang file.](#)
2. [Convert the selang file to a selang XML file.](#)
3. [Convert the selang XML file to a CA EEM XML file.](#)
4. [Apply security policy changes to Unicenter AutoSys JM 4.5.1 policies.](#)
5. [Apply regular expression resource name changes to policies for the current release.](#)

#### NOTE

Policies cannot be imported until a repository (the application) exists in CA EEM. This task does not happen until you register the product. After you complete these steps, register the product with CA EEM and import your XML files.

## **Export CA AC Policies to a selang File**

To migrate your policies, first export all Unicenter AutoSys JM 4.5.1 resources from CA AC into a script file containing the selang commands that are required to duplicate the database. Export resources only from the following user-defined classes:

- as-calendar
- as-control
- as-cycle
- as-gvar
- as-job
- as-list
- as-machine
- as-owner

CA AC provides the dbmgr utility to export the necessary resources into a script file containing the selang commands that are required to duplicate the database.

### **NOTE**

For more information about how to use the dbmgr utility to export resources from the listed classes into a selang file, see the *CA ControlMinder Reference documentation*.

## **Convert the selang File to a selang XML File**

The JRE is required to convert the selang file to a selang XML file. You must identify the selang commands from the exported file and generate an equivalent XML file containing the selang commands as XML tags. After the script file is converted to an XML file, you can use an XML parser to translate the CA AC XML tags to the equivalent CA EEM XML tags.

To convert the selang file to a selang XML file, go to the \$AUTOSYS/EEMmigrate (on UNIX) or %AUTOSYS%\EEMmigrate (on Windows) directory and enter the following command:

```
java -jar se2xml.jar exported_selang_file_name
```

- **exported\_selang\_file\_name**  
Identifies the name of the selang file to convert.

An XML file with the name *exported\_selang\_file\_name.xml* is generated.

## **Convert the selang XML File to a CA EEM XML File**

The JRE and the Xalan and Serializer jar files are required to convert the selang XML file to a CA EEM XML file. An XML parser is used to identify the CA AC tags from the selang XML file and generate an XML file containing the equivalent CA EEM tags.

To convert the selang XML file to a Unicenter AutoSys JM 4.5.1 CA EEM XML file, go to the \$AUTOSYS/EEMmigrate (on UNIX) or %AUTOSYS%\EEMmigrate (on Windows) directory on the computer where AutoSys Workload Automation is installed, and enter the following command:

### **Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears.

2. Enter the following command:

```
java -cp {"EEMmigrate_Directory\xalan.jar";|EEMmigrate_Directory/
xalan.jar:}{"EEMmigrate_Directory\serializer.jar"|EEMmigrate_Directory.jar}
org.apache.xalan.xslt.Process -IN exported_selang_file_name.xml -XSL selang2eem.xsl
-OUT EEM_file_name.xml -PARAM ApplicationName WorkloadAutomationAE -PARAM
PoliciesFolder EEM_backend_server_policy_folder_name
```

– **EEMmigrate\_Directory**

Identifies the Windows or UNIX directory where the Serializer and xalan jar files are located:

- (UNIX) \$AUTOSYS/EEMmigrate
- (Windows) %AUTOSYS%\EEMmigrate

**WARNING**

Ensure that you use the correct syntax for your platform; otherwise the command will be invalid. On Windows, use a backslash (\) to separate the directory name from the file name, enclose the full path for each file in quotes ("), and separate the two file names with a semicolon. On UNIX, use a slash (/) to separate the directory name from the file name and separate the two file paths with a colon (:).

- **exported\_selang\_file\_name.xml** Identifies the selang XML file to convert.
- **EEM\_file\_name.xml** Defines the name of the CA EEM XML file to create.
- **EEM\_backend\_server\_policy\_folder\_name** Specifies the path on the CA EEM back-end server where the policies are imported. **Limits:** Precede this value with a slash. For example, you could use /MigratedPolicies.

The selang file is converted to an XML file with the name *EEM\_file\_name.xml*.

### **Apply Security Policy Changes to Unicenter AutoSys JM 4.5.1 Policies**

The JRE and the Xalan and Serializer jar files are required to convert the Unicenter AutoSys JM 4.5.1 CA EEM XML file to a CA EEM XML file for the current release. Apply the security policy changes required to work with the current release.

**Follow these steps:**

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:

- (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
- (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt appears

2. Enter the following command:

```
java -cp {"EEMmigrate_Directory\xalan.jar";|EEMmigrate_Directory/xalan.jar:}
{"EEMmigrate_Directory\serializer.jar"|EEMmigrate_Directory.jar} -Xmx128M
org.apache.xalan.xslt.Process -IN EEM_file_name.xml -XSL AutoSys.xsl -OUT
WAAE_EEM_file_name.xml
```

– **EEMmigrate\_Directory**

Identifies the Windows or UNIX directory where the Serializer and xalan jar files are located:

- (UNIX) \$AUTOSYS/EEMmigrate
- (Windows) %AUTOSYS%\EEMmigrate

**WARNING**

Ensure that you use the correct syntax for your platform; otherwise the command will be invalid. On Windows, use a backslash (\) to separate the directory name from the file name, enclose the full path

for each file in quotes ("), and separate the two file names with a semicolon. On UNIX, use a slash (/) to separate the directory name from the file name and separate the two file paths with a colon (:).

- **EEM\_file\_name.xml** Identifies the CA EEM XML file generated from the selang file.
- **WAAE\_EEM\_file\_name.xml** Defines the name of the CA EEM XML file to create. This file contains the policy changes for Unicenter AutoSys JM Release 11.

The policy changes are applied and an XML file with the name *WAAE\_EEM\_file\_name.xml* is generated.

### Apply Regular Expression Resource Name Changes to Policies for the Current Release

The JRE is required to convert the Unicenter AutoSys JM 4.5.1 CA EEM XML file to a CA EEM XML file with regular expression policies for Release 11. This migration step scans the converted CA EEM policies for any resource names containing asterisks in positions other than first or last. If a policy with these asterisks is found, the regular expression attribute of the policy is set. Every asterisk in the resource name is prefixed with a period to conform to a Perl 5 regular expression.

Open the EEMmigrate directory and enter the xalan and serializer commands:

- (UNIX)

```
{java -cp "%AUTOSYS%\EEMmigrate\xalan.jar";"%AUTOSYS%\EEMmigrate\serializer.jar"
-Xmx128M org.apache.xalan.xslt.Process -IN WAAE_EEM_file_name.xml -XSL
PostRegex.xsl -OUT final_EEM_file_name.xml|java -cp $AUTOSYS/EEMmigrate/xalan.jar:
$AUTOSYS/EEMmigrate/serializer.jar -Xmx128M org.apache.xalan.xslt.Process -
IN WAAE_EEM_file_name.xml -XSL PostRegex.xsl -OUT final_EEM_file_name.xml}
```

#### NOTE

Ensure that the Xalan and Serializer jar file names are in quotes and separated by a colon (UNIX) or a semicolon (Windows).

- **WAAE\_EEM\_file\_name.xml** Identifies the CA EEM XML file to modify.
- **final\_EEM\_file\_name.xml** Defines the name of the final CA EEM XML file to create.

An XML file with the name *final\_EEM\_file\_name.xml* is generated.

### Change CA EEM Backup XML File Values for Unicenter AutoSys JM 4.5.1

To ensure that your XML file containing policies created after migrating them from Unicenter AutoSys JM 4.5.1 work with more current AutoSys Workload Automation releases, first clean the file of invalid data for legacy Unicenter AutoSys JM Release 11.

#### Follow these steps:

1. Open the XML file in a text editor.
  2. Make the following changes:
    - a. In the Attach label, replace UnicenterAutoSysJM with WorkloadAutomationAE.
    - b. Search for the Folder tag and replace UnicenterAutoSysJM\_Common with WorkloadAutomationAE\_Common and UnicenterAutoSysJM\_XXX with WorkloadAutomationAE\_XXX.
    - c. For each Policy tag, replace /UnicenterAutoSysJM\_XXX with /WorkloadAutomationAE\_XXX and UnicenterAutoSysJM\_Common with WorkloadAutomationAE\_Common.
      - **XXX** Specifies the AutoSys Workload Automation application instance.
    - d. In the Policy tag, search for UnicenterAutoSysJM and replace it with WorkloadAutomationAE.
    - e. Search all other tags and replace UnicenterAutoSysJMAAdmin with WorkloadAutomationAEAdmin, UnicenterAutoSysJM with WorkloadAutomationAE, and Unicenter AutoSys JM with AutoSys Workload Automation.
- The XML file is now clean of obsolete strings, labels, and values specific to Unicenter AutoSys JM Release 11.

## **Register the Product in the Target CA EEM Instance Using as\_safetool**

The AutoSys Workload Automation as\_safetool command is used to register AutoSys Workload Automation instances in the CA EEM instance. Use as\_safetool to register new CA Workload Automation AE Release 11.3.6 instances in the CA EEM instance; it cannot be done using safex. The as\_safetool command is also used to complete the migration process by upgrading the WorkloadAutomationAE application instance from a previous release to Release 11.3.6. If you are using as\_safetool to upgrade the WorkloadAutomationAE application instance to the current release, only register one of the instance names that are represented in the previous release of the WorkloadAutomationAE application instance.

### **WARNING**

If you are migrating security policy to Release 11.3.6, run as\_safetool to upgrade the application instance. If the CA EEM XML file you are using contains an XML tag named "Register", do this step **after** completing the registration process using safex. This requirement is because the CA EEM XML file you are using to register the product in the target CA EEM is a captured copy of a Release 11.3 application instance. If the CA EEM XML file does not contain the XML tag named "Register", **first** register the current release of the product with CA EEM using as\_safetool. This requirement is because the application instance import it to does not yet exist in CA EEM.

### **NOTE**

Before you perform this task, use the setenv or export command to set the ASSAFETOOLPW environment variable to the password of the EiamAdmin user, the CA EEM server administrative account. For example, export ASSAFETOOLPW=mypass.

### **Follow these steps:**

1. Open a AutoSys Workload Automation command prompt.
2. Enter the following command:

```
as_safetool -b EEM_server_host_name -s
```

A list of AutoSys Workload Automation instances that are already registered with the CA EEM server appear.

3. Enter the following command for the instance that is added to that list:

```
as_safetool -b EEM_server_host_name -i instance
```

4. (Optional) Repeat Step 3 for any other AutoSys Workload Automation instances you register. When you have deregistered all AutoSys Workload Automation instances, the product is automatically registered in the source CA EEM instance. The product is registered in the target CA EEM instance.

## **Import the CA EEM XML File Using safex**

You can use the backup security policy file to migrate objects from a previous release of the product on the target CA EEM server.

### **WARNING**

To import a CA EEM XML file into the target CA EEM instance using safex, **first** register the current release of the product with CA EEM using as\_safetool. This requirement is because the application instance you import it to does not yet exist in CA EEM.

To import the XML file, open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

- **EEM\_servername**

Specifies the IP address or host name of the computer on which CA EEM is installed.

- **admin\_user**

Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.

- **password**

Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.

- **XML\_file**

Specifies the path and name of the CA EEM XML file.

- **-fips**

Specifies the path and name of the CA EEM XML file.

### **Verify Imported Security Policies from Unicenter AutoSys JM 4.5.1**

Resource applications function properly only when the security policies in place comply with the requirements of the AutoSys Workload Automation release that you are using. After you migrate security policies, we recommend that you verify that the access policies comply with the requirements of the current release.

To verify imported security policies for AutoSys Workload Automation, review the following topic to ensure that your imported policies comply:

[Security Policy Changes from Unicenter AutoSys JM 4.5.1](#)

## **CA EEM Security Data Replication, Backup, and Migration**

When using CA EEM security, follow the data replication, backup, and migration scenarios in this section to achieve any of the following goals:

- Generate a backup copy of your security policies every time you modify data.

### **NOTE**

In this case, no additional steps are required after the export except for copying the backup to a secure location.

- Simplify replication of up-to-date security policies onto replacement or reserve CA EEM servers.
- Quickly recreate your security policies on a replacement CA EEM instance in a critical failure situation.
- Migrate security policies from previous product releases when you upgrade AutoSys Workload Automation and Web UI to a new release.

Besides CA EEM, the tools used to migrate your existing security policies to the current release are as follows:

- **AutoSys Workload Automation as\_safetool Utility**

Installs the new and updated resource classes and default policies for all instances that require migration. AutoSys Workload Automation installs the as\_safetool utility.

- **CA EEM safex Utility**

Imports the final generated XML file containing the migrated policies to the CA EEM server.

### **NOTE**

The safex utility is only available on a computer where the CA EEM back-end server is installed. This utility is located in the bin subdirectory of the CA EEM installation path. The safex utility directs all output to stderr. We recommend that you capture this output and save it to a file so you can examine errors.

- **AutoSys Workload Automation autosys\_secure Utility**

Manages security-related settings including security certificate creation.

Where applicable, the scenarios provide you with a choice to perform the individual tasks using CA EEM or the command-line utilities (safex, as\_safetool, and autosys\_secure). You export security policies from CA EEM instances as XML files.

This article contains six scenarios:

- [Capture a Copy of Your CA EEM Security Policies](#)

Follow the instructions in this scenario when you want to export AutoSys Workload Automation or Web UI security policies for backup and replication purposes.

- [Configure a New CA EEM Instance With Existing AutoSys Workload Automation AE Security Policies](#)  
Follow the instructions in this scenario when you want to import saved AutoSys Workload Automation security policies to a CA EEM instance. For example, complete this scenario when you replace a failed CA EEM server, or want to use a different CA EEM server.
- [Configure a New CA EEM Instance With Existing Web UI Security Policies](#)  
Follow the instructions in this scenario when you want to import saved Web UI security policies to a CA EEM instance. For example, complete this scenario when you replace a failed CA EEM server, or you want to use a different CA EEM server.
- [Migrate CA Workload Automation AE Security Policies from Unicenter AutoSys JM Release 11](#)  
Follow the instructions in this scenario when you want to clean up and reuse Unicenter AutoSys JM Release 11 security policies with a new AutoSys Workload Automation release.
- [Migrate AutoSys Workload Automation Security Policies from CA Workload Automation AE Release 11.3](#)  
Follow the instructions in this scenario when you want to clean up and reuse AutoSys Workload Automation Release 11.3 security policies with a new AutoSys Workload Automation release.
- [Migrate Web UI Security Policies from CA WCC Release 11.1 Service Packs or Release 11.3](#)  
Follow the instructions in this scenario when you want to clean up and reuse CA WCC Release 11.1 SP1 (and later service packs) or Release 11.3 security policies with a new Web UI release.

#### NOTE

When you upgrade CA Workload Automation AE from Release 11.3.5 to Release 11.3.6, the security policies are migrated to Release 11.3.6. We did not change the security policies from Release 11.3.5 to Release 11.3.6.

## Capture a Copy of Your CA EEM Security Policies

This procedure explains how to capture a copy of your CA EEM security policies.

As an administrator, you sometimes operate in a complex environment that uses CA EEM security and has many AutoSys Workload Automation and AutoSys Web UI (Web UI) security policies defined. CA EEM database or server failures can cause you to lose the security policy data, and recreating the policies without a backup copy is complicated and time consuming. To ensure that you can restore your security policies in such cases, you can set up your system to generate a backup copy of the policies regularly.

You can also capture a copy of your security policies for the following purposes:

- To migrate the current policies from a previous release when you upgrade to a new release
- To import the existing policies to a new instance when you change CA EEM servers
- To update the backup copy after you modify policies

#### NOTE

We recommend that you manually capture a copy of policies each time that you modify the policies, before you upgrade to a new release, and before you change CA EEM servers even if your system is configured to generate backup copies at regular intervals. Creating a backup manually ensures that the file that you use to restore, replicate, or migrate your policies has the most up-to-date policies.

To capture a copy of your CA EEM security policies, follow these steps:

1. Export the security policies to an XML file using one of the following tools:
  - [CA EEM](#)
  - [safex utility](#)
2. [Copy the XML file to the appropriate location.](#)

#### NOTE

You can store the file in a directory on your CA EEM server or in any other secure location.



## **Export the Security Policies to an XML File Using CA EEM**

The CA EEM web interface enables you to export security policies to an XML file without the complex scripting requirements that are required when you use the safex utility to export the policies.

### **Follow these steps:**

1. Open a browser and go to the following URL:

`http://EEM_servername:5250/spin/eiam`

#### **EEM\_servername**

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

2. Select your Web UI instance (for example, WCC0004) from the Application drop-down list, enter EiamAdmin or the name of another user with CA EEM Administrative privileges in the User Name field, enter the appropriate password in the Password field, and click Log In.  
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.  
Multiple links are displayed, with the Applications link selected by default.
4. Select the CA EEM Server link.  
Multiple links are displayed in the left pane.
5. Select the Export Application link.  
The Export Application dialog is displayed in the right pane.
6. Select or clear the check boxes as appropriate, and click Export.

#### **NOTE**

Clear all Global objects check boxes if you are using CA EEM Active Directory.

The File Download dialog is displayed.

7. Select Save.  
The Save As dialog is displayed.
8. Specify a file name and a location to save the file to, and click Save.

#### **NOTE**

CA EEM saves the XML file as a GZIP archive.

The security policies are exported to the specified XML file.

#### **NOTE**

After you export from CA EEM Release 12.6, open the exported xml file and replace all instances of "GroupName" with "Name".

## **Export the Security Policies to an XML File Using safex**

You can export CA EEM data to an XML file using the safex utility. The CA EEM web interface does not require scripting, but you can only use the web interface to export data when you are logged in to the CA EEM server machine through a web browser.

### **Follow these steps:**

1. Log in to the CA EEM back-end server and navigate to the following directory:

#### **Windows:**

CA EEM Release 12.6 — EEM\_installation\_root\bin\

CA EEM r8.4 — c:\Program Files\CA\SharedComponents\iTechnology

#### **UNIX:**

CA EEM Release 12.6 — EEM\_installation\_root/bin/

CA EEM r8.4 — /opt/CA/SharedComponents/iTechnology

## 2. Enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

### NOTE

You must use the safex version of the same CA EEM installed version to export. For example, the safex version shipped with Release 11.3.6 does not work when exporting policies from CA EEM 8.x.

- **EEM\_servername**  
Specifies the IP address or host name of the computer on which EEM is installed.
- **admin\_user**  
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- **password**  
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- **XML\_file**  
Specifies the name of the XML file that contains the commands for exporting policies. The input safex file can include options for globalfolders, global usergroups, globalusers, and so on.
- **-fips**  
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

For Web UI , the content of the input safex XML file is similar to the following:

```
<Safex>
<Attach label="WCC0003"/>
<Export file="WCC0003-backup.xml" globalfolders="y" globalusergroups="y" globalusers="y" globalsettings="y"
folders="y" usergroups="y" users="y" calendars="y" policies="y" appobjects="y"/>
<Detach/>
</Safex>
```

### NOTE

- WCC0003 is the value for CA WCC r11.3. If you are migrating from a CA WCC r11.1 service pack, the value is WCC0002.
- If you do not want to export an object in the file, you must change "y" to "n". For example, if you are using Active Directory or an external LDAP with CA EEM, you would enter "n" for the global entries

The XML file is created in the location specified by the Export file tag of the input safex XML file. All the security policies for the specified product are retrieved from the CA EEM instance. This XML file contains the policies for all instances of the specified product.

## Copy the Exported XML File to the Appropriate Location

After exporting the existing security policies from the source CA EEM server, copy the file to a secure location. We recommend that you copy the entire GZIP archive containing the XML file to a location on the appropriate server. For example, you can copy the WCC0004-backup.xml file to a directory on the Web UI server that accesses the target CA EEM instance.

## Configure a New CA EEM Instance with AutoSys Workload Automation Security Policies

This article explains how to configure a new CA EEM instance with AutoSys Workload Automation security policies.

As an administrator, you sometimes operate in CA EEM-enabled environments where CA EEM does not use High Availability. Therefore, you keep a backup CA EEM server on standby. You can minimize downtime by configuring the backup CA EEM instance to use existing AutoSys Workload Automation security policies. To configure the backup server to use the existing policies, replicate the security policies to the backup CA EEM server by importing the XML file that contains the current policies.

Importing existing AutoSys Workload Automation security policies to a new CA EEM instance is also useful in these situations:

- To replace your CA EEM instance with another physical server. For example, when your environment outgrows the hardware capabilities of the existing CA EEM server and you need to configure a freshly added server as your CA EEM instance.
- To reset security policies to their default values when you revert to a fresh instance from a test environment.

As a prerequisite, you must have a current XML file containing the CA EEM security policies for each AutoSys Workload Automation instance.

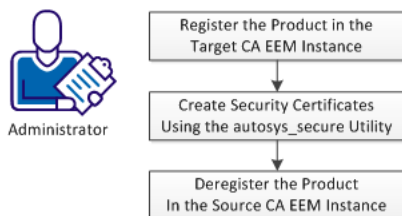
### NOTE

An XML file containing default security policies, UWCCRegister.xml, is created in the safex subdirectory of the root Web UI installation directory during installation. You can also use this file to register Web UI with a CA EEM instance, for example, if you want to reset all your security policies to their default values. The following considerations apply:

- You must recreate all previously defined custom Web UI security policies.
- The default XML file already contains the correct certificate definitions. If you use this file in place of a backed-up XML file, skip the procedure described in Edit the CA EEM Backup XML File.

This scenario explains the process to restore your AutoSys Workload Automation security policies into a new CA EEM server instance. This diagram illustrates the process to configure a new CA EEM instance with existing AutoSys Workload Automation security policies:

**How to Configure a New CA EEM Instance with Existing Autosys Workload Automation Security Policies**



To configure a new CA EEM instance with existing AutoSys Workload Automation security policies, follow these steps:

1. [Register your AutoSys Workload Automation instance in the target CA EEM instance.](#)
2. [Create certificate files for your AutoSys Workload Automation instance.](#)
3. Deregister your AutoSys Workload Automation instance in the source CA EEM instance using one of the following methods:
  - [CA EEM](#)
  - [as\\_safetool utility](#)

After you complete these tasks, you can remove the intermediate XML files created during the process.

### **Register the Product in the Target CA EEM Instance Using as\_safetool**

The AutoSys Workload Automation `as_safetool` command is used to register AutoSys Workload Automation instances in the CA EEM instance. Use `as_safetool` to register new CA Workload Automation AE Release 11.3.6 instances in the CA

EEM instance; it cannot be done using safex. The `as_safetool` command is also used to complete the migration process by upgrading the AutoSysWorkloadAutomation application instance from a previous release to Release 11.3.6. If you are using `as_safetool` to upgrade the AutoSysWorkloadAutomation application instance to the current release, only register one of the instance names that are represented in the previous release of the AutoSysWorkloadAutomation application instance.

#### NOTE

Before you perform this task, you must use the `setenv` or `export` command to set the `ASSAFETOOLPW` environment variable to the password of the EiamAdmin user, the CA EEM server administrative account. For example, `export ASSAFETOOLPW=myspass`.

#### Follow these steps:

1. Open a AutoSys Workload Automation command prompt.
2. Enter the following command:

```
as_safetool -b EEM_server_host_name -s
```

A list of AutoSys Workload Automation instances that are registered with the CA EEM server appear.

3. Enter the following command for the instance that should be added to that list:

```
as_safetool -b EEM_server_host_name -i instance
```

The AutoSys Workload Automation instance is registered in the target CA EEM instance.

#### NOTE

The `as_safetool` command installs some default CA EEM policies for each AutoSys Workload Automation instance. We recommend that you review these policies and update them accordingly.

### Create Security Certificates Using `autosys_secure`

To finalize the registration of your AutoSys Workload Automation instance with a CA EEM instance, create security certificates for AutoSys Workload Automation. This task is done using the `autosys_secure` utility.

#### Follow these steps:

1. Log in to AutoSys Workload Automation as the EDIT superuser and enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosys_secure
```

The following menu appears:

```
Please select from the following options:
```

- ```
[1] Activate EEM instance security.
[2] Manage EDIT/EXEC superusers.
[3] Change database password.
[4] Change remote authentication method.
[5] Manage user@host users.
[6] Get Encrypted Password.
[0] Exit CA WAAE Security Utility.
```

2. Enter 1 and press the Enter key.
3. Enter the CA EEM server details when prompted.

NOTE

You can specify multiple CA EEM host names in a comma-separated format. However, each host name must be fewer than 80 characters and the entire list must be fewer than 255 characters.

The security certificates are created, and the following message is displayed:

```
CAUAJM_I_60201 EEM instance security successfully set.
```

Deregister the Product in the Source CA EEM Instance Using CA EEM

To use the target CA EEM server for authentication, deregister the product in the source CA EEM instance.

Follow these steps:

1. Open a browser and go to the following URL:

```
http://EEM_servername:5250/spin/eiam
```

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

2. Select Global from the Application drop-down list, enter the name of a CA EEM administrative user in the User Name field, enter the corresponding password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Do one of the following tasks:
 - For AutoSys Workload Automation, select WorkloadAutomationAE.
 - For Web UI, select WCC0004.
 The Application Instance options are displayed in the right pane.
5. Click Unregister.
The selected product is deregistered in the source CA EEM instance.

Deregister the Product in the Source CA EEM Instance Using as_safetool

To use the target CA EEM server for authentication, deregister the product in the source CA EEM instance.

NOTE

Before you perform this task, use the setenv or export command to set the ASSAFETOOLPW environment variable to the password of the EiamAdmin user, the CA EEM server administrative account. For example, export ASSAFETOOLPW=mypass.

Follow these steps:

1. Open a AutoSys Workload Automation command prompt.
2. Enter the following command:

```
as_safetool -b EEM_server_host_name -s
```


A list of AutoSys Workload Automation instances that are registered with the CA EEM server appear.
3. Enter the following command for the instance that should be removed from that list:

```
as_safetool -b EEM_server_host_name -u instance
```
4. (Optional) Repeat Step 3 for any other AutoSys Workload Automation instances that you need to deregister. When you have deregistered all AutoSys Workload Automation instances, the product is automatically deregistered in the source CA EEM instance.
The AutoSys Workload Automation instance or instances are deregistered in the source CA EEM instance.

Configure a New CA EEM Instance with Existing Web UI Security Policies

As an administrator, you sometimes operate in environments that do not support CA EEM failover. You can replicate existing Web UI security policies to a new CA EEM server following a critical failure to configure the new instance quickly with the same security policy as the failed instance.

Importing existing Web UI security policies to a new CA EEM instance is also useful in the following situations:

- To replace your CA EEM instance with another physical server. For example, when your environment outgrows the hardware capabilities of the existing CA EEM server and you need to configure a freshly added server as your CA EEM instance.
- To reset security policies to their default values when you revert to a fresh instance from a test environment.

As a prerequisite, you must have a current XML file containing the CA EEM security policies for each Web UI instance.

NOTE

An XML file containing default security policies, UWCCRegister.xml, is created in the safex subdirectory of the root Web UI installation directory during installation. You can also use this file to register Web UI with a CA EEM instance, for example, if you want to reset all your security policies to their default values. The following considerations apply:

- You have to recreate all previously defined custom Web UI security policies.
- The default XML file already contains the correct certificate definitions. If you use this file in place of a backed-up XML file, skip the procedure described in [Edit the CA EEM Backup XML File](#).

To configure a new CA EEM instance with existing Web UI security policies, follow these steps:

1. [Edit the XML file that contains the backup CA EEM security policies for your Web UI instance](#).
2. (Optional) Deregister the product in the target CA EEM instance using one of the following tools:
 - [CA EEM](#)
 - [safex utility](#)

Use this option when you are importing the policies to a system that contains previously configured Web UI security policies.
3. [Register the product in the target CA EEM instance](#)
4. [Copy the certificate files onto your Web UI server](#).
5. (Optional) [Configure a new CA EEM instance using the CECL utility](#).
Use this option when your current CA EEM server cannot be used for authentication because it has suffered a critical failure.
6. [Update the target CA EEM server definition on your Web UI instance](#).
7. Deregister the product in the source CA EEM instance using one of the following tools:
 - [CA EEM](#)
 - [safex utility](#)

Edit the CA EEM Backup XML File

Edit the backup XML file so that you create the correct certificate.

Follow these steps:

1. Open the file in a text editor.
2. Locate the following line:


```
<Register certfile="certfile.pl2" password="abc123">
```
3. Modify the line as follows:


```
<Register certtype="pem" certfile="wcc_new.pem" keyfile="wcc_new.key">
```
4. Save the XML file.

The correct certificates are created when you use this XML file to register your Web UI instance with a CA EEM server.

Deregister the Product in the Target CA EEM Instance Using CA EEM

Sometimes you want to configure a CA EEM to use existing Web UI policies on a server that contains previously defined security policies. You cannot define the new policies on a server until you clear the previously defined policies from it.

We also recommend clearing the policies from the source server before you define policies on a target server, but it is not required. Clearing the obsolete policies from the source server helps to ensure that the server is not overloaded with unnecessary data. To clear previously defined policies from a CA EEM instance, deregister the product on that instance.

Follow these steps:

1. Open a browser and go to the following URL:

```
http://EEM_servername:5250/spin/eiam
```

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

2. Select Global from the Application drop-down list, enter the name of a CA EEM administrative user in the User Name field, enter the corresponding password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Do one of the following tasks:
 - For AutoSys Workload Automation, select WorkloadAutomationAE.
 - For Web UI, select WCC0004.
 The Application Instance options are displayed in the right pane.
5. Click Unregister.
The selected product is deregistered in the source CA EEM instance.

Deregister the Product in the Target CA EEM Instance Using safex

Sometimes you want to configure a CA EEM to use existing Web UI policies on a server that contains previously defined security policies. You cannot define the new policies on a server until you clear the previously defined policies from it.

We also recommend clearing the policies from the source server before you define policies on a target server, but it is not required. Clearing the obsolete policies from the source server helps to ensure that the server is not overloaded with unnecessary data. To clear previously defined policies from a CA EEM instance, deregister the product on that instance.

To deregister the current product, open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f UWCCUnRegister.xml -fips
```

- ***EEM_servername***
Specifies the IP address or host name of the computer on which CA EEM is installed.
- ***admin_user***
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- ***password***
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- ***-fips***
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

The product is deregistered in the source CA EEM instance.

Register the Product in the Target CA EEM Instance Using safex

You can use the backup security policy XML file to register the product in the target CA EEM instance.

To register the product in the target CA EEM instance, open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

- ***EEM_servername***
Specifies the IP address or host name of the computer on which CA EEM is installed.
- ***admin_user***
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- ***password***
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- ***XML_file***
Specifies the path and name of the CA EEM XML file.
- ***-fips***
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

The product is registered in the target CA EEM instance.

Copy the Certificate Files to the Web UI Config Directory

To enable your Web UI instance to use the target CA EEM instance for authentication, copy the certificate file and key file into the config directory on the Web UI server.

To copy the certificate files, copy the *wcc_new.pem* certificate file and *wcc_new.key* file to *WCC_installation_root\data\config* file on Windows or the *WCC_installation_root/data/config* file on UNIX.

Your Web UI instance now has the correct certificate files set up and can use the target CA EEM server for authentication.

Disaster Recovery - Configure a New CA EEM Instance Using the CECL Utility

If the CA EEM server that is configured for your Web UI instance cannot be used for authentication, you are unable to change the CA EEM server in Configuration. When your CA EEM server has suffered a critical failure, you must install a new CA EEM instance.

Use the Change CA EEM Command Line (CECL) utility to configure a new CA EEM server for Web UI when your current CA EEM server cannot be used to authorize such a change.

The CECL executable is located in the bin folder under the Web UI installation root. On Windows, it is the batch file (*change_eem.bat*); on UNIX, it is the shell script (*change_eem.sh*). Before you run the CECL, change your current directory to the following directory which contains the *change_eem* executable:

- Windows:
WCC_installation_root\bin
- UNIX
WCC_installation_root/bin/

To configure a new CA EEM instance, use the following command syntax:

```
change_eem -eemadmin username -eempassword password -eemappid WCC0004  
-eemhostname eem_host_name - eemcertname eem_certificate_name -  
eemcertkey eem_certificate_key [-v] [-?]
```

- ***eem_host_name***

Specifies the name of the new CA EEM host.

- **eem_administrator**
Specifies the administrator name for the new instance of CA EEM.
- **eem_password**
Specifies the password for the administrator of the new instance of CA EEM.
- **eem_application_id**
(Optional) Specifies the application ID that contains the Web UI policies, for example, WCC0004.
- **eem_certificate_name**
Specifies the CA EEM certificate file name. This file is in the following location:
 - Windows
`WCC_installation_root\data\config`
 - UNIX
`WCC_installation_root/data/config`
- **eem_certificate_key**
Specifies the CA EEM certificate key file name. This file is in the following location:
 - Windows
`WCC_installation_root\data\config`
 - UNIX
`WCC_installation_root/data/config`
- **-v**
(Optional) Displays detailed processing messages.

The new CA EEM instance is configured.

Example: Change Your CA EEM Server to Instance wcc-eem4

The following command changes the CA EEM instance used for Web UI authentication to wcc-eem4:

```
change_eem -eemadmin admin_user -eempassword admin01 -eemappid WCC0004 -eemhostname wcc-  
eem4 - eemcertname wcc_new.pem - eemcertkey wcc_new.key
```

WARNING

After you update the CA EEM server definition, restart the services CA-wcc and CA-wcc-services on Windows, or the service CA-wcc-services on UNIX. Use the Services tool in the Control Panel to restart the services on Windows. Use the `unisrvctr` command to restart the services on UNIX. For more information, see [Web UI Installation](#).

Update the Target CA EEM Server Using `wcc_config.bat`

To use the target CA EEM instance for Web UI authentication, you must update the CA EEM server definition in the configuration of your Web UI instance.

To update the CA EEM server definition in Web UI, use the following command syntax:

```
wcc_config -u user_name -p password --eemhostname eem_host_name --  
eemadmin eem_administrator --eempassword eem_password [--eemappid eem_application_id] --  
eemcertname eem_certificate_name --eemcertkey eem_certificate_key
```

- **user_name**
Specifies the authorized user name.
- **password**

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

- **eem_host_name[eem_host_name2,eem_host_name3...]**

Specifies one or more host names for the new CA EEM servers that you want the Web UI instance to reference. This host is used to authenticate the user name and password.

NOTE

If CA EEM is configured for failover, you can optionally specify a list of the servers in the failover setup. Separate server names by commas and list them in the order of reliability.

- **eem_administrator**

Specifies the administrator name for the new instance of CA EEM.

- **eem_password**

Specifies the password for the administrator of the new instance of CA EEM.

- **eem_application_id**

(Optional) Specifies the application ID that contains the Web UI policies, for example, WCC0004.

- **eem_certificate_name**

Specifies the CA EEM certificate file name. This file is in the following location:

- Windows

`WCC_installation_root\data\config`

- UNIX

`WCC_installation_root/data/config`

- **eem_certificate_key**

Specifies the CA EEM certificate key file name. This file is in the following location:

- Windows

`WCC_installation_root\data\config`

- UNIX

`WCC_installation_root/data/config`

The CA EEM server definition in Web UI is updated.

WARNING

After you update the CA EEM server definition, restart the services CA-wcc and CA-wcc-services on Windows, or the service CA-wcc-services on UNIX. Use the Services tool in the Control Panel to restart the services on Windows. Use the `unisrvctr` command to restart the services on UNIX. For more information, see [Web UI Installation](#).

Example: Update the CA EEM Server Definition

The following command updates the CA EEM server definition in Web UI when you use the optional CA EEM certificate name and keyfile:

```
wcc_config -u ejmcommander -p ejmcommander --eemhostname EEM_01 --eemadmin admin_user --
eempassword admin --eemcertname certificate_01.pem --eemcertkey eem_cert_key_01.key - -
eemappid WCC0004
```

After you run the command, restart the services CA-wcc and CA-wcc-services on Windows, or the service CA-wcc-services on UNIX.

Deregister the Product in the Source CA EEM Instance Using CA EEM

We recommend that you deregister the product in the source CA EEM instance before you register it in the target instance, although it is not required. Deregistering the product cleans up the source server.

Follow these steps:

1. Open a browser and go to the following URL:

`http://EEM_servername:5250/spin/eiam`

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

2. Select Global from the Application drop-down list, enter the name of a CA EEM administrative user in the User Name field, enter the corresponding password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Do one of the following tasks:
 - For AutoSys Workload Automation, select WorkloadAutomationAE.
 - For Web UI, select WCC0004.
The Application Instance options are displayed in the right pane.
5. Click Unregister.

The selected product is deregistered in the source CA EEM instance.

Deregister the Product in the Source CA EEM Instance Using safex

We recommend that you deregister the product in the source CA EEM instance before you register it in the target instance, although it is not required. Deregistering the product cleans up the source server.

To deregister the current product, open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f UWCCUnRegister.xml -fips
```

- ***EEM_servername***
Specifies the IP address or host name of the computer on which CA EEM is installed.
- ***admin_user***
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- ***password***
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- ***-fips***
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

The product is deregistered in the source CA EEM instance.

Configure CA EEM for Root Certificates Generated with Different Key Lengths

From Release 11.4 SP7, CA WCC, requires root certificates with a key length of 2048 or more. By default, CA EEM generates root certificates with a key length of 1024. CA Workload Automation AE uses 1024 key length certificates.

You can generate root certificates with various key lengths (2048 or more) and use them on a EEM Server.

Also, in some instances, previous and current releases of CA WCC using root certificates with key lengths of 1024 and 2048 may point to the same CA EEM server. Before you install or upgrade WCC, you must modify the CA EEM server certificate.

For CA EEM to work with key lengths of 1024 and 2048 (or more) follow these steps:

Follow these steps:**On the EEM server, complete the following steps:**

1. Copy the rootcert.cer and rootcert.key file, found in the iTechnology directory, in the same folder and keep them in a drive of your choice, for example c:\1024_certificate\.

NOTE

After 2048 key length certificates are generated, if you want to use 1024 and 2048 keys, you have to import both 1024 and 2048 certificates.

The iTechnology directory is at the following locations:

Windows: C:\Program Files\CA\SC\iTechnology

Linux: /opt/CA/SharedComponents/iTechnology

2. Generate a certificate with custom key length (for example, 2048).

NOTE

For information about how to generate the certificate, see the section [How to Generate the Certificates](#).

3. (Optional) Copy the rootcert.cer and rootcert.key file to the c:\2048_certificate\ folder on the computer where you will open the browser to import the certificates.

NOTE

- The rootcert.cer and rootcert.key files must be copied to the same folder on the computer where you will open the browser.
- If you want to use 2048 key length certificates, import those certificates into the above mentioned folder.

On your computer, complete the following steps:

1. Open a browser after successful verification and type the following URL in the address field, and click Go:

https://<EEMBOX MACHINE NAME>:5250/spin

The CA iTechnology SPIN log in page opens.

2. Select iTech Administrator and click Go.

The iTechnology Administrator page opens.

3. Click Login.

The CA iTechnology log in window opens.

4. Enter the Eiamadmin username and password, select Type: iAuthority, and click Log In.

Select the Configure tab and complete the **Add Trusted Root** information.

5. Complete the following options:

Label: 1024_certs

PEM Certificate File: Click Choose File and upload the rootcert.cert backup file, for example, c:\1024_certificate\.

NOTE

In Step 3 of the previous procedure on the EEM server, you imported 2048 certificates; now you are importing 1024 certificates so that EEM works with both 1024 and 2048 key length certificates.

6. Click Add Trusted Root.

The successful invocation is displayed.

CA EEM is now configured to work with both the 1024 and 2048 key length options; the client generated certificates for the 1024 and 2048 key lengths are displayed in the Label column of the window.

Migrate AutoSys Workload Automation Security Policies from Unicenter AutoSys JM Release 11

As an administrator, you upgrade the environment when new releases become available. You want to migrate your existing security policies to minimize the workload intensity of the upgrade process.

The following scenario explains the process of migrating your existing AutoSys Workload Automation security policies from Unicenter AutoSys JM Release 11 to CA Workload Automation AE Release 11.3.5 or Release 11.3.6.

NOTE

When you upgrade CA Workload Automation AE from Release 11.3.5 to Release 11.3.6, the security policies are migrated to Release 11.3.6. We did not change the security policies from Release 11.3.5 to Release 11.3.6.

To migrate AutoSys Workload Automation security policies, follow these steps:

1. Export the security policies to an XML file using one of the following tools:
 - [CA EEM](#)
 - [Safex utility](#)
2. [Change CA EEM backup XML file values.](#)
3. [Register the product in the target CA EEM instance using safex.](#)
4. [Register the product in the target CA EEM instance using as_safetool.](#)
5. [Verify imported security policies.](#)

Security Policy Changes from Unicenter AutoSys JM Release 11

This topic describes the changes to the security policy implementation from Unicenter AutoSys JM Release 11 to CA Workload Automation AE Release 11.3.5 or Release 11.3.6. Until you complete the migration process to apply these changes, you cannot import the Unicenter AutoSys JM Release 11 security policies into CA Embedded Entitlements Manager (CA EEM) servers that are linked with CA Workload Automation AE Release 11.3.5 or Release 11.3.6.

This list details the security policy changes from Unicenter AutoSys JM Release 11 to CA Workload Automation AE Release 11.3.5 or Release 11.3.6:

- CA Workload Automation AE Release 11.3.5 or Release 11.3.6 does not use the UnicenterAutoSysJM CA EEM application instance. Migrate all policies for the deprecated CA EEM application instance to the WorkloadAutomationAE CA EEM application instance.
- The addition of the as-base-jobtype resource class with EXECUTE access mode supports the authorization of jobs based on their job types. This new resource class includes a default policy that grants EXECUTE access for all jobs in the AutoSys Workload Automation instance.
- The addition of the as-resource resource class supports the authorization of AutoSys Workload Automation resources.
- Updates to the as-control resource class support the new policy resources named *AGGREGATE* and *SENDEVENT_GRPAPP* for the sendevent command.
- Updates to the as-control resource class support the new policy resource named *ARCHIVE_JOBS* for the archive_jobs command.
- Updates to the as-list resource class support a new policy resource named *FORECAST* for the forecast command.
- Updates to the as-appl and as-group resource classes support the READ and WRITE access levels.

Export the Security Policies to an XML File Using CA EEM

The CA EEM web interface enables you to export security policies to an XML file without the complex scripting requirements that are required when you use the safex utility to export the policies.

Follow these steps:

1. Open a browser and go to the following URL:

`http://EEM_servername:5250/spin/eiam`

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.
The CA EEM login page appears.

2. Select your Web UI instance (for example, WCC0003) from the Application drop-down list, enter EiamAdmin or the name of another user with CA EEM Administrative privileges in the User Name field, enter the appropriate password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Select the EEM Server link.
Multiple links are displayed in the left pane.
5. Select the Export Application link.
The Export Application dialog is displayed in the right pane.
6. Select or clear the check boxes as appropriate, and click Export.

NOTE

Clear all Global objects check boxes if you are using CA EEM Active Directory.

The File Download dialog is displayed.

7. Select Save.
The Save As dialog is displayed.
8. Specify a file name and a location to save the file to, and click Save.

NOTE

CA EEM saves the XML file as a GZIP archive.

The security policies are exported to the specified XML file.

NOTE

After you export from CA EEM Release 12.6, open the exported xml file and replace all instances of "GroupName" with "Name".

Export the Security Policies to an XML File Using safex

You can export CA EEM data to an XML file using the safex utility. The CA EEM web interface does not require scripting. You can only use the web interface to export data when you are logged in to the CA EEM server machine through a web browser.

Follow these steps:

1. Log in to the CA EEM back-end server and navigate to the following directory:

Windows:

CA EEM Release 12.6 — EEM_installation_root\bin\

CA EEM r8.4 — c:\Program Files\CA\SharedComponents\iTechnology

UNIX:

CA EEM Release 12.6 — EEM_installation_root/bin/

CA EEM r8.4 — /opt/CA/SharedComponents/iTechnology

2. Enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

NOTE

Use the safex version of the same CA EEM installed version to export. For example, the safex version that is shipped with Release 11.3.6 does not work when exporting policies from CA EEM 8.x.

- **EEM_servername**
Specifies the IP address or host name of the computer on which CA EEM is installed.
- **admin_user**
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- **password**

Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.

- **XML_file**

Specifies the name of the XML file that contains the commands for exporting policies. The input safex file can include options for globalfolders, global usergroups, globalusers, and so on.

- **-fips**

Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

For Web UI, the content of the input safex XML file is similar to the following sample:

```
<Safex>
  <Attach label="WCC0003"/>
  <Export file="WCC0003-backup.xml" globalfolders="y" globalusergroups="y" globalusers="y" globalsettings="y"
  folders="y" usergroups="y" users="y" calendars="y" policies="y" appobjects="y"/>
  <Detach/>
</Safex>
```

NOTE

- WCC0003 is the value for CA WCC r11.3. If you are migrating from a CA WCC r11.1 service pack, the value is WCC0002.
- If you do not want to export an object in the file, change "y" to "n". For example, if you are using Active Directory or an external LDAP with CA EEM, enter "n" for the global entries.

The XML file is created in the location specified by the Export file tag of the input safex XML file. All the security policies for the specified product are retrieved from the CA EEM instance. This XML file contains the policies for all instances of the specified product.

Change CA EEM Backup XML File Values for Unicenter AutoSys JM Release 11

To ensure that your XML file containing policies created by Unicenter AutoSys JM Release 11 will work with current AutoSys Workload Automation releases, clean the file of invalid data for legacy Unicenter AutoSys JM Release 11.

Follow these steps:

1. Open the XML file in a text editor.
2. Search for the ApplicationInstance tag and make the following changes:
 - a. In the name and label attributes, replace Unicenter AutoSys JM with AutoSys Workload Automation and UnicenterAutoSysJM with WorkloadAutomationAE.
 - b. In the Brand tag, replace Unicenter with Workload Automation.
 - c. Change the value of the MinorVersion tag to 3.
 - d. In the Translations tag, make the following changes:
 - Replace UnicenterAutoSysJM with WorkloadAutomationAE, Unicenter with Workload Automation, and Unicenter AutoSys JM with AutoSys Workload Automation.
 - Add the following tag after the as-owner string tag:


```
&lt;string&gt;
&lt;key&gt;as-resource&lt;/key&gt;
&lt;/string&gt;
```
 - e. In the ResourceClass tag, search for as-appl and as-group tags, and add the following tags if they are missing:


```
<Action>read</Action>
<Action>write</Action>
```
 - f. Add the ResourceClass tag as follows:

```

<ResourceClass>
  <Name>as-resource</Name>
  <BestMatchEvaluation>true</BestMatchEvaluation>
  <Action>read</Action>
  <Action>write</Action>
  <Action>create</Action>
  <Action>delete</Action>
  <Action>execute</Action>
</ResourceClass>

```

- g. In the Attach label, replace UnicenterAutoSysJM with WorkloadAutomationAE.
- h. Search for the Folder tag and replace UnicenterAutoSysJM_Common with WorkloadAutomationAE_Common and UnicenterAutoSysJM_XXX with WorkloadAutomationAE_XXX.
- i. For each Policy tag, replace /UnicenterAutoSysJM_XXX with /WorkloadAutomationAE_XXX and UnicenterAutoSysJM_Common with WorkloadAutomationAE_Common.
 - **XXX**
Specifies the AutoSys Workload Automation application instance.
- j. In the Policy tag, search for UnicenterAutoSysJM and replace it with WorkloadAutomationAE.
- k. Search all other tags and replace UnicenterAutoSysJMAAdmin with WorkloadAutomationAEAdmin, UnicenterAutoSysJM with WorkloadAutomationAE, and Unicenter AutoSys JM with AutoSys Workload Automation.
- l. Add the FORECAST policy as follows:

```

<Policy folder="/WorkloadAutomationAE_Common" name="FORECAST Policy">
  <Description>Gives permission to list all jobs using the WorkloadAutomationAE
  forecast program</Description>
  <ResourceClassName>as-list</ResourceClassName>
  <PolicyType>identityacl</PolicyType>
  <Disabled>False</Disabled>
  <ExplicitDeny>False</ExplicitDeny>
  <PreDeployment>False</PreDeployment>
  <RegexCompare>False</RegexCompare>
  <Resource>*.FORECAST</Resource>
  <Action>read</Action>
</Policy>

```

- m. Add the ARCHIVE_JOBS and SENDEVENT_GRPAPP policies only if they are not included in the XML file as follows:

```

<Policy folder="/WorkloadAutomationAE_Common" name="ARCHIVE_JOBS Policy">
  <Description>Controls who can use the WorkloadAutomationAE Archive_jobs
  utility</Description>
  <ResourceClassName>as-control</ResourceClassName>
  <PolicyType>identityacl</PolicyType>
  <Disabled>False</Disabled>
  <ExplicitDeny>False</ExplicitDeny>
  <PreDeployment>False</PreDeployment>
  <RegexCompare>False</RegexCompare>
  <Resource>*.ARCHIVE_JOBS</Resource>
  <Action>execute</Action>
</Policy>
<Policy folder="/WorkloadAutomationAE_Common" name="SENDEVENT_GRPAPP Policy">

```



```

    <Description>Controls who can use the WorkloadAutomationAE Sendevent utility
    with Group (-B) and Application (-i) options.</Description>
    <ResourceClassName>as-control</ResourceClassName>
    <PolicyType>identityacl</PolicyType>
    <Disabled>False</Disabled>
    <ExplicitDeny>False</ExplicitDeny>
    <PreDeployment>False</PreDeployment>
    <RegexCompare>False</RegexCompare>
    <Resource>*.SENDEVENT_GRPAPP</Resource>
    <Action>execute</Action>
  </Policy>

```

- n. Add default policies for the as-resource resource class for each AutoSys Workload Automation instance by replacing XXX with the AutoSys Workload Automation instance name in the following tags:

```

<Policy folder="/WorkloadAutomationAE_XXX" name="XXX: Default Resource Policy">
  <ResourceClassName>as-resource</ResourceClassName>
  <PolicyType>identityacl</PolicyType>
  <Disabled>False</Disabled>
  <ExplicitDeny>False</ExplicitDeny>
  <PreDeployment>False</PreDeployment>
  <RegexCompare>False</RegexCompare>
  <Resource>XXX.*</Resource>
  <Action>read</Action>
  <Action>write</Action>
  <Action>execute</Action>
</Policy>

```

- o. Modify all the policies for as-appl and as-group resource classes by adding the following tags if they are missing:

```

<Action>read</Action>
<Action>write</Action>

```

NOTE

These policies are not common to other instances. Add the tags to each corresponding AutoSys Workload Automation instance policy as follows:

```

<Policy folder="/WorkloadAutomationAE_XXX" name="XXX: Default Groups Policy">
  <ResourceClassName>as-group</ResourceClassName>
  <PolicyType>identityacl</PolicyType>
  <Disabled>False</Disabled>
  <ExplicitDeny>False</ExplicitDeny>
  <PreDeployment>False</PreDeployment>
  <RegexCompare>False</RegexCompare>
  <Resource>XXX.*</Resource>
  <Action>read</Action>
  <Action>write</Action>
  <Action>execute</Action>
</Policy>

```

The XML file is now clean of obsolete strings, labels, and values specific to Unicenter AutoSys JM Release 11.

Register the Product in the Target CA EEM Instance Using safex

Use the backup security policy file to register the current release of the product on the target CA EEM server.

WARNING

You can only use safex to register the product in the target CA EEM instance with a CA EEM XML file that contains an XML tag named "Register". If you are migrating security policy to Release 11.3.5 or Release 11.3.6, run `as_safetool` to upgrade the application instance *after* completing the registration process using safex. This requirement is because the CA EEM XML file you are using to register the product in the target CA EEM is a captured copy of a Release 11.3 application instance.

To register the product or import the XML file, open a command prompt, navigate to the `safex` directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

- **EEM_servername** Specifies the IP address or host name of the computer on which CA EEM is installed.
- **admin_user** Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- **password** Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- **XML_file** Specifies the path and name of the CA EEM XML file.
- **-fips** Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

Register the Product in the Target CA EEM Instance Using as_safetool

The AutoSys Workload Automation `as_safetool` command is used to register AutoSys Workload Automation instances in the CA EEM instance. Use `as_safetool` to register new CA Workload Automation AE Release 11.3.5 or Release 11.3.6 instances in the CA EEM instance; it cannot be done using safex. The `as_safetool` command is also used to complete the migration process by upgrading the WorkloadAutomationAE application instance from a previous release to Release 11.3.5 or Release 11.3.6. If you are using `as_safetool` to upgrade the WorkloadAutomationAE application instance to the current release, only register one of the instance names that are represented in the previous release of the WorkloadAutomationAE application instance.

WARNING

If you are migrating security policy to Release 11.3.5 or Release 11.3.6, run `as_safetool` to upgrade the application instance. If the CA EEM XML file you are using contains an XML tag named "Register", complete this task *after* completing the registration process using safex. This requirement is because the CA EEM XML file you are using to register the product in the target CA EEM is a captured copy of a Release 11.3 application instance. If the CA EEM XML file does not contain the XML tag named "Register", *first* register the current release of the product with CA EEM using `as_safetool`. This requirement is because the application instance that you need to import it to does not exist in CA EEM.

NOTE

Before you perform this task, use the `setenv` or `export` command to set the `ASSAFETOOLPW` environment variable to the password of the EiamAdmin user, the CA EEM server administrative account. For example, `export ASSAFETOOLPW=myspass`.

Follow these steps:

1. Open a AutoSys Workload Automation command prompt.
2. Enter the following command:

```
as_safetool -b EEM_server_host_name -s
```

A list of AutoSys Workload Automation instances that are already registered with the CA EEM server appear.

3. Enter the following command for the instance to be added to that list:

```
as_safetool -b EEM_server_host_name -i instance
```

4. (Optional) Repeat Step 3 for any other AutoSys Workload Automation instances that you need to register. When you have deregistered all AutoSys Workload Automation instances, the product is automatically registered in the source CA EEM instance.

The product is registered in the target CA EEM instance.

Verify Imported Security Policies for Unicenter AutoSys JM Release 11

Resource applications function properly only when the security policies in place comply with the requirements of the AutoSys Workload Automation release that you are using. After you migrate security policies, we recommend that you verify that the access policies comply with the requirements of the current release.

To verify imported security policies for AutoSys Workload Automation, review the following topic to ensure that your imported policies comply:

Migrate CA Workload Automation AE Security Policies from Release 11.3

As an administrator, you upgrade the environment when new releases become available. You want to migrate your existing security policies to minimize the workload intensity of the upgrade process.

The following scenario explains the process to migrate your existing AutoSys Workload Automation security policies from CA Workload Automation AE Release 11.3 to CA Workload Automation AE Release 11.3.5 or Release 11.3.6.

NOTE

When you upgrade CA Workload Automation AE from Release 11.3.5 to Release 11.3.6, the security policies are migrated to Release 11.3.6.

To migrate AutoSys Workload Automation security policies, follow these steps:

1. (Optional) Export the security policies to an XML file using one of the following tools:
 - [CA EEM](#)
 - [safex utility](#)

NOTE

This step is required only if you are migrating policies from a CA Embedded Entitlements Manager (CA EEM) instance on one computer to a CA EEM instance on another computer. To determine whether you need to install a CA EEM instance on another computer as part of the upgrade process, see [Upgrading to the current release](#).

2. (Optional) [Register the product in the target CA EEM instance using safex](#).

NOTE

This step is required only if you are migrating policies from a CA Embedded Entitlements Manager (CA EEM) instance on one computer to a CA EEM instance on another computer. To determine whether you need to install a CA EEM instance on another computer as part of the upgrade process, see [Upgrading to the Current Release](#).

3. [Register the product in the target CA EEM instance using as_safetool](#).
4. [Verify imported security policies](#).
5. [Update migrated security policies](#).

Security Policy Changes from CA Workload Automation AE Release 11.3

This topic describes the changes to the security policy implementation in CA Workload Automation AE Release 11.3.5 or Release 11.3.6. Until you complete the migration process to apply these changes, you cannot import CA Workload Automation AE Release 11.3 security policies into CA EEM servers that are linked with CA Workload Automation AE Release 11.3.6.

This list details the security policy changes from CA Workload Automation AE Release 11.3 to Release 11.3.5 or Release 11.3.6:

- The addition of the WorkloadAutomationAEWebService CA EEM user group.
- The addition of as-base-jobtype resource class with EXECUTE access mode supports the authorization of jobs based on their job types. This new resource class includes a default policy that grants EXECUTE access for all jobs in the AutoSys Workload Automation instance.
- The addition of the policy resource named *AGGREGATE* for the sendevent command in the as-control resource class.
- Updates to the as-owner resource class to control whether a user has permissions to include the owner attribute in the job definition. In addition, it also controls whether a user has permissions to the existing job owner value to update, override, or delete any attribute of a job. The as-owner resource class is also updated to include the *SENDEVENT_JOBEXECUTE* access mode that controls whether you can execute the sendevent command.
- The addition of the as-sendevent resource class and its default CA EEM policy with actions to control access to execute the sendevent command.
- The addition of the as-connectionprofile resource class and its default CA EEM policy with READ, CREATE, DELETE, EXECUTE, and WRITE access modes to control access to a connection profile.

NOTE

- We recommend that you migrate the security policies created in CA Workload Automation AE r11.3 to the current release before activating CA EEM for the AutoSys Workload Automation instance. When you upgrade from CA Workload Automation AE r11.3 to Release 11.3.5 or Release 11.3.6, repeat the migration process for each CA Workload Automation AE r11.3 instance.
- CA Workload Automation AE r11.3, Release 11.3.5, and Release 11.3.6 all use the WorkloadAutomationAE CA EEM application instance. The instance version depends on the release you are using. When you migrate policies from r11.3 to Release 11.3.5 or Release 11.3.6, AutoSys Workload Automation updates the instance version.
- CA Workload Automation AE Release 11.3.5 or Release 11.3.6 security policies use the same CA EEM application instance as CA Workload Automation AE r11.3 security policies. The CA Workload Automation AE r11.3, Release 11.3.5, and Release 11.3.6 security policies coexist within the WorkloadAutomationAE CA EEM application instance of CA Workload Automation AE Release 11.3.5 or Release 11.3.6. You can continue to install security policies for one or more CA Workload Automation AE r11.3 instances in the WorkloadAutomationAE CA EEM application instance of CA Workload Automation AE Release 11.3.5 or Release 11.3.6.

Export the Security Policies to an XML File Using CA EEM

The CA EEM web interface enables you to export security policies to an XML file without the complex scripting requirements that are required when you use the safex utility to export the policies.

Follow these steps:

1. Open a browser and go to the following URL:

`http://EEM_servername:5250/spin/eiam`

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.
The CA EEM login page appears.

2. Select your Web UI instance (for example, WCC0004) from the Application drop-down list, enter EiamAdmin or the name of another user with CA EEM Administrative privileges in the User Name field, enter the appropriate password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Select the EEM Server link.
Multiple links are displayed in the left pane.
5. Select the Export Application link.
The Export Application dialog is displayed in the right pane.
6. Select or clear the check boxes as appropriate, and click Export.

NOTE

Clear all Global objects check boxes if you are using CA EEM Active Directory.

The File Download dialog is displayed.

7. Select Save.
The Save As dialog is displayed.
Specify a file name and a location to save the file to, and click Save.

NOTE

CA EEM saves the XML file as a GZIP archive.

The security policies are exported to the specified XML file.

NOTE

After you export from CA EEM Release 12.6, open the exported xml file and replace all instances of "GroupName" with "Name".

Export the Security Policies to an XML File Using safex

You can export CA EEM data to an XML file using the safex utility. The CA EEM web interface does not require scripting, but you can only use the web interface to export data when you are logged in to the CA EEM server machine through a web browser.

Follow these steps:

1. Log in to the CA EEM back-end server and navigate to the following directory:

Windows:

CA EEM Release 12.6 — EEM_installation_root\bin\

CA EEM r8.4 — c:\Program Files\CA\SharedComponents\iTechnology

UNIX:

CA EEM Release 12.6 — EEM_installation_root/bin/

CA EEM r8.4 — /opt/CA/SharedComponents/iTechnology

2. Enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

NOTE

You must use the safex version of the same CA EEM installed version to export. For example, the safex version shipped with Release 11.3.6 does not work when exporting policies from CA EEM 8.x.

- **EEM_servername**
Specifies the IP address or host name of the computer on which CA EEM is installed.
- **admin_user**

Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.

- **password**
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- **XML_file**
Specifies the name of the XML file that contains the commands for exporting policies. The input safex file can include options for globalfolders, global usergroups, globalusers, and so on.
- **-fips**
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

For Web UI, the content of the input safex XML file is similar to the following:

```
<Safex>
  <Attach label="WCC0003"/>
  <Export file="WCC0003-backup.xml" globalfolders="y" globalusergroups="y" globalusers="y" globalsettings="y"
    folders="y" usergroups="y" users="y" calendars="y" policies="y" appobjects="y"/>
  <Detach/>
</Safex>
```

NOTE

- WCC0003 is the value for CA WCC r11.3. If you are migrating from a CA WCC r11.1 service pack, the value is WCC0002.
- If you do not want to export an object in the file, you must change "y" to "n". For example, if you are using Active Directory or an external LDAP with CA EEM, you would enter "n" for the global entries

The XML file is created in the location specified by the Export file tag of the input safex XML file. All the security policies for the specified product are retrieved from the CA EEM instance. This XML file contains the policies for all instances of the specified product.

Register the Product in the Target CA EEM Instance Using safex

You can use the backup security policy file to register the current release of the product on the target CA EEM server.

WARNING

Important: You can only use safex to register the product in the target CA EEM instance with a CA EEM XML file that contains an XML tag named 'Register'. If you are migrating security policy to Release 11.3.5 or Release 11.3.6, run as `_safetool` to upgrade the application instance after completing the registration process using safex. This requirement is because the CA EEM XML file you are using to register the product in the target CA EEM is a captured copy of a r11.3 release of the application instance.

To register the product or import the XML file, open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

- **EEM_servername**

Specifies the IP address or host name of the computer on which CA EEM is installed.

- **admin_user**

Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.

- **password**

Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.

- ***XML_file***

Specifies the path and name of the CA EEM XML file.

- ***-fips***

Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

Register the Product in the Target CA EEM Instance Using as_safetool

The `as_safetool` command is used to register AutoSys Workload Automation instances in the CA EEM instance. You must use `as_safetool` to register new CA Workload Automation AE Release 11.3.5 or Release 11.3.6 instances in the CA EEM instance; it cannot be done using `safex`. The `as_safetool` command is also used to complete the migration process by upgrading the `WorkloadAutomationAE` application instance from a previous release to Release 11.3.5 or Release 11.3.6. If you are using `as_safetool` to upgrade the `WorkloadAutomationAE` application instance to the current release, you only need to register one of the instance names that are represented in the previous release of the `WorkloadAutomationAE` application instance.

WARNING

If you are migrating security policy to Release 11.3.5 or Release 11.3.6, you must run `as_safetool` to upgrade the application instance. If the CA EEM XML file you are using contains an XML tag named "Register", you must do this *after* completing the registration process using `safex`. This is because the CA EEM XML file you are using to register the product in the target CA EEM is a captured copy of an r11.3 release of the application instance. If the CA EEM XML file does not contain the XML tag named "Register", you must *first* register the current release of the product with CA EEM using `as_safetool`. This is because the application instance you need to import it to does not yet exist in CA EEM.

NOTE

Before performing this task, you must use the `setenv` or `export` command to set the `ASSAFETOOLPW` environment variable to the password of the `EiamAdmin` user, the CA EEM server administrative account. For example, `export ASSAFETOOLPW=mypass`.

Follow these steps:

1. Open a AutoSys Workload Automation command prompt.
2. Enter the following command:

```
as_safetool -b EEM_server_host_name -s
```

A list of AutoSys Workload Automation instances that are already registered with the CA EEM server appear.

3. Enter the following command for the instance that should be added to that list:

```
as_safetool -b EEM_server_host_name -i instance
```

4. (Optional) Repeat Step 3 for any other AutoSys Workload Automation instances you need to register. When you have deregistered all AutoSys Workload Automation instances, the product is automatically registered in the source CA EEM instance.

The product is registered in the target CA EEM instance.

Verify Imported Security Policies from CA Workload Automation AE Release 11.3

Resource applications function properly only when the security policies in place comply with the requirements of the AutoSys Workload Automation release that you are using. After you migrate security policies, we recommend that you verify that the access policies comply with the requirements of the current release.

To verify imported security policies for AutoSys Workload Automation, review the following topic to ensure that your imported policies comply:

[Security Policy Changes from CA Workload Automation AE r11.3](#)

Update Migrated Security Policies

You can use the [as_safetool](#) command to update the migrated security policies. To update the security policies, open a command prompt and enter the following command:

```
as_safetool -b EEM_server_name - a EEM_admin_name -m -v version
```

The security policies are updated. The actions (that are added in a later release level than the release level that you specified in the `-v version` option) under each CA EEM policy are set to the default.

NOTE

Do not use the `-m` option on a machine that has customized policies as this option resets the actions to the default.

Migrate CA WCC Security Policies to Release 11.3.6

As an administrator, you upgrade the environment when new releases become available. You want to migrate your existing security policies to minimize the workload intensity of the upgrade process.

The following scenario provides an example of how to migrate existing CA WCC security policies. The example explains the process to migrate default policies from CA WCC Release 11.1 SP1 (and later service packs) or Release 11.3 or Release 11.3.5 to CA WCC Release 11.3.6.

WARNING

You cannot use these procedures to migrate customized security policies. If you customized your security policies and you want to upgrade to CA WCC Release 11.3.6, contact CA Support.

To migrate CA WCC security policies, follow these steps:

1. Export the security policies to an XML file using one of the following tools:
 - [CA EEM](#)
 - [safex utility](#)
2. Change CA Embedded Entitlements Manager (CA EEM) backup XML file values:
 - [Common to both the r11.1 service packs and r11.3 releases](#)
 - [Release-specific for r11.1 service packs](#)
 - [Release-specific for r11.3](#)

NOTE

No action is required for Release 11.3.5.

3. [Register the product in the target CA EEM instance.](#)
4. [Import security policies and verify.](#)

Export the Security Policies to an XML File Using CA EEM

The CA EEM web interface enables you to export security policies to an XML file without the complex scripting requirements that are required when you use the safex utility to export the policies.

Follow these steps:

1. Open a browser and go to the following URL:
http://EEM_servername:5250/spin/eiam

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

2. Select your CA Workload Control Center (CA WCC) instance (for example, WCC0004) from the Application drop-down list, enter EiamAdmin or the name of another user with CA EEM Administrative privileges in the User Name field, enter the appropriate password in the Password field, and click Log In.
The CA EEM web interface opens. The Home tab is displayed by default.
3. Select the Configure tab.
Multiple links are displayed, with the Applications link selected by default.
4. Select the EEM Server link.
Multiple links are displayed in the left pane.
5. Select the Export Application link.
The Export Application dialog is displayed in the right pane.
6. Select or clear the checkboxes as appropriate, and click Export.

NOTE

If you are using CA EEM Active Directory, clear all Global objects check boxes.

The File Download dialog is displayed.

7. Select Save.
The Save As dialog is displayed.
8. Specify a file name and a location to save the file to, and click Save.

NOTE

CA EEM saves the XML file as a GZIP archive.

The security policies are exported to the specified XML file.

NOTE

After you export from CA EEM Release 12.6, open the exported xml file and replace all instances of "GroupName" with "Name".

Export the Security Policies to an XML File Using safex

You can export CA EEM data to an XML file using the safex utility. The CA EEM web interface does not require scripting, but you can only use the web interface to export data when you are logged in to the CA EEM server machine through a web browser.

Follow these steps:

1. Log in to the CA EEM back-end server and navigate to the following directory:

Windows:

CA EEM Release 12.6 — EEM_installation_root\bin\

CA EEM r8.4 — c:\Program Files\CA\SharedComponents\iTechnology

UNIX:

CA EEM Release 12.6 — EEM_installation_root/bin/

CA EEM r8.4 — /opt/CA/SharedComponents/iTechnology

2. Enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

NOTE

You must use the safex version of the same CA EEM installed version to export. For example, the safex version shipped with Release 11.3.6 does not work when exporting policies from CA EEM 8.x.

- **EEM_servername**

Specifies the IP address or host name of the computer on which CA EEM is installed.

- **admin_user**
Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.
- **password**
Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.
- **XML_file**
Specifies the name of the XML file that contains the commands for exporting policies. The input safex file can include options for globalfolders, global usergroups, globalusers, and so on.
- **-fips**
Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

For CA WCC, the content of the input safex XML file is similar to the following:

```
<Safex>
  <Attach label="WCC0003"/>
  <Export file="WCC0003-backup.xml" globalfolders="y" globalusergroups="y" globalusers="y" globalsettings="y"
    folders="y" usergroups="y" users="y" calendars="y" policies="y" appobjects="y"/>
  <Detach/>
</Safex>
```

NOTE

- WCC0003 is the value for CA WCC r11.3. If you are migrating from a CA WCC r11.1 service pack, the value is WCC0002.
- If you do not want to export an object in the file, you must change "y" to "n". For example, if you are using Active Directory or an external LDAP with CA EEM, you would enter "n" for the global entries

The XML file is created in the location specified by the Export file tag of the input safex XML file. All the security policies for the specified product are retrieved from the CA EEM instance. This XML file contains the policies for all instances of the specified product.

Change Common CA EEM Backup XML File Values for CA WCC

To ensure that your XML file containing policies created by CA WCC Release 11.1 service packs or Release 11.3 works with CA WCC Release 11.3.6, clean the file of invalid data common for legacy CA WCC releases Release 11.1 service packs and Release 11.3.

Follow these steps:

1. Open the XML file in a text editor.
2. Change the value of the ApplicationInstance label element to WCC0004.
3. Change the value of the <Brand> node to CA Technologies.
4. Change the value of the <MinorVersion> node to 3.6.
5. Change the value of the string CA in the Translations node to CA Technologies.
6. Change the value of the string WCC0002 or WCC0003 in the Translations node to WCC0004.
7. Change the value of the administrative user identity element in the DefaultViewEventPolicy and AdministerObjects Policy nodes to CERT-WCC0004.
8. Change the value of the Attach label element preceding the AddIfMissing parent node to WCC0004.
9. Delete all strings from the Translations node:

NOTE

To delete a string, delete the following lines in the XML file:

```
<string>
<key>node_name</key>
</string>
```

10. Delete the obsolete DisableJSCServerLinks Policy folder node and its subnodes.
11. Delete the following obsolete Action elements from the ApplicationAccess ResourceClass, and ApplicationAccessDefaultPolicy folder nodes:
 - EventConsole
 - HighAvailability
 - JobStatusConsole
 - Tutorials
 - WccMain (r11.1 SP1 to Release 11.3.6 migration only)
12. Delete the following obsolete NamedAttr elements from the ApplicationAccess ResourceClass node:
 - component
 - category
 - name
13. Delete the following obsolete Action elements from the ConfigurationControl ResourceClass node:
 - Apply
 - Publish
 - Set
14. Delete the following strings from the Filter logic elements of the ApplicationAccessDefault Policy folder node:
 - EventConsole
 - HighAvailability
 - JobStatusConsole
 - Tutorials
 - WccMain (r11.1 SP1 to Release 11.3.6 migration only)
15. Change the value of the Resource element of the ObjectControlDefault Policy folder node from */AlertPolicy to AlertPolicy.
16. Delete all CreateTimestamp Attribute element nodes.

Your XML file is now clean of obsolete strings, labels, and values common for CA WCC Release 11.1 service packs and Release 11.3.

Change CA EEM Backup XML File Values Specific for CA WCC Release 11.1 Service Packs

To ensure that your XML file containing policies created by the CA WCC Release 11.1 service packs work with CA WCC Release 11.3.6, clean it of invalid data that was specific to the CA WCC Release 11.1 Service Pack on which the policies were created.

Follow these steps:

1. Change all instances of Credential to Credentials.
2. Delete the following obsolete ResourceClass nodes and all their subnodes:
 - PortletAccess
 - JobActionCA7
 - JobActionAutoSys
3. Delete the following obsolete Policy folder nodes and all their subnodes:

-
- ConfigurationControlCPMDefault
 - ConfigurationControlCredentialUpdateAdminDefault
 - ConfigurationControlCredentialUpdateUserDefault
 - ConfigurationControlHighAvailabilityDefault
 - PortletAccessDefault
 - JobActionCA7Default
 - JobActionAutoSysDefault
4. Delete the following obsolete Action elements from the ApplicationAccess ResourceClass and ApplicationAccessDefault Policy folder nodes:
 - CPM
 - Framework
 - JobEditor
 - JobFlowDesign
 - JobFlowMonitoring
 - WebServicesAdvanced
 - WebServicesBasic
 - Reporting
 5. Delete the following obsolete Action elements from the MonitorViewControl ResourceClass and MonitorViewControlDefault Policy folder nodes:
 - Publish
 - Resync
 - SessionModify
 6. Delete the EventProcessorLog Action element from the LogAccess ResourceClass and LogAccessDefault Policy folder nodes.
 7. Delete the EventProcessorLog string from the Filter logic elements of the LogAccessDefault Policy folder node.
 8. Delete the following obsolete Resource elements from the ObjectAccessDefault Policy folder node:
 - Cycle
 - Calendar
 - ExtendedCalendar
 - GlobalVariable
 - Job
 - Jobset
 - Report
 9. Change the value of the AlertPolicy Resource element in the ObjectAccessDefault Policy folder node to AlertPolicy/*.
 10. Delete the following obsolete Resource elements from the ObjectControlDefault Policy folder node:
 - ARFDef
 - ARFSet
 - Dataset
 11. Verify that the value of the Resource element in the ObjectControlDefault Policy folder node is AlertPolicy and not */AlertPolicy.
 12. Delete the following obsolete Resource elements from the MonitorViewControlDefault Policy folder node:
 - Priorrun
 - Runlog
 13. Delete the following strings from the Filter logic elements of the ApplicationAccessDefault Policy folder node:

- CPM
- Framework
- JobEditor
- JobFlowDesign
- JobFlowMonitoring
- WebServicesAdvanced
- WebServicesBasic

Your XML file is now clean of obsolete strings, labels, and values specific to CA WCC Release 11.1 service packs.

Change CA EEM Backup XML File Values Specific for CA WCC Release 11.3

To ensure that your XML file containing policies created by CA WCC Release 11.3 works with CA WCC Release 11.3.6, clean it of invalid data specific for CA WCC Release 11.3.

Follow these steps:

1. Change all instances of ReportsForecast to Forecast.
2. Delete the following obsolete Policy folder nodes and all their subnodes:
 - ConfigurationControlFrameworkThemesDefault
 - ConfigurationControlHighAvailabilityDefault
 - DisableJSCServerLinks
3. Delete the following obsolete Action elements from the ApplicationAccess ResourceClass and ApplicationAccessDefault Policy folder nodes:
 - CommandLineConsole
 - JobStatusViews
 - HighAvailability
 - Tutorials
 - JobStatusConsole
 - EventConsole
 - MonitoringViews
4. Delete the JobStatus Resource element from the ObjectAccessDefault Policy folder node.
5. Delete the following items from the Filter logic in the ApplicationAccessDefault policy:
 - CommandLineConsole
 - JobStatusViews
6. Change */AlertPolicy/* to AlertPolicy/* in the ObjectAccessDefault policy for r11.3.

Your XML file is now clean of obsolete strings, labels, and values specific for CA WCC r11.3.

Register the Product in the Target CA EEM Instance Using safex

To register the product in the target CA EEM instance, the target EEM instance contains the Application WCC0004 that is created in one of the following ways:

- Installing CA WCC Release 11.3.6 and specifying the target CA EEM instance
- Copying the UWCCRegister.xml file located in DVD_ROOT/Catalog.UWCC/uwcc/root/safex to the EIAM_HOME/bin directory on the CA EEM server and then running the safex tool

The correct certificates are created during registration. The certificate files are not created when CA EEM has the WCC0004 application that is registered by CA WCC Release 11.3.5 or CA WCC Release 11.3.6.

Import Security Policies for CA WCC and Verify

Resource applications function properly only when the security policies in place comply with the requirements of the CA WCC release that you are using. After you import the security policies, we recommend that you verify that the access policies comply with the Release 11.3.6 requirements.

Follow these steps:

1. Open the XML file that contains the policies you want to import in a text editor.

2. Locate the following line:

```
<Register certfile="certfile.pl2" password="abc123">
```

3. Modify the line as follows:

```
<Register certtype="pem" certfile="wcc_new.pem" keyfile="wcc_new.key">
```

4. Save the XML file.

The correct certificates are created during registration.

NOTE

The certificate files are not created when CA EEM has the WCC0004 application that is registered by CA WCC Release 11.3.5 or CA WCC Release 11.3.6.

5. Open a command prompt, navigate to the safex directory, and enter the following command:

```
safex -h EEM_servername -u admin_user -p password -f XML_file -fips
```

– ***EEM_servername***

Specifies the IP address or host name of the computer on which CA EEM is installed.

– ***admin_user***

Specifies the name of a CA EEM user with administrative privileges, such as EiamAdmin.

– ***password***

Specifies the password of the CA EEM administrative user, the server administrative account with permissions to update the CA EEM server.

– ***XML_file***

Specifies the path and name of the CA EEM XML file.

– ***-fips***

Specifies that the CA EEM server on which the policies are defined is running in FIPS mode. This parameter is required when you are importing policies to or exporting policies from a CA EEM server that is configured to run in FIPS mode; otherwise, it is not valid.

6. Verify the policies by doing the following steps:

- a. Open a browser and to the following URL:

```
http://EEM_servername:5250/spin/eiam
```

EEM_servername

Specifies the IP address or host name of the computer on which CA EEM is installed.

The CA EEM login page appears.

- b. Click the Manage Access Policies tab.

The CA EEM Access Policies policy tree appears.

- c. Verify that the policy tree does not contain the JobActionAutoSys, JobActionCA7, or PortletAccess resource classes.

- d. Double-click the WidgetAccess folder and verify that the resource class defines the access rights to the Dashboard tab widgets.

NOTE

In Release 11.3.6, the WidgetAccess resource class controls access to the Dashboard tab widgets instead of the PortletAccess resource class.

- e. Verify that the policy tree contains the ConfigurationControl resource class.

- f. Double-click the ConfigurationControl folder and verify that the resource class does not contain the following resources:
 - Framework/theme
 - HighAvailability/server
 - CPM/MyFilter
 - CPM/GlobalFilter
 - CPM/ActiveFilter
- g. Verify that the ConfigurationControl resource class contains the following newly added resources:
 - ExternalApplication
 - Widget

NOTE

- Use the resource that is named Event to set up access rights for sendevent functionality. Use the policies that are defined in the WorkloadAutomationAE CA EEM application to control scheduler access rights. The obsolete r11.3 resource class that is named JobActionAutosys previously defined these access rights.
- JAWS is considered an external application that is no longer supported by CA WorkloadAutomationAE. Use the resource class that is named ExternalApplication to define access rights that were defined using CPM filters and the Framework/theme resource in previous releases.
- Use the resource that is named Widget to define create and delete access rights for widgets on the Dashboard tab.

- h. Double-click the ApplicationAccess folder and verify that the resource class does not contain the following obsolete applications:
 - CommandLineConsole
 - CPM
 - EventConsole
 - Framework
 - HighAvailability
 - JobEditor
 - JobFlowDesign
 - JobFlowMonitoring
 - JobStatusConsole
 - JobStatusViews
 - MonitoringViews
 - ReportsForecast
 - Tutorials
 - WCCMain

NOTE

- The ApplicationAccessDefault policy is overwritten after migration. Verify that the following actions are listed in that policy: Reporting, Dashboard, and Launcher (which is necessary for users to log in to CA WCC).
- The QuickEdit, ApplicationEditor, and Monitoring applications have the same access rights that the JobEditor, JobStatusConsole, JobFlowDesign, and JobFlowMonitoring applications had.

- i. Verify that any new or modified CA WCC modules and applications have valid access rights.

NOTE

- The Forecast application has the same access rights that the ReportsForecast application had.
- The Launcher application now controls access to the CA WCC user interface.

- j. Double-click the ObjectAccess folder and verify that the AlertPolicy resource name does not contain the server name.
- k. Verify that the ObjectAccess resource class does not contain the following resources:
 - JobStatus
 - GlobalVariable
 - Calendar
 - ExtendedCalendar
 - Cycle
 - Job
 - JobSet
- l. (Optional) To use the ObjectControl resource class to control the resources that were removed from the ObjectAccess resource class, double-click the ObjectControl folder, and add the resources to that resource class.

NOTE

You can also use the WorkloadAutomationAE application to control these resources.

- m. Double-click the ReportTemplateAccess folder and verify that the resource class contains the policies that define who has access to the reporting templates that are contained in the Reporting application.

WARNING

The resource named Report that was contained in ObjectAccess resource class in previous releases defined these access rights. That resource is obsolete, the policies are no longer valid, and the resource removed from the resource class. If you do not define access rights to reporting templates in the ReportTemplateAccess resource class, the templates are not accessible.

- n. Double-click the ExternalApplicationAccess folder and verify that the resource class defines access rights rules for JAWS.

NOTE

In Release 11.3.6, the JAWS server type and JAWS instances are considered external applications. The policies in the ServerAccess resource class that contain access rights to these applications are no longer valid. We recommend that you also verify that those policies have been deleted from the ServerAccess resource class.

- o. Open the folders for resource classes that contain policy filters that are associated with applications that handle Job Status Console data and verify that the value of the component that is named attribute for these policy filters has been updated from JobStatusConsole to Monitoring.
- p. Open the folders for resource classes that contain policy filters that are associated with Forecast data and verify that the value of the component that is named attribute for these filters has been updated from ReportsForecast to Forecast.
- q. Open the resource classes that contain policy filters that are associated with resources that are obsolete in Release 11.3.6 and verify that all the obsolete applications have been removed. The resource classes do not contain the following obsolete applications:
 - JobEditor
 - JobFlowDesign
 - JobFlowMonitoring

You have verified that the imported security policies are valid for Release 11.3.6.

7. Verify that the MonitoringJobs action exists in the WidgetAccess resource class.

NOTE

The new action is not set by default to any user. The customer is required to set policies for existing users.

You have successfully migrated the security policies to Release 11.3.6.

Configure SSA to Use a Valid Certificate

This article describes how to configure SSA to use a valid certificate.

WARNING

These procedures apply only to the legacy agent. They do not apply to the native agents or clients.

NOTE

These procedures are only reference examples; adjust them according to your needs and specifications.

SSL Certificates

SSA is installed and configured automatically during the AutoSys Workload Automation installation. During the installation, SSA installs a default certificate that can be used to install and run SSL without any additional configuration. This certificate is the same on all installations and does not provide any level of authentication. To authenticate this certificate, replace it with a valid certificate that is issued and signed by your own Certificate Authority (CA).

The following csamconfigedit command parameters authenticate SSL connections:

- **CertificateFile**

Defines the location of the standard X.509 certificate file in PKCS# 12 or PEM format. The certificate file contains the public key and a Certificate Authority must sign it.

Default: \$CSAM_SOCKADAPTER/cfg/cert.pem (on UNIX) or %CSAM_SOCKADAPTER%\cfg\cert.pem (on Windows)

NOTE

You can create this certificate file using third-party tools such as OpenSSL.

- **PrivateKeyFile**

Defines the location of the file that includes the client private key. This file is used with the client certificate and public key to authenticate SSL connections.

Default: \$CSAM_SOCKADAPTER/cfg/cert.pem (on UNIX) or %CSAM_SOCKADAPTER%\cfg\cert.pem (on Windows)

NOTE

By default, SSA expects the private key to be stored in the same file as the certificate.

- **CaFile**

Holds the location of the certificate file which includes one or more certificate authority keys that are used to sign the certificates of peers you connect with. If a peer connects to you with a certificate that was not signed by an authority in the file, the connection is denied.

Default: \$CSAM_SOCKADAPTER/cfg/root.pem (on UNIX) or %CSAM_SOCKADAPTER%\cfg\root.pem (on Windows)

Changing the Certificate

You can create a valid certificate and can use it to authenticate SSL connections. Use *one* of the following methods to change the default certificate:

- Replace the default certificate with the valid certificate.

NOTE

You do not need to perform additional configuration provided you ensure that the valid certificate uses the same name and location as the default certificate.

- Configure SSA to use the valid certificate.

Prerequisites

To change the certificate, the following prerequisites apply:

- Install OpenSSL on the AutoSys Workload Automation server.
- A working directory must be created. This directory must contain the .cnf configuration file.

NOTE

For information about creating the working directory and the .cnf file, see the OpenSSL documentation.

(UNIX) Install OpenSSL

OpenSSL for UNIX can be obtained through your distribution repository.

The following commands can be used to install OpenSSL:

- On a Debian-based distribution:
apt-get install openssl
- On an RPM-based distribution:
yast -i openssl

(Windows) Install and Configure OpenSSL**NOTE**

Before you request a new certificate or import an existing certificate, install and configure OpenSSL on the AutoSys Workload Automation server.

Follow these steps:

1. (Optional) Download and install Visual C++ 2008 Redistributable from the following URL:
<http://www.microsoft.com>
2. Download and install OpenSSL from the following URL:
<http://www.openssl.org>

NOTE

The recommended version is OpenSSL v0.9.8r Light.

3. Select Run from the Start menu, enter the following command in the dialog that appears, and click OK:
SYSDM.CPL
The System Properties dialog opens.
4. Click Advanced tab, Environment Variables.
The Environment Variables dialog opens.
5. Select Path Variable in the System Variables list, and click Edit.
The Edit System Variable dialog opens.
6. Add the following text at the end of the string in the Variable value field:
;C:\OpenSSL\bin
7. Click OK to close each dialog that appears.
OpenSSL is installed and configured on the AutoSys Workload Automation server.

TEC586204/TEC586205 - How to Configure SSA to Use the Valid Certificate

You can create a valid certificate and can use it to authenticate SSL connections.

To configure SSA to use the valid certificate on UNIX, follow these steps:

1. [Create the authority key and self-signed authority certificate](#)
2. [Generate a private key and sign certificate with an authority](#)
3. Configure SSA to use the valid certificate [UNIX](#) or [Windows](#).

Create the Authority Key and Self-Signed Authority Certificate

You create an encrypted form of the authority key and the self-signed authority certificate to distribute to every host.

NOTE

- For more information about the certificate information, see the openssl documentation.
- This procedure uses the openssl.cnf configuration file in the command as an example.

Follow these steps:

1. Run the following command in the working directory:

```
openssl req -newkey rsa:1024 -sha1 -keyout rootkey.pem -out rootreq.pem -config path/openssl.cnf
```

– **path/**

Specifies the path of the .cnf file based on your operating system.

Example: /etc/pki/tls/openssl.cnf (on UNIX)

You are prompted for a pass phrase.

2. Enter the pass phrase.

The rootreq.pem and rootkey.pem files are created. The rootkey.pem file stores the authority's private key in an encrypted form.

NOTE

The passphrase is used to encrypt the private key and is required when you sign a certificate.

3. Enter the information for each of the prompts as appropriate.

4. Run the following command in the working directory:

```
openssl x509 -req -in rootreq.pem -sha1 path/openssl.cnf -extensions certificate_extensions -signkey rootkey.pem -out root.pem
```

– **path/**

Specifies the path of the .cnf file based on your operating system.

Example: /etc/pki/tls/openssl.cnf (on UNIX)

You are prompted for a pass phrase.

5. Enter the pass phrase that you entered in Step 2.

The root.pem file that contains the self-signed authority certificate is created. Distribute this file to every host that must be authenticated.

Generate a Private Key and Sign Certificate with an Authority

You can generate a private key and a valid self-signed certificate that can be used to authenticate SSL connections.

For each certificate generated, a .cnf file must be provided. This file is similar to the .cnf file that is used to create the self-signed authority certificate, except that the commonName setting must be unique for each host. No restrictions apply to the name as it is a unique identifier. The value is usually the fully qualified name of the host for which the certificate is generated.

NOTE

This procedure uses the host.cnf configuration file in the command as an example.

Follow these steps:

1. Run the following command in the working directory:

```
openssl req -newkey rsa:1024 -sha1 -keyout hostkey.pem -out hostreq.pem -config path/
host.cnf -reqexts req_extensions
```

– **path/**

Specifies the path of the .cnf file based on your operating system.

Example: /etc/pki/tls/openssl.cnf (on UNIX)

You are prompted for a pass phrase.

2. Enter the pass phrase.

The hostreq.pem and hostkey.pem files are created. The hostkey.pem file stores the private key in an encrypted form.

NOTE

The passphrase is used to encrypt the private key and is required when an application loads the private key using OpenSSL.

3. Run the following command in the working directory:

```
openssl x509 -req -in hostreq.pem -sha1 -extfile path/host.cnf -extensions
certificate_extensions -CA root.pem -CAkey rootkey.pem -CAcreateserial -out
hostcert.pem
```

– **path/**

Specifies the path of the .cnf file based on your operating system.

Example: /etc/pki/tls/openssl.cnf (on UNIX)

You are prompted for a pass phrase., .

4. Enter the pass phrase that you entered in Step 2.

The hostcert.pem file is created. The certificate is generated and you now have the authority to sign it.

5. Append the contents of the hostkey.pem file to the hostcert.pem file.

The data in the hostkey.pem and hostcert.pem files is now contained in one file.

(UNIX) Configure SSA to Use the Valid Certificate

After you create a valid certificate, you must configure SSA to use it.

Follow these steps:

1. Log in to AutoSys Workload Automation as the root user. Run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following command at the operating system prompt:

```
unisrvctr stop CA-WAAE
```

The scheduler, application server, web server, and the agent stop.

3. Enter the following command at the operating system prompt:

```
csampmux stop
```

The csampmux process stops.

4. Enter the following commands at the operating system prompt:

```
cd $CSAM_SOCKETADAPTER/bin
configedit CertificateFile=PathtoCertificateFile
configedit PrivateKeyFile=PathtoPrivateKeyFile
configedit CaFile=PathtoCaFile
PathtoCertificateFile
Specifies the path to the valid CertificateFile.
PathtoPrivateKeyFile
```

Specifies the path to the valid PrivateKeyFile.

PathToCaFile

Specifies the path to the valid CaFile.

5. Enter any *one* of the following commands at the operating system prompt:

```
configedit PassBlank=true
```

```
configedit Pass=passphrase
```

```
configedit PassFile=PathToPassphraseFile
```

```
configedit UsePassCallback=true
```

PassBlank=true

Specifies the password is set to blank. By default, this parameter is set to blank.

Pass=passphrase

Specifies the actual value of the password.

PassFile=PathToPassphraseFile

Specifies the file name that contains the password.

UsePassCallback=true

Uses a callback function to return the password.

The pass phrase for the PrivateKeyFile is set.

NOTE

- The PassBlank, Pass, PassFile, and UsePassCallBack parameters are mutually exclusive.
- If you set the UsePassCallBack parameter to true, you must create a shared library or DLL with a function that returns the password. This function is used to authenticate SSL connections.

6. Enter the following command at the operating system prompt:

```
unisrvctr start CA-WAAE
```

The scheduler, application server, web server, and the agent start. SSA is configured to use the valid certificate.

Example: Configure SSA to Use the Valid Certificate

NOTE

This example uses the OpenSSL utility to create the valid certificate

This example configures SSA to use the valid certificate. The PrivateKeyFile and the certificate are stored in the same file and encrypted with a password named *test*.

```
configedit CertificateFile=c:\openssl-0.9.8b\outdll\belje05.pem
```

```
configedit PrivateKeyFile=c:\openssl-0.9.8b\outdll\belje05.pem
```

```
configedit Pass=test
```

```
configedit CaFile=c:\openssl-0.9.8b\outdll\rootcert.pem
```

To verify the SSL settings, enter the following command at the AutoSys Workload Automation operating system prompt:

```
configedit display
```

The output is displayed as follows:

```
SSL Authentication and Encryption:
```

```
EnableSSL=False
```

```
CertificateFile=C:\openssl-0.9.8b\outdll\belje05.pem
```

```
PrivateKeyFile=C:\openssl-0.9.8b\outdll\ belje05.pem
```

```

Pass=test
PassFile=<no value>
UsePassCallback=<no value>
PassBlank=<no value>
AllowDefaultCa=False
CaFile=C:\openssl-0.9.8b\outdll\rootcert.pem
CaDir=<no value>
CrlFile=<no value>
CrlDir=<no value>
CertificateCallback=<no value>
AuthenticateServer=True
AuthenticateClient=True
AuthenticateCipherSuite=ALL:!DH:!ADH:!LOW:!EXP:!MD5:@STRENGTH
AnonymousCipherSuite=ALL:ADH:!LOW:!EXP:!MD5:@STRENGTH
EnableSelectBeforeSSLRead=False
EnableMinimalReadNotifications=False

```

(Windows) Configure SSA to Use the Valid Certificate

After you create a valid certificate, you must configure SSA to use it.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and complete the following steps:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, web server, and the agent service, and click Stop.
The scheduler, application server, web server, and the agent stop.
2. Stop the CA Connection Broker service from the Windows Service Control Manager.
3. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt is displayed.
4. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:


```

cd $CSAM_SOCKADAPTER/bin
configedit CertificateFile=PathtoCertificateFile
configedit PrivateKeyFile=PathtoPrivateKeyFile
configedit CaFile=PathtoCaFile
PathtoCertificateFile
Specifies the path to the valid CertificateFile.
PathtoPrivateKeyFile
Specifies the path to the valid PrivateKeyFile.
PathtoCaFile
Specifies the path to the valid CaFile.

```
5. Enter any *one* of the following commands as appropriate at the instance command prompt:


```

configedit PassBlank=true

```

```

configedit Pass=passphrase
configedit PassFile=PathToPassphraseFile
configedit UsePassCallback=true
PassBlank=true
Specifies the password is set to blank. By default, this parameter is set to blank.
Pass=passphrase
Specifies the actual value of the password.
PassFile=PathToPassphraseFile
Specifies the file name that contains the password.
UsePassCallback=true
Uses a callback function to return the password.
The pass phrase for the PrivateKeyFile is set.

```

NOTE

- The PassBlank, Pass, PassFile, and UsePassCallBack parameters are mutually exclusive.
- If you set the UsePassCallBack parameter to true, you must create a shared library or DLL with a function that returns the password. This function is used to authenticate SSL connections.

6. Start the CA Connection Broker service from the Windows Service Control Manager.
7. Complete the following steps:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select an instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, web server, and the agent service, and click Start.
The scheduler, application server, web server, and the agent start. SSA is configured to use the valid certificate.

Example: Configure SSA to Use the Valid Certificate

NOTE

This example uses the OpenSSL utility to create the valid certificate.

This example configures SSA to use the valid certificate. The PrivateKeyFile and the certificate are stored in the same file and encrypted with a password named *test*.

```

configedit CertificateFile=c:\openssl-0.9.8b\outdll\belje05.pem
configedit PrivateKeyFile=c:\openssl-0.9.8b\outdll\belje05.pem
configedit Pass=test
configedit CaFile=c:\openssl-0.9.8b\outdll\rootcert.pem

```

To verify the SSL settings, enter the following command at the AutoSys Workload Automation instance command prompt:

```
configedit display
```

The output is displayed as follows:

```

SSL Authentication and Encryption:
EnableSSL=False
CertificateFile=C:\openssl-0.9.8b\outdll\belje05.pem
PrivateKeyFile=C:\openssl-0.9.8b\outdll\ belje05.pem

```

```

Pass=test
PassFile=<no value>
UsePassCallback=<no value>
PassBlank=<no value>
AllowDefaultCa=False
CaFile=C:\openssl-0.9.8b\outdll\rootcert.pem
CaDir=<no value>
CrlFile=<no value>
CrlDir=<no value>
CertificateCallback=<no value>
AuthenticateServer=True
AuthenticateClient=True
AuthenticateCipherSuite=ALL:!DH:!ADH:!LOW:!EXP:!MD5:@STRENGTH
AnonymousCipherSuite=ALL:ADH:!LOW:!EXP:!MD5:@STRENGTH
EnableSelectBeforeSSLRead=False
EnableMinimalReadNotifications=False

```

Passphrase Callback

If you set the UsePassCallBack parameter to true, you must create a shared library or DLL with a function that returns a password. This function is used to authenticate SSL connections.

The name of the shared library must be as follows:

- Windows -- csamsslext.dll
- UNIX -- libcsamsslext.so

The shared library must have at least one exported function for the library with the following prototype:

```

int password_cb(char *buf, int size, int rwflag, void *userdata)
{
    strcpy(buf, "test");
    return(strlen(buf));
}

```

- ***password_cb***
Specifies the password that is used during decryption. This password is provided by the application.
- ***buf***
Specifies the buffer into which the password is written.
- ***size***
Specifies the size of the buffer. The calling function returns the actual length of the password.
- ***rwflag=0 | 1***
Indicates whether the callback option is used for decryption (reading) or encryption (writing).
 - **0**
Specifies the callback option is used for decryption or reading.
 - **1**
Specifies the callback option is used for encryption or writing.
- ***userdata***
Indicates a pointer to userdata when the function is invoked.

Customize SSL for Web Services

Security in Web Services is implemented using basic authentication over a customized Secure Sockets Layer (SSL) connection. Authentication is performed using the Tomcat JAAS realm with an CA EEM LoginModule. Using basic authentication, the CA EEM LoginModule authenticates the login credentials. You must have an SSL connection to secure the login credentials. To ensure the security of a web service-related transaction, use a self-signed certificate or a certificate that a trusted certificate authority issues.

NOTE

The procedures in this section are only reference examples; adjust them according to your needs and specifications.

Change the Certificate

The following options are available for changing the certificate:

- Generate and use a self-signed certificate.
- Request and use a certificate from a trusted certificate authority.

Generate and Use a Self-Signed Certificate

You can generate and use a self-signed certificate. Self-signed certificates cannot be revoked if they become compromised and do not provide an acceptable security level. You can use self-signed certificates for testing and non-commercial environments, such as labs. We recommend that you do not use self-signed certificates in a production environment.

To generate and use a self-signed certificate, follow these steps:

1. [Generate a private key and self-signed certificate.](#)
2. [Restart the AutoSys Workload Automation web server.](#)

Generate a Private Key and Self-Signed Certificate

You can create a private key and self-signed certificate from the AutoSys CA Workload Automation command line. You can then add them to the keystore.

Follow these steps:

1. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to remove the previous key from the keystore:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -delete -alias tomcat \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath $AUTOSYS/lib/bc-fips.jar \
  -storetype BCFKS
```

– On Windows:

```
# keytool -delete -alias tomcat \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath %AUTOSYS%\bin/bc-fips.jar \
```

```
-storetype BCFKS
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -delete -alias tomcat \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit
```

– On Windows:

```
# keytool -delete -alias tomcat \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit
```

NOTE

The default name of the key is tomcat. To access the keystore, you also need the keystore password. The default password is changeit.

If you want to use a different keystore password in the server.xml file, add the java environment-variable in the wrapper.conf file, as follows:

```
wrapper.java.additional.<number>=-Djavax.net.ssl.trustStorePassword=<keystorepassword>
```

Example: To use 'myCustomPassword' as the new password for keystore:

```
wrapper.java.additional.15=-Djavax.net.ssl.trustStorePassword=myCustomPassword
```

If you do not add the java environment-variable, you may repeatedly receive the following text in the Web UI log file:

“InvalidAlgorithmParameterException: the trustAnchors parameter must be non-empty “

2. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to generate a new key:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \
  -keypass changeit \ -providerclass
org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath $AUTOSYS/lib/bc-fips.jar \
  -storetype BCFKS \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

– On Windows:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -keypass changeit \ -providerclass
org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath %AUTOSYS%\bin/bc-fips.jar \
  -storetype BCFKS \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore $AUTOUSER/webserver/conf/.keystore \
```

```
-storepass changeit \
-keypass changeit \
-keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

– On Windows:

```
# keytool -genkey -alias tomcat -keyalg RSA \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit \
-keypass changeit \
-keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

3. In the preceding command:

- Replace SERVER_NAME with the name of your server.
- The keysize argument lets you specify the key size. Typical values are 1024 or 2048.
- The validity argument lets you specify the certificate validity period in days.

A new key and certificate are created and added to the keystore.

Restart the AutoSys Workload Automation Web Server

After you generate the private key and self-signed certificate, restart the AutoSys Workload Automation web server to activate the new configuration.

Follow these steps:

- On UNIX:

```
# unicntrl restart waae_webserver.$AUTOSERV
```

- **\$AUTOSERV**

Specifies the name of the AutoSys Workload Automation instance.

- On Windows:

Restart the AutoSys Workload Automation Web Server service.

Your server is updated and the new configuration is active.

Request and Use a Certificate from a Trusted Certificate Authority

You can request and use a certificate from a trusted certificate authority. If a certificate that is issued by a trusted certificate authority is compromised, you can revoke it, providing better security than self-signed certificates. For production environments, we strongly recommend this option.

NOTE

Obtaining a certificate from a trusted certificate authority can take hours or days.

For Releases 11.3.6 SP8 and higher, complete the following step and proceed. For all other releases, you can ignore this step.

Update \$JAVA_HOME/lib/security/java.security file with the following changes. These changes are required for Bouncy Castle.

```
security.provider.1=org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider
security.provider.2=com.sun.net.ssl.internal.ssl.Provider
security.provider.3=sun.security.provider.Sun
security.provider.4=sun.security.rsa.SunRsaSign
security.provider.5=sun.security.ec.SunEC
security.provider.6=com.sun.crypto.provider.SunJCE
security.provider.7=sun.security.jgss.SunProvider
```

```
security.provider.8=com.sun.security.sasl.Provider
security.provider.9=org.jcp.xml.dsig.internal.dom.XMLDSigRI
security.provider.10=sun.security.smartcardio.SunPCSC
```

To request and use a certificate from a trusted certificate authority, follow these steps:

1. [Generate a Private Key and Self-Signed Certificate.](#)
2. [Request a Certificate.](#)
3. [Download the Certificate Chain.](#)
4. [Add the Certificates to the Keystore.](#)
5. [Restart the AutoSys Workload Automation Web Server.](#)

Generate a Private Key and Self-Signed Certificate

You can create a private key and self-signed certificate from the AutoSys Workload Automation command line. You can then add them to the keystore.

Follow these steps:

1. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to remove the previous key from the keystore:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -delete -alias tomcat \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath $AUTOSYS/lib/bc-fips.jar \
-storetype BCFKS
```

– On Windows:

```
# keytool -delete -alias tomcat \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath %AUTOSYS%\bin/bc-fips.jar \
-storetype BCFKS
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -delete -alias tomcat \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit
```

– On Windows:

```
# keytool -delete -alias tomcat \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit
```

NOTE

The default name of the key is tomcat. To access the keystore, you also need the keystore password. The default password is changeit.

If you want to use a different keystore password in the server.xml file, add the java environment-variable in the wrapper.conf file, as follows:

```
wrapper.java.additional.<number>=-Djavax.net.ssl.trustStorePassword=<keystorepassword>
```

Example: To use 'myCustomPassword' as the new password for keystore:

```
wrapper.java.additional.15=-Djavax.net.ssl.trustStorePassword=myCustomPassword
```

If you do not add the java environment-variable, you may repeatedly receive the following text in the Web UI log file:

"InvalidAlgorithmParameterException: the trustAnchors parameter must be non-empty "

2. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to generate a new key:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \

  -keypass changeit \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath $AUTOSYS/lib/bc-fips.jar \
  -storetype BCFKS \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

– On Windows:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -keypass changeit \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath %AUTOSYS%\bin/bc-fips.jar \
  -storetype BCFKS \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \
  -keypass changeit \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

– On Windows:

```
# keytool -genkey -alias tomcat -keyalg RSA \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -keypass changeit \
  -keysize 1024 -dname "cn=SERVER_NAME" -validity 825
```

3. In the preceding command:

- Replace SERVER_NAME with the name of your server.
 - The keysize argument lets you specify the key size. Typical values are 1024 or 2048.
 - The validity argument lets you specify the certificate validity period in days.
- A new key and certificate are created and added to the keystore.

Request a Certificate

To use a certificate that a trusted certificate authority issued, request that the authority issue a certificate to you.

Follow these steps:

1. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to generate a certificate request file (certreq.csr):

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -certreq -alias tomcat \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \
  -storetype BCFKS \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath $AUTOSYS/lib/bc-fips.jar \
  -file certreq.csr
```

– On Windows:

```
# keytool -certreq -alias tomcat \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -storetype BCFKS \
  -providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
  -providerpath %AUTOSYS%\bin/bc-fips.jar \
  -file certreq.csr
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -certreq -alias tomcat \
  -keystore $AUTOUSER/webserver/conf/.keystore \
  -storepass changeit \
  -file certreq.csr
```

– On Windows:

```
# keytool -certreq -alias tomcat \
  -keystore %AUTOUSER%\webserver\conf\.keystore \
  -storepass changeit \
  -file certreq.csr
```

2. Request a certificate in PEM format from a certificate authority. Contact the certificate authority that you chose for specific instructions.

NOTE

AutoSys Workload Automation uses certificates in PEM format.

Download the Certificate Chain

To validate the certificate that is issued to you by a trusted certificate authority, download the public certificate of the authority. Insert it into your keystore. Usually, this process means downloading the certificate of the authority that issued your personal certificate, and any certificates that this authority has been issued by higher-level certificate authorities. This activity is referred to as the certificate chain.

To download the certificate chain, go to the website of the trusted certificate authority you chose and follow the instructions to download the certificates in the certificate chain.

You now have the public certificate chain of your certificate authority.

NOTE

AutoSys Workload Automation uses certificates in PEM format.

Add the Certificates to the Keystore

When you have received your private certificate, downloaded the certificate chain of your certificate authority, and converted or cleaned all certificates, insert the certificates into your keystore.

Follow these steps:

1. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to insert the certificate of the top-level issuing authority into your keystore:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -importcert -alias RootCA \
-file RootCA.cer \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath $AUTOSYS/lib/bc-fips.jar \
-storetype BCFKS
```

– On Windows:

```
# keytool -importcert -alias RootCA \
-file RootCA.cer \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath %AUTOSYS%\bin/bc-fips.jar \
-storetype BCFKS
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -importcert -alias RootCA \
-file RootCA.cer \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit
```

– On Windows:

```
# keytool -importcert -alias RootCA \
-file RootCA.cer \
-keystore %AUTOUSER%\webserver\conf\.keystore \
```

```
-storepass changeit
```

2. (Optional) Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a sourced shell window (on UNIX) to insert an extra lower-level issuer certificate into your keystore:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -importcert -alias SubCA \

-file SubCA.cer \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath $AUTOSYS/lib/bc-fips.jar \
-storetype BCFKS
```

– On Windows:

```
# keytool -importcert -alias SubCA \

-file SubCA.cer \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath %AUTOSYS%\bin/bc-fips.jar \
-storetype BCFKS
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -importcert -alias SubCA \
-file SubCA.cer \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit
```

– On Windows:

```
# keytool -importcert -alias SubCA \
-file SubCA.cer \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit
```

3. Repeat Step 2 for each lower-level issuer certificate you have downloaded.
4. Run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to insert your private certificate into your keystore:

For Releases 11.3.6 SP8 and higher:

– On UNIX:

```
# keytool -importcert -trustcacerts \
-file certificate.cer -alias tomcat \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath $AUTOSYS/lib/bc-fips.jar \
-storetype BCFKS
```

– On Windows:


```
# keytool -importcert -trustcacerts \
-file certificate.cer -alias tomcat \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath %AUTOSYS%/bin/bc-fips.jar \
-storetype BCFKS
```

For Releases 11.3.6 SP7 and lower:

– On UNIX:

```
# keytool -importcert -trustcacerts \
-file certificate.cer -alias tomcat \
-keystore $AUTOUSER/webserver/conf/.keystore \
-storepass changeit
```

– On Windows:

```
# keytool -importcert -trustcacerts \
-file certificate.cer -alias tomcat \
-keystore %AUTOUSER%\webserver\conf\.keystore \
-storepass changeit
```

The certificates are inserted into the keystore.

Restart the AutoSys Workload Automation Web Server

After you generate the private key and self-signed certificate, restart the AutoSys Workload Automation web server to activate the new configuration.

Follow these steps:

- On UNIX:

```
# unicntrl restart waee_webserver.$AUTOSERV
```

- **\$AUTOSERV**

Specifies the name of the AutoSys Workload Automation instance.

- On Windows:

Restart the AutoSys Workload Automation Web Server service.

Your server is updated and the new configuration is active.

Reuse an Existing Keystore File

You can reuse a keystore file from an existing AutoSys Workload Automation web server to enable SSL in your environment. An existing keystore file can contain self-signed certificates or certificates issued by a certificate authority.

To reuse an existing keystore file, follow these steps:

1. [Copy an existing keystore file.](#)
2. [Restart the AutoSys Workload Automation web server.](#)

Copy an Existing Keystore File

You can reuse the keystore file from an existing AutoSys Workload Automation web server with a new AutoSys Workload Automation web server installation.

To reuse the keystore file, copy the keystore file in the following directory from the existing AutoSys Workload Automation web server to the new AutoSys Workload Automation web server:

- UNIX:
\$AUTOUSER/webserver/conf/.keystore
- Windows:
\$AUTOUSER\webserver\conf\keystore

Restart the AutoSys Workload Automation Web Server

After you generate the private key and self-signed certificate, restart the AutoSys Workload Automation web server to activate the new configuration.

Follow these steps:

- On UNIX:
unicntrl restart waae_webserver.\$AUTOSERV
– **\$AUTOSERV**
Specifies the name of the AutoSys Workload Automation instance.
- On Windows:
Restart the AutoSys Workload Automation Web Server service.

Your server is updated and the new configuration is active.

Import an Existing Private Key and Certificate

You can use a certificate that you already have, or you can import a private key and certificate from a PKCS12 file.

To import an existing private key and certificate, follow these steps:

1. [Import the certificate from PKCS12.](#)
2. [Restart the AutoSys Workload Automation web server.](#)

Import Certificate from PKCS12

You can import existing keys and certificates you have packaged as PKCS12 files.

To import the certificate from PKCS12, run the following commands in a AutoSys Workload Automation command prompt window (on Windows) or in a AutoSys Workload Automation sourced shell window (on UNIX) to import the certificate from PKCS12 and insert it into the keystore:

For Releases 11.3.6 SP8 and higher:

- On UNIX:
keytool -importkeystore \
-srckeystore myp12.p12 -srcstoretype PKCS12 -srcstorepass my-password \
-deststorepass changeit -destkeypass changeit \
-destkeystore \$AUTOUSER/webserver/conf/.keystore \
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath \$AUTOSYS/lib/bc-fips.jar \
-deststoretype BCFKS \
-alias 1 -destalias tomcat
- On Windows:
keytool -importkeystore \
-srckeystore myp12.p12 -srcstoretype PKCS12 -srcstorepass my-password \
-deststorepass changeit -destkeypass changeit \
-destkeystore %AUTOUSER%\webserver\conf\keystore \

```
-providerclass org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider \
-providerpath %AUTOSYS%/bin/bc-fips.jar \
-deststoretype BCFKS \
-alias 1 - destalias tomcat
```

For Releases 11.3.6 SP7 and lower:

- On UNIX:

```
# keytool -importkeystore \
-srckeystore myp12.p12 -srcstoretype PKCS12 -srcstorepass my-password \
-deststorepass changeit -destkeypass changeit \
-destkeystore $AUTOUSER/webserver/conf/.keystore \
-alias 1 - destalias tomcat
```

- On Windows:

```
# keytool -importkeystore \
-srckeystore myp12.p12 -srcstoretype PKCS12 -srcstorepass my-password \
-deststorepass changeit -destkeypass changeit \
-destkeystore %AUTOUSER%\webserver\conf\.keystore \
-alias 1 - destalias tomcat
```

NOTE

The certificate chain that is found in the PKCS12 package is also imported (although it is not visible with the keytool program). If the package did not contain the certificate chain of the issuing authority, you can import the chain.

Restart the AutoSys Workload Automation Web Server.

After you generate the private key and self-signed certificate, restart the AutoSys Workload Automation web server to activate the new configuration.

Follow these steps:

- On UNIX:

```
# unicntrl restart waae_webserver.$AUTOSERV
```

- **\$AUTOSERV**

Specifies the name of the AutoSys Workload Automation instance.

- On Windows:

Restart the AutoSys Workload Automation Web Server service.

Your server is updated and the new configuration is active.

Troubleshoot Certificate Issues

Invalid Certificate Security Alert

Symptom:

I get the following Security Alert message in my browser:

The name on the security certificate is invalid or does not match the name of the site.

This problem occurs in the following cases:

- When I try to access the AutoSys Workload Automation web server using localhost. For example, `https://localhost:9443/AEWS`.
- When I try to access the AutoSys Workload Automation web server through a URL that does not match the CN attribute of the certificate Issued To field exactly. For example, `https://server01:9443/AEWS` causes a Security Alert if the CN attribute of the certificate Issued To field is `server01.ca.com`.

Solution:

Use the name of the server as it is specified in the CN attribute of the certificate Issued To field. For example, `https://server01.ca.com:9443/AEWS`.

Security Alert with Genuine Certificate**Symptom:**

I get a Security Alert in my browser even though I am using a genuine Certificate Authority generated certificate. The CN attribute of the certificate Issued To field matches the hostname/URL that I used to access the AutoSys Workload Automation Web Server.

Solution:

Check with your certificate authority if a chain certificate is needed in the keystore to establish full authenticity of the certificate.

Error When Migrating the JKS Keystore Certificate File**Symptom:**

I see the following error in the upgrade log when I upgrade from any previous release to Release 11.3.6 SP8 or later.

```
Error migration of keystore certificate file
```

This error occurs in the following situations:

- When any failure occurs when the installer is migrating the JKS keystore file to the BCFKS keystore file.
- When the keystore includes multiple aliases and one of these aliases that is mentioned in the `server.xml` file and is used by the web server is converted to BCFKS format.

Solution:

Run the following commands in a AutoSys Workload Automation sourced shell window (on UNIX) or in a AutoSys Workload Automation command prompt window (on Windows) and restart the web server:

- On UNIX

```
# keytool -importkeystore -srckeystore $AUTOUSER/webserver/conf/.keystore_backup /
-destkeystore $AUTOUSER/webserver/conf/.keystore -srcstoretype JKS -deststoretype
BCFKS /
-srcalias tomcat -destalias tomcat -srcstorepass changeit -deststorepass changeit /
-srckeypass changeit -destkeypass changeit
```

- On Windows

```
# keytool -importkeystore -srckeystore %AUTOUSER%\webserver\conf\.keystore_backup \
-destkeystore %AUTOUSER%\webserver\conf\.keystore -srcstoretype JKS -deststoretype
BCFKS \
-srcalias tomcat -destalias tomcat -srcstorepass changeit -deststorepass changeit \
-srckeypass changeit -destkeypass changeit
```

Trusted Certificate Import Fails (for Releases 11.3.6 SP8 and higher)**Symptom:**

I get the following exception when importing a signed trusted certificate to a keystore.
keytool error: java.lang.Exception: Public keys in reply and keystore don't match

Solution:

Update the \$JAVA_HOME/lib/security/java.security file with the following changes. These changes are required for Bouncy Castle.

- Make BouncyCastleFipsProvider as the first line.
- Also, move com.sun.net.ssl.internal.ssl.Provider to the second line.

Before change

```
security.provider.1=sun.security.provider.Sun
security.provider.2=sun.security.rsa.SunRsaSign
security.provider.3=sun.security.ec.SunEC
security.provider.4=com.sun.net.ssl.internal.ssl.Provider
security.provider.5=com.sun.crypto.provider.SunJCE
security.provider.6=sun.security.jgss.SunProvider
security.provider.7=com.sun.security.sasl.Provider
security.provider.8=org.jcp.xml.dsig.internal.dom.XMLDSigRI
security.provider.9=sun.security.smartcardio.SunPCSC
security.provider.10=org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider
```

After change

```
security.provider.1=org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider
security.provider.2=com.sun.net.ssl.internal.ssl.Provider
security.provider.3=sun.security.provider.Sun
security.provider.4=sun.security.rsa.SunRsaSign
security.provider.5=sun.security.ec.SunEC
security.provider.6=com.sun.crypto.provider.SunJCE
security.provider.7=sun.security.jgss.SunProvider
security.provider.8=com.sun.security.sasl.Provider
security.provider.9=org.jcp.xml.dsig.internal.dom.XMLDSigRI
security.provider.10=sun.security.smartcardio.SunPCSC
```

Now, follow the procedure in the Request and Use a Certificate from a Trusted Certificate Authority page at [Customize SSL for Web Services](#)

DER Length Error When Listing Keystore (for Releases 11.3.6 SP8 and higher)**Symptom:**

I get the following exception when the keystore type is JKS and I had mentioned BCFKS in the command.
keytool -list -keystore .keystore -storepass changeit -storetype BCFKS
keytool error: [java.io.IOException](#): DER length more than 4 bytes: 109

Solution:

Ensure that you are using the correct storetype in the command.

Example:

The following keytool command shows a keystore list of type JKS and then lists the JKS store with BCFKS as storetype.

```
keytool -list -keystore .keystore -storepass changeit -storetype JKS
```

Keystore type: jks

Keystore provider: SUN

Your keystore contains 1 entry

tomcat, Jul 18, 2019, PrivateKeyEntry,

Certificate fingerprint (SHA1): 23:0F:07:A4:48:80:A8:E0:64:7C:50:8D:A7:14:07:FA:AB:28:38:1F

Invalid Keystore Format Error When Listing Keystore (for Releases 11.3.6 SP8 and higher)

Symptom:

I get the following exception when keystore type is BCFKS and I had mentioned JKS in the command.

```
keytool -list -keystore .keystore -storepass changeit
```

keytool error: [java.io.IOException](#): Invalid keystore format

Resolution:

Ensure that you are using the correct storetype in the command.

```
keytool -list -keystore .keystore -storepass changeit -storetype BCFKS
```

Keystore type: BCFKS

Keystore provider: BCFIPS

Your keystore contains 1 entry

tomcat, Jul 18, 2019, PrivateKeyEntry,

Certificate fingerprint (SHA1): 27:A7:15:EF:1D:69:0B:63:7E:88:1F:DF:FB:C5:7D:D8:34:82:56:2D

BCFKS Not Found Error When Running Keytool Command (for Releases 11.3.6 SP8 and higher)

Symptom:

I get the following exception when the storetype is BCFKS and bc-fips.jar is not in path.

keytool error: java.security.KeyStoreException: BCFKS not found

Resolution:

Ensure that the bc-fips.jar is in the path. Run the keytool command on AE prompt. As AE sources JAVA_HOME paths, bc-fips.jar is available for the keytool.

MAC Calculation Failed Error When Using Keytool Command

Symptom:

I get the following exception when storepass is not provided as part of the keytool command.

```
keytool -list -keystore .keystore -storetype BCFKS
```

keytool error: [java.io.IOException](#): BCFKS KeyStore corrupted: MAC calculation failed.

Resolution:

Ensure that storepass is used whenever required in the keytool command.

```
keytool -list -keystore .keystore -storetype BCFKS -storepass changeit
```

Migration of Keystore from JKS to BCFKS Failure (for Releases 11.3.6 SP8 and higher)

Symptom:

When I of upgrade from "Release 11.3.6 SP7 or lower" to "Release 11.3.6 SP8 or higher", Keystore migration from JKS to BCFKS may fail in the following cases or Go to Success without all Certificates.

- Keystore having first alias not as Tomcat.
- Keystore with CA Trusted Certificates. In this case, only first alias from the store is migrated.

Resolution:

Import JKS to BCFKS.

- Import the backed up JKS keystore to BCFKS
`keytool -importkeystore -srckeystore $AUTOUSER/webserver/conf/.keystore_backup -destkeystore $AUTOUSER/webserver/conf/.keystore -srcstoretype JKS -deststoretype BCFKS -srcstorepass changeit -deststorepass changeit`
- List the .keystore and ensure the alias are correctly present and the storetype is BCFKS

Keytool Hangs**Symptom:**

The keytool command takes more time, generally in minutes, to perform operations such as list, delete, and genkey.

Resolution:

The time taken is expected. It depends on the "cryptographically-secure pseudo-random number generator (CSPRNG)". Also, it is a one time activity.

For more information, see <https://www.redhat.com/en/blog/understanding-red-hat-enterprise-linux-random-number-generator-interface>

Use a Different Alias Other than the One Specified in the server.xml File

If the keystore includes multiple aliases and you want to use a different alias other than the one that is specified in the server.xml file, follow these steps.

1. Run the following commands in a AutoSys Workload Automation sourced shell window (on UNIX) or in a AutoSys Workload Automation command prompt window (on Windows):
 - On UNIX


```
# keytool -importkeystore -srckeystore $AUTOUSER/webserver/conf/.keystore_backup /
  -destkeystore $AUTOUSER/webserver/conf/.keystore -srcstoretype JKS -deststoretype
  BCFKS /
  -srcalias tomcat -destalias tomcat -srcstorepass changeit -deststorepass changeit /
  -srckeypass changeit -destkeypass changeit
```
 - On Windows


```
# keytool -importkeystore -srckeystore %AUTOUSER%\webserver\conf\.keystore_backup \
  -destkeystore %AUTOUSER%\webserver\conf\.keystore -srcstoretype JKS -deststoretype
  BCFKS \
  -srcalias tomcat -destalias tomcat -srcstorepass changeit -deststorepass changeit \
  -srckeypass changeit -destkeypass changeit
```
2. Update the keyAlias and keyPass in the server.xml file as appropriate.
3. Restart the web server for the changes to take effect, as follows:
 - On UNIX, run the following command:


```
# unicntrl restart waae_webserver.$AUTOSERV
```
 - On Windows, restart the AutoSys Workload Automation Web Server service.

Administrating

This section includes administration procedures and troubleshooting procedures AutoSys Workload Automation and Web UI.

WARNING

This section contains confidential subsections that are not viewable to anonymous users. To view all subsections, select Sign In in the upper-right corner. Enter your CA Support Online credentials. If you are logged in, ignore this message.

This section contains information about the following sections:

- **AutoSys Administration** Includes procedures to manage AutoSys. For example, procedures to maintain the server, maintain agents, control the server, and manage email/alarm notifications.
- **Web UI Administration** Includes procedures to manage batch interfaces in Web UI.
- **Troubleshooting Web UI** Includes steps to troubleshoot Web UI.

After you log in, use the navigation tree on the left or search for Administrating content:

Administrating

AutoSys Workload Automation Administration

This section contains the following topics:

Configuring AutoSys Workload Automation

You can configure AutoSys Workload Automation to control the run-time behavior of each instance, including which database to connect to and how to react to error conditions. You can also set up the Notification feature to communicate problems to users in your enterprise.

You can define environment variables on AutoSys Workload Automation to customize logging, network communication, and the behavior of the scheduler, application server, client utilities, and SDK.

On UNIX, you configure AutoSys Workload Automation to control the run-time behavior by modifying the parameters in the configuration file and the agentparm.txt file, and by modifying the environment variables in the /etc/auto.profile file and the WAAE.txt file.

On Windows, you configure AutoSys Workload Automation to control the run-time behavior by modifying the configuration parameters in either the Administrator utility or the configuration file, modifying the environment variables in either the Administrator utility or the WAAE.txt file, and modifying the agent parameters in the agentparm.txt file.

NOTE

On Windows, the parameters in the configuration file have corresponding fields in the Administrator utility. We recommend that you use the Administrator utility instead of the configuration file to configure AutoSys Workload Automation on Windows.

Modifying Configuration Parameters

You can configure AutoSys Workload Automation by editing the parameters in the configuration file (UNIX) or the equivalent fields on the Administrator utility (Windows). The parameter values are set during the AutoSys Workload Automation installation. You can modify these values as required.

You can configure the agent by editing the parameters in the agentparm.txt file.

agentparm.txt File

You can configure the agent by editing the parameters in the agentparm.txt file. When you install the agent, the installation program adds commonly configured agent parameters to the agentparm.txt file. Other agent parameters exist, which you must manually add to the agentparm.txt file to configure the agent. You can modify these parameter values as required.

The agentparm.txt file is located in the following directory:

installation_directory/SystemAgent/agent_name

NOTE

If the agent was installed using a program that was not provided with AutoSys Workload Automation (for example, the CA Workload Automation Standalone Agent DVD), the path to the agentparm.txt might be different. In this case, the agentparm.txt file is located in the root directory where the agent is installed.

For information about the parameters in the agentparm.txt file and how to configure them to work with the scheduling manager, see [Workload Automation Agents](#).

Configuration File

You can configure AutoSys Workload Automation by setting the parameters in the configuration file. The configuration file is specific to an instance. On startup, AutoSys Workload Automation reads the configuration file to verify its behavior, including which database to connect to and how to react to certain error conditions.

On Windows, you can configure AutoSys Workload Automation by either editing the fields in the Administrator utility or setting the parameters in the configuration file. The parameters in the configuration file have corresponding fields in the Administrator utility. We recommend that you use the Administrator utility instead of the configuration file to set the configuration parameters on Windows.

WARNING

The scheduler and the application server read the settings in the configuration file only on startup. Therefore, if you make a change that you want to implement immediately, restart the scheduler and application server or pause and resume the scheduler and application server.

The configuration file has the following name:

- On UNIX -- \$AUTOUSER/config.\$AUTOSERV
- On Windows -- %AUTOUSER\config.%AUTOSERV

AUTOSERV

Defines the name of the instance (an uppercase, three-character alphanumeric name) that is associated with the configuration file.

Default: ACE

AUTOUSER

Identifies the path to the directory that is associated with a specific instance. This directory contains the instance-wide configuration files, scheduler or application server output files, encryption files, archive output files that are generated during database maintenance, and sound files (for operating environments supporting audio functionality).

A sample configuration file is available in the \$AUTOSYS/install/data/config.ACE (on UNIX) or %AUTOSYS%\install\data\config.ACE (on Windows) directory. We recommend that you make a copy of the sample configuration file before you modify it.

NOTE

- Events are associated with a specific instance. They have a unique ID, named an eoid, which is prefixed to the instance name. This naming convention ensures the uniqueness and traceability of an event across multiple instances.
- Before you can issue commands at the UNIX operating system prompt, the AutoSys Workload Automation environment must be sourced in the shell and your UNIX user ID and password must be defined on AutoSys

Workload Automation. For more information about sourcing the environment and defining user IDs, see [Install AutoSys Workload Automation](#).

(Windows) AutoSys Workload Automation Administrator

On Windows, you can view or modify the configuration parameters or the environment variables of all the AutoSys Workload Automation instances that you have installed by using the AutoSys Workload Automation Administrator utility. The configuration parameters are also stored in the configuration file. The environment variables are also stored in the Windows Registry. The only exceptions are the LOGROLLOVER and ISDBGACTIV environment variables, which are stored in the configuration file.

You can also use the Administrator utility to create job profiles that contain the environment variables that must be set for a job to run.

WARNING

The scheduler and the application server read the settings in the configuration file only on startup. Therefore, if you make a change that you want to implement immediately, restart the scheduler and application server or pause and resume the scheduler and application server using the Services - AutoSys Workload Automation Administrator window of the Administrator utility.

To open the Administrator utility, Click Start, Programs, AutoSys, Administrator. The Administrator utility starts and the Instance - AutoSys Workload Automation Administrator window opens. This window lets you select an installed instance that you want to view or modify the configuration settings for.

NOTE

- You must have Windows Administrators group privileges to view or modify the configuration parameters of a AutoSys Workload Automation instance using the Administrator utility.
- In Unicenter AutoSys JM r11, you can use the Administrator to connect to and administer a remote registry. In AutoSys Workload Automation r11.3, Release 11.3.5, remote registry connection and administration are not supported.
- On Windows, you can set the configuration parameters by using the Administrator utility or the configuration file. The parameters in the configuration file have a corresponding field on the Administrator utility. We recommend that you use the Administrator utility to set the configuration parameters on Windows.

Environment Variables

You can define environment variables on AutoSys Workload Automation to customize:

- How the AutoSys Workload Automation components generate and manage the output log files
- How the AutoSys Workload Automation components communicate with each other over the network
- The behavior of the scheduler, application server, client utilities, or SDK

On UNIX, you can define the environment variables by issuing either the setenv or the export command (depending on your UNIX operating system) at the operating system prompt.

On Windows, you can define the environment variables using *one* of the following ways:

- Using the System - AutoSys Workload Automation Administrator window of the Administrator utility. Use this method when you define environment variables to customize the scheduler or application server behavior.
- Issuing the set command at the instance command prompt - Use this method when you define environment variables to configure the client utilities.

WAAE.txt File

The WAAE.txt file defines the environment settings for jobs that are started on behalf of all managers for all instances of AutoSys Workload Automation. You can define the environment variables on a single line as a variable=value pair. The jobs that the agent runs inherit these environment variables.

NOTE

The WAAE.txt file applies to the Workload Automation Agent for UNIX, Linux, or Windows.

The WAAE.txt file is located in the following directory:

- Windows -- %AUTOROOT%\SystemAgent\agent_name\Profiles
- UNIX -- \$AUTOROOT/SystemAgent/agent_name/profiles

NOTE

For more information about how to set up the environment variables for the agent, see [Workload Automation Agents](#).

(UNIX) auto.profile File

The /etc/auto.profile file is one of the several objects that source the environment for a job. The /etc/auto.profile file is automatically created during installation and contains variable definitions such as AUTOUSER. The file is on the computer where AutoSys Workload Automation is installed.

A sample auto.profile file is available in the \$AUTOSYS/install/data/auto.profile directory. We recommend that you make a copy of this file before you modify it.

System environment variables are automatically set in the environment for a job. When a job is submitted, the agent processes the following information to source the environment, in the following order:

1. /etc/auto.profile
2. The environment variables that are defined using the envvars attribute in the job definition (if specified).
3. The job profile that is defined using the profile attribute (if specified).

(Windows) Environment Variables in the Administrator Utility

You can view the environment variables on the System - AutoSys Workload Automation Administrator window of the Administrator utility. These settings are defined when you open the AutoSys Workload Automation instance command prompt. You can modify or add to these standard environment variables. These settings reflect the basic information that is used when you run the AutoSys Workload Automation components. You can prevent potential problems by properly setting the required environment variables in every user's environment and configuring AutoSys Workload Automation correctly.

The System - AutoSys Workload Automation Administrator window displays the following environment variables:

• **ASB_INSTANCE**

Identifies a unique AutoSys Workload Automation broker instance name. This environment variable is used by the cross-platform interface of AutoSys Workload Automation. CA, Inc. Common Communications Interface (CAICCI) uses this name to identify the scheduling manager or agent when communicating with remote mainframe instances.

• **AUTOROOT**

Identifies the top-level AutoSys Workload Automation directory.

NOTE

On UNIX, the equivalent environment variable is \$AUTOROOT.

• **AUTOSERV**

Identifies the instance ID, which is a three-character uppercase alphanumeric ID that identifies a AutoSys Workload Automation server installation on a particular computer. By default, the instance ID is ACE. You cannot change this ID

using the Administrator utility. To change the instance ID, uninstall AutoSys Workload Automation and reinstall it using the new ID.

NOTE

On UNIX, the equivalent environment variable is \$AUTOSERV.

- **AUTOSYS**

Identifies the path of the AutoSys Workload Automation software.

NOTE

On UNIX, the equivalent environment variable is \$AUTOSYS.

- **AUTOUSER**

Identifies the path of the AutoSys Workload Automation user's directory. This directory contains the user configuration files, scheduler output files, encryption files, and archive output files that are generated during database maintenance.

- **ISDBGACTIV**

Identifies the debug level to use. The valid values are LIGHT, HEAVY, JOB, DUMP, AFM, DBQUERY, EXTVJ, COMM, MS, GBE, OFF, and UTC.

Default: OFF

- **LOGROLLOVER**

Identifies when the scheduler, application server, and aggregator logs roll over or are purged. The log can roll over at midnight or when the log file size is equal to the specified size. The log is purged when the log file becomes older than the specified number of days. The valid values are OFF, SIZE(x), MIDNIGHT, and PURGE(x).

Default: MIDNIGHT,PURGE(7)

NOTE

You can specify multiple values; separate each value with a comma. For example, SIZE(x),PURGE(x),MIDNIGHT.

Specify the log file size in megabytes. If you do not specify a value for the SIZE(x) parameter, it is set to SIZE(100) by default. The scheduler, application server, and aggregator logs roll over when the log file size reaches 100 MB.

By default, all log files that are older than 7 days are purged at midnight. If you do not specify a value for the PURGE(x) parameter or you set this value to PURGE(0), the log files are not purged. The maximum value that you can set the PURGE(x) parameter to is the total numbers of days since the epoch date (January 01, 1970).

- **PATH**

Identifies the path of the AutoSys Workload Automation utilities. This path is appended to the system path at runtime.

- **PRIMARYCCISYSID**

Identifies the aliased CAICCI system ID specified in the ccirmttd configuration file. This environmental variable is used by the cross-platform interface of AutoSys Workload Automation. It allows the agents to transfer event data to either a primary or a shadow scheduler when running in high availability mode.

(Windows) Manage System Environment Variables Using the Administrator Utility

You can add a system environment variable that AutoSys Workload Automation can use.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
2. Select the instance to which you want to add the system environment variable from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
4. Right-click the scheduler and select Stop. Repeat this action for the application server.
5. Click the System icon on the toolbar.
6. Enter a variable name in the Variable field and its value in the Value field, and click Set.

The environment variable details are displayed in the Environment Variables pane.

7. Click the Services icon on the toolbar.
8. Right-click the scheduler and select Start. Repeat this action for the application server.
The scheduler and application server start. The system environment variable is added to the selected instance.

NOTE

- To modify a system environment variable, double-click the system environment variable to modify in the Environment Variables pane and modify the variable name or its value, or both, and click Set. After you restart the scheduler and the application server, the system environment variable definition is modified.
- To delete a system environment variable, double-click the system environment variable to delete in the Environment Variables pane and click Delete. After you restart the scheduler and the application server, the system environment variable definition is deleted from the selected instance.

ISDBGACTIV

The ISDBGACTIV parameter specifies the debug level to use to control the display of trace messages. This setting is different for the various AutoSys Workload Automation components as follows:

- For the client utilities, it is an operating system environment variable.
- For the scheduler and application server, it is a parameter in the configuration file.
- For the agent, it is a parameter in the agentparm.txt file.

NOTE

For more information about the parameters in the agentparm.txt file, see [Workload Automation Agent Parameters](#).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`ISDBGACTIV=value,value,...`
- Windows -- Set the ISDBGACTIV environment variable using the System - AutoSys Workload Automation Administrator window of the Administrator utility. The ISDBGACTIV environment variable is stored in the configuration file.

NOTE

For information about adding, modifying, or deleting environment variables using the System - AutoSys Workload Automation Administrator window of the Administrator utility, see [Manage System Environment Variables Using the Administrator Utility](#).

AutoSys Workload Automation interprets the ISDBGACTIV values as follows:

- **AFM**
Traces Automation Framework Messages (AFM).
- **COMM**
Traces network communication activity at the sockets level.
- **DBQUERY**
Traces and measures the elapsed time of calls to the database.
- **DUMP**
Traces data that AutoSys Workload Automation sends or receives when communication occurs through the internal communication interface or the cross-platform interface.

NOTE

The dump files are located in the \$AUTOUSER/out (on UNIX) or %AUTOUSER%\out (on Windows) directory. The dump files with a prefix libmsg.* are created when communication occurs through the

internal communication interface. The dump files with a prefix `event_demon_dump.*` are created when communication occurs through the cross-platform interface.

- **EXTVJ**
Generates trace data pertaining to user-defined job validation.
- **GBE**
Traces scheduler events as they are read from the `ujo_event` table.
- **HEAVY**
Returns full trace information.
- **JOB**
Traces the run time of a job.
- **LIGHT**
Returns light trace information.
- **MS**
Adds milliseconds to the time in the log output.
- **OFF**
Returns no trace information. This is the default.
- **UTC**
Displays the time that the event is marked to be picked up in the scheduler log output.
- **NOTE**
 - *n* specifies the level of tracing. This value can be a number in the range 1 through 10, where 1 is the lowest level of tracing and 10 is the highest.
 - The `dcam_appsvr.log` and `dcam_scheduler.log` log files are created in the `$AUTOUSER/out` (on UNIX) or `%AUTOUSER%/out` (on Windows) directory.
 - To combine trace settings, separate each setting with a comma (,). If you use the OFF setting with other settings, the traceable applications do not display a trace.
 - After you set the `ISDBGACTIV` parameter, pause and resume the scheduler and application server for the new settings to take effect.
 - You can modify the `ISDBGACTIV` value at any time while the traceable applications are running. On startup, the traceable applications search for the specified `ISDBGACTIV` value and output the appropriate trace messages

LOGROLLOVER

The LOGROLLOVER parameter specifies when the scheduler, aggregator, and the application server logs roll over or are purged. The log can roll over at midnight or when the log file size is equal to the specified size. The log is purged when the log file becomes older than the specified number of days.

NOTE

The aggregator logs are generated only when the `AggregateStatistics` parameter is set to 1 in the configuration file or the `Aggregate Statistics` check box is selected on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility (Windows).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`LOGROLLOVER=value`
- Windows -- Set the LOGROLLOVER environment variable using the System - AutoSys Workload Automation Administrator window of the Administrator utility. The LOGROLLOVER environment variable is stored in the configuration file.

NOTE

For information about adding, modifying, or deleting environment variables using the System - AutoSys Workload Automation Administrator window of the Administrator utility, see [Manage System Environment Variables Using the Administrator Utility](#).

You can set the LOGROLLOVER value to any *one* of the following values:

- **OFF**
Disables the roll over and purging of log files.
- **SIZE(x)**
Specifies that the log rolls over when the log file size is equal to the specified size.

NOTE

Specify the log file size in megabytes. The default value is SIZE(100); the log rolls over when the file size reaches 100 MB.

- **MIDNIGHT**
Specifies that the log rolls over at midnight.
- **PURGE(x)**
Specifies that all log files that are older than the specified number of days are purged at midnight.

NOTE

- The default value is PURGE(7); all log files that are older than 7 days are purged at midnight.
- If you upgrade to the current release, the log files are not purged by default. You must set this value.
- If you do not specify a value for the PURGE(x) parameter or you set this value to PURGE(0), the log files are not purged.
- The maximum value that you can set this value to is the total numbers of days since the epoch date (January 01, 1970). Default: MIDNIGHT,PURGE(7).

NOTE

- The default value is MIDNIGHT,PURGE(7).
- You can specify multiple values; separate each value with a comma. For example, SIZE(x),PURGE(x),MIDNIGHT.
- After you set the LOGROLLOVER parameter, pause and resume the scheduler and application server for the new settings to take effect.

Example: Specify the Log Rollover

This example rolls over the scheduler, aggregator, and application server logs at midnight and when the log file size is equal to 100 MB. The logs are not purged.

```
LOGROLLOVER=SIZE(100),MIDNIGHT
```

Environment Variables -- Logging

You can define the following environment variables to customize how the AutoSys Workload Automation components generate and manage the output log files:

- **APIJNI_LOGPATH**
Specifies the full path to the directory where the libapijni.JNI_process_ID.out output log file is created.
 - **JNI_process_ID**
Specifies the ID of the Java process that invokes the Java Native Interface (JNI) to execute the routines that the AutoSys Workload Automation Java SDK uses.
- **AUTOSYS_LOG**

Specifies that the client utilities direct all output to the *client_name.process_ID.out* log file in the \$AUTOUSER/out (UNIX) or %AUTOUSER%\out (Windows) directory.

– **process_ID**

Specifies the process ID of the client process.

- **ISDBGACTIV**

Identifies the debug level to use. The valid values are LIGHT, HEAVY, JOB, DUMP, AFM, DBQUERY, EXTVJ, COMM, MS, GBE, OFF, and RESOURCE*n*.

Default: OFF

- **LOGROLLOVER**

Identifies when the scheduler, application server, and aggregator logs roll over or are purged. The log can roll over at midnight or when the log file size is equal to the specified size. The log is purged when the log file becomes older than the specified number of days. The valid values are OFF, SIZE(x), MIDNIGHT, and PURGE(x).

Default: MIDNIGHT,PURGE(7)

NOTE

You can specify multiple values; separate each value with a comma. For example, SIZE(x),PURGE(x),MIDNIGHT

Specify the log file size in megabytes. If you do not specify a value for the SIZE(x) parameter, it is set to SIZE(100) by default. The scheduler, application server, and aggregator logs roll over when the log file size reaches 100 MB.

By default, all log files that are older than 7 days are purged at midnight. If you do not specify a value for the PURGE(x) parameter or you set this value to PURGE(0), the log files are not purged. The maximum value that you can set the PURGE(x) parameter to is the total numbers of days since the epoch date (January 01, 1970).

Environment Variables -- Scheduler

You can define the following environment variables to customize the behavior of the scheduler:

- **AGENT_RECV_TIMEOUT**

Specifies the time (in seconds) that the scheduler waits for a response from the agent. Increasing this value might delay the scheduler in placing a machine in the offline state when the corresponding agent becomes unresponsive. Decreasing the value might result in more time outs.

Default: 20

Limits: 1 or higher

NOTE

- For the communication between the scheduler and the agent, this variable applies to the agent autoping, chase, and CPU utilization (load balancing jobs) requests.
- If you define this variable, the CONNECTION_TIMEOUT variable is ignored for communications between the scheduler and the agent.
- This variable does not affect the following job requests to the agent:
 - Starting a job.
 - Killing a job.
 - Issuing a UNIX signal to a job or signaling a Windows event that a job is listening on.
 - Restarting a job.
 - Replying to a job that has paused execution.

- **ASNOTIFY_SMTP_TIMEOUT**

Specifies the time (in seconds) that the scheduler waits for a response from the SMTP server when the email notification job is running and there is a network problem connecting to the SMTP server.

Default: 300

- **BOX_NO_SUCCESS**

Specifies whether the scheduler keeps the box in the RUNNING state if all the jobs in the box enter the INACTIVE state. The valid values are 0 and 1.

Default: 0; the scheduler does not keep the box in the RUNNING state if all the jobs in the box enter the INACTIVE state.

NOTE

If you do not define this variable, the scheduler evaluates the status of the box to SUCCESS when all jobs in the box enter the INACTIVE state.

- **CHASE_SLEEP**

Specifies the time (in seconds) that the chase utility pauses:

- Between successive sends of chase data to an agent following a connection failure.

Default: 5

Limits: 1 or higher; increasing this value might delay recovery during a network failure. Decreasing this value might result in premature chase failures.

- Before raising the CHASE_ALARM alarm; during periods of network load or high workload activity, you can set this variable to let the chase utility extend the time that it waits for the scheduler to receive delayed job statuses from the agent; thus preventing the chase utility from raising false alarms.

Default: 10

- **CONTINUE_JOB_COND_JOID_MISSING**

Specifies that the scheduler continues to evaluate the starting conditions of other downstream dependent jobs even when it fails to evaluate the starting conditions of one of the downstream dependent jobs.

- **DB_CONNECTIONS**

Defines the maximum number of database connections that the scheduler can connect to. Increasing this value might increase the ability of the scheduler to perform simultaneous database operations.

Default: 16

Limits: 1-128

- **DB_CONNECTIONS_SEND**

Defines the maximum number of database connections that the scheduler can connect to in order to update the network communication-based database tables for every message that the scheduler sends to an agent. Increasing this value might increase the ability of the scheduler to process simultaneous outbound network data to numerous agents.

Default: 2

Limits: 1-128

Recommended: Under normal circumstances, use the default value. The value should not exceed the value of SCHED_SCALE*2.

- **DB_CONNECTIONS_RECV**

Defines the maximum number of database connections that the scheduler can connect to in order to update the network communication-based database tables for every message that the scheduler receives from an agent. Increasing this value might increase the ability of the scheduler to process simultaneous inbound network data from numerous agents.

Default: 2

Limits: 1-128

Recommended: Under normal circumstances, use the default value. The value should not exceed the value of SCHED_SCALE*2.

- **EMSEC_EXIT_CRYPT**

Specifies that the scheduler dynamically loads the \$CAIGLBL0000/sche/lib/libcacexit (UNIX) or %CAIGLBL0000%\bin\cacexit.dll (Windows) library. This library contains routines to encrypt or decrypt user passwords for cross-platform job submissions to and from the remote managers using CAICCI. The valid values are ON and OFF.

Default: OFF; the scheduler does not dynamically load the library.

NOTE

If you set this variable to ON, ensure that you run the scheduler in an environment where the CAIGLBL0000 variable is defined.

- **PE_AUTO_DISABLE**

Specifies whether the scheduler disables the auto offline machine feature. The valid values are 0 and 1.

Default: 0; the scheduler enables the auto offline machine feature.

NOTE

- When the auto offline feature is enabled, the scheduler takes the following actions:
 - Places all machines in the unqualified or missing state when it fails to connect to the corresponding agents.
 - Places the machines in the offline state when it receives a message from an agent that is shutting down.
 - Periodically tests the agents corresponding to the machines in the unqualified and missing states to determine if it can automatically bring the machine online when the agent becomes responsive.
- When the auto offline feature is disabled, the scheduler takes the following actions:
 - Maintains all machines in the online state; regardless of whether the machines can connect to the agent or not.
 - Restarts jobs that are scheduled to agents with connection problems until it exhausts the number of retries that are allotted to the job.
 - Generates an automatic MACH_OFFLINE event when it receives a message from an agent that is shutting down. Although, the scheduler generates an automatic MACH_OFFLINE event, it keeps the machine in online state.
 - Places jobs that are scheduled to machines that are manually placed offline in the PEND_MACH state. You can manually send a MACH_OFFLINE event to place a machine in the offline state. Jobs remain in the PEND_MACH state until you manually send a MACH_ONLINE event to bring the machine back online.

- **RESTRICT_FORCE_STARTJOB**

Specifies that the scheduler restricts you from running multiple instances of a job while the job is in the STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.

NOTE

If you set this variable to any value, the scheduler restricts you from running multiple instances of a job while the job is in the STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.

- **SCHED_SCALE**

Defines the maximum limit of process threads that can be started dynamically. This value does not represent the total number of process threads that are active. Rather, it is a scale value that the scheduler uses to calculate the maximum limit of process threads that can be started dynamically as the workload demands. Therefore, a SCHED_SCALE value of 1 represents the lowest ceiling of extra dynamic threads that can be started to process job events (some arbitrary ceiling not necessarily equal to one). Increasing this value might increase the ability of the scheduler to process job events.

Default: 5

Limits: 1-64

- **UA_PORT**

Specifies the port number of the r11 legacy agent.

Default: 49152 (SSA virtual port)

Limits: 1-50176

Environment Variables -- Application Server

You can define the following environment variables to customize the behavior of the application server:

- **AGENT_RECV_TIMEOUT**

Specifies the time (in seconds) that the application server waits for a response from the scheduler or the agent. Increasing this value might delay the application server when the scheduler or agent becomes unresponsive. Decreasing the value might result in more time outs.

Default: 20

Limits: 1 or higher.

NOTE

- If you set this variable, the CONNECTION_TIMEOUT variable is ignored.
- For the communication between the application server and the scheduler, this variable applies to the chk_auto_up request. For the communication between the application server and the agent, this variable applies to the autoping and chase requests.

- **AS_IAM_CACHE_UPDATE**

Specifies the time (in seconds) that the application server polls the external security server for modified security policies.

Default: -2

Limits: -2, -1, or higher.

NOTE

If you set this variable to -1, the application server does not poll the external security server for policies. If you set this variable to -2, the application server obtains the poll interval from the external security server configuration.

- **AS_IAM_MAX_POLICIES**

Specifies the maximum number of security policies that the application server collects from the external security server.

Default: 10,000

Limits: 2001 or higher.

NOTE

If you do not define this variable or you set this variable to a value less than 2001, the application server uses the default value.

- **ASQMAX**

Specifies the maximum length of the application server queue of outstanding client requests. If a client request arrives when the queue is full, the application server discards the request and the client times out. Increasing this value might place a greater demand on the system memory. Decreasing this value might result in more client time outs when the application server is busy.

Default: 32,768

Limits: 0 or higher.

NOTE

If you set this variable to 0 (zero), the queue length is set to the default.

- **DB_CONNECTIONS**

Defines the maximum number of database connections that the application server can connect to. Increasing this value might increase the ability of the application server to process simultaneous AutoSys Workload Automation client and agent requests.

Default: 32

Limit: 1-128

- **DB_CONNECTIONS_SEND**

Defines the maximum number of database connections that the application server can connect to in order to update the network communication-based database tables for every message that the application server sends to an agent. Increasing this value might increase the ability of the application server to process simultaneous outbound network data to numerous agents.

Default: 2

Limits: 1-128

Recommended: Under normal circumstances, use the default value. The value should not exceed the value of DB_CONNECTIONS.

- **DB_CONNECTIONS_RECV**

Defines the maximum number of database connections that the application server can connect to in order to update the network communication-based database tables for every message that the application server receives from an agent. Increasing this value might increase the ability of the application server to process simultaneous inbound network data from numerous agents.

Default: 2

Limits: 1-128

Recommended: Under normal circumstances, use the default value. The value should not exceed the value of DB_CONNECTIONS.

- **RESTRICT_DELETE_DEPENDENT_JOB**

Specifies that the application server restricts you from deleting a job with dependencies.

NOTE

If you set this variable to any value, the application server restricts you from deleting a job with dependencies.

- **RESTRICT_DELETE_JOB**

Specifies that the application server restricts you from deleting a job when the job is in the ACTIVATED, STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.

NOTE

If you set this variable to any value, the application server restricts you from deleting a job when the job is in the ACTIVATED, STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.

- **SKIP_CHASE_PENDING_CHECK**

Specifies that the application server executes a single request to the agent instead of a two-step request to get the list of running jobs as part of the chase operation. Setting this environment variable helps the application server to reduce the amount of network traffic for each chase request. However, skipping the execution of the second step of the chase request might result in acquiring an incomplete list of running jobs and raising more chase alarms. This variable is not set by default. The valid values are y, Y, or 1.

NOTE

Workload Automation Agent for UNIX, Linux, Windows, or i5/OS Release 11.3 SP4 (Incremental 1) or lower ignores the second step of the chase request that is made by the application server. If your environment consists solely of Workload Automation Agent for UNIX, Linux, Windows, or i5/OS Release 11.3 SP4 (Incremental 1) or lower, set the environment variable to avoid making two requests.

Environment Variables -- Client Utilities or SDK

You can define the following environment variables to customize the behavior of the client utilities or the SDK:

- **AS_HOSTALIAS**

Specifies the name of the client as an alternative to the value obtained by the operating system. Use this name in environments where a local hostname is valid only on the client and not on the server. In these environments, this variable also applies to the agent as it uses the SDK libraries to communicate with the application server.

NOTE

If the client has access to the configuration file, the hostname resolves to the value you specified in the ManagerHostAlias parameter.

- **AS_FC_THREADS**

Specifies the maximum number of worker threads that the forecast utility uses at any given time. When all worker threads are occupied, the forecast utility postpones new work until the worker thread completes its processing and becomes available. Increasing this value might place a greater demand on system resources. Decreasing this value might delay the processing of the forecast request.

Default: 7

Limits: 1 or higher.

- **CHASE_SLEEP**

Specifies the time (in seconds) that the chase utility pauses:

- Between successive sends of chase data to an agent following a connection failure.

Default: 5

Limits: 1 or higher; increasing this value might delay recovery during a network failure. Decreasing this value might result in premature chase failures.

- Before raising the CHASE_ALARM alarm. During periods of network load or high workload activity, you can set this variable to let the chase utility extend the time that it waits for the scheduler to receive delayed job statuses from the agent; thus preventing the chase utility from raising false alarms.

Default: 10

- **CHASE_STARTING_WAIT_INTERVAL**

Specifies the minimum time (in minutes) that the chase utility allows jobs to remain in the STARTING state before raising the CHASE_ALARM alarm. During periods of network load or high workload activity, it might take more than the default two minutes for the agent to notify the scheduler that a job is executing. Setting this environment variable helps reduce the number of alarms that the chase utility generates during this time. Set this environment variable on the computer where you are running the chase command.

Default: 2

Limits: 1 to 35791394

- **DBSPACE_ALARM_SPACE**

Specifies the maximum space (in MB) that all the database tables AutoSys Workload Automation uses can occupy before the dbspace utility raises an alarm. If the total space used by all the database tables exceeds the value that you specify in this variable, the dbspace utility raises the DB_PROBLEM alarm. Increasing this value might result in the delayed notification of potential database issues due to lack of space. Decreasing this value might result in the premature raising of alarms.

Default: 1000

Limits: 1 or higher.

Environment Variables -- Network Communication

You can define the following environment variables to customize how the AutoSys Workload Automation components communicate with each other over the network:

- **AS_RESOLVEHOST_TIMEOUT**

(UNIX only) Specifies the time (in seconds) the AutoSys Workload Automation component waits for the system to resolve a hostname into an IP address. The AutoSys Workload Automation component waits for an extra 15 seconds to the time you specify in this variable. For example, if you set this variable to 20 seconds, the AutoSys Workload Automation component waits for 35 seconds to resolve a hostname into an IP address.

Default: 15

Limits: 1-120

- **CONNECTION_TIMEOUT**

Specifies the time (in seconds) the AutoSys Workload Automation component waits to complete the initial connection with another AutoSys Workload Automation component. Increasing this value might delay recovery during a network failure. Decreasing this value might result in connection failures.

Default: 20

Limits: 1 or higher.

- **MAX_BUFFER**

Specifies the maximum number of bytes to allocate for buffered responses in the application server to client communications. Increasing this value might place a greater demand on the system memory. Decreasing this value might result in more network activity.

Default: 65,536 (UNIX); 32,768 (Windows)

Limits: 1 or higher.

- **MB4Q**

Specifies the maximum memory (in megabytes) that is used by each internal queue in the communications engine to store incoming network data. Exceeding the memory for queued incoming network data indicates a potentially severe performance issue that might necessitate the shut down of the product. Increasing the limit might allow the product to run longer but might worsen the performance issue in the long run.

Default: 8

Limits: 4095

- **USE_SINGLETHREADED_AGENTLISTENER**

Specifies that the scheduler and application server are running in single threaded listener mode where a single thread processes connections from the agents in a serial manner. By default, the scheduler and application server run in multi-threaded listener mode where multiple threads process connections from agents simultaneously.

When the application server and scheduler run in single threaded listener mode, they serialize the processing of inbound communications from every agent in the instance. The scheduler and application server acknowledge one set of messages from one agent and do not acknowledge the receipt of messages from other agents. If many agents send messages to the scheduler or application server, they might experience frequent receive-based time outs because the scheduler or application server is busy with other agents.

When the application server and scheduler run in multi-threaded listener mode, they asynchronously manage communications from multiple agents at a time. While the scheduler and application server acknowledge one set of messages from one agent, they simultaneously acknowledge the receipt of messages from other agents. If many agents send messages to the scheduler or application server, they might experience receive-based time outs less frequently because the scheduler or application server is managing messages from multiple agents at a time.

NOTE

- If you set this variable to any value, the scheduler runs in single threaded listener mode.
- When the scheduler runs in multi-threaded listener mode, by default the number of listener threads is three if the SCHED_SCALE parameter is set to the default value of five. The formula that is used to calculate the number of threads is $(\text{SCHED_SCALE} + 1)/2$. To increase the number of threads, set the SCHED_SCALE value accordingly.

- **AGENTLISTENER_THREAD_CEILING**

Specifies the number of agent listener threads.

Default: 3

Limits: Number of threads supported by your operating environment. Negative values and a value of zero are ignored.

Recommendation: Under normal circumstances, use the default value. Change this value only when the difference in the number of seconds between the STARTING and RUNNING events of multiple jobs continue to increase significantly over time resulting in delays in the workflow. Tune this value to a number at which the delay in the workflow is optimal in your environment. Do not set this value to exceed 32. You can set the AGENTLISTENER_THREAD_CEILING environment variable as follows:

- On UNIX, use the export command to export it in the \$AUTOUSER/autosys.*sh.hostname file.
- On Windows, set this environment variable using the System - AutoSys Workload Automation Administrator window of the Administrator utility.

You must restart the AutoSys Workload Automation services for the new settings to take effect.

- **MSGSVCLISTEN_BACKLOG**

Specifies the maximum length of the queue of outstanding connections that are associated with a socket listen operation. The underlying service provider responsible for the socket sets the backlog to a maximum reasonable value for the operating system. Increasing this value might silently reduce to a value imposed by the operating system. Decreasing this value might result in a loss of socket connections.

Limits: 1 or higher.

NOTE

- For more information about the backlog parameter of the listen function, see the appropriate operating system sockets documentation.
- Set the MSGSVC_LISTEN_BACKLOG environment variable only when the scheduler is running in single threaded listener mode. Unset it when the scheduler is running in multi-threaded listener mode.

- **MSGSVC_LISTEN_TIMEOUT**

Specifies the time (in seconds) that AutoSys Workload Automation waits for incoming network data to arrive on a socket connection before failing and closing its end of the connection.

Default (when the environment variable is not set): 20

Limits: 1 or higher.

NOTE

Set the MSGSVC_LISTEN_TIMEOUT environment variable only when the scheduler is running in single threaded listener mode. Unset it when the scheduler is running in multi-threaded listener mode.

- **RECV_TIMEOUT**

Specifies the time (in seconds) that the AutoSys Workload Automation component waits to complete a request-response exchange with another AutoSys Workload Automation component. The AutoSys Workload Automation component doubles the time that you specify in this variable to wait for each response to arrive. If no network activity is detected during this time, the waiting component times out the request. Increasing this value might delay the execution time of the component during a network failure. Decreasing this value might result in more component time outs.

Default: 30

Limits: 1 or higher.

NOTE

- For the communication between the application server and the scheduler, this variable applies to the scheduler log retrieval and autoping requests. For the communication between the application server and the agent, this variable applies to the agent job log retrieval and user authentication requests and to the requests made on behalf of Web UI.
- If CA Technical Support previously instructed you to set the RECV_TIMEOUT variable to a low value to time out poor socket connections sooner due to frequent network delays and you upgraded from Release 11.3.6 SP2 or lower, unset the RECV_TIMEOUT variable and set the MSGSVC_LISTEN_TIMEOUT variable to the same low value instead. Restart the scheduler or application server for the changes to take effect.
- There is no relationship between RECV_TIMEOUT and MSGSVC_LISTEN_TIMEOUT. After upgrading to Release 11.3.6 SP3 or higher, you might set both without consequence.

Configure an Instance

You can configure a AutoSys Workload Automation instance by editing the following instance parameters in the configuration file (UNIX) or the equivalent fields on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility (Windows).

(UNIX) Follow these steps:

1. Log in to AutoSys Workload Automation as a user authorized to stop the scheduler and the application server and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt to stop the scheduler and application server:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
```


3. Edit the instance parameters in the configuration file as appropriate and save the file.
4. Enter the following commands at the operating system prompt to start the scheduler and application server:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The new parameter settings take effect.

(Windows) Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
 2. Select the instance that you want to configure from the Instance drop-down list in the Settings pane.
 3. Click the Services icon on the toolbar.
 4. Right-click the scheduler and select Stop. Repeat this action for the application server.
 5. Click the Instance icon on the toolbar.
 6. Edit the fields on the Instance - AutoSys Workload Automation window as appropriate.
 7. Click the Services icon on the toolbar.
 8. Right-click the scheduler and select Start. Repeat this action for the application server.
- The scheduler and the application server start and the new parameter settings take effect.

DateFormat

The DateFormat parameter specifies the date format for entering and displaying dates. The default value is MM/DD/YYYY.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
DateFormat=date_format
```
- Windows -- Select a value from the Date Format drop-down list on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

After you set the DateFormat parameter, restart the scheduler and application server for the new settings to take effect.

EnableFIPSMODE

The EnableFIPSMODE parameter specifies whether you want AutoSys Workload Automation to operate in FIPS mode.

AutoSys Workload Automation uses FIPS-approved and non-FIPS approved security algorithms unless you enable FIPS mode. When AutoSys Workload Automation operates in FIPS mode, the system only uses security algorithms that comply with the standards in the FIPS 140-2 publication.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
EnableFIPSMODE=0|1
```
- **0**
 Specifies that AutoSys Workload Automation uses both FIPS-approved and non-FIPS approved security algorithms.
- **1**
 Specifies that AutoSys Workload Automation uses only FIPS-approved security algorithms.
- Windows -- Click the FIPS Mode tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Enable FIPS Mode check box.

NOTE

For more information about enabling FIPS mode, see [FIPS Mode](#).

EnableIPCaching

The EnableIPCaching parameter specifies whether you want to enable IP address caching.

By default, AutoSys Workload Automation caches the IP addresses of the computers that it connects to for running jobs or any other communication. AutoSys Workload Automation automatically recovers from a dynamic IP address when the old IP address change in the cache becomes invalid. AutoSys Workload Automation does not function properly when a cached IP address is still valid but points to another computer with a AutoSys Workload Automation installation on it. To avoid potential loss of productivity in dynamic environments, disable IP address caching.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

NOTE

AutoSys Workload Automation verifies the IP address on every network request when IP address caching is disabled. These verifications impact system performances. We recommend that you disable IP address caching only in dynamic environments.

```
EnableIPCaching=0|1
```

- **0**
Disables IP address caching.
- **1**
Enables IP address caching. This is the default.
- Windows -- Click the Communication tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility and select the Enable IP Address Caching check box. If you clear the box, IP address caching is disabled.

NOTE

After you set the EnableIPCaching parameter, pause and resume the scheduler and application server for the new setting to take effect.

IPAddressPreference

The IPAddressPreference parameter specifies the criteria that AutoSys Workload Automation uses to select an IP address from the list of IP addresses that are returned by the operating system after translating the host name.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
IPAddressPreference=0|1|2|3
```

- **0**
Specifies that AutoSys Workload Automation selects the first IPv6 address that is returned by the operating system if the operating system lets it create a TCP/IP socket for the IPv6 protocol or selects the final IPv4 address if no IPv6 addresses are returned. This is the default.
- **1**
Specifies that AutoSys Workload Automation always selects the first IP address that is returned by the operating system.
- **2**

- Specifies that AutoSys Workload Automation always selects the first IPv4 address that is returned by the operating system or returns an error if no IPv4 addresses are returned.
- **3**
 - Specifies that AutoSys Workload Automation selects the first IPv6 address that is returned by the operating system or selects the final IPv4 address if no IPv6 addresses are returned.
- Windows -- Click the Communication tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility and select any *one* of the following options in the IP Address Preference pane as appropriate:
 - **Prefer Socket Test**
 - Specifies that AutoSys Workload Automation selects the first IPv6 address that is returned by the operating system if the operating system lets it create a TCP/IP socket for the IPv6 protocol or selects the final IPv4 address if no IPv6 addresses are returned. This is the default.
 - **Prefer First**
 - Specifies that AutoSys Workload Automation always selects the first IP address that is returned by the operating system.
 - **Prefer IPv4**
 - Specifies that AutoSys Workload Automation always selects the first IPv4 address that is returned by the operating system or returns an error if no IPv4 addresses are returned.
 - **Prefer IPv6**
 - Specifies that AutoSys Workload Automation selects the first IPv6 address that is returned by the operating system or selects the final IPv4 address if no IPv6 addresses are returned.

ManagerHostAlias

The ManagerHostAlias parameter specifies the host name of the scheduler and application server as an alternative to the value obtained by the operating system. If you specify the manager host name alias, the host name of the scheduler and application server resolves to the specified value.

In a cluster environment, AutoSys Workload Automation components can maintain communication with the active server components that are under the control of the cluster manager when you specify a manager host name alias. Use the same manager host name alias on each node in the cluster.

NOTE

The cluster manager restarts AutoSys Workload Automation server components when they fail. Other AutoSys Workload Automation components can lose communication with server components that the cluster manager restarts when you do not use the same manager host name alias for every node in the cluster. For more information about specifying the manager host name alias in a cluster environment, see [\(UNIX\) Improve Workload Performance: Highly-Available Environments](#) or [\(Windows\) How to Improve Workload Performance: Highly-Available Environments](#).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:


```
ManagerHostAlias=value
```
- Windows -- Click the Communication tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Manager Hostname Alias check box. If you select the check box, then enter the cluster host name.

(Windows) MSSQLDriverConnString

The MSSQLDriverConnString variable lets you connect AutoSys Workload Automation with the SQL Server that is TLS enabled.

You can configure this variable as follows:

1. Add the MSSQLDriverConnString configuration variable in the configuration file (%AUTOUSER%/config.%AUTOSERV%) and set its value (within curly brackets) to the driver that are available on the AutoSys Workload Automation server computer.

NOTE

- By default, the value is set to {SQL Server}.
- You can download the latest drivers from the SQL Server support site.
- To view the list of drivers that are installed on the AutoSys Workload Automation server, follow these steps:
 - a. Navigate to the Windows Control Panel.
 - b. Double-click Administrative Tools.
 - c. Double-click ODBC (32-bit).
The ODBC Data Source Administrator (32-bit) panel is displayed.
 - d. Click the Drivers tab.

2. Save the configuration file.

Once the AutoSys Workload Automation services are restarted, AutoSys Workload Automation connects to the SQL Server that is TLS enabled.

For example, if the Microsoft ODBC Driver 11.0 is available on the AutoSys Workload Automation server computer, set the value of the MSSQLDriverConnString variable as follows:

```
MSSQLDriverConnString={ODBC Driver 11 for SQL Server}
```

UseCommAliasEncryption

The UseCommAliasEncryption parameter specifies whether encryption is used for all communication between the AutoSys Workload Automation components and z/OS managers.

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES 128-bit encryption, generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory. The cryptkey_alias.txt file stores the communication alias key. You can specify the communication alias encryption key using the -a option of the as_config command.

WARNING

Set the AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, disable AES 128-bit encryption and use no encryption for z/OS communication.
- If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting. For more information about configuring the AGENTDEF data sets, see [Generate an Instance-Wide Communication Alias Encryption File](#).
- For more information about generating the instance-wide communication alias encryption file using the as_config command, see [Generate an Instance-Wide Communication Alias Encryption File](#).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`UseCommAliasEncryption=0|2`
 - **0**
Specifies that no encryption is used.
 - **2**
Specifies that AES 128-bit encryption is used to encrypt data.
- Windows -- Click the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility, and select or clear the Use AES 128-bit encryption when communicating with zOS managers check box. If you select the check box, then enter the encryption key.

NOTE

- The encryption key must be a hexadecimal string of 32 characters.
- The zOS Encryption Status field in the Instance Encryption Status pane displays the current encryption set for the selected AutoSys Workload Automation instance.

UseEncryption

The UseEncryption parameter defines the instance-specific encryption key for all communication between the AutoSys Workload Automation components of the same instance. For AutoSys Workload Automation to work properly, the encryption key must be the same on all client computers. This encryption key is stored in the \$AUTOUSER/cryptkey.txt (on UNIX) or %AUTOUSER%/cryptkey.txt (on Windows) file.

NOTE

During the AutoSys Workload Automation installation, a default encryption key is created. However, if you perform a custom installation, you can specify the encryption key during the installation. You can modify the encryption key after the AutoSys Workload Automation installation.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`UseEncryption=0|1|2`
 - **0**
Specifies that no encryption is used.
 - **1**
Specifies that the default encryption key is used to encrypt data. This is the default.
 - **2**
Specifies that a user-specified encryption key is used to encrypt data. Specify the encryption key using the `as_config` command.
- Windows -- Click the Instance Wide Encryption tab on the Instance - AutoSys Workload Automation Administrator window of the Administrator utility, and select or clear the Use AES 128-bit encryption check box. If you select the check box, then select any *one* of the following encryption options:
 - **Use Default Key**
Encrypts data using the default encryption key. By default, this option is selected.
 - **Specify a Custom Key**
Encrypts data using a user-specified encryption key. If you select this option, select the key format from the Key Format drop-down list, enter the encryption key in the Key field, and confirm the encryption key in the Verify Key field.

NOTE

The encryption key can be a hexadecimal string of 32 characters (starting with 0x) or a passphrase string of 16 characters.

NOTE

The Instance Wide Encryption Status field in the Instance Encryption Status pane displays the current encryption set for the selected AutoSys Workload Automation instance.

Configure a Scheduler

You can configure a scheduler by editing the following scheduler parameters in the configuration file (UNIX) or the equivalent fields on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility (Windows):

(UNIX) Follow these steps:

1. Log in to AutoSys Workload Automation as a user authorized to stop the scheduler and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt to stop the scheduler:

```
sendevent -E STOP_DEMON
```

3. Edit the scheduler parameters in the configuration file as appropriate and save the file.
4. Enter the following command at the operating system prompt to start the scheduler:

```
eventor
```

The new parameter settings take effect.

NOTE

You can [pause and resume](#) the scheduler to read the modified values of some of the parameters in the configuration file at runtime.

(Windows) Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
2. Select an instance for which you want to configure the scheduler from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
4. Right-click the scheduler and select Stop.
5. Click the Scheduler icon on the toolbar.
6. Edit the fields on the Scheduler - AutoSys Workload Automation window as appropriate.
7. Click the Services icon on the toolbar.
8. Right-click the scheduler and select Start.

The scheduler starts and the new parameter settings take effect.

NOTE

You can [pause and resume](#) the scheduler to read the modified values of some of the fields on the Administrator utility at runtime.

RoleDesignator

The RoleDesignator parameter specifies whether the scheduler is the primary scheduler, a shadow scheduler, or a tie-breaker scheduler.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
RoleDesignator=1|2|3
```

- **1** Specifies that the scheduler is the primary scheduler. This is the default.
- **2**
Specifies that the scheduler is the shadow scheduler.
- **3**
Specifies that the scheduler is the tie-breaker scheduler.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator and select the scheduler role.

HAPollInterval

The HAPollInterval parameter defines the time interval (in seconds) between status polls when the scheduler runs in high availability mode. The default value is 15.

In high availability mode, the primary, shadow, and tie-breaker schedulers update the database with their heartbeats at regular intervals. If a scheduler does not update the database after two intervals, that scheduler is deemed unavailable and the system either fails over to the available scheduler or leaves high availability mode. You can configure the length of each interval.

NOTE

If AutoSys Workload Automation runs in high availability mode with dual event servers, the specified high availability poll interval value doubles. The increased interval provides enough time for both databases to be updated before the schedulers poll for one another's status. For example, when you set the interval to 15 seconds, the schedulers refresh their status every 15 seconds in single event server mode and every 30 seconds in dual event server mode.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`HAPollInterval=value`
- Windows -- Enter the time interval in the HA Poll Interval field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- In single event server mode, AutoSys Workload Automation enters high availability mode when both the primary and shadow schedulers are running. If the shadow scheduler does not update the database for two consecutive intervals after entering high availability mode, the primary scheduler issues an EP_HIGH_AVAIL alarm with a message to indicate that the shadow scheduler has not updated its status. If the shadow scheduler returns and posts updates at regular intervals, AutoSys Workload Automation reenters high availability mode. If the primary scheduler does not update the database for two consecutive intervals, the shadow scheduler issues an EP_ROLLOVER alarm with a message to indicate that the primary scheduler has not updated its status. It proceeds to failover and starts processing events. If the original primary scheduler returns, it detects that the shadow scheduler has failed over and shuts down. AutoSys Workload Automation remains in failover status until the shadow scheduler is shut down. If the primary or shadow scheduler loses its connection to the event server, the high availability evaluations stop until the scheduler restores its connection to the event server.
- In dual event server mode, AutoSys Workload Automation enters high availability mode when the primary, shadow, and tie-breaker schedulers are running. The detection and failover procedure is the same as in single event server mode. However, before either of the schedulers make the final decision to failover, AutoSys Workload Automation verifies the tie-breaker scheduler has sent regular updates. If either the primary or shadow scheduler fails to detect two consecutive updates from both its counterparts and the tie-breaker scheduler, the scheduler shuts down. If the primary, shadow, or tie-breaker scheduler loses its connection to one or both event servers, the high availability evaluations stop until the scheduler restores

its connection to the event server or rolls over to single event server mode. In the meantime, the scheduler continues to update the accessible database at regular intervals.

FileSystemThreshold

The FileSystemThreshold parameter defines the minimum amount of disk space that must be available to write to the scheduler log file. The default value is 20480 KB (20 MB). You can specify a value in the range 8192 KB (8 MB) through 102400 KB (100 MB). If the specified value is not in the valid range, the scheduler issues a warning message and resets the value to the default.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
FileSystemThreshold=value
```
- Windows -- Enter the minimum scheduler log disk space in the FileSystem Threshold KB field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

If the available disk space falls below the specified value, the scheduler issues warning messages (every minute) similar to the following:

```
CAUAJM_W_40358 The disk partition containing the CA WAAE Scheduler log file is full
CAUAJM_W_40359 The CA WAAE Scheduler will shutdown if partition has less than 8,388,608
bytes available.
```

If the disk space falls below 8192 KB (8 MB), the scheduler issues an EP_SHUTDOWN alarm, shuts down, and displays messages similar to the following:

```
CAUAJM_W_40360 Error: No disk space left to write the CA WAAE Scheduler log file.
CAUAJM_I_40247 CA WAAE Scheduler processing of events complete.
CAUAJM_I_40248 CA WAAE Scheduler shutdown complete. Exiting.
```

EvtTransferWaitTime

The EvtTransferWaitTime parameter defines the time-out delay (in seconds) for transferring events in dual event server mode. An event that is missing from one event server is copied over to the second event server after this time-out delay. The default value is 5.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
EvtTransferWaitTime=value
```
- Windows -- Enter the time-out delay in the Event Transfer Wait field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

ChaseOnStartup

The ChaseOnStartup parameter specifies whether to invoke the chase functionality when the scheduler starts.

The chase command verifies whether jobs and agents are running. If you run the chase command at regular intervals, you can track network problems. For example, if a computer is unreachable while running a job, the chase command detects that the computer is down and sends an alarm to alert you about the problem.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`ChaseOnStartup=0|1`
 - **0**
Specifies that the chase functionality is not invoked when the scheduler starts. This is the default.
 - **1**
Specifies that the chase functionality is invoked when the scheduler starts. The chase functionality is invoked with the `-A` and `-E` parameters.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Chase on Startup check box. Select the check box if you want to run the chase command when the scheduler starts.

GlobalAutoHold

The GlobalAutoHold parameter specifies whether to start the scheduler in Global Auto Hold mode. If you restart a scheduler after a period of downtime, you might want to start it in Global Auto Hold mode. Starting the scheduler in Global Auto Hold mode prevents the system from being flooded with jobs that were scheduled to run during the downtime. The scheduler evaluates all the jobs whose starting conditions are met and eligible to run. Instead of starting the jobs, the scheduler puts them in ON_HOLD status.

This approach lets you decide which jobs must run and selectively start them by using the sendevent command to send a FORCE_STARTJOB event. The only way to start a job when you start the scheduler in Global Auto Hold mode is to send a FORCE_STARTJOB event.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`GlobalAutoHold=0|1`
 - **0**
Specifies that the scheduler does not start in Global Auto Hold mode. This is the default.
 - **1**
Specifies that the scheduler starts in Global Auto Hold mode.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Global Auto Hold check box. Select the check box if you want to start the scheduler in Global Auto Hold mode.

NOTE

- To send a FORCE_STARTJOB event, enter the following command at the UNIX operating system prompt or the Windows instance command prompt

```
sendevent -E FORCE_STARTJOB -J job_name
```

- The Global Auto Hold mode affects jobs in ON_NOEXEC status. On startup, AutoSys Workload Automation places these jobs in ON_HOLD status. The same applies to any child jobs in boxes that use the auto_hold job attribute.

This video describes how to enable Global Auto Hold mode:

NOTE

Click the YouTube logo to open this video in a new tab with all normal YouTube functionality.

LocalMachineDefinition

The LocalMachineDefinition parameter specifies the host name of the type 'a' machine that acts as the localhost. This machine must be defined in the event server.

On startup, the scheduler checks the value of the LocalMachineDefinition parameter. If a value is specified, the scheduler uses the specified machine to run jobs if the job's machine attribute is localhost.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`LocalMachineDefinition=name`
- Windows -- Enter the host name in the Local Machine Definition field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- An empty value is allowed.
- If the LocalMachineDefinition parameter has an empty value, AutoSys Workload Automation performs system calls to get the host name of the machine. The clients can add a machine definition using the resolved name as the machine name while the scheduler is running. The scheduler detects the presence of this machine definition and uses this machine definition when it starts running jobs on the localhost.
- When you define a job, if you set the machine attribute to localhost and the LocalMachineDefinition parameter has an empty value and no machine definition exists for the resolved host name of the scheduler, a warning message is issued and AutoSys Workload Automation fails to start the job on the localhost.
- The LocalMachineDefinition parameter value may differ on the primary scheduler and the shadow scheduler. If the shadow scheduler fails over due to the loss of the primary scheduler, the jobs that are scheduled to run on the localhost are redirected to run either on the machine defined in the LocalMachineDefinition parameter or on the machine with the resolved host name of the shadow scheduler (if defined).

ResourceWaitPollInterval

The ResourceWaitPollInterval parameter defines how frequently (in seconds) the scheduler polls for resource availability. The default value is 15.

When you send a STARTJOB event to a job that is defined to use a real or virtual resource, the scheduler queries for the availability of the required resources. If the required resource is unavailable, the job enters RESWAIT state because the job's resource condition is not met. The job remains in RESWAIT state until the resources are available. The scheduler polls at the specified intervals to determine the resource availability for all jobs in RESWAIT state. The eligible jobs are started when the resources are available.

NOTE

More virtual resources can be made available by increasing their quantity using jil. Otherwise, resources become available when other jobs that are using specific resources go into SUCCESS, FAILURE, or TERMINATED state and the resources are released.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`ResourceWaitPollInterval=value`

- Windows -- Enter a value in the Res Wait Poll Interval field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

AppendEventMessageText

The AppendEventMessageText parameter specifies whether the text associated with an event is appended to the corresponding event message or printed as a standalone message in the scheduler log file.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
AppendEventMessageText=1|0
```

 - **1**
Specifies that the event message text is appended to the corresponding event message in the scheduler log file. In the event message, the text is displayed after the keyword TEXT.
 - **0**
Specifies that the event message text is printed as a standalone message. This is the default.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Append Event Message Text check box. Select the check box to append the event message text to the corresponding event message in the scheduler log. In the event message, the text is displayed after the keyword TEXT.

SetJobAttributeEnvironmentals

The SetJobAttributeEnvironmentals parameter specifies whether AutoSys Workload Automation automatically sets the supported job definition JIL attributes as environment variables. The scheduler prepares the environment variables for the agent to source based on the job definition JIL attributes. The agent sources the environment variables to provide applications with knowledge of the supported job definition JIL attributes.

The SetJobAttributeEnvironmentals parameter supports setting the following environment variables that are based on job definition JIL attribute values:

```
__job_name=job_name
__box_name=box_name
__machine=machine
__run_machine=run_machine
__max_exit_success=max_exit_success
```

- **machine**
Specifies the value of the machine attribute in the job definition.
- **run_machine**
Specifies the AutoSys Workload Automation machine name that the scheduler resolves after processing the machine attribute.
- **max_exit_success**
Specifies the value of the max_exit_success job attribute in the job definition.

NOTE

- The `__box_name` environment variable is set only if the job is contained within a box.
- The `__max_exit_success` environment variable is set only for command jobs.
- Custom job applications that require knowledge of more job attributes must be rewritten to invoke the `GetJobsWithFilter` class of the C++ or Java SDK.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
SetJobAttributeEnvironmentals=0|1
```

– **0**

Specifies that AutoSys Workload Automation does not automatically set the supported job definition JIL attributes as environment variables.

– **1**

Specifies that AutoSys Workload Automation automatically sets the supported job definition JIL attributes as environment variables.

- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Set Job Attribute Environmentals check box. Select the check box if you want AutoSys Workload Automation to automatically set the supported job definition JIL attributes as environment variables.

NOTE

- The `SetJobAttributeEnvironmentals` parameter applies only to CA Workload Automation AE Release 11.3.5 and later. The legacy agent supports the setting of job definition JIL attributes as environment variables, which are sourced as part of running a job.
- After you set the `SetJobAttributeEnvironmentals` parameter, pause and resume the scheduler for the new setting to take effect.

SchedAuxiliaryListeningPort

The `SchedAuxiliaryListeningPort` parameter defines the port number the scheduler uses to listen for inbound Automation Framework Message (AFM) protocol data using non-SSA communication. The default port number is 7507.

Network data between the scheduler and the agent or the scheduler and the CA Workload Automation EE manager is prepared using the proprietary AFM protocol. The scheduler auxiliary listening port is used primarily to receive inbound messages from agents running on non-SSA ports. The scheduler auxiliary listening port is also used to receive job dependencies and statuses from the mainframe scheduling managers. For example, the scheduler uses this port to receive messages from the CA Workload Automation EE manager or Workload Automation Agent for z/OS (a mainframe manager with minimal agent capabilities).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
SchedAuxiliaryListeningPort=port
```

- Windows -- Click the Communication Ports tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and enter the port number in the Auxiliary Listening Port field.

NOTE

- The scheduler auxiliary listening port must be different from the application server auxiliary listening port. The scheduler and application server auxiliary listening ports must be physical ports. So, we recommend that you configure the SSA port setting to disable port multiplexing (`EnablePmux=False`), otherwise AutoSys Workload Automation uses the virtual ports that are provided by SSA. If `EnablePmux` is set to True or the

scheduler or application server auxiliary listening port is not defined, AutoSys Workload Automation does not initiate communication with agents that are configured to run on physical ports.

For more information about configuring AutoSys Workload Automation to work with SSA, see [Configure AutoSys Workload Automation to Work with Other Broadcom products](#).

- By default, AutoSys Workload Automation does not enable SSL communication for any of its ports (physical or virtual). When you configure the SSA port settings of the scheduler auxiliary listening port, ensure that you do not enable SSL communication.
- During the AutoSys Workload Automation installation, the scheduler auxiliary listening port is set to 7507 by default. If you are performing a custom installation, you can specify a different scheduler auxiliary listening port using the Scheduler Properties page. AutoSys Workload Automation uses both the virtual ports and the auxiliary ports simultaneously. The scheduler log displays the ports that are used by AutoSys Workload Automation as follows:

```
CAUAJM_I_20366 CA WAAE Scheduler operational on agent listener port 49161.
CAUAJM_I_20367 CA WAAE Scheduler operational on auxiliary agent listener port
7507.
```

GlobalPendMachInterval

The GlobalPendMachInterval parameter controls the starting of jobs in PEND_MACH status to reduce the load on the scheduler and agent computers.

The scheduler puts a machine in the offline state if it is unable to contact the agent to run a job. You can manually put a machine in the offline state by issuing the sendevent command to send a MACH_OFFLINE event. Jobs that are scheduled to start on offline machines are placed in PEND_MACH status by default.

When an offline machine returns to service, the scheduler immediately starts all jobs in PEND_MACH status on that machine. Starting too many jobs in PEND_MACH status on the machine at a time places a heavy demand for resources on both the scheduler and agent computers. This might introduce performance problems that affect all scheduled workload.

Use the GlobalPendMachInterval parameter to define the following values:

- The time interval (in seconds) that the scheduler waits before starting jobs in PEND_MACH status when an offline machine returns to service. The default value is 0. You can specify a value in the range 0 through 3600.
- The burst value, that is, the number of jobs in PEND_MACH status that the scheduler starts after waiting for the specified interval.

NOTE

Use caution when you specify the burst value. For example, if you set the GlobalPendMachInterval value to 1,100 and three offline machines return to service, the scheduler tries to start 300 jobs per second, which might not be supported in all environments due to system constraints.

The scheduler starts the specified number of jobs, waits for the specified interval, starts the specified number of jobs, waits for the specified interval, and so on. This process repeats until all jobs in PEND_MACH status for that machine are started.

When an offline machine returns to service, the scheduler updates the status of all jobs that are scheduled for that machine from PEND_MACH to ACTIVATED unless the jobs fail their starting condition checks. If a job fails its starting condition checks, the scheduler updates its status to INACTIVE, unless it is in a box. All jobs that are contained in boxes are placed in the ACTIVATED state, even if they fail their starting condition checks.

NOTE

You can use the EvaluateQueuedJobStarts parameter to configure AutoSys Workload Automation to skip reevaluation of starting conditions for queued jobs. If you use this parameter, the scheduler immediately updates jobs in the PEND_MACH state to ACTIVATED when the required machine returns to service.

If the ACTIVATED jobs meet their starting conditions or if AutoSys Workload Automation is configured to skip starting condition evaluation for queued jobs, the scheduler verifies the following before starting the jobs:

- Manual changes to the status of the job -- If you send a manual CHANGE_STATUS event for a job and the status of the job is updated, the scheduler detects the status change and does not run the job on the machine. A message is written in the scheduler log and the scheduler proceeds with the next job start.
- Issuing a FORCE_STARTJOB event -- If you issue a FORCE_STARTJOB event for a job, the scheduler detects the status change and does not run the job on the machine. A message is written in the scheduler log and the scheduler proceeds with the next job start.
- Reevaluation of the scheduled machine -- The scheduler runs a job on a different machine (any machine other than the machine that was used to bring the job out of the PEND_MACH status) during the following situations:
 - If a job is defined to run on a virtual machine and one or more component machines in the virtual machine return to service, the scheduler determines the best machine for the job from one of the component machines in the virtual machine. The same applies for jobs that are defined to run against a comma-separated list of machines. When one or more machines in the comma-separated list of machines return to service, the scheduler determines the best machine for the job from one of the machines in the list.
 - If a job is initially defined against a single machine and you define a one-time override against the machine attribute while the job is in PEND_MACH status and the initial single machine returns to service, the scheduler starts the job on the machine specified by the override. If the scheduler cannot contact the machine that is specified by the override, it puts the job in PEND_MACH status until the machine specified by the override becomes available.
 - If a job is initially defined against a virtual machine and you define a one-time override against the machine attribute while the job is in PEND_MACH status and one or more component machines in the initial virtual machine return to service, the scheduler starts the job on the machine specified by the override. If the scheduler cannot contact the machine that is specified by the override, it puts the job in PEND_MACH status until the machine specified by the override becomes available.
 - If a job is initially defined against a comma-separated list of machines and you define a one-time override against the machine attribute while the job is in PEND_MACH status, the scheduler behaves as follows:
 - If you define a one-time override to a single machine while the job is in PEND_MACH status, the scheduler starts the job when the overridden single machine returns to service.
 - If you define a one-time override to a virtual machine while the job is in PEND_MACH status, the scheduler starts the job when one or more component machines in the overridden virtual machine return to service. The scheduler determines the best machine for the job from one of the component machines in the overridden virtual machine.
 - If you define a one-time override to a different list of machines while the job is in PEND_MACH status, the scheduler starts the job when one or more machines in the overridden list return to service. The scheduler determines the best machine for the job from one of the machines in the overridden list.

If the scheduler cannot contact the machine that is specified by the override, it puts the job in PEND_MACH status until the machine specified by the override becomes available.

NOTE

Overriding the machine attribute of a job in PEND_MACH status does not cause the scheduler to start the job. The job remains in PEND_MACH status regardless of the status of the machine that is specified by the override. The scheduler only evaluates jobs in PEND_MACH status when it processes the MACH_ONLINE events.

- Online machine status -- If the scheduler loses contact with the machine, the machine is put in offline state and all remaining jobs are put back in PEND_MACH status.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`GlobalPendMachInterval=interval,burst`
- Windows -- Enter a value in the Global Pend Mach Interval field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The GlobalPendMachInterval parameter is applied at the global level and not at the machine level or job level. That is, this parameter value applies to all machines that return to service and have jobs in PEND_MACH status that are scheduled to start on them.
- If you set the interval to 0 (the default value), the scheduler starts all jobs in PEND_MACH status with no delay between job starts and the burst value is ignored.
- Use a comma to separate the interval and burst values.
- The order in which the jobs are started depends on the job priority and amount of time the job has been in PEND_MACH status. For example, if JOB1, JOB2, and JOB3 have the same priority and they enter the PEND_MACH status at 08:00:00 a.m., 08:00:01a.m., and 08:00:02 a.m. respectively, the order in which they are started is JOB1, JOB2, and JOB3. Once the scheduler determines the starting order of jobs that are coming out of the PEND_MACH status, you cannot modify the starting order by using the sendevent command to send the CHANGE_PRIORITY event. If you send the CHANGE_PRIORITY event, the job priority changes do not apply to the run of the job exiting in PEND_MACH status. The job priority changes apply to the next run of the job.
- If jobs enter the PEND_MACH status simultaneously and have the same priority, their starting order is not guaranteed. If the job priority is set to 0, it overrides the duration that the job has been in PEND_MACH status and starts immediately.
- If the GlobalPendMachInterval parameter is set to a new value while the scheduler is starting jobs in PEND_MACH status, the scheduler applies the new interval only after it completes the current wait cycle.
- After you set the GlobalPendMachInterval parameter, pause and resume the scheduler for the new setting to take effect.
- For more information about controlling jobs in PEND_MACH status, see [Manage Your Jobs](#).

Example: Start 100 Jobs in PEND_MACH Status with an Interval of 30 Seconds on UNIX

Suppose that you want the scheduler to wait for 30 seconds before starting each of the 100 jobs in PEND_MACH status when an offline machine returns to service. Specify the GlobalPendMachInterval parameter as follows:

```
GlobalPendMachInterval=30
```

The scheduler takes 50 minutes (100 jobs that are multiplied by 1/2 minute each) to start all the jobs.

Example: Incrementally Start 600 Jobs in PEND_MACH Status on Windows

Suppose that you have 600 jobs in PEND_MACH status for a machine that is offline. When the machine returns to service, you want the scheduler to start five jobs at 5-second intervals. Enter 5,5 in the Global Pend Mach Interval field on the Scheduler - AutoSys Workload Automation Administrator window. The scheduler takes 10 minutes to start all the jobs.

GlobalPendMachStatus

The GlobalPendMachStatus parameter defines the completion status that the scheduler assigns to jobs that are scheduled on an offline machine. The jobs temporarily remain in PEND_MACH status before the scheduler assigns the status that is specified in the GlobalPendMachStatus parameter.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
GlobalPendMachStatus=value
```
- Windows -- Enter a value in the Global Pend Mach Status field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The valid values are FAILURE, INACTIVE, ONICE, PEND_MACH, SUCCESS, and TERMINATED. The default value is PEND_MACH.
- The GlobalPendMachStatus parameter is applied at the global level and not at the machine level or job level. That is, this parameter value applies to all jobs that are scheduled on an offline machine.
- If you set the GlobalPendMachStatus parameter to a valid status (other than PEND_MACH), it results in a CHANGE_STATUS event being sent for the desired completion status. The scheduler records this event in its log as an event that was the result of setting the GlobalPendMachStatus parameter.
- If the scheduler changes the status of a job from PEND_MACH to any other valid status because of the GlobalPendMachStatus setting or a manual CHANGE_STATUS event, the job is not run when the machine returns to service. Only jobs in the PEND_MACH status are eligible to start on the machine that returns to service.
- After you set the GlobalPendMachStatus parameter, pause and resume the scheduler for the new setting to take effect.

GlobalPendMachDelay

The GlobalPendMachDelay parameter defines the time interval (in seconds) that the scheduler waits before updating the status of the job to the status specified in the GlobalPendMachStatus parameter.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`GlobalPendMachDelay=value`
- Windows -- Enter a value in the Global Pend Mach Delay field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The GlobalPendMachDelay parameter is applied at the global level and not at the machine level or job level. That is, this parameter value applies to all jobs that are scheduled on an offline machine.
- The default value is 0. You can specify a value in the range 0 through 3600.
- If you use the default value, the scheduler immediately sends a CHANGE_STATUS event to update the status of jobs in PEND_MACH status to the status specified in the GlobalPendMachStatus parameter. If you specify a value other than the default, jobs remain in PEND_MACH status until the delay interval expires, and are then assigned the status that is specified in the GlobalPendMachStatus parameter. If the machine returns to service within the delay interval, the scheduler does not set the status of the job to the status specified in the GlobalPendMachStatus parameter. Because the job is in PEND_MACH status when the machine returns to service, the scheduler reschedules it to run on the online machine that is based on the interval that is specified in the GlobalPendMachInterval parameter.
- After you set the GlobalPendMachDelay parameter, pause and resume the scheduler for the new setting to take effect.

The following table shows job status that is based on some sample values set for the GlobalPendMachDelay and GlobalPendMachStatus parameters:

GlobalPendMachDelay Value	GlobalPendMachStatus Value	Job Status
0 (default)	PEND_MACH (default)	The scheduler puts the jobs in PEND_MACH status when the machine is offline.

0 (default)	INACTIVE	The scheduler puts the jobs in PEND_MACH status when the machine is offline. The scheduler immediately sends a CHANGE_STATUS event to put the jobs in INACTIVE state. Jobs are put in INACTIVE status when the scheduler processes the CHANGE_STATUS event.
30	INACTIVE	The scheduler puts the jobs in PEND_MACH status when the machine is offline. If the machine returns to service within 30 seconds, jobs are started based on the interval that is specified in the GlobalPendMachInterval parameter. If the machine does not return to service within 30 seconds, the scheduler sends a CHANGE_STATUS event to put the jobs in INACTIVE state. Jobs are put in INACTIVE status when the scheduler processes the CHANGE_STATUS event.

AgentStatusPollInterval

The AgentStatusPollInterval parameter specifies the time interval (in minutes) between status polls that the scheduler waits before polling all the agents. The default value is 1, that is, the scheduler polls the agents every one minute.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`AgentStatusPollInterval=value`
- Windows -- Enter the time interval in the Agent Status Poll Interval field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- If you specify a value that is less than 1, the scheduler polls the agents every one minute.
- After you set the AgentStatusPollInterval parameter, pause and resume the scheduler for the new settings to take effect.
- When you define machines, if the agent_name and node_name attributes are the same and the ports are different, the machine goes to a Missing or Unqualified state. We recommend that you look for machines that are accessible from autoping but are placed in the Missing or Unqualified state by the scheduler and correct the machine definitions to specify the correct port.

PrimaryFailbackMode

The PrimaryFailbackMode parameter defines the mode that the primary scheduler enters when you restart AutoSys Workload Automation after a failover.

High-availability mode is disabled when the primary scheduler fails over to the shadow scheduler. The shadow scheduler takes over processing events when the primary scheduler fails. But, you can return to high-availability mode by restoring the primary scheduler. How the primary scheduler resumes processing events after you restore it depends on the primary failback mode.

By default, the primary failback mode is set to Off. The primary scheduler cannot resume processing events while the shadow scheduler is processing events. In this case, you can restart the primary scheduler only after you shut down

the shadow scheduler. To minimize downtime, we recommend that you set the primary failback mode to Dormant or Immediate. When the primary failback mode is set to Dormant or Immediate, the primary scheduler restarts while the shadow scheduler runs.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:


```
PrimaryFailbackMode=0|1|2
```

 - **0**
Specifies that the primary scheduler cannot restart while the shadow scheduler processes events. In this mode, stop the shadow scheduler and restart the primary scheduler.
 - **1**
Specifies that the primary scheduler runs dormant and resumes processing events only when a failback occurs. A failback occurs when the shadow scheduler fails or when you initiate a manual failback. If a failback occurs because the shadow scheduler fails, high-availability mode is disabled. In this case, return to high-availability mode by restoring the shadow scheduler.
 - **2**
Specifies that the primary scheduler resumes processing events as soon as it detects activity from the other schedulers. AutoSys Workload Automation automatically reenters high availability mode and the primary scheduler resumes processing events.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility, select Primary in the Scheduler Role pane, and then select any *one* of the following primary failback mode options:
 - **Off**
Specifies that the primary scheduler cannot restart while the shadow scheduler processes events. In this mode, stop the shadow scheduler and restart the primary scheduler.
 - **Dormant**
Specifies that the primary scheduler runs dormant and resumes processing events only when a failback occurs. A failback occurs when the shadow scheduler fails or when you initiate a manual failback. If a failback occurs because the shadow scheduler fails, high-availability mode is disabled. In this case, return to high-availability mode by restoring the shadow scheduler.
 - **Immediate**
Specifies that the primary scheduler resumes processing events as soon as it detects activity from the other schedulers. AutoSys Workload Automation automatically reenters high availability mode and the primary scheduler resumes processing events.

NOTE

The Primary Failback Mode options are only available when the Scheduler Role is set to Primary.

AggregateStatistics

The AggregateStatistics parameter specifies whether AutoSys Workload Automation aggregates the job, alarm, and scheduler statistics automatically at a specified interval. The job, alarm, and scheduler statistics are aggregated into the ujo_rep_hourly, ujo_rep_daily, ujo_rep_weekly, and ujo_rep_monthly database tables. You can display these aggregated statistics in a report format in AutoSys Workload Automation. Other applications, such as Web UI, can also retrieve these aggregated statistics to generate predefined and custom reports.

The scheduler automatically aggregates the job, alarm, and scheduler statistics as follows:

- Hourly -- At the start of every hour
- Daily -- At midnight every day
- Weekly -- At midnight on the first day of every week, which is taken as Sunday
- Monthly -- At midnight on the first day of every month

NOTE

- For information about how to retrieve aggregated statistics, see [How to Retrieve Aggregated Job, Alarm, and Scheduler Statistics](#).
- In all cases, the scheduler aggregates statistics in smaller intervals before starting the specified aggregation. For example, if you schedule a monthly aggregation process, the scheduler aggregates the hourly, daily, and weekly statistics before starting the monthly aggregation.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
AggregateStatistics=0|1
```

 - **0**
Specifies that the job, alarm, and scheduler statistics are not aggregated.
 - **1**
Specifies that the job, alarm, and scheduler statistics are aggregated automatically. This is the default.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Aggregate Statistics check box. By default, the check box is selected and the job, alarm, and scheduler statistics are aggregated automatically.

NOTE

- The scheduler does not aggregate monthly statistics for the current month, week, day, or hour. For example, if the oldest event in the `ujo_proc_event` table is 09/27/2011 00:05:10 and you schedule the aggregation process on 10/11/2011 at 18:30:00, the statistics are aggregated as follows:
 - Hourly statistics -- From 09/27/2011 00:00:00 through 10/11/2011 17:59:59
 - Daily statistics -- From 09/27/2011 00:00:00 through 10/10/2011 23:59:59 (14 days)
 - Weekly statistics -- From 09/25/2011 00:00:00 through 10/08/2011 23:59:59 (two weeks)
 - Monthly statistics -- From 09/01/2011 00:00:00 through 09/30/2011 17:59:59 (one month)
- You can aggregate statistics manually by sending an aggregation event using the `sendevent` command. You do not need to set the `AggregateStatistics` parameter to aggregate statistics manually. For more information about the `sendevent` command, see [Reference](#).
- When the aggregation is scheduled automatically, starting and completed messages are written in the scheduler log file. Detailed messages are written in the aggregation log file (`aggregator.instance`) located in the `$AUTOUSER/out` (on UNIX) or `%AUTOUSER%\out` (on Windows) directory. The aggregation log file has the same properties as the scheduler log file and it rolls over based on the `LOGROLLOVER` parameter value (on UNIX) or `LOGROLLOVER` environment variable setting (on Windows).
- If the scheduler is shut down while the aggregation process is active, the aggregation process is terminated. A message is displayed to indicate the same.
- After you set the `AggregateStatistics` parameter, pause and resume the scheduler for the new setting to take effect.

EvaluateQueuedJobStarts

The `EvaluateQueuedJobStarts` parameter specifies whether AutoSys Workload Automation skips starting condition evaluation for jobs leaving a queued state. When AutoSys Workload Automation skips starting condition evaluation, queued jobs start immediately upon leaving a queued state.

AutoSys Workload Automation starts a job only when it meets the starting conditions that are specified in the job definition. Some external conditions might prevent jobs that meet their starting conditions from running. Depending on the reason, AutoSys Workload Automation places the job in one of the following queued states:

- When an offline machine prevents the job from running, the job enters PEND_MACH status.
- When held resources prevent the job from running, the job enters RES_WAIT status.
- When unavailable load units prevent the job from running, the job enters QUE_WAIT status.

When queued jobs leave the queued state, they might no longer meet their starting conditions. Starting jobs that no longer meet their starting conditions might result in scheduling conflicts or related problems. However, skipping job starts might result in the loss of critical work.

By default, AutoSys Workload Automation re-evaluates the starting conditions for these jobs other than the date condition check for the day of evaluation before starting them. If a job fails its starting condition checks after leaving a queued state, the scheduler places the job in an INACTIVE state. If the job meets its starting conditions, the scheduler starts the job. If the job is in a box that is running and it fails its starting condition checks, the scheduler places the job in the ACTIVATED state. If the job is in a box that is not running and it fails its starting condition checks, the scheduler places the job in an INACTIVE state. If a job with date conditions fails its starting condition checks after leaving a queued state, the scheduler re-schedules the job to its next start time after resetting its status.

You can configure AutoSys Workload Automation to schedule jobs leaving a queued state. When you use the EvaluateQueuedJobStarts parameter, queued jobs perform one of the following actions upon leaving a queued state:

- Start immediately on the day of evaluation.
- Start on the day of evaluation regardless of their date conditions unless other starting conditions apply and are not satisfied.
- Start on the day of evaluation after they meet their date conditions unless other starting conditions apply and are not satisfied.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
EvaluateQueuedJobStarts=0|1|2
```

– 0

Specifies that the scheduler immediately starts the job without evaluating the starting conditions of the job.

– 1

Specifies that the scheduler evaluates the starting conditions of the job other than the date condition check for the day of evaluation before starting the job. This is the default.

NOTE

- The scheduler does not re-evaluate the following job attributes: run_calendar, days_of_week, start_times, and start_mins.
- Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.

– 2

Specifies that the scheduler evaluates the starting conditions of the job including the date condition check for the day of evaluation before starting the job.

NOTE

- The scheduler does not re-evaluate the following job attributes: start_times and start_mins.
- Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select any one of the following options in the Evaluate Queued Job Starts pane as appropriate:
 - **Off**
Specifies that the scheduler immediately starts the job without evaluating the starting conditions of the job.
 - **Skip same day check**
Specifies that the scheduler evaluates the starting conditions of the job other than the date condition check for the day of evaluation before starting the job. This is the default.

NOTE

- The scheduler does not re-evaluate the following job attributes: run_calendar, days_of_week, start_times, and start_mins.
- Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- **Include same day check**
Specifies that the scheduler evaluates the starting conditions of the job including the date condition check for the day of evaluation before starting the job.

NOTE

- The scheduler does not re-evaluate the following job attributes: start_times and start_mins.
- Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.

RepeatAlarmNotification

The RepeatAlarmNotification parameter specifies how email notifications are sent for alarms that are repeatedly generated for a job. In a job run, for each alarm type, you can specify whether you want to send an email notification for all alarms, just the first alarm, or only the last alarm.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
RepeatAlarmNotification=ALL|FIRST|LAST
```

 - **ALL**
Specifies that an email notification is sent for all alarms of a particular alarm type. This is the default.
 - **FIRST** Specifies that an email notification is sent for just the first alarm of a particular alarm type.
 - **LAST**
Specifies that an email notification is sent for only the last alarm of a particular alarm type. This email notification is sent when the job completes.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select any one of the following options in the Repeat Alarm Notification pane as appropriate:
 - **ALL**
Specifies that an email notification is sent for all alarms of a particular alarm type. This is the default.
 - **FIRST**
Specifies that an email notification is sent for just the first alarm of a particular alarm type.
 - **LAST**

Specifies that an email notification is sent for only the last alarm of a particular alarm type. This email notification is sent when the job completes.

NOTE

After you set the RepeatAlarmNotification parameter, pause and resume the scheduler for the new settings to take effect.

MachineMethod

The MachineMethod parameter defines the method to achieve load balancing. This parameter only applies to virtual machines of type "n" and "v".

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
MachineMethod=cpu_mon|rstatd|job_load|roundrobin|roundrobin_jobload
```

– **cpu_mon**

Specifies that the cpu_mon method is used to determine the percentage of available CPU cycles on the agent. The scheduler runs the CPU Monitoring (OMCPU) job on the target computer to get the available CPU cycles. This is the default.

NOTE

- The cpu_mon method does not apply to z/OS machines (Workload Automation Agent on z/OS) because the OMCPU job is not supported on z/OS.

– **rstatd**

Specifies that the rstatd method is used by the UNIX schedulers to determine the percentage of available CPU cycles on UNIX computers. The scheduler makes a UNIX remote procedure call to the remote kernel statistics daemon (rstatd) on the target computer to get the available CPU cycles.

NOTE

- The rstatd method does not apply to type "n" machines.
- You cannot configure the rstatd method for the Windows scheduler.
- Ensure that the rstatd daemon is running on the UNIX scheduler and on all target UNIX computers.
- *Set the value of the opsys attribute for type 'a' machines to one of the following values that support the rstatd method: aix, hpux, linux, or openvms.*
- If the load balancing request is sent to an agent with an opsys attribute value that does not support the rstatd method, AutoSys Workload Automation uses the cpu_mon method to obtain the available CPU cycles.

– **job_load**

Specifies that the job_load method is used to determine the machine where the job runs. The scheduler uses the job_load and max_load attributes to calculate the available load for a computer.

– **roundrobin**

Specifies that the roundrobin method is used to determine the real machine where the job runs from the list of available real machines belonging to a virtual machine. The jobs are run on the available real machines in a round robin manner.

– **roundrobin_jobload**

Specifies that the job_load method is used first and then the roundrobin method is used on the resulting real machines to determine the real machine where the job runs. However, if none of the machines have sufficient load units, AutoSys Workload Automation puts the job in QUE_WAIT status. The job stays in QUE_WAIT status until one of the machines has sufficient load units available. When sufficient load units are available, the job runs on that machine. In this situation, the roundrobin method is not considered.

NOTE

For information about how load balancing works, see [How Load Balancing Works](#). For information about how AutoSys Workload Automation queues jobs, see [Queuing Jobs](#) and [How AutoSys Workload Automation Queues Jobs](#).

- Windows -- Enter a value in the Machine Method field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility. The valid values are `cpu_mon`, `job_load`, `roundrobin`, and `roundrobin_jobload`. The default value is `cpu_mon`.

NOTE

- The `cpu_mon` or `rstatd` settings verify the CPU usage statistics of candidate machines, whereas the `job_load` setting checks only the job load and no real-time usage of machine is required.
- If you set the `MachineMethod` parameter value and also specify the [machine_method Attribute -- Specify the Method to Achieve Load Balancing](#) attribute in your job definition, the `machine_method` attribute value overrides the value you specified for the `MachineMethod` parameter.

(UNIX) Verify that the Remote Kernel Statistics Daemon is Running

If you set the `MachineMethod` parameter to `rstatd`, verify that the remote kernel statistics daemon is running on the scheduler and on all the target computers.

Follow these steps:

1. Edit the internet services daemon (`inetd`) configuration file (`/etc/inetd.conf`) on all client computers, and uncomment the `rstatd` entry.
2. Send a `SIGHUP` signal (`kill -1`).
The running `inetd` process is reset.

NOTE

Sometimes a `kill -1` command is not sufficient to reset the `inetd`. If `rstatd` fails, you might have to issue a `kill -9` command, and restart `inetd`. If necessary, verify with your systems administrator.

UseMachineMethodOnForceStartJob

The `UseMachineMethodOnForceStartJob` parameter specifies whether the scheduler uses the `MachineMethod` parameter value to select the machine where the job must run when you force start a job and multiple non-zero factor online machines are available.

NOTE

The `MachineMethod` parameter defines the method that determines the percentage of CPU cycles available on a real machine belonging to a virtual machine. This method is used to achieve load balancing. For more information about the `MachineMethod` parameter, see [MachineMethod](#).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`UseMachineMethodOnForceStartJob=0|1`
 - **0**
Specifies that the scheduler ignores the `MachineMethod` parameter value and selects a random machine to run the job when you force start a job and multiple machines are available.
 - **1**

Specifies that the scheduler uses the MachineMethod parameter value to select the machine where the job must run when you force start a job and multiple non-zero factor online machines are available. This is the default.

- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select or clear the Machine Method On Force Starts check box. The check box is selected by default and the scheduler uses the Machine Method parameter value to select the machine where the job must run when you force start a job and multiple non-zero factor online machines are available.

RestartConstant, RestartFactor, and MaxRestartWait

The RestartConstant, RestartFactor, and MaxRestartWait parameters define the values to calculate the maximum amount of time (in seconds) that AutoSys Workload Automation waits before it tries to restart a job.

The RestartConstant parameter defines the Restart Constant value that is used for calculating the wait time (in seconds) between attempts to restart a job. The default value is 10.

The RestartFactor parameter defines the Restart Factor value that is used for calculating the wait time (in seconds) between attempts to restart a job. This value multiplies with every job restart and is used to gradually increase the number of seconds per retry attempt. The default value is 5.

The MaxRestartWait parameter defines the maximum amount of time (in seconds) that AutoSys Workload Automation waits before it tries to restart a job. The default value is 300.

The following formula is used to calculate the wait time:

```
WaitTime=RestartConstant+(Num_of_Trys*RestartFactor)
if WaitTime > MaxRestartWait,
then WaitTime = MaxRestartWait
```

The Num_of_Trys value is specified by the internal job starter counter, which indicates the number of times AutoSys Workload Automation has already tried to start the job. If the calculated wait time is greater than the specified value for the MaxRestartWait parameter, then the wait time is set to the value of the MaxRestartWait parameter.

You can configure these parameters as follows:

- UNIX -- Edit the following entries in the configuration file:

```
RestartConstant=constant
RestartFactor=factor
MaxRestartWait=wait_time
```

- Windows -- Enter a value in the Restart Constant, Restart Factor, and Max Restart Wait fields on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

KillSignals

The KillSignals parameter defines a comma-separated list of signals to send to a job whenever the KILLJOB event is sent. The default value is 2,9.

NOTE

The KillSignals parameter applies to UNIX agents only.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

`KillSignals=value, value, ...`

- Windows -- Enter a value in the Legacy Kill Signals field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- On agents (r11.3, Release 11.3.5, or Release 11.3.6), you can send multiple signals. Separate each value (specify only positive integers) with a comma. The signals are sent in the order listed, with a three second interval between each call.
- We recommend that you set the KillSignals parameter to 2,9. Usually, this configures AutoSys Workload Automation to return a TERMINATED state for the target job. If it does not, set the KillSignals parameter to 9.
- The signals are referenced by numeric process IDs (PIDs). Each signal has a default set of effects on a program.
- The KillSignals listed in the configuration file are overridden when you issue the sendevent command with the -k option.
- If the job is running on a UNIX computer, this parameter specifies a single or comma-delimited list of UNIX signals to send to the job. If a list of signals is specified, the signals are sent in the order listed, with a five-second interval between each call.
- If the job to be killed is running on a Windows computer, this list is ignored and the job is terminated.
- Microsoft Windows does not support the concept of process groups. If the job being run is not an EXE file (for example, it is a BAT, CMD, or COM file), the legacy agent uses CMD.EXE to launch the job, and KILLJOB kills only the CMD.EXE process. The job status is then set according to the return code of the killed CMD.EXE process; this status can be SUCCESS, FAILURE, or TERMINATED. Processes that were launched by user applications or batch (*.bat) files cannot be killed.

MaxRestartTrys

The MaxRestartTrys parameter defines the maximum number of times the scheduler tries to restart a job. AutoSys Workload Automation might be unable to start a job due to system problems including computer unavailability, a timed-out socket connect, an inability to create new processes, or failure of the file system space resource check.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`MaxRestartTrys=value`
- Windows -- Enter a value in the Max Restart Trys field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The default value is 10. When the MaxRestartTrys value is zero, the job is not restarted.
- The MaxRestartTrys parameter is instance-specific. You can configure it to be agent-specific for a AutoSys Workload Automation instance.
- The MaxRestartTrys parameter governs retries due to system or network problems. This value is different from the n_retrys job definition attribute, which controls restarts when a job fails due to application failure (for example, when AutoSys Workload Automation cannot find a file or a command, or permissions are not properly set).
- For more information about the n_retrys job definition attribute, see [n_retrys Attribute—Define the Number of Times to Restart a Job After a Failure](#).

Configure the MaxRestartTrys Parameter to Be Machine-Specific

The MaxRestartTrys parameter is instance-specific. You can configure the MaxRestartTrys parameter to be machine-specific for a AutoSys Workload Automation instance.

Follow these steps:

1. Create a configuration file named MaxRestartTrys.%AUTOSERV% (on Windows) or MaxRestartTrys.\$AUTOSERV (on UNIX) in the AUTOUSER directory on the scheduler machine.
2. In the configuration file, add the following entry:

```
machine_name: value
```

- **machine_name**

Specifies the name of the AutoSys Workload Automation machine where you want to define the MaxRestartTrys value.

- **value**

Defines the maximum number of times the scheduler tries to restart a job after a system or network problem occurs.

3. Repeat Step 2 for each AutoSys Workload Automation machine where you want to define the MaxRestartTrys value.

NOTE

Each entry must begin on a new line.

4. Save the file.

The MaxRestartTrys parameter is defined and is now machine-specific for a AutoSys Workload Automation instance.

NOTE

- If you do not create the MaxRestartTrys.%AUTOSERV% (on Windows) or MaxRestartTrys.\$AUTOSERV (on UNIX) configuration file and add entries for AutoSys Workload Automation machines where you want to define the MaxRestartTrys value, the instance-wide MaxRestartTrys parameter is applied.
- After you define the machine-specific MaxRestartTrys value, restart the scheduler for the settings to take effect.
- When the MaxRestartTrys value is zero, the job is not restarted.
- If the entry for the machine does not follow the correct syntax or the value is either non-numerical or less than zero, the entry is ignored.
- If multiple entries exist for a machine, the last occurring entry takes effect.

Check_Heartbeat

The Check_Heartbeat parameter defines the time interval (in minutes) that the scheduler uses when checking for late or missing heartbeats from jobs that are running. The default value is 2.

A *heartbeat* is a job event that is sent at regular intervals to the application server by a user application started by an agent. User applications can be programmed to issue heartbeats by linking with the Autosys Workload Automation Software Development Kit (SDK) library. Thus, heartbeats can be used to monitor the progress of user applications. The scheduler checks that the HEARTBEAT event has occurred during the heartbeat interval that is specified in the heartbeat_interval attribute in a job definition. The scheduler issues the MISSING_HEARTBEAT alarm for every job that has not issued the HEARTBEAT event within the job's heartbeat interval.

NOTE

The scheduler (not the agent) checks for heartbeats. If there is a problem between the user application and the application server, the HEARTBEAT event is not sent and the scheduler issues an alarm at the next heart beat interval. Therefore, the HEARTBEAT event can indicate whether the network is working properly.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
Check_Heartbeat=value
```

- Windows -- Enter a value in the Job Heartbeat Interval field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

The Check_Heartbeat is an optional parameter. If there are no applications that send heartbeats, do not set this parameter.

Heartbeat

A *heartbeat* is a job event that is sent at regular intervals to the application server by a user application started by an agent. User applications can be programmed to issue heartbeats by linking with the AutoSys Workload Automation Software Development Kit (SDK) library. Thus, heartbeats can be used to monitor the progress of user applications. The scheduler checks that the HEARTBEAT event has occurred during the heartbeat interval specified in the heartbeat_interval attribute in a job definition. The scheduler issues the MISSING_HEARTBEAT alarm for every job that has not issued the HEARTBEAT event within the job's heartbeat interval.

NOTE

The scheduler (not the agent) checks for heartbeats. If there is a problem between the user application and the application server, the HEARTBEAT event is not sent and the scheduler issues an alarm at the next check heartbeat interval. Therefore, the HEARTBEAT event can indicate whether the network is working properly.

AutoInstWideAppend

The AutoInstWideAppend parameter specifies whether the agent overwrites or appends job command output or error information to the standard error and standard output files.

NOTE

If you are running jobs across operating environments, the scheduler of the issuing AutoSys Workload Automation instance controls the default behavior. For Windows, the default scheduler behavior is to overwrite the standard error and standard output files.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
AutoInstWideAppend=0|1|2
```

- **0**
Specifies that the agent overwrites information to the standard output and standard error files.
- **1**
Specifies that the agent appends new information to the standard output and standard error files.
- **2**
Specifies that the information is forcefully appended to the standard output and standard error files; the agent setting is overridden.
- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select any *one* of the following options in the Append STDOUT/STDERR pane:
 - **Overwrite**
Specifies that the agent overwrites information to the standard output and standard error files.
 - **Append**
Specifies that the agent appends new information to the standard output and standard error files.
 - **Append Forcefully**
Specifies that the information is forcefully appended to the standard output and standard error files; the agent setting is overridden.

To determine whether the information is appended to the files or the files are overwritten, AutoSys Workload Automation checks the settings in the following order:

- Checks the AutoSys Workload Automation job definition for append or overwrite notation. If there is a notation, AutoSys Workload Automation uses the indicated behavior and does not verify other settings.

To set the behavior at the job definition level, place the appropriate notation as the first characters in the `std_err_file` and `std_out_file` specification in JIL. Use the following notation to specify whether the files must be appended or overwritten:

```
> Overwrite file
>> Append file
```

- Checks the `AutoMachWideAppend` variable setting in the `/etc/auto.profile` file (on UNIX) or on the System - AutoSys Workload Automation Administrator window of <UAJM Administrator (on Windows). In the `/etc/auto.profile` file, the `AutoMachWideAppend` variable is set as follows:

```
#AUTOENV#AutoMachWideAppend=TRUE
```

If this variable is set for the computer where the job runs, AutoSys Workload Automation uses the indicated behavior and does not verify other settings.

NOTE

This applies to legacy agents only.

- Checks the `AutoInstWideAppend` parameter value and uses this setting.

Example: Append New Information to the Standard Error File

This example appends new information to the standard error file.

```
std_err_file: >>/tmp/test.err
```

CrossPlatformScheduling

The `CrossPlatformScheduling` parameter specifies whether a AutoSys Workload Automation instance can submit job requests to or receive job requests from an external scheduling manager or agent.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
CrossPlatformScheduling=0|1|2
```

- **0**
Disables cross-platform scheduling. This is the default.
- **1**
Enables outbound cross-platform scheduling (manager only). When you select this option, a AutoSys Workload Automation instance can dispatch job requests to an agent.
- **2**
Enables outbound and inbound cross-platform scheduling (manager and agent). When you select this option, a AutoSys Workload Automation instance can dispatch job requests to an agent and receive job start requests from a manager.

NOTE

This option takes effect only when you initialize the scheduler.

- Windows -- Click the Options tab on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility and select any *one* of the following options in the Cross Platform Scheduling pane:
 - **Off**
Disables cross-platform scheduling. This is the default.
 - **Manager Only**

Enables outbound cross-platform scheduling (manager only). When you select this option, a AutoSys Workload Automation instance can dispatch job requests to an agent.

– **Manager and Agent**

Enables outbound and inbound cross-platform scheduling (manager and agent). When you select this option, a AutoSys Workload Automation instance can dispatch job requests to an agent and receive job start requests from a manager.

NOTE

This option takes effect only when you initialize the scheduler.

NOTE

After you set the CrossPlatformScheduling parameter, pause and resume the scheduler for the new settings to take effect.

Configure an Event Server

You can configure an event server by editing the following event server parameters in the configuration file (UNIX) or the equivalent fields on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility (Windows):

(UNIX) Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt to stop the scheduler, application server, and web server:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
unisrvcntr stop waae_webserver.$AUTOSERV
```

3. Edit the event server parameters in the configuration file as appropriate and save the file.
4. Enter the following commands at the operating system prompt to start the scheduler, application server, and web server:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
unisrvcntr start waae_webserver.$AUTOSERV
```

The new parameter settings take effect.

(Windows) Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
2. Select an instance for which you want to configure the event server from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
4. Right-click the scheduler and select Stop. Repeat this action for the application server and web server.
5. Click the Event Server icon on the toolbar.
6. Edit the fields on the Event Server - AutoSys Workload Automation window as appropriate.
7. Click the Services icon on the toolbar.
8. Right-click the scheduler and select Start. Repeat this action for the application server and web server. The scheduler, application server, and the web server start and the new parameter settings take effect.

Provider

The Provider parameter specifies the database provider of the Relational Database Management System (RDBMS) that is used by AutoSys Workload Automation.

You can define the event server information that is used by AutoSys Workload Automation during or after installation. To define the event server information after the installation, modify the Provider, DBAccess, EventServer_1, or EventServer_2 parameters. The scheduler, application server, web server, and some client utilities such as dbstatistics, archive_events, and archive_jobs use the event server information to connect to the event server.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
Provider=ORA|SYB|MSQ
```

- **ORA**

Identifies Oracle as the database provider.

- **SYB**

Identifies Sybase as the database provider.

- **MSQ**

Identifies Microsoft SQL Server as the database provider.

- Windows -- Select a value from the Provider drop-down list on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

When you switch to a new database provider, the information about the previous database provider is not cleared. Ensure that the fields on the Event Server - AutoSys Workload Automation Administrator window are properly updated to support the new database provider.

DBAccess

The DBAccess parameter defines the user name and password (in encrypted format) the scheduler, application server, or the web server uses to connect to the database. This database user name and password are defined during the AutoSys Workload Automation installation.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
DBAccess=username/password
```

- Windows -- Enter a user name in the User field and its associated password (up to 32 characters) in the Password field on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- When you install AutoSys Workload Automation, a database user is added. Typically, this database user is named *autosys*. The database user is granted rights to the AutoSys Workload Automation objects and can make changes to specific information in the database.
- You can generate the DBAccess parameter password in AES encrypted format using the *autosys_secure* command (using option 6 and then option 1). However, if you change the password using option 3 of the *autosys_secure* command, the local configuration file is automatically updated. To update the configuration files on other machines with server installations, either copy the new password in all configuration files or issue the *autosys_secure* command on all server machines.

DBAccessMode

The DBAccessMode parameter specifies the access mode that AutoSys Workload Automation is using to establish connections with the database.

You can configure this parameter as follows:

```
DBAccessMode=0 | 1
```

- **0**
Specifies that password based authentication is used to access the database. This is the default. AutoSys Workload Automation uses the user name and password that you specified using the DBAccess parameter to connect to the database.
- **1**
Specifies that external authentication is used. The DBAccess parameter is ignored.

NOTE

- On Oracle, if you set the value to 1, SSL certificate is used to access the database. You can configure AutoSys Workload Automation to use single sign-on wallets with certificates that are present on the file system to access the Oracle database. For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

WARNING

IMPORTANT! Use the switchDBAccessMode.pl script to switch the database access mode from password to SSL certificate mode or vice versa. For information about how to switch the database access mode, see [\(Oracle Only\) Switch the Database Access Mode](#).

- On Sybase, if you set the value to 1, Kerberos authentication is used. Kerberos is a network authentication protocol that provides strong authentication and encryption using the secret-key cryptography. The required ticket granting ticket (TGT) must be configured properly for the product to work. For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)

WARNING

IMPORTANT! Use the switchDBAccessMode.pl script to switch the database access mode from password to Kerberos authentication or vice versa. For information about how to switch the database access mode, see [\(Sybase Only\) Switch the Database Access Mode](#)

On Windows, the Access Mode field on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility displays EXTERNALAUTH if AutoSys Workload Automation is using external authentication to access the database. By default, PASSWORD is displayed in the Access Mode field as AutoSys Workload Automation uses password to access the database. When AutoSys Workload Automation uses password to access the database, it uses the user name and password that you specified in the User and Password fields to connect to the database.

EventServer_1 and EventServer_2

The EventServer_1 parameter specifies the database for a specific event server. When AutoSys Workload Automation runs in single event server mode, only one event server is required. If you want to run AutoSys Workload Automation in dual event server mode, configure the EventServer_2 parameter by specifying the database that the second event server must be connected to.

You can configure these parameters as follows:

- UNIX -- Edit the following entries in the configuration file:

```

EventServer_1=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST,SPN
| ORACLE_SVR,SCHEMA_OWNER,DBPORT,DBHOST | MSSQL_SVR:MSSQL_DB,DBPORT,DBHOST
EventServer_2=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST,SPN |
ORACLE_SVR,SCHEMA_OWNER,DBPORT,DBHOST | MSSQL_SVR:MSSQL_DB,DBPORT,DBHOST

```

– **SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST,SPN**

Identifies the Sybase database for a specific event server.

NOTE

- The EventServer_1 and EventServer_2 parameter values are defined by the sybase server name:database name,database TCP/IP listener port,database host name, service principal name combination. The service principal name is optional.
- For Sybase, the database name is defined in the interface file. In dual event server mode, the value for both event servers can be different.

– **ORACLE_SVR,SCHEMA_OWNER,DBPORT,DBHOST**

Identifies the Oracle database for a specific event server.

NOTE

The EventServer_1 and EventServer_2 parameter values are defined by the Oracle system identifier (ORACLE_SID),schema owner,database TCP/IP listener port,database host name combination.

– **MSSQL_SVR:MSSQL_DB,DBPORT,DBHOST**

Identifies the Microsoft SQL Server database for a specific event server.

NOTE

- The EventServer_1 and EventServer_2 parameter values are defined by the Microsoft SQL Server server name:database name, database TCP/IP listener port,database host name combination.
- For Microsoft SQL Server, use the Client Network Utility to define the database name. In dual event server mode, the value for both event servers can be different.

- Windows -- Complete the following fields in the Event Server 1 and Event Server 2 panes on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility:

– **Name or Alias**

Defines the name of the database server on Sybase or Microsoft SQL Server. On Oracle, it defines the alias for the event server. When performing database queries, AutoSys Workload Automation uses this value to locate the event server. If you are using multiple instances and cross-instance job dependencies, do the following:

- For Oracle, supply a unique alias for each event server.
- For Sybase or Microsoft SQL Server, you can define multiple databases on the same event server.

NOTE

Depending on the database that you are using, you can find the name in the following locations:

- For Sybase, this name is stored in the Sybase SQL.INI interface file.
- For Oracle, this alias is stored in the Oracle TNSNAMES.ORA file.
- For Microsoft SQL Server, this name is the AutoSys Workload Automation instance name. For example, machine_name:instance_name, where machine_name defines the machine name or the host name of the machine and instance_name defines the name of the AutoSys Workload Automation instance. The host name is valid when the default instance is used as the event server. You can check the database settings using the Microsoft SQL Enterprise Manager.

– **Database**

Defines the database name for a specific event server. The default value is AEDB. In dual event server mode, the value for both event servers can be different. This field is not displayed for Oracle installations.

NOTE

The entire value of the Name and Database fields can be up to 64 characters.

- **Host Name**
Defines the database host name.
- **Port**
Defines the database TCP/IP listener port.
- **Status**
Indicates the status of the event server.
- **Service Principal Name**
(Optional) Specifies the service principal name to be used for Kerberos authentication. **Limits:** Up to 255 characters

DBLibWaitTime

The DBLibWaitTime parameter specifies the time (in seconds) the scheduler, application server, or web server waits before breaking the connection with an event server in an unknown state. That is, the scheduler, application server, or web server maintain and verify connections with the databases, and if an event server is in an unknown state, the connection is broken after the specified time.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`DBLibWaitTime=value`
- Windows -- Enter a value in the Wait Time field on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

The Wait Time field does not accept a negative value.

NOTE

- The default value is 90 seconds. If you set the DBLibWaitTime parameter to 0 (zero), the scheduler, application server, or web server does not time out. They wait until the database responds. We do not recommend setting this parameter to 0 because the scheduler might stop responding.
- Typically, the database never times out. However, if it does, AutoSys Workload Automation tries to reconnect to the database the number of times that is specified in the DBEventReconnect parameter. If the database connections are frequently timing out, it probably indicates a system or event server contention problem.

DBEventReconnect

When the scheduler or application server fails to update one of the event servers while running in dual event server mode, AutoSys Workload Automation stops processing events while it tries to re-establish the connection with the event server. The DBEventReconnect parameter controls the number of times the scheduler or application server tries to connect (or reconnect) to an event server before shutting down or before rolling over to single event server mode. This parameter is used on startup and when there is a connection problem at run time.

When the Oracle client environment is configured for TAF, the Oracle client library proactively notifies the scheduler or application server when it detects a loss of connection to the database. When notified, the scheduler or application server sleeps for five seconds and instructs the Oracle client library to either resume checking the status of the database connection or end all further status checks. The DBEventReconnect parameter controls the number of times the scheduler or application server instructs the Oracle client library to resume checking the status of the database connection before shutting down.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

DBEventReconnect=value [,value2]

– **value**

Defines the number of times the scheduler or application server tries to connect (or reconnect) to the event server before shutting down or before rolling over to single event server mode. When the Oracle client environment is configured for TAF, it defines the number of times the Oracle client library rechecks the failover status.

– **value2**

(Optional) Defines the number of times the scheduler or application server tries to connect (or reconnect) to the second event server before shutting down or before rolling over to single event server mode. Specify this value when you configure AutoSys Workload Automation to run in dual event server mode.

- Windows -- Enter the values in the Event Reconnect field on the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

The Event Reconnect field does not accept a negative value.

NOTE

- The default value is 50 in single event server mode and 50,5 in dual event server mode. For dual event server mode, specify two numbers that are separated by a comma.
- In single event server mode, the default setting specifies that the scheduler tries to connect to the event server 50 times before shutting down. That is, the scheduler tries to reconnect 50 times both on startup or when there is a connection problem.
- In dual event server mode, the default setting specifies that the scheduler tries to connect to the event server (that is not responding) five times before switching over to single event server mode. On startup, AutoSys Workload Automation makes 50 attempts to create a pool of connections to both event servers. If AutoSys Workload Automation exhausts all its attempts to either create a pool of connections or restore its lost connections to both event servers, it assumes that there is a connection or configuration problem and shuts down.
- In dual event server mode, the DBEventReconnect parameter is set to the default only if you initially install dual event servers. If you add a second event server after the AutoSys Workload Automation installation, set the DBEventReconnect value appropriately.
- Only the primary and shadow schedulers roll over to single event server mode when the number of reconnection attempts is exceeded. The primary or shadow scheduler performs the following actions during a database rollover:
 - Sends a DB_ROLLOVER alarm to the event server.
 - Updates the event server to reflect that AutoSys Workload Automation is running in single event server mode.
 - A copy of the current configuration file is saved as config.rollover.\$AUTOSERV (on UNIX), where AUTOSERV defines the name of the instance. The EventServer_1 or EventServer_2 parameter in the configuration file is updated to include the active event server.
 - On Windows, the status of the failed event server is updated in the configuration file. This configuration file entry activates the Enable button corresponding to the failed event server on the Event Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
- The tie-breaker scheduler and the application server do not automatically roll over to single event server mode. They maintain both their connections to the event server and try to reconnect until they receive notification from either the primary or shadow scheduler to roll over. The application server does not service API requests from AutoSys Workload Automation clients from the time the application server detects the failure of one of the event servers until the time it receives notification to roll over.
- If any of the AutoSys Workload Automation components lose their database connectivity to all event servers, either before or after the database rollover occurs, the components shut down. If the scheduler or the application server receives a request to shut down, the database reconnection process is interrupted immediately after the active connection attempt is completed.

DBMaintTime and DBMaintCmd

The DBMaintTime defines the time when the database maintenance command runs the maintenance script. The DBMaintCmd parameter defines the location of the DBMaint script (on UNIX) or the DBMaint.bat batch file (on Windows).

The DBMaintTime and DBMaintCmd parameters let you automate the database maintenance. The scheduler performs internal database maintenance once a day. It does not process any events during maintenance, and it waits for the maintenance activities to complete before resuming normal operations.

You can configure these parameters as follows:

- UNIX -- Edit the following entries in the configuration file:

```
DBMaintTime=HH:MM
DBMaintCmd=pathed_command
```
- Windows -- Enter the time in the Start Time field and the location in the Command field on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The default value for the DBMaintTime parameter is 3:30 a.m. Specify the time in a 24-hour format.
- The default value for the DBMaintCmd parameter is \$AUTOSYS/bin/DBMaint (on UNIX) or %AUTOSYS%\bin (on Windows).
- After you configure the DBMaintTime or DBMaintCmd parameter, restart the scheduler for the new settings to take effect.
- Schedule the maintenance command to run when the system activity is minimal. We recommend that you configure your system to back up the database during the maintenance cycle.

Configure an Application Server

You can configure an application server by editing the following application server parameters in the configuration file (UNIX) or the equivalent fields on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility (Windows):

(UNIX) Follow these steps:

1. Log in to AutoSys Workload Automation as a user authorized to stop the application server and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt to stop the application server:

```
unisrvcntr stop waae_server.$AUTOSERV
```
3. Edit the application server parameters in the configuration file as appropriate and save the file.
4. Enter the following command at the operating system prompt to start the application server:

```
unisrvcntr start waae_server.$AUTOSERV
```

The new parameter settings take effect.

NOTE

You can [pause and resume](#) the application server to read the modified values of some of the parameters in the configuration file at runtime.

(Windows) Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
2. Select an instance for which you want to configure the application server from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
4. Right-click the application server and select Stop.

5. Click the Application Server icon on the toolbar.
6. Edit the fields on the Application Server - AutoSys Workload Automation window as appropriate.
7. Click the Services icon on the toolbar.
8. Right-click the application server and select Start.

The application server starts and the new parameter settings take effect.

NOTE

You can [pause and resume](#) the application server to read the modified values of some of the fields on the Administrator utility at runtime.

AppSrvAuxiliaryListeningPort

The AppSrvAuxiliaryListeningPort parameter defines the port number the application server uses to listen for inbound AFM protocol data using non-SSA communication. The default value is 7500.

Network data between the scheduler and the agent or the scheduler and the CA Workload Automation EE manager is prepared using the proprietary AFM protocol. The application server auxiliary listening port is used primarily to receive inbound messages from agents running on non-SSA ports.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`AppSrvAuxiliaryListeningPort=port`
- Windows -- Enter the port number in the Auxiliary Listening Port field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The application server auxiliary listening port must be different from the scheduler auxiliary listening port. The scheduler and application server auxiliary listening ports must be physical ports. So, we recommend that you configure the SSA port setting to disable port multiplexing (EnablePmux=False), otherwise AutoSys Workload Automation uses the virtual ports that are provided by SSA. If EnablePmux is set to True or the scheduler or application server auxiliary listening port is not defined, AutoSys Workload Automation does not initiate communication with agents that are configured to run on physical ports.
- For more information about configuring AutoSys Workload Automation to work with SSA, see [Configure AutoSys Workload Automation to Work with Other Broadcom products](#).
- By default, AutoSys Workload Automation does not enable SSL communication for any of its ports (physical or virtual). When you configure the SSA port settings of the application server auxiliary listening port, ensure that you do not enable SSL communication.
- During the AutoSys Workload Automation installation, the application server auxiliary listening port is set to 7500 by default. If you are performing a custom installation, you can specify a different application server auxiliary listening port using the Application Server Information (on UNIX) or Application Server Properties (on Windows) page. AutoSys Workload Automation uses both the virtual ports and the auxiliary ports simultaneously. The application server log displays the ports that are used by AutoSys Workload Automation as follows:

```
CAUAJM_I_20366 CA WAAE Application Server operational on agent listener port
49169.
```

```
CAUAJM_I_20367 CA WAAE Application Server operational on auxiliary agent
listener port 7500.
```

AutoServer

The AutoServer parameter specifies the host name(s) of the application server to which all clients in the instance connect to.

NOTE

You can enter multiple application server host names in a comma-separated list.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`AUTOSERVER=hostname1,hostname2,hostname3...`
- Windows -- Enter the host name(s) in the Server Host field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

After you define the application server host name(s), restart the scheduler as it propagates the application server host name value(s) to the agent.

AutoServerAliasId

The AutoServerAliasId parameter defines a unique communication alias that the application server uses to communicate with CA Workload Automation EE and the Workload Automation Agent on z/OS. The communication alias is set to *INSTANCENAME_ABBREVIATEDHOSTNAME* during the AutoSys Workload Automation installation.

If the AutoSys Workload Automation instance has multiple application servers, the communication alias for each application server must be unique. If an alias is not unique, define another alias for that application server. The application server does not start if it detects another application server with the same communication alias.

NOTE

- The scheduler also requires a communication alias to communicate with CA Workload Automation EE and the Workload Automation Agent on z/OS. However, the communication alias for the scheduler is automatically set to *INSTANCENAME_AGT* (in uppercase). You cannot change this value.
- For information about configuring CA Workload Automation EE or the Workload Automation Agent on z/OS to work with AutoSys Workload Automation, see [Configure Cross-Platform Scheduling for AutoSys](#).

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:
`AutoServerAliasId=unique_ID`
- Windows -- Enter the communication alias in the Communication Alias Identifier field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

Specify the communication alias identifier in the *INSTANCENAME_ABBREVIATEDHOSTNAME* format. You can specify up to 16 uppercase characters. If you specify the value in lowercase or mixed case, the value is automatically changed to uppercase. The abbreviated hostname consists of the last 12 characters of the node name excluding the domain name. For example, the communication alias of the application server on myhost.example.com is set to ACE_MYHOST, where ACE is the name of the AutoSys Workload Automation instance.

AutoServerId

The AutoServerId parameter defines a unique identifier that the application server uses to communicate with the agent. The default value is *INSTANCENAME_APP_MachineName*.

If the AutoSys Workload Automation instance has multiple application servers, the identifier for each application server must be unique. If an identifier is not unique, define another identifier for that application server. The application server does not start if it detects another application server with the same identifier.

NOTE

The scheduler also requires an identifier to communicate with the agent. However, the identifier for the scheduler is automatically set to *INSTANCE_NAME_SCH* (in uppercase). You cannot change this value.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
AutoServerId=unique_ID
```

- Windows -- Enter the identifier in the Communication Identifier field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

AutoServerPort

The AutoServerPort parameter defines the application server listening port for all SSA communication. The default value is 9000.

You can configure the application server to listen on a different virtual port. Both the AutoSys Workload Automation application server and the agent require a port to listen to for incoming connections. By default, the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. You might want to reconfigure the application server to listen on a different virtual port if another Broadcom product is using the default virtual port and you want that product to continue using that port.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
AutoServer=port
```

- Windows -- Enter the port number in the Client Communication Port field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

LogMaxEndLines

The LogMaxEndLines parameter defines the maximum number of lines to retrieve from the Agent log file, Job log file, STDOUT file, and STDERR file. You can retrieve these logs either from Web UI or AutoSys Workload Automation.

You can configure this parameter as follows:

- UNIX -- Edit the following entry in the configuration file:

```
LogMaxEndLines=value
```

- Windows -- Enter a value in the Log Max End Lines field on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility.

NOTE

- The default value is 2000. If the LogMaxEndLines parameter is set to 0 (zero), the application server retrieves the entire contents of the Agent log file, Job log file, STDOUT file, and STDERR file.
- The value of the LogMaxEndLines parameter can be a number in the range 0 through 10000. If the specified value is not in the valid range, the application server resets the value to the default. If the specified value is in the valid range, the application server passes the request to the agent. The agent retrieves the specified number of lines starting from the end of the Agent log file, Job log file, STDOUT file, and STDERR file.
- The LogMaxEndLines parameter does not apply to the scheduler or application server log files.
- After you set the LogMaxEndLines parameter, pause and resume the scheduler and application server for the new settings to take effect.

MaxBinaryTransferSize

The MaxBinaryTransferSize parameter specifies the maximum size (in megabytes) for a blob.

You can configure this parameter by editing the following entry in the configuration file:

```
MaxBinaryTransferSize=value
```

NOTE

- The default value is 1024.
- By default, this parameter is not included in the configuration file. If you want to set the maximum size for a blob, add the MaxBinaryTransferSize parameter in the configuration file.
- You must set the MaxBinaryTransferSize parameter on all application servers that are present in an AutoSys Workload Automation instance.

Differences Between the Legacy Agent and the Agent

Starting in r11.3, the Workload Automation Agent for UNIX, Linux, or Windows replaced the Remote Agent (auto_remote) that was provided with Unicenter AutoSys JM 4.5.1 and r11. The documentation refers to auto_remote as the *legacy agent* and to Workload Automation Agent for UNIX, Linux, or Windows as the *agent*.

The agent provides additional job types, including monitoring, Micro Focus, and FTP jobs. In addition to the job types supported by the agents and agent plug-ins, other agent features and behaviors were also changed.

This section describes the differences between the legacy agent and the agent:

Installing the Agent

You can install the agent using the AutoSys Workload Automation DVD or the AutoSys Workload Automation Standalone Agent DVD. If you install the agent using either of these DVDs, the installer automatically configures the agent to work with AutoSys Workload Automation.

Starting in Release 11.3.5, if you select the Configure Agent with Legacy Remote Agent Compatibility check box on the Agent Attributes panel during the installation, the installer automatically configures the agent to behave like the legacy agent. That is, the job processing behavior of the agent closely matches the job processing behavior of the legacy agent. We recommend that you select the Configure Agent with Legacy Remote Agent Compatibility check box if you want to configure the agent to behave like the legacy agent.

If you did not select the Configure Agent with Legacy Remote Agent Compatibility check box during the installation, but later decide to configure the agent to behave like the legacy agent, manually add or edit the compatibility parameters in the agent's agentparm.txt file.

NOTE

- If you are using the Workload Automation Agent DVD to install the agent, we recommend that you select to install the agent in the AutoSys Workload Automation compatibility mode as the installer configures the agent to work with AutoSys Workload Automation. If you do not select to install the agent in the AutoSys Workload Automation compatibility mode, take the following actions as the installer does not configure the agent to use all the available features of AutoSys Workload Automation:
 - a. Back up the agentparm.txt file.
 - b. Uninstall the agent.
 - c. Use the AutoSys Workload Automation DVD or the AutoSys Workload Automation Standalone Agent DVD to install the agent.
 - d. Compare the backed up agentparm.txt file with the new agentparm.txt file and merge the appropriate parameters into the new agentparm.txt file.

If you are installing the agent for the first time, use the AutoSys Workload Automation DVD or the AutoSys Workload Automation Standalone Agent DVD.

- For more information, see [Configure the Agent to Behave like a Legacy Remote Agent](#).

AutoSys Workload Automation Configuration on the Agent

To communicate with the agent, your AutoSys Workload Automation instance must be specified in the agent's agentparm.txt configuration file. Certain parameters defined on AutoSys Workload Automation and the agent must match.

Environment Variables

In 4.5.1 and r11, the legacy agent's environment was set by sourcing the environment variables specified in the /etc/auto.profile file. The variables are preceded by #AUTOENV#.

Starting in r11.3, the agent sources the environment variables specified in the following locations:

- The WAAE.txt file—The agent-wide environment variables are defined in the WAAE.txt file located in the \$AUTOROOT/SystemAgent/agent_name/profiles (on UNIX) or %AUTOROOT%\SystemAgent\agent_name\Profiles (on Windows) directory.

NOTE

The oscomponent.environment.variable parameter in the agent's agentparm.txt file specifies the path to the WAAE.txt file.

- *The instance_name.txt file—The manager-specific environment variables are defined in the instance_name.txt file located in the \$AUTOROOT/SystemAgent/agent_name/profiles (on UNIX) or %AUTOROOT%\SystemAgent\agent_name\Profiles (on Windows) directory.*

NOTE

The oscomponent.environment.variable_manager_instance_name_SCH parameter in the agent's agentparm.txt file specifies the path to the instance_name.txt file.

- The profile JIL attribute in a job definition
- The envvars JIL attribute in a job definition

NOTE

For more information about the parameters in the agentparm.txt file, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#). For more information about setting profiles and environment variables in a job definition, see [Job Profiles](#).

(UNIX) Sourcing Job Profiles

The legacy agent sourced environment variables specified in several locations as part of running a job. The legacy agent can source environments from the profile specified in the job definition, the UNIX /etc/profile file, or the UNIX global profile (/etc/auto.profile file).

Starting in r11.3, the agent sources environment variables from new locations and applies them in a different order than the legacy agents. The agent sources environments from new locations such as the files specified in the agentparm.txt file (replacing profile environments stored in the Windows registry) and the envvars job attribute. Also, the agent behaves differently than the legacy agent for sourcing job profiles and global profiles from within a script when a user profile is not specified in the job definition. You can configure the agent to skip sourcing the user's profile, UNIX /etc/profile file, and the UNIX global profile by setting the following parameters in the agent's agentparm.txt file to true:

- **oscomponent.noforceprofile**

The agent executes a temporary shell script without sourcing the user's profile. The agent does not source the UNIX /etc/profile file, the user's login profile, the UNIX global profile, or the job profile as part of running the job.

NOTE

This is a global setting and applies to all job executions.

- **oscomponent.profiles.src.delay**
The agent drops sourcing of the profiles into the temporary shell script and passes the temporary shell script file into the shell interpreter. The agent does not add the UNIX /etc/profile file or the user's login profile to the temporary script.
- **oscomponent.profiles.src.verify**
The agent verifies the existence of a job profile prior to execution. If the job profile does not exist, the job terminates and an error is returned.
- **oscomponent.profiles.src.order.global.first**
The agent sources the UNIX global profile first prior to execution.

NOTE

If you set the `oscomponent.profiles.src.order.global.first` parameter to false, the agent sources the UNIX global profile last.

- **oscomponent.profiles.global.override**
 - If the job profile is present, the agent sources the job profile. The UNIX global profile is not sourced.
 - If the job profile is not present, the agent sources the UNIX global profile.

You can configure the agent to source job or global profiles relative to the initial working directory when the specified profile does not contain a relative or full path by setting the following parameter in the agent's `agentparm.txt` file:

- **oscomponent.profiles.src.location.iwd**
Specifies whether the initial working directory is used to source the specified profile when it does not contain a relative or full path.
 - **true**
Indicates that the initial working directory is used to source the specified profile when it does not contain a relative or full path.
 - **false**
Indicates that the `installDir/profiles` directory is used to source the specified profile when it does not contain a relative or full path. This is the default.

NOTE

- If you set the `oscomponent.profiles.src.order.global.override` parameter to false, the agent always sources the UNIX global profile.
- When the `oscomponent.profiles.src.order.global.override` parameter is set to true, if you run a command job that executes a AutoSys Workload Automation client utility using a job profile that does not source the environments required by the AutoSys Workload Automation client utility, the job fails.
- For more information about job profiles and environment variables, see [Job Profiles](#). For more information about the parameters in the agent's `agentparm.txt` file, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Job Attribute Environment Variables

The legacy agent supports the setting of job definition JIL attributes as environment variables, which are sourced as part of running a job.

The agent supports the setting of only the following environment variables based on job definition JIL attribute values:

- `__job_name`
- `__box_name`
- `__machine`
- `__run_machine`
- `__max_exit_success`

NOTE

- You can set these job definitions JIL attributes as environment variables only for command and file watcher jobs.
- The `__box_name` environment variable is set only if the job is contained within a box.
- The `__max_exit_success` environment variable is set only for command jobs.

In the current release, you can configure AutoSys Workload Automation to automatically set the supported job definition JIL attributes as environment variables using the `SetJobAttributeEnvironmentals` parameter in the configuration file (on UNIX) or the Set Job Attribute Environmentals check box on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (on Windows).

NOTE

For information about setting job definition JIL attributes as environment variables on UNIX, see [Configure a Scheduler](#).

Custom job applications that require knowledge of additional job attributes should be rewritten to invoke the `GetJobsWithFilter` class of the C++ or Java SDK.

Communication Port

The configuration required to communicate with the agent is different from the configuration for the legacy agent.

In 4.5.1 and r11, the scheduler used the following port setting to communicate with the legacy agent:

- UNIX—The `AutoRemPort` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The Remote Agent Port field in the Administrator utility.

NOTE

In the current release, this field is named *Legacy Remote Agent Port*.

Starting in r11.3, you can use those port values to communicate with the legacy agent. Those port values do not apply to the agent.

NOTE

For more information about the remote agent port settings, see [Configure a Scheduler](#).

Minimum Disk Space Used for Logging

In 4.5.1 and r11, the following settings specified the minimum amount of disk space that must be available to write to the scheduler log:

- UNIX—The `FileSystemThreshold` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The File System Threshold KB field in the Administrator utility.

Starting in r11.3, those settings control the scheduler and application server log file settings, but are only used when running jobs on the legacy agents. The agent has parameters in the `agentparm.txt` file that control the log file settings.

NOTE

For more information about the log file settings, see [Log Settings](#). For more information about the `FileSystemThreshold` parameter, see [Configure a Scheduler](#).

Location of Log Files

In 4.5.1 and r11, the legacy agent's log files were written to the following locations:

- UNIX—The directory specified in the `AutoRemoteDir` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The directory specified in the Enterprise Wide Logging Directory field in the Administrator utility.

NOTE

In the current release, this field is named *Legacy Enterprise Wide Logging Directory*.

Starting in r11.3, those logging directories are only used when running jobs on the legacy agents.

The agent writes all log files to the following directories:

- On UNIX
 - *installation_directory/SystemAgent/agent_name/log*
 - *installation_directory/SystemAgent/agent_name/spool* (for job spool files)
- On Windows
 - *installation_directory\SystemAgent\agent_name\log*
 - *installation_directory\SystemAgent\agent_name\spool* (for job spool files)

NOTE

- In 4.5.1 and r11, you had to override the default log file directory on operating systems that do not support the locking of files in the /tmp directory. This is because the legacy agent used the locks to check whether a job was running. You no longer have to change the default log file directory because the agent stores the job spool files in the *installation_directory/SystemAgent/agent_name/spool* directory (UNIX) or the *installation_directory\SystemAgent\agent_name\spool* directory (Windows) by default. However, you must change the default log file directory if you run jobs on legacy agents and the operating system on any of the legacy agent computers does not support the locking of files in the /tmp directory.
- For more information about the logging directory settings, see [Configure a Legacy Agent](#).

Log File Maintenance

In 4.5.1 and r11, the following settings specified whether the legacy agent's temporary log files were automatically removed:

- UNIX—The CleanTmpFiles parameter in the \$AUTOUSER/config.\$AUTOSERV configuration file.
- Windows—The Clean Temporary Files field in the Administrator utility.

NOTE

In the current release, this field is named *Legacy Clean Temp Files*.

Starting in r11.3, those settings are only used for legacy agent log files.

The agent has parameters in the agentparm.txt file that control how log files and job spool files are maintained.

NOTE

For more information about maintaining agent log files and clearing job spool files, see [Log File Maintenance](#).
For more information about the temporary log file settings, see [Configure a Legacy Agent](#).

(UNIX) Support Chained Commands

The agent generates a wrapper script to support chained commands (commands separated by a semi-colon). To support chained commands, set the following parameters in the agent's agentparm.txt file:

```
oscomponent.wrapper.exec.force=false
oscomponent.cmdprefix.force=true
```

NOTE

- The `oscomponent.wrapper.exec.force` parameter applies to UNIX agents only.
- The `oscomponent.wrapper.exec.force` parameter is applicable only if the `oscomponent.cmdprefix.force` parameter is set to true.
- When the `oscomponent.wrapper.exec.force` parameter is set to false (the default), the `SEND_SIGNAL` event is not supported.
- If you set the `oscomponent.wrapper.exec.force` parameter to true, the `exec` function is supported and chained commands are not supported.
- To support both, the chained commands and the `SEND_SIGNAL` event, take the following actions:
 1. a. Set the `oscomponent.wrapper.exec.force` parameter to true and move the chained commands to a script file.
 - b. Update the command attribute in the job definition to execute the script file instead of the chained command.

Remote Profile Files

In 4.5.1 and r11, the following settings specified whether the scheduler redirects all standard error and standard output information to the `auto.rem*` log file:

- UNIX—The `RemoteProFiles` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The Remote Profile Logging check box in the Administrator utility.

NOTE

In the current release, this field is named *Legacy Remote Profile Logging*.

The output information is generated when the `/etc/auto.profile` file is sourced.

Starting in r11.3, those settings are only used when running jobs on the legacy agents.

The agent does not use these settings or writes any output generated by the `/etc/auto.profile` file.

NOTE

For more information about the remote profile files settings, see [Configure a Scheduler](#).

Signals for a KILLJOB Event

In 4.5.1 and r11, you can specify a comma-separated list of signals to send to a job whenever the `KILLJOB` event is sent. The following settings specified the signals:

- UNIX—The `KillSignals` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The Kill Signals field in the Administrator utility.

NOTE

In the current release, this field is named *Legacy Kill Signals*.

Starting in r11.3, those settings are only used when running jobs on the legacy agents.

NOTE

For more information about the `KillSignals` parameter, see [Configure a Scheduler](#).

Calculating Machine Load

In 4.5.1 and r11, you can define the method used to determine the percentage of CPU cycles available on a real machine that belongs to a virtual machine. The following settings specified the method:

- UNIX—The `MachineMethod` parameter in the `$AUTOUSER/config.$AUTOSERV` configuration file.
- Windows—The Machine Method field in the Administrator utility.

Starting in r11.3, the agent does not use the UNIX vmstat utility or Windows performance counters to determine the percentage of available CPU. Instead, the agent runs a CPU Monitor job to determine the current load on the machine.

The rstatd method continues to be supported by the UNIX scheduler. However, for this method to be used, the value of the opsys attribute for a type-a machine definition must be set to an operating system that supports rstatd ('aix', 'hpux', 'linux', or 'openvms'). If the value of the opsys attribute is not set or is set to an operating system that does not support rstatd, the UNIX scheduler will use a CPU Monitor job to calculate the available machine load.

NOTE

For more information about the machine method setting, see [Configure a Scheduler](#).

Debugging Logs

In 4.5.1 and r11, the ISDBGACTIV setting controlled the display of trace messages for debugging.

Starting in r11.3, the administrator for the agent can set the log.level parameter in the agent's agentparm.txt file. This parameter controls the type of debugging logs that are generated.

NOTE

For more information about the log.level parameter, see [log.level](#). For more information about the ISDBGACTIV settings, see [Configuring AutoSys Workload Automation](#).

Running Windows Commands

You can define Command jobs to run Windows operating system commands, such as dir and echo. In 4.5.1 and r11, you specified only the command and arguments in the command attribute (for example, command: "dir c:\temp\"). The legacy agent prefixed "*path*\cmd.exe /c" to the command before running the process.

Starting in r11.3, the agent does not automatically prefix the command with the path to the command interpreter. To automatically prefix the command, set the following parameters in the agent's agentparm.txt file to true:

```
oscomponent.lookupcommand=true
oscomponent.cmdprefix.force=true
oscomponent.cmdprefix.force.quotes.full=true
```

When you set the oscomponent.cmdprefix.force.quotes.full parameter to true, commands that have spaces in their path run without error. The agent wraps the entire command in double quotes before the Windows command interpreter runs the command.

If these agent parameters are not set, you must explicitly invoke the command interpreter in the command attribute, as shown in the following example:

```
command: "c:\winnt\system32\cmd.exe /c dir c:\temp\"
```

NOTE

For more information about the agent parameters, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Specifying a Command or Script Name Without the Full Path

In 4.5.1 or r11, you specified the full path to the command or script name you want to run in the Command job definition.

Starting in r11.3, you can specify the command or script name you want to run without the full path in the Command job definition. To specify the command or script name without the full path, set the following parameter in the agent's agentparm.txt file to true:

```
oscomponent.lookupcommand=true
```

On UNIX, the agent looks up the path to the script or command name for the specified user ID. On Windows, the agent looks up the path to the command file in the following directories, in the order listed: agent installation directory, WINDOWS\system32 directory on the agent computer, WINDOWS\system directory on the agent computer, WINDOWS directory on the agent computer, any directory whose path is set in the system path or user path on the agent computer.

NOTE

- If the `oscomponent.lookupcommand` parameter is set to true, verify that the agent on UNIX is running under the root account.
- For more information about specifying a command or script name without the full path, see [.Specify a Command or Script Name Without the Full Path vAE_12](#). For more information about the agent parameters, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Directory the Jobs Runs Under

In 4.5.1 or r11, the legacy agent jobs ran from the directory where the agent is installed.

Starting in r11.3, you can configure the directory the Command jobs run under using the `oscomponent.initialworkingdirectory` parameter in the agent's `agentparm.txt` file.

NOTE

- You can override the initial working directory on UNIX by specifying a value for the HOME and PWD environment variables.
- For more information about the agent parameters, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Evaluation of Job's Termination Time

If a job runs on a machine that has the agent installed, the scheduler evaluates the `term_run_time` attribute and automatically generates a `CHK_TERM_RUNTIME` event to instruct the scheduler to check whether a job has ended by the specified time. If the job does not end by the specified time, the scheduler sends the agent a request to kill the job.

If a job runs on a machine that has the legacy agent installed, the legacy agent evaluates the `term_run_time` attribute and no `CHK_TERM_RUNTIME` event is generated. If the job does not end by the specified time, the legacy agent terminates the job.

Exit Code for Job Failures

AutoSys Workload Automation reports the exit codes of jobs as they complete. Exit codes typically indicate reasons for job failures or command failures in the case of command jobs. A command exit code of zero typically indicates successful execution of the command. If a command job completes successfully, AutoSys Workload Automation reports an exit code of zero. When a job encounters an error and does not complete successfully, the log files for that job display an exit code identifying the error that caused the failure.

In 4.5.1 and r11, the legacy agent generates custom error codes to indicate certain job failures that occur before it can execute the command. For example, the legacy agent sends a custom exit code of 256 to indicate that it failed to locate the command. When the legacy agent reports a custom error code, it logs an entry in the legacy agent log file. This entry includes a message indicating an internal return code with the value of the custom error code. In the current release, the agent does not generate custom exit codes to indicate job failures and only reports operating system exit codes for job failures that occur before it executes the command. In all cases, when the agent executes the command successfully and the command fails, the agent reports the exit code returned by the command, which depends on the implementation of the command that is executed.

In 4.5.1 and r11, when a job encounters an error and does not complete successfully, the legacy agent log file for that job displayed custom error messages indicating the problem. For a specific set of errors, custom error messages were

displayed in the detailed report that is generated using the autorep command. In the current release, the exit codes and messages are displayed in the spool and job log files located in the agent spool directory.

NOTE

For more information about the exit codes that the agent generates, see [Exit Codes](#).

Legacy Agent Exit Codes

In 4.5.1 and r11, the legacy agent generates custom error codes to indicate certain job failures that occur before it can execute the command. When the legacy agent reports a custom error code, it logs an entry in the legacy agent log file. This entry includes a message indicating an internal return code with the value of the custom error code. When the legacy agent reports a command error code, it logs an entry in the legacy agent log file indicating that it obtained an error code from the job process and displays the value of the error code.

NOTE

On UNIX, when the legacy agent reports a command error code, it displays a log entry containing three exit code values formatted as a single number followed by two numbers in parentheses; for example, 1 (0 1) or 15 (15 0). The three values represent the following:

- The unmasked exit code that is returned by the UNIX process.
- The first exit code value that is masked to show the UNIX signal value if the job process exited as a result of processing a UNIX signal. If the job process did not exit as a result of processing a UNIX signal, this value is undefined. The UNIX signal values are documented in the UNIX operating system manual pages that are on the computer where the job is executed. If the job process exited as a result of processing a UNIX signal, the legacy agent changes the status of the job to TERMINATED. You can terminate the job process by issuing the SEND_SIGNAL or the KILLJOB event.
- The first exit code value that is masked to show command exit code if the job process exited normally. If the process did not exit normally, this value is undefined. The command exit code depends on the implementation of the command that is executed.

The legacy agent log file for a job can display the following custom exit codes:

- **121**
Indicates that the job cannot run because the input file specified in the std_in_file attribute does not exist or is inaccessible. Check permissions.
- **122**
Indicates that the job cannot run because the output directory specified in the std_out_file attribute does not exist or is inaccessible. Check permissions and verify that the file system is not full.
- **123**
Indicates that the job cannot run because the output directory specified in the std_err_file attribute does not exist or is inaccessible. Check permissions and verify that the file system is not full.
- **124**
Indicates that the Windows job cannot run because the job profile is missing.
- **223**
Indicates that the UNIX 4.5.1 legacy agent cannot run the job because the chk_file conditions of the job are not met.
- **256**
Indicates that AutoSys Workload Automation cannot execute the command. Typically due to one of the following situations:
 - The command or file to execute does not exist.
 - The file to execute is not executable. Check permissions.
 - The file to execute is in a directory that is not accessible. Check permissions (ls -ld). If the directory is NFS-mounted, verify that the mount exists.
 - You are using a job environment file with an incorrect PATH variable. Typically due to *one* of the following situations:

- The file to execute is in a directory that is specified in PATH.
- The PATH command uses an undefined variable (which translates as blank space, causing the PATH command to terminate before all the directories are defined to it).
- The job environment file uses non-Bourne shell commands (for example, alias), and the expected settings (or aliases) do not exist.
- The command contains incorrect options (for example, date - JASD).
- **512**
Indicates that the job cannot complete because the command contains incorrect options (for example, awk 'junk').
- **998**
Indicates that the UNIX job cannot run because the job profile is missing.
- **-4**
Indicates that the Windows 4.5.1 or r11 legacy agent cannot run the job because the chk_file conditions of the job are not met.

(UNIX) PAM Service for User Authentication

PAM (Pluggable Authentication Modules) is used for security to check whether a service should be used. A service is a program that provides a function that requires authentication. Examples of services are login, sshd, pam, and sudo.

By default, the agent uses login service. AutoSys Workload Automation uses sshd service to authenticate users. To avoid user authentication errors, we recommend that you configure the agent to use sshd service using the oscomponent.auth.pam.svc and oscomponent.auth.pam.lib parameters in the agent's agentparm.txt file.

NOTE

For more information about the parameters in the agentparm.txt file, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Polling Interval for File Watcher Jobs

Starting in r11.3, you can define a File Watcher job to run on a machine that has the legacy agent or the agent installed. The behavior of the agent for the polling interval is different from the legacy agent.

The legacy agent uses a default polling interval of 60 seconds for a File Watcher job. You can override this value by specifying the watch_interval attribute in a job definition.

The agent uses two polling intervals for a File Watcher job: a global default of 30 seconds set on the agent and the watch_interval attribute value specified in the job definition. If the condition the File Watcher job is monitoring is not satisfied, the agent sleeps for 30 seconds based on its global polling interval. After 30 seconds, the agent checks the condition of the file again. If the condition is satisfied, the agent waits for the second polling interval specified by the watch_interval attribute to ensure the file remains steady. If the file has not changed after the second polling interval elapses, the agent returns the status. If the file does change, the agent goes back to sleep for the duration of the second polling interval until the file eventually stabilizes.

Monitoring Jobs

Starting in r11.3, depending on the type of file activity the agent monitors, you can configure the agent to skip the first scan of a monitored file by setting the following parameter in the agent's agentparm.txt file to true:

```
filemon.firstscan.skip=true
```

NOTE

If you set this parameter to true, it is backward compatible with the legacy agent.

If the File Watcher job runs on the legacy agent and the file already exists when the job runs, the job waits for the interval specified in the watch_interval attribute, checks for the watched file size, and goes into success when the file reaches the specified minimum size and is no longer growing in size.

If the File Watcher job runs on the agent, the agent uses the `watch_interval` as a second polling interval. The first polling interval is set by default on the agent for 30 seconds. If the job does not complete after the first polling interval, the agent waits for the second polling interval specified by the `watch_interval` attribute to ensure the file remains steady. If the file has not changed after the second polling interval elapses, the agent returns the status. If the file does change, the agent goes back to sleep for the duration of the second polling interval until the file eventually stabilizes.

NOTE

For more information about the agent parameters, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

Monitoring Available Disk Space

By default, jobs do not check the available file space. You can verify the file space that is available on the machine where the job runs using the `chk_files` attribute. At run time, the legacy agent or the agent checks whether the required space is available on the machine where the job runs. If the requirements are met, the job starts. If the requirements are not met, an alarm is generated and the job does not start.

Starting in r11.3, you can configure the agent to monitor the amount of available disk space for the database directory and send notifications to warn you when the space is too low by setting the following parameter in the agent's `agentparm.txt` file to true:

```
agent.resourcemon.enable=true
```

Starting in Release 11.3.6 SP1, the agent is configured to monitor the amount of available disk space and send notifications to warn you when the space is too low. When you install AutoSys Workload Automation, the installer adds the following parameters in the `agentparm.txt` file to enable the agent to monitor the available disk space:

```
core.health.monitor.enable=true
agent.resourcemon.enable=true
agent.resourcemon.threshold.disk.warning.notice=21M
agent.resourcemon.threshold.disk.warning.severe=20M
agent.resourcemon.threshold.disk.critical=10M
```

NOTE

For more information about the agent parameters, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#). For information about verifying the file space before a job starts, see [Verify File Space Before a Job Starts](#).

(Windows) User Requirements for Interactive Jobs

User requirements for interactive jobs differ depending on the release. User requirements for interactive jobs are as follows:

- In Unicenter AutoSys JM 4.5.1, an interactive job ran to success without requiring a user to be logged into the system.
- Starting with AutoSys Workload Automation r11.3, the user specified to run the job must be logged in with an active session to schedule an interactive job. The user can have only one active session. If more than one session exists for the specified user, the agent cannot determine which session to use and therefore fails the job.

Configure AutoSys Workload Automation to Work with Workload Automation Agent for UNIX, Linux, Windows, or i5/OS

After you install and define the agent, you can run jobs with that agent. This section provides an overview about the `agentparm.txt` file and describes how the agent connects to the AutoSys Workload Automation instance.

The following are the optional procedures you can perform to configure AutoSys Workload Automation to work with the agent installed on a UNIX, Linux, Windows, or i5/OS computer:

NOTE

For information about advanced agent configuration tasks (for example, setting up the agent as an FTP server), agent parameters, security settings, limitations, and known issues, see [Configure the Agent](#).

agentparm.txt File

You can configure the agent by editing the parameters in the agentparm.txt file. When you install the agent, the installation program adds commonly-configured agent parameters to the agentparm.txt file. Other agent parameters exist, which you must manually add to the agentparm.txt file to configure the agent. You can modify these parameter values as required.

NOTE

For more information about the agentparm.txt file, see [WA Agent for UNIX, Linux, Windows, or iSeries Agent Parameters](#).

The agentparm.txt file is located in the following directory:

- On UNIX

install_directory/SystemAgent/agent_name

- On Windows

install_directory\SystemAgent\agent_name

install_directory

Specifies the root directory where AutoSys Workload Automation is installed.

agent_name

Specifies the name of the agent.

NOTE

- If you select the Configure Agent with Legacy Remote Agent Compatibility check box on the Agent Attributes (on UNIX) or Agent Properties (on Windows) panel during the installation, the installer adds the compatibility parameters in the agent's agentparm.txt file to configure the agent to behave like the legacy agent.
- For information about the parameters in the agentparm.txt file and how to configure them to work with the scheduling manager, see Workload Automation Agents.
- When you make a change to an agent parameter that is also defined on AutoSys Workload Automation, such as the agent name, you must configure the corresponding parameter on AutoSys Workload Automation. Similarly, when you configure an agent parameter on AutoSys Workload Automation, the agentparm.txt file must be updated to include the change.
- If the agent was installed using a program that was not provided with AutoSys Workload Automation (for example, the installation program provided on the Workload Automation Agent DVD), the path to the agentparm.txt may be different. In this case, the agentparm.txt file is located in the root directory where the agent is installed.

How the Agent Connects to the AutoSys Workload Automation Instance

The agent stores the connection properties of the AutoSys Workload Automation instance that it works with in the agent's agentparm.txt file using the following parameters:

- `communication.managerid_n`
- `communication.manageraddress_n`
- `communication.managerport_n`
- `communication.socket_n`

These parameters are automatically updated when AutoSys Workload Automation sends a message to the agent. You do not have to modify these parameters.

However, when you define the agent as a machine to the AutoSys Workload Automation instance, you must specify the following agent properties in the machine definition:

- **Agent name**—Specifies the name of the agent. This value must match the `agentname` parameter in the `agentparm.txt` file.
- **Agent port**—Specifies the port of the agent. This value must match the `communication.inputport` parameter in the `agentparm.txt` file.
- **Encryption key**—Specifies the path to the `cryptkey.txt` file that stores the encryption key for the agent. This value must match the `security.cryptkey` parameter in the `agentparm.txt` file.

You must ensure that these values in the machine definition match the `agentparm.txt` values. Otherwise, the agent and the AutoSys Workload Automation instance cannot communicate.

Configure AutoSys Workload Automation to Work with Agent

As an administrator, your responsibilities include configuring AutoSys Workload Automation to work with agents. After an agent is installed, you define the agent on AutoSys Workload Automation to enable communication between AutoSys Workload Automation and the agent. You also define users that a job runs under and run jobs on the agent.

NOTE

This information assumes that an agent is already installed on the server, client, or other computer that you use to run jobs.

The following diagram illustrates how to configure AutoSys Workload Automation to work with agents:

Figure 15: How to Configure AutoSys Workload Automation to Work with Agents

To configure AutoSys Workload Automation to work with agents, follow these steps:

1. [Collect information about the agent.](#)
2. [Define an agent on AutoSys Workload Automation.](#)
3. [Define a user on AutoSys Workload Automation.](#)
4. (Optional for i5/OS only) [Run UNIX workload on a System i5 computer.](#)
5. [Verify that the agent works with AutoSys Workload Automation.](#)
 - a. (UNIX) [Set up the environment](#)
 - b. [Test communication between AutoSys Workload Automation and the agent.](#)
 - c. [Define a test job.](#)
 - d. [Run the test job.](#)
 - e. [Monitor the test job.](#)

NOTE

- Repeat these steps for each agent where you want to run jobs.
- These steps apply to agents installed on Windows, UNIX, or i5/OS.

After you complete these steps, you can run jobs on the agent.

Collecting Information About the Agent

To define an agent on AutoSys Workload Automation, you use the same values specified on the agent in the agentparm.txt file.

NOTE

When you install an agent, the agentparm.txt file is installed in the agent installation directory. You can configure the agent by editing the parameters in the agentparm.txt file.

Before you define an agent on AutoSys Workload Automation, collect the following required information from the agentparm.txt file:

- Agent name—The agentname parameter specifies the name of the agent.
- Agent port—The communication.inputport parameter specifies the port of the agent.
- Encryption key—The security.cryptkey parameter specifies the path to the cryptkey.txt file that stores the encryption key for the agent.

Define an Agent on AutoSys Workload Automation

After you install an agent, define that agent on AutoSys Workload Automation to enable communication between AutoSys Workload Automation and the agent.

WARNING

When you define an agent on AutoSys Workload Automation, the parameter values you specify in the machine definition must match the corresponding parameters in the agentparm.txt file. If the values do not match, AutoSys Workload Automation cannot communicate with the agent.

NOTE

When you install additional agents by performing an agent-only installation on other computers (other than the client or server), define these agents on the computer where the AutoSys Workload Automation server or client is installed.

Follow these steps:

1. Do *one* of the following:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).

2. Enter **jil**.

The JIL command prompt is displayed for the local AutoSys Workload Automation instance.

3. Enter the following commands:

```
insert_machine: machine_name
type: a
node_name: address
agent_name: agent_name
port: port_number
encryption_type: NONE | DEFAULT | AES
key_to_agent: key
```

– ***machine_name***

Defines a unique name for the agent. When defining jobs, specify this name in the machine attribute.

– ***a***

Specifies that the machine is a Workload Automation Agent for UNIX, Linux, Windows, or iOS.

– ***address***

(Optional) Defines the IP address or DNS name of the computer where the agent is installed.

Default: The value specified in the `insert_machine: machine_name` command.

NOTE

If you do not specify the `node_name` attribute, `insert_machine: machine_name` (the default) must be the DNS name of the agent machine. Otherwise, AutoSys Workload Automation cannot connect to the agent.

– ***agent_name***

(Optional) Specifies the name of an agent.

Default: `WA_AGENT`

NOTE

- This name must match the `agentname` parameter in the `agentparm.txt` file.
- You can specify the alias name for the agent in the `agent_name` parameter to configure the alias on AutoSys Workload Automation.

– ***port_number***

(Optional) Specifies the port that the agent uses to listen for traffic.

Default: 7520

NOTE

This port number must match the `communication.inputport` parameter in the `agentparm.txt` file.

– ***NONE | DEFAULT | AES***

(Optional) Specifies the encryption type the agent uses to communicate with AutoSys Workload Automation. You can set the encryption type to one of the following:

• ***NONE***

Specifies that the agent uses no encryption.

• ***DEFAULT***

Specifies that the agent uses the default encryption key and type.

• ***AES***

Specifies that the agent uses AES 128-bit encryption.

NOTE

If you set the encryption type to AES, specify a key using the `key_to_agent` attribute.

– ***key***

(Optional) Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. Specify *one* of the following:

- A 32-digit hexadecimal key
- A passphrase with up to 16 characters

NOTE

This key must match the encryption key that is stored in the cryptkey.txt file. The security.cryptkey parameter in the agentparm.txt file specifies the path to the cryptkey.txt file that stores the encryption key for the agent. You can encrypt a key using the as_config command or the keygen utility.

4. (Optional) Specify optional machine attributes:

- character_code
- description
- opsys
- max_load
- factor
- heartbeat_attempts
- heartbeat_freq

5. Enter exit.

The data is loaded into the database. The agent is defined on AutoSys Workload Automation.

Example: Define an Agent Using the Default Values

This example defines an agent using the default values for the following attributes: type: a, node_name: sysagt, port: 7520, and agent_name: WA_AGENT.

```
insert_machine: sysagt
```

Example: Define an Agent

This example defines a machine named *eagle* where the agent WA_AGENT runs on the node myagenthostname and uses 49154 as its main input port.

```
insert_machine: eagle
type: a
agent_name: WA_AGENT
node_name: myagenthostname
port: 49154
max_load: 100
factor: 1.0
```

Define a User on AutoSys Workload Automation

You cannot run jobs on an agent machine until you define user IDs and passwords for the users that own those jobs.

NOTE

- To define the owner of a job, specify the owner attribute in the job definition.
- (UNIX) We recommend that you use the root account to run a job on the agent computer. Using the root account lets you run jobs under different user accounts.
- You can define all jobs on the agent computer to run under a default user ID by specifying that user ID in the agentparm.txt file. For more information, see [Define a Default User ID](#).
- You must define user IDs and passwords on AutoSys Workload Automation for Database, FTP, PeopleSoft, or SAP jobs. You do not need to define user IDs and passwords on AutoSys Workload Automation for Oracle jobs or Command jobs on UNIX.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EDIT superuser.
2. Do *one* of the following:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
3. Enter the following command:

```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

4. Enter option [5] Manage users.
The Manage users submenu appears.
5. Enter option [1] Manage users with password.
The Manage users with password menu appears.
6. Enter option [1] Create a user. The utility prompts you to enter the user name.
7. Enter the user name in the following format:

```
{user@host|user@domain}
```

- **user**
Specifies the name of the user.
- **host**
Specifies the name of the local host machine on which the user is defined.
- **domain**
Specifies the name of the domain on which the user is defined.

The utility prompts you to enter a password.

8. Enter a password.
The utility prompts you to enter the password again.
9. Re-enter the password.
A message appears to indicate whether the user was successfully added to the database.

NOTE

For the owner of a Hadoop job, you can specify a key file or key string using the autosys_secure menu options.

10. Make any required changes to the user definition, depending on the message that you received, and then enter option [9] Exit from "Manage users with password" menu.
The utility redirects to the main menu. The user is saved in the database. You can continue managing user IDs and security settings or exit the utility.

Run UNIX Workload on a System i5 Computer

Workload Automation Agent for i5/OS lets you schedule jobs on the i5/OS operating system. In addition to scheduling native i5/OS jobs, you can schedule most UNIX workload, such as UNIX scripts, in the PASE environment on i5/OS.

To run both native and UNIX jobs on the same i5/OS computer, you must install two Workload Automation Agents for i5/OS on that computer. On the agent that runs native i5/OS jobs, set the following parameter in the agentparm.txt file:

```
oscomponent.targetenvironment=I5
```

On the agent that runs UNIX jobs, set the following parameter in the agentparm.txt file:

```
oscomponent.targetenvironment=UNIX
```

NOTE

- For more information about setting the `oscomponent.targetenvironment` parameter, see [Installation Considerations](#) in the Workload Automation Agents documentation.
- For more information about UNIX workload that can run in the PASE environment, see the IBM i5/OS documentation.

Verify that the Agent Works with AutoSys Workload Automation

To verify that the agent works with AutoSys Workload Automation, you define, run, and monitor a test job that runs on the agent. If the job completes successfully, your agent setup is successful.

To verify that the agent works with AutoSys Workload Automation, follow these steps:

1. (UNIX) Set up the environment.
2. Test communication between AutoSys Workload Automation and the agent.
3. Define a test job.
4. Run the test job.
5. Monitor the test job.

You have successfully configured AutoSys Workload Automation to work with agents; you can now run jobs on the agent.

Setting Up Security Permissions on AutoSys Workload Automation

You must set up the following security permissions on AutoSys Workload Automation to control agent access:

- Permission to run work on the agent—By defining the agent using the `insert_machine` subcommand and specifying that agent in the job definition.
- Permission to run a job on the agent under a user ID—By defining the user IDs and passwords using the `autosys_secure` command.
- Permission for the agent to control which AutoSys Workload Automation user IDs can perform FTP transfers or submit jobs under a specific agent user ID—The `security.txt` file contains the local security rules that allow or deny the AutoSys Workload Automation user IDs the authority to perform FTP transfers or submit jobs under a specific agent user ID.

NOTE

For more information about the `security.txt` file and setting up local security on the agent, see [Setting up Local Security on the Agent](#). The following agent local security rules work differently on AutoSys Workload Automation as described in that topic.

- The security rule that controls which scheduling manager user IDs can issue control commands and send messages to an agent:

```
c a | d manager_userID CONTROL command
```

- The security rule that controls which users are allowed to submit jobs on behalf of other users:

```
x a | d manager_userID agent_userID path
```

On AutoSys Workload Automation, jobs are always submitted to run under the user specified in the owner attribute. If local security is enabled on the agent, the agent checks the permissions of the job owner only. The agent does not check the AutoSys Workload Automation user who submits the job. Therefore, if local security is enabled on the agent, you can define security rules as follows:

```
x a | d job_owner agent_userID path
```


Modify the Encryption Type and Encryption Key on AutoSys Workload Automation

(UNIX) Modify the Encryption Type and Encryption Key

You can specify the encryption type and encryption key to be used for each agent during the agent installation. However, after you install the agent, you can modify the encryption type and the encryption key using the `encryption_type` and `key_to_agent` JIL attributes. On the agent, the encryption key is stored in the `cryptkey.txt` file, which is located in the agent installation directory. The `security.cryptkey` parameter in the `agentparm.txt` file specifies the path to the `cryptkey.txt` file.

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr stop waae_agent-WA_AGENT
```

– **WA_AGENT**

Defines the name of the agent to stop.

The agent stops.

3. Enter **jil** at the operating system prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
4. Enter the following commands:

```
update_machine: machine_name
agent_name: agent_name
encryption_type: NONE | DEFAULT | AES
key_to_agent: key
```

– **machine_name**

Specifies the name of the agent to update.

– **agent_name**

Specifies the name of an agent.

– **NONE | DEFAULT | AES**

Specifies the type of encryption to be used by the agent. You can set the encryption type to *one* of the following:

- **NONE**
Specifies that the agent uses no encryption.
- **DEFAULT**
Specifies that the agent uses the default encryption key and type. This is the default.
- **AES**
Specifies that the agent uses AES 128-bit encryption.

NOTE

You must specify a key using the `key_to_agent` attribute.

– **key**

Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. This value must match the `security.cryptkey` parameter in the agent's `agentparm.txt` file, without the prefix 0x. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. You must specify *one* of the following:

- A 32-digit hexadecimal key
- A passphrase with up to 16 characters

5. Enter exit.
The data is loaded into the database.

6. If you specify the encryption type as NONE, open the agentparm.txt file, set the security.cryptkey parameter to no value as follows, and save the file:

```
security.cryptkey=
```

7. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_agent-WA_AGENT
```

- – **WA_AGENT**
Defines the name of the agent to start.

The agent starts. The encryption key is modified on AutoSys Workload Automation.

Example: Define a User-specific Encryption Key on AutoSys Workload Automation

This example defines a user-specific encryption key on AutoSys Workload Automation. The encryption key you specify must match the encryption key specified in the cryptkey.txt file.

```
update_machine: machine3
agent_name: WA_MACH3
node_name: machine3
encryption_type: AES
key_to_agent: 0x000102030405060708090A0B0C0D0E0F1
```

(Windows) Modify the Encryption Type and Encryption Key

You can specify the encryption type and encryption key to be used for each agent during the agent installation. However, after you install the agent, you can modify the encryption type and the encryption key using the encryption_type and key_to_agent JIL attributes. On the agent, the encryption key is stored in the cryptkey.txt file, which is located in the agent installation directory. The security.cryptkey parameter in the agentparm.txt file specifies the path to the cryptkey.txt file.

Follow these steps:

1. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click Stop.
The agent stops.
2. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
3. Enter **jil** at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
4. Enter the following commands:

```
update_machine: machine_name
agent_name: agent_name
encryption_type: NONE | DEFAULT | AES
key_to_agent: key
```

- ***machine_name***
Specifies the name of the agent to update.
- ***agent_name***
Specifies the name of an agent.
- **NONE | DEFAULT | AES**
Specifies the type of encryption to be used by the agent. You can set the encryption type to *one* of the following:
 - **NONE**
Specifies that the agent uses no encryption.
 - **DEFAULT**
Specifies that the agent uses the default encryption key and type. This is the default.
 - **AES**
Specifies that the agent uses AES 128-bit encryption.

NOTE

You must specify a key using the `key_to_agent` attribute.

- ***key***
Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. This value must match the `security.cryptkey` parameter in the agent's `agentparm.txt` file, without the prefix `0x`. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. You must specify *one* of the following:
 - A 32-digit hexadecimal key
 - A passphrase with up to 16 characters
5. Enter `exit`
The data is loaded into the database.
 6. If you specify the encryption type as `NONE`, open the `agentparm.txt` file, set the `security.cryptkey` parameter to no value as follows, and save the file:

```
security.cryptkey=
```
 7. Do the following:
 - a. Click **Start, Programs, AutoSys, Administrator**.
The **Instance - AutoSys Workload Automation Administrator** window opens.
 - b. Select your AutoSys Workload Automation instance from the **Instance** drop-down list.
 - c. Click the **Services** icon on the toolbar.
The **Services - AutoSys Workload Automation Administrator** window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click **Start**.
The agent starts.

The encryption key is modified on AutoSys Workload Automation.

Example: Define a User-specific Encryption Key on AutoSys Workload Automation

This example defines a user-specific encryption key on AutoSys Workload Automation. The encryption key you specify must match the encryption key specified in the `cryptkey.txt` file.

```
update_machine: machine3
agent_name: WA_MACH3
node_name: machine3
encryption_type: AES
key_to_agent: 0x000102030405060708090A0B0C0D0E0F1
```

Configure Agent Parameters

(UNIX) Configure Agent Parameters

When you make a change to an agent parameter in the agentparm.txt file that is also defined on AutoSys Workload Automation, such as the agent name, you must configure the agent parameter on AutoSys Workload Automation.

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr stop waee_agent-WA_AGENT
```

– **WA_AGENT**

Defines the name of the agent to stop.

The agent stops.

3. Open the agentparm.txt file located in the agent installation directory.
4. Edit the parameters to make the required changes.
5. Save and close the agentparm.txt file.
6. Enter **jil** at the operating system prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
7. Enter the following commands to make the same changes on AutoSys Workload Automation:

```
update_machine: machine_name
agent_name: agent_name
port: port_number
max_load: load_units
factor: real_number
encryption_type: NONE | DEFAULT | AES
key_to_agent: key
heartbeat_attempts: number_of_signals
heartbeat_freq: minutes
```

– **machine_name**

Specifies the name of the agent to update.

– **agent_name**

(Optional) Specifies the name of an agent.

Default: WA_AGENT

– **port_number**

(Optional) Specifies the port that the agent uses to listen for traffic.

Default: 7520

– **load_units**

(Optional) Defines how many load units are allowed on the agent simultaneously. This number can be any value in the user-defined range of possible values. The range is also arbitrary.

– **real_number**

(Optional) Defines a real number from a user selected range of values.

Default: 1.0

– **NONE | DEFAULT | AES**

(Optional) Specifies the type of encryption to be used by the agent. You can set the encryption type to *one* of the following:

- **NONE**

Specifies that the agent uses no encryption.

- **DEFAULT**

Specifies that the agent uses the default encryption key and type. This is the default.

- **AES**

Specifies that the agent uses AES 128-bit encryption.

NOTE

You must specify a key using the `key_to_agent` attribute.

- **key**

(Optional) Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. This value must match the `security.cryptkey` parameter in the agent's `agentparm.txt` file, without the prefix `0x`. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. You must specify *one* of the following:

- A 32-digit hexadecimal key
- A passphrase with up to 16 characters

- **number_of_signals**

(Optional) Specifies the number of heartbeat signals the scheduler tries to detect before it sends an SNMP message indicating inactivity.

Default: 1

- **minutes**

(Optional) Specifies how frequently the scheduler sends the heartbeat signal (in minutes).

Default: 5

8. Enter exit.

The data is loaded into the database.

9. If you specify the encryption type as NONE, open the `agentparm.txt` file, set the `security.cryptkey` parameter to no value as follows, and save the file:

```
security.cryptkey=
```

10. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_agent-WA_AGENT
```

- **WA_AGENT**

Defines the name of the agent to start.

The agent starts. The agent parameters are configured.

Example: Configure the Agent Name on AutoSys Workload Automation

Suppose that you want to change the agent name to `WA_MACH2`. You must edit the `agentname` parameter in the `agentparm.txt` file to `WA_MACH2` and enter the following commands at the operating system prompt:

```
update_machine: machine1
agent_name: WA_MACH2
```

NOTE

You must stop and restart the agent for the changes to take effect.

(Windows) Configure Agent Parameters

When you make a change to an agent parameter in the `agentparm.txt` file that is also defined on AutoSys Workload Automation, such as the agent name, you must configure the agent parameter on AutoSys Workload Automation.

Follow these steps:

1. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click Stop.
The agent stops.
2. Open the agentparm.txt file located in the agent installation directory.
3. Edit the parameters to make the required changes.
4. Save and close the agentparm.txt file.
5. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
6. Enter **jil** at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
7. Enter the following commands to make the same changes on AutoSys Workload Automation:

```
update_machine: machine_name
agent_name: agent_name
port: port_number
max_load: load_units
factor: real_number
encryption_type: NONE | DEFAULT | AES
key_to_agent: key
heartbeat_attempts: number_of_signals
heartbeat_freq: minutes
```

- **machine_name**
Specifies the name of the agent to update.
- **agent_name**
(Optional) Specifies the name of an agent.
Default: WA_AGENT
- **port_number**
(Optional) Specifies the port that the agent uses to listen for traffic.
Default: 7520
- **load_units**
(Optional) Defines how many load units are allowed on the agent simultaneously. This number can be any value in the user-defined range of possible values. The range is also arbitrary.
- **real_number**
(Optional) Defines a real number from a user selected range of values.**Default:** 1.0
- **NONE | DEFAULT | AES**
(Optional) Specifies the type of encryption to be used by the agent. You can set the encryption type to *one* of the following:
 - **NONE**
Specifies that the agent uses no encryption.
 - **DEFAULT**
Specifies that the agent uses the default encryption key and type. This is the default.
 - **AES**

Specifies that the agent uses AES 128-bit encryption.

NOTE

You must specify a key using the `key_to_agent` attribute.

- **key** (Optional) Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. This value must match the `security.cryptkey` parameter in the agent's `agentparm.txt` file, without the prefix `0x`. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. You must specify *one* of the following:
 - A 32-digit hexadecimal key
 - A passphrase with up to 16 characters
 - **number_of_signals** (Optional) Specifies the number of heartbeat signals the scheduler tries to detect before it sends an SNMP message indicating inactivity.
Default: 1
 - **minutes** (Optional) Specifies how frequently the scheduler sends the heartbeat signal (in minutes).
Default: 5
8. Enter exit.
The data is loaded into the database.
 9. If you specify the encryption type as `NONE`, open the `agentparm.txt` file, set the `security.cryptkey` parameter to no value as follows, and save the file:


```
security.cryptkey=
```
 10. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click Start.
The agent starts.
The agent parameters are configured.

Example: Configure the Agent Name on AutoSys Workload Automation

Suppose that you want to change the agent name to `WA_MACH2`. You must edit the `agentname` parameter in the `agentparm.txt` file to `WA_MACH2` and enter the following commands at the instance command prompt:

```
update_machine: machine1
agent_name: WA_MACH2
```

NOTE

You must stop and restart the agent for the changes to take effect.

Configure the Agent to Communicate with AutoSys Workload Automation

(UNIX) Configure the Agent to Communicate with AutoSys Workload Automation

You can configure the agent to communicate with AutoSys Workload Automation by editing or adding the communication parameters in the `agentparm.txt` file.

NOTE

- You can configure the agent to work with multiple scheduling managers by adding additional definitions in the agentparm.txt file.
- You must add the following parameter to the agentparm.txt file to configure the agent to communicate with AutoSys Workload Automation that supports Internet Protocol version 6 (IPv6) and dual Internet Protocol (dual IP) environments:

```
java.net.preferIPv6Addresses=true
```

- On HP-UX, you must add the following parameter to the agentparm.txt file to enable IPv6 on Java.

```
java.net.preferIPv4Stack=false
```

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr stop waee_agent-WA_AGENT
```

– **WA_AGENT**

Defines the name of the agent to stop.

The agent stops.

3. Open the agentparm.txt file located in the agent installation directory.
4. Edit or add the communication parameters as appropriate, and save the file.
5. Enter the following command at the operating system prompt:

```
unisrvcntr start waee_agent-WA_AGENT
```

– **WA_AGENT**

Defines the name of the agent to start.

The agent starts, and is configured to communicate with AutoSys Workload Automation.

Example: Configure the Agent to Communicate with AutoSys Workload Automation

This example shows the configuration parameters that are set in the agentparm.txt file for the AutoSys Workload Automation instance “ACE” at address 172.31.255.255. AutoSys Workload Automation listens for incoming messages from the agent on port 7500:

```
communication.manageraddress_1=172.31.255.255
communication.managerid_1=ACE_SCH
communication.managerport_1=7500
communication.inputport=7520
communication.receiver.socket.main=plain
communication.socket_1=plain
communication.single_connection_attempts_1=1
communication.single_connection_hold_1=100
```

(Windows) Configure the Agent to Communicate with AutoSys Workload Automation

You can configure the agent to communicate with AutoSys Workload Automation by editing or adding the communication parameters in the agentparm.txt file.

NOTE

- You can configure the agent to work with multiple scheduling managers by adding additional definitions in the agentparm.txt file.
- You must add the following parameter to the agentparm.txt file to configure the agent to communicate with AutoSys Workload Automation that supports Internet Protocol version 6 (IPv6) and dual Internet Protocol (dual IP) environments:

```
java.net.preferIPv6Addresses=true
```

- On HP-UX, you must add the following parameter to the agentparm.txt file to enable IPv6 on Java.

```
java.net.preferIPv4Stack=false
```

Follow these steps:

1. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click Stop.
The agent stops.
2. Open the agentparm.txt file located in the agent installation directory.
3. Edit or add the communication parameters as appropriate, and save the file.
4. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the agent service, and click Start.
The agent starts.

The agent is configured to communicate with AutoSys Workload Automation.

Example: Configure the Agent to Communicate with AutoSys Workload Automation

This example shows the configuration parameters that are set in the agentparm.txt file for the AutoSys Workload Automation instance "ACE" at address 172.31.255.255. AutoSys Workload Automation listens for incoming messages from the agent on port 7500:

```
communication.manageraddress_1=172.31.255.255
communication.managerid_1=ACE_SCH
communication.managerport_1=7500
communication.inputport=7520
communication.receiver.socket.main=plain
communication.socket_1=plain
communication.single_connection_attempts_1=1
communication.single_connection_hold_1=100
```

Communication Parameters in the agentparm.txt File

When the scheduler communicates with the agent, the communication parameters are added to the agentparm.txt file.

The communication parameters are added to the agentparm.txt file in the following situations:

- When you issue the autoping command or run client utilities, such as the jil and autorep commands.
- When you start the scheduler or the application server.

NOTE

The agentparm.txt file includes the communication parameters corresponding to all the AutoSys Workload Automation instances that communicate with the agent.

The following communication parameters are added to the agentparm.txt file:

- **communication.manageraddress_1**

Defines the IP address or host name of AutoSys Workload Automation that the agent works with. You can specify a list of addresses for AutoSys Workload Automation.

Example: 172.24.36.107 (IPv4) or 0:0:0:0:FFFF:192.168.00.00 (IPv6)

NOTE

- The communication.manageraddress_1 value specified in the agentparm.txt file must match the AutoServer parameter value in the configuration file. For more information about the AutoServer parameter, see [Configure an Application Server](#).
- You can specify a DNS name instead of the IP address for AutoSys Workload Automation. However, your agent computer must be able to resolve the DNS name at all times. If there is a DNS outage and your agent computer cannot resolve DNS names, the agent cannot communicate with AutoSys Workload Automation.
- If the AutoSys Workload Automation IP address never changes, enter the DNS name for AutoSys Workload Automation in your agent computer's hosts file. This entry helps ensure that the IP address can be resolved after DNS disruptions.

- **communication.managerid_1**

Specifies the name of the AutoSys Workload Automation instance that the agent works with.

Default: *instance_SCH*

Example: ACE_SCH

NOTE

The communication.managerid_1 value specified in the agentparm.txt file must match the AutoServerId parameter value in the configuration file. For more information about the AutoServer parameter, see [Configure an Application Server](#).

- **communication.managerport_1**

Specifies the port that AutoSys Workload Automation listens on for communication from agents. The valid port range is 1024-65534.

NOTE

The communication.managerport_1 value specified in the agentparm.txt file must match the AutoServerPort or AppSrvAuxiliaryListeningPort parameter value in the configuration file.

- **communication.inputport**

(Optional) Specifies the main port number the agent uses to listen for incoming messages from AutoSys Workload Automation.

Default: 7520

Limits: 1024-65534

NOTE

On UNIX, ports 1 - 1023 are reserved ports and require root access.

- **communication.single_connection_attempts_1**

Specifies the number of times to check if the transmitter queue contains data to send.

- **communication.single_connection_hold_1**

Specifies the time (in milliseconds) to hold the connection between checks after the last message is sent.

- **communication.socket_1**

Defines the socket type the agent and AutoSys Workload Automation use for communication. The following socket types are available:

- plain
- dylan

Default: plain

Optional Communication Parameters

You can add the following optional communication parameters to the agentparm.txt file to configure the communication between AutoSys Workload Automation and the agent:

- **communication.inputport.aux**

(Optional) Specifies the auxiliary port number the agent uses to listen for incoming messages from AutoSys Workload Automation.

- **communication.receiver.socket.aux**

(Optional) Specifies the type of socket the agent uses for its auxiliary port. The value of this parameter must be different than the communication.receiver.socket.main parameter. You can specify the following socket types:

- plain
- dylan

- **communication.receiver.socket.main**

(Optional) Specifies the type of socket the agent uses for its main port. The value of this parameter must be different than the communication.receiver.socket.aux parameter. You can specify the following socket types:

- plain
- dylan

Default: plain

NOTE

If you are using the agent with two scheduling managers that require different socket types for communication, you can specify a main and auxiliary socket for the agent.

Configure the Agent to Communicate Using SSA Ports

NOTE

- This procedure only applies to agents that communicate with AutoSys Workload Automation using SSA ports.
- By default the agent uses plain socket ports for communication. Although you can change to SSA communication, we do not recommend it.

SSA lets AutoSys Workload Automation and the agent use a single multiplexed communication port to ease firewall administration.

To configure the agent to communicate using SSA ports, follow these steps:

1. Configure the agent to communicate using an SSA-enabled port on [UNIX](#) or [Windows](#).
2. Define the agent SSA port on AutoSys Workload Automation on [UNIX](#) or [Windows](#).
3. [Test communication between AutoSys Workload Automation and the agent.](#)

(UNIX) Configure the Agent to Communicate Using an SSA-Enabled Port

By default, the agent is configured to use plain socket ports for communication. You can use the `csamconfigedit` utility (installed with SSA) to enable SSA communication between the agent and AutoSys Workload Automation.

WARNING

The port number that is defined in the machine definition for the agent must match the `communication.inputport` parameter in the `agentparm.txt` file.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following command at the operating system prompt:

```
unisrvctr stop CA-WAAE
```

The scheduler, application server, web server, and the agent stop.

3. Enter the following command at the operating system prompt:

```
csampmux stop
```

The `csampmux` process stops.

4. Change to the `$CSAM_SOCKADAPTER/bin` directory, and enter the following command at the operating system prompt:

```
csamconfigedit Port=value EnableSSL=False EnablePmux=True
```

– value

Specifies the SSA port number of the agent. This port is not to be used by another application.

NOTE

The AutoSys Workload Automation installer automatically configures SSA to register 49154-50176 as virtual ports. These ports are known as the *ephemeral port range* and are used for short-term communications for application server, scheduler, and the agent.

5. Open the `agentparm.txt` file located in the agent installation directory.

6. Edit the following parameters, and save the file:

```
communication.inputport=port
```

```
communication.receiver.socket.main=dylan
```

```
oscomponent.classpath=jars/*.jar:jars/ext/*:common_components_installation_path/Csam/
SockAdapter/lib/casocket.jar
```

– port

Specifies the SSA port number configured using the `csamconfigedit` command.

– common_components_installation_path

Specifies the path to the directory where the CA common components are installed.

Default: `/opt/CA/SharedComponents`

NOTE

Append the location of the `casocket.jar` file to the classpath to specify the location of SSA.

7. Enter the following command at the operating system prompt:

```
unisrvctr start CA-WAAE
```

The scheduler, application server, web server, and the agent start. The agent is configured to communicate using an SSA-enabled port.

(Windows) Configure the Agent to Communicate Using an SSA-Enabled Port

By default, the agent is configured to use plain socket ports for communication. You can use the `csamconfigedit` utility (installed with SSA) to enable SSA communication between the agent and AutoSys Workload Automation.

WARNING

The port number that is defined in the machine definition for the agent must match the `communication.inputport` parameter in the `agentparm.txt` file.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Stop.
The scheduler, application server, and the agent stop.
2. Stop the CA Connection Broker service from the Windows Service Control Manager.
3. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt is displayed.
4. Change to the `%CSAM_SOCKADAPTER%\bin` folder, and enter the following command:


```
csamconfigedit Port=value EnableSSL=False EnablePmux=True
```

 - **value**
Specifies the SSA port number of the agent. This port is not to be used by another application.

NOTE

The AutoSys Workload Automation installer automatically configures SSA to register 49154-50176 as virtual ports. These ports are known as the *ephemeral port range* and are used for short-term communications for application server, scheduler, and the agent.

5. Start the CA Connection Broker service from the Windows Service Control Manager.
6. Open the `agentparm.txt` file located in the agent installation directory.
7. Edit the following parameters, and save the file:


```
communication.inputport=port
communication.receiver.socket.main=dylan
oscomponent.classpath=jars/*.jar;jars/ext/*;common_components_installation_path/Csam/
SockAdapter/bin/casocket.jar
```

 - **port**
Specifies the SSA port number configured using the `csamconfigedit` command.
 - **common_components_installation_path**
Specifies the path to the directory where the CA common components are installed.
Default: `C:\Program Files\CA\SC`

NOTE

Append the location of the `casocket.jar` file to the classpath to specify the location of SSA.

8. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Start.
The scheduler, application server, and the agent start.

The agent is configured to communicate using an SSA-enabled port.

(UNIX) Define the Agent SSA Port on AutoSys Workload Automation

To communicate with the agent using an SSA-enabled port, you must change the port defined in the machine definition for the agent on AutoSys Workload Automation.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
 2. Enter **jil** at the operating system prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
 3. Enter *one* of the following subcommands:
 - If you are creating an agent machine definition:
`insert_machine: machine_name`
 - If you are updating an existing agent machine definition:
`update_machine: machine_name`
 - **machine_name**
Defines a unique name for the agent. When defining jobs, specify this name in the machine attribute.
 4. Specify the following attribute:
`port: port_number`
 - **port_number**
Specifies the port that the agent uses to listen for traffic. If you configured SSA on the agent, this value is the agent port number configured using the csamconfigedit command. This value must match the communication.input port parameter in the agentparm.txt file for the agent.
- The SSA port is defined in the machine definition for the agent on AutoSys Workload Automation.

(Windows) Define the Agent SSA Port on AutoSys Workload Automation

To communicate with the agent using an SSA-enabled port, you must change the port defined in the machine definition for the agent on AutoSys Workload Automation.

Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt is displayed.
2. Enter **jil** at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Enter *one* of the following subcommands:
 - If you are creating an agent machine definition:
`insert_machine: machine_name`
 - If you are updating an existing agent machine definition:
`update_machine: machine_name`
 - **machine_name**
Defines a unique name for the agent. When defining jobs, specify this name in the machine attribute.
4. Specify the following attribute:
`port: port_number`
 - **port_number**

Specifies the port that the agent uses to listen for traffic. If you configured SSA on the agent, this value is the agent port number configured using the `csamconfigedit` command. This value must match the `communication.input port` parameter in the `agentparm.txt` file for the agent.

The SSA port is defined in the machine definition for the agent on AutoSys Workload Automation.

Configure AutoSys Workload Automation to Work with Workload Automation Agent for z/OS

After you install the agent on z/OS, you must perform additional configuration tasks on AutoSys Workload Automation and the agent so that they can communicate with each other. After you perform these tasks, you can define and run jobs on the mainframe.

AGENTDEF Data Set

You can configure the agent on z/OS by editing the parameters in the AGENTDEF data set. The parameter values in the AGENTDEF data set are set during the agent installation. You can modify these parameter values as required.

NOTE

- For information about the parameters in the AGENTDEF data set and how to configure them to work with the scheduling manager, see the *Workload Automation Agent for z/OS Installation and Configuration documentation*.
- When you make a change to an agent parameter that is also defined on AutoSys Workload Automation, such as the agent name, you must configure the corresponding parameter on AutoSys Workload Automation. Similarly, when you configure an agent parameter on AutoSys Workload Automation, the AGENTDEF data set must be updated to include the change.

Encryption Between AutoSys Workload Automation and the Agent on z/OS

Depending on the encryption types that the agent on z/OS supports, data can be transferred between AutoSys Workload Automation and the agent with no encryption or with AES 128-bit encryption. The encryption settings on AutoSys Workload Automation and the agent must match.

Encryption occurs in two ways:

- The data received from the agent
- The data sent to the agent

Encryption of Data Received from the Agent on z/OS

The encryption setting for AutoSys Workload Automation is determined as follows:

- On UNIX—By the `UseCommAliasEncryption` parameter in the configuration file.
- On Windows—By the `Use AES 128-bit encryption when communicating with zOS managers` check box under the `zOS Encryption` tab on the `Instance - AutoSys Workload Automation Administrator` window of AutoSys Workload Automation Administrator.

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with the agent on z/OS. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (`cryptkey_alias.txt`) in the `$AUTOUSER.instance_name` (on UNIX) or `%AUTOUSER%.instance_name` (on Windows) directory. The `cryptkey_alias.txt` file stores the communication alias encryption key. AutoSys Workload Automation expects the agent on z/OS to encrypt the data using the key specified in the `cryptkey_alias.txt` file.

If you are using no encryption, AutoSys Workload Automation expects the data it receives from the agent on z/OS to be unencrypted.

WARNING

You must set AES encryption only if AES encryption is also configured on the agent on z/OS. For more information about the encryption types that the agent on z/OS supports, see the Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, you must disable AES 128-bit encryption and use no encryption for z/OS communication.
- If AutoSys Workload Automation works with other agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting.

Example: Using No Encryption When Receiving Data from the Agent on z/OS

Suppose that you do not want the data received from the agent on z/OS to be encrypted. To use no encryption, you must set the UseCommAliasEncryption parameter to 0 in the configuration file (on UNIX) or clear the Use AES 128-bit encryption when communicating with zOS managers check box under the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (on Windows).

When you define your AutoSys Workload Automation instance to the agent, you must specify the NOENCRYPT operand, as follows:

```
COMMCHAN instancename_AGT ADDRESS(address) PORT(sched_aux_port) -
UNIX ASCII TCPIP PREF(2) NOENCRYPT
EXTSCHED AUTO SAFUSER(user) EVENTPREFIX(prefix) -
APPLPREFIX(ZOS) MAXACTIVE(3) ID(agent_name)
```

Encryption of Data Sent to the Agent on z/OS

The encryption setting for the agent on z/OS is determined by the ZOSAGENT initialization parameter in the AGENTDEF data set, as shown in the following example:

```
ZOSAGENT NAME(manager_name) TCPIP
```

In this example, the ENCRYPT operand is excluded from the parameter, so no encryption is used. The agent expects the data it receives from AutoSys Workload Automation to be unencrypted.

If encryption is specified in the ZOSAGENT parameter, the agent expects AutoSys Workload Automation to encrypt the data.

WARNING

On AutoSys Workload Automation, you must specify the same agent encryption setting using the encryption_type and key_to_agent attributes. Therefore, to use encryption, the agent on z/OS and AutoSys Workload Automation must support the same encryption type. For more information about the encryption types that the agent on z/OS supports, see the Workload Automation Agent for z/OS documentation.

Example: Using No Encryption to Send Data to the Agent on z/OS

Suppose that agent on z/OS does not need the data transferred to be encrypted. To use no encryption, the ENCRYPT operand is excluded from the ZOSAGENT initialization parameter of the AGENTDEF data set, as follows:

```
ZOSAGENT NAME(manager_name) TCPIP
```


When you define the agent on z/OS to AutoSys Workload Automation, you must specify the `encryption_type`: NONE attribute, as follows:

```
insert_machine: machine_name
type: a
opsys: zos
node_name: address
agent_name: agent_name
port: port_number
encryption_type : NONE
```

Configure AutoSys Workload Automation to Work with the Agent on z/OS

Workload Automation Agent for z/OS submits and tracks z/OS jobs.

To configure AutoSys Workload Automation to work with the agent on z/OS, follow these steps:

1. Configure the scheduler and application server auxiliary listening ports on [UNIX](#) or [Windows](#).
2. Define a unique communication alias for the application server on [UNIX](#) or [Windows](#).
3. Set encryption for z/OS communication on [UNIX](#) or [Windows](#).
4. (AES 128-bit encryption only) [Generate an instance-wide communication alias encryption file](#).
5. Configure the AGENTDEF data set on the agent on z/OS on [UNIX](#) or [Windows](#).
6. [Define the agent on z/OS on AutoSys Workload Automation](#).
7. [Verify that the agent on z/OS works with AutoSys Workload Automation](#).

(UNIX) Configure the Scheduler and Application Server Auxiliary Listening Ports

The scheduler and the application server communicate with CA Workload Automation EE and Workload Automation Agent for z/OS using non-SSA ports. Therefore, you must disable port multiplexing and SSL encryption for the scheduler and application server auxiliary listening ports.

NOTE

If the ports are already configured, skip this procedure.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Edit the following parameters in the configuration file as follows, and save the file:

```
SchedAuxiliaryListeningPort=sch_port
AppSrvAuxiliaryListeningPort=appsrv_port
```

– **sch_port**

Defines the port number the scheduler uses to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.

– **appsrv_port**

Defines the port number the application server uses to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.

The scheduler and application server auxiliary listening ports are defined.

3. Enter the following commands at the operating system prompt:

```
cd $CSAM_SOCKADAPTER/bin
```

```
csamconfigedit Port=sch_port EnableSSL=False EnablePmux=False
```

– ***sch_port***

Specifies the port number to configure. You must specify the same scheduler auxiliary listening port that you specified in the SchedAuxiliaryListeningPort parameter in the configuration file.

Port multiplexing and SSL encryption are disabled for the specified scheduler auxiliary listening port.

4. Enter the following command at the operating system prompt:

```
csamconfigedit Port=appsrv_port EnableSSL=False EnablePmux=False
```

– ***appsrv_port***

Specifies the port number to configure. You must specify the same application server auxiliary listening port that you specified in the AppSrvAuxiliaryListeningPort parameter in the configuration file.

Port multiplexing and SSL encryption are disabled for the specified application server auxiliary listening port.

5. Enter the following commands at the operating system prompt:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
```

The scheduler and the application server stop.

6. Enter the following command at the operating system prompt:

```
csampmux stop
```

The csampmux process stops.

7. Enter the following commands at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. The scheduler and application server auxiliary listening ports are configured.

NOTE

For more information about the SchedAuxiliaryListeningPort and AppSrvAuxiliaryListeningPort parameters, see [Configure a Scheduler](#).

(Windows) Configure the Scheduler and Application Server Auxiliary Listening Ports

The scheduler and the application server communicate with CA Workload Automation EE and Workload Automation Agent for z/OS using non-SSA ports. Therefore, you must disable port multiplexing and SSL encryption for the scheduler and application server auxiliary listening ports.

NOTE

If the ports are already configured, skip this procedure.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Click the Scheduler icon on the toolbar.
The Scheduler - AutoSys Workload Automation Administrator window appears.
3. Enter the port number in the Auxiliary Listening Port field in the Communication Ports pane, and click Apply.
The scheduler auxiliary listening port is defined. The scheduler uses this port to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.

4. Click the Application Server icon on the toolbar.
The Application Server - AutoSys Workload Automation Administrator window appears.
5. Enter the port number in the Communication Alias Identifier field, and click Apply.
The application server auxiliary listening port is defined. The application server uses this port to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.
6. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
7. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:

```
csamconfigedit Port=sch_port EnableSSL=False EnablePmux=False
```

 - **sch_port** Specifies the port number to configure. You must specify the same scheduler auxiliary listening port that you specified in the Auxiliary Listening Port field on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
Port multiplexing and SSL encryption are disabled for the specified scheduler auxiliary listening port.
8. Enter the following command:

```
csamconfigedit Port=appsrv_port EnableSSL=False EnablePmux=False
```

 - **appsrv_port** Specifies the port number to configure. You must specify the same application server auxiliary listening port that you specified in the Communication Alias Identifier field on the Application Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
Port multiplexing and SSL encryption are disabled for the specified application server auxiliary listening port.
9. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select an instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler service, and click Stop.
The scheduler stops.
 - e. Right-click the application server service, and click Stop.
The application server stops.
10. Stop the CA Connection Broker service from the Windows Service Control Manager.
11. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select an instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler service, and click Start.
The scheduler starts.
 - e. Right-click the application server service, and click Start.
The application server starts.

The scheduler and application server auxiliary listening ports are configured.

NOTE

For more information about the AutoSys Workload Automation Administrator, see [Configuring AutoSys Workload Automation](#).

(UNIX) Define a Unique Communication Alias for an Application Server

The application server requires an additional communication alias to communicate with Workload Automation Agent for z/OS. The communication alias is set to *INSTANCENAME_ABBREVIATEDHOSTNAME* during the AutoSys Workload Automation installation.

If the AutoSys Workload Automation instance has multiple application servers, the communication alias for each application server must be unique. If an alias is not unique, you must define another alias for that application server.

NOTE

The scheduler also requires a communication alias to communicate with the agent on z/OS. However, the communication alias for the scheduler is automatically set to *INSTANCENAME_AGT* (in uppercase). You cannot change this value.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr stop waae_server.$AUTOSERV
```

The application server stops.

3. Edit the following parameter in the configuration file as follows, and save the file:

```
AutoServerAliasId=unique_alias
```

– *unique_alias*

Defines a unique communication alias that the application server uses to communicate with the agent on z/OS.

Default: *INSTANCENAME_ABBREVIATEDHOSTNAME*. The abbreviated hostname consists of the last 12 characters of the node name excluding the domain name. For example, the communication alias of the application server on myhost.ca.com is set to ACE_MYHOST, where ACE is the name of the AutoSys Workload Automation instance.

Limits: Up to 16 uppercase characters

NOTE

If you specify the value in lowercase or mixed case, the value is automatically changed to uppercase.

4. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_server.$AUTOSERV
```

The application server starts. The unique communication alias is defined for the application server. The application server uses this alias to communicate with the agent on z/OS.

(Windows) Define a Unique Communication Alias for an Application Server

The application server requires an additional communication alias to communicate with CA Workload Automation EE and Workload Automation Agent for z/OS. The communication alias is set to *INSTANCENAME_ABBREVIATEDHOSTNAME* during the AutoSys Workload Automation installation.

If the AutoSys Workload Automation instance has multiple application servers, the communication alias for each application server must be unique. If an alias is not unique, you must define another alias for that application server.

The default value is *INSTANCENAME_ABBREVIATEDHOSTNAME*. The abbreviated hostname consists of the last 12 characters of the node name excluding the domain name. For example, the communication alias of the application server on myhost.ca.com is set to ACE_MYHOST, where ACE is the name of the AutoSys Workload Automation instance.

NOTE

The scheduler also requires a communication alias to communicate with CA Workload Automation EE and the agent on z/OS. However, the communication alias for the scheduler is automatically set to *INSTANCENAME_AGT* (in uppercase). You cannot change this value.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list.
3. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the application server service, and click Stop.
The application server stops.
5. Click the Application Server icon on the toolbar.
The Application Server - AutoSys Workload Automation Administrator window appears.
6. Enter a unique communication alias in the Communication Alias Identifier field, and click Apply.

NOTE

You can enter a maximum of 16 characters (in uppercase). If you specify this value in lowercase or mixed case, it is automatically changed to uppercase.

7. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
8. Right-click the application server service, and click Start.
The application server starts. The unique communication alias is defined for the application server. The application server uses this alias to communicate with CA Workload Automation EE and the agent on z/OS.

(UNIX) Set Encryption for z/OS Communication

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (*cryptkey_alias.txt*) in the *\$AUTOUSER.instance_name* (on UNIX) or *%AUTOUSER%.instance_name* (on Windows) directory. The *cryptkey_alias.txt* file stores the communication alias encryption key.

WARNING

You must set AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, you must disable AES 128-bit encryption and use no encryption for z/OS communication. To disable AES encryption, you must set the *UseCommAliasEncryption* parameter to 0 in the configuration file.
- If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting.

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following commands at the operating system prompt:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
```

The scheduler and the application server stop.

3. Edit the following parameter in the configuration file, and save the file:

```
UseCommAliasEncryption=0|2
```

- **0**
Specifies that no encryption is used.
- **2**
Specifies that AES encryption is used to encrypt data.

NOTE

If you set the UseCommAliasEncryption parameter to 2, you must generate the cryptkey_alias.txt file and specify the communication alias encryption key using the -a option of the as_config command.

4. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. The encryption for z/OS communication is set.

NOTE

On Windows, you can select the equivalent value using the Use AES 128-bit encryption when communicating with zOS managers check box under the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator. For information about setting encryption for z/OS communication on Windows, see [Configure an Instance](#).

(Windows) Set Encryption for z/OS Communication

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory. The cryptkey_alias.txt file stores the communication alias encryption key.

WARNING

You must set AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, you must disable AES 128-bit encryption and use no encryption for z/OS communication. To disable AES encryption, you must clear the Use AES 128-bit encryption when communicating with zOS managers check box.
- If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select the instance you want to set encryption for from the Instance drop-down list.

3. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the scheduler and select Stop. Repeat this action for the application server.
The scheduler and application server stop.
5. Click the Instance icon on the toolbar.
The Instance - AutoSys Workload Automation Administrator window appears.
6. Click the zOS Encryption tab, and select the Use AES 128-bit encryption when communicating with zOS managers check box.
The Hexadecimal Key and Verify Hexadecimal Key fields are enabled.

NOTE

If you clear the Use AES 128-bit encryption when communicating with zOS managers check box, AutoSys Workload Automation uses no encryption for z/OS communication.

7. Enter the encryption key, confirm the encryption key in the Verify Hexadecimal Key field, and click Apply.

NOTE

The encryption key must be a hexadecimal string of 32 characters

The zOS Encryption Status field displays the current encryption set for the instance.

8. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
9. Right-click the scheduler and select Start. Repeat this action for the application server.
The scheduler and application server start. The encryption for z/OS communication is set.

Generate an Instance-Wide Communication Alias Encryption File

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate the instance-wide communication alias encryption file (cryptkey_alias.txt).

The cryptkey_alias.txt file stores the communication alias encryption key. The cryptkey_alias.txt file is located in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory.

WARNING

Do this procedure only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- A AutoSys Workload Automation instance can have only one cryptkey_alias.txt file. Before you do this procedure, check whether the file already exists. If the file exists, skip this procedure. You must provide the key associated with that file to the CA Workload Automation EE or agent administrator. They need the key to configure the AGENTDEF data set.
- If you do not know the key associated with the existing cryptkey_alias.txt, you can regenerate the file using a new key. If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the new key.

Follow these steps:

1. Do *one* of the following:

- (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
 - (UNIX) Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the instance command prompt (Windows) or the operating system prompt (UNIX):

```
as_config -a key
```

- **key**
Specifies the communication alias encryption key. You must prefix the hexadecimal identifier 0x to this value.
Limits: Must contain 32 characters; valid characters are 0-9 and A-F.

NOTE

This key must match the key stored in the ENCRYPT KEYNAME(keyname) parameter in the AGENTDEF data set of CA Workload Automation EE or the agent on z/OS.

The communication alias encryption file (cryptkey_alias.txt) is generated with the encryption key. AES 128-bit encryption is used.

(UNIX) Configure the AGENTDEF Data Set on the Agent on z/OS

For communication to occur between AutoSys Workload Automation and the agent on z/OS, you must configure the AGENTDEF data set on the agent on z/OS. The parameters in the AGENTDEF data set must match the settings defined on AutoSys Workload Automation.

To configure the AGENTDEF data set, add the following entries:

```
COMMCHAN INSTANCENAME_AGT ADDRESS(sch_ip_address) PORT(sch_aux_port) platform +
ASCII TCPIP PREF(2) encryption_setting
```

```
COMMCHAN AutoServerAliasId ADDRESS(appsrv_ip_address) PORT(appsrv_aux_port) +
platform ASCII TCPIP PREF(2) encryption_setting
```

The following table describes the operands that are not self-explanatory and their corresponding AutoSys Workload Automation settings:

AGENTDEF Operand	Description	Corresponding AutoSys Workload Automation Setting
COMMCHAN INSTANCENAME_AGT	Specifies the name associated with the encryption data between AutoSys Workload Automation and the agent on z/OS.	INSTANCENAME_AGT This is the communication alias for the AutoSys Workload Automation scheduler. The value must be in uppercase. INSTANCENAME is the name of the AutoSys Workload Automation instance.
COMMCHAN AutoServerAliasId	Specifies the name associated with the encryption data between AutoSys Workload Automation and the agent on z/OS.	AutoServerAliasId parameter in the configuration file.
ADDRESS(sch_ip_address)	Specifies the host name or the IP address of the computer where the AutoSys Workload Automation scheduler is installed.	None
ADDRESS(appsrv_ip_address)	Specifies the host name or the IP address of the computer where the AutoSys Workload Automation application server is installed.	None

<code>PORT(sch_aux_port)</code>	Specifies the port number that the AutoSys Workload Automation scheduler uses for all non-SSA communication.	SchedAuxiliaryListeningPort parameter in the configuration file.
<code>PORT(appsrv_aux_port)</code>	Specifies the port number that the AutoSys Workload Automation application server uses for all non-SSA communication.	AppSrvAuxiliaryListeningPort parameter in the configuration file.
<code>platform</code>	Specifies whether the AutoSys Workload Automation instance is installed on UNIX or Windows. Options are UNIX or NT.	None
<code>encryption_setting</code>	Specifies the type of encryption used.	If encryption is <i>not</i> configured on the agent, <code>encryption_type: NONE</code> must be defined in the machine definition, and the <code>cryptkey_alias.txt</code> file must not exist. If encryption is configured on the agent, <code>encryption_type: AES</code> must be defined in the machine definition. The same key must be stored in the <code>cryptkey_alias.txt</code> file in the <code>\$AUTOUSER.instance_name</code> directory. Note: To use encryption, both the agent on z/OS and AutoSys Workload Automation must support AES encryption. For more information about the encryption types that the agent supports and how to specify the encryption setting, see the Workload Automation Agent for z/OS documentation.

NOTE

For more information about the AGENTDEF data set on the agent on z/OS, see the *Workload Automation Agent for z/OS Installation and Configuration documentation*.

(Windows) Configure the AGENTDEF Data Set on the Agent on z/OS

For communication to occur between AutoSys Workload Automation and the agent on z/OS, you must configure the AGENTDEF data set on the agent on z/OS. The parameters in the AGENTDEF data set must match the settings defined on AutoSys Workload Automation.

To configure the AGENTDEF data set, add the following entries:

```
COMMCHAN INSTANCENAME_AGT ADDRESS(sch_ip_address) PORT(sch_aux_port) platform +
ASCII TCPIP PREF(2) encryption_setting
```

```
COMMCHAN AutoServerAliasId ADDRESS(appsrv_ip_address) PORT(appsrv_aux_port) +
platform ASCII TCPIP PREF(2) encryption_setting
```

The following table describes the operands that are not self-explanatory and their corresponding AutoSys Workload Automation settings:

AGENTDEF Operand	Description	Corresponding AutoSys Workload Automation Setting
COMMCHAN <i>INSTANCENAME_AGT</i>	Specifies the name associated with the encryption data between AutoSys Workload Automation and the agent on z/OS.	<i>INSTANCENAME_AGT</i> This is the communication alias for the AutoSys Workload Automation scheduler. The value must be in uppercase. <i>INSTANCENAME</i> is the name of the AutoSys Workload Automation instance.
COMMCHAN <i>AutoServerAlias/d</i>	Specifies the name associated with the encryption data between AutoSys Workload Automation and the agent on z/OS.	Communication Alias Identifier field on the Application Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
ADDRESS(<i>sch_ip_address</i>)	Specifies the host name or the IP address of the computer where the AutoSys Workload Automation scheduler is installed.	None
ADDRESS(<i>appsrv_ip_address</i>)	Specifies the host name or the IP address of the computer where the AutoSys Workload Automation application server is installed.	None
PORT(<i>sch_aux_port</i>)	Specifies the port number that the AutoSys Workload Automation scheduler uses for all non-SSA communication.	Auxiliary Listening Port field on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
PORT(<i>appsrv_aux_port</i>)	Specifies the port number that the AutoSys Workload Automation application server uses for all non-SSA communication.	Auxiliary Listening Port field on the Application Server - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
<i>platform</i>	Specifies whether the AutoSys Workload Automation instance is installed on UNIX or Windows. Options are UNIX or NT.	None
<i>encryption_setting</i>	Specifies the type of encryption used.	If encryption is <i>not</i> configured on the agent, encryption_type: NONE must be defined in the machine definition, and the cryptkey_alias.txt file must not exist. If encryption is configured on the agent, encryption_type: AES must be defined in the machine definition. The same key must be stored in the cryptkey_alias.txt file in the \$AUTOUSER. <i>instance_name</i> directory. Note: To use encryption, both the agent on z/OS and AutoSys Workload Automation must support AES encryption. For more information about the encryption types that the agent supports and how to specify the encryption setting, see the Workload Automation Agent for z/OS documentation.

NOTE

For more information about the AGENTDEF data set on the agent on z/OS, see the Workload Automation Agent for z/OS Installation and Configuration *documentation*.

Define the Agent on z/OS on AutoSys Workload Automation

You must define the agent on z/OS on AutoSys Workload Automation to enable communication between the agent and the server.

You must ensure that the parameters you specify when you define the agent on z/OS on AutoSys Workload Automation match the corresponding parameters in the AGENTDEF data set.

Follow these steps:

1. Do *one* of the following:
 - Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
 - Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter **jil** at the instance command prompt (Windows) or the operating system prompt (UNIX):. The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Enter the following commands:

```
insert_machine: machine_name
type: a
node_name: address
agent_name: agent_name
port: port_number
encryption_type: NONE | AES
```

– **machine_name**

Defines a unique name for the agent on z/OS. When defining jobs, specify this name in the machine attribute.

– **a**

Specifies that the machine is a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS.

– **address**

(Optional) Defines the IP address or DNS name of the computer where the agent on z/OS is installed.

Default: The value specified in the insert_machine: *machine_name* command.

NOTE

If you do not specify the node_name attribute, insert_machine: *machine_name* (the default) must be the DNS name of the agent machine. Otherwise, AutoSys Workload Automation cannot connect to the agent on z/OS.

– **agent_name**

(Optional) Specifies the name of the agent on z/OS.

Default: WA_AGENT

NOTE

This name must match the ZOSAGENT(name) parameter in the AGENTDEF data set (on Workload Automation Agent for z/OS).

– **port_number**

(Optional) Specifies the port that the agent on z/OS uses to listen for traffic.

Default: 7520

NOTE

This port number must match the COMMCHAN PORT(port) parameter in the AGENTDEF data set (on Workload Automation Agent for z/OS).

– **NONE | AES**

(Optional) Specifies the type of encryption to be used by the agent on z/OS. You can set the encryption type to *one* of the following:

- **NONE**
Specifies that the agent on z/OS uses no encryption.
- **AES**
Specifies that the agent on z/OS uses AES 128-bit encryption.

NOTE

You must generate the instance-wide communication alias encryption file (cryptkey_alias.txt) using the as-config command.

NOTE

To use encryption, both the agent on z/OS and AutoSys Workload Automation must support AES encryption. You must specify the same AutoSys Workload Automation encryption setting in the AGENTDEF data set of the agent. For more information about the encryption types that the agent supports, see the Workload Automation Agent for z/OS documentation.

4. (Optional) Specify optional machine attributes:

- character_code
- description
- opsys
- max_load
- factor
- heartbeat_attempts
- heartbeat_freq
- Enter exit.

The data is loaded into the database. The agent on z/OS is defined on AutoSys Workload Automation.

NOTE

For more information about the AGENTDEF data set on the agent on z/OS, see the Workload Automation Agent for z/OS Installation and Configuration *documentation*.

Example: Define the agent on z/OS on AutoSys Workload Automation

This example defines the agent on z/OS on AutoSys Workload Automation.

```
insert_machine: zagentl13
type: a
opsys: zos
port: 7520
encryption_type: NONE
character_code: EBCDIC
```

How to Verify the Agent on z/OS Works With AutoSys Workload Automation

You can verify communication between the agent on z/OS and AutoSys Workload Automation by defining, running, and monitoring a test job.

To verify the agent on z/OS works with AutoSys Workload Automation, follow these steps:

1. [Test communication between AutoSys Workload Automation and the agent.](#)
2. [Define a z/OS job.](#)
3. [Run the test job.](#)
4. [Monitor the test job.](#)

Define a z/OS Job

You can define a z/OS job to test communication between AutoSys Workload Automation and the agent on z/OS.

Follow these steps:

1. Do *one* of the following:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
2. Enter **jil** at the operating system prompt (UNIX) or instance command prompt (Windows).
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Enter the following commands:


```
insert_job: job_name
job_type: ZOS
machine: machine_name
jcl_library: library
jcl_member: member
owner: user@host
```

A z/OS job is defined. The following message appears:

```
CAUAJM_I_50323 Inserting/Updating job: job_name
CAUAJM_I_50205 Database Change WAS Successful!
```

4. Enter **exit**.
The data is loaded into the database.

Configure AutoSys Workload Automation to Work with Other Broadcom Products

You can configure AutoSys Workload Automation to integrate with other Broadcom products by editing the following integration parameters in the configuration file (UNIX) or the equivalent fields on the Integration - AutoSys Workload Automation Administrator window of the Administrator utility (Windows):

- OneAutomationEvents, EventType, RestURL, AuthorizationKey—Configure these parameters to [integrate AutoSys Workload Automation with Automic Automation](#).
- ServiceDeskURL, ServiceDeskUser, and ServiceDeskCust—Configure these parameters to [integrate AutoSys Workload Automation with CA Service Desk](#).
- NotifyMethod, NotifySMTPHost, UseSMTPAuthentication, NotifySMTPUser, and NotifySMTPFromAddress—Configure these parameters to [send email notifications](#).
- [SnmpManagerHosts and SnmpCommunity](#)

CA Embedded Entitlements Manager (CA EEM)

AutoSys Workload Automation includes features that let you secure objects such as jobs, calendars, cycles, global variables, machines, and resources. You can delegate administrative privileges to these objects to specific users or user groups. AutoSys Workload Automation provides security in the following ways:

- System-level security
- Native security
- External security

External security is the default mode under which AutoSys Workload Automation runs. The external security mode is robust and provides better flexibility than the native security mode.

When AutoSys Workload Automation is running in external security mode, CA EEM is used to assign administrative rights to a user to define policies and to check whether a given user can switch the security mode of AutoSys Workload Automation back to native. CA EEM lets you manage your user base, create roles for your enterprise, and assign roles to users. It also maintains security policies that govern what objects can be accessed by which users.

NOTE

While external security mode is enabled, native security is not enforced. For more information about system-level security, native security, or external security, see [Securing](#).

Integrate with Automic Automation

You can integrate AutoSys Workload Automation with Automic Automation to automate manual problem resolution tasks, filter, monitor for unusual conditions, and take proper corrective action.

The Automic Automation Event Engine lets you send events using the REST API. You can filter or process the content of the events using rules. The rules map events to the executable objects and their variables or promptsets. You can identify events that are important to your organization and define rules and actions that specify the processing that Automic Automation performs when the events occur. For example, you can specify which AutoSys Workload Automation object you want to trigger if an event passes the rules successfully.

NOTE

- To access the Automic Automation Event Engine, contact [Broadcom Support](#).
- For information about API Keys, see the Analytics REST API Key Management section in the Automic Administration Guide.

Messages

For Automic Automation to receive messages from the AutoSys Workload Automation server, import the VARA.IA.CA_WA_AE.XML file in the Automic Automation. The VARA.IA.CA_WA_AE.XML file is located in the opt/CA/WorkloadAutomationAE/autosys/install/Integrations/OneAutomation (UNIX) or CA\AutoSys Workload Automation\autosys\install\Integrations\OneAutomation (Windows) directory.

The AutoSys Workload Automation messages are in the following format: *source_severity_messageid message*. For example, CAUAJM_I_10116 Localhost machine definition has been successfully set to ae-auxfs1/autosys.

The VARA.IA.CA_WA_AE.XML file contains the following objects:

- **application**
Specifies the AutoSys Workload Automation instance name in the WAAE_Autosys,*instance_name* format. For example, WAAE_AutoSys,ORA.
- **user**
Specifies the user name under which the AutoSys Workload Automation scheduler runs in the *system_name/user_name* format. For example, auxfs1/autosys, where auxfs1 is the system name and autosys is the user name.
- **machine**
Specifies the agent machine name.
- **box_name**
Specifies the name of the box that is associated with the job.
- **eoid**
Specifies the object identifier of the event.
- **messageid** Specifies the message identifier of the event. For example, 10116
- **message**

Specifies the message that is associated with the event. For example, Localhost machine definition has been successfully set to ae-auxfs1/autosys.

- **severity**
Specifies the severity level of the event. The severity level can be I (informational), W (warning), or E (error).
- **status**
Specifies the status of the job.
- **event**
Specifies the type of the event.
- **joid**
Specifies the job identifier.
- **job_name**
Specifies the name of the job.
- **ntry**
Specifies the number of job retries.
- **run_num**
Specifies the job run number.
- **source**
Specifies the name of the product, that is, CAUAJM.

Integrate AutoSys Workload Automation With Automic Automation

To integrate AutoSys Workload Automation with Automic Automation, follow these steps:

1. Install Automic Automation.

NOTE

For information about installing Automic Automation Platform, see the [Automic documentation](#).

2. Import the VARA.IA.CA_WA_AE.XML file in Automic Automation Platform.
3. Configure message forwarding on [UNIX](#) or [Windows](#).

(UNIX) Configure Message Forwarding

After installing Automic Automation Platform, you must configure the AutoSys Workload Automation server to activate its message forwarding interface so that messages are forwarded to Automic Automation Platform.

Follow these steps:

1. Log in to AutoSys Workload Automation as a user authorized to stop the scheduler and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt to stop the scheduler:


```
sendevent -E STOP_DEMON
```
3. Edit the following parameters in the configuration file, and save the file:


```
OneAutomationEvents=1|0
EventType=VARA.IA.CA_WA_AE
AuthorizationKey=authorization_key
RestURL=oneautomation_restURL
```

 - **1**
Specifies that the AutoSys Workload Automation messages are forwarded to the Automic Automation Platform. Any messages written to the scheduler log also appear in the Automic Automation Platform.
 - **0**

Specifies that the AutoSys Workload Automation messages are not forwarded to the Automic Automation Platform. This is the default.

– **VARA.IA.CA_WA_AE**

Specifies the type of event that is defined in the Automic Automation Platform.

- **authorization_key** Specifies the key to authorize the post request that is sent to the URL. This is the client-specific API key to publish events to Automic Automation Platform.

NOTE

- On UNIX, enter the encrypted authorization key in the configuration file. For information about encrypting a key, see the Encrypt a Password section in the [AutoSys Workload Automation Data Encryption](#) page.
- For information about how to generate client-specific API keys, see the Analytics REST API Key Management section in the CA Automic Administration Guide.

– **oneautomation_restURL**

Specifies the URL of the Automic Automation Platform server.

Example: `http://automic_machine_hostname:8090/analytics/api/v1/events/`

4. Enter the following command at the operating system prompt to start the scheduler:

```
eventor
```

The scheduler starts. Any messages that are written to the scheduler log now also appear in the Automic Automation. After AutoSys Workload Automation starts publishing events to Automic Automation, you can define Rule Events to filter the messages and to perform actions. For more information, see the [Automic documentation](#).

(Windows) Configure Message Forwarding

After installing Automic Automation, you must configure the AutoSys Workload Automation server to activate its message forwarding interface so that messages are forwarded to Automic Automation.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
4. Right-click the scheduler and select Stop.
5. Click the Integration icon on the toolbar.
The Integration - AutoSys Workload Automation Administrator window appears.
6. Click the Messaging tab and select the Forward All AutoSys Scheduler Log Messages to OneAutomation check box.
7. Enter **VARA.IA.CA_WA_AE** in the Event Type field.
8. Enter the authorization key in the Authorization Key field.
9. Enter the URL of the Automic Automation server in the Rest URL field, and click Apply.
The AutoSys Workload Automation messages are forwarded to the Automic Automation.
10. Click the Services icon on the toolbar.
11. Right-click the scheduler service, and select Start.
The scheduler starts. Any messages written to the scheduler log now also appear in Automic Automation. After AutoSys Workload Automation starts publishing events to Automic Automation, you can define Rule Events to filter the messages and to perform actions. For more information, see the [Automic documentation](#).

Integrate with SSA

CA Secure Socket Adapter (SSA) is an application that lets CA components use a single multiplexed communication port to ease firewall administration and minimize conflicts with other applications.

SSA consists of a Connection Broker that receives incoming connections from the physical port and redirects it to the corresponding network application (such as AutoSys Workload Automation) that is listening on a virtual port. Similarly, the Connection Broker redirects all outgoing connections sent by network applications using different virtual ports through the same physical port. The Connection Broker must recognize the virtual ports to redirect network traffic to the correct application.

SSA is installed automatically during the AutoSys Workload Automation installation. During installation, AutoSys Workload Automation configures the default virtual ports it intends to use. However, you can configure AutoSys Workload Automation to listen on a different virtual port.

SSA provides the following features:

- Port multiplexing (PMUX) -- Increases security and the efficient use of physical ports available on any given host by restricting all AutoSys Workload Automation traffic to a single physical port. The only exception is CAICCI, which is not PMUX-enabled and uses its own physical ports, the agent, scheduler, and application server auxiliary ports.

NOTE

- If port multiplexing is enabled, AutoSys Workload Automation uses virtual ports and traffic through those ports is restricted to a single physical port named *PmuxServerPort*. The default value is 7163 and is set during the AutoSys Workload Automation installation. We recommend that you do not change the *PmuxServerPort* value; however if you want to enable other ports after the AutoSys Workload Automation installation, you must configure those additional ports.
- To accommodate port multiplexing, AutoSys Workload Automation uses a daemon broker process named *csampmuxf*. You do not need to start the *csampmuxf* process because it starts automatically with the first AutoSys Workload Automation binary.
- The virtual ports can have only one process bound to them. The bound process is generally considered a tcp-server. Any number of remote or local clients can connect to the tcp-server process bound to a port.
- Secure Sockets Layer (SSL) -- Provides an added layer of protection by encrypting network data before transmitting it over the network. SSL also decrypts the data upon receipt. By default, AutoSys Workload Automation is not SSL-enabled.

WARNING

The PMUX and SSL settings on the AutoSys Workload Automation server and clients (such as the jil and autorep commands and SDKs) must match. If AutoSys Workload Automation has PMUX or SSL enabled, its clients must also have PMUX or SSL enabled. If PMUX or SSL encryption is turned on for any server process, it must be turned on for all client and server processes that communicate with it. AutoSys Workload Automation processes depend on the PMUX and SSL settings of the host.

By default, the agents use plain socket ports to communicate with AutoSys Workload Automation. They do not use SSA ports. However, agents that have been configured to use SSA PMUX and SSL setting must also follow these requirements. This includes Unicenter AutoSys JM r11 legacy agents.

The csamconfigedit Command -- Configure the Port Settings

AutoSys Workload Automation listens to incoming data using virtual ports. SSA redirects the data sent and received from the virtual ports to a single physical port.

The *csamconfigedit* command lets you configure the settings for the port used by AutoSys Workload Automation. This command is located in the bin directory that is referenced by the *CSAM_SOCKADAPTER* environment variable.

NOTE

- SSA is installed and configured automatically during the AutoSys Workload Automation installation. After installation, you can configure the ports that AutoSys Workload Automation listens to. To change the port numbers or the settings of existing ports, you must use the `csamconfigedit` command. Before you configure the ports, you must stop the AutoSys Workload Automation processes and the `csampmxf` process on all hosts. You must ensure that the port settings must be the same on both the client and server processes.
- This topic explains only those `csamconfigedit` command parameters that are used to configure the port settings used by AutoSys Workload Automation. The SSA configuration settings must be the same on both the client and server processes.

This command has the following format when used to configure AutoSys Workload Automation:

- To specify a port number to configure:
`csamconfigedit Port[=value] [display|delete] [EnablePmux=True|False] [EnableSSL=True|False] [PmuxConnectionTimeout=value]`
- To specify a range of port values to configure:
`csamconfigedit PortRange=49152-50176 [display|delete] [EnablePmux=True|False] [EnableSSL=True|False] [PmuxConnectionTimeout=value]`
- To display the command help:
`csamconfigedit usage`

Port[=*value*]

Defines the port number to configure.

PortRange=49152-50176

Specifies the range of port values to configure. You cannot change this value.

NOTE

AutoSys Workload Automation uses only a few ports in this range. These ports are virtual because port multiplexing is enabled by default (recommended). If you virtualize these ports, any processes other than AutoSys Workload Automation processes that are using these ports are not affected.

display|delete

Displays or deletes the current configuration settings of the port or port range.

EnablePmux=True|False

(Optional) Enables port multiplexing.

Default: True

NOTE

If `EnablePmux` is set to True, AutoSys Workload Automation uses virtual ports provided by SSA. If `EnablePmux` is set to False, AutoSys Workload Automation runs on physical ports.

EnableSSL=True|False

Enables SSL encryption.

Default: False

NOTE

- To successfully communicate with Unicenter AutoSys JM r11 (scheduling jobs and exchanging cross-instance dependencies) the AES encryption must be set to NONE or OFF. To secure message transport within the AutoSys Workload Automation Release 11.3.6 environment and to Unicenter AutoSys JM r11, we recommend that you enable SSL encryption (`EnableSSL=True`). If your environment consists of instances that all support AES 128-bit encryption, you do not need to use SSA's SSL encryption. If you enable SSL encryption and are using AES 128-bit instance-wide encryption, the message payload is

encrypted twice (once at the application level using AES 128-bit instance-wide encryption and again in the messaging layer using SSL encryption). This incurs additional overhead.

- If you enable SSL encryption (EnableSSL=True), you can also specify the following keyword and value:
ServerStyle=Passive|Active|Deny|Negotiate|Mandate
 Defines the style that the AutoSys Workload Automation clients use to handle incoming SSL connections. It also applies to outward connections in deciding whether the server can override the client's decisions about using SSL. The ServerStyle parameter applies only if port multiplexing is enabled (EnablePmux=True) and is used even if SSL is not enabled (EnableSSL=False). You can set the ServerStyle parameter to *one* of the following values:
 - **Passive**
 Accepts both SSL and non-SSL connections based on the SSA configuration settings (EnableSSL and EnablePmux). This is the default.
 - **Active**
 Accepts both the SSL and non-SSL connections, but the SSL connections must have matching authentication methods.
 - **Deny**
 Accepts non-SSL connections only. All SSL connections are rejected.
 - **Negotiate**
 Accepts SSL connections only, but the client can select the authentication methods instead of the server.
 - **Mandate**
 Accepts SSL connections only. If you apply this style, you must enable SSL (EnableSSL=True) and port multiplexing (EnablePmux=True), and the SSA configuration settings must be the same on both the client and server processes.

Example: EnableSSL=True EnablePmux=True ServerStyle=Mandate

NOTE

When establishing a connection, the client and the server processes are authenticated based on the authentication methods defined in SSA. AutoSys Workload Automation uses the default authentication methods that are defined in SSA.

PmuxConnectionTimeout=value

Specifies the time (in seconds) the SSA Connection Broker holds a connection for AutoSys Workload Automation to accept it. If the demand placed on the AutoSys Workload Automation scheduler and its agent is high, we recommend that you set the Connection Broker time-out period to 30 seconds.

Default: 5

usage

Displays the help for the csamconfigedit command.

(UNIX) Configure AutoSys Workload Automation to Run with SSL

Typically, a client process is remote from the server process. However, a client can be on the same computer that hosts a server process. The clients communicate with the servers across operating environments with no additional configuration. By default, AutoSys Workload Automation uses AES algorithm to encrypt data and all messages (whether they are local or across the network).

When SSL is enabled, additional overhead incurs at process startup time. Persistent processes (such as the scheduler, application server, and the agent) incur this one-time cost at startup and function normally after. Client processes (such as JIL, autorep, or sendevent), which are not persistent or are invoked repetitively, incur this cost for each time the process is invoked.

NOTE

To successfully communicate with Unicenter AutoSys JM r11 (scheduling jobs and exchanging cross-instance dependencies) the AES encryption must be set to NONE or OFF. To secure message transport within the

AutoSys Workload Automation Release 11.3.6 environment and to Unicenter AutoSys JM r11, we recommend that you enable SSL encryption (EnableSSL=True). If your environment consists of instances that all support AES 128-bit encryption, you do not need to use SSA's SSL encryption. If you enable SSL encryption and are using AES 128-bit instance-wide encryption, the message payload is encrypted twice (once at the application level using AES 128-bit instance-wide encryption and again in the messaging layer using SSL encryption). This incurs additional overhead.

Follow these steps:

1. Log in to AutoSys Workload Automation as the root user and run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following command at the operating system prompt:

```
unisrvctr stop CA-WAAE
```

The scheduler, application server, web server, and the agent stop.

3. Enter the following command at the operating system prompt:

```
csampmux stop
```

The csampmux process stops.

4. Enter the following commands at the operating system prompt:

```
cd $CSAM_SOCKADAPTER/bin
```

```
csamconfigedit Port=value EnableSSL=True EnablePmux=True display
```

NOTE

- If you defined an application server port by editing the AutoServerPort parameter in the configuration file, you must use the same port number for Port=value. For more information about defining the communication ports for the application server, see [Configure an Application Server](#).
- By default, the ServerStyle parameter value is set to Passive. You must specify the ServerStyle parameter if you want to set the value to Mandate.

5. Enter the following command at the operating system prompt:

```
unisrvctr start CA-WAAE
```

The scheduler, application server, web server, and the agent start. AutoSys Workload Automation is configured to run with SSL, and the configuration settings of the port are displayed.

NOTE

- If you run multiple application servers, you must enable each application server with the same settings.
- If you enable an application server port other than the default, you must also consider how you want that port to behave under the PMUX feature and enable it accordingly.
- You must set the same port configurations on both the AutoSys Workload Automation server and client to ensure that the two components can communicate.
- If you enable SSL on one host in the AutoSys Workload Automation network, you must enable SSL on all the other hosts in the AutoSys Workload Automation network. After you enable SSL for a given host, you must stop and start all the AutoSys Workload Automation processes on that host. After all hosts are enabled, all AutoSys Workload Automation network traffic is encrypted under SSL.

Example: Enable PMUX and SSL for Port 5101

This example enables PMUX and SSL for port 5101.

```
cd $CSAM_SOCKADAPTER/bin
```

```
csamconfigedit Port=5101 ServerStyle=Mandate EnablePmux=True EnableSSL=True
```

Example: Enable PMUX and SSL for the Port Range 49152-50176

This example enables PMUX and SSL for the port range 49152-50176.

```
cd $CSAM_SOCKADAPTER/bin
csamconfigedit PortRange=49152-50176 ServerStyle=Mandate EnablePmux=True EnableSSL=True
```

(Windows) Configure AutoSys Workload Automation to Run with SSL

Typically, a client process is remote from the server process. However, a client can be on the same computer that hosts a server process. The clients communicate with the servers across operating environments with no additional configuration. By default, AutoSys Workload Automation uses AES algorithm to encrypt data and all messages (whether they are local or across the network).

When SSL is enabled, additional overhead incurs at process startup time. Persistent processes (such as the scheduler, application server, and the agent) incur this one-time cost at startup and function normally after. Client processes (such as JIL, autorep, or sendevent), which are not persistent or are invoked repetitively, incur this cost for each time the process is invoked.

NOTE

To successfully communicate with Unicenter AutoSys JM r11 (scheduling jobs and exchanging cross-instance dependencies) the AES encryption must be set to NONE or OFF. To secure message transport within the AutoSys Workload Automation Release 11.3.6 environment and to Unicenter AutoSys JM r11, we recommend that you enable SSL encryption (EnableSSL=True). If your environment consists of instances that support AES 128-bit encryption, you do not need to use SSA's SSL encryption. If you enable SSL encryption and are using AES 128-bit instance-wide encryption, the message payload is encrypted twice (once at the application level using AES 128-bit instance-wide encryption and again in the messaging layer using SSL encryption). This incurs additional overhead.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Stop.
The scheduler, application server, and the agent stop.
2. Stop the CA Connection Broker service from the Windows Service Control Manager.
3. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt is displayed.
4. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:


```
csamconfigedit Port=value EnableSSL=True EnablePmux=True display
```

NOTE

- If you defined an application server port using the Administrator utility, you must use the same port number for Port=value. For more information about defining the communication ports for the application server, see [Configure an Application Server](#).
 - By default, the ServerStyle parameter value is set to Passive. You must specify the ServerStyle parameter if you want to set the value to Mandate.
5. Start the CA Connection Broker service from the Windows Service Control Manager.
 6. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select an instance from the Instance drop-down list.

- c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Start.
The scheduler, application server, and the agent start.
- AutoSys Workload Automation is configured to run with SSL, and the configuration settings of the port are displayed.

NOTE

- If you run multiple application servers, you must enable each application server with the same settings.
- If you enable an application server port other than the default, you must also consider how you want that port to behave under the PMUX feature and enable it accordingly.
- You must set the same port configurations on both the AutoSys Workload Automation server and client to ensure that the two components can communicate.
- If you enable SSL on one host in the AutoSys Workload Automation network, you must enable SSL on all the other hosts in the AutoSys Workload Automation network. After you enable SSL for a given host, you must stop and start all the AutoSys Workload Automation processes on that host. After all hosts are enabled, all AutoSys Workload Automation network traffic is encrypted under SSL.

Example: Enable PMUX and SSL for Port 5101

This example enables PMUX and SSL for port 5101.

```
csamconfigedit Port=5101 ServerStyle=Mandate EnablePmux=True EnableSSL=True
```

Example: Enable PMUX and SSL for the Port Range 49152-50176

This example enables PMUX and SSL for the port range 49152-50176.

```
csamconfigedit PortRange=49152-50176 ServerStyle=Mandate EnablePmux=True EnableSSL=True
```

Virtual Ports Used by AutoSys Workload Automation

The AutoSys Workload Automation application server and the agent require a port to listen for incoming connections. By default, the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. You can configure the application server to listen on a different virtual port.

However, for the AutoSys Workload Automation application server, scheduler, and the agent to communicate with one another, virtual ports are dynamically assigned values in the 49152 - 50176 range. This range is known as the *ephemeral port range* and is reserved for short-term communications. The AutoSys Workload Automation installation also configures SSA to register the ephemeral port range as virtual ports.

(UNIX) Configure the Application Server to Listen on a Different Virtual Port

You might want to reconfigure the port that the AutoSys Workload Automation application server listens to in the following situations:

- Another Broadcom product is using the default virtual port and you want that product to continue using that port.
- You want to enable more than one application server to run on the same host. You must specify a unique virtual port for each application server.

NOTE

By default, port multiplexing is enabled on AutoSys Workload Automation, and the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. If you install multiple AutoSys Workload Automation instances on the same computer, subsequent installations use incremental virtual port numbers, such as 9001, 9002, and so on.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:


```
unisrvctr stop CA-WAAE
```

The scheduler, application server, web server, and the agent stop.
3. Enter the following command at the operating system prompt:


```
csampmux stop
```

The csampmux process stops.
4. Change to the \$CSAM_SOCKADAPTER/bin directory, and enter the following command at the operating system prompt:


```
csamconfigedit Port=value EnablePmux=True
```

 - **value**
Defines the port number to configure.
5. Enter the following command at the operating system prompt:


```
csamconfigedit Port=value display
```

The configuration settings of the specified virtual port are displayed.
6. Set the AutoServerPort parameter to the specified virtual port in the configuration file.
7. Enter the following command at the operating system prompt:


```
unisrvctr start CA-WAAE
```

The scheduler, application server, web server, and the agent start. A virtual port is defined for the application server. The application server now listens on the specified virtual port.

(Windows) Configure the Application Server to Listen on a Different Virtual Port

You might want to reconfigure the port that the AutoSys Workload Automation application server listens to in the following situations:

- Another CA product is using the default virtual port and you want that product to continue using that port.
- You want to enable more than one application server to run on the same host. You must specify a unique virtual port for each application server.

NOTE

By default, port multiplexing is enabled on AutoSys Workload Automation, and the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. If you install multiple AutoSys Workload Automation instances on the same computer, subsequent installations use incremental virtual port numbers, such as 9001, 9002, and so on.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Stop.
The scheduler, application server, and the agent stop.
2. Stop the CA Connection Broker service from the Windows Service Control Manager.
3. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).

The AutoSys Workload Automation instance command prompt is displayed.

4. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:

```
csamconfigedit Port=value EnablePmux=True
```

– **value**

Defines the port number to configure.

5. Enter the following command at the instance command prompt:

```
csamconfigedit Port=value display
```

The configuration settings of the specified virtual port are displayed.

6. Start the CA Connection Broker service from the Windows Service Control Manager.

7. Do the following:

- a. Click Start, Programs, AutoSys, Administrator.

The Instance - AutoSys Workload Automation Administrator window opens.

- b. Select your AutoSys Workload Automation instance from the Instance drop-down list.

- c. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

- d. Right-click the scheduler, application server, and the agent service, and click Start.

The scheduler, application server, and the agent start.

A virtual port is defined for the application server. The application server now listens on the specified virtual port.

(UNIX) Configure the Connection Broker Time-Out Period

SSA consists of a Connection Broker that receives incoming connections from the physical port and redirects it to the corresponding network application (such as AutoSys Workload Automation) that is listening on a virtual port. All the connections to the Connection Broker are managed through the connection queue. Under typical conditions, the Connection Broker hands over the connection to AutoSys Workload Automation in 5 seconds (the default). However, under a large load, the Connection Broker queues up the connection requests and is not able to service a connection within the default time-out period, and the connection is broken. As a result, if you are running jobs on an agent that uses SSA ports, you may notice performance issues and connection failures between the AutoSys Workload Automation scheduler and the agent. Therefore, to handle large loads, we recommend that you set the Connection Broker time-out period to 30 seconds.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.

2. Enter the following command at the operating system prompt:

```
unisrvctr stop CA-WAAE
```

The scheduler, application server, web server, and the agent stop.

3. Enter the following command at the operating system prompt:

```
csampmux stop
```

The csampmuxf process stops.

4. Change to the \$CSAM_SOCKADAPTER/bin directory, and enter the following command at the operating system prompt:

```
csamconfigedit Port=value EnablePmux=True PmuxConnectionTimeout=30
```

– **value**

Defines the port number to configure.

– **30**

Specifies that the SSA Connection Broker time-out period is 30 seconds.

5. Enter the following command at the operating system prompt:


```
unisrvcntr start CA-WAAE
```

The scheduler, application server, web server, and the agent start. The Connection Broker time-out period is set to 30 seconds.

(Windows) Configure the Connection Broker Time-Out Period

SSA consists of a Connection Broker that receives incoming connections from the physical port and redirects it to the corresponding network application (such as AutoSys Workload Automation) that is listening on a virtual port. All the connections to the Connection Broker are managed through the connection queue. Under typical conditions, the Connection Broker hands over the connection to AutoSys Workload Automation in 5 seconds (the default). However, under a large load, the Connection Broker queues up the connection requests and is not able to service a connection within the default time-out period, and the connection is broken. As a result, if you are running jobs on an agent that uses SSA ports, you may notice performance issues and connection failures between the AutoSys Workload Automation scheduler and the agent. Therefore, to handle large loads, we recommend that you set the Connection Broker time-out period to 30 seconds.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Stop.
The scheduler, application server, and the agent stop.
2. Stop the CA Connection Broker service from the Windows Service Control Manager.
3. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt is displayed.
4. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:


```
csamconfigedit Port=value EnablePmux=True PmuxConnectionTimeout=30
```

 - **value**
Defines the port number to configure.
 - **30**
Specifies that the SSA Connection Broker time-out period is 30 seconds.
5. Start the CA Connection Broker service from the Windows Service Control Manager.
6. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select your AutoSys Workload Automation instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler, application server, and the agent service, and click Start.
The scheduler, application server, and the agent start.

The Connection Broker time-out period is set to 30 seconds.

Integrate with CA Service Desk

You can integrate AutoSys Workload Automation with CA Service Desk to let you open a service desk ticket (request or incident) when a job fails.

CA Service Desk is an enterprise-level service desk solution that lets you automate IT processes and provide audit trails for regulatory compliance. CA Service Desk is installed as a standalone product. CA Service Desk provides a self-service web interface that helps customers resolve their own issues. From this web interface, they can submit tickets, check status, and browse the knowledge base. To initiate a service desk ticket to CA Service Desk, AutoSys Workload Automation requires the web service desk URL, login identifier, and password. If you use a web service customer to access CA Service Desk, the web service desk customer can be provided to use as the customer in the service desk request instead of the web service desk login identifier.

The default data files that are added to the CA Service Desk database during configuration are used to create requests through the web services. In addition to the data files, CA Service Desk provides the default templates and contacts for AutoSys Workload Automation. That is, the default users and templates use the names of AutoSys Workload Automation.

NOTE

Only the CA Service Desk administrator is authorized to modify the default templates that are provided during the configuration process.

Integrate AutoSys Workload Automation with CA Service Desk

This topic provides an overview of the steps that you must perform to integrate AutoSys Workload Automation with CA Service Desk.

To integrate AutoSys Workload Automation with CA Service Desk, follow these steps:

1. Configure AutoSys Workload Automation to work with CA Service Desk on [UNIX](#) or [Windows](#).
2. [Initiate a Service Desk ticket using AutoSys Workload Automation](#).

(UNIX) Configure AutoSys Workload Automation to Work with CA Service Desk

AutoSys Workload Automation works with CA Service Desk to let you open a service desk ticket (request or incident) when a job fails. To integrate AutoSys Workload Automation with CA Service Desk, you must issue an integration command on CA Service Desk and activate the Service Desk interface on AutoSys Workload Automation.

The ServiceDeskCust, ServiceDeskURL, and ServiceDeskUser parameters let you activate the Service Desk interface. You must set either the ServiceDeskURL and ServiceDeskUser parameters, or the ServiceDeskURL and ServiceDeskCust parameters to activate the Service Desk interface. When you set these service desk parameters, the scheduler initiates the opening of a service desk ticket through CA Service Desk if the job the scheduler is processing was defined with the appropriate service desk attributes.

NOTE

- AutoSys Workload Automation supports CA Service Desk r12.9 and 14.1.
- By default, CA Service Desk integration is inactive.

Follow these steps:

1. Open the CA Service Desk application and verify that it operates properly.
2. Access the following location:

```
C:\Program Files\CA\Service Desk\data\integrations
```

3. Run the following commands as appropriate:

```
pdm_load -f integAutoSys.dat
pdm_load -f itil_integAutoSys.dat
```

NOTE

- If you perform an ITIL configured CA Service Desk installation, you must first load the integAutoSys.dat file and then load the itil_integAutoSys.dat file. However, if you perform a non-ITIL configured CA Service Desk installation, you must load only the integAutoSys.dat file.
- The roadmap for CA Service Desk is to drop support for non-ITIL configuration in future releases.

4. Log in to AutoSys Workload Automation as a user authorized to stop the scheduler and run the shell that is sourced to use AutoSys Workload Automation.

5. Enter the following commands at the operating system prompt:

```
unisrvcntr status waae_sched.$AUTOSERV
unisrvcntr status waae_server.$AUTOSERV
```

The scheduler process ID is displayed as follows:

```

CA Services Status Report
Component Name      Pid      Status
-----
WAAE Scheduler (ACE)      32220    running
```

The application server process ID is displayed as follows:

```

CA Services Status Report
Component Name      Pid      Status
-----
WAAE Application Server (ACE)      33330    running
```

6. Edit the following parameters in the configuration file, and save the file:

```
ServiceDeskURL=web_server_site
ServiceDeskUser=user/password
ServiceDeskCust=web_server_customer
```

– **web_server_site**

Defines the address for the CA Service Desk web service. You can specify a HTTP or HTTPS URL. Before you specify a HTTPS URL, you must enable SSL communication between AutoSys Workload Automation and CA Service Desk.

Examples: `http://servicedeskhost:port_number/axis/services/USD_R11_WebService`, `https://sdm-host.domain.com:port/axis/services/USD_R11_WebService`

port_number

Specifies the port that CA Service Desk Manager uses to communicate with AutoSys Workload Automation.

port

Specifies the SSL port that CA Service Desk Manager uses to communicate with AutoSys Workload Automation.

– **user/password**

Defines the user name and its associated password that is used to connect to the CA Service Desk web service.

– **web_server_customer**

(Optional) Defines the customer name for the CA Service Desk web service.

NOTE

The ServiceDeskCust parameter can be substituted for the ServiceDeskUser parameter in the service desk request. That is, if you use a web service customer to access CA Service Desk, you can provide the ServiceDeskCust parameter (that displays as the customer in the request) instead of the ServiceDeskUser parameter.

7. Enter the following command at the operating system prompt:

```
kill -HUP scheduler_pid
kill -HUP applicationserver_pid
```

– **scheduler_pid**

Defines the process ID of the scheduler to pause and resume.

– ***applicationserver_pid***

Defines the process ID of the application server to pause and resume.

The scheduler and application server resume. AutoSys Workload Automation is configured to work with CA Service Desk, and you can now open a service desk ticket.

Enable SSL Communication Between AutoSys Workload Automation and CA Service Desk

You can enable SSL communication between AutoSys Workload Automation and CA Service Desk.

NOTE

On HP-UX, you cannot enable SSL communication between AutoSys Workload Automation and CA Service Desk.

Follow these steps:

1. Configure CA Service Desk Manager to support SSL communication.

NOTE

For information about configuring CA Service Desk Manager to enable SSL communication, see the CA Service Desk Manager documentation.

2. Export the signed certificate from your certificate authority into the \$AUTOUSER directory and name it as cawaae_sd.pem.

NOTE

AutoSys Workload Automation supports certificates only in .pem format. If the certificate is not in the .pem format, use the OpenSSL tool to convert your .cer or .p12 certificate file to .pem format and then place the certificate file in .pem format in the \$AUTOUSER directory. For more information about using OpenSSL tool to convert files from one format to another, see the OpenSSL documentation.

NOTE

You can export the signed certificate from the certificate authority from your browser provided the certificate is already imported. For example, if you exported the signed certificate as cawaae_sd.cer, run the following command to convert it to .pem format

```
openssl x509 -inform der -in cawaae_sd.cer -out cawaae_sd.pem:
```

OpenSSL is an open source tool that is available on all UNIX platforms by default. Using this tool, you can convert a certificate on a UNIX machine and transfer it to a Windows machine in binary format and use it.

(Windows) Configure AutoSys Workload Automation to Work with CA Service Desk

AutoSys Workload Automation works with CA Service Desk to let you open a service desk ticket (request or incident) when a job fails. To integrate AutoSys Workload Automation with CA Service Desk, you must issue an integration command on CA Service Desk and activate the Service Desk interface on AutoSys Workload Automation.

The Web Service Login ID, Web Service Login Password, Web Service Customer, and URL Location parameters let you activate the Service Desk interface. You must set either the URL Location, Web Service Login ID, and Web Service Login Password parameters, or the URL Location and Web Service Customer parameters to activate the Service Desk interface. When you set these service desk parameters, the scheduler initiates the opening of a service desk ticket through CA Service Desk if the job the scheduler is processing was defined with the appropriate service desk attributes.

NOTE

- AutoSys Workload Automation supports CA Service Desk r12.9 and 14.1.
- By default, CA Service Desk integration is inactive.

Follow these steps:

1. Open the CA Service Desk application and verify that it operates properly.
2. Access the following location:

`C:\Program Files\CA\Service Desk\data\integrations`

3. Run the following commands as appropriate:

`pdm_load -f integAutoSys.dat`

`pdm_load -f itil_integAutoSys.dat`

NOTE

- If you perform an ITIL configured CA Service Desk installation, you must first load the integAutoSys.dat file and then load the itil_integAutoSys.dat file. However, if you perform a non-ITIL configured CA Service Desk installation, you must load only the integAutoSys.dat file.
- The roadmap for CA Service Desk is to drop support for non-ITIL configuration in future releases.

4. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
5. Select an instance from the Instance drop-down list in the Settings pane.
6. Click the Integration icon on the toolbar.
The Integration - AutoSys Workload Automation Administrator window appears.
7. Click the Service Desk tab, complete the following fields as appropriate, and click Apply:
 - **Web Service Login ID**
Defines the user name that is used to connect to the CA Service Desk web service.

NOTE

On UNIX, the equivalent configuration parameter is ServiceDeskUser.

- **Web Service Login Password**
Defines the password associated with the Web Service Login ID.
Limits: Up to 32 characters

NOTE

On UNIX, the equivalent configuration parameter is ServiceDeskUser.

- **Web Service Customer**
(Optional) Defines the customer name for the CA Service Desk web service.

NOTE

- The Web Service Customer field can be substituted for the Web Service Login ID field in the service desk request. That is, if you use a web service customer to access CA Service Desk, the Web Service Customer field is used in the service desk request instead of the Web Service Login ID field.
- On UNIX, the equivalent configuration parameter is ServiceDeskCust.

- **URL Location**
Defines the address for CA Service Desk web service. You can specify a HTTP or HTTPS URL. Before you specify a HTTPS URL, you must enable SSL communication between AutoSys Workload Automation and CA Service Desk. **Examples:** `http://servicedeskhost:port_number/axis/services/USD_R11_WebService`, `https://sdm-host.domain.com:port/axis/services/USD_R11_WebService`

port_number

Specifies the port that CA Service Desk Manager uses to communicate with AutoSys Workload Automation.

port

Specifies the SSL port that CA Service Desk Manager uses to communicate with AutoSys Workload Automation.

NOTE

On UNIX, the equivalent configuration parameter is ServiceDeskURL.

8. Right-click the scheduler service, and click Pause.

The scheduler pauses

9. Right-click the scheduler service, and click Resume.

The scheduler resumes.

10. Right-click the application server service, and click Pause.

The application server pauses.

11. Right-click the application server service, and click Resume.

The application server resumes. AutoSys Workload Automation is configured to work with CA Service Desk, and you can now open a service desk ticket.

Initiate a Service Desk Ticket Using AutoSys Workload Automation

The integration of CA Service Desk and AutoSys Workload Automation lets you open a service desk ticket (request or incident) when a job fails. When a job is defined to open a service desk ticket, the AutoSys Workload Automation scheduler initiates the opening of the ticket during terminal status processing. The scheduler prepares and sends the ticket. Messages are written to the scheduler log indicating whether the ticket was sent and processed successfully.

To initiate a service desk ticket using AutoSys Workload Automation, specify the following attributes in the job definition:

- service_desk
- (Optional) svcdesk_desc
- (Optional) svcdesk_pri
- (Optional) svcdesk_imp
- (Optional) svcdesk_sev
- (Optional) svcdesk_attr

NOTE

Only the service_desk attribute is required. If the optional attributes are not set, the job uses the AutoSys Workload Automation service desk template values set in CA Service Desk. This template is included in the CA Service Desk installation. Before initiating a service desk ticket, make sure that the web services for CA Service Desk are active.

Example: Initiate a Service Desk Incident

This example initiates a service desk incident with a priority of 1 for a job named service_desk_on_failure_1.

```
insert_job: service_desk_on_failure_1
machine: localhost
command: false
owner: user@localhost
service_desk: y
svcdesk_pri: 1
svcdesk_desc: "service_desk_on_failure_1 has failed."
```

Example: Initiate a Service Desk Request

This example initiates a service desk request with an impact of 3 and a severity of 4 for a job named service_desk_on_failure_2.

```
insert_job: service_desk_on_failure_2
machine: localhost
command: false
owner: user@localhost
service_desk: y
svcdesk_imp: 3
```

```
svcdesk_sev: 4
svcdesk_desc: "service_desk_on_failure_2 has failed."
```

Integrate with CAICCI

CA, Inc. Common Communications Interface (CAICCI) is the communication layer that lets the AutoSys Workload Automation scheduler, which handles cross-platform events, communicate with legacy agents on the distributed, mid-range, and mainframe platforms.

CAICCI consists of several daemon processes (UNIX) or services (Windows) and a library. On UNIX, the scheduler communicates with the CAICCI API through the shared library. On Windows, the Cross-Platform Interface in AutoSys Workload Automation accesses the CAICCI API through the shared library.

On UNIX, CAICCI consists of the following four daemon processes:

- **caiccid**
Builds the CAICCI resources and starts the other CAICCI daemon processes. Referred to as the main CAICCI daemon because it is started first.
- **cciclnd**
Maintains the CAICCI Inter-Process Communication (IPC) resources. Referred to as the clean daemon process because of that responsibility.
- **ccirmtd**
Transmits data across the network remotely.
- **ccijimd**
Acts as a proxy between CAICCI (multi-threaded application) and AutoSys Workload Automation (single-threaded application).

On Windows, CAICCI consists of the following four services:

- **quenetd**
Sends or receives messages to or from other Windows hosts and AS/400 hosts.
- **ccirmtd**
Transmits data across the network remotely.
- **ccinrsd**
Defines the name resolution service that provides a unique location where applications may go to make inquiries about the location of other applications using CAICCI.
- **ccinrcd**
Defines the name resolution client. It is the client counterpart to the ccinrsd service.

AutoSys Workload Automation uses CAICCI to communicate with the following products:

- (Windows only) Notification Services component of CA NSM
- CA Workload Automation SE
- CA Workload Automation EE
- AutoSys Workload Automation Connect Option
- CA Jobtrac
- CA Scheduler

(UNIX) caiccid.prf File -- Specify Max_Recvrs Value

The caiccid.prf file specifies the Max_Recvrs value and starts the CAICCI clean and remote daemon processes when you start CAICCI.

NOTE

We recommend that you do not edit the caiccid.prf file, unless the file hits the Max_Recvrs limit.

The caiccid.prf file is located in the following directory:

\$CAIGLBL0000/cci/config/*nodename*/caiccid.prf

- ***nodename***

Identifies the computer where the CAICCI daemons run.

The caiccid.prf file has the following format:

```
CLN_Demon = cciclnd startup
RMT_Demon = ccirmtd startup
Max_Recvrs = nn,mm
```

- **cciclnd startup**

Starts the CAICCI clean daemon process (cciclnd) when you start CAICCI.

- **ccirmtd startup**

Starts the CAICCI remote daemon process (ccirmtd) when you start CAICCI.

- **Max_Recvrs = *nn,mm***

- ***nn***

Defines the number of CAICCI receivers that determine the size of the shared memory segment for RVT lists. This has the effect of limiting the number of application receivers.

Default: 48. You may need to change the default value to match your installation.

NOTE

- Each application requires at least one RVT and each unprocessed CAICCI message requires another RVT. On a busy server, you may need to increase this value to 200 or 300. If the CAICCI_E_FREERVTs error message is displayed in the system log, you must increase this value.
- After you increase this value, asbll may not start because CAICCI has not started. This sometimes happens because CAICCI must protect access to this shared memory using a semaphore group. CAICCI must create a semaphore group with a semaphore identifier for each RVT plus three extras. On most UNIX platforms, the number of semaphore identifiers in a semaphore group is governed by the SEMMSL kernel parameter. You may need to increase the SEMMSL value. You must ensure that you follow the following rule when increasing this value in the Max_Recvrs parameter:

$$\text{SEMMSL} \geq nn + 3$$

CAICCI requires two distinct semaphore identifiers at most. If CAICCI gets ID=0, it holds this group and requests another. CA NSM has requirements for XXXMNI, which must be added to the requirements for other products using the IPC resources.

- ***mm***

Indicates the number of messages that CAICCI queues up for each application.

Limits: 1-700

NOTE

- The maximum value is 700. However, we recommend specifying 699 as the maximum value because CAICCI shut downs when the 700 buffer is filled to avoid problems with possible limited resources on the local machine, but not for the remote daemon.
- If you set this value to a value higher than the recommended maximum value, it defaults to the maximum.
- Sometimes an application stops responding or is too busy to pick up its messages. In these situations, the following error message is displayed in the system log:


```
CAICCI_E_RECVBUSY Target [ ] queue is full, sender [ ]
```

The application sleeps while it waits for room on the buffer (the default behavior).

Shared Memory for RVTs

Information about a receiver is stored in a structure named *RVT*. Since all applications must have access to this information, it is stored in shared memory as an RVT list. On UNIX, you can view this information by using the `cci show` command, which produces blocks of data. The first block contains global data and each subsequent block contains information specific to a particular receiver. This information includes the following details:

- The CAICCI address
- The process ID of the application
- Whether the application is ready to accept data or has a message pending.

On UNIX, the `caiccid` daemon process creates the shared memory segment for RVT lists. When CAICCI starts, it creates the shared memory segment. Therefore, CAICCI must know beforehand how large a memory segment to create.

In addition to storing an application's CAICCI address, the shared memory is used to store the data that the application is sending. Sometimes messages arrive too quickly for the target application to dispose of them. In this scenario, an application may request that CAICCI queues up messages.

(UNIX) ccicInd.prf File -- Define the Time to Sleep Between System Scans

The `ccicInd.prf` file defines the number of seconds to sleep between system scans for communications buffer and connections cleaning.

WARNING

The default value is one second. Do not change this value unless instructed by CA Technical Support.

The `ccicInd.prf` file is located in the following directory:

```
$CAIGLBL0000/cci/config/nodename/ccicInd.prf
```

- ***nodename***
Identifies the computer where the CAICCI daemons run.

ccirmtd.rc File -- Identify Local and Remote Parameters

The `ccirmtd` file identifies the local CAICCI node name, the UNIX or Windows host name, and the block size for the local and remote computers.

The `ccirmtd` file is located as follows:

- (UNIX) `$CAIGLBL0000/cci/config/nodename/ccirmtd.prf`
 - ***nodename***
Identifies the computer where the CAICCI daemons run.
- (Windows) `C:\Program Files\CA\SC\CA_APPSW\ccirmtd.rc` or `C:\CA_APPSW\ccirmtd.rc`

NOTE

This path assumes that AutoSys Workload Automation is installed on your C drive.

To enable CAICCI communication between computers, you must add the CAICCI REMOTE parameter, so that CAICCI knows how to connect to the `nodename` and which port to connect to.

The local and remote parameters have the following format:

```
LOCAL = nodename cciname max_msg_size [startup | nostart][port=p retry=x]
```

```
REMOTE = nodename cciname max_msg_size [startup | nostart][port=p retry=x]
```

- **nodename**
Defines the host name that is passed to `gethostbyname`. This value can be any name that can be resolved to the correct IP address and does not require logical connection to `cciname`.
 - **cciname**
Defines the logical name that CAICCI uses to identify the local host. This name is determined by the `ca_uname` function during installation and by the `ca_nodename` function at run time. These functions are the equivalent of `uname -n`.
Limits: Up to 64 characters.
- NOTE**
An alias must be used for names with more than eight characters.
- **max_msg_size**
Defines the maximum buffer that CAICCI uses to send or receive messages over the socket. We recommend that you do not edit this value. Each side of the connection may have this set to different values (up to 32 KB). The lesser of the two values is used.
 - **startup | nostart**
(Optional) Indicates whether or not to initiate a connection. Sometimes you may only want one side to initiate the connection. Not having the server start connections eliminates a succession of messages when CAICCI is recycled. STARTUP tells CAICCI to attempt a remote connection when activated, whereas NOSTART implies that the remote system will be initiating the connection to the node.
 - **retry=x**
(Optional) Determines how the `ccirmttd` service behaves if the connection is dropped. Options are the following:
 - **0**
Does not retry the connection.
 - **-1**
Starts with a two-second retry interval and doubles after each unsuccessful retry attempt.
 - **n**
Waits for *n* seconds between retry attempts, where *n* is any number greater than 0 (zero).
- NOTE**
Retry interval is mainly used in conjunction with the `nostart` option to allow the server to sit passively and wait for incoming connection requests. If a client host goes down, the server will not attempt to reconnect.
- **port=p**
(Optional) Specifies another port for this specific connection only.
Default: 1721

Example: Identify Local and Remote Parameters

This example specifies the local and remote parameters.

```
LOCAL = abcdef31 abcdef31 32768 startup
REMOTE = abcdef33 abcdef33 32768 startup
REMOTE = abcdef33 abcdef33 32768 startup port=7000
```

(UNIX) Required CAICCI Daemon Processes

The following three daemon processes must be running for CAICCI:

```
$ ps -ef|grep cci
root 17733 17731 0 Jan 18 ? 30:36 /uni/cci_kit/cci/bin/ccirmttd
root 17732 17731 0 Jan 18 ? 1:20 /uni/cci_kit/cci/bin/cciclnd
root 17731 1 0 Jan 18 ? 1:25 /uni/cci_kit/cci/bin/caiccid
```

NOTE

You must have administrator privileges to start or stop CAICCI.

Important Considerations

The following are important considerations about CAICCI:

- You install CAICCI using the CA Common Components DVD.

NOTE

For more information about how to install CAICCI, see [CA Common Components Installation](#).

- CAICCI must be installed on the AutoSys Workload Automation server if you want to perform cross-platform scheduling with the following products:
 - CA UJMA
 - AutoSys Workload Automation Connect Option

Start CAICCI

After you configure CAICCI, you must start CAICCI for the configuration settings to take effect.

NOTE

On UNIX, you must log in as the root user to start or stop CAICCI. On Windows, you must have administrator privileges to start or stop CAICCI.

(UNIX) To start CAICCI, run the following script:

```
$CAIGLBL0000/cci/scripts/CA-cci start
```

(Windows) To start CAICCI, enter *one* of the following commands:

```
C:\Program Files\CA\SC\CA_APPSW\ccicntrl start
```

or

```
C:\CA_APPSW\ccicntrl start
```

Stop CAICCI

On UNIX, you must log in as the root user to start or stop CAICCI. On Windows, you must have administrator privileges to start or stop CAICCI.

(UNIX) To stop CAICCI, run the following script:

```
$CAIGLBL0000/cci/scripts/cshut
```

(Windows) To stop CAICCI, enter *one* of the following commands:

```
C:\Program Files\CA\SC\CA_APPSW\ccicntrl stop
```

or

```
C:\CA_APPSW\ccicntrl stop
```

On UNIX, if CAICCI stops responding and you cannot shut it down, take the following actions:

1. Issue the following command:

```
kill -9 ccirmtd_pid cciclnl_pid caiccid_pid
```

The CAICCI daemon processes stop running.

2. Issue the following command:

```
ipcs -a|grep 0000d
```

The shared memory is searched for the caiccid process.

3. Issue the following command:

```
ipcrm -q message_queue_id
```

The message queue for the caiccid process returned from Step 2 is removed.

4. Issue the following command:

```
ipcrm -m shared_memory_id
```

The shared memory for the caiccid process returned from Step 2 is removed.

5. Issue the following command:

```
ipcrm -s semaphore_id
```

The semaphore for the caiccid process returned from Step 2 is removed.

(Windows) Check the CAICCI Status

To check the status of CAICCI, enter *one* of the following commands:

```
C:\Program Files\CA\SC\CA_APPSW\ccicntrl status
```

or

```
C:\CA_APPSW\ccicntrl status
```

(UNIX) Enable CAICCI Remote Communications

After you install CAICCI, you can use the \$CAIGLBL0000/cci/scripts/cci.config script to enable CAICCI remote communications.

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt:

```
CCI_RemoteHost=<hostname>; export CCI_RemoteHost
$CAIGLBL0000/cci/scripts/cci.config
```

The following message appears:

```
During the initial installation, several CCI parameter files were created.
Do you want to recreate these files as part of this reinstallation?
(y/n) (default: n)
```

3. Enter **y**.

You are prompted for the node names that the CAICCI remote daemon must communicate with, as follows:

```
Please enter the name of the remote host or RETURN to end:
```

4. Enter a single node name.
The prompt is repeated.
5. Enter additional node names, one at a time, until you specified all the node names, and press the Enter key twice.
CAICCI remote communications are enabled.

(UNIX) CAICCI Environment Variables

On UNIX, several CAICCI environment variables are used. You must set these environment variables in the \$CAIGLBL0000/cci/scripts/CA-cci file before the main CAICCI daemon process (caiccid) runs, unless otherwise noted.

CAI_CCI_DEBUG Environment Variable

The CAI_CCI_DEBUG environment variable specifies whether the CAICCI traces are enabled or disabled. CAI_CCI_DEBUG affects all processes and applications using CAICCI.

You can use the CAI_CCI_DEBUG environment variable to enable or disable CAICCI traces.

The CAI_CCI_DEBUG environment variable has the following format:

```
CAI_CCI_DEBUG=y|n
```

- **y**
Enables CAICCI traces.
- **n**
Disables CAICCI traces.

CAI_CCI_LOG Environment Variable

The CAI_CCI_LOG environment variable specifies the directory where the CAICCI trace file is written to. CAI_CCI_LOG affects all processes and applications using CAICCI.

You can use the CAI_CCI_LOG environment variable when a larger volume is required for trace output.

The CAI_CCI_LOG environment variable has the following format:

```
CAI_CCI_LOG=path
```

- **path**
Specifies the directory where the CAICCI trace file is written to.

CAI_CCI_CONFIG Environment Variable

The CAI_CCI_CONFIG environment variable specifies the path to the CAICCI configuration directory. CAI_CCI_CONFIG affects all processes using CAICCI.

You can use the CAI_CCI_CONFIG environment variable to set the path to the CAICCI configuration directory.

The CAI_CCI_CONFIG environment variable has the following format:

```
CAI_CCI_CONFIG=path
```

- **path**
Specifies the path to the CAICCI configuration directory. The configuration files are located at \$path\ - n.

CAI_CCI_SHMMIN Environment Variable

The CAI_CCI_SHMMIN environment variable specifies the minimum size of shared memory segment that CAICCI requests. CAI_CCI_SHMMIN affects all processes and applications using CAICCI.

You can use the CAI_CCI_SHMMIN environment variable when the value of the SHMMIN kernel parameter is greater than 1.

The CAI_CCI_SHMMIN environment variable has the following format:

```
CAI_CCI_SHMMIN=SHMMIN
```

- **SHMMIN**
Defines the value of the SHMMIN kernel parameter.

CAI_CCI_PORT1 Environment Variable

The CAI_CCI_PORT1 environment variable specifies the port that the ccirmt.d.prf file binds to before connect calls. CAI_CCI_PORT1 affects the remote daemon process.

You can use the CAI_CCI_PORT1 environment variable when there is a firewall or a multi Network Interface Card (NIC) situation.

The CAI_CCI_PORT1 environment variable has the following format:

```
CAI_CCI_PORT1=n
```

- ***n***
Specifies the port (greater than 256) that the ccirmt.d.prf file binds to before connect calls.

CCI_SELECT_TIME Environment Variable

The CCI_SELECT_TIME environment variable determines the time-out period for the TCP/IP handshake to complete. CCI_SELECT_TIME affects the remote daemon process.

You can use the CCI_SELECT_TIME environment variable when the network conditions cause the TCP/IP handshake to take a long time to complete.

The CCI_SELECT_TIME environment variable has the following format:

```
CCI_SELECT_TIME=n
```

- ***n***
Determines the time-out period for the TCP/IP handshake to complete. This value must be greater than 1.

(Windows) CAICCI Environment Variables

On Windows, several CAICCI environment variables are used. These environment variables are usually set as system variables using the Control Panel in Windows. You must restart the computer for the settings to take effect.

SERVERNODE Environment Variable

The SERVERNODE environment variable specifies the node where the NR-server resides. SERVERNODE affects all applications using CAICCI registered with the NR-server.

You usually set the SERVERNODE environment variable during installation.

RMTHOSTS Environment Variable

The RMTHOSTS environment variable specifies the hosts that can be contacted only through the remote service. RMTHOSTS affects only the CAICCI remote service.

You can use the RMTHOSTS environment variable if the local host is a bridgenode and all UNIX hosts are listed.

CCIDOMAINS Environment Variable

The CCIDOMAINS environment variable links other CAICCI servers to communicate inquiries across domains. CCIDOMAINS affects the CAICCI NR-server.

This variable can help reduce the exposure of one CAICCI server. You can break up the CAICCI domain into smaller domains and use the CCIDOMAINS environment variable to link these domains.

CA_CCITRACE Environment Variable

The CA_CCITRACE environment variable enables CAICCI tracing. CA_CCITRACE affects all applications using CAICCI services.

CA_PROTEPINDX Environment Variable

The CA_PROTEPINDX environment variable specifies the protocol where a Remote Procedure Call (RPC) runs. You can set this value to *one* of the following:

- 0 for NetBEUI
- 1 for Transmission Control Protocol (TCP). This is the default.
- 2 for User Datagram Protocol (UDP)

CA_PROTEPINDX affects the communication service. All hosts must use the same protocol.

QUENTB_PORT Environment Variable

The QUENTB_PORT environment variable specifies the listening port for RPC over NetBEUI. The default value is 202.

QUENTB_PORT affects the communication service. You can use the QUENTB_PORT environment variable to avoid opening published ports or when testing.

QUETCP_PORT Environment Variable

The QUETCP_PORT environment variable specifies the listening port for RPC over TCP. The default value is 7003.

QUETCP_PORT affects the communication service. You can use the QUETCP_PORT environment variable to avoid opening published ports or when testing.

QUEUDP_PORT Environment Variable

The QUEUDP_PORT environment variable specifies the listening port for RPC over UDP. The default value is 7001.

QUEUDP_PORT affects the communication service. You can use the QUEUDP_PORT environment variable to avoid opening published ports or when testing.

CCI_BRIDGENODE Environment Variable

The CCI_BRIDGENODE environment variable specifies the Windows node through which you can reach the UNIX hosts. CCI_BRIDGENODE affects the remote service.

You can use the CCI_BRIDGENODE environment variable if a host other than the CAICCI server is a bridgenode.

CA_QUE_RETRIES Environment Variable

The CA_QUE_RETRIES environment variable controls how long a sender waits for a target to pick up a message. This value is set in 30-second increments. The default value is 3.

CA_QUE_RETRIES affects all applications using CAICCI. You can use the CA_QUE_RETRIES environment variable if an application is busy for long periods of time (for example, when database access takes a long time to handle CAICCI messages).

SnmpManagerHosts and SnmpCommunity

AutoSys Workload Automation uses Simple Network Management Protocol (SNMP) to send alarms and signals to SNMP managers. The SNMP trap mechanism is used to post alarms and signals.

When you configure the `SnmpManagerHosts` and `SnmpCommunity` parameters, AutoSys Workload Automation sends traps to the SNMP manager. You can monitor the alarms and signals that AutoSys Workload Automation generates.

For example, you can integrate AutoSys Workload Automation with Hewlett-Packard Node Manager software, versions 4.10 through 7.0x. This enables OpenView users to do the following tasks:

- Monitor all alarms that AutoSys Workload Automation generates.
- Monitor all signals that the scheduler receives.
- Specify that certain commands be issued when OpenView receives an alarm or signal.

Suppose that the scheduler receives a `STOP_DEMON` event. The scheduler posts the SNMP event to OpenView. This can be useful if the `STOP_DEMON` event is sent to shut down the scheduler. The event is posted to OpenView before the scheduler shuts down.

You can configure these parameters as follows:

- UNIX -- Edit the following entries in the configuration file:
`SnmpManagerHosts=hostname1,hostname2,...`
`SnmpCommunity=public`
 - **hostname1,hostname2,...**
 Defines the host names to which the scheduler sends SNMP traps. You can specify a list of host names on the network that are running as the SNMP managers, such as HP's OpenView or IBM's NetView, and to which you want to send SNMP traps (for example, post SNMP events).
 - **public**
 Defines the SNMP community that is associated with all the SNMP traps sent by the scheduler.

NOTE

- The `SnmpCommunity` parameter is effectively a keyword that is used to filter the SNMP traps that AutoSys Workload Automation sends to SNMP managers.
- The SNMP community is mostly public, and so the value of this parameter is set to public in the configuration file.
- Windows -- Click the SNMP tab on the Integration - AutoSys Workload Automation Administrator window of the Administrator utility and complete the following fields as appropriate:
 - **SNMP Manager Host**
 Defines the host names to which the scheduler sends SNMP traps. You can specify a list of host names on the network that are running as the SNMP managers, such as HP's OpenView or IBM's NetView, and to which you want to send SNMP traps (for example, post SNMP events).
 - **SNMP Community**
 Defines the SNMP community that is associated with all the SNMP traps sent by the scheduler.
Default: public

NOTE

The SNMP Community parameter is effectively a keyword that is used to filter the SNMP traps that AutoSys Workload Automation sends to SNMP managers.

SNMP Traps

AutoSys Workload Automation uses the SNMP trap mechanism to post alarms and signals. The SNMP traps notify you about events or alarms that are generated during job processing.

The SNMP trap passes the following values:

- **alarmName** -- The name of the alarm generated.
- **alarmJobName** -- The job that generated the alarm.
- **alarmText** -- The message that describes the cause of the alarm.
- **alarmCode** -- The integer code for the alarm.
- **alarmExitCode**—The integer code that represents the exit code of the job.
- **alarmMachineName**—The name of the machine where the job runs. If the machine does not exist, NO MACHINE is displayed.
- **alarmBoxName**—The name of the box that includes the job. If the box does not exist, NO BOX is displayed.
- **alarmApplicationName**—The name of the application that is associated with the job. If the application does not exist, NO APPLICATION is displayed.
- **alarmGroupName**—The name of the group that is associated with the job. If the group does not exist, NO GROUP is displayed.
- **alarmRunNum**—The run number of the job.
- **alarmNtry**—The number of times the job was restarted.
- **alarmInstanceName**—The name of the instance.
- **trapMessage** -- The trap description. This description includes the instance name and the machine name where the trap is generated from. If the machine does not exist, NO MACHINE is displayed.
- **trapDate** -- The date and time the trap is generated.

AutoSys Workload Automation generates the following SNMP traps:

- **trapEventProcessor**

Indicates that the scheduler received a fatal signal.

Trap number: 1

NOTE

The trapEventProcessor trap only passes the trapMessage and trapDate values.

- **trapForkFail**

Indicates that the legacy agent cannot start a user process because no process slots are available.

Trap number: 501

- **trapMinRunAlarm**

Indicates that the job completed in less than the minimum run time.

Trap number: 502

NOTE

You can specify the minimum run time for a job using the min_run_alarm attribute.

- **trapJobFailure**

Indicates that the job failed.

Trap number: 503

- **trapJobTerminated**

Indicates that the job terminated.

Trap number: 552

- **trapMaxRetrys**

Indicates that the number of application restarts exceeded the n_retrys limit for the job.

Trap number: 505

NOTE

The n_retrys attribute specifies the number of times to restart the job after it exits with a FAILURE status.

- **trapStartJobFail**

Indicates that the job cannot start.

Trap number: 506

- **trapEventHdlrError**

Indicates that the scheduler generated an error while processing an event.

Trap number: 507

- **trapEventQueueError**

Indicates that the event cannot be marked as processed.

Trap number: 508

- **trapJobNotOnHold**

Indicates that the job cannot be placed on hold even after the JOB_ON_HOLD event occurred.

Trap number: 509

- **trapMaxRunAlarm**

Indicates that the job exceeded the maximum run time limit.

Trap number: 510

NOTE

You can specify the maximum run time for a job using the max_run_alarm attribute.

- **trapResource**

Indicates that the resources required to run the job are not available.

Trap number: 512

- **trapMissingHeartbeat**

Indicates that the job did not send a heartbeat within the interval that is specified for the job.

Trap number: 513

- **trapChaseAlarm**

Indicates that a chase alarm has been generated.

Trap number: 514

- **trapDatabaseCommAlarm**

Indicates that the legacy agent cannot send an event to the database.

Trap number: 516

- **trapAppServerComm**

Indicates that the autoping command is not successful. The agent cannot communicate with the application server.

Trap number: 517

- **trapVersionMismatch**

Indicates that the legacy agent version is different from the version of the routine or process calling it.

Trap number: 518

- **trapDbRollover**

Indicates that AutoSys Workload Automation rolled over from dual event server mode to single event server mode.

Trap number: 519

- **trapEpRollover**

Indicates that the shadow scheduler is taking over.

Trap number: 520

- **trapEpShutdown**

Indicates that the scheduler is shutting down.

Trap number: 521

- **trapEpHighAvail**

Indicates that AutoSys Workload Automation running in high availability mode detected a system or network problem.

Trap number: 522

- **trapDbProblem**

Indicates a problem with one of the AutoSys Workload Automation databases.

Trap number: 523

- **trapDuplicateEvent**

Indicates that the event server processed a duplicate event.

Trap number: 524

- **trapInstanceUnavailable**

Indicates that the event server of the receiving AutoSys Workload Automation instance cannot be reached.

Trap number: 525

- **trapAutoPing**
Indicates that the autoping -M -A command cannot connect to the client.
Trap number: 526
- **trapExternalDepsError**
Indicates that the cross-platform interface cannot send external dependencies to the remote node.
Trap number: 529
- **trapMachineUnavailable**
Indicates that the machine the scheduler communicates with is not responding or has been deleted.
Trap number: 532
- **trapServiceDeskFail**
Indicates that the scheduler cannot open a service desk ticket for a failing job.
Trap number: 533
- **trapUninotifyFailure**
Indicates that the scheduler cannot send a notification to the requesting job.
Trap number: 534
- **trapBadCPIJobName**
Indicates that the cross-platform interface received a job name that is not valid for the agent it is submitting the job to.
Trap number: 535
- **trapCPIUnavailable**
Indicates that the AutoSys Workload Automation cross-platform interface is not running.
Trap number: 536
- **trapMustStartAlarm**
Indicates that the must start alarm has been generated.
Trap number: 537
- **trapMustCompleteAlarm**
Indicates that the must complete alarm has been generated.
Trap number: 538
- **trapKillJobFail**
Indicates that the KILLJOB event failed for a job.
Trap number: 540
- **trapSendSigFail**
Indicates that the SEND_SIGNAL event failed.
Trap number: 541
- **trapReplyResponseFail**
Indicates that the REPLY_RESPONSE event failed because the job status is not WAIT_REPLY.
Trap number: 542
- **trapReturnResourceFail**
Indicates that the job failed to release resources.
Trap number: 543
- **trapRestartJobFail**
Indicates that the RESTARTJOB event failed.
Trap number: 544
- **trapDBMaintFail**
Indicates that the DBMaint utility failed to complete successfully.
Trap number: 548
- **trapMachineDiskThreshold**
Indicates that the disk space is low on the agent machine.
Trap number: 549
- **trapSuspendJobFail**
Indicates that the job is not in the RUNNING state and so it cannot be put in the SUSPENDED state.

Trap number: 550

- **trapResumeJobFail**

Indicates that the job is not in the SUSPENDED state and so it cannot be resumed.

Trap number: 551

- **trapTelemetryFail**

Indicates that Telemetry failed to complete successfully.

Trap number: 553

Configure Cross-Instance Dependencies with AutoSys Workload Automation

A AutoSys Workload Automation instance is one licensed version of AutoSys Workload Automation software running as a server and as one or more clients, on one or more computers. An instance uses its own scheduler, one or more application servers, and event server, and operates independently of other instances.

Different instances can run from the same executables and can have the same value for \$AUTOSYS. However, each instance must have different values for \$AUTOUSER and \$AUTOSERV. Different instances can also be run on the same computer.

Multiple AutoSys Workload Automation instances are not connected, but they can communicate with one another. This communication lets you schedule workload across instances in your enterprise. You can define jobs that have dependencies on jobs running on other instances (*cross-instance job dependencies*). A AutoSys Workload Automation job with these dependencies conditionally starts based on the status of the job on the other instance. In this situation, the local instance scheduler acts as a client and issues sendevent commands to the external instance. The application server of the other instance processes the sendevent request and stores the dependency request or status update in its database. You can also manually send events from one instance to another.

When the status of a job with cross-instance dependencies changes, the scheduler sends a CHANGE_STATUS event to the remote instance event server while the job in the local instance runs.

NOTE

The use of the sendevent command to manually send events from one instance to a 4.5.1 external instance is not supported.

Sending Events to External Instances

The following condition applies for jobs with cross-instance dependencies:

When a local scheduler sends events to external instances, it populates the Event text field to indicate the action in the local instance that resulted in the external event. The event text can be as follows:

- Job *job name* entered *status*
- Job *job name* was placed ON_HOLD|ON_ICE|ON_NOEXEC
- Job *job name* bypassed execution and was placed ON_NOEXEC
- Job *job name* was reset to INACTIVE|RUNNING|ACTIVATED|SUCCESS in non-execution mode
- Job *job name* was taken off ON_HOLD|ON_ICE
- Job *job name* was placed in RESTART due to application failure

Examples:

- When a job in the local instance is reset to the ON_NOEXEC status and the job is not in the INACTIVE, ACTIVATED, or SUCCESS status, the scheduler does not send the JOB_ON_NOEXEC event to the external instance. Instead, the scheduler sends the CHANGE_STATUS event to INACTIVE with the following text:

Job *job name* was placed ON_NOEXEC

Otherwise the completion or current status remains in its current status and no event is sent to the external instance.

- When a box job in the local instance is in the ON_NOEXEC status and the scheduler schedules the box to run, it sends a CHANGE_STATUS event to the external instance with the following text:
Job job name was reset to RUNNING in non-execution mode
- When a job in a box in the local instance is in the ON_NOEXEC status and the scheduler schedules the box to run, it sends a CHANGE_STATUS event to the external instance with the following text:
Job job name was reset to ACTIVATED in non-execution mode
- When a job in the local instance is bypassed and placed in the ON_NOEXEC status, the scheduler does not send the BYPASS event to the external instance. Instead, the scheduler sends the CHANGE_STATUS event to SUCCESS with the following text:
Job job name bypassed execution and was placed ON_NOEXEC
- When a job in the local instance is in the ON_NOEXEC status and you manually send a CHANGE_STATUS event to INACTIVE or SUCCESS, the scheduler sends a CHANGE_STATUS event to the external instance with the following text:
Job job name was reset to INACTIVE|SUCCESS in non-execution mode

Reporting Status Changes for Jobs with Cross-Instance Dependencies

The cross-instance interface design supports reporting status changes to the remote instance for jobs with cross-instance dependencies when those changes result from one of the following:

- The scheduler changes the status of the job when unavailable machine load units, resources, or agents prevent a job from running.
- The user changes the status of the job by issuing a sendevent command for one of the following events: JOB_ON_HOLD, JOB_OFF_HOLD, JOB_ON_ICE, JOB_OFF_ICE, JOB_ON_NOEXEC

If the local instance scheduler does not report these status changes to the remote instance scheduler, downstream jobs dependent on the remote jobs may not run when they should, or may run when they should not.

The scheduler internally generates an equivalent CHANGE_STATUS event to report the status change to the remote instance. This ensures that the remote scheduler accurately evaluates downstream jobs dependent on the remote jobs, including the job status and exit code conditions of the dependent jobs.

The equivalent CHANGE_STATUS event represents the actual status change that occurs in the local instance, and the event includes text specifying the actual status change. The remote scheduler log records this information. The remote scheduler reports a representative status and exit code for the dependent job that is based on the actual status change that occurs in the local instance:

Actual Status Change (Local Instance)	Reported Status (Remote Instance)
A job is placed on hold by the JOB_ON_HOLD event	INACTIVE with exit code -656
A job is taken off hold by the JOB_OFF_HOLD event	INACTIVE with the local job's exit code
A job is placed on ice by the JOB_ON_ICE event	SUCCESS with exit code -656
A job is taken off ice by the JOB_OFF_ICE event	INACTIVE with the local job's exit code
A job is bypassed by the BYPASS event	SUCCESS with exit code 0
A job is placed in the ON_NOEXEC status by the JOB_ON_NOEXEC event and the job is not in the INACTIVE, ACTIVATED, or SUCCESS status	INACTIVE with the local job's exit code
An internal update changes the job status to PEND_MACH	INACTIVE with the local job's exit code
An internal update changes the job status to QUE_WAIT	INACTIVE with the local job's exit code
An internal update changes the job status to RESWAIT	INACTIVE with the local job's exit code

NOTE

The scheduler automatically sends a BYPASS event when you place a job in the ON_NOEXEC status and that job starts. In cross-instance dependencies, the local scheduler reports the translated status to the remote instance when it bypasses the dependent job.

AutoSys Workload Automation External Instance Type

To use external job dependencies, the scheduling manager or remote machine must be defined as an *external instance* in the AutoSys Workload Automation database.

When you define AutoSys Workload Automation as an external instance, you must specify the **xtype: a** JIL attribute in the definition. This attribute indicates that the external scheduling manager is a AutoSys Workload Automation application server instance.

Configure Cross-Instance Dependencies for an r11, r11.3, Release 11.3.5, or Release 11.3.6 Instance

You can create cross-instance job dependencies between your local instance and external AutoSys Workload Automation r11, r11.3, Release 11.3.5, or Release 11.3.6 instances. You can also start jobs that are defined on the external instance. Before the instances can communicate with each other, you must configure them.

Note: For information about configuring cross-instance scheduling for 4.5.1 external instances, see [Configure Cross-Instance Dependencies for an 4.5.1 Instance](#).

To configure cross-instance scheduling between your local r11.3, Release 11.3.5, or Release 11.3.6 instance and another r11, r11.3, Release 11.3.5, or Release 11.3.6 instance, follow these steps:

1. Do the following:
 - a. [Define the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.](#)
 - b. [Define the local r11.3, Release 11.3.5, or Release 11.3.6 instance on the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance.](#)

Cross-instance dependencies are configured. You can define jobs on one instance that depends on jobs on the other instance.

2. (Optional) [Install a client](#) on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.

NOTE

When prompted for the application server properties, enter the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance's application server host name and port number.

The client is installed and you can issue the sendevent command to start a job on the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance.

NOTE

For more information about defining cross-instance jobs and job dependencies, see [Schedule Cross-Instances](#).

Define the External r11, r11.3, Release 11.3.5, or Release 11.3.6 Instance on the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance

Before the local AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6 instance can communicate with an external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance, you must define the external instance on the local instance. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the autorep command.

Follow these steps:

1. Log in to the local instance and do *one* of the following:

- Issue JIL in interactive mode.
- Open a JIL script in a text editor.

2. Specify the following definition:

```
insert_xinst: external_AE_instance_name
xtype: a
xmachine: external_appsrv_host_name[,external_appsrv_host_name2,...]
xcrypt_type : NONE | DEFAULT | AES
xkey_to_manager: encryption_key_for_external_instance
xport: port_number_for_external_appservers
```

NOTE

- You can specify up to four external application server computers in the xmachine attribute. Separate each external application server computer with a comma.
- External r11 instances do not support AES encryption. To support external instance job dependencies with r11, AES encryption must be disabled on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.

3. Repeat Step 2 for every external instance that you want to communicate with.

4. Do *one* of the following:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the jil command if you are using a script.

The external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance is defined on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.

Define the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance on the External r11, r11.3, Release 11.3.5, or Release 11.3.6 Instance

You must define the local AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6 instance on the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance so that the external instance can send requests or status updates to it.

Follow these steps:

1. Log in to the external instance and do *one* of the following:

- Issue JIL in interactive mode.
- Open a JIL script in a text editor.

2. Do *one* of the following:

- If the external instance is r11.3, Release 11.3.5, or Release 11.3.6, specify the following definition:

```
insert_xinst: local_AE_instance_name
xtype: a
xmachine: local_appserver_host_name[,local_appserver_host_name2,...]
xcrypt_type : NONE | DEFAULT | AES
xkey_to_manager: encryption_key_for_local_instance
xport: port_number_for_local_appservers
```

NOTE

You can specify up to four local application server computers in the xmachine attribute. Separate each external application server computer with a comma.

- If the external instance is r11, specify the following definition:

```
insert_xinst: local_AE_instance_name
```

```
xtype: a
xmachine: host_name\:port
```

- If the external instance is r11 and the local instance uses multiple application servers, add the following attribute for each application server:

```
xmachine: appserver_host_name\:port
```

NOTE

External r11 instances do not support AES encryption. To support external instance job dependencies with r11, AES encryption must be disabled on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.

3. Do *one* of the following:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the jil command if you are using a script.

The local r11.3, Release 11.3.5, or Release 11.3.6 instance is defined on the external r11, r11.3, Release 11.3.5, or Release 11.3.6 instance.

Example: Configure Cross-Instance Dependencies for Release 11.3.6 and r11 External Instances

Suppose that you want to create cross-instance job dependencies between the following AutoSys Workload Automation instances:

Figure 16: Example, Configure Cross-Instance Dependencies for Release 11_3_6 and r11 External Instances

To enable communication, you must define the external instances on the local instance and the local instance on the external instances.

On AC1, define the AC2 and AC3 external instances as follows:

```
insert_xinst: AC2
xtype: a
xmachine: machine2
xcrypt_type: AES
xkey_to_manager: 0x0102030405060708090A0B0C0D0E0F10
xport: 9002
```

```
insert_xinst: AC3
xtype: a
xmachine: machine3
xcrypt_type: NONE
xport: 9003
```

On AC2, define the AC1 local instance as follows:

```
insert_xinst: AC1
xtype: a
xmachine: machine1
xcrypt_type: NONE
xport: 9001
```

On AC3, define the AC1 local instance as follows:

```
insert_xinst: AC1
xtype: a
```



```
xmachine: machine1\:9001
```

AC3 is an r11 external instance and does not support AES encryption. To communicate with AC3, AES encryption cannot be configured on the AC1 Release 11.3.6 instance. However, without encryption, AC1 can still communicate with AC2, which is also a Release 11.3.6 instance and has AES encryption configured.

Example: Configure Cross-Instance Dependencies for a Release 11.3.6 External Instance with Multiple Application Servers

This example configures the local instance to support cross-instance dependencies with an external AutoSys Workload Automation Release 11.3.6 instance named ACE. One application server for the ACE instance resides on machineA on port 9000. Another application server resides on machineB on port 9000. ACE is defined on the local instance as follows:

```
insert_xinst: ACE
xtype: a
xmachine: machineA,machineB
xcrypt_type: AES
xkey_to_manager: 0102030405060708090A0B0C0D0E0F10
xport: 9000
```

Configure Cross-Instance Dependencies for an 4.5.1 Instance

You can create cross-instance job dependencies between your local r11.3, Release 11.3.5, or Release 11.3.6 instance and external Unicenter AutoSys JM 4.5.1 instances. You can also start jobs that are defined on the external instance. To configure cross-instance dependencies, install the r11.3, Release 11.3.5, or Release 11.3.6 application server that connects to the event server used by the 4.5.1 instance and the 4.5.1 external instance connects to the event server used by the r11.3, Release 11.3.5, or Release 11.3.6 instance. Before the instances can communicate with each other, perform a custom server installation, run the required SQL statements, and define the external instances.

NOTE

The use of the sendevent command to manually send events from one instance to a 4.5.1 external instance is not supported.

To configure cross-instance scheduling between your local r11.3, Release 11.3.5, or Release 11.3.6 instance and an 4.5.1 instance, follow these steps:

1. Prepare the 4.5.1 external instance for cross-instance dependency with an r11.3, Release 11.3.5, or Release 11.3.6 instance by installing the r11.3, Release 11.3.5, or Release 11.3.6 lightweight application server, running the required SQL statements, and defining the local r11.3, Release 11.3.5, or Release 11.3.6 instance on the external 4.5.1 instance on [UNIX](#) or [Windows](#).
2. Prepare the r11.3, Release 11.3.5, or Release 11.3.6 local instance for cross-instance dependency with an 4.5.1 instance, as follows:
 - a. Run the Required SQL Statements on the local r11.3, Release 11.3.5, or Release 11.3.6 instance on [UNIX](#) or [Windows](#).
 - b. [Define the external 4.5.1 instance on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.](#)

Cross-instance dependencies are configured. You can define jobs on one instance that depends on jobs on the other instance.

NOTE

- For more information about defining cross-instance jobs and job dependencies, see [Schedule Cross-Instances](#).
- Refer to the readme at \$AUTOSYS/dbobj/ExtDep/README.txt (UNIX) or %AUTOSYS%\dbobj\ExtDep\README.txt (Windows) to configure cross-instance dependencies for a 4.5.1, r11.3, Release 11.3.5, or Release 11.3.6 instance. However, we recommend that you refer to this article as it includes detailed instructions on how to configure cross-instance dependencies for a 4.5.1, r11, r11.3, Release 11.3.5, or Release 11.3.6 instance.

Lightweight Application Server

When you install an r11.3, Release 11.3.5, or Release 11.3.6 application server and connect it to the event server used by an external 4.5.1 instance, that application server is named *lightweight application server*. The lightweight application server detects the presence of 4.5.1 event server data and runs with limited functionality. It processes external dependency and sendevent requests from the local 11.3, Release 11.3.5, or Release 11.3.6 instance and writes job events directly into the 4.5.1 event server.

NOTE

The use of the sendevent command to manually send events from one instance to a 4.5.1 external instance is not supported.

(UNIX) Install the r11.3, Release 11.3.5, or Release 11.3.6 Lightweight Application Server, Run the Required SQL Statements, and Define the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance on the External 4.5.1 Instance

Before you can start jobs that are defined on an external 4.5.1 instance or create cross-instance job dependencies, you must run the required SQL statements on the event server the external 4.5.1 instance uses. These SQL statements create the r11.3, Release 11.3.5, or Release 11.3.6 database views and stored procedures that the lightweight application server requires to work with the 4.5.1 event server.

WARNING

Ensure that you do not install the r11.3, Release 11.3.5, or Release 11.3.6 lightweight application server and the external 4.5.1 instance on the same computer.

Follow these steps:

1. [Install the server](#).

NOTE

During the server installation, do the following:

- Perform a custom server installation and select to install only the application server component.
- The instance name of the application server must match the instance name of the 4.5.1 instance.
- Select the Install as a Lightweight Application Server check box on the Application Server Properties page.

If you select the Install as a Lightweight Application Server check box, the Set the Application Server to Start Automatically at System Startup and Start the Application Server Following the Installation check boxes are disabled.

- Enter the database information for the 4.5.1 database.

2. Run the SQL statement as follows:

- On the r11.3 server, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory and run the 11.3_Support_In_4.5.sql SQL statement on the 4.5.1 event server using native database tools like isql or sqlplus based on your database.
- On the Release 11.3.5 server, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory and run the 11.3.5_Support_In_4.5.1.sql SQL statement on the 4.5.1 event server using native database tools like isql or sqlplus based on your database.
- On the Release 11.3.6 server, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory and run the 11.3.x_Support_In_4.5.1.sql SQL statement on the 4.5.1 event server using native database tools like isql or sqlplus based on your database.

NOTE

On Oracle, you must manually replace the schema owner password and the tablespace value in the sql file.

3. Add the r11.3, Release 11.3.5, or Release 11.3.6 external instance definition, as follows:

- a. Run the shell that is sourced to use the Unicenter AutoSys JM 4.5.1 instance.
- b. Enter the following command at the operating system prompt:

```
sendevent -E STOP_DEMON
```

The scheduler completes any processing that it is performing and stops:

- c. Create a configuration file named config.EXTERNAL file in the \$AUTOUSER directory on the scheduler machine.
- d. In the configuration file, enter the following and save the file:

```
INSTANCE:EventServer=database_conn_details
```

- **INSTANCE**

Defines the name of the r11.3, Release 11.3.5, or Release 11.3.6 external instance.

- **database_conn_details**

Identifies the database for the r11.3, Release 11.3.5, or Release 11.3.6 event server.

NOTE

For Sybase, this value is defined by the database server name: database combination where the database server name is defined in the interface file. For Oracle, this value is the Oracle system identifier (ORACLE_SID).

Example: ORA:EventServer=ORCL1

4. Enter the following command at the operating system prompt:

```
eventor
```

The scheduler starts. The local r11.3, Release 11.3.5, or Release 11.3.6 instance is defined on the external 4.5.1 instance.

(Windows) Install the r11.3, Release 11.3.5, or Release 11.3.6 Lightweight Application Server, Run the Required SQL Statements, and Define the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance on the External 4.5.1 Instance

Before you can start jobs that are defined on an external 4.5.1 instance or create cross-instance job dependencies, you must run the required SQL statements on the event server the external 4.5.1 instance uses. These SQL statements create the r11.3, Release 11.3.5, or Release 11.3.6 database views and stored procedures that the lightweight application server requires to work with the 4.5.1 event server.

WARNING

Ensure that you do not install the r11.3, Release 11.3.5, or Release 11.3.6 lightweight application server and the external 4.5.1 instance on the same computer.

Follow these steps:

1. [Install the server.](#)

NOTE

During the server installation, do the following:

- Perform a custom server installation and select to install only the application server component.
- The instance name of the application server must match the instance name of the 4.5.1 instance.
- Select the Install as a Lightweight Application Server check box on the Application Server Properties page.

If you select the Install as a Lightweight Application Server check box, the Set the Application Server to Start Automatically at System Startup and Start the Application Server Following the Installation check boxes are disabled.

- Enter the database information for the 4.5.1 database.
- Validate that you are connected to a 4.5.1 database by clicking Test on the Database Test panel.

2. Run the SQL statement as follows:

- On the r11.3 server, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory and run the 11.3_Support_In_4.5.sql SQL statement on the 4.5.1 event server using native database tools like isql, osql, or sqlplus based on your database.
- On the Release 11.3.5 server, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory and run the 11.3.5_Support_In_4.5.1.sql SQL statement on the 4.5.1 event server using native database tools like isql, osql, or sqlplus based on your database.
- On the Release 11.3.6 server, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory and run the 11.3.x_Support_In_4.5.1.sql SQL statement on the 4.5.1 event server using native database tools like isql, osql, or sqlplus based on your database.

NOTE

On Oracle, you must manually replace the schema owner password and the tablespace value in the sql file.

3. Add the r11.3, Release 11.3.5, or Release 11.3.6 external instance definition, as follows:

- Log on to the machine where the Unicenter AutoSys JM 4.5.1 instance is installed and do the following:
 - Click Start, Programs, Computer Associates, Unicenter, AutoSys Job Management, Administrator. The Unicenter AutoSys Instance window opens.
 - Select the Unicenter Autosys JM 4.5.1 external instance from the AutoSys Instance drop-down list.
 - Select the Services option from the AutoSys menu. The Unicenter AutoSys Services window appears.
 - Select the scheduler service from the Service drop-down list, and click Stop Service. The scheduler stops.
- Create a configuration file named config.EXTERNAL file in the %AUTOUSER% directory on the scheduler machine.
- In the configuration file, enter the following and save the file:

```
INSTANCE:EventServer=database_conn_details
```

- **INSTANCE**

Defines the name of the r11.3, Release 11.3.5, or Release 11.3.6 external instance.

- **database_conn_details**

Identifies the database for the r11.3, Release 11.3.5, or Release 11.3.6 event server.

NOTE

For Sybase and Microsoft SQL Server, this value is defined by the database server name: database combination; however on Sybase, the database server name is defined in the interface file. For Oracle, this value is the Oracle system identifier (ORACLE_SID).

Examples: ORA:EventServer=ORCL1, MSQ:EventServer=Mach1:AEDB

4. Follow the steps mentioned here:

- a. Click Start, Programs, Computer Associates, Unicenter, AutoSys Job Management, Administrator.
The Unicenter AutoSys Instance window opens.
 - b. Select the Unicenter AutoSys JM 4.5.1 external instance from the AutoSys Instance drop-down list.
 - c. Select the Services option from the AutoSys menu.
The Unicenter AutoSys Services window appears.
 - d. Select the scheduler service from the Service drop-down list, and click Start Service.
The scheduler starts.
1. The local r11.3, Release 11.3.5, or Release 11.3.6 instance is defined on the external 4.5.1 instance.

(UNIX) Run the Required SQL Statements on the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance

Before you can start jobs that are defined on an external 4.5.1 instance or can create cross-instance job dependencies, run the required SQL statements on the event server of the local r11.3, Release 11.3.5, or Release 11.3.6 instance. These SQL statements add the 4.5.1 database views and stored procedures that the 4.5.1 scheduler requires to work with the r11.3, Release 11.3.5, or Release 11.3.6 event server.

To run the required SQL statements on the local r11.3 instance, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory on the r11.3 server and run the 45_Support_In_11.3.sql SQL statement on the r11.3 event server using native database tools like isql or sqlplus based on your database.

To run the required SQL statements on the local Release 11.3.5 instance, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory on the Release 11.3.5 server and run the 4.5.1_Support_In_11.3.5.sql SQL statement on the Release 11.3.5 event server using native database tools like isql or sqlplus based on your database.

To run the required SQL statements on the local Release 11.3.6 instance, go to \$AUTOSYS/dbobj/ExtDep/ORA or \$AUTOSYS/dbobj/ExtDep/SYB directory on the Release 11.3.6 server and run the 4.5.1_Support_In_11.3.x.sql SQL statement on the Release 11.3.6 event server using native database tools like isql or sqlplus based on your database.

NOTE

On Oracle, you must manually replace the tablespace value in the sql file.

(Windows) Run the Required SQL Statements on the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance

Before you can start jobs that are defined on an external 4.5.1 instance or can create cross-instance job dependencies, run the required SQL statements on the event server of the local r11.3, Release 11.3.5, or Release 11.3.6 instance. These SQL statements add the 4.5.1 database views and stored procedures that the 4.5.1 scheduler requires to work with the r11.3, Release 11.3.5, or Release 11.3.6 event server.

To run the required SQL statements on the local r11.3 instance, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory on the r11.3 server and run the 45_Support_In_11.3.sql SQL statement on the r11.3 event server using native database tools like isql, osql, or sqlplus based on your database.

To run the required SQL statements on the local Release 11.3.5 instance, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory on the Release 11.3.5 server and run the 4.5.1_Support_In_11.3.5.sql SQL statement on the Release 11.3.5 event server using native database tools like isql, osql, or sqlplus based on your database.

To run the required SQL statements on the local Release 11.3.6 instance, go to %AUTOSYS%\dbobj\ExtDep\ORA, %AUTOSYS%\dbobj\ExtDep\SYB, or %AUTOSYS%\dbobj\ExtDep\MSQ directory on the Release 11.3.6 server and run the 4.5.1_Support_In_11.3.x.sql SQL statement on the Release 11.3.6 event server using native database tools like isql, osql, or sqlplus based on your database.

NOTE

- On Oracle, you must manually replace the tablespace value in the sql file.
- On Microsoft SQL Server 2005 or greater, you must manually replace the following lines in the 45_Support_In_11.3.sql, 4.5.1_Support_In_11.3.5.sql, or 4.5.1_Support_In_11.3.x.sql file:

```
sp_addlogin 'anyone', 'anything'
go
sp_adduser 'anyone', 'anyone'
go
with
CREATE LOGIN anyone WITH PASSWORD = 'anything', CHECK_POLICY = OFF
go
CREATE USER anyone FOR LOGIN anyone
go
```

Define the External 4.5.1 Instance on the Local r11.3, Release 11.3.5, or Release 11.3.6 Instance

Before the local r11.3, Release 11.3.5, or Release 11.3.6 instance can communicate with an external 4.5.1 instance, define the 4.5.1 instance on the r11.3, Release 11.3.5, or Release 11.3.6 instance. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the autorep command.

Follow these steps:

1. Log in to the local r11.3, Release 11.3.5, or Release 11.3.6 instance and do one of the following:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.

2. Specify the following definition:

```
insert_xinst: external_AE_r45_instance_name
xtype: a
xmachine: lightweight_appserver_host_name[,lightweight_appserver_host_name2,...]
xport: port_number
xcrypt_type: NONE
```

NOTE

- You can specify up to four external application server computers in the xmachine attribute. Separate each external application server computer with a comma.
 - Use the AutoSys Web UI Release 11.3.5 DVD to install the Release 11.3.5 lightweight application server, which enables communication between the external 4.5.1 and Release 11.3.5 instances.
 - Use the AutoSys Web UI Release 11.3.6 DVD to install the Release 11.3.6 lightweight application server, which enables communication between the external 4.5.1 and Release 11.3.6 instances.
 - If you are using the r11 lightweight application server that was installed using the Unicenter AutoSys JM r11 DVD, AES encryption is not supported. Therefore, to support external instance job dependencies with r11, the xcrypt_type attribute of the external instance definition must be set to NONE. If you are using the r11 lightweight application server, we recommend that you upgrade it to the current release to support AES encryption.
3. Repeat Step 2 for every external 4.5.1 instance that you want to communicate with.
 4. Do *either* of the following:
 - Enter **exit** if you are using interactive mode.

OR

- Redirect the script to the jil command if you are using a script.

The external 4.5.1 instance is defined on the local r11.3, Release 11.3.5, or Release 11.3.6 instance.

Configure Cross-Instance Dependencies with CA Workload Automation EE

You can define jobs that have cross-instance dependencies on CA Workload Automation EE jobs. A AutoSys Workload Automation job with these dependencies conditionally starts based on the status of the job running on the CA Workload Automation EE instance. These dependencies are jobs that execute on an external instance but were not initiated on behalf of the local AutoSys Workload Automation instance. When the dependent job completes, status information is sent to the local AutoSys Workload Automation instance and recorded in the database.

NOTE

Bi-directional scheduling is currently not supported between AutoSys Workload Automation and CA Workload Automation EE.

CA Workload Automation EE External Instance Type

To use external job dependencies, the scheduling manager or remote machine must be defined as an *external instance* in the AutoSys Workload Automation database.

When you define CA Workload Automation EE as an external instance, you must specify **xtype: e** JIL attribute in the definition. This attribute indicates that the external scheduling manager is CA Workload Automation EE.

Encryption Between AutoSys Workload Automation and CA Workload Automation EE

Data can be transferred between AutoSys Workload Automation and CA Workload Automation EE with no encryption or with AES 128-bit encryption. Encryption occurs in two ways:

- The data received from CA Workload Automation EE
- The data sent to CA Workload Automation EE

The encryption settings on the scheduling managers must match.

Encryption of Data Received from CA Workload Automation EE

The encryption setting for AutoSys Workload Automation is determined as follows:

- On UNIX—By the UseCommAliasEncryption parameter in the configuration file.
- On Windows—By the Use AES 128-bit encryption when communicating with zOS managers check box under the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory. The cryptkey_alias.txt file stores the communication alias encryption key. AutoSys Workload Automation expects CA Workload Automation EE to encrypt the data using the key specified in the cryptkey_alias.txt file.

If you are using no encryption, AutoSys Workload Automation expects the data it receives from CA Workload Automation EE to be unencrypted.

You must specify the same AutoSys Workload Automation encryption setting in the CA Workload Automation EE AGENTDEF data set.

WARNING

You must set AES encryption only if AES encryption is also configured on CA Workload Automation EE. For more information about the encryption types that CA Workload Automation EE supports, see the CA Workload Automation EE documentation.

NOTE

If you do not know the key associated with the existing cryptkey_alias.txt file, you can regenerate the file using a new key. If AutoSys Workload Automation works with other CA Workload Automation EE external instances, the administrator must update all AGENTDEF data sets with the new key.

Example: Using No Encryption When Receiving Data from CA Workload Automation EE

Suppose that you do not want the data received from CA Workload Automation EE to be encrypted. To use no encryption, you must set the UseCommAliasEncryption parameter to 0 in the configuration file (on UNIX) or clear the Use AES 128-bit encryption when communicating with zOS managers check box under the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (on Windows).

When you define your AutoSys Workload Automation instance to CA Workload Automation EE, you must specify the NOENCRYPT operand, as follows:

```
COMMCHAN instancename_AGT ADDRESS(address) PORT(sched_aux_port) UNIX ASCII TCPIP
      PREF(2) NOENCRYPT
```

Example: Using AES 128-Bit Encryption When Receiving Data from CA Workload Automation EE

Suppose that you want CA Workload Automation EE to encrypt data with AES 128-bit encryption. To use AES encryption, you must set the UseCommAliasEncryption parameter to 2 in the configuration file (on UNIX) or select the Use AES 128-bit encryption when communicating with zOS managers check box under the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator (on Windows). After you set the communication alias encryption setting, you must generate the cryptkey_alias.txt file. When you generate the cryptkey_alias.txt file, you specify a key to secure and store in that file. On CA Workload Automation EE, you must issue the CRYPTKEY command using that same key. The CRYPTKEY command secures and stores the key using a key name on CA Workload Automation EE.

When you define your AutoSys Workload Automation instance to CA Workload Automation EE, you must specify the key name using the ENCRYPT operand, as follows:

```
COMMCHAN ACE_AGT ADDRESS(address) PORT(sched_aux_port) UNIX ASCII TCPIP PREF(2) ENCRYPT
      KEYNAME(keyname)
```

Encryption of Data Sent to CA Workload Automation EE

The encryption setting for CA Workload Automation EE is determined by the MANAGER initialization parameter in the AGENTDEF data set. The following example shows the options:

```
MANAGER NAME(manager_name) {NOENCRYPT | ENCRYPT KEYNAME(keyname) }
```

If no encryption is specified, CA Workload Automation EE expects the data it receives from AutoSys Workload Automation to be unencrypted.

If encryption is specified, CA Workload Automation EE expects AutoSys Workload Automation to encrypt the data using the key name specified in the ENCRYPT operand.

On AutoSys Workload Automation, you must specify the CA Workload Automation EE encryption setting using the xcrypt_type and xkey_to_manager attributes.

Example: Using No Encryption to Send Data to CA Workload Automation EE

Suppose that CA Workload Automation EE does not need the data transferred to be encrypted. To use no encryption, the NOENCRYPT operand is specified in the MANAGER initialization parameter of the AGENTDEF data set, as follows:


```
MANAGER NAME(manager_name) NOENCRYPT
```

When you define CA Workload Automation EE as an external instance to AutoSys Workload Automation, you must specify the `xcrypt_type`: NONE attribute, as follows:

```
insert_xinst: external_instance_name
xtype: e
xmachine: host_name
xcrypt_type : NONE
xmanager: manager_name
xport: port_number
```

Example: Using AES 128-Bit Encryption to Send Data to CA Workload Automation EE

Suppose that CA Workload Automation EE needs the data transferred to be encrypted using AES 128-bit encryption. To use AES encryption, the ENCRYPT operand is specified in the MANAGER initialization parameter of the AGENTDEF data set, as follows:

```
MANAGER NAME(manager_name) ENCRYPT KEYNAME(keyname)
```

When you define CA Workload Automation EE as an external instance to AutoSys Workload Automation, you must specify the `xcrypt_type`: AES and `xkey_to_manager` attributes, as follows:

```
insert_xinst: external_instance_name
xtype: e
xmachine: host_name
xcrypt_type: AES
xmanager: manager_name
xport: port_number
xkey_to_manager: key /* For AES encryption only */
```

Contact the CA Workload Automation EE administrator to get the key specified by ENCRYPT KEYNAME(*keyname*). You must enter the same key in the `xkey_to_manager` attribute. The *key* value must be prefixed with the hexadecimal identifier “0x” and must contain 32 characters. Valid characters are 0-9 and A-F, as shown in the following example:

```
key_to_manager: 0x0123456789ABCDEF0123456789ABCDEF
```

Configure Dependencies with CA Workload Automation EE

You can create job dependencies between AutoSys Workload Automation and CA Workload Automation EE. Before the scheduling managers can communicate with each other, you must configure them.

To configure dependencies with CA Workload Automation EE, follow these steps:

1. Configure the scheduler auxiliary listening port on [UNIX](#) or [Windows](#).
2. Set encryption for z/OS communication on [UNIX](#) or [Windows](#).
3. (AES 128-bit encryption only) [Generate an instance-wide communication alias encryption file](#).
4. Define CA Workload Automation EE as an external instance on [UNIX](#) or [Windows](#).
5. [Configure the AGENTDEF data set on CA Workload Automation EE](#).

6. [Verify the setup.](#)

After you set up this configuration, you can create job dependencies between the instances.

(UNIX) Configure the Scheduler Auxiliary Listening Port

The scheduler communicates with CA Workload Automation EE and Workload Automation Agent for z/OS using non-SSA ports. Therefore, you must disable port multiplexing and SSL encryption for the scheduler auxiliary listening port.

NOTE

If the port is already configured, skip this procedure.

Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Edit the following parameter in the configuration file as follows, and save the file:


```
SchedAuxiliaryListeningPort=sch_port
```

 - ***sch_port***
Defines the port number the scheduler uses to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.
The scheduler auxiliary listening port is defined.
3. Enter the following commands at the operating system prompt:


```
cd $CSAM_SOCKADAPTER/bin
csamconfigedit Port=sch_port EnableSSL=False EnablePmux=False
```

 - ***sch_port***
Specifies the port number to configure. You must specify the same scheduler auxiliary listening port that you specified in the SchedAuxiliaryListeningPort parameter in the configuration file.
Port multiplexing and SSL encryption are disabled for the specified scheduler auxiliary listening port.
4. Enter the following command at the operating system prompt:


```
unisrvcntr stop waae_sched.$AUTOSERV
```

The scheduler stops.
5. Enter the following command at the operating system prompt:


```
csampmux stop
```

The csampmux process stops.
6. Enter the following command at the operating system prompt:


```
unisrvcntr start waae_sched.$AUTOSERV
```

The scheduler starts. The scheduler auxiliary listening port is configured.

NOTE

For more information about the SchedAuxiliaryListeningPort parameter, see [Configure a Scheduler](#).

(Windows) Configure the Scheduler Auxiliary Listening Port

The scheduler communicates with CA Workload Automation EE and Workload Automation Agent for z/OS using non-SSA ports. Therefore, you must disable port multiplexing and SSL encryption for the scheduler auxiliary listening port.

NOTE

If the port is already configured, skip this procedure.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Click the Scheduler icon on the toolbar.
The Scheduler - AutoSys Workload Automation Administrator window appears.
3. Enter the port number in the Auxiliary Listening Port field in the Communication Ports pane, and click Apply.
The scheduler auxiliary listening port is defined. The scheduler uses this port to communicate with CA Workload Automation EE and the agent on z/OS. This port is used for all non-SSA communication.
4. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
5. Change to the %CSAM_SOCKADAPTER%\bin folder, and enter the following command:

```
csamconfigedit Port=sch_port EnableSSL=False EnablePmux=False
```

– ***sch_port***

Specifies the port number to configure. You must specify the same scheduler auxiliary listening port that you specified in the Auxiliary Listening Port field on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.

Port multiplexing and SSL encryption are disabled for the specified scheduler auxiliary listening port.

6. Do the following:
 - a. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 - b. Select an instance from the Instance drop-down list.
 - c. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
 - d. Right-click the scheduler service, and click Stop.
The scheduler stops.
 - e. Right-click the scheduler service, and click Start.
The scheduler starts.

The scheduler auxiliary listening port is configured.

NOTE

For more information about the AutoSys Workload Automation Administrator, see [Configuring AutoSys Workload Automation](#).

(UNIX) Set Encryption for z/OS Communication

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory. The cryptkey_alias.txt file stores the communication alias encryption key.

WARNING

You must set AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, you must disable AES 128-bit encryption and use no encryption for z/OS communication. To disable AES encryption, you must set the UseCommAliasEncryption parameter to 0 in the configuration file.
- If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting.

Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
```

The scheduler and the application server stop.

3. Edit the following parameter in the configuration file, and save the file:

```
UseCommAliasEncryption=0|2
```

- **0**
Specifies that no encryption is used.
- **2**
Specifies that AES encryption is used to encrypt data.

NOTE

If you set the UseCommAliasEncryption parameter to 2, you must generate the cryptkey_alias.txt file and specify the communication alias encryption key using the -a option of the as_config command.

4. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. The encryption for z/OS communication is set.

NOTE

On Windows, you can select the equivalent value using the Use AES 128-bit encryption when communicating with zOS managers check box under the zOS Encryption tab on the Instance - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator. For information about setting encryption for z/OS communication on Windows, see [Configure an Instance](#).

(Windows) Set Encryption for z/OS Communication

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate an instance-wide communication alias encryption file (cryptkey_alias.txt) in the \$AUTOUSER.instance_name (on UNIX) or %AUTOUSER%.instance_name (on Windows) directory. The cryptkey_alias.txt file stores the communication alias encryption key.

WARNING

You must set AES encryption for z/OS communication only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload

Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- The current release of Workload Automation Agent for z/OS is r2.0. This release of the agent does not support AES 128-bit encryption. To run z/OS jobs using this agent, you must disable AES 128-bit encryption and use no encryption for z/OS communication. To disable AES encryption, you must clear the Use AES 128-bit encryption when communicating with zOS managers check box.
- If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the encryption setting.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select the instance you want to set encryption for from the Instance drop-down list.
3. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the scheduler and select Stop. Repeat this action for the application server.
The scheduler and application server stop.
5. Click the Instance icon on the toolbar.
The Instance - AutoSys Workload Automation Administrator window appears.
6. Click the zOS Encryption tab, and select the Use AES 128-bit encryption when communicating with zOS managers check box.
The Hexadecimal Key and Verify Hexadecimal Key fields are enabled.

NOTE

If you clear the Use AES 128-bit encryption when communicating with zOS managers check box, AutoSys Workload Automation uses no encryption for z/OS communication.

7. Enter the encryption key, confirm the encryption key in the Verify Hexadecimal Key field, and click Apply.

NOTE

The encryption key must be a hexadecimal string of 32 characters.

The zOS Encryption Status field displays the current encryption set for the instance.

8. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
9. Right-click the scheduler and select Start. Repeat this action for the application server.
The scheduler and application server start. The encryption for z/OS communication is set.

Generate an Instance-Wide Communication Alias Encryption File

AutoSys Workload Automation uses no encryption or AES 128-bit encryption to exchange data with CA Workload Automation EE and Workload Automation Agent for z/OS. If you are using AES encryption, you must generate the instance-wide communication alias encryption file (cryptkey_alias.txt).

The cryptkey_alias.txt file stores the communication alias encryption key. The cryptkey_alias.txt file is located in the \$AUTOUSER.*instance_name* (on UNIX) or %AUTOUSER%.*instance_name* (on Windows) directory.

WARNING

Do this procedure only if AES encryption is also configured on CA Workload Automation EE or the agent on z/OS. For more information about the encryption types that CA Workload Automation EE and the agent on z/OS support, see the CA Workload Automation EE and Workload Automation Agent for z/OS documentation.

NOTE

- A AutoSys Workload Automation instance can have only one `cryptkey_alias.txt` file. Before you do this procedure, check whether the file already exists. If the file exists, skip this procedure. You must provide the key associated with that file to the CA Workload Automation EE or agent administrator. They need the key to configure the AGENTDEF data set.
- If you do not know the key associated with the existing `cryptkey_alias.txt`, you can regenerate the file using a new key. If AutoSys Workload Automation works with other CA Workload Automation EE external instances or the agents on z/OS, the administrator must update all AGENTDEF data sets with the new key.

Follow these steps:

1. Do *one* of the following:
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
 - (UNIX) Log in to AutoSys Workload Automation as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the instance command prompt (Windows) or the operating system prompt (UNIX):

```
as_config -a key
```

– **key**

Specifies the communication alias encryption key. You must prefix the hexadecimal identifier 0x to this value.

Limits: Must contain 32 characters; valid characters are 0-9 and A-F.

NOTE

This key must match the key stored in the ENCRYPT KEYNAME(keyname) parameter in the AGENTDEF data set of CA Workload Automation EE or the agent on z/OS.

The communication alias encryption file (`cryptkey_alias.txt`) is generated with the encryption key. AES 128-bit encryption is used.

(UNIX) Define CA Workload Automation EE as an External Instance

Before you can create job dependencies between AutoSys Workload Automation and CA Workload Automation EE, you must define CA Workload Automation EE as an external instance. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the `autorep` command.

Follow these steps:

1. Log in to the AutoSys Workload Automation instance and do *one* of the following:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.
2. Specify the following definition:

```
insert_xinst: external_instance_name
xtype: e
xmachine: host_name
xcrypt_type : NONE | AES
xmanager: manager_name
xport: port_number
```

NOTE

The `xcrypt_type` value must match the encryption setting specified in the MANAGER initialization parameter in the CA Workload Automation EE AGENTDEF data set.

3. If `xcrypt_type` is set to AES, specify the following additional attribute:

```
xkey_to_manager: key
```

- **key**

Specifies the encryption key defined on CA Workload Automation EE. This value must match the key specified by ENCRYPT KEYNAME(keyname) in the CA Workload Automation EE AGENTDEF data set. You must prefix the hexadecimal identifier 0x to this value.

Limits: Must contain 32 characters; valid characters are 0-9 and A-F

Example: 0x0123456789ABCDEF0123456789ABCDEF

4. Do *one* of the following:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the `jil` command if you are using a script.

The external instance is defined on AutoSys Workload Automation.

NOTE

- Do this procedure for every CA Workload Automation EE instance that you want to create external job dependencies for.
- If you modify external instance entries while the AutoSys Workload Automation scheduler is active, the scheduler handles the modifications in real time. You do not have to recycle the scheduler to create, update, and delete external instance entries.

(Windows) Define CA Workload Automation EE as an External Instance

Before you can create job dependencies between AutoSys Workload Automation and CA Workload Automation EE, you must define CA Workload Automation EE as an external instance. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the `autorep` command.

Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
2. Enter **jil** at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Specify the following definition:

```
insert_xinst: external_instance_name
xtype: e
xmachine: host_name
xcrypt_type : NONE | AES
xmanager: manager_name
xport: port_number
```

NOTE

The `xcrypt_type` value must match the encryption setting specified in the MANAGER initialization parameter in the CA Workload Automation EE AGENTDEF data set.

4. If `xcrypt_type` is set to AES, specify the following additional attribute:

```
xkey_to_manager: key
```

– **key**

Specifies the encryption key defined on CA Workload Automation EE. This value must match the key specified by ENCRYPT KEYNAME(*keyname*) in the CA Workload Automation EE AGENTDEF data set. You must prefix the hexadecimal identifier 0x to this value.

Limits: Must contain 32 characters; valid characters are 0-9 and A-F

Example: 0x0123456789ABCDEF0123456789ABCDEF

5. Enter **exit**.

The data is loaded into the database. The external instance is defined on AutoSys Workload Automation.

NOTE

- Do this procedure for every CA Workload Automation EE instance that you want to create external job dependencies for.
- If you modify external instance entries while the AutoSys Workload Automation scheduler is active, the scheduler handles the modifications in real time. You do not have to recycle the scheduler to create, update, and delete external instance entries.

Configure the AGENTDEF Data Set on CA Workload Automation EE

For communication to occur between AutoSys Workload Automation and CA Workload Automation EE, you must configure the AGENTDEF data set on CA Workload Automation EE. The parameters in the AGENTDEF data set must match the settings defined on AutoSys Workload Automation.

To configure the AGENTDEF data set, add the following entry:

```
COMMCHAN alias_name ADDRESS(sch_ip_address) PORT(sch_aux_port) platform +
ASCII TCPIP PREF(2) {NOENCRYPT|ENCRYPT KEYNAME(keyname) }
```

The following table describes the operands that are not self-explanatory and their corresponding AutoSys Workload Automation settings:

AGENTDEF Operand	Description	Corresponding AutoSys Workload Automation Setting
COMMCHAN <i>alias_name</i>	Specifies the name associated with the encryption data between AutoSys Workload Automation and CA Workload Automation EE.	<i>INSTANCENAME_AGT</i> This is the communication alias for the AutoSys Workload Automation scheduler. The value must be in uppercase. <i>INSTANCENAME</i> is the name of the AutoSys Workload Automation instance.
ADDRESS(<i>sch_ip_address</i>)	Specifies the host name or the IP address of the computer where the AutoSys Workload Automation scheduler is installed.	None
PORT(<i>sch_aux_port</i>)	Specifies the port number that the AutoSys Workload Automation scheduler uses for all non-SSA communication.	(Windows) Auxiliary Listening Port field on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator. (UNIX) SchedAuxiliaryListeningPort parameter in the configuration file.

<i>platform</i>	Specifies whether the AutoSys Workload Automation instance is installed on UNIX or Windows. Options are UNIX or NT.	None
NOENCRYPT ENCRYPT KEYNAME(<i>keyname</i>)	Specifies the type of encryption used and the key that AutoSys Workload Automation expects the data to be encrypted with. For AES encryption, you must secure and store the key in a key name on CA Workload Automation EE. To generate the key name, use the CRYPTKEY command.	If NOENCRYPT is specified, xcrypt_type: NONE must be defined in the external instance definition, and the cryptkey_alias.txt file must not exist. If ENCRYPT KEYNAME(<i>keyname</i>) is specified, xcrypt_type: AES must be defined in the external instance definition. The same key must be stored in the cryptkey_alias.txt file in the \$AUTOUSER. <i>instance_name</i> (UNIX) or %AUTOUSER%. <i>instance_name</i> (Windows) directory.

NOTE

For more information about the AGENTDEF data set on CA Workload Automation EE, see the *CA Workload Automation EE Installation and Configuration Guide*.

Verify the Setup

After you configure cross-instance support on AutoSys Workload Automation and CA Workload Automation EE, verify the configuration is set up properly.

Follow these steps:

1. Enter the following command at the operating system prompt:

```
autorep -X EE_external_instance [-q] [-n]
```

– **-X EE_external_instance**

Specifies the external instance that you defined for CA Workload Automation EE.

The autorep command is issued and a report is generated. If CA Workload Automation EE is successfully defined as an external instance, the report output is similar to the following:

Name	Type	Server	Port
AES	e	server1	16190

2. Enter the following command:

```
autosyslog -e
```

The scheduler log file is displayed. If the configuration is set up properly, the report output is similar to the following:

```
[10/21/2009 10:18:18] CAUAJM_I_40245 EVENT: REFRESH_EXTINST
[10/21/2009 10:18:18] CAUAJM_I_50407 Reading external instance information
[10/21/2009 10:18:18] CAUAJM_I_50408 Instance=[AES]: Type=[e] Server=[server1]
Port=[16190] Manager Alias=[WAEEMGR]
```

Configure Cross-Instance Dependencies with CA UJMA and AutoSys Workload Automation Connect Option

You can define jobs that have dependencies on jobs running on external machines (*external job dependencies*). These machines must be defined as *external instances* on AutoSys Workload Automation. A AutoSys Workload Automation job with these dependencies conditionally starts based on the status of the job running on the other instance. These dependencies are jobs that execute on an external instance but were not initiated on behalf of the local AutoSys Workload Automation instance. When the dependent job completes, status information is sent to the local AutoSys Workload Automation instance and recorded in the database. For example, a job in one instance can be defined to start based on the status of jobs running on a mainframe system.

Cross-Platform Scheduling Requirements

To work with CA UJMA and AutoSys Workload Automation Connect Option, the following components must be installed on the AutoSys Workload Automation computer:

- AutoSys Workload Automation scheduler (the component that communicates with AutoSys Workload Automation Connect Option and CA UJMA)
- CAICCI

NOTE

For more information about installing CAICCI, see [CA Common Components Installation](#).

You must have one of the following Broadcom products installed on the external machine that AutoSys Workload Automation works with:

Software on the External Machine	Required Integration Software	Environment
CA UJMA	None	Distributed
CA Job Management Option	CA UJMA	Distributed
CA Jobtrac Job Management	CA UJMA or AutoSys Workload Automation Connect Option	Mainframe
CA Scheduler Job Management	CA UJMA or AutoSys Workload Automation Connect Option	Mainframe
CA Workload Automation EE	None	Mainframe
CA Workload Automation SE	CA UJMA or AutoSys Workload Automation Connect Option	Mainframe

NOTE

- CA UJMA requires TCP/IP.
- For more information about configuring cross-instance support on the external machine, see the documentation for the Broadcom product installed on that machine.

CA UJMA

CA UJMA (CA Universal Job Management Agent) can run on distributed platforms as a standalone agent that executes binaries and scripts in a method similar to the agent for AutoSys Workload Automation.

Additionally, all CA mainframe scheduling products can behave as CA UJMA agents, which lets the AutoSys Workload Automation scheduler run jobs through those mainframe schedulers. In this scenario, the job being run is a named job known to the scheduler and is not a command or script.

Similarly, the AutoSys Workload Automation scheduler can appear as a CA UJMA agent to other Broadcom scheduler managers in the enterprise. In this scenario, the job submitted to the CA UJMA interface is a job defined to AutoSys Workload Automation and is not a command or script.

You cannot create job dependencies on the mainframe using CA UJMA. To use mainframe job dependencies, you must install AutoSys Workload Automation Connect Option on the same computer as the mainframe scheduling manager.

The scheduler uses CAICCI to communicate with CA UJMA agents.

NOTE

CA UJMA was formerly named Unicenter Universal Job Management Agent (UUJMA).

AutoSys Workload Automation Connect Option

AutoSys Workload Automation Connect Option lets the AutoSys Workload Automation scheduler run jobs on mainframe scheduling managers. It also lets you create dependencies between jobs running on AutoSys Workload Automation and the mainframe scheduling manager.

CA UJMA and AutoSys Workload Automation Connect Option Considerations

Consider the following points when you schedule jobs across platforms:

Maintaining cross-platform data in high availability mode:

If you are running AutoSys Workload Automation in high availability mode, ensure the following so that job statuses and dependencies are not lost when the shadow scheduler takes over:

- – The PRIMARYCCISYSID environment variable is correctly set on the primary and secondary schedulers and on the CA UJMA computers.
- Jobs and external dependencies were sent to the external instance with proper release levels to support PRIMARYCCISYSID.

Exit codes in CA Job Management Option:

When running a job from CA Job Management Option, you may need to modify the default fail codes currently set for the CA Job Management Option-defined job. Exit codes 2 through 99 are defined as the default fail codes for CA Job Management Option jobs. Therefore, an exit code of 0 to 1 indicates success. When you run a job from CA Job Management Option that executes a job in AutoSys Workload Automation and the job fails with an exit code 1 (for example, bad command), the AutoSys Workload Automation job ends with a status of FAILURE. However, the CA Job Management Option-defined job ends with a status of SUCCESS or COMPLETE. You must modify the fail codes to accommodate the differences in how success and failure are interpreted between the two scheduling managers. That is, you must define exit codes 1 through 99 as the fail codes for the CA Job Management Option-defined job and define only an exit code of 0 to indicate success.

Multiple AutoSys Workload Automation instances running on one computer:

If more than one instance of AutoSys Workload Automation runs on a single computer and you plan to activate the Cross-Platform Interface, only one instance of AutoSys Workload Automation can run with the Cross-Platform Scheduling option set to a value of 2. Only one instance can function as an agent. That is, only one instance can accept job submissions from an external scheduling manager.

chase and autoping commands:

The chase and autoping commands return limited information about AutoSys Workload Automation Connect Option and CA UJMA jobs and computers.

Remote user authentication:

Remote user authentication is not supported for jobs running on AutoSys Workload Automation Connect Option. For CA UJMA jobs, remote user authentication is performed using the owner name associated with the job.

CHANGE_PRIORITY and SEND_SIGNAL events:

You cannot execute the CHANGE_PRIORITY and SEND_SIGNAL events on AutoSys Workload Automation Connect Option and CA UJMA jobs and computers.

CA UJMA and AutoSys Workload Automation Connect Option External Instance Types

To use external job dependencies, the scheduling manager or remote machine must be defined as an *external instance* in the AutoSys Workload Automation database.

When you define CA UJMA or AutoSys Workload Automation Connect Option as an external instance, you must specify one of the following JIL attributes in the definition:

- **xtype: c**

Indicates that AutoSys Workload Automation Connect Option is installed with the external scheduling manager. AutoSys Workload Automation Connect Option can be installed on the mainframe and supports cross-platform jobs and job dependencies. It lets you submit job requests to and receive job submissions from the following mainframe scheduling managers:

- CA Jobtrac Job Management
- CA Scheduler Job Management
- CA Workload Automation SE

The AutoSys Workload Automation scheduler uses CAICCI to communicate with AutoSys Workload Automation Connect Option.

- **xtype: u**

Indicates that CA UJMA is installed with the external scheduling manager or on the remote machine. CA UJMA can be installed on the mainframe, UNIX, and Windows. It lets you submit job requests to the remote machine where CA UJMA is installed. It lets you submit job requests to and receive job submissions from the following scheduling managers:

- CA Job Management Option
- CA Jobtrac Job Management
- CA Scheduler Job Management
- CA Workload Automation SE

The AutoSys Workload Automation scheduler uses CAICCI to communicate with CA UJMA.

NOTE

Unlike AutoSys Workload Automation Connect Option, CA UJMA does not let you define cross-instance job dependencies on the mainframe. To define job dependencies on the mainframe, you must install AutoSys Workload Automation Connect Option on the same computer as the mainframe scheduling manager.

Configure Dependencies with CA UJMA and AutoSys Workload Automation Connect Option

You can create external job dependencies between AutoSys Workload Automation and a machine running another Broadcom scheduling manager. That external machine can run on a different platform, including mainframe.

Before you can create external job dependencies, you must configure AutoSys Workload Automation.

To configure dependencies with CA UJMA and AutoSys Workload Automation Connect Option, follow these steps:

1. Ensure that the external machine that the dependent job runs on meets the [requirements for cross-instance scheduling](#).
2. Enable bi-directional scheduling on AutoSys Workload Automation on [UNIX](#) or [Windows](#).
3. Configure and start CAICCI on [UNIX](#) or [Windows](#).

4. (High availability environments only) Configure failover support for cross-instance scheduling on [UNIX](#) or [Windows](#).
5. Define the machine as an external instance on AutoSys Workload Automation on [UNIX](#) or [Windows](#).

After you set up this configuration, you can create job dependencies between AutoSys Workload Automation and the external instance.

(UNIX) Enable Bi-Directional Scheduling on AutoSys Workload Automation

To schedule jobs or create job dependencies on another Broadcom scheduling manager, you must enable the AutoSys Workload Automation instance to support bi-directional scheduling.

Follow these steps:

1. On the AutoSys Workload Automation instance, log in as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command:

```
sendevent -E STOP_DEMON
```

The scheduler completes any processing it is currently performing and stops.

3. Edit the following parameter in the configuration file, and save the file:

```
CrossPlatformScheduling=2
```

4. Enter the following command:

```
eventor
```

The scheduler starts.

5. Open the scheduler log (event_demon.\$AUTOSERV) and check that it contains the following messages:

```
CAUAJM_I_40005 Cross Platform Interface Initialization in progress
```

```
CAUAJM_I_40015 Cross Platform Interface is now active
```

The cross-platform scheduling interface is active and bi-directional (outbound and inbound) scheduling is enabled. The instance can send job requests to an agent and receive job requests from a scheduling manager.

NOTE

For more information about the CrossPlatformScheduling parameter, see [Configure a Scheduler](#). For more information about the configuration file (\$AUTOUSER/config.\$AUTOSERV), see [Configuring AutoSys Workload Automation](#).

(Windows) Enable Bi-Directional Scheduling on AutoSys Workload Automation

To schedule jobs or create job dependencies on another Broadcom scheduling manager, you must enable the AutoSys Workload Automation instance to support bi-directional scheduling.

Follow these steps:

1. On the local AutoSys Workload Automation instance, log in as the EXEC superuser and enter the following command at the instance command prompt:

```
sendevent -E STOP_DEMON
```

The scheduler completes any processing it is currently performing and stops.

2. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
3. Select an instance from the Instance drop-down list.
4. Click the Scheduler icon on the toolbar.
The Scheduler - AutoSys Workload Automation Administrator window appears.
5. Select Manager and Agent in the Cross Platform Scheduling pane, and click Apply.
6. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears.

7. Right-click the scheduler service, and click Start.

The scheduler starts.

8. Open the scheduler log (event_demon.%AUTOSERV%) and check that it contains the following messages:

```
CAUAJM_I_40005 Cross Platform Interface Initialization in progress
CAUAJM_I_40015 Cross Platform Interface is now active
```

The cross-platform scheduling interface is active and bi-directional (outbound and inbound) scheduling is enabled. The instance can send job requests to an agent and receive job requests from a scheduling manager.

NOTE

For more information about the Cross Platform Scheduling pane on the Scheduler - AutoSys Workload Automation Administrator window, see [Configure a Scheduler](#). For more information about the scheduler log file, see [Maintain the Scheduler](#).

(UNIX) Configure and Start CAICCI

CAICCI is the communication layer that connects applications running on mainframe, UNIX, Windows, and other operating systems. You must configure and start CAICCI on AutoSys Workload Automation before you can schedule jobs on the other systems.

Follow these steps:

1. Log in as root on the AutoSys Workload Automation instance.
2. Open the \$CAIGLBL0000/cci/config/local_host/ccirmtd.prf file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:


```
LOCAL = local_host local_host 32768 startup
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CAICCI is configured.

4. Issue the following command:

```
unicntrl start cci
```

CAICCI restarts. The updated configuration settings are applied.

(Windows) Configure and Start CAICCI

CAICCI is the communication layer that connects applications running on mainframe, UNIX, Windows, and other operating systems. You must configure and start CAICCI on AutoSys Workload Automation before you can use cross-platform scheduling.

NOTE

You do not need to configure the ccirmtd file on a Windows to Windows platform environment. However, you must configure the ccirmtd file on any cross-platform environment (for example, Windows to UNIX, UNIX to mainframe, and mainframe to Windows).

Follow these steps:

1. On the AutoSys Workload Automation instance, log in with an account that has administrative privileges.
2. Locate and open the ccirmtd.rc file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:

```
LOCAL = local_host local_host 32768 startup
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CAICCI is configured.

4. Issue the following command:

```
ccicntrl start
```

CAICCI restarts. The updated configuration settings are applied.

(UNIX) Configure Failover Support for Cross-Instance Scheduling

If you are using high availability mode, you can define the aliased CAICCI system ID on the local AutoSys Workload Automation instance. The cross-platform interface of AutoSys Workload Automation uses this CAICCI system ID to communicate with remote mainframe or UJMA nodes during failover. If the primary scheduler shuts down or becomes unreachable, all communication on the secondary scheduler proceeds as normal. Any statuses currently residing on the remote computers (mainframe or UJMA) are dispatched to the secondary scheduler computer for processing.

Follow these steps:

1. Open all the AutoSys Workload Automation environment scripts, add the following variable, and save the file.

```
PRIMARYCCISYSID = cci_system_id
export PRIMARYCCISYSID
```

– *cci_system_id*

Defines the aliased CAICCI system ID specified in the ccirmt configuration file.

NOTE

This variable is initially configured during scheduler installation.

NOTE

The environment scripts are as follows:

- \$AUTOUSER/autosys.sh.hostname—for Bourne shell users
- \$AUTOUSER/autosys.csh.hostname—for C shell users
- \$AUTOUSER/autosys.ksh.hostname—for Korn shell users
- \$AUTOUSER/autosys.bash.hostname—for Bash shell users

2. Restart the scheduler.

Failover support is configured for cross-instance scheduling.

(Windows) Configure Failover Support for Cross-Instance Scheduling

If you are using high availability mode, you can define the aliased CAICCI system ID on the local AutoSys Workload Automation instance. The cross-platform interface of AutoSys Workload Automation uses this CAICCI system ID to communicate with remote mainframe or UJMA nodes during failover. If the primary scheduler shuts down or becomes unreachable, all communication on the secondary scheduler proceeds as normal. Any statuses currently residing on the agent computers (mainframe or UJMA) are sent to the secondary scheduler computer for processing.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select your instance from the Instance drop-down list.
3. Click the System icon on the toolbar.
The System - AutoSys Workload Automation Administrator window appears.
4. Double-click the PRIMARYCCISYSID variable in the Environment Variables pane.

The window refreshes to display PRIMARYCCISYSID variable in the Variable field and its value in the Value field.

5. Enter the aliased CAICCI system ID in the Value field, and click Set.

NOTE

You can find the aliased CAICCI system ID in the ccirmtcd configuration file. The PRIMARYCCISYSID variable is initially configured during scheduler installation.

The PRIMARYCCISYSID variable is modified and displayed in the Environment Variables pane.

NOTE

For more information about the Environment Variables pane on the System - AutoSys Workload Automation Administrator window, see [Configuring AutoSys Workload Automation](#).

(UNIX) Define the Machine as an External Instance on AutoSys Workload Automation

Before you can create external job dependencies, you must define the machine where the dependent job runs as an external instance on AutoSys Workload Automation. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the autorep command.

Follow these steps:

1. Log in to the AutoSys Workload Automation instance and do one of the following:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.

2. Specify the following definition:

```
insert_xinst: external_instance_name
xtype: c | u
xmachine: host_name
xcrypt_type : NONE | DEFAULT | AES
xport: port_number
```

– **xtype: c | u**

Specifies the external instance type. Options include the following:

- c—Identifies a AutoSys Workload Automation Connect Option instance.
- u—Identifies a CA UJMA or CA NSM instance.

3. If xcrypt_type is set to AES, specify the following additional attribute:

```
xkey_to_manager: key
```

4. Repeat Steps 2 and 3 for each external instance that you want to communicate with.
5. Do *one* of the following:
 - Enter *exit* if you are using interactive mode.
 - Redirect the script to the jil command if you are using a script.

The external instance is defined on AutoSys Workload Automation.

NOTE

If you modify external instance entries while the AutoSys Workload Automation scheduler is active, the scheduler handles the modifications in real time. You do not have to recycle the scheduler to create, update, and delete external instance entries.

(Windows) Define the Machine as an External Instance on AutoSys Workload Automation

Before you can create external job dependencies, you must define the machine where the dependent job runs as an external instance on AutoSys Workload Automation. All external instance entries are stored in the AutoSys Workload Automation event server and can be reported using the `autorep` command.

Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
2. Enter `jil` at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Specify the following definition:

```
insert_xinst: external_instance_name
xtype: c | u
xmachine: host_name
xcrypt_type : NONE | DEFAULT | AES
xport: port_number
```

– xtype: c | u

Specifies the external instance type. Options include the following:

- `c`—Identifies a AutoSys Workload Automation Connect Option instance.
- `u`—Identifies a CA UJMA or CA NSM instance.

4. If `xcrypt_type` is set to AES, specify the following additional attribute:

```
xkey_to_manager: key
```

5. Repeat Steps 2 and 3 for each external instance that you want to communicate with.
6. Enter `exit`.
The data is loaded into the database. The external instance is defined on AutoSys Workload Automation.

NOTE

If you modify external instance entries while the AutoSys Workload Automation scheduler is active, the scheduler handles the modifications in real time. You do not have to recycle the scheduler to create, update, and delete external instance entries.

Example Configure Cross-Instance Scheduling for CA Workload Automation SE

Suppose that you want to start jobs between a local AutoSys Workload Automation instance and an external CA Workload Automation SE instance. You also want to create job dependencies between the instances, so AutoSys Workload Automation Connect Option is installed on the mainframe.

Follow these steps:

1. Enable bi-directional scheduling on the local AutoSys Workload Automation instance.
2. Configure and start CAICCI.
3. Define the CA Workload Automation SE machine on AutoSys Workload Automation as follows, where `remote3` is the CA Workload Automation SE host name:

```
insert_machine: remote3
type: c
```

The type `c` indicates that AutoSys Workload Automation uses AutoSys Workload Automation Connect Option installed on the mainframe to submit jobs to the mainframe and create cross-instance dependencies.

NOTE

Alternatively, if you do not need to create cross-instance dependencies, you can submit jobs from AutoSys Workload Automation directly to the mainframe. In this situation, you define the CA Workload Automation SE machine as follows:

```
insert_machine: remote3
type: u
```

4. Define a corresponding external instance for CA Workload Automation SE on AutoSys Workload Automation as follows:

```
insert_xinst: SE7
xtype: c
xmachine: remote3
```

5. Define the AutoSys Workload Automation instance on CA Workload Automation SE.

After cross-platform scheduling is configured, you can define and submit jobs as follows:

- Suppose that you want to run a job named SE7JOBNM that is defined on CA Workload Automation SE. The job has no starting conditions and you want to submit it directly to CA Workload Automation SE. You can define the following Command job on AutoSys Workload Automation:

```
insert_job: SE_job1
job_type: CMD
command: SE7JOBNM
machine: remote3 *
date_conditions: 1
days_of_week: all
start_mins: 25
```

NOTE

The machine definition for remote3 must have type **u**.

- Suppose that you want to run a job named SE7JOBNM that is defined on CA Workload Automation SE. The job has no starting conditions and you want to submit the job through AutoSys Workload Automation Connect Option, which is installed on CA Workload Automation SE. You can define the following Command job on AutoSys Workload Automation:

```
insert_job: SE_job2
job_type: CMD
command: auto_cnct -a remote3 -j SE7JOBNM -s CA7 -c RUN -d
machine: remote3 *
date_conditions: 1
days_of_week: all
start_mins: 25
```

NOTE

The machine definition for remote3 must have type **c**, which indicates AutoSys Workload Automation Connect Option.

- Suppose that you want to submit two jobs with external dependencies. The first job depends on the JB5MINS job on the mainframe. The second job depends on the JB5HRS job on the mainframe. The jobs do not have starting conditions. You can define the following Command jobs on AutoSys Workload Automation:

```
insert_job: test_dep1
job_type: CMD
command: sleep 100
condition: success(JB5MINS^RMT)
machine: remote3
```

```
insert_job: test_dep2
job_type: CMD
command: sleep 100
condition: success(JB5HRS^RMT)
machine: remote3
```

NOTE

RMT is defined as an external instance on AutoSys Workload Automation, and the machine definition for remote3 must have type **c**, which indicates AutoSys Workload Automation Connect Option. You can also define jobs as a combination of both Command jobs and external dependencies.

- Suppose that you want to run a job named ASYS7002 that is defined on CA Workload Automation SE. The job must run after the JB5HRS completes. The job has no starting conditions, and you want to submit it through AutoSys Workload Automation Connect Option, which is installed on CA Workload Automation SE. You can define the following Command job on AutoSys Workload Automation:

```
insert_job: SEjob4
job_type: CMD
command: auto_cnct -a remote3 -j ASYS7002 -s CA7 -c RUN -d
machine: remote3
condition: success(JB5HRS^RMT)
```

NOTE

RMT is defined as an external instance on AutoSys Workload Automation, and the machine definition for remote3 must have type **c**, which indicates AutoSys Workload Automation Connect Option.

Configuring Cross-Platform Scheduling

Cross-Platform Scheduling

Cross-platform scheduling lets you schedule and reroute jobs between AutoSys Workload Automation and other machines running on different platforms, including mainframe.

To use cross-platform scheduling, required components must be installed on the AutoSys Workload Automation computer and on the external machine that AutoSys Workload Automation works with. The scheduling manager or remote machine are defined as an *external instance* in the AutoSys Workload Automation database.

When the scheduler receives the cross-platform start job requests, it requests the application server to authenticate and authorize that request. For the job to execute successfully, you must have EXECUTE access for the following resource classes:

- as-appl
- as-group
- as-machine
- as-job

Bi-Directional Scheduling

AutoSys Workload Automation supports *bi-directional scheduling*, which lets you start jobs from remote machines (inbound) or submit jobs on remote machines (outbound).

With *inbound job scheduling*, AutoSys Workload Automation acts as an agent and accepts job submissions from remote machines or other scheduling managers (such as CA Jobtrac Job Management and CA Workload Automation SE). The jobs are defined and run on the AutoSys Workload Automation instance that is acting as an agent.

With *outbound job scheduling*, AutoSys Workload Automation acts as a scheduling manager and sends job submissions to remote machines. The jobs are defined on the AutoSys Workload Automation instance that is acting as a scheduling manager. The jobs run on the remote machine or other scheduling manager.

For example, a Linux Oracle instance can initiate jobs in a Windows Microsoft SQL Server instance, or a Windows Microsoft SQL Server instance can initiate jobs in a AIX Oracle instance. You can add additional instances, such as Sybase, AIX Oracle, or HP Oracle instance, to the environment.

The AutoSys Workload Automation cross-platform interface controls the bi-directional scheduling mode. You can configure the cross-platform interface to enable the following modes:

- Outbound job scheduling
- Inbound and outbound job scheduling (bi-directional scheduling)
- No cross-platform scheduling (the default)

NOTE

There are no restrictions on platforms, event servers, or number of instances when running in bi-directional scheduling mode.

Configure Cross-Platform Scheduling

From AutoSys Workload Automation, you can submit jobs on a machine that is running on CA UJMA or another Broadcom scheduling manager. That machine can run on a different platform, including mainframe. Similarly, AutoSys Workload Automation can receive job submissions from the other machine.

Before you can submit jobs, you must configure cross-platform support.

NOTE

This process does not apply to CA Workload Automation EE. Bi-directional scheduling is not supported between AutoSys Workload Automation and CA Workload Automation EE.

To configure cross-platform scheduling, follow these steps:

1. Ensure that the external machine that the job runs on meets the [scheduling requirements](#).
2. Enable bi-directional scheduling on AutoSys Workload Automation on [UNIX](#) or [Windows](#).
3. Configure and start CAICCI on [UNIX](#) or [Windows](#).
4. (High availability environments only) Configure failover support for cross-platform scheduling on [UNIX](#) or [Windows](#).
5. Define the external machine on AutoSys Workload Automation on [UNIX](#) or [Windows](#).

6. (CA UJMA only) [Define CA UJMA user IDs and passwords on AutoSys Workload Automation](#).
7. Configure cross-platform support on the external machine, if needed.

NOTE

For more information about configuring cross-platform support on the external machine, see the documentation for the Broadcom product installed on that machine.

After you configure cross-platform scheduling, you can define and submit jobs between AutoSys Workload Automation and the defined external machine.

NOTE

- To submit a job from AutoSys Workload Automation to the mainframe, the job (specified in the command attribute in the job definition) must be defined as a valid job on the mainframe scheduling system.
- To submit a job from the mainframe to AutoSys Workload Automation, the job (specified by the SUBFILE parameter of the mainframe job) must be defined as a valid job on AutoSys Workload Automation.

(UNIX) Enable Bi-Directional Scheduling on AutoSys Workload Automation

To schedule jobs or create job dependencies on another Broadcom scheduling manager, you must enable the AutoSys Workload Automation instance to support bi-directional scheduling.

Follow these steps:

1. On the AutoSys Workload Automation instance, log in as the EXEC superuser and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command:

```
sendevent -E STOP_DEMON
```

The scheduler completes any processing that it is currently performing and stops.

3. Edit the following parameter in the configuration file, and save the file:

```
CrossPlatformScheduling=2
```

4. Enter the following command:

```
eventor
```

The scheduler starts.

5. Open the scheduler log (event_demon.\$AUTOSERV) and check that it contains the following messages:

```
CAUAJM_I_40005 Cross Platform Interface Initialization in progress
```

```
CAUAJM_I_40015 Cross Platform Interface is now active
```

The cross-platform scheduling interface is active and bi-directional (outbound and inbound) scheduling is enabled. The instance can send job requests to an agent and can receive job requests from a scheduling manager.

NOTE

For more information about the CrossPlatformScheduling parameter and the configuration file (\$AUTOUSER/config.\$AUTOSERV), see [Configure a Scheduler](#).

(Windows) Enable Bi-Directional Scheduling on AutoSys Workload Automation

To schedule jobs or create job dependencies on another Broadcom scheduling manager, you must enable the AutoSys Workload Automation instance to support bi-directional scheduling.

Follow these steps:

1. On the local AutoSys Workload Automation instance, log in as the EXEC superuser and enter the following command at the instance command prompt:

```
sendevent -E STOP_DEMON
```

- The scheduler completes any processing that it is performing and stops.
2. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
 3. Select an instance from the Instance drop-down list.
 4. Click the Scheduler icon on the toolbar.
The Scheduler - AutoSys Workload Automation Administrator window appears.
 5. Select Manager and Agent in the Cross Platform Scheduling pane, and click Apply.
 6. Click the Services icon on the toolbar.
The Services - AutoSys Workload Automation Administrator window appears.
 7. Right-click the scheduler service, and click Start.
The scheduler starts.
 8. Open the scheduler log (event_demon.%AUTOSERV%) and check that it contains the following messages:


```
CAUAJM_I_40005 Cross Platform Interface Initialization in progress
CAUAJM_I_40015 Cross Platform Interface is now active
```

The cross-platform scheduling interface is active and bi-directional (outbound and inbound) scheduling is enabled. The instance can send job requests to an agent and can receive job requests from a scheduling manager.

NOTE

For more information about the Cross Platform Scheduling pane on the Scheduler - AutoSys Workload Automation Administrator window, see [Configure a Scheduler](#). For more information about the scheduler log file, see [Maintain the Scheduler](#).

(UNIX) Configure and Start CAICCI

CAICCI is the communication layer that connects applications running on mainframe, UNIX, Windows, and other operating systems. You must configure and start CAICCI on AutoSys Workload Automation before you can schedule jobs on the other systems.

Follow these steps:

1. Log in as root on the AutoSys Workload Automation instance.
2. Open the \$CAIGLBL0000/cci/config/local_host/ccirmtd.prf file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:


```
LOCAL = local_host local_host 32768 startup
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CAICCI is configured.

4. Issue the following command:


```
unicntrl start cci
```

CAICCI restarts. The updated configuration settings are applied.

(Windows) Configure and Start CAICCI

CAICCI is the communication layer that connects applications running on mainframe, UNIX, Windows, and other operating systems. You must configure and start CAICCI on AutoSys Workload Automation before you can use cross-platform scheduling.

NOTE

You do not need to configure the `ccirmtd` file on a Windows to Windows platform environment. However, you must configure the `ccirmtd` file on any cross-platform environment (for example, Windows to UNIX, UNIX to mainframe, and mainframe to Windows).

Follow these steps:

1. On the AutoSys Workload Automation instance, log in with an account that has administrative privileges.
2. Locate and open the `ccirmtd.rc` file.
3. Edit the LOCAL and REMOTE parameters as follows, and save the file:

```
LOCAL = local_host local_host 32768 startup
REMOTE = remote_host cci_system_id 32768 startup port=7000
```

CAICCI is configured.

4. Issue the following command:

```
ccicntrl start
```

CAICCI restarts. The updated configuration settings are applied.

(UNIX) Configure Failover Support for Cross-Instance Scheduling

If you are using high availability mode, you can define the aliased CAICCI system ID on the local AutoSys Workload Automation instance. The cross-platform interface of AutoSys Workload Automation uses this CAICCI system ID to communicate with remote mainframe or UJMA nodes during failover. If the primary scheduler shuts down or becomes unreachable, all communication on the secondary scheduler proceeds as normal. Any statuses currently residing on the remote computers (mainframe or UJMA) are dispatched to the secondary scheduler computer for processing.

Follow these steps:

1. Open all the AutoSys Workload Automation environment scripts, add the following variable, and save the file.

```
PRIMARYCCISYSID = cci_system_id
export PRIMARYCCISYSID
```

– ***cci_system_id***

Defines the aliased CAICCI system ID specified in the `ccirmtd` configuration file.

NOTE

This variable is initially configured during scheduler installation.

NOTE

The environment scripts are as follows:

- `$AUTOUSER/autosys.sh.hostname`—for Bourne shell users
- `$AUTOUSER/autosys.csh.hostname`—for C shell users
- `$AUTOUSER/autosys.ksh.hostname`—for Korn shell users
- `$AUTOUSER/autosys.bash.hostname`—for Bash shell users

2. Restart the scheduler.

Failover support is configured for cross-instance scheduling.

(Windows) Configure Failover Support for Cross-Instance Scheduling

If you are using high availability mode, you can define the aliased CAICCI system ID on the local AutoSys Workload Automation instance. The cross-platform interface of AutoSys Workload Automation uses this CAICCI system ID to

communicate with remote mainframe or UJMA nodes during failover. If the primary scheduler shuts down or becomes unreachable, all communication on the secondary scheduler proceeds as normal. Any statuses currently residing on the agent computers (mainframe or UJMA) are sent to the secondary scheduler computer for processing.

Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select your instance from the Instance drop-down list.
3. Click the System icon on the toolbar.
The System - AutoSys Workload Automation Administrator window appears.
4. Double-click the PRIMARYCCISYSID variable in the Environment Variables pane.
The window refreshes to display PRIMARYCCISYSID variable in the Variable field and its value in the Value field.
5. Enter the aliased CAICCI system ID in the Value field, and click Set.

NOTE

You can find the aliased CAICCI system ID in the ccirmtcd configuration file. The PRIMARYCCISYSID variable is initially configured during scheduler installation.

The PRIMARYCCISYSID variable is modified and displayed in the Environment Variables pane.

NOTE

For more information about the Environment Variables pane on the System - AutoSys Workload Automation Administrator window, see [Configure a Scheduler](#).

(UNIX) Define the External Machine on AutoSys Workload Automation

Before you can submit jobs on an external machine running CA UJMA or another Broadcom scheduling manager, you must define the machine on AutoSys Workload Automation.

Follow these steps:

1. Log in to the AutoSys Workload Automation instance and do one of the following:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.
2. Specify the following definition:


```
insert_machine: host_name
type: machine_type
```

 - **insert_machine: *host_name***
Specifies the host name of the external machine.
 - **type: *machine_type***
Specifies the type of machine you are defining. Options include the following:
 - c—Specifies a AutoSys Workload Automation Connect Option machine.
 - u—Specifies a CA UJMA or CA NSM machine.
3. Do *one* of the following:
 - Enter **exit** if you are using interactive mode.
 - Redirect the script to the jil command if you are using a script.

The external machine is defined on AutoSys Workload Automation.

NOTE

Computers that are managed by CA UJMA cannot be part of a virtual machine. The job_load, max_load, and factor attributes are not supported for these types of computers.

(Windows) Define the External Machine on AutoSys Workload Automation

Before you can submit jobs on an external machine running CA UJMA or another Broadcom scheduling manager, you must define the machine on AutoSys Workload Automation.

Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation instance command prompt opens.
2. Enter **jil** at the instance command prompt.
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
3. Specify the following definition:


```
insert_machine: host_name
type: machine_type
```

 - **insert_machine: host_name**
Specifies the host name of the external machine.
 - **type: machine_type**
Specifies the type of machine you are defining. Options include the following:
 - c—Specifies a AutoSys Workload Automation Connect Option machine.
 - u—Specifies a CA UJMA or CA NSM machine.
4. Enter **exit**.
The data is loaded into the database. The external machine is defined on AutoSys Workload Automation.

NOTE

Computers managed by CA UJMA cannot be part of a virtual machine. The job_load, max_load, and factor attributes are not supported for these types of computers.

Define CA UJMA User IDs and Passwords on AutoSys Workload Automation

After you define a CA UJMA machine to AutoSys Workload Automation, you can define jobs to run on that machine. In a job definition, you can specify the CA UJMA machine using the machine definition. You specify the user ID that the job runs under using the owner attribute. The following example runs a job on the ZASYS400 computer under the user, bob:

```
insert_job: as400ji
owner: bob@ZASYS400
machine: ZASYS400
command: DLYJOB DLY(16)
```

The user who is specified in the owner attribute must have an account on the target CA UJMA computer. The account must match the owner value exactly for the job to run. Specify the owner value as *user@host* or *user@domain*. Before you can specify a user in a job definition, you must define the user and its password on the local AutoSys Workload Automation instance.

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The instance command prompt appears.
2. Enter the following command:


```
autosys_secure
```

The AutoSys Security Utility interactive menu appears.

3. Enter [5] Manage users.
The Manage users submenu appears.
4. Enter option [1] Manage users with password.
The Manage users with the password menu appears.
5. Enter option [1] Create a user.
The utility prompts you to enter the user name, host or domain name, and password.
6. Enter the required information.

NOTE

Where the operating system permits, CA UJMA user IDs can contain up to 30 alphanumeric characters. The user IDs can contain both uppercase and lowercase characters (when the operating system permits mixed case). You cannot use blank spaces and tab characters.

A message appears to indicate whether the user was added successfully.

7. If an error message appears and the user is not added, navigate the interactive menu to make the required changes. Repeat this action until the message indicates that the user is successfully added.
8. (Optional) Create and define any additional users that you want to add.
9. Enter option [0] Exit the CA WAAE Security Utility.
The utility closes, the users are added, and the user definitions are saved in the database.

(UNIX) Improve Workload Performance: Highly-Available Environments

As an administrator, you want to set up an environment that promotes optimal workload performance. Sometimes the scheduler fails, interrupting the execution of the workflow. Highly-available environments provide at least one back-up scheduler to help minimize these interruptions. In some highly-available environments, you can enable additional options that further improve workload performance and prevent data loss. AutoSys Workload Automation provides the following high-availability options:

- **Highly-Available Cluster Environments**

Provides a failover solution for the scheduler and the application server. We recommend using this option when you need a failover solution unless you do not have cluster management software installed on your system.

The cluster manager does not provide a failover solution for the database unless you also configure your database to operate on the cluster. You can configure your database to operate on the cluster only when you are using database software that is cluster aware.

Operating in a highly-available cluster environment without a highly-available database increases the risk of downtime and is not recommended. If your database is cluster aware, configure it to operate on the cluster. If your database is not cluster aware, enable dual event server mode.

AutoSys Workload Automation can also cooperate with cluster management software to form a clustered agent. A clustered agent improves load balancing and promotes the optimal use of system resources for job flows that require them. We recommend that you set up a highly-available cluster environment before you set up a clustered agent. To run AutoSys Workload Automation in a highly-available cluster environment with a clustered agent, set up a clustered server on one cluster and set up a clustered agent on a separate cluster that exists within the same AutoSys Workload Automation instance.

- **High-Availability Mode**

Provides an internal failover solution for the scheduler. We recommend using this option to set up a highly-available environment only when you cannot install cluster management software on your system.

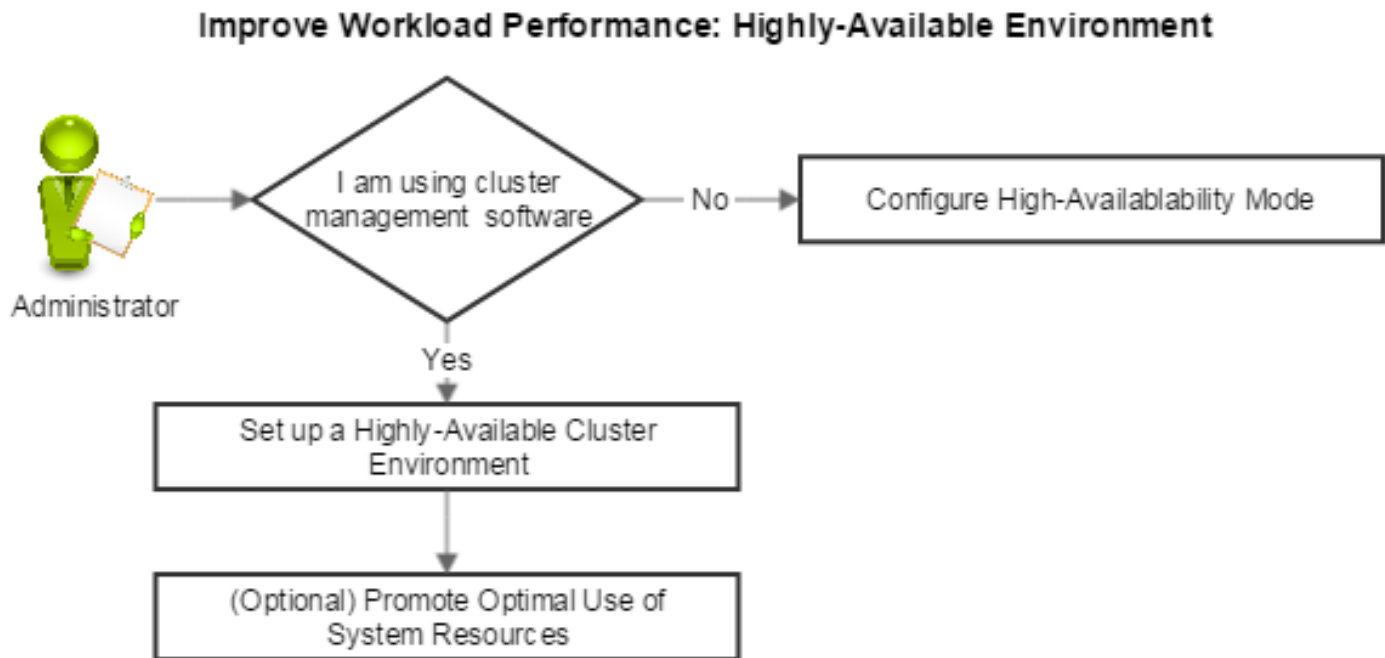
- **Dual Event Server Mode**

Provides a failover solution for the database. Use this option only when *one* of the following situations applies:

- You have configured AutoSys Workload Automation to run in a highly-available cluster environment but you are using database software that is not cluster aware.
- You are not using cluster software, you have configured AutoSys Workload Automation to run in high-availability mode, and you are using database vendor that does not provide an internal failover solution.

The following diagram illustrates how you can improve workload performance by setting up a highly-available environment:

Figure 17: how to improve workload performance by setting up a highly-available environment



Follow these steps:

1. [Set up a highly-available cluster environment.](#)
2. (Optional) [Promote optimal use of system resources.](#)
Use this option when the scheduler is running jobs that use shared resources, are CPU-intensive, or execute client utilities. Ensure that you set up the clustered agent on a separate cluster than the clustered server and that both of these clusters exist within the same AutoSys Workload Automation instance.
3. [Configure high-availability mode.](#)

AutoSys Workload Automation(UNIX) Set Up a Highly-Available Cluster Environment

As an administrator, your responsibilities include configuring the AutoSys Workload Automation environment. Highly-available environments minimize down time and data loss. The most stable highly-available environment is a highly-available cluster environment.

AutoSys Workload Automation and cluster-aware databases can cooperate with cluster management software to create a highly-available application server, scheduler, and database. Configure cluster-aware databases to operate on the cluster before completing this scenario. Enable dual event servers to provide a failover solution when using a database that is not cluster-aware.

In a highly-available cluster environment, you can further improve workload performance by setting up a clustered agent or clustered client.

WARNING

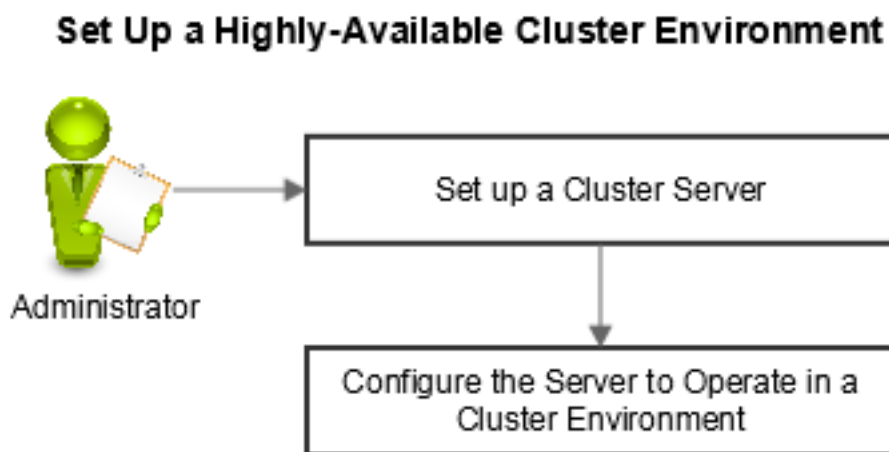
Transparent Application Failover (TAF) mode prevents AutoSys Workload Automation from accessing its database schema on the active database node following a failover on the Oracle Server. So, for AutoSys Workload Automation to work with the Oracle server that is running in the TAF mode, ensure that the Oracle client on the AutoSys Workload Automation server is configured for the Oracle RAC failover. No configuration changes are required on the Oracle server. For more information about configuring the AutoSys Workload Automation server to work with Oracle RAC, see [Install AutoSys Workload Automation with Oracle](#).

NOTE

- Running AutoSys Workload Automation in a highly-available cluster environment requires that you have cluster management software installed on your system. There are many brands of UNIX compatible cluster management software, such as Red Hat Cluster Suite. For more information about installing and configuring a cluster manager, see the documentation for your cluster management software.
- For information about how to configure your database to operate on the cluster, see the documentation for your database vendor.

The following diagram illustrates how you can set up a highly-available cluster environment:

Figure 18: set up a highly-available cluster environment



Follow these steps:

1. [Set Up a Clustered Server](#)

WARNING

The database cannot operate on the cluster when AutoSys Workload Automation is operating in dual event server mode. You can enable or disable dual event server mode when you set up a clustered server. Enable dual event server mode only when your database software is not cluster aware.

2. [Configure AutoSys Workload Automation to Operate in a Cluster Environment](#)

(UNIX) Set Up a Clustered Server

In a highly-available cluster environment, the AutoSys Workload Automation server components are installed on each node of the cluster and run on the active node. When a component fails, the cluster manager restarts that component on one of the passive nodes.

Follow these steps:

1. Verify that a cluster manager is installed.
2. Verify that your database is configured to operate on the cluster.

NOTE

Some database software is not cluster aware. In this case, use dual event server mode as a failover solution for your database.

3. Install AutoSys Workload Automation on the node that you designated as active when you installed the cluster manager.
 - a. Log in as the root user, mount the AutoSys Workload Automation media, change directories to the mounted file system, and run the following command:

```
./wa_setup.sh
```

The Welcome page appears.

- b. Click Next. The License Agreement page appears.
- c. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
- d. Select New and click Next. The Installation Type page appears.
- e. Select Custom and click Next. The Components page appears.
- f. Select the Application Server, Scheduler, Client and Agent, and any optional components that you want to install and click Next. The Application Server Information page appears.
- g. Enter the cluster host name that you used when you installed and configured the cluster manager in the Application Server Host field, complete all other required fields, and click Next.

NOTE

If you do not know the cluster host name, view the configuration settings for your cluster manager. For more information about how to display these settings, see the documentation for your cluster management software.

The Application Server Properties page appears.

- h. Select the Configure this Instance for Use in a Clustered Environment check box, enter all other required information, and click Next. The default behavior to automatically start the application server and scheduler following installation and at system startup is disabled, the installer automatically clears and disables the corresponding check boxes on the Application Server Properties page and Scheduler Properties page, and the Database Type page appears.
- i. If you configured your database software to operate on the cluster, clear the Enable Dual Event Servers check box, and click Next. If your database software is not cluster aware, select the Enable Dual Event Servers check box, and click Next.

WARNING

The database cannot operate on the cluster when AutoSys Workload Automation operates in dual event server mode. Always disable dual event servers when you configure your database to operate on the cluster.

Operating in single event server mode without a clustered database limits the benefits of a highly-available cluster environment. If your database software is not cluster aware, ensure that you enable dual event servers.

The Primary Event Server Properties pages appear.

- j. Provide the information that the installer needs to create the database during installation by selecting the relevant check boxes, and then enter all other required information and click Next. The check boxes that appear depend on your database provider.
 - Oracle:
 - Create or the Refresh the Database
 - Create the Tablespaces, Database Users, and Role
 - Sybase:
 - Create or the Refresh the Database
 - Create New Database Devices

The next page in the Primary Event Server Properties page series appears.

- k. Enter the required information and click Next on each page of the series. If you enabled dual event servers, specify the second event server properties by entering the required information and clicking Next on each page in the Secondary Event Servers Properties page series. Specify a different name and database server for the second event server than you did for the first event, and ensure that all of the other properties are identical for both event servers. The Database Test page appears.
- l. Click Test, wait for the test to complete successfully, and then click Next. If the database test fails, read the error message to determine what caused the failure, use the Back button to navigate to the installation pages that contain the errors, correct the errors, click Next until you return to the Database Test page, and click Test to run the test again. Repeat this action until the test completes successfully, and click Next.
- m. Enter the required information on each page and click Next until the Agent Properties page appears.
- n. Enter the required information on the Agent Properties page, ensuring that it matches the information that you entered when you installed the server on the active node, and click Next. The Scheduler Properties page appears.
- o. Clear the Configure High-Availability check box, enter all other required information, and click Next.

WARNING

The Configure High-availability check box applies to using internal AutoSys Workload Automation options to set-up a failover solution for the scheduler. You cannot use a cluster manager to set up a highly-available environment when this check box is selected during installation.

- p. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
 - q. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server is installed on the active node.
4. Repeat the server installation on the passive nodes.
- a. Log in as the root user on the a passive node, mount the AutoSys Workload Automation media, change directories to the mounted file system, and run the following command:


```
./wa_setup.sh
```

The Welcome page appears.

- b. Continue with the installation by entering the required information on each page and clicking Next until the Primary Event Server Properties page appears. Ensure that the information that you enter on these pages matches the information that you entered when you installed the server on the active node.
- c. Enter the required information and click Next. Ensure that you select the Enable Dual Event Servers check box if you enabled dual event servers when you installed the server on the active node, and that you clear that check box if you are configured your database to operate on the cluster. Ensure that the installer does not create the database on the passive nodes during installation by clearing the relevant check boxes, and then enter all other required information and click Next. The check boxes that appear depend on your database provider.

- Oracle:
 - Create or the Refresh the Database
 - Create the Tablespaces, Database Users, and Roles
- Sybase:
 - Create or the Refresh the Database

- d. Create New Database Devices.
 - e. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. Ensure that the information that you enter on these pages matches the information that you entered when you installed the server on the active node. The page displays the information that you entered in the installation wizard.
 - f. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server is installed on the first passive node.
 - g. Repeat the server installation on each of the other passive nodes.
- The server components are installed on all nodes in the cluster, and you can configure the server to operate in a cluster environment.

(UNIX) Configure the Server to Operate in Cluster Environment

The cluster manager cannot restart failed components on passive nodes until you configure the cluster manager and the server to operate cooperatively.

Follow these steps:

1. Log in as the root user on the active node, run the shell that is sourced to use AutoSys Workload Automation. The environment is defined and the operating command prompt opens.
2. Open the \$AUTOUSER/config.\$AUTOSERV file. The configuration file opens.
3. Locate and edit the following parameter:

```
ManagerHostAlias=cluster_host_name
```

NOTE

If you are connecting from Web UI (Web UI), ensure that you also locate and set the AutoServer parameter in the configuration file to the cluster host name. Open the configuration file on all nodes (active and passive) in the cluster and set the AutoServer parameter to the same cluster host name. If the AutoSys Workload Automation cluster fails over, Web UI connects to the cluster host you specified in the AutoServer parameter.

4. Open the configuration file on each of the passive nodes, and edit the ManagerHostAlias parameter so that the value on all nodes in the cluster is the same on each node in the cluster.

WARNING

To specify the value for the ManagerHostAlias, use the same cluster host name that you used when you installed and configured the cluster manager. For information about how to display the configuration settings for your cluster manager, see the documentation for your cluster management software.

5. Enter the cluster configuration scripts in the location that is specified in the documentation for your cluster management software.
 - a. Enter one of the following scripts in the cluster configuration file, depending on your operating environment:
 - AIX


```
/etc/rc.d/init.d/waae_server.$AUTOSERV stop
```
 - HP-UX

```
/sbin/init.d/waae_server.$AUTOSERV stop
```

- Linux

```
/etc/init.d/waae_server.$AUTOSERV stop
```

The cluster manager is configured to force the application server to shutdown in the event of a failover.

b. Enter one of the following scripts in the cluster configuration file, depending on your operating environment:

- AIX

```
/etc/rc.d/init.d/waae_server.$AUTOSERV monitor
```

- Linux

```
/etc/init.d/waae_server.$AUTOSERV monitor
```

The cluster manager is configured to monitor the activity of the application server.

c. Enter one of the following scripts in your cluster configuration, depending on your operating environment:

- AIX

```
/etc/rc.d/init.d/waae_server.$AUTOSERV start
```

- HP-UX

```
/sbin/init.d/waae_server.$AUTOSERV start
```

- Linux

```
/etc/init.d/waae_server.$
```

The cluster manager is configured to restore the application server after a failover.

d. Enter one of the following scripts in the cluster configuration file, depending on your operating environment:

- AIX

```
/etc/rc.d/init.d/waae_sched.$AUTOSERV stop
```

- HP-UX

```
/sbin/init.d/waae_sched.$AUTOSERV stop
```

- Linux

```
/etc/init.d/waae_sched.$AUTOSERV stop
```

The cluster manager is configured to force the scheduler to shutdown in the event of a cluster failover.

e. Enter one of the following scripts in your cluster configuration file, depending on your operating environment:

- AIX

```
/etc/rc.d/init.d/waae_sched.$AUTOSERV start
```

- HP-UX

```
/sbin/init.d/waae_sched.$AUTOSERV start
```

- Linux

```
/etc/init.d/waae_sched.$AUTOSERV start
```

The cluster manager is configured to restore the scheduler after a failover.

f. Enter one of the following scripts in the cluster configuration file, depending on your operating environment:

- AIX

```
/etc/rc.d/init.d/waae_sched.$AUTOSERV monitor
```

- HP-UX

```
/sbin/init.d/waae_sched.$AUTOSERV monitor
```

- Linux

```
/etc/init.d/waae_sched.$AUTOSERV monitor
```

The cluster manager is configured to monitor the activity of the scheduler.

6. Start the server components on the active node.

- Login on the active node as the root user and run the shell that is sourced to use AutoSys Workload Automation. The environment is defined and the operating system command prompt appears.

- b. Enter the following commands:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and application server start on the active node.

7. If you want to run jobs on the local host machine, define one manually.

WARNING

Setting up a clustered server requires that you disable the server installation option to start the application server automatically. Disabling this option prevents the installer from automatically inserting the local host machine definition. If you do not insert the definition manually on each node, jobs that are defined to run on the local host machine fail to start.

- a. Open the operating system command prompt on the active node, and enter the following command:

```
jil
```

The JIL command prompt opens.

- b. Enter the following command and attributes:

```
insert_machine: cluster_host_name
agent_name: agent_name
encryption_type: {NONE | AES | DEFAULT}
key_to_agent: key_value
port: agent_port
```

- **cluster_host_name**
Specifies the name of the node for which you are defining the local host machine.
- **agent_name**
Specifies the name of the agent.
- **NONE | AES | DEFAULT**
Specifies the encryption type for the agent.
- **key_value**
Specifies the encryption key. This parameter is only required for agents that use AES encryption.
- **agent_port**
Specifies the port that the agent listens to.

WARNING

When you install a server on a node, the installer prompts you to enter agent definition information. The agent definition information is identical across all nodes on the cluster. Ensure that the agent definition on a node and the local host machine definition specify identical values for the following attributes:

```
agent_name: agent_name
encryption_type: {NONE | AES | DEFAULT} key_to_agent: key_value
port: agent_port
```

The local host machine is defined, and the definition is stored in the database.

- c. Enter the following command:

```
exit
```

The JIL command prompt closes, and the local host machine is defined as the machine that represents the clustered server.

The server is configured to operate in a cluster environment.

You have set up a highly-available cluster environment. Failover solutions are in place for the scheduler and application server. You can also set up an environment that promotes optimal use of system resources.

(UNIX) Promote Optimal Use of System Resources

As an administrator, you want to set up an environment that enables the scheduler to improve workload performance. With cluster management software installed, you can set up a load-balancing cluster. A load-balancing cluster consists of a clustered agent. Optionally, you can also set up a clustered client.

With a load-balancing cluster and a machine that is defined to represent that cluster, the scheduler can improve load-balancing and decrease job failures by defining certain jobs to run on the cluster. We recommend this method for the following jobs:

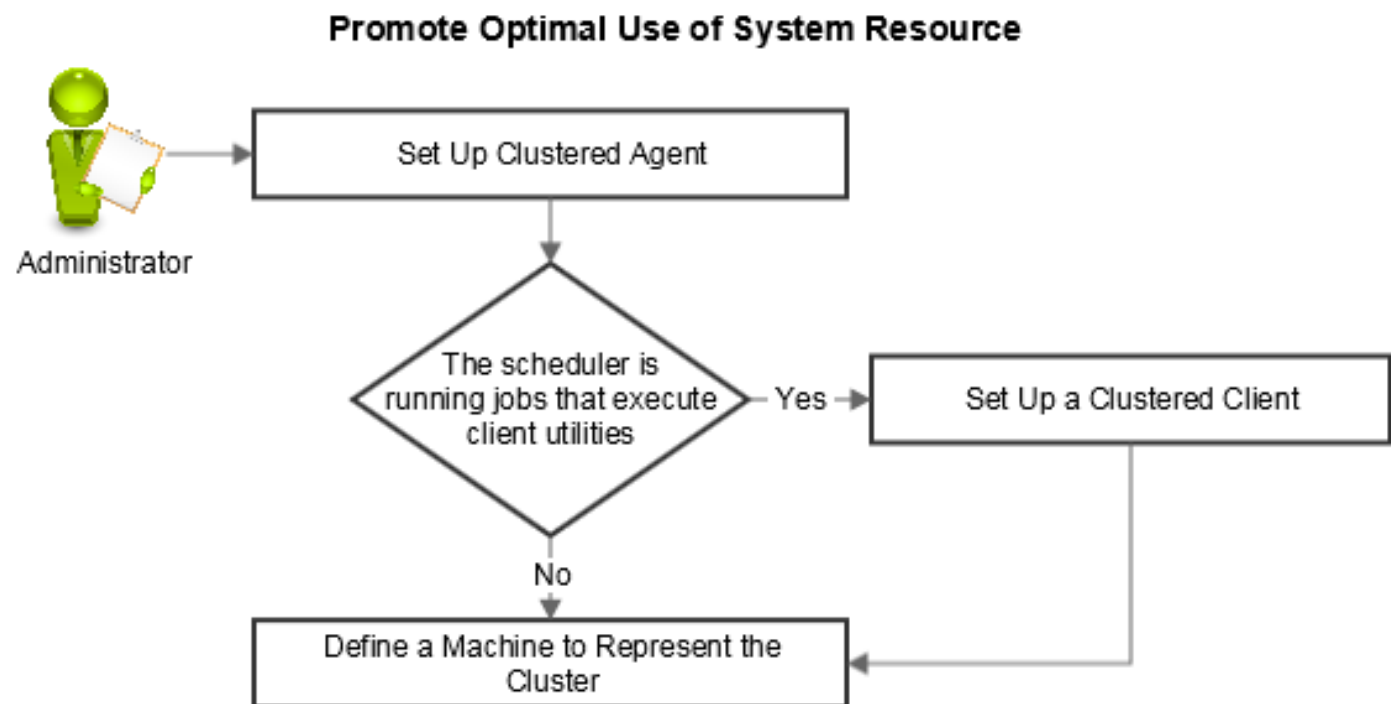
- CPU intensive jobs
- Jobs that use shared resources
- Command jobs that execute client utilities

NOTE

Multiple AutoSys Workload Automation instances can utilize the same load-balancing cluster.

The following diagram illustrates how you can promote use of system resources:

Figure 19: promote use of system resources



Follow these steps:

1. [Set up a clustered agent.](#)

WARNING

Ensure that you set up the clustered agent on a different cluster than the clustered server.

2. [Set up a clustered client.](#)
Use this option to decrease the risk of failure for jobs that execute client utilities.
3. [Define a machine to represent the cluster.](#)

(UNIX) Set Up a Clustered Agent

A cluster manager complements the load-balancing capabilities of the scheduler when jobs are defined to run on the cluster. The scheduler cannot define jobs to run on a cluster until you set up a clustered agent.

Follow these steps:

1. Verify that a UNIX compatible cluster manager is installed within the AutoSys Workload Automation instance, and that at least two separate clusters exist.
2. Verify that the AutoSys Workload Automation instance is operating in a highly-available cluster environment and that the client is installed on the active node of the highly-available cluster.

WARNING

Do not set up the clustered agent on the highly-available cluster. Designate another cluster as the load-balancing cluster and set up the clustered agent on that cluster.

3. Configure the cluster manager to load-balance clustered resources.

NOTE

For information about configuring your cluster manager to load balance clustered resources, see the documentation for your cluster management software.

4. Log in as the root user on any node of the load-balancing cluster, mount the AutoSys Workload Automation Standalone Media, change directories to the mounted file system, and execute the following commands:

```
cd Agent
./agent_setup.sh -I agent_name
```

– **agent_name**

Defines a unique name for the agent. You can install multiple agents on the same machine or node if you define a unique name for each of those agents. Agents that you install on different nodes on one cluster for the purpose of improving load-balancing can have the same name or unique names. If you use unique names, define a common agent name using an agent alias.

Limits: A string of up to 16 characters, where the first character is a letter and the remaining characters are any of the following characters:

- Alphanumeric characters
- The following special characters: @, \$, and underscore (_).

- a. Click Next. The License Agreement page appears.
 - b. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Type page appears.
 - c. Select Custom, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the default settings.
 - d. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The agent with the specified name is installed on the active node.
5. Repeat the agent installation on every node in the cluster using a common agent name. Define a common agent name using one of the following methods:

- Install one agent on each node, and use the same name for all of these agents.
- Install a uniquely named agent on each node, and define an alias agent name that is common to all of the agents on the cluster. To define an alias agent name, open the agentparam.txt file and add the following parameters:

```
communication.alias1=common_agent_name
[communication.alias2=second_agent_alias]
...
[communication.aliasnnn=additional_agent_alias]
```

- **common_agent_name** Specifies the common agent name for the cluster and defines it as the first alias for the agent.
 - **second_agent_alias** (Optional) Specifies a second alias for the agent. Do not specify the same value that you used for the unique agent name or the common agent name.
 - **nnn** (Optional) Specifies the numeral that corresponds with the additional agent alias.
 - **additional_agent_alias** (Optional) Specifies an additional alias for the agent. Do not specify the same value that you used for the unique agent name, common agent name, or any other agent alias.
6. If you used an agent alias name to define the common agent name, configure AutoSys Workload Automation to recognize the individual agent on each node. Ensure that you specify the same common agent name as the first agent alias for this agent in the agentparm.txt file on each node.
- a. Open the operating system command prompt on the active node and enter the following command:
- ```
jil
```
- The JIL command prompt for the local AutoSys Workload Automation instance appears.
- b. Insert a AutoSys Workload Automation machine definition for each uniquely named agent that should operate on the cluster by entering the following JIL command and attributes:
- ```
insert_machine: machine_name
node_name: cluster_node
agent_name: unique_agent_name
encryption_type: {NONE | AES | DEFAULT}
key_to_agent: key_value
port: common_agent_port
```
- **machine_name** Specifies the name of the AutoSys Workload Automation machine.
 - **cluster_node** Specifies the name of the node on which you are installing the agent.
 - **unique_agent_name** Specifies the unique name of the agent. Do not use the same name that you use as the common agent name for the cluster.
 - **NONE | AES | DEFAULT** Specifies the encryption type. Specify the same encryption type for all agents.
 - **key_value** Specifies the encryption key. This parameter is only required for agents that use AES encryption.
 - **common_agent_port** (Optional) Specifies the port that the agent listens to. Specify the same port number for every agent.
7. If you want to run jobs on a AutoSys Workload Automation machine that references an agent by one of its additional agent aliases, insert a AutoSys Workload Automation machine definition for each additional agent alias.
- a. Log in on the active node of highly-available cluster and click Start, Programs, AutoSys Workload Automation, Command Prompt (*instance_name*). The instance command prompt appears.
- b. Enter the following command:
- ```
jil
```
- The JIL command prompt appears.
- c. Enter the following JIL subcommand and attributes:
- ```
insert_machine: machine_name
node_name: cluster_node
agent_name: second_agent_alias|additional_agent_alias
encryption_type: {NONE | AES | DEFAULT}
key_to_agent: key_value
port: common_agent_port
```
- d. Enter the following command:
- ```
exit
```
- The AutoSys Workload Automation definitions are saved in the database and the JIL command prompt closes. AutoSys Workload Automation is operating with a clustered agent.

## (UNIX) Set Up a Clustered Client

Command jobs that execute client utilities are more likely to succeed when they run on a load-balancing cluster that consists of both a clustered agent and a clustered client. A load-balancing cluster without a clustered client provides no improvement in workload performance for these types of jobs. If the scheduler is running these types of jobs, set up a clustered client.

### Follow these steps:

1. Verify that the AutoSys Workload Automation instance is operating in a highly-available cluster environment, that the client is installed on the active node of the highly-available cluster, and that a separate load-balancing cluster with a clustered agent exists within the same instance.
2. Log in as the root user on the active node of the load-balancing cluster and install the client.
  - a. Mount the AutoSys Workload Automation media, change directories to the mounted file system, and run the following command:

```
./wa_setup.sh
```

The Welcome page appears.

- b. Click Next. The License Agreement page appears.
- c. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
- d. Select New and click Next. The Installation Type page appears.
- e. Select Custom and click Next. The Components page appears.
- f. Select the Client check box, click Next, and then continue with the installation by entering the required information on each page and clicking Next until the Instance Information page appears.
- g. Enter the AutoSys Workload Automation instance name that you used when you installed the server, click Next, and then continue with the installation by entering the required information and clicking Next on each page until the Application Server Information page appears.
- h. Enter the host name of the highly-available cluster in the Application Server Host field, enter all other required information, and click Next. Use the same cluster host name that you used when you installed and configured the cluster manager for the highly-available cluster.

### WARNING

If you do not remember the host name of the highly-available cluster, check the configuration settings for your cluster manager. For information about how to display these settings, see the documentation for your cluster management software.

The Owner and Group Settings page appears.

- i. Continue with the installation by entering the required information and clicking Next on each page until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
- j. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The client is installed on the active node.
3. Repeat the client installation on all nodes in the cluster that define, run, or monitor jobs.
4. If jobs that execute client utilities are defined within other AutoSys Workload Automation instances and you want to run those jobs on the load-balancing cluster, configure the clustered client to operate within those instances.
  - a. Log in on any node of the load-balancing cluster, insert and mount the installation media, start the installation, enter the appropriate instance name in the Instance Name field on the Instance Information page, and complete the installation.
  - b. Repeat the installation, entering the same instance name, on all nodes in the cluster that define, run, or monitor jobs.
  - c. Repeat these actions for every applicable AutoSys Workload Automation instance.

AutoSys Workload Automation is operating with a clustered client.

## (UNIX) Define a Machine to Represent the Cluster

A machine that represents the cluster can run jobs on any agent that is installed as part of the cluster. The scheduler can define a machine to represent the cluster when a cluster manager is installed on the system and multiple agents are installed as part of the cluster. Typically, the administrator defines a machine to represent the cluster when setting up an environment that promotes optimal use of system resources.

Running jobs on a cluster improves workload performance. Load balancing improves for jobs that use shared resources or have high CPU consumption. Job failures decrease for jobs that execute client utilities when you have also set up a clustered client. You can define jobs to run on a cluster only when a machine is defined to represent the cluster.

### Follow these steps:

1. Log in on the active node of the highly-available cluster and run the shell that is sourced to use AutoSys Workload Automation. The environment is defined and the operating system command prompt appears.
2. Enter the following command:

```
jil
```

The JIL command prompt for the local AutoSys Workload Automation instance appears.

3. Enter the following command and attributes:

```
insert_machine: cluster_machine_name
node_name: cluster_host_name
agent_name: common_agent_name
port: common_agent_port
```

- **cluster\_machine\_name** Specifies the name of the real AutoSys Workload Automation machine that represents all agents in the cluster.

#### WARNING

You cannot define a virtual machine or a machine pool to represent the cluster.

- **cluster\_host\_name** Specifies the name of the host for the cluster.

#### WARNING

The cluster machine name is not the cluster host name. To specify the cluster host name, use the same cluster host name that you used when you installed and configured the cluster manager. For information about how to display the configuration settings for your cluster manager, see the documentation for your cluster management software.

- **common\_agent\_name** Specifies the name or alias that is common to all agents installed as part of the cluster.

#### WARNING

Use the same common agent name or alias that you used when you installed multiple agents as part of the cluster.

- **common\_port\_name** Specifies the port that is common to all agents that are installed as part of the cluster.

#### WARNING

Use the same port that you specified when you installed multiple agents as part of the cluster.

4. (Optional) Specify optional machine attributes.
5. Enter the following command:

```
exit
```

The machine that represents the cluster is defined, and the definition is saved on the database.

You have set up an environment that promotes optimal use of system resources. The scheduler can define jobs to run on the cluster.

## (UNIX) Configure High-Availability Mode

As an administrator, you want to promote optimal workload performance without installing cluster management software on your system. You can set-up a highly-available environment using internal options.

You can activate high-availability mode by installing a shadow scheduler to act as a failover solution for the primary scheduler. In this mode, the primary scheduler processes events and the shadow scheduler is available to take over this work in the event of a failover.

The primary scheduler is installed during the initial server installation. To operate AutoSys Workload Automation in high-availability mode, repeat the server installation on another machine designate the scheduler on that machine as the shadow scheduler. The shadow scheduler is designed to take over processing events when the primary scheduler fails.

Sometimes you want to force the shadow scheduler to take over, such as when you need to perform maintenance on the machine where the primary scheduler runs. In this case, you can initiate a manual failover using the `sendevent` command. The primary scheduler shuts down, and the shadow scheduler takes over.

High-availability mode is disabled and the shadow scheduler becomes the active scheduler when a failover occurs, including when you initiate a manual failover. To return to high-availability mode following a failover, restore the primary scheduler.

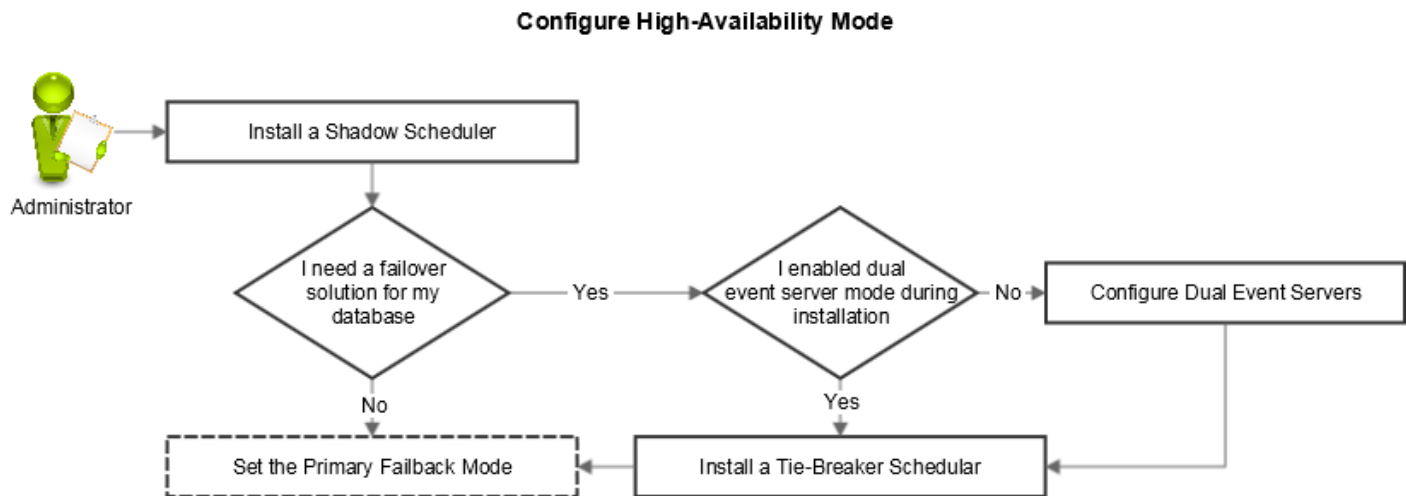
By default, the primary scheduler automatically shuts down when it detects that the shadow scheduler is processing events. In this case, you can return to high-availability mode only if you shut down the shadow scheduler, restore the primary scheduler, and then restart the shadow scheduler. You can avoid this downtime by configuring the primary failback mode.

High-availability mode does not provide a failover solution for the database. Most database vendors provide internal failover solution. You can also provide a failover solution by configuring AutoSys Workload Automation to run in dual event server mode. We recommend configuring dual event server mode only when your database vendor does not support an internal failover solution. High-availability mode does not operate properly with dual event servers unless you install a tie-breaker scheduler.

### NOTE

- High-availability mode does not support optimal use of system resources. You can promote optimal use of system resources only when you have cluster management software installed.
- For more information about how to initiate a manual failover or failback using the `sendevent` command, see [sendevent Command—Stop the Schedulers or Application Servers](#).

The following diagram illustrates how you can configure high-availability mode:

**Figure 20: configure high-availability mode****Follow these steps:**

1. [Install the shadow scheduler.](#)
2. [Configure dual event servers.](#)
3. [Install the tie-breaker scheduler.](#)
4. (Optional) [Set the primary failback mode.](#)

**(UNIX) Install the Shadow Scheduler**

You can install a shadow scheduler to take over processing events when the primary scheduler fails. Starting the shadow scheduler enables high availability mode.

**NOTE**

The shadow scheduler cannot run on the same computer as the primary scheduler. The primary scheduler and the shadow scheduler can run on computers with different operating systems as long as those computers use the same database platform and are defined within the same AutoSys Workload Automation instance. Ensure that the primary, shadow, and tie-breaker schedulers are at the same release level.

**Follow these steps:**

1. Install the server on the local drive of the computer where the scheduler should run.

**WARNING**

Do not install the server on a network drive.

- a. Log in as the root user and mount the AutoSys Workload Automation media.
- b. Change directories to the mounted file system and run the following command:

```
./wa_setup.sh
```

The Welcome page appears.

- c. Click Next. The License Agreement page appears.
- d. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
- e. Select New and click Next. The Installation Type page appears.



- f. Select Server and click Next. The Instance Information page appears.
  - g. Continue with the installation by entering the required information on each page and clicking Next until the Scheduler Properties page appears.
  - h. Clear the Start the Scheduler Following the Installation check box, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Primary Event Server Properties page appears. The check boxes that appear depend on your database provider.
    - Oracle:
      - Create or the Refresh the Database
      - Create the Tablespaces, Database Users, and Roles
    - Sybase:
      - Create or the Refresh the Database
      - Create New Database Devices
  - i. Ensure that the installer does not create the database during installation by clearing the relevant check boxes, and then enter all other required information and click Next.
  - j. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
  - k. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server installation is complete.
2. Designate the scheduler on this computer as the shadow scheduler by editing the \$AUTOUSER/config.\$AUTOSERV file as follows:
 

```
RoleDesignator=2
```
  3. Open the operating system prompt and enter the following command:
 

```
eventor -q
```

The Shadow scheduler becomes active. If you did not enable dual event servers during installation, AutoSys Workload Automation enters high-availability mode as soon as the primary scheduler detects the activity of the shadow scheduler.

## (UNIX) Configure Dual Event Servers

We recommend using the internal failover solution that is provided by your database vendor, but some databases do not provide one. In this case, use dual event server mode as the failover solution for your database.

AutoSys Workload Automation automatically installs a second event server that is identical to the primary event server when you enable dual event server mode during installation. You can also switch from single event mode to dual event server mode by copy the database and configuring dual event servers.

In dual event server mode, the secondary active scheduler performs a database rollover to the secondary event server when the primary event server fails. In this case, AutoSys Workload Automation begins operating in single event server mode.

Configure dual event servers in the following situations:

- When you want to operate in dual event server mode but did not enable dual event servers during installation
- When you want to resume operating in dual event server mode after a database rollover

### NOTE

When a rollover occurs, AutoSys Workload Automation backs up the configuration file before commenting out the failed event server. You can use the backup file to restore pre-rollover configuration settings when you reconfigure dual event servers.

**Follow these steps:**

1. Login to AutoSys Workload Automation machine in the instance with a client installation, and run the shell that is sourced to use AutoSys Workload Automation. The environment is defined and the operating system command prompt opens.
2. Enter the following command:
 

```
sendevent -E STOP_DEMON -v ALL
```

 All server processes that are running on the instance stop.
3. If you did not enable dual event servers during installation and want to switch to dual event server mode, install and configure the second event server:
  - a. Install database software on another machine within the AutoSys Workload Automation instance. Ensure that the two separate database servers are the same size and type but that each one resides on different machines and has a unique name. A second database server is installed in the instance. You can install another event server on this database server.
  - b. Run the [CreateAEDB](#) script.

**WARNING**

Dual event server mode functions properly only when the event servers have unique names. When you run the script to create the secondary event server, specify a name that is different from the name of the primary event server.

The secondary event server is installed.

- c. Open the configuration file on the machine where the primary scheduler is installed and locate the following parameter:
 

```
#EventServer_1|#EventServer_2
```
- d. Edit the parameter as follows:
 

```
Event_Server_1|EventServer_2
```
- e. Specify the primary event server and the secondary event server:
  - To make the new database the secondary event server, add the following parameter:
 

```
EventServer_2=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST | ORACLE_SVR,DBPORT,DBHOST
```

    - **SYBASE\_DB,DBPORT,DBHOST** Identifies the Sybase database for the second event server.
    - **ORACLE\_SVR,DBPORT,DBHOST** Identifies the Oracle database for the second event server.
  - To make the restored database the secondary event server, specify the active database in the EventServer\_1 parameter and the restored database in the EventServer\_2 parameter.

**NOTE**

Run the CA WAAE BCP database scripts to synchronize data from one event server to the other.

- To make the new database the primary event server, specify the existing database that is defined in the EventServer\_1 parameter as the secondary event server by changing it to EventServer\_2, then add the following parameter:
 

```
EventServer_1=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST | ORACLE_SVR,DBPORT,DBHOST
```
  - To make the restored database the primary event server, verify that the restored database is specified in the EventServer\_1 parameter and the active database is specified in the EventServer\_2 parameter.
- f. Specify the database reconnect behavior for the second event server by modifying the following parameter in the configuration file:
 

```
DBEventReconnect=value, value2
```

    - **value** Identifies the database reconnect behavior for the first event server. **Limits:** 0-99
    - **value2** Identifies the database reconnect behavior for the second event server. **Limits:** 0-99

**NOTE**

During typical installation, AutoSys Workload Automation sets the reconnect value for the single event server to 50 by default. During a custom installation in which you enable dual event server mode, AutoSys Workload Automation sets the reconnect value for both event servers to 50, 5 by default. Ensure that you add a reconnect value for the second event server when you configure AutoSys Workload Automation to run in dual event server mode after running it in single event server mode. Optionally, you can modify the default reconnect value for the first event server.

The secondary event server is configured on the primary scheduler machine. Repeat the secondary event server configuration on every server machine in the instance. Ensure that the event server information in the configuration file is the same on each of these machines.

4. If you want to return to dual event server mode following a database failure, take the following actions:
  - a. Review the scheduler log file to determine the problem that caused the failure.
  - b. Consult your database software documentation, and follow the instructions for manually resolving the problem that caused the database failure. The status of the failed database changes and you can reconfigure dual event servers. If you want to restore pre-rollover configuration settings, delete the modified configuration file and rename the backup configuration file:
    - a. Delete \$AUTOUSER/config.\$AUTOSERV.
    - b. Locate \$AUTOUSER/config.\$AUTOSERV.rollover and change the name to \$AUTOUSER/config.\$AUTOSERV.
    - c. Repeat these actions on every server machine that is installed in the instance. If you want to specify new configuration settings when you restore the failed event server, modify the configuration file, open the \$AUTOUSER/config.\$AUTOSERV file on the machine where the primary scheduler is installed and modify the event server configuration parameters. Repeat this action on every server machine that is installed in the instance. Ensure that the event server information is the same in the configuration files on all of these machines.
5. Run the bulk copy script ([Configure AutoSys Workload Automation to Run in Dual Event Server Mode](#)) based on your database vendor to synchronize the event servers.
6. Restart the server processes for the instance.
  - a. On the machine where the primary scheduler is running, open the operating system command prompt and enter the following commands:
 

```
unisrvcntr start waee_sched.$AUTOSERV
unisrvcntr start waee_server.$AUTOSERV
```
  - b. Repeat this action on all other server machines in the instance.

All schedulers and application servers that are running on a machine in the instance start. AutoSys Workload Automation is configured to run in dual event server mode.

**(UNIX) Install the Tie-Breaker Scheduler**

In dual event server mode, high-availability mode functions properly only with a tie-breaker scheduler installed. The tie-breaker scheduler updates the database to confirm that it is still running and it logs changes in the state of high-availability. At all other times, the tie-breaker scheduler runs idle. The primary and shadow schedulers rely on database updates from the tie-breaker scheduler to determine the state of high-availability.

**NOTE**

- The tie-breaker scheduler cannot run on the same computer as the primary scheduler or the shadow scheduler. The schedulers can run on computers with different operating systems as long as those computers use the same database server.
- The tie-breaker scheduler updates the database at regular intervals and it responds when you execute the `chk_auto_up` command utility. The interval at which the scheduler sends automatic updates depends on the

number of seconds that are specified in the HAPollInterval configuration parameter. Ensure that the primary, shadow, and tie-breaker schedulers are at the same release level.

### Follow these steps:

1. Install the server on the local drive of the computer where the scheduler should run.

#### **WARNING**

Do not install the server on a network drive.

- a. Log in as the root user and mount the AutoSys Workload Automation media.
- b. Change directories to the mounted file system and run the following command:  

```
./wa_setup.sh
```

The Welcome page appears.
- c. Click Next. The License Agreement page appears.
- d. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
- e. Select New and click Next. The Installation Type page appears.
- f. Select Server and click Next. The Instance Information page appears.
- g. Continue with the installation by entering the required information on each page and clicking Next until the Scheduler Properties page appears.
- h. Clear the Start the Scheduler Following the Installation check box, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Primary Event Server Properties page appears. The check boxes that appear depend on your database provider.
  - Oracle:  

```
Create or the Refresh the Database
```

```
Create the Tablespaces, Database Users, and Roles
```
  - Sybase:  

```
Create or the Refresh the Database
```

```
Create New Database Devices
```
- i. Ensure that the installer does not create the database during installation by clearing the relevant check boxes, and then enter all other required information and click Next.
- j. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
- k. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server installation is complete.

2. Designate the scheduler on this computer as the tie-breaker scheduler by editing the \$AUTOUSER/config.\$AUTOSERV file as follows:

```
RoleDesignator=3
```

3. Open the operating system prompt and enter the following command:

```
eventor -q
```

The Tie-Breaker scheduler becomes active and sends updates to the database based on the interval that is specified in the configuration file. The server is installed on three computers. The primary scheduler is running on the first computer, the shadow scheduler is active on the second computer, and the tie-breaker scheduler is active on the third computer. When the primary scheduler detects the activity of the other two schedulers, AutoSys Workload Automation begins operating in high-availability mode with dual event servers.

## (UNIX) Set the Primary Failback Mode

High-availability mode is disabled when the primary scheduler fails over to the shadow scheduler, but you can return to high-availability mode by restoring the primary scheduler. How the primary scheduler resumes processing events after you restore it depends on the primary failback mode.

By default, the primary failback mode is off and cannot resume processing events while the shadow scheduler is processing events. In this case, you can restart the primary scheduler only after you shutdown the shadow scheduler. To minimize downtime, we recommend that set one of the following primary failback mode options:

- **1**  
Specifies that the primary scheduler runs dormant and resumes processing events only when a failback occurs. A failback occurs when the shadow scheduler fails or when you initiate a manual failback. If a failback occurs because the shadow scheduler fails, high-availability mode is disabled. In this case, return to high-availability mode by restoring the shadow scheduler.
- **2** Specifies that the primary scheduler resumes processing events as soon as it detects activity from the other schedulers.

### Follow these steps:

1. Log in as the root user on the machine where the primary scheduler is installed, and run the shell that is sourced to use AutoSys Workload Automation. The environment is defined and the operating system command prompt appears.
2. Enter the following command:

```
unisrvcntr stop waae_sched.$AUTOSERV
```

The primary scheduler stops.

3. Change the primary failback mode.
  - a. Open the \$AUTOUSER/config.\$AUTOSERV configuration file and locate parameter:

```
PrimaryFailbackMode=0
```

- b. Modify the value of the parameter as follows:

```
PrimaryFailbackMode=2 | 1
```

- c. Save and close the configuration file. The primary failback mode is set.

4. Open the operating system prompt and enter the following command:

```
unisrvcntr start waae_sched.$AUTOSERV
```

The primary scheduler starts. In the event of a failover, the primary scheduler resumes processing events according to the mode that you specified.

You have configured AutoSys Workload Automation to run in high-availability mode. Downtime and interruptions in work flow decrease. Scheduler and database failures adversely affect and sometimes disable high-availability. The operator can maintain a highly-available environment by monitoring the environment and restoring from scheduler or database schedulers as needed.

### NOTE

For more information about maintaining highly-available environments, see [How to Maintain Highly-Available Environment](#).

## (Windows) Improve Workload Performance: Highly-Available Environments

As an administrator, you want to set up an environment that promotes optimal workload performance. Sometimes the scheduler fails, interrupting the execution of the workflow. Highly-available environments provide at least one back-up scheduler to help minimize these interruptions. In some highly-available environments, you can enable additional options that further improve workload performance and prevent data loss. AutoSys Workload Automation provides the following high-availability options:

- **Highly-Available Cluster Environments**

Provides a failover solution for the scheduler and the application server. We recommend using this option when you need a failover solution unless you do not have cluster management software installed on your system.

The cluster manager does not provide a failover solution for the database unless you also configure your database to operate on the cluster. You can configure your database to operate on the cluster only when you are using database software that is cluster aware.

Operating in a highly-available cluster environment without a highly-available database increases the risk of downtime and is not recommended. If your database is cluster aware, configure it to operate on the cluster. If your database is not cluster aware, enable dual event server mode.

AutoSys Workload Automation can also cooperate with cluster management software to form a clustered agent. A clustered agent improves load balancing and promotes the optimal use of system resources for job flows that require them. We recommend that you set up a highly-available cluster environment before you set up a clustered agent. To run AutoSys Workload Automation in a highly-available cluster environment with a clustered agent, set up a clustered server on one cluster and set up a clustered agent on a separate cluster that exists within the same AutoSys Workload Automation instance.

- **High-Availability Mode**

Provides an internal failover solution for the scheduler. We recommend using this option to set up a highly-available environment only when you cannot install cluster management software on your system.

- **Dual Event Server Mode**

Provides a failover solution for the database. Use this option only when *one* of the following situations applies:

- You have configured AutoSys Workload Automation to run in a highly-available cluster environment but you are using database software that is not cluster aware.
- You are not using cluster software, you have configured AutoSys Workload Automation to run in high-availability mode, and you are using database vendor that does not provide an internal failover solution.

We recommend using Microsoft Clustering Solutions, but there are also third-party Windows compatible cluster managers available. Microsoft Clustering Solutions includes the following programs:

- **Windows Failover Clustering**

Enables you to set up a highly-available cluster environment

- **Network Load Balancing Cluster**

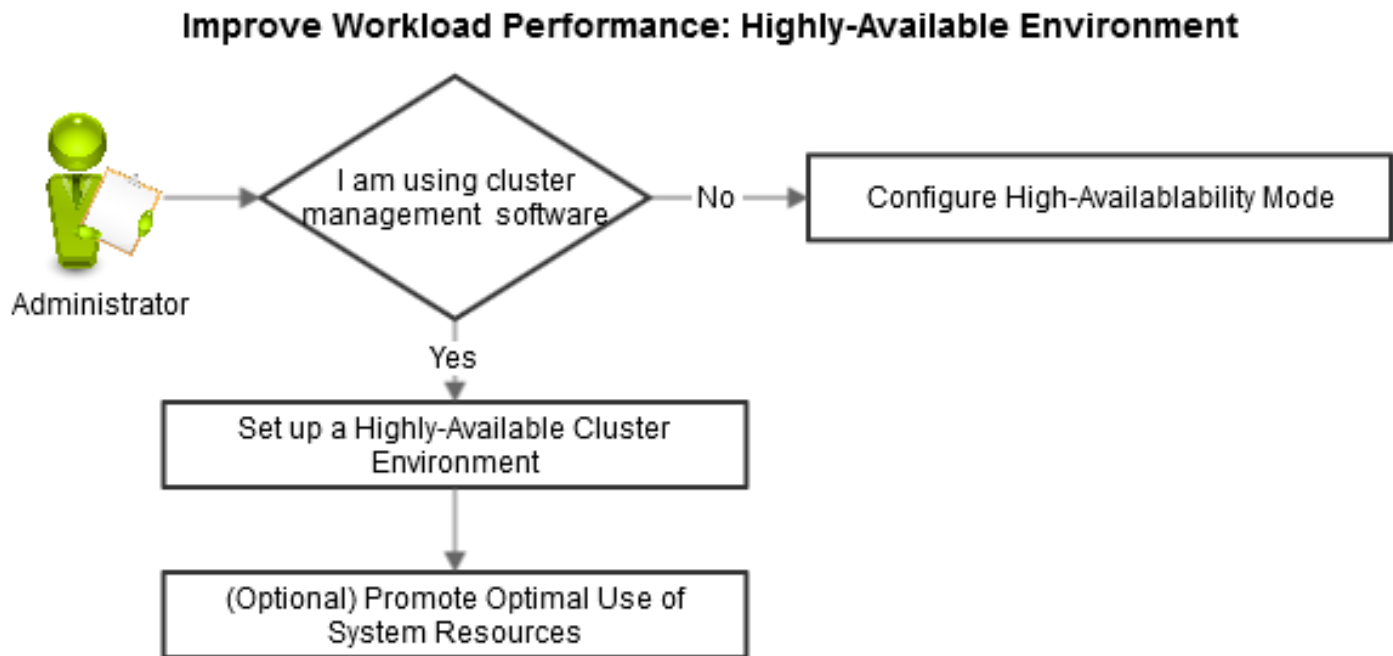
Enables you to promote optimal use of system resources

### NOTE

- Some tasks that are outlined in this scenario are accomplished using the recommended Microsoft Clustering Solutions. For information about how to accomplish these tasks using a third-party Windows compatible cluster manager, consult the documentation for that product.
- For information about how to configure your database to operate on the cluster, see the documentation for your database vendor.

The following diagram illustrates how you can set up a highly available environment:

**Figure 21: how to improve workload performance by setting up a highly-available environment on windows**



1. [Set up a highly available cluster environment.](#)
2. (Optional) [Promote optimal use of system resources.](#)
3. [Configure high availability mode.](#)

### **(Windows) Set Up a Highly Available Cluster Environment**

As an administrator, your responsibilities include configuring the AutoSys Workload Automation environment. Highly-available environments minimize down time and data loss. The most stable highly-available environment is a highly-available cluster environment.

AutoSys Workload Automation and cluster-aware databases can cooperate with cluster management software to create a highly-available application server, scheduler, and database. Configure cluster-aware databases to operate on the cluster before completing this scenario. Enable dual event servers to provide a failover solution when using a database that is not cluster-aware.

In a highly-available cluster environment, you can further improve workload performance by setting up a clustered agent or clustered client.

#### **WARNING**

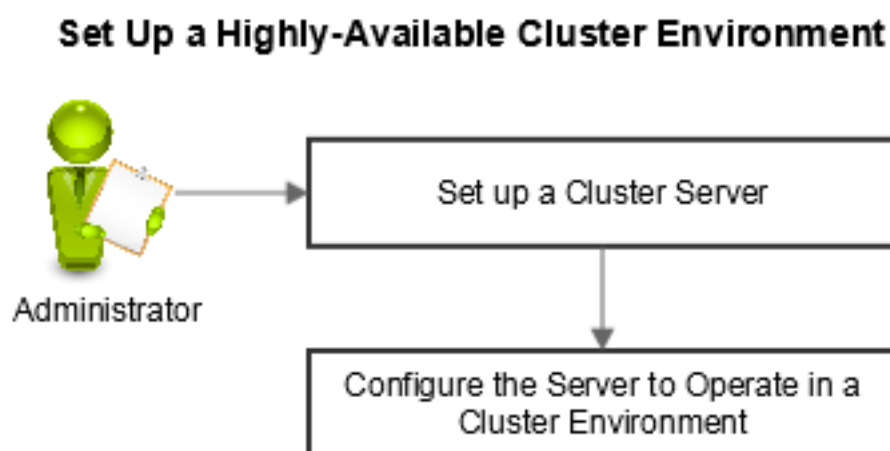
Transparent Application Failover (TAF) mode prevents AutoSys Workload Automation from accessing its database schema on the active database node following a failover on the Oracle Server. So, for AutoSys Workload Automation to work with the Oracle server that is running in the TAF mode, ensure that the Oracle client on the AutoSys Workload Automation server is configured for the Oracle RAC failover. No configuration changes are required on the Oracle server. For more information about configuring the AutoSys Workload Automation server to work with Oracle RAC, see [Install AutoSys Workload Automation with Oracle](#).

#### **NOTE**

- Running AutoSys Workload Automation in a highly-available cluster environment requires that you have cluster management software installed on your system. There are many brands of Windows compatible cluster management software, such as Microsoft Cluster Service. For more information about installing and configuring a cluster manager, see the documentation for your cluster management software.
- For information about how to configure your database to operate on the cluster, see the documentation for your database vendor.

This diagram illustrates how you can set up a highly-available cluster environment:

**Figure 22: set up a highly-available cluster environment**



**Follow these steps:**

1. [Set up a clustered server.](#)

**WARNING**

The database cannot operate on the cluster when AutoSys Workload Automation is operating in dual event server mode. You can enable or disable dual event server mode when you set up a clustered server. Enable dual event server mode only when your database software is not cluster aware.

2. [Configure the server to operate in a cluster environment.](#)

### (Windows) Set Up a Clustered Server

Command jobs that execute client utilities are more likely to succeed when they run on a load-balancing cluster that consists of both a clustered agent and a clustered client. A load-balancing cluster without a clustered client provides no improvement in workload performance for these types of jobs. If the scheduler is running these types of jobs, set up a clustered client.

**Follow these steps:**

1. Verify that Microsoft Clustering Solutions is installed.
2. If your database is cluster aware, configure it to operate on the cluster.

**NOTE**

For more information about how to configure your database to operate on the cluster, see the documentation for your database vendor and cluster management software.

3. Determine which node on the cluster is active.



**NOTE**

This step applies to identifying the active node when you are using Microsoft Clustering Solutions. If you are using another Windows compatible cluster manager, identify the active node using the steps that are outlined in the documentation for your cluster management software.

- a. Click Start, Programs, Administrative Tools, Failover Cluster Management. The Failover Cluster Management console appears.
- b. Click the plus (+) icon next to Services and applications. A list of cluster resource groups that represent the clusters in the instance appears.
- c. Select the resource group that represents the cluster that you are using to set up your highly-available environment.

**NOTE**

You specify a resource group to represent each cluster in an instance when you install and configure the cluster manager for that instance. To determine which resource group represents a particular cluster, view the cluster manager configuration settings. For more information about how to view these settings, see the documentation for your cluster management software.

The cluster resource group pane appears displaying summary information about the selected resource group. The node name that is displayed in the Current Owner field of the cluster resource group pane is the active node.

4. Install the server on the active node.
  - a. Log in as a user with Windows Administrators group privileges, insert the installation media into the drive, and mount the media.
  - b. If the Product Explore does not appear, run setup.exe. The Product Explorer appears.
  - c. Select AutoSys Workload Automation, and click Install. The Welcome page appears.
  - d. Click Next. The License Agreement page appears.
  - e. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
  - f. Select New and click Next. The Installation Type page appears.
  - g. Select Custom and click Next. The Components page appears.
  - h. Select Scheduler, Application Server, Agent, and any optional components that you want to install, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Application Server Properties page appears.
  - i. Select the Configure this Instance for Use in a Clustered Environment check box, enter all other required information, and click Next. The installer automatically disables the Configure High-Availability check box on the Scheduler Properties page. The default behavior to automatically start the application server and scheduler following installation and at system startup is disabled. The installer automatically clears and disables the corresponding check boxes on the Application Server Properties page and Scheduler Properties page. Depending on the components that you selected on the Components page, the CA Embedded Entitlements Manager (CA EEM) Properties page or the Database Type page appears.
  - j. If the CA EEM Properties page appears, enter the required information, and click Next. The Database Type page appears.
  - k. Select the database system that you are using from the Database type drop-down list.
  - l. If your database is configured to operate on the cluster, clear the Employ Dual Event Servers check box, and click Next. If your database is not cluster-aware, select the Employ Dual Event Servers check box, and click Next.

**WARNING**

The database cannot operate on the cluster when AutoSys Workload Automation operates in dual event server mode. Always disable dual event servers when you configure your database to operate on the cluster.

Operating in single event server mode without a clustered database limits the benefits of a highly-available cluster environment. If your database software is not cluster aware, ensure that you enable dual event servers.

The first of the Primary Event Server Properties pages appears.

- m. Provide the information that the installer needs to create the database during installation by selecting the relevant check boxes, enter all other required information, and click Next. The check boxes that appear depend on your database provider.
  - **Microsoft SQL:**
    - Create or the Refresh Database
  - **Oracle:**
    - Create or the Refresh Database
    - Create the Tablespaces, Database Users, and Roles
  - **Sybase:**
    - Create or the Refresh Database
    - Create New Database Devices

The next page in the Primary Event Server Properties page series appears.

- n. Complete the required information and click Next on each page in the series. If you enabled dual event servers, specify the second event server properties by entering the required information and clicking Next on each page in the Secondary Event Servers Properties page series. Ensure that you specify different properties for each event server but that both event servers have the same user name and password. The Database Test page appears and the database test starts.
- o. If the database test completes successfully, click Next. If the database test fails, read the error message to determine what caused the failure, use the Back button to navigate to the installation pages that contain the errors, correct the errors, click Next until you return to the Database Test page, and wait for the test to complete. Repeat this action until the test completes successfully, and click Next. The Scheduler Properties page appears.
- p. Enter the required information and click Next.

#### NOTE

If you selected Configure this Instance for Use in a Clustered Environment check box, the installer disabled the following check boxes:

- Configure High-Availability
- Start the Scheduler Following Installation
- Start the Scheduler at System Startup

If these check boxes are enabled, use the back button to navigate to the Application Server Properties page, verify that the Configure this Instance for Use in a Clustered Environment check box is selected, use the Next button to return to the Scheduler Properties page, enter the required information and click Next. The Cluster Resource Group page appears.

- q. Provide the information that the installer needs to add the scheduler and application server to the appropriate resource group during installation by selecting the cluster resource group from the Select Cluster Resource Group drop down list and click Next. If you need to verify the cluster resource group for a particular node, open the Failover Cluster Management console, click Services and applications, and locate the cluster resource group that displays that node in the Current Owner field. The Agent Properties page appears.
- r. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
- s. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server is installed on the active node.

5. Repeat the server installation on the passive nodes.
  - a. Click Start, Administrative Tools, Failover Cluster Management, click the plus (+) icon next to Services and applications, select the cluster resource group, click Move this service or application to another node, and select one of the passive nodes from the drop-down list. The cluster resource group moves to the selected node and the cluster resource group pane displays the name of that node in the Current Owner field.
  - b. Open the AutoSys Workload Automation installation wizard on the node where the cluster resource group resides, enter the required information on each page and click Next until the first of the Primary Event Server Properties pages appears.
  - c. Enter the required information and click Next. Ensure that you select the Employ Dual Event Servers check box if you enabled dual event servers when you installed the server on the active node, and that you clear that check box if you are configured your database to operate on the cluster. Ensure that the installer does not create the database on the passive nodes by clearing the relevant check boxes. The check boxes that appear on the first page of each series depend on your database provider.
    - Microsoft SQL:  
Create or the Refresh Database
    - Oracle:  
Create or the Refresh Database  
Create the Tablespaces, Database Users, and Roles
    - Sybase:  
Create or the Refresh Database  
Create New Database Devices

The next page in the Primary Event Server Properties page appears.
  - d. Continue with the installation by entering the required information on each page in the Primary Event Servers properties page series, ensuring that it matches the information that you entered when you installed the server on the active node, and clicking Next until you complete the series. If the Secondary Event Server Properties page series appears, ensure that the backup database is not created on the active nodes by clearing the appropriate check boxes on the first page in the series, complete all other required information on the page, click Next, and specify the other second event server information by entering the required information on each page of the Secondary Event Server Properties page series and clicking Next until you complete the series. The Database Test Page appears and the database test starts.
  - e. If the test completes successfully click Next. If the test fails, click Skip Test to bypass the test. The Agent Properties page appears.
  - f. Enter the required information on the Agent Properties page, ensuring that it matches the information that you entered when you installed the server on the active node, and click Next. Repeat this action on each subsequent page until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
  - g. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server components are installed on all nodes in the cluster.

The server components are installed on all nodes in the cluster. You have set up a clustered server, and you can configure the server to operate in a cluster environment.

## (Windows) Configure the Server to Operate in Cluster Environment

The cluster manager cannot restart failed components on passive nodes until you configure the cluster manager and the server to operate cooperatively.

### Follow these steps:

1. Verify that the server components are offline.

2. Define the Manager Host Alias on each node, beginning with the active node.
  - a. Click Start, Programs, AutoSys, Administrator. The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select the appropriate instance from the Instance drop down list in the settings pane.
  - c. Click the Communication tab, select the Manager Hostname Alias check box, and enter the cluster host name in the Manager Hostname Alias field.

**WARNING**

Ensure that you specify the same manager host alias name on all nodes.

The Manager Hostname Alias is defined.

**NOTE**

If you are connecting from the Web UI, ensure that on each node you also set the Server Host field to the cluster host name on the Application Server - AutoSys Workload Automation Administrator window of the Administrator utility. If the AutoSys Workload Automation cluster fails over, Web UI connects to the cluster host you specified in the Server Host field.

3. Use the AutoSys Workload Automation Administrator to start the server components on the active node.
  - a. Click the Services icon on the toolbar. The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - b. Right click the Scheduler service, and select Start. Repeat this action for the application server.
4. If you want to schedule jobs to run on the local host machine, define one manually.

**WARNING**

Setting up a clustered server requires that you disable the server installation option to start the application server automatically. Disabling this option prevents the installer from automatically inserting the local host machine definition. If you do not insert the definition manually on each node, jobs that are defined to run on the local host machine fail to start.

- a. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*). The instance command prompt appears.
- b. Enter the following command:

```
jil
```

The JIL command prompt appears.

- c. Enter the following command and attributes:

```
insert_machine:cluster_host_name
agent_name: agent_name
encryption_type: {NONE | AES |DEFAULT}
key_to_agent: key_value
port: agent_port
```

**cluster\_host\_name** Specifies the name of the node for which you are defining the local host machine.

**agent\_name** Specifies the name of the agent.

**NONE | AES |DEFAULT** Specifies the encryption type for the agent.

**key\_value** Specifies the encryption key. This parameter is only required for agents that use AES encryption.

**agent\_port** Specifies the port that the agent listens to.

**WARNING**

When you install a server on a node, the installer prompts you to enter agent definition information. The agent definition information is identical across all nodes on the cluster. Ensure that the agent definition on a node and the local host machine definition specify identical values for the following attributes:

```
agent_name: agent_name
encryption_type: {NONE | AES |DEFAULT}
key_to_agent: key_value
```

```
port: agent_port
```

The local host machine is defined.

d. Enter the following command:

```
exit
```

The JIL command prompt closes, and the local host machine is defined as the machine that represents the clustered server.

The server is configured to operate in a cluster environment.

You have set up a highly-available cluster environment. Failover solutions are in place for the scheduler and application server. You can also set up an environment that promotes optimal use of system resources.

## **(Windows) Promote Optimal Use of System Resources**

As an administrator, you want to set up an environment that enables the scheduler to improve workload performance. With cluster management software installed, you can set up a load-balancing cluster. A load-balancing cluster consists of a clustered agent. Optionally, you can also set up a clustered client.

With a load-balancing cluster and a machine that is defined to represent that cluster, the scheduler can improve load-balancing and decrease job failures by defining certain jobs to run on the cluster. We recommend this method for the following jobs:

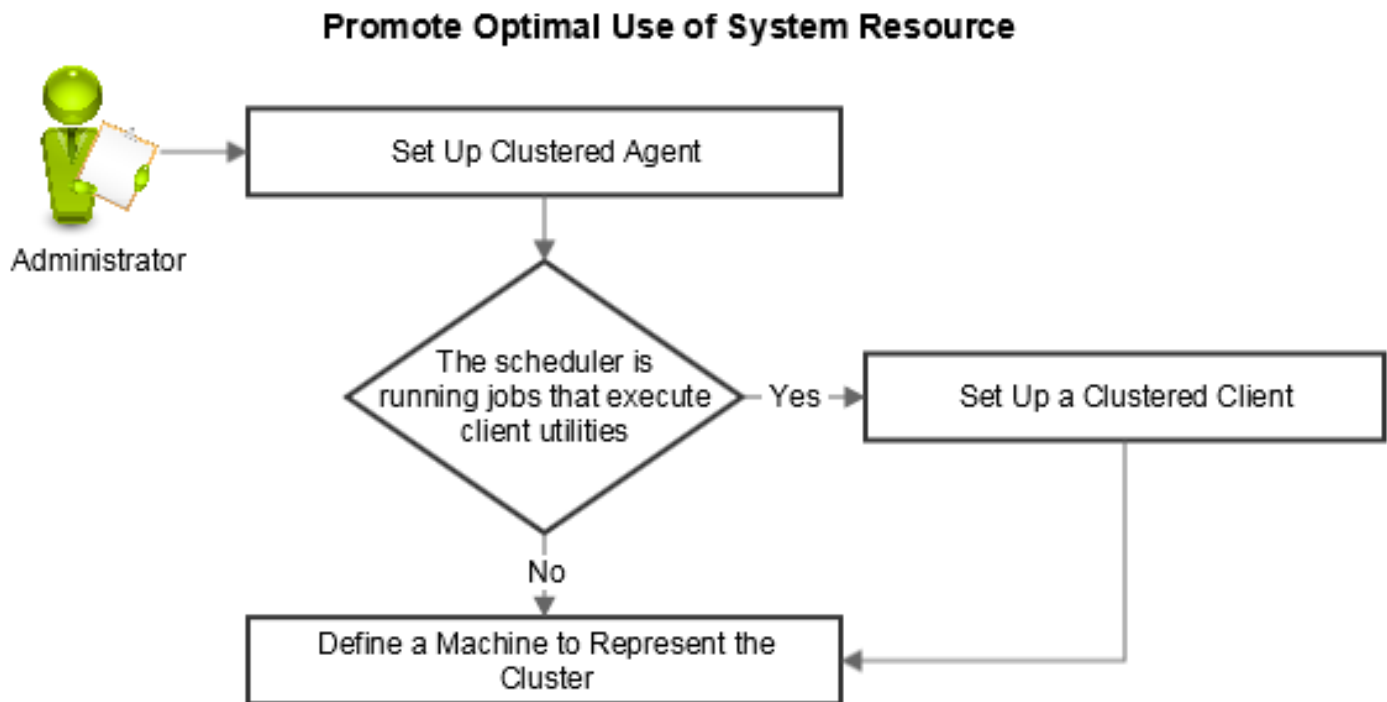
- CPU intensive jobs
- Jobs that use shared resources
- Command jobs that execute client utilities

### **NOTE**

Multiple AutoSys Workload Automation instances can utilize the same load-balancing cluster.

The following diagram illustrates how you can promote use of system resources:

Figure 23: promote use of system resources

**Follow these steps:**

1. [Set up a clustered agent.](#)

**WARNING**

Ensure that you set up the clustered agent on a different cluster than the clustered server.

2. (Optional) [Set up a clustered client.](#) Use this option to decrease the risk of failure for jobs that execute client utilities.
3. [Define a machine to represent the cluster.](#)

**(Windows) Set Up a Clustered Agent**

A cluster manager complements the load-balancing capabilities of the scheduler when jobs are defined to run on the cluster. The scheduler cannot define jobs to run on a cluster until you set up a clustered agent.

**Follow these steps:**

1. Verify that Microsoft Clustering Solutions is installed within the AutoSys Workload Automation instance and that two separate clusters exist.
2. Verify that the AutoSys Workload Automation instance is operating in a highly-available cluster environment and that the client is installed on the active node of the highly-available cluster.

**WARNING**

Do not set up the clustered agent on the highly-available cluster. Designate another cluster as the load-balancing cluster and set up the clustered agent on that cluster.

3. Configure the cluster manager to load-balance clustered resources.

**NOTE**

For information about configuring your cluster manager to load-balance clustered resources, see the documentation for your cluster management software.

4. Install an additional agent on any node of the load-balancing cluster.
  - a. Log in as a user with Windows Administrators group privileges, insert the installation media into the drive, and mount the media.
  - b. If the Product Explore does not appear, run setup.exe. The Product Explorer appears.
  - c. Select AutoSys Workload Automation, and click Install. The Welcome page appears.
  - d. Click Next. The License Agreement page appears.
  - e. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
  - f. Select New and click Next. The Installation Type page appears.
  - g. Select Custom and click Next. The Components page appears.
  - h. Select the Agent check box and click Next. The Installation Path page appears.
  - i. Continue the installation by entering the required information on each page and clicking Next until the Review Settings page appears.

**NOTE**

If you use the same agent name for one agent on each node in the cluster, these commonly named agents are automatically configured to operate as part of the cluster. If you want each agent to have a unique name, then the agents cannot operate on the cluster until you define a common agent name for them.

The page displays the information that you entered in the installation wizard.

- j. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The agent with the specified name is installed.
5. Repeat the agent installation on every node in the cluster using a common agent name. Define a common agent name using one of the following methods:
    - Install one agent on each node, and use the same name for all of these agents.
    - Install a uniquely named agent on each node, and define an alias agent name that is common to all of the agents on the cluster. To define an alias agent name, open the agentparam.txt file and add the following parameters:
 

```
communication.alias_n=common_agent_name
[communication.alias2=second_agent_alias]
...
[communication.aliasnnn=additional_agent_alias]
```
    - **common\_agent\_name** Specifies the common agent name for the cluster and defines it as the first alias for the agent.
    - **second\_agent\_alias** (Optional) Specifies a second alias for the agent. Do not specify the same value that you used for the unique agent name or the common agent name.
    - **nnn** (Optional) Specifies the numeral that corresponds with the additional agent alias.
    - **additional\_agent\_alias** (Optional) Specifies an additional alias for the agent. Do not specify the same value that you used for the unique agent name, common agent name, or any other agent alias.
  6. If you used an agent alias name to define the common agent name, configure AutoSys Workload Automation to recognize the individual agent on each node. Ensure that you specify the same common agent name as the first agent alias for this agent in the agentparm.txt file on each node.
    - a. Log in on the active node of the highly-available cluster and click Start, Programs, AutoSys Workload Automation, Command Prompt (*instance\_name*). The instance command prompt appears.
    - b. Enter the following command:

```
jil
```

The JIL command prompt for the local AutoSys Workload Automation instance appears.

- c. Insert a AutoSys Workload Automation machine definition for each uniquely named agent that should operate on the cluster by entering the following JIL command and attributes:

```
insert_machine: machine_name
node_name: cluster_node
agent_name: unique_agent_name
encryption_type: {NONE | AES |DEFAULT}
key_to_agent: key_value
port: common_agent_port
```

- d. If you want to run jobs on a machine that references an agent by one of its additional agent aliases, insert a AutoSys Workload Automation machine definition for each additional agent alias by entering the following JIL command and attributes:

```
insert_machine: machine_name
node_name: cluster_node
agent_name: {second_agent_alias|additional_agent_alias}
encryption_type: {NONE | AES |DEFAULT}
key_to_agent: key_value
port: common_agent_port
```

- **machine\_name** Specifies the name of the AutoSys Workload Automation machine.
- **cluster\_node** Specifies the name of the node on which you are installing the agent.
- **unique\_agent\_name** Specifies the unique name or one of the alias names of the agent. Use a different name for every subsequent agent installation. Do not use the same name that you use as the common agent name for the cluster.
- **second\_agent\_alias** (Optional) Specifies a second alias for the agent. Do not specify the same value that you used for the unique agent name or the common agent name.
- **additional\_agent\_alias** (Optional) Specifies an additional alias for the agent. Do not specify the same value that you used for the unique agent name, common agent name, or any other agent alias.
- **NONE | AES |DEFAULT** Specifies the encryption type. Specify the same encryption type for all agents.
- **key\_value** Specifies the encryption key. This parameter is only required for agents that use AES encryption.
- **common\_agent\_port** (Optional) Specifies the port that the agent listens to. Specify the same port number for every agent.

- e. Enter the following command:

```
exit
```

The AutoSys Workload Automation machine definitions are stored and the JIL command prompt closes.

The common agent, all uniquely named agents, and all agent aliases on the cluster are defined and AutoSys Workload Automation is operating with a clustered agent.

## (Windows) Set Up a Clustered Client

Command jobs that execute client utilities are more likely to succeed when they run on a load-balancing cluster that consists of both a clustered agent and a clustered client. A load-balancing cluster without a clustered client provides no improvement in workload performance for these types of jobs. If the scheduler is running these types of jobs, set up a clustered client.

### Follow these steps:

1. Verify that the AutoSys Workload Automation instance is operating in a highly-available cluster environment, that the client is installed on the active node of the highly-available cluster, and that a separate load-balancing cluster with a clustered agent exists.



2. Install the client on any node of the load-balancing cluster.
  - a. Log in as a user with Windows Administrators group privileges, insert the installation media into the drive, and mount the media. If the Product Explorer does not appear, run setup.exe. The Product Explorer appears.
  - b. Select AutoSys Workload Automation, click Install, and then start the installation by clicking Next. The License Agreement page appears.
  - c. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next.

#### NOTE

The Detected Software page appears only when the installation wizard finds at least one AutoSys Workload Automation component or CA AC installed on the computer. Use the Add or Remove Programs tool in the Windows Control Panel to remove unwanted components.

The Installation Option page appears.

- d. Select New and click Next. The Installation Type page appears.
- e. Select Custom and click Next. The Components page appears.
- f. Select the Client check box, click Next, and then continue with the installation by entering the required information and clicking Next on each page until the Instance Information page appears.
- g. Enter the AutoSys Workload Automation instance name that you used when you installed the server, click Next, and then continue with the installation by entering the required information and clicking Next on each page until the Application Server Information page appears.
- h. Enter cluster host name that you used when you installed and configured the cluster manager for the highly-available cluster in the Application Server Host field, enter all other required information, and click Next.

#### NOTE

You can verify the cluster host name by viewing the configuration settings for your cluster manager. For more information about how to view these settings, see the documentation for your cluster management software.

- i. Continue with the installation by entering the required information on each page and clicking Next until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
  - j. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The client is installed on the node.
  3. Repeat the client installation on all nodes in the cluster that define, run, or monitor jobs.
  4. If jobs that execute client utilities are defined within other AutoSys Workload Automation instances and you want to run those jobs on the load-balancing cluster, configure the clustered client to operate within those instances.
    - a. Log in on any node of the load-balancing cluster, insert and mount the installation media, start the installation, enter the appropriate instance name in the Instance Name field on the Instance Information page, and complete the installation.
    - b. Repeat the installation, entering the same instance name, on all nodes in the cluster that define, run, or monitor jobs.
    - c. Repeat these actions for every applicable AutoSys Workload Automation instance.
- AutoSys Workload Automation is operating with a clustered client.

## (Windows) Define a Machine to Represent the Cluster

A machine that represents the cluster can run jobs on any agent that is installed as part of the cluster. The scheduler can define a machine to represent the cluster when a cluster manager is installed on the system and multiple agents are installed as part of the cluster. Typically, the administrator defines a machine to represent the cluster when setting up an environment that promotes optimal use of system resources.

Running jobs on a cluster improves workload performance. Load balancing improves for jobs that use shared resources or have high CPU consumption. Job failures decrease for jobs that execute client utilities when you have also set up a clustered client. You can define jobs to run on a cluster only when a machine is defined to represent the cluster.

### Follow these steps:

1. Log in on the active node of the highly-available cluster and Click Start, Programs, AutoSys, Command Prompt (*instance\_name*). The instance command prompt appears.

2. Enter the following command:

```
jil
```

The JIL command prompt for the local AutoSys Workload Automation instance appears.

3. Enter the following command and attributes:

```
insert_machine: cluster_machine_name
node_name: cluster_host_name
agent_name: common_agent_name
port: common_agent_port
```

- **cluster\_machine\_name** Specifies the name of the real AutoSys Workload Automation machine that represents all agents in the cluster.

#### WARNING

You cannot define a virtual machine or a machine pool to represent the cluster.

- **cluster\_host\_name** Specifies the name of the host for the cluster.

#### WARNING

The cluster machine name is not the cluster host name. To specify the cluster host name, use the same cluster host name that you used when you installed and configured the cluster manager. For information about how to display the configuration settings for your cluster manager, see the documentation for your cluster management software.

- **common\_agent\_name** Specifies the name or alias that is common to all agents installed as part of the cluster.

#### WARNING

Use the same common agent name or alias that you used when you installed multiple agents as part of the cluster.

- **common\_port\_name** Specifies the port that is common to all agents that are installed as part of the cluster.

#### WARNING

Use the same port that you specified when you installed multiple agents as part of the cluster.

4. (Optional) Specify optional machine attributes.

5. Enter the following command:

```
exit
```

The machine that represents the cluster is defined, and the definition is saved on the database.

You have set up an environment that promotes optimal use of system resources. The scheduler can define jobs to run on the cluster.

## (Windows) Configure High-Availability Mode

As an administrator, you want to promote optimal workload performance without installing cluster management software on your system. You can set-up a highly-available environment using internal options.

You can activate high-availability mode by installing a shadow scheduler to act as a failover solution for the primary scheduler. In this mode, the primary scheduler processes events and the shadow scheduler is available to take over this work in the event of a failover.

The primary scheduler is installed during the initial server installation. To operate AutoSys Workload Automation in high-availability mode, repeat the server installation on another machine designate the scheduler on that machine as the shadow scheduler. The shadow scheduler is designed to take over processing events when the primary scheduler fails.

Sometimes you want to force the shadow scheduler to take over, such as when you need to perform maintenance on the machine where the primary scheduler runs. In this case, you can initiate a manual failover using the `sendevent` command. The primary scheduler shuts down, and the shadow scheduler takes over.

High-availability mode is disabled and the shadow scheduler becomes the active scheduler when a failover occurs, including when you initiate a manual failover. To return to high-availability mode following a failover, restore the primary scheduler.

By default, the primary scheduler automatically shuts down when it detects that the shadow scheduler is processing events. In this case, you can return to high-availability mode only if you shut down the shadow scheduler, restore the primary scheduler, and then restart the shadow scheduler. You can avoid this downtime by configuring the primary fallback mode.

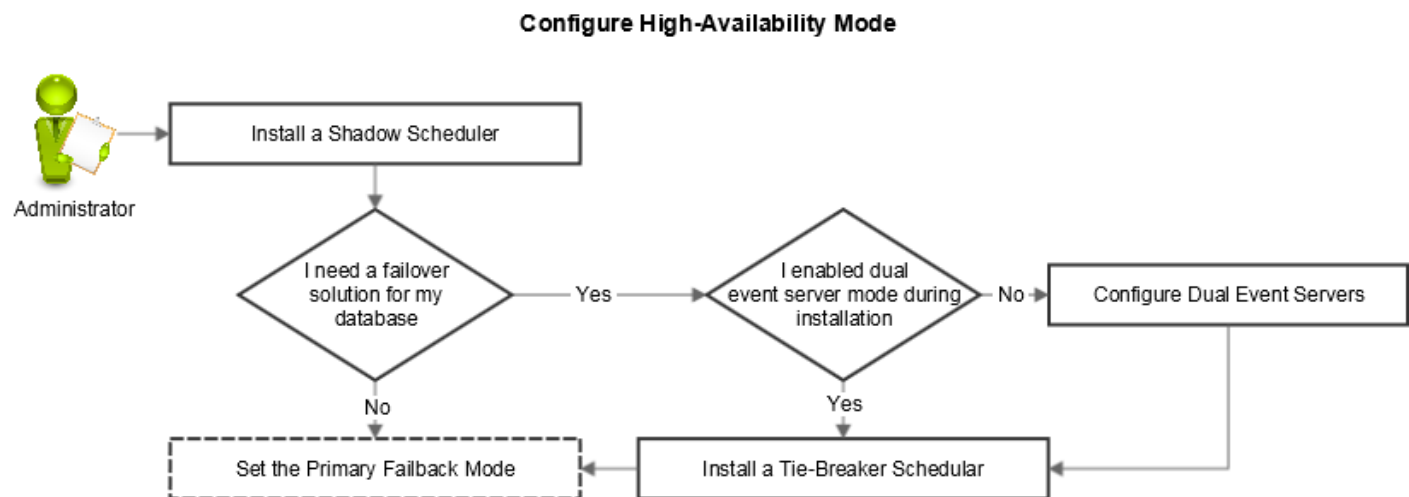
High-availability mode does not provide a failover solution for the database. Most database vendors provide internal failover solution. You can also provide a failover solution by configuring AutoSys Workload Automation to run in dual event server mode. We recommend configuring dual event server mode only when your database vendor does not support an internal failover solution. High-availability mode does not operate properly with dual event servers unless you install a tie-breaker scheduler.

#### NOTE

- High-availability mode does not support optimal use of system resources. You can promote optimal use of system resources only when you have cluster management software installed.
- For more information about how to initiate a manual failover or fallback using the `sendevent` command, see [sendevent Command—Stop the Schedulers or Application Servers](#).

The following diagram illustrates how you can configure high-availability mode:

**Figure 24: configure high-availability mode**



#### Follow these steps:

1. [Install the shadow scheduler.](#)
2. [Configure dual event servers.](#)
3. [Install the tie-breaker scheduler.](#)

4. (Optional) [Set the primary failback mode.](#)

## (Windows) Install the Shadow Scheduler

You can install a shadow scheduler to take over processing events when the primary scheduler fails. Starting the shadow scheduler enables high availability mode.

### NOTE

The shadow scheduler cannot run on the same computer as the primary scheduler. The primary scheduler and the shadow scheduler can run on computers with different operating systems as long as those computers use the same database platform and are defined within the same AutoSys Workload Automation instance. Ensure that the primary, shadow, and tie-breaker schedulers are at the same release level.

### Follow these steps:

1. Install the server on the computer where the shadow scheduler should run.
  - a. Log in as a user with Windows Administrators group privileges, insert the installation media into the drive, and mount the media.
  - b. If the Product Explore does not appear, run setup.exe. The Product Explorer appears.
  - c. Select AutoSys Workload Automation, and click Install. The Welcome page appears.
  - d. Click Next. The License Agreement page appears.
  - e. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
  - f. Select New and click Next. The Installation Type page appears.
  - g. Select Custom and click Next. The Components page appears.
  - h. Select Scheduler, Application Server, Agent, and any optional components that you want to install, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Primary Event Server Properties pages appear.
  - i. Ensure that the installer does not create the database during installation by clearing the relevant check boxes, and then enter all other required information and click Next. The check boxes that appear depend on your database provider.
    - Microsoft SQL:  
Create or the Refresh Database
    - Oracle:  
Create or the Refresh Database  
Create the Tablespaces, Database Users, and Roles
    - Sybase:  
Create or the Refresh Database  
Create New Database Devices

The next page in the Primary Event Server Properties page series appears.

  - j. Enter the required information and click Next on each page in the series. If you enabled dual event servers, specify the second event server properties by entering the required information and clicking Next on each page in the Secondary Event Servers Properties page series until you complete the series. Ensure that you specify different properties for each event server but that both event servers have the same user name and password. The Database Test page appears and the database test starts.
  - k. If the database test is successful, click Next. If the database test fails, click Skip Test to bypass the test. The Agent Properties page appears.
  - l. Continue with the installation by entering the required information on each page and clicking Next until the Scheduler Properties page appears.

- m. Clear the Start the Scheduler Following the Installation check box, click Next, and continue with the installation by entering the required information and clicking Next on each page until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
  - n. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server installation is complete.
2. Define the scheduler role.
    - a. Click Start, Programs, AutoSys, Administrator. The Instance - AutoSys Workload Automation Administrator window opens.
    - b. Select the instance on which the Scheduler runs from the Instance drop-down list on the Settings pane and Click the Scheduler icon on the toolbar. The Scheduler - AutoSys Workload Automation Administrator window appears.
    - c. Select the Shadow option in the Scheduler Role pane and click Apply. The scheduler role is defined.
  3. Click the Services icon, right click Scheduler, and select Start. The Shadow scheduler becomes active. If you did not enable dual event servers during installation, AutoSys Workload Automation enters high-availability mode as soon as the primary scheduler detects the activity of the shadow scheduler.

## (Windows) Configure Dual Event Servers

We recommend using the internal failover solution that is provided by your database vendor, but some databases do not provide one. In this case, use dual event server mode as the failover solution for your database.

AutoSys Workload Automation automatically installs a second event server that is identical to the primary event server when you enable dual event server mode during installation. You can also switch from single event mode to dual event server mode by copy the database and configuring dual event servers.

In dual event server mode, the secondary active scheduler performs a database rollover to the secondary event server when the primary event server fails. In this case, AutoSys Workload Automation begins operating in single event server mode.

Configure dual event servers in the following situations:

- When you want to operate in dual event server mode but did not enable dual event servers during installation
- When you want to resume operating in dual event server mode after a database rollover

### Follow these steps:

1. Log in to AutoSys Workload Automation machine in the instance with a client installation, and Click Start, Programs, AutoSys, Command Prompt (instance\_name). The instance command prompt appears.
2. Enter the following command:
 

```
sendevent -E STOP_DEMON -v ALL
```

All server processes that are running on the instance stop.
3. If you did not enable dual event servers during installation and want to switch to dual event server mode, install and configure the second event server:
  - a. Install database software on another machine within the AutoSys Workload Automation instance. Ensure that the two separate database servers are the same size and type but that each one resides on different machines and has a unique name. A second database server is installed in the instance. You can install another event server on this database server.
  - b. Run the [CreateAEDB](#) script.

### WARNING

Dual event server mode functions properly only when the event servers have unique names. When you run the script to create the secondary event server, specify a name that is different from the name of the primary event server.

The secondary event server is installed.

4. Stop the scheduler and the application server services.
  - a. Click Start, Programs, AutoSys, Administrator. The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane, and then click the Services icon on the toolbar. The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - c. Right-click Scheduler and select Stop, and then right-click Application Server and select Stop. The scheduler and application server stop.
5. Click the Event Server icon on the toolbar of any AutoSys Workload Automation Administrator window. The Event Server - AutoSys Workload Automation Administrator window appears.
6. Complete the required information for Event Server 1 and click Apply. Ensure that the information that you enter for Event Server 2 matches the information that you entered for Event Server 1. The database configuration settings are saved to the configuration file.
7. Run the bulk copy script ([Configure AutoSys Workload Automation to Run in Dual Event Server Mode](#)) based on your database vendor to synchronize the event servers.
8. Enable the second event server.
  - a. Click the Event Server icon on the toolbar of any AutoSys Workload Automation Administrator window. The Event Server - AutoSys Workload Automation Administrator window appears.
  - b. Click the Event Server 1 or Event Server 2 tab based on the event server you want to enable and click Enable. The second event server is enabled. The status of the second event server is displayed as UP and the A Database Rollover Has Occurred check box is cleared and disabled.
9. If you want to return to dual event server mode following a database failure, recover the failed event server, and then enable and reconfigure dual event servers.
  - a. Review the scheduler log file to determine the problem that caused the failure.
  - b. Consult your database software documentation, and follow the instructions for manually resolving the problem that caused the database failure. The status of the failed database changes and you can reconfigure dual event servers.
  - c. Click the Event Server icon on the toolbar in any AutoSys Workload Automation Administrator window. The Event Server - AutoSys Workload Automation Administrator window appears.
  - d. Clear the A Database Rollover Has Occurred check box, verify the event server configuration settings and make any desired modifications, and then click Apply.

### **WARNING**

Ensure that the configuration settings for Event Server 2 are identical to the configuration settings for Event Server 1.

The failed event server is restored and the configuration settings are saved, but AutoSys Workload Automation does not return to dual event server mode until you synchronize the event servers.

- e. Run the AutoSys Workload Automation bulk copy script. The directory and script that you synchronize the event servers following a database failure are the same as the directory and script that you use to switch to dual event server mode when you did not enable dual event servers during installation. The event servers are synchronized and AutoSys Workload Automation is configured to run in dual event server mode.
10. Restart the server processes for the instance.
  - a. Click the Services icon on the toolbar in any AutoSys Workload Automation Administrator window. A list of services installed on the AutoSys Workload Automation instance appears.
  - b. Right-click the scheduler and select Start, and then right-click the application server and select Start. All schedulers and application servers that are running on a machine in the instance start. AutoSys Workload Automation is configured to run in dual event server mode.

## (Windows) Install the Tie-Breaker Scheduler

In dual event server mode, high-availability mode functions properly only with a tie-breaker scheduler installed. The tie-breaker scheduler updates the database to confirm that it is still running and it logs changes in the state of high-availability. At all other times, the tie-breaker scheduler runs idle. The primary and shadow schedulers rely on database updates from the tie-breaker scheduler to determine the state of high-availability.

### NOTE

- The tie-breaker scheduler cannot run on the same computer as the primary scheduler or the shadow scheduler. The schedulers can run on computers with different operating systems as long as those computers use the same database server.
- The tie-breaker scheduler updates the database at regular intervals and it responds when you execute the `chk_auto_up` command utility. The interval at which the scheduler sends automatic updates depends on the number of seconds that are specified in the `HAPollInterval` configuration parameter. Ensure that the primary, shadow, and tie-breaker schedulers are at the same release level.

### Follow these steps:

1. Install the server on the computer where the tie-breaker scheduler should run.
  - a. Log in as a user with Windows Administrators group privileges, insert the installation media into the drive, and mount the media.
  - b. If the Product Explore does not appear, run `setup.exe`. The Product Explorer appears.
  - c. Select AutoSys Workload Automation, and click Install. The Welcome page appears.
  - d. Click Next. The License Agreement page appears.
  - e. Read the license text, select the I Accept the Terms of the License Agreement option, and click Next. The Installation Option page appears.
  - f. Select New and click Next. The Installation Type page appears.
  - g. Select Custom and click Next. The Components page appears.
  - h. Select Scheduler, Application Server, Agent, and any optional components that you want to install, click Next, and continue with the installation by entering the required information on each page and clicking Next until the Primary Event Server Properties pages appear.
  - i. Ensure that the installer does not create the database during installation by clearing the relevant check boxes, and then enter all other required information and click Next. The check boxes that appear depend on your database provider.
    - Microsoft SQL:  
Create or the Refresh Database
    - Oracle:  
Create or the Refresh Database  
Create the Tablespaces, Database Users, and Roles
    - Sybase:  
Create or the Refresh Database  
Create New Database Devices

The next page in the Primary Event Server Properties page series appears.

  - j. Enter the required information and click Next on each page in the series. The Secondary Event Server Properties pages appear.
  - k. Enter the required information and click Next on each page in the Secondary Event Servers Properties page series. Specify a different name and database server for the second event server than you did for the first event, and ensure that all of the other properties are identical for both event servers. The Database Test page appears and the database test starts.



- l. If the database test is successful, click Next. If the database test fails, click Skip Test to bypass the test. The Agent Properties page appears.
  - m. Continue with the installation by entering the required information on each page and clicking Next until the Scheduler Properties page appears.
  - n. Clear the Start the Scheduler Following the Installation check box, click Next, and continue with the installation by entering the required information and clicking Next on each page until the Review Settings page appears. The page displays the information that you entered in the installation wizard.
  - o. Review the information, use the back button to navigate to any incorrect entries, make the necessary corrections, use the Next button to navigate back to the Review Settings page, click Install, wait for the Installation Complete page to appear, and click Finish. The server installation is complete.
2. Define the scheduler role.
    - a. Click Start, Programs, AutoSys, Administrator. The Instance - AutoSys Workload Automation Administrator window opens.
    - b. Select the instance on which the Scheduler runs from the Instance drop-down list on the Settings pane and Click the Scheduler icon on the toolbar. The Scheduler - AutoSys Workload Automation Administrator window appears.
    - c. Select the Tie-Breaker option in the Scheduler Role pane and click Apply. The scheduler role is defined.
    - d. Click the Services icon, right-click the scheduler, and select Start.

The Tie-Breaker scheduler becomes active and sends updates to the database based on the interval that is specified in the configuration file. The server is installed on three computers. The primary scheduler is running on the first computer, the shadow scheduler is active on the second computer, and the tie-breaker scheduler is active on the third computer. When the primary scheduler detects the activity of the other two schedulers, AutoSys Workload Automation begins operating in high-availability mode with dual event servers.

## (Windows) Set the Primary Failback Mode

High-availability mode is disabled when the primary scheduler fails over to the shadow scheduler, but you can return to high-availability mode by restoring the primary scheduler. How the primary scheduler resumes processing events after you restore it depends on the primary failback mode.

By default, the primary failback mode is off and cannot resume processing events while the shadow scheduler is processing events. In this case, you can restart the primary scheduler only after you shutdown the shadow scheduler. To minimize downtime, we recommend that set one of the following primary failback mode options:

- **Dormant** Specifies that the primary scheduler runs dormant and resumes processing events only when a failback occurs. A failback occurs when the shadow scheduler fails or when you initiate a manual failback. If a failback occurs because the shadow scheduler fails, high-availability mode is disabled. In this case, return to high-availability mode by restoring the shadow scheduler.
- **Immediate** Specifies that the primary scheduler resumes processing events as soon as it detects activity from the other schedulers.

### Follow these steps:

1. Log in as the root user on the machine where the primary scheduler runs and Click Start, Programs, AutoSys, Administrator. The Instance-AutoSys Workload Automation Adminstrator window opens.
2. Select the appropriate the Instance drop-down list on the Settings pane, and then click the Services icon on the toolbar. The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services that are installed on the selected instance.
3. Right-click the scheduler and select Stop. The primary scheduler stops.
4. Select Scheduler from the WAAE drop-down list in the file menu. The Scheduler - AutoSys Workload Automation Administrator window opens.
5. Click the Options tab and select *one* of the following Primary Failback Mode options:



- Dormant
  - Immediate
6. Click Apply, and then click the Services icon on the toolbar. The primary fallback mode is set, and the Services - AutoSys Workload Automation window appears, displaying a list of services that are installed on the instance.
  7. Right-click the scheduler and select Start.  
The primary scheduler starts. In the event of a failover, the primary scheduler resumes processing events according to the mode that you specified.

You have configured AutoSys Workload Automation to run in high-availability mode. Downtime and interruptions in work flow decrease. Scheduler and database failures adversely affect and sometimes disable high-availability. The operator can maintain a highly-available environment by monitoring the environment and restoring from scheduler or database schedulers as needed.

#### NOTE

For more information about maintaining highly available environment, see [How to Maintain Highly-Available Environment](#).

## Send Email Notifications

You can configure AutoSys Workload Automation to send email notifications to operators or administrators who resolve problems or attend to emergencies. When you define a job to send an email notification, the scheduler sends the email notification during terminal status processing. Messages are written to the scheduler log indicating whether the email notification was sent successfully.

To send an email notification using AutoSys Workload Automation, follow these steps:

1. Configure AutoSys Workload Automation on [UNIX](#) or [Windows](#).
2. [Specify the notification attributes in your job definition](#).

### (UNIX) Configure AutoSys Workload Automation to Send Email Notifications

You can configure AutoSys Workload Automation to send email notifications to operators or administrators who resolve problems or attend to emergencies.

#### Follow these steps:

1. Log in to AutoSys Workload Automation as a user authorized to stop the scheduler and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr status waae_sched.$AUTOSERV
```

The scheduler process ID is displayed as follows:

```

CA Services Status Report
Component Name Pid Status

WAAE Scheduler (ACE) 32220 runnin
```

3. Edit the following parameters in the configuration file, and save the file:

```
NotifyMethod=0|1
```

- **0**  
Specifies that AutoSys Workload Automation does not send notifications to operators or administrators even if you specify the notification attributes in the job definition. This is the default.
- **1**  
Specifies that AutoSys Workload Automation sends email notifications to operators or administrators.

**NOTE**

If you set this parameter to 1, specify the Simple Mail Transfer Protocol (SMTP) server host name and port number in the NotifySMTPHost parameter.

NotifySMTPHost=*hostname:port*

– **hostname**

Defines the host name of the SMTP server.

– **port**

Defines the port number the SMTP server uses to send email notifications. Do not specify this value if the SMTP server is using the default port.

**Default:** 25

**Example:** SMTPserver:333

UseSMTPAuthentication=0|1

NotifySMTPUser=*user@email\_domain.com/password*

NotifySMTPFromAddress=*email\_address*

– **0**

Specifies that the SMTP server does not require authentication to send an email. This is the default.

– **1**

Specifies that the SMTP server requires authentication to send an email.

**NOTE**

Check with your administrator and set this parameter to 1 only if the SMTP server requires authentication to send an email.

– ***user@email\_domain.com/password***

Defines the user name and the associated password that is used to connect to the SMTP server.

**NOTE**

- Specify the user name in the form of an email address. For example, specify user@example.com.
- Use the autosys\_secure interactive utility to get the encrypted password and then add this encrypted password in the configuration file. Follow these steps to get the encrypted password:
  - a. Open the UNIX operating system and run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - b. Enter the following command:  
autosys\_secure  
The interactive menu appears.
  - c. Type 6 and press the Enter key.  
The Get encrypted password submenu appears.
  - d. Type 1 and press the Enter key.  
The utility prompts you to specify your password.
  - e. Type your password in plain text and press the Enter key. Confirm your password and press the Enter key.  
The utility displays the encrypted password.

– **email\_address**

Defines a valid SMTP from email address.

**Default:** CA WAAE.XXX.DO\_NOT\_REPLY; XXX specifies the name of the AutoSys Workload Automation instance.

**NOTE**

You can use \$\$XXX\$\$ to specify the name of the AutoSys Workload Automation instance. For example, if you set the value to Mail.From.\$\$XXX\$\$@xyz.com, the SMTP from email address is displayed as Mail.From.ACE@xyz.com, where ACE is the name of the AutoSys Workload Automation instance.

4. Enter the following command at the operating system prompt:

```
kill -HUP scheduler_pid
```

– ***scheduler\_pid***

Defines the process ID of the scheduler to pause and resume.

The scheduler resumes. AutoSys Workload Automation is configured to send email notifications.

**NOTE**

- If you specify an invalid email address, AutoSys Workload Automation does not send the email notification. A login denied error message is displayed in the scheduler log after the terminal status events are processed. AutoSys Workload Automation does not send the email notification even if one of the multiple email addresses you specified is invalid.
- For better readability, view the email notification as plain text. This may require a change to the font used by your email editor.

### **(Windows) Configure AutoSys Workload Automation to Send Email Notifications**

You can configure AutoSys Workload Automation to send email notifications to operators or administrators who resolve problems or attend to emergencies.

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Integration icon on the toolbar.  
The Integration - AutoSys Workload Automation Administrator window appears. The Notification tab is selected by default.
4. Select Standard Email from the Method drop-down list in the Settings pane.  
The SMTP Host, SMTP Server Requires Authentication, SMTP User, SMTP Password, and SMTP From Address fields appear.

**NOTE**

- If you select None from the Method drop-down list, AutoSys Workload Automation does not send notifications even if you specify the notification attributes in the job definition.
- The equivalent configuration parameter is NotifyMethod.

5. Complete the following fields as appropriate, and click Apply.

– **SMTP Host**

Specifies the Simple Mail Transfer Protocol (SMTP) server host name and port number the scheduler sends the email to.

**Example:** SMTPserver:333

**NOTE**

- The default port number is 25. If the SMTP server is using the default port, you do not need to enter the port number in the SMTP Host field. However, if the SMTP server is using a port other than the default, enter the port number in the SMTP Host field. Use a colon to separate the SMTP server host name and port number values.
- The equivalent configuration parameter is NotifySMTPHost.

– **SMTP Server Requires Authentication**

Specifies whether the SMTP server requires authentication to send an email.

**NOTE**

- The SMTP Server Requires Authentication check box is not selected by default. Verify with your administrator and select this check box only if the SMTP server requires authentication to send an email.
- The equivalent configuration parameter is UseSMTPAuthentication.

– **SMTP User**

Defines the user name that is used to connect to the SMTP server.

**Example:** user@example.com

**NOTE**

- Enter the user name in the form of an email address.
- The SMTP User field is enabled only if you select the SMTP Server Requires Authentication check box.
- The equivalent configuration parameter is NotifySMTPUser.

– **SMTP Password**

Defines the password that is associated with the SMTP User.

**NOTE**

- The SMTP Password field is enabled only if you select the SMTP Server Requires Authentication check box.
- The equivalent configuration parameter is NotifySMTPUser.

– **SMTP From Address**

Defines a valid SMTP from email address.

**Default:** CA WAAE.XXX.DO\_NOT\_REPLY; XXX specifies the name of the AutoSys Workload Automation instance.

**NOTE**

You can use \$\$\$\$ to specify the name of the AutoSys Workload Automation instance. For example, if you enter Mail.From.\$\$\$\$@xyz.com, the SMTP from email address is displayed as Mail.From.ACE@xyz.com, where ACE is the name of the AutoSys Workload Automation instance.

6. Click the Services icon on the toolbar.
  7. Right-click the scheduler service, and click Pause.
  8. Right-click the scheduler service, and click Resume.
- The scheduler resumes. AutoSys Workload Automation is configured to send email notifications

**NOTE**

- If you specify an invalid email address, AutoSys Workload Automation does not send the email notification. A login denied error message is displayed in the scheduler log after the terminal status events are processed. AutoSys Workload Automation does not send the email notification even if one of the multiple email addresses you specified is invalid.
- For better readability, view the email notification as plain text. This may require a change to the font used by your email editor.

**Specify the Notification Attributes**

To send an email notification, specify the following notification attributes in your job definition:

- `send_notification`
- `notification_msg`
- `notification_emailaddress`
- `notification_template`
- `notification_alarm_types`
- `notification_emailaddress_on_success` (to send an email notification using the email notification template when a job completes successfully)
- `notification_emailaddress_on_failure` (to send an email notification using the email notification template when a job fails)
- `notification_emailaddress_on_terminated` (to send an email notification using the email notification template when a job terminates)
- `notification_emailaddress_in_alarm` (to send an email notification using the email notification template when an alarm is raised)

#### NOTE

To send an email notification using an email notification template, set the `send_notification` attribute value to **y** or **f** and specify the `notification_template` and the `notification_emailaddress`, `notification_emailaddress_on_success`, `notification_emailaddress_on_failure`, `notification_emailaddress_on_terminated`, or `notification_emailaddress_on_alarm` attributes in your job definition based on the job status or alarm you want to send the email notification for.

#### Example: Send an Email Notification When the Job Completes

This example specifies that an email notification must be sent to `admin@example.com` when job `email_job_1` completes.

```
insert_job: email_job_1
machine: localhost
job_type: c
command: as_test -t 2
send_notification:1
notification_emailaddress:admin@example.com
notification_msg:"email_job_1 has completed"
```

#### Example: Send an Email Notification When the Job Fails

This example specifies that an email notification must be sent to `admin@example.com` when job `email_job_fails` fails.

```
insert_job: email_job_fails
machine: localhost
job_type: c
command: as_test -t 2 -e 2
send_notification:F
notification_emailaddress:admin@example.com
notification_msg:"email_job_fails has failed"
```

#### Example: Send an Email Notification Using the Template When the Job Terminates

This example specifies that an email notification is sent to `admin@example.com` and `terminate@example.com` using the `emailtemplate` email notification template when `email_job_2` terminates.

```
insert_job: email_job_2
machine: localhost
job_type: c
```

```

command: as_test -t 2
send_notification: y
notification_template: emailtemplate
notification_emailaddress: terminate@example.com
notification_emailaddress_on_terminate: admin@example.com

```

## Set Alarm Notifications

You can configure AutoSys Workload Automation to call user-defined routines that communicate alarms to specific users who are outside of the AutoSys Workload Automation event system. For example, by using email or a command-line pager utility, you can notify the administrator when there is a database problem, when the scheduler shuts down, or when the scheduler notifies all application servers to shut down.

You can configure AutoSys Workload Automation to call user-defined routines for the following system alarms:

- APP\_SERVER\_SHUTDOWN (UNIX) or Application Server Shutdown (Windows)
- DB\_PROBLEM (UNIX) or Database Problem (Windows)
- DB\_ROLLOVER (UNIX) or Database Rollover (Windows)
- EP\_HIGH\_AVAIL (UNIX) or Scheduler High Availability (Windows)
- EP\_ROLLOVER (UNIX) or Scheduler Rollover (Windows)
- EP\_SHUTDOWN (UNIX) or Scheduler Shutdown (Windows)

### (UNIX) Follow these steps:

1. Create a file named notify.\$AUTOSERV in the \$AUTOUSER directory of the computer where the primary or shadow scheduler is running.
2. Copy the sample notification file "\$AUTOSYS/install/data/notify.ACE" to the \$AUTOUSER directory and rename it to notify.\$AUTOSERV.
3. Edit the following parameters in the notify.\$AUTOSERV file as appropriate:
  - **APP\_SERVER\_SHUTDOWN**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when all the application servers are shutting down because of a shutdown request received when the sendevent -E STOP\_DEMON command with the -v ALL or -v ROLE=A option is issued.
  - **DB\_PROBLEM**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when there is a problem with one of the event servers.
  - **DB\_ROLLOVER**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when it rolls over from dual event server mode to single event server mode.
  - **EP\_HIGH\_AVAIL**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the scheduler is shutting down and the shadow scheduler does not see an update in the event server from the primary scheduler, or when other scheduler takeover problems occur.
  - **EP\_ROLLOVER**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the shadow scheduler takes over processing.
  - **EP\_SHUTDOWN** Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the active scheduler (the primary or shadow scheduler after failing over in high availability mode) is shutting down because of a normal shutdown process or an error condition.

The alarm notifications are set.

### Example: Call the /usr/local/bin/pager Program when the Scheduler Shuts Down

Suppose that you want AutoSys Workload Automation to call the program `/usr/local/bin/pager` when the scheduler shuts down. Copy the sample notification file from `$AUTOSYS/install/data/notify.ACE` to the `$AUTOUSER/notify.$AUTOSERV` directory and modify the `EP_SHUTDOWN` line in the notification file as follows:

```
EP_SHUTDOWN /usr/local/bin/pager
```

If the scheduler shuts down, AutoSys Workload Automation passes a numeric code and a text message to the pager program. Code the pager program to accept these parameters.

**(Windows) Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator on the computer where the primary or shadow scheduler is running.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.
4. Right-click the scheduler and select Stop. Repeat this action for the application server.
5. Click the Alarm Notification icon on the toolbar.
6. Complete the following fields as appropriate:
  - **Scheduler High Availability**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the scheduler is shutting down and the shadow scheduler does not see an update in the event server from the primary scheduler, or when other scheduler takeover problems occur.  
  
**NOTE**  
On UNIX, the equivalent configuration parameter is `EP_HIGH_AVAIL`.
  - **Scheduler Rollover**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the shadow scheduler takes over processing.  
  
**NOTE**  
On UNIX, the equivalent configuration parameter is `EP_ROLLOVER`.
  - **Scheduler Shutdown**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when the active scheduler (the primary or shadow scheduler after failing over in high availability mode) is shutting down because of a normal shutdown process or an error condition.  
  
**NOTE**  
On UNIX, the equivalent configuration parameter is `EP_SHUTDOWN`.
7. Click the Database tab and complete the following fields as appropriate:
  - **Database Problem**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when there is a problem with one of the AutoSys Workload Automation databases.  
  
**NOTE**  
On UNIX, the equivalent configuration parameter is `DB_PROBLEM`.
  - **Database Rollover**  
Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when it rolls over from dual event server mode to single event server mode.  
  
**NOTE**  
On UNIX, the equivalent configuration parameter is `DB_ROLLOVER`.
8. Click the Application Server tab, complete the following field as appropriate, and click Apply:
  - **Application Server Shutdown**

Defines the user-defined routine (complete path and executable name) that AutoSys Workload Automation calls when all the application servers are shutting down because of a shutdown request received when the sendevent -E STOP\_DEMON command with the -v ALL or -v ROLE=A option is issued.

#### NOTE

On UNIX, the equivalent configuration parameter is APP\_SERVER\_SHUTDOWN.

9. Click the Services icon on the toolbar.
10. Right-click the scheduler and select Start. Repeat this action for the application server.  
The scheduler and application server start. The alarm notifications are set.

#### Example: Call the C:\utils\pager.bat Program when the Scheduler Shuts Down

Suppose that you want AutoSys Workload Automation to call the program C:\utils\pager.bat when the scheduler shuts down. Type the following value in the Scheduler Shutdown field on the Alarm Notification - AutoSys Workload Automation Administrator window:

```
C:\utils\pager.bat
```

If the scheduler shuts down, AutoSys Workload Automation passes a numeric code and a text message to pager.bat. Code pager.bat to accept these parameters.

## (Windows) Create an Instance Feedback Report Using the Administrator Utility

You can create a report that includes the environment and operating system information for each installed AutoSys Workload Automation instance. This report also includes information about all the agents that are installed for the instance.

#### Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Click the Feedback icon on the toolbar.  
The Feedback - AutoSys Workload Automation Administrator window appears.
3. Select an instance from the CA WAAE Instance drop-down list.  
The components that are installed, environment variables, database provider, application server host and port number, database mode (single or dual event server mode), and the database details for the selected instance are displayed in the CA WAAE Environment pane. The operating system, current user, administrative rights, domain, Windows and system directories, and the port details are displayed in the Operating System pane.
4. Click Create Report.  
A report is created in C:\WorkloadAutomationAE-*instancename*\_Report.txt. This report includes the environment and operating system information for the selected instance.

## (Windows) Configure the Local Agent Using the Administrator Utility

You can perform the following tasks to configure the local agent using the Administrator utility. You can also perform these tasks by modifying the parameters in the agentparm.txt file, which is located in the SystemAgent directory.

- Define the JRE path or Java classpath for the local agent.
- Modify the port settings. You might want to configure the local agent to listen on a different port in the following situations:



- The default local agent port (7520) is in use by another Broadcom product and you want that product to continue using that port.
- You want the local agent to communicate with SSA sockets instead of plain sockets.
- You installed multiple agents on the same computer and one of the agents is already using the default agent port.
- Add an agent plug-in to work with AutoSys Workload Automation. After the agent plug-in is added, you can automate, monitor, and manage workload on the computer where that plug-in and its agent are installed.

**NOTE**

You can also add an agent plug-in using the agentparm.txt file. You must ensure that you install the agent plug-in before you add it.

- Add, modify, or delete an authorized manager. When a scheduling manager tries to communicate with the local agent using a proper encryption key, the local agent automatically adds that scheduling manager to its authorized manager list. However, you can manually add a scheduling manager that is authorized to communicate with the local agent.

**NOTE**

- For information about configuring AutoSys Workload Automation to work with agents, see [Configure AutoSys Workload Automation to Work with Workload Automation Agent for UNIX, Linux, Windows, or i5/OS](#).
- For information about modifying the parameters in the agentparm.txt file, see [Workload Automation Agent Parameters](#).

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.
2. Click the Services icon on the toolbar.
3. Right-click the agent and select Stop.
4. Click the Agent icon on the toolbar.
5. Select the local agent that you want to configure from the Local Agent Name drop-down list.  
The Agent - AutoSys Workload Automation Administrator window displays the settings of the selected local agent.
6. Perform any of the following tasks as appropriate and click Apply:
  - Enter the JRE path in the JRE Path field or the Java classpath in the Java Classpath field.

**NOTE**

The Java classpath must include the path to casocket.jar if the local agent is communicating over SSA.

The JRE path or the Java classpath for the local agent is defined.

- Right-click the port to modify in the Local Agent Port Settings pane, and click Settings. Complete the fields on the CA WAAE Administrator - Local Agent Communication Settings dialog as appropriate, and click OK.

**NOTE**

You can specify a primary port and an optional auxiliary port.

The CA WAAE Administrator - Local Agent Communication Settings dialog closes. The modified port settings are displayed in the Local Agent Port Settings pane.

- Right-click in the Plug-in List pane, and click New. Enter the agent plug-in name in the Name field, and click OK. The CA WAAE Administrator - Plugin Settings dialog closes. The new agent plug-in is listed in the Plug-in List pane. If an agent plug-in is no longer supported by the local agent, you can delete it using the Delete button.
- Perform any one of the following actions based on whether you want to add, modify, or delete an authorized manager:

- Right-click in the Authorized Manager List pane, and click New. Complete the fields on the CA WAAE Administrator - Authorized Manager Settings dialog as appropriate, and click OK.
- Right-click the authorized scheduling manager to modify in the Authorized Manager List pane, and click Update. Complete the fields on the CA WAAE Administrator - Authorized Manager Settings dialog as appropriate, and click OK.
- Right-click the authorized scheduling manager to delete in the Authorized Manager List pane, and click Delete.

The CA WAAE Administrator - Authorized Manager Settings dialog closes. The new scheduling manager is listed in the Authorized Manager List pane or the modified settings are displayed in the Authorized Manager List pane.

#### NOTE

If you delete a scheduling manager from the Authorized Manager List pane on the Agent - AutoSys Workload Automation Administrator window, it does not help ensure that the scheduling manager cannot run jobs on the local agent. When the local agent receives the first job request from the deleted scheduling manager, the deleted scheduling manager gets listed in the Authorized Manager List pane on the Agent - AutoSys Workload Automation Administrator window. To restrict an unauthorized scheduling manager, you must change the encryption key of the local agent using the `as_config` command and JIL statements for machine definitions. For more information about how to set the encryption key, see [Modify the Encryption Type and Encryption Key on AutoSys Workload Automation](#).

7. Click the Services icon on the toolbar.
8. Right-click the agent and select Start.  
The agent starts and the new settings take effect.

## (Windows) Configure an SSA Port Using the Administrator Utility

You can configure an SSA port. There are several reasons why you might want to reconfigure the port that the application server listens to. For example, you might want to reconfigure the port in the following situations:

- The default virtual port is in use by another Broadcom product and you want that product to continue using that virtual port.
- You want to enable more than one application server to run on the same host, which requires you to specify a unique virtual port for each application server.

#### NOTE

By default, port multiplexing is enabled on AutoSys Workload Automation, and the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. If you install multiple AutoSys Workload Automation instances on the same computer, subsequent installations use incremental virtual port numbers, such as 9001, 9002, and so on.

#### Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the scheduler and select Stop. Repeat this action for the application server and agent.  
The scheduler, application server, and agent stop.
5. Stop the CA Connection Broker service from the Windows Service Control Manager.
6. Configure the SSA port as follows:
  - a. Click the SSA icon on the toolbar.  
The SSA - AutoSys Workload Automation Administrator window appears, displaying the details of the AutoSys Workload Automation communication ports.

- b. Double-click the port number to configure.  
The port details are displayed in the Port Configuration pane.
  - c. Select the following options in the Port Configuration pane as appropriate, and click Set.
    - • **Enable Port Multiplexing (PMUX)**  
Enables port multiplexing.
    - **Enable Secure Socket Layer (SSL)**  
Enables SSL.
  7. Start the CA Connection Broker service from the Windows Service Control Manager.
  8. Start the scheduler, application server, and agent services as follows:
    - a. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
    - b. Right-click the scheduler and select Start. Repeat this action for the application server and agent.  
The scheduler, application server, and agent start.
- The SSA port is configured.

### **View the Properties of a Communication Port**

You can view the properties of any communication port used by AutoSys Workload Automation.

#### **NOTE**

By default, port multiplexing is enabled on AutoSys Workload Automation, and the AutoSys Workload Automation installation configures SSA to recognize virtual port 9000 for the application server. If you install multiple AutoSys Workload Automation instances on the same computer, subsequent installations use incremental virtual port numbers, such as 9001, 9002, and so on.

#### **Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the SSA icon on the toolbar.  
The SSA - AutoSys Workload Automation Administrator window appears, displaying the details of the AutoSys Workload Automation communication ports.
4. Right-click a port number, and select Properties.  
The CA WAAE Administrator - SSA Port Configuration Information dialog opens displaying the current properties of the selected port. If the port is controlled by SSA, the ephemeral port range is also displayed.

## **(Windows) Manage Job Profiles Using the Administrator Utility**

A *job profile* defines the non-system environment variables for a job. Before running a job, the agent sets the assigned job profile on the job's target computer.

Job profiles apply to the following job types:

- Command jobs
- File Watcher (FW) jobs that are submitted to the legacy agent

You can use the Job Profiles - AutoSys Workload Automation Administrator window to define a job profile that contains the environment variables that must be set for a job to run. After you define a profile, you can use the profile attribute or the Web UI to assign that job profile to one or more Command or File Watcher jobs.

The r11.3 or later release agents can only interpret the job profiles represented as flat files. To support the jobs containing job profiles, all the job profiles in the Windows Registry are converted into flat files. The autoprofm tool is used for this conversion process and is available only on Windows. This tool can be invoked either manually from the console or by the

installer during the upgrade, and it creates the flat files at the location specified by the user. You must include the full path when specifying the location of the flat files.

In Unicenter AutoSys JM 4.0, 4.5, 4.5.1, job profiles are instance-specific. In Unicenter AutoSys JM r11, job profiles are product-specific. Therefore, the same job profile name may exist under two different instances. In these situations, the autoprofm tool renames the 4.x files to *profilename\_instancename* or the r11 files to *profilename\_11.0\_Base* to avoid overwriting information and sends a notification message to the administrator.

Similarly, if the r11.3 or later release instance already has job profiles with the same name, the autoprofm tool renames the migrated files to *profilename\_instancename* or *profilename\_11.0\_Base*. The files are renamed to avoid overwriting the existing job profiles.

#### NOTE

- For more information about job profiles and how to assign a job profile to a Command job, see [Job Profiles](#).
- On the agent, you can define environment variables that apply to all jobs at a global agent level, scheduling manager level, or user level. For example, suppose that you want to set an environment variable for all jobs that run on an agent under a specific user (owner). Instead of defining that variable in every job definition using the `envvars` attribute or in a job profile, you can define the variable on the agent using the `oscomponent.environment.variable_user_userid` parameter. For more information about setting environment variables on the agent, see [Workload Automation Agent Parameters](#).

### Manage Job Profiles

You can create a job profile that contains the environment variables required for a specific job to run.

#### NOTE

You must add the new job profile on the machine where the job runs. If you want to use the same job profile on more than one machine, you can manually copy the job profile on the destination machine or create the job profile on all the machines.

#### Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.
2. Click the Job Profiles icon on the toolbar.
3. Select the local agent for which you want to create a job profile from the Local Agent Name drop-down list. The Job Profiles - AutoSys Workload Automation Administrator window lists the job profiles of the selected local agent in the Profile Name drop-down list.
4. Enter the name of the job profile in the Profile Name drop-down list.  
A new job profile is created.
5. Enter the variable name in the Variable field and its value in the Value field, and click Set.  
An environment variable is created and added to the new job profile. The environment variable definition is displayed in the Environment Variables pane.

#### NOTE

- Repeat this step to define additional environment variables. Variable definitions are case-insensitive on Windows. The Administrator utility replicates the case you enter in the Variable and Value fields in the job profile.
- By default, the DEFAULT profile is selected in the Profile Name field and the Sort By Name check box is selected and so the environment variables are sorted by name in the Environment Variable pane. If you want to view the environment variables in the order in which they are created, clear the Sort By Name check box.

**NOTE**

- To view the current settings of a job profile, select a profile from the Profile Name drop-down list. The current settings of the selected profile are displayed in the Environment Variables pane.
- To delete a job profile that is no longer required for a job to run, select the profile from the Profile Name drop-down list and click Delete Profile. The job profile and its environment variables are deleted.

**Manage an Environment Variable in a Job Profile**

You can modify the environment variable definition of a job profile, if the job requires a different value to run.

**Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.
2. Click the Job Profiles icon on the toolbar.
3. Select the local agent for which you want to modify an environment variable in a job profile from the Local Agent Name drop-down list.  
The Job Profiles - AutoSys Workload Automation Administrator window lists the job profiles of the selected local agent in the Profile Name drop-down list.
4. Select a job profile from the Profile Name drop-down list.
5. Double-click the environment variable to modify in the Environment Variables pane.  
The window refreshes to display the name of the selected variable in the Variable field and its value in the Value field.
6. Modify the variable name or its value, and click Set.  
The environment variable definition is modified. The modified environment variable definition is displayed in the Environment Variables pane.

**NOTE**

To delete an environment variable from the job profile, double-click the environment variable to delete in the Environment Variables pane, and click Delete.

**Maintain the Scheduler**

The scheduler (the event\_demon binary) is the engine of AutoSys Workload Automation.

You must start the scheduler to schedule and run jobs. If the scheduler is not running, you cannot initiate new job flows. If you stop the scheduler, any job flows that have already started run to completion.

**NOTE**

The event server must be available, running, and properly identified before you can start the scheduler.

After you start the scheduler, it performs the following tasks before it begins processing:

- Verifies that no other scheduler is running on that computer.
- Runs the chase command with the -A and -E parameters. The chase command verifies whether jobs and agents are running. For each client computer, the chase command passes a list of jobs that are supposed to be running on the agent. The agent then verifies that the processes are running. If the chase command detects errors, it sends an alarm. If a job is not running as expected, the scheduler sends the necessary corrective event for the job, if the job definition allows it.
- If a STARTJOB event is being processed and the job it started is still active, the scheduler does not restart the job. The purpose of running the chase command is to guarantee that the scheduler starts with all the processes in a known state. Problems are detected on scheduler startup. This method is similar to a database checkpointing and rolling forward or back upon recovery.

**NOTE**

For information about running the chase command or starting the scheduler on Windows, see [Configure a Scheduler](#).

This section contains the following topics:

## Back Up and Restore Definitions

We recommend that you back up the following definitions periodically so you have files to restore from in the event of a system failure:

- Calendar definitions
- Machine definitions
- Resource definitions
- User-defined job type definitions
- Job definitions
- Monitor report definitions
- Global variables

### Back Up Definitions

To back up definitions, follow these steps:

- [Back Up Calendar Definitions](#)
- [Back Up Machine, Resource, User-defined Job Type, Job, and Monitor Report Definitions](#)
- [Back Up Global Variable Values](#)

### Back Up Calendar Definitions

We recommend that you back up your calendar definitions periodically so you have files to restore from in the event of a system failure.

To back up calendar definitions, enter the following commands at the UNIX operating system prompt or the Windows instance command prompt:

```
autocal_asc - E /directory/autosys.ecal - e ALL
autocal_asc - E /directory/autosys.ccal - c ALL
autocal_asc - E /directory/autosys.scal - s ALL
```

- **directory**  
Defines a directory outside of the AutoSys Workload Automation directory structure.

A backup of the calendar definitions is created in the specified directory.

### Back Up Machine, Resource, User-defined Job Type, Job, and Monitor Report Definitions

We recommend that you back up your machine, resource, user-defined job type, job, and monitor report definitions periodically so you have files to restore from in the event of a system failure.

#### Follow these steps:

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -M ALL -q > /directory/autosys.jil
```

- **directory**  
Defines a directory outside of the AutoSys Workload Automation directory structure. We recommend that you use the same directory where you saved your calendar definitions.  
Your machine definitions are saved to a file named autosys.jil in the specified directory.

**NOTE**

To append definitions to an existing file, you must enter >> (instead of >) in the command. We recommend that you append your machine, resource, user-defined job type, job, and monitor report definitions to the same file so you have only one file to restore following a system failure.

2. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -V ALL -q >> /directory/autosys.jil
```

Your resource definitions are saved to a file named autosys.jil in the specified directory.

3. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -Y ALL -q >> /directory/autosys.jil
```

Your user-defined job type definitions are saved to a file named autosys.jil in the specified directory.

4. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -J ALL -q >> /directory/autosys.jil
```

Your job definitions are saved to a file named autosys.jil in the specified directory.

5. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
monbro -N ALL -q >> /directory/autosys.jil
```

Your monitor report definitions are appended to the file that contains your backed-up machine, resource, user-defined job types, and job definitions. A backup of the machine, resource, user-defined job types, jobs, and monitor report definitions is created.

### **Back Up Global Variable Values**

We recommend that you back up your global variable values periodically so you have files to restore from in the event of a system failure.

To back up global variable values, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -G ALL > /directory/globals.txt
```

- ***directory***

Defines a directory outside of the AutoSys Workload Automation directory structure. We recommend that you use the same directory where you saved your calendar, machine, resource, user-defined job type, job, and monitor report definitions.

A backup of the global variable values is created. Your global variable values are saved to a file named globals.txt in the specified directory. This file is a record of what you must redefine after a system failure.

### **Restore Definitions**

You must restore backed-up definitions if you have lost data during a system failure or you want to reset the definitions in your database to a previous level. This procedure assumes that you have previously backed up your global variables and your calendar, machine, resource, user-defined job type, job, and monitor report definitions.

#### **Follow these steps:**

1. Log in to AutoSys Workload Automation and enter the following commands at the UNIX operating system prompt or the Windows instance command prompt:

```
autocal_asc -I /directory/autosys.ecal
```

```
autocal_asc - I /directory/autosys.ccal
autocal_asc - I /directory/autosys.scal
```

– **directory**

Defines the directory where you previously backed up the definitions.

Your calendar definitions are restored to the database.

2. Enter the following command at the operating system prompt:

```
jil < /directory/autosys.jil
```

Your machine, resource, user-defined job type, job, and monitor report definitions are restored to the database.

3. Open the globals.txt file that contains your backed-up global variables and manually redefine any global variables according to the values in the globals.txt file by entering the following command for each global variable:

```
sendevent - E SET_GLOBAL - g VARIABLE=VALUE
```

Your global variables are restored.

## Scheduler Log File

The scheduler log file contains a record of all the actions taken by the scheduler, including startup and shutdown information.

### View the Scheduler Log File

To view the scheduler log file, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosyslog -e
```

The last ten lines of the scheduler log file are displayed and all subsequent additions to the log are automatically displayed as they occur.

#### **NOTE**

To terminate autosyslog, press Ctrl+C.

### Scheduler Log File Location

When the scheduler encounters starting problems, it logs errors to a location that is dependent on when the starting process fails. You can find the error description in one of the following locations:

- If the scheduler fails early in startup, it writes errors to the Windows Event Log.
- If the scheduler fails during startup or encounters problems while running, it writes errors to the following location:
  - On UNIX -- \$AUTOUSER/out/event\_demon.\$AUTOSERV

#### **NOTE**

If the \$AUTOUSER directory is NFS mounted, you can view the output from any computer on the network.

- On Windows -- %AUTOUSER%/out/event\_demon.%AUTOSERV%

### Specify the Scheduler, Aggregator, and Application Server Log Rollover or Purge on UNIX

You can use the [LOGROLLOVER](#) parameter to specify when the scheduler, aggregator, and the application server logs roll over or are purged. The log can roll over at midnight or when the log file size is equal to the specified size. The log is purged when the log file becomes older than the specified number of days.



**NOTE**

The aggregator logs are generated only when the `AggregateStatistics` parameter is set to 1 in the configuration file or the `Aggregate Statistics` check box is selected on the `Scheduler - AutoSys Workload Automation Administrator` window of the `Administrator` utility (Windows).

## How Shadow Scheduler Backup Works

You can configure a shadow scheduler to use as a backup scheduler. In this scenario, both the primary and shadow schedulers periodically update their heartbeats in the event server to indicate that they are active. The shadow scheduler remains dormant, checking the event server for heartbeats from the primary scheduler. These heartbeats indicate that the primary scheduler is running. If the primary scheduler fails to update the event server, the shadow scheduler takes over and processes events.

If the primary scheduler and the event server are on the same computer, the scheduler failure could also mean an event server failure. In this case, if dual event servers are configured, AutoSys Workload Automation rolls over to single event server mode and fails over to the shadow scheduler. AutoSys Workload Automation uses the tie-breaker scheduler to resolve contentions and eliminates situations where one scheduler takes over because of network problems. However, the shadow scheduler is not guaranteed to take over in every case. For example, in the case of network problems, AutoSys Workload Automation might not be able to determine which scheduler works and might shut down both the schedulers. In such cases, you must resolve the network problems so that the primary, shadow, and tie-breaker schedulers can update both event servers, and start AutoSys Workload Automation.

### Restore the Primary Scheduler After a Failover

If you run AutoSys Workload Automation with a shadow scheduler, the shadow scheduler takes over processing events if the primary scheduler fails. You can configure the primary scheduler to do one of the following after a failover:

- Automatically failback
- Failback only after you do one of the following:
  - a. Manually restore the primary scheduler
  - b. Restart AutoSys Workload Automation
- Failback only after you manually restore the primary scheduler

#### **(UNIX) Follow these steps:**

1. Log in to a shadow scheduler as a user authorized to stop the scheduler and run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
sendevent -E STOP_DEMON
```

The shadow scheduler completes any processes it is currently performing and stops.

**NOTE**

If you are running with dual event servers, the tie-breaker scheduler must also be stopped.

3. On the primary scheduler, enter the following command at the operating system prompt:

```
eventor
```

The primary scheduler is restored.

4. On the shadow scheduler, enter the following command at the operating system prompt:

```
eventor
```

The shadow scheduler is restarted.

**NOTE**

If you are running with dual event servers, the tie-breaker scheduler must also be restarted.

**(Windows) Follow these steps:**

1. Log in to a shadow scheduler as a user authorized to stop the scheduler and enter the following command at the instance command prompt:

```
sendevent -E STOP_DEMON
```

The shadow scheduler completes any processes it is currently performing and stops.

**NOTE**

If you are running with dual event servers, the tie-breaker scheduler must also be stopped.

2. On the primary scheduler, Click Start, Programs, AutoSys, Administrator.
3. Select an instance from the Instance drop-down list.
4. Click the Services icon on the toolbar.
5. Right-click the scheduler service and click Start.  
The primary scheduler is restored. The Status column indicates the status.
6. On the shadow scheduler, Click Start, Programs, AutoSys, Administrator.
7. Select an instance from the Instance drop-down list.
8. Click the Services icon on the toolbar.
9. Right-click the scheduler service, and click Start.  
The shadow scheduler is restarted. The Status column indicates the status.

**NOTE**

If you are running with dual event servers, the tie-breaker scheduler must also be restarted.

**Run the Scheduler in Test Mode**

You can run the scheduler in test mode to troubleshoot problems and check your configuration. For example, you can check whether the scheduler and the agent are installed and configured properly. Running in test mode uses the same mechanisms of starting jobs and sending events that AutoSys Workload Automation uses in normal mode.

You can also test the setup and execution of the jil command without running the defined jobs. For example, you can check whether the conditional logic for jobs, including nested boxes, is functioning correctly. In test mode, the scheduler runs a simple test job instead of the defined jobs.

**(UNIX) Follow these steps:**

1. Run the shell that it sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
setenv AUTOTESTMODE=1|2
```

**– 1**

Runs each job with the following test mode variations:

- The `as_test` command runs on the remote computer instead of the command specified in the job definition.

**NOTE**

If you performed an agent-only install on the remote computer, you can run the `as_test` command on the remote computer only if you installed the agent using the AutoSys Workload Automation media. If you installed the agent using the Workload Automation Agent DVD, the `as_test` command is not available on the remote computer. We recommend that you install the agent using the AutoSys

Workload Automation media as it configures the agent specifically for communication with AutoSys Workload Automation.

- The scheduler redirects standard output and standard errors for the command to the /tmp/autotest.\$AUTO\_JOB\_NAME file, where \$AUTO\_JOB\_NAME is the job name as defined to AutoSys Workload Automation.
- If the type of the job being run in test mode is not a command job, the job is not disabled. The scheduler runs it as it would in normal mode.
- If the opsys attribute of the type 'a' machine is set to a value other than aix, hpux, linux, or windows, the job is not disabled. The scheduler runs it as it would in normal mode. If the opsys attribute of the type 'a' machine is not set, the job runs as if it is running on a machine with a UNIX opsys attribute value.

This test mode disables the following functions:

- Minimum and maximum run alarms
- Job terminations after job run exceeds the maximum runtime
- Sourcing a user-defined job profile file
- Minimum disk space verifications

## – 2

Runs each job with the same behaviors as \$AUTOTESTMODE = 1, and also includes the following functions:

- Minimum disk space verifications are performed.
- Alarms are sent if job run completes either before the minimum runtime or after the maximum runtime.
- Job runs exceeding the maximum runtime are terminated.
- A user-defined job profile is sourced.
- The scheduler redirects output from the as\_test command to the user-defined standard output and standard error files (if they are defined). Otherwise, the scheduler redirects output to the /tmp/autotest.\$AUTO\_JOB\_NAME file.

The level the test mode must run in is set.

### NOTE

- You must use either the setenv command or the export command (depending on your UNIX operating system) to set the \$AUTOTESTMODE variable.
- The as\_test command is new in r11.3 and obsoletes the ntgetdate command from previous releases.
- When a test mode job runs on a machine of type 'r' or 'n', the output is written to the enterprise-wide logging directory.

### 3. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
```

The scheduler starts and runs in test mode.

### NOTE

The scheduler cannot run partially in test mode, and AutoSys Workload Automation does not provide a test mode for the database. You must use caution when you run the scheduler in test mode on a live production system.

### (Windows) Follow these steps:

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. In the Instance drop-down list, select the instance that you want to add the %AUTOTESTMODE% environment variable to.
3. Click the System icon on the toolbar.  
The System - AutoSys Workload Automation Administrator window appears.

4. Enter AUTOTESTMODE in the Variable field and its value in the Value field. You can set the value to *one* of the following:

– 1

Runs each job with the following test mode variations:

- The `as_test` command runs on the remote computer instead of the command specified in the job definition.

**NOTE**

If you performed an agent-only install on the remote computer, you can run the `as_test` command on the remote computer only if you installed the agent using the AutoSys Workload Automation media. If you installed the agent using the Workload Automation Agent DVD, the `as_test` command is not available on the remote computer. We recommend that you install the agent using the AutoSys Workload Automation media as it configures the agent specifically for communication with AutoSys Workload Automation.

- The scheduler redirects standard output and standard errors for the command to the `%TEMP%\autotest.%AUTO_JOB_NAME%` file, where `%AUTO_JOB_NAME%` is the job name as defined to AutoSys Workload Automation.
- If the type of the job being run in test mode is not a command job, the job is not disabled. The scheduler runs it as it would in normal mode.
- If the `opsys` attribute of the type 'a' machine is set to a value other than `aix`, `hpux`, `linux`, or `windows`, the job is not disabled. The scheduler runs it as it would in normal mode. If the `opsys` attribute of the type 'a' machine is not set, the job runs as if it is running on a machine with a UNIX `opsys` attribute value.

This test mode disables the following functions:

- Minimum and maximum run alarms
- Job terminations after job run exceeds the maximum runtime
- Sourcing a user-defined job profile file
- Minimum disk space verifications

– 2

Runs each job with the same behaviors as `%AUTOTESTMODE% = 1`, and also includes the following functions:

- Minimum disk space verifications are performed.
- Alarms are sent if job run completes either before the minimum runtime or after the maximum runtime.
- Job runs exceeding the maximum runtime are terminated.
- A user-defined job profile is sourced.
- The scheduler redirects output from the `as_test` command to the user-defined standard output and standard error files (if they are defined). Otherwise, the scheduler redirects output to the `%TEMP%\autotest.%AUTO_JOB_NAME%` file.

5. Click Set.

The `%AUTOTESTMODE%` environment variable is listed in the Environment Variables pane. The level the test mode must run in is set.

**NOTE**

- The `as_test` command is new in r11.3 and obsoletes the `ntgetdate` command from previous releases.
- To ensure that the `as_test` command runs properly, set the value of the `opsys` attribute of the type 'a' machine to `windows` for each Windows agent.
- When a test mode job runs on a machine of type 'a' with the `opsys` attribute value set to `windows`, the output log files are written to the `%TEMP%` location as defined by the job owner's environment.

6. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

7. Right-click the scheduler service, and click Start.

The scheduler starts and runs in test mode.

**NOTE**

The scheduler cannot run partially in test mode, and AutoSys Workload Automation does not provide a test mode for the database. You must use caution when you run the scheduler in test mode on a live production system.

## Maintain the Event Server

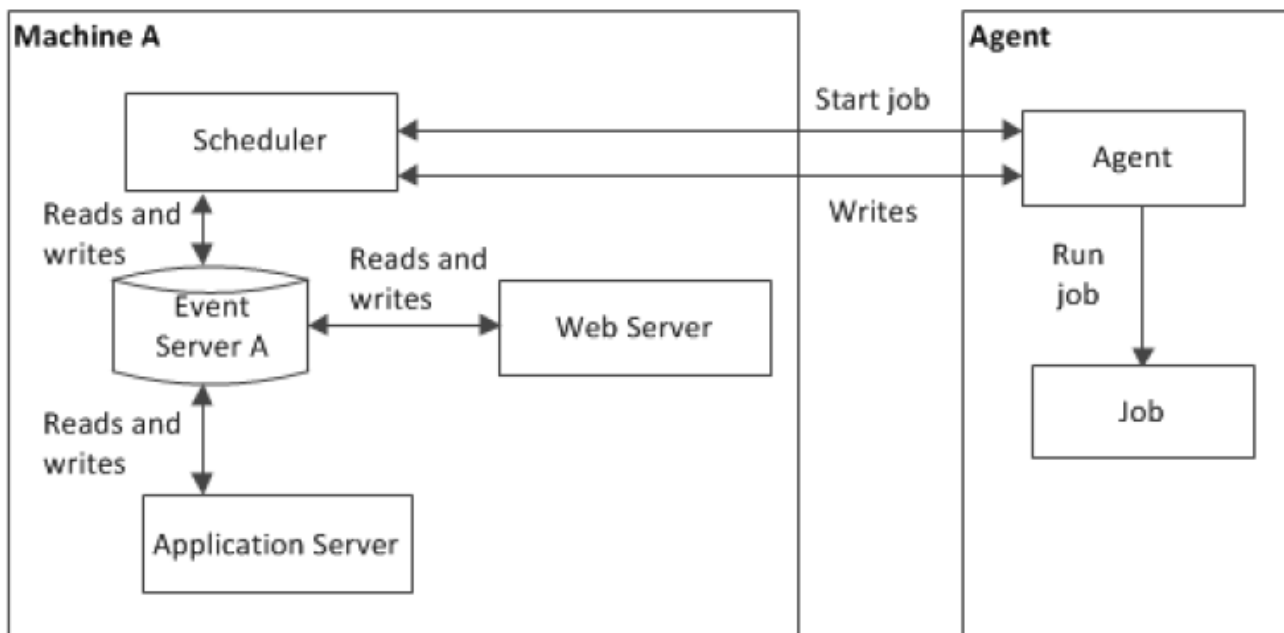
### Single Event Server Mode

By default, AutoSys Workload Automation is configured to run with one event server (database). This configuration is named *single event server mode*. You can configure AutoSys Workload Automation to run with two event servers either during installation or after a single event server mode installation.

When AutoSys Workload Automation is running in dual event server mode and the scheduler detects an unrecoverable error condition on one of the event servers, it automatically rolls over to single event server mode using the other event server.

The following illustration shows how the primary components (the scheduler, the application server, the event server, the web server, and the agent) interact in single event server mode:

**Figure 25: Single Event Server Mode**



### Dual Event Server Mode

AutoSys Workload Automation can run in *dual event server mode*, which means that it runs with two event servers or databases. These two event servers contain identical data, including object definitions and events. AutoSys Workload Automation reads from one event server and writes to both the event servers simultaneously.

We recommend that you synchronize the event servers when you configure AutoSys Workload Automation to run in dual event server mode. Event server synchronization ensures that AutoSys Workload Automation recovers after one of the event servers fails.

The scheduler reads from both event servers when it processes events. Sometimes the scheduler detects an event on one event server but not on the other event server. The scheduler copies any missing events to the other event server to prevent temporary problems from interrupting the event processing.

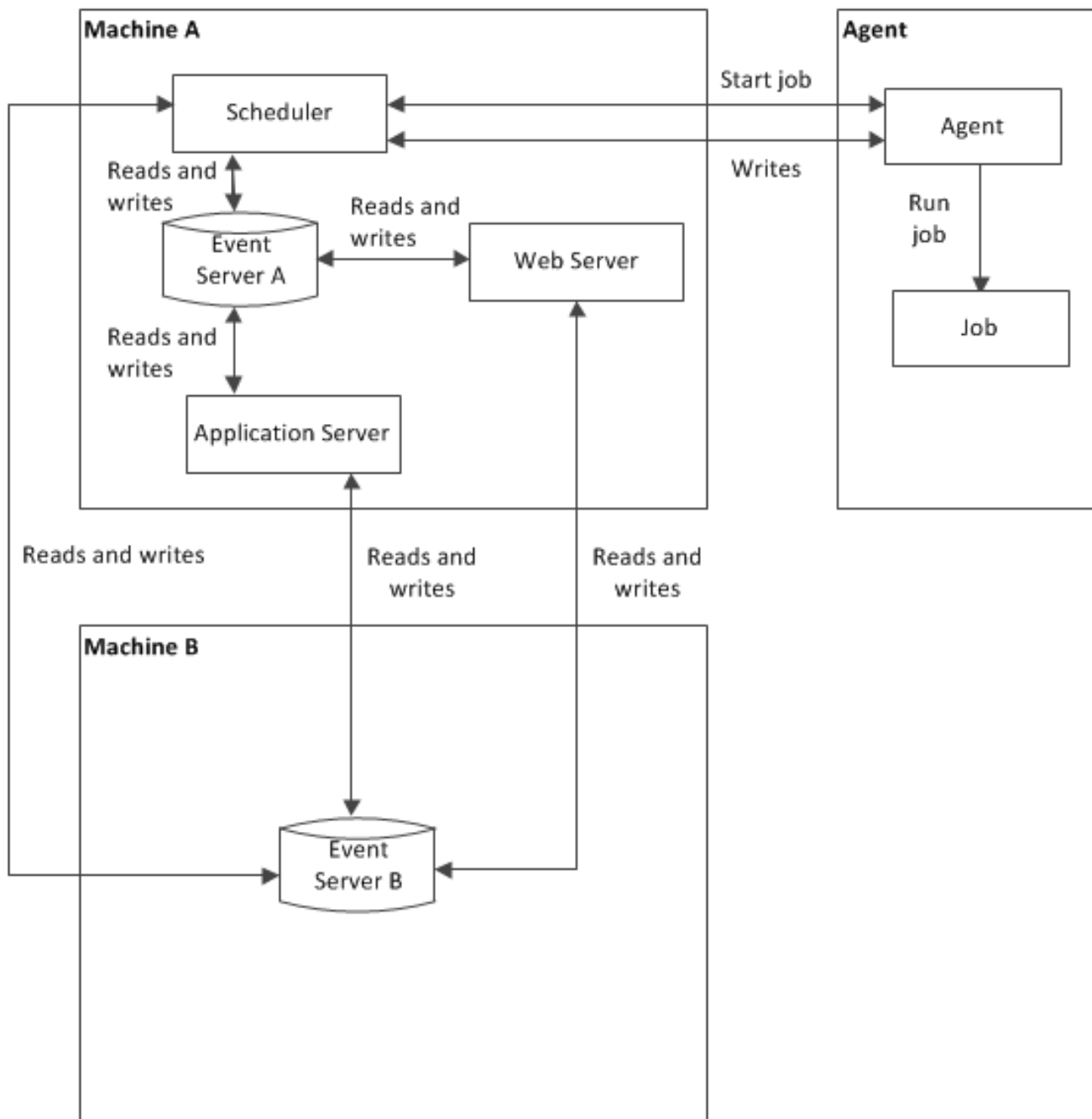
**NOTE**

To avoid a single point of failure, ensure that the two event servers reside on two different data servers that are running on two different computers.

The following diagram illustrates how the dual event server mode operates:

- How the databases are laid out
- How AutoSys Workload Automation verifies which database to use
- How the primary components (the scheduler, the application server, the event server, the agent, and the web server) interact

**Figure 26: The scheduler the application server the event server the agent and the web server interact**



### **Define Event Server Information**

You can define the event server information used by AutoSys Workload Automation during or after installation. To define the event server information after the installation, modify the [Provider](#), [DBAccess](#), [EventServer\\_1](#), or [EventServer\\_2](#) parameter. The scheduler, application server, web server, and some client utilities such as `dbstatistics`, `archive_events`, and `archive_jobs` use the event server information to connect to the event server.

## Configure the Event Server Time-Out Period

You can use the [DBLibWaitTime](#) parameter to specify the time (in seconds) the scheduler, application server, or web server waits before breaking the connection with an event server in an unknown state. The scheduler, application server, or web server maintain and check connections with the databases, and if an event server is in an unknown state, the connection is broken after the specified time.

This section contains the following topics:

## Configure AutoSys Workload Automation to Run in Dual Event Server Mode

By default, AutoSys Workload Automation is configured to run in single event server mode during installation. You can configure AutoSys Workload Automation to run in dual event server mode during installation or later by using the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility (on Windows) or by modifying the parameters in the configuration file (on UNIX).

If you configured AutoSys Workload Automation to run in dual event server mode and one event server goes down, AutoSys Workload Automation automatically rolls over to the second event server and continues running in single event server mode. After you recover the event server that failed, you can reconfigure AutoSys Workload Automation to run in dual event server mode.

### NOTE

Do not try to run AutoSys Workload Automation in dual event server mode if it was previously running in single event server mode or if it rolled over to single event server mode. You must synchronize the two event servers before configuring AutoSys Workload Automation to run in dual event server mode.

### NOTE

- On Windows, the A Database Rollover Has Occurred check box indicates whether a database rollover has occurred from dual event server mode to single event server mode. If there has been a rollover and a switch to single event server mode, the A Database Rollover Has Occurred check box is selected and the Status field displays which event server is down.
- For more information about how to install and configure dual event servers, see [\(Windows\) Improve Workload Performance: Highly-Available Environments](#) or [\(UNIX\) Improve Workload Performance: Highly-Available Environments](#).

### (Windows) Follow these steps:

- Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
- Select an instance from the Instance drop-down list in the Settings pane.
- Click the Event Server icon on the toolbar.  
The Event Server - AutoSys Workload Automation Administrator window appears.
- Verify that the A Database Rollover Has Occurred check box is disabled.

### NOTE

By default, the A Database Rollover Has Occurred check box is disabled.

- Stop the scheduler and the application server services as follows:
  - Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - Right-click the scheduler and select Stop. Repeat this action for the application server.  
The scheduler and application server stop.
- Enter the following command at the instance command prompt:
 

```
cd %AUTOSYS%\dbobj\dbtype
```



- **dbtype**  
Specifies the type of database in use: MSQ (Microsoft SQL Server), ORA (Oracle), or SYB (Sybase).  
The database type is identified.
- 7. Run the bulk copy script (autobcp*DB*) based on your database vendor to synchronize the event servers.
- 8. Enable the second event server as follows:
  - a. Click the Event Server icon on the toolbar.  
The Event Server - AutoSys Workload Automation Administrator window appears.
  - b. Click the Event Server 1 or Event Server 2 tab based on the event server you want to enable and click Enable.  
The second event server is enabled. The status of the second event server is displayed as UP and the A Database Rollover Has Occurred check box is cleared and disabled.
- 9. Start the scheduler and the application server services as follows:
  - a. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - b. Right-click the scheduler and select Start. Repeat this action for the application server.  
The scheduler and application server start.

AutoSys Workload Automation is configured to run in dual event server mode.

**(UNIX) Follow these steps:**

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt:
 

```
unisrvctr stop waae_sched.$AUTOSERV
unisrvctr stop waae_server.$AUTOSERV
```

The scheduler and the application server stop.
3. Open the configuration file and add the following parameter after the EventServer\_1 parameter that corresponds to the first event server:
 

```
EventServer_2=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST | ORACLE_SVR,DBPORT,DBHOST
```

  - **SYBASE\_DB,DBPORT,DBHOST**  
Identifies the Sybase database for the second event server.
  - **ORACLE\_SVR,DBPORT,DBHOST**  
Identifies the Oracle database for the second event server.

**NOTE**

When the scheduler automatically rolls over to single event server mode, it creates a backup of the configuration file and modifies the existing file by commenting out the EventServer\_1 or EventServer\_2 parameter based on the event server that experienced the unrecoverable error. Uncomment the parameter to recover the event server. Alternatively, you can delete the modified configuration file and rename the backed up copy to config.*INSTANCENAME*.

4. Specify the database reconnect behavior for the second event server by modifying the following parameter in the configuration file:
 

```
DBEventReconnect=value, value2
```

  - **value**  
Identifies the database reconnect behavior for the first event server.  
**Limits:** 0-99
  - **value2**  
Identifies the database reconnect behavior for the second event server.  
**Limits:** 0-99

**NOTE**

During typical installation, AutoSys Workload Automation sets the reconnect value for the single event server to 50 by default. During a custom installation in which you enable dual event server mode, AutoSys Workload Automation sets the reconnect value for both event servers to 50.5 by default. Ensure that you add a reconnect value for the second event server when you configure AutoSys Workload Automation to run in dual event server mode after running it in single event server mode. Optionally, you can modify the default reconnect value for the first event server.

5. Save and exit the configuration file.

The database information and reconnect behavior for the second event server is defined. The database configuration changes from single event server mode to dual event server mode.

6. Run the AutoSys Workload Automation bulk copy script (autobcpORA or autobcpSYB) based on your database type. The event servers are synchronized.

7. Enter the following commands at the operating system prompt:

```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. AutoSys Workload Automation is configured to run in dual event server mode.

**autobcpDB Script -- Synchronize Databases**

The autobcpDB script synchronizes data servers on different computers to prepare them for dual event server mode. This script uses the information on the source data server to create two identical servers.

**NOTE**

- The autobcpDB script deletes all of the data in the target database and replaces it with the data in the source database. If you want to save the data in the target database, archive it before you run the autobcpDB script.
- You must stop the scheduler and application server before you run the autobcpDB script.
- You can enter the autobcpDB script on a single line or in interactive mode which prompts you for the required information line by line.
- On Oracle, you can also use the autobcpORAdp.pl script to move data from the source database to the target database quickly; thereby improving performance. The autobcpORAdp script is located at \$AUTOSYS/dbobj/ORA/autobcpORAdp.pl (UNIX) or %AUTOSYS%\dbobj\ORA\autobcpORAdp.pl (Windows).

An autobcpDB script for each database vendor is included in the following directories:

- Oracle  
\$AUTOSYS/dbobj/ORA/autobcpORA.pl
- Sybase  
\$AUTOSYS/dbobj/SYB/autobcpSYB.pl
- Microsoft SQL Server  
\$AUTOSYS/dbobj/MSQ/autobcpSQL.pl

The autobcpDB script that you use to synchronize the event servers depends on your database platform:

- Oracle; if using password to access the database  
perl autobcpORA.pl *source\_server target\_server source\_userid source\_password target\_userid target\_password dump\_file oracle\_directory*
- Oracle; if using an SSL certificate to access the database  
perl autobcpORA.pl-CERTMODE *source\_server target\_server source\_tnsadmin target\_tnsadmin dump\_file oracle\_directory*
- Sybase; if using password to access the database

```
perl autobcpSYB.pl source_server source_db target_server target_db source_userid
source_password target_userid target_password dump_file blk_size
```

- Sybase; if using Kerberos authentication to access the database

```
perl autobcpSYB.pl-EXTAUTH source_server source_db target_server target_db dump_file
SPN_1 SPN_2 blk_size
```

#### NOTE

To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login which is the owner of the database schema.

- Microsoft SQL Server

```
perl autobcpMSQ.pl source_server source_db target_server target_db source_userid
source_password target_userid target_password dump_file
```

- **CERTMODE** Specifies that an SSL certificate is used to access the database.

#### NOTE

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- **EXTAUTH** Specifies that Kerberos authentication is used to access the database.

#### NOTE

For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)

- **source\_server** Defines the name of the source Oracle System ID (for example, AEDB), Sybase server (for example, SourceServer), or Microsoft SQL Server server (for example, SourceServer). For Sybase, the source server name is defined in the interfaces file. For Microsoft SQL Server, you can view the source server name using Microsoft SQL Enterprise Manager.

#### NOTE

On Windows, the entire value of the database server name and the database name can be up to 64 characters.

- **source\_db**  
Defines the source Microsoft SQL Server or Sybase database (for example, AEDB).
- **source\_userid**  
Defines the user ID that is used to connect to the source Oracle System ID, Microsoft SQL Server server, or Sybase server.

#### NOTE

On Oracle, use aebadmin as the source user ID.

- **source\_password**  
Defines the password that corresponds to the user ID that is used to connect to the source Oracle System ID, Microsoft SQL Server server, or Sybase server.
- **target\_server**  
Defines the target Oracle System ID (for example, AEDB2), Microsoft SQL Server server (for example, DestinationServer), or Sybase server (for example, DestinationServer). For Sybase, the target server name is defined in the interfaces file. For Microsoft SQL Server, you can view the target server name using Microsoft SQL Enterprise Manager.

**NOTE**

- For Oracle, the source server must be different from the target server.
- On Windows, the entire value of the database server name and the database name can be up to 64 characters.
- **target\_db** Defines the target Microsoft SQL Server or Sybase database (for example, AEDB2).

**NOTE**

The autobcpDB script deletes all of the data in the target database and replaces it with the data in the source database. If you want to save the data in the target database, archive it before you run the autobcpDB script.

- **target\_userid**  
Defines the user ID that is used to connect to the target Oracle System ID, Microsoft SQL Server server, or Sybase server.

**NOTE**

On Oracle, use schema owner as the target user ID.

- **target\_password**  
Defines the password that corresponds to the user ID that is used to connect to the target Oracle System ID, Microsoft SQL Server server, or Sybase server.
- **dump\_file**  
Defines the temporary file that is used in the transfer of data from one database to the other database.

**NOTE**

Specify a file that is local to the computer where this script is running.

- **oracle\_directory**  
Defines the path to the Oracle home directory.
- **SPN\_1** (Optional) Specifies the source service principal name to be used for Kerberos authentication. If you do not want to specify the service principal name, include an empty space for this parameter.
- **SPN\_2** (Optional) Specifies the target service principal name to be used for Kerberos authentication. If you do not want to specify the service principal name, include an empty space for this parameter. **Example:** perl autobcp.pl -EXTAUTH WAKRB1 AEDBHP1 WAKRB1 AEDBHP dump.fil "" "" 5000
- **blk\_size**  
(Optional) Specifies the number of rows that can be inserted from the dump\_file to the destination database at a time.  
**Default:** 5000

**NOTE**

The default value is used if you run the autobcpSYB.pl script in the interactive mode or you do not specify the blk\_size value. Do not specify a large value because the transaction log encounters problems when it becomes too full.

**NOTE**

While running the autobcpSYB.pl script on Sybase, ensure the following:

- Both event servers use the same 'Character set'.
- The 'LANG' environment variable is unset from the shell or the command prompt window (from which the autobcpSYB.pl script is executed) using the following commands for Windows and UNIX.
  - On UNIX, use the following command:  
\$ unset LANG
  - On Windows, use the following command:  
C:\PROGRA~1\CA\UNICEN~1> set LANG=

The autobcpSYB.pl script might have problems while copying data from one event server to another, and might fail with errors if the environment variables are different. For more information, see the Sybase documentation.

### Example: Synchronize Databases on Sybase

This example copies data from the source database (AEDB) to the target database (AEDB2) on the source server (AUTOSYSDB) and the target server (AUTOSYSDB2).

#### NOTE

You can copy data faster and reduce the database log requirements by using the target user ID with the truncate command.

```
perl $AUTOSYS/dbobj/SYB/autobcpSYB.pl AUTOSYSDB AEDB AUTOSYSDB2 AEDB2 autosys autosys sa
autosys /tmp/dumpfile | tee /tmp/autobcp.out
```

### Synchronize the Event Servers on Oracle

#### NOTE

<https://communities.ca.com/videos/3239>: This video covers how to use the autobcp Oracle data pump option for fast HADS rollover recovery.

You can use the autobcpORAdp.pl script to move data from the source database to the target database quickly; thereby improving performance.

#### NOTE

Before you run the autobcpORAdp.pl script, ensure that you perform the following tasks:

- Create an Oracle dump directory with read and write permissions granted to the schema owner on the target database.
- Create a database link pointing to the source database on the target database so that data from the source server is accessed by the target server.

### Follow these steps:

1. Stop the AutoSys Workload Automation services as follows:

UNIX—Run the following commands from the operating system prompt:

```
unisrvctr stop waae_sched.$AUTOSERV
unisrvctr stop waae_server.$AUTOSERV
```

Windows—Navigate to the Services - AutoSys Workload Automation Administrator window of the Administrator utility, right-click the service, and select Stop.

#### NOTE

Repeat this step on the primary, shadow, and tie-breaker schedulers for all instances.

2. Log in as a user with SYSDBA permissions on the target database and run the following commands:

```
CREATE OR REPLACE DIRECTORY autobcpdump as 'path_to_oradata';
GRANT READ, WRITE ON DIRECTORY autobcpdump to aedbadmin;
```

#### – **autobcpdump**

Specifies the name of the Oracle dump directory.

#### – **path\_to\_oradata**

Defines the absolute directory path on the computer where you run the command to create the database link.

The Oracle dump directory with read and write permissions granted to the schema owner is created on the target database.

3. Run the following command:

If using password to access the database:

```
CREATE DATABASE LINK dblink_to_src CONNECT TO aedbadmin IDENTIFIED BY aedbadmin_pwd USING
'SOURCE_INSTANCE_SID';
```

If using an SSL certificate to access the database:

```
CREATE DATABASE LINK dblink_to_src USING 'SOURCE_INSTANCE_SID';
```

- ***dblink\_to\_src***  
Defines the name of the database link that is created between the source database and the target database.
- ***schema owner\_pwd***  
Specifies the password that is associated with the schema owner.
- ***SOURCE\_INSTANCE\_SID***  
Specifies the name of the source Oracle instance; enclose this value in single quotes.

A network link or database link pointing to the source database is created on the target server so that data from the source server is accessed by the target server.

4. Log in as the schema owner and verify if the database link is created.

If using password to access the database, run the following commands on the destination server:

```
$ sqlplus aedbadmin/aedbadmin_pwd@dest_sid
1> SELECT * FROM ujo_alamode@DATABASE_LINK_NAME;
```

If using an SSL certificate to access the database, set the TNS\_ADMIN value for the Oracle schema owner of the destination server and run the following commands on the destination server:

```
$ sqlplus /@dest_sid
1> SELECT * FROM aedbadmin.ujo_alamode@db_link
```

The contents of the ujo\_alamode table in the source database are displayed. If you encounter any errors, fix your database link or the source database before you run the autobcpORAdp.pl script.

5. Run the following command from the UNIX operating system prompt or the Windows instance command prompt:

If using password to access the database:

```
perl autobcpORAdp.pl TARGET_SID TARGETusername
TARGETUserpassword aedbadmin_pwd dblink_dest_SRC autobcpdump oracle_directory
```

If using an SSL certificate to access the database:

```
perl autobcpORAdp.pl -CERTMODE TARGET_SID target_tnsadmin dblink_dest_SRC autobcpdump
oracle_directory
```

- **CERTMODE** Specifies that an SSL certificate is used to access the database.

#### NOTE

For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).

- **TARGET\_SID**  
Specifies the name of the target Oracle instance; enclose this value in single quotes.
- **schema owner\_pwd**  
Specifies the password that is associated with the schema owner.
- **target\_tnsadmin**  
Specifies the TNS\_ADMIN value for the Oracle schema owner of the target database.
- **dblink\_dest\_SRC**  
Defines the name of the database link that is created between the source database and the target database.
- **autobcpdump**  
Specifies the name of the Oracle dump directory.
- **oracle\_directory**  
Defines the path to the Oracle home directory.

6. Start the AutoSys Workload Automation services as follows:

UNIX—Run the following commands from the operating system prompt:

```
unisrvctr start waee_sched.$AUTOSERV
```

```
unisrvcntr start waae_server.$AUTOSERV
```

Windows—Navigate to the Services - AutoSys Workload Automation Administrator window of the Administrator utility, right-click the service, and select Start.

**NOTE**

Repeat this step on the primary, shadow, and tie-breaker schedulers for all instances.

The event servers are synchronized.

**NOTE**

If you want the schema owner to create the database\_link and directory, your SYSDBA must grant permissions to the schema owner by running the following command:

```
GRANT CREATE ANY DIRECTORY TO aedbadmin;
GRANT CREATE DATABASE LINK TO aedbadmin;
```

**Example: (UNIX) Synchronize Databases on Oracle**

This example copies the data to the target database (AEDB) using the database link (dblink) that is created between the source database and the target database.

```
perl $AUTOSYS/dbobj/ORA/autobcpORAdp.pl AEDB password dblink autobcpdump $ORACLE_HOME
```

**Event Server Synchronization**

AutoSys Workload Automation provides the AutoSys Workload Automation bulk copy scripts (autobcpORA, autobcpORAdp, autobcpSYB, and autobcpMSQ) to synchronize the event servers. These scripts identify one event server as the source and the other event server as the target for the synchronization process.

**NOTE**

- You must synchronize the event servers before enabling dual event server mode.
- The greater the data that must be synchronized, the longer the AutoSys Workload Automation bulk copy script runs.

Before you synchronize the event servers, do the following:

- Check that both event servers are running.
- Check that no AutoSys Workload Automation schedulers, application servers, or client utilities like archive\_events, sendevent, and so on are running.
- For Microsoft SQL Server, check that both databases are defined correctly. Use the Microsoft SQL Enterprise Manager to view the information.
- For Oracle, check that the \$TNS\_ADMIN/tnsnames.ora (on UNIX) or %TNS\_ADMIN%\tnsnames.ora (on Windows) file contains valid entries for both event servers.

**NOTE**

If you installed Oracle Instant Client with the SQL\*Plus package, you need not configure the tnsnames.ora file. Check that you can sqlplus to both the Oracle databases using the proper Oracle Net connection identifier.

- For Sybase, check that the \$SYBASE/interfaces (on UNIX) or %SYBASE%\ini\sql.ini (on Windows) file contains entries for both event servers.
- Note the path to the database software so you can provide it when you run the AutoSys Workload Automation bulk copy script.
- Check that you have at least as much free disk space as the size of your database to store the temporary file that the AutoSys Workload Automation bulk copy script creates. The script deletes this temporary file after the synchronization process is complete.

**NOTE**

When you stop the scheduler, any jobs that are running on the agent run to completion. Although it is recommended that you stop all jobs before synchronizing the databases, you can run the AutoSys Workload Automation bulk copy script while the jobs are running on the agent.

**Handle Event Server Synchronization Errors**

Sometimes the autobcpDB script encounters errors when it is running. In these cases, the script exits and AutoSys Workload Automation displays the following message:

```
The CA WAAE data server is not accessible.
Please check the data server and rerun this script.
```

You can handle errors by correcting data server problems and rerunning the script.

To handle errors, verify the following and rerun the autobcpORA, autobcpORAdp, autobcpSYB, or autobcpMSQ script:

- Are both event servers running?  
To verify this, ensure that you can connect to the event server.
  - For Microsoft SQL Server, look for the MSSQLSERVER and SQLSERVERAGENT services.
  - For Oracle, look for the OracleService\*, OracleStart\*, and OracleTNSListener services (where \* indicates the Oracle SID).
  - For Sybase, the service name is user-configurable.
- Did you specify the source and the target event servers correctly in the autobcpORA, autobcpORAdp, autobcpSYB, or autobcpMSQ script?
- Did you enter the passwords correctly in the autobcpORA, autobcpORAdp, autobcpSYB, or autobcpMSQ script?
- Did you set the Sybase or Oracle environment variables correctly?
  - The Oracle environment variable, ORACLE\_HOME, defines the path to the top-level Oracle directory.
  - The Sybase environment variables are DSQUERY and SYBASE. The DSQUERY variable defines the name of the Sybase event server. The SYBASE variable defines the complete path to the Sybase software directory.
- Did you specify the event server names and ports correctly?
  - For Microsoft SQL Server, you can view this information using the Microsoft SQL Enterprise Manager.
  - For Oracle, this information is located in the TNSNAMES.ORA file.
  - For Sybase, this information is located in the interfaces file.

**Switch to Single Event Server Mode**

You can configure AutoSys Workload Automation to run in single event server mode. This is useful after a database rollover. When AutoSys Workload Automation is running in dual event server mode and the scheduler detects an unrecoverable error condition on one of the event servers, it automatically rolls over to single event server mode on the other event server. To keep that event server active (UP status), you can switch to single event server mode.

**(UNIX) Follow these steps:**

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following commands at the operating system prompt:

```
unisrvcntr stop waae_sched.$AUTOSERV
unisrvcntr stop waae_server.$AUTOSERV
```

The scheduler and the application server stop.

3. Open the configuration file, comment out the EventServer\_1 or EventServer\_2 parameter (corresponding to the event server that you do not want to use), and save the file.
4. Enter the following commands at the operating system prompt:



```
unisrvcntr start waae_sched.$AUTOSERV
unisrvcntr start waae_server.$AUTOSERV
```

The scheduler and the application server start. AutoSys Workload Automation is now configured to run in single event server mode.

**(Windows) Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Event Server icon on the toolbar.  
The Event Server - AutoSys Workload Automation Administrator window appears.
4. Verify that the A Database Rollover Has Occurred check box is cleared.

**NOTE**

By default, the A Database Rollover Has Occurred check box is disabled.

5. Stop the scheduler and the application server services as follows:
  - a. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - b. Right-click the scheduler and select Stop. Repeat this action for the application server.  
The scheduler and application server stop.
6. Disable the second event server as follows:
  - a. Click the Event Server icon on the toolbar.  
The Event Server - AutoSys Workload Automation Administrator window appears.
  - b. Click the Event Server 1 or Event Server 2 tab based on the event server you want to disable and Click Disable.  
The second event server is disabled. The status of the second event server is displayed as DOWN.
7. Start the scheduler and the application server services as follows:
  - a. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
  - b. Right-click the scheduler and select Start. Repeat this action for the application server.  
The scheduler and application server start.

AutoSys Workload Automation runs in single event server mode. You can view the relevant messages in the scheduler and the application server log files.

## Event Server Rollover Recovery

When AutoSys Workload Automation is running in dual event server mode and the scheduler detects an unrecoverable error condition on one of the event servers, it automatically rolls over to single event server mode on the other event server.

An unrecoverable error is defined as one of the following:

- The connection to the database is lost, and after the configured number of reconnect attempts, the database remains unconnected.
- A database has an unrecoverable error (for example, database corruption or media failure).

**NOTE**

- On Sybase, a full transaction log is considered as an unrecoverable error. If you do not allocate sufficient log space for the level of database activity, the transaction log fills up under heavy load resulting in a severe

error. In dual event server mode, <UAJM> rolls over to single event server mode using the database with available transaction log space. In single event server mode, <UAJM> shuts down.

- On Oracle, if you do not allocate sufficient log space for the level of database activity, the transactions are suspended indefinitely without any error until the transaction log space becomes available. <UAJM> does not change the event server mode, but may halt until Oracle releases the control back to it.

When an event server rollover occurs, AutoSys Workload Automation does the following:

- On UNIX, the configuration file indicates whether a database rollover has occurred from dual event server mode to single event server mode by commenting out (prefixes #AUTO-ROLLOVER#) the EventServer\_1 or EventServer\_2 parameter that defines the event server that went offline.

#### NOTE

- A backup of the original configuration file is saved in \$AUTOUSER/config.\$AUTOSERV.rollover.
- The configuration file is modified on the primary or shadow scheduler or both. The configuration file on the client computers is not modified.
- On Windows, the Event Server - AutoSys Workload Automation Administrator window of the Administrator utility indicates whether a database rollover has occurred from dual event server mode to single event server mode. If there has been a database rollover and a switch to single event server mode, the A Database Rollover Has Occurred check box is selected and the Status field displays which event server is DOWN.

#### NOTE

For more information about switching to single event server mode on Windows, see [Switch to Single Event Server Mode](#).

AutoSys Workload Automation makes these changes so that the scheduler or the application server trying to access the database is aware that it is now running in single event server mode.

## High Availability Recovery

Running AutoSys Workload Automation with high availability and dual event server options helps protect the service from being interrupted due to application, network, and database failures. This section describes the behavior of the scheduler and the application server when a failure is detected and how AutoSys Workload Automation tries to recover.

#### NOTE

For more information about the high availability options and how to configure them, see [\(UNIX\) Improve Workload Performance: Highly-Available Environments](#) or [\(Windows\) Improve Workload Performance: Highly-Available Environments](#)

### Set the Number of Scheduler or Application Server Connection Attempts

When the scheduler or application server fails to update one of the event servers while running in dual event server mode, AutoSys Workload Automation stops processing events while it tries to re-establish the connection with the event server. You can use the [DBEventReconnect](#) parameter to set the number of times the scheduler or application server tries to connect (or reconnect) to an event server before shutting down or switching over to single event server mode.

### Configure the Scheduler Heartbeat Interval

In high availability mode, the primary, shadow, and tie-breaker schedulers update the database with their statuses at regular intervals. If a scheduler does not update the database after two intervals, that scheduler is unavailable and the system leaves high availability mode. You can configure the length of each interval using the [HAPollInterval](#) parameter.

## **Recovery Scenarios**

The following sections describe the recovery behavior of AutoSys Workload Automation after a point of failure. The recovery scenarios apply to single event server mode and dual event server mode, as well as to non-high availability and high availability modes.

### **NOTE**

In the dual event server mode scenarios documented in this section, the primary or shadow scheduler notifies the tie-breaker scheduler and the application server to roll over by updating the accessible database. The tie-breaker scheduler and the application server receive the notification when they fetch the updated database entry. If both the databases are unavailable, the notification cannot be written and the tie-breaker scheduler and the application server do not roll over.

### **Non-High Availability in Single Event Server Mode**

If the connection to the single event server is lost, AutoSys Workload Automation does the following:

- The scheduler tries to reconnect to the event server for the configured number of times. If the scheduler cannot reconnect, it shuts down.
- The application server tries to reconnect to the event server for the configured number of times. If the application server cannot reconnect, it shuts down.

### **Non-High Availability in Dual Event Server Mode**

If the connection to one of the event servers is lost, AutoSys Workload Automation does the following:

- The scheduler tries to reconnect to the event server for the configured number of times. If the scheduler cannot reconnect, it rolls over and notifies the application server.
- The application server tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:
  - It re-establishes a connection.
  - It receives notification to roll over.
  - It receives a shutdown request.

If the connections to both event servers are lost, AutoSys Workload Automation does the following:

- The scheduler tries to reconnect to the event server for the configured number of times. If the scheduler cannot reconnect, it rolls over and fails to notify the application server. If the scheduler fails to connect to the second event server after the configured number of times, it shuts down.
- The application server tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:
  - It re-establishes a connection.
  - It receives a shutdown request.
  - It detects the loss of connection to the second event server and shuts down.

### **High Availability in Single Event Server Mode**

If the primary scheduler becomes unavailable, the shadow scheduler issues an EP\_ROLLOVER alarm, fails over, and starts processing events.

If the shadow scheduler becomes unavailable, the primary scheduler issues an EP\_HIGH\_AVAIL alarm and continues to run.

If the event server connection is lost, AutoSys Workload Automation does the following:

- The scheduler tries to reconnect to the event server for the configured number of times. If the scheduler cannot reconnect, it shuts down.
- The application server tries to reconnect to the event server for the configured number of times. If the application server cannot reconnect, it shuts down.

### **High Availability in Dual Event Server Mode**

If the primary scheduler becomes unavailable, the shadow scheduler issues an EP\_ROLLOVER alarm, fails over, and starts processing events.

If the shadow scheduler becomes unavailable, the primary scheduler issues an EP\_HIGH\_AVAIL alarm and continues to run.

If the tie-breaker scheduler becomes unavailable, the primary scheduler issues an EP\_HIGH\_AVAIL alarm and continues to run.

If the connection to one of the event servers is lost, AutoSys Workload Automation does the following:

- The primary scheduler tries to reconnect to the event server for the configured number of times. If the primary scheduler cannot reconnect, it rolls over and notifies the tie-breaker scheduler and the application server. The primary scheduler then checks for status updates from the shadow and tie-breaker schedulers. If the shadow and tie-breaker schedulers have updated the event server, the primary scheduler continues to run. If neither the shadow scheduler nor the tie-breaker scheduler has updated the event server in two consecutive poll intervals, the primary scheduler shuts down. If only the shadow scheduler has not updated the event server in two consecutive poll intervals, the primary scheduler issues an EP\_HIGH\_AVAIL alarm and continues to run.
- The shadow scheduler tries to reconnect to the event server for the configured number of times. If the shadow scheduler cannot reconnect, it rolls over and notifies the tie-breaker scheduler and the application server. The shadow scheduler then checks for status updates from the primary and tie-breaker schedulers. If the primary and tie-breaker schedulers have updated the event server, the shadow scheduler continues to run. If neither the primary scheduler nor the tie-breaker scheduler has updated the event server in two consecutive poll intervals, the shadow scheduler shuts down. If only the primary scheduler has not updated the event server in two consecutive poll intervals, the shadow scheduler fails over and starts processing events.
- The tie-breaker scheduler tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:
  - It re-establishes a connection.
  - It receives notification to roll over.
  - It receives a shutdown request.
 In the meantime, it continues to update the accessible event server with its heartbeat.
- The application server tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:
  - It re-establishes a connection.
  - It receives notification to roll over.
  - It receives a shutdown request.

If the connections to both event servers are lost, AutoSys Workload Automation does the following:

- The primary and shadow schedulers try to reconnect to the event server for the configured number of times. If the primary and shadow schedulers cannot reconnect, they roll over and fail to notify the tie-breaker scheduler and the application server. If the primary and shadow schedulers fail to connect to the second event server after the configured number of times, they shut down.
- The tie-breaker scheduler tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:

- It re-establishes a connection.
- It receives a shutdown request.
- It detects the loss of connection to the second event server and shuts down.
- The application server tries to reconnect to the event server for the configured number of times. It continues to try to reconnect to the event server at regular intervals until one of the following occurs:
  - It re-establishes a connection.
  - It receives a shutdown request.
  - It detects the loss of connection to the second event server and shuts down.

## Database Storage Requirements

The limit on how much disk space a database can use is based on the underlying operating system and its file size limitations. Databases need disk space for more than just the database tables and stored procedures. They require sufficient disk space for sorting temporary and transient files. In addition, product operation and database backups can require a lot of space.

The size requirements for your database depend on the following:

- The number of jobs you define.
- The number of jobs that have dependencies.
- How often the jobs run.
- How often the database is cleaned.

### NOTE

Every time a job runs, it generates at least three events and an entry in both the `ujo_job_runs` and `ujo_extented_jobrun_info` tables.

The standard sizes for databases are as follows:

- Microsoft SQL Server -- 800 MB
- Oracle -- 800 MB for the data tablespace and 80 MB for an index tablespace.
- Sybase -- 800 MB for the data device and 100 MB for the log device.

The database tables are created with the option that automatically extends as long as there is space in the file system. The database sizes specified are the recommended initial size. If your job load is large, create a larger database.

## General Database Maintenance

Periodic database maintenance helps ensure that AutoSys Workload Automation is working correctly. Each run of each job generates several events. If you do not remove these events from the database periodically, the database eventually reaches its size limit, bringing AutoSys Workload Automation and its jobs to a halt. Therefore, periodic database maintenance is recommended.

### Automate Database Maintenance

You can automate the database maintenance using the [DBMaintTime](#) and [DBMaintCmd](#) parameters. The scheduler performs internal database maintenance once a day. Event processing is slowed down during maintenance. We recommend that you schedule the maintenance during low processing times. For environments where DBMaint starts becoming disruptive, we recommend that you minimize the impact by running DBMaint multiple times per day so that the processing is spread over multiple shorter periods as opposed to a single longer duration.

## How the DBMaint.bat Batch File or DBMaint Script Runs

By default, AutoSys Workload Automation runs the DBMaint script on UNIX or the DBMaint.bat batch file on Windows during the daily maintenance cycle. The DBMaint command runs the dbstatistics, archive\_events, and archive\_jobs commands to perform maintenance on the AutoSys Workload Automation database.

The DBMaint command runs the dbstatistics command to perform the following tasks:

- Update statistics in the database for optimal performance. For Oracle and Sybase databases, it computes statistics for all the tables.
- Run the dbspace command to check the available space in the database. If the amount of free space is insufficient, the dbspace command issues warning messages and generates a DB\_PROBLEM alarm.

### NOTE

The DB\_PROBLEM alarm is issued if the database space exceeds the value specified in the DBSPACE\_ALARM\_SPACE environment variable. The default value is 1000 MB.

- Calculate and update the average job run statistics in the ujo\_avg\_job\_run table. When the dbstatistics command runs, it overwrites old data with the new data.

The DBMaint command runs the archive\_events command to remove old information from various database tables. Specifically, the archive\_events command removes the following:

- Events and associated alarms from the ujo\_event table
- Job run information from the ujo\_job\_runs table
- autotrack log information from the ujo\_audit\_info and ujo\_audit\_msg tables

The DBMaint command runs the archive\_jobs command to delete obsolete job versions from the database tables. It specifically removes the obsolete information from the job type database tables.

The output from the DBMaint command reports the amount of space remaining in your database so you can monitor whether the event tables are filling up. By monitoring these values, you can calculate how many events you can safely maintain in a day before archiving.

### NOTE

For more information about the DBMaint, dbspace, dbstatistics, archive\_events, and archive\_jobs commands, see [AutoSys Commands](#).

## (UNIX) Modify the DBMaint Script

You can modify the \$AUTOSYS/bin/DBMaint script. For example, you might want to modify the script to perform database backups.

### Follow these steps:

1. Make a copy of the \$AUTOSYS/bin/DBMaint script and modify the copied version.
2. Log in to AutoSys Workload Automation and run the shell that is sourced to use AutoSys Workload Automation.
3. Check that the modified DBMaint script is placed in the location specified by the following parameter of the configuration file:

`DBMaintCmd=pathed_command`

#### – ***pathed\_command***

Defines the location of the DBMaint script.

**Default:** \$AUTOSYS/bin/DBMaint

AutoSys Workload Automation uses the modified DBMaint script to perform database maintenance.

### NOTE

When you upgrade from Unicenter AutoSys JM r11 to AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6 you will not lose the changes you made in the copied version. You can modify the DBMaint script

that is installed when you upgrade to AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6 to match your copied version.

### **(Windows) Modify the DBMaint.bat File**

You can modify the %AUTOSYS%\bin\DBMaint.bat file. For example, you might want to modify the batch file to perform database backups.

#### **Follow these steps:**

1. Make a copy of the DBMaint file and modify the copied version.
2. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
3. Select an instance from the Instance drop-down list.
4. Click the Scheduler icon on the toolbar.  
The Scheduler - AutoSys Workload Automation Administrator window appears.
5. Click the Database Maintenance tab, enter the location of the modified DBMaint.bat batch file in the Command field, and click Apply.  
AutoSys Workload Automation uses the modified DBMaint script to perform database maintenance.

#### **NOTE**

When you upgrade from Unicenter AutoSys JM r11 to AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6, you will not lose the changes you made in the copied version. You can modify the DBMaint file that is installed when you upgrade to AutoSys Workload Automation r11.3, Release 11.3.5, or Release 11.3.6 to match your copied version.

## **Oracle Errors**

When Oracle returns any of the following error values while executing a query, AutoSys Workload Automation considers these errors as fatal database errors and stops using the database. If AutoSys Workload Automation is running in single event server mode, it shuts down gracefully. If AutoSys Workload Automation is running in dual event server mode, it rolls over to single event server mode. You must troubleshoot and fix the cause of these fatal database errors to run AutoSys Workload Automation.

```
ORA-00257
ORA-00600
ORA-01035
ORA-01650 - ORA-01655
ORA-01680
ORA-01683
ORA-01688
ORA-01691 - ORA-01692
ORA-02700 - ORA-02855
ORA-02875 - ORA-02899
ORA-03233 - ORA-03234
ORA-03238
ORA-04031
ORA-06400 - ORA-06422
ORA-06430 - ORA-06457
ORA-06600 - ORA-06622
ORA-07200 - ORA-07999
ORA-09700 - ORA-09999
```

---

ORA-12315 - ORA-12499  
ORA-12526  
ORA-30036

**NOTE**

Fatal errors include internal Oracle failures like read or write IO errors, internal driver errors, or other internal software errors within Oracle.

When Oracle returns any of the following error values while executing a query, AutoSys Workload Automation considers these errors as recoverable database errors and retries the query execution after refreshing the database connection:

ORA-00018  
ORA-00020  
ORA-00025  
ORA-00028  
ORA-00051  
ORA-00601 - ORA-00604  
ORA-01089 - ORA-01090  
ORA-03113 - ORA-03114  
ORA-03123  
ORA-03126  
ORA-03135  
ORA-06105 - ORA-06108  
ORA-06117 - ORA-06118  
ORA-06121  
ORA-06402  
ORA-06409 - ORA-06410  
ORA-06413  
ORA-06422 - ORA-06423  
ORA-06705  
ORA-06707  
ORA-06713  
ORA-06722  
ORA-06731  
ORA-06734  
ORA-06736  
ORA-06737  
ORA-06754  
ORA-06756  
ORA-06792  
ORA-06793  
ORA-06970 - ORA-06979  
ORA-12150 - ORA-12314  
ORA-12500 - ORA-12629  
ORA-25402  
ORA-25405  
ORA-25408  
ORA-25425



When Oracle returns any error value that is not listed above while executing a query, AutoSys Workload Automation considers these errors as unknown database errors and does not retry the query execution.

#### NOTE

For more information about Oracle error codes, see the Oracle documentation.

## Rebuild Table Indexes for a AutoSys Workload Automation Database

Over time, the database table indexes can become inefficient while you run jobs and update them. You can rebuild the table indexes of a specified AutoSys Workload Automation database to renew the efficiency.

#### NOTE

We recommend that you run the `reindexDB` script when the system activity is minimal. Otherwise, AutoSys Workload Automation might experience a slow down or time-out condition while performing database transactions.

To rebuild indexes for a AutoSys Workload Automation database, you must run the `reindexDB` script at the UNIX operating system prompt or the Windows instance command prompt.

### Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Sybase

This example rebuilds table indexes for a AutoSys Workload Automation database on Sybase where `SYBASESRV` is the name of the Sybase server.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl SYB SYBASESRV sa password
AEDB
```

### Example: Rebuild Tables Indexes for a AutoSys Workload Automation database on Oracle

This example rebuilds table indexes for a AutoSys Workload Automation database on Oracle where `ORACLESRV` is the Oracle System ID.

```
perl /opt/CA/WorkloadAutomationAE/autosys/dbobj/reindexDB.pl ORA ORACLESRV aedbadmin
password
```

### reindexDB Script -- Rebuild Table Indexes

The `reindexDB` script rebuilds the table indexes of a specified AutoSys Workload Automation database.

The `reindexDB` script is located as follows:

- On UNIX -- `$AUTOSYS/dbobj`
- On Windows -- `%AUTOSYS%\dbobj`

This script has the following format:

```
reindexDB.pl database_type server_name server_userid server_password database_name
```

On Sybase, if you are using Kerberos authentication to access the database, this script has the following format:

```
reindexDB.pl -EXTAUTH SYB server_name database_name SPN
```

#### NOTE

To use Kerberos authentication, ensure that the logged in OS user running this script is already authenticated with Kerberos and is mapped to a Sybase login with `sa_role` privileges.

On Oracle, if you are using SSL certificate to access the database, this script has the following format:

```
reindexDB.pl -EXTAUTH ORA server_name aedbadmin_tnsadmin
```

- **database\_type**  
Specifies the database type. This value can be one of the following:
  - **ORA**  
Identifies Oracle as the database.
  - **SYB**  
Identifies Sybase as the database.
  - **MSQ**  
Identifies Microsoft SQL Server as the database.
- **EXTAUTH**  
Specifies that an SSL certificate (Oracle) or Kerberos authentication (Sybase) is used to access the database.
 

**NOTE**

  - On Oracle, you can also use the CERTMODE parameter (instead of the EXTAUTH parameter) to specify that SSL certificate is used to access the database.
  - For information about creating Oracle wallets with certificates, see [Create Oracle Wallets with Certificates](#).
  - For information about configuring MIT Kerberos on AutoSys Workload Automation, see [Configure MIT Kerberos on AutoSys Workload Automation](#)
- **aedbadmin\_tnsadmin**  
Specifies the TNS\_ADMIN value for the schema owner.
- **server\_name**  
Defines the name of the Oracle System ID, Sybase server, or Microsoft SQL Server server.
- **server\_userid**  
Defines the user ID that is used to connect to the Oracle, Sybase, or Microsoft SQL Server server.  
**Default:** sa (Sybase, Microsoft SQL Server)
 

**NOTE**

For Oracle, you must use schema owner as the server user ID.
- **server\_password**  
Defines the password that corresponds to the user ID that is used to connect to the Oracle, Sybase, or Microsoft SQL Server server.  
**Default:** autosys
- **database\_name**  
Defines the name of the Sybase or Microsoft SQL Server database.
 

**NOTE**

The database\_name parameter does not apply to Oracle.
- **SPN** (Optional) Specifies the service principal name to be used for Kerberos authentication.

## Tune the Oracle Server

If you run a large number of jobs every day in your enterprise, you must tune the Oracle database to prevent database errors and improve the performance.

To tune the Oracle database, do the following:

1. [Configure the Oracle database](#).
2. [Tune the Oracle database](#).

### Configure the Oracle Database

When you install AutoSys Workload Automation, you must configure the database size, data file size, and index file size based on the number of jobs that run every day. For example, if you run 50,000 jobs every day, you must set the following values:

- Database size -- 2000 MB
- Data file size (AEDB\_DATA) -- 2000 MB
- Index file size (AEDB\_INDEX) -- 200 MB

### **Tune the Oracle Database**

To tune the Oracle database to run a large number of jobs every day, you must increase the default value of the processes parameter that is installed with Oracle.

#### **NOTE**

The processes parameter specifies the maximum number of operating system processes that can be connected to the Oracle database concurrently. For more information about the processes parameter that is installed with Oracle, see the Oracle documentation.

#### **Follow these steps:**

1. Issue the following commands:

```
sqlplus /nolog
SQL> connect sys/sys_password as sysdba
SQL> shutdown
SQL> exit
```

#### **– sys\_password**

Defines the password that corresponds to the Oracle system user ID.

The Oracle database stops.

2. Issue the following commands:

```
cd $ORACLE_HOME/dbs
cp -rfp spfileORACLE_SID.ora spfileORACLE_SID.ora.orig
```

A backup of the SPFILE binary file is created.

3. Issue the following commands:

```
sqlplus /nolog
SQL> connect sys/sys_password as sysdba
SQL> create pfile from spfile;
SQL> exit
```

The PFILE text file is created from the SPFILE binary file.

4. Issue the following commands:

```
cd $ORACLE_HOME/dbs
cp -rfp initORACLE_SID.ora initORACLE_SID.ora.orig
```

A backup of the initORACLE\_SID.ora text file is created.

5. Edit the initORACLE\_SID.ora file to make the following changes, and save the file:

```
vi initORACLE_SID.ora
*.processes=value
```

#### **– value**

Defines the number of processes. The recommended value is 300.

6. Issue the following commands:

```
sqlplus /nolog
```

```
SQL> connect sys/sys_password as sysdba
SQL> create spfile from pfile;
```

The SPFILE binary file is created from the PFILE text file.

7. Issue the following command:

```
SQL> startup
```

The database starts. The default value for the processes parameter is increased to 300.

8. Issue the following command:

```
SQL> show parameter processes;
```

An output similar to the following is displayed:

| NAME                      | TYPE    | VALUE |
|---------------------------|---------|-------|
| -----                     | -----   | ----- |
| aq_tm_processes           | integer | 0     |
| db_writer_processes       | integer | 1     |
| gcs_server_processes      | integer | 0     |
| job_queue_processes       | integer | 10    |
| log_archive_max_processes | integer | 2     |
| processes                 | integer | 300   |

You can verify the number of processes has been changed to 300.

## Tune the Sybase Server

If you run a large number of jobs every day in your enterprise, you must tune the Sybase server to prevent database errors and improve the performance.

To tune the Sybase server, do the following:

1. [Configure the Sybase server.](#)
2. [Tune the Sybase server.](#)

### Configure the Sybase Server

When you install AutoSys Workload Automation or create a new Sybase server, you must configure the database size, data file size, and log device size based on the number of jobs that run every day. For example, if you run 50,000 jobs every day, you must set the following values:

- Database size -- 2000 MB
- Data file size (AEDB\_DATA) -- 1760 MB
- Log device size (AEDB\_LOG) -- 240 MB

### Tune the Sybase Server

You must tune the Sybase server to prevent database errors and improve the performance.

#### **NOTE**

You can tune the Sybase server based on the number of jobs that run every day in your enterprise. In this procedure, the Sybase server is tuned to run 50,000 jobs every day.

**Follow these steps:**

1. Select an 8 KB page size when you install the Sybase server.
2. Create a 2000 MB AutoSys Workload Automation database.
3. Run the following SQL commands:

```
sp_configure 'max memory',120000
go
sp_configure 'user connections',250
go
sp_configure "procedure cache size",30000
go
sp_configure 'max online engines',2
go
sp_configure 'number of engines at startup',2
go
```

**NOTE**

- The max online engines and number of engines at startup parameters specify the number of CPUs on the database server computer.
- You must increase the kernel shared memory if it is not sufficient to increase the Adaptive Server Enterprise (ASE) memory. Kernel shared memory is an operating system specific variable. For more information about modifying the kernel shared memory value, contact your UNIX administrator.
- If you increase the number of user connections, you must increase the ASE physical memory that is allocated to the server.

4. Increase the database (tempdb) size from 12 MB (default) to 100 MB as follows:

- a. Issue the following commands:

```
disk resize
name="master",
size="180M"
go
```

The master device size is extended from 120 MB (default) to 300 MB.

- b. Issue the following commands:

```
sp_helpdevice
go
```

The master device size is displayed.

- c. Issue the following commands:

```
alter database tempdb on master=88
go
```

The database (tempdb) size is extended from 12 MB (default) to 100 MB.

- d. Issue the following commands:

```
sp_helpdb
go
```

The database size is displayed.

- e. Issue the following commands:

```
sp_cacheconfig 'default data cache', '16M'
go
```

The default data cache size is increased to 16 M.

- f. Issue the following commands:

```
sp_helpcache
go
```

The cache size is displayed.

#### NOTE

You must increase the database size because you installed the Sybase server with 8 KB page size.

5. Stop and restart the Sybase server.

The Sybase server is tuned to run 50,000 jobs every day.

## Maintain the Agent

The agent writes all log files to the following directories:

- *installation\_directory/SystemAgent/agent\_name/log*
- *installation\_directory/SystemAgent/agent\_name/spool* (for job spool files)

#### ***installation\_directory***

Specifies the directory where the agent is installed.

#### ***agent\_name***

Specifies the name of the agent.

This section contains the following topics:

## Log or Spool File Maintenance

### Log File Maintenance

The agent keeps a set of logs that must be cleared periodically to maintain disk space availability. The log files contain records of all messages between the agent and AutoSys Workload Automation as well as internal messages. These files are located in the log directory by default and are updated continually while the agent is running. The types and number of logs that are generated depend on the log.level parameter set in the agentparm.txt file.

You can configure agent log file properties that control the log file size, the types and number of log files that are generated, and how the agent archives the log files.

#### NOTE

For information about configuring the agent log file properties or enabling or disabling job logs, see [Log File Maintenance](#).

### Spool File Maintenance

The output for workload is stored in spool files that the agent software generates. Depending on the type of workload the agent runs, the spool files are stored in and accessed from different locations.

Spool files are limited in size by the available space on the file system where they reside. To maintain storage space, the agent immediately clears the spool files for successfully completed jobs. After seven days, the agent clears the spool files for failed jobs. You can change these default settings.

## Clean Spool and Job Log Files

The spool and job log files are stored in the agent spool directory. To maintain disk space availability, the agent immediately clears the spool and log files for successfully completed jobs. After seven days, the agent clears the files for failed jobs. You can change these default settings on the agent.

### Follow these steps:

1. Do **one** of the following steps based on your operating system:

**UNIX:** Run the shell that is sourced to use AutoSys Workload Automation and enter the following command at the operating system prompt:

```
unisrvcntr stop waee_agent-WA_AGENT
```

#### – **WA\_AGENT**

Defines the name of the agent.

The agent stops.

#### **Windows:**

- a. Click Start, Programs, AutoSys, Administrator.

The Instance - AutoSys Workload Automation Administrator window opens.

- b. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

- c. Right-click the agent service and click Stop.

The agent stops.

2. Open the agentparm.txt file located in the agent installation directory.

3. Add or edit the following parameters:

```
oscomponent.joblog.success.autocleanup=true
```

```
agent.spool.success.autocleanup=true
```

```
runnerplugin.spool.clean.enable=true
```

These settings configure the agent to immediately clear log and spool files for successfully completed jobs.

4. Add or edit the following parameter:

```
runnerplugin.spool.expire=expire_time
```

#### – **expire\_time**

Specifies how long to keep the spool files for. The files are cleared after the specified period. The options are as follows:

#### – **nd**

Specifies that spool files are kept for n days.

**Default:** 7d (Seven days)

#### – **nh**

Specifies that the spool files are kept for n hours.

**Example:** 10h

#### – **nm**

Specifies that the spool files are kept for n minutes.

**Example:** 50m

#### – **ns**

Specifies that the spool files are kept for n seconds.

**Example:** 30s

### **NOTE**

You cannot specify combinations of time periods. For example, 12d3h is not valid. If you specify a number only, the agent uses days by default.

This setting configures the agent to clear the spool files for failed jobs after the specified expiration time.

5. Save the file.
6. Do **one** of the following steps:  
**UNIX:** Enter the following command at the operating system prompt:

```
unisrvctr start waae_agent-WA_AGENT
```

The agent starts.

**Windows:**

- a. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
- b. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
- c. Right-click the agent service and click Start.  
The agent starts.

The agent is configured to clean the spool and job log files.

**NOTE**

- Spool and log files of jobs that completed successfully before the cleanup are not affected.
- For more information about the agent spool or job log files or the parameters in the [agentparm.txt](#) file, see [Maintain Spool and Log Files](#).

## Obtain the Job Log ID

The job log ID is used to track a job in the spool file.

To obtain the job log ID, follow these steps:

1. [Obtain the job run number and job ID](#).
2. [Obtain the job log ID](#).

## Obtain the Job Run Number and Job ID

You can use this procedure to obtain the job run number and job ID, which you require to locate the job log ID.

**Follow these steps:**

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -J job_name -d
```

A detailed report is generated. This report displays the job run number in the Run column.

2. Connect to the database, and run the following query:

```
select joid from ujo_job where job_name='job_name'
```

– **job\_name**

Defines the name of the job.

The job ID is displayed.

**NOTE**

- You can also obtain the job run number for command jobs by viewing the log returned by the autosyslog command.
- You can also obtain the job run number and the job ID by extracting the most recent CAUAJM\_I\_10082 message for the job name from the scheduler log file. The CAUAJM\_I\_10082 message is displayed as follows:



```
CAUAJM_I_10082 [machine_name connected
for job_namejob_ID.run_number.retry_number]
```

### Example: Obtain the Job Run Number and Job ID

This example obtains the job run number and job ID of the payload job.

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -J payload -d
```

A detailed report is generated. The payload job run number (50130) is displayed in the Run column, as follows:

| Job Name | Last Start          | Last End            | ST | Run/Ntry | Pri/Xit |
|----------|---------------------|---------------------|----|----------|---------|
| payload  | 07/16/2009 10:45:09 | 07/16/2009 10:45:09 | FA | 50130/1  | 20005   |

2. Connect to the database, and run the following query:

```
select joid from ujo_job where job_name='payload'
```

The payload job ID is displayed, as follows:

```
joid

172
```

### NOTE

In the scheduler log file, the CAUAJM\_I\_10082 message for the payload job is displayed as follows:

```
CAUAJM_I_10082 [mymachine connected for payload 172.50130.1]
```

### Obtain the Job Log ID

You must obtain the job log ID to track the job in the spool file.

To obtain the job log ID, connect to the database, and run the following query:

```
select run_info from ujo_extended_jobrun_info where joid=job_ID and run_num=run_num and
type=1 and seq_num=1
```

- **job\_ID**  
Defines the job ID.
- **run\_num**  
Defines the job's run number.

The job log ID is displayed. You can now use the job log ID to track the job.

### Example: Obtain the Job Log ID

This example obtains the job log ID of the payload job. Connect to the database and run the following query:

```
select run_info from ujo_extended_jobrun_info where joid=172 and run_num=50130 and
type=1 and seq_num=1
```

The job log ID of the payload job is displayed, as follows:

```
WobId(172.50130_1/WAAE_WF0.1/MAIN) JobLogId(9D7F247C63C2D21061CB83AB7AADDFFAD9563A10)
```

## Control the Services

AutoSys Workload Automation runs the following services:

- Agent
- Application server
- Scheduler
- Web server
- SOAP web server
- Web UI
- Web UI Collector

You can start, stop, restart, or pause and resume a AutoSys Workload Automation service. You can also verify the status of a AutoSys Workload Automation service.

During the installation, if the appropriate check boxes are selected on the corresponding installation panels, the services start automatically on system startup. After you installed AutoSys Workload Automation, you stop and start a service, restart a service, or pause and resume a service for the configuration changes to take effect.

## (UNIX) Control a Service

Use the `unisrvcntr` command to start, stop, restart, or verify the status of a service on UNIX.

### NOTE

- We do *not* recommend manually updating the `casrvc.auth` file located at `/opt/CA/SharedComponents/csutils/config/`, when you want to provide service management (such as start, stop, and restart) access to other users.
- If you have to manually update the `casrvc.auth` file, you must also manually make similar updates to the `assrvc.auth` file located at `/opt/CA/WorkloadAutomationAE/autosys/csutils/config`.

To control a service, log in as the root user and enter the following command at the operating system command prompt:

```
unisrvcntr start|stop|restart|status service_name
```

- **start**  
Starts a service.
- **stop**  
Stops a service.
- **restart**  
Restarts a service.
- **status**  
Verifies the status of a service.
- **service\_name**  
Specifies the name of the service. Specify the following values as appropriate based on the service you want to control:
  - `waee_agent-WA_AGENT` -- Controls the agent. `WA_AGENT` specifies the name of the agent.
  - `waee_server.instance_name` -- Controls the application server that is associated with the AutoSys Workload Automation instance.
  - `waee_sched.instance_name` -- Controls the scheduler that is associated with the AutoSys Workload Automation instance.
  - `waee_webserver.instance_name` -- Controls the web server that is associated with the AutoSys Workload Automation instance.

You can also use the `sendevent` command to stop the scheduler or application server as follows:

- (scheduler only) Using the `sendevent` command:

```
sendevent -E STOP_DEMON
```

The STOP\_DEMON event is sent to the database. The scheduler reads the STOP\_DEMON event, enters an orderly shutdown cycle by completing any processing it is performing, and stops.

#### NOTE

There might be a delay between when you send the STOP\_DEMON event and when the scheduler reads it and it shuts down. If the scheduler does not stop immediately, do not send another STOP\_DEMON event because the scheduler will process that event the next time it starts and promptly shuts down. To assign a high priority to the sendevent command, include the -P 1 option as follows:

```
sendevent -E STOP_DEMON -P 1
```

- Using the sendevent -E STOP\_DEMON -v option:

```
sendevent -E STOP_DEMON -v ROLE=value
```

#### – **ROLE=value**

Stops the application server or the scheduler according to the role or combination of roles specified. Specify the values as follows:

- **A**  
Stops the application server.
- **P**  
Stops the primary scheduler.
- **S**  
Stops the shadow scheduler.
- **T**  
Stops the tie-breaker scheduler.

#### NOTE

You can specify any combination of roles. For example, if you specify ROLE=PST, the primary, shadow, and tie-breaker scheduler are shut down.

#### NOTE

- You can also start the scheduler using the eventor command.
- The event server must be available, running, and properly identified before you can start the scheduler or application server.
- Jobs that are processing when you stop the agent continue to run, but the agent cannot track the process of those jobs.
- Stopping the scheduler does not affect jobs that are already running. They continue to run to completion, when their exit events are stored until the scheduler is restarted. When you stop the scheduler, jobs dependent on the events that are processed at the time of shutdown are scheduled but are not run until you start the scheduler.
- Do not attempt to stop the scheduler by terminating the process. This method stops the scheduler immediately, even if it is processing an event. Also, if you are using dual event servers and you terminate the process in any way other than issuing the sendevent command, the databases can lose synchronization.

## (Windows) Control a Service

Use the Windows Services dialog or the Administrator utility to start, stop, or verify the status of a service on Windows. To restart a service, use the Windows Services dialog.

#### Follow these steps:

1. Log in as a user with Windows Administrators group privileges.

#### NOTE

If you are not an administrator, launch the Administrator utility using the Run as administrator option.

2. Click Start, Programs, AutoSys, Administrator.
3. Select an instance from the Instance drop-down list in the Settings pane and click Apply.
4. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance. The Status column indicates the status.

**NOTE**

If you installed multiple agents, the status of all the agents is displayed on the Services - AutoSys Workload Automation Administrator window.

5. Right-click the service and select the Start or Stop option.  
The service starts or stops based on the option you selected.

**NOTE**

You can also start or stop a service by selecting the service and clicking the Start Service or Stop Service icon on the toolbar.

**NOTE**

- The event server must be available, running, and properly identified before you can start the scheduler or application server.
- Jobs that are processing when you stop the agent continue to run, but the agent cannot track the process of those jobs.
- Stopping the scheduler does not affect jobs that are already running. They continue to run to completion, when their exit events are stored until the scheduler is restarted. When you stop the scheduler, jobs dependent on the events that are processed at the time of shutdown are scheduled but are not run until you start the scheduler.
- Do not attempt to stop the scheduler by terminating the process. This method stops the scheduler immediately, even if it is processing an event. Also, if you are using dual event servers and you terminate the process in any way other than issuing the sendevent command, the databases can lose synchronization.

## Pause and Resume a Service

You can pause and resume the scheduler or the application server to read the modified values of the following parameters in the configuration file at runtime:

- CrossPlatformScheduling
- GlobalPendMachDelay
- GlobalPendMachStatus
- GlobalPendMachInterval
- AgentStatusPollInterval
- RepeatAlarmNotification
- AggregatorStatistics
- EnableIPCaching
- ISDBGACTIV
- LOGROLLOVER
- NotifyMethod
- NotifySMTPHost
- ServiceDeskUser
- ServiceDeskCust
- ServiceDeskURL
- DCAUser
- DCAURL
- UseSMTPAuthentication
- NotifySMTPUser
- NotifySMTPFromAddress
- SetJobAttributeEnvironmentals

**(UNIX) Follow these steps:**

1. Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
2. Enter the following commands at the operating system prompt:

```
unisrvcntr status waae_sched.$AUTOSERV
unisrvcntr status waae_server.$AUTOSERV
```

The scheduler and application server process IDs are displayed as follows:

| CA Services Status Report |       |         |
|---------------------------|-------|---------|
| Component Name            | Pid   | Status  |
| -----                     | ----- | -----   |
| WAAE Scheduler (ACE)      | 32220 | running |

| CA Services Status Report     |       |         |
|-------------------------------|-------|---------|
| Component Name                | Pid   | Status  |
| -----                         | ----- | -----   |
| WAAE Application Server (ACE) | 33330 | running |

3. Enter the following command:

```
kill -HUP {scheduler_pid|applicationserver_pid}
```

- ***scheduler\_pid***  
Defines the process ID of the scheduler that you want to pause and resume.
- ***applicationserver\_pid***  
Defines the process ID of the application server that you want to pause and resume.

The scheduler or application server pause and resume.

**(Windows) Follow these steps:**

1. Log in as a user with Windows Administrators group privileges.

**NOTE**

If you are not an administrator, launch the Administrator utility using the *Run as administrator* option.

2. Click Start, Programs, AutoSys, Administrator.
3. Select an instance from the Instance drop-down list in the Settings pane and click Apply.
4. Click the Services icon on the toolbar.
5. Right-click the service, and click Pause.  
The selected service pauses. The Status column indicates the status.
6. Right-click the service, and click Resume.  
The selected service resumes.

## Retrieve Aggregated Job, Alarm, and Scheduler Statistics

As an administrator, you can retrieve aggregated job, alarm, and scheduler statistics from the AutoSys Workload Automation database and can display them in a report format.

You can use the statistical data to determine, for example, how many of your jobs are ending in a failure status each month, alarms that get acknowledged or closed on average in an hour, the average daily latency in event processing, and so on.

When you aggregate statistics for the first time, depending on the hardware the product is installed on, the current active workload, and the total number of events that are stored in the database, the aggregation process might take a considerable amount of time. We recommend that you schedule a manual aggregation process and then configure the scheduler to aggregate statistics automatically at a specified interval. When you configure the scheduler to aggregate statistics automatically after the manual aggregation process is complete, the scheduler aggregates new statistics for hourly aggregation from the point where the manual aggregation process was completed. The scheduler uses the existing aggregated statistics for the daily, weekly, and monthly aggregation.

To retrieve the aggregated statistics, follow these steps:

1. [Aggregate statistics manually.](#)  
Perform a one-time aggregation of the data manually.

**NOTE**

Schedule the manual aggregation process at a time when the system activity is minimal.

2. [Configure AutoSys Workload Automation to aggregate statistics automatically.](#)  
Set up AutoSys Workload Automation to do a regular aggregation (hourly, daily, weekly, or monthly).
3. [Verify the resulting statistical data.](#)  
Generate the reports and verify the output.

### Aggregation Considerations

Consider the following points when you aggregate statistics:

- The scheduler does not aggregate statistics for the current month, week, day, or hour. For example, if the oldest event in the `ujo_proc_event` table is 09/27/2011 00:05:10 and you schedule the aggregation process on 10/11/2011 at 18:30:00, the statistics are aggregated as follows:

- Hourly statistics -- From 09/27/2011 00:00:00 through 10/11/2011 23:59:59
- Daily statistics -- From 09/27/2011 00:00:00 through 10/10/2011 23:59:59 (14 days)
- Weekly statistics -- From 09/25/2011 00:00:00 through 10/08/2011 23:59:59 (two weeks)
- Monthly statistics -- From 09/01/2011 00:00:00 through 09/30/2011 23:59:59 (one month)
- When the aggregation is scheduled, starting and completed messages are written in the scheduler log file. Detailed messages are written in the aggregation log file (*aggregator.instance*) located in the \$AUTOUSER/out (on UNIX) or %AUTOUSER%\out (on Windows) directory. The aggregation log file has the same properties as the scheduler log file and it rolls over based on the LOGROLLOVER parameter value (on UNIX) or LOGROLLOVER environment variable setting (on Windows).
- In all cases, the scheduler aggregates statistics in smaller intervals before starting the specified aggregation. For example, if you schedule a monthly aggregation process, the scheduler aggregates the hourly, daily, and weekly statistics before starting the monthly aggregation.
- You can aggregate statistics manually by sending an aggregation event using the sendevent command. Do not set the AggregateStatistics parameter in the configuration file (on UNIX) or select the Aggregate Statistics check box in the AutoSys Workload Automation Administrator utility (on Windows) to aggregate statistics manually.
- If the scheduler is shut down while the aggregation process is active, the aggregation process is terminated. A message is displayed to indicate this.

### Aggregate Statistics Manually

You can aggregate the job, alarm, and scheduler statistics manually by sending an AGGREGATE event using the sendevent command. The job, alarm, and scheduler statistics are aggregated into the ujo\_rep\_hourly, ujo\_rep\_daily, ujo\_rep\_weekly, and ujo\_rep\_monthly database tables. You can display these aggregated statistics in a report format in AutoSys Workload Automation. Other applications, such as Web UI, can also retrieve these aggregated statistics to generate predefined and custom reports.

To aggregate statistics manually, run the following command in a AutoSys Workload Automation Windows instance command prompt (on Windows) or in a shell where AutoSys Workload Automation environment is sourced (on UNIX):

```
sendevent -E AGGREGATE -l option
```

- **-l *option***  
Specifies the type of statistics to aggregate, where *option* is as follows:
- **HOURLY**  
Aggregates hourly statistics.
- **DAILY**  
Aggregates daily statistics.
- **WEEKLY**  
Aggregates weekly statistics.
- **MONTHLY**  
Aggregates monthly statistics.

#### NOTE

- The HOURLY, DAILY, WEEKLY, and MONTHLY options are mutually exclusive. Specify *one* of these options.
- You can also aggregate automatically by setting the AggregateStatistics parameter to 1 (on UNIX) or selecting the Aggregate Statistics check box (on Windows).

### Example: Aggregate Hourly Statistics

This example sends a manual aggregation event to aggregate hourly statistics.

```
sendevent -E AGGREGATE -l HOURLY
```

### Example: Aggregate Weekly Statistics

This example sends a manual aggregation event to aggregate weekly statistics.

```
sendevent -E AGGREGATE -l WEEKLY
```

### **Configure AutoSys Workload Automation to Aggregate Statistics Automatically**

You can configure AutoSys Workload Automation to aggregate the job, alarm, and scheduler statistics automatically at a specified interval. The job, alarm, and scheduler statistics are aggregated into the `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_weekly`, and `ujo_rep_monthly` database tables. You can display these aggregated statistics in a report format in AutoSys Workload Automation. Other applications, such as Web UI, can also retrieve these aggregated statistics to generate predefined and custom reports.

The scheduler automatically aggregates the job, alarm, and scheduler statistics as follows:

- Hourly -- At the start of every hour
- Daily -- At midnight every day
- Weekly -- At midnight on the first day of every week, which is taken as Sunday
- Monthly -- At midnight on the first day of every month

#### **(UNIX) Follow these steps:**

1. Use the shell script and source AutoSys Workload Automation environment.
2. Enter the following command at the operating system prompt:

```
unisrvcntr status waae_sched.$AUTOSERV
```

The scheduler process ID is displayed as follows:

| CA Services Status Report |       |         |
|---------------------------|-------|---------|
| Component Name            | Pid   | Status  |
| -----                     | ----- | -----   |
| WAAE Scheduler (ACE)      | 32220 | running |

3. Set `AggregateStatistics` to 1 in the configuration file and save the file.

#### **NOTE**

By default, the `AggregateStatistics` parameter value is set to 1 and so AutoSys Workload Automation aggregates statistics automatically.

4. Enter the following command at the operating system prompt:

```
kill - HUP scheduler_pid
```

#### **– scheduler\_pid**

Defines the process ID of the scheduler to pause and resume.

The scheduler resumes. The statistics are aggregated automatically.

#### **(Windows) Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.
2. Select an instance from the Instance drop-down list in the Settings pane.
3. Click the Scheduler icon on the toolbar.
4. Click the Options tab, select the Aggregate Statistics check box in the Settings pane, and click Apply.

#### **NOTE**

By default, the Aggregate Statistics check box is selected and so AutoSys Workload Automation aggregates statistics automatically.

5. Click the Services icon on the toolbar.
6. Right-click the scheduler service, and click Pause.
7. Right-click the scheduler service, and click Resume.



The scheduler resumes. The statistics are aggregated automatically.

### **Verify the Resulting Statistical Data**

You can generate a report of each database table (ujo\_rep\_hourly, ujo\_rep\_daily, ujo\_rep\_weekly, and ujo\_rep\_monthly) and can verify that the resulting statistics are aggregated correctly using the autoaggr command.

The autoaggr command is a client component utility that generates reports that are based on the aggregated job, alarm, and scheduler statistics that are retrieved from the database.

To generate a report, log in to AutoSys Workload Automation and enter *one* of the following commands at the UNIX operating system prompt or the Windows instance command prompt:

- For daily statistics:

```
autoaggr [-d]
 [-A] [-J] [-S]
 [-o filename]
 [-F from_date_time]
 [-T to_date_time]
 [-c] [-x] [?]
```

- For hourly statistics:

```
autoaggr [-h]
 [-A] [-J] [-S]
 [-o filename]
 [-F from_date_time]
 [-T to_date_time]
 [-c] [-x] [?]
```

- For weekly statistics:

```
autoaggr [-w]
 [-A] [-J] [-S]
 [-F from_date_time]
 [-T to_date_time]
 [-o filename]
 [-c] [-x] [?]
```

- For monthly statistics:

```
autoaggr [-m]
 [-A] [-J] [-S]
 [-o filename]
 [-F from_date_time]
 [-T to_date_time]
 [-c] [-x] [?]
```

- **-d**  
(Optional) Generates a report on daily aggregated statistics.
- **-h**  
(Optional) Generates a report on hourly aggregated statistics.
- **-m**  
(Optional) Generates a report on monthly aggregated statistics.
- **-w**  
(Optional) Generates a report on weekly aggregated statistics.
- **-A**

- (Optional) Generates a report only on alarm statistics.
- **-J**  
(Optional) Generates a report only on job statistics.
- **-S**  
(Optional) Generates a report only on scheduler statistics.
- **-o "filename"**  
(Optional) Defines the path and file name where you want to direct all standard output.

**NOTE**

- You can specify only the file name in this option. If you specify only the file name, the file is written to the current directory.
  - For readability purposes, we recommend that you direct the output to a file by using either the `-o` parameter or the command shell redirection command (`>` or `>>`).
  - Enclose the file name in double quotation marks.
- **-F "from\_date\_time"**  
(Optional) Specifies the aggregation start date and time in `"MM/dd/yyyy hh: mm"` format. The report displays the aggregated statistics starting from this specified date and time.

**NOTE**

- If you do not specify the `-F from_date_time` parameter, it defaults to the oldest processed event in the database.
  - Enclose the start date and time in double quotation marks.
- **-T "to\_date\_time"**  
(Optional) Specifies the aggregation end date and time in `"MM/dd/yyyy hh: mm"` format. The report displays the aggregated statistics up to this specified date and time.

**NOTE**

- If you do not specify the `-T to_date_time` parameter, it defaults to the current time.
  - Enclose the end date and time in double quotation marks.
- **-c**  
(Optional) Generates the output in a comma-separated format.
  - **-x**  
(Optional) Returns the autoaggr version number to standard output.
  - **-?**  
(Optional) Displays help for the command.

**NOTE**

- The **-d**, **-h**, **-m**, and **-w** options are mutually exclusive. Specify *one* of these options.
- The **-A**, **-J**, and **-S** options can be specified in any combination. If you do not specify the **-A**, **-J**, or **-S** options, the report includes job, alarm, and scheduler statistics.
- Verify that you issue the autoaggr command after the time range (specified using the **-F** and **-T** parameters) for which you want to generate a report has passed. For example, to generate a report on aggregated hourly statistics from 10/13/2011 16:00 to 10/13/2011 16:59:59, issue the autoaggr command after 17:00 on 10/13/2011.
- You can run these reports at any later time to view this output regularly.

**Example: Generate a Report for Hourly Statistics on Windows**

This example generates a report that displays the hourly job, alarm, and scheduler statistics from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the `c:\all.out` file.

```
autoaggr -h -o "c:\all.out" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

### Example: Generate a Report for Hourly Job Statistics in Comma-separated Format on UNIX

This example generates a report that displays the hourly job statistics in a comma-separated format from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the /tmp/job.csv file.

```
autoaggr -h -c -J -o "/tmp/job.csv" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

### Example: Generate a Report for Daily Alarm Statistics on Windows

This example generates a report that displays the daily alarm statistics from 09/26/2011 00:00 to 09/29/2011 00:00. The output is directed to the c:\alarm.out file.

```
autoaggr -d -A -o "c:\alarm.out" -F "09/26/2011 00:00" -T "09/29/2011 00:00"
```

## Statistics Reported by the autoaggr Command

The *autoaggr* command generates reports on the aggregated job, alarm, and scheduler statistics.

### NOTE

The time column in the report represents the hourly time after the reported time. For example, if the reported time is 10:00:00, it represents the aggregated statistics that are generated from the activity that occurred from 10:00:00 to 10:59:59.

The reports display job statistics that show the total number of jobs in the following statuses:

- STARTING
- RUNNING
- SUCCESS
- BYPASS
- FAILURE
- TERMINATED
- ON\_HOLD
- OFF\_HOLD
- ON\_ICE
- OFF\_ICE
- ON\_NOEXEC
- OFF\_NOEXEC
- INACTIVE
- RESTART
- SUSPENDED

The reports display the following alarm statistics:

- **AVERAGE RESPONSE TIME**  
Displays the average time that is taken to respond to an alarm.
- **TOTAL ALARMS**  
Displays the total number of alarms irrespective of alarm status.

### NOTE

The statistics that are displayed in the report includes the count of all the alarms that AutoSys Workload Automation generates.

- **UNANSWERED**  
Displays the total number of alarms that are open. This does not include alarms that are acknowledged or closed.
- **JOB FAILURE**

Displays the total number of JOBFAILURE alarms due to jobs that are in FAILURE or TERMINATED state.

- **START FAILURE**  
Displays the total number of STARTJOBFAIL alarms.
- **MAX RETRY**  
Displays the total number of MAX\_RETRY alarms.
- **MAX RUNTIME**  
Displays the total number of MAXRUNALARM alarms.
- **MIN RUNTIME**  
Displays the total number of MINRUNALARM alarms.
- **DATABASE ROLLOVER**  
Displays the total number of DB\_ROLLOVER alarms.
- **SCHEDULER FAILOVER**  
Displays the total number of EP\_ROLLOVER alarms.
- **SCHEDULER SHUTDOWN**  
Displays the total number of EP\_SHUTDOWN alarms.
- **SUSPENDJOB FAILURE**  
Displays the total number of SUSPENDJOBFAIL alarms.
- **RESUMEJOB FAILURE**  
Displays the total number of RESUMEJOBFAIL alarms.

The reports display the following scheduler statistics:

- **JOB RUNS**  
Displays the total number of job runs.
- **JOB FAILURES**  
Displays the total number of jobs that are in FAILURE or TERMINATED status.
- **JOB FORCE START**  
Displays the total number of jobs that were force started.
- **KILL JOB**  
Displays the total number of jobs that are killed using the KILLJOB event.
- **SERVICE DESK REQUESTS**  
Displays the total number of jobs for which service desk issues were opened.
- **TOTAL EVENTS**  
Displays the total number of events (including internal events) that AutoSys Workload Automation generates.
- **AVERAGE LATENCY**  
Displays the average latency (in seconds) in processing events, which is the average processing time that is required for all the events. The latency for an event is calculated by subtracting the time that the event was inserted into the database from the time it was picked up and processed by the scheduler. The average latency value is calculated by adding all the event latencies and then dividing the total latency by the number of events.
- **MAX LATENCY**  
Displays the highest latency (in seconds) for processed events within a given time range. The latency for an event is calculated by subtracting the time that the event was inserted into the database from the time it was picked up and processed by the scheduler.
- **AVERAGE LAG TIME**  
Displays the average lag time (in seconds) in processing events, which is the average processing time that is required for all the events. The lag time for an event is calculated by subtracting the time that the internal processing starts for the event from the time the scheduler completes processing it. Unlike average latency, the average lag time does not include the time that is required to fetch the event from the database. The average lag time value is calculated by adding all the event lag times and then dividing the total lag time by the number of events.
- **MAX LAG TIME**  
Displays the highest lag time (in seconds) for processed events within a given time range. The lag time for an event is calculated by subtracting the time that the internal processing starts for the event from the time the scheduler

completes processing it. Unlike max latency, the max lag time does not include the time that is required to fetch the event from the database.

## Delete Aggregated Statistics

You can delete the aggregated job, alarm, and scheduler statistics in the following situations:

- When you want the aggregation process to regenerate all statistics.
- When you no longer want to store the aggregated statistics in the database.

When you delete aggregated statistics, the statistics in the `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_weekly`, and `ujo_rep_monthly` tables are deleted.

### Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation. The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*). The instance command prompt appears.

2. Enter the following command:

```
sendevent - E AGGREGATE [-l {HOURLY|DAILY|WEEKLY|MONTHLY} -d
```

### Example: Delete All Statistics from the Database

This example sends a manual aggregation event to delete all statistics from the database.

```
sendevent - E AGGREGATE - d
```

### Example: Delete Statistics Before Aggregating Hourly Statistics

This example sends a manual aggregation event to delete the statistics from the `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_weekly`, and `ujo_rep_monthly` tables and then aggregates the hourly statistics.

```
sendevent - E AGGREGATE - l HOURLY -d
```

## Define Tuning Parameters for the Scheduler

AutoSys Workload Automation supports scalability. If run on high-end computers, you can tune the scheduler to make efficient use of the CPU, memory, and database connections of the computer to increase the overall productivity.

You can define the `SCHED_SCALE` and `DB_CONNECTIONS` parameters to tune the scheduler. On UNIX, the `SCHED_SCALE` and `DB_CONNECTIONS` parameters are operating system environment variables. Depending on your UNIX operating system, you can use either the `setenv` or the `export` command to define these tuning parameters. On Windows, the `SCHED_SCALE` and `DB_CONNECTIONS` parameters are registry keys.

### (UNIX) Follow these steps:

1. Log in to AutoSys Workload Automation as the EXEC superuser and enter the following command at the operating system prompt:

```
sendevent -E STOP_DEMON
```

The scheduler completes any current processing and stops.

2. Enter the following commands at the operating system prompt based on your UNIX shell command interpreter:

```
setenv SCHED_SCALE=limit
setenv DB_CONNECTIONS=max_number
```

Or

```
export SCHED_SCALE=limit
export DB_CONNECTIONS=max_number
```

– ***limit***

Defines the maximum limit of process threads that can be started dynamically. This value does not represent the total number of process threads that are active. This value is a scale value that is used by the scheduler to calculate the maximum limit of process threads that can be started dynamically as the workload demands. Therefore, a SCHED\_SCALE value of 1 represents the lowest ceiling of additional dynamic threads that can be started to process job events (some arbitrary ceiling not necessarily equal to one). Increasing this value increases the ability of the scheduler to process job events.

**Default:** 5

**Limits:** 1-64

– ***max\_number***

Defines the maximum number of database connections that the scheduler can connect to. Increasing this value increases the ability of the scheduler to perform simultaneous database operations.

**Default:** 16

**Limits:** 1-128

3. Enter the following command at the operating system prompt:

```
eventor
```

The scheduler starts. The tuning parameters for the scheduler are defined. On startup, the scheduler reads and applies the tuning parameters and these parameter values persist throughout the life of the process.

**NOTE**

- You can also define the SCHED\_SCALE and DB\_CONNECTIONS parameters in any of the environment script files (*autosys.UNIX shell.computer name*) located in the \$AUTOUSER directory.
- The application server also shares the same DB\_CONNECTIONS tuning parameter. The DB\_CONNECTIONS parameter value is applied to both the scheduler and the application server on start up under the following conditions:
  - You set this value in the environment script files located in the \$AUTOUSER directory.
  - You start the scheduler and the application server on the same computer after sourcing the environment.

**Example: Set the Maximum Limit of Process Threads That the Scheduler Can Start Dynamically**

This example sets the value that is used in the algorithm to control the maximum number of process threads that the scheduler can start dynamically to seven.

```
setenv SCHED_SCALE 7
```

**Example: Set the Maximum Number of Database Connections That the Scheduler Can Connect To**

This example sets the maximum number of database connections that the scheduler can have to 22.

```
export DB_CONNECTIONS=22
```

**(Windows) Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select the instance to which you want to define the tuning parameters for the scheduler from the Instance drop-down list.
3. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

4. Right-click the scheduler service, and click Stop.

The scheduler stops.

5. Click the System icon on the toolbar.

The System - AutoSys Workload Automation Administrator window appears.

6. Enter the following variable in the Variable field and its value in the Value field, and click Set.

– **SCHED\_SCALE**

Defines the maximum limit of process threads that can be started dynamically. This value does not represent the total number of process threads that are active. This value is a scale value that is used by the scheduler to calculate the maximum limit of process threads that can be started dynamically as the workload demands. Therefore, a SCHED\_SCALE value of 1 represents the lowest ceiling of additional dynamic threads that can be started to process job events (some arbitrary ceiling not necessarily equal to one). Increasing this value increases the ability of the scheduler to process job events.

**Default:** 5

**Limits:** 1-64

The SCHED\_SCALE variable is listed in the Environment Variables pane.

7. Enter the following variable in the Variable field and its value in the Value field, and click Set.

– **DB\_CONNECTIONS**

Defines the maximum number of database connections that the scheduler can connect to. Increasing this value increases the ability of the scheduler to perform simultaneous database operations.

**Default:** 16

**Limits:** 1-128

The DB\_CONNECTIONS variable is listed in the Environment Variables pane.

8. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

9. Right-click the scheduler service, and click Start.

The scheduler starts. The tuning parameters for the scheduler are defined. On startup, the scheduler reads and applies the tuning parameters and these parameter values persist throughout the life of the process.

**NOTE**

The application server also shares the same DB\_CONNECTIONS tuning parameter. On start up, if the scheduler and application server are running on the same computer, the DB\_CONNECTIONS parameter value is applied to both the scheduler and the application server.

**Example: Set the Maximum Limit of Process Threads That the Scheduler Can Start Dynamically**

Suppose that you want to set the value used in the algorithm for controlling the maximum number of process threads that the scheduler can start dynamically to seven, enter SCHED\_SCALE in the Variable field and 7 in the Value field and click Set on the System - AutoSys Workload Automation Administrator window of the Administrator utility.

**Example: Set the Maximum Number of Database Connections That the Scheduler Can Connect To**

Suppose that you want to set the maximum number of database connections that the scheduler can have to 22, enter DB\_CONNECTIONS in the Variable field and 22 in the Value field and click Set on the System - AutoSys Workload Automation Administrator window of the Administrator utility.

## Define Tuning Parameters for the Application Server

You can define the DB\_CONNECTIONS parameter to tune the application server. On UNIX, the DB\_CONNECTIONS parameter is an operating system environment variable. Depending on your UNIX operating system, you can use either the setenv or the export command to define the DB\_CONNECTIONS parameter. On Windows, the DB\_CONNECTIONS parameter is a registry key.

**(UNIX) Follow these steps:**

1. Log in to AutoSys Workload Automation as the EXEC superuser and enter the following command at the operating system prompt:

```
unisrvcntr stop waae_server.$AUTOSERV
```

The application server stops.

2. Enter the following command at the operating system prompt based on your UNIX shell command interpreter:

```
setenv DB_CONNECTIONS=max_number
```

Or

```
export DB_CONNECTIONS=max_number
```

– **max\_number**

Defines the maximum number of database connections that the application server can connect to. Increasing this value increases the ability of the application server to process simultaneous AutoSys Workload Automation client and agent requests.

**Default:** 32

**Limit:** 1-128

3. Enter the following command at the operating system prompt:

```
unisrvcntr start waae_server.$AUTOSERV
```

The application server starts. The application server tuning parameter is defined. On startup, the application server reads and applies the DB\_CONNECTIONS parameter and this parameter value persists throughout the life of the process.

**NOTE**

- You can also define the DB\_CONNECTIONS tuning parameter in any of the environment script files (*autosys.UNIX shell.computer name*) located in the \$AUTOUSER directory.
- The scheduler also shares the same DB\_CONNECTIONS tuning parameter. The DB\_CONNECTIONS parameter value is applied to both the scheduler and the application server on start up under the following conditions:
  - You set this value in the environment script files located in the \$AUTOUSER directory.
  - You start the scheduler and the application server on the same computer after sourcing the environment.

**Example: Set the Maximum Number of Database Connections That the Application Server Can Connect To**

This example sets the maximum number of database connections that the application server can have to 42.

```
export DB_CONNECTIONS=42
```

**(Windows) Follow these steps:**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Select the instance to which you want to define the tuning parameter for the application server from the Instance drop-down list.
3. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
4. Right-click the application server service, and click Stop.  
The application server stops.
5. Click the System icon on the toolbar.  
The System - AutoSys Workload Automation Administrator window appears.
6. Enter the following variable in the Variable field and its value in the Value field, and click Set.
  - **DB\_CONNECTIONS**



Defines the maximum number of database connections that the application server can connect to. Increasing this value increases the ability of the application server to process simultaneous AutoSys Workload Automation client and agent requests.

**Default:** 32

**Limit:** 1-128

The DB\_CONNECTIONS variable is listed in the Environment Variables pane.

7. Click the Services icon on the toolbar.

The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.

8. Right-click the application server service, and click Start.

The application server starts. The tuning parameter for the application server is defined. On start up, the application server reads and applies the DB\_CONNECTIONS parameter and this parameter value persists throughout the life of the process.

#### NOTE

The scheduler also shares the same DB\_CONNECTIONS tuning parameter. On start up, if the scheduler and application server are running on the same computer, the DB\_CONNECTIONS parameter value is applied to both the scheduler and the application server.

### Example: Set the Maximum Number of Database Connections That the Application Server Can Connect To

Suppose that you want to set the maximum number of database connections that the application server can have to 42, enter DB\_CONNECTIONS in the Variable field and 42 in the Value field and click Set on the System - AutoSys Workload Automation Administrator window of the Administrator utility.

## General Debugging

### Contents

#### Trace Settings

The scheduler, application server, agent, client utilities, and communication and database infrastructure routines generate trace messages.

For the scheduler, application server, or the agent that generate their own log files, trace messages are added to the log files when the components encounter them.

For applications (such as jil, autorep, and sendevent client utilities) that are executed interactively or in batches, trace messages are written to the active window or to a file if streamed.

#### Configure the Client Utilities to Generate Run-time Traces

You can configure the client utilities to generate run-time traces.

To configure the client utilities to generate run-time traces, set the **ISDBGACTIV** value as follows:

- On UNIX, issue either the setenv or export command (depending on your UNIX operating system) at the operating system prompt.
- On Windows, issue the set command at the instance command prompt.

#### NOTE

- You must set the ISDBGACTIV value before initiating the client utilities.
- On startup, the traceable applications search for the specified ISDBGACTIV value and output the appropriate trace messages according to the value assigned.

### Example: Configure the Client Utilities to Generate Light Traces on UNIX

This example configures the client utilities to generate light trace information on UNIX.

```
export ISDBGACTIV=LIGHT
```

### Example: Configure the Client Utilities to Generate Light Traces on Windows

This example configures the client utilities to generate light trace information on Windows.

```
set ISDBGACTIV=LIGHT
```

### Configuring Agent Log File Properties

The agent keeps the log files that contain records of communication with AutoSys Workload Automation, as well as internal messages. By default, these files are located in the log directory and are updated continually while the agent is running. You can configure the agent log file properties by editing the log.level, log.archive, and log.maxsize parameters in the agentparm.txt file.

#### NOTE

For more information about the log.level, log.archive, and log.maxsize parameters and configuring agent log file properties, see [Workload Automation Agent Logs](#).

### Configuring the Scheduler and Application Server to Generate Run-time Traces

You can use the [ISDBGACTIV](#) parameter to configure the scheduler and application server to generate run-time traces.

## Performance Improvements in AutoSys Workload Automation Release 11.3.6 SP1

This article describes the latency improvements in AutoSys Workload Automation Release 11.3.6 SP1 when compared to Unicenter AutoSys JM r11 SP5.

### Test Details

The latency for an event is calculated by subtracting the time the event was inserted into the database from the time it was picked up and processed by the scheduler.

The same test is run on both r11 SP5 and Release 11.3.6 SP1. To get accurate results, the same hardware configuration including the operating systems and databases of same versions are used.

The details of the test are as follows:

- A sample job stream from a customer is used for this test; The job stream is modified as required and then run in the test environment.
- Around 100k jobs are run per day.
- Date/time conditions and calendars are used for jobs
- The Release 11.3.6 agents are used for testing in Release 11.3.6 SP1 ; r11 agents are used for testing in r11 SP5 testing.
- Jobs are scheduled to run on eight virtual machines. Each virtual machine has two physical machines in a specific combination. Only five physical machines are used in total. Different combinations of physical machines are used to create multiple virtual machines.

For example:

```
Virtual_machine1: Physical_mac1 & Physical_mac5
Virtual_machine2: Physical_mac2 & Physical_mac5
Virtual_machine3: Physical_mac1 & Physical_mac2
.....
```

- The jobs are run for 6 days and all performance metrics are captured.

### **Artifacts/Configuration**

This table lists the artifacts and configurations that are used for testing:

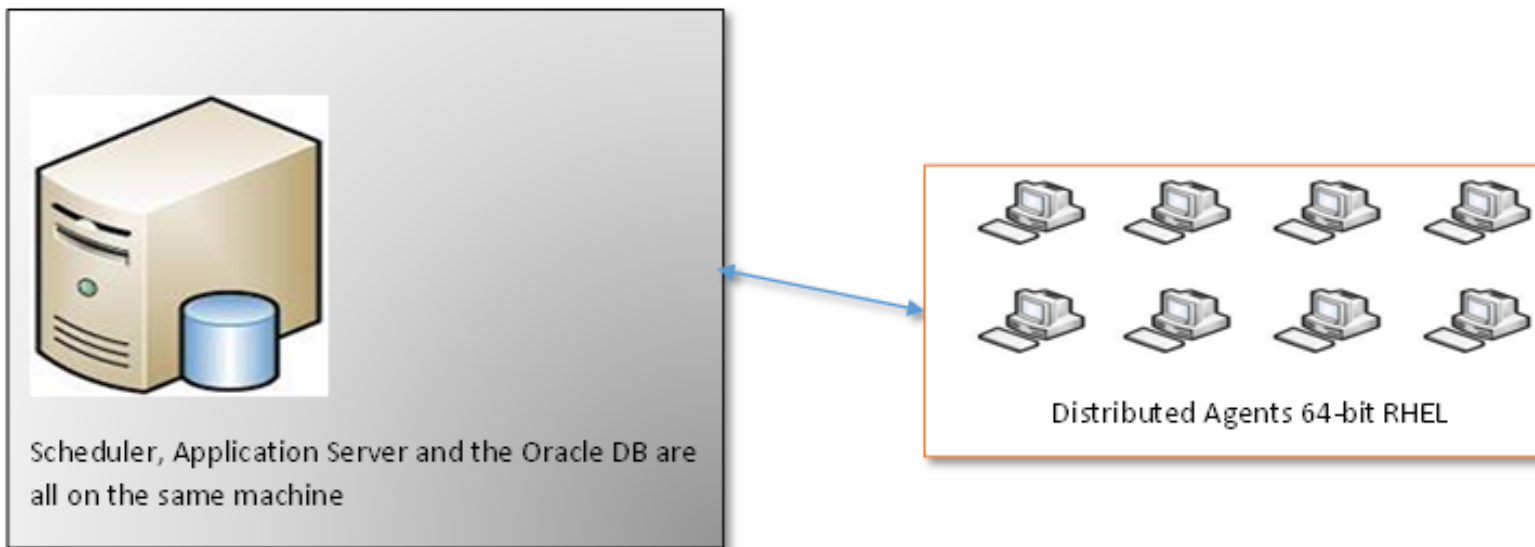
| <b>Artifacts/Configurations</b> |                                                              |
|---------------------------------|--------------------------------------------------------------|
| Number of Jobs in the database  | 45,000                                                       |
| Number of calendars             | 347                                                          |
| Number of agents or machines    | 5                                                            |
| Global variables                | 0                                                            |
| CA EEM enabled                  | Yes                                                          |
| Cross-platform scheduling       | No                                                           |
| Encryption                      | Default<br>r11 SP5: NONE<br>Release 11.3.6 SP1: Default /AES |

The SCHED\_SCALE and DB\_CONNECTIONS parameters are set to the default values as follows:

- SCHED\_SCALE: 5
- DB\_CONNECTIONS: 16

### **Test Systems Architecture**

The following diagram describes the test system architecture:



### **Performance Metrics Captured**

The following performance metrics are captured during the test:

- Number of jobs run per day
- Number of events processed per day
- Average latency per day
- Scheduler memory and CPU usage

## Hardware Specification

### Server Machine Information

| Hardware Components              | Hardware Specifications                                                                      |
|----------------------------------|----------------------------------------------------------------------------------------------|
| Machine Hardware Configuration   | Physical Machine : Intel(R) Xeon(R) CPU E5410 @ 2.33GHz<br>+ 8 GB RAM + 12 GB Virtual Memory |
| Operating System                 | RHEL 6.5 64-bit                                                                              |
| Database                         | Oracle 11g R2                                                                                |
| AutoSys Workload Automation mode | Single Server Mode                                                                           |

### Agent Machines Information

| Hardware Components             | Hardware Specifications                                                            |
|---------------------------------|------------------------------------------------------------------------------------|
| Operating System                | RHEL 6.1 64-bit                                                                    |
| Machine Hardware Configurations | 4 CPU and 4 GB RAM (4 Virtual machines)<br>8 CPU and 8 GB RAM (1 Physical machine) |

## Product Releases

| Release        | Server                 | Agent                |
|----------------|------------------------|----------------------|
| r11            | r11 SP5 INC3 + T42B290 | r11 SP5 Agent        |
| Release 11.3.6 | Release 11.3.6 SP1     | Release 11.3.6 Agent |

## Test Result

The test result summary is captured in separate tables for both r11 SP5 and Release 11.3.6 SP1. For detailed information about CPU usage and graphs, see the [Appendix](#).

### r11 SP5

| Date      | Number of Events | Number of Job Runs | Average Daily Latency (in Seconds) | Scheduler CPU Usage (in Percentage) | Schedule Memory Usage (in MB) |
|-----------|------------------|--------------------|------------------------------------|-------------------------------------|-------------------------------|
| 4/10/2015 | 324128           | 110728             | 245                                | 0.60                                | 50                            |
| 4/11/2015 | 183767           | 62243              | 18.34                              | 0.60                                | 50                            |
| 4/12/2015 | 171021           | 57898              | 19.95                              | 0.60                                | 50                            |
| 4/13/2015 | 280401           | 96196              | 71.17                              | 0.60                                | 50                            |
| 4/14/2015 | 289814           | 99412              | 69.08                              | 0.60                                | 50                            |
| 4/15/2015 | 290172           | 99542              | 70.56                              | 0.60                                | 50                            |

**Release 11.3.6 SP1**

| Date      | Number of Events | Number of Job Runs | Average Daily Latency (in Seconds) | Scheduler CPU Usage (in Percentage) | Schedule Memory Usage (in MB) |
|-----------|------------------|--------------------|------------------------------------|-------------------------------------|-------------------------------|
| 4/10/2015 | 319511           | 109529             | 0.9                                | 0.50                                | 35                            |
| 4/11/2015 | 188966           | 63982              | 0.4                                | 0.50                                | 35                            |
| 4/12/2015 | 179579           | 60749              | 0.39                               | 0.50                                | 35                            |
| 4/13/2015 | 308606           | 105734             | 0.924                              | 0.50                                | 35                            |
| 4/14/2015 | 317963           | 108901             | 0.98                               | 0.50                                | 35                            |
| 4/15/2015 | 318209           | 108978             | 0.97                               | 0.50                                | 35                            |

**Observations**

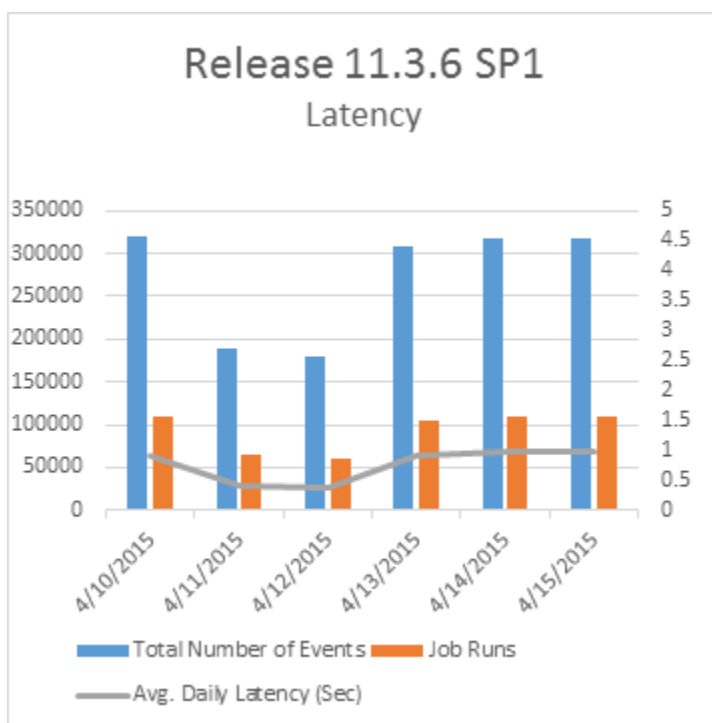
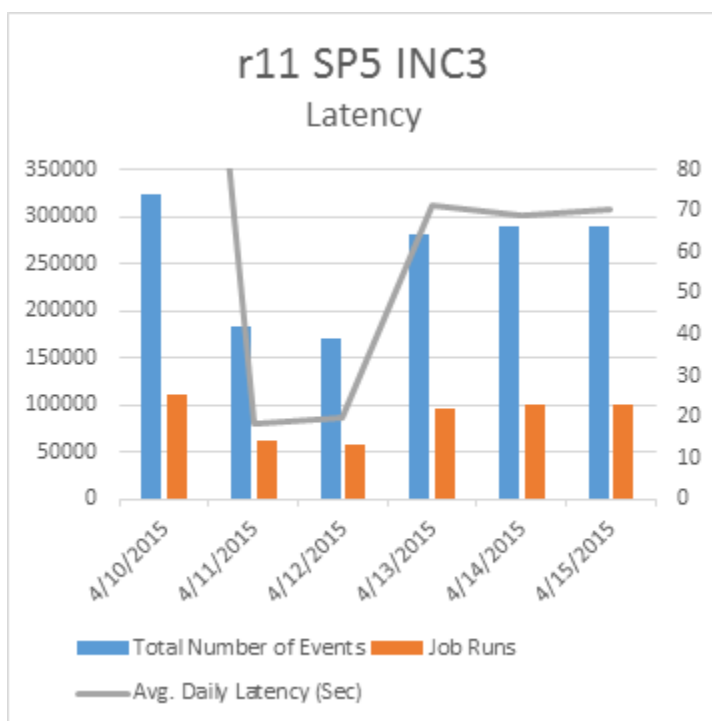
When the test results of both the releases are compared, Release 11.3.6 SP1 processed around eight to ten percent more jobs with 70-80 times lower latency at peak workload consistently. Also, a remarkable improvement is seen in the CPU and memory usages of the Release 11.3.6 SP1 scheduler.

**Recommendation**

We recommend that you migrate your AutoSys Workload Automation environment to Release 11.3.6 SP1 at the earliest so that you can use the benefits of the performance improvements done in Release 11.3.6 SP1.

**Appendix****Latency Charts:**

The following diagrams show the latency charts in r11 SP5 and Release 11.3.6 SP1:

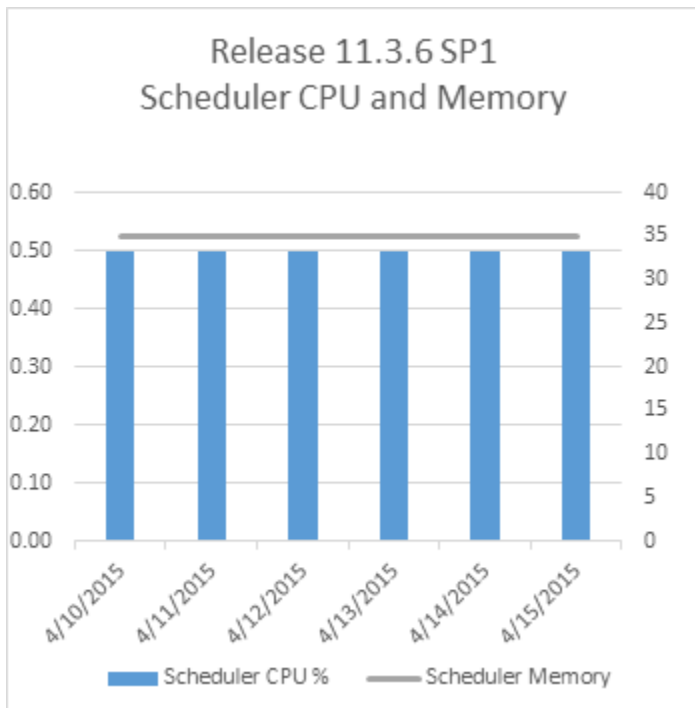
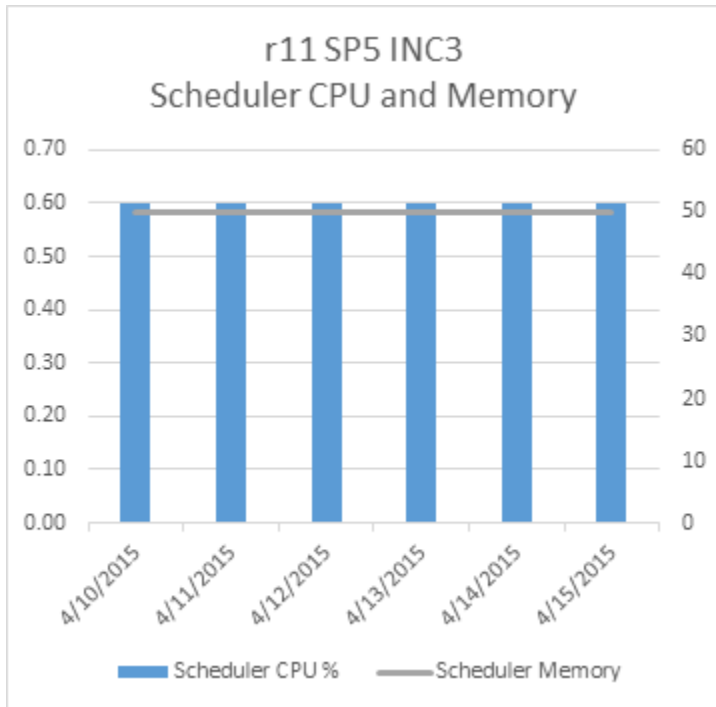


#### NOTE

The latency scale in the charts is different in r11 SP5 and Release 11.3.6 SP1.

**CPU and Memory Charts:**

The following diagrams show the CPU and memory charts of the scheduler in r11 SP5 and Release 11.3.6 SP1:

**Performance Improvement with Multi-threaded Listener From Release 11.3.6 SP3****Contents**

From AutoSys Workload Automation Release 11.3.6 SP3 onwards, the scheduler runs in multi-threaded listener mode where connections from multiple agents are processed simultaneously as opposed to the serial processing of statuses from agents earlier; thus providing increased throughput of jobs and decrease in event latency. Also, the number of autoping timeouts and false chase alarms are reduced.

By default, the scheduler runs in multi-threaded listener mode where multiple threads process connections from agents simultaneously. If you want the scheduler to run in single threaded listener mode, set the `USE_SINGLETHREADED_AGENTLISTENER` environment variable to any value.

When the scheduler runs in multi-threaded listener mode, by default, the number of listener threads is three if the `SCHED_SCALE` parameter is set to the default value of five. The formula used to calculate the number of threads is  $(\text{SCHED\_SCALE} + 1)/2$ . To increase the number of threads, set the `SCHED_SCALE` value accordingly.

The multi-threaded listener feature fixed the following problem:

Latency issues were seen in the scheduler as it processes incoming messages serially using a single thread that accepts a connection from the agent and processes the AFM; throttling the flow of communication from agents to the scheduler.

**Symptoms:** Timeouts, restarting of jobs, latency between STARTING and RUNNING events, scheduler being affected by a slower agent and so on are some of the problems that were observed due to serial processing of received messages.

### **Performance Metrics in Release 11.3.6 SP3**

The following performance metrics were calculated while testing the performance improvement from Release 11.3.6 SP2 and 11.3.6 SP3:

- **AVERAGE LAG TIME**

Indicates the average lag time (in seconds) in processing events, which is the average processing time that is required for all the events. The lag time for an event is calculated by subtracting the time the internal processing starts for the event from the time the scheduler completes processing it. Unlike average latency, the average lag time does not include the time that is required to fetch the event from the database. The average lag time value is calculated by adding all the event lag times and then dividing the total lag time by the number of events.

- **AVERAGE LATENCY OR HORIZONTAL LATENCY**

Indicates the average latency (in seconds) in processing events, which is the average processing time that is required for all the events. The latency for an event is calculated by subtracting the time the event was inserted into the database from the time it was picked up and processed by the scheduler. The average latency value is calculated by adding all the event latencies and then dividing the total latency by the number of events.

The autorep command displays the horizontal latency for the STARTING event as follows:

```
QA_4389 11/02/2015 23:53:11 11/02/2015 23:53:11 SU 1673/1 0

Status/[Event] Time Ntry ES ProcessTime Machine

STARTING 11/02/2015 23:53:10 1 PD 11/02/2015 23:53:10 machine_name
RUNNING 11/02/2015 23:53:11 1 PD 11/02/2015 23:53:11 machine_name
```

- **VERTICAL LATENCY**

Indicates the vertical latency (in seconds), which is the difference in time between the STARTING and RUNNING event.

The autorep command displays the vertical latency as follows:

```
QA_4389 11/02/2015 23:53:11 11/02/2015 23:53:11 SU 1673/1 0
Status/[Event] Time Ntry ES ProcessTime Machine
```



```

STARTING 11/02/2015 23:53:10 1 PD 11/02/2015 23:53:10 machine_name
RUNNING 11/02/2015 23:53:11 1 PD 11/02/2015 23:53:11 machine_name
<Executing at WA_AGENT>
SUCCESS 11/02/2015 23:53:11 1 PD 11/02/2015 23:53:11 machine_name

```

#### • **JOB LATENCY**

Indicates the job latency (in seconds), which is the difference in time between the STARTJOB event and the STARTING event processed.

The autorep command displays the job latency as follows:

```

[STARTJOB] 10/28/2015 14:56:00 0 UP
STARTING 10/28/2015 14:58:00 1 PD 10/28/2015 14:58:00 machine_name
<Executing at WA_AGENT>
SUCCESS 11/02/2015 23:53:11 1 PD 11/02/2015 23:53:11 machine_name

```

The following points were considered while testing the performance improvement from Release 11.3.6 SP2 and 11.3.6 SP3:

- 100K jobs were defined in the database running to 600 agents.
- The average latency or horizontal latency was captured for all STARTING, RUNNING, and SUCCESS events on an hourly and daily basis.
- For job latency, in case STARTJOB fails and the job retries, then only the difference between STARTJOB and the first STARTING event processed was considered.
- The average latency, horizontal latency, vertical latency, or job latency was captured on a daily basis.

The following metrics were captured with the SCHED\_SCALE and DB\_CONNECTIONS parameters set to the default value:

#### **Release 11.3.6 SP2 Single Server Unpatched Environment**

| Total Events<br>Aggregator | Job Runs<br>Aggregator | Aggregator<br>Average<br>Latency(in<br>seconds) | Aggregator<br>Average<br>Lag Time(in<br>seconds) | Vertical<br>Latency(in<br>seconds) | Job<br>Latency(in<br>seconds) | Scheduler<br>CPU Usage | Scheduler<br>Memory<br>Usage | Application<br>CPU Usage | Application<br>Memory<br>Usage |
|----------------------------|------------------------|-------------------------------------------------|--------------------------------------------------|------------------------------------|-------------------------------|------------------------|------------------------------|--------------------------|--------------------------------|
| 665921                     | 111826                 | 3.12                                            | 1.78                                             | 1366.24                            | 597.96                        | 0.4%                   | ~60MB                        | 0.3%                     | ~50MB                          |

#### **Release 11.3.6 SP3 Single Server Multi-threaded Listener Environment**

| Total Events<br>Aggregator | Job Runs<br>Aggregator | Aggregator<br>Average<br>Latency(in<br>seconds) | Aggregator<br>Average<br>Lag Time(in<br>seconds) | Vertical<br>Latency(in<br>seconds) | Job<br>Latency(in<br>seconds) | Scheduler<br>CPU Usage | Scheduler<br>Memory<br>Usage | Application<br>CPU Usage | Application<br>Memory<br>Usage |
|----------------------------|------------------------|-------------------------------------------------|--------------------------------------------------|------------------------------------|-------------------------------|------------------------|------------------------------|--------------------------|--------------------------------|
| 1157514                    | 172598                 | 6.47                                            | 2.32                                             | 299.65                             | 170.62                        | 0.4%                   | ~63MB                        | 0.2%                     | ~38MB                          |

#### **Conclusion:**

Although there is a slight increase in the horizontal latency, the overall throughput is increased and the vertical latency and job latency are decreased.

- Vertical latency is reduced by 21% (1366s to 300s).

- Job latency is reduced by 28% (597s to 170s).
- Horizontal latency is slightly increased (3s to 6.5s).
- Throughput (number of job runs per day) is increased by 35%.
- Total events is increased by 42%.

### Performance Metrics in Release 11.3.6 SP4

The following points were considered while testing the performance in Release 11.3.6 SP4 in single server mode:

- 100K jobs were defined in the database running to 600 agents.
- The average latency or horizontal latency was captured for all STARTING, RUNNING, and SUCCESS events on an hourly and daily basis.
- For job latency, in case STARTJOB fails and the job retries, then only the difference between STARTJOB and the first STARTING event processed was considered.
- The average latency, horizontal latency, vertical latency, or job latency was captured on a daily basis.

The following metrics were captured with the SCHED\_SCALE and DB\_CONNECTIONS parameters set to the default value:

| Total Events<br>Aggregator | Job Runs<br>Aggregator | Aggregator<br>Average<br>Latency(in<br>seconds) | Aggregator<br>Average<br>Lag Time(in<br>seconds) | Vertical<br>Latency(in<br>seconds) | Job<br>Latency(in<br>seconds) | Scheduler<br>CPU Usage | Scheduler<br>Memory<br>Usage | Application<br>CPU Usage | Application<br>Memory<br>Usage |
|----------------------------|------------------------|-------------------------------------------------|--------------------------------------------------|------------------------------------|-------------------------------|------------------------|------------------------------|--------------------------|--------------------------------|
| 1315152                    | 212147                 | 4.48                                            | 1.32                                             | 351.85                             | 136.39                        | 0.4%                   | ~70MB                        | 0.3%                     | ~55MB                          |

## Web UI Administration

This section contains the following topics:

### Administer Agent Inventory

Agent information is retrieved periodically from AutoSys Workload Automation and displayed in Agent Inventory. For systems with many machines, the interval should not be too short or an extra load is placed on the system. Longer intervals reduce the load, but the Agent List table containing the updated agents information and changed agents status may not immediately appear on the Agent Inventory display.

#### NOTE

Agent Inventory supports only the agents that are in AutoSys compatible mode.

You must specify the Web Services URL in the Application Server category under the AutoSys Workload Automation Server Properties for use by Agent Inventory to collect data.

#### NOTE

Add the monitoring user to the WorkloadAutomationAEWebService group as the monitoring user is used to connect to the web server.

Because Typical installation does not install web services, select Custom installation for the web server installation and specify the Web Services URL.

#### NOTE

For more information about the Web Services URL, see the [Server Properties and Application Preferences](#) section.

CA Workload Automation AE Release 11.3.6 SP7 and higher transfers only changed agent data to Agent Inventory. This Refresh Time(Delta) specifies the duration by which Agent Inventory must fetch the updated or modified data from the AutoSys Workload Automation server. The Refresh Time(Full) specifies the duration by which Agent Inventory must fetch the entire data and must refresh the Agent Inventory. CA Workload Automation AE Release 11.3.6 SP6 and previous releases always refresh the entire agent list.

Typically changes to the refresh rates should not be required. However they can be changed if necessary to tune resource consumption and are listed in the table:

| Name                | Default   | Min       | Max   | Description                                                                                                                                                  |
|---------------------|-----------|-----------|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Refresh Time(Delta) | 1 minute  | 5 seconds | 1 day | How often to initiate a delta transfer when available (collects data from Autosys Workload Automation Release 11.3.6 SP7 or higher)                          |
| Refresh Time(Full)  | 5 minutes | 1 minute  | 1 day | How often to initiate a full refresh API when delta transfer is not available (collects data from Autosys Workload Automation Release 11.3.6 SP6 or earlier) |

#### NOTE

For more information about Agent Inventory Preferences, see the [Preferences Section in the Configuration Application](#) page.

#### NOTE

##### More information:

- [Agent Inventory](#)

### Datasources Page

The Datasources page displays the AutoSys Workload Automation servers and their status.

The Datasources columns display the following information:

- **Server Name**  
Indicates the name of the AutoSys Workload Automation server.
- **Status**  
Indicates the status of the AutoSys Workload Automation server. The possible states are:
  - **Ok**  
Indicates that the scheduler is connected and providing agent data.
  - **Error**  
Indicates that the scheduler is either not connected or not providing agent data.
  - **Disabled**  
Indicates that the Agent Inventory connection to the scheduler is disabled. Agent Inventory does not connect to a disabled scheduler or gather agent data. A scheduler connection is disabled if the "Web Services URL" is blank in the AutoSys Workload Automation Server Configuration in Web UI.
- **Status Details**

Indicates a description of the condition causing an error.

- **Actions**

Indicates some information about how to resolve the error or disabled status.

- **Last Update**

Indicates one of the following times, which is based on the Status:

- When Status is OK, this time is the last time agents were synchronized from AutoSys Workload Automation web server to Agent Inventory.
- When Status is Error, this time is the last time Agent inventory attempted to connect to the AutoSys Workload Automation web server or the Agent Inventory attempted to synchronize agents.

## **Security**

Users must have agent inventory access to retrieve information from agents and perform the required actions. To configure access, log in to EEM, click ApplicationAccess under Access Policies.

### **Manage Access Policies in CA Embedded Entitlements Manager**

Web UI security is implemented through CA Embedded Entitlements Manager (EEM). You can manage access policies in EEM for various users and groups, and grant permissions to perform various actions on the Agent Inventory.

#### **NOTE**

For more information about access policies for Web UI in CA EEM, see the section [Set Up and Verify CA EEM Security Policies](#).

## **AgentAccess**

The AgentAccess policy provides permissions to the users and groups to perform various tasks in the Agent List tab against the machine name, such as Download Logs, Status Details, Stop or Restart Agents.

#### **NOTE**

If you try to perform any of the following actions for which you do not have access permissions, an error message displays.

To provide access to users and groups through **AgentAccess**, follow these steps:

1. Log in to EEM.
2. Click the Manage Access Policies tab.
3. Select AgentAccess under Access Policies.
4. Select the Policy from the Policy Table. For example, select AgentAccessDefault.
5. Select the user, click Search Identities to search the identity.
6. Select the identity and actions as required, and resources as appropriate.  
By default, the Identities Administrator has View, Fetch, and Upgrade actions and Commander has View, Fetch, Configure, and Control and Upgrade actions selected. You can view the following actions in the Web UI interface and their corresponding permissions as follows:
  - a. View
    - Plugin Details
    - Status Details
  - b. Fetch
    - Download Logs
    - Download AgentParams
    - View Agent Details
  - c. Configure
    - Update Agent Parameters

- d. Control
    - Stop Agent
    - Restart Agent
    - Reset Action Status
  - e. Upgrade
    - Stage Agent
    - Upgrade Agent
    - Stage/Upgrade Status Details
    - Cancel Scheduled Upgrade
    - Rollback Upgrade
  - f. Control or Upgrade
    - Quiesce Agents
    - Unquiesce Agents
7. Customize the permissions for identities against actions that you want them to perform.
  8. Click Save to complete the access permissions.

**NOTE**

If you select more than one machine name in the Agent List and select any action, Agent Inventory performs the selected action on the machine for which you have permissions and displays an error message for the machines that you do not have permissions.

**ApplicationAccess**

The ApplicationAccess policy provides permissions to the users and groups access to the Agent Inventory tab in the Web UI.

To provide access to the Agent Inventory application using **ApplicationAccess**, follow these steps:

1. Log in to EEM.
2. Click the Manage Access Policies tab.
3. Select ApplicationAccess under Access Policies.
4. Select the Policy from the Policy Table. For example, select ApplicationAccessDefault.
5. Select the user, click Search Identities to search the identity, and add as required.
6. Select the identity and AgentInventory action as required, and resources as appropriate.
7. Click Save to complete the access permissions to the Agent Inventory tab.

**ServerAccess**

The ServerAccess policy provides permissions to the users and groups access the servers. You can access machines that are available on those servers.

To provide access to the application using **ServerAccess**, follow these steps:

1. Log in to EEM.
2. Click the Manage Access Policies tab.
3. Select ServerAccess under Access Policies.
4. Select the Policy from the Policy Table. For example, select ServerAccessDefault.
5. Select the user, click Search Identities to search the identity and add as required.
6. Customize the permissions for the resources against actions that you want them to perform.
7. Click Save to complete the access permissions.

## Administer the Configuration Batch Interface

The Configuration Batch Interface lets you export and import these configurations, encrypt command line arguments, run encrypted commands, and change the CA EEM server definition. You can run the Configuration Batch Interface from a Windows command line or UNIX console as an alternative to using the Configuration application.

You can use the Configuration Batch Interface to do the following procedures:

- Export configurations to a file. After exporting the configurations, you can do the following procedures using a text editor:
  - Add multiple objects to an exported configuration file using the copy and paste feature of your text editor. For example, if you want to add several AutoSys Workload Automation servers, you can create one server definition in Configuration. Next, export the appropriate configurations, open the text file, and copy the server properties for it. Paste an instance of them for each new server you want to add, and modify the server names and other properties as necessary.
  - Modify or delete multiple objects in one export operation. For example, you can export the appropriate configurations, and can change all non-encrypted passwords.
- Import modified configurations back to the same server or to another Web UI server.

### NOTE

The Configuration Batch Interface utility provides access to the same server and application properties that are available within Configuration.

- Propagate the configurations from one Web UI server to other Web UI servers. For example, you can set up all the Web UI servers in your environment to run with the same configurations by exporting the configurations on the server that you want the other servers to run with, copying the configurations to the other servers, and then importing the configurations on those servers.
- Archive configurations by exporting the current configurations and saving the resulting configuration file. The configurations can be imported if you need to run Web UI with previous configurations.
- Create a file that contains encrypted `wcc_config` arguments. The encrypted file can be distributed, and a job or script can be created so that another user with lower security access can run the `wcc_config` command with the encrypted file at a convenient time. For example, you can create a file to add a new or modified AutoSys Workload Automation configuration on a server.
- Display the CA EEM server configuration and update the CA EEM Server definition in Web UI.

### NOTE

You cannot update the CA EEM Server definition in Web UI using Configuration.

- Display the Web UI or Reporting database server information.
- Specify a different database for Web UI or for Reporting.
- Change the Web UI or Reporting database credentials.
- Change the event bus port.
- Change the SSL mode to SSL or non-SSL.
- Change the Tomcat server port.
- Change the Tomcat stop port.

The Configuration Batch Interface Help contains procedures and examples that explain the `wcc_config` command options. Many options for the `wcc_config` command can be specified using either the short or long form of the option. The syntax for all short forms uses one preceding hyphen, while the syntax for all long forms uses two. For example, the export option can be specified using either of the following formats:

- Short form: `wcc_config -x`
- Long form: `wcc_config --export`

**NOTE**

- The Configuration Batch Interface is a batch file (.bat) on Windows and a shell script (.sh) on UNIX.
- To use this batch interface, the user that is specified in the command must have rights to the associated application in the CA EEM ApplicationAccess policy.
- On UNIX, if Web UI was installed by the root user, only the root user has permission to run this batch interface.
- On UNIX, if Web UI was installed with the Run As User option, the run as user has permission to run this batch interface. In that case, the root user should not run this batch interface because the run as user cannot then run it successfully. However, if the root user does run the batch interface, change the group and owner of the file and folder in \$CA\_WCC\_INSTALL\_LOCATION/log to the run as user.
- Some options require a list of items. When listing items, they should be separated by a comma only; a space should not follow the comma.
- Batch Interface command arguments are case-sensitive. You must use the case that is shown in this help for each argument.
- The Configuration Batch Interface lets you configure the subset of the enterprise configuration that you can also configure using the Configuration. The Configuration Batch Interface does not let you configure any other aspects of Web UI.
- The wcc\_config.sh log file is located in the %CA\_WCC\_INSTALL\_LOCATION%/log/application/cli directory.
- On a (Remote) Web UI Collector machine, the wcc\_config script can be used only to run the following three commands; all the other commands are supported only on the Web UI machine.
  - Specify a Different Collector Database
  - Specify a Different Collector Database for Reporting
  - Change the Port for the Collector

**Propagate Configurations to Other Servers**

After you set up one Web UI server, you can use the export and import options of the Configuration Batch Interface to set up other Web UI servers with the same configurations.

**NOTE**

Configurations for all Web UI instances in a High Availability group share a common database and are automatically synchronized. Therefore, it is not necessary to copy configurations from one Web UI instance to another instance.

The following is the process for propagating configurations from one Web UI server to other Web UI servers:

1. Export all configurations from the Web UI server that has been configured.
2. Copy the text file created by the export to the Web UI server that you want to run with the same configurations.
3. Change the server name where it appears in the text file to the name of the server you have copied the file to.
4. Import the text file on the server that you want to run with the same configurations.
5. Repeat Steps 2-4 for each Web UI server that you want to run with the same configurations.

**Update CA EEM on All High Availability Servers**

When you update the CA EEM server defined to one Web UI instance in a High Availability group, you must also update the definition on the other Web UI instances in the group. You can do this by distributing the necessary commands in encrypted files to all Web UI instances in the group, and then the appropriate users can update their Web UI instances at a convenient time.

**NOTE**

Authorized users can alternatively run `wcc_config` with the appropriate arguments to update the CA EEM server definition and restart services; however, the following process lets users with lower security access update their Web UI instance while keeping passwords secure.

The following is the process for creating the encrypted files, updating one Web UI instance, and distributing the files:

1. Create an encrypted file containing the arguments to update the CA EEM server definition.
2. Create an encrypted file containing the arguments to restart the services CA-wcc and CA-wcc-services.
3. Run `wcc_config` with the encrypted file to update the CA EEM server definition on one Web UI instance.
4. Run `wcc_config` with the encrypted file to restart the services CA-wcc and CA-wcc-services on the same Web UI instance.
5. Distribute the two encrypted files to the appropriate users.

The appropriate users can perform Steps 3 and 4 on their Web UI instance during a period of low activity. User name and password arguments are not required when `wcc_config` is run with an encrypted file.

**Delete a User**

When you delete a user in CA EEM, you must also delete the user and the templates associated with that user in Web UI.

The following is the process for deleting a user and their templates in Web UI:

1. Create a new file in your text editor that includes the users to be deleted, using the following format:

```
delete_user: user_to_delete
```

**NOTE**

If you are deleting more than one user, enter the command for each user on a separate line.

2. Import the file in Configuration Batch Interface using the following command:

```
wcc_config -u user_name -p password -i input_file.txt
```

3. Delete each template associated with each user in Job Template Batch Interface using the following command:

```
job_template -u user_name -p password -f deleted_user -r template_name
```

**NOTE**

You can specify a list of templates to delete.

**NOTE**

You can verify that the user has been deleted in an exported configuration.

**Export Configurations**

You can export Web UI configurations from the server to a text file. You can then modify the resulting file using a text editor and then import it back to the server, or copy it to another server and import it on that server to propagate the configurations.

To export configurations, use the following command syntax:

```
wcc_config -u user_name -p password -x output_file [-c config1,config2...] [-plainpwd] [-v]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **-x, --export output\_file**



Specifies the fully-qualified file name to which the configurations are exported. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-x) or the long name (--export).

- **-c, --component *config1,config2...***

(Optional) Specifies the configurations to export. Configuration names in the list must be separated by a comma; a space should not follow the comma. You can use either the short name (-c) or the long name (--component).

Valid values are as follows:

- APPEDITOR
- CONFIG
- CREDSD
- ECLI
- MONITOR
- FORECAST
- QUICKVIEW
- QUICKEDIT
- REPORTING
- RESOURCES

#### NOTE

- If you do not specify any configurations, the exported file contains server definitions and preferences for all applications.
- If you specify a list of configurations, the exported file contains preferences for the specified applications only.

- **-plainpwd**

(Optional) Exports passwords as plain text (non-encrypted).

- **-v, --verbose**

(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

### Example: Export all configurations

The following command exports all configurations to the file named output.txt.

```
wcc_config -u ejmcommander -p ejmcommander -x output.txt
```

### Example: Export selected configurations with non-encrypted passwords

The following command exports the CONFIG, and MONITOR configurations to the file named output.txt with no password encryption:

```
wcc_config -u ejmcommander -p ejmcommander --export output.txt -c CONFIG,MONITOR -
plainpwd
```

### Import Configurations

You can import the text file that contains the Web UI configurations you have modified. You can also copy the file to other servers, and then import the file on those servers so that all run with the same configurations.

To import a configuration file, use the following command syntax:

```
wcc_config -u user_name -p password -i input_file [v]
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

- **-i, --import *input\_file***  
Specifies the fully-qualified name of the text file to import. You can use either the short name (-i) or the long name (--import).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

#### Example: Import configurations from the specified file

The following command imports configurations in the file named output.txt to the local Web UI server.

```
wcc_config -u ejmcommander -p ejmcommander-i output.txt
```

### Encrypt Command Line Arguments

You can create a file that contains encrypted arguments for the wcc\_config command. This lets you distribute a file with arguments, such as the user name and password, that cannot be viewed in plain text by the recipient. For example, you can create a job or script so that another user with lower security access can run the wcc\_config command with the encrypted file at a convenient time.

To encrypt command line arguments, use the following command syntax:

```
wcc_config -e encrypted_file -u user_name -p passwordargument1 argument2...
```

- **-e, --encrypt *encrypted\_file***  
Specifies the fully-qualified file name to which the arguments should be encrypted and saved. The file is created by the encryption; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch utility. You can use either the short name (-e) or the long name (--encrypt).
- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- ***argument1 argument2*...**  
Specifies the arguments that you would normally use to perform the action.

#### Example: Create a file of encrypted arguments

The following command encrypts and stores the user name, password, and import command arguments to a file named arg\_file:

```
wcc_config -e arg_file -u ejmcommander -p ejmcommander -i in_file
```

### Run wcc\_config With Encrypted Arguments

You can run the wcc\_config command with a previously created file that contains encrypted arguments, including the user name and password.

To run wcc\_config with encrypted arguments, use the following command syntax:

```
wcc_config encrypted_file
```

- ***encrypted\_file***  
Specifies the fully-qualified file name that contains the encrypted arguments.

#### Example: Run wcc\_config with an encrypted arguments file

The following command runs `wcc_config` and decrypts the arguments in the file named `arg_file`:

```
wcc_config arg_file
```

### **Display the Database Server Information**

You can display the following information for the Web UI or Reporting database server:

- Host name
- Port
- User
- Platform
- Database name
- Password (You will see only asterisks because the password is encrypted.)

When using the TNS name configuration with an Oracle database, the following information is displayed:

- TNS name of the database connection
- Path to the `tnsnames.ora` file
- Platform
- User
- Password (You will see only asterisks because the password is encrypted.)

#### **NOTE**

The Configuration Batch Interface must be able to connect to the database using the parameters provided.

To display the database server information, use the following command syntax:

```
wcc_config -u user_name -p password --displaydb [--dbapp application]
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--dbapp *application***  
(Optional) Specifies the application that the database server is associated with.  
Valid values are as follows:
  - REPORTING
  - Web UI

#### **Example: Display the Web UI database server information**

The following command displays the Web UI database server information:

```
wcc_config -u ejmcommander -p ejmcommander --displaydb
```

#### **Example: Display the Reporting database server information**

The following command displays the server information for the Reporting database:

```
wcc_config -u ejmcommander -p ejmcommander --displaydb --dbapp REPORTING
```

### **Specify a Different Database**

You can specify a different database for Autosys Web UI or for Reporting after Autosys Web UI is installed and the database is accessible.

**NOTE**

- From Version 12.0, to switch from one database to another database, use the installer to create the schema and then use the wcc\_config utility to switch the database.
- If additional collectors are installed, then use the wcc\_config utility with --collectorname to change the database pointed by the collector.
- The Configuration Batch Interface must be able to connect to the database using the parameters provided.
- All Web UI instances in a High Availability group must reference the same database. When you specify a different database for one Web UI instance in the High Availability group, you must also specify the same database on the other Web UI instances in the group.

To specify a different database, use the following command syntax:

```
wcc_config -u user_name -p password --dbplatform db_platform --dbuser db_user
--dbpassword db_password --dbhost db_host --dbport db_port --dbname db_name [--
dbapp application] [--dbtnsname tns_name] [- - dbtnspath tns_path]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--dbplatform db\_platform**  
Specifies the database platform.  
Valid values are as follows:
  - DERBY
  - MSSQL
  - ORACLE
- **--dbuser db\_user**  
Specifies the database user name.
- **--dbpassword db\_password**  
Specifies the password for the database user name.
- **--dbhost db\_host**  
Specifies the computer that hosts the database.
- **--dbport db\_port**  
Specifies the database port.  
Default port numbers are as follows:
  - DERBY -- 1543
  - MSSQL -- 1433
  - ORACLE -- 1521
- **--dbname db\_name**  
Specifies the database name.
- **--dbapp application**  
(Optional) Specifies the application that the database is associated with.  
Valid values are as follows:
  - REPORTING
  - WCC

**NOTE**

If the dbapp argument is not included in the command string, the Web UI database is specified by default.

- **--dbtnsname tns\_name**

(Optional) Specifies the TNS name of the Oracle database connection.

**NOTE**

This parameter applies to Oracle only.

- **--dbtnspath *tns\_path***

**NOTE**

This parameter applies to Oracle only.

(Optional) Specifies the Oracle tnsnames.ora file, including its full path.

**Example:** c:\oracle\tnsnames.ora

**WARNING**

For the current release Autosys Web UI 11.4 SP3 and upcoming releases, the path needs to be without the file name.

**Example:** c:\oracle\

- **--dbtruststorepath *trustStorePath***(Optional) Path to the trust store or wallet.
- **--dbtruststoretype *trustStoreType***(Optional) Type of the trust store or wallet [SSO].

**NOTE**

SSO is the only value.

- **--dbserverdnmatch *serverDNmatch***(Optional) SSL server DN match [true|false].

**NOTE**

- To specify the TNS name and path for Oracle RAC, you must use the **--dbtnspath** and **--dbtnsname** parameters.
- For Oracle non-RAC, you can specify the **--dbtnspath** and **--dbtnsname** parameters, or specify the following parameters and values:
  - **--dbhost** localhost
  - **--dbport** 1521
  - **--dbname** XE

**WARNING**

After you specify a different Web UI or Reporting database, you must restart the services CA-wcc and CA-wcc-services on Windows, and the service CA-wcc-services on UNIX. On Windows, you must use the Services tool in the Control Panel to start or stop the services and, on UNIX, you must use the `unisrvctr` command to start or stop the services.

**Example: Specify a different Web UI database**

```
wcc_config -u ejmcommander -p ejmcommander --dbplatform ORACLE --dbuser user_a --
dbpassword my_password --dbhost localhost --dbport 1521 --dbname xe
```

**Example: Specify a different Reporting database**

```
wcc_config -u ejmcommander -p ejmcommander --dbplatform MSSQL --dbuser user_a --
dbpassword my_password --dbhost localhost --dbport 3306 --dbname xe --dbapp REPORTING
```

**Example: Specify a different Web UI database for Oracle RAC**

```
wcc_config -u ejmcommander -p ejmcommander --dbplatform ORACLE --dbuser user_a --
dbpassword my_password --dbtnspath c:\oracle\tnsnames.ora --dbtnsname MY_RAC
```

**Example: Specify a different Reporting database for Oracle RAC**

```
wcc_config -u ejmcommander -p ejmcommander --dbplatform ORACLE --dbuser user_a --
dbpassword my_password --dbtnspath /opt/oracle/tnsnames.ora --dbtnsname MY_RAC --dbapp
Reporting
```

### Example: Specify a Web UI database to an Oracle database running on remote server using TNS definition and certificate authentication

```
wcc_config.sh -u username -p password -dbapp WCC --dbplatform oracle --dbtnspath /opt/CA/WorkloadCC/data/config/tns/ --dbtnsname CNT --dbtruststorepath /opt/CA/WorkloadCC/data/config/wallet/cwallet.sso --dbtruststoretype SSO --dbserverdnmatch false
```

### Change the Database Credentials

You can change the user name, password, or both for the Web UI or Reporting database.

#### NOTE

The Configuration Batch Interface must be able to connect to the database using the parameters provided.

To change the database credentials, use the following command syntax:

```
wcc_config -u user_name -p password --dbuser db_user --dbpassword db_password [--dbapp application]
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--dbuser *db\_user***  
Specifies the new user name for the database.
- **--dbpassword *db\_password***  
Specifies the new password for the database user name.
- **--dbapp *application***  
(Optional) Specifies the application that the database is associated with.  
Valid values are as follows:
  - REPORTING
  - Web UI

#### NOTE

If the dbapp argument is not included in the command string, the specified credentials for the Web UI database are changed by default.

**Example: Specify a new password for the Web UI database** The following example specifies my\_password as the password for the Web UI database.

```
wcc_config -u ejmcommander -p ejmcommander --dbpassword my_password
```

**Example: Specify a new user and password for the Reporting database** The following example specifies my\_password as the password for user\_b for the Reporting database.

```
wcc_config -u ejmcommander -p ejmcommander --dbuser user_b --dbpassword my_password --dbapp REPORTING
```

### Specify a Different Collector Database

You can specify a different database for Web UI Collectors after Web UI is installed and the database is accessible.

The wcc\_config script can only be used to run following commands from a (Remote) Web UI Collector machine; all the other commands are supported only on the Web UI machine.

#### NOTE

You need to change the database for Web UI before changing the database for the collector.

To specify a different collector database, use the following command syntax:

```
wcc_config --dbplatform ORACLE --dbuser user_a --dbpassword my_password --dbhost
localhost --dbport db_port --dbname xe --collectorname collector_name
```

#### Example: Specify a Different Collector Database

```
wcc_config --dbplatform ORACLE --dbuser user_a --dbpassword my_password --dbhost
localhost --dbport 1521 --dbname xe --collectorname collector1
```

#### Example: Specify a Different Collector Database for Reporting

```
wcc_config --dbplatform ORACLE --dbuser user_a --dbpassword my_password --dbhost localhost --dbport 1521 --
dbname xe --collectorname collector1 --dbapp REPORTING
```

#### Change the Port for the Collector

To change the collector port number, use the following command syntax:

```
wcc_config --collectorname collector_name -- collectorport collector_port
```

#### Example:

```
wcc_config --collectorname collector1 --collectorport 5707
```

#### Display the CA EEM Server Configuration

You can display the current CA EEM server configuration. To display the CA EEM server configuration, use the following command:

```
wcc_config -u ejmcommander -p ejmcommander --displayeem
```

#### Change EiamAdmin Password Using the CA EEM User Interface

You can change the EiamAdmin password, when required, using the CA EEM User Interface (UI). **Follow these steps:**

1. Log in to the EEM UI using a browser ([https://<EEM\\_Server\\_host>:5250/spin/eiam/](https://<EEM_Server_host>:5250/spin/eiam/)) as EiamAdmin user and choose the Global application.
2. On the Home (page/tab), click "Change Password" under the Self-Administration menu.
3. Enter "EiamAdmin" in the User field. Enter the old (current) password and new password (twice, to confirm) and click OK.
4. Click Close.
5. Log out and log in again to the EEM UI to ensure the password is changed.

After successfully changing the EiamAdmin password, update the Web UI configuration using the [Update the CA EEM Server Definition in Web UI](#) section.

#### NOTE

Perform this procedure on each Web UI server, if you have multiple Web UI HA servers.

#### Reset Missing or Forgotten EiamAdmin Password

The EiamAdmin user is defined to an internal datastore for EEM, and external to the application itself, so you must reset this password if it is lost or forgotten. Access to the itechpoz DSA is required, so you must modify the current security for this DSA and then restart the itechpoz DSA. **Follow these steps: Linux**

1. Open the file `/opt/CA/SharedComponents/CADirectory/dxserver/config/access/itechpoz.dxc`  
Change "set access-controls = **false**;" and save the file.
2. Open the file `/opt/CA/SharedComponents/CADirectory/dxserver/config/settings/itechpoz.dxc`  
Change "set min-auth = **none**;" and save the file.
3. Open the file `/opt/CA/SharedComponents/CADirectory/dxserver/config/knowledge/itechpoz.dxc`  
Change the following and save the file:

"set auth-levels = anonymous""address = ipv4 "<IP address>" port 509, ipv4 "<hostname>" port 509"

4. Reinitialize itechpoz DSA from the command line using `su - dsa -c "dxserver init all"`

## Windows

1. Open the file `C:\Program Files (x86)\CA\Directory\dxserver\config\access\itechpoz.dxc`  
Change "set access-controls = false;" and save the file.
2. Open the file `C:\Program Files (x86)\CA\Directory\dxserver\config\settings\itechpoz.dxc`  
Change "set min-auth = none;" and save the file.
3. Open the file `C:\Program Files (x86)\CA\Directory\dxserver\config\knowledge\itechpoz.dxc`  
Change "set auth-levels = anonymous" and save the file.
4. Restart the "CA iTechnology iGateway x.x" service and CA Directory Services.

Complete the following steps that are common to all platforms:

1. Install and run any LDAP browser on your EEM server. We recommend JXplorer 3.2.2 or higher.
2. Connect to the itechpoz DSA using your EEM server information.

**Open LDAP/DSML Connection**

Host:  Port:

Protocol:

DSML Service:

Optional Values

Base DN:

Security

Level:

User DN:

Password:

Use a Template

Save  Delete Default

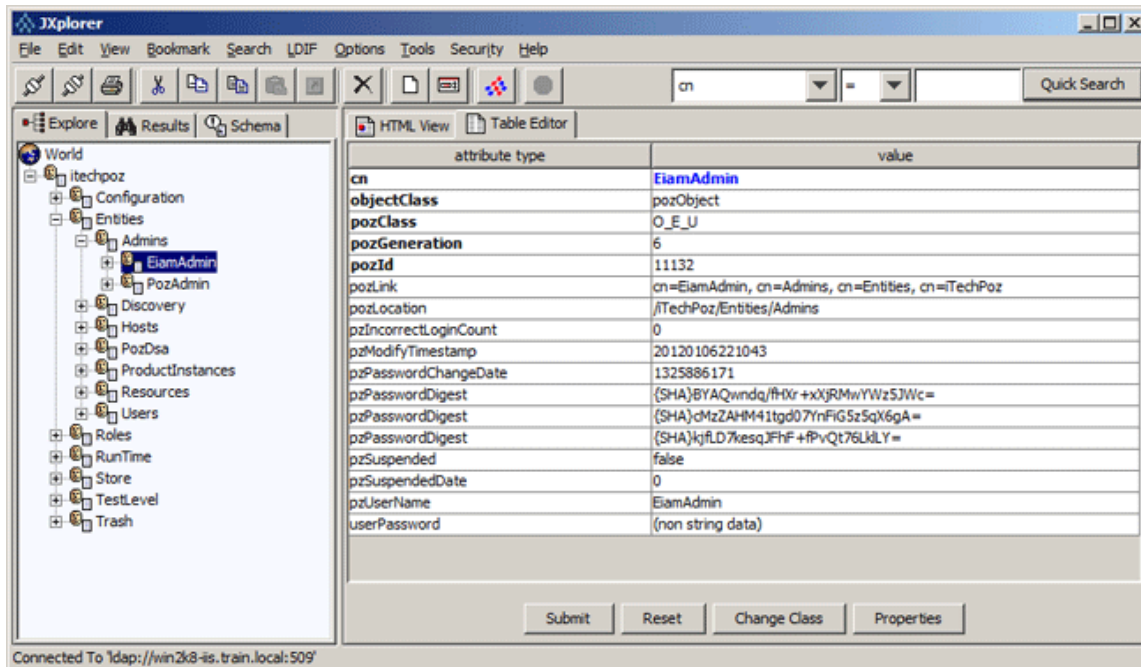
OK Cancel Help

## NOTE

If required, you can save the template.

3. Follow the tree down to the EiamAdmin ID.





4. Double-click on the userPassword value, enter the new password and then re-enter to confirm.
5. Click OK and then click Submit.

### Update the CA EEM Server Definition in Web UI

You can update the CA EEM server definition in Web UI after installation.

#### NOTE

If CA EEM is not configured for failover, all Web UI instances in a High Availability group must reference the same instance of CA EEM. When you change the CA EEM server definition on one Web UI instance in the High Availability group and the original CA EEM instance was not configured for failover, you must also change the definition on the other Web UI instances in the group to reference the new instance of CA EEM.

To update the CA EEM server definition in Web UI, use the following command syntax:

```
wcc_config -u user_name -p password --eemhostname eem_host_name --eemadmin eem_administrator --
eempassword eem_password [--eemappid eem_application_id] [--eemcertname eem_certificate_name] [--
eemcertkey eem_certificate_key]
```

- **-u, --user user\_name**

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password password**

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

- **--eemhostname eem\_host\_name[,eem\_host\_name2,eem\_host\_name3...]**

Specifies the host names of the new CA EEM servers that you want the Web UI instance to reference. The user name and password are authenticated against this host.

#### NOTE

If CA EEM is configured for failover, you can optionally specify a list of the servers in the failover setup. Server names should be separated by a comma only and listed in the order of reliability.

--eemadmin eem\_administrator Specifies the administrator name for the new instance of CA EEM.--eempassword eem\_password Specifies the password for the administrator of the new instance of CA EEM.--eemappid eem\_application\_id (Optional) Specifies the application ID that contains the Web UI policies, for example, WCC0004. --eemcertname eem\_certificate\_name (Optional) Specifies the CA EEM certificate file name. This is required only if you are using the EEM certificate. This file is in the following location: \$CA\_WCC\_INSTALL\_LOCATION/data/config --eemcertkey eem\_certificate\_key(Optional) Specifies the CA EEM certificate key file name. This is required only if you are using the EEM certificate. This file is in the following location: \$CA\_WCC\_INSTALL\_LOCATION/data/config

#### NOTE

After you update the CA EEM server definition, you must restart the services CA-wcc and CA-wcc-services on Windows, or the service CA-wcc-services on UNIX. On Windows, you must use the Services tool in the Control Panel to start or stop the services and, on UNIX, you must use the unisrvcntr command to start or stop the services.

Example: Update the CA EEM server definition

The following command updates the CA EEM server definition in Web UI when you use the optional CA EEM certificate name and keyfile:

```
wcc_config -u ejmcommander -p ejmcommander --eemhostname EEM_01 --eemadmin EiamAdmin --eempassword admin --eemcertname certificate_01.pem --eemcertkey eem_cert_key_01.key - -eemappid WCC0004
```

### Change the Event Bus Port Value

You can change the Autosys Web UI event bus port value. Changing the value is necessary when a conflict with the event bus port occurs. To change the event bus port value, use the following command syntax:

```
wcc_config -u user_name -p password --eventport port_value
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--eventport port\_value**  
Specifies the event bus port value.

#### NOTE

**Limits:** 1025-65535

**Example: Change the event bus port value** The following example changes the event bus port value to 5064.

```
wcc_config -u ejmcommander -p ejmcommander --eventport 5064
```

#### NOTE

- If the port value you specify in the command is already in use or the value is not within the limits, an error message is displayed. However, only ports assigned to applications that are running or used by Web UI are detected.
- Do not specify the port that is assigned to Web UI.
- The Event Bus port must not be in the dynamic port range of the machine.
- After you change the event bus port value, restart Web UI services.

### Change the SSL Mode

You can change the SSL mode of a Web UI server to SSL or to non-SSL. Changing the SSL mode from SSL to non-SSL is required when you add a Web UI server to a High Availability group. Changing the SSL mode from non-SSL to SSL is required before users can access Web UI using a secure URL (https).

**NOTE**

All Web UI servers in a High Availability group must run in either non-SSL mode or SSL mode.

To change the SSL mode, use the following command syntax:

```
wcc_config -u user_name -p password --ssl on|off
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--ssl on|off**  
Specifies whether the Web UI server runs in SSL or non-SSL mode. Valid values are as follows:
  - on -- The Web UI server runs in SSL mode.
  - off -- The Web UI server runs in non-SSL mode.

**Example: Change the SSL mode to non-SSL** The following example changes the SSL mode of a Web UI server to non-SSL mode.

```
wcc_config -u ejmcommander -p ejmcommander --ssl off
```

**NOTE**

- List of servers and their SSL status is displayed.
- When you change the mode to SSL, a self-signed certificate is generated.
- If Web UI is using port number 80 for the Tomcat server, do not change the mode from non-SSL to SSL. You can change the port to a different port as described in [Change the Tomcat Server Port Values](#), and then change the mode to SSL.
- If Web UI is using port number 443 for the Tomcat server, do not change the mode from SSL to non-SSL. You can change the port to a different port as described in [Change the Tomcat Server Port Values](#), and then change the mode to non-SSL.
- After you change the SSL mode, restart Web UI services.

**Change the Tomcat Server Port Value**

You can change the Tomcat server port value. Changing the value is necessary when a conflict with another port occurs. To change the Tomcat server port value, use the following command syntax:

```
wcc_config -u user_name -p password --port port_value
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--port *port\_value***  
Specifies the Tomcat server port value.  
**Limits:** 80 (non-SSL only), 443 (SSL only), 1025-65535 (SSL or non-SSL)  
**Defaults:** 8080 (non-SSL), 8443 (SSL)

**Example: Change the Tomcat server port value** The following example changes the Tomcat server port value to 8090.

```
wcc_config -u ejmcommander -p ejmcommander --port 8090
```

**NOTE**

- If the port value you specify in the command is already in use or the value is not within the limits, an error message is displayed. However, only ports assigned to applications that are running or used by Web UI are detected.
- If Web UI is running in SSL mode, do not specify port number 80 for the Tomcat server port.
- If Web UI is running in non-SSL mode, do not specify port number 443 for the Tomcat server port.
- If a run as (non-root) user was specified during Web UI installation on UNIX and you intend to specify port 80 or 443, the port must be set up for non-root user access before you change the port value.
- After you change the Tomcat server port value, restart Web UI services.

**Change the Tomcat Stop Port Value**

You can change the Tomcat stop port value. Changing the value is necessary when a conflict with another port occurs. To change the Tomcat stop port value, use the following command syntax:

```
wcc_config -u user_name -p password --stopport stop_port_value
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **--stopport *stop\_port\_value***  
Specifies the Tomcat stop (shutdown) port value.  
**Limits:** 1025-65535  
**Default:** 8005

**Example: Change the Tomcat stop port value** The following example changes the Tomcat stop port value to 8015.

```
wcc_config -u ejmcommander -p ejmcommander --stopport 8015
```

**NOTE**

- If the port value you specify in the command is already in use or the value is not within the limits, an error message is displayed. However, only ports assigned to applications that are running or used by Web UI are detected.
- After you change the Tomcat stop port value, restart Web UI services.

**Display Component Short Names**

You can display a list of the short names for configuration components. Short names are used with the -c option, which can be used with the -x option to specify the components to export. To display component short names, use the following command syntax:

```
wcc_config -u ejmcommander -p ejmcommander -components
```

**Enable or Disable Reporting**

You can enable the Reporting option to see the Reporting configuration. If the Reporting option is disabled, you cannot see the Reporting tab and also the "Connection status" wizard of the Web UI Dashboard does not show the Reporting database entry. Reporting collection and cleanup will not start. To enable or disable Reporting, use the following command syntax:

```
wcc_config -u ejmcommander -p ejmcommander -enablereporting true|false
```

Valid values are:

True - enables Reporting. With this option, you can also Change the Database Credentials, for Reporting.

False - disables Reporting

Based on your selection, Collectors and Database Cleanup are enabled or disabled respectively.

#### NOTE

For more information, see Reporting in the Configuration Properties section. When you change the Reporting database using *enablereporting=true*, you need not specify *--dbapp*, because it is specific to Reporting.

#### Examples:

Example usage to enable or disable Reporting (UNIX command syntax is used):

To disable Reporting:

```
./wcc_config.sh -u username -p password -enablereporting false
```

When you enable Reporting, you can additionally specify Reporting database details if you want to change the Reporting database (example given to change database to MS SQL).

```
./wcc_config.sh -u username -p password -enablereporting true --dbplatform mssql --
dbname WCC_MAIN --dbhost my_hostname --dbport 1433 --dbuser wcc_user --dbpassword
wcc_pwd
```

#### Display Help

You can display the inline help, which describes the options for the *wcc\_config* command. To display help, use one of the following commands:

```
wcc_config -?
wcc_config --help
```

#### Configuration Short Names

The following table lists the Web UI configuration short names and the property settings they contain. Short names are used with the *-c* option, which can be used with the *-x* option to specify the configurations to export.

| Configuration Short Name | Contains                            |
|--------------------------|-------------------------------------|
| AGENTINVENTORY           | Agent Inventory preferences         |
| APPEDITOR                | Application Editor preferences      |
| CONFIG                   | Configuration preferences           |
| CREDS                    | Credentials preferences             |
| ECLI                     | Enterprise Command Line preferences |
| FORECAST                 | Forecast preferences                |
| MONITOR                  | Monitoring preferences              |
| QUICKEDIT                | Quick Edit preferences              |
| QUICKVIEW                | Quick View preferences              |
| REPORTING                | Reporting preferences               |
| RESOURCES                | Resources preferences               |

#### Configuration Properties

##### Preferences Names

Tables are provided that map the application preference fields in Configuration to the attributes listed in the file exported by the Configuration Batch Interface.

### Application Editor

The following table maps Application Editor preference categories and fields in Configuration to the attributes in the APPEDITOR configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category  | Preference Field Name                      | Exported Attribute Name                        | Default Value |
|----------------------|--------------------------------------------|------------------------------------------------|---------------|
| <b>User Defaults</b> | Rows per Page for SAP Filters              | appedit_env_sapdnd_filters_row<br>sperpage     | 20            |
|                      | Rows per Page for SAP Filter Results       | appedit_env_sapdnd_results_ro<br>wsperpage     | 50            |
|                      | Rows per Page for Import Jobs Dialog       | appedit_env_jobimport_rowspe<br>rpage          | 50            |
|                      | Rows per Page for Search Results in Dialog | appedit_env_dialog_searchtable<br>_rowsperpage | 20            |
|                      | Hide SAP Drag and Drop Panes               | appedit_env_sapdnd_panel                       | false         |
|                      | Description Column Length                  | appedit_env_searchtable_descr<br>iptionlength  | 500           |
|                      | Maximum Number of Imported Jobs            | appedit_env_jobimport_maximu<br>mjobs          | 1000          |
|                      | New Job Name Default                       | appedit_env_flow_graph_jobna<br>metemplate     | {1} Job {0}   |
|                      | Maximum Flow Height                        | appedit_env_flow_graph_maxh<br>eight           | 570           |
|                      | Left Panes Width                           | appedit_env_leftpanels_width                   | 250           |
| <b>System</b>        | Owner Field Mandatory                      | appedit_env_owner_mandatory                    | false         |
|                      | Graph Tooltip Length                       | appedit_env_flow_tooltip_value<br>length       | 64            |
|                      | Import All Button                          | appedit_env_jobimport_importa<br>ll            | true          |
|                      | Commit Threads                             | appedit_env_commit_threads                     | 20            |
|                      | Commit Dialog Refresh Interval             | appedit_env_commit_refreshra<br>te             | 2000          |
|                      | Submit Owner on Commit                     | appedit_env_commit_submit_o<br>wner            | true          |
|                      | Proxy Server Name                          | appedit_env_proxy_server                       |               |
|                      | Proxy Port                                 | appedit_env_proxy_port                         |               |
|                      | Proxy User Name                            | appedit_env_proxy_user                         |               |
|                      | Proxy Password                             | appedit_env_proxy_password                     |               |

### Configuration

The following table maps Configuration preference categories and fields in Configuration to the attributes in the CONFIG configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category | Preference Field Name         | Exported Attribute Name       | Default Value |
|---------------------|-------------------------------|-------------------------------|---------------|
| <b>EEM</b>          | EEM Connection Retry Interval | config_env_eem_retry_interval | 1             |

|                       |                                       |                                                  |       |
|-----------------------|---------------------------------------|--------------------------------------------------|-------|
|                       | Maximum EEM Connection Retries        | config_env_eem_retry_tries                       | 1     |
| <b>Autosys</b>        | Enable SSL                            | config_env_autosys_enable_ssl                    | false |
|                       | Enable PMUX                           | config_env_autosys_enable_pmux                   | true  |
| <b>Credentials</b>    | Validation Thread Pool Minimum Size   | config_env_cred_validation_thread_pool_core_size | 2     |
|                       | Validation Thread Pool Maximum Size   | config_env_cred_validation_thread_pool_max_size  | 4     |
|                       | Validation Thread Pool Timeout        | config_env_cred_validation_thread_pool_timeout   | 600   |
|                       | Persist Default User                  | config_env_cred_persist_default_user             | false |
|                       | Validation Refresh Interval           | config_env_cred_validation_refresh               | 5000  |
| <b>Single Sign-On</b> | Single Sign-On                        | config_env_single_sign_on                        | none  |
|                       | Force Kerberos Compatibility          | config_env_single_sign_on_force_ntlm             | true  |
| <b>Security</b>       | Session Timeout                       | config_env_sso_session_timeout                   | 30    |
|                       | Auto Refresh prevents session timeout | config_env_auto_refresh_prevents_session_timeout | false |
|                       | Single Session per User               | config_env_single_session_per_user               | false |
|                       | URL Parameters Encryption             | config_env_encrypt_query_parameters              | false |
| <b>Environment</b>    | Default Time Zone                     | config_user_timezone                             |       |
| <b>Integrations</b>   | AM base URL                           | config_env_integration_cam_base_url              |       |

#### Override Kerberos Compatibility

In the current release, you can configure single sign-on capability when you are using a Kerberos compatible browser. The Enable NTLM preference in Configuration forces Web UI to override Kerberos compatibility when NTLM authentication is enabled. By default, NTLM protocol is enabled and Web UI is configured to override Kerberos in favor of NTLM. Web UI does not support Kerberos authentication protocol. If single sign-on is active and Web UI is not configured to override Kerberos, you are required to authenticate your session by entering login credentials.

#### Enterprise Command Line

The following table maps Enterprise Command Line preference categories and fields in Configuration to the attributes in the ECLI configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category         | Preference Field Name         | Exported Attribute Name                     | Default Value |
|-----------------------------|-------------------------------|---------------------------------------------|---------------|
| <b>My Commands Settings</b> | Maximum My Commands Saved     | commandline_env_max_number_mycommands_saved | 25            |
|                             | Rows per Page for My Commands | commandline_env_table_mycommands_rows       | 10            |

|                                 |                                   |                                                  |            |
|---------------------------------|-----------------------------------|--------------------------------------------------|------------|
| <b>Global Commands Settings</b> | Maximum Global Commands Saved     | commandline_env_max_number_global_commands_saved | 25         |
|                                 | Rows per Page for Global Commands | commandline_env_table_global_commands_rows       | 10         |
| <b>General Settings</b>         | Maximum Number of Servers         | commandline_env_max_number_server_selection      | 7          |
|                                 | Command Timeout                   | commandline_env_max_command_processing_wait_time | 300000     |
|                                 | Maximum Auto Complete Items       | commandline_env_max_number_history               | 10         |
|                                 | Default Export File Name          | commandline_env_export_default_file_name         | export.txt |

### Forecast

The following table maps Forecast preference categories and fields in Configuration to the attributes in the FORECAST configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category       | Preference Field Name           | Exported Attribute Name                  | Default Value |
|---------------------------|---------------------------------|------------------------------------------|---------------|
| <b>User Interface</b>     | Rows per Page for Forecast      | forecast_env_definitions_table_row_count | 50            |
|                           | Rows per Page for Table View    | forecast_env_results_table_row_count     | 50            |
|                           | Rows per Page for Timeline View | forecast_env_results_gantt_row_count     | 50            |
| <b>Command Parameters</b> | Execution Timeout               | forecast_env_command_execution_timeout   | 300000        |

### Monitoring

The following table maps Monitoring preference categories and fields in Configuration to the attributes in the MONITOR configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category | Preference Field Name       | Exported Attribute Name                          | Default Value                                                             |
|---------------------|-----------------------------|--------------------------------------------------|---------------------------------------------------------------------------|
| <b>UI</b>           | Auto Refresh Interval       | monitor_env_viewdetail_auto_refresh_interval     | 15                                                                        |
|                     | Views Table Columns         | monitor_env_views_table_column_list              | name,job_status,server_status,alert_severity,jobs,collections,status_time |
|                     | Servers Table Columns       | monitor_env_servers_table_column_list            | name,status,status_time,type,filters                                      |
|                     | Jobs Table Columns          | monitor_env_serverdetail_job_table_column_list   | job_name,job_type,status,server,run_machine,status_time,duration          |
|                     | Alerts Table Columns        | monitor_env_serverdetail_alert_table_column_list | alert,severity,status,job,server,create_time,status_time                  |
|                     | Scheduler Log Default Lines | monitor_env_scheduler_log_lines_default          | 50                                                                        |



|                            |                                          |                                                   |          |
|----------------------------|------------------------------------------|---------------------------------------------------|----------|
|                            | Scheduler Log Auto Refresh Interval      | monitor_env_scheduler_log_auto_refresh_interval   | 15       |
| <b>Job Level Security</b>  | Use AutoSys Workload Automation Security | monitor_env_jobLevelSecurity_active               | Selected |
| <b>Back-end Services</b>   | Collector Mode                           | monitor_env_collectorsentry_mode                  | all      |
|                            | Job Collector Run Interval               | monitor_env_job_collector_run_interval            | 5        |
|                            | Alarm Collector Run Interval             | monitor_env_alarm_collector_run_interval          | 5        |
|                            | Alerts Deletion Threshold                | monitor_env_alertscleaner_stale_alerts_threshold  | 7        |
|                            | Alert Cleaner Run Interval               | monitor_env_alertscleaner_run_interval            | 24       |
|                            | Security Cache Timeout                   | monitor_env_securitycache_timeout                 | 60       |
|                            | DB Statistics Update Threshold           | monitor_env_dbstatsupdater_threshold              | 5000     |
| <b>Server Status Order</b> | Failure                                  | monitor_env_status_server_severity_failure        | 0        |
|                            | Disconnected                             | monitor_env_status_server_severity_disconnected   | 1        |
|                            | Unknown                                  | monitor_env_status_server_severity_unknown        | 2        |
|                            | Not Configured                           | monitor_env_status_server_severity_not_configured | 3        |
|                            | Connected                                | monitor_env_status_server_severity_connected      | 5        |
| <b>Job Status Order</b>    | Failure                                  | monitor_env_status_job_severity_failure           | 111      |
|                            | Waiting on Reply                         | monitor_env_status_job_severity_wait_reply        | 112      |
|                            | Pending Machine                          | monitor_env_status_job_severity_pend_mach         | 113      |
|                            | Waiting on Resources                     | monitor_env_status_job_severity_reswait           | 114      |
|                            | Queue Wait                               | monitor_env_status_job_severity_que_wait          | 115      |
|                            | Restart                                  | monitor_env_status_job_severity_restart           | 116      |
|                            | Terminated                               | monitor_env_status_job_severity_terminated        | 117      |
|                            | On Ice                                   | monitor_env_status_job_severity_on_ice            | 118      |

|  |                 |                                            |     |
|--|-----------------|--------------------------------------------|-----|
|  | On Hold         | monitor_env_status_job_severity_on_hold    | 119 |
|  | Unknown         | monitor_env_status_job_severity_unknown    | 120 |
|  | Success         | monitor_env_status_job_severity_success    | 121 |
|  | Running         | monitor_env_status_job_severity_running    | 122 |
|  | Starting        | monitor_env_status_job_severity_starting   | 123 |
|  | Activated       | monitor_env_status_job_severity_activated  | 124 |
|  | Inactive        | monitor_env_status_job_severity_inactive   | 125 |
|  | On No Execution | monitor_env_status_job_severity_on_no_exec | 126 |
|  | Suspended       | monitor_env_status_job_severity_suspended  | 127 |

### Quick Edit

The following table maps Quick Edit preference categories and fields in Configuration to the attributes in the QUICKEDIT configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category  | Preference Field Name                       | Exported Attribute Name                             | Default Value |
|----------------------|---------------------------------------------|-----------------------------------------------------|---------------|
| <b>User Defaults</b> | Rows per Page for Search Results            | quickedit_env_searchtable_row_sperpage              | 50            |
|                      | Rows per Page for Search Results in Dialog  | quickedit_env_dialog_searchtable_rowsperpage        | 20            |
|                      | Search History Items                        | quickedit_env_searchhistory_maxitems                | 30            |
|                      | Description Column Length                   | quickedit_env_searchtable_descriptionlength         | 500           |
|                      | Single Result Redirect                      | quickedit_env_search_singleresultredirect           | true          |
|                      | Display Calendar with 24 Months             | quickedit_env_calendar_numberofmonths               | false         |
| <b>System</b>        | Owner Field Mandatory                       | quickedit_env_owner_mandatory                       | false         |
|                      | Copy Job Owner                              | quickedit_env_copy_jobowner                         | true          |
|                      | User Defined Job Description Maximum Length | quickedit_env_job_userdefined_maxlength_description | 50            |
|                      | User Defined Job Command Maximum Length     | quickedit_env_job_userdefined_maxlength_command     | 50            |
|                      | Proxy Server Name                           | quickedit_env_proxy_server                          |               |
|                      | Proxy Port                                  | quickedit_env_proxy_port                            |               |

|  |                 |                              |  |
|--|-----------------|------------------------------|--|
|  | Proxy User Name | quickedit_env_proxy_user     |  |
|  | Proxy Password  | quickedit_env_proxy_password |  |

### Quick View

The following table maps Quick View preference categories and fields in Configuration to the attributes in the QUICKVIEW configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category  | Preference Field Name            | Exported Attribute Name              | Default Value |
|----------------------|----------------------------------|--------------------------------------|---------------|
| <b>User Defaults</b> | Rows per Page for Search Results | quickview_env_pageSize               | 50            |
|                      | Rows per Page for Reports        | quickview_env_reports                | 50            |
|                      | Search History Items             | quickview_env_searchHistorySize      | 30            |
|                      | Job Details View                 | quickview_env_showTreeView           | TREE_VIEW     |
|                      | Maximum Flow Height              | quickview_env_flowScreenSize         | 400           |
|                      | Job Flow Initial State           | quickview_env_flowInitialState       | EXPANDED      |
|                      | Show Dependent Jobs              | quickview_env_showDependentJobs      | true          |
|                      | Show Starting Conditions         | quickview_env_showStartingConditions | true          |
|                      | Show Job Runs                    | quickview_env_showJobRuns            | true          |

### Reporting

The following table maps Reporting preference categories and fields in Configuration Manager to the attributes in the REPORTING configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category     | Preference Field Name  | Exported Attribute Name                            | Default Value |
|-------------------------|------------------------|----------------------------------------------------|---------------|
| <b>Collectors</b>       | Enable Collector       | reporting_env_collectors_collectorsActive          | true          |
|                         | Collector Run Interval | reporting_env_collectors_frequencyCollectorSeconds | 60            |
| <b>Database Cleanup</b> | Enable Cleanup         | reporting_env_collectors_cleanupActive             | true          |
|                         | Cleanup Interval       | reporting_env_collectors_frequencyCleanupDays      | 2             |
|                         | Purge Data             | reporting_env_collectors_cleanupDataOlderThanDays  | 30            |

### Resources

The following table maps Resources preference categories and fields in Configuration Manager to the attributes in the RESOURCES configuration file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category   | Preference Field Name        | Exported Attribute Name       | Default Value |
|-----------------------|------------------------------|-------------------------------|---------------|
| <b>Table Controls</b> | Rows per Page for All Tables | resources_env_table_row_count | 50            |

|                                |                                     |                                      |    |
|--------------------------------|-------------------------------------|--------------------------------------|----|
|                                | Rows per Page for All Dialog Tables | resources_env_table_dialog_row_count | 20 |
| <b>Search History Controls</b> | Search History Items                | resources_env_searchhistory_maxitems | 30 |

### Server Preference Names

Tables are provided that map the server preference fields in Configuration Manager to the attributes listed in the file exported by the Configuration Batch Interface.

### Prerequisites for AutoSys Workload Automation Server Definition in Web UI Configuration

When you define a AutoSys Workload Automation server in Configuration, note the following:

- An agent must be installed and running on the application server machine. This is not required for core AutoSys Workload Automation. If the application server is on the AutoSys Workload Automation Scheduler machine, an agent is already installed (with the Scheduler) and the requirement is met.
- There must be a machine definition in AutoSys Workload Automation for the agent whose name matches the DNS alias associated with the application server host name.
- In Configuration, the name specified in the Application Server Host Address field is case-sensitive and must match the name in the AutoSys Workload Automation application server agent machine definition. If the machine definition in AutoSys Workload Automation is a fully-qualified host name (machine.domain.com) then the host name in Configuration must also use the fully-qualified host name.
- If multiple host names are specified in the Application Server Host Address field, each host name specified must meet these requirements.

#### NOTE

For more information, see *Configuration Help*.

### AutoSys Workload Automation Server

The following table maps AutoSys Workload Automation server preference categories and fields in Configuration to the attributes in the file exported by the Configuration Batch Interface. Default values are also listed.

| Preference Category       | Preference Field Name           | Exported Attribute Name | Default Value | Required |
|---------------------------|---------------------------------|-------------------------|---------------|----------|
| <b>General</b>            | Server Name                     | <command value>         |               | yes      |
|                           | Instance Name                   | instance_name           |               | yes      |
|                           | Cluster Name                    | cluster_name            | Default       | yes      |
|                           | Product Version                 | version                 | 11.3          | no       |
| <b>Application Server</b> | Application Server Host Address | host_address            |               | yes      |
|                           | Application Server Host Port    | port                    | 9000          | yes      |
|                           | Web Services URL                | web_services_url        |               | no       |
| <b>Compatibility</b>      | Legacy Compatibility Port       | legacy_compat_port      | 4444          | yes      |
| <b>Monitor User</b>       | Monitor ID                      | monitor_id              |               | yes      |
|                           | Monitor Password                | monitor_password        |               | yes      |
| <b>Preference</b>         | EEM Enabled                     | enable_eem              | true          | no       |

|                   |                 |                 |         |    |
|-------------------|-----------------|-----------------|---------|----|
|                   | Time Zone       | time_zone_id    |         | no |
| <b>Encryption</b> | Encryption Key  | encryption_key  |         | no |
|                   | Encryption Type | encryption_type | DEFAULT | no |

## Administer the Filter Job Types Batch Interface

The Filter Job Types Batch Interface lets you create an authorized list of job types using the Windows command line or UNIX console for all users. The default authorized list provides a list of all job types supported in AutoSys Workload Automation. Because your enterprise may not use all of the supported AutoSys Workload Automation job types, as an administrator, you can remove unused job types from the list, and then add them at a later time if required. You can also review the list at any time and make modifications as needed. This list is used to populate the Filter Job Types tab in the Customize dialog in Quick Edit and Application Editor, where the list of job types can be further limited by individual users.

The command syntax for the Filter Job Types Batch Interface is as follows:

```
filter_jobtypes -u user_name -p password -t application -l | -r [all|
job_type1,job_type2] | -a [all|job_type1,job_type2]
```

### NOTE

- You must use the list (-l), add (-a), or remove (-r) attribute in a single command, but you can only use one.
- You must use the application (-t) attribute in a single command.

*Filter Job Types Batch Interface Help* contains procedures and examples that explain the command options. Many options for the filter\_jobtypes command can be specified using either the short or long form of the option. The syntax for all short forms uses one preceding hyphen, while the syntax for all long forms uses two. For example, the list option can be specified using either of the following:

- Short form: filter\_jobtypes -l
- Long form: filter\_jobtypes --list

### NOTE

- The Filter Job Types Batch interface executable is a batch file (.bat) on Windows and a shell script (.sh) on UNIX.
- On UNIX, if Web UI was installed by the root user, only the root user has permission to run this batch interface.
- On UNIX, if Web UI was installed with the Run As User option, the run as user has permission to run this batch interface. In that case, the root user should not run this batch interface because the run as user cannot then run it successfully. However, if the root user does run the batch interface, change the group and owner of the file and folder in \$CA\_WCC\_INSTALL\_LOCATION/log to the run as user.
- Batch Interface command arguments are case-sensitive. You must use the case shown in this help for each argument.

## List the Job Types in the Job Types Filtered List

You can use the Filter Job Types Batch Interface to list the job types available, so you can decide if the current access is appropriate.

### NOTE

- Certain job types are only supported by Autosys Workload Automation Release 11.3.5 and Release 11.3.6.
- If this is the first time you are using the batch interface, all job types supported in AutoSys Workload Automation are available in the list.

To view the authorized list of job types, use the following command syntax:

```
filter_jobtypes -u user_name -p password -t application_name -l
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-t, --target-application *application\_name***  
Specifies to access only those job types contained in the authorized list of job types for the specified application. You can use either the short name (-t) or the long name (--target-application). The options for *application\_name* are either qedit for Quick Edit or appedit for Application Editor.
- **-l, --list**  
Specifies to display the contents of the authorized list of job types. You can use either the short name (-l) or the long name (--list).

#### Example: Display all job types in the authorized list for Quick Edit

The following command produces a list of all job types in the authorized list for Quick Edit:

```
filter_jobtypes -u ejmcommander -p ejmcommander -t qedit -l
```

The results of the above command will display a list similar to the following:

Available job types: cmd, box, fw, sql, ftp, entybean, etc.

Unavailable types: dbproc, dbmon, dbtrig, jmsspub

#### Add Job Types to the Job Types Filtered List

You can use the Filter Job Types Batch Interface to add job types to the authorized list of job types when previously-removed job types become supported in your enterprise, or when a user requires the addition of a specific job type to complete his tasks.

#### NOTE

- If this is the first time you are using the batch interface, all job types supported in AutoSys Workload Automation are available in the list.
- In a High Availability environment, you must manually propagate any changes that you have made to job types filtered list. To do so, copy the following directories from the Web UI server on which you have made the changes to all other Web UI servers in the High Availability group:
  - *WCC\_installation\_root\data\config\application\ae*
  - *WCC\_installation\_root\data\config\application\qe*

To add job types to the authorized list, use the following command syntax:

```
filter_jobtypes -u user_name -p password -t application_name -a
[all|job_type1,job_type2]
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-t, --target-application *application\_name***

Specifies to access only those job types contained in the authorized list of job types for the specified application. You can use either the short name (-t) or the long name (--target-application). The options for *application\_name* are either qedit for Quick Edit or appedit for Application Editor.

- **-a, --add *all|job\_type1, job\_type2***

Specifies the job types to add to the authorized list of job types. You can use either the short name (-a) or the long name (--add).

#### Example: Add specific job types to the authorized list in Quick Edit

The following command adds the command, box, and file watcher job types to the authorized list for Quick Edit:

```
filter_jobtypes -u ejmcommander -p ejmcommander -t qedit -a cmd,box,fw
```

#### Remove Job Types from the Filtered List

You can use the Filter Job Types Batch Interface to remove job types from the authorized list of job types when those job types are not supported in your enterprise.

#### NOTE

- If this is the first time you are using the batch interface, all job types supported in AutoSys Workload Automation are available in the list.
- In a High Availability environment, you must manually propagate any changes that you have made to job types filtered list. To do so, copy the following directories from the Web UI server on which you have made the changes to all other Web UI servers in the High Availability group:
  - *WCC\_installation\_root\data\config\application\ae*
  - *WCC\_installation\_root\data\config\application\qe*

To remove job types from the authorized list, use the following command syntax:

```
filter_jobtypes -u user_name -p password -t application_name -r
[all|job_type1, job_type2]
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-t, --target-application *application\_name***

Specifies to access only those job types contained in the authorized list of job types for the specified application. You can use either the short name (-t) or the long name (--target-application). The options for *application\_name* are either qedit for Quick Edit or appedit for Application Editor.

- **-r, --remove *all/job-type1, job-type2***

Specifies the job types to be removed from the authorized list of job types. You can use either the short name (-r) or the long name (--remove).

### Example: Remove specific job types from the authorized list of job types in Application Editor

The following command removes the command, box, and file watcher job types from the authorized list of job types in Application Editor:

```
filter_jobtypes -u ejmcommander -p ejmcommander -t appedit -r cmd,box,fw
```

### Display Help

You can display the inline help, which describes the options for the filter\_jobtypes command.

To display help, use the following command syntax:

```
filter_jobtypes -?
```

#### NOTE

You can use either the short name (-?) or the long name (--help).

## Administer the Job Template Batch Interface

The Job Template Batch Interface lets the administrator of your enterprise create and store job templates containing common property values for specified job types. These templates can then be used in Quick Edit and Application Editor to create new jobs. A job template can be created from a text file, written in a JIL command-like structure that contains a job property name followed by a colon, space, and value (for example, dependency: v(DEFAULT\_GLOBAL\_VAR)), or it can be created from an exported JIL file. The Job Template Batch Interface is run from the Windows command line or UNIX console.

#### NOTE

Job templates appear in the Create Object and Create Job dialogs in Quick Edit and in the Palette in Application Editor only if the associated job type is available.

As an administrator, you can use the Job Template Batch Interface to do the following:

- Import a file to Web UI as a named job template, and make it available for a specified user or all users.
- Delete a template for a specified user or all users.
- View a list of templates available for a specified user or all users.
- Export a job template by name from a server and save it as a text file.
- Export a template from one server and import it to another server.
- Remove a job template from a server, and then import a text file to the same server as a job template with the same name.

The command syntax for the Job Templates Batch Interface is as follows:

```
job_template -u user_name -p password -g | -f user_id -l | -r template_name | -
i template_name file_name | -e template_name file_name [-o] [-?]
```



**NOTE**

- You must use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.
- You must use the list (-l), remove (-r), import (-i), or export (-e) attribute in a single command, but you can only use one.

*Job Template Batch Interface Help* contains procedures and examples that explain the command options. Many options for the `job_template` command can be specified using either the short or long form of the option. The syntax for all short forms uses one preceding hyphen, while the syntax for all long forms uses two. For example, the list option can be specified using either of the following:

- Short form: `job_template -l`
- Long form: `job_template --list`

**NOTE**

- The Job Template Batch Interface executable is a batch file (.bat) on Windows and a shell script (.sh) on UNIX.
- On UNIX, if Web UI was installed by the root user, only the root user has permission to run this batch interface.
- On UNIX, if Web UI was installed with the Run As User option, the run as user has permission to run this batch interface. In that case, the root user should not run this batch interface because the run as user cannot then run it successfully. However, if the root user does run the batch interface, change the group and owner of the file and folder in `$CA_WCC_INSTALL_LOCATION/log` to the run as user.
- Batch Interface command arguments are case-sensitive. You must use the case shown in this help for each argument.

**Import a Text File for Use as a Template**

You can use the Job Template Batch Interface to upload a specified text file and make it available as a job template for given job type.

**NOTE**

In a High Availability environment, you must manually propagate any changes that you have made to job templates. To do so, copy the following directory from the Web UI server on which you have made the changes to all other Web UI servers in the High Availability group:

```
WCC_installation_root\data\config\application\editor
```

To save a file as a job template, use the following command syntax:

```
job_template -u user_name -p password -g | -f user_id [-o] -i template_name file_name
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-g, --global**  
Specifies to access the job templates for all users. You can use either the short name (-g) or the long name (--global).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-f, --for-user *user\_id***

Specifies to access the job templates assigned to a particular user. You can use either the short name (-f) or the long name (--for-user).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-o, --overwrite**

Specifies to overwrite an existing job template with an imported job template that has the same name. You can use either the short name (-o) or the long name (--overwrite).

**NOTE**

If you do not specify the -o option, and you attempt to import a text file as a job template to a server that already contains a template with the same name, an error message is displayed.

- **-i, --import *template\_name file\_name***

Specifies to import the text file and make it available as a template with a specified named. You can use either the short name (-i) or the long name (--import).

### **Example: Import a text file to create a job template for all users and overwrite the existing template**

The following command imports the Tmpl1.txt file located on hard drive c:\ to all users as a job template named POJO\_Tmpl1:

```
job_template -u ejmcommander -p ejmcommander -g -o - i POJO_Tmpl1 c:\Tmpl1.txt
```

### **Example: Import a text file to create a job template for all users with no overwrite option**

The following command imports the Tmpl1.txt file located on hard drive c:\ to all users as a job template named POJO\_Tmpl1:

```
job_template -u ejmcommander -p ejmcommander -g - i POJO_Tmpl1 c:\Tmpl1.txt
```

### **Example: Import a text file to create a job template for a specific user**

The following command imports the Tmpl2.txt file located on hard drive c:\ to User\_A as a job template named POJO\_Tmpl2:

```
job_template -u ejmcommander -p ejmcommander -f User_A - i POJO_Tmpl2 c:\Tmpl2.txt
```

### **Export a Job Template to a Text File**

You can use the Job Template Batch Interface to retrieve a job template with a specified name and save it to a file with a new name. You can edit the file and use it to create a new job template as needed.

To export an existing job template and save it to a text file, use the following command syntax:

```
job_template -u user_name -p password -g | -f user_id -e template_name file_name
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-g, --global**

Specifies to access the job templates for all users. You can use either the short name (-g) or the long name (--global).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-f, --for-user *user\_id***

Specifies to access the job templates assigned to a particular user. You can use either the short name (-f) or the long name (--for-user).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-e, --export *template\_name file\_name***

Specifies to export the job template and save it as a text file with a new name. You can use either the short name (-e) or the long name (--export).

**Example: Export a job template from a single user and save it as a text file**

The following command exports the Template1\_Command job template assigned to User\_A, and saves it to a text file named User\_Command\_Tmpl1.txt:

```
job_template -u ejmcommander -p ejmcommander - f User_A - e Template1_Command c:
\User_Command_Tmpl1.txt
```

**Example: Export a job template from all users and save it as a text file**

The following command exports the Template2\_Command job template assigned to all users, and saves it to a text file named User\_Command\_Tmpl2.txt:

```
job_template -u ejmcommander -p ejmcommander - g - e Template2_Command c:
\User_Command_Tmpl2.txt
```

**List the Job Templates**

You can use the Job Template Batch Interface to list the job templates available to all users or to a specified user, so you can decide if the current availability is appropriate.

To view the list of job templates, use the following command syntax:

```
job_template -u user_name -p password -g |-f user_id -l
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-g, --global**

Specifies to access the job templates for all users. You can use either the short name (-g) or the long name (--global).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-f, --for-user *user\_id***

Specifies to access the job templates assigned to a particular user. You can use either the short name (-f) or the long name (--for-user).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-l, --list**

Specifies to display the list of the available job templates. You can use either the short name (-l) or the long name (--list).

**Example: Display the job templates for all users**

The following command produces a list of the job templates created for all users:

```
job_template -u ejmcommander -p ejmcommander -g -l
```

**Example: Display the job templates for a specified user**

The following command produces a list of job templates available for User\_A:

```
job_template -u ejmcommander -p ejmcommander -f User_A -l
```

**Remove a Job Template**

When you no longer need a specific job template, you can use the Job Template Batch Interface to remove that job template from the templates available to all users or to a specified user.

**NOTE**

In a High Availability environment, you must manually propagate any changes that you have made to job templates. To do so, copy the following directory from the Web UI server on which you have made the changes to all other Web UI servers in the High Availability group:

```
WCC_installation_root\data\config\application\editor
```

To remove job templates that are no longer needed, use the following command syntax:

```
job_template -u user_name -p password -g | -f user_id -r template_name |
template_name1,template_name2
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-g, --global**

Specifies to access the job templates for all users. You can use either the short name (-g) or the long name (--global).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-f, --for-user *user\_id***

Specifies to access the job templates assigned to a particular user. You can use either the short name (-f) or the long name (--for-user).

**NOTE**

You can use either the global (-g) or the for-user (-f) attribute in a single command, but you cannot use both.

- **-r, --remove *template\_name|template\_name1, template\_name2***

Specifies to remove the job template with the specified name. You can remove multiple templates in a single command by entering the template names in a comma-separated list. You can use either the short name (-r) or the long name (--remove).

**Example: Remove a specific job template for all users**

The following command removes the User\_Defined\_CMD\_Tmpl1 job template for all users:

```
job_template -u ejmcommander -p ejmcommander - -global --remove User_Defined_CMD_Tmpl1
```

**Example: Remove a specific job template for a specified user**

The following command removes the User\_Defined\_CMD\_Tmpl2 job template for User\_A:

```
job_template -u ejmcommander -p ejmcommander --for-user User_A --remove
User_Defined_CMD_Tmpl2
```

**Display Help**

You can display the inline help, which describes the options for the job\_template command.

To display help, use the following command syntax:

```
job_template -?
```

**NOTE**

You can use either the short name (-?) or the long name (--help).

**Administer the Monitoring Batch Interface**

The Monitoring Batch Interface lets you insert, modify, and delete Web UI monitoring views and alert policies from the Windows command line or UNIX console as an alternative to using the Monitoring application. You can export and import existing views and alert policies from one Web UI server to another, or create a text file that lets you insert, modify, or delete views and alert policies. An exported file can be modified using a text editor.

You can use the Monitoring Batch Interface to do the following:

- Propagate the views and alert policies from one Web UI server to other Web UI servers. For example, you can export the views from the server that contains them, copy the exported text file to the servers on which you want to use the

same views, and then import them on each of those servers. You can use the same process to migrate Monitoring views from a Autosys Web UI Release 11.3, Autosys Web UI Release 11.3.5, and Autosys Web UI Release 11.3.6 servers to a Web UI server for this release.

- Add, modify, or delete multiple views or alert policies in one operation. For example, you can export the views, modify the columns to be displayed, and import the modified views.
- Add multiple objects to an exported file using the copy and paste feature of your text editor. For example, if you want to add several views, you can create one view in Monitoring, export it, open the text file and copy the properties for it, paste an instance of them for each new view you want to add, and then modify the properties as necessary.

*Monitoring Batch Interface Help* contains procedures and examples that explain the command options. Many options for the `wcc_monitor` command can be specified using either the short or long form of the option. The syntax for all short forms uses one preceding hyphen, while the syntax for all long forms uses two. For example, the import option can be specified using either of the following:

- Short form: `wcc_monitor -i`
- Long form: `wcc_monitor --import`

#### NOTE

- The Monitoring Batch Interface executable is a batch file (.bat) on Windows and a shell script (.sh) on UNIX.
- To use this batch interface, the user that is specified in the command must have rights to the associated application in the CA EEM ApplicationAccess policy.
- On UNIX, if Web UI was installed by the root user, only the root user has permission to run this batch interface.
- On UNIX, if Web UI was installed with the Run As User option, the run as user has permission to run this batch interface. In that case, the root user should not run this batch interface because the run as user cannot then run it successfully. However, if the root user does run the batch interface, change the group and owner of the file and folder in `$CA_WCC_INSTALL_LOCATION/log` to the run as user.
- Batch Interface command arguments are case-sensitive. You must use the case shown in this help for each argument.

### **Propagate View and Alert Policy Definitions to Other Servers**

After you create view and alert policy definitions on one Web UI server, you can use the export and import options of the Monitoring Batch Interface to set up other Web UI servers with the same definitions.

The process for propagating views and alert policies from one Web UI server to other Web UI servers is as follows:

1. Export all view and alert policy definitions from the Web UI server where they were created.
2. Copy the text file created by the export to the Web UI server where you want the same definitions.
3. Import the text file on the server.
4. Repeat Steps 2 and 3 for each Web UI server to which you want the views and policies copied.

### **Export View and Alert Policy Definitions**

You can export Web UI view and alert policy definitions from the server to a text file. You can modify the resulting file using a text editor, and then import it back to the server. By default, all view and alert policy definitions are exported.

To export view and alert policy definitions, use the following command syntax:

```
wcc_monitor -u user_name -p password -x output_file [-v]
```

- **-u, --user user\_name**

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-x, --export *output\_file***

Specifies the .txt file to which the view definitions are exported and, optionally, its path. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-x) or the long name (--export).

**NOTE**

If a file with the specified name already exists, the file is overwritten.

- **-w, --views [*view1*][*view2*]...**

(Optional) Specifies which views are to be exported. You can use either the short name (-w) or the long name (--view).

**NOTE**

You can enter multiple view names in a comma-separated list.

- **-P, --policies [*policy1*][*policy2*]...**

Specifies which policies are to be exported. You can use either the short name (-P) or the long name (--policy).

**NOTE**

You can enter multiple alert policy names in a comma-separated list.

- **-v, --verbose**

(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

**NOTE**

When logging is enabled, the output is written to the standard out and to the log file. The log file is located in *install\_location/Logs/application/cli/wcc\_monitor.log*.

### Example: Export all views and policies

The following command exports all views, their filters, alert policies, and jobs to the file named output.txt:

```
wcc_monitor -u ejmcommander -p ejmcommander -x output.txt
```

### Example: Export selected views

The following example exports the specified views to the file named export.txt:

```
wcc_monitor -u ejmcommander -p ejmcommander -x export.txt --views View1,View2
```

### Example: Export selected alert policies

The following command exports the specified alert policies to a text file:

```
wcc_monitor -u ejmcommander -p ejmcommander -x export.txt --policies Policy1,Policy2
```

## Import View and Alert Policy Definitions

You can import view and alert policy definitions to the Web UI server.

**NOTE**

When a view references an alert policy, the alert policy must be imported before the view. If not, the filter in the view will not be imported.

To import the view or alert policy export file, use the following command syntax:

```
wcc_monitor -u user_name -p password -i input_file [-v]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-i, --import input\_file**  
Specifies the fully-qualified name of the text file to import. You can use either the short name (-i) or the long name (--import).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

**Example: Import a text file**

The following command imports all views and filters included in the file input.txt to the Web UI server:

```
wcc_monitor -u ejmcommander -p ejmcommander -i input.txt
```

**Clear the Server Cache**

You can clear all the jobs and alerts in the database cache for one or more servers. The cache will then be automatically rebuilt.

**NOTE**

- AutoSys Workload Automation alarms are backed up by the AutoSys Workload Automation server and are automatically rebuilt. However, new job status change alerts are generated only if the job status matches the alert policy conditions. These alerts have a new create date and time and an open status. Job status change alerts, not matching the current job status, are removed and cannot be regenerated.
- After the cache has been deleted, the run interval defined in Configuration is ignored, and the collectors for the jobs and alarms run immediately.

To clear the cache, use the following command syntax:

```
wcc_monitor -u user_name -p password -d server1[,server2] [-v]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).



**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-d, --delete ALL, *server1*[,*server2*]**

Specifies the names of the servers for which to clear the cache. You can use either the short name (-d) or the long name (--delete).

**NOTE**

- You can enter multiple server names in a comma-separated list.
- You can use the percent (%) in a server name to match more than one server.
- You can use ALL to clear the cache for all servers provided ALL is not a valid server name.

- **-r, --reload**

Specifies when the monitoring data (such as job statuses) is inconsistent with data in the flow. Reload is an action from the internal database to an internal flow model.

- **-v, --verbose**

(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

**Example: Clear the cache for a specified server**

The following command clears the cache for the server named serverAutoSys:

```
wcc_monitor -u ejmcommander -p ejmcommander -d serverAutoSys
```

**Encrypt Command Line Arguments**

You can create a file that contains encrypted arguments for the wcc\_monitor command. This lets you distribute a file with arguments, such as the user name and password, that cannot be viewed in plain text by the recipient. For example, you can create a job or script so that another user with lower security access can run the wcc\_monitor command with the encrypted file at a convenient time.

To encrypt command line arguments, use the following command syntax:

```
wcc_monitor -u user_name -p password -e encrypted_file -i input_file | -x output_file |
-d server[,server2]
```

- **-u, --user *user\_name***

Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).

- **-p, --password *password***

Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

**NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-e, --encrypt *encrypted\_file***

(Optional) Specifies the fully-qualified file name to which the arguments should be encrypted and saved. The file is created by the encryption; however, the path must exist. If you do not specify a file name, the file output will be written to the console. You can use either the short name (-e) or the long name (--encrypt).

- **-i, --import *input\_file***

Specifies the fully-qualified name of the text file to import. You can use either the short name (-i) or the long name (--import).

#### NOTE

You can specify either the import option or the export option, but not both.

- **-x, --export *output\_file***

Specifies the file to which the views are exported and, optionally, its path. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch utility. You can use either the short name (-x) or the long name (--export).

#### NOTE

You can specify either the import option or the export option, but not both.

- **-d, --delete ALL, *server1*[,*server2*]**

Specifies the names of the servers for which to clear the cache. You can use either the short name (-d) or the long name (--delete).

#### NOTE

- You can enter multiple server names in a comma-separated list.
- You can use the percent (%) in a server name to match more than one server.
- You can use ALL to clear the cache for all servers provided ALL is not a valid server name.

### Example: Create a file of encrypted arguments

The following command encrypts and stores the user name, password, and export command arguments to a file named *arg\_file*:

```
wcc_monitor -u ejmcommander -p ejmcommander -e arg_file -x export.txt
```

#### NOTE

The user name and password are required when using -x or -i. They must be specified in the command line or included in the encrypted file.

### Run a Command from an Encrypted File

You can run the *wcc\_monitor* command with a previously-created file that contains encrypted arguments, including the user name and password.

#### NOTE

Arguments specified in the command line override the arguments specified in the encrypted file.

To run *wcc\_monitor* with encrypted arguments, use the following command syntax:

```
wcc_monitor encrypted_file
```

- ***encrypted\_file***

Specifies the fully-qualified file name that contains the encrypted arguments.

### Example: Run *wcc\_monitor* with an encrypted arguments file

The following command runs *wcc\_monitor* and decrypts the arguments in the file named *arg\_file*:

```
wcc_monitor arg_file
```

## Display Help

You can display the inline help, which describes the options for the `wcc_monitor` command.

To display help, use the following command syntax:

```
wcc_monitor -?
```

### NOTE

You can use either the short name (-?) or the long name (--help).

## Work with View Definitions in a Text File

You can create a new text file or edit an existing one to insert or delete monitoring view definitions and their filters.

The following commands are available:

- `insert_view: view_name`
- `modify_view: view_name`
- `delete_view: view_name`
- `insert_filter: server_name`
- `modify_filter: server_name`
- `delete_filter: server_name`

### NOTE

- Text is case-sensitive.
- If an invalid attribute is encountered during the import or export of a view definition file, then the command on which the error occurred fails, and the script continues with the next command in the definition file.

## Insert a View Definition

You can insert a view definition in a text file and then import the file back to the Web UI server.

Use the following syntax to insert a view definition:

```
insert_view: view_name attributes
```

- **view\_name**

Specifies a name for the view. The name is required.

The following attributes are available:

- **render\_flow: true|false**

Specifies whether the Flow section is displayed.

**Default:** true

- **job\_columns: text**

Specifies the columns to display in the Jobs table. Multiple names can be entered, separated by commas.

The following options are available:

- agent\_status
- agent\_lstatus
- application
- box\_name
- description
- duration
- end\_time
- exit\_code
- group
- instance
- job\_name
- job\_type
- joid
- machine
- next\_start
- owner
- run\_machine
- server
- start\_time
- status
- status\_time
- timezone

**Default:** job\_name,status,server,run\_machine,status\_time

– **alert\_columns**

Specifies the columns to display in the Alerts table. Multiple names can be entered, separated by commas. The following options are available:

- alert
- alert\_type
- severity
- status
- job
- server
- create\_time
- status\_time
- response
- comment
- text
- eoid
- modified\_user
- alert\_policy
- url

**Default:** alert, severity, status, job, server, create\_time, status\_time

– **description: text**

Specifies a description for the view.

**Limits:** 0-255 alphanumeric characters

**Example: Insert a view**

The following example inserts a view named East-Accounting, which includes all jobs with names starting with EAcct that run on the AE1 server:

```
insert_view: East-Accounting
insert_filter: AE1
view: East-Accounting
filter: job_name like EAcct%
```

### **Modify a View Definition**

You can modify a view definition in a text file and then import the file back to the Web UI server.

Use the following syntax to modify a view definition:

```
modify_view: view_name attributes
```

- ***view\_name***  
Specifies a name for the view. The name is required.  
The following attributes are available:
  - **render\_flow: *true|false***  
Specifies whether the Flow section is displayed.  
**Default:** true
  - **job\_columns: *text***  
Specifies the columns to display in the Jobs table. Multiple names can be entered, separated by commas.  
The following options are available:
    - agent\_status
    - agent\_lstatus
    - application
    - box\_name
    - description
    - duration
    - end\_time
    - exit\_code
    - group
    - instance
    - job\_name
    - job\_type
    - joid
    - machine
    - next\_start
    - owner
    - run\_machine
    - server
    - start\_time
    - status
    - status\_time
    - timezone**Default:** job\_name,status,server,run\_machine,status\_time
  - **alert\_columns**

Specifies the columns to display in the Alerts table. Multiple names can be entered, separated by commas.

The following options are available:

- alert
- alert\_type
- severity
- status
- job
- server
- create\_time
- status\_time
- response
- comment
- text
- eoid
- modified\_user
- alert\_policy
- url

**Default:** alert, severity, status, job, server, create\_time, status\_time

– **description: text**

Specifies a description for the view.

**Limits:** 0-255 alphanumeric characters

### **Delete a View Definition**

You can delete a view definition in a text file and then import the file back to the Web UI server.

Use the following syntax to delete a view definition:

```
delete_view: view_name
```

- **view\_name**  
Specifies a name for the view. The name is required.

### **Example: Delete a view definition**

The following command deletes the SampleView definition:

```
delete_view: SampleView
```

### **Insert a Filter**

You can insert a view filter in a text file and then import the file back to the Web UI server.

Use the following syntax to insert a filter.

- **server\_name**  
Specifies the name of the server. The name is required.

The following attributes are available:

- **view: view\_name**  
Specifies the monitoring view to which this filter is attached. The view name must match a previously-defined monitoring view. The name is required.
- **filter: string** Specifies a logical expression against the specified server. All criteria must be represented by a job property, followed by an operator and a value. Multiple defined criteria must be linked with either AND or OR. Balanced

parentheses are used to enforce interpretation. The following operators are allowed: <, =, >, <=, >=, !=, LIKE, and NOT\_LIKE.

The following properties are available:

- *job\_name operator name* -- Includes all jobs with the specified name. A wildcard can be used.
- *owner operator name* -- Includes all jobs with the specified owner. A wildcard can be used.
- *job\_type operator text* -- Includes all jobs with the specified job type. A wildcard can be used.
- *application operator name* -- Includes all jobs that belong to the application. A wildcard can be used.
- *group operator name* -- Includes all jobs that belong to the group. A wildcard can be used.
- *box operator name* -- Includes all jobs in the named box, including all child boxes and their contents. Only = and != are allowed.
- *box\_name operator name* -- Includes only the jobs directly contained in the named box, not including child boxes. A wildcard can be used.
- *timezone operator timezone* -- Includes all jobs with the specified time zone. A wildcard can be used.
- *machine operator name* -- Includes all jobs that have been defined to run on the specified machine.
- *status operator status* -- Includes all jobs with the specified job status. A wildcard can be used.
- *started\_in\_past operator time* -- Includes all jobs that started within the past number of days and hours. You can specify days, hours, or both. For example, you can enter '1d 10h'.
- *run\_machine operator name* -- Includes all jobs that are running or that last ran on the specified machine.

#### NOTE

- Where indicated, you can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represent zero or more characters. You can also use the question mark (?) to represent any single character.
- Jobs are removed from the started\_in\_past range in 15 minute intervals. If a job has a start time within the refresh range, they are listed in the view until the next refresh.
- **alert\_policy: *policy name***  
(Optional) Specifies the name of the alert policy to attach to this view.
- **active: *true|false***  
(Optional) Specifies whether the filter is active. If the filter is active, then the jobs and collections that match the filter are displayed in the view. If the filter is not active, then the jobs and box jobs are not displayed in the view, but the filter definition remains as part of the view.  
**Default:** true

#### Modify a Filter

You can modify a view filter in a text file and then import the file back to the Web UI server.

Use the following syntax to modify a filter:

```
modify_filter: server_name attributes
```

- **server\_name**  
Specifies the name of the server. the name is required.

The following attributes are available:

- **view: *view\_name***

Specifies the monitoring view to which this filter is attached. The view name must match a previously-defined monitoring view. The name is required.

- **filter: *string*** Specifies a logical expression against the specified server. All criteria must be represented by a job property, followed by an operator and a value. Multiple defined criteria must be linked with either AND or OR. Balanced parentheses are used to enforce interpretation. The following operators are allowed: <, =, >, <=, >=, !=, LIKE, and NOT\_LIKE.

The following properties are available:

- **job\_name operator name** Includes all jobs with the specified name. A wildcard can be used.
- **owner operator name** Includes all jobs with the specified owner. A wildcard can be used.
- **job\_type operator text** Includes all jobs with the specified job type. A wildcard can be used.
- **application operator name** Includes all jobs that belong to the application. A wildcard can be used.
- **group operator name** Includes all jobs that belong to the group. A wildcard can be used.
- **box operator name** Includes all jobs in the named box, including all child boxes and their contents. Only = and != are allowed.
- **box\_name operator name** Includes only the jobs directly contained in the named box, not including child boxes. A wildcard can be used.
- **timezone operator timezone** Includes all jobs with the specified time zone. A wildcard can be used.
- **machine operator name** Includes all jobs that have been defined to run on the specified machine.
- **status operator status** Includes all jobs with the specified job status. A wildcard can be used.
- **started\_in\_past operator time** Includes all jobs that started within the past number of days and hours. You can specify days, hours, or both. For example, you can enter '1d 10h'.
- **run\_machine operator name** Includes all jobs that are running or that last ran on the specified machine.

#### NOTE

- Where indicated, you can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represent zero or more characters. You can also use the question mark (?) to represent any single character.
- Jobs are removed from the started\_in\_past range in 15 minute intervals. If a job has a start time within the refresh range, they are listed in the view until the next refresh.
- **alert\_policy: *policy name***
- (Optional) Specifies the name of the alert policy to attach to this view.
- **active: *true|false***  
(Optional) Specifies whether the filter is active. If the filter is active, then the jobs and collections that match the filter are displayed in the view. If the filter is not active, then the jobs and box jobs are not displayed in the view, but the filter definition remains as part of the view.  
**Default:** true

#### Delete a Filter

You can delete a filter in a text file and then import the file back to the Web UI server.

Use the following syntax to delete a filter:

```
delete_filter: server_name
```

- **server\_name**  
Specifies the name of the server. The name is required.

The following attribute is required:



- **view: *view\_name***  
Specifies the monitoring view to which this filter is attached. The view name must match a previously-defined monitoring view.

#### Example: Delete a filter

The following command deletes a filter on server autosys-nc3 for the view SampleView:

```
delete_filter: autosys-nc3
view: SampleView
```

### Work with Alert Policy Definitions in a Text File

You can create a new alert policy file or edit an existing one to insert or delete alert policy definitions.

The following commands are available:

- insert\_alert\_policy: *policy\_name*
- modify\_alert\_policy: *policy\_name*
- delete\_alert\_policy: *policy\_name*

#### NOTE

- Text is case-sensitive.
- If an invalid attribute is encountered in the definition file, then the command on which the error occurred fails, and the script continues with the next command in the definition file.

### Insert an Alert Policy Definition

Use the following syntax to insert an alert policy definition:

```
insert_alert_policy: policy_name attributes
```

- ***policy\_name***  
Specifies the name of the alert policy. The name is required and must be unique.  
The following attributes are available:
  - **job\_status: *status***  
Indicates a job status with which to associate the policy.  
The following options are available:

- Running
  - Starting
  - Success
  - Failure
  - Terminated
  - Suspended
  - On Ice
  - Inactive
  - Activated
  - Restart
  - On Hold
  - Queue Wait
  - Wait Reply
  - Pending Machine
  - Waiting for Resource
  - On No Execution
  - Unknown
- **alarm\_type: *alarm\_name***  
Specifies one or more alarm types to include in the policy.  
The following options are available:

- \*
- APP\_SERVER\_SHUTDOWN (545)
- AUTO\_PING (526)
- DATABASE\_COMM (516)
- DB\_PROBLEM (523)
- DB\_ROLLOVER (519)
- CPI\_UNAVAILABLE (536)
- DUPLICATE\_EVENT (524)
- EP\_HIGH\_AVAIL (522)
- EP\_ROLLOVER (520)
- EP\_SHUTDOWN (521)
- KILLJOBFAIL (540)
- MACHINE\_UNAVAILABLE (532)
- OB\_SHUTDOWN (527)
- EXTERN\_DEPS\_ERROR (529)
- INSTANCE\_UNAVAILABLE (525)
- VERSION\_MISMATCH (518)
- CHASE (514)
- CPI\_JOB\_NAME\_INVALID (535)
- FORKFAIL (501)
- JOBFailure (503)
- JOBNOT\_ONICEHOLD (509)
- EVENT\_HDLR\_ERROR (507)
- EVENT\_QUE\_ERROR (508)
- MAX\_RETRYS (505)
- MAXRUNALARM (510)
- MINRUNALARM (502)
- MISSING\_HEARTBEAT (513)
- MUST\_COMPLETE\_ALARM (538)
- MUST\_START\_ALARM (537)
- REPLY\_RESPONSE\_FAIL (542)
- RESOURCE (512)
- RESTARTJOBFAIL (544)
- RETURN\_RESOURCE\_FAIL (543)
- SENDSIGFAIL (541)
- SERVICEDESK\_FAILURE (533)
- STARTJOBFAIL (506)
- UNINOTIFY\_FAILURE (534)
- WAIT\_REPLY\_ALARM (539)

**Note:** Entering \* indicates to associate the policy with all alarm types.

– **job\_name:** *job\_pattern*

Specifies a job or group of jobs with which to associate the policy.

**Note:** You can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represents zero or more characters. You can also use the question mark (?) to represent any single character.

– **box\_name:** *boxjob\_pattern*

Specifies a box job or group of box jobs with which to associate the policy.

**Note:** You can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represents zero or more characters. You can also use the question mark (?) to represent any single character.

- **severity: *state***  
Specifies the severity level for the alert, such as Critical, High, Medium, or Low.
- **server: *server\_name***  
(Optional) Specifies the name of the server or servers to which to monitor global job alerts. If this attribute is specified, the alerts are displayed in the Alerts table on the Servers details page.
- **active: *true|false***  
Indicates whether the policy will be applied.  
**Default:** true
- **status: *event\_state***  
Specifies one or more alert states to include, such as Open, Acknowledged, or Closed.
- **text: *text***  
(Optional) Describes the alert.  
**Limits:** 255 characters
- **url\_name: *url\_name***  
(Optional) Specifies the display name of the link to content related to the alert.
- **url\_link: *url\_link***  
(Optional) Indicates the URL at which content related to the alert resides.

### Example: Insert an Alert Policy

The following definition creates an alert policy named FailedJobs that lets you monitor job status alerts for failed jobs and JOBFAILURE and STARTJOBFAIL alerts for all jobs on servers AS1 and AS2. Only those alerts with a status of Open or Acknowledge will be displayed in the Alerts table with an assigned severity of High.

```
insert_alert_policy: FailedJobs
job_status: Failure
alarm_type: JOBFAILURE, STARTJOBFAIL
job_name: *
severity: High
server: AS1, AS2
status: OPEN, ACKNOWLEDGED
active: true
```

### Modify an Alert Policy

You can modify an alert policy in a text file and then import the file back to the Web UI server.

Use the following syntax to modify an alert policy:

```
modify_alert_policy: policy_name attributes
```

- ***policy\_name***  
Specifies the name of the alert policy. The name is required.  
The following attributes are available:
  - **job\_status: *status***  
Indicates a job status with which to associate the policy.  
The following options are available:

- Running
  - Starting
  - Success
  - Failure
  - Terminated
  - Suspended
  - On Ice
  - Inactive
  - Activated
  - Restart
  - On Hold
  - Queue Wait
  - Wait Reply
  - Pending Machine
  - Waiting for Resource
  - On No Execution
  - Unknown
- **alarm\_type: *alarm\_name***  
Specifies one or more alarm types to include in the policy.  
The following options are available:

- \*
- APP\_SERVER\_SHUTDOWN (545)
- AUTO\_PING (526)
- DATABASE\_COMM (516)
- DB\_PROBLEM (523)
- DB\_ROLLOVER (519)
- CPI\_UNAVAILABLE (536)
- DUPLICATE\_EVENT (524)
- EP\_HIGH\_AVAIL (522)
- EP\_ROLLOVER (520)
- EP\_SHUTDOWN (521)
- KILLJOBFAIL (540)
- MACHINE\_UNAVAILABLE (532)
- OB\_SHUTDOWN (527)
- EXTERN\_DEPS\_ERROR (529)
- INSTANCE\_UNAVAILABLE (525)
- VERSION\_MISMATCH (518)
- CHASE (514)
- CPI\_JOB\_NAME\_INVALID (535)
- FORKFAIL (501)
- JOBFailure (503)
- JOBNOT\_ONICEHOLD (509)
- EVENT\_HDLR\_ERROR (507)
- EVENT\_QUE\_ERROR (508)
- MAX\_RETRYS (505)
- MAXRUNALARM (510)
- MINRUNALARM (502)
- MISSING\_HEARTBEAT (513)
- MUST\_COMPLETE\_ALARM (538)
- MUST\_START\_ALARM (537)
- REPLY\_RESPONSE\_FAIL (542)
- RESOURCE (512)
- RESTARTJOBFAIL (544)
- RETURN\_RESOURCE\_FAIL (543)
- SENDSIGFAIL (541)
- SERVICEDESK\_FAILURE (533)
- STARTJOBFAIL (506)
- UNINOTIFY\_FAILURE (534)
- WAIT\_REPLY\_ALARM (539)

**Note:** Entering \* indicates to associate the policy with all alarm types.

– **job\_name:** *job\_pattern*

Specifies a job or group of jobs with which to associate the policy.

**Note:** You can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represents zero or more characters. You can also use the question mark (?) to represent any single character.

– **box\_name:** *boxjob\_pattern*

Specifies a box job or group of box jobs with which to associate the policy.

**Note:** You can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represents zero or more characters. You can also use the question mark (?) to represent any single character.

- **severity: *state***  
Specifies the severity level for the alert, such as Critical, High, Medium, or Low.
- **server: *server\_name***  
(Optional) Specifies the name of the server or servers to which to monitor global job alerts. If this attribute is specified, the alerts are displayed in the Alerts table on the Servers details page.
- **active: *true|false***  
Indicates whether the policy will be applied.  
**Default:** true
- **status: *event\_state***  
Specifies one or more alert states to include, such as Open, Acknowledged, or Closed.
- **text: *text***  
(Optional) Describes the alert.  
**Limits:** 255 characters
- **url\_name: *url\_name***  
(Optional) Specifies the display name of the link to content related to the alert.
- **url\_link: *url\_link***  
(Optional) Indicates the URL at which content related to the alert resides.

### Delete an Alert Policy

You can delete an alert policy definition in a text file and then import the file back to the Web UI server.

Use the following syntax to delete an alert policy:

```
delete_alert_policy:policy_name
```

- **policy\_name**  
Specifies a name for the alert policy. The name is required.

### Example: Delete an event policy definition

The following command deletes the JobFailure alert policy definition:

```
delete_alert_policy: JobFailure
```

## Configure AutoSys Web UI Using Configuration Application

Use the tasks in this section to configure AutoSys Web UI using the Configuration application. For more information about related concepts or user interface components, see the *Configuration Help*. These tasks are duplicated in *Configuration Help*, which can be opened in a separate browser window to have them available while you perform your configuration.

### Set Up Servers

Initial AutoSys Web UI setup is performed using the Setup section of Configuration, which contains options for performing the following tasks:

- Add, edit, or delete servers for AutoSys Web UI

#### NOTE

For information about editing or deleting servers, see *Configuration Help*.

- Verify the status of all scheduling manager servers currently configured in AutoSys Web UI

## **Create a Server Using the Property Fields**

If you want information that is collected from a specific back-end server to be displayed in AutoSys Web UI, you must define that server to AutoSys Web UI.

### **Follow these steps:**

1. Click Create.
2. Complete the fields in the Create Server section as appropriate, and click Save.

#### **NOTE**

The server configuration is saved and is available for information retrieval.

#### **NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to AutoSys Web UI.

## **Create One or More Servers Using the SMC**

As an alternative to completing property fields to define a server, you can use the Server Management Console (SMC) to define servers with preset commands.

### **Follow these steps:**

1. Click SMC in the Servers section.
2. Click the `insert_ae_server` link in the Commands pane to add a AutoSys Workload Automation server definition. The command and its required attributes are displayed in the Command pane.
3. Edit the server definition as appropriate.

#### **NOTE**

- Every server must have a unique name.
- The Cluster Name and Instance Name combination must be unique for each AutoSys Workload Automation server.

4. Repeat Steps 2 and 3 for each server definition you want to create.

#### **NOTE**

To add additional server definitions, you can click the `insert_ae_server` link in the Commands pane for the next server you want to create, use the browser copy and paste commands to create additional server definitions, or do both.

5. Click Execute.  
AutoSys Web UI verifies the validity of the server definitions. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed. The servers are created and are available for information retrieval.

#### **NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to AutoSys Web UI.

## **Copy a Server**

After you set up one server definition for a certain server type, you can use that definition as the basis for creating new definitions of that type to reduce data entry.

### **Follow these steps:**

1. Select the Select check boxes to the left of the servers that you want to copy. Next, select Edit in SMC from the Select and drop-down list, and click Go.



2. Change the `modify_ae_server` command to `insert_ae_server` for each server and edit the server definitions as appropriate.

**NOTE**

- Every server must have a unique name.
- The Cluster Name and Instance Name combination must be unique for each AutoSys Workload Automation server.
- The `instance_name` parameter must be added to each AutoSys Workload Automation server definition you modify.

3. Click Execute.

AutoSys Web UI verifies the validity of the server definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed. The server is created and is available for information retrieval.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to AutoSys Web UI.

### **Verify a Server's Status**

To update the current status of a server, or to view the error message for a server when a problem is encountered, we recommend that you validate the status of the server definition in Configuration.

To verify a server status, select the check box to the left of the server in the Server Results section. Next, select Validate from the Select and drop-down list, and click Go.

A status message is displayed at the top of the page and the Server Results section refreshes to display the status of the selected server.

**Follow these steps:**

1. Find the server that you want to verify.
2. Select the check box to the left of that server in the Server Results section.
3. Select Validate from the Select and drop-down list, and click Go.  
A status message is displayed at the top of the page and the Server Results section refreshes to display the status of the selected server.

### **Edit Application Preferences**

You can use the Preferences section to change the general properties of the applications for this AutoSys Web UI instance.

**Follow these steps:**

1. Select the appropriate application from the View/Modify Preferences drop-down list in the Preferences section, and click Go.
2. Edit the fields as appropriate, and click Save.  
The properties are saved and are now in effect.

## **Configure the Web UI Server**

This article explains how to configure the AutoSys Web UI server on Windows. After the installation is complete, you must prepare AutoSys Web UI for use by configuring it.

To successfully configure AutoSys Web UI, follow these steps:

1. Add users to AutoSys Web UI through the CA EEM interface. You can add users by referencing an Active Directory as the source.
2. Customize your CA EEM policies as necessary to assign the appropriate user access rights.
3. Open Configuration and do the following:
  - a. Create AutoSys Web UI server definitions for each back-end server you want AutoSys Web UI to access, assigning the appropriate values for the properties of that server.

**NOTE**

- The Server Name you use must be unique for each server.
- Each AutoSys Workload Automation server definition must specify a user ID and password for the monitor user. A monitor user gathers job data from the AutoSys Workload Automation instance on behalf of all AutoSys Web UI users.

- b. Edit the configuration preferences for the various AutoSys Web UI applications in Configuration as appropriate.

**NOTE**

You can also use the Configuration Batch Interface to perform many of the Configuration functions.

4. (Optional) [Add credentials for the appropriate back-end servers using the Configuration Batch Interface, the Credentials application, or both as appropriate.](#)

**NOTE**

If you want to define a global user for enterprise-wide access to an AutoSys Workload Automation instance, enter “\_GLOBAL\_: *username*” in the Credentials application and assign that user a password. Only one global user can be added for each server.

5. (Optional) [Customize the Session Timeout value to maximize session security.](#)

## **Configure AutoSys Web UI to Work with CA EEM**

CA EEM is used for security. The three key objects of CA EEM are identities (users and user groups), resource classes, and policies. CA EEM is used to secure AutoSys Web UI objects, providing the following capabilities:

- **Authentication**  
CA EEM authenticates the user. The authenticated user can then be used in subsequent authorization processing.
- **Authorization**  
CA EEM provides access to a user for a particular resource. A resource can be any logical or physical entity. In AutoSys Web UI, the typical resource is a user interface component (for example, tab, command, drop-down list, and so on). Authorization is controlled by a set of policies associated with a resource class. These policies are the primary way to integrate CA EEM with AutoSys Web UI.

**NOTE**

For more information about configuring AutoSys Web UI to work with CA EEM, see the *Security section*.

## **Users**

The following CA EEM users are created by default during the AutoSys Web UI installation:

- **EJADMIN**  
Provides system configuration access for AutoSys Web UI. Typically, these users are system administrators or security administrators.  
**User name:** ejmadmin  
**Password:** ejmadmin
- **EJMEEXEC**

Provides access to summary information on job status for AutoSys Web UI. Typically, these users are executives and managers from other disciplines.

**User name:** ejmexec

**Password:** ejmexec

- **EJMSCHEDULER**

Provides a basic level of access for AutoSys Web UI. Typically, these users are console operators.

**User name:** ejmoperator

**Password:** ejmoperator

- **EJMSUPERVISOR**

Provides a level of access that includes the ability to create and modify jobs and job objects. Typically, these users are job administrators.

**User name:** ejmscheduler

**Password:** ejmscheduler

- **EJMSUPERVISOR**

Provides a higher level of access than that granted to the scheduler. Typically, these users are scheduling or monitoring supervisors and are responsible for creating views.

**User name:** ejmsupervisor

**Password:** ejmsupervisor

- **EJMCOMMANDER**

Provides the highest level of access. These users have access to all in AutoSys Web UI features. Typically, these users have a combination of the rights of the system administrator and the scheduling supervisor.

**User name:** ejmcommander

**Password:** ejmcommander

These users are set up in CA EEM, each with a default display in AutoSys Web UI, letting them access the applications to which they have rights.

Typically, you would use either the system administrator (ejmadmin) or the superuser (ejmcommander) to perform system configuration.

## User Groups

Although CA EEM can provide access authentication that is set up for individuals, the best practice for policy creation is to assign groups to policies and include individual users in those groups, instead of assigning individual users directly to the policies. This enables you to maintain stable policies even when individuals' access rights change.

You can assign any of the following types of groups:

- **Global Group**

Includes users across all applications in the current CA EEM instance.

- **Application Group**

Includes users in the currently selected application only. For example, you can create an application group based on a specific business function, such as Payroll. The users will be included in the group only if they are part of this function.

- **Dynamic Groups**

Includes users based on their characteristics. For example, you can create a dynamic group based on parameters such as office, city, and title. The users will be automatically included in the group only if they meet the parameters set. Dynamic groups are only available for the currently selected CA EEM application.

**Note:** For more information about user groups, see the CA Embedded Entitlements Manager documentation.

## User Roles

A number of preconfigured user roles provided with AutoSys Web UI are used in the default CA EEM policy definitions. These roles are defined as follows:

- **Console Operator** The *console operator* is responsible for monitoring job streams, correcting minor errors that are resolved through forcing job status or through one-time overrides of data. The console operator also reports significant application errors to other operators and to the application owners. To complete these tasks, console operators use Monitoring to monitor the jobs for which they are responsible. Optionally, they may use Reporting to generate server statistics and job run activity reports. Console operators may also use Quick View to view the detailed properties for a particular job in the flow and Forecast to create reports that show jobs that started or that are scheduled to start during a specific time period.
- **Scheduler**  
The *scheduler* is responsible for the creation and maintenance of the job definitions. The schedulers use Quick Edit and Application Editor to create, modify, and delete machines, jobs, calendars, and other job-related information such as constraints. Schedulers may also monitor job streams to make sure that they are running properly. Schedulers receive reports about problems with jobs that have already run, and they might need to look at historical information to analyze and correct a job stream.
- **Supervisor**  
The *supervisor* role performs all the tasks of a scheduler, and an additional level of tasks to help oversee and assist all the schedulers. The supervisor also performs other tasks such as defining job streams and monitoring views.
- **Administrator**  
The *administrator* is responsible for configuring and maintaining the AutoSys Web UI environment. This includes defining the scheduling manager servers, configuring software-specific properties, and defining host connection links for mainframe systems (if applicable). System administrators may also be responsible for security, and for creating users and user groups assigned to specific roles.
- **Executive**  
The *executive*, or *manager*, has a very different set of needs compared to the other roles. The executive's tasks include monitoring, at the highest level, the overall status of the scheduling managers operating in their area. They can use Reporting to generate and view reports. The executive also uses the Views widget to track the status of their views.
- **Security Administrator**  
The *security administrator* is responsible for configuring and maintaining security, and creating users and user groups assigned to specific roles. The security administrator also creates and maintains policies for CA EEM.
- **Commander**  
The *commander* is a superuser created as a convenience, and is a member of all the application groups.

The following table shows the application groups and default users for each role:

| Role                   | Application Group      | Default User              |
|------------------------|------------------------|---------------------------|
| Console Operator       | Console Operator       | ejmoperator               |
| Supervisor             | Supervisor             | ejmsupervisor             |
| Scheduler              | Scheduler              | ejmscheduler              |
| Administrator          | Administrator          | ejmadmin                  |
| Executive              | Executive              | ejmexec                   |
| Security Administrator | Security Administrator | ejmadmin and ejmcommander |
| Commander              | Commander              | ejmcommander              |

### **Configure CA EEM to Reference Active Directory Global Users**

Configuring CA EEM to reference the Active Directory global users enables the applications using CA EEM, such as AutoSys Workload Automation and AutoSys Web UI, to share the user information from the Active Directory.

#### **NOTE**

You may want to configure filters so that users do not access all entries in the global Active Directory.

**Follow these steps:**

1. Access the CA EEM Home page.
2. Click the Configure tab.  
The Configure page opens.
3. Click the User Store link.  
The User Store page appears.
4. Click User Store.  
The User Store Configuration page appears in the right pane.
5. Select the Reference from an external LDAP directory option.  
The Directory Information section appears.
6. Leave the default configuration type as Basic LDAP Directory and click Add external LDAP directory.  
Additional fields relating to the external directory appear.
7. Leave the default attribute map as Microsoft Active Directory, and complete the following fields:
  - a. Enter a name for the configuration.
  - b. Enter the host computer on which Active Directory is running.
  - c. Enter the port on which Active Directory is listening and click



The default port to enter is 389.

**NOTE**

Enter a port number other than 389 only if the Active Directory Server administrator has reconfigured the default Active Directory Server port.

The host and port appear in the Selected Hostnames section.

- d. (Optional) Add other host computers and ports to the Selected Hostnames section as needed.

**NOTE**

Other host computers can be added only if Active Directory failover is configured.

- e. (Optional) Select the appropriate protocol from the Protocol drop-down list. The default is LDAP.

**NOTE**

You can select a different protocol if TLS or SSL is enabled in the Active Directory.

- f. Enter the Base DN. Only global users and groups discovered under this DN are mapped to CA EEM.
  - g. Enter the User DN that is used to attach to the Active Directory server.
  - h. Enter the password for the user specified in the User DN field. This value is required.
  - i. Re-enter the password in the Confirm Password field.
  - j. (Optional) Click Show Advanced Configuration and enter the values in the fields based on the enterprise configuration.
8. Click Save.  
The values you have specified are saved, and CA EEM is configured to the Active Directory.

**NOTE**

- You can configure CA EEM to reference users from any external LDAP directory. To reference users from another external directory, edit the attribute map.
- For more information about referencing users from an external LDAP directory, see the *CA Embedded Entitlements Manager Online Help*.

**AutoSys Web UI Resource Classes**

Resource classes are used in CA EEM to classify logical or physical entities. A resource class contains the following information:

- Name
- Actions (List)
- Named Attributes (List)

The following resource classes are available for the AutoSys Web UI application. The classes can be categorized into the following types:

- **Access Resource Classes**

The ApplicationAccess and ExternalApplicationAccess classes control access to applications. The WidgetAccess class controls access to widgets. The ServerAccess class controls access to servers. The ConfigurationControl class lets you configure additional levels of access to certain application features, such as those in Credentials.

- **Action or Log Resource Classes**

The AlertAction, LogAccess, CommandSetup, and CommandExecute classes control access to alert actions, job and scheduler logs, and Enterprise Command Line commands respectively.

- **Object Access Resource Classes**

The MonitorViewControl, ReportTemplateAccess, ObjectAccess, and ObjectControl classes are used to control specific types of access to a variety of objects that are displayed and maintained by AutoSys Web UI. Each of the actions that can be performed on these objects is enumerated in the resource actions. The name of the object is specified for the resource name. There are typically attributes specified for server type and server name. These resource classes are only implemented in server-side code.

## **CA EEM Policies**

Policies define the access rights of a particular user or user group to a particular resource. They associate identities with resources. There are one or more default policies for each resource class. The default policy for each class is delivered with the AutoSys Web UI application in CA EEM.

## **Configure AutoSys Web UI Using Configuration**

Configuration is an AutoSys Web UI application that lets you configure workload automation servers. In addition to this initial setup, Configuration lets you set application preferences for the individual AutoSys Web UI applications.

You can access Configuration by clicking the Configuration tab on the AutoSys Web UI main page.

### **NOTE**

For more information, such as conceptual information or field-level descriptions, see *Configuration Help*.

## **Set Up a Monitor User**

A monitor user gathers job information in the background from an AutoSys Workload Automation server on behalf of all AutoSys Web UI users. To do this, special user authentication is used with AutoSys Workload Automation.

A monitor user must be set up on the AutoSys Workload Automation server and in the AutoSys Web UI definition for the AutoSys Workload Automation server.

A Monitor user is created when your system administrator adds a server using Configuration. Only a user with system administrator rights can change the Monitor user name and password, either in Configuration, or using the Credentials application.

The monitor user must have permissions to retrieve or access jobs and machines from AutoSys Workload Automation.

When you set up the monitor user on the AutoSys Workload Automation server, one of the following is required, depending on the type of security used:

- If AutoSys Workload Automation native security is used, the password for *user@host* specified for the monitor user must be defined in *autosys\_secure*, and the user must be specified as a SUPER EDIT user.
- If CA EEM is used for security, the monitor user specified must be a valid user on the CA EEM server (*@host* has no meaning in the context of CA EEM). The CA EEM server defined to AutoSys Workload Automation will be used for authentication.

The monitor user must have at least Read access to the following policies in CA EEM:

- – as-job
- as-appl
- as-group
- as-list
- as-machine

#### **NOTE**

For more information about Configuration, see *Configuration Help*. For more information about the Credentials application, see *Credentials Help*.

### **Add New Credentials for a Server**

To set up credentials for a new user on a back-end server, you can enter the credentials directly into the Credentials application.

#### **NOTE**

You can add multiple credentials to a server.

#### **Follow these steps:**

1. Select the option button next to the appropriate server.
2. Enter a new user ID in the User Name field.
3. Enter the password for that user ID in the Password and Confirm Password fields.
4. Click Add.

The new credentials for the back-end server are saved.

### **Configure Session Timeout**

A login timeout is the time that a user session can remain idle before timing out and releasing the session. When your session times out, you are redirected to the session page. To modify the session timeout period, use the Configuration application. Sessions also expire if the CA-wcc service is stopped or restarted. The user is then required to log in again to access AutoSys Web UI applications.

If you are experiencing a significant number of session time-outs, modify the Session Timeout value to accommodate expected usage.

#### **NOTE**

For more information about setting session time-outs, see *Configuration Help*.

### **Batch Interface Utilities**

Several batch interface utilities are available for performing AutoSys Web UI functions from the Windows command line or UNIX console. The batch interface utilities replicate the features of the associated applications and also let you perform batch actions.

#### **NOTE**

To use a batch interface, you must be authorized to use the associated application.

Help for each batch interface can be found in the following locations:

## Windows

`WCC_installation_root\LauncherServer\webapps\wcc\Documentation\Help\ BatchInterfaces`

## UNIX

`WCC_installation_root/LauncherServer/webapps/wcc/Documentation/Help/ BatchInterfaces`

The batch interfaces included with AutoSys Web UI are as follows:

- **Configuration Batch Interface**

Lets you modify AutoSys Web UI configurations containing application preferences. You can also add, modify, or delete server definitions from the Windows command line or UNIX console as an alternative to using the Configuration application to define the configuration. You can export, import, and deploy one or more AutoSys Web UI configurations. An exported configuration can be modified using a text editor. Additionally, this batch interface lets you update the CA EEM server definition in AutoSys Web UI, change the Tomcat server port, change the Tomcat stop port, change the event bus port, and change the SSL mode, after installation.

*Configuration Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Configuration_Batch_Interface_Help.html`.

- **Filter Job Types Batch Interface**

Lets you create a list of authorized job types for all users. The list is used to populate the Filter Job Types tab in the Customize dialog that you can access in Quick Edit and Application Editor.

*Filter Job Types Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Filter_Job_Types_Batch_Interface_Help.html`.

- **Job Template Batch Interface**

Lets you create job templates for use with Quick Edit that contain common job properties. You can then select a job template from the Create Job dialog in Quick Edit and create a job without the need to enter common job properties.

*Job Template Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Job_Template_Batch_Interface_Help.html`.

- **Monitoring Batch Interface**

Lets you insert, modify, or delete AutoSys Web UI monitoring views from the Windows command line or UNIX console as an alternative to using the Monitoring application. You can export and import existing views from one AutoSys Web UI r11.3.6 server to another, or create a text file that lets you insert, modify, or delete view definitions and their filters. An exported file can be modified using a text editor. Additionally, you can clear all the Monitoring jobs in the AutoSys Web UI database cache for one or more servers. The AutoSys Web UI database cache will then be automatically rebuilt.

*Monitoring Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Monitoring_Batch_Interface_Help.html`.

## Configure Upload File Size Limits on Windows

You can configure the size limit of the files that are uploaded to AutoSys Web UI. To set the parameters and avoid possible upload file failures in AutoSys Web UI, you can set limitations for importing files from the View Management Console (VMC), Enterprise Command Line (ECLI), or Quick Edit.

### NOTE

- After each configuration change, restart AutoSys Web UI services.
- Specify the file size limit in bytes.

### Follow these steps:

1. Configure the size limit for importing files from VMC as follows:
  - a. Open the `web.xml` file that is located in the `WCC_INSTALLATION_ROOT/tomcat/webapps/jsc-rest/WEB-INF` directory.
  - b. Locate the following section:



```

<!-- File upload configuration parameters -->
<context-param>
<param-name>com.ca.wcc.jsc.vmc.FILE_UPLOAD_LIMIT</param-name>
<!-- Use 10,000K -->
<param-value>10240000</param-value>
</context-param>

```

- c. Edit the value within `<param-value>` and `</param-value>` elements to specify the file size limit and save the file.
2. Configure the size limit for importing files from ECLI as follows:
  - a. Open the `web.xml` file that is located in the `WCC_INSTALLATION_ROOT/tomcat/webapps/eccli/WEB-INF` directory.
  - b. Locate the following section:

```

<!-- File upload configuration parameters -->
<context-param>
<!-- Maximum memory per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_MEMORY</param-name>
<!-- Use 500K -->
<param-value>512000</param-value>
</context-param>
<context-param>
<!-- Maximum disk space per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_DISK_SPACE</param-name>
<!-- Use 50,000K -->
<param-value>51200000</param-value>
</context-param>

```

- c. Edit both values within `<param-value>` and `</param-value>` elements to specify the file size limits and save the file.
3. Configure the size limit for importing files from Quick Edit (WSDL Wizard) as follows:
  - a. Open the `web.xml` file that is located in the `WCC_INSTALLATION_ROOT/tomcat/webapps/quick-edit/WEB-INF` directory.
  - b. Locate the following section:

```

<!-- File upload configuration parameters -->
<context-param>
<!-- Maximum memory per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_MEMORY</param-name>
<!-- Use 500K -->
<param-value>512000</param-value>
</context-param>
<context-param>
<!-- Maximum disk space per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_DISK_SPACE</param-name>
<!-- Use 20,000K -->
<param-value>20480000</param-value>
</context-param>

```

- c. Edit both values within `<param-value>` and `</param-value>` elements to specify the file size limits and save the file.

**NOTE**

The value that is used for the maximum memory size parameter must be lower than the value used for the maximum disk size parameter.

## Configure the AutoSys Web UI Server on UNIX

This article explains how to configure the AutoSys Web UI server on UNIX. After the installation is complete, you must prepare AutoSys Web UI for use by configuring it.

To successfully configure AutoSys Web UI, follow these steps:

1. Add users to AutoSys Web UI through the CA EEM interface. You can add users by referencing an Active Directory as the source.
2. Customize your CA EEM policies as necessary to assign the appropriate user access rights.
3. Open Configuration and do the following:
  - a. Create AutoSys Web UI server definitions for each back-end server you want AutoSys Web UI to access, assigning the appropriate values for the properties of that server.

**NOTE**

- The Server Name you use must be unique for each server.
- Each AutoSys Workload Automation server definition must specify a user ID and password for the monitor user. A monitor user gathers job data from the AutoSys Workload Automation instance on behalf of all AutoSys Web UI users.

- b. Edit the configuration preferences for the various AutoSys Web UI applications in Configuration as appropriate.

**NOTE**

You can also use the Configuration Batch Interface to perform many of the Configuration functions.

4. (Optional) [Add credentials for the appropriate back-end servers using the Configuration Batch Interface, the Credentials application, or both as appropriate.](#)

**NOTE**

If you want to define a global user for enterprise-wide access to an AutoSys Workload Automation instance, enter “\_GLOBAL\_: *username*” in the Credentials application and assign that user a password. Only one global user can be added for each server.

5. (Optional) [Customize the Session Timeout value to maximize session security.](#)

### Configure AutoSys Web UI to Work with CA EEM

CA EEM is used for security. The three key objects of CA EEM are identities (users and user groups), resource classes, and policies. CA EEM is used to secure AutoSys Web UI objects, providing the following capabilities:

- **Authentication**  
CA EEM authenticates the user. The authenticated user can then be used in subsequent authorization processing.
- **Authorization**  
CA EEM provides access to a user for a particular resource. A resource can be any logical or physical entity. In AutoSys Web UI, the typical resource is a user interface component (for example, tab, command, drop-down list, and so on). Authorization is controlled by a set of policies associated with a resource class. These policies are the primary way to integrate CA EEM with AutoSys Web UI.

**NOTE**

For more information about configuring AutoSys Web UI to work with CA EEM, see the *Security section*.

### Users

The following CA EEM users are created by default during the AutoSys Web UI installation:

- **EJADMIN**  
Provides system configuration access for AutoSys Web UI. Typically, these users are system administrators or security administrators.  
**User name:** ejmadmin  
**Password:** ejmadmin
- **EJMEEXEC**  
Provides access to summary information on job status for AutoSys Web UI. Typically, these users are executives and managers from other disciplines.  
**User name:** ejmexec  
**Password:** ejmexec
- **EJMOPERATOR**  
Provides a basic level of access for AutoSys Web UI. Typically, these users are console operators.  
**User name:** ejmoperator  
**Password:** ejmoperator
- **EJMSCHEDULER**  
Provides a level of access that includes the ability to create and modify jobs and job objects. Typically, these users are job administrators.  
**User name:** ejmscheduler  
**Password:** ejmscheduler
- **EJMSUPERVISOR**  
Provides a higher level of access than that granted to the scheduler. Typically, these users are scheduling or monitoring supervisors and are responsible for creating views.  
**User name:** ejmsupervisor  
**Password:** ejmsupervisor
- **EJMCOMMANDER**  
Provides the highest level of access. These users have access to all in AutoSys Web UI features. Typically, these users have a combination of the rights of the system administrator and the scheduling supervisor.  
**User name:** ejmcommander  
**Password:** ejmcommander

These users are set up in CA EEM, each with a default display in AutoSys Web UI, letting them access the applications to which they have rights.

Typically, you would use either the system administrator (ejmadmin) or the superuser (ejmcommander) to perform system configuration.

## User Groups

Although CA EEM can provide access authentication that is set up for individuals, the best practice for policy creation is to assign groups to policies and include individual users in those groups, instead of assigning individual users directly to the policies. This enables you to maintain stable policies even when individuals' access rights change.

You can assign any of the following types of groups:

- **Global Group**  
Includes users across all applications in the current CA EEM instance.
- **Application Group**  
Includes users in the currently selected application only. For example, you can create an application group based on a specific business function, such as Payroll. The users will be included in the group only if they are part of this function.
- **Dynamic Groups**  
Includes users based on their characteristics. For example, you can create a dynamic group based on parameters such as office, city, and title. The users will be automatically included in the group only if they meet the parameters set. Dynamic groups are only available for the currently selected CA EEM application.

## NOTE

For more information about user groups, see the CA Embedded Entitlements Manager documentation.

### User Roles

A number of preconfigured user roles provided with AutoSys Web UI are used in the default CA EEM policy definitions. These roles are defined as follows:

- **Console Operator** The *console operator* is responsible for monitoring job streams, correcting minor errors that are resolved through forcing job status or through one-time overrides of data. The console operator also reports significant application errors to other operators and to the application owners. To complete these tasks, console operators use Monitoring to monitor the jobs for which they are responsible. Optionally, they may use Reporting to generate server statistics and job run activity reports. Console operators may also use Quick View to view the detailed properties for a particular job in the flow and Forecast to create reports that show jobs that started or that are scheduled to start during a specific time period.
- **Scheduler**  
The *scheduler* is responsible for the creation and maintenance of the job definitions. The schedulers use Quick Edit and Application Editor to create, modify, and delete machines, jobs, calendars, and other job-related information such as constraints. Schedulers may also monitor job streams to make sure that they are running properly. Schedulers receive reports about problems with jobs that have already run, and they might need to look at historical information to analyze and correct a job stream.
- **Supervisor**  
The *supervisor* role performs all the tasks of a scheduler, and an additional level of tasks to help oversee and assist all the schedulers. The supervisor also performs other tasks such as defining job streams and monitoring views.
- **Administrator**  
The *administrator* is responsible for configuring and maintaining the AutoSys Web UI environment. This includes defining the scheduling manager servers, configuring software-specific properties, and defining host connection links for mainframe systems (if applicable). System administrators may also be responsible for security, and for creating users and user groups assigned to specific roles.
- **Executive**  
The *executive*, or *manager*, has a very different set of needs compared to the other roles. The executive's tasks include monitoring, at the highest level, the overall status of the scheduling managers operating in their area. They can use Reporting to generate and view reports. The executive also uses the Views widget to track the status of their views.
- **Security Administrator**  
The *security administrator* is responsible for configuring and maintaining security, and creating users and user groups assigned to specific roles. The security administrator also creates and maintains policies for CA EEM.
- **Commander**  
The *commander* is a superuser created as a convenience, and is a member of all the application groups.

The following table shows the application groups and default users for each role:

| Role                   | Application Group      | Default User              |
|------------------------|------------------------|---------------------------|
| Console Operator       | Console Operator       | ejmoperator               |
| Supervisor             | Supervisor             | ejmsupervisor             |
| Scheduler              | Scheduler              | ejmscheduler              |
| Administrator          | Administrator          | ejmadmin                  |
| Executive              | Executive              | ejmexec                   |
| Security Administrator | Security Administrator | ejmadmin and ejmcommander |
| Commander              | Commander              | ejmcommander              |

## Configure CA EEM to Reference Active Directory Global Users

Configuring CA EEM to reference the Active Directory global users enables the applications using CA EEM, such as AutoSys Workload Automation and AutoSys Web UI, to share the user information from the Active Directory.

### NOTE

You may want to configure filters so that users do not access all entries in the global Active Directory.

### Follow these steps:

1. Access the CA EEM Home page.
2. Click the Configure tab.  
The Configure page opens.
3. Click the User Store link.  
The User Store page appears.
4. Click User Store.  
The User Store Configuration page appears in the right pane.
5. Select the Reference from an external LDAP directory option.  
The Directory Information section appears.
6. Leave the default configuration type as Basic LDAP Directory and click Add external LDAP directory.  
Additional fields relating to the external directory appear.
7. Leave the default attribute map as Microsoft Active Directory, and complete the following fields:
  - a. Enter a name for the configuration.
  - b. Enter the host computer on which Active Directory is running.
  - c. Enter the port on which Active Directory is listening and click



The default port to enter is 389.

### NOTE

Enter a port number other than 389 only if the Active Directory Server administrator has reconfigured the default Active Directory Server port.

The host and port appear in the Selected Hostnames section.

- d. (Optional) Add other host computers and ports to the Selected Hostnames section as needed.

### NOTE

Other host computers can be added only if Active Directory failover is configured.

- e. (Optional) Select the appropriate protocol from the Protocol drop-down list. The default is LDAP.

### NOTE

You can select a different protocol if TLS or SSL is enabled in the Active Directory.

- f. Enter the Base DN. Only global users and groups discovered under this DN are mapped to CA EEM.
  - g. Enter the User DN that is used to attach to the Active Directory server.
  - h. Enter the password for the user specified in the User DN field. This value is required.
  - i. Re-enter the password in the Confirm Password field.
  - j. (Optional) Click Show Advanced Configuration and enter the values in the fields based on the enterprise configuration.
8. Click Save.  
The values you have specified are saved, and CA EEM is configured to the Active Directory.

**NOTE**

- You can configure CA EEM to reference users from any external LDAP directory. To reference users from another external directory, edit the attribute map.
- For more information about referencing users from an external LDAP directory, see the *CA Embedded Entitlements Manager Online Help*.

**AutoSys Web UI Resource Classes**

Resource classes are used in CA EEM to classify logical or physical entities. A resource class contains the following information:

- Name
- Actions (List)
- Named Attributes (List)

The following resource classes are available for the AutoSys Web UI application. The classes can be categorized into the following types:

- **Access Resource Classes**

The `ApplicationAccess` and `ExternalApplicationAccess` classes control access to applications. The `WidgetAccess` class controls access to widgets. The `ServerAccess` class controls access to servers. The `ConfigurationControl` class lets you configure additional levels of access to certain application features, such as those in `Credentials`.

- **Action or Log Resource Classes**

The `AlertAction`, `LogAccess`, `CommandSetup`, and `CommandExecute` classes control access to alert actions, job and scheduler logs, and Enterprise Command Line commands respectively.

- **Object Access Resource Classes**

The `MonitorViewControl`, `ReportTemplateAccess`, `ObjectAccess`, and `ObjectControl` classes are used to control specific types of access to a variety of objects that are displayed and maintained by AutoSys Web UI. Each of the actions that can be performed on these objects is enumerated in the resource actions. The name of the object is specified for the resource name. There are typically attributes specified for server type and server name. These resource classes are only implemented in server-side code.

**CA EEM Policies**

Policies define the access rights of a particular user or user group to a particular resource. They associate identities with resources. There are one or more default policies for each resource class. The default policy for each class is delivered with the AutoSys Web UI application in CA EEM.

**Configure AutoSys Web UI Using Configuration**

Configuration is a AutoSys Web UI application that lets you configure workload automation servers. In addition to this initial setup, Configuration lets you set application preferences for the individual AutoSys Web UI applications.

You can access Configuration by clicking the Configuration tab on the AutoSys Web UI main page.

**NOTE**

For more information, such as conceptual information or field-level descriptions, see *Configuration Help*.

**Set Up a Monitor User**

A monitor user gathers job information in the background from an AutoSys Workload Automation server on behalf of all AutoSys Web UI users. To do this, special user authentication is used with AutoSys Workload Automation.

A monitor user must be set up on the AutoSys Workload Automation server and in the AutoSys Web UI definition for the AutoSys Workload Automation server.

A Monitor user is created when your system administrator adds a server using Configuration. Only a user with system administrator rights can change the Monitor user name and password, either in Configuration, or using the Credentials application.

The monitor user must have permissions to retrieve or access jobs and machines from AutoSys Workload Automation.

When you set up the monitor user on the AutoSys Workload Automation server, one of the following is required, depending on the type of security used:

- If AutoSys Workload Automation native security is used, the password for *user@host* specified for the monitor user must be defined in *autosys\_secure*, and the user must be specified as a SUPER EDIT user.
- If CA EEM is used for security, the monitor user specified must be a valid user on the CA EEM server (*@host* has no meaning in the context of CA EEM). The CA EEM server defined to AutoSys Workload Automation will be used for authentication.

The monitor user must have at least Read access to the following policies in CA EEM:

- – as-job
- as-appl
- as-group
- as-list
- as-machine

#### NOTE

For more information about Configuration, see *Configuration Help*. For more information about the Credentials application, see *Credentials Help*.

### **Add New Credentials for a Server**

To set up credentials for a new user on a back-end server, you can enter the credentials directly into the Credentials application.

#### NOTE

You can add multiple credentials to a server.

#### **Follow these steps:**

1. Select the option button next to the appropriate server.
2. Enter a new user ID in the User Name field.
3. Enter the password for that user ID in the Password and Confirm Password fields.
4. Click Add.

The new credentials for the back-end server are saved.

### **Configure Session Timeout**

A login timeout is the time that a user session can remain idle before timing out and releasing the session. When your session times out, you are redirected to the session page. To modify the session timeout period, use the Configuration application. Sessions also expire if the CA-wcc service is stopped or restarted. The user is then required to log in again to access AutoSys Web UI applications.

If you are experiencing a significant number of session time-outs, modify the Session Timeout value to accommodate expected usage.

#### NOTE

For more information about setting session time-outs, see *Configuration Help*.

## Batch Interface Utilities

Several batch interface utilities are available for performing AutoSys Web UI functions from the Windows command line or UNIX console. The batch interface utilities replicate the features of the associated applications and also let you perform batch actions.

### NOTE

To use a batch interface, you must be authorized to use the associated application.

Help for each batch interface can be found in the following locations:

### Windows

`WCC_installation_root\LauncherServer\webapps\wcc\Documentation\Help\ BatchInterfaces`

### UNIX

`WCC_installation_root/LauncherServer/webapps/wcc/Documentation/Help/ BatchInterfaces`

The batch interfaces included with AutoSys Web UI are as follows:

- **Configuration Batch Interface**

Lets you modify AutoSys Web UI configurations containing application preferences. You can also add, modify, or delete server definitions from the Windows command line or UNIX console as an alternative to using the Configuration application to define the configuration. You can export, import, and deploy one or more AutoSys Web UI configurations. An exported configuration can be modified using a text editor. Additionally, this batch interface lets you update the CA EEM server definition in AutoSys Web UI, change the Tomcat server port, change the Tomcat stop port, change the event bus port, and change the SSL mode, after installation.

*Configuration Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Configuration_Batch_Interface_Help.html`.

- **Filter Job Types Batch Interface**

Lets you create a list of authorized job types for all users. The list is used to populate the Filter Job Types tab in the Customize dialog that you can access in Quick Edit and Application Editor.

*Filter Job Types Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Filter_Job_Types_Batch_Interface_Help.html`.

- **Job Template Batch Interface**

Lets you create job templates for use with Quick Edit that contain common job properties. You can then select a job template from the Create Job dialog in Quick Edit and create a job without the need to enter common job properties.

*Job Template Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Job_Template_Batch_Interface_Help.html`.

- **Monitoring Batch Interface**

Lets you insert, modify, or delete AutoSys Web UI monitoring views from the Windows command line or UNIX console as an alternative to using the Monitoring application. You can export and import existing views from one AutoSys Web UI r11.3.6 server to another, or create a text file that lets you insert, modify, or delete view definitions and their filters. An exported file can be modified using a text editor. Additionally, you can clear all the Monitoring jobs in the AutoSys Web UI database cache for one or more servers. The AutoSys Web UI database cache will then be automatically rebuilt.

*Monitoring Batch Interface Help* contains procedures and examples that explain the command arguments. The name of the help file is `Monitoring_Batch_Interface_Help.html`.

## Configure Upload File Size Limits on UNIX

You can configure the size limit of the files that are uploaded to AutoSys Web UI. To set the parameters and avoid possible upload file failures in AutoSys Web UI, you can set limitations for importing files from the View Management Console (VMC), Enterprise Command Line (ECLI), or Quick Edit.



**NOTE**

- After each configuration change, restart AutoSys Web UI services.
- Specify the file size limit in bytes.

**Follow these steps:**

1. Configure the size limit for importing files from VMC as follows:
  - a. Open the web.xml file that is located in the *WCC\_INSTALLATION\_ROOT*/tomcat/webapps/jsc-rest/WEB-INF directory.
  - b. Locate the following section:
 

```
<!-- File upload configuration parameters -->
<context-param>
<param-name>com.ca.wcc.jsc.vmc.FILE_UPLOAD_LIMIT</param-name>
<!-- Use 10,000K -->
<param-value>10240000</param-value>
</context-param>
```
  - c. Edit the value within `<param-value>` and `</param-value>` elements to specify the file size limit and save the file.
2. Configure the size limit for importing files from ECLI as follows:
  - a. Open the web.xml file that is located in the *WCC\_INSTALLATION\_ROOT*/tomcat/webapps/ecli/WEB-INF directory.
  - b. Locate the following section:
 

```
<!-- File upload configuration parameters -->
<context-param>
<!-- Maximum memory per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_MEMORY</param-name>
<!-- Use 500K -->
<param-value>512000</param-value>
</context-param>
<context-param>
<!-- Maximum disk space per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_DISK_SPACE</param-name>
<!-- Use 50,000K -->
<param-value>51200000</param-value>
</context-param>
```
  - c. Edit both values within `<param-value>` and `</param-value>` elements to specify the file size limits and save the file.
3. Configure the size limit for importing files from Quick Edit (WSDL Wizard) as follows:
  - a. Open the web.xml file that is located in the *WCC\_INSTALLATION\_ROOT*/tomcat/webapps/quick-edit/WEB-INF directory.
  - b. Locate the following section:
 

```
<!-- File upload configuration parameters -->
<context-param>
<!-- Maximum memory per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_MEMORY</param-name>
<!-- Use 500K -->
<param-value>512000</param-value>
</context-param>
<context-param>
<!-- Maximum disk space per request (in bytes) -->
<param-name>org.apache.myfaces.trinidad.UPLOAD_MAX_DISK_SPACE</param-name>
```

```
<!-- Use 20,000K -->
<param-value>20480000</param-value>
</context-param>
```

- c. Edit both values within `<param-value>` and `</param-value>` elements to specify the file size limits and save the file.

#### NOTE

The value that is used for the maximum memory size parameter must be lower than the value used for the maximum disk size parameter.

## CA EEM Policy Migration

AutoSys Web UI security is implemented through CA EEM. You can migrate security policies from one CA EEM instance to a different CA EEM instance. For CA EEM policy migration details, see [CA EEM Security Data Replication, Backup, and Migration](#).

#### NOTE

For information about how to export and import EEM policies using the GUI, see [Export and Import EEM Policies \(using the GUI\)](#)

## Migrate ECLI Command Definitions

As an administrator, you may be responsible for migrating existing AutoSys Web UI r11.1 or r11.3 ECLI command definitions to AutoSys Web UI Release 11.4.

The ECLI Command Migration Utility (ECMU) lets you import commands defined in ECLI Global Commands and My Commands for all users from an AutoSys Web UI r11.1 or r11.3 server into an AutoSys Web UI Release 11.4 server database. Authorized users can then use the commands in ECLI.

#### NOTE

You must first copy the `cmdline` directory in the following path from the source AutoSys Web UI server to the directory of your choice on the target AutoSys Web UI server:

- **Windows**

```
%CA_WCC_INSTALL_LOCATION%\ConfigServer\config\application\
```

- **UNIX**

```
$CA_WCC_INSTALL_LOCATION/ConfigServer/config/application/
```

The ECMU executable is located in the `bin` folder under the AutoSys Web UI installation root. On Windows, it is the batch file (`ecli_migrate.bat`); on UNIX, it is the shell script (`ecli_migrate.sh`). Before you run the ECMU, change your current directory to the following directory that contains the `ecli_migrate` executable.

- **Windows**

```
%CA_WCC_INSTALL_LOCATION%\bin\
```

- **Unix**

```
$CA_WCC_INSTALL_LOCATION/bin/
```

To migrate ECLI command definitions, use the following command syntax:

```
ecli_migrate -u user -p password -s source_directory [-f] [-t] [-v] [?]
```

- **-u user**

Specifies the authorized user name.

**NOTE**

The specified user must have access rights to Configuration.

- **-p *password***  
Specifies the password for the authorized user name.
- **-s *source\_directory***  
Specifies the fully qualified directory containing the commands to import into the database.
- **-f, *force-update***  
(Optional) Overwrites existing commands with the commands in the specified source directory.
- **-t, *test***  
(Optional) Runs the migration in test mode. The changes are not saved.
- **-v, *verbose***  
(Optional) Print detailed processing messages.
- **-?, *help***  
(Optional) Displays a list of the available arguments.

**WARNING**

To ensure security, delete the cmdline directory from the target server after you migrate the command definitions.

**Example: Migrate All Definitions to an AutoSys Web UI Release 11.4 Server**

The following command migrates all ECLI commands from the directory c:\import\cmdline on the server on which the command is run:

```
ecli_migrate -u ejmcommander -p ejmcommander -s c:\import\cmdline
```

**Example: Migrate ECLI Commands****NOTE**

Suppose your company has upgraded AutoSys Web UI r11.1 or r11.3 to AutoSys Web UI Release 11.4. Because you do not want to recreate all of the ECLI command definitions, you can migrate them from the older AutoSys Web UI version to the current version.

**Migrate Job Flow Monitoring Objects to Monitoring**

As an administrator, you may be responsible for converting your existing AutoSys Web UI r11.1 SP1 through r11.1 SP4 Job Flow Monitoring views and filters into valid monitoring definitions in the current version of AutoSys Web UI Monitoring. To convert the information, you can run the available command line utilities from the Windows command line or UNIX console.

To migrate Job Flow Monitoring objects to Monitoring, follow these steps:

1. [Export the Job Flow Monitoring view definitions from AutoSys Web UI r11.1 SP1 using the wcc\\_jfd utility.](#)
2. [Copy the output file created in Step 1 to the current AutoSys Web UI server and run the jfd\\_migrate utility.](#)
3. [Import the file created in Step 2 into this version of AutoSys Web UI using the wcc\\_monitor utility.](#)

**NOTE**

- The `wcc_jfd` utility is not included with this release of AutoSys Web UI.
- For information about the `wcc_jfd` utility, see *Job Flow Batch Interface Help* in the AutoSys Web UI r11.1 SP1 documentation.
- Conversion of Job Flow Monitoring view definitions in AutoSys Web UI r11.1 SP1 through r11.1 SP4 are supported.
- Job Flows containing a single quote in the name cannot be migrated.
- Migration from other sources is not supported, even if the objects can be imported with the AutoSys Web UI r11.1 SP1 `wcc_jfd` utility.

**Example: Migrate Job Flow Monitoring Views from AutoSys Web UI r11.1 SP1 to Monitoring Views**

Suppose that your company has upgraded AutoSys Web UI r11.1 SP1 to AutoSys Web UI Release 11.4. Because you do not want to recreate all of the Job Flow Monitoring views, you can migrate them from the older AutoSys Web UI version to the current version.

The following table shows how objects are migrated from Job Flow Monitoring r11.1 SP1 through r11.1 SP4 to Monitoring in Release 11.4:

AutoSys Web UI Job Flow Monitoring Object	AutoSys Web UI Monitoring Object	Comments
Flow	View	
Filter	Filter	Multiple filters for the same server and view are migrated into one filter with filter condition containing AND or OR logical operators.
Jobs	Filter	Multiple jobs for the same server and view are migrated into one filter with the filter condition containing AND or OR logical operators.

**NOTE**

- Objects, other than those in the above table, are ignored.
- No manual translations are required for migrating Job Flow Monitoring view definitions.

**Export Job Flow Definitions**

You can export AutoSys Web UI job flow definitions from the server to a text file. You can modify the resulting file using a text editor, and then import it back to the server. By default, all job flow definitions are exported.

To export job flow definitions, use the following command syntax:

```
wcc_jfd -u user_name -p password -x output_file
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **-x, --export output\_file**  
Specifies the .txt file to which the job flow definitions are exported and, optionally, its path. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-x) or the long name (--export).

### Example: Export all job flows

The following command exports all job flows, their filters, and jobs to the file named output.txt:

```
wcc_jfd -u ejmcommander -p ejmcommander -x output.txt
```

### Migrate Job Flow Monitoring Objects

You can use the jfd\_migrate utility to convert a file of exported Job Flow Monitoring objects into valid Monitoring objects for this release.

#### NOTE

- You must first copy the output file to the current AutoSys Web UI server to convert the objects.
- To access help using the jfd\_migrate utility, enter the command `jfd_migrate -?` in the command line.

To migrate Job Flow Monitoring objects, use the following command syntax:

```
jfd_migrate -i input_file -o output_file [-v]
```

- **-i, --input *input\_file***  
Specifies the fully-qualified name of the file of Job Flow Monitoring objects to import. You can use either the short name (-i) or the long name (--input).
- **-o, --output *output\_file***  
Specifies the fully-qualified file name to which the translated objects are migrated. The file is created by the output option; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-o) or the long name (--output).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

### Example: Migrate all job flow definitions contained in the exported file

The following command migrates all job flow definitions contained in the export.txt file and creates an export file named migrate.txt:

```
jfd_migrate -i export.txt -o migrate.txt
```

### Import Translated Monitoring Objects

You can use the wcc\_monitor utility to import a file of translated view definition objects into AutoSys Web UI.

To import a translated monitoring object file, use the following command syntax:

```
wcc_monitor -u user_name -p password -i input_file [-v]
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-i, --import *input\_file***

Specifies the fully-qualified name of the translated Monitoring Release 11.4 view definition file. You can use either the short name (-i) or the long name (--import).

- **-v, --verbose**

(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

### Example: Import the converted objects into current monitoring view definitions

The following command imports all translated definitions contained in the migrate.txt file:

```
wcc_monitor -u ejmcommander -p ejmcommander -i migrate.txt -v
```

## Migrate Job Status Console Objects to Monitoring

As an administrator, you may be responsible for converting your existing AutoSys Web UI Job Status Console views, filters, and alert policies into valid monitoring definitions in the current version of AutoSys Web UI Monitoring. To convert the information, you can run the available command line utilities from the Windows command line or UNIX console.

To migrate Job Status Console objects to Monitoring, follow these steps:

1. [Export the Job Status Console view, filter, and alert policy definitions using the wcc\\_jsc utility.](#)
2. [Copy the output file created in Step 1 to the current AutoSys Web UI server and run the jsc\\_migrate utility.](#)
3. [Import the file created in Step 2 into this version of AutoSys Web UI using the wcc\\_monitor utility.](#)

### NOTE

- The wcc\_jsc utility is not included with this release of AutoSys Web UI.
- For information about the wcc\_jsc utility, see *Job Status Console Batch Interface Help* in the AutoSys Web UI r11.3 documentation.
- Migration from other sources is not supported, even if the objects can be imported with the AutoSys Web UI r11.3 wcc\_jsc utility.
- Conversion of Job Status Console view definitions in AutoSys Web UI r11.1 SP1 through r11.1 SP4 and r11.3 are supported.

### Example: Migrate Job Status Console Views from AutoSys Web UI r11.3 to Monitoring Views

Suppose that your company has upgraded AutoSys Web UI r11.3 to AutoSys Web UI Release 11.4. Because you do not want to recreate all of the Job Status Console views, you can migrate them from the older AutoSys Web UI version to the current version.

The following table shows how objects are migrated from Job Status Console r11.1 SP1 through r11.1 SP4 and r11.3 to Monitoring in Release 11.4:

AutoSys Web UI Job Status Console Object	AutoSys Web UI Monitoring Object	Comments
Job Status view	View	
Job Status filter	Filter	Multiple Job Status filters for the same server and view are migrated into one filter with filter condition containing AND or OR logical operators.
Alert filter Alert policy	Alert policy	The Release 11.4 alert policy is created with attributes from the Job Status Console Alert Filter and Alert Policy.
Alarm filter	Alert policy	

## Export Job Status Console Definitions

You can export AutoSys Web UI job status console definitions from the server to a text file. You can modify the resulting file using a text editor and then import it back to the server, or copy it to another server and import it there to propagate the views and alert policies. By default, all job status console definitions are exported.

To export job status console definitions, use the following command syntax:

```
wcc_jsc -u user_name -p password -x output_file [-w view_list] [-v]
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).
- **-x, --export *output\_file***  
Specifies the file to which the job status views are exported and, optionally, its path. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch utility. You can use either the short name (-x) or the long name (--export).
- **-w, --views *view\_list***  
(Optional) Specifies the set of views and alert policies to be exported. You can enter a list of views, alert policies, or a combination of both in a single command. You can use either the short name (-w) or the long name (--views).

### NOTE

- For AutoSys Web UI r11.3 only, separate the names in the *view\_list* with commas. For AutoSys Web UI r11.1 SP1 through r11.1 SP4, separate the names in the view with a space.
- For AutoSys Web UI r11.3 only, you can use percent (%) or question mark (?) with a partial name, where % represents zero or more characters and ? represents a single character, to match all views and alert policies that contain the partial name.
- If a view name in the list contains a space, the entire list must be enclosed in double quotes. For example, to include the views named view1, view test, and view2, use the following syntax:

```
-w "view1,view test,view2"
```

- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

### Example: Export all job status console definitions

The following command exports all job status console definitions to the file named export.txt:

```
wcc_jsc -u ejmcommander -p ejmcommander -x export.txt
```

### Example: Export selected job status console views

The following example exports the specified job status console views and associated filters to the file named export.txt:

```
wcc_jsc -u ejmcommander -p ejmcommander -x export.txt -w View1,View2
```

### Example: Export selected alert policies

The following example exports the specified alert policies to the file named export.txt:

```
Wcc_jsc - u ejmcommander - p ejmcommander - x export.txt - w
AlertPolicy1,AlertPolicy2
```

## Migrate Job Status Console Objects

You can use the `jsc_migrate` utility to convert a file of exported Job Status Console objects into valid Monitoring objects for this release.

### NOTE

- You must first copy the output file to the current AutoSys Web UI server to convert the objects.
- The `jsc_migrate` command creates a separate alert policy for every view containing an alert filter defined with that policy.
- The `jsc_migrate` command does not translate the following in an alarm filter: `broker_unavailable`, `multiple_ep_shutdown`, and `already_running`. These alarms are not supported by AutoSys Workload Automation r11.3 and Release 11.4.
- The `jsc_migrate` command does not automatically translate an alarm filter that contains a comma-separated list of jobs; it requires manual intervention.
- To access help using the `jsc_migrate` utility, enter the command `jsc_migrate -?` in the command line.

To migrate Job Status Console objects, use the following command syntax:

```
jsc_migrate -i input_file -o output_file [-v]
```

- **-i, --input *input\_file***  
Specifies the fully-qualified name of the file of Job Status Console objects to import. You can use either the short name (-i) or the long name (--input).
- **-o, --output *output\_file***  
Specifies the fully-qualified file name to which the translated objects are migrated. The file is created by the output option; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-o) or the long name (--output).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

### Example: Migrate all job status console definitions contained in the exported file

The following command migrates all job status console definitions contained in the `export.txt` file and creates an export file named `migrate.txt`:

```
jsc_migrate -i export.txt -o migrate.txt
```

### Examples

The `jsc_migrate` utility translates r11.1 SP1 through r11.1 SP4 and r11.3 Job Status Console attributes to Release 11.4 attributes for Monitoring. Most, but not all of the Job Status Console view definition properties are automatically migrated to the new monitoring terminology. Some properties require manual intervention. When you run `jsc_migration`, you may be prompted to make a selection from the option menu. When this occurs, you can decide whether to apply the chosen selection to all other occurrences of the same option.

### Example: Run Script for the Command: `jsc_migrate - I exportFiles\export.txt - o exportFiles\migrate.txt - v`

The following displays an example of a script requiring manual intervention and of the output file that is generated:

```
jsc_migrate - I exportFiles\export.txt - o exportFiles\migrate.txt - v
View "View1" generated.
Filter "ServerAE1" generated.
Choose alarm severity for alert filter "Default Alarm Filter" for view "View1"
1> Critical
2> High
3> Medium
```



```

4> Low
Please choose one option.
2 <Enter>
Apply to all?
1> Yes
2> No
2 <Enter>
Alert Policy "View1-DefaultAlarmFilter" generated.
Job Filter pattern <C?> for alert policy "ABC" and job pattern <A?> for Alert filter
"Default Alert Filter" in view "View1" cannot be merged.
Choose one of them
1> Alert filter
2> Alert policy
Please choose one option.
1 <Enter>
Apply to all?
1> Yes
2> No
Please choose one option
1 <Enter>
Alert Policy "View1-Default Alert Filter - ABC" generated.

```

The resulting migrate.txt file displays similar to the following example:

```

//Event policy View1 - Default Alarm Filter
insert_alert_policy: "
alarm_type: "
job_name: "
severity: HIGH
status: OPEN,ACKNOWLEDGED
system_log: false
active: false

//Event policy View1 - Default Alert Filter - ABC
insert_alert_policy: 'View1-Default Alert Filter=ABC'
job_status: TERMINATED
job_name: A?
severity: CRITICAL
status: OPEN,ACKNOWLEDGED, CLOSED
system_log: false
url_link: "http:ABC"
url_name: ABC
text: ABC
active: false

//view View1
insert_view: View1
render_flow: true

```

```

render_collections: true
render_jobs: true
job_columns: job_name,status,box_name,server,start_time,end_time
alert_columns: alert,alert_type,severity,status,server,job,create_time,status_time

Insert_filter:AEServer1
View: View1
Filter: job_name like *
Alert_policy: 'View1-DefaultAlarmFilter' "View1-DefaultAlertFilter-ABC"
Active: true

```

### **Translation of Job Status Console Attributes**

The jsc\_migrate utility translates r11.1 SP1 through r11.1 SP4 and r11.3 Job Status Console attributes to Release 11.4 attributes for Monitoring. The tables that follow show the translations.

#### **Job Status View Attributes**

<b>Job Status Console Attribute Name</b>	<b>Monitoring Attribute Name</b>	<b>Comments</b>
Name	View name	
Server	n/a	Default view Server is not used in Release 11.4
scroll_size	n/a	User preference
update_interval	n/a	Global preference (auto refresh interval)
autorefresh	n/a	Auto refresh can be enabled for any view, job, server list, or dashboard
time_format	n/a	User preference
Description	Description	
background_color	n/a	Icon used
foreground_color	n/a	Icon used
Property	job_columns	
dashboard_update_interval	n/a	Global preference (auto refresh interval)
dashboard_auto_refresh	n/a	Auto refresh can be enabled for any view, job, server list, or dashboard
dashboard_color	n/a	Icon used
alert_scroll_size	n/a	User preference
alert_update_interval	n/a	Global preference (auto refresh interval)
alert_auto_refresh	n/a	Auto refresh can be enabled for any view, job, server list, or dashboard
alert_color	n/a	Icon used
alert_property	alert_columns	

#### **Job Status Filter Attributes**

<b>Job Status Console Attribute Name</b>	<b>Monitoring Attribute Name</b>	<b>Comments</b>
name	N/A	The filter name is the name of the server.
view	view	

server	filter name	
active	active	
include_jobset	Filter	job_name = jobset pattern
job	Filter	job_name
jobset	Filter	box_name
status	Filter	status
past_days	Filter	started_in_past
past_hours	Filter	started_in_past
group	Filter	group
application	Filter	application
owner	Filter	owner
target_machine	Filter	machine
actual_machine	Filter	run_machine

**NOTE**

- When active and inactive filters are defined to the same server in a view, you can either ignore the inactive filter or create another view for the inactive filter.
- In previous versions of AutoSys Web UI Job Status Console, some filter attributes supported comma-separated values. In the current version, these values are combined using an OR expression. For example, job: A\*,B\* is translated to filter: (job\_name like A\* OR job\_name like B\*). Additionally, a filter containing multiple attributes are combined using the AND condition. For example, job:A\*, owner:user1 translates to filter: job\_name like A\* AND owner = user1.

**Alert Filter Attributes**

Job Status Console Attribute Name	Monitoring Attribute Name	Comments
name	policy (name)	part of policy name
view	N/A	combination of view_name/policy name
server	filter (policy)	The policy name is added to the view filter policy attribute
active	active (policy)	
job	job_name	
jobset	box_name	
severity	severity (policy)	
state	status (policy)	
alert	N/A	

**NOTE**

- The remaining attributes are supplied from the AutoSys Web UI r11.1 SP1 through r11.1 SP4 and r11.3 alert policies.
- In the Monitoring Attribute Name column, policy refers to the Alert policy in AutoSys Web UI Monitoring Release 11.4.
- In the current AutoSys Web UI version, an alert policy is applied only to the jobs in the view. To migrate an alert policy from AutoSys Web UI r11.1 SP1 through r11.1 SP4 or r11.3, the server associated with the alert policy must match one of the view filter servers.

## Alarm Filter Attributes

Job Status Console Attribute Name	Monitoring Attribute Name	Comments
name	policy (name)	.part of policy name
view	filter (policy)	The policy name is added to the view filter policy attribute.
server	filter (policy)	The policy name is added to the view filter policy attribute.
active	policy (active)	
job	policy (job_name)	Defined by the server/view filter
state	policy (status)	Defined by the server/view filter
start_time	N/A	UI Filter
end_time	N/A	UI Filter
past	N/A	UI Filter
alarm	alarm_type	

### NOTE

- The alert policy severity must be selected manually because the alarm filter in AutoSys Web UI r11.1 SP1 through r11.1 SP4 and r11.3 Job Status Console does not have a severity.
- If an alarm filter does not have a specific job status filter for the server, the alarm filter is not translated.
- The migration process automatically creates a Monitoring alert policy from the alarm filter and connects the policy to the view filter.

## Alert Policy Attributes

Job Status Console Attribute Name	Monitoring Attribute Name	Comments
name	name	part of the name (view policy)
server	filter (policy)	The policy name is added to the view filter policy attribute.
job	job_name	
jobset	box_name	
active	N/A	The alert filter defines the active attribute.
status	job_status	
system_logging	system_log	
alert_name	N/A	The alert name is not used.
alert_severity	severity	
alert_description	text	
alert_url_name	url_name	
alert_url_link	url_link	

### NOTE

- If a job attribute in the Job Status Console r11.1 SP1 through r11.1 SP4 or r11.3 Alert Filter and Alert Policy is not '\*', the migration utility attempts to choose a more restrictive value. For example, if A\* and AB\* are

defined, the value changes to AB\*. If a value cannot be automatically selected, you must choose a job attribute from the Alert Policy or Alert Filter.

- If an alert policy specific to the alert filter is not included in the exported Job Status Console view definition file, the alert filter for the view is not migrated.

## Dashboard Filter

Dashboard filter migration is not supported.

## Import Translated Monitoring Objects

You can use the `wcc_monitor` utility to import a file of translated view definition objects into AutoSys Web UI.

To import a translated monitoring object file, use the following command syntax:

```
wcc_monitor -u user_name -p password -i input_file [-v]
```

- **-u, --user *user\_name***  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password *password***  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-i, --import *input\_file***  
Specifies the fully-qualified name of the translated Monitoring Release 11.4 view definition file. You can use either the short name (-i) or the long name (--import).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

## Example: Import the converted objects into current monitoring view definitions

The following command imports all translated definitions contained in the `migrate.txt` file:

```
wcc_monitor -u ejmcommander -p ejmcommander -i migrate.txt -v
```

## Migrate Monitoring Views

As an administrator, you may be responsible for migrating your existing AutoSys Web UI r11.3 Monitoring views and filters into the current version of AutoSys Web UI Monitoring. To migrate the views, you can run the available command line utilities from the Windows command line or UNIX console.

To migrate AutoSys Web UI r11.3 Monitoring views to the current version of AutoSys Web UI, follow these steps:

1. [Export the views on the AutoSys Web UI r11.3 server to a text file.](#)
2. Copy the exported text file to the new AutoSys Web UI server.
3. [Import the text file on the new AutoSys Web UI server.](#)

### NOTE

For more information about using the export and import features, see [Administer the Monitoring Batch Interface](#).

## Example: Migrate AutoSys Web UI Monitoring Views from AutoSys Web UI r11.3 to the Current Version

Suppose that your company has upgraded AutoSys Web UI r11.3 to AutoSys Web UI Release 11.4. Because you do not want to recreate all of the Monitoring views, you can migrate them from the older AutoSys Web UI version to the current version.

### **Export View Definitions**

You can export AutoSys Web UI view definitions from the server to a text file. You can modify the resulting file using a text editor, and then import it back to the server. By default, all view definitions are exported.

To export view definitions, use the following command syntax:

```
wcc_monitor -u user_name -p password -x output_file [-v]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### **NOTE**

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-x, --export output\_file**  
Specifies the .txt file to which the view definitions are exported and, optionally, its path. The file is created by the export; however, the path must exist. If you do not specify a path, the file will be created in the folder that contains the batch interface utility. You can use either the short name (-x) or the long name (--export).
- **-w, --view [view1][,view2]...**  
(Optional) Specifies which views are to be exported. You can use either the short name (-w) or the long name (--view).

#### **NOTE**

You can enter multiple view names in a comma-separated list.

- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

#### **NOTE**

When logging is enabled, the output is written to the standard out and to the log file. The log file is located in *install\_location/Logs/application/cli/wcc\_monitor.log*.

### **Example: Export all views**

The following command exports all views, their filters, and jobs to the file named output.txt:

```
wcc_monitor -u ejmcommander -p ejmcommander -x output.txt
```

### **Example: Export selected views**

The following example exports the specified views to the file named export.txt:

```
wcc_monitor -u ejmcommander -p ejmcommander -x export.txt --view View1,View2
```

### **Import View and Alert Policy Definitions**

You can import view and alert policy definitions to the AutoSys Web UI server.

#### **NOTE**

When a view references an alert policy, the alert policy must be imported before the view. If not, the filter in the view will not be imported.

To import the view or alert policy export file, use the following command syntax:

```
wcc_monitor -u user_name -p password -i input_file [-v]
```

- **-u, --user user\_name**  
Specifies the authorized user name. You can use either the short name (-u) or the long name (--user).
- **-p, --password password**  
Specifies the password for the authorized user name. You can use either the short name (-p) or the long name (--password).

#### NOTE

The password is optional. If you choose not to enter it, you will be prompted for the password after executing the command. This option lets you enter the password so that it is not echoed on the console, making it more secure.

- **-i, --import input\_file**  
Specifies the fully-qualified name of the text file to import. You can use either the short name (-i) or the long name (--import).
- **-v, --verbose**  
(Optional) Displays debug messages. You can use either the short name (-v) or the long name (--verbose).

#### Example: Import a text file

The following command imports all views and filters included in the file input.txt to the AutoSys Web UI server:

```
wcc_monitor -u ejmcommander -p ejmcommander -i input.txt
```

## Migrate Application Editor SAP Filter Definitions

As an administrator, you may be responsible for migrating existing AutoSys Web UI r11.3 Application Editor SAP filter definitions to AutoSys Web UI Release 11.4. The Editor Migration Utility (EMU) lets you import Application Editor SAP filters from an AutoSys Web UI r11.1 or r11.3 server into an AutoSys Web UI Release 11.4 server database. Authorized users can then use the commands in Application Editor.

#### NOTE

You must first copy the ae directory in the following path from the source AutoSys Web UI server to the directory of your choice on the target AutoSys Web UI server:

- **Windows**  
%CA\_WCC\_INSTALL\_LOCATION%\ConfigServer\config\application\
- **UNIX**  
\$CA\_WCC\_INSTALL\_LOCATION/ConfigServer/config/application/

The EMU executable is located in the bin folder under the AutoSys Web UI installation root. On Windows, it is the batch file (editor\_migrate.bat); on UNIX, it is the shell script (editor\_migrate.sh). Before you run the EMU, change your current directory to the following directory that contains the editor\_migrate executable.

- **Windows**  
%CA\_WCC\_INSTALL\_LOCATION%\bin\
- **UNIX**  
\$CA\_WCC\_INSTALL\_LOCATION/bin/

To migrate SAP filter definitions, use the following command syntax:

```
editor_migrate -u user -p password -s source_directory [-f] [-t] [-v] [?]
```

- **-u user**

Specifies the authorized user name.

#### NOTE

The specified user must have access rights to Configuration.

- **-p *password***  
Specifies the password for the authorized user name.
- **-s *source\_directory***  
Specifies the fully qualified directory containing the SAP filters to import into the database.
- **-f, *force-update***  
(Optional) Overwrites existing SAP filters with the SAP filters in the specified source directory.
- **-t, *test***  
(Optional) Runs the migration in test mode. The changes are not saved.
- **-v, *verbose***  
(Optional) Print detailed processing messages.
- **-?, *help***  
(Optional) Displays a list of the available arguments.

#### WARNING

To ensure security, delete the ae directory from the target server after you migrate the SAP filters definitions.

### Example: Migrate All SAP Filters to an AutoSys Web UI Release 11.4 Server

The following command migrates all SAP commands from the directory c:\import\ae on the server on which the command is run:

```
editor_migrate -u ejmcommander -p ejmcommander -s c:\import\ae
```

### Example: Migrate Application Editor SAP filters

#### NOTE

Suppose your company has upgraded AutoSys Web UI r11.3 to AutoSys Web UI Release 11.4. Because you do not want to recreate all of the Application Editor SAP filters, you can migrate them from the older AutoSys Web UI version to the current version.

## Export and Import CA EEM Policies (using the GUI) - Web UI

You can export your AutoSys Web UI policies from one environment and then import them into a different environment. By completing your export, you have created a backup copy of your CA EEM policies for disaster recovery.

#### NOTE

If your CA EEM instances are in failover or multi-write, export the policies only from *one* instance.

To export and import policies, see the following sections:

### Create a Backup of Web UI Policies (Complete Policy Export)

By export, you can create a back up of your Web UI policies.

#### Follow these steps:

1. Go to your Source CA EEM GUI from where you want to export your policies.
2. Select the 'WCC000X' application from the drop-down box and log in.
3. Click the Configure tab.
4. Click the 'CA EEM Server' link.



- On the left, click the 'Export Application' link.  
By default, everything except the 'Override The Max Search Size' check box is selected.

**NOTE**

If you are using CA CA EEM Active Directory, clear all the Global objects' checkboxes.

- Click Export.  
A xml.gz file is created.
- Unzip that file, extract the .xml file, and save it to a location of your choice; for example, a temp directory.

**NOTE**

For disaster recovery, save this file in a location that is *not* on the servers where you have AutoSys Workload Automation applications installed.

**Import Web UI Policies**

You can import the Web UI policies by completing this procedure.

**Follow these steps:**

- Open Windows command prompt (Run as Administrator) and navigate to your \$CA\_WCC\_INSTALL\_LOCATION\safex directory [or %CA\_WCC\_INSTALL\_LOCATION%\safex directory]
- Run the following command:

**Linux**

```
./safex -h target_EEM_servername -u EiamAdmin -p EiamAdmin_Password -f
full_path_where_your_xml_file_is_located
```

**Windows**

```
safex -h target_EEM_servername -u EiamAdmin -p EiamAdmin_Password -f
full_path_where_your_xml_file_is_located
```

**Regenerate Security Certificates**

After completing the policy import process, you can regenerate the Security certificates.

**Follow these steps:**

- Run the following command:

**Linux**

```
./safex -h target_EEM_servername -u EiamAdmin -p EiamAdmin_Password -f
IssueCertificate.xml
```

**Windows**

```
safex -h target_EEM_servername -u EiamAdmin -p EiamAdmin_Password -f
IssueCertificate.xml
```

- Log in to the target CA EEM GUI (under the Web UI application) where you have imported your policies.
- Verify that your Web UI policies have been imported.
- Move the wcc.pem and wcc.key files to the \$CA\_WCC\_INSTALL\_LOCATION/data/config directory [or %CA\_WCC\_INSTALL\_LOCATION%\data\config directory].
- For Linux, complete the following:
  - Change the permissions of the wcc.pem and wcc.key files {chmod 750 wcc.pem wcc.key}.
  - Change the owner and group of the wcc.pem and wcc.key files to the owner and group of the AutoSys Web UI installation {chown wcc:wcc wcc.pem wcc.key}.

## Server Properties and Application Preferences

This article includes information to configure properties for servers you have defined in Web UI, and to configure user and environmental preferences for Web UI applications. For more information, see *Configuration Help*.

### Servers Section

#### **NOTE**

The Servers section appears at the top of the Configuration page.

The Servers section is comprised of the following subsections:

- **Server Search**  
Lets you find a specific back-end server by name or type.
- **Server Results**  
Lets you view the server definitions set up to access back-end servers through the Web UI interface.
- **Create Server**  
Lets you define your back-end servers to Web UI. Clicking the Create button in the Servers subsection opens the Create Server subsection, in which you can enter the necessary properties for configuring a server definition.
- **Edit Server**  
Lets you edit your server definitions. Clicking a server name from the Servers subsection opens the Edit Server subsection, in which you can modify the appropriate server properties.

#### **NOTE**

For information about the Server Search and Search Results subsections, see *Configuration Help*.

### AutoSys Workload Automation Server Properties

The Create Server subsection for AutoSys Workload Automation displays when you select AutoSys Workload Automation Server from the drop-down list in the Server Results subsection and click Create; the Edit Server subsection displays when you click the link for a specific AutoSys Workload Automation server in the Server Results subsection.

Use the Create Server subsection to define the characteristics of a new AutoSys Workload Automation server, or use the Edit Server subsection to modify the properties of an existing server.

The AutoSys Workload Automation properties in the Create Server and Edit Server subsections are displayed by category.

The General category contains the following fields:

- **Server Name**  
Defines a "friendly" name for the server. This name must be unique.  
**Limits:** 64 or fewer alphanumeric characters; must contain no spaces and begin with a letter

#### **NOTE**

This field is read-only when editing the properties of an existing server.

- **Product Version**  
Specifies the AutoSys Workload Automation release installed on the target server.
- **Instance Name**  
Defines the name of the instance that corresponds to the AutoSys Workload Automation scheduler. The combination of the Instance Name and Cluster Name must be unique for each server defined  
**Limits:** Exactly three uppercase alphanumeric characters; must begin with a letter

**NOTE**

- Web UI automatically converts lowercase entries to uppercase.
- This field is read-only when editing the properties of an existing server.

- **Cluster Name**

Specifies the cluster name that uniquely identifies the external AutoSys Workload Automation application server. The combination of the Instance Name and Cluster Name must be unique for each server defined.

**Default:** Default

**NOTE**

This field is read-only when editing the properties of an existing server.

The Application Server category contains the following fields:

- **Application Server Host Address**

Specifies the Host Address of the computer on which the AutoSys Workload Automation application server resides. When entering more than one host address, each address must be separated by a comma.

**Limits:** 255 or fewer characters

- **Application Server Host Port**

Specifies the virtual port that is assigned to the AutoSys Workload Automation application server host computer. For more information about the Application Server Host Port, see the *AutoSys Workload Automation Reference* section.

**Default:** 9000

**Limits:** 1-65535

- **Web Services URL**

Specifies the AutoSys REST endpoint used by Agent Inventory to collect data, ECLI to execute commands, and Forecast to generate reports. When entering more than one Web Services URL, each URL must be separated by a comma. All the entered Web Services must be of the same version. If the AutoSys REST endpoint is unable to connect with the specified first url, it moves to the next url.

**Default:** empty

**Limit:** 4000 characters

**NOTE**

- For AutoSys Version 12.0 and higher, typical installation installs web services. For previous releases, select Custom installation for the web server installation and specify the Web Services URL.
- Add the monitoring user to the WorkloadAutomationAEWebService group as the monitoring user is used to connect to the web server.
- If the Web Services URL is empty, the server is disabled for Agent Inventory. Agent Inventory does not display agents associated with this server's machines.
- ECLI commands and Forecast reports will fail if we do not specify the Web Services URL.
- Monitor user should have access to all the machines so that Agent Inventory can display all the machines.
- If Monitor ID is modified to provide access permissions to all the machines, you must remove Web Services URL and add it again for Agent Inventory to collect all the machines. Otherwise, new machines for which access permissions have been provided are not displayed.

The Compatibility category contains the following field:

- **Legacy Compatibility Port**

Specifies the port assigned to the listener installed on the AutoSys Workload Automation agent machine.

**Default:** 4444

**Limits:** 1-65535

The Monitor User category contains the following fields:

- **Monitor ID**

Defines the AutoSys Workload Automation user ID assigned by the system administrator for use by the common monitoring functions. The name must be entered as follows:

`user_name@node`

– **user\_name**

Specifies the authorized user name.

– **node**

Specifies a name such as the Application Server Host Name or any machine name. The name does not have to be a valid machine name. It is used to create a key in the AutoSys Workload Automation database in which to look up the associated password. Any AutoSys Workload Automation application server can look up the `user_name@node` key in the AutoSys Workload Automation password database.

**Limits:** 80 or fewer characters; must not contain the < or > characters

**NOTE**

- The specified Monitor ID should have view rights to a superset of AutoSys Workload Automation jobs to provide common monitoring functionality. The Monitor ID does not require update or create authority.
- If AutoSys Workload Automation native security is used, the password for `user_name@node` specified for the Monitor ID must be defined in `autosys_secure`, and the user must be specified as a SUPER EDIT user.
- If CA EEM is used for security, the Monitor ID specified must be a valid user on the CA EEM server (`@node` has no meaning in the context of CA EEM).
- When AutoSys Workload Automation is configured for High Availability and the Application Server Host Name field includes a group of application servers, user credentials must specify the explicit application server for which each user is authorized. For example, if "SERVER1,SERVER2,SERVER3" is entered in the Application Server Host Name field, the user must have a user name of "user@SERVERn", where n is 1, 2, or 3. The application server used to define the credentials for this user can be any available computer that has a AutoSys Workload Automation remote agent installed. Users that do not have an explicit application server specified will not be able to log into the back-end AutoSys Workload Automation server.

- **Monitor Password**

Defines the password for the Monitor ID account.

The Preference category contains the following field:

- **EEM Enabled**

Indicates whether to enable CA EEM pass-through security for this server.

**NOTE**

Web UI and AutoSys Workload Automation must be running in the same CA EEM instance for pass-through security to work.

**Default:** true

- **Time Zone**

Specifies the time zone in which AutoSys Workload Automation instance is running.

**Default:** empty

The Encryption category contains the following fields:

- **Encryption Type**

Specifies the type of encryption: AES, NONE, or DEFAULT. For more information about AES encryption, see the *AutoSys Workload Automation UNIX Implementation* or *Windows Implementation* sections.

**Default:** DEFAULT

- **Encryption Key**

Specifies the encryption key. This key must be either a 32-digit hexadecimal key or a 16 character or less pass phrase.

**Limits:** 32 or fewer alphanumeric characters

**NOTE**

This field is required when AES is selected from the Encryption Type field.

**Configure AutoSys Workload Automation Application Server Host Port**

When AutoSys Workload Automation services are configured within a Web UI environment, the default application server port is 9000. The user may need to select a different port dependent upon security restrictions or if another port is used by a different application.

**NOTE**

The following procedure must be executed only when Web UI is operating on Linux or Solaris by a nonroot user and the AutoSys Workload Automation server is using a different port other than the default (9000) setting.

**Follow these steps:**

1. Log in as the root user on the Web UI server or active node in the Web UI cluster.
2. Enter the following command from the UNIX system command prompt:
 

```
$CSAM_SOCKADAPTER/bin/csamconfigedit port=port_number
enableSSL=[true|false] enablePMUX=[true|false]
```

  - **port\_number**  
Specifies the port for the application server host.
  - **[enableSSL=true|false]**  
Specifies whether to use SSL security.
  - **[enablePMUX=true|false]**  
Specifies whether to use port multiplexing.
3. (Linux or Solaris based High Availability Cluster Environments) Repeat the procedure on each node within the Web UI cluster as follows:
  - a. Log in to the node as the root user.
  - b. Run the csamconfigedit command, ensuring that you specify the same port number and connection attributes that you specified on the server machine.

**WARNING**

Nodes where the command does not run cannot connect to the AutoSys Workload Automation application server.

The application server host port settings are configured.

For more information about the Application Server Host Port, see the *AutoSys Workload Automation Reference section*.

**NOTE**

The following procedure must be executed when AutoSys Workload Automation server is using a different port other than the default 9000 setting and AutoSys Web UI uses more HA nodes.

**Unix****Follow these steps:**

1. Log in as the root user on the other node in the Web UI cluster.
2. Enter the following command from the UNIX system command prompt:
 

```
$CSAM_SOCKADAPTER/bin/csamconfigedit port=port_number
enableSSL=[true|false] enablePMUX=[true|false]
```

  - **port\_number**  
Specifies the port for the application server host.
  - **[enableSSL=true|false]**

Specifies whether to use SSL security.

– **[enablePMUX=true|false]**

Specifies whether to use port multiplexing.

3. Repeat the procedure on each node within the Web UI cluster as follows:

- a. Log in to the node as the root user.
- b. Run the `csmaconfigedit` command, ensuring that you specify the same port number and connection attributes that you specified on the server machine.

**WARNING**

Nodes where the command does not run cannot connect to the AutoSys Workload Automation application server.

The application server host port settings are configured.

## Windows

### Follow these steps:

1. Log in as the root user on the other node in the Web UI cluster.
2. Enter the following command from the Windows system command prompt:

```
%CSAM_SOCKADAPTER%\bin\csamconfigedit port=port_number
enableSSL=[true|false] enablePMUX=[true|false]
```

– **port\_number**

Specifies the port for the application server host.

– **[enableSSL=true|false]**

Specifies whether to use SSL security.

– **[enablePMUX=true|false]**

Specifies whether to use port multiplexing.

3. Repeat the procedure on each node within the Web UI cluster as follows:

- a. Log in to the node as the root user.
- b. Run the `csmaconfigedit` command, ensuring that you specify the same port number and connection attributes that you specified on the server machine.

**WARNING**

Nodes where the command does not run cannot connect to the AutoSys Workload Automation application server.

The application server host port settings are configured.

For more information about the Application Server Host Port, see the *AutoSys Workload Automation Reference section*.

## Preferences Section

You can use the Preferences section to view and modify environment properties for the instance of Web UI you are using. The environment properties are displayed by category. Each category displayed is determined by the selection you make in the View/Modify Preferences drop-down list.

This section contains the following fields:

**View/Modify Preferences** Lets you select the environment properties to view or edit.

The following options are available:

- **Agent Inventory**  
Displays the Agent Inventory Preferences, so you can view and edit properties for the Agent Inventory application.
- **Application Editor**

Displays the Application Editor Preferences, so you can view and edit properties for the Application Editor application.

- **Configuration**  
Displays the Configuration Preferences, so you can view and edit properties for the Configuration application.
- **Enterprise Command Line**  
Displays the Enterprise Command Line Preferences, so you can view and edit properties for the Enterprise Command Line application
- **Monitoring**  
Displays the Monitoring Preferences, so you can view and edit properties for the Monitoring application.
- **Quick Edit**  
Displays the Quick Edit Preferences, so you can view and edit properties for the Quick Edit application.
- **Quick View**  
Displays the Quick View Preferences, so you can view and edit properties for the Quick View application.
- **Forecast**  
Displays the Forecast Preferences, so you can view and edit properties for the Forecast application.
- **Reporting**  
Displays the Reporting Preferences, so you can view and edit properties for the Reporting application.
- **Resources**  
Displays the Resources Preferences, so you can view and edit properties for the Resources application.

This section contains the following columns:

- **Property Name**  
Displays the properties for the selected option.
- **Property Value**  
Displays the assigned values for the associated properties.

This section contains the following buttons:

- **Go**  
Performs the action selected in the View/Modify Preferences drop-down list.
- **Save**

Saves your changes.

#### **NOTE**

If an incorrect value is entered into any field in the Preferences section, the action saves the valid entries and flags invalid entries, letting you correct them as necessary.

- **Reset**  
Discards your changes.

### **Agent Inventory Preferences**

The Agent Inventory properties display when you select Agent Inventory from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Agent Inventory Preferences subsection to specify properties for the Agent Inventory application for this Web UI instance.

The properties in the Agent Inventory Preferences subsection are displayed by category.

The Agent Table category contains the following fields:

- **Table Columns**  
Specifies the comma-separated values for the columns to be displayed in the Agent List table. **Valid Values:** name, status, type, agentName, address, port, version, platform, schedulerId, actionStatus, actionTimestamp, plugins

**NOTE**

The column value 'name' is a mandatory.

- **Refresh Time(Delta)** Specifies the maximum duration by which Agent Inventory must fetch the updated or modified data from the AutoSys Workload Automation server. Collects data from AutoSys Workload Automation Release 11.3.6 SP7 or higher.  
**Limits** (in seconds): 5-86400  
**Default:** 60
- **Refresh Time(Full)**  
 Specifies the maximum duration by which Agent Inventory must fetch the entire data and must refresh the Agent Inventory page. Collects data from AutoSys Workload Automation Release 11.3.6 SP6 or earlier.  
**Limits** (in seconds): 60-86400  
**Default:** 300 seconds
- **Stop Timeout**  
 Specifies the maximum duration by which the agent is stopped.  
**Default:** 300 seconds
- **Restart Timeout**  
 Specifies the maximum duration by which the agent is restarted.  
**Default:** 300 seconds
- **Update Agent Parameters Timeout**  
 Specifies the maximum duration by which the agent parameters file is updated and the Agent Inventory page is refreshed.  
**Default:** 300 seconds
- **Agent Access Policy Resource Type**  
 Specifies the resource type options for agent access policy evaluation.  
**Options:** HostName:Port or MachineName:ServerName  
**Default:** HostName:Port
- **Installer Directory on Server**  
 Specifies the location where the Agent installer is available on the server.
- **Staging Directory (Windows)**  
 Specifies the relevant Agent installer file location for the Windows operating system.
- **Staging Directory (UNIX)**  
 Specifies the relevant Agent installer file location for the UNIX operating system.
- **Post-Quiesce Timeout Action for Upgrade/Rollback**  
 Select the required action after the post-quiesce timeout. Options are:
  - **Cancel Action**  
 Cancels the upgrade.
  - **Continue Action**  
 Continues with the upgrade.
- **Delete staged installer after successful upgrade**  
**Default:** Selected checkbox
- **Delete backup directory after successful upgrade**  
**Default:** Blank checkbox
- **Agent Upgrade using Java VM**  
 Specifies the Java VM path against the operating system and while upgrading, the same path is replicated in the Agent inventory tab.
- **Stage Timeout**  
 Specifies the duration after which the staging times out.  
**Default:** 300
- **Quiesce Timeout**  
 Specifies the duration to complete running jobs during upgrade or rollback.



**Default:** 600

- **Upgrade/Rollback Timeout**

Specifies the duration after which the upgrade or rollback times out.

**Default:** 600

## **Application Editor Preferences**

The Application Editor properties display when you select Application Editor from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Application Editor Preferences subsection to specify properties for the Application Editor application for this Web UI instance.

The properties in the Application Editor Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for SAP Filters** Specifies the maximum number of rows to display per page in the SAP Filters table.  
**Default:** 20
- **Rows per Page for SAP Filter Results** Specifies the maximum number of rows to display per page in the SAP Filter Results table.  
**Default:** 50
- **Rows per Page for Import Jobs Dialog** Specifies the maximum number of rows to display per page in the Search Results table in the Import Jobs dialog.  
**Default:** 50
- **Rows per Page for Search Results in Dialog** Specifies the maximum number of rows to display per page in the Search Results table in search dialogs.  
**Default:** 20
- **Hide SAP Drag and Drop Panes** Indicates whether to hide the SAP drag and drop panes.  
**Default:** Not selected
- **Description Column Length** Specifies the maximum number of characters to display in the Description column of the search results in the Import Jobs dialog.  
**Default:** 500  
**Limits:** 1-500, or 0 to not display a description
- **Maximum Number of Imported Jobs**  
Specifies the maximum number of jobs to import into Application Editor at any given time.  
**Default:** 1000  
**Limits:** 1-5000
- **New Job Name Default**  
Specifies the default naming convention to use for new jobs dropped on to the Graph.  
**Default:** {1} Job {0}

### **NOTE**

- In the default expression, {0} is required; {1} is optional.
- When the job is dropped on to the Graph:
  - {0} is replaced by a unique number.
  - {1} is replaced by the SAP job name if a job is dragged from the SAP Filter Results pane.
  - {1} is replaced by the job type if a non-template SAP job is dragged from the Palette.
  - {1} is replaced by the template name if a job template is dragged from the Palette.
- **Maximum Flow Height**  
Specifies the maximum height of the Graph in pixels.  
**Default:** 570  
**Limits:** 507-9999
- **Left Panes Width**

Specifies the width of the Palette, Delete List, and Overview panes in pixels.

**Default:** 250

**Limits:** 250-500

The System category contains the following fields:

- **Owner Field Mandatory**

Indicates whether the owner field is mandatory when creating a job. Selecting this check box requires the user to enter a value in the owner field; clearing this check box specifies that the AutoSys Workload Automation default should be used.

**Default:** Not selected

- **Graph Tooltip Length**

Specifies the maximum number of characters to be displayed for each value in the Show tooltips feature.

**Default:** 64

**Limits:** 0-256

- **Import All Button**

Indicates whether the Import buttons should be displayed in Application Editor's job import dialog. Selecting this check box displays the Import All and Import Selected buttons; clearing this check box only displays the OK and Cancel buttons.

**Default:** Selected

- **Commit Threads**

Specifies the number of threads that should be used to commit jobs to the back-end server.

**Default:** 20

**Limits:** 1-50

- **Commit Dialog Refresh Interval**

Specifies the interval (in milliseconds) at which the page refreshes after a job is committed.

**Default:** 2000

**Limits:** 2000-10000

- **Submit Owner on Commit**

Indicates whether the owner field should be submitted by default when the flow is committed.

**Default:** Selected

- **Proxy Server Name**

(Optional) Specifies the name of the proxy server to use for Web Service jobs.

**Limits:** 128 or fewer characters

- **Proxy Port**

(Optional) Specifies the port assigned to the proxy server for Web Service jobs.

**Limits:** 80-65535

- **Proxy User Name**

(Optional) Specifies the user name to use to connect to the proxy server for Web Service jobs

**Limits:** 128 or fewer characters

- **Proxy Password**

(Optional) Specifies the password associated with the proxy user name entered.

**Limits:** 128 or fewer characters

## **Configuration Preferences**

The Configuration properties display when you select Configuration from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Configuration Preferences subsection to specify additional properties for the Configuration application for this Web UI instance.

The properties in the Configuration Preferences subsection are displayed by category.

The EEM category contains the following fields:

- **EEM Connection Retry Interval**

Specifies the interval (in seconds) at which Web UI attempts to re-connect to the CA EEM server.

**Default:** 1

**Limits:** 1-100

- **Maximum EEM Connection Retries**

Specifies the maximum number of times that Web UI attempts to connect to the CA EEM server.

**Default:** 1

**Limits:** 1-100

The AutoSys Workload Automation category contains the following fields:

- **Enable SSL**

Indicates whether Secure Socket Layer (SSL) is enabled for communications with the AutoSys Workload Automation application server.

**Default:** Not selected

**NOTE**

The value displayed here must match the SSL setting for the AutoSys Workload Automation agents that communicate with the target AutoSys Workload Automation application server.

For more information about the SSL setting, see the CA Secure Socket Adapter (SSA) topics in the *AutoSys Workload Automation UNIX Implementation* or *AutoSys Workload Automation Windows Implementation* sections.

- **Enable PMUX**

Indicates whether port multiplexing (PMUX) is enabled for communications with the AutoSys Workload Automation application server.

**Default:** Selected

**NOTE**

The value displayed here must match the PMUX setting for the AutoSys Workload Automation agents that communicate with the target AutoSys Workload Automation application server.

For more information about the PMUX setting, see the CA Secure Socket Adapter (SSA) topics in the *AutoSys Workload Automation UNIX Implementation* or *AutoSys Workload Automation Windows Implementation* sections.

The Credentials category contains the following fields:

- **Validation Thread Pool Minimum Size**

Specifies the minimum size to use for the Credentials application validation thread pool.

**Default:** 2

**Limits:** 2-10

- **Validation Thread Pool Maximum Size**

Specifies the maximum size to use for the Credentials application validation thread pool.

**Default:** 4

**Limits:** 2-10

- **Validation Thread Pool Timeout**

Specifies the interval (in seconds) before the Credentials application validation thread pool times out.

**Default:** 600

**Limits:** 600-5000

- **Persist Default User**

Indicates whether to keep the default user across sessions.

**Default:** Not selected

- **Validation Refresh Interval**

Specifies the interval (in milliseconds) at which to refresh the Credentials application when a validation is in progress.

**Default:** 5000

**Limits:** 1000-60000

The Single Sign-On category contains the following fields:

- **Single Sign-On**

Indicates whether single sign-on using NTLM authentication protocol or integration with CA Siteminder is enabled. Web UI uses NTLM protocol or CA Siteminder to authenticate your session when single sign-on is active. To disable single sign-on, select None. If you disable single sign-on, Web UI requires you to authenticate your session by entering login credentials.

**Default:** None

- **Force Kerberos Compatibility**

Indicates whether to force NTLM authentication when the browser is Kerberos capable. When this check box is selected and NTLM single sign-on is active, Web UI overrides Kerberos protocol and uses NTLM protocol to authenticate your session.

**NOTE**

If NTLM is not selected, Force Kerberos Compatibility is ignored.

**Default:** Selected

The Security category contains the following fields:

- **Session Timeout**

Specifies the time (in minutes) that a user session can remain idle before timing out.

**Default:** 30

**Limits:** 1-1440

- **Auto Refresh Prevents Session Timeout** Indicates that Auto refresh is no longer a user action when you are idle in the application and remain logged in. **Default:** Not Selected

- **Single Session per User** Indicates whether to allow only one active session per user. The change applies only to the new session and does not affect a user that is already logged in to Web UI.

**Default:** Not selected

- **URL Parameters Encryption**

Specifies if the URL parameters are encrypted to allow protection of potentially sensitive information. **Default:** Not selected.

**NOTE**

The affected URL parameters are jobName, serverName, server, and objectName.

The Environment Category contains the following fields:

- **Default Time Zone** Indicates which time zone is used as a default for AutoSys Web UI users.

**Default:** Empty Selection

The Integrations Category contains the following fields:

- **AM base URL** Specifies that the base URL where AM is running. AM base URL must include HTTP or HTTPS protocol prefix and port.

**NOTE**

For example, the AM running on http://camhostname:8080/wla\_am/ui, the preference value should be http://camhostname:8080/.

## **Enterprise Command Line Preferences**

The Enterprise Command Line properties display when you select Enterprise Command Line from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Enterprise Command Line Preferences subsection to view and edit properties for the Enterprise Command Line application for this Web UI instance.

The properties in the Enterprise Command Line Preferences subsection are displayed by category.

The My Commands Settings category contains the following fields:

- **Maximum My Commands Saved**  
Specifies the maximum number of commands that can be saved in the My Commands table.  
**Default:** 100
- **Rows per Page for My Commands**  
Specifies the maximum number of rows to display per page in the My Commands table.  
**Default:** 10

The Global Commands Settings category contains the following fields:

- **Maximum Global Commands Saved**  
Specifies the maximum number of commands that can be saved in the Global Commands table.  
**Default:** 100
- **Rows per Page for Global Commands**  
Specifies the maximum number of rows to display per page in the Global Commands table.  
**Default:** 10

The General Settings category contains the following fields:

- **Maximum Number of Servers**  
Specifies the maximum number of target servers available for a command.  
**Default:** 30
- **Command Timeout**  
Specifies the maximum time (in milliseconds) allowed for a command to complete processing before it times out.  
**Default:** 86400000
- **Maximum Auto Complete Items**  
Specifies the maximum number of items to save in the auto-complete history table.  
**Default:** 25
- **Default Export File Name**  
Specifies the default name to use for the exported output file.  
**Default:** export.txt

## **Monitoring Preferences**

The Monitoring properties are displayed when you select Monitoring from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Monitoring Preferences subsection to view and edit properties for the Monitoring application for this Web UI instance.

### **NOTE**

The environment properties in the Monitoring Preferences subsection are displayed by category.

The UI category contains the following fields:

- **Auto Refresh Interval**  
Specifies the interval (in seconds) at which the display on the page refreshes.  
**Default:** 15  
**Limits:** 15-2147483647
- **Views Table Columns**  
Specifies the names of the columns to be displayed in the Views table. Multiple names can be entered, separated by commas. Valid values are name, job\_status, server\_status, alert\_severity\_status, jobs, collections, filters, inactive\_filters, unbound\_filters, status\_time, last\_edit\_time, last\_edit\_user, description, and servers.  
**Default:** name,job\_status,server\_status,alert\_severity\_status,jobs,collections,status\_time
- **Servers Table Columns**

Specifies the names of the columns to be displayed in the Servers table. Multiple names can be entered, separated by commas. Valid values are name, status, status\_time, status\_detail, filters, version, and type.

**Default:** name,status,status\_time,type,filters

- **Jobs Table Columns**

Specifies the names of the columns to be displayed in the Jobs table. Multiple names can be entered, separated by commas. Valid values are agent\_status, agent\_lstatus, application, box\_name, description, end\_time, exit\_code, group, instance, job\_name, job\_type, joid, machine, next\_start, owner, run\_machine, server, start\_time, status, status\_time, and timezone.

**Default:** job\_name,job\_type,status,server,run\_machine,status\_time

- **Alerts Table Columns**

Specifies the names of the columns to be displayed in the Alerts table. Multiple names can be entered, separated by commas. Valid values are alert, severity, status, job, server, create\_time, status\_time, response, comment, text, eiod, modified\_user, alert\_policy, url, and alert\_type.

**Default:** alert,severity,status,job,server,create\_time,status\_time

- **Scheduler Log Default Lines**

Specifies the number of lines to retrieve for the Scheduler Log.

**Default:** 50

**Limits:** 1-99999

- **Scheduler Log Auto Refresh Interval**

Specifies the interval (in seconds) at which the display on the page refreshes.

**Default:** 15

**Limits:** 15-2147483647

The Job Level Security category contains the following field:

- **Use AutoSys Workload Automation Security**

Indicates to use the AutoSys Workload Automation as-job, as-appl, and as-group policies in CA EEM to control access to jobs.

**Default:** Selected

**NOTE**

If changes are made in the as-job, as-appl, and as-group policies after the Monitoring collector has collected the data from the AutoSys Workload Automation server, changes to existing jobs contained in the views and monitoring database are not reflected.

The Back-end Services category contains the following fields:

- **Collector Mode**

Specifies whether to collect data for all defined servers or only the servers that are part of the active view filters.

**Default:** all

**Limits:** all or filter

- **Job Collector Run Interval**

Specifies the interval (in seconds) at which to run the collectors (get data from servers) and synchronize server changes from configuration.

**Default:** 5

**Limits:** 5-2147483647

- **Alarm Collector Run Interval**

Specifies the interval (in seconds) at which to run the alarm collector.

**Default:** 5

**Limits:** 5-2147483647

- **Alerts Deletion Threshold**

Specifies the number of days to store alerts in the database before deleting them.

**Default:** 7

**Limits:** 1-365

- **Alert Cleaner Run Interval**

Specifies the interval (in hours) in which to run database cleanup activity.

**Default:** 24

**Limits:** 1-2147483647

- **Security Cache Timeout**

Specifies the time (in minutes) after which the security cache is invalidated.

**Default:** 60

**Limits:** 0,1-2147483647

**NOTE**

When any CA EEM policy that affects Monitoring is changed, by default, it takes 60 minutes to be reflected in Web UI. The default timeout value reduces the overhead of checking CA EEM for every request. However, you can set the timeout to a lower value to reflect CA EEM security changes in Monitoring more quickly. Setting the value to 0 allows security changes to be reflected immediately.

- **DB Statistics Update Threshold**

(Oracle only) Specifies the number of records to be updated in the Oracle database before the DB statistics updater starts.

**Default:** 5000

**Limits:** 1-1000000, or 0 to disable the updater.

The Server Status Order category contains the fields that specify the severity value that should be assigned to this server status. This value is used to prioritize the overall server status for the view. The lowest number assigned indicates the highest severity. The following fields are available:

- **Failure**

Specifies the Failure severity value

**Default:** 0

**Limits:** 0-99

- **Disconnected**

Specifies the Disconnected severity value.

**Default:** 1

**Limits:** 0-99

- **Unknown**

Specifies the Unknown severity value.

**Default:** 2

**Limits:** 0-99

- **Not Configured**

Specifies the Not Configured severity value.

**Default:** 3

**Limits:** 0-99

- **Connected**

Specifies the Connected severity value.

**Default:** 50

**Limits:** 0-99

The Job Status Order category contains the fields that define the severity value that should be assigned to this job status. This value is used to prioritize the overall job status for the view. The lowest number assigned indicates the highest severity. If the priority is changed, the view status is updated on the next job run. The following fields are available:

- **Failure**

Defines the Failure severity value.

**Default:** 111

---

**Limits:** 100-127

- **Waiting on Reply**  
Defines the Waiting on Reply severity value.  
**Default:** 112  
**Limits:** 100-127
- **Pending Machine**  
Defines the Pending Machine severity value.  
**Default:** 113  
**Limits:** 100-127
- **Waiting on Resources**  
Defines the Waiting on Resources severity value.  
**Default:** 114  
**Limits:** 100-127
- **Queue Wait**  
Defines the Queue Wait severity value.  
**Default:** 115  
**Limits:** 100-127
- **Restart**  
Defines the Restart severity value.  
**Default:** 116  
**Limits:** 100-127
- **Terminated**  
Defines the Terminated severity value.  
**Default:** 117  
**Limits:** 100-127
- **On Ice**  
Defines the On Ice severity value.  
**Default:** 118  
**Limits:** 100-127
- **On Hold**  
Defines the On Hold severity value.  
**Default:** 119  
**Limits:** 100-127
- **Unknown**  
Defines the Unknown severity value.  
**Default:** 120  
**Limits:** 100-127
- **Success**  
Defines the Success severity value.  
**Default:** 121  
**Limits:** 100-127
- **Running**  
Defines the Running severity value.  
**Default:** 122  
**Limits:** 100-127
- **Starting**  
Defines the Starting severity value.  
**Default:** 123  
**Limits:** 100-127
- **Activated**  
Defines the Activated severity value.



**Default:** 124

**Limits:** 100-127

- **Inactive**

Defines the Inactive severity value.

**Default:** 125

**Limits:** 100-127

- **On No Execution**

Defines the On No Execution severity value.

**Default:** 126

**Limits:** 100-127

## **Quick Edit Preferences**

The Quick Edit properties display when you select Quick Edit from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Quick Edit Preferences subsection to specify properties for the Quick Edit application for this Web UI instance.

The properties in the Quick Edit Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for Search Results**

Specifies the maximum number of results per page to display in the Search Results section.

**Default:** 50

- **Rows per Page for Search Results in Dialog**

Specifies the maximum number of rows per page to display in the Search Results table in search dialogs.

**Default:** 20

- **Search History Items**

Specifies the maximum number of entries to display in the auto-complete history for the Name field in the Search section.

**Default:** 30

**Limits:** 1-100

- **Description Column Length**

Specifies the maximum number of characters to display for the Description field in the Search Results section.

**Default:** 500

**Limits:** 0-500

- **Single Result Redirect**

Indicates whether to automatically display the search result in the Properties section if the search returns a single item. Selecting this check box specifies that a single result displays directly in the Properties section instead of in the Search Results section; clearing this check box specifies that the result is displayed in the Search Results section.

**Default:** Selected

- **Display Calendar with 24 Months**

Indicates whether to display a 24-month calendar view instead of a 12-month calendar view. Selecting this check box specifies that the calendar view displays 24 months; clearing this check box specifies that the calendar view displays 12 months.

**Default:** Not selected

The System category contains the following fields:

- **Owner Field Mandatory**

Indicates whether the owner field is mandatory when creating a job. Selecting this check box requires the user to enter a value in the owner field; clearing this check box specifies that the AutoSys Workload Automation default should be used.

**Default:** Not selected

- **Copy Job Owner**

Indicates whether to copy the owner field when a job is copied. Selecting this check box retains the same owner name as the original job. If this field is not selected and the Owner Field Mandatory option is selected, the user must enter a value in the owner field; if this field is not selected and the Owner Field Mandatory option is cleared, the default user specified by the AutoSys Workload Automation instance is used.

**Default:** Selected

- **User Defined Job Description Maximum Length**

Specifies the maximum number of characters to display in the Description field for a User Defined job.

**Default:** 50

**Limits:** 1-256

- **User Defined Job Command Maximum Length**

Specifies the maximum number of characters to display in the Command field for a User Defined job.

**Default:** 50

**Limits:** 1-510

- **Proxy Server Name**

(Optional) Specifies the name of the proxy server to use for Web Service jobs.

**Limits:** 128 or fewer characters

- **Proxy Port**

(Optional) Specifies the port assigned to the proxy server for Web Service jobs.

**Limits:** 80-65535

- **Proxy User Name**

(Optional) Specifies the user name to use to connect to the proxy server for Web Service jobs.

**Limits:** 128 or fewer characters

- **Proxy Password**

(Optional) Specifies the password associated with the proxy user name entered.

**Limits:** 128 or fewer characters

## **Quick View Preferences**

The Quick View properties display when you select Quick View from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Quick View Preferences subsection to view and edit properties for the Quick View application for this Web UI instance.

The properties in the Quick View Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for Search Results**

Specifies the number of rows to display per page in the Search Results table.

**Default:** 50

- **Rows per Page for Reports**

Specifies the number of rows to display per page in a report table.

**Default:** 50

- **Search History Items**

Specifies the maximum number of entries to display in the auto-complete history for the Name field in the Search section.

**Default:** 30

**Limits:** 1-100

- **Job Details View**

Specifies whether the job details should be displayed as a tree or as JIL in the Job Details section.

**Default:** Tree

- **Maximum Flow Height**

Specifies the maximum height of the Job Flow section in pixels.

**Default:** 400

**Limits:** 150-1000

- **Show Dependent Jobs**

Indicates whether to display the Dependent Jobs report section.

**Default:** Selected

- **Show Starting Conditions**

Indicates whether to display the Starting Conditions report section, which includes the Job Dependencies and Job Resource Dependencies subsections.

**Default:** Selected

- **Show Job Runs**

Indicates whether to display the Job Runs report section, which includes job run events and logs.

**Default:** Selected

## **Forecast Preferences**

The Forecast properties display when you select Forecast from the View/Modify Preference drop-down list in the Preference section, and click Go.

Use the Forecast Preferences subsection to view and edit properties for the Forecast application for this Web UI instance.

The properties in the Forecast Preferences subsection are displayed by category.

The User Interface category contains the following fields:

- **Rows per Page for Forecast**

Specifies the maximum number of rows to display per page in the Forecast table.

**Default:** 50

- **Rows per Page for Table View**

Specifies the maximum number of rows to display per page in the Forecast Output Table View.

**Default:** 50

- **Rows per Page for Timeline View**

Specifies the maximum number of rows to display per page in the Forecast Output Timeline view.

**Default:** 50

The Command Parameters category contains the following field:

- **Execution Timeout**

Specifies the interval (in milliseconds) allowed for the forecast command to run and the report being generated.

**Default:** 300000

**Limits:** 60000-2147483647

## **Reporting Preferences**

The Reporting properties are displayed when you select Reporting from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Reporting Preferences subsection to view and edit properties for the Reporting application for this Web UI instance.

The properties in the Reporting Preferences subsection are displayed by category.

The Collectors category contains the following fields:

- **Enable Collector**

Indicates to automatically run the collectors for all servers.

**Default:** Selected

- **Collector Run Interval**  
Specifies the time interval between collector runs, in seconds.  
**Default:** 60

The Database Cleanup category contains the following fields:

- **Enable Cleanup**  
Indicates to automatically perform a periodic database cleanup.  
**Default:** Selected
- **Cleanup Interval**  
Specifies the number of days between database cleanup activity.  
**Default:** 2
- **Purge Data**  
Specifies the number of days that the data can exist in the database before being removed during database cleanup.  
**Default:** 30

### **Resources Preferences**

The Resources properties display when you select Resources from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Resources Preferences subsection to view and edit properties for the Resources application for this Web UI instance.

The properties in the Resources Preferences subsection are displayed by category.

The Table Controls category contains the following fields:

- **Rows per Page for All Tables**  
Specifies the maximum number of rows to display per page in the section tables.  
**Default:** 50
- **Rows per Page for All Dialog Tables**  
Specifies the number of rows to display per page in the dialog tables. **Default:** 20

The Search History Controls category contains the following field:

- **Search History Items**  
Specifies the maximum number of entries to display in the auto-complete history for the Name, Amount, and Description fields in the Search section, the Name and Amount fields in the Properties section, and the Name field in Resource Usage Search section.  
**Default:** 30  
**Limits:** 1-100

Add the monitoring user to the WorkloadAutomationAEWebService group as the monitoring user is used to connect to the web server.

## **Manage AutoSys Web UI Default Ports and Services**

This section describes AutoSys Web UI default ports and services.

## Tomcat Application Servers and Default Ports

The following table lists the default Tomcat application servers installed by AutoSys Web UI, the related AutoSys Web UI service and applications running on the server, and default port settings:

Server	AutoSys Web UI Service	Applications Running	Port	SSL	Shutdown
Tomcat	CA-wcc	AutoSys Web UI 64-bit User Interface	8080	8443	8005
Services Tomcat	CA-wcc-db		1527		

The following table lists other default ports:

Name	Port
CA EEM	5250
CA EEM Failover Port	509
Event Bus	7004
Internal database	1527
SSA Port	7163

### NOTE

If a port conflict is detected during the installation of AutoSys Web UI, an error message is displayed. However, port conflict detection is effective only for the port currently in use at the time of installation. If another application is configured to use a port and that application is not running when you install AutoSys Web UI, the port conflict cannot be detected. Ensure that ports specified for AutoSys Web UI do not conflict with other applications. To continue the installation, select a different port number. Note the default port assignments and consider the following when you select a different port:

- The range of Tomcat and Event Bus ports is 1025-65535. Additionally, port 80 is available for the Tomcat port and port 443 is available for the Tomcat SSL port. However, if you are installing AutoSys Web UI on UNIX, specify a run as (non-root) user, and you intend to use port 80 or 443, the port must be set up for non-root user access before you begin the installation.
- The Event Bus port must not be in the dynamic port range of the machine.
- The range of database ports is 1025-65535 for the internal database and for the port on which an external database is running.
- The range for the SSA port is 49152-50176.
- The range of system or well-known ports is 0-1023.
- The range of user or user (registered) ports is 1024-49151.
- The range of dynamic and private ports is 49152-65535.
- If the firewall is enabled and AutoSys Web UI is using the internal database, and you are using AutoSys Workload Automation in standalone mode and CA EEM with no failover installed on different machines, then you can open all outgoing ports and can close all incoming ports except the following for AutoSys Web UI:
  - AutoSys Web UI Tomcat port (the default for non-SSL is 8080; the default for SSL is 8443)
  - SSA physical port 7163
- The firewall must accept all traffic from the loopback (lo) interface.
- After enabling the firewall, the firewall rules must ensure that the AutoSys Web UI server can ping the computer on which the AutoSys Web UI database, Reporting database, and CA EEM are hosted.
- The Event Bus port 7004 is to be opened in an enabled firewall if you are using HA mode.

## **AutoSys Web UI Services**

The following are the services that AutoSys Web UI runs:

- CA-wcc
- CA-wcc-services
- CA-wcc-db

### **NOTE**

- CA-wcc-db runs only if the internal database was selected in the Database Settings panel when AutoSys Web UI was installed.
- When you start or restart CA-wcc, CA-wcc-services is automatically started if it is not yet running.
- When you stop CA-wcc-services, CA-wcc is automatically stopped.
- On Windows, you can manage services using the Services window.
- On UNIX, use the `unisrvcntr` command to manage services.

## **unisrvcntr Command**

On UNIX, use the `unisrvcntr` command to stop, start, or restart a AutoSys Web UI service. The syntax for the `unisrvcntr` command is as follows:

```
unisrvcntr {start|stop|restart|status} service_name
```

- **start**  
Starts the service.
- **stop**  
Stops the service.
- **restart**  
Stops and then starts the service.
- **status**  
Reports the status of the service.

### **NOTE**

- The `unisrvcntr` command must be run as a root user.
- The *service\_name* is optional when you run the command with the status argument. With no service name specified, the status of each service that is managed by `unisrvcntr` is displayed.

## **Restart AutoSys Web UI Services**

After you update the CA EEM server definition, change ports, or change the SSL mode using the Configuration Batch Interface, restart AutoSys Web UI services.

UNIX:

To restart AutoSys Web UI services on UNIX, run the following command in a shell window:

```
unisrvcntr restart CA-wcc-services
```

Windows:

### **Follow these steps:**

1. Open the Services window.
2. Select CA-wcc-services and click the Restart button.  
The Restart Other Services dialog opens.
3. Click Yes.

The dialog closes and AutoSys Web UI services are restarted.

**NOTE**

After you restart AutoSys Web UI services, verify that the AutoSys Web UI servers have started before you use or perform operations with AutoSys Web UI. You display AutoSys Web UI server startup status by running the following command:

UNIX:

```
grep "startup in" $CA_WCC_INSTALL_LOCATION/log/*.log
```

Windows:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

## Customize Secure Access to AutoSys Web UI

This article explains how to customize secure access to AutoSys Web UI. This product can run in SSL mode or non-SSL mode. When the product is running in SSL mode, users log into AutoSys Web UI using a secure URL (https). For SSL mode, a self-signed certificate or a certificate issued by a trusted certificate authority is required on the AutoSys Web UI server. This section contains guidelines for customizing secure access to AutoSys Web UI.

**NOTE**

- The examples in this section are for reference only and can be adjusted according to your requirements.
- This section applies only if you selected the Enable SSL option during AutoSys Web UI installation or you have changed the mode to SSL using the Configuration Batch Interface.

### Section Conventions

The following conventions are used in this section:

\$CA\_WCC\_INSTALL\_LOCATION is the root directory of the AutoSys Web UI installation on UNIX.

%CA\_WCC\_INSTALL\_LOCATION% is the root directory of the AutoSys Web UI installation on Windows.

The default AutoSys Web UI installation directories are as follows:

- UNIX:

```
/opt/CA/WorkloadAutomationAE/wcc
```

- Windows:

```
C:\Program Files\CA\WorkloadAutomationAE\wcc
```

**NOTE**

- The backslash character (\) at the end of a line indicates that the UNIX command continues on the next line.
- The caret character (^) at the end of a line indicates that the Windows command continues on the next line.

### Change the Certificate

The following options are available for changing the certificate:

- Generate and use a self-signed certificate.
- Request and use a certificate from a trusted certificate authority.
- Reuse an existing keystore file.
- Import an existing private key and certificate from a PKCS12 file.

## **Generate and Use a Self-Signed Certificate**

You can generate and use a self-signed certificate to enable SSL in your environment. Self-signed certificates cannot be revoked if they become compromised and do not provide an acceptable security level. You can use self-signed certificates for testing and non-commercial environments, such as labs. In a production environment, using a self-signed certificate is not recommended.

### **Follow these steps:**

1. [Generate a private key and self-signed certificate.](#)
2. [Restart AutoSys Web UI services.](#)

## **Generate a Private Key and Self-Signed Certificate**

You can generate a private key and self-signed certificate and then add them to the keystore.

### **Follow these steps:**

1. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows):
  - UNIX:
 

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -delete -alias tomcat -keystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit
```
  - Windows:
 

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -delete -alias tomcat -keystore "%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass changeit
```

#### **NOTE**

The default name of the key is tomcat. To access the keystore, you also need the keystore password. The default password is changeit.

The previous key is removed from the keystore.

2. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows):
  - UNIX:
 

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -genkey -alias tomcat -keyalg RSA -keystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit -keypass changeit -keysize 1024 -dname "cn=WCC_SERVER_NAME" -validity 825
```
  - Windows:
 

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -genkey -alias tomcat -keyalg RSA -keystore "%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass changeit -keypass changeit -keysize 1024 -dname "cn=WCC_SERVER_NAME" -validity 825
```

#### **NOTE**

- The keysize argument lets you specify the key size. Typical values are 1024 or 2048.
- *WCC\_SERVER\_NAME* is the name of your AutoSys Web UI server.
- The validity argument lets you specify the certificate validity period in days.
- If a publicly trusted certificate is used, then the validity must be 398 in place of 825.

A new private key and self-signed certificate are generated.

## **Restart AutoSys Web UI Services**

After you generate, obtain, or import the key and certificate, restart AutoSys Web UI services.

UNIX:



To restart AutoSys Web UI services on UNIX, run the following command in a shell window:

```
unisrvcntr restart CA-wcc-services
```

Windows:

**Follow these steps:**

1. Open the Services window.
2. Select CA-wcc-services and click the Restart button.  
The Restart Other Services dialog opens.
3. Click Yes.  
The dialog closes and AutoSys Web UI services are restarted.

**NOTE**

After you restart AutoSys Web UI services, verify that the AutoSys Web UI servers have started before you use or perform operations with AutoSys Web UI. You display AutoSys Web UI server startup status by running the following command:

UNIX:

```
grep "startup in" $CA_WCC_INSTALL_LOCATION/log/*.log
```

Windows:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

### **Request and Use a Certificate From a Trusted Certificate Authority**

You can request and use a certificate from a trusted certificate authority to enable SSL in your environment. Certificates issued by trusted certificate authorities, if compromised, can be revoked, providing better security than self-signed certificates. In a production environment, using a certificate from a trusted authority is recommended.

**NOTE**

This section describes how to generate the private key and certificate request through the keystore used by AutoSys Web UI. Alternatively, many certificate authorities let you generate the private key and request the certificate through your browser's keystore. The private key and certificate you obtain can be exported from your browser to a PKCS12 file which can be then imported into AutoSys Web UI.

**Follow these steps:**

1. [Generate a private key and self-signed certificate.](#)
2. [Request a certificate from the trusted authority.](#)
3. [Add the certificates to the keystore.](#)
4. [Restart AutoSys Web UI services.](#)

### **Generate a Private Key and Self-Signed Certificate**

You can generate a private key and self-signed certificate and then add them to the keystore.

**Follow these steps:**

1. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows):
  - UNIX:
 

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -delete -alias tomcat -keystore
$CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit
```
  - Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -delete -alias tomcat -keystore
"%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass changeit
```

**NOTE**

The default name of the key is tomcat. To access the keystore, you also need the keystore password. The default password is changeit.

The previous key is removed from the keystore.

2. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows):

- UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -genkey -alias tomcat -keyalg RSA -
keystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit -
keypass changeit -keysize 1024 -dname "cn=WCC_SERVER_NAME" -validity 825
```

- Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -genkey -alias tomcat -keyalg RSA -
keystore "%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass changeit -
keypass changeit -keysize 1024 -dname "cn=WCC_SERVER_NAME" -validity 825
```

**NOTE**

- The keysize argument lets you specify the key size. Typical values are 1024 or 2048.
- *WCC\_SERVER\_NAME* is the name of your AutoSys Web UI server.
- The validity argument lets you specify the certificate validity period in days.
- If a publicly trusted certificate is used, then the validity must be 398 in place of 825.

A new private key and self-signed certificate are generated.

**Request a Certificate**

To use a certificate from a trusted certificate authority, create a certificate request file and send it to the certificate authority. The certificate authority will send you the certificate.

**Follow these steps:**

1. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows):

- UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -certreq -alias tomcat -keystore
$CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit -file
certreq.csr
```

- Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -certreq -alias tomcat -keystore
"%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass changeit -file
certreq.csr
```

A certificate request file (certreq.csr) is generated.

2. Send the certificate request file to the certificate authority. Contact the certificate authority for specific instructions.

**NOTE**

AutoSys Web UI uses certificates in PEM or DER format. When using the PEM format, the certificate must not contain any information before the BEGIN CERTIFICATE marker or after the END CERTIFICATE marker.

**Add the Certificates to the Keystore**

Add the certificates to the keystore after you receive your private certificate.

**Follow these steps:**

1. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows) to add the certificate of the root certificate authority to the keystore:

## – UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -importcert -alias RootCA -file RootCA.cer
-keystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit
```

## – Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -importcert -alias RootCA -file
RootCA.cer -keystore "%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass
changeit
```

The certificate of the root certificate authority is added to the keystore.

2. (Optional) Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows) to add a certificate of a subordinate authority to the keystore:

## – UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -importcert -alias SubCA -file SubCA.cer
-keystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore -storepass changeit
```

## – Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -importcert -alias SubCA -file
SubCA.cer -keystore "%CA_WCC_INSTALL_LOCATION%\data\config\.keystore" -storepass
changeit
```

The certificate of the subordinate authority is added to the keystore.

3. Repeat Step 2 for each certificate of the subordinate authority you have downloaded.
4. Run the following commands in a shell window (on UNIX) or from a command prompt (on Windows) to add your private certificate to the keystore:

## – UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -importcert -trustcacerts -file
certificate.cer - alias tomcat -keystore $CA_WCC_INSTALL_LOCATION/data/
config/.keystore -storepass changeit
```

## – Windows

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -importcert -trustcacerts -file
certificate.cer - alias tomcat -keystore "%CA_WCC_INSTALL_LOCATION%\data\config
\.keystore" -storepass changeit
```

The private certificate is added to the keystore.

**Restart AutoSys Web UI Services**

After you generate, obtain, or import the key and certificate, restart AutoSys Web UI services.

## UNIX:

To restart AutoSys Web UI services on UNIX, run the following command in a shell window:

```
unisrvcntr restart CA-wcc-services
```

## Windows:

**Follow these steps:**

1. Open the Services window.
2. Select CA-wcc-services and click the Restart button.

The Restart Other Services dialog opens.

3. Click Yes.

The dialog closes and AutoSys Web UI services are restarted.

### NOTE

After you restart AutoSys Web UI services, verify that the AutoSys Web UI servers have started before you use or perform operations with AutoSys Web UI. You display AutoSys Web UI server startup status by running the following command:

UNIX:

```
grep "startup in" $CA_WCC_INSTALL_LOCATION/log/*.log
```

Windows:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

### Reuse an Existing Keystore File

You can reuse a keystore file from an existing AutoSys Web UI server to enable SSL in your environment. An existing keystore file can contain self-signed certificates or certificates issued by a certificate authority.

#### Follow these steps:

1. [Copy the keystore file to the AutoSys Web UI server.](#)
2. [Restart AutoSys Web UI services.](#)

### Copy an Existing Keystore File

You can reuse the keystore file from an existing AutoSys Web UI server with a new AutoSys Web UI server installation.

To reuse the keystore file, copy the keystore file in the following directory from the existing AutoSys Web UI server to the new AutoSys Web UI server:

- UNIX:  
\$CA\_WCC\_INSTALL\_LOCATION/data/config/.keystore
- Windows:  
%CA\_WCC\_INSTALL\_LOCATION%\data\config\keystore

### Restart AutoSys Web UI Services

After you generate, obtain, or import the key and certificate, restart AutoSys Web UI services.

UNIX:

To restart AutoSys Web UI services on UNIX, run the following command in a shell window:

```
unisrvctr restart CA-wcc-services
```

Windows:

#### Follow these steps:

1. Open the Services window.
2. Select CA-wcc-services and click the Restart button.  
The Restart Other Services dialog opens.
3. Click Yes.  
The dialog closes and AutoSys Web UI services are restarted.

**NOTE**

After you restart AutoSys Web UI services, verify that the AutoSys Web UI servers have started before you use or perform operations with AutoSys Web UI. You display AutoSys Web UI server startup status by running the following command:

UNIX:

```
grep "startup in" $CA_WCC_INSTALL_LOCATION/log/*.log
```

Windows:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

**Import an Existing Private Key and Certificate from a PKCS12 File**

You can import an existing private key and certificate packaged in a PKCS12 file to enable SSL in your environment.

**NOTE**

Most certificate authorities let you generate the private key and request the certificate from their website. Once you have obtained the certificate and have installed it in your browser, you can export the private key and certificate to a PKCS12 file, and then import the private key and certificate into AutoSys Web UI.

**Follow these steps:**

1. [Import the private key and certificate from a PKCS12 file.](#)
2. [Restart AutoSys Web UI services.](#)

**Import the Key and Certificate from a PKCS12 File**

You can import a private key and certificate from a PKCS12 file. The private key and certificate are then added to the keystore.

**Follow these steps:**

1. Run the following command in a shell window (on UNIX) or from a command prompt (on Windows) to determine the alias name inside the PKCS12 file:

– UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -list -storetype PKCS12 -keystore
myp12.p12
```

– Windows:

```
%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool -list -storetype PKCS12 -keystore
myp12.p12
```

The following is an example of the text that is displayed:

```
Keystore type: PKCS12
Keystore provider: SunJSSE
Your keystore contains 1 entry
alias, Jan 18, 2012, PrivateKeyEntry,
Certificate fingerprint (MD5): 5B:7E:A1:64:EA:1F:3A:63:74:41:30:FD:32:CF:BF:77
```

**NOTE**

The name of the alias, *alias*, is used in the next step.

2. Run the following command in a shell window (on UNIX) or from a command prompt (on Windows) to import the private key and add it to the keystore:
  - UNIX:

```
$CA_WCC_INSTALL_LOCATION/jre/bin/keytool -importkeystore -srckeystore myp12.p12
-srcstoretype PKCS12 -srcstorepass my-password -deststorepass changeit -
destkeypass changeit -destkeystore $CA_WCC_INSTALL_LOCATION/data/config/.keystore
-alias alias -destalias tomcat
```

– Windows:

```
"%CA_WCC_INSTALL_LOCATION%\jre\bin\keytool" -importkeystore -srckeystore
myp12.p12 -srcstoretype PKCS12 -srcstorepass my-password -deststorepass changeit
-destkeypass changeit -destkeystore "%CA_WCC_INSTALL_LOCATION%\data\config
\keystore" -alias alias -destalias tomcat
```

The private key and certificate are added to the keystore.

**NOTE**

- The certificate chain found in the PKCS12 file is also imported, although it is not visible with the keytool program. If the PKCS12 file does not contain the certificate chain of the issuing authority, you can add the certificate chain to the keystore.
- A PKCS12 file can have either a .pfx or .p12 extension.

### **Restart AutoSys Web UI Services**

After you generate, obtain, or import the key and certificate, restart AutoSys Web UI services.

UNIX:

To restart AutoSys Web UI services on UNIX, run the following command in a shell window:

```
unisrvctr restart CA-wcc-services
```

Windows:

**Follow these steps:**

1. Open the Services window.
2. Select CA-wcc-services and click the Restart button.  
The Restart Other Services dialog opens.
3. Click Yes.  
The dialog closes and AutoSys Web UI services are restarted.

**NOTE**

After you restart AutoSys Web UI services, verify that the AutoSys Web UI servers have started before you use or perform operations with AutoSys Web UI. You display AutoSys Web UI server startup status by running the following command:

UNIX:

```
grep "startup in" $CA_WCC_INSTALL_LOCATION/log/*.log
```

Windows:

```
findstr /C:"startup in" "%CA_WCC_INSTALL_LOCATION%\log*.log
```

### **Troubleshoot Certificate Issues**

This section contains solutions to problems you can encounter when changing keys and certificates.

## **Invalid Certificate Security Alert**

### **Symptom:**

I get the following SecurityAlert message in my browser: The name on the security certificate is invalid or does not match the name of the site.

This problem occurs in the following cases:

- When I try to access AutoSys Web UI using localhost. For example, `https://localhost:8443/wcc`.
- When I try to access AutoSys Web UI through a URL that does not match the CN attribute of the certificate Issued To field exactly. For example, using `https://wcc01:8443/wcc` causes a SecurityAlert if the CN attribute of the certificate Issued To field is `wcc01.ca.com`.

### **Solution:**

Use the name of the server as it is specified in the CN attribute of the certificate Issued To field. For example, use `https://wcc01.ca.com:8443/wcc` if the CN attribute of the certificate Issued To field is `wcc01.ca.com`.

## **Security Alert with Genuine Certificate**

### **Symptom:**

I get a SecurityAlert in my browser even though I am using a genuine Certificate Authority generated certificate. The CN attribute of the certificate Issued To field matches the hostname/URL that I used to access AutoSys Web UI.

### **Solution:**

Ask your certificate authority whether a chain certificate is needed in the keystore to establish full authenticity of the certificate.

## **Disable Weak Ciphers in SSL Mode**

The following procedure describes how to disable weak ciphers in AutoSys Web UI in SSL mode.

### **NOTE**

- Browsers that do not support secure encryption, and those that have secure encryption disabled, will be denied access to AutoSys Web UI.
- You can change the list of ciphers based on your requirements.

### **Follow these steps:**

1. Open the `server.xml` file in a text editor.
2. Locate the `<Connector .../>` XML element.
3. Insert the following attribute:

```
ciphers="TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA,
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256,
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256,
TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256"
```

The `<Connector .../>` XML element should display as follows:

### **NOTE**

"..." illustrates other attributes

```

<Connector ...
...
compression="on"
keyPass="changeit"
keyAlias="tomcat"
compressableMimeType="text/html,text/xml,text/plain,text/jav
ascript,text/css,application/x-javascript"
server="WCC"
ciphers="TLS_ECDHE_ECDSA_WITH_AES_128_GCM_SHA256,
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA,
TLS_ECDHE_ECDSA_WITH_AES_128_CBC_SHA256,
TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256,
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA,
TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256,
TLS_DHE_RSA_WITH_AES_128_GCM_SHA256,
TLS_DHE_RSA_WITH_AES_128_CBC_SHA,
TLS_DHE_RSA_WITH_AES_128_CBC_SHA256"

```

**NOTE**

Disable TLS 1.0 and 1.1 protocols by adding: `sslEnabledProtocols="TLSv1.2"` to the `server.xml` 9443 connector.

- Repeat steps 1-3 for each of the following `server.xml` files:

**WINDOWS:**

`%CA_WCC_INSTALL_LOCATION%\tomcat\conf\server.xml`

**UNIX:**

`$CA_WCC_INSTALL_LOCATION/tomcat/conf/server.xml`

- Restart the AutoSys Web UI services.

**Control Access to the AutoSys Web UI Web Interface**

To protect against Cross Site Request Forgeries (XSRF), every HTTP request within AutoSys Web UI contains a token, whose value is unique for each session. An unsupported method in the previous release allowed the user to directly access a specific component of AutoSys Web UI after logging in.

AutoSys Web UI protection against XSRF can be configured to allow direct access to some components; not all of the AutoSys Web UI components or functionality may work properly. For example, to add an embedded link from your internal website to the view details in a specific AutoSys Web UI view, configure AutoSys Web UI to bypass the `AntiXSRFToken` check for this view.

**Example: Allow direct access to View Details**

Suppose that you want to allow direct access to the url that displays the `flow_line` view details in AutoSys Web UI. To do this, you must configure the following url: `WCC_Server/wcc/monitoring/pages/viewDetail0.faces?displayMode=inContext&viewName=flow_line`

**Follow these steps:**

- Locate the following file:

```

$CA_WCC_INSTALL_LOCATION/data/config/permission-filter/custom-allowed-redirect-after-
login.txt

```

- Add the following line to the end of the file:



```
^/wcc/monitoring/pages/showViewDetail.faces\?displayMode=inContext&viewName=flow_line
$
```

**NOTE**

The format of the line is written in Java regexp, therefore it is necessary to escape the ? character with a \.

3. Locate the following file:

```
$CA_WCC_INSTALL_LOCATION/data/config/permission-filter/custom-bypass-anti-xsrf-
token.txt
```

4. Add the following line to the end of the file:

```
^/wcc/monitoring/pages/showViewDetail.faces.*
```

**NOTE**

Only the path is required.

5. Restart the AutoSys Web UI services.

You can now access WCC\_Server/wcc/monitoring/pages/showViewDetail.faces?  
displayMode=inContext&viewName=flow\_line

**NOTE**

- If the user is already logged in to AutoSys Web UI, and the url is accessed from the same browser on the same machine, the 'flow\_line' view details are displayed. If the user is not logged in, the user will be sent to the login page. After login, the 'flow\_line' view details will appear.
- When AutoSys Web UI is configured in High Availability, the procedure must be repeated on every node.

**Configure NTLM Single Sign-On**

You can use NTLM protocol to configure single sign-on capability. AutoSys Web UI does not support Kerberos authentication protocol but compatibility with Kerberos can be enabled by selecting the Force Kerberos compatibility checkbox. If you do not enable NTLM, AutoSys Web UI requires you to authenticate your session by entering login credentials.

**NOTE**

- If you change your domain password while NTLM authentication is enabled, clear the browser cache.
- For more information about authentication protocol configuration settings, see the *Configuration Help*.

**Configure Integration with CA Siteminder**

In the current release, you can use CA Siteminder single sign-on capabilities. If you do not enable integration with CA Siteminder, AutoSys Web UI requires you to authenticate your session by entering login credentials.

**NOTE**

- The minimum versions of CA Siteminder and EEM that are needed for this solution are:
  - CA Siteminder - 12.5 CR2 (12.05.0002.843)
  - EEM - r12.5 (12.5.0.7)
- CA Siteminder has an extensive installation and configuration process that you can find in the CA Siteminder support related documentation.
- For more information about authentication protocol configuration settings, see the *Configuration Help*.

1. Example:

- UNIX

```
cd $CA_WCC_INSTALL_LOCATION/data/config
```

- Windows

```
cd "%CA_WCC_INSTALL_LOCATION%\data\config"
```

## Tune Component Performance

This article includes information about component performance and recommendations for improving performance.

### Component Performance and Recommendations

AutoSys Web UI component performance depends on preference settings, the database, the environment, and the workload.

### Monitoring

Monitoring performance is affected by factors such as monitoring preference settings, whether job level security is enabled, and the number of alerts.

### Monitoring Recommendations

Monitoring performance is affected by many factors. To improve performance, consider the following recommendations:

- The Monitoring application depends heavily on database performance. The following methods can be used to improve Monitoring performance:
  - Use an appropriate server for the external database. We recommend that the database server meets the same requirements as specified for an AutoSys Web UI server.
  - Use an external database for AutoSys Web UI that is in the same physical location as the AutoSys Web UI server.
  - Verify that there is sufficient network bandwidth and latency between the AutoSys Web UI server and the database server.
  - Use an Oracle RAC or Microsoft SQL cluster.
- Job flow details performance depends on the AutoSys Web UI server CPU and RAM. To improve job flow details performance, use AutoSys Web UI in High Availability mode.

#### **NOTE**

- Job details performance depends heavily on the <UAJM> application server response time.
- For information about Monitoring preferences, see Configuration Help.
- Alerts can affect performance. The following methods can be used to improve alert performance:
  - Keep the number of alerts that you define low.
  - Use alerts only if alert states such as open or acknowledged are processed.
  - Do not use the closed alert state in alert policy definitions.
  - Use the Reporting application for alert statistics (alarms, job status changes).
  - Control alert history by adjusting the Alerts Deletion Threshold property in Monitoring preferences in Configuration. The value should be equal to or greater than the AutoSys Workload Automation alarm maintenance interval.
- The number of AutoSys Workload Automation servers referenced by one instance of AutoSys Web UI can affect performance. The following methods can be used to improve performance:
  - Add only AutoSys Workload Automation servers that are running in Configuration. Delete AutoSys Workload Automation servers that are not used.
  - Increase collection time. You can adjust the following parameters in Monitoring preferences in Configuration:
    - Job Collector Run Interval
    - Alarm Collector Run Interval

**NOTE**

Status updates are slower when collection time is increased; however, the AutoSys Web UI instance can reference more AutoSys Workload Automation servers.

- Use AutoSys Web UI in High Availability mode with an external database.

**Optimize Monitoring**

You can optimize Monitoring by modifying Monitoring preferences in Configuration.

**Follow these steps:**

1. Click the Configuration tab, select Monitoring from the View/Modify Preferences drop-down list in the Preferences section, and click Go.  
The Monitoring preferences are displayed.

2. Modify the following fields according to the recommended settings:

- **Collector Mode**

Specifies whether to collect data for all defined servers or only the servers that are part of the active view filters. The recommended setting is all.

**Example:** AutoSys Workload Automation servers defined in Configuration are: A1, A2, A3, A4, and A5. Only A1 and A2 are defined in active monitoring view filters.

The effects of the settings are:

- all -- collect data from servers A1, A2, A3, A4, A5.
- filter -- collect data only from servers A1, A2.

**NOTE**

Selecting filter instead of all has an adverse effect on performance. Selecting filter enables data collection from servers defined in active Monitoring view filters.

- **Job Collector Run Interval**

Specifies the interval (in seconds) between collector runs and server changes synchronization.

**NOTE**

If the collector service generates a high load on the server, you can increase this value (10 seconds or 15 seconds is sufficient). This situation can occur if AutoSys Web UI is attached to many AutoSys Workload Automation servers that generate a high load (many updates during a 5-second interval).

3. Click Save.  
Your changes are saved.

**NOTE**

For more information about Monitoring preferences, see *Configuration Help*.

**Job Level Security**

Enabling job level security in AutoSys Web UI can decrease view performance because CA EEM security policies must be checked for each job that a user accesses. View performance depends on the number of jobs that must be retrieved from the database to display one page of data. View Performance also depends on the percentage of jobs in a view or on a server to which the user has access.

The following table shows that view access defined in CA EEM policies affects performance. For example, assume that 1000 jobs are available to a view and the user wants to display the first 100 jobs to which they have access. User A has access to all of the jobs in view (or server) and performance is optimal. That is, only 100 jobs must be scanned to display the first 100 jobs. However, user B has access to only half of the jobs. Performance decreases because 200 jobs must be scanned to display the first 100 jobs. Performance decreases most significantly for user D; all of the jobs must be scanned

to display the first 100 jobs. The same comparison applies to alert policies. The following table shows the performance comparison for different users:

User	Number of Jobs Allowed by CA EEM Policies	Number of Jobs Scanned to Display the First 100 Jobs
User A	1000	100
User B	500	200
User C	200	500
User D	100	1000

We recommend that you verify whether job level security for Monitoring is required at your site. Monitoring contains the job flow and fields such as job name, status, and job type. View level security is always applied, and a user can see only jobs or job flows in views to which they have access. Therefore, job level security in Monitoring is not always necessary. Job level security for Monitoring can be enabled or disabled in Monitoring preferences in Configuration. To do so, select or clear the Use AutoSys Workload Automation policies in EEM check box.

Access to the job details in Quick View is controlled by the job level security policies in CA EEM, even if the job level security is disabled for Monitoring. For example, a user can see the command attribute value in Quick View only if they have read or full access to the job.

### **Flow Rendering Performance**

The time it takes to render the flow is dependent on the complexity value of the flow. The complexity value is determined by the number of jobs and connections within a flow. Performance is also affected by the client browser and hardware that is used to generate a flow.

#### **NOTE**

There is no limit for the number of jobs in the flow that can be defined on the server.

The following table displays the performance factor by number of jobs and browser:

Browser	200 Jobs*	500 Jobs*	1500 Jobs*	3500 Jobs*	7500 Jobs*
Firefox 33	1 seconds	4 seconds	7seconds	13 seconds	25 seconds
Chrome 38	1 seconds	2.5 seconds	4 seconds	7 seconds	13 seconds
IE 9	1 seconds	5 seconds	10 seconds	25 seconds	75 seconds
IE 11	1 seconds	2.5 seconds	5.5 seconds	14 seconds	30 seconds

\*Specifies the number of visible jobs in the flow. This number is typically lower than the total number of jobs in a view.

#### **NOTE**

The above results are based on tests that are performed on an Intel Core Duo computer with 4 GB memory.

### **Reporting**

This section describes resource requirements for reports and contains test data.

#### **Resource Requirements for Reporting**

Creating reports requires CPU cycles, RAM, and disk space. The type of report and the number of job runs included in the report affect the amount of resources that are required.

##### **Memory Usage**

RAM memory for reports is required when a report is created or exported. The amount of memory usage depends on the number of pages in the report. Depending on the report, the peak memory usage is approximately 20 KB to 40 KB per page. A user can have several reports open and displayed in the browser. However, if a report is not currently being processed, the memory usage is minimal.

The following table shows the various reports and their memory usage:

Report	Memory Usage Depends On
Job Run Summary, Job Statistics	Number of jobs; one job requires approximately 2 KB of memory
Job Run Details	Number of job runs; one job run requires approximately 2 KB of memory

For example, a Job Run Details report on 50,000 jobs with a total of 500,000 job runs requires approximately 1 GB of memory. A Job Run Details report on 100,000 jobs with a total of 1,000,000 job runs requires approximately 2 GB of memory.

#### NOTE

Memory usage for other reports depends on the number of AutoSys Workload Automation servers and the selected date range. However, the impact on memory usage is minimal.

#### Disk Usage for Temporary Files

Reporting creates temporary files to reduce memory usage. Depending on the report, the peak required disk space is approximately 40 KB to 80 KB per page. Approximately half of the disk space remains for the file until the user logs out of AutoSys Web UI. The following table shows the temporary files and their locations on UNIX:

Temporary File	Comments
\$WCC /temp/swap/ <i>filename</i>	Exists only during the building of the report.
\$WCC /temp/serializedRefs/ <i>task_ID</i>	Created after building the report and is used for the report. <b>Note:</b> This file remains on the AutoSys Web UI server hard drive during the user session. Large reports occupy a significant amount of disk space until the user logs out of AutoSys Web UI.
\$WCC /tomcat/webapps/reporting-rest/output/ <i>task_ID</i>	Used for exporting the report. <b>Note:</b> This file remains on the AutoSys Web UI server hard drive during the user session. Large reports occupy a significant amount of disk space until the user logs out of AutoSys Web UI.

#### CPU Usage

For Reporting, CPU usage depends on the number of pages in a report and the amount of data that must be gathered to create the report. The following table shows the factors that affect the amount of CPU time that is required to create a report:

Report	CPU Usage Depends On
Summary, Job Statistics	Number of jobs
Details	Number of job runs
Throughput, Event Summary, Event Status	Number of job runs

**NOTE**

- CPU usage for other reports depends only on the number of AutoSys Workload Automation servers and the selected date range.
- When you export a report, only the number of pages in the report affects CPU usage.

**Reporting Database Size**

The Reporting database size depends on many factors. The following table shows the approximate size of the Reporting database for various databases, jobs, and job runs:

Database	Jobs	Job Runs	Size
Derby (internal)	50,000	100,000	500 MB
Derby (internal)	50,000	1,000,000	3.5 GB
Derby (internal)	50,000	3,000,000	7 GB
Microsoft SQL Server	50,000	150,000	650 MB
Microsoft SQL Server	50,000	1,000,000	3.2 GB
Microsoft SQL Server	150,000	10,000,000	15 GB
Oracle	50,000	150,000	2 GB
Oracle	150,000	10,000,000	12 GB
Oracle	150,000	30,000,000	31 GB

**NOTE**

For Oracle, an additional 4 GB is required for undo and temp tablespaces.

**Application Editor**

Application Editor performance is affected by the number of concurrent users and the value of the Commit Threads property.

**Optimize Application Editor**

You can optimize Application Editor by modifying the value of the Commit Threads field in Configuration. The Commit Threads value specifies the number of threads used to commit jobs to the back-end server.

**Follow these steps:**

1. Click the Configuration tab, select Application Editor from the View/Modify Preferences drop-down list in the Preferences section, and click Go.  
The Application Editor preferences are displayed.
2. Modify the value of the Commit Threads field in the System category.

**NOTE**

The default value is 20; specify a higher value for a faster commit speed. Valid values are 1-50. Specifying a higher value affects only the commit speed of jobs that are not included in a box job container. AutoSys Workload Automation box jobs have priority over other jobs, and use a single commit thread. Therefore, this setting only improves performance on systems with many jobs in the flow that are not included in a box.

3. Click Save.  
Your changes are saved.

**NOTE**

For more information about Application Editor preferences, see [AutoSys Web UI Configuration Application](#).

## **CA EEM**

CA EEM Event auditing and the order that CA EEM servers are listed in the failover setup can affect performance.

### **Event Auditing**

CA EEM event auditing can affect the performance of AutoSys Web UI applications. If you are experiencing performance problems, review the event policy settings for AutoSys Web UI and AutoSys Workload Automation applications. You can limit EEM event auditing by removing some resources from policies with submit action. You can disable CA EEM event auditing by removing all resources from policies with submit action or by disabling all policies with submit action.

The policies are located in the Event Policies folder on the Manage Access Policies tab in CA EEM.

The DefaultSubmitEvent policy should not contain the resource authorizeWith\* or any matching resources such as authorizeWithSession. If that resource is present, remove it from the policy. The same rule applies to custom policies with filters that include any AutoSys Web UI application that operates with AutoSys Workload Automation job list or views. If the removal of authorizeWith\* resource from Event Policies does not improve performance, consider disabling those policies.

### **CA EEM Server Failover**

CA EEM servers in a failover setup can affect the load time of the initial AutoSys Web UI login page and the startup time of batch interface utilities. The initial AutoSys Web UI login page is loaded after the CA-wcc service starts or restarts. Failed attempts to start that service occur when servers at the beginning of the CA EEM server list are unavailable. Failed attempts to start the CA-wcc service result in timeouts. The AutoSys Web UI login page or batch interface utility does not load until an available CA EEM server is found.

CA EEM servers in a failover setup are listed in the --eemhostname eem\_host\_name parameter in the Configuration Batch Interface. We recommend that you enter the CA EEM servers with the highest availability (lowest probability of failure) at the beginning of the list.

#### **NOTE**

For information about adding, changing, and deleting the CA EEM servers that AutoSys Web UI can access, see *Configuration Batch Interface Help*.

### **Example: Define the CA EEM Server List**

Suppose that the following CA EEM servers are available to AutoSys Web UI:

- Server A is dedicated to CA EEM and has good performance characteristics.
- Server B is shared by many applications. Frequent reboots are expected, due to the demands of other applications.
- Server C is activated only when servers A or B are inactive.

For best performance, list the servers in the order A,B,C when you define the CA EEM servers in the Configuration Batch Interface. Dedicated server A is listed first, so that the first attempt to connect to CA EEM is successful. Listing the servers in the order of C, B, A can delay the successful connection to CA EEM.

### **CA EEM Recommendations**

Recommendations for CA EEM are as follows:

- Review event policies in both AutoSys Workload Automation and AutoSys Web UI applications to verify that they are correctly defined.
- If you use Active Directory as your CA EEM user repository, set the appropriate EEM/AD connection filters. Allow only groups and individuals that actually use AutoSys Web UI, as opposed to retrieving the entire Active Directory.

## General Recommendations

General recommendations for performance tuning are as follows:

- Verify that the AutoSys Web UI server meets the system requirements in the *Release Notes*.
- If AutoSys Web UI is installed on a shared or virtual server, resolve any resource contention problems. An example of resource contention is virtual AutoSys Web UI servers that host multiple environments. If such a server experiences periodic performance drops, verify that the other environments are not causing spikes in resource usage.
- On UNIX operating systems, use `ULIMIT - n = 16000` to increase the number of file descriptors per process.
- As a baseline, perform a ping test to determine the transit time between the AutoSys Workload Automation servers and the AutoSys Web UI server.
- On UNIX operating systems, ensure the entropy is above 500. If the entropy is running low, AutoSys Web UI performance can be degraded. To check the current entropy, execute the following command:

```
cat /proc/sys/kernel/random/entropy_avail
```

Additionally, you can configure AutoSys Web UI to use a less strong random number generator by modifying both `wrapper.conf` configuration files.

### Follow these steps:

- a. Open the `wrapper.conf` files in the following locations:

```
$CA_WCC_INSTALL_LOCATION/tomcat/conf/
```

- b. Remove the pound sign (#) at the beginning of the following line:

```
wrapper.java.additional.101=-Djava.security.egd=file:/dev/./urandom
```

- c. Restart AutoSys Web UI services.

## Database Recommendations

The internal Derby database can be used for AutoSys Web UI and Reporting. If your workload or environment exceeds any of the following limitations, we recommend that you use a supported external database instead of the internal Derby database:

- 50,000 jobs when all AutoSys Workload Automation servers are referenced to one instance of AutoSys Web UI
- 50,000 job runs per day when all AutoSys Workload Automation servers are referenced to one instance of AutoSys Web UI
- 100 concurrent users; dependant on the load generated by the users
- 200 views; the number of job runs also affects this limitation
- 500 alerts
- Up to 1,000,000 job runs in the Reporting database.
- Five AutoSys Workload Automation servers; the number of job runs on those servers also affects this limitation

### NOTE

- The maximum number of days that data is kept in the Reporting database is defined by the `Purge Data` property in Reporting preferences. You can change the value depending on your requirements. The recommended value is determined by dividing the total number of job runs by the number of job runs per day. For example, with 1,000,000 total job runs and 50,000 job runs per day, the `Purge Data` value should be set to 20 days. For information about modifying Reporting preferences, see *Configuration Help*.
- The internal Derby database is not supported with AutoSys Web UI running in High Availability mode.
- For information about supported external databases, see the *Release Notes*.
- For information about configuring AutoSys Web UI to use an external database, see *Configuration Batch Interface Help*.

### Improve Derby Performance

As an AutoSys Web UI Administrator, you can improve performance of the Derby database.



**Follow these steps:****1. Stop the following AutoSys Web UI services:**

```

unisrvctr stop CA-wcc
unisrvctr stop CA-wcc-services
unisrvctr stop CA-wcc-db

```

**2. Increase Derby memory in file /opt/CA/WorkloadCC/derby/conf/wrapper.conf**

```

Java Application
wrapper.java.command=../../jre/bin/java
#wrapper.java.initmemory=64
wrapper.java.initmemory=1024
#wrapper.java.maxmemory=512
wrapper.java.maxmemory=3072

```

**3. Restart the AutoSys Web UI services:**

```

unisrvctr stop CA-wcc
unisrvctr stop CA-wcc-services
unisrvctr stop CA-wcc-db

```

**Compress Derby**

As an AutoSys Web UI Administrator, you can compress the AutoSys Web UI Derby database and the disk space will revert to the operating system.

**NOTE**

During the purge operation, it is possible that Derby may not release the disk space to the operating system. It may be necessary to compress the Derby database to release disk space to the OS.

**Follow these steps:****1. Create a file on the AutoSys Web UI Derby host called "derby\_compress.table" that contains the following using text-only format:**

```

call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_ACTIVE_CRED', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_APPLICATION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_APPLICATION_INSTANCE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_AUTOSYS_11_3', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_AUTOSYS_NOT_DEPLOYED', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_CREDENTIAL', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_EXT_APPLICATION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_GLOBAL_PREF', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_HA_GROUP', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_HA_INSTANCE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_JAWS_APPLICATION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_SERVER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_SERVER_PREF', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_SERVER_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_SERVER_STATUS_NON_DEPLOYED', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_USER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_USER_PREF', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_WIDGET', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'CFG_WIDGET_INSTANCE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'ECLI_HISTORY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'ECLI_SAVED_COMMAND', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_ALARM_TYPE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT', 1);

```

```

call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_EVENT_POLICY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_FILTER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_FILTER_ALARM_TYPE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_POLICY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_POLICY_ALARM_TYPE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_POLICY_EVENT_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_POLICY_JOB_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_POLICY_SERVER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_SEVERITY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_EVENT_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_JOB', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_JOB_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_SERVER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_SERVER_STATUS', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW_FILTER', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW_FILTER_EVENT_POLICY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW_JOB', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW_STATUS_DEFINITION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'MON_VIEW_STATUS_SEVERITY', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_EVENT_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_DEF_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_DESC_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_OWNER_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_RUN_EVENT_FACT', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_RUN_EVENT_STAGE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_RUN_FACT', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_JOB_RUN_STAGE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_MACHINE_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_STATISTIC_FACT', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_STATISTIC_STAGE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_AS_STATUS_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_DATE_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_PROFILE', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_SERVER_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'RPT_TIME_DIMENSION', 1);
call SYSCS_UTIL.SYSCS_COMPRESS_TABLE('WCC', 'SEQUENCE', 1);
commit;

```

2. Run **'ustat'** to check if the AutoSys Web UI is operational with the CA Service Status Report.

Component Name	Pid	Status
-----	-----	-----
CA-wcc-services Server	29036	running
CA-wcc-db Server	28960	running
CA-wcc Server	29178	running

3. Stop all functions with the following commands:

```

unisrvcntr stop CA-wcc
unisrvcntr stop CA-wcc-services
unisrvcntr stop CA-wcc-db

```

4. Save the Derby database with a tar command.

```
cd /opt/CA/WorkloadCC/derby
tar cvf /tmp/DB_FILES.tar ./DB_FILES/*
```

5. Copy the "derby\_compress.table" file to /opt/CA/WorkloadCC/derby/lib

**NOTE**

- Ensure the OS owner/group permissions are same as the other files.
- If you don't have write access to /opt/CA/WorkloadCC/derby/lib, you can copy it to another location.

6. Increase Derby memory in file /opt/CA/WorkloadCC/derby/conf/wrapper.conf

```
Java Application
wrapper.java.command=../../jre/bin/java
#wrapper.java.initmemory=64
wrapper.java.initmemory=1024
#wrapper.java.maxmemory=512
wrapper.java.maxmemory=3072
```

**NOTE**

Keep these values to improve Derby performance.

7. Run the following to acquire the current size of the Derby database:

```
du -sk /opt/CA/WorkloadCC/derby/DB_FILES/WCC_MAIN/seg0
```

8. Start Derby with this line:

```
unisrvcntr start CA-wcc-db
```

The following output appears:

```
Executed CA-wcc-db start.....OK
cd /opt/CA/WorkloadCC/derby/lib
java -jar derbyrun.jar ij
ij version 10.8
ij> connect 'jdbc:derby://localhost:1527/WCC_MAIN' user 'wcc' password 'wcc';
```

**NOTE**

- If wcc is not the password, replace with your password
- When logged on to Derby, run this command to get the list of derby tables

9. Verify that Derby responds properly and run the following:

```
ij> show tables;
```

10. Run the procedure with the following:

```
run '/opt/CA/WorkloadCC/derby/lib/derby_compress.table';
```

**NOTE**

If you have never run ECLI commands in AutoSys Web UI, errors may occur with tables 'ECLI\_HISTORY' and 'WCC.ECLI\_SAVED\_COMMAND' because they don't exist.

11. Exit Derby by typing **quit**.

12. Verify new disk space with the following line:

```
du -sk /opt/CA/WorkloadCC/derby/DB_FILES/WCC_MAIN/seg0
```

13. Start the two other AutoSys Web UI services to complete the process:

```
unisrvcntr start CA-wcc
unisrvcntr start CA-wcc-services
```

**NOTE**

If there is a problem during this compress operation, rename '/opt/CA/WorkloadCC/derby/DB\_FILES' and restore the tar file taken at the beginning of this procedure. You have compressed Derby.

You have compressed Derby.

## **HA Recommendations**

HA helps improve performance. Consider using HA in the following situations:

Job level security for monitoring is enabled in an environment with >300 concurrent users

- Job level security for monitoring is enabled and you use autorefresh heavily in an environment with >100 concurrent users
- Job level security is disabled in an environment with >600 concurrent users
- You are operating in an environment with >10 AutoSys Workload Automation servers connected to a single AutoSys Web UI server

### **NOTE**

- If using Oracle RAC or MSSQL in cluster mode, two or more AutoSys Web UI servers can be used in HA mode.
- Use AutoSys Web UI servers in HA mode in the same rack or datacenter.

## **Logging Recommendations**

AutoSys Web UI includes logging for all applications.

Log files are located in the following directory:

- **Windows**  
%CA\_WCC\_INSTALL\_LOCATION%\log\*application*
- **UNIX**  
\$CA\_WCC\_INSTALL\_LOCATION/log/*application*

Each *application* subfolder contains a log4j.properties file that controls logging settings. Logging levels are set to INFO or ERROR by default. To optimize performance, verify that the logging values are not set to DEBUG.

The wrapper.logfile.maxfiles parameter in wrapper.conf controls number of retained rolled log files. A value of 0 specifies to retain an unlimited number of logs.

The number of rolled log files kept can be specified by setting the wrapper.logfile.maxfiles parameter in wrapper.conf. The log is rolled based on size and the value 0 indicates no limit on the number of logs.

### **WARNING**

Before you change any logging settings, make a backup copy of this file. After you change settings in the log4j.properties file, save the file, then stop and start the CA-wcc service.

## **Scheduling Engine Recommendations**

AutoSys Workload Automation r11.3, Release 11.3.5, and Release 11.4 use an application server to manage secure communication with the AutoSys Workload Automation database. The application server is typically installed on the AutoSys Workload Automation scheduler computer. If the computer on which the application server is installed is heavily used, the computer can become CPU-bound. To maximize performance, install the instance of the application server that is used by AutoSys Web UI on a computer with a low load. You can maximize performance using the following methods:

- Install the AutoSys Workload Automation application server on the AutoSys Web UI server that has more than 10 GB of free RAM and a CPU with at least five cores.
- Install a new AutoSys Workload Automation application server on a computer other than the AutoSys Workload Automation Scheduler.
- Configure AutoSys Web UI to reference the shadow server when the AutoSys Workload Automation instance has a shadow scheduler installed.

**NOTE**

Application servers are utilized in the order specified in the Application Server Host Address field in the Server Properties section of Configuration. Therefore, application servers that are dedicated to AutoSys Web UI must be specified first in the list.

**Adjust Memory for Services**

Sufficient memory must be available for AutoSys Web UI services. The Java Virtual Machine (JVM) does not load when the memory specified by the maxmemory parameter cannot be allocated. You can change the maxmemory value to meet the memory requirements.

**Follow these steps:**

1. Open the wrapper.conf file in the following location:  
 UNIX  
`$CA_WCC_INSTALL_LOCATION/tomcat/conf/`  
 Windows  
`%CA_WCC_INSTALL_LOCATION%\tomcat\conf\`
2. Change the value of the wrapper.java.maxmemory parameter in the Settings section. The value is specified in MB.

**NOTE**

This value is associated with the Tomcat server for the AutoSys Web UI user interface. The recommended value for Tomcat 64-bit is 4GB or greater.

3. Save the wrapper.conf file.
4. Restart AutoSys Web UI services.

**Control EEM Caching Behavior**

To control the EEM SDK caching behavior, you can switch on/off full cache updates and set the interval between the updates. The group memberships are evaluated on every cache update if the full cache update is enabled. This behavior can decrease EEM server performance.

The caching behavior can be controlled by the connection.properties configuration file in the AutoSys Web UI root directory at the following location:

**UNIX**

```
$CA_WCC_INSTALL_LOCATION/data/config/application/config/resources/connection.properties
```

**Windows**

```
%CA_WCC_INSTALL_LOCATION%\data\config\application\config\resources\connection.properties
```

The properties that control the EEM SDK caching behavior are as follows:

- eem.fullCacheUpdate - set to true to enable full cache update; set to false to disable full cache update  
Default value: true
- eem.cacheUpdateInterval - set the cache update refresh interval (in seconds)  
Default value: 30

If the connection.properties file does not contain these properties, default values will be used.

**NOTE**

- The connection with the EEM server is established during AutoSys Web UI startup, AutoSys Web UI services have to be restarted before the new settings take effect.
- The group memberships are evaluated only during user login if full cache update is disabled. This requires logout and login again to see the latest changes in group memberships.

## Embed AutoSys Web UI in an HTML Frame

AutoSys Web UI contains security measures to prevent AutoSys Web UI from being embedded into an HTML frame. If your environment requires AutoSys Web UI to be embedded, disable the security measures.

### Follow these steps:

1) Locate the following *web.xml* files in the following folders:

```
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\app-editor\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\configuration\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\ecli\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\forecast\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\jsc-rest\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\quick-edit\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\quickview\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\reporting-rest\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\resources\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\rest\WEB-INF
%CA_WCC_INSTALL_LOCATION%\tomcat\webapps\wcc\WEB-INF
```

2) Change the *<param-value>* to ANY.

```
<filter>
 <description>Define X-Frame-Options header to prevent clickJacking vulnerability
</description>
 <filter-name>XFrameOptionsFilter</filter-name>
 <filter-class>com.ca.wcc.common.XFrameOptionsFilter</filter-class>
 <init-param>
 <param-name>x-frame-options</param-name>
 <param-value>SAMEORIGIN</param-value>
 </init-param>
</filter>

<filter-mapping>
 <filter-name>XFrameOptionsFilter</filter-name>
 <url-pattern>/*</url-pattern>
</filter-mapping>
```

3) Restart CA-wcc service in order to allow changes to take effect.  
Security measures are disabled to allow embedding into an HTML frame.

## User Login Audit

AutoSys Web UI administrators have a log file containing user login and logout events. The log file, located inside `$CA_WCC_INSTALL_LOCATION/log/audit`, contains only the events that happened on the particular AutoSys Web UI instance. The log file contains information about the following user events:

- Successful login, logout, and failed login attempts
- Logout caused by session expiration
- Failed login caused by duplicate login in a “Single session per user” mode
- Successful Single Sign-on login for NTLM or SiteMinder

#### NOTE

- When AutoSys Web UI is running in HA setup, each HA node has its own audit log
- Audit logger configuration is located in \$CA\_WCC\_INSTALL\_LOCATION/log/application/rest/log4j.properties file in section “audit appender”

## Agent Inventory Audit

The following are the Agent Inventory audit and log information:

- All user actions on the Agent Inventory tab are audited and logged in the <WCC\_INSTALLED\_DIRECTORY>/log/audit/asi-audit.log file.
- To modify the Agent Inventory audit settings, configure the audit appender properties in the <WCC\_INSTALLED\_DIRECTORY>/log/application/asi/log4j.properties file.  
You can include the logs written to the asi-audit.log file as part of the asi.log file. To enable this option, change the audit value to 'true', as follows, in the log4j.properties file:  
log4j.additivity.com.ca.workload.services.cam.audit=true  
**Default:** false

## Web UI Collectors

Presently, AutoSys Workload Automation Release 11.3.6 SP8 uses 32-bit architecture and so Web UI needs to use 32-bit Java to communicate with AutoSys Workload Automation, leading to resource constraints. This architecture impacted Web UI performance.

#### Current Challenges:

The following are some of the current challenges along with 32-bit architecture:

- The CA-wcc-services Tomcat was the single point source for collection and if it failed, the entire collection process failed.
- When Web UI is configured with a large number of Application servers with increased load, the collection process slowed down resulting in delayed search results.
- The inter-process communication between multiple Tomcats (32-bit and 64-bit) slowed the request process, resulting in delayed rendering of information on the GUI.

#### Solution:

AutoSys Web UI Version 12.0 is now reorganized such that it improves performance, enables handling of increased workload, and reuse the system resources better.

The Web UI is the user interface to manage and monitor the AutoSys artifacts. This Web UI has two different components:

- A Web server provides the GUI functionality using Tomcat and serves the web application requests. Also, this communicates with the AutoSys Application Servers to perform operations over the AutoSys artifacts.
- A Server component “Collector” collects the AutoSys artifacts like Job definitions, Job runs, Alarms, and Machines from AutoSys and populates the Web UI database. The collected information is used to monitor and manage the workload. In Web UI Collector, processes are now dedicated to the collection but not for user workload.

When Web UI is installed, a default 64-bit collector gets installed and collects the artifacts from all the configured AutoSys servers. You can increase the memory for the collector process.

The Web UI environment supports multiple installations of the collector. Web UI allows the Collector to be horizontally scalable, based on the number of AutoSys servers, load across each of the servers like job runs, alarms, etc. The new instances of the Collector can be installed within the same or across different machines across the Web UI supported platforms. As new Collectors are added, the load distribution of collecting the AutoSys artifacts across all the AutoSys servers is seamlessly taken care of. At least one Collector must be running to ensure that the AutoSys artifacts are collected.

## NOTE

You can execute the following command if the Web UI collector has to communicate to an application server running on a port other than 9000.

### **Example: Enabling EnablePmux for port 9001**

**Windows:** "%CSAM\_SOCKADAPTER%\bin64\csamconfigedit64" port=9001 EnablePmux=True

**Linux:** \$CSAM\_SOCKADAPTER/bin/csamconfigedit port=9001 EnablePmux=True

Based on the number of configured Application servers and the artifacts present in them (for example, number of Job definitions, Job runs, Machines, and Alarms), you may have to increase the heap memory and accommodate the user load as follows:

- Increase the heapMemory of Web UI Collector under the respective Web UI Collector installed folder. For instance, in the file <AutoSys installed location>/wcc/collectors/<Collector name>/conf/wrapper.conf increase the value of the wrapper.java.maxmemory property, which defaults to 1024 (1GB).
- Increase the heapMemory of Web UI under the respective Web UI installed folder. For instance, in the file <AutoSys installed location>/wcc/tomcat/conf/wrapper.conf increase the value of the wrapper.java.maxmemory property, which defaults to 2048 (2GB).



# Scheduling

---

This article includes AutoSys Workload Automation and Web UI scheduling information for various jobs.

- **AutoSys Scheduling**  
Explains how to manage your calendars, connection profiles, jobs, and resources with AutoSys Workload Automation. To search for content specific to AutoSys Workload Automation, go to AutoSys Workload Automation Scheduling and use the search box. The results will only show AutoSys Workload Automation content.
- **Schedule with Web UI**  
Explains how to manage your calendars, connection profiles, jobs, and resources with Web UI. To search for content specific to Web UI, go to Web UI Scheduling and use the search box. The results will only show Web UI content.

After you log in, use the navigation tree on the left or search for Scheduling content:

Scheduling

## AutoSys Workload Automation Scheduling

AutoSys Workload Automation is an automated job control system that lets you define, manage, and report on workload.

After you log in, use the navigation tree on the left or search for AutoSys Workload Automation Scheduling content:

AutoSys Workload Automation Scheduling

## Manage JIL

Job Information Language (JIL) is a scripting language that lets you define and modify assets such as jobs, global variables, machines, job types, external instances, and blobs. For more detailed information about JIL and specific JIL objects, see [AutoSys Workload Automation Job Information Language](#).

The jil command runs the language processor that interprets JIL. You use JIL subcommands and attributes to specify the asset definitions that you want to create or modify. The jil command loads the data that you specify into the database.

You can define and modify assets using the following methods:

- Issue the jil command, which displays the JIL command prompt. You enter subcommands interactively one line at a time. When you exit the command prompt, the language processor interprets the asset definition and loads it into the database.
- Create a JIL script by entering subcommands and attribute statements in a text file. You redirect the JIL script to the jil command. The jil command activates the language processor, interprets the information in the script, and loads this information in the database.

### NOTE

You can also submit job definitions using Web UI. For more information about using Web UI to define jobs, see [Web UI Scheduling](#).

This section contains the following topics:

## JIL Subcommands

JIL subcommands let you define and modify asset definitions. You specify JIL subcommands using the jil command.

**The following JIL subcommands define and modify jobs and boxes:**

- [delete\\_box](#)

Deletes an existing box job and all the jobs in that box from the database.

- **delete\_job**  
Deletes a job from the database. If the specified job is a box job, the box job is deleted and the jobs in the box become stand-alone jobs.
- **insert\_job**  
Adds a new job definition to the database.
- **override\_job**  
Defines a one-time override for an existing job definition. This override affects the job for the next run only.
- **update\_job**  
Updates an existing job definition in the database.

**The following JIL subcommands define and modify machines:**

- **delete\_machine**  
Deletes an existing real or virtual machine definition from the database.
- **insert\_machine**  
Inserts a new real or virtual machine definition in the database. A machine must be defined before it can be used in a job definition.
- **update\_machine**  
Updates an existing machine in the database.

**The following JIL subcommands define and modify monitor or report definitions:**

- **delete\_monbro**  
Deletes the specified monitor or report definition from the database.
- **insert\_monbro**  
Adds a new monitor or report definition to the database.
- **update\_monbro**  
Updates an existing monitor or report definition in the database.

**The following JIL subcommands define and modify job types:**

- **delete\_job\_type**  
Verifies that no jobs are currently defined with the specified job type, then deletes the specified job type definition from the database.
- **insert\_job\_type**  
Adds a new user-defined job type definition to the database. This is the only way to create a user-defined job type.
- **update\_job\_type**  
Updates an existing user-defined job type definition in the database. You can use `update_job_type` to change the values of the command and description attributes.
- **rename\_job**  
Renames an existing job, and all dependencies reflect the changed name for that specific job.

**The following JIL subcommands define and modify blobs and globs:**

- **insert\_blob**  
Adds a new blob definition that is associated with an existing job.
- **insert\_glob**  
Adds a new glob definition that is referenced by a given name.
- **delete\_blob**  
Decouples a blob definition from an existing job and deletes the blob from the database.
- **delete\_glob**  
Deletes the specified glob definition from the database.

**The following JIL subcommands define and modify external instances:**

- **delete\_xinst**  
Deletes the specified external instance definition from the database.
- **insert\_xinst**  
Adds a new external instance definition to the database.
- **update\_xinst**  
Updates an existing external instance definition in the database.

The following JIL subcommands define and modify resources:

- **delete\_resource**  
Deletes a virtual resource from the database.
- **insert\_resource**  
Adds a new virtual resource definition to the database.
- **update\_resource**  
Updates an existing virtual resource definition in the database.

The following JIL subcommands define and modify connection profiles:

- **delete\_connectionprofile**  
Deletes an existing connection profile from the database.
- **insert\_connectionprofile**  
Adds a new connection profile to the database. After you create a connection profile, you can assign it to a Hadoop (HDFS, Hive, Oozie, Sqoop, or Pig) job.
- **update\_connectionprofile**  
Updates an existing connection profile in the database.

## JIL Syntax Rules

JIL scripts contain one or more JIL subcommands and one or more attribute statements. These elements constitute a job definition. You can create job definitions in interactive mode by opening the operating system or instance command prompt and entering the following command:

```
jil
```

Alternatively, you can enter the subcommands and attribute statements for all your job definitions in a text (.txt) file and then import the job definitions by redirecting the file as input to JIL. To import job definitions from a text file, open the operating system or instance command prompt and enter the following command:

```
jil < file_name.txt
```

A JIL script is valid only when it is written according to the JIL syntax rules. If you create the job definitions in a text file import them, ensure that you follow the JIL syntax rules. To verify the syntax automatically, enter the following command:

```
jil -v syntax < file_name.txt
```

### WARNING

The syntax checker does not commit the job definitions in the file to the database. To commit the definitions, ensure that you correct any syntax errors that the checker identifies and then run the following command:

```
jil < file_name.txt
```

### NOTE

Running the syntax checker adds time to the import process.

A valid job definition adheres to the following syntax rules:

- **Rule 1**

Each subcommand uses the following form:

```
sub_command:object_name
```

- **sub\_command**

Defines a JIL subcommand.

- **object\_name**

Defines the name of the object (for example, a job or a machine) to act on.

- **Rule 2**

You can follow each subcommand with one or more attribute statements. These statements can occur in any order and are applied to the object specified in the preceding subcommand. A subsequent subcommand begins a new set of attributes for a different object. The attribute statements have the following form:

```
attribute_keyword:value
```

- **attribute\_keyword**

Defines a valid JIL attribute.

- **value**

Defines the setting to apply to the attribute.

### NOTE

- The entire *attribute\_keyword:value* statement can be up to 4096 characters.
- The following subcommands do not accept attributes: `delete_box`, `delete_job`, `delete_job_type`, `delete_xinst`, `delete_monbro`, and `delete_glob`.

- **Rule 3**

You can enter multiple attribute statements on the same line, but you must separate the attribute statements with at least one space.

- **Rule 4**

A box job definition must exist before you can add jobs to it.

- **Rule 5**

Valid *value* settings can include any of the following characters:

- Uppercase and lowercase letters (A-Z, a-z)
- Hyphens (-)
- Underscores (\_)
- Pound signs (#)
- Numbers (0-9)
- Colons (:), if the colon is escaped with quotation marks (" ") or a preceding backslash (\)
- The at character (@)

**NOTE**

Object names can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#). Do not include embedded spaces or tabs.

- **Rule 6**

Because JIL parses on the combination of a keyword followed by a colon, you must use escape characters (a backslash) or must enclose the value in quotation marks with any colons that are used in the value of an attribute statement. For example, to define the start time for a job, specify 10\:00 or "10:00".

**NOTE**

When specifying drive letters in commands, you must use escape characters with the colon (:). For example, "C:\tmp" and C:\:tmp are valid; C:\tmp is not.

- **Rule 7**

Use one of the following methods to indicate comments in a JIL script:

- To comment an entire line, put a pound sign (#) in the first column.
- To comment one or more lines, you can use the C programming syntax for beginning a comment with a forward slash and asterisk (/\*) and ending it with an asterisk and a forward slash (\*). For example:

```
/* this is a comment */
```

- **Rule 8**

You can use the blob\_input attribute to enter multiline text manually. This attribute is only valid for the insert\_job, update\_job, insert\_blob, and insert\_glob subcommands. The blob\_input attribute has the following form:

```
blob_input:<auto_blobt> this is a multi-
line text

</auto_blobt>
```

**NOTE**

Use the auto\_blobt meta-tags to indicate the beginning and end of multiline text. JIL interprets every character input between the auto\_blobt meta-tags literally. This behavior implies that JIL does not enforce any of the previously discussed rules for text that is entered in an open auto\_blobt meta-tag.

- **Rule 9**

To specify a comma in *keyword=value* combinations, you must do *one* of the following steps:

- Escape the comma with a backslash, as follows:

```
j2ee_parameter: String=Hello1\, World
```

- Enclose the entire value in quotation marks, as follows:

```
j2ee_parameter: String="Hello1, World"
```

If the *keyword=value* combination contains commas and quotation marks, you must escape the commas and quotation marks as follows:

```
j2ee_parameter: String=\"Hello1\, World\"
```

## Issue JIL in Interactive Mode

### Issue JIL in Interactive Mode on UNIX

You can issue the JIL command and subcommands at the operating system prompt. The operating system prompt lets you enter data interactively. You can also create aliases for the commands that you use frequently. The interactive mode is helpful when you do not need to issue JIL commands in the batch. For example, you can use the interactive mode when you want to test specific job definitions.

#### NOTE

Before you can issue commands at the operating system prompt, the AutoSys Workload Automation environment must be sourced in the shell and your UNIX user ID and password must be defined on AutoSys Workload Automation. For more information about sourcing the environment and defining user IDs, see [Environment and Database Connection](#).

#### Follow these steps:

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Do *one* of the following steps:
3. • Enter **jil** at the operating system prompt.  
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
- Enter **jil -S instance** at the operating system prompt.  
The JIL command prompt is displayed for the specified AutoSys Workload Automation instance.
  - **instance** Specifies the AutoSys Workload Automation instance where you want to issue the JIL subcommand.  
For single-instance environments, the jil command defaults to the only available instance.
4. Enter a JIL subcommand and follow the prompts.
5. Enter **exit** when you are done entering the data.  
JIL interactive mode ends and the data is loaded into the database.

### Issue JIL in Interactive Mode on Windows

You can issue the JIL command and sub-commands at the instance command prompt. The instance command prompt lets you enter data interactively. The interactive mode is helpful when you do not need to issue JIL commands in the batch. For example, you can use the interactive mode when you want to test specific job definitions.

#### Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt is displayed.
2. Do *one* of the following steps:
3. • Enter **jil**.  
The JIL command prompt is displayed for the local AutoSys Workload Automation instance.
- Enter **jil -S instance**.  
The JIL command prompt is displayed for the specified AutoSys Workload Automation instance.
  - **instance** Specifies the AutoSys Workload Automation instance where you want to issue the JIL subcommand.  
For single-instance environments, the jil command defaults to the only available instance.
4. Enter a JIL subcommand and follow the prompts.
5. Enter **exit** when you are done entering the data.  
JIL interactive mode ends and the data is loaded into the database.

**NOTE**

- You cannot use the MS-DOS command prompt to issue JIL subcommands.
- For the jil command to work properly, the correct environment variables must be assigned.

**Issue JIL Using a Script**

You can embed JIL sub-commands and attribute statements in a shell script and redirect the script to the jil command. The jil command activates the language processor, interprets the information in the script, and loads this information in the database. Using a script is helpful when you want to issue JIL sub-commands in the batch.

**Issue JIL Using a Script on UNIX**

You can embed JIL sub-commands and attribute statements in a shell script and redirect the script to the jil command. The jil command activates the language processor, interprets the information in the script, and loads this information in the database. Using a script is helpful when you want to issue JIL sub-commands in the batch.

**NOTE**

Before you can run the JIL script, the AutoSys Workload Automation environment must be sourced in the shell and your UNIX user ID and password must be defined on AutoSys Workload Automation. For more information about sourcing the environment and defining user IDs, see [Environment and Database Connection](#).

**Follow these steps:**

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Do *one* of the following steps:
  - Enter **jil < script** at the operating system prompt.  
The sub-commands in the JIL script are issued on the local AutoSys Workload Automation instance and the data is loaded into the database.
  - Enter **jil -S instance < script** at the operating system prompt.  
The sub-commands in the JIL script are issued on the specified AutoSys Workload Automation instance and the data is loaded into the database.
    - **script**  
Specifies the name of the script that contains the JIL sub-commands and attributes.
    - **instance**  
Specifies the AutoSys Workload Automation instance where you want to run the JIL script. For single-instance environments, the jil command defaults to the only available instance.

**Example: Submit a Job Definition Using a JIL Script**

This example redirects a file that is called my\_jil\_script to the jil command on the PRD instance of AutoSys Workload Automation.

```
jil - S PRD < my_jil_script
```

**Issue JIL Using a Script on Windows**

You can embed JIL sub-commands and attribute statements in a script and redirect the script to the jil command. The jil command activates the language processor, interprets the information in the script, and loads this information in the database. Using a script is helpful when you want to issue JIL sub-commands in the batch.

**Follow these steps:**

1. Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt is displayed.
2. Do *one* of the following steps:

- Enter **jil < script**.  
The sub-commands in the JIL script are issued on the local AutoSys Workload Automation instance and the data is loaded into the database.
- Enter **jil -S instance < script**.  
The sub-commands in the JIL script are issued on the local AutoSys Workload Automation instance and the data is loaded into the database.

**Where:**

- **script**  
Specifies the name of the script that contains the JIL subcommands and attributes.
- **instance**  
Specifies the AutoSys Workload Automation instance where you want to run the JIL script. For single-instance environments, the jil command defaults to the only available instance.

**NOTE**

- You cannot use the MS-DOS command prompt to redirect a JIL script to the jil command.
- To issue commands that run on a UNIX computer from Windows, use the syntax appropriate to the operating system where the job runs.
- For the jil command to work properly, the correct environment variables must be assigned. For more information about environment variables, see [Environment and Database Connection](#).

**Example: Submit a Job Definition to the Local Instance Using a JIL Script**

This example redirects a file called newjobdef to the jil command on the local instance of AutoSys Workload Automation.

```
jil < newjobdef
```

**Example JIL Script**

The following JIL script creates a command job, a file watcher job, and a box job to do the following steps:

1. You expect a file that is named /DOWNLOAD/MAINFRAME/SALE.RAW to arrive from the mainframe some time after 2:00 a.m.
2. When the file arrives, the command file that is named filter\_mainframe\_info processes it, and the results are sent as an output to the file named /DOWNLOAD/MAINFRAME/SALES.SQL.
3. When the actions are complete, the SQL commands in the file named /DOWNLOAD/MAINFRAME/SALES.SQL runs.

```
Example of a Machine
insert_machine: lowgate
type: a
Example of Jobs
insert_job: Nightly_Download
job_type: box
date_conditions: yes
days_of_week: all
start_times: "02:00"
insert_job: Watch_4_file
job_type: ft
box_name: Nightly_Download
watch_file: /DOWNLOAD/MAINFRAME/SALES.RAW
watch_file_type: generate
watch_no_change: 1
machine: lowgate
```



```

insert_job: filter_data
job_type: cmd
box_name: Nightly_Download
condition: success(Watch_4_file)
command: filter_mainframe_info
machine: lowgate
std_in_file: /DOWNLOAD/MAINFRAME/SALES.RAW
insert_job: parse_data
job_type: cmd
box_name: Nightly_Download
condition: success(filter_data)
machine: lowgate
command: isql -U mutt -P jeff
std_in_file: /DOWNLOAD/MAINFRAME/SALES.SQL
std_out_file: /LOG/parse_data.out
std_err_file: /LOG/LOG/parse_data.err

```

## Real and Virtual Machines

Before you can schedule jobs to run on a machine, you must define the machine to AutoSys Workload Automation. You can define real machines, virtual machines, or real machine pools.

### NOTE

Do not define `localhost` using the `insert_machine: localhost` command. The `localhost` machine name is a reserved name.

### Real Machines

A *real machine* is any single machine that meets the following criteria:

- It has been identified in the appropriate network so that AutoSys Workload Automation can access it.
- An agent, AutoSys WA Connect Option, CA NSM, or CA UAJM is installed, which lets AutoSys Workload Automation run jobs on it.
- It is defined to AutoSys Workload Automation as a real machine using JIL.

A real machine must meet these conditions to run jobs. However, for AutoSys Workload Automation to perform intelligent load balancing and queuing while running jobs, it must know the relative processing power of the real machines. AutoSys Workload Automation uses the virtual machines to provide load balancing and queuing.

### Virtual Machines

A *virtual machine* is a machine definition that references one or more existing real machine definitions. By defining virtual machines to AutoSys Workload Automation and submitting jobs to run on those machines, you can specify the following:

- Run-time resource policies (or constraints) at a high level.
- That AutoSys Workload Automation automatically runs those policies in a multi-machine environment.

### NOTE

Previous releases of AutoSys Workload Automation required that all machines in a virtual machine are of the same type. In the current release, the component real machines in a virtual machine definition can be UNIX or Windows machines or a mix of both.

## Real Machine Pools

A real machine pool is similar to a virtual machine in that it references one or more existing real machine definitions.

This section contains the following topics:

### The localhost Definition

The localhost machine name is a reserved name. You cannot define a machine for localhost by creating an **insert\_machine: localhost** definition.

By default, the localhost value is resolved to the name of the machine where the AutoSys Workload Automation scheduler was installed. You can override the reserved localhost value to the name of another real machine using the local machine definition setting. On UNIX, you can configure this setting using the [LocalMachineDefinition](#) parameter in the configuration file. On Windows, you can configure this setting using the Local Machine Definition field in the Scheduler window of AutoSys Workload Automation Administrator (autosysadmin).

Create a machine definition in the database for the machine resolved from the localhost. To create a machine definition, use the **insert\_machine JIL** command.

As part of the AutoSys Workload Automation installation process, your administrator must have created a machine definition for the default localhost (the machine where the scheduler was installed) in the database. If you configure the local machine definition setting to another machine, you must create a definition for that machine in the database. For example, if you configure the local machine definition setting to a machine named prod, you must define machine prod using the **insert\_machine: prod** command.

### How the Localhost Value Is Resolved

If the **machine: localhost** attribute is specified in a job definition, the scheduler tries to resolve the localhost value when it runs the job. The localhost value is resolved as follows:

- The scheduler checks the value of the LocalMachineDefinition parameter (UNIX) or the Local Machine Definition field (Windows).
- If the local machine definition setting is set to a value other than "localhost", the scheduler searches the database for a machine definition with that name. For example, suppose that LocalMachineDefinition is set to agentmach. If an agentmach machine definition is found and all conditions are satisfied, the job runs on agentmach. If the scheduler cannot find an agentmach machine definition, or if it finds multiple agentmach machine definitions, the scheduler does not resolve localhost. All jobs that are defined to run on the localhost machine fail.
- If the local machine definition is not defined or is set to "localhost", the scheduler searches the database for a machine definition corresponding to the machine where the scheduler was started (the default localhost). For example, suppose that the scheduler was started on a machine that is named prodserver and LocalMachineDefinition is not defined. When the job runs, the scheduler searches for a machine definition named prodserver. If the scheduler cannot find the prodserver definition, or if it finds multiple prodserver definitions, the scheduler does not resolve localhost. All jobs that are defined to run on the localhost machine fail.
- In a High Availability failover where the shadow scheduler takes over from the primary scheduler, the localhost is resolved in the same way. To run a job on the localhost, the shadow scheduler first checks its local machine definition setting, which may be different from the setting for the primary scheduler. If the local machine definition is not defined, the localhost is resolved to the machine where the shadow scheduler was started.

## Define a Machine

Before you can schedule jobs to run on a machine, you must define the machine to AutoSys Workload Automation.

### Follow these steps:

1. Do one of the following tasks:

- Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following definition:
- ```
insert_machine: machine_name: node_name: address: type
```
- **machine_name** Defines a unique name for the machine to add.
 - **address** Specifies the IP address or DNS name of the machine. The default is the machine_name specified on the insert_machine statement.
 - **type** Specifies the type of machine you are defining. This list details the options:
 - a** Specifies a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS machine. This value is the default.
 - c** Specifies a CA Workload Automation AutoSys Edition Connect Option machine.
 - I** Specifies a 4.5.1 real UNIX machine. You must specify a lowercase **I**.
 - L** Specifies a 4.5.1 real Windows machine. Specify a capital **L**.
 - n** Specifies an r11 real Windows machine or a virtual machine that consists only of r11 real Windows machines (type **n**).
- NOTE**
- In the documentation, the type "p" machine is referred to as the real machine pool.
- r** Specifies an r11 real UNIX machine. **u** Specifies a CA NSM or a CA Universal Job Management Agent (CA UJMA) machine. **v** Specifies a virtual machine. The virtual machine can consist of Workload Automation Agent machines (type **a**), r11 real UNIX machines (type **r**), and r11 real Windows machines (type **n**). Specify
3. (Virtual type only) Specify the following attribute:
- **machine** Specifies a real machine as a component of the virtual machine or real machine pool. The specified machine must have been defined to AutoSys Workload Automation as a real machine.
4. (Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS machine only) Specify the following attribute:
- **opsys** Specifies the operating system where the Workload Automation Agent is installed. This list details the options:
 - aix** Specifies a Workload Automation Agent for UNIX installed on an AIX computer.
 - hpux** Specifies a Workload Automation Agent for UNIX installed on an HP-UX computer.
 - linux** Specifies a Workload Automation Agent for LINUX.
 - i5os** Specifies a Workload Automation Agent for i5/OS.
 - windows** Specifies a Workload Automation Agent for Windows.
 - zos** Specifies a Workload Automation Agent for z/OS.
5. Specify additional optional attributes as required:
- **agent_name**
 - **character_code**
 - **description**
 - **encryption_type**
 - **factor**
 - **heartbeat_attempts** (Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS only)
 - **heartbeat_freq** (Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS only)
 - **key_to_agent**
 - **max_load**
 - **opsys**
 - **port**
6. Do *one* of the following tasks:
- – Enter exit if you are using interactive mode.
 - Redirect the script to the jil command if you are using a script.

The data is loaded into the database and the machine is defined.

NOTE

- You can only use JIL to define real and virtual machines.
- Without CA Automation Suite for Data Centers, load balancing and queuing is possible only if real and virtual machines have been defined to AutoSys Workload Automation using the machine attributes. The max_load

and factor attributes, which are used when defining real machines, are important for load balancing and queuing.

Examples Defining Real Machines

The following examples define real machines:

Example: Define a Workload Automation Agent

This example defines a machine that is named eagle where the agent WA_AGENT runs on the node myagenthostname and uses 49154 as its main input port.

```
insert_machine: eagle
type: a
agent_name: WA_AGENT
node_name: myagenthostname
port: 49154
max_load: 100
factor: 1.0
```

Example: Define an r11 Real Windows Machine

This example defines a Windows real machine named jaguar.

```
insert_machine: jaguar
type: n
max_load: 100
factor: 1.0
```

Example: Define an r11 Real UNIX Machine

This example defines a UNIX real machine named jaguar.

```
insert_machine: jaguar
type: r
max_load: 100
factor: 1.0
```

Examples Defining Virtual Machines

The following examples define virtual machines:

Example: Define a Virtual Machine to Include Two Real Windows Machines

This example defines a virtual machine that is named giraffe to include two real Release 11.3.6 Windows machines (cheetah with a factor value of 5.0 and a max_load value of 400, and lily with a factor value of 2 and a max_load value of 15).

```
insert_machine: cheetah
type: a
opsys: windows
insert_machine: lily
type: a
opsys: windows
insert_machine: giraffe
type: v
```

```

machine: cheetah
max_load: 400
factor: 5.0
machine: lily
max_load: 15
factor: 2

```

NOTE

In this example, the two real machines (when specified in a job definition through the virtual machine) vary considerably in capacity from a scheduling standpoint. However, when these machines are specified by name explicitly in a job definition, the factor and max_load attributes defined here have no effect; they only have these values when used through the virtual machine.

Example: Define a Windows Virtual Machine with Subsets of r11 Real Machines

This example defines two r11 Windows real machines (lion and lotus), and a virtual machine (gorilla), which is composed of slices, or subsets, of the max_load specified for the real machines. Although the real machines were defined with specific max_load values (100 and 80), the virtual machine only uses the reduced loads specified in the virtual machine definition (10 and 9).

```

insert_machine: lion
type: n
max_load: 100
factor: 1
insert_machine: lotus
type: n
max_load: 80
factor: .8
insert_machine: gorilla
type: v
machine: lion
max_load: 10
machine: lotus
max_load: 9

```

Example: Define a Virtual Machine with Default Real Machines

This example defines a virtual machine (sheep), which is composed of two Release 11.3.6 UNIX real machines (warthog and camel). Because the max_load and factor attributes are not defined for the real machines, they use the default values for these attributes (a factor of 1.0 and a max_load of none, indicating unlimited load units).

```

insert_machine: warthog
opsys: linux
insert_machine: camel
insert_machine: sheep
type: v
machine: warthog
machine: camel

```

Example: Define a Virtual Machine with r11 Real Machines

This example defines two r11 UNIX real machines (lion and lotus), and a virtual machine (zebra), which is composed of the two real machines. The virtual machine is a superset of the two real machines and uses the `max_load` and `factor` attributes defined for them.

```
insert_machine: lion
type: r
max_load: 100
factor: 1
insert_machine: lotus
type: r
max_load: 80
factor: .9
insert_machine: zebra
type: v
machine: lion
machine: lotus
```

Delete a Machine

With the following topics, this article explains how to delete a machine:

Delete a Real Machine

To delete a real machine definition, specify the following subcommand in the JIL script:

```
delete_machine: name_of_real_machine
```

```
[remove_references: y]
```

```
[force: y]
```

- ***name_of_real_machine***
Specifies the name of the machine to delete.
- **remove_references: y**
(Optional) Instructs JIL to remove references to the specified machine from the definitions of machine pools and virtual machines. We recommend that you use this option when you delete real machines that are referenced in the definitions of any machine pools or virtual machines. If you do not instruct JIL to remove the references, you cannot delete the real machine until you delete all the references manually.
- **force: y**
(Optional) Use this option to delete a machine that is in use.

Example: Delete a Real Machine Not Referenced in Virtual Machines or Real Machine Pools

This example deletes the real machine definition for the computer named jaguar:

```
delete_machine: jaguar
```

Example: Delete a Real Machine Currently Referenced via a Virtual Machine:

This example explicitly deletes the real machine hyena reference from the virtual machine carnivores followed by the real machine definition itself.

```
delete_machine: carnivores

machine: hyena

delete_machine: hyena
```

Example: Delete a Real Machine and All Virtual Machines References Implicitly:

This example deletes the real machine panther and implicitly deletes all references to it that may be in any virtual machines or real machine pools.

```
delete_machine: panther

remove_references: y
```

Delete a Virtual Machine

To delete a virtual machine, specify the delete_machine: *machine_name* subcommand without the machine attribute in the JIL script. When you delete a virtual machine, the definitions for its component real machines are not deleted.

You can delete all real machine references in a virtual machine until there is only one reference remaining. You cannot delete the last reference. To delete all real machine references in a virtual machine, delete the virtual machine itself.

Example: Delete a Virtual Machine

This example deletes the virtual machine definition named gorilla:

```
delete_machine: gorilla
```

Delete a Real Machine Pool

To delete a real machine pool, specify the delete_machine: *machine_name* subcommand without the machine attribute in the JIL script. When you delete a real machine pool, the definitions for its component real machines are not deleted.

You can delete all real machine references in a real machine pool until there is only one reference remaining. You cannot delete the last reference. To delete all real machine references in a real machine pool, delete the real machine pool itself.

Example: Delete a Real Machine Pool

This example deletes the real machine pool definition named gorilla:

```
delete_machine: gorilla
```

Delete a Real Machine from a Virtual Machine or Real Machine Pool

To delete a virtual machine or real machine pool reference to a real machine, specify the following subcommand in the JIL script:

```
delete_machine: virtual_machine_name
```

```
machine: real_machine_name_referenced
```

Example: Delete a Real Machine from a Virtual Machine

This example deletes the real machine named camel from the virtual machine named sheep. The machine definitions for sheep and camel are *not* deleted from the database.

```
delete_machine: sheep
```

```
machine: camel
```

Specify Machine Load (max_load) or Job Load (job_load)

Specify Machine Load (max_load)

You can use the max_load attribute to define the maximum load (in load units) that a machine can reasonably handle. The max_load attribute is valid in a real machine definition or component machines defined to virtual machines.

Load units are arbitrary values, the range of which is user-defined. You can use any weighting scheme that you prefer. For example, a load unit with a range of 10 to 100 would specify that machines with limited processing power are expected to carry a load of only 10, while machines with ample processing power can carry a load of 100. There is no direct relationship between the load unit value and any of the machine's physical resources. Therefore, we recommend that you use conventions that are meaningful to you. You cannot use zero (0) or negative numbers as load units.

The max_load attribute is primarily used to limit the load on a machine. As long as a load of job does not exceed a machine's maximum load, the max_load attribute does not influence which machine a job runs on.

If you do not define the max_load attribute in a machine definition, AutoSys Workload Automation does not limit the load on the machine.

Example: Set the Maximum Load for a Real Machine

Suppose that the range of possible load values is 1 through 100. This example sets the maximum load for a relatively low-performance real machine.

```
max_load: 20
```

Specify Job Load (job_load)

For load balancing to work, you must assign a job_load value to every job that impacts the load on a machine. The job_load attribute in a job definition defines the relative amount of processing power the job consumes (the relative load the job places on a machine).

Load units are arbitrary values, and the range is user-defined. You can use any weighting scheme that you prefer. You can use the max_load attribute to assign a real machine a maximum job load. Then, you can use the job_load attribute in the

job definition to assign the job a load value that indicates the relative amount of the machine's load that the job consumes. These attributes let you control machine loading and prevent a machine from being overloaded.

Example: Define the Relative Processing Load for a Job

Suppose that the range of possible load values is 1 through 100. This example sets the load for a job that typically uses 10 percent of the CPU.

```
job_load: 10
```

Specify Queuing Priority

This article explains how to specify queuing priority. When a job is ready to run on a designated machine but the current load on that machine is too large to accept the new job's load or the job has resource dependencies that are not satisfied, AutoSys Workload Automation queues the job for that machine so it runs when sufficient resources are available.

For job queuing to take place, you must define the priority attribute in the job definition. The queue priority establishes the relative priority of all jobs queued for a given machine. A lower number attribute value indicates a higher priority. If you do not set the priority attribute or the priority is set to 0, the job is not queued behind other jobs and runs immediately on a machine if resource dependencies permit or if the job has no resource dependencies. The scheduler ignores any load unit values that are defined for the job or machine when the job has a priority value of zero.

AutoSys Workload Automation follows these rules to run jobs with priority, load balancing attributes, and resource attributes:

- Jobs with the highest priority and load balancing attributes run ahead of jobs with lower priority for the same machine-based queue.
- Jobs with the highest priority and resource attributes run ahead of jobs with lower priority for the same virtual resource-based queue.
- Jobs with lower priority and load balancing attributes enter the QUE_WAIT state when higher priority jobs for the same machine-based queue are in the QUE_WAIT state.
- Jobs with lower priority and resource attributes enter the RESWAIT state when higher priority jobs for the same resource-based queue are in the RESWAIT state.
- Jobs with equal or higher priority and both load balancing and resource attributes run if there is sufficient load on the machine and sufficient resource units to execute.
- The scheduler tries to evaluate the starting conditions of jobs that are associated with a group or application in order of their priority; however it does not ensure that jobs execute strictly in order of their priority.

NOTE

For more information about how AutoSys Workload Automation queues jobs, see [How AutoSys Workload Automation Queues Jobs](#).

Example: Set the Job to Run with Highest Priority

This example sets the job to run with the highest priority without overriding the machine load control mechanism.

```
priority: 1
```

Example: Set the Job to Run in the Background

This example sets the job to run in the background when the machine load is low.

```
priority: 100
```

Specify Relative Processing Power (factor)

You can use the factor attribute to determine the relative processing power for a machine. To calculate the relative processing power for each machine, the scheduler multiplies the available CPU cycles by the factor attribute value: (Available CPU Cycles) x (Factor Attribute Value) = Relative Processing Power

The scheduler determines which of the agent machines that are specified in the job definition have the best calculated usage (highest relative processing power). The scheduler starts the job on the agent machine with the best calculated usage (highest relative processing power).

Setting the factor value to zero (0) results in a calculated usage of zero (0) but does not disqualify the machine. The scheduler selects an agent machine with a factor value of zero (0) only when all other available machines specified in the job definition also have a factor value of zero (0).

NOTE

- Factor units are arbitrary, user-defined values. The value consists of a real number, typically between 0.0 and 1.0. You can set factor units to a value containing a decimal, such as 0.5. If you do not define the factor attribute in a machine definition, the value defaults to 1.0.
- The factor attribute is valid in a real machine definition or component machines defined to virtual machine.
- Sometimes the scheduler identifies multiple machines as having the best calculated usage (highest relative processing power). In such cases, the scheduler randomly selects one of those machines and starts the job on it. To allow the scheduler to start the job on any machine specified in the job definition, set the factor attribute value for all those machines to zero (0).

Example: Set the Factor for a Low-Performance Real Machine

This example sets the factor for a low-performance real machine, on a scale of 0.0 to 1.0.

```
factor: 0.1
```

Example: Set the Factor for a High-Performance Real Machine

This example sets the factor for a high-performance real machine, on a scale of 0.0 to 1.0.

```
factor: 1.0
```

Machine Status

Real machines have a run-time status attribute designed to reflect the availability of the machine. The machine status lets the scheduler run more efficiently by not wasting time trying to contact machines that are out of service. If a job is scheduled for a machine that is offline, it is set to PEND_MACH status until the machine comes back online. For a virtual machine, offline machines are not considered as possible candidates for running a job.

A machine can have one of following statuses:

- **Online**
Indicates that the machine is available and accepting jobs to run.
- **Offline**

Indicates that the machine has been manually removed from service and does not accept jobs to run.

- **Missing**
Indicates that the scheduler has verified that the machine is not responding and has automatically removed it from service. The machine does not accept jobs to run.
- **Unqualified**
Indicates that the scheduler is attempting to qualify the status of an agent before switching the machine from an online to missing status. The machine does not accept jobs to run.
- **Empty**
Indicates that a virtual machine or real machine pool does not contain any component machines. Jobs that are scheduled to machines in this status do not run.
- **Blocked**
Indicates that the agent blocks communication. All jobs that are scheduled to start on the agent are put in a PEND_MACH status.

AutoSys Workload Automation web services use the following character codes to represent the machine status:

- o—Online
- m—Offline
- a—Missing
- u—Unqualified
- e—Empty
- n—Blocked

Take a Machine Offline Manually

To take a machine offline manually (for example, during hardware service), use the sendevent command to send a MACH_OFFLINE event.

When you send a MACH_OFFLINE event, the jobs that are currently running, run to completion even though the status of the machine is offline. You can use the autorep command to monitor running jobs.

If you shut down a machine for servicing, you might want to let the running jobs complete before continuing. With the machine offline, you can service the machine while the scheduler continues running. All jobs that are scheduled to start on the offline machine are put in PEND_MACH status until the machine returns to service.

Example: Manually Take a Machine Offline

This example takes the machine cheetah offline:

```
sendevent -E MACH_OFFLINE -n cheetah
```

The scheduler log displays a message similar to the following message when the machine is offline:

```
[11/28/2005 15:38:21] CAUAJM_I_40245 EVENT: MACH_OFFLINE MACHINE: cheetah
```

Put a Machine Online Manually

To put a machine online manually, use the sendevent command to send a MACH_ONLINE event. The MACH_ONLINE event places the machine in the online status after the scheduler calibrates that the machine is actually available and accepts jobs to run. If the machine is not available, the machine status is set to missing or blocked.

When you send a MACH_ONLINE event for a machine, jobs with a status of PEND_MACH on that machine are automatically started.

The online status of a machine affects how the scheduler evaluates jobs with load balancing attributes or resource dependencies. When taking a machine offline, the scheduler places the job in the PEND_MACH status when all candidate machines are offline. While evaluating a job that is queued in the QUE_WAIT or RESWAIT status, the scheduler considers machine load units or machine-based resources as unavailable if the machine is offline. In that case, the job remains in the QUE_WAIT or RESWAIT status and does not transition into the PEND_MACH status. While evaluating a job that is queued in the RESWAIT status, the scheduler evaluates global virtual resources even if the machine is offline. When global virtual resources become available while the machine is offline, the job exits the RESWAIT status and transitions into the PEND_MACH status. When the job enters the PEND_MACH status, it releases all held global virtual resources. When bringing the machine back online, the scheduler evaluates jobs in the QUE_WAIT or RESWAIT status first followed by the jobs in the PEND_MACH status.

Example: Manually Put a Machine Online

This example returns the machine cheetah to online status:

```
sendevent - E MACH_ONLINE - n cheetah
```

The scheduler log displays a message, similar to the following message, when the machine is online:

```
[11/28/2005 15:38:21] CAUAJM_I_40245 EVENT: MACH_ONLINE MACHINE: cheetah
```

How Status Changes Automatically

When the scheduler verifies that a real machine is not reachable, it uses the following process to manage machine and job status:

- If the scheduler fails to contact an agent of the machine, the scheduler marks the machine as unqualified and logs a message similar to the following message:

```
[11/28/2005 16:01:46] CAUAJM_W_40290 Machine cheetah is in question. Placing machine in t
```

- The scheduler puts all jobs that are scheduled to start on the unqualified machine in PEND_MACH status. The scheduler verifies the [GlobalPendMachStatus](#) parameter (on UNIX) or Global Pend Mach Status field (on Windows) value. If the status is set to any valid value other than the default (PEND_MACH), the scheduler verifies the [GlobalPendMachDelay](#) parameter (on UNIX) or Global Pend Mach Delay field (on Windows) value. If the delay interval is set to the default value (zero), the scheduler immediately sends a CHANGE_STATUS event for the job. If the delay interval is set to a value other than the default, the scheduler waits for the specified interval before sending the CHANGE_STATUS event.
- The scheduler attempts to qualify the status of that machine by pinging the agent every 10 seconds if there are pending jobs. If there are no pending jobs, the scheduler verifies the AgentStatusPollInterval parameter (on UNIX) or Agent Status Poll Interval field (on Windows) value. If the interval is set to the default value (1); the scheduler pings the agent every one minute. If the interval is set to a value other than the default, the scheduler waits for the specified interval before pinging the agent.
- If the agent responds, the scheduler sends a MACH_ONLINE event and the machine returns to service.
- When the machine returns to service, the scheduler starts all jobs in PEND_MACH status for that machine. The scheduler verifies the [GlobalPendMachInterval](#) parameter (on UNIX) or Global Pend Mach Interval field (on Windows) value. If the interval is set to the default value (zero), the scheduler does not wait between job starts. If the interval is

set to a value other than the default, the scheduler waits for the specified interval before starting jobs in PEND_MACH status, and then repeats that cycle until all the jobs are restarted.

- If the agent fails to respond after three attempts (in case there are pending jobs) or after one attempt (in case there are no pending jobs and the AgentStatusPollInterval parameter (on UNIX) or Agent Status Poll Interval field (on Windows) value is set to the default. If you specified any other value other than the default, the scheduler waits for those number of times before marking the machine as missing), the scheduler marks the machine as missing, issues a MACHINE_UNAVAILABLE alarm, and logs a message similar to the following message:

```
[11/28/2005 16:01:46]      CAUAJM_I_40253 Machine cheetah is not responding. Taking offline.
```

- The scheduler puts all jobs that are scheduled to start on the missing machine in PEND_MACH status based on the values set for the GlobalPendMachStatus and GlobalPendMachDelay parameters (on UNIX) or Global Pend Mach Status and Global Pend Mach Delay fields (on Windows). These values control the status of jobs that are scheduled on a machine that is offline.
- If the machine definition is updated, the scheduler marks the machine as unqualified, logs the following message, and pings the agent until the machine returns to service or is marked missing:

```
[11/28/2005 16:01:46]      CAUAJM_W_40291 Machine cheetah has been updated. Placing machine in
```

- Otherwise, the scheduler pings the missing agent of the machine based on the time interval you specified in the AgentStatusPollInterval parameter (on UNIX) or Agent Status Poll Interval field (on Windows) to verify its availability.
- If the agent responds, the scheduler sends a MACH_ONLINE event and the machine returns to service.
- When the machine returns to service, the scheduler starts all jobs in PEND_MACH status for that machine based on the value set for the GlobalPendMachInterval parameter (on UNIX) or Global Pend Mach Interval field (on Windows). This parameter controls the starting of jobs in PEND_MACH status.

NOTE

If you understand the cause of a missing machine and intervene to correct it, you can use the sendevent command to send a MACH_ONLINE event to bring the machine back online instead of waiting for the scheduler to do so.

How Status Affects Jobs on Virtual Machines

If a job is defined to run on a virtual machine or a list of machines and one of those machines is offline, the job runs on another available machine with which it is associated.

If, however, all machines in the virtual list are offline, the scheduler puts the job in PEND_MACH status. If any of the machines with which the job is associated comes back online, the scheduler removes the job from PEND_MACH status and runs it on the online machine, subject to the queuing criteria.

How Machine Status Changes When Disk Threshold is Breached

The agent is configured to monitor the amount of available disk space and send notifications to warn you when the space is too low. The agent has three disk space warning thresholds:

- **Notice**
The agent logs a warning notice when the available disk space is less than the size specified in the agent.resourcemon.threshold.disk.warning.notice parameter in the agentparm.txt file. The agent continues to run; it accepts new and eligible pending jobs requests.
- **Severe**

The agent logs a severe warning when the available disk space is less than the size specified in the `agent.resourcemon.threshold.disk.warning.severe` parameter in the `agentparm.txt` file. The agent stops accepting new job requests. All jobs that are scheduled to start on this agent are put in a `PEND_MACH` status.

- **Critical**

The agent logs a critical warning and shuts down when the available disk space is less than the size specified in the `agent.resourcemon.threshold.disk.critical` parameter in the `agentparm.txt` file.

NOTE

For information about configuring the agent to monitor the available disk space, see the [Workload Automation Agents](#) documentation.

The following table describes the scheduler behavior and the machine status changes that happen when the agent logs notice, severe, or critical warnings when the disk threshold is breached:

| Disk Threshold Breach | Scheduler Behavior | Machine Status |
|-----------------------|---|---|
| Notice | Raises an ALERT event.
Raises <code>MACHINE_DISKTHRESHOLD</code> alarm.
Continues to communicate with the agent. | Online |
| Severe | Raises ALERT event.
Raises <code>MACHINE_DISKTHRESHOLD</code> alarm.
Changes the machine status to Blocked as the agent blocks communication.
Raises <code>MACHINE_UNAVAILABLE</code> alarm. | Blocked
When the available disk space is greater than the size specified in the <code>agent.resourcemon.threshold.disk.warning.severe</code> parameter in the <code>agentparm.txt</code> file, the scheduler brings the agent back to Online status. |
| Critical | Raises ALERT event.
Raises <code>MACHINE_DISKTHRESHOLD</code> alarm.
Changes the machine status to Missing as the agent shuts down.
Raises <code>MACHINE_UNAVAILABLE</code> alarm. | Missing
When the available disk space is greater than the size specified in the <code>agent.resourcemon.threshold.disk.critical</code> parameter in the <code>agentparm.txt</code> file, free enough disk space and then manually restart the agent. After the agent is restarted, the scheduler puts the agent in the Blocked or Online status based on the disk space that is available. |

NOTE

- When sufficient disk space is available and the agent comes out of all the threshold breaches, the scheduler raises an ALERT event and the scheduler brings the agent back to Online status.
- When you manually take a machine offline, the machine status remains in the Offline status irrespective of the threshold breaches.
- If `PE_AUTO_DISABLE` environment variable is set to 1, the machine status remains in the Online status irrespective of the threshold breaches.

How Load Balancing Works

Load balancing can be implemented to use inherent features of AutoSys Workload Automation.

You can implement load balancing (where the workload is spread across multiple machines that are based on capabilities of each machine) by using the machine attribute to specify a virtual machine or multiple real machines in a job definition. This is also an easy way to ensure reliable job processing. For example, the scheduler can use load balancing to select which of the machines in a job definition is best suited to run the job, and automatically start it on that machine.

The advantages of building a virtual machine are as follows:

- Its definition can be changed and the new construct is immediately applied globally.
- The max_load and factor values can vary between machines.

Alternatively, you can specify a list of real machines in the job's machine attribute. The system configuration includes machines of varying processing power. AutoSys Workload Automation chooses a real machine based on the load balancing method ([MachineMethod](#)) you specified in the configuration file or in the machine_method attribute:

- **cpu_mon** load method—AutoSys Workload Automation chooses which machine to run the job on based on the available processing power obtained from the agent.
- **job_load** method—AutoSys Workload Automation chooses which machine to run the job on based on the max_load and factor attributes for each real machine with the job definition's priority and job_load attributes.
- (UNIX only) **rstatd** method—AutoSys Workload Automation chooses which machine to run the job on based on the information obtained from the remote UNIX computer's remote kernel statistics daemon.
- **roundrobin** method—AutoSys Workload Automation chooses which machine to run the job on in a round robin manner.
- **roundrobin_jobload**—AutoSys Workload Automation chooses which machine to run the job on based on the job_load method and then the roundrobin method is used on the resulting machines to choose the real machine where the job runs. However, if none of the machines have sufficient load units, AutoSys Workload Automation puts the job in QUE_WAIT status. The job stays in QUE_WAIT status until one of the machines has sufficient load units available. When sufficient load units are available, the job runs on that machine. In this situation, the roundrobin method is not considered.

NOTE

For information about how AutoSys Workload Automation queues jobs, see [How AutoSys Workload Automation Queues Jobs](#).

In either case, AutoSys Workload Automation uses the following process to verify the available relative processing cycles for each machine:

1. AutoSys Workload Automation calculates the number of load units available on each real machine in the specified virtual machine. To do this, AutoSys Workload Automation uses the load balancing method specified in the configuration file.

NOTE

- For the Workload Automation Agent on UNIX, Linux, Windows, or iOS (machine type: a), AutoSys Workload Automation uses cpu_mon, rstatd, or the job_load method. If the machine method that is specified in the configuration file is set to the cpu_mon method, the scheduler runs a CPU Monitoring (OMCPU) job to determine the available CPU cycles. This is the default. For the Workload Automation Agent on UNIX and Linux only (opsys: aix, hpux, or linux), AutoSys Workload Automation supports the rstatd method.

2. AutoSys Workload Automation performs the following calculation:

$$\text{Machine Usage} = \text{Available Load Units} * \text{Factor value}$$

3. AutoSys Workload Automation chooses the machine with the most relative load units available, based on the calculation in Step 2.

NOTE

- If a real machine in the virtual machine is not online, the scheduler does not attempt to contact it and it is not considered in the load balancing algorithm.
- If the machines have equal max_load and factor values, it is equivalent to defining a job and specifying the following in the machine field:

```
machine: cheetah, camel
```

- If the factor attribute is not specified for a machine, AutoSys Workload Automation assumes the default factor value for each machine (1.0).
- On UNIX, the load balancing method is specified using the MachineMethod parameter in the AutoSys Workload Automation configuration file. On Windows, the method is specified using the Machine Method field on the Scheduler window of AutoSys Workload Automation Administrator. For more information, see [Configure a Scheduler](#).
- The cpu_mon machine method does not apply to z/OS machines (Workload Automation Agent on z/OS) because the OMCPU job is not supported on z/OS.

Example: Load Balancing With a Virtual Machine

This example defines a virtual machine (marmot) with three real machines (cheetah, hippogriff, and camel):

```
insert_machine: marmot
machine: cheetah
factor: 1
machine: hippogriff
factor: .8
machine: camel
factor: .3
```

To start a job on this virtual machine, specify marmot in the job's machine attribute. The scheduler performs the necessary calculations to verify on which machine to run the job, and reflects these calculations in its output log. The output is similar to the following:

```
[11/22/2005 10:16:53] CAUAJM_I_40245 EVENT: STARTJOB JOB: tvn
[11/22/2005 10:16:54] CAUAJM_I_10208 Machine usages obtained for job <tvn>:
[11/22/2005 10:16:59] CAUAJM_I_10258 Machine <cheetah>: Method <CPU monitor>.
Available usage <78*[1.00]=78>
[11/22/2005 10:17:02] CAUAJM_I_10258 Machine <hippogriff>: Method <CPU monitor>.
Available usage <80*[.80]=64>
[11/22/2005 10:17:07] CAUAJM_I_10258 Machine <camel>: Method <CPU monitor>.
Available usage <20*[.30]=6>
[11/22/2005 10:17:07] CAUAJM_I_10209 Machine <cheetah> selected for job <tvn>.
[11/22/2005 10:17:11] CAUAJM_I_40245 EVENT: CHANGE_STATUS STATUS: STARTING
JOB: tvn
[11/22/2005 10:17:11] CAUAJM_I_10082 [cheetah connected for tvn]
```

NOTE

Even though the machine usage on cheetah was less than that of machine hippogriff, machine cheetah was picked because of the result of the factor calculation (machine cheetah had 78% of its processing power available, while machine hippogriff only had 64% available). Thus, the factors weigh each machine to account for variations in processing power.

Load Balancing Using Virtual Resource Dependencies

Load balancing can be performed using virtual resources combined with virtual machines or machine lists for basic throttling and serialization. If you assign a virtual machine or list of real machines to a job with the virtual resource dependencies, the jobs can be dispatched to the various machines based on resource availability. If two or more

machines satisfy the virtual resource requirements, the job is scheduled to run on the machine with the most available CPU resources.

For example, if both machine1 and machine2 satisfy the virtual resource requirements and have equal amount of CPU resources available, the scheduler randomly selects a machine and runs the job on the selected machine. If both machine1 and machine2 satisfy the virtual resource requirements and machine1 has less amount of CPU resource available, the job runs on machine2.

Concurrent Jobs Processing

The number of jobs that can be run concurrently at a given time varies with the installation set up and the available hardware and software. If the installation does not have enough resources to handle the load efficiently, it would result in a delay in job processing.

We recommend that you stagger your jobs instead of starting all jobs at the same time to avoid delays in batch cycle executions. All jobs must first get a status of STARTING before any of them go into a RUNNING state. The staggering of jobs helps in avoiding a bottleneck in the environment and improves the system performance and processing time.

Jobs with Load Balancing and Resource Attributes

CA Workload Automation AE 11.3.6 SP5 introduced an unintended change in the scheduling behavior for jobs having a job_load of 0 (zero) that specify both a priority and a virtual resource dependency. Due to the strict priority enforcement, jobs with resource requirements and no load can now delay or even block lower priority jobs having load units that are scheduled to the same machine, although these jobs do not utilize virtual resources.

Example: Load Balancing Using Machine Virtual Resources

Suppose that you have three machines (sloth, tiger, and leopard) capable of running several applications but they vary in capacity or physical resources, for example CPU speed, memory, or utilization. You also have various jobs that use different runtime resources. Jobs with low resource usage can run anywhere. Jobs that use more resources are limited to where they can run. The number of concurrent jobs and their requirements must be controlled to avoid overburdening any machine.

- Define the real machine definitions for sloth, tiger, and leopard. Then define a virtual machine, domain, that references all the machines where the jobs should be allowed to run.

```
insert_machine: sloth
insert_machine: tiger
insert_machine: leopard
insert_machine: domain
type: v
machine: sloth
machine: tiger
machine: leopard
```

- Define the maximum amount of virtual resources available to each machine. Remember, they are virtual resources. They do not really exist. The virtual resource amounts are approximations that are based on estimated capabilities of the machines. In this example, the sloth machine has the fewest capabilities while the leopard machine has the most.

```
insert_resource: job_weight
res_type: r
machine: sloth
amount: 2
insert_resource: job_weight
res_type: r
machine: tiger
```

```

amount: 10
insert_resource: job_weight
res_type: r
machine: leopard
amount: 30

```

- Define jobs to run on one of the real machines referenced by the virtual machine domain. Identify the virtual resource units each job consumes. Similar to the resource definition, these values are approximations that are based on perceptions or expectations of what the job consumes while running. The quantity required for the job determines where it can run and the mix of jobs that can run concurrently.

```

insert_job: quick_job
machine: domain
command: efficient_report
resources: (job_weight,quantity=1,free=y)
insert_job: heavy_job
machine: domain
command: analytical_report
resources: (job_weight,quantity=5,free=y)
insert_job: beastly_job
machine: domain
command: quarterly_update
resources: (job_weight,quantity=10,free=y)

```

Based on the above definitions, all machines have at least one unit of the `job_weight` virtual resource defined to it. Hence, the job `quick_job` could run on any of the machines that is defined to the virtual machine named `domain` and that has the most available CPU resources. The jobs `heavy_job` and `beastly_job` can only be scheduled to real machines `tiger` and `leopard`. The two jobs cannot be scheduled concurrently to the `tiger` machine as that would exceed the virtual resources defined to it. If the job `heavy_job` is already running on the `tiger` machine when job `beastly_job` is being scheduled, the job `beastly_job` would be scheduled to run on the `leopard` real machine.

You can also implement load balancing using virtual resources as dependencies to a job. Virtual resource dependencies can be defined to both virtual machines and real machine pools.

If you assign a virtual machine or real machine pool to a job with either virtual resource dependencies, the job runs on the machine that satisfies the resource dependencies. If the job has virtual resource dependencies and two or more machines satisfy the specified metrics, the job runs on the machine that satisfies the virtual resource requirements and has the most available CPU resources.

NOTE

If AutoSys Workload Automation is not integrated with CA Automation Suite for Data Centers the job is submitted on the machine that satisfies the virtual resource requirements and has the most available CPU resources. If the job does not have virtual resources, it runs on the machine as determined by load balancing using the `max_load` and `factor` values.

Example: Load Balancing Using Virtual Resource Dependencies

Suppose that you want to define a job that gets submitted on a machine that satisfies the virtual resource dependencies, you can do the following:

- Define a renewable resource `ren_glb1` at the global level:

```

insert_resource: ren_glb1
res_type: R

```

```
amount: 10
```

- Define a job `job_load` with virtual resource dependencies:

```
insert_job: job_load
job_type: CMD
command: sleep 1
machine: DCAPOOL
owner: autosys
date_condition: 0
alarm_if_fail: 1
resources: (ren_glb1, quantity=2, free=y) and (MEM_INUSE_PCT, VALUE=30, VALUEOP=LTE)
```
- Generate a report for the job (`job_load`) to view whether the virtual resource dependencies are satisfied on the MWIN, MLIN, and MSOL machines:

```
job_depends -J job_load -r
```

The report might resemble the following:

```
Job Name      Machine
-----
job_load      MLIN
  Virtual Resources
  -----
Resource      Type    Amount  Satisfied?
-----
ren_glb1      R      2      YES
  Real Resources
  -----
Resource                               Satisfied?
-----
MEM_INUSE_PCT, VALUE=30, VALUEOP=LTE    NO
-----
```

```
Job Name      Machine
-----
job_load      MSOL
  Virtual Resources
  -----
Resource      Type    Amount  Satisfied?
-----
ren_glb1      R      2      YES
  Real Resources
  -----
Resource                               Satisfied?
-----
MEM_INUSE_PCT, VALUE=30, VALUEOP=LTE    YES
-----
```

```
Job Name      Machine
-----
```

```

job_load          MWIN
  Virtual Resources
  -----
Resource          Type    Amount  Satisfied?
-----
ren_glb1          R      2      YES
  Real Resources
  -----
Resource                                Satisfied?
-----
MEM_INUSE_PCT, VALUE=30, VALUEOP=LTE      YES

```

The report displays that the virtual resource (ren_glb1) is satisfied on all the machines.

- Start the job (job_load):

```
sendevent - E STARTJOB - J job_load
```
- Generate a detailed report for the job (job_load) to view the machine on which the job runs:

```
autorep -J job_load -d
```

The resulting report might resemble the following:

| Job Name | Last Start | Last End | ST Run/Ntry |
|----------|---------------------|---------------------|-------------|
| Pri/Xit | | | |
| job_load | 10/07/2010 15:06:21 | 10/07/2010 15:06:23 | SU 689/1 0 |

| Status/[Event] | Time | Ntry | ES | ProcessTime | Machine |
|-------------------------|---------------------|------|----|---------------------|---------|
| ----- | ----- | -- | -- | ----- | |
| STARTING | 10/07/2010 15:06:21 | 1 | PD | 10/07/2010 15:06:22 | MSOL |
| RUNNING | 10/07/2010 15:06:22 | 1 | PD | 10/07/2010 15:06:22 | MSOL |
| <Executing at WA_AGENT> | | | | | |
| SUCCESS | 10/07/2010 15:06:23 | 1 | PD | 10/07/2010 15:06:23 | MSOL |

Force a Job to Start

The sendevent command lets you override defined starting conditions (including capacity requirements) and start a job immediately. This feature is useful when you encounter unexpected interruptions to the work flow, such as hardware failures.

The following conditions apply:

- When a job is defined to run on an offline machine and you force start it, the scheduler repeatedly restarts the job until the scheduler makes contact with the agent or exhausts the maximum restart attempts specified in the job definition.
- When you force start a job that requires unavailable (held) resources, the scheduler starts the job as soon as the resources become available. If the job is not already queued, the scheduler places it in the RESWAIT state. It is not necessary to force start a job that is already in the RESWAIT state, and the scheduler ignores attempts to do so.
- When you force start a job that is in a non-executable state (ON_ICE, ON_HOLD), the scheduler places the job in an executable state. The job starts immediately, runs, and remains in an executable state. Force start a job that is in a non-executable state only when you need to start the job immediately. To resume the regular run schedule, change the executable status of the job.
- Jobs defined to run on virtual machines or pools run on the machine with most available load units.
- When you force start a job and multiple non-zero factor online machines are available, the scheduler uses the MachineMethod parameter value to select the machine where the job must run when the UseMachineMethodOnForceStartJob parameter in the configuration file is set to 1 (UNIX) or the Machine Method On Force Starts check box is selected on the Scheduler - AutoSys Workload Automation Administrator window of the Administrator utility (Windows).
- When you force start a job with load balancing attributes or resource attributes or both:
 - The scheduler ignores the load balancing attributes and never places the job in the QUE_WAIT state.
 - The scheduler always checks for available resources but ignores the job priority:
 - If the resources are not available, the scheduler places the job in the RESWAIT state. When the resources are available, the scheduler resumes force-starting jobs that entered the RESWAIT state (ignoring priorities or load balancing attributes if any).
 - If the resources are available, the scheduler executes the job regardless of its priority. The scheduler never places a job in the RESWAIT state if a higher priority job (a job having a lower priority attribute value) is queued in either the RESWAIT or QUE_WAIT state.
 - The scheduler can force start a job that is already in the RESWAIT state regardless of other higher priority jobs (jobs having a lower priority value) in either the RESWAIT or QUE_WAIT state as long as resources are available.

Do not issue the sendevent command to force start a box while the jobs in the box itself or jobs belonging to boxes within the box (at higher box levels) are executing. In case you need to re-execute jobs in a box run, we recommend that you follow any one of the following methods as appropriate after all scheduled jobs at all levels in the box have executed:

- To fix the current run of the box, force start the individual jobs that are stuck or otherwise did not execute properly in the order of the highest box level (bottom of the box tree) to the lowest box level (top of the box tree). This ensures that the scheduler propagates the completed box statuses all the way to the top of the box tree.
- To execute a new clean run of the entire box, issue the CHANGE_STATUS event to INACTIVE against the level zero box to reset the status of all jobs in all levels of the box and then use the sendevent command to send a FORCE_STARTJOB event against the level zero box.

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The instance command prompt appears.
2. Enter the following command:

```
sendevent -E FORCE_STARTJOB {-J job_name [-o step_name] | -B group [-I application] | -I application} [options]
```

– **-J job_name [-o step_name] | -B group [-I application] | -I application**

Specifies which jobs to start. Specify a single job or all jobs that are associated with a particular group or application. To start a multi-step job at the step where it failed on the previous run, use the `-o` option.

– **options**

Specifies optional sendevent parameters.

The job starts or is queued.

Queuing Jobs

Queuing is a mechanism that is used in AutoSys Workload Automation to verify the run order of jobs that cannot run immediately. There is no actual physical queue. Instead, AutoSys Workload Automation uses queuing policies, which are based on the following criteria:

- The use and subsequent interaction of the `job_load` and `priority` attributes in a job definition and the `max_load` and `factor` attributes in a machine definition.
- The use and subsequent interaction of the `resource` and `priority` attributes in a job definition and the attributes in a resource definition.
- The use of both the load balancing and resource attributes outlined above in a job definition.

Jobs that are in a queued state already meet their starting conditions, but cannot start due to external conditions.

Jobs that meet their starting conditions but are in the `ON_HOLD` or `ON_ICE` state also do not start; however, these jobs are not considered queued jobs. To place a job on hold or on ice, send a `JOB_ON_HOLD` or `JOB_ON_ICE` event using the `sendevent` command. Jobs in these states do not start until you take them off hold or off ice.

When a job leaves a queued state, the scheduler determines whether to start the job by re-evaluating starting conditions for that job unless you configure AutoSys Workload Automation to skip starting condition evaluation for queued jobs. If a job that is contained in a box fails its starting condition checks when leaving the queue, the scheduler places that job in the `ACTIVATED` state. If a job that is not contained in a box fails its starting condition checks when leaving the queue, the scheduler places that job in the `INACTIVE` state. If the job meets its starting conditions, or if you configure the system to skip starting condition evaluation for queued jobs, the job starts.

When you take a job off hold, the scheduler re-evaluates starting conditions for that job. When you take a job off ice, the scheduler does not restart the job until its starting conditions recur, even if those conditions were met while the job was on ice. If a job is contained in a running box when it is taken off ice, the scheduler does not restart the job until the following run of the box, even if its starting conditions recur during the existing run of the box.

When you instruct the scheduler to bypass execution of a job, the scheduler starts the job when it meets its starting conditions. The scheduler simulates running these jobs, but the agent does not execute commands associated with the jobs. Bypassed jobs evaluate as successfully completed on the scheduler machine. When you issue the `JOB_OFF_NOEXEC` event, the agent executes commands associated with the specified job the next time the job starts. Changing the executable status of a job does not affect evaluation of starting conditions.

NOTE

- You can configure AutoSys Workload Automation to perform one of the following actions when scheduling jobs leaving a queued state:

- Skip starting condition evaluation.
- Evaluate the starting conditions of queued jobs other than the date condition check for the day of evaluation.
- Evaluate the starting conditions of queued jobs including the date condition check for the day of evaluation.
- If you configure AutoSys Workload Automation to skip starting condition evaluation for queued jobs, those jobs start immediately upon leaving a queued state.
- If you configure AutoSys Workload Automation to evaluate the starting conditions of queued jobs other than the date condition check for the day of evaluation, the scheduler does not re-evaluate date and time conditions. Jobs that meet their original date and time conditions before entering a queued state start immediately after they leave the queued state unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- If you configure AutoSys Workload Automation to evaluate the starting conditions of queued jobs including the date condition check for the day of evaluation, the scheduler re-evaluates date conditions but not time conditions. Jobs that meet their date conditions after leaving a queued state start unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- When you take jobs off hold, the scheduler does not re-evaluate date and time conditions. Jobs that meet their date and time conditions while they are on hold start immediately after they are taken off hold unless other starting conditions apply and are not satisfied. Jobs that are taken off hold on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- For more information about configuring AutoSys Workload Automation to schedule jobs leaving a queued state, see [EvaluateQueuedJobStarts](#).

The following sections discuss queuing jobs and give examples of how to use load balancing and queuing to optimize job processing in your environment.

How AutoSys Workload Automation Queues Jobs

For queuing to be effective, set the priority attribute for all jobs. By default, the priority attribute is set to 0, indicating that the job should not be queued behind other jobs and should run immediately if resource dependencies allow it to run. When you let the priority attribute default for a job, it runs even if its job load would push the machine over its load limit. However, even when jobs have a priority of 0, AutoSys Workload Automation tracks job loads on each machine so that jobs with non-zero priorities can be queued. If the job has resource dependencies that are not met, it is queued until resources are available even when the priority attribute is set to 0.

NOTE

If the job has resource dependencies, AutoSys Workload Automation does not use the following process to limit the job load on machines and to queue jobs for processing. Instead, the resource manager (AutoSys Workload Automation) is used to select the best machine to run the job.

The scheduler follows these rules when queuing jobs:

- If the priority of a job is set to 0 (zero), the scheduler does not evaluate the `job_load` attribute value.
- The scheduler queues jobs with load balancing attributes (`job_load`, `priority`, `max_load`, `factor`) based on “like” machine attribute values. A job in the `QUE_WAIT` state for one machine attribute value automatically blocks all the lower priority jobs that specify the same machine attribute value. It does not automatically block higher or equal priority jobs that

specify the same machine attribute value or a job that specifies a different machine attribute value. The scheduler schedules the job as long as the machine has available load units to execute the job.

- Jobs with both load balancing and resource attributes that enter the QUE_WAIT state do not consume load units or resources. These jobs do not automatically block lower priority jobs that specify the same resource attribute and a different machine attribute value.
- The resource dependencies are evaluated after the load balancing attributes are evaluated and the machine has available load units to execute the job.
- If you did not specify a value for the resource attribute in your job definition, the resource dependencies are not evaluated.
- The scheduler queues jobs with virtual resource dependencies (whether global or machine-based) based on “like” resource names. A job in the RESWAIT state for one resource name automatically blocks all the lower priority jobs that specify the same resource name. It does not automatically block higher or equal priority jobs that specify the same resource name or a job that specifies a different resource name. The scheduler schedules the job as long as the resources are available.
- The scheduler queues jobs with real resource dependencies based on “like” node_name machine attribute values corresponding to the job’s machine attribute value. A job in the RESWAIT state for one machine attribute value automatically blocks all lower priority jobs that specify a machine attribute value with a matching node_name attribute value. It does not automatically block higher or equal priority jobs that specify the same machine attribute value with a matching node_name attribute value or a job that specifies a machine attribute value with a different node_name attribute value. The scheduler schedules job as long as the node has available system resources to execute the job.
- Jobs with both load balancing and resource attributes that enter the RESWAIT state after the load balancing attributes are successfully evaluated do not consume any load units. These jobs do not automatically block lower priority jobs that specify the same machine attribute value and either do not specify the resource attribute or specify a different resource attribute value.
- When a job completes and load balancing units on the machine become available, the scheduler evaluates the resource condition of a job that is coming out of the QUE_WAIT state. Thus, it is possible for a job to leave the QUE_WAIT state and enter the RESWAIT state.
- When resource becomes available, the scheduler re-evaluates the load balancing attributes of a job that is coming out of the RESWAIT state. Thus, it is possible for a job to leave the RESWAIT state and enter the QUE_WAIT state.
- When evaluating queued jobs, the scheduler evaluates jobs in the order of priority first. Among the jobs with the same priority, the scheduler evaluates jobs in the order of the time the job entered a queued state. When the job leaves a queued state (QUE_WAIT or RESWAIT) to enter the other queued state (RESWAIT or QUE_WAIT respectively), the scheduler preserves the initial time that the job entered the first queued state.

AutoSys Workload Automation uses the following process to limit the job load on machines and to queue jobs for the processing:

- If you set a job_load value for a job and you assigned a max_load for every real machine comprising a virtual machine, AutoSys Workload Automation checks if each machine has sufficient available load units before running the job. When more than one job is queued, the priority value is considered first when deciding which job to run next. If insufficient load units are available to run the highest priority job, no other priority jobs are considered subsequently.
- If each real machine has sufficient load units, AutoSys Workload Automation employs the load balancing and factor algorithms to verify on which machine the job should start.
- If only one of the machines has sufficient load units, the job runs on that machine.
- If none of the machines have sufficient load units, AutoSys Workload Automation puts the job in QUE_WAIT status for all the machines. The job stays in QUE_WAIT status until one of the machines has sufficient load units available.

NOTE

- If a job is in QUE_WAIT status and you want it to run immediately, do not force the job to start. Instead, use the sendevent command to send a CHANGE_PRIORITY event that changes the job's priority to 0.
- You can send a CHANGE_PRIORITY event to change the priority of RESWAIT jobs.

Example: Job Queuing

This example shows a simple job queuing scenario that uses a previously defined machine named lion with a max_load of 100:

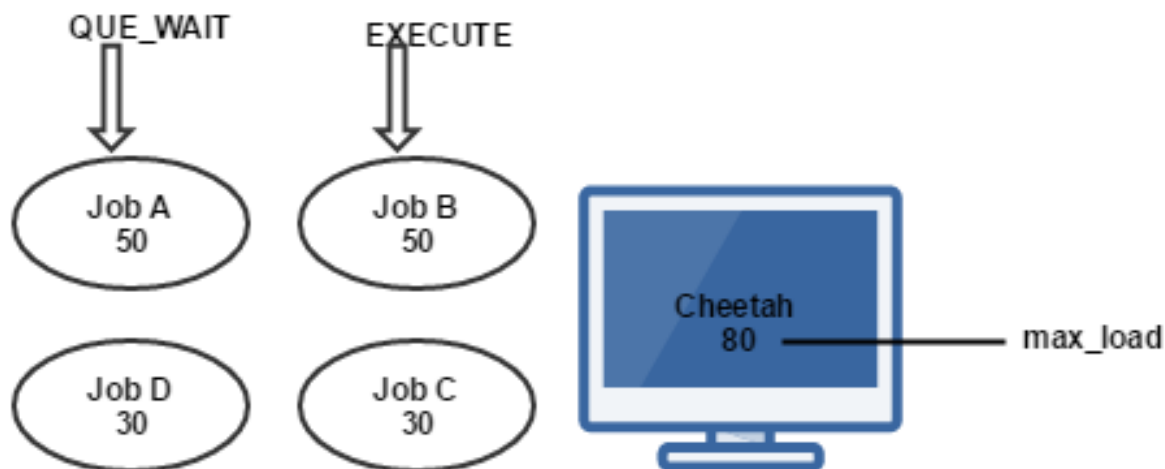
```
insert_job: jobA
machine: lion
job_load: 80
priority: 1
insert_job: jobB
machine: lion
job_load: 90
priority: 1
```

In this example, if JobA was running when JobB started, AutoSys Workload Automation would put JobB in QUE_WAIT status until JobA completes, at which point JobB can run.

Example: Job Queuing and Load Balancing

This example shows a situation in which a machine has 80 load units and multiple jobs are waiting to start. In this example, JobB and JobC are executing, while JobA and JobD are queued (in the QUE_WAIT state) and waiting for available load units. The numbers in the following illustration indicate the job_load assigned to each job, and the max_load set for the machine.

Figure 27: Queing Jobs



The following JIL statements define the machine and the jobs in this example:

```
insert_machine: cheetah
max_load: 80
insert_job: JobA
machine: cheetah
job_load: 50
priority: 70
insert_job: JobB
```

```

machine: cheetah
job_load: 50
priority: 50
insert_job: JobC
machine: cheetah
job_load: 30
priority: 60
insert_job: JobD
machine: cheetah
job_load: 30
priority: 80

```

In this example, JobB and JobC are already running because their starting conditions were satisfied first. After JobB or JobC completes, JobA is considered to start before JobD because JobA has a higher priority.

How soon JobA starts is determined by a combination of its priority and job_load attributes, and the max_load machine attribute. The result differs based on whether JobB or JobC finishes first, as follows:

- If JobB finishes first, 50 load units become available, so JobA runs. After JobA or JobB complete, sufficient load units become available, so JobD runs.
- If JobC finishes first, only 30 load units become available, so JobA and JobD remain queued until JobB completes.
- After JobB completes, a total of 80 load units become available, so JobA and JobD are eligible to run. Because JobA has a higher priority, it runs first. JobD runs shortly after.

Use a Virtual Machine as a Subset of a Real Machine

One variety of virtual machine can be considered a subset of a real machine. Typically, you would use this type of virtual machine to construct an individual queue on a given machine. One use for this construct might be to limit the number of jobs of a certain type that run on a machine at any given time.

Example: Define a Virtual Machine as a Subset of a Real Machine

This example shows how to define a virtual machine that functions as a subset of a real machine, thereby acting as a queue.

In this example, cheetah is a real machine with a max_load value of 80. If you create three different print jobs, but you want only one job to run on a machine at a time, you can use a combination of the max_load attribute for a virtual machine and the job_load attributes for the jobs themselves to control how the jobs run.

To implement this scenario, first create the virtual machine that is named cheetah_printQ as follows:

```

insert_machine: cheetah_printQ

machine: cheetah max_load: 15

```

Next, you would define the three print jobs as follows:

```

insert_job: Print1

machine: cheetah_printQ

job_load: 15

```

```

priority: 1

insert_job: Print2

machine: cheetah_printQ

job_load: 15

priority: 1

insert_job: Print3

machine: cheetah_printQ

job_load: 15

priority: 2

```

Although the real machine cheetah has a max_load value of 80, meaning that all three jobs (with their job_load values of 15) could run on it simultaneously, the virtual machine cheetah_printQ effectively resets the max_load of the real machine to 15. Because each job is defined to run on cheetah_printQ, not cheetah, only one of the jobs can run at a time because each job requires all the load units available on the specified machine.

NOTE

The load units that are associated with a virtual machine have no interaction with the load units for the real machine. This example implies that the virtual load of 15 does not subtract from the load units of 80 for the real machine. Load units are simply a convention that lets the user restrict concurrent jobs running on any one machine.

Use a Virtual Machine to Combine Subsets of Real Machines

You can also define virtual machines to combine subsets (or slices) of real machines into one virtual machine. For example, you might take this step if you have two machines that are print servers and you want only one print job to run at a time on each.

Example: Define a Virtual Machine to Combine Subsets of Real Machines

This example defines a virtual machine (printQ) that uses subsets of the loads available on two real machines to control where jobs run.

To implement this option, create the virtual machine that is named printQ, and specify two real machines (cheetah and camel), as shown in the following JIL statements:

```

insert_machine: printQ

type: v

machine: cheetah

max_load: 15

```

```
machine: camel

max_load: 15
```

When a job is ready to start on printQ, AutoSys Workload Automation checks if the component real machine (cheetah or camel) has enough load units available to run the job.

- If neither machine has enough available load units, the product puts the job in QUE_WAIT status and the product starts it when there are enough load units.
- If only one machine has enough available load units, the product starts the job on that machine.
- If both machines have enough available load units, the product checks the usage on each, and the product starts the job on the machine with the most available CPU resources.

User-Defined Load Balancing

Instead of using the load balancing methods that AutoSys Workload Automation provides, you can write your own programs or batch files to select which machine to use at run time. If you specify the name of a program or batch file as the value of the machine attribute in the job definition, the scheduler runs the batch file at job run time, and substitutes its output for the machine name.

If the machine returned by the script is offline, the product puts the job in PEND_MACH status for that machine. When the missing machine returns to service, the pending job runs on it regardless of whether the script would return a different machine name then. Because a machine must be defined for the scheduler to run a job on it, you must have previously defined the machine returned by the script to AutoSys Workload Automation.

Example: User-Defined Load Balancing

This example shows how you would specify a user-defined program or batch file in place of a real or virtual machine for processing a job.

For example, you might supply the following code:

```
insert_job: run_free
machine: '/usr/local/bin/pick_free_mach'
command: $HOME/DEL_STUFF
```

At run time, the script /usr/local/bin/pick_free_mach runs on the scheduler machine. The standard output is substituted for the name of the machine, and the job runs on that machine.

WARNING

The escape character in the machine value above is the back-tic character (`), not an apostrophe ('). Escape a program or batch file that is used as the machine attribute value with back-tic characters as shown for the scheduler to recognize that the machine value specifies a script. The apostrophe and quotation mark characters do not work in this case.

Manage Your Jobs

All activity controlled by AutoSys Workload Automation is based on jobs. A *job* is any single command or executable, UNIX shell script, or Windows batch file. Other objects, such as monitors, reports, and the Job Status Console, track job progress. A job is the foundation for the entire operations cycle.

You define jobs to AutoSys Workload Automation by creating job definitions. Each job definition contains attributes that specify the job's properties and behavior. For example, you can specify conditions that determine when and where a job runs.

You can define jobs using the following methods:

- **JIL**
A scripting language that lets you define and modify assets such as jobs, global variables, machines, user-defined job types, external instances, and blobs.
- **Web UI**
A tool that lets you interactively set the attributes that describe when, where, and how a job should run. The fields in the Web UI correspond to the JIL subcommands and attributes. In addition, from the Web UI, you can define calendars, global variables, and monitor and manage jobs. Web UI is supplied with AutoSys Workload Automation.

Both methods set the same attributes and the job definition is always stored in the database. You can also update and delete existing jobs.

NOTE

- The scheduler must be running before you start any processes, so you should start it before performing the tasks described in this chapter.
- For information about using the Web UI to define jobs, see the Web UI documentation.
- Before you update or delete an existing job, ensure that the job is not running. To help avoid losing job definitions in the event of a system failure, we recommend that you back up your job definitions periodically.

This section contains the following topics:

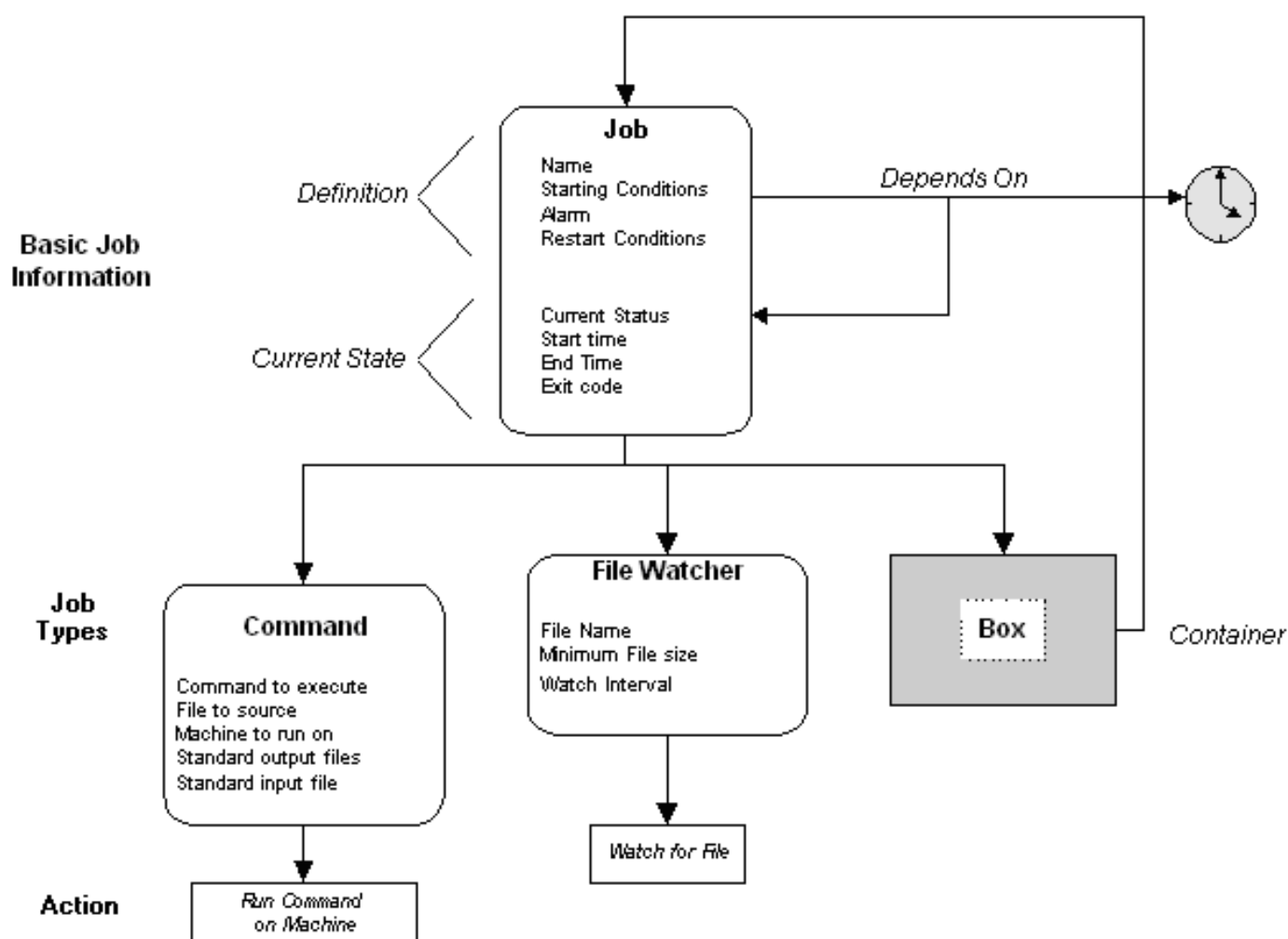
Job Types

Job types define the type of work to be scheduled. For example, you can create a CMD job to run a Windows command, an FTP job to download a file from a server, or an SAPEM job to monitor for the triggering of an SAP event. You can also define box jobs, which are containers that hold other jobs or box jobs. You can define your own job type.

When you create a job definition, you must specify the job type. If you do not specify a job type in the definition, the job type is set to CMD by default. Each job type has required and optional attributes that define the job. The job types have many common attributes and AutoSys Workload Automation treats them all similarly. The primary differences between them are the actions taken when the jobs run.

The structure of a job depends on the job type. For example, the following illustration shows the structure of a Command, File Watcher, and Box job:

Figure 28: Job Types Overview



Common Optional Attributes

All valid job definitions specify the job name and the machine that the job runs as follows:

```
insert_job: job_name
```

```
machine: machine_name
```

Optionally, a job definition may include statements that use any of the following attributes:

- [alarm_if_fail](#)
- [application](#)

- [auto_delete](#)
- [auto_hold](#)
- [avg_runtime](#)
- [box_name](#)
- [box_terminator](#)
- [condition](#)
- [date_conditions](#)
- [days_of_week](#)
- [description](#)
- [exclude_calendar](#)
- [group](#)
- [job_class](#)

NOTE

This attribute is not valid for z/OS job types.

- [job_load](#)
- [job_type](#)
- [max_run_alarm](#)
- [min_run_alarm](#)
- [must_complete_times](#)
- [must_start_times](#)
- [n_retrys](#)
- [notification_msg](#)
- [owner](#)

NOTE

This attribute is not valid for file trigger jobs.

- [permission](#)
- [priority](#)
- [resources](#)
- [run_calendar](#)
- [run_window](#)
- [send_notification](#)
- [service_desk](#)
- [start_mins](#)
- [start_times](#)
- [svcdesk_attr](#)
- [svcdesk_desc](#)

- [svcdesk_imp](#)
- [svcdesk_pri](#)
- [svcdesk_sev](#)
- [term_run_time](#)
- [timezone](#)

Job States

AutoSys Workload Automation tracks the current state, or status, of every job. AutoSys Workload Automation displays job status in the job report that the autorep command generates and in the Web UI.

The scheduler records job run information, including job status. The ujo_job_status table stores job status for the current run, and the ujo_job_runs table stores job status for previous runs. These tables use numeric codes to represent job status. The numeric codes are resolved using the ujo_intcodes table. A job can have one of the following statuses:

ACTIVATED (9)

Indicates that job is contained in a box job with a status of RUNNING but that the job itself is waiting to start.

FAILURE (5)

Indicates that a job fails to complete successfully. The scheduler issues this alarm when the alarm_if_fail attribute in a job definition is set to Y and that job fails.

INACTIVE (8)

Indicates that a newly created job has not yet run for the first time.

PEND_MACH (14)

Indicates that the offline status of the machine to which a job is assigned prevents the job from starting. These jobs can otherwise logically start, so the scheduler attempts to start them when the offline machine returns to service.

NOTE

You can use the GlobalPendMach configuration options to control how jobs in PEND_MACH status start and what happens to new jobs scheduled to currently offline machines. For more information about the GlobalPendMach configuration options, see [Configure a Scheduler](#).

ON_HOLD (11)

Indicates that the job is on hold and cannot run until you take it off hold.

NOTE

- You can place a job in this status only by sending the JOB_ON_HOLD event.
- To take a job off hold, send the JOB_OFF_HOLD event.

ON_ICE (7)

Indicates that the job is removed from the job stream but is still defined.

NOTE

- You can place a job in this status only by sending the JOB_ON_ICE event.
- You cannot manually change the status of a job from ON_ICE to INACTIVE. To return a job that is on ice to the job stream and resume running it, send the JOB_OFF_ICE event.

ON_NOEXEC (16)

Indicates that the scheduler bypasses execution of the job.

NOTE

- You can place a job in this status only by sending the JOB_ON_NOEXEC event.
- When you instruct the scheduler to bypass the execution of a job, the scheduler places the job in the ON_NOEXEC status and the effect is the same as sending the CHANGE_STATUS event to INACTIVE for the job. The scheduler does not immediately schedule downstream jobs that have dependency on the NOEXEC job nor does it evaluate their success conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the INACTIVE status.
- When you instruct the scheduler to bypass the execution of a box job, the scheduler automatically places all jobs in the box job (including all jobs that are contained in lower-level boxes within the box job) in ON_NOEXEC status. The effect is the same sending a CHANGE_STATUS event to INACTIVE for a job in a box.
- When you instruct the scheduler to change the status of an ACTIVATED job that is within a running box. The job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job are met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.
- When the starting conditions of a NOEXEC job are met, the scheduler bypasses the execution of the job and places the job in the ON_NOEXEC status. The effect is the same as sending a CHANGE_STATUS event to SUCCESS with an exit code of 0. The scheduler schedules downstream jobs that have a dependency on the NOEXEC job and evaluates their success conditions to success.
- To resume executing bypassed jobs, send the JOB_OFF_NOEXEC event.

QUE_WAIT (12)

Indicates that a lack of available load units on the machines to which a job is assigned prevents the job from starting. It also indicates that the load balancing job is blocked behind one or more higher priority jobs that are queued in the QUE_WAIT or RESWAIT state. When all higher priority queued jobs exit their queued state and the required load units become available, the job starts.

RESTART (10)

Indicates that the scheduler is attempting to restart a job that failed to start as scheduled. The scheduler attempts to restart the job at periodic intervals until the job starts or the maximum number of restart attempts is exceeded.

NOTE

You can set the maximum number of restart attempts to zero (0). If the scheduler reaches the maximum number of restart attempts without successfully starting the job, the job fails. The maximum number of restart attempts depends on the following global settings:

- Max Restart Trys (on the Windows Administrator) or MaxRestartTrys (in the UNIX configuration file)
- Max Restart Wait (on the Windows Administrator) or MaxRestartWait (in the UNIX configuration file)

RESWAIT (15)

Indicates that a lack of available resources to which a job is assigned prevents the job from starting. It also indicates that the resource job is blocked behind one or more higher priority jobs that are queued in the QUE_WAIT or RESWAIT state. When all higher priority queued jobs exit their queued state and the resource is available, the job starts.

RUNNING (1)

Indicates *one* of the following situations:

- The agent is executing the job.
- The scheduler has instructed the agent to start jobs that are contained in the box job.

NOTE

A box job acts as a container for other jobs but performs no action itself. The agent does not execute box jobs. The scheduler places box jobs in RUNNING status to send a message to the agent that it can start jobs that are contained in the box job. These jobs start when they meet the starting conditions that are specified in the job definitions.

STARTING (3)

Indicates that the scheduler initiated the start job procedure with the agent. This status does not apply to box jobs.

SUCCESS (4)

Indicates that the job exits with a code equal to or less than the maximum exit code for success that is specified in the job definition. A box job enters this status in the following situations:

- All jobs that are contained in the box job succeed.
- The exit condition for the box job evaluates to TRUE.

SUSPENDED (17)

Indicates that the Hadoop Oozie workflow is suspended.

TERMINATED (6)

Indicates that the job ends while it is still in the RUNNING state. The scheduler issues an alarm when a job is terminated. A job is terminated when *one* of the following situations occurs:

- You send a KILLJOB event.
- You issue the kill command (UNIX).
- The job definition specifies that the scheduler should terminate the job under specific circumstances and those circumstances occur. For example, you can specify that the scheduler should terminate a job when the box job containing it fails.
- The job does not complete within the maximum run time that is specified in the job definition.

WAIT_REPLY (13)

Indicates that the job cannot continue running without manual intervention. The scheduler issues an alarm when a job requires a user response to continue running.

NOTE

The job continues running only when you reply to this alarm by sending the REPLY_RESPONSE event.

The job state reflects the most recent event processed. A job enters one of the completed states (such as SUCCESS) when all of the events that are associated with that job are processed, and the job remains in that state until the job starts

again. Sometimes displays a status that does not reflect reality; for example, the system displays completed jobs as in the **RUNNING** state when the scheduler is still processing events that are associated with that job. You can use the `autorep` command to view all the events (including unprocessed events) that are associated with a job.

NOTE

The scheduler determines whether to start jobs that are based on a few factors. These factors depend on the job type and the state of the job:

- The scheduler re-evaluates starting conditions for jobs that are in the **ON_HOLD** state when you issue the **JOB_OFF_HOLD** event using the `sendevent` command. The scheduler also re-evaluates starting conditions for jobs that are leaving one of the following queued states:
 - **QUE_WAIT**
 - **RESWAIT**
 - **PEND_MACH**

The scheduler does not re-evaluate date and time conditions for these jobs if they meet those conditions while they are on hold or queued. Jobs that meet date and time conditions while they are on hold or queued start *unless* they do not meet other starting conditions. Jobs that did not meet date and time conditions while they were on hold or queued and still do not meet those conditions do not start. You can configure AutoSys Workload Automation to skip starting condition evaluation for queued jobs. In this case, the jobs start immediately upon leaving the queue, even if their starting conditions are no longer satisfied.

- The scheduler does *not* start jobs that are in the **ON_ICE** state when you issue the **JOB_OFF_ICE** event using the `sendevent` command. These jobs start the next time that their starting conditions recur unless they are contained in boxes that are currently running. Jobs that are contained in running boxes start the next time their starting conditions recur and a new run of their containing box begins.
- The scheduler starts box jobs that are in the **ON_NOEXEC** state but does not start jobs of other types that are in this state. The scheduler places these box jobs in **RUNNING** status but individual **ON_NOEXEC** jobs that are contained in the box job remain in the **ON_NOEXEC** state. The scheduler automatically generates one **BYPASS** event against jobs that are in the **ON_NOEXEC** state. The **BYPASS** event is sent in place of the completion event for box jobs and in place of the **RUNNING**, **STARTING** and completion events for individual jobs. After the scheduler places an **ON_NOEXEC** box job in **RUNNING** status, the scheduler waits for all jobs in the box job to bypass execution before returning the box job to the **ON_NOEXEC** state.
- Box jobs in **ON_NOEXEC** status start and all of the jobs that are contained in the box job enter the **ACTIVATED** state. The scheduler immediately bypasses and marks as complete the jobs that are contained in the box job unless other conditions apply. Jobs that are not marked complete by the time the box job completes enter or remain in the **ON_NOEXEC** state.
- All jobs that are contained in a box job enter the **ACTIVATED** when the box job starts. Jobs run immediately unless other conditions apply. Jobs that are contained in a box job that do not complete by the time that the box job completes enter **INACTIVE** status. These jobs do not retain their statuses from previous box job processing cycles when a new box job cycle begins.

Define Jobs

Job definitions specify the work that individual jobs do.

When you define jobs, you can optionally schedule those jobs using the date and time condition attributes. Scheduling jobs to run on customized schedules requires defining custom calendars that you can reference in job definitions. Jobs that do not have date and time conditions that are specified in their definitions only run when you start them manually.

To improve workload performance for certain types of jobs, you can define them to run on a cluster.

Insert a Job Definition

Follow these steps:

1. Do *one* of the following steps:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.
2. Specify the following definition:

```
insert_job: job_name
```

```
machine: machine_name
```

```
job_type: type
```

```
required_attribute: value
```

```
[attribute: value...]
```

- **job_name** Defines a unique name for the job.
 - **machine_name** Specifies the name of the machine on which the job runs.
 - **type** Specifies the type of job you are defining.
 - **required_attribute: value** Specifies the name of a required JIL attribute and the corresponding value. The attributes that are required in a job definition depend on the type of job that you are defining.
 - **optional_attribute: value** (Optional) Specifies that name of an optional JIL attribute and the corresponding value. The optional attributes that you can specify in a job definition depend on the type of job that you are defining.
 - **value** Defines the value of the corresponding attribute.
3. Do *one* of the following:
 - Enter **exit** if you are using interactive mode.
 - Redirect the script to the jil command if you are using a script.
 The data is loaded into the database and the job is defined.

NOTE

You can also define a job using Web UI. For more information about using Web UI, see [Web UI Scheduling](#).

Example: Define a Command Job

This example runs the `/bin/touch` command on the file named `/tmp/test_run.out`. The job runs on the UNIX client computer named `unixagent`.

```
insert_job: test_run
```

```
job_type: CMD /* This attribute is optional for Command jobs. CMD is the default. */
```

```
machine: unixagent
```

```
command: /bin/touch /tmp/test_run.out
```

Update a Job Definition

To make minor adjustments to a job definition, such as scheduling changes, update the existing definition.

WARNING

The `update_job` subcommand succeeds only when the job is not in the `STARTING` state.

To update a job in a box when AutoSys Workload Automation is running in native security mode, you must have edit permissions on the job with the execute permissions on the box that contains the job. By default the `EDIT` and `EXEC` Superusers have full permissions on all the jobs regardless of the job owner.

Follow these steps:

1. Take one of the following actions steps:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.
2. Specify the following definition:

```
update_job: job_name
```

```
attribute: value
```

```
[attribute: value...]
```

- **job_name**
Specifies the name of the job you want to update.
 - **attribute**
Specifies the name of a JIL attribute that applies to the job type that you are updating. You can specify one or more attributes.
 - **value**
Defines the value of the corresponding attribute.
3. Take one of the following actions:
 - Enter **exit** if you are using interactive mode.
 - Redirect the script to the `jil` command if you are using a script.
 The data is loaded into the database and the job is updated.

NOTE

- When you update a job, you cannot update the resources attribute in the existing job definition if the job has a resource dependency and has held the resource.
- You can also update a job using Web UI. For more information about using Web UI, see [Web UI Scheduling](#).

Improve Workload Performance

Defining jobs to run on a cluster helps improve workload performance. We recommend this method for the following types of jobs:

- Jobs that use shared resources
- Jobs with high CPU consumption
- Command jobs that execute client utilities

You can define a job to run on a cluster when you are running AutoSys Workload Automation in a highly available cluster environment that meets *all* of the following conditions:

- The agent is clustered.
- A machine is defined to represent the cluster.

NOTE

You can schedule command jobs that execute client utilities to run on a cluster only when the client is also clustered.

A machine is defined to represent a cluster when the machine definition meets the following requirements:

- The machine representing the cluster is a real machine (not a virtual machine or a machine pool).
- The node name that is specified for the machine is the same as the host name of the cluster that the machine represents.
- The agent name that is specified for the machine is the same as the common agent name for the agents that are installed on the cluster that the machine represents.
- The agent port that is specified for the machine is the same as the common agent port for the agents that are installed on the cluster that the machine represents.
- The encryption type and key that are specified for the machine are the same as the encryption type and key that are specified for the agents that are installed on the cluster that the machine represents.

The cluster manager performs load balancing for jobs that use shared resources or have high CPU consumption when you define those jobs to run on the cluster. Command jobs that execute client utilities are more likely to succeed when you define them to run on a cluster with multiple agents and clients that are installed on it.

NOTE

For more information about setting up a clustered agent and a clustered client, see [\(UNIX\) How to Improve Workload Performance: Highly Available Environments](#) or [\(Windows\) How to Improve Workload Performance: Highly Available Environments](#).

Delete a Job

When you no longer need a job definition, you can delete it from the database.

Follow these steps:

1. Do *one* of the following steps:
 - Issue JIL in interactive mode.
 - Open a JIL script in a text editor.
2. Specify the following subcommand:

```
delete_job: job_name
```

– **job_name**

Specifies the name of the job you want to delete.

3. Do *one* of the following steps:
 - Enter **exit** if you are using interactive mode.
 - Redirect the script to the jil command if you are using a script.

The delete request is issued. When JIL is in job verification mode (the default), the `delete_job` subcommand scans the `ujo_job_cond` table and notifies you of any dependent conditions for the deleted job before deleting it.

NOTE

You can also delete a job using Web UI. For more information about using Web UI, see [Web UI Scheduling](#).

Example: Delete a Job

This example deletes the `test_run` job.

```
delete_job: test_run
```

Run a Job After Using JIL

After you submit a job definition to the database, it runs according to the starting parameters specified in its JIL script. That is, the scheduler continually polls the database, and when it verifies that the starting parameters are met it runs the job.

If a JIL script does not specify any starting parameters for a job, the scheduler does not start the job automatically; the job starts only if you issue the `sendevent` command.

Example: Run a Job with the sendevent Command

This example assumes that `test_install` job has no starting parameters that are specified in its JIL script. The only way to start it is to issue the following command:

```
sendevent -E STARTJOB -J test_install
```

This command tells the scheduler to start the `test_install` job.

Specify the Job Owner

By default, the operating system user who invokes `jil` to define a job is the owner of that job. You can change the owner a job by specifying a different user ID in the job definition.

To change the owner of a job, specify the user ID of a AutoSys Workload Automation user in the owner attribute job definition. AutoSys Workload Automation accesses the associated password from the database.

WARNING

Jobs that run on other software, such as PeopleSoft and databases, can run only when the owner that is specified in the job definition is defined as a user on AutoSys Workload Automation and on the authenticated software. For example, to run an SAP job, specify an SAP user that is also defined as a AutoSys Workload Automation user.

You can use the AutoSys Security Utility to define users on AutoSys Workload Automation.

To specify the job owner, add the owner attribute to your job definition.

NOTE

- The owner attribute does not apply to File Trigger jobs.
- If AutoSys Workload Automation is running in native security mode, you can change the owner value only if you have EDIT superuser permissions.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you can specify the owner value only if as-owner security policies grant you access to the owner value. You can update the job to specify a different owner value only if as-owner security policies grant you access to both the existing and new owner values.
- AutoSys Workload Automation uses the owner value for all job types except for File Trigger. AutoSys Workload Automation does *not* use the `oscomponent.default.user` parameter that is located in the `agentparm.txt` file of the agent.
- Other application-specific uses of the owner attribute apply to certain job types. For more information about these job types specific uses, see the documentation on individual job types.

Example: Specify a Job Owner

Suppose that AutoSys Workload Automation is running in external security mode and as-owner security policies grant you access to the “prod@unixagent” value as defined in CA EEM. You can specify prod@unixagent as the value of the owner attribute for the new job definition. The following job runs under the prod user on the unixagent computer:

```
insert_job: jobA

job_type: CMD
machine: unixagent
command: /bin/touch /tmp/test_run.out
owner: prod@unixagent
```

Example: Specify a Different Job Owner

Suppose that AutoSys Workload Automation is running in external security mode and as-owner security policies grant you access to both the “prod@unixagent” and “newprod@unixagent” values as defined in CA EEM. You can change the value of the owner attribute for the existing job definition from “prod@unixagent” to “newprod@unixagent”. The following job runs under the newprod user on the unixagent computer:

```
update_job: jobA
owner: newprod@unixagent
```

Global Variables

Define global variables that you can use as a job dependency. The job dependency is satisfied only when the value of the expression evaluates to TRUE.

You can define global variables using the `sendevent` command. After you define a global variable to AutoSys Workload Automation, you can use the variable as a job dependency. The job dependency is satisfied only when the value of the expression evaluates to TRUE. You can reference a global variable as part of the syntax of any of the following attributes:

- [command](#)
- [condition](#)
- [connection_factory](#)
- [connect_string](#)
- [destination_file](#)
- [destination_name](#)
- [ftp_local_name](#)
- [ftp_remote_name](#)
- [ftp_server_name](#)
- [hadoop_hdfs_destination](#)
- [hadoop_hdfs_path](#)
- [hadoop_hdfs_source](#)
- [i5_library_list](#)
- [i5_name](#)
- [i5_params](#)
- [informatica_folder_name](#)
- [informatica_param_file](#)
- [informatica_repository_name](#)
- [informatica_security_domain](#)
- [informatica_target_name](#)
- [informatica_task_name](#)
- [informatica_workflow_inst_name](#)
- [informatica_workflow_name](#)
- [initial_context_factory](#)
- [j2ee_parameter](#)
- [monitor_cond](#)
- [provider_url](#) (EJB and JMS Jobs)
- [provider_url](#) (HTTP Jobs)
- [sap_recipients](#) (only recipient parameter)
- [sap_step_parms](#) (only variant and step_user parameters)
- [sap_target_sys](#)
- [scp_local_name](#)
- [scp_remote_dir](#)
- [scp_remote_name](#)
- [sp_arg](#) (only name and value parameters)
- [sp_name](#)
- [sql_command](#)
- [sqlagent_jobname](#)
- [sqlagent_server_name](#)
- [sqlagent_step_name](#)
- [sqlagent_target_db](#)
- [std_err_file](#)
- [std_in_file](#)
- [std_out_file](#)
- [success_criteria](#)
- [tablename](#)
- [text_file_name](#)
- [trigger_cond](#)
- [watch_file](#)
- [ws_parameter](#)

NOTE

- If the length of the attribute value exceeds the limits after the global variable expansion, the job goes into a RESTART state. The job restarts based on the value specified in the MaxRestartTrys (on UNIX) or Max Restart Trys (on Windows) parameter.
- For the std_in_file, std_out_file, and std_err_file attributes, the length of the attribute value after the global variable expansion can exceed the limits by four characters. For the command attribute, the length of the attribute value after the global variable expansion can be up to 1024 characters.

Alerts Based on Conditions

This article explains how alerts are written based on monitored conditions. You can define the following job types to monitor a condition continuously:

- CPU Monitoring (OMCPU)
- Database Monitor (DBMON)
- Database Trigger (DBTRIG)
- Disk Monitoring (OMD)
- File Trigger (FT)
- Text File Reading and Monitoring (OMTF)
- Windows Event Log Monitoring (OMEL)
- Windows Services monitoring (OMS)

Each time the specified condition occurs, an ALERT event is written to the scheduler log file (event_demon.\$AUTOSERV on UNIX and event_demon.%AUTOSERV% on Windows). An alert helps you track and report each time that a monitored condition occurs.

NOTE

An alert is only generated for triggers that occur during continuous monitoring. Alerts are not generated for non-continuous monitoring (NOW and WAIT).

To stop a continuous monitor, complete the job manually by issuing the following command:

```
sendevent - E KILLJOB - J job_name
```

For non-continuous monitors the proper event order is reflected as:

1. STARTING
2. RUNNING
3. SUCCESS

For a continuous monitor the event order is reflected as:

1. STARTING
2. RUNNING
3. ALERT
4. ALERT
5. ALERT...
6. KILLJOB
7. TERMINATED
8. JOBFailure (ALARM)

You can view the text of alerts using the following methods:

- View the scheduler log file
- Issue the following command:

```
autorep - J job_name - d
```

- Web UI

NOTE

You cannot manually send the ALERT event using the sendevent command.

Example: Trigger Alerts When Monitoring CPU Usage Continuously

This example continuously monitors used CPU on the unixagent computer. When the job runs, it goes into a RUNNING status. When the job detects that the used CPU is within 70 and 100 percent, an ALERT event is raised (an alert is written to the scheduler log file). The available, used CPU and load averages are reported as part of the ALERT event and the status message is reported with the RUNNING event. Then, each time the job detects that the use CPU meets the monitored condition, an alert is triggered. The job only ends when it is complete manually.

```
insert_job: cpu_monitoring_used
```

```
job_type: OMCPU
```

```
machine: unixagent
```

```
lower_boundary: 70
```

```
cpu_usage: USED
```

```
inside_range: TRUE
```

```
monitor_mode: CONTINUOUS
```

In contrast, the following example monitors used CPU in WAIT monitor mode. When the job runs, it goes into a RUNNING status. When the job detects that the used CPU is within 70 and 100 percent, the job completes. No alert is triggered. The available, used CPU and load averages are reported on the SUCCESS event.

```
insert_job: cpu_monitoring_used_wait
```

```
job_type: OMCPU
```

```
machine: unixagent
```

```
lower_boundary: 70
```

```
cpu_usage: USED
```

```
inside_range: TRUE
```

```
monitor_mode: WAIT
```

Start Conditions

AutoSys Workload Automation verifies whether it starts a job by evaluating the starting conditions that are defined for the job. All defined starting conditions must be *true* for a job to start.

The product starts all jobs that meet their starting conditions, unless *one* of the following conditions apply:

You place the job on hold or on ice. If you put a job on hold or on ice, the job does not start automatically until you take it off hold or off ice.

NOTE

- To put a job on hold or on ice, issue the JOB_ON_HOLD or JOB_ON_ICE event using the sendevent command. To take off a job hold or off ice, issue the JOB_OFF_HOLD or JOB_OFF_ICE event.
- You can also instruct the scheduler to bypass execution of a job by issuing the JOB_ON_NOEXEC event.
- Jobs in the ON_NOEXEC state start immediately when they meet their starting conditions. The scheduler simulates running these jobs, but the agent does not execute commands that are associated with the jobs. Jobs in this state immediately return an evaluation of successfully completed on the Scheduler machine. When you issue the JOB_OFF_NOEXEC event against a job in the ON_NOEXEC state, the agent resumes executing commands that are associated with that job the next time the job runs.
- External conditions prevent the job from running. Depending on the reason, AutoSys Workload Automation places the job in *one* of the following queued states:
 - When an offline machine prevents the job from running, the job enters PEND_MACH status.
 - When held resources prevent the job from running, the job enters RESWAIT status.
 - When unavailable load units prevent the job from running, the job enters QUE_WAIT status.

When a job leaves a queued state, the scheduler determines whether to start the job by re-evaluating starting conditions for that job unless you configure AutoSys Workload Automation to skip starting condition evaluation for queued jobs. If a job that is contained in a box fails its starting condition checks when leaving the queue, the scheduler places that job in the ACTIVATED state. If a job that is not contained in a box fails its starting condition checks when leaving the queue, the scheduler places that job in the INACTIVE state. If the job meets its starting conditions, or if you configure the system to skip starting condition evaluation for queued jobs, the job starts.

When you take a job off hold, the scheduler re-evaluates starting conditions for that job. When you take a job off ice, the scheduler does not restart the job until its starting conditions recur, even if those conditions were met while the job was on ice. If a job is contained in a running box when it is taken off ice, the scheduler does not restart the job until the following run of the box, even if its starting conditions recur during the existing run of the box.

When you instruct the scheduler to bypass execution of a job, the scheduler starts the job when it meets its starting conditions. The scheduler simulates running these jobs, but the agent does not execute commands associated with the jobs. Bypassed jobs evaluate as successfully completed on the scheduler machine. When you issue the JOB_OFF_NOEXEC event, the agent executes commands associated with the specified job the next time the job starts. Changing the executable status of a job does not affect evaluation of starting conditions.

NOTE

- You can configure AutoSys Workload Automation to perform one of the following actions when scheduling jobs leaving a queued state:
 - Skip starting condition evaluation.
 - Evaluate the starting conditions of queued jobs other than the date condition check for the day of evaluation.

- Evaluate the starting conditions of queued jobs including the date condition check for the day of evaluation.
- If you configure AutoSys Workload Automation to skip starting condition evaluation for queued jobs, those jobs start immediately upon leaving a queued state.
- If you configure AutoSys Workload Automation to evaluate the starting conditions of queued jobs other than the date condition check for the day of evaluation, the scheduler does not re-evaluate date and time conditions. Jobs that meet their original date and time conditions before entering a queued state start immediately after they leave the queued state unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- If you configure AutoSys Workload Automation to evaluate the starting conditions of queued jobs including the date condition check for the day of evaluation, the scheduler re-evaluates date conditions but not time conditions. Jobs that meet their date conditions after leaving a queued state start unless other starting conditions apply and are not satisfied. Jobs that leave the queued state on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- When you take jobs off hold, the scheduler does not re-evaluate date and time conditions. Jobs that meet their date and time conditions while they are on hold start immediately after they are taken off hold unless other starting conditions apply and are not satisfied. Jobs that are taken off hold on a day that is defined in an exclusion calendar or at a time outside their run window do not start and are re-scheduled to their next start time.
- For more information about configuring AutoSys Workload Automation to schedule jobs leaving a queued state, see [Configure a Scheduler](#).

Evaluate Start Conditions for Downstream Dependent Jobs

When you put a job on ice or you instruct the scheduler to bypass execution of the job, it affects starting conditions of downstream dependent jobs. How the product evaluates the downstream dependent jobs depends on the condition of the job that you bypass or put on ice.

The following conditions apply when you send the JOB_ON_NOEXEC event to a job:

- When you send the JOB_ON_NOEXEC event and the job is in the INACTIVE, SUCCESS, or ACTIVATED status, the job retains its current status; otherwise the effect is the same as if the job enters the INACTIVE status. The scheduler does not immediately schedule jobs that have a dependency on the NOEXEC job. However, the scheduler evaluates the success conditions of downstream dependent jobs to true if the predecessor job retains its SUCCESS status.
- When the starting conditions of the job are met and the scheduler sends the BYPASS event, the effect is the same as if the job enters the SUCCESS status with an exit code of 0. The scheduler schedules jobs that have a dependency on the NOEXEC job and evaluates their success conditions to success.

Suppose that you put JobA in ON_NOEXEC status. When AutoSys Workload Automation bypasses JobA, downstream jobs dependent upon JobA are evaluated when starting conditions of JobA are met, based on the condition of JobA, as follows:

| Condition | Evaluates to |
|-------------------|---|
| success (JobA) | TRUE, <ul style="list-style-type: none"> • When the job retains its current SUCCESS status • When the starting conditions of jobA are met and the job execution is bypassed FALSE, when the starting conditions of jobA are not met |
| failure (JobA) | FALSE |
| terminated (JobA) | FALSE |

| | |
|-------------------|--|
| done (JobA) | TRUE, <ul style="list-style-type: none"> When the job retains its current SUCCESS status When the starting conditions of jobA are met and the job execution is bypassed FALSE, when the starting conditions of jobA are not met |
| notrunning (JobA) | TRUE, <ul style="list-style-type: none"> If jobA is not a box If jobA is a box and either the starting conditions for the box are not met or the execution for all jobs in the box are bypassed FALSE, when jobA is a box, starting conditions for the box are met, and starting conditions for all the jobs in the box are not met. |
| exitcode | TRUE, when the starting conditions of jobA are met, job execution is bypassed, and the expression of jobA evaluates to true with an exit code of 0
FALSE, <ul style="list-style-type: none"> When the starting conditions of jobA are met, job execution is bypassed, and the expression of jobA evaluates to false with an exit code of 0 When the starting conditions of jobA are not met |

Suppose that you put JobA in ON_ICE status. The downstream jobs dependent upon JobA are immediately evaluated based on the condition of JobA, as follows:

| Condition | Evaluates to |
|-------------------|--------------|
| success (JobA) | TRUE |
| failure (JobA) | FALSE |
| terminated (JobA) | FALSE |
| done (JobA) | TRUE |
| notrunning (JobA) | TRUE |
| exitcode | FALSE |

Date and Time Dependencies

You can use JIL statements to schedule AutoSys Workload Automation jobs to start at a specific date and time. AutoSys Workload Automation then calculates a matrix of specified day, date, and time values and starts jobs accordingly. A time range cannot span more than 24 hours.

You can specify days of the week or actual dates, but you cannot specify both. For example, you can define a job to start on Monday, Wednesday, and Friday at 8:00 a.m. *or* on the 15th and the 30th of the month at 8:00 a.m. but *not* on Monday the 15th at 8:00 a.m.

You can specify days of the week using JIL, but you can only specify actual dates using custom calendars. You can also specify a time zone to apply to your starting times, and you can define a job to start at one specific time of day or hourly, denoted in minutes past the hour.

TZ Environment Variable

Valid on UNIX

By default, jobs with time-based starting conditions that do not specify a time zone have their start event scheduled based on the time zone under which the database runs.

Before you start the scheduler or application server, ensure that the TZ environment variable is set. The scheduler or application server references this setting to determine the default time zone. After you upgrade your database, you must start the scheduler to insert a time zone offset value (calculated from the value of the TZ environment variable) into the database. Do this before executing jil or autorep.

WARNING

Ensure that the database is running in the same time zone that the scheduler starts up with.

Job Dependencies Based on Job Status

You can define start conditions to start jobs that are based on the status of one or more jobs that exist in the database. In this way, you can program simple or complex prerequisites for starting a job.

For example, you can implement a single-threaded, batch queue-like set of job dependencies so that JobB starts when JobA achieves a SUCCESS status and JobC starts when JobB achieves a SUCCESS status.

You can configure more complex conditions by combining a series of conditions with the AND and OR logical operators. You can use the pipe symbol (|) instead of the word OR and the ampersand symbol (&) instead of the word AND. Spaces between conditions and delimiters are optional. You can specify even more complex conditions by grouping the expressions in parentheses, which force precedence. The equation is evaluated from left to right.

For example, in the following set of starting conditions, either both A and B must be successful or both D and E must be successful for the statement to evaluate as TRUE:

```
(success(JobA) and success(JobB)) or (success(JobD) AND success(Job E))
```

NOTE

If you specify a condition for an undefined job, the condition evaluates as FALSE, and any jobs dependent on this condition do not run. You can use the job_depends command to check for this type of invalid condition statement.

The syntax for defining job dependencies is the same whether the job is being defined using JIL or the Web UI, except that the JIL statement begins with the JIL condition keyword.

The syntax for conditions that are based on job status, is as follows:

```
status(job_name)
```

- **status**

Indicates *one* of the following statuses:

- **success**

Indicates that the status condition for *job_name* is SUCCESS, ON_ICE or that the scheduler bypasses the execution of a NOEXEC job. You can abbreviate this value to s.

- **failure**

Indicates that the status condition for *job_name* is FAILURE. You can abbreviate this value to f.

- **done**

Indicates that the status condition for *job_name* is SUCCESS, FAILURE, TERMINATED, ON_ICE or that the scheduler bypasses the execution of a NOEXEC job. You can abbreviate this value to d.

- **terminated**

Indicates that the status condition for *job_name* is TERMINATED. You can abbreviate this value to t.

- **notrunning**

Indicates that the status condition for *job_name* is anything except STARTING, RUNNING, WAIT_REPLY, RESTART, or SUSPENDED. You can abbreviate this value to n.

- **job_name**

Identifies the job on which the new job is dependent.

You can also abbreviate the dependency specification EXIT CODE to e and VALUE (of a global variable) to v.

You can use the max_exit_success (maximum exit code for success) attribute set for a job to control the value of the SUCCESS status. If you specify this attribute, any job that exits with an exit code less than or equal to the specified value is treated as a success. A FAILURE status means that the job exited with an exit code higher than this value. The default exit code for a normal job completion is 0. A TERMINATED status means that the job was killed.

NOTE

You can use either uppercase or lowercase letters to specify a status. However, you cannot use mixed case.

Example: Job Dependencies

For a job that runs only when the job named DB_BACKUP succeeds, you would specify the job dependency as follows:

```
success(DB_BACKUP)
```

If only JobC starts when JobA and JobB complete successfully or when JobD and JobE complete (regardless of whether JobD and JobE failed, succeeded, or terminated), specify the following dependency in the job definition for JobC:

```
(success(JobA) AND success(JobB)) OR (done(JobD) AND done(JobE))
```

As indicated in this example, you can use any job status as part of the starting condition specifications for a specific job. With this latitude, you can program branching paths that must be taken and must provide alternate actions for error conditions.

For example, if JobB fails after partially processing, you might call a routine that is called Backout that reverses the changes that were made. You would specify the following job dependency in the job definition for Backout:

```
failure(JobB)
```

You can use the notrunning operator to keep multiple jobs from running simultaneously. For example, assume that you do not want to run a database dump (DB_DUMP) and a file backup (BACKUP) simultaneously because such processing would adversely impact performance. However, you might have a smaller job that can run as long as both of these resource-intensive jobs are not running. You would specify the dependency of a smaller job as follows:

```
notrunning(DB_DUMP) AND notrunning(BACKUP)
```

Manage Job Status

Start conditions that are based on job status use the current (or most recent) completion status of the job. The job run defines the current completion status, regardless of when that run occurred.

However, to enforce the concept of time-based processing cycles, where the completion status of a job for some previous time period should not affect the processing of this time cycle, several options are available.

When a box job starts, the status of all the jobs in the box changes to ACTIVATED. Therefore, subsequent jobs in the box that depend on the completion of jobs that were performed earlier in the same box only use the completion statuses from this box run. Placing the jobs in one processing cycle inside a top-level box and setting the box to start at the beginning of the processing cycle prevents invalid information from affecting time-critical jobs.

When a job is first entered into the database, and before it runs for the first time, its status is set to INACTIVE. By changing the status of jobs that have completed but whose completion status should no longer be used in dependent job conditions to INACTIVE, the completion status from the last run is no longer the current status and it is not used.

Use the sendevent command to change a job status to INACTIVE. Alternatively, you could create a AutoSys Workload Automation job to accomplish this goal. If you change the status of a top-level box to INACTIVE, all the jobs in the box also change to INACTIVE.

Deleting and reinserting the job using JIL accomplishes the same thing. However, the past reporting history on the job is no longer available. Updating a job using JIL does not change the status of the job.

Job Dependencies Based on Exit Codes

You can use the following syntax to base job dependencies on exit codes that indicate completed tasks. In this way, you can implement even more specific branching logic for recovering from job failures.

This method of defining job dependencies has the following format:

```
exitcode (job_name) operator value
```

- **job_name**
Defines the name of the job upon which the new job depends.
- **operator**
Specifies one of the following exit code comparison operators:
 - **=**
Equal to.
 - **!=**
Not equal to.
 - **<**
Less than.
 - **>**
Greater than.
 - **<=**
Less than or equal to.
 - **>=**
Greater than or equal to.
- **value**
Defines the numeric exit code value on which to base the dependency.

For example, if a broken communication line results in JobA failing with an exit code of 4, and you want the system to run a script (JobB) that redials the line when this code is encountered, enter the following code for the job dependency specification for the JobB redial job:

```
exitcode (JobA) = 4
```

You can use any job status or exit codes as part of the specification for starting conditions. You can abbreviate the dependency specification exitcode with the letter e (uppercase or lowercase).

Exit Codes and Batch Files in Jobs Running on Windows

When you define jobs to run batch files on Windows, account for Windows-specific behavior. Windows programs return any exit values that are programmed in the executable code. This exit value is the last thing that is returned to Windows when the program terminates.

Generally, a zero (0) exit code indicates success, while a non-zero exit code indicates an error. The expected error values should be documented with each individual program, but some programs can return unexpected exit codes. Modify these programs so that they return expected values, and use these values when specifying exit code dependencies.

Jobs are created using standard Windows process creation techniques. After the job is created, the agent waits for the job to complete. When the job completes, AutoSys Workload Automation gets the program exit code from Windows and stores it in the database for later use.

When launching programs directly, the exit codes are returned and put in the database. However, there are some exit code behaviors that you must consider when using a job to start *.BAT batch files.

The exit code returned from a batch file is the return code from the last operation that is executed in that particular batch file. Consider the following example:

```
REM test batch file

test

if errorlevel 1 goto bad

goto good

:bad

del test.tmp

:good

exit
```

This sample batch file returns a 0 exit code when the test program exits with a 1 exit code as long as test.tmp exists. If test.tmp does not exist, the return code is from the del line and not from the line that runs the test. Therefore, this batch file returns a 0 (successful) exit code, even if test failed to execute as intended.

To help handle situations like this, AutoSys Workload Automation supplies a program that is named FALSE.EXE. This program resides in the Windows %AUTOSYS/bin directory and takes only one parameter, which is the exit code you want FALSE.EXE to return on completion. You can use FALSE.EXE as follows:

```
REM test batch file

test

if errorlevel 1 goto bad
```

```

exit

:bad

del test.tmp

false 1

```

When the test fails with error level 1, this batch file returns an exit code of 1 from FALSE.EXE, whether the test.tmp file exists or not.

Job Dependencies Based on Global Variables

You can base job dependencies on global variables set using the sendevent command. When using global variables in this way, the job dependency is satisfied only when the value of the expression evaluates to TRUE.

This method of defining job dependencies has the following format:

```
VALUE(global_name) operator value
```

- ***global_name***

Defines the name of the global variable upon which the job depends.

Limits: This value can be up to 64 characters in length. The following characters are valid: a-z, A-Z, 0-9, period (.), underscore (_), and hyphen (-). You can include spaces in a global variable name.

- ***operator***

Specifies one of the following exit code comparison operators:

- **=**
Equal to.
- **!=**
Not equal to.
- **<**
Less than.
- **>**
Greater than.
- **<=**
Less than or equal to.
- **>=**
Greater than or equal to.

- ***value***

Defines the numeric or text value of the global variable on which to base the dependency.

Limits: This value can be up to 100 characters in length and cannot contain quotation marks. The following characters are valid: a-z, A-Z, 0-9, period (.), underscore (_), and hyphen (-).

NOTE

When using JIL, use the condition attribute to enter the above expression in the appropriate JIL script.

For example, assume that a set of jobs in a box should only run with approval from manager. In this case, use the following syntax to set the global variable that is named manager-ok to OK, and make the top-level box job dependent on this global variable:

```
VALUE(manager-ok) = OK
```

You can abbreviate the dependency specification VALUE with the letter v (uppercase or lowercase).

Start Conditions and Boxes

When you put a job in a box, it inherits all the start conditions of the box. Therefore, all start conditions that are defined for the box must be met. Additionally, the box must enter the RUNNING state before the job can run. If there are no additional conditions on the job, it starts when the box starts. A job runs only once for each box execution.

By default, there is no sequential job processing in a box. For example, if three jobs are in a box, all three jobs start when the box starts if they have no additional conditions.

To implement a processing sequence for jobs in a box, specify more starting conditions for each job. For example, you could specify that Job1 has no starting conditions, Job2 depends on the completion of Job1, and Job3 depends on the completion of Job2.

NOTE

Jobs that depend on a job that is ON_ICE status run immediately as if the starting condition of the ON_ICE has instantly been satisfied. Jobs that depend on a job that is in ON_NOEXEC status only run when the starting condition of the ON_NOEXEC job has been satisfied. In this scenario:

- If Job2 enters the ON_ICE status, Job1 and Job3 start simultaneously when the box they are in starts running.
- If Job2 enters the ON_NOEXEC status, AutoSys Workload Automation bypasses Job2 and starts Job3 after Job1 completes.
- If Job2 is in the ON_ICE state, the box completes successfully after Job1 and Job3 complete.
- If Job2 is in ON_NOEXEC status, the box completes successfully after AutoSys Workload Automation bypasses Job2 and Job1 and Job3 complete.

Control Jobs in PEND_MACH Status

This article explains how to control jobs in the PEND_MACH status. The scheduler puts a machine in the offline state if it is unable to contact the agent to run a job. You can manually put a machine in the offline state by issuing the sendevent command to send a MACH_OFFLINE event. Jobs that are scheduled to start on offline machines are placed in PEND_MACH status by default.

When an offline machine returns to service, the scheduler immediately starts all jobs in PEND_MACH status on that machine. Starting too many jobs in PEND_MACH status on the machine at a time places a heavy demand for resources on both the scheduler and agent computers. This may introduce performance problems that affect all scheduled workload.

You can use the GlobalPendMachInterval, GlobalPendMachStatus, or GlobalPendMachDelay parameters (on UNIX) or the Global Pend Mach Interval, Global Pend Mach Status, or Global Pend Mach Delay fields (on Windows) to:

- Control the starting of jobs in PEND_MACH status when an offline machine returns to service.
- Control the status of jobs that are scheduled on a machine that is offline.

NOTE

- The values that are specified in the GlobalPendMachInterval, GlobalPendMachStatus, or GlobalPendMachDelay parameters (on UNIX) or the Global Pend Mach Interval, Global Pend Mach Status, or Global Pend Mach Delay fields (on Windows) to:

or Global Pend Mach Delay fields (on Windows) are applied at the global level and not at the machine level or job level. That is, these values apply to all machines that return to service and have jobs in PEND_MACH status that are scheduled to start on them.

- If the scheduler changes the status of a job from PEND_MACH to any other valid status because of the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows) setting or a manual CHANGE_STATUS event, the job is not run when the machine returns to service. Only jobs in PEND_MACH status are eligible to start on the machine that returns to service.

Control the Starting of Jobs in PEND_MACH Status

You can control the starting of jobs in PEND_MACH status in the following ways:

- By defining the time interval (in seconds) that the scheduler waits before starting jobs in PEND_MACH status when an offline machine returns to service.
- By defining the burst value. The burst value defines the number of jobs in PEND_MACH status that the scheduler starts after waiting for the specified interval.

The scheduler starts the specified number of jobs, waits for the specified interval, starts the specified number of jobs, waits for the specified interval, and so on. This process repeats until all jobs in PEND_MACH status for that machine are started.

On UNIX, you can configure this setting using the [GlobalPendMachInterval](#) parameter in the configuration file. On Windows, you can configure this setting using the Global Pend Mach Interval field on the Scheduler window of AutoSys Workload Automation Administrator (autosysadmin).

NOTE

- If you set the interval to 0 (the default value), the scheduler starts all jobs in PEND_MACH status with no delay between job starts and the burst value is ignored.
- The order in which the jobs are started depends on the job priority and amount of time the job has been in PEND_MACH status. For example, if JOB1, JOB2, and JOB3 have the same priority and they enter the PEND_MACH status at 08:00:00 a.m., 08:00:01a.m., and 08:00:02 a.m. respectively, the order in which they are started is JOB1, JOB2, and JOB3. Once the scheduler determines the starting order of jobs that are coming out of the PEND_MACH status, you cannot modify the starting order by using the sendevent command to send the CHANGE_PRIORITY event. If you send the CHANGE_PRIORITY event, the job priority changes do not apply to the run of the job exiting in PEND_MACH status. The job priority changes apply to the next run of the job.
- If jobs enter PEND_MACH status simultaneously and have the same priority, their starting order is not guaranteed. If the job priority is set to 0, it overrides the duration that the job has been in PEND_MACH status and starts immediately.

Control the Status of Jobs Scheduled on an Offline Machine

You can control the status of jobs that are scheduled on a machine that is offline in the following ways:

- By defining the completion status that the scheduler assigns to jobs that are scheduled on an offline machine. On UNIX, you can configure this setting using the [GlobalPendMachStatus](#) parameter in the configuration file. On Windows, you can configure this setting using the Global Pend Mach Status field on the Scheduler window of AutoSys Workload Automation Administrator (autosysadmin). The jobs temporarily remain in PEND_MACH status before the scheduler assigns the status that is specified in the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows).

NOTE

If the scheduler changes the status of a job from PEND_MACH to any other valid status because of the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows) setting or a manual CHANGE_STATUS event, the job is not run when the machine returns to service. Only jobs in PEND_MACH status are eligible to start on the machine that returns to service.

- By defining the time interval (in seconds) that the scheduler waits before updating the status of the job to the status specified in the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows). On UNIX, you can configure this setting using the [GlobalPendMachDelay](#) parameter in the configuration file. On Windows, you can configure this setting using the Global Pend Mach Delay field on the Scheduler window of AutoSys Workload Automation Administrator (autosysadmin).

NOTE

If you use the default value, the scheduler immediately sends a CHANGE_STATUS event to update the status of jobs in PEND_MACH status to the status specified in the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows). If you specify a value other than the default, jobs remain in PEND_MACH status until the delay interval expires, and are then assigned the status specified in the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows). If the machine returns to service within the delay interval, the scheduler does not set the status of the job to the status specified in the GlobalPendMachStatus parameter (on UNIX) or the Global Pend Mach Status field (on Windows). Since the job is in PEND_MACH status when the machine returns to service, the scheduler reschedules it to run on the online machine that is based on the interval that is specified in the GlobalPendMachInterval parameter (on UNIX) or the Global Pend Mach Interval field (on Windows).

Job Run Numbers and Names

AutoSys Workload Automation uses run numbers for jobs. The *run number* is a unique integer that is associated with every run of a job.

Consecutive run numbers are assigned every time a top-level job starts. A top-level job is a job that is not contained in a box, and every job inherits these run numbers in a box. This means that all jobs in a top-level box have the same run number as the number used for the run of the box. This design permits runs of nested jobs to be associated together in the same run.

If a job restarts, the run number remains the same and the ntrys field is incremented. In the standard reports (autorep command), these two values are displayed in the run column as run_num/ntry.

The run_num/ntry value is defined in the run-time environment for the job, and is accessible to shell scripts or executables run as the command of the job. This value is contained in the variable \$AUTORUN.

AutoSys Workload Automation also maintains a value for each job name, which is defined in the runtime environment for the job.

As with \$AUTORUN, this value is accessible to shell scripts or executables run as the UNIX command of the job. The value is contained in the variable \$AUTO_JOB_NAME.

On Windows, the environment variables are %AUTORUN% and %AUTO_JOB_NAME%.

How Time Dependencies Are Set

If you do not define starting conditions for a job, it runs only when you start it manually. To schedule a job to run automatically on specific days and at specific times, set time dependencies.

Example: Set Time Dependencies for a Job

This example shows how to use the date_conditions, days_of_week, and start_times attributes to set time dependencies for a job.

To set the existing job `test_run` to run automatically on certain days at a certain time (such as 10:00 a.m. and 2:00 p.m. on Mondays, Wednesdays, and Fridays), you could modify the job using the following JIL script:

```
update_job: test_run

date_conditions: y

days_of_week: mo, we, fr

start_times: "10:00, 14:00"
```

This JIL script instructs AutoSys Workload Automation to do the following steps:

- Update the `test_run` job.
- Activate the conditions that are based on date.
- Set the job to run on Mondays, Wednesdays, and Fridays.
- Start the job at 10:00 a.m. and 2:00 p.m. on each of the specified days.

The times that are shown in the sample script are surrounded by quotation marks because they contain a colon. You can also use a backslash (\) as an escape character for the colon, as the following example shows:

```
start_times: 10\:00, 14\:00
```

NOTE

If you save the updated definition less than a few seconds before the scheduled start time of the job, the application server may not process the update by the scheduled start time. In this case, the job runs at the next scheduled start time.

Example: Base Time Settings on a Specific Time Zone

Use the `timezone` attribute to base the time settings for a job on a specific time zone. If you specify a time zone that includes a colon, you must surround the time zone name with quotation marks, as in the following example:

```
timezone: "IST-5:30"
```

Example: Run a Job Every Day

To run the job every day, instead of only on specific days, specify the `all` value instead of listing the individual day values. For example:

```
days_of_week: all
```

Example: Schedule a Job to Run on Specific Dates

To schedule the job for specific dates, instead of specific days of the week, specify a custom calendar. Use the `autocal_asc` command to define the calendar, and then use the `run_calendar` attribute to specify the calendar name (for example, `weekday_cal`) in the job definition. For example:

```
run_calendar: weekday_cal
```

Example: Exclude a Job from Running on Specific Dates

To specify a custom calendar that defines the days on which the job should not run, use the `autocal_asc` command to define the calendar, and use the `exclude_calendar` attribute to specify the calendar name (for example, `holiday_cal`) in the job definition. For example:

```
exclude_calendar: holiday_cal
```

Example: Schedule a Job to Run at Specific Times Every Hour

To run the job at specific times every hour instead of at specific times of the day, use the `start_mins` attribute to specify the minutes past every hour that the job should run. For example, to run a job at 15 minutes after and 15 minutes before each hour, add the following statement to the job definition:

```
start_mins: 15, 45
```

Dependent Jobs

Jobs can depend on the successful completion of other jobs. The only difference between a dependent job and a simple job is its dependency on another job. To define job dependencies, specify the `condition` attribute in the job definition.

AutoSys Workload Automation lets you specify a time limit in the `condition` attribute that applies in job dependency evaluations. The profile exclusively verifies the execution environment of the job, which is sourced immediately before the job starts. On UNIX, by default the `/etc/auto.profile` file on the client computer is sourced. On Windows, the variables set by the installer in the profile directory of the agent are sourced or set. You can use the `profile` attribute to override the default profile.

Example: Create a Dependent Command Job

This example shows a JIL script that defines a dependent command job named `EOD_post`. `EOD_post` depends on the successful completion of the File Watcher job named `EOD_watch`.

```
insert_job: EOD_post

job_type: cmd

machine: prod

condition: success(EOD_watch)

command: $HOME/POST
```

This JIL script instructs AutoSys Workload Automation to do the following steps:

- Add a new job named EOD_post.
- Define the job as a command job.
- Run the job on the client computer named prod.
- Run the job only if the file watcher job named EOD_watch completes with a SUCCESS status.
- Source the /etc/auto.profile file (AutoSys Workload Automation sources this file by default), and run the POST job that is located in the home directory of the job owner.

Look-Back Conditions

AutoSys Workload Automation supports look-back conditions. You can use look-back conditions to base dependencies for a job on the last run of another job. The ending time of the last successful run of a job defines the *last run*. If the job has run with the specified result, the condition or predecessor is satisfied and the job starts. If not, the condition is not satisfied and the job for which the look-back condition is defined does not start.

To specify a look-back dependency, enter the job name followed by a comma (,) then *HH* (hours), period (.) and *MM* (minutes).

Example: Specifying Look-Back Conditions

This example shows a job definition with look-back conditions.

In the following job definition, the command job test_sample_04 can only start if all the following conditions are met:

- The last run of test_sample_01 completed successfully during the last 12 hours.
- The last run of test_sample_02 completed with a FAILURE status during the last 24 hours.
- The last run of test_sample_03 completed successfully at any time.

```
insert_job: test_sample_04
```

```
machine: localhost
```

```
command: sleep 10
```

```
condition: success(test_sample_01,12.00) AND failure(test_sample_02,24.00) AND success(test_sam
```

Specify One-Time Job Overrides

You can use the override_job subcommand to specify an override that changes the behavior of a specific job during its next run. Job overrides are applied only once. If a RESTART event is generated because of system problems, AutoSys Workload Automation reissues a job override until the job actually runs once, or until the maximum number of retries limit is met. After this, AutoSys Workload Automation discards the override.

You can modify the following attributes in a job override:

- auto_hold
- command
- condition
- date_conditions
- days_of_week
- exclude_calendar
- machine
- max_run_alarm
- min_run_alarm
- n_retrys
- profile
- run_calendar
- run_window
- start_mins
- start_times
- std_err_file
- std_in_file
- std_out_file
- term_run_time
- watch_file
- watch_file_min_size
- watch_interval

If it results in an invalid job definition, JIL does not accept an override. For example, if a job definition has only one starting condition, start_times, JIL does not let you set the start_times attribute to NULL because removing the start condition makes the job definition invalid (no start time could be calculated).

One-time job overrides are applied to jobs restarted due to system problems, but are not applied to jobs restarted because of application failures.

System problems include the following problems:

- Machine unavailability
- Media failures
- Insufficient disk space

Application failures include the following failures:

- Inability to read or write a file
- Command not found
- Exit status greater than the defined maximum exit status for success
- Various syntax errors

NOTE

- Overriding the machine attribute of a job in PEND_MACH status does not cause the scheduler to start the job. The job remains in PEND_MACH status regardless of the status of the machine that is specified by the

override. The scheduler only evaluates jobs in PEND_MACH status when it processes the MACH_ONLINE events.

- AutoSys Workload Automation does not execute overridden jobs in ON_NOEXEC status. If you issue a one-time override for a job in this status or put a job in this status after overriding, the override remains in effect until you issue the JOB_OFF_NOEXEC event.
- If you are overriding the start_times attribute of a job and your job definition includes the must_start_times or must_complete_times attribute, ensure that the start_times value is earlier than the must_start_times or must_complete_times value to avoid a false MUST_START_ALARM alarm.

How Job Overrides Are Set

To set job overrides, use the override_job subcommand to specify the job and attributes to override. You can also temporarily delete a job attribute in this manner.

Example: Define a One-time Override for a Job

This example shows how to define a one-time job override. The following script runs the job RunData with no conditions (where some had been previously specified) and outputs the results to a different output file:

UNIX:

```
override_job: RunData

condition: NULL

std_out_file: "tmp\SpecialRun.out"
```

Windows:

```
override_job: RunData

condition: NULL

std_out_file: "C:\tmp\SpecialRun.out"
```

Example: Cancel a Job Override Before it Runs

This example shows how to cancel a job override before it runs. To cancel overrides for a job, enter the override_job subcommand followed by the job name and the delete parameter. For example:

```
override_job: RunData delete
```

NOTE

After you submit a JIL script to the database, you cannot view the script or cannot edit an override. To change the override values, you must submit another JIL script with new values or must use the Web UI Quick Edit. However, the original override remains stored in the ujo_overjob table in the database.

Date and Time Attributes and Time Changes

Your operating system might automatically change the system clock to reflect the switch to standard time (ST) or daylight time (DT), and the scheduling of time-dependent AutoSys Workload Automation jobs might shift to adjust for the time change. Jobs that are not time-dependent run as appropriate.

Two types of time dependencies exist: *absolute* and *relative*.

Jobs with absolute time dependencies are defined to run at a specific time of the day (for example, 9:30 on Thursday or 12:00 on December 25). The following attributes define absolute time dependencies:

- `days_of_week`
- `exclude_calendar`
- `must_start_times`
- `must_complete_times`
- `run_calendar`
- `run_window`
- `start_times`

Relative time dependencies are based either on the current time or relative to the start of the hour (for example, start a job at 10 and 20 minutes after the hour, or terminate a job after it has run for 90 minutes). The following attributes define relative time dependencies:

- `auto_delete`
- `max_run_alarm`
- `min_run_alarm`
- `must_start_times`
- `must_complete_times`
- `start_mins`
- `term_run_time`
- `watch_interval`

During the time change, absolute time attributes behave differently than relative time attributes.

Daylight Time Changes

Because the clock loses an hour during the change from standard time to daylight time in the spring, AutoSys Workload Automation cannot schedule any jobs using time-dependent attributes during that time.

The solution is to schedule jobs with absolute time dependencies for the missing hour to start during the first minute of the next hour. In this case, because the time change automatically occurs at 2:00 a.m., a job that is scheduled to run on Sundays at 2:05 runs at 3:00:05 that day; a job that is scheduled to run every day at 2:45 runs at 3:00:45. Although it might not be possible to start many jobs during the first minute of the hour, this feature does preserve the scheduling order.

If you schedule a job to run more than once during the missing hour (for example, at 2:05 and 2:25), only the first scheduled job run occurs. More start times for the same job in the missing hour are ignored.

Jobs with relative time dependencies run as expected. For example, a job that is specified to run at 0, 20, and 40 minutes after the hour is scheduled for 1:00 ST, 1:20 ST, 1:40 ST, 3:00 DT, 3:20 DT, and 3:40 DT.

Jobs that specify absolute must start or must complete times behave similarly to jobs with absolute time dependencies. When the specified must start or must complete times falls during the missing hour, AutoSys Workload Automation re-evaluates the time the job must start or complete to a time during the first minute of the next hour. For example, a job that must start by 2:05 and must complete by 2:45 generates an alarm if the job does not start by 3:00:05 or if it does not complete by 3:00:45.

Specifying an absolute must start or must complete time during the first minute of the hour following the missing hour when a job has an absolute time dependency for the missing hour is a special case. Suppose a job is scheduled to run at 2:45 with a must complete time of 3:00. In this case, because the time change automatically occurs at 2:00 a.m., the job that is scheduled at 2:45 runs at 3:00:45, a time that takes place after the scheduled must complete time. To prevent scheduling a job after its must start or must complete time, AutoSys Workload Automation also re-evaluates the time the job must start or must complete to the final second of the first minute of the hour following the missing hour. In the previous example, the job generates an alarm if it does not complete by 3:00:59.

Jobs that specify must start or must complete times as a time relative to its time dependency evaluate as expected. For example, a job that is specified to run at 2:45 with a relative must start time of 5 minutes and a relative must complete time of 15 minutes, runs at 3:00:45 and generates an alarm if it does not start by 3:05:45 or if it does not complete by 3:15:45.

Run windows are treated differently. When the specified end of the run window falls during the missing hour, AutoSys Workload Automation recalculates its end time, so that the effective duration of the run window remains the same. For example, the product recalculates a run window of 1:00 - 2:30 so that the window ends at 3:30 and the run window remains open for 90 minutes.

When the specified start time of the run window falls during the missing hour, AutoSys Workload Automation moves the start time to 3:00. The end time does not change, so the run window is shortened. For example, a run window of 2:45 - 3:45 becomes 3:00 - 3:45, shortening the run window by 15 minutes.

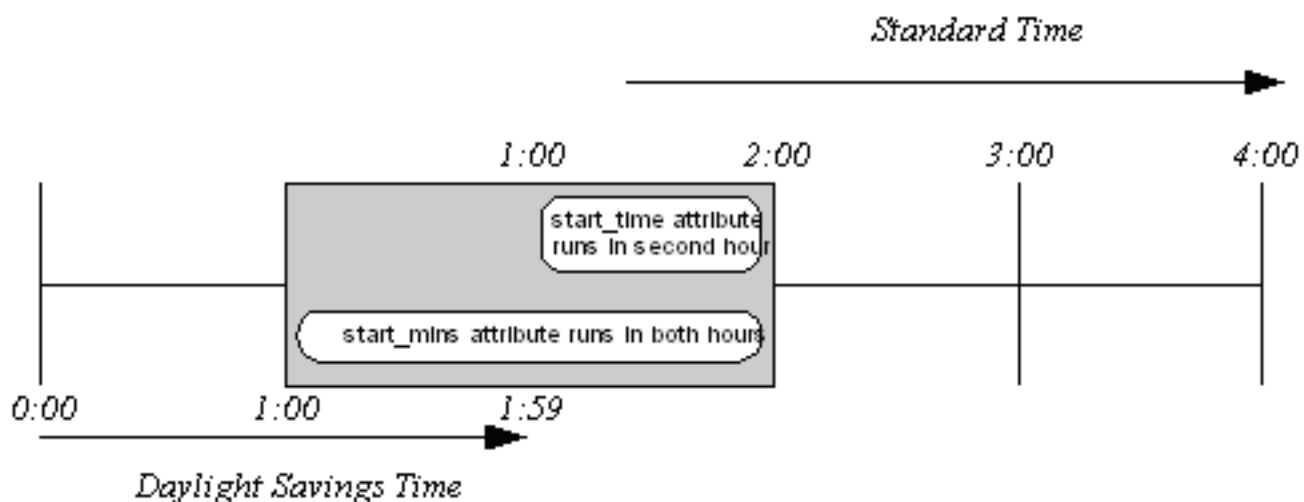
When both the start time and the end time of the run window, fall during the missing hour, AutoSys Workload Automation moves the start time to the first minute after 3:00 and the end time to one hour later. Therefore, the resulting run window might be lengthened. For example, a run window of 2:15 - 2:45 becomes 3:00 - 3:45, or 15 minutes longer.

Standard Time Changes

Because the clock gains one hour during the change from daylight time to standard time in autumn, there are two 1:00-1:59 hours. AutoSys Workload Automation only runs jobs for which the `start_time` attribute is set to between 1:00 and 1:59 during the second (standard time) hour. Jobs for which the `start_mins` attribute is set run in both hours.

For example, a job that is scheduled to run on Sundays at 1:05 runs only at the second 1:05. A job that is scheduled to run every 30 minutes runs at 1:00 DT and 1:30 DT, then again at 1:00 ST and 1:30 ST, and so on, as the following illustration shows:

Figure 29: Standard Time Changes



Jobs that are not time-based but have other dependencies still run during the first hour.

Jobs with relative time dependencies run as expected. For example, if a job is scheduled to run on Sunday at 0:30 and its `term_run_time` attribute is set to 120 minutes, the job would normally terminate at 2:30. On the day of the autumn time change, the job terminates at 1:30 standard time, which is 120 minutes after the job started.

When testing how the change from daylight time to standard time affects your jobs, you must set the system clock to a time before 1:00 a.m. and must allow the entire hour to pass before you can observe the time change. If you manually set the time to a period during the 1:00 a.m. to 2:00 a.m. window, the system assumes that the time change has already occurred and does not reset at 2:00 a.m.

Run windows are treated differently. When the specified start of a run window is before the time change and its specified end occurs during the repeated hour, the run window closes during the daylight time period (the first hour). For example, a run window of 11:30 - 1:30 ends at 1:30 DT, not 1:30 ST (that is, the run window remains open for its specified two hours, not for three hours). A problem might occur if there are also associated start times for the job that occur during the repeated hour. If the job in our example also had a start time of 1:15, the start time would be calculated for 1:15 ST and the job would not run on the day of the time change.

When the specified opening of the run window falls during the repeated hour, AutoSys Workload Automation moves its start time to the second, standard time hour. The end time does not change, so the length of the run window remains the same. For example, a run window of 1:45 - 2:45 becomes 1:45 ST - 2:45 ST.

When both the specified start and end of the run window occur during the repeated hour, the run window opens during the second, standard time hour.

Jobs that specify absolute must start or must complete times behave similarly to the way the end time of run windows behave. That is, when the specified start of the job is before the time change and either the must start or must complete times or both occur during the repeated hour, CA Workload Automation raises alarms during the daylight time period (the first hour). For example, a job that is scheduled to run at midnight and must complete at 1:30 generates an alarm if the job has not completed by 1:30 DT, not 1:30 ST.

When the specified start of the job and either the must start or must complete times or both occur during the repeated hour, CA Workload Automation raises alarms during the second standard time hour.

Job Profiles

A *job profile* defines the non-system environment variables for a job. When you define a job, you can assign a job profile to it. Only one profile can be sourced for a job.

Job profiles apply to the following job types:

- Command jobs
- File Watcher (FW) jobs that are submitted to the legacy agent

On UNIX, job profiles are shell scripts that typically include the definitions and exports of environment variables. The command that the job runs can reference these variables. You can store job profiles on UNIX in any directory.

Job profiles on UNIX are always sourced using the job default shell of the owner, which is set for the user in the `etc/passwd` file. Therefore, when you create a job profile, you must use the syntax of the default shell of the owner. For example, if the default shell of the owner is the Korn shell, you must use Korn syntax in the profile script.

On Windows, you create job profiles using the Job Profiles window in the AutoSys Workload Automation Administrator utility. These profiles contain `variable=value` pairs that define the environment variables. The profiles are stored in the `SystemAgent\agent_name\profiles` directory of the AutoSys Workload Automation computer. If you move a job profile to another location, you must specify the full path when you assign the profile to a job.

The agent uses the `variable=value` pairs in the job profile to set the environment variables.

Environment Variables

System environment variables are automatically set in the environment of a job. However, user environment variables are not automatically set. Ensure that you define all other required environment variables using one or both of the following methods:

- envvars attribute in the job definition
- Job profile

If a job profile is assigned to a job, the agent sources the profile before running the job. When the agent reads the profile, the environment variables in the profile are expanded. For example, if Path is a variable in the profile, the following results occur:

- Any environment variables that are specified in the value of Path are expanded.
- That path is used to search for the command.
- The new value for the %Path% variable is set before running the command.

If you want to specify the full path name, you can use variables set from the job profile in the path name specification.

The agent reads profile variables in alphabetical order. Therefore, if you plan to expand variables in the profile itself, you must define the variables so that they are in the appropriate order when read alphabetically.

NOTE

On the agent, you can define environment variables that apply to all jobs at a global agent level, scheduling manager level, or user level. For example, suppose that you want to set an environment variable for all jobs that run on an agent under a specific user (owner). Instead of defining that variable in every job definition using the envvars attribute or in a job profile, you can define the variable on the agent using the `oscomponent.environment.variable_user_userid` parameter. For more information about setting environment variables on the agent, see the [Workload Automation Agent](#) documentation.

How the Environment for a Job is Sourced

System environment variables are automatically set in the environment for a job. When the job is submitted, the agent processes the following additional information to source the environment, in the following order:

1. Variables set during the AutoSys Workload Automation installation.
 - **On UNIX** The `auto.profile` file is automatically created during AutoSys Workload Automation installation and contains variable definitions such as `AUTOUSER`. This file is in the AutoSys Workload Automation computer.
 - **On Windows** The installer sets the variables in the profile directory of the agent. The `WAAE.txt` file contains a set of variables for the product, and an `<instance_name>.txt` file contains a set of variables for each instance. Typically, these files contain the `%AUTOROOT%`, `%AUTOSYS%`, `%AUTOUSER%`, and `%PATH%` variables.
2. Environment variables are defined using the envvars attribute in the job definition (if specified).
3. The job profile is defined using the profile attribute (if specified).

NOTE

The environment variables are set before the job profile variables. Therefore, you can reference system environment variables in job profiles. However, if a variable is set more than once, the last value read is used.

Create a Job Profile

You can create a job profile to define the non-system environment variables that must be set for a Command job or for a File Watcher (FW) job that is submitted to a legacy agent.

To create a job profile, do *one* of the following steps:

- **On UNIX** Create a shell script file that contains the environment variables you want to source. Use the syntax for the default shell of the job owner. You can store this script in any directory.
- **On Windows**
Use the Job Profiles window in the AutoSys Workload Automation Administrator utility.
The job profile is stored in the SystemAgent\agent_name\profiles directory of the AutoSys Workload Automation computer.

NOTE

For more information about creating, viewing, and deleting job profiles on Windows, see [\(Windows\) Manage Job Profiles Using the Administrator Utility](#).

Assign a Job Profile to a Job

You can assign a job profile to a job to source the non-system environments that must be set before the job can run.

Follow these steps:

1. Do one of the following steps:
 - [Define a Command job](#).
 - [Define a File Watcher job on the legacy agent](#).
2. Add the following attribute to the job definition:
 - **profile: path_name**
Specifies a profile that defines the non-system environment variables for the job.

NOTE

- On UNIX, specify the name of the profile script. Alternatively, you can specify *path/profile_name*.
- On Windows, specify the name of the profile that you created in the Administrator utility. Alternatively, you can specify *path/profile_name*.
- You can specify both the computer name and the profile name, which lets you run the job on one computer while using a job profile defined on another computer. For example, you can specify the following path on Windows: \\machine_name\share_name\profile_name.txt

3. Run the job.
The job profile is assigned to Command or File Watcher job.

NOTE

- Job profiles are instance-specific. You cannot assign a profile that is defined in one AutoSys Workload Automation instance to a job defined in another.
- You can also use Web UI to assign a profile to one or more jobs.

Convert Job Profiles to the New Format (Windows Only)

In Unicenter AutoSys JM 4.5.1 and r11 on Windows, job profile information was stored in the Windows registry. To upgrade to r11.3, Release 11.3.5, or Release 11.3.6, the job profile information in the registry must be converted to a file format that is compatible with the new Workload Automation Agent for Windows. When you upgrade AutoSys Workload Automation, the upgrade process automatically converts the job profiles. You can also issue the autopfrm command to convert profiles manually.

Follow these steps:

1. Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation command prompt opens. The command prompt presets all the environment variables for the instance.
2. Enter the following command:

```
autoprofm -P directory [-N agent_name] [-x] [-?]
```

The job profiles in the Windows registry are converted to the new format and stored in text files. The new files are stored in the directory that is specified by the -P option.

When a job is submitted in r11.3, Release 11.3.5, or Release 11.3.6, the agent refers to the converted profile specified in the profile attribute and sources the environment variables.

Must Start Times and Must Complete Times

You can define the time that a job must start or complete by. If the job does not start by the starting time or complete by the completing time, an alert is issued. Defining must start times and must complete times is helpful when you want to be notified that a job has not started or completed on time. For example, a must start alarm can alert you to investigate whether the starting conditions of the job have not been satisfied or whether the job is stuck in the STARTING state.

You can specify more than one must start time or must complete time for a job. If the job has multiple start times, you must specify the same number of must start times or must complete times. For example, if the job runs at three different times during the day, you must specify three must start times, corresponding to each run of the job.

To define must start times, add the `must_start_times` attribute to your job definition.

To define must complete times, add the `must_complete_times` attribute to your job definition.

You can specify both `must_start_times` and `must_complete_times` attributes in the same job definition.

After the job is defined, you can issue the `autorep -q` command to display start times and complete times. You can issue the `autorep -d` command to view the alarms generated. You can also view the scheduler log file (`event_demon.$AUTOSERV` on UNIX and `event_demon.%AUTOSERV%` on Windows) to see which alarms were issued.

How Must Start Times and Must Complete Times Work

When you define a job with the `must_start_times` or `must_complete_times` attribute, the job definition and corresponding events are added to the database. The scheduler checks the events to determine whether to issue alarms.

AutoSys Workload Automation uses the following process to track the start and complete times:

- The job is inserted in the event table in the database.
- If the job definition includes the `must_start_times` attribute, a `CHK_START` event corresponding to the next must start time is inserted in the event table.
- If the job definition includes the `must_complete_times` attribute, the `CHK_COMPLETE` event corresponding to the next must complete time is inserted in the event table.
- The scheduler checks the `CHK_START` event to see whether the job has started successfully.
- If the job has not started, a `MUST_START_ALARM` is issued. An alert is written to the scheduler log file to indicate that the `CHK_START` criteria has not been satisfied.
- The scheduler checks the `CHK_COMPLETE` event to see whether the job has completed successfully. The scheduler checks for the `SUCCESS`, `FAILURE`, or `TERMINATED` events.
- If the job has not completed, a `MUST_COMPLETE_ALARM` is issued. An alert is written to the scheduler log file to indicate that the `CHK_COMPLETE` criteria has not been satisfied.
- After the job completes, the scheduler calculates the next must start time and must complete time for the job and inserts the following events in the event table:
 - A new `STARTJOB` event
 - A new `CHK_START` event for the next must start time
 - A new `CHK_COMPLETE` event for the next must complete time

Specify Must Start Times and Must Complete Times Examples

The following examples are jobs that have must start times and must complete times that are defined:

Example: Specify Absolute Must Start and Must Complete Times

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. The job must start by 10:02 a.m., 11:02 a.m., and 12:02 p.m. respectively. The job must complete by 10:08 a.m., 11:08 a.m., and 12:08 p.m. respectively. Otherwise, an alarm is issued for each missed start or complete time.

```
insert_job: test_must_start_complete

command: /opt/StartTransactions.sh

machine: localhost

date_conditions: y

days_of_week: all

start_times: "10:00, 11:00, 12:00"

must_start_times: "10:02, 11:02, 12:02"

must_complete_times: "10:08, 11:08, 12:08"
```

NOTE

The number of must start times and must complete times must match the number of start times. Otherwise, the job cannot be defined. For example, the job cannot be defined if it has one start time and two must start and complete times, as follows:

```
start_times: "10:00"
must_start_times: "10:02, 12:02"
must_complete_times: "10:08, 12:08"
```

Example: Specify Absolute Must Start and Must Complete Times on the Next Day

This example defines a job to run every day at 12:00 a.m. Suppose that you want each job that is run to start by 10:10 a.m. the next day and end by 10:12 a.m. the next day. Specify 34:10 in the `must_start_times` attribute and 34:12 in the `must_complete_times` attribute.

```
insert_job: job3

command: echo "hello"

machine: localhost

date_conditions: y

days_of_week: all

start_times: "12:00"

must_start_times: "34:10"

must_complete_times: "34:12"
```

The start time is calculated as follows:

```
must start time + 24 hours
= 10:10 + 24 hours
= 34:10
```

The complete time is calculated as follows:

```
must complete time + 24 hours
= 10:12 + 24 hours
```

= 34:12

If a job run does not start by the start time, a `MUST_START_ALARM` is issued to notify you that the job has not started on time. If a job run does not complete by the complete time, a `MUST_COMPLETE_ALARM` is issued to notify you that the job has not completed on time.

Example: Specify Relative Must Start and Must Complete Times

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. Each job run must start within 3 minutes after each start time (10:03 a.m., 11:03 a.m., and 12:03 p.m.). Each job run must complete within 8 minutes after each start time (10:08 a.m., 11:08 a.m., and 12:08 p.m.). Otherwise, an alarm is issued for each missed start or complete time.

```
insert_job: test_must_start_complete

job_type: CMD

machine: localhost

command: /opt/StartTransactions.sh

date_conditions: y

days_of_week: all

start_times: "10:00, 11:00, 12:00"

must_start_times: +3

must_complete_times: +8
```

Example: Specify Relative Must Start and Must Complete Times With start_mins

This example defines a job to run every day at 10-minute intervals every hour (for example, 2:00 p.m., 2:10 p.m., 2:20 p.m.). Each job run starts within 2 minutes after the specified start times and complete within 7 minutes after the specified start times. Otherwise, an alarm is issued for each missed start or complete time. For instance, the 2:10 p.m. job run must start by 2:12 p.m. and must complete by 2:17 p.m.

```
insert_job: test_must_start_complete

job_type: CMD

machine: localhost
```

```

command: /opt/StartTransactions.sh

date_conditions: y

days_of_week: all

start_mins: 10, 20, 30, 40, 50, 00

must_start_times: +2

must_complete_times: +7

```

Delete Obsolete Job Versions

When you update or delete a job definition, previous versions of the definitions are stored in the database. To prevent the database from being overloaded with job versions, you can delete obsolete job versions. Job versions are obsolete when the job is inactive and the database no longer refers to it.

NOTE

We recommend that you issue the `archive_events` command before issuing the `archive_jobs` command. We also recommend that you run `archive_jobs` as part of your usual database maintenance.

Follow these steps:

1. Do *one* of the following steps:
 - On UNIX, run the shell that is sourced to use AutoSys Workload Automation. The UNIX operating system prompt is displayed.
 - On Windows, Click Start, Programs, AutoSys, Command Prompt (*instance_name*). The AutoSys Workload Automation command prompt opens. The command prompt presets all the environment variables for the instance.
2. Enter the following command:

```
archive_jobs -j number_of_days [-d "directory_name"] [-A] [-x] [-?]
```

The obsolete job versions are deleted.

Example: Delete Obsolete Job Versions

- This example deletes obsolete job versions older than 7 days:

```
archive_jobs -j 7
```

This example deletes obsolete job versions older than 7 days and creates the archive flat file in the \$AUTOUSER/ archive file (the default):

```
archive_jobs -j 7 -A
```

This example deletes obsolete job versions older than 7 days and creates the archive flat file in the /tmp/archive directory:

```
archive_jobs -j 7 -A -d "/tmp/archive"
```

Restrict the Runtime Behavior of Jobs

You can define the following environment variables on AutoSys Workload Automation to restrict the runtime behavior of a job:

- **RESTRICT_FORCE_STARTJOB=1**
This variable restricts a user from running multiple instances of a job while the job is in the STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.
Default: 0
- **RESTRICT_DELETE_JOB=1**
This variable restricts a user from deleting a job when the job is in the ACTIVATED, STARTING, RUNNING, WAIT_REPLY, or SUSPENDED state.
Default: 0
- **RESTRICT_DELETE_DEPENDENT_JOB=1**
This variable restricts a user from deleting a job if the job has dependencies.
Default: 0

NOTE

For more information about adding system environment variables, see [Environment Variables](#).

Manage Your Calendars

Calendars help you schedule jobs as follows:

- Exclude fixed dates, such as holidays, from a job's regular run schedule.
- Run a job only on certain fixed dates.
- Generate a job schedule based on set of complex criteria.

AutoSys Workload Automation includes three types of calendar objects to facilitate the above scheduling options:

- **Standard Calendar**
Lists fixed dates.
- **Cycle**
Lists date ranges (*periods*) that complex criteria in an extended calendar apply to.
- **Extended Calendar**
Specifies complex criteria to generate a schedule based on.

NOTE

You can reference a cycle when you define an extended calendar but not when you define a job. To schedule a job based on standard or extended calendar, reference the calendar in the job definition.

Define Standard Calendars

A standard calendar defines a list of specific dates, typically holidays. You can use a standard calendar for any of the following purposes:

- To schedule a job to run only on the specified dates
- To specify that a job should not run on the specified dates even if the dates meet the other scheduling conditions
- To specify the dates as holidays when defining an extended calendar

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, CA, WAAE, Command Prompt (*instance_name*).
The instance command prompt appears.
2. Enter the following command:

```
autocal_asc
```

The CA WAAE Calendar Utility menu appears.

3. Type 1 and press Enter.
The Administer Standard Calendar submenu appears.
4. Take one of the following actions:
 - Type 1 and press Enter, then specify a calendar name and description and press Enter again.
The calendar is created and the utility prompts you to enter a date to add.
 - Type 2 and press Enter, then specify a calendar name and press Enter again.
The calendar is selected and the utility prompts you to enter a date to add.
 - Type 3 and press Enter, then specify a calendar name and press Enter again.
The calendar is selected and the utility prompts you to enter a date to delete.

NOTE

For the calendar name, you can specify up to 64 characters. Calendar description is optional and you can specify up to 1024 characters.

5. Specify a date and press Enter. Repeat this action until you are finished specifying dates, then press Enter without specifying a date.
The utility redirects to Administer Standard Calendar menu.
6. (Optional) Repeat the procedure as necessary.
7. Type 9 and press Enter.
The standard calendars are defined or updated, and the CA WAAE Calendar Utility redirects to the main menu.

Define Extended Calendars

You can define a job to run on a complex schedule only after you define an extended calendar that generates that schedule. Extended calendar definitions typically reference a holiday calendar, a cycle, or both, but it is not required.

A holiday calendar specifies a list of fixed dates to treat as holidays. Specify a holiday calendar when you want to treat the dates in that calendar as non-workdays or when you want to apply special rules to those dates. When you specify an action at the Holiday Action prompt, the utility applies that action to all of the dates listed in the calendar that you specify at the Holiday Calendar prompt. When you do not specify an action at the Holiday Action prompt, the utility treats the dates listed in the calendar that you specify at the Holiday Calendar prompt as non-workdays according to the value that you specify at the Non-workday Action prompt.

A cycle specifies a series of date ranges (or *periods*), such as fiscal quarters. Specify a cycle when you want to apply rules to the periods that are listed in that cycle.

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, CA, WAAE, Command Prompt (*instance_name*).

The instance command prompt appears.

2. Enter the following command:

```
autocal_asc
```

The CA WAAE Calendar Utility menu appears.

3. (Optional) Create or modify a cycle as follows:

- a. Type 3 and press Enter.
The Administer cycle submenu appears.
- b. Take one of the following actions:
 - Type 1 and press Enter, then specify a unique name and description and press Enter again.
The cycle is created and the utility prompts you to define a period to add.
 - Type 2 and press Enter, then specify the cycle name and press Enter again.
The cycle is selected and the utility prompts you to define a period add.
 - Type 3 and press Enter, then specify the cycle name and press Enter again.
The cycle is selected and the utility prompts you to specify a period to delete.

NOTE

For the cycle name, you can specify up to 64 characters. The cycle description is optional and you can specify up to 1024 characters.

- c. Specify the start date and press Enter, then specify the end date and press Enter again. Repeat this action until you are finished adding or deleting periods and then press Enter without specifying a value.
The periods are defined and the utility redirects to the Administer Cycle submenu.
- d. Type 9 and press Enter.
4. The cycle is defined or updated, and the CA WAAE Calendar Utility redirects to the main menu.
5. Type 2 and press Enter.
The Administer Extended Calendar submenu appears.
6. Take one of the following actions:
 - Type 1 and press Enter, then specify a unique calendar name and press Enter again.
The extended calendar is created and the utility prompts you to enter a set of criteria beginning with the description.
 - Type 2 and press Enter, then specify the calendar name and press Enter again.
The extended calendar is selected and the utility displays the current criteria beginning with the description.

NOTE

For the extended calendar name, you can specify up to 64 characters. The extended calendar description is optional and you can specify up to 1024 characters.

7. Take one of the following actions:
 - Specify or modify workdays by specifying 7-character string where the first character corresponds with Monday, an x represents a workday, and a period (.) represents a non-workday.
 - Maintain the existing workday criteria by pressing Enter without specifying a value.
The utility prompts you to enter a non-workday action.
8. Take one of the following actions:
 - Specify or modify the action by entering *one* of the following values:
 - O

- Specifies to include only non-workdays that also meet all other criteria.
- **SS** Specifies to include both workdays and non-workdays that also meet other criteria.
- **NS** Specifies to include the next workday that also meets all other criteria.
- **W** Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the next work day.
- **P** Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the previous work day.
- Maintain the current criteria or continue without specifying a non-workday action by pressing Enter without specifying a value.

The utility prompts you to enter a holiday action.

9. Take one of the following actions:

- Specify or modify the action by entering one of the following values:
 - **O**
Specifies to include only holidays that also meet all other criteria.
 - **SS** Specifies to include both holidays and non-holidays that also meet other criteria.
 - **NS** Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the next day. If you enter this option, the job runs on the next day even when the next day is also a non-workday.
 - **W** Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the next work day.
 - **P** Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the previous work day.
- Maintain the current criteria or continue without specifying a holiday action by pressing Enter without specifying a value.

The utility prompts you to specify a holiday calendar.

10. Take one of the following actions:

- Specify or modify which dates to treat as holidays by typing the name of a standard calendar and pressing Enter.
- Continue without modifying holiday dates or without specifying holidays by pressing Enter without specifying a value.

WARNING

The utility continues without a specified holiday calendar only when no holiday action is specified; otherwise, the holiday calendar prompt appears again.

The utility prompts you to enter a cycle name.

11. If you want to apply rules to a series of periods in a cycle, enter the name of that cycle and press the Enter key. Ensure that these periods are specified in a cycle that has been defined as described in step 3. If you do not want to apply rules to a series of periods, press the Enter key without specifying a value.

The utility prompts you to enter a date adjustment.

12. If you want to replace non-workdays or holidays with dates that cannot be specified by one of the non-workday action or holiday action values, enter a plus (+) or minus (-) followed by a numeral (between 1 and 9) and press the Enter key. For example, enter -3 to include dates that are three days prior to the excluded dates or +2 to include dates that are two days after the excluded dates. If you do not need to replace the excluded days or if you specified replace days using non-workday action or holiday action values, enter 0.

The date adjustment is set and the utility prompts you to enter date conditions.

13. (Optional) Specify additional rules with the appropriate [date condition keywords](#) and then press the Enter key. The utility prompts you to indicate whether or not you want to preview the calendar.

14. Enter 1 and press the Enter key.

The utility displays a preview of the schedule and prompts you to indicate whether or not you want to create the calendar.

15. If you are not satisfied with the schedule displayed in the preview, enter 0, press the Enter key, and then enter option [9] Exit from the "Administer Extended Calendar" menu.

The calendar is discarded, the utility redirects to the main menu, and you can create a new calendar.

16. If you are satisfied with the schedule, enter 1 and press the Enter key.

The extended calendar is created and the schedule that is displayed in the preview is automatically generated.

17. Repeat the procedure until you have defined an extended calendar for each job that runs on a complex schedule.
18. Enter option [9] Exit from the "Administer Extended Calendar" menu, and then enter option [0] Exit CA WAAE Calendar Utility.

The utility closes and the extended calendar is defined and stored in the database.

Example: Define an Extended Calendar Using the Calendar Utility Interactive Menu

Suppose that you want to create an extended calendar named FIRST-WED-MINUS-1 for the day before the first Wednesday of the month.

This example creates the extended calendar FIRST-WED-MINUS-1. Issue the `autocal_asc` command with no additional options to open the Calendar Utility Interactive menu, and then follow these steps:

1. Type 2 and press Enter.
2. Type 1 and press Enter.
3. Enter FIRST-WED-MINUS-1 when the utility prompts you to enter the calendar name.
4. Enter "Calendar for the day before the first Wednesday of the month" when the utility prompts you to enter the calendar description.
5. Press Enter when the utility prompts you to enter the workdays.
6. Press Enter when the utility prompts you to enter the non-workday actions.
7. Press Enter when the utility prompts you to enter the holiday actions.
8. Press Enter when the utility prompts you to enter the holiday calendar.
9. Press Enter when the utility prompts you to enter the cycle name.
10. Enter -1 when the utility prompts you to enter date adjustments.
11. Enter WED#1 when the utility prompts you to enter the date conditions.
12. Enter 1 to preview the extended calendar.
13. Enter 1 to create the extended calendar FIRST-WED-MINUS-1.
14. Enter 0 to exit the Calendar Utility interactive menu.

The WED#1 date condition indicates the first Wednesday of the month and the -1 date adjustment sets it to the day before that Wednesday.

Example: Define an Extended Calendar to get the US Holidays

Suppose that you want to create an extended calendar named `ext_cal_US` to get the list of US holidays from the 04/09/2018 (the current date) to 01/18/2038.

This example creates the extended calendar `ext_cal_US`. Issue the `autocal_asc` command with no additional options to open the Calendar Utility Interactive menu, and then follow these steps:

1. Type 3 and press Enter.
2. Type 1 and press Enter.
3. Enter `week_after` when the utility prompts you to enter the cycle name.
4. Enter "Cycle for US Calendar" when the utility prompts you to enter the cycle description.
5. Enter 11/23/1972 when the utility prompts you to enter the start date.
6. Enter 11/29/1972 when the utility prompts you to enter the end date.
7. Press Enter to create the cycle named `week_after`.
8. Type 9 and press Enter to exit from the Administer Cycle submenu.
9. Type 1 and press Enter.
10. Type 1 and press Enter.
11. Enter `pago_pago` when the utility prompts you to enter the standard calendar name.

12. Enter "Us Holiday Calendar" when the utility prompts you to enter the standard calendar description.
13. Enter 01/18/2038 16:14 when the utility prompts you to enter the date.
14. Press Enter to create the standard calendar named pago_pago.
15. Type 9 and press Enter to exit from the Administer Standard Calendar submenu.
16. Type 2 and press Enter.
17. Type 1 and press Enter.
18. Enter ext_cal_US when the utility prompts you to enter the extended calendar name.
19. Enter "US Holiday Ext Calendar" when the utility prompts you to enter the extended calendar description.
20. Enter mo,tu,th,fr when the utility prompts you to enter the workdays.
21. Press Enter when the utility prompts you to enter the non-workday actions.
22. Enter s when the utility prompts you to enter the holiday actions.
23. Enter pago_pago when the utility prompts you to enter the holiday calendar.
24. Enter week_after when the utility prompts you to enter the cycle name.
25. Enter 0 when the utility prompts you to enter date adjustments.
26. Enter the following conditions when the utility prompts you to enter the date conditions:

```
(jan&workd#1) | ((jan|feb) &mon#3) | (may&monm1) | (jul#3&fri) | (jul#4&weekdays) | (jul#5&mon) |
(sep&mon#1) | (oct&mon#2) | (nov&thu#4) | (cycp#1&fri) | (dec#23&(thu|fri) | (dec#24&workdays) |
(dec#25&weekdays) | (dec#26&(mon|tue|fri))
```

27. Enter 1 to preview the extended calendar.

You can preview the US holidays from 04/09/2018 (the current date) to 01/18/2038. Below, only the dates generated for the year 2019 are displayed:

2019 (MM/DD/YYYY)

2019 January

2019 February

2019 March

01/01/2019 00:00:00

02/18/2019 00:00:00

01/21/2019 00:00:00

2019 April

2019 May

2019 June

05/27/2019 00:00:00

2019 July

2019 August

2019 September

07/04/2019 00:00:00

09/02/2019 00:00:00

2019 October

2019 November

2019 December

10/14/2019 00:00:00

11/28/2019 00:00:00

12/24/2019 00:00:00

11/29/2019 00:00:00

12/25/2019 00:00:00

28. Enter 1 to create the extended calendar ext_cal_US.

29. Enter 9 to exit the Administer Extended Calendar submenu.

30. Enter 0 to exit the CA WAAE Calendar Utility.

Manage Calendars

As a workload scheduler, you manage the calendars that you use to define jobs with customized schedules.

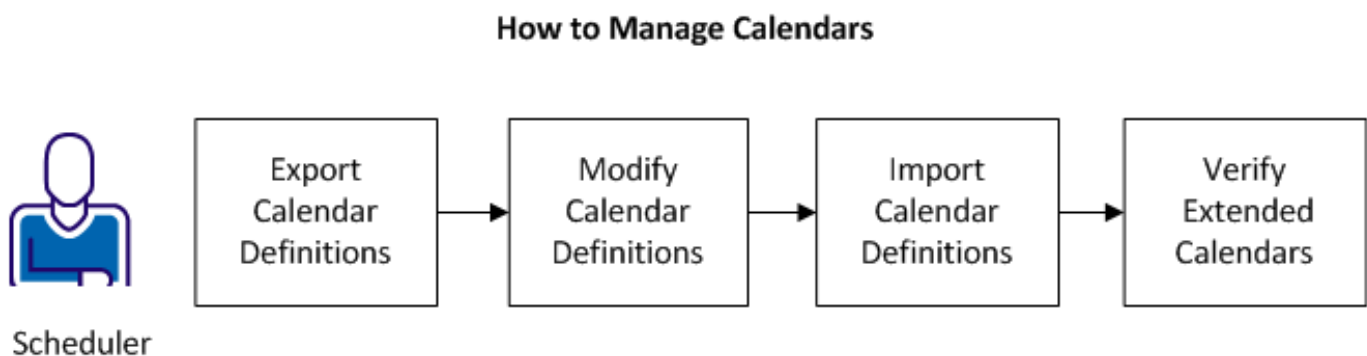
The autocal_asc command lets you export calendar definitions to text files, so that you can modify multiple calendar objects simultaneously. To update calendar objects after you modify a text file, import the text file.

NOTE

The interactive calendar utility lets you modify calendars one at a time. To use this method, select the appropriate interactive menu options and specify the required information. To launch the interactive utility, issue the autocal_asc command with no options specified. We recommend this method only when you need to make simple modifications to a single calendar.

The following diagram illustrates how a workload scheduler manages calendars:

Figure 30: How to manage calendars



To manage calendars, follow these steps:

1. [Export calendar definitions to text files](#)
2. [Modify calendar definitions](#)
3. [Import calendar definitions](#)

4. [Verify extended calendars](#)

Export Calendar Definitions to Text Files

If you use the interactive calendar utility to create new calendars and want to modify your calendar definitions in batch, export the calendars to a single text file.

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The instance command prompt appears.
2. Enter the following commands:

```
autocal_asc -s ALL -E export_filename_s.txt
```

```
autocal_asc -c ALL -E export_filename_c.txt
```

```
autocal_asc -e ALL -E export_filename_e.txt
```

- **export_filename_s.txt**
Specifies a file to export standard calendars to.
- **export_filename_c.txt**
Specifies a file to export cycles to.
- **export_filename_e.txt**
Specifies a file to export extended calendars to.

Modify Calendar Definitions

To update calendar objects in batch, modify the text file that contains calendar definitions.

Follow these steps:

1. Merge the three text files, so that all calendar definitions are stored in the same file.
2. Open the text file.
The calendar definitions appear as follows:

```
calendar: std_cal_name
```

```
[description: text]
```

```
MM/DD/YYYY [hh:mm]
```

```
MM/DD/YYYY [hh:mm]
```

```
...
```

```
cycle: cycle_name
```

```

[description: text]

start_date: MM/DD/YYYY

end_date: MM/DD/YYYY

start_date: MM/DD/YYYY

end_date: MM/DD/YYYY

...

ext_calendar: ext_cal_name

[description: text]

[workday: {X|.}{X|.}{X|.}{X|.}{X|.}{X|.}{X|.}]

non_workday: {O|S|N|W|P}

[holiday: {O|S|N|W|P}]

[holcal: std_cal_name]

[cyccal: cycle_name]

[adjust: {+|-n}]

[condition: keyword]

[condition: keyword]

...

```

- **calendar: *std_cal_name***
Specifies the name of the standard calendar. A standard calendar is a list of specific dates, usually holidays.
- **description: *text***
(Optional) Specifies a description for the standard calendar.
- **MM/DD/YYYY [hh:mm]**
Specifies a date and, optionally a time, to include in the standard (holiday) calendar.
- **cycle: *cycle_name***
Specifies the name of the cycle. A cycle is a list of date ranges or periods. A period is between two and 365 days. A cycle contains a minimum of two periods and a maximum of 30.
- **description: *text***
(Optional) Specifies a description for the cycle.
- **start_date: MM/DD/YYYY**
Specifies the start date for the cycle.
- **end_date: MM/DD/YYYY**

Specifies the end date for the cycle.

- **ext_calendar: *ext_cal_name***
Specifies the name of an extended calendar. An extended calendar generates a list of dates based on criteria or rules specified in the calendar definition.
- **description: *text***
(Optional) Specifies a description for the extended calendar.
- **workday: {X|.}{X|.}{X|.}{X|.}{X|.}{X|.}{X|.}**
(Optional) Specifies non-default workdays and non-workdays as follows:
 - X specifies a workday
 - A period (.) specifies a non-workday
 - The first character in the string corresponds with Monday

NOTE

By default, Mon-Fri are workdays and Sat-Sun are non-workdays.

- **non_workday: {O|S|N|W|P}**
(Optional) Specifies that non-workday action as follows:
 - **O**
Includes only non-workdays that also meet all other criteria.
 - **S**
Includes workdays and non-workdays that also meet all other criteria.
 - **N**
Specifies to include the next workday that also meets all other criteria.
 - **W**
Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the next work day.
 - **P**
Specifies to adjust the schedule so that days that meet all criteria and fall on non-workdays run on the previous work day.
- **holiday: {O|S|N|W|P}**
(Optional) Specifies the holiday action as follows:
 - **O**
Include only holidays that also meet all other criteria.
 - **S**
Include holidays and non-holidays that also meet all other criteria.
 - **N**
Excludes the holiday and includes the next day. This applies even if the next day is a holiday or non-workday.
Example: December 25th is a holiday but meets all other criteria. If you specify this option, the calendar includes December 26th instead of December 25th. This applies even if December 26th is a holiday.
 - **W**
Excludes the holiday and includes the next non-holiday workday.
Example: December 25th is a holiday but meets all other criteria. December 26th a holiday, December 27th is a non-workday, and December 28th is a non-holiday workday. If you specify this option, the calendar includes December 28th instead of December 25th.
 - **P**
Excludes the holiday and includes the previous non-holiday workday.
Example: December 25th is a holiday but meets all other criteria. December 23rd and 24th are non-workdays, and December 22nd is a non-holiday workday. If you specify this option, the calendar includes December 22nd instead of December 25th.
- **holcal: *std_cal_name***
(As required) Specifies the standard calendar containing dates to treat as holidays.

The utility applies the holiday action to the dates in this calendar. Ensure that you specify this argument when you specify a holiday action.

- **cyccal: *cycle_name***
(As required) Specifies the cycle that any cycle-related conditions apply to.
Ensure that you specify this argument when you specify any cycle-related conditions.
 - **adjust: {+|-}n**
(Optional) Specifies an adjustment to apply to holidays and non-workdays that meet all other criteria. Specify this option when none of the non-work action and holiday action options indicate the desired adjustment.
 - **condition: *keyword***
(Optional) Specifies an additional condition. To specify conditions, use [date condition keywords](#).
3. Modify argument values as necessary, then save and close the file.
The definitions are updated in the text file.

Import Calendar Definitions

To update calendar objects after you modify definitions in a text file, import the calendars.

Follow these steps:

1. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
The operating system command prompt appears.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The instance command prompt appears.
2. Enter the following commands:

```
autocal_asc -I import_filename_s.txt
```

```
autocal_asc -I import_filename_c.txt
```

```
autocal_asc -I import_filename_e.txt
```

- **import_filename_s.txt**
Specifies a file to import standard calendars from.
- **import_filename_c.txt**
Specifies a file to import cycles from.
- **import_filename_e.txt**
Specifies a file to import extended calendars from.

The utility imports the calendar definitions.

NOTE

If you specify a calendar name that already exists, the import fails. In such cases, use the `autocal_asc` command with the `-F` option to force overwrite the calendar definitions from the file to the calendar.

Verify Extended Calendars

The Preview an Extended Calendar option in the interactive calendar utility lets you verify that an extended calendar definition generates the desired dates.

Follow these steps:

1. Open the operating system or instance command prompt as follows:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.

The operating system command prompt appears.
 (Windows) Click Start, Programs, CA, WAAE, Command Prompt (*instance_name*).
 The instance command prompt appears.

2. Enter the following command:

```
autocal_asc
```

The CA WAAE Calendar Utility menu appears.

3. Type 2 and press Enter.
 The Administer Extended Calendar submenu appears.
4. Type 5 and press Enter.
5. Specify the calendar name and press Enter.
6. Verify the displayed dates.
7. (As needed) Update the rules in the appropriate text file.
8. Repeat the procedure for all modified extended calendars.
9. (As needed) Import calendar definitions and verify the dates.
10. Repeat the procedure as necessary.

Date Condition Keywords

The extended calendar prompts in the interactive calendar utility menu include a request for date conditions. You can specify these optional conditions using date condition keywords.

When you export extended calendars to a text file, the rules are specified as a list of date condition keywords. To modify the rules, change the values of these keywords and then import the text file.

When you specify date conditions, they represent additional criteria. The dates that are specified are included in the schedule that an extended calendar generates only when the dates also meet all the criteria specified at other criteria prompts.

Dates that are excluded by workday or holiday-related criteria are replaced with alternative dates according to the values specified at the Holiday Action prompt, Workday Action prompt, or Date Adjustment prompt. Consider the following examples:

- Your date conditions specify that the schedule includes the last day of each period in cycle that you specified at the Cycle Name prompt, but the last day of one of the periods falls on a holiday. The last day is adjusted according to the value that you specified at the Holiday Action prompt or at the Date Adjustment prompt.
- Your date conditions specify that the schedule includes the 15th day of each month but excludes holidays and non-workdays, but the 14th is a holiday and the 15th and 16th are both non-workdays. The Holiday Action and Non-workday Action prompts do not provide the option to specify a valid date adjustment, so the job runs according to the value that you specified at the Date Adjustment prompt, as long as that value is not -1, 0, or +1.

Some date condition keywords in the below list use variables. The keywords are not valid unless you enter valid values in place of these variables:

- Replace *n* with a 1-digit number between 1 and 9.
- Replace *nn* with a 2-digit number between 01 and 31.
- Replace *nnn* with a 3-digit number between 001 and 365.
- Replace *ddd* with one of the following 3-letter abbreviations: mon, tue, wed, thu, fri, sat, sun.
- Replace *mmm* with one of the following 3-letter abbreviations: jan, feb, mar, apr, may, jun, jul, aug, sep, oct, nov, dec.

Some date condition keywords are inclusive and others are exclusive. If you want to exclude dates for which there is no exclusive date condition keyword, enter NOT before the inclusive date condition keyword that specifies the dates that you want to exclude.

You can use logical operators between two date condition keywords to specify compound conditions:

- Use AND when you want to specify only dates that meet both conditions.
- Use OR to specify dates that meet at least one of the conditions.

The below list of keywords uses all capital letters; however, the date condition keywords are *not* case-sensitive. To specify multiple rules, use a comma-separated list.

- **Daily date condition keywords**

Specify rules that apply to specific days.

- **DAILY**

(Default) Specifies that the schedule that is generated by the extended calendar includes all days. If you press Enter without specifying any values at the date condition prompt, the calendar defaults to this value.

- **WORKDAYS**

Specifies that the schedule that is generated by the extended calendar includes all workdays.

NOTE

The utility automatically considers holidays to be non-workdays if you specified a holiday calendar at the Holiday Calendar prompt.

- **WORKD#nn** Specifies that the schedule that is generated by the extended calendar includes the *nnth* workday of the month.

- **WORKDMnn** Specifies that the schedule that is generated by the extended calendar includes the *nnth* workday of the month, counting backwards.

- **WORKDXnn**

Specifies that the schedule that is generated by the extended calendar excludes the *nnth* workday of the week.

- **FOMWORK**

Specifies that the schedule that is generated by the extended calendar includes the first workday of the month.

- **EOMWORK**

Specifies that the schedule that is generated by the extended calendar includes the last workday of the month.

- **XFOMWORK**

Specifies that the schedule that is generated by the extended calendar excludes the first workday of the month.

- **XEOMWORK**

Specifies that the schedule that is generated by the extended calendar excludes the last workday of the month.

- **ddd**

Specifies that the schedule that is generated by the extended calendar includes the specified day of the week.

- **ddd#n**

Specifies that the schedule that is generated by the extended calendar includes the *nth* monthly occurrence of the specified day of the week.

- **Xddd#n**

Specifies that the schedule that is generated by the extended calendar excludes the *nth* monthly occurrence of the specified day of the week.

Example: To exclude the first Friday of the month but include all other Fridays, enter the following keyword:

```
Xfri#1
```

- **dddMn**

Specifies that the schedule that is generated by the extended calendar includes the *nth* monthly occurrence of the specified day of the week counting backward.

- **XdddMn**

Specifies that the schedule that is generated by the extended calendar excludes the *nth* monthly occurrence of the specified day of the week counting backward.

- **WEEKDAYS**

Specifies that the schedule that is generated by the extended calendar includes Monday through Friday.

NOTE

The utility automatically excludes all dates that are listed in the calendar that you specify in the holiday calendar field.

- **WEEKD#n**
Specifies that the schedule that is generated by the extended calendar includes the *n*th weekday, where 1 corresponds with the first day of the week.
- **Weekly date condition keywords**
Specifies rules that apply to specific weeks.
 - **WEEKDMn**
Specifies that the schedule that is generated by the extended calendar includes the *n*th weekday, counting backwards, where 1 corresponds with the last day of the week.
 - **WEEKDXn**
Specifies that the schedule that is generated by the extended calendar excludes *n*th day of the week. The numeral 1 corresponds with the first day of the week. All weeks in the year begin on the same weekday as January 1 of that year.

NOTE

You can specify a different start day by using the **WEEKDstartdayXn** keyword.

- **FOMWEEK**
Specifies that the schedule that is generated by the extended calendar includes the first weekday of the month.
- **EOMWEEK**
Specifies that the schedule that is generated by the extended calendar includes the last weekday of the month.
- **XFOMWEEK**
Specifies that the schedule that is generated by the extended calendar excludes the first weekday of the month.
- **XEOMWEEK**
Specifies that the schedule that is generated by the extended calendar excludes the last weekday of the month.
- **WEEK#nn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th week of the year.
- **WEEK#E** Specifies that the schedule that is generated by the extended calendar includes all even weeks of the year.
- **WEEK#O** Specifies that the schedule that is generated by the extended calendar includes all odd weeks of the year.
- **WEEKXnn**
Specifies that the schedule that is generated by the extended calendar excludes the *nn*th week of the year.
- **WEEKMnn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th week of the year, counting backwards where 01 corresponds with the last week of the year.

By default, all weeks in the year begin on the same weekday as January 1 of the same year, but you can specify a different start day by modifying the keywords as follows:

- **WEKRddd#nn**
- **WEKRdddXnn**
- **WEKRdddMnn**

Example: Suppose that you want to consider full weeks as those that start on a Monday and you are defining an extended calendar that generates a schedule for 2014. Because January 1, 2014 falls on a Wednesday, AutoSys Workload Automation starts all weeks during 2014 on Wednesday unless you specify Monday in the weekly keyword word as follows:

- WEKRMOn#nn
- WEKRMOnXnn
- WEKRMOnMnn
- **Monthly date condition keywords**
Specify rules that apply to specific months.
 - **MNTHD#nn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th day of the month.
 - **MNTHDMnn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th day of the month, counting backwards.
 - **MNTHDXnn**
Specifies that the schedule that is generated by the extended calendar excludes the *nn*th day of the month.
 - **FOM**
Specifies that the schedule that is generated by the extended calendar includes the first day of the month.
 - **EOM**
Specifies that the schedule that is generated by the extended calendar includes the last day of the month.
 - **XFOM**
Specifies that the schedule that is generated by the extended calendar excludes the first day of the month.
 - **XEOM**
Specifies that the schedule that is generated by the extended calendar excludes the last day of the month.
 - **mmm#nn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th day the specified month.

NOTE
To include every day in the specified month, omit the #nn.
 - **Xmmm#nn**
Specifies that the schedule that is generated by the extended calendar excludes the *nn*th day the specified month.

NOTE
To exclude every day in the specified month, omit the #nn.
 - **mmm M nn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th day of the specified month, counting backwards.
 - **XmmmMnn**
Specifies that the schedule that is generated by the extended calendar excludes the *nn*th day of the month counting backwards.
- **Cycle date condition keywords**
Specify rules that apply to the cycle specified at the Cycle Name prompt.
 - **CYCLE**
Specifies that the schedule that is generated by the extended calendar includes dates that fall within any of the periods defined in the cycle.
 - **CYCL#nnn**
Specifies that the schedule that is generated by the extended calendar includes the *nnn*th day of each period in the cycle.
 - **CYCLMnnn**
Specifies that the schedule that is generated by the extended calendar includes the *nnn*th from the last day of each period in the cycle that you specified at the Cycle Name prompt as long as that day meets the criteria specified by the values indicated at the other criteria prompts.
 - **CYCLXnnn**

- Specifies that the schedule that is generated by the extended calendar excludes the *nn*th day of each period in the cycle that you specified at the Cycle Name prompt.
- **CYCP#nn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th period in the cycle.
 - **CWEEK#nn**
Specifies that the schedule that is generated by the extended calendar includes dates that fall within the *nn*th week of each period in the cycle.
 - **CWEEK#E** Specifies that the schedule that is generated by the extended calendar includes all even weeks of the cycle.
 - **CWEEK#O**
Specifies that the schedule that is generated by the extended calendar includes all odd weeks of the cycle.
 - **CWEEK#L**
Specifies that the schedule that is generated by the extended calendar includes dates that fall within the last week of each period in the cycle.
 - **CWEEKMnn**
Specifies that the schedule that is generated by the extended calendar includes the *nn*th week from the end of each period in the cycle that you specified at the Cycle Name prompt.
 - **CWEEKXnn**
Specifies that the schedule that is generated by the extended calendar excludes the *nn*th week of each period in the cycle that you specified at the Cycle Name prompt.
 - **CWRK#nnn**
Specifies that the schedule that is generated by the extended calendar includes the *nnn*th workday of each period in the cycle that you specified at the Cycle Name prompt as long as that day meets the criteria that are specified by the values indicated at other criteria prompts.
 - **CWRK#L**
Specifies that the schedule that is generated by the extended calendar includes the last workday of each period in the cycle that you specified at the Cycle Name prompt as long as that day meets the criteria that are specified by the values indicated at the other criteria prompts... Dates that are excluded by other criteria are replaced according to the date adjustment or holiday action values that are specified at the corresponding prompts.
 - **CWRKMnnn**
Specifies that the schedule that is generated by the extended calendar includes the *nnn*th workday, counting backwards, of each period in the cycle.
 - **CWRKXnnn**
Specifies that the schedule that is generated by the extended calendar excludes the *nnn*th workday of each period in the cycle.
 - **CWRKXL**
Specifies that the schedule that is generated by the extended calendar excludes the last workday of each period in the cycle.
 - **CWEK#n**
Specifies that the schedule that is generated by the extended calendar includes the *n*th of weekday of each week in each period in the cycle.
 - **CWEK#L**
Specifies that the schedule that is generated by the extended calendar includes the last weekday of each week in each period in the cycle.
 - **CWEKMn**
Specifies that the schedule that is generated by the extended calendar includes the *n*th of weekday, counting backwards, of each week in each period in the cycle.
 - **CWEKXn**
Specifies that the schedule that is generated by the extended calendar includes the *n*th day of each period in the cycle.
 - **Cddd#nn**

Specifies that the schedule that is generated by the extended calendar includes the *nn*th occurrence of the specified weekday in each period in the cycle.

Example: To include the second Tuesday in each period, enter the following keyword:

```
Ctue#02
```

– **Cddd#L**

Specifies that the schedule that is generated by the extended calendar includes the last occurrence of the specified day of each period in the cycle.

Example: To include the last Tuesday in each period, enter the following keyword:

```
Ctue#L
```

– **CdddMnn**

Specifies that the schedule that is generated by the extended calendar includes the *nn*th occurrence of the specified weekday, counting backwards, in the cycle.

Example: To include the last Tuesday in each period, enter the following keyword:

```
Ctue#M01
```

– **XCddd#nn**

Specifies that the schedule that is generated by the extended calendar excludes the *nn*th occurrence of the specified day in each period in the cycle.

– **XCdddMnn**

Specifies that the schedule that is generated by the extended calendar excludes *nn*th occurrence of the specified day in each period in the cycle.

Example: To exclude the last Friday in each period but include all other Fridays, enter the following keyword:

```
XCfriM01
```

Connection Profiles

A *connection profile* is a set of attributes that are required for connecting to environments where the job runs. For example, to run a job in the Hadoop environment, you can create a connection profile with the appropriate parameters to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database.

A connection profile lets you save the connection information and you can use the connection profile in the job definition. For example, when you define a Hadoop HDFS job, you can assign a connection profile to it.

A connection profile applies to the following job types:

- HDFS
- Hive
- Oozie
- Sqoop
- Pig

Create a Connection Profile

Before you define a Hadoop job, create a connection profile. You can create one or more connection profiles specific to the Hadoop environment.

Follow these steps:

1. Do one of the following tasks:
 - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.
 - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance_name*).
The AutoSys Workload Automation command prompt opens.
2. Enter the following JIL subcommand and attribute:value combinations (as appropriate) based on whether you want to connect to the Hadoop cluster, Oozie server, Hive database, or Scoop database:

```
insert_connectionprofile: conn_name
connectionprofile_type: conn_type
description: texthadoop_edgenode: edgenodename
hadoop_edgenode_port: SSH_port
hadoop_binpath: path_to_CLI
hadoop_hdfs_namenode: nodename
hadoop_oozie_server_url: location
hadoop_hive_jdbcurl: hive_jdbc_url
hadoop_hive_driverclass: hive_driverclass
hadoop_sqoop_jdbcurl: sqoop_jdbc_url
hadoop_sqoop_driverclass: sqoop_driverclass
exit
```

conn_name

Defines the name of the connection profile.

Limits: Up to 64 alphanumeric characters; the value cannot contain embedded blanks or tabs.

conn_type

Specifies the type of the connection profile.

text

(Optional) Specifies a description for the connection profile.

Limits: Up to 256 characters

edgenodename

Defines the name of the Hadoop edge node where the specified operation is performed. This can be a simple name, fully qualified domain name, or IP address.

Limits: Up to 80 characters

SSH_port

(Optional) Specifies the Secure Shell (SSH) port on the Hadoop edge node.

Default: 22

Limits: 1-65535

path_to_CLI

(Optional) Specifies the path to the Hadoop client binaries on the Hadoop edge node.

Limits: Up to 255 characters

NOTE

If the path includes a colon, enclose it in double quotation marks.

Specify this attribute only if the Hadoop client binaries are not in the user's path. The `hadoop_binpath` value is prepended to the client name at execution time.

nodename

(Optional) Specifies the namenode of the Hadoop HDFS cluster.

Limits: Up to 255 characters

location

Specifies the Oozie server URL.

Limits: Up to 255 characters

NOTE

If you are creating a connection profile to connect to the Oozie server, you must specify the `hadoop_oozie_server_url` attribute in the connection profile.

hive_jdbc_url

Specifies the JDBC URL that you provide to the Hadoop Beeline client to connect to the database.

Limits: Up to 255 characters

NOTE

If you are creating a connection profile to connect to the Hive database, you must specify the `hadoop_hive_jdbcurl` attribute in the connection profile.

hive_driverclass

(Optional) Specifies the name of the JDBC driver class that you provide to the Hadoop Beeline client.

Limits: Up to 255 characters

sqoop_jdbc_url

Specifies the JDBC URL that you provide to the Hadoop Sqoop client to connect to the database.

Limits: Up to 255 characters

NOTE

If you are creating a connection profile to connect to the Sqoop database, you must specify the `hadoop_sqoop_jdbcurl` attribute in the connection profile.

sqoop_driverclass

(Optional) Specifies the name of the JDBC driver class to execute on the database.

Limits: Up to 255 characters

The connection profile is created and added in the database.

NOTE

- You can update a connection profile using the [update_connectionprofile](#) subcommand.
- If you no longer need a connection profile, you can delete it using the [delete_connectionprofile](#) subcommand. Use the [force](#) attribute to forcefully delete a connection profile that is currently in use.

Examples: Add a Connection Profile

This example creates a connection profile *conn1* that includes the parameters required to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database. You can specify this connection profile in your job definition when you define a HDFS, Hive, Sqoop, Pig, or Oozie job.

```
insert_connectionprofile: conn1
connectionprofile_type: HADOOP
hadoop_edgenode: HADOOP_EDGE_NODE
hadoop_edgenode_port: 22
hadoop_binpath: /usr/bin
hadoop_hdfs_namenode: "hdfs://HADOOP_EDGE_NODE:8020"
hadoop_oozie_server_url: "http://HADOOP_EDGE_NODE:11000/oozie"
hadoop_hive_jdbcurl: "jdbc:hive2://HADOOP_EDGE_NODE:10000"
hadoop_hive_driverclass: org.apache.hive.jdbc.HiveDriver
```



```
hadoop_sqoop_jdbcurl: "jdbc:sqlserver://
DATABASE_SERVER_MACHINE:1433\;databaseName=SAMPLE"
hadoop_sqoop_driverclass: com.microsoft.sqlserver.jdbc.SQLServerDriver
```

This example creates a connection profile *conn2* that includes the parameters required to connect to the Hadoop cluster. You can specify this connection profile in your job definition when you define a HDFS job.

```
insert_connectionprofile: conn2
connectionprofile_type: HADOOP
hadoop_edgenode: HADOOP_EDGE_NODE
```

Examples: Update a Connection Profile

This example updates the path to the Hadoop client binaries on the Hadoop edge node in the *conn1* connection profile.

```
update_connectionprofile: conn1
hadoop_binpath: /user/root/bin
```

This example updates the *conn2* connection profile to include the *hadoop_oozie_server_url* parameter that is required to connect to the Oozie server. You can now specify this connection profile in your job definition when you define a HDFS or Oozie job.

```
update_connectionprofile: conn2
hadoop_oozie_server_url: "http://HADOOP_EDGE_NODE:11000/oozie"
```

Examples: Delete a connection profile

This example deletes the *conn1* connection profile.

```
delete_connectionprofile: conn1
```

This example force deletes the *conn2* connection profile that is currently being used.

```
delete_connectionprofile: conn2
force: Y
```

AgentCLI_To_JobConverter Utility

The AgentCLI_To_JobConverter utility lets you convert:

- Workload Automation Agent for Informatica CLI command jobs to AutoSys Workload Automation Informatica jobs— This allows you to use the inbuilt capabilities of the Informatica job type.
- CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs— This allows you to use the inbuilt capabilities of the SQL Server Agent job type.

The AgentCLI_To_JobConverter utility is installed automatically when you perform a server installation and is available in the following location:

- UNIX—/opt/CA/WorkloadAutomationAE/autosys/utls/AGTCLIJobConv
- Windows—C:\Program Files(x86)\CA\WorkloadAutomationAE\autosys\utls\AGTCLIJobConv

NOTE

The AgentCLI_To_JobConverter utility does not update the jobs automatically.

You can use AgentCLI_To_JobConverter utility to perform the following tasks:

- [Convert Agent CLI command jobs to AutoSys Workload Automation jobs](#)
- [Import users into AutoSys Workload Automation](#)

Prerequisites

To run the AgentCLI_To_JobConverter utility, the following must be installed on your computer:

- JRE 1.6 (32-bit) or higher
- Perl 5.8 or greater

Convert Agent CLI Command Jobs to AutoSys Workload Automation Jobs

The AgentCLI_To_JobConverter utility lets you convert:

- Workload Automation Agent for Informatica CLI command jobs to AutoSys Workload Automation Informatica jobs.
- CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs.

NOTE

- When you convert Workload Automation Agent for Informatica CLI command jobs to AutoSys Workload Automation Informatica jobs, the following notes apply:
 - The AgentCLI_To_JobConverter utility converts only the startWorkflow (--command=startWorkflow) CLI command jobs.
 - The JIL input file must include all the Workload Automation Agent for Informatica startWorkflow CLI command job definitions that you want to convert to AutoSys Workload Automation Informatica job definitions. If the JIL input file includes any other job definitions other than the Workload Automation Agent for Informatica startWorkflow CLI command jobs, the utility does not convert those job definitions.
- When you convert CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs, the following notes apply:
 - The AgentCLI_To_JobConverter utility converts only the RunJob (--operation=RunJob) CLI command jobs.
 - The JIL input file must include all the CA WA Agent for Microsoft SQL Server RunJob CLI command job definitions that you want to convert to AutoSys Workload Automation SQL Server Agent job definitions. If the JIL input file includes any other job definitions other than the CA WA Agent for Microsoft SQL Server RunJob CLI command jobs, the utility does not convert those job definitions.

Follow these steps:

1. Create a JIL input file that includes the Agent CLI command jobs that you want to convert to AutoSys Workload Automation jobs.
2. Run the AgentCLI_To_JobConverter.pl script.
The following interactive menu appears:
Please select from the following options:
[1] Convert Informatica Agent CLI jobs to INFORMATICA jobs
[2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs
[3] Import users into CA WAAE
[0] Exit
3. Enter any *one* of the following options:
 - [1] Convert Informatica Agent CLI jobs to INFORMATICA jobs to convert Workload Automation Agent for Informatica CLI command jobs to AutoSys Workload Automation Informatica jobs.
 - [2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs to convert CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs.
 The utility prompts you to provide the JIL input file path.
4. Enter the JIL input file path.
The Agent CLI command jobs are converted to AutoSys Workload Automation jobs. A log file that is named AgentCLI_To_JobConverter.log is created in the /opt/CA/WorkloadAutomationAE/autosys/utills/AGTCLJobConv/log (UNIX) or C:\Program Files(x86)\CA\WorkloadAutomationAE\autosys\utills\AGTCLJobConv\log (Windows) directory.

NOTE

The AgentCLI_To_JobConverter.log file includes information about the location of the following output files:

- JobDefinitions_ *JOBTYPE_JIL_input_filename*.txt—This file includes the new AutoSys Workload Automation job definitions.
- JobDeletions_ *JOBTYPE_JIL_input_filename*.txt—This file includes the job definition deletions that correspond to the new job definitions in the JobDefinitions_ *JOBTYPE_JIL_input_filename* file.
- Import_ *JOBTYPE_JIL_input_filename_users*.txt—This file includes the user entries that can be imported into AutoSys Workload Automation.

where *JOBTYPE* specifies INFORMATICA for AutoSys Workload Automation Informatica jobs or SQLAGENT for AutoSys Workload Automation SQL Server Agent jobs.

The contents of the AgentCLI_To_JobConverter.log file are similar to the following:

```
Jan 02, 2017 4:22:37 AM INFO: Parsing the input file: C:\install_test\auto\integration
\cli2job\InformaticaJobs.jil
Jan 02, 2017 4:22:37 AM INFO: Processing job: job_name
Jan 02, 2017 4:22:37 AM INFO: Generating the CLI equivalent JIL attributes
Jan 02, 2017 4:22:37 AM INFO: Writing the job definition to file
Jan 02, 2017 4:22:37 AM INFO: Writing the job deletion definition to file
Jan 02, 2017 4:22:37 AM INFO: Writing the import users to file
Jan 02, 2017 4:22:37 AM INFO: Reached end of the input file
Jan 02, 2017 4:22:37 AM INFO: Converted 1 jobs out of 1 jobs provided
Jan 02, 2017 4:22:37 AM INFO: The new jobs definition are available in C:\Program
Files (x86)\CA\Workload Automation AE\autosys\install\AgentCLI_To_JobConverter\output
\JobDefinitions_INFORMATICA_InformaticaJobs.jil
Jan 02, 2017 4:22:37 AM INFO: The jobs deletion definition are available in C:\Program
Files (x86)\CA\Workload Automation AE\autosys\install\AgentCLI_To_JobConverter\output
\JobDeletions_INFORMATICA_InformaticaJobs.jil
Jan 02, 2017 4:22:37 AM INFO: The import users definition are available in C:\Program
Files (x86)\CA\Workload Automation AE\autosys\install\AgentCLI_To_JobConverter\output
\Import_INFORMATICA_InformaticaJobs.jil_users.txt
```

Import Users into AutoSys Workload Automation

You can use the AgentCLI_To_JobConverter utility to import the Informatica or SQL Server Agent users into AutoSys Workload Automation. This utility stores the encrypted passwords that are specified in the Agent CLI command jobs so that they can be used by the AutoSys Workload Automation jobs.

NOTE

- Before you perform this procedure, ensure that the Import_INFORMATICA_ *JIL_input_filename_users*.txt or Import_SQLAGENT_ *JIL_input_filename_users*.txt file is available in the /opt/CA/WorkloadAutomationAE/autosys/utlis/AGTCLIJobConv/output (UNIX) or C:\Program Files(x86)\CA\WorkloadAutomationAE\autosys\utlis\AGTCLIJobConv/output (Windows) directory.
- The Import_INFORMATICA_ *JIL_input_filename_users*.txt file is created when you convert Workload Automation Agent for Informatica CLI command jobs to AutoSys Workload Automation Informatica jobs using the AgentCLI_To_JobConverter utility.
- The Import_SQLAGENT_ *JIL_input_filename_users*.txt file is created when you convert CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs using the AgentCLI_To_JobConverter utility.

Follow these steps:

1. Log in as the EDIT Superuser and run the AgentCLI_To_JobConverter.pl script from the Windows instance command prompt or the UNIX operating system prompt:
The following interactive menu appears:
Please select from the following options:
[1] Convert Informatica Agent CLI jobs to INFORMATICA jobs
[2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs
[3] Import users into CA WAAE
[0] Exit
2. Enter [3] Import users into CA WAAE.
The utility prompts you to provide the path to the input file (*Import_JOBTYPE_
JIL_input_filename_users.txt*) containing user information.
3. Enter the file path.
The users are imported into AutoSys Workload Automation.

Parameters

The following table lists the Workload Automation Agent for Informatica CLI command job parameters and their corresponding AutoSys Workload Automation Informatica job attributes:

| Workload Automation Agent for Informatica CLI Command Job Parameters | AutoSys Workload Automation Informatica Job Attributes |
|--|--|
| folderName | informatica_folder_name |
| workflowName | informatica_workflow_name |
| workflowInstanceName | informatica_workflow_inst_name |
| taskName | informatica_task_name |
| passOnSuccessOnly | informatica_pass_on_success |
| paramFile | informatica_param_file |
| infaTarget | informatica_target_name |
| repositoryName | informatica_repository_name |
| user | informatica_user |
| Password | -- |

The following table lists the Workload Automation Agent for Microsoft SQL Server job parameters and their corresponding AutoSys Workload Automation SQL Server Agent job attributes:

| Workload Automation Agent for Microsoft SQL Server Jobs Parameters | AutoSys Workload Automation SQL Server Agent Job Attributes |
|--|---|
| • JobName | sqlagent_jobname |
| • ServerName | sqlagent_server_name |
| • JobId | sqlagent_jobid |
| • TargetDB | sqlagent_target_db |
| • Domain | sqlagent_domain_name |
| • StepName | sqlagent_step_name |
| • UserName | sqlagent_user_name |
| • Password | -- |

The following is the list of unsupported parameters for AutoSys Workload Automation:

AutoSys Workload Automation Informatica Jobs

- verbose
- track

AutoSys Workload Automation SQL Server Agent Jobs

- track

Example: Convert CA WA Agent for Microsoft SQL Server CLI Command Jobs to AutoSys Workload Automation SQL Server Agent Jobs on Windows

This example converts CA WA Agent for Microsoft SQL Server CLI command jobs to AutoSys Workload Automation SQL Server Agent jobs.

Follow these steps:

1. Include the following two Workload Automation Agent for Microsoft SQL Server CLI RunJob command jobs in a JIL input file that is named SQLAGENTJOBS.txt:

```
insert_job : Job1
command : wrappers/mssql_call.bat --operation=RunJob --UserName=sa --
Password=0C6C30073DF47232 --JobName=SqlJob --StepName=Step1 --TargetDB=master --
ServerName=mssqlserver --Domain=mydomain
machine : sqlagent

insert_job : Job2
command : wrappers/mssql_call.bat --operation=RunJob --JobID=308C3291-2917-4A92-9A4D-
F6B6D89E5AD8 --Track=Y
machine : sqlagent
description : Autosys Job with SQL Job ID
```

2. Run the AgentCLI_To_JobConverter.pl script.

The following interactive menu appears:

Please select from the following options:

- ```
[1] Convert Informatica Agent CLI jobs to INFORMATICA jobs
[2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs
[3] Import users into CA WAAE
[0] Exit
```

3. Enter [2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs.

The utility prompts you to provide the path to the SQLAGENTJOBS.txt file.

4. Enter the path to the SQLAGENTJOBS.txt file.

The two Workload Automation Agent for Microsoft SQL Server CLI RunJob command jobs are converted to AutoSys Workload Automation jobs. A log file that is named AgentCLI\_To\_JobConverter.log is created in the C:\Program Files(x86)\CA\WorkloadAutomationAE\autosys\utils\AGTCLIJobConv\log directory. The contents of the log file are as follows:

```
Jan 13, 2017 4:21:19 AM INFO: Parsing the input file: SQLAGENTJOBS.txt
Jan 13, 2017 4:21:19 AM INFO: Processing job: Job1
Jan 13, 2017 4:21:19 AM INFO: Generating the CLI equivalent JIL attributes
Jan 13, 2017 4:21:19 AM INFO: Writing the job definition to file
Jan 13, 2017 4:21:19 AM INFO: Writing the job deletion definition to file
Jan 13, 2017 4:21:19 AM INFO: Writing the import users to file
```

```

Jan 13, 2017 4:21:19 AM INFO: Processing job: Job2
Jan 13, 2017 4:21:19 AM INFO: Generating the CLI equivalent JIL attributes
Jan 13, 2017 4:21:19 AM INFO: CLI keyword TRACK ignored!
Jan 13, 2017 4:21:19 AM INFO: Writing the job definition to file
Jan 13, 2017 4:21:19 AM INFO: Writing the job deletion definition to file
Jan 13, 2017 4:21:19 AM INFO: Writing the import users to file
Jan 13, 2017 4:21:19 AM INFO: Reached end of the input file
Jan 13, 2017 4:21:19 AM INFO: Converted 2 job(s) out of 2 job(s) provided
Jan 13, 2017 4:21:19 AM INFO: The new jobs definition are available in C:\Program
Files (x86)\CA\Workload Automation AE\autosys\utils\AGTCLIJobConv\output
\JobDefinitions_SQLAGENT_SQLAGENTJOBS.txt
Jan 13, 2017 4:21:19 AM INFO: The jobs deletion definition are available in C:
\Program Files (x86)\CA\Workload Automation AE\autosys\utils\AGTCLIJobConv\output
\JobDeletions_SQLAGENT_SQLAGENTJOBS.txt
Jan 13, 2017 4:21:19 AM INFO: The import users definition are available in C:
\Program Files (x86)\CA\Workload Automation AE\autosys\utils\AGTCLIJobConv\output
\Import_SQLAGENT_SQLAGENTJOBS.txt_users.txt

```

The AgentCLI\_To\_JobConverter.log file includes information about the location of the following output files:

- **JobDefinitions\_SQLAGENT\_SQLAGENTJOBS.txt**—This file includes the new AutoSys Workload Automation job definitions, as follows:
 

```

insert_job: Job1
job_type: SQLAGENT
machine: sqlagent
sqlagent_server_name: mssqlserver
sqlagent_step_name: Step1
sqlagent_user_name: sa@mydomain
sqlagent_jobname: SqlJob
sqlagent_target_db: master
sqlagent_domain_name: mydomain
insert_job: Job2
job_type: SQLAGENT
machine: sqlagent
sqlagent_jobid: 308C3291-2917-4A92-9A4D-F6B6D89E5AD8
description: Autosys Job with SQL Job ID
sqlagent_user_name: user@machine_0

```
- **JobDeletions\_SQLAGENT\_SQLAGENTJOBS.txt**—This file includes the job definition deletions that correspond to the new job definitions in the JobDefinitions\_SQLAGENT\_SQLAGENTJOBS.txt file, as follows:
 

```

delete_job: Job1
delete_job: Job2

```
- **Import\_SQLAGENT\_SQLAGENTJOBS.txt\_users.txt**—This file includes the user entries that can be imported into AutoSys Workload Automation, as follows:
 

```

sa@mydomain:0C6C30073DF47232
user@machine_0:0C6C30073DF47232

```

5. Log in as the EDIT Superuser and run the AgentCLI\_To\_JobConverter.pl script from the Windows instance command prompt:

The following interactive menu appears:

Please select from the following options:

- [1] Convert Informatica Agent CLI jobs to INFORMATICA jobs
  - [2] Convert SQL Server Agent CLI jobs to SQLAGENT jobs
  - [3] Import users into CA WAAE
  - [0] Exit
6. Enter [3] Import users into CA WAAE.  
The utility prompts you to provide the path to the Import\_SQLAGENT\_SQLAGENTJOBS.txt\_users.txt file.
  7. Enter C:\Program Files(x86)\CA\WorkloadAutomationAE\autosys\utils\AGTCLIJobConv\output  
Import\_SQLAGENT\_SQLAGENTJOBS.txt\_users.txt.  
The users are imported into AutoSys Workload Automation.
  8. Enter the following command at the Windows instance command prompt:  
`jil < JobDeletions_SQLAGENT_SQLAGENTJOBS.txt`  
Job1 and Job2 are deleted.
  9. Enter the following command at the Windows instance command prompt:  
`jil < JobDefinitions_SQLAGENT_SQLAGENTJOBS.txt`  
Job1 and Job2 jobs are defined in AutoSys Workload Automation.

## Application Services Jobs Overview

Application Services jobs let you manage entity beans, session beans, and MBeans, publish and consume JMS messages, invoke programs over HTTP, and run other types of Java-based workload.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

You can define the following Application Services jobs:

- **Entity Bean**Lets you create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database.
- **HTTP**Lets you invoke a program over HTTP or HTTPS in a similar way to a web browser. For example, you can use the HTTP job to invoke a CGI script, a Perl script, or a servlet. The HTTP job sends a URL over HTTP using the GET method or a form over HTTP using the POST method.
- **JMS Publish**Lets you send a message to a queue or publish a message to a topic on a JMS server.
- **JMS Subscribe**Lets you consume messages from a queue or topic on a JMS server.
- **JMX-MBean Attribute Get**Lets you query a JMX server for the value of an MBean attribute. The returned value is stored on the computer where the Application Services agent plug-in resides.
- **JMX-MBean Attribute Set**Lets you change the value of an MBean attribute on a JMX server.
- **JMX-MBean Create Instance**Lets you create an MBean on a JMX server.
- **JMX-MBean Operation**Lets you invoke an operation on an MBean on a JMX server.
- **JMX-MBean Remove Instance**Lets you remove an MBean from a JMX server.
- **JMX-MBean Subscribe**Lets you monitor an MBean for a single notification or monitor continuously for notifications.
- **POJO**Lets you instantiate a class to create a Java object and invoke a method on it. The job is restricted to classes that take constructors with no arguments (default constructors). You can use the POJO job to invoke custom Java code on a local computer.
- **RMI**Lets you set up interaction between Java objects on different computers in a distributed network. Using an RMI job, you can access a remote server and can invoke a method on a Java object.
- **Session Bean**Lets you access a session bean on an application server. This job type can make a Remote Procedure Call (RPC) to the session bean, invoke a method that defines the business logic, pass parameters to the method, and return the results as serialized Java output. You can access stateless and stateful session beans using the Session Bean job.

---

## Payload Producing and Payload Consuming Jobs

A payload producing job is a job that produces binary output that is persisted as a serialized Java object.

The following job types are payload producing jobs:

- HTTP
- JMS Subscribe

### NOTE

Ensure that the `appservices.jms.subscribe.persist` parameter is set to true in the `agentparm.txt` file of the agent, for JMS Subscribe jobs to be payload producing jobs.

- JMX-MBean Attribute Get
- JMX-MBean Attribute Set
- JMX-MBean Operation
- POJO
- RMI
- Session Bean
- Web Service

By default, the serialized Java object is stored on the agent computer in the spool directory, using the job name and a numeric suffix as the file name. You can redirect the output to a destination file.

A payload consuming job is a job that uses the output from a payload producing job as an input value for the parameter.

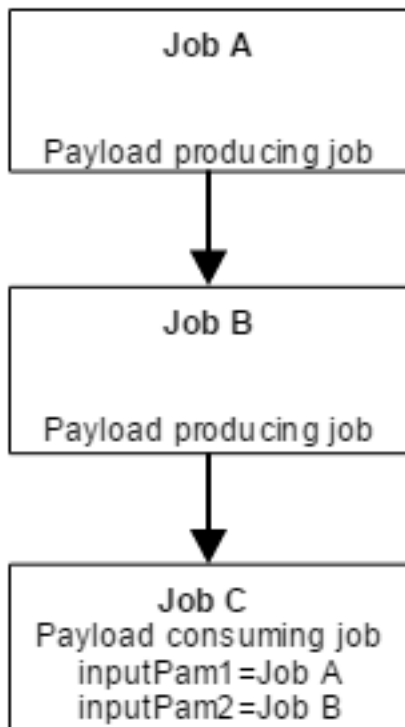
The following job types are payload consuming jobs:

- HTTP
- Entity Bean
- JMS Publish
- JMX-MBean Attribute Set
- JMX-MBean Create Instance
- JMX-MBean Operation
- POJO
- RMI
- Session Bean
- Web Service

We recommend that the payload producing job is a predecessor job to the payload consuming job although it does not need to be an immediate predecessor.

The following diagram illustrates a job flow in which a payload consuming job named Job C uses the output that the payload produces, payload producing jobs that are named Job A and Job B. In this example, Job B depends on Job A and Job C is defined to take the output from Job A as the value for the input parameter that is named `inputParm1` and the output from Job B as the value for the input parameter named `inputParm2`.



**Figure 31: Payload Producing and Payload Consuming Job Flow**

## Entity Bean Jobs Overview

An entity bean represents a data object, such as a customer, an order, or a product. Entity beans may be stored in a relational database, where each instance of the bean corresponds to a row in a database table. Each entity bean has a unique identifier that is known as a primary key, which is used to find a specific instance of the bean within the database. For example, a customer entity bean may use the customer number as its primary key.

Unlike session beans, which are destroyed after use, entity beans are persistent. You can use an entity bean under the following conditions:

- The bean represents a business entity, not a procedure. For example, you use an entity bean to represent an order and use a session bean to represent the procedure to process the order.
- The state of the bean must be stored. For example, if the bean instance terminates or the application server shuts down, the state of the bean still exists in a database.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and an entity bean residing on an application server:

**Figure 32: Entity Bean Jobs Overview**

The Entity Bean job lets you create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database. To find the entity bean, the agent uses the bean's Java Naming and Directory Interface (JNDI) name along with its finder method.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

To define an Entity Bean job, you require the following information:

- Initial context factory that is supplied by the JNDI service provider
- Service provider URL for accessing the JNDI services
- Entity bean JNDI name
- Operation type (CREATE, UPDATE, or REMOVE)
- Finder method name (UPDATE and REMOVE operation types only)

**Define an Entity Bean Job**

You can define an Entity Bean (ENTYBEAN) job to create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: ENTYBEAN**  
This attribute specifies that the job type is Entity Bean.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **bean\_name**  
This attribute specifies the JNDI name of the entity bean.
  - **initial\_context\_factory Attribute -- Specify an Initial Context Factory** This attribute specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **operation\_type Attribute -- Specify the Operation Type** This attribute specifies the operation to perform on the entity bean: CREATE, UPDATE, REMOVE.
  - **provider\_url**  
This attribute specifies the JNDI service provider URL.

2. Specify the following attributes if the `operation_type` attribute is set to CREATE:
  - **`create_name`**  
(Optional) Specifies the name of the create method.
  - **`create_parameter`**  
(Optional) Specifies create parameters to create an entity bean in a relational database on your application server.
3. Specify the following attributes if the `operation_type` attribute is set to UPDATE:
  - **`finder_name` Attribute -- Specify a Finder Method** This attribute specifies the name of the finder method.
  - **`finder_parameter` Attribute -- Specify Finder Parameters** This attribute specifies the finder parameters.
  - **`method_name` Attribute -- Specify the Method or Operation to be Invoked** This attribute specifies the method to be invoked on the application server.
  - **`modify_parameter`**  
(Optional) This attribute specifies the modify parameters.
4. Specify the following attributes if the `operation_type` attribute is set to REMOVE:
  - **`finder_name` Attribute -- Specify a Finder Method** This attribute specifies the name of the finder method.
  - **`finder_parameter`**  
(Optional) This attribute specifies the finder parameters.
5. (Optional) Specify optional Entity Bean attributes:
  - **`j2ee_user`**
  - **`job_class`**
6. (Optional) Specify common attributes that apply to all job types.  
The Entity Bean job is defined. When the job runs, it creates an entity bean, updates the property values of an existing entity bean, or removes an entity bean from the database.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

#### Example: Create an Entity Bean

Suppose that you want to create an entity bean that stores information about a customer such as the customer ID and phone number. The initial context factory that is supplied by the JNDI service provider is `weblogic.jndi.WLInitialContextFactory`. The service provider's URL is `t3://localhost:7001`, where `localhost` is the domain name of the WebLogic application server and `7001` is the ORB port. When the job runs, the entity bean instance is created.

```
insert_job: create
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
```

```

provider_url: "t3://localhost:7001"
bean_name: customer
create_name: createcustomer
operation_type: CREATE
create_parameter: String="customerid", String="800-555-0100"

```

### Example: Update an Entity Bean

Suppose that you want to update the phone number for the Acme company to 800-555-0199. The customer entity bean stores the customer ID and phone number. The primary key for the customer is the customer ID. To find the entity bean, the job uses the customer ID of Acme. When the job runs, the phone number of Acme company is changed.

```

insert_job: update
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: UPDATE
method_name: changephone
finder_name: findByPrimaryKey
finder_parameter: String="customerid"
modify_parameter: String="800-555-0199"

```

### Example: Remove an Entity Bean

Suppose that you want to remove the customer record for the Acme customer. The record is stored in the database by the customer ID. When the job runs, the row in the customer table that corresponds to the Acme customer ID is removed.

```

insert_job: remove
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: REMOVE
finder_name: findByPrimaryKey
finder_parameter: String="customerid"

```

## HTTP Jobs Overview

The HTTP job invokes a program over HTTP in a similar way to a web browser. For example, you can use the HTTP job to invoke a CGI script, a Perl script, or a servlet.

The HTTP job sends a URL over HTTP. The output of the invocation is returned in the spool file of the job and also sent back to the manager. HTTP methods GET, POST, PUT, and DELETE are supported.

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

The GET method requests data and sends the data as part of the URL. The POST method submits data and is the preferred method for sending lengthy form data.

To define an HTTP job, you require the following information:

- URL of the application server
- Program or servlet to invoke

If your company has a firewall and you must communicate through a proxy server to access a computer outside the firewall, agent configuration is required. For more information on configuring the agent for a proxy, see the [Workload Automation Agents](#) documentation.

## Consuming the HTTP Response

The response from a HTTP job can be consumed by the successor jobs for their execution. The response can contain payload, header, cookies or a combination of all. The payload is in the form of XML, JSON, or Plain Text. The header and cookies are in the form of serialized JSON.

The successor job can consume the response in either of the following two ways:

1. By directly referring the xpath, jsonpath, regex, or key\_index.  
The persisted payload and header can be consumed directly from the successor jobs by specifying them in any of the following formats.
  - `$$payload^jobname@<x>xpath</x>`
    - *jobname*  
Specifies the name of the job whose payload needs to be consumed.
    - *xpath*  
Specifies the xpath expression to be used to fetch value from the XML payload.
  - `$$payload^jobname@<j>jsonpath</j>`
    - *jobname*  
Specifies the name of the job whose payload needs to be consumed.
    - *jsonpath*  
Specifies the jsonpath expression to be used to fetch the value from the JSON payload.
  - `$$payload^jobname@<r>regex</r><g>groups</g>`
    - *jobname*  
Specifies the name of the job whose payload needs to be consumed.
    - *regex*  
Specifies the regular expression to be used to extract value from the output payload.
    - *groups*  
(Optional) Specifies the numerical group ID in any order. This group ID enables the extraction of the groups that are matched while applying the regex. The `var_group` variable applies to regex and `key_index` formats.
  - `$$header^jobname@<k>keyname</k><i>index</i>`
    - *jobname*  
Specifies the name of the job whose persisted header data is consumed.
    - *keyname*  
Specifies the header name.

### Example:

The job `test_header` when invoking the `provider_url`: `https://www.google.com` persists with the header and the output is as follows:

```
[{"Date":"Tue, 04 Feb 2020 05:12:59 GMT"}, {"Expires":"-1"}, {"Cache-Control":"private, max-age=0"}, {"Content-Type":"text/html; charset=ISO-8859-1"}, {"P3P":"CP=\"This is not a P3P policy! See g.co/vp3phelp for more info.\""}, {"Server":"gws"}, {"X-XSS-Protection":"0"}]
```

`$$header^test_header@<k date /k> $0 >>` will cause the substitution with "Tue, 04 Feb 2020 05:15:03 GMT".

2. By referring output variables which store the extracted values from the payload and header.
  - `$$jobname@<v>name</v>`

- *jobname*  
Specifies the name of the job, as specified for the \$\$payload expression.
- *name*  
Specifies the name of the output variable job blob.

### **Define Output Variables**

To define output variables, specify the [output\\_var](#) attribute in your job definition. Based on the output format (xpath for XML, jsonpath for JSON, key\_index for JSON, or regex for plain text), you can specify the output\_var attribute value in any of the following ways and the corresponding variables are extracted when the response is received by the manager:

- output\_var: var\_name="varx", var\_value="/store/book[1]/title", var\_type=payload, var\_format="xpath"
- output\_var: var\_name="vart", var\_value="[a-zA-Z]+", var\_type=payload, var\_format="regex", var\_group="<<\$1>>"

#### **NOTE**

- The var\_group specifies the numerical group ID in any order. This group ID enables the extraction of the groups that are matched while applying the regex. The var\_group variable applies to regex and key\_index formats.
- If you do not specify the var\_group variable, the entire value that is matched by the regex specified against the var\_value variable is stored as the output\_var value.
- In a job definition, you can specify the output\_var attribute multiple times. Ensure that the var\_name value is different and the var\_format value is either the same or a valid combination of values. The header and cookies var\_types do not apply to the xpath format. For example, you can specify the values as follows:

```
output_var: var_name="varx", var_value="/store/book[1]/title",
var_type=payload, var_format="xpath"
```

or

```
output_var: var_name="varj", var_value="x.store.book[0].title",
var_format="jsonpath"
output_var: var_name="vart", var_value="[a-zA-Z]+", var_type=header,
var_format="regex", var_group="<<$1>>"
```

When a HTTP job that contains the output\_var attribute completes successfully, the scheduler parses the incoming response for the elements whose value is specified in the var\_value variable and the resultant is stored as a job blob with the name that is specified in the var\_name variable.

#### **NOTE**

- While updating a job definition, if you update the output\_var attribute value, the already defined ones in the previous version are deleted from the database and the newer ones are populated. If you update the output\_var attribute to include an empty value, all the previously defined ones are deleted.
- Only after a successful job run with the current job definition, the outputpayload or outputpayloadvar type job blobs are available for consumption by the successor jobs.

You can persist the payload, header and cookies from the response in the database as a blob so that it can be used to extract values during subsequent job processing. To do this, set the [persist\\_output\\_payload](#), [persist\\_output\\_header](#) and [persist\\_output\\_cookies](#) attribute values to true in your job definition.

### **Output Payload Variable Substitution**

You can reference an output payload variable as part of the syntax of any of the following attributes:

- blob\_input
- destination\_file
- headers
- header\_filter
- j2ee\_conn\_domain
- j2ee\_conn\_origin
- j2ee\_conn\_user
- j2ee\_parameter
- j2ee\_proxy\_domain
- j2ee\_proxy\_host
- j2ee\_proxy\_origin\_host
- j2ee\_proxy\_user
- media\_type
- payload\_uri
- provider\_url
- results\_file
- success\_criteria\_json
- success\_criteria\_regex
- success\_criteria\_value
- success\_criteria\_xpath

### Examples: Define a HTTP Job to Perform a HTTP Query

The following example performs a HTTP query using the HTTP POST method. The job's output is written "\$payload^test@<r>.\*(temp).\*(dir1).\*</r><g>\<<\$1>>\<<\$2>></g>", which is \temp\dir1 directory. When the job runs, the payload is sent to the agent in plain text format as: *multi-lined text data for job \temp\dir1 blob*.

The regex ".\*(temp).\*(dir1).\*" is run on the output payload response of the already run HTTP job named test. The specified format of the groups is assigned to the destination\_file before sending it to the agent. In case of payload data, since the group is not specified, the entire text of the regular expression ".\*(temp).\*(dir1).\*" is returned.

```
insert_job: subvar1
job_type: HTTP
machine: localhost
invocation_type: POST

provider_url: "http://www.google.com/$$payload^test@<r>.*(temp).*(dir1).*</r><g>\<<$1>>\<<$2>></g>"
persist_output_payload: 1
headers: Accept-Language="en-US,en;q=0.8"
headers: Referer="catest.google.com"
headers: User-Agent="Mozilla"
blob_input: <auto_blobt>multi-lined text data for job "$
$payload^test@<r>regex=".*(temp).*(dir1).*</r>" blob</auto_blobt>compatibility: 12
```

The following example performs a HTTP query using the GET method. The job queries http://www.google.com every 30 seconds and it does wait between retrying HTTP requests. The output of the invocation is returned in the spool file of the job. The job's output is written to the /tmp/testHTTPGetJob file, the results file is overwritten when a file with the same name exists, and the global proxy defaults that are specified in the agentparm.txt file are overridden. The headers that are used in this query are: Accept-Language="en-US,en;q=0.8", "catest.google.com", and User-Agent="Mozilla".

The payload output variables are stored in regex format; the variable name is var1, the variable value is [a-zA-Z]+, and the group ID is 0. It picks the entire string that is matched by the regular expression [a-zA-Z]+ as zero group represents the entire matched string. This value is persisted as a blob in the database.

```
insert_job: HTTPQuery
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://www.google.com"
j2ee_no_global_proxy_defaults: Y
j2ee_proxy_port: 80
retry_on_failure_details: 0
results_file: /tmp/testHTTPGetJob
results_file_overwrite: true
j2ee_parameter: interval=30
headers: Accept-Language="en-US,en;q=0.8"
headers: Referer="catest.google.com"
headers: User-Agent="Mozilla"
output_var: var_name="var1",var_value="([a-zA-Z]+)", var_format="regex",
 var_type=payload, var_group="<<$0>>"
persist_output_payload: true
connection_retry: 4
```

## Generating Reports

You can use the autorep -jb command to generate reports on the outputpayload or outputpayloadvar type job blobs.

## Define an HTTP Job

You can define a HTTP job to invoke a program over HTTP.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

There are two versions of HTTP job types; namely, Pre-Version 12.0 and Version 12.0.

The jil attribute “compatibility” defines to which version HTTP job definition belongs to. By default, the compatibility value is 11 which allows only keywords from before Version 12.0. If the specified value is 12, all new keywords are allowed.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:

#### Mandatory attributes for Pre-Version 12.0 and Version 12.0:

- **job\_type: HTTP**  
This attribute specifies that the job type is HTTP.
- **machine**  
This attribute specifies the name of the machine on which the job runs.
- **provider\_url**  
This attribute specifies the host where the program you want to invoke resides.

#### Mandatory attributes for Version 12.0:

- **compatibility**  
Specifies the version of the HTTPJob. Options are as follows:



**Default: 11**

- **11**  
Specifies releases before Version 12.0
- **12**  
Specifies Version 12.0 and higher

2. (Optional) Specify optional HTTP attributes:

**Optional attributes for Pre-Version 12.0 and Version 12.0.**

- invocation\_type
- j2ee\_authentication\_order
- j2ee\_conn\_domain
- j2ee\_conn\_origin
- j2ee\_conn\_user
- j2ee\_no\_global\_proxy\_defaults
- j2ee\_proxy\_domain
- j2ee\_proxy\_host
- j2ee\_proxy\_origin\_host
- j2ee\_proxy\_port
- j2ee\_proxy\_user
- job\_class

**Optional attributes specific to Pre-Version 12.0.**

- filter
- j2ee\_parameter
- method\_name

**NOTE**

If method\_name is specified, the method and query parameters specified in the provider\_url are replaced by the value mentioned in the method\_name while running the job at the agent side.

**Optional attributes specific to Version 12.0.**

- blob\_file
- blob\_input
- compatibility
- connection\_retry
- connection\_timeout
- fail\_codes
- form\_parameter
- header\_filter
- headers
- input\_cookies
- media\_type
- output\_var
- payload\_uri
- persist\_output\_cookies
- persist\_output\_payload
- persist\_output\_header
- preemptive\_authentication

**NOTE**

If **preemptive\_authentication** is set to true, it is mandatory to specify **j2ee\_conn\_user** and **j2ee\_authentication\_order** must be empty or should contain BASIC as one of the authentication protocols.

- [results\\_file](#)
  - [results\\_file\\_overwrite](#)
  - [retry\\_on\\_failure\\_count](#)
  - [retry\\_on\\_failure\\_delay](#)
  - [retry\\_on\\_failure\\_details](#)
  - [success\\_codes](#)
  - [success\\_criteria\\_json](#)
  - [success\\_criteria\\_regex](#)
  - [success\\_criteria\\_value](#)
  - [success\\_criteria\\_xpath](#)
3. Optional) Specify common attributes that apply to all job types.  
The HTTP job is defined. When the job runs, it invokes a program over HTTP.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Examples - Define an HTTP Job**

The following examples are for the HTTP Job, mainly using various payload output formats and how you can use different attributes to match and extract content from the payload.

The following JSON payload is used for testing the JSONPath sample jobs.

```
{
 "store": {
 "book": [
 {
 "category": "reference",
 "author": "Nigel Rees",
 "title": "Sayings of the Century",
 "price": 8.95
 },
 {
 "category": "fiction",
 "author": "Evelyn Waugh",
```

```

 "title": "Sword of Honour",
 "price": 12.99
 },
 {
 "category": "fiction",
 "author": "Herman Melville",
 "title": "Moby Dick",
 "isbn": "0-553-21311-3",
 "price": 8.99
 },
 {
 "category": "fiction",
 "author": "J. R. R. Tolkien",
 "title": "The Lord of the Rings",
 "isbn": "0-395-19395-8",
 "price": 22.99
 }
],
 "bicycle": {
 "color": "red",
 "price": 19.95
 }
},
 "expensive": 10
}

```

The following XML payload is used for testing the XPATH sample jobs.

```

<?xml version="1.0" encoding="UTF-8"?>

<bookstore>

<book category="cooking">
 <title lang="en">Everyday Italian</title>
 <author>Giada De Laurentiis</author>
 <year>2005</year>
 <price>30.00</price>
</book>

<book category="children">
 <title lang="en">Harry Potter</title>
 <author>J K. Rowling</author>
 <year>2005</year>
 <price>29.99</price>
</book>

<book category="web">
 <title lang="en">XQuery Kick Start</title>
 <author>James McGovern</author>
 <author>Per Bothner</author>
 <author>Kurt Cagle</author>
 <author>James Linn</author>
 <author>Vaidyanathan Nagarajan</author>
 <year>2003</year>

```

```

 <price>49.99</price>
 </book>

 <book category="web">
 <title lang="en">Learning XML</title>
 <author>Erik T. Ray</author>
 <year>2003</year>
 <price>39.95</price>
 </book>

</bookstore>

```

The examples are classified into the following two categories:

### **Evaluating Success Criteria**

There are three different ways of evaluating the success criteria for a job. If the response is XML document, you can use `success_criteria_xpath` and if it is a JSON document you can use `success_criteria_json`. The third one `success_criteria_regex` can be used on any kind of response payload.

#### **success\_criteria\_json**

The following job is checking if the response from the web service has "Nigel Rees" as the author of the first book. If the author by the said name is found, the job is evaluated to success; else, to failure.

```

insert_job: sample1
job_type: HTTP
invocation_type: GET
machine: localhost
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstore"
compatibility: 12
success_criteria_json: store.book[0].author
success_criteria_value: "Nigel Rees"

```

The following example checks if the response contains any book authored by Nigel Rees:

```

insert_job: sample2
job_type: HTTP
invocation_type: GET
machine: localhost
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstore"
compatibility: 12
success_criteria_json: $..book[?(@.author=='Nigel Rees')].author
success_criteria_value: "Nigel Rees"

```

#### **success\_criteria\_xpath**

The following job is checking in the XML response if the author of the book in the cooking category is "Giada De Laurentiis". If such an element is found, the job goes to success; else, it fails.

```

insert_job: sample3
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstorexml"
compatibility: 12
success_criteria_xpath: //bookstore/book[@category='cooking']/author

```

```
success_criteria_value: "Giada De Laurentiis"
```

### **success\_criteria\_regex**

The following example checks if the JSON response contains any book authored by Nigel Rees.

```
insert_job: sample4
job_type: HTTP
invocation_type: GET
machine: localhost
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstore"
compatibility: 12
success_criteria_regex: "\"author\": \"Nigel Rees\""
```

The following job is checking in the XML response if there is an author by the name "Giada De Laurentiis" in the cooking category. If such an element is found, the job goes to success; else, it fails.

```
insert_job: sample5
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstorexml"
compatibility: 12
success_criteria_regex: "<book category=\"cooking\">\s*.*\s*<author>Giada De Laurentiis<\/author>"
```

### **NOTE**

Special consideration for `success_criteria_regex` must be taken for characters which are to be escaped both in Regular expression (RegEx) and JIL. If the RegEx data requires a backward slash, then it must be escaped so that it can be treated as a literal backward slash character. Similarly, backward slashes are treated as escape characters in JIL. Hence, the backward slashes must be escaped twice. A single backward slash in the data becomes four backward slashes.

- If there is a forward slash found in the JSON data, then that forward slash must be escaped with four backward slashes when specified from JIL. Otherwise the job fails.
- Essentially if the data contains "\", then it must be replaced with "\\\"."
- If the response payload is JSON document and data contains "/", it must be replaced with "\\\\", otherwise "\V".

In summary, each backslash or slash character must be immediately preceded by three backslashes (for example, \\V or \\\) when used by JIL.

### **Consuming Response Payload**

The generated response payload from the web service can be consumed using the following options:

- Output variables
- Substitution variables

### **Output Variables**

The output variables against a job can be queried by running autorep with jb option.

**Example:** autorep -jb jobname

These variables can be used just like the way global variables are used. The notation to be followed is \$`jobname@<v>title</v>`, where title here refers to the name of the output variable defined in the job "jobname".

### **Output variables using JSONPath**

In the following example we are capturing the title of the book in the variable “title” whose price is less than \$ 8.99. In case more than one title satisfies the criteria, only the first title found is stored in the variable title.

```
insert_job: sample6
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstore"
compatibility: 12
output_var:var_name="title", var_value="store.book[?(@.price < 8.99)].title" var_format="jsonpath",
var_type="payload"
```

### Output variables using XPATH

In the following example, we are capturing the author of the book in the variable “author” whose price is \$ 29.99.

```
insert_job: sample7
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstorexml"
compatibility: 12
output_var:var_name="author", var_value="//bookstore/book[price=29.99]/author", var_format="xpath",
var_type="payload"
```

### Output variables using Regular Expression

Here the regular expression specified against var\_value matches element called book belonging to "children" category. Instead of storing the entire matched text, we are interested in storing the value as "The title of the book is Harry Potter". The group "1" gives "Harry Potter" from the matched text.

```
insert_job: sample8
job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "http://wabld-linux-itcl.ibn.broadcom.net:8082/vmc/bookstorexml"
compatibility: 12
output_var:var_name="title", var_value="<book category=\"children\">\s*<title lang=\"en\">(.*?)<\/title>",
var_format="regex", var_type="payload", var_group="The title of the book is <<$1>>"
```

### Substitution Variables

We can also consume an element from the persisted response directly by specifying the JSONPath, XPATH, or RegEx. These variables can be used anywhere the way global variables are used.

#### Consuming the persisted JSON data

If we wanted to extract from the response the title of a book whose price is less than \$8.99, then the notation used is `$payload^sample6@<j>store.book[?(@.price < 8.99)].title</j>`. Here payload specifies that we are interested in parsing the response of type “payload”. This payload is produced by the job sample6 and the JSONPath expression to be applied on the response is `store.book[?(@.price < 8.99)].title`. The job sample6 needs to have `persist_output_payload` set to 1.

#### Consuming the persisted XML data

If we wanted to extract from the response the author of a book whose price is \$29.99, then the notation used is `$payload^sample7@<x>//bookstore/book[price=29.99]/author</x>`. The job sample7 must have `persist_output_payload` set to 1.

#### Consuming the persisted RegEX data

If we wanted to extract from the response the title of the book in children category, then the notation used is `$payload^sample8@<r><book category=\"children\">\s*<title lang=\"en\">(.*?)\\title</r>`. The job sample8 needs to have `persist_output_payload` set to 1.

## HTTP Job Attributes with Default Values

This article explains HTTP job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the `agentparm.txt` file.

If you specify the attribute in a job definition, it overrides the default.

The following HTTP job attributes have default values:

- **invocation\_type**  
This attribute specifies whether to send the URL over HTTP using the GET or POST method.  
**Default:** POST
- **j2ee\_conn\_origin**  
This attribute specifies the domain for NTLM connection authentication.  
**Default:** The computer name where the agent is running
- **j2ee\_no\_global\_proxy\_defaults**  
This attribute specifies whether to use the global proxy configuration specified by the proxy parameters in the `agentparm.txt` file.  
**Default:** Y (The job does not use the global proxy configuration specified by the proxy parameters in the `agentparm.txt` file.)
- **j2ee\_proxy\_domain**  
This attribute specifies the domain for proxy authentication.  
**Default:** `http.proxyDomain` agent parameter, if specified
- **j2ee\_proxy\_host**  
This attribute specifies the proxy host name to use for the request.  
**Default:** `http.proxyHost` agent parameter, if specified
- **j2ee\_proxy\_origin\_host**  
This attribute specifies the origin host name for proxy authentication.  
**Default:** The computer name where the agent is running.
- **j2ee\_proxy\_port**  
This attribute specifies the proxy port to use for the request.
- **j2ee\_proxy\_user**  
This attribute specifies the user name required for proxy authentication.  
**Default:** `http.proxyUser` agent parameter, if specified
- **persist\_output\_cookies**  
This attribute specifies whether you want to persist the cookies output in the database as a blob so that it can be consumed by the successor jobs for their execution.  
**Default:** False
- **persist\_output\_header**  
This attribute specifies whether you want to persist the header output in the database as a blob so that it can be consumed by the successor jobs for their execution.  
**Default:** False
- **persist\_output\_payload**  
This attribute specifies whether you want to persist the payload output in the database as a blob so that it can be consumed by the successor jobs for their execution.  
**Default:** False
- **preemptive\_authentication**

This attribute specifies whether or not to authenticate the application credentials using preemptive authentication.

**Default:** False

- **results\_file\_overwrite**

This attribute specifies whether you want to overwrite the results file when a file with the same name exists.

**Default:** False

- **retry\_on\_failure\_details**

The `retry_on_failure_details` specifies whether to send the failure details (indicating the failure and retry count) to the manager when the `retry_on_failure_count` attribute is set to a value that is greater than zero.

**Default:** False

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Define an HTTP Job to Send a URL Over HTTP

Several attributes in the following job definition override the default values.

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## JMS Publish and JMS Subscribe Jobs Overview

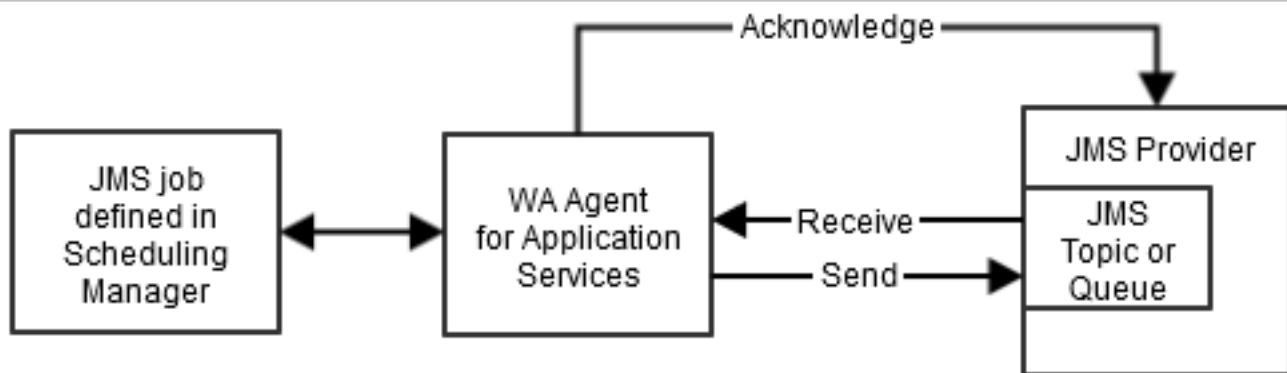
Java Message Service (JMS) is the standard for enterprise messaging that lets a Java program or component (JMS client) produce and consume messages. Messages are the objects that communicate information between JMS clients.

In a JMS system, a messaging server known as the JMS provider acts between two JMS clients (the publisher and the subscriber). Publishers send messages to the JMS provider while subscribers receive messages from the JMS provider.

The following diagram shows the functional relationship between the scheduling manager, the Workload Automation Agent for Application Services, and a JMS provider:



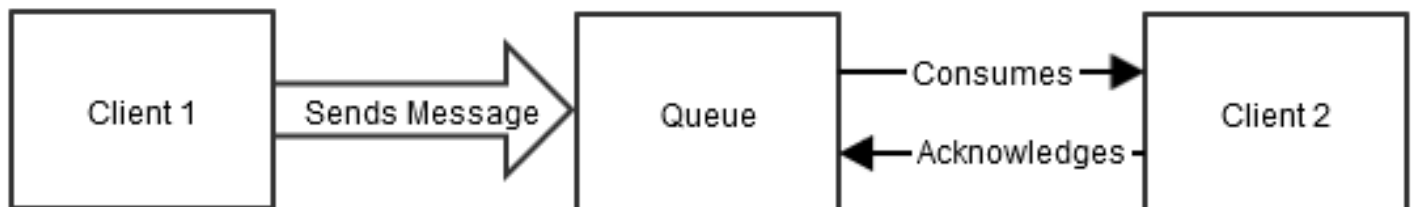
**Figure 33: Scheduling Manager sends JMS job to Queue and receives acknowledgement through Workload Automation Agent**



A queue is an object on the JMS server that holds messages sent by a client that are waiting to be consumed by another client. The queue retains a message until the message is consumed or the message expires.

The following diagram shows Client 2 (the subscriber) consuming a message that Client 1 (the publisher) sends to a queue:

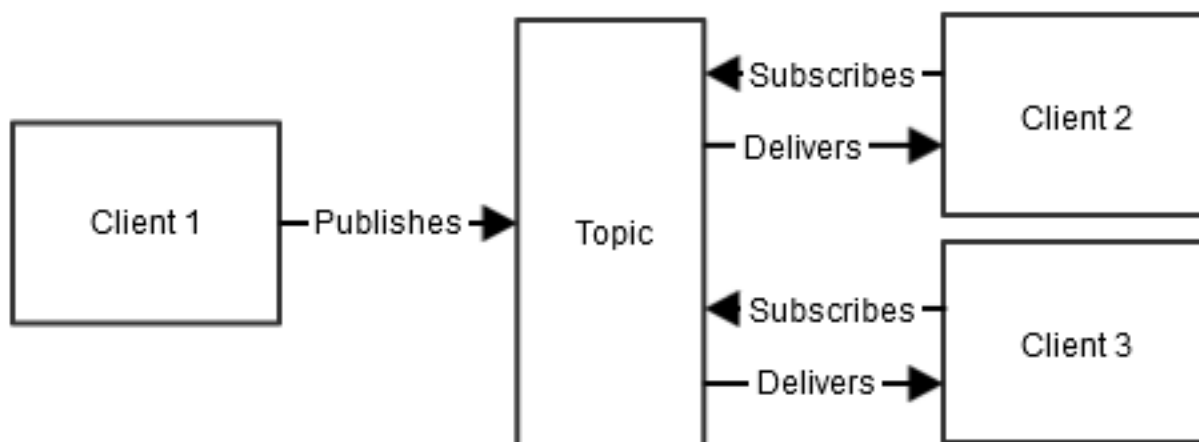
**Figure 34: Client 1 sends message to Queue, Client 2 consumes message from Queue**



A topic is an object a client uses to specify the target of the messages it produces and the source of the messages it consumes. A client acquires a reference to a topic on a JMS server, and sends messages to that topic. When messages arrive for that topic, the JMS provider is responsible for notifying all clients.

The following diagram shows two subscribers, Client 2 and Client 3, subscribed to a topic that the publisher, Client 1, publishes to:

**Figure 35: Client 1 publishes a Topic to which Clients 2 and 3 subscribe**



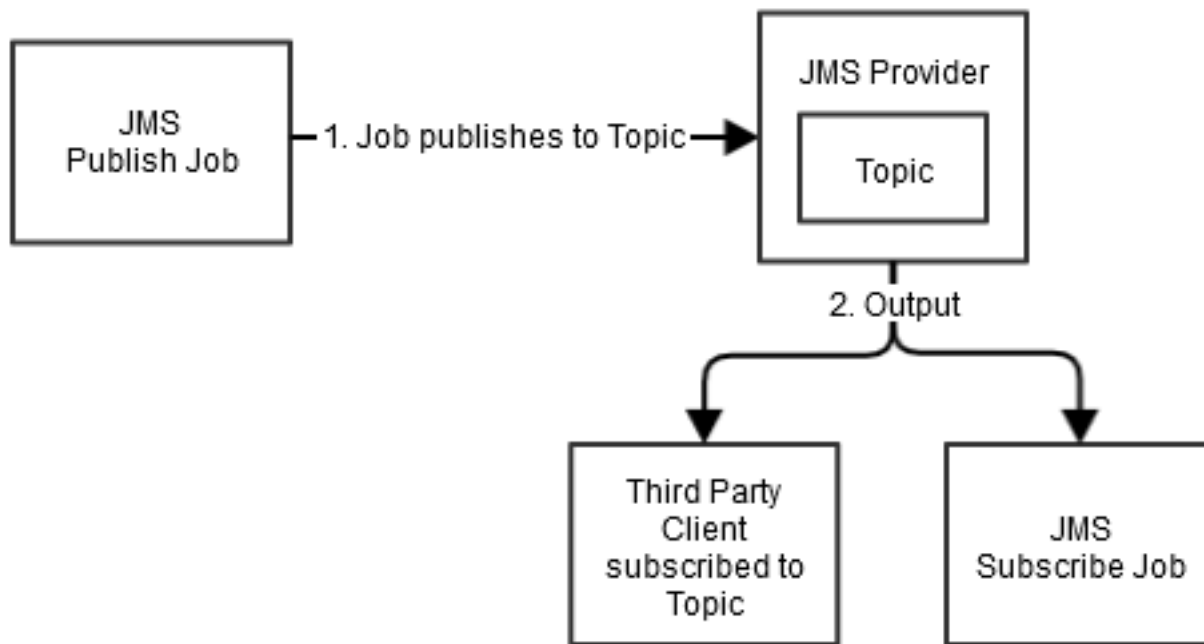
A JMS Publish job lets you send a message to a queue or publish a message to a topic. Using a JMS Publish job to publish to a topic, you can broadcast a message to any topic subscriber. A third-party client can consume this message, or a JMS Subscribe job can listen for a particular message (using a filter).

**NOTE**

To run these jobs, your system requires CA WA Agent for UNIX, Linux, or Windows and CA WA Agent for Application Services.

The following diagram shows a JMS Publish job scenario:

**Figure 36: Topic delivered to third party subscriber and JMS subscribe job**

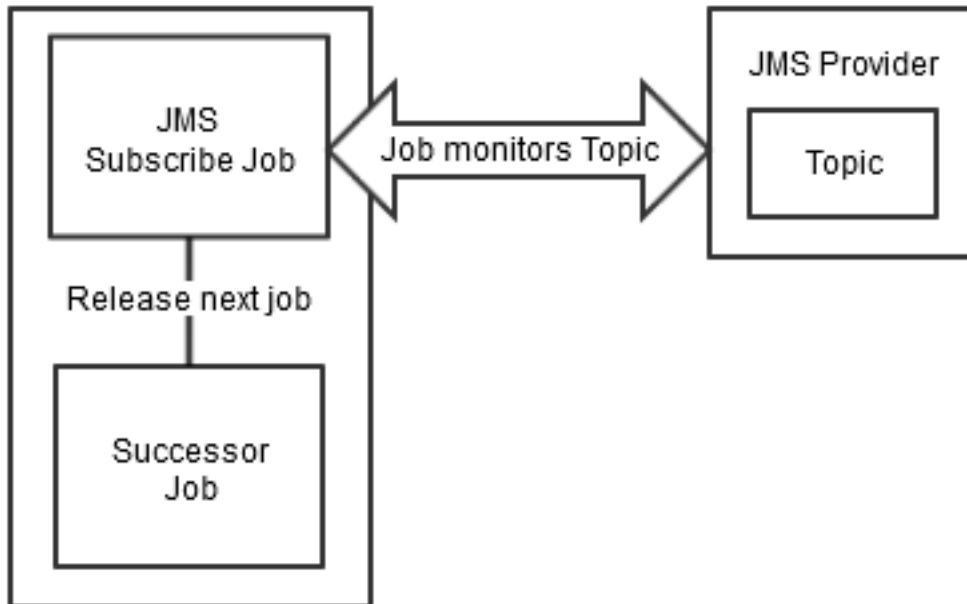


A JMS Subscribe job lets you consume messages from a queue or topic. Using a filter that you define within the job definition, the agent monitors the topic or queue output for specific data. The scheduling manager then sends the message that meets the filter criteria to a destination file you specify. You can define the job to continuously monitor JMS messages.

**NOTE**

To run these jobs, your system requires CA WA Agent for UNIX, Linux, or Windows and CA WA Agent for Application Services.

The following diagram shows a JMS Subscribe job scenario:

**Figure 37: When JMS subscriber receives Topic, it releases successor job**

To define a JMS Publish or JMS Subscribe job, you require the following information:

- Initial context factory supplied by the Java Naming and Directory Interface (JNDI) service provider
- JMS provider URL for accessing the JNDI services
- Connection factory JNDI name that looks up the referenced topic or queue
- JNDI name of the topic or queue on the JMS server
- (JMS Publish jobs only) Java class of the JMS message to send or publish
- (JMS Publish jobs only) Message to send to a queue or publish to a topic

## Define a JMS Publish Job

You can define a JMS Publish (JMSPUB) job to send a message to a queue or publish a message to a topic.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

JMS Publish jobs support the TextMessage and ObjectMessage message types.

### NOTE

The StreamMessage, BytesMessage, and MapMessage message types are not supported.

## Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMSPUB**  
This attribute specifies that the job type is JMS Publish.
  - **machine** This attribute specifies the name of the machine on which the job runs.
  - **connection\_factory** This attribute specifies the connection factory JNDI name. The connection factory contains all the bindings that are needed to look up the referenced topic or queue. JMS jobs use the connection factory to create a connection with the JMS provider.

- **destination\_name** This attribute specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.
  - **initial\_context\_factory** This attribute specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory that is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **j2ee\_parameter** This attribute specifies the message to send to a queue or publish to a topic.
  - **message\_class** This attribute specifies the Java class of the JMS message.
  - **provider\_url** This attribute specifies the JNDI service provider URL.
2. (Optional) Specify optional JMS Publish attributes:
- **j2ee\_user**
  - **job\_class**
  - **use\_topic**
  - **correlation\_id**
  - **reply\_to**
  - **destination\_user**
3. (Optional) Specify common attributes that apply to all job types.
- The JMS Publish job is defined. When the job runs, it sends a message to a queue or publishes a message to a topic.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and then import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

#### Example: Publish a Message to a WebSphere MQ server

This example publishes the message "this is my message" to the queue that is named Queue1. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The URL of the service provider is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the cyberuser@host JNDI user name to gain access to the connection factory named ConnectionFactory.

```
insert_job: publish
job_type: JMSPUB
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue1
use_topic: FALSE
```

```
message_class: String
j2ee_parameter: java.lang.String="this is my message"
j2ee_user: cyberuser@host
```

**NOTE**

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

**Define a JMS Subscribe Job**

You can define a JMS Subscribe (JMSSUB) job to consume messages from a queue or topic.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

JMS Subscribe jobs support the TextMessage and ObjectMessage message types.

**NOTE**

The StreamMessage, BytesMessage, and MapMessage message types are not supported.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMSSUB**  
This attribute specifies that the job type is JMS Subscribe.
  - **machine** This attribute specifies the name of the machine on which the job runs.
  - **connection\_factory** This attribute specifies the connection factory JNDI name. The connection factory contains all the bindings that are required to look up the referenced topic or queue. JMS jobs use the connection factory to create a connection with the JMS provider.
  - **destination\_name** This attribute specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.
  - **initial\_context\_factory** This attribute specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **provider\_url** This attribute specifies the JNDI service provider URL.
2. (Optional) Specify optional JMS Subscribe attributes:
  - **continuous**
  - **destination\_file**
  - **filter**
  - **j2ee\_user**
  - **job\_class**
  - **job\_terminator**
  - **use\_topic**
  - **correlation\_id**
  - **destination\_user**
3. (Optional) Specify common attributes that apply to all job types.  
The JMS Subscribe job is defined. When the job runs, it consumes messages from a queue or topic.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and then import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Example: Monitor a Queue on a WebLogic Application Server**

This example continuously monitors the queue named Queue1 (residing on WebLogic) for a message matching the filter criteria. The consumed messages from the queue are stored in the file /export/home/user1/outputfile1. The URL of the service provider is t3://172.24.0.0:7001, where 172.24.0.0 is the IP address of the WebLogic Application server and 7001 is the port.

```
insert_job: monitor
job_type: JMSSUB
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://172.24.0.0:7001"
connection_factory: ConnectionFactory
destination_name: Queue1
continuous: Y
filter: abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
use_topic: FALSE
destination_file: /export/home/user1/outputfile1
j2ee_user: cyberuser@host
```

In this example, the regular expression that is used as the filter criteria can be defined as follows:

```
abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
```

- **abc\s**  
Specifies the text abc, followed by white space.
- **...\s**  
Specifies any three characters, followed by white space.
- **[a-zA-Z]+\s**  
Specifies at least one letter, followed by white space.
- **Filter![\sa-z0-9]+**  
Specifies the text Filter!, followed by at least one of the following characters: white space or digit or lowercase letter.

**Example:** abc vvv B Filter! 95

## JMS Publish and JMS Subscribe Job Attributes with Default Values

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the `agentparm.txt` file.

If you specify the attribute in a job definition, it overrides the default.

The following JMS job attributes have default values:

- **continuous (JMS Subscribe jobs only)** This attribute specifies whether the job monitors the topic or queue continuously for messages. **Default:** N (The job immediately checks for the condition and completes.)
- **job\_terminator (JMS Subscribe jobs only)** This attribute specifies whether to terminate the job if its containing box fails or terminates. **Default:** n (The job does not terminate if its containing box fails or terminates.)
- **destination\_file (JMS Subscribe jobs only)** This attribute specifies the output destination file for the consumed messages. **Default:** spooldir agent parameter, if specified
- **use\_topic**  
This attribute specifies whether to send or publish messages to a topic or queue. **Default:** FALSE (The job sends or publishes messages to a queue.)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Specify Optional Attributes in a JMS Subscribe Job

The `continuous` and `destination_file` attributes in the following job definition override the default values.

This example continuously monitors the queue named `Queue` (residing on `WebLogic`) for a message matching the filter criteria. The consumed messages from the queue are stored in the file `/export/home/user1/outputfile1`. The URL of the service provider is `t3://172.24.0.0:7001`, where `172.24.0.0` is the IP address of the `WebLogic` Application server and `7001` is the ORB port.

```
insert_job: monitor

job_type: JMSSUB

machine: appagent

initial_context_factory: weblogic.jndi.WLInitialContextFactory

provider_url: "t3://172.24.0.0:7001"

connection_factory: ConnectionFactory

destination_name: Queue

continuous: Y

filter: abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+

use_topic: FALSE
```

```
destination_file: /export/home/user1/outputfile1

j2ee_user: cyberuser
```

In this example, the regular expression that is used as the filter criteria can be defined as follows:

```
abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
```

- **abc\s**  
This attribute specifies the text abc, followed by white space.
- **...\s**  
This attribute specifies any three characters, followed by white space.
- **[a-zA-Z]+\s**  
This attribute specifies at least one letter, followed by white space.
- **Filter![\sa-z0-9]+**  
This attribute specifies the text Filter!, followed by at least one of the following: white space or digit or lowercase letter.

**Example:** abc vvv B Filter! 95

## JMX Jobs Overview

Java Management Extension (JMX) technology is included in the Java Standard Edition (SE) platform, version 5 and higher. JMX lets you remotely access applications, using a Remote Method Invocation (RMI) connector, for monitoring and management purposes.

JMX jobs let you access a remote JMX server that advertises MBeans. An MBean is a managed bean (Java object) that represents an application, a device, or any resource that you want to manage. An MBean contains a set of attributes and a set of operations that can be invoked. Some MBeans can send out notifications, for example, when an attribute changes.

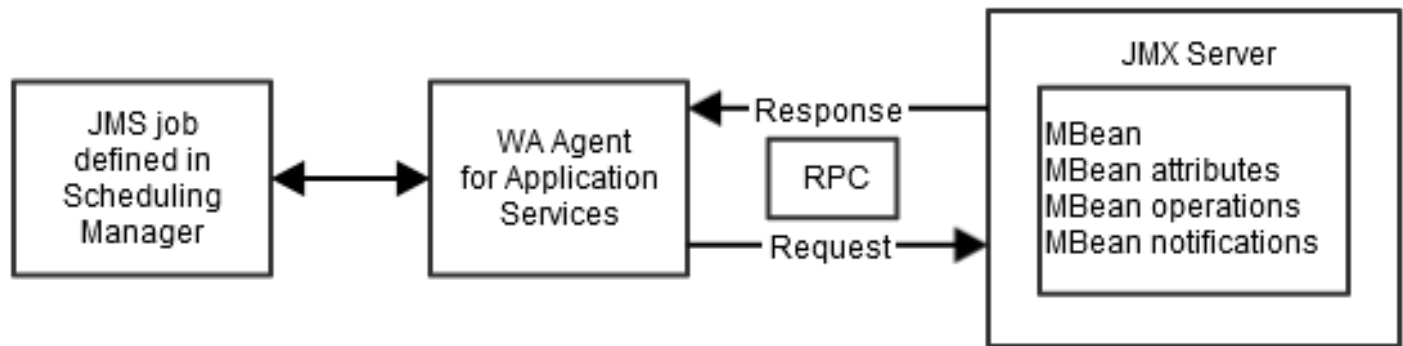
### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

Consider an MBean named Config that represents configuration of an application. The configuration parameters within that application are represented in Config by a set of attributes. Getting the attribute that is named cachesize, for example, returns the current value of the cachesize. Setting the value updates the cachesize. The Config MBean can send out a notification every time the cachesize changes. An operation that is named update, for example, can save changes to the configuration parameters.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and the JMX server:



**Figure 38: JMX jobs access remote JMX server that advertises MBeans**

The JMX jobs provide support for getting and setting JMX MBean attributes, invoking JMX MBean operations, subscribing to MBean notifications, and creating and removing instances of MBeans on a JMX server.

You can define the following six types of JMX jobs:

- JMX-MBean Attribute Get
- JMX-MBean Attribute Set
- JMX-MBean Create Instance
- JMX-MBean Operation
- JMX-MBean Remove Instance
- JMX-MBean Subscribe

The JMX-MBean Attribute Set, JMX-MBean Create Instance, and JMX-MBean Operation jobs support calls to MBeans that can involve passing parameters. Each parameter can be an actual value or a serialized Java object passed by another job. When the JMX-MBean Operation job invokes an operation on an MBean that passes parameters, the parameters are passed to the MBean and the returned serialized Java object is stored on the agent computer in the spool directory or in a destination file you specify.

To define JMX jobs, you require a URL to connect to the JMX server using an RMI connector.

### Define a JMX-MBean Attribute Get Job

You can define a JMX-MBean Attribute Get (JMXMAG) job to query a JMX server for the value of an MBean attribute. The returned value is stored on the computer where the agent resides. You can specify a success pattern to determine the success or failure of the job. If the returned attribute value matches the success pattern, the job completes successfully; otherwise, it fails.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXMAG**  
This attribute specifies that the job type is JMX-MBean Attribute Get.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **mbean\_attr Attribute -- Specify the MBean Attribute to Query or Set** This attribute specifies the name of the MBean attribute that you want to query.

- **mbean\_name Attribute -- Specify the Name of the MBean** This attribute specifies the full object name of an MBean.
- **URL**

This attribute specifies the URL to connect to the JMX server using an RMI connector.

2. (Optional) Specify optional JMX-MBean Attribute Get attributes:

- [job\\_class](#)
- [jmx\\_user](#)
- [success\\_pattern](#)

3. (Optional) Specify common attributes that apply to all job types.

The JMX-MBean Attribute Get job is defined. When the job runs, it queries a JMX server for the value of an MBean attribute.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

#### Example: Query a JMX Server for the Value of an MBean Attribute

Suppose that you want to know the value of the cachesize attribute for the Config MBean. The URL for the JMX server is service:jmx:rmi:///jndi/rmi://localhost:9999/server, where localhost is the host name and 9999 is the port number.

```
insert_job: query
job_type: JMXMAG
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_attr: cachesize
```

#### Define a JMX-MBean Attribute Set Job

You can define a JMX-MBean Attribute (JMXMAS) Set job to change the value of an MBean attribute on a JMX server. You can specify a set value for the attribute or can use the serialized Java object passed by another job. When the attribute is set, the job returns the original attribute value as output. You can specify a success pattern to determine the success or failure of the job. If the output of the job matches the success pattern, the job completes successfully; otherwise, it fails.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXMAS**  
This attribute specifies that the job type is JMX-MBean Attribute Set.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **mbean\_attr Attribute -- Specify the MBean Attribute to Query or Set** This attribute specifies the name of the MBean attribute that you want to set.
  - **mbean\_name**  
This attribute specifies the full object name of an MBean.
  - **URL**  
This attribute specifies the URL to connect to the JMX server using an RMI connector.
2. (Optional) Specify optional JMX-MBean Attribute Set attributes:
  - **jmx\_parameter**
  - **jmx\_user**
  - **job\_class**
  - **success\_pattern**
3. (Optional) Specify common attributes that apply to all job types.  
The JMX-MBean Attribute Set job is defined. When the job runs, it changes the value of an MBean attribute on a JMX server.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Change the Value of an MBean Attribute**

Suppose that you want to set the value of the State attribute of the cdc.jmx.SimpleDynamic MBean to the serialized Java object returned by a JMX-MBean Attribute Set job named size.

```
insert_job: change

job_type: JMXMAS

machine: appagent

URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"
```

```
mbean_name: "DefaultDomain:index=1,type=cdc.jmx.SimpleDynamic"

mbean_attr: State

jmx_parameter: payload_job=size

condition: success(size)
```

## Define a JMX-MBean Create Instance Job

You can define a JMX-MBean Create Instance (JMXMC) job to create an MBean on a JMX server.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXMC**  
This attribute specifies that the job type is JMX-MBean Create Instance.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **class\_name**  
This attribute specifies the fully qualified Java class of the MBean object.
  - **mbean\_name**  
This attribute specifies the full object name of an MBean.
  - **URL**  
This attribute specifies the URL to connect to the JMX server using an RMI connector.
2. (Optional) Specify optional JMX-MBean Create Instance attributes:
  - **jmx\_parameter**
  - **jmx\_user**
  - **job\_class**
3. (Optional) This attribute specifies the common attributes that apply to all job types.  
The JMX-MBean Create Instance job is defined. When the job runs, it creates an MBean on a JMX server.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

### Example: Create an MBean Instance on a JMX Server

Suppose that you want to create an MBean instance on a JMX server. The job uses the `cdc.jmx.SimpleDynamic` class. The constructor of the class takes a single string parameter with the value "Hello".

```
insert_job: create

job_type: JMXMC

machine: appagent

URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"

mbean_name: "DefaultDomain:index=CreateIns1,type=cdc.jmx.SimpleDynamic"

class_name: cdc.jmx.SimpleDynamic

jmx_parameter: java.lang.String="Hello"
```

### Define a JMX-MBean Operation Job

You can define a JMX-MBean Operation (JMXMOP) job to invoke an operation on an MBean. You can specify one or more parameter values to pass to the operation. You can specify a success pattern to determine the success or failure of the job. If the output of the operation matches the success pattern, the job completes successfully; otherwise, it fails.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXMOP**  
This attribute specifies that the job type is JMX-MBean Operation.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **mbean\_name**  
This attribute specifies the full object name of an MBean.
  - **mbean\_operation**

This attribute specifies the operation to be invoked.

– **URL**

This attribute specifies the URL to connect to the JMX server using an RMI connector.

2. (Optional) This attribute specifies the optional JMX-MBean Operation attributes:

- [jmx\\_parameter](#)
- [jmx\\_user](#)
- [job\\_class](#)
- [success\\_pattern](#)

3. (Optional) This attribute specifies the common attributes that apply to all job types.

The JMX-MBean Operation job is defined. When the job runs, it invokes an operation on an MBean.

## NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

## Example: Invoke an Operation on an MBean

Suppose that you want to invoke the resetmem operation on the config MBean to reset the value of the memory parameter to 50.

```
insert_job: reset
job_type: JMXMOP
machine: agent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_operation: resetmem
jmx_parameter: Integer=50
```

## Example: Pass Payload Producing Output as Input to Payload Consuming Job

Suppose that you want to use a JMX-MBean Operation job to invoke a method on an MBean and pass the output of the method as input to another JMX-MBean Operation job.

In this example, the first job, test\_JMXMOP2a, is a payload producing job. It takes a single input parameter and invokes the reset method on the MBean. The output of this job is stored as a serialized Java object on the computer where the agent resides.

The second job, test\_JMXMOP2b, is a payload consuming job. It takes two input parameters: the string "Hello" and the serialized Java object produced by the first job. The two input parameters are passed to the reset method, which is invoked on the MBean.

```
insert_job: test_JMXMOP2a
machine: localhost
job_type: JMXMOP
url: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:type=SimpleStandard,index=1"
mbean_operation: reset
jmx_parameter: String="Hello"
insert_job: test_JMXMOP2b
machine: localhost
job_type: JMXMOP
url: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:type=SimpleStandard,index=1"
mbean_operation: reset
jmx_parameter: String="Hello", payload_job=test_JMXMOP2a
condition: S(test_JMXMOP2a)
```

## Define a JMX-MBean Remove Instance Job

You can define a JMX-MBean Remove Instance (JMXMREM) job to remove an MBean from a JMX server.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXMREM**  
This attribute specifies that the job type is JMX-MBean Remove Instance.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **mbean\_name**  
This attribute specifies the full object name of an MBean.
  - **URL**  
This attribute specifies the URL to connect to the JMX server using an RMI connector.
2. (Optional) Specify optional JMX-MBean Remove Instance attributes:
  - **jmx\_user**
  - **job\_class**
3. (Optional) This attribute specifies the common attributes that apply to all job types.  
The JMX-MBean Remove Instance job is defined. When the job runs, it removes an MBean from a JMX server.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Remove an MBean Instance from a JMX Server

Suppose that you want to remove an MBean instance.

```
insert_job: remove
job_type: JMXMREM
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"
mbean_name: "DefaultDomain:index=CreateIns1,type=cdc.jmx.SimpleDynamic"
```

### Define a JMX-MBean Subscribe Job

You can define a JMX-MBean Subscribe (JMXSUB) job to monitor an MBean for a single notification or monitor continuously for notifications. You can filter the notifications the job monitors by attributes or by type of notifications.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JMXSUB**  
This attribute specifies that the job type is JMX-MBean Subscribe.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **mbean\_name**  
This attribute specifies the full object name of an MBean
  - **URL**  
This attribute specifies the URL to connect to the JMX server using an RMI connector.
2. (Optional) This attribute specifies the optional JMX-MBean Subscribe attributes:



- [continuous](#)
  - [filter](#)
  - [filter\\_type](#)
  - [jmx\\_user](#)
  - [job\\_class](#)
  - [job\\_terminator](#)
3. (Optional) This attribute specifies the common attributes that apply to all job types. The JMX-MBean Subscribe job is defined. When the job runs, it monitors an MBean for a single notification or continuously for notifications.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Monitor for a Change to a Specific MBean Attribute

Suppose that you want to monitor for a change to the cachesize attribute of the MBean named Config. The job filters the notifications the MBean sends by attribute. When the cachesize attribute changes, the job completes.

```
insert_job: change
job_type: JMXSUB
machine: agent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
filter_type: Attributes
filter: cachesize
```

#### Example: Monitor for Changes to Any MBean Attribute

Suppose that you want to set up continuous monitoring for changes to any attribute of the MBean named Config. Each time an attribute changes, an alert is written to the scheduler log file.

```
insert_job: change
job_type: JMXSUB
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
```

```
continuous: Y
filter_type: Types
filter: jmx.attribute.change
```

## JMX Subscribe Job Attributes with Default Values

This article explains JMX Subscribe job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following JMX Subscribe job attributes have default values:

- **continuous**  
This attribute specifies whether the job monitors the MBean continuously for notifications.  
**Default:** N (The job immediately checks for the condition and completes.)
- **filter\_type**  
This attribute specifies whether to filter notifications by attribute or by notification type.  
**Default:** Attributes (The job filters notifications by attribute.)
- **job\_terminator**  
This attribute specifies whether to terminate the job if its containing box fails or terminates.  
**Default:** n (The job does not terminate if its containing box fails or terminates.)
- **destination\_file** This attribute specifies the output destination file for the Java serialized object produced by the job.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the following topic: [Manage JIL](#).

**Default:** spooldir agent parameter, if specified

## Example: Monitor for Changes to Any MBean Attribute

The continuous attribute in the following job definition overrides the default value.

Suppose that you want to set up continuous monitoring for changes to any attribute of the MBean named Config. Each time an attribute changes, an alert is written to the scheduler log file.

```
insert_job: change
job_type: JMXSUB
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
continuous: Y
filter_type: Types
filter: jmx.attribute.change
```

## POJO Jobs Overview

A Plain Old Java Object (POJO) is a Java object that follows the Java Language Specification only. All Java objects are POJOs.

The POJO job lets you instantiate a class to create a Java object and invoke a method on it. The job is restricted to classes that take constructors with no arguments (default constructors).

**NOTE**

To run these jobs, your system requires the following agents:

- Workload Automation Agent for UNIX, Linux, or Windows
- Workload Automation Agent for Application Services or Workload Automation Agent for Web Services

You can use the POJO job to invoke custom Java code on a local computer. POJO jobs support method calls that can involve passing parameters. The parameters can be actual values or a serialized Java object passed by another job. When the POJO job invokes a method on an object, the parameters, if any, are passed to the object and the returned values are stored in a Java serialized object file.

To define a POJO job, you require the class name and method you want to call on the instantiated object.

**NOTE**

If you use custom Java code, verify that the required JAR file is in the jars subdirectory of the agent installation directory.

**Define a POJO Job**

You can define a POJO job to create a Java object instance with no arguments, invoke a method on the object instance, and store the output of the method.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and either Workload Automation Agent for Application Services or Workload Automation Agent for Web Services.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: POJO**  
This attribute specifies that the job type is POJO.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **class\_name**  
This attribute specifies the Java class to instantiate.
  - **method\_name**  
This attribute specifies the Java method to call on the instance of the Java object.
2. (Optional) Specify optional POJO attributes:
  - **destination\_file**
  - **j2ee\_parameter**
  - **job\_class**
3. (Optional) Specify common attributes that apply to all job types.  
The POJO job is defined. When the job runs, it creates a Java object instance with no arguments, invokes a method on the object instance, and stores the output of the method.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes

specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.

- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Invoke a Method on a Java Object Instance

Suppose that you want to define a POJO job that creates a Java String with value "5" and calls the parseInt method with the created Java String object as an argument. The parseInt method returns a Java Integer object.

```
insert_job: ignore
job_type: POJO
machine: appagent
class_name: java.lang.Integer
method_name: parseInt
j2ee_parameter: java.lang.String=5
```

## RMI Jobs Overview

Remote Method Invocation (RMI) is the Java version of a Remote Procedure Call (RPC), which is a technology that lets a program request a service from another program located in another address space. That address space could be on the same computer or on a different one.

RMI jobs let you set up interaction between Java objects on different computers in a distributed network. Using an RMI job, you can access a remote server and can invoke a method on a Java object. A method is a programmed procedure that is defined as a part of a Java class.

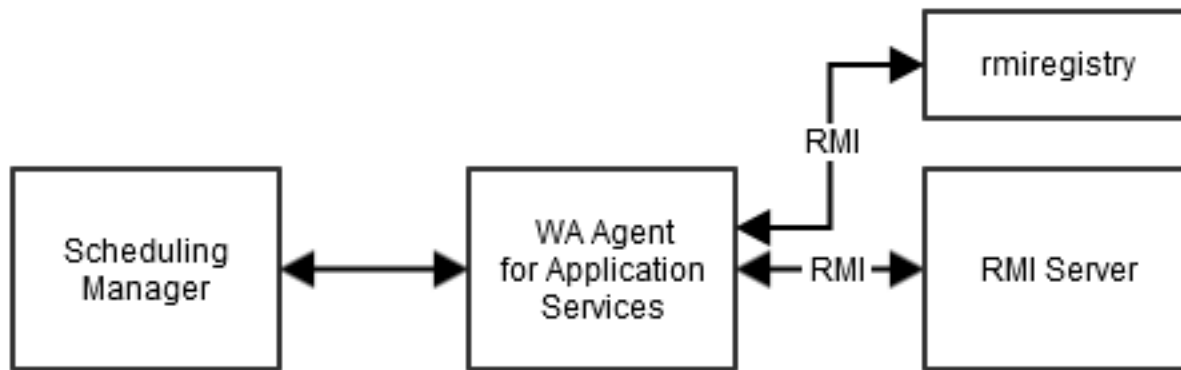
### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

RMI jobs support method calls to remote objects that can involve passing parameters. The parameters can be actual values or a serialized Java object passed by another job. When the RMI job invokes a method on an object that passes parameters, the parameters are passed to the remote object and the returned serialized Java object is stored on the agent computer in the spool directory or in a destination file you specify.

RMI uses a naming or directory service to locate the remote object on the remote server. To define an RMI job, you require the naming class of the Java object you want to invoke a method on. That naming class takes a name that is a java.lang.String in URL format.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and an RMI Server:

**Figure 39: RMI jobs that run on agents make RMI requests to remote servers**

## Define a RMI Job

You can define an RMI job to call a method on a remote server and store the output of the method.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: JAVARMI**  
This attribute specifies that the job type is RMI.
  - **machine**  
This attribute specifies the name of the machine on which the job runs.
  - **method\_name**  
This attribute specifies the method of the remote Java class to invoke.
  - **remote\_name**  
This attribute specifies the reference location of the object you want to invoke a method on.
2. (Optional) Specify optional RMI attributes:
  - **destination\_file**
  - **j2ee\_parameter**
  - **job\_class**
3. (Optional) This attribute specifies common attributes that apply to all job types.  
The RMI job is defined. When the job runs, it calls a method on a remote server and stores the output of the method.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Define a Job to Start a Remote Server Immediately

Suppose that you want to invoke a method that starts a remote server using remote object activation. You want the server to start immediately.

```
insert_job: start
job_type: JAVARMI
machine: appagent
remote_name: "rmi://remotehost/Test"
method_name: startserver
j2ee_parameter: String="now"
```

## Session Bean Jobs Overview

A session bean represents business logic or action to be taken (for example, charging a credit card or adding items to an online shopping cart).

Unlike entity beans, which are stored in a database, session beans might be destroyed after each use. For example, when a session bean is invoked to perform credit card validation, the application server creates an instance of that session bean, performs the business logic to validate the credit card transaction, and then destroys the session bean instance after the credit card transaction has been validated.

You can use a session bean under the following conditions:

- The bean represents a procedure and not a business entity. For example, you use a session bean to encrypt data or add items to an online shopping cart.
- The state of the bean does not have to be kept in permanent storage. For example, when the bean instance terminates or the application server shuts down, the state of the bean is no longer required.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and a session bean residing on an application server:

**Figure 40: Session bean job uses RPC to access session bean on remote server**



The Session Bean job lets you access a session bean on an application server. This job type can make a Remote Procedure Call (RPC) to the session bean, invoke a method that defines the business logic, pass parameters to the

method, and have the results returned as serialized Java output. The output can be stored on the agent computer as text in the spool file or as a serialized Java object in the spool directory or a destination file you specify.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

You can access stateless and stateful session beans using the Session Bean job. The job acts in a similar way for both types of beans. For both stateful and stateless beans, you can specify parameters to pass to the method. When you define a stateful session bean, however, you must specify parameters to define the bean. After the method is invoked, the agent destroys the stateful bean.

Use a stateless Session Bean job to invoke a single instance of a method on the bean, such as encrypting data or sending an email to confirm an order. Use a stateful Session Bean job to invoke the same method on the bean multiple times, such as adding multiple items to an online shopping cart.

A Session Bean job requires a dedicated connection between the agent and the application server. To define a Session Bean job, you require the following information:

- Initial context factory supplied by the Java Naming and Directory Interface (JNDI) service provider
- Service provider URL for accessing the JNDI services
- Session bean JNDI name
- Method to be invoked

## Define a Session Bean Job

You can define a Session Bean (SESSBEAN) job to access a stateless or stateful session bean, invoke a method on the bean, and return the results.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SESSBEAN**  
This attribute specifies that the job type is Session Bean.
  - **machine** This attribute specifies the name of the machine on which the job runs.
  - **bean\_name** This attribute specifies the JNDI name of the session bean.
  - **initial\_context\_factory** This attribute specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **method\_name** This attribute specifies the method to be invoked on the application server.
  - **provider\_url** This attribute specifies the JNDI service provider URL.
2. Specify the following attributes to access a stateful session bean:
  - **create\_method** This attribute specifies the name of the create method.
  - **create\_parameter** This attribute specifies the create parameters.
3. (Optional) Specify optional Session Bean attributes:

- [destination\\_file](#)
- [j2ee\\_parameter](#)
- [j2ee\\_user](#)
- [job\\_class](#)

4. (Optional) Specify common attributes that apply to all job types.

The Session Bean job is defined. When the job runs, it accesses a stateless or stateful session bean, invokes a method on the bean, and returns the results.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Invoke a Method on a Stateless Session Bean

Suppose that you want to invoke the reverse method on the CybEJBTestBean stateless session bean. The reverse method has one parameter, with type java.lang.String and value a23. The output from the reverse method is saved in the C:\Makapt15 file. The initial context factory that is supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The URL of the service provider is corbaloc:iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere application server and 2809 is the ORB port. When the job runs, the output of the reverse method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"
```

#### Example: Invoke a Method on a Stateful Session Bean

Suppose that you want to access a stateful session bean for an online shopping cart. The createaddbook method creates the Shoppingcart stateful bean during the job. The addbook method adds books to the shopping cart using the ISBN number of the book. In this example, the Session Bean job adds two books to the shopping cart with ISBN numbers 1551929120 and 1582701709. When the job runs, two books are added to the shopping cart.

```
insert_job: addbook
job_type: SESSBEAN
```



```

machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
bean_name: Shoppingcart
create_method: createaddbook
method_name: addbook
create_parameter: String="ISBN"
j2ee_parameter: Integer[2]="1551929120,1582701709"

```

## Session Bean Job Attributes with Default Values

This article explains Session Bean job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Session Bean job attributes have default values:

- **create\_method**This attribute specifies the name of the create method.  
**Default:** create
- **destination\_file**This attribute specifies the output destination file for the Java serialized object produced by the job. **Default:** spooldir agent parameter, if specified.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

## Example: Invoke a Method on a Stateful Session Bean

The create\_method attribute in the following job definition overrides the default value.

Suppose that you want to access a stateful session bean for an online shopping cart. The createaddbook method creates the Shoppingcart stateful bean during the job. The addbook method adds books to the shopping cart using the ISBN number of the book. In this example, the Session Bean job adds two books to the shopping cart with ISBN numbers 1551929120 and 1582701709. When the job runs, two books are added to the shopping cart.

```

insert_job: addbook
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
bean_name: Shoppingcart
create_method: createaddbook
method_name: addbook
create_parameter: String="ISBN"
j2ee_parameter: Integer[2]="1551929120,1582701709"

```

## Box Jobs Overview

A Box job (or box) is a container of other jobs. You can use it to organize and control process flow. The box itself performs no actions, although it can trigger other jobs to run. An important feature of this type of job is that boxes can contain other

boxes. You can use boxes to contain other boxes that contain jobs that are related by starting conditions or other criteria. This feature lets you group the jobs and operate on them in a logical manner.

Box jobs are powerful tools for organizing, managing, and administering large numbers of jobs that have similar starting conditions or complex logic flows. Knowing how and when to use boxes is often the result of some experimentation.

For example, assume that you want to schedule a group of jobs to start running when a File Watcher job completes successfully. Instead of making each job dependent on the File Watcher job, you can create a Box job that depends on the File Watcher job, remove the File Watcher job dependency from the individual jobs, and put all of those jobs in the box. When the File Watcher job completes successfully, the Box job starts, which in turn starts all the jobs it contains.

## Start Conditions for Box Jobs

When no other start conditions are specified at the job level, a job in a box runs when the starting conditions for the box are satisfied. When several jobs in a box do not have job-level starting conditions, they all run in parallel. When any job in a box changes state, other jobs verify if they are eligible to start running.

When the [priority](#) attribute is set for jobs in a box, they are processed in order of priority, highest to lowest.

Jobs in boxes run only once for each box execution. If you specify multiple start times for a job during one box processing cycle, only the first start time is used. This behavior prevents jobs in boxes from inadvertently running multiple times.

AutoSys Workload Automation starts a job when the current time matches, or is later than, the specified start time.

In addition to explicit starting conditions, jobs in boxes have the implicit condition that the box job itself is running.

This condition means that jobs in a box start only if the box job is running. However, if a job in a box starts and the box job is stopped, the started job runs to completion.

### NOTE

Use caution when putting a job with more than one time-related starting condition in a box. For example, assume that a job that runs at 15 and 45 minutes past the hour is put in a box that runs at the start of every hour. The first time the box starts, the job runs at 15 minutes past the hour. A future start is then issued for 45 minutes past the hour, by which time the box has completed. As a result, the job will not run until the box runs again at the start of the next hour. Then, the job runs when the box starts because it is past its start time. The job runs, another future start job is issued for 15 minutes past the hour, the box completes, and the cycle repeats itself.

## Basic Box Job Concepts

A *box* is a container of jobs with similar starting conditions (date and time conditions or job dependency conditions). Use boxes to group jobs with similar scheduling parameters, not to group jobs organizationally. For example, you can group jobs that run daily at 1:00 a.m. in a box and assign them a daily start condition. However, you should not group various account processing jobs with diverse starting conditions in the same box.

### Default Box Job Behavior

The following default rules apply to boxes:

- Jobs in a box run only once for each box execution.
- Jobs in a box start only if the box itself has a status of RUNNING.
- Boxes are used primarily for jobs with the same starting conditions.
- A box that is used to group sequential jobs can contain up to 1,000 jobs.
- A box remains in RUNNING state until all the jobs it contains have run.
- A box returns a status of SUCCESS when all the jobs it contains have run and returned a status of SUCCESS.
- A box returns a status of FAILURE when all the jobs it contains have run and one or more of the jobs has returned a status of FAILURE.
- A box runs until it reaches a status of SUCCESS or FAILURE.
- Using the sendevent command to change the state of a box to INACTIVE changes the state of all the jobs it contains to INACTIVE.
- Using the sendevent command to change the state of a job in a box to INACTIVE affects the box's completion status as if the INACTIVE job returned a status of SUCCESS.

### **Box Job Recommendations**

Because all jobs in a box change status when a box starts running, you may want to use boxes to implement job cycle behavior. However, placing jobs in a box to achieve this behavior can affect your system adversely because the job status changes put a larger load on the scheduler when the box starts running.

Do not put jobs in a box solely to run reports on all of them. When you run autorep on a box, the command generates a report about the box and all the jobs it contains (unless you use the -L0 option).

#### **NOTE**

Job names can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen (-). You cannot include spaces in a job name.

### **How a Box Runs**

When a box starts running, the status of all the jobs it contains (including subboxes) changes to ACTIVATED, which means they are eligible to run. Because of this status change, jobs in boxes do not retain their statuses from previous box cycles.

When a box starts running, the system performs the following actions:

- Analyzes each job for additional starting conditions.
- Starts all jobs with no additional starting conditions and without any implied order or priority.
- Maintains jobs with additional starting conditions in the ACTIVATED state until those additional dependencies are met.
- Maintains the box in the RUNNING state as long as there are jobs in it with ACTIVATED or RUNNING status.
- Changes the status of the job directly from ACTIVATED to INACTIVE if its containing box is terminated before the job starts.

#### **NOTE**

- Jobs in a box cannot start unless the box has a status of RUNNING. However, after a job starts running, it runs to completion even if the box is stopped.
- When the box is scheduled to run, the statuses of ON\_NOEXEC jobs in the box change to ACTIVATED. If the box is terminated before the jobs start, the jobs return to ON\_NOEXEC status.
- When a box that is in ON\_NOEXEC status is scheduled to run, the status of the box changes to RUNNING. Jobs in the box with more starting conditions remain in the ACTIVATED state until those additional conditions are met. Once AutoSys Workload Automation bypasses the last scheduled job in the box, the box status returns to ON\_NOEXEC.
- You can issue the JOB\_ON\_NOEXEC event to an ACTIVATED job that is within a running box. The job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting

conditions of the job are met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.

- You can issue the JOB\_OFF\_NOEXEC event for a job, it places the job in the INACTIVE, ACTIVATED, or SUCCESS status based on the job's existing status while in non-execution mode. If the job is in a box, the scheduler does not evaluate the overall box status. If you send the JOB\_OFF\_NOEXEC to a box, all jobs in the box (including all jobs contained in lower level boxes within the box) are reset to the INACTIVE, ACTIVATED, or SUCCESS status.
- You can issue the sendevent command to change the status of a job that is in the ON\_NOEXEC status to either the INACTIVE or SUCCESS status while remaining in non-execution mode. When you change the status of a non-execution job to SUCCESS, the scheduler completes the job's run and schedules the downstream dependent jobs. Changing the status of a non-execution box job to INACTIVE also changes the status of all the jobs in that box job to INACTIVE in non-execution mode. Changing the status of a non-execution box job to the RUNNING status starts the box job.

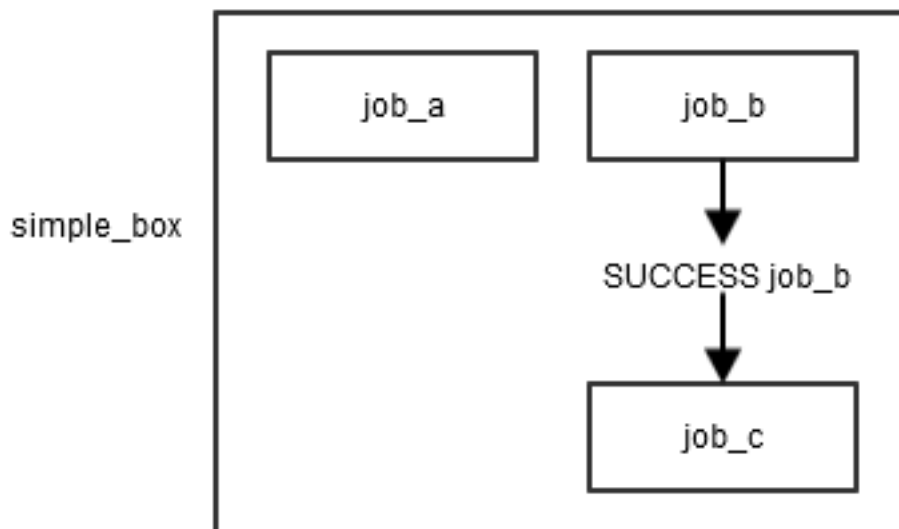
After all the jobs in a box have completed successfully, the box completes with a status of SUCCESS. The status of the box and the jobs it contains remain unchanged until the next time the box runs.

If a box changes to TERMINATED state (for example, if a user sends a KILLJOB event), it stays in TERMINATED state until the next time it is started, regardless of any later state changes of the jobs it contains.

### Example: Simple Box Job

The following illustration shows a box that is named `simple_box` that contains three jobs (`job_a`, `job_b`, and `job_c`). `job_a` and `job_b` have no starting conditions. The starting condition for `job_c` is the success of `job_b`.

**Figure 41: simple box job with no defined contingencies**



When `simple_box` starts running, the status of all the jobs changes to ACTIVATED. Because `job_a` and `job_b` have no additional starting conditions, they start running. When `job_b` completes successfully, `job_c` starts. When `job_c` completes successfully, the box completes with a SUCCESS status.

If `job_b` fails, `job_c` does not start but remains in the ACTIVATED state. Because no contingency conditions have been defined, `simple_box` continues running, waiting for the default completion criteria (that all jobs in the box have run) to be met.

### **How Job Status Changes Affect Box Status**

If a box that is not running contains a job that changes status because of a `FORCE_STARTJOB` or `CHANGE_STATUS` event, the new job status could change the status of its containing box. A status change for the box could then trigger the start of downstream jobs that depend on the box.

If a box contained only one job, and the job changed status, the box status would change as shown in the following table:

Current Box Status	New Job Status	New Box Status
SUCCESS	TERMINATED or FAILURE	FAILURE
FAILURE	INACTIVE or SUCCESS	SUCCESS
FAILURE	FAILURE	Box status does not change
INACTIVE	INACTIVE or SUCCESS	SUCCESS
INACTIVE	TERMINATED or FAILURE	FAILURE
TERMINATED	Any change	Box status does not change

If another job depends on the status of the box, the status change could trigger the job to start. If the box status does not change, dependent jobs are not affected.

#### **WARNING**

If the box contains other jobs in addition to the job that changed status, the status of the box is evaluated again according to the success or failure conditions assigned to the box (either the default or user-assigned). Any jobs in the box with a status of `INACTIVE` are ignored when the status of the box is being re-evaluated. For example, consider an `INACTIVE` box that contains four jobs, all with a status of `INACTIVE` (this is typical of a newly created box). If one of the jobs is forced to start and completes successfully, the status of the box changes to `SUCCESS` even though none of the other jobs ran. Simultaneously, if the status of the same job is being updated to `INACTIVE` and all the other jobs inside the box are already in `INACTIVE` status, the box status is re-evaluated and returns a `SUCCESS` status as it ignores all the jobs that are in `INACTIVE` status.

### **How the Status of a Job Changes When Added Into a Box**

The following conditions apply when you add jobs into a box:

- When you insert a job to add it into a box that is not in the `ON_NOEXEC` status, AutoSys Workload Automation places the job in the `INACTIVE` status.
- When you update the `box_name` attribute of a job to add it into a box that is not in the `ON_NOEXEC` status, AutoSys Workload Automation does not change the status of the job.
- When you insert a job to add it into a box that is in the `ON_NOEXEC` status, AutoSys Workload Automation places the job in the `ON_NOEXEC` status.
- When you update the `box_name` attribute of a job to add it into a box that is in the `ON_NOEXEC` status, AutoSys Workload Automation places the job in the `ON_NOEXEC` status.
- If a job is in the `ON_ICE` status, you cannot update the `box_name` attribute of the job to add it into a box that is in the `ON_NOEXEC` status. You must first send the `JOB_OFF_ICE` event to ensure that the scheduler calculates the next start time of the job.
- When you insert a job to add it into a box that is in the `RUNNING` status (whether it was previously set in the `ON_NOEXEC` status or not), AutoSys Workload Automation does the following:
  - Issues the `ALERT` event for the box. AutoSys Workload Automation populates the event text with the name of the job that was added into the running box.

**NOTE**

AutoSys Workload Automation sends the ALERT event only when you add jobs into a box that is in the RUNNING status.

- Sets the run number of the job equal to the run number of the box job
- Sets the job status to ACTIVATED
- Sends the STARTJOB event to schedule the job
- Sends a warning to the client indicating that the job was added to a running box and the STARTJOB event was issued
- When you update the box\_name attribute of a job to add it into a box that is in the RUNNING status, AutoSys Workload Automation does the following:
- Issues the ALERT event for the box. AutoSys Workload Automation populates the event text with the name of the job that was added into the running box.

**NOTE**

AutoSys Workload Automation sends the ALERT event only when you add jobs into a box that is in the RUNNING status.

- Updates the status and run number of the job and sends the STARTJOB event to schedule the job if:
  - The job is not in the STARTING, RUNNING, or ON\_ICE status
  - The current run number of the job does not exceed the run number of the box
- Sends a warning to the client indicating that the job was added to a running box and whether the STARTJOB event was issued. If no STARTJOB event was issued, the warning specifies the reason.

**How the Status of a Job Changes When Removed Out of a Box**

The following conditions apply when you remove jobs out of a box:

- When you update the box\_name attribute of a job that is in the ACTIVATED status to remove it out of a box that is not in the ON\_NOEXEC status, AutoSys Workload Automation places the job in the INACTIVE status.
- When you update the box\_name attribute of a job that is in the ACTIVATED status to remove it out of a box that is in the ON\_NOEXEC status, AutoSys Workload Automation places the job in the ON\_NOEXEC status with the equivalent behavior of being set to the INACTIVE status.
- When you delete a job or update the box\_name attribute of a job to remove it from a box that is in the RUNNING status, AutoSys Workload Automation issues an ALERT event for the box. AutoSys Workload Automation populates the event text with the name of the job that was removed from the running box.

**Box Job Attributes and Terminators**

The following sections describe how to use various job attributes to control the behavior of box jobs and the jobs they contain.

**Control How AutoSys Workload Automation Evaluates a Box Job Completion State**

By default, AutoSys Workload Automation evaluates the completion state of a box job when all of the jobs that are contained in that box job complete. You can control how AutoSys Workload Automation evaluates the completion state of box jobs by specifying the optional [box\\_success](#) and [box\\_failure](#) attributes when you define box jobs. These attributes

define the conditions under which a box job succeeds (enters the SUCCESS state) and the conditions under which it fails (enters the FAILURE state). You can define conditions that meet *any* of the following criteria:

- **Specify that a job that is contained in the box job enters a certain state**  
AutoSys Workload Automation evaluates the overall success or failure of the box job when the specified job enters the specified state, regardless of the states of other jobs that are contained in the box job. The condition is not met if the specified job is not scheduled. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.
- **Specify that a job that is not contained in the box job enters a certain state**  
AutoSys Workload Automation evaluates the overall success or failure of the box job when *one* of the jobs that is contained in the box job completes after the specified job (which is not contained in the box) enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.
- **Specify that a job that is defined on an external instance enters a certain state**  
AutoSys Workload Automation evaluates the overall success or failure of the box job when one of the jobs that is contained in the box completes after the specified external job enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified external job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the box job and it remains in the RUNNING state.
- **Specify a global variable**  
AutoSys Workload Automation evaluates the overall success or failure of the box job when *one* of the jobs that is contained in the box job completes *after* the global variable is defined and set to the value that is specified in the condition.  
The condition is not met if all of the jobs that are contained in the box job complete before the global variable is set to the specified value. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.

#### NOTE

To define a global variable and store it in the database, issue a [SET\\_GLOBAL event](#) using the sendevent command.

#### Example: Set the Success of a Specific Job in a Box Job as the Success Condition for that Box Job

This example defines a box job that is named box\_a, sets the success of the job that is named job\_a as the success condition for box\_a, and defines the jobs that are named job\_a, and job\_b as jobs that are contained in box\_a.

```
insert_job: box_a

job_type: b

box_success: success(job_a)

insert_job: job_a

box_name: box_a

command: sleep 15

machine: machine1

insert_job: job_b

box_name: box_a
```

```
command: sleep 60

machine: machine1
```

AutoSys Workload Automation evaluates the success of box\_a when job\_a completes, regardless of the state of job\_b. box\_a enters the SUCCESS state when job\_a enters the SUCCESS state.

The success condition is not met when job\_a enters a completion state other than SUCCESS (such as FAILURE or TERMINATED). In this case, AutoSys Workload Automation evaluates the overall failure according to the default behavior (after job\_b also completes) because the job definition does not specify a failure condition. box\_a enters the FAILURE state because the default failure condition for box\_a was met when job\_a entered the FAILURE state.

### Example: Set the Failure of an External Job as the Failure Condition for a Box Job

This example defines a box job that is named box\_a, sets the failure of the external job that is named job\_c^ACE as the failure condition for box\_a, and defines the jobs that are named job\_a and job\_b as jobs that are contained in box\_a.

```
insert_job: box_a

job_type: b

box_failure: failure(job_c^ACE)

insert_job: job_a

box_name: box_a

command: sleep 60

machine: machine1

insert_job: job_b

box_name: box_a

command: sleep 300

machine: machine1
```

AutoSys Workload Automation evaluates the overall failure of box\_a when either job\_a or job\_b completes. box\_a enters the FAILURE state when job\_c from the external instance that is named ACE enters the FAILURE state before AutoSys Workload Automation evaluates box\_a.

The failure condition is not met when *one* of the following situations occur:

- Job\_c enters a completion state other than SUCCESS (such as FAILURE or TERMINATED), regardless of the completion state of job\_a or job\_b.
- Job\_c enters the FAILURE state *after* both job\_a and job\_b complete.

In these cases, AutoSys Workload Automation evaluates the overall success of box\_a according to the default behavior (after job\_a and job\_b complete) because the job definition does not specify a the box\_success attribute. box\_a enters



the SUCCESS state when job\_a and job\_b both complete and enter the SUCCESS state. box\_a remains in a RUNNING status when either job\_a or job\_b enters a state other than success because the neither the failure condition specified in the box\_failure attribute nor the default success condition were met.

**Example: Set the Failure of a Specific Job that is not contained in the Box Job as the Success Condition for that Box Job**

This example defines a box job named box\_a, sets the success of a job that is outside of box\_a and is named job\_d as the success condition for box\_a, and defines the jobs that are named job\_a and job\_b as jobs that are contained in box\_a.

```
insert_job: box_a

job_type: b

box_success: failure(job_d)

insert_job: job_a

box_name: box_a

command: sleep 60

machine: machine1

insert_job: job_b

box_name: box_a

command: sleep 300

machine: machine1

insert_job: job_d

command: sleep 5

machine: machine2
```

AutoSys Workload Automation evaluates the overall success of box\_a when either job\_a or job\_b completes. box\_a enters the SUCCESS state when job\_d enters the FAILURE before the job that triggers the evaluation completes.

The success condition is not met when job\_d enters a terminal state other than FAILURE, regardless of whether job\_a or job\_b completes. In this case AutoSys Workload Automation evaluates the overall failure of box\_a according to the default behavior (after job\_a and job\_b complete) because the job definition does not specify the box\_failure attribute. box\_a enters the FAILURE state when the completion state of either job\_a or job\_b is not SUCCESS. box\_a remains in the RUNNING state when both job\_a and job\_b complete and enter the SUCCESS state because the success condition was not met.

**Example: Set a Global Variable as the Failure Condition for a Box Job**

This example defines a box job that is named box\_a, sets the global variable that is named TEST with a value of ABC as the failure condition for box\_a, and defines the jobs that are named job\_a and job\_b as jobs that are contained in box\_a.

```

insert_job: box_a

job_type: b

box_failure: v(TEST) = ABC

insert_job: job_a

box_name: box_a

command: sleep 300

machine: machine

insert_job: job_b

box_name: box_a

command: sleep 600

machine: machine

```

AutoSys Workload Automation evaluates the overall failure of box\_a when either job\_a or job\_b completes, regardless of the state of the other job that is contained in the box. box\_a enters the FAILURE state when the global variable named TEST evaluates to value ABC before job\_a or job\_b completes.

The failure condition is not met when one of the following situations occurs:

- TEST evaluates to a value other than ABC, regardless of the completion state of job\_a or job\_b.
- TEST evaluates to ABC after both job\_a and job\_b complete.

In these cases, AutoSys Workload Automation evaluates the overall success of box\_a according to default behavior (after job\_a and job\_b complete) because the job definition does not specify the box\_success attribute. box\_a enters the SUCCESS state when job\_a and job\_b complete and enter the SUCCESS state. box\_a remains in the RUNNING state when either job\_a or job\_b enter a completion state other than SUCCESS because neither the failure condition specified in the box\_failure attribute nor the default success condition were met.

## Time Conditions in a Box

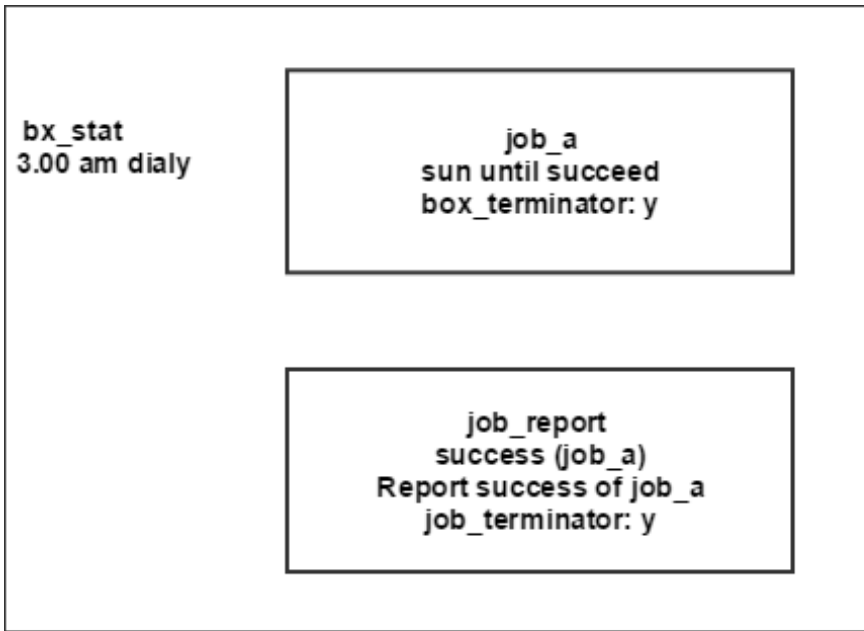
Each job in a box runs only once each time the box runs. Therefore, do not define more than one time attribute for any job in a box because the job only runs the first time. If you want to put a job in a box, but you also want it to run more than once, you must define multiple start time conditions for the box itself, and must define no time conditions for the job.

### NOTE

If the box is running simultaneously with the job, assign the start time for a job. When the box starts next time, the job also starts immediately.

### Example: Define Time Conditions for a Box Job

The following example shows a scenario with time conditions in a box. Specifically, this example shows that at 3:00 a.m. a box job bx\_stat starts running, which causes job\_a to start running. If job\_a is successful, job\_report runs and is successful.



In the illustration, `job_a` is defined to run repeatedly until it succeeds; `job_report` has one starting condition, the success of `job_a`.

At 3:00 a.m., `bx_stat` starts running, which causes `job_a` to start running. If `job_a` is successful, `job_report` runs and also succeeds.

If `job_a` fails, it will not be able to run again until the next time the box starts because jobs run only once per box execution. In this situation, the following conditions occur:

- Job `job_report` is still ACTIVATED while it waits for the success of `job_a`, and the status of the box is RUNNING.
- Because `job_a` is defined as a box terminator, the box then enters into a TERMINATED state.
- This change also terminates job `job_report` because its `job_terminator` attribute is set to y.
- Box `bx_stat` is now in a state that permits it to run again at 3:00 a.m. the following day.

If `job_a` was not defined as a box terminator, the box remains in RUNNING state indefinitely.

## Force the Job or the Box to Stop Running

You can use the following attributes in the job definition of a job in a box to force the job or the box to stop running:

- **box\_terminator: y**  
This attribute specifies that if the job completes with a FAILURE or TERMINATED status, the box terminates. Define more conditions for the other jobs in the box in case the box is terminated.
- **job\_terminator: y**  
This attribute specifies that if the box that contains the job completes with a FAILURE or TERMINATED status, the job terminates. Ensure that you add this attribute to each job definition that you want to terminate upon box failure.

### NOTE

- If a job defined with the `job_terminator` attribute is in ON\_NOEXEC status, the job does not terminate when the box fails.
- If a job defined with the `box_terminator` attribute is in ON\_NOEXEC status, then AutoSys Workload Automation bypasses the job and the box that contains the job does not terminate.

## Force Jobs in a Box to Start

You can use the sendevent command to send a FORCE\_STARTJOB event to force a job to start, even if its starting conditions have not been met.

You can also execute the FORCE\_STARTJOB command by selecting the Force Start Job button in the Job Activity Console, which is part of the Web UI.

Do not issue the sendevent command to force start a box while the jobs in the box itself or jobs belonging to boxes within the box (at a higher box level) are executing. If you want to re-execute jobs in a box run, we recommend that you follow any one of the following methods as appropriate after all scheduled jobs at all levels in the box have executed:

- To fix the current run of the box, force start the individual jobs that are stuck or otherwise did not execute properly in the order of the highest box level (bottom of the box tree) to the lowest box level (top of the box tree). This ensures that the scheduler propagates the completed box statuses all the way to the top of the box tree.
- To execute a new clean run of the entire box, issue the CHANGE\_STATUS event to INACTIVE against the level zero box to reset the status of all jobs in all levels of the box and then use the sendevent command to send a FORCE\_STARTJOB event against the level zero box.

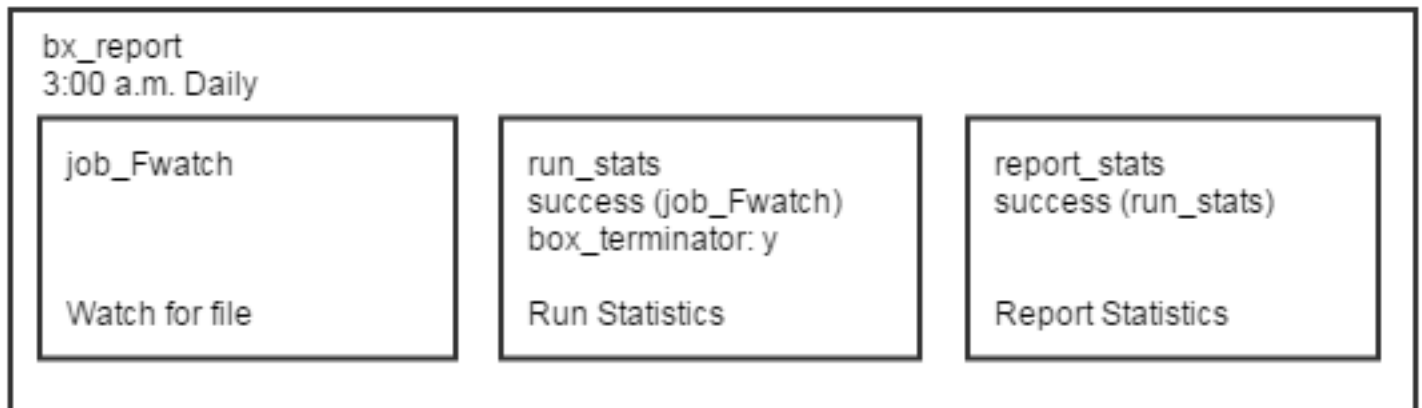
### Example: Force a Job in a Box to Start

This example defines a sendevent command that sends a FORCE\_STARTJOB event to force a job in a box to run. You could use the following command to force the job run\_stats to start:

```
sendevent -E FORCE_STARTJOB -J run_stats
```

In the following illustration, the box job bx\_report contains three jobs (job\_Fwatch, run\_stats, and report\_stats). If the job run\_stats fails, the bx\_report box job terminates because run\_stats has a box\_terminator attribute. If you force start run\_stats, and it completes successfully, report\_stats would still not start because the box it is in is not running.

**Figure 42: bx\_report box fails if run\_stats job fails**



## Box Job Flow Examples

This section contains examples that explain the flow of box jobs and the jobs they contain. These scenarios provide a clearer understanding of box job flow concepts.

---

## Default Box Success and Box Failure

This scenario describes the default job flow for box job success and failure.

The box job `do_statistics` runs every day at 3:00 a.m. It contains three jobs:

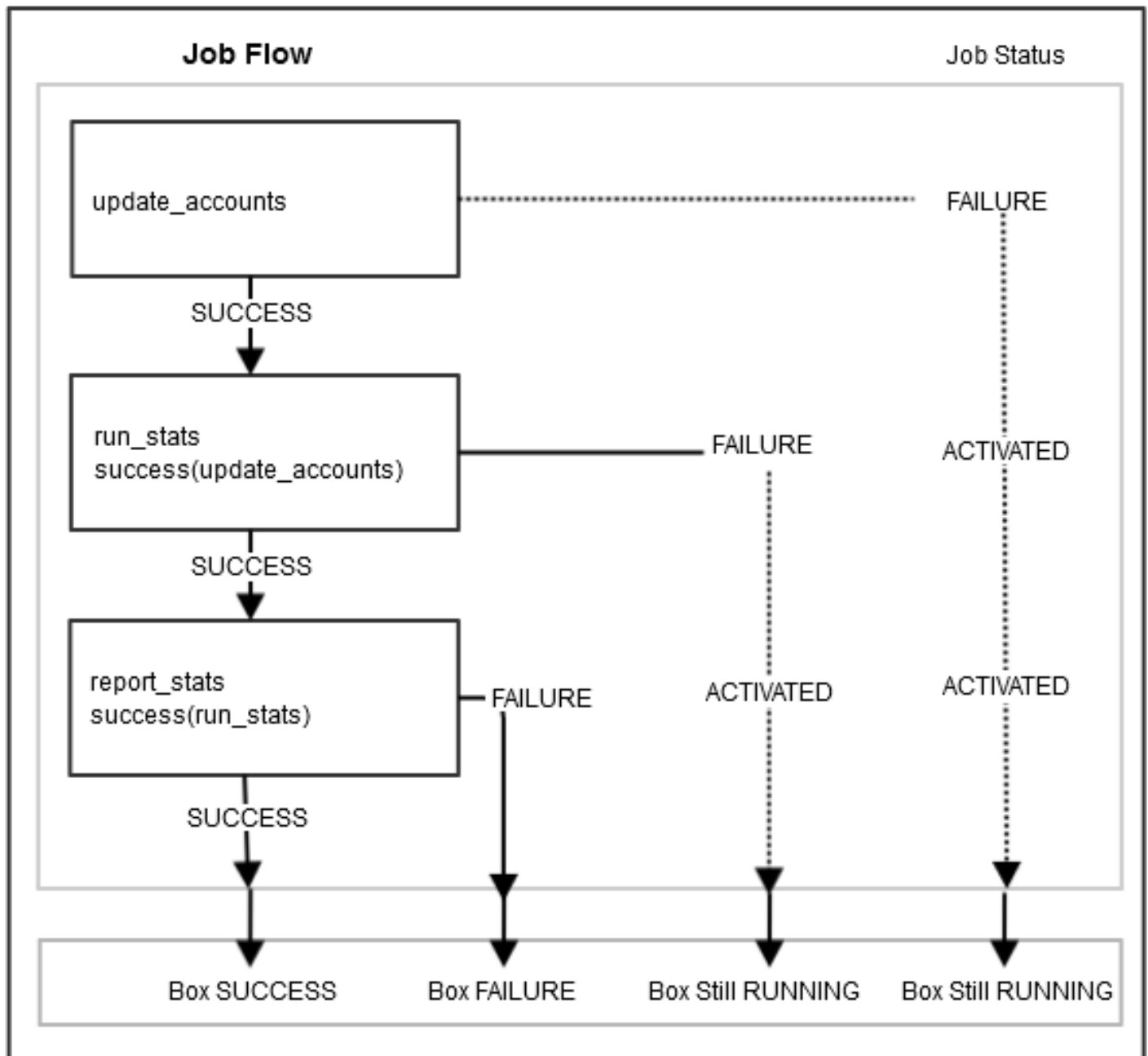
- **update\_accounts**  
Updates files. This job starts when `do_statistics` starts running. It has no other starting conditions.
- **run\_stats**  
Runs statistics. This job starts when `update_accounts` completes successfully. It has no other starting conditions.
- **report\_stats**  
Reports statistics. This job starts when `run_stats` completes successfully. It has no other starting conditions.

No conditions for success or failure have been defined for `do_statistics`; therefore the default conditions are applied. The box job completes successfully when all the jobs it contains have run and completed successfully. The box job fails when all jobs in the box have run and at least one has failed. The box job remains in the `RUNNING` state until all the jobs it contains have run.

The following illustration shows this job flow:

box\_name "do\_statistics"

Figure 43: do\_statistics box job lacks conditions for success or failure



### Explicit Box Success and Box Failure

This scenario provides an example job flow in which specific conditions are defined for the success or failure of a box job. The box job do\_statistics runs every day at 3:00 a.m. It contains three jobs:

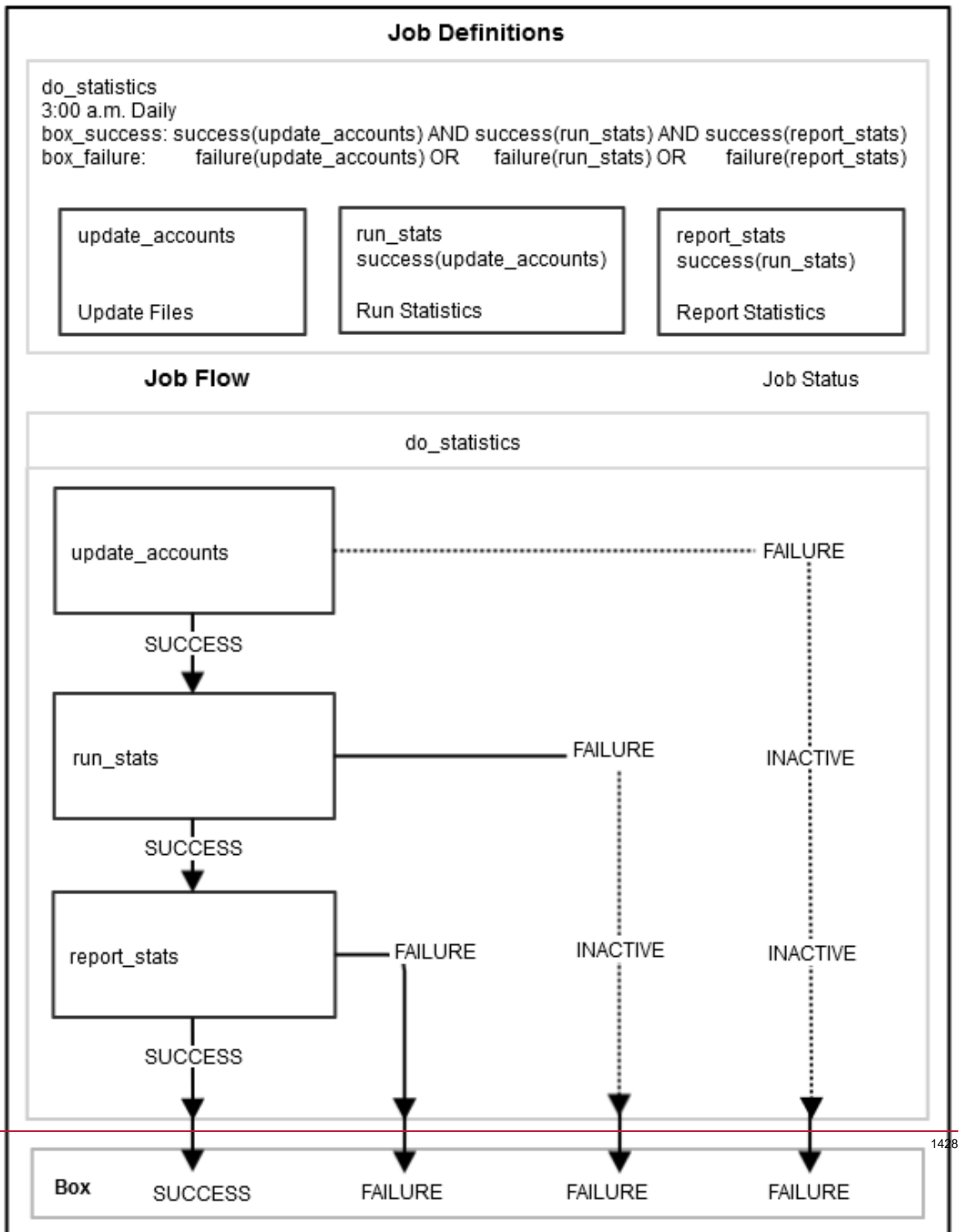
- **update\_accounts**  
Updates files. This job starts when do\_statistics starts running. It has no other starting conditions.
- **run\_stats**  
Runs statistics. This job starts when update\_accounts completes successfully. It has no other starting conditions.
- **report\_stats**  
Reports statistics. This job starts when run\_stats completes successfully. It has no other starting conditions.

The following conditions define the criteria for success or failure of the box job do\_statistics:

- The box job can complete successfully only when all the jobs it contains have completed successfully.
- The box job fails if any of the jobs it contains fails.

The following illustration shows the job definitions and the job flow:

Figure 44: do\_statistics box job contains conditions for box success and failure





## Job Flow with Job Terminator Attribute

This scenario provides an example job flow in which the `job_terminator` attribute is defined for a job in a box job.

The box job `daily_accounts` runs every day at 3:00 a.m. It contains two jobs:

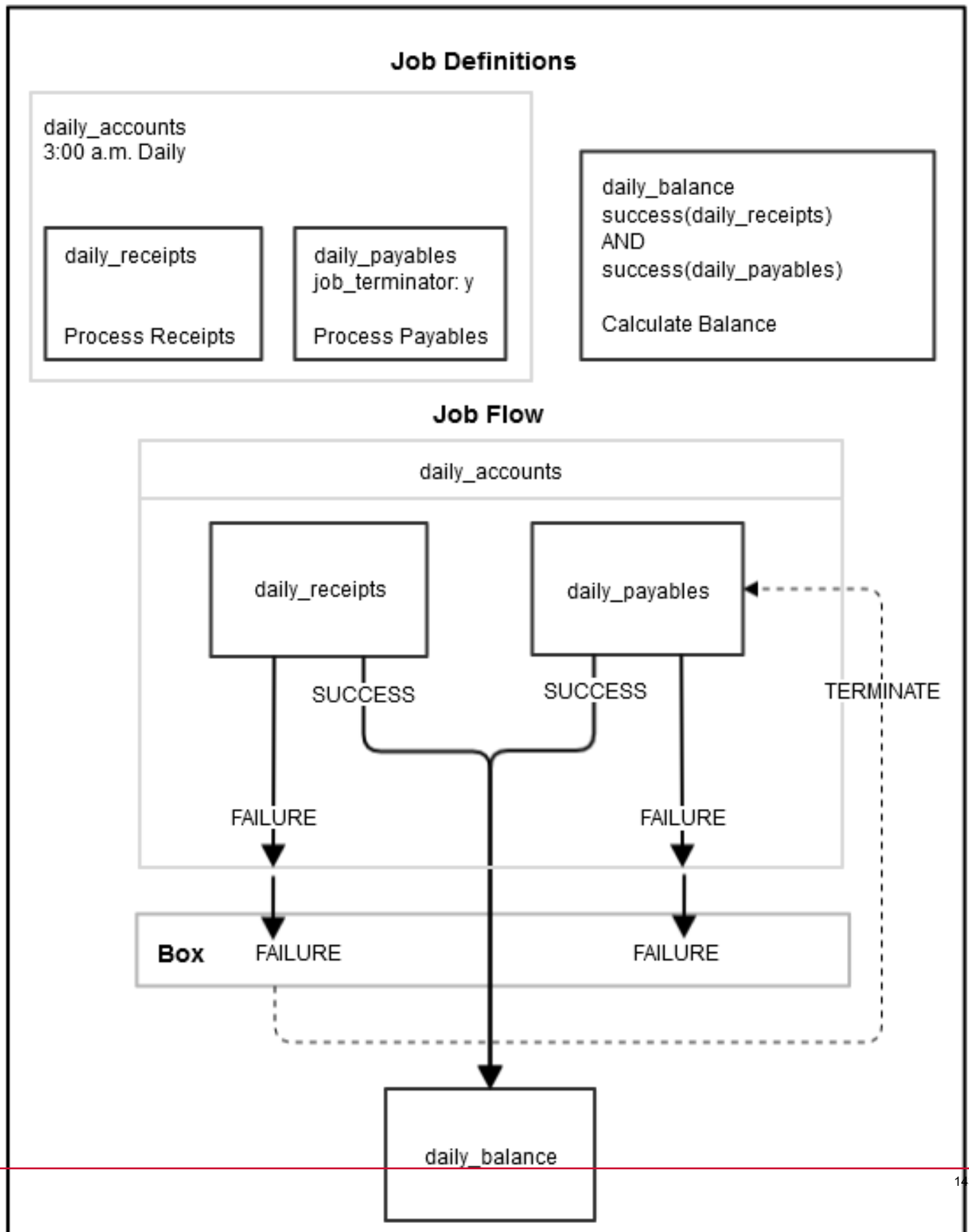
- **daily\_receipts**  
Processes receipts. This job runs when `daily_accounts` starts because it has no other starting conditions.
- **daily\_payables**  
Processes payables. This job runs when `daily_accounts` starts because it has no other starting conditions. Because `daily_payables` includes a `job_terminator` attribute, `daily_account` is terminated if this job fails.

A third job, `daily_balance`, is not contained in `daily_accounts` and runs only if both `daily_receipts` and `daily_payables` complete successfully.

Because `daily_accounts` can only complete successfully if both of the jobs it contains complete successfully, the failure of `daily_receipts` causes `daily_accounts` to fail. This in turn triggers the `job_terminator` attribute in `daily_payables`, which terminates the job if the box that contains it fails.

The following illustration shows the job definitions and the job flow:

Figure 45: job flow with job terminator attribute



## Job Flow with Box Terminator Attribute

This scenario provides an example job flow in which the `box_terminator` attribute is defined for jobs in a box job.

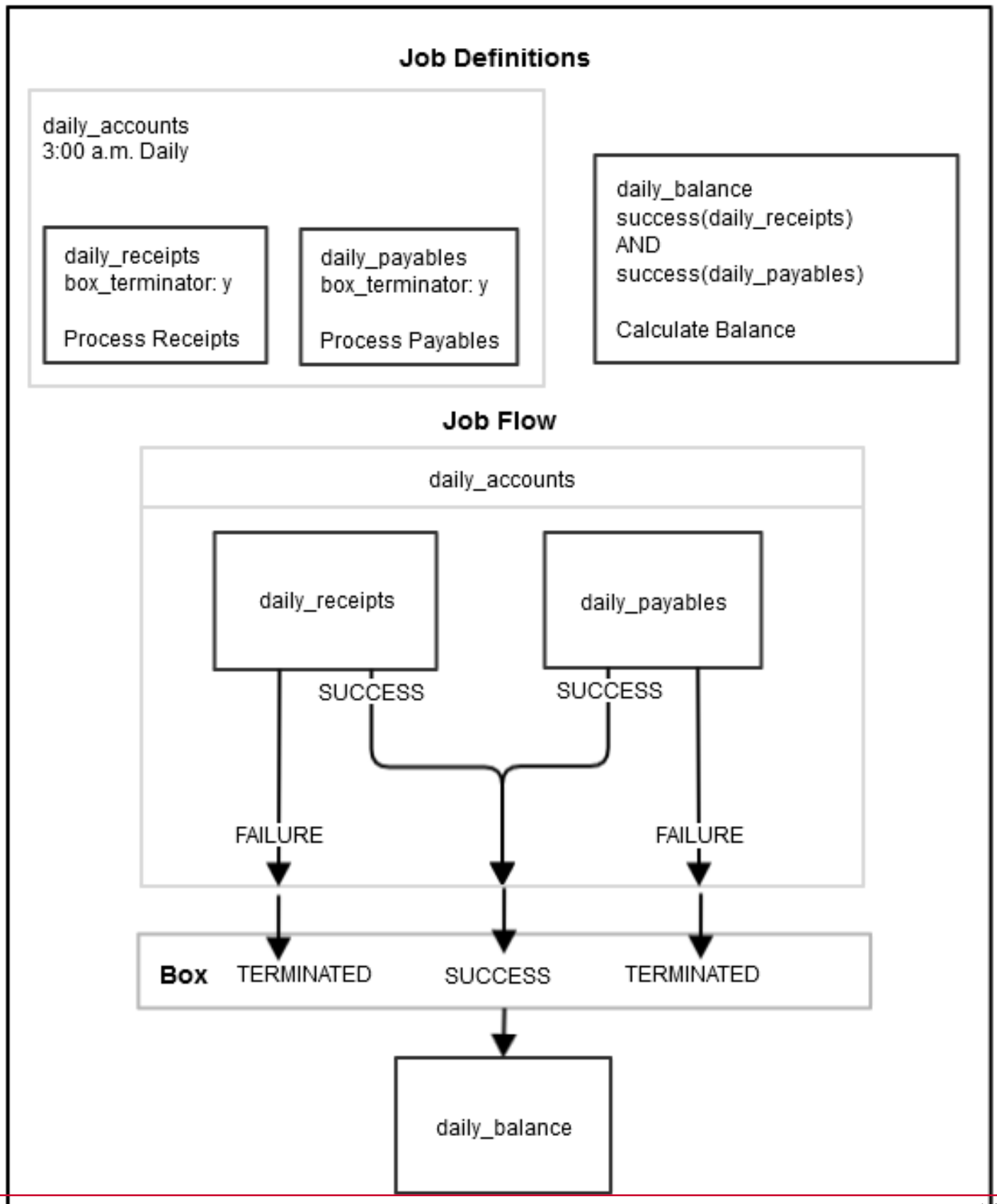
The box job `daily_accounts` runs every day at 3:00 a.m. It contains two jobs:

- **daily\_receipts**  
Processes receipts. This job runs when `daily_accounts` starts because it has no other starting conditions. Because `daily_receipts` includes a `box_terminator` attribute, `daily_accounts` will be terminated if this job fails.
- **daily\_payables**  
Processes payables. This job runs when `daily_accounts` starts because it has no other starting conditions. Because `daily_payables` includes a `box_terminator` attribute, `daily_accounts` will be terminated if this job fails.

A third job, `daily_balance`, is not contained in `daily_accounts` and will run only if both `daily_receipts` and `daily_payables` complete successfully.

The following illustration shows the job definitions and the job flow in which the box terminator attribute is defined for jobs in the box job.

Figure 46: job flow with box terminator attribute



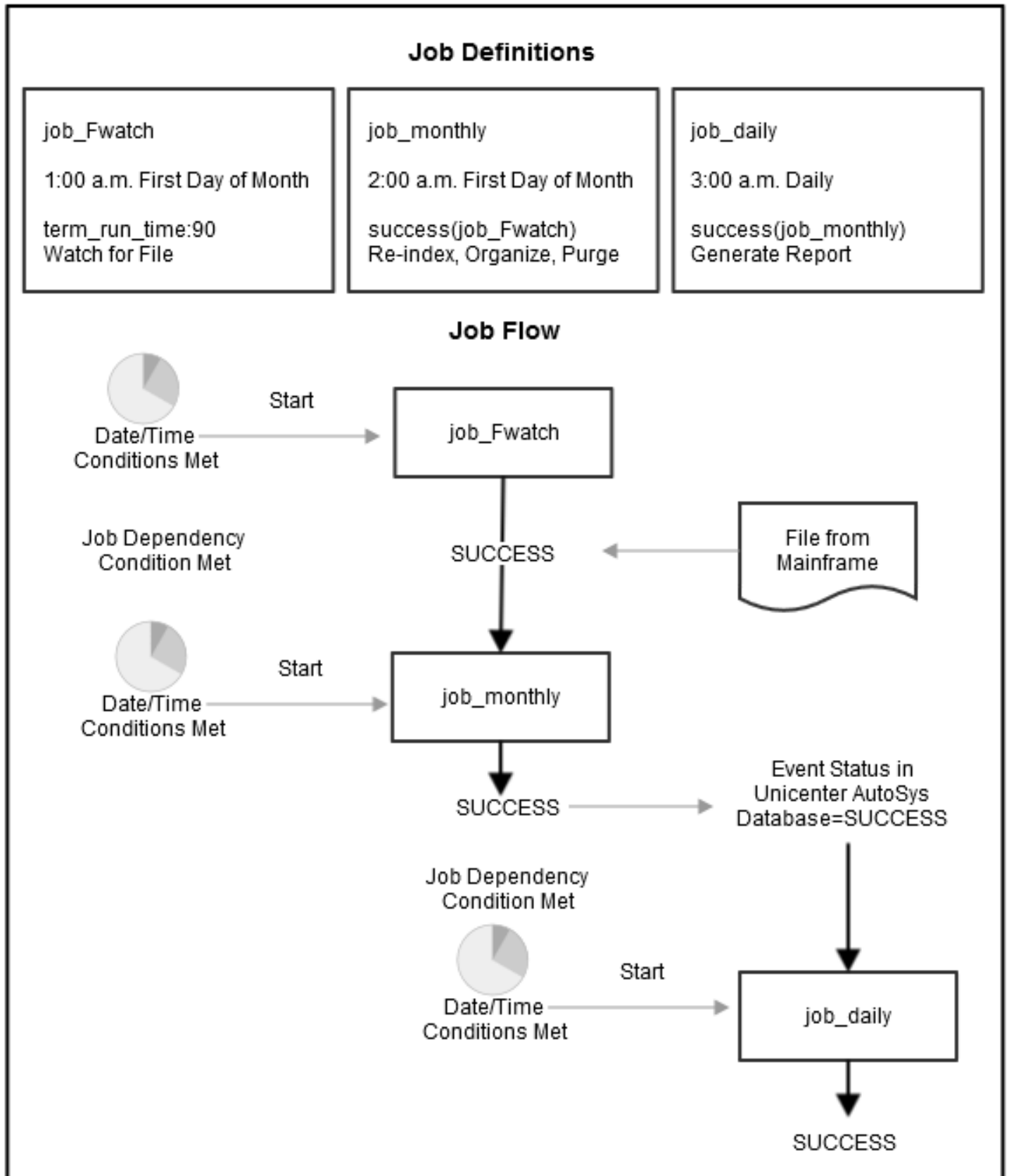
## **Advanced Box Job Flow Examples**

This section contains examples that explain the flow of box jobs and the jobs they contain in advanced situations. These scenarios provide a clearer understanding of advanced job flow concepts.

### **Job Flow with Time Conditions Running on the First of the Month**

This scenario is an example of a job flow that begins on the first of every month.

Figure 47: job flow with time conditions running on the first of the month



The job flow consists of three jobs:

- **job\_Fwatch**  
Waits for a specific file to be created by some mainframe process. This job runs at 1:00 a.m. on the first of every month and waits for 90 minutes before giving up.
- **job\_monthly**  
Re-indexes, organizes, and purges its records based on the file created by the mainframe. This job runs at 2:00 a.m. on the first of the month only when job\_Fwatch completes successfully.
- **job\_daily**  
Generates a report. This job runs daily at 3:00 a.m. when job\_monthly completes successfully.

The failure of job\_Fwatch causes job\_monthly to skip its scheduled run because job\_monthly can only complete successfully if job\_Fwatch completes successfully. Job job\_daily only runs if job\_monthly completes successfully. By the same logic, job\_daily always runs if job\_monthly was able to successfully run at least once.

**NOTE**

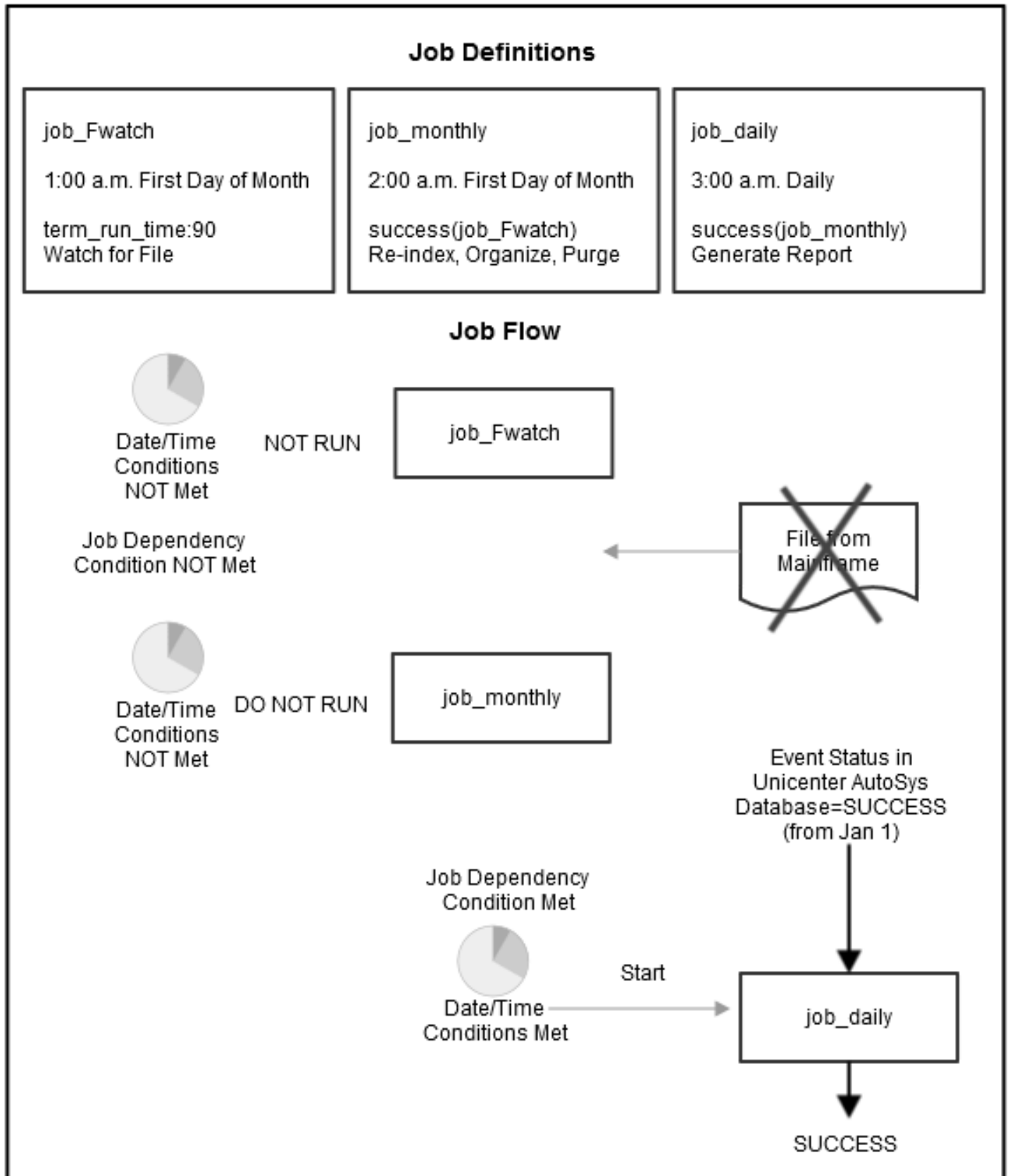
The first time the cycle is run (for example, January 1), statuses behave as expected.

### **Job Flow with Time Conditions Running on the Second of the Month**

This scenario builds upon the previous scenario and takes place on the following day.

On days of the month other than the first, job\_Fwatch and job\_monthly do not run. They still have a status of SUCCESS in the database from the previous run on the first day of the month. As a result, job\_daily still runs.

Figure 48: job flow with time conditions running on the second of the month



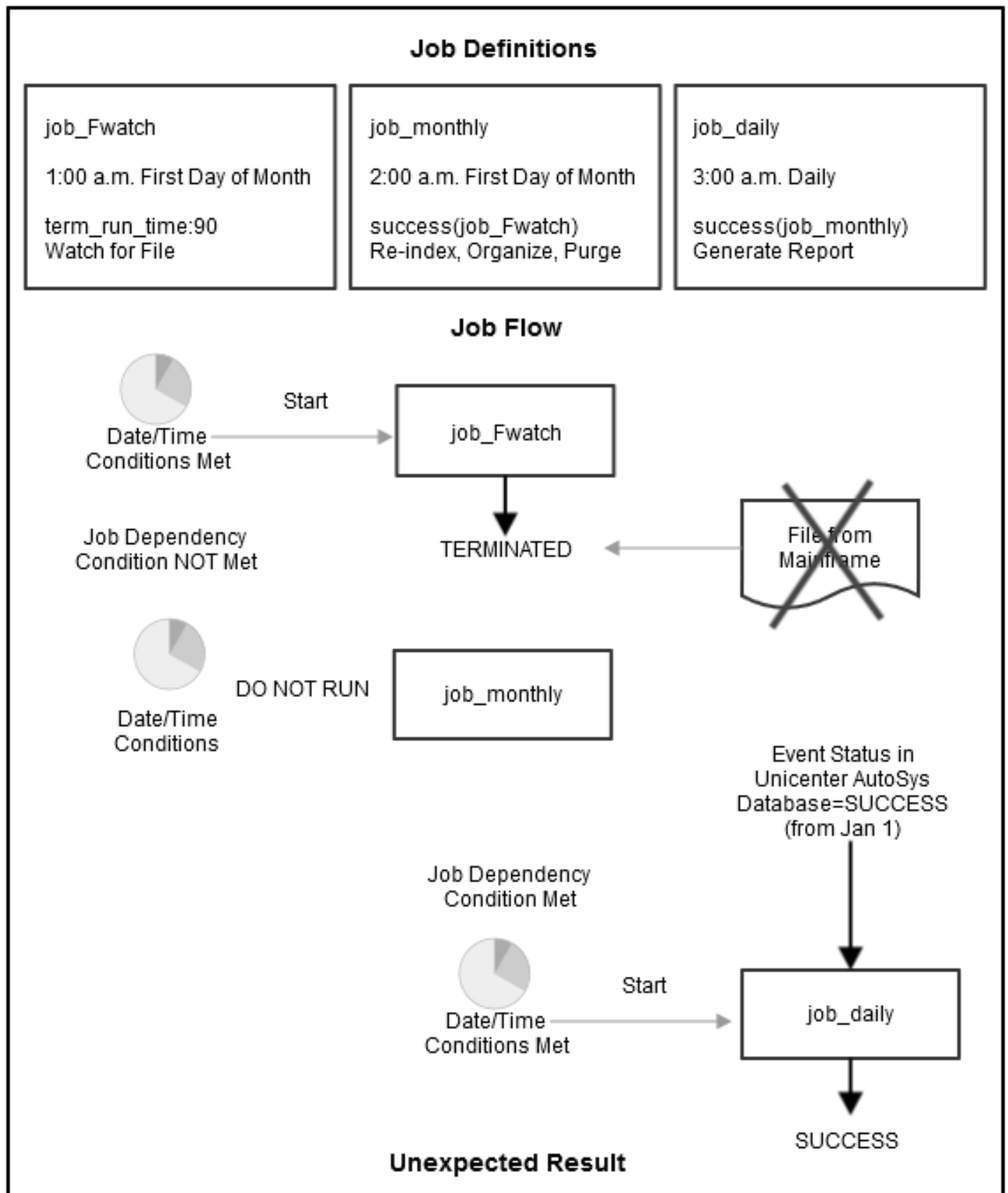


## **Job Flow with Time Conditions Running on the First of the Following Month**

This scenario builds upon the previous scenario and takes place on the first day of the following month.

On the first day of the next month (for example, February 1), the file from the mainframe fails to arrive in the 90-minute wait time; therefore, job\_Fwatch self-terminates. As a result, job\_monthly misses its run for the month. However, because its event status in the database is still SUCCESS from the previous month, job\_daily is able to run every day this month. When job\_daily runs, it uses the data from the previous month leading to invalid reports for the month.

Figure 49: job flow with time conditions running on the first of the following month



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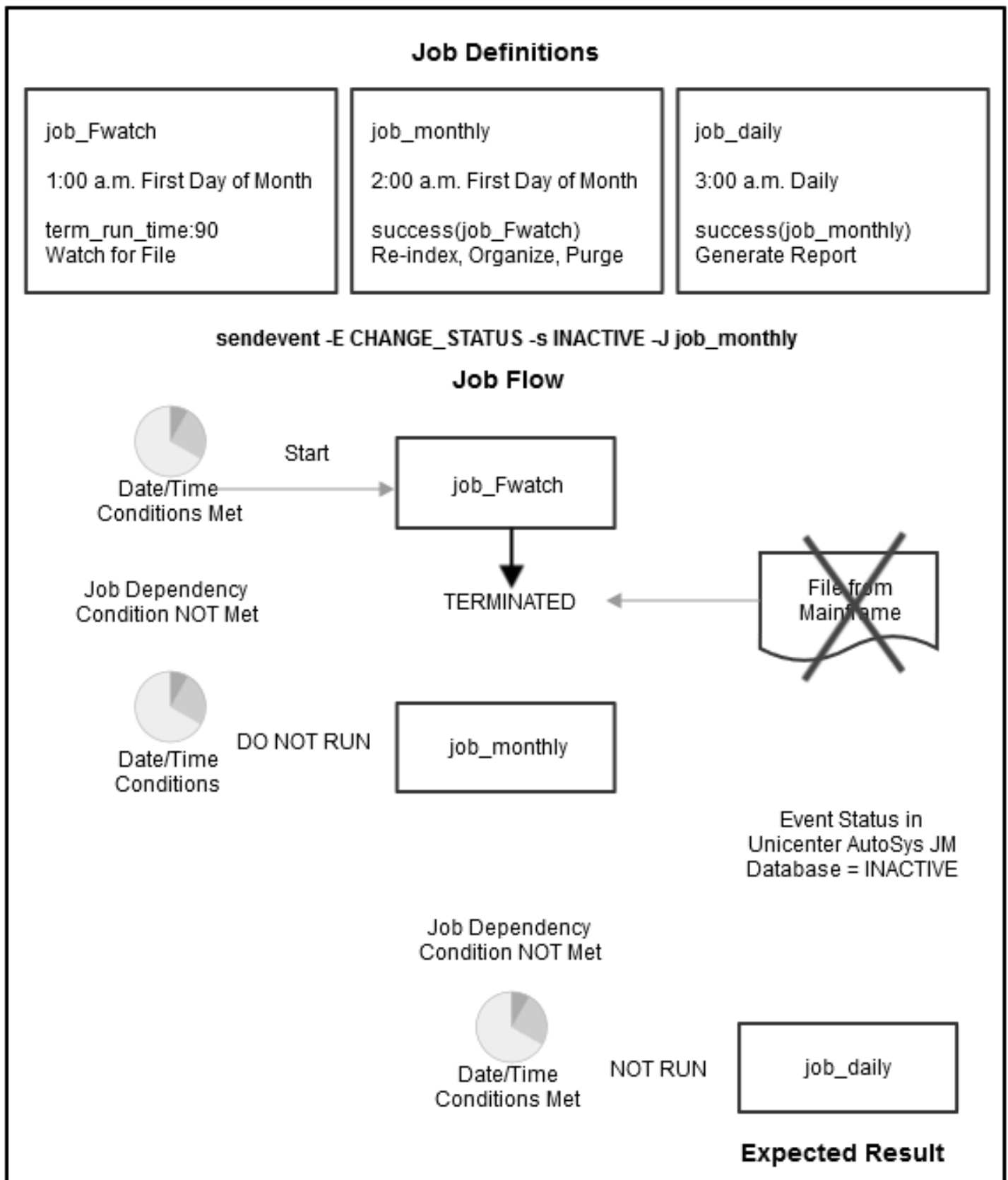
## Reset a Job Flow with Time Conditions through INACTIVE Status Change

This scenario builds upon the previous scenario and takes place on the last day of the month.

To fix time-related statuses, you can use a sendevent command to change them to INACTIVE at the end of their valid interval. You can create another job to do this automatically.

Changing the status of job\_monthly to INACTIVE at the end of every month allows job\_daily to run only in the months that job\_monthly completes successfully. In the following example, when job\_Fwatch fails, job\_monthly and job\_daily will not run because its status has been reset to INACTIVE.

Figure 50: resetting a job flow with time conditions through INACTIVE status change



---

## Reset a Job Flow with Time Conditions through Box Job

This scenario builds upon the previous scenarios and takes place on the first day of the month.

Instead of issuing a sendevent command to change the status of the jobs, you can put the monthly process in a box, and can set the box\_failure or box\_terminator attribute appropriately.

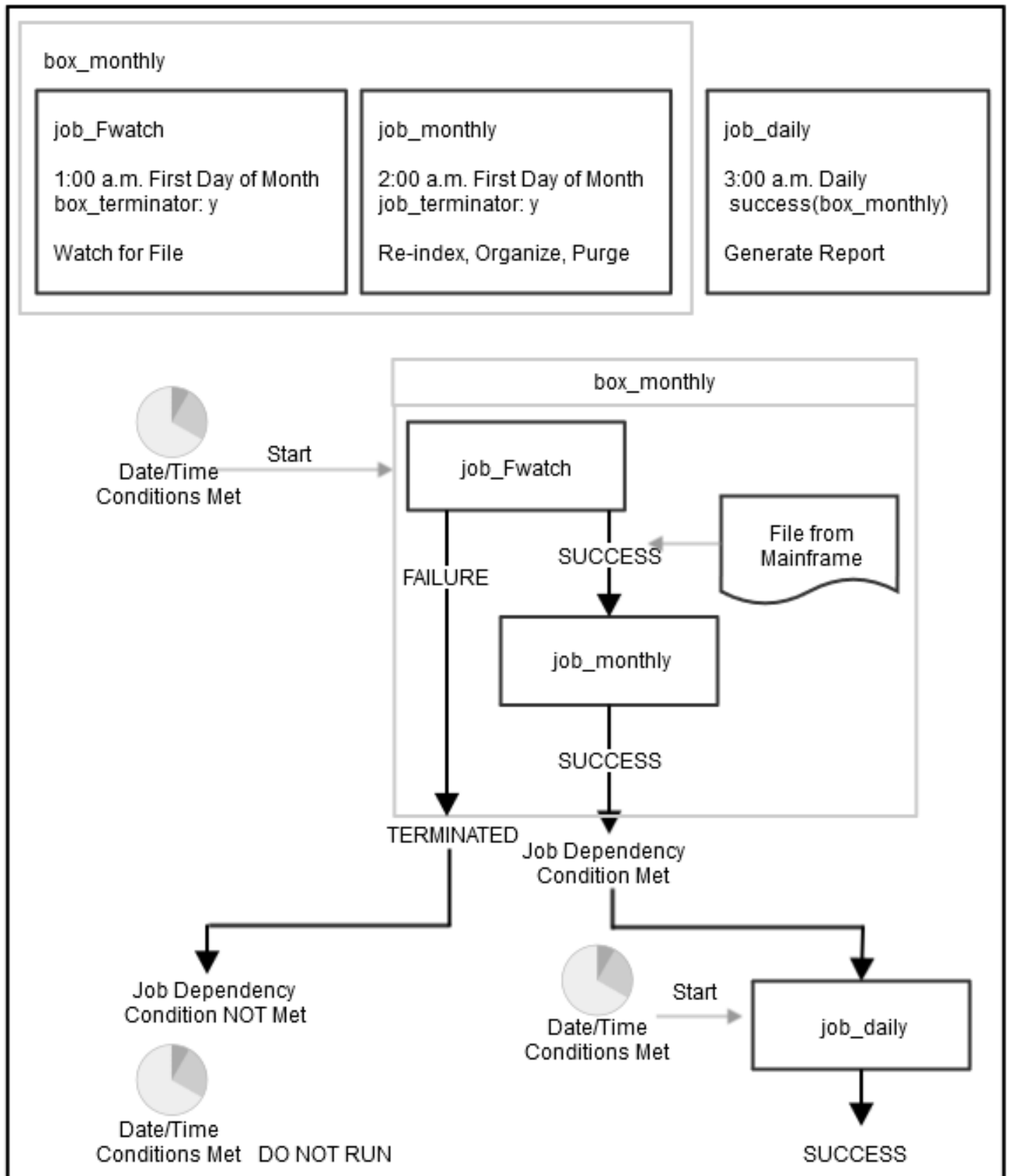
The job flow now consists of a box that is called box\_monthly that runs at 1:00 a.m. on the first day of every month with the following jobs:

- **job\_Fwatch**  
Waits for some mainframe process to create a specific. This job runs at 1:00am on the first of every month and waits for 90 minutes before giving up.
- **job\_monthly**  
Re-indexes, organizes, and purges its records based on the file created by the mainframe. This job runs at 2:00 a.m. on the first of the month only when job\_Fwatch completes successfully.

A third job, job\_daily, is not contained in box\_monthly and runs only if job\_Fwatch and job\_monthly complete successfully.

The failure of job\_Fwatch causes box\_monthly to terminate because box\_monthly can only complete successfully if both of the jobs it contains complete successfully. This in turn triggers the job\_terminator attribute in job\_monthly, which terminates the job if the box that contains it fails.

Figure 51: resetting job flow with time conditions through a box job



---

## Define a Box Job

Box jobs are a convenient way to start multiple jobs. When you put jobs in a box, you only have to start a single job (the box) for all the jobs in the box to start running.

Assume that you want to schedule a group of jobs to start running when a file watcher job completes successfully. Instead of making each job dependent on the file watcher job, you can create a box job that depends on the file watcher job, remove the file watcher job dependency from the individual jobs, and put all of those jobs in the box. When the file watcher job completes successfully, the box job starts, which in turn starts all of the jobs it contains.

### Follow these steps:

1. Insert a job and specify the following attribute in the definition:
  - **job\_type: BOX**  
Specifies that the job type is BOX.
2. (Optional) Specify optional BOX attributes:

- alarm\_if\_fail
- application
- auto\_delete
- auto\_hold
- avg\_runtime
- box\_failure
- box\_name
- box\_success
- box\_terminator
- condition
- date\_conditions
- days\_of\_week
- description
- exclude\_calendar
- group
- job\_terminator
- max\_run\_alarm
- min\_run\_alarm
- must\_complete\_times
- must\_start\_times
- n\_retrys
- notification\_msg
- owner
- permission
- priority
- run\_calendar
- run\_window
- send\_notification
- service\_desk
- start\_mins
- start\_times
- svcdesk\_attr
- svcdesk\_desc
- svcdesk\_imp
- svcdesk\_pri
- svcdesk\_sev
- term\_run\_time
- timezone

3. (Optional) Specify common attributes that apply to all job types.  
The box job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about



common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

### Example: Create a Box Job

This example shows how to define a box job that is named EOD\_box that depends on the success of a file watcher job to run:

```
insert_job: EOD_box
job_type: box
condition: success(EOD_watch)
```

This JIL script instructs AutoSys Workload Automation to do the following:

- Add a new job named EOD\_box.
- Define the job as a box job.
- Run the job only if the file watcher job named EOD\_watch completes with a SUCCESS status.

## Box Job Attributes

The following attributes are required for all box job definitions:

- **box\_name**  
Defines the name that is used to identify the job to AutoSys Workload Automation. This name is used by other jobs as the name of their parent box.
- **job\_type: BOX**  
This attribute specifies that the job type is box.

### NOTE

If you omit this attribute, the job type defaults to cmd (Command job).

- **condition**  
This attribute defines the dependency conditions that must be met for the job to run.

### NOTE

The condition attribute is not always required, for example, when a job is always started manually.

## How Job Groupings Are Created

Box jobs provide one method of grouping jobs, but are typically used when all the jobs in the box share starting condition. AutoSys Workload Automation provides the [group](#) and [application](#) attributes so you can logically group sets of jobs and boxes with unrelated starting conditions or dependencies. By specifying both group and application attributes in a job definition, you can make the job belong to both a group logical set and an application logical set.

### Example: Associate Jobs with Groups and Applications

This example shows how you can associate jobs with specific groups and applications to control processing.

Assume that you want to create a set of jobs that run a suite of applications that are called EmployeePay that is used to manage employee salaries. The Accounting and Human Resources groups each have their own jobs that are defined to use the EmployeePay applications. The following JIL script defines two jobs (HR\_payroll and ACCT\_salaryreport):

```
insert_job: HR_payroll

job_type: cmd

...

group: HumanResources

application: EmployeePay

insert_job: ACCT_salaryreport

job_type: cmd

...

group: Accounting

application: EmployeePay
```

This JIL script instructs AutoSys Workload Automation to do the following steps:

- Add two new command jobs (HR\_payroll and ACCT\_salaryreport).
- Associate job HR\_payroll with the HumanResources group and the EmployeePay application.
- Associate job ACCT\_salaryreport with the Accounting group and the EmployeePay application.

To run a job that is associated with the EmployeePay application, enter the following:

```
sendevent -e STARTJOB -I EmployeePay
```

To run a job that is associated with the Accounting group, enter the following code:

```
sendevent -e STARTJOB -B Accounting
```

To run a job that is associated with both the EmployeePay application and Accounting group (intersection of both sets), enter the following code:

```
sendevent -e STARTJOB -I EmployeePay -B Accounting
```

## How an Existing Job Is Put in a Box

To place an existing job in a box, verify that the job is not running, and do either of the following steps:

- Use the [update\\_job](#) subcommand to change the current job definition.
- Use the [delete\\_job](#) subcommand to delete the current job definition, and use the [insert\\_job](#) subcommand to redefine the job.  
This method is useful when the job definition contains many non-default attributes that you want to deactivate instead of resetting them. However, if you delete and redefine the job, you must redefine any non-default attributes that you want to keep from the previous definition.

### Example: Put an Existing Job in a Box

This example shows how to update the definition of an existing job to include it in a box.

The following JIL script uses the `update_job` subcommand to change the `EOD_post` job to put it in the `EOD_box` job:

```
update_job: EOD_post

box_name: EOD_box
```

This JIL script instructs AutoSys Workload Automation to do the following steps:

- Update the job named `EOD_post`.
- Put the job named `EOD_post` in the box named `EOD_box`.

## Delete a Box Job

To delete a box and every job it contains, enter the `delete_box` subcommand followed by the name of the box job to delete. For example, to delete the box `EOD_box` and every job in it, you would enter the following code:

```
delete_box: EOD_box
```

To delete a box without deleting the jobs that it contains, enter the `delete_job` command followed by the name of the box job to delete. The jobs in the box become stand-alone jobs. For example, to delete the box `EOD_box` without deleting the jobs in it, you would enter the following code:

```
delete_job: EOD_box
```

## Command Jobs Overview

Command jobs let you run workload on UNIX and Windows client computers. On UNIX, you can define jobs to run scripts and commands. On Windows, you can define jobs to run command files and batch files.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

When you define a Command job, you can specify settings including the following conditions:

- **Start Conditions**

This setting defines conditions (for example, date, time, job state, and machine state) that must be met before a job can start.

- **Disk Space Criteria**

This setting defines the amount of free space that is required before a process can start. If the free space is not available, an alarm is sent and the job is rescheduled.

- **Job Profile**

This setting specifies a script to be sourced that defines the environment where the command runs.

**NOTE**

On Windows, you can define a job profile using the Job Profiles - AutoSys Workload Automation Administrator window in the Administrator utility.

- **Environment Variables**

This setting specifies variables that define the local environment where the job runs.

- **User-defined Exit Codes**

This setting defines exit codes to indicate job success or job failure. By default, an exit code of 0 (zero) indicates job success and any other code indicates job failure. When the job completes, the exit event (either SUCCESS or FAILURE) and the exit code of the program exit code are stored in the database.

- **Standard I/O Files**

This setting specifies the standard input, output, and error files.

**NOTE**

To [support chained commands](#) (commands that are separated by a semicolon) on UNIX, ensure that the `oscomponent.wrapper.exec.force` parameter in the `agentparm.txt` file is set to false (the default).

## The Directory the Job Runs Under

The following settings determine the directory that a Command job runs under:

- **oscomponent.initialworkingdirectory Agent Parameter** This setting specifies the default initial working directory for all scripts. The options are as follows:

- **SCRIPT**

Sets the path to where the script resides.

- **USER**

Sets the path to the home directory of the owner of the script.

- **PATH**

Sets the path to an absolute path to where the script runs.

- **USER\_CONT\_NOT\_AVAIL**

Specifies that the script should run even if a home directory for the user is not defined.

If you do not specify a value, the parameter defaults to the path where the running cybAgent resides.

You can specify the `oscomponent.initialworkingdirectory` parameter in the `agentparm.txt` file of the agent.

- **HOME and PWD Environment Variables (UNIX only)** Overrides the `oscomponent.initialworkingdirectory` default directory. The directory depends on the following settings:
  - If you specify a value for PWD (Present Working Directory) in the job definition, the job runs under the PWD you specified.
  - If you specify a value for HOME in the job definition, but you do not specify a value for PWD, the job runs under the HOME directory.
  - If you specify values for HOME and PWD in the job definition, the job runs under the PWD directory.

## Determine Which Shell is Used (UNIX)

A shell is a program that provides an environment for users to interact with the system. Shell programs are files that contain a series of user commands that are grouped, using logic that is provided by the shell, to accomplish a task.

You can define shells, the programmer of the script, your agent administrator, and your UNIX administrator. When you define a job that runs on UNIX, you might want to know which shell is used to run the script because different shells have different facilities and characteristics. Some functions are specific to certain shells and might be incompatible with other shells.

To run a UNIX script, the agent selects the shell to use as follows:

1. Use the shell from agent parameter `oscomponent.defaultshell` if and only if the agent parameter `oscomponent.defaultshell.force` is set. These parameters are found in the `agentparam.txt` file.
2. Use the shell specified in the job definition through the shell job attribute.
3. Use the shell specified in the shell script itself. Normally this is specified as the shell directive or the first line of the script.
4. The value from the `oscomponent.defaultshell` parameter found in the `agentparm.txt` file.
5. Shell from the user's account profile.

#### NOTE

- The agent administrator must define all of the shells that the agent uses in the `agentparm.txt` file, unless the `oscomponent.checkvalidshell` parameter is set to false. The shells are defined using the `oscomponent.validshell` parameter.
- If the shell is defined in the first line of the script, you do not need to include the shell attribute in the job definition. The shell attribute tells the agent which shell interpreter to use when launching the command. When the command executes a script that specifies a shell, the script runs from that point forward under the shell specified in the script.
- If the `oscomponent.checkvalidshell` parameter in the `agentparm.txt` file of the agent is set to true (the default), all shells that are used must be specified using the `oscomponent.validshell` parameter. The path that is defined in the first line of the script or in the job definition must match the corresponding path in the `oscomponent.validshell` parameter. If the shell you want to use is not defined on the agent, the job fails. For more information about specifying valid shells, see the `oscomponent.checkvalidshell` and `oscomponent.validshell` parameters in [Workload Automation Agents](#).

## (UNIX) Shell Initialization Files

When you log in to the UNIX operating system, the operating system reads a series of initialization files depending on the shell you use.

### C Shell Initialization Files

When you log in to the C shell, it automatically runs few files. The first file run is a system file that is named `/etc/.login`, which contains systemwide configuration information (such as your default path). After these files are run, the C shell reads and runs the commands from the following files in your home directory:

- **.cshrc**

Establishes variables and operands that are local to a specific shell. Each time that you create a shell, the C shell reinitializes these variables for the new shell.

The following code is a sample `.cshrc` file:

```
set noclobber

set ignoreeof

set path = (~bin $path /usr/games)
```

```
alias h history
```

- **.login**

Contains commands that you want to run once at the beginning of each session. If the C shell is running as a login shell, it reads and runs the commands in the .login file in your home directory after login.

The following code is a sample .login file:

```
setenv history 20
```

```
setenv MAIL /usr/spool/mail/$user
```

- **.logout**

Runs when you exit from your login shell.

### **Korn Shell Initialization Files**

The Korn shell supports three startup scripts:

- **/etc/profile**

Contains systemwide startup commands. This file is a login script and runs when you log in.

- **\$HOME/.profile**

Runs when you log in. Use this login script to set options, set the path, and set and export variable values.

The following code is a sample \$HOME/.profile file:

```
set -o allexport
```

```
PATH=./bin:/usr/bin:$HOME/bin
```

```
CDPATH=./$HOME:$HOME/games
```

```
PS1='! $PWD> '
```

```
ENV=$HOME/.kshrc
```

```
TIMEOUT=0
```

```
set +o allexport
```

- **The script named in the Korn shell variable ENV**

Runs when you create a Korn shell or run a Korn shell script. Use this file to define aliases and functions and to set default options and variables that you want to apply to the Korn shell invocation.

### **Bourne Shell Initialization Files**

When you log in to your system, the Bourne Shell looks for the /etc/profile file. This file contains systemwide startup commands for the Bourne Shell. Only the system administrator can change this file.

After running the commands in the /etc/profile file, the Bourne Shell runs the commands in the \$HOME/.profile file.

Therefore, you can override the systemwide commands in the /etc/profile file with commands in the \$HOME/.profile file.

## Define a Command Job

You can define a Command (CMD) job to run workload on UNIX and Windows client computers. The job can run a script, execute a UNIX command, or run a Windows command file.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:

- **job\_type: CMD**

This attribute specifies that the job type is Command.

### NOTE

For backwards compatibility with previous versions of AutoSys Workload Automation, you can specify **job\_type: c**.

- **machine** This attribute specifies the name of the machine on which the job runs.
  - **command** This attribute specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.
2. (Optional) The following specify optional Command attributes:
    - blob\_file
    - blob\_input
    - chk\_files
    - elevated
    - envvars
    - fail\_codes
    - heartbeat\_interval
    - interactive
    - job\_class
    - job\_terminator
    - max\_exit\_success
    - profile
    - shell
    - std\_err\_file
    - std\_in\_file
    - std\_out\_file
    - success\_codes
    - ulimit
  3. (Optional) This attribute specifies the common attributes that apply to all job types. The Command job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- All UNIX jobs that run a command (a binary file) require a user ID that has the authority to run the job on the agent computer. The owner attribute in the job definition specifies this user ID (the default is the user who invokes jil to define the job).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.
- To [support chained commands](#) (commands that are separated by a semicolon) on UNIX, ensure that the `oscomponent.wrapper.exec.force` parameter in the `agentparm.txt` file is set to `false` (the default).

### Example: Run a Command on UNIX

This example runs the `/bin/touch` command on the file named `/tmp/test_run.out`. The job runs on the UNIX client computer named `unixagent`.

```
insert_job: test_run
job_type: CMD /* This attribute is optional for Command jobs. CMD is the default. */
machine: unixagent
command: /bin/touch /tmp/test_run.out
```

### Example: Run a Command File on Windows

This example runs the `c:\bin\test.bat` command on the Windows client computer named `winagent`. The command is enclosed in quotation marks because the path contains a colon.

```
insert_job: test_run
job_type: CMD /* This attribute is optional for Command jobs. CMD is the default. */
machine: winagent
command: "c:\bin\test.bat"
```

## Command Job Attributes with Default Values

This article explains Command job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the `agentparm.txt` file.

If you specify the attribute in a job definition, it overrides the default.

The following Command job attributes have default values:

- **fail\_codes** This attribute defines which exit codes indicate job failure.  
**Default:** Any exit code other than 0 (zero) indicates job failure
- **interactive**  
(Windows only) This attribute specifies whether to run a Command job in interactive mode or in batch mode on Windows.  
**Default:** `n` (The job runs in batch mode.)
- **max\_exit\_success** This attribute defines the maximum exit code that the job can exit with and can be considered a success.  
**Default:** 0 (The job interprets only zero as job success.)
- **owner**  
This attribute specifies the user ID that the job runs under.



**Default:** The default owner (the user ID who invokes jil to define the job)

- **shell**

(UNIX only) This attribute specifies the name of the shell that is used to execute the script or command file.

**NOTE**

Alternatively, you can override the default shell by specifying the shell in the first line of the script. If a shell is specified in the script and in the job definition, the job uses the shell specified in the job definition.

**Default:**

- `oscomponent.defaultshell` agent parameter, if specified
- The user default shell that is defined in the user profile, if the shell is not specified in the job definition, script, or `oscomponent.defaultshell` parameter

- **std\_err\_file** This attribute defines the path and file name where you want to redirect all standard error output.

**Workload Automation Agent Default:** The spool directory of the agent (*installation\_directory/*  
*SystemAgent/agent\_name/spool*)

- **std\_out\_file** This attribute defines the path and file name where you want to redirect all standard output.

**Workload Automation Agent Default:** The spool directory of the agent (*installation\_directory/*  
*SystemAgent/agent\_name/spool*)

- **success\_codes**

This attribute defines which exit codes indicate job success.

**Default:** 0 (The job interprets zero as job success.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Override the Default Shell Using the shell Attribute

This example overrides the default shell using the shell attribute. The job uses the C shell run the test1.csh script.

```
insert_job: unix_job1
job_type: CMD
machine: unixagent
command: /mfg/test1.csh
shell: /bin/csh
```

**NOTE**

This job succeeds only if the user's default shell is csh. To set the default shell, define the following parameters in the `agentparm.txt` file of the agent:

```
oscomponent.defaultshell=/bin/csh
oscomponent.defaultshell.force=true
```

### Example: Override the Default Shell by Specifying the Shell in a Script

This example overrides the default shell with the C shell specified in the following test1.csh script:

```
#!/bin/csh -f
if ($LOGNAME != guest) then
 echo "User is not guest"
endif
echo $LOGNAME logon
```

```
exit 0
```

The following job definition runs the test1.csh script:

```
insert_job: unix_job2
job_type: CMD
machine: unixagent
command: /mfg/test1.csh
owner: guest
```

The shell attribute is not required in the job definition because the shell is specified in the first line of the script.

### Example: Override the Default Background Mode Using the interactive Attribute

This example opens the configuration text file (config.txt) in the Windows Desktop application named Notepad.

```
insert_job: edit_file
job_type: CMD
machine: winagent
description: "Edit/review a configuration file"
command: notepad.exe "c:\run_info\config.txt"
interactive: y
```

## Verify File Space Before a Job Starts

You can define a Command job to verify if one or more UNIX file systems or Windows drives have the required amount of available space.

At run time, the agent checks whether the required space is available on the machine where the job runs. If the requirements are met, the job starts. If the requirements are not met, the agent generates an alarm and the job does not start. The system tries to verify the file space again and start the job. The number of tries is determined using the `n_retrys` attribute. If the `n_retrys` attribute is not specified in the job definition, the `MaxRestartTrys` parameter determines the number of tries in the configuration file (UNIX) or the `Max Restart Trys` field in the Administrator utility (Windows). If the required space is still not available after all the restart attempts, the job fails.

By default, jobs do not check the available file space.

### Follow these steps:

1. [Define a Command job](#).
2. Add the following attribute to the job definition:
  - **chk\_files**  
This attribute specifies the required amount of space on one or more file systems (UNIX) or drives (Windows) in kilobytes (KB).
3. Run the job.  
The file space is verified before the job starts.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Verify the Available File Space on UNIX

This example checks whether the file system named roots has 100 KB of available space. This example also checks whether the file system named auxfs1 has 120 KB of available space. The specified file space must be available before the job can start.

```
insert_job: unix_chk

job_type: CMD

machine: unixagent

command: /u1/procrun.sh

chk_files: /roots 100 /auxfs1 120
```

### Example: Verify the Available File Space on Windows

This example checks whether the C: drive has 100 KB of available space and the D: drive has 120 KB of available space. The specified file space must be available before the job can start.

```
insert_job: win_chk

job_type: CMD

machine: winagent

command: "C:\Programs\Payroll\pay.exe"

chk_files: "C: 100 D: 120"
```

## Pass Positional Arguments in a Command Job

When running workload, you might need to pass data between jobs and across platforms. You can pass positional arguments to a command or script in your job definition. Positional arguments are variables that can be passed to a program at the time the program is invoked. The arguments are assigned in the order they are passed.

### Follow these steps:

1. [Define a Command job.](#)
2. Add the following attribute to the job definition:

```
command: file argument...
```

- **file**

This attribute specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.

- **argument...**

This attribute specifies one or more arguments to pass to the command or script at run time.

### NOTE

Separate each argument with a space. Ensure that you specify each argument in the order it is expected in the program.

3. Run the job.  
The positional arguments are passed to the program.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

**Example: Pass Positional Arguments to a UNIX Script**

This example passes three arguments to a UNIX script. The first argument that is passed is "user 1". This argument is enclosed with quotation marks because it contains a space. The second argument that is passed is 905-555-1212, and the third argument is 749.

```
insert_job: cmd_job1
job_type: CMD
machine: unixprod
command: addinfo.sh "user 1" 905-555-1212 749
```

**Example: Pass Positional Arguments to a Windows Program**

This example passes two data files to a Windows program. The arguments are enclosed with quotation marks because they contain spaces.

```
insert_job: cmd_job2
job_type: CMD
machine: winprod
command: "c:\Programs\Payroll\pay.exe" "C:\Pay Data\salary.dat" "C:\Pay Data
\benefits.dat"
```

**Pass Environment Variables in a Command Job**

You can specify environment variables to define the local environment that the program runs in. You can modify existing environment variables or can create your own.

**Follow these steps:**

1. [Define a Command job](#).
2. Add the following attribute to the job definition:
  - **envvars**  
This attribute specifies the environment variables that define the local environment.
3. Run the job.  
The environment variables are passed to the program.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

**Example: Pass Environment Variables to a UNIX Script**

This example includes two envvars attributes that pass environment variables to a script and a third envvars attribute that defines the Present Working Directory (PWD). The parameter "user 1" is enclosed with quotation marks because it contains a space.

```
insert_job: unix_job
job_type: CMD
machine: unixprod
```

```
command: /home/scripts/pay
envvars: NAME="user 1"
envvars: JOB=PAYROLL
envvars: PWD=/usr/scripts/dailyrun
```

In this example, the pay script can reference these variables:

Environment Variable	Value Passed
NAME	user 1
JOB	PAYROLL
PWD	/usr/scripts/dailyrun

### NOTE

If the parameter `oscomponent.loginshell` is set to true in the `agentparm.txt` file of the agent, the agent invokes the user environment while running the script. To override the value of a shell variable that is already defined in the user login file, reassign a value to this variable in the script.

### Example: Pass Two Environment Variables to a Windows Command File

This example runs the `processrun.exe` command file on a Windows computer named `winprod`. The job uses two local environment variables named `PATH` and `TEMP`.

```
insert_job: cmd_job1
job_type: CMD
machine: winprod
command: "c:\cmds\processrun.exe"
envvars: PATH="c:\windows\system32"
envvars: TEMP="c:\temp"
```

### UNIX Environment Variables

When a Command job runs under a specific user account on UNIX, the agent can pass the environment variables of the user to the script or program. You can also set up the running environment of a script by overriding the environment variables in the job definition. For example, you can override the `HOME` environment variable to run the script under the login directory of a user.

You can pass the following UNIX environment variables in a job definition to override the variable values:

- **HOME** Identifies the login directory of the user. You can override the `HOME` value to set up a user-specific environment by specifying a different login directory in the job definition.

**Example:** `HOME=/home/guest/bin`

### NOTE

- You can set up the running environment of the script in the `.profile` and `.login` files.
- Set the `oscomponent.loginshell` parameter to true in the `agentparm.txt` file of the agent to run the login scripts located in the `HOME` directory.
- **PATH** Provides a list of directories that the shell searches when it needs to find a command. Each directory is separated by a colon, and the shell searches the directories in the order listed. The `PATH` variable is the most important environment variable. You can override the `PATH` value to set up a user-specific environment by specifying a different `PATH` in the job definition.

**NOTE**

Overriding the default system path can result in the *command not found* error.

- **ENV** Contains the name of the initialization file to run when a new Korn shell starts. You can override the ENV value to set up a user-specific environment by specifying a different ENV value in the job definition.

**Example:** ENV=/home/guest/bin/myenv

**NOTE**

The name of the file used to set up the script-running environment must be .profile. The .profile must be the same file used with the HOME variable.

- **PWD**  
Contains the absolute path name of your current directory.

**Define Alternative Error, Input, and Output Sources**

By default, the standard output and standard error output are redirected to the spool directory of the agent. You can define alternative error, input, and output sources in your job definition that override these defaults.

**Follow these steps:**

1. [Define a Command job](#).
2. Add one or more of the following attributes to the job definition:
  - **std\_err\_file**  
This attribute specifies where you want to redirect the standard error output. The output can be redirected to a file or a blob.
  - **std\_in\_file**  
This attribute specifies where you want to redirect standard input from. The input can be a file or a blob.
  - **std\_out\_file**  
This attribute specifies where you want to redirect the standard output. The output can be redirected to a file or a blob.
3. Run the job.  
The alternative sources are defined.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

**Example: Specify the Standard Input File and Output on UNIX**

This example specifies the /tmp/test.in file as the standard input file and /tmp/test.out as the standard output file .

```
insert_job: unix_stdin

job_type: CMD

machine: unixagent

command: /usr/common/backup

std_in_file: /tmp/test.in

std_out_file: /tmp/test.out
```

## Create a Job Blob

You can create a job blob that is associated with the Command job you are defining. The blob can contain the data that you specify in the job definition (input job blob), the output of the job (output job blob), or the error messages of the job (error job blob). Input job blobs are uploaded to the database when the job is defined. Output and error job blobs are uploaded to the database after the job runs.

Other jobs on different computers can use the input job blobs and output job blobs. Jobs cannot use error job blobs as input.

### Follow these steps:

1. [Define a Command job](#).
2. Add one or more of the following attributes to the job definition:
  - **blob\_file**  
This attribute specifies the name of the file that contains the input data.
  - **blob\_input**  
This attribute specifies the text to insert for the blob.
  - **std\_err\_file**  
This attribute specifies that the job blob contains the standard error messages of the job in textual or binary data format.
  - **std\_out\_file**  
This attribute specifies that the job blob contains the standard output messages of the job in textual or binary data format.
3. Run the job.  
The job blob is created.

### NOTE

- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.
- You can also use the `insert_blob` subcommand to create an input job blob after a job is defined.

### Example: Create an Input Job Blob at Job Definition Time

This example creates a job blob from the text that is specified in the `blob_input` attribute statement and uses the blob as input data for the command job named `test_blob`.

```
insert_job: test_blob
job_type: CMD
command: cat
machine: unixagent
blob_input: <auto_blobt>multi-lined text data for job blob
</auto_blobt>
std_in_file: $$blobt
owner: produser@unixagent
```

### Example: Create an Input Job Blob Using a File at Job Definition Time

This example creates a job blob from the data that is contained in the file named `blob_input_file.txt` and uses the blob data as input for the command job named `test_blob`.

```
insert_job: test_blob
```

```

job_type: CMD
machine: unixagent
command: cat
blob_file: /blob_input_file.txt
std_in_file: $$blobt
owner: produser@unixagent

```

### Example: Create an Error Input Blob

This example captures the standard error output and stores it as a textual format blob associated with the job.

```

insert_job: report_job1
job_type: CMD
machine: localhost
command: myapplication
std_err_file: $$blobt

```

### Example: Create an Output Blob

This example captures the standard output and stores it as a textual format blob associated with the job.

```

insert_job: report_job1
job_type: CMD
machine: localhost
command: myapplication
std_out_file: $$blobt

```

## Send a User-Defined Exit Code

By default, the scheduling manager interprets an exit code of 0 (zero) as job success and any other number as job failure. However, you can map exit codes other than 0 as job success and you can map specific codes as job failure.

### Follow these steps:

1. [Define a Command job](#).
2. Add one or more of the following attributes to the job definition:
  - **fail\_codes**  
This attribute defines which exit codes indicate job failure.
  - **success\_codes**  
This attribute defines which exit codes indicate job success.
3. Run the job.  
The user-defined exit code is sent.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Define a Range of Exit Codes to Indicate Job Failure

Suppose that you want a job that is named CMDJOB to run the procjob.exe file. The job is considered to have failed if the exit code is in the 40-50 range.

```
insert_job: CMDJOB
```



```
job_type: CMD

machine: winagt

command: "c:\temp\procjob.exe"

fail_codes: 40-50
```

### Example: Define a Exit Code to Indicate Job Success

This example shows the first and last lines of the payroll.sh script. The script returns the self-defined exit code 100 to the scheduling manager.

```
#!/usr/bin/sh

.

.

.

exit 100
```

The following job definition runs the script. The success\_codes attribute defines exit code 100 as success, indicating successful completion of the script.

```
insert_job: test_blob

job_type: CMD

machine: unixagent

command: /home/esp/payroll.sh

success_codes: 100
```

### Specify a Command or Script Name Without the Full Path

When defining a Command job, the agent usually requires the full path to the command or script name you want to run. However, you can specify the command or script name without the full path if the proper conditions are met.

#### Follow these steps:

1. [Define a Command job](#).
2. Add the following attribute to the job definition:
  - **command**

This attribute specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.

**UNIX:** To specify a name without the full path, the following conditions must be met:

- The agent is running under the root account.
- The agent is configured to resolve environment variables.
- The user ID you enter in the owner attribute has the authority to run the job on the agent. The user default shell is used.
- The path to the script or command name is set in the PATH system environment variable for the specified user ID.

**Windows:** To specify a name without the full path, the following conditions must be met:

- The agent is configured to search for paths to command or script files.
- The script or command file is located in *one* of the following directories:
  - The directory the agent is installed in
  - WINDOWS\system32 directory on the agent computer
  - WINDOWS\system directory on the agent computer
  - WINDOWS directory on the agent computer
  - Any other directory whose path is set in the system path or user path on the agent computer

### 3. Run the job.

The command or script name is specified.

#### NOTE

- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.
- To configure the agent to resolve environment variables, ask your agent administrator to refer to the information about the `oscomponent.lookupcommand` parameter in [Workload Automation Agent Parameters](#).

### Example: Run a Script that is Located in a Path Set in the PATH Variable (UNIX)

This example runs a script named `procscript.sh`. The job runs under the user ID `jsmith`, which has the authority to run the script. The path to `procscript.sh` is set in the PATH system environment variable for `jsmith` on the agent computer and the agent is configured to search for paths to command and script files.

```
insert_job: unix_job
job_type: CMD
machine: unixagent
command: procscript.sh
owner: jsmith
```

### Example: Specify a Script Name Without the Full Path (Windows)

This example runs a script named `procscript.bat`. The path to `procscript.bat` is set in the system path on the agent computer and the agent is configured to search for paths to command and script files.

```
insert_job: win_job
job_type: CMD
machine: winagent
command: procscript.bat
```

### Specify a Command or Script Name Using an Environment Variable (UNIX)

When defining a Command job on UNIX, you can specify the command or script name using an environment variable (for example, `$MY_PATH/myscript.sh`).

**Follow these steps:**

1. [Define a Command job.](#)
2. Add the following attribute to the job definition:
  - **command**  
This attribute specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met. Include the environment variable in the name.

**NOTE**

To use an environment variable, the following conditions must be met:

- The agent is running under the root account.
- The agent is configured to resolve environment variables.
- The user ID you enter in the owner attribute has the authority to run the job on the agent computer. The user default shell is used.
- The environment variable that you use, such as \$MY\_PATH, is set in the specified user ID's profile file.

3. Run the job.  
The command or script name is specified.

**NOTE**

- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.
- To configure the agent to resolve environment variables, ask your agent administrator to refer to the information about the `oscomponent.lookupcommand` parameter in [Workload Automation Agent Parameters](#).

**Example: Run a Script that is Located in a Path Set in a User Environment Variable**

In this example, an agent that is named `unixagent` runs a script named `myscript.sh`. The job runs under the user ID `jsmith`, which has the authority to run the script. The path to `myscript.sh` is set in the user environment variable `$MY_PATH`, which is defined in the profile file for `jsmith`. The agent is configured to search for paths to command and script files.

```
insert_job: unix_job
job_type: CMD
machine: unixagent
command: $MY_PATH/myscript.sh
owner: jsmith
```

**Run a Script Under a Specific User Account (UNIX)**

You can define a Command job to run a UNIX command or script under a specific account of a user.

**Follow these steps:**

1. On the client computer where you want to run the job, do *one* of the following steps:
  - Start the agent as root.
  - Start the agent as a user other than root.

**NOTE**

This user account must have permissions to access the resources that the job requires. If the job is defined to run under a different user (owner), the user account the agent is started under must have permissions to switch to that user. Otherwise, the job fails.

2. [Define a Command job.](#)
3. Add the following attribute to the job definition:
  - **owner**  
This attribute specifies the user ID that the job runs under.

**Default:** The default owner (the user ID who invokes jil to define the job)

#### NOTE

The user ID must have the permissions to access the required directories and run the commands and scripts on the agent computer.

#### 4. Run the job.

The script runs under the specified user.

#### NOTE

- If the shell is specified in the first line of the script and its path matches the path that is defined in the `oscomponent.validshell` parameter, you do not have to specify the shell attribute in the job definition to define the shell that you want the agent to use. For example, you can run a script using the environment defined in a specific account of a user.
- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Run a Script Under a Specific User Account

In this example, if the first line of the script is `#!/bin/ksh` and the path `/bin/ksh` is defined using the `oscomponent.validshell` parameter, you can use the owner attribute to run the script under a specific user account, as follows:

```
insert_job: unix_job
job_type: CMD
machine: unixagent
command: /home/guest/bin/cmd1.ksh
owner: guest
```

In this example, the agent runs the `cmd1.ksh` Korn script using the environment defined in either the `$HOME/.login` file or the

`$HOME/.profile` file, depending on which login shell is defined for the guest user:

- If `csh` is defined, the `$HOME/.login` file is used.
- If `ksh` is defined, the `$HOME/.profile` file is used.

The login shell for the guest user does not have to be the Korn shell. For example, the agent can pick up the following environment variables for the guest user:

```
HOME=/home/guest
LOGNAME=guest
USER=guest
SHELL=/usr/bin/csh
PWD=/home/guest
```

In this example, user `guest` has specified the login shell as the C shell. The agent, therefore, runs the `cmd1.ksh` script using the environment defined in the `$HOME/.login` file.

## Modify Resource Limits (UNIX)

When you define a Command job to run UNIX workload, you can set the job to modify resource limits on the agent computer for a given job. For example, you can define a job that modifies the maximum core filesize and CPU times on the UNIX computer.

### Follow these steps:

1. [Define a Command job.](#)

2. Add the following attribute to the job definition:

– **ulimit**

This attribute specifies one or more resource types and their soft and hard limits. Specify the attribute as follows:

```
ulimit: resource_type="soft_value,hard_value"
 [,resource_type="soft_value,hard_value"...]
```

3. Run the job.

The resource limits on the UNIX computer are modified.

**NOTE**

- If the job runs under a non-root user ID, the hard limit will not be modified if the value in the job definition is greater than the hard limit on the UNIX computer.
- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

**Example: Modify Multiple Resource Limits**

This example runs the `procrun.sh` script on the `unixagent` computer. The job modifies the following resource limits on the UNIX computer:

- The core filesize limit is 100 KB (soft limit). The size can be increased to 200 KB (hard limit).
- The stack size limit is 250 KB (soft limit). The size can be increased to 300 KB (hard limit).
- The CPU time can be up to 4000 seconds.
- The process virtual size limit is 3332 KB (soft limit). The size can be increased to an unlimited value.

```
insert_job: cmd_job
job_type: CMD
machine: unixagent
command: /u1/procrun.sh
ulimit: c="100,200", s="250,300", t="4000,unlimited", m="3332,unlimited"
```

**NOTE**

The resource limits are modified for the current job definition only. When you run another job, the default values are used.

## Customize the Run-time Environment for a Korn Shell Script (UNIX)

When you define a Command job to run a Korn shell script under the account of a user, you can customize the run-time environment of the job. The agent runs the job using the login environment of the specified user and the run-time environment you specify.

**Follow these steps:**

1. [Define a Command job](#).
2. Ensure that the `oscomponent.loginshell` parameter in the `agentparm.txt` file of the agent is set to `true`.
3. Add the following attributes to the job definition:

```
envvars: HOME=/directory_name

envvars: ENV=/directory_name/myenv
```

4. Run the job.

The run-time environment for the Korn shell script is customized. The agent runs the `/directory_name/.profile` file and the `/directory_name/myenv` file, before he runs the Korn script,

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

## Customize the Run-time Environment for a Bourne Shell Script (UNIX)

When you define a Command job to run a Bourne Shell script under an account of the user, you can customize the run-time environment of the job. The agent runs the job using the specified login environment of the user and the run-time environment you specify.

#### Follow these steps:

1. [Define a Command job](#).
2. Ensure that the `oscomponent.loginshell` parameter in the `agentparm.txt` file of the user is set to true.
3. Define a specific environment for the Korn shell.
4. Ask your agent administrator to configure the agent to run the Bourne Shell script using the Korn shell
5. Run the job.

The run-time environment for the Bourne Shell script is customized.

#### NOTE

- The ENV variable only works for the Korn shell. The Korn shell is a superset of the Bourne Shell. Anything that runs under the Bourne Shell runs without modification under the Korn shell.
- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

#### Example: Customize the Run-time Environment for a Bourne Shell Script

This example runs the `cmd1.sh` Bourne Shell script. The agent runs the following code, before he runs the script:

- The login file for the guest user
- The `/esp/myenv` file

```
insert_job: cmd_job
```

```
job_type: CMD
```

```
machine: unixagent
```

```
command: /home/bin/cmd1.sh
```

```
shell: /bin/ksh
```

```
owner: guest
```

```
envvars: ENV=/esp/myenv
```

## Customize the Run-time Environment for a C Shell Script (UNIX)

When you define a Command job to run a C shell script under an account of the user, you can customize the run-time environment of the job. The agent runs the job using the specified login environment of the user and the run-time environment you specify.

### Follow these steps:

1. [Define a Command job](#).
2. Ensure that the `oscomponent.loginshell` parameter in the `agentparm.txt` file of the agent is set to `true`.
3. Set the environment in a `.login` file.
4. Add the following attribute to the job definition:

```
envvars: HOME=login_file
```

#### – ***login\_file***

This attribute specifies the path to the `.login` file.

5. Run the job.  
The run-time environment for the C shell script is customized.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Customize the Run-time Environment for a C Shell Script

In this example, the agent picks up the environment for user `root` and runs the `/home/bin/.login` file before running the `cmd1.csh` C shell script. The agent is running as `root`.

### NOTE

The login shell for `root` is not the C shell.

```
insert_job: cmd_job
job_type: CMD
machine: unixagent
command: /home/guest/bin/cmd1.csh
envvars: HOME=/home/bin/
```

## Define a Command Job to Run a Perl Script (UNIX)

You can define a Command job to run a Perl script on UNIX.

You can use the following ways to run a Perl script:

- **Using the script**

When running a Perl script as a Command, the syntax must be as follows:

```
command: command argument..
```

Assumes that the `/usr/bin/perl` has been added to the `oscomponent.validshell` parameter in `agentparm.txt` file.

- **Using the `/usr/bin/perl` binary**

When the path to the script itself must be passed as argument.

```
command: /usr/bin/perl/command argument..
```

You need not add `/usr/bin/perl` to the `oscomponent.validshell` parameter in `agentparm.txt` file.

- You cannot use the optional shell attribute to force the command job to run under perl as it will cause an error. This is because Perl is not a true shell. For example the following syntax will display an error.

```
command: command argument..
shell: /usr/bin/perl
```

### Follow these steps:

1. Ask your agent administrator to add the path of the Perl interpreter to the `oscomponent.validshell` parameter in the `agentparm.txt` file of the agent, as shown in the following example:

```
oscomponent.validshell=/usr/bin/sh,/bin/csh,/bin/ksh,/usr/local/bin/perl,/usr/local/bin/bash
```

#### NOTE

If the `oscomponent.checkvalidshell` parameter is set to false, you do not need to perform the first step.

2. [Define a Command job.](#)
3. Add the command attribute to the job definition using the following syntax:

```
command: /usr/script
```

#### – **script**

This attribute specifies the path to the script you want to run.

**Example:** `cmd1.pl`

4. Do *one* of the following steps:

- Specify the path to the Perl interpreter in the first line of the script, as shown in the following example:

```
Perl Script cmd1.pl
#!/usr/local/bin/perl
print " $0 @ARGV\n";
while (($var, $value) = each %ENV) {
 print "$var = $value\n";
}
$live=$ENV{pick};
print " user variable pick = $live\n";
```

- Specify the path to the Perl interpreter in the job definition, using the shell attribute, as shown in the following example:

```
insert_job: perl_job
job_type: CMD
machine: unixagent
command: /bin/cmd1.pl Hello world
shell: /usr/local/bin/perl
owner: guest
```

5. Run the job.  
The job runs the Perl script.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

## Run the Windows Command Interpreter (Windows)

You can schedule a job to run a command, a batch script, program executable using the Windows command interpreter (`cmd.exe`).



**Follow these steps:**

1. Verify that the following parameters are set in the agentparm.txt file of the agent:

```
oscomponent.lookupcommand=true
oscomponent.cmdprefix.force=true
```

2. [Define a Command job](#).

3. Add the command attribute to the job definition using the following syntax:

```
command: command argument...
```

- **command**

This attribute specifies the cmd.exe command to run.

**Examples:** copy, dir, echo

- **argument...**

This attribute specifies one or more arguments to pass to the cmd.exe command.

**NOTE**

Separate each argument with a space. Ensure that you specify each argument in the order it is expected in the program.

4. Run the job.

The job runs the specified cmd.exe command.

**NOTE**

- If the agent parameters specified in Step 1 are not set to true, you must explicitly invoke the command interpreter in the command attribute, as follows:

```
command: "path\cmd.exe /C command argument..."
```

- **path**

Specifies the path to cmd.exe. The path to cmd.exe depends on your Windows operating system version. For example, on Windows NT, the path is C:\WINNT\system32\cmd.exe.

- To allow commands that have spaces in their path that is run without error, set the oscomponent.cmdprefix.force.quotes.full parameter to true. When you set the oscomponent.cmdprefix.force.quotes.full parameter to true, the agent wraps the entire command in double quotes before the Windows command interpreter runs the command.
- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

**Example: Run a Windows Operating System Command**

This example lists the contents of the c:\temp directory. The oscomponent.lookupcommand and oscomponent.cmdprefix.force parameters in the agentparm.txt file are set to true, so the path to the command interpreter (*path*\cmd.exe /c) is automatically prefixed to the command before running the process.

```
insert_job: cmd_job
job_type: CMD
machine: winprod
command: dir "c:\temp\"
```

**Example: Explicitly Invoke the Command Interpreter**

This example uses cmd.exe to copy a file to another location. The oscomponent.lookupcommand and oscomponent.cmdprefix.force parameters in the agentparm.txt file are not set to true, so the path to the command interpreter must be explicitly invoked in the command attribute.

```
insert_job: cmd_job
job_type: CMD
machine: winprod
```

```
command: "C:\Windows\system32\cmd.exe /C copy C:\env.txt C:\test\env.txt"
```

## Access Network Resources (Windows)

When your agent runs as a Windows service, you can schedule Windows workload that accesses network resources. For example, you can specify UNC names and can share names in your job definition.

Usually, when running a Windows program as a service, you are restricted to how you can access data on remote computers. For example, to access data on a remote computer as a specified user ID, you must run the Windows service with that user ID.

With the agent, however, those restrictions do not apply. Instead of running the agent service with a specific user ID, you can specify the user ID with the owner attribute in your job definition. To use the owner attribute, your agent must run as a Windows service under the local system account (the default configuration).

### Follow these steps:

1. Verify with your agent administrator that the agent is running as a Windows service under the local system account.
2. Ask your scheduling manager administrator to define a user ID and password on the scheduling manager that has access to the file on the remote Windows system.
3. [Define a Command job](#).
4. Add the following attributes to the job definition:
  - **command**  
This attribute specifies a command, executable, application, or batch file to run when all the starting conditions are met.

#### NOTE

- You can specify UNC (Universal Naming Convention) names. A UNC name is the name of a file or other resource that begins with two backslashes (\\), indicating that it exists on a remote computer.
- You can specify share names. A share name is an alias for the path the resource exists in.
- You can specify the share names C\$ and ADMIN\$ if the agent service logs in to a remote Windows server as a user with administrative authority. The agent can then access remote resources that are not marked as shared.
- You can specify a network mapped drive if you have set `oscomponent.restoredrives` parameter to true in the `agentparam.txt` file. Prior to the job execution, the agent restores the mapped drive letters and their respective network shares that are defined by the external user. It internally maps them as the job user and accesses the external files on the network mapped drive. We recommend that you specify UNC names as the network mapped drive exhibits low performance based on the type of the network share mapped, the windows operating system, and the available system and network resources.

#### – owner

This attribute specifies the Windows user ID and the domain the user ID belongs to.

**Default:** The default owner (the user ID who invokes jil to define the job)

5. Run the job.  
The job accesses the specified Windows network resources.

#### NOTE

- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.
- Verify that you are complying with the terms of your agent license agreement, before accessing network resources with your agent, . In most situations, you are permitted to access data

on remote computers; however, scripts or executable files run by an agent should use the CPU and memory of the computer where the agent resides.

- Although not recommended, your agent administrator can run the agent as a Windows service under a local user account (the This Account option). When you run the service under a local user account, when the service starts, it runs using the security context of the specified user account. If the user account and password are valid, the service process has access to network resources.
- When you access a remote computer from an agent on Windows, the user ID defined in the owner attribute or in the This Account option is a domain user. If the local and remote servers are standalone servers, you must have the same user IDs and passwords defined on both servers.
- For more information about configuring and running the agent as a Windows service, see the [Workload Automation Agents](#) documentation.

### Example: Run a Command on a Remote Server

In this example, the path c:\WINNT\Profiles\Visitor\Desktop\ has the share name MyDesktop. The command notify.cmd is in that path on the CYBNT server. JDOE is a user ID in the CYBDOM domain and has access to the notify.cmd command. JDOE's password is defined on the scheduling manager.

```
insert_job: exe_job1
job_type: CMD
machine: NT20
command: \\CYBNT\MyDesktop\notify.cmd
owner: CYBDOM\JDOE
```

### Example: Run an Executable in Public Folder on a Remote Server

This example runs calc.exe on the CYBNT server. CYBUSER is a user ID in the CYBDOM domain. CYBUSER is defined on the scheduling manager and has access permission to the public folder.

```
insert_job: exe_job2
job_type: CMD
machine: NT30
command: \\CYBNT\public\calc.exe
owner: CYBDOM\CYBUSER
```

### Example: Access a Remote Resource Using the C\$ Share Name

In this example, an administrator accesses the drive C over the network through an agent. The agent is running under the System Account option. The agent runs the test application in the c:\working directory on the server CYBNT. The directory c:\working is not a shared resource. The user admin1 is a valid user on both the local and remote computers and belongs to the Administrators group. admin1 is also in the CYBDOM domain.

```
insert_job: exe_job3
job_type: CMD
machine: NT30
command: \\CYBNT\C$\working\test
owner: CYBDOM\admin1
```

### Specifying a Password for a User ID (Windows)

You can define a Command job to access Windows network resources by specifying the user ID and the domain name the user ID belongs to using the owner attribute. The resources are accessed under the specified user ID.

**NOTE**

To use the owner attribute in a Command job that runs on Windows, your agent must run as a Windows service under the local system account.

If the user ID requires a password, your administrator must define the password on the scheduling manager. For security reasons, you do not define the password in the job definition. Your administrator must define and store the password in the AutoSys Workload Automation database using the `autosys_secure` command.

When you specify a user ID that requires a password in a job definition, the scheduling manager sends the user ID and password pair (the password is encrypted) to the agent. The scheduling manager searches the repository for an entry matching the specified owner. The results from the search, the encrypted password or null, are provided to the agent to run the job.

The following job types require a user password:

- Database
- FTP
- PeopleSoft
- SAP
- Command (for Windows)

## Database Jobs Overview

Database jobs let you automate common database tasks on Oracle, Microsoft SQL Server, Sybase, and IBM DB2 databases.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

You can define the following database jobs:

- **SQL**  
This job lets you run an SQL query against a database.
- **Database Stored Procedure**  
This job lets you run a stored procedure
- **Database Trigger**  
This job lets you monitor for added, deleted, and updated rows in a database table.
- **Database Monitor**  
This job lets you monitor for added and deleted rows in a database table.

**NOTE**

You can also run jobs in the Microsoft SQL Server Agent using the `mssql_call` script. For more information about the Workload Automation Agent for Microsoft SQL Server or about running the `mssql_call` script using AutoSys Workload Automation, see [Workload Automation Agents documentation](#).

## Database Trigger Job and Database Monitor Job Differences

You can monitor database changes by using a Database Trigger job or a Database Monitor job. The Database Trigger job offers the following advantages over Database Monitor jobs:

- You can monitor for more conditions. With Database Trigger jobs, you can monitor for rows added, deleted, and updated. With Database Monitor jobs, you can only monitor for rows added and deleted. Database Trigger jobs define

triggers to the underlying database being monitored. Database Monitor jobs poll the database table for changes in row counts or column values.

- Database Trigger jobs detect all changes that are made to the database; Database Monitor jobs monitor for changes only in 10-second intervals, by default.

Suppose that you want to send a notification when a new row is added. Within a 10-second interval, assume that a row is added while another row is deleted. A Database Trigger job that monitors for an INSERT would complete and send the notification when the new row is detected. A Database Monitor job that monitors for an INCREASE would not complete or send a notification because no change in the total number of rows was detected.

Each Database Trigger job creates a database trigger on the database. The database trigger templates are provided with the agent. Consult with your database administrator before using a Database Trigger job.

#### NOTE

- If a table is dropped, the table being monitored is not dropped because the Database Trigger or Database Monitor job remains in the RUNNING status .
- For more information about the database trigger templates, see the db.trig.profile parameter in [Configure the Agent for Monitoring File Activity](#).

## User IDs and Passwords for Database Jobs

All database jobs require a user ID that has the appropriate permissions to access the information in the database. The job runs under that user ID. You specify a user ID using the owner attribute.

These database user IDs and passwords must be defined on AutoSys Workload Automation by using the autosys\_secure command. Specify a database user ID using the owner attribute, or use the default owner value when you define a database job.

## Define a Database Monitor Job

You can define a Database Monitor (DBMON) job to monitor a database table for an increase or decrease in the number of rows. To monitor the database table for specific changes, you can add a monitor condition to the job definition. When the condition is met, the job completes.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: DBMON**  
This attribute specifies that the job type is Database Monitor.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **tablename**  
This attribute specifies the name of the database table to monitor for the changes.
2. Do *one* of the following steps:
  - Ensure that a default database resource location is defined in the agentparm.txt file of the agent using the db.default.url parameter.
  - Add the following attribute to the definition:
    - **connect\_string**  
This attribute specifies the database resource location.

**NOTE**

This attribute overrides the db.default.url agent parameter.

3. (Optional) Specify the following attribute:

- **owner Attribute -- Define the Owner of the Job** This attribute specifies the user ID that the job runs under. This value overrides the default owner of the job.

**Default:** The user ID who invokes jil to define the job

**NOTE**

Database jobs do not support Windows authentication.

4. (Optional) Specify optional Database Monitor attributes:

- [continuous](#)
- [job\\_class](#)
- [job\\_terminator](#)
- [monitor\\_cond](#)
- [monitor\\_type](#)
- [user\\_role](#)

5. (Optional) Specify common attributes that apply to all job types.

The Database Monitor job is defined.

**NOTE**

- Database jobs do not support Windows authentication.
- All database jobs require the job owner to have access to the information in the database. Ensure that you assign ownership to a user with the appropriate database access permissions or that the default owner (the user who defines the job) has those permissions.
- All database jobs require a database resource location. Ensure that you define this value; otherwise, the job fails. Your agent administrator can specify a default location using the db.default.url parameter on the agent, or you can specify the connect\_string attribute in the job definition.
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

**Example: Monitor a Table for Added Rows**

This example monitors for an increase in the number of rows in the staff table. The table is in a SQL Server database named ORDERS.

```
insert_job: dbmon_add
job_type: DBMON
machine: dbagent
owner: dbuser@MSSQL
```

```
tablename: staff
monitor_type: INCREASE
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=ORDERS"
```

### Example: Monitor a Table for Added or Deleted Rows

This example monitors the STAFF table for a change in the number of rows. When a row that contains the name Jonson is added or deleted, the job completes. The job connects to the default database resource location defined on the agent.

```
insert_job: dbmon_job
job_type: DBMON
machine: dbagent
owner: entadm@myhost
monitor_type: VARIANCE
tablename: staff
monitor_cond: NAME='Jonson'
```

### Example: Monitor a Table for Added Rows With a Condition

This example monitors the emp table for an increase in the number of rows. When a new row has a sal greater than 100000, the job completes. The job connects to a SQL Server database named ORDERS. The database user ID is set to the user who invokes jil to define the job (the default owner).

```
insert_job: dbmon1
job_type: DBMON
machine: DB_agent
owner: dbuser@MSSQL
monitor_type: INCREASE
monitor_cond: sal>100000
tablename: emp
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=ORDERS"
```

### Example: Monitor for an Increase in the Number of Rows in a Table Using a Condition

This example defines a job to monitor the table named Inventory\_List and complete if the number of rows increases when the number of units of ProductA is below 1000.

```
insert_job: dbmon_job
job_type: DBMON
machine: dbagt
tablename: Inventory_List
monitor_type: INCREASE
monitor_cond: "ProductA < 1000"
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

## Define a Database Trigger Job

You can define a Database Trigger (DBTRIG) job to monitor a database table for added, deleted, or updated rows. To monitor the database table for specific changes, you can add a condition to the job definition. When the condition is met, the job completes.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: DBTRIG**  
This attribute specifies that the job type is Database Trigger.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **dbtype**  
This attribute specifies the type of the database that the job monitors.
  - **tablename**  
This attribute specifies the name of the database table to monitor for the changes.
2. Do *one* of the following steps:
  - Ensure that a default database resource location is defined in the agentparm.txt file of the agent using the db.default.url parameter.
  - Add the following attribute to the definition:
    - **connect\_string**  
This attribute specifies the database resource location.

**NOTE**

This attribute overrides the db.default.url agent parameter.

3. (Optional) Specify the following attribute:
  - **owner Attribute -- Define the Owner of the Job** This attribute specifies the user ID that the job runs under. This value overrides the default owner of the job.  
**Default:** The user ID who invokes jil to define the job.

**NOTE**

This user ID must be authorized to create triggers on the database or schema the table belongs to. For Microsoft SQL Server, this user ID must own the database table identified by the tablename attribute. The password for the user must be defined in the database using the autosys\_secure command.

4. (Optional) Specify the optional Database Trigger attributes:
  - **continuous**
  - **job\_class**
  - **job\_terminator**
  - **trigger\_cond**
  - **trigger\_type**
  - **user\_role**
5. (Optional) Specify common attributes that apply to all job types.  
The Database Trigger job is defined.

**NOTE**

- Database jobs do not support Windows authentication.
- All database jobs require a database resource location. Ensure that you define this value; otherwise, the job fails. Your agent administrator can specify a default location using the db.default.url parameter on the agent, or you can specify the connect\_string attribute in the job definition.
- All database jobs require the job owner to have access to the information in the database. Ensure that you assign ownership to a user with the appropriate database access permissions or that the default owner (the user who defines the job) has those permissions.
- All database jobs require a user ID. This user ID must be authorized to create triggers on the database or schema the table belongs to. For Microsoft SQL Server, this user ID must own the database table identified



by the tablename attribute. The job runs under the user ID. The owner attribute in the job definition specifies the user ID (the default is the user who invokes jil to define the job).

- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

### Example: Monitor a Table for Added Rows

This example monitors the emp table. The trigger\_type attribute is not specified, so the job monitors for added rows by default. When a row is added, the job completes. The database resource location is defined on the agent, so the connect\_string attribute is not required in the job definition.

```
insert_job: dbtrig_job
job_type: DBTRIG
machine: DB_agent
owner: dbuser@ORA
dbtype: Oracle
tablename: emp
```

### Example: Specify a Trigger Type for SQL Server

This example monitors the stores table for an added row or a deleted row. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job remains in a RUNNING state waiting for an added or deleted row. When a row is either added or deleted, the job completes.

```
insert_job: dbtrig1
job_type: DBTRIG
machine: DB_agent
trigger_type: DELETE,INSERT
tablename: stores
dbtype: MSSQL
owner: sa@myhost
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

## Examples: Monitoring Oracle Database Tables

The following examples are Database Trigger Jobs that monitor Oracle database tables:

### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

### Example: Monitor an Oracle Database Table for Deleted Rows

This example monitors the emp table for deleted rows. The job runs under the user who is named scott, who has the authority to create triggers on the database or schema the table belongs to. When a row is deleted, the job completes.

```
insert_job: dbtrig2
job_type: DBTRIG
machine: DB_agent
dbtype: Oracle
trigger_type: DELETE
tablename: emp
owner: scott@orcl
connect_string: "jdbc:oracle:thin:@myhost:1521:orcl"
```

### Example: Monitor an Oracle Database Table for an Added or Deleted Row

This example monitors the emp table for an added row or a deleted row. The job runs under the user who is named scott, who has the authority to create triggers on the database or schema the table belongs to. The job remains in a RUNNING state while waiting for an added or deleted row. When a row is either added or deleted, the job completes.

```
insert_job: dbtrig_ora
job_type: DBTRIG
machine: dbagent
dbtype: Oracle
trigger_type: DELETE,INSERT
tablename: emp
owner: scott@orcl
connect_string: "jdbc:oracle:thin:@myhost:1521:orcl"
```

### Examples: Specify Trigger Conditions for Deleted Rows in an Oracle Database

- This example monitors the emp table for deleted rows. The job runs under a user who has the authority to create triggers on the database or schema the table belongs to. The job connects to the default database resource location defined on the agent. When a row containing deptno 75 is deleted, the job completes.

```
insert_job: dbtrig_delete
job_type: DBTRIG
machine: dbagent
dbtype: Oracle
trigger_type: DELETE
trigger_cond: old.deptno=75
tablename: emp
owner: scott@orcl
```

- This example monitors the emp table for added rows. The job runs under a user who has the authority to create triggers on the database or schema the table belongs to. When a row containing an ename beginning with the letter g is added, the job completes.

```
insert_job: dbtrig_insert
job_type: DBTRIG
machine: dbagent
dbtype: Oracle
trigger_type: INSERT
trigger_cond: new.ename like 'g%%'
```

```
tablename: emp
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

- This example monitors the emp table for added or updated rows. The job runs under a user who has the authority to create triggers on the database or schema the table belongs to. The job completes when a new or updated row does not contain a job field equal to sales.

#### NOTE

The <> symbol indicates not equal to.

```
insert_job: dbtrig_insertupdate
job_type: DBTRIG
machine: dbagent
dbtype: Oracle
trigger_type: INSERT,UPDATE
trigger_cond: new.job<>'sales'
tablename: emp
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

## Examples: Monitoring Microsoft SQL Server Database Tables

The following examples are Database Trigger jobs that monitor Microsoft SQL Server database tables:

#### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

### Example: Monitor a SQL Server Database Table for a New or Deleted Row

This example monitors the stores table for an added row or a deleted row. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job remains in a RUNNING state waiting for an added or deleted row. When a row is either added or deleted, the job completes.

```
insert_job: dbtrig1
job_type: DBTRIG
machine: DB_agent
trigger_type: DELETE,INSERT
tablename: stores
dbtype: MSSQL
owner: sa@myhost
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

### Example: Monitor a SQL Server Database Table for Two Changes

This example monitors the sales table for changes to the ord\_date and qty columns. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job completes only when both columns have changed.

```
insert_job: dbtrig_sqlsvr
job_type: DBTRIG
machine: dbagent
```

```
dbtype: MSSQL
owner: sa@myhost
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
tablename: sales
trigger_type: UPDATE
trigger_cond: UPDATE(ord_date) and UPDATE(qty)
```

### Example: Monitor a SQL Server Database Table for Added Rows with a Trigger Condition

This example monitors the sales table for added rows. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job connects to the default database resource location defined on the agent. When the qty for inserted title ID TC7777 is greater than or equal to 20, the job completes.

```
insert_job: dbtrig3
job_type: DBTRIG
machine: DB_agent
dbtype: MSSQL
trigger_type: INSERT
trigger_cond: (select QTY from INSERTED where TITLE_ID='TC7777')>=20
tablename: sales
owner: sa@myhost
```

### Example: Monitor a SQL Server Database Table for Deleted Rows

This example monitors the sales table for deleted rows. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job completes when a row is deleted.

```
insert_job: dbtrig4
job_type: DBTRIG
machine: DB_agent
dbtype: MSSQL
trigger_type: DELETE
tablename: sales
owner: sa@myhost
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

## Examples: Monitoring IBM DB2 Database Tables

The following examples are Database Trigger jobs that monitor IBM DB2 Server database tables:

### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition sections](#).

### Example: Monitor an IBM DB2 Database Table for Added Rows with a Trigger Condition

This example monitors the STAFF table for added rows. The job runs under the entadm user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. When the total number of rows is greater than or equal to 37, the job completes.

```
insert_job: dbtrig1
job_type: DBTRIG
```

```

machine: DB_agent
dbtype: DB2
trigger_type: INSERT
trigger_cond: (select count(*) from STAFF)>=37
tablename: STAFF
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"

```

### Example: Monitor an IBM DB2 Database Table for Changes

This example monitors the STAFF table for changes. The job connects to the SAMPLE database and runs under the entadm user, who is authorized to create triggers on the database or schema the table belongs to. The job completes when the table has changed.

```

insert_job: dbtrig_db2
job_type: DBTRIG
machine: dbagent
dbtype: DB2
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
tablename: STAFF
dbtype: DB2
trigger_type: UPDATE

```

### Example: Monitoring a Sybase Database Table

The following example is a Database Trigger Job that monitors a Sybase database table:

#### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

### Example: Monitor a Sybase Database Table for Changes

This example monitors the ap\_invoices table for changes. The job connects to the Sybase database named APDB that is reachable on port 5001 on a host named myhost. The job completes when the table has changed.

```

insert_job: db_sqltrig_upd
job_type: DBTRIG
machine: localhost
dbtype: Sybase
connect_string: "jdbc:sybase:Tds:myhost:5001/APDB"
trigger_type: UPDATE
tablename: ap_invoices
owner: admin@myhost

```

### Define a Database Stored Procedure Job

You can define a Database Stored Procedure (DBPROC) job to invoke a procedure stored in a database. You can add criteria to the job definition to test the output procedure. If the result matches the criteria, the job completes successfully. When the procedure executes, the output parameter values from the database are returned to AutoSys Workload

Automation. You can view the output parameter values in the spool file of the job. By default, the agent separates the output parameter values in the return string with a vertical bar ( | ).

If you are using Oracle or SQL Server database, you can also define a Database Stored Procedure job to run a stored function.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: DBPROC**  
This attribute specifies that the job type is Database Stored Procedure.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **sp\_name**  
This attribute specifies the database stored procedure to run.
2. Do *one* of the following steps:
  - Ensure that a default database resource location is defined in the agentparm.txt file of the agent using the db.default.url parameter.
  - Add the following attribute to the definition:
    - **connect\_string**  
This attribute specifies the database resource location.

#### NOTE

This attribute overrides the db.default.url agent parameter.

3. (Optional) Specify the following attribute:
  - **owner Attribute -- Define the Owner of the Job** This attribute specifies the user ID that the job runs under. This value overrides the default owner of the job.  
**Default:** The user ID who invokes jil to define the job

#### NOTE

Database jobs do not support Windows authentication.

4. (Optional) Specify optional Database Stored Procedure attributes:
  - **job\_class**
  - **job\_terminator**
  - **result\_type**
  - **sp\_arg**
  - **success\_criteria**
  - **user\_role**
5. (Optional) Specify common attributes that apply to all job types.  
The Database Stored Procedure job is defined.

**NOTE**

- Database jobs do not support Windows authentication.
- All database jobs require the job owner to have access to the information in the database. Ensure that you assign ownership to a user with the appropriate database access permissions or that the default owner (the user who defines the job) has those permissions.
- All database jobs require a database resource location. Ensure that you define this value; otherwise, the job fails. Your agent administrator can specify a default location using the `db.default.url` parameter on the agent, or you can specify the `connect_string` attribute in the job definition.
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

**Example: Invoke a Stored Procedure from a Database**

This example invokes the `calcproc` stored procedure. The default database resource location is defined on the agent, so the `connect_string` attribute is not required in the job definition.

```
insert_job: sp_default
job_type: DBPROC
machine: DB_agent
owner: dbuser@dbhost
sp_name: calcproc
```

**Example: Invoke a Stored Procedure from a SQL Server Database**

This example invokes the `byroyalty` stored procedure located in the `pubs` database. When the job runs, a value of 40 is passed to the input parameter named `percentage`.

```
insert_job: sp1_job
job_type: DBPROC
machine: DB_agent
owner: sa
sp_name: byroyalty
sp_arg: name=percentage, argtype=IN, datatype=INTEGER, value=40
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

**Example: Invoke a Stored Procedure with Input and Output Parameters from a SQL Server Database**

This example invokes the following stored procedure located in the `pubs` database. The procedure returns a value from the `emp` table.

```
CREATE PROCEDURE EMPLOY
(@f_name VARCHAR(20),
```

```

@l_name VARCHAR(30),
@pubid CHAR(4) OUTPUT)
AS BEGIN
SELECT
@pubid=pub_id
FROM emp
WHERE
fname=@f_name
and
lname=@l_name
print @l_name+@f_name+@pubid
END
GO

```

The job returns the pubid that matches the employee named John Doe. The pubid is recorded in the job's spool file.

```

insert_job: sp2_job
job_type: DBPROC
machine: DB_agent
sp_name: EMPLOY
sp_arg: name=f_name, argtype=IN, datatype=VARCHAR, value=John
sp_arg: name=l_name, argtype=IN, datatype=VARCHAR, value=Doe
sp_arg: name=pubid, argtype=OUT, datatype=CHAR
owner: sa
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"

```

### Example: Invoke a Stored Procedure with Input and Output Parameters from an IBM DB2 Database

This example invokes the following stored procedure under the user entadm:

```

CREATE PROCEDURE DEPT_MEDIAN
 (IN deptNumber SMALLINT, OUT medianSalary DOUBLE)
LANGUAGE SQL
BEGIN
 DECLARE v_numRecords INTEGER DEFAULT 1;
 DECLARE v_counter INTEGER DEFAULT 0;
 DECLARE c1 CURSOR FOR
 SELECT CAST(salary AS DOUBLE) FROM staff
 WHERE DEPT = deptNumber
 ORDER BY salary;
 DECLARE EXIT HANDLER FOR NOT FOUND
 SET medianSalary = 6666;
-- initialize OUT parameter
 SET medianSalary = 0;
 SELECT COUNT(*) INTO v_numRecords FROM staff
 WHERE DEPT = deptNumber;
 OPEN c1;
 WHILE v_counter < (v_numRecords / 2 + 1) DO
 FETCH c1 INTO medianSalary;

```



```

 SET v_counter = v_counter + 1;
 END WHILE;
 CLOSE c1;
END

```

DEPT\_MEDIAN returns the median salary for the department with deptNumber 20 from the STAFF table. The median salary, 18171.25, is recorded in the job's spool file.

```

insert_job: deptmed
job_type: DBPROC
machine: DB_agent
sp_name: ENTADM.DEPT_MEDIAN
sp_arg: name=deptNumber, argtype=IN, datatype=SMALLINT, value=20
sp_arg: name=medianSalary, argtype=OUT, datatype=DOUBLE
owner: entadm
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"

```

The spool file for this job contains the following output:

```

Output of messages for workload object DEPTMED/DBAPPL.7/MAIN
Start date Thu Aug 31 15:23:44 EDT 2006

{ call ENTADM.DEPT_MEDIAN(?, ?) }
medianSalary=18171.25

```

## Define a SQL Job

You can define an SQL job to run an SQL query against an Oracle, SQL Server, Sybase, or DB2 database. When the job runs, the SQL statement is invoked and the results are stored in an output file or job spool file. You can also add criteria to the job definition to test the query result. If the result matches the criteria, the job completes successfully. Otherwise, the job fails.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SQL**  
This attribute specifies that the job type is SQL.
  - **machine Attribute -- Define the Client Where a Job Runs** This attribute specifies the name of the machine on which the job runs.
  - **sql\_command**  
This attribute specifies the SQL statement to run against a database table.
2. Do *one* of the following steps:
  - Ensure that a default database resource location is defined in the agentparm.txt file of the agent using the db.default.url parameter.
  - Add the following attribute to the definition:
    - **connect\_string**  
This attribute specifies the database resource location.

**NOTE**

This attribute overrides the db.default.url agent parameter.

3. (Optional) Specify the following attribute:

- **owner Attribute -- Define the Owner of the Job** This attribute specifies the user ID that the job runs under. This value overrides the default owner of the job.

**Default:** The user ID who invokes jil to define the job

**NOTE**

Database jobs do not support Windows authentication.

4. (Optional) Specify optional SQL attributes:

- [destination\\_file](#)
- [job\\_class](#)
- [job\\_terminator](#)
- [success\\_criteria](#)
- [user\\_role](#)

5. (Optional) Specify common attributes that apply to all job types.

The SQL job is defined.

**NOTE**

- All database jobs require a database resource location. Ensure that the agent parameter file (agentparm.txt) specifies a default location or that the job definition includes a connect\_string attribute statement.
- All database jobs require the job owner to have access the information in the database. Ensure that you assign ownership to a user with the appropriate database access permissions or that the default owner (the user who defines the job) has those permissions.
- Database jobs do not support Windows authentication.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Example: Delete a Row from a Table**

This example deletes the row for stor\_id 6523 from the stores table. The default database resource location is defined on the agent, so the connect\_string attribute is not required in the job definition.

```
insert_job: deletejob
job_type: SQL
machine: DB_agent
sql_command: DELETE FROM stores WHERE stor_id='6523'
owner: scott@orcl
```

**Example: Delete a Row from an Oracle Database Table**

This example deletes a row from the emp table for the employee with ename robert.

```
insert_job: deletejob
job_type: SQL
machine: DB_agent
sql_command: DELETE FROM EMP WHERE ENAME='robert'
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

**Examples: Running SQL Queries Against Oracle Database Tables**

The following examples are jobs that run SQL queries against the Oracle database tables:

**NOTE**

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition sections](#).

**Example: Add a Row to an Oracle Database Table**

This example adds a row of data to the emp table.

```
insert_job: insertjob
job_type: SQL
machine: DB_agent
sql_command: INSERT INTO EMP(EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO)
VALUES(2476, 'robert', 'sales', 435, '01-OCT-2011', 65000, 10, 75)
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

**Example: Update a Row in an Oracle Database Table**

This example updates a record in the emp table and changes the sal to 75,000 for the employee with ename Robert.

```
insert_job: updatejob
job_type: SQL
machine: DB_agent
sql_command: UPDATE EMP SET SAL=75000 where ENAME='robert'
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

**Example: Delete a Row from an Oracle Database Table**

```
insert_job: deletejob
job_type: SQL
machine: DB_agent
sql_command: DELETE FROM EMP WHERE ENAME='robert'
owner: scott@orcl
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
```

**Example: Return Data from an Oracle Database Table that Match a Condition**

This example queries the emp table for enames that have salaries greater than 40,000. If the query returns an ename that begins with the letter d, the job completes:

```
insert_job: selectjob
job_type: SQL
machine: DB_agent
sql_command: SELECT ename FROM emp WHERE sal > 40000
owner: scott@orcl
success_criteria: ENAME=d.*
connect_string:"jdbc:oracle:thin:@myhost:1521:orcl"
destination_file: /emp/salary.txt
```

For example, the salary.txt file contains the following output:

```
Output for: SELECT ename FROM emp WHERE sal > 40000
ENAME

donald
```

## Examples: Running SQL Queries Against Microsoft SQL Server Database Tables

The following examples are jobs that run SQL queries against Microsoft SQL Server database tables:

### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

### Example: Add a Row to a SQL Server Database Table

This example adds a row for a new store to the stores table.

```
insert_job: insertjob
job_type: SQL
machine: DB_agent
sql_command: INSERT INTO stores(stor_id, stor_name, stor_address, city, state, zip)
VALUES('6523', 'BooksMart', '6523 Main St.', 'San Diego', 'CA', '93223')
owner: sa@myhost
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

### Example: Delete a Row from a SQL Server Database Table

This example deletes the row for stor\_id 6523 from the stores table.

```
insert_job: deletejob
job_type: SQL
machine: DB_agent
sql_command: DELETE FROM stores WHERE stor_id='6523'
owner: sa@myhost
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

### Example: Update a Row in a SQL Server Database Table

This example updates the row in the sales table that matches ord\_num 6871 and changes the values for the ord\_date and qty.

```

insert_job: updatejob
job_type: SQL
machine: DB_agent
sql_command: UPDATE sales SET ord_date='6/15/2006', qty=10 WHERE ord_num='6871'
owner: sa@myhost
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"

```

### Example: Return Data from a SQL Server Database Table that Match a Condition

This example queries the sales table for ord\_num that have a qty greater than 20. The ord\_num that match the query appear in the output file ordnum.txt.

```

insert_job: selectjob
job_type: SQL
machine: DB_agent
sql_command: SELECT ord_num FROM sales WHERE qty > 20
owner: sa@myhost
success_criteria: ord_num=A2976
connect_string:"jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
destination_file: "C:\sales\ordnum.txt"

```

The ordnum.txt file contains the following ord\_num:

```

A2976
QA7442.3
P2121
N914014
P3087a
P3087a
X999
P723
QA879.1

```

The job completes because the query returns an ord\_num that matches the job criteria A2976.

Suppose that we change the success\_criteria attribute to the following code:

```
success_criteria: B+[0-9]
```

In this case, the query would return the same order numbers, but the job fails because it cannot find a matching ord\_num containing the letter B and followed by a number.

### Examples: Running SQL Queries Against Sybase Database Tables

The following examples are jobs that run SQL queries against the Sybase database tables:

#### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

#### Example: Add a Row to a Sybase Database Table

This example adds a row for a new store to the stores table.

```
insert_job: insertjob
```

```
job_type: SQL
```

```
machine: DB_agent
```

```
sql_command: INSERT INTO stores(stor_id, stor_name, stor_address, city, state, zip) VALUES('652
```

```
owner: sa@myhost
```

```
connect_string:"jdbc:sybase:Tds:myhost:5001/APDB"
```

### **Example: Delete a Row from a Sybase Table**

This example deletes the row for stor\_id 6523 from the stores table.

```
insert_job: deletejob
```

```
job_type: SQL
```

```
machine: DB_agent
```

```
sql_command: DELETE FROM stores WHERE stor_id='6523'
```

```
owner: sa@myhost
```

```
connect_string:"jdbc:sybase:Tds:myhost:5001/APDB"
```

### **Example: Update a Row in a Sybase Table**

This example updates the row in the sales table that matches ord\_num 6871 and changes the values for the ord\_date and qty.

```
insert_job: updatejob
```

```
job_type: SQL
```

```
machine: DB_agent
```

```
sql_command: UPDATE sales SET ord_date='6/15/2006', qty=10 WHERE ord_num='6871'
```

```
owner: sa@myhost
```

```
connect_string:"jdbc:sybase:Tds:myhost:5001/APDB"
```

## Examples: Running SQL Queries Against IBM DB2 Tables

The following examples are jobs that run SQL queries against the IBM DB2 database tables:

### NOTE

These examples use optional database attributes. For more information about the optional attributes and their JIL syntax, see the [Common Job Attributes](#) and [JIL Job Definition](#) sections.

### Example: Add a Row to an IBM DB2 Database Table

This example adds a row of data to the STAFF table under the user entadm.

```
insert_job: insertjob
job_type: SQL
machine: DB_agent
sql_command: INSERT into ENTADM.STAFF(ID, NAME, DEPT, JOB, YEARS, SALARY, COMM)
VALUES(556, 'Jonson', 84, 'Sales', 1, 40500.50, 100)
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
```

### Example: Update a Row in an IBM DB2 Database Table

This example updates a record in the STAFF table under the user entadm. The job changes the years to 3 for the employee with the name Jonson.

```
insert_job: updatejob
job_type: SQL
machine: DB_agent
sql_command: UPDATE ENTADM.STAFF SET YEARS=3 where NAME="Jonson"
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
```

### Example: Delete a Row from an IBM DB2 Database Table

This example deletes a row from the STAFF table under the user entadm for the employee with the name Jonson.

```
insert_job: deletejob
job_type: SQL
machine: DB_agent
sql_command: DELETE FROM ENTADM.STAFF where NAME="Jonson"
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
```

### Example: Return Data from an IBM DB2 Database Table that Match a Condition

This example queries the STAFF table under the user entadm for names that have salaries greater than 40,000. If the query returns a name that begins with the letter J, the job completes.

```
insert_job: selectjob
job_type: SQL
machine: DB_agent
sql_command: SELECT NAME FROM ENTADM.STAFF where SALARY > 40000
owner: entadm@myhost
success_criteria: NAME=J.*
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
```

```
destination_file: /staff/salary.txt
```

For example, the salary.txt file contains the following output:

```
Output for: SELECT NAME FROM ENTADM.STAFF where SALARY > 40000
NAME

Jonson
```

## Database Job Attributes with Default Values

This article explains Database job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Database job attributes have default values:

- **connect\_string**

This attribute specifies the database resource location.

**Default:** db.default.url agent parameter, if specified

**NOTE**

If you do not specify the connect\_string attribute, the default database resource location must be defined in the db.default.url parameter in the agentparm.txt file of the agent. Otherwise, the job fails.

- **destination\_file (SQL jobs only)**

This attribute specifies the output destination file that stores the SQL query results.

**Default:** spooldir agent parameter, if specified

- **monitor\_type (Database Monitor jobs only)**

This attribute specifies the type of database change to monitor for.

**Default:** VARIANCE (The job monitors for an increase or a decrease in the number of rows in the table.)

- **owner**

This attribute specifies the user ID that the job runs under.

**Default:** The user ID who invokes jil to define the job

**NOTE**

Database jobs do not support Windows authentication.

- **trigger\_type (Database Trigger jobs only)**

This attribute specifies the type of database change to monitor for.

**Default:** INSERT (The job monitors for an insertion of a row in the table.)

- **user\_role**

This attribute specifies the Oracle database user type.

**Default:** db.default.userType agent parameter, if specified

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Manage JIL](#) and [Reference](#) sections.

### Example: Override Default Values

Suppose that you want to run an SQL job that queries the NEWWORDS table. This job overrides the default database resource location defined on the agent using the connect\_string attribute. This job also overrides the default owner with the dbuser1 user ID, who is logged in with sysdba privileges. The output is stored in the spool file of the job by default.



```
insert_job: QRY1
job_type: SQL
machine: dbagent
owner: dbuser1@myhost
user_role: as sysdba
sql_command: SELECT * from NEWORDS
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

## File Trigger Jobs Overview

File Trigger jobs let you monitor file activity. You can define File Trigger jobs for UNIX, Linux, Windows, or i5/OS systems.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

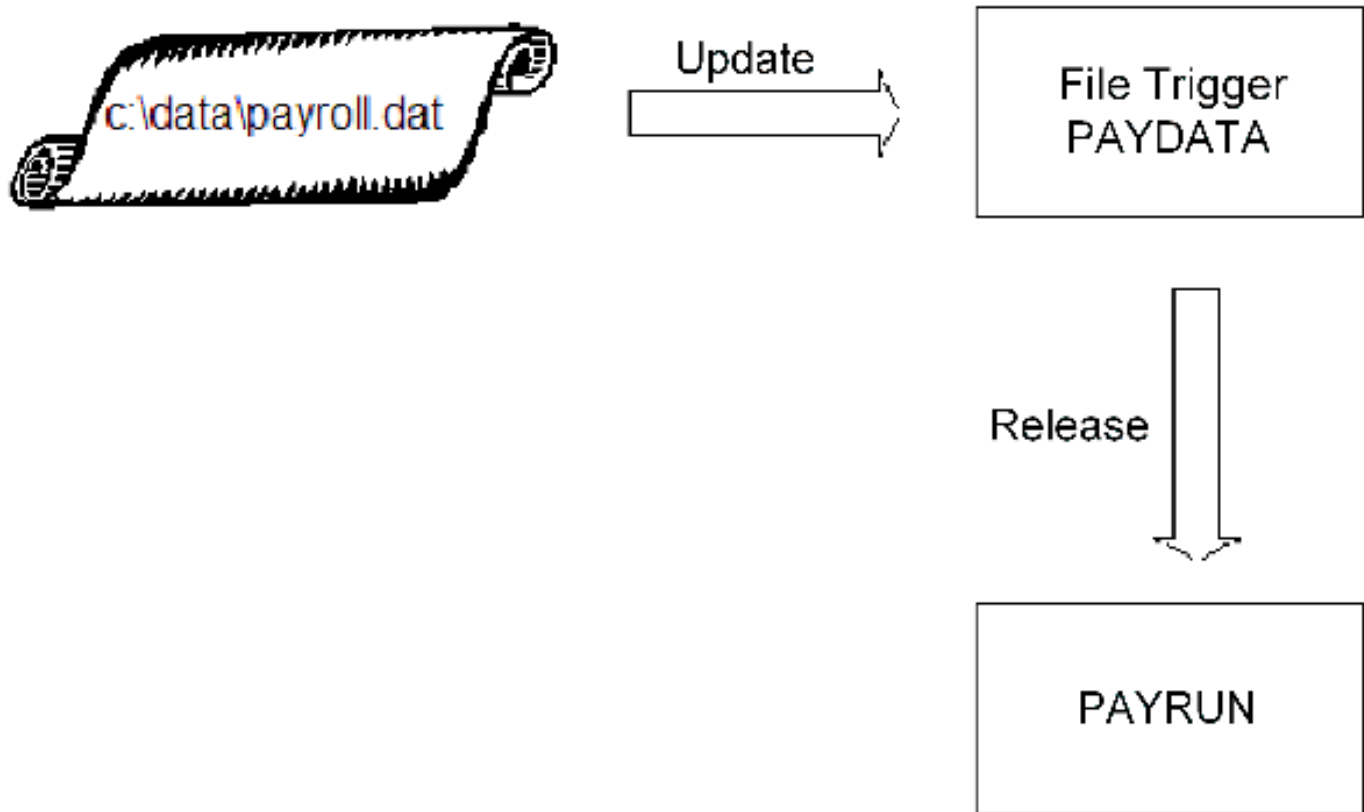
The File Trigger job can monitor when a file is created, updated, deleted, expanded, or shrunk, and when a file exists or does not exist.

### Example: Monitor for an Update to a File

Suppose that a File Trigger job named PAYDATA monitors for an update to the payroll.dat file on a Windows computer. When the file is updated, the job completes and the scheduling manager releases a job named PAYRUN.

The following diagram shows the scenario:

Figure 52: File Trigger Jobs Overview



## Define a File Trigger Job

You can define a File Trigger (FT) job to monitor when a file is created, updated, deleted, expanded, or shrunk, and when a file exists or does not exist. By default, File Trigger jobs monitor for the existence of a file. It monitors file activity for UNIX, Windows, or i5/OS systems.

### NOTE

- To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.
- Difference between File Watcher and File Trigger jobs on Windows: File Watcher and File Trigger jobs produce different results when the dot (.) character is used as a literal. The pattern "\*.\*)" is treated differently depending on which job is used:
  - File Trigger jobs treat a dot character (.) as a literal. To be considered a match, a file must contain a dot and characters on either side of it. For example, if two files exist in the same directory, one named "abc" and the other "abc.txt," only "abc.txt" is considered a match.
  - File Watcher jobs use Windows native APIs, which assume the dot character can be implicit. For example, File Watcher jobs consider "abc" to be the same as "abc." So, in this case, either "abc" or "abc.txt" is a match.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: FT**  
Specifies that the job type is File Trigger.
  - **machine Attribute -- Define the Client Where a Job Runs** Specifies the name of the machine on which the job runs.
  - **watch\_file**  
Specifies the path to and name of one or more files to monitor.
2. (Optional) Specify optional File Trigger attributes:
  - **continuous**
  - **job\_class**
  - **job\_terminator**
  - **watch\_file\_change\_type**
  - **watch\_file\_change\_value**
  - **watch\_file\_groupname**
  - **watch\_file\_owner**
  - **watch\_file\_recursive**
  - **watch\_file\_type**
  - **watch\_file\_win\_user**
  - **watch\_no\_change**
3. (Optional) Specify common attributes that apply to all job types.  
The File Trigger job is defined. When the job runs, it monitors for the existence of the specified file.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values specified in attribute statements within a job definition override the default values.

**Example: Monitor for the Existence of a File**

This example monitors for the existence of the batch.input file in the tmp directory. If the file exists in the directory, the job completes. If the file does not exist in that directory, the job fails.

```
insert_job: ft_job
job_type: FT
machine: unixagt
watch_file: /tmp/batch.input
watch_file_type: EXIST
```

**Example: Monitor for an Increase in File Size to a Certain Amount**

This example monitors the test file in the data directory. When that file size reaches 100 bytes or more, the job completes.

```
insert_job: ft_unix_size
job_type: FT
machine: unixagt
watch_file: /data/test
watch_file_type: EXPAND
watch_file_change_type: SIZE
watch_file_change_value: 100
```

## Monitor for Other Types of File Activity

A File Trigger job checks for the existence of a file by default. You can define the job to monitor for other types of file activity.

### Follow these steps:

1. [Define a File Trigger job.](#)
2. Add the following attribute to the job definition:
  - **watch\_file\_type**  
Specifies the type of file activity that a File Trigger job monitors for. Choose one of the following options:
    - **CREATE**  
Indicates that the file trigger occurs when the file is located and remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is located. A file that exists when the job starts is located during the first scan, which occurs 30 seconds after the job starts. Subsequent scans occur every 30 seconds, so a file that does not exist when the job starts is located a maximum of 30 seconds after it is created.
    - **DELETE**  
Indicates that the file trigger occurs when the file is deleted.
    - **EXIST**  
Indicates that the file trigger occurs if the file exists. If the file does not exist, the job fails.
    - **EXPAND**  
Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirement is met when the file is equal to or larger than the specified size or expands by at least the specified amount, depending on the type of size change that you specify in the job definition.
    - **SHRINK**  
Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirements are met when the file is equal to or smaller than the specified size or shrinks by at least the specified amount.
    - **UPDATE**  
Indicates that the file trigger occurs when the file is updated and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is updated.
    - **GENERATE**

Indicates that the trigger occurs when the file is created or updated and then remains unchanged for the amount of time specified in the `watch_no_change` attribute. When the `watch_no_change` attribute is not specified, the trigger occurs as soon as the file is created or updated.

#### NOTE

If the file exists when the job starts and is not updated after the job starts, no trigger occurs. In this case, the job does not complete and remains in the **RUNNING** state indefinitely.

3. (As applicable) Specify the size change information. Ensure that you specify this information for any job that monitors for the creation of a file or for a change in the size of file; otherwise, it is not applicable. To specify size change information, enter the following attribute arguments:

```
[watch_file_change_type: {DELTA|PERCENT|SIZE}]
```

```
watch_file_change_value: value
```

- **[watch\_file\_change\_type: {DELTA|PERCENT}]**

(Optional) Specifies the type of size as follows:

By default, the trigger occurs when the file expands to or shrinks to the specified size. To issue a trigger when a file expands or decreases by a specified amount, include this argument. You can measure expansion or shrinkage as a percentage of total size or as a specific amount.

- **watch\_file\_change\_value: value**

Specifies a number of bytes or a percentage depending on the type of size change you specified. The trigger occurs when the file expands to at least the specified size or by at least the specified amount, when the file shrinks by at least the specified amount, or when the file shrinks to the specified size or less.

When monitoring for a specific size or for a change in size that is not a percentage change, specify the value in bytes. When measuring size change as a percentage of the total size, specify a value between 1 and 100.

4. Run the job.

The job monitors the file for the specified activity.

#### NOTE

- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.
- By default, the agent scans for the monitored conditions every 30 seconds. If the file changes more than once between scans, the trigger occurs only once or not at all. For example, suppose that your job monitors for the creation of a file and that file is created and deleted between scans. The trigger does not occur because the file does not exist when the directory is scanned.

### Example: Monitor the Creation of a File

This example monitors for the creation of a file named `monthly.log`. The job completes when the file is created.

```
insert_job: ft_job
job_type: FT
machine: ftagt
watch_file: "c:\data\monthly.log"
watch_file_type: CREATE
```

### Example: Monitor the Creation and Stability of a File

This example monitors for the creation of a file named `monthly.log`. The job completes when the file is created and has not changed for consecutive `watch_no_change` polling intervals.

```
insert_job: ft_job
job_type: FT
machine: ftagt
watch_file: "c:\data\monthly.log"
watch_file_type: GENERATE
```

```
watch_no_change: 2
```

### Example: Monitor for the Deletion of Files with Names Beginning with Pay

This example continuously monitors the /usr/data/ directory for files that have names beginning with pay. When all files that have a name beginning with pay are deleted, the job completes successfully.

```
insert_job: ftjob
job_type: FT
machine: ftagt
watch_file: /usr/data/pay*
watch_file_type: DELETE
continuous: Y
```

### Example: Monitor for an Increase in File Size by a Specific Amount

This example defines a job named ft\_unix\_delta to monitor the file named record in the credit directory and issue a trigger when the file increases in size by at least 200000 bytes and then remains unchanged for 30 minutes.

```
insert_job: ft_unix_delta
job_type: FT
machine: unixagt
watch_file: /credit/record
watch_file_type: EXPAND
watch_file_change_type: DELTA
watch_file_change_value: 200000
watch_no_change: 30
```

### Example: Check that a File Does Not Exist in a Directory

This example defines a job named ft\_unix\_notexist to search the /start/term/ directory for a file named vacation and to issue a trigger if the file does not exist.

```
insert_job: ft_unix_notexist
job_type: FT
machine: unixagt
watch_file: /start/term/vacation
watch_file_type: NOTEXIST
```

### Example: Monitor for a Decrease in File Size to a Specific Amount

This example defines a job named ft\_unix\_shrink to issue a trigger when the file named distribute that is located in the /cash/items directory is 1000 bytes or smaller.

```
insert_job: ft_unix_shrink
job_type: FT
machine: unixagt
watch_file: /cash/items/distribute
watch_file_type: SHRINK
watch_file_change_type: SIZE
watch_file_change_value: 1000
```

### Example: Monitor for Updates to Any of the Files in a Directory and its Subdirectories

This example defines a job named `ft_unix_update` to monitor the `/usr/data/` directory and all of its subdirectories and to issue a trigger when any file that is located in those directories is updated and then is not modified again during the next 30 minutes.

```
insert_job: ft_unix_update
job_type: FT
machine: unixagt
watch_file: "/usr/data/*"
watch_file_type: UPDATE
watch_file_recursive: Y
watch_no_change: 30
```

## File Trigger Job Attributes with Default Values

This article explains File Trigger job attributes with default values. Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the `agentparm.txt` file.

If you specify the attribute in a job definition, it overrides the default.

The following File Trigger job attributes have default values:

- **continuous** Specifies whether the job monitors the file continuously for a specified condition. **Default:** N (The job immediately checks for the condition and completes.)
- **watch\_file\_recursive**  
Specifies whether the job monitors for file activity in the specified directory only or in the specified directory and all of its subdirectories.  
**Default:** N (The job does not monitor subdirectories.)
- **watch\_file\_change\_type**  
Specifies the type of change to detect when monitoring a file for an increase or decrease in size..  
**Default:** SIZE (The job monitors the file to reach a specified size.)
- **watch\_file\_type**  
Specifies the type of file activity to monitor for.  
**Default:** CREATE (The job monitors for the creation of a file.)
- **watch\_no\_change** Defines the number of minutes the file must remain unchanged to satisfy the monitor condition.  
**Default:** 1 (The file must remain unchanged for 1 minute.)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Override Default Values

The `watch_file_type` and `watch_no_change` attributes in the following job definition override the default values.

In this example, the `unixagt` agent monitors the analysis file in the `/research` directory. If the file size expands to 1 byte or more and remains unchanged for 120 minutes or more, the job completes.

```
insert_job: ft_unix_nochange
job_type: FT
machine: unixagt
watch_file: /research/analysis
watch_file_type: EXPAND
watch_file_change_type: SIZE
```

```
watch_file_change_value: 1
watch_no_change: 120
```

## Monitor a File Continuously

You can define a File Trigger job to monitor a file continuously for a CREATE, GENERATE, EXPAND, UPDATE, DELETE, or SHRINK condition. Each time the condition occurs, an alert is written to the scheduler log file (event\_demon.\$AUTOSERV on UNIX and event\_demon.%AUTOSERV% on Windows).

If a job monitors for the deletion of files, the job completes when all the monitored files are deleted or it is completed manually. For all other conditions, the job continues to monitor until it is completed manually.

### NOTE

You cannot define File Trigger jobs to monitor a file continuously for the EXIST or NOTEXIST conditions.

### Follow these steps:

1. [Define a File Trigger job.](#)
2. Add the following attribute to the job definition:
  - **continuous** Specifies whether the job monitors the file continuously for a specified condition.
3. Run the job.  
The specified file is monitored continuously.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Continuously Monitor the Same File Using Multiple File Trigger Jobs

In this example, two File Trigger jobs monitor the same file for a change in size by 10 KB. One job monitors for an increased change in size and the other job monitors for a decreased change in size. The jobs are independent and do not relate to each other in any way.

```
insert_job: ftjob1

job_type: FT

machine: ftagent

watch_file: "c:\data\totals"

watch_file_type: EXPAND

continuous: Y

watch_file_change_type: DELTA

watch_file_change_value: 10240 /
* The value must be entered in bytes (10 x 1024 bytes = 10240) */

insert_job: ftjob2

job_type: FT
```



```

machine: ftagent

watch_file: "c:\data\totals"

watch_file_type: SHRINK

continuous: Y

watch_file_change_type: DELTA

watch_file_change_value: 10240 /
* The value must be entered in bytes (10 x 1024 bytes = 10240) */

```

Suppose that the initial file size of totals (c:\data\totals) is 100 KB and it changes as follows:

100 KB (initial size), 80 KB, 90 KB, 110 KB, 50 KB, 60 KB

The following triggers occur:

- Because the file trigger type is EXPAND, the first job (ftjob1) writes an alert to the scheduler log file once when the file size changes from 90 KB to 110 KB.
- Because the file trigger type is SHRINK, the second job (ftjob2) writes an alert to the scheduler log file twice. The triggers occur when the file shrinks from 100 KB to 80 KB and then again when the file shrinks from 110 KB to 50 KB.

The jobs end when they are forced to complete.

## Monitor a File that is Owned by a UNIX Owner or Group

You can define a File Trigger job to monitor a file that is owned by a specific UNIX owner or group. If the file is not owned by the specified owner or group, the following occurs:

- The job continues monitoring if the file trigger type is CREATE or GENERATE.
- The job completes if the file trigger type is DELETE or NOTEXIST.
- The job fails if the file trigger type is EXPAND, EXIST, SHRINK, or UPDATE.

### NOTE

You can monitor a file owned by a specific owner or group on i5/OS if the file is not a QSYS object.

### Follow these steps:

1. [Define a File Trigger job.](#)
2. Add one or more of the following attributes to the job definition:
  - **watch\_file\_groupname** Specifies the name of the group that owns the file to be monitored.
  - **watch\_file\_owner** Specifies the user ID that owns the file to be monitored.
3. Run the job.  
The specified file is monitored.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Monitor for the Creation of a File that Is Owned by a Specified UNIX User ID

This example defines a job named ft\_job to issue a trigger in the following situations:

- When a file named payroll with attributes that specify JDOE as the file owner is created and is not modified within the first 30 minutes of existence.
- When a file named payroll with attributes that specify JDOE as the file owner exists before the job starts, is located during the first scan, and is not modified within the following 30 minutes.
- When the owner of the file named payroll changes to JDOE and the file is not modified within the next 30 seconds.

```
insert_job: ft_job

job_type: FT

machine: ftagt

watch_file: /usr/data/payroll

watch_file_type: CREATE

watch_file_owner: JDOE

watch_no_change: 30
```

### Example: Monitor for the Existence of a File that Is Owned by a Specified UNIX Group

Suppose that you want a job to monitor for the existence of the /data/payroll.dat file that is owned by the UNIX group ACCTS:

- If the file exists and the file is owned by the group ACCTS when the job is readied, the job completes immediately.
- If the file exists and the file is not owned by the group ACCTS when the job is readied, the job fails.

```
insert_job: ft_unixgroup

job_type: FT

machine: unixagent

watch_file: /data/payroll.dat

watch_file_type: EXIST

watch_file_groupname: ACCTS
```

## Configure the Agent to Run File Trigger Jobs as an External Process

File Trigger jobs typically run as threads under the Workload Automation Agent process. To monitor files with names that contain variables, or to monitor a file on a remote computer, run the job as an external process.

To configure the agent to run File Trigger jobs as an external process, ask your agent administrator to set the following parameters in the agent's agentparm.txt file:

```
filemonplugin.runexternal=true
```

```
oscomponent.default.user=user
```

```
oscomponent.default.password=password
```

- **filemonplugin.runexternal=true**  
Runs a File Trigger job as an external process, but it is limited to Create.
- **oscomponent.default.user=user**  
(Optional) Specifies the default operating system user ID that all jobs on the agent computer run under. This user ID must have access to all files monitored by all File Trigger jobs.

#### NOTE

If this parameter is not set, File Trigger jobs run under the user ID that started the agent.

- **oscomponent.default.password=password**  
(Optional) Specifies the password for the default user ID.

## Resolve File Names That Contain Variables

You can define a File Trigger job that monitors a file whose name contains variables. To resolve the variables in the file name, the agent runs the File Trigger job as an external process.

### Follow these steps:

1. [Configure the agent to run File Trigger jobs as an external process.](#)
2. [Define a File Trigger job.](#)
3. Add the following attribute:
  - **watch\_file** Specifies the path to and name of one or more files to monitor. You can specify file names that contain variables.  
**Example:** \$AUTOSYS/reports/august2011.out
4. Run the job.  
The specified files are monitored.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Monitor for the Existence of a File

This example monitors for the existence of the august2011.out file on the unixagt machine. The job monitors the directory specified by the \$MY\_PATH variable. The job runs under the user that started the unixagt agent. The filemonplugin.runexternal agent parameter is set to true. If the file exists in the directory, the job completes. If the file does not exist in that directory, the job fails.

```
insert_job: ft_job
```

```
job_type: FT
```

```
machine: unixagt
```

```
watch_file: $MY_PATH/reports/august2011.out

watch_file_type: EXIST
```

## Monitor a File on a Remote UNIX Computer

You can define a File Trigger job to monitor remote files across a UNIX network using a specified user ID.

### Follow these steps:

1. [Configure the agent to run File Trigger jobs as an external process.](#)
2. [Define a File Trigger job.](#)
3. Add the following attribute:
  - **watch\_file** Specifies the path to and name of one or more files to monitor. You can specify file names that contain variables and remote files on a UNIX network.  
**Example:** \$AUTOSYS/reports/august2011.out
4. Run the job.  
 The specified files are monitored.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Monitor for the Existence of a File

This example monitors for the existence of the august2011.out file on the unixagt machine. The job monitors the directory specified by the \$MY\_PATH variable. The job runs under the user that started the unixagt agent. The filemonplugin.runexternal agent parameter is set to true. If the file exists in the directory, the job completes. If the file does not exist in that directory, the job fails.

```
insert_job: ft_job

job_type: FT

machine: unixagt

watch_file: $MY_PATH/reports/august2011.out

watch_file_type: EXIST

owner: jsmith
```

## Monitor a File on a Remote Windows Computer

You can define a File Trigger job to monitor a file on a remote Windows system if the agent runs as a Windows service under the local system account.

### Follow these steps:

1. Verify with your agent administrator that the agent is running as a Windows service under the local system account.
2. (Optional) Ask your agent administrator to set the following parameters in the agent's agentparm.txt file:

```
oscomponent.default.user=user
oscomponent.default.password=password
```

– **oscomponent.default.user=user**

Specifies the default operating system user ID that all jobs on the agent computer run under. This user ID must have access to the monitored files.

**NOTE**

- The `watch_file_win_user` attribute overrides this parameter.
- If this parameter is not set, specify the `watch_file_win_user` attribute in the File Trigger job definition.

– **oscomponent.default.password=password**

Specifies the password for the default user ID.

**NOTE**

If this parameter is not set, specify the `watch_file_win_user` attribute in the File Trigger job definition and define the corresponding user ID and password via `autosys_secure`.

3. Ask your AutoSys Workload Automation administrator to define a user ID and password on AutoSys Workload Automation that has access to the file on the remote Windows computer.
4. [Define a File Trigger job.](#)
5. Add the following attributes to the job definition:
  - **watch\_file** Specifies the path to and name of one or more files to monitor. Specify a UNC (Universal Naming Convention) name. A UNC name is the name of a file or other resource that begins with two backslashes (\\), indicating that it exists on a remote computer.
  - **watch\_file\_win\_user** Specifies the user ID and the domain the user ID belongs to. This user ID must have access to the remote files in the UNC path. This attribute overrides the `oscomponent.default.user` parameter on the agent.
6. Run the job.  
The specified file on the remote Windows computer is monitored.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see the [Reference](#) and [Manage JIL](#) sections.

### Example: Monitor for an Update to a File on a Remote Windows Computer

This example monitors for an update to the `payroll.dat` file on a remote Windows computer named `CYBNT`. The job runs under `JSMITH`, which is a user ID on `CYBNT` and is in the `CYBDOM` domain. `JSMITH` is defined on the scheduling manager and has access to the `AccountingFiles` directory. The winagent machine must have been defined to AutoSys Workload Automation with the **opsys: windows** attribute.

```
insert_job: ft_remotefile
job_type: FT
machine: winagent
watch_file: \\CYBNT\AccountingFiles\payroll.dat
watch_file_type: UPDATE
watch_file_win_user: CYBDOM\JSMITH@CYBDOM
```

## File Watcher Jobs Overview

A File Watcher job is similar to a Command job. However, instead of starting a user-specified command on a client computer, a File Watcher job starts a process that monitors for the existence and size of a specific operating system

file. When that file reaches the specified minimum size and is no longer growing in size, the File Watcher job completes successfully, indicating that the file has arrived.

Using File Watcher jobs provides a means of integrating events that are external to AutoSys Workload Automation into the processing conditions of jobs. For example, assume a file must be downloaded from a mainframe, and it is expected to arrive after 2:00 a.m. After it arrives, a batch job is run to process it, possibly even starting a whole sequence of jobs.

You could set up a File Watcher job to start at 2:00 a.m., wait for the arrival of the specified file, and exit. You could also set up the batch job so that the completion of the File Watcher job is its only starting condition.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

## Define a File Watcher Job

You can define a File Watcher (FW) job to monitor for the existence and size of a file. AutoSys Workload Automation considers the watched file complete when the following events occur:

- The file reaches the minimum file size that is specified in the `watch_file_min_size` attribute.
- The file reaches a “steady state” during the polling interval.

A steady state indicates that the watched file has not grown during the specified interval.

#### NOTE

- To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS
- Difference between File Watcher and File Trigger Jobs on Windows: File Watcher and File Trigger jobs produce different results when the dot (.) character is used as a literal. The pattern `"*.txt"` is treated differently depending on which job is used:
  - File Trigger jobs treat a dot character (.) as a literal. To be considered a match, a file must contain a dot and characters on either side of it. For example, if two files exist in the same directory, one named `"abc"` and the other `"abc.txt"`, only `"abc.txt"` is considered a match.
  - File Watcher jobs use Windows native APIs, which assume the dot character can be implicit. For example, File Watcher jobs consider `"abc"` to be the same as `"abc."` So, in this case, either `"abc"` or `"abc.txt"` is a match.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: FW**  
Specifies that the job type is File Watcher.
  - **machine** Specifies the name of the machine on which the job runs.
  - **watch\_file** Specifies the path to and name of one or more files to monitor.
2. (Optional) Specify optional File Watcher attributes:
  - **job\_class**
  - **job\_terminator**
  - **profile**
  - **watch\_file\_min\_size**

#### NOTE

The default size to monitor is 0 (zero). If you do not specify the `watch_file_min_size` attribute, the job completes if the file exists. You can specify this attribute to override the default setting.

- **watch\_interval**

**NOTE**

- You can specify the `watch_interval` attribute to override the default setting.
- On the Workload Automation Agent, if you do not specify the `watch_interval` attribute, the job checks the file every 30 seconds (or the time specified in the agent's `filemonplugin.sleepperiod` parameter). If you specify this attribute for a FW job that is submitted to an the Workload Automation Agent, the agent uses the value as a “no-change” or steady interval. The steady interval means that once the file condition is satisfied the file must remain steady for the duration specified by the `watch_interval` attribute.

3. (Optional) Specify common attributes that apply to all job types.  
The File Watcher job is defined.

**NOTE**

- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values specified in attribute statements within a job definition override the default values.

**Example: Monitor a File Every 120 Seconds**

This example monitors the `watch_file.log` file on the `winagent` computer. The job completes when the file reaches 10,000 bytes and maintains a steady state for at least 120 seconds (30 seconds for the agent's global poll interval plus 90 seconds for the `watch_interval`).

```
insert_job: fw_job
job_type: FW
machine: winagent
watch_file: "c:\tmp\watch_file.log"
watch_file_min_size: 10000
watch_interval: 90
```

**FTP Jobs Overview**

Using your agent, you can automate File Transfer Protocol (FTP) transfers with an FTP job. The FTP job can upload data to or download data from an existing FTP server or another agent running as an FTP server. The FTP job always acts as an FTP client.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

You can use an FTP job to automate the following:

- Download ASCII, binary, or EBCDIC (i5/OS only) files from a remote FTP server to your agent computer.
- Upload ASCII, binary, or EBCDIC (i5/OS only) files from your agent computer to a remote FTP server.

Your agent administrator can set up the agent to run as an FTP client, FTP server, or both.

**EBCDIC File Transfers**

The EBCDIC transfer type applies to Workload Automation Agent for i5/OS only.

For the QSYS file system on i5/OS systems, you can only transfer members of FILE objects.

**NOTE**

For more information about FTP restrictions on i5/OS systems, see the IBM documentation.

**Wildcard Characters in File Names**

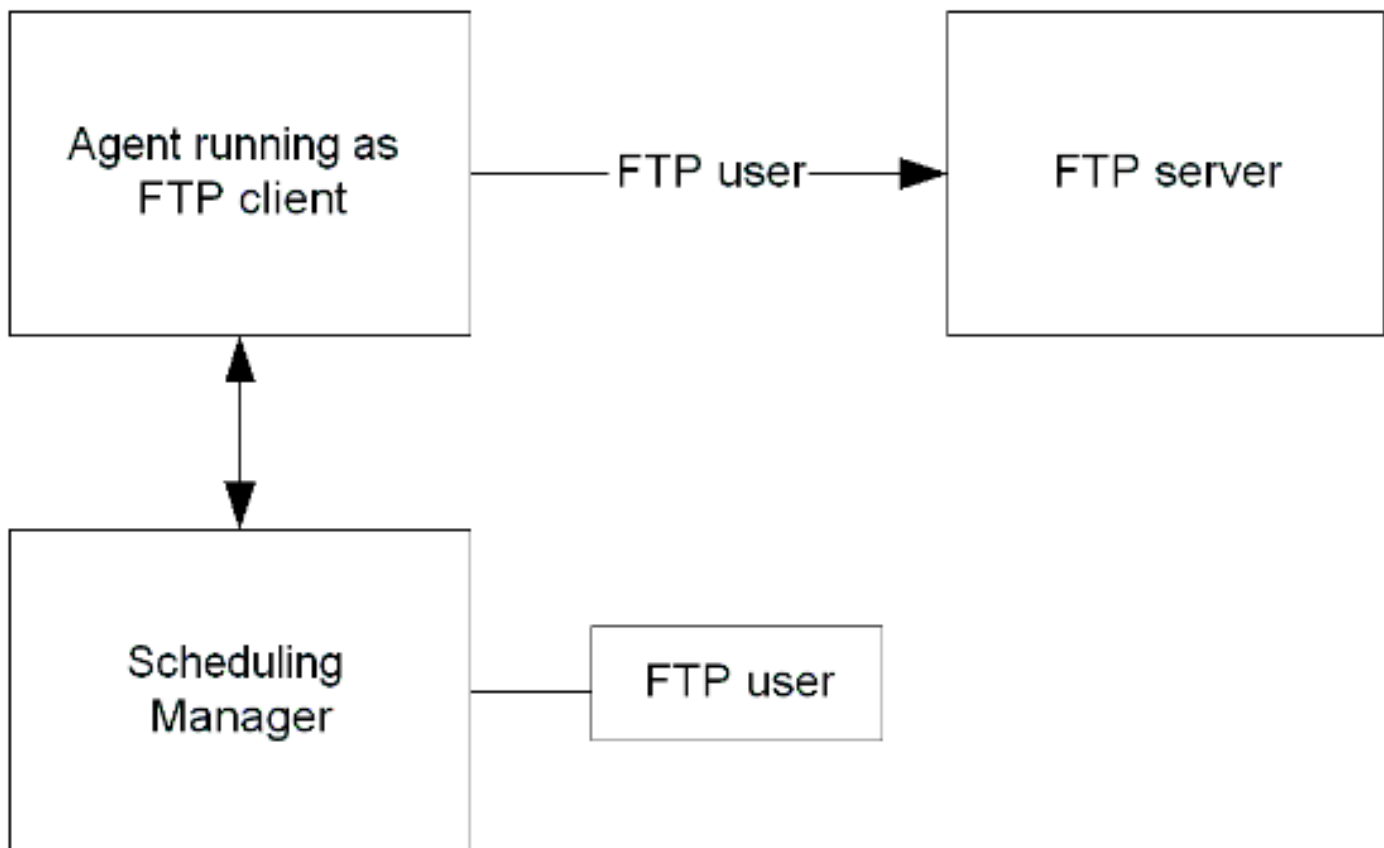
You can use wildcards in file names for ASCII, binary, and EBCDIC transfers. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.

**Running the Agent as an FTP Client**

If the agent runs as an FTP client, the agent can log in to remote FTP servers and download files from and upload files to those servers.

The following diagram shows the relationship between an agent running as an FTP client, the scheduling manager, and an FTP server:

**Figure 53: Relationship between an Agent running as an FTP client, the Scheduling Manager, and an FTP Server**

**NOTE**

The FTP user ID used to connect to the FTP server must be defined on the scheduling manager.

When the agent runs as an FTP client only, other FTP clients (such as other agents) cannot log in to the agent to transfer files. To let other FTP clients log in and transfer files, the agent administrator needs to set up the agent to run as an FTP server.



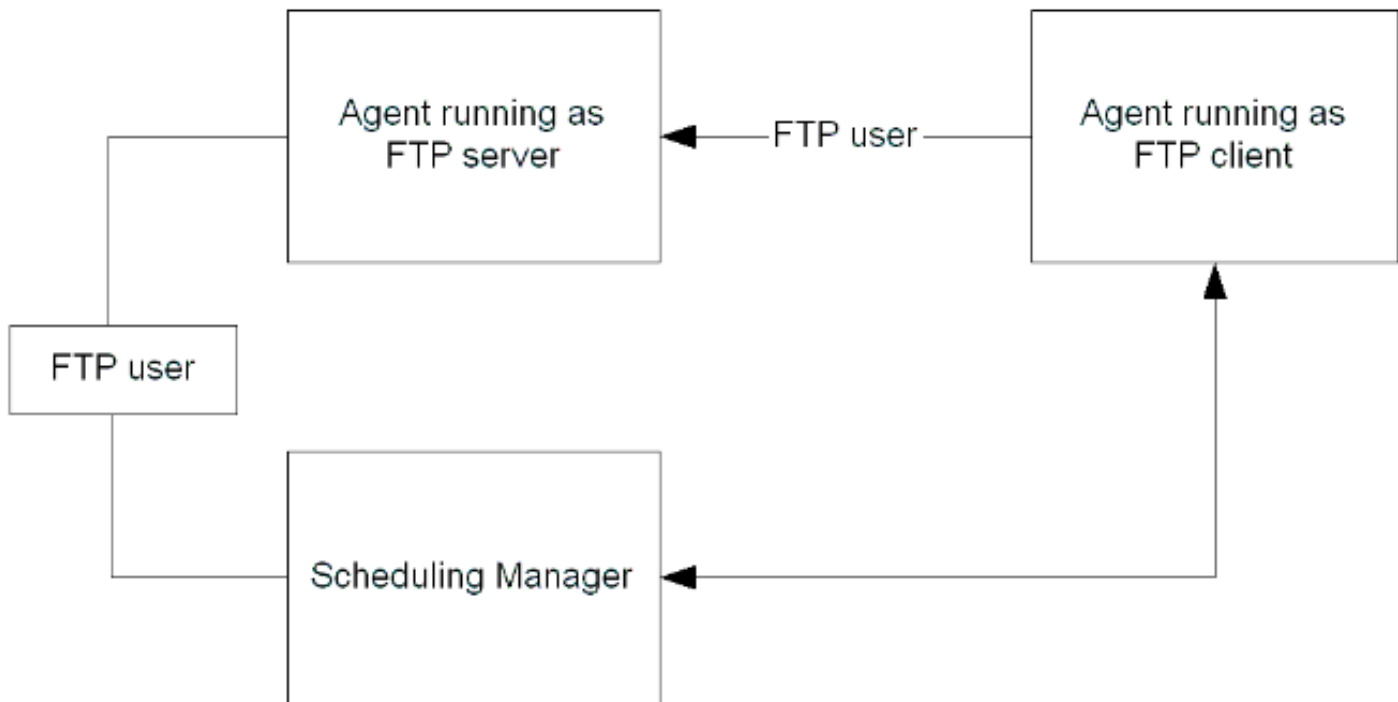
### Running the Agent as an FTP Server

The agent supports a built-in FTP server capability. The agent administrator can enable the agent to act as a generic FTP server in addition to its other roles. This server adheres to the security rules established for the agent.

If the agent runs as an FTP server, clients can log in to the agent and transfer files.

The following diagram shows the relationship between an agent running as an FTP server, the scheduling manager, and another agent running as an FTP client:

**Figure 54: Relationship between an agent running as an FTP server, the Scheduling Manager, and another agent running as an FTP Client.**



#### NOTE

The FTP user ID used to connect to the agent running as an FTP server must be defined on that agent and the scheduling manager.

If the agent is configured as an FTP server, the agent can handle ASCII, binary, and EBCDIC file transfers, wildcard requests, simple GET and PUT requests for single files, and MGET and MPUT requests for multiple files.

The agent has a secure store of FTP server user IDs and associated passwords. The `ftpusers.txt` file, located in the directory where the agent is installed, stores these user IDs and their corresponding hashed passwords.

The agent running as an FTP server does not support anonymous FTP requests. For audit purposes, the agent provides a detailed log of all FTP requests.

#### NOTE

For more information about enabling the agent to act as a generic FTP server, contact your agent administrator and see [Workload Automation Agents](#).

## FTP User IDs and Passwords

All FTP user IDs and passwords must be defined on AutoSys Workload Automation by using the `autosys_secure` command. When you define an FTP job, you must specify an FTP user ID using the `owner` attribute, or use the default owner value. This user ID is used to connect to the FTP server for the file transfer.

If an agent runs as an FTP server, the FTP user ID and password must also be defined on that agent.

### NOTE

For more information about the `autosys_secure` command, see [autosys\\_secure Command](#). For more information about defining user IDs on the agent, see [Workload Automation Agents](#).

## Define a FTP Job

You can define an FTP job to automate File Transfer Protocol (FTP) transfers. The FTP job can download a file from a remote FTP server to your agent computer or upload a file from your agent computer to a remote FTP server. The FTP job can use an existing FTP server or another agent as an FTP server. The FTP job always acts as an FTP client. The output is directed to the spool file through an FTP server.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: FTP**  
Specifies that the job type is FTP.
  - **machine** Specifies the name of the machine on which the job runs.
  - **ftp\_local\_name** Specifies the destination of the file (if downloading) or the source location of the file (if uploading).
  - **ftp\_remote\_name** Specifies the source location of the file (if downloading) or the destination of the file (if uploading).
  - **ftp\_server\_name** Specifies the DNS name or IP address of a remote server.
2. (Optional) Specify optional FTP attributes:
  - **ftp\_command**
  - **ftp\_compression**
  - **ftp\_local\_user**
  - **ftp\_server\_port**
  - **ftp\_transfer\_direction**
  - **ftp\_transfer\_type**
  - **ftp\_use\_SSL**
  - **ftp\_user\_type**
  - **job\_class**
  - **run\_external**
  - **external\_os\_user**
3. (Optional) Specify the following attribute:
  - **owner**  
Specifies the user ID with the authority to download the file from the remote FTP server or upload the file to the remote FTP server.

### NOTE

The job runs under the owner of the job. The password associated with the owner must be defined using `autosys_secure`.

4. (Optional) Specify common attributes that apply to all job types.

The FTP job is defined.

#### NOTE

- All FTP jobs require an FTP user ID. The owner attribute in the job definition specifies this user ID (the default is the user who invokes jil to define the job).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values specified in attribute statements within a job definition override the default values.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files and then import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

#### Example: Download a File from a UNIX Computer to a Windows Computer

The FTP job in this example downloads a file from a UNIX server with IP address 172.16.0.0 and port 21 to a Windows computer.

```
insert_job: FTPT1A
job_type: FTP
machine: ftpagent
ftp_server_name: 172.16.0.0
ftp_server_port: 21
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/ftpdata/textfile
ftp_local_name: "c:\ftpfiles"
owner: ftpuser@172.16.0.0
```

#### Example: Download a Binary File on UNIX

The FTP job in this example uses a binary transfer.

```
insert_job: FTPBINARY
job_type: FTP
machine: unixagent
ftp_server_name: hpunix
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: B
ftp_remote_name: /u1/qatest/ftpdata/binaryfile
ftp_local_name: /export/home/qatest/ftpdata/transf.bin
owner: test@hpunix
```

#### Example: Download a QSYS EBCDIC-encoded File

This example downloads an EBCDIC-encoded file named DATAFILE in the QSYS file system from an i5/OS system to another i5/OS system. The file names are specified in the path format.

```
insert_job: EBCDIC_FILE
job_type: FTP
machine: I5AGENT
ftp_server_name: i5agent
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: E
ftp_remote_name: /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
ftp_local_name: /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
owner: test@i5agent
```

### Example: Upload a File to a Remote Windows Computer

In this example, a file named textfile is uploaded from a local UNIX computer and copied to a remote Windows computer. Both locations include a complete path in the ftp\_remote\_name and ftp\_local\_name attributes. The remote Windows computer uses an agent as the FTP server. Its IP address is 172.16.0.0.

```
insert_job: FTP_REMOTE
job_type: FTP
machine: unixagent
ftp_server_name: 172.16.0.0
ftp_server_port: 123
ftp_transfer_direction: UPLOAD
ftp_transfer_type: A
ftp_remote_name: "/C:/Temp/textfile"
ftp_local_name: /export/home/ftpfiles/ftpdata/textfile
owner: test@172.16.0.0
```

#### NOTE

To transfer a file from a Windows computer, substitute a Windows path in the ftp\_local\_name attribute and enclose the entire path in quotation marks.

### FTP Job Attributes with Default Values

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following FTP job attributes have default values:

- **ftp\_compression**  
Specifies the data compression level (0-9).  
**Default:**
  - 0 (The data is not compressed.)
  - ftp.data.compression agent parameter. This parameter overrides the default 0 setting.
- **ftp\_transfer\_direction**  
Specifies the file transfer direction.

**Default:** DOWNLOAD (The job transfers files from the remote server to the agent computer.)

- **ftp\_transfer\_type**

Specifies the type of data involved in an FTP transfer (B for binary, A for ASCII, or E for EBCDIC).

**Default:** B (The job performs a binary data transfer.)

- **ftp\_use\_ssl**

Specifies whether to transfer the data with Secure Sockets Layer (SSL) communication or regular communication.

**Default:**

- FALSE (The job uses regular communication.)
- ftp.client.ssl agent parameter. This parameter overrides the default FALSE setting.

- **ftp\_user\_type**

Specifies how the value of the owner attribute is passed to the FTP server for authentication.

**Default:** Simple (The job passes only the user part of the owner attribute value to the FTP server for authentication.)

- **run\_external**

Specifies whether to define a user ID that runs the job on behalf of the agent user.

**Default:** F

- **owner**

Specifies the FTP user ID that the job runs under.

**Default:** The user defining the job (The FTP transfer runs under the owner of the job.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Perform an ASCII Data Transfer

The ftp\_transfer\_type and owner attributes in the following job definition override the default values. The job transfers an ASCII file under the testuser account.

The FTP job in this example downloads an ASCII file named textfile from the remote UNIX server to the /export/home/ftpfiles/ftpdata/textfile\_dn directory on the agent computer.

```
insert_job: DOWNLOAD_SINGLE

job_type: FTP

machine: RAGENT

ftp_server_name: hprsupp

ftp_server_port: 5222

ftp_transfer_direction: DOWNLOAD

ftp_transfer_type: A

ftp_remote_name: /u1/files/ftpdata/textfile

ftp_local_name: /export/home/ftpfiles/ftpdata/textfile_dn/textfile

owner: testuser@hprsupp
```

## Transfer Files Using SSL FTP

If the agent FTP server supports SSL FTP and the agent FTP client has SSL FTP configured, you can securely transfer data using Secure Sockets Layer (SSL) communication.

### Follow these steps:

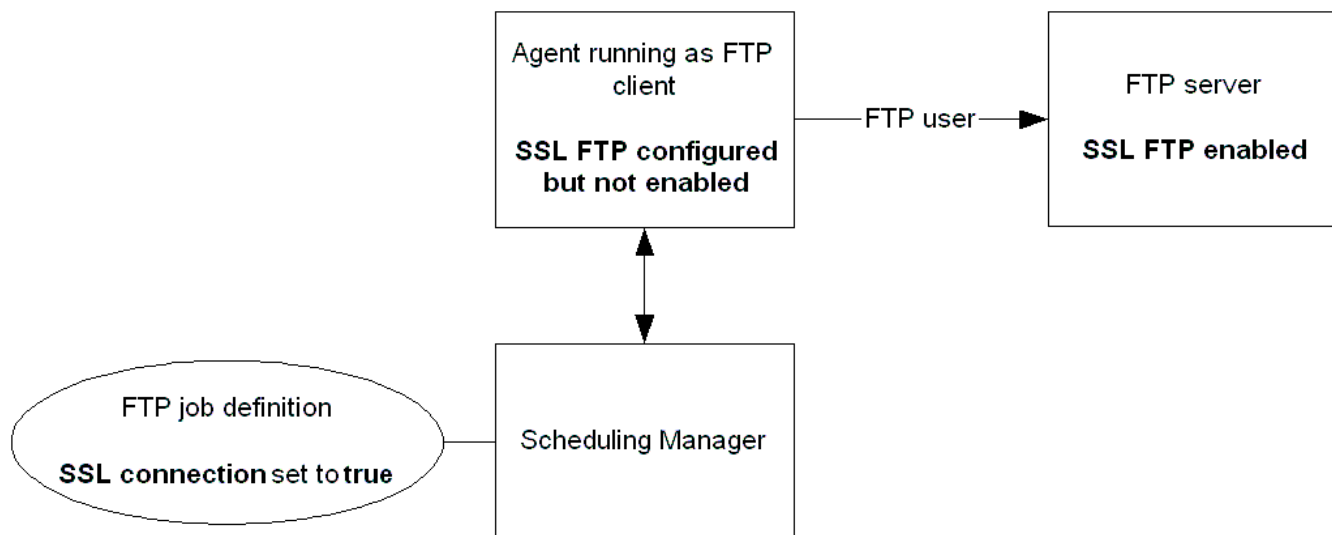
1. [Define an FTP job.](#)
2. Add the following attribute to the job definition:
  - **ftp\_use\_ssl**  
Specifies whether to transfer the data with Secure Sockets Layer (SSL) communication or regular communication.
3. Run the job.  
The files are transferred using the specified communication setting.

### NOTE

- To transfer data using SSL, the FTP server must have SSL FTP enabled and the FTP client must have SSL configured (SSL FTP can be enabled or disabled).
- If you do not specify the `ftp_use_ssl` attribute, the data is transferred using the default FTP setting (regular FTP or SSL FTP) set on the agent FTP client as follows:
  - If the agent FTP client has SSL FTP enabled, all FTP jobs on that agent automatically use SSL FTP. If the FTP server does not support SSL FTP, set the `ftp_use_ssl` attribute to `FALSE`. Otherwise, the job fails.
  - If the agent FTP client has SSL FTP disabled, all FTP jobs on that agent automatically use regular FTP.

### Example: Upload a File from a Local Computer to a Remote Windows Server Using SSL FTP

This example defines a job named `FTP_UPLOAD` to upload the `filename.txt` file from the agent FTP client to the `c:\uploaded_files` directory on the remote Windows server named `winserver` using SSL FTP.



```

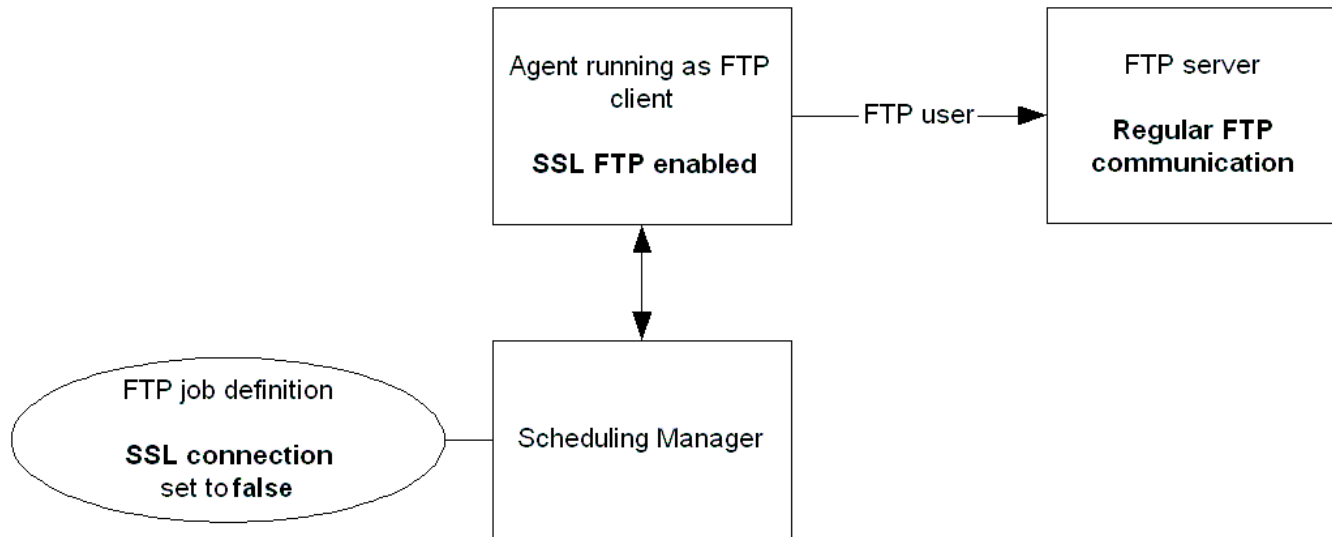
insert_job:FTP_UPLOAD
job_type: FTP
machine: winagent
owner: user1@winserver
ftp_server_name: winserver
ftp_server_port: 21
ftp_transfer_direction: UPLOAD

```

```
ftp_use_SSL: TRUE
ftp_remote_name: "c:\uploaded_files\filename.txt"
ftp_local_name: "d:\files_to_upload\filename.txt"
```

### Example: Download a File from a Remote UNIX Server That Does Not Support SSL FTP to a Local Computer That Supports SSL FTP

In this example, the agent runs on a local computer as an FTP client and has SSL FTP enabled (all FTP jobs on the agent computer run using SSL FTP by default). The remote UNIX server does not support SSL FTP.



The following FTP job downloads the filename.txt file from the remote UNIX server to the d:\downloaded\_files directory on the local computer. Because the FTP server does not support SSL FTP, the ftp\_use\_SSL attribute is set to FALSE so the job does not fail.

```
insert_job: FTP_DOWNLOAD
job_type: FTP
machine: winagent
owner: user1@hpunix
ftp_server_name: hpunix
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_use_SSL: FALSE
ftp_remote_name: /files_to_download/filename.txt
ftp_local_name: "d:\downloaded_files\filename.txt"
```

### Compress Data for FTP

When you run FTP workload between an FTP client and FTP server that both run on agent software, you can compress the data for the transfer by specifying the compression level in the job definition.

#### NOTE

If the compression level is specified and the FTP server or the FTP client does not run on the agent, the data will be transferred without compression.

**Follow these steps:**

1. [Define an FTP job.](#)
2. Add the following attribute to the job definition:
  - **ftp\_compression**  
Specifies the data compression level. The compression level is a one-digit value from 0 to 9, where 0 is no data compression and 9 is the best data compression.

**NOTE**

The effectiveness of the compression is dependent upon the data. Compressing the data may not result in faster transfer times. The overhead of compressing and uncompressing the data may exceed the time saved from sending smaller amount of data.

3. Run the job.  
The files are transferred using SSL FTP.

**Example: Compress a File**

The local computer in this example has an agent running as an FTP client. The remote server has an agent running as an FTP server. Both computers operate on a low bandwidth network.

The following FTP job downloads a large file named largefile.txt from the remote server to the FTP client. The computers are on a low bandwidth network, so the data is compressed at compression level 3.

```
insert_job: FTPJOB

job_type: FTP

machine: ftpagent

ftp_server_name: aixunix

ftp_server_port: 5222

ftp_transfer_direction: DOWNLOAD

ftp_compression: 3

ftp_remote_name: /files_to_download/largefile.txt

ftp_local_name: "c:\downloaded_files\largefile.txt"

owner: ftpuser@aixunix
```

**Send Site-Specific FTP Commands to FTP Servers**

When you define an FTP job, you can specify one or more commands to execute before file transfer. You can use this feature to send site-specific FTP commands to FTP servers.

**Follow these steps:**

1. [Define an FTP job.](#)
2. Add the following attribute to the job definition:



**– ftp\_command**

Defines a command that is to be executed before file transfer.

**NOTE**

This value includes commands such as locsite but does not include commands such as cd or lcd.

3. (Optional) Specify additional ftp\_command attributes for each command that you want to run.

4. Run the job.

The site-specific FTP command is sent to the FTP server.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Send FTP Commands to an FTP Server**

This example sends two FTP commands to the FTP server prior to transferring a file.

```
insert_job: CYBJK.FTP

job_type: FTP

machine: ftpagent

ftp_server_name: ftp.ca.com

ftp_server_port: 21

ftp_transfer_direction: DOWNLOAD

ftp_transfer_type: B

ftp_compression: 8

ftp_remote_name: /pub/cazip.exe

ftp_local_name: /tmp/bla

ftp_command: locsite blksize=FB

ftp_command: locsite automount

owner: user1@ftp.ca.com
```

**Verify the FTP Job Status**

You can verify that the transfer completed successfully without file corruption.

To verify that the transfer completed successfully, check the job's spool file for the following responses:

- If the data was transferred using SSL FTP, the spool file contains the following response:

```
AUTH TLS
```

```
234 AUTH command OK. Initializing SSL connection.
```

- If the data was compressed and transferred without file corruption, the spool file contains a response as follows:

```
Downloaded 81920/26119 bytes (original/compressed) in 0.161 seconds, 496.89 Kbytes/sec.
```

- If the file was downloaded successfully, the spool file contains the following response:

```
Download successful
```

## WA Advanced Integration for Hadoop

Similar to Agents, Advanced Integrations are the key integration components of AutoSys Workload Automation products. Advanced Integrations let you automate, monitor, and manage workload on advanced systems such as Hadoop distributed environments. For example, if you want to automate Hadoop workload, you can install and configure WA Advanced Integration for Hadoop. To run workload on a particular system, you can install the advanced integration on any other system.

WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define the following Hadoop job types using WA Advanced Integration for Hadoop:

- HDFS (Hadoop Distributed File System)
- OOZIE
- HIVE
- SQOOP
- PIG

### NOTE

- To run Hadoop jobs, your system requires WA Advanced Integration for Hadoop. For more information about WA Advanced Integration for Hadoop, see the [WA Advanced Integration for Hadoop](#) documentation.
- WA Advanced Integration for Hadoop is supported only on Linux operating system.

WA Advanced Integration for Hadoop uses the standard agent functionality including encryption, control, configuration, security, logs, and so on.

## Define an HDFS Job

Hadoop Distributed File System (HDFS) is the primary distributed storage for Hadoop applications. You can define an HDFS job to perform the following file operations on HDFS:

- Create one or more zero-byte files.
- Create one or more directories.
- Delete one or more files or directories.
- Copy one or more files or directories within HDFS, from the local file system to HDFS, or from HDFS to the local file system.
- Move one or more files or directories within HDFS or from the local file system to HDFS.

**NOTE**

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: HDFS**  
Specifies that the job type is HDFS.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **owner**  
Specifies the user ID that is required for connecting to an edge node.

**NOTE**

Specify the password or key credentials that is associated with the user ID using the `autosys_secure` command.

- **hadoop\_pollinterval**  
(Optional) Specifies the number of seconds to wait between job status checks.
- **hadoop\_hdfs\_type**  
Specifies the file operation that the job performs on HDFS. The valid values are as follows:
  - **TOUCH**—Creates one or more zero-byte files.
  - **MKDIR**—Creates one or more directories.
  - **REMOVE**—Deletes one or more files or directories.
  - **COPY**—Copies one or more files or directories within HDFS.
  - **COPYFROMLOCAL**—Copies one or more files or directories from the local file system to HDFS.
  - **COPYTOLOCAL**—Copies one or more files or directories from HDFS to the local file system.
  - **MOVE**—Moves one or more files or directories within HDFS.
  - **MOVEFROMLOCAL**—Moves one or more files or directories from the local file system to HDFS.
- **connectionprofile**  
Specifies the name of a connection profile.

**NOTE**

The connection profile must already be defined in the database before you can assign it to a HDFS job.

2. Specify the following attribute if the `hadoop_hdfs_type` attribute is set to **TOUCH**:
  - **hadoop\_hdfs\_path**  
Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.
3. Specify the following attributes if the `hadoop_hdfs_type` attribute is set to **MKDIR**:
  - **hadoop\_hdfs\_path**  
Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.
  - **hdfs\_create\_parentdir**  
(Optional) Specifies whether you want to create a parent directory.
4. Specify the following attributes if the `hadoop_hdfs_type` attribute is set to **REMOVE**:
  - **hadoop\_hdfs\_path**

Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.

- [hadoop\\_hdfs\\_skiptrash](#)  
(Optional) Specifies whether you want to bypass the trash and delete the file or directory permanently.
  - [hadoop\\_hdfs\\_recursive](#)  
(Optional) Specifies whether you want to delete the specified file or directory and all its sub-directories.
5. Specify the following attributes if the `hadoop_hdfs_type` attribute is set to COPY, COPYFROMLOCAL, or COPYTOLOCAL:
- [hadoop\\_hdfs\\_source](#)  
Specifies the path of one or more files or directories that are to be copied from the source location.
  - [hadoop\\_hdfs\\_destination](#)  
Specifies the path of a file or directory that is copied to the destination location.
  - [hadoop\\_hdfs\\_preserve\\_status](#)  
(Optional) Specifies whether you want to retain the timestamps, ownership, and permission attributes for the file or directory after it is copied from the source location to the destination location.
  - [hadoop\\_hdfs\\_overwrite Attribute – Specify Whether to Overwrite a File or Directory](#)  
(Optional) Specifies whether you want to overwrite an existing file or directory in the destination location when a file or directory with the same name exists.

#### NOTE

The `hadoop_hdfs_overwrite` attribute does not apply for COPYTOLOCAL operation.

6. Specify the following attributes if the `hadoop_hdfs_type` attribute is set to MOVE or MOVEFROMLOCAL:
- [hadoop\\_hdfs\\_source](#)  
Specifies the path of one or more files or directories that are to be copied from the source location.
  - [hadoop\\_hdfs\\_destination](#)  
Specifies the path of a file or directory that is moved to the destination location.
7. (Optional) Specify optional Hadoop HDFS job attributes:
- [job\\_class](#)
8. (Optional) Specify common attributes that apply to all job types.  
The Hadoop HDFS job is defined.

#### NOTE

- If you specify the attribute in the job definition, it overrides the default value. For more information about possible default values, see the syntax and notes for the attributes you are using.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the `JIL` command and `JIL` syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

#### Example: Create a File

This example creates a zero-byte file named *file1*. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile

```

insert_job: HDFS_TOUCH_JOB
job_type: HDFS
machine: HADOOPAI
hadoop_hdfs_type: TOUCH
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_path: /tmp/file1
connectionprofile: conn1

```

### Example: Create a Directory

This example creates a directory *HDFS\_DIR* under the parent directory *HDFS\_PRT\_DIR*. If the parent directory does not exist, it creates the parent directory. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```

insert_job: HDFS_MKDIR_JOB
job_type: HDFS
machine: HADOOPAI
hadoop_hdfs_type: MKDIR
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_path: /HDFS_PRT_DIR/HDFS_DIR
hadoop_hdfs_create_parentdir: true
connectionprofile: conn1

```

### Example: Remove Files or Directories

This example deletes the path1, path2, path3, path 4, and path5 files or directories (and its sub-directories) permanently. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```

insert_job: HDFS_REMOVE_JOB
job_type: HDFS
machine: HADOOPAI
hadoop_hdfs_type: REMOVE
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_skiptrash: true
hadoop_hdfs_recursive: true
hadoop_hdfs_path: /tmp/path1
hadoop_hdfs_path: tmp/path2, ../../path3
hadoop_hdfs_path: "/tmp/path 4", /tmp/path5
connectionprofile: conn1

```

### Example: Copy Files or Directories Within HDFS

This example copies path4 and path2 which can be files or directories within the hadoop file system to path5. If a file or directory with the same name exists in path5, it is overwritten. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```

insert_job: HDFS_COPY_JOB
job_type: HDFS
machine: HADOOPAI
hadoop_hdfs_type: COPY
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_source: /tmp/path4, /test/path2

```

```
hadoop_hdfs_destination: /tmp/path5
hadoop_hdfs_overwrite: true
connectionprofile: conn1
```

### Example: Copy Files or Directories from the Local File System to HDFS

This example copies path2 which can be a file or directory on the local file system to path5. If a file or directory with the same name exists in path5, it is overwritten. The timestamps, ownership, and permission attributes for the file or directory in path2 are retained after it is copied to path5. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```
insert_job: HDFS_COPYFROMLOCAL_JOB
job_type: HDFS
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_type: COPYFROMLOCAL
hadoop_hdfs_source: /tmp/file/path2
hadoop_hdfs_destination: /tmp/dir/path5
hadoop_hdfs_overwrite: 1
hadoop_hdfs_preserve_status: true
connectionprofile: conn1
```

### Example: Copy Files or Directories from the Local File System to HDFS

This example copies path3 which can be a file or directory in the hadoop file system to path6. The timestamps, ownership, and permission attributes for the file or directory in path3 are retained after it is copied to path6. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```
insert_job: HDFS_COPYTOLOCAL_JOB
job_type: HDFS
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_type: COPYTOLOCAL
hadoop_hdfs_source: /tmp/file/path3
hadoop_hdfs_destination: /tmp/dir/path6
hadoop_hdfs_preserve_status:
 true
connectionprofile: conn1
```

### Example: Move Files or Directories from the Local File System to HDFS

This example moves path2 which can be a file or directory on the local file system to path7. The attributes that are required for connecting to the Hadoop environment are specified in the *conn1* connection profile.

```
insert_job: HDFS_MOVEFROMLOCAL_JOB
job_type: HDFS
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_type: MOVEFROMLOCAL
hadoop_hdfs_source: /tmp/file/path2
hadoop_hdfs_destination: /tmp/dir/path7
connectionprofile: conn1
```

## HDFS Job Attributes with Default Values

If you specify the attribute in a job definition, it overrides the default.

The following HDFS job attributes have default values:

- **hadoop\_hdfs\_create\_parentdir**  
Specifies whether you want to create a parent directory.  
**Default:** false (The job does not create a parent directory)
- **hadoop\_hdfs\_recursive**  
Specifies whether you want to invoke the file operation on the file or directory and all its sub-directories.  
**Default:** false (The job does not invoke the file operation on the file or directory and all its sub-directories)
- **hadoop\_hdfs\_skiptrash**  
Specifies whether you want to bypass the trash and delete the file or directory permanently.  
**Default:** false (The job does not bypass the trash)
- **hadoop\_hdfs\_preserve\_status**  
Specifies whether you want to retain the timestamps, ownership, and permission attributes for the file or directory after it is copied from the source location to the destination location.  
**Default:** false (The job does not retain the timestamps, ownership, and permission attributes for the file or directory)
- **hadoop\_hdfs\_overwrite**  
Specifies whether you want to overwrite an existing file or directory in the destination location when a file or directory with the same name is copied from the source location.  
**Default:** false (The job does not overwrite the file or directory)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

## Define an Oozie Job

Oozie is a workflow scheduler system to manage Apache Hadoop jobs. Oozie Workflow jobs are Directed Acyclical Graphs (DAGs) of actions.

Oozie is integrated with the rest of the Hadoop stack supporting several types of Hadoop jobs out of the box (such as Java map-reduce, Streaming map-reduce, Pig, Hive, Sqoop, and Distcp) and system-specific jobs (such as Java programs and shell scripts).

Oozie jobs let you run an Oozie workflow and monitor the workflow status until termination. You can define an Oozie job to perform the following tasks:

- Start a workflow
- Stop a running workflow
- Suspend a running workflow
- Resume a suspended workflow

### NOTE

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type:** **OOZIE**  
Specifies that the job type is Oozie.
  - **machine**

Specifies the name of the machine on which the job runs.

– **owner**

Specifies the user ID that is required for connecting to an edge node.

**NOTE**

Specify the password or key credentials that is associated with the user ID using the `autosys_secure` command.

– **hadoop\_pollinterval**

(Optional) Specifies the number of seconds to wait between job status checks.

– **connectionprofile**

Specifies the name of a connection profile.

**NOTE**

The connection profile is already defined in the database before you can assign it to a Oozie job.

– **hadoop\_oozie\_cfgfile**

Specifies the path to the job configuration file (.xml or .properties) on the Hadoop edge node. This can be a fully qualified path name or a path that is relative to the home directory of the user on the edge node.

2. (Optional) Specify optional Hadoop Oozie job attributes:

– **job\_class**

3. (Optional) Specify common attributes that apply to all jobs.

The Oozie job is defined.

**NOTE**

- If you specify the attribute in the job definition, it overrides the default value. For more information about possible default values, see the syntax and notes for the attributes you are using.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

### Example: Run a Oozie Workflow

This example runs the `workflow.job.properties` Oozie workflow; consuming the attribute values specified in the `conn1` connection profile.

```
insert_job: OOZIE_JOB
job_type: OOZIE
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_oozie_cfgfile: /tmp/oozie/workflow.job.properties
hadoop_pollinterval: 20
connectionprofile: conn1
```



## Define a Hive Job

Hive is a data warehousing infrastructure based on Hadoop. Hive lets you query data in HDFS using HiveQL, an SQL-like query language. CA WA Advanced Integration for Hadoop supports the following Hive operations:

- Use Beeline to run a Hive job
- Use Beeline to cancel a Hive job

### NOTE

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:

- **job\_type**: **HIVE**  
Specifies that the job type is Hive.
- **machine** Specifies the name of the machine on which the job runs.
- **owner**  
Specifies the user ID that is required for connecting to an edge node.

### NOTE

Specify the password or key credentials that is associated with the user ID using the `autosys_secure` command.

- **hadoop\_pollinterval**  
(Optional) Specifies the number of seconds to wait between job status checks.
- **connectionprofile**  
Specifies the name of a connection profile.

### NOTE

The connection profile must already be defined in the database before you can assign it to a Hive job.

- **hadoop\_hive\_initfile** (Optional) Specifies the name of the initialization SQL script file.
- **hadoop\_hive\_user**  
(Optional) Defines the user name to connect to the database. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command.

**Default:** anonymous (the Hive default user)

### NOTE

Specify the password that is associated with the Hive user using the `autosys_secure` command.

- **hadoop\_hive\_scriptfile**  
Specifies the name of the SQL script file to execute on the database.
- **hadoop\_hive\_configparams**  
(Optional) Specifies a Hive configuration property and its value in the `property=value` format. If you specify multiple configuration properties, separate each configuration property with a comma.
- **hadoop\_hive\_variables**  
(Optional) Specifies a Hive variable and its value in the `variable=value` format. These variables are set at session level and used in Hive commands or queries. If you specify multiple variables, separate each variable with a comma.
- **hadoop\_hive\_additional\_params**  
(Optional) Specifies an additional Hadoop Beeline command line argument and its value.

**Example:** `--color=true --showHeader=true`

**NOTE**

You can specify multiple additional parameters. Separate each additional parameter with a blank space.

2. (Optional) Specify optional Hadoop Hive job attributes:
  - [job\\_class](#)
3. (Optional) Specify common attributes that apply to all job types.  
The Hadoop Hive job is defined.

**NOTE**

- If you specify the attribute in the job definition, it overrides the default value. For more information about possible default values, see the syntax and notes for the attributes you are using.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files and then import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

**Example: Run a Hive Job**

This example connects to the Hive default database using *user1* user and its associated password. The attributes that are required for connecting to the Hive environment are specified in the *conn1* connection profile. The required initialization file, configuration parameters, variables, and additional command line arguments are provided to the Hadoop Beeline client. For every 20 seconds, the agent checks the status of the Hive job.

```
insert_job: HIVE_JOB
job_type: HIVE
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_hive_initfile: /tmp/initfile
hadoop_hive_configparams: mapred.job.queue.name=default
hadoop_hive_scriptfile: /tmp/apps/hive/script.q
hadoop_hive_variables: VAR1=/tmp/examples/input-data/table, VAR2=/tmp/examples/output-
data/hive
hadoop_hive_additional_params: --color=true
hadoop_hive_user: user1@token
hadoop_pollinterval: 20
connectionprofile: conn1
```

**Hive Job Attributes with Default Values**

If you specify the attribute in a job definition, it overrides the default.

The following Hive job attribute has a default value:

- **hadoop\_hive\_user**  
(Optional) Defines the user name to connect to the database.  
**Default:** anonymous (the Hive default user)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

**Define a Sqoop Job**

Sqoop is a tool that is designed to transfer data between Hadoop and relational databases, such as MySQL and Oracle. You can define a Sqoop job in AutoSys Workload Automation to perform the following operations:

- Import data from a relational database to Hadoop HDFS.
- Export data from Hadoop HDFS to a relational database.
- Terminate a running Sqoop job.

**NOTE**

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SQOOP**  
Specifies that the job type is SQOOP.
  - **machine:**  
Specifies the name of the machine on which the job runs.
  - **owner**  
Specifies the user ID that is required for connecting to an edge node.

**NOTE**

Specify the password or key credentials that is associated with the user ID using the `autosys_secure` command.

- **connectionprofile**  
Specifies the name of a connection profile.

**NOTE**

The connection profile must already be defined in the database before you can assign it to a Sqoop job.

- **hadoop\_sqoop\_type**  
Specifies the operation that the Sqoop job performs. The valid values are as follows:
  - **IMPORT**  
Imports data from a relational database (such as MySQL and Oracle) to Hadoop HDFS.
  - **EXPORT**  
Exports data from Hadoop HDFS to a relational database.
- **hadoop\_sqoop\_jdbcuser-specify-a-JDBC User Name**  
Defines the JDBC user name to connect to the database. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command.

**NOTE**

Specify the password that is associated with the JDBC user using the `autosys_secure` command.

- **hadoop\_sqoop\_additional\_params** (Optional) Specifies an additional Hadoop Sqoop command line argument and its value.

**NOTE**

- You can specify multiple additional parameters. Separate each additional parameter with a blank space. For example, `-m 1 --outdir directorypath --validate`.
  - When you specify multiple additional parameters, use the same format that you use while specifying Hadoop Sqoop command line arguments.
- **hadoop\_sqoop\_jdbctable** Specifies the database table from which the data is imported or to which the data is exported.
  - **hadoop\_sqoop\_nullstring** (Optional) Specifies the string that indicates a null value for string columns.
  - **hadoop\_sqoop\_nullnonstring** (Optional) Specifies the string that indicates a null value for non-string columns.
2. **hadoop\_pollinterval** (Optional) Specifies the number of seconds to wait between job status checks.
  3. Specify the following attributes if the `hadoop_sqoop_type` attribute is set to `IMPORT`:
    - **hadoop\_sqoop\_destination** Specifies the HDFS destination directory to which the data is imported.
    - **hadoop\_sqoop\_columns** (Optional) Specifies the columns to import from the database table. This value is passed to the `--columns` Hadoop Sqoop import control argument.
    - **hadoop\_sqoop\_whereclause** (Optional) Specifies the `WHERE` clause to use during import. This value is passed to the `--where` Hadoop Sqoop import control argument.
    - **hadoop\_sqoop\_boundary\_query** (Optional) Specifies the boundary query to use for creating splits. This value is passed to the `--boundary-query` Hadoop Sqoop import control argument.
  4. Specify the following attribute if the `hadoop_sqoop_type` attribute is set to `EXPORT`:
    - **hadoop\_sqoop-source** Specifies the HDFS source directory from where the data is exported.
    - **hadoop\_sqoop\_stagingtable** (Optional) Specifies the table in which the data is staged before being inserted into the destination table.
    - **hadoop\_sqoop\_clearstagingtable** (Optional) Specifies whether you want to delete the data in the staging table. When you define the `hadoop_sqoop_clearstagingtable` attribute, you must also specify the `hadoop_sqoop_stagingtable` attribute in your job definition.
  5. (Optional) Specify optional Hadoop Sqoop job attributes:
    - **job\_class**
  6. (Optional) Specify common attributes that apply to all job types.  
The Hadoop Sqoop job is defined.

**NOTE**

- If you specify the attribute in the job definition, it overrides the default value. For more information about possible default values, see the syntax and notes for the attributes you are using.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the `JIL` command and `JIL` syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

**Example: Import Data**

This example imports data from the *TABLE\_NAME* table to the *output\_data* file.

```
insert_job: SQOOP_IMPORT_JOB
job_type: SQOOP
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_sqoop_type: IMPORT
hadoop_sqoop_jdbcuser: jdbcuser@token
hadoop_sqoop_destination: /tmp/sqoop/input/output_data
hadoop_sqoop_additional_params: --outdir /tmp/sqoop/output
hadoop_sqoop_jdbctable: TABLE_NAME
hadoop_sqoop_columns: COLUMN1, COLUMN2
hadoop_sqoop_whereclause: COLUMN1 < 100
hadoop_sqoop_boundary_query: "SELECT MAX(COLUMN1) FROM TABLE_NAME"
hadoop_sqoop_nullstring: 'NULLVALUE'
hadoop_sqoop_nullnonstring: '999'
hadoop_pollinterval: 20
connectionprofile: conn1
```

### Example: Export Data

This example exports data from the *output\_data* file to the *TABLE\_NAME* table.

```
insert_job: SQOOP_EXPORT_JOB
job_type: SQOOP
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_sqoop_type: EXPORT
hadoop_sqoop_jdbcuser: jdbcuser@token
hadoop_sqoop_source: /tmp/sqoop/input/output_data
hadoop_sqoop_additional_params: --outdir /tmp/sqoop/output
hadoop_sqoop_jdbctable: TABLE_NAME
hadoop_sqoop_columns: COLUMN1, COLUMN2
hadoop_sqoop_nullstring: UNDEF
hadoop_sqoop_nullnonstring: UNDEF
hadoop_sqoop_clearstagingtable: 1
hadoop_sqoop_stagingtable: tmp_results_updated
hadoop_pollinterval: 20
connectionprofile: conn1
```

## Sqoop Job Attributes with Default Values

If you specify the attribute in a job definition, it overrides the default.

The following Sqoop job attribute has a default value:

- **hadoop\_sqoop\_clearstagingtable**  
(Optional) Specifies whether you want to delete the data in the staging table.  
**Default:** false (The job does not delete the data in the staging table)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Working with JIL](#) and [Reference](#).

**Define a Pig Job**

You can define a Pig job in AutoSys Workload Automation to perform the following operations:

- Run a Pig job
- Terminate a running Pig job

Pig is a high-level scripting language that is used with Hadoop. Pig lets data workers write complex data transformations using SQL-like scripting language named *Pig Latin*.

**NOTE**

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: PIG**  
Specifies that the job type is PIG.
  - **machine** Specifies the name of the machine on which the job runs.
  - **owner**  
Specifies the user ID that is required for connecting to an edge node.

**NOTE**

Specify the password or key credentials that is associated with the user ID using the `autosys_secure` command.

- **hadoop\_pollinterval**  
(Optional) Specifies the number of seconds to wait between job status checks.
- **connectionprofile**  
Specifies the name of a connection profile.

**NOTE**

The connection profile must already be defined in the database before you can assign it to a Pig job.

- **hadoop\_pig\_scriptfile**  
Specifies the path to the Hadoop Pig script file to execute on the database.
- **hadoop\_pig\_parameterfile** (Optional) Specifies the path to the Hadoop Pig parameter file.
- **hadoop\_pig\_propertyfile**  
(Optional) Specifies the path to the Hadoop Pig property file.
- **hadoop\_pig\_additional\_params** (Optional) Specifies an additional Hadoop Pig command line argument and its value.

**NOTE**

You can specify multiple additional parameters. Separate each additional parameter with a blank space.  
For example, `-verbose -warning -stop_on_failure -printCmdDebug`

2. (Optional) Specify optional Hadoop Pig job attributes:
  - **job\_class**
3. (Optional) Specify common attributes that apply to all job types.  
The Hadoop Pig job is defined.

**NOTE**

- If you specify the attribute in the job definition, it overrides the default value. For more information about possible default values, see the syntax and notes for the attributes you are using.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

**Example: Run a Pig Job**

This example executes the `run_pig` script. The required parameter file, property file, and additional command line arguments are provided for the script to execute successfully. The attributes that are required for connecting to the Hadoop environment are specified in the `conn1` connection profile.

```
insert_job: PIG_JOB
job_type: PIG
machine: HADOOPAI
owner: user1@HADOOP_EDGE_NODE
hadoop_pig_parameterfile: /tmp/pig/pig_paramfile.txt
hadoop_pig_propertyfile: /tmp/pig/pig_job.properties
hadoop_pig_scriptfile: /tmp/pig/run_pig.pig
hadoop_pig_additional_params: --color=true
connectionprofile: conn1
```

**i5/OS Jobs Overview**

The i5/OS job lets you run a program or issue a command on an i5/OS system. You can run i5/OS jobs in the following file systems:

- Root file system
- Open systems file system (QOpenSys)
- Library file system (QSYS)

**NOTE**

To run these jobs, your system requires Workload Automation Agent for i5/OS.

You can specify the following details in an i5/OS job definition:

- Library name, library list, and current library for running a program
- The i5/OS job name, options under which the job runs, where it runs, and which user runs it
- Ending exit value of the program, such as a severity code

You can define parameter values that you want to pass to a program at the time the program is invoked.

**NOTE**

Default values may be set for certain parameters, such as the i5/OS user ID that the jobs run under. Contact your agent administrator about the parameters set in the agentparm.txt file.

**Run UNIX Workload on a System i5 Computer**

In addition to scheduling native i5/OS jobs, you can schedule most UNIX workload, such as UNIX scripts, in the PASE environment on i5/OS.

To run both native and UNIX jobs on the same i5/OS computer, you must install two i5/OS agents and must configure the oscomponent.targetenvironment parameter in the agentparm.txt file to handle the appropriate job type. For more information about configuring the parameter, see [AE Installation](#).

**NOTE**

For more information about UNIX workload that can run in the PASE environment, see the IBM i5/OS documentation.

**i5/OS Naming Conventions**

When specifying i5/OS paths and names in your workload, you can use the following naming conventions, depending on where the file is located on the i5/OS system:

- Root file system  
To specify a file in the root file system, use UNIX path and file formats.
- Open systems file system (QOpenSys)  
To specify a file in QOpenSys, use UNIX path and file formats. QOpenSys file names are case-sensitive.
- Library file system (QSYS)  
To specify an object in QSYS, use one of the following formats (unless described differently in the job definition syntax):
  - Path format

*/QSYS.LIB/library.LIB/object.type/*

To specify \*FILE objects, use the following format:

*/QSYS.LIB/library.LIB/object.FILE/member.MBR*

- i5/OS standard format

*library/object/type*

To specify \*FILE objects, use the following format:

*library/object/\*FILE(member)*

**NOTE**

\*FILE is optional when *member* is specified. That is, you can specify a file member using the following format:

*library/object(member)*

**NOTE**



- The values for *library*, *object*, *type*, and *member* can be up to 10 characters each.
- You can use \*ALL to match any name.
- You can use \*FIRST for *member*.
- You can use generic names for *library* and *object*.

## Define an i5/OS Job

You can define an i5/OS job to schedule workload to run on an i5/OS system. The job can run a program or an i5/OS command. You can run i5/OS jobs in the root file system, open systems file system (QOpenSys), and library file system (QSYS).

### NOTE

To run these jobs, your system requires Workload Automation Agent for i5/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: I5**  
Specifies that the job type is i5/OS.
  - **machine** Specifies the name of the machine on which the job runs.
  - **i5\_name**  
Specifies the program, the source file for the program, or the command that you want to run.

### NOTE

The value must correspond to the i5\_action value. If you do not specify the i5\_action attribute, the job interprets the corresponding i5\_name value as a command by default.

2. (Optional) Specify optional i5/OS attributes:
  - fail\_codes
  - i5\_action
  - i5\_cc\_exit
  - i5\_curr\_lib
  - i5\_job\_desc
  - i5\_job\_name
  - i5\_job\_queue
  - i5\_lda
  - i5\_lib
  - i5\_library\_list
  - i5\_others
  - i5\_params
  - i5\_process\_priority
  - job\_class
  - max\_exit\_success
  - success\_codes
3. (Optional) Specify common attributes that apply to all jobs.  
The i5/OS job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes

specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.

- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

### Example: Run an i5/OS Command

This example runs the command that is named DSPJOBLOG on the i5agent computer.

```
insert_job: i5job_runcmd
job_type: I5
machine: i5agent
i5_name: DSPJOBLOG
```

### Example: Run a CL Program

This example runs a program that is named MFGDATA on an i5/OS system. The MFGDATA program is contained in the library named PRODJOB.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: MFGDATA
i5_lib: PRODJOB
```

## i5/OS Job Attributes with Default Values

This article explains i5/OS job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following i5/OS job attributes have default values:

- **i5\_action**  
Specifies whether to run a program or issue a command.  
**Default:** COMMAND (The job interprets the i5\_name value to be a command.)
- **i5\_cc\_exit**  
Specifies the type of exit code returned by an i5/OS job.  
**Default:** \*SEVERITY (The job sends the ending severity code as the exit code.)
- **i5\_job\_desc**  
Specifies the job description for the submitted program.  
**Default:** os400.default.jobdname agent parameter, if specified
- **i5\_job\_queue**

Specifies the i5/OS job queue for the submitted program.

#### NOTE

os400.default.jobqname agent parameter, if specified

- **i5\_process\_priority**  
Specifies the process priority of the i5/OS job.  
**Default:** NORMAL
- **owner**  
Specifies the user ID that the job runs under.  
**Default:** The user ID who invokes jil to define the job (the owner of the job)

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Override Default Values

This example runs an i5/OS program. A default job queue is defined in the agent's agentparm.txt file. The i5\_action and i5\_job\_queue attributes in this job definition override the default values.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYLOAD
i5_job_queue: /QYS.LIB/QBASE.LIB/JQUEUE.JOBQ
```

## Pass Positional Parameters

When running workload, you might need to pass data between jobs and across platforms. You can pass positional parameters to an i5/OS program in your job definition. Positional parameters are variables that can be passed to a program at the time the program is invoked. The parameters are assigned in the order they are passed.

### Follow these steps:

1. [Define an i5/OS job](#).
2. Add the following attribute to the job definition:
  - **i5\_params**  
Defines the parameter values that you want to pass to the program at the time the program is invoked.
3. Run the job.  
The specified positional parameters are passed to the i5/OS program.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Pass Multiple Parameters to an i5/OS Job

This example passes six parameters to an i5/OS program named PAYJOB. The parameter VALUE C is enclosed with double quotation marks because it contains a space.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
```

```
i5_name: PAYJOB
i5_params: ABC 1 P "VALUE C" X r
```

## Use a User Library List

The agent uses the library list in the job description by default. However, if the user is defined, you can set up your job definition to use the user's library list instead.

### Follow these steps:

1. [Define an i5/OS job.](#)
2. Do *one* of the following tasks:
  - Add the following attribute to the job definition:
    - **i5\_curr\_lib: (\*USRPRF)**  
Specifies that the job uses the user's current library when it runs.
  - Add the following attributes to the job definition:
    - **i5\_job\_desc: (\*JOBDD)**  
Specifies that the job uses the job description that is assigned to the user to access the library list.
    - **i5\_library\_list: (\*USRPRF)**  
Specifies that the job accesses the user's library list when it runs.
3. Run the job.  
The job uses the user's library list.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

## Pass Keyword Parameters to SBMJOB

When AutoSys Workload Automation submits a job to the i5/OS system, the job must pass through the SBMJOB command to execute. The following JIL attributes map to parameters for the SBMJOB command:

JIL Attributes	SBMJOB Parameters
i5_user	USER
i5_job_desc	JOBDD
i5_library_list	INLLIBL
i5_job_queue	JOBQ
i5_curr_lib	CURLIB
i5_job_name	JOB

You can also pass additional keyword parameters, such as OUTQ(\*JOBDD), to the SBMJOB command.

### Follow these steps:

1. [Define an i5/OS job.](#)
2. Add the following attribute to the job definition:
  - **i5\_others**  
Specifies SBMJOB command keyword and value combinations.
3. Run the job.  
The specified keywords and values are passed to the SBMJOB command.

**NOTE**

- The special values for these SBMJOB parameters, such as \*USRPRF and \*JOBID, also apply to the JIL attributes. You can use these special values in your job definitions. For more information about the SBMJOB parameters and their special values, see the IBM documentation.
- For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

**Example: Specify the Printer and Output Queue for an i5/OS Job**

This example runs a program named PAYJOB on an i5/OS system. The printer and output queue information is taken from the job definition.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYJOB
i5_others: PRTDEV=*JOBID,OUTQ=*JOBID
```

**Respond to Suspended Jobs that Require Manual Intervention**

This article explains how to respond to suspended jobs that require manual intervention.

A program that is run on an i5/OS system can temporarily suspend itself until it receives more feedback from the end user. In this scenario, Workload Automation Agent for i5/OS notifies the scheduler that a manual response is required. The scheduler sets the job status to the WAIT\_REPLY state. A WAIT\_REPLY\_ALARM is raised with text containing the query of the i5/OS program and the set of expected responses. For the i5/OS job to resume program execution, you must send a response to the job.

To respond to suspended jobs that require manual intervention, issue the following command:

```
sendevent -J job_name -E REPLY_RESPONSE -r response
```

The response is sent to the Workload Automation Agent for i5/OS and the job resumes running.

**Example: Save Data**

Suppose that you schedule an i5/OS job to save data in a file. If the file contains data, the i5/OS program prompts you to confirm that the data in the file can be overwritten.

**Return a Job Exit Status to AutoSys Workload Automation**

A job exit code indicates whether the job completed successfully. By default, the agent sends the job ending severity code to AutoSys Workload Automation when a job completes. AutoSys Workload Automation interprets an exit code of zero (0) as job success and any other number as job failure.

In addition to sending the job ending severity code, you can return a job exit status in other ways. For example, you can send the return code of an ILE program or module as the exit status or you can specify a user-defined exit code of 100 as success.

You can return a job exit status to AutoSys Workload Automation using the following methods:

- Send a program return code using the `i5_cc_exit` attribute
- Send a user-defined exit code using the `success_codes` or `fail_codes` attribute
- Return an exit status using the `EXIT_SUCCESS` and `EXIT_FAILURE` macros

This article details two options to send exit codes to the product:

### **Send a Program Return Code**

When a job completes, the agent sends the job exit code to AutoSys Workload Automation. By default, the agent sends the job ending severity code as the job exit code.

Instead of sending the job ending severity code, the agent can send the return code of one of the following entities:

- An ILE program or module
- An OPM program

For example, if your job runs an ILE C, ILE RPG, OPM RPG, or OPM COBOL program that contains an exit or return statement, the agent can send that return code as the exit code.

#### **Follow these steps:**

1. [Define an i5/OS job](#).
2. Add *one* of the following attributes to the job definition:
  - **`i5_cc_exit: *USER`**  
Specifies that the return code of an ILE program or module is sent as the exit code.
  - **`i5_cc_exit: *PROGRAM`**  
Specifies that the return code of an OPM program is sent as the exit code.
3. Run the job.  
The program return code is sent instead of the job ending severity code.

#### **NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

### **Example: Send an OPM COBOL Program Return Code as the Job's Exit Code**

This example runs an OPM COBOL program named PAYROLL. The agent sends the PAYROLL program's return code to AutoSys Workload Automation.

```
insert_job: i5job_returnOPM
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYROLL
i5_cc_exit: *PROGRAM
```

### **Example: Send an ILE C Program Return Code as the Job's Exit Code**

This example runs a C language program named SALARY. The agent sends the SALARY program return code to AutoSys Workload Automation. Ending severity codes of 1 or 10 indicate job success.

```
insert_job: i5job_returnC
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
```

```
i5_name: SALARY
i5_cc_exit: *USER
success_codes: 1,10
```

### NOTE

The i5 system always writes the job ending severity code to the job's spool file. You can check the spool file for the job ending severity code for more information about the job status. For example, suppose that the C program return code indicates failure, but the ending severity code that is shown in the spool file is 10, which might indicate that the job ran with a minor issue. Assuming that this issue can be ignored, you can indicate ending severity codes of 10 as job success using the `success_codes` attribute.

### Send a User-Defined Exit Code

By default, AutoSys Workload Automation interprets an exit code of 0 (zero) as job success and any other number as job failure. However, you can map exit codes other than 0 as job success.

#### Follow these steps:

1. [Define an i5/OS job](#).
2. Add *one* of the following attributes to the job definition:
  - **success\_codes**  
Defines which exit codes indicate job success.  
**Default:** 0 (zero)
  - **fail\_codes**  
Defines which exit codes indicate job failure.  
**Default:** Any non-zero exit code
3. Run the job.  
The specified user-defined exit code is sent.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

### Example: Send Exit Code 100 as Success

This example runs the PAYPROG program. If it returns an exit code of 1 or 100, the program is considered to have completed successfully.

```
insert_job: i5job_succ
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYPROG
i5_cc_exit: *PROGRAM
success_codes: 1,100
```

### Example: Send Exit Code 40 as Failure

This example runs the RECPROG program. If it returns an exit code of 40, the program is considered to have failed. All other exit codes in the range from 50 to 255 indicate job success.

```
insert_job: i5job_fail
job_type: I5
machine: i5agent
```

```
i5_action: RUN_PROGRAM
i5_name: RECPROG
i5_cc_exit: *PROGRAM
fail_codes: 40
success_codes: 50-255
```

## Specify Data for a Local Data Area

The local data area is a temporary 1024-byte storage area that exists for the duration of an i5/OS job. You can use the local data area to pass data to the job and to other programs that run as part of the job. When the job is submitted, the agent initializes the local data area with the specified data. When the job completes, the local data area is destroyed automatically by the operating system.

### Follow these steps:

1. [Define an i5/OS job](#).
2. Add the following attribute to the job definition:
  - **i5\_lda**  
Specifies data for the local data area in an i5/OS job.
3. Run the job.  
The data is specified for a local data area.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Working with JIL](#) and [Reference](#).

### Example: Specify Data for the Local Data Area in Hexadecimal Format

This example defines an i5/OS job with data for the local data area. When the job is submitted, the agent initializes the local data area with the hexadecimal data abcd. When the job completes, the local data area is destroyed automatically by the operating system.

```
insert_job: i5job_lda
job_type: I5
machine: i5agent
i5_action: COMMAND
i5_name: IVP
i5_lda: x'abcd'
```

## Informatica Jobs Overview

Informatica jobs let you start predefined workflows on Informatica PowerCenter. In Informatica, jobs are known as workflows. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

Informatica PowerCenter provides an environment that lets you load data into a centralized location, such as a data warehouse or operational data store (ODS). You can extract data from multiple sources, transform the data according to business logic you build in the client application, and load the transformed data into file and relational targets.



**NOTE**

To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.

**Define an Informatica Job**

You can define an Informatica job to start a predefined workflow on Informatica PowerCenter. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

**NOTE**

- To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.
- Due to a known issue in Informatica, you cannot run workflow instances in parallel.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: INFORMATICA**  
Specifies that the job type is Informatica.
  - **machine** Specifies the name of the machine on which the job runs.
  - **informatica\_folder\_name** Specifies the Informatica PowerCenter repository folder name where the specified workflow resides.
  - **informatica\_workflow\_name** Specifies the name of the Informatica PowerCenter workflow to be started.
2. (Optional) Specify optional Informatica attributes:
  - **job\_class**
  - **informatica\_workflow\_inst\_name**
  - **informatica\_task\_name**
  - **informatica\_pass\_on\_success**
  - **informatica\_user**
  - **informatica\_param\_file**
  - **informatica\_repository\_name**
  - **informatica\_target\_name**
  - **informatica\_security\_domain**
3. (Optional) Specify common attributes that apply to all job types.  
The Informatica job is defined. When the job runs, it starts a predefined workflow on Informatica PowerCenter.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- For information about required attributes and job type specific optional attributes, see the procedure topics that provide instructions for [defining jobs](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [the jil](#)

[command](#) and JIL (Job Information Language). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

#### Example: **Start a Task in a Workflow on Informatica PowerCenter**

This example starts the copy task in the WF\_1 workflow on the Informatica PowerCenter. The workflow is started from the copy task. Based on the status of tasks, the workflow status is reported.

```
insert_job: informatica
job_type: INFORMATICA
machine: 160_infor
owner: root@informatica
informatica_folder_name: FSTL
informatica_workflow_name: WF_1
informatica_pass_on_success: 1
informatica_task_name: copy
```

## Micro Focus Jobs Overview

The Micro Focus Net Express JCL engine lets you run your mainframe JCL and COBOL programs in a Windows environment.

You can schedule and control your Micro Focus jobs using AutoSys Workload Automation and Workload Automation Agent for Micro Focus. The agent provides an interface to the Micro Focus Net Express JCL engine.

### NOTE

- To run a Micro Focus job, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus.
- You must install the Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus on the same computer where the Micro Focus Enterprise server is installed.
- You must set the `oscomponent.cmdprefix.force` parameter in the `agentparam.txt` file, as follows:

```
oscomponent.cmdprefix.force=false
```

## Define a Micro Focus Job

You can define a Micro Focus (MICROFOCUS) job to run your mainframe JCL and COBOL programs in a Windows environment.

### NOTE

- To run a Micro Focus job, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus.
- Install the Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus on the same computer where the Micro Focus Enterprise server is installed.  
Set the `oscomponent.cmdprefix.force` parameter in the `agentparam.txt` file, as follows:

```
oscomponent.cmdprefix.force=false
```

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: MICROFOCUS**  
Specifies that the job type is Micro Focus.
  - **machine** Specifies the name of the machine on which the job runs.
  - **mf\_jcl\_name Attribute -- Specify the Path to and Name of a JCL File** Specifies the path to and name of a JCL file to run.
  - **mf\_server**  
Specifies the server name or the IP address and port of the Micro Focus Enterprise Server.
2. (Optional) Specify optional Micro Focus attributes:
  - **envvars**
  - **fail\_codes**
  - **job\_class**
  - **job\_terminator**
  - **max\_exit\_success**
  - **mf\_user**
  - **mf\_version**
  - **os\_user**
  - **success\_codes**
3. (Optional) Specify common attributes that apply to all jobs.  
The Micro Focus job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Example: Pass the JCL by Content**

This example passes the contents of the JCL file to the Micro Focus Enterprise Server with IP address 172.31.255.255 and port 2222. The contents of the if.jcl file is read and passed to the server.

```
insert_job: mf_job_by_content
```

```
job_type: MICROFOCUS
```

```
machine: mfagent

mf_server_address_type: ADDRESS

mf_server: "tcp:172.31.255.255:2222"

mf_jcl_type: CONTENT

mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
```

### Example: Pass the JCL by Reference

This example passes the name of the JCL file to the Micro Focus Enterprise Server. The server reads and submits the named file. The job runs under the Micro Focus user ID named MFUSER01.

```
insert_job: mf_job_by_reference

job_type: MICROFOCUS

machine: mfagent

mf_server_address_type: NAME

mf_server: mfprod

mf_jcl_type: REFERENCE

mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"

mf_user: MFUSER01@mfprod
```

## Micro Focus Job Attributes with Default Values

This article explains Micro Focus job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Micro Focus job attributes have default values:

- **fail\_codes**  
Defines which exit codes indicate job failure.  
**Default:** Any exit code other than 0 (The job interprets any code other than zero as failure.)
- **job\_terminator**  
Specifies whether to terminate the job if its containing box fails or terminates.  
**Default:** n (The job does not terminate if its containing box fails or terminates.)
- **max\_exit\_success**

Defines the maximum exit code that the job can exit with and be considered a success.

**Default:** 0 (The job interprets only zero as job success.)

- **mf\_jcl\_type**

Specifies whether to pass the JCL by content or by reference to the Micro Focus Enterprise Server.

**Default:** CONTENT (The job passes the contents of the JCL file physically to the server.)

- **mf\_server\_address\_type**

Specifies whether the mf\_server attribute value is the server name or the IP address and port of the Micro Focus Enterprise Server.

**Default:** NAME (The job assumes that the name of the Micro Focus Enterprise Server is specified.)

- **mf\_version**

Specifies the environment on the mainframe that manages the job you want to run.

**Default:** JES2 (The job assumes that the environment is JES2.)

- **success\_codes**

Defines which exit codes indicate job success.

**Default:** 0 (The job interprets zero as success.)

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

### Example: Specify the JES3 Environment

This example runs the JCL file in the JES3 environment. The mf\_version attribute overrides JES2, the default environment.

```
insert_job: mf_job_with_version_jes3
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: NAME
mf_server: mfprod
mf_jcl_type: REFERENCE
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
mf_version: JES3
```

## Pass Variables as Parameters to a JCL File

You can pass variables as parameters to a JCL file.

### Follow these steps:

1. [Define a Micro Focus job](#).
2. Add the following attribute to the job definition:

```
envvars: parm_name=parm_value[, parm_name=parm_value...]
```

- **parm\_name**

Defines the name of a new environment variable or specifies the name of an existing environment variable.

- **parm\_value**

Specifies the value of the environment variable.

3. (Optional) Add an envvars attribute for each additional variable that you want to pass.

4. Run the job.

The specified variables are passed as parameters to the Micro Focus program.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#) and [Reference](#).

#### Example: Pass SORT and MAX as Parameters to the JCL file

Suppose that you want to pass SORT for symbolic variable VAR1 and MAX for symbolic variable VAR2 in the following JCL file named mfprog.jcl:

```
//MFIDSAL1 JOB 'DAVINDER',CLASS=A,MSGCLASS=A,NOTIFY=MFIDSA
//*-----
//*
//*-----
//STEP1 EXEC PGM=%VAR1,PARM='%VAR2'
//STEPLIB DD DSN=CEE.SCEERUN,DISP=SHR
// DD DSN=MFIDSA.DSA.LOAD,DISP=SHR
//SYSPRINT DD SYSOUT=* 00510000
//CEEDUMP DD SYSOUT=* 00520000
//SYSUDUMP DD SYSOUT=* 00530000
//OLDLIB DD DISP=(OLD),DSN=MFE.PROCLIB.L01
```

The parameter values are specified in the job definition using the envvars attribute as follows:

```
insert_job: mf_job_with_envvars
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: NAME
mf_server: mfprod
mf_jcl_type: CONTENT
mf_jcl_name: "C:\Program Files\microfocus\es-jcldemo\mfprog.jcl"
mf_user: MFUSER01@mfprod
envvars: VAR1=sort
envvars: VAR2=MAX
```

The following JCL is submitted to the Micro Focus Enterprise Server when the job runs:

```
//MFIDSAL1 JOB 'DAVINDER',CLASS=A,MSGCLASS=A,NOTIFY=MFIDSA
//*-----
//*
//*-----
//STEP1 EXEC PGM=sort,PARM='MAX'
//STEPLIB DD DSN=CEE.SCEERUN,DISP=SHR
// DD DSN=MFIDSA.DSA.LOAD,DISP=SHR
//SYSPRINT DD SYSOUT=* 00510000
//CEEDUMP DD SYSOUT=* 00520000
//SYSUDUMP DD SYSOUT=* 00530000
```

---

```
//OLDLIB DD DISP=(OLD),DSN=MFE.PROCLIB.L01
```

## Monitor Jobs Overview

Monitoring jobs let you monitor different aspects of your system.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

You can define the following monitoring jobs:

- **CPU Monitoring**  
Lets you monitor CPU usage.
- **Disk Monitoring**  
Lets you monitor disk space.
- **IP Monitoring**  
Lets you monitor an IP address.
- **Process Monitoring**  
Lets you monitor process execution.
- **Text File Reading and Monitoring**  
Lets you search a text file for a string.
- **Windows Event Log Monitoring**  
Lets you monitor a Windows event log.
- **Windows Service Monitoring**  
Lets you monitor the status of Windows services.

## Define a CPU Monitoring Job

You can define a CPU Monitoring (OMCPU) job to monitor the CPU usage of the computer where the specified agent is installed. By default, the job monitors for available CPU and completes when the specified conditions are met.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMCPU**  
Specifies that the job type is CPU Monitoring.
  - **machine** Specifies the name of the machine on which the job runs.
2. (Optional) Specify optional CPU Monitoring attributes:
  - **cpu\_usage**
  - **inside\_range**
  - **job\_class**
  - **job\_terminator**
  - **monitor\_mode**
  - **no\_change**
  - **poll\_interval**
3. Specify *at least one* of the following attributes if **monitor\_mode** is set to WAIT (the default) or CONTINUOUS:

- [lower\\_boundary](#)
- [upper\\_boundary](#)

**NOTE**

- If the `monitor_mode` is set to `WAIT`, the job monitors for available CPU on the specified machine and it completes when the CPU usage value falls within the lower and upper boundaries.
- If the `monitor_mode` is set to `CONTINUOUS`, the job monitors for the conditions continuously and an alert is written to the scheduler log file.
- If the `monitor_mode` is set to `WAIT` or `CONTINUOUS`, specify the `lower_boundary` attribute, the `upper_boundary` attribute, or both.

4. (Optional) Specify common attributes that apply to all job types.

The CPU Monitoring job is defined. When the job runs, it monitors for available CPU on the specified machine and it completes when the CPU usage value falls within the lower and upper boundaries.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Example: Check Available CPU Immediately**

This example monitors available CPU usage on the unixagent computer. The job checks the CPU usage immediately and reports the value in percent.

```
insert_job: omcpu_job
job_type: OMCPU
machine: unixagent
monitor_mode: NOW
```

**Example: Monitor When the Available CPU Reaches 50 Percent**

This example monitors the CPU available on the unixagent computer. The default monitor mode is `WAIT`, so a lower boundary, upper boundary, or both boundaries must be specified. In this example, the lower boundary is specified and the default upper boundary (100 percent) is used. When the available CPU usage is within 50 and 100 percent, the job completes.

```
insert_job: omcpu_job
job_type: OMCPU
machine: unixagent
lower_boundary: 50
```

**Example: Monitor Available CPU**



This example continuously monitors the CPU available on the unixagent computer. The job polls the CPU usage every 60 seconds. Each time the available CPU is within 75 and 95 percent, an alert is written to the scheduler log file. The job continues monitoring the CPU usage until it is ended manually.

```
insert_job: omcpu_job
job_type: OMCPU
machine: unixagent
cpu_usage: FREE
inside_range: TRUE
poll_interval: 60
monitor_mode: CONTINUOUS
lower_boundary: 75
upper_boundary: 95
```

## CPU Monitoring Modes

When you define a CPU monitoring job, you can define the mode it runs in. Depending on the mode, CPU monitoring jobs can do the following:

- Monitor CPU usage and complete when the specified conditions are met (WAIT mode). This is the default.
- Record the CPU usage at the time the job runs. In this case, the job runs only once (NOW mode).
- Monitor CPU usage and trigger an alert if the CPU usage meets the defined criteria (CONTINUOUS mode). For example, a job can trigger an alert when the computer is using between 80 and 100 percent of CPU, and the job continues to run until you manually terminate it.

## CPU Monitoring Job Attributes with Default Values

This article explains CPU Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following CPU Monitoring job attributes have default values:

- **cpu\_usage**  
Specifies whether the job monitors the available or used CPU processing capacity.  
**Default:** FREE (The job monitors for available CPU.)
- **inside\_range**  
Specifies whether the job completes (or triggers if monitoring continuously) when the value of CPU usage is inside or outside the specified boundaries.  
**Default:** TRUE (The job completes or triggers if the value of the CPU usage is within the lower and upper boundaries.)
- **lower\_boundary**  
Defines the minimum amount of CPU usage to monitor for in percent.  
**Default:** 0 (The job monitors the CPU usage between zero and the upper boundary.)
- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** WAIT (The job waits until the specified conditions are met before completing.)
- **poll\_interval**  
Defines the interval (in seconds) between successive scans of the CPU usage.

**Default:** objmon.scaninterval agent parameter (This parameter is automatically set to 10. The job polls the CPU usage every 10 seconds during continuous monitoring.)

- **upper\_boundary**

Defines the maximum amount of CPU usage to monitor for in percent.

**Default:** 100 (The job monitors the CPU usage between the lower boundary and 100 percent.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Monitor CPU Usage Until Condition Successful**

This example checks the CPU usage and completes successfully when the used CPU is within 80 and 95 percent. The `cpu_usage`, `monitor_mode`, `lower_boundary`, and `upper_boundary` attributes in this job definition override the default values.

```
insert_job: omcpu_job
job_type: OMCPU
machine: winagent
cpu_usage: USED
inside_range: TRUE
monitor_mode: WAIT
lower_boundary: 80
upper_boundary: 95
```

## Monitor CPU Usage Examples

The following examples are CPU Monitoring Jobs:

**Example: Monitor Used CPU**

This example monitors the used CPU on the unixagent computer. The job completes when the used CPU is less than 20 percent or greater than 80 percent.

```
insert_job: omcpu_job

job_type: OMCPU

machine: unixagent

lower_boundary: 20

upper_boundary: 80

inside_range: FALSE

cpu_usage: USED
```

**Example: Monitor CPU Availability Within a Range**

This example continuously monitors the CPU available on the unixagent computer. The job polls the CPU usage every 60 seconds. Each time the available CPU is within 75 and 95 percent, an alert is written to the scheduler log file. The job continues monitoring the CPU usage until it is ended manually.

```
insert_job: omcpu_job

job_type: OMCPU

machine: unixagent

cpu_usage: FREE

inside_range: TRUE

poll_interval: 60

monitor_mode: CONTINUOUS

lower_boundary: 75

upper_boundary: 95
```

The following table shows when alerts would be triggered with and without the no\_change value:

Time	CPU	Is the Alert Triggered When No Change is not specified?	Is the Alert Triggered When No Change is specified at 10 percent?
14:00:01	25 percent	No. Available CPU must be below 25 percent or above 75 percent.	No. Available CPU must be below 25 percent or above 75 percent.
14:00:02	20 percent	Yes. Available CPU is below 25 percent.	Yes. Available CPU is below 25%.
14:00:03	19 percent	Yes. Available CPU is below 25 percent.	No. Available CPU remains below 25 percent, but the change from the last reading is only 1 percent.
14:00:04	8 percent	Yes. Available CPU is below 25 percent.	Yes. CPU usage has changed 12 percent from the last time the alert was triggered.
14:00:05	19 percent	Yes. Available CPU is below 25 percent.	Yes. CPU usage has changed 11 percent from the last time the alert was triggered.
14:00:06	32 percent	No. Available CPU must be below 25 percent or above 75 percent.	No. Although CPU usage changed by more than 10 percent, it no longer falls within the range that is defined by the lower_boundary and upper_boundary fields. Usage is not below 25 percent or above 75 percent.

## Define a Disk Monitoring Job

On UNIX and Windows systems, you can define a Disk Monitoring (OMD) job to monitor the available or used space on a disk or logical partition. On i5/OS systems, you can define a Disk Monitoring job to monitor storage space in the file systems that are mounted on the i5/OS operating system.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMD**  
Specifies that the job type is Disk Monitoring.
  - **machine** Specifies the name of the machine on which the job runs.
  - **disk\_drive** Specifies the path to the disk, logical partition, or auxiliary storage pool to be monitored.
2. (Optional) Specify optional Disk Monitoring attributes:
  - **disk\_format**
  - **disk\_space**
  - **inside\_range**
  - **job\_class**
  - **job\_terminator**
  - **monitor\_mode**
  - **no\_change**
  - **poll\_interval**
3. Specify *one or both* of the following attributes if **monitor\_mode** is set to WAIT (the default) or CONTINUOUS:
  - **lower\_boundary**
  - **upper\_boundary**

### NOTE

If the **monitor\_mode** attribute is set to WAIT or CONTINUOUS and the **disk\_format** attribute is set to any value other than PERCENT, specify the **lower\_boundary** attribute, the **upper\_boundary** attribute, or both.

4. (Optional) Specify common attributes that apply to all job types.  
The Disk Monitoring job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the **jil** command, or use Web UI to define them. For more information about the **JIL** command and **JIL** syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

### Example: Check Available Disk Space Immediately

This example monitors available disk space on a local UNIX partition. The job checks the disk usage immediately and reports the value in megabytes.

```
insert_job: unix_freemb
job_type: OMD
machine: unixagent
disk_drive: /export/home
disk_format: MB
monitor_mode: NOW
```

### Example: Monitor When the Available Disk Space Reaches 50 Percent

This example monitors the disk space available in /export/home on the unixagent computer. The default monitor mode is WAIT, so a lower boundary, upper boundary, or both boundaries must be specified. In this example, the lower boundary is specified and the default upper boundary (100 percent) is used. When the available disk space is within 50 and 100 percent, the job completes.

```
insert_job: omd_job
job_type: OMD
machine: unixagent
disk_drive: /export/home
lower_boundary: 50
```

### Example: Continuously Monitor Used Space on a UNIX Partition

This example continuously monitors used disk space on a local UNIX partition. Each time the used disk space falls between 90 and 100 percent, an alert is written to the scheduler log file. The job continues monitoring the disk space until it is ended manually.

```
insert_job: unix_used
job_type: OMD
machine: unixagent
disk_drive: /export/home
disk_space: USED
disk_format: PERCENT
monitor_mode: CONTINUOUS
lower_boundary: 90
```

## Disk Monitoring Job Attributes with Default Values

This article explains Disk Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Disk Monitoring job attributes have default values:

- **disk\_format**  
Specifies the unit of measurement used to monitor available or used disk space.  
**Default:** PERCENT (The job monitors disk usage by percent.)
- **disk\_space**  
Specifies whether the job monitors for available or used disk space.

**Default:** FREE (The job monitors for available disk space.)

- **inside\_range**

Specifies whether the job completes (or triggers if monitoring continuously) when the value of disk usage is within or outside the specified boundaries.

**Default:** TRUE (The job completes or triggers if the value of the disk usage is within the lower and upper boundaries.)

- **monitor\_mode**

Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.

**Default:** WAIT (The job waits until the specified conditions are met before completing.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Monitor for Used Disk Space Outside of a Range

The `inside_range` and `disk_space` attributes in the following job definition override the default values.

This example monitors the C drive on a Windows computer for used space. When the value of disk space used falls below 16 percent or exceeds 95 percent, the job completes.

```
insert_job: omd_win
job_type: OMD
machine: winagent
disk_drive: C
disk_format: PERCENT
lower_boundary: 16
upper_boundary: 95
inside_range: FALSE
disk_space: USED
```

## Monitor Disk Space Examples

The following examples are Disk Monitoring Jobs:

### Example: Monitor Available Space on a UNIX Partition

This example monitors available disk space on a local UNIX partition. The job checks the disk usage immediately and reports the value in megabytes.

```
insert_job: unix_freemb

job_type: OMD

machine: unixagent

disk_drive: /export/home

disk_format: MB
```

---

```
monitor_mode: NOW
```

### Example: Continuously Monitor Used Space on a UNIX Partition

This example continuously monitors used disk space on a local UNIX partition. Each time the used disk space falls between 90 and 100 percent, an alert is written to the scheduler log file. The job continues monitoring the disk space until it is ended manually.

```
insert_job: unix_used

job_type: OMD

machine: unixagent

disk_drive: /export/home

disk_space: USED

disk_format: PERCENT

monitor_mode: CONTINUOUS

lower_boundary: 90
```

### Example: Monitor Used Space on a Windows Drive

This example monitors the used disk space on a local Windows C drive. When the used disk space falls below 16 percent or exceeds 95 percent, the job completes.

```
insert_job: win_used

job_type: OMD

machine: winagent

disk_drive: C

disk_format: PERCENT

disk_space: USED

monitor_mode: WAIT

lower_boundary: 16

upper_boundary: 95

inside_range: FALSE
```

---

**Example: Monitor the System Auxiliary Storage Pool on an i5/OS Computer**

This example continuously monitors the system auxiliary storage pool on an i5/OS computer. Each time the used disk space falls between 90 and 100 percent, an alert is written to the scheduler log file. The job continues monitoring the disk space until it is ended manually.

```
insert_job: i5_used

job_type: OMD

machine: i5agent

disk_drive: /

disk_space: USED

disk_format: PERCENT

monitor_mode: CONTINUOUS

lower_boundary: 90

upper_boundary: 100
```

**Example: Monitor Available Disk Space That Changes by At Least 100 KB**

This example continuously monitors the available disk space in kilobytes (KB) on the local Windows C drive. When the available space is in the 35000000 to 36000000 KB range, the first alert is written to the scheduler log file.

Subsequently, an alert is triggered each time the available disk space is within the specified boundaries and the disk usage changes by more than 100 KB. If the amount of change is less than or equal to 100 KB, the job does not register a change.

```
insert_job: omd_job

job_type: OMD

machine: winagent

disk_drive: C

disk_space: FREE

disk_format: KB

lower_boundary: 35000000

upper_boundary: 36000000

inside_range: TRUE
```



```
no_change: 100
```

```
monitor_mode: CONTINUOUS
```

The following table shows four sequential scans:

Scan	Scanned Amount (Kilobytes)	Does the Trigger Occur?
1	35018896	Yes.
2	35018900	No. Comparing scan 2 to scan 1, the delta value is only 4 KB. This scanned amount will not be included in the next calculation.
3	35018795	Yes. Comparing scan 3 to scan 1, the delta value is greater than 100 KB. The delta value of the next scan will be calculated using the scan 3 value of 35018795.
4	36000001	No. The scanned amount is outside the lower and upper boundary range.

## Define an IP Monitoring Job

You can define an IP Monitoring (OMIP) job to monitor an IP address or a port at an IP address. An IP Monitoring job monitors for a running or stopped status. The job can return the result immediately or can wait for the specified device to reach the specified state.

If the port is not specified or it is specified as zero, then while monitoring an IP host, the agent pings the machine. If the ping is successful, it indicates that the host is **RUNNING**. If the ping is unsuccessful, it indicates that the host is **STOPPED**.

If you specify a host and a non-zero port, the agent tries to connect to the port to check if an application is listening on it. If the connection is successful, it indicates that the application on the host is **RUNNING**. If the connection is unsuccessful, it indicates that the application on the host is **STOPPED**.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

To monitor remote IP addresses through the agent, the agent must run as root (on the Workload Automation Agent for UNIX, Linux, or Windows) or under a profile with sufficient authority to use the system ping command (on the Workload Automation Agent for i5/OS). If it runs as a user without root privileges, the job shows complete but with the message "Ping (ip address) insufficient privilege" in the status field and transmitter.log.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMIP**  
Specifies that the job type is IP Monitoring.
  - **machine** Specifies the name of the machine on which the job runs.
  - **ip\_host** Specifies the DNS name or IP address.
2. (Optional) Specify optional IP Monitoring attributes:

- `ip_port`
  - `ip_status`
  - `job_class`
  - `job_terminator`
  - `monitor_mode`
3. (Optional) Specify common attributes that apply to all job types.  
The IP Monitoring job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

#### Example: Monitor an IP Address for a Stopped Status

This example monitors a device with DNS name `myhost`. When the device stops running, the job completes.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: myhost
ip_status: STOPPED
monitor_mode: WAIT
```

#### Example: Monitor a Device Until it Stops

This example monitors the device `APPARCL`. When the device stops running, the job completes.

```
insert_job: omip_stop
job_type: OMIP
machine: SYSAG
ip_host: APPARCL
monitor_mode: WAIT
```

### Monitor Remote IP Addresses on UNIX

To monitor remote IP addresses through the agent, the agent must run as root (on the Workload Automation Agent for UNIX, Linux, or Windows) or under a profile with sufficient authority to use the system ping command (on the Workload Automation Agent for i5/OS). If the agent runs as a user without root privileges, a job that monitors a remote IP address shows complete but with the following message in the status field and `transmitter.log`:

---

```
Ping (ip address) insufficient privilege
```

## IP Monitoring Job Attributes with Default Values

This article explains IP Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following IP Monitoring job attributes have default values:

- **ip\_status**  
Specifies the status of the IP address to monitor.  
**Default:** STOPPED (The job monitors the IP address for a stopped status.)
- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** NOW (The job checks for the conditions immediately and completes.)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Monitor for a RUNNING Status Immediately

The ip\_status attribute in the following job definition overrides the default value.

This example monitors a device at IP address 172.31.255.255 and port 7510. When the job runs, it immediately checks if the device is running. If the device is running, the job completes successfully.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: 172.31.255.255
ip_port: 7510
monitor_mode: NOW
ip_status: RUNNING
```

## Monitor an IP Address Examples

The following examples are IP Monitoring Jobs:

### Example: Monitor an IP Address for a Stopped Status

This example monitors a device with DNS name myhost. When the device stops running, the job completes.

```
insert_job: omip_job

job_type: OMIP

machine: monagt

ip_host: myhost
```

```
ip_status: STOPPED

monitor_mode: WAIT
```

### Example: Monitor an Agent IP Address Specified in Dotted Decimal Format

This example checks whether an application at a specific IP address using a specific port is running. The IP address is 172.24.2.20 and the input port is 9401. When the job runs, it checks the status immediately. If the application is running, the job completes.

```
insert_job: omip_job

job_type: OMIP

machine: SYSAG

ip_host: 172.24.2.20

ip_port: 9401

ip_status: RUNNING

monitor_mode: NOW
```

## Define a Process Monitoring Job

You can define a Process Monitoring (OMP) job to monitor the status of a process on the computer where the agent is installed.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMP**  
Specifies that the job type is Process Monitoring.
  - **machine** Specifies the name of the machine on which the job runs.
  - **process\_name** Specifies the name of the process to be monitored.
2. (Optional) Specify optional Process Monitoring attributes:
  - **job\_class**
  - **job\_terminator**
  - **monitor\_mode**
  - **process\_status**
3. (Optional) Specify common attributes that apply to all job types.  
The Process Monitoring job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Monitor a Process**

This example monitors the nlnotes process. By default, the job checks if the process is stopped. If the process is stopped, the job completes successfully. If the process is running, the job continues monitoring it until it stops.

```
insert_job: omp_unix
job_type: OMP
machine: unixagt
process_name: nlnotes
monitor_mode: WAIT
```

**Example: Monitor a Running Process**

This example monitors the process with ID 2568. If the process is running, the job completes successfully. If the process is not running, the job continues monitoring it until the process starts running.

```
insert_job: omp_unix
job_type: OMP
machine: unixagt
process_name: 2568
process_status: RUNNING
monitor_mode: WAIT
```

**Process Monitoring Job Attributes with Default Values**

This article explains Process Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Process Monitoring job attributes have default values:

- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** NOW (The job checks for the conditions immediately and completes.)
- **process\_status**

Specifies the status of the process to be monitored.

**Default:** STOPPED (The job checks if the process is stopped.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Monitor for a STOPPED Status**

The monitor\_mode attribute in the following job definition overrides the default value.

This example monitors the server.exe process. The job checks the process status immediately and completes successfully if the process is running. If the process is running, the job continues monitoring until the process starts.

```
insert_job: omp_win

job_type: OMP

machine: winagt

process_name: "c:\Program files\Web_Server\server.exe"

process_status: RUNNING

monitor_mode: WAIT
```

## Monitor Processes Examples

The following examples are Process Monitoring Jobs:

**Example: Monitor Multiple Instances**

This example monitors the Microsoft SQL Server processes of two instances using the full path name. When the server process stops, the job monitoring that instance completes successfully. The first job monitors the sqlserver.exe process in the ...\\MSSQL.1\\MSSQL\\Binn directory. The second job monitors the sqlserver.exe process in the ...\\MSSQL.2\\MSSQL\\Binn directory.

```
insert_job: mon_sql_server_instance1

job_type: OMP

machine: mssqlserver

process_name: "C:\Program Files\Microsoft SQL Server\MSSQL.1\MSSQL\Binn\sqlservr.exe"

process_status: stopped

monitor_mode: wait

insert_job: mon_sql_server_instance2
```

```
job_type: OMP

machine: mssqlserver

process_name: "C:\Program Files\Microsoft SQL Server\MSSQL.2\MSSQL\Binn\sqlservr.exe"

process_status: stopped

monitor_mode: wait
```

### Example: Monitor the Agent Process on i5/OS

This example monitors the APPLPGM process on an i5/OS computer. The job checks the process status immediately and completes successfully if the process is running.

```
insert_job: omp_i5_onejob

job_type: OMP

machine: i5agt

process_name: 123456/PROD/APPLPGM

process_status: RUNNING

monitor_mode: NOW
```

### Example: Monitor Multiple i5/OS Processes That Have Similar Names

This example monitors all processes running on an i5/OS computer under the JDOE user profile and whose names start with CALC. When all of these processes stop running, the job completes successfully.

```
insert_job: omp_i5

job_type: OMP

machine: i5agt

process_name: *ALL/JDOE/CALC*

process_status: STOPPED

monitor_mode: WAIT
```

## Define a Text File Reading and Monitoring Job

You can define a Text File Reading and Monitoring (OMTF) job to search a text file on a Windows, UNIX, or i5/OS computer for a text string. For example, you can monitor a log file for an error message after a script executes.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMTF**  
Specifies that the job type is Text File Reading and Monitoring.
  - **machine** Specifies the name of the machine on which the job runs.
  - **text\_file\_filter** Defines the text string to search for. You can specify the text string as a regular expression.
  - **text\_file\_name** Specifies the path to and name of the text file to search.
2. (Optional) Specify optional Text File Reading and Monitoring attributes:
  - **encoding**
  - **job\_class**
  - **job\_terminator**
  - **lower\_boundary**
  - **monitor\_mode**
  - **text\_file\_filter\_exists**
  - **text\_file\_mode**
  - **time\_format**
  - **time\_position**
  - **upper\_boundary**
3. (Optional) Specify common attributes that apply to all job types.  
The Text File Reading and Monitoring job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and iimport them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Monitor a File for a Specified String**

This example monitors the entire transactions.log file for the ERROR MESSAGE string. The job waits for the string to be found and then completes successfully.

```
insert_job: textfile_job
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transactions.log
text_file_filter: ERROR MESSAGE
```



### Example: Monitor a File for a Text String

This example searches for a text string in the log text file on a Windows computer. The path to the text file is enclosed in quotation marks because the path contains a colon. The job searches for the string "ERROR MESSAGE" between lines 1234 and 1876. If the string is found, the job completes successfully.

```
insert_job: textfile_job1
job_type: OMTF
machine: monagt
text_file_name: "c:\program files\agent\log"
text_file_filter: ERROR MESSAGE
text_file_mode: LINE
lower_boundary: 1234
upper_boundary: 1876
monitor_mode: NOW
```

### Text File Reading and Monitoring Job Attributes with Default Values

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Text File Reading and Monitoring job attributes have default values:

- **encoding** Specifies the name of the character set used to encode the data in the file.  
**Default:** US-ASCII (The job monitors the file as US-ASCII.)
- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** WAIT (The job waits until the specified conditions are met before completing.)
- **text\_file\_mode**  
Specifies the search mode when monitoring a text file.  
**Default:** LINE (The job searches for the text in the specified line boundaries.)
- **text\_file\_filter\_exists**  
Specifies whether the job monitors the text file to check if the text string exists or does not exist.  
**Default:** TRUE (The job checks if the text string exists.)

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Search for Text Between Regular Expressions

The text\_file\_mode attribute in the following job definition overrides the default LINE monitor mode.

This example searches the /export/home/systemagent/agentparm.txt file. The search starts at the first line that contains the word "agent" at the beginning of the line (as specified by \A in the regular expression specified for lower\_boundary) and until it finds the string "level=2" at the end of a line (as specified by \Z in the regular expression specified for upper\_boundary).

```
insert_job: omtf_unix_line

job_type: OMTF
```

```
machine: monagt

text_file_name: /export/home/systemagent/agentparm.txt

text_file_filter: \.0/MAIN$

text_file_mode: REGEX

lower_boundary: \Aagent

upper_boundary: level=2\Z

monitor_mode: NOW
```

### Example: Specify a Data Encoding Value

The encoding attribute in the following job definition overrides the default US-ASCII data encoding value.

This example monitors a text file that contains data encoded in the ISO Latin Alphabet No. 1 (also named ISO-LATIN-1). The job checks the text file immediately and completes successfully if the specified string is found. If the string is not found, the job fails.

```
insert_job: textfile_job

job_type: OMTF

machine: monagt

text_file_name: /export/home/logs/transactions.log

text_file_filter: ERROR MESSAGE

text_file_mode: LINE

lower_boundary: 1

upper_boundary: 50

encoding: ISO-8859-1

monitor_mode: NOW
```

### Monitor a Text File for Specified Text Examples

The following examples are Text File Reading and Monitoring Jobs:

#### Example: Search for a Regular Expression

In this example, the text string contains regular expression pattern matching syntax. The search range is also a regular expression as indicated by the `text_file_mode` attribute.

```

insert_job: omtf2

job_type: OMTF

machine: monagt

text_file_name: /export/home/agentdir/agentparm.txt

text_file_filter: ^\w{4,10}\.

text_file_mode: REGEX

lower_boundary: "log*.*"

monitor_mode: now

```

The regular expression can be interpreted as follows:

- ^ or \A -- match only at the beginning of string (line)
  - \Z or \$ -- match only at the end of string
  - \w -- a word character [a-zA-Z0-9]
  - \W -- a non-word character
  - \s -- a whitespace character
  - {4,10} -- match at least 4 times but not more than 10 times
- To illustrate the last item (4, 10), consider the syntax:

```
text_file_filter: b1{1,3}c
```

Evaluating this expression yields the following conditions:

- The line contains the text b1.
- Numeric 1 should exist at least once, but not more than three times.
- The specified text string must be followed by the letter c.

### Example: Search for a String in a File Starting at a Specified Line

This example searches the c:\callog file in line mode. The job starts searching the content from line 143 of the file. The upper boundary is not defined, so the job searches to the last line of the file. The job completes successfully if the ERROR MESSAGE string is found.

```

insert_job: omtf_line

job_type: OMTF

machine: monagt

text_file_name: "c:\ca\log"

text_file_filter: ERROR MESSAGE

```

```
text_file_mode: LINE

lower_boundary: 143

monitor_mode: NOW
```

**Example: Search for a String in a File When the Search Mode is REGEX**

This example searches the c:\callog file in regular expression mode. The lower boundary is not defined, so the job searches the content from the first line of the file to the upper boundary (a line that contains the word service). The job completes successfully if the ARCHIVE string is found.

```
insert_job: omtf_regex

job_type: OMTF

machine: monagt

text_file_name: "c:\ca\log"

text_file_filter: ARCHIVE

text_file_mode: REGEX

upper_boundary: service

monitor_mode: NOW
```

**Example: Search for a String in a File When the Search Mode is DATETIME**

This example searches the /export/home/logs/transmitter.log file in date and time mode. The job searches the content between May 20, 2010 at midnight and May 27, 2010 at 11:59 p.m. The date and time values are defined using the format specified in the time\_format attribute. The job completes successfully if the transmitted string is found.

```
insert_job: omtf_timedate

job_type: OMTF

machine: monagt

text_file_name: /export/home/logs/transmitter.log

text_file_filter: transmitted

text_file_mode: DATETIME

lower_boundary: "Thu May 20 00:00:00.000 EDT 2010"
```

```
upper_boundary: "Thu May 27 23:59:59.999 EDT 2010"

time_format: "EEE MMM dd HH:mm:ss.SSS zzz yyyy"

time_position: 12

monitor_mode: NOW
```

### Example: Monitor a Text File Continuously

This example searches the transmitter.log file for the text string "Warning".

```
insert_job: textfile_job

job_type: OMTF

machine: monagt

text_file_name: /export/home/log/transmitter.log

text_file_filter: Warning

text_file_mode: LINE

lower_boundary: 25

text_file_filter_exists: TRUE

monitor_mode: CONTINUOUS
```

When the job first runs, it searches the content between line 25 and the end of the file. An alert is written to the scheduler log file the first time that the string is found. In other words, suppose that the file contains multiple occurrences of "Warning" between lines 25 to the end of the file, as follows:

```
.

.

.

25

26 Warning

27

28 Warning
```

```

29
30
31 Warning
.
.
.
EOF

```

When the job first runs, the trigger only occurs at the first occurrence of the text (line 26). Subsequently, the job continues monitoring only the new data that is *appended* to the file. An alert is triggered each time the string is found in the appended data.

#### NOTE

Alerts are not triggered for new occurrences of the "Warning" string in the data that has already been searched. For example, suppose that the job has already searched lines 25 to 100 of the file. The file is then modified to include "Warning" on line 30. During continuous monitoring, an alert is *not* triggered for that occurrence.

This job runs until it is completed manually.

#### Example: Search for a Text String on an i5/OS Computer

This example searches for a text string in the DATA member of a QSYS file object on an i5/OS computer. The job searches the content between lines 1 and 20. The job completes successfully if the string is found.

```

insert_job: textfile_job3

job_type: OMTF

machine: monagt

text_file_name: /QSYS.LIB/LIBRARY.LIB/RESULTS.FILE/DATA.MBR

text_file_filter: Create file failed

text_file_mode: LINE

lower_boundary: 1

upper_boundary: 20

monitor_mode: NOW

```

#### Example: Check If a String Does Not Exist in a File

This example searches lines 1 to 200 of the transmitter.log file for the text string "Warning". If the string is *not* found, the job completes successfully. If the string is found, the job fails.

```
insert_job: textfile_job

job_type: OMTF

machine: monagt

text_file_name: /export/home/log/transmitter.log

text_file_filter: Warning

text_file_mode: LINE

lower_boundary: 1

upper_boundary: 200

text_file_filter_exists: FALSE

monitor_mode: NOW
```

## Define a Windows Event Log Monitoring Job

You can define a Windows Event Log Monitoring (OMEL) job to monitor a Windows event log in the computer where the agent is running. The monitor returns the most recent event available or continuously monitors for events in a particular Windows event log.

The Windows Event Log Monitoring job only monitors event logs maintained by the operating system and available in the Event Viewer. For more information about the Windows event properties that are stored in Windows event logs, view the logs in the Event Viewer on your local machine.

### NOTE

To run these jobs, your system requires Workload Automation Agent for Windows.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMEL**  
Specifies that the job type is Windows Event Log Monitoring.
  - **machine Attribute -- Define the Client Where a Job Runs** Specifies the name of the machine on which the job runs.
  - **win\_log\_name Attribute -- Specify the Name of a Windows Event Log** Specifies the name of the event log.
2. (Optional) Specify optional Windows Event Log Monitoring attributes:

- [job\\_class](#)
  - [job\\_terminator](#)
  - [monitor\\_mode](#)
  - [win\\_event\\_category](#)
  - [win\\_event\\_computer](#)
  - [win\\_event\\_datetime](#)
  - [win\\_event\\_description](#)
  - [win\\_event\\_id](#)
  - [win\\_event\\_op](#)
  - [win\\_event\\_source](#)
  - [win\\_event\\_type](#)
3. (Optional) Specify common attributes that apply to all job types.  
The Windows Event Log Monitoring job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Monitor an Application Event Log

This example monitors an Application event log. By default, the job waits until an Error event occurs before it completes.

```
insert_job: eventlog_job

job_type: OMEL

machine: monagt

win_log_name: Application
```

### Example: Monitor an Application Log That Occurs On or After a Specified Date

This example monitors an application log that occurs any time on or after January 12, 2010, 6:30 a.m. When the job finds an application log that occurs any time on or after that date and time, the job completes successfully.

```
insert_job: win_eventlog
job_type: OMEL
```



```

machine: winagent
win_log_name: Application
win_event_type: info
win_event_category: None
win_event_source: LLDSAPNT223
win_event_datetime: "20100112 06:30:00"

```

## Event Log Types

The Windows Event Log Monitoring job only monitors event logs maintained by the operating system and available in the Event Viewer. Windows operating systems record events in at least three types, including the following logs:

- **Application log**  
The application log contains events that are logged by applications or programs. For example, a database program might record a file error in the application log.
- **System log**  
The system log contains events that are logged by the Windows system components. For example, the failure of a driver or other system component to load during startup is recorded in the system log.
- **Security log**  
The security log can record security events (such as valid and invalid logon attempts) and events that are related to resource use (such as creating, opening, or deleting files).

For more information about Windows logs, select Start, Settings, Control Panel, Administrative Tools, Event Viewer. Select any of the three log categories and double-click to view its property page.

## Event Log Monitoring Job Attributes with Default Values

This article explains Event Log Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Windows Event Log Monitoring job attributes have default values:

- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** WAIT (The job waits until the specified conditions are met before completing.)
- **win\_event\_type**  
Specifies the event type to monitor in the Windows event log.  
**Default:** ERROR (The job monitors for the Error event type.)
- **win\_event\_op**  
Specifies a comparison operator against the value of a Windows Event ID.  
**Default:** EQ (The job monitors for an Event ID that is equal to the specified value.)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

## Example: Monitor an Event Log Continuously for Info Events

The monitor\_mode, win\_event\_op, and win\_event\_type attributes in the following job definition override the default values.

This example monitors the event log for applications continuously for all instances of an INFO event type, where the event source is LLDSAPNT223, the event description contains the word started, and the event ID is less than or equal to 4000. Each time the specified conditions occur, an alert is written to the scheduler log file.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_source: LLDSAPNT223
win_event_category: None
win_event_type: INFO
win_event_op: LE
win_event_id: 4000
win_event_description: started
monitor_mode: CONTINUOUS
```

## Monitor a Windows Event Log Examples

The following examples are Windows Event Log Monitoring Jobs:

### Example: Monitor an Application Log That Occurs on or after a Specified Date

This example monitors an application log that occurs any time on or after January 12, 2010, 6:30 a.m. When the job finds an application log that occurs any time on or after that date and time, the job completes successfully.

```
insert_job: win_eventlog
job_type: OMEL
machine: winagent
win_log_name: Application
win_event_type: info
win_event_category: None
win_event_source: LLDSAPNT223
win_event_datetime: "20100112 06:30:00"
```

### Example: Monitor Events with IDs Equal to 0

This example checks for an event ID number less than 1. The job returns the first application event from the application log that has an event ID equal to 0.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
```

```
win_log_name: Application
win_event_op: LT
win_event_id: 1
```

### Example: Monitor a System Event Log

This example monitors a system event log for an event type of WARN, event source of MrxSmb, and event category of None.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: System
win_event_type: WARN
win_event_source: MrxSmb
win_event_category: None
```

### Example: Monitor a Security Event Log for Audit Success Events

In this example, the security log is monitored for a successful audit of a security access attempt. The event category is System Event, the term succeeded is excluded, but the words Audit and log are included in the event description.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Security
win_event_type: AUDITS
win_event_category: System Event
win_event_source: Service Control Manager
win_event_description: "-succeeded +Audit log"
```

### Example: Monitor a System Event Log for Particular Errors

In this example, the event description must include the words conflict and state as indicated by the plus signs but must exclude the words deny, master, or browser as indicated by the minus sign. The plus sign is the default and is optional.

```
insert_job: eventlog_job1
job_type: OMEL
machine: monagt
win_log_name: System
win_event_type: ERROR
win_event_description: "+conflict +state -deny -master -browser"
```

### Example: Monitor an Application Log for Events Indicating Normal Shutdown

In this example, the event description must include the words Normal shutdown. This example does not use the plus sign, and by default, the specified words are included in the search.

```

insert_job: eventlog_job2
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_type: INFO
win_event_description: "Normal shutdown"

```

## Define a Windows Service Monitoring Job

You can define a Windows Service Monitoring (OMS) job to monitor a service on a Windows computer where the agent is running.

### NOTE

To run these jobs, your system requires Workload Automation Agent for Windows.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OMS**  
Specifies that the job type is Windows Service Monitoring.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **win\_service\_name** Specifies the name of the local Windows service to be monitored.
2. (Optional) Specify optional Windows Service Monitoring attributes:
  - **job\_class**
  - **job\_terminator**
  - **monitor\_mode**
  - **win\_service\_status**
3. (Optional) Specify common attributes that apply to all job types.  
The Windows Service Monitoring job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Monitor a Windows Service

This example monitors a Windows service named App Server. The `win_service_status` attribute is not specified in the job definition, so the job monitors for a `RUNNING` status by default. The job completes when the service is running.

```
insert_job: oms_job2
job_type: OMS
machine: winagt
win_service_name: App Server
monitor_mode: WAIT
```

### Example: Check a Service Status Immediately

This example monitors the `schedmanager` service for a status of `RUNNING`. The job checks the status immediately and completes successfully if the service is running. If the service is not running, the job fails.

```
insert_job: oms_job3
job_type: OMS
machine: winagt
win_service_name: schedmanager
win_service_status: RUNNING
monitor_mode: NOW
```

## Windows Service Monitoring Job Attributes with Default Values

This article explains Windows Service Monitoring job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the `agentparm.txt` file.

If you specify the attribute in a job definition, it overrides the default.

The following Windows Service Monitoring job attributes have default values:

- **monitor\_mode**  
Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.  
**Default:** `NOW` (The job checks for the conditions immediately and completes.)
- **win\_service\_status**  
Specifies the status of the Windows Service to be monitored.  
**Default:** `RUNNING` (The job checks if the Windows service is running.)

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Monitor for a PAUSED Status Immediately

The `win_service_status` attributes in the following job definition overrides the default value.

This example monitors a Windows service named Proc Server. The job checks the status immediately and completes successfully if the service is paused. If the service is not paused, the job fails.

```
insert_job: oms_job1
job_type: OMS
machine: winagt
win_service_name: Proc Server
win_service_status: PAUSED
monitor_mode: NOW
```

---

## Monitoring Windows Services Examples

The following examples are Windows Services Monitoring Jobs:

### Example: Monitor for the Existence of a Windows Service

This example monitors a Windows service named Proc Server. The job completes successfully if the service exists. By default, the job checks for the condition immediately and completes, so the `monitor_mode` attribute is not required in the job definition.

```
insert_job: oms_job1
job_type: OMS
machine: winagt
win_service_name: Proc Server
win_service_status: EXISTS
```

### Example: Specify a Path to a Windows Service Executable

This example monitors a Windows service named Log App. The job waits until the service status is `CONTINUE_PENDING` before it completes.

```
insert_job: oms_job2
job_type: OMS
machine: winagt
win_service_name: "C:\Program Files\Log App\apptask.exe"
win_service_status: CONTINUE_PENDING
monitor_mode: WAIT
```

### Example: Check a Service Status Immediately

This example monitors the `schedmanager` service for a status of `RUNNING`. The job checks the status immediately and completes successfully if the service is running. If the service is not running, the job fails.

```
insert_job: oms_job3

job_type: OMS

machine: winagt

win_service_name: schedmanager

win_service_status: RUNNING

monitor_mode: NOW
```

## Oracle E-Business Suite Jobs

You can define jobs to run Oracle E-Business Suite workload.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Oracle E-Business Suite.

You can define the following Oracle E-Business Suite jobs:

- **Copy Single Request Job**  
Copies an existing single request defined on Oracle Applications and runs it under the agent.
- **Request Set Job**  
Runs multiple programs in an Oracle Applications application.
- **Single Request Job**  
Runs a single program in an Oracle Applications application.

**Define an Oracle E-Business Suite Copy Single Request Job**

You can define an Oracle E-Business Suite Copy Single Request (OACOPY) job to copy an existing single request that is defined on Oracle E-Business Suite and run it under the agent. When the job runs, it can override values in the original definition with values that are specified on the agent or in the job definition. The OACOPY job is useful when you want to reuse existing job definitions.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Oracle E-Business Suite.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OACOPY**  
Specifies that the job type is Oracle E-Business Suite Copy Single Request.
  - **machine** Specifies the name of the machine on which the job runs.
  - **request\_id**  
Specifies the request ID of the Oracle E-Business Suite request you want to copy.

**WARNING**

All Oracle E-Business Suite jobs require a responsibility name. Define this value; otherwise, the job fails. To define this value, you can specify a default responsibility name using the `oa.default.responsibility` parameter in the agent's `agentparm.txt` file. Alternatively, you can specify the `oracle_resp` attribute in a job definition, which overrides the default responsibility name.

All Oracle E-Business Suite jobs require a user name. Define this value; otherwise, the job fails. To define this value, you can specify a default user name using the `oa.default.user` parameter in the agent's `agentparm.txt` file. Alternatively, you can specify the `oracle_user` attribute in a job definition, which overrides the default user name.

2. Do *one* of the following tasks:
  - Ensure that a default Oracle E-Business Suite user name and responsibility name are defined in the agent's `agentparm.txt` file using the `oa.default.user` and `oa.default.responsibility` parameters, respectively.
  - Add the following attributes to the definition:
    - **oracle\_user**  
Specifies the Oracle E-Business Suite user name that the job runs under.  
**Note:** This attribute overrides the `oa.default.user` agent parameter.
    - **oracle\_resp**  
Specifies an Oracle E-Business Suite responsibility name.

**NOTE**

This attribute overrides the `oa.default.responsibility` agent parameter.

3. (Optional) Specify optional Oracle E-Business Suite Copy Single Request attributes:

- [job\\_class](#)
- [job\\_terminator](#)
- [oracle\\_mon\\_children](#)
- [oracle\\_mon\\_children\\_delay](#)

4. (Optional) Specify common attributes that apply to all jobs.

The Oracle E-Business Suite Copy Single Request job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Copy a Single Request**

Suppose that you want to copy an existing single request that is defined on Oracle E-Business Suite. In this example, the job copies the single request job with request ID 2255470 and overrides the Oracle EBusiness Suite user name and responsibility name defined in the `agentparm.txt` file.

```
insert_job: oacopy_single
job_type: oacopy
machine: oaagent
request_id: 2255470
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

**Define an Oracle E-Business Suite Request Set Job**

You can define an Oracle E-Business Suite Request Set (OASET) job to run a request set program. You must get the following information from the original Oracle E-Business Suite request set:

- Display name or short name of the Oracle E-Business Suite application the request set belongs to
- Request set short name or display name
- User name that the job runs under
- Responsibility name



**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Oracle E-Business Suite.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OASET**  
Specifies that the job type is Oracle E-Business Suite Request Set.
  - **machine** Specifies the name of the machine on which the job runs.
  - **oracle\_appl\_name**  
Specifies the name of the Oracle E-Business Suite application that the request set belongs to.
  - **oracle\_req\_set**  
Specifies the request set short name or display name depending on the oracle\_req\_set\_type attribute.

**WARNING**

All Oracle E-Business Suite jobs require a responsibility name. Define this value; otherwise, the job fails. To define this value, you can specify a default responsibility name using the oa.default.responsibility parameter in the agent's agentparm.txt file. Alternatively, you can specify the oracle\_resp attribute in a job definition, which overrides the default responsibility name.

All Oracle E-Business Suite jobs require a user name. Define this value; otherwise, the job fails. To define this value, you can specify a default user name using the oa.default.user parameter in the agent's agentparm.txt file. Alternatively, you can specify the oracle\_user attribute in a job definition, which overrides the default user name.

2. Do *one* of the following tasks:
  - Ensure that a default Oracle E-Business Suite user name and responsibility name are defined in the agent's agentparm.txt file using the oa.default.user and oa.default.responsibility parameters, respectively.
  - Add the following attributes to the definition:
    - **oracle\_user**  
Specifies the Oracle E-Business Suite user name that the job runs under.

**NOTE**

This attribute overrides the oa.default.user agent parameter.

- **oracle\_resp**  
Specifies an Oracle E-Business Suite responsibility name.

**NOTE**

This attribute overrides the oa.default.responsibility agent parameter.

3. (Optional) Specify optional Oracle E-Business Suite Request Set attributes:

- [job\\_class](#)
  - [job\\_terminator](#)
  - [oracle\\_appl\\_name\\_type](#)
  - [oracle\\_custom\\_property](#)
  - [oracle\\_mon\\_children](#)
  - [oracle\\_mon\\_children\\_delay](#)
  - [oracle\\_print\\_copies](#)
  - [oracle\\_print\\_style](#)
  - [oracle\\_printer](#)
  - [oracle\\_programdata](#)
  - [oracle\\_req\\_set\\_type](#)
  - [oracle\\_use\\_arg\\_def](#)
  - [oracle\\_use\\_set\\_defaults\\_first](#)
4. (Optional) Specify common attributes that apply to all job types.  
The Oracle E-Business Suite Request Set job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Run a Request Set

This example runs a request set named FNDRSSUB1310 on the oaagent agent. The request set belongs to the application with the short name BIS in Oracle E-Business Suite. The job runs under the SYSADMIN user with Business Intelligence Super User, Progress S&L responsibility. The job's output is saved.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_user: SYSADMIN
oracle_resp: "Business Intelligence Super User, Progress S&L"
oracle_save_output: Y
```

## Specify Data for an Individual Program in a Request Set

When you define an Oracle E-Business Suite Request Set (OASET) job, you can specify data for an individual program in the request set. This data overrides the arguments and print parameters specified for the entire request set. You can specify the following data for a program:

- Program arguments- (using the `oracle_args` attribute)
- Printer- (using the `oracle_printer` attribute)
- Print style- (using the `oracle_print_style` attribute)
- Number of copies to print- (using the `oracle_print_copies` attribute)
- Whether to save the output from the program- (using the `oracle_save_output` attribute)
- List of user names to notify using short or display names- (using the `oracle_notify_users` and `oracle_notify_display_users` attributes)
- Output format- (using the `oracle_output_format` attribute)
- Operating unit name- (using the `oracle_oprunit` attribute)
- Template name, template language, and template territory- (using the `oracle_template_name`, `oracle_template_language`, and `oracle_template_territory` attributes)
- Whether to quote resolved expressions in default values- (using the `oracle_quote_in_default` attribute)

### Follow these steps:

1. [Define an Oracle E-Business Suite Request Set job.](#)
2. Add the following attribute to the job definition:
  - **oracle\_programdata**  
Specifies data for an individual program in an Oracle E-Business Suite request set.
3. Run the job  
The data is specified for the program.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Specify Values for Individual Programs in a Request Set

Suppose that you want to run an Oracle E-Business Suite request set named `EXTRACTS` on the `oaagent` agent. The request set belongs to the application with the display name `Application Object Library` in Oracle E-Business Suite. The job definition specifies the following default settings for all programs in the request set:

- The number of copies to be printed is 1.
- The print style is `LANDSCAPE`.
- The printer is `\\printer path\Q8`.

The first `oracle_programdata` attribute overrides the default values for the first program in the request set. The first program uses the arguments `T23` and `R1` and prints two copies using the `\\printer path\Q1` printer in `PORTRAIT` style.

The second `oracle_programdata` attribute overrides the default values for the fifth program in the request set. The fifth program uses arguments `R` and `R1` and prints three copies using the `\\printer path\Q2` printer in `PORTRAIT` style.

The other programs in the request set use the default values.

```
insert_job: oaset_prog
job_type: oaset
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_req_set: EXTRACTS
```

```

oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_printer: "\\printer path\Q8"
oracle_print_style: LANDSCAPE
oracle_print_copies: 1
oracle_programdata: index=1, args="T23,,R1", printer="
\\printer\Q1", print_style=PORTRAIT, print_copies=2, saveop=Y
oracle_programdata: index=5, args="R,R1,", printer="
\\printer\Q2", print_style=PORTRAIT, print_copies=3, saveop=N

```

Suppose that you want to run an Oracle E-Business Suite request set named FNDRSSUB1310 on the local host. The request set belongs to the application with the short name BIS in Oracle EBusiness Suite. The SYSADMIN and ASGUEST users are notified when the single request completes on QATEST1.

The first oracle\_programdata attribute overrides the default values for the first program in the request set. The first program uses the output format as PDF, the template language is English, and template territory is US.

The second oracle\_programdata attribute overrides the default values for the fifth program in the request set. The second program uses the output format as EXCEL, the template language is English, and template territory is US.

The other programs in the request set use the default values.

```

delete_job: test_OASET
insert_job: oaset_fmt
job_type: OASET
machine: localhost
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_name_type: DISPLAY
oracle_req_set: FNDRSSUB1310
oracle_use_arg_def: Y
oracle_programdata: index=1, output_format=PDF, template_language=en, template_ter
ritory=US, notify_users="'SYSADMIN','ASGUEST','QATEST1'"
oracle_programdata: index=2, output_format=EXCEL, template_language=en, template_t
erritory=US, notify_users="'SYSADMIN','ASGUEST','QATEST1'"
group: OA

```

## Specify Argument Values to Pass to a Program in a Request Set

You can pass argument values to an individual program in an Oracle E-Business Suite Request Set (OASET) job. The job can also use the default values that are defined by the registered Oracle Applications Concurrent Manager program. When an argument value is defined in both the job definition and as a default, the argument value in the job definition overrides the default.

### Follow these steps:

1. [Define an Oracle E-Business Suite Request Set job.](#)
2. Add the following attribute to the job definition:
  - **oracle\_programdata**  
Specifies data for an individual program in an Oracle E-Business Suite request set.

3. (Optional) Add the following attribute:

– **oracle\_use\_arg\_def**

Specifies whether to use default values for arguments that are not defined using the `oracle_programdata` attribute.

4. Run the job.

The argument values are passed to the program in the request set.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Specify Argument Values for a Program in a Request Set**

This example runs an Oracle E-Business Suite Request Set job that uses the argument defaults in Oracle E-Business Suite and the argument values defined in the job definition. The second program in the request set has four arguments and you want to pass T23 as the first value and R1 as the fourth value. The argument string specifies placeholders for the second and third arguments, so the job uses the default values for those arguments.

```
insert_job: oaset_prog

job_type: oaset

machine: oaagent

oracle_appl_name_type: DISPLAY

oracle_appl_name: Application Object Library

oracle_req_set: EXTRACTS

oracle_user: SYSADMIN

oracle_resp: System Administrator

oracle_use_arg_def: Y

oracle_programdata: index=2, args="T23,,R1", printer="\printer
\Q1", print_style=PORTRAIT, print_copies=2, saveop=Y
```

**Define an Oracle E-Business Suite Single Request Job**

You can define an Oracle E-Business Suite Single Request (OASG) job to run a single request program. You must get the following information from the original Oracle E-Business Suite single request:

- Display name or short name of the Oracle E-Business Suite application the single request belongs to
- Program short name or display name
- User name that the job runs under
- Responsibility name

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Oracle E-Business Suite.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: OASG**  
Specifies that the job type is Oracle E-Business Suite Single Request.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **oracle\_appl\_name**  
Specifies the name of the Oracle E-Business Suite application that the single request belongs to.
  - **oracle\_program**  
Specifies the single request program short name or display name depending on the oracle\_program\_name\_type attribute.

**WARNING**

All Oracle E-Business Suite jobs require a responsibility name. Define this value; otherwise, the job fails. To define this value, you can specify a default responsibility name using the oa.default.responsibility parameter in the agent's agentparm.txt file. Alternatively, you can specify the oracle\_resp attribute in a job definition, which overrides the default responsibility name.

All Oracle E-Business Suite jobs require a user name. Define this value; otherwise, the job fails. To define this value, you can specify a default user name using the oa.default.user parameter in the agent's agentparm.txt file. Alternatively, you can specify the oracle\_user attribute in a job definition, which overrides the default user name.

2. Do *one* of the following tasks:
  - Ensure that a default Oracle E-Business Suite user name and responsibility name are defined in the agent's agentparm.txt file using the oa.default.user and oa.default.responsibility parameters, respectively.
  - Add the following attributes to the definition:
    - **oracle\_user**  
Specifies the Oracle E-Business Suite user name that the job runs under.

**NOTE**

This attribute overrides the oa.default.user agent parameter.

- **oracle\_resp**  
Specifies an Oracle E-Business Suite responsibility name.

**NOTE**

This attribute overrides the oa.default.responsibility agent parameter.

3. (Optional) Specify optional Oracle E-Business Suite Single Request attributes:

- [job\\_class](#)
  - [job\\_terminator](#)
  - [oracle\\_appl\\_name\\_type](#)
  - [oracle\\_args](#)
  - [oracle\\_custom\\_property](#)
  - [oracle\\_desc](#)
  - [oracle\\_mon\\_children](#)
  - [oracle\\_mon\\_children\\_delay](#)
  - [oracle\\_notify\\_display\\_users](#)
  - [oracle\\_notify\\_users](#)
  - [oracle\\_output\\_format](#)
  - [oracle\\_oprunit](#)
  - [oracle\\_print\\_copies](#)
  - [oracle\\_print\\_style](#)
  - [oracle\\_printer](#)
  - [oracle\\_program\\_name\\_type](#)
  - [oracle\\_quote\\_in\\_default](#)
  - [oracle\\_save\\_output](#)
  - [oracle\\_template\\_name](#)
  - [oracle\\_template\\_language](#)
  - [oracle\\_template\\_territory](#)
  - [oracle\\_user](#)
  - [oracle\\_use\\_arg\\_def](#)
4. (Optional) Specify common attributes that apply to all job types.  
The Oracle E-Business Suite Single Request job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Run a Single Request

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility.

```
insert_job: oasg_short
job_type: oasg
```

```

machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator

```

### Example: Run a Single Request Program

This example runs a single request program named FNDSCARU. The single request belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The history of the program is included in the description.

```

insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212

```

### Specify Argument Values to Pass to a Program in a Single Request vAE\_11.3.6\_SP4

You can pass argument values to a program in an Oracle E-Business Suite Single Request (OASG) job. The job can also use the default values that are defined by the registered Oracle Applications Concurrent Manager program. When an argument value is defined in both the job definition and as a default, the argument value in the job definition overrides the default.

#### Follow these steps:

1. [Define an Oracle E-Business Suite Single Request job.](#)
2. Add the following attribute to the job definition:
  - **oracle\_args**  
Defines the argument values to pass to an Oracle E-Business Suite single request.
3. (Optional) Add the following attribute:
  - **oracle\_use\_arg\_def**  
Specifies whether to use default values for arguments that are not defined using the oracle\_args attribute.
4. Run the job.  
The argument values are passed to the program in the single request.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Specify Argument Values for a Single Request

Suppose that you want to pass the argument values, T,DefArg2,X23,,DefArg5, to a single request program. The job uses the argument values that are specified in the job definition and the default values defined in Oracle E-Business Suite as follows:



- The first argument, T, and the third argument, X23, are specified in the job definition.
- The second argument, *DefArg2*, and the fifth argument, *DefArg5*, are defined as defaults in Oracle E-Business Suite.
- The fourth argument is not specified in the job definition or defined as a default on Oracle E-Business Suite, so the agent passes an empty string for that argument.

```
insert_job: oasg_args
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_program: FNDSCARU
oracle_use_arg_def: Y
oracle_args: T,,X23,,
```

## Oracle E-Business Suite Job Attributes with Default Values

This article explains Oracle E-Business Suite job attributes and default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Oracle E-Business Suite job attributes have default values:

- **oracle\_appl\_name\_type**  
Specifies whether the name of the Oracle E-Business Suite application is the short name or the display name.  
**Default:** SHORT
- **oracle\_desc (OASG jobs only)**  
Defines a description for the Oracle E-Business Suite Single Request job, which is displayed in the Oracle Concurrent Manager.  
**Default:** oa.default.desc agent parameter, if specified
- **oracle\_mon\_children**  
Specifies whether the children of the Oracle E-Business Suite programs are monitored.  
**Default:** N (The job does not monitor children programs.)
- **oracle\_output\_format**  
Specifies the output format for a single request or request set.  
**Default:** oa.default.outputFormat agent parameter, if specified
- **oracle\_print\_style**  
Specifies an Oracle Applications print style.  
**Default:** oa.default.printStyle agent parameter, if specified
- **oracle\_printer**  
Specifies the name of a printer to be used by Oracle Applications.  
**Default:** oa.default.printer agent parameter, if specified
- **oracle\_program\_name\_type (OASG jobs only)**  
Identifies whether the name of the Oracle E-Business Suite program is the short name or the display name.  
**Default:** SHORT
- **oracle\_quote\_in\_default**  
Specifies whether to quote resolved expressions in default values.

**Default:** N (The resolved expressions in default values are not quoted.)

- **oracle\_req\_set\_type (OASET jobs only)**

Identifies whether the name of the Oracle E-Business Suite request set is the short name or the display name.

**Default:** SHORT

- **oracle\_resp**

Specifies an Oracle Applications responsibility name. Specify an Oracle Applications user name using the `oracle_user` attribute or by setting the `oa.default.user` parameter in the `agentparm.txt` file.

**Default:** `oa.default.responsibility` agent parameter, if specified

**NOTE**

All Oracle Applications jobs require a responsibility name. If you do not specify this attribute in the job definition, a default responsibility name must be defined in the agent's `agentparm.txt` file using the `oa.default.responsibility` parameter. Otherwise, the job fails.

- **oracle\_save\_output (OASG and OASET jobs only)**

Specifies whether to save the output from an Oracle E-Business Suite Single Request or Request Set job.

**Default:** N (The job does not save the output.)

- **oracle\_template\_language**

Specifies the template language for a single request or request set.

**Default:** `oa.default.templateLanguage` agent parameter, if specified

- **oracle\_template\_territory**

Specifies the template territory for a single request or request set.

**Default:** `oa.default.templateTerritory` agent parameter, if specified

- **oracle\_use\_arg\_def (OASG and OASET jobs only)**

Specifies whether to use default values for arguments that not defined using the `oracle_args` attribute or the `oracle_programdata` attribute. The default arguments are defined in Oracle E-Business Suite.

**Default:** N (The job does not use the default values for the arguments.)

- **oracle\_use\_set\_defaults\_first (OASET job only)**

Specifies whether request set defaults take precedence over concurrent program defaults in Oracle Applications.

**Default:** N (The concurrent program defaults take precedence over request set defaults.) or `oa.default.useSetDefaultsFirst` agent parameter, if specified.

- **oracle\_user**

Specifies an Oracle Applications user name that the job runs under.

**Default:** `oa.default.user` agent parameter, if specified

**NOTE**

All Oracle Applications jobs require a user name. If you do not specify this attribute in the job definition, a default user must be defined in the agent's `agentparm.txt` file using the `oa.default.user` parameter. Otherwise, the job fails.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Override a Default Value in an OASET Job

This example overrides the default responsibility name using the `oracle_resp` attribute. The job also overrides the default user that the job runs under, using the `oracle_user` attribute.

```
insert_job: oaset_resp
```

```
job_type: oaset
```

```
machine: oaagent

oracle_appl_name_type: SHORT

oracle_appl_name: BIS

oracle_req_set: FNDRSSUB1310

oracle_user: SYSADMIN

oracle_resp: System Administrator
```

## PeopleSoft Jobs Overview

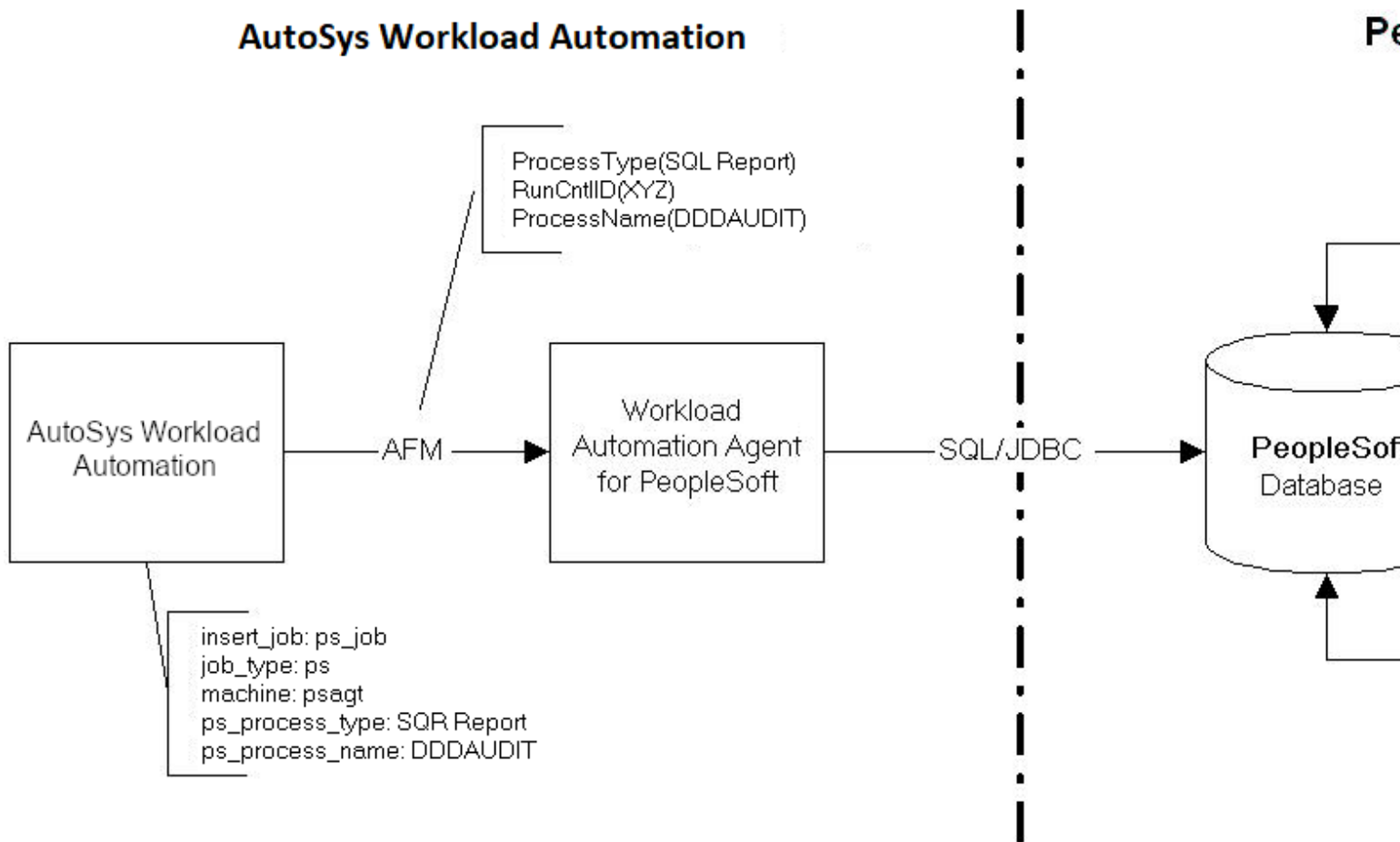
PeopleSoft jobs let you run different types of PeopleSoft processes defined in your PeopleSoft system. For example, you can define PeopleSoft jobs to execute PeopleSoft programs and report the program status.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for PeopleSoft.

When you define a PeopleSoft job, you can set the output type and format of a report. For email and web output types, you can set various distribution properties such as the recipients and message text. You can also pass run control parameter values that will be stored in the corresponding run control table.

When a PeopleSoft program runs, it modifies its run status (RUNSTATUS) in the PSPRCRQST table in the PS database. The following diagram shows the functional relationship between the scheduling manager, the agent, and the PeopleSoft system:



## PeopleSoft Exit Codes

A PeopleSoft (PS) job can return exit codes 9 and 17, which indicate success. When a PS job terminates with either of these exit codes, AutoSys Workload Automation changes the status of the job to SUCCESS and logs the non-zero exit code.

## PeopleSoft Operator ID and Password

The operator ID sets the authority for running PeopleSoft reports. Your agent administrator can set defaults for an operator ID and corresponding password using the `ps.default.oprId` and `ps.default.oprPassword` parameters in the `agentparm.txt`. You can override the default operator ID by specifying the `ps_operator_id` attribute in a PeopleSoft job definition. The `ps_operator_id` must be defined on AutoSys Workload Automation using the `autosys_secure` command unless the `ps.skipOprPswdValidation=true` parameter is configured in the `agentparm.txt` file.

## Define a PeopleSoft Job

You can define a PeopleSoft (PS) job to schedule workload to run in PeopleSoft. The job runs a PeopleSoft process request or a collection of process requests. In the job definition, you can set the output type and format of a report. For email and web output types, you can set various distribution properties such as the recipients and message text. You can add parameters to the run control table that contains the PeopleSoft run parameters.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for PeopleSoft.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: PS**  
Specifies that the job type is PeopleSoft.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **ps\_process\_name**  
Specifies the name of the PeopleSoft report to run. This value corresponds to the Process Name field in PeopleSoft.
  - **ps\_process\_type**  
Specifies the type of PeopleSoft report that you want the job to run.
2. Do *one* of the following:
  - Ensure that a default run control ID is defined in the agent's agentparm.txt file using the ps.default.runCntlId parameter.
  - Add the following attribute to the definition:
    - **ps\_run\_cntrl\_id**  
Specifies the value that is assigned to the run control identifier. This value corresponds to the Run Control ID field in PeopleSoft.

**NOTE**

This attribute overrides the ps.default.runCntlId agent parameter.

3. (Optional) Specify optional PeopleSoft attributes:

- **envvars**
- **job\_class**
- **job\_terminator**
- **ps\_args**
- **ps\_dest\_format**
- **ps\_dest\_type**
- **ps\_detail\_folder**
- **ps\_dlist\_roles**
- **ps\_dlist\_users**
- **ps\_email\_address**
- **ps\_email\_address\_expanded**
- **ps\_email\_log**
- **ps\_email\_subject**
- **ps\_email\_text**
- **ps\_email\_web\_report**
- **ps\_operator\_id**
- **ps\_output\_dest**
- **ps\_restarts**
- **ps\_run\_cntrl\_args**
- **ps\_run\_control\_table**
- **ps\_server\_name**
- **ps\_skip\_parm\_updates**
- **ps\_time\_zone**

**NOTE**

A PeopleSoft job must have a valid operator ID to run successfully. Check with your agent administrator to determine whether a default operator ID is set on the agent. If a default is not set, specify this attribute.

4. (Optional) Specify common attributes that apply to all jobs.  
The PeopleSoft job is defined.

**NOTE**

- All PeopleSoft jobs require a run control ID. Define this value; otherwise, the job fails. Your agent administrator can specify a default run control ID using the `ps.default.runCtrlId` parameter on the agent, or you can specify the `ps_run_cntrl_id` attribute in the job definition.
- All PeopleSoft jobs require a valid operator ID. The operator ID and corresponding password can be defined on the agent using the `ps.default.oprId` and `ps.default.oprPassword` parameters in the `agentparm.txt` file. You can specify the operator ID in a job definition using the `ps_operator_id` attribute.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Run a PeopleSoft Process**

This example runs an Application Engine process named DDDAUDIT. The job runs on the agent named `psagt`.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
ps_process_name: DDDAUDIT
ps_process_type: Application Engine
```

**NOTE**

The job uses the default operator ID, output destination format, output destination type, and run control ID defined on the agent. The PeopleSoft Server that runs the job is not defined in the job definition or as a default on the agent. Therefore, the PeopleSoft Process Scheduler determines the PeopleSoft Server that runs the job.

**Example: Run a PeopleSoft Request**

This example runs an SQR Report. The report is formatted as PDF and sent to a file.

```
insert_job: ps_file
job_type: ps
machine: psagent
ps_process_name: XRFWIN
ps_process_type: SQR Report
ps_dest_type: FILE
```

```

ps_dest_format: PDF
ps_output_dest: /export/home/PSoutput/report1.pdf
ps_run_cntrl_id: test
owner: VP1@ps1

```

## PeopleSoft Job Attributes with Default Values

This article explains PeopleSoft job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following PeopleSoft job attributes have default values:

- **ps\_dest\_format**  
Specifies the type of format for the report output.  
**Default:** ps.default.outDestFormat agent parameter, if specified
- **ps\_dest\_type**  
Specifies the output destination type for the PeopleSoft report.  
**Default:** ps.default.outDestType agent parameter, if specified
- **ps\_email\_log**  
Specifies whether to email job logs with the PeopleSoft report to recipients on a distribution list.  
**Default:** No (Job logs are not emailed to recipients.)
- **ps\_email\_web\_report**  
Specifies whether to email a web report to recipients on a distribution list.  
**Default:** No (A web report is not emailed to recipients.)
- **ps\_operator\_id**  
Specifies the operator ID under whose authority the PeopleSoft reports run.  
**Default:** ps.default.oprld agent parameter, if specified
- **ps\_output\_dest**  
Specifies the output destination for the PeopleSoft request. The destination can be a file directory or a printer.  
**Default:** If ps\_dest\_type is PRINTER, the default is one of the following values, in the following order:
  - ps.default.printer parameter in the agent's agentparm.txt file, if specified
  - The default PeopleSoft printer, lpt1
- **ps\_restarts**  
Specifies whether to disable a restart feature for previously failed jobs from the point where the job failed.  
**Default:** No (The restart feature is not disabled.)
- **ps\_run\_cntrl\_id**  
Specifies a set of PeopleSoft run parameters for a given PeopleSoft process.  
**Default:** ps.default.runCntlId agent parameter, if specified

### NOTE

All PeopleSoft jobs require a run control ID. If you do not specify this attribute in the job definition, a default run control ID must be defined in the ps.default.runCntlId parameter in the agent's agentparm.txt file. Otherwise, the job fails.

- **ps\_server\_name**  
Specifies the target server that runs the PeopleSoft job.  
**Default:** ps.default.serverName

**NOTE**

If the PeopleSoft Server is not specified as a default on the agent or in the job definition, the PeopleSoft Process Scheduler determines the PeopleSoft Server that runs the job.

- **ps\_skip\_parm\_updates**

Specifies whether you want the agent to update job parameters with data in the PS\_PRCSEFN table.

**Default:** NO (The agent updates job parameters with data in the PS\_PRCSEFN table.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Specify a Run Control ID and a Server Name**

This example overrides the default run control ID using the ps\_run\_cntrl\_id attribute and server name using the ps\_server\_name attribute. The job uses the Application Engine process type. The job uses the default output destination format and type that is defined on the agent. The job overrides the default operator ID using the ps\_operator\_id attribute.

```
insert_job: ps_cst
job_type: ps
machine: psagt
ps_process_name: PAYROLL
ps_server_name: PSPR
ps_process_type: Application Engine
ps_run_cntrl_id: PS_ALL
ps_operator_id: vp1@ps1
```

**PeopleSoft Fields to Job Attributes Mapping**

When you define a PeopleSoft job, you specify JIL attributes that map to your PeopleSoft process request. The following table maps the PeopleSoft fields to the attributes:

PeopleSoft Field Name	Attribute Name
Format	ps_dest_format
Type	ps_dest_type
Folder Name	ps_detail_folder
ID Type (Role selected) Distribution ID	ps_dlist_roles
ID Type (User selected) Distribution ID	ps_dlist_users
Email Address List	ps_email_address
Email With Log	ps_email_log
Email Subject	ps_email_subject
Message Text	ps_email_text
Email Web Report	ps_email_web_report
Output Destination	ps_output_dest
Process Name	ps_process_name
Process Type	ps_process_type
Run Control Arguments	ps_run_cntrl_args



Run Control ID	ps_run_cntrl_id
Run Control Table	ps_run_control_table
Server Name	ps_server_name
Time Zone	ps_time_zone

## Distribute a PeopleSoft Report

If you specify EMAIL as the output destination type (ps\_dest\_type), you can distribute a PeopleSoft report electronically to operators, groups of people, or individuals.

### Follow these steps:

1. [Define a PeopleSoft job.](#)
2. Add the following attributes to the job definition:
  - **ps\_dest\_type: EMAIL**  
Sends the output of the PeopleSoft report as an email message.  
  
This attribute corresponds to the Type field in PeopleSoft.
  - **ps\_dest\_format**  
  
Specifies the field name of the output destination format. PeopleSoft stores the list of output destination formats in the PSXLATITEM table. This value corresponds to the Format field in PeopleSoft.
3. Add *one or both* of the following attributes:
  - **ps\_dlist\_roles**  
Specifies a distribution list of the roles that represent the individuals who are receiving the PeopleSoft report. This value corresponds to the ID Type field (with Role selected) and the Distribution ID field in PeopleSoft.
  - **ps\_dlist\_users**  
Specifies a distribution list of operator IDs to send a PeopleSoft report to. This value corresponds to the ID Type field (with User selected) and the Distribution ID field in PeopleSoft.
4. (Optional) Add the following attributes:
  - **ps\_email\_address**  
Specifies the email addresses of the recipients on a distribution list. This value corresponds to the Email Address List field in PeopleSoft.
  - **ps\_email\_subject**  
Defines an email subject to include in the email. This value corresponds to the Email Subject field in PeopleSoft.
  - **ps\_email\_text**  
Defines the body text of the email. This value corresponds to the Message Text field in PeopleSoft.
5. Run the job.  
The PeopleSoft report is sent to the specified distribution lists and email addresses.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Distribute a Report to Users

This example runs a Crystal report under the VP3 operator ID. The report is formatted as PDF and distributed in an email to the VP1, VP2, and VP3 operator IDs.

```
insert_job: ps_users
```

```
job_type: PS
```

```
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_dlist_users: VP1,VP2,VP3
ps_operator_id: VP3@ps1
ps_run_cntrl_id: test
```

**Example: Email a PeopleSoft Report**

This example runs a Crystal report and emails the output to recipients. The Crystal report runs under the VP2 operator ID. The output is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title.

```
insert_job: ps_email
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_operator_id: VP2@ps1
ps_email_address: user1@example.com;user2@example.com

ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.
```

## Store PeopleSoft Job Output as a Web Report

You can define a PeopleSoft job to run a process and store the output as a web report to view later. You can also define the job to email the web report to one or more recipients.

### Follow these steps:

1. [Define a PeopleSoft job.](#)
2. Add the following attributes to the job definition:
  - **ps\_dest\_type: WEB**  
Posts the output of the PeopleSoft report on a website.  
This attribute corresponds to the Type field in PeopleSoft.
  - **ps\_dest\_format**  
Specifies the field name of the output destination format. PeopleSoft stores the list of output destination formats in the PSXLATITEM table. This value corresponds to the Format field in PeopleSoft.
3. (Optional) Add the following attributes:
  - **ps\_email\_web\_report: YES**  
Specifies that the job emails a web report to the recipients on the distribution list. This attribute corresponds to the Email Web Report field in PeopleSoft.
  - **ps\_email\_address**  
Specifies the email addresses of the recipients on a distribution list. This value corresponds to the Email Address List field in PeopleSoft.
  - **ps\_email\_subject**  
Defines an email subject to include in the email. This value corresponds to the Email Subject field in PeopleSoft.
  - **ps\_email\_text**  
Defines the body text of the email. This value corresponds to the Message Text field in PeopleSoft.
4. Run the job.  
The output of the PeopleSoft job is stored as a web report. The job emails the web report to recipients if the email attributes are specified.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Format a PeopleSoft Job Output as an HTML Web Report

This example runs the XRFWIN process. The process type is SQR Report and the run control ID is PS\_ALL. The server that is named PSPR runs the job, and the output is stored as an HTML web report.

```
insert_job: ps_htmfile
job_type: ps
machine: psagt
ps_process_name: XRFWIN
ps_process_type: SQR Report
ps_server_name: PSPR
ps_dest_type: WEB
ps_dest_format: HTM
ps_run_cntrl_id: PS_ALL
```

### Example: Distribute a PeopleSoft Web Report in an Email

This example stores the output as a PDF web report. The web report is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title and body text.

```
insert_job: ps_web

job_type: PS

machine: psagt

ps_process_name: XRFWIN

ps_process_type: Crystal

ps_dest_type: WEB

ps_dest_format: PDF

ps_operator_id: VP2@ps1

ps_email_web_report: YES

ps_email_address: user1@example.com;user2@example.com

ps_email_subject: PeopleSoft Report Status

ps_email_text: This report is available for distribution.
```

## Send PeopleSoft Job Output to a Printer

You can define a PeopleSoft job to run a process and send the output to a printer.

### Follow these steps:

1. [Define a PeopleSoft job.](#)
2. Add the following attributes to the job definition:
  - **ps\_dest\_type: PRINTER**  
Sends the output of the PeopleSoft report to a printer.  
This attribute corresponds to the Type field in PeopleSoft.
  - **ps\_output\_dest**  
Specifies the network location of the printer including the printer server and shared printer name. This value corresponds to the Output Destination field in PeopleSoft.
3. Run the job.  
The output of the PeopleSoft job is sent to a printer.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Send a Job's Output to a Specified Printer

This example runs an SQR Report. The report is formatted as PS and outputted to a printer.

```
insert_job: ps_printer

job_type: ps

machine: psagent

ps_process_name: XRFWIN

ps_process_type: SQR Report

ps_dest_type: PRINTER

ps_dest_format: PS

ps_output_dest: \\printers\PRINTER1

ps_run_cntrl_id: test

ps_operator_id: VP1@ps1
```

## Process Automation Jobs

Process Automation jobs let you launch CA Process Automation processes and monitor them to completion.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

You can define the following Process Automation jobs:

- **Process Automation Process Execution**  
Lets you directly execute a CA Process Automation process.
- **Process Automation Start Request Form**  
Lets you submit a Start Request Form to execute a CA Process Automation process. The Start Request Form sets the values for the current execution of the process.

## Define a Process Automation Process Execution Job

You can define a Process Automation Process Execution (PAPROC) job to execute a CA Process Automation process.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: PAPROC**

- Specifies that the job type is Process Automation Process Execution.
  - **machine** Specifies the name of the machine on which the job runs.
  - **endpoint\_URL**  
Specifies the target endpoint address URL in a Web Service or Process Automation job.
  - **pa\_name**  
Specifies the fully qualified name of the process or Start Request Form in a Process Automation job.
2. (Optional) Specify optional Process Automation Process Execution attributes.
    - **job\_class**
    - **job\_terminator**
    - **owner**
    - **pa\_monitor\_progress**
    - **pa\_parameter**
    - **pa\_trace**
  3. (Optional) Specify common attributes that apply to all jobs.  
The Process Automation Process Execution job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Execute a CA Process Automation Process

This example executes a CA Process Automation process that is named /CAWAAE/Test Process using the agent on the localhost machine and connecting to a CA Process Automation installation on capamach. It uses the credentials from pamadmin@capamach to authenticate on the Process Automation installation and passes values for the process variable named strTestName. It monitors the process until the process completes and report success or failure.

```
insert_job: execproc
job_type: PAPROC
machine: localhost
endpoint_URL: "http://capamach:8080/itpam/soap"
pa_name: "/CAWAAE/Test Process"
owner: pamadmin@capamach
pa_monitor_progress: Y
pa_parameter: param_name="strTestName", param_value="PA Test"
```

## Define a Process Automation Start Request Form Job

You can define a Process Automation Start Request Form (PAREQ) job to submit a Start Request Form to execute a CA Process Automation process. The Start Request Form sets the values for the current execution of the process.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: PAREQ**  
Specifies that the job type is Process Automation Start Request Form.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **endpoint\_URL**  
Specifies the target endpoint address URL in a Web Service or Process Automation job.
  - **pa\_name**  
Specifies the fully qualified name of the process or Start Request Form in a Process Automation job.
  - **pa\_path**  
Specifies the path of the Start Request Form.

### NOTE

The path specified must end in a / character.

2. (Optional) Specify optional Process Automation Start Request Form attributes.
  - **job\_class**
  - **owner**
  - **pa\_monitor\_progress**
  - **pa\_parameter**
  - **pa\_trace**
3. (Optional) Specify common attributes that apply to all jobs.  
The Process Automation Start Request Form job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Submit a Start Request Form

This example submits a Start Request Form that is named Process A in the /CAWAE/ path. It uses the credentials of the pamadmin@capamach user to authenticate the user on the Process Automation installation on capamach. The Start Request Form has a field that is named strTestName that is populated with the value "PA Test" before the form is submitted. The process is tracked to completion.

```
insert_job: runsrf
job_type: PAREQ
machine: localhost
endpoint_URL: "http://capamach:8080/itpam/soap"
pa_name: "Process A"
pa_path: /CAWAE/
owner: pamadmin@capamach
pa_monitor_progress: Y
pa_parameter: param_name="strTestName", param_value="PA Test"
```

### Example: Specify the Path to the Start Request Form

This example specifies the Start Request Form located at /ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/. It is being passed values for three parameters that are defined in that form, and it continues to monitor the progress of the job until the job completes.

```
insert_job: srfl
job_type: PAREQ
endpoint_URL: "http://paserver:8080/itpam/soap"
machine: paagent
owner: pamadmin@paserver
pa_path: "/ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/"
pa_name: "Start Request Form"
pa_parameter: Name="AlarmProcessingProduct", Value="Test Alarm Product"
pa_parameter: Name="Server", Value="paserver"
pa_parameter: Name="Port", Value=80
pa_monitor_progress: Y
pa_trace: N
```

## Process Automation Job Attributes with Default Values

This article explains Process Automation job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Process Automation job attributes have default values:

- **owner**  
Specifies the Process Automation user. These credentials will be validated on the Process Automation server.  
**Default:** The user who invokes JIL to define the job
- **pa\_monitor\_progress**  
Indicates whether the process is tracked to completion.  
**Default:** Y; the process is tracked to completion.
- **pa\_trace**  
Indicates whether a trace of SOAP messages is written to the spool file.



**Default:** N (A trace of SOAP messages is not written to the spool file.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Track a Process Automation Process to Completion**

The `pa_monitor_progress` attribute in the following job definition overrides the default value.

```
insert_job: pam1

job_type: PAPROC

endpoint_url: "http://canor01-vm2k3c:8080/itpam/soap"

machine: paagent

owner: pamadmin@capamach

pa_name: "/ITPAM Tutorials/S01 Getting started/L01 Creating a Simple Process/
L01-2 Running the Process/Running the Process"

pa_monitor_progress: Y

pa_trace: N
```

## Remote Execution Jobs Overview

Remote Execution jobs let you execute commands to a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Secure Shell (SSH2) or Telnet.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Remote Execution. The Workload Automation Agent for Remote Execution supports only remote target environments of UNIX, HP Integrity NonStop, and OpenVMS.

## Define a Remote Execution Job

You can define a Remote Execution (PROXY) job to run commands to a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Telnet or Secure Shell (SSH2).

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Remote Execution. The Workload Automation Agent for Remote Execution supports only remote target environments of UNIX, HP Integrity NonStop, and OpenVMS.

With the Workload Automation Agent for Remote Execution, you can define and run remote execution jobs.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type:** [PROXY](#)

Specifies that the job type is Remote Execution.

- **machine**

Specifies the name of the machine on which the job runs.

- **remote\_command**

Specifies a command or script to run on a remote computer.

- **remote\_target**

Specifies the name of the custom properties (remote\_target.properties) file that is created on the agent for the remote system. The file contains the default user credentials that are used to monitor all jobs run on the target machine by the agent. The name should not include the .properties extension.

2. (Optional) Specify optional Remote Execution attributes:

- **envvars**
- **fail\_codes**
- **job\_class**
- **spool\_file**
- **submit\_modifier**
- **success\_codes**

3. (Optional) Specify the following attribute:

- **owner** Specifies a valid user ID that is used to run the job on the target machine, for example, user@host. The credentials for this user ID must be valid on the remote target machine. Define this user ID and its password using the autosys\_secure command.

4. (Optional) Specify common attributes that apply to all jobs.

The Remote Execution job is defined.

#### NOTE

- To run Remote Execution jobs, provide the following user credentials:
  - A valid user ID that is used to run the job on the target machine, for example, user@host. The credentials for this ID must be valid on the target machine. The owner attribute in the job definition specifies this user ID.
  - A valid default user credential specified in the custom properties file for the target machine (in the target.properties file). This is required as a monitoring ID but is not used for submitting jobs. This credential is used to monitor all jobs run by the agent on the target machine. The default user ID for job submission is the user who invokes jil to define the job.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes are defined on the agent in the custom properties file for the remote system. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Execute the list command on a remote machine

Suppose that you want to list the contents of /opt on the remote agent machine.

```
insert_job: proxy_job
job_type: PROXY
machine: agentmachine
owner: root@remoteagent
remote_target: remoteagent
spool_file: /opt/spool/commandout.out
remote_command: ls /opt
```

### Example: Run a Command on a Remote System

This example defines a Remote Execution job.

```
insert_job: ls_cmd
job_type: PROXY
machine: proxyagent
remote_target: hpserver
remote_command: ls
spool_file: >>/opt/rootdev/CA_USER/test/ls.out
owner: user1@proxymtest
```

## Remote Execution Job Attributes with Default Values

This article explains Remote Execution job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Default values for some attributes can be defined on the agent in the custom properties file for the remote system. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

If you specify the attribute in a job definition, it overrides the default.

The following Remote Execution job attributes have default values:

- **fail\_codes**  
Defines which exit codes indicate job failure.  
**Default:** Any exit code other than 0 (The job interprets any code other than zero as failure.)
- **spool\_file**  
Specifies the path to the spool file.  
**Default:** spoolHome custom property
- **owner**  
Specifies a valid user ID that is used to run the job on the target machine, for example, user@host. The credentials for this user ID must be valid on the remote target machine. Define this user ID and its password using the autosys\_secure command.
- **success\_codes**  
Defines which exit codes indicate job success.  
**Default:** 0 (The job interprets zero as success.)

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Override the Default Path to the Spool File

This example overrides the default path to the spool file specified in the custom properties file for the remote system.

```
insert_job: ls_cmd

job_type: PROXY

machine: proxyagent

remote_target: hpserver

remote_command: ls

spool_file:>>/opt/rootdev/CA_USER/test/ls.out
owner: user1@hpserver
```

## SAP Jobs

SAP jobs let you run SAP workload.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

You can define the following SAP jobs:

- **SAP Batch Input Session**  
Imports data from external systems to the SAP system.
- **SAP Business Warehouse (BW) InfoPackage**  
Transfers data from a data source to an SAP Business Warehouse system.
- **SAP Business Warehouse (BW) Process Chain**  
Creates Process Chains on the SAP system.
- **SAP Data Archiving**  
Stores information in an SAP Archiving Object.
- **SAP Event Monitor**  
Monitors and triggers SAP events.
- **SAP Job Copy**  
Copies an existing SAP R/3 job.
- **SAP Process Monitor**  
Monitors for a specific SAP process status.
- **SAP R/3**  
Schedules an SAP R/3 job on your SAP system.

## SAP Connection Attributes

The following connection attributes are common to SAP job types and may be required in your job definitions:

- **sap\_rfc\_dest**  
Specifies the SAP system to connect to. This value corresponds to the `destination@properties` agent configuration file name containing the SAP connection information. The `sap_rfc_dest` value overrides the `sap.default.destination` configuration value in the `agentparm.txt` file.
- **sap\_client**  
Specifies the client number that is associated with an SAP instance. An SAP instance can have multiple clients that are defined for it. Each client has its own data. Your agent administrator can define a default SAP client number using the `jco.client.client` property in the connection properties file.
- **sap\_lang**  
Specifies the language that is required to run a job. Your agent administrator can define a default language using the `jco.client.lang` property in the connection properties file.
- **sap\_target\_sys**  
Specifies the host name of an SAP application server where the job is to run.  
**Note:** If the `sap_target_sys` attribute is not defined, the SAP system selects an application server based on available resources.

**NOTE**

- If a default value is not defined on the agent, specify the attribute in the job definition.
- For more information about specifying attributes in a job definition, see [Define Jobs](#).

## SAP User IDs and Passwords

When you define an SAP job, specify the SAP user ID using the `owner` attribute. The job runs under this user ID.

If you defined the default SAP user ID in the `destname.properties` file while configuring the Workload Automation Agent for SAP and you specified the same user ID as the `owner` attribute value, you can run the SAP jobs using this default SAP user ID. You do not need to define this default SAP user ID on AutoSys Workload Automation using the `autosys_secure` command. However, if you want to run the SAP jobs using a different user ID other than the default SAP User ID, you must define this user ID and password on AutoSys Workload Automation by using the `autosys_secure` command.

**NOTE**

For more information, see [Configuring Workload Automation Agent for SAP](#).

## Define a SAP Batch Input Session Job

You can define an SAP Batch Input Session (SAPBDC) job to import large amounts of data from external systems to the SAP system.

To schedule a BDC job, you first define an ABAP that creates a Batch Input Session (BDC ABAP) on the SAP system. Next, you schedule an SAP Batch Input Session job in AutoSys Workload Automation to run the BDC ABAP on the SAP system. After the job runs, the BDC job starts the data transfer.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAPBDC**  
Specifies that the job type is SAP Batch Input Session.
  - **machine**

Specifies the name of the machine on which the job runs.

- **sap\_job\_name**  
Specifies the name of the ABAP job that creates the BDC session on the SAP system.
- **sap\_step\_parms**  
Defines the SAP R/3 step specifications.

#### NOTE

We recommend that you limit the number of steps (ABAPs) to one per job. If you run a job and one of the ABAPs fails, the job is marked as failed. If the ABAP fails, you cannot re-run the ABAP without re-running the entire job.

#### 2. (Optional) Specify optional SAP Batch Input Session attributes:

- **bdc\_err\_rate**
- **bdc\_ext\_log**
- **bdc\_proc\_rate**
- **bdc\_system**
- **job\_class**
- **job\_terminator**
- **sap\_client**
- **sap\_job\_class**
- **sap\_lang**
- **sap\_office**
- **sap\_recipients**
- **sap\_release\_option**
- **sap\_rfc\_dest**
- **sap\_step\_parms**
- **sap\_target\_sys**

#### 3. (Optional) Specify common attributes that apply to all job types.

The SAP Batch Input Session job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Define an SAP Batch Input Session Job

This example runs the ZBDCTEST ABAP that creates a Batch Input Session (BDC ABAP) on the default SAP system that is defined to the agent. The job runs as soon as possible after the job is defined. After this job runs, the BDC job starts the data transfer.

```

insert_job: bdcjob
job_type: SAPBDC
machine: sapagent
owner: WAAESAP@sapagent
sap_job_name: ZBDCTEST
sap_release_option: A
sap_step_parms: abap_name="ZBDCTEST"

```

### Example: Define an SAP Batch Input Session Job with Step Parameters

This example runs the SAP Batch Input Session job named ZBDCTEST. When SAPBDC\_job runs, it releases ZBDCTEST immediately on the SAP system. If no free background processing is available, the ZBDCTEST is not released and stays in the Scheduled SAP job state.

```

insert_job: SAPBDC_job
job_type: SAPBDC
machine: localhost
owner: WAAESAP@sapserver
sap_job_name: ZBDCTEST
sap_release_option: I
sap_step_parms: abap_name="ZBDCTEST", banner_page=no, release=no, print_imm=no,
new_spool=no, footer=no

```

### Define a SAP BW InfoPackage Job

You can define an SAP BW InfoPackage (SAPBWIP) job to transfer data from any data source into an SAP Business Warehouse system. When the job runs, the data is transferred.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAPBWIP**  
Specifies that the job type is SAP BW InfoPackage.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **sap\_info\_pack**  
Specifies the name of the Business Warehouse InfoPackage.
2. (Optional) Specify optional SAP BW InfoPackage attributes:
  - **job\_class**
  - **job\_terminator**
  - **sap\_client**
  - **sap\_ext\_table**
  - **sap\_job\_name**
  - **sap\_lang**
  - **sap\_rfc\_dest**
3. (Optional) Specify common attributes that apply to all job types.

The SAP BW InfoPackage job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Define an SAP BW InfoPackage Job

This example runs the Business Warehouse InfoPackage 0PAK\_D2XZMZ1HD5WFL3EN1NVIT4V at the SAP destination, SM1.

```
insert_job: InfoBW2
job_type: SAPBWIP
machine: sapagent
owner: WAAESAP@sapagent
sap_job_name: InfoBW2
sap_rfc_dest: SM1
sap_info_pack: 0PAK_D2XZMZ1HD5WFL3EN1NVIT4V
```

#### Example: Define an SAP BW InfoPackage Job

This example runs the Business Warehouse InfoPackage ZPAK\_DIBX41NKYK0S7FB1FOV0FQ7BV on the SAP system.

```
insert_job: bwip_job
job_type: SAPBWIP
machine: localhost
owner: user@sapserver
sap_job_name: BI_BTCHSAP_TEST
sap_rfc_dest: mts
sap_info_pack: ZPAK_DIBX41NKYK0S7FB1FOV0FQ7BV
sap_lang: EN
sap_client: 001
job_class: a
```

#### Define a SAP BW Process Chain Job

You can define an SAP BW Process Chain (SAPBWPC) job to run a sequence of background processes on the SAP system. Some SAP processes trigger events that can start other processes. An SAPBWPC job runs the individual processes in the chain as job steps.



**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAPBWPC**  
Specifies that the job type is SAP BW Process Chain.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **sap\_chain\_id**  
Specifies the name of the Business Warehouse Process Chain.
2. (Optional) Specify optional SAP BW Process Chain attributes:
  - **job\_class**
  - **job\_terminator**
  - **sap\_client**
  - **sap\_lang**
  - **sap\_rfc\_dest**
3. (Optional) Specify common attributes that apply to all job types.  
The SAP BW Process Chain job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Define an SAP BW Process Chain Job**

This example defines the SAPBWPC3 job that runs the DEMO\_CHAIN1 Process Chain on the SAP system. The language that is used to log in to the SAP system is English.

```
insert_job: SAPBWPC3
job_type: SAPBWPC
machine: sapagent
owner: WAAESAP@sapagent
sap_chain_id: DEMO_CHAIN1
sap_lang: EN
sap_rfc_dest: SM1
```

## Define a SAP Data Archiving Job

You can define an SAP Data Archiving (SAPDA) job to store information that is described in an SAP Archiving Object into an SAP data archive.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAPDA**  
Specifies that the job type is SAP Data Archiving.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **arc\_obj\_name**  
Specifies the name of the archiving object.
  - **arc\_obj\_variant**  
Specifies the name of the archiving object variant.
2. (Optional) Specify optional SAP Data Archiving attributes:
  - **arc\_parms**
  - **job\_class**
  - **job\_terminator**
  - **sap\_client**
  - **sap\_lang**
  - **sap\_print\_parms**
  - **sap\_rfc\_dest**
  - **sap\_target\_sys**
3. (Optional) Specify common attributes that apply to all job types.  
The SAP Data Archiving job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Define an SAP Data Archiving Job

This example defines a job that stores information that is described in the BC\_ARCHIVE Archiving Object into an SAP data archive. The archiving object variant is BC\_ARCVARIANT.

```
insert_job: SAPDA_job
job_type: SAPDA
machine: sapagent
owner: WAAESAP@sapagent
arc_obj_name: BC_ARCHIVE
arc_obj_variant: BC_ARCVARIANT
sap_print_parms: dest=LP01,prt_arc_mode=PRINT
```

## Define a SAP Event Monitor Job

You can define an SAP Event Monitor (SAPEVT) job to schedule workload based on the activity of an SAP event or trigger an SAP event at the appropriate time in your schedule.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:

- **job\_type: SAPEVT**  
Specifies that the job type is SAP Event Monitor.
- **machine**  
Specifies the name of the machine on which the job runs.
- **sap\_event\_id**  
Specifies the name of the SAP event to monitor or trigger.

2. (Optional) Specify optional SAP Event Monitor attributes:

- **continuous**
- **job\_class**
- **job\_terminator**
- **sap\_client**
- **sap\_event\_parm**
- **sap\_is\_trigger**
- **sap\_lang**
- **sap\_rfc\_dest**

3. (Optional) Specify common attributes that apply to all job types.  
The SAP Event Monitor job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Trigger an SAP Event

This example defines an SAPEVT job. The job triggers the SAP\_TEST event and the SAP job dependencies waiting on the SAP\_TEST event are satisfied.

```
insert_job: SAPEVT_job
job_type: SAPEVT
machine: sapagent
owner: WAAESAP@sapagent
sap_event_id: SAP_TEST
sap_rfc_dest: BI1
sap_is_trigger: Y
```

### Example: Monitor an SAP Event

This example defines an SAPEVT job. The sap\_is\_trigger attribute is set to N, so the job monitors the SAP\_TEST event and remains in RUNNING status.

```
insert_job: trigger_evt
job_type: SAPEVT
machine: localhost
owner: WAAESAP@sapserver
sap_rfc_dest: BI1
sap_event_id: SAP_TEST
sap_is_trigger: N
sap_event_parm: L L
sap_client: 001
sap_lang: EN
continuous: y
```

## Define a SAP Process Monitor Job

You can define an SAP Process Monitor (SAPPM) job to monitor for a specific SAP process status and end after detecting a process. You can also use SAP Process Monitor jobs to set up predecessor or dependent job relationships with other jobs or SAP processes.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAPPM**

- Specifies that the job type is SAP Process Monitor.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **sap\_process\_status**  
Specifies the SAP process status to monitor (RUNNING, STOPPED, or WAITING).
2. (Optional) Specify the following optional attributes if `sap_process_status` is set to RUNNING or STOPPED:
    - **sap\_abap\_name**
    - **sap\_proc\_user**
    - **sap\_process\_client**
  3. (Optional) Specify optional SAP Process Monitor attributes:
    - **continuous**
    - **job\_class**
    - **job\_terminator**
    - **sap\_client**
    - **sap\_lang**
    - **sap\_proc\_type**
    - **sap\_rfc\_dest**
    - **sap\_target\_sys**
  4. (Optional) Specify common attributes that apply to all job types.  
The SAP Process Monitor job is defined.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Define an SAP Process Monitor Job

This example defines an SAP Process Monitor job that monitors for the ABAP program ZMYABAP to change to the RUNNING status.

```
insert_job: test_SAPPM
job_type: SAPPM
machine: sapagent
sap_abap_name: ZMYABAP
sap_process_status: RUNNING
sap_rfc_dest: BI1
owner: WAAESAP@sapagent
```

## Define a SAP Job Copy Job

You can define an SAP Job Copy (SAPJC) job to copy an existing SAP R/3 job.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:

- **job\_type: SAPJC**  
Specifies that the job type is SAP Job Copy.
- **machine**  
Specifies the name of the machine on which the job runs.
- **sap\_job\_name**  
Specifies the name of the SAP R/3 job to be copied.

2. (Optional) Specify optional SAP Job Copy attributes:

- **job\_class**
- **job\_terminator**
- **sap\_client**
- **sap\_fail\_msg**
- **sap\_job\_count**
- **sap\_lang**
- **sap\_mon\_child**
- **sap\_release\_option**
- **sap\_rfc\_dest**
- **sap\_step\_num**
- **sap\_success\_msg**
- **sap\_target\_jobname**
- **sap\_target\_sys**

3. (Optional) Specify common attributes that apply to all job types.  
The SAP Job Copy job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Define an SAP Job Copy Job

This example defines an SAP Job Copy job that copies the AM job with job count 113315AM and runs the new copy.

```
insert_job: SAPJC_job
job_type: SAPJC
owner: WAAESAP@sapagent
machine: sapagent
sap_job_name: AM
sap_job_count: 113315AM
```

## Define a SAP R/3 Job

You can define an SAP R/3 job (SAP) to schedule an SAP R/3 job on your SAP system.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SAP**  
Specifies that the job type is SAP R/3.
  - **machine** Specifies the name of the machine on which the job runs.
  - **sap\_job\_name**  
Specifies the SAP job name that identifies the workload in the SAP system.
  - **sap\_step\_parms**  
Defines the SAP R/3 step specifications.

### NOTE

We recommend that you limit the number of steps (ABAPs) to one per job. If you run a job and one of the ABAPs fails, the job is marked as failed. If the ABAP fails, you cannot re-run the ABAP without re-running the entire job.

2. (Optional) Specify optional SAP R/3 attributes:
  - **job\_class**
  - **job\_terminator**
  - **sap\_client**
  - **sap\_fail\_msg**
  - **sap\_job\_class**
  - **sap\_lang**
  - **sap\_mon\_child**
  - **sap\_office**
  - **sap\_recipients**
  - **sap\_release\_option**
  - **sap\_rfc\_dest**
  - **sap\_success\_msg**
  - **sap\_target\_sys**
3. (Optional) Specify common attributes that apply to all job types.  
The SAP R/3 job is defined.

**NOTE**

- We recommend that you limit the number of steps (ABAPs) to one per job. If you run a job and one of the ABAPs fails, the job is marked as failed. If the ABAP fails, you cannot re-run the ABAP without re-running the entire job.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values that are specified in attribute statements within a job definition override the default values.

**Example: Define an SAP R/3 Job With One Step**

This example runs the SAP R/3 job named BI\_WRITE\_PROT\_TO\_APPLLOG. The job runs the SAP program (ABAP) named RSBATCH\_WRITE\_PROT\_TO\_APPLLOG. The owner is defined in AutoSys Workload Automation using the autosys\_secure command.

```
insert_job: test_SAP
job_type: SAP
machine: sapagent
sap_job_name: BI_WRITE_PROT_TO_APPLLOG
sap_rfc_dest: BI1
sap_step_parms: abap_lang=EN,abap_name=RSBATCH_WRITE_PROT_TO_APPLLOG
owner: WAAESAP@sapagent
```

**Example: Define an SAP R/3 Job With Multiple Steps**

This example runs an SAP R/3 job that is named SAP 3 STEPS\_@#\$. The job runs three steps. Each step runs the same ABAP but with different parameters.

```
insert_job: TEST_1
job_type: SAP
machine: localhost
job_class: c
sap_job_name: "SAP 3 STEPS_@# $"
sap_release_option: I
sap_step_parms: abap_name=BTCTEST,variant=test,arc_printer=LP01,copies=2,
prt_arc_mode=BOTH,arc_obj_type=ARCHIVE,arc_doc_type=ARCHIVE,arc_info=inf,num_lines=65,
num_columns=80
sap_step_parms: abap_name=BTCTEST,variant=test2,arc_printer=LP01,copies=3,
prt_arc_mode=BOTH,print_imm=N,release=N,sap_banner=Y,banner_page=Y,
recipient_name=cyber,num_lines=65,num_columns=80,arc_obj_type=ARCHIVE,
arc_doc_type=ARCHIVE,arc_info=inf
sap_step_parms: abap_name=BTCTEST,variant=test,arc_printer=LP01,copies=1,
```



```
prt_arc_mode=BOTH,sap_banner=Y,banner_page=Y,recipient_name=cyber,num_lines=65,
num_columns=80,authorization=string2,arc_obj_type=ARCHIVE,arc_doc_type=ARCHIVE,
arc_info=inf
```

## SAP Job Attributes with Default Values

This article explains SAP job attributes with default values.

Some attributes have default values that automatically apply to all job definitions. Your agent administrator can also define default values in the agent's agentparm.txt file or the SAP agent connection properties file. If a default value exists, you do not have to specify the corresponding attribute in the job definition. However, you can specify the attribute to override the default.

The following SAP job attributes have default values:

- **bdc\_ext\_log (SAPBDC jobs only)**  
Specifies whether to generate advanced logging of the Batch Input Session (BDC) running on the SAP system.  
**Default:** N (The job does not generate advanced logging.)
- **sap\_client**  
Specifies the SAP client within the SAP system.  
**Default:** jco.client.client connection properties parameter, if specified
- **sap\_is\_trigger (SAPEVT jobs only)**  
Specifies whether to trigger or monitor an SAP event.  
**Default:** N (The job monitors an SAP event.)
- **sap\_lang**  
Specifies a character code representing a valid language for SAP.  
**Default:**
  - EN (The default language is English.)
  - jco.client.lang connection properties parameter, if specified. This parameter overrides the default type (EN).
- **sap\_mon\_child (SAPJC and SAP jobs only)**  
Specifies whether to monitor children jobs.  
**Default:** N (The job does not monitor children jobs.)
- **sap\_office**  
Specifies whether to save outgoing documents to the SAPoffice outbox of the SAP user who is associated with the job.  
**Default:** N (The job does not save outgoing documents.)
- **sap\_release\_option (SAPBDC, SAPJC, and SAP jobs only)**  
Specifies the action to take with a job after it is defined.  
**Default:** A (The job releases the job as soon as possible.)
- **sap\_rfc\_dest**  
Specifies the destination value for the Remote Function Call (RFC) connection and gateway information.  
**Default:** sap.default.destination agent parameter, if specified
- **sap\_target\_jobname (SAPJC jobs only)**  
Specifies the name of the target job to copy.  
**Default:** The name of the source job

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Monitor an SAP Event and Override the Default Client ID

The sap\_client attribute in the following job definition overrides the default client that is specified in the agent connection properties file.

This example defines an SAPEVT job. The `sap_is_trigger` attribute is set to N, so the job monitors the SAP\_TEST event and remains in RUNNING status.

```
insert_job: trigger_evt
job_type: SAPEVT
machine: localhost
owner: WAAESAP@sapserver
sap_rfc_dest: BI1
sap_event_id: SAP_TEST
sap_is_trigger: N
sap_event_parm: L L
sap_client: 001
sap_lang: EN
continuous: y
```

## Email an SAP Job's Spool File

You can email the SAP spool file for all steps in an SAP R/3 (SAP) job to recipients. A copy of the spool file is emailed on job completion or failure. The recipient can be an email address, an SAPoffice distribution list, or an SAP user.

### Follow these steps:

1. Define an [SAP R/3 job](#).
2. Add the following attribute to the job definition:
  - **sap\_step\_parms: abap\_name=abap, sap\_maillist=value**  
Specifies one or more recipients on a distribution list and the ABAP name.
3. (Optional) Add one or both of the following attributes:
  - **sap\_fail\_msg**  
Specifies a string that indicates the job failed. If the string is found in the job's spool file, the job is considered failed even if the job succeeds on the SAP system.
  - **sap\_success\_msg**  
Specifies a string that indicates the job completed successfully. If the string is found in the job's spool file, the job is considered successfully completed even if the job fails on the SAP system.
4. (Optional) Add the following attribute:
  - **sap\_office**  
Specifies whether to save outgoing documents to the SAPoffice outbox of the SAP user associated with the job.
5. Run the job.  
The job's spool file is emailed to the recipients.

### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

## Example: Email an SAP Job's Spool File

This example emails the SAP spool file for an SAP job to user1@example.com.

```
insert_job: test_SAP_1
job_type: SAP
machine: sapagent
owner: WAAESAP@sapserver
sap_job_name: BI_WRITE_PROT_TO_APPLLOG
```

```
sap_rfc_dest: BI1
sap_step_parms: abap_name=RSBATCH_WRITE_PROT_TO_APPLLOG, sap_maillist="user1@example.com", variant
```

## Email the Spool File of a Single Step in an SAP Job

You can email the SAP spool file for a single step in an SAP R/3 (SAP) or SAP Batch Input Session (SAPBDC) job to recipients. A copy of the spool file is emailed on step completion or failure. The recipient can be an email address, an SAPoffice distribution list, or an SAP user.

### Follow these steps:

1. Define an [SAP R/3 job](#) or [SAP Batch Input Session job](#).
2. Add the following attribute to the job definition:

```
sap_step_parms: abap_name=abap, sap_maillist=address
```

```
[, sap_fail_msg= message]
```

```
[, sap_success_msg=message]
```

- **abap\_name=abap**  
Specifies the valid SAP system ABAP name. This keyword corresponds to the SAPGUI ABAP Program Name field on the Create Step dialog.
- **sap\_maillist=address**  
Specifies one or more recipients to send the spool list results to.
- **sap\_fail\_msg=message**  
(Optional)

Specifies a string that indicates the failure of the step. If the string matches the SAP ABAP output for the step, the step is considered failed even if the step succeeds on the SAP system.

#### NOTE

This keyword does not apply to SAPBDC jobs.

- **sap\_success\_msg=message**  
(Optional)

Specifies a string that indicates the success of the step. If the string matches the SAP ABAP output for the step, the step is considered successfully completed even if the step fails on the SAP system.

#### NOTE

This keyword does not apply to SAPBDC jobs.

3. (Optional) Add optional parameters to the sap\_step\_parms attribute.
4. (Optional) Add the following attribute:

- **sap\_office**  
Specifies whether to save outgoing documents to the SAPoffice outbox of the SAP user associated with the job.

5. Run the job.

The spool file of a single step is emailed to the recipients.

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

## Use Success and Failure Messages within an SAP Job Definition

You can check an SAP job's output for specific text strings to determine whether the job is a success or a failure. You specify the text string in the `sap_success_msg` or `sap_fail_msg` attributes in the job definition. For example, suppose when you cancel an SAP job you want it to complete to release its successor job. You can specify the text string 'Job canceled' as a success message in the job definition. When you cancel the job, the agent checks the job's spool file, finds a match for 'Job canceled', and marks the job as complete.

You can also check the output of a step (ABAP) to determine whether the step is a success or failure. You specify the text string in the `sap_fail_msg=value` and `sap_success_msg=value` keyword value pairs in the `sap_step_parms` attribute.

### NOTE

The `sap_step_parms: sap_fail_msg` and `sap_step_parms: sap_success_msg` keywords do not apply to SAP Batch Input Session (SAPBDC) jobs.

For more flexibility, specify regular expressions instead of simple text strings within the success message and failure message fields. For example, you can use a regular expression to search for multiple strings simultaneously. To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules using a Google search for `java pattern`.

### NOTE

To enable regular expression processing, you must configure the agent for the following parameter:  
`sap.useRegularExpressions=true`.

### Examples: Using Regular Expressions

- This expression checks for "TEST" in the job output file. The first `.*` indicates that any number of characters can precede TEST. The second `.*` indicates that any number of characters can follow it.  
`.*TEST.*`
- This expression checks whether "not found" or "started" appears in the job output file.  
`.*(not found|started).*`
- This expression checks whether "Job canceled" appears in the job output file.  
`.*Job\scanceled.*`

## Secure Copy Jobs Overview

You can define a Secure Copy job to transfer binary files between an agent computer and a remote computer. The Secure Copy job can upload data to or download data from a remote server. The data is encrypted during the transfer. By default, a Secure Copy job uses the Secure File Transfer Protocol (SFTP) protocol. However, you can define the job to use the SCP protocol. The SCP protocol does not support wildcard transfers.

SFTP protocol lets you securely transfer binary or ASCII files using SFTP. The SFTP protocol supports wildcard transfers, so you can upload multiple files to a remote FTP server or download multiple files to the agent computer.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS. The agent must be configured as an FTP client using the Secure Copy Protocol or the Secure File Transfer Protocol.

The SFTP job supports the following types of authentication for file transfer:

- User authentication—This authentication requires the FTP user ID and password for authentication to the SFTP server.
- Public-key authentication—This authentication requires the private key and passphrase for authentication to the SFTP server. If you create the private key using a blank passphrase, the passphrase is not required for the authentication.

**NOTE**

The SFTP job does not support public or private keys that are generated using Putty Gen or that are encrypted in DES3 format.

- Multifactor authentication—This authentication requires both the FTP user ID and password and the private key and passphrase for authentication to the SFTP server.

**Define a Secure Copy Job**

You can define a Secure Copy (SCP) job to transfer binary files using the Secure Copy Protocol.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SCP**  
Specifies that the job type is Secure Copy.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **scp\_local\_name**  
Specifies a file on the agent computer to be downloaded or uploaded.
  - **scp\_remote\_dir**  
Specifies the file's remote source directory (if downloading) or the file's remote destination directory (if uploading).
  - **scp\_remote\_name**  
Specifies the file's source location (if downloading) or the file's destination (if uploading).
  - **scp\_server\_name**  
Specifies a remote server name.
2. (Optional) Specify optional Secure Copy attributes:
  - **job\_class**
  - **scp\_local\_user**
  - **scp\_protocol**
  - **scp\_server\_port**
  - **scp\_target\_os**
  - **scp\_transfer\_direction**
  - **scp\_transfer\_mode**
  - **run\_external**
  - **external\_os\_user**
3. (Optional) Specify optional Secure Copy attributes if you are using SFTP protocol:
  - **scp\_delete\_sourcefile**
  - **scp\_delete\_sourcedir**
  - **scp\_rename\_sourcefile**
  - **scp\_create\_targetdir**
4. (Optional) Specify common attributes that apply to all job types.  
The Secure Copy job is defined.

**NOTE**

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes

specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.

- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files and then import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.
- When a SFTP job fails, the details are logged in the spool file that is located in the `installation_directory/SystemAgent/agent_name/spool/instance_name.SCH/MAIN/WAAE_WF0.1` (UNIX) or `installation_directory\SystemAgent\agent_name\spool\instance_name.SCH\MAIN\WAAE_WF0.1` (Windows) directory.
  - **installation\_directory**  
Specifies the directory where the agent is installed.
  - **agent\_name** Specifies the name of the agent.
  - **instance\_name**  
Specifies the name of the AutoSys Workload Automation instance.

The spool file name is `JOID.run_num_ntry`. For example, `151.30_1`.

- **JOID**  
Defines the unique job object ID associated with the job.
- **run\_num** Defines the run number of the job.
- **ntry** Defines the number of tries or restarts.

The agent logs are not created for the SFTP jobs as the commands are not executed on the agent machine and it is just a file transfer.

Using AutoSys Workload Automation, you can view the spool file by executing the following command from the UNIX operating system prompt or Windows instance command prompt:

```
autosyslog -j job_name -t S
```

### Example: Upload a File Using the Secure File Transfer Protocol

This example uploads the `logs.tar` file to the `/u/tmp` directory on the `hpsupport` server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_upload

job_type: SCP

machine: WINAGENT

scp_transfer_direction: UPLOAD

scp_server_name: hpsupport

scp_remote_dir: /u/tmp
```

```
scp_remote_name: logs.tar

scp_local_name: "D:\temp\logs.tar"

scp_protocol: SFTP

owner: causer@WINAGENT
```

### NOTE

The owner must be defined on AutoSys Workload Automation using the [autosys\\_secure](#) command.

### Example: Upload Multiple Files Using the Secure File Transfer Protocol

This example uploads the files in the c:\temp\upload directory to the /u1/build/uploaded directory on the aixunix server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_up_mult
job_type: SCP
machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: aixunix
scp_remote_dir: /u1/build/uploaded
scp_remote_name: "*"
scp_local_name: "c:\temp\upload*"
scp_protocol: SFTP
owner: causer@WINAGENT
```

## Secure Copy Job Attributes with Default Values

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Secure Copy job attributes have default values:

- **scp\_local\_user**  
Specifies a user ID on the computer where the agent is installed. This user ID determines the access permissions on the agent computer.  
**Default:** User that defined the job
- **scp\_protocol**  
Specifies whether the SCP data transfer uses Secure File Transfer Protocol (SFTP) or regular Secure Copy (SCP).  
**Default:** SFTP
- **scp\_server\_port**  
Specifies the port number of the remote server.  
**Default:** 22
- **scp\_target\_os**  
Specifies the remote operating system type, which is used to determine the path separator on the remote system.  
**Default:** UNIX
- **scp\_transfer\_direction**

Specifies the file transfer direction between the agent computer and the remote server.

**Default:** DOWNLOAD

- **scp\_delete\_sourcefile**

Specifies whether you want to delete the local source files (if uploading) after they are uploaded from the agent computer or the remote source files (if downloading) after they are downloaded from the remote SFTP server.

**Default:** F

- **scp\_delete\_sourcedir**

Specifies whether to delete the directory that contains the local source files (if uploading) or the remote source files (if downloading) after the files are uploaded or downloaded.

**Default:** F

- **scp\_create\_targetdir** Specifies whether to create a target directory before you upload or download a file. **Default:** F

- **scp\_transfer\_mode**

Specifies the type of data you are transferring.

**Default:** B

- **run\_external**

Specifies whether to define a user ID that runs the job on behalf of the agent user.

**Default:** F

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Override the Transfer Direction

The `scp_transfer_direction` attribute in the following job definition overrides the default transfer direction.

This example uploads the `logs.tar` file to the `/u/tmp` directory on the `hpsupport` server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_upload

job_type: SCP

machine: WINAGENT

scp_transfer_direction: UPLOAD

scp_server_name: hpsupport

scp_remote_dir: /u/tmp

scp_remote_name: logs.tar

scp_local_name: "D:\temp\logs.tar"

scp_protocol: SFTP

owner: causer@WINAGENT
```



## SNMP Jobs Overview

The agent supports a built-in SNMP manager capability. You can enable the agent to act as an SNMP manager to emit and listen for SNMP traps in addition to its other roles. The agent supports SNMP v1, v2, and v3. After the agent is configured, you can define and run SNMP job types on the agent.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

You can define the following types of SNMP jobs:

- Simple Network Management Protocol Value Get (SNMPGET)
- Simple Network Management Protocol Value Set (SNMPSET)

The SNMP Value Get job queries a network device for the value of a variable that is assigned to a Management Information Base (MIB) address. You can use the SNMP Value Get job to retrieve information about a network device to determine whether an administrator is required to be notified.

The SNMP Value Set job modifies a variable on a network device. The variable is assigned to the MIB address that you specify. You can use the SNMP Value Set job to update a variable that reports on the failure or success of a mission-critical policy.

## Define a SNMP Value Get Job

You can define a Simple Network Management Protocol Value Get (SNMPGET) job to retrieve the value of an SNMP variable.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent as an SNMP manager. For more information about configuring the agent as an SNMP manager, see [Workload Automation Agents](#).

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SNMPGET**  
Specifies that the job type is SNMP Value Get.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **snmp\_host**  
Specifies the host name or IP address of the network device.
  - **snmp\_oid Attribute -- Specify an SNMP OID** Specifies the SNMP OID (object identifier) in numeric or string format.

### NOTE

If you specify the snmp\_oid attribute in string format, you must also specify the snmp\_mib attribute for the job to complete successfully

2. If your SNMP version is v3, you must specify the following attribute:
  - **snmp\_privacy\_user**  
Specifies the user name whose credentials are used for authentication.
3. (Optional) If your SNMP version is v1 or v2, specify the following attribute:
  - **snmp\_comm\_string**  
Specifies the community string that is used to authenticate against the network device.

4. (Optional) If your SNMP version is v3, you can specify the following attributes:
  - **snmp\_auth\_protocol**  
Specifies the SNMP v3 authentication protocol to use when connecting with the user specified in the `snmp_privacy_user` attribute.
  - **snmp\_context\_engine\_id**  
Specifies the context engine ID in hexadecimal format.
  - **snmp\_context\_name**  
Specifies the name of the context that the variable belongs to.
  - **snmp\_privacy**  
Specifies the SNMP v3 privacy protocol to use.
5. (Optional) Specify optional SNMP Value Get attributes:
  - **job\_class**
  - **destination\_file**
  - **port**
  - **snmp\_mib**
  - **snmp\_subtree**
  - **snmp\_table\_view**
  - **snmp\_version**
6. (Optional) Specify common attributes that apply to all job types.  
The SNMP Value Get job is defined. When the job runs, it retrieves the value of the SNMP variable specified in the `snmp_oid` attribute.

#### NOTE

- For the SNMP job to complete successfully, the SNMP attribute values specified in the job definition must match the SNMP configuration parameter values in the `agentparm.txt` file. For example, if the value of the `management.snmp.agent.version` parameter in the `agentparm.txt` file is set to 2, you must set the value of the `snmp_version` attribute in the job definition to 2. Similarly, if the value of the `management.snmp.agent.port` parameter in the `agentparm.txt` file is set to 161, you must set the `port` attribute in the job definition to 161.
- If your network does not support Internet Protocol version 6 (IPv6), but the IPv6 stack is active on the Windows computer where the agent is running, the IPv6 stack must be disabled. For more information about disabling IPv6 on Windows, see the Microsoft Knowledge Base.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

#### Example: Query a Network Device for the Value of an SNMP Variable using SNMP v2

Suppose that you want to know the value of the `agentVersion` variable hosted by a network device. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the read community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```

insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public

```

### Example: Query a Network Device for the Value of an SNMP Variable using SNMP v3

Suppose that you want to know the value of a variable hosted by a network device using SNMP v3. In this example, the job specifies the OID in numeric format, the AES privacy protocol, and the SHA authentication protocol.

#### NOTE

The owner attribute value is the user ID associated with the authentication password. The snmp\_privacy\_user attribute value is the user ID associated with the privacy password. The owner and snmp\_privacy\_user users and their corresponding passwords are specified using the autosys\_secure command.

```

insert_job: getvarv3
job_type: SNMPGET
machine: localhost
owner: Administrator@localhost
snmp_oid: .1.3.1.1.3.1.0.0
snmp_host: snmp
snmp_version: 3
snmp_auth_protocol: SHA
snmp_privacy: AES
snmp_privacy_user: user1

```

### Define an SNMP Value Set Job

You can define a Simple Network Management Protocol Value Set (SNMPSET) job to set the value of an SNMP variable.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent as an SNMP manager. For more information about configuring the agent as an SNMP manager, see [Workload Automation Agents](#).

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SNMPSET**  
Specifies that the job type is SNMP Value Set.
  - **machine Attribute -- Define the Client Where a Job Runs** Specifies the name of the machine on which the job runs.
  - **snmp\_host Attribute -- Specify an SNMP Host Name or IP Address** Specifies the host name or IP address of the network device.
  - **snmp\_oid Attribute -- Specify an SNMP OID** Specifies the SNMP OID (object identifier) in numeric or string format.

**NOTE**

If you specify the `snmp_oid` attribute in string format, specify the `snmp_mib` attribute for the job to complete successfully.

- **`snmp_value`**  
Specifies the type and value of the variable that you are changing.
- 2. If your SNMP version is v3, you must specify the following attribute:
  - **`snmp_privacy_user` Attribute -- -- Specify the SNMP v3 User Name** Specifies the user name whose credentials are used for authentication.
- 3. (Optional) If your SNMP version is v1 or v2, specify the following attribute:
  - **`snmp_comm_string`**  
Specifies the community string that is used to authenticate against the network device.
- 4. (Optional) If your SNMP version is v3, you can specify the following attributes:
  - **`snmp_auth_protocol`**  
Specifies the SNMP v3 authentication protocol to use when connecting with the user specified in the `snmp_privacy_user` attribute.
  - **`snmp_context_engine_id`**  
Specifies the context engine ID in hexadecimal format.
  - **`snmp_context_name`**  
Specifies the name of the context that the variable belongs to.
  - **`snmp_privacy`**  
Specifies the SNMP v3 privacy protocol to use.
- 5. (Optional) Specify optional SNMP Value Set attributes:
  - **`job_class`**
  - **`destination_file`**
  - **`port`**
  - **`snmp_mib`**
  - **`snmp_version`**
- 6. (Optional) Specify common attributes that apply to all job types.  
The SNMP Value Set job is defined. When the job runs, it sets the SNMP variable specified in the `snmp_oid` attribute to the value specified in the `snmp_value` attribute.

**NOTE**

- For the SNMP job to complete successfully, the SNMP attribute values specified in the job definition must match the SNMP configuration parameter values in the `agentparm.txt` file. For example, if the value of the `management.snmp.agent.version` parameter in the `agentparm.txt` file is set to 2, set the value of the `snmp_version` attribute in the job definition to 2. Similarly, if the value of the `management.snmp.agent.port` parameter in the `agentparm.txt` file is set to 161, you must set the `port` attribute in the job definition to 161.
- If your network does not support Internet Protocol version 6 (IPv6), but the IPv6 stack is active on the Windows computer where the agent is running, the IPv6 stack must be disabled. For more information about disabling IPv6 on Windows, see the Microsoft Knowledge Base.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

### Example: Change the Value of an SNMP Variable Using an Integer Value

Suppose that you want to set the value of the agentLogLevel variable to its highest level to diagnose a problem. In this example, the host name of the network device is host.example.com and its port is 161. The SNMP version is v2 and the write community string is public. The name of the MIB file is cybermation.mib, which is located in the agent installation directory.

```
insert_job: setvarint
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentLogLevel
port: 161
snmp_version: 2
snmp_comm_string: public
snmp_value: int=8
```

### Example: Change the Value of an SNMP Variable Using an IP Address Value

Suppose that you want to set the value of a variable to an IP address. In this example, the job specifies the OID in numeric format, the DES privacy protocol, and the MD5 authentication protocol. The credentials of user user1 are used for authorization.

```
insert_job: setvarip
job_type: SNMPSET
machine: localhost
owner: Administrator@localhost
snmp_oid: .1.3.1.1.2.1.9.3
snmp_host: snmp
snmp_value: addr=172.24.2.20
snmp_version: 3
snmp_auth_protocol: MD5
snmp_privacy: DES
snmp_privacy_user: user1
```

## SNMP Job Attributes with Default Values

This article explains SNMP job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following SNMP job attributes have default values:

- **port**

Specifies the port of the network device.

**Default:** 161

- **snmp\_auth\_protocol** (SNMP v3 only)

Specifies the SNMP v3 authentication protocol to use when connecting with the user specified in the `snmp_privacy_user` attribute.

**Default:** MD5 (The job uses the Message Digest 5 authentication protocol.)

- **snmp\_comm\_string** (SNMP v1 and v2 only)

Specifies the community string that is used to authenticate against the network device.

**Default:** public

- **snmp\_privacy** (SNMP v3 only)

Specifies the SNMP v3 privacy protocol to use.

**Default:** DES (The job uses the Data Encryption Standard privacy protocol.)

- **snmp\_subtree**

Specifies that the job walks the SNMP subtree.

**Default:** n (The job does not walk the SNMP subtree.)

- **snmp\_table\_view**

Specifies that the job retrieves the MIB data in table format.

**Default:** n (The job retrieves the MIB data in the oid=value format.)

- **snmp\_version**

Specifies the SNMP version used when connecting to the network device.

**Default:** 1

#### NOTE

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

#### Example: Override the Privacy Protocol and the SNMP Version in an SNMP Value Set Job

The `snmp_privacy` and `snmp_version` attributes in the following job definition override the default values.

Suppose that you want to set the value of the `out456789` variable using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The credentials of user `user1` are used for authorization.

```
insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8
```

#### Example: Use Defaults in an SNMP Value Get Job

The port and snmp\_version attributes are not specified in the following job definition, so default values are used. The port is set to 161 and snmp\_version is set to 1. Since snmp\_version is set to 1, snmp\_comm\_string is also set to public (the default read community string).

#### NOTE

The snmp\_auth\_protocol, snmp\_privacy, and snmp\_privacy\_user attributes are not specified because they only apply to SNMP v3.

```
insert_job: getvarv1
job_type: SNMPGET
machine:localhost
owner: Administrator@localhost
snmp_oid: .1.3.6.1.2.1.1.6.0
snmp_host: snmp
```

## SQL Server Agent Jobs

SQL Server Agent jobs let you execute predefined jobs in the SQL Server Agent to perform administrative tasks. The jobs can be Transact-SQL scripts, command prompt applications, Microsoft ActiveX scripts, Integration Services packages, Analysis Services commands and queries, or Replication tasks.

#### NOTE

- To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.
- Due to a Microsoft SQL Server restriction, the agent cannot run multiple SQL Server Agent jobs with the same name simultaneously. If a SQL Server Agent job is running, you cannot rerun the job until its previous execution finishes.
- SQL Server Agent jobs do not produce a spool file.

## Define a SQL Server Agent Job

You can define a SQL Server Agent job to execute predefined jobs in the SQL Server Agent to perform administrative tasks.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: SQLAGENT**  
Specifies that the job type is SQL Server Agent.
  - **machine** Specifies the name of the machine on which the job runs.
  - **sqlagent\_jobname** Specifies the name for a SQL Server Agent job.
  - **sqlagent\_jobid** Specifies the SQL Server Agent job identification number.

#### NOTE

sqlagent\_jobname and sqlagent\_jobid are mutually exclusive. Specify any one of these two attributes in your definition.

2. (Optional) Specify optional SQL Server Agent job attributes:

- [job\\_class](#)
  - [sqlagent\\_step\\_name](#)
  - [sqlagent\\_user\\_name](#)
  - [sqlagent\\_domain\\_name](#)
  - [sqlagent\\_target\\_db](#)
  - [sqlagent\\_server\\_name](#)
3. (Optional) Specify common attributes that apply to all job types.  
The SQL Server Agent job is defined. When the job runs, it executes predefined jobs in the SQL Server Agent to perform administrative tasks

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- For information about required attributes and job type specific optional attributes, see the procedure topics that provide instructions for [defining jobs](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [the jil command and JIL \(Job Information Language\)](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).

#### Example: Run a Predefined Job on a Target Server

This example runs a predefined job that is named mssql\_job on the myserver target server.

```
insert_job: sqlagent_job
job_type: SQLAGENT
machine: SQL_AGENT
sqlagent_jobname: mssql_job
sqlagent_user_name: mydomainuser
sqlagent_domain_name: mydomain
sqlagent_target_db: mytargetDB
sqlagent_server_name: myserver
```

## Wake on LAN Jobs Overview

You can save energy using the agent's Wake on LAN (WOL) feature to automate the startup and shutdown of your computers. WOL lets you define and schedule WOL jobs to send a signal to a server to turn it on. When the server is no longer needed, you can schedule a different command job to power it down.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent to support WOL. For more information about configuring the agent to support WOL, see [Set Wake up on LAN](#).



Wake on LAN (WOL) is a hardware and software solution that lets you wake up a computer remotely. The solution requires an ACPI-compliant computer and a special software program that sends a signal to the computer's network card to wake it up. The agent provides the AMD magic packet to broadcast the signal to a computer that has been soft-powered-down (ACPI D3-warm state).

## Define a Wake on LAN Job

You can define a Wake on LAN (WOL) job to send a signal to a server to turn it on. The job can wake up a remote computer that has been soft-powered-down (ACPI D3-warm state). You can schedule another job to power-down the server when it is no longer needed. WOL jobs save energy by turning on the server only when needed.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent to support WOL. For more information about configuring the agent to support WOL, see [Set up Wake on LAN](#).

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: WOL**  
Specifies that the job type is Wake on LAN.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **broadcast\_address**  
Specifies the IP address of the LAN or subnet of the computer that receives the Wake on LAN (WOL) signal.
  - **mac\_address**  
Specifies the Media Access Control (MAC) address of the computer that receives the Wake on LAN (WOL) signal.
2. (Optional) Specify optional Wake on LAN attributes:
  - **job\_class**
  - **ping\_host**
  - **ping\_ports**
  - **ping\_timeout**
  - **wake\_password**
3. (Optional) Specify common attributes that apply to all job types.  
The Wake on LAN job is defined.

### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Wake Up a Server and Ping the Default Port

This example broadcasts the WOL signal to the subnet with the 172.16.255.255 IP address and the 00-11-43-73-38-DC MAC address. After the agent sends the WOL signal, the agent pings the default ports at the 172.16.1.101 IP address to verify if these ports are available. If at least one of these ports is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_default_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-11-43-73-38-DC
ping_host: 172.16.1.101
```

### Example: Wake Up a Server and Ping a Port

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server with the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
```

### Example: Broadcast the WOL Signal Including a Password

This example broadcasts the WOL signal including a password to the subnet with the 172.16.255.255 broadcast IP address and the server with the 11-22-33-44-55-66 MAC address. If the specified password matches the password that is stored on the server's network card, the server wakes up. The job completes successfully after the scheduler sends the WOL signal without verifying that the machine starts.

```
insert_job: wol_pwd_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 11-22-33-44-55-66
wake_password: AA-BB-CC-DD-EE-FF
```

## Wake on LAN Job Attributes with Default Values

This article explains Wake on LAN (WOL) job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following WOL job attributes have default values:

- **port**

Specifies the ports to contact after the agent sends the WOL signal. If you do not enter a value, the agent contacts the defaults ports.

**Defaults:** 21 (ftp), 22 (ssh), 23 (telnet), 80 (http), 111 (sunrpc), 135 (epmap), 139 (netbios-ssn), 445 (microsoft-ds)

- **timeout**

Specifies the timeout for the ping in seconds.

**Default:** 120 seconds

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

### Example: Specify a Timeout Limit When you Broadcast the WOL Signal to a MAC address

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server identified by the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the WOL signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 becomes available within 5 minutes (300 seconds), the job completes successfully. Otherwise, it exceeds the timeout limit and fails.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
ping_timeout: 300
```

## Web Service Jobs Overview

The term web service describes a standardized method for exchanging data between applications and systems. Web services use XML to code and decode the data and Simple Object Access Protocol (SOAP) to transfer it.

Web Service Description Language (WSDL) is an XML-based language that describes a web service and how to access it. A WSDL document specifies the location of the service and the operations the service exposes.

Universal Description, Discovery, and Integration (UDDI) is an XML-based registry for businesses to list their available web services on the Internet. You can use the UDDI to access the WSDL.

Web services provide access to applications written in Java and Microsoft®.NET. A web service lets you invoke operations such as currency conversion, stock exchange quotes, or product pricing. In an enterprise workload automation environment, a web service might be used to invoke a business process such as posting accounts payable to the General Ledger. Some scheduling manager functions are also available as web services.

The following are the web service job types:

- **Web Service RPC/Encoded**

Lets you call an operation within a web service and pass parameters to the operation using RPC/encoded style binding. The parameters can be actual values or a serialized Java object passed by another job.

- **Web Service Document/Literal**

Lets you call an operation within a web service and pass parameters to the operation using document/literal style binding. The parameters represent a flattened view of the XML document the agent constructs. The values passed into the XML document can be literal values or a serialized Java object passed by another job.

When the job invokes the web service, the parameters are passed to the operation. The job's output is stored by default as a serialized Java object in the job's spool directory. You can also specify a destination file for the output.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Web Services, and a web service residing on a web server:

**Figure 55: Web Service Jobs Overview**

**NOTE**

If your company has a firewall and you must communicate through a proxy server to access a computer outside the firewall, agent configuration is required. For more information, see [Configure the Agent Plug-in for a Proxy](#).

## Define a Web Service Document/Literal Job

The Web Service Document/Literal (WSDOC) job lets you call an operation within a web service and pass parameters to the operation using document/literal style binding.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

There are two versions of WSDOC job types; namely, Pre-Version 12.0 and Version 12.0.

The `jil` attribute “compatibility” defines to which version WSDOC job definition belongs to. By default, the compatibility value is 11 which allows only keywords from before Version 12.0. If the specified value is 12, all new keywords are allowed.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:  
**Mandatory attributes for Pre-Version 12.0 and Version 12.0:**
  - **`job_type: WSDOC`**  
Specifies that the job type is Web Service Document/Literal.
  - **`machine`**  
Specifies the name of the machine on which the job runs.
  - **`endpoint_URL`**  
Specifies the target endpoint address URL.**Mandatory attributes for Pre-Version 12.0:**
  - **`port_name`**  
Specifies the WSDL port name.
  - **`service_name`**  
Specifies the web service name.
  - **`wsdl_operation`**

Specifies the operation to be invoked.

- **wsdl\_URL**

Specifies the URL to the Web Service Description Language (WSDL) of the web service to invoke.

**Mandatory attributes for Version 12.0:**

- **blob\_input**

Specifies one or more lines of text to contain in a blob.

**NOTE**

For blob\_input, the type can be only XML.

- **compatibility**

Specifies the version of the HTTPJob. Options are as follows:

**Default:** 11

- **11**

Specifies releases before Version 12.0

- **12**

Specifies Version 12.0 and higher

2. (Optional) Specify optional Web Service Document/Literal attributes.

**Optional attributes for Pre-Version 12.0 and Version 12.0:**

- **job\_class**
- **ws\_authentication\_order**
- **ws\_conn\_domain**
- **ws\_conn\_origin**
- **ws\_conn\_user**
- **ws\_global\_proxy\_defaults**
- **ws\_proxy\_domain**
- **ws\_proxy\_host**
- **ws\_proxy\_origin**
- **ws\_proxy\_port**
- **ws\_proxy\_user**
- **ws\_security**

**Optional attributes specific to Pre-Version 12.0.**

- **destination\_file**
- **job\_criteria**
- **ws\_parameter**

**Optional attributes specific to Version 12.0:**

- **connection\_retry**
- **connection\_timeout**
- **fail\_codes**
- **output\_var**

**NOTE**

For output\_var, var\_format must be only of type RegEx or XPATH, and var\_type must be only payload.

- **persist\_output\_payload**
- **preemptive\_authentication**

**NOTE**

If the **preemptive\_authentication** attribute is set to true, it is mandatory to specify **ws\_conn\_user**.

**ws\_authentication\_order** must be empty or contain BASIC as one of the authentication protocols.

- [results\\_file](#)
- [results\\_file\\_overwrite](#)
- [retry\\_on\\_failure\\_count](#)
- [retry\\_on\\_failure\\_delay](#)
- [retry\\_on\\_failure\\_details](#)
- [soap\\_action](#)
- [soap\\_version](#)

#### NOTE

You can specify the attributes `soap_action` and `soap_version` only when `blob_input` is specified.

- [success\\_codes](#)
- [success\\_criteria\\_regex](#)
- [success\\_criteria\\_value](#)
- [success\\_criteria\\_xpath](#)

### 3. Optional) Specify common attributes that apply to all jobs.

The Web Service Document/Literal job is defined. When the job runs, it calls an operation within a web service.

#### NOTE

- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Optional Attributes](#).
- You can define jobs interactively, create job definitions in script files, and import them using the `jil` command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.
- You can cancel a running Web Service Document/Literal Job workflow using the `KILLJOB` event.

## Examples - Web Service Document/Literal Job

The following examples are for the Web Service Document/Literal Job, mainly using various payload output formats and how you can use different attributes to match and extract content from the payload.

The examples are based on the assumption that the response payload is an XML document with the following structure:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
 <soapenv:Body>
 <ns1:MathsOutput xmlns:ns1="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <ns1:Result>3</ns1:Result>
 </ns1:MathsOutput>
 </soapenv:Body>
</soapenv:Envelope>
```

The examples are classified into the following two categories:

### **Evaluating Success Criteria**

The success criteria for a job can be evaluated using the following options:

- `success_criteria_xpath`
- `success_criteria_regex`

#### **success\_criteria\_xpath**

The following example is checking if the SOAP web service meant for adding two numbers when invoked with input parameters 1 and 2 yields the result as 3.

```
insert_job: WSDOC1
job_type: WSDOC
compatibility: 12
machine: localhost
endpoint_url: http\://wabld-agent-ws.lvn.broadcom.net\:8080/SampleSoapWS/services/MathsServiceService
blob_input: <auto_blobt><soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:sch="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope></auto_blobt>
persist_output_payload: true
success_criteria_xpath: //soapenv\:Envelope/soapenv\:Body/ns1\:MathsOutput/ns1\:Result
success_criteria_value: 3
```

#### **success\_criteria\_regex**

The following example is checking if the SOAP web service meant for adding two numbers when invoked with input parameters 1 and 2 yields the result as 3.

```
insert_job: WSDOC2
job_type: WSDOC
compatibility: 12
machine: localhost
endpoint_url: http\://wabld-agent-ws.lvn.broadcom.net\:8080/SampleSoapWS/services/MathsServiceService
blob_input: <auto_blobt><soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:sch="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope></auto_blobt>
persist_output_payload: true
success_criteria_regex: ns1\:Result=3
success_criteria_value: 3
```

```

 </soapenv:Body>
 </soapenv:Envelope></auto_blobt>
persist_output_payload: true
success_criteria_regex: <ns1\:Result>3<\\\/ns1\:Result

```

## NOTE

Special consideration for success\_criteria\_regex must be taken for characters which are to be escaped both in Regular expression (RegEx) and JIL. If the RegEx data requires a backward slash, then it must be escaped so that it can be treated as a literal backward slash character. Similarly, backward slashes are treated as escape characters in JIL. Hence, the backward slashes must be escaped twice. A single backward slash in the data becomes four backward slashes.

## Consuming Response Payload

The generated response payload from the web service can be consumed using the following options:

- Output variables
- Substitution variables

## Output Variables

The output variables against a job can be queried by running autorep with jb option.

**Example:** autorep -jb jobname

These variables can be used just like the way global variables are used. The notation to be followed is \$ \$jobname@<v>title</v>, where title here refers to the name of the output variable defined in the job "jobname".

## Output variables using XPATH

In the following example, we are capturing the result of the addition of two numbers in the variable result.

```

insert_job: WSDOC3
job_type: WSDOC
compatibility: 12
machine: localhost
endpoint_url: http\://wabld-agent-ws.lvn.broadcom.net\:8080/SampleSoapWS/services/MathsServiceService
blob_input: <auto_blobt><soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:sch="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope></auto_blobt>
persist_output_payload: true
output_var:var_name="result", var_value="//soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result ",
var_format="xpath", var_type="payload"

```

## Output variables using Regular Expression

In the following example, we are capturing the result of the two numbers using groups in the variable result.



```

insert_job: WSDOC4
job_type: WSDOC
compatibility: 12
machine: localhost
endpoint_url: http\://wabld-agent-ws.lvn.broadcom.net\:8080/SampleSoapWS/services/MathsServiceService
blob_input: <auto_blobt><soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/"
 xmlns:sch="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope></auto_blobt>
persist_output_payload: true
output_var:var_name="result", var_value="<ns1:Result>(.*?)</ns1:Result>", var_format="regex",
var_type="payload" var_group="<<$1>>"

```

### **Substitution Variables**

We can also consume an element from the persisted response directly by specifying the XPATH or RegEx. These variables can be used anywhere the way global variables are used.

### **Consuming the persisted XML data**

If we wanted to extract from the response the result of two numbers, then the notation used is `$payload^WSDOC3@<x>//soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result </x>`. Here payload specifies that we are interested in parsing the response of type "payload". This payload is produced by the job WSDOC3 and the xpath expression to be applied on the response is `//soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result`. The job WSDOC3 needs to have `persist_output_payload` set to 1.

### **Consuming the persisted RegEx data**

If we wanted to extract from the response the result of two numbers, then the notation used is `$payload^WSDOC4@<r><ns1:Result>(.*?)</ns1:Result></r><g><<$1>></g>`. The job WSDOC4 needs to have `persist_output_payload` set to 1.

### **Other Examples:**

#### **Pre-Version 12.0 Example: Invoke a Web Service to List All Amazon Buckets Operation**

Suppose that you want to invoke an Amazon web service that lists all your Amazon "buckets". The URL for the WSDL that describes the web service and its location is `http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wSDL`. The target endpoint address URL is `https://s3.amazonaws.com/soap`. The job calls the operation `ListAllMyBuckets` within the AmazonS3 web service. When the job invokes the web service, information about the list of my buckets is passed to the operation. The `ListAllMyBuckets` operation assigns different parameter values.

```

insert_job: execws
job_type: WSDOC
machine: wsagent
endpoint_URL: "https://s3.amazonaws.com/soap"
wsdl_url: "http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wSDL"
service_name: AmazonS3

```

```

port_name: AmazonS3
wsdl_operation: ListAllMyBuckets
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: user@domain
ws_proxy_domain: domain
ws_parameter: Name="/ListAllMyBuckets", Value=""
ws_parameter: Name="/ListAllMyBuckets/AWSAccessKeyId", Value="0x0102030405060708"
ws_parameter: Name="/ListAllMyBuckets/Timestamp", Value="2011-08-25T02:24:21"
ws_parameter: Name="/ListAllMyBuckets/Signature", Value="0x0102030405060708"

```

### Pre-Version 12.0 Example: Return the Maximum of an Arbitrary Number of Integers

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2. The expected return is 4. The job attempts to authenticate the myuser@mydomain credentials using NTLM authentication.

```

insert_job: getmax1
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://wsmach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://wsmach:8080/axis2/services/TestWS?wsdl"
ws_conn_user: myuser@mydomain
ws_conn_domain: mydomain
ws_conn_origin: myhost.example.com
ws_authentication_order: NTLM
ws_parameter: param_name="/intGetMax", param_value=""
ws_parameter: param_name="/intGetMax/i[3]", param_value="1,4,2"
job_criteria: criteria_xpath="//ns:return", criteria_filter="4"

```

### Define a Web Service RPC/Encoded Job

You can define a Web Service RPC/Encoded (WBSVC) job to call an operation within a web service.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: WBSVC**  
Specifies that the job type is Web Service RPC/Encoded.
  - **machine** Specifies the name of the machine on which the job runs.
  - **target\_namespace**  
Specifies the target namespace used for the names of messages, port type, binding, and services defined in the WSDL for the web service. Complex data types such as arrays require the target namespace.
  - **wsdl\_operation**

Specifies the operation to be invoked.

2. (Optional) Specify optional Web Service RPC/Encoded attributes:

- [destination\\_file](#)
- [endpoint\\_URL](#)
- [job\\_class](#)
- [one\\_way](#)
- [port\\_name](#)
- [return\\_class\\_name](#)
- [return\\_namespace](#)
- [return\\_xml\\_name](#)
- [service\\_name](#)
- [success\\_pattern](#)
- [web\\_parameter](#)
- [web\\_user](#)
- [WSDL\\_URL](#)

**NOTE**

- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, you must specify both the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL\_URL and endpoint\_URL attributes, then the service\_name and port\_name attributes are optional.
- The agent does not support document/literal styles of web services.

3. (Optional) Specify common attributes that apply to all job types.

The Web Service RPC/Encoded job is defined. When the job runs, it calls an operation within a web service.

**NOTE**

- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, specify the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify the WSDL\_URL and endpoint\_URL attributes, the service\_name and port\_name attributes are optional.
- The one\_way attribute is set to FALSE by default. If you do not specify this attribute in your job definition, the job waits for a response after the agent invokes the operation before completing. You can override this default setting by specifying the one\_way attribute in your job definition.
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- The agent parameter file might specify default values for some attribute statements, depending on the parameters that the agent administrator defines during agent configuration. Values specified in attribute statements within a job definition override the default values.

**Example: Get a Company Stock Quote**

**NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is <http://host:8080/axis/services/StockQuote?wsdl>. The WSDL port name within the target namespace <http://ws.ca.com> is StockQuote. The target endpoint address URL is <http://host:8080/axis/services/StockQuote>. The job calls the operation GetQuote within the StockQuoteService web service. When the job invokes the

web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

## Web Services Job Attributes with Default Values

This article explains Web Services job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the attribute in a job definition, it overrides the default.

The following Web Service Document/Literal Job attributes have default values:

- **ws\_global\_proxy\_defaults**

Specifies whether to use the global proxy configuration specified by the proxy parameters in the agentparm.txt file.

**Default:** 0 (The job uses the global proxy configuration specified by the proxy parameters in the agentparm.txt file.)

- **ws\_proxy\_domain**

Specifies the domain for proxy authentication.

**Default:** http.proxyDomain agent parameter, if specified

- **ws\_proxy\_host**

Specifies the proxy host name to use for the request.

**Default:** http.proxyHost agent parameter, if specified

- **ws\_proxy\_origin**

Specifies the origin host name for proxy authentication.

**Default:** http.proxyOrigin agent parameter, if specified

- **ws\_proxy\_port**

Specifies the proxy port to use for the request.

**Default:** http.proxyPort agent parameter, if specified

- **ws\_proxy\_user**

Specifies the user name required for proxy authentication.

**Default:** http.proxyUser agent parameter, if specified

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Override the Proxy Host, Proxy User, and the Proxy Domain in a Web Service Document/Literal Job**

Several attributes in the following job definition override the default values.

Suppose that the service name and the port name are AmazonS3. In this example, ws\_proxy\_host, ws\_proxy\_port, ws\_proxy\_user and ws\_proxy\_domain attributes override the global proxy defaults specified in the agentparm.txt file. The credentials of user user@domain are used for authorization to the proxy server.

```
insert_job: execws
job_type: WSDOC
machine: wsagent
endpoint_URL: "https://s3.amazonaws.com/soap"
wsdl_url: "http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wsdl"
service_name: AmazonS3
port_name: AmazonS3
wsdl_operation: ListAllMyBuckets
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: user@domain
ws_proxy_domain: domain
ws_parameter: Name="/ListAllMyBuckets", Value=""
ws_parameter: Name="/ListAllMyBuckets/AWSAccessKeyId", Value="0x0102030405060708"
ws_parameter: Name="/ListAllMyBuckets/Timestamp", Value="2011-08-25T02:24:21"
ws_parameter: Name="/ListAllMyBuckets/Signature", Value="0x0102030405060708"
```

## z/OS Jobs Overview

You can use z/OS jobs to run mainframe workload.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

Workload Automation Agent for z/OS submits and tracks the z/OS jobs. You can define the following three types of z/OS jobs:

- **z/OS Regular**  
Schedules z/OS jobs.
- **z/OS Manual**  
Creates dependencies on z/OS jobs that are submitted outside of the scheduling manager.
- **z/OS Data Set Trigger**  
Creates dependencies on data set activities.

You can customize trigger conditions to define the conditions in which the z/OS Data Set Trigger job completes. You can specify trigger conditions for the following data set activities:

- When a data set is created or updated
- When a specific job, group of jobs, or user ID creates a data set
- When an explicit data set notification is received (used when the data set activity does not generate an SMF record)
- When an FTP file is sent or received successfully

#### NOTE

Each data set must have its own individual z/OS Data Set Trigger job. To create dependencies on multiple data sets, create multiple z/OS Data Set Trigger jobs.

## Define a z/OS Data Set Trigger Job

You can define a z/OS Data Set Trigger (ZOSDST) job to create dependencies on data set activities.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for z/OS.

#### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: ZOSDST**  
Specifies that the job type is z/OS Data Set Trigger.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **zos\_dataset**  
Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job. The JCLLIB is a z/OS data set name.
2. (Optional) Specify optional z/OS Data Set Trigger attributes:
  - **zos\_dsn\_renamed**
  - **zos\_dsn\_updated**
  - **zos\_explicit\_dsn**
  - **zos\_ftp\_direction**
  - **zos\_ftp\_host**
  - **zos\_ftp\_userid**
  - **zos\_trigger\_by**
  - **zos\_trigger\_on**
  - **zos\_trigger\_type**
3. (Optional) Specify common attributes that apply to all job types.  
The z/OS Data Set Trigger job is defined.

#### NOTE

- All z/OS jobs require a z/OS user ID. This user owns the job on z/OS. The owner attribute in the job definition specifies this user ID (the default is the user who invokes jil to define the job).
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Attributes](#).

- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

### Example: Define a z/OS Data Set Trigger Job

This example triggers when the data set PROD.CICS.FILE1602 is closed (created or updated).

```
insert_job: PROD.NIGHTLY
job_type: ZOSDST
machine: ZOS1
zos_dataset: PROD.CICS.FILE1602
owner: zosuser
```

### Example: Run a Compress Job After 100 Closures of a Data Set

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release a compress job named COPYJCL.COMPRESS after every 100 closures of the data set CYBER.COPY.JCL. The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.COPY.JCL
zos_trigger_on: 100
```

#### NOTE

Define the compress job, COPYJCL.COMPRESS, as a successor to the z/OS Data Set Trigger job.

### z/OS Job Attributes with Default Values

This article explains z/OS job attributes with default values.

Attributes that have a default value automatically apply to the job definition. Therefore, you do not have to specify those attributes in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file. If you specify the attribute in a job definition, it overrides the default.

The following z/OS Data Set Trigger job attributes have default values:

- **zos\_explicit\_dsn**  
Specifies whether the job monitors for an explicit data set.  
**Default:** FALSE (The job does not monitor for an explicit data set.)
- **zos\_dsn\_renamed**  
Specifies whether the job monitors when a data set is renamed.  
**Default:** N (The job does not monitor when the data set is renamed.)
- **zos\_dsn\_updated**  
Specifies whether the job monitors for updates to a data set.

**Default:** N (The job does not monitor for updates to the data set.)

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see *Manage JIL*.

## Monitor Data Set Activity by a User or Job

You can define a z/OS Data Set Trigger job to monitor when a specific job, group of jobs, or user ID creates a data set. When the specified condition is met, the job completes.

**Follow these steps:**

1. [Define a z/OS Data Set Trigger job.](#)
2. Add the following attributes to the job definition:
  - **zos\_trigger\_type**  
Specifies whether the job monitors data set activity by a job or a user ID.
  - **zos\_trigger\_by**  
Specifies the name of the job or user who performs the data set activity that triggers the job.
3. Run the job.  
The job monitors for data set activity by the specified user or job.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see *Manage JIL*.

**Example: Restrict the Trigger to Specific Data Sets Created by a Particular User**

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release its successors when the user CYB1 creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.Gzos_
trigger_type: zos_user_id
zos_trigger_by: CYB1
```

**Example: Restrict the Trigger to Specific Data Sets Created by a Particular Job**

Suppose that you want a z/OS Data Set Trigger job named PROD.PAY\_DATA to release its successors when job ABC creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.Gzos_
trigger_type: zos_job_name
zos_trigger_by: ABC
```

## Monitor an FTP Transfer on z/OS

You can define a z/OS Data Set Trigger job to monitor when an FTP file is sent or received successfully. When the specified condition is met, the job completes.



**Follow these steps:**

1. [Define a z/OS Data Set Trigger job.](#)
2. Add the `zos_explicit_dsn` attribute to the job definition using the following syntax:  
`zos_explicit_dsn: FALSE`
3. Add the following attributes:
  - **zos\_ftp\_direction**  
Specifies whether the job monitors for an FTP transfer to a remote computer or from a remote computer.
  - **zos\_ftp\_host**  
Specifies the name of the remote computer involved in the FTP transfer. The data is transferred to or from the local mainframe computer.
  - **zos\_ftp\_userid**  
Specifies the FTP user ID used to connect to a remote computer.
4. Run the job.  
The job monitors for the specified FTP transfer.

**NOTE**

For more information about JIL job types and other job definition attributes, the values that you can specify for those attributes, and JIL syntax, see [Manage JIL](#).

**Example: Monitor for a Data Set Sent to a Remote FTP partner**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when data set CYBER.XFER.001 is successfully sent from the local mainframe partner to a remote FTP

partner. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
```

```
job_type: ZOSDST
```

```
machine: ZOS1
```

```
owner: CYBDL01
```

```
zos_dataset: CYBER.XFER.001
```

```
zos_ftp_direction: SEND
```

**Example: Restrict Triggering to a Specific Host**  
Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a

remote FTP partner with IP address 172.16.0.0 successfully transfers a file creating the data set

CYBER.XFER.001. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
```

```
job_type: ZOSDST
```

```
machine: ZOS1
```

```
owner: CYBDL01
```

```
zos_dataset: CYBER.XFER.001
```

```
zos_ftp_direction: RECEIVE
```

```
zos_ftp_host: 172.16.0.0
```

```
zos_ftp_userid: CYB1
```

**Example: Restrict Triggering to a Specific Login ID**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a

remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming

that the remote FTP partner logged on to the FTP server with the CYBER005 user ID. The agent

ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
```

```
job_type: ZOSDST
```

```
machine: ZOS1
```

```

owner: CYBDL01
zos_dataset: CYBER.XFER.001
zos_ftp_direction: RECEIVE
zos_ftp_host: 172.16.0.0
zos_ftp_userid: CYBER005

```

**Example: Restrict the Trigger to an FTP Transfer from a Specific User ID**

This example releases the job's successors when a remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming that the user ID prefix of the local FTP partner is CYB (CYB-). The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```

insert_job: CYBER.XFER
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.XFER.001
zos_trigger_type: zos_user_id
zos_trigger_by: CYBzos_
ftp_direction: RECEIVE
zos_ftp_userid: CYB

```

## Define a z/OS Manual Job

You can define a z/OS Manual (ZOSM) job to create dependencies on z/OS jobs that are submitted outside the scheduling manager, such as a job that a user submits manually.

### NOTE

To run these jobs, your system requires Workload Automation Agent for z/OS.

### Follow these steps:

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: ZOSM**  
Specifies that the job type is z/OS Manual.
  - **machine** Specifies the name of the machine on which the job runs.
  - **zos\_jobname**  
Specifies the name of the z/OS job that is submitted outside of AutoSys Workload Automation.
2. (Optional) Specify optional z/OS Manual attributes:
  - **auth\_string**
  - **job\_terminator**
  - **search\_bw**
3. (Optional) Specify common attributes that apply to all job types.  
The z/OS Manual job is defined.

**NOTE**

- All z/OS jobs require a z/OS user ID. This user owns the job on z/OS. The owner attribute in the job definition specifies this user ID (the default is the user who invokes jil to define the job).
- The job\_terminator attribute is set to N by default. If you do not specify this attribute in your job definition, the job does not terminate if its containing box completes with a FAILURE or TERMINATED status. You can override this default setting by specifying the job\_terminator attribute in your job definition.
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the agentparm.txt file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

**Example: Post a z/OS Manual Job as Complete Based on the User ID**

This example posts a z/OS Manual job as complete when the manually submitted job ABC runs under user CYBER. The ZOS1 agent monitors job ABC.

```
insert_job: ABC_job
job_type: ZOSM
machine: ZOS1
zos_jobname: ABC
owner: zosuser
auth_string: CYBER
```

**Define a z/OS Regular Job**

You can define a z/OS Regular (ZOS) job to schedule a z/OS job.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: ZOS**  
Specifies that the job type is z/OS Regular.
  - **machine**  
Specifies the name of the machine on which the job runs.
  - **jcl\_library**  
Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job.
2. Specify the following attribute if jcl\_library is a partitioned data set (PDS):
  - **jcl\_member**  
Specifies the JCL member that contains the JCL for your job. If the jcl\_library attribute specifies a fully qualified data set name, the jcl\_member attribute specifies the z/OS job name.

**NOTE**

If you specify a partitioned data set (PDS) in the jcl\_library attribute, the jcl\_member attribute is required.

3. (Optional) Specify optional z/OS Regular attributes:

- [condition\\_code](#)
  - [copy\\_jcl](#)
  - [envvars](#)
  - [job\\_terminator](#)
4. (Optional) Specify common attributes that apply to all job types.  
The z/OS Regular job is defined.

#### NOTE

- All z/OS jobs require a z/OS user ID. This user owns the job on z/OS. The owner attribute in the job definition specifies this user ID (the default is the user who invokes jil to define the job).
- The `job_terminator` attribute is set to N by default. If you do not specify this attribute in your job definition, the job does not terminate if its containing box completes with a FAILURE or TERMINATED status. You can override this default setting by specifying the `job_terminator` attribute in your job definition.
- You can define jobs interactively, create job definitions in script files, and import them using the jil command, or use Web UI to define them. For more information about the JIL command and JIL syntax, see [Manage JIL](#). For more information about using Web UI to define the job, see [Web UI Scheduling](#).
- Default values for some attributes can be defined on the agent in the `agentparm.txt` file. If you specify the attribute in the job definition, it overrides the default value that is defined on the agent. For more information about possible default values, see the syntax and notes for the attributes you are using.

#### Example: Store a Working Copy of the JCL that You Submitted

Suppose that the agent ZOS1 submits the JCL in member CYBDL01A in the CYBDL01.JCLLIB library. If the job fails, you can modify a working copy of the JCL in the CYBDL01.COPY.JCLLIB data set, and can resubmit the job without affecting the JCL source.

```
insert_job: CYBDL01A
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB
jcl_member: CYBDL01A
owner: CYBDL01
copy_jcl: CYBDL01.COPY.JCLLIB
```

#### Example: Define a z/OS Regular Job

This example submits the JCL in member CYBDL01A in the CYBDL01.JCLLIB library.

```
insert_job: zos_job
job_type: ZOS
machine: zosagt
jcl_library: CYBDL01.JCLLIB
jcl_member: CYBDL01A
```

## Manage User-Defined Job Types

This section contains the following topics:

### User-Defined Job Types

AutoSys Workload Automation lets you define simple user-defined jobs. A user-defined job type is similar to a command job except that each user-defined job type is associated with a custom program/script/adaptor. For example, a new job

type can be defined to perform FTP. For this step to work, a custom program/script/adaptor has to be provided which does FTP by taking a few arguments. Jobs using the user-defined job type may optionally specify arguments to the command defined in the user-defined job type. For example, if we define FTP job type as '2', a custom program, script, or adaptor must be provided as part of the definition of type 2.

```
insert_job_type: 2
command: /home/scripts/myftp
```

When jobs of type 2 are defined, all of them execute /home/scripts/myftp when those jobs are run.

```
insert_job: ftp_test
job_type: 2
machine: localhost
std_in_file: /tmp/ftp_params
```

When a new version of FTP script is used, only the definition of job type has to be modified.

You can use the following jil commands to create, update, and delete user-defined job types.

- `insert_job_type`
- `delete_job_type`
- `update_job_type`

Only three attributes are associated with user-defined job types:

- **job\_type**  
Defines the user-specified job type.  
**Limits:** This value can be a single-digit number (0-9).
- **command**  
Defines the command to associate with the job type.  
**Limits:** This value can be up to 510 characters in length.  
The command attribute in the job definition is optional. If you do not specify the command attribute in the job definition, the job type uses the command attribute of the job definition. If you specify the command attribute in the job definition, it appends the command attribute to the job type command.
- **description**  
Defines a description of the job type.  
**Limits:** This value can be up to 256 characters in length.

Any other attribute is rejected and JIL fails.

#### NOTE

Define a job type before you can use it to define a job. For more information, see [Reference](#).

#### Example: Use `insert_job_type` to Add a User-Defined Job Type

This example creates an association between a user-defined job type and an executable.

```
insert_job_type: 5
description: Web Service Adapter
command:ws.exe
```

#### Example: Use `delete_job_type` to Delete a User-Defined Job Type

This example verifies that no jobs are currently using the specified job type, and deletes the job type.

```
delete_job_type: 5
```

#### Example: Use `update_job_type` to Modify a User-Defined Job Type

This example modifies an existing job type, changing the values of the description and command attributes.

```
update_job_type: 5
description: WorldView Adapter
command: wv.exe
```

### Example: Pass Arguments To a User-Defined Job Type Command

This example creates a job type. Two jobs are defined that run with different arguments. The user-defined job is defined to run the command `/bin/sleep`. Each job specifies the sleep time as an argument that is appended to the command. For example, the job `sleep_5` would be executed with a command `/bin/sleep 5`.

```
update_job_type: 7
description: sleep for a spell
command: /bin/sleep
insert_job: sleep_5
description: sleep for 5 seconds
job_type: 7
machine: localhost
command: 5
insert_job: sleep_30
description: sleep for 30 seconds
job_type: 7
machine: localhost
command: 30
```

## Create a Job Type

You can create job types. For example, you have an adapter binary and you want to create 300 jobs that invoke the adapter. You can create 300 command jobs and specify the command each time, or you can define a single job type (for example '0') that represents the adapter command and define 300 jobs of type '0'.

### Follow these steps:

1. Insert `job_type:0`.
2. Enter the following command: `special_adapter`.
3. Enter the following description: This is a job type to run special adapter commands.  
The job type is created.

## Use a New Job Type

After you create a job type, define a job to use it.

### Follow these steps:

1. Insert `job:test`.
2. Enter `job_type:0`.
3. Enter machine name: `localhost`.  
The newly created job type can be used.

AutoSys Workload Automation also supports `delete_job_type` and `update_job_type`. You can use the `delete_job_type` to delete a user-defined job type, and the `update_job_type` when you want to modify an existing user-defined job type.

## Manage Resources

This section contains the following topics:

### Define Virtual Resource Types

Virtual resources can help you control job execution and can improve your environment's performance. They represent values that can be quantified, but they are not directly tied to a physical system. AutoSys Workload Automation does not check whether a virtual resource is an actual device that exists or whether it is a device that is being used by another process.

You can use virtual resources to prevent jobs from running simultaneously and ensure that a job is submitted only when the minimum number of resources is available. For example, you can define a virtual resource to represent the maximum number of floating product licenses available in your enterprise. Each time a qualified job runs, a unit of that resource is used. When all the units are used, no more jobs can run.

You can define the following types of virtual resources on AutoSys Workload Automation:

- Depletable
- Renewable
- Threshold

A virtual resource can be defined for a specific machine, or it can be defined at the global level. A resource defined for a specific machine is available to jobs submitted to that machine. A resource may be defined to more than one specific machine. A resource at the global level is available to all machines controlled by AutoSys Workload Automation. Global resources can help control workload balancing across all machines.

#### NOTE

You can define resources for distributed machines only. Before you can define a resource on a machine, the machine must be defined on the database.

You can specify virtual resources as dependencies to jobs. A job with resource dependencies is submitted only when the resources required are available. If the required resources are not available, the job goes into a RESWAIT state and is not submitted until the resources are available.

Virtual resources are associated with corresponding resource pools. When jobs with virtual resource dependencies run, the used resources are temporarily or permanently removed from the resource pool, depending on the virtual resource type.

AutoSys Workload Automation is the resource manager for virtual resources.

#### NOTE

- When you force start a job in FAILURE or TERMINATED status that has a virtual resource dependency with free=Y or free=N and has not released the virtual resources, the FORCE\_STARTJOB event verifies if the job's current status is FAILURE or TERMINATED and schedules the job using the held virtual resources. Before force starting the job, the scheduler does not re-evaluate other resource dependencies. For more information about the FORCE\_STARTJOB event, see [Reference](#).
- When you update a job, you cannot update the resources attribute in the existing job definition if the job has a resource dependency and has held the resource.
- Virtual resources are secured using CA EEM (external security mode). The native security in AutoSys Workload Automation does not secure virtual resources. For more information about securing virtual resources using CA EEM, see [Securing](#).

## Define Depletable Resources

A depletable resource is a consumed resource. When a job that uses this resource is submitted, the used resource units are permanently removed from the resource pool. When the resource is completely depleted or jobs require more units than what is currently available, jobs that need it go into a RESWAIT state and are not submitted until the resource is available. You can manually replenish the resource using the `update_resource` JIL subcommand. After the resource is replenished, other jobs can use it.

Depletable resources are helpful when you want to represent values that have a limit, such as the maximum times to run a job. Depletable resources are also helpful when you want to control how many times a job can run in a specified time period.

### Example: Run a Job Only Once a Day

Suppose that a bank wants to post daily transactions to its master database only once a day at midnight to ensure data integrity and system performance. To run this critical job only once a day, you can do the following:

1. Define a depletable resource with an amount of 1, as follows:

```
insert_resource: depletable1
res_type: D
machine: hostname
amount: 1
```

2. Define the critical job with the following conditions:

- The job requires one unit of the depletable resource before it can start.
- The job is scheduled to start at midnight.

The job is defined as follows:

```
insert_job: ResDepJob
job_type: CMD
command: /tmp/DBIntensiveApp
machine: hostname
owner: root@hostname
resources: (depletable1,QUANTITY=1)
```

When the job is submitted, one unit of the resource is permanently removed from the resource pool. If the job fails, AutoSys Workload Automation restarts the job (based on the `n_retrys` attribute), but the job goes into RESWAIT state because no units of the resource are available. The RESWAIT state indicates that a potential problem occurred. You must run the job manually.

### NOTE

To add additional resources to a depletable resource, use the `update_resource` subcommand. For example, the following command adds 15 units to the `depletable1` resource:

```
update_resource: depletable1
machine: hostname
amount: 15
```

## Define Renewable Resources

A renewable resource is a borrowed resource. When a job that uses this resource is submitted, the used resource units are temporarily removed from the resource pool. When the job completes, the resource units are returned to the pool, or the units are held until they are manually released back to the pool. Renewable resources are helpful when you want to control jobs that run concurrently or serially.

When the resource is being used, other jobs that need more units than what is currently available go into a RESWAIT state and are not submitted until the resource is available. You can change the amount of resource units available using



the `update_resource` JIL subcommand. After the amount is changed, a greater number of jobs that need the resource can run concurrently.

### Example: Control the Maximum Number of Licenses Used

Suppose that your enterprise has 10 floating licenses for a program. Multiple licenses can be used on one machine, or they can be used on up to 10 machines. At any time, you want to ensure that the maximum number of licenses used is 10. To control the maximum number of licenses being used, you can do the following:

1. Define a renewable resource at the global level with an amount of 10, as follows:

```
insert_resource: r1
res_type: R
amount: 10
```

2. Define each job that requires the license with the following conditions:
  - The job requires one unit of the renewable resource before it can start.
  - The job frees the renewable resource whether it completes successfully or not.

The job is defined as follows:

```
insert_job: jrl
command: sleep 500
machine: hostname
resources: (r1,quantity=1, FREE=A)
```

When a job is submitted, one unit of the resource is temporarily removed from the resource pool. Because there are only 10 units, only 10 of these jobs can run simultaneously on any machine in the enterprise. Other jobs that require a license cannot be submitted because no resources are available. When a job that is running completes, one unit of the resource is returned to the resource pool. Another job can be submitted because a unit is now available.

### Define Threshold Resources

A threshold resource is a sizing resource. For example, if the threshold resource is set to 2, AutoSys Workload Automation submits the jobs that require 2 or fewer units. Threshold resources are helpful when you want to define a boundary that controls which jobs are submitted to run. The used resource units are not removed from the resource pool.

A job that has a dependency on a threshold resource is only submitted if it requires the available resource units or fewer. Otherwise, the job goes into a `RESWAIT` state and is not submitted until the threshold amount is increased. You can change the threshold amount using the `update_resource` JIL subcommand. After the amount is changed, the jobs that meet the threshold can run.

### Example: Prevent Jobs from Running When a Critical Resource is Offline

Suppose that multiple jobs read and write to a disk on a machine named `mach1`. When the disk needs to be formatted, you want the jobs to stop running. To prevent the jobs from running when the disk is offline, you can do the following:

1. Define a threshold resource on the `mach1` machine with an amount of 1, as follows:

```
insert_resource: t1
machine: mach1
res_type: T
amount: 1
```

2. Define each job that reads and writes to the disk and requires one unit of the threshold resource before it can start, as follows:

```
insert_job: jtl
command: readnawritetodisk
machine: mach1
resources: (t1, quantity=1)
```

3. Define a job named TRIGGER\_JOB that resets the threshold amount to 0 to indicate that the disk is offline. The trigger job runs the update\_resource command as follows:

```
update_resource: t1
machine: mach1
amount: 0
```

When the disk is online, the threshold resource amount is 1, so all jobs that need to read and write to the disk are submitted (the jobs meet the threshold requirement). When the disk needs to be formatted, you can run TRIGGER\_JOB, which resets the threshold resource amount to 0. All the jobs that need to read and write to the disk go into a RESWAIT state and are not submitted until the threshold is set to 1.

## Define a Virtual Resource

To use virtual resource dependencies in your jobs, first add the virtual resource definition to the database. You can define virtual resources on distributed machines only.

### Follow these steps:

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following subcommand and attributes:
  - **insert\_resource: resource\_name**  
Specifies the virtual resource to be defined.
  - **amount**  
Defines the number of units to assign to the virtual resource.
  - **res\_type**  
Specifies the virtual resource type (D for depletable, R for renewable, or T for threshold).
3. Specify the following attribute if you want to define the resource for a machine:
  - **machine**  
Specifies the name of the machine to define the virtual resource for. The machine is defined on the database, and the type of the machine cannot be v, w, or p.

### NOTE

If you do not specify the machine attribute, the resource is available to all machines (global level resource).

4. (Optional) Specify the following attribute:
  - **description**  
Defines a free-form text description of the virtual resource.
5. Do *one* of the following tasks:
  - Enter **exit** if you are using interactive mode.
  - Redirect the script to the jil command if you are using a script.

The insert\_resource subcommand is issued and the specified virtual resource is defined.

### NOTE

- The virtual resource name must be unique across all resource types.
- You cannot define the same virtual resource at the machine level and global level.
- You can define the same virtual resource on multiple machines.
- For more information about the syntax for the insert\_resource subcommand and related attributes, see [Reference](#).

### Example: Define a Global Depletable Resource

This example defines a virtual depletable resource named `glob_res`. The machine attribute is not specified, so the resource is available to all machines.

```
insert_resource: glob_res

res_type: D

amount: 50

description: "This resource is permanently consumed."
```

### Example: Define a Machine-Level Threshold Resource

This example defines a threshold resource for the `unixagent` machine. The resource is assigned 10 units, so only jobs that require 10 or fewer units of this resource are submitted to the `unixagent` machine.

```
insert_resource: threshold_res

res_type: T

machine: unixagent

amount: 10
```

## Update a Virtual Resource

You can update the amount and description properties of a virtual resource definition in the database. Updating the resource amount is helpful when you want to change the number of jobs that can run concurrently or serially.

### Follow these steps:

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following subcommand:
  - **update\_resource: resource\_name**  
Specifies the virtual resource to be updated. This resource must be defined in the database.
3. Specify the following attribute if the resource is defined for a machine:
  - **machine**  
Identifies the name of the machine that the virtual resource is defined for.

#### NOTE

If you do not specify the machine attribute, AutoSys Workload Automation assumes that the resource is a global resource. If the resource is machine-level, but the machine attribute is not specified, you get an error.

4. (Optional) Specify the following attributes:
  - **amount**  
Defines the number of units to assign to the virtual resource. The number can be an absolute value or a relative value.
  - **description**

Defines a free-form text description of the virtual resource.

5. Do *one* of the following tasks:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the jil command if you are using a script.

The `update_resource` subcommand is issued and the specified virtual resource is updated.

#### NOTE

- You cannot update resource types or machine names using the `update_resource` subcommand. To update resource types or machine names, delete the resource and add it to database again with the new properties.
- For more information about the syntax for the `update_resource` subcommand and related attributes, see [Reference](#).

### Example: Update a Machine-Level Virtual Resource

This example updates a virtual resource that is associated with the `unixagent` machine. The number of units is changed to 35.

```
update_resource: mach_res
machine: unixagent
amount: 35
```

Suppose that the `amount` attribute is defined as follows:

```
amount: +20
```

In this situation, the command adds 20 units to the available resource count. For example, if the resource already has 35 units, the command adds 20 units and the total would be 55.

Similarly, suppose that the `amount` attribute is defined as follows:

```
amount: -20
```

If the resource has 35 units, the command removes 20 units and the total would be 15.

### Delete a Virtual Resource

You can delete a virtual resource that you no longer use.

#### NOTE

You cannot delete a virtual resource if it is referenced as a dependency in a job. To delete the resource, you must delete all the related job dependencies first. If a resource is deleted while a job that references it as a dependency is active or running, the job continues to run and is not affected.

#### Follow these steps:

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following subcommand:
  - **`delete_resource: resource_name`**  
Specifies the virtual resource that you want to delete.
3. Specify the following attribute if the resource is defined for a machine:
  - **`machine`**  
Identifies the name of the machine that the virtual resource is defined for.

**NOTE**

If you do not specify the machine attribute, AutoSys Workload Automation assumes that the resource is a global resource. If the resource is machine-level, but the machine attribute is not specified, you will get an error.

4. Do *one* of the following tasks:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the jil command if you are using a script.

The delete\_resource subcommand is issued and the specified virtual resource is deleted.

**NOTE**

For more information about the syntax for the delete\_resource subcommand and related attributes, see [Reference](#).

**Example: Delete a Global Virtual Resource**

This example deletes the global virtual resource named glob\_resource.

```
delete_resource: glob_resource
```

**Define Virtual Resource Dependencies in a Job**

You can define a job to have dependencies on virtual resources. A job with resource dependencies is submitted only when the resources required are available. Using resource dependencies can help you control job execution and can improve your environment's performance.

**NOTE**

You can specify only global resources on a box job. However, the resource specified for a box applies only to the box and not to the jobs within the box.

**Follow these steps:**

## 1. Insert a job and add the following attribute to the job definition:

- **resources**  
Defines one or more virtual resource dependencies.

**NOTE**

To define a virtual resource dependency, the virtual resource must already be defined on the database.

## 2. (Optional) Add the following attribute:

- **priority**  
Defines the queue priority of the job. The queue priority establishes the relative priority of all jobs queued for a given machine. A lower number indicates a higher priority. Jobs with a higher priority get the required resources first and run before lower priority jobs.

## 3. Run the job.

The job is defined with the specified resource dependencies. The job is submitted when the required resources are available.

**NOTE**

For detailed information about the syntax for the resources and priority attributes, see [Reference](#).

**Example: Define a Virtual Resource Dependency that Does Not Free the Resources After Job Completion**

This example defines a Command job that has a dependency on a virtual renewable resource. Before the job can start running, it needs all the units of the renew\_res resource.

```
insert_job: no_free_job
job_type: CMD
```

```
machine: unixagent
command: /u1/procrun.sh
resources: (renew_res, QUANTITY=ALL, free=N)
```

After the job completes, the units of the `renew_res` resource are not freed from the job. To return the units back to the available resource pool, you must issue the following command:

```
sendevent -E RELEASE_RESOURCE -J no_free_job
```

Suppose that the job is defined with the following attribute:

```
resources: (renew_res, QUANTITY=5, free=Y)
```

If the job completes successfully, the resource is added back to the resource pool. This behavior is the default.

Suppose that the job is defined with the following attribute:

```
resources: (renew_res, QUANTITY=5, free=A)
```

The resource is released back to the pool unconditionally.

### Example: Define Virtual Resource Dependencies

This example defines a Command job that has virtual resource dependencies. Before the job can start running, it needs a machine that satisfies all the following dependencies:

- 1 unit of the depletable resource named D1
- 3 units of the threshold resource named T1
- 4 units of the renewable resource named R1
- `SYSTEM_OS_TYPE` is AIX 5.3
- `SYSTEM_PHYSICAL_MEMORY` is greater than 2 GB

```
insert_job: res_dep_job
job_type: CMD
machine: unixagent
command: /u1/procrun.sh
resources: (D1, QUANTITY=1) AND (T1, QUANTITY=3) AND
(R1, QUANTITY=4, FREE=Y) AND
(SYSTEM_OS_TYPE, VALUEOP=EQ, VALUE=AIX, VERSION=5.3) AND
(SYSTEM_PHYSICAL_MEMORY, VALUEOP=GT, VALUE=2048)
```

### Example: Define a Global Resource on a Box Job

This example defines a global resource that is named *glb\_resource1* on the box that is named *box1\_resource\_Y*. The global resource applies only to the box and not to the jobs within the box.

```
insert_job: box1_resource_Y
job_type: box
resources: (glb_resource1, QUANTITY=1, FREE=Y)
```

Issue the following command to generate a detailed report for the *glb\_resource1* virtual resource. The resource is global, so it is available to all machines.

```
autorep -V glb_resource1 -d
```

The resulting report might resemble the following examples:

Resource Name	Type	Machine
Defined Available		

glb_resource1	R	---
10 9		

/\*\*\*\* Current Resource Usage \*\*\*\*/

ResName	JobName
Run/Ntry	Amount in Use
Status	Machine
glb_resource1	box1_resource_Y
1608/1	1
RUNNING	**NOMACHINE**

## Update Virtual Resource Dependencies in a Job

You can update the "Not an ITPAM application user."virtual resource dependencies defined in a job.

### Follow these steps:

- Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
- Specify the following subcommand and attribute:
  - update\_job: job\_name**  
Specifies the job to update.
  - resources**  
Specifies one or more virtual resource dependencies to update.
- Do *one* of the following tasks:
  - Enter **exit** if you are using interactive mode.
  - Redirect the script to the jil command if you are using a script.

The update\_job subcommand is issued and the specified virtual resource dependencies are updated.

### NOTE

For detailed information about the syntax for the resources attribute, see [Reference](#).

### Example: Update Virtual Resource Dependencies

This example updates the virtual resources in the proc\_daily job. RENEW3 virtual resource dependency does not have the free keyword defined, so the resource units are freed if the job completes successfully (the default).

```
update_job: proc_daily
resources: (RENEW3, QUANTITY=3) AND (THRESHOLD2, QUANTITY=3) AND (SYSTEM_CPU_COUNT,
VALUEOP=GT, VALUE=2)
```

## Release Renewable Resources

When you define a job to have a virtual renewable resource dependency, you can specify whether the units of the resource are freed after the job completes. If the units are not freed, you can manually release them back to the resource pool so that other jobs can use them.

To release renewable resources manually, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
sendevent -E RELEASE_RESOURCE -J job_name
```

- ***job\_name***  
Specifies the job that you want to release the renewable resources from.

The sendevent command is issued and the resources are released from the job.

### NOTE

For more information about the sendevent command, see [Reference](#).

### Example: Release Renewable Resources

Suppose that you defined a virtual renewable resource named ren1 that has two units. JobA and JobB are also defined with a dependency on ren1. Each job requires one unit of ren1 before it can run. The jobs release the resource units only if they complete successfully.

When the jobs are submitted, the required units are temporarily removed from the resource pool. Suppose that JobA completes successfully and JobB fails. JobA returns one unit of resource to the resource pool. The other unit used by JobB is held, so the resource pool only has one unit. Other jobs that require more than one unit go into a RESWAIT state until the required units are available.

The following command manually releases the one unit of resource held by JobB so that other jobs can use it:

```
sendevent -E RELEASE_RESOURCE -J JobB
```

## Generate a Report on Current Resource Definitions

You can generate a report that displays the current resource definitions in a database. Generating a report is helpful when you want to determine the properties of a resource and the values that can be specified when defining resource dependencies in jobs.

To generate a report on the current resource definitions, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -V resource_name -q
```

- ***resource\_name***  
Specifies the virtual resource definition that you want to generate a report on. You can use wildcard characters.

The autorep command is issued and the report is generated.

### NOTE

For more information about the autorep command, see [Reference](#).

## Generate a Report to Display a Job's Resource Dependencies

You can generate a report that displays a job's current definition, including its resource dependencies.



To generate a report to display a job's resource dependencies, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep - J job_name -q
```

- ***job\_name***

Specifies the job that you want to generate a report for.

The autorep command is issued and the report is generated. The job's resource dependencies are included in the report.

**NOTE**

For more information about the autorep command, see [Reference](#).

## Generate a Report to Monitor Virtual Resource Usage

You can generate a report that monitors virtual resource usage. This report is helpful when you want to check how many units a virtual resource has defined, how many units are currently available, and which machines are using the resource during run time.

**NOTE**

The detail report (-d option) only displays the jobs that are *currently* using the resource.

To generate a report to monitor virtual resource usage, enter *one* of the following commands at the UNIX operating system prompt or the Windows instance command prompt:

- To generate a summary report:

```
autorep - V resource_name -s
```

- To generate a summary report and detail report:

```
autorep - V resource_name -d [-M machine_name]
```

- ***-V resource\_name***

Specifies the virtual resource definition that you want to generate a report for. You can use wildcard characters.

**NOTE**

You can specify ALL to generate a report on all resource definitions.

- ***-s***  
Generates a summary report.
- ***-d***  
Generates a summary report and detail report.
- ***[-M machine\_name]***  
(Optional) Filters the detail report to display only the specified machine. You can use wildcard characters to match machine names.  
**Default:** ALL (If you do not specify the -M option, the report displays all machines that are using the resource.)

The autorep command is issued and the report is generated.

**NOTE**

For more information about the autorep command, see [Reference](#).

### Example: Generate a Summary Report for a Global Virtual Resource

This example generates a summary report for the virtual resource named ren1. The resource is global, so it is available to all machines.

```
autorep -V ren1 -s
```

The report displays the following information:

Name	Machine	Defined	Available
----	-----	-----	-----
ren1	--	4	1

The resource has a total of 4 units defined. It has 1 unit available, so 3 units are currently being used by jobs.

#### Example: Generate Summary and Detail Reports for a Global Virtual Resource

This example generates a summary report and a detail report for the ren1 virtual resource. The resource is global, so it is available to all machines.

```
autorep -V ren1 -d
```

The resulting report might resemble the following:

Resource Name	Type	Machine	Defined	Available
-----	-----	-----	-----	-----
ren1	V	---	4	1

```
/**** Current Resource Usage ****/
```

ResName	JobName	Run/			
Ntry	Status	Machine	Amount	in Use	
-----	-----	-----	-----	-----	-----
ren1	job1	1/1	RUNNING	machine1	1
ren1	job2	1/1	RUNNING	machine2	2

The -M option is not specified in the command, so the report displays all the jobs that are currently using the resource.

#### Example: Generate Summary and Detail Resource Reports for Individual Machines

Suppose that the virtual renewable resource named `res_count` is defined on three different machines (`amachine1`, `amachine2`, and `bmachine3`). This example generates a summary report and a detail report for the `res_count` resource.

```
autorep -V res_count -d -M amachine%
```

The report displays the following information:

Name	Machine	Defined	Available
----	-----	-----	-----
<code>res_count</code>	<code>amachine1</code>	4	0
<code>res_count</code>	<code>amachine2</code>	3	1

```
/**** Current Resource Usage ****/
```

ResName	JobName	Run/Ntry	Status	Machine	Amount in Use
-----	-----	-----	-----	-----	-----
<code>res_count</code>	<code>Job1</code>	1/1	RUNNING	<code>amachine1</code>	4
<code>res_count</code>	<code>Job2</code>	2/1	RUNNING	<code>amachine2</code>	2

The report displays the jobs that are currently running. The `-M` option is specified with a wildcard, so the report only displays the jobs on the machines whose names match the `-M` value.

## Generate a Report to Monitor Resource Dependencies

You can generate a report that monitors virtual resource dependencies. This report is helpful when you want to check which resources a job depends on and which of the resource dependencies are satisfied.

To generate a report to monitor virtual resource dependencies, enter the following commands at the UNIX operating system prompt or the Windows instance command prompt:

```
job_depends -J job_name -r
```

- ***job\_name***

Specifies the job that you want to generate a dependency report for.

The `job_depends` command is issued and the report is generated.

### NOTE

For more information about the `job_depends` command, see [Reference](#).

## Example: Generate a Report on Virtual Resource Dependencies

Suppose that you define three virtual resources (dep1, ren1, and thr1). The resources are global, so they are available to all machines.

```
insert_resource: dep1
res_type: D
amount: 0
insert_resource: ren1
res_type: R
amount: 0
insert_resource: thr1
res_type: T
amount: 1
```

You also define a job named job1 that belongs to a machine group that includes the M1 and M2 machines. job1 depends on all three virtual resources.

```
insert_job: job1
job_type: CMD
command: &sleep 500
machine: machineGroup1 /* This machine group includes machines M1 and M2. */
resources: (ren1, QUANTITY=1, FREE=A) AND (dep1, QUANTITY= 1) AND (thr1, QUANTITY= 1)
```

The following command generates a report on job1's resource dependencies on the M1 and M2 machines:

```
job_depends -J job1 -r
```

The report displays the following information:

Job Name	Machine		
-----	-----		
job1	M1		
Virtual Resources			
-----			
Resource	Type	Amount	Satisfied?
-----	----	----	-----
ren1	R	1	NO
dep1	D	1	NO
thr1	T	1	YES

Job Name	Machine		
-----	-----		
job1	M2		
Virtual Resources			
-----			
Resource	Type	Amount	Satisfied?
-----	----	----	-----
ren1	R	1	NO
dep1	D	1	NO
thr1	T	1	YES

## Manage Binary Large Objects (Blobs)

This section contains the following topics:

### Binary Large Objects

Binary Large Objects (blobs) are binary data of variable length. AutoSys Workload Automation supports blobs in job definitions, and after they are defined, they are stored in the database. This behavior lets the blob data be shared by jobs running on multiple computers.

To understand the advantages of using blobs in AutoSys Workload Automation environment, refer to the following example, which explains the process that is used to share data among the jobs that are running on a single computer:

1. When the jobs are running on a single computer, you can define a command job to run a program that outputs the data to a file using the `std_out_file` attribute.
2. When the job is completed, a file is created in the location that is specified by the `std_out_file` attribute.
3. All the other jobs that depend on this output data can access this file.
4. You can also define a second command job to run a program that reads the output data of the previous job, by specifying the file name in the `std_in_file` attribute.
5. This second command job opens the file that is specified by the `std_in_file` attribute and passes the data to the program, allowing it to complete successfully.

Based on this example, as the output data is stored in a file on one computer, it is not available to all the other jobs that are scheduled to run on other computers. However, the use of blobs allows the data that is saved as output by a job on one computer to be shared by all the other jobs that are running across multiple computers.

Also, you can define a command job to run a program that uploads the output data to the database as a blob using the `std_out_file` attribute. You can also define a second command job to run a program that reads the blob data of the previous job using the `std_in_file` attribute. The second command job downloads the blob data that is specified by the `std_in_file` attribute from the database and passes the data to the program, allowing it to complete successfully.

Blob data can be of the following types:

- **Binary Data**

Requires a program that understands the format of the data to interpret the bytes in binary data. For example: Multimedia files which include the following items:

- Images
- Video files
- Audio files

- **Textual Data**

Requires an operating system that can interpret the bytes in textual data, which contains the characters that conform to the ASCII standard.

**NOTE**

Some operating systems handle the specification of a new line in the textual data differently. In this instance, you must convert the necessary textual data when it is copied across operating systems.

AutoSys Workload Automation allows you to specify the type of blob data that is being used and converts the textual data when it is downloaded across multiple operating systems.

### Blob Types

AutoSys Workload Automation supports the following types of blobs:

- Job blobs
- Global blobs

**NOTE**

If you install the 64-bit agent using the Workload Automation Agent for UNIX, Linux, or Windows DVD, you cannot run jobs that contain globs or blobs as input or output. The native AutoSys Workload Automation client or SDK runs only in 32-bit mode and does not support being called by a 64-bit application.

## Job Blobs

Job blobs are associated with an existing AutoSys Workload Automation job and are referenced by the job name. Job blobs can be created at the time of the job definition or after the job has been defined.

There are three types of job blobs:

- **Input**  
Contains the input data that is reserved for the job to which they are associated in textual data format. This also includes the inputpayload (an input payload job blob) job blobs.
- **Output**  
Stores the program output messages of a running job in textual or binary data format. This also includes the outputpayload (an output payload job blob) and outputpayloadvar (an output payload var job blob) job blobs.
- **Error**  
Stores the error messages of a running job in textual or binary data format.

Job blobs are deleted when the job is deleted.

### Input Job Blobs

Input blobs are uploaded to the database using JIL. You can insert an input job blob multiple times. Each time it is inserted, it acquires a new version number.

When the job starts, the most recent version of the job input blob is used. All the earlier versions of the blob remain in the database until they are manually deleted. If you delete an input job blob, only the active version of the input job blob is deleted. The version which was prior to the deleted version becomes the new active version.

When you run a job, the AutoSys Workload Automation agent downloads the active version of job's input blob from the database into a temporary file on the computer. This file is then passed into the standard input of the program that is executed by the job. When the job completes, the temporary file containing the input blob data is deleted. The blob in the database, however, is not deleted and remains as the active version for subsequent job runs.

### Output and Error Job Blobs

Output and error job blobs store the program output and error messages of a running job. When you run a job, the AutoSys Workload Automation agent creates temporary files on the computer that are used to capture the standard output and standard error messages from the program that was executed by the job. After the job has completed its run, the agent uploads the files containing the output data as blobs into the database, overwriting the existing files, and deletes the temporary files. An output job blob can be used as input by another job. An error job blob, on the other hand, cannot be used as input by another job.

## Global Blobs

Global blobs are general purpose blobs in textual or binary data format. Like the AutoSys Workload Automation global variables, they are referenced by a unique name. You can either upload the global blobs to the database using JIL or they can be uploaded by the AutoSys Workload Automation agent, after a job has completed its run. After a global blob is created, it is available to any job as input. Global blobs remain in the database until they are deleted using JIL.

## Blob Attributes

The following table lists the subcommands and attributes associated with the definition or destruction of a blob:

Task	Subcommands	Attributes
Create input job blob	<a href="#">insert_job</a> , <a href="#">update_job</a>	<a href="#">blob_input</a> or <a href="#">blob_file</a> (Optional) <a href="#">blob_type</a> ; only in and inputpayload blob types are supported.
Create input job blob	<a href="#">insert_blob</a>	<a href="#">blob_input</a> or <a href="#">blob_file</a> (Optional) <a href="#">blob_type</a> ; only in and inputpayload blob types are supported.
Delete job blob	<a href="#">delete_blob</a>	<a href="#">blob_type</a> ; only in, out, and inputpayload blob types are supported.
Create global blob	<a href="#">insert_glob</a>	<a href="#">blob_mode</a> , <a href="#">blob_input</a> , or <a href="#">blob_file</a> (Optional) <a href="#">blob_type</a>
Delete global blob	<a href="#">delete_glob</a>	

The [blob\\_input](#) attribute lets you manually input the contents of a blob containing textual data. The [blob\\_input](#) attribute has the following format:

```
blob_input: <auto_blobt>textual data</auto_blobt>
```

### NOTE

The textual data begins immediately after the `auto_blobt` XML-style open tag and might span multiple lines. JIL recognizes the end of the textual data when it reads the `auto_blobt` XML-style end tag. This implies that the literal character string `</auto_blobt>` cannot form part of the [blob\\_input](#) value. If you want to include this character string as part of the textual blob data, use the [blob\\_file](#) attribute.

The [blob\\_file](#) attribute lets you specify the location and name of a file on the computer that serves as the input job blob or global blob file. The [blob\\_file](#) attribute has the following format:

```
blob_file: filename
```

### NOTE

If the [blob\\_file](#) attribute is used to specify an input job blob through the [insert\\_job](#) or [insert\\_blob](#) subcommand, the file is interpreted as a text-based file.

## Create Input Job Blobs

To create an input job blob in the database using JIL, do the following:

- Upload an input job blob at the time of the definition of the associated job.
- Upload an input job blob after you have defined the job.

### NOTE

Input job blobs are referenced by the name of the job.

To create an input job blob at the time of the definition of the associated job, use the [insert\\_job](#) JIL subcommand and specify either the [blob\\_input](#) or [blob\\_file](#) attributes, as follows. You can also specify the blob type using the [blob\\_type](#) attribute; however only in and inputpayload blob types are supported.

```
insert_job: test_job_with_blob
```

```
job_type: cmd
```

```
command: sleep 60

machine: juno

owner: jerry@juno

std_in_file: $$blobt

blob_input: <auto_blobt>multi-lined text data for job blob</auto_blobt>

blob_type: inputpayload
```

or

```
insert_job: test_job_with_blob

job_type: cmd

command: sleep 60

machine: juno

owner: jerry@juno

std_in_file: $$blobt

blob_file: /test_job_with_blob_file.txt

blob_type: in
```

To create an input job blob after you have defined the job, use the `insert_blob` JIL subcommand and specify either the `blob_input` or `blob_file` attributes, as follows:

```
insert_blob: test_job_with_blob

blob_type: inputpayload

blob_input: <auto_blobt>multi-lined text data for job blob</auto_blobt>
```

or

```
insert_blob: test_job_with_blob

blob_type: in

blob_file: /test_job_with_blob_file.txt
```



JIL interprets the file name that is specified in the `blob_file` attribute as a file that contains the textual data and performs a conversion of the new line character. JIL also displays the version number of the most recent input job blob.

## Delete Job Blobs

You can use the JIL `delete_blob` subcommand to delete the following items:

- Active version of the input job blob
- Output and error job blobs

Specify whether to delete the job input or output blob data using the `blob_type` attribute.

### NOTE

Job blobs are referenced by the name of the job. JIL displays the version number of the most recent job input blob.

To delete the most recent version of the input job blob, use the `delete_blob` JIL subcommand and specify the `blob_type` attribute with the value of *input*, as follows:

```
delete_blob: test_job_with_blob
```

```
blob_type: input
```

To delete the *inputpayload* job blob, use the `delete_blob` JIL subcommand and specify the `blob_type` attribute with the value of *inputpayload*, as follows:

```
delete_blob: test_job_with_blob
```

```
blob_type: inputpayload
```

To delete the output and error job blobs, use the `delete_blob` JIL subcommand and specify the `blob_type` attribute with the value of *output*, as follows:

```
delete_blob: test_job_with_blob
```

```
blob_type: output
```

## Create Global Blobs

You can use the JIL `insert_glob` subcommand to upload blobs containing textual or binary data.

As the global blobs are not associated with a job, you must specify the following items:

- A unique identifier.
- Mode of the blob data that is being used in the `blob_mode` attribute.

### NOTE

If you use the `insert_glob` JIL subcommand using the same name as an existing global blob, the blob data is reinserted into the database. In this case, the original blob data is deleted, and the new blob data takes its place.

To create a global blob containing textual data, use the `insert_glob` JIL subcommand and specify the `blob_mode` attribute with a value of *text* and either the `blob_input` or `blob_file` attributes, as follows:

```
insert_glob: my_text_global_blob

blob_mode: text
blob_input: <auto_blobt>multi-lined text data for job blob</auto_blobt>
```

or

```
insert_glob: my_text_global_blob

blob_mode: text

blob_file: /my_text_global_blob_file.txt
```

**NOTE**

JIL interprets the file name that is specified in the `blob_file` attribute as a file that contains textual data and performs a conversion of the new line character.

To create a global blob containing binary data, use the `insert_glob` subcommand and specify the `blob_mode` attribute with a value of *binary* and the `blob_file` attribute, as follows:

```
insert_glob: my_binary_global_blob

blob_mode: binary
blob_file: /my_binary_global_blob_file
```

**NOTE**

You cannot use the `blob_input` attribute to create a global blob that contains the binary data.

## Delete Global Blobs

You can use the JIL `delete_glob` subcommand to delete the existing global blobs.

**NOTE**

You must provide a unique identifier because global blobs are not associated with a job.

To delete a global blob, use the `delete_glob` JIL subcommand and provide the name of an existing global blob, as follows:

```
delete_glob: my_global_blob
```

## Use Blobs in Job Definitions

You can use the `std_in_file`, `std_out_file`, and `std_err_file` attributes of the JIL `insert_job`, `update_job`, or `override_job` subcommands to reference blobs in addition to files. Based on the keyword values you specify for these attributes, AutoSys Workload Automation downloads a blob for input or uploads a job's output as blob to meet the job's needs. You can now use substitution variables which also consume blobs.

### std\_in\_file Attribute

The keywords that are supported by the `std_in_file` attribute include the following:

- **\$\$blobt**  
Uses the input job blob of the current job as input and treats the blob data as textual data.
- **\$\$blob.<job name>**  
Uses the output job blob of the specified job as input and treats the blob data as binary data.
- **\$\$blobt.<job name>**  
Uses the output job blob of the specified job as input and treats the blob data as textual data.
- **\$\$glob.<global blob name>**  
Uses the specified global blob as input and treats the blob data as binary data.
- **\$\$globt.<global blob name>**  
Uses the specified global blob as input and treats the blob data as textual data.

#### NOTE

You cannot use the keyword `$$blob` to specify the use of the current job's input blob.

### Example: Define a job that uses the output blob of its previous run as input

1. Define the job so that the job's name is in the `std_in_file` attribute using either the `$$blob.<job name>` or `$$blobt.<job name>` keyword.
2. Apply a one-time override of the `std_in_file` attribute, so that the job reads from a local file on the computer on its first run.

### std\_out\_file and std\_err\_file Attributes

The keywords that are supported by the `std_out_file` and `std_err_file` attributes include the following:

- **\$\$blob**  
Uploads the output or error of the current job as a job blob and treats the data as binary data.
- **\$\$blobt**  
Uploads the output or error of the current job as a job blob and treats the data as textual data.
- **\$\$glob.<global blob name>**  
Uploads the output or error of the current job as a global blob with the specified name and treats the data as binary data.
- **\$\$globt.<global blob name>**  
Uploads the output or error of the current job as a global blob with the specified name and treats the data as textual data.

#### NOTE

- You cannot append data to an existing job or global blob.
- AutoSys Workload Automation does not support the use of `>` or `>>` character strings in the `std_out_file` or `std_err_file` attributes.
- Existing blob data is overwritten with the new data after the job run is completed.

## Generate Blob Reports Using autorep

You can use the autorep utility to report on and download the input job blobs and global blobs. To export the job definition using the autorep -J *job\_name* -q option includes exporting all versions of that job's input blob. If a download path is not specified, the contents of all input job blobs are displayed with the job definition. Otherwise, autorep downloads the input blob to the specified directory and displays the input blob file names that are numbered by version with the job definition. Reports that are generated against one or more global blobs are extracted in binary format unless otherwise specified using the -a command line parameter. If a download path is not specified, autorep downloads the global blob into a temporary directory.

Options specific to blob and glob data are as follows:

```
[-z globalblob_name] [-type blob_type] [[-f blobfilepath] [-a] | [-n | -w] | -q]
```

```
[-jb job_name] [-type blob_type [-name blob_name]] [[-f blobfilepath] [-a] [-n]]
```

### NOTE

For information about the autorep options, see [autorep Command -- Report Job, Machine, and Variable Information](#).

### Example: Export Job Definition with Input Blobs

This example uses the autorep command to export a job definition:

```
autorep -J ALL -q
```

The output might resemble the following example:

```
insert_job: test_job

job_type: cmd

command: cat

machine: juno

owner: jerry@ca

permission: gx,ge,wx

alarm_if_fail: 1
```

If the job has one or more input blobs tied to it, in addition to the job definition, the autorep command extracts each of the job blob definitions, and the output might resemble the following example:

```
insert_job: test_job_with_blob job_type: cmd

command: cat

machine: juno
```

```

owner: jerry@juno

permission:

std_in_file: $$blobt

alarm_if_fail: 1

/* -- test_job_with_blob:insert_blob #1 -- */

insert_blob: test_job_with_blob

blob_input: <auto_blobt>multi-lined text data for job blob 1

</auto_blobt>

/* -- test_job_with_blob:insert_blob #2 -- */

insert_blob: test_job_with_blob

blob_input: <auto_blobt> multi-lined text data for job blob 2

</auto_blobt>

/* -- test_job_with_blob:insert_blob #3 -- */

insert_blob: test_job_with_blob

blob_input: <auto_blobt> multi-lined text data for job blob 3

</auto_blobt>

```

You can also specify a location to download the blobs using the `-f` parameter as follows:

```
autorep -J ALL -q -f /myblobsdir
```

The output might resemble the following example:

```

insert_job: test_job_with_blob job_type: cmd

command: cat

machine: juno

owner: jerry@juno

permission:

```

```

std_in_file: $$blobt

alarm_if_fail: 1

/* -- test_job_with_blob:insert_blob #1 -- */

insert_blob: test_job_with_blob

blob_file: /myblobsdir/test_job_with_blob_1.txt

/* -- test_job_with_blob:insert_blob #2 -- */

insert_blob: test_job_with_blob

blob_file: /myblobsdir/test_job_with_blob_2.txt

/* -- test_job_with_blob:insert_blob #3 -- */

insert_blob: test_job_with_blob

blob_file: /myblobsdir/test_job_with_blob_3.txt

```

### Example: Generate a Report for All Global Blobs

This example generates a report that downloads the contents of all global blobs to the location */myblobsdir* as binary data:

```
autorep -z ALL -f /myblobsdir
```

The report might resemble the following example:

Glob Name	File Name
MYGLOB	/myblobsdir/MYGLOB
REPORT_CHART	/myblobsdir/REPORT_CHART
ARCHIVED_DATA	/myblobsdir/ARCHIVED_DATA
JOB_SNAPSHOT	/myblobsdir/JOB_SNAPSHOT

This example generates a report that downloads the contents of all global blobs to the location */myblobsdir* as text data:

```
autorep -z ALL -f /myblobsdir -a
```

The report might resemble the following example:

Glob Name	File Name
MYGLOB	/myblobsdir/MYGLOB.txt
REPORT_CHART	/myblobsdir/REPORT_CHART.txt
ARCHIVED_DATA	/myblobsdir/ARCHIVED_DATA.txt
JOB_SNAPSHOT	/myblobsdir/JOB_SNAPSHOT.txt

[ -z GlobalBlobName ] [ -type BlobType ] [[ -f BlobDownloadDirectory ] [ -a ] | [ -n | -w ] | -q ]

## Schedule Cross-Instances

This section contains the following topics:

### Bi-Directional Scheduling

AutoSys Workload Automation supports *bi-directional scheduling*, which lets you start jobs from remote machines (inbound) or submit jobs on remote machines (outbound).

With *inbound job scheduling*, AutoSys Workload Automation acts as an agent and accepts job submissions from remote machines or other scheduling managers (such as CA Jobtrac Job Management and CA Workload Automation SE). The jobs are defined and run on the AutoSys Workload Automation instance that is acting as an agent.

With *outbound job scheduling*, AutoSys Workload Automation acts as a scheduling manager and sends job submissions to remote machines. The jobs are defined on the AutoSys Workload Automation instance that is acting as a scheduling manager. The jobs run on the remote machine or other scheduling manager.

For example, a Linux Oracle instance can initiate jobs in a Windows Microsoft SQL Server instance, or a Windows Microsoft SQL Server instance can initiate jobs in a AIX Oracle instance. You can add additional instances, such as Sybase, AIX Oracle, or HP Oracle instance, to the environment.

The AutoSys Workload Automation cross-platform interface controls the bi-directional scheduling mode. You can configure the cross-platform interface to enable the following modes:

- Outbound job scheduling
- Inbound and outbound job scheduling (bi-directional scheduling)
- No cross-platform scheduling (the default)

#### NOTE

There are no restrictions on platforms, event servers, or number of instances when running in bi-directional scheduling mode.

## Cross-Instance Job Dependencies

Multiple AutoSys Workload Automation instances are not connected, but they can communicate with one another. This communication lets you schedule workload across instances in your enterprise. You can define jobs that have dependencies on jobs running on other instances (*cross-instance job dependencies*). A AutoSys Workload Automation job with these dependencies conditionally starts based on the status of the job on the other instance. In this situation, the local instance scheduler acts as a client and issues sendevent commands to the external instance. The application server of the other instance processes the sendevent request and stores the dependency request or status update in its database. You can also manually send events from one instance to another.

When the status of a job with cross-instance dependencies changes, the scheduler sends a CHANGE\_STATUS event to the remote instance event server while the job in the local instance runs. The scheduler processes incoming events and stores the status changes in the ujo\_ext\_job table on the remote instance event server. The scheduler evaluates condition dependencies for external jobs based on the stored status.

The scheduler also sends an equivalent CHANGE\_STATUS event to the remote instance for status changes not resulting from a CHANGE\_STATUS event, specifically status changes resulting from one of the following actions:

- Unavailable machine load units, resources or agents prevent a job from running and the scheduler change the status of the job.
- The user changes the status by issuing a sendevent command for one of the following events: JOB\_ON\_HOLD, JOB\_OFF\_HOLD, JOB\_ON\_ICE, JOB\_OFF\_ICE, JOB\_ON\_NOEXEC, JOB\_OFF\_NOEXEC.

The equivalent CHANGE\_STATUS event may also result in changes to job exit codes stored on the remote instance. This behavior ensures that the scheduler accurately evaluates downstream jobs dependent on the remote jobs, including the job status and exit code conditions of the dependent jobs.

### NOTE

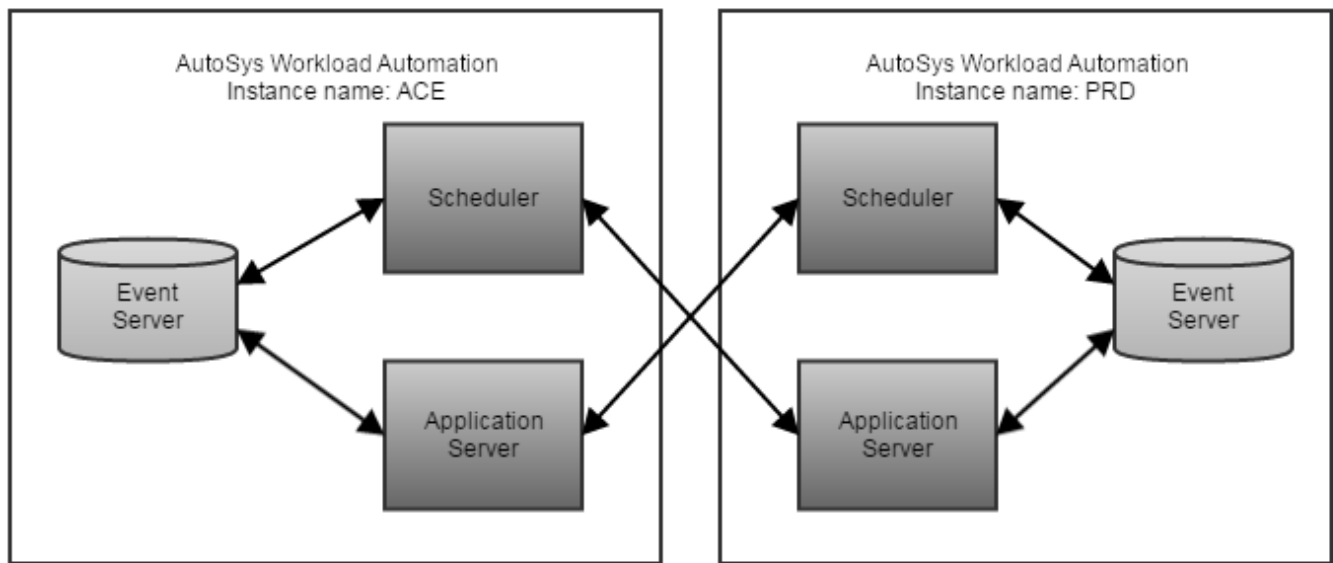
- The equivalent CHANGE\_STATUS event represents the actual status change that occurs in the local instance, and the event includes text specifying the actual status change. The remote scheduler log records this information.
- For more information about the translated status that the local scheduler sends to the remote instance, see [Administering](#).
- Before you can submit jobs on other AutoSys Workload Automation instances, you must define the instances to each other. For more information about configuring the product to support cross-instance scheduling, see [Configure Cross-Instance Dependencies with AutoSys Workload Automation](#).

## How Cross-Instance Job Dependencies are Processed

This article explains how cross-instance job dependencies are processed. You can associate jobs with more than one AutoSys Workload Automation instance. For example, you can define a job to conditionally start based on the status of a job on another instance.

The following illustration shows two instances exchanging cross-instance job dependencies:



**Figure 56: exchanging cross-instance job dependencies****NOTE**

If instance ACE's application server runs on port 9001 and instance PRD's application server runs on port 9002, you must verify that both ports are configured using SSA on both machines. Otherwise, communication with the remote application server will fail.

Each time a job definition with cross-instance job dependency is submitted to the database, the scheduler sends the `EXTERNAL_DEPENDENCY` event to the external instance. While processing the external instance, the scheduler stores the required details in the database to send job statuses to the issuing instance. The issuing instance does not send the `EXTERNAL_DEPENDENCY` event to the external instance if the same job status is required for another job. The `EXTERNAL_DEPENDENCY (DELETE)` event is sent only when the last dependency on the external job is removed.

The use of multiple databases is independent of instances using cross-instance dependencies. You can have multiple instances that each use dual event servers.

When AutoSys Workload Automation encounters a cross-instance dependency, it sends an `EXTERNAL_DEPENDENCY` event from the requesting instance. The following process occurs when one instance cannot send status updates (events) to the other instance:

- An `INSTANCE_UNAVAILABLE` alarm is issued.
- While the external instance is offline, the scheduler stores the external events to be sent to the external instance and forwards the same when the external instance is back online.
- The local instance periodically tries to connect to the external instance and marks online after it connects successfully.

**External Instance Types**

To use external job dependencies, the scheduling manager or remote machine must be defined as an *external instance* in the AutoSys Workload Automation database.

When you define the external instance, you must identify the type using the `xtype` JIL attribute. Options are the following:

**xtype: a**

Indicates that the external scheduling manager is a AutoSys Workload Automation application server instance.

**xtype: c**

Indicates that CA AutoSys WA Connect Option is installed with the external scheduling manager. CA AutoSys WA Connect Option can be installed on the mainframe and supports cross-instance jobs and job dependencies. This value lets you submit job requests to and receive job submissions from the following mainframe scheduling managers:

- – CA Jobtrac Job Management
- CA Scheduler Job Management
- CA Workload Automation SE

The AutoSys Workload Automation scheduler uses CAICCI to communicate with CA AutoSys WA Connect Option.

**xtype: u**

Indicates that CA UJMA is installed with the external scheduling manager or on the remote machine. CA UJMA can be installed on the mainframe, UNIX, and Windows. It lets you submit job requests to the remote machine where CA UJMA is installed. It lets you submit job requests to and receive job submissions from the following scheduling managers:

- CA Job Management Option
- CA Jobtrac Job Management
- CA Scheduler Job Management
- CA Workload Automation SE

The AutoSys Workload Automation scheduler uses CAICCI to communicate with CA UJMA.

**NOTE**

Unlike CA AutoSys WA Connect Option, CA UJMA does not let you define cross-instance job dependencies on the mainframe. To define cross-instance job dependencies on the mainframe, you must install CA AutoSys WA Connect Option on the same computer as the mainframe scheduling manager.

**xtype: e**

Indicates that the external scheduling manager is CA Workload Automation EE. You can define cross-instance job dependencies.

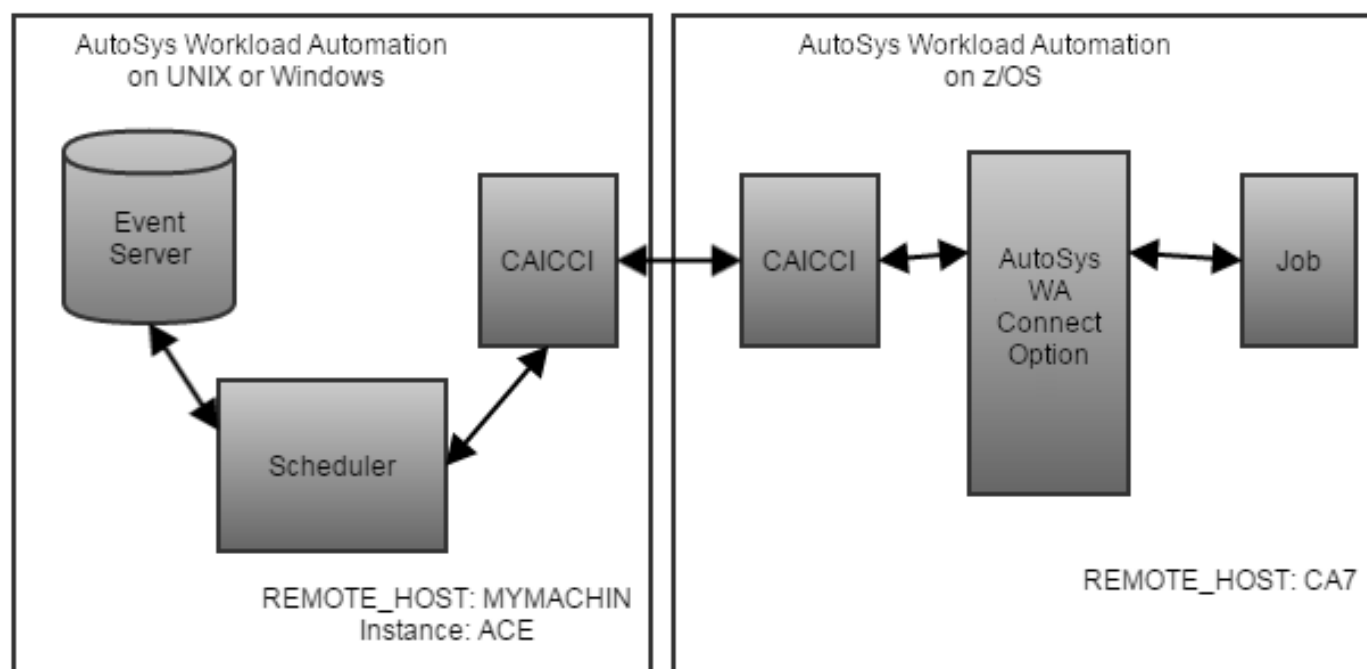
**NOTE**

Bi-directional scheduling is not supported between AutoSys Workload Automation and CA Workload Automation EE.

## Create Cross-Instance Job Dependencies Using AutoSys WA Connect Option

AutoSys Workload Automation jobs can have dependencies on jobs managed by an AutoSys WA Connect Option and a Broadcom scheduling manager running on the mainframe. The mainframe scheduling manager uses AutoSys WA Connect Option and CAICCI to communicate with AutoSys Workload Automation. The AutoSys Workload Automation scheduler also uses its cross-platform scheduling interface for communication.

For example, the following illustration shows a AutoSys Workload Automation job defined on a UNIX or Windows computer. The job's starting condition is the successful completion of a job running on the mainframe.

**Figure 57: Cross-Instance Job Dependencies with WA Connect Option**

AutoSys Workload Automation jobs can be dependent on the status of external jobs managed by CA AutoSys WA Connect Option, and external jobs can be dependent on the status of AutoSys Workload Automation jobs. AutoSys Workload Automation uses the following process to create cross-instance dependencies:

1. The AutoSys Workload Automation scheduler sends a request for the status of a CA AutoSys WA Connect Option job.
2. CA AutoSys WA Connect Option registers the request.
3. The CA AutoSys WA Connect Option job runs on the mainframe.
4. CA AutoSys WA Connect Option sends the job status to the AutoSys Workload Automation scheduler.
5. The AutoSys Workload Automation scheduler communicates the status to the event server.
6. The AutoSys Workload Automation scheduler processes the status and starts the job that is dependent on the completion of the CA AutoSys WA Connect Option job, if appropriate.

**NOTE**

For more information about the CAICCI components, see [Configure AutoSys Workload Automation to Work with Other Broadcom products](#).

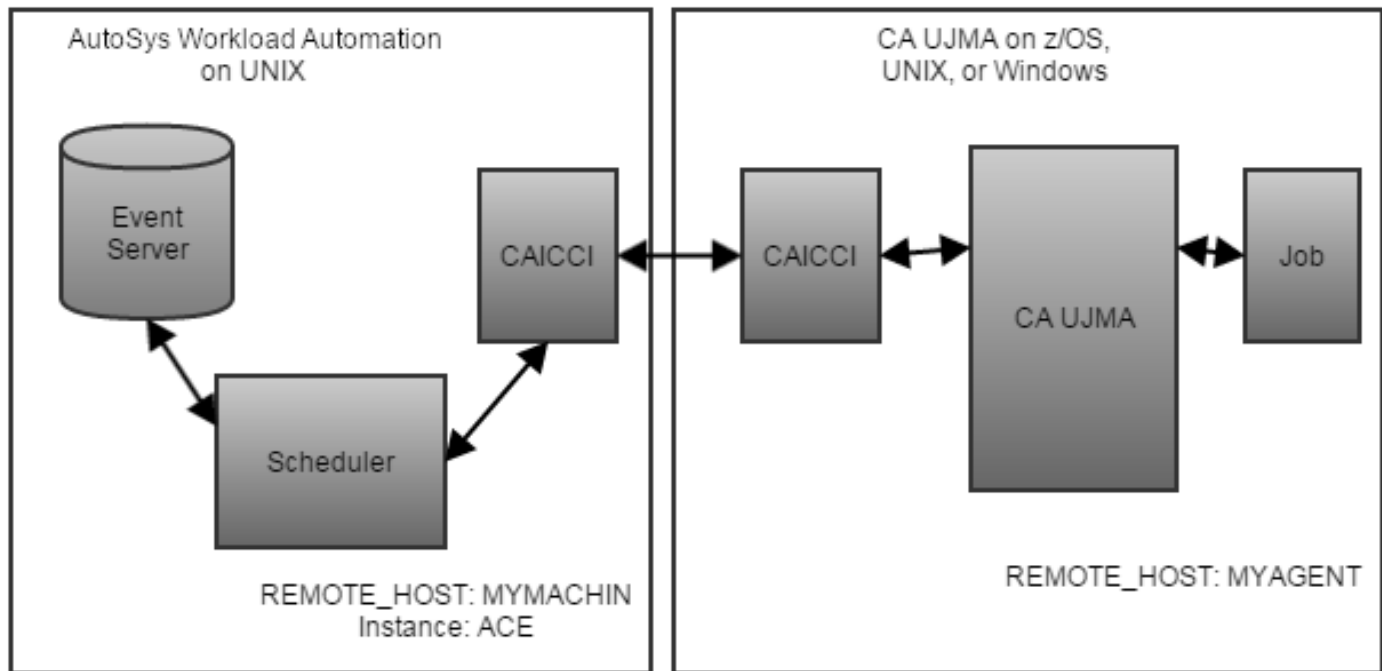
**Submit a Job to and from Another Computer Using CA UJMA**

AutoSys Workload Automation can schedule jobs on a computer that has CA UJMA installed on it. As job submission requests are processed, the scheduler log file records events sent between AutoSys Workload Automation and CA UJMA. The following descriptions help you understand the events recorded in the scheduler log file.

**NOTE**

AutoSys Workload Automation can directly schedule jobs on a computer that is running a supported workload automation agent. This topic only discusses the communication with CA UJMA.

The following diagram shows the components involved in the communication:

**Figure 58: Submitting to and from computer with CA UJMA**

The AutoSys Workload Automation scheduler communicates directly with CA UJMA using CAICCI and the AutoSys Workload Automation Cross-Platform Interface. The communication components running on the CA UJMA computer receive information from the AutoSys Workload Automation scheduler and pass it to CA UJMA. Similarly, CA UJMA passes information through the communication components to the AutoSys Workload Automation scheduler.

CA UJMA does the following:

- Receives job requests from one or more Broadcom scheduling managers (such as CA Job Management Option (JMO), AutoSys Workload Automation, and CA Workload Automation SE). CA UJMA initiates the requested program, script, JCL, or other unit of work. If you are scheduling to the mainframe, the command or program to initiate is the job name of the job as defined in the mainframe scheduling system.
- Collects status information about job runs.
- Sends status information to the requesting scheduling manager.

**NOTE**

Unlike CA AutoSys WA Connect Option, CA UJMA does not support cross-instance job dependencies.

The following process is used to start a job on a CA UJMA computer:

- You define a job on AutoSys Workload Automation that specifies the job to start on the CA UJMA computer. The mainframe job to start (specified in the command attribute of the AutoSys Workload Automation job definition) must be

a named job known to the AutoSys Workload Automation scheduler. The mainframe job to start cannot be a command or script.

- The AutoSys Workload Automation scheduler sends the job information to CA UJMA.
- The job changes to STARTING status.
- CA UJMA starts the job and sends an event to the scheduler. The event sent is JOBINITU if the job started or JOBFAILU if the job could not start.
- The scheduler converts the JOBINITU event to RUNNING, puts it in the database, and updates the job's status to RUNNING. If CA UJMA sent a JOBFAILU event, the scheduler converts the event to FAILURE and processes it accordingly.
- If the job completes successfully, CA UJMA sends a JOBTERMU event to the scheduler.
- The scheduler converts the JOBTERMU event to SUCCESS, FAILURE, or TERMINATED based on the exit code of the job. If the job exited with a normal end of job code (EOJ) the scheduler converts JOBTERMU to SUCCESS or FAILURE. If the job exited with an abnormal end of job code (AEOJ), the scheduler converts JOBTERMU to TERMINATED.

AutoSys Workload Automation can also receive job submission requests from a CA UJMA computer. The job to start must be a job defined on AutoSys Workload Automation and cannot be a command or script. If AutoSys Workload Automation receives a job submission request from the mainframe, the job to run (specified by the SUBFILE parameter of the mainframe job) must be defined as a valid job on the AutoSys Workload Automation computer.

For the AutoSys Workload Automation scheduler to communicate with CA UJMA computers, the scheduler must convert AutoSys Workload Automation-based events to events that CA UJMA can interpret. Similarly, the AutoSys Workload Automation scheduler must convert events returned from CA UJMA back to events the scheduler can interpret.

The following table lists the AutoSys Workload Automation and CA UJMA events:

Operation	AutoSys Workload Automation	CA UJMA
Starting a job	STARTJOB	SUBMITU
Job has started and is running	RUNNING	JOBINITU
Job has terminated successfully with an exit code	SUCCESS or FAILURE	JOBTERMU
Job has failed to start	FAILURE	JOBFAILU

## Unsupported Attributes for CA AutoSys WA Connect Option or CA UJMA Jobs

The following table lists attributes that are not supported for CA AutoSys WA Connect Option and CA UJMA jobs. If you specify these attributes, they are ignored.

JIL Attribute	Web UI Field
chk_files	File system check
heartbeat_interval	Heartbeat interval
job_load	Job load
job_terminator	Terminate on failure of containing box
job_type:f	File Watcher (FW) job in either Quick Edit or Application Editor
n_retrys	Times to restart job after failure
priority	Priority
profile	Profile
std_err_file	Standard error file
std_in_file	Standard input file

std_out_file	Standard output file
term_run_time	Minutes to wait before terminating
watch_file	Files to watch
watch_file_min_size	Minimum file size
watch_interval	Watch interval

**NOTE**

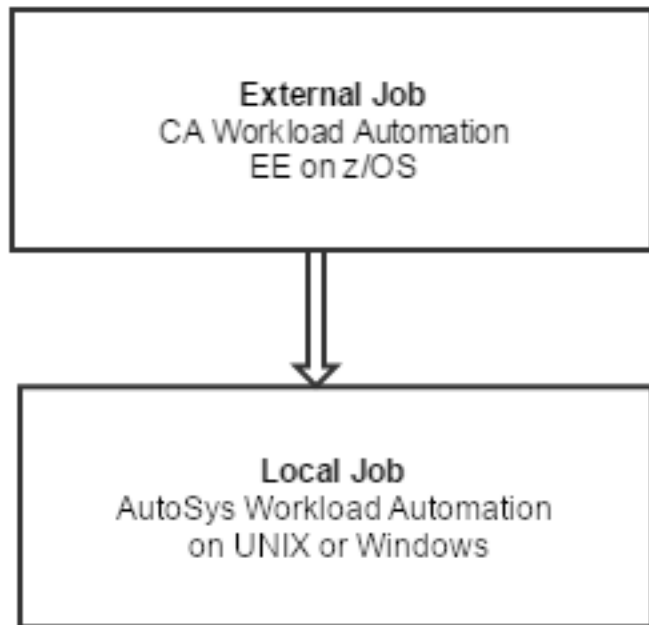
Computers that are managed by CA UJMA computers cannot be part of a virtual machine. The max\_load and factor attributes are also not supported when defining an external CA UJMA instance.

## How CA Workload Automation EE Processes Job Dependencies

You can define job dependencies between AutoSys Workload Automation and an external CA Workload Automation EE instance running on the mainframe. As job dependencies are processed, the scheduler log file records events communicated between AutoSys Workload Automation and CA Workload Automation EE. The following descriptions help you understand the events recorded in the scheduler log file.

A AutoSys Workload Automation job can depend on another job on CA Workload Automation EE as shown in the following diagram:

**Figure 59: How Job Dependencies Are Processed**



The job dependency is processed as follows:

- The job dependency is specified in the condition attribute of the local AutoSys Workload Automation job definition. For example, the following job depends on the success of the ZOS01 job defined on the CA Workload Automation EE instance named MSR:

```
insert_job: MSR1
```

```
job_type: CMD
```

```

command: sleep 10

machine: localhost

owner: root@hostname.ca.com

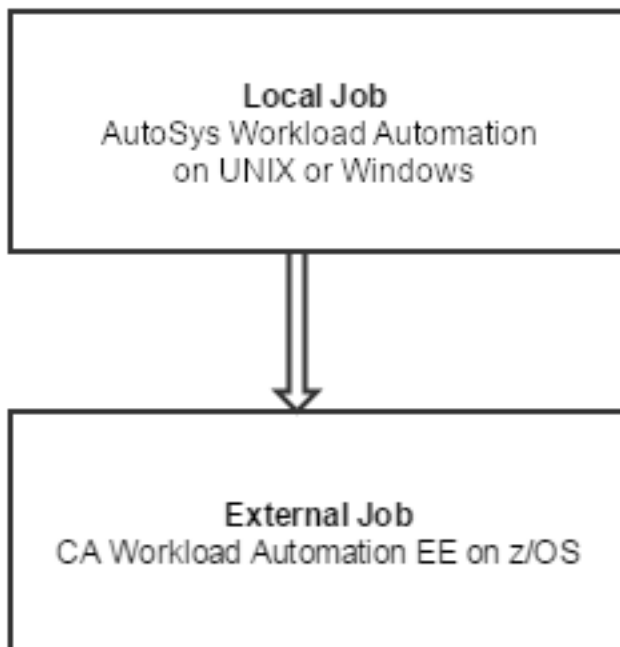
condition: s(ZOS01^MSR)

```

- When the AutoSys Workload Automation job is created or updated, an `EXTERNAL_DEPENDENCY` event is created in the AutoSys Workload Automation database. The event contains the information in the condition attribute.
- The AutoSys Workload Automation scheduler sends a job dependency request to CA Workload Automation EE.
- As the job runs on the external instance, CA Workload Automation EE sends status updates to AutoSys Workload Automation.
- The AutoSys Workload Automation scheduler updates the `ujo_ext_job` table with the status updates from CA Workload Automation EE.
- When the AutoSys Workload Automation job is deleted or its condition attribute is changed, an `EXTERNAL_DEPENDENCY` event is created in the AutoSys Workload Automation database. The event contains the information required to remove the external job dependency from the `ujo_ext_job` table.
- The AutoSys Workload Automation scheduler sends a request to CA Workload Automation EE to deregister the job dependency on the external instance.

A job on CA Workload Automation EE can depend on a AutoSys Workload Automation job as shown in the following diagram:

**Figure 60: Local To External**



The job dependency is processed as follows:

- When a new generation of an application is started in the external instance, CA Workload Automation EE sends a job dependency request to AutoSys Workload Automation for a single run of a job.
- The AutoSys Workload Automation scheduler creates an EXTERNAL\_DEPENDENCY event on its behalf.
- The AutoSys Workload Automation scheduler updates the ujo\_req\_job database table to record that a CA Workload Automation EE job is dependent on the local job.
- As the job runs on the local AutoSys Workload Automation instance, the AutoSys Workload Automation scheduler sends status updates to CA Workload Automation EE.
- When the job has its run, the AutoSys Workload Automation scheduler deletes the dependent job entry from the ujo\_req\_job table.

The following process occurs when AutoSys Workload Automation cannot send status updates (events) to CA Workload Automation EE:

- An INSTANCE\_UNAVAILABLE alarm is issued.
- The ujo\_aset\_config table is updated to indicate that the external instance is offline.
- While the job continues to run, all events to be sent to the external instance are stored in the ujo\_ext\_event table.
- The AutoSys Workload Automation scheduler periodically tries to connect to CA Workload Automation EE.
- When the AutoSys Workload Automation scheduler successfully re-connects to CA Workload Automation EE, all the events are sent to CA Workload Automation EE, and the events are deleted from the ujo\_ext\_event table.

To communicate with CA Workload Automation EE, the AutoSys Workload Automation scheduler converts AutoSys Workload Automation-based events to events that CA Workload Automation EE can interpret. Similarly, the scheduler must convert events returned from CA Workload Automation EE back to events that the scheduler can interpret. All converted events are recorded in the scheduler log file to indicate that the events are from an external instance.

The following table lists the AutoSys Workload Automation and CA Workload Automation EE events:

AutoSys Workload Automation Event	CA Workload Automation EE Event
STARTING	READY
RUNNING	EXEC
SUCCESS	COMPLETED
FAILURE	FAILED
TERMINATED	FAILED with a Job Terminated status
ON_ICE	COMPLETED with a BYPASSED status
INACTIVE	HELD

#### NOTE

- AutoSys Workload Automation communicates directly with CA Workload Automation EE. You do not have to install additional products to integrate the scheduling managers.
- Bi-directional scheduling is currently not supported between AutoSys Workload Automation and CA Workload Automation EE. You cannot submit jobs to or receive job requests from CA Workload Automation EE at this time.

## Cross-Platform Scheduling

*Cross-platform scheduling* lets you schedule and reroute jobs between AutoSys Workload Automation and other machines running on different platforms, including mainframe.

To use cross-platform scheduling, required components must be installed on the AutoSys Workload Automation computer and on the external machine that AutoSys Workload Automation works with. The scheduling manager or remote machine is also defined as an *external instance* in the AutoSys Workload Automation database.



When the scheduler receives the cross-platform start job requests, it requests the application server to authenticate and authorize that request. For the job to execute successfully, you must have EXECUTE access for the following resource classes:

- as-appl
- as-group
- as-machine
- as-job

#### NOTE

Before you can submit jobs on other scheduling managers, activate the AutoSys Workload Automation cross-platform interface and define the scheduling manager as an external instance on AutoSys Workload Automation. For more information about configuring the product to support cross-platform scheduling, see [Configure Cross-Instance Dependencies with AutoSys Workload Automation](#).

## Submit a Job to and from the Mainframe Using AutoSys WA Connect Option

AutoSys Workload Automation lets you schedule workload across distributed and mainframe platforms. You can submit jobs on the mainframe and receive job submission requests from the mainframe. Depending on the mainframe scheduling manager, the mainframe must have the following software installed on it:

- CA Jobtrac Job Management and AutoSys WA Connect Option
- CA Scheduler Job Management and AutoSys WA Connect Option
- CA Workload Automation EE
- CA Workload Automation SE and AutoSys WA Connect Option

AutoSys Workload Automation and the mainframe scheduling manager use CAICCI to communicate.

When submitting a job to the mainframe, the following process occurs:

- The AutoSys Workload Automation scheduler on the distributed platform interrogates the start job request, processes it, and transmits it to the scheduling manager installed on the mainframe.
- After the scheduling manager receives the request on the mainframe, it submits and tracks the mainframe job. The job submission and tracking is completed in one of the following ways:
  - Directly by the XPS-enabled mainframe scheduling product
  - Through AutoSys WA Connect Option

When AutoSys Workload Automation accepts a job submitted from a mainframe system, AutoSys Workload Automation reports its status back to the scheduling manager on the mainframe.

#### NOTE

For more information about submitting the job from an external mainframe scheduling manager, see the appropriate product documentation.

## Cross-Platform Interface Messages Logged for CA UJMA

On UNIX, all messages that are produced by the cross-platform interface are written to the AutoSys Workload Automation scheduler log, which is located in the \$AUTOUSER/out directory.

On Windows, all messages that are produced by the cross-platform interface are written to the AutoSys Workload Automation scheduler log, which is located in the %AUTOUSER%\out directory.

The following message indicates that the scheduler has transferred a job to the cross-platform interface for submission to a CA UJMA computer:

---

```
CAUAJM_I_10073 AutoSys --> Cross Platform Interface:
```

```
machine=machine_name job_name= job_name
```

- ***machine\_name***  
Identifies the CA UJMA computer to which the job is being submitted.
- ***job\_name***  
Identifies the AutoSys Workload Automation job name as defined to the event server.

The following message indicates that an event status has been received from CA UJMA. The event status is converted to the appropriate AutoSys Workload Automation event status and inserted in the event server.

```
CAUAJM_I_40263 EVENTU: event_name
```

```
EXITCODE: exitcode/dbcode JOB: job_name
```

- ***event\_name***  
Identifies one of the following events:
  - **JOBINITU**  
Indicates that a job has started on a CA UJMA computer.
  - **JOBTERMU**  
Indicates that a job has completed on a CA UJMA computer.
  - **JOBFAILU**  
Indicates that a job has failed to start on a CA UJMA computer.
  - **SUBMITU**  
Indicates that a job has been submitted to AutoSys Workload Automation.
- ***exitcode/dbcode***  
Identifies the actual job exit code that CA UJMA returns.
- ***job\_name***  
Identifies the AutoSys Workload Automation job name as defined to the event server.

## Define an External Instance

To use cross-instance scheduling, define the external instance to AutoSys Workload Automation. The external instance can be another AutoSys Workload Automation instance or a scheduling manager running on another platform.

### Follow these steps:

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following definition:
 

```
insert_xinst: instance_name
xtype: a | c | u | e
xmachine: host_name
```

  - **xtype: a | c | u | e**  
Specifies the external instance type. Options include the following:

- **a**  
Identifies a remote AutoSys Workload Automation application server instance.
- **c**  
Identifies a CA AutoSys Workload Automation Connect Option instance.
- **u**  
Identifies a CA Universal Job Management Agent or CA NSM instance.
- **e**  
Identifies a CA Workload Automation EE instance.

3. (AutoSys Workload Automation instances only) Specify the following additional attributes:

```
xcrypt_type: NONE | DEFAULT | AES
xkey_to_manager: encryption_key
xport: port_number
```

4. (CA Workload Automation EE instances only) Specify the following additional attributes:

```
xmanager: manager_name
xport: port_number
xcrypt_type: NONE | AES
xkey_to_manager: encryption_key
```

5. Do *one* of the following:

- Enter **exit** if you are using interactive mode.
- Redirect the script to the jil command if you are using a script.

The insert\_xinst subcommand is issued and the specified external instance is defined.

#### NOTE

- For CA Workload Automation EE, the xcrypt\_type value must match the encryption type specified in the AGENTDEF data set.
- For more information about the syntax for the insert\_xinst subcommand and related attributes, see [AE Job Information Language](#).

### Example: Define a CA Workload Automation EE Instance

This example defines the CA Workload Automation EE instance named CYB to AutoSys Workload Automation. CYB runs on the CYBHOST computer at port 7550. The CM manager name is unique to this CA Workload Automation EE instance. The AutoSys Workload Automation scheduler communication alias is defined in the AGENTDEF data set on CA Workload Automation EE. CA Workload Automation EE is also configured to point to the AutoSys Workload Automation computer on the auxiliary listening port.

```
insert_xinst: CYB
xtype: e
xmanager: CM
xmachine: CYBHOST
xcrypt_type: NONE
xport: 7550
```

### Update an External Instance

You can update an external instance defined in AutoSys Workload Automation. The external instance can be another AutoSys Workload Automation instance or a scheduling manager running on another platform. You can update the instance definition when the connection information changes. The connection information must be accurate so that the scheduling managers can communicate with each other.

**Follow these steps:**

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following JIL subcommand:

```
update_xinst: instance_name
```

3. Specify the following attributes as appropriate:

```
xmachine: host_name
```

```
xcrypt_type: NONE | DEFAULT | AES
```

```
xkey_to_manager: encryption_key
```

```
xport: port_number
```

```
xmanager: manager_name
```

4. Do *one* of the following:
  - Enter **exit** if you are using interactive mode.
  - Redirect the script to the jil command if you are using a script.The update\_xinst subcommand is issued and the specified external instance is updated.

**NOTE**

For more information about the syntax for the update\_xinst subcommand and related attributes, see [AE Job Information Language](#).

**Example: Update a CA Workload Automation EE Instance**

Suppose that an external instance is defined on AutoSys Workload Automation for the CA Workload Automation EE instance named CYB. This example updates the external instance definition so that CYB points to the CYBHOST2 computer at port 7551.

```
update_xinst: CYB
```

```
xmachine: CYBHOST2
```

**Delete an External Instance**

You can delete an external instance that you no longer use from AutoSys Workload Automation.

**Follow these steps:**

1. Do *one* of the following tasks:
  - Issue JIL in interactive mode.
  - Open a JIL script in a text editor.
2. Specify the following JIL subcommand:

```
delete_xinst: instance_name
```

3. Do *one* of the following:
  - Enter **exit** if you are using interactive mode.
  - Redirect the script to the jil command if you are using a script.

The insert\_xinst subcommand is issued and the specified external instance is deleted from AutoSys Workload Automation.

**NOTE**

For more information about the syntax for the delete\_xinst subcommand, see [AE Job Information Language](#).

**Example: Delete a CA Workload Automation EE Instance**

This example deletes the definition of the CA Workload Automation EE instance named CYB.

```
delete_xinst: CYB
```

**Start a Job on an External AutoSys Workload Automation Instance**

This article explains how to start a job on an external AutoSys Workload Automation instance. AutoSys Workload Automation instances are not connected, but they can communicate with one another. You can send a STARTJOB event directly from one instance to another. This helps you integrate your workload across AutoSys Workload Automation instances in your enterprise.

To start a job on an external AutoSys Workload Automation instance, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
sendevent -E STARTJOB -J job_name -S autoserv
```

***job\_name***

Specifies a job defined on the *autoserv* instance.

***autoserv***

Specifies the instance's unique, capitalized three-character identifier.

**Example: ACE**

The sendevent command is issued and the job request is submitted on the external AutoSys Workload Automation instance.

**NOTE**

- To use the sendevent -S option, the instance from where you issue the sendevent command must have a client installed for the external instance. For more information about configuring cross-instance communication, see [Configure Cross-Instance Dependencies with AutoSys Workload Automation](#).
- For more information about the syntax for the sendevent command and related attributes, see [AE Commands](#).

**Define a Job to Run on an External Instance**

From your local AutoSys Workload Automation instance, you can submit a job to run on an external instance. The external instance is a different scheduling manager that is installed on another platform, including mainframe. Submitting external jobs helps you integrate workload across platforms in your enterprise.

**NOTE**

Before you can submit jobs on other scheduling managers, activate the AutoSys Workload Automation cross-platform interface and define the scheduling manager as an external instance on AutoSys Workload Automation. For more information about configuring AutoSys Workload Automation to support cross-platform scheduling, see [Configure Cross-Instance Dependencies with AutoSys Workload Automation](#).

**Follow these steps:**

1. Insert a job and specify the following attributes in the definition:
  - **job\_type: CMD**  
Specifies that the job type is Command.
  - **machine**  
Specifies the name of the external instance.  
**Note:** The external instance must be defined on AutoSys Workload Automation using the insert\_xinst subcommand.
  - **owner**  
Specifies the user on the external instance that the job runs under. The owner must be specified as *owner@machine*.  
**CA UJMA Note:** The specified owner must have an account on the external CA UJMA computer. The account must match the owner name exactly, and the owner (user ID and password) must be defined using the autosys\_secure command.
  - **command**  
Specifies the job to run on the external instance when all the starting conditions are met.  
**CA UJMA Limits (UNIX or Windows):** Where the operating system permits, CA UJMA job names on distributed systems can contain up to 64 alphanumeric characters and can contain both uppercase and lowercase characters. You cannot use blank spaces and tab characters.  
**CA UJMA Limits (mainframe):** CA UJMA job names on the mainframe must follow these guidelines:
    - The first character of a job name must be an uppercase letter (A-Z), a pound sign (#), an at sign (@), or a dollar sign (\$).
    - The remaining characters in the job name can be any combination of uppercase letters (A-Z), numbers (0-9), pound signs (#), at signs (@), and dollar signs (\$).
    - All letters (A-Z) must be in uppercase.
    - Job names can be up to eight characters in length.
2. (Optional) Specify common attributes that apply to all jobs.  
The external job is defined.

**NOTE**

For more information about the syntax for the JIL attributes, see [AE Job Information Language](#).

**Example: Define a Job to Run on an AS/400 Computer**

This example defines a command job to run on an AS/400 computer.

```
insert_job: as400_a1
job_type: CMD
command: DLYJOB DLY(15)
machine: usprncax
owner: user1@usprncax
permission: gx,wx
date_conditions: 1
days_of_week: all
start_mins: 30
```

### Example: Define a Job to Run Through CA AutoSys WA Connect Option

This example defines a command job to run in CA Workload Automation CA 7 Edition through AutoSys Workload Automation Connect.

```
insert_job: ca71
job_type: CMD
command: auto_cnct -a A87SOENF -j RYAKEJ01 -c RUN -p SCHEDULE=RYAKE01 -s CA7
machine: A87SOENF
owner: user1@A87SOENF
permission: gx,wx
date_conditions: 1
days_of_week: all
start_mins: 45
```

### Example: Define a Job to Run Directly in CA Workload Automation SE

This example defines a command job to run directly in CA Workload Automation SE.

```
insert_job: ca72
job_type: CMD
command: RYAKEJ01
machine: A87SOENF
owner: user1@A87SOENF
permission: gx,wx
date_conditions: 1
days_of_week: all
start_mins: 45
```

### Example: Define a Job to Run in Another AutoSys Workload Automation Instance

This example defines a command job to run in another AutoSys Workload Automation instance.

```
insert_job: ca72
job_type: cmd
command: job_in_other_instance
machine: othermachine
owner: user1@othermachine
permission: gx,wx
date_conditions: 1
days_of_week: all
```

```
start_mins: 45
```

This example uses the following configuration:

- In the machine definition for othermachine, *machine\_type* was set to u, indicating a machine that runs CA UJMA.
- The cross-platform interface was activated on the server (where job ca72 is being submitted) by setting the CrossPlatformScheduling parameter to 1 (run jobs directly on a CA UJMA agent). If this instance receives job submissions from any scheduling manager in the enterprise, set the CrossPlatformScheduling parameter to 2 (enable bi-directional scheduling support) instead.
- The CrossPlatformScheduling parameter for the machine othermachine was set to 2 (enable bi-directional scheduling support).

### Example: Define a Job to Run on a UNIX Computer

This example defines a command job to run on a UNIX computer.

```
insert_job: soljob1
command: "@SYS$LOGIN:SCHEDULE_WAIT.COM "
```

```
machine: mylinux
owner: user1@mylinux
max_exit_success: 1
```

#### NOTE

A job that runs successfully on a UNIX computer returns an exit code of 1. By default, AutoSys Workload Automation interprets an exit code of 1 as a failure unless the max\_exit\_success attribute is properly set in the job definition.

## Define a Cross-Instance Job Dependency

You can define a job to depend on another job that runs on an external instance. The external instance can be another AutoSys Workload Automation instance or a scheduling manager running on another platform.

To define a cross-instance job dependency, add the following attribute to your job definition:

```
condition: status(JOB_NAME^INS) [AND|OR status(JOB_NAME^INS) ...]
```

- **status**

Specifies the status that the external job must have before your job starts. Status options are as follows:

- done
- failure
- notrunning
- success
- terminated

- **JOB\_NAME**

Specifies the name of the external job that the local job depends on.

**Limits:** The names of jobs that are specified as job dependencies between AutoSys Workload Automation and CA AutoSys WA Connect Option must follow these guidelines:



- The first character of a job name must be an uppercase letter (A-Z), a pound sign (#), an at sign (@), or a dollar sign (\$).
- The remaining characters in the job name can be any combination of uppercase letters (A-Z), numbers (0-9), pound signs (#), at signs (@), and dollar signs (\$).
- All letters (A-Z) must be in uppercase.
- Job names can be up to eight characters in length.
- **INS**  
Specifies the name of the external instance that *JOB\_NAME* runs on. The name must be three uppercase alphanumeric characters. The first character must be a letter (A-Z).

**NOTE**

You can specify multiple job dependencies in a definition.

**Example: Specify a Cross-Instance Job Dependency**

This example defines a job that runs only when the following starting conditions are met:

- jobA on the same instance returns SUCCESS
- jobB on the AutoSys Workload Automation instance PRD returns SUCCESS

```
insert_job: dep_job1

machine: localhost

job_type: CMD

command: sleep 100

condition: success(jobA) AND success(jobB^PRD)
```

**Generate a Report on an External Instance**

You can generate a report on an external instance to verify that the instance configuration is correct. Generating a report is helpful when you want to check the status of an external instance during failover. The event details in a report can also help you determine whether the correct events are waiting for updates from the external instance.

To generate a report on an external instance, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autorep -X external_instance [-q]
```

**-X external\_instance**

Specifies the external instance that you want to generate a report on.

**-q**

(Optional) Generates a query report, which contains the current job or machine definition.

The autorep command is issued and the report is generated and displayed.

**Example: Generate a Report on an External Instance**

This example generates a report on the external machine named PRD.

```
autorep -X PRD
```

Name	Type	Server	Port
PRD	a	nyc-wall-04	9001

### Example: Generate a Report on an External Instance

This example generates a query report on the external machine named PRD. The report displays the machine definition.

```
autorep -X PRD -q
```

```
/* ----- PRD ----- */
```

```
insert_xinst: PRD
```

```
xtype: a
```

```
xmachine: nyc-wall-04
```

```
xport: 9001
```

```
xcrypt_type: DEFAULT
```

### Example: Export External Instance Definitions

This example uses the autorep command to export all external instance definitions.

```
autorep -X ALL -q
```

The output might resemble the following example:

```
/* ----- CCT ----- */
```

```
insert_xinst: CCT
```

xtype: c

xmachine: WACNCTHOST

/\* ----- NSM ----- \*/

insert\_xinst: NSM

xtype: u

xmachine: NSMHOST

/\* ----- WAE ----- \*/

insert\_xinst: WAE

xtype: a

xmachine: WAAEHOST

xport: 9001

xcrypt\_type: DEFAULT

/\* ----- WEE ----- \*/

insert\_xinst: WEE

xtype: e

xmachine: WAEHOST

xport: 7550

```
xcrypt_type: NONE
```

```
xmanager: WAAEMGR
```

### Example: Generate a Report for All External Instances

This example uses the autorep command to generate a report of all external instances:

```
autorep -X ALL
```

The report might resemble the following example:

Name	Type	Server	Port
CCT	c	WACNCTHOST	0
NSM	u	NSMHOST	0
WAE	a	WAAEHOST	9001
WEE	e	WAAEHOST	7550

## Web UI Scheduling

Workload Scheduling is intended for schedulers, supervisors, application developers and users, console operators -- anyone who is responsible for defining and scheduling workload using Web UI. This information applies to the Application Editor, Quick Edit, and Quick View applications of Web UI, all of which show workload object properties; however, the tasks and examples apply to Application Editor and Quick Edit only because they relate to scheduling, not monitoring. Refer to the related documentation for other Web UI applications.

### Documentation Conventions

The following documentation conventions are used in this section:

- In the properties topics, fields marked as “Optional” do not require input in Web UI and are not required by AutoSys Workload Automation to create the job. If there are no default values displayed in Web UI for these fields and they are left blank, no values are passed to AutoSys Workload Automation for these properties. Fields marked as “Defaulted” are assigned values by AutoSys Workload Automation when creating the job but do not require input in Web UI. If these fields are left blank, the values documented in the Default: descriptions in this section are used to create the job. Fields not marked as either “Optional” or “Defaulted” do require input in Web UI.
- If a dialog is available to assist you in completing a field, it is documented in this section. However, in most cases, you are not required to use the dialog to complete your entry.

## **Application Editor**

Application Editor lets you work with a graphical representation of jobs in a job flow. The job flow shows the relationships between jobs and their dependencies. You can manage individual jobs using Application Editor; however, the focus is on the job flow.

## **Quick Edit**

Quick Edit lets you create, edit, delete, and search for individual objects: jobs, machines, calendars, and global variables.

## **Agents and Agent Plug-ins**

Agents are the key integration components of Broadcom workload automation products. Agents let you automate, monitor, and manage workload on all major platforms, applications, and databases. To run workload on a particular system, you install an agent on that system. If your workload must run on a UNIX computer, for example, you can install and configure the Workload Automation Agent for UNIX. The agent can run UNIX scripts, execute UNIX commands, transfer files using FTP, monitor file activity on the agent computer, and perform many other tasks.

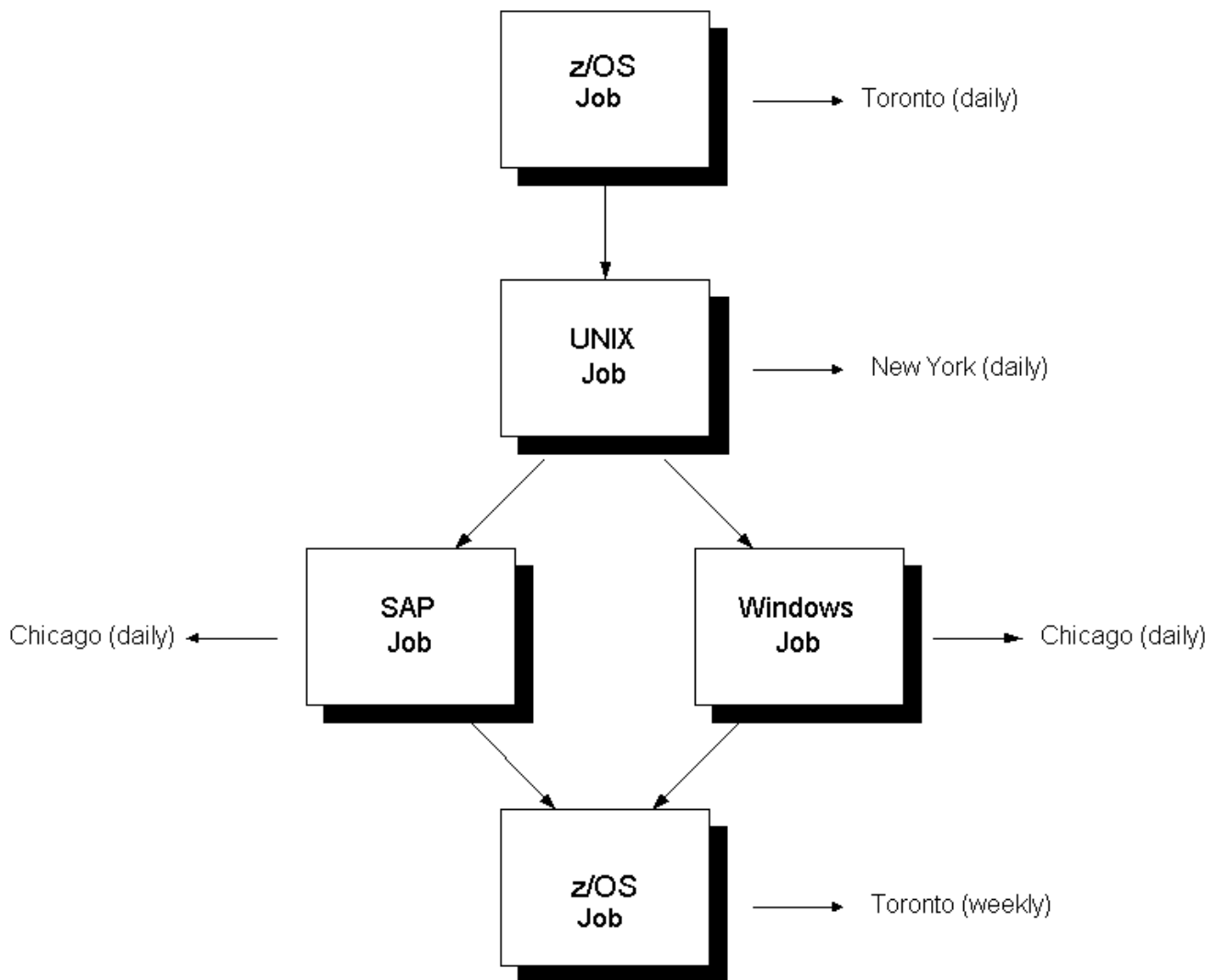
You can extend the functionality of the agent by installing one or more agent plug-ins in the agent installation directory. If you have a relational database such as Oracle, you can install a database agent plug-in to query and monitor the database. Other agent plug-ins are also available. For more information about agent plug-ins, see [Implementation](#) for the appropriate agent plug-in.

### **NOTE**

The agent plug-ins are only available for UNIX, Linux, and Windows operating environments.

## **Example: Workload with Different Types of Jobs**

The following workload contains z/OS jobs, a UNIX job, an SAP job, and a Windows job, running on different computers, in different locations, and at different times:

**Figure 61: Workload with Different Types of Jobs**

## Manage Calendars Using Web UI

Calendars are used for scheduling jobs to identify the days when jobs should or should not process. You can define a custom calendar to include any number of dates. You can then associate any calendar with one or more jobs. A custom calendar can be created using a standard calendar or an extended calendar.

A *standard calendar* is a set of manually selected, user-specified dates. There are two types of standard calendars: run and exclude. A *run calendar* specifies the days on which a job should run, such as the first weekday of every month. Conversely, an *exclude calendar* specifies the days on which a job should not run. For example, to help ensure that a job never runs on any corporate holiday, you could create a calendar named Holidays that has each corporate holiday selected.

An *extended calendar* is a dynamic calendar that is generated automatically based on a combination of holiday calendars, cycles, and conditions defined.

Each calendar is stored in the database as a separate object with a unique name.

Cycles are used as processing periods in an extended calendar definition. These user-defined date ranges make it easy to define rules for non-standard processing periods, such as business cycles, quarters, and accounting periods. Cycles can be set to repeat every year.

## **Define a Standard Calendar**

You can create a standard calendar to define specified dates on which jobs should run.

### **Follow these steps:**

1. Open Quick Edit.
2. Click Create.
3. Select the server where you want to store the calendar from the Server drop-down list.
4. Click the Calendar link.
5. Specify the following required property:

- **Name**  
Specifies a name for the calendar.  
**Limits:** Up to 30 characters

### **NOTE**

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

6. (Optional) Specify when the calendar starts.
  - a. Select the starting month.
  - b. Select the starting year.
  - c. Click Go.
7. Perform one of the following options:
  - Click individual days in the calendar to set the processing days.
  - Define rules in the Generate Days section for determining processing and blocked days in the calendar and for rescheduling processing when schedule conflicts arise.
8. (Optional) Provide a description
  - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

### **NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

9. Click Commit.  
The standard calendar is defined.

## **Standard Calendar Properties**

The Calendar Properties section for standard calendars includes the following properties:

### **Calendar**

Defines days on which jobs associated with the calendar can or cannot run. You can select a specific year and month, and then click the appropriate days to select or clear them.

#### **Name**

Specifies the name of the calendar.

**Limits:** Up to 30 alphanumeric characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-)

**NOTE**

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

**Description**

(Optional) A non-mandatory description that contains any text.

**Limits:** Up to 1024 characters.

**NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

**Starting Month**

(Optional) Specifies the starting month for the selected year.

**Default:** Either the current month or the month specified

**Starting Year**

(Optional) Specifies the year for the selected Starting Month.

**Default:** Either the current year or the year specified

**Generate Days Properties**

You can use the Generate Days section to define and apply rules that more specifically control processing calendars.

The Primary, Recurrence Pattern, and Reschedule Rule categories include the following properties:

**Action**

(Optional) Specifies whether the generation rule you are creating defines set, unset, or blocked dates on the calendar.

The following options are available:

- **Set Processing days**

Indicates that the current settings should be used to set processing days on the calendar.

- **Set Blocked days**

Indicates that the current settings should be used to clear processing days on the calendar. Unset days may be used for processing, for example, when a rescheduling rule is used to reschedule processing in the event of a conflict.

- **Clear Processing days**

Indicates that the current settings should be used to clear processing days on the calendar. Unset days may be used for processing, for example, when a rescheduling rule is used to reschedule processing in the event of a conflict.

- **Clear Blocked days**

Indicates that the current settings should be used to clear blocked days on the calendar.

- **Default:** Set Processing days

**Days**

(Optional) Specifies the days on which the rule applies, based on your selections for Occurrence and Periods.

The following options are available:

- Any day
- Weekday
- Specific day



The name of the associated field changes to reflect your selection, letting you select one or more specific days if Specific day is selected.

- **Default:** Any day

### Date range

(Optional) Specifies whether the rule applies to the specified year or to a specified date range.

The following options are available:

- **All days in year**  
Indicates that the rule applies to the year you select from the associated drop-down list.
- **All days in range**  
Indicates that the rule applies to the interval between the dates you specify in the associated Start and End fields.

The name of the associated field changes to reflect your selection, letting you enter the range or select a year as appropriate.

- **Default:** All days in year, with the current year selected from the associated list

### Occurrence

(Optional) Specifies the occurrences of the value selected for Days on which to apply the rule for the duration selected for Periods.

The following check boxes are available:

- First
- Second
- Third
- Fourth
- Every
- Next to last
- Last

You can make multiple selections.

The following fields are also available:

- **Nth**  
Specifies a numeric value to set a one-time occurrence, such as 5 to indicate on the 5th occurrence.
- **Every Nth**  
Specifies a numeric value to set a repeating occurrence, such as 6 to indicate every 6th occurrence.

### Periods

(Optional) Specifies the interval to which the rule applies.

The following options are available:

- No period
- Monthly
- Quarterly
- Number of weeks -- Specifies the number of weeks between 1 and 99 to which this rule will apply.
- In calendar -- Specifies the name of the calendar to which this rule will apply.

**Default:** No period

The name of the associated field changes to reflect your selection, letting you enter a number or name as appropriate.

**Example:** To define a rule to run a job on the first Monday of each quarter, select First for Occurrence, Specific day and Monday for Days, and Quarterly for Periods.

### Reschedule

(Optional) Specifies whether to use the settings in this category. To apply rescheduling modifications, this check box should be selected.

**Default:** not selected

### Reschedule Day Type

(Optional) Specifies the period to which to move processing (in the direction specified by the Reschedule direction value) when a conflict occurs.

The following options are available:

- **Any day**

Moves processing to the first non-blocked day.

- **Weekday**

Moves processing to the first non-blocked weekday.

- **Day in calendar**

Moves processing to the first processing day in the calendar selected from the In calendar list.

- **Day not in calendar**

Moves processing to the first non-processing day that is in the calendar selected from the Not in calendar list.

- **Default:** Any day (when Apply rescheduling rule is selected)

The name of the associated field changes to reflect your selection, letting you select a day as appropriate.

### Reschedule direction

(Optional) Specifies the direction in which to move processing days to reschedule processing in the event of a conflict.

The following options are available:

- **Previous**

Moves processing to the previous period (as determined by the Reschedule Day Type value) when a conflict occurs.

- **Next**

Moves processing to the next period (as determined by the Reschedule Day Type value) when a conflict occurs.

- **Default:** Previous (when Reschedule is selected)

### Generate Days Dialog

The Generate Days dialog opens when you click Apply Rules, located in the Generate Days section for a standard calendar. You can use the Generate Days dialog to view and apply the rules set in the Generate Days section, or to edit and define additional processing or blocked days.

This dialog contains the following properties:

#### Calendar

(Optional) Defines days on which jobs associated with the calendar can or cannot run. You can select a specific year and month, and then click the appropriate days to select or clear them. Clicking the Processing Days button lets you select days on which a job should run; clicking the Blocked Days button lets you select days on which a job should not run.

## Clear Days

(Optional) Specifies whether you want to clear processing or blocked days from the calendar.

The following options are available:

- **Clear all days**  
Removes all processing and blocked days in the calendar.
- **Clear processing days**  
Removes all processing days selected in the calendar.
- **Clear blocked days**  
Removes all blocked days from the calendar, making them available as potential processing days.

## Starting Month

(Optional) Specifies the first month to display in the calendar.

## Starting Year

(Optional) Specifies the year for the selected Starting Month.

## Define a Cycle

You can create a cycle to define date ranges to be used as input to an extended calendar.

### Follow these steps:

1. Open Quick Edit.
2. Click Create.
3. Select the server where you want to store the cycle from the Server drop-down list.
4. Click the Cycle link.
5. Specify the following required properties:
  - **Name**  
Specifies the name for a cycle.  
**Limits:** Up to 30 characters

### NOTE

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

6. (Optional) Select the Repeat every year check box to repeat the cycle every year.
7. Add one or more periods to the cycle, as follows:
  - a. Select Add in the Periods field.
  - b. Click Go.
  - c. Specify a Start and End date for the period.

### NOTE

You must add at least one period, but you cannot exceed 30 periods per cycle.

8. (Optional) Provide a description
  - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

### NOTE

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

9. Click Commit.  
The cycle is defined.

## Example: Create a Cycle and Associate it with an Extended Calendar

To associate a cycle with an extended calendar, you first create the cycle with the periods defined. Then, when you create the extended calendar with which you want to associate it, enter CYCLE in the Conditions field, enter the name of the cycle calendar in the Cycle name field, and click Generate Preview to view the processing dates selected. You can also click Date Picker icon to the right of the Cycle name field to select a cycle from the Search Cycle dialog.

### **Cycle Properties**

#### **Name**

Defines the name of the cycle.

**Limits:** 30 or fewer characters

#### **NOTE**

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

#### **Description**

(Optional) A non-mandatory description that contains any text.

**Limits:** Up to 1024 characters.

#### **NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

#### **Periods**

(Optional) Specifies the start and end dates for each period in the cycle. The periods in a cycle cannot overlap.

The following fields are available:

- Start
- End

**Default:** The current month and day

#### **NOTE**

- One cycle has a maximum of 30 periods.
- The following options are available from the Select an action drop-down list: Add, Add Before, Move Up, Move Down, and Delete.

#### **Repeat every year**

(Optional) Specifies whether the cycle should run every year.

### **Define an Extended Calendar**

You can create an extended calendar to define processing days, using a combination of a holiday calendar, cycle, and conditions, for scheduled jobs.

#### **Follow these steps:**

1. Open Quick Edit.
2. Click Create.
3. Select the server where you want to store the calendar from the Server drop-down list.
4. Click the Extended Calendar link.
5. Specify the following required properties:
  - **Name**  
Specifies the name for an extended calendar.

**Limits:** Up to 30 characters

**NOTE**

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- **Adjustment**  
Specifies an integer by which to adjust the days generated by the other criteria. This lets you include days that are the specified number of days before or after the generated days instead.
6. (Optional) Specify any of the following properties in the Primary section:
- **Conditions**  
Specifies date conditions for the extended calendar.
- Note:**
- Clicking  opens an Edit Text dialog, which lets you specify the date conditions..
- Examples:**
- DAILY - Selects all dates after the current date
  - MON - Selects all Mondays after the current date
  - FOMWEEK - Selects the first weekday of each month after the current date
  - EOMWEEK - Selects the last weekday of each month after the current date
  - XFOM - Selects all days except the first day of each month after the current date
- **Cycle name**  
Specifies the name of an existing cycle.
- Note:**
- Clicking  opens the Search Cycle dialog, which lets you search for a cycle.
7. (Optional) Specify the following properties in the Holiday Action section:
- **Reschedule rule**  
Specifies a rule for holidays, as follows:
    - Do not schedule the job on holidays
    - Schedule only on a holiday
    - Schedule anyway, ignoring holidays
    - Schedule on the next day
    - Schedule on the next workday
    - Schedule on the previous workday
  - **Calendar**  
Specifies the name of a standard calendar that defines the holidays.
- Note:**
- Clicking  opens the Search Calendar dialog, which lets you search for a calendar.
8. (Optional) Specify any of the following properties in the Non Workday Action section:
- **Non-workday action**  
Specifies a rule for non-workdays, as follows:

- Schedule anyway, ignoring workdays
  - Schedule only on non-workdays
  - Schedule only on workdays
  - Schedule on the next workday
  - Schedule on the previous workday
- **Workdays**  
Specifies the workdays.
9. (Optional) Click Generate Preview to open the Extended Calendar Preview dialog so you can preview the dates generated by the rules in the extended calendar.
10. (Optional) Provide a description
- **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

**NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

11. Click Commit.  
The extended calendar is defined.

**NOTE**

If you create an extended calendar by entering a value in the Name field only and saving it, all days are selected from the current day forward, for a total of 366 days.

**Example: Create an Extended Calendar Based on Field Adjustment**

To create an extended calendar that processes associated jobs one day after each holiday, you would first create a calendar defining the holidays for the year, then create an extended calendar with the following options selected:

- Your holidays calendar in the Calendar field
- Schedule only on a holiday in the Holiday Action list
- 1 in the Adjustment field

**Extended Calendar Properties**

The Primary, Holiday Action, and Non-Workday Action categories for extended calendars include the following properties:

**Adjustment**

Defines the number of days before (negative number) or after (positive number) the other properties to run an associated job. You can use adjustments if the other properties do not define the necessary days. The Web UI default is 0, signifying no adjustment

**Limits:** -100 - 100, including 0

**Example:** To run a job one day before the calendar settings specify, enter -1 in the Adjustment field.

**Calendar**

(Optional) Defines the calendar name to use for holiday dates.

**Limits:** Up to 30 characters

**NOTE**

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

**Conditions**

(Optional) Defines when to use the holiday calendar and cycle.

**Limits:** Up to 1024 characters

**Example:** DAILY uses the calendar or cycle every day

### Cycle name

(Optional) Defines the cycle name to use and is only valid when the cycle keyword appears in the Conditions field.

**Limits:** Up to 30 characters

#### NOTE

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

### Name

Specifies the extended calendar name.

**Limits:** Up to 30 characters

#### NOTE

If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

### Description

(Optional) A non-mandatory description that contains any text.

**Limits:** Up to 1024 characters.

#### NOTE

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

### Non-workday action

(Optional) Indicates the action to take when the selected day falls on a non-workday, as follows:

- Schedule anyway, ignoring workdays
- Schedule only on non-workdays
- Schedule only on workdays
- Schedule on the next workday
- Schedule on the previous workday

### Reschedule rule

(Optional) Indicates the action to take if the selected day is a holiday, as follows:

- Do not schedule the job on holidays
- Schedule only on a holiday
- Schedule anyway, ignoring holidays
- Schedule on the next day
- Schedule on the next workday
- Schedule on the previous workday

### Workdays

(Optional) Specifies whether the days of the week are workdays. The checkboxes for Mo, Tu, We, Th, and Fr are selected by default.

## Manage Machines Using Web UI

Before you can schedule jobs to run on a machine, you must define the machine. You can define real machines and virtual machines.

### Real Machines

A *real machine* is any single physical machine that meets the following criteria:

- It has been identified in the appropriate network so that AutoSys Workload Automation and Web UI can access it.
- An agent, CA AutoSys Workload Automation Connect Option, CA Network and Systems Management, or AutoSys Workload Automation is installed, which lets AutoSys Workload Automation run jobs on it.
- It is defined to AutoSys Workload Automation as a real machine using JIL or Web UI.

A real machine must meet these criteria to run jobs.

### Virtual Machines

A *virtual machine* is a machine definition that references one or more existing real machine definitions. By defining virtual machines and submitting jobs to run on those machines, you can specify run-time resource policies (or constraints) at a high level and ensure that AutoSys Workload Automation automatically runs those policies in a multi-machine environment.

### Define a Real Machine

You can define real machines to run the jobs that you create.

#### **Follow these steps:**

1. Open Quick Edit.
2. Click Create.
3. Select the appropriate server from the Server drop-down list and click the Real Machine link.
4. Specify the following required properties:
  - **Name**  
Specifies the name of the machine.  
**Limits:** Up to 80 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), and hash (#); do not include embedded spaces or tabs
  - **Type**  
Specifies the type of the machine.
5. (Optional) Specify optional [real machine properties](#).
6. Click Commit.  
The machine definition is saved in the scheduler database.

### Real Machines Properties

The Primary section for real machines includes the following properties:

#### **Name**

Specifies the name of the machine.

**Limits:** Up to 80 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), and hash (#); do not include embedded spaces or tabs

#### **Type**



Specifies the type of machine to create. You can select one of the following machine types:

- UNIX (r)
- Windows (n)
- CA NSM or CA Universal Job Management Agent (u)
- CA AutoSys Workload Automation Connect Option (c).
- Legacy UNIX (l)
- Legacy Windows (L)
- AutoSys Workload Automation Agent (a)

### Description

(Optional) Specifies a description of the machine.

**Limits:** Up to 256 alphanumeric characters (including spaces)

### Node name

(Optional) Specifies the IP address or DNS name of the machine.

**Limits:** Up to 80 characters; for machine type u and c, 64 characters

### Max load

(Optional) Specifies the maximum load for the machine in load units.

**Limits:** 0-2147483647

### Factor

(Optional; Unix, Windows, Legacy Unix and Legacy windows machine types only.) Specifies the multiplication factor to apply to the available CPU cycles of the machine. The resulting number determines the relative available CPU cycles for the machine.

**Limits:** 0.01-1024

The AutoSys Workload Automation Agent section includes the following properties:

### Agent provisioning

(Optional) Indicates whether to request the Data Center Automation Manager to provision a AutoSys Workload Automation agent for this machine.

### Administrator ID

(Optional) Specifies the administrator user ID that is used to request a AutoSys Workload Automation agent from the DCA (Data Center Automation) Manager.

**Limits:** Up to 64 alphanumeric characters.

### Agent name

(Optional) Specifies the AutoSys Workload Automation agent instance name.

**Limits:** Up to 16 alphanumeric characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen (-); do not include embedded spaces or tabs

### Port number

(Optional) Specifies the IP port number of the AutoSys Workload Automation traffic listener.

**Limits:** 1-65535

### Character encoding

(Optional) Specifies the character encoding to use. You can select either of the following options:

- ASCII
- EBCDIC

### Heartbeat attempts

(Optional) Specifies the number of failed heartbeat signals that should occur before the scheduler sends an SNMP message indicating AutoSys Workload Automation inactivity.

**Limits:** 0-2147483647

### Heartbeat frequency

(Optional) Specifies the scheduler heartbeat signal frequency.

**Limits:** 0-2147483647

### Operating system

(Optional) Specifies the agent host operating system. You can select one of the following operating systems:

- aix
- hpux
- linux
- openvms
- i5os
- tandem
- windows
- zos

### Encryption type

(Optional) Specifies the encryption type used. You can select the following types:

- NONE
- DEFAULT
- AES

### Agent security key

(Optional) Specifies the agent security key. To obtain the security key, copy the security.cryptkey parameter from the agentparm.txt file. Do not include the 0x prefix.

**Limits:** Up to 32 characters

### Define a Virtual Machine

You can define virtual machines to run the jobs that you create.

#### Follow these steps:

1. Open Quick Edit.
2. Click Create.
3. Select the appropriate server from the Server drop-down list and click the Virtual Machine link.
4. Specify the following required properties:
  - **Name**  
Specifies the name of the machine.  
**Limits:** Up to 80 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), and hash (#); do not include embedded spaces or tabs
  - **Type**  
(Optional) Specifies the type of the machine.

**NOTE**

If you do not specify a type for the virtual machine, a type will be set automatically based on the type of associated real machines.

- **Machines**

Specifies one or more real machines associated with this virtual machine.

5. (Optional) Specify optional [virtual machine properties](#).

6. Click Commit.

The machine definition is saved in the scheduler database.

## **Virtual Machines Properties**

The Primary section for virtual machines includes the following properties:

### **Name**

Specifies the name of the machine.

**Limits:** Up to 80 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), and hash (#); do not include embedded spaces or tabs

### **Type**

(Optional) Specifies the type of machine to create. You can select one of the following machine types:

- v - Virtual
- n - Windows
- p - Pool of DCA managed machines

**NOTE**

If you do not specify a type for the virtual machine, a type will be set automatically based on the type of associated real machines.

### **Description**

(Optional) Specifies a description of the machine.

**Limits:** Up to 256 alphanumeric characters (including spaces)

### **Machines**

Specifies one or more real machines associated with this virtual machine. The following field is available:

- **Name**

Specifies the name of the real machine.

**Limits:** Up to 80 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), and hash (#); do not include embedded spaces or tabs

### **Machine Method**

(Optional) Specifies the [method to achieve load balancing](#). You can select any one of the following values:

- rstatd
- job\_load
- cpu\_mon
- roundrobin
- roundrobin\_jobload

The following additional fields are available for virtual machines type v (virtual) and n (Windows):

- **Max load**

(Optional) Specifies the maximum load for the machine in load units.

**Limits:** 0-2147483647

- **Factor**

(Optional) Specifies the multiplication factor to apply to the available CPU cycles of the machine. The resulting number determines the relative available CPU cycles for the machine.

**Limits:** 0.01-1024

## Specify the Method to Achieve Load Balancing

Load balancing can be performed using virtual resources combined with virtual machines or machine lists for basic throttling and serialization. If you assign a virtual machine or list of real machines to a job with the virtual resource dependencies, the jobs can be dispatched to the various machines based on resource availability. If two or more machines satisfy the virtual resource requirements, the job is scheduled to run on the machine with the most available CPU resources.

You can use Machine Method in the Virtual Machine properties section to specify the method to achieve load balancing.

### Follow these steps:

1. Open the Virtual Machine properties in Quick Edit.
2. Select v - Virtual against Type.
3. Select one of the following options from the Machine Method drop-down list:
  - **rstatd** Specifies that the rstatd method is used by the UNIX schedulers to determine the percentage of available CPU cycles on UNIX computers. The scheduler makes a UNIX remote procedure call to the remote kernel statistics daemon (rstatd) on the target computer to get the available CPU cycles.

#### NOTE

- You cannot configure the rstatd method for the Windows scheduler.
  - Ensure that the rstatd daemon is running on the UNIX scheduler and on all target UNIX computers.
  - Set the Operating system property value for Type 'a' machines to one of the following values that support the rstatd method: aix, hpux, linux, or openvms.
  - If the load balancing request is sent to an agent with an Operating System value that does not support the rstatd method, AutoSys Workload Automation uses the cpu\_mon method to obtain the available CPU cycles.
- **job\_load** Specifies that the job\_load method is used to determine the percentage of available CPU cycles. The scheduler uses the Job load and Max load properties to calculate the available load for a computer.
  - **cpu\_mon** Specifies that the cpu\_mon method is used to determine the percentage of available CPU cycles on the agent. The scheduler runs the CPU Monitoring (OMCPU) job on the target computer to get the available CPU cycles. This is the default.

#### NOTE

- The cpu\_mon method does not apply to z/OS machines (Workload Automation Agent on z/OS) because the OMCPU job is not supported on z/OS.
- **roundrobin** Specifies that the roundrobin method is used to determine the real machine where the job runs from the list of available real machines belonging to a virtual machine. The jobs are run on the available real machines in a round robin manner.
  - **roundrobin\_jobload** Specifies that the job\_load method is used to determine the percentage of available CPU cycles and then the roundrobin method is used on the resulting real machines to determine the real machine where the job runs.
4. Commit the machine.
- Based on the machine method you selected, AutoSys Workload Automation determines the machine to run the job.

## Manage Jobs and Job Flows Using Web UI

All activity controlled by AutoSys Workload Automation is based on jobs. For example, a *job* is any single command or executable, UNIX shell script, or Windows batch file. Other objects, such as monitors, reports, and the Job Status Console, track job progress. A job is the foundation for the entire operations cycle.

You define jobs to AutoSys Workload Automation by creating job definitions that specify the job's properties and behavior. For example, you can specify conditions that determine when and where a job runs.

When you create a job definition, you must specify the job type. Job types define the type of work to be scheduled. For example, you can create a Command job to run a Windows command file, an FTP job to download a file from a server, or an SAP Event Monitor job to monitor for the triggering of an SAP event. You can also define Box jobs, which are containers that hold other jobs or Box jobs. You can define your own job type with a User Defined job.

Each job type has required and optional properties that define the job. The job types have many common properties and AutoSys Workload Automation treats them all similarly. The primary differences between them are the actions taken when the jobs run.

You can define jobs using the Web UI Quick Edit application. The job definition and properties in Quick Edit is always stored in the database. You can also modify and delete existing job definitions.

### **Scheduling Jobs and Job Dependencies**

Web UI lets you interactively set the properties that describe when, where, and how a job should run. In addition, Web UI lets you define calendars, global variables, and box jobs, and monitor and manage jobs. The fields in Web UI correspond to AutoSys Workload Automation JIL subcommands and attributes.

You can define job properties to do the following:

- Specify starting conditions for jobs within your job definitions.
- Set date and time dependencies for a job so that it runs automatically on specific days and at specific times.
- Base job-starting conditions on a calendar event (basic time/day or custom calendar), job dependency status, or file arrival.
- Specify resource dependencies so that a job will not start execution until the required resources are available.

Additionally, you can specify job termination conditions within your job definitions. For example, you can specify exit codes that determine the success or failure of a job or the number of minutes to wait before terminating the job.

### **Define a Job in Quick Edit**

You can create jobs to manage and schedule workload in your enterprise.

#### **Follow these steps:**

1. Open Quick Edit.
2. Click Create.
3. Select the server on which you want to create a job from the Server drop-down list.
4. Click the link for the type of job you want to create.

#### **NOTE**

You can select a specific job type or, if available, use a job template containing pre-defined properties to create a new job.

5. Perform the following actions to define your job:
  - a. Enter a name for the new job in the Name field.
  - b. Complete the fields in each category as appropriate.
6. Click Commit.

The job is saved in the scheduler database.

### **Define a Global Variable in Quick Edit**

You can use Quick Edit to define and store global variables in the AutoSys Workload Automation database. You can then use the global variable within a job definition to set up a dependency condition.

#### **Follow these steps:**

1. Open Quick Edit.
2. Click Create.
3. Select the server on which you want to create a job from the Server drop-down list.
4. Click the Global Variable link.
5. Specify the following properties:
  - **Name**  
Specifies a name for the global variable.  
**Limits:** Up to 64 characters
  - **Value**  
(Optional) Specifies a value for the global variable.  
**Limits:** Up to 100 characters
6. Click Commit.  
The global variable is saved in the scheduler database.

### **Commit Changes in Quick Edit**

You must commit job property and job flow changes to the appropriate AutoSys Workload Automation server to make the changes permanent on that server.

To commit changes in Quick Edit, make the required changes to the job properties, and click Commit.

A confirmation message appears.

#### **NOTE**

If you save a job to the AutoSys Workload Automation server, and a patch has not been applied for that job type, unsupported attributes are ignored.

## **Manage Common Job Properties**

This article explains how to manage common job properties such job owner, starting conditions, and job status dependencies.

### **Specify the Job Owner**

By default, all jobs are created with the owner *authenticated\_user@host*. If you are using a shared instance of CA EEM between AutoSys Workload Automation and Web UI, the *authenticated\_user* is the Web UI logged-in user name; however, if you are not sharing the same instance of CA EEM, the *authenticated\_user* is the operating system user who defined the job. This user must be defined in the Credentials application. If your job requires a different user ID to run successfully, you must override the default owner value. For example, the default owner ID and associated password typically do not correlate to an operating system user ID for jobs that are run by subsystems (environments other than command line interfaces), such as ERP or database subsystems. For these jobs, the owner property should be overridden with an ID that correlates to the subsystem. Attempts made to execute a job with an incorrect owner will result in job execution failures.

**WARNING**

If AutoSys Workload Automation is running in native security mode, you can change the owner value only if you have EDIT superuser permissions. If AutoSys Workload Automation is running in external security mode using CA EEM, you can change the owner value only if you have as-owner authority.

To specify the job owner, use the Owner property, located under the Primary category in the Properties section of a job definition.

**Example: Specify a Job Owner**

Suppose that AutoSys Workload Automation is running in external security mode and you have as-owner authority as defined in CA EEM. You can specify the owner property in job definitions. The job runs under the prod user on the unixagent computer.

**Follow these steps:**

1. Create a job.
2. Enter the following Primary properties:
  - Name -- jobA
  - Send to machine -- UNIXAGENT
  - Owner -- prod@unixagent
3. Enter the following Command property:
  - Command -- /bin/touch /tmp/test\_run.out
4. Save the job.

**Define Starting Conditions for a Job**

You can define dependencies to use as starting conditions for a job. AutoSys Workload Automation uses Boolean logic to evaluate these dependencies. The job cannot run until all specified dependencies evaluate to true. You can specify one or more of the following dependencies:

- [Job status dependencies](#)
- [Cross-instance job dependencies](#)
- [Exit code dependencies](#)
- [Look-back dependencies](#)
- [Global variable dependencies](#)

You specify job dependencies using the Condition property, located under the Primary category in the Properties section of a job definition.

**NOTE**

When you specify a condition for an undefined job, the condition evaluates as false and any jobs dependent on the condition do not run.

**Job Status Dependencies**

You can define a starting condition for a job so that it starts when another job returns a specific status. For example, you might specify that JobB starts when JobA returns a SUCCESS status and JobC starts when JobB returns a SUCCESS status.

To define a starting condition based on a job's AutoSys Workload Automation status, use the following format in the Condition property:

```
status(job_name)
```

- **status**  
Specifies one of the following:

- **Done**  
Indicates that the job you are defining may run when *job\_name*'s status is SUCCESS, FAILURE, or TERMINATED.
- **Failure**  
Indicates that the job you are defining may run when *job\_name*'s status is FAILURE.
- **Notrunning**  
Indicates that the job you are defining may run when *job\_name*'s status is anything except RUNNING. Set status to notrunning to keep the dependent job from running at the same time as *job\_name*.
- **Success**  
Indicates that the job you are defining may run when *job\_name*'s status is SUCCESS.  
**Note:** You can use the Max exit code for success property, located under the Termination category, to control the value of the SUCCESS status for a specific job. When you define the Max exit code for success, a job that exits with an exit code less than or equal to the specified value is treated as a success. FAILURE means the job exited with an exit code greater than the Max exit code for success value.
- **Terminated**  
Indicates that the job you are defining may run when *job\_name*'s status is TERMINATED. A status of TERMINATED means the job was killed.

#### NOTE

- You can use either uppercase or lowercase characters to define status condition identifiers.
- You can abbreviate the status condition identifiers with the first letter (d, f, n, s, and t). You can specify these abbreviations in uppercase or lowercase.
- If you specify a condition for an undefined job, the condition evaluates as FALSE, and any jobs dependent on this condition do not run.
- You can configure more complex conditions by combining a series of conditions with the AND and OR logical operators. The equation is evaluated from left to right. For example, in the following set of starting conditions, either both A and B must be successful or both D and E must be successful for the statement to evaluate as TRUE:  
(success(JobA) and success(JobB)) or (success(JobD) AND success(Job E))
- ***job\_name*** identifies a job on which the job you are defining depends.

#### Cross-Instance Job Dependencies

To specify a cross-instance job dependency, enter the job name followed by a caret (^) and the name of another instance in the Condition property. For example, the following condition statement indicates that the dependent job can start when jobB on the AutoSys Workload Automation instance PRD and jobA both return a status of SUCCESS:

```
success(jobA) AND success(jobB^PRD)
```

#### Exit Code Dependencies

You can base job dependencies on exit codes that indicate completed tasks. For example, you can specify that the system should run JobB if JobA fails with an exit code of 4.

To define a starting condition based on a job's exit code, use the following format in the Condition property:

```
exitcode(job_name) operatorvalue
```

- ***job\_name***

Identifies a job on which the job you are defining depends.

- ***operator***  
Specifies one of the following comparison operators:  
= Equals



- != Does not equal
- < Less than
- > Greater than
- <= Less than or equal to
- >= Greater than or equal to

- **value**  
Defines the exit code value to compare against.

#### NOTE

- You can use either uppercase or lowercase characters to define the exitcode starting condition identifier.
- You can abbreviate the starting condition for exitcode with the first letter (e). You can specify this abbreviations in uppercase or lowercase.

### Look-Back Dependencies

The look-back feature is defined as the last run of the condition or predecessor job. The last run is equivalent to the ending time of the last successful run of the job. If the condition or predecessor job has run successfully after the job for which that condition or predecessor job is defined, then the dependency is satisfied and the job starts. Otherwise, the dependency is not satisfied and the job is not started.

To support the look-back feature, you must add an additional time operand to the Condition statement. You can apply the look-back feature to job status, cross-instance or external dependency job status, and exit code condition types only. You cannot apply it to the global variable condition type. To define a starting condition based on a look-back value, use one of the following formats in the Condition property:

```
status(job_name, hhhh:mm)
status(job_name^INS, hhhh:mm)
exitcode(job_name, hhhh:mm) operator value
exitcode(job_name^INS, hhhh:mm) operator value
```

#### NOTE

- The *status* can be any starting condition, such as success, failure, terminated, notrunning, or done.
- You can use a period instead of a colon between the specified hours and minutes.

- **hhhh**

Indicates the hours taken for the last run of the condition or predecessor job.

**Limits:** One to four numeric characters from 0 - 9999, which makes it possible to look back approximately 416.58 days

#### NOTE

When specifying minutes also, the valid values range from 0 - 9998 instead.

- **mm**

Indicates the minutes taken for the last run of the condition or predecessor job. Periods or escaped colons are used to separate minutes from hours.

**Limits:** One to two numeric characters from 0 to 59, representing minutes

**Examples:** 0, 00.15, 06.30, 23.30, 24, 98.30, 720, 9998.59, and 9999

#### NOTE

- When the elapsed time to be specified is less than one hour, you must specify the hourly value as "00". For example, "00.30" represents 30 minutes. If hhhh.mm is specified as "30", it implies an elapsed time of 30 hours. Specifying an elapsed time of ".30" is not valid.
- When you specify 0, AutoSys Workload Automation examines the last end time of the job first. It then examines the last end time of the condition or predecessor job. If the condition or predecessor job has run since the last run of the job for which it is a dependency, that job is allowed to start. If the condition or

predecessor job has not run since the last run of the job for which it is a dependency, that job is not allowed to start.

### **Global Variable Dependencies**

You can base job dependencies on global variables set using the sendevent command. When using global variables, the value of the variable must evaluate to true to satisfy the job dependency.

To define a starting condition based on a global variable, use the following format in the Condition property:

```
VALUE(global_name) operatorvalue
```

- **global\_name**  
Identifies a global variable that is already defined.  
**Limits:** Up to 64 alphanumeric characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-)
- **operator**  
Specifies one of the following comparison operators:  
= Equals  
!= Does not equal  
< Less than  
> Greater than  
<= Less than or equal to  
>= Greater than or equal to
- **value**  
Defines the global variable value to compare against.  
**Limits:** Up to 100 alphanumeric characters; cannot contain quotation marks or spaces

### **Starting Condition Examples**

The following examples provide sample starting conditions for jobs.

#### **Example: Define Starting Conditions Based on Job Success**

This example (specified in the job definition for JobA) runs JobA if JobB succeeds:

```
success(JobB)
```

#### **Example: Define Starting Conditions Based on the Status of Multiple Jobs**

This example (specified in the job definition for JobC) starts JobC only when both JobA and JobB complete successfully or when both JobD and JobE complete (regardless of whether they failed, succeeded, or terminated):

```
(success(JobA) AND success(JobB)) OR (done(JobD) AND done(JobE))
```

#### **Example: Define Starting Conditions Based on Job Failure**

This example (specified in the job definition for JobA) runs JobA if JobB fails:

```
failure(JobB)
```

#### **Example: Define Starting Conditions Based on Job Success in a Given Duration (Look-Back dependency)**

This example (specified in the job definition for JobA) runs JobA if JobB has completed successfully in the last 12 hours:

```
success(JobB,12)
```

#### **Example: Define Starting Conditions Based on the Status of Multiple Jobs in a Given Duration (Look-Back dependency)**

This example (specified in the job definition of JobA) runs JobA if JobB has failed in the last 30 minutes and JobC has completed successfully on external instance XYZ in the last 90 minutes:

```
failure(JobA,00.30) AND success(JobC^XYZ,01.30)
```

#### Example: Define Starting Conditions with the notrunning Status Condition

You may want to use the notrunning status condition to avoid a database dump (DB\_DUMP) and a file backup (BACKUP) from running at the same time, which would cause frequent hard disk accesses. This example (specified in the job definition for JobA) runs JobA only when both of these resource-intensive jobs are **not** running:

```
notrunning(DB_DUMP) AND notrunning(BACKUP)
```

#### Example: Define Starting Conditions Based on Multiple Condition Types

This example (specified in the job definition for JobA) runs JobA only if the job BACKUP completes with a SUCCESS status and the global variable HOLD has a value of No:

```
success(BACKUP) AND value(HOLD)=No
```

#### Example: Define Starting Conditions Based on a Cross-Instance Job Status

This example (specified in the job definition for JobA) runs JobA only when the job DB\_BACKUP (which resides on another AutoSys Workload Automation instance named PRD) succeeds:

```
success(DB_BACKUP^PRD)
```

#### Example: Define Starting Conditions Based on Job Exit Code

This example (specified in the job definition for JobB) runs JobB when JobA completes with an exit code of 4:

```
exitcode (JobA) = 4
```

#### Example: Specify a Look-Back Dependency

The following Condition statements let the dependent job start if JobA is successful during the last hour and JobB failed during the last 2 hours and 15 minutes:

```
success(JobA,01.00) AND failure(JobB,02.15)
success(JobA,01:00) AND failure(JobB,02:15)
```

#### Example: Define Starting Conditions Based on a Global Variable Value

This example (specified in the job definition for JobA) runs JobA only if the value of the global variable OK\_TO\_RUN is greater than 2:

```
VALUE(OK_TO_RUN)>2
```

### Primary Properties

The Primary category includes the following properties:

#### Application

(Optional) Defines the name of the application with which to associate the job. You can only associate a job with one application, but an application may have many jobs associated with it.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

**JIL attribute:** application

#### Box

(Optional) Identifies an existing Box job in which to put the job.

Clicking the Search icon opens the Search Box Job dialog, from which you can select a box job for the job definition. For Quick Edit, the box job must reside on the selected server.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen (-)

**JIL attribute:** box\_name

### Condition

(Optional) Defines any combination of dependencies based on job status, job exit codes, or global variables.

**Limits:** Up to 4096 characters

#### NOTE

- To configure complex conditions, combine a series of conditions with the AND or the OR logical operators. You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following condition statement lets the dependent job start when either JobA and JobB return SUCCESS or JobD and JobE return SUCCESS, FAILURE, or TERMINATED:

```
(success(JobA) AND success(JobB)) OR (done(JobD) AND done(Job E))
```

**JIL attribute:** condition

### Description

(Optional) Defines an optional free-form text description of the job.

**Limits:** Up to 500 alphanumeric characters (including spaces)

**JIL attribute:** description

### Group

(Optional) Defines the name of the group with which to associate the job. You can only associate a job with one group, but a group may have many jobs associated with it.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

**JIL attribute:** group

### Name

Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

JIL attribute: insert\_job

### Owner

(Defaulted) Defines the owner of the job. By default, all jobs are created with the owner *authenticated\_user@host*. If you are using a shared instance of CA EEM between AutoSys Workload Automation and Web UI, the *authenticated\_user* is the Web UI logged-in user name; however, if you are not sharing the same instance of CA EEM, the *authenticated\_user* is the operating system user who defined the job. This user must be defined in the Credentials application. If your job requires a different user ID to run successfully, you must override the default owner value. For example, the default owner ID and associated password typically do not correlate to an operating system user ID for jobs that are run by subsystems

(environments other than a command line interfaces), such as ERP or database subsystems. For these jobs, the owner property should be overridden with an ID that correlates to the subsystem. Attempts made to execute a job with an incorrect owner will result in job execution failures.

#### NOTE

- If AutoSys Workload Automation is running in native security mode, you can change the owner value only if you have EDIT superuser permissions.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you can change the owner value only if you have as-owner authority.

**Default:** *authenticated\_user@host*

**Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** owner

#### Resources

(Optional) Specifies virtual resource dependencies.

#### NOTE

To specify a virtual resource dependency, the virtual resource must already be defined on the database. Otherwise, you receive an error.

**Limits:** Up to 4096 characters

**JIL attribute:** resources

#### Send to machine

Specifies the name of the machine on which the job runs. Web UI defaults this as localhost.

**JIL attribute:** machine

#### Define Resource Dependencies for a Job

In the AutoSys Workload Automation environment, a resource is a system component that can be quantified. Resources govern job execution and play a vital role in scheduling. Before starting a job, AutoSys Workload Automation compares the resources required with the resources available. For example, you may have specified resource requirements that must be met before a job can run. If the required resources are not available, AutoSys Workload Automation defers the job run until resources are available. The only type of resource is virtual.

*Virtual resources* are only representative and are not tied to a physical system. The number of licenses available to all machines controlled by AutoSys Workload Automation is an example of a virtual resource. As each job that requires a license is submitted to the system, the number of licenses in the pool is decremented. When the number of licenses reaches zero, no more jobs can be submitted until the license pool is replenished.

Based on their properties and behavior, virtual resources are of the following types:

- **Depletable**  
Defines a resource where the amount allocated to the job decreases as it is used, for example, the disk space. When AutoSys Workload Automation starts a job, the depletable resources needed for the job are permanently removed from the resource pool.
- **Renewable**  
Defines a resource that is returned to the resource pool after use, for example, write access to a database. When AutoSys Workload Automation starts a job, the job borrows the renewable resource from the resource pool, and then returns it after use.
- **Threshold**

Defines a resource that is used as a limit. For example, if the threshold resource quantity is set to two, AutoSys Workload Automation submits only those jobs that require two or fewer units of that resource.

You can define a job to have dependencies on virtual resources. A job with resource dependencies is submitted only when the resources required are available. Using resource dependencies can help you control job execution and improve your environment's performance.

To specify virtual resource dependencies, use the following format for the Resources property in the Primary section of a job definition:

```
(virtual_resource_name, quantity=amount|ALL[, free=Y|N|A]) [AND ...]
```

- **virtual\_resource\_name**

Specifies the name of a virtual resource.

**Note:** The virtual resource must already be defined in the database. For more information about defining virtual resources using Web UI, see the *Resources Help*.

- **quantity=amount | ALL**

Specifies the amount of virtual resource required. The following options are available:

- *amount* -- Specifies the number of units that the job requires. The number must be a positive integer.
- ALL -- Specifies that the job requires all the units of the resource.

- **free=Y | N | A**

(Optional for renewable virtual resources only) Specifies whether the units of the virtual resource are freed from the job. The following options are available:

- Y -- Frees the units only if the job completes successfully. This is the default.
- N -- Does not free the units.

#### NOTE

To free the resources, you must issue the following command:

```
sendevent -E RELEASE_RESOURCE -J job_name
```

- A -- Frees the units unconditionally when the job completes.

## Resource Dependency Examples

The following examples provide sample resource dependencies for jobs.

### Example: Define a Virtual Resource Dependency

This example (specified in the Resources property of a job definition) defines a dependency on a virtual renewable resource. Before the job can start running, it needs all the units of the `renew_res` resource. After the job completes, the units of the `renew_res` resource are not freed from the job.

```
(renew_res, QUANTITY=ALL, free=N)
```

### Example: Define Real and Virtual Resource Dependencies

This example (specified in the Resources property of a job definition) defines real and virtual resource dependencies. Before the job can start running, it needs a machine that satisfies all the following dependencies:

- 1 unit of the depletable resource named D1
- 3 units of the threshold resource named T1
- 4 units of the renewable resource named R1
- SYSTEM\_OS\_TYPE is AIX
- SYSTEM\_PHYSICAL\_MEMORY is greater than 2 GB

The `free` keyword is not specified for the virtual resource dependencies, so the resource units are freed only if the job completes successfully.

```
(D1, QUANTITY=1) AND (T1, QUANTITY=3) AND (R1, QUANTITY=4, FREE=Y) AND (SYSTEM_OS_TYPE,
VALUEOP=EQ, VALUE=AIX) AND (SYSTEM_PHYSICAL_MEMORY, VALUEOP=GT, VALUE=2097152)
```

## **Defining Date and Time Dependencies**

You can schedule jobs to start at a specific date and time. AutoSys Workload Automation then calculates a matrix of specified day, date, and time values and starts jobs accordingly. A time range cannot span more than 24 hours. For example, you can define a job to start on Monday, Wednesday, and Friday at 8:00 a.m. and again at 5:00 p.m.

You can specify days of the week or actual dates, but you cannot specify both. You can only specify actual dates using custom calendars. You can also specify a time zone to apply to your starting times, and you can define a job to start at a specific time of day or hourly at a designated number of minutes past the hour.

You can set up one of the following types of time dependencies: absolute or relative.

Jobs with absolute time dependencies are defined to run at a specific time of the day (for example, 9:30 on Thursday or 12:00 on December 25th). The following Schedule properties define absolute time dependencies:

- Days: Run calendar
- Days: Run days
- Exclude calendar
- Run window
- Time: Times of day

Relative time dependencies are based either on the current time or relative to the start of the hour (for example, start a job at 10 and 20 minutes after the hour, or terminate a job after it has run for 90 minutes).

The following Schedule, Termination, and Notification properties define relative time dependencies:

- (Schedule) Time: Minutes after each hour
- (Termination) Hours after successful completion to delete job
- (Termination) Minutes to wait before terminating
- (Notification) Alarms: Maximum run time (minutes)
- (Notification) Alarms: Minimum run time (minutes)

Your operating system might automatically change the system clock to reflect the switch to either standard time (ST) or daylight savings time (DT), and the scheduling of time-dependent jobs might shift to adjust for the time change. Jobs that are not time-dependent run as appropriate. During the time change, absolute times behave differently than relative times.

## **Time Zone**

By default, jobs with time-based starting conditions that do not specify a time zone are scheduled to start based on the time zone under which the AutoSys Workload Automation server is running.

The Time zone property under the Schedule category defines the time zone on which to base scheduling for the job you are defining. For example, if you define a start time of 01:00 for a job running on a machine in Denver, and set the Time zone property to San Francisco (which is in the Pacific time zone, one hour earlier than Denver), the job starts at 2:00 a.m. Denver time.

## **Custom Calendars**

You can use Quick Edit to define any number of calendars, each with a unique name and containing any number of dates. You can use these calendars to specify days on which to run the jobs with which they are associated or to specify days on which jobs with which they are associated should not run. Calendars exist independently of any jobs associated with them and are referenced by jobs through job definitions.

## **Must Start Times and Must Complete Times**

You can define the time, or a list of times, that a job must start or complete by. If the job does not start or complete by a specified time, an alarm is issued. Defining must start times and must complete times is helpful when you want to be notified when a job has not started or completed. For example, a must start alarm can alert you to investigate whether the job's starting conditions are not satisfied or the job is stuck in the STARTING state. You use the Must start times and Must complete times properties under the Schedule category to set these times.

### **NOTE**

- To define absolute times, you must select the Use date/time conditions property and specify times in the Times of day property.
- You cannot define absolute times when Minutes after each hour is specified in the job definition. You will get an error if you define absolute times with Minutes after each hour.
- You can specify more than one must start time or must complete time for a job. If the job has multiple start times, you must specify the same number of must start times or must complete times. For example, if the job runs at three different times during the day, you must specify three must start times, corresponding to each run of the job.

### **Example: Specify Absolute Must Start Times**

This example defines a job to run at 10:00 a.m., 11:00 a.m., and 12:00 p.m. every day. The job must start by 10:02 a.m., 11:02 a.m., and 12:02 p.m. respectively. Otherwise, an alarm is issued for each missed start time.

```
Use date/time conditions -- selected
Days -- Run days
Run days -- Su,Mo,Tu,We,Th,Fr,Sa
Time -- Times of day
Times of day -- 10:00, 11:00, 12:00
Must start times -- 10:02, 11:02, 12:02
Must complete times -- 10:08, 11:08, 12:08
```

### **NOTE**

The number of must start times must match the number of start times. For example, an error is returned if you try to define a job with one start time and two must start times, as follows:

```
Times of day -- 10:00
Must start times -- 10:02, 12:02
```

### **Example: Specify an Absolute Must Start Time on the Next Day**

This example defines a job to run at 12:00 a.m. Suppose that you want the job to start by 10:10 a.m. the next day. You must specify 34:10 in the Must start times field.

```
Use date/time conditions -- selected
Run days -- Su,Mo,Tu,We,Th,Fr,Sa
Time -- Times of day
Times of day -- 12:00
Must start times -- 34:10
Must complete times -- 34:12
```

The must start time is calculated as follows:

```
Times of day in hh:mm format + time difference between the Times of day and the Must
start times
= 0:00 + (time difference between 12:00 a.m. and 10:10 a.m. the next day)
```



```
= 0:00 + 34:10
= 34:10
```

### Example: Specify an Absolute Must Start Time that Crosses Two Days

This example defines a job to run at 16:00 (4:00 p.m.). Suppose that you want the job to start by 6:10 a.m. two days later. You must specify 54:10 in the Must start times field.

```
Name -- job3
Send to machine -- localhost
Command -- echo "hello"
Use date/time conditions -- selected
Days -- Run days
Run days -- Su,Mo,Tu,We,Th,Fr,Sa
Times of day -- 16:00
Must start times -- 54:10
Must complete times -- 54:12
```

The Must start time is calculated as follows:

```
Times of day in hh:mm format + time difference between the Times of day and the Must
start times
= 16:00 + (time difference between 16:00 and 6:10 a.m. two days later)
= 16:00 + 38:10
= 54:10
```

### Example: Specify Relative Must Start Times

This example defines a job to run at 10:00 a.m., 11:00 a.m., and 12:00 p.m. The job must start within 3 minutes after each start time (10:03 a.m., 11:03 a.m., and 12:03 p.m.). Otherwise, an alarm is issued for each missed start time.

```
Name -- test_must_start_complete
Send to machine -- localhost
Command -- /opt/StartTransactions.sh
Use date/time conditions -- selected
Days -- Run days
Run days -- Su,Mo,Tu,We,Th,Fr,Sa
Time -- Times of day
Times of day -- 10:00, 11:00, 12:00
Must start times -- 3
Must complete times -- 8
```

### Example: Specify Relative Must Start Times with Minutes After Each Hour

This example defines a job to run at every 10 minute interval (for example, 2:00 p.m., 2:10 p.m., 2:20 p.m., and so on). The job must start within 2 minutes after the specified start times. Otherwise, an alarm is issued for each missed start time. For instance, the job must start by 2:12 p.m.

```
Name -- test_must_start_complete
Send to machine -- localhost
Command -- /opt/StartTransactions.sh
Use date/time conditions -- selected
Days -- Run days
```

```
Run days -- Su,Mo, Tu, We, Th, Fr, Sa
Time -- Minutes after each hour
Minutes after each hour -- 10, 20, 30, 40, 50, 00
Must start times -- 2
Must complete times: 7
```

### Example: Specify a Relative Must Start Time on the Next Day

This example specifies a relative must start time of 12:01 a.m. the next day. The must start time is calculated by adding three minutes to the start time. The calculated time crosses into the next day.

```
Name -- job2
Send to machine -- localhost
Command -- echo "Hello"
Use date/time conditions -- selected
Days -- Run days
Run days -- Su,Mo,Tu,We,Th,Fr,Sa
Times -- Times of day
Times of day -- 23:58
Must start times -- 3
```

### Date and Time Dependency Examples

The following examples set time dependencies using Schedule properties within a job definition.

#### Example: Set Time Dependencies for a Job

This example sets the existing job test\_run to run automatically at 10:00 a.m. and 2:00 p.m. on Mondays, Wednesdays, and Fridays.

#### Follow these steps:

1. Open the text\_run job.
2. Specify the following Schedule properties:
  - Use date/time conditions -- selected
  - Days -- Run days
  - Run days -- Mo, We, Fr
  - Times of day -- 10:00, 14:00
3. Commit the job.

#### Example: Base Time Settings on a Specific Time Zone

You can use the Time zone field to base the time settings for a job on a time zone other than the default. If you specify a time zone that includes a colon, you must surround the time zone name with quotation marks, as in the following example:

```
Time zone -- "IST-5:30"
```

#### Example: Run a Job Every Day

To run the job every day, instead of only on specific days, select all of the days for the Run days property as in the following example:

```
Run days: Su,Mo,Tu,We,Th,Fr,Sa
```

#### Example: Schedule a Job to Run on Specific Dates

To schedule a job for specific dates, instead of days of the week, you can specify a run calendar. Define the calendar and then use the Run calendar field to specify the calendar name in the job definition.

For example, specify the following Schedule properties to schedule a job to run on the days defined in the weekday\_cal calendar:

- Days -- Run calendar
- Run calendar -- weekday\_cal

### **Example: Exclude a Job from Running on Specific Dates**

You can specify a custom calendar that defines days on which a job should not run, such as business holidays. Define the calendar and use the Exclude calendar field to specify the calendar name in the job definition. For example, specify the following Schedule property to exclude days defined in the holiday\_cal calendar:

Exclude calendar -- holiday\_cal

### **Example: Schedule a Job to Run at Specific Times Every Hour**

To run a job at specific times every hour instead of at specific times of the day, you can use the Time and Minutes after each hour fields. For example, enter the following Schedule properties to run a job at 15 minutes after and 15 minutes before each hour:

- Time -- Minutes after each hour
- Minutes after each hour -- 15, 45

## **Schedule Properties**

The Schedule category includes the following properties:

### **Auto hold job when box starts**

(Optional) Indicates whether to place an automatic hold on the job when its containing Box job starts. Selecting the check box places an automatic hold on the job. Clearing the check box removes the auto hold from the job.

**JIL attribute:** auto\_hold

### **Average run time**

(Optional) Specifies the average run time (in minutes) to use for a job that is newly submitted to the database.

**Limits:** 0-527040

**JIL attribute:** avg\_runtime

### **Use date/time conditions**

(Optional) Indicates whether to use date or time conditions specified to determine when to run the job.

**JIL attribute:** date\_conditions

### **Days: Run calendar**

(Optional) Specifies the calendar to define the schedule for running the job.

**JIL attribute:** run\_calendar

### **Days: Run days**

(Optional) Specifies the days to run the job.

**JIL attribute:** days\_of\_week

### **Exclude calendar**

(Optional) Specifies whether to use the calendar selected from the associated list as an *exclude* calendar at run time. An exclude calendar specifies days on which the job should not run.

**JIL attribute:** exclude\_calendar

**Job load**

(Defaulted) Defines the relative load of the job and can be any user-defined value. The value should have some relationship to the value defined by the `max_load` attribute.

**Default:** 0 (no load is assigned)

**NOTE**

The `max_load` attribute is an attribute for a machine, not a job.

**JIL attribute:** `job_load`

**Must complete times**

(Optional) Defines the amount of time that a job is allowed to run or a specific time by which a job must run before completing. You can specify a relative value to the start time (Minutes after each hour) or an absolute value. If the job has not completed before the time expires, an alarm is issued. Relative times must be preceded by a plus sign (+). Absolute times must be specified in `hh:mm` format, where `hh` denotes hours (in 24-hour format) and `mm` denotes minutes. To specify multiple absolute times, separate each value by a comma.

**Limits:** Up to 255 characters; 71:59 maximum absolute value

**Examples:** **+5 (relative time), 11:30, 12:30, 15:30 (absolute times)**

**JIL attribute:** `must_complete_times`

**Must start times**

(Optional) Defines the amount of time that a job is allowed to wait or a specific time until which a job must wait before starting. You can specify a relative value to the start time (Minutes after each hour) or an absolute value. If the job has not started before the time expires, an alarm is issued. Relative times must be preceded by a plus sign (+). Absolute times must be specified in `hh:mm` format, where `hh` denotes hours (in 24-hour format) and `mm` denotes minutes. To specify multiple absolute times, separate each value by a comma.

**Limits:** Up to 255 characters; 71:59 maximum absolute value

**Examples:** **+5 (relative time), 11:30, 12:30, 15:30 (absolute times)**

**JIL attribute:** `must_start_times`

**Priority**

(Defaulted) Defines the queue priority of the job. The lower the value, the higher the priority; therefore, 1 is the highest possible queued priority. 0 signifies to run the job immediately, regardless of the current machine load.

**Limits:** 0-2147483647

**Default:** 0

**JIL attribute:** `priority`

**Run window**

(Optional) Specifies the interval during which a job may start in the format `hh:mm-hh:mm`, where `hh` denotes hours (in 24-hour format) and `mm` denotes minutes. The specified interval can overlap midnight, but it cannot encompass more than 24 hours.

**Limits:** Up to 20 characters

**JIL attribute:** `run_window`

**Time: Times of day**

(Optional) Specifies the times to start the job in `hh:mm` format, where `hh` denotes hours (in 24-hour format) and `mm` denotes minutes. To specify multiple times, separate each value by a comma.

**Limits:** Up to 255 characters

**Example:** 12:00, 00:00 runs the job at noon and midnight

**JIL attribute:** start\_times

**Time: Minutes after each hour**

(Optional) Specifies the number of minutes after each hour, in *mm* format, at which to start the job. To specify multiple values, separate each value by a comma.

**Limits:** Up to 255 characters

**Example:** 00, 15, 30, 45 runs the job every 15 minutes

**JIL attribute:** start\_mins

**Time zone**

(Optional) Specifies the time zone for the job. The job's time settings are based on this time zone, regardless of the server location. Specify a string that corresponds to an entry in the ujo\_timezones table of the AutoSys Workload Automation database. On UNIX, you can specify any time zone recognized by the operating system.

**UNIX Limits:** Up to 50 characters; valid characters are a-z, A-Z, decimal digits, slash (/), hyphen (-), and underscore (\_); not case-sensitive

**Windows Limits:** Do not include spaces or periods; not case-sensitive

**Examples:** IST-5:30, MountainTime (instead of Mountain Time), SolomonIs (instead of Solomon Is)

**JIL attribute:** timezone

**Times to restart job after failure**

(Defaulted) Specifies how many attempts should be made to restart the job after failure.

**Limits:** 0-200

**Default:** 0

**JIL attribute:** n\_retrys

### **Specifying Environment Variables**

Environment variables define the local environment where a script, command, or batch file runs. A profile defines the non-system environment variables that a job uses. You specify environment variables and profiles using the Environment variables and Profile properties, located under the Environment category in the Properties section of a job definition.

### **Environment Variables Examples**

The following examples provide samples of using environment variables on UNIX and Windows.

#### **Example: Pass UNIX Environment Variables to a Script**

This example passes two system environment variables to a script and a third environment variable that defines the Present Working Directory (PWD). In this example, the pay script can reference these variables:

Environment Variable	Value Passed
NAME	user 1
JOB	PAYROLL
PWD	/usr/scripts/dailyrun

**Follow these steps:**

1. Create a job.
2. Enter the following properties:
  - Name -- UNIX\_JOB
  - Send to machine -- UNIX\_NY
  - Command -- /home/scripts/pay
  - Environment variables -- NAME=user 1,JOB=PAYROLL,PWD=/usr/scripts/dailyrun
3. Commit the job.

**Example: Set the User's Profile on UNIX**

This example sets the user's profile my\_profile in the home directory /usr/home:

**Follow these steps:**

1. Create a job.
2. Enter the following Environment property:
  - Job environment -- /usr/home/my\_profile
3. Commit the job.

**Example: Pass Windows Environment Variables to a Batch File**

This example passes three environment variables to a Windows batch file and a fourth environment variable that defines the home directory. In this example, the command command1 can reference these variables:

Environment Variable	Value Passed
A	B
C	'X Y'
E	pay
HOME	c:\export\u1

**Follow these steps:**

1. Create a job.
2. Enter the following properties:
  - Name -- NT\_JOB
  - Send to machine -- NT\_NY
  - Command -- c:\payroll\command1
  - Environment variables -- A=B,C=X Y,E=pay,HOME=c:\export\u1
3. Commit the job.

**Example: Set the Windows Environment from a Profile**

This example sets the environment from the profile eng (which was previously defined in the Job Profiles Manager):

**Follow these steps:**

1. Create a job.
2. Enter the following Environment property:
  - Job environment -- eng
3. Commit the job.

**Example: Set the Environment from a Profile on a Machine Other than the Machine on which the Job Runs**

This example sets the environment from the profile eng that was defined on the machine dev, which is different from the machine on which the job runs:

**Follow these steps:**

1. Create a job.
2. Enter the following Environment property:
  - Job environment -- \\dev\eng
3. Commit the job.

**Environment Properties**

**NOTE**

This section will not display for certain job types that do not have Environment properties.

The Environment category includes the following properties:

**Environment variables**

(Optional) Specifies environment variables to define the local environment where the script, command, or batch file runs. You can modify existing environment variables.

This field uses the following format:

```
parm_name=parm_value[, parm_name=parm_value...]
```

- ***parm\_name***

Specifies the name of an existing environment variable.

**NOTE**

Enclose names that contain spaces in apostrophes.

**Example: 'exceptions 1'**

- ***parm\_value***

Specifies the value of the environment variable.

**NOTE**

Enclose values that contain spaces in quotation marks.

**Example: "cmd1 cmd2 cmd3"**

**JIL attribute:** envvars

**File system check**

(Optional) Defines the minimum amount of file space that must be available on designated drives for the job to start. Windows: Specify one or more drives (by drive letter) and their corresponding sizes in kilobytes (KB). Only drives are checked; directories, if specified, are ignored. When specifying drive letters, you must escape the colon with quotation marks or backslashes.

**Example: "C: 100 D: 120"** UNIX: Specify one or more file systems (with full path names or directory names) and their corresponding sizes in kilobytes (KB).

**Example: /roots 100 /auxfs1 120**

**Limits:** Up to 255 characters for the entire value

**NOTE**

If you specify multiple drive or file system/size pairs, separate each with a single space.

**JIL attribute:** chk\_files

### Profile

(Optional) Specifies a job profile that defines environment variables to set before executing the specified command. Enter the name of a profile located on the job's client or a path name using the following format that includes the machine on which the profile resides:

*\\machine\_name\profile\_name*

**Limits:** Up to 255 characters

**Default:** If you omit this field, the scheduling manager uses the "default" profile on Windows and the /etc/auto.profile on UNIX.

### NOTE

This attribute is only available for Command, User Defined, and File Watcher job types.

**JIL attribute:** profile

### Ulimit

(Optional) Specifies the UNIX shell's resource usage limit using the following format:

*resource\_type="soft\_value,hard\_value"[, resource\_type="soft\_value,hard\_value" ...]*

- **resource\_type**

Specifies the resource type. The following options are available:

- c -- Specifies the core file size in blocks.
- d -- Specifies the data segment size in kilobytes (KB).
- f -- Specifies the maximum file size in blocks.
- m -- Specifies the process virtual size in kilobytes (KB).
- n -- Specifies the number of files.
- s -- Specifies the stack size in kilobytes (KB).
- t -- Specifies the CPU time in seconds.
- **soft\_value**

Defines the usage (soft) limit for the specified resource type. The soft limit can be increased up to the specified hard limit and can be changed without root authority.

**Limits:** Any numeric digits or you can specify **unlimited**; the value must be less than or equal to the hard limit

- **hard\_value**

Defines the maximum usage (hard) limit for the specified resource type. The hard limit can only be increased by the root user. Any user ID can decrease a hard limit.

**Limits:** Any numeric digits or you can specify **unlimited**; the value must be greater than or equal to the soft limit

### NOTE

If the job runs under a non-root user ID and the value specified in the job definition is greater than the current hard limit on the UNIX system, the hard limit will not be increased.

**Example:** c="100,200", s="250,300"

**JIL attribute:** ulimit



## Defining Job Termination Conditions

You can specify job termination conditions using the Termination category of a job definition. For example, you can specify exit codes that AutoSys Workload Automation interprets as job success or the maximum number of minutes a job can run before it is terminated.

You can specify one or more of the following job termination conditions:

- [Exit codes to indicate job failure or success](#)
- [Maximum runtime](#)
- [Automatic deletion when a job completes](#)
- [Termination of a failed box job or containing box](#)

## Exit Codes to Indicate Job Failure or Success

By default, the scheduling manager interprets an exit code of zero (0) as job success and any other number as job failure. However, you can map exit codes other than 0 as job success and you can map specific codes as job failure. You can also specify a maximum exit code the job can finish with to still be considered successful.

To define exit codes to indicate job failure or success use the following Termination properties within a job definition:

- Failure exit codes
- Success exit codes
- Max exit code for success

### NOTE

These properties apply to Command, i5/OS, Micro Focus, z/OS Regular, and z/OS Manual jobs.

## Example: Define a Range of Exit Codes to Indicate Job Failure

Suppose that you want a job named CMDJOB to run the procjob.exe file. The job is considered to have failed if the exit code is in the range of 40-50.

### Follow these steps:

1. Create a job.
2. Enter the following Primary properties:
  - Name -- cmdjob
  - Send to machine -- AGENT
3. Enter the following Command property:
  - Command -- "c:\temp\procjob.exe"
4. Enter the following Termination property:
  - Failure exit codes -- 40-50
5. Save the job.

## Example: Define an Exit Code to Indicate Job Success

This example shows the first and last lines of the payroll.sh script. The script returns the self-defined exit code 100 to the scheduling manager.

```
#!/usr/bin/sh
.
.
.
exit 100
```

The following job definition runs the script. The Success exit codes property defines exit code 100 as success, indicating successful completion of the script.

**Follow these steps:**

1. Create a job.
2. Enter the following Primary properties:
  - Name -- test\_blob
  - Send to machine -- AGENT
3. Enter the following Command property:
  - Command -- /home/esp/payroll.sh
4. Enter the following Termination property:
  - Success exit codes -- -0,100
5. Save the job.

### **Maximum Runtime**

The Minutes to wait before terminating property under the Termination category specifies the maximum run time (in minutes) that the job you are defining requires to finish normally. If the job runs longer than the specified time, AutoSys Workload Automation terminates it.

#### **Example: Terminate the Job if it Runs Beyond the Maximum Run Time**

This example specifies that AutoSys Workload Automation should terminate the job if it runs for more than two hours:

```
Minutes to wait before terminating -- 120
```

### **Automatic Deletion When a Job Completes**

The Hours after successful completion to delete job property specifies whether to automatically delete the job after completion. If the job is a Box job, the product deletes the Box job and all the jobs it contains.

This property is useful for scheduling and running a one-time batch job. You can specify the interval after the job's completion at which to delete the job.

#### **Example: Delete a Job on Successful Completion**

This example specifies that AutoSys Workload Automation should automatically delete the job five hours after successful completion:

```
Hours after successful completion to delete job -- 5
```

### **Termination of a Failed Box Job or Containing Box**

The Terminate containing box on job failure property in the Termination category of a job definition specifies whether to terminate the Box job containing the job you are defining when the job completes with a FAILURE or TERMINATED status. This property only applies to jobs that are in a box.

The Terminate on failure of containing box property specifies whether to use a KILLJOB event to terminate the job if its containing Box job completes with a FAILURE or TERMINATED status.

Use these two Termination properties to control how nested jobs react when a job fails.

### **Termination Properties**

The Termination category includes the following properties:

#### **Failure exit codes**

(Optional) Defines which exit codes indicate job failure. Specify a single exit code (for example, 4) a range of exit codes (for example, 0-9999), a list of multiple exit codes separated by commas, or a list of ranges separated by commas.

**Limits:** Up to 256 characters. The exit code must be an integer in the range 0 to 2147483647.

**Default:** Any exit code other than 0 (zero) indicates job failure

**Examples:** 1,150,20-30

**NOTE**

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

**JIL attribute:** fail\_codes

**Hours after successful completion to delete job**

(Defaulted) Specifies the maximum time (in hours) after job completion that the job should exist before being automatically deleted.

**Limits:** -1-17520

**Default:** -1 (job will never be deleted automatically)

**JIL attribute:** auto\_delete

**Max exit code for SUCCESS**

(Defaulted) Defines the maximum exit code that the job can exit with and be considered a success.

**Limits:** 0-2,147,483,647

**Default:** 0

**NOTE**

- This field will not display for certain job types.
- AutoSys Workload Automation reserves exit codes 120 and greater for internal use.

**JIL attribute:** max\_exit\_success

**Minutes to wait before terminating**

(Defaulted) Specifies the maximum run time (in minutes) for the job. If the job runs longer than the specified time, it is automatically terminated.

**Default:** 0

**JIL attribute:** term\_run\_time

**Success exit codes**

(Optional) Defines which exit codes indicate job success. Specify a single exit code (for example, 4), a range of exit codes (for example, 0-9999), or a list of multiple exit codes or ranges separated by commas.

**Example:** 1,20-30,150

**NOTE**

- This field will not display for certain job types.
- If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

**JIL attribute:** success\_codes

**Terminate containing box on job failure**

(Optional) Indicates whether to terminate the Box job containing the job if the job fails or terminates.

**JIL attribute:** box\_terminator

## Terminate on failure of containing box

(Optional) Indicates whether to terminate the job if the Box job it is in fails or terminates.

**JIL attribute:** job\_terminator

## Setting Up Notifications

Notifications let you automatically monitor a job's progress and notify users or take action when the job reaches a certain state or condition. You can specify properties within a job definition to generate an alarm, send an email notification, or create a service desk request. Notifications let you automate your workload processing.

You can specify one or more of the following notification actions:

- [Specify to Send an Alarm Under Certain Conditions](#)
- [Send a notification when a job completes](#)
- [Create a service desk request when a job fails](#)

You can specify properties within a job definition to generate an alarm that alerts an operator to take corrective action. You can set up an alarm for when a job completes with a FAILURE or TERMINATED status, or when a job completes in less than or greater than a specified time.

### Follow these steps:

1. Create a job.
2. Expand the Notification category.
3. Specify the following optional Alarm properties:
  - Minimum run time (minutes)**  
Specifies the minimum run time for the job in minutes. When the job ends before the specified interval elapses, Web UI generates an alarm.
  - Maximum run time (minutes)**  
Specifies the maximum run time for the job in minutes. When the job does not finish before the specified interval elapses, Web UI generates an alarm.
  - Send alarm on failure**  
Specifies whether to generate an alarm if the job fails for any reason other than the minimum or maximum run time elapses.
  - Send alarm on termination**  
Specifies whether to generate an alarm when the job terminates before completion.
4. Commit the job.  
When the job completes under one of the defined conditions, Web UI generates an alarm.

## Specify to Send a Notification When a Job Completes

You can specify properties within a job definition to send a notification to someone when the job completes. You can specify whether Web UI sends a notification if the job completes with a FAILURE status or when the job completes with any status.

### Follow these steps:

1. Create a job.
2. Expand the Notification category.
3. Specify the following Notification services properties:
  - **Notification**  
Specifies whether a notification is sent when a job completes and if so, the condition under which a notification is sent. Options are:
    - **Do not notify**
    - No notification is sent.
    - **Send notification on selected alarms and any job status**

Displays all notification options for the job status: Success, Failure, Terminate, and when an alarm is raised.

- **Send notification on selected alarms and job failure**

Displays only failed jobs for notification.

- **ID** Specifies the notification ID or email address of the user who receives the notification.
- **Default Email address** Specifies one or more email addresses to which the notification is sent.

#### NOTE

This field is valid only for AutoSys Workload Automation Release 11.3.5 and later servers.

- **Email template** Specifies the email template that is used for notifications.
  - **Email address for job success notifications**  
Specifies the email address to which notifications of successfully completed jobs are sent.
  - **Email address for job failure notifications**  
Specifies the email address to which notifications of failed jobs are sent.
  - **Email address for job terminate notifications** Specifies the email address to which notifications of terminated jobs are sent.
  - **Email address for alarm notifications** Specifies the email address of the recipient of the email notification when an alarm is raised.
  - **Alarm type** Specifies the alarm for which to send the email notification. You can select any of the following alarms: MINRUNALARM, JOBFAILURE, JOBTERMINATED, MAX\_RETRY, STARTJOBFAIL, EVENT\_HDLR\_ERROR, EVENT\_QUE\_ERROR, JOBNOT\_ONICEHOLD, MAXRUNALARM, RESOURCE, MISSING\_HEARTBEAT, CHASE, DATABASE\_COMM, VERSION\_MISMATCH, DUPLICATE\_EVENT, INSTANCE\_UNAVAILABLE, AUTO\_PING, EXTERN\_DEPS\_ERROR, SERVICEDesk\_FAILURE, UNINOTIFY\_FAILURE, CPI\_JOBNAME\_INVALID, CPI\_UNAVAILABLE, MUST\_START\_ALARM, MUST\_COMPLETE\_ALARM, WAIT\_REPLY\_ALARM, KILLJOBFAIL, SENDSIGFAIL, REPLY\_RESPONSE\_FAIL, RETURN\_RESOURCE\_FAIL, RESTARTJOBFAIL, JOBNOT\_ONNOEXEC, QUEUEDJOB\_STARTFAIL, SUSPENDJOBFAIL, RESUMEJOBFAIL. You can select multiple check boxes. If you select ALL, an email notification is sent when any of the above listed alarms are raised.

#### NOTE

In releases till 11.4 SP5, the JOBFAILURE alarm specified whether to post an alarm when a job completes with FAILURE or TERMINATED status. After r11.4 SP5, you can specify a Alert policy individually for JOBTERMINATED or JOBFAILURE. This helps you distinguish the alarms based on the FAILURE and TERMINATED job statuses.

Before you select any of the above options, you must select an email template. However, for sending an email notification when an alarm is raised, you can select either an email template or define the message in the Message property.

**To select an email template, follow these steps:**

- Click the  icon next to the Email template field. The Search for Email Template dialog opens.
- Enter the search criteria or the template name and click Go. The Search Results are displayed.

#### NOTE

You can sort the data in the table by clicking on a table column heading.

- Select the required email template. The template properties are displayed at the bottom of the dialog.
- Click OK to complete selecting your email template.

#### NOTE

You can specify either the ID or the Email address in your job definition. If you specify both, an error message is displayed.

- **Message** Defines a message to include in the notification.

4. Commit the job.  
When the job completes with the status specified, AutoSys Workload Automation sends a notification to the ID or notification email addresses specified.

### **Specify Whether to Create a Service Desk Request When a Job Fails**

You can specify properties within a job definition to create a service desk request when the job completes with a FAILURE status.

#### **Follow these steps:**

1. Create a job.
2. Expand the Notification category.
3. Select the Service desk request check box.
4. Specify the following optional Service desk request properties:
  - **Description**  
Specifies a description to include in the CA Service Desk request generated when the job fails.
  - **Priority**  
Specifies the priority level of the request.
  - **Impact**  
Specifies the impact level of the request
  - **Severity**  
Specifies the severity level of the request.

#### **NOTE**

Alternatively you can use the Additional information property to set attribute values. For more information, see the CA Service Desk documentation.

5. Commit the job.  
When the job fails, AutoSys Workload Automation sends a service desk request.

### **Notification Properties**

The Notification category includes the following properties:

#### **Alarms: Maximum run time (minutes)**

(Defaulted) Specifies the maximum run time for the job in minutes. When the job does not finish before the specified interval elapses, AutoSys Workload Automation generates an alarm.

**Limits:** 0-65535

**Default:** 0

**JIL attribute:** max\_run\_alarm

#### **Alarms: Minimum run time (minutes)**

(Defaulted) Specifies the minimum run time for the job in minutes. When the job ends before the specified interval elapses, AutoSys Workload Automation generates an alarm.

**Limits:** 0-65535

**Default:** 0

**JIL attribute:** min\_run\_alarm

#### **Alarms: Send alarm on failure**

(Defaulted) Specifies whether to generate an alarm if the job fails for any reason other than the minimum or maximum run time elapses.

**Default:** selected (generates alarm if job fails)

**JIL attribute:** alarm\_if\_fail

#### **Notification Services: Message**

(Optional) Defines a message to include in the notification.

This property supports Global Variable substitution.

**Limits:** Up to 255 characters

**JIL attribute:** notification\_msg

#### **Notification Services: Notification**

(Defaulted) Specifies the conditions under which a notification is sent when a job completes, as follows:

- Do not notify -- Does not send a notification.
- Notify only if job fails -- Sends a notification only if the job completes with a FAILURE status.
- Notify unconditionally -- Sends a notification when the job completes with any status.

**Default:** Do not notify

**JIL attribute:** send\_notification

#### **Notification Services: Default Email address**

(Optional) Specifies the email address or addresses to which the notification is sent.

This property supports Global Variable substitution.

**JIL attribute:** notification\_emailaddress

#### **NOTE**

- This field is valid only for AutoSys Workload Automation Release 11.3.5 and later servers.
- You can specify either the ID or the Email address in your job definition. If you specify both, an error message is displayed.

#### **Service Desk Request: Additional information**

(Optional) Specifies additional information to include in the service desk request generated when the job fails.

**Limits:** Up to 255 characters

**JIL attribute:** svcdesk\_attr

#### **Service Desk Request: Description**

(Optional) Specifies a description to include in the service desk request generated when the job fails.

**Limits:** Up to 255 characters

**JIL attribute:** svcdesk\_desc

#### **Service Desk Request: Impact**

(Optional) Specifies the impact level of the request, as follows:

- 0 - None
- 1 - High
- 2 - Medium-high
- 3 - Medium
- 4 - Medium-low
- 5 - Low

**JIL attribute:** svcdesk\_imp

#### **Service Desk Request: Priority**

(Optional) Specifies the priority level of the request, as follows:

- 0 - None
- 1 - High
- 2 - Medium-high
- 3 - Medium
- 4 - Medium-low
- 5 - Low

**JIL attribute:** svcdesk\_pri

#### **Service Desk Request: Service desk request**

(Optional) Indicates whether to open a service desk request when the job fails.

**JIL attribute:** service\_desk

#### **Service Desk Request: Severity**

(Optional) Specifies the severity level of the request, as follows:

- 1 - Low
- 2 - Medium-low
- 3 - Medium
- 4 - Medium-low
- 5 - High

**JIL attribute:** svcdesk\_sev

#### **Specify the Users with Edit and Execute Permissions**

Native security permissions specify which users have edit and execute permissions for the job you are defining.

#### **NOTE**

The job's owner always has full edit and execute permissions.

#### **Follow these steps:**

1. Create a job.
2. Select one or more of the Native security permissions properties:
  - Group (execute)
  - Group (edit)
  - World (execute)
  - World (edit)
  - All hosts (execute)
  - All hosts (edit)



3. Save the job.

### **Example: Set Permissions to Execute a Job on Windows**

This example sets permissions so that any authorized user can execute the job regardless of the machine they are logged on to. A Command job running on Windows is defined with the following native security permission selected:

- All hosts (execute)

### **Example: Set Permissions to Execute and Edit a Job on UNIX**

This example sets permissions so that any user can execute the job, but only members of your group can edit it. A Command job running on UNIX is defined with the following native security permissions selected:

- Group (edit)
- World (execute)

### **Native Security Permissions Properties**

The Native Security Permissions category includes the following properties:

#### **Permissions**

(Optional) Indicates the permission levels to associate with the job, as follows:

- **Group (execute)**  
Assigns group execute permissions to the job.  
**JIL attribute:** permission: gx
- **Group (edit)**  
Assigns group edit permissions to the job.  
**JIL attribute:** permission: ge
- **World (execute)**  
Assigns world execute permissions to the job.  
**JIL attribute:** permission: wx
- **World (edit)**  
Assigns world edit permissions to the job.  
**JIL attribute:** permission: we
- **All hosts (execute)**  
Indicates that any authorized user may execute the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).  
**JIL attribute:** permission: mx
- **All hosts (edit)**  
Indicates that any authorized user may edit the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).  
**JIL attribute:** permission: me

### **Specifying One-Time Job Overrides**

You can use the Create Override action in Quick Edit to specify an override that changes the behavior of a specific job during its next run. Job overrides are applied only once. If a RESTART event is generated because of system problems, AutoSys Workload Automation reissues a job override until the job actually runs once, or until the maximum number of retries limit is met. After this, AutoSys Workload Automation discards the override.

You can modify the following properties in a job override:

- Auto hold job when box starts
- Command
- Condition
- Use date/time conditions
- Exclude Calendar
- File(s) to watch
- Send to machine
- Maximum run time (minutes)
- Minimum file size
- Minimum run time (minutes)
- Minutes after each hour
- Minutes to wait before terminating
- Profile
- Run calendar
- Run days
- Run window
- Standard error file
- Standard input file
- Standard output file
- Times of day
- Times to restart job after failure
- Watch interval

You cannot save a job if an override results in an invalid job definition. For example, if a job definition has only one starting condition, you cannot set the Times of day property to NULL because removing the start condition makes the job definition invalid (no start time could be calculated).

**NOTE**

The maximum number of job restarts after system or network failure is specified in the Times to restart job after failure property in the Schedule category.

One-time job overrides are applied to jobs restarted due to system problems, but are not applied to jobs restarted because of application failures.

System problems include the following:

- Machine unavailability
- Media failures
- Insufficient disk space

Application failures include the following:

- Inability to read or write a file
- Command not found
- Exit status greater than the defined maximum exit status for success
- Various syntax errors

**How Job Overrides are Set**

You can specify the properties to override for a selected job, or temporarily delete a job property, using the Create Override action in Quick Edit.

**Follow these steps:**

1. Find the appropriate job in Quick Edit.
2. Do one of the following:
  - Click  in the Actions column to the right of that job.
  - Click the job name in the Search Results section, select Create Override from the Select an action drop-down list, and then click Go.

The Properties section displays with only certain fields that can be modified in each category.
3. Expand the appropriate categories and modify the properties as necessary.
4. Commit the job.  
The job override for the next run is saved.

**NOTE**

You can use Quick Edit to view the most current override created for a job by selecting View Override from the Select an action drop-down list, and then clicking Go. However, the original override remains stored in the AutoSys Workload Automation database.

When the job completes with the status specified, AutoSys Workload Automation sends a notification to the ID or notification email addresses specified.

**Provide Read-Only Access to Jobs**

You can provide read-only access to group members so that they can view the jobs in Web UI, but cannot perform any actions like Send Event.

**Example:**

There are three users already created and assigned to the application group and can access Web UI with default permissions. Now, we can restrict those users to have read-only access to these jobs.

**Follow these steps:**

1. Log in to EEM and select the WorkloadAutomationAE as the application.
2. Create a group App\_Team\_1 and users Appteam\_user1, Appteam\_user2 & Appteam\_user3 as part of this group.
3. Go to the as\_job policy under "Manage Access Policies" and create a as\_job policy.
4. Under the Identities section, select Type as Global Group and search for your app group. In this case it is App\_Team\_1, click the right arrow.
5. Under the "Access Policy Configuration" select everything except "read" and under add resource, add your instance name.\* (for example, ACE.\*) as you are applying this instance wide.
6. Scroll to the top, select "Explicit Deny" and click Save.

**NOTE**

Your new policy will be available under "Explicit Denies" after you save, and not under "Explicit Grants" .

7. Verify the access using one of the following methods:
  - Go to Permissions and check in EEM to see if the user has *Allow* for read and *Deny* for the rest of the actions.
  - Log in to Web UI as an application user, filter jobs, and try to run jobs or put them on hold. The following message is displayed:  
"Error: E142004 Error occurred while sending On Hold event to BOX1 job. Reason:CAUAJM\_W\_10419 Job Execute Access Denied! "

## Manage Common Job Properties (12.0 UI)

This article explains how to manage common job properties such job owner, starting conditions, and job status dependencies in the modified Graphic User Interface in Version 12.0

### General

The General category includes the following properties:

#### Condition

(Optional) Defines any combination of dependencies based on job status, job exit codes, or global variables.

**Limits:** Up to 4096 characters

#### NOTE

- To configure complex conditions, combine a series of conditions with the AND or the OR logical operators. You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following condition statement lets the dependent job start when either JobA and JobB return SUCCESS or JobD and JobE return SUCCESS, FAILURE, or TERMINATED:

```
(success (JobA) AND success (JobB)) OR (done (JobD) AND done (Job E))
```

**JIL attribute:** condition

#### Resources

(Optional) Specifies virtual resource dependencies.

#### NOTE

To specify a virtual resource dependency, the virtual resource must already be defined on the database. Otherwise, you receive an error.

**Limits:** Up to 4096 characters

**JIL attribute:** resources

#### Owner

(Defaulted) Defines the owner of the job. By default, all jobs are created with the owner *authenticated\_user@host*. If you are using a shared instance of CA EEM between AutoSys Workload Automation and Web UI, the *authenticated\_user* is the Web UI logged-in user name; however, if you are not sharing the same instance of CA EEM, the *authenticated\_user* is the operating system user who defined the job. This user must be defined in the Credentials application. If your job requires a different user ID to run successfully, you must override the default owner value. For example, the default owner ID and associated password typically do not correlate to an operating system user ID for jobs that are run by subsystems (environments other than a command line interfaces), such as ERP or database subsystems. For these jobs, the owner property should be overridden with an ID that correlates to the subsystem. Attempts made to execute a job with an incorrect owner will result in job execution failures.

#### NOTE

- If AutoSys Workload Automation is running in native security mode, you can change the owner value only if you have EDIT superuser permissions.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you can change the owner value only if you have as-owner authority.

**Default:** *authenticated\_user@host*

**Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** owner

## Box

(Optional) Identifies an existing Box job in which to put the job.

Clicking the Search icon opens the Search Box Job dialog, from which you can select a box job for the job definition. For Quick Edit, the box job must reside on the selected server.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen (-)

**JIL attribute:** box\_name

## Group

(Optional) Defines the name of the group with which to associate the job. You can only associate a job with one group, but a group may have many jobs associated with it.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

**JIL attribute:** group

## Application

(Optional) Defines the name of the application with which to associate the job. You can only associate a job with one application, but an application may have many jobs associated with it.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

**JIL attribute:** application

## Agent job class

(Optional) Specifies the name of an initiator class defined in the agentparm.txt file. If the class name is not available on the target machine, the job will fail to be submitted.

**Limits:** Up to 32 characters

### NOTE

The class must be a string without blanks in it.

**JIL attribute:** job\_class

## Initial Status

Sets an initial status for a job during insertion.

**Valid values:** INACTIVE, SUCCESS, FAILURE, TERMINATED, ON ICE, ON HOLD, or ON NO EXECUTION.

**JIL attribute:** status

## Termination

The Termination category includes the following properties:

### Failure exit codes

(Optional) Defines which exit codes indicate job failure. Specify a single exit code (for example, 4) a range of exit codes (for example, 0-9999), a list of multiple exit codes separated by commas, or a list of ranges separated by commas.

**Limits:** Up to 256 characters. The exit code must be an integer in the range 0 to 2147483647.

**Default:** Any exit code other than 0 (zero) indicates job failure

**Examples:** 1,150,20-30

**NOTE**

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

**JIL attribute:** fail\_codes

**Hours after successful completion to delete job**

(Defaulted) Specifies the maximum time (in hours) after job completion that the job should exist before being automatically deleted.

**Limits:** -1-17520

**Default:** -1 (job will never be deleted automatically)

**JIL attribute:** auto\_delete

**Max exit code for SUCCESS**

(Defaulted) Defines the maximum exit code that the job can exit with and be considered a success.

**Limits:** 0-2,147,483,647

**Default:** 0

**NOTE**

- This field will not display for certain job types.
- AutoSys Workload Automation reserves exit codes 120 and greater for internal use.

**JIL attribute:** max\_exit\_success

**Minutes to wait before terminating**

(Defaulted) Specifies the maximum run time (in minutes) for the job. If the job runs longer than the specified time, it is automatically terminated.

**Default:** 0

**JIL attribute:** term\_run\_time

**Success exit codes**

(Optional) Defines which exit codes indicate job success. Specify a single exit code (for example, 4), a range of exit codes (for example, 0-9999), or a list of multiple exit codes or ranges separated by commas.

**Example:** 1,20-30,150

**NOTE**

- This field will not display for certain job types.
- If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

**JIL attribute:** success\_codes

**Terminate containing box on job failure**

(Optional) Indicates whether to terminate the Box job containing the job if the job fails or terminates.

**JIL attribute:** box\_terminator

**Terminate on failure of containing box**

(Optional) Indicates whether to terminate the job if the Box job it is in fails or terminates.

**JIL attribute:** job\_terminator

## **Schedule**

The Schedule category includes the following properties:

### **NOTE**

#### **General considerations:**

- If "Run window" and "Times of day/Minutes after each hour" are specified, any of the values in the "Times of day" or "Minutes after each hour" fields must be in the Run window range.  
**Example:** When Run window is 09:00-09:30 and Times of day is 07:30, 08:00, 09:01, here 09:01 is in run window range; if we specify 09:31 in place of 09:01 it is not valid or if "Minutes after each hour" is specified as 01,32,43,55 and if we enter 31 in place of 01 it is not valid.
- If the "Run window" start time is greater than the "Run window" end time, then the end time is considered the next day's time.
- If you specify "Times of day" or "Minutes after each hour", you must specify "Run calendar" or "Run days".
- Start times must be after the previous must start and complete times.  
**Example:** When "Times of day" is 07:30, 08:00, 09:01 and "Must Start times" is 07:45, 08:20, 09:30, in "Times of day" if we replace 08:00 with 07:35 it displays an error as it is less than previous must start time which is 07:45.

### **Times to restart job after failure**

(Defaulted) Specifies how many attempts should be made to restart the job after failure.

**Limits:** 0-200

**Default:** 0

**JIL attribute:** n\_retrys

### **Average run time**

(Optional) Specifies the average run time (in minutes) to use for a job that is newly submitted to the database.

**Limits:** 0-527040

**JIL attribute:** avg\_runtime

### **Priority**

(Defaulted) Defines the queue priority of the job. The lower the value, the higher the priority; therefore, 1 is the highest possible queued priority. 0 signifies to run the job immediately, regardless of the current machine load.

**Limits:** 0-2147483647

**Default:** 0

**JIL attribute:** priority

### **Job load**

(Defaulted) Defines the relative load of the job and can be any user-defined value. The value should have some relationship to the value defined by the max\_load attribute.

**Default:** 0 (no load is assigned)

### **NOTE**

The max\_load attribute is an attribute for a machine, not a job.

**JIL attribute:** job\_load

### **Auto hold job when box starts**

(Optional) Indicates whether to place an automatic hold on the job when its containing Box job starts. Selecting the check box places an automatic hold on the job. Clearing the check box removes the auto hold from the job.

**JIL attribute:** auto\_hold

### **Use date/time conditions**

(Optional) Indicates whether to use date or time conditions specified to determine when to run the job.

#### **NOTE**

Ensure there are no errors when you specify.

**JIL attribute:** date\_conditions

### **Time zone**

(Optional) Specifies the time zone for the job. The job's time settings are based on this time zone, regardless of the server location. Specify a string that corresponds to an entry in the ujo\_timezones table of the AutoSys Workload Automation database. On UNIX, you can specify any time zone recognized by the operating system.

**UNIX Limits:** Up to 50 characters; valid characters are a-z, A-Z, decimal digits, slash (/), hyphen (-), and underscore (\_); not case-sensitive

**Windows Limits:** Do not include spaces or periods; not case-sensitive

**Examples:** IST-5:30, MountainTime (instead of Mountain Time), SolomonIs (instead of Solomon Is)

**JIL attribute:** timezone

### **Run window**

(Optional) Specifies the interval during which a job may start in the format *hh:mm-hh:mm*, where *hh* denotes hours (in 24-hour format) and *mm* denotes minutes. The specified interval can overlap midnight, but it cannot encompass more than 24 hours.

#### **NOTE**

Times of day and Minutes after each hour must be within the specified Run window interval.

**Limits:** Up to 20 characters

**JIL attribute:** run\_window

### **Days: Run calendar**

(Optional) Specifies the calendar to define the schedule for running the job.

**JIL attribute:** run\_calendar

### **Days: Run days**

(Optional) Specifies the days to run the job.

**JIL attribute:** days\_of\_week

### **Exclude calendar**

(Optional) Specifies whether to use the calendar (Standard, Extended, or All) selected from the associated list as an *exclude* calendar at run time. An exclude calendar specifies days on which the job should not run.

**JIL attribute:** exclude\_calendar

### **Time: Times of day**



(Optional) Specifies the times to start the job in *hh:mm* format, where *hh* denotes hours (in 24-hour format) and *mm* denotes minutes. To specify multiple times, separate each value by a comma.

**Limits:** Up to 255 characters

**Example:** 12:00, 00:00 runs the job at noon and midnight

**JIL attribute:** start\_times

**Time: Minutes after each hour**

(Optional) Specifies the number of minutes after each hour, in *mm* format, at which to start the job. To specify multiple values, separate each value by a comma.

**Limits:** Up to 255 characters

**Example:** 00, 15, 30, 45 runs the job every 15 minutes

**JIL attribute:** start\_mins

**Must start times**

(Optional) Defines the amount of time that a job is allowed to wait or a specific time until which a job must wait before starting. You can specify a relative value to the start time (Minutes after each hour) or an absolute value. If the job has not started before the time expires, an alarm is issued. Relative times must be preceded by a plus sign (+). Absolute times must be specified in *hh:mm* format, where *hh* denotes hours (in 24-hour format) and *mm* denotes minutes. To specify multiple absolute times, separate each value by a comma.

**Limits:** Up to 255 characters; 71:59 maximum absolute value, 4319 maximum relative value

**Examples:** +5 (relative time), 11:30, 12:30, 15:30 (absolute times)

**JIL attribute:** must\_start\_times

**Must complete times**

(Optional) Defines the amount of time that a job is allowed to run or a specific time by which a job must run before completing. You can specify a relative value to the start time (Minutes after each hour) or an absolute value. If the job has not completed before the time expires, an alarm is issued. Relative times must be preceded by a plus sign (+). Absolute times must be specified in *hh:mm* format, where *hh* denotes hours (in 24-hour format) and *mm* denotes minutes. To specify multiple absolute times, separate each value by a comma.

**Limits:** Up to 255 characters; 71:59 maximum absolute value, 4319 maximum relative value

**Examples:** +5 (relative time), 11:30, 12:30, 15:30 (absolute times)

**JIL attribute:** must\_complete\_times

## **Notifications**

Notifications let you automatically monitor a job's progress and notify users or take action when the job reaches a certain state or condition. You can specify properties within a job definition to send an email notification, generate an alarm, or create a service desk request. Notifications let you automate your workload processing.

You can specify one or more of the following notification actions:

- **Notification Services**

You can specify properties within a job definition to send a notification to someone when the job completes. You can specify whether Web UI sends a notification if the job completes with a FAILURE status or when the job completes with any status.

- **Alarms**

You can specify properties within a job definition to generate an alarm that alerts an operator to take corrective action. You can set up an alarm for when a job completes with a FAILURE or TERMINATED status, or when a job completes in less than or greater than a specified time.

- **Service Desk Requests**

You can specify properties within a job definition to create a service desk request when the job completes with a FAILURE status.

## **Notification Services**

You can specify properties within a job definition to send a notification to someone when the job completes. You can specify whether Web UI sends a notification if the job completes with a FAILURE status or when the job completes with any status.

### **WARNING**

Unicenter notification is discontinued from AutoSys Workload Automation Version 12.0. Unicenter notification which was available in previous releases, continue the feature.

### **Follow these steps:**

1. Create a job or edit a job for which you want to create notification services.
2. Expand the Notification services category.
3. Specify the following Notification services properties:

- **Send notification**

Specifies whether a notification is sent when a job completes and if so, the condition under which a notification is sent. Options are:

- **Do not notify**

No notification is sent.

### **NOTE**

Ensure there are no errors when you specify.

- **On any job status**

Notification is sent on job completion, whatever the job status.

- **On job failure**

Notification is sent when the job fails.

- **Email notification** Specifies whether a email notification is sent when a job completes and if so, the condition under which a notification is sent. The options under Email options are Email message and Email template:

- **Email options**

Select one of the following options from the drop-down:

- **Email message**

Defines a message to include in the notification. Specifies whether a email notification message is sent when a job completes and if so, the condition under which a notification is sent.

Select from the following options on when you want notifications:

Notify on

- **Any job status**

Displays all notification options for the job status: Success, Failure, and Terminated.

- **Job failure**

Displays only failed jobs for notification.

Enter the following fields as required:

- **Default email address**

Specifies one or more email addresses to which the notification is sent.

**Limits:** Valid characters are letters, numbers, at sign (@), period (.), hyphen (-), and underscore (\_)

**JIL attribute:** notification\_emailaddress

- **Message**

Defines a message to include in the email notification.

**Limits:** Up to 255 alphanumeric characters

**JIL attribute:** notification\_msg

- **Email template**

Specifies the email template that is used for notifications. You can specify whether the email notification message is to be based on a template and sent when a job completes and if so, the condition under which a notification is sent.

- **Template**

Select the required email template.

**Limits:** 256 characters

**JIL attribute:** notification\_template

#### NOTE

Click the Preview template to view the selected template.

Select from the following options on when you want notifications:

Notify on

- **Selected alarms and on any job status**

Displays all notification options for the job status: Success, Failure, Terminated, and when an alarm is raised.

- **Selected alarms and on job failure**

Displays only failed jobs for notification and when an alarm is raised.

#### NOTE

The email address for job success and the email address for job termination fields are not available for this option.

Enter the following fields as required:

- **Default email address**

Specifies one or more email addresses to which the notification is sent.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -) for each line.

**JIL attribute:** notification\_emailaddress

#### NOTE

This field is valid only for AutoSys Workload Automation Release 11.3.5 and later servers.

- **Email address for job success**

Specifies the email address to which notifications of successfully completed jobs are sent.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -) for each line.

**JIL attribute:** notification\_emailaddress\_on\_success

- **Email address for job failure**

Specifies the email address to which notifications of failed jobs are sent.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -) for each line.

**JIL attribute:** notification\_emailaddress\_on\_failure

- **Email address for job termination**

Specifies the email address to which notifications of terminated jobs are sent.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -) for each line.

**JIL attribute:** notification\_emailaddress\_on\_terminated

- **Alarm type**

Specifies the alarm for which to send the email notification. You can select any of the following alarms: AUTO\_PING, CHASE, CPI\_JOBNAME\_INVALID, CPI\_UNAVAILABLE, DATABASE\_COMM, DUPLICATE\_EVENT, EVENT\_HDLR\_ERROR, EVENT\_QUEUE\_ERROR, EXTERN\_DEPS\_ERROR, INSTANCE\_UNAVAILABLE, JOBFailure, JOBNOT\_ONICEHOLD, JOBNOT\_ONNOEXEC, JOBTERMINATED, KILLJOBFAIL, MAX\_RETRYs, MAXRUNALARM, MINRUNALARM, MISSING\_HEARTBEAT, MUST\_COMPLETE\_ALARM, MUST\_START\_ALARM,

QUEUEDJOB\_STARTFAIL, REPLY\_RESPONSE\_FAIL, RESOURCE, RESTARTJOBFAIL, RESUMEJOBFAIL, RETURN\_RESOURCE\_FAIL, SENDSIGFAIL, SERVICEDESK\_FAILURE, STARTJOBFAIL, SUSPENDJOBFAIL, UNINOTIFY\_FAILURE, VERSION\_MISMATCH, WAIT\_REPLY\_ALARM.

You can select multiple check boxes. If you select ALL, an email notification is sent when any of the above listed alarms are raised.

**JIL attribute:** notification\_alarm\_types

- **NOTE**

In releases till 11.4 SP5, the JOBFAILURE alarm specified whether to post an alarm when a job completes with FAILURE or TERMINATED status. After r11.4 SP5, you can specify a Alert policy individually for JOBTERMINATED or JOBFAILURE. This helps you distinguish the alarms based on the FAILURE and TERMINATED job statuses.

- **Email address for alarm notifications**

Specifies the email address of the recipient to which notifications are sent when an alarm is raised.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -) for each line.

**JIL attribute:** notification\_emailaddress\_on\_alarm

4. Click Save job.

When the job completes with the status specified, AutoSys Workload Automation sends a notification to the specified email addresses.

## **Native Permissions**

Native security permissions specify which users have edit and execute permissions for the job you are defining.

### **NOTE**

The job's owner always has full edit and execute permissions.

### **Follow these steps:**

1. Create a job.
2. Select one or more of the Native security permissions properties:
  - Group (execute)
  - Group (edit)
  - World (execute)
  - World (edit)
  - All hosts (execute)
  - All hosts (edit)
3. Save the job.

### **Example: Set Permissions to Execute a Job on Windows**

This example sets permissions so that any authorized user can execute the job regardless of the machine they are logged on to. A Command job running on Windows is defined with the following native security permission selected:

- All hosts (execute)

### **Example: Set Permissions to Execute and Edit a Job on UNIX**

This example sets permissions so that any user can execute the job, but only members of your group can edit it. A Command job running on UNIX is defined with the following native security permissions selected:

- Group (edit)
- World (execute)

---

## **Native Security Permissions Properties**

The Native Security Permissions category includes the following properties:

### **Permissions**

(Optional) Indicates the permission levels to associate with the job, as follows:

- **Group (execute)**  
Assigns group execute permissions to the job.  
**JIL attribute:** permission: gx
- **Group (edit)**  
Assigns group edit permissions to the job.  
**JIL attribute:** permission: ge
- **World (execute)**  
Assigns world execute permissions to the job.  
**JIL attribute:** permission: wx
- **World (edit)**  
Assigns world edit permissions to the job.  
**JIL attribute:** permission: we
- **All hosts (execute)**  
Indicates that any authorized user may execute the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).  
**JIL attribute:** permission: mx
- **All hosts (edit)**  
Indicates that any authorized user may edit the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).  
**JIL attribute:** permission: me

### **Specifying One-Time Job Overrides**

You can use the Create Override action in Quick Edit to specify an override that changes the behavior of a specific job during its next run. Job overrides are applied only once. If a RESTART event is generated because of system problems, AutoSys Workload Automation reissues a job override until the job actually runs once, or until the maximum number of retries limit is met. After this, AutoSys Workload Automation discards the override.

You can modify the following properties in a job override:

- Auto hold job when box starts
- Command
- Condition
- Use date/time conditions
- Exclude Calendar
- File(s) to watch
- Send to machine
- Maximum run time (minutes)
- Minimum file size
- Minimum run time (minutes)
- Minutes after each hour
- Minutes to wait before terminating
- Profile
- Run calendar
- Run days
- Run window
- Standard error file
- Standard input file
- Standard output file
- Times of day
- Times to restart job after failure
- Watch interval

You cannot save a job if an override results in an invalid job definition. For example, if a job definition has only one starting condition, you cannot set the Times of day property to NULL because removing the start condition makes the job definition invalid (no start time could be calculated).

**NOTE**

The maximum number of job restarts after system or network failure is specified in the Times to restart job after failure property in the Schedule category.

One-time job overrides are applied to jobs restarted due to system problems, but are not applied to jobs restarted because of application failures.

System problems include the following:

- Machine unavailability
- Media failures
- Insufficient disk space

Application failures include the following:

- Inability to read or write a file
- Command not found
- Exit status greater than the defined maximum exit status for success
- Various syntax errors

**How Job Overrides are Set**

You can specify the properties to override for a selected job, or temporarily delete a job property, using the Create Override action in Quick Edit.

**Follow these steps:**

1. Find the appropriate job in Quick Edit.
2. Do one of the following:
  - Click  in the Actions column to the right of that job.
  - Click the job name in the Search Results section, select Create Override from the Select an action drop-down list, and then click Go.

The Properties section displays with only certain fields that can be modified in each category.
3. Expand the appropriate categories and modify the properties as necessary.
4. Commit the job.  
The job override for the next run is saved.

**NOTE**

You can use Quick Edit to view the most current override created for a job by selecting View Override from the Select an action drop-down list, and then clicking Go. However, the original override remains stored in the AutoSys Workload Automation database.

When the job completes with the status specified, AutoSys Workload Automation sends a notification to the ID or notification email addresses specified.

**Provide Read-Only Access to Jobs**

You can provide read-only access to group members so that they can view the jobs in Web UI, but cannot perform any actions like Send Event.

**Example:**

There are three users already created and assigned to the application group and can access Web UI with default permissions. Now, we can restrict those users to have read-only access to these jobs.

**Follow these steps:**

1. Log in to EEM and select the WorkloadAutomationAE as the application.
2. Create a group App\_Team\_1 and users Appteam\_user1, Appteam\_user2 & Appteam\_user3 as part of this group.
3. Go to the as\_job policy under "Manage Access Policies" and create a as\_job policy.
4. Under the Identities section, select Type as Global Group and search for your app group. In this case it is App\_Team\_1, click the right arrow.
5. Under the "Access Policy Configuration" select everything except "read" and under add resource, add your instance name.\* (for example, ACE.\*) as you are applying this instance wide.
6. Scroll to the top, select "Explicit Deny" and click Save.

**NOTE**

Your new policy will be available under "Explicit Denies" after you save, and not under "Explicit Grants" .

7. Verify the access using one of the following methods:
  - Go to Permissions and check in EEM to see if the user has *Allow* for read and *Deny* for the rest of the actions.
  - Log in to Web UI as an application user, filter jobs, and try to run jobs or put them on hold. The following message is displayed:  
"Error: E142004 Error occurred while sending On Hold event to BOX1 job. Reason:CAUAJM\_W\_10419 Job Execute Access Denied! "

## Manage Actions

You can use the Refresh button and the Actions drop-down in the Web UI title bar to perform various actions on an existing job definition.

These options are currently available for HTTP and WSDOC jobs.

### **Refresh**

You can click the Refresh button to refresh the already opened job definition and also display the fields with the latest filled values.

### **Copy a Job**

You can copy an existing job to create a new job with the same properties. You must specify a different name for the new job; otherwise, the job you copied is simply updated.

#### **Follow these steps:**

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click Copy.
3. Define your job by doing the following:
  - a. Enter a name for the new job in the Job Name field.

#### **NOTE**

By default, the job owner attribute and the job definition are copied.

- b. Edit the job definition as necessary.
4. Click Save Job.  
The new job is saved.

#### **NOTE**

Copying a job also copies all its dependencies.

### **Edit a Job**

You can modify date and time conditions, change dependencies or triggers, manage permissions, or change notification settings for a specific job.

#### **Follow these steps:**

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click Edit.
3. Edit the job definition as necessary.

#### **NOTE**

The Name field is read-only.

4. Click Save Job.  
The changes are saved.



## **Rename a Job**

You can rename an existing job, and all dependencies reflect the changed name for that specific job.

### **NOTE**

To rename a job, the user must have write permissions on the existing job which is to be renamed.

### **Follow these steps:**

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click Rename job.
3. Enter the new name of the job.

**Limits:** Up to 64 characters; valid characters are letters, numbers, period (.), underscore (\_), hyphen (-), hash (#), colon (:), and backslash followed by colon (\:); do not include embedded spaces or tabs.

### **NOTE**

- If the specified job name exists, you must specify a different name.
  - When you try to rename a job which has dependencies, a dialog presenting the dependent jobs (internal dependencies and cross-references) is displayed; you can either proceed to rename the job or cancel the rename operation
4. Click Save.  
The job is saved with the new name.

## **Delete a Job**

If you no longer need to manage and schedule a job in your enterprise, you can delete it.

### **Follow these steps:**

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click Delete.  
The following message displays when the job selected for deletion has dependent job(s).  
– The selected job has the following dependent(s), displaying a list of dependent job(s).
3. Click Delete.  
The selected job is deleted.

## **Create a Job Override**

As a scheduler, for one run only, you want to modify the properties of a job that runs on a regular basis. An override lets you modify the properties of a job for the next run only.

For example, you have several jobs that are scheduled to run on a specific day of the week, but you want one of those jobs not to run this week. You can create an override in which you clear the run calendar option for the job. The job does not run the next day that it is normally scheduled to run; however it runs on its scheduled day in the weeks that follow.

### **Follow these steps:**

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.

2. Click the Actions drop-down, and click Create override.  
The Properties section displays with only certain fields that can be modified in each category.
3. Edit the properties as necessary.
4. Click Save override.  
The job override for the next run is saved.

### **Example: Create an Override for a Job**

Suppose that you have some jobs that specify a calendar named RunFridays in their properties and run every Friday. You plan to be out of the office on Friday of this week and you do not want the print jobs to run that day. You want them to run on the following Monday instead. To do this task, you can find the jobs that specify the RunFridays calendar. You can then create an override for just the print jobs, in which you remove the specified calendar and specify Monday as the run day instead. Once you commit the override, you can search for the jobs and view the override details to verify that they are correct. The jobs do not run on the Friday after the day the override is created but run on Fridays as usual thereafter.

### **View a Job Override**

If a one-time override has been defined for a job or box job, you can view the properties and values for the latest override set for the next run.

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click View override.  
The Properties section for the specific job displays the latest properties and values set for the next run.

### **Delete a Job Override**

If a one-time override has been defined for a job or box job and you no longer need it, you can delete it.

1. Find and open the appropriate job.  
The Properties section for the specific job type is displayed, with the relevant categories. For example, the Primary details and Properties section including the categories such as General, Termination, Schedule, Notifications, and Permissions are displayed.
2. Click the Actions drop-down, and click View override.  
The Properties section for the specific job displays the latest properties and values set for the next run.
3. Click the Actions drop-down, and click Delete override.  
A confirmation prompt is displayed.
4. Click OK.  
The job override is deleted.

## **Manage Box Jobs Using Web UI**

A Box job (or box) is a container of other jobs. You can use it to organize and control process flow. The box itself performs no actions, although it can trigger other jobs to run. An important feature of this type of job is that boxes can contain other boxes. You can use boxes to contain other boxes that contain jobs related by starting conditions or other criteria. This feature let you group the jobs and operate on them in a logical manner.

Box jobs are powerful tools for organizing, managing, and administering large numbers of jobs that have similar starting conditions or complex logic flows. Knowing how and when to use boxes is often the result of some experimentation.

For example, assume you want to schedule a group of jobs to start running when a File Watcher job completes successfully. Instead of making each job dependent on the File Watcher job, you can create a Box job that is dependent

on the File Watcher job, remove the File Watcher job dependency from the individual jobs, and put all of those jobs in the box. When the File Watcher job completes successfully, the Box job starts, which in turn starts all the jobs it contains.

#### NOTE

- A box used to group sequential jobs can contain up to 1,000 jobs. However, you should not add a large number of jobs to a single box job. Instead, you should use multiple box jobs and split the dependent jobs between them. Defining box jobs that contain fewer jobs improves performance.
- Jobs in a box start only if the box itself has a status of RUNNING.
- When no other starting conditions are specified at the job level, a job in a box runs when the starting conditions for the box are satisfied. When several jobs in a box do not have job-level starting conditions, they all run in parallel. When any job in a box changes state, other jobs check if they are eligible to start running.
- Jobs in boxes run only once for each box execution. If you specify multiple start times for a job during one box processing cycle, only the first start time is used. This prevents jobs in boxes from inadvertently running multiple times.

### Define a Box Job

You can create a Box job to contain multiple jobs that are related by starting conditions or other criteria. When you put jobs in a box, you only have to start a single job (the box) for all the jobs in the box to start running.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

#### Follow these steps:

1. Create a Box (BOX) job.
2. Specify the following required property:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
3. (Optional) Specify optional [Box properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Box job is defined.

### Box Job Properties

The Box category includes the following properties:

#### Failure

(Optional) Specifies the job failure conditions using a logical expression. The Box job is evaluated against these conditions in addition to all job statuses of jobs within the box. If no job failure condition is specified, the Box job fails if at least one job in the box fails.

**Limits:** Up to 4096

charactersClicking 

the Edit Text dialog, which lets you edit the job failure conditions.

**NOTE**

- You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following expression indicates that the Box job is considered to have failed when either JobA and JobB return FAILURE or JobD returns FAILURE (for example, if you set up JobD to attempt to recover a failed job):

```
(f(JobA) AND f(JobB)) OR f(JobD)
```

**JIL attribute:** box\_failure

**Success**

(Optional) Specifies the job success conditions using a logical expression. The Box job is evaluated against these conditions. If no job success conditions are specified, the Box job succeeds only if all the jobs in the box complete successfully.

**Limits:** Up to 4096

charactersClicking 

the Edit Text dialog, which lets you edit the job success conditions.

**NOTE**

- You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following expression indicates that the Box job is considered successful when either JobA and JobB return SUCCESS or JobD and JobE return SUCCESS, FAILURE, or TERMINATED:

```
(s(JobA) & s(JobB)) | (d(JobD) & d(Job E))
```

**JIL attribute:** box\_success

**Group Jobs and Boxes**

Box jobs provide one method of grouping jobs, but are typically used when all the jobs in the Box job share the same starting condition. AutoSys Workload Automation provides the Group and Application properties so you can logically group sets of jobs and Box jobs with unrelated starting conditions or dependencies. By specifying both Group and Application properties in a job definition, you can make the job belong to both a group logical set and an application logical set.

**NOTE**

For more information, see the *AutoSys Workload Automation Reference section*.

**Example: Associate Jobs with Groups and Applications**

Assume you want to create a set of jobs that run a suite of applications called EmployeePay that is used to manage employee salaries. The Accounting and Human Resources groups each have their own jobs defined to use the EmployeePay applications. The following defines two jobs (HR\_payroll and ACCT\_salaryreport):

```
Name -- HR_payroll
Group -- HumanResources
Application -- EmployeePay
Name -- ACCT_salaryreport
```

```
Group -- Accounting
Application -- EmployeePay
```

These jobs instruct AutoSys Workload Automation to do the following:

- Associate job HR\_payroll with the HumanResources group and the EmployeePay application.
- Associate job ACCT\_salaryreport with the Accounting group and the EmployeePay application.

### **Add a Job to a Box Job**

You can use a Box job to group jobs with like starting conditions such as date/time conditions or job dependency conditions. For example, if you have a number of jobs that run daily at 1:00 a.m., you can add all these jobs to a Box job and assign a daily start condition to the box.

#### **Follow these steps:**

1. Open the job you want to add.
2. Specify the following Primary property to identify the Box job:

#### **– Box**

Identifies an existing Box job in which to put the job.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen (-)

#### **NOTE**

Clicking the Search icon opens the Search Box Job dialog, from which you can select a box job for the job definition. For Quick Edit, the box job must reside on the selected server.

3. Commit the job.  
The job is added to the Box job.

## **Manage Command Jobs Using Web UI**

Command jobs let you run workload on UNIX and Windows client computers. On UNIX, you can define jobs to run scripts and commands. On Windows, you can define jobs to run command files and batch files.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

When you define a Command job, you can specify settings including the following:

- **Starting Conditions**  
Defines conditions (for example, date, time, job state, and machine state) that must be met before a job can start.
- **Disk Space Criteria**  
Defines the amount of free space required before a process can start. If the free space is not available, an alarm is sent and the job is rescheduled.
- **Job Profile**  
Specifies a script to be sourced that defines the environment where the command runs.

#### **NOTE**

On Windows, you can define a job profile using the Job Profiles - AutoSys Workload Automation Administrator window in the Administrator utility.

- **Environment Variables**  
Specifies variables that define the local environment where the job runs.
- **User-defined Exit Codes**

Defines exit codes to indicate job success and job failure. By default, an exit code of 0 (zero) indicates job success and any other code indicates job failure. When the job completes, the exit event (either SUCCESS or FAILURE) and program's exit code are stored in the database.

- **Standard I/O Files**

Specifies the standard input, output, and error files.

**NOTE**

To support chained commands (commands are separated by a semicolon) on UNIX, ensure that the `oscomponent.wrapper.exec.force` parameter in the `agentparm.txt` file is set to false (the default). For more information about supporting chained commands, see the UNIX Implementation section.

## **Define a User Defined Job**

You can define a Command job to run workload on UNIX and Windows computers. The job can run a script, execute a UNIX command, or run a Windows command file.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### **Follow these steps:**

1. Create a Command (CMD) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.

**NOTE**  
The machine must be already defined to AutoSys Workload Automation for the job to be created.

  - **Command**  
Specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.  
**Limits:** Up to 1000 characters; case-sensitive. Up to 512 characters if the AutoSys Workload Automation server that is configured in Web UI is at Release 11.3.6 SP6 or below.  
**Note (UNIX only):** When you define a job that runs a script that calls a second script, the fully qualified path of the called script must be provided.
3. (Optional) Specify optional [Command properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Command job is defined.

## **Command Job Examples (UNIX)**

The following examples describe sample Command jobs that run UNIX scripts and commands.

### **Example: Run a Command on UNIX**

This example runs the `/bin/touch` command on the file named `/tmp/test_run.out`. The job runs on the UNIX client computer named AGENT.

### **Follow these steps:**

1. Create a Command job.

2. Enter the following properties:
  - Name -- test\_run
  - Send to machine -- AGENT
  - Command -- /bin/touch /tmp/test\_run.out
3. Commit the job.

**Example: Run a Script that is Located in a Path Set in the PATH Environment Variable**

Suppose that the job PROCSCRIPT runs a script named procscrip.sh. The job runs under jsmith, the user who has the authority to run the script. The path to procscrip.sh is set in the PATH system environment variable for jsmith. This job runs on the default agent computer.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- PROCSCRIPT
  - Owner -- jsmith
  - Send to machine -- AGENT
  - Command -- procscrip.sh
3. Commit the job.

**Example: Run a Script that is Located in a Path Set in a User Environment Variable**

Suppose that the job MYSCRIPTJOB runs a script named myscrip.sh. The job runs under jsmith, the user who has the authority to run the script. The path to myscrip.sh is set in the user environment variable \$MY\_PATH, which is defined in the profile file for jsmith. This job runs on the default agent computer.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- MYSCRIPTJOB
  - Owner -- jsmith
  - Send to machine -- AGENT
  - Command -- \$MY\_PATH/myscrip.sh
3. Commit the job.

**Example: Run a Script that Uses the C Shell**

In the following example, the C shell is used to run the sort script on the UNIX\_LA agent computer.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- MFGSORT
  - Send to machine -- UNIX\_LA
  - Command -- /mfg/test/sort
  - Shell -- /bin/csh
3. Commit the job.

**Example: Send a User-Defined Exit Code**

The following example shows the first and last lines of the payroll.sh script, which returns the self-defined exit code 100 to the scheduling manager:

```
#!/usr/bin/sh
.
.
.
exit 100
```

In this job, the Success exit codes field defines exit code 100 as SUCCESS, indicating successful completion of the script.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- SUCCESSJOB
  - Owner -- jsmith
  - Send to machine -- SUN\_NY
  - Command -- /home/esp/payroll.sh
  - Success exit codes -- 0,100
3. Commit the job.

**Command Job Examples (Windows)**

The following examples describe sample Command jobs that run Windows command files.

**Example: Run a Command File on Windows**

This example runs the c:\bin\test.bat command on the Windows client computer named AGENT.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- test\_run
  - Send to machine -- AGENT
  - Command -- "c:\bin\test.bat"
3. Commit the job.

**Example: Access a Windows Network Resource with Universal Naming Convention (UNC) and Share Names**

Suppose that the path c:\WINNT\Profiles\Visitor\Desktop\ has the share name MyDesktop. The command notify.cmd is in that path on the CYBNT server, and is accessed by the UNC name and share name in the job JOBC, which runs on the agent WINAGENT.

**NOTE**

A share name is an alias for the path in which the resource exists. A UNC name is the name of a file or other resource that begins with two backslashes (\\), indicating that it exists on a remote computer.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- JOBC
  - Send to machine -- WINAGENT
  - Command -- \\CYBNT\MyDesktop\notify.cmd
3. Commit the job.

**Example: Use Administrative Authority to Access a Remote Windows Resource that is Not Shared**



The agent service can log on to a remote Windows server as a user with administrative authority. Using this authority lets the agent access remote resources that are not marked as shared using the share names C\$ and ADMIN\$. Suppose that drive C is accessed by an administrator over the network through the WINAGENT agent. The agent runs under the System Account option, and it runs the test Application in the c:\working directory on the server CYBNT. The directory c:\working is not a shared resource. The user admin1 is a valid user on both the local and remote computers, and belongs to the Administrators group.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- SHAREFILE
  - Owner -- admin1
  - Send to machine -- WINAGENT
  - Command -- \\CYBNT\C\$\working\test
3. Commit the job.

**Example: Pass Arguments to a Visual Basic Script**

Suppose that you want the job VBS to pass three arguments (one, two, and three) to the Visual Basic script located at D:\temp\vbs\params.vbs. To pass arguments to a Visual Basic script, specify the location of the Visual Basic script and its arguments in the Command field. This job runs on the default agent computer.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- VBS
  - Send to machine -- WINAGENT
  - Command -- c:\winnt\system32\wscript.exe D:\temp\vbs\params.vbs one two three
3. Commit the job.

**Command Properties**

The Command category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

**Blob input**

(Optional) Specifies the text to insert for the blob, using the following format:

```
Blob_input:
<auto_blobt>input</auto_blobt>
```

- **input**  
Specifies the text to insert for the blob.

Clicking



opens the Edit Text dialog, which lets you edit the text for the blob.

**JIL attribute:** blob\_input

### Command

Specifies a command, executable, UNIX shell script, application, or batch file to run on the client using the following format:

This property supports Global Variable substitution.

*file argument...*

- **file**  
Specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.
- **argument...**  
Specifies one or more arguments to pass to the command or script at run time.

### NOTE

Separate each argument with a space. You must specify each argument in the order it is expected in the program.

Clicking



opens the Edit Text dialog, which lets you edit the text for the files and arguments.

**Limits:** Up to 1000 characters; case-sensitive. Up to 512 characters if the AutoSys Workload Automation server that is configured in Web UI is at Release 11.3.6 SP6 or below.

### NOTE

When you define a job that runs a script that calls a second script, the fully qualified path of the called script must be provided.

**JIL attribute:** command

### Heartbeat interval

(Defaulted) Defines the frequency (in minutes) with which the job sends a heartbeat. The heartbeat is a signal that helps the scheduling manager monitor the progress of a job.

**Limits:** 0-2147483647

**Default:** 0 (the agent does not listen for heartbeats)

**JIL attribute:** heartbeat\_interval

### Interactive

(Optional) Indicates whether the specified command needs to be user interactive.

**JIL attribute:** interactive

### Shell

(Optional) Specifies the name of the shell used to execute the script or command file. You can specify any of the following UNIX shells: /bin/ksh (Korn shell)/bin/sh (Bourne shell)/bin/bash (Bourne again shell)/bin/csh (C shell)/usr/local/bin/perl (Perl shell)

**Limits:** Up to 20 characters

**JIL attribute:** shell

### Standard error file

(Optional) Defines the path and file name where you want to redirect all standard error output. If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter >> as the first characters in the value. To overwrite the file, enter > as the first character in the value.

This property supports Global Variable substitution.

**Limits:** Up to 255 characters; cannot contain spaces between the > or >> characters and the full path or file name

#### NOTE

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location.
- Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables. Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

#### UNIX:

- This value overrides the instance-wide setting for the AutoInstWideAppend parameter in the configuration file.
- This value overrides the machine-specific setting for the AutoMachWideAppend parameter in the /etc/auto.profile file.

#### Windows:

- This value overrides the instance-wide setting for the Append stdout/stderr field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.
- If the Windows command specified in the job definition does not exist, the job does not run. The standard error file is not created and the job log indicates that the error file is not found.

**Workload Automation Agent Default:** The agent directory (*installation\_directory/SystemAgent/agent\_name/*)

**JIL attribute:** std\_err\_file

### Standard input file

(Optional) Specifies the path and file name where you want to redirect standard input from.

This property supports Global Variable substitution.

**Limits:** Up to 255 characters

#### NOTE

- The job owner must have read permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location.
- Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables. Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

#### Windows:

- If you do not specify a full path name, the path defaults to the job owner's home directory.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**JIL attribute:** std\_in\_file

### Standard output file

(Optional) Defines the path and file name where you want to redirect all standard output. If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter >> as the first characters in the value. To overwrite the file, enter > as the first character in the value.

This property supports Global Variable substitution.

**Limits:** Up to 255 characters; cannot contain spaces between the > or >> characters and the full path or file name

### NOTE

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location.
- Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \$\$ (global\_name), or \$\$ {global\_name} format to reference global variables. Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### UNIX:

- This value overrides the instance-wide setting for the AutoInstWideAppend parameter in the configuration file.
- This value overrides the machine-specific setting for the AutoMachWideAppend parameter in the /etc/auto.profile file.

### Windows:

- This value overrides the instance-wide setting for the Append stdout/stderr field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Default:** /dev/null (UNIX) or NULL (Windows)

**JIL attribute:** std\_out\_file

### Run in elevated mode

(Optional) Indicates whether to run the Command job in elevated mode on Microsoft Windows operating systems. In elevated mode, the User Account Control (UAC) feature is enabled. UAC improves the security of Microsoft Windows by limiting application software to standard user privileges until an administrator authorizes an elevation.

### NOTE

- This field is valid only for AutoSys Workload Automation Release 11.3.5 and later servers.
- If the agent administrator sets the oscomponent.logon.elevated parameter on the AutoSys Workload Automation server to true in the agentparm.txt file, the agent runs all Command jobs in elevated mode on Windows, regardless of the elevated attribute. To use the elevated attribute in a job definition, the oscomponent.logon.elevated parameter must be set to false.

**Default:** Not selected

**JIL attribute:** elevated

## **Verify File Space Before a Job Starts**

By default, jobs do not check the available file space. However, you can define a Command job to check if one or more file systems (UNIX) or drives (Windows) have the required amount of available space. At run time, the agent checks whether the required space is available on the machine where the job runs. If the requirements are met, the job starts. If the requirements are not met, the agent generates an alarm and the job does not start. The system tries to verify the file space again and start the job. The number of tries is determined by the value entered in the Times to restart job after failure field. If no value is specified in the job definition, the number of tries is determined by the default value set in the AutoSys Workload Automation server configuration file. If the required space is still not available after all the restart attempts, the job fails.

### **Follow these steps:**

1. Define a Command job.
2. Specify the following Environment property in the job definition:
  - **File system check**  
Specifies the required amount of space on one or more file systems (UNIX) or drives (Windows) in kilobytes (KB).  
**Limits:** Up to 255 characters
3. Commit the job.  
The file space is verified before the job starts.

### **Example: Verify the Available File Space on UNIX**

This example checks if the file system named roots has 100 KB of available space. This example also checks if the file system named auxfs1 has 120 KB of available space. The specified file space must be available before the job can start.

### **Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- unix\_chk
  - Send to machine -- unixagent
  - Command -- /u1/procrun.sh
  - File system check -- /roots 100 /auxfs1 120
3. Commit the job.

### **Example: Verify the Available File Space on Windows**

This example checks if the C: drive has 100 KB of available space and if the D: drive has 120 KB of available space. The specified file space must be available before the job can start.

### **Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- win\_chk
  - Send to machine -- winagent
  - Command -- "C:\Programs\Payroll\pay.exe"
  - File system check -- "C: 100 D: 120"
3. Commit the job.

## **Pass Positional Arguments in a Command Job**

When running workload, you might need to pass data between jobs and across platforms. You can pass positional arguments to a command or script in your job definition. Positional arguments are variables that can be passed to a program at the time the program is invoked. The arguments are assigned in the order they are passed.

**Follow these steps:**

1. Define a Command job.
2. Specify the Command property using the following format:

```
file argument...
```

- **file**  
Specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.
- **argument...**  
Specifies one or more arguments to pass to the command or script at run time.

**NOTE**

Separate each argument with a space. You must specify each argument in the order it is expected in the program.

3. Commit the job.  
The positional arguments are passed to the program when the job runs.

**Example: Pass Positional Arguments to a UNIX Script**

This example passes three arguments to the UNIX script addinfo.sh. The first argument passed is "user 1". This argument is enclosed with quotation marks because it contains a space. The second argument passed is 905-555-1212, and the third argument is 749.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- cmd\_job1
  - Send to machine -- unixprod
  - Command -- addinfo.sh "user 1" 905-555-1212 749
3. Commit the job.

**Example: Pass Positional Arguments to a Windows Program**

This example passes two data files to the Windows program pay.exe. The arguments are enclosed with quotation marks because they contain spaces.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:
  - Name -- cmd\_job2
  - Send to machine -- winprod
  - Command -- c:\Programs\Payroll\pay.exe "C:\Pay Data\salary.dat" "C:\Pay Data\benefits.dat"
3. Commit the job.

**Example: Run cmd.exe**

Suppose that you want the job COPYFILE to use the Windows command prompt (cmd.exe) to copy the file c:\env.txt to the test directory. To pass an argument to cmd.exe, you must enclose the argument in quotation marks and precede the argument with the /C switch.

**Follow these steps:**

1. Create a Command job.
2. Enter the following properties:

- Name -- COPYFILE
- Send to machine -- WINAGENT
- Command -- c:\Windows\system32\cmd.exe "/C copy c:\env.txt c:\test\env.txt"

3. Commit the job.

### **UNIX Environment Variables**

When a Command job runs under a specific user account on UNIX, the agent can pass the user's environment variables to the script or program. You can also set up a script's running environment by overriding the environment variables in the job definition. For example, you can override the HOME environment variable to run the script under a user's login directory.

You can pass the following UNIX environment variables in a job definition to override the variable values:

- **HOME**

Identifies the user's login directory. You can override the HOME value to set up a user-specific environment by specifying a different login directory in the job definition.

**Example:** HOME=/home/guest/bin

**NOTE**

- You can set up the script's running environment in the .profile and .login files.
- You must also set the oscomponent.loginshell parameter to true in the agent's agentparm.txt file to run the login scripts located in the HOME directory.

- **PATH**

Provides a list of directories that the shell searches when it needs to find a command. Each directory is separated by a colon, and the shell searches the directories in the order listed. The PATH variable is the most important environment variable. You can override the PATH value to set up a user-specific environment by specifying a different PATH in the job definition.

**NOTE**

Overriding the default system path can result in the "command not found" error.

- **ENV**

Contains the name of the initialization file to run when a new Korn shell starts. You can override the ENV value to set up a user-specific environment by specifying a different ENV value in the job definition.

**Example:** ENV=/home/guest/bin/myenv

**NOTE**

The name of the file used to set up the script-running environment must be .profile. The .profile must be the same file used with the HOME variable.

- **PWD**

Contains the absolute path name of your current directory.

### **Define Alternative Error, Input, and Output Sources and Destinations**

By default, the standard output and standard error output are redirected to /dev/null (UNIX) or NULL (Windows). You can define alternative error, input, and output sources and destinations in your job definition that override the defaults.

#### **Follow these steps:**

1. Define a Command job.
2. Add one or more of the following Command properties to the job definition:
  - **Standard error file**  
Defines the path and file name where you want to redirect all standard error output.
  - **Standard input file**

Specifies the path and file name where you want to redirect standard input from.

- **Standard output file**

Defines the path and file name where you want to redirect all standard output.

3. Commit the job.

The alternative sources are defined.

## Manage File Watcher Jobs Using Web UI

A File Watcher job is similar to a Command job. However, instead of starting a user-specified command on a client computer, a File Watcher job starts a process that monitors for the existence and size of a specific operating system file. When that file reaches the specified minimum size and is no longer growing in size, the File Watcher job completes successfully, indicating that the file has arrived.

Using File Watcher jobs provides a means of integrating events that are external to AutoSys Workload Automation into the processing conditions of jobs. For example, assume a file must be downloaded from a mainframe, and it is expected to arrive after 2:00 a.m. After it arrives, a batch job is run to process it, possibly even starting a whole sequence of jobs.

You could set up a File Watcher job to start at 2:00 a.m., wait for the arrival of the specified file, and exit. You could also set up the batch job so that the completion of the File Watcher job is its only starting condition.

### NOTE

Difference between File Watcher and File Trigger Jobs on Windows

On Windows, File Watcher and File Trigger jobs produce different results when the dot (.) character is used as a literal. The pattern "\*.txt" is treated differently depending on which job is used:

- File Trigger jobs treat a dot character (.) as a literal. To be considered a match, a file must contain a dot and characters on either side of it. For example, if two files exist in the same directory, one named "abc" and the other "abc.txt," only "abc.txt" is considered a match.
- File Watcher jobs use Windows native APIs, which assume the dot character can be implicit. For example, File Watcher jobs consider "abc" to be the same as "abc." So, in this case, either "abc" or "abc.txt" is a match.

### Define a File Watcher Job

You can define a File Watcher job to start a process that monitors for the existence and size of a specific operating system file. When that file reaches the specified minimum size and is no longer growing in size, the File Watcher job completes successfully, indicating that the file has arrived.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### Follow these steps:

1. Create a File Watcher (FW) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **File(s) to watch**  
Specifies the path to and name of one or more files to monitor.
3. (Optional) Specify optional [File Watcher job properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).



5. Commit the job.  
The File Watcher job is defined.

### **File Watcher Job Examples**

The following example describes a sample File Watcher job that monitors file activity.

#### **Example: Creating a File Watcher Job**

This example creates a File Watcher job named EOD\_watch. The job watches every 60 seconds for a file named EodTransFile, located in the /tmp directory. When the watched file reaches the specified minimum size and does not change between check intervals, it is considered complete. When this occurs, the File Watcher job ends with a SUCCESS status.

#### **Follow these steps:**

1. Create a File Watcher job.
2. Enter the following properties:
  - Name -- EOD\_watch
  - Send to machine -- SYSAGENT
  - File(s) to watch -- /tmp/EodTransFile
  - Minimum file size -- 50000
  - Watch interval -- 60
3. Commit the job.

### **File Watcher Properties**

The File Watcher category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **File(s) to watch**

Specifies the path to and name of one or more files to monitor. To specify multiple files, use the asterisk (\*) and question mark (?) wildcard characters. Use \* for any number of characters and ? for any single character. Do not use the \* and ? in the path.

This property supports Global Variable substitution.

**Limits:** Up to 255 characters; case-sensitive

#### **UNIX Notes:**

- You can specify variables exported from the profile script or global variables in the name.
- Wildcard characters are expanded according to the wildcard expansion rules of the Bourne Shell.
- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, \${PATH}). Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

#### **Windows Notes:**

- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, %{PATH}%). Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Example:** /tmp/a\*.data

**JIL attribute:** watch\_file

#### Minimum file size

(Defaulted) Specifies the minimum file size (in bytes) that must be reached to cause this job to complete.

**Default:** 0- 9223372036854775807

**JIL attribute:** watch\_file\_min\_size

#### Watch interval

(Defaulted) Specifies the frequency (in seconds) the File Watcher job checks for the existence and size of the watched file.

**Limits:** 0-2147483647

**Default:** 60

#### NOTE

- The job is considered complete when the watched file reaches the minimum size and remains in a steady state (has not grown or shrunk ) during the specified interval. We recommend that you specify a reasonable interval to help ensure that the file does not appear to be in a steady state when it is still being modified over a longer period of time.
- If you are monitoring for the existence of a file (not the size) and the file already exists when the job runs, the job completes immediately. The value in the Watch interval field is ignored.
- On Workload Automation Agent, if you do not specify this attribute, the job checks the file every 30 seconds (or the time specified in the agent's filemonplugin.sleeperperiod parameter). Additionally, if the file was created or updated between polls, the job checks whether the file remained in a steady state (did not grow) for 60 seconds (the default watch\_interval value).

**JIL attribute:** watch\_interval

## Manage User-Defined Jobs Using Web UI

A User-Defined job is similar to a Command job; however, the command is predefined on AutoSys Workload Automation and not on a per-job basis. If your organization uses a command often in your workload definitions, you can predefine the command on AutoSys Workload Automation and then make it available as a User-Defined job type in Web UI.

For example, suppose you often use the payroll.sh script to run UNIX workload. You can define the following job type 1 on AutoSys Workload Automation:

```
insert_job_type: 1
command: /home/scripts/payroll.sh
```

Once the job type 1 is defined, you can use it to create a User-Defined job using Quick Edit by selecting 1 (/home/scripts/payroll.sh) in the Type drop-down list for that job. Within the User-Defined job, you can add optional arguments to pass to the script. Jobs of type 1 execute /home/scripts/payroll.sh when those jobs are run. When a new version of the payroll.sh script is used, only the definition of job type 1 has to be modified in AutoSys Workload Automation.

## **Define a User-Defined Job**

You can define a User-Defined job to run commands that are predefined on AutoSys Workload Automation. You can define up to nine User-Defined jobs.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### **Follow these steps:**

1. Create a User-Defined job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
Limits: Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
3. (Optional) Specify optional [User-Defined properties](#).

### **NOTE**

Verify that the appropriate AutoSys Workload Automation job type is selected in the Type drop-down list

4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The User-Defined job is defined.

## **User-Defined Job Examples**

The following example describes a sample User-Defined job that runs a script.

### **Example: Create a User-Defined Job to Run a Script**

This example creates a User-Defined job named Run\_Payroll. This job uses job type 1 created in AutoSys Workload Automation that contains the payroll.sh script.

### **Follow these steps:**

1. Create a User-Defined job.
2. Enter the following properties:
  - Name -- Run\_Payroll
  - Send to machine -- localhost
  - Type -- 1 (/home/scripts/payroll.sh)
3. Commit the job.

## **User-Defined Properties**

The User-Defined category includes the following properties:

### **Agent job class**

(Optional) Specifies the name of an initiator class defined in the agentparm.txt file. If the class name is not available on the target machine, the job will fail to be submitted.

**Limits:** Up to 32 characters

### **NOTE**

The class must be a string without blanks in it.

**JIL attribute:** job\_class

### Blob input

(Optional) Specifies the text to insert for the blob, using the following format:

```
Blob_input:
<auto_blobt>input</auto_blobt>
```

#### *input*

Specifies the text to insert for the blob.

Clicking



opens the Edit Text dialog, which lets you edit the text to insert for the blob.

**JIL attribute:** blob\_input

### Heartbeat interval

(Defaulted) Defines the frequency (in minutes) with which the job sends a heartbeat. The heartbeat is a signal that helps the scheduling manager monitor the progress of a job.

**Limits:** 0-21474836470

**Default:** 0 (the agent does not listen for heartbeats)

**JIL attribute:** heartbeat\_interval

### Interactive

(Optional) Indicates whether the specified command needs to be user interactive.

**JIL attribute:** interactive

### Optional arguments

(Optional) Specifies one or more numeric or alphabetic strings of data to pass to a script or command when the job runs. The arguments must be listed in the order expected in the script or command and separated by blank spaces.

Clicking



opens the Edit Text dialog, which lets you edit the argument.

**Limits:** Up to 512 alphanumeric characters

#### **NOTE**

To pass an argument containing spaces, you must enclose its value in quotation marks, for example, 362 "629 630" 748.

**JIL attribute:** command

### Shell

(Optional) Specifies the name of the shell used to execute the script or command file. You can specify any of the following UNIX shells: /bin/ksh (Korn shell)/bin/sh (Bourne shell)/bin/bash (Bourne again shell)/bin/csh (C shell)/usr/local/bin/perl (Perl shell)

**Limits:** Up to 20 characters

**JIL attribute:** shell

## Standard error file

(Optional) Defines the path and file name where you want to redirect all standard error output. If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter >> as the first characters in the value. To overwrite the file, enter > as the first character in the value.

**Limits:** Up to 255 characters; cannot contain spaces between the > or >> characters and the full path or file name

### NOTE

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location. Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.
- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### UNIX:

- This value overrides the instance-wide setting for the AutoInstWideAppend parameter in the configuration file.
- This value overrides the machine-specific setting for the AutoMachWideAppend parameter in the /etc/auto.profile file.

### Windows:

- This value overrides the instance-wide setting for the Append stdout/stderr field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.
- If the Windows command specified in the job definition does not exist, the job does not run. The standard error file is not created and the job log indicates that the error file is not found.

**Workload Automation Agent Default:** The agent directory (*installation\_directory/SystemAgent/agent\_name/*)

**JIL attribute:** std\_err\_file

## Standard input file

(Optional) Specifies the path and file name where you want to redirect standard input from.

**Limits:** Up to 255 characters

### NOTE

- The job owner must have read permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location. Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.
- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### Windows:

- If you do not specify a full path name, the path defaults to the job owner's home directory.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**JIL attribute:** std\_in\_file

### Standard output file

(Optional) Defines the path and file name where you want to redirect all standard output. If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter >> as the first characters in the value. To overwrite the file, enter > as the first character in the value.

**Limits:** Up to 255 characters; cannot contain spaces between the > or >> characters and the full path or file name

### NOTE

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location.
- Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables. Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### UNIX:

- This value overrides the instance-wide setting for the AutoInstWideAppend parameter in the configuration file.
- This value overrides the machine-specific setting for the AutoMachWideAppend parameter in the /etc/auto.profile file.

### Windows:

- This value overrides the instance-wide setting for the Append stdout/stderr field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Default:** /dev/null (UNIX) or NULL (Windows)

**JIL attribute:** std\_out\_file

### Run in elevated mode

(Optional) Indicates whether to run the User-Defined job in elevated mode on Microsoft Windows operating systems. In elevated mode, the User Account Control (UAC) feature is enabled. UAC improves the security of Microsoft Windows by limiting application software to standard user privileges until an administrator authorizes an elevation.

### NOTE

- This field is valid only for AutoSys Workload Automation Release 11.3.5 and later servers.
- If the agent administrator sets the oscomponent.logon.elevated parameter on the AutoSys Workload Automation server to true in the agentparm.txt file, the agent runs all Command jobs in elevated mode on Windows, regardless of the elevated attribute. To use the elevated attribute in a job definition, the oscomponent.logon.elevated parameter must be set to false.

**Default:** Not selected

**JIL attribute:** elevated

### Type

(Optional) Specifies the job type and name of the predefined User-Defined job that was created in AutoSys Workload Automation. All existing AutoSys Workload Automation User Defined jobs are available from the drop-down list.

**JIL attribute:** job\_type

## Manage Entity Bean and Session Bean Jobs Using Web UI

An entity bean represents a data object, such as a customer, an order, or a product. Entity beans may be stored in a relational database, where each instance of the bean corresponds to a row in a database table. Each entity bean has a unique identifier known as a primary key, which is used to find a specific instance of the bean within the database. For example, a customer entity bean may use the customer number as its primary key.

Unlike session beans, which are destroyed after use, entity beans are persistent. You can use an entity bean under the following conditions:

- The bean represents a business entity, not a procedure. For example, you use an entity bean to represent an order and use a session bean to represent the procedure to process the order.
- The state of the bean must be stored. For example, if the bean instance terminates or the application server shuts down, the bean's state will still exist in a database.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and an entity bean residing on an application server:

**Figure 62: functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and an entity bean**



The Entity Bean job lets you create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database. To find the entity bean, the agent uses the bean's Java Naming and Directory Interface (JNDI) name along with its finder method.

Note: To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

To define an Entity Bean job, you require the following information:

- Initial context factory supplied by the JNDI service provider
- Service provider URL for accessing the JNDI services
- Entity bean JNDI name
- Operation type (CREATE, UPDATE, or REMOVE)
- Finder method name (UPDATE and REMOVE operation types only)

A session bean represents business logic or action to be taken (for example, charging a credit card or adding items to an online shopping cart).

Unlike entity beans, which are stored in a database, session beans may be destroyed after each use. For example, when a session bean is invoked to perform credit card validation, the application server creates an instance of that session bean, performs the business logic to validate the credit card transaction, and then destroys the session bean instance after the credit card transaction has been validated.

You can use a session bean under the following conditions:

- The bean represents a procedure and not a business entity. For example, you use a session bean to encrypt data or add items to an online shopping cart.
- The state of the bean does not have to be kept in permanent storage. For example, when the bean instance terminates or the application server shuts down, the bean's state is no longer required.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and a session bean residing on an application server:

**Figure 63: Functional Relationship between Scheduling Manager, Workload Automation Agent for Application Services and Session Bean**



The Session Bean job lets you access a session bean on an application server. This job type can make a Remote Procedure Call (RPC) to the session bean, invoke a method that defines the business logic, pass parameters to the method, and have the results returned as serialized Java output. The output can be stored on the agent computer as text in the spool file or as a serialized Java object in the spool directory or a destination file you specify.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

You can access stateless and stateful session beans using the Session Bean job. The job acts in a similar way for both types of beans. For both stateful and stateless beans, you can specify parameters to pass to the method. When you define a stateful session bean, however, you must specify parameters to define the bean. After the method is invoked, the agent destroys the stateful bean.

Use a stateless Session Bean job to invoke a single instance of a method on the bean, such as encrypting data or sending an email to confirm an order. Use a stateful Session Bean job to invoke the same method on the bean multiple times, such as adding multiple items to an online shopping cart.

A Session Bean job requires a dedicated connection between the agent and the application server. To define a Session Bean job, you require the following information:

- Initial context factory supplied by the Java Naming and Directory Interface (JNDI) service provider
- Service provider URL for accessing the JNDI services
- Session bean JNDI name
- Method to be invoked

### Define an Entity Bean Job

You can define an Entity Bean job to create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.



**Follow these steps:**

1. Create an Entity Bean (ENTYBEAN) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Initial context factory**  
Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **Provider URL**  
Specifies the JNDI service provider URL.
  - **Bean name**  
Specifies the JNDI name of the session or entity bean.
  - **Operation type**  
Specifies the operation to perform on the entity bean. The following options are available:
    - CREATE -- Create an entity bean
    - UPDATE -- Update the property values of an existing entity bean
    - REMOVE -- Remove an entity bean from the database

**NOTE**

- To create an entity bean, you use the Create name and Create parameters properties.
- To update an entity bean, you use the Finder name, Finder parameters, Method name, and Modify parameter properties.
- To remove an entity bean, you use the Finder name and Finder parameters properties.

3. (Optional) Specify optional [Entity Bean properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Entity Bean job is defined.

**Entity Bean Job Examples**

The following examples describe sample Entity Bean jobs.

**Example: Create an Entity Bean**

Suppose that you want to create an entity bean that stores information about a customer such as the customer ID and phone number. The initial context factory supplied by the JNDI service provider is weblogic.jndi.WLInitialContextFactory. The service provider's URL is t3://localhost:7001, where localhost is the domain name of the WebLogic application server and 7001 is the ORB port. When the job runs, the entity bean instance is created.

**Follow these steps:**

1. Create an Entity Bean job.
2. Enter the following properties:

- Name -- newbean
  - Send to machine -- APP\_AGENT
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- t3://localhost:7001
  - Bean name -- customer
  - Operation type -- CREATE
  - Create name -- createcustomer
  - Create parameters -- String=customerid, String=800-555-0100
3. Commit the job.

### Example: Update an Entity Bean

Suppose that you want to update the phone number for the Acme company to 800-555-0199. The customer entity bean stores the customer ID and phone number. The primary key for the customer is the customer ID. To find the entity bean, the job uses the Acme's customer ID. When the job runs, the Acme company's phone number is changed.

#### Follow these steps:

1. Create an Entity Bean job.
2. Enter the following properties:
  - Name -- upbean
  - Send to machine -- APP\_AGENT
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- t3://localhost:7001
  - Bean name -- customer
  - Operation type -- UPDATE
  - Finder name -- acme
  - Finder parameters -- String=customerid, String=800-555-0100
  - Method name -- changephone
  - Modify parameters -- String=customerid, String=800-555-0199
3. Commit the job.

### Example: Remove an Entity Bean

Suppose that you want to remove the customer record for the Acme customer. The record is stored in the database by the customer ID. When the job runs, the row in the customer table that corresponds to the Acme customer ID is removed.

#### Follow these steps:

1. Create an Entity Bean job.
2. Enter the following properties:
  - Name -- delbean
  - Send to machine -- APP\_AGENT
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- t3://localhost:7001
  - Bean name -- customer
  - Operation type -- REMOVE
  - Finder name -- acme
  - Finder parameters -- String=customerid
3. Commit the job.

## **Entity Bean Properties**

The Entity Bean category includes the following properties:

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Bean name**

Specifies the JNDI name of the session or entity bean.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** bean\_name

### **Create name**

(Optional) Specifies the name of the create method.

**Limits:** Up to 256 characters

**JIL attribute:** create\_name

### **Create parameters**

(Optional) Specifies the [create parameters](#) to create an entity bean in a relational database on your application server.

Selecting Add from the Select an action drop-down list and clicking Go displays a row below the Create parameters field.

In that field,

clicking 

the Edit Parameter dialog, which lets you specify the create parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

### **NOTE**

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** create\_parameter

### **Finder name**

(Optional) Specifies the name of the finder method to update the property values of an existing entity bean or to remove an entity bean from the database. To find the entity bean, the agent uses the bean's Java Naming and Directory Interface (JNDI) name along with its finder method.

**Limits:** Up to 256 characters; case-sensitive

### **NOTE**

This field is required when the Operation type is UPDATE or REMOVE.

**JIL attribute:** finder\_name

### **Finder parameters**

(Optional) Specifies the [finder parameters](#) to update the properties of an entity bean or remove an entity bean from a relational database on your application server.

Selecting Add from the Select an action drop-down list and clicking Go displays a row below the Finder parameters field.

In that field,

clicking 

the Edit Parameter dialog, which lets you specify the finder parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

#### NOTE

- Specify the parameters in the same order as they appear in the method. The job fails if the parameters are listed in the wrong order.
- This field is required when the Operation type is UPDATE or REMOVE.

**JIL attribute:** finder\_parameter

#### Initial context factory

Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters

**JIL attribute:** initial\_context\_factory

#### J2EE user name

(Optional) Specifies the JNDI user ID to be used to gain access to the connection factory. **Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces) **JIL attribute:** j2ee\_user

#### Method name

(Optional) Specifies the method to be invoked on the application server.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

This field is required when the Operation type is UPDATE.

**JIL attribute:** method\_name

#### Modify parameters

(Optional) Specifies the [modify parameters](#) to update the property values of an existing entity bean. The modify parameters are used by the setter method specified by the Method name property.

Selecting Add and clicking Go displays a row below the Modify parameters field.

In that field,

clicking 

the Edit Parameter dialog, which lets you specify the modify parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

#### NOTE

Specify the parameters in the same order as they appear in the method. The job fails if the parameters are listed in the wrong order.

**JIL attribute:** modify\_parameter

## Operation type

Specifies the operation to perform on the entity bean. The following options are available:

- **CREATE** -- Create an entity bean
- **REMOVE** -- Remove an entity bean from the database
- **UPDATE** -- Update the property values of an existing entity bean

**JIL attribute:** operation\_type

## Provider URL

Specifies the JNDI service provider URL. Both HTTP and HTTPS are supported. For WebLogic servers, use the following format:

```
t3://WLIPAddress[:ORBPort]
```

- **WLIPAddress**  
Specifies the IP address or domain name of the WebLogic Application Server.
- **ORBPort**  
(Optional) Specifies the ORB port.  
**Default:** 7001  
**Example:** t3://localhost:7001

For WebSphere servers, use the following format:

```
iiop://WSIPAddress[:ORBPort]
```

- **WSIPAddress**  
Specifies the IP address or domain name of the WebSphere Application Server.
- **ORBPort**  
(Optional) Specifies the ORB port.  
**Default:** 2809  
**Example:** iiop://172.24.0.0:2809

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** provider\_url

## Define a Session Bean Job

You can define a Session Bean job to access a stateless or stateful session bean, invoke a method on the bean, and return the results.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

## Follow these steps:

1. Create a Session Bean (SESSBEAN) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Initial context factory**

Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.

- **Provider URL**

Specifies the JNDI service provider URL.

- **Bean name**

Specifies the JNDI name of the session or entity bean.

- **Method name**

Specifies the methods to be invoked on the application server.

3. (Optional) Specify optional [Session Bean properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Session Bean job is defined.

### **Session Bean Job Examples**

The following examples describe sample Session Bean jobs.

#### **Example: Invoke a Method on a Stateless Session Bean**

Suppose that you want to invoke the reverse method on the CybEJBTestBean stateless session bean. The reverse method has one parameter, with type java.lang.String and value a23. The output from the reverse method saves to the file C:\Makapt15. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere application server and 2809 is the ORB port. When the job runs, the output of the reverse method is stored in the output destination file.

#### **Follow these steps:**

1. Create a Session Bean job.
2. Enter the following properties:
  - Name -- reverse
  - Send to machine -- APP\_AGENT
  - Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
  - Provider URL -- iiop://172.24.0.0:2809
  - Bean name -- CybEJBTestBean
  - Method name -- reverse
  - J2EE parameters -- java.lang.String=a23
  - Destination file -- C:\Makapt15
3. Commit the job.

#### **Example: Invoke a Method on a Stateful Session Bean**

Suppose that you want to access a stateful session bean for an online shopping cart. The Createaddbook method creates the Shoppingcart stateful bean for the duration of the job. The addbook method adds books to the shopping cart using the book's ISBN number. In this example, when the Session Bean job runs, it adds two books to the shopping cart with ISBN numbers 1551929120 and 1582701709.

#### **Follow these steps:**

1. Create a Session Bean job.
2. Enter the following properties:

- Name -- addbook
  - Send to machine -- APP\_AGENT
  - Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
  - Provider URL -- iiop://172.24.0.0:2809
  - Bean name -- Shoppingcart
  - Method name -- addbook
  - Create method -- createaddbook
  - Create parameters -- String=ISBN
  - J2EE parameters -- integer=1551929120, integer=1582701709
3. Commit the job.

### **Session Bean Properties**

The Session Bean category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Bean name**

Specifies the JNDI name of the session or entity bean.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** bean\_name

#### **Create method**

(Optional) Specifies the name of the create method.

**Limits:** Up to 256 characters; must always begin with create

**JIL attribute:** create\_method

#### **Create parameters**

(Optional) Specifies the [create parameters](#) to create an entity bean in a relational database on your application server.

Selecting Add and clicking Go displays a row below the Create parameters field.

In that field,

clicking 

the Edit Parameter dialog, which lets you specify the create parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

#### **NOTE**

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** create\_parameter

#### **Destination file**

(Optional) Specifies the output destination file to store the Java serialized object. If you omit this statement, the output is stored in the job's spool file or in the file referenced by the job's URL.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

### Initial context factory

Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters

**JIL attribute:** initial\_context\_factory

### J2EE parameters

(Optional) Specifies the [input parameters](#) of the EJB Method to be invoked remotely.

Selecting Add and clicking Go displays a row below the J2EE parameters field.

In that field,

clicking 

the Edit Parameter dialog, which lets you specify the input parameters.

This property supports Global Variable substitution.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

#### NOTE

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** j2ee\_parameter

### J2EE user name

(Optional) Specifies the JNDI user ID to be used to gain access to the connection factory.

**Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** j2ee\_user

### Method name

Specifies the method to be invoked on the application server.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** method\_name

### Provider URL

Specifies the JNDI service provider URL. Both HTTP and HTTPS are supported. For WebLogic servers, use the following format:

```
t3://WLIPAddress[:ORBPort]
```

- **WLIPAddress**



Specifies the IP address or domain name of the WebLogic Application Server.

- **ORBPort**  
(Optional) Specifies the ORB port.  
**Default:** 7001

**Example:** t3://localhost:7001

For WebSphere servers, use the following format:

```
iiop://WSIPAddress[:ORBPort]
```

- **WSIPAddress**  
Specifies the IP address or domain name of the WebSphere Application Server.
- **ORBPort**  
(Optional) Specifies the ORB port.  
**Default:** 2809

**Example:** iiop://172.24.0.0:2809

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** provider\_url

### Edit Parameters

You can use the Edit Parameter dialog to define parameters for the following job types:

- Entity Bean
- Session Bean
- JMS Publish
- JMX-MBean Attribute Set
- JMX-MBean Create Instance
- POJO
- RMI
- HTTP
- Web Service RPC/Encoded
- Web Service Document/Literal

### **Follow these steps:**

1. Open the Properties section for the job.

#### **NOTE**

For JMX-MBean Attribute Set jobs, skip to Step 3.

2. Select Add from the Select an action drop-down list at the end of the J2EE parameter field, and click Go.  
A blank row appears below the property field to add a parameter.
3. Click   
the end of the row of the parameter you want to edit.  
The Edit Parameter dialog opens.
4. Select one of the following options in the Parameter Type drop-down list:
  - Value -- Identifies a single value
  - Payload producing job -- Identifies the name of a predecessor job
  - Array -- Identifies more than one value
  - Name-Value -- Identifies a parameter name/value pair
  - Name-Payload -- Identifies a parameter name/payload producing job pair

**NOTE**

- HTTP jobs provide only the Value option.
- Web Service RPC/Encoded jobs provide only the Value and Payload producing job options.
- Web Service Document/Literal jobs provide only the Name-Value and Name-Payload options.

5. The Edit Parameter dialog displays new fields based on the chosen parameter type.

6. Complete the following fields as appropriate:

– **Array**

Specifies a set of string values for the method parameter. Select Add Before or Add After from the Select an action drop-down list and click Go for each value you want to enter.

**NOTE**

This field is used with the Array parameter type.

– **Array Type**

Specifies the Java class of the parameter.

**NOTE**

This field is used with the Array parameter type only.

– **Parameter Name**

Specifies the name of the parameter.

**NOTE**

This field is used with the Name-Value and Name-Payload parameter types only.

– **Parameter Value**

Specifies the value assigned to the parameter name.

**NOTE**

This field is used with the Name-Value parameter type only.

– **Payload Producing Job**

Specifies the name of the predecessor job that produced the serialized Java object to be used as an input parameter.

**NOTE**

This field is used with the Payload producing job and Name-Payload parameter types only.

– **Value**

Specifies the String value for the method parameter.

**NOTE**

This field is used with the Value parameter type.

– **Value Type**

Specifies the Java class of the parameter.

**NOTE**

This field is used with the Value parameter type only.

7. Click OK.

The parameter is added to the job definition.

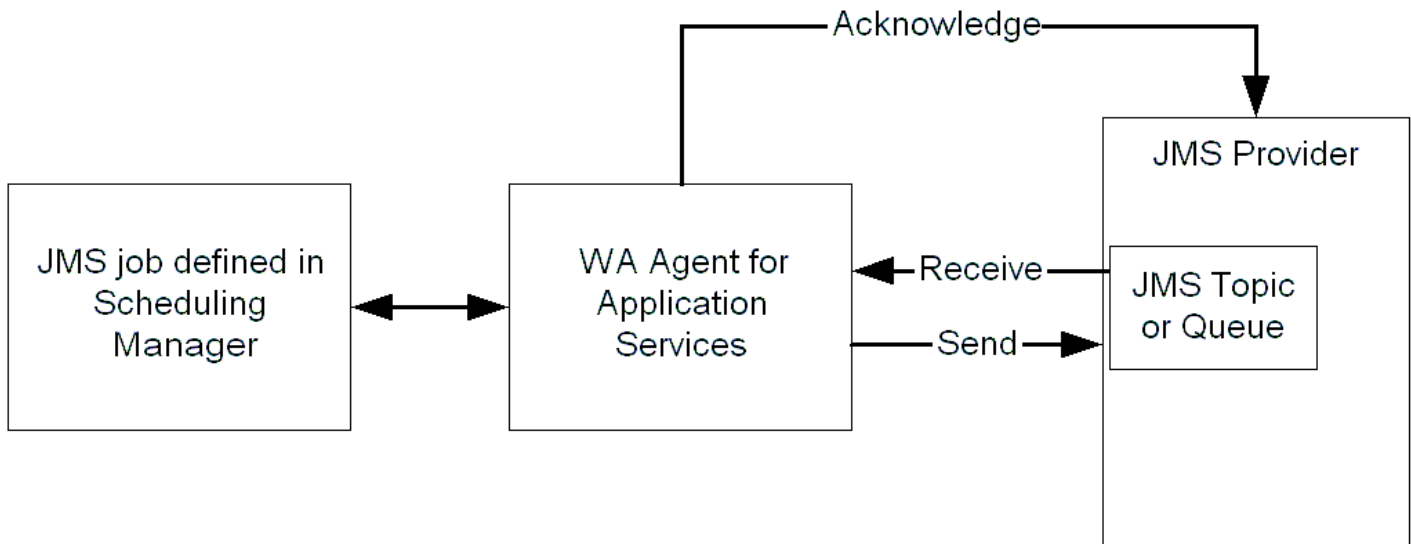
## Manage JMS Publish and JMS Subscribe Jobs Using Web UI

Java Message Service (JMS) is the standard for enterprise messaging that lets a Java program or component (JMS client) produce and consume messages. Messages are the objects that communicate information between JMS clients.

In a JMS system, a messaging server known as the JMS provider acts between two JMS clients (the publisher and the subscriber). Publishers send messages to the JMS provider while subscribers receive messages from the JMS provider.

The following diagram shows the functional relationship between the scheduling manager, the Workload Automation Agent for Application Services, and a JMS provider:

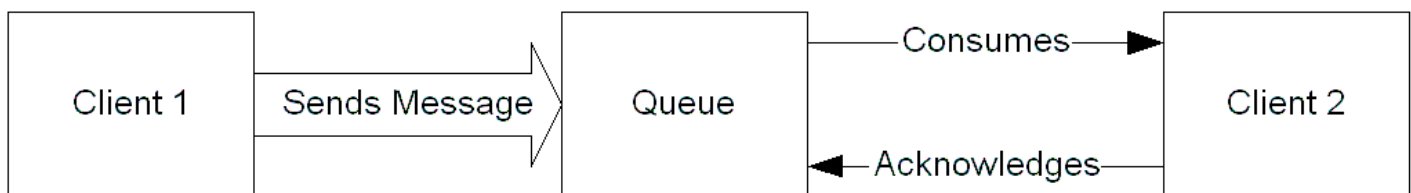
**Figure 64: JMS Publish Jobs**



A queue is an object on the JMS server that holds messages sent by a client that are waiting to be consumed by another client. The queue retains a message until the message is consumed or the message expires.

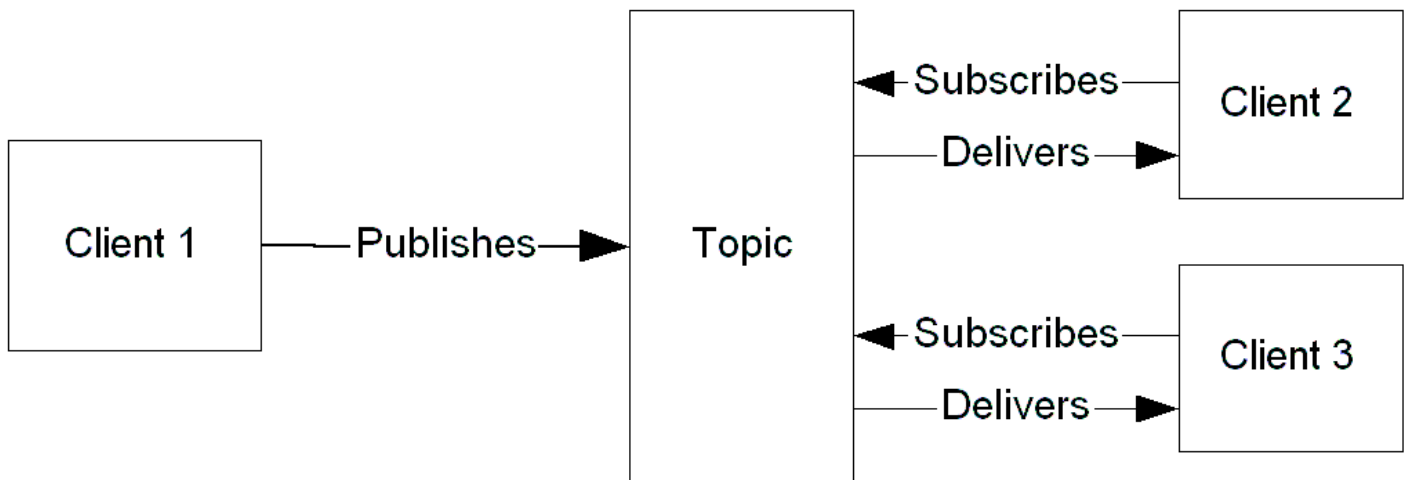
The following diagram shows Client 2 (the subscriber) consuming a message that Client 1 (the publisher) sends to a queue:

**Figure 65: JMS Queue**



A topic is an object a client uses to specify the target of the messages it produces and the source of the messages it consumes. A client acquires a reference to a topic on a JMS server, and sends messages to that topic. When messages arrive for that topic, the JMS provider is responsible for notifying all clients.

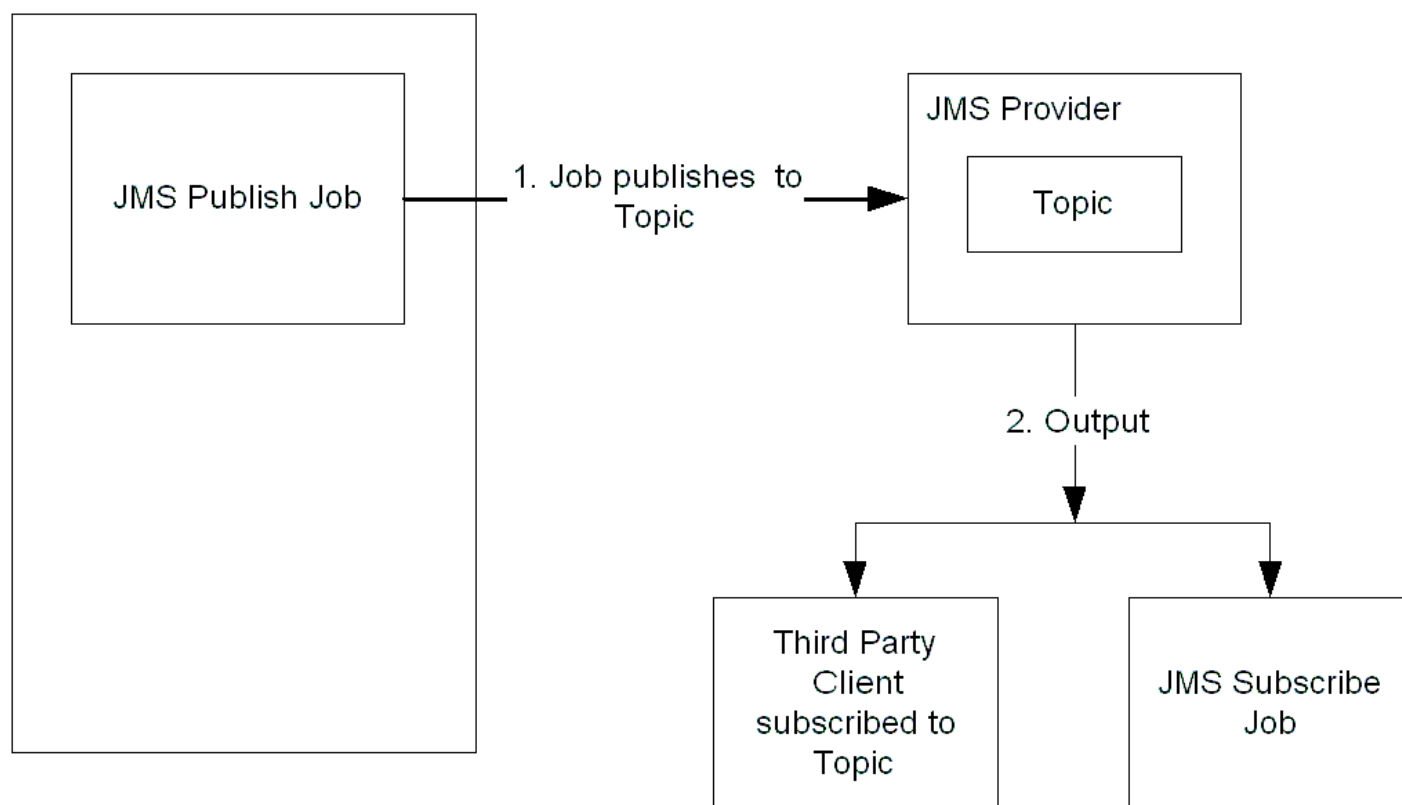
The following diagram shows two subscribers, Client 2 and Client 3, subscribed to a topic that the publisher, Client 1, publishes to:

**Figure 66: JMS Topic**

A JMS Publish job lets you send a message to a queue or publish a message to a topic. Using a JMS Publish job to publish to a topic, you can broadcast a message to any topic subscriber. A third-party client can consume this message, or a JMS Subscribe job can listen for a particular message (using a filter).

**Note:** To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

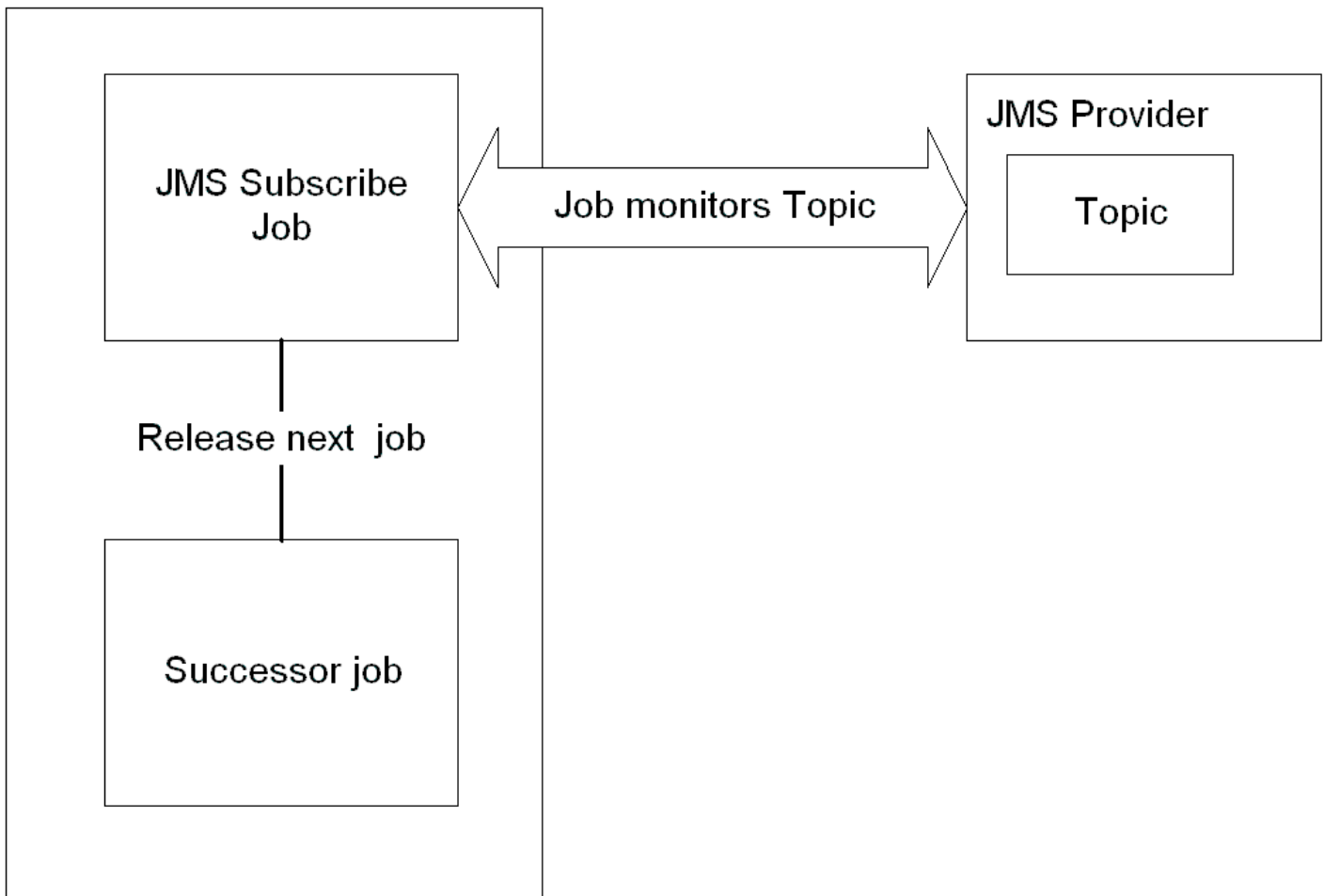
The following diagram shows a JMS Publish job scenario:

**Figure 67: Sample Use of JMS Publish Job**

A JMS Subscribe job lets you consume messages from a queue or topic. Using a filter that you define within the job definition, the agent monitors the topic or queue output for specific data. The scheduling manager then sends the message that meets the filter criteria to a destination file you specify. You can define the job to continuously monitor JMS messages.

Note: To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

The following diagram shows a JMS Subscribe job scenario:

**Figure 68: Sample Use of JMS Subscribe Job**

To define a JMS Publish or JMS Subscribe job, you require the following information:

- Initial context factory supplied by the Java Naming and Directory Interface (JNDI) service provider
- JMS provider URL for accessing the JNDI services
- Connection factory JNDI name that looks up the referenced topic or queue
- JNDI name of the topic or queue on the JMS server
- (JMS Publish jobs only) Java class of the JMS message to send or publish
- (JMS Publish jobs only) Message to send to a queue or publish to a topic

### **Define a JMS Publish Job**

You can define a JMS Publish job to send a message to a queue or publish a message to a topic.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### **Follow these steps:**

1. Create a JMS Publish (JMSPUB) job in either Quick Edit or Application Editor. The Properties section for the JMS Publish job appears.

2. Specify the following required properties:

- **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Initial context factory** Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
  - **Provider URL**  
Specifies the JNDI service provider URL.
  - **Connection factory**  
Specifies the connection factory JNDI name that contains all the bindings needed to look up the referenced topic or queue.
  - **Destination name**  
*Specifies either the JNDI name of the topic or queue or the reply\_to name of the queue. Select one of the following options:*
    - • JNDI name—Specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received. This is the default.
    - • Reply to—Specifies the reply\_to name of the queue. For example: subscriber; the Destination name field is populated with the queue name that is specified in the Reply to field of the publisher job, which is consumed and stored by the subscriber job that is named subscriber.
  - **Message class**  
Specifies the Java class of the JMS message.
  - **J2EE parameters**  
Specifies [Manage Entity Bean and Session Bean Jobs Using Web UI](#) of the JMS message to send to a queue or publish to a topic.
3. (Optional) Specify optional [JMSPublishProperties](#).
4. (Optional) Specify [Manage Common Job Properties](#).
5. Commit the job.  
The JMS Publish job is defined.

## JMS Publish Job Examples

### Example: Publish a Message to the IBM WebSphere MQ server

This example publishes the message "this is my message" to the queue that is named Queue1 whose owner is queuser@host. The Java class of the message is String. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the IBM WebSphere Application server and 2809 is the ORB port. The job uses the connection factory named ConnectionFactory.

#### Follow these steps:

1. Create a JMS Publish job.
2. Enter the following properties:

- Name -- publish
- Send to machine -- APP\_AGENT
- Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
- Provider URL -- iiop://172.24.0.0:2809
- Connection factory -- ConnectionFactory
- Destination name -- Queue1
- Destination user name -- queuser@host
- Use topic -- not selected
- Message class -- String
- J2EE parameters -- java.lang.String='this is my message'

3. Commit the job.

### Example: Specify the Message Correlation ID in a JMS Job

This example defines a JMS Publish job that is named jmssub3 and a JMS Subscribe job that is named jmssub3.

The jmssub3 job publishes the message "Hello World" to the weblogic.examples.jms.exampleQueue queue. The Java class of the message is String. The correlation ID of the message is user\_defined\_CorrelationID. The initial context factory that is supplied by the JNDI service provider is weblogic.jndi.WLInitialContextFactory. The service provider's URL is t3://jms\_server:7001, where jms\_server:7001 is the IP address of the WebLogic Application server and 7001 is the port. The job uses the connection factory named weblogic.examples.jms.QueueConnectionFactory.

#### Follow these steps:

1. Create a JMS Publish job.
2. Enter the following properties:
  - Name -- jmssub3
  - Send to machine -- jms
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- "t3://jms\_server:7001"
  - Connection factory -- weblogic.examples.jms.QueueConnectionFactory
  - Destination name -- weblogic.examples.jms.exampleQueue
  - Use topic -- not selected
  - Message class -- String
  - J2EE parameters -- String="Hello World"
  - Correlation Id -- user\_defined\_CorrelationID
3. Commit the job.

The jmssub3 consumes the message with the user\_defined\_CorrelationID correlation ID from the weblogic.examples.jms.exampleQueue queue.

#### Follow these steps:

1. Create a JMS Subscribe job.
2. Enter the following properties:
  - Name -- jmssub3
  - Send to machine -- jms
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- "t3://jms\_server:7001"
  - Connection factory -- weblogic.examples.jms.QueueConnectionFactory
  - Destination name -- weblogic.examples.jms.exampleQueue
  - Use topic -- not selected
  - Correlation Id -- user\_defined\_CorrelationID



3. Commit the job.

### Example: Specify the Job Name as the Correlation ID

This example defines a JMS Publish job that is named jmssub4 (the correlation\_id is not specified in this job definition) and a JMS Subscribe job that is named jmssub4 (this job uses the correlation ID that is generated by the jmssub4 job).

The jmssub4 job publishes the message "Hello World" to the weblogic.examples.jms.exampleQueue queue. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is weblogic.jndi.WLInitialContextFactory. The service provider's URL is t3://jms\_server:7001, where jms\_server:7001 is the IP address of the WebLogic Application server and 7001 is the port. The job uses the connection factory that is named weblogic.examples.jms.QueueConnectionFactory.

#### Follow these steps:

1. Create a JMS Publish job.
2. Enter the following properties:
  - Name -- jmssub4
  - Send to machine -- jms
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- "t3://jms\_server:7001"
  - Connection factory -- weblogic.examples.jms.QueueConnectionFactory
  - Destination name -- weblogic.examples.jms.exampleQueue
  - Use topic -- not selected
  - Message class -- String
  - J2EE parameters -- String="Hello World"
3. Commit the job.

The jmssub4 job uses the correlation ID that is generated by the jmssub4 job to get the message from the weblogic.examples.jms.exampleQueue queue. The jmssub4 job runs after the jmssub4 job completes successfully.

#### Follow these steps:

1. Create a JMS Subscribe job.
2. Enter the following properties:
  - Name -- jmssub4
  - Send to machine -- jms
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- "t3://jms\_server:7001"
  - Connection factory -- weblogic.examples.jms.QueueConnectionFactory
  - Destination name -- weblogic.examples.jms.exampleQueue
  - Use topic -- not selected
  - Correlation Id -- jmssub4
  - Condition -- s(jmssub4)
3. Commit the job.

### Example: Specify the reply\_to Name of the Queue

To use the Reply to value, the requestor must define a JMS Publish job and a JMS Subscribe job and the responder must define a JMS Subscribe job and a JMS Publish job so it can receive the message, update the message, and publish it. In this example, the requestor and responder defined the jobs as follows:

- publisher1—This is a request publisher job, which expects the reply to be sent to the queue that is specified in the Reply to field.
- subscriber1—This is a responder subscriber job, which receives the message from the publisher1 job.
- publisher2—This is a responder publisher job, which publishes the reply to the requestor job using the reply to value that is specified in the Destination name field.
- subscriber2—This is a request subscriber job, which listens for the reply from the publisher2 job.

The publisher1 job publishes the message "Hello World" to the queue that is named Queue1. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0:2809 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the connection factory that is named ConnectionFactory. The queue to which the reply must be sent is Queue2.

**Follow these steps:**

1. Create a JMS Publish job.
2. Enter the following properties:
  - Name -- publisher1
  - Send to machine -- jms
  - Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
  - Provider URL -- "iiop://172.24.0.0:2809"
  - Connection factory -- ConnectionFactory
  - Destination name -- Queue1
  - Use topic -- not selected
  - Message class -- String
  - J2EE parameters -- java.lang.String='Hello World'
  - Reply to -- Queue2
3. Commit the job.

The subscriber1 job uses the correlation ID that is generated by the publisher1 job name to get the message from the queue that is named Queue1. The subscriber1 job runs only after the publisher1 job completes successfully.

**Follow these steps:**

1. Create a JMS Subscribe job.
2. Enter the following properties:
  - Name -- subscriber1
  - Send to machine -- jms
  - Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
  - Provider URL -- "iiop://172.24.0.0:2809"
  - Connection factory -- ConnectionFactory
  - Destination name -- Queue1
  - Use topic -- not selected
  - Correlation Id -- publisher1
  - Condition -- s(publisher1)
3. Commit the job.

The publisher2 job acts on the message that is received by subscriber1 and sends its response to the reply\_to queue that is specified by publisher1, which is Queue2. The publisher2 job runs only after the subscriber1 job completes successfully.

**Follow these steps:**

1. Create a JMS Publish job.
2. Enter the following properties:

- Name -- publisher2
- Send to machine -- jms
- Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
- Provider URL -- "iiop://172.24.0.0:2809"
- Connection factory -- ConnectionFactory
- Destination name -- reply\_to=subscriber1
- Use topic -- not selected
- Correlation Id -- subscriber1
- Message class -- String
- J2EE parameters -- java.lang.String='Hello World Updated'
- Condition -- s(subscriber1)

3. Commit the job.

The subscriber2 job uses the publisher2 job name as its correlation id to consume the publisher2 message from the queue that is named Queue2. The subscriber2 job runs only after the publisher2 job completes successfully.

**Follow these steps:**

1. Create a JMS Subscribe job.
2. Enter the following properties:
  - Name -- subscriber2
  - Send to machine -- jms
  - Initial context factory -- com.ibm.websphere.naming.WsnInitialContextFactory
  - Provider URL -- "iiop://172.24.0.0:2809"
  - Connection factory -- ConnectionFactory
  - Destination name -- Queue2
  - Use topic -- not selected
  - Correlation Id -- publisher2
  - Condition -- s(publisher2)
3. Commit the job.

### **JMS Publish Properties**

The JMS Publish category includes the following properties:

- **Agent job class**(Optional) Specifies the name of an initiator class defined in the agentparm.txt file. If the class name is not available on the target machine, the job fails to be submitted.  
**Limits:** Up to 32 characters  
**Note:** The class must be a string without blanks in it.
- **Connection factory** Specifies the connection factory JNDI name. The connection factory contains all the bindings needed to look up the referenced topic or queue. JMS jobs use the connection factory to create a connection with the JMS provider.  
 This property supports Global Variable substitution.  
**Limits:** Up to 256 characters; case-sensitive  
**JIL attribute:** connection\_factory
- **Correlation Id**(Optional) Specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue.  
**Limits:** Up to 64 characters; case-sensitive  
**JIL attribute:** correlation\_id

**NOTE**

- Correlation Id can be used only when the Use Topic checkbox is not selected.
  - If you do not specify the correlation id in your job definition, the system generates a unique correlation id. The job that wants to consume the message must refer to the correlation id value of the predecessor job. That is, the correlation id of the message that was last posted by the predecessor job is fetched and the same is consumed by the successor job. In this case, the predecessor job must run first and then the successor.
  - If you specified the correlation id in your job definition and it does not refer to a job, it is treated as a literal string that the consumer consumes directly if the correlation id value matches one of its predecessors. In this case, the order of the predecessor and successor does not matter as we are referring to a literal value and not a system generated id that can change with each run of the job.
- **Destination name** Specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.  
This property supports Global Variable substitution.  
**Limits:** Up to 256 characters; case-sensitive  
**JIL attribute:** destination\_name

**NOTE**

You can specify the reply to value in the Destination name field, as follows:

```
Destination name -- subscriber
```

The Destination name is populated with the queue name that is specified in the Reply to field of the publisher job, which is consumed and stored by the subscriber job that is named subscriber.

- **Destination user name** Specifies the user name that is required to connect to the topic or queue in a JMS job.  
**Limits:** Up to 145 characters  
**JIL attribute:** destination\_user

**NOTE**

If you specify the destination user name in your job definition, you must set the password that is associated with the destination user using the autosys\_secure command.

- **Initial context factory** Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.  
**Limits:** Up to 256 characters  
**JIL attribute:** initial\_context\_factory
- **J2EE parameters** Specifies [Manage Entity Bean and Session Bean Jobs Using Web UI](#) of the EJB Method to be invoked remotely.  
Selecting Add and clicking Go displays a row below the J2EE parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the input parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

**NOTE**

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** j2ee\_parameter

- **J2EE user name** (Optional) Specifies the JNDI user ID to be used to gain access to the connection factory.  
**Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** j2ee\_user

- **Message class** Specifies the Java class of the JMS message.  
**Limits:** Up to 256 characters; case-sensitive

#### NOTE

If the package is not specified, java.lang is assumed.

**JIL attribute:** message\_class

- **Provider URL** Specifies the JNDI service provider URL. Both HTTP and HTTPS are supported. For WebLogic servers, use the following format:

This property supports Global Variable substitution.

```
t3://WLIPAddress[:ORBPort]
```

#### – **WLIPAddress**

Specifies the IP address or domain name of the WebLogic Application Server.

#### – **ORBPort**

(Optional) Specifies the ORB port.

**Default:** 7001

**Example:** t3://localhost:7001. For WebSphere servers, use the following format:

```
iiop://WSIPAddress[:ORBPort]
```

#### – **WSIPAddress**

Specifies the IP address or domain name of the WebSphere Application Server.

#### – **ORBPort**

(Optional) Specifies the ORB port.

**Default:** 2809

**Example:** iiop://172.24.0.0:2809

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** provider\_url

- **Reply to** (Optional) Specifies the JNDI name of the reply to a queue. This is applicable only to the message ID pattern in a queue. **Limits:** 256 characters; case-sensitive **JIL attribute:** reply\_to

#### NOTE

You can also specify the reply to value in the Destination name field, as follows:

```
Destination name -- subscriber
```

The Destination name is populated with the queue name that is specified in the Reply to field of the publisher job, which is consumed and stored by the subscriber job that is named subscriber.

- **Use topic** (Optional) Indicates whether the job publishes messages to a topic or queue. When selected, the agent publishes messages to a topic.  
**Default:** not selected (job publishes to a queue)  
**JIL attribute:** use\_topic

### Define a JMS Subscribe Job

You can define a JMS Subscribe job to consume messages from a queue or topic.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### Follow these steps:

1. Create a JMS Subscribe (JMSSUB) job in either Quick Edit or Application Editor.

The Properties section for the JMS Subscribe job appears.

2. Specify the following required properties:

- **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
- **Send to machine**  
Specifies the name of the machine on which the job runs.
- **Initial context factory** Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.
- **Provider URL**  
Specifies the JNDI service provider URL.
- **Connection factory**  
Specifies the connection factory JNDI name that contains all the bindings needed to look up the referenced topic or queue.
- **Destination name**  
Specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.

3. (Optional) Specify optional [JMSSubscribeProperties](#).

4. (Optional) Specify [Manage Common Job Properties](#).

5. Commit the job.  
The JMS Subscribe job is defined.

## **JMS Subscribe Job Example**

The following example describes a sample JMS Subscribe job.

### **Example: Monitor a Queue on a WebLogic Application Server**

This example continuously monitors the queue named Queue (residing on WebLogic) for messages matching the filter criteria. The consumed messages from the queue are stored in the file /export/home/user1/outputfile1. The service provider's URL is t3://172.24.0.0:7001, where 172.24.0.0 is the IP address of the WebLogic Application server and 7001 is the port.

#### **Follow these steps:**

1. Create a JMS Subscribe job.
2. Enter the following properties:
  - Name -- monitor
  - Send to machine -- APP\_AGENT
  - Initial context factory -- weblogic.jndi.WLInitialContextFactory
  - Provider URL -- t3://172.24.0.0:7001
  - Connection factory -- ConnectionFactory
  - Correlation Id -- correlation\_id
  - Filter -- abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
  - Destination file -- /export/home/user1/outputfile1
  - Destination name -- Queue
  - Use topic -- not selected
3. Commit the job.

## JMS Subscribe Properties

The JMS Subscribe category includes the following properties:

- **Agent job class**(Optional) Specifies the name of an initiator class defined in the agentparm.txt file. If the class name is not available on the target machine, the job will fail to be submitted.

**Limits:** Up to 32 characters

### NOTE

The class must be a string without blanks in it.

**JIL attribute:** job\_class

- **Connection factory** Specifies the connection factory JNDI name. The connection factory contains all the bindings needed to look up the referenced topic or queue. JMS jobs use the connection factory to create a connection with the JMS provider.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** connection\_factory

- **Continuous**(Defaulted) Indicates whether a job continuously monitors for a condition. When selected, the agent monitors for the conditions continuously and writes an alert to the scheduler log file each time a condition occurs.

**Default:** not selected (the job does not monitor for the conditions)

### NOTE

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

- **Correlation Id**(Optional) Specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue.

**Limits:** Up to 64 characters; case-sensitive

**JIL attribute:** correlation\_id

### NOTE

- Correlation Id can be used only when the Use Topic checkbox is not selected.
- If you do not specify the correlation id in your job definition, the system generates a unique correlation id. The job that wants to consume the message must refer to the correlation id value of the predecessor job. That is, the correlation id of the message that was last posted by the predecessor job is fetched and the same is consumed by the successor job. In this case, the predecessor job must run first and then the successor.
- If you specified the correlation id in your job definition and it does not refer to a job, it is treated as a literal string that the consumer consumes directly if the correlation id value matches one of its predecessors. In this case, the order of the predecessor and successor does not matter as we are referring to a literal value and not a system generated id that can change with each run of the job.
- **Destination file**(Optional) Specifies the output destination file that stores the consumed messages.  
**Limits:** Up to 256 characters; case-sensitive  
**JIL attribute:** destination\_file
- **Destination name** Specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.  
This property supports Global Variable substitution.  
**Limits:** Up to 256 characters; case-sensitive  
**JIL attribute:** destination\_name
- **Destination user name** Specifies the user name that is required to connect to the topic or queue in a JMS job.  
**Limits:** Up to 145 characters  
**JIL attribute:** destination\_user

**NOTE**

If you specify the destination user name in your job definition, you must set the password that is associated with the destination user using the `autosys_secure` command.

- **Filter**(Optional) Specifies a regular expression to filter messages from the topic or queue. Any message not matching the filter is ignored. The scheduling manager uses the % character for JavaScript variables. To code a % character in a regular expression, use \% in your expression.

Clicking



opens the Edit Text dialog, which lets you edit the filter text.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** filter

- **Initial context factory** Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server.

**Limits:** Up to 256 characters

**JIL attribute:** initial\_context\_factory

- **J2EE user name**(Optional) Specifies the JNDI user ID to be used to gain access to the connection factory. **Limits:** Up to 145 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** j2ee\_user

- **Provider URL** Specifies the JNDI service provider URL. Both HTTP and HTTPS are supported. For WebLogic servers, use the following format:

This property supports Global Variable substitution.

```
t3://WLIPAddress[:ORBPort]
```

- **WLIPAddress**

Specifies the IP address or domain name of the WebLogic Application Server.

- **ORBPort**

(Optional) Specifies the ORB port.

**Default:** 7001

**Example:** t3://localhost:7001. For WebSphere servers, use the following format:

```
iiop://WSIPAddress[:ORBPort]
```

- **WSIPAddress**

Specifies the IP address or domain name of the WebSphere Application Server.

- **ORBPort**

(Optional) Specifies the ORB port.

**Default:** 2809

**Example:** iiop://172.24.0.0:2809

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** provider\_url

- **Use topic**(Optional) Specifies whether the job publishes messages to a topic or queue. When selected, the agent publishes messages to a topic.

**Default:** not selected (the job publishes to a queue)

**JIL attribute:** use\_topic

## Manage JMX Jobs Using Web UI

This article explains how to manage JMX jobs.



Java Management Extension (JMX) technology is included in the Java Standard Edition (SE) platform, version 5 and higher. JMX lets you remotely access applications, using a Remote Method Invocation (RMI) connector, for monitoring and management purposes.

JMX jobs let you access a remote JMX server that advertises MBeans. An MBean is a managed bean (Java object) that represents an application, a device, or any resource that you want to manage. An MBean contains a set of attributes and a set of operations that can be invoked. Some MBeans can send out notifications, for example, when an attribute changes.

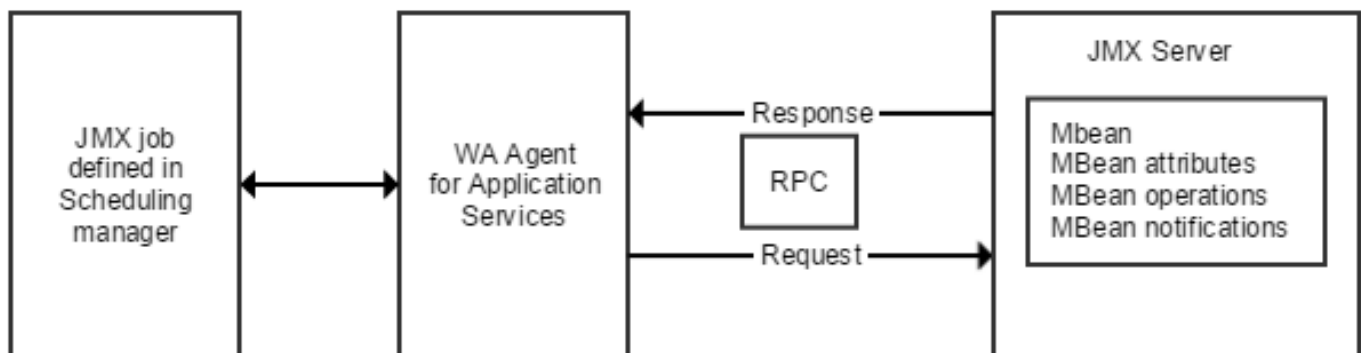
#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

Consider an MBean named Config that represents an application's configuration. The configuration parameters within that application are represented in Config by a set of attributes. Getting the attribute named cachesize, for example, returns the current value of the cachesize. Setting the value updates the cachesize. The Config MBean can send out a notification every time the cachesize changes. An operation named update, for example, can save changes to the configuration parameters.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and the JMX server:

**Figure 69: JMX Jobs**



The JMX jobs provide support for getting and setting JMX MBean attributes, invoking JMX MBean operations, subscribing to MBean notifications, and creating and removing instances of MBeans on a JMX server.

You can define the following six types of JMX jobs:

- JMX-MBean Attribute Get
- JMX-MBean Attribute Set
- JMX-MBean Create Instance
- JMX-MBean Operation
- JMX-MBean Remove Instance
- JMX-MBean Subscribe

The JMX-MBean Attribute Set, JMX-MBean Create Instance, and JMX-MBean Operation jobs support calls to MBeans that can involve passing parameters. Each parameter can be an actual value or a serialized Java object passed by another job. When the JMX-MBean Operation job invokes an operation on an MBean that passes parameters, the parameters are passed to the MBean and the returned serialized Java object is stored on the agent computer in the spool directory or in a destination file you specify.

To define JMX jobs, you require a URL to connect to the JMX server using an RMI connector.

---

## **Payload Producing and Payload Consuming Jobs**

A payload producing job is a job that produces binary output that is persisted as a serialized Java object or as a binary large object (BLOB). A BLOB is a collection of binary data stored in a relational database.

The following job types are payload producing jobs:

- HTTP
- JMX-MBean Attribute Get, JMX-MBean Attribute Set, JMX-MBean Operation
- POJO
- RMI
- Session Bean
- Web Service RPC/Encoded
- Web Service Document/Literal

A payload consuming job describes a job that uses the output from a payload producing job as a parameter's input value. You can use the following job types as payload consuming jobs:

- Entity Bean
- JMS Publish
- JMX-MBean Attribute Set, JMX-MBean Operation, JMX-MBean Create Instance
- POJO
- RMI
- Session Bean
- Web Service RPC/Encoded
- Web Service Document/Literal

The payload producing job must be a predecessor job to the payload consuming job, although it does not need to be an immediate predecessor.

The following diagram illustrates a job flow in which a payload consuming job named Job C uses the output produced by payload producing jobs named Job A and Job B. In this example, Job B is dependent on Job A and Job C is defined to take the output from Job A as the value for the input parameter named inputParm1 and the output from Job B as the value for the input parameter named inputParm2.

**Figure 70: Payload Producing and Consuming Job****Define a JMX-MBean Attribute Get Job**

You can define a JMX-MBean Attribute Get job to query a JMX server for the value of an MBean attribute. The returned value is stored on the computer where the agent resides. You can specify a success pattern to determine the job's success or failure. If the returned attribute value matches the success pattern, the job completes successfully; otherwise, it fails.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

**Follow these steps:**

1. Create a JMX-MBean Attribute Get (JMXMAG) job.
2. Specify the following required properties
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**

Specifies the name of the machine on which the job runs.

- **URL**

Specifies the URL to connect to the JMX server using an RMI connector.

- **MBean name**

Specifies the full object name of an MBean.

Clicking  the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

- **MBean attribute** Specifies the name of the MBean attribute you want to query or set.

Note: Clicking



opens the Browse MBean Attributes dialog, from which you can locate an MBean attribute on the JMX server for the job definition.

3. (Optional) Specify optional [JMX-MBean Attribute Get properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The JMX-MBean Attribute Get job is defined.

### **JMX-MBean Attribute Get Job Examples**

The following example describes a sample JMX-MBean Attribute Get job.

#### **Example: Query a JMX Server for the Value of an MBean Attribute**

Suppose that you want to know the value of the cachesize attribute for the Config MBean. The URL for the JMX server is service:jmx:rmi:///jndi/rmi://localhost:9999/server, where localhost is the host name and 9999 is the port number.

#### **Follow these steps:**

1. Create a JMX-MBean Attribute Get job.
2. Enter the following properties:
  - Name -- jmxag
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- 'DefaultDomain:index=1,type=Config'
  - MBean attribute -- cachesize
3. Commit the job.

### **JMX-MBean Attribute Get Properties**

The JMX-MBean Attribute Get category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

**JMX user**

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with `autosys_secure`.

**Limits:** Up to 145 characters

**Example:** `webuser@webhost`

**JIL attribute:** `jmx_user`

### MBean attribute

Specifies the name of the MBean attribute you want to query or set.

Clicking



opens the Browse MBean Attributes dialog, from which you can locate an MBean attribute on the JMX server for the job definition.

**Limits:** Up to 256 characters

**JIL attribute:** `mbean_attr`

### MBean name

Specifies the full object name of an MBean. The format is as follows:

```
domain_name: key=value[,key=value...]
```

- ***domain\_name***  
Specifies the default domain name.
- ***key=value[,key=value...]***  
Specifies one or more key value pairs.

Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** `DefaultDomain:type=SimpleDynamic,index=3`

**JIL attribute:** `mbean_name`

### Success pattern

(Optional) Specifies a regular expression to use as a filter.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**JIL attribute:** `success_pattern`

### URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the port number of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

### Define a JMX-MBean Attribute Set Job

You can define a JMX-MBean Attribute Set job to change the value of an MBean attribute on a JMX server. You can specify a set value for the attribute or use the serialized Java object passed by another job. When the attribute is set, the job returns the original attribute value as output. You can specify a success pattern to determine the job's success or failure. If the job's output matches the success pattern, the job completes successfully; otherwise, it fails.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### **Follow these steps:**

1. Create a JMX-MBean Attribute Set (JMXMAS) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **URL** Specifies the URL to connect to the JMX server using an RMI connector.
  - **MBean name** Specifies the full object name of an MBean.

Clicking 

the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

- **MBean attribute** Specifies the name of the MBean attribute you want to query or set.

Note: Clicking



opens the Browse MBean Attributes dialog, from which you can locate an MBean attribute on the JMX server for the job definition.

- **JMX parameters** Specifies a [JMX parameter](#), as a type value pair, that the MBean attribute should be set to.
3. (Optional) Specify optional [JMX-MBean Attribute Set properties](#).
  4. (Optional) Specify [common properties that apply to all job types](#).
  5. Commit the job.  
The JMX-MBean Attribute Set job is defined.

## JMX-MBean Attribute Set Job Examples

The following example describes a sample JMX-MBean Attribute Set job.

### Example: Change the Value of an MBean Attribute

Suppose that you want to set the value of the cachesize attribute of the Config MBean to the serialized Java object returned by a JMX-MBean Attribute Set job named size.

#### Follow these steps:

1. Create a JMX-MBean Attribute Set job.
2. Enter the following properties:
  - Name -- jmxas
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- DefaultDomain:index=1,type=Config
  - MBean attribute -- cachesize
  - JMX parameters -- payload\_job=size
3. Commit the job.

## JMX-MBean Attribute Set Properties

The JMX-MBean Attribute Set category includes the following properties:

### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### JMX parameters

Specifies a [JMX parameter](#), as a type value pair, that the MBean attribute should be set to.

Clicking



opens the Edit Parameter dialog, which lets you edit the JMX parameters.

**Limits:** Up to 1024 characters

**JIL attribute:** jmx\_parameter

### JMX user

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with autosys\_secure.

**Limits:** Up to 145 characters

**Example:** webuser@webhost

**JIL attribute:** jmx\_user

### MBean attribute

Specifies the name of the MBean attribute you want to query or set.

### Clicking



opens the Browse MBean Attributes dialog, from which you can locate an MBean attribute on the JMX server for the job definition.

**Limits:** Up to 256 characters

**JIL attribute:** mbean\_attr

### MBean name

Specifies the full object name of an MBean. The format is as follows:

```
domain_name:key=value[,key=value...]
```

- **domain\_name**  
Specifies the default domain name.
- **key=value[,key=value...]**  
Specifies one or more key value pairs.

### Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** DefaultDomain:type=SimpleDynamic,index=3

**JIL attribute:** mbean\_name

### Success pattern

(Optional) Specifies a regular expression to use as a filter.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**JIL attribute:** success\_pattern

### URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the port number of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive



**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

### **Define a JMX-MBean Create Instance Job**

You can define a JMX-MBean Create Instance job to create an MBean on a JMX server.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

#### **Follow these steps:**

1. Create a JMX-MBean Create Instance (JMXMC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **URL** Specifies the URL to connect to the JMX server using an RMI connector.
  - **MBean name** Specifies the full object name of an MBean.

Clicking  the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

  - **Class name** Specifies fully-qualified Java class of the MBean object.
3. (Optional) Specify optional [JMX-MBean Create Instance properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The JMX-MBean Create Instance job is defined.

### **JMX-MBean Create Instance Job Examples**

The following example describes a sample JMX-MBean Create Instance job.

#### **Example: Create an MBean Instance on a JMX Server**

Suppose that you want to create an MBean instance on a JMX server. The job uses the class SimpleDynamic. The constructor of the class takes a single string parameter with the value "Hello".

#### **Follow these steps:**

1. Create a JMX-MBean Create Instance job.
2. Enter the following properties:
  - Name -- jmxmc
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- DefaultDomain:type=SimpleDynamic,index=3
  - Class name -- SimpleDynamic
  - JMX parameters -- java.lang.String=Hello
3. Commit the job.

## JMX-MBean Create Instance Properties

The JMX-MBean Create Instance category includes the following properties:

### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### Class name

Specifies fully-qualified Java class of the MBean object.

**Limits:** Up to 1024 characters; case-sensitive; cannot contain delimiters (such as spaces)

**JIL attribute:** class\_name

### JMX parameters

(Optional) Specifies a [JMX parameter](#), as a type value pair, that the MBean attribute should be set to.

Selecting Add and clicking Go displays a row below the JMX parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the JMX parameters.

**Limits:** Up to 1024 characters

**JIL attribute:** jmx\_parameter

### JMX user

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with autosys\_secure.

**Limits:** Up to 145 characters

**Example:** webuser@webhost

**JIL attribute:** jmx\_user

### MBean name

Specifies the full object name of an MBean. The format is as follows:

```
domain_name:key=value[,key=value...]
```

- **domain\_name**  
Specifies the default domain name.
- **key=value[,key=value...]**  
Specifies one or more key value pairs.

Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** DefaultDomain:type=SimpleDynamic,index=3

**JIL attribute:** mbean\_name

## URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the host name or IP address of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

## Define a JMX-MBean Operation Job

You can define a JMX-MBean Operation job to invoke an operation on an MBean. You can specify one or more parameter values to pass to the operation. You can specify a success pattern to determine the job's success or failure. If the operation's output matches the success pattern, the job completes successfully; otherwise, it fails.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Create a JMX-MBean Operation (JMXMOP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **URL** Specifies the URL to connect to the JMX server using an RMI connector.
  - **MBean name**  
Specifies the full object name of an MBean.

**Note:**

Clicking  the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

  - **MBean operation** Specifies the operation to be invoked.

Clicking  the Browse MBean Operations dialog, from which you can locate an operation for the job definition.
3. (Optional) Specify optional [JMX-MBean Operation properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).

5. Commit the job.  
The JMX-MBean Operation job is defined.

### **JMX-MBean Operation Job Examples**

The following example describes a sample JMX-MBean Operation job.

#### **Example: Invoke an Operation on an MBean**

Suppose that you want to invoke the resetmem operation on the Config MBean to reset the value of the memory parameter to 50.

#### **Follow these steps:**

1. Create a JMX-MBean Operation job.
2. Enter the following properties:
  - Name -- jmxmop
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- 'DefaultDomain:index=1,type=Config'
  - MBean operation -- resetmem
  - JMX parameters -- integer=50
3. Commit the job.

### **JMX-MBean Operation Properties**

The JMX-MBean Operation category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **JMX parameters**

(Optional) Specifies a [JMX parameter](#), as type value pair, of the MBean operation.

Selecting Add and clicking Go displays a row below the JMX parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the JMX parameters.

**Limits:** Up to 1024 characters

**JIL attribute:** jmx\_parameter

#### **JMX user**

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with autosys\_secure. **Limits:** Up to 145 characters **Example:** webuser@webhost **JIL attribute:** jmx\_user

#### **MBean name**

Specifies the full object name of an MBean. The format is as follows:

```
domain_name:key=value[,key=value...]
```

- **domain\_name**  
Specifies the default domain name.
- **key=value[,key=value...]**  
Specifies one or more key value pairs.

Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** DefaultDomain:type=SimpleDynamic,index=3

**JIL attribute:** mbean\_name

### MBean operation

Specifies the name of the MBean operation to invoke.

Clicking



opens the Browse MBean Operations dialog, from which you can locate an operation for the job definition.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** mbean\_operation

### Success pattern

(Optional) Specifies a regular expression to use as a filter.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**JIL attribute:** success\_pattern

### URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the port number of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

## Define a JMX-MBean Remove Instance Job

You can define a JMX-MBean Remove Instance job to remove an MBean from a JMX server.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### Follow these steps:

1. Create a JMX-MBean Remove Instance (JMXMREM) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **URL** Specifies the URL to connect to the JMX server using an RMI connector.
  - **MBean name** Specifies the full object name of an MBean.

Clicking 

the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

3. (Optional) Specify optional [JMX-MBean Remove Instance properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The JMX-MBean Remove Instance job is defined.

## JMX-MBean Remove Instance Job Examples

The following example describes a sample JMX-MBean Remove Instance job.

### Example: Remove an MBean Instance from a JMX Server

Suppose that you want to remove an MBean instance created by a JMX-MBean Create Instance job.

### Follow these steps:

1. Create a JMX-MBean Remove Instance job.
2. Enter the following properties:
  - Name -- jmxrem
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- 'DefaultDomain:type=SimpleDynamic,index=3'
3. Commit the job.

## JMX-MBean Remove Instance Properties

The JMX-MBean Remove Instance category includes the following properties:

### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### JMX user

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with autosys\_secure. **Limits:** Up to 145 characters **Example:** webuser@webhost **JIL attribute:** jmx\_user

### MBean name

Specifies the full object name of an MBean. The format is as follows:

```
domain_name:key=value[,key=value...]
```

- **domain\_name**  
Specifies the default domain name.
- **key=value[,key=value...]**  
Specifies one or more key value pairs.

Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** DefaultDomain:type=SimpleDynamic,index=3

**JIL attribute:** mbean\_name

### URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the port number of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

### Define a JMX-MBean Subscribe Job

You can define a JMX-MBean Subscribe job to monitor an MBean for a single notification or monitor continuously for notifications. You can filter the notifications the job monitors by attributes or by type of notifications.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

**Follow these steps:**

1. Create a JMX-MBean Subscribe (JMXSUB) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **URL** Specifies the URL to connect to the JMX server using an RMI connector.
  - **MBean name** Specifies the full object name of an MBean.
1. Clicking  the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.
2. (Optional) Specify optional [JMX-MBean Subscribe properties](#).
3. (Optional) Specify [common properties that apply to all job types](#).
4. Commit the job.  
The JMX-MBean Subscribe job is defined.

**JMX-MBean Subscribe Job Examples**

The following examples describe sample JMX-MBean Subscribe jobs.

**Example: Set Up Notifications for Change to an MBean Attribute**

Suppose that you want to set up continuous monitoring for changes to the cachesize attribute of the MBean named Config. The job filters the notifications the MBean sends by attribute. Each time the cachesize attribute changes, an alert is written to the scheduler log file.

**Follow these steps:**

1. Create a JMX-MBean Subscribe job.
2. Enter the following properties:
  - Name -- jmxsub1
  - Send to machine -- APPAGENT
  - URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
  - MBean name -- 'DefaultDomain:index=1,type=Config'
  - Filter type -- Attributes
  - Filter -- cachesize
  - Continuous -- selected
3. Commit the job.

**Example: Set Up Notifications for Changes to Any MBean Attribute**

Suppose that you want to set up continuous monitoring for changes to any attribute of the MBean named Config. Each time an attribute changes, an alert is written to the scheduler log file.

**Follow these steps:**

1. Create a JMX-MBean Subscribe job.
2. Enter the following properties:



- Name -- jmxsub2
- Send to machine -- APPAGENT
- URL -- service:jmx:rmi:///jndi/rmi://localhost:9999/server
- MBean name -- 'DefaultDomain:index=1,type=Config'
- Filter type -- Attributes
- Filter -- jmx.attribute.change
- Continuous -- selected

3. Commit the job.

### **JMX-MBean Subscribe Properties**

The JMX-MBean Subscribe category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Continuous**

(Defaulted) Indicates whether a job continuously monitors for a condition. When selected, the agent monitors for the conditions continuously and writes an alert to the scheduler log file each time a condition occurs.

**Default:** not selected (the job does not monitor for the conditions)

#### **NOTE**

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

#### **Filter**

(Optional) Specifies the name of attributes or the type of notifications to filter depending on the filter type specified.

**Limits:** Up to 256 characters

**JIL attribute:** filter

#### **Filter type**

(Optional) Specifies whether to filter notifications by attribute or by notification type, as follows:

- Attributes -- Filters notifications by attribute.
- Types -- Filters notifications by notification type.

**Default:** Attributes

**JIL attribute:** filter\_type

#### **JMX user**

(Optional) Specifies the JMX user name formatted as follows: *user@token*. This field is required only when the JMX job requires an ID. The password is resolved by the value stored with autosys\_secure.

**Limits:** Up to 145 characters

**Example:** webuser@webhost

**JIL attribute:** jmx\_user

### MBean name

Specifies the full object name of an MBean. The format is as follows:

```
domain_name:key=value[,key=value...]
```

- **domain\_name**  
Specifies the default domain name.
- **key=value[,key=value...]**  
Specifies one or more key value pairs.

Clicking



opens the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

**Limits:** Up to 715 characters; case-sensitive

**Example:** DefaultDomain:type=SimpleDynamic,index=3

**JIL attribute:** mbean\_name

### URL

Specifies the URL to connect to the JMX server using an RMI connector. The format is as follows:

```
service:jmx:rmi:///jndi/rmi://host_name:port_number/jmxserver_name
```

- **host\_name**  
Specifies the host name or IP address of the JMX server.
- **port\_number**  
Specifies the port number of the JMX server.
- **jmxserver\_name**  
Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** service:jmx:rmi:///jndi/rmi://localhost:9999/server

**JIL attribute:** url

**Note:** Clicking 

the Search MBeans dialog, from which you can locate an MBean on the JMX server for the job definition.

## Manage POJO, RMI, and Web Service RPC/Encoded Jobs Using Web UI

This article explains how to manage HTTP, POJO, RMI, and Web Service RPC/Encoded jobs.

### POJO Jobs

A Plain Old Java Object (POJO) is a Java object that follows the Java Language Specification only. All Java objects are POJOs.

The POJO job lets you instantiate a class to create a Java object and invoke a method on it. The job is restricted to classes that take constructors with no arguments (default constructors).

- Note: To run these jobs, your system requires the following agents:
- Workload Automation Agent for UNIX, Linux, or Windows
- Workload Automation Agent for Application Services or Workload Automation Agent for Web Services

You can use the POJO job to invoke custom Java code on a local computer. POJO jobs support method calls that can involve passing parameters. The parameters can be actual values or a serialized Java object passed by another job. When the POJO job invokes a method on an object, the parameters, if any, are passed to the object and the returned values are stored in a Java serialized object file.

To define a POJO job, you require the class name and method you want to call on the instantiated object.

#### NOTE

- If you use custom Java code, verify that the required JAR file is in the jars subdirectory of the agent installation directory.
- By default, the `oscomponent.classpath` and `oscomponent.javapath` environment variables are set in the `agentparm.txt` file when you install the agent. You cannot modify environment variables using a POJO job. But you can run a batch file using a Windows job to set the `CLASSPATH` and `JAVA_HOME` environment variables, and then run the Java code.
- After you change or redeploy a class or JAR file in the agent installation directory, restart the agent. For more information about restarting the agent, see the *Workload Automation Agent for UNIX, Linux, or Windows Implementation documentation*.

The JRE version of the agent that you use to run POJO jobs and the Java compiler on your computer must be the same. To find the JRE version of the agent, follow these steps:

- a. Change to the following directory at the command prompt:
  - On Windows:
 

```
agent_install_dir\jre\bin
agent_install_dir/jre/bin
```
  - On UNIX:
    - **`agent_install_dir`**  
Specifies the agent installation directory.
- b. Enter the following command:
 

```
java.exe - version
```

## RMI Jobs

Remote Method Invocation (RMI) is the Java version of a Remote Procedure Call (RPC), which is a technology that lets a program request a service from another program located in another address space. That address space could be on the same computer or on a different one.

RMI jobs let you set up interaction between Java objects on different computers in a distributed network. Using an RMI job, you can access a remote server and invoke a method on a Java object. A method is a programmed procedure that is defined as a part of a Java class.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

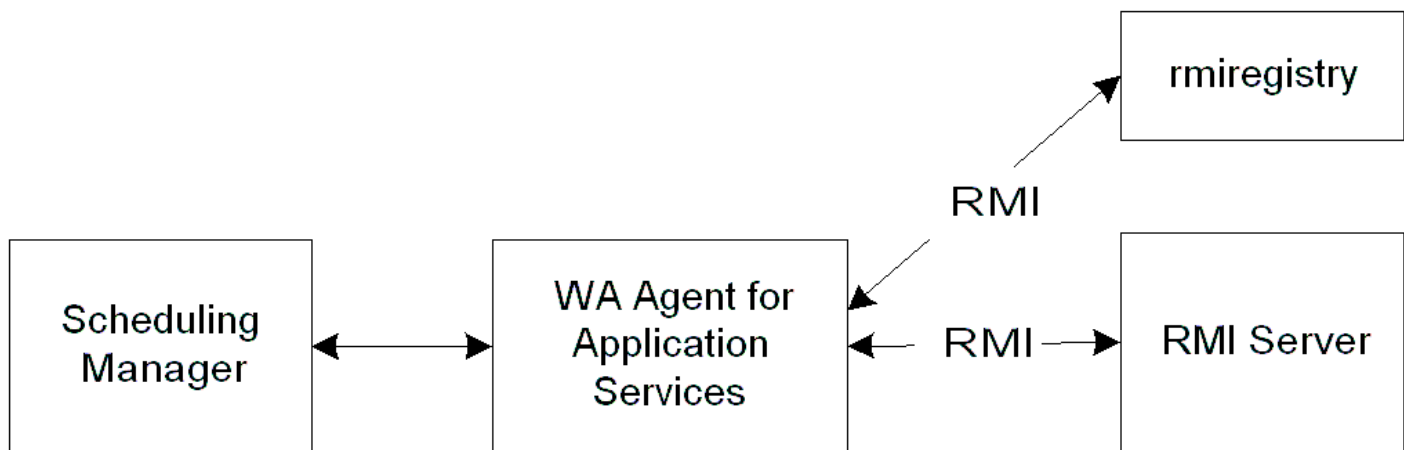
RMI jobs support method calls to remote objects that can involve passing parameters. The parameters can be actual values or a serialized Java object passed by another job. When the RMI job invokes a method on an object that passes

parameters, the parameters are passed to the remote object and the returned serialized Java object is stored on the agent computer in the spool directory or in a destination file you specify.

RMI uses a naming or directory service to locate the remote object on the remote server. To define an RMI job, you require the naming class of the Java object you want to invoke a method on. That naming class takes a name that is a `java.lang.String` in URL format.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Application Services, and an RMI Server:

**Figure 71: RMI Job**



### **Web Service Jobs**

The term web service describes a standardized method for exchanging data between applications and systems. Web services use XML to code and decode the data and Simple Object Access Protocol (SOAP) to transfer it.

Web Service Description Language (WSDL) is an XML-based language that describes a web service and how to access it. A WSDL document specifies the location of the service and the operations the service exposes.

Universal Description, Discovery and Integration (UDDI) is an XML-based registry for businesses to list their available web services on the Internet. You can use the UDDI to access the WSDL.

Web services provide access to applications written in Java and Microsoft®.NET. A web service lets you invoke operations such as currency conversion, stock exchange quotes, or product pricing. In an enterprise workload automation environment, a web service might be used to invoke a business process such as posting accounts payable to the General Ledger. Some scheduling manager functions are also available as web services.

The following are the web service job types:

- **Web Service RPC/Encoded**  
Lets you call an operation within a web service and pass parameters to the operation using RPC/encoded style binding. The parameters can be actual values or a serialized Java object passed by another job.
- **Web Service Document/Literal**  
Lets you call an operation within a web service and pass parameters to the operation using document/literal style binding. The parameters represent a flattened view of the XML document the agent constructs. The values passed into the XML document can be literal values or a serialized Java object passed by another job.

When the job invokes the web service, the parameters are passed to the operation. The job's output is stored by default as a serialized Java object in the job's spool directory. You can also specify a destination file for the output.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

The following diagram shows the functional relationship between the scheduling manager, Workload Automation Agent for Web Services, and a web service residing on a web server:

**Figure 72: Web Service Jobs**

**NOTE**

If your company has a firewall and you must communicate through a proxy server to access a computer outside the firewall, agent configuration is required. For more information on configuring the agent for a proxy, see the *Workload Automation Agent for Web Services Implementation documentation*.

### Define a POJO Job

You can define a POJO job to create a no-argument Java object instance, invoke a method on the object instance, and store the method's output.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

### **Follow these steps:**

1. Create a POJO (POJO) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Class name**  
Specifies the name of the Java class to use.
  - **Method name**  
Specifies the Java method to call.
3. (Optional) Specify optional [POJO properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The POJO job is defined.

### POJO Job Examples

The following example describes a sample POJO job.

#### **Example: Invoke a Method on a Java Object Instance**

Suppose that you want to define a POJO job that creates an empty Java string and calls the method `parseInt` with the argument "5". The job instantiates the `java.lang.Integer` class.

**Follow these steps:**

1. Create a POJO job.
2. Enter the following properties:
  - Name -- parse
  - Send to machine -- APP\_AGENT
  - Class name -- java.lang.Integer
  - Method name -- parseInt
  - J2EE parameters -- java.lang.String=5
3. Commit the job.

**POJO Properties**

The POJO category includes the following properties:

**Agent job class**

(Optional) Specifies the name of an initiator class defined in the `agentparm.txt` file. If the class name is not available on the target machine, the job will fail to be submitted.

**Limits:** Up to 32 characters

**NOTE**

The class must be a string without blanks in it.

**JIL attribute:** job\_class

**Class name**

Specifies the name of the Java class to use.

**Limits:** Up to 1024 characters; case-sensitive

**JIL attribute:** class\_name

**Destination file**

(Optional) Specifies the output destination file.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

**J2EE parameters**

(Optional) Specifies [input parameters](#) of the EJB Method to be invoked remotely.

Selecting Add and clicking Go displays a row below the J2EE parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the J2EE parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

**NOTE**

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** j2ee\_parameter

**Method name**

Specifies the Java method to call.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** method\_name

**Define an RMI Job**

You can define an RMI job to call a method on a remote server and store the method's output.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

**Follow these steps:**

1. Create an RMI (JAVARMI) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Remote name**  
Specifies the reference location of the object you want to invoke a method on.
  - **Method name**  
Specifies a method name to execute on a remote object defined in the Remote name field.
3. (Optional) Specify optional [RMI properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The RMI job is defined.

**RMI Job Examples**

The following example describes a sample RMI job.

**Example: Define a Job to Start a Remote Server Immediately**

Suppose that you want to invoke a method that starts a remote server. You want the server to start immediately.

**Follow these steps:**

1. Create an RMI job.
2. Enter the following properties:
  - Name -- startsvr
  - Send to machine -- APP\_AGENT
  - Remote name -- rmi://remotehost/Test
  - Method name -- startserver
  - J2EE parameters -- string=now
3. Commit the job.

**RMI Properties**

The RMI category includes the following properties:

**Agent job class**

(Optional) Specifies the name of an initiator class defined in the agentparm.txt file. If the class name is not available on the target machine, the job will fail to be submitted.

**Limits:** Up to 32 characters

**NOTE**

The class must be a string without blanks in it.

**JIL attribute:** job\_class

**Destination file**

(Optional) Specifies the output file where the method execution output is stored. If you do not specify a file, the output will be written to the spool file by default.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

**J2EE parameters**

(Optional) Specifies [input parameters](#) of the EJB Method to be invoked remotely.

Selecting Add and clicking Go displays a row below the J2EE parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the J2EE parameters.

**Limits:** Up to 1024 characters for one J2EE parameter; up to 64 *type=value* pairs

**NOTE**

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** j2ee\_parameter

**Method name**

Specifies the method of the remote Java class to invoke.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** method\_name

**Remote name**

Specifies the reference location of the object you want to invoke a method on. The format is the following:

```
rmi://[host:port/]name
```

- **host**  
(Optional) Specifies the name of the local or remote computer where the RMI registry that hosts the remote object runs.  
**Default:** localhost
- **port**  
(Optional) Specifies the RMI registry port number the host uses to listen for requests.  
**Default:** 1099, the default port of the host's RMI registry
- **name**  
Specifies the name of a remote object.

**Limits:** Up to 256 characters; case-sensitive



**JIL attribute:** remote\_name

### **Define a Web Service RPC/Encoded Job Using the Wizard**

The Web Service RPC/Encoded job features a wizard that lets you locate the WSDL document for a web service, select the operation, and enter the parameter values. You must use the wizard if you need to specify your company's proxy settings to access the Internet. Whenever possible, use the wizard to prevent entry errors.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

#### **Follow these steps:**

1. Create a Web Service RPC/Encoded (WBSVC) job.

2. Click



in the WSDL location field to open the WSDL wizard.

The Browse WSDL dialog opens displaying "Step 1 of 5".

3. Complete the following fields and click Next:

#### – **WSDL location**

Indicates whether the WSDL file location is local or remote, as follows:

- Local WSDL File
- WSDL URL

#### **NOTE**

- When you select WSDL URL, the wizard displays a Proxy section to specify a proxy server. If your company uses a proxy server to connect to the Internet, you must complete the Proxy fields.
- In a Web Service RPC/Encoded job, if you specify the WSDL location property but not the End point URL property, you must specify both the Service name and Port name properties. For the job to run successfully without the End point URL property, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL Location and End point URL properties, then the Service name and Port name properties are optional.
- The agent does not support document/literal styles of web services.

#### – **WSDL**

Specifies the local path or remote URL for the WSDL file.

**Limits:** Up to 256 characters

"Step 2 of 5" appears.

4. Select a web service from the List of services and click Next.

"Step 3 of 5" appears.

5. Select a port from the List of ports and click Next.

"Step 4 of 5" appears.

6. Select an operation from the List of operations and click Next.

"Step 5 of 5" appears.

7. Specify values for the input parameters of the operation.

8. Click Finish to save your input.

The Browse WSDL dialog closes and values for the web service appear in the job definition.

9. (Optional) Specify optional [Web Service RPC/Encoded properties](#).

(Optional) Specify [common properties that apply to all job types](#).

10. Commit the job.

The Web Service RPC/Encoded job is defined.

## Define a Web Service RPC/Encoded Job Using Manual Entries

You can define a Web Service RPC/Encoded job to call an operation within a web service. The easiest way to define a Web Service RPC/Encoded job is to use the wizard. However, you may want to enter the job definition values manually. The target namespace and the name of the operation you want to invoke are required. Depending on the web service provider, other values may also be required.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

### Follow these steps:

1. Create a Web Service RPC/Encoded (WBSVC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Target namespace**  
Specifies the target namespace used for the names of messages, port type, binding, and services defined in the WSDL for the web service. Complex data types such as arrays require the target namespace.
  - **Operation**  
Specifies the operation to be invoked.
3. (Optional) Specify optional [Web Service RPC/Encoded properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Web Service RPC/Encoded job is defined.

## Web Service RPC/Encoded Job Examples

The following examples describe sample Web Service RPC/Encoded jobs.

### Example: Get a Company Stock Quote

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is <http://www.websvcex.net/stockquote.asmx?WSDL>. The WSDL port name within the target namespace <http://www.websvcex.net/> is StockQuoteSoap. The target endpoint address URL is <http://www.websvcex.net/stockquote.asmx>. The job calls the operation GetQuote within the StockQuote web service. When the job invokes the web service, the company's stock symbol is passed to the operation. The GetQuote operation returns a java.lang.String object, which maps to the XML type string in the return namespace <http://www.websvcex.net/>. When the job completes, the stock quote for the company is stored as a serialized Java object in the job's spool directory. In this example, the company is CA.

### Follow these steps:

1. Create a Web Service RPC/Encoded job.
2. Enter the following properties:

- Name -- quote
- Send to machine -- WSAGENT
- WSDL location -- "http://www.websvcicex.net/stockquote.asmx?WSDL"
- Target namespace -- "http://www.websvcicex.net/"
- Service name -- StockQuote
- Port name -- StockQuoteSoap
- Operation -- GetQuote
- End point URL -- "http://www.websvcicex.net/stockquote.asmx"
- Operation parameters -- xsd:string="CA"
- Return class name -- java.lang.String
- Return xml type -- string
- Return namespace -- "http://www.websvcicex.net/"
- One way -- not selected

3. Commit the job.

### Example: Validate an Email Address in a Web Service RPC/Encoded Job

Suppose that you want to invoke a web service that validates an email address. The URL for the WSDL that describes the web service and its location is <http://www.websvcicex.net/ValidateEmail.asmx?wsdl>. The job calls the operation `IsValidEmail` within the `ValidateEmail` web service. When the job invokes the web service, the email address is passed to the operation. If the email address is valid, the operation returns `true` and the job completes successfully. If the email address is invalid, the operation returns `false` and the job fails. In this example, you are validating John Smith's email address.

#### Follow these steps:

1. Create a Web Service RPC/Encoded job.
2. Enter the following properties:
  - Name -- subscribe
  - Send to machine -- WSAGENT
  - WSDL location -- "http://www.websvcicex.net/ValidateEmail.asmx?wsdl"
  - Target namespace -- "http://www.websvcicex.net/"
  - Service name -- ValidateEmail
  - Port name -- ValidateEmailSoap
  - Operation -- IsValidEmail
  - End point URL -- "http://www.websvcicex.net/ValidateEmail.asmx"
  - Operation parameters -- xsd:string="john.smith@example.com"
  - Return class name -- java.lang.Boolean
  - Return xml type -- boolean
  - Return namespace -- "http://www.websvcicex.net/"
  - Job criteria -- true
3. Commit the job.

### Web Service RPC/Encoded Properties

The Web Service RPC/Encoded category includes the following properties:

#### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### Destination file

(Optional) Specifies the output destination file. In a payload producing job, the file stores the serialized Java object produced by a job.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

### End point URL

(Optional) Specifies the target endpoint address URL. In a published WSDL file, the URL defining the target endpoint address is found in the location attribute of the port's soap:address element.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** endpoint\_url

### Job criteria

(Optional) Specifies a regular expression to use as a filter.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**JIL attribute:** success\_pattern

### One way

(Defaulted) Indicates whether to invoke the web service one way. When selected, the agent invokes the service one way and the Web Service RPC/Encoded job completes without waiting for a response.

**Default:** not selected

**JIL attribute:** one\_way

### Operation

Specifies the operation to be invoked.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** wsdl\_operation

### Operation parameters

(Optional) Specifies the type=value pairs of [parameters](#) that are passed to the operation invoked by the Web Service RPC/Encoded job.

Selecting Add and clicking Go displays a row below the Operation parameters field. In that field, clicking



opens the Edit Parameter dialog, which lets you specify the Operation parameters.

**Limits:** Up to 1024 characters for one operation parameter; up to 64 type=value pairs

#### NOTE

Specify the parameters in the same order as they appear in the method. The job will fail if the parameters are listed in the wrong order.

**JIL attribute:** web\_parameter

### Port name

(Optional) Specifies the WSDL port name within the target namespace. A WSDL port describes the operations exposed by a web service and defines the connection point to the web service.

**Limits:** Up to 100 characters; case-sensitive

**JIL attribute:** port\_name

### Return class name

(Optional) Specifies the Java class name of the return value.

**Limits:** Up to 1024 characters; case-sensitive

**JIL attribute:** return\_class\_name

### Return namespace

(Optional) Specifies the XML namespace for the XML type that maps to the Java class name of the return value.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** return\_namespace

### Return XML type

(Optional) Specifies the XML type that maps to the Java class name of the return value.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** return\_xml\_name

### Service name

(Optional) Specifies the web service name within the target namespace.

**Limits:** Up to 255 characters; case-sensitive

**JIL attribute:** service\_name

### Target namespace

Specifies the target namespace used for the names of messages, port type, binding, and services defined in the WSDL for the web service. Complex data types such as arrays require the target namespace.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** target\_namespace

### User

(Optional) Specifies the user name.

**Limits:** Up to 145 characters

**JIL attribute:** web\_user

### WSDL location

(Optional) Specifies the local path or remote URL for the WSDL file.

Clicking



opens the Browse WSDL wizard, which lets you locate the WSDL file.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** wsdl\_url

## Manage HTTP Jobs Using Web UI

The HTTP job type is now with a new look and feel, and an easy to use graphical user interface.

The HTTP job invokes a program over HTTP in a similar way to a web browser. For example, you can use the HTTP job to invoke a RESTful web service, CGI script, a Perl script, or a servlet.

The HTTP job sends a URL over HTTP using the GET method or a form over HTTP using the POST method, along with the DELETE and PUT methods. The output of the invocation is returned in the spool file of the job and also sent back to the manager.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services Version 12.0.

The GET and DELETE methods request data and send the data as part of the URL. The POST and PUT methods submit data in the message body and are the preferred methods for sending lengthy form data.

To define an HTTP job, you require the following information:

- URL of the application server
- Program or servlet to invoke

### NOTE

If your company has a firewall and you must communicate through a proxy server to access a computer outside the firewall, agent configuration is required. For more information on configuring the agent for a proxy, see the *Workload Automation Agent for Application Services Implementation documentation*.

## Define a HTTP Job (Web UI)

You can define a HTTP job to invoke a program or REST web service over HTTP.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Application Services.

There are two versions of HTTP job types; namely, Pre-Version 12.0 and Version 12.0. The “compatibility” option defines to which version HTTP job definition belongs to. By default, the compatibility value is 12 which allows all new properties. If the specified value is 11, only properties from before Version 12.0 are allowed.

The following properties can be specified for HTTP jobs from Pre-Version 12.0 only:

- Filter
- Form parameters (JIL attribute: j2ee\_parameter)
- Method name

The following properties can be specified for HTTP jobs only from AutoSys Version 12 or higher:

- Use cookies from job (JIL attribute: input\_cookies)
- Payload content type
- File path
- Header parameters
- Form Parameters (JIL attribute: form\_parameter)
- Response persist options fields:
  - Payload file path

- Overwrite payload file
- Persist headers
- Persist payload output
- Persist cookies
- Response variables fields:
  - Response type
  - Expression type
  - Name
  - Expression/Header keyword
  - RegEx group/Header index
- Authentication parameters field:
  - Preemptive authentication
- Success criteria fields:
  - Success exit code
  - Failure exit code
  - Header filter
  - Expression type
  - Expression
  - RegEx expression
- Connection parameters fields:
  - Send failure details for retries
  - Connection retry count
  - Connection timeout (in seconds)
  - Failure retry count
  - Failure retry interval (in seconds)

The following properties are common to all AutoSys releases:

- Type
- URL
- Authentication parameters fields:
  - Connection user
  - Connection origin
  - Connection domain
  - Authentication order
- Proxy details fields:
  - Use agent global proxy (J2EE no global proxy defaults)
  - Proxy user
  - Proxy origin
  - Proxy domain
  - Proxy host
  - Proxy port

### **Define an HTTP Job**

You can define an HTTP job to invoke a program over HTTP.

#### **Follow these steps:**

1. Create an HTTP (HTTP) job.
2. Specify the following required properties:
  - **Job name**  
Defines the name of the job that you want to schedule.

**Limits:** 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

– **Select machine**

Specifies the name of the machine on which the job runs.

3. (Optional) Enter a **Description** for the job.

Defines an optional free-form text description of the job.

**Limits:** 500 alphanumeric characters (including spaces)

**JIL attribute:** description

4. Specify HTTP Properties under the HTTP tab.

– **Select the Workload Automation Agents version for HTTP web service:**

**NOTE**

After saving the job, you cannot change the selected compatibility.

• **12 Compatibility**

Works with Workload Automation Agents Version 12.0 and higher.

**NOTE**

**12 Compatibility Considerations**

- The GET and DELETE Type do not support Form parameters. Query parameters must be specified directly in the URL.
- Method name is not supported. Specify method name directly in the URL.
- In Form parameters, you must not encode the "Name" and "Value"; the product encodes them (including substitution-variables) during job-execution.

• **11 Compatibility**

Works with Workload Automation Agents Version 11.0 and higher.

**NOTE**

**11 Compatibility Considerations**

- If Action, Method name, or query parameters is provided in the URL under GET/POST it is replaced by the Method name property and query parameters if specified in the URL are ignored.
- In Form parameters, you must specify encoded values for the "Name" and "Value"; the product does not encode them during job-execution. Only the "resolved substitution-variable values" are encoded as part of the name and value; the user must encode any other required specific characters.

– **Type**

(Defaulted) Specifies the HTTP method type, as follows:

• **GET**

Sends the URL over HTTP using the GET method. The GET method requests data and sends the data as part of the URL.

• **POST**

Sends the URL over HTTP using the POST method. The POST method submits data and is the preferred method for creating resource identified by the URL as the operation is idempotent and for sending lengthy form data.

• **PUT**

Sends the URL over HTTP using the PUT method. The PUT method submits data and is the preferred method for updating resource identified by the URL as the operation is idempotent.

• **DELETE**

Sends the URL over HTTP using the DELETE method. The DELETE method removes the data.

**NOTE**

- PUT and DELETE options are available under 12 Compatibility and only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later.



**Default:** POST

**JIL attribute:** invocation\_type

– **URL\***

\*This is a mandatory field. Specifies the host where the application to be invoked resides.

The URL has the following format:

`https://host_name:port_number/action`

- **host\_name**

Specifies the name of the computer running the application server or where the REST web server is hosted.

- **port\_number**

Specifies the port the host uses to listen for HTTP requests.

**Default:** 80

- **action**

(Optional for 11 compatibility) Specifies the name of the program or servlet to be invoked. If you omit the action, then you must enter a value in the Method name field.

**Limits:** 4000 characters; case-sensitive

**JIL attribute:** provider\_url

– **Use cookies from job**

Enter or select the job from the editable dropdown, the cookies of which can be used in the new job.

**Limits:** 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

**JIL attribute:** input\_cookies

– **Payload content type**

(Optional) Specifies the type of payload data for a POST or PUT request in an HTTP job.

**NOTE**

- The Payload content type property is not applicable if you have set the Type property value to GET or Delete.
- You can use global variables to specify the type of payload. Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.
- When Payload content type is specified as "application/x-www-form-urlencoded", only then data can be specified as name-value pairs using Form parameters.
- When Payload content type is specified other than "application/x-www-form-urlencoded", raw data can only be specified using Payload section in the Payload content field.
- Payload content field is not displayed when the Payload content type is "application/x-www-form-urlencoded".
- If the Payload content type field is blank, Form parameters are not displayed.

**Limits:** 128 characters

**Examples:** application/JSON, application/x-www-form-urlencoded, application/XML, text/CSS, text/CSV, text/HTML, text/Plain and so on.

**JIL attribute:** media\_type

– **Payload**

This option is available under 12 compatibility and only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later.

**Load from Local File** - Specifies the payload data in text format. Allows to load payload data from the local file where Web UI is accessed through the browser. The input data is stored in the AutoSys database as blob.

- **Content**

To select a file, follow these steps:

- a. Click Browse. The Select File window displays.
- b. Navigate across the directories to select an existing file or enter the file name.

**Load from Agent File** - Specifies the payload data for the request.

- **File Path**

Displays the selected Local File or Agent File path.

Click INSERT PATH, to display the selected or entered File Path.

The file path must begin with File://. **Windows:** file:/// and **Linux:** file://

**Limits:** 256 characters, case-sensitive

**Example:** file://foldername/filename

**JIL attribute:** payload\_uri

The Load from Local File and Load from Agent File properties are mutually exclusive.

**JIL attribute:** blob\_input

5. (Optional) Add form data (name-value pairs) when Payload content type is “application/x-www-form-urlencoded” under Form Parameters.
6. (Optional) You can input headers (name-value pairs) under Header parameters.
7. (Optional) Define response variables (name-expression pairs) under Response variables to extract portions of data from the job output/payload.
8. (Optional) Specify success-criteria based on Xpath/JSON/RegEx to govern job success using Success criteria.
9. (Optional) Specify [Manage Common Job Properties \(12.0 UI\)](#) for General, Termination, Schedule, Notifications, and Native Permissions.

#### NOTE

You can use search on demand for the Machine, Box, Email Template, and Calendar properties. Enter the appropriate criteria in the relevant field. You can also select the value from the drop-down list wherever available or start typing to prompt options.

10. Save the job.

The HTTP job is defined.

#### NOTE

- To use the Refresh button and Actions drop-down in the Web UI title bar and perform various actions on an existing job definition, see the [Manage Actions](#) section.
- For the Refresh button and Actions drop down job retrievals, the last version is only accessible; previous versions are disabled

## Examples - HTTP Jobs (Web UI)

The following examples are for the HTTP Job, mainly using various payload output formats and how you can use different attributes to match and extract content from the payload.

The following JSON payload is used for testing the JSONPath sample jobs.

```
{
 "store": {
 "book": [
 {
 "category": "reference",
 "author": "Nigel Rees",
 "title": "Sayings of the Century",
 "price": 8.95
 },
 {
 "category": "fiction",
 "author": "Evelyn Waugh",
 "title": "Sword of Honour",
 "price": 12.99
 }
]
 }
}
```

```

 {
 "category": "fiction",
 "author": "Herman Melville",
 "title": "Moby Dick",
 "isbn": "0-553-21311-3",
 "price": 8.99
 },
 {
 "category": "fiction",
 "author": "J. R. R. Tolkien",
 "title": "The Lord of the Rings",
 "isbn": "0-395-19395-8",
 "price": 22.99
 }
],
 "bicycle": {
 "color": "red",
 "price": 19.95
 }
},
"expensive": 10
}

```

The following XML payload is used for testing the XPATH sample jobs.

```

<?xml version="1.0" encoding="UTF-8"?>

<bookstore>

<book category="cooking">
 <title lang="en">Everyday Italian</title>
 <author>Giada De Laurentiis</author>
 <year>2005</year>
 <price>30.00</price>
</book>

<book category="children">
 <title lang="en">Harry Potter</title>
 <author>J K. Rowling</author>
 <year>2005</year>
 <price>29.99</price>
</book>

<book category="web">
 <title lang="en">XQuery Kick Start</title>
 <author>James McGovern</author>
 <author>Per Bothner</author>
 <author>Kurt Cagle</author>
 <author>James Linn</author>
 <author>Vaidyanathan Nagarajan</author>
 <year>2003</year>
 <price>49.99</price>
</book>

```

```
<book category="web">
 <title lang="en">Learning XML</title>
 <author>Erik T. Ray</author>
 <year>2003</year>
 <price>39.95</price>
</book>

</bookstore>
```

The examples are classified into the following two categories:

### **Evaluating Success Criteria**

You can evaluate the response content using success criteria fields. If the response is a XML document, you can use XPath as the Expression type and if it is a JSON document you can use JSONPath as the Expression type. You can also use Regex as the expression type to evaluate response for all response content types.

### **Using JSONPath**

The following job is checking if the response from the web service has "Nigel Rees" as the author of the first book. If the author by the said name is found, the job is evaluated to success; else, to failure.

```
Job name -- sample1
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itcl1.ibn.broadcom.net:8082/vmc/bookstore
Workload Automation Agents version compatibility -- 12
Expression type -- JSONPath
Expression -- store.book[0].author
Regex Expression -- Nigel Rees
```

The following example checks if the response contains any book authored by Nigel Rees:

```
Job name -- sample2
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itcl1.ibn.broadcom.net:8082/vmc/bookstore
Workload Automation Agents version compatibility -- 12
Expression type -- JSONPath
Expression -- $..book[?(@.author=='Nigel Rees')].author
Regex Expression -- Nigel Rees
```

### **Using XPATH**

The following job is checking in the XML response if the author of the book in the cooking category is "Giada De Laurentiis". If such an element is found, the job goes to success; else, it fails.

```
Job name -- sample3
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itcl1.ibn.broadcom.net:8082/vmc/bookstorexml
Workload Automation Agents version compatibility -- 12
Expression type -- XPath
Expression -- //bookstore/book[@category='cooking']/author
Regex Expression -- Giada De Laurentiis
```

## Using RegEx

The following example checks if the JSON response contains any book authored by Nigel Rees.

```
Job name -- sample4
Type -- GET
Select machine -- localhost
URL -- http://wabld-linux-itc1.ibn.broadcom.net:8082/vmc/bookstore
Workload Automation Agents version compatibility -- 12
Expression type -- RegEx
Expression -- "author": "Nigel Rees"
```

The following job is checking in the XML response if there is an author by the name "Giada De Laurentiis" in the cooking category. If such an element is found, the job goes to success; else, it fails.

```
Job name -- sample5
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itc1.ibn.broadcom.net:8082/vmc/bookstorexml
Workload Automation Agents version compatibility -- 12
Expression type -- RegEx
Expression -- <book category="cooking">\s*.*\s*<author>Giada De Laurentiis</author>
```

## Consuming Response Payload

The generated response payload from the web service can be consumed using the following options:

- Response variables
- Substitution variables

## Response Variables

The response variables against a job can be queried by running autorep with jb option.

**Example:** autorep -jb jobname

These variables can be used just like the way global variables are used. The notation to be followed is \$  
\$jobname@<v>title</v>, where title here refers to the name of the response variable defined in the job "jobname".

## Response variables using JSONPath

In the following example we are capturing the title of the book in the variable "title" whose price is less than \$ 8.99. In case more than one title satisfies the criteria, only the first title found is stored in the variable title.

```
Job name -- sample6
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itc1.ibn.broadcom.net:8082/vmc/bookstore
Workload Automation Agents version compatibility -- 12
Response variable -- Response type=Payload, Expression type=JSONPath, Name=title, Expression =store.book[?
(@.price < 8.99)].title
```

## Response variables using XPATH

In the following example, we are capturing the author of the book in the variable "author" whose price is \$ 29.99.

```
Job name -- sample7
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itc1.ibn.broadcom.net:8082/vmc/bookstorexml
Workload Automation Agents version compatibility -- 12
```

```
Response variable -- Response type=Payload, Expression type=XPath Name=author, Expression=//bookstore/
book[price=29.99]/author
```

## Response variables using Regular Expression

Here the regular expression specified against Expression keyword matches element called book belonging to "children" category. Instead of storing the entire matched text, we are interested in storing the value as "The title of the book is Harry Potter". The group "1" gives "Harry Potter" from the matched text.

```
Job name - sample8
Select machine -- localhost
Type -- GET
URL -- http://wabld-linux-itcl1.ibn.broadcom.net:8082/vmc/bookstorexml
Workload Automation Agents version compatibility -- 12
Response variable -- Response type=Payload, Expression type=Regex, Name=title, Expression = <book
category="cooking">\s*.*\s*<author>Giada De Laurentiis</author>, Regex group=The title of the book is <<$1>>
```

## Substitution Variables

We can also consume an element from the persisted response directly by specifying the JSONPath, XPATH, or RegEx. These variables can be used anywhere the way global variables are used.

We can use the Generate Expression dialog and consume the persisted data as follows:

### Consuming the persisted JSON data

If we wanted to extract from the response the title of a book whose price is less than \$8.99, then the notation used is `$payload^sample6@<j>store.book[?(@.price < 8.99)].title</j>`. Here payload specifies that we are interested in parsing the response of type "payload". This payload is produced by the job sample6 and the JSONPath expression to be applied on the response is `store.book[?(@.price < 8.99)].title`. The job sample6 must have the Persist payload output checkbox selected under the Response persist options.

### Consuming the persisted XML data

If we wanted to extract from the response the author of a book whose price is \$29.99, then the notation used is `$payload^sample7@<x>//bookstore/book[price=29.99]/author</x>`. The job sample7 needs to have the Persist payload output checkbox selected under the Response persist options.

### Consuming the persisted RegEX data

If we wanted to extract from the response the title of the book in children category, then the notation used is `$payload^sample8@<r><book category="cooking">\s*.*\s*<author>Giada De Laurentiis</author></r>`. The job sample8 needs to have the Persist payload output checkbox selected under the Response persist options.

## Consume the HTTP Response (Web UI)

The response from a HTTP job can be consumed by the successor jobs for their execution. The response can contain payload, header, cookies or a combination of all. The payload is in the form of XML, JSON, or Plain Text. The header and cookies are in the form of serialized JSON. The Response persist options let you specify the path to the payload file which stores the output (in text format) of the invoked web service. The response variables can be specified for job execution.

For more information, see the following sections:

- Response persist options
- Response variable

---

## **Generate expression**

You can use generate expression and consume the payload of a job in successor jobs.

### **Follow these steps:**

1. Click Generate Expressions.  
The Generate expression window opens.
2. Enter or select the following information:
  - Job name**  
Displays the existing job name which has the payload data persisted using Response variables or Response persist options.
  - Data source**  
Select one of the following data sources:
    - Payload - Specifies the output payload of selected job. Use, when the selected job has stored the completed output using the persist payload option.
    - Header - Specifies the response headers of the selected job. Use, when the selected job has stored the headers using the persist headers field.
    - Response variable - Specifies the response variables defined in the selected job.
  - Expression type**  
Select or enter the following fields for the selected data source:
    - For Payload, XPath, JSONPath, or RegEx.
    - For Header, Header keywords\* and Header index.
  - Variable name**
    - For Response variable, select or enter the Variable name.

Enter the **Expression** field for JSONPath and XPath expression type.  
Enter the **Regular expression** and **Regular expression group** fields for RegEx expression type.

- Generated expression**  
Displays the generated expression based on the selected values.
- 3. Click Copy generated expression, to copy the Generated Expression for use in the current job properties.

## **Global Variables**

The following properties support Global Variable substitution:

- j2ee\_parameter
- provider\_url
- j2ee\_conn\_user
- j2ee\_conn\_origin
- j2ee\_conn\_domain
- j2ee\_proxy\_host
- j2ee\_proxy\_user
- j2ee\_proxy\_origin\_host
- method\_name
- j2ee\_proxy\_domain
- headers
- header\_filter
- payload\_uri
- media\_type
- success\_criteria\_xpath
- success\_criteria\_json
- success\_criteria\_value
- blob\_input
- results\_file
- success\_criteria\_regex
- form\_parameter

## HTTP Job Properties (Web UI)

This article explains HTTP job properties of the Web UI. Properties that have a default value automatically apply to the job definition. Therefore, you do not have to specify those properties in the definition. Your agent administrator can define some default values on the agent in the agentparm.txt file.

If you specify the property in a job definition, it overrides the default.

The HTTP category includes the following properties:

### Header parameters

Specifies the header to use for the request in an HTTP job. It has the following Custom Keywords parameters:

#### **Keyword**

(Optional) Specifies the keyword to use for the request in an HTTP job.

Limits: 4000 characters for each combined Keyword and Value.

JIL attribute: headers

#### **Value**

(Optional) Specifies the value of the keyword to use for the request in an HTTP job.

Limits: 4000 characters for each combined Keyword and Value.

JIL attribute: headers

### Form parameters

You can use the Form parameters to define J2EE parameters for the HTTP job type.



**NOTE**

12 compatibility - You can use the Form parameters for POST and PUT options to pass form data when it is url-form-encoded.

11 compatibility - Used as j2ee parameter, for GET and POST to pass query parameters /form data to HTTP jobs.

Limits: 4000 characters for each combined values of Name and Value for one J2EE parameter; up to 64 type=value pairs

**Name**

Specifies the name of the parameter.

**Value**

Specifies the value of the parameter.

JIL attribute: form\_parameter/j2ee\_parameter

**Response persist options****Payload file path**

(Optional) Specifies the path to the payload file. The payload file stores the output (in text format) of the invoked web service. It contains HTTPResponse only.

To save a file, follow these steps:

1. Click Browse.
2. Select the folder where the Payload output file will be created.
3. Click Insert file path.
4. Append or enter the file name to the Payload file path.

Limits: 256 characters; case-sensitive

JIL attribute: results\_file

**Examples:** /tmp/myoutput, "D:\Stock output\STOCK3"

**NOTE**

To enable the Overwrite payload file checkbox, you must specify the Payload file path.

**Overwrite payload file**

Specifies whether you want to overwrite the results file when a file with the same name exists.

Default: Blank checkbox

JIL attribute: results\_file\_overwrite

**Persist headers**

Specifies whether you want to persist the headers output in the database as a blob so that it can be consumed by the successor jobs for their execution.

Default: Blank checkbox

JIL attribute: persist\_output\_header

**Persist payload output**

Specifies whether you want to persist the payload output in the database as a blob so that it can be consumed by the successor jobs for their execution.

Default: Blank checkbox

JIL attribute: persist\_output\_payload

### **Persist cookies**

Specifies whether you want to persist the cookies output in the database as a blob so that it can be consumed by the successor jobs for their execution.

Default: Blank checkbox

JIL attribute: persist\_output\_cookies

### **Response variables**

You can specify the response variables to persist the payload output that can be captured by JSONPath, XPath, or RegEx.

### **Parameters**

Specifies the response variables (response type, expression type, name, expression, regex group) that can be consumed by the successor jobs for their execution.

An HTTP job on its execution produces payload that can be consumed by the successor jobs for their execution. The payload is in the form of XML, JSON, or Plain Text. When a job that contains the response variables properties completes successfully, the scheduler parses the incoming payload for the elements whose path is specified in the expression and the resultant is stored as a job blob with the name that is specified in the name.

JIL\_attribute: output\_var

Example

```
output_var: var_name="var1", var_value="x.store.book[0].title" var_type=payload, var_format="jsonpath"
```

### **Response type**

Specifies the type of the output variable. Valid values are payload and header.

JIL sub-keyword: var\_type

### **Expression type**

Specifies the format of the response variable. Valid values are XPath, JSONPath, or RegEx.

JIL sub-keyword: var\_format

### **Name**

Specifies the name of the response variable.

Limits: 256 characters

JIL sub-keyword: var\_name

### **NOTE**

The variable names must be unique for each job definition.

### **Expression/Header keyword**

Specifies the value of the response variable. You can specify the xpath for XML, jsonpath for JSONPath, or an ordinary regex expression for Plain Text.

Limits: 3950 characters

JIL sub-keyword: var\_value

### **RegEx Group / Header index**

Specifies the numerical group ID in any order. This group ID enables the extraction of the groups that are matched while applying the regex. Applies to RegEx format only.

Limits: 3950 characters

JIL sub-keyword: var\_group

**Example:** If the payload contains a string in the form “This is a sample string” and you want “This string” as a variable, specify two group IDs <<\$1>> <<\$5>> to pick the value from group 1 and 5.

#### NOTE

You cannot specify JSONPath and XPath together for response type Payload.

### Evaluate All

You can use the Evaluate All option to validate the entered XPath, JSONPath, or RegEx for the given text, so that you can use the XPath, JSONPath, or RegEx for subsequent job parameters.

#### Follow these steps:

1. Click Evaluate All.  
The Evaluate response type window opens with the following options, based on the added response variables:
  - a. Payload response content.
  - b. Header response content.
2. Enter or copy and paste the response content, or select the browse file, for validation.
3. Click Evaluate.  
The View result button is enabled. You can click to view the evaluation result of using XPATH/JSON/REGEX provided against the response content.

#### NOTE

For more information, see [JSON parser](#).

### Examples:

Specify the response variables in XPath format, as follows:

Response variables:

Response Type-Payload

Expression Type-XPath

Name-PAR1

Expression header/keyword-/store/book[1]/title

Specify the response variables in JSONPath format, as follows:

Response variables:

Response Type-Header

Expression Type-key\_index

Name-PAR1

Expression-x.store.book[0].title

Specify the response variables in RegEx format, as follows:

Response variables:

Response Type-PayloadExpression Type-RegEx

Name-PAR1

Expression-[a-zA-Z]+

RegEx Group-<<\$1>>

---

## **Authentication parameters**

### **Connection user**

Specifies the user name for connection authentication.

Limits: 145 characters

#### **NOTE**

- If you specify the connection user property in your job definition, you must set the password that is associated with the connection user using the `autosys_secure` command.

JIL attribute: `j2ee_conn_user`

### **Connection origin**

Specifies the origin host name for NTLM connection authentication.

Default: The computer name where the agent is running

Limits: 80 characters; case-sensitive. Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: `j2ee_conn_origin`

### **Connection domain**

Specifies the domain for NTLM connection authentication.

Limits: 80 characters; case-sensitive. Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: `j2ee_conn_domain`

### **Authentication order**

Specifies a list of protocols to be used by a web server for authentication in an HTTP job. You can use the following protocols in any order of your choice:

- **BASIC**  
Indicates the BASIC protocol.
- **DIGEST**  
Indicates the DIGEST protocol.
- **NTLM**  
Indicates the NTLM protocol.

#### **NOTE**

- You can specify any of the following protocols in any order: BASIC, DIGEST, and NTLM.
- If you are connecting to an old web server that cannot negotiate authentication protocols, enter the following list in the specified order: BASIC,DIGEST,NTLM.

JIL attribute: `j2ee_authentication_order`

### **Preemptive authentication**

Specifies whether or not to authenticate the application credentials using preemptive authentication. In preemptive authentication, credentials are submitted along with the initial request.

#### **NOTE**

To enable preemptive authentication, you must specify Connection user; you can also specify BASIC, either alone or in a combination, in the Authentication order.

Default: Blank checkbox

JIL attribute: preemptive\_authentication

### **Proxy details**

Specifies a list of defaults to be used by a web server for authentication in an HTTP job. Specify the following proxy details:

#### **Use agent global proxy**

(Optional) Specifies whether the agent uses the global proxy defaults specified in the agentparm.txt file

- Leave the checkbox blank to ignore the global proxy configuration specified by the proxy parameters in the agentparm.txt file (the default). Do not select this option if you want to ignore the proxy defaults for this job only. For example, you might ignore the proxy defaults if the request is going to a server on the LAN that does not require the proxy. This is the default.
- If the checkbox is selected, the application uses the global proxy configuration specified by the proxy parameters in the agentparm.txt file.

Default: Blank checkbox

JIL attribute: j2ee\_no\_global\_proxy\_defaults

#### **Proxy user**

(Optional) Specifies the user name required for proxy authentication. If you omit this property, the user name defaults to the value specified in the http.proxyUser parameter in the agentparm.txt file.

Limits: 80 characters; case-sensitive; cannot contain delimiters such as spaces

#### **NOTE**

- Your agent administrator can specify a default proxy user name for all HTTP jobs by setting the http.proxyUser parameter in the agent's agentparm.txt file.
- The proxy user property overrides the default proxy user name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: j2ee\_proxy\_user

#### **Proxy origin**

Specifies the origin host name for proxy authentication used for NTLM authentication. If you omit this property, the host name defaults to the value specified in the http.proxyOrigin parameter in the agentparm.txt file.

Limits: 256 characters; case-sensitive

Default: The computer name where the agent is running

JIL attribute: j2ee\_proxy\_origin\_host

#### **Proxy domain**

Specifies the domain for proxy authentication required for NTLM authentication. If you omit this property, the domain defaults to the value specified in the http.proxyDomain parameter in the agentparm.txt file.

Limits: 80 characters; case-sensitive

#### **NOTE**

- Your agent administrator can specify a default proxy domain name for all HTTP jobs by setting the http.proxyDomain parameter in the agent's agentparm.txt file.
- The proxy domain property overrides the default proxy domain name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: j2ee\_proxy\_domain

### Proxy host

Specifies the proxy host name to use for the request. If you omit this property, the host name defaults to the value specified in the http.proxyHost parameter in the agentparm.txt file.

Limits: 256 characters; case-sensitive

#### NOTE

- Your agent administrator can specify a default proxy host name for all HTTP jobs by setting the http.proxyHost parameter in the agent's agentparm.txt file.
- The proxy host property overrides the default proxy host name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: j2ee\_proxy\_host

### Proxy port

(Defaulted) Specifies the port of the proxy server to use for the request. If you omit this property, the port defaults to the value specified in the http.proxyPort parameter in the agentparm.txt file.

Default: 80

Limits: 0-65535

JIL attribute: j2ee\_proxy\_port

### Success criteria

The following response codes indicate and provide detailed information of the success or failure of the job.

#### Success exit code

Defines which response codes indicate job success. For example, you can define the Success exit code to be 20-30. If a job returns an response code between 20 and 30, the job is considered to have completed successfully.

#### NOTE

If you specify multiple response codes, enter the most specific codes first followed by the ranges.

Specify a single response code (for example, 4), a range of response codes (for example, 0-9999), or a list of multiple response codes or ranges separated by commas.

**Examples:** 1,150,20-30

Default: 100 - 299

Limits: 256 characters. The response code must be an integer in the range 0 to 2147483647.

JIL attribute: success\_codes

#### Failure exit code

Defines which response codes indicate job failure. For example, you can define the Failure exit code to be 40-50. If a job returns an response code in the range 40 to 50, the job is considered to have failed.

#### NOTE

If you specify multiple response codes, enter the most specific codes first followed by the ranges.

Specify a single response code (for example, 4), a range of response codes (for example, 0-9999), a list of multiple response codes separated by commas, or a list of ranges separated by commas.

**Examples:** 1,150,20-30

Default: Responses less than 100 and above 299

Limits: 256 characters. The response code must be an integer in the range 0 to 2147483647.

JIL attribute: fail\_codes

### Header filter

Specifies a filter (a regular expression) that is applied on the header of a response in an HTTP job. If any line from the server response matches the filter, the job completes. Else, the job fails.

Limits: 256 characters

JIL attribute: header\_filter

### Expression type

Specifies the *<filtertype>* (XPath, JSONPath, or Regex). The XPath or JSONPath evaluated value is validated against the value that you specify in the RegEx expression.

### Expression

Specifies the value of the JSONPath, XPath, or Regex expression type. For XPath and JSONPath, complete the RegEx expression.

Limits: 256 characters

JIL attribute: success\_criteria\_*<filtertype>*

### RegEx expression

Defines a regular expression that is used to evaluate the XPath expression (that you specify using the success\_criteria\_xpath) or the JSONPath (that you specify using the success\_criteria\_json). If the evaluated XPath or JSONPath expression value matches the regular expression, the job completes successfully. Otherwise, the job fails.

Limits: 256 characters

JIL attribute: success\_criteria\_value

### Filter

(This option is available only for 11 compatibility) Specifies a regular expression to use as a filter. In an HTTP job, this attribute filters the output of the invoked servlet.

Limits: 256 characters; case-sensitive

JIL attribute: filter

### Connection parameters

The following options provide you with information and attempts to retry a failed job.

#### Send failure details for retries

Specifies whether to send the failure details (indicating the failure and retry count) to the manager when the Failure Retry Count is set to a value that is greater than zero.

- Select the checkbox to send the failure details to the manager when the Failure Retry Count is set to a value that is greater than zero.
- If the checkbox is not selected failure details are not sent to the manager when the Failure Retry Count is set to a value that is greater than zero.

Default: Blank checkbox

JIL attribute: retry\_on\_failure\_details

**Connection retry count**

Specifies the number of attempts to be made to restart the failed job.

Limits: 1-200

Default: 3

JIL attribute: connection\_retry

**Connection timeout** (in seconds)

Specifies the duration, in seconds, after which the request times out.

Limits: 0-2147483647

JIL attribute: connection\_timeout

**Failure retry count**

Specifies the number of times an HTTP request is retried when the HTTP exit code is a failure code or the specified filter criteria does not match the response.

Default: 0 (no retries)

JIL attribute: retry\_on\_failure\_count

**Failure retry interval** (in seconds)

Specifies the number of seconds to wait between retrying HTTP requests if the Failure Retry Count is set to a value that is greater than zero..

Default: 10

**NOTE**

Your agent administrator can set a default value by setting the Failure Retry Interval parameter in the agent's agentparm.txt file. The Failure Retry Interval value overrides the default value that is specified in the agent's agentparm.txt file.

JIL attribute: retry\_on\_failure\_delay

## Manage Database Jobs Using Web UI

Database jobs let you automate common database tasks on Oracle, Microsoft SQL Server, and IBM DB2 databases.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

You can define the following database jobs:

- **Database Monitor**  
Lets you monitor for an increase or decrease in the number of rows in a database table.
- **Database Stored Procedure**  
Lets you run a stored procedure.
- **Database Trigger**  
Lets you monitor for added, deleted, and updated rows in a database table.
- **SQL**  
Lets you execute an SQL statement.



## How Database Trigger Jobs Differ from Database Monitor Jobs

You can monitor database tables using Database Monitor or Database Trigger jobs.

A Database Monitor job can monitor a database table for an increase or decrease in the number of rows. To monitor the database for specific changes, you can add a monitor condition to the job definition. Database Monitor jobs count the number of rows that satisfy the monitor condition using a polling interval, which is every 10 seconds by default. Database Monitor jobs do not detect other updates to a row or changes to the number of rows that cancel each other out during the polling interval. For example, suppose that within a 10-second interval, a row is added while another row is deleted. Since the total number of rows did not change, the Database Monitor job does not detect the row addition and deletion.

Alternatively, a Database Trigger job can monitor a database table for added rows, deleted rows, or updated rows. To monitor the database for specific changes, you can add a trigger condition to the job definition. Each Database Trigger job creates a database trigger on the database. The database trigger templates that the agent uses are located in the directory where the agent is installed. The template files are named `dbtrigdatabase_type.properties`, for example, `dbtrigOracle.properties`. Contact your database administrator before choosing to use a Database Trigger job.

For either job type, you can set up continuous monitoring so that each time a database change occurs, an alert or an event is triggered.

### NOTE

Not all scheduling managers support alerts and events.

### WARNING

You should not drop a table that is being referenced by either a Database Monitor or a Database Trigger job. The job will remain active, even though the table has been dropped.

## Define a Database Monitor Job

You can define a Database Monitor job to monitor a database table for an increase or decrease in the number of rows. To monitor the database table for specific changes, you can add a monitor condition to the job definition. When the condition is met, the job completes.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

### Follow these steps:

1. Create a Database Monitor (DBMON) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Table name**  
Specifies the name of the database table to monitor for changes.
3. (Optional) Specify optional [Database Monitor properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Database Monitor job is defined.

## **Database Monitor Job Examples**

The following examples describe sample Database Monitor jobs.

### **Example: Monitor a Table for a Change in the Number of Rows**

Suppose that you want a job to monitor the STAFF table for a change in the number of rows. When a row that has the name Jonson is added or deleted, the job completes. The job uses the database resource location default defined on the agent.

#### **Follow these steps:**

1. Create a Database Monitor job.
2. Enter the following properties:
  - Name -- staff
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - Table name -- staff
  - Monitor type -- VARIANCE
  - Monitor condition -- Jonson
3. Commit the job.

### **Example: Monitor a Table for Increases in the Number of Rows**

Suppose that you want a job to monitor the EMP table for increases in the number of rows. When a new row has a salary greater than 100000, the scheduling manager sends an alert. The job remains in a monitoring state until it is forced complete or cancelled. The job uses the user name and database resource location defaults defined on the agent.

#### **Follow these steps:**

1. Create a Database Monitor job.
2. Enter the following properties:
  - Name -- emp
  - Send to machine -- DB\_AGENT
  - Table name -- emp
  - Monitor type -- INCREASE
  - Monitor condition -- select count(\*) from emp where sal>100000
3. Commit the job.

## **Database Monitor Properties**

The Database Monitor category includes the following properties:

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Continuous**

(Defaulted) Indicates whether to monitor for the conditions continuously. When selected, each time the specified conditions occur, an alert is written to the scheduler log file.

**Default:** not selected (the job does not monitor for the conditions)

**NOTE**

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

**Database connect string**

(Optional) Specifies a JDBC database resource location (URL) for a job. Use the appropriate format for your database type, as follows: For an Oracle database: "jdbc:oracle:thin:@host:port:database\_name" For a Microsoft SQL Server database: "jdbc:sqlserver://host:port;DatabaseName=database\_name" For an IBM DB2 database: "jdbc:db2://host:port/database\_name" For a Sybase database: connect\_string: "jdbc:sybase:Tds:host:port / dbname

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

Your agent administrator can also set a default value for this property using the db.default.url parameter in the agentparm.txt file.

**JIL attribute:** connect\_string

**Monitor condition**

(Optional) Specifies the condition to monitor in the database. This condition is equivalent to an SQL where clause.

Clicking



opens the Edit Text dialog, which lets you enter the condition.

This property supports Global Variable substitution.

**Limits:** Up to 500 characters; case-sensitive

**JIL attribute:** monitor\_cond

**Monitor type**

(Defaulted) Specifies the type of database change to monitor for, as follows:

- **DECREASE (Row decrease)**

Monitors for a decrease in the number of rows in the database table.

- **INCREASE (Row increase)**

Monitors for an increase in the number of rows in the database table.

- **VARIANCE (Both of the above)**

Monitors for an increase or a decrease in the number of rows in the database table.

**Default:** VARIANCE (Both of the above)

**JIL attribute:** monitor\_type

**Oracle user role**

(Optional) Specifies the role of the Oracle user to log in as. The user role must be defined in the Oracle database. The user role uses the following syntax:

```
as userrole
```

**Limits:** Up to 30 characters; case-sensitive

**Example:** as sysdba

**JIL attribute:** user\_role

### Table name

Specifies the name of the database table to monitor for changes.

This property supports Global Variable substitution.

**Limits:** Up to 300 characters

**JIL attribute:** tablename

## Define a Database Stored Procedure Job

You can define a Database Stored Procedure job to invoke a procedure stored in a database. You can add criteria to the job definition to test the procedure's output. If the result matches the criteria, the job completes successfully. When the procedure executes, the output parameter values from the database are returned to the scheduling manager. You can view the output parameter values in the job's spool file. By default, the agent separates the output parameter values in the return string with a vertical bar ( | ).

If you are using Oracle or Microsoft SQL Server, you can also define a Database Stored Procedure job to run a stored function.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

### Follow these steps:

1. Create a Database Stored Procedure (DBPROC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Stored procedure**  
Specifies the name of the database stored procedure to run.
3. (Optional) Specify optional [Database Stored Procedure properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Database Stored Procedure job is defined.

## Database Stored Procedure Job Examples

The following examples describe sample Database Stored Procedure jobs.

### Example: Invoke a Simple Stored Procedure from a Microsoft SQL Server Database

Suppose that you want a job to run the byroyalty stored procedure located in the pubs sample database. When the job runs, the agent passes the input parameter percentage a value of 40.

### Follow these steps:

1. Create a Database Stored Procedure job.

2. Enter the following properties:
  - Name -- staff
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - Stored procedure -- byroyalty
  - Parameters list:
    - Argtype -- IN
    - Name -- percentage
    - Datatype -- INTEGER
    - Value -- 40
    - Ignore -- No
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
3. Commit the job.

### **Example: Invoke a Stored Procedure with Input and Output Parameters from a Microsoft SQL Server Database**

Suppose that you want a job to run the emp stored procedure, which returns a value from the emp table. The pubid returned matches the employee named Ann Devon. The pubid, 9952, appears in the job's spool file.

#### **Follow these steps:**

1. Create a Database Stored Procedure job.
2. Enter the following properties:
  - Name -- staff
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - Stored procedure -- emp
  - Parameters list:
    - Argtype -- IN
    - Name -- f\_name
    - Datatype -- VARCHAR
    - Value -- Ann
  - Parameters list:
    - Argtype -- IN
    - Name -- l\_name
    - Datatype -- VARCHAR
    - Value -- Devon
  - Parameters list:
    - Argtype -- OUT
    - Name -- pubid
    - Datatype -- CHAR
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
3. Commit the job.

To run the job in this example, create the emp stored procedure in the pubs database using the following script:

```
CREATE PROCEDURE EMP
 (@f_name VARCHAR(20),
 @l_name VARCHAR(30),
 @pubid CHAR(4) OUTPUT)
AS BEGIN
 SELECT
 @pubid=pub_id
```

```

FROM emp
WHERE
fname=@f_name
and
lname=@l_name
print @l_name+@f_name+@pubid
END
GO

```

### Example: Invoke a Stored Procedure with Input and Output Parameters from an IBM DB2 Database

Suppose that you want a job to run the DEPT\_MEDIAN stored procedure under the user entadm. DEPT\_MEDIAN returns the median salary of department 20 from the STAFF table. The median salary, 18171.25, appears in the job's spool file.

#### Follow these steps:

1. Create a Database Stored Procedure job.
2. Enter the following properties:
  - Name -- staff
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - Stored procedure -- ENTADM.DEPT\_MEDIAN
  - Parameters list:
    - Argtype -- IN
    - Name -- deptNumber
    - Datatype -- SMALLINT
    - Value -- 20
  - Parameters list:
    - Argtype -- OUT
    - Name -- medianSalary
    - Datatype -- DOUBLE
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
3. Commit the job.

The spool file for this job contains the following output:

```

{ call ENTADM.DEPT_MEDIAN(?, ?) }
medianSalary=18171.25

```

The job in this example runs the following stored procedure in the SAMPLE database:

```

CREATE PROCEDURE DEPT_MEDIAN
 (IN deptNumber SMALLINT, OUT medianSalary DOUBLE)
LANGUAGE SQL
BEGIN
 DECLARE v_numRecords INTEGER DEFAULT 1;
 DECLARE v_counter INTEGER DEFAULT 0;
 DECLARE c1 CURSOR FOR
 SELECT CAST(salary AS DOUBLE) FROM staff
 WHERE DEPT = deptNumber
 ORDER BY salary;
 DECLARE EXIT HANDLER FOR NOT FOUND

```

```

 SET medianSalary = 6666;
-- initialize OUT parameter
 SET medianSalary = 0;
 SELECT COUNT(*) INTO v_numRecords FROM staff
 WHERE DEPT = deptNumber;
 OPEN c1;
 WHILE v_counter < (v_numRecords / 2 + 1) DO
 FETCH c1 INTO medianSalary;
 SET v_counter = v_counter + 1;
 END WHILE;
 CLOSE c1;
END

```

### **Database Stored Procedure Properties**

The Database Stored Procedure category includes the following properties:

#### **Database connect string**

(Optional) Specifies a JDBC database resource location (URL) for a job. Use the appropriate format for your database type, as follows: For an Oracle database: "jdbc:oracle:thin:@host:port:database\_name" For a Microsoft SQL Server database: "jdbc:sqlserver://host:port;DatabaseName=database\_name" For an IBM DB2 database: "jdbc:db2://host:port/database\_name" For a Sybase database: connect\_string: "jdbc:sybase:Tds:host:port / dbname

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

#### **NOTE**

Your agent administrator can also set a default value for this property using the db.default.url parameter in the agentparm.txt file.

**JIL attribute:** connect\_string

#### **Oracle user role**

(Optional) Specifies the role of the Oracle user to log in as. The user role must be defined in the Oracle database. The user role uses the following syntax:

```
as userrole
```

**Limits:** Up to 30 characters; case-sensitive

**Example:** as sysdba

**JIL attribute:** user\_role

#### **Parameters list: Argtype**

(Optional) Specifies the parameter type, as follows:

This property supports Global Variable substitution.

- IN -- Specifies that the parameter is an input parameter.
- OUT -- Specifies that the parameter is an output parameter.
- INOUT -- Specifies that the parameter is an input-output parameter.

**JIL attribute:** sp\_arg: argtype

**Parameters list: Datatype**

(Optional) Specifies the database [data type](#) of the parameter.

This property supports Global Variable substitution.

**JIL attribute:** sp\_arg: datatype

**Parameters list: Ignore**

(Optional) Specifies whether the value is ignored, as follows:

This property supports Global Variable substitution.

- Yes -- Specifies that the value is ignored.
- No -- Specifies that the value is returned.

This property is useful if you want to hide data such as a password.

**Default:** No

**NOTE**

This argument applies to output parameters only.

**JIL attribute:** sp\_arg: ignore

**Parameters list: Name**

(Optional) Specifies the name of the parameter.

This property supports Global Variable substitution.

**Limits:** Up to 64 characters

**JIL attribute:** sp\_arg: name

**Parameters list: Value**

(Optional) Specifies the value of the parameter.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

- This argument applies to input parameters only.
- Enclose values that contain spaces in apostrophes.

**JIL attribute:** sp\_arg: value

**Result type**

(Optional) Specifies the data type to return from a stored procedure.

**JIL attribute:** result\_type

**Stored procedure**

Specifies the database stored procedure to run.

This property supports Global Variable substitution.

**Limits:** Up to 300 characters

**JIL attribute:** sp\_name

**Success criteria**



(Optional) Defines a regular expression that is used to evaluate a return string. If the return string matches the regular expression, the job completes successfully. Otherwise, the job fails.

This property supports Global Variable substitution.

**Limits:** Up to 500 characters; case-sensitive

#### NOTE

Each return string includes the field name from the SELECT statement and its value, separated by an equal sign (=). For example, consider the query SELECT ORD\_NUM FROM SALES. To match order number A2976, specify the regular expression ORD\_NUM=A2976. Specifying the regular expression A2976 does not match any return string causing the job to fail.

**JIL attribute:** success\_criteria

### Supported Data Types

The following table lists the supported data types of different databases. The data types listed in the database columns are defined in the procedure definition within the database. In the job definition, use the corresponding JDBC data type from the first column.

JDBC Data Type	Valid Input Format	Oracle	MS SQL Server	DB2
CHAR	Plain text	CHAR	CHAR	CHAR
VARCHAR	Plain text	VARCHAR2(x) VARCHAR(x)	VARCHAR(x)	VARCHAR(x)
LONGVARCHAR		Not supported	Not supported	Not supported
NUMERIC	Number	NUMBER, NUMERIC	NUMERIC	NUMERIC
DECIMAL	Number	DECIMAL	DECIMAL	DECIMAL
BIT		Not supported	BIT	Not supported
BOOLEAN		Not supported	Not supported	Not supported
TINYINT		Not supported	TINYINT	Not supported
SMALLINT	Number	SMALLINT	SMALLINT	SMALLINT
INTEGER	Number	INTEGER	INTEGER INT	INTEGER
BIGINT	Number	Not supported	BIGINT	BIGINT
REAL	Number [dot Number]	REAL	REAL	REAL
FLOAT	Number [dot Number]	FLOAT	FLOAT	FLOAT
DOUBLE			FLOAT	DOUBLE
DATE	yyyy-mm-dd	DATE	Not supported	DATE
String (e.g., SYSDATE)				
TIME	hh:mm:ss		Not supported	TIME
TIMESTAMP	yyyy-mm-dd hh:mm:ss.ffffff	TIMESTAMP	datetime smalldatetime	TIMESTAMP

**NOTE**

Although JDBC provides functionality for Booleans, the Oracle database driver does not. To handle unsupported data types such as Booleans, you can call a wrapper procedure in the job definition. The wrapper procedure converts the input values to values supported by the database driver and calls the appropriate stored procedure from inside the database.

**Example: Run a Wrapper Procedure**

In the following example, the agent runs a wrapper procedure called boolwrap, which converts the inputted integer value to a Boolean and calls the boolproc stored procedure:

```
CREATE OR REPLACE PROCEDURE boolproc(x boolean) AS BEGIN
[...]
END;
CREATE OR REPLACE PROCEDURE boolwrap(x int) AS BEGIN
IF (x=1) THEN
 boolproc(TRUE);
ELSE
 boolproc(FALSE);
END IF;
END;
```

**Define a Database Trigger Job**

You can define a Database Trigger job to monitor a database table for added, deleted, or updated rows. To monitor the database table for specific changes, you can add a condition to the job definition. When the condition is met, the job completes.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

**Follow these steps:**

1. Create a Database Trigger (DBTRIG) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Table name**  
Specifies the name of the database table to monitor for changes.
  - **Database type**  
Specifies whether the database is an Oracle, MSSQL, DB2, SYBASE, POSTGRESQL, or TERADATA database.
  - **Trigger type**  
Specifies the type of database change to monitor for.
3. (Optional) Specify optional [Database Trigger properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Database Trigger job is defined.

## **Database Trigger Job Examples for Oracle**

The following examples describe sample Database Trigger jobs that monitor Oracle database tables.

### **Example: Monitor an Oracle Database Table for an Added or Deleted Row**

Suppose that you want a job to monitor the emp table for an added row or a deleted row. The job runs under the user named scott, who has the authority to create triggers on the database. The job remains in a running state while waiting for an added or deleted row. When a row is either added or deleted, the job completes.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- adddelrow
  - Send to machine -- DB\_AGENT
  - Table name -- emp
  - Trigger type -- INSERT,DELETE
  - Database connect string -- "jdbc:oracle:thin:@myhost:1521:orcl"
  - Database type -- ORACLE
  - Oracle user role -- scott
3. Commit the job.

### **Example: Monitor an Oracle Database Table for Deleted Rows with a Trigger Condition**

Suppose that you want a job to monitor the emp table for deleted rows. The job runs under the user named scott, who has the authority to create triggers on the database. When a row containing department 75 is deleted, the job completes.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- delrow
  - Send to machine -- DB\_AGENT
  - Table name -- emp
  - Trigger type -- DELETE
  - Trigger conditions -- old.deptno=75
  - Database connect string -- "jdbc:oracle:thin:@myhost:1521:orcl"
  - Database type -- ORACLE
  - Oracle user role -- scott
3. Commit the job.

### **Example: Monitor an Oracle Database Table for Added Rows with a Trigger Condition**

Suppose that you want a job to monitor the emp table for added rows. The job runs under the user named scott, who has the authority to create triggers on the database. When a row containing a name beginning with the letter g is added, the job completes.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:

- Name -- addrow
  - Send to machine -- DB\_AGENT
  - Table name -- emp
  - Trigger type -- INSERT
  - Trigger conditions -- 'new.ename like "g%%"
  - Database connect string -- jdbc:oracle:thin:@myhost:1521:orcl
  - Database type -- ORACLE
  - Oracle user role -- scott
3. Commit the job.

### **Example: Monitor an Oracle Database Table for Added or Updated Rows with a Trigger Condition**

Suppose that you want a job to monitor the emp table for added or updated rows. The job runs under the user named scott, who has the authority to create triggers on the database. The job completes when a new or updated row does not contain a job field equal to sales.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- adduprow
  - Send to machine -- DB\_AGENT
  - Table name -- emp
  - Trigger type -- INSERT,UPDATE
  - Trigger conditions -- new.job<>'sales'
  - Database connect string -- "jdbc:oracle:thin:@myhost:1521:orcl"
  - Database type -- ORACLE
  - Oracle user role -- scott
3. Commit the job.

#### **NOTE**

The <> symbol indicates not equal to.

### **Database Trigger Job Examples for Microsoft SQL Server**

The following examples describe sample Database Trigger jobs that monitor Microsoft SQL Server database tables.

#### **Example: Monitor a Microsoft SQL Server Database Table for a New or Deleted Row**

Suppose that you want a job to monitor the stores table for an added row or a deleted row. The job runs on the DB\_AGENT agent computer under the sa user. The job remains in a running state waiting for an added or deleted row. When a row is either added or deleted, the job completes.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- newdelrow
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
  - Table name -- stores
  - Trigger type -- INSERT,DELETE
3. Commit the job.

**Example: Monitor a Microsoft SQL Server Database Table for Specific Changes**

Suppose that you want a job to monitor the sales table for changes to the ord\_date and qty columns. The job runs under the sa user and completes only when both columns have changed.

**Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- changes
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
  - Table name -- sales
  - Trigger type -- UPDATE
  - Trigger condition -- 'UPDATE(ord\_date) and UPDATE(qty)'
3. Commit the job.

**Example: Monitor a Microsoft SQL Server Database Table for Added Rows with a Trigger Condition**

Suppose that you want a job to monitor the sales table for added rows. When the quantity for inserted title ID TC7777 is greater than or equal to 20, the job completes.

**Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- insert
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
  - Table name -- sales
  - Trigger type -- INSERT
  - Trigger condition -- '(select QTY from INSERTED where TITLE\_ID="TC7777")>=20'
3. Commit the job.

**Database Trigger Job Examples for IBM DB2**

The following examples describe sample Database Trigger jobs that monitor IBM DB2 database tables.

**Example: Monitor an IBM DB2 Database Table for Added Rows with a Trigger Condition**

Suppose that you want a job to monitor the STAFF table for added rows. When the total number of rows is greater than or equal to 37, the job completes.

**Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:

- Name -- addrow
- Send to machine -- DB\_AGENT
- Owner -- entadm
- Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
- Table name -- staff
- Trigger type -- INSERT
- Trigger condition -- '(select count(\*) from STAFF)>=37'

3. Commit the job.

### **Example: Monitor an IBM DB2 Database Table for Specific Changes**

Suppose that you want to monitor the STAFF table for changes to the DEPT and JOB columns. The job runs on the DB\_AGENT agent computer and connects to the SAMPLE database. The job runs under the entadm user and completes once DEPT or JOB is updated.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- deleted
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
  - Table name -- staff
  - Trigger type -- UPDATE
  - Trigger condition -- 'UPDATE of DEPT, JOB'
3. Commit the job.

### **Database Trigger Job Example for Sybase**

#### **Example: Monitor a Sybase Database Table for Specific Changes**

This example monitors the ap\_invoices table for changes. The job connects to the Sybase database named AEDB that is reachable on port 5001 on a host named myhost. The job completes when the table has changed.

#### **Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- db\_sqltrig\_upd
  - Owner -- admin@myhost
  - Database connect string -- "jdbc:sybase:Tds:myhost:5001/AEDB"
  - Send to machine -- localhost
  - Database type -- Sybase
  - Trigger type -- UPDATE
  - Table name -- ap\_invoices
3. Commit the job.

### **Database Trigger Job Example for PostgreSQL**

#### **Example: Monitor a PostgreSQL Database Table for Specific Changes**

Suppose that you want a job to monitor the PGS table for added rows. The job runs under the user named Trevor, who has the authority to create triggers on the database. When a new row is added, the job completes.

**Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- DatabaseTrigger524623
  - Owner -- postgres@mydb
  - Database connect string -- "jdbc:postgresql://waaeqa-jms:5433/mydb"
  - Send to machine -- Postgres\_Agent
  - Database type -- PostgreSQL
  - Trigger type -- INSERT
  - Table name -- PGS
3. Commit the job.

**Database Trigger Job Example for Teradata****Example: Monitor a Teradata Database Table for Specific Changes**

Suppose that you want a job to monitor the TD table for added rows. The job runs under the user named John, who has the authority to create triggers on the database. When a new row is added, the job completes.

**Follow these steps:**

1. Create a Database Trigger job.
2. Enter the following properties:
  - Name -- DatabaseTrigger624623
  - Owner -- terauser@kolon04-UI89684
  - Database connect string -- "jdbc:teradata://kolon04-UI89684/dbc"
  - Send to machine -- Teradata\_Agent
  - Database type -- Teradata
  - Trigger type -- INSERT
  - Table name -- TERAUSER.TD
3. Commit the job.

**Database Trigger Properties**

The Database Trigger category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

**Continuous**

(Defaulted) Indicates whether to monitor for the conditions continuously if selected. Each time the specified conditions occur, an alert is written to the scheduler log file.

**Default:** not selected (the job does not monitor for the conditions)

**NOTE**

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

### Database connect string

(Optional) Specifies a JDBC database resource location (URL) for a job. Use the appropriate format for your database type, as follows: For an Oracle database: "jdbc:oracle:thin:@host:port:database\_name" For a Microsoft SQL Server database: "jdbc:sqlserver://host:port;DatabaseName=database\_name" For an IBM DB2 database: "jdbc:db2://host:port/database\_name" For a Sybase database: connect\_string: "jdbc:sybase:Tds:host:port / dbname

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** connect\_string

### Database type

Specifies whether the database is an Oracle, MSSQL, Sybase, or DB2 database.

**JIL attribute:** dbtype

### Oracle user role

(Optional) Specifies the role of the Oracle user to log in as. The user role must be defined in the Oracle database. The user role uses the following syntax:

```
as userrole
```

**Limits:** Up to 30 characters; case-sensitive

**Example:** as sysdba

**JIL attribute:** user\_role

### Table name

Specifies the name of the database table to monitor for changes.

**Limits:** Up to 300 characters

**JIL attribute:** tablename

### Trigger condition

(Optional) Specifies the condition to monitor in the database, as follows:

- For Oracle and DB2, this condition is the WHEN clause.
- For SQL Server, this condition is the IF clause.

This property supports Global Variable substitution.

**Limits:** Up to 500 characters; case-sensitive

### NOTE

For the specific database syntax, refer to your database vendor's documentation.

**JIL attribute:** trigger\_cond

### Trigger type

Specifies the type of database change to monitor for, as follows:

- **DELETE**

Monitors for the deletion of a row in the database table.



- **INSERT**

Monitors for the insertion of a row in the database table.

- **UPDATE**

Monitors for an update to any of the rows in the database table.

**Default:** INSERT

**Example:** Selecting INSERT and UPDATE monitors for the insertion of a row or an update to any of the rows.

**NOTE**

You can select multiple trigger types.

**JIL attribute:** trigger\_type

### Define an SQL Job

You can define an SQL job to run an SQL query against an Oracle, Microsoft SQL Server, or IBM DB2 database. When the job runs, the SQL statement is invoked and the results are stored in an output file or job spool file. You can also add criteria to the job definition to test the query result. If the result matches the criteria, the job completes. Otherwise, the job fails.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Databases.

**Follow these steps:**

1. Create an SQL job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **SQL command**  
Specifies an SQL command to run against database tables.
3. (Optional) Specify optional [SQL properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SQL job is defined.

### SQL Job Examples for Oracle

The following examples describe sample SQL jobs that run SQL commands against Oracle database tables.

**Example: Add a Row to an Oracle Database Table**

Suppose that you want a job to add a row of data to the emp table in an Oracle database.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:

- Name -- addrow
  - Send to machine -- DB\_AGENT
  - SQL command -- 'INSERT INTO EMP(EMPNO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO) VALUES(2476, "robert", "sales", 435, "01-OCT-2005", 65000, 10, 75)'
  - Database connect string -- "jdbc:oracle:thin:@myhost:1521:orcl"
  - Oracle user role -- as sysdba
3. Commit the job.

#### **Example: Update a Row in an Oracle Database Table**

Suppose that you want a job to update a record in the emp table and change the salary to 75,000 for the employee with name robert. The table is stored in a Microsoft SQL Server database.

##### **Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- updatetow
  - Send to machine -- DB\_AGENT
  - SQL command -- 'UPDATE EMP SET SAL=75000 where ENAME="robert"'
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=docs"
  - Oracle user role -- scott
3. Commit the job.

#### **Example: Delete a Row from an Oracle Database Table**

Suppose that you want a job to delete a row from the emp table for the employee with name robert. The table is stored in an IBM DB2 database.

##### **Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- deleterow
  - Send to machine -- DB\_AGENT
  - SQL command -- 'DELETE FROM EMP WHERE ENAME="robert"'
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
  - Oracle user role -- scott
3. Commit the job.

#### **Example: Return Data from an Oracle Database Table that Matches a Condition**

Suppose that you want a job to query the emp table for names that have salaries greater than 40,000. If the query returns a name that begins with the letter d, the job completes.

##### **Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:

- Name -- query
- Send to machine -- DB\_AGENT
- SQL command -- 'SELECT ENAME FROM EMP WHERE SAL > 40000'
- Destination file -- /emp/salary.txt
- Success criteria -- ENAME=d.\*
- Database connect string -- "jdbc:oracle:thin:@myhost:1521:orcl"
- Oracle user role -- as sysdba

### 3. Commit the job.

For example, the salary.txt file contains the following output:

```
Output for: SELECT ENAME FROM EMP WHERE SAL > 40000
```

```
ENAME

donald
```

## **SQL Job Examples for Microsoft SQL Server**

The following examples describe sample SQL jobs that run SQL commands against Microsoft SQL Server database tables.

### **Example: Add a Row to a Microsoft SQL Server Database Table**

Suppose that you want a job to add a row for a new store to the stores table.

#### **Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- addrow
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - SQL command -- 'INSERT INTO stores(stor\_id, stor\_name, stor\_address, city, state, zip) VALUES("6523", "Chapters", "6523 Main St.", "San Diego", "CA", "93223")'
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=docs"
3. Commit the job.

### **Example: Update a Row in a Microsoft SQL Server Database Table**

Suppose that you want a job to update the row in the sales table that matches order number 6871 and change the values for the date and quantity of that order.

#### **Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- updatetrow
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - SQL command -- 'UPDATE sales SET ord\_date="6/15/2006", qty=10 WHERE ord\_num="6871"'
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=docs"
3. Commit the job.

### **Example: Delete a Row from a Microsoft SQL Server Database Table**

Suppose that you want a job to delete the row for store ID 6523 from the stores table.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- delrow
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - SQL command -- 'DELETE FROM stores WHERE stor\_id="6523"'
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=docs"
3. Commit the job.

**Example: Return Data from a Microsoft SQL Server Database Table that Matches a Condition**

Suppose that you want a job to query the sales table for orders that have a quantity greater than 20. The order numbers that match the query appear in the output file ordnum.txt.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- delrow
  - Send to machine -- DB\_AGENT
  - Owner -- sa
  - SQL command -- 'SELECT ORD\_NUM FROM SALES WHERE QTY > 20'
  - Destination file -- C:\sales\ordnum.txt
  - Success criteria -- ORD\_NUM=A2976
  - Database connect string -- "jdbc:sqlserver://myhost:1433;DatabaseName=docs"
3. Commit the job.

The ordnum.txt file contains the following order numbers:

```
A2976
QA7442.3
P2121
N914014
P3087a
P3087a
X999
P723
QA879.1
```

Because the query returns an order number that matches the job criteria A2976, the job completes.

Suppose we change the job criteria statement in the above example to the following:

```
JOB_CRITERIA .*B[0-9]
```

In this case, the query would still return the same order numbers, but the job would fail because it would not find a matching order number containing the letter B and followed by a number.

**SQL Job Examples for IBM DB2**

The following examples describe sample SQL jobs that run SQL commands against IBM DB2 database tables.

**Example: Add a Row to an IBM DB2 Database Table**

Suppose that you want a job to add a row of data to the STAFF table under the user entadm.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- addrow
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - SQL command -- 'INSERT into ENTADM.STAFF(ID, NAME, DEPT, JOB, YEARS, SALARY, COMM) VALUES(556, "Jonson", 84, "Sales", 1, 40500.50, 100)'
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
3. Commit the job.

**Example: Update a Row in an IBM DB2 Database Table**

Suppose that you want a job to update a record in the STAFF table under the user entadm. The job changes the years to 3 for the employee with the name Jonson.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- update
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - SQL command -- 'UPDATE ENTADM.STAFF SET YEARS=3 where NAME="Jonson"'
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
3. Commit the job.

**Example: Delete a Row from an IBM DB2 Database Table**

Suppose that you want a job to delete a row from the STAFF table under the user entadm for the employee with the name Jonson.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:
  - Name -- delrow
  - Send to machine -- DB\_AGENT
  - Owner -- entadm
  - SQL command -- 'DELETE FROM ENTADM.STAFF where NAME="Jonson"'
  - Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"
3. Commit the job.

**Example: Return Data from an IBM DB2 Database Table that Matches a Condition**

Suppose that you want a job to query the STAFF table under the user entadm for employees that have salaries greater than 40,000. If the query returns an employee name that begins with the letter J, the job completes.

**Follow these steps:**

1. Create an SQL job.
2. Enter the following properties:

- Name -- data
- Send to machine -- DB\_AGENT
- Owner -- entadm
- SQL command -- 'SELECT NAME FROM ENTADM.STAFF where SALARY > 40000'
- Destination file -- /staff/salary.txt
- Success criteria -- NAME=J.\*
- Database connect string -- "jdbc:db2://10.1.4.131:50000/SAMPLE"

### 3. Commit the job.

For example, the salary.txt file contains the following output:

```
Output for: SELECT NAME FROM ENTADM.STAFF where SALARY > 40000
```

```
NAME

Jonson
```

## **SQL Properties**

The SQL category includes the following properties:

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Database connect string**

(Optional) Specifies a JDBC database resource location (URL) for a job. Use the appropriate format for your database type, as follows: For an Oracle database: "jdbc:oracle:thin:@host:port:database\_name" For a Microsoft SQL Server database: "jdbc:sqlserver://host:port;DatabaseName=database\_name" For an IBM DB2 database: "jdbc:db2://host:port/database\_name" For a Sybase database: connect\_string: "jdbc:sybase:Tds:host:port / dbname"

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** connect\_string

### **Destination file**

(Optional) Specifies the output destination file that stores the SQL query results.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

### **SQL command**

Specifies an SQL command to run against database tables.

This property supports Global Variable substitution.

**Limits:** Up to 1000 characters

**NOTE**

The value can contain any SQL command including SELECT, DELETE, UPDATE, or INSERT.

**JIL attribute:** sql\_command

**Success criteria**

(Optional) Defines a regular expression that is used to evaluate a return string. If the return string matches the regular expression, the job completes successfully. Otherwise, the job fails.

This property supports Global Variable substitution.

**Limits:** Up to 500 characters; case-sensitive

**NOTE**

Each return string includes the field name from the SELECT statement and its value, separated by an equal sign (=). For example, consider the query SELECT ORD\_NUM FROM SALES. To match order number A2976, specify the regular expression ORD\_NUM=A2976. Specifying the regular expression A2976 does not match any return string causing the job to fail.

**JIL attribute:** success\_criteria

**Oracle user role**

(Optional) Specifies the role of the Oracle user to log in as. The user role must be defined in the Oracle database. The user role uses the following syntax:

*as userrole*

**Limits:** Up to 30 characters; case-sensitive

**Example:** as sysdba

**JIL attribute:** user\_role

## Manage Monitoring and File Trigger Jobs Using Web UI

This article explains how to manage monitoring and file trigger jobs.

### Monitoring Jobs

Monitoring jobs let you monitor different aspects of your system.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

You can define the following monitoring jobs:

- **CPU Monitoring**  
Lets you monitor CPU usage.
- **Disk Monitoring**  
Lets you monitor disk space.
- **IP Monitoring**  
Lets you monitor an IP address.
- **Process Monitoring**

Lets you monitor process execution.

- **Text File Reading and Monitoring**

Lets you search a text file for a string.

- **Windows Event Log Monitoring**

Lets you monitor a Windows event log.

- **Windows Service Monitoring**

Lets you monitor the status of Windows services.

### **File Trigger Jobs**

File Trigger jobs let you monitor file activity. You can define File Trigger jobs for UNIX, Linux, Windows, or i5/OS systems.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

The File Trigger job can monitor when a file is created, updated, deleted, expanded, or shrunk, and when a file exists or does not exist.

#### **NOTE**

Difference between File Watcher and File Trigger Jobs on Windows

On Windows, File Watcher and File Trigger jobs produce different results when the dot (.) character is used as a literal. The pattern "\*.\*)" is treated differently depending on which job is used:

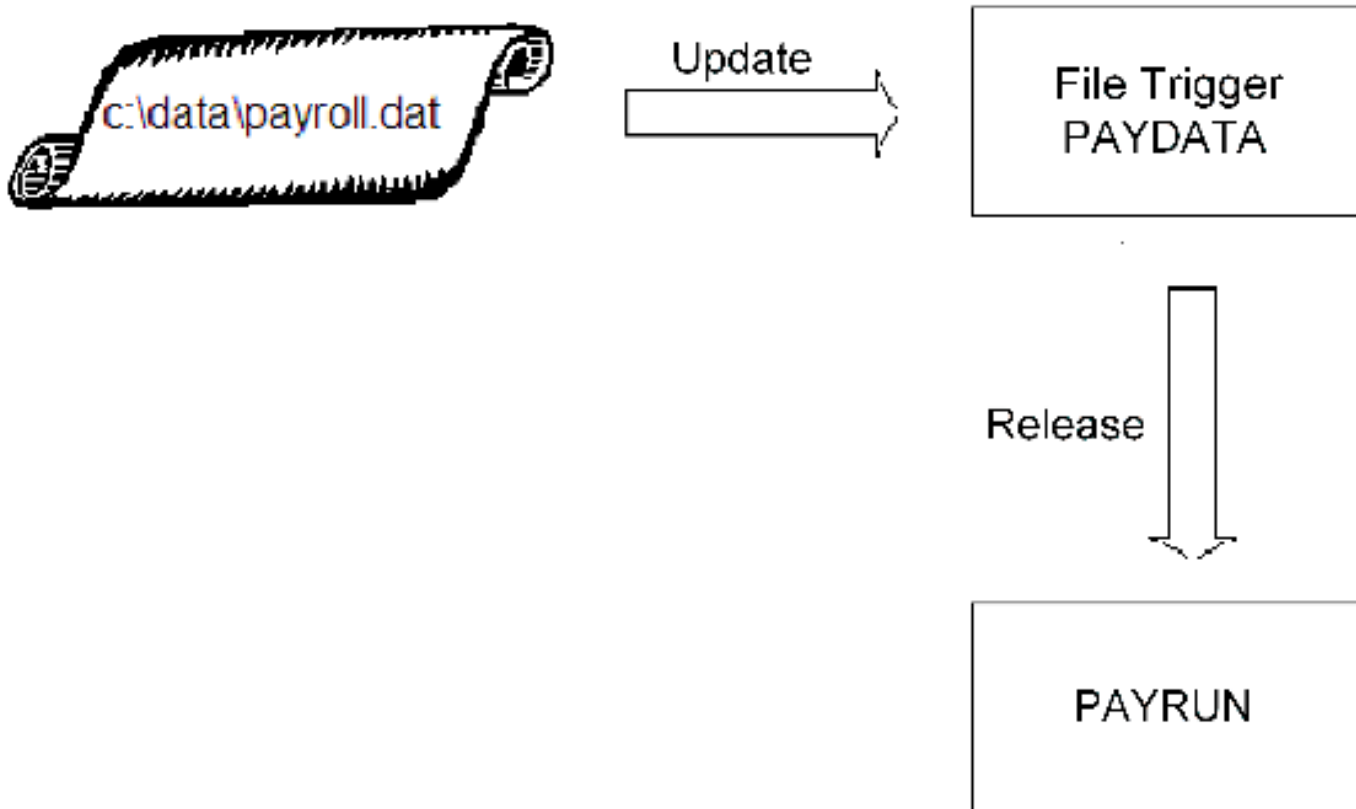
- File Trigger jobs treat a dot character (.) as a literal. To be considered a match, a file must contain a dot and characters on either side of it. For example, if two files exist in the same directory, one named "abc" and the other "abc.txt," only "abc.txt" is considered a match.
- File Watcher jobs use Windows native APIs, which assume the dot character can be implicit. For example, File Watcher jobs consider "abc" to be the same as "abc." So, in this case, either "abc" or "abc.txt" is a match.

### **Example: Monitor for an Update to a File**

Suppose that a File Trigger job named PAYDATA monitors for an update to the payroll.dat file on a Windows computer. When the file is updated, the job completes and the scheduling manager releases a job named PAYRUN.

The following diagram shows the scenario:



**Figure 73: File TiggerConcept**

### **Windows Event Log Monitoring Jobs**

The Windows Event Log Monitoring job only monitors event logs maintained by the operating system and available in the Event Viewer. Windows operating systems record events in at least three types of logs, including the following:

- **Application log**  
The application log contains events logged by applications or programs. For example, a database program might record a file error in the application log.
- **System log**  
The system log contains events logged by the Windows system components. For example, the failure of a driver or other system component to load during startup is recorded in the system log.
- **Security log**  
The security log can record security events (such as valid and invalid logon attempts) and events related to resource use (such as creating, opening, or deleting files).

### **Define a CPU Monitoring Job**

You can define a CPU Monitoring job to monitor the CPU usage of the computer the specified agent is installed on.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

**Follow these steps:**

1. Create a CPU Monitoring (OMCPU) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
3. (Optional) Specify optional [CPU Monitoring properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The CPU Monitoring job is defined.

**CPU Monitoring Job Examples**

The following example describes a sample CPU Monitoring job.

**Example: Monitor CPU Availability Continuously**

In this example, the agent returns information on the CPU capacity that is used within the range specified by the lower and upper boundaries.

**Follow these steps:**

1. Create a CPU Monitoring job.
2. Enter the following properties:
  - Name -- unix\_textfile
  - Send to machine -- AGENT
  - CPU Usage -- USED
  - Monitor mode -- CONTINUOUS
  - Lower boundary (%) -- 75
  - Upper boundary (%) -- 100
3. Commit the job.

**CPU Monitoring Properties**

The CPU Monitoring category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.  
**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)  
**JIL attribute:** job\_class

**CPU usage**

(Defaulted) Specifies whether the job monitors the available or used CPU processing capacity. The following values are available:

- **FREE**

Specifies that the job monitors the available CPU processing capacity.

- **USED**

Specifies that the job monitors the used CPU processing capacity.

**Default:** FREE

**NOTE**

To use this property, you must specify WAIT or CONTINUOUS for the Monitor mode.

**JIL attribute:** cpu\_usage

**Inside range**

(Defaulted) Indicates whether the job triggers when the value of CPU usage is within the range specified by the lower and upper boundary properties.

**Default:** selected (inside the range)

**NOTE**

To use this property, you must specify WAIT or CONTINUOUS for the Monitor mode.

**JIL attribute:** inside\_range

**Lower boundary (%)**

(Optional) Defines the lower boundary of CPU usage in percent.

**Limits:** 0-99

**Default:** 0

**NOTE**

- If you specify a lower boundary without an upper boundary, the monitoring takes place between the lower boundary and 100 percent.
- An inside range is used for the boundary unless you specify outside for the range.
- To use this property, you must specify WAIT or CONTINUOUS for the Monitor mode.

**JIL attribute:** lower\_boundary

**Monitor mode**

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately. The following values are available:

- **CONTINUOUS**

Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the log file.

**NOTE**

- When using the CONTINUOUS value in CPU Monitoring jobs, you must also specify the lower boundary and the upper boundary.
- To end continuous monitoring, you must complete the job manually.

- **NOW**

- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.**

**WAIT**  
Waits for the conditions to occur. When the conditions are met, the job completes.

**NOTE**

When using the WAIT value in CPU Monitoring jobs, you must also specify the lower boundary and the upper boundary.

**Default:** WAIT

**JIL attribute:** monitor\_mode

**No change (%)**

(Optional) Specifies the percentage change in CPU usage that must occur to trigger the job.

**Limits:** 0-100

**JIL attribute:** no\_change

**Poll interval (secs)**

(Optional) Specifies the interval (in seconds) between successive scans for a job that uses the CONTINUOUS Monitor mode.

**NOTE**

The default is set on the agent by the objmon.scaninterval parameter.

**JIL attribute:** poll\_interval

**Upper boundary (%)**

(Optional) Defines the upper boundary of CPU usage in percent.

**Limits:** 1-100

**Default:** 100

**NOTE**

- If you specify an upper boundary without a lower boundary, the monitoring takes place between zero percent and the upper boundary.
- An inside range is used for the boundary unless you specify outside for the range.
- To use this property, you must specify WAIT or CONTINUOUS for the Monitor mode.

**JIL attribute:** upper\_boundary

**Define a Disk Monitoring Job**

On Windows and UNIX systems, you can define a Disk Monitoring job to monitor the available or used space on a disk or logical partition. On i5/OS systems, you can define a Disk Monitoring job to monitor storage space in an auxiliary storage pool (ASP).

**Note:** To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

**Follow these steps:**

1. Create a Disk Monitoring (OMD) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Disk drive**  
Specifies the path to the disk, logical partition, or auxiliary storage pool to be monitored.
3. (Optional) Specify optional [Disk Monitoring properties](#).

4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Disk Monitoring job is defined.

### **Disk Monitoring Job Examples**

The following examples describe sample Disk Monitoring jobs.

#### **Example: Monitor Available Space on a Windows Drive**

In this example, the local C drive is monitored. The space available is monitored and the value is expressed in megabytes.

##### **Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:
  - Name -- cdrive
  - Send to machine -- AGENT
  - Disk drive -- C
  - Disk format -- MB
  - Disk space -- FREE
  - Monitor mode -- NOW
3. Commit the job.

#### **Example: Monitor a Local Partition Continuously**

In this example, a local partition is continuously monitored. The space used is monitored and the value is expressed as a percentage. When the value for disk space used falls between 90 and 100 percent, an alert (DISK) is issued.

##### **Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:
  - Name -- partition
  - Send to machine -- AGENT
  - Disk drive -- /export/home
  - Disk format -- PERCENT
  - Disk space -- USED
  - Inside range -- selected
  - Monitor mode -- CONTINUOUS
  - Lower boundary -- 90
3. Commit the job.

#### **Example: Monitor a Local Drive Continuously**

In this example, the local C drive is continuously monitored. The space used is monitored and the value is expressed as a percentage. When the value of disk space used is less than 16 percent or greater than 95 percent, an alert (DISK) is issued.

##### **Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:

- Name -- local
  - Send to machine -- AGENT
  - Disk drive -- C
  - Disk format -- PERCENT
  - Disk space -- USED
  - Monitor mode -- CONTINUOUS
  - Inside range -- NOT SELECTED
  - Lower boundary -- 16
  - Upper boundary -- 95
3. Commit the job.

### **Example: Monitor the System Auxiliary Storage Pool (ASP)**

In this example, the agent monitors the system ASP. As specified in the DISK statement, the space available is monitored and the value is expressed in megabytes. The upper and lower boundaries are not specified, so the job returns immediately with the current status of the space being monitored.

The system ASP is denoted by the forward slash (/). You can also specify the system ASP as /dev/QASP01.

#### **Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:
  - Name -- ASP
  - Send to machine -- AGENT
  - Disk drive -- /
  - Disk format -- MB
  - Disk space -- FREE
  - Monitor mode -- NOW
3. Commit the job.

### **Example: Monitor an ASP for a Storage Space Value within a Range**

In this example, the agent monitors the auxiliary storage pool (ASP) named QASP02. The space used is monitored and the value is expressed as a percentage. When 90 to 100 percent of the storage space is used, the job completes.

#### **Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:
  - Name -- AUXASP
  - Send to machine -- AGENT
  - Disk drive -- /dev/QASP02
  - Disk format -- PERCENT
  - Disk space -- USED
  - Monitor mode -- WAIT
  - Lower boundary -- 90
  - Upper boundary -- 100
3. Commit the job.

### **Example: Monitor a Mounted File System for a Storage Space Value Outside of a Range**

In this example, the agent monitors a Network File System (NFS) server mounted at /u1. The space used is monitored and the value is expressed as a percentage. When the value of storage space used is less than 16 percent or greater than 95 percent, the job completes.

**Follow these steps:**

1. Create a Disk Monitoring job.
2. Enter the following properties:
  - Name -- NFS
  - Send to machine -- AGENT
  - Disk drive -- /u1
  - Disk format -- PERCENT
  - Disk space -- USED
  - Monitor mode -- WAIT
  - Inside range -- NOT SELECTED
  - Lower boundary -- 16
  - Upper boundary -- 95
3. Commit the job.

**Disk Monitoring Properties**

The Disk Monitoring category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

**Disk drive**

Specifies the path to the disk, logical partition, or auxiliary storage pool to be monitored.

**Limits:** Up to 256 characters; case-sensitive

**Example:** /dev/QASP02

**UNIX/Windows:** Specify the path to the disk or logical partition to monitor.

**i5/OS:**

- Specify the path to a file system mounted on the i5/OS operating system.
- To specify the system ASP, type the forward slash (/).
- To specify any other ASP, use the following format, where *digit1* and *digit2* indicate the ASP number:

/dev/QASP*digit1digit2*

**JIL attribute:** disk\_drive

**Disk format**

(Defaulted) Specifies the unit of measurement used to monitor available or used disk space. The following values are available:

- B -- Monitors disk usage in bytes.
- GB -- Monitors disk usage in gigabytes.
- KB -- Monitors disk usage in kilobytes.
- MB -- Monitors disk usage in megabytes.

PERCENT -- Monitors disk usage by percentage.

**Default:** PERCENT

**JIL attribute:** disk\_format

### Disk space

(Defaulted) Specifies whether the job monitors the available or used disk space. The following values are available:

- **FREE**

Specifies that the job reads the available disk space for monitoring.

- **USED**

Specifies that the job reads the used disk space for monitoring.

**Default:** FREE

**JIL attribute:** disk\_space

### Inside range

(Defaulted) Indicates whether the job completes when the disk space value is within the range specified by the lower and upper boundary properties.

**Default:** selected (inside the range)

**JIL attribute:** inside\_range

### Lower boundary

(Optional) Defines the start of the range to be searched. The Disk format property specifies the units for this value.

**Limits:** Up to 20 characters

**JIL attribute:** lower\_boundary

### Monitor mode

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately. The following values are available:

- **CONTINUOUS**

Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the log file.

**Note:** To end continuous monitoring, you must complete the job manually.

- **NOW**

- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.WAIT**

Waits for the conditions to occur. When the conditions are met, the job completes.

**Default:** WAIT

**JIL attribute:** monitor\_mode

### No change (%)

(Optional) Specifies the percentage value that the disk usage must change to complete the job. If the change in disk usage is within this specified value, the job does not complete.

**JIL attribute:** no\_change

### Poll interval (secs)

(Optional) Specifies the interval (in seconds) between successive scans when the monitor mode is CONTINUOUS.



**NOTE**

This property can be set on the agent using the `objmon.scaninterval` parameter. The default value set on the agent is 10 seconds.

**JIL attribute:** `poll_interval`

**Upper boundary**

(Optional) Defines the end of the range to be searched. The Disk format property specifies the units for this value.

**Limits:** Up to 20 characters

**NOTE**

This value is required when the Monitor mode is set to WAIT or CONTINUOUS.

**JIL attribute:** `upper_boundary`

**Define an IP Monitoring Job**

You can define an IP Monitoring job to monitor an IP address or a port on an IP address.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

**Follow these steps:**

1. Create an IP Monitoring (OMIP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **IP host**  
Specifies the DNS name or IP address.
3. (Optional) Specify optional [IP Monitoring properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The IP Monitoring job is defined.

**IP Monitoring Job Examples**

The following example describes a sample IP Monitoring job.

**Example: Monitor an Agent IP Address**

You can use an IP Monitoring job to find out if an agent is running. In this example, the agent IP address is 172.24.2.20 and the input port is 9401. The following job completes if the agent is running.

**Follow these steps:**

1. Create an IP Monitoring job.
2. Enter the following properties:

- Name -- agent\_running
- Send to machine -- AGENT
- IP host -- 172.24.2.20
- IP port -- 9401
- IP status -- RUNNING

3. Commit the job.

### **IP Monitoring Properties**

The IP Monitoring category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **IP host**

Specifies the DNS name or IP address.

**Limits:** Up to 100 characters; case-sensitive

**JIL attribute:** ip\_host

#### **IP port**

(Optional) Specifies the port number for the IP address being monitored. The agent attempts to connect to this port.

**Limits:** 0-65535

#### **NOTE**

If you specify 0 or no value for the port, the agent uses ping for monitoring.

**JIL attribute:** ip\_port

#### **IP status**

(Defaulted) Specifies the status of the IP address to monitor, as follows:

- **RUNNING** Specifies that the job monitors the IP address for a running status.
- **STOPPED** Specifies that the job monitors the IP address for a stopped status.

**Default:** STOPPED

**JIL attribute:** ip\_status

#### **Monitor mode**

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.

- **NOW**
- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.** **WAIT**  
Waits for the conditions to occur. When the conditions are met, the job completes.

**Default:** NOW

**JIL attribute:** monitor\_mode

## Define a Process Monitoring Job

You can define a Process Monitoring job to monitor process execution on the computer where the agent is installed.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Create a Process Monitoring (OMP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Process name**  
Specifies the name of the process to be monitored.
3. (Optional) Specify optional [Process Monitoring properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Process Monitoring job is defined.

## Process Monitoring Job Examples

The following examples describe sample Process Monitoring jobs.

### Example: Monitor the cybAgent.exe Process on Windows

In this example, the agent monitors the cybAgent.exe process. If the cybAgent.exe process is running, the job completes successfully.

### Follow these steps:

1. Create a Process Monitoring job.
2. Enter the following properties:
  - Name -- cybAgent\_Win
  - Send to machine -- AGENT
  - Process name -- 'c:\Program Files\CA\WA System Agent\cybAgent.exe'
  - Process status -- RUNNING
3. Commit the job.

### Example: Monitor the cybAgent Process on UNIX

In this example, the agent monitors the cybAgent process. If the cybAgent process is running, the job completes successfully.

### NOTE

Alternatively, you can also specify a PID number in place of the process name.

### Follow these steps:

1. Create a Process Monitoring job.
2. Enter the following properties:

- Name -- cybAgent\_UNIX
- Send to machine -- AGENT
- Process name -- cybAgent
- Process status -- RUNNING

3. Commit the job.

### **Example: Monitor a Process that Has Any Job Number and Is Running Under Any User Name**

In this example, the agent monitors for any process named CYBAGENT, regardless of the first part (job number) and the second part (user name) of the process ID. The job completes successfully if at least one process named CYBAGENT is running.

#### **Follow these steps:**

1. Create a Process Monitoring job.
2. Enter the following properties:
  - Name -- CYBAGENT
  - Send to machine -- i5AGENT
  - Process name -- '\*ALL/\*ALL/CYBAGENT'
  - Process status -- RUNNING
3. Commit the job.

### **Example: Monitor Multiple Processes that Have Similar Names**

In this example, the agent monitors all processes running under the JDOE user profile with names that start with CALC. When all of these processes stop running, the job completes successfully.

#### **Follow these steps:**

1. Create a Process Monitoring job.
2. Enter the following properties:
  - Name -- CALC
  - Send to machine -- i5AGENT
  - Process name -- '\*ALL/JDOE/CALC'
  - Process status -- STOPPED
3. Commit the job.

## **Process Monitoring Properties**

The Process Monitoring category includes the following properties:

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Monitor mode**

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.

- **NOW**
- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.WAIT**

Waits for the conditions to occur. When the conditions are met, the job completes.

**Default:** NOW

**JIL attribute:** monitor\_mode

### Process name

Specifies the name of the process to be monitored.

**Limits:** Up to 256 characters; case-sensitive

**UNIX:** Specify a PID or process name.

**Windows:** Specify a full path or process name.

**JIL attribute:** process\_name

### Process status

(Defaulted) Specifies the status of the process to be monitored, as follows:

- **RUNNING** Specifies that the job monitors the process for a running status.
- **STOPPED** Specifies that the job monitors the process for a stopped status.

**Default:** STOPPED

**JIL attribute:** process\_status

## Define a Text File Reading and Monitoring Job

You can define a Text File Reading and Monitoring job to search a text file on a Windows or UNIX computer for a text string. For example, you can monitor a log file for an error message after a script executes.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

Create a Text File Reading and Monitoring (OMTF) job.

1. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Text file name**  
Specifies the path to and name of the text file to search.
  - **Text file filter**  
Defines the text string to search for. You can specify the text string as a regular expression.
  - **Monitor mode** Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.
  - **Text file mode**  
Specifies the search mode when monitoring a text file.
2. (Optional) Specify optional [Text File Reading and Monitoring properties](#).
3. (Optional) Specify [common properties that apply to all job types](#).
4. Commit the job.

The Text File Reading and Monitoring job is defined.

### **Text File Reading and Monitoring Job Examples**

The following examples describe sample Text File Reading and Monitoring jobs.

#### **Example: Search for a Text String on a Windows System**

In this example, the job searches the text file component.txt, between lines 1 and 20, for the text string "CreateService failed". When the text string is found, the job completes.

##### **Follow these steps:**

1. Create a Text File Reading and Monitoring job.
2. Enter the following properties:
  - Name -- component
  - Send to machine -- AGENT
  - Text file name -- 'c:\Program Files\CA\component.txt'
  - Text file filter -- 'CreateService failed'
  - Text file filter exists -- selected
  - Monitor mode -- NOW
  - Text file mode -- LINE
  - Lower boundary -- 1
  - Upper boundary -- 20
3. Commit the job.

#### **Example: Search for a Text String on an i5/OS System**

In this example, the job searches the DATA member, between lines 1 and 20, for the text string "Create file failed". When the text string is found, the job completes.

##### **Follow these steps:**

1. Create a Text File Reading and Monitoring job.
2. Enter the following properties:
  - Name -- DATA
  - Send to machine -- I5AGENT
  - Text file name -- /QSYS.LIB/LIBRARY.LIB/RESULTS.FILE/DATA.MBR
  - Text file filter -- 'Create file failed'
  - Text file filter exists -- selected
  - Monitor mode -- NOW
  - Text file mode -- Line
  - Lower boundary -- 1
  - Upper boundary -- 20
3. Commit the job.

#### **Example: Search for a Regular Expression**

In this example, the text string contains regular expression pattern matching syntax. The search range is also a regular expression as indicated by the operand REGEX in the SEARCHRANGE statement.

##### **Follow these steps:**

1. Create a Text File Reading and Monitoring job.
2. Enter the following properties:

- Name -- regex
  - Send to machine -- AGENT
  - Text file name -- '/home/cybesp/file.txt'
  - Text file filter -- '^\\w{4,10}\\.'
  - Text file filter exists -- selected
  - Monitor mode -- NOW
  - Text file mode -- REGEX
  - Lower boundary -- (\\A\\W\\sE)
3. Commit the job.

The regular expression can be interpreted as follows:

- ^ or \\A -- match only at the beginning of string (line)
- \\Z or \$ -- match only at the end of string
- \\w -- a word character [a-zA-Z0-9]
- \\W -- a non-word character
- \\s -- a whitespace character
- {4,10} -- match at least 4 times but not more than 10 times

To illustrate the last item {4, 10}, consider the following syntax:

```
TEXTSTRING 'b1{1,3}c'
```

Evaluating this expression yields the following conditions:

- The line contains the text b1.
- Numeric 1 should exist at least once, but not more than three times.
- The specified text string must be followed by the letter c.

### **Text File Reading and Monitoring Properties**

The Text File Reading and Monitoring category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Encoding**

(Defaulted) Specifies the name of the character set used to encode the data in the file.

**Default:** US-ASCII

**Limits:** Up to 42 characters

**JIL attribute:** encoding

#### **Lower boundary**

(Optional) Defines the start of the range to be searched. The format of this value depends on the text file search mode. The formatting options are as follows: Numeric -- Specify a numeric value if the search mode is LINE.Regular expression -- Specify a regular expression if the search mode is REGEX.Date and time -- Specify date and time values if the search mode is DATETIME. Define the date and time using the format specified by the time format.

**Limits:** Up to 256 characters; case-sensitive. When the search mode is LINE, the upper limit is 9223372036854775807.

**JIL attribute:** lower\_boundary

### Monitor mode

Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately. The options are as follows:

- **CONTINUOUS**

Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the log file.

#### NOTE

- When using the CONTINUOUS value, you can specify a lower boundary but you cannot specify an upper boundary.
- To end continuous monitoring, you must complete the job manually.

- **NOW**

- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.**

**WAIT**  
Waits for the conditions to occur. When the conditions are met, the job completes.

#### NOTE

When using the WAIT value, you can specify a lower boundary but you cannot specify an upper boundary.

**Default:** WAIT

**JIL attribute:** monitor\_mode

### Text file filter

Defines the text string to search for. You can specify the text string as a regular expression.

**Limits:** Up to 1024 characters; case-sensitive

Clicking 

the Edit Text dialog, which lets you enter the text string for which to search.

**JIL attribute:** text\_file\_filter

### Text file filter exists

(Defaulted) Indicates how the job monitors a specified text file for a text string specified in the Text file filter field.

- Selected -- Monitors whether a text string exists in a specified text file. If the text string exists, the job completes successfully or an alert is triggered (if monitoring continuously).
- Not selected -- Monitors whether a text string does not exist in a specified text file. If the text string does not exist, the job completes successfully. If the text string exists, the job fails.

**Default:** selected (monitors for a text string)

**JIL attribute:** text\_file\_filter\_exists

### Text file mode

Specifies the search mode when monitoring a text file.

- **LINE** Searches for a text string between line numbers that define the upper and lower boundaries to search in the text file.
- **REGEX** Searches for a text string between regular expressions that define the lower and upper boundaries to search in the text file.



- **DATETIME** Searches for a text string between a date range specified by the upper and lower boundaries.

**JIL attribute:** text\_file\_mode

#### Text file name

Specifies the path to and name of the text file to search.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** text\_file\_name

#### Time format

(Optional) Defines the date and time pattern to use when the upper and lower boundaries are specified as date and time. The upper and lower boundaries are used to search inside a log file.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- This field is required when Text file mode is DATETIME.
- To specify the time format, construct a time pattern string that is used as a mask for searching the particular time stamp you want to find. In this pattern, all ASCII letters are reserved as pattern letters.
- For a list of symbols you can use to build a time format pattern and examples, refer to the *AutoSys Workload Automation Reference* section for this JIL attribute.

**JIL attribute:** time\_format

#### Time position

(Optional) Specifies the first column of the time stamp in the log file.

**Limits:** 1-99999

#### NOTE

This property is valid only when Text file mode is DATETIME.

**JIL attribute:** time\_position

#### Upper boundary

(Optional) Defines the end of the range to be searched. The format of this value depends on the text file search mode. The formatting options are as follows: Numeric -- Specify a numeric value if the search mode is LINE. Regular expression -- Specify a regular expression if the search mode is REGEX. Date and time -- Specify date and time values if the search mode is DATETIME. Define the date and time using the format specified by the time format.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

You cannot specify the TO operand when using the WAITMODE statement with the WAIT value.

#### NOTE

The upper boundary can only be set when the Monitor mode is NOW.

**JIL attribute:** upper\_boundary

### **Define a Windows Event Log Monitoring Job**

You can define a Windows Event Log Monitoring job to monitor a Windows event log on a local computer. The monitor returns the most recent event available or continuously monitors for events in a particular Windows event log.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for Windows.

**Follow these steps:**

1. Create a Windows Event Log Monitoring (OMEL) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Log name** Specifies the name of the event log.
3. (Optional) Specify optional [Windows Event Log Monitoring properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Windows Event Log Monitoring job is defined.

**Windows Event Log Monitoring Job Examples**

The following examples describe sample Windows Event Log Monitoring jobs.

**Example: Monitor an Application Log Continuously**

In this example, the event log for applications is monitored continuously and the agent issues an alert ELOG for all instances of the Information event type where the event source is LLDSAPNT223, the event description contains the word 'started', and the event ID is less than or equal to 4000.

**Follow these steps:**

1. Create a Windows Event Log Monitoring job.
2. Enter the following properties:
  - Name -- info
  - Send to machine -- AGENT
  - Log name -- ELOG
  - Type -- INFO
  - Event ID -- Less than or equal to (<=) 4000
  - Monitor mode -- CONTINUOUS
  - Description -- started
  - Source -- LLDSAPNT223
3. Commit the job.

**Example: Monitor an Application Log that Occurs on or after a Specified Date**

The following job monitors an application log that occurs any time on or after January 11, 2009, 6:30 a.m. When the job finds an application log that occurs any time on or after that date and time, the job completes successfully.

**Follow these steps:**

1. Create a Windows Event Log Monitoring job.
2. Enter the following properties:

- Name -- jan11
- Send to machine -- AGENT
- Log name -- Application
- Type -- INFO
- Start date/time -- 20090111 06:30:00
- Source -- LLDSAPNT223

3. Commit the job.

### **Windows Event Log Monitoring Properties**

The Windows Event Log Monitoring category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Category**

(Optional) Specifies the event category as displayed in the Windows Event Viewer.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** win\_event\_category

#### **Computer**

(Optional) Specifies the local computer name where the event log is monitored.

**Limits:** Up to 80 characters; case-sensitive

**JIL attribute:** win\_event\_computer

#### **Description**

(Optional) Specifies the event description as displayed in the Windows Event Viewer.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** win\_event\_description

#### **Event ID**

(Optional) Specifies both an operator to compare against the Windows Event ID to monitor and a value for that ID. The following operators are available:

- Greater than (>)
- Less than (<)
- Less than or equal to (<=)
- Equal to (=)
- Greater than or equal to (>=)

**Default** (for operator): Equal to (=)

**Limits** (for Windows Event ID): Up to 10 digits

**Example:** >4000

**JIL attributes:** win\_event\_op, win\_event\_id

### Log name

Specifies the name of the event log.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** win\_log\_name

### Monitor mode

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.

- **CONTINUOUS**

Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the log file.

**Note:** To end continuous monitoring, you must complete the job manually.

- **NOW**

- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.**

**WAIT**  
Waits for the conditions to occur. When the conditions are met, the job completes.

**Default:** WAIT

**JIL attribute:** monitor\_mode

### Source

(Optional) Specifies the event source as displayed in the Windows Event Viewer.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** win\_event\_source

### Start date/time

(Optional) Specifies the date and time of the Windows event log. This property uses the following format:

*yyyymmdd hh:mm:ss*

**Limits:** Up to 20 characters

**Example:** 20090501 06:30:00

**JIL attribute:** win\_event\_datetime

### Type

(Defaulted) Specifies the event type to monitor in the Windows event log, as follows:

- **AUDITF** Specifies the Failure Audit event type.
- **AUDITSS** Specifies the Success Audit event type.
- **ERROR** Specifies the Error event type.
- **INFO** Specifies the Information event type.
- **WARN** Specifies the Warning event type.

**Default:** ERROR

**JIL attribute:** win\_event\_type

## **Define a Windows Service Monitoring Job**

You can define a Windows Service Monitoring job to monitor services on a local Windows computer.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for Windows.

**Follow these steps:**

1. Create a Windows Service Monitoring (OMS) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Service name**  
Specifies the name of the local Windows service to be monitored.
3. (Optional) Specify optional [Windows Service Monitoring properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Windows Service Monitoring job is defined.

**Windows Service Monitoring Job Examples**

The following example describes a Windows Service Monitoring job.

**Example: Monitor a Service for a Status of RUNNING**

In this example, the job monitors the service Event Log for a status of RUNNING. The job waits until this status occurs before it completes.

**Follow these steps:**

1. Create a Windows Service Monitoring job.
2. Enter the following properties:
  - Name -- running
  - Monitor mode -- WAIT
  - Send to machine -- AGENT
  - Service name -- Event Log
  - Service status -- RUNNING
3. Commit the job.

**Windows Service Monitoring Properties**

The Windows Service Monitoring category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

**Monitor mode**

(Defaulted) Specifies whether the job waits until the monitor conditions are met or tries to verify them immediately. The following values are available:

- **NOW**
- **Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails.****WAIT**  
Waits for the conditions to occur. When the conditions are met, the job completes.

**Default:** NOW

**JIL attribute:** monitor\_mode

### Service name

Specifies the name of the local Windows service to be monitored.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

You can specify the service name only or the full path to the service executable.

**JIL attribute:** win\_service\_name

### Service status

(Defaulted) Specifies the status of the Windows service to be monitored, as follows:

- **CONTINUE PENDING** Specifies that the job monitors the Windows service for a continue pending status.
- **EXISTSS** Specifies that the job checks whether the Windows service exists.
- **NOT EXISTSS** Specifies that the job checks whether the Windows service does not exist.
- **PAUSED** Specifies that the job monitors the Windows service for a paused status.
- **PAUSE PENDING** Specifies that the job monitors the Windows service for a pause pending status.
- **RUNNING** Specifies that the job monitors the Windows service for a running status.
- **START PENDING** Specifies that the job monitors the Windows service for a start pending status.
- **STOPPED** Specifies that the job monitors the Windows service for a stopped status.
- **STOP PENDING** Specifies that the job monitors the Windows service for a stop pending status.

**Default:** RUNNING

**JIL attribute:** win\_service\_status

## Define a File Trigger Job

You can define a File Trigger job to monitor when a file is created, updated, deleted, expanded, or shrunk, and when a file exists or does not exist. By default, File Trigger jobs monitor for the existence of a file.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### Follow these steps:

1. Create a File Trigger (FT) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

- **Send to machine**  
Specifies the name of the machine on which the job runs.
- **File(s) to watch**  
Specifies the path to and name of one or more files to monitor.

**NOTE**

To specify a variable in the path or file name, your administrator must configure the agent to run the job as an external process. For more information about this configuration, see the AutoSys Workload Automation *documentation*, or contact the AutoSys Workload Automation administrator.

3. (Optional) Specify optional [File Trigger properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The File Trigger job is defined.

### **File Trigger Job Examples**

The following examples describe sample File Trigger jobs that monitor file activity.

#### **Example: Monitor Daily Update to Payroll File on a Windows Computer**

Suppose that you want a job to monitor a daily update to a payroll.dat file on a Windows computer. When the file is updated, the job completes.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- PAYDATA
  - Send to machine -- NT\_NY
  - Application -- PAYROLL
  - File(s) to watch -- c:\data\payroll.dat
  - Trigger type -- Update
3. Commit the job.

#### **Example: Monitor Daily Update to Payroll File on a UNIX Computer**

Suppose that you want a job to monitor a daily update to a payroll file on a UNIX computer. When the file is updated, the job completes.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- PAYDATA
  - Send to machine -- UNIX\_NY
  - Application -- PAYROLL
  - File(s) to watch -- /usr/data/payroll
  - Trigger type -- Update
3. Commit the job.

#### **Example: Monitor for the Creation of a File that Is Owned by a Specified UNIX User ID**

Suppose that you want a job to monitor for the creation of the /data/payroll.dat file that is owned by the UNIX user ID JDOE:

- If the file does not exist when the job is readied, the job does not complete until the file is created.
- If the file exists and the owner of the file is JDOE when the job is readied, the job completes immediately.
- If the file exists or is created, but the owner of the file is not JDOE, the job does not complete. It waits until all specified criteria are satisfied, including the USER criteria. If the owner of the file changes to JDOE, the job completes.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- PAYDATA
  - Send to machine -- SYSAGENT
  - Application -- PAYROLL
  - File(s) to watch -- /data/payroll.dat
  - File owner -- JDOE
  - Trigger type -- Create
3. Commit the job.

**Example: Monitor for the Existence of a File that Is Owned by a Specified UNIX Group**

Suppose that you want a job to monitor for the existence of the /data/payroll.dat file that is owned by the UNIX group ACCTS:

- If the file exists and the file is owned by the group ACCTS when the job is readied, the job completes immediately.
- If the file exists and the file is not owned by the group ACCTS when the job is readied, the job fails.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- PAYDATA
  - Send to machine -- SYSAGENT
  - Application -- PAYROLL
  - File(s) to watch -- /data/payroll.dat
  - File group -- ACCTS
  - Trigger type -- Exist
3. Commit the job.

**Example: Monitor a Daily Update to a Payroll File Object on an i5/OS System**

Suppose that you want a job to monitor a daily update to a payroll file object on an i5/OS computer. When the file is updated, the job completes.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- JOBDATA
  - Send to machine -- I5AGENT
  - Application -- PAYROLL
  - File(s) to watch -- /QSYS.LIB/LIBRARY.LIB/DEPT.FILE/PAYROLL.MBR
  - Trigger type -- Update
3. Commit the job.

**Example: Use Wildcards to Specify a File Name on the Root File System**



In this example, the job monitors for files that are created in the /home/cybesp/ directory in the root file system. The job completes if a file is created with a file name that matches the following criteria:

- Starts with PID
- Ends with four characters
- Has any extension

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- JOBDATA
  - Send to machine -- I5AGENT
  - Application -- NEWFILE
  - File(s) to watch -- '/home/cybesp/PID????.\*'
  - Trigger type -- Create
3. Commit the job.

**Example: Use a Generic Name to Specify a QSYS File Object**

In this example, the job completes when any file object with a file name that starts with PAY and ends with any characters is created in the QSYS file system. Note that the file name is enclosed in apostrophes so that the text following /\* is not interpreted as a comment.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:
  - Name -- JOBDATA
  - Send to machine -- I5AGENT
  - Application -- NEWFILE
  - File(s) to watch -- 'LIB/PAY/\*FILE'
  - Trigger type -- Create
3. Commit the job.

**Example: Monitor the Size of a QSYS File Object**

In this example, the job completes when the EXPOBJ file object in the QSYS file system increases by 10 bytes or more. Since no member is specified in the file name, the job monitors the entire object size. The entire object size includes the size of the file object itself and the total size of the file object's members.

**NOTE**

- The file name is enclosed in apostrophes so that the text following /\* is not interpreted as a comment.
- To monitor only the total size of the file object's members, you would specify \*ALL as the member name.

**Follow these steps:**

1. Create a File Trigger job.
2. Enter the following properties:

- Name -- JOBDATA
- Send to machine -- I5AGENT
- Application -- NEWFILE
- File(s) to watch -- 'LIB/EXPOBJ/\*FILE'
- Trigger type -- Expand
- Change type -- Size
- Change value -- 10

3. Commit the job.

### **File Trigger Properties**

The File Trigger category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Change type**

(Defaulted) Specifies the type of change to detect when monitoring a file for an increase or decrease in size, as follows:

- **Delta** Specifies that the job monitors for a change in the file size.
- **Percent** Specifies that the job monitors for a change in the file size by percentage.
- **Size** Specifies that the job monitors the file to reach a specified size.

#### **NOTE**

Some file systems will create files with extra space in anticipation of further file expansion requests. Therefore, specifying exact byte counts will not always work.

**Default:** Size

#### **NOTE**

- This field is required when the Trigger type is CREATE, EXPAND or SHRINK.
- The change type works in concert with the change value; if one is specified, the other must be also.

**JIL attribute:** watch\_file\_change\_type

#### **Change value**

(Optional) Defines one of the following:

- The change in file size when the file Change type is Delta or Percent.
- The limit of the file size when the file Change type is Size. If the type of file activity is Expand, the file trigger occurs when the file size is equal to or is greater than the specified size. If the type of file activity is Shrink, the file trigger occurs when the file size is equal to or less than the specified size.

#### **NOTE**

- This field is required when the Trigger type is CREATE, EXPAND or SHRINK.
- This value corresponds to the type of file activity.

**JIL attribute:** watch\_file\_change\_value

**Continuous**

(Defaulted) Indicates whether the job should wait for the condition to be satisfied before it ends.

**Default:** not selected (the job does not wait)

**NOTE**

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

**File(s) to watch**

Specifies the path to and name of one or more files to monitor. To specify multiple files, use the asterisk (\*) and question mark (?) wildcard characters. Use \* for any number of characters and ? for any single character. Do not use the \* and ? in the path.

**Limits:** Up to 255 characters; case-sensitive

**UNIX Notes:**

- You can specify variables exported from the profile script or global variables in the name.
- Wildcard characters are expanded according to the wildcard expansion rules of the Bourne Shell.
- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, \${PATH}). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

**Windows Notes:**

- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, %{PATH}%). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Example:** /tmp/a\*.data

**JIL attribute:** watch\_file

**File group**

(Optional) Specifies the name of the group that owns the file to be monitored.

**Limits:** Up to 80 characters

**JIL attribute:** watch\_file\_groupname

**File owner**

(Optional) Specifies the user ID that owns the file to be monitored.

**Limits:** Up to 80 characters

**JIL attribute:** watch\_file\_owner

**Trigger type**

(Defaulted) Specifies the type of file activity that a File Trigger job monitors for, as follows:

- **Create** Indicates that the file trigger occurs when the file is located and remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is located. A file that exists when the job starts is located during the first scan, which occurs 30 seconds after the job starts. Subsequent scans occur every 30 seconds, so a file that does not exist when the job starts is located a maximum of 30 seconds after it is created.

- **Delete** Indicates that the file trigger occurs when the file is deleted.
- **Exist**  
Completes the job if the file exists when the file trigger is set.
- **Expand** Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirement is met when the file is equal to or larger than the specified size or expands by at least the specified amount, depending on the type of size change that you specify in the job definition.
- **Generate**  
Indicates that the file trigger occurs when the file is created and stops changing.
- **Notexist**  
Completes the job if the file does not exist when the file trigger is set.
- **Shrink** Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirements are met when the file is equal to or smaller than the specified size or shrinks by at least the specified amount.
- **Update** Indicates that the file trigger occurs when the file is updated and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is updated.

**Default:** CREATE

**JIL attribute:** watch\_file\_type

#### **Watch no change (min)**

(Defaulted) Specifies the interval (in minutes) the File Trigger job should check for the existence and size of the watched file.

**Default:** 1

#### **NOTE**

This field is required when the Trigger type is CREATE, EXPAND, SHRINK, or UPDATE.

**JIL attribute:** watch\_no\_change

#### **Watch recursive**

(Optional) Indicates whether a File Trigger job monitors for file activity in the specified directory only or in the specified directory and all of its subdirectories.

**Default:** not selected (the job monitors for file activity in the specified directory only)

**JIL attribute:** watch\_file\_recursive

#### **Windows user**

(Optional) Specifies the user ID that is used for monitoring UNC files (Windows only).

**Limits:** Up to 80 characters

**JIL attribute:** watch\_file\_win\_user

## Manage Oracle E-Business Suite Jobs Using Web UI

Oracle E-Business Suite jobs let you run Oracle E-Business Suite workload.

### NOTE

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for Oracle E-Business Suite.

You can define the following Oracle E-Business Suite jobs:

- **Oracle E-Business Suite Single Request**  
Runs a single program in an Oracle Applications application.
- **Oracle E-Business Suite Copy Single Request Job**  
Copies an existing single request defined on Oracle Applications and runs it under the agent.
- **Oracle E-Business Suite Request Set**  
Runs multiple programs in an Oracle Applications application.

### Define an Oracle E-Business Suite Single Request Job

You can define an Oracle E-Business Suite Single Request job to run a single program in an Oracle E-Business Suite application.

### NOTE

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for Oracle E-Business Suite.

### Follow these steps:

1. Create an Oracle E-Business Suite Single Request (OASG) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Program name**  
Specifies the short name of the Oracle E-Business Suite single request program.
  - **Application name**  
Specifies the short name or display name of the Oracle E-Business Suite application that the request set belongs to.
3. Specify the following properties if your agent administrator has not set up defaults for them in the agentparm.txt file:
  - **Responsibility**  
Specifies an Oracle E-Business Suite responsibility name.
  - **Application user**  
Specifies the Oracle E-Business Suite user name that the job runs under.

### NOTE

Your agent administrator can also set default values for these properties using the oa.default.responsibility and oa.default.user parameters.

4. (Optional) Specify optional [Oracle E-Business Suite Single Request properties](#).
5. (Optional) Specify [common properties that apply to all job types](#).
6. Commit the job.  
The Oracle E-Business Suite Single Request job is defined.

## **Oracle E-Business Suite Single Request Job Examples**

The following example describes a sample Oracle E-Business Suite Single Request job.

### **Example: Run an Oracle E-Business Suite Single Request**

In this example, the Oracle E-Business Suite Single Request job runs the single request XXCOFI on the Oracle E-Business Suite system. The single request belongs to the application with the display name Application Object Library in Oracle E-Business Suite.

#### **Follow these steps:**

1. Create an Oracle E-Business Suite Single Request job.
2. Enter the following properties:
  - Name -- REQ\_XXCOFI
  - Send to machine -- CYB0A
  - Program name -- XXCOFI
  - Application name -- 'Application Object Library'
  - Application name type -- DISPLAY
  - Application user -- SYSADMIN
  - Responsibility -- System Administrator
3. Commit the job.

## **Oracle E-Business Suite Single Request Properties**

The Oracle E-Business Suite Single Request category includes the following properties:

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Application name**

Specifies the short name or display name of the Oracle E-Business Suite application that the request set belongs to.

**Limits:** Up to 256 characters

**JIL attribute:** oracle\_appl\_name

### **Application name type**

(Defaulted) Specifies whether the name of the Oracle E-Business Suite application is the short name or the display name, as follows:

- SHORT -- The short name is defined by Oracle E-Business Suite.
- DISPLAY -- The display name is part of the request definition in Oracle E-Business Suite and is found in the Application field.

**Default:** SHORT

**JIL attribute:** oracle\_appl\_name\_type

### **Application user**

Specifies the Oracle E-Business Suite user name that the job runs under.

**Limits:** Up to 30 characters; case-sensitive

**NOTE**

- You must also specify the Responsibility property if you specify Application user.
- Your agent administrator can set a default value for this property using the oa.default.user parameter.
- If a default user name is not specified in the oa.default.user parameter in the agent's agentparm.txt file, you must specify the oracle\_user attribute in the job definition.

**JIL attribute:** oracle\_user

**Custom property**

(Optional) Specifies the custom property to be passed to the agent in the following format: key=value.

**Limits:** Up to 256 characters for each key and each value; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Separate each additional key/value pair by a comma or enter it on a new line.
- Enclose key/value pairs that contain special characters in double quotation marks.
- Within quoted strings, double quotation marks and backslashes must be prefixed with a backslash as an escape character.
- If you want the agent to quote the resolved value, enter the default value in single quotes, for example, 'SELECT '1' FROM DUAL'.
- The agent does not resolve Field or Segment validation value sets.

**JIL attribute:** oracle\_custom\_property

**Monitor children**

(Optional) Indicates whether the children of the Oracle E-Business Suite programs are monitored. When selected, the agent monitors children. Program children are programs that are released by the parent program.

**Default:** Not selected (The job does not monitor children programs.)

**NOTE**

You must also specify the Monitor children delay property if you specify Monitor children.

**JIL attribute:** oracle\_mon\_children

**Monitor children delay**

(Optional) Defines the number of seconds to wait after an Oracle E-Business Suite program completes before monitoring its children.

**Limits:** 0-99999

**NOTE**

You must also specify the Monitor children property if you specify Monitor children delay.

**JIL attribute:** oracle\_mon\_children\_delay

**Notify users (by display name)**

(Optional) Specifies a list of users to notify using display names when the request completes.

**Limits:** Up to 1024 characters; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Enclose each user name in single quotes.
- To specify multiple users, separate each user name with a comma.

**JIL attribute:** oracle\_notify\_display\_users

**Notify users (by short name)**

(Optional) Specifies a list of users to notify using short names when the request completes.

**Limits:** Up to 1024 characters; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Enclose each user name in single quotes.
- To specify multiple users, separate each user name with a comma.

**JIL attribute:** oracle\_notify\_users

**Operating unit**

(Optional) Specifies the operating unit name for a single request or request set.

**NOTE**

This field is valid only for CA Workload Automation AE Release 11.3.6 SP5 and later servers.

**JIL attribute:** oracle\_oprunit

**Output format**

(Optional) Specifies the output format for a single request or request set. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example: PDF****NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- You must also specify the Template language property if you specify Output format.
- Your agent administrator can specify a default description for all Oracle E-Business Suite jobs by setting the oa.default.outputFormat parameter in the agent's agentparm.txt file.
- The Output format value overrides the default output format specified in the agent's agentparm.txt.

**JIL attribute:** oracle\_output\_format

**Print copies**

(Optional) Specifies the number of copies to print. In Oracle E-Business Suite, the number of copies is specified as a request definition option and is found in the Copies column of the Upon Completion dialog.

**Limits:** 0-9999999

**JIL attribute:** oracle\_print\_copies

**Print style**

(Optional) Specifies a print style. The value must match the name of an Oracle E-Business Suite print style. In Oracle E-Business Suite, the print style is specified as a request definition option and is found in the Style field of the Upon Completion dialog.



**Limits:** Up to 30 characters

**Example:** PORTRAIT

**JIL attribute:** oracle\_print\_style

### Printer

(Optional) Specifies the name of a printer that Oracle E-Business Suite can print to. This printer must be defined in Oracle E-Business Suite. In Oracle E-Business Suite, the printer name is specified as a request definition option and is found in the Printer column of the Upon Completion dialog.

**Limits:** Up to 30 characters

**Example:** "\\printer path\K6700"

### NOTE

Enclose paths that contain delimiters (such as spaces) in quotation marks.

**JIL attribute:** oracle\_printer

### Program arguments

(Optional) Defines each argument value to pass to an Oracle E-Business Suite single request. In Oracle E-Business Suite, arguments are part of the program definition and are found in the Concurrent Program Parameters dialog.

**Limits:** Up to 500 characters; up to 100 arguments

### Notes:

- Separate each argument with a comma. Do not add a space after a comma unless the argument value starts with a space. If a value starts with a space, enclose the value in apostrophes.
- Clicking



opens the Edit Text dialog, which lets you enter the argument to pass.

**JIL attribute:** oracle\_args

### Program name

Specifies the short name of the Oracle E-Business Suite single request program. The program short name must be registered in the Oracle E-Business Suite Concurrent Manager. In Oracle E-Business Suite, the program short name is part of the program definition and is found in the Short Name field of the Concurrent Programs dialog.

**Limits:** Up to 30 characters

**JIL attribute:** oracle\_program

### Program name type

(Optional) Specifies whether the name of the Oracle E-Business Suite program is the short name or the display name.

**Default:** SHORT

**JIL attribute:** oracle\_program\_name\_type

### Quote resolved expressions

(Optional) Specifies whether to quote resolved expressions in default values.

**Default:** N

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- In an Oracle E-Business Suite Request Set job, you can specify whether to quote resolved expressions in default values in the request set by using the Program data list property. This value overrides the Quote resolved expressions property value for that particular program.

**JIL attribute:** oracle\_quote\_in\_default

**Request description**

(Optional) Defines a description for an Oracle E-Business Suite Single Request job. The description displays in the Name column of the Requests dialog.

**Limits:** Up to 30 characters

**JIL attribute:** oracle\_desc

**Responsibility**

(Optional) Specifies an Oracle E-Business Suite responsibility name.

**Limits:** Up to 80 characters; case-sensitive

**NOTE**

You must also specify the Application user property if you specify Responsibility. Your agent administrator can set a default value for this property using the oa.default.responsibility parameter.

**JIL attribute:** oracle\_resp

**Save output**

(Optional) Indicates whether to save output from the job execution. When Yes is selected, the output from an Oracle E-Business Suite program is saved.

**Default:** No

**JIL attribute:** oracle\_save\_output

**Template language**

(Optional) Specifies the template language. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** en

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the oa.default.templateLanguage parameter in the agent's agentparm.txt file.
- The Template language value overrides the default setting specified in the oa.default.templateLanguage parameter in the agent's agentparm.txt.
- You must also specify the Output format property if you specify Template language.

**JIL attribute:** oracle\_template\_language

**Template name**

(Optional) Specifies the template name for a single request. In Oracle Applications, the template name is specified as a request definition option and is found in the Template Name column of the Upon Completion dialog.

**Limits:** Up to 120 characters; case-sensitive

**Example:** OraTemplate

**NOTE**

This field is valid only for CA Workload Automation AE Release 11.3.6 SP5 and later servers.

**JIL Attribute:** oracle\_template\_name

**Template territory**

(Optional) Specifies the template territory. In Oracle Applications, the template territory is specified as a request definition option and is found in the For Territory column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** en

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the oa.default.templateTerritory parameter in the agent's agentparm.txt file.
- The Template territory value overrides the default setting specified in the oa.default.templateTerritory parameter in the agent's agentparm.txt.

**JIL attribute:** oracle\_template\_territory

**Use argument defaults**

(Optional) Indicates whether the agent uses default values for arguments. When selected, the agent uses default argument values.

**NOTE**

The default arguments are defined in Oracle E-Business Suite.

**JIL attribute:** oracle\_use\_arg\_def

**Define an Oracle E-Business Suite Copy Single Request Job**

You can define an Oracle E-Business Suite Copy Single Request Job to copy an existing single request defined on Oracle E-Business Suite and run it under the agent. When the job runs, it can override values in the original definition with values specified on the agent or in the job definition.

**NOTE**

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for Oracle E-Business Suite.

**Follow these steps:**

1. Create an Oracle E-Business Suite Copy Single Request (OACOPY) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Request ID**  
Specifies the request ID of the single request you want to copy.

3. Specify the following properties if your agent administrator has not set up defaults for them in the agentparm.txt file:

- **Responsibility**  
Specifies an Oracle E-Business Suite responsibility name.
- **Application user**  
Specifies the Oracle E-Business Suite user name that the job runs under.

**NOTE**

Your agent administrator can also set default values for these properties using the oa.default.responsibility and oa.default.user parameters.

- 4. (Optional) Specify optional [Oracle E-Business Suite Copy Single Request properties](#).
- 5. (Optional) Specify [common properties that apply to all job types](#).
- 6. Commit the job.  
The Oracle E-Business Suite Copy Single Request job is defined.

### **Oracle E-Business Suite Copy Single Request Job Examples**

The following example describes a sample Oracle E-Business Suite Copy Single Request job.

#### **Example: Copy a Single Request**

Suppose that you want to copy the Oracle E-Business Suite single request with request ID 2255470 and override the Oracle E-Business Suite user name and responsibility name. You also want to monitor the children of the single request program 60 seconds after the program completes.

#### **Follow these steps:**

- 1. Create an Oracle E-Business Suite Copy Single Request job.
- 2. Enter the following properties:
  - Name -- REQ\_2255470
  - Send to machine -- CYB0A
  - Request ID -- 2255470
  - Application user -- SYSADMIN
  - Responsibility -- System Administrator
  - Monitor children -- selected
  - Monitor children delay -- 60
- 3. Commit the job.

### **Oracle E-Business Suite Copy Single Request Properties**

The Oracle E-Business Suite Copy Single Request category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators

**.Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Application user**

(Optional) Specifies the Oracle E-Business Suite user name that the job runs under.

**Limits:** Up to 30 characters; case-sensitive

**NOTE**

You must also specify the Responsibility property if you specify Application user. Your agent administrator can set a default value for this property using the `oa.default.user` parameter.

**JIL attribute:** `oracle_user`

**Monitor children**

(Optional) Indicates whether the children of the Oracle E-Business Suite programs are monitored. When selected, the agent monitors the children. Program children are programs that are released by the parent program.

**NOTE**

You must also specify the Monitor children delay property if you specify Monitor children.

**JIL attribute:** `oracle_mon_children`

**Monitor children delay**

(Optional) Defines the number of seconds to wait after an Oracle E-Business Suite program completes before monitoring its children.

**Limits:** 0-99999

**NOTE**

You must also specify the Monitor children property if you specify Monitor children delay.

**JIL attribute:** `oracle_mon_children_delay`

**Request ID**

Specifies the request ID of the single request you want to copy. In Oracle Applications, the request ID is found in the Request ID column of the Requests dialog.

**Limits:** Up to 20 numeric digits

**Example:** **493895459709656750**

**JIL attribute:** `request_id`

**Responsibility**

(Optional) Specifies an Oracle E-Business Suite responsibility name.

**Limits:** Up to 80 characters; case-sensitive

**NOTE**

You must also specify the Application user property if you specify Responsibility. Your agent administrator can set a default value for this property using the `oa.default.responsibility` parameter.

**JIL attribute:** `oracle_resp`

**Define an Oracle E-Business Suite Request Set Job**

You can define an Oracle E-Business Suite Request Set job to run multiple programs in an Oracle E-Business Suite application.

**NOTE**

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for Oracle E-Business Suite.

**Follow these steps:**

1. Create an Oracle E-Business Suite Request Set (OASET) job.

2. Specify the following required properties:
    - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
    - **Send to machine**  
Specifies the name of the machine on which the job runs.
    - **Request set**  
Specifies the short name of the Oracle E-Business Suite request set.
    - **Application name**  
Specifies the short name or display name of the Oracle E-Business Suite application that the request set belongs to.
  3. Specify the following properties if your agent administrator has not set up defaults for them in the agentparm.txt file:
    - **Responsibility**  
Specifies an Oracle E-Business Suite responsibility name.
    - **Application user**  
Specifies the Oracle E-Business Suite user name that the job runs under.
- NOTE**  
Your agent administrator can also set default values for these properties using the oa.default.responsibility and oa.default.user parameters.
4. (Optional) Specify optional [Oracle E-Business Suite Request Set properties](#).
  5. (Optional) Specify [common properties that apply to all job types](#).
  6. Commit the job.  
The Oracle E-Business Suite Request Set job is defined.

### **Specify Data for an Individual Program in a Request Set**

When you define an Oracle E-Business Suite Request Set job, you can specify data such as arguments, the number of copies to print, and whether to save the output for an individual program in the request set. This data overrides the arguments and print parameters specified for the entire request set.

#### **Follow these steps:**

1. Define an Oracle E-Business Suite Request Set job.
2. Select Add in the Program data list field, and click Go.
3. Specify the order number of the program in the following field:
  - **Number**  
Specifies a number to identify the program sequence. This number should be a unique integer greater than zero.  
**JIL attribute:** oracle\_programdata: index
4. Specify one or more values in the following fields:
  - **Arguments**  
(Optional) Defines the argument values to override the default values that are defined by the registered Oracle E-Business Suite Concurrent Manager program.

#### **NOTE**

Separate each argument with a comma. Do not add a space after a comma unless the argument value starts with a space. If a value starts with a space, enclose the value in apostrophes.

**Limits:** Up to 500 characters; up to 100 arguments

**JIL attribute:** oracle\_programdata: args

- **Notify users (by display name)**(Optional) Specifies a list of users to notify using display names when the request completes.

**Limits:** Up to 1024 characters; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Enclose each user name in single quotes.
- To specify multiple users, separate each user name with a comma.

**JIL attribute:** oracle\_notify\_display\_users

- **Notify users (by short name)**(Optional) Specifies a list of users to notify using short names when the request completes.

**Limits:** Up to 1024 characters; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Enclose each user name in single quotes.
- To specify multiple users, separate each user name with a comma.

**JIL attribute:** oracle\_notify\_users

- **Operating unit**(Optional) Specifies the operating unit name for a single request or request set.

**NOTE**

This field is valid only for CA Workload Automation AE Release 11.3.6 SP5 and later servers.

**JIL attribute:** oracle\_oprunit

- **Output Format**(Optional) Specifies the output format for a single request or request set. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** OutputFor1

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- You must also specify the Template language property if you specify Output format.
- Your agent administrator can specify a default description for all Oracle E-Business Suite jobs by setting the oa.default.outputFormat parameter in the agent's agentparm.txt file.
- The Output format value overrides the default output format specified in the agent's agentparm.txt.

**JIL attribute:** oracle\_output\_format

- **Printer**  
(Optional) Specifies the name of a printer that Oracle E-Business Suite can print to. This printer must be defined in Oracle E-Business Suite. In Oracle E-Business Suite, the printer name is specified as a request definition option and is found in the Printer column of the Upon Completion dialog.

**NOTE**

Enclose paths that contain delimiters (such as spaces) in quotation marks.

**Limits:** Up to 30 characters

**Example:** "\\printer path\K6700"

**JIL attribute:** oracle\_programdata: printer

- **Print copies**  
(Optional) Specifies the number of copies to print. In Oracle E-Business Suite, the number of copies is specified as a request definition option and is found in the Copies column of the Upon Completion dialog.

**Limits:** 0-999999

**JIL attribute:** oracle\_programdata: print\_copies

– **Print style**

(Optional) Specifies a print style. The value must match the name of an Oracle E-Business Suite print style. In Oracle E-Business Suite, the print style is specified as a request definition option and is found in the Style field of the Upon Completion dialog.

**Limits:** Up to 30 characters

**JIL attribute:** oracle\_programdata: print\_style

– **Quote resolved expressions**

(Optional) Specifies whether to quote resolved expressions in default values.

**JIL attribute:** oracle\_programdata: quote\_in\_default

– **Save output**

(Optional) Indicates whether to save output from the job execution. When Yes is selected, the agent saves output from an Oracle E-Business Suite program.

**JIL attribute:** oracle\_programdata: saveop

– **Template language**

(Optional) Specifies the template language. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** en

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the oa.default.templateLanguage parameter in the agent's agentparm.txt file.
- The Template language value overrides the default setting specified in the oa.default.templateLanguage parameter in the agent's agentparm.txt.
- You must also specify the Output format property if you specify Template language.

**JIL attribute:** oracle\_template\_language

– **Template name**(Optional) Specifies the template name for a single request. In Oracle Applications, the template name is specified as a request definition option and is found in the Template Name column of the Upon Completion dialog.

**Limits:** Up to 120 characters; case-sensitive**Example:** OraTemplate

**NOTE**

This field is valid only for CA Workload Automation AE Release 11.3.6 SP5 and later servers.

**JIL attribute:** oracle\_template\_name

– **Template territory**(Optional) Specifies the template territory. In Oracle Applications, the template territory is specified as a request definition option and is found in the For Territory column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** en

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the oa.default.templateTerritory parameter in the agent's agentparm.txt file.
- The Template territory value overrides the default setting specified in the oa.default.templateTerritory parameter in the agent's agentparm.txt.

**JIL attribute:** oracle\_template\_territory

These properties override the arguments and print parameters specified for the entire request set.



### Example: Specify Values for Individual Programs in a Request Set

Suppose that you want to define an Oracle E-Business Suite Request Set job to run a request set named EXTRACTS on the OAAGENT agent. The request set belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job definition specifies the following default settings for all programs in the request set:

- The number of copies to be printed is 1.
- The print style is LANDSCAPE.
- The printer is Q8.

The first program data list identifies the first program in the request set. Subsequent statements override the default values for this program. The first program uses the arguments T23 and R1 and prints two copies using the Q1 printer in PORTRAIT style.

The second program data list identifies the fifth program in the request set. Subsequent statements override the default values for this program. The fifth program uses arguments R and R1 and prints three copies using the Q2 printer in PORTRAIT style.

The other programs in the request set use the default values.

#### Follow these steps:

1. Create an Oracle E-Business Suite Request Set job.
2. Enter the following properties:
  - Name -- oaset\_resp
  - Send to machine -- oaagent
  - Request set -- EXTRACTS
  - Application name -- Application Object Library
  - Application name type -- DISPLAY
  - Application user -- SYSADMIN
  - Responsibility -- System Administrator
  - Printer -- Q8
  - Print style -- LANDSCAPE
  - Print copies -- 1
3. Add a program data list with the following values:
  - Number -- 1
  - Arguments -- T23,R1
  - Printer -- Q1
  - Print style -- PORTRAIT
  - Print copies -- 2
  - Save output -- selected
4. Add another program data list with the following values:
  - Number -- 5
  - Arguments -- R,R1
  - Printer -- Q2
  - Print style -- PORTRAIT
  - Print copies -- 3
  - Save output -- not selected
5. Commit the job.

### Oracle E-Business Suite Request Set Job Examples

The following example describes a sample Oracle E-Business Suite Request Set job.

## Example: Run a Request Set

This example runs a request set named FNDRSSUB1310. The request set belongs to the application with the short name BIS in Oracle Applications. The job uses the default Oracle E-Business Suite responsibility name and user name defined on the agent.

### Follow these steps:

1. Create an Oracle E-Business Suite Request Set job.
2. Enter the following properties:
  - Name -- oaset\_resp
  - Send to machine -- oaagent
  - Request set -- FNDRSSUB1310
  - Application name -- BIS
  - Application name type -- SHORT
  - Application user -- SYSADMIN
  - Responsibility -- System Administrator
3. Commit the job.

## Oracle E-Business Suite Request Set Properties

The Oracle E-Business Suite Request Set category includes the following properties:

### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### Application name

Specifies the short name or display name of the Oracle E-Business Suite application that the request set belongs to.

**Limits:** Up to 256 characters

**JIL attribute:** oracle\_appl\_name

### Application name type

(Defaulted) Specifies whether the name of the Oracle E-Business Suite application is the short name or the display name, as follows:

- SHORT -- The short name is defined by the Oracle E-Business Suite Concurrent Manager (ACM).
- DISPLAY -- The display name is part of the request definition in Oracle E-Business Suite and is found in the Application field.

**Default:** SHORT

**JIL attribute:** oracle\_appl\_name\_type

### Application user

(Optional) Specifies the Oracle E-Business Suite user name that the job runs under.

**Limits:** Up to 30 characters; case-sensitive

**NOTE**

You must also specify the Responsibility property if you specify Application user. Your agent administrator can set a default value for this property using the `oa.default.user` parameter.

**JIL attribute:** `oracle_user`

**Custom property**

(Optional) Specifies the custom property to be passed to the agent in the following format: `key=value`.

**Limits:** Up to 256 characters for each key and each value; case sensitive

**NOTE**

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- Separate each additional key/value pair by a comma or enter it on a new line.
- Enclose key/value pairs that contain special characters in double quotation marks.
- Within quoted strings, double quotation marks and backslashes must be prefixed with a backslash as an escape character.
- If you want the agent to quote the resolved value, enter the default value in single quotes, for example, `'SELECT '1' FROM DUAL'`.
- The agent does not resolve Field or Segment validation value sets.

**JIL attribute:** `oracle_custom_property`

**Monitor children**

(Defaulted) Indicates whether the children of the Oracle E-Business Suite programs are monitored. When selected, the agent monitors the children. Program children are programs that are released by the parent program.

**Default:** not selected

**NOTE**

You must also specify the Monitor children delay property if you specify Monitor children.

**JIL attribute:** `oracle_mon_children`

**Monitor children delay**

(Optional) Defines the number of seconds to wait after an Oracle E-Business Suite program completes before monitoring its children.

**Limits:** 0-99999

**NOTE**

You must also specify the Monitor children property if you specify Monitor children delay.

**JIL attribute:** `oracle_mon_children_delay`

**Print copies**

(Optional) Specifies the number of copies to print. In Oracle E-Business Suite, the number of copies is specified as a request definition option and is found in the Copies column of the Upon Completion dialog.

**Limits:** 0-999999

**JIL attribute:** `oracle_print_copies`

**Print style**

(Optional) Specifies a print style. The value must match the name of an Oracle E-Business Suite print style. In Oracle E-Business Suite, the print style is specified as a request definition option and is found in the Style field of the Upon Completion dialog.

**Limits:** Up to 30 characters

**Example:** **PORTRAIT**

**JIL attribute:** oracle\_print\_style

### Printer

(Optional) Specifies the name of a printer that Oracle E-Business Suite can print to. This printer must be defined in Oracle E-Business Suite. In Oracle E-Business Suite, the printer name is specified as a request definition option and is found in the Printer column of the Upon Completion dialog.

**Limits:** Up to 30 characters

**Example:** "\\printer path\K6700"

### NOTE

Enclose paths that contain delimiters (such as spaces) in quotation marks.

**JIL attribute:** oracle\_printer

### Program data list

(Optional) Indicates whether to use data for an individual program in an Oracle E-Business Suite request set.

**Default:** Not selected

### NOTE

To add program data, select Add and click Go. A row of data fields appears below the Program data list field.

**JIL attribute:** oracle\_programdata

### Request set

Specifies the short name of the Oracle E-Business Suite request set. In Oracle E-Business Suite, the request set short name is part of the request set definition and is found in the Set Code field of the Request Set dialog.

### NOTE

- For CA Workload Automation AE r11.3, use short names.
- For CA Workload Automation AE Release 11.3.5 and later, specify the name format in the Request set type field.

**Limits:** Up to 30 characters

**JIL attribute:** oracle\_req\_set

### Request set type

(Optional) Specifies whether the name of the Oracle E-Business Suite request set is the short name or the display name.

**Default:** SHORT

### NOTE

- This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.
- You must also specify the Request set property if you specify Request set type.

**JIL attribute:** oracle\_req\_set\_type

### Responsibility

(Optional) Specifies an Oracle E-Business Suite responsibility name.

**Limits:** Up to 80 characters; case-sensitive

**NOTE**

You must also specify the Application user property if you specify Responsibility. Your agent administrator can set a default value for this property using the `oa.default.responsibility` parameter.

**JIL attribute:** `oracle_resp`

**Save output**

(Defaulted) Indicates whether to save output from the job execution.

**Default:** Not selected

**JIL attribute:** `oracle_save_output`

**Use argument defaults**

(Defaulted) Indicates whether the agent uses default values for arguments. When selected, the agent uses default argument values.

**Default:** not selected

**NOTE**

The default arguments are defined in Oracle E-Business Suite.

**JIL attribute:** `oracle_use_arg_def`

**Use set defaults first**

(Optional) Specifies whether to use default values for arguments that are not defined using the Program arguments property or the Program data list property. The default arguments are defined in Oracle E-Business Suite.

**NOTE**

This field is valid only for CA Workload Automation AE Release 11.3.5 and later servers.

**Default:** N

**JIL attribute:** `oracle_use_set_defaults_first`

**NOTE**

Enclose paths that contain delimiters (such as spaces) in quotation marks.

## Manage SAP Jobs Using Web UI

This article explains how to manage SAP jobs, which let you run SAP workload.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

You can define the following SAP jobs:

- **SAP Batch Input Session**  
Imports data from external systems to the SAP system.
- **SAP Business Warehouse (BW) InfoPackage**  
Transfers data from a data source to an SAP Business Warehouse system.
- **SAP Business Warehouse (BW) Process Chain**

Creates Process Chains on the SAP system.

- **SAP Data Archiving**  
Stores information in an SAP Archiving Object.
- **SAP Event Monitor**  
Monitors and triggers SAP events.
- **SAP Job Copy**  
Copies an existing SAP R/3 job.
- **SAP Process Monitor**  
Monitors for a specific SAP process status.
- **SAP R/3**  
Schedules an SAP R/3 job on your SAP system.

### **Define an SAP Batch Input Session Job**

You can define an SAP Batch Input Session job to import large amounts of data from external systems to the SAP system. You create Batch Input Session jobs on the SAP system. This procedure lets you create a job from scratch; however, you can use an SAP filter and copy an existing job from the SAP system to save time.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

#### **Follow these steps:**

1. Create an SAP Batch Input Session (SAPBDC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **SAP job name**  
Specifies the name of the ABAP job that creates the BDC session on the SAP system.
3. (Optional) Specify optional [SAP Batch Input Session properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP Batch Input Session job is defined.

### **SAP Batch Input Session Job Examples**

The following example describes a sample SAP Batch Input Session job.

#### **Example: Define an SAP Batch Input Session Job**

This example runs the ZBDCTEST ABAP that creates a Batch Input Session (BDC ABAP) on the SAP system. The job runs as soon as possible after the job is defined. After this job runs, the BDC job starts the data transfer.

#### **Follow these steps:**

1. Create an SAP Batch Input Session job.
2. Enter the following properties:

- Name -- bdcjob
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - SAP job name -- ZBDCTEST
  - Release option -- As soon as possible
3. Click  to open the Step Parameters dialog.
  4. Enter the following property on the Command tab:
    - ABAP Name -- ZBDCTEST
  5. Commit the job.

### **SAP Batch Input Session Properties**

The SAP Batch Input Session category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Error rate (%)**

(Optional) Specifies the maximum acceptable error rate as a percentage of the total transactions. When set to 100, all errors are acceptable. When set to zero (0), the job fails if any transaction contains an error. When set to 5, for example, the job fails if more than five percent of the transactions contain errors.

**Limits:** 0-100

**Default:** 0

**JIL attribute:** bdc\_err\_rate

#### **Extended logging**

(Defaulted) Indicates whether to generate advanced logging of the Batch Input Session (BDC) running on the SAP system. When selected, the agent uses extended logging.

**Default:** not selected

**JIL attribute:** bdc\_ext\_log

#### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

#### **Logon Parameters: Language**

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

### Logon Parameters: RFC destination

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

### Office output

(Defaulted) Indicates whether to save outgoing documents to the SAPoffice outbox of the SAP user associated with the job. When selected, the agent saves outgoing documents to the SAPoffice outbox.

**Default:** not selected

**JIL attribute:** sap\_office

### Processed rate (%)

(Optional) Specifies the minimum acceptable process rate as a percentage of the total transactions. When set to 100, the job fails if all transactions are not processed. When set to zero (0), no minimum processed transactions are required. When set to 90, for example, the job fails if less than 90 percent of the transactions are processed.

**Limits:** 0-100

**JIL attribute:** bdc\_proc\_rate

### Release option

(Defaulted) Specifies the action to take with a job after it is defined, as follows:

- Immediately -- Releases the job immediately
- As soon as possible -- Releases the job as soon as possible

**Default:** As soon as possible

**JIL attribute:** sap\_release\_option

### SAP job class

(Optional) Specifies a valid SAP system job class.

**Limits:** One character

**JIL attribute:** sap\_job\_class

### SAP job name

Specifies the name of the ABAP job that creates the BDC session on the SAP system.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** sap\_job\_name

### Spool list recipients

(Optional) Specifies the [recipient properties](#) for the job's output (spool file).

This property supports Global Variable substitution.

Note: To add spool list recipients, select Add and click Go. A row appears below the Spool list recipients field. Click



to open the Spool List Recipient dialog to specify the recipients.

**JIL attribute:** sap\_recipients



## Step parameters

(Optional) Specifies the [SAP R/3 step specifications](#).

This property supports Global Variable substitution.

Note: The SAP Batch Input Session job can have only one step. Click



to open the Step Parameters dialog to specify the parameters for the job definition.

**JIL attribute:** sap\_step\_parms

## System

(Optional) Specifies the system the BDC session should be executed on.

**Limits:** Up to 256 characters

**JIL attribute:** bdc\_system

## Target system

(Optional) Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

Note: Clicking



opens the Browse SAP Target Systems dialog, which lets you locate the host computer.

**JIL attribute:** sap\_target\_sys

## Define an SAP Business Warehouse InfoPackage Job

You can define an SAP Business Warehouse InfoPackage job to transfer data from any data source into an SAP Business Warehouse system. This procedure lets you create a job from scratch; however, you can use an SAP filter and copy an existing job from the SAP system to save time.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Create an SAP BW InfoPackage (SAPBWIP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **InfoPackage name**  
Specifies the name of the Business Warehouse InfoPackage.

Note: Clicking



opens the Search SAP InfoPackage dialog, which lets you locate an SAP InfoPackage for the job definition.

3. (Optional) Specify optional [SAP BW InfoPackage properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.

The SAP BW InfoPackage job is defined.

### **SAP BW InfoPackage Job Examples**

The following examples describe sample SAP BW InfoPackage jobs.

#### **Example: Define an SAP BW InfoPackage Job**

This example runs the Business Warehouse InfoPackage 0PAK\_D2XZMZ1HD5WFVFL3EN1NVIT4V in the SAP destination SM1.

#### **Follow these steps:**

1. Create an SAP BW InfoPackage job.
2. Enter the following properties:
  - Name -- InfoBW2
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - Logon Parameters: RFC destination -- SM1
  - SAP job name -- InfoBW2
  - InfoPackage name -- 0PAK\_D2XZMZ1HD5WFVFL3EN1NVIT4V
3. Commit the job.

### **SAP BW InfoPackage Properties**

The SAP BW InfoPackage category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **InfoPackage details**

(Optional) Specifies the SAP Business Warehouse (BW) InfoPackage data selection criteria, as follows:

- **Field Name**  
Specifies the BW InfoPackage Technical Name.  
**Limits:** Up to 100 characters
- **High**  
Specifies the BW InfoPackage Technical To Value.  
**Limits:** Up to 256 characters
- **Low**  
Specifies the BW InfoPackage Technical From Value.

**Limits:** Up to 256 characters

- **Object Name**

Specifies the BW InfoPackage Technical InfoObject.

**Limits:** Up to 100 characters

**NOTE**

To assign no value to this data selection criterion, use apostrophes.

- **Operation**

Specifies the BW InfoPackage Technical Operation, as follows:

- BT -- Between
- EQ -- Equal

- **Sign**

Specifies the BW InfoPackage Technical Sign, as follows:

- E -- Exclude
- I -- Include

**Limits:** Up to 4096 characters

Note: You must click



to complete the InfoPackage details. The agent returns values for the InfoPackage based on details that are configured in the SAP system.

**JIL attribute:** sap\_ext\_table

**InfoPackage name**

Specifies the name of the Business Warehouse InfoPackage.

**Limits:** Up to 30 valid SAP characters; case-sensitive

Note: You can click



to open the Search SAP InfoPackage dialog to locate the SAP InfoPackage.

**JIL attribute:** sap\_info\_pack

**Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

**Logon Parameters: Language**

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

**Logon Parameters: RFC destination**

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

**SAP job name**

(Optional) Specifies the SAP R/3 job name.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** sap\_job\_name

**Define an SAP Business Warehouse Process Chain Job**

You can define an SAP Business Warehouse Process Chain job to run a sequence of background processes on the SAP system. Some SAP processes trigger events that can start other processes. A job runs the individual processes in the chain as job steps. This procedure lets you create a job from scratch; however, you can use an SAP filter and copy an existing job from the SAP system to save time.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

**Follow these steps:**

1. Create an SAP BW Process Chain (SAPBWPC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Chain identifier**  
Specifies the name of the Business Warehouse Process Chain.

Note: Clicking  opens the Browse Process Chain Identifiers dialog, which lets you locate a BW Process Chain for the job definition.
3. (Optional) Specify optional [SAP BW Process Chain properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP BW Process Chain job is defined.

**SAP BW Process Chain Job Examples**

The following example describes a sample SAP BW Process Chain job.

**Example: Define an SAP BW Process Chain Job**

This example defines the SAPBWPC3 job, which runs the DEMO\_CHAIN1 Process Chain on the SAP system. The language used to log on to the SAP system is English.

**Follow these steps:**

1. Create an SAP BW Process Chain job.
2. Enter the following properties:

- Name -- SAPBWPC3
- Send to machine -- sapagent
- Owner -- WAAESAP@sapagent
- Logon Parameters: Language -- EN
- Logon Parameters: RFC destination -- SM1
- Chain identifier -- DEMO\_CHAIN1

3. Commit the job.

### **SAP BW Process Chain Properties**

The SAP BW Process Chain category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Chain identifier**

Specifies the name of the Business Warehouse Process Chain.

**Limits:** Up to 30 characters

Note: Clicking



opens the Browse Process Chain Identifiers dialog, letting you locate the process chain.

**JIL attribute:** sap\_chain\_id

#### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

#### **Logon Parameters: Language**

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

#### **Logon Parameters: RFC destination**

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

### **Define an SAP Data Archiving Job**

You can define an SAP Data Archiving job to store information described in an SAP archiving object in an SAP data archive.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

**Follow these steps:**

1. Create an SAP Data Archiving (SAPDA) job.
2. Specify the following required properties:
  - **Name** Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Archiving object name**  
Specifies the name of the archiving object.

**NOTE**

Clicking  opens the Search Archiving Objects dialog, which lets you locate an archiving object for the job definition.

- **Archiving object variant** Specifies the name of the archiving object variant.

Note: Clicking



opens the Browse Archiving Objects Variants dialog, which lets you locate an archiving object variant for the job definition.

- **Print parameters**  
Specifies [print parameters](#).

Note: Clicking



opens the Print Parameters dialog, which lets you specify the parameters for the job definition.

3. (Optional) Specify optional [SAP Data Archiving properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP Data Archiving job is defined.

**SAP Data Archiving Job Examples**

The following example describes a sample SAP Data Archiving job.

**Example: Define an SAP Data Archiving Job**

This example defines a job that stores information described in the BC\_ARCHIVE Archiving Object in an SAP data archive. The archiving object variant is BC\_ARCVARIANT.

**Follow these steps:**

1. Create an SAP Data Archiving job.
2. Enter the following properties:
  - Name -- SAPDA\_job
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - Archiving object name -- BC\_ARCHIVE
  - Archiving object variant -- BC\_ARCVARIANT

## 3. Click



to open the Print Parameters dialog.

## 4. Enter the following properties:

- Archiving Mode -- Print
- Output Device -- LP01

## 5. Save the job.

### **SAP Data Archiving Properties**

The SAP Data Archiving category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Archiving object name**

Specifies the name of the archiving object.

Clicking



opens the Search Archiving Objects dialog, which lets you locate an archiving object for the job definition.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** arc\_obj\_name

#### **Archiving object variant**

Specifies the name of the archiving object variant.

**Limits:** Up to 256 characters; case-sensitive

Note: Clicking



opens the Browse Archiving Objects Variants dialog, which lets you locate an archiving object variant for the job definition.

**JIL attribute:** arc\_obj\_variant

#### **Archiving parameters**

(Optional) Specifies [archiving parameters](#).

**Limits:** Up to 4096 characters

Note: Clicking



opens the Archiving Parameters dialog, which lets you specify the parameters for the job definition.

**JIL attribute:** arc\_parms

#### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

### Logon Parameters: Language

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

### Logon Parameters: RFC destination

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

### Print parameters

Specifies [print parameters](#).

**Limits:** Up to 4096 characters

Note: Clicking



opens the Print Parameters dialog, which lets you specify the parameters for the job definition.

**JIL attribute:** sap\_print\_parms

### Target system

(Optional) Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

Clicking



opens the Browse SAP Target Systems dialog, which lets you locate the host computer.

**JIL attribute:** sap\_target\_sys

### Print Parameters

You can access the Print Parameters dialog by clicking



to the right of the Print parameters field for an SAP Data Archiving job. The Print Parameters dialog includes the following parameters:

### Archiving Mode

Specifies whether the spool file is printed, archived, or both.

#### NOTE

When Print is selected, the Archiving Parameters dialog is disabled.

**JIL attribute:** prt\_arc\_mode

### Authorization



(Optional) Specifies the password for viewing the print spool list.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** authorization

### Dataset Name

(Optional) Specifies the print spool data set name.

**Limits:** Up to six characters; case-sensitive

**JIL attribute:** dataset

### Delete After Output

(Optional) Indicates whether the spool request associated with the ABAP is deleted after printing. When selected, the agent deletes the spool request after printing. This parameter corresponds to the SAPGUI Spool options, Delete after output field on the Background Print Parameters dialog.

**JIL attribute:** release

### Department

(Optional) Specifies the department name to print on the cover page.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** dept

### Format

(Optional) Specifies an output format for an SAP report.

**Limits:** Up to 16 characters; case-sensitive

Clicking



opens the Search Print Format dialog, which lets you locate the output format for the report.

**JIL attribute:** format

### New Spool Request

(Optional) Indicates whether to create a new spool request or append the spool request to an existing spool request with similar attributes (if any). When selected, the agent creates a new spool request.

**JIL attribute:** new\_spool

### Number of Columns

(Optional) Specifies the number of columns to be printed on the page.

**Limits:** 1-255

**JIL attribute:** num\_columns

### Number of Copies

(Optional) Specifies the number of print copies.

**Limits:** 0-255

**JIL attribute:** copies

### Number of Lines

(Optional) Specifies the number of lines to be printed on the page.

**Limits:** 0-486

**JIL attribute:** num\_lines

### OS Cover Page

(Optional) Specifies whether to print a host page, as follows:

- Yes -- Prints the host page
- No -- Does not print the host page
- Default -- Uses the SAP default

**JIL attribute:** host\_page

### Output Device

Specifies the output device name. This parameter corresponds to the SAPGUI Output device field on the Background Print Parameters dialog.

**Limits:** Up to four characters; case-sensitive

Clicking



opens the Search Output Devices dialog, which lets you locate the output device.

**JIL attribute:** dest

### Print Immediately

(Optional) Indicates whether to print the job immediately after completion. When selected, the agent prints the job immediately. This parameter corresponds to the SAPGUI Spool options, Print immediately field on the Background Print Parameters dialog.

**JIL attribute:** print\_imm

### Print Cover Page

(Optional) Indicates whether to include an SAP cover page with the report output. When selected, the agent prints the report with a cover page.

**JIL attribute:** banner\_page

### Print Footer

(Optional) Indicates whether to print a footer on an SAP report. When selected, the agent prints a report with a footer.

**JIL attribute:** footer

### Print Priority

(Optional) Specifies the priority of the SAP spool request. The priority is a number between 1 (highest priority) and 9 (lowest priority).

**Limits:** 0-9

**JIL attribute:** print\_priority

### Recipient Name

(Optional) Specifies the spool request recipient's name to appear on the SAP cover page, if applicable. This parameter corresponds to the SAPGUI Cover sheets options, Recipient field on the Background Print Parameters dialog.

**Limits:** Up to 12 characters; case-sensitive

Clicking



opens the Search SAP Office Users dialog, which lets you locate the spool request recipient name.

**JIL attribute:** recipient\_name

### Request Type

(Optional) Specifies the print request type.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** req\_type

### SAP Cover Page

(Optional) Indicates whether to include an SAP cover page containing information such as recipient name, department name, and format used, as follows:

- Yes -- Prints the SAP cover page
- No -- Does not print the SAP cover page
- Default -- Uses the SAP default

**JIL attribute:** sap\_banner

### Spool Name

(Optional) Specifies the name of the print spool.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** spool\_name

### Spool Retention

(Optional) Specifies the number of days before the spool request is deleted.

**Limits:** 0-9

**JIL attribute:** expiration

### Title

(Optional) Specifies the spool request description.

**Limits:** Up to 68 characters

**JIL attribute:** title

### Archiving Parameters

You can access the Archiving Parameters dialog by clicking



to the right of the Archiving parameters field for an SAP Data Archiving job. The Archiving Parameters dialog includes the following parameters:

#### ArchiveLink Host

(Optional) Specifies the SAP archive link RPC host.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** arc\_host\_link

#### ArchiveLink Information

(Optional) Specifies the SAP archive link information.

**Limits:** Up to three characters; case-sensitive

**NOTE**

This parameter is required if the print parameters Archiving mode is set to Archive or Both.

**JIL attribute:** arc\_info

**ArchiveLink Object Type**

(Optional) Specifies the type of external system archive object. This parameter corresponds to the Obj. type field on the Define Background Archive Parameters dialog.

**Limits:** Up to 10 characters; case-sensitive

**NOTE**

This parameter is required if the print parameters Archiving mode is set to Archive or Both.

**JIL attribute:** arc\_obj\_type

**Archiving Path**

(Optional) Specifies the standard archive path.

**Limits:** Up to 70 characters; case-sensitive

**JIL attribute:** arc\_path

**Client**

(Optional) Specifies the SAP archive link client.

**Limits:** 0-999

**JIL attribute:** arc\_client

**Connection Name**

(Optional) Specifies the SAP archive link communication connection.

**Limits:** Up to 14 characters; case-sensitive

**JIL attribute:** arc\_connect

**Date**

(Optional) Specifies the date to do the archiving in YYYYMMDD format, using the SAP agent time zone.

**Limits:** Up to eight characters

**JIL attribute:** arc\_date

**Document Class**

(Optional) Specifies the SAP archive link document class.

**Limits:** Up to 20 characters; case-sensitive

**JIL attribute:** arc\_doc\_class

**Document Type**

(Optional) Specifies the name of the document type. This parameter corresponds to the Doc type field on the Define Background Archive Parameters dialog.

**Limits:** Up to 10 characters; case-sensitive

**Note:** This parameter is required if the print parameters Archiving mode is set to Archive or Both.

**JIL attribute:** arc\_doc\_type

#### **Format**

(Optional) Specifies the SAP archive link output format.

**Limits:** Up to 16 characters; case-sensitive

**JIL attribute:** arc\_format

#### **Printer**

(Optional) Specifies the SAP archive link target printer.

**Limits:** Up to four characters; case-sensitive

**JIL attribute:** arc\_printer

#### **Protocol**

(Optional) Specifies the SAP archive storage connection protocol.

**Limits:** Up to eight characters; case-sensitive

**JIL attribute:** arc\_protocol

#### **Report**

(Optional) Specifies the SAP archive link report name.

**Limits:** Up to 30 characters; case-sensitive

**JIL attribute:** arc\_report

#### **Service Destination**

(Optional) Specifies the SAP archive link RPC service/destination.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** arc\_service

#### **Target Storage System**

(Optional) Specifies the SAP archive link target storage system.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** arc\_storage

#### **Text**

(Optional) Specifies the SAP archive link Text Information field.

**Limits:** Up to 30 characters; case-sensitive

**JIL attribute:** arc\_text

#### **User**

(Optional) Specifies the archive data element for the user.

**Limits:** Up to 12 characters

**JIL attribute:** arc\_userid

#### **Version**

(Optional) Specifies the archive version.

**Limits:** Up to four characters

**JIL attribute:** arc\_version

### **Define an SAP Event Monitor Job**

You can define an SAP Event Monitor job to schedule workload based on the activity of an SAP event or to trigger an SAP event at the appropriate time in your schedule.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

#### **Follow these steps:**

1. Create an SAP Event Monitor (SAPEVT) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Event ID**  
Specifies the name of the SAP event to monitor or trigger.

Clicking



opens the Browse Background Event Identifiers dialog, which lets you locate an SAP event for the job definition.

3. (Optional) Specify optional [SAP Event Monitor properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP Event Monitor job is defined.

### **SAP Event Monitor Job Examples**

The following examples describe sample SAP Event Monitor jobs.

#### **Example: Trigger an SAP Event**

This example defines an SAPEVT job with the Event trigger property selected, so the job triggers the SAP\_TEST event by default.

#### **Follow these steps:**

1. Create an SAP Event Monitor job.
2. Enter the following properties:
  - Name -- SAPEVT\_job
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - Logon Parameters: RFC destination -- BI1
  - Event ID -- SAP\_TEST
  - Event trigger -- selected
3. Commit the job.

### Example: Monitor an SAP Event

This example defines an SAPEVT job without the Event trigger property selected, so the job monitors the SAP\_TEST event and remains in RUNNING status.

#### Follow these steps:

1. Create an SAP Event Monitor job.
2. Enter the following properties:
  - Name -- trigger\_evt
  - Send to machine -- localhost
  - Owner -- WAAESAP@sapagent
  - Logon Parameters: Client -- 001
  - Logon Parameters: Language -- EN
  - Logon Parameters: RFC destination -- BI1
  - Event ID -- SAP\_TEST
  - Event parameters -- L L
  - Event trigger -- not selected
3. Commit the job.

### SAP Event Monitor Properties

The SAP Event Monitor category includes the following properties:

#### Agent job class

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### Continuous

(Defaulted) Indicates whether to monitor for the conditions continuously. When selected, each time the specified conditions occur, an alert is written to the scheduler log file.

**Default:** not selected (the job does not monitor for the conditions)

#### NOTE

- To end continuous monitoring, you must complete the job manually.
- Either Continuous or Event trigger can be selected, but not both.

**JIL attribute:** continuous

#### Event ID

Specifies the name of the SAP event to monitor or trigger.

**Limits:** Up to 32 valid SAP characters; case-sensitive

Clicking



opens the Browse Background Event Identifiers dialog, which lets you locate an SAP event.

**JIL attribute:** sap\_event\_id

#### Event parameter

(Optional) Specifies the name of the SAP event parameter.

**Limits:** Up to 64 characters

**JIL attribute:** sap\_event\_parm

### Event trigger

(Defaulted) Indicates whether to trigger or monitor an SAP event. When selected, the agent triggers an SAP event.

**Default:** not selected (the job monitors for an SAP event)

### NOTE

Either Continuous or Event trigger can be selected, but not both.

**JIL attribute:** sap\_is\_trigger

### Logon Parameters: Client

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

### Logon Parameters: Language

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

### Logon Parameters: RFC destination

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

## Define an SAP Job Copy Job

You can define an SAP Job Copy job to schedule an existing SAP R/3 job.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Create an SAP Job Copy (SAPJC) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **SAP job name**  
Specifies the SAP R/3 job name.
  - **SAP job count**



Specifies the ID of the job to be copied.

3. (Optional) Specify optional [SAP Job Copy properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP Job Copy job is defined.

### **SAP Job Copy Job Examples**

The following example describes a sample SAP Job Copy job.

#### **Example: Define an SAP Job Copy Job**

This example defines an SAP Job Copy job that copies the AM job with job count 11331500.

#### **Follow these steps:**

1. Create an SAP Job Copy job.
2. Enter the following properties:
  - Name -- SAPJC\_job
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - SAP job count -- 11331500
  - SAP job name -- AM
3. Save the job.

### **SAP Job Copy Properties**

The SAP Job Copy category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Failure message**

(Optional) Specifies a string that indicates the job failed. If the string is found in the job's spool file, the job is considered failed even if the job succeeds on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive

**JIL attribute:** sap\_fail\_msg

#### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

#### **Logon Parameters: Language**

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

### **Logon Parameters: RFC destination**

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

### **Monitor children**

(Optional) Indicates whether to monitor a job's children. Children jobs are jobs produced by a parent job. When selected, the agent monitors the job's children.

**Default:** Not selected (The job does not monitor children jobs.)

**JIL attribute:** sap\_mon\_child

### **Release option**

(Defaulted) Specifies the action to take with an SAP R/3 job after it is defined, as follows:

- Immediately -- Releases the job immediately
- As soon as possible -- Releases the job as soon as possible

**Default:** As soon as possible

**JIL attribute:** sap\_release\_option

### **SAP job count**

(Optional) Specifies the ID of the job to be copied.

**Limits:** Up to eight digits

**JIL attribute:** sap\_job\_count

### **SAP job name**

Specifies the SAP R/3 job name.

**Limits:** Up to 32 characters; case-sensitive

Clicking



opens the Search SAP Job Name and Job Count dialog, which lets you locate an SAP job.

**JIL attribute:** sap\_job\_name

### **Step number**

(Optional) Specifies the number of the first step to start copying job data from.

**Limits:** 0- $n$ , where  $n$  is the highest step number

#### **NOTE**

If you specify 0 or 1, all steps are copied.

**JIL attribute:** sap\_step\_num

### **Success message**

(Optional) Specifies a string that, when found in the condition output of an SAP job, indicates the success of the job.

**Limits:** Up to 128 characters; case-sensitive

**JIL attribute:** sap\_success\_msg

### Target job name

(Optional) Specifies the name of the target job to copy. If unspecified, the target job will have the same name as the source job.

**Limits:** Up to 32 characters

**JIL attribute:** sap\_target\_jobname

### Target system

(Optional) Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

Clicking



opens the Browse SAP Target Systems dialog, which lets you locate the host computer.

**JIL attribute:** sap\_target\_sys

## Define an SAP Process Monitor Job

You can define an SAP Process Monitor job to monitor for a specific SAP process status and end after detecting a process. You can also use SAP Process Monitor jobs to set up predecessor or dependent job relationships with other jobs or SAP processes.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

### Follow these steps:

1. Create an SAP Process Monitor (SAPPM) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Process status**  
Specifies the SAP process status to monitor, as follows:
    - Waiting
    - Running
    - Stopped
3. (Optional) Specify optional [SAP Process Monitor properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP Process Monitor job is defined.

## **SAP Process Monitor Job Examples**

The following example describes a sample SAP Process Monitor job.

### **Example: Define an SAP Process Monitor Job**

This example defines an SAP Process Monitor job that monitors the RUNNING status.

#### **Follow these steps:**

1. Create an SAP Process Monitor job.
2. Enter the following properties:
  - Name -- test\_SAPPM
  - Send to machine -- sapagent
  - Logon Parameters: RFC destination -- BI1
  - Process status -- Running
3. Commit the job.

## **SAP Process Monitor Properties**

The SAP Process Monitor category includes the following properties:

### **ABAP name**

(Optional) Starts a step definition within the job definition and identifies the ABAP step to be run. This property corresponds to the SAPGUI ABAP program Name field on the Create Step dialog.

**Limits:** Up to 40 characters

#### **NOTE**

This property is not valid if the Process status is set to Waiting.

**JIL attribute:** sap\_abap\_name

### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

### **Continuous**

(Optional) Indicates whether to monitor for the conditions continuously. When selected, each time the specified conditions occur, an alert is written to the scheduler log file.

**Default:** not selected (the job does not monitor for the conditions)

#### **NOTE**

To end continuous monitoring, you must complete the job manually.

**JIL attribute:** continuous

### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

**Logon Parameters: Language**

(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

**Logon Parameters: RFC destination**

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

**Process client**

(Optional) Specifies the client monitored for the process.

**Limits:** Up to three digits

**NOTE**

This property is not valid if the Process status is set to Waiting.

**JIL attribute:** sap\_process\_client

**Process status**

Specifies the SAP process status to monitor.

- Running
- Stopped
- Waiting

**JIL attribute:** sap\_process\_status

**Process type**

(Optional) Specifies the SAP business process type to monitor. BGD -- backgroundDIA -- dialogENQ -- lockSPO -- spoolUPD -- updateUP2 -- update2

**JIL attribute:** sap\_proc\_type

**Process user**

(Optional) Specifies an SAP user name. When a process matches the specified process status, the job the process runs is checked for a match to this user.

**Limits:** Up to 12 characters; case-sensitive

**NOTE**

This property is not valid if the Process status is set to Waiting.

**JIL attribute:** sap\_proc\_user

**Target system**

(Optional) Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.

**Limits:** Up to 256 characters; case-sensitive

Clicking



opens the Browse SAP Target Systems dialog, which lets you locate the host computer.

**JIL attribute:** sap\_target\_sys

### **Define an SAP R/3 Job**

You can define an SAP R/3 job to schedule an SAP R/3 job on your SAP system. This procedure lets you create a job from scratch; however, you can use an SAP filter and copy an existing job from the SAP system to save time.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for SAP.

To schedule an SAP R/3 job, you must define at least one ABAP (SAP program) in the job definition.

#### **Follow these steps:**

1. Create an SAP R/3 (SAP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **SAP job name**  
Specifies the SAP R/3 job name.
  - **Step parameters**  
Specifies the [SAP R/3 step specifications](#).

#### **NOTE**

We recommend that you limit the number of steps (ABAPs) to one per job. If you run a job and one of the ABAPs fails, the job is marked as failed. If the ABAP fails, you cannot re-run the ABAP without re-running the entire job.

3. (Optional) Specify optional [SAP R/3 properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The SAP R/3 job is defined.

### **SAP R/3 Job Examples**

The following examples describe sample SAP R/3 jobs.

#### **Example: Define an SAP R/3 Job with One Step**

This example runs the SAP R/3 job named BI\_WRITE\_PROT\_TO\_APPLLOG, which in turn runs the SAP program (ABAP) named RSBATCH\_WRITE\_PROT\_TO\_APPLLOG.

#### **Follow these steps:**

1. Create an SAP R/3 job.
2. Enter the following properties:

- Name -- test\_SAP
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - SAP job name -- BI\_WRITE\_PROT\_TO\_APPLLOG
  - Logon Parameters: RFC destination -- BI1
3. Select Add from the Select an Action drop-down list to the right of the Step parameters field, and click Go.
  4. Click  to open the Step Parameters dialog.
  5. Enter the following properties on the Command tab:
    - ABAP Name -- RSBATCH\_WRITE\_PROT\_TO\_APPLLOG
    - ABAP Language -- EN
  6. Click OK.  
The step specifications are added to the job definition.
  7. Commit the job.

### Example: Define an SAP R/3 Job with Multiple Steps

This example runs an SAP R/3 job named SAP\_2\_STEPS, which in turn runs two steps. Each step runs the same ABAP but with different parameters.

#### Follow these steps:

1. Create an SAP R/3 job.
2. Enter the following properties:
  - Name -- test\_1
  - Send to machine -- sapagent
  - Owner -- WAAESAP@sapagent
  - SAP job name -- SAP\_2\_STEPS
  - Release option -- Immediately
3. Select Add from the Select an action drop-down list to the right of the Step Parameters field, and click Go.
4. Click  to open the Step Parameters dialog.
5. Enter the following properties on the Command tab:
  - ABAP Name -- BTCTEST
  - Variant -- test
  - ABAP Language -- EN
6. Enter the following properties on the Output tab:
  - Archiving Mode -- archived
  - Number of Copies -- 2
  - Number of Columns -- 80
  - Number of Lines -- 65
7. Enter the following properties on the Archive tab:
  - Printer -- LP01
  - ArchiveLink Object Type -- ARCHIVE
  - Document Type -- ARCHIVE
  - ArchiveLink Information -- inf,num
8. Click OK.  
The step specifications are added to the job definition.

9. Select Add from the Select an Action drop-down list, and click Go.

10. Click



to open the Step Parameters dialog.

11. Enter the following properties on the Command tab:

- ABAP Name -- BTCTEST
- Variant -- test
- ABAP Language -- EN

12. Enter the following properties on the Output tab:

- Archiving Mode -- archived
- Number of Copies -- 1
- Number of Columns -- 80
- Number of Lines -- 65
- SAP Cover Page -- Yes
- Print Cover Page -- selected
- Recipient Name -- cyber

13. Enter the following properties on the Archive tab:

- Printer -- LP01
- ArchiveLink Object Type -- ARCHIVE
- Document type -- ARCHIVE
- ArchiveLink Information -- inf,num

14. Click OK.

The step specifications are added to the job definition.

15. Commit the job.

### **SAP R/3 Properties**

The SAP R/3 category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Failure message**

(Optional) Specifies a string that, when found in the condition output of an SAP job, indicates the failure of the job.

**Limits:** Up to 128 valid SAP characters; case-sensitive

**JIL attribute:** sap\_fail\_msg

#### **Logon Parameters: Client**

(Optional) Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**JIL attribute:** sap\_client

#### **Logon Parameters: Language**



(Optional) Specifies a character code representing a valid language for SAP.

**Limits:** Up to two characters; case-sensitive

**JIL attribute:** sap\_lang

### Logon Parameters: RFC destination

(Optional) Indicates the destination specified in the connection properties file during installation.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** sap\_rfc\_dest

### Monitor children

(Optional) Indicates whether to monitor a job's children. Children jobs are jobs spawned by a parent job. When selected, the agent monitors the job's children.

**Default:** Not selected (The job does not monitor children jobs.)

**JIL attribute:** sap\_mon\_child

### Office output

(Optional) Indicates whether to save outgoing documents to the SAPoffice outbox of the SAP user associated with the job. When selected, the agent saves outgoing documents.

**JIL attribute:** sap\_office

### Release option

(Defaulted) Specifies the action to take with a job after it is defined, as follows:

- Immediately -- Releases the job immediately
- As soon as possible -- Releases the job as soon as possible

**Default:** As soon as possible

**JIL attribute:** sap\_release\_option

### SAP job class

(Optional) Specifies a valid SAP system job class.

**Limits:** One character

**JIL attribute:** sap\_job\_class

### SAP job name

Specifies the SAP R/3 job name.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** sap\_job\_name

### Spool list recipients

(Optional) Specifies the [recipient properties](#) for the job's output (spool file).

Selecting Add and clicking Go displays a row below the Spool list recipients field. In that field, clicking



opens the Spool List Recipient dialog, which lets you specify the recipients.

**JIL attribute:** sap\_recipients

### Step parameters

Specifies the [SAP R/3 step specifications](#).

Selecting Add and clicking Go displays a row below the Step parameter field. In that field, clicking



opens the Step Parameters dialog, which lets you specify the parameters for the job definition.

**JIL attribute:** sap\_step\_parms

### Success message

(Optional) Specifies a string that, when found in the condition output of an SAP job, indicates the success of the job.

**Limits:** Up to 128 characters; case-sensitive

**JIL attribute:** sap\_success\_msg

### Target system

(Optional) Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.

**Limits:** Up to 256 characters; case-sensitive

Clicking



opens the Browse SAP Target Systems dialog, which lets you locate the host computer.

**JIL attribute:** sap\_target\_sys

## Using Success and Failure Messages within an SAP Job

You can check the output of an SAP R/3 job for specific text strings to determine whether the job is a success or a failure. You specify the text string in the Success message and Failure message SAP R/3 properties in the job definition. For example, suppose that when you cancel an SAP R/3 job, you want it to complete to release its successor job. You can specify the text string 'Job canceled' as a success message in the job definition. When you cancel the job, the agent checks the job's spool file, finds a match for 'Job canceled', and marks the job as complete.

You can also check the output of a step (ABAP) to determine whether the step is a success or failure. You specify the text string in either the Success message property or Failure message property within the Step parameter description on the Step parameters - Command tab.

For more flexibility, you can specify regular expressions instead of simple text strings within the Success message and Failure message properties. For example, you can use a regular expression to search for multiple strings at the same time. To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules using an Internet search for java pattern.

### NOTE

To enable regular expression processing, the agent administrator must configure the agent, setting the `sap.useRegularExpressions` parameter to true.

## Examples: Using Regular Expressions

- The following expression checks for "TEST" in the job's output file:

```
(?s) .*TEST.*
```

### NOTE

To check multiple lines in the file, you must use `(?s)`. The first `.*` indicates that any number of characters can precede TEST. The second `.*` indicates that any number of characters can follow it.

- The following expression checks whether "not found" or "started" appears in the job's output file:

```
<(?s).*[not found|started].*>
```

- The following expression checks whether "Job canceled" appears in the job's output file:

```
(?s).*Job\scanceled.*
```

### **Define Recipients for the Job's Output**

In an SAP R/3 or a Batch Input Session job, you can define recipients for the job's output (spool file). The recipient can be an email address, an SAPOffice distribution list, or an SAPOffice user.

#### **Follow these steps:**

1. Open the SAP R/3 or Batch Input Session job for which you want to define recipients for the job's output.
2. Select Add and click Go in the Spool list recipients field to add a row.
3. Click



beside the field to open the Spool List Recipient dialog.

4. Select a recipient type, as follows:
  - INT -- Specifies an email address
  - DLI -- Specifies an SAPOffice distribution list
  - SU -- Specifies an SAP user
5. Select whether to send the job's output using the To, cc, or bcc option in the Recipient field.
6. Specify the recipient name in the Recipient field.
7. (Optional) Select one or more of the following SAPOffice options to control delivery of the job output:
  - No forward -- Lets SAPOffice users forward the document.
  - No print -- Lets SAPOffice users print the document.
  - Express -- Sends instant messages to recipients if they are logged in to SAP.

#### **NOTE**

These options apply to the DLI and SU recipient types only.

8. Click OK.  
The spool list recipients are added to the job definition.

### **Example: Send a Spool File to a Distribution List**

This example sends the spool file to the users on the payroll distribution list.

#### **Follow these steps:**

1. Open the Spool List Recipient dialog.
2. Edit the following properties:
  - Recipient Type -- DLI
  - Recipient -- To, payroll
3. Click OK.

### **Example: Send a Carbon Copy of a Spool File**

This example sends a carbon copy of a job spool file to the email address jsmith@company1.com.

#### **Follow these steps:**

1. Open the Spool List Recipient dialog.
2. Enter the following properties:
  - Recipient Type -- INT
  - Recipient -- cc, jsmith@company1.com
3. Click OK.

### Example: Save an Outgoing Document to an SAPOffice Outbox

This example saves outgoing documents to the SAPOffice outbox of the SAP user J01Prod.

#### Follow these steps:

1. Open the Spool List Recipient dialog.
2. Enter the following properties:
  - Recipient Type -- SU
  - Recipient -- To, J01PROD
3. Select SAPOffice Options.
4. Click OK.

### Spool List Recipient Properties

The Spool List Recipient dialog includes the following properties:

#### Recipient

Specifies the target recipient's email address. This property is required if the spool list recipient exists.

**Limits:** Up to 256 characters

**JIL attribute:** recipient

#### Recipient Type

Specifies the recipient type, as follows:

- INT -- Specifies an email address
- DLI -- Specifies an SAPOffice distribution list
- SU -- Specifies an SAP user
- If the Recipient Type is DLI, clicking  opens the Search Distribution List dialog, which lets you locate a recipient.
- If the Recipient Type is Su, clicking  opens the Search SAP Office Users dialog, which lets you locate the spool request recipient name.

**JIL attribute:** rcp\_type

#### SAPOffice Options

(Optional) Specifies recipient options when the Recipient type is set to DLI or SU, as follows:

- Express -- Sends instant messages to recipients if they are logged in to SAP.  
**JIL attribute:** express
- No forward -- Lets SAPOffice users forward the document.  
**JIL attribute:** no\_forward
- No print -- Lets SAPOffice users print the document.  
**JIL attribute:** no\_print

### Define Print or Archive Parameters for a Step

In an SAP R/3 or a Batch Input Session job, you can define print or archive parameters, or both, for a step.

#### Follow these steps:

1. Open the SAP R/3 or Batch Input Session job for which you want to define print or archive parameters.

2. Click



beside the Step parameters field.

3. Specify the following required property on the Command tab:

- **ABAP Name**

Defines the valid SAP system ABAP name.

4. (Optional) Specify optional [command parameters](#) for the step.
5. Click the Output tab.
6. (Optional) Specify optional for the step.
7. Click the Archive tab.

#### NOTE

The Archive tab is only available if Archiving Mode on the Output tab is either Archive or Both.

8. (Optional) Specify optional [archive parameters](#) for the step.
9. Click OK.  
The parameters for the step are added to the job definition.

#### Example: Specify Archive Parameters

In this example, the archive target printer is PRT1. The archive mode, document type, and external archive object are Archive. The archive link information is inf and the archive link client is 800.

#### Follow these steps:

1. Open the Step Parameters dialog for the step.
2. Enter the following property on the Output tab:
  - Archive Mode -- Archive
3. Enter the following properties on the Archive tab:
  - ArchiveLink Object Type -- ARCHIVE
  - Document Type -- ARCHIVE
  - ArchiveLink Information -- info
  - Printer -- PRT1
  - Client -- 800
4. Click OK.

#### Example: Specify Print Parameters

In this example, the print destination for the RSPARAM ABAP is LP01. The number of lines per list page is 65 and the line width of the list is 80. One copy of the report associated with the ABAP is printed.

#### Follow these steps:

1. Open the Step Parameters dialog for the step.
2. Enter the following property on the Command tab:
  - ABAP Name -- RSPARAM
3. Enter the following properties on the Output tab:
  - Archiving Mode -- archived
  - Number of Copies -- 1
  - Number of Columns -- 80
  - Number of Lines -- 65
4. Enter the following properties on the Archive tab:

- ArchiveLink Object Type -- Archive
- Document Type -- Archive
- ArchiveLink Information -- info
- Printer -- LP01
- Client -- 800

5. Click OK.

### **Step Parameters - Command Properties**

The Step Parameters dialog includes the following properties on the Command tab, which define the ABAP (step):

#### **ABAP Language**

(Optional) Specifies a language for the ABAP.

**Limits:** Up to two characters

#### **NOTE**

The SAP system default logon language is the default.

**JIL attribute:** abap\_lang

#### **ABAP Name**

Defines the valid SAP system ABAP name. The ABAP name starts a step definition within the job definition and identifies the ABAP step to be run.

**Limits:** Up to 40 characters

- This parameter corresponds to the SAPGUI ABAP program name field on the Create Step dialog.
- Clicking



opens the Search ABAP Name dialog, which lets you locate a valid ABAP name.

**JIL attribute:** abap\_name

#### **Failure Message**

(Optional) Specifies a string that indicates the job failed. If the string is found in the job's spool file, the job is considered failed even if the job succeeds on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive

#### **NOTE**

This keyword does not apply to SAP Batch Input Session (SAPBDC) jobs.

**JIL attribute:** sap\_fail\_msg

#### **Mail List**

(Optional) Specifies a comma-separated list of email addresses.

**Limits:** Up to 2048 total characters; each list can have maximum of 1024

**JIL attribute:** sap\_maillist

#### **Step User**

(Optional) Specifies the SAP R/3 system user under whose authorization the ABAP program runs.

**Limits:** Up to 12 characters

**JIL attribute:** step\_user

**Success Message**

(Optional) Specifies a string that indicates the job completed successfully. If the string is found in the job's spool file, the job is considered successfully completed even if the job fails on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive

**NOTE**

This keyword does not apply to SAP Batch Input Session (SAPBDC) jobs.

**JIL attribute:** sap\_success\_msg

**Variant**

(Optional) Specifies the variant to pass.

**Limits:** Up to 14 characters

Clicking



opens the Browse ABAP Variant dialog, which lets you locate a variant to pass.

**JIL attribute:** variant

**Step Parameters - Output Properties**

The Step Parameters dialog includes the following properties on the Output tab:

**Archiving Mode**

(Optional) Specifies whether the spool file is printed, archived, or both.

**JIL attribute:** prt\_arc\_mode

**Authorization**

(Optional) Specifies the password for viewing the print spool list.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** authorization

**Dataset Name**

(Optional) Specifies the print spool data set name.

**Limits:** Up to six characters; case-sensitive

**JIL attribute:** dataset

**Delete After Output**

(Optional) Indicates whether to delete the print spool file after output. When selected, the agent deletes the print spool file.

**JIL attribute:** release

**Department**

(Optional) Specifies the department name to print on the cover page.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** dept

**Format**

(Optional) Specifies an output format for an SAP report.

**Limits:** Up to 16 characters; case-sensitive

Clicking



opens the Search Print Format dialog, which lets you locate an output format.

**JIL attribute:** format

### New Spool Request

(Optional) Indicates whether to create a new spool request. When selected, the agent creates a new spool request. SAP does not append the spool request to an existing spool request.

**JIL attribute:** new\_spool

### Number of Columns

(Optional) Specifies the number of columns for the SAP report.

**Limits:** 1-255

**JIL attribute:** num\_columns

### Number of Copies

(Optional) Specifies the number of copies to print of the SAP report.

**Limits:** 0-255

**JIL attribute:** copies

### Number of Lines

(Optional) Specifies the number of lines for the SAP report.

**Limits:** 0-486

**JIL attribute:** num\_lines

### OS Cover Page

(Optional) Specifies whether to print the operating system page, as follows:

- No
- Yes
- Default

**JIL attribute:** host\_page

### Output Device

(Optional) Specifies the name of the output device.

**Limits:** Up to four characters; case-sensitive

Clicking



opens the Search Output Devices dialog, which lets you locate the output device.

**JIL attribute:** dest

### Print Cover Page

(Optional) Indicates whether to print the selection cover page. When selected, the agent prints the selection cover page.

**JIL attribute:** banner\_page



---

**Print Footer**

(Optional) Indicates whether to print a footer on the SAP report. When selected, the agent prints a footer.

**JIL attribute:** footer

**Print Immediately**

(Optional) Indicates whether to print the job immediately after completion. When selected, the agent prints the job immediately.

**JIL attribute:** print\_imm

**Print Priority**

(Optional) Specifies the priority number of the print spool request.

**Limits:** 0-9

**JIL attribute:** print\_priority

**Recipient Name**

(Optional) Specifies the recipient on the SAP cover page.

**Limits:** Up to 12 characters; case-sensitive

Clicking



opens the Search SAP Office Users dialog, which lets you locate the spool request recipient name.

**JIL attribute:** recipient

**Request Type**

(Optional) Specifies the print request type.

**Limits:** Up to 12 characters; case-sensitive

**JIL attribute:** req\_type

**SAP Cover Page**

(Optional) Specifies whether to print the SAP cover page, as follows:

- No
- Yes
- Default

**JIL attribute:** sap\_banner

**Spool Name**

(Optional) Specifies the spool request name.

**Limits:** Up to 12 characters

**JIL attribute:** spool\_name

**Spool Retention**

(Optional) Specifies the number of days before the spool request expires and is deleted.

**Limits:** 0-9

**JIL attribute:** expiration

**Title**

(Optional) Specifies the spool request description.

**Limits:** Up to 68 characters

**JIL attribute:** title

### **Step parameters - Archive Properties**

The Step Parameters dialog includes the following properties on the Archive tab:

#### **ArchiveLink Host**

(Optional) Specifies the SAP ArchiveLink RPC host.

**Limits:** Up to 32 characters

**JIL attribute:** arc\_host\_link

#### **ArchiveLink Information**

Specifies the SAP ArchiveLink RPC information.

**Limits:** Up to three characters

**JIL attribute:** arc\_info

#### **ArchiveLink Object Type**

Specifies the SAP ArchiveLink object type.

**Limits:** Up to 10 characters

**JIL attribute:** arc\_obj\_type

#### **Archiving Path**

(Optional) Specifies the standard archive path.

**Limits:** Up to 70 characters; case-sensitive

**JIL attribute:** arc\_path

#### **Client**

(Optional) Specifies the SAP ArchiveLink client.

**Limits:** 0-999

**JIL attribute:** arc\_client

#### **Connection Name**

(Optional) Specifies the SAP ArchiveLink communication connection.

**Limits:** Up to 14 characters; case-sensitive

**JIL attribute:** arc\_connect

#### **Date**

(Optional) Specifies the date to do the archiving in YYYYMMDD format.

**Limits:** Up to eight characters

**JIL attribute:** arc\_date

#### **Document Class**

(Optional) Specifies the SAP ArchiveLink document class.

**Limits:** Up to 20 characters

**JIL attribute:** arc\_doc\_class

### **Document Type**

Specifies the SAP ArchiveLink document type.

**Limits:** Up to 10 characters

**JIL attribute:** arc\_doc\_type

### **Format**

(Optional) Specifies the SAP ArchiveLink output format.

**Limits:** Up to 16 characters; case-sensitive

**JIL attribute:** arc\_format

### **Printer**

(Optional) Specifies the SAP ArchiveLink target printer.

**Limits:** Up to four characters; case-sensitive

**JIL attribute:** arc\_printer

### **Protocol**

(Optional) Specifies the archive storage connection protocol.

**Limits:** Up to eight characters; case-sensitive

**JIL attribute:** arc\_protocol

### **Report**

(Optional) Specifies the SAP ArchiveLink report name.

**Limits:** Up to 30 characters; case-sensitive

**JIL attribute:** arc\_report

### **Service Destination**

(Optional) Specifies the SAP ArchiveLink RPC service/destination.

**Limits:** Up to 32 characters; case-sensitive

**JIL attribute:** arc\_service

### **Target Storage System**

(Optional) Specifies the SAP ArchiveLink target storage system.

**Limits:** Up to two characters

**JIL attribute:** arc\_storage

### **Text**

(Optional) Specifies the archiving text.

**Limits:** Up to 30 characters; case-sensitive

**JIL attribute:** arc\_text

### **User**

(Optional) Specifies the archive data element for the user.

**Limits:** Up to 12 characters

**JIL attribute:** arc\_userid

### Version

(Optional) Specifies the archive version.

**Limits:** Up to four characters

**JIL attribute:** arc\_version

## Send the SAP Spool File by Email on Step Completion or Failure

In an SAP R/3 or a Batch Input Session job, you can send the SAP spool file by email on step completion or failure.

### Follow these steps:

1. Specify the ABAPNAME statement to start the step definition within the job definition.
2. Specify the EMAILADDR statement after the ABAPNAME statement.  
To specify multiple email addresses, separate each address with a comma and enclose the entire list of addresses in apostrophes.
3. (Optional) Specify the following optional statements to determine whether the step is a success or failure based on a text string in the job's output:
  - SUCCESSMSG
  - FAILUREMSG

### NOTE

This step does not apply to Batch Input Session jobs.

### Example: Email Spool File to Two Addresses

This example emails the spool file to jsmith and djones upon completion or failure of the BTCTEST ABAP.

```
AGENT SAPHTAGENT
SAPUSER J01PROD
ABAPNAME BTCTEST
 VARIANT TEST
 EMAILADDR 'jsmith@company1.com,djones@company1.com'
```

## Manage PeopleSoft Jobs Using Web UI

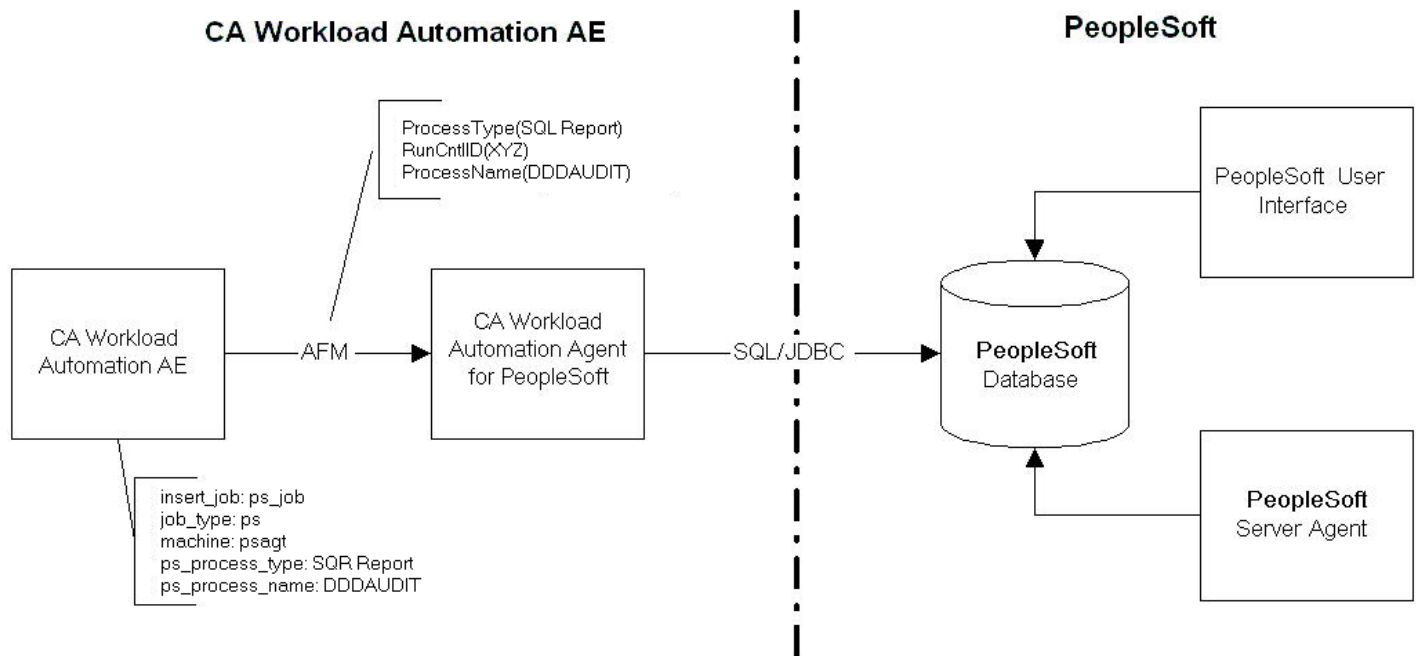
PeopleSoft jobs let you run different types of PeopleSoft processes defined in your PeopleSoft system. For example, you can define PeopleSoft jobs to execute PeopleSoft programs and report the program status.

### NOTE

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for PeopleSoft.

When you define a PeopleSoft job, you can set the output type and format of a report. For email and web output types, you can set various distribution properties such as the recipients and message text. You can also pass run control parameter values that will be stored in the corresponding run control table.

When a PeopleSoft program runs, it modifies its run status (RUNSTATUS) in the PSPRCRQST table in the PS database. The following diagram shows the functional relationship between the scheduling manager, the agent, and the PeopleSoft system:

**Figure 74: People Soft Jobs****Define a PeopleSoft Job**

You can define a PeopleSoft job to schedule workload to run in PeopleSoft. The job runs a PeopleSoft process request or a collection of process requests.

**NOTE**

To run these jobs, your system requires WA Agent for UNIX, Linux, or Windows and WA Agent for PeopleSoft.

**Follow these steps:**

1. Create a PeopleSoft (PS) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Process name**  
Specifies the name of the PeopleSoft report to run. This value corresponds to the Process Name field in PeopleSoft.
  - **Process type**  
Specifies the PeopleSoft process type corresponding to the process that the job runs. PeopleSoft stores the list of process types in the PS\_PRCSTYPEDEFN table. This value corresponds to the Process Type field in PeopleSoft.
3. (Optional) Specify optional [PeopleSoft properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The PeopleSoft job is defined.

## Mapping of PeopleSoft Field Names to PeopleSoft Job Properties

When you define a PeopleSoft job, you specify PeopleSoft job properties that map to your PeopleSoft process request. The following table maps the PeopleSoft fields to the properties:

PeopleSoft Field Name	PeopleSoft Job Property
Format	Output format
Type	Output type
Folder Name	Folder name
ID Type (Role selected) Distribution ID	Roles distribution list
ID Type (User selected) Distribution ID	User distribution list
Email Address List	Email address, Expanded email address
Email With Log	Email with log
Email Subject	Email subject
Message Text	Message text
Email Web Report	Email web report
Output Destination	Output destination path
Process Name	Process name
Process Type	Process type
Run Control Arguments	Run control arguments
Run Control ID	Run control ID
Run Control Table Name	Run control table name
Server Name	Server name
Time Zone	Time zone

## PeopleSoft Job Examples

The following examples describe sample PeopleSoft jobs.

### Example: Run a PeopleSoft Process

This example runs an Application Engine process named DDDAUDIT. The job runs on the agent named psagt.

#### Follow these steps:

1. Create a PeopleSoft job.
2. Enter the following properties:
  - Name -- audit
  - Send to machine -- psagt
  - Process name -- DDDAUDIT
  - Process type -- Application Engine
3. Commit the job.

#### NOTE

The job uses the default output destination format, output destination type, and run control ID defined on the agent. The PeopleSoft server that runs the job is not defined in the job definition or as a default on the agent, so the PeopleSoft process scheduler determines the PeopleSoft server that will run the job.

**Example: Distribute a Report to Users**

This example runs a Crystal report under the VP3 operator ID. The report is formatted as PDF and distributed in an email to the VP1, VP2, and VP3 operator IDs.

**Follow these steps:**

1. Create a PeopleSoft job.
2. Enter the following properties:
  - Name -- ps\_users
  - Send to machine -- psagt
  - Process name -- XRFWIN
  - Process type -- Crystal
  - Run control ID -- test
  - Output destination type -- EMAIL
  - Output destination format -- PDF
  - User distribution list -- VP1,VP2,VP3
3. Commit the job.

**Example: Email a PeopleSoft Report**

This example runs a Crystal report and emails the output to recipients. The Crystal report runs under the VP3 operator ID. The output is sent to the email addresses specified in the Email address field. The email includes a subject title and body text.

**Follow these steps:**

1. Create a PeopleSoft job.
2. Enter VP3@ps1 in the Owner field.
3. Enter the following properties:
  - Name -- ps\_email
  - Send to machine -- psagt
  - Process name -- XRFWIN
  - Process type -- Crystal
  - Run control ID -- test
  - Operator ID -- VP3@ps1
  - Output destination type -- EMAIL
  - Output destination format -- PDF
  - Email subject -- PeopleSoft Report Status
  - Message text -- This report is available for distribution
  - Email address -- user1@example.com;user2@example.com
4. Commit the job.

**Example: Format a PeopleSoft Job Output as an HTML Web Report**

This example runs the XRFWIN process. The process type is SQR Report and the run control ID is PS\_ALL. The server named PSPR runs the job, and the output is stored as an HTML web report.

**Follow these steps:**

1. Create a PeopleSoft job.
2. Enter VP3@ps1 in the Owner field.
3. Enter the following properties:

- Name -- ps\_htmfile
  - Send to machine -- psagt
  - Process name -- XRFWIN
  - Process type -- SQR Report
  - Operator ID -- VP3@ps1
  - Server name -- PSPR
  - Output destination type -- WEB
  - Output destination format -- HTM
  - Run control ID -- PS\_ALL
4. Commit the job.

### **Example: Send a Job's Output to a Specified Printer**

This example runs an SQR Report. The report is formatted as PS and sent as output to a printer.

#### **Follow these steps:**

1. Create a PeopleSoft job.
2. Enter VP3@ps1 in the Owner field.
3. Enter the following properties:
  - Name -- ps\_printer
  - Send to machine -- psagt
  - Process name -- XRFWIN
  - Process type -- SQR Report
  - Operator ID -- VP3@ps1
  - Output destination type -- PRINTER
  - Output destination format -- PS
  - Output destination path -- \\printers\PRINTER1
  - Run control ID -- test
4. Commit the job.

### **PeopleSoft Properties**

The PeopleSoft category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Arguments**

(Optional) Specifies additional arguments to append to the parameter list in the PeopleSoft database in the PSPRCSPARMS.PARMLIST field.

**Limits:** Up to 254 characters

**JIL attribute:** ps\_args

#### **Enable restart**



(Defaulted) Indicates whether to enable a restart feature for previously-failed jobs. When selected, previously-failed jobs restart from the beginning. When not selected, a failed job that is resubmitted restarts from where it stopped.

**Default:** selected

**NOTE**

- This option is used primarily in Application Engine programs.
- Ensure that the Enable restart checkbox is cleared so that the restart property for the job is disabled. Web UI does not support the use of RESTARTJOB events to resubmit PeopleSoft (PS) jobs. Peoplesoft application developers, system administrators, or application users who want to resubmit the batch processes that are associated with Peoplesoft jobs that enable the restart feature must use the PeopleSoft Process Scheduler directly.

**JIL attribute:** ps\_restarts

**Distribution: Email address**

Specifies the email addresses of the recipients on a distribution list. The PeopleSoft report is emailed to the recipients. You can specify multiple email addresses.

**Limits:** Up to 256 characters; separate each address with a semi-colon

**NOTE**

This property applies only if you specify EMAIL or WEB in the Output destination type field.

**JIL attribute:** ps\_email\_address

**Distribution: Email subject**

(Optional) Defines an email subject to include in the email to distribute to recipients of a PeopleSoft report. This value corresponds to the Email Subject field in PeopleSoft.

**Limits:** Up to 256 characters

**NOTE**

This property applies only if you specify EMAIL or WEB in the Output destination type field.

**JIL attribute:** ps\_email\_subject

**Distribution: Expanded email address**

(Optional) Specifies additional email addresses of the recipients on a distribution list. The PeopleSoft report is emailed to the recipients. You can specify multiple email addresses.

**Limits:** Up to 256 characters; separate each address with a semi-colon

**NOTE**

This property applies only if you specify EMAIL or WEB in the Output destination type field.

**JIL attribute:** ps\_email\_address\_expanded

**Distribution: Folder name**

(Optional) Specifies the name of a distribution detail folder. This value corresponds to the Folder Name field in PeopleSoft.

**Limits:** Up to 18 characters

**NOTE**

This property applies only if you specify Email or Web in the Output destination type field.

**JIL attribute:** ps\_detail\_folder

**Distribution: Message text**

(Optional) Defines the body text of the email. This value corresponds to the Message Text field in PeopleSoft.

**Limits:** Up to 1000 characters

**NOTE**

This property applies only if you specify Email or Web in the Output destination type field.

**JIL attribute:** ps\_email\_text

**Distribution: Roles distribution list**

(Optional) Specifies a role or a list of roles to send the report to. This value corresponds to the ID Type field (with Role selected) and the Distribution ID field in PeopleSoft.

**Limits:** Up to 256 characters; separate multiple roles with a comma

**Example:** Role1, Role2

**NOTE**

This property applies only if you specify EMAIL or WEB in the Output destination type field.

**JIL attribute:** ps\_dlist\_roles

**Distribution: User distribution list**

(Optional) Specifies a distribution list of operator IDs to send a PeopleSoft report to. This value corresponds to the ID Type field (with User selected) and the Distribution ID field in PeopleSoft.

**Limits:** Up to 256 characters; separate multiple users with a comma

**Example:** User1, User2

**NOTE**

This property applies only if you specify EMAIL or WEB in the Output destination type field.

**JIL attribute:** ps\_dlist\_users

**Distribution: Email web report**

(Defaulted) Indicates whether to email a web report to recipients on a distribution list. When selected, the agent emails the web report.

**Default:** not selected (the web report is not emailed)

**JIL attribute:** ps\_email\_web\_report

**Distribution: Email with log**

(Defaulted) Indicates whether to email job logs along with the PeopleSoft report to recipients on a distribution list. When selected, the agent emails the job logs.

**Default:** not selected (the job does not email job logs)

**JIL attribute:** ps\_email\_log

**Operator ID**

(Optional) Specifies the operator ID under whose authority the PeopleSoft reports run, formatted as follows:

*PeopleSoft\_operator\_ID@tag*

**Limits:**

- For CA Workload Automation AE r11.3, up to 30 characters total; the *PeopleSoft\_operator\_ID* is limited to five characters
- For CA Workload Automation AE Release 11.3.5 and later, up to 111 characters total; the *PeopleSoft\_operator\_ID* is limited to 30 characters

**Example:** ptdmo@peoplesoft

**JIL attribute:** ps\_operator\_id

### Output destination path

(Optional) Specifies the output destination for the PeopleSoft request. The destination can be a file directory or a printer using the following format:

`file_path | printer_path`

- ***file\_path***

Specifies the path to the output directory and the output file name.

**Limits:** Up to 127 characters; case-sensitive

**Default:** The PeopleSoft log/output directory

**NOTE**

Specify this path when the Output destination type is FILE.

- ***printer\_path***

Specifies the network location of the printer, including the printer server and shared printer name.

**Limits:** Up to 127 characters; case-sensitive

**NOTE**

- Specify this path when the Output destination type is PRINTER.
- The specified printer must be set up on the PeopleSoft system.
- Your agent administrator can specify a default printer by setting the ps.default.printer parameter in the agent's agentparm.txt file; however, a value specified here overrides the default printer specified in the agent's agentparm.txt.

**JIL attribute:** ps\_output\_dest

### Output destination format

(Optional) Specifies the type of format for the report output, as follows:

- ANY -- Any
- NONE -- (None)
- PDF -- Acrobat (\*.pdf)
- CSV -- Comma-delimited (\*.csv)
- HP -- HP Format (\*.lis)
- HTM -- HTML Documents (\*.htm)
- LP -- Line printer format (\*.lis)
- WKS -- Lotus 1-2-3 files
- XLS -- Microsoft Excel Files (\*.xls)
- DOC -- Microsoft Word (\*.doc)
- PS -- Postscript (\*.lis)
- RPT -- Crystal Report (\*.rpt)
- RTF -- Rich Text File (\*.rtf)
- SPF -- SQR Portable Format (\*.spf)
- TXT -- Text Files (\*.txt)
- OTHER -- Other
- Default -- Default
- XML -- XML Format (\*.xml)
- DAT -- Data Mover Data File (\*.dat)

**JIL attribute:** ps\_dest\_format

#### **Output destination type**

(Optional) Specifies the output destination type for a PeopleSoft report, as follows:

- NONE -- Specifies no output destination type.
- FILE -- Sends the output of the PeopleSoft report to a file.
- PRINTER -- Sends the output of the PeopleSoft report to a printer.
- EMAIL -- Sends the output of the PeopleSoft report as an email message.
- WEB -- Posts the output of the PeopleSoft report on a website.

**JIL attribute:** ps\_dest\_type

#### **Process name**

Specifies the name of the PeopleSoft report to run. This value corresponds to the Process Name field in PeopleSoft. PeopleSoft stores the list of process names in the PS\_PRCSEDEFN table.

**Limits:** Up to 12 characters

**JIL attribute:** ps\_process\_name

#### **Process type**

Specifies the PeopleSoft process type corresponding to the process that the job runs. PeopleSoft stores the list of process types in the PS\_PRCSTYPEDEFN table. This value corresponds to the Process Type field in PeopleSoft.

The following types are available:

- Application Engine
- COBOL SQL
- Crw Online
- Crystal
- Crystal Check
- Cube Builder
- Database Agent
- Demand Planning Upload
- Essbase
- HyperCube Builder
- Message Agent API
- nVision
- nVision-Report
- nVison-ReportBook
- Optimization Engine
- PSJob
- SQR PO-Special Process
- SQR Process
- SQR Report
- SQR Report For WF Delivery
- Winword

**Limits:** Up to 30 characters; case-sensitive

Clicking



opens the Select Process Type dialog, which lets you select the process type.

**JIL attribute:** ps\_process\_type

### Run control arguments

(Optional) Specifies the run control arguments for a run control table on the PeopleSoft system. You can specify multiple arguments.

**Limits:** Up to 1024 characters for each argument; separate each argument with a comma

- You must also specify the run control table using the Run control table name property.
- The arguments must match the structure in the run control table.
- Clicking



opens the Edit Text dialog, which lets you enter the run control arguments.

**JIL attribute:** ps\_run\_cntrl\_args

### Run control ID

(Optional) Specifies the value assigned to the run control identifier. This value corresponds to the Run Control ID field in PeopleSoft.

**Limits:** Up to 30 characters; case-sensitive

**Example:** FLOOR8\_COLOR

**NOTE**

- In PeopleSoft, some processes spawn children using the CreateProcessRequest PeopleCode function. PeopleSoft may identify the parent process as successfully completed even if one or more of its children fail. If you want the agent to check the status of children when determining the completion status of the parent, add ".\*" to the end of the Run Control ID, for example, PS\_ALL.\*.
- Your agent administrator can also set a default value for this property using the ps.default.runCntlId parameter in the agentparm.txt file.

**JIL attribute:** ps\_run\_cntrl\_id

**Run control table name**

(Optional) Specifies the name of the table that contains the run parameters for a given PeopleSoft process.

**Limits:** Up to 50 characters

**NOTE**

You must specify run control arguments using the Run control arguments field.

**JIL attribute:** ps\_run\_control\_table

**Server name**

(Optional) Specifies the name of the target server that runs the PeopleSoft job. PeopleSoft stores the list of server names in the PS\_SERVERDEFN table. This value corresponds to the Server Name field in PeopleSoft.

**Limits:** Up to eight characters, cannot contain delimiters (such as spaces)

**Example: PSNT****NOTE**

If the PeopleSoft Server is not specified as a default on the agent or in the job definition, the PeopleSoft Process Scheduler determines the PeopleSoft Server that will run the job.

**JIL attribute:** ps\_server\_name

**Skip parameter updates**

(Optional) Indicates whether you want the agent to update job parameters with data in the PS\_PRCSEFN table. When selected, the agent does not update job parameters. When the updates are skipped, you can use the Run control arguments field to pass additional argument values.

**JIL attribute:** ps\_skip\_parm\_updates

**Time zone**

(Optional) Specifies a different time zone for the report being run. This value corresponds to the Time Zone field in PeopleSoft.

**Limits:** Up to nine characters

**NOTE**

You can view the time zone settings in PeopleSoft.

**JIL attribute:** ps\_time\_zone

**Manage FTP and Secure Copy Jobs Using Web UI**

This article explains how to manage FTP and secure copy jobs.

## FTP Jobs

Using your agent, you can automate File Transfer Protocol (FTP) transfers with an FTP job. The FTP job can upload data to or download data from an existing FTP server or another agent running as an FTP server. The FTP job always acts as an FTP client.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

You can use an FTP job to automate the following:

- Download ASCII, binary, or EBCDIC (i5/OS only) files from a remote FTP server to your agent computer.
- Upload ASCII, binary, or EBCDIC (i5/OS only) files from your agent computer to a remote FTP server.

Your agent administrator can set up the agent to run as an FTP client, FTP server, or both.

## Secure Copy Jobs

You can define a Secure Copy job to transfer binary files between an agent computer and a remote computer. The Secure Copy job can upload data to or download data from a remote server. The data is encrypted during the transfer. By default, a Secure Copy job uses the SFTP protocol. However, you can define the job to use the SCP protocol.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS. The agent must be configured as an FTP client using the Secure Copy Protocol or the Secure File Transfer Protocol.

## Define a File Transfer Protocol (FTP) Job

You can define an FTP job to automate FTP transfers. The output is directed to the spool file through an FTP server.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Create a File Transfer Protocol (FTP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Server name**  
Specifies the DNS name or IP address of a remote server.
  - **Local file name**  
Specifies the destination of the file (if downloading) or the source location of the file (if uploading).  
Clicking  the Browse Agent File System dialog, which lets you locate the local file destination or location.
3. **Remote file name**
4. Specifies the source location of the file (if downloading) or the destination of the file (if uploading).
5. (Optional) Specify the following property in the Primary category:
  - **Owner**  
Specifies the user ID with the authority to download the file from the remote FTP server or upload the file to the remote FTP server.

**NOTE**

The job runs under the owner of the job by default. The Owner property overrides the default value.

6. (Optional) Specify optional [FTP properties](#).
7. (Optional) Specify [common properties that apply to all job types](#).
8. Commit the job.  
The FTP job is defined.

**FTP Job Examples**

The following examples describe sample FTP jobs.

**Example: Download a File from a UNIX System to a Windows System**

In this example, a file named textfile is downloaded from a UNIX system and then copied to a local Windows system. Note that each of the two locations includes a complete path statement. After the download is complete, the job completes.

**Follow these steps:**

1. Create an FTP job.
2. Enter the following properties:
  - Name -- unix\_textfile
  - Send to machine -- AGENT
  - Server name -- HPUNIX
  - Server port -- 5222
  - Transfer direction -- DOWNLOAD
  - Transfer type -- ASCII
  - Local file name -- c:\qatest\ftpdata\textfile
  - Remote file name -- /u1/qatest/ftpdata/textfile
3. Commit the job.

**Example: Upload an ASCII-Encoded File in the Root File System from an i5/OS System to a UNIX System**

In this example, a file named textfile in the root file system is uploaded from an i5/OS system to a UNIX system. Note that each of the two locations includes a complete path statement. After the upload is complete, the job completes.

**Follow these steps:**

1. Create an FTP job.
2. Enter the following properties:
  - Name -- i5\_textfile
  - Send to machine -- I5AGENT
  - Server name -- HPUNIX
  - Server port -- 5222
  - Transfer direction -- UPLOAD
  - Transfer type -- ASCII
  - Local file name -- /home/cybesp/textfile
  - Remote file name -- /u1/qatest/ftpdata/textfile
3. Commit the job.

**Example: Download a QSYS EBCDIC-Encoded File**

In this example, an EBCDIC-encoded file named datafile in the QSYS file system is downloaded from an i5/OS system to another i5/OS system. Note that the file names are specified in the path format.



**Follow these steps:**

1. Create an FTP job.
2. Enter the following properties:
  - Name -- datafile
  - Send to machine -- I5AGENT
  - Server name -- i5UNIX
  - Server port -- 5222
  - Transfer direction -- DOWNLOAD
  - Transfer type -- EBCDIC
  - Local file name -- /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
  - Remote file name -- /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
3. Commit the job.

**Example: Upload a Windows File to a Mainframe Dataset**

This example uploads a file named tests.txt to the mainframe. By default, the Remote file name uses the user ID of the logged-on user for the dataset high-level prefix.

**NOTE**

To specify a different high-level prefix, fully qualify the value of the Remote file name statement and wrap quotation marks around it.

**Follow these steps:**

1. Create an FTP job.
2. Enter the following properties:
  - Name -- mainframe\_push
  - Send to machine -- localhost
  - Local file name -- "C:\My Documents\Microsoft Office Directories WordPad\tests.txt"
  - Remote file name -- TESTDATA
  - Server name -- MAINFRAME.COM
  - Transfer type -- ASCII
  - Transfer direction -- UPLOAD
  - Owner -- causer@mainframe
3. Commit the job.

**Example: Download a Mainframe Dataset to a Windows File**

This example downloads the dataset member CA07XX01 to a Windows computer. The value in the Remote file name field is fully qualified by wrapping quotation marks around the apostrophes.

**Follow these steps:**

1. Create an FTP job.
2. Enter the following properties:
  - Name -- mainframe\_pull
  - Send to machine -- localhost
  - Local file name -- "C:\My Documents\File Transfers\CA07XX01.txt"
  - Remote file name -- SYS4.CA7.JCLLIB(CA07XX01)
  - Server name -- MAINFRAME.COM
  - Transfer type -- ASCII
  - Transfer direction -- DOWNLOAD
  - Owner -- lmla@mainframe

3. Commit the job.

### **File Transfer Protocol Properties**

The File Transfer Protocol category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Compression**

(Optional) Specifies the data compression level.

**Limits:** 0-9 (0 is no data compression and 9 is the highest data compression)

#### **NOTE**

If the Compression operand is not specified, the data is compressed using the level set in the ftp.data.compression parameter in the agentparam.txt file on the agent FTP client. This parameter is automatically set to 0 by default (no data compression).

**JIL attribute:** ftp\_compression

#### **Initialization command**

(Optional) Defines a comma-separated list of commands that are to run prior to file transfer.

**Limits:** Up to 1024 characters

- You can specify any number of initialization commands for an FTP job.
- Clicking  the Edit Text dialog, which lets you enter the commands for an FTP job.
- For more information about using the initialization command, see the *AutoSys Workload Automation documentation*.

**JIL attribute:** ftp\_command

#### **Local file name**

Specifies the destination of the file (if downloading) or the source location of the file (if uploading).

This property supports Global Variable substitution.

**Limits:** The total of all files specified cannot exceed 256 characters; case-sensitive

#### **UNIX/Windows:**

- If you are downloading a file, specify the full path and file name.
- If you are uploading files, you can use wildcards in the file name. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.
- You cannot use wildcards in the path.

#### **i5/OS:**

- To specify a file in the root file system, use UNIX path and file formats.
- To specify a \*FILE object in QSYS, use the following format:  
/QSYS.LIB/libraryname.LIB/objectname.FILE/membername.MBR

**NOTE**

- Clicking opens the Browse Agent File System dialog, which lets you locate the local file destination or location.
- To specify multiple names, separate each value by a semi-colon (;).

**JIL attribute:** ftp\_local\_name

**Local user**

(Defaulted) Specifies the user ID to be used to determine ownership of the downloaded file. The user ID must have super-EDIT authority if using native security or as-owner authority if using CA EEM; otherwise, the value specified is ignored.

**Limits:** Up to 80 characters; case-sensitive

**Default:** *authenticated\_user@host*

**NOTE**

- This property applies only to UNIX.
- If you do not specify a value, the authenticated user ID is used by default.

**JIL attribute:** ftp\_local\_user

**Run as user**

Specifies the user ID that runs the job on behalf of the agent user. You can use this field to access remote resources that the agent user does not have access to. You are restricted to how you can access data on remote computers. To access restricted remote resources, you can run the job under a user ID that has access to those resources. The user must be defined in the Topology. This field supports the use of a namespace for a user that has more than one password. Contact your administrator for the user name defined in the Topology.

**Examples:** Bob, Production:Bob

**JIL attribute:** external\_os\_user

**Remote file name**

Specifies the source location of the file (if downloading) or the destination of the file (if uploading).

This property supports Global Variable substitution.

**Limits:** The total of all files specified, with separators, cannot exceed 256 characters; case-sensitive

**UNIX/Windows:**

- If you are uploading a file, specify the full path and file name.
- If you are downloading files, you can use wildcards in the file name. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.
- You cannot use wildcards in the path.
- If you want to use a Windows file as a remote file, use a forward slash at the beginning of the path statement and between the directories and file name. For example:

*remotefilename /TEMP/textfile*

**i5/OS:**

- To specify a file in the root file system, use UNIX path and file formats.
  - To specify a \*FILE object in QSYS, use the following format:
- /QSYS.LIB/libraryname.LIB/objectname.FILE/membername.MBR*

**NOTE**

To specify multiple names, separate each value by a semi-colon (;).

---

**JIL attribute:** ftp\_remote\_name

**Server name**

Specifies the DNS name or IP address of a remote server.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**Example:** 172.24.36.107 (IPv4) or 0:0:0:0:FFFF:192.168.00.00 (IPv6)

**JIL attribute:** ftp\_server\_name

**Server port**

(Optional) Specifies the port number of the remote server.

**Limits:** 0-65535

**JIL attribute:** ftp\_server\_port

**Transfer direction**

(Defaulted) Specifies the file transfer direction between the agent computer and the remote server, as follows:

- **UPLOAD** -- Transfers files from the agent computer to the remote server.
- **DOWNLOAD** -- Transfers files from the remote server to the agent computer.

**Default:** DOWNLOAD

**JIL attribute:** ftp\_transfer\_direction

**Transfer type**

(Defaulted) Specifies the type of data involved in an FTP transfer: ASCII, Binary, or EBCDIC (i5/OS remote host only).

**Default:** Binary

**JIL attribute:** ftp\_transfer\_type

**Use SSL**

(Defaulted) Indicates whether the FTP data transfer uses Secure Sockets Layer (SSL) FTP or regular FTP.

**Default:** not selected (uses regular FTP)

**JIL attribute:** ftp\_use\_ssl

**User Type**

Specifies how the value of the owner attribute is passed to the FTP server for authentication, as follows:

- **Simple** -- Transfers only the user part of the owner attribute value. For example, if you set the owner attribute to userxyz@domain, userxyz is passed to the FTP server for authentication.
- **Windows** -- Transfers the value of the owner attribute in the form of domain\user. For example, if you set the owner attribute to userxyz@domain, domain\userxyz is passed to the FTP server for authentication.
- **Qualified** -- Transfers the value of the owner attribute as an email address or fully-qualified domain name. For example, if you set the owner attribute to userxyz@domain, userxyz@domain is passed to the FTP server for authentication.

**Default:** Simple

**JIL attribute:** ftp\_user\_type

## Define a Secure Copy Job

You can define a Secure Copy job to transfer binary files using the Secure Copy Protocol (SCP) or the Secure File Transfer Protocol (SFTP).

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, Windows, or i5/OS.

### Follow these steps:

1. Create a Secure Copy (SCP) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Server name**  
Specifies the DNS name or IP address of a remote server.
  - **Remote directory**  
Specifies the file's remote source directory (if downloading) or the file's remote destination directory (if uploading).
  - **Remote file name**  
Specifies the file's source location (if downloading) or the file's destination (if uploading).
  - **Local file name**  
Specifies the file's destination (if downloading) or the file's source location (if uploading).

Clicking  the Browse Agent File System dialog, which lets you locate the local file destination or location.
3. (Optional) Specify optional [Secure Copy properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Secure Copy job is defined.

## Secure Copy Job Examples

The following examples describe sample Secure Copy jobs.

### Example: Download a File from a UNIX System to a Windows System

In this example, a file named textfile is downloaded from a UNIX system and then copied to a local Windows system. Note that each of the two locations includes a complete path statement. After the download is complete, the job completes.

### Follow these steps:

1. Create a Secure Copy job.
2. Enter the following properties:

- Name -- unix\_textfile
  - Send to machine -- AGENT
  - Server name -- HPUNIX
  - Transfer direction -- DOWNLOAD
  - Remote OS type -- UNIX
  - Remote directory -- /u1/qatest/ftpdata
  - Remote file name -- /u1/qatest/ftpdata/textfile
  - Local file name -- c:\qatest\ftpdata\textfile
3. Commit the job.

### **Example: Upload a File Using Secure File Transfer Protocol**

In this example, all files located in the ftpdata directory on a remote UNIX server are uploaded to the temporary directory on a Windows computer using SFTP.

#### **Follow these steps:**

1. Create a Secure Copy job.
2. Enter the following properties:
  - Name -- wildcard
  - Send to machine -- AGENT
  - Server name -- UNIX
  - Transfer direction -- UPLOAD
  - Remote OS type -- UNIX
  - Remote directory -- /u1/qatest/ftpdata
  - Remote file name -- /u1/qatest/ftpdata/\*
  - Local file name -- c:\temp\upload\\*
  - Protocol -- SFTP
3. Commit the job.

### **Secure Copy Properties**

The Secure Copy category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Local file name**

Specifies the file's destination (if downloading) or the file's source location (if uploading). The path should be included. The local file or files are located on the computer where the agent is installed.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**Note:**

- Clicking  the Browse Agent File System dialog, which lets you locate the local file destination or location.
- You can use wildcards in the file name only if the Use SFTP property is selected. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character. You cannot rename files if wildcards are used. You must specify the full path and file name. You cannot use wildcards in the path.

**JIL attribute:** scp\_local\_name

### Local user

(Optional) Specifies a user ID on the computer where the agent is installed.

**Limits:** Up to 80 characters; case-sensitive

**Default:** the user ID that defined the Secure Copy job

**JIL attribute:** scp\_local\_user

### Run as user

Specifies the user ID that runs the job on behalf of the agent user. You can use this field to access remote resources that the agent user does not have access to. You are restricted to how you can access data on remote computers. To access restricted remote resources, you can run the job under a user ID that has access to those resources. The user must be defined in the Topology. This field supports the use of a namespace for a user that has more than one password. Contact your administrator for the user name defined in the Topology.

**Examples:** Bob, Production:Bob

**JIL attribute:** external\_os\_user

### Protocol

(Defaulted) Specifies whether the SCP data transfer uses Secure File Transfer Protocol (SFTP) or regular Secure Copy (SCP).

**Default:** SFTP

**JIL attribute:** scp\_protocol

### Remote directory

Specifies the file's remote source directory (if downloading) or the file's remote destination directory (if uploading). The remote directory is not located on the computer where the agent is installed.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

You cannot use wildcards in the path.

**JIL attribute:** scp\_remote\_dir

### Remote file name

Specifies the file's source location (if downloading) or the file's destination (if uploading). The remote file or files are not located on the computer where the agent is installed.

This property supports Global Variable substitution.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

You can use wildcards in the file name only if the Use SCPv2 property is selected. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character. You cannot rename files if wildcards are used.

**JIL attribute:** scp\_remote\_name

**Transfer type**

(Defaulted) Specifies the type of data involved in an SCP transfer: ASCII, Binary, or /tmp.

**Default:** Binary

**JIL attribute:** scp\_transfer\_mode

**Remote OS type**

(Defaulted) Specifies the remote OS type, which is used to determine the path separator on the remote system. The following remote OS types are supported:

- UNIX
- Windows
- OpenVMS
- Tandem
- I5/OS
- z/OS

**Default:** UNIX

**JIL attribute:** scp\_target\_os

**Server name**

Specifies the DNS name or IP address of a remote server.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** scp\_server\_name

**Server port**

(Defaulted) Specifies the port number of the remote server.

**Limits:** 0-65535

**Default:** 22

**JIL attribute:** scp\_server\_port

**Transfer direction**

(Defaulted) Specifies the file transfer direction between the agent computer and the remote server, as follows:

- UPLOAD -- Transfers files from the agent computer to the remote server.
- DOWNLOAD -- Transfers files from the remote server to the agent computer.

**Default:** DOWNLOAD

**JIL attribute:** scp\_transfer\_direction

Clicking [opens](#) the Browse Agent File System dialog, which lets you locate the local file destination or location. To specify multiple names, separate each value by a semi-colon (;).

**File operation**

Specifies if you want to rename, delete or not delete files.



Options are :

- Do not delete
- Delete
- Rename

### Delete source directory

Indicates the deletion of the directory containing the local source file (if uploading) or the remote source file (if downloading) after the file is uploaded or downloaded. To delete the source directory along with the source file, select this checkbox.

**JIL attribute:** scp\_delete\_sourcedir

### Delete source file(s)

Indicates the deletion of the local source files (if uploading) after they are uploaded from the agent computer or the remote source files (if downloading) after they are downloaded from the remote SFTP server.

**JIL attribute:** scp\_delete\_sourcefile

### Rename source file

Specifies the new source file name entered by you. When renaming a file, you cannot use wildcards (\* or ?) in the local file name field (if uploading) or remote file name field (if downloading).

**JIL attribute:** scp\_rename\_sourcefile

### Create target directory

Indicates the creation of a new directory, if no directory is present.

**JIL attribute:** scp\_create\_targetdir

## Manage i5/OS Jobs Using Web UI

The i5/OS job lets you run a program or issue a command on an i5/OS system. You can run i5/OS jobs in the following file systems:

- Root file system
- Open systems file system (QOpenSys)
- Library file system (QSYS)

### NOTE

To run these jobs, your system requires Workload Automation Agent for i5/OS.

You can specify the following details in an i5/OS job definition:

- Library name, library list, and current library for running a program
- The i5/OS job name, options under which the job will run, where it will run, and which user will run it
- Ending exit value of the program, such as a severity code

You can define parameter values that you want to pass to a program at the time the program is invoked.

### NOTE

Default values may be set for certain parameters, such as the i5/OS user ID that the jobs run under. Contact your agent administrator about the parameters set in the agentparm.txt file.

## Running UNIX Workload on a System i5 Computer

In addition to scheduling native i5/OS jobs, you can schedule most UNIX workload, such as UNIX scripts, in the PASE environment on i5/OS.

To run both native and UNIX jobs on the same i5/OS computer, you must install two i5/OS agents and configure the `oscomponent.targetenvironment` parameter in the `agentparm.txt` file to handle the appropriate job type. For more information about configuring the parameter, see the *AutoSys Workload Automation UNIX Implementation section* or *Windows Implementation section*.

#### NOTE

For more information about UNIX workload that can run in the PASE environment, see the IBM i5/OS documentation.

### **i5/OS Naming Conventions**

When specifying i5/OS paths and names in your workload, you can use the following naming conventions, depending on where the file is located on the i5/OS system:

- Root file system  
To specify a file in the root file system, use UNIX path and file formats.
- Open systems file system (QOpenSys)  
To specify a file in QOpenSys, use UNIX path and file formats. QOpenSys file names are case-sensitive.
- Library file system (QSYS)  
To specify an object in QSYS, use one of the following formats (unless described differently in the job definition syntax):

- Path format

`/QSYS.LIB/library.LIB/object.type/`

To specify \*FILE objects, use the following format:

`/QSYS.LIB/library.LIB/object.FILE/member.MBR`

- i5/OS standard format

`library/object/type`

To specify \*FILE objects, use the following format:

`library/object/*FILE (member)`

#### NOTE

\*FILE is optional when *member* is specified. That is, you can specify a file member using the following format:

`library/object (member)`

#### NOTE

- The values for *library*, *object*, *type*, and *member* can be up to 10 characters each.
- You can use \*ALL to match any name.
- You can use \*FIRST for *member*.
- You can use generic names for *library* and *object*.

### **Define an i5/OS Job**

You can define an i5/OS job to schedule workload to run on an i5/OS system. The job can run a program or an i5/OS command.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for i5/OS.

### **Follow these steps:**

1. Create an i5/OS (I5) job.
2. Specify the following required properties:

- **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
- **Send to machine**  
Specifies the name of the machine on which the job runs.
- **Action (CMD/Program) name**  
Specifies the program, the source file for the program, or the command that you want to run.
  - The value must correspond to the action value. If you do not specify the action value, the job interprets the corresponding Action (CMD/Program) name value as a command by default.
  - Clicking  the Browse Agent File System dialog, which lets you locate the program, source file for the program, or the command on the agent computer.
- 3. (Optional) Specify optional [i5/OS properties](#).
- 4. (Optional) Specify [common properties that apply to all job types](#).
- 5. Commit the job.  
The i5/OS job is defined.

### **i5/OS Job Examples**

The following examples describe sample i5/OS jobs.

#### **Example: Run an i5/OS Command**

This example runs the command named CALC on the i5agent computer.

##### **Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_runcmd
  - Send to machine -- i5agent
  - Action (CMD/Program) name -- CALC
3. Commit the job.

#### **Example: Pass Multiple Parameters to an i5/OS Job**

This example passes six parameters to an i5/OS program named PAYJOB. The parameter VALUE C is enclosed within apostrophes because it contains a space.

##### **Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_lib
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- PAYJOB
  - Positional parameters -- "ABC 1 P 'VALUE C' X r"
3. Commit the job.

#### **Example: Specify the Printer and Output Queue for an i5/OS Job**

This example runs a program named PAYJOB on an i5/OS system. The printer and output queue information is taken from the job definition.

**Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_lib
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- PAYJOB
  - Keyword parameters -- PRTDEV=\*JOB,OUTQ=\*JOB

**NOTE**

You must select Add from the Select an action drop-down list, and then click Go for each Keyword parameter you want to add.

3. Commit the job.

**Example: Send an OPM COBOL Program's Return Code as the Job's Exit Code**

This example runs an OPM COBOL program named PAYROLL. The agent sends the PAYROLL program's return code to AutoSys Workload Automation.

**Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_returnOPM
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- PAYROLL
  - i5/OS exit program -- \*PROGRAM
3. Commit the job.

**Example: Send an ILE C Program's Return Code as the Job's Exit Code**

This example runs a C language program named SALARY. The agent sends the SALARY program's return code to AutoSys Workload Automation. Ending severity codes of 0 (zero) or 10 indicate job success.

**Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_returnOPM
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- SALARY
  - i5/OS exit program -- \*USER
3. Enter the following Termination property:
  - Success exit codes -- 0,10
4. Commit the job.

**Example: Send Exit Code 100 as Success**

This example runs the PAYPROG program. The program is considered to have completed successfully if it returns an exit code of 0 or 100.

**Follow these steps:**

1. Create an i5/OS job.

2. Enter the following properties:
  - Name -- i5job\_succ
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- PAYPROG
  - i5/OS exit program -- \*PROGRAM
3. Enter the following Termination property:
  - Success exit codes -- 0,100
4. Commit the job.

#### **Example: Send Exit Code 40 as Failure**

This example runs the RECPROG program. The program is considered to have failed if it returns an exit code of 40. All other exit codes in the range from 0 to 255 indicate job success.

#### **Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_fail
  - Send to machine -- i5agent
  - Action -- RUN\_PROGRAM
  - Action (CMD/Program) name -- RECPROG
  - i5/OS exit program -- \*PROGRAM
3. Enter the following Termination properties:
  - Success exit codes -- 0-255
  - Failure exit codes -- 40
4. Commit the job.

#### **Example: Specify Data for the Local Data Area in Hexadecimal Format**

This example defines an i5/OS job with data for the local data area. When the job is submitted, the agent initializes the local data area with hexadecimal data. When the job completes, the local data area is destroyed automatically by the operating system.

#### **Follow these steps:**

1. Create an i5/OS job.
2. Enter the following properties:
  - Name -- i5job\_lda
  - Send to machine -- i5agent
  - Action -- COMMAND
  - Action (CMD/Program) name -- IVP
  - Local data area -- X'C1C2C3C4'
3. Commit the job.

### **i5/OS Properties**

The i5/OS category includes the following properties:

#### **Action**

(Defaulted) Specifies whether to run a program or issue a command on an i5/OS system, as follows:

- **RUN\_PROGRAM** -- Runs a program on an i5/OS system.
- **RUN\_PROG\_IN\_FILE** -- Runs a program from a database file member on an i5/OS system. This is similar to running a shell script on a UNIX system or a batch file on a Windows system.
- **COMMAND** -- Issues a command on an i5/OS system when the i5/OS job runs.

**Default:** COMMAND

#### **NOTE**

When the **RUN\_PROG\_IN\_FILE** option is selected, the Positional parameters field is disabled.

**JIL attribute:** i5\_action

#### **Action (CMD/Program) name**

Specifies the program, the source file for the program, or the command that you want to run.

This property supports Global Variable substitution.

**Limits:** Up to 120 characters for the **COMMAND** action and up to 60 characters for the other i5 actions; case-sensitive

Clicking 

the Browse Agent File System dialog, which lets you locate the program, source file for the program, or the command on the agent computer.

**JIL attribute:** i5\_name

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Current library**

(Optional) Specifies the current library for the job.

**Limits:** Up to 46 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** i5\_curr\_lib

#### **i5/OS exit program**

(Defaulted) Specifies the type of exit code returned by an i5/OS job, as follows:

- **\*SEVERITY** -- Sends the job's ending severity code.
- **\*USER** -- Sends the return code of an ILE program or module. For example, if your job runs an ILE C or ILE RPG program that contains an exit or return statement, that return code is sent as the exit code.
- **\*PROGRAM** -- Sends the return code of an OPM program. For example, if your job runs an OPM RPG or OPM COBOL program that contains an exit or return statement, that return code is sent as the exit code.

**Default:** \*SEVERITY

**JIL attribute:** i5\_cc\_exit

#### **i5/OS job description**

(Optional) Specifies the job description for the submitted program using one of the following formats:

*description*  
*library/description*

---

```
/QSYS.LIB/library.LIB/description.JOBD
```

- **description** Specifies the name of the job description for the program submitted. The value must be a valid i5/OS job description name.
- **library** Specifies the name of the library that contains the job description. The value must be a valid i5/OS library name.

**Limits:** Up to 46 characters; cannot contain delimiters (such as spaces)

Clicking 

the Browse Agent File System dialog, which lets you locate the job description on the agent computer.

**JIL attribute:** i5\_job\_desc

### i5/OS job name

(Optional) Specifies the name of the i5/OS job.

**Limits:** Up to 10 characters; the first character must be one of the following: A-Z, \$, #, or @; the second to tenth characters can be A-Z, 0-9, \$, #, \_, ., or @

**JIL attribute:** i5\_job\_name

### i5/OS job queue

(Optional) Specifies the i5/OS job queue for the submitted program using one of the following formats:

```
jobqueue
library/jobqueue
/QSYS.LIB/library.LIB/jobqueue.JOBQ
```

- **jobqueue** Specifies the name of the job queue for the submitted program. The value must be a valid i5/OS job queue name.
- **library** Specifies the name of the library that contains the job description. The value must be a valid i5/OS library name.

**Limits:** Up to 46 characters; case-sensitive

Clicking 

the Browse Agent File System dialog, which lets you locate the i5/OS job queue on the agent computer.

**JIL attribute:** i5\_job\_queue

### i5/OS process priority

(Defaulted) Specifies the order in which processes are scheduled on the System i5 processor, as follows:

- High -- Indicates processes that must be executed immediately. These processes can use nearly all available CPU time.
- Above normal -- Indicates processes that have priority above the Normal level, but below the High level.
- Normal -- Indicates processes without special scheduling needs.
- Below normal -- Indicates processes that have priority above the Idle level, but below the Normal level.
- Idle -- Indicates processes that will run only when the system is idle.

Depending on the priority level, process priority can speed up or slow down a process.

**Default:** Normal

**JIL attribute:** i5\_process\_priority

### Keyword parameters

(Optional) Specifies any valid SBMJOB command keyword and value combination. The value corresponds to the SBMJOB command keyword (the value that would appear in brackets after the keyword in the SBMJOB command).

**Limits:** Up to 4096 characters; enclose values that contain spaces in apostrophes

- You can specify multiple keyword=value combinations by selecting Add from the Select an action drop-down list, and clicking Go. A new row displays below the Keyword parameters field.
- Clicking  the Edit Parameter dialog, which lets you specify the command keyword and value combination

**JIL attribute:** i5\_others

### Library

(Optional) Specifies the name of the library that contains the program, the source file for the program, or the command that you want to run. The value must be a valid i5/OS library name.

**Limits:** Up to 46 characters; cannot contain delimiters (such as spaces)

- Instead of using the Library field, you can specify the library name in the i5/OS (CMD/Program) name field, or you can specify the library name using the Current library or Library list fields.
- Clicking  the Edit Text dialog, which lets you enter the library name.

**JIL attribute:** i5\_lib

### Library list

(Optional) Specifies the name of a library that you want to add to the library list.

This property supports Global Variable substitution.

**Limits:** Up to 274 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** i5\_library\_list

### Local data area

(Optional) Specifies data for the local data area in an i5/OS job using the following format:

```
"data" | X'hh...'
```

- **"data"**  
Specifies the data in character format.  
**Example:** "My data"
- **X'hh...'**  
Specifies the data in hexadecimal format, with each pair of hexadecimal digits representing one byte of data.  
**Example:** X'C1C2C340C1C2C3'

**Limits:** Up to 2051 characters; case-sensitive

- The local data area is a temporary 1024-byte storage area that exists for the duration of the job. You can use the local data area to pass data to the job and to other programs that run as part of the job.
- Clicking  the Edit Text dialog, which lets you enter the data for the local data area.

**JIL attribute:** i5\_lda

### Positional parameters

(Optional) Defines the parameter values that you want to pass to the program at the time the program is invoked.



This property supports Global Variable substitution.

**Limits:** Up to 1024 characters for each parameter; up to 64 parameters

- Enclose parameters containing a space in apostrophes; otherwise, enclose parameters in quotation marks.
- Clicking  the Edit Text dialog, which lets you enter the parameters to pass to the program.

**JIL attribute:** i5\_params

### **Pass Positional Parameters**

When running workload, you might need to pass data between jobs and across platforms. You can pass positional parameters to an i5/OS program in your job definition. Positional parameters are variables that can be passed to a program at the time the program is invoked. The parameters are assigned in the order they are passed.

**Follow these steps:**

1. [Define an i5/OS job.](#)
2. Add the following property to the job definition:
  - **Positional parameters**  
Defines the parameter values that you want to pass to the program at the time the program is invoked.
3. Commit the job.  
The specified positional parameters are passed to the i5/OS program.

### **Use a User's Library List**

The agent uses the library list in the i5/OS job description by default. However, if the user is defined, you can set up your job definition to use the user's library list instead.

**Follow these steps:**

1. [Define an i5/OS job.](#)
2. Specify one of the following i5/OS properties:
  - **Current library**  
Specifies that the job uses the user's current library when it runs.
  - **Library**  
Specifies the name of the library that contains the program, the source file for the program, or the command that you want to run.
  - **Library list**  
Specifies the name of a library that you want to add to the library list.

#### **NOTE**

If you specify Library list, you must also specify the following property:

3. **i5/OS job description**  
Specifies the job description for the submitted program.
4. Commit the job.  
The job uses the user's library list.

## **Pass Keyword Parameters to SBMJOB**

When AutoSys Workload Automation submits a job to the i5/OS system, the job must pass through the SBMJOB command to execute. The following i5/OS properties map to parameters for the SBMJOB command:

i5/OS Properties	SBMJOB Parameters
i5/OS job description	JOBID
Library list	INLLIBL
i5/OS job queue	JOBQ
Current library	CURLIB
i5/OS job name	JOB

You can also pass additional keyword parameters, such as OUTQ(\*JOBID), to the SBMJOB command.

### **Follow these steps:**

1. [Define an i5/OS job.](#)
2. Add the following attribute to the job definition:
  - **Keyword parameters**  
Specifies SBMJOB command keyword and value combinations.
3. Commit the job.  
The specified keywords and values are passed to the SBMJOB command.

### **NOTE**

The special values for these SBMJOB parameters, such as \*USRPRF and \*JOBID, also apply to the properties. You can use these special values in your job definitions. For more information about the SBMJOB parameters and their special values, see the IBM documentation.

## **Returning a Job's Exit Status to AutoSys Workload Automation**

A job's exit code indicates whether the job completed successfully or failed. By default, the agent sends the job's ending severity code to AutoSys Workload Automation when a job completes. AutoSys Workload Automation interprets an exit code of zero (0) as job success and any other number as job failure.

You can return a job's exit status in other ways. For example, you can send the return code of an ILE program or module as the exit status, or you can specify a user-defined exit code of 100 as success.

You can return a job's exit status to AutoSys Workload Automation using the following methods:

- Send a program's return code using the i5/OS property i5/OS exit program
- Send a user-defined exit code using the Termination property Success exit codes or Failure exit codes

### **Send a Program's Return Code**

When a job completes, the agent sends the job's exit code to AutoSys Workload Automation. By default, the agent sends the job's ending severity code as the job's exit code. Instead of sending the job's ending severity code, the agent can send the return code of an ILE program or module or an OPM program. For example, if your job runs an ILE C, ILE RPG, OPM RPG, or OPM Cobol program that contains an exit or return statement, the agent can send that return code as the exit code.

### **Follow these steps:**

1. [Define an i5/OS job.](#)
2. Select *one* of the following i5/OS exit program options:
  - **\*SEVERITY**

Sends the job's ending severity code.

- **\*USER**

Specifies that the return code of an ILE program or module is sent as the exit code.

- **\*PROGRAM**

Specifies that the return code of an OPM program is sent as the exit code.

3. Commit the job.

The program's return code is sent instead of the job's ending severity code.

#### **NOTE**

The i5 system always writes the job's ending severity code to the job's spool file. You can check the spool file for the job's ending severity code for more information about the job status. For example, suppose that the C program's return code indicates failure, but the ending severity code shown in the spool file is 10, which might indicate that the job ran with a minor issue. Assuming that this issue can be ignored, you can indicate ending severity codes of 10 as job success using the Success exit codes property in the Termination section of the job definition.

### **Send a User-defined Exit Code**

By default, AutoSys Workload Automation interprets an exit code of 0 (zero) as job success and any other number as job failure. However, you can map exit codes other than 0 as job success.

#### **Follow these steps:**

1. [Define an i5/OS job.](#)
2. Add *one* of the following Termination properties to the job definition:
  - **Success exit codes**  
Defines which exit codes indicate job success.  
**Default:** 0 (zero)
  - **Failure exit codes**  
Defines which exit codes indicate job failure.  
**Default:** Any non-zero exit code
3. Commit the job.  
The specified user-defined exit code is sent to AutoSys Workload Automation.

### **Specify Data for a Local Data Area**

The local data area is a temporary 1024-byte storage area that exists for the duration of an i5/OS job. You can use the local data area to pass data to the job and to other programs that run as part of the job. When the job is submitted, the agent initializes the local data area with the specified data. When the job completes, the local data area is destroyed automatically by the operating system.

#### **Follow these steps:**

1. [Define an i5/OS job.](#)
2. Add the following property to the job definition:
  - **Local data area**  
Specifies data for the local data area in an i5/OS job.
3. Commit the job.  
The data is specified for a local data area.

## **Manage z/OS Jobs Using Web UI**

You can use z/OS jobs to run mainframe workload.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

Workload Automation Agent for z/OS submits and tracks the z/OS jobs. You can define the following three types of z/OS jobs:

- **z/OS Regular**  
Schedules z/OS jobs.
- **z/OS Manual**  
Creates dependencies on z/OS jobs that are submitted outside of the scheduling manager.
- **z/OS Data Set Trigger**  
Creates dependencies on data set activities.

**z/OS Data Set Trigger Jobs**

You can define z/OS Data Set Trigger jobs to create dependencies on data set activities. You can customize trigger conditions to define the conditions in which the z/OS Data Set Trigger job completes.

You can specify trigger conditions for the following data set activities:

- When a data set is created or updated
- When a specific job, group of jobs, or user ID creates a data set
- When an explicit data set notification is received (used when the data set activity does not generate an SMF record)
- When an FTP file is sent or received successfully

**NOTE**

Each data set must have its own individual z/OS Data Set Trigger job. To create dependencies on multiple data sets, you must create multiple z/OS Data Set Trigger jobs.

**Define a z/OS Data Set Trigger Job**

You can define a z/OS Data Set Trigger job to create dependencies on data set activities.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

**Follow these steps:**

1. Create a z/OS Data Set Trigger (ZOSDST) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Data set name**  
Specifies the Job Control Language (JCL) library name.
3. (Optional) Specify optional [z/OS Data Set Trigger properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The z/OS Data Set Trigger job is defined.

## **z/OS Data Set Trigger Job Examples**

The following examples describe sample z/OS Data Set Trigger jobs.

### **Example: Monitor for Data Set Creation and Update**

Suppose that you want a z/OS Data Set Trigger job named PROD.NIGHTLY to release its successors when the data set PROD.CICS.FILE1602 is closed (created or updated). The agent ZOS1 monitors the data set under user CYBDL01.

#### **Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- PROD.NIGHTLY
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- PROD.CICS.FILE1602
3. Save the job.

### **Example: Restrict the Trigger to Specific Data Sets Created by a Particular Job**

Suppose that you want a z/OS Data Set Trigger job named PROD.PAY\_DATA to release its successors when job ABC creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

#### **Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- PROD.PAY\_DATA
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- USER1.PAYROLL.G-
  - Trigger by -- Job name, ABC
3. Save the job.1. ■ 3.

### **Example: Restrict the Trigger to Specific Data Sets Created by a Particular User**

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release its successors when the user CYB1 creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

#### **To restrict the trigger to specific data sets created by a particular user**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- PROD.PAY\_DATA
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- USER1.PAYROLL.G-
  - Trigger by -- user ID, CYB1
3. Save the job.2. 3.

### **Example: Run a Compress Job after 100 Closures of a Data Set**

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release a compress job named COPYJCL.COMPRESS after every 100 closures of the data set CYBER.COPY.JCL. The agent ZOS1 monitors the data set under user CYBDL01.

**Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- PROD.PAY\_DATA
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- CYBER.COPY.JCL
  - Trigger on -- 100
3. Save the job.

**NOTE**

Define the compress job, COPYJCL.COMPRESS, as a successor to the z/OS Data Set Trigger job.

**Example: Monitor for a Data Set Received from a Remote FTP Partner**

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release its successors when a file is successfully received from a remote FTP partner, creating generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the FTP transfer under user CYBDL01.

**To monitor for a data set received from a remote FTP Partner**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- PROD.PAY\_DATA
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- USER1.PAYROLL.G-
  - FTP direction -- RECEIVE
3. Save the job.

**Example: Monitor for a Data Set Sent to a Remote FTP partner**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when data set CYBER.XFER.001 is successfully sent from the local mainframe partner to a remote FTP partner. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

**Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- CYBER.XFER
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- CYBER.XFER.001
  - FTP direction -- SEND
3. Save the job.

**Example: Restrict Triggering to a Specific Host**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a remote FTP partner with IP address 172.16.0.0 successfully transfers a file creating the data set CYBER.XFER.001. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

**Follow these steps:**

1. Create a z/OS Data Set Trigger job.

2. Enter the following properties:
  - Name -- CYBER.XFER
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- CYBER.XFER.001
  - FTP direction -- RECEIVE
  - FTP host -- 172.16.0.0
3. Save the job.

### **Example: Restrict Triggering to a Specific User ID**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming that the user ID prefix of the local FTP partner is CYB (CYB-). The agent ZOS1 monitors the FTP transfer under user CYBDL01.

#### **Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- CYBER.XFER
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- CYBER.XFER.001
  - Trigger by -- user ID, CYB-
  - FTP direction -- RECEIVE
3. Save the job.

### **Example: Restrict Triggering to a Specific Logon ID**

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming that the remote FTP partner logged on to the FTP server with the CYBER005 user ID. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

#### **Follow these steps:**

1. Create a z/OS Data Set Trigger job.
2. Enter the following properties:
  - Name -- CYBER.XFER
  - Send to machine -- ZOS1
  - Owner -- CYBDL01
  - Data set name -- CYBER.XFER.001
  - FTP direction -- RECEIVE
  - FTP userid -- CYBER005
3. Save the job.

### **z/OS Data Set Trigger Properties**

The z/OS Data Set Trigger category includes the following properties:

#### **Dataset name**

Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job. The JCLLIB is a z/OS data set name. Data set names typically have up to four positions; positions must each be eight characters or less and separated by periods.

**Limits:** Up to 44 characters

**Example: prod.jcllib****JIL attribute:** zos\_dataset**FTP direction**

(Optional) Specifies the direction of FTP transfer between a remote computer and a local mainframe computer, as follows:

- RECEIVE -- Transfers from a remote computer to a local mainframe computer.
- SEND -- Transfers to a remote computer from a local mainframe computer.

**NOTE**

This field is valid only when Trigger on explicit data set is set to False.

**JIL attribute:** zos\_ftp\_direction**FTP host**

(Optional) Specifies the host name used to restrict the activation of the data set trigger from transferring to or from the specified remote host.

**Limits:** Up to 128 characters**NOTE**

This field is valid only when Trigger on explicit data set is set to False.

**JIL attribute:** zos\_ftp\_host**FTP userid**

(Optional) Specifies the user ID for the FTP connection.

**Limits:** Up to 8 characters**NOTE**

This field is valid only when Trigger on explicit data set is set to False.

**JIL attribute:** zos\_ftp\_userid**Trigger by**

(Optional) Specifies whether the job monitors data set activity by a job or a user ID. You can enter the name of the job or user responsible for the data set activity that triggers the job in the Type value field. When a data set is updated or renamed by the specified job or user ID, the job triggers.

**Limits:** Up to 8 characters**JIL attribute:** zos\_trigger\_type, zos\_trigger\_by**Trigger on**

(Optional) Specifies the number of actions that must occur before the job triggers.

**Limits:** 0-100**JIL attribute:** zos\_trigger\_on**Trigger on data set renamed**

(Optional) Indicates whether the job monitors when a data set is renamed. If selected, the job triggers when the data set is renamed.

**Default:** not selected**NOTE**

You can set Trigger on data set renamed or Trigger on data set updated, but you cannot set both.



**JIL attribute:** zos\_dsn\_renamed

### Trigger on data set updated

(Optional) Indicates whether the job monitors when a data set is updated. If selected, the job triggers when the data set is updated.

**Default:** not selected

#### NOTE

You can set Trigger on data set renamed or Trigger on data set updated, but you cannot set both.

**JIL attribute:** zos\_dsn\_updated

### Trigger on explicit dataset

(Optional) Indicates whether the job monitors for an explicit data set notification (used when the data set activity does not generate an SMF record). If selected, the job triggers when it receives notification that the specified data set is being updated.

**Default:** not selected

**JIL attribute:** zos\_explicit\_dsn

## Define a z/OS Manual Job

You can define a z/OS Manual job to create dependencies on z/OS jobs that are submitted outside the scheduling manager, such as a job that is submitted manually by a user.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for z/OS.

### Follow these steps:

1. Create a z/OS Manual (ZOSM) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job submitted outside the scheduling manager.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **z/OS job name**  
Defines the name of the job being monitored.
3. (Optional) Specify optional [z/OS Manual properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The z/OS Manual job is defined.

## z/OS Manual Job Examples

The following example describes a sample z/OS Manual job.

### Example: Post a z/OS Manual Job as Complete Based on the User ID

Suppose that you want to post a z/OS Manual job to complete when the manually-submitted job ABC runs under user CYBER. The ZOS1 agent monitors job ABC.

**Follow these steps:**

1. Create a z/OS Manual job.
2. Enter the following properties:
  - Name -- ABC
  - Send to machine -- ZOS1
  - Authorization string -- CYBER
  - z/OS job name -- watch\_j2
3. Commit the job.

**z/OS Manual Properties**

The z/OS Manual category includes the following properties:

**Authorization string**

(Optional) Defines the authorization string the scheduling manager uses to post and track the correct job.

**Limits:** Up to 80 characters

**NOTE**

Contact your agent administrator to confirm whether the agent checks the authorization string. Depending on the scenario, the action to take differs as follows: If authorization checking (AUTHSTR) is not set up for this agent, do not specify the authorization string. If authorization checking (AUTHSTR) is set to RACUSER, specify the user ID that owns the job. If authorization checking (AUTHSTR) is set to ACCOUNT1, ACCOUNT2, ACCOUNT3, or ACCOUNT4, specify the value found in the corresponding position of the job's Job Control Language (JCL). For example, if AUTHSTR is set to check the ACCOUNT2 field, specify the value found in the second accounting position in the job's JCL.

**JIL attribute:** auth\_string

**Backward search (hours)**

(Optional) Defines the number of hours that the agent does a backward search for a manually-submitted job.

**Limits:** 0-999

**JIL attribute:** search\_bw

**z/OS job name**

Defines the name of the job being monitored.

**Limits:** Up to eight characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

**JIL attribute:** zos\_jobname

**Define a z/OS Regular Job**

You can define a z/OS Regular job to schedule a z/OS job.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

**Follow these steps:**

1. Create a z/OS Regular (ZOS) job.
2. Specify the following required properties:
  - **Name**

Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

– **Send to machine**

Specifies the name of the machine on which the job runs.

– **JCL library**

Specifies the library name containing the Job Control Language (JCL) for this job.

**Note:** If you specify a partitioned data set, you must also specify the JCL Member attribute in your job definition.

3. (Optional) Specify optional [z/OS Regular properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The z/OS Regular job is defined.

### **z/OS Regular Job Examples**

The following example describes a sample z/OS Regular job.

#### **Example: Store a Working Copy of the JCL that You Submitted**

Suppose that the agent ZOS1 submits the JCL in member CYBDL01A in the CYBDL01.JCLLIB library. If the job fails, you can modify a working copy of the JCL in the CYBDL01.COPY.JCLLIB data set, and resubmit the job without affecting the JCL source.

#### **Follow these steps:**

1. Create a z/OS Regular job.
2. Enter the following properties:
  - Name -- CYBDL01A
  - Send to machine -- ZOS1
  - JCL library -- CYBDL01.JCLLIB
  - JCL member -- CYBDL01A
  - Copy JCL library -- CYBDL01.COPY.JCLLIB
3. Commit the job.

### **z/OS Regular Job Properties**

The z/OS Regular category includes the following properties:

#### **Condition code: Return code**

Specifies the return code using one of the following values:

- A condition code for one number or a range of numbers between 1 and 4095 (ranges are separated by colons, such as 1:4095).
- A system abend code (Sccc), such as S0C1 or SB37.
- A user abend code (Unnnn), such as U0001 or U0462. The nnnn must be exactly four decimal digits and cannot exceed 4095.

**Limits:** Up to nine characters

#### **NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** condition\_code: rc

**Condition code: Result**

(Defaulted) Specifies whether to consider the job as failed based on the condition code, as follows:

- SUCCESS -- Considers the job not failed if the condition code is matched.
- FAILURE -- Considers the job failed if the condition code is matched.

**Default:** SUCCESS

**NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** result

**Condition code: Action**

(Defaulted) Specifies whether the job continues with the next step regardless of whether the job is considered as failed.

- CONTINUE -- Continues with the next step if the condition code is matched.
- STOP -- Stops the job if the condition code is matched.

**Default:** CONTINUE

**NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** action

**Condition code: Procstep**

(Optional) Specifies a valid z/OS procedure name.

**Limits:** Up to eight characters; first character must not be numeric

**NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** proc\_step

**Condition code: Program**

(Optional) Specifies a valid z/OS program name.

**Limits:** Up to eight characters; first character must not be numeric

**NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** program

**Condition code: Stepname**

(Optional) Specifies a valid z/OS step name.

**Limits:** Up to eight characters; first character must not be numeric

**NOTE**

- Selecting Add from the Select an action drop-down list, and then clicking Go displays the Condition code subfields.
- If you specify a Condition code, you must also specify a Return code.

**JIL attribute:** step\_name

**Copy JCL library**

(Optional) Specifies the data set name that receives the working copy of the JCL you submitted. Data set names typically have up to four positions; positions must each be eight characters or less and separated by periods.

**Limits:** Up to 44 characters

**Example:** prod.copyjcl.weekly

**JIL attribute:** copy\_jcl

**JCL library**

Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job. The JCLLIB is a z/OS data set name. Data set names typically have up to four positions. Each position can be up to eight characters. Separate the positions using periods. The JCL library can be any one of the following:

- A valid z/OS partitioned data set (PDS) name.
- A valid z/OS data set name, prefixed with lib- to indicate Librarian input.
- A valid z/OS data set name, prefixed with pan- to indicate Panvalet input.
- A valid z/OS data set name, prefixed with seq- to indicate sequential input.

**Limits:** Up to 44 characters

**NOTE**

If you specify a partitioned data set, you must also specify the JCL member field in your job definition.

**JIL attribute:** jcl\_library

**JCL member**

(Optional) Specifies the JCL member that contains the JCL for your job. If the jcl\_library attribute specifies a fully qualified data set name, the jcl\_member attribute specifies the zOS job name.

**Limits:** Up to eight characters

**JIL attribute:** jcl\_member

## Manage Micro Focus Jobs

The Micro Focus Net Express JCL engine lets you run your mainframe JCL and COBOL programs in a Windows environment.

You can schedule and control your Micro Focus jobs using AutoSys Workload Automation and Workload Automation Agent for Micro Focus. The agent provides an interface to the Micro Focus Net Express JCL engine.

**NOTE**

- The Micro Focus job type is supported with AutoSys Workload Automation Release 11.3.5 and later only.
- To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus.

## Define a Micro Focus Job

You can define a Micro Focus job to run your mainframe JCL and COBOL programs in a Windows environment.

### NOTE

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus.

### Follow these steps:

1. Create a Micro Focus (MICROFOCUS) job in Quick Edit.  
Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Server**  
Specifies the server name or the IP address and port of the Micro Focus Enterprise Server.
  - **JCL name**  
Specifies the path to and name of a JCL file to run.
2. (Optional) Specify optional [Micro Focus properties](#).
3. (Optional) Specify [common properties that apply to all job types](#).
4. Commit the job.  
The Micro Focus job is defined.

## Micro Focus Job Examples

The following examples describe sample Micro Focus jobs.

### Example: Pass the JCL by Content

This example passes the contents of the JCL file to the Micro Focus Enterprise Server with IP address 172.31.255.255 and port 2222. The contents of the if.jcl file is read and passed to the server.

### Follow these steps:

1. Create a Micro Focus job.
2. Enter the following properties:
  - Name -- mf\_job\_by\_by\_content
  - Send to machine -- mfagent
  - Server address type -- ADDRESS
  - Server -- tcp:172.31.255.255:2222
  - JCL type -- CONTENT
  - JCL name -- C:\Program Files\micro focus\es-jcldemo\if.jcl
3. Commit the job.

### Example: Pass the JCL by Reference

This example passes the name of the JCL file to the Micro Focus Enterprise Server. The server reads and submits the named file. The job runs under the Micro Focus user ID named MFUSER01.

**Follow these steps:**

1. Create a Micro Focus job.
2. Enter the following properties:
  - Name -- mf\_job\_by\_reference
  - Send to machine -- mfagent
  - Server address type -- NAME
  - Server -- mfprod
  - JCL type -- REFERENCE
  - JCL name -- C:\Program Files\micro focus\es-jcldemo\if.jcl
  - Micro Focus user -- MFUSER01@mfprod
3. Commit the job.

**Example: Run a Micro Focus Job Under an Operating System User ID**

This example runs a Micro Focus job under the Micro Focus user ID named MFUSER01, and the operating system user ID named OSUSER.

**Follow these steps:**

1. Create a Micro Focus job.
2. Enter the following properties:
  - Name -- mf\_job\_with\_os\_user
  - Send to machine -- mfagent
  - Server address type -- NAME
  - Server -- mfprod
  - JCL type -- REFERENCE
  - JCL name -- C:\Program Files\micro focus\es-jcldemo\if.jcl
  - Micro Focus user -- MFUSER01@mfprod
  - O/S user -- OSUSER@osserver
3. Commit the job.

**Example: Specify the JES3 Environment**

This example runs the JCL file in the JES3 environment. The value in the Version field overrides JES2, the default environment.

**Follow these steps:**

1. Create a Micro Focus job.
2. Enter the following properties:
  - Name -- mf\_job\_with\_version\_jes3
  - Send to machine -- mfagent
  - Server address type -- NAME
  - Server -- mfprod
  - JCL type -- REFERENCE
  - JCL name -- C:\Program Files\micro focus\es-jcldemo\if.jcl
  - Version -- JES3
3. Commit the job.

**Micro Focus Properties**

The Micro Focus category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** `job_class`

### JCL name

Specifies the path to and name of a JCL file to run. Depending on the value specified in the JCL type field, the contents or the name of the JCL file is passed to the Micro Focus Enterprise Server.

**Limits:** Up to 256 characters

**JIL attribute:** `mf_jcl_name`

### JCL type

(Defaulted) Specifies whether to pass the JCL by content or by reference to the Micro Focus Enterprise Server. Selecting **CONTENT** passes the contents of the JCL file to the Micro Focus Enterprise Server. The submitted JCL is read from the file specified in the JCL name field, and the contents of the JCL file are physically sent to the server. Selecting **REFERENCE** passes the name of the JCL file to the Micro Focus Enterprise Server as specified in the JCL name field. The server then reads and submits the named file.

**Default:** **CONTENT**

#### NOTE

To edit the JCL before submitting the job, or to restart the job from a particular step, you must select **CONTENT**.

**JIL attribute:** `mf_jcl_type`

### Micro Focus user

(Optional) Specifies the Micro Focus user ID that the job runs under. The *user\_id* must be defined on the Micro Focus Enterprise server where *identifier* specifies the name of the server. The format is as follows: *user\_id@identifier*

**Limits:** Up to 80 characters

**JIL attribute:** `mf_user`

### O/S user

(Optional) Specifies the Operating System user ID that the job runs under. The *os\_userid* must meet the operating system requirements and should be defined on the Micro Focus Enterprise server where *identifier* specifies the name of the server. The format is as follows: *os\_userid@identifier*

**Limits:** Up to 80 characters

#### NOTE

If the agent runs as a service under the system account, the Micro Focus job is executed under the system account. If the Micro Focus Enterprise Server runs under a particular user ID, and you have specified a Micro Focus Enterprise Server name for the job, you must specify the same operating system user ID for the job to find the server.

**JIL attribute:** `os_user`

### Server

Specifies the server name or the IP address and port of the Micro Focus Enterprise Server. The format of the IP address is as follows: *tcp:ip\_address:port*

**Limits:** Up to 256 characters for the server name

**JIL attribute:** `mf_server`



## Server address type

(Defaulted) Specifies whether the Server field value is the server name or the IP address and port of the Micro Focus Enterprise Server.

**Default:** NAME (The job assumes that the name of the Micro Focus Enterprise Server is specified.)

**JIL attribute:** mf\_server\_address\_type

## Version

(Defaulted) Specifies the environment on the mainframe that manages the job you want to run.

**Default:** JES2 (The job assumes that the environment is JES2.)

**JIL attribute:** mf\_version

## Pass Variables as Parameters to a JCL File

You can pass variables as parameters to a JCL file.

### Follow these steps:

1. [Define a Micro Focus job.](#)
2. Specify the following environment variables in the Environment section of the job definition:  
`parm_name=parm_value[, parm_name=parm_value...]`
  - **parm\_name** Defines the name of a new environment variable or specifies the name of an existing environment variable.
  - **parm\_value** Specifies the value of the environment variable.
3. (Optional) Add additional variables that you want to pass.
4. Commit the job.  
 The specified variables are passed as parameters to the Micro Focus program.

### Example: Pass SORT and MAX as Parameters to the JCL file

Suppose that you want to pass SORT for symbolic variable VAR1 and MAX for symbolic variable VAR2 in the following JCL file named mfprog.jcl:

```
//MFIDSAL1 JOB 'DAVINDER',CLASS=A,MSGCLASS=A,NOTIFY=MFIDSA
//*-----
//*
//*-----
//STEP1 EXEC PGM=%VAR1, PARM='%VAR2'
//STEPLIB DD DSN=CEE.SCEERUN,DISP=SHR
// DD DSN=MFIDSA.DSA.LOAD,DISP=SHR
//SYSPRINT DD SYSOUT=* 00510000
//CEEDUMP DD SYSOUT=* 00520000
//SYSUDUMP DD SYSOUT=* 00530000
//OLDLIB DD DISP=(OLD),DSN=MFE.PROCLIB.L01
```

The parameter values are specified in the job definition using the Environment variables field as follows:

- Name: mf\_job\_with\_envvars
- Send to machine: mfagent
- Server address type: NAME
- Server: mfprod
- JCL type: CONTENT
- JCL name: "C:\Program Files\microfocus\es-jcldemo\mfprog.jcl"
- Micro Focus user: MFUSER01@mfprod
- Environment variables: VAR1=SORT, VAR2=MAX

The following JCL is submitted to the Micro Focus Enterprise Server when the job runs:

```
//MFIDSAL1 JOB 'DAVINDER',CLASS=A,MSGCLASS=A,NOTIFY=MFIDSA
//*-----
//*
//*-----
//STEP1 EXEC PGM=**SORT**, PARM= '**MAX**'
//STEPLIB DD DSN=CEE.SCEERUN,DISP=SHR
// DD DSN=MFIDSA.DSA.LOAD,DISP=SHR
//SYSPRINT DD SYSOUT=* 00510000
//CEEDUMP DD SYSOUT=* 00520000
//SYSUDUMP DD SYSOUT=* 00530000
//OLDLIB DD DISP=(OLD),DSN=MFE.PROCLIB.L01
```

## Manage Process Automation Jobs Using Web UI

Process Automation jobs let you launch CA Process Automation processes and monitor them to completion.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

You can define the following Process Automation jobs:

- **Process Automation Process Execution**  
Lets you directly execute a CA Process Automation process.
- **Process Automation Start Request Form**  
Lets you submit a Start Request Form to execute a CA Process Automation process. The Start Request Form sets the values for the current execution of the process.

### Define a Process Automation Process Execution Job

You can define a Process Automation Process Execution (PAPROC) job to execute a CA Process Automation process.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

### Follow these steps:

1. Create a Process Automation Process Execution (PAPROC) job.
2. Specify the following required properties:
  - **Name**

Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

- **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Process name**  
Specifies the fully qualified name of the process.
  - **Target end-point URL** Specifies the target endpoint address URL on the CA Process Automation server.
3. (Optional) Specify optional [Process Automation Process Execution properties](#).
  4. (Optional) Specify [common properties that apply to all job types](#).
  5. Commit the job.  
The Process Automation Process Execution job is defined.

### **Process Automation Process Execution Job Examples**

The following examples describe sample Process Automation Process Execution jobs.

#### **Example: Execute a Process Automation Process**

This example executes the process named Running the Process on the machine named paagent.

##### **Follow these steps:**

1. Create a Process Automation Process Execution job.
2. Enter the following properties:
  - Name -- pam1
  - Send to machine -- paagent
  - Owner -- pamadmin@paserver
  - Process name -- /ITPAM Tutorials/S01 Getting started/L01 Creating a Simple Process/L01-2 Running the Process/Running the Process
  - Target end-point URL -- http://paserver:8080/itpam/soap
3. Commit the job.

#### **Example: Write a Trace of SOAP Messages to the Spool File**

This example writes a trace of SOAP messages to the spool file. The process named Test Process is tracked to completion.

##### **Follow these steps:**

1. Create a Process Automation Process Execution job.
2. Enter the following properties:
  - Name -- execproc
  - Send to machine -- localhost
  - Owner -- pamadmin@capamach
  - Process name -- Test Process
  - Target end-point URL -- http://capamach:8080/itpam/soap
  - Track process to completion -- selected
  - Write SOAP trace -- selected
  - Parameter -- strTestName=PA Test
3. Commit the job.

## **Process Automation Process Execution Properties**

The Process Automation Process Execution category includes the following properties:

### **Agent job class**

- (Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators. **Limits:** Up to 32 characters; cannot contain delimiters (such as spaces) **JIL attribute:** job\_class

### **Parameter**

(Optional) Specifies the parameters in a Process Automation job. When the job runs, the parameters are passed to the executing process.

**Limits:** Up to 1024 characters for the name, up to 2048 characters for the value; case-sensitive

**Example: Port=80**

#### **NOTE**

- Specify multiple parameters by entering each name and value pair on a separate line.
- If a space precedes or follows the value, enclose the value in quotation marks, for example: Server="paserver "

**JIL attribute:** pa\_parameter

### **Process name**

Specifies the fully qualified name of the process.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** pa\_name

### **Target end-point URL**

Specifies the target endpoint address URL on the CA Process Automation server.

**Limits:** Up to 2048 characters; case-sensitive

**JIL attribute:** endpoint\_url

### **Track process to completion**

(Defaulted) Specifies whether the process is tracked to completion.

**Default:** selected (tracks the process to completion)

**JIL attribute:** pa\_monitor\_progress

### **Write SOAP trace**

(Defaulted) Indicates whether a trace of SOAP messages is written to the spool file.

**Default:** not selected (does not write a trace of SOAP messages to the spool file)

**JIL attribute:** pa\_trace

## **Define a Process Automation Start Request Form Job**

You can define a Process Automation Start Request Form (PAREQ) job to submit a Start Request Form to execute a CA Process Automation process. The Start Request Form sets the values for the current execution of the process.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

**Follow these steps:**

1. Create a Process Automation Start Request Form (PAREQ) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Process name**  
Specifies the name of the Start Request Form.
  - **Target end-point URL**  
Specifies the target endpoint address URL on the CA Process Automation server.
  - **Start request form path** Specifies the path of the Start Request Form.
3. (Optional) Specify optional [Process Automation Start Request Form properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job. The Process Automation Start Request Form job is defined.

**NOTE**

The specified path must end in a slash (/) character.

**Process Automation Start Request Form Job Examples**

The following examples describe sample Process Automation Start Request Form jobs.

**Example: Submit a Start Request Form**

This example submits a Start Request Form named Process A in the /CAWAE/ path. The example uses the credentials of the pamadmin@capamach user to authenticate the user on the Process Automation installation on capamach. The Start Request Form has a field called strTestName that is populated with the value PA Test before the form is submitted. The process is tracked to completion.

**Follow these steps:**

1. Create a Process Automation Start Request Form job.
2. Enter the following properties:
  - Name -- runsrf
  - Send to machine -- localhost
  - Owner -- pamadmin@capamach
  - Process name -- Process A
  - Target end-point URL -- http://capamach:8080/itpam/soap
  - Start request form path -- /CAWAAE/
  - Track process to completion -- selected
  - Parameter -- strTestName=PA Test
3. Commit the job.

**Example: Specify the Path to the Start Request Form**

This example specifies the Start Request Form located at /ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/. Values are passed for three parameters defined in that form and the progress of the job is monitored until the job completes.

#### Follow these steps:

1. Create a Process Automation Start Request Form job.
2. Enter the following properties:
  - Name -- srf1
  - Send to machine -- paagent
  - Owner -- pamadmin@paserver
  - Process name -- Start Request Form
  - Target end-point URL -- http://paserver:8080/itpam/soap
  - Start request form path -- /ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/
  - Track process to completion -- selected
  - Write SOAP trace -- not selected
  - Parameter -- enter each of the following parameters on a separate line:
    - AlarmProcessingProduct=Test Alarm Product
    - Server=paserver
    - Port=80
3. Commit the job.

#### Process Automation Start Request Form Properties

The Process Automation Start Request Form category includes the following properties:

##### Agent job class

- (Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators. **Limits:** Up to 32 characters; cannot contain delimiters (such as spaces) **JIL attribute:** job\_class

##### Parameter

(Optional) Specifies the parameters in a Process Automation job. When the job runs, the parameters are passed to the executing process.

**Limits:** Up to 1024 characters for the name, up to 2048 characters for the value; case-sensitive

**Example: Port=80**

##### NOTE

- Specify multiple parameters by entering each name and value pair on a separate line.
- If a space precedes or follows the value, enclose the value in quotation marks, for example: Server="paserver "

**JIL attribute:** pa\_parameter

##### Process name

Specifies the name of the Start Request Form.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** pa\_name

##### Start request form path

Specifies the path of the Start Request Form.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

The specified path must end in a slash (/) character.

**JIL attribute:** pa\_path

**Target end-point URL**

Specifies the target endpoint address URL on the CA Process Automation server.

**Limits:** Up to 2048 characters; case-sensitive

**JIL attribute:** endpoint\_url

**Track process to completion**

(Defaulted) Specifies whether the process is tracked to completion.

**Default:** selected (tracks the process to completion)

**JIL attribute:** pa\_monitor\_progress

**Write SOAP trace**

(Defaulted) Indicates whether a trace of SOAP messages is written to the spool file.

**Default:** not selected (does not write a trace of SOAP messages to the spool file)

**JIL attribute:** pa\_trace

Specify multiple parameters by entering each name and value pair on a separate line. If a space precedes or follows the value, enclose the value in quotation marks, for example: Server=" paserver "

## Manage Remote Execution Jobs Using Web UI

Remote Execution jobs let you run commands on a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Secure Shell (SSH2) or Telnet.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Remote Execution. The Workload Automation Agent for Remote Execution supports only remote target environments of UNIX, HP Integrity NonStop, and OpenVMS.

### Define a Remote Execution Job

You can define a Remote Execution (PROXY) job to run commands to a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Telnet or Secure Shell (SSH2).

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Remote Execution. The Workload Automation Agent for Remote Execution supports only remote target environments of UNIX, HP Integrity NonStop, and OpenVMS.

**Follow these steps:**

1. Create a Remote Execution job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

- **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Remote command**  
Specifies a command or script to run on a remote computer.
  - **Remote target** Specifies the name of the custom properties (remote\_target.properties) file that is created on the agent for the remote system. The file contains the default user credentials that are used to monitor all jobs run on the target machine by the agent. The name should not include the .properties extension.
3. (Optional) Specify optional [Remote Execution properties](#).
  4. (Optional) Specify [common properties that apply to all job types](#).
  5. Commit the job.  
The Remote Execution job is defined.

### **Remote Execution Job Examples**

The following examples describe sample Remote Execution jobs.

#### **Example: Run the ls Command on a Remote System**

This example executes the ls command on the hpserver remote system. Additionally, it specifies a spool file location other than the agent default without overwriting previous spool files with the same name.

#### **Follow these steps:**

1. Create a Remote Execution job.
2. Enter the following properties:
  - Name -- ls\_cmd
  - Send to machine -- proxyagent
  - Remote command -- ls
  - Remote target -- hpserver
  - Spool file -- >>/home/user1/test/ls.out
3. Commit the job.

### **Remote Execution Properties**

The Remote Execution category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Remote command**

Specifies a command or script to run on a remote computer.

**Limits:** Multiple lines allowed, up to 1024 characters per line; case-sensitive

#### **NOTE**

If the command operand contains delimiters, the command operand must be enclosed in single quotes to denote the start and end of the operand. If the command operand contains a single quote, double up that single quote so it does not denote the end of the operand. Do not use double quotes; it must be two single quotes.



**JIL attribute:** remote\_command

### Remote target

Specifies the name of the custom properties (remote\_target.properties) file that is created on the agent for the remote system. The file contains the default user credentials that are used to monitor all jobs run on the target machine by the agent. The name should not include the .properties extension.

**Limits:** Up to 128 characters; case-sensitive; valid characters are letters, numbers, and underscore (\_); the first character must be a letter

**JIL attribute:** remote\_target

### Spool file

(Defaulted) Specifies the path to the spool file when you do not want to use the default location. If a spool file is specified, it overwrites a previous spool file with the same name unless you specify to append by beginning the name with >>.

**Default:** The spoolHome property in the custom properties file for the remote system

**Limits:** Up to 256 characters; case-sensitive; valid characters are a-z, A-Z, 0-9, and delimiters

#### NOTE

- The operating system or file system may impose additional requirements on the value, such as using valid path and file name notation.
- For more information about the custom properties file, see the *Workload Automation Agent for Remote Execution Implementation documentation*.

**JIL attribute:** spool\_file

### Submit modifier

(Optional) Specifies the commands you want to run or the options to set prior to executing a job on a remote system. Running these setup commands provides additional flexibility to pass additional information to commands executed on the target system.

**Limits:** Up to 600 characters per line; case-sensitive; valid characters are a-z, A-Z, 0-9, and delimiters

**Example: submit\_modifier:** Option1="HIGHPIN ON,INSPECT ON",option2="2",option3="ABENDONABEND"

#### NOTE

The syntax and option requirements of the target system job apply.

**JIL attribute:** submit\_modifier

## Manage SNMP Jobs Using Web UI

The agent supports a built-in SNMP manager capability. You can enable the agent to act as an SNMP manager to emit and listen for SNMP traps in addition to its other roles. The agent supports SNMP v1, v2, and v3. After the agent is configured, you can define and run SNMP job types on the agent.

#### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent as an SNMP manager. For more information about configuring the agent as an SNMP manager, see the *Workload Automation Agent for UNIX, Linux, or Windows Implementation documentation*.

You can define the following types of SNMP jobs:

- Simple Network Management Protocol Value Get (SNMPGET)
- Simple Network Management Protocol Value Set (SNMPSET)

The SNMP Value Get job queries a network device for the value of a variable that is assigned to a Management Information Base (MIB) address. You can use the SNMP Value Get job to retrieve information about a network device to determine whether an administrator is required to be notified.

The SNMP Value Set job modifies a variable on a network device. The variable is assigned to the MIB address that you specify. You can use the SNMP Value Set job to update a variable that reports on the failure or success of a mission-critical policy.

### **Define an SNMP Value Get Job**

You can define a Simple Network Management Protocol Value Get (SNMPGET) job to retrieve the value of an SNMP variable.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent as an SNMP manager. For more information about configuring the agent as an SNMP manager, see the Workload Automation Agent for UNIX, Linux, or Windows Implementation *documentation*.

#### **Follow these steps:**

1. Create an SNMP Value Get (SNMPGET) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **SNMP object ID**  
Specifies the SNMP object identifier in numeric or string format. In an SNMP Value Get job, it identifies the variable whose value you want to retrieve.
  - **MIB** Specifies the name of the MIB (management information base) file.

#### **NOTE**

This field is required only when the SNMP object ID is entered in string format.

- **Host name** Specifies the host name or IP address of the network device.
  - **SNMP value**  
Specifies the type and value of the variable that you are changing in an SNMP Value Set job. The value is converted to the corresponding type.
3. Specify the following field if your SNMP version is v3:
    - **SNMP user**  
Specifies the user name whose credentials are used for authentication.
  4. (Optional) Specify optional [SNMP Value Get properties](#).
  5. (Optional) Specify [common properties that apply to all job types](#).
  6. Commit the job.  
The SNMP Value Get job is defined.

### **SNMP Value Get Job Examples**

The following examples describe sample SNMP Value Get jobs.

#### **Example: Query a Network Device for the Value of an SNMP Variable using SNMP v2**

Suppose that you want to know the value of the agentVersion variable hosted by a network device. In this example, the host name of the network device is host.example.com and its port is 161. The SNMP version is v2 and the default value of the community string is public. The name of the MIB file is cybermation.mib, which is located in the agent installation directory.

**Follow these steps:**

1. Create an SNMP Value Get job.
2. Enter the following properties:
  - Name -- getvarv2
  - Send to machine -- snmpagent
  - MIB -- cybermation.mib
  - Host name -- host.example.com
  - SNMP object ID -- agentVersion
  - Port -- 161
  - Version -- 2
3. Commit the job.

**Example: Query a Network Device for the Value of an SNMP Variable using SNMP v3**

Suppose that you want to know the value of a variable hosted by a network device using SNMP v3. In this example, the job specifies the SNMP object ID in numeric format, the AES privacy protocol, and the SHA authentication protocol.

**NOTE**

The Owner value is the user ID associated with the authentication password. The SNMP user value is the user ID associated with the privacy password. The SNMP user and Owner users and their corresponding passwords are configured in AutoSys Workload Automation using the autosys\_secure command.

**Follow these steps:**

1. Create an SNMP Value Get job.
2. Enter the following properties:
  - Name -- getvarv3
  - Send to machine -- localhost
  - Owner -- Administrator@localhost
  - SNMP object ID -- .1.3.1.1.3.1.0.0
  - Host name -- snmp
  - Version -- 3
  - Authentication protocol -- SHA
  - Privacy protocol -- AES
  - SNMP user -- user1
3. Commit the job.

**Example: Specify a Table Format to Retrieve MIB Data in an SNMP Value Get Job**

Suppose that you want to specify the format to retrieve the MIB data in a table format. In this example, the job walks the whole SNMP subtree starting with the object ID specified in the SNMP object ID field and MIB data is returned in table format.

**Follow these steps:**

1. Create an SNMP Value Get job.
2. Enter the following properties:

- Name -- getsubtree
  - Send to machine -- snmpagent
  - MIB -- cybermation.mib
  - Host name -- host.example.com
  - SNMP object ID -- agentVersion
  - Port -- 161
  - Version -- 2
  - Community -- public
  - Subtree -- Y
  - Table view -- Y
3. Commit the job.

### **SNMP Value Get Properties**

The SNMP Value Get category includes the following properties:

#### **Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** job\_class

#### **Authentication protocol**

(Defaulted) Specifies the authentication protocol to use when connecting with the user specified in the SNMP user field. Valid values are: MD5 (Message Digest 5) and SHA (Secure Hash).

**Default:** MD5

#### **NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** snmp\_auth\_protocol

#### **Community**

(Defaulted) Specifies the community string that is used to authenticate against the network device.

**Limits:** Up to 256 characters; case sensitive

**Default:** public

#### **NOTE**

This field applies to SNMP v1 and v2 only.

**JIL attribute:** snmp\_comm\_string

#### **Context engine ID**

(Optional) Specifies the context engine ID that uniquely identifies an SNMP entity within an administrative domain.

**Limits:** Up to 24 hexadecimal characters

**Example:** X'cab0'

#### **NOTE**

This field applies to SNMP v3 only.

---

**JIL attribute:** snmp\_context\_engine\_id

**Context name**

(Optional) Specifies the context name that the SNMP object ID variable belongs to.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** snmp\_context\_name

**Destination file**

(Optional) Specifies the output destination file. In a payload producing job, the file stores the serialized Java object produced by a job.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

**Host name**

Specifies the host name or IP address of the network device.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** snmp\_host

**MIB**

(Optional) Specifies the name of the MIB (management information base) file.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

This field is required only when the SNMP object ID is entered in string format.

**JIL attribute:** snmp\_mib

**Port**

(Defaulted) Specifies the port of the SNMP agent.

**Limits:** 0-2,147,483,647

**Default:** 161

**JIL attribute:** port

**Privacy protocol**

(Defaulted) Specifies the SNMP v3 privacy protocol to use. Valid values are: DES (Data Encryption Standard) and AES (Advance Encryption Standard).

**Default:** DES

**NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** snmp\_privacy

**SNMP object ID**

Specifies the SNMP object identifier in numeric or string format. In an SNMP Value Get job, it identifies the variable whose value you want to retrieve.

**Limits:** Up to 80 characters; case-sensitive

**Examples:** .1.2.3.1.2.1.1.6.0, agentLogLevel

**NOTE**

You must also specify the MIB when this field is specified in string format.

**JIL attribute:** snmp\_oid

**SNMP user**

Specifies the user name whose credentials are used for authentication.

**Limits:** Up to 145 characters, case-sensitive, cannot contain delimiters (such as spaces)

**NOTE**

This field is required for SNMP v3 only.

**JIL attribute:** snmp\_privacy\_user

**Subtree**

(Defaulted) Specifies whether to walk the SNMP subtree starting with the SNMP object ID.

**Default:** N

**JIL attribute:** snmp\_subtree

**Table view**

(Defaulted) Specifies whether the job retrieves the MIB data in a table format.

**Default:** N

**NOTE**

This field is valid only when Y is selected in the Subtree field.

**JIL attribute:** snmp\_table\_view

**Version**

(Defaulted) Specifies the SNMP version used when connecting to the network device. Valid values are: 1, 2, 3.

**Default:** 1

**JIL attribute:** snmp\_version

**Define an SNMP Value Set Job**

You can define a Simple Network Management Protocol Value Set (SNMPSET) job to set the value of an SNMP variable.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

**Follow these steps:**

1. Create an SNMP Value Set (SNMPSET) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs

- **Send to machine**  
Specifies the name of the machine on which the job runs.
- **SNMP object ID**  
Specifies the SNMP object identifier in numeric or string format. In an SNMP Value Set job, it identifies the variable whose value you want to change.
- **MIB** Specifies the name of the MIB (management information base) file.

#### NOTE

This field is required only when the SNMP object ID is entered in string format.

- **Host name** Specifies the host name or IP address of the network device.
  - **SNMP value**  
Specifies the type and value of the variable that you are changing in an SNMP Value Set job. The value is converted to the corresponding type.
3. Specify the following field if your SNMP version is v3:  
**SNMP user**  
 Specifies the user name whose credentials are used for authentication.
  4. (Optional) Specify optional [SNMP Value Set properties](#).
  5. (Optional) Specify [common properties that apply to all job types](#).
  6. Commit the job.  
 The SNMP Value Set job is defined.

### SNMP Value Set Job Examples

The following examples describe sample SNMP Value Set jobs.

#### **Example: Change the Value of an SNMP Variable with an Integer Value**

Suppose that you want to set the value of the agentLogLevel variable to its highest level to diagnose a problem. In this example, the host name of the network device is host.example.com and its port is 161. The SNMP version is v2 and the write community string is public. The name of the MIB file is cybermation.mib, which is located in the agent installation directory.

#### **Follow these steps:**

1. Create an SNMP Value Set job.
2. Enter the following properties:
  - Name -- setvarint
  - Send to machine -- snmpagent
  - SNMP object ID -- agentLogLevel
  - MIB -- cybermation.mib
  - Host name -- host.example.com
  - Port -- 161
  - Version -- 2
  - Community -- public
  - SNMP value -- int=8
3. Commit the job.

#### **Example: Change the Value of an SNMP Variable with an IP Address Value**

Suppose that you want to set the value of a variable to an IP address. In this example, the job specifies the SNMP object ID in numeric format, the DES privacy protocol, and the MD5 authentication protocol.

**NOTE**

The Owner value is the user ID associated with the authentication password. The SNMP user value is the user ID associated with the privacy password. The SNMP user and Owner users and their corresponding passwords are configured in AutoSys Workload Automation using the `autosys_secure` command.

**Follow these steps:**

1. Create an SNMP Value Set job.
2. Enter the following properties:
  - Name -- setvarip
  - Send to machine -- localhost
  - Owner -- Administrator@localhost
  - SNMP object ID -- .1.3.1.1.2.1.9.3
  - Host name -- snmp
  - Version -- 3
  - SNMP user -- user1
  - Authentication protocol -- MD5
  - Privacy protocol -- DES
  - SNMP value -- addr=172.24.2.20
3. Commit the job.

**SNMP Value Set Properties**

The SNMP Value Set category includes the following properties:

**Agent job class**

(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** `job_class`

**Authentication protocol**

(Defaulted) Specifies the authentication protocol to use when connecting with the user specified in the SNMP user field. Valid values are: MD5 (Message Digest 5) and SHA (Secure Hash).

**Default:** MD5

**NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** `snmp_auth_protocol`

**Community**

(Defaulted) Specifies the community string that is used to authenticate against the network device.

**Limits:** Up to 256 characters; case-sensitive

**Default:** public

**NOTE**

This field applies to SNMP v1 and v2 only.

**JIL attribute:** `snmp_comm_string`

**Context engine ID**



(Optional) Specifies the context engine ID that uniquely identifies an SNMP entity within an administrative domain.

**Limits:** Up to 24 hexadecimal characters

**Example:** X'cab0'

**NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** snmp\_context\_engine\_id

**Context name**

(Optional) Specifies the context name that the SNMP object ID variable belongs to.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

This field applies to SNMP v3 only.

**JIL attribute:** snmp\_context\_name

**Destination file**

(Optional) Specifies the output destination file. In a payload producing job, the file stores the serialized Java object produced by a job.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** destination\_file

**Host name**

Specifies the host name or IP address of the network device.

**Limits:** Up to 256 characters; case-sensitive

**JIL attribute:** snmp\_host

**MIB**

(Optional) Specifies the name of the MIB (management information base) file.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

This field is required only when the SNMP object ID is entered in string format.

**JIL attribute:** snmp\_mib

**Port**

(Defaulted) Specifies the port of the SNMP agent.

**Limits:** 0-2,147,483,647

**Default:** 161

**JIL attribute:** port

**Privacy protocol**

(Defaulted) Specifies the SNMP v3 privacy protocol to use. Valid values are: DES (Data Encryption Standard) and AES (Advance Encryption Standard).

**Default:** DES

**Note:** This field applies to SNMP v3 only.

**JIL attribute:** snmp\_privacy

### SNMP object ID

Specifies the SNMP object identifier in numeric or string format. In an SNMP Value Set job, it identifies the variable whose value you want to change.

**Limits:** Up to 80 characters; case-sensitive

**Examples:** .1.2.3.1.2.1.1.6.0, agentLogLevel

#### NOTE

You must also specify the MIB when this field is specified in string format.

**JIL attribute:** snmp\_oid

### SNMP user

Specifies the user name whose credentials are used for authentication.

**Limits:** Up to 145 characters, cannot contain delimiters (such as spaces); case-sensitive

**JIL attribute:** snmp\_privacy\_user

### SNMP value

Specifies the type and value of the variable that you are changing in an SNMP Value Set job. The value is converted to the corresponding type. Options for the type include the following:

- **int**  
Specifies an integer value.
- **uint**  
Specifies an unsigned integer value.
- **ticks**  
Specifies a time ticks (measured in hundredths of a second) value.
- **addr**  
Specifies an IP address value.  
**Note:** SNMP only supports IPv4 addresses. IPv6 addresses are not supported.
- **oid**  
Specifies an OID (object identifier) value.
- **string**  
Specifies an octet string value.
- **string.x**  
Specifies an octet string value represented in hexadecimal format.
- **counter32**  
Specifies a counter32 value. A counter32 value is a non-negative integer that increases until it reaches a maximum value of 232 - 1 (4294967295). It then wraps around and starts increasing again from zero.
- **gauge32**  
Specifies a gauge32 value. A gauge32 value is a non-negative integer that can increase or decrease, but cannot exceed a maximum value. The maximum value cannot exceed 232 - 1 (4294967295). The value of a Gauge has its maximum value whenever the information being modeled is greater or equal to that maximum value; if the information being modeled then decreases below the maximum value, the Gauge also decreases.

**Limits:** Up to 1024 characters; case-sensitive

**Examples:** int=123, addr=172.24.2.20

**NOTE**

- Specify multiple parameters by entering each parameter type and value on a new line.
- Enclose values that contain spaces in quotation marks.

**JIL attribute:** snmp\_value

**Version**

(Defaulted) Specifies the SNMP version used when connecting to the network device. Valid values are: 1, 2, 3.

**Default:** 1

**JIL attribute:** snmp\_version

SNMP user Specifies the user name whose credentials are used for authentication.

## Define a Job Flow in Application Editor

This article explains how to define a Job flow in the Application Editor.

### Define a Job Flow in Application Editor

A job flow is a graphical representation of the relationships between Web UI jobs and their dependencies. The job flow displays the conditions of the relationships that must be met before a job can run, and the successor jobs are triggered when a job finishes running.

This list provides a high-level overview of the tasks to define an editor.

1. Open Application Editor.
2. Add jobs to the Graph using one of the following methods:
  - Add a job using the Palette
  - Import jobs
3. Specify job properties for each job
4. (Optional) Add one or both of the following dependent objects:
  - Add global variables
  - Add logical operators
5. Draw dependency links
6. Set values for those dependencies, as follows:
  - Set a value for a job dependency
  - Set a value for a global variable dependency
7. Commit changes in Application Editor

### Add a Job to a Job Flow Using the Palette

The Palette in Application Editor provides icons for all the available job types.

#### **Follow these steps:**

1. Open Application Editor.  
A blank Graph appears in the workspace.

**NOTE**

You can open the last Graph you worked on by selecting Load Flow from the Select an action drop-down list.

2. Locate the job you want to add from the Palette.

**NOTE**

You can change which jobs appear in the Palette using the Filter Job Types tab of the Customize dialog.

3. Drag and drop the job icon onto the workspace
4. Repeat Steps 2 and 3 for each job you want to add to the job flow. The jobs are added to the job flow.

**NOTE**

We recommend that you perform the following actions:

- If a job must run after another job, place it below that job on the workspace.
  - If a job must run at the same time as another job, place it beside that job on the workspace.
  - If a job must run before another job, place it above the job on the workspace.
5. (Optional) Select Save Flow from the Select an action drop-down list and click Go.  
The workspace is saved.

**NOTE**

The job is not created on the AutoSys Workload Automation server until you commit the workspace.

### **Import Jobs**

You can import one or more jobs into Application Editor to view the flow, add jobs, edit job properties, add dependencies, or modify existing dependencies.

**Follow these steps:**

1. Open Application Editor.
2. Select Import from Scheduler from the Select an action drop-down list in the Graph section and click Go.
3. Select the server from which to import jobs from the Server drop-down list.
4. Specify one or more of the following properties and click Go:
  - Name
  - Group
  - Application

**NOTE**

- The following are permitted:
  - Valid job name, group, or application—Finds the specified jobs on the selected server that match the specified criteria.
  - Asterisk (\*), percent (%), or question mark (?), used as a wildcard with a partial name, where the \* or % represents zero or more characters, or ? represents a single character—Finds all jobs on the selected server that contain the partial job, group, or application name. For example, entering n\* in the Name field only would find all jobs on the server that begin with the letter n.
  - Asterisk (\*) or percent (%)—Finds all jobs on the selected server when the wildcard is specified for the Name field and the other two fields are blank.
- With the Name field blank, specifying a question mark and an asterisk (?\*) or percent (?%) in the Group or Application field will return a list of all jobs that have the Group or Application attribute specified.
- With the Name field blank, specifying an asterisk (\*) or percent (%) in the Group and Application fields finds all jobs on the selected server.

A list of jobs appears in the Search Results section.

**NOTE**

- The Search Results section will include all jobs and box jobs that match the criteria entered.
- No more than 5000 jobs can be imported into the Graph section.

5. Perform one of the following actions:

- Select the check box for the jobs and box jobs you want to import, and click Import Selected Jobs.
- Click Import All.

The Import Jobs dialog closes, and the specified jobs, box jobs, and their dependencies appear in the Graph.

6. (Optional) Select Save Flow from the Select an action drop-down list and click Go.

The workspace is saved.

**Specify Job Properties in Application Editor**

You specify properties to define a job. Each job has required properties, which are identified with a red dot.

**NOTE**

For specific job details, see the related job section in this space.

**Follow these steps:**

1. Open the job flow in Application Editor.
2. Right-click the job icon in the Graph and select Edit.
3. Specify the required properties.
4. Specify other optional properties as necessary.
5. Click Apply in the Properties section.  
The job properties are saved to the Application Editor workspace and the Properties section closes.
6. (Optional) Select Save Flow from the Select an action drop-down list and click Go.  
The workspace is saved.

**Add a Global Variable to a Job Flow**

You can add a global variable to a job flow if you have a set of jobs you want to run based on specific criteria.

When you add a global variable to a job flow in Application Editor, it is actually a global variable condition, not a global variable object. That is, the global variable is part of the dependency string, and processing will proceed when the variable is set with that value. These global variables do not necessarily exist on the AutoSys Workload Automation server before being set to satisfy the condition. Adding a global variable in the job flow automatically meets its requirement and triggers its dependent jobs.

**NOTE**

- Global variables you add to the flow do not have to be defined in the AutoSys Workload Automation database to be used in the flow.
- You cannot assign a value to a global variable that you add to the flow; however, you can define the properties of the dependency link that connects the global variable to another object. The dependency link defines the value that the global variable must be equal to, not equal to, less than, or greater than for the dependency condition to be met.
- The global variable you add to the flow is included in the dependency statement of the job that depends on it.

You can set a global variable value in a sendevent command. AutoSys Workload Automation keeps it in memory so it can be used.

**Follow these steps:**

1. Open the job flow in Application Editor.
2. Drag the Global Variable icon from the Palette and drop it onto the workspace.

3. Right-click the Global Variable icon and select Edit.
4. Specify the following global variable property:
  - Name**  
Specifies the name of the global variable.
  - Limits:** Up to 64 characters
5. Click OK.  
The name of the global variable changes in the Graph.
6. (Optional) Select Save Flow from the Select an action drop-down list and click Go.  
The global variable is added and the workspace is saved.

### **Add a Logical Operator to a Job Flow**

A logical operator uses the logically-combined result of two or more objects in the job flow to control subsequent processing. Objects are connected to logical operators using dependency links. You must set the value of the dependency link, such as the result of a job run from objects that precede the logical operator. The dependency link from a logical operator to the next object in the job flow does not have a value; the operator itself controls subsequent processing.

The following logical operators are available:



AND

Subsequent processing occurs as the result of the logical AND of two or more preceding objects. For example, if you want Job-C to run when Job-A and Job-B both complete successfully, the dependency statement for Job-C is: s(Job-A) & s(Job-B).



OR

Subsequent processing occurs as the result of the logical OR of two or more preceding objects. For example, if you want Job-C to run when either Job-A or Job-B completes successfully, the dependency statement for Job-C is: s(Job-A) | s(Job-B).

#### **Follow these steps:**

1. Open the job flow in Application Editor.
2. Drag the Logical Operator icon from the Palette and drop it onto the workspace.
3. (Optional) Select Save Flow from the Select an action drop-down list and click Go.  
The workspace is saved.

### **Draw Dependency Links**

You can create dependencies between jobs, global variables, and logical operators by drawing dependency links in the job flow. A dependency is created by linking two objects in the job flow. The dependency link appears as an arrow pointing from the dependent object to the object that depends on it. The dependency link property defines the condition of the object that must be met before the next step in the job flow is performed. You can select the properties for a dependency link originating from a job or a global variable by right-clicking the link and selecting Edit.

#### **Follow these steps:**

1. Open the job flow in Application Editor.
2. Click   
the Graph toolbar.
3. Hold down the mouse key and draw a line between two objects on the graph.

on

**NOTE**

- To set up a dependency between two jobs, select a predecessor job and drag the mouse to the next job to be released (successor).
- To set up a dependency between a global variable and a job or logical operator, draw the line starting from the global variable.

4. Repeat Step 3 to draw all the dependency links.
5. Click  to organize the layout of the job flow.
6. (Optional) Select Save Flow from the Select an action drop-down list, and click Go. The workspace is saved.

**Set a Value for a Job Dependency**

You set a value for a dependency link from a job in the job flow. The value sets the condition that must be met before the next step in the job flow is performed. For example, if JOB\_A runs successfully, processing will continue.

**Follow these steps:**

1. Open the job flow in Application Editor.
2. Right-click the dependency link from a job in the job flow and select Edit.
3. Specify the following fields:

**Dependency type**

Specifies the job status for the dependent job, as follows:

- • **SUCCESS**—If the job completes successfully, processing continues with the next step in the flow. The link color is green.
- **FAILURE**—If the job fails, processing continues with the next step in the flow. The link color is red.
- **DONE**—If the job indicates that it is done, processing continues with the next step in the flow. The link color is blue.
- **TERMINATED**—If the job is terminated, processing continues with the next step in the flow. The link color is magenta.
- **NOTRUNNING**—If the job is not running, processing continues with the next step in the flow. The link color is gray.
- **EXITCODE**—If the job returns the exit code you select, processing continues with the next step in the flow. The link color is black.

**[Operator]**

Specifies the operator to use when comparing the specified value with the exit code (=, !=, >, <)

**NOTE**

This field is only active when EXITCODE is selected for the dependency type.

**Value**

Specifies the value to which the exit code is compared.

**NOTE**

This field is only active when EXITCODE is selected for the dependency type.

4. Click OK.  
The job dependency value is set and the color of the dependency link changes to match the dependency type.

## **Set a Value for a Global Variable Dependency**

You set a value for a dependency link from a global variable in the job flow. The value sets the condition that must be met before the next step in the job flow is performed. For example, if the global variable `run_count` is equal to 5, processing will continue.

### **Follow these steps:**

1. Open the job flow in Application Editor.
2. Right-click the dependency link from a global variable in the job flow and select Edit.
3. Specify the following fields:

#### **Value**

Specifies the value to which the global variable is compared.

**Limits:** Up to 100 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), and hyphen; do not include quotation marks (")

#### **[Operator]**

Specifies the operator to use when comparing the specified value with the current value of the global variable, as follows:

- =**  
Equal. If the global variable is equal to the value specified, processing continues with the next step in the flow.
- !=**  
Not equal. If the global variable is not equal to the value specified, processing continues with the next step in the flow.
- <**  
Less than. If the global variable is less than the value specified, processing continues with the next step in the flow.
- >**  
Greater than. If the global variable is greater than the value specified, processing continues with the next step in the flow.

4. Click OK.  
The global variable dependency value is set.

## **Commit Changes in Application Editor**

You must commit job property and job flow changes to the appropriate AutoSys Workload Automation server to make the changes permanent on that server.

### **NOTE**

You must apply job property changes before you perform a commit; however, it is not necessary to save the Application Editor workspace before you perform a commit.

### **Follow these steps:**

1. Make the required changes in Application Editor.
2. Select Commit to Scheduler from the Select an action drop-down list, and click Go.
3. Select the server you want to commit the changes to, select the appropriate options, and click OK.

### **NOTE**

- If you save a job to the AutoSys Workload Automation server, any unsupported attributes are ignored.
- If you have deleted jobs in the flow, you must select Permanently Delete Listed Jobs to delete the jobs and box jobs in the Delete List on the specified server.

The Job Commit Progress dialog appears and the changes are committed to the scheduler.

4. Click Close when the commit process completes.  
The Job Commit Progress dialog closes.



## Manage the Advanced Integration for Hadoop Using WA

WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define the following job types using WA Advanced Integration for Hadoop:

- HDFS (Hadoop Distributed File System)
- Hive
- Oozie
- Pig
- Sqoop

### NOTE

To run these jobs, your system requires WA Advanced Integration for Hadoop.

WA Advanced Integration for Hadoop uses the standard agent functionality including encryption, control, configuration, security, logs, and so on.

## Define HDFS Jobs

Hadoop Distributed File System (HDFS) is the primary distributed storage for Hadoop applications. You can define an HDFS job to perform the following file operations on HDFS:

- Create one or more zero-byte files.
- Create one or more directories.
- Delete one or more files or directories.
- Copy one or more files or directories within HDFS, from the local file system to HDFS, or from HDFS to the local file system.
- Move one or more files or directories within HDFS or from the local file system to HDFS.

### Follow these steps:

1. Create an HDFS job.
2. Specify the following properties:
  - **Connection profile** Specifies the name of a connection profile.
  - **Job status refresh** (Optional) Specifies the pull interval.
  - **Agent job class** (Optional) Specifies the agent interpretation of the job class against the job classes defined.
  - **File operation**  
Specifies the file operation that the job performs on HDFS. The valid values are as follows:
    - Create zero size file (TOUCH)
    - Copy file or directory within HDFS (COPY)
    - Copy file or directory to HDFS (COPYFROMLOCAL)
    - Copy file or directory from HDFS (COPYTOLOCAL)
    - Make directory (MKDIR)
    - Move file or directory within HDFS (MOVE)
    - Move file or directory from local file system to HDFS (MOVEFROMLOCAL)
    - Remove file or directory (REMOVE)
3. Specify the following attribute if the type attribute is set to TOUCH:
  - **Path**  
Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.
4. Specify the following attributes if the type attribute is set to MKDIR:

5. • **Path**  
Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.
  - **Create parent directories**  
(Optional) Specifies whether you want to create a parent directory.  
**Default:** false (The job does not create a parent directory)
6. Specify the following attributes if the type attribute is set to REMOVE:
  - **Path**  
Specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.
  - **Remove recursively**  
(Optional) Specifies whether you want to delete the specified file or directory and all its sub-directories.  
**Default:** false (The job does not invoke the file operation on the file or directory and all its sub-directories)
  - **Skip trash**  
(Optional) Specifies whether you want to bypass the trash and delete the file or directory permanently.  
**Default:** false (The job does not bypass the trash).
7. Specify the following attributes if the type attribute is set to COPY, COPYFROMLOCAL, or COPYTOLOCAL:
  - **Source files and directories**  
Specifies the path of a file or directory that is to be copied from the source location.
  - **Destination**  
Specifies the path of a file or directory that is copied to the destination location.
  - **Overwrite existing**  
(Optional) Specifies whether you want to overwrite an existing file or directory in the destination location when a file or directory with the same name exists.**Default:** false (The job does not overwrite the file or directory).
  - **Preserve status flags**  
(Optional) Specifies whether you want to retain the timestamps, ownership, and permission attributes for the file or directory after it is copied from the source location to the destination location.**Default:** false (The job does not retain the timestamps, ownership, and permission attributes for the file or directory)

#### NOTE

The overwrite existing attribute does not apply for COPYTOLOCAL operation.

8. Specify the following attributes if the type attribute is set to MOVE or MOVEFROMLOCAL:
  - **Source files and directories**  
Specifies the path of a file or directory that is to be moved from the source location.
  - **Destination**  
Specifies the path of a file or directory that is moved to the destination location.
9. (Optional) Specify common attributes that apply to all job types.

The Hadoop HDFS job is defined.

#### NOTE

- (Optional) You can remotely browse the agent file system or the Hadoop file system using the FIND icon.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about

common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Properties](#).

- For information about required attributes and job specific optional attributes, see [Working with Jobs and Job Flows Using Web UI](#).

## Define Hive Jobs

Hive is a data warehousing infrastructure based on Hadoop. It allows you to query data in HDFS using HiveQL, an SQL-like query language. CA WA Advanced Integration for Hadoop supports the following Hive operations:

- Use Beeline to run a Hive job
- Use Beeline to cancel a Hive job

### Follow these steps:

1. Create a Hive job.
2. Specify the following properties:
  - **Connection profile** Specifies the name of a connection profile.
  - **Hive user**  
(Optional) Defines the user name to connect to the database.  
**Default:** anonymous (the Hive default user)
  - **Initialization script** (Optional) Specifies the name of the SQL script file to execute on the database.
  - **Script to execute**  
Specifies the name of the SQL script file to execute on the database.
  - **Additional parameters**  
(Optional) Specifies an additional Hadoop Beeline command line argument and its value.  
**Example:** --color=true --showHeader=true

### NOTE

You can specify multiple additional parameters. Separate each additional parameter with a blank space.

- **Hive variables**  
(Optional) Specifies a Hive variable and its value in the variable=value format. These variables are set at session level and used in Hive commands or queries. If you specify multiple variables, separate each variable with a comma.
  - **Configuration parameters**  
(Optional) Specifies a Hive configuration property and its value in the property=value format. If you specify multiple configuration properties, separate each configuration property with a comma.
  - **Job status refresh** (Optional) Specifies the pull interval.
  - **Agent job class** (Optional) Specifies the agent interpretation of the job class against the job classes defined.
3. (Optional) Specify common attributes that apply to all job types.

The Hadoop Hive job is defined.

### NOTE

- (Optional) You can remotely browse the Hadoop file system using the FIND icon.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Properties](#).

- For information about required attributes and job specific optional attributes, see [Working with Jobs and Job Flows Using Web UI](#).

## Define Oozie Jobs

Oozie jobs let you run an Oozie workflow and monitor the workflow status until termination. You can define an Oozie job to perform the following tasks:

- Start a workflow
- Stop a running workflow
- Suspend a running workflow
- Resume a suspended workflow

### Follow these steps:

1. Create an Oozie job.
2. Specify the following properties
  - **Connection profile** Specifies the name of a connection profile.
  - **Configuration file**  
Specifies the path to the job configuration file (.xml or .properties) on the Hadoop edge node. This can be a fully qualified path name or a path that is relative to the home directory of the user on the edge node.
  - **Job status refresh**(Optional) Specifies the pull interval.
  - **Agent job class**(Optional) Specifies the agent interpretation of the job class against the job classes defined.
3. (Optional) Specify common attributes that apply to all jobs.

The Hadoop Oozie job is defined.

### NOTE

- (Optional) You can remotely browse the Hadoop file system using the FIND icon.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as [common optional attributes](#). For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Properties](#).
- For information about required attributes and job specific optional attributes, see [Manage Jobs and Job Flows Using Web UI](#).

## Define Pig Jobs

Pig is a high-level scripting language that is used with Hadoop. Pig enables data workers to write complex data transformations using SQL-like scripting language named *Pig Latin*.

You can define a Pig job in AutoSys Workload Automation to perform the following operations:

- Run a Pig job
- Terminate a running Pig job

### Follow these steps:

1. Create a Pig job.

2. Specify the following properties:

- **Connection profile** Specifies the name of a connection profile.
- **Script to execute**  
Specifies the path to the Hadoop Pig script file to execute on the database.
- **Parameters file** (Optional) Specifies the path to the Hadoop Pig parameter file.
- **Property file**  
(Optional) Specifies the path to the Hadoop Pig property file.
- **Additional parameters** (Optional) Specifies an additional Hadoop Pig command line argument and its value.

**NOTE**

You can specify multiple additional parameters. Separate each additional parameter with a blank space.  
For example, `-verbose -warning -stop_on_failure -printCmdDebug`

- **Job status refresh** (Optional) Specifies the pull interval.
  - **Agent job class** (Optional) Specifies the agent interpretation of the job class against the job classes defined.
3. (Optional) Specify common attributes that apply to all job types.

The Hadoop Pig job is defined.

**NOTE**

- (Optional) You can remotely browse the Hadoop file system using the FIND icon.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Properties](#).
- For information about required attributes and job specific optional attributes, see [Working with Jobs and Job Flows Using Web UI](#).

## Define Sqoop Jobs

Sqoop is a tool that is designed to transfer data between Hadoop and relational databases, such as MySQL and Oracle. You can define a Sqoop job in AutoSys Workload Automation to perform the following operations:

- Import data from a relational database to Hadoop HDFS.
- Export data from Hadoop HDFS to a relational database.
- Terminate a running Sqoop job.

**Follow these steps:**

1. Create a Sqoop job.
2. Specify the following properties:
  - **Connection profile** Specifies the name of a connection profile.
  - **Sqoop user** Specifies the user ID that is required for connecting to an edge node.
  - **Operation type** Specifies the operation that the Sqoop job performs. The valid values are as follows:

- **IMPORT**—Imports data from a relational database (such as MySQL and Oracle) to Hadoop HDFS.
- **EXPORT**—Exports data from Hadoop HDFS to a relational database.
- **Table** Specifies the database table from which the data is imported or to which the data is exported.
- **Null string interpretation** (Optional) Specifies the string that indicates a null value for string columns.
- **Nonstring null interpretation** (Optional) Specifies the string that indicates a null value for non-string columns.
- **Additional parameters** (Optional) Specifies an additional Hadoop Sqoop command line argument and its value.

#### NOTE

- You can specify multiple additional parameters. Separate each additional parameter with a blank space. For example, `-m 1 --outdir directorypath --validate`.
  - When you specify multiple additional parameters, use the same format that you use while specifying Hadoop Sqoop command line arguments.
  - **Job status refresh** (Optional) Specifies the pull interval.
  - **Agent job class** (Optional) Specifies the agent interpretation of the job class against the job classes defined.
3. Specify the following attributes if the sqoop type attribute is set to **IMPORT**:
    - **Destination directory** Specifies the HDFS destination directory to which the data is imported.
    - **Columns** (Optional) Specifies the columns to import from the database table. This value is passed to the `--columns` Hadoop Sqoop import control argument.
    - **SQL WHERE clause** (Optional) Specifies the WHERE clause to use during import. This value is passed to the `--where` Hadoop Sqoop import control argument.
    - **Boundary query** (Optional) Specifies the boundary query to use for creating splits. This value is passed to the `--boundary-query` Hadoop Sqoop import control argument.
  4. Specify the following attribute if the sqoop type attribute is set to **EXPORT**:
    - **Source directory** Specifies the HDFS source directory from where the data is exported.
    - **Staging table** (Optional) Specifies the table in which the data is staged before being inserted into the destination table.
    - **Clearstaging table** (Optional) Specifies whether you want to delete the data in the staging table. When you define the sqoop clearstagingtable attribute, you must also specify the sqoop stagingtable attribute in your job definition. **Default:** false (The job does not delete the data in the staging table)
  5. (Optional) Specify common attributes that apply to all job types.

The Hadoop Sqoop job is defined.

#### NOTE

- (Optional) You can remotely browse the Hadoop file system using the **FIND** icon.
- Attributes that have a default value automatically apply to the job definitions; therefore, they are optional. For example, jobs with no specified job type are defined as command jobs by default. Other optional attributes specify information that is not required but affects how or when a job runs, such as attributes that specify scheduling conditions.
- Some optional attributes are common to all job types but others apply to certain jobs types only. Optional attributes that apply to all job types are known as *common optional attributes*. For more information about common optional attributes and the values that you can specify for them (including their default values when applicable), see [Common Job Properties](#).
- For information about required attributes and job-specific optional attributes, see [Working with Jobs and Job Flows Using Web UI](#).

## Manage Wake on LAN Jobs Using Web UI

You can save energy using the agent's Wake on LAN (WOL) feature to automate the startup and shutdown of your computers. WOL lets you define and schedule WOL jobs to send a signal to a server to turn it on. When the server is no longer needed, you can schedule a different command job to power it down.

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows. Your agent administrator must configure the agent to support WOL. For more information about configuring the agent to support WOL, see the *Workload Automation Agent for UNIX, Linux, or Windows Implementation documentation*.

Wake on LAN (WOL) is a hardware and software solution that lets you wake up a computer remotely. The solution requires an ACPI-compliant computer and a special software program that sends a signal to the computer's network card to wake it up. The agent provides the AMD magic packet to broadcast the signal to a computer that has been soft-powered-down (ACPI D3-warm state).

### Define a Wake on LAN Job

You can define a Wake on LAN (WOL) job to send a signal to a server to turn it on. The job can wake up a remote computer that has been soft-powered-down (ACPI D3-warm state).

### NOTE

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

### Follow these steps:

1. Create a Wake on LAN (WOL) job.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Broadcast address**  
Specifies the IP address of the LAN or subnet of the computer that receives the Wake on LAN (WOL) signal.
  - **MAC address**  
Specifies the Media Access Control (MAC) address of the computer that receives the Wake on LAN (WOL) signal.
3. (Optional) Specify optional [Wake on LAN properties](#).
4. (Optional) Specify [common properties that apply to all job types](#).
5. Commit the job.  
The Wake on LAN job is defined.

### Wake on LAN Job Examples

The following examples describe sample Wake on LAN jobs.

#### Example: Broadcast the WOL Signal and Ping a Port

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server with the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails.

### Follow these steps:

1. Create a Wake on LAN job.

2. Enter the following properties:
  - Name -- wol\_ping\_job
  - Send to machine -- agentnme
  - Broadcast address -- 172.16.255.255
  - MAC address -- 00-1E-4F-C1-0F-FE
  - Ping host -- 172.16.1.101
  - Ping ports -- 7
3. Commit the job.

### **Example: Broadcast the WOL Signal Including a Password**

This example broadcasts the WOL signal including a password to the subnet with the 172.16.255.255 broadcast IP address and the server with the 11-22-33-44-55-66 MAC address. If the specified password matches the password stored on the server's network card, the server wakes up. The job completes successfully after the scheduler sends the WOL signal without verifying that the machine starts.

#### **Follow these steps:**

1. Create a Wake on LAN job.
2. Enter the following properties:
  - Name -- wol\_pwd\_job
  - Send to machine -- agentnme
  - Broadcast address -- 172.16.255.255
  - MAC address -- 11-22-33-44-55-66
  - Wake password -- AA-BB-CC-DD-EE-FF
3. Commit the job.

### **Example: Specify a Timeout Limit When you Broadcast the WOL Signal to a MAC address**

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server identified by the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 becomes available within 5 minutes (300 seconds), the job completes successfully; otherwise, it exceeds the timeout limit and fails.

#### **Follow these steps:**

1. Create a Wake on LAN job.
2. Enter the following properties:
  - Name -- wol\_timeout\_job
  - Send to machine -- agentnme
  - Broadcast address -- 172.16.255.255
  - MAC address -- 00-1E-4F-C1-0F-FE
  - Ping host -- 172.16.1.101
  - Ping ports -- 7
  - Ping timeout -- 300
3. Commit the job.

### **Wake on LAN Properties**

The Wake on LAN category includes the following properties:

#### **Agent job class**



(Optional) Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process a greater number of jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.

**Limits:** Up to 32 characters; cannot contain delimiters (such as spaces)

**JIL attribute:** `job_class`

### Broadcast address

Specifies the IP address of the LAN or subnet of the computer that receives the Wake on LAN (signal). On UNIX, you can use `ifconfig` to obtain the broadcast address. On Windows, you must calculate the broadcast address by performing a bitwise OR operation on the IP address and the bit complement of the subnet mask. You can obtain the IP address and subnet mask using `ipconfig`. The special broadcast address of 255.255.255.255 broadcasts to the subnet that is local to the agent machine.

**JIL attribute:** `broadcast_address`

### MAC address

Specifies the Media Access Control (MAC) address of the computer that receives the Wake on LAN (WOL) signal. The MAC address consists of six 2-digit hexadecimal values separated by dashes (-) or colons (:). On some UNIX versions, you can obtain the MAC address using `ifconfig` (listed under `HWaddr`). On Windows, you can obtain the MAC address using `ipconfig /all` (listed under `Physical Address`).

**Example:** 00-1E-4F-C1-0F-FE

**JIL attribute:** `mac_address`

### Ping host

(Optional) Specifies the host name or IP address of the target host that receives the Wake on LAN (WOL) signal. Specify a value for this field to enable the Ping timeout and Ping ports fields.

**Example:** 172.16.1.101

#### NOTE

If you do not specify the Ping host, the job completes immediately after the WOL signal is sent without verifying whether the signal worked.

**JIL attribute:** `ping_host`

### Ping ports

(Defaulted) Specifies the ports to contact after the agent sends the Wake on LAN (WOL) signal. If you do not enter a value, the agent contacts the defaults ports. After the agent sends the Wake on LAN (WOL) signal, it attempts to connect to the ports specified in the Ping ports field. The agent connects to the ports to confirm that the computer is awake by verifying that an application is listening on the port. The agent creates a TCP connection; it does not send and receive ICMP ping packets. If at least one of the specified ports is available the job completes successfully; otherwise it fails.

**Limits:** 0-65535

#### NOTE

- To specify multiple ports, separate each port with a comma.
- If a firewall blocks the specified port or the machine is not addressable on the network, the connection attempt fails.

**JIL attribute:** `ping_ports`

### Ping timeout

(Defaulted) Specifies the maximum amount of time in seconds that the agent attempts to connect to the target host before a timeout occurs.

**Limits:** 0-99999

**NOTE**

Also specify the Ping host value when you use this field.

**JIL attribute:** ping\_timeout

**Wake password**

(Optional) Specifies the Wake on LAN (WOL) password for network cards that support the Secure On security feature. After the agent broadcasts the WOL signal, the network card wakes up the computer only if the specified password is correct. The WOL password is formatted as either a set of four 2-digit or six 2-digit hexadecimal values separated by dashes (-), periods (.), or colons (:).

**Examples:** AA-BB-CC-DD, AA-BB-CC-DD-EE-FF

**NOTE**

The password is not encrypted on the scheduling manager.

**JIL attribute:** wake\_password

## Manage Informatica Jobs

Informatica jobs let you start predefined workflows on Informatica PowerCenter. In Informatica, jobs are known as workflows. A workflow can contain multiple tasks. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

**NOTE**

- To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.
- Due to a known issue in Informatica, you cannot run workflow instances in parallel.

Informatica PowerCenter provides an environment that lets you load data into a centralized location, such as a data warehouse or operational data store (ODS). You can extract data from multiple sources, transform the data according to business logic you build in the client application, and load the transformed data into file and relational targets.

### Define an Informatica Job

You can define an Informatica job to start a predefined workflow on Informatica PowerCenter. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

**Follow these steps:**

1. Create an Informatica (INFORMATICA) job in either Quick Edit or Application Editor.  
The Properties section for the Informatica job appears.
2. Specify the following required properties:
  - **Name**  
Defines the name of the job that you want to schedule.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - **Folder name** Specifies the Informatica PowerCenter repository folder name where the specified workflow resides. You can obtain this value from the Informatica server.  
**Limits:** Up to 79 characters.
  - **Workflow name** Specifies the name of the Informatica PowerCenter workflow to be started.

**Limits:** Up to 79 characters.

Click

the 

to search for an Informatica workflow.

3. (Optional) Specify optional [InformaticaProperties](#).
4. (Optional) Specify [Manage Common Job Properties](#).
5. Commit the job.  
The Informatica job is defined.

## **Informatica Job Examples**

### **Example: Start a Task in a Workflow on Informatica PowerCenter**

This example starts the copy task in the WF\_1 workflow on the Informatica PowerCenter. The workflow is started from the copy task. Based on the status of tasks, the workflow status is reported.

#### **Follow these steps:**

1. Create an Informatica job.
2. Enter the following properties:
  - Name -- INFORMATICA
  - Send to machine -- 160\_infor
  - Folder name – FSTL
  - Workflow name – WF\_1
  - Task name – copy
  - Pass when successful – 1
3. Commit the job.

### **Example: Process a Parameter File in a Workflow**

This example processes the wf2.prm parameter file on the Informatica server in the WF\_2 workflow. Based on the status of tasks, the workflow status is reported.

The wf2.prm parameter file includes symbolic variables that override the default log directory and file name:

```
;Workflow Parameters [INFAPLUGIN.WF:wf_3] ;Session parameters $PMSessionLogFile=wf3.log
$PMSessionLogDir=C:\Infa\Log ;mapping parameters
```

#### **Follow these steps:**

1. Create an Informatica job.
2. Enter the following properties:
  - Name -- INFORMATICA
  - Send to machine -- 160\_infor
  - Folder name – FSTL
  - Workflow name – WF\_2
  - Parameter file – C:\Infa\wf2.prm
  - Pass when successful – 1
3. Commit the job.

## **Informatica Properties**

The Informatica category includes the following properties:

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- **Folder name** Specifies the Informatica PowerCenter repository folder name where the specified workflow resides. You can obtain this value from the Informatica server.  
**Limits:** Up to 79 characters.
- **Workflow name** Specifies the name of the Informatica PowerCenter workflow to be started.  
**Limits:** Up to 79 characters.

Click

the 

to search for an Informatica workflow.

- **Informatica target** (Optional) Specifies the name of the Informatica target (infaTarget). The corresponding directory (install\_dir/config/informatica/infaTarget) stored on the agent contains the properties for connecting to the Informatica server and one or more database repositories.  
**Limits:** Up to 128 characters.

#### NOTE

If you do not specify any value in the Informatica target field, the agent uses the default value specified in the informatica.server.target.default parameter in the agentparm.txt file.

- **Repository name** (Optional) Specifies the name of the repository that is created on the Informatica server. The repository can be obtained from the Informatica Administrative console. The corresponding property file (repositoryName.properties) that is stored on the agent contains the database connection properties for a specified Informatica target (infaTarget).  
**Limits:** Up to 128 characters

#### NOTE

If you do not specify any value in the Repository name field, the agent uses the default value specified in the informatica.repository.name.default parameter in the informatica.properties file stored on the agent.

- **Informatica domain** (Optional) Specifies the name of the LDAP security domain that is configured with the Informatica Server. Specify this attribute in your job definition, to authenticate the user against the specified LDAP security domain.  
**Limits:** Up to 80 characters
- **Informatica user** (Optional) Specifies the Informatica server user name. **Limits:** Up to 145 characters

#### NOTE

- If you specify the Informatica user in your job definition, you must set the password that is associated with the informatica user using the autosys\_secure command.
- If you do not specify any value in the Informatica user field, the agent uses the default value that is specified in the informatica.user.id.default parameter in the informatica.properties file that is stored on the agent.

- **Workflow instance** (Optional) Specifies the name of the workflow instance to be started. When the task name and the workflow instance name are specified, the workflow instance is started from the specified task. When the task name is not specified, the workflow instance starts from the beginning. If the instance name is not specified, the workflow is started.  
**Limits:** Up to 79 characters

#### NOTE

Before you specify the workflow instance name, you must specify the workflow using the Workflow name field.

- **Task name** (Optional) Specifies the name of the task to start the workflow from. If the task name is not specified, the workflow starts from the beginning.  
**Limits:** Up to 79 characters.

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Click

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to search for a task in an Informatica workflow.

- **Parameter file**(Optional) Specifies the path and name of a parameter file on the Informatica server. Informatica PowerCenter processes the parameter file when running a workflow. The parameter file contains values for symbolic variables and must adhere to the format that Informatica PowerCenter expects. A property value that is set in a parameter file overrides a value for this property that is set at the session level or elsewhere.  
**Limits:** Up to 256 characters
- **Pass when successful**(Optional) Specifies whether the status of the workflow is retrieved directly from Informatica or inferred depending on the successful execution of all the tasks.
  - **0**Returns the workflow status as reported by Informatica. In Informatica, a workflow can succeed even if one or more tasks in the workflow fail.
  - **1**Returns the workflow status based on the status of tasks in the Informatica workflow. If all tasks in the workflow complete successfully, the agent reports the status of the workflow as successful. If any task in the workflow fails, the agent reports the status of the workflow as failed.**Default:** 0

#### NOTE

If you do not specify any value in the Pass when successful field, the agent uses the default value that is specified in the `informatica.passonsuccessonly.default` parameter in the `agentparm.txt` file.

### Search for an Informatica Workflow

You can use the Search Informatica Workflows dialog to search for and select a workflow that is defined on Informatica PowerCenter.

#### Follow these steps:

1. Click the  next to the Workflow name field in the Properties section for the Informatica job. The Search Informatica Workflows dialog opens.  
**Note:** The fields in this dialog are populated with the information that is specified in the Properties section for the Informatica job.
2. (Optional) Specify the information in the Informatica Server Info section.  
**Note:** The fields in this section are populated with the information that is specified in the Properties section for the Informatica job, if any.
3. (Optional) Edit the folder name in the Folder name field.
4. (Optional) Specify the filter criteria in the following field:
  - **Matching Pattern**  
Specifies the matching pattern string that is used to retrieve the names of workflows. Only workflows whose names contain the search pattern are displayed. % is the only wildcard character that is supported for the pattern matching.  
**Limits:** Up to 79 characters  
**Example:** If you specify "A%ab%" as the matching pattern, all the workflows whose names start with A and contain ab in the middle or at the end are displayed. If you specify "%file%" as the matching pattern, all the workflows whose names contain file are displayed, for example: file\_task, X\_file\_Task and T\_file. If you specify "%inf%\_" as the matching pattern, all workflows whose names contains INF and underscore ( ) are displayed.

#### NOTE

If you do not specify a matching pattern, all workflows that belong to the specified repository folder are displayed.

- Click Go.  
The Workflows table is populated with the list of workflows that meet the matching pattern string.

**NOTE**

You can sort the data in the table by clicking on a table column heading.

- Select a workflow and click Select Workflow to specify the workflow name in the job definition.  
The selected workflow details and the Informatica properties are populated in the Properties section for the Informatica job.

### **Search for Informatica Tasks in an Informatica Workflow**

You can specify the Folder Name and Workflow Name as filtering criteria and use the Search Informatica Tasks dialog to retrieve the workflow tasks that are defined in an Informatica Workflow.

**NOTE**

You must define the workflow name before you start searching for a workflow task.

**Follow these steps:**

- Specify Send to machine (Agent name) and Folder name in the Properties section for the Informatica job.
- Complete the steps in the [Search for an Informatica Workflow](#) section.
- Click  
the  next to the Task name field.  
The Browse Informatica Tasks dialog displays a list of workflow tasks in the selected Informatica workflow.

**NOTE**

You can sort the data in the table by clicking on a table column heading.

- (Optional) Select a task and click Task Details to display the details of the selected task in a tabular format.
- (Optional) Select a task and click Object Hierarchy to display the hierarchy of the selected task within the workflow in a graphical format.**Note:** The Object Hierarchy button does not display in the Internet Explorer and Microsoft Edge browsers.
- Select a task and click Select Task to specify the task name in the job definition.  
The selected task name is set on the Basic page of the job definition.

### **View Workflow Details**

You can now view the Informatica workflow details in a tabular or graphical format.

**Follow these steps:**

- Create an Informatica (INFORMATICA) job in either Quick Edit or Application Editor.  
The Properties section for the Informatica job appears.
- Specify the Folder name and click  
the  to search for an Informatica workflow.The Search Informatica Workflows dialog opens.
- (Optional) Specify the filter criteria.
- Click Go.The Search Results display a list of workflows.
- (Optional) Select a workflow and click Workflow Details to display the sequence of tasks for the selected workflow in a tabular format.
  - The Workflow Details window displays the Workflow Name, Workflow ID, and the following Task Sequences:

- • FromInstance
  - ToInstance
  - Link
6. (Optional) Select a workflow and click Object Hierarchy to display the hierarchy of all tasks within the selected workflow in a graphical format.**Note:** The Object Hierarchy button does not display in the Internet Explorer and Microsoft Edge browsers.
  7. Click OK to return to the Search Informatica Workflows dialog.

### **View Workflow Task Details**

You can now view the Informatica workflow task details in a tabular or graphical format.

#### **Follow these steps:**

1. Select the task in the Browse Informatica tasks dialog and click Task Details.  
The Task Details window displays the Task Instance Name and the following properties' Parameters and Values in a tabular format:
  - Task Type
  - Mapping Name
  - Note:** This parameter is displayed when the selected Task Type is SESSION.
  - Folder Id
  - Folder Name
  - Workflow Id
  - Workflow Name
  - Task Instance Id
  - Task Instance Name
2. Click Close to close the Task Details window.
3. (Optional) Select a task and click Object Hierarchy to display the hierarchy of the selected task within the workflow in a graphical format.**Note:** The Object Hierarchy button does not display in the Internet Explorer and Microsoft Edge browsers.
4. Click OK to close the window.

### **View Workflow Status**

You can view the workflow status of an Informatica job.

#### **Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Get Workflow Status icon under Actions.

#### **NOTE**

The Actions column is displayed only after a job is run. If the job is not yet run, you can see no information in the Start Time and End Time columns and the Actions column is not displayed.

The Informatica Workflow status with the following information is displayed.

#### **Workflow Id**

Specifies the workflow id created with the job.

#### **Workflow Run Id**

Specifies the run id of the workflow.

#### **Workflow Status**

Specifies the workflow status. **Example:** succeeded

---

## **View Workflow Run Details**

You can view the workflow run details of an Informatica job.

### **NOTE**

Only the latest run details are displayed.

### **Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Get Workflow Run Details icon under Actions.

### **NOTE**

The Actions column is displayed only after a job is run.

The Informatica Workflow Run Details window displays the workflow and tasks details in a tabular format.

4. Click OK to close the window.

## **Restart an Informatica Workflow**

You can restart a failed workflow or its instance from the point of failure or from the beginning in recovery mode. For example, if a workflow is terminated abnormally, you can restart the workflow.

### **NOTE**

To restart a job, it must be in Failed state.

### **Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Restart icon under the Actions column.

### **NOTE**

You can also restart a workflow from Job Flow or Send Events.

The Restart Workflow dialog opens.

4. (Optional) Select a SessionTask that you want to restart the workflow from.

### **NOTE**

To restart a workflow or its instance from a point of failure, select a session task. If a session task is not selected, the workflow is restarted from the beginning in recovery mode.

5. Click Restart.  
The server restarts the specified workflow or its instance in recovery mode from the specified session task or from the beginning.

You can also restart an Informatica workflow from the Dashboard tab.

### **Follow these steps:**

1. Select Dashboard, enter Name under the Jobs pane, and click Go.  
The search results are displayed.
2. Right-click on the job and select Restart.



**NOTE**

You can also restart a workflow from Send Events.

3. (Optional) Select a SessionTask that you want to restart the workflow from.

**NOTE**

To restart a workflow or its instance from a point of failure, select a session task. If a session task is not selected, the workflow is restarted from the beginning in recovery mode.

4. Click Restart.  
The server restarts the specified workflow or its instance in recovery mode from the specified session task or from the beginning.

You can also restart an Informatica workflow from the Monitoring tab.

**Follow these steps:**

1. Select Monitoring, Views.  
Select a job from the displayed list.
2. Right-click on the job and select Restart.

**NOTE**

You can also search for a job from the Jobs and Alerts tab.

You can also restart a workflow from Send Events, and Flow tab.

3. (Optional) Select a SessionTask that you want to restart the workflow from.

**NOTE**

To restart a workflow or its instance from a point of failure, select a session task. If a session task is not selected, the workflow is restarted from the beginning in recovery mode.

4. Click Restart.  
The server restarts the specified workflow or its instance in recovery mode from the specified session task or from the beginning.

**Retrieve the Log for an Informatica Workflow**

You can retrieve the log of a completed or failed Informatica workflow or its sessions. The workflow log details include the workflow run ID, log type (workflow or session), log output, and so on.

**Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Select the required Job.
4. Expand Job Runs, select the required Job Run.
5. Select Job Workflow Log against Action and click Go.  
The Informatica Workflow Log dialog opens.
6. (Optional) Select the Session Name from the drop-down list.
7. Click OK.  
The Job Workflow Log section displays the log details for the specified workflow or session.

## Manage Web UI Connection Profiles

A *connection profile* is a set of attributes that are required for connecting to environments where the job runs. For example, to run a job in the Hadoop environment, you can create a connection profile with the appropriate parameters to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database.

A connection profile lets you save the connection information and you can use the connection profile in the job definition. For example, when you define a Hadoop (HDFS, Hive Oozie, Sqoop, or Pig) job, you can assign a connection profile to it. A connection profile applies to the following Hadoop job types:

- HDFS
- Hive
- Oozie
- Sqoop
- Pig

### Create a Connection Profile

Before you define a Hadoop job, create a connection profile. You can create one or more connection profiles specific to the Hadoop environment.

#### Follow these steps:

1. Create a Connection Profile.
2. Specify the following properties: value combinations (as appropriate) based on whether you want to connect to the Hadoop cluster, Oozie server, Hive database or Scoop database:

**Connection name** Defines the name of the connection profile.

**Description** (Optional) Specifies a description for the connection profile.

**Host** Defines the name of the Hadoop edge node where the specified operation is performed. This can be a simple name, fully qualified domain name, or IP address.

**Port** (Optional) Specifies the Secure Shell (SSH) port on the Hadoop edge node.

**Default:** 22**Hadoop bin directory path**(Optional) Specifies the path to the Hadoop client binaries on the Hadoop edge node.

#### NOTE

- If the path includes a colon, enclose it in double quotation marks.
- Specify this attribute only if the Hadoop client binaries are not in the user's path. The hadoop bin directory path value is prepended to the client name at execution time.

**Hadoop name node** (Optional) Specifies the namenode of the Hadoop HDFS cluster. **Oozie server URL** (Optional) Specifies the Oozie server URL.

3. Specify the following attributes for Hive database connection:**JDBC URL** Specifies the JDBC URL that you provide to the Hadoop Beeline client to connect to the database.

#### NOTE

If you are creating a connection profile to connect to the Hive database, you must specify the JDBC URL attribute in the connection profile.

**Drive class** (Optional) Specifies the name of the JDBC driver class that you provide to the Hadoop Beeline client.

4. Specify the following attributes for Sqoop database connection:

**JDBC URL** Specifies the JDBC URL that you provide to the Hadoop Sqoop client to connect to the database.

**NOTE**

If you are creating a connection profile to connect to the Sqoop database, you must specify the JDBC URL attribute in the connection profile.

**Drive class** (Optional) Specifies the name of the JDBC driver class to execute on the database.

The connection profile is created and added in the database.

**NOTE**

The availability to create, modify, and delete a connection profile depends on the user's access rights defined in CA EEM.

## Manage SQL Server Agent Jobs Using Web UI

SQL Server Agent jobs let you execute predefined jobs in the SQL Server Agent to perform administrative tasks.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

### Define a Microsoft SQL Server Agent Job

You can define a MSSQL Server Agent job to execute predefined jobs in the SQL Server Agent to perform administrative tasks.

Follow these steps:

1. Create a MSSQL Server Agent job in either Quick Edit or Application Editor.  
The Properties section for the SQL Server Agent job appears.
2. Specify the Job details field:
  - Job name**  
Specifies the name for a SQL Server Agent job.  
**Limits:** Up to 128 characters.
  - Job ID**  
Specifies the SQL Server Agent job identification number.  
**Limits:** Up to 36 characters.
3. (Optional) Specify optional [SQL Server Agent Properties](#).
4. (Optional) Specify common properties that apply to all job types.
5. Commit the job.  
The SQL Server Agent job is defined.

### Microsoft SQL Server Agent Job Example

#### **Example: Run a Predefined Job on a Target Server**

This example runs a predefined job that is named MSSQL on the MyServer target server.

#### **Follow these steps:**

1. Create a SQL Server Agent job.
2. Enter the following properties:

- Target DB: MyTargetDB
  - Job name: MSSQL
  - MSSQL server name: MyServer
  - User name: MyDomainUser
  - Domain: MyDomain
3. Commit the job.

### **Microsoft SQL Server Agent Properties**

The SQL Server Agent category includes the following properties:

#### **Target DB**

Specifies the target server database.

**Limits:** Up to 128 alphanumeric or underscore characters. The first character cannot be a number.

#### **MSSQL server name**

Specifies the MSSQL server to start the job on.

**Limits:** Up to 128 characters.

#### **Step name**

Specifies the name of the job step to start the job from. The default is to start the job from the first step.

**Limits:** Up to 128 characters.

#### **User name**

Specifies the user that the agent uses to connect to the database.

**Limits:** Up to 145 characters; up to 64 characters for user and up to 80 characters for token.

#### **Domain**

Specifies the Windows domain name when Windows authentication is used to connect to the database server.

**Limits:** Up to 24 alphanumeric characters.

### **Search for a Microsoft SQL Server Agent Job**

You can use the Search MSSQL Jobs dialog to search for and select a job that is defined in Microsoft SQL Server Agent.

#### **Follow these steps:**

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Enter the machine name (agent name) in the Send to machine field.
3. Click the



icon next to the Job name field in the Properties section for the MSSQL Server job.  
The MSSQL Job Search dialog opens.

#### **NOTE**

The fields in this dialog are populated with the information that is specified on the basic page of the job definition, if any.

4. (Optional) Specify the filter criteria in the following fields in the Filter Options section:
  - **Job Type**  
Specifies the job type to filter on. Options are LOCAL and MULTI-SERVER. The default is to return all jobs regardless of job type.
  - **Owner Login Name**

Specifies the login name of the owner of the job to filter on. The default is to return all jobs regardless of the owner login name.

– **Category**

Specifies the category name to filter on. Options are as follows:

- Uncategorized (Local)
- Data Collector
- Database Engine Tuning Advisor
- Database Maintenance
- Full-Text
- Jobs From MSX
- Log Shipping
- REPL-Alert Response
- REPL-Checkup
- REPL-Distribution
- REPL-Distribution Cleanup
- REPL-History Cleanup
- REPL-LogReader
- REPL-Merge
- REPL-QueueReader
- REPL-Snapshot
- REPL-Subscription Cleanup

The default is to return all jobs regardless of category.

– **Subsystem**

Specifies the name of the subsystem to filter on. Options are as follows:

- TSQL
- ActiveScripting
- CmdExec
- Snapshot
- LogReader
- Distribution
- Merge
- QueueReader
- AnalysisQuery
- AnalysisCommand
- SSIS
- PowerShell

**NOTE**

When this field is specified, only jobs with job steps that belong to the specified subsystem are listed. The default is to return all jobs regardless of subsystem.

– **Execution Status**

Specifies the execution status to filter on. Options are as follows:

- **Suspended**  
Lists only jobs that are not idle or suspended.
- **Executing**  
Lists only jobs that are executing.
- **WaitingForThread**

Lists only jobs that are waiting for a thread.

- **BetweenRetries**

Lists only jobs that are between retries.

- **Idle**

Lists only jobs that are idle.

- **PerformingCompletionActions**

Lists only jobs that are performing completion options.

The default is to return all jobs regardless of execution status.

- **Date Created**

Specifies a date range based on when the job was created or last modified. To filter by date, specify the following:

- Date comparator (After or Before)
- Date (using the calendar)
- Time

**Example:** DateCreated After Sept 13, 2013 12:00:00 AM (returns jobs that were created since midnight on September 13, 2013).

- **Enabled**

Indicates whether to list only enabled or disabled jobs.

- **TRUE**

Lists only enabled jobs.

- **FALSE**

Lists only disabled jobs.

The default is to return enabled and disabled jobs.

- **Description**

Specifies the job description to filter on. The Description field can include the wildcard characters \*, ?, ^, and [] for pattern matching. The \* wildcard character is equivalent to the % SQL wildcard character. The ? wildcard character is equivalent to the \_ SQL wildcard character. The default is to return all jobs regardless of description.

5. Click Go.

The Search Results table is populated with the list of jobs that meet the search criteria.

### NOTE

You can sort the data by clicking on a table column heading.

6. Select the job you want from the table.

7. Click Select Job Name to specify the job name or click Select Job ID to specify the job ID in the job definition.  
The specified job name or job ID of the selected job is set on the basic page of the job definition.

### Example: Retrieve a List of Microsoft SQL Server Agent Jobs Based on Job Description Criteria

Suppose that you want to retrieve a list of all jobs with the job description that starts with sqljob.

#### To list the Microsoft SQL Server Agent jobs based on job description criteria

1. Open the basic page of the MSSQL Server job definition.

2. Enter **AGENT\_MSSQL** in the Send to machine field.

3. Click the



icon next to the Job name field in the Properties section for the MSSQL Server job..  
The Search MSSQL Jobs dialog opens.

4. Enter **sqljob\*** in the Description field.

5. Click Go.

The List of Jobs table is populated with the list of jobs that meet the search criteria.

## **Retrieve the Details of a Microsoft SQL Server Agent Job**

You can retrieve the details of an MSSQL Server job. The job details include the name of the server the job belongs to, job category, job owner, date that the job was last modified, and so on.

### **Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Job Details link under Actions.  
The MSSQL Server Job Detail window displays the following Parameters and their Values in a tabular format:  
**Note:** The date and time are displayed in Greenwich Mean Time (GMT).
  - Job ID
  - Originating server
  - Job name
  - Enabled
  - Description
  - Category
  - Owner
  - Date modified
4. Click OK to close the MSSQL Server Job Detail window.

## **Search for a Target Server that the Microsoft SQL Server Agent Job Can Run on**

You can use the Select MSSQL Job Target Server dialog to search for and select a server that the Microsoft SQL Server Agent job can run on.

### **NOTE**

To list the target servers that the job can run on, you must enlist the target servers into the master server in the Microsoft SQL Server database.

### **Follow these steps:**

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Complete the following required fields:
  - **Send to machine**  
Specifies the name of the machine on which the job runs.
  - Specify the Job details from the drop-down and enter the Job name or Job ID field:
    - Job name**  
Specifies the name for a SQL Server Agent job.
    - Job ID**  
Specifies the SQL Server Agent job identification number.
3. Click the  icon next to the MSSQL server name field in the Properties section for the MSSQL job.  
The Select MSSQL Job Target Server dialog displays the list of target servers that the job can run on.

### **NOTE**

You can sort the data in the List of Target Servers table by clicking on a table column heading.

4. Select the server that you want to run the job on from the table.
5. Click OK to specify the target server in the job definition.  
The selected target server is set on the basic page of the job definition.

### Example: Retrieve a List of Target Servers that the Microsoft SQL Server Agent Job Can Run on

Suppose that you want to retrieve a list of all target servers that the job named Test job monitoring can run on.

#### To list the target servers that the Microsoft SQL Server Agent job can run on

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Enter **AGENT\_MSSQL** in the Send to machine field.
3. Select the Job name option under Job details and enter **Test job monitoring**.
4. Click Go.  
The Select MSSQL Job Target Servers dialog displays the list of target servers that the job can run on.

### Search for a Job Step in a Microsoft SQL Server Agent Job

You can use the MSSQL Job Steps Search dialog to search for and select a job step that is defined in the Microsoft SQL Server Agent job. You can start a job from the job step that you want.

#### Follow these steps:

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Complete the following required fields:
  - **Send to machine** Specifies the name of the machine on which the job runs.
  - Specify the Job details from the drop-down and enter the Job name or Job ID field: **Job name** Specifies the name for a SQL Server Agent job. **Job ID** Specifies the SQL Server Agent job identification number.
3. Click the  icon next to the Step name field in the Properties section for the MSSQL job.  
The MSSQL Job Steps Search dialog opens.
4. (Optional) Specify the search criteria in *one* of the following fields:
  - **Step Name**  
Specifies the name of the job step to filter on.
  - **Step ID**  
Specifies the ID of the job step to filter on.

The default is to include all job steps for the specified field.
5. Click Go.  
The List of Job Steps table is populated with the job steps that meet the search criteria for the specified job.

#### NOTE

You can sort the data in the List of Job Steps table by clicking on a table column heading.

6. Select the job step that you want to start the job from in the table.  
The default is to start the job from the first step.
7. Click OK to specify the step name in the job definition.  
The specified step name of the selected job step is set on the basic page of the job definition.

### Example: Display Details of All Job Steps

Suppose that you want to display details of all job steps for a job named Test job monitoring.

#### To display details of all job steps

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Enter **AGENT\_MSSQL** in the Send to machine field.
3. Select the Job name option button under Job details and enter **Test job monitoring**.
4. Click Go.  
The MSSQL Job Steps Search dialog opens.



5. Click Go.

The List of Job Steps table is populated with the details of all the job steps that are defined in the job.

### **Example: Display Details of a Job Step**

Suppose that you want to display details of a job step named Step 2 for a job named Test job monitoring.

#### **To display details of a job step**

1. Click Quick Edit and open the basic page of the MSSQL Server job definition.
2. Enter **AGENT\_MSSQL** in the Send to machine field.
3. Select the Job name option button under Job details and enter **Test job monitoring**.
4. Click Go.  
The Search MSSQL Job Steps dialog opens.
5. Enter **Step 2** in the Job Step Name field.
6. Click Go.  
The List of Job Steps table is populated with the details of only Step 2.

### **View Microsoft SQL Server Agent Job History**

You can view the job history of a MSSQL Server Agent job.

#### **Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Job History link under Actions.  
**Note:** The Actions column is displayed only after a job is run. If the job is not yet run, you can see no information in the Start Time and End Time columns and the Actions column is not displayed.  
The MSSQL Job History dialog is displayed.
4. Select any one of the filter criteria and specify the information.  
**Default:** Show recent  
**Note:** If no filter is specified, the most recent job history is displayed.
5. Click Go.  
The MSSQL Job History table displays the following information:
  - Job Name
  - Step ID
  - Step Name
  - Message Id
  - Severity
  - Message
  - Status
  - Run Date
  - Run Time
  - Run Duration
6. Click Close to close the window.

### **Restart a MSSQL Server Agent Job**

You can restart a failed MSSQL Server Agent job from the point of failure or from the beginning. For example, if a job is terminated abnormally, you can restart the job.

**NOTE**

To restart a job, it must be in Failed state.

**Follow these steps:**

1. Select Quick View.
2. Select the Server from the drop-down, specify the Name, and click Go.  
The Search Results are displayed.
3. Click the Restart icon under the Actions column.

**NOTE**

You can also restart a job from Send Events.

The Restart dialog opens.

4. (Optional) Select a Job Step that you want to restart the job from.

**NOTE**

To restart a job from a point of failure, select a job step. If a job step is not selected, the job is restarted from the beginning.

5. Click Restart.  
The server restarts the specified job from the specified job step or from the beginning.

You can also restart a MSSQL Server Agent Job from the Dashboard tab.

**Follow these steps:**

1. Select Dashboard, enter Name under the Jobs pane, and click Go.  
The search results are displayed.
2. Right-click on the job and select Restart.

**NOTE**

You can also restart a job from Send Events.

3. (Optional) Select a Job Step that you want to restart the job from.

**NOTE**

To restart a job from a point of failure, select a job step. If a job step is not selected, the job is restarted from the beginning.

4. Click Restart.  
The server restarts the specified job from the job step or from the beginning.

You can also restart a MSSQL Server Agent Job from the Monitoring tab.

**Follow these steps:**

1. Select Monitoring, Views.  
Select a job from the displayed list.
2. Right-click on the job and select Restart.

**NOTE**

You can also search for a job from the Jobs and Alerts tab.

You can also restart a job from Send Event.

3. (Optional) Select a Job Step that you want to restart the job from.

**NOTE**

To restart a job from a point of failure, select a job step. If a job step is not selected, the job is restarted from the beginning.

4. Click Restart.

The server restarts the specified job from the specified job step or from the beginning.

## Manage Web Service Jobs

The term web service describes a standardized method for exchanging data between applications and systems. Web services use XML to code and decode the data and Simple Object Access Protocol (SOAP) to transfer it.

Web Service Description Language (WSDL) is an XML-based language that describes a web service and how to access it. A WSDL document specifies the location of the service and the operations the service exposes.

Universal Description, Discovery and Integration (UDDI) is an XML-based registry for businesses to list their available web services on the Internet. You can use the UDDI to access the WSDL.

Web services provide access to applications written in Java and Microsoft®.NET. A web service lets you invoke operations such as currency conversion, stock exchange quotes, or product pricing. In an enterprise workload automation environment, a web service might be used to invoke a business process such as posting accounts payable to the General Ledger. Some scheduling manager functions are also available as web services.

The following are the web service job types:

- **Web Service RPC/Encoded**

Lets you call an operation within a web service and pass parameters to the operation using RPC/encoded style binding. The parameters can be actual values or a serialized Java object passed by another job.

- **Web Service Document/Literal**

There are two ways to define a WSDOC job type. One is specifying "webservice parameters" (existing approach) and the other is by specifying envelope-xml. Envelope is the recommended approach to define the jobtype. The job lets you call an operation within a web service and pass parameters to the operation using document/literal style binding. For SOAP Envelope, you can provide xml content which contains all the parameters that are required to execute the operation.

For Web service parameter, the parameters represent a flattened view of the XML document the agent constructs. The values passed into the XML document can be literal values or a serialized Java object passed by another job. When the job invokes the web service, the parameters are passed to the operation. The job's output is stored by default as a serialized Java object in the job's spool directory. You can also specify a destination file for the output.

**NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

If your company has a firewall and you must communicate through a proxy server to access a computer outside the firewall, agent configuration is required. For more information on configuring the agent for a proxy, see the *Workload Automation Agent for Web Services Implementation documentation*.

## Manage Web Service Document/Literal Jobs

You can define a Web Service Document/Literal Job using the wizard or using manual entries.

**NOTE**

- To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.
- Supports SOAP envelopes with Workload Automation Agents Version 12.0 and higher and Web service parameters with Workload Automation Agents Version 11.0 and higher.
- SOAP envelope is not supported, if you select Web service parameters. We recommend SOAP envelope to define WSDOC.

The following properties can be specified for Document/Literal jobs for Web service parameters only.

- WSDL location
- Service name
- Port name
- Operation
- Destination file
- Job criteria
- Web service parameters (ws\_parameter) fields:
  - Type
  - Name
  - Payload value/Job
- Web service security options (such as the following:)
  - USERTOKEN
  - TIMESTAMP
  - SIGNATURE
  - ENCRYPTION
- Success criteria fields:
  - Success exit code
  - Failure exit code
  - Expression type
  - Expression

The following properties can be specified for Document/Literal jobs only for SOAP envelope.

- SOAP web service input format fields:
  - Envelope XML
  - SOAP action
  - SOAP version
- Authentication parameters field:
  - Preemptive authentication
- Response persist options fields:
  - Payload file path
  - Overwrite payload file
  - Persist payload output
- Response variables fields:
  - Expression type
  - Name
  - Expression
  - RegEx Group
- Connection parameters fields:
  - Send failure details for retries
  - Connection retry count
  - Connection timeout (in seconds)
  - Failure retry count

Failure retry interval (in seconds)

The following properties are common to both input formats.

- End point URL
- Authentication parameters fields:
  - Connection user
  - Connection origin
  - Connection domain
  - Authentication order
- Proxy details fields
  - Use agent global proxy
  - Proxy user
  - Proxy origin
  - Proxy domain
  - Proxy host
  - Proxy port

### **Define a Web Service Document/Literal Job Using the Wizard**

The Web Service Document/Literal (WSDOC) job lets you call a WSDL operation with given inputs to the operation parameters in the form of SOAP envelope or Web service parameters. AutoSys Web UI provides 'WSDL Wizard' helper dialog to auto-generate SOAP envelopes or Web service parameters with user given inputs. WSDOC job type also allows users to copy-paste or specify SOAP envelopes generated from external tools in the 'Envelope XML' text area from Web UI.

#### **Follow these steps:**

1. Create a Web Service Document/Literal (WSDOC) job.
2. Select the input format for the SOAP web service:

#### **NOTE**

After saving the job, you cannot change the selected input format.

- SOAP envelope
 

Works with Workload Automation Agents Version 12.0 and higher. We recommend SOAP envelopes to define WSDOC job type.
  - Web service parameters
 

Works with Workload Automation Agents Version 11.0 and higher.
3. Open the WSDL wizard.
 

The Browse WSDL dialog for the selected format opens.
  4. Specify the following required properties:
    - **Job name**

Defines the name of the job that you want to schedule.

**Limits:** 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), colon(:), and pound (#); do not include embedded spaces or tabs
    - **Select machine**

Specifies the name of the machine on which the job runs.
  5. Click the Browse WSDL button to open the wizard. Click Next to proceed.
  6. Complete the following fields as required:
    - **WSDL location**

Indicates whether the WSDL file location is local or remote, as follows:

      - URL
      - Local file

**NOTE**

When you select URL, the wizard displays an option to use proxy details. If your company uses a proxy server to connect to the Internet, you must complete the Proxy fields.

- **WSDL URL**  
Specifies the remote URL for the WSDL file.  
**Limits:** Up to 2048 characters
  - **WSDL Local file**  
Specifies the local path for the WSDL file.  
**Limits:** Up to 256 characters
7. Click Next to proceed and select Service.
  8. Select a web service from the list of services and Click Next to proceed to Port.
  9. Select a port name from the list of ports for the selected service and Click Next to proceed to Operation.
  10. Select an operation from the list of operations for the selected port and Click Next to proceed to Web service parameters.
  11. Specify values for the web service parameters for the selected operation.
  12. Click Finish to temporarily retain your data.  
The Browse WSDL dialog closes and values for the web service appear in the job definition.
  13. (Optional) Specify optional [Web Service Document/Literal properties](#).
  14. (Optional) Specify [Manage Common Job Properties \(12.0 UI\)](#) for General, Termination, Schedule, Notifications, and Native Permissions.

**NOTE**

You can use search on demand for the Machine, Box, Email Template, and Calendar properties. Enter the appropriate criteria in the relevant field. You can also select the value from the drop-down list wherever available or start typing to prompt options.

15. Save the job.  
The Web Service Document/Literal job is defined.

**Define a Web Service Document/Literal Job Using Manual Entries**

You can define a Web Service Document/Literal job to call an operation within a web service and pass parameters to the operation using document/literal style binding. The easiest way to define a Web Service Document/Literal job is to use the wizard. However, you can enter the job definition values manually. The end point URL, name of the operation you want to invoke, port name, web service name, and WSDL URL fields are required. Depending on the web service provider, other values may also be required.

**NOTE**

When the selected WSDL Location is "Local" and the Browse WSDL wizard is finished, the "WSDL Location" field is not auto-populated in the job page. You must manually provide the location of the WSDL file for the agent file system.

From Version 12.0, you can define WSDOC using a SOAP envelope or using WS parameters.

**Follow these steps:**

1. Create a Web Service Document/Literal (WSDOC) job.
2. Specify the following required properties:
  - **Job name**  
Defines the name of the job that you want to schedule.  
**Limits:** 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); do not include embedded spaces or tabs
  - **Select machine**

Specifies the name of the machine on which the job runs.

3. (Optional) Enter a **Description** for the job.

- Defines an optional free-form text description of the job.  
**Limits:** 500 alphanumeric characters (including spaces)  
**JIL attribute:** description

4. Select the input format for SOAP web service:

**NOTE**

After saving the job, you cannot change the selected input format.

- SOAP envelope  
Works with Workload Automation Agents Version 12.0 and higher.
  - Web service parameters  
Works with Workload Automation Agents Version 11.0 and higher.
5. Specify [Web Service Document/Literal properties](#) under the SOAP tab.
6. (Optional) Specify [Manage Common Job Properties \(12.0 UI\)](#) for Notifications and General properties.
7. (Optional) Specify [common properties that apply to all job types](#) under the Schedule, Termination, and Permissions tabs.

**NOTE**

You can use search on demand for the Machine, Box, Email Template, and Calendar properties. Enter the appropriate criteria in the relevant field. You can also select the value from the drop-down list wherever available or start typing to prompt options.

8. Save the job.

The Web Service Document/Literal job is defined.

**NOTE**

- To use the Refresh button and Actions drop-down in the Web UI title bar and perform various actions on an existing job definition, see the [Manage Actions](#) section.
- For the Refresh button and Actions drop down job retrievals, the last version is only accessible; previous versions are disabled

## **Web Service Document/Literal properties**

### **SOAP envelope**

Complete the following fields:

#### **End point URL\***

\*This a mandatory field. Specifies the target endpoint address URL. In a published Web Service Description Language (WSDL) file, the URL defining the target endpoint address is found in the location attribute of the port's soap:address element.

Limits: Up to 2048 characters; case-sensitive

JIL attribute: endpoint\_URL

#### **Envelope XML\***

\*This a mandatory field. Specifies one or more lines of text to contain in an envelope. It contains XML data that is used to create an input for the job. This property lets you type the envelope's data directly in the job definition.

When you specify this property in a job definition, the envelope is uploaded to the database when the job is defined. This input job envelope is associated with the job. The job itself can then use the envelope as input at run time.

Envelope can be generated for a webservice operation using the "WSDL Wizard" helper dialog or can be copy-pasted from an external soap tool.

Limits: Up to 2048 characters; case-sensitive

JIL attribute: blob\_input

### **SOAP version**

(Optional) Displays the version of the SOAP message.

JIL attribute: soap\_version

### **SOAP action**

(Optional) Specifies the HTTP request header field that indicates the intent of the SOAP HTTP request.

JIL attribute: soap\_action

### **Web service parameters**

Complete the following fields (all fields are mandatory):

#### **WSDL location (URL/Agent file path)\***

\*This a mandatory field. Specifies the URL (local path or remote) to the Web Service Description Language (WSDL) file of the web service to invoke. Click Browse WSDL to locate the WSDL file and load.

Limits: Up to 2048 characters; case-sensitive

JIL attribute: wsdl\_url

#### **NOTE**

- When the selected WSDL Location is "Local" and the Browse WSDL wizard is finished, the "WSDL Location" field is not auto-populated in the job page. You must manually provide the location of the WSDL file for the agent file system.
- The file specified through the wizard is from your local computer in which the browser is launched, so it will not work at runtime. You must specify the local file location with respect to the agent file system.
- When you select Browse WSDL, the wizard displays a Proxy section to specify a proxy server. If your company uses a proxy server to connect to the Internet, you must complete the Proxy fields.

#### **End point URL\***

\*This a mandatory field. Specifies the target endpoint address URL. In a published WSDL file, the URL defining the target endpoint address is found in the location attribute of the port's soap:address element.

Limits: Up to 2048 characters; case-sensitive

JIL attribute: endpoint\_url

#### **Service name\***

\*This a mandatory field. Specifies the web service name within the target namespace.

Limits: Up to 255 characters; case-sensitive

JIL attribute: service\_name

#### **Port name\***

\*This a mandatory field. Specifies the WSDL port name within the target namespace. A WSDL port describes the operations exposed by a web service and defines the connection point to the web service.

Limits: Up to 256 characters; case-sensitive

JIL attribute: port\_name

#### **Operation\***

\*This a mandatory field. Specifies the web service operation to be invoked.

Limits: Up to 256 characters; case-sensitive

JIL attribute: wsdl\_operation

#### **Add Web service parameters (properties)**



(Optional) Specifies the name/value pairs of [parameters](#) that are passed to the web service in the form of flattened XML. When the job runs, the parameters are passed to the executing process as a literal XML document.

You can add new web service parameters for use in the literal XML document.

Click Add new.

Complete the following options:

- **Type**  
Specifies the name or path of the element. The following options are available.
  - **Value**  
Specifies the value type (scalar string value).
  - **Payload**  
Specifies the payload type.
- **Value**  
Specifies the parameter name and array of values.
  - **Name**  
Specifies the name or path of the element. When the parameter name contains `[/]`, it indicates an array of values.
  - **Payload value**  
Specifies an array of scalar string values. The individual array values are separated by commas and the entire value is enclosed in double quotes.  
**Limits:** Up to 2048 characters for Name and Value/Payload together; case-sensitive
- **Payload**  
Specifies the parameter name and the name of a payload producing job.
  - **Name**  
Specifies the name or path of the element.
  - **Payload job**  
Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

To specify an XML hierarchy of the document, use multiple entries of Web service parameters.

In each Web service parameter entry, separate the name/value pair with a comma. The entire value can be up to 2048 characters.

In the XML document, these parameters are represented in a flattened view. Consider the following example:

```
<tag1>
 <tag2>value</tag2>
</tag1>
```

In this example, the full path to the value is `</tag1/tag2>`.

For tags that do not hold any value, the full path is required because the no-value tags could represent arrays of structures and appear in the multiple levels. Consider the following example:

```
ws_parameter: param_name=/tag1,param_value=
```

```
ws_parameter: param_name=/tag1/tag2, param_value=value
```

#### NOTE

If a value is not specified for `param_value`, empty double quotes (`""`) following the equal sign are not required.

For parameters that contain attributes, the attributes are added to the full path of the parameter separating them with the '@' symbol. Consider the following example:

```
<tag1>
<tag2 key="thetag">value</tag2>
</tag1>
```

In this example, the list of parameters would be as follows:

```
ws_parameter: param_name=/tag1,param_value=
ws_parameter: param_name=/tag1/tag2, param_value=value
ws_parameter: param_name=/tag1/tag2@key, param_value=thetag
```

### NOTE

If a value is not specified for param\_value, empty double quotes (") following the equal sign are not required.

For Web Service Document/Literal payload producing jobs, only the portion of the returned data that satisfies the Job criteria specification will be stored in the output file for use as payload.

**JIL attribute:** ws\_parameter

The Web Service Document/Literal category includes the following properties:

- Web service security options
- Authentication parameters
- Response persist options
- Response variables
- Proxy details
- Success criteria
- Connection parameters

### **Web service security options**

(Optional) Specifies SOAP-level Web service security options that the agent uses to communicate with the web service.

1. Select the Security type, complete the options as required, and click Add.

The specified Security type and values are displayed in the right pane.

### NOTE

You can select the added security type and its values and edit or delete them by using the Edit or Delete buttons.

You can specify the following Web service security parameters:

- **Usertoken**  
Adds a user name and password to the SOAP message that is validated in the server's SOAP layer.
  - **Username**  
Specifies the user name that is added to the SOAP message. The autosys\_secure command stores the password encrypted in the database for the given user name.
  - **Digest - True | False**  
Specifies whether you want to pass the password in digest form. Valid values are:
    - **True**  
Sends the password in the encrypted form to the SOAP request.
    - **False**  
Sends the password in the plain form to the SOAP request.
- **Timestamp**  
Adds a timestamp to the SOAP message with an optional time to live option to prevent reuse of old messages.

- **Time to live**  
Specifies the time to live, in seconds.

- **Signature**

Adds a digital signature to the SOAP message, using the certificate specified in the agentparm.txt file.

**NOTE**

Your agent administrator must specify the ws.wss.keystore, ws.wss.keystore.alias, and ws.wss.keystore.password parameters in the agentparm.txt file.

- **Encryption**

Encrypts the message. The server will decrypt the message in the SOAP layer.

**NOTE**

Your agent administrator must specify the ws.wss.keystore, ws.wss.keystore.alias, and ws.wss.keystore.password parameters in the agentparm.txt file, if wss\_keytype is Issuer serial.

- **Service alias**

Specifies the alias in the keystore defined by properties for the Web Service certificate.

**Limits:** Up to 512 characters; case-sensitive

- **Issuer serial**

Key type for this Encryption security type format is Issuer serial. Uses X509 certificate for encryption. It is obtained from the keystore defined by the properties specified in the agentparm.txt file.

- **Embedded Keynote**

Key type for this Encryption security type format is Embedded Keynote. Uses the following key specified in the job definition for encryption:

- **Keyname**

Specifies the key name as defined by the web service to be invoked.

- **Keyvalue**

Specifies the key in hexadecimal format as defined by the web service to be invoked. The agent sends the key encrypted.

**NOTE**

For an existing job, asterisks are displayed in this field. Do not delete any asterisks unless you are changing the value. If you change the value, overwrite all the asterisks.

**NOTE**

- The entire value can be up to 2048 characters.
- The Web service security options provide the following additional SOAP-level security for the web services messages:
  - Transport-level authentication
    - Usertoken -- a user name token and password are added to each message and validated in the web service's SOAP layer.
    - Timestamp -- a timestamp, with a time-to-live parameter, is added to each message to prevent message hijacking and reuse.
  - Message-level security
    - Signature -- messages are signed with a certificate.
    - Encryption -- messages are encrypted at the message level, increasing security when going through proxies that may require decrypting or re-encrypting of the transport stream.
- When properties for an existing job are displayed, the Keyvalue field displays asterisks. You can overwrite the asterisks with a new value.
- For more information about controlling the agent's use of certificates for the Signature and Encryption Web service security options, see the *Workload Automation Agent for Web Services Implementation documentation*.

JIL attribute: ws\_security

### **Authentication parameters**

#### **Connection user**

Specifies the user name for connection authentication.

Limits: 145 characters

#### **NOTE**

- If you specify the connection user property in your job definition, you must set the password that is associated with the connection user using the autosys\_secure command.
- To enable preemptive authentication, you must specify Connection user; you can leave the Authentication order blank or specify BASIC, either alone or in a combination.

JIL attribute: ws\_conn\_user

#### **Connection origin**

Specifies the origin host name for NTLM connection authentication.

Default: The computer name where the agent is running.

Limits: 128 characters; case-sensitive. Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: ws\_conn\_origin

#### **Connection domain**

Specifies the domain for NTLM connection authentication.

Limits: 128 characters; case-sensitive. Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: ws\_conn\_domain

#### **Authentication order**

Specifies a list of protocols to be used by a web server for authentication in an WSDOC job.

- **BASIC**  
Indicates the BASIC protocol.
- **DIGEST**  
Indicates the DIGEST protocol.
- **NTLM**  
Indicates the NTLM protocol.

#### **NOTE**

- You can specify any of the following protocols in any order: BASIC, DIGEST, and NTLM.
- If you are connecting to an old web server that cannot negotiate authentication protocols, enter the following list in the specified order: BASIC,DIGEST,NTLM.

JIL attribute: ws\_authentication\_order

#### **Preemptive authentication**

Specifies whether or not to authenticate the application credentials using preemptive authentication. In preemptive authentication, credentials are submitted along with the initial request. No authentication process occurs between the target server and the client.

#### **NOTE**

The Preemptie authentication checkbox is enabled only as follows:

- Connection user is specified and AuthenticationOrder is blank.
- Connection user is specified and AuthenticationOrder contains BASIC.

Default: Blank checkbox

JIL attribute: preemptive\_authentication

### **Response persist options**

#### **NOTE**

The Payload file path, Overwrite payload, Persist payload output options are available only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later. The Destination file path is applicable for Web service parameters, Workload Automation Agents Version 11.0 and higher releases.

### **Destination file path**

(Optional) Specifies the output destination file for the job when the input format is Web service parameters. The file stores the output of the invocation, which if not specified would have been stored in a spool file.

To select a file, follow these steps:

1. Click Browse.  
The Select File window displays.
2. Navigate across the directories to select an existing file or enter the file name, click INSERT PATH, to display the selected or entered File Path.

Limits: 256 characters; case-sensitive

Examples: /tmp/myoutput, "D:\Stock output\STOCK3"

#### **NOTE**

The destination file path property is not applicable if you have set the method property value to delete.

JIL attribute: destination\_file

### **Payload file path**

(Optional) Specifies the path to the payload file when the input format is SOAP envelope. The payload file stores the output (in text format) of the invoked web service. It contains SOAP webservice Response only.

To save a file, follow these steps:

1. Click Browse.  
The Select File window displays.
2. Enter a file name for the payload output file and click Save.  
The payload output is saved as a results file.

Limits: 256 characters; case-sensitive

Examples: /tmp/myoutput, "D:\Stock output\STOCK3"

#### **NOTE**

- If you omit to specify the payload file path in a WSDOC job, the output is stored in the job's spool file.
- To enable the Overwrite payload file checkbox, you must specify the Payload file path.

JIL attribute: results\_file

### **Overwrite payload file**

(Optional) Specifies whether you want to overwrite the results file when a file with the same name exists.

Default: Blank checkbox

JIL attribute: results\_file\_overwrite

## Persist payload output

(Optional) Specifies whether you want to persist the payload output in the database as a blob so that it can be consumed by the successor jobs for their execution.

Default: Blank checkbox

JIL attribute: persist\_output\_payload

## Response variables

The following options are available for SOAP envelopes only. A WSDOC/SOAP Doc Literal job on its execution can produce payload that can be consumed by the successor jobs for their execution. The payload is in the form of XML or Plain Text. When a job that contains the response variables properties completes successfully, the scheduler parses the incoming payload for the elements whose path is specified in the expression and the resultant is stored as a job blob with the name that is specified in the name.

### NOTE

This category is introduced in Version 12.0 and is available only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later.

## Parameters

Specifies the response variables (expression type, name, expression, regex group) that can be consumed by the successor jobs for their execution.

All the fields are optional.

### Expression type

Specifies the format of the response variable. Valid values are XPath or RegEx.

JIL attribute: var\_format

### Name

Specifies the name of the response variable.

Limits: 256 characters

JIL attribute: var\_name

### NOTE

The variable names must be unique for each job definition.

## Expression

Specifies the value of the response variable. You can specify the xpath for XML or an ordinary regex expression for Plain Text.

Limits: 3950 characters

JIL attribute: var\_value

## RegEx Group

Specifies the numerical group ID in any order. This group ID enables the extraction of the groups that are matched while applying the regex. Applies to RegEx format only.

Limits: 3950 characters

JIL attribute: var\_group

Example: If the payload contains a string in the form "This is a sample string" and you want "This string" as a variable, specify two group IDs <<\$1>> <<\$5>> to pick the value from group 1 and 5.

## Evaluate All

You can use the Evaluate All option to validate the entered XPath or RegEx for the given text, so that you can use the XPath or RegEx for subsequent job parameters.

**Follow these steps:**

1. Click Evaluate All. The Evaluate response type window opens with the following options, based on the added response variables:  
Payload response content.
2. Enter or copy and paste the response content, or select the browse file, for validation.
3. Click Evaluate.  
A validation success message is displayed.

**Examples:**

Specify the response variables in XPath format, as follows:

Response Variables:

Expression Type - XPath

Name-PAR1

Expression-/store/book[1]/title

Specify the response variables in RegEx format, as follows:

Response Variables:

Expression Type-RegEx

Name-PAR1

Expression-[a-zA-Z]+

RegEx Group-<<\$1>>

**Proxy details**

Specifies a list of defaults to be used by a web server for authentication in an WSDOC job

**Use agent global proxy**

(Optional) Specifies whether the agent uses the global proxy defaults specified in the agentparm.txt file.

- Leave the checkbox blank to ignore the global proxy configuration specified by the proxy parameters in the agentparm.txt file (the default). Specify this option if you want to apply the proxy defaults for this job only. For example, you might ignore the proxy defaults if the request is going to a server on the LAN that does not require the proxy.
- If the checkbox is selected, the application uses the global proxy configuration specified by the proxy parameters in the agentparm.txt file.

Default: Blank checkbox

JIL attribute: ws\_global\_proxy\_defaults

**Proxy user**

(Optional) Specifies the user name required for proxy authentication. If you omit this property, the user name defaults to the value specified in the http.proxyUser parameter in the agentparm.txt file.

Limits: 128 characters; case-sensitive; cannot contain delimiters such as spaces

**NOTE**

- Your agent administrator can specify a default proxy user name for all WSDOC jobs by setting the http.proxyUser parameter in the agent's agentparm.txt file.
- The proxy user property overrides the default proxy user name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: ws\_proxy\_user

### Proxy origin

Limits: 128 characters; case-sensitive

JIL attribute: ws\_proxy\_origin\_host

### Proxy domain

(Optional) Specifies the domain for proxy authentication required for NTLM authentication. If you omit this property, the domain defaults to the value specified in the http.proxyDomain parameter in the agentparm.txt file.

Limits: 128 characters; case-sensitive

#### NOTE

- Your agent administrator can specify a default proxy domain name for all WSDOC jobs by setting the http.proxyDomain parameter in the agent's agentparm.txt file.
- The proxy domain property overrides the default proxy domain name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: ws\_proxy\_domain

Limits: 128 characters; case-sensitive

#### NOTE

- Your agent administrator can specify a default proxy host name for all WSDOC jobs by setting the http.proxyHost parameter in the agent's agentparm.txt file.
- The proxy host property overrides the default proxy host name specified in the agent's agentparm.txt file.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

JIL attribute: ws\_proxy\_host

### Proxy port

Limits: 0-65535

JIL attribute: ws\_proxy\_port

### Success criteria

All the fields are optional.

#### NOTE

The Exit code, Criteria expression options are introduced in Version 12.0 and is available only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later. These options are available for SOAP envelope.

The **Exit code, Criteria Expression** section contains the following success criteria:

The following Exit codes indicate and provide detailed information of the success or failure of the job.

### Success exit code

(Optional) Defines which exit codes indicate job success. For example, you can define the Success Exit Codes to be 20-30. If a job returns an exit code between 20 and 30, the job is considered to have completed successfully.

#### NOTE

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.



Specify a single exit code (for example, 4), a range of exit codes (for example, 0-9999), or a list of multiple exit codes or ranges separated by commas.

**Examples:** 1,150,20-30. ExitCode(201,s) Ignored, because 201 is a default success response.ExitCode(400,s) 100-299, 400 are a success. The default 100-299 and the specified 400 are successful.

ExitCode(200,f 400,s) Expressions separated by space rather than comma. Success: 100-199, 201-299, 400. The default 100-199 and 201-299 and the specified 400 are successful. The specified 200 is left out as a failure.

Limits: 256 characters. The exit code must be an integer in the range 0 to 2147483647.

Default: By default, responses between 100 and 299 are treated as success.

JIL attribute: success\_codes

### Failure exit code

(Optional) Defines which exit codes indicate job failure. For example, you can define the Failure Exit Codes to be 40-50. If a job returns an exit code in the range 40 to 50, the job is considered to have failed.

#### NOTE

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

Limits: 256 characters. The exit code must be an integer in the range 0 to 2147483647.

Default: By default, responses less than 100 and above 299 are treated as failure.

JIL attribute: fail\_codes

### Expression type

(Optional) Specifies the *<filtertype>* (XPath or Regex).

#### Expression

(Optional) Specifies the value of the XPath or Regex expression type.

Limits: 256 characters

JIL attribute: success\_criteria\_*<filtertype>*

The following options are available for Web service parameters only:

#### Regex:

#### Expression type

(Optional) Specifies the *<filtertype>* (Regex).

#### Expression

(Optional) Specifies the value of the Regex expression type.

Limits: 2048

JIL attribute: criteria\_filter

#### NOTE

When the input format is Web service parameters, criteria\_filter and criteria\_xpath are used.

Regex case --> criteria\_filter

Xpath case --> criteria\_filter and criteria\_xpath

When the input format is SOAP envelope, success\_criteria\_*<filtertype>*, and success\_criteria\_value are used.

Regex case --> success\_criteria\_regex

Xpath case --> success\_criteria\_xpath, and success\_criteria\_value

#### XPath:

#### Expression type

(Optional) Specifies the *<filtertype>* (XPath). The XPath evaluated value is validated against the value that you specify in the Expected expression value.

**Expression**

Limits: 2048

JIL attribute: criteria\_filter

**RegEx expression**

(Optional) Defines a regular expression that is used to evaluate the XPath expression. If the evaluated XPath expression value matches the regular expression, the job completes successfully. Otherwise, the job fails.

Limits: 1024

JIL attribute: criteria\_xpath

**Connection parameters**

The following options are available for SOAP envelopes only. The following options provide you with information and attempts to retry a failed job.

All the fields are optional.

**NOTE**

This category is introduced in Version 12.0 and is available only if the AutoSys Workload Automation server that is configured in Web UI is at Version 12.0 and later.

**Send failure details for retries**

(Optional) Specifies whether to send the failure details (indicating the failure and retry count) to the manager when the Failure retry count is set to a value that is greater than zero.

- Select the checkbox to send the failure details to the manager when the Failure retry count is set to a value that is greater than zero.
- If the checkbox is not selected failure details are not sent to the manager when the Failure retry count is set to a value that is greater than zero.

Default: Blank checkbox

JIL attribute: retry\_on\_failure\_details

**Connection retry count**

(Optional) Specifies the number of attempts to be made to restart the failed job.

Default: 3

JIL attribute: connection\_retry

**Connection timeout (in seconds)**

Specifies the duration, in seconds, after which the request times out.

Limits: 0-2147483647

JIL attribute: connection\_timeout

**Failure retry count**

(Optional) Specifies the number of times a request is retried when the exit code is a failure code or the specified filter criteria does not match the response.

Default: 0 (no retries)

JIL attribute: retry\_on\_failure\_count

**Failure retry interval (in seconds)**

(Optional) Specifies the number of seconds to wait between retrying requests if the Failure retry count is set to a value that is greater than zero.

Default: 10

**NOTE**

Your agent administrator can set a default value by setting the Failure retry interval parameter in the agent's agentparm.txt file. The Failure retry interval value overrides the default value that is specified in the agent's agentparm.txt file.

JIL attribute: retry\_on\_failure\_delay

**Global Variables**

The following properties support Global Variable substitution:

- wsdl\_url
- service\_name
- port\_name
- endpoint\_url
- ws\_parameter
- job\_criteria
- ws\_conn\_origin
- ws\_conn\_domain
- ws\_proxy\_host
- ws\_proxy\_user
- ws\_proxy\_origin
- ws\_proxy\_domain
- ws\_conn\_user
- ws\_security
- destination\_file
- results\_file
- success\_criteria\_xpath
- success\_criteria\_value
- success\_criteria\_regex
- blob\_input

**Example: Use SOAP Envelope to Return the Number of Two Integers**

Using SOAP envelope you can add and return the sum of two integers. This job calls a soap web service "MathsServiceService" which will add two given numbers using soap request as part of Envelope XML property.

**Follow these steps:**

1. Create a Web Service Document/Literal job.
2. Enter the following properties:
  - Name: exampleWithEnvelopeXML
  - Send to machine: localhost
  - Select SOAP envelope
  - End point URL: http://ExampleSoapHost:8080/SampleSoapWS/services/MathsServiceService
  - Envelope XML:
 

```
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:sch="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
```

```

<sch:Param1>1</sch:Param1>
<sch:Param2>2</sch:Param2>
</sch:MathsInput>
</sch:AddMessageInput>
</soapenv:Body>
</soapenv:Envelope>

```

3. Save the job.

### Example: Return the Maximum of an Arbitrary Number of Integers

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2. The expected return is 4. The job attempts to authenticate the myuser@mydomain credentials using NTLM authentication.

#### Follow these steps:

1. Create a Web Service Document/Literal job.
2. Enter the following properties:
  - Name: getmax1
  - Send to machine: wsmach
  - Select Web service parameters
  - End point URL: http://wsmach:8080/axis2/services/TestWS
  - Port name: TestWSHttpSoap11Endpoint
  - Service name: TestWS
  - WSDL location: http://wsmach:8080/axis2/services/TestWS?wsdl
  - Operation: intGetMax
  - Use global proxy defaults -- Selected
  - Connection user: myuser@mydomain
  - Connection domain: mydomain
  - Connection origin: myhost.example.com
  - Authentication order: NTLM
  - Web service parameter 1:
    - Type -- Name-Value
    - Name -- /intGetMax
    - Payload value/Job -- (No value; leave empty)
  - Web service parameter 2:
    - Type -- Name-Value
    - Name -- /intGetMax/i[3]
    - Payload value/Job -- 1,4,2
  - Job criteria: criteria\_xpath=//ns:return, criteria\_filter=.\*4.\*
3. Save the job.

### Example: Use Web Service Parameters to Invoke a Web Service and List All Amazon Buckets Operation

Suppose that you want to invoke an Amazon web service that lists all your Amazon "buckets". The URL for the WSDL that describes the web service and its location is <http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wsdl>. The target endpoint address URL is <https://s3.amazonaws.com/soap>. The job calls the operation ListAllMyBuckets within the AmazonS3 web service. When the job invokes the web service, information about the list of my buckets is passed to the operation. The ListAllMyBuckets operation assigns different parameter values.

#### Follow these steps:

1. Create a Web Service Document/Literal job.
2. Enter the following properties:

- Name: execws
  - Send to machine: wsagent
  - Select Web service parameters
  - End point\_URL: "https://s3.amazonaws.com/soap"
  - WSDL location: "http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wsdl"
  - Service name: AmazonS3
  - Port name: AmazonS3
  - Operation: ListAllMyBuckets
  - Use global proxy defaults -- Not selected
  - HTTP proxy host: 172.31.255.255
  - HTTP proxy port: 80
  - Proxy authentication user : user@domain
  - Proxy authentication domain: domain
  - Web service parameter 1:
    - Type -- Name-Value
    - Name -- /ListAllMyBuckets
    - Payload value/Job -- (No value; leave empty)
  - Web service parameter 2:
    - Type -- Name-Value
    - Name -- /ListAllMyBuckets/AWSAccessKeyId
    - Payload value/Job -- 0x0102030405060708
  - Web service parameter 3:
    - Type -- Name-Value
    - Name -- /ListAllMyBuckets/Timestamp
    - Payload value/Job -- 2011-08-25T02:24:21
  - Web service parameter 4:
    - Type -- Name-Value
    - Name -- /ListAllMyBuckets/Signature
    - Payload value/Job -- 0x0102030405060708
3. Save the job.

## Examples - WSDOC Jobs (Web UI)

The following examples are for the Web Service Document/Literal Job, mainly using payload output format and how you can use different attributes to match content from the payload.

The examples are based on the assumption that the response payload is an XML document with the following structure:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/">
 <soapenv:Body>
 <ns1:MathsOutput xmlns:ns1="http://www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <ns1:Result>3</ns1:Result>
 </ns1:MathsOutput>
 </soapenv:Body>
</soapenv:Envelope>
```

The examples are classified into the following two categories:

## Evaluating Success Criteria

The success criteria for a job can be evaluated using the following options:

- Using XPATH
- Using RegEX

### Using XPATH

The following example is checking if the SOAP web service meant for adding two numbers when invoked with input parameters 1 and 2 yields the result as 3.

```
Job name - WSDOCSample1
Select machine -- localhost
Input format for SOAP web service -- SOAP envelope
End point URL -- http://wabld-agent-ws.lvn.broadcom.net:8080/SampleSoapWS/services/MathsServiceService
Envelope XML -- <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:sch="http://
www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope>
Persist payload output - Checked
Expression type -- XPath
Expression -- //soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result
RegEx expression -- 3
```

### Using RegEx

The following example is checking if the SOAP web service meant for adding two numbers when invoked with input parameters 1 and 2 yields the result as 3.

```
Job name -- WSDOCSample2
Select machine -- localhost
Input format for SOAP web service -- SOAP envelope
End point URL -- http://wabld-agent-ws.lvn.broadcom.net:8080/SampleSoapWS/services/MathsServiceService
Envelope XML -- <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:sch="http://
www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope>
Persist payload output - Checked
Expression type -- RegEx
```

```
Expression -- <ns1:Result>3</ns1:Result>
```

## Consuming Response Payload

The generated response payload from the web service can be consumed using the following options:

- Response variables
- Substitution variables

## Response Variables

The response variables against a job can be queried by running autorep with jb option.

**Example:** autorep -jb jobname

These variables can be used just like the way global variables are used. The notation to be followed is \$  
\$jobname@<v>title</v>, where title here refers to the name of the response variable defined in the job "jobname".

## Response variables using XPATH

In the following example, we are capturing the result of the addition of two numbers in the variable result.

```
Job name - WSDOCSample3
Select machine -- localhost
Input format for SOAP web service -- SOAP envelope
End point URL -- http://wabld-agent-ws.lvn.broadcom.net:8080/SampleSoapWS/services/MathsServiceService
Envelope XML -- <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:sch="http://
www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope>
Response variable -- Expression type=XPath, Name=result, Expression = //soapenv:Envelope/soapenv:Body/
ns1:MathsOutput/ns1:Result
```

## Response variables using Regular Expression

In the following example, we are capturing the result of the two numbers using groups in the variable result.

```
Job name - WSDOCSample4
Input format for SOAP web service -- SOAP envelope
Select machine -- localhost
End point URL -- http://wabld-agent-ws.lvn.broadcom.net:8080/SampleSoapWS/services/MathsServiceService
Envelope XML -- <soapenv:Envelope xmlns:soapenv="http://schemas.xmlsoap.org/soap/envelope/" xmlns:sch="http://
www.tibco.com/schemas/Maths/MathsService/Schema.xsd">
 <soapenv:Header/>
 <soapenv:Body>
 <sch:AddMessageInput>
 <sch:MathsInput>
 <sch:Param1>1</sch:Param1>
 <sch:Param2>2</sch:Param2>
 </sch:MathsInput>
 </sch:AddMessageInput>
 </soapenv:Body>
</soapenv:Envelope>
```

```

 </sch:MathsInput>
 </sch:AddMessageInput>
</soapenv:Body>
</soapenv:Envelope>
Response variable -- Expression type= RegEx, Name = result, Expression = <ns1:Result>(.*?)</ns1:Result> , RegEx
group = <<$1>>

```

### **Substitution Variables**

We can also consume an element from the persisted response directly by specifying the XPATH or RegEx. These variables can be used anywhere the way global variables are used.

We can use the Generate Expression dialog and consume the persisted data as follows:

#### **Consuming the persisted XML data**

If we wanted to extract from the response the result of two numbers, then the notation used is `$payload^WSDOCSample3@<x>//soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result </x>`. Here `payload` specifies that we are interested in parsing the response of type “payload”. This payload is produced by the job `WSDOCSample3` and the xpath expression to be applied on the response is `//soapenv:Envelope/soapenv:Body/ns1:MathsOutput/ns1:Result`. The job `WSDOCSample3` must have the Persist payload output checkbox selected under the Response persist options.

#### **Consuming the persisted RegEx data**

If we wanted to extract from the response the result of two numbers, then the notation used is `$payload^WSDOCSample4@<r><ns1:Result>(.*?)</ns1:Result></r><g><<$1>></g>`. The job `WSDOCSample4` must have the Persist payload output checkbox selected under the Response persist options.



# Monitoring and Reporting

This section contains information on the following topics:

## AutoSys Monitoring and Reporting

To identify problems with the workflow, use any combination of the following monitoring and reporting tools:

- **Forecast Reports**

Generate reports that display information about the predicted workflow to identify problems before they occur.

**NOTE**

You can also use forecast reports to plan changes to your workflow in a test environment.

- **Monitors**

Track events to identify problems as they occur.

- **Browsers**

Generate reports that display information about past events to identify recurring problems.

You can solve problems before they occur or as they occur when you can identify the issue that is associated with the problem. When you cannot determine the cause of a problem, notify the administrator.

To solve a problem that you identify in real time using a monitor, correct the associated issue and restart the job. To address recurring problems or problems with predicted workflow that you identify using browsers and forecast reports, correct the associated issues and use monitors to track the progress of the workflow.

Correcting issues that cause jobs to fail requires modifying workflow objects (job definitions, machine definitions, and calendar object definitions). You can modify workflow objects only when you have write access to those objects. When you cannot solve a problem without modifying a workflow object and you do not have write access to the problematic object, notify the scheduler.

**WARNING**

Modifying workflow objects sometimes has unexpected impacts on the rest of the workflow. We recommend that you plan changes to the workflow in a test environment before you implement the changes in the live instance.

**NOTE**

- You can also use the Web UI Forecast and Monitoring applications to monitor workflow.
- You can use the sendevent command to restart a job. You can use jil to update job or machine definitions and the autocal\_asc command to update calendar definitions.

## Define and Run a Monitor or Browser

Monitors and browsers help you identify problems with your workflow as follows:

- **Monitors**

Track events in real time.

- **Browsers**

Generate reports that display historical information about events.

**NOTE**

By default, AutoSys Workload Automation is configured to specify the date and time using the "MM/dd/yyyy HH:mm:ss" format. You can use a different format by changing the value of the DateFormat parameter in the configuration file.

**Limits:**

- *MM*: 01-12
- *dd*: 01-31
- *yyyy*: 1900-current year
- *HH*: 00-23
- *mm*: 00-59
- *ss*: 00-59

**Follow these steps:**

1. Open the operating system or instance command prompt:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation. The UNIX operating system command prompt opens. The shell that is sourced to use AutoSys Workload Automation presets all the environment variables for the instance.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*). The AutoSys Workload Automation instance command prompt opens. The command prompt presets all the environment variables for the instance.

2. Enter the following command:

```
jil
```

The JIL command prompt for the local AutoSys Workload Automation instance opens.

3. Define the monitor or browser by entering the following jil command and attributes:

```
insert_monbro: monbro_name
mode: monitor | browser
[alarm:y|n]
[status_keyword:y]
[status_keyword:y]
[...]
all_status: y|n
all_events: y|n
currun|after_time: date_time
alarm_verif: y|n
```

**NOTE**

The utility also accepts values of m (for monitor) and b (for browser) in the mode attribute of the jil monbro definition.

- **status\_keyword: y**  
(Optional) Specifies whether to track status change events that are associated with a particular job state. You can set multiple individual status filters to track status change events that are associated with any combination of the following job states:
  - RUNNING
  - SUCCESS
  - FAILURE
  - TERMINATED
  - STARTING
  - RESTART

The status keyword is the name of the job state in lower-case letters. To track status change events that are associated with one of these job states, enter the corresponding status keyword as the jil attribute and set the value of that attribute to y.

**Example:** To track status change events that are associated with either the FAILURE state or the TERMINATED state, specify the following attributes and values:

```
failure: y
terminated: y
```

1. – **all\_status: y|n**  
Specifies whether to track or report on all status change events. To track or report on only status change events that are associated with certain job states, set the value of the attribute to n and specify other status filters. Setting the value of this attribute to y overrides other status filters.
- **all\_events: y|n**  
Specifies whether to track or report on all events. To track or report on only certain types of events, set the value of the attribute to n and specify other event filters. Setting the value of this attribute to y overrides other event filters.
- **currn: y|after\_time: date\_time**  
(Browsers) Specifies the historical time frame on which the browser reports. To display only events that occurred during the most recent job run, set the value of the currn attribute to y. To display all events that occurred after a particular date and time, specify the after\_time attribute. To specify the after\_time attribute, ensure that the currn attribute is set to the default value of n.

#### NOTE

By default, AutoSys Workload Automation deletes archived event data that is older than seven days, but you can configure the product to store data for a longer period by editing the DBMaint script. The location of this script is specified in the DBMaintCmd parameter of the configuration file. Specify a value for the after\_date attribute that is within the past seven days or edit the script.

**Example:** To generate a report that displays a list of all processed events that occurred after January 2, 2012 at 5 p.m., specify the following jil attributes:

```
currn: n
after_time: "01/02/2012 17:00:00"
```

- **alarm\_verif: y|n**(Monitors) Specifies whether the monitor waits for you to acknowledge alarms that the scheduler issues. This attribute is valid only for monitors that are defined to track alarms. When you set the value of this attribute to y, you are prompted to acknowledge any alarms that the scheduler sends.

A monitor is defined to track alarms when the value of the alarms attribute is set to y or when the value of the all\_events attribute is set to y. For monitors that are defined to prompt you to acknowledge alarms, the monbro utility pauses, displays the name of the alarm, and issues the following message and prompts:

```
Alarm: alarm_name issue MM/dd/yyyyHH:mm:ss Run# run_num
Message Acknowledged by: user_name
Comment:[your_comment]
```

The utility displays the alarm information and leaves the following two fields blank. To acknowledge the alarm, enter your user name in the Message Acknowledged by field. Optionally, you can enter information in the Comment field (such as "Reported to DB Administrator for follow-up").

#### WARNING

The monitor cannot resume until you acknowledge the alarm by entering your user name.

2. 4. Enter the following jil command:

```
exit
```

The monitor or browser is inserted into the database, and the jil command prompt closes.

5. Open the operating system or instance command prompt and enter the following command:

```
{monbro -N monbro_name|monbro -N monbro_name -P ss}
```

- **-P ss**  
(Optional, Monitors only) Specifies the frequency, in seconds, at which the monitor polls for events. The monitor polls for events at the default frequency when you do not specify this option. The option is not valid with browsers.  
**Limits:** Integers greater than 0

**Default: 10**

The monitor or browser runs. A monitor polls for events at the specified interval. A browser generates a report that displays the event information for the specified period based on the specified event filters.

**Example: Define a Browser**

This example defines the browser that is named `job_restarts` to generate a report listing all events that changed the status of a job to RESTART after September 10, 2012 at midnight.

```
insert_monbro: job_restarts
mode: browser
restart: y
all_status:n
all_events:n
after_time: "09/10/2012 00:00:00"
```

**Example: Define a Monitor**

This example defines the monitor that is named `track_alarms` to track alarm events and to prompt the user to acknowledge all alarms as they occur.

```
insert_monbro: track_alarms
mode: monitor
alarm:y
all_status: n
all_events: n
alarm_verif: y
```

**Example: Acknowledge an Alarm**

This example shows that the operator acknowledged notification of a `JOBFAILURE` alarm when the job with the run number `33:2` failed at `20:15 29` on `09/24/2012` and that the operator reported the job failure to the scheduler for follow-up and resolution.

```
Alarm: JOBFAILURE fail 09/20/2012 20:15:29 Run# 33:2
Message Acknowledged by: operator
Comment:Reported to the scheduler for follow up.
```

**Example: Run a Monitor**

This example runs the monitor that is named `track_alarms` so that it polls the database for alarm events at 30-second intervals.

```
monbro -N track_alarms -P 30
```

## Generate a Forecast Report

Forecast reports display information about predicted workflow. Forecast reports help you identify problems with the predicted workflow to resolve them before they occur or to plan changes in the workflow.

**Note:** By default, AutoSys Workload Automation is configured to specify the date and time using the "`MM/dd/yyyy HH:mm[:ss]`" format. You can use a different format by changing the value of the `DateFormat` parameter in the configuration file.

**Limits:**

- *MM*: 01-12
- *dd*: 01-31
- *yyyy*: 1900-*current year*
- *HH*: 00-23
- *mm*: 00-59
- *ss*: 00-59

### Follow these steps:

1. Open the operating system or instance command prompt:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation.  
The UNIX operating system command prompt opens. The shell that is sourced to use AutoSys Workload Automation presets all the environment variables for the instance.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The AutoSys Workload Automation instance command prompt opens. The command prompt presets all the environment variables for the instance.
2. Enter the following command:
 

```
forecast {-M machine_name| -J job_name [-M machine_name]} -F "mm/dd/yyyy HH:MM [:ss]"
[-T "mm/dd/yyyy HH:MM [:ss]]" " [OPTIONS]
```

  - **-M *machine\_name* | -J *job\_name* [-M *machine\_name*]**  
Specifies what predicted workflow information the forecast report displays. To specify predicted workflow for a particular machine, use the -M parameter. To specify predicted workflow for a particular job, use the -J parameter.

**NOTE**

  - The report displays predicted workflow for all jobs that are scheduled to run on the specified machine when you specify the -M parameter alone.
  - The report displays a predicted workflow that lists runs of the specified job on all machines when you specify the -J parameter alone.
  - The report displays a predicted workflow that lists runs of the specified job on the specified machine and excludes runs on other machines when you specify the -J and -M parameters together.
  - **-F "*MM*/*dd*/*yyyy* *HH:mm* [:*ss*]"**  
Specifies that the forecast report predicts workflow starting on the specified date.
  - **-T "*MM*/*dd*/*yyyy* *HH:mm* [:*ss*]"**  
(Optional) Specifies that the forecast report predicts workflow ending on the specified date. This option is required only when you want to forecast workflow for multiple days.
  - **OPTIONS**  
(Optional) Specifies optional parameters that you can use to control what information the forecast report displays. The command generates a forecast report that displays information about predicted workflow.
3. (Optional) Depending on the information that you need, repeat the procedure for other jobs or machines that are defined within the instance.  
The command generates extra forecast reports.

## Maintain Highly-Available Environments

As an operator, it is your responsibility to monitor the environment and address problems that occur. In highly available environments, scheduler and database failures adversely affect, and sometimes disable, high-availability. To ensure that you maintain a properly functioning highly available environment, monitor the database and scheduler and resolve failures.

In highly available cluster environments, use the monitoring tools that are provided by your cluster management software and database vendor to monitor the state of the cluster and the state of the database. We recommend that you also

monitor the scheduler log on the active node. When your cluster or clustered database is not functioning properly, consult the documentation for your cluster management software or your database vendor and follow the instructions to restore the environment.

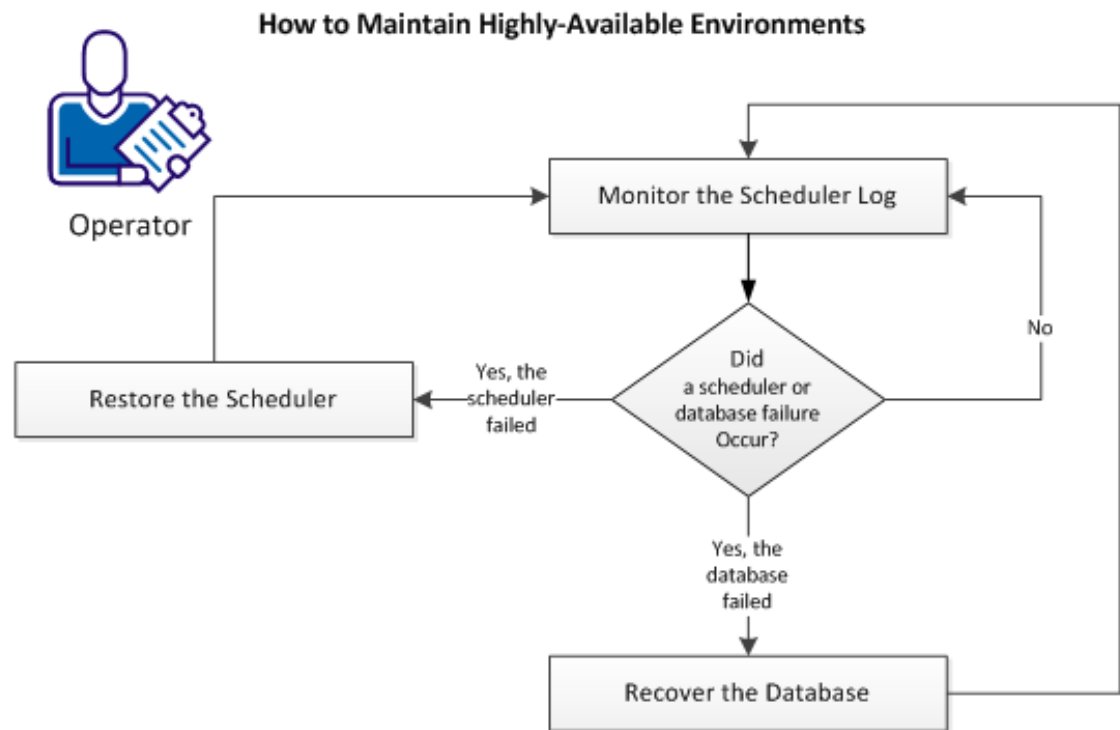
In high-availability and dual event server modes, monitor the scheduler logs for database and scheduler failures and take appropriate action to return to high-availability mode when failures occur.

#### NOTE

For more information about monitoring tools that are provided by your cluster management software and database vendor, see the documentation for those products.

The following diagram demonstrates how you can maintain a highly available environment:

**Figure 75: How to Maintain High-Available Environments**



#### Follow these steps:

1. [Monitor the Scheduler Log](#)
2. Restore the failed scheduler on [UNIX](#) or [Windows](#)
3. Recover the failed database on [UNIX](#) or [Windows](#)

#### Monitor the Scheduler Log

The scheduler log displays information about alarms and related error messages so that you know when you take action to resolve problems with your AutoSys Workload Automation environment. For example, the scheduler log displays alarm information when the scheduler issues a database rollover alarm.

- If you are operating in a highly available cluster environment, monitor the scheduler log on the active node. The scheduler log does not display all the information that you monitor the environment. Ensure that you also use the monitoring tools that are provided by your cluster manager to monitor the state of the cluster.
- If you are operating in high-availability mode, monitor the state of the dormant scheduler.
- The primary scheduler is typically the active scheduler. In this case, the shadow scheduler begins processing events and issues an alarm when the primary scheduler fails (EP\_ROLLOVER).
- If you restore the primary scheduler when the primary failback mode is set to 1, the shadow scheduler remains active and the primary scheduler runs dormant. In this case, the primary scheduler resumes processing events and issues an alarm when the shadow scheduler fails (EP\_ROLLOVER).
- If you are operating in dual event server mode, monitor the active scheduler log.
- When a database rollover occurs, AutoSys Workload Automation begins operating in single event server mode and the active scheduler issues a database rollover alarm (DB\_ROLLOVER).
- If you are operating with a clustered database, use the monitoring tools that are provided by your cluster manager and database vendor to monitor the state of the database.
- In all AutoSys Workload Automation environments, monitor the scheduler log for alarm information and error messages so that you can take appropriate action to resolve problems with the environment as they occur.

#### NOTE

For information about the monitoring tools that are provided by your cluster manager and database vendor, see the documentation for those products.

#### Follow these steps:

1. Log in as the root user on the machine or node where the scheduler is installed.
2. Open the UNIX operating system prompt or the Windows instance command prompt as follows:
  - (UNIX) Run the shell that is sourced to use AutoSys Workload Automation. The operating system command prompt appears.
  - (Windows) Click Start, Programs, AutoSys, Command Prompt (*instance\_name*). The instance command prompt appears.
3. Enter the following command:

```
autosyslog -e
```

The scheduler log appears. When AutoSys Workload Automation encounters a problem, the scheduler issues an alarm. The log displays the alarm information and related error messages so that you can resolve the problem.

#### Restore the Failed Scheduler on UNIX

If you are operating AutoSys Workload Automation in high-availability mode and the active scheduler fails, high-availability is disabled. If you are operating in a highly available cluster environment, the action that you take to return to a highly available environment depends on your cluster management software.

In highly available cluster environments, manual intervention following a failover is not required. The cluster manager can issue informational messages. When the cluster manager issues messages indicating that manual intervention is required, perform *one* of the following actions:

- Review messages and follow the instructions that they provide.
- Consult the documentation for your cluster management software and the instructions for manually recovering the failed cluster node.
- Contact your network administrator.

If you are operating in high-availability mode and high-availability is disabled because the active scheduler fails, return to high-availability mode by restoring the failed scheduler.

**Follow these steps:**

1. If the primary failback mode is set to 0 and you are restoring the primary scheduler, stop the shadow scheduler.

**NOTE**

By default, the primary failback mode is set to 0 and this step is required. When the primary failback mode is set to 1 or 2, restart the primary scheduler without stopping the shadow scheduler.

- a. Log in to a machine in the instance with a client installation as the root user.  
You can issue commands that execute client utilities.
  - b. Run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
  - c. Enter the following command:  

```
sendevent -E STOP_DEMON -v ROLE=S
```
2. If high-availability mode was disabled because a failover occurred, restore the primary scheduler.
    - a. Log in to the machine where the primary scheduler is installed and run the shell that is sourced to use AutoSys Workload Automation.  
The operating system command prompt appears.
    - b. Enter the following command:  

```
eventor
```

  
The primary scheduler is restored.  
The primary scheduler either runs dormant or resumes processing events, depending on the primary failback mode. AutoSys Workload Automation returns to high-availability mode.

**WARNING**

If the primary failback mode is set to 0, the primary scheduler resumes processing events when it is restored but AutoSys Workload Automation does not return to high-availability mode. To return to high-availability mode, restore the shadow scheduler.

If the primary failback mode is set to 1 or 2 AutoSys Workload Automation returns to high-availability mode.

3. If high-availability mode was disabled because of a failback that was not automatic or manual, use the eventor command to restore the shadow scheduler.

**NOTE**

- Automatic failbacks occur when primary failback mode is set to 2 and you restore the primary scheduler. You can control when a failback occurs by setting primary failback mode to 1 and initiating a manual failback only when you want the primary scheduler to resume processing events.
- The shadow scheduler process restarts itself in a dormant state when an automatic failback occurs or when you initiate a manual failback. The shadow scheduler does not stop running when you initiate a manual failback. In both cases, no action is required to return to high-availability mode.
- Failbacks also occur when the shadow scheduler fails. In this case, high-availability mode is disabled until you restore the shadow scheduler.

The primary scheduler continues processing events, the shadow scheduler is restored but runs dormant, and AutoSys Workload Automation returns to high-availability mode.

**Restore the Failed Scheduler on Windows**

If you are operating AutoSys Workload Automation in high-availability mode and the active scheduler fails, high-availability is disabled. If you are operating in a highly available cluster environment, the action that you take to return to a highly available environment depends on your cluster management software.



In highly available cluster environments, manual intervention following a failover is not required. The cluster manager can issue informational messages. When the cluster manager issues messages indicating that manual intervention is required, perform *one* of the following actions:

- Review messages and follow the instructions that they provide.
- Consult the documentation for your cluster management software and the instructions for manually recovering the failed cluster node.
- Contact your network administrator.

If you are operating in high-availability mode and high-availability is disabled because the active scheduler fails, return to high-availability mode by restoring the failed scheduler.

#### Follow these steps:

1. If the primary failback mode is set to Off and you are restoring the primary scheduler, stop the shadow scheduler.

#### NOTE

By default, the primary failback mode is set to Off and this step is required. When the primary failback mode is set to Immediate or Dormat, restart the primary scheduler without stopping the shadow scheduler.

- a. Log in to a machine in the instance with a client installation as the root user.  
You can issue commands that execute client utilities.
  - b. Click Start, Programs, AutoSys, Command Prompt (instance\_name).  
The instance command prompt appears.
  - c. Enter the following command:  

```
sendevent -E STOP_DEMON -v ROLE=S
```
2. If high-availability mode was disabled because a failover occurred, recover the primary scheduler.
    - a. Log in to the machine where the scheduler is installed and Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
    - b. Select an instance from the Instance drop-down list in the Settings pane, and then click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance.
    - c. Right-click the Scheduler service, and select Start.

#### WARNING

If the primary failback mode is set to Off, the primary scheduler resumes processing events when it is restored but AutoSys Workload Automation does not return to high-availability mode. To return to high-availability mode, restore the shadow scheduler.

If the primary failback mode is set to Dormant or Immediate, AutoSys Workload Automation returns to high-availability mode. If high-availability mode was disabled because a failback occurred, restore the shadow scheduler.

#### NOTE

- Automatic failbacks occur when primary failback mode is set to Immediate and you restore the primary scheduler. You can control when a failback occurs by setting primary failback mode to Dormant and initiating a manual failback only when you want the primary scheduler to resume processing events.
- The shadow scheduler process restarts itself in a dormant state when an automatic failback occurs or when you initiate a manual failback. The shadow scheduler does not stop running when you initiate a manual failback. In both cases, no action is required to return to high-availability mode.
- Failbacks also occur when the shadow scheduler fails. In this case, high-availability mode is disabled until you restore the shadow scheduler.

The primary scheduler continues processing events, the shadow scheduler is restored but runs dormant, and AutoSys Workload Automation returns to high-availability mode. The scheduler recovers and AutoSys Workload Automation returns to high-availability mode.

### **Recover the Failed Database on UNIX**

If you are using dual event server mode as your database failover solution and the scheduler initiates a database rollover, dual event server mode is disabled. Highly-available cluster environments do not function properly without a highly-available database. A database rollover does not disable high-availability mode, but the risk of down-time and data loss increases without a failover solution for your database.

Operating without a failover solution to the database increases the risk of downtime and data loss. To mitigate this risk, restore the failed database.

If you are operating with a clustered database and a database failure occurs, some cluster managers require manual intervention to recover the failed database. In this case, perform *one* of the following actions:

- Review messages issued by the cluster manager and database vendor, and follow the instructions that the messages provide.
- Consult the documentation for your cluster management software and your database software, and the instructions for manually recovering from a database failure.
- Contact your database or network administrator.

#### **NOTE**

- You can operate with a clustered database only when you have cluster management software installed and are using a cluster aware database. Some cluster managers restore a failed database automatically, so no action is required. In this case, your cluster manager issues messages indicating that the database is restored.
- If you have cluster management software installed, we recommend that you set up a highly-available cluster environment instead of configuring high-availability mode.
- When a rollover occurs, AutoSys Workload Automation backs up the configuration file before commenting out the failed event server. You can use the backup file to restore pre-rollover configuration settings when you reconfigure dual event servers

If you are using dual event server mode as your database failover solution and the scheduler rolls over the database, AutoSys Workload Automation begins operating in single event server mode. To return to dual event server mode, recover from the failure and reconfigure dual event servers.

#### **Follow these steps:**

1. Review the scheduler log file to determine the problem that caused the failure.
2. Consult your database software documentation, and follow the instructions for manually resolving the problem that caused the database failure.  
The failed database is recovered, and you can reconfigure dual event servers.
3. Log in to AutoSys Workload Automation machine in the instance with a client installation, and click Start, CA, Workload Automation AE, Command Prompt.  
The operating system command prompt opens.
4. Enter the following command:  

```
sendevent -E STOP_DEMON -v ALL
```

  
All server processes that are running on the instance stop.
5. If you want to restore the pre-rollover configuration settings, delete the modified configuration file and rename the backup file.
  - a. Delete \$AUTOUSER/config.\$AUTOSERV.
  - b. Locate \$AUTOUSER/config.\$AUTOSERV.rollover and change the name to \$AUTOUSER/config.\$AUTOSERV.

- c. Repeat these actions on every server machine that is installed in the instance.
6. If you want to specify new configuration settings when you restore the failed event server, modify the \$AUTOUSER/config.\$AUTOSERV file.
  - a. Open the configuration file on the machine where the primary scheduler is installed and locate the following parameter:
 

```
#EventServer 1|#EventServer 2
```
  - b. Edit the parameter as follows:
 

```
Event_Server_1|EventServer_2
```
  - c. Specify the primary event server and the secondary event server:
    - To make the new database the secondary event server, add the following parameter:
 

```
EventServer_2=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST | ORACLE_SVR:SCHEMAOWNER,DBPORT,DBHOST
```

**SYBASE\_DB,DBPORT,DBHOST**  
Identifies the Sybase database for the second event server.

**ORACLE\_SVR:SCHEMAOWNER,DBPORT,DBHOST**  
Identifies the Oracle database for the second event server.
    - To make the restored database the secondary event server, specify the active database in the EventServer\_1 parameter and the restored database in the EventServer\_2 parameter.
    - To make the new database the primary event server, specify the existing database that is defined in the EventServer\_1 parameter as the secondary event server by changing it to EventServer\_2, then add the following parameter:
 

```
EventServer_1=SYBASE_SVR:SYBASE_DB,DBPORT,DBHOST |
ORACLE_SVR:SCHEMAOWNER,DBPORT,DBHOST
```
    - To make the restored database the primary event server, verify that the restored database is specified in the EventServer\_1 parameter and the active database is specified in the EventServer\_2 parameter.
  - d. Specify the database reconnect behavior for the second event server by modifying the following parameter in the configuration file:
 

```
DBEventReconnect=value, value2
```

**Value**

    - Identifies the database reconnect behavior for the first event server.
    - **Limits:** 0-99

**Value 2**

    - Identifies the database reconnect behavior for the second event server.
    - **Limits:** 0-99

**NOTE**

During typical installation, AutoSys Workload Automation sets the reconnect value for the single event server to 50 by default. During a custom installation in which you enable dual event server mode, AutoSys Workload Automation sets the reconnect value for both event servers to 50, 5 by default. Ensure that you add a reconnect value for the second event server when you configure AutoSys Workload Automation to run in dual event server mode after running it in single event server mode. Optionally, you can modify the default reconnect value for the first event server.

The secondary event server is configured on the primary scheduler machine.
7. Repeat the secondary event server configuration on every server machine in the instance. Ensure that the event server information in the configuration file is the same on each of these machines.
8. Run the AutoSys Workload Automation bulk copy script ([Configure AutoSys Workload Automation to Run in Dual Event Server Mode](#)) based on your database vendor to synchronize the event servers. Restart the server processes for the instance. If you are operating in a highly available cluster environment, restart the services on the active node.
  1. a. Open the operating system command prompt, and enter the following commands:

```
unisrvcntr start waee_sched.$AUTOSERV
unisrvcntr start waee_server.$AUTOSERV
```

- b. Repeat this action on every server machine in the instance. If you are operating in a highly available cluster environment, restart the components on the active node only.

#### NOTE

In highly available cluster environments, the scheduler and application server actively execute work on the active node only. The cluster manager prompts the scheduler or application server on one of the passive nodes to begin executing tasks only when it detects a failure of the same component on the active node.

The database is restored and High-availability is restored. To maintain your highly available environment, continue monitoring the scheduler log and recovering from scheduler and database failures when they occur.

### **Recover the Failed Database on Windows**

If you are using dual event server mode as your database failover solution and the scheduler initiates a database rollover, dual event server mode is disabled. Highly-available cluster environments do not function properly without a highly-available database. A database rollover does not disable high-availability mode, but the risk of down-time and data loss increases without a failover solution for your database.

Operating without a failover solution to the database increases the risk of downtime and data loss. To mitigate this risk, restore the failed database.

If you are operating with a clustered database and a database failure occurs, some cluster managers require manual intervention to recover the failed database. In this case, perform *one* of the following actions:

- Review messages issued by the cluster manager and database vendor, and follow the instructions that the messages provide.
- Consult the documentation for your cluster management software and your database software, and the instructions for manually recovering from a database failure.
- Contact your database or network administrator.

#### NOTE

- You can operate with a clustered database only when you have cluster management software installed and are using a cluster aware database. Some cluster managers restore a failed database automatically, so no action is required. In this case, your cluster manager issues messages indicating that the database is restored.
- If you have cluster management software installed, we recommend that you set up a highly-available cluster environment instead of configuring high-availability mode.
- When a rollover occurs, AutoSys Workload Automation backs up the configuration file before commenting out the failed event server. You can use the backup file to restore pre-rollover configuration settings when you reconfigure dual event servers

If you are using dual event server mode as your database failover solution and the scheduler rolls over the database, AutoSys Workload Automation begins operating in single event server mode. To return to dual event server mode, recover from the failure and reconfigure dual event servers.

### **Follow these steps:**

1. Review the scheduler log file to determine the problem that caused the failure.
2. Consult your database software documentation, and follow the instructions for manually resolving the problem that caused the database failure.

The failed database is recovered, and you can reconfigure dual event servers.

3. Log in to AutoSys Workload Automation machine in the instance with a client installation and Click Start, Programs, AutoSys, Command Prompt (*instance\_name*).  
The instance command prompt opens.
4. Enter the following command:  

```
sendevent -E STOP_DEMON -v ALL
```

All server processes that are running in the instance stop.
5. Restore, enable, and reconfigure the failed database (event server).
  - a. Review the scheduler log file to determine the problem that caused the failure.
  - b. Consult your database software documentation, and follow the instructions for manually resolving the problem that caused the database failure.  
The status of the failed database changes and you can reconfigure dual event servers.
  - c. Click the Event Server icon on the toolbar in any AutoSys Workload Automation Administrator window.  
The Event Server - AutoSys Workload Automation Administrator window appears.
  - d. Clear the A Database Rollover Has Occurred check box, verify the event server configuration settings and make any desired modifications, and then click Apply.

### WARNING

Ensure that the configuration settings for Event Server 2 are identical to the configuration settings for Event Server 1.

The failed event server is restored and the configuration settings are saved, but AutoSys Workload Automation does not return to dual event server mode until you synchronize the event servers.

6. Run the AutoSys Workload Automation bulk copy script ([Configure AutoSys Workload Automation to Run in Dual Event Server Mode](#)) based on your database vendor. The directory and script that you synchronize the event servers following a database failure are the same as the directory and script that you use to switch to dual event server mode when you did not enable dual event servers during installation.  
The event servers are synchronized.
7. Restart the server processes for the instance. If you are operating in a highly-available cluster environment, restart the services on the active node.
  - a. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
  - b. Select an instance from the Instance drop-down list in the Settings pane.
  - c. Click the Services icon on the toolbar.  
A list of the services that are running on the instance appears.
  - d. Right-click the scheduler service and select Start, then right-click the application server service and select start.
8. If you are operating in a highly-available cluster environment, restart the components on the active node only. Otherwise, repeat these actions on every server machine in the instance. If you are operating in a highly-available cluster environment, restart the components on the active node only.

### NOTE

In highly-available cluster environments, the scheduler and application server actively execute work on the active node only. The cluster manager prompts the scheduler or application server on one of the passive nodes to begin executing tasks only when it detects a failure of the same component on the active node.

The database is recovered and AutoSys Workload Automation returns to dual event server mode. High-availability is restored. To maintain your highly-available environment, continue monitoring the scheduler log and recovering from scheduler and database failures when they occur.

# Web UI Monitoring and Reporting

## Web UI Monitoring Application

The Monitoring application lets you create, manage, and monitor job-based and server-based views and alerts. You can access the Monitoring application by clicking the Monitoring tab on the main Web UI page.

## Web UI Reporting Application

The Reporting application, accessed by clicking the Reporting tab from the Web UI Dashboard, lets you create, view, modify, and run reports on multiple servers.

### **NOTE**

Access to Reporting is granted or denied based on CA EEM policies. The tasks available are based on the rights granted to you by the security administrator.

## Web UI Monitoring Application

The Monitoring application lets you create, manage, and monitor job-based and server-based views and alerts. A *view* comprises the jobs, box jobs, servers, and any alerts or alarms that are associated with a business activity you want to monitor. The Monitoring page displays a list of views and server instances from which you can view the active jobs and most important properties of each. Additionally, you can define filters to configure specialized views and alerts, and save the views for later use.

You can access the Monitoring application by clicking the Monitoring tab on the main Web UI page.

For more information about job administration and scheduling, see the *Quick Edit Help* and *Workload Scheduling sections*.

## Monitoring Overview

Monitoring is a high-performance application that provides information about job status, alerts, and AutoSys Workload Automation alarms. Monitoring provides the ability to view the enterprise by servers (workload automation server instances) or by user-defined business views.

Views can be created based on various criteria. Through creation of custom views, Monitoring provides the ability to monitor by exception across the enterprise. For example, users can create views that show failed jobs, running jobs, terminated or canceled jobs, or milestone jobs. Monitoring can also be used to monitor job streams by business relevance. Views can be created based on job name, box job name, application name, or group name. Additionally, views can be created based on other criteria such as owner, machine, status, job type, time zone, run machine, and started in the past number of days or hours. You can open a view and can see a summary by job status (each view has the status of the most critical job status within the view). A status for views by server status is also available letting you identify problems with server connectivity. For example, a valid status could be Connected Successfully. When you detect a problem in a particular view, such as a failure status, you can open the view to display more detailed analysis.

Alerts and AutoSys Workload Automation alarms are reported in the Alerts table. Alerts are similar to AutoSys Workload Automation alarms. Alerts can be created for AutoSys Workload Automation servers, and are designed to reflect the requirements and operations of a particular enterprise. Alerts are created for each workload automation server accessed through Web UI, and do not have to be the same across all the servers of a particular job manager. The information on the Alerts table can be filtered by one or more criteria, and the properties that are displayed can be modified and reordered. From the Alerts table, you can open, acknowledge, and close individual or multiple alerts and AutoSys Workload Automation alarms.

The Summary tab displays a summary of the current view or server. This tab displays summary statistics on job status, alerts, and alarms. The Summary tab also contains a summary of the filters applied in the current view, and displays an overall summary of the status of the view or server, which are based on user-defined criteria. The overall status is also

displayed in the Views table on the Dashboard tab. From this tab, you can monitor the status of jobs across the enterprise, and respond immediately if there is a change in status.

The Flow tab monitors a view using a graphical representation of interconnected workload objects. From the Flow, you can quickly see the status of each job and job type by its icon, status color, review the jobs contained in box jobs, and see the job predecessors and successors. Additionally, the job object provides information about connected resources through an attached icon in the Flow.

### **Job Type Icons**

The following job type icons are used in Web UI:

 Box (BOX)	 Secure Copy (SCP)
 Command (CMD)	 User Defined (specify 0-9)
 File Watcher (FW)	 z/OS Regular (ZOS)
 File Trigger (FT)	 z/OS Data Set Trigger (ZOSDST)
 FTP (FTP)	 z/OS Manual (ZOSM)
 i5/OS (I5)	
<b>Application and Web Services Jobs</b>	
 Entity Bean (ENTYBEAN)	 JMX-MBean Operation (JMXMOP)
 HTTP (HTTP)	 JMX-MBean Remove Instance (JMXMREM)
 JMS Publish (JMSPUB)	 JMX-MBean Subscribe (JMXSUB)
 JMS Subscribe (JMSSUB)	 POJO (POJO)
 JMX-MBean Attribute Get (JMXMAG)	 RMI (JAVARMI)
 JMX-MBean Attribute Set (JMXMAS)	 Session Bean (SESSBEAN)
 JMX-MBean Create Instance (JMXMC)	 Web Service RPC/Encoded (WBSVC)

 Document/Literal Style Web Service (WSDOC)	
<b>Database Jobs</b>	
 Database Monitor (DBMON)	 Database Trigger (DBTRIG)
 Database Stored Procedure (DBPROC)	 SQL (SQL)
<b>Object Monitoring Jobs</b>	
 CPU Monitoring (OMCPU)	 Text File Reading and Monitoring (OMTF)
 Disk Monitoring (OMD)	 Windows Event Log Monitoring (OMEL)
 IP Monitoring (OMIP)	 Windows Service Monitoring (OMS)
 Process Monitoring (OMP)	
<b>ERP Jobs</b>	
 Oracle E-Business Suite Copy Single Request (OACOPY)	 SAP BW Process Chain (SAPBWPC)
 Oracle E-Business Suite Request Set (OASET)	 SAP Data Archiving (SAPDA)
 Oracle E-Business Suite Single Request (OASG)	 SAP Event Monitor (SAPEVT)
 PeopleSoft (PS)	 SAP Job Copy (SAPJC)
 SAP Batch Input Session (SAPBDC)	 SAP Process Monitor (SAPPM)
 SAP BW InfoPackage (SAPBWIP)	 SAP R/3 (SAP)
<b>HADOOP Jobs</b>	
 Hadoop Distributed File System (HDFS)	 Oozie
 Hive	 Pig
 Sqoop	
<b>SNMP Jobs</b>	



 SNMP Value Get (SNMPGET)	 SNMP Value Set (SNMPSET)
<b>Other Jobs</b>	
 Micro Focus (MICROFOCUS)	 Remote Execution (PROXY)
 Process Automation Process Execution (PAPROC)	 Wake on LAN (WOL)
 Process Automation Start Request Form (PAREQ)	 Informatica
 MSSQL Server	

## Monitoring Navigation

Monitoring lets you create and manage views on multiple servers, view detailed information about all currently defined views, define new views as needed, and view their flows graphically. This application is accessed by clicking the Monitoring tab on the main Web UI page.

Additionally, the Dashboard tab, which is displayed by default when you log into Web UI, displays the Views table, Servers table, and Alerts table.

You can click



in the Navigation bar on the Monitoring page to hide the left pane, expanding the sections in the right pane.

## Dashboard Tab

The Dashboard tab, which is displayed by default when you log in to Web UI, contains a list of views, servers, jobs, and alerts to which you have access. From this tab, you can quickly view the status of a view or server, and open a view to display more details. For example, opening a view takes you to the Monitoring main page, with the open view displayed in the left navigation pane and the Summary View displayed in the right pane.

## Monitoring Tab

The Monitoring tab contains a high-level overview of all the views to which you have access. From this tab, you can quickly search for specific views, edit, delete, import, or export selected views. You can also filter your views by server, job-based status, and view name, or view a statistical summary of job and alert status.

The Monitoring tab is comprised of two panes:

- The left pane provides navigation capability and displays a list of views, servers, and alert policies in their respective sections.
- The right pane provides search capability and detailed information about your views, servers, alert policies, and associated alerts. You can use the right pane to manage your views, to create, edit, import, export, or delete views. The Jobs and Alerts tab lets you view jobs by server and jobs contained within the views, and monitor alerts and alert policies. You can also access the Summary tab to view statistical data of jobs and alerts for the current server or view. You can select the Flow tab to graphically monitor the jobs within the current view.

## Navigation Pane

The left navigation pane is comprised of three tabs that include Views, Servers, and Alert Policies.

- Views - Provides a tree structure of views and their job contents.
- Servers - Provides information about available servers and their job contents
- Alert Policies - Provides a list of alert policies

### **View Management Console (VMC)**

You can use the VMC button to open the View Management Console dialog. The View Management Console dialog lets you create, modify, delete, import, or export view definitions and create, modify, or delete alert policies. You can also access the View Management Console by selecting the edit, delete, or export actions available from the pop-up menu in the Views table.

The View Management Console is comprised of four panes: Command, View Commands, Filter Commands, and Alert Policy Commands. Depending on how you open the View Management Console determines the content displayed. For example, clicking the VMC button opens the View Management Console with a blank Command pane in which to create your views and alert policies. You can use the commands in the View Commands, Filter Commands, and Alert Policy Commands panes to select valid commands and attributes to define your views and alert policies. Clicking any command link populates the Commands section with the correct command syntax.

### **Monitoring Job Tree Navigation**

The structure of the monitoring navigation provides a tree hierarchy of boxes and jobs. This enables supervision of the job status and allows for quick access to the job flow. When you select a job or a box job in the flow, it will be reflected in the tree and vice versa. When you select a job or a box job in the Jobs table, it will be reflected in the tree.

Expand/Collapse All functionality, in a particular view, is supported through the flow toolbar buttons and is synced when these flow actions are triggered.

#### **NOTE**

Multi- selection functionality is not available.

### **Quick Edit and Quick View In Context**

If you are authorized to use Quick Edit, selecting the appropriate option on the Monitoring tab opens Quick Edit in context. A new window that contains a subset of the functionality of the Quick Edit application is displayed. The Search and Search Results sections and the Create and Customize buttons are not available. If you need the full functionality of Quick Edit, you can access the application by clicking the Quick Edit tab. For information about using Quick Edit, see *Quick Edit Help*.

If you are authorized to use Quick View, selecting the appropriate option on the Monitoring tab opens Quick View in context. A new window that contains a subset of the functionality of the Quick View application is displayed. The Search section and the Customize button are not available. If you need the full functionality of Quick View, you can access the application by clicking the Quick View tab. For information about using Quick View, see *Quick View Help*.

#### **NOTE**

Authorization to access Quick Edit and Quick View is checked when you click any of the links or buttons.

### **Security**

Depending on the CA EEM security rights assigned by your administrator, you may not have access to all functionality in Monitoring. The following are examples of functionality that is not available:

- If you are not authorized to create and modify a view, you cannot add, modify, or delete a filter associated with the view. If you are not authorized to create, modify, copy, or delete views, clicking Execute simply displays an error message in the View Management Console dialog.
- If you are not authorized to use Quick Edit or Quick View in context, selecting the Edit, Open, or Show Details options or icons simply displays an error message at the top of the page.
- If you are not authorized to send events, the send event and job actions functions are not available from Monitoring.

**NOTE**

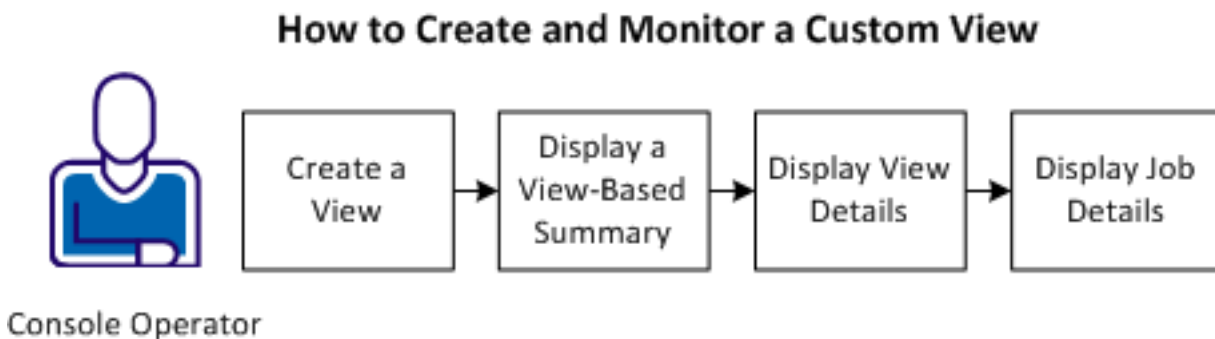
Any authorization changes to views and servers become effective only when you log in or perform a manual refresh. All other policy changes take effect within the time specified in the Security Cache Timeout Interval field set in Configuration Manager.

**Create and Monitor a Custom View**

This article explains how to create and monitor a custom view. As a console operator, you are responsible for creating and managing the views and alerts for your enterprise.

This scenario walks you through the process of creating and monitoring a view.

**Figure 76: How to Create and Monitor a Custom View**



To create and monitor a custom view, follow these steps:

1. Do one of the following tasks:
  - [Create a view using VMC.](#)
  - [Create a view using the View Editor.](#)
2. [Display a view-based summary.](#)
3. [Display view details.](#)
4. [Display AutoSys Workload Automation job details.](#)

**Example: Create and Monitor a Custom View Containing AutoSys Workload Automation Jobs**

Suppose that you are responsible for monitoring all of the payroll jobs in your enterprise. You can create a view that displays only those AutoSys Workload Automation jobs and properties you want to monitor. After the view is created, you can open the view to display its overall status. You can then display all of the jobs contained in the view and view the overall properties of the failed jobs in Quick View.

**Create a View Using VMC**

In Monitoring, you can use existing scheduling objects in the database to define a view. The command language used in the View Management Console is modeled on the AutoSys Workload Automation JIL language. JIL uses a command and a set of attributes for that particular command to comprise the complete definition. You can use the View Management Console to create one or more views.

**Follow these steps:**

1. Click VMC.  
The View Management Console dialog opens with the available commands and command attributes listed in their respective panes.
2. Click the insert\_view command in the View Commands pane.

3. Edit the view definition as appropriate.

**NOTE**

- Clicking a command link populates the Commands section with the command.
- At least one filter must be defined to create a usable view.
- View names can contain up to 80 characters; valid characters are a-z, A-Z, 0-9, underscore (\_), and hyphen (-). View names with embedded spaces must be enclosed in single quotes. View names cannot include any of the following: #, \$, %, \*, ., ?, or @.

4. Click Execute.

Web UI verifies the validity of the view definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the view is created.

5. Close the View Management Console dialog.

**Example: Create the View PayrollView**

When executed, the following command creates a view with the name PayrollView. The view includes jobs on ServerAE1 beginning with payroll and belonging to the Pay application:

```
insert_view: PayrollView
job_columns: job_name,status,server,run_machine,status_time
alert_columns: alert,severity,status,job,server,status_time,create_time
description: 'this view contains the payroll jobs'

insert_filter: ServerAE1
view: PayrollView
filter: (application LIKE PAY AND job_name LIKE Payroll*)
active: true
```

**Create a View Using the View Editor**

In Monitoring, you can create a view using the form-based View Editor.

**Follow these steps:**

1. Click Add.  
The View Editor dialog opens.
2. Do the following steps in the View section:
  - a. Enter a name for the view in the View Name field.
  - b. Select the appropriate columns and the order in which to display them from the Job Columns selection pane.
  - c. Select the appropriate columns and the order in which to display them from the Alert Columns selection pane.
  - d. (Optional) Enter a description in the Description field.

**NOTE**

View names can contain up to 80 characters; valid characters are a-z, A-Z, 0-9, underscore (\_), and hyphen (-). View names with embedded spaces must be enclosed in single quotes. View names cannot include any of the following: #, \$, %, \*, ., ?, or @.

3. Click Add Filter in the navigation pane in the Filter section.  
The Create Filter section expands to display the necessary fields.

**NOTE**

- To create a usable view, at least one filter must be defined.
- You can only have one active filter per server. Use the Active field to indicate which filters you want to be active at one time.

4. Select the server for the view from the Server drop-down list.
5. (Optional) Select the alert policy to attach to the view from the Alert Policy drop-down list.
6. Do one of the following:
  - Select the Text control option and enter the filter definition in the Job Filter field.
  - Select the Interactive control option to define the filter definition using the Add Filter Expression table, and then do the following:
    - a. Click Add Filter Expression.
    - b. Select the appropriate value in the Attribute field. The following options are available: job\_name, owner, job\_type, application, group, box, box\_name, timezone, status, started\_in\_past, system, run\_machine, and machine.
    - c. Select the appropriate value in the Operator field. The following options are available: =, !=, LIKE and NOT\_LIKE.
    - d. Enter the appropriate criteria in the Value field.
    - e. (Optional) Select AND or OR from the Logic drop-down list to create a complex filter definition and enter additional filter criteria in the appropriate fields.
    - f. Click Save Expression.
 The filter is added to the Job Filter field.

**NOTE**

When adding a complex filter using the interactive method, you must enter the parentheses manually. For example, you can enter the filter (job\_name like JOB\*) AND ((application like APP\*) OR (group like GRP\*)). This filter specifies to include all jobs beginning with JOB that are included in either the applications beginning with APP or the groups beginning with GRP.

7. Click Save Filter.  
The filter is added to the Filter section.
8. (Optional) Repeat steps 3-7 to add additional filters.
9. Click Save.  
The view is created.

**Display a View-Based Summary**

You can display a view-related summary of job and box job status from the Dashboard tab or the Monitoring page.

To display a view-based summary, right-click a view name and select Open from the pop-up menu or double-click the view name in the Views table.

**NOTE**

- When you display a view summary, the Open Views folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open views.
- To switch between multiple open views, you can click a different view name in the Navigation pane, displaying an updated Summary tab.
- To view multiple summaries at a single time, you must open multiple instances of Web UI.

**Display View Details**

You can display a table of jobs or box jobs represented in the view from the Dashboard tab or the Monitoring page.

**Follow these steps:**

1. Right-click a view name and select Open from the pop-up menu or double-click a view name in the Views table.
2. Select the Jobs and Alerts tab.  
The Jobs and Alerts tables open in Monitoring, displaying the jobs and box jobs contained in the view and all associated alerts.

**NOTE**

- When you display view details, the Open Views folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open views.
- When job level security is enabled in Configuration, only the jobs to which you have access and the alerts associated with those jobs are displayed in the tables.

**Display AutoSys Workload Automation Job Details**

You can open Quick View in context to display details, dependencies, starting conditions, and associated alerts for a AutoSys Workload Automation job.

**Follow these steps:**

1. Find the appropriate job or box job.
2. Do one of the following tasks:
  - Right-click that job or box job in the Jobs table and select Open from the pop-up menu.
  - Double-click that job or box job name in the Jobs table.
  - Right-click that job or box job in the flow. Next, select Open from the context menu.
 Quick View opens in context in a separate browser window.
3. Scroll down as necessary to view all of the job information available.
4. Click Close when you are finished viewing the details.  
The Quick View application closes.

**NOTE**

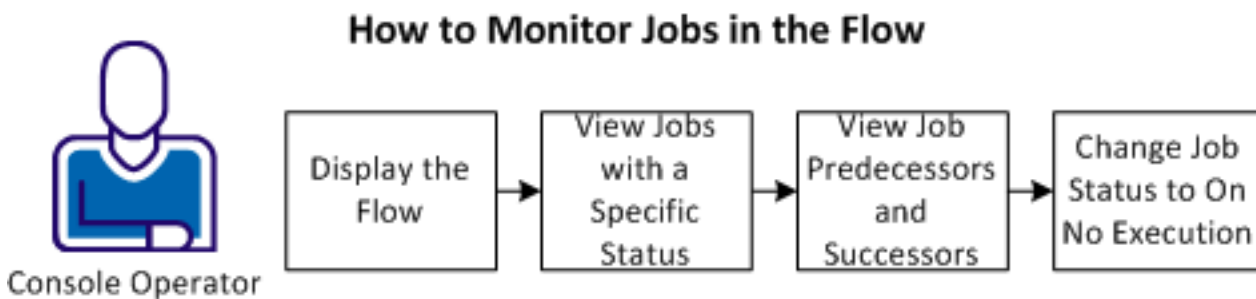
For more information about viewing job properties in Quick View, see the *Quick View Help*.

**Monitor Views Using the Flow**

This article explains how to monitor views using the Flow. As a console operator, you can use the Flow to monitor a view using a graphical representation of interconnected workload objects. From the Flow, you can quickly see the status of each job by its icon, review the jobs contained in box jobs, and see the job predecessors and successors.

This scenario walks you through the process of monitoring jobs using the Flow.

**Figure 77: How to Monitor the Jobs in the Flow**



To monitor the jobs in the Flow, follow these steps:

1. [Display the flow.](#)
2. [View jobs with a specific status.](#)
3. [View job predecessors and successors.](#)
4. [Change job status to On No Execution.](#)

### Example: Monitor PayrollView in the Flow

Suppose that you want to monitor the jobs in PayrollView using the Flow. You can select the Flow tab to quickly see a graphical representation of jobs (including box jobs) with their predecessors and successors. Perhaps you are interested in locating only jobs that have a status of failure so you can take the appropriate action. For example, you see that Payroll003 failed to run, and PayrollDone is a successor to Payroll003. You can send an On No Execution event from the pop-up menu to bypass Payroll003 and then restart the job flow. Payroll003 is not executed, but the remaining dependent jobs run successfully.

### Display the Flow

You can use the Flow to monitor the view graphically.

#### Follow these steps:

1. Select the appropriate view in the Views table, and select the Flow tab.
2. (Optional)



Click

open the Overview dialog and drag the boundary box to display various sections of the job flow.

#### NOTE

You can move the Overview dialog to any location on the page.

### View Jobs with a Specific Status in the Flow

The Flow toolbar helps you locate problem jobs by status when corrective action is required. To locate a problem job, use the drop-down function on the toolbar and select a job status.

#### Follow these steps:

1. Click Select by status in the Flow toolbar.
2. Select job status.

#### NOTE

The jobs in the specified state are selected when the Flow refreshes.

### View Job Predecessors and Successors in the Flow

Job predecessors, which must occur before a job can start, and job successors, which are initiated once a job has completed, help position a job in a job flow.

#### Follow these steps:

1. Locate that job in the Flow.
2. Verify that an arrow connects the job with another object, which indicates a predecessor or successor relationship.
3. Move your mouse pointer over the dependency line to view a tooltip describing the relationship between the predecessor and successor object.

#### NOTE

You can also open the Legend window to see what dependency conditions the different line colors signify.

### Put a Job on No Execution

You can use the On No Execution command to send an ON\_NOEXEC event to the selected job or box job, changing the job status to ON\_NOEXEC. Putting a job on no execution is the functional equivalent of bypassing the job during the scheduled run.

**NOTE**

To select multiple jobs at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Find the appropriate job or box job.
2. Perform one of the following actions:
  - Select one or more jobs in the Jobs table, right-click, and select On No Execution from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select On No Execution from the pop-up menu.
3. Click Yes to confirm.  
The selected job is bypassed during the next scheduled run.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

## Create and Assign an Alert Policy to a Custom View

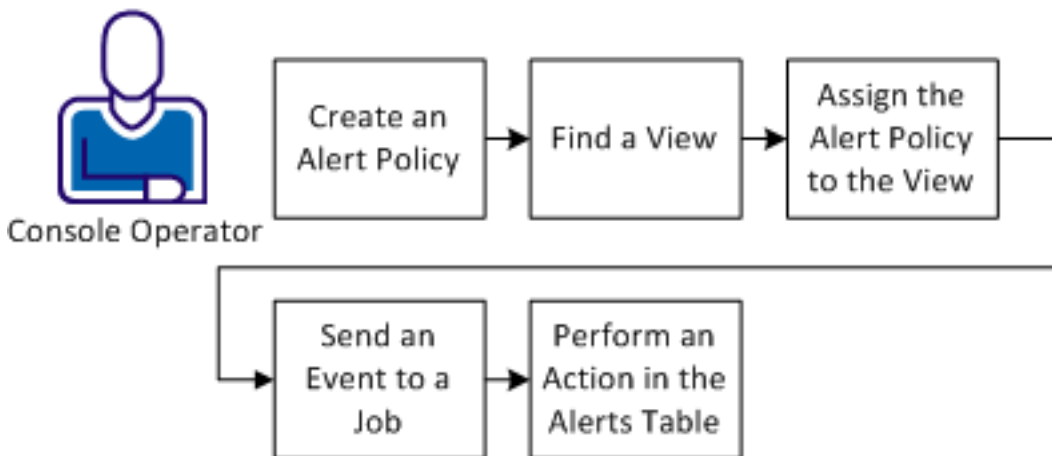
This article explains how to create and assign an Alert policy to a custom view. For AutoSys Workload Automation servers, you can utilize both alerts and alarms. As a console operator, you can create alert policies that include only alerts of interest based on any of the following:

- A particular set of jobs (filtered by job name)
- One or more alarm types
- Job status changes
- Alert status (OPEN, ACKNOWLEDGED, CLOSED)

This scenario walks you through the process of creating an alert policy, assigning that alert policy to a specific view, and monitoring the alerts generated.

**Figure 78: How to Create and Assign an Alert Policy to a Custom View**

### How to Create and Assign an Alert Policy to a Custom View



To create and assign an alert policy to a view and monitor alerts, follow these steps:

1. Perform one of the following actions:
  - [Create an alert policy in VMC.](#)
  - [Create an alert policy using the Alert Policy Dialog.](#)



2. [Find a view.](#)
3. Perform one of the following actions:
  - [Assign an alert policy to the view in VMC.](#)
  - [Assign an alert policy to the view using the View Editor.](#)
4. [Send an event to a job.](#)
5. [View the Alerts table and perform an action.](#)

### Example: Create an Alert Policy and Assign it to the WorldJobs View

Suppose that you are responsible for monitoring the jobs in a view named WorldJobs. However, you want to assign an alert policy that issues an alert when a job in the view fails. You can create an alert policy based on job failures, and can apply that policy to the WorldJobs view by adding the policy to the view definition. For example, you can apply the policy to the view definition by adding the following line to the filter definition in VMC:

```
alert_policy: FailedJobs
```

To test the view, send a FORCE\_STARTJOB event to the box job WORLD01BOX, for example. If WORLD005, a job in WORLD01BOX, contains an invalid command, it fails and issues an alert. You can view that alert in the Alerts table and can take the appropriate action.

### Create an Alert Policy in VMC

You can create an alert policy to generate alerts when certain criteria are met.

#### Follow these steps:

1. Click VMC.
2. Click the insert\_alert\_policy command in the Alert Policy Commands pane.

#### NOTE

Clicking a command link populates the Commands section with the command.

3. Edit the template definition as appropriate.
4. Click Execute.
 

Web UI verifies the validity of the alert policy definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the policy is created.
5. Close the View Management Console dialog.

### Example: Create an Alert Policy in VMC and Assign it to a Specific Server

The following definition creates an alert policy named FailedJobs that lets you monitor job status alerts for failed jobs and JOBFAILURE and STARTJOBFAIL alerts for all jobs on servers AS1 and AS2. Only those alerts with a status of Open or Acknowledge are displayed in the Alerts table with an assigned severity of High.

```
insert_alert_policy: FailedJobs
job_status: Failure
alarm_type: JOBFAILURE, STARTJOBFAIL
job_name: *
severity: High
server: AS1, AS2
status: OPEN, ACKNOWLEDGED
active: true
```

## Create an Alert Policy Using the Alert Policy Dialog

In addition to creating an alert policy using the View Management Console dialog, you can also use the form-based Alert Policy dialog.

### Follow these steps:

1. Click Add.  
The Alert Policy dialog opens.
2. Define your policy by doing the following steps:
  - a. Enter a name for the policy in the Name field.

#### NOTE

Alert policy names can contain up to 80 characters; valid characters are a-z, A-Z, 0-9, underscore (\_), and hyphen (-). Names with embedded spaces must be enclosed in single quotes. Names cannot include any of the following: #, \$, %, \*, ., ?, or @.

- b. Enter a name for the jobs to associate with the policy in the Job Name field.

#### NOTE

You can use the asterisk (\*) or percent (%) as a wildcard for the entire name or for a partial name, where the \* and % represent zero or more characters. You can also use the question mark (?) to represent any single character.

- c. Select one or more alert states to include in the policy from the Status drop-down list.
  - d. Select one or both of the following items:
    - Select one or more alarm types to include in the policy from the Alarm Type selection pane.
    - Select one or more job statuses with which to associate the alert policy from the Job Status selection pane.
  - e. Select the severity level for the alert from the Severity drop-down list.
3. (Optional) Specify any of the following information:
  - Clear the Active check box to indicate that the policy should not be applied.
  - Select one or more servers to monitor global job alerts and alarms from the Servers drop-down list. If this attribute is specified, the alerts are displayed in the Alerts table for the specified server.
  - Enter a description for the alert in the Text field.
  - Enter the name of the link to the content related to the alert in the URL Name field.
  - Enter the URL at which the content related to the alert resides in the URL Link field.
  - Select one or more views to include the created alert from the View selection pane.
4. Click Save.  
The alert policy is created and the dialog closes.

## Find a View

You can use the filter section in the Views table to find one or more specific views defined in Web UI.

### Follow these steps:

1. Click the Display Filter icon to display the filter.
2. Perform one or more of the following actions:
  - Enter the view name in the Name field.
  - Enter the name of the server in the Servers field or select the server from the Servers drop-down list.

#### NOTE

You can enter any of the following in the Name and Servers fields:

- Valid name—Finds the specified view.
- Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all views or servers in the enterprise

that contain the partial name. For example, entering n\* in the Name field would find all views that begin with the letter n, and entering \*n would find all views that end with the letter n.

- Asterisk (\*), percent (%), or blank—Finds all views or servers.
  - Multiple names, separated by a comma.
- Select one or more job-based statuses from the Job Status drop-down list.
  - Select one or more of the following options from the Filter drop-down list:
    - Active -- Finds all views that have an active filter.
    - Inactive -- Finds all views that have an inactive filter.
    - Unbound -- Finds all views that have a filter specifying a server that is no longer configured.
    - None -- Finds all views that do not have a filter defined.
3. Click Go.  
The views that meet your search criteria are listed in the Views table.

#### NOTE

- The title changes to indicate that a filter has been applied to the table.
  - The filter is persisted through user sessions until you change or clear the filter criteria.
4. (Optional) Click the Display Filter icon to hide the search filter.

### **Assign an Alert Policy to a View in VMC**

You can assign an alert policy to a view in the View Management Console. When the alert policy is added to a view filter definition, only job alerts belonging to the view are monitored. An alert policy can be assigned to one or more views.

#### NOTE

If an alert policy is defined to a specific server and to a specific view, only the alerts for jobs in the view are displayed in the view Alerts table. The server defined in the alert policy is ignored. However, all alerts for the server defined in the alert policy are displayed in the server Alerts table.

#### **Follow these steps:**

1. Right-click a view name and select Edit in VMC from the pop-up menu.
2. Add the alert\_policy attribute to the end of the filter definition and enter one or more policy names. Multiple names must be separated by commas.  
For example, alert\_policy: FailedJobs
3. Click Execute.  
Web UI verifies the validity of the view definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the view is created.
4. Click Close.  
The View Management Console dialog closes.

### **Example: Assign an Alert Policy to a View Using the View Management Console**

The following filter definition assigns two alert policies to the MyView1 view on server AS1. All alerts matching the policies FailedJobs and TerminatedJobs are displayed in the view.

```
insert_filter: AS1
view: MyView1
filter: job_name like Backup*
alert_policy: FailedJobs, TerminatedJobs
active: true
```

### **Assign an Alert Policy to a View Using the View Editor**

You can assign an alert policy to a view using the View Editor. When the alert policy is added to a view filter definition, only job alerts that belong to the view are monitored. An alert policy can be assigned to one or more views.

#### **NOTE**

If an alert policy is defined to a specific server and to a specific view, only the alerts for jobs in the view are displayed in the view Alerts table. The server defined in the alert policy is ignored. However, all alerts for the server defined in the alert policy are displayed in the server Alerts table.

#### **Follow these steps:**

1. Right-click a view name and select Edit from the pop-up menu.
2. Click the name of the filter in the navigation pane in the Filter section to display the fields in the Edit Filter section.
3. Select one or more alert policies to attach to the view from the Alert Policy drop-down list.
4. Click Save Filter and then click Save.  
The alert policy is assigned to the view.

### **Send an Event to a AutoSys Workload Automation Job**

You can send an event of the appropriate type to change the state, status, or priority for one or more AutoSys Workload Automation jobs or box jobs. You can also send an event to have a UNIX signal sent for one or more jobs or box jobs.

#### **NOTE**

To select multiple jobs from the Jobs table at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate job or box jobs in the Jobs table, right-click, and select Send Event from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Send Event from the pop-up menu.
2. Select an event to send from the Event drop-down list.
3. Perform one of the following actions, if applicable:
  - If you selected the Reply option in Step 3, enter the response in the Reply field.
  - If you selected the Send Signal option in Step 3, enter the UNIX signal to send in the Signal field.
  - If you selected the Change Priority option in Step 3, enter the new priority for the job in the Priority field.
  - If you selected the Change Status option in Step 3, select the new status for the job from the Status drop-down list.
  - If you selected the Set Global option in Step 3, enter a name for the global variable to set in the Global Name field.  
Next, enter a value for the variable in the Global Value field.
4. (Optional) Enter a comment describing the event and your reasons for sending it in the Comment field.
5. Enter a value in the Send Priority field.
6. (Optional) Specify a date and time at which to send the selected event.  
The event is not processed until the specified date and time.
7. Click OK.  
When the required criteria for the selected event are met, the event is sent and the Send Event dialog closes.

### **Initiate an Action on an Alert**

You can acknowledge, open, or close alerts from the Alerts table on the Monitoring or Dashboard tabs.

#### **Follow these steps:**

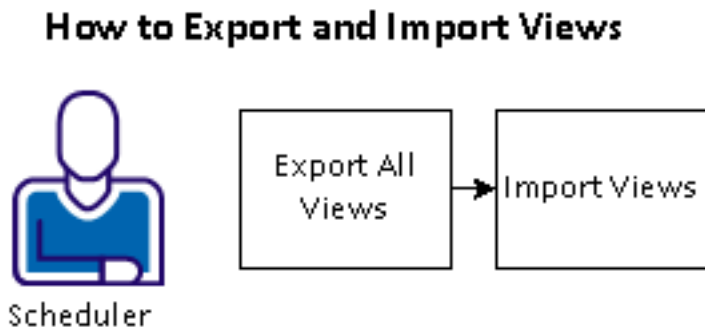
1. Select the appropriate alert or alerts, right-click, and select the appropriate action from the pop-up menu.
2. (Optional) Enter a response for the Acknowledge, Open, or Close actions as necessary, and click OK.  
The action is performed.

## Export and Import Views in the Monitoring Application

This article explains how to export and import views in the Monitoring application. As a workload scheduler, you can copy existing views from one Web UI server to other Web UI servers. You can export the views on one server as a file, and then import that file on the other servers.

This scenario walks you through the process of exporting and importing views from one Web UI server to another Web UI server.

**Figure 79: How to Export and Import Views**



To export all views from one Web UI server and import those views to another Web UI server, follow these steps:

1. [Export all views.](#)
2. [Import views.](#)

### Example: Move Views from One Web UI Server to Another Web UI Server

Suppose that you created several views on the Web UI test server and you want to put those views into production. You do not want to create the views manually. You can export these views to an output file and then use that file to import the views to your Web UI production server.

#### Export All Views

You can export all views from the Views table and can import the output file on other Web UI servers in your enterprise to copy the views to them.

##### **Follow these steps:**

1. Click VMC and click Export.
2. Click Save.  
A browser dialog opens, prompting you for a location in which to save the file.
3. Select the appropriate file location and click OK.  
The export file is saved.

#### Import a View

You can import view definitions using the output file that is exported from other Web UI servers in your enterprise.

##### **Follow these steps:**

1. Click VMC to open the View Management Console and click Import.  
The Import Views dialog opens, prompting for the location of the import text file.
2. Locate the import text file and click Open.
3. (Optional) Edit the view definitions as appropriate.

4. Click Execute.  
Web UI verifies the validity of the view definitions. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the views are saved.

## Manage Views in the Monitoring Application

Manage the Views that are available in the Monitoring application:

### Customize the Views Table

You can customize the columns that display in the Views table by selecting from the following options:

- Name
- Alert Severity
- Job Status
- Server Status
- Jobs
- Box Jobs
- Modified by
- Modified Time
- Status Time
- Active Filters
- Inactive Filters
- Description
- Servers

#### **NOTE**

- You can customize the Views table in the Dashboard tab and the Monitoring tab.
- Only the columns specified in the Views Table Columns property field in Configuration are available for selection.

#### **Follow these steps:**

1. Click  in any column in the Views table.
2. Select  from the drop-down list.
3. Either select the check boxes for the columns you want to display, clear the check boxes for the columns you do not want to display, or do both.
4. Drag the columns to the left or right to order the columns in the table.
5. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### Find a View

You can use the filter section in the Views table to find one or more specific views defined in Web UI.

#### **Follow these steps:**

1. Click the Display Filter icon to display the filter.

2. Perform one or more of the following actions:

- Enter the view name in the Name field.
- Enter the name of the server in the Servers field or select the server from the Servers drop-down list.

**NOTE**

You can enter any of the following in the Name and Servers fields:

- Valid name—Finds the specified view.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all views or servers in the enterprise that contain the partial name. For example, entering n\* in the Name field would find all views that begin with the letter n, and entering \*n would find all views that end with the letter n.
  - Asterisk (\*), percent (%), or blank—Finds all views or servers.
  - Multiple names, separated by a comma.
- Select one or more job-based statuses from the Job Status drop-down list.
  - Select one or more of the following options from the Filter drop-down list:
    - Active -- Finds all views that have an active filter.
    - Inactive -- Finds all views that have an inactive filter.
    - Unbound -- Finds all views that have a filter specifying a server that is no longer configured.
    - None -- Finds all views that do not have a filter defined.

3. Click Go.

The views that meet your search criteria are listed in the Views table.

**NOTE**

- The title changes to indicate that a filter has been applied to the table.
- The filter is persisted through user sessions until you change or clear the filter criteria.

4. (Optional) Click the Display Filter icon to hide the search filter.

### **Edit a View Using VMC**

Editing an existing view is similar to creating a view. You can edit a single view or multiple views to modify the properties, or to add, delete, or modify filtering criteria.

**NOTE**

- To select multiple views at one time, use the Shift or Ctrl key and the mouse.
- You cannot modify a view's filter to change the name of the server to which it points. Delete the current filter first and insert a new filter.

### **Follow these steps:**

1. Select one or more views in the Views table, right-click, and select edit in VMC from the pop-up menu.
2. Edit the view definition as appropriate.
3. Click Execute.  
Web UI verifies the validity of the view definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the view is modified.
4. Close the View Management Console dialog.

### **Edit a View Using the View Editor**

You can use the View Editor dialog to modify the properties of a single view.

**NOTE**

- To edit multiple views at one time, you must select the views to edit, and select Edit in VMC from the pop-up menu.
- The view name cannot be modified.
- You cannot modify a view's filter to change the name of the server to which it points. Right-click the filter, select Delete from the pop-up menu, and insert a new filter.

**Follow these steps:**

1. Right-click a view and select Edit from the pop-up menu.
2. Edit the view definition as appropriate.
3. Click the filter you want to modify in the Filter section. Alternatively, you can right-click a filter in the navigation pane and select Edit from the pop-up menu.
4. Select the Text control option and edit the filter definition in the Job Filter field, or select the Interactive control option to edit the filter definition using the Add Filter Expression table.:
5. Click Save Filter.  
The changes to the selected filter are saved.
6. Click Save.  
The view is modified.

**Copy a View**

You can copy an existing view. The new view has the same commands and attributes as the copied view; however, the information entered in the New Name field for the copied view must be unique. All commands and attributes can be modified after the new view is created.

**Follow these steps:**

1. Right-click a view name and select Copy from the pop-up menu.
2. (Optional) Enter a name for the new view in the New Name field or enter a new suffix in the Select and append suffix field and click Go.
3. Click Next.
4. Edit the view definition as appropriate and click Execute.  
Web UI verifies the validity of the view definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the view is created.
5. Close the View Management Console dialog.

**Export a View**

You can export a specified view and import the output file on other Web UI servers in your enterprise to copy the view to them.

**Follow these steps:**

1. Right-click a view and select Export from the pop-up menu.  
The File Download dialog opens, displaying the default name for the text file.
2. Click Save.
3. Select the appropriate file location and click OK.  
The export file is saved.

**Delete a View**

If you no longer need to monitor a view in your enterprise, you can delete it.



**NOTE**

To select multiple views at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Select one or more views from the Views table, right-click, and select Delete from the pop-up menu.
2. Click Execute.  
Web UI verifies the validity of the view definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the view is deleted.
3. Close the View Management Console dialog.

**Manage Servers in the Monitoring Application**

This article explains how to manage servers in the Monitoring application.

**Customize the Servers Table**

You can customize the columns that display in the Servers table by selecting from the following options:

- Name
- Status
- Type
- Version
- Views
- Status Time
- Status Details

**NOTE**

- You can customize the Servers table in the Dashboard and Monitoring tabs.
- Only the columns specified in the Servers Table Columns property field in Configuration are available for selection.

**Follow these steps:**

1. Click All Servers in the Servers section of the Navigation pane.  
The Servers table is displayed in the right pane.
2. Click  in any column in the Servers table.
3. Select  Columns from the drop-down list.
4. Either select the check boxes for the columns you want to display, clear the check boxes for the columns you do not want to display, or do both.
5. Drag the columns to the left or right to order the columns in the table.
6. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

**Find a Server**

You can use the filter in the Server table to find one or more specific servers.

**Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Either enter the server name in the Server Name field, select the version of the server from the Version drop-down list, or do both.

**NOTE**

You can enter any of the following characters in the Server Name field:

- Valid name—Finds the specified server.
- Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all servers in the enterprise that contain the partial name. For example, entering n\* in the Server Name field would find all servers that begin with the letter n, and entering \*n would find all servers that end with the letter n.
- Asterisk (\*), percent (%), or blank—Finds all servers.
- Multiple names, separated by a comma.

3. Click Go.

The servers that meet your search criteria are listed in the Server table.

**Display Server Details**

You can display server-related jobs or box jobs and the associated alerts on a server from the Dashboard tab or the Monitoring page.

**Follow these steps:**

1. Find the appropriate server.
2. Perform one of the following actions:
  - Click the name of the server in the Servers table on the Monitoring tab.
  - Right-click a server name in the Servers table, select Open from the pop-up menu, and select the Jobs and Alerts tab.
  - Double-click the server name in the Servers table, and select the Jobs and Alerts tab.

The Jobs and Alerts tables open in Monitoring, displaying the jobs, box jobs, and all alerts that are associated with the selected server.

**NOTE**

- When you display server details, the Open Servers folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open servers.
- When job level security is enabled in Configuration, only the jobs to which you have access and the alerts that are associated with those jobs are displayed in the tables.

**Display a Server-Based Summary**

You can display a server-related summary of job and box job status from the Dashboard tab or the Monitoring page.

To display a server-related summary, right-click a server name in the Servers table and select Open from the pop-up menu, or double-click the server name in the Servers table.

The Summary tab is displayed in the right pane.

**NOTE**

- When you display a server summary, the Open Servers folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open servers.
- To switch between servers, you can open multiple servers and click the name of a server in the Navigation pane to display the Summary tab in the right pane.
- To view multiple summaries at a single time, you must open multiple instances of Web UI.

**Display Scheduler Log Details**

You can view the scheduler log for the selected AutoSys Workload Automation instance from the Servers table on the Monitoring page.

**Follow these steps:**

1. Right-click the appropriate server in the Servers table or tree and select Open Scheduler Log from the pop-up menu.
2. Complete the fields in the Search section as appropriate and click Go.
3. (Optional) Click



to open the scheduler log in a separate window.

4. (Optional) Click Download to save the log to a file.  
A dialog opens so you can save or open the log as a text file.

**Send an Event to the AutoSys Workload Automation Server**

You can change the state, status, or priority of all jobs and box jobs associated with a specific AutoSys Workload Automation server, group, or application. For example, you might want to put all jobs associated with an application on hold. You can send an event to the server from the Servers table on the Monitoring page.

**NOTE**

To select multiple servers at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Select one or more servers in the Servers table, right-click, and select Send Server Event from the pop-up menu.
2. Select an event to send from the Event drop-down list.
3. (Optional) Enter a value in the Send Priority field.
4. Select the appropriate option button and enter the group or application with which the relevant jobs are associated in the Group or Application field. At least one must be specified. However, Group and Application cannot be specified when sending a Set Global event.
5. Perform one of the following actions, if applicable:
  - If you selected the Send Signal option in Step 2, enter the UNIX signal to send in the Signal field. Multiple signals can be entered in a comma-separated list.
  - If you selected the Change Priority option in Step 2, enter the new queue priority for the job in the Priority field.
  - If you selected the Change Status option in Step 2, select the new status for the job from the Status drop-down list.
  - If you selected the Set Global option in Step 2, enter a name for the global variable to set in the Global Name field and a value for the variable in the Global Value field.
6. (Optional) Enter a comment describing the event and your reasons for sending it in the Comment field.
7. (Optional) Specify a date and time at which to send the selected event.

**NOTE**

The event will not be processed until the specified date and time.

8. Click OK.  
When the required criteria for the selected event are met, the event is sent and the Send Server Event dialog closes.

---

### **Cancel an Event for a AutoSys Workload Automation Server**

You can cancel an event for all jobs and box jobs associated with a specific AutoSys Workload Automation server, group, or application. For example, you might want to prevent all jobs associated with an application from being taken off hold. You can cancel an event for a server from the Monitoring page.

#### **NOTE**

To select multiple servers at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Select one or more servers in the Servers table, right-click, and select Cancel Server Event from the pop-up menu.
2. Select an event to cancel from the Event drop-down list.

#### **NOTE**

When you select Set Global, you must also specify the name of the global variable for which to cancel the Set Global event in the Global field.

3. Enter the group or application with which the relevant jobs are associated in the Group and Application fields. At least one must be specified. However, Group and Application cannot be specified when canceling a Set Global event.
4. (Optional) Specify the date and time for the event you want to cancel, if that event was scheduled.
5. Click OK.

When the required criteria for the selected event are met, the event is canceled and the Cancel Server Event dialog closes.

### **Disable Filters for All Views Associated with a Server**

You can disable the filters for all views that are associated with a server at any given time. For example, if there is a problem with the server used by the views, you may want to disable all of the view filters and then correct the problem with the server. When the problem is corrected, you can use the Enable Server Filters to enable access to your views.

#### **NOTE**

To select multiple servers at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Select one or more servers in the Servers table, right-click the server name, and select Disable Server Filters from the pop-up menu.
2. Click Confirm.  
The dialog closes and the filters are disabled.

### **Enable Filters for All Views Associated with a Server**

You can enable the filters for all views that are associated with a server at any given time. For example, if a server is removed in Configuration, the filters for the views referencing that server are automatically marked as inactive. When the server is reconfigured in Configuration, you can use the Enable Server Filters to reactivate the views.

#### **NOTE**

- View filters are not activated automatically when a server problem is corrected.
- To select multiple servers at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Select one or more servers from the Servers table, right-click and select Enable Server Filters from the pop-up menu.
2. Click Confirm.  
The dialog closes and the filters are enabled.

## Manage Alerts and Alarms in the Monitoring Application

Manage Alerts and Alarms that are available in the Monitoring application:

### Customize the Alerts Table

You can customize the columns that display in the Alerts table by selecting from the following options:

- Alert
- Type
- Job
- Status
- Severity
- Server
- Alert Policy
- Response
- Create Time
- Status Time
- Comment
- Text
- Eoid
- User
- URL

#### **NOTE**

- You can customize the Alerts table in the Dashboard and Monitoring tabs.
- If you open the Alerts table from the Views table, only the columns specified in the view definition are available for selection.
- If you open the Alerts table from the Servers table, only the columns specified in the Alerts Table Columns property field in Configuration are available for selection.
- Additionally, you can select the columns displayed in the Alerts table and order them using the Alert Columns selection pane in the View Editor.

#### **Follow these steps:**

1. Open the Alerts table in the right pane.
2. Click  in any column in the Alerts table.
3. Select  from the drop-down list.
4. Either select the check boxes for the columns you want to display, clear the check boxes for the columns you do not want to display, or do both.
5. Drag the columns to the left or right to order the columns in the table.
6. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### Find an Alert

You can use the filter section in the Alerts table to find one or more specific alerts.

**Follow these steps:**

1. Click



to display the filter.

2. Perform one or more of the following actions:

- Select an Alarm from the Alarm drop-down list.
- Enter the name of the server in the Servers field or select the server from the Servers drop-down list.
- Enter the name of one or more jobs in the Jobs field.

**NOTE**

You can enter any of the following in the Servers and Jobs fields:

- Select one or more severities from the Severity drop-down list.
- Select one or more alarms from the Alarms drop-down list.
- Select one or more statuses from the Status drop-down list.
- Select one or more job statuses from the Job Status drop-down list.
- Enter a start time in the Start Time field or select a time from the Start Time drop-down list.
- Enter an end time in the End Time field or select a time from the End Time drop-down list.
- Enter a number of days from the current date back in the Past N Days in which an alert occurred.

3. Click Go.

The alerts that meet your search criteria are listed in the Alerts table.

**NOTE**

- The title changes to indicate that a filter has been applied to the table.
- The filter is persisted through user sessions until you change or clear the filter criteria.

4. (Optional) Click



to hide the search filter.

**Display Alert Details**

You can display a table of server-related alerts or job-related alerts, depending on whether the Alerts table is opened from the Servers table or from the Views table.

To display the details for an alert, find the appropriate alert and click



The information is displayed directly below the selected alert.

**NOTE**

When job level security is enabled in Configuration, only the alerts associated with the jobs to which you have access are displayed in the Alerts table.

**Customize the Alert Policies Table**

You can customize the columns that display in the Alert Policies table by selecting from the following options:

**NOTE**

- In releases till 11.4 SP5, the JOBFailure alarm specified whether to post an alert when a job completes with FAILURE or TERMINATED status. After r11.4 SP5, you can specify a Alert policy individually for

JOBTERMINATED or JOBFAILURE. This helps you distinguish the alarms and alerts based on the FAILURE and TERMINATED job statuses.

- If you created a Alert policy for JOBFAILURE alarm in previous releases, and upgrade to the current release, JOBTERMINATED alarm will also be added to the Alert policy.

- Name
- Alarm Type
- Job Status
- Severity
- Job Name
- Box Name
- Server

#### Follow these steps:

1. Open the Alert Policies table in the right pane.
2. Click



in any column in the Alert Policies table.

3. Select



from the drop-down list.

4. Either select the check boxes for the columns you want to display, clear the check boxes for the columns you do not want to display, or do both.
5. Drag the columns to the left or right to order the columns in the table.
6. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

#### Find an Alert Policy

You can use the filter in the Alert Policies table on the Monitoring tab to find one or more specific alert policies.

#### Follow these steps:

1. Click



to display the filter in the Alert Policies table.

2. Perform one or more of the following actions:
  - Specify one or more names for the alert in the Name field.
  - Specify the name for one or more servers in the Servers field or select the servers from the Servers drop-down list.

#### NOTE

You can enter any of the following in the Name or Servers field:

- Select one or more severity levels for the alert from the Severity drop-down list.

3. Click Go.

The policies that meet your search criteria are listed in the Alert Policies table.

#### NOTE

The filter is persisted through user sessions until you change or clear the filter criteria.

## **Edit an Alert Policy**

Editing an existing alert policy is similar to creating a policy. You can use the VMC to modify the properties of one or more alert policies, or you can edit a single policy using the Alert Policy dialog.

### **NOTE**

To select multiple policies in the Alert Policy table at one time, use the Shift or Ctrl key and the mouse.

### **Follow these steps:**

1. Perform one of the following actions:
  - Select a one or more policies in the Alert Policy table, right-click, and select Edit in VMC from the pop-up menu.
  - Right-click a policy and select Edit from the pop-up menu to modify the policy using the Alert Policy dialog.
2. Edit the alert policy definition as appropriate.
3. Click Execute in the View Management Console to modify the policy, and click Close. Click Save in the Alert Policy dialog to save the changes to the policy definition.  
The changes to the alert policy definition are saved.

## **Delete an Alert Policy**

If you no longer need the alert policy, you can delete it.

### **NOTE**

To select multiple policies in the Alert Policy table at one time, use the Shift or Ctrl key and the mouse.

### **Follow these steps:**

1. Select one or more alert policies in the Alert Policy table, right-click, and select Delete from the pop-up menu.
2. Click Execute.  
Web UI verifies the validity of the command. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the command is executed and the alert policy is deleted.
3. Close the View Management Console dialog.  
The View Management Console dialog closes.

## **Manage Jobs in Monitoring Help**

You customize and manage jobs and events to use the Web UI Monitoring and Reporting functionality. Use the following procedures to manage jobs and events:

### **Customize the Jobs Table**

Select the following options to customize the columns that display in the Jobs table:



- Agent Status
- Agent Status Long
- Application
- Box Name
- Description
- End Time
- Exit Code
- Group
- Instance
- Job Type
- Joid
- Machine
- Name
- Next Start
- Owner
- Run Machine
- Server
- Start Time
- Status
- Status Time
- TimeZone

**NOTE**

- If you open the Jobs table from the Views table, only the columns specified in the view definition are available for selection.
- If you open the Jobs table from the Servers table, only the columns specified in the Jobs Table Columns property field in the Monitoring Preferences section in Configuration are available for selection.
- Additionally, you can select the columns displayed in the view and order them using the Job Columns selection pane in the View Editor.

**Follow these steps:**

1. Click



in any column in the Jobs table.

2. Select



Columns from the drop-down list.

3. Either select the check boxes for the columns you want to display, clear the check boxes for the columns you do not want to display, or do both.
4. Drag the columns to the left or right to order the columns in the table.
5. Select the number of rows to display in the table from the Show drop-down list.

The changes are saved.

**Find a Job from the Monitoring Tab**

You can use the filter in the Jobs table to find one or more specific jobs.

**Follow these steps:**

1. Click  to display the filter.
2. Perform one or more of the following actions:
  - Enter the job name in the Name field.
  - Select a job status from the Job Status drop-down list.
  - Select a job type from the Job Type drop-down list.
  - Select one or more servers from the Servers drop-down list.
  - Enter the application name in the Application field.
  - Enter the group name in the Group field.

**NOTE**

You can enter any of the following characters in the Name, Servers, Application, or Group field:

- Valid name—Finds the specified job, server, application, or group.
- Asterisk ( \* ), percent (%), or question mark ( ? ) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all jobs in the enterprise that contain the partial name. For example, entering n\* in the Job Name field would find all jobs that begin with the letter n, and entering \*n would find all jobs that end with the letter n.
- Asterisk ( \* ), percent (%), or blank—Finds all jobs.
- Multiple names, separated by a comma.

3. Click Go.  
The jobs that meet your search criteria are listed in the Jobs table.

**NOTE**

- The title changes to indicate that a filter has been applied to the table.
- The filter is persisted through user sessions until you change or clear the filter criteria.

4. (Optional) Click  to hide the search filter.

**Find a Job or Box Job in the Flow**

You can find a specific job or box job in the current view from the Flow section.

**Follow these steps:**

1. Click  in the Flow toolbar.
2. Enter the value to match for the job or box job name in the Name field.

**NOTE**

This value is case-sensitive and can contain wildcard characters. You can use an asterisk ( \* ) or a percentage sign (%) as a wildcard to represent multiple characters, or use a question mark ( ? ) as a wildcard to represent any single character. You can use these wildcard characters anywhere in the name. For example, specifying "" results in a filter on any name. Similarly, specifying "JOB%" results in a filter on any name that begins with the string JOB (JOB1, JOB15, JOBS, and so on). Specifying "?JOB" results in a filter on any name that begins with any one character and ends with the string JOB (1JOB, 2JOB, AJOB, and so on). You can also use an exclamation ( ! ) with a partial name to find results that do not match the entered criteria. For example, entering !n specifies to find the results that do not begin with n. Multiple names can be entered in a comma-separated list.

3. Press the Enter key.  
Web UI searches for the specified jobs, highlights the first job found in the flow, and displays an informational message letting you know how many jobs were found. If no jobs are found, a message is returned.

4. (Optional) Click



Previous or Next  to cycle through the list of jobs matching the criteria, bringing them into focus and highlighting them.

5. (Optional) Click Select All



to highlight all the jobs in the flow that match the search criteria.

6. Click  when you finish searching for jobs.

The Find toolbar is removed from the view.

### **Find a Job in the Flow by Name**

The Monitoring flow toolbar lets you search the job flow by job name. Use this option to locate a problem job and take appropriate corrective action.

#### **Follow these steps:**

1. Enter the value to match for the job or box job name in the Search field.

#### **NOTE**

This value is case-sensitive and can contain wildcard characters. You can use an asterisk ( \* ) or a percentage symbol ( % ) as a wildcard to represent multiple characters, or use a question mark ( ? ) as a wildcard to represent any single character. For example, you can enter \*JOB which returns all jobs that end with JOB. If no job matches this criteria, then nothing is returned. Alternatively, you can enter JOB in the Search field, and all jobs that contain JOB in the name are returned.

2. Select the appropriate job.  
The job is selected in the flow.

#### **NOTE**

- Alternatively, if you do not know the name of the job, you can search for the job using the Select by status drop-down to find a job by status.
- The Search Results section contains a maximum of 100 jobs. If more results are available, you need to refine the search to limit the number of results.

### **Send an Event to a AutoSys Workload Automation Job**

You can send an event of the appropriate type to change the state, status, or priority for one or more AutoSys Workload Automation jobs or box jobs. You can also send an event to have a UNIX signal sent for one or more jobs or box jobs.

#### **NOTE**

To select multiple jobs from the Jobs table at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Perform one of the following actions:

- Select the appropriate job or box jobs in the Jobs table, right-click, and select Send Event from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Send Event from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Select an event to send from the Event drop-down list.
  3. Perform one of the following actions, if applicable:
    - If you selected the Reply option in Step 3, enter the response in the Reply field.
    - If you selected the Send Signal option in Step 3, enter the UNIX signal to send in the Signal field.
    - If you selected the Change Priority option in Step 3, enter the new priority for the job in the Priority field.
    - If you selected the Change Status option in Step 3, select the new status for the job from the Status drop-down list.
    - If you selected the Set Global option in Step 3, enter a name for the global variable to set in the Global Name field. Next, enter a value for the variable in the Global Value field.
  4. (Optional) Enter a comment describing the event and your reasons for sending it in the Comment field.
  5. Enter a value in the Send Priority field.
  6. (Optional) Specify a date and time at which to send the selected event.  
The event is not processed until the specified date and time.
  7. Click OK.  
When the required criteria for the selected event are met, the event is sent and the Send Event dialog closes.

### **Cancel an Event for a AutoSys Workload Automation Job**

You can cancel an event associated with one or more jobs or box jobs, for example, to prevent a job from being taken off hold.

#### **NOTE**

To select multiple jobs from the Jobs table at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click , and select Cancel Event from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Cancel Event from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Select the event to cancel from the Event drop-down list.
3. (Optional) Specify the date and time for the event you want to cancel, if that event was scheduled.
4. Click OK.  
When the required criteria for the selected event are met, the event is canceled and the Cancel Event dialog closes.

### **Start a Job**

You can use the Start command to send a STARTJOB event to start the selected job or top-level box job when its starting conditions are satisfied. The STARTJOB event ignores date and time conditions, but it enforces all other starting conditions such as job dependencies.

#### **NOTE**

- You cannot send a STARTJOB event to a job that is contained in a box job. Jobs in box jobs inherit the starting conditions of these box jobs; as a result, they start when the box job starts. If the job is contained in a box job, the STARTJOB event has no effect.
- To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

#### **Follow these steps:**

1. Perform one of the following actions:

- Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Start from the pop-up menu
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Start from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is scheduled to start when its starting conditions, other than date and time, are satisfied.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.  
For more information, see [Find a Job from the Monitoring Tab](#).

**Force Start a Job**

You can use the Force Start command to send a FORCE\_STARTJOB event to the selected job or box job. The manually-generated FORCE\_STARTJOB event starts the specified job, regardless of whether its starting conditions are satisfied. Because you can use this command to start multiple instances of the same job, we recommend that you use the FORCE\_STARTJOB event only in extreme situations.

**NOTE**

- If you assign a resource as a dependency to a job, and that resource is not available, a FORCE\_STARTJOB event will not run.
- To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Force Start from the pop-up menu
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Force Start from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is started.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

**Put a Job on Hold**

You can use the On Hold command to send a JOB\_ON\_HOLD event to the selected AutoSys Workload Automation job or box job. Jobs that are on hold do not start, temporarily preventing the remainder of the job flow from executing. If the job's status is STARTING or RUNNING, the On Hold command has no effect.

**NOTE**

- A box job cannot successfully complete if a job in it is on hold.
- To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select On Hold from the pop-up menu
  - Select the appropriate jobs or box jobs in the flow, right-click, and select On Hold from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The job is put on hold.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

**Take a Job off Hold**

You can use the Off Hold command to send a JOB\_OFF\_HOLD event to the selected job or box job. The JOB\_OFF\_HOLD event instructs the scheduling manager to remove the specified job from ON\_HOLD status. If the job's starting conditions are met, the job starts.

**NOTE**

To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Off Hold from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Off Hold from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is taken off hold.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

For more information, see [Find a Job from the Monitoring Tab](#).

**Put a Job on Ice**

You can use the On Ice command to send a JOB\_ON\_ICE event to the selected job or box job, changing the job's status to ON\_ICE. Putting a job on ice is the functional equivalent of deactivating the job definition. If the job's status is STARTING or RUNNING, the On Ice command has no effect.

**NOTE**

To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select On Ice from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select On Ice from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is put on ice.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

For more information, see [Find a Job from the Monitoring Tab](#).

**Take a Job off Ice**

You can use the Off Ice command to send a JOB\_OFF\_ICE event to the selected job or box job. The JOB\_OFF\_ICE event instructs the scheduling manager to remove the specified job from ON\_ICE status. Jobs that are taken off ice do not start until the next time their starting conditions are met. If the specified job is in a box job with a RUNNING status, the scheduling manager attempts to start it, conditions permitting.

**NOTE**

To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Off Ice from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Off Ice from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is taken off ice.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

For more information, see [Find a Job from the Monitoring Tab](#).

**Take a Job off No Execution**

You can use the Off No Execution command to send an OFF\_NOEXEC event to the selected job or box job. The OFF\_NOEXEC event instructs the scheduling manager to remove the specified job from ON\_NOEXEC status. Jobs that are taken off no execution status start the next time their starting conditions are met.

**NOTE**

To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Off No Execution from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Off No Execution from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is taken off no execution.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

For more information, see [Find a Job from the Monitoring Tab](#).

**Kill a Job**

You can use the Kill command to send a KILLJOB event to the selected job or box job. The manually-generated KILLJOB event instructs the scheduling manager to kill the specified job if it is RUNNING and change its status to TERMINATED. If the job's status is not RUNNING, the Kill command has no effect.

**NOTE**

- If the selected job is a box job, the scheduling manager changes the box job's status to TERMINATED and (if configured to do so) also kills the jobs it contains. The result of this command depends on the type of job selected and the type of server (UNIX or Windows) on which the job is running.
- This event is not supported with all job types.
- To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Kill from the pop-up menu.
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Kill from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The selected job is terminated.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

**Release a Resource**

You can define renewable resources using the Free attribute with one of the following options: Y (free resource on success only), A (free resource unconditionally always), or N (do not free resources).

If you assign a resource as a dependency to a job with the attribute Free=N, and that job is terminated, the resource continues to be held by that job. For this resource to be available to other dependent jobs waiting to run, you must use the RELEASE\_RESOURCE sendevent command to free the resource.

**NOTE**

To select multiple jobs in the Jobs table at one time, use the Shift or Ctrl key and the mouse.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the appropriate jobs or box jobs in the Jobs table, right-click, and select Release Resource from the pop-up menu
  - Select the appropriate jobs or box jobs in the flow, right-click, and select Release Resource from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm. The resources for the selected job are released.

**NOTE**

- The Free attribute is only valid for renewable resources.
- The Release Resources action is not available for box jobs.
- Alternatively, you can use the Send Event or Send Server Event dialog to send this event. For more information, see [Find a Job from the Monitoring Tab](#).

**Send a Reply to an i5/OS Job**

To send a reply to an i5/OS agent notification requesting information required for an i5/OS job run to continue, you can use the Reply command.

**Follow these steps:**

1. Perform one of the following actions:
  - Select the waiting i5/OS jobs in the Jobs table, right-click, and select Reply from the pop-up menu
  - Select the waiting i5/OS jobs in the flow, right-click, and select Reply from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The reply is sent.

**NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.



### **Resume Suspended Hadoop Oozie Job**

To resume a suspended Oozie job you can use the Resume command.

#### **Follow these steps:**

1. Perform one of the following actions:
  - Select the suspended Oozie jobs in the Jobs table, right-click, and select Resume from the pop-up menu
  - Select the suspended Oozie jobs in the flow, right-click, and select Resume from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The job is resumed.

#### **NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

### **Suspend a Running Hadoop Oozie Job**

To suspend a running Oozie job you can use the Suspend command.

#### **Follow these steps:**

1. Perform one of the following actions:
  - Select the running Oozie jobs in the Jobs table, right-click, and select Suspend from the pop-up menu
  - Select the running Oozie jobs in the flow, right-click, and select Suspend from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
2. Click Yes to confirm.  
The job is suspended.

#### **NOTE**

Alternatively, you can use the Send Event or Send Server Event dialog to send this event.

### **Display the Jobs Contained in a Box Job**

You can quickly display and manage all of the jobs contained in a box job from the Flow. To display the jobs contained in a box job, find the box job in the flow and expand it.

### **Display Job Dependencies External to the View**

You can display a list of all the jobs that have dependencies to jobs or box jobs in a selected view, but that are not contained in the view definition.

#### **Follow these steps:**

1. Right-click the view name in the Views table and select Show External Dependencies from the pop-up menu.
2. (Optional) Select Successor from the Show Dependency Type drop-down list.  
The dependencies for the jobs and box jobs in the view are displayed in the table.

### **Edit the Properties of a Job**

If the properties of a job are incorrect, you can edit the job from Monitoring.

#### **Follow these steps:**

1. Perform one of the following actions:

- Right-click a job or box job in the Jobs table on the Monitoring page and select Edit from the pop-up menu.
  - Right-click a job in the flow and select Edit from the pop-up menu.
  - Select the appropriate jobs or box jobs in the tree view, right-click, and select Send Event from the pop-up menu.
  - Select the job and press Enter.
2. Edit the job definition as necessary, and click Commit.  
The changes to the job definition are saved.
  3. Close Quick Edit when you are finished.

**NOTE**

For more information about editing job definitions in Quick Edit, see the [Web UI Quick Edit Application](#).

**Save the Job Flow Image**

You can download the job flow as a .png image or .pdf document and save it to a file.

**Follow these steps:**

1. Click  on the toolbar and click Save.
2. Select a folder, enter a file name in the Save As dialog, and click Save.  
The image is saved.

**Print the Job Flow Image**

The Flow toolbar lets you zoom in on and print portions of the flow. This is helpful with complex flows because it lets you examine each portion of the flow in detail.

**Follow these steps:**

1. Click  on the toolbar.
2. Select File, Print from the menu options.
3. Select a printer, modify its settings if necessary, and click Print.

**NOTE**

- You may need to select the Print background colors and images option in your internet browser for the page to print correctly.
- If your browser is blocking popup windows, the print dialog may not appear.

The job flow image is printed.

**Navigate the Tree for Views**

The structure of the monitoring tab provides for easy navigation of jobs, machines, and alerts. The options in the tab help you to monitor job status and also allow for quick access to the job flow. You can log in to the Web UI application and click the Monitoring tab to view the following sub menu options:

- – Jobs
- Machines
- Alerts

**NOTE**

The Jobs view is the default monitoring page. If you were viewing or editing a job in Server Views before moving from the Monitoring tab, the next time you open the Monitoring tab, it opens with Server Views.

The Monitoring page is unique to each user and defaults to the previously used sub menu option.

- Select the required submenu option and you can see the respective views for the selected submenu option. The pan window opens with a split and with a tree structure on one side. You can monitor the jobs that are based on the views defined. The left side panel displays the following views:
  - My Views - Displays the jobs that are created in the current version and the user views as the first-level child.
  - Shared Views - Displays the existing jobs that were created in previous releases and the shared views as the first-level child.
  - Server Views - Displays the Web UI Servers and the jobs inside them and the server views as first-level child.
- Job names in the tree view are color coded based on the job status and you can perform various actions on the jobs. The following are the different colors against various jobs' status:
  - Red - Failure/Terminated
  - Blue - Running
  - Green - Success
- Expand the tree which may have further jobs and box jobs that can further contain child jobs, based on the filter criteria. You can now view more jobs in the tree. The link appears for all the children if it exceeds the configured value (default value 25). Navigate to the "Display next 100 jobs..." link using the Down arrow key and then press ENTER. Expand any view or box in the tree structure to see the number of jobs in different colors that are based on the status of the jobs.
- You can select multiple objects in any one of the tree views, namely My Views, Shared Views, and Server Views.
  - Hold the Ctrl key in the keyboard to select the objects one by one.
- Right-click on any item in the tree structure to see the relevant context menu for selected item.
- Click on the autorefresh button on top of the tree structure.
- You can enable and disable the autorefresh.
- You can also see the "Refresh Now" link and the Last Updated date/time above the tree.
- Click on any object in the tree structure. The selected object opens on the right side and also expands the object.
- Click on any object in the tree structure and use the following keys from the keyboard for navigation across the tree structure:
  - Up Arrow  
Navigate upwards the tree
  - Down Arrow  
Navigate downwards the tree
  - Left Arrow  
Collapse the expanded node
  - Right Arrow  
Expand the node
  - ENTER  
Refresh the table and open the selected object

## Table for Views

This section contains information on the Views table and the following topics:

You can select any one of the views, My Views, Shared Views, or Server Views, in the left pane and the associated views or servers are displayed in a table.

- Click **My Views** to display the associated views created by the user in the table. You can click a view under My Views to display the associated jobs table.
- Click **Shared Views** to display the associated views shared across the organization in the table. You can click a view under Shared Views to display the associated jobs table.
- Click **Server Views** to display the associated servers configured in Web UI in the table. You can click a view under Server Views to display the associated jobs table.

#### NOTE

You can customize the table columns by adding new column headings or clearing the selection of the default columns, and the table displays the customized column headings.

For more information about modifying the table, see the [Customize the Table](#) section.

If more than one page is displayed, you can use the arrows at the bottom of the page to move back and forth across the pages.

#### Follow these steps:

1. Log in to Web UI, click Monitor.  
The monitoring default view opens.
2. Click My Views, Shared Views, or Server Views.

##### My Views

For My Views, the associated views are displayed in the table. You can click a view under My Views to display the associated jobs list. By default, the following columns are displayed:

- View Name
- Job Status
- Server Status
- Alert Severity Status
- Jobs
- Box Jobs
- Status Time

##### Shared Views

For Shared Views, the associated views are displayed in the table. You can click a view under Shared Views to display the associated jobs list. By default, the following columns are displayed:

- – View Name
- Job Status
- Server Status
- Alert Severity Status
- Jobs
- Box Jobs
- Status Time

##### Server Views

For Server Views, the associated servers are displayed in the table. You can click a server under Server Views to display the associated jobs list. By default, the following columns are displayed:

- – Server Name
- Status
- Status Time
- Type
- Views

The following options are also available for your use:

- Click the job link in the table to open or edit the job details.
- Select jobs across multiple pages and the Selected Count is displayed at the bottom of the page.

### **Table for Jobs**

You can click the required view or server in the tree in the left panel or in the respective Views table to display the associated jobs list in a table.

The Jobs table contains the following default columns. You can customize the table columns by adding new column headings or clearing the selection of the default columns, and the Jobs table displays the customized column headings.

- Job Name
- Job Type
- Status
- Server
- Run Machine
- Status Time
- Duration (D HH:MM:SS)

### **Customize the Table**

You can customize the table using the following options:

- Sort all the table columns under various headings.
- Modify the displayed number of rows. The table displays 15 rows per page, by default. You can modify the number of rows, by selecting from the drop-down list (25, 50, 100, and 500) at the bottom of the page or by manually entering the required number of rows. For the number of rows, minimum value is 15 and any value above 500 is limited to 500.
- Select the table columns to be displayed by using the dynamic selection of columns (at the top right corner).

## **Create View**

You can now create a view definition to monitor the jobs from the Monitoring tab.

### **Follow these steps:**

1. Log in to Web UI, click Monitor.  
The monitoring default view opens.
2. Right-click My View, click Create View.  
The Create View window opens with all the default attributes.
3. Complete the following information:
  - a. (Mandatory) Enter Name.  
**Limit:** 80 characters

#### **NOTE**

Valid characters are a-z, A-Z, 0-9, hash (#), dot (.), underscore (\_), and hyphen (-). View names cannot include any of the following: \$, %, \*, ?, or @.

- b. Enter Description.  
**Limit:** 255 characters

**NOTE**

Valid characters are a-z, A-Z, 0-9, comma(,), hash(#), dot(.), underscore (\_), hyphen (-), question mark (?), ampersand (&), asterisk (\*), and percent(%).

- c. Select Job Columns from the drop-down list in the order in which to display in the Jobs table.  
The options are as follows:  
Name, Job Type, Status, Server, Run Machine, Status Time, Agent Status, End Time, Machine, Next Start Time, Start Time, Agent Status Long, Application, Box Name, Description, Group, Instance, Joid, Owner, Timezone, and Exit Code.
  - d. Select Alert Columns from the drop-down list in the order in which to display in the Alerts table.  
The options are as follows:  
Alert, Severity, Status, Job, Server, Create Time, Status Time, Type, Response, Text, User, Alert Policy, URL, Comment, and Eoid.  
You can select the Add All checkbox to select all the Jobs columns or Alerts columns.
4. Click FILTER DETAILS.
    - a. Select the Active checkbox.

**NOTE**

- To create a usable view, at least one filter must be defined.
- You can only have one active filter per server. Use the Active field to indicate which filters you want to be active at one time.

- b. (Mandatory) Select the server for the view from the Server drop-down list.  
The filter is applied to the selected server.
- c. Select the alert policy to attach to the view from the Alert Policy drop-down list.  
The selected policies are associated with the filter.

**NOTE**

Select a server to enable the Expressions tab.

5. Click EXPRESSIONS, Add Expression and enter filter criteria to filter the jobs for the selected server in FILTER DETAILS.  
By default, the first filter expression Logic is disabled.
  - a. Select the Interactive option to define the filter definition using Add Expression, and then do the following:
    - a. Select AND or OR from the Logic drop-down list to create a complex filter definition and enter additional filter criteria in the appropriate fields.  
**Default:** AND
    - b. Select Attribute from the drop-down.  
**Default:** job\_name
    - c. Select Operator from the drop-down.  
**Default:** LIKE
    - d. Enter the appropriate criteria in the Value field. You can also select the value from the drop-down list or start typing to prompt options.  
For example, if you select 'box' in the Attribute column, you can either select the value from the drop-down list or type Box\_job in the Value field to list box job names that contain the term Box\_job.
  - b. Select the Text control option and enter the filter definition in the text box.
6. Click SAVE FILTER.  
The filter is saved and added to the Filters table.
7. (Optional) Repeat steps 4-7 to add additional filters. You can also add new filters using the *Create new* link in the Filters table.
8. Click Save.

The view is created.

### **Add an Expression**

You can add an expression in the Create View window.

#### **Follow these steps:**

1. Click EXPRESSIONS, Add Expression.
2. Enter the following fields from the drop-down lists as required:
  - a. Select AND or OR from the Logic drop-down list to create a complex filter definition and enter additional filter criteria in the appropriate fields.
  - b. Select Attribute.  
The options are: job\_name, box, box\_name, status, job\_type, application, group, machine, owner, Timezone, started\_in\_past, and run\_machine.
  - c. Select Operator.  
Displays all or some of the following options that are based on the selected attribute: LIKE, NOT Like, =, and !=.
  - d. Enter the appropriate criteria in the Value field. You can also select the value from the drop-down list or start typing to prompt options.  
For example, if you select 'box' in the Attribute column, you can either select the value from the drop-down list or type Box\_job in the Value field to list box job names that contain the term Box\_job.
3. Click SAVE FILTER.  
The filter is saved and added to the Filters table.

### **Edit, Delete Saved Filters**

Saved Filters are displayed in the Filters table in the Create View page with the Edit and Delete links in the Actions column.

Click the Delete link to delete the filter.

#### **Follow these steps:**

1. Click Edit.  
The form of that particular filter opens with the fields that are filled in with appropriate values. The server field is non-editable.
2. (Optional) Modify the Alerts to associate alert policies and click Continue.
3. (Optional) Modify the criteria as required and click Save Filter.  
The modified values are applied to the filter.
4. Click Save.  
The view is saved.

## **Web UI Reporting Application**

The Reporting application, accessed by clicking the Reporting tab from the Web UI Dashboard, lets you create, view, modify, and run reports on multiple servers.

### **NOTE**

Access to Reporting is granted or denied based on CA EEM policies. The tasks available are based on the rights granted to you by the security administrator.

The Reporting application uses *collectors* to obtain job run and server statistical data from the AutoSys Workload Automation scheduling manager. It then reformats that data and stores it in the reporting *data warehouse*, a central and consolidated database repository of that collected data. The data is stored in a format that facilitates flexible reporting and high-performance retrieval. You can run any of a set of predefined reports against selected information stored in the data

warehouse. The Reporting application offers you the ability to run reports on demand and save them in different formats based on your needs.

The main page of Reporting is comprised of two panes:

- The left pane provides navigation capability and displays a list of reporting templates and reports that you can manage.
- The right pane displays selected reports and the Reporting Dashboard. You can use the Reporting Dashboard to create reports and view the status of the collectors.

You can click



in the navigation bar to hide the left pane, expanding the sections in the right pane.

To display only those templates and reports that match your filtering criteria, enter the criteria on which to filter in the Filter field. For example, you can enter Stat in the Filter field. Only those templates and reports containing the word Statistics in the name are displayed. Clicking



clears your entry and displays all available templates and reports.

## Reporting Considerations

The following list displays important considerations for using the Reporting application:

- The data warehouse can reside on the Web UI server or a separate, independent server.
- Collectors can retrieve data from various AutoSys Workload Automation servers (r11.3 and later releases) and store it all in a single data warehouse.
- Only the previous 48 hours of data is collected and imported into the data warehouse the first time you connect to a server. This is also true for server connections that occur 48 hours or more since the last connection.

## Predefined Server Statistical and Job Run Activity Reports

This article describes predefined server statistical and job run activity reports.

The Reporting application installs a set of predefined reports. These reports include server statistical reports and job run activity reports. The data for these reports is periodically collected and stored in the statistical and job run detail tables in the data warehouse.

### NOTE

- On CA Workload Automation AE r11.3, the autoaggr utility must be run periodically to calculate the server statistics that are used for AutoSys Workload Automation reports. The command job created as a result of running the following jil script calculates the AutoSys Workload Automation server statistics with the necessary resolution:
 

```
insert_job: do_aggr
command: autoaggr -h
machine: localhost
date_conditions: y
days_of_week: all
start_mins: 5
```
- On CA Workload Automation AE Release 11.3.5 and later, aggregation is controlled by a property defined for each AutoSys Workload Automation scheduler.

The following report types are included with the Reporting application:



## • **Server Statistical Reports**

Statistical reports are based on job management activity for one or more scheduling servers. The data for the statistical reports is accumulated by the scheduling servers and then collected on a periodic basis. The collected data is stored in statistical tables in the data warehouse and is used to generate the statistical reports.

## • **Job Run Activity Reports**

Job run activity reports are based on scheduling manager logs and provide detailed reporting down to the job detail level. The data for these reports is periodically collected from the scheduling servers and stored in the job run detail tables in the data warehouse.

The following predefined reports are available:

- Job Run Reports
  - [Job Run Summary](#)
  - [Job Run Details](#)
  - [Job Run Throughput](#)
  - [Job Run Event Summary](#)
  - [Job Run Event Status](#)
- Server Statistical Reports
  - [Server Statistics by Day](#)
  - [Job Statistics](#)
  - Job Status Reports
    - [Job Status Success/Failure](#)
    - [Job Status Starting/Running](#)
    - [Job Status Inactive/Terminated](#)
  - Job Run Reports
    - [Job Run Failure/Total Runs](#)
    - [Job Run Force Start/Restart](#)
    - [Job Run Event Statistics](#)

### **Job Run Summary Report**

The Job Run Summary report displays the job run summary and job run details for a specified period in table and graphical formats. The table displays all jobs per server, and the number of job runs and average run delay times for each job. The graph displays the total number of job runs for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Run Details Report**

The Job Run Details report displays the job run details per server for each job run for a specified period in table format. Duration time is displayed in seconds.

The report inputs are server, job (name pattern), and date range. The results displayed are based on the inputs.

### **Job Run Throughput Report**

The Job Run Throughput report displays the remote agent usage statistics for a specified period in table and graphical formats. The table displays the number of job runs per server and run machine, and the total number of job runs per server and run machine. The graph displays a summary of the number of times a run machine was used to execute jobs.

The report inputs are server, machine, and date range. The results displayed are based on the inputs.

---

### **Job Run Event Summary Report**

The Job Run Event Summary report displays the job run event summary and details for a specified period in table and graphical formats. The table displays all job run events per server, and the number of job run events for each job. The graph displays the total number of job run events for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Run Event Status Report**

The Job Run Event Status report displays the job run event statuses for a specified period in table and graphical formats. The table displays the number of job run events by status per server, and the total number of job run events per server. The graph displays the total number of job run events by status for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Server Statistics by Day Report**

The Server Statistics by Day report displays server level statistics aggregated by day for a specified period in table format.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Statistics Report**

The Job Statistics report displays job level statistics aggregated by server and job name in table format.

The report inputs are server, job (name pattern), and date range. The results displayed are based on the inputs.

### **Job Run Event Statistics Report**

The Job Run Event Statistics report displays the number of job run events for a server for a specified period in table and graphical formats. The table displays the number of job run events per server, the average event latency per server, and the total number of job run events and average event latency per server. The graph displays the total number of job run events for each server and the total number of job runs for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Status Success/Failure Report**

The Job Status Success/Failure report displays the number of success and failure job change status events for a server for a specified period in table and graphical formats. The table displays the number of job change status events per server with success and failure statuses, and the total number of job change status events with success and failure statuses. The graph displays the total number of job change status events with success and failure statuses for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Status Starting/Running Report**

The Job Status Starting/Running report displays the number of starting and running job change status events for a server for a specified period in table and graphical formats. The table displays the number of job change status events per server with starting and running statuses, and the total number of job change status events with starting and running statuses. The graph displays the total number of job change status events with starting and running statuses for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Status Inactive/Terminated Report**

The Job Status Inactive/Terminated report displays the number of inactive and terminated job change status events for a server for a specified period in table and graphical formats. The table displays the number of job change status

events per server with inactive and terminated statuses, and the total number of job change status events with inactive and terminated statuses. The graph displays the total number of job change status events with inactive and terminated statuses for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Run Failure/Total Runs Report**

The Job Run Failure/Total Runs report displays job run totals and failures or terminations for a server for a specified period in table and graphical formats. The table displays the number of job runs and job failures or terminations per server, and the total number of job runs and job failures or terminations. The graph displays the total number of job runs and failures or terminations for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

### **Job Run Force Start/Restart Report**

The Job Run Force Start/Restart report displays the number of force start and restart job runs for a server for a specified period in table and graphical formats. The table displays the number of force start and restart job runs per server, the total number of job runs per server, and the total number of job force start and restart job runs. The graph displays the total number of job force start and restart job runs for each server.

The report inputs are server and date range. The results displayed are based on the inputs.

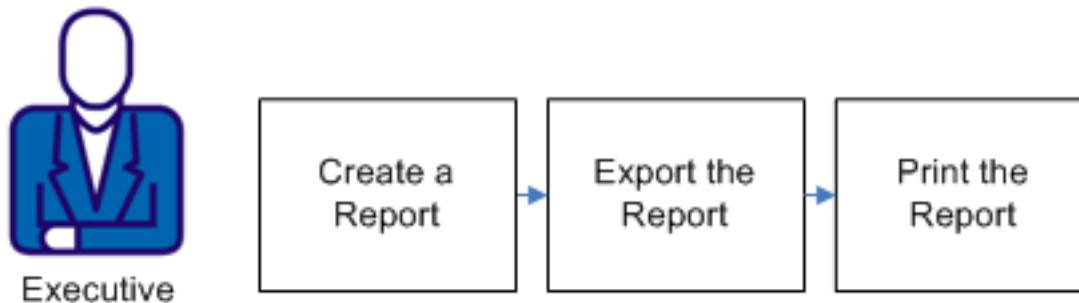
## **Generate a Report in the Reporting Application**

This article explains how to generate a report in the Reporting application. The Reporting application lets you produce high-level reports to understand the overall status of the job streams and applications operating in your functional areas.

Predefined report templates are provided. You simply have to set the available parameters to generate specific report output in real-time. The report output can also be exported to a variety of formats and saved for later viewing or printing.

The following scenario walks you through the process of generating a report using the Reporting application.

### **How to Create and Print a Report**



To generate a job run or server statistical information report, you can follow these steps:

1. [Define your report.](#)
2. [Export and save the report with the appropriate output format.](#)
3. (Optional) [Print the report.](#)

### **Example: Provide Statistical Information for Monthly Management Meetings**

As an executive, you review the following report information for all servers on a monthly basis:

- remote agent usage statistics sorted by server for the past month
- job run failures for the past month displaying server name, job failures, and job runs

This information is updated for the monthly management meetings and presented in a printed format. To provide the information, you create a Job Run Throughput report and a Job Run Status Failure/Total Job Run report. Next, export the output to Adobe PDF format, and print the reports as needed.

### **Define a Report**

You can define a report that is based on an existing template and then export that report to a printable format. Once you have defined the report and clicked Finish, the report is run and displayed immediately.

#### **NOTE**

- If an error is encountered while creating or exporting a report, a message is displayed in the right pane.
- The maximum report size is approximately 10,000 pages when 1 GB of available memory has been allocated to the java heap memory. When creating a report, if the limit is exceeded, an error message is displayed and the report being generated is canceled.

### **Follow these steps:**

1. Perform one of the following actions:
    - Click Create Report in the Reporting Dashboard, select the appropriate template from the Create Report: Select Template dialog, and then click OK.
    - Click the appropriate report template in the Navigation pane.
    - Right-click the appropriate report template in the Navigation pane and select Create from the pop-up menu. The Create Report dialog opens.
  2. Select the appropriate server or servers from the Servers in Reporting DB section.
  3. Perform one of the following actions:
    - Select the Predefined option in the Date Time Range section and select a range from the drop-down list.
    - Select the Custom option in the Date Time Range section and enter date and time specifications.

Clicking  the Date Picker, so you can select a date to populate the Start Date and End Date fields.
  4. Click Finish to complete the report if no other parameters are required, or click Next to continue.
  5. Select the appropriate machine or machines from the Machines in Reporting DB section, if applicable.
  6. (Optional) Enter the appropriate job name pattern in the Job Name Filter field, if applicable.
  7. (Optional) Select the appropriate field by which to sort the report from the Sort by drop-down list, and select ascending or descending.
  8. Click Finish.
- The dialog closes. The generated report is displayed in the Reporting pane and listed in the Navigation pane in the Reports section.

#### **NOTE**

- An underscore and sequential number is appended to each report listed in the Reports section in the Navigation pane. For example, you can have Summary\_1 and Summary\_2 listed in the reports section if the report was run twice.
- Reports are available for the current session only. To save the report, export the report to the appropriate format and save it.
- You can use the pagination field to view specific pages or ranges of pages in the report.

---

## **Export a Report to the Appropriate Output Format**

Once you have created a report in the Reporting application, exporting it generates a format that you can view or print.

### **NOTE**

- If an error is encountered while creating or exporting a report, a message is displayed in the right pane.
- When exporting a report to PDF or Excel format, if the java heap memory limit is exceeded, an error message is displayed and the export is canceled.

### **Follow these steps:**

1. Perform one of the following options:
  - Right-click the report name in the Reports section in the Navigation pane and select Export.
  - Click Export in the right pane.
2. Select one of the following options:
  - **Pdf**  
Converts the report to an Adobe PDF file.
  - **Excel**  
Converts the report to a Microsoft Excel XLS file.
  - **Html**  
Converts the report to an HTML file.  
The Export Report dialog opens.
3. Click the link to open the report.  
The report is displayed in the selected format.

## **Print a Report**

After you have created a report and have exported the report to the a printable format, you can view the report. If you opt to, you can also print the report.

### **Follow these steps:**

1. With the report open, select File, Print from the menu.  
The Print dialog appears.
2. Select the appropriate printer and click Print.  
The report is printed.

## Reference

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This article includes AutoSys Workload Automation and Web UI reference information for web services, commands, Job Information Language (JIL), and system states.

### WARNING

This section contains confidential subsections that are not viewable to anonymous users. To view all subsections, select Sign In in the upper-right corner. Enter your CA Support Online credentials. If you are logged in, ignore this message.

This section contains the following articles:

- [AE Web Services](#)  
Includes web services information to help you perform tasks such as retrieving a job status and performing job actions such as start, cancel, hold, and so on.
- [AE Commands](#)  
Includes information to help you run multiple commands in scripts (UNIX) or batch files (Windows) and create aliases for frequently used commands.
- [AE Job Information Language](#)  
Includes information about the JIL subcommands and attribute statements. JIL is a scripting language that lets you define jobs, machines, resources, external instances, monitors, reports, blobs, and job types (or *JIL objects*).
- [AE System States](#)  
Describes the system events, alarms, statuses, and exit codes that AutoSys Workload Automation processes.

After you log in, use the navigation tree on the left or search for Reference content:

Reference

## AutoSys Web Services

Web services in AutoSys Workload Automation focus on job monitoring and control. Using web services, you can perform tasks such as retrieving a job status and performing job actions such as start, cancel, hold, and so on.

Web-enabled applications use web services to integrate with AutoSys Workload Automation. In AutoSys Workload Automation, the web services are based on the Representational State Transfer (REST) type of architecture. The security aspect of web services is based on standard Hypertext Transfer Protocol (HTTP), which is also the base model for REST.

### NOTE

- For all web-enabled transactions that use web services, standard HTTP /1.1 response codes are supported. Status codes conform to section 10 of RFC 2616. For more information about these codes, refer to <http://www.w3.org>.
- All time is represented in ISO 8601 format.
- This content is for Application Developers who are responsible for providing basic monitoring and control type services using AutoSys Workload Automation web services. To use this content, you must be familiar with AutoSys Workload Automation, REpresentational State Transfer (REST) architecture, and the Apache Tomcat web server. We assume that you have already installed and are running AutoSys Workload Automation.

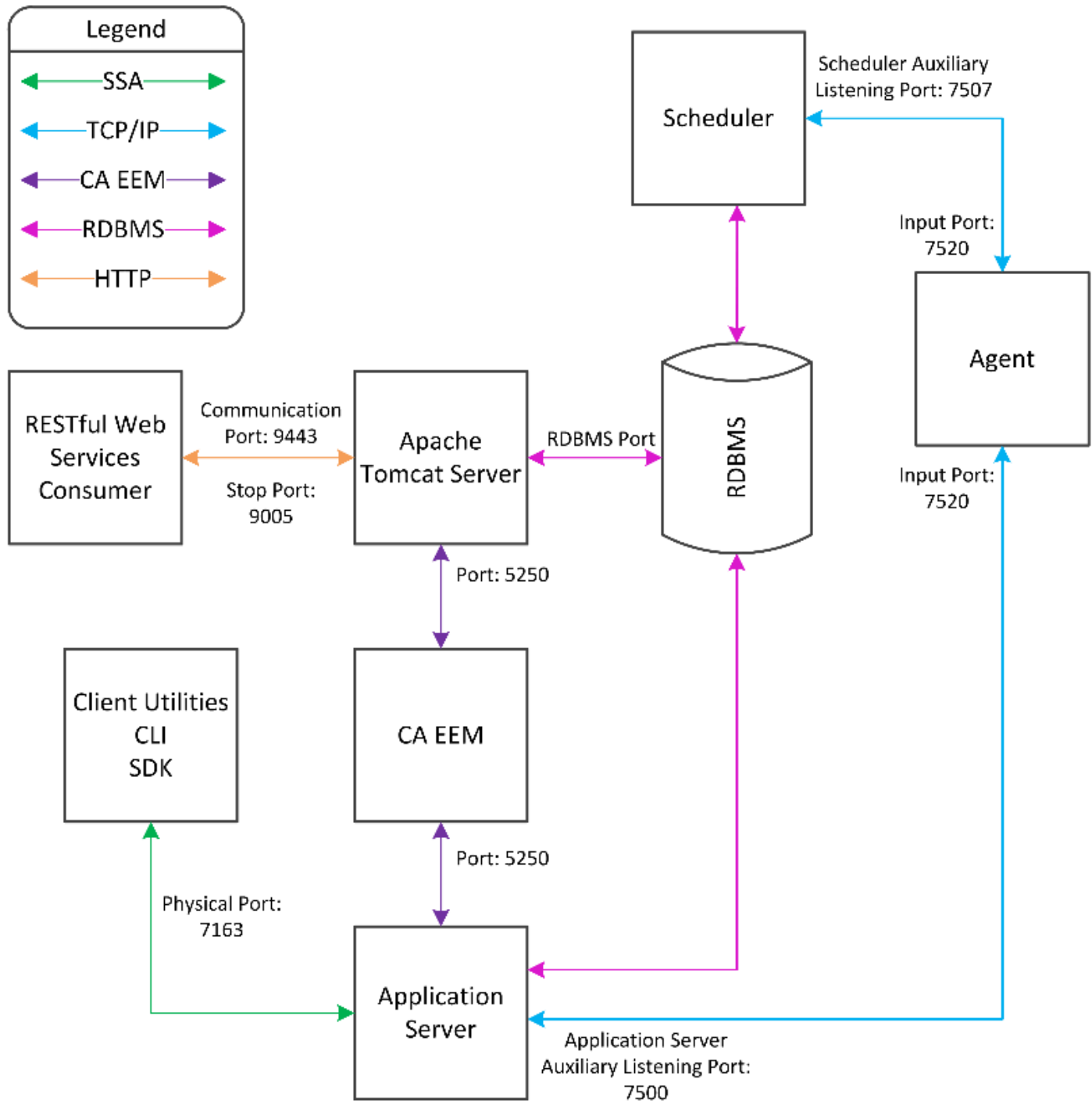
### Web Services Components

A web services transaction interfaces with the following components:

- Web server (Apache Tomcat)
- RDBMS (database)
- CA EEM (security)

The following illustration shows the communication path between the components of web services that are part of the AutoSys Workload Automation architecture:

**Figure 80: webservices**



## **Web Server**

Apache Tomcat is the designated web server that is used to host web services. To install the web server, perform a custom AutoSys Workload Automation installation. The web server also retrieves the agent information from AutoSys Workload Automation and displays it in the [Agent Inventory](#) tab in Web UI.

Apache Tomcat shares the configuration parameters that are provided for AutoSys Workload Automation database access and security. Web services use port 9443 to communicate with the Apache Tomcat server. The Apache Tomcat server uses port 5250 to communicate with CA EEM, which is the default (non-native) security provider for AutoSys Workload Automation.

## **RDBMS**

In AutoSys Workload Automation, access to the relational database is enabled as a RESTful web service. Entities are queried, created, and modified using HTTP methods such as GET and POST.

## **CA EEM**

Session authentication in web services is secured using CA EEM. CA EEM is the default (non-native) security provider for AutoSys Workload Automation. CA EEM ensures the security of a web services transaction by providing identity authentication, authorization, and access management services. CA EEM communicates with the designated web server, Apache Tomcat, using port 5250.

## **Security Implementation**

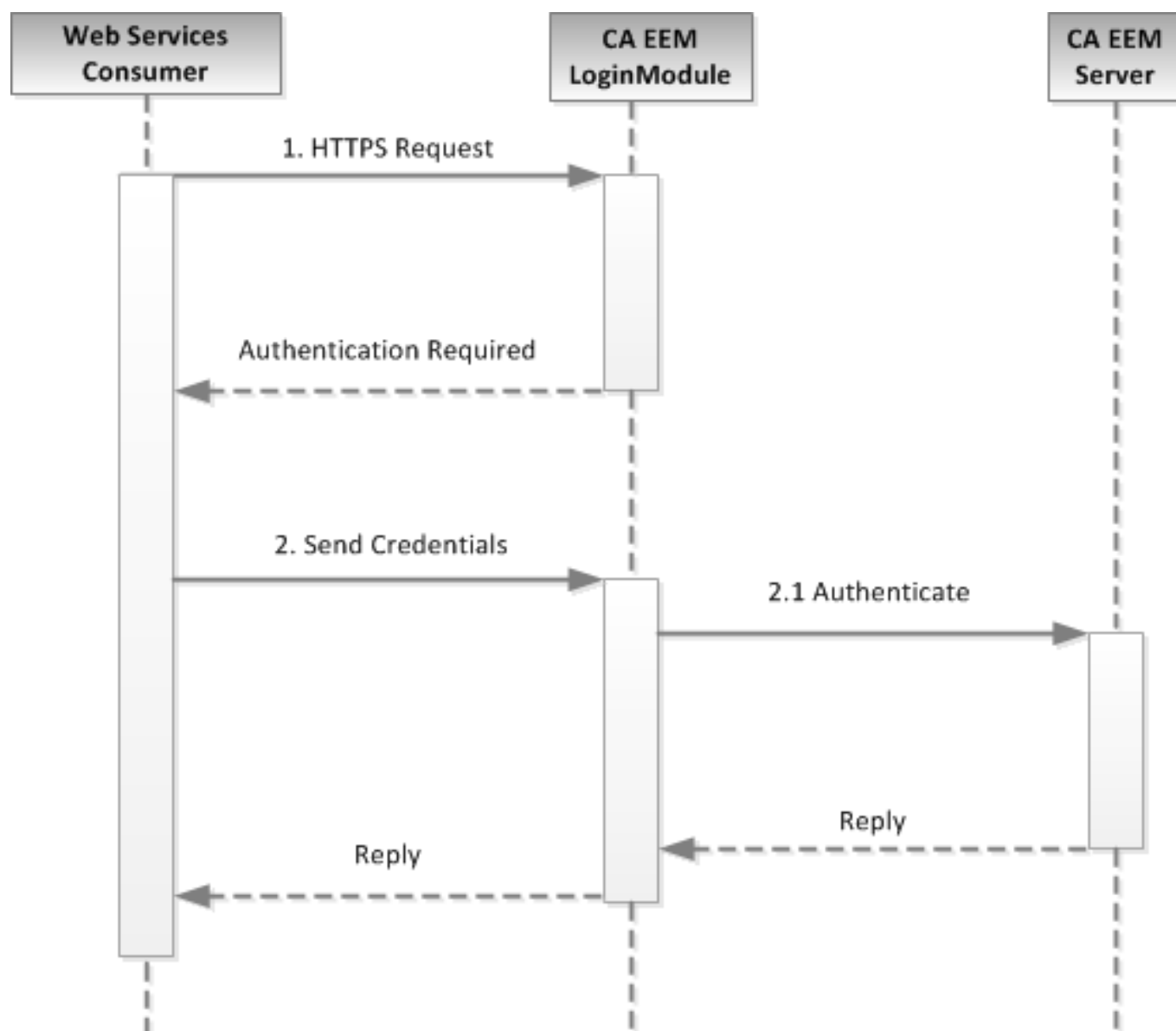
In AutoSys Workload Automation, the security of a web services transaction is implemented using an HTTPS connection, which is HTTP basic authentication over a Secure Sockets Layer (SSL) connection.

Authentication is performed using the Tomcat Java™ Authentication and Authorization Service (JAAS) realm with CA EEM security. The CA EEM-specific JAAS LoginModule authenticates the login credentials. Data authorization checks are performed based on the authenticated user ID and CA EEM authorization policies that are already implemented by AutoSys Workload Automation. A CA EEM local application group, for example, WorkloadAutomationAEWebService, controls access to web services. The use of web services is restricted to members of this group.

## **Authentication**

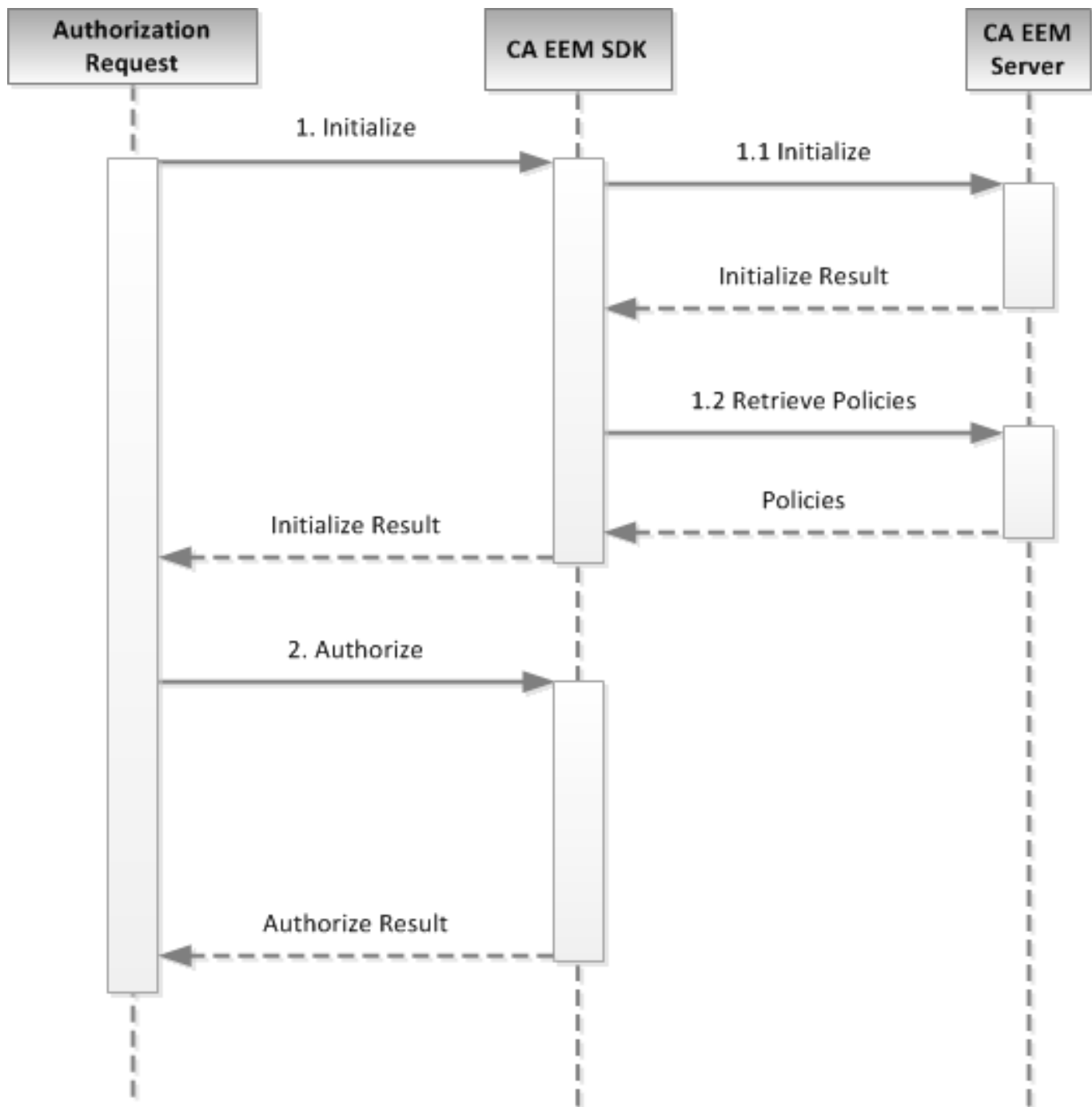
The following illustration shows the authentication process of a web services transaction in a basic configuration:



**Figure 81: Authentication**

### **Authorization**

The following illustration shows the authorization process of a web services transaction in a basic configuration:

**Figure 82: Authorization**

## ComponentRegister Utility

The ComponentRegister utility registers the REST web server that is installed with AutoSys Workload Automation and updates the database. The utility is installed automatically if you select the Web server check box on the Components panel when you perform a server installation.

The ComponentRegister utility extends the functionality of the asapi API to retrieve the AutoSys Workload Automation REST configuration.

The ComponentRegister utility is available in the following location:

- UNIX—\$AUTOSYS/utls/ ComponentRegister/ComponentRegister.pl -register -component webserver
- Windows—%AUTOSYS%\utls\ ComponentRegister\ComponentRegister.pl -register -component webserver

## Service Directory

The AutoSys Workload Automation web service generates a Web Application Definition Language (WADL) dynamically at runtime. The WADL is generated in XML format and it provides a machine-readable description of a HTTP-based web application. The web application is typically a Representational State Transfer (REST) web service. The AutoSys Workload Automation web service lets you access either the full WADL, in case of an application, or a single resource WADL. To access the WADL, the following Uniform Resource Identifiers (URIs) are available:

- [GET /AEWS/application.wadl](#) -- full WADL
- [OPTIONS /AEWS/resource](#) -- single resource WADL

### **GET /AEWS/application.wadl -- Retrieve Details of an Application**

The GET AEWS/application.wadl URI retrieves the full WADL for the specified application.

#### **HTTP Method**

GET

#### **URL Resource**

```
https://webserver:port/AEWS/application.wadl
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Example: Retrieve the application WADL**

The GET https://WSServer:9443/AEWS/application.wadl command returns the following response:

#### **NOTE**

This example is an extract of the full WADL. The contents of the WADL are subject to change.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<application xmlns="http://wadl.dev.java.net/2009/02">

 <doc xmlns:jersey="http://
jersey.java.net/" jersey:generatedBy="Jersey: 1.11 12/09/2011 11:05 AM"/>

 <grammars>

 <include href="application.wadl/xsd0.xsd">

 <doc xml:lang="en" title="Generated"/>
```

```
</include>

</grammars>

<resources base="https://webserver:9443/AEWS/">

 <resource path="/job/{name}">

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="template" name="name"/>

 <method name="GET" id="getJob">

 <request>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="locale"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="version"/>

 </request>

 <response>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/json" element="job"/>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/xml" element="job"/>

 </response>

 </method>

 </resource>

 <resource path="/job">

 <method name="GET" id="getJob">

 <request>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="count"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="index"/>
```

```

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="filter"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="sortBy"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="startIndex"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="locale"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="version"/>

 </request>

 <response>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/json" element="jobs"/>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/xml" element="jobs"/>

 </response>

</method>

</resource>

...

</resources>

</application>

```

### **OPTIONS AEWS/resource -- Retrieve Details of a Resource**

The **OPTIONS AEWS/resource** URI retrieves the WADL for the specified resource.

#### **HTTP Method**

**OPTIONS**

#### **URL Resource**

`https://webserver:port/AEWS/resource`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### Parameters

- **resource** Specifies the name of the resource for which to retrieve the WADL. The name is required.

#### Example: Retrieve the job resource WADL

The OPTIONS `https://WSServer:9443/AEWS/job` command returns the following response:

#### NOTE

This example is an extract of the resource WADL. The contents of the WADL are subject to change.

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>

<application xmlns="http://wadl.dev.java.net/2009/02">

 <doc xmlns:jersey="http://
jersey.java.net/" jersey:generatedBy="Jersey: 1.11 12/09/2011 11:05 AM"/>

 <grammars>

 <include href="https://webserver:9443/AEWS/application.wadl/xsd0.xsd">

 <doc xml:lang="en" title="Generated"/>

 </include>

 </grammars>

 <resources base="https://webserver:9443/AEWS/">

 <resource path="job">

 <method name="GET" id="getJob">

 <request>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="count"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="index"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="filter"/>

 </request>

 </method>

 </resource>

 </resources>

</application>
```

```

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="sortBy"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="startIndex"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="locale"/>

 <param xmlns:xs="http://www.w3.org/2001/
XMLSchema" type="xs:string" style="query" name="version"/>

 </request>

 <response>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/json" element="jobs"/>

 <ns2:representation xmlns:ns2="http://
wadl.dev.java.net/2009/02" xmlns="" mediaType="application/xml" element="jobs"/>

 </response>

</method>

</resource>

</resources>

</application>

```

## Data Schema

The data schema defines the attributes that are used to generate the client classes and to organize the input and output data for AutoSys Workload Automation RESTful web services. Each schema is defined using multiple base elements, referred to as a complex type schema. The data schema is defined as follows:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes" ?>
- <xs:schema version="1.0" xmlns:xs="http://www.w3.org/2001/XMLSchema">
 <xs:element name="command" type="command" />
 <xs:element name="dependency" type="dependency" />
 <xs:element name="event" type="event" />
 <xs:element name="gvar" type="gvar" />
 <xs:element name="gvars" type="gvars" />
 <xs:element name="job" type="job" />
 <xs:element name="jobRun" type="jobRun" />

```

```

<xs:element name="jobRuns" type="jobRuns" />
<xs:element name="jobs" type="jobs" />
<xs:element name="link" type="atomLink" />
<xs:element name="machine" type="machine" />
<xs:element name="machines" type="machines" />
<xs:element name="predecessor" type="predecessor" />
<xs:element name="predecessors" type="predecessors" />
<xs:element name="successor" type="successor" />
<xs:element name="successors" type="successors" />
- <xs:complexType name="atomLink">
 <xs:sequence />
 <xs:attribute name="href" type="xs:string" />
 <xs:attribute name="rel" type="xs:string" />
 <xs:attribute name="type" type="xs:string" />
</xs:complexType>
- <xs:complexType name="command">
 - <xs:sequence>
 <xs:element name="application" type="xs:string" minOccurs="0" />
 <xs:element name="comment" type="xs:string" minOccurs="0" />
 <xs:element name="group" type="xs:string" minOccurs="0" />
 <xs:element name="gvarValue" type="xs:string" minOccurs="0" />
 <xs:element name="jobName" type="xs:string" minOccurs="0" />
 <xs:element name="response" type="xs:string" minOccurs="0" />
 <xs:element name="signal" type="xs:string" minOccurs="0" />
 <xs:element name="stepName" type="xs:string" minOccurs="0" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="dependency">
 - <xs:sequence>
 <xs:element ref="predecessors" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element ref="successors" minOccurs="0" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="predecessors">
 - <xs:sequence>
 <xs:element name="predecessor" type="predecessor" nillable="true" minOccurs="0"
maxOccurs="unbounded" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="predecessor">
 - <xs:sequence>
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="type" type="xs:string" minOccurs="0" />
 <xs:element name="xinst" type="xs:string" minOccurs="0" />

```



```

 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="successors">
- <xs:sequence>
 <xs:element name="successor" type="successor" nillable="true" minOccurs="0"
maxOccurs="unbounded" />
 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="successor">
- <xs:sequence>
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="type" type="xs:string" minOccurs="0" />
 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="event">
- <xs:sequence>
 <xs:element name="alarm" type="xs:int" />
 <xs:element name="evtNum" type="xs:int" />
 <xs:element name="gvar" type="xs:string" minOccurs="0" />
 <xs:element name="gvarValue" type="xs:string" minOccurs="0" />
 <xs:element name="machine" type="xs:string" minOccurs="0" />
 <xs:element name="ntry" type="xs:int" />
 <xs:element name="processedTime" type="xs:string" minOccurs="0" />
 <xs:element name="queuedTime" type="xs:string" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="status" type="xs:int" />
 <xs:element name="stepName" type="xs:string" minOccurs="0" />
 <xs:element name="text" type="xs:string" minOccurs="0" />
 <xs:element name="type" type="xs:int" />
 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="gvar">
- <xs:sequence>
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="updateTime" type="xs:string" minOccurs="0" />
 <xs:element name="value" type="xs:string" minOccurs="0" />
 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="gvars">
- <xs:sequence>
 <xs:element ref="gvar" minOccurs="0" maxOccurs="unbounded" />
 <xs:element name="next" type="atomLink" minOccurs="0" />
 <xs:element name="prev" type="atomLink" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />

```

```

 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="job">
- <xs:sequence>
 <xs:element name="application" type="xs:string" minOccurs="0" />
 <xs:element name="boxName" type="xs:string" minOccurs="0" />
 <xs:element name="description" type="xs:string" minOccurs="0" />
 <xs:element name="group" type="xs:string" minOccurs="0" />
 <xs:element name="jobType" type="xs:string" minOccurs="0" />
 <xs:element name="machine" type="xs:string" minOccurs="0" />
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="status" type="xs:int" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="jobRun">
- <xs:sequence>
 <xs:element name="application" type="xs:string" minOccurs="0" />
 <xs:element name="avgRunTime" type="xs:int" />
 <xs:element name="boxName" type="xs:string" minOccurs="0" />
 <xs:element name="machine" type="xs:string" minOccurs="0" />
 <xs:element name="runEndTime" type="xs:string" minOccurs="0" />
 <xs:element name="exitCode" type="xs:int" />
 <xs:element name="group" type="xs:string" minOccurs="0" />
 <xs:element name="events" minOccurs="0">
- <xs:complexType>
- <xs:sequence>
 <xs:element ref="event" minOccurs="0" maxOccurs="unbounded" />
 </xs:sequence>
</xs:complexType>
</xs:element>
<xs:element name="name" type="xs:string" minOccurs="0" />
<xs:element name="runMachine" type="xs:string" minOccurs="0" />
<xs:element name="runnum" type="xs:int" />
<xs:element name="self" type="atomLink" minOccurs="0" />
<xs:element name="runStartTime" type="xs:string" minOccurs="0" />
<xs:element name="status" type="xs:int" />
</xs:sequence>
</xs:complexType>
- <xs:complexType name="jobRuns">
- <xs:sequence>
 <xs:element name="jobRun" type="jobRun" nillable="true" minOccurs="0"
maxOccurs="unbounded" />
 <xs:element name="next" type="atomLink" minOccurs="0" />
 <xs:element name="prev" type="atomLink" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />

```

```

 </xs:sequence>
 </xs:complexType>
- <xs:complexType name="jobs">
- <xs:sequence>
 <xs:element ref="job" minOccurs="0" maxOccurs="unbounded" />
 <xs:element name="next" type="atomLink" minOccurs="0" />
 <xs:element name="prev" type="atomLink" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="machine">
- <xs:sequence>
 <xs:element name="agentBuild" type="xs:string" minOccurs="0" />
 <xs:element name="agentMaintenanceLevel" type="xs:string" minOccurs="0" />
 <xs:element name="agentName" type="xs:string" minOccurs="0" />
 <xs:element name="agentRelease" type="xs:string" minOccurs="0" />
 <xs:element name="agentServicePack" type="xs:string" minOccurs="0" />
 <xs:element name="characterCode" type="xs:string" minOccurs="0" />
 <xs:element name="description" type="xs:string" minOccurs="0" />
 <xs:element name="factor" type="xs:float" />
 <xs:element name="heartBeatAttempts" type="xs:int" />
 <xs:element name="heartBeatFrequency" type="xs:int" />
 <xs:element name="maxLoad" type="xs:int" />
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="nodeName" type="xs:string" minOccurs="0" />
 <xs:element name="parentName" type="xs:string" minOccurs="0" />
 <xs:element name="platform" type="xs:string" minOccurs="0" />
 <xs:element name="pluginList" type="xs:string" minOccurs="0" />
 <xs:element name="port" type="xs:int" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="status" type="xs:string" minOccurs="0" />
 <xs:element name="type" type="xs:string" minOccurs="0" />
 </xs:sequence>
</xs:complexType>
- <xs:complexType name="machines">
- <xs:sequence>
 <xs:element ref="machine" minOccurs="0" maxOccurs="unbounded" />
 <xs:element name="next" type="atomLink" minOccurs="0" />
 <xs:element name="prev" type="atomLink" minOccurs="0" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 </xs:sequence>
</xs:complexType>
</xs:schema>

```

The data schema includes the following individual complex type schemas:

- [atomLink](#)
- [command](#)
- [dependency](#)
- [event](#)
- [gvar](#)
- [gvars](#)
- [job](#)
- [jobRun](#)
- [jobRuns](#)
- [machine](#)
- [predecessors](#)
- [predecessor](#)
- [successors](#)
- [successor](#)

### **Event Schema**

The event complex type is returned as an element in the jobRun complex type. The event complex type schema is created as follows:

```
<xsd:complexType name="event">

 <xsd:sequence>

 <xsd:element name="alarm" type="xsd:int" minOccurs="0"/>

 <xsd:element name="evtNum" type="xsd:int"/>

 <xsd:element name="gvar" type="xsd:string" minOccurs="0"/>

 <xsd:element name="gvarValue" type="xsd:string" minOccurs="0"/>

 <xsd:element name="machine" type="xsd:string" minOccurs="0"/>

 <xsd:element name="ntry" type="xsd:int"/>

 <xsd:element name="processedTime" type="xsd:string" minOccurs="0"/>

 <xsd:element name="queuedTime" type="xsd:string" minOccurs="0"/>

 <xsd:element name="self" type="atomLink" minOccurs="0"/>

 <xsd:element name="status" type="xsd:int" minOccurs="0"/>

 <xsd:element name="stepName" type="xsd:string" minOccurs="0"/>

 <xsd:element name="text" type="xsd:string" minOccurs="0"/>

 <xsd:element name="type" type="xsd:int" minOccurs="0"/>

 </xsd:sequence>

</xsd:complexType>
```

```
</xsd:sequence>
```

```
</xsd:complexType>
```

### **Command Schema**

The command complex type is sent with the the following POST action URIs:

- POST /AEWS/event/start-job
- POST /AEWS/event/force-start-job
- POST /AEWS/event/restart-job
- POST /AEWS/event/hold-job
- POST /AEWS/event/off-hold-job
- POST /AEWS/event/ice-job
- POST /AEWS/event/off-ice-job
- POST /AEWS/event/noexec-job
- POST /AEWS/event/off-noexec-job
- POST /AEWS/event/cancel-job
- POST /AEWS/event/release-resources-job
- POST /AEWS/event/reply-job
- POST /AEWS/event/set-gvar/{name}
- POST /AEWS/event/delete-gvar/{name}

The command complex type schema is created as follows:

```
<xsd:complexType name="command">

 <xsd:sequence>

 <xsd:element name="application" type="xsd:string" minOccurs="0"/>

 <xsd:element name="comment" type="xsd:string" minOccurs="0"/>

 <xsd:element name="group" type="xsd:string" minOccurs="0"/>

 <xsd:element name="gvarValue" type="xsd:string" minOccurs="0"/>

 <xsd:element name="jobName" type="xsd:string" minOccurs="0"/>

 <xsd:element name="response" type="xsd:string" minOccurs="0"/>

 <xsd:element name="signal" type="xsd:string" minOccurs="0"/>

 <xsd:element name="stepName" type="xsd:string" minOccurs="0"/>

 </xsd:sequence>

</xsd:complexType>
```

### **gvar Schema**

The gvar complex type is returned from the retrieve specific global variable URI. The gvar complex type schema is created as follows:

```
GET /AEWS/gvar/variableName
```

```
<xsd:complexType name="gvar">
 <xsd:sequence>
 <xsd:element name="name" type="xsd:string" minOccurs="0"/>
 <xsd:element name="self" type="atomLink" minOccurs="0"/>
 <xsd:element name="updateTime" type="xsd:string" minOccurs="0"/>
 <xsd:element name="value" type="xsd:string" minOccurs="0"/>
 </xsd:sequence>
</xsd:complexType>
```

### **gvars Schema**

The gvars complex type is returned from the retrieve list global variables URI. The gvars complex type schema is created as follows:

```
GET /AEWS/gvar
```

```
<xsd:complexType name="gvars">
 <xsd:sequence>
 <xsd:element name="gvar" type="gvar" minOccurs="0" maxOccurs="unbounded"/>
 <xsd:element name="next" type="atomLink" minOccurs="0"/>
 <xsd:element name="prev" type="atomLink" minOccurs="0"/>
 <xsd:element name="self" type="atomLink" minOccurs="0"/>
 </xsd:sequence>
</xsd:complexType>
```

## **Job Schema**

The job complex type is returned from the retrieve job definition URI. The job complex type schema is created as follows:

```
GET /AEWS/job

<xsd:complexType name="job">

 <xsd:sequence>

 <xsd:element name="application" type="xsd:string" minOccurs="0"/>

 <xsd:element name="boxName" type="xsd:string" minOccurs="0"/>

 <xsd:element name="description" type="xsd:string" minOccurs="0"/>

 <xsd:element name="group" type="xsd:string" minOccurs="0"/>

 <xsd:element name="jobType" type="xsd:string" minOccurs="0" />

 <xsd:element name="machine" type="xsd:string" minOccurs="0"/>

 <xsd:element name="name" type="xsd:string" minOccurs="0"/>

 <xsd:element name="self" type="atomLink" minOccurs="0"/>

 <xsd:element name="status" type="xsd:int"/>

 </xsd:sequence>

</xsd:complexType>
```

## **jobRun Schema**

The jobRun complex type is returned as elements in the jobRuns output for the job run-time URIs. The jobRun complex type schema is created as follows:

```
<xsd:complexType name="jobRun">

 <xsd:sequence>

 <xsd:element name="application" type="xsd:string" minOccurs="0"/>

 <xsd:element name="avgRunTime" type="xsd:int"/>

 <xsd:element name="boxName" type="xsd:string" minOccurs="0"/>
```

```

<xsd:element name="machine" type="xsd:string" minOccurs="0"/>

<xsd:element name="runEndTime" type="xsd:string" minOccurs="0"/>

<xsd:element name="exitCode" type="xsd:int"/>

<xsd:element name="group" type="xsd:string" minOccurs="0"/>

<xsd:element name="events" minOccurs="0">

 <xsd:complexType>

 <xsd:sequence>

 <xsd:element name="event" type="event" minOccurs="0" maxOccurs="unbounded"/>

 </xsd:sequence>

 </xsd:complexType>

</xsd:element>

<xsd:element name="name" type="xsd:string" minOccurs="0"/>

<xsd:element name="runMachine" type="xsd:string" minOccurs="0"/>

<xsd:element name="runnum" type="xsd:int"/>

<xsd:element name="self" type="atomLink" minOccurs="0"/>

<xsd:element name="runStartTime" type="xsd:string" minOccurs="0"/>

<xsd:element name="status" type="xsd:int"/>

</xsd:sequence>

</xsd:complexType>

```

### **jobRuns Schema**

The jobRuns complex type is returned from the retrieve list of job run-times or retrieve specific job run-time URIs. The jobRuns complex type schema is created as follows:

```
GET /AEWS/job-run-info/jobName
```

```
GET /AEWS/job-run-info
```



```

<xsd:complexType name="jobRuns">

 <xsd:sequence>

 <xsd:element name="jobRun" type="jobRun" minOccurs="0" maxOccurs="unbounded" nillable="true" />

 <xsd:element name="next" type="atomLink" minOccurs="0"/>

 <xsd:element name="prev" type="atomLink" minOccurs="0"/>

 <xsd:element name="self" type="atomLink" minOccurs="0"/>

 </xsd:sequence>

</xsd:complexType>

```

### **machine Schema**

The machine complex type is returned from the retrieve machine definition URI. The machine complex type schema is created as follows:

```

GET /AEWS/machine
- <xs:complexType name="machine">
 - <xs:sequence>
 <xs:element name="agentBuild" type="xs:string" minOccurs="0" />
 <xs:element name="agentMaintenanceLevel" type="xs:string" minOccurs="0" />
 <xs:element name="agentName" type="xs:string" minOccurs="0" />
 <xs:element name="agentRelease" type="xs:string" minOccurs="0" />
 <xs:element name="agentServicePack" type="xs:string" minOccurs="0" />
 <xs:element name="characterCode" type="xs:string" minOccurs="0" />
 <xs:element name="description" type="xs:string" minOccurs="0" />
 <xs:element name="factor" type="xs:float" />
 <xs:element name="heartBeatAttempts" type="xs:int" />
 <xs:element name="heartBeatFrequency" type="xs:int" />
 <xs:element name="maxLoad" type="xs:int" />
 <xs:element name="name" type="xs:string" minOccurs="0" />
 <xs:element name="nodeName" type="xs:string" minOccurs="0" />
 <xs:element name="parentName" type="xs:string" minOccurs="0" />
 <xs:element name="platform" type="xs:string" minOccurs="0" />
 <xs:element name="pluginList" type="xs:string" minOccurs="0" />
 <xs:element name="port" type="xs:int" />
 <xs:element name="self" type="atomLink" minOccurs="0" />
 <xs:element name="status" type="xs:string" minOccurs="0" />
 <xs:element name="type" type="xs:string" minOccurs="0" />
 </xs:sequence>
</xs:complexType>

```

**atomLink Schema**

The atomLink complex type is used for self, next, and previous URI pointers in the returned output. The atomLink complex type schema is created as follows:

```
<xsd:complexType name="atomLink">
 <xsd:sequence/>
 <xsd:attribute name="href" type="xsd:string"/>
 <xsd:attribute name="rel" type="xsd:string"/>
 <xsd:attribute name="type" type="xsd:string"/>
</xsd:complexType>
```

**Dependency Schema**

The dependencies complex type is returned from the retrieve job dependency URI. The dependency complex type schema is created as follows:

```
GET /AEWS/job-dependencies/jobname
```

```
<xsd:complexType name="dependency">
 <xsd:sequence>
 <xsd:element name="predecessors" type="predecessors" minOccurs="0"/>
 <xsd:element name="self" type="atomLink" minOccurs="0"/>
 <xsd:element name="successors" type="successors" minOccurs="0"/>
 </xsd:sequence>
</xsd:complexType>
```

**Predecessors Schema**

The prececessors complex type is returned as an element in the dependency output for the job dependency URI. The predecessors complex type schema is created as follows:

```
<xsd:complexType name="predecessors">
```

```

<xsd:sequence>

 <xsd:element name="predecessor" type="predecessor" minOccurs="0" maxOccurs="unbounded" nill
>

</xsd:sequence>

</xsd:complexType>

```

### **Predecessor Schema**

The predecessor complex type is returned as an element in the predecessors output for the job dependency URI. The predecessor complex type schema is created as follows:

```

<xsd:complexType name="predecessor">

 <xsd:sequence>

 <xsd:element name="name" type="xsd:string" minOccurs="0"/>

 <xsd:element name="self" type="atomLink" minOccurs="0"/>

 <xsd:element name="type" type="xsd:string" minOccurs="0"/>

 <xsd:element name="xinst" type="xsd:string" minOccurs="0"/>

 </xsd:sequence>

</xsd:complexType>

```

### **Successors Schema**

The successors complex type is returned as an element in the dependency output for the job dependency URI. The successors complex type schema is created as follows:

```

<xsd:complexType name="successors">

 <xsd:sequence>

 <xsd:element name="successor" type="successor" minOccurs="0" maxOccurs="unbounded" nillable
>

 </xsd:sequence>

</xsd:complexType>

```

## Successor Schema

The successor complex type is returned as an element in the successors output for the job dependency URI. The successor complex type schema is created as follows:

```
<xsd:complexType name="successor">

 <xsd:sequence>

 <xsd:element name="name" type="xsd:string" minOccurs="0"/>

 <xsd:element name="self" type="atomLink" minOccurs="0"/>

 <xsd:element name="type" type="xsd:string" minOccurs="0"/>

 <xsd:element name="xinst" type="xsd:string" minOccurs="0"/>

 </xsd:sequence>

</xsd:complexType>
```

## Manage Calendars (AEWS)

Following are the procedures for managing Calendars:

### POST AEWS/calendar method-- Imports calendar definitions

The POST AEWS/calendar method imports calendar definitions given in the request body and supports standard, extended, and cycle calendars.

#### HTTP Method

POST

#### URL Resource

`https://webserver:port/AEWS/calendar?parameters`

- **webserver**  
Specifies the name of the web server host computer.
- **port**  
Specifies the port number for the web server host computer.

#### Parameters

- **force**  
Force overwrites calendars on import.  
Corresponds to autocal\_asc -F option.  
**Default:** False
- **date\_format**  
Date format used in import calendar.  
Corresponds to autocal\_asc -f option.

**Default:** DateFormat value specified in autosys config file.

- **timeout**

Specifies the request timeout period in seconds. By default, the request times out if the operation is not completed in 300secs.

**Default:** 300sec

## Request Formats

- TEXT\_PLAIN

## Response Formats

- xml -- Specifies to return the data in XML format.
- json – Specifies to return the data in JSON format.

Response contains:

**Output:** Specifies the stdout and stderr of the autocall\_asc command with the given input request definitions. It is an array of strings where each line of stdout and stderr is an array element.

**Rc:** Specifies the exit code of the autocall\_asc operation.

**Status:** Specifies the status corresponding to the operation.

- success - If all the given calendar definitions are successfully imported.
- failed – If atleast one calendar definition fails to import.

### Example1: POST https://WSServer:9443/AEWS/calendar

Import a standard calendar with default query parameters. The request body should be as follows:

calendar: Cal1

description:

20/06/2020 20:53:00

**Response Code: 200**

### Response in XML Format:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
 <output>CAUAJM_I_50430 Insert Successful for Calendar: Cal1.The following dates are inserted. </output>

 <output> 02/06/2020 20:53</output>
 <rc>0</rc>
 <status>success</status>
</cliResponse>
```

### Response in JSON Format:

```
{
 "output": [
 "CAUAJM_I_50430 Insert Successful for Calendar: Cal1.The following dates are inserted.",
 " 02/06/2020 20:53"
],
 "rc": "0",
 "status": "success"
}
```

### Example2: POST https://WSServer:9443/AEWS/calendar

Import an already existing standard calendar. Import should fail. The request body should be as follows:

calendar: Cal1

description:

20/06/2020 20:53:00

**Response Code: 406**

**Response in XML Format:**

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
 <output>CAUAJM_I_50431 Insert Failed for Calendar: Cal1 </output>
 <output>CAUAJM_E_10601 Cannot insert calendar Cal1, calendar name already exists.</output>
 <rc>1</rc>
 <status>failed</status>
</cliResponse>
```

**Response in JSON Format:**

```
{
 "output": [
 "CAUAJM_I_50431 Insert Failed for Calendar: Cal1 ",
 "CAUAJM_E_10601 Cannot insert calendar Cal1, calendar name already exists."
],
 "rc": "1",
 "status": "failed"
}
```

**Example3: POST https://WSServer:9443/AEWS/calendar?force=true**

Import an existing standard calendar with force as true. Import should succeed because force option is set to true. The request body should be as follows:

calendar: Cal1

description:

20/06/2020 20:53:00

**Response Code: 200**

**Response in XML Format:**

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><cliResponse>
 <output>CAUAJM_I_50430 Insert Successful for Calendar: Cal1.The following dates are inserted.</output>
 <output> 02/06/2020 20:53</output>
 <rc>0</rc>
 <status>success</status>
</cliResponse>
```

**Response in JSON Format:**

```
{
 "output": [
 "CAUAJM_I_50430 Insert Successful for Calendar: Cal1.The following dates are inserted.",
 " 02/06/2020 20:53"
],
 "rc": "0",
}
```

```

 "status": "success"
 }

```

#### Example4: POST https://WSServer:9443/AEWS/calendar?date\_format=DD/MM/YYYY

Import a calendar with date\_format as DD/MM/YYYY. Import should be successful. The request body should be as follows:

calendar: Cal1

description:

20/06/2020 20:53:00

**Response Code: 200**

#### Response in XML Format:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
 <output>CAUAJM_I_50430 Insert Successful for Calendar: Cal1. The following dates are inserted.</output>
 <output> 06/20/2020 20:53</output>
 <rc>0</rc>
 <status>success</status>
</cliResponse>

```

#### Response in JSON Format:

```

{
 "output": [
 "CAUAJM_I_50430 Insert Successful for Calendar: Cal1. The following dates are inserted.",
 " 06/20/2020 20:53"
],
 "rc": "0",
 "status": "success"
}

```

### GET AEWS/calendar/{type} method – Exports calendar definitions of the given type

The GET AEWS/calendar/{type} method exports calendar definitions of the given type based on the query parameters.

#### HTTP Method

GET

#### URL Resource

https://webserver:port/AEWS/calendar/{type}?parameters

- **webserver**  
Specifies the name of the web server host computer.
- **port**  
Specifies the port number for the web server host computer.

#### type

Specifies the type of the calendar to export.

**Valid Values:** standard, extended, and cycle

#### Parameter

- **name**  
Specifies the name of the calendar for export. If not specified, exports all the calendar definitions of the given type.  
**Default:** ALL
- **date\_format**  
Date format used in export calendar.  
Corresponds to autocal\_asc -f option.  
**Default:** DateFormat value specified in autosys config file.
- **timeout**  
Specifies the request timeout period in seconds. By default, the request times out if the operation is not completed in 300secs.  
**Default:** 300sec

### Response Formats

- TEXT\_PLAIN

**Example1:** Get all the standard calendar definitions

**GET** https://WSServer:9443/AEWS/calendar/standard

**Response Code: 200**

**Response body:**

calendar: Cal1

description:

06/20/2020 20:53:00

calendar: Cal2

description:

20/06/2020 20:53:00

**Example2:** Get a specific standard calendar Cal1 definition

**GET** https://WSServer:9443/AEWS/calendar/standard?name=Cal1

**Response Code: 200**

**Response body:**

calendar: Cal1

description:

06/20/2020 20:53:00

**Example3:** Get a non existing standard calendar definition

**GET** https://WSServer:9443/AEWS/calendar/standard?name=invalid

**Response Code: 404**

**Response body:** CAUAJM\_E\_50308 Standard calendar invalid does not exist.

**Example4:** Get all the standard calendar definitions in the specified date\_format

**GET** https://WSServer:9443/AEWS/calendar/standard?date\_format=DD/MM/YYYY

**Response Code: 200**

**Response body:**

calendar: Cal1



description:

20/06/2020 20:53:00

calendar: Cal2

description:

06/20/2020 20:53:00

## Manage Connection Profiles

Following are the procedures for managing connection profiles:

### GET AEWS/connection-profile/connectionprofilename method—Retrieve Information of a Specific Connection Profile

The GET AEWS/connection-profile/connectionprofilename method returns information for a specific connection profile.

#### HTTP Method

GET

#### URL Resource

`https://webserver:port/AEWS/connection-profile/connectionprofilename?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.
- **connectionprofilename** Specifies the name of the connection profile to retrieve. The name is required.

#### Parameters

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### Response Formats

- **xml**—Returns the data in XML format.
- **json**—Returns the data in JSON format.

#### Example: GET https://WSServer:9443/AEWS/connection-profile/sample\_connectionprofile

The command returns the following response that is based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/connection-profile/sample_connectionprofile
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
```

```

Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 11 Sep 2015 13:01:05 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?
><connectionProfile><name>sample_connectionprofile</name>
<connectionProfileType>501</connectionProfileType><description>Connection profile for
hadoop</description>
<self href="https://WSServer:9443/AEWS/connection-profile/sample_connectionprofile"
rel="self"/></connectionProfile>

```

#### • JSON Format

```

-- request --
GET https://WSServer:9443/AEWS/connection-profile/sample_connectionprofile
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 11 Sep 2015 13:04:55 GMT
{"name":"sample_connectionprofile","connectionProfileType":"501","description":"Connection
profile for hadoop",
"self":{"@href":"https://WSServer:9443/AEWS/connection-profile/
sample_connectionprofile","@rel":"self"}}

```

### **GET AEWS/connection-profile method—Retrieve Information of a Single or Multiple Connection Profiles**

The GET AEWS/connection-profile method returns information for one or more connection profiles that are based on the results of a specified query.

#### **HTTP Method**

GET

#### **URL Resource**

`https://webserver:port/AEWS/connection-profile?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **filter=FilterCriteria** (Optional) Specifies one or more expressions that are used to limit the data returned. Multiple expressions are represented by a connection profile property, followed by an operator and a value. The connection profile properties included in the response such as name, connectionProfileType, and description are allowed. The following operators are allowed:

- – == Match exactly
- – != Not an exact match
- – =@ Match the pattern. A wildcard can be used.
- – !@ Not match the pattern. A wildcard can be used.
- – < Less than
- – <= Less than or equal to
- – > Greater than
- – >= Greater than or equal to

Multiple expressions can be linked with either the and (;) and or (||) Boolean operators. For example, entering `name==conn1;description==HadoopProfile` returns all connection profiles that have a connection profile name of `conn1` and a description of `HadoopProfile`. Multiple values can also be specified to create an or comparison. For example, entering `name==conn1,conn2` returns names that match either `conn1` or `conn2`.

#### NOTE

- When using `=@` or `!@` in the expression, you can use the asterisk (\*) as a wildcard for the entire name or for a partial name, where the \* represents zero or more characters. You can also use the question mark (?) to represent any single character.
- From AutoSys Workload Automation Release 11.3.6 SP7 onwards, you can perform a case insensitive search for the description property.
- **count=*n*** (Optional) Specifies the maximum number of records to return. If 0 is specified, previous and next references are not returned.
- **startIndex=*n*** (Optional) Specifies the starting logical page number, beginning with 1. A logical page contains the number of items specified by count. This option is ignored if count is set to 0.

#### NOTE

Although supported, we do not recommend using this parameter. All records must be read and processed before the requirement is fulfilled or cannot be met.

- **sortBy=*col1[,col2]...***  
(Optional) Specifies to order the results in the order of the specified elements. Multiple elements can be entered, separated by commas.  
**Default:** ascending order

#### NOTE

To specify descending order, prefix the element name with the hyphen (-).

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

### Response Formats

- `xml`—Returns the data in XML format.
- `json`—Returns the data in JSON format.

**Example: GET `https://WSServer:9443/AEWS/connection-profile?filter=name==@sample*`**

The command returns the following response that is based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/connection-profile?filter=name=@sample*
-- response --
200 OK
Server: Apache-Coyote/1.1
```

```

Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 11 Sep 2015 13:04:55 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?
><connectionProfiles><connectionProfile>
<name>sample_connectionprofile</name><connectionProfileType>501</connectionProfileType>
<description>Connection profile for hadoop</description>
<self href="https://WSServer:9443/AEWS/connection-profile/sample_connectionprofile"
rel="self"/></connectionProfile>
<connectionProfile><name>sample_connectionprofile2</name><connectionProfileType>501</
connectionProfileType>
<description>Connection profile for hadoop</description>
<self href="https://WSServer:9443/AEWS/connection-profile/sample_connectionprofile2"
rel="self"/></connectionProfile>
<self href="https://WSServer:9443/AEWS/connection-profile?filter=name=@sample*"
rel="self"/></connectionProfiles>

```

#### • JSON Format

```

-- request --
GET https://WSServer:9443/AEWS/connection-profile?filter=name=@sample*
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 11 Sep 2015 13:04:55 GMT
{"connectionProfile":
[{"name":"sample_connectionprofile","connectionProfileType":"501","description":"Connection
profile for hadoop",
"self":{"@href":"https://WSServer:9443/AEWS/connection-profile/
sample_connectionprofile","@rel":"self"}},
{"name":"sample_connectionprofile2","connectionProfileType":"501","description":"Connection
profile for hadoop",
"self":{"@href":"https://WSServer:9443/AEWS/connection-profile/
sample_connectionprofile2","@rel":"self"}},
"self":{"@href":"https://WSServer:9443/AEWS/connection-profile?
filter=name=@sample*","@rel":"self"}}

```

## Manage JIL (AEWS)

Following are the procedures for managing Job Information Language:

### POST AEWS/jil method-- Imports or Verifies JIL Definitions

The POST AEWS/jil method imports or verifies jil definitions given in the request body.

#### HTTP Method

POST

#### URL Resource

`https://webserver:port/AEWS/jil?parameters`

- **webserver**  
Specifies the name of the web server host computer.
- **port**  
Specifies the port number for the web server host computer.

#### Parameter

- **verify**  
If true, verifies the syntax of the given jil definitions. Corresponds to jil -V syntax.  
**Default:** False
- **timeout**  
Specifies the request timeout period in seconds. By default, the request times out if the operation is not completed in 300secs.  
**Default:** 300secs

#### Request Formats

- TEXT\_PLAIN

#### Response Formats

- xml -- Specifies to return the data in XML format.
- json -- Specifies to return the data in JSON format.

Response contains:

**Output:** Specifies the stdout and stderr of the JIL command with the given input request definitions.

**Rc:** Specifies the exit code of the jil operation.

**Status:** Specifies the status corresponding to the operation.

- success - If all the given jil definitions are successfully imported.
- failed -- If atleast one jil definition fails to import.

#### Example1: POST `https://WSServer:9443/AEWS/jil`

Insert a machine and then insert a job to run on the machine. The request body must look as follows:

```
/* ----- SampleMachine ----- */
insert_machine: SampleMachine
type: a
factor: 1.00
port: 7520
node_name: SampleMachine
```

```

agent_name: WA_AGENT
encryption_type: DEFAULT
character_code: ASCII

```

```

/* ----- SampleJob ----- */
insert_job: SampleJob job_type: CMD
command: as_test -T 2
machine: SampleMachine
owner: Administrator@SampleJob
permission:
date_conditions: 0
alarm_if_fail: 1
heartbeat_interval: 5
alarm_if_terminated: 1

```

- **Response Code: 200**
- **Response in XML Format**

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_50323 Inserting/Updating machine: SampleMachine</output>
 <output>CAUAJM_I_50205 Database Change WAS Successful!</output>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_50323 Inserting/Updating job: SampleJob</output>
 <output>CAUAJM_I_50205 Database Change WAS Successful!</output>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_52301 Exit Code = 0</output>
<output>_____</output>
 <output></output>
 <rc>0</rc>
 <status>success</status>
</cliResponse>

```

### Response in JSON Format:

```

{
 "output": [
 "_____",
 "",
 "CAUAJM_I_50323 Inserting/Updating machine: SampleMachine",
 "CAUAJM_I_50205 Database Change WAS Successful!",
 "_____",
 "",
 "CAUAJM_I_50323 Inserting/Updating job: SampleJob",
 "CAUAJM_I_50205 Database Change WAS Successful!",
 "_____",
 "",
 "CAUAJM_I_52301 Exit Code = 0",
 "_____"
]
}

```

```

],
 "rc": "0",
 "status": "success"
}

```

### Example2: POST https://WSServer:9443/AEWS/jil

Insert an existing machine and then insert an existing job to run on the machine. The request body must look as follows:

```

/* ----- SampleMachine ----- */
 insert_machine: SampleMachine
 type: a
 factor: 1.00
 port: 7520
 node_name: SampleMachine
 agent_name: WA_AGENT
 encryption_type: DEFAULT
 character_code: ASCII
/* ----- SampleJob ----- */
 insert_job: SampleJob job_type: CMD
 command: as_test -T 2
 machine: SampleMachine
 owner: Administrator@SampleJob
 permission:
 date_conditions: 0
 alarm_if_fail: 1
 heartbeat_interval: 5
 alarm_if_terminated: 1

```

### Response Code: 406

### Response in XML Format:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_50323 Inserting/Updating machine: SampleMachine</output>
 <output>CAUAJM_E_10305 *** CANNOT INSERT Machine: SampleMachine, because it ALREADY EXISTS ***</output>
 <output>CAUAJM_E_10302 Database Change WAS NOT successful.</output>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_50323 Inserting/Updating job: SampleJob</output>
 <output>CAUAJM_E_10288 *** CANNOT INSERT Job: SampleJob, because it ALREADY EXISTS ***</output>
 <output>CAUAJM_E_10302 Database Change WAS NOT successful.</output>
<output>_____</output>
 <output></output> <output>CAUAJM_E_50198 Exit Code = 2</output>
<output>_____</output>
 <output></output>
 <rc>2</rc>
 <status>failed</status>
</cliResponse>

```

### Response in JSON Format:

```

{

```

```

 "output": [
 "
 ",
 "CAUAJM_I_50323 Inserting/Updating machine: SampleMachine",
 "CAUAJM_E_10305 *** CANNOT INSERT Machine: SampleMachine, because it ALREADY EXISTS ***",
 "CAUAJM_E_10302 Database Change WAS NOT successful.",
 "
 ",
 "CAUAJM_I_50323 Inserting/Updating job: SampleJob",
 "CAUAJM_E_10288 *** CANNOT INSERT Job: SampleJob, because it ALREADY EXISTS ***",
 "CAUAJM_E_10302 Database Change WAS NOT successful.",
 "
 ",
 "CAUAJM_E_50198 Exit Code = 2",
 "
 "
],
 "rc": "2",
 "status": "failed"
 }

```

### Example3: POST https://WSServer:9443/AEWS/jil?verify=true

Verify syntax to insert an existing machine and then insert an existing job to run on the machine. The request body must look as follows:

```

/* ----- SampleMachine ----- */
insert_machine: SampleMachine
type: a
factor: 1.00
port: 7520
node_name: SampleMachine
agent_name: WA_AGENT
encryption_type: DEFAULT
character_code: ASCII
/* ----- SampleJob ----- */
insert_job: SampleJob job_type: CMD
command: as_test -T 2
machine: SampleMachine
owner: Administrator@SampleJob
permission:
date_conditions: 0
alarm_if_fail: 1
heartbeat_interval: 5
alarm_if_terminated: 1

```

### Response Code: 406

### Response in XML Format:

```

<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<cliResponse>
<output>_____</output>
 <output></output>
 <output>CAUAJM_I_50322 Checking Syntax for Inserting/Updating machine: SampleMachine</output>

```



```

<output>CAUAJM_W_10701 Machine 'SampleMachine' is defined.</output>
<output>CAUAJM_W_10701 Machine 'SampleMachine' is defined.</output>
<output>CAUAJM_I_50255 JIL syntax check PASSED WITH WARNINGS</output>
<output>_____</output>
<output></output>
<output>CAUAJM_I_50322 Checking Syntax for Inserting/Updating job: SampleJob</output>
<output>CAUAJM_W_10282 WARNING for Job: SampleJob. Job SampleJob is defined </output>
<output>CAUAJM_E_50246 JIL syntax check FAILED</output>
<output>_____</output>
<output></output>
<output>CAUAJM_W_50433 Warnings = 1</output>
<output>_____</output>
<output></output> <output>CAUAJM_E_50198 Exit Code = 1</output>
<output>_____</output>
<output></output>
<rc>1</rc>
<status>failed</status>
</cliResponse>

```

### Response in JSON Format:

```

{
 "output": [
 "_____",
 "",
 "CAUAJM_I_50322 Checking Syntax for Inserting/Updating machine: SampleMachine",
 "CAUAJM_W_10701 Machine 'SampleMachine' is defined.",
 "CAUAJM_W_10701 Machine 'SampleMachine' is defined.",
 "CAUAJM_I_50255 JIL syntax check PASSED WITH WARNINGS",
 "_____",
 "",
 "CAUAJM_I_50322 Checking Syntax for Inserting/Updating job: SampleJob",
 "CAUAJM_W_10282 WARNING for Job: SampleJob. Job SampleJob is defined ",
 "CAUAJM_E_50246 JIL syntax check FAILED",
 "_____",
 "",
 "CAUAJM_W_50433 Warnings = 1",
 "_____",
 "",
 "CAUAJM_E_50198 Exit Code = 1",
 "_____",
 ""
],
 "rc": "1",
 "status": "failed"
}

```

### GET AEWS/jil/{type} method – Exports JIL Definitions of the Given Type

The GET AEWS/jil/{type} method exports jil definitions of the given type based on the query parameters.

#### HTTP Method

GET

#### URL Resource

`https://webserver:port/AEWS/jil/{type}?parameters`

- **webserver**  
Specifies the name of the web server host computer.
- **port**  
Specifies the port number for the web server host computer.

### type

Specifies the type of the object to export the jil definition.

**Valid Values:** job, machine, resource, connection-profile, and xinst

- **job** Gets the job definitions
- **machine** Gets the machine definitions
- **resource** Gets the resource definitions
- **connection-profile** Gets the connection-profile definitions
- **xinst** Gets the xinst definitions

### Parameter

- **name**  
Specifies which object definition to export. If not specified, exports all the definitions of the given type.  
**Default:** ALL
- **timeout**  
Specifies the request timeout period in seconds. By default, the request times out if the operation is not completed in 300secs.  
**Default:** 300sec

### Request Formats

- TEXT\_PLAIN

**Example1:** Get all the job definitions

**GET** `https://WSServer:9443/AEWS/jil/job`

**Response Code:** 200

**Response body:**

```
/* ----- Sample1Box ----- */
 insert_job: Sample1Box job_type: BOX
 owner: Administrator@testmachine
 permission:
 date_conditions: 0
 description: "Box"
 alarm_if_fail: 1
 alarm_if_terminated: 1
/* ----- TESTJOB_00001 ----- */
 insert_job: TESTJOB_00001 job_type: CMD
 command: as_test -T 2
 machine: localhost
 owner: Administrator@testmachine
 permission:
 date_conditions: 0
 alarm_if_fail: 1
 heartbeat_interval: 5
 alarm_if_terminated: 1
```

```

/* ----- TESTJOB_00002 ----- */
insert_job: TESTJOB_00002 job_type: CMD
command: as_test -T 2
machine: localhost
owner: Administrator@testmachine
permission:
date_conditions: 0
alarm_if_fail: 1
heartbeat_interval: 5
alarm_if_terminated: 1

```

**Example2:** Get a specific job TESTJOB\_00001 definition

**GET** https://WSServer:9443/AEWS/jil/job?name=TESTJOB\_00001

**Response Code: 200**

**Response body:**

```

/* ----- TESTJOB_00001 ----- */
insert_job: TESTJOB_00001 job_type: CMD
command: as_test -T 2
machine: localhost
owner: Administrator@testmachine
permission:
date_conditions: 0
alarm_if_fail: 1
heartbeat_interval: 5
alarm_if_terminated: 1

```

**Example3:** Get a non existing TESTJOB\_0000100 job definition

**GET** https://WSServer:9443/AEWS/jil/job?name=TESTJOB\_0000100

**Response:**

**Response Code: 404**

**Response body:** CAUAJM\_E\_50027 Invalid Job Name: TESTJOB\_0000100

## **Troubleshooting**

This section describes a few common problems, their symptoms, and possible solutions.

- **Out of memory**

When you see the error message : **java.lang.OutOfMemoryError: Java heap space**, increase the Maximum Java Heap Size in the wrapper.conf web server file, per requirements.

**Example**

```

Maximum Java Heap Size (in MB)
wrapper.java.maxmemory=1024

```

- **CLI component missing on the Web Server**

The REST JIL requires the CLI component to be present on the Web Server. If not, the following error displays:

**UNIX:** bash: autorep: command not found.

**Windows:** 'autorep' is not recognized as an internal or external command, operable program, or batch file.

## Manage Jobs

Following are the procedures for managing jobs:

### GET AEWS/job/jobname method-- Retrieve Information of a Specific Job

The GET AEWS/job/*jobname* method returns information for a specific job.

#### HTTP Method

GET

#### URL Resource

```
https://webserver:port/AEWS/job/jobname?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### Parameters

- **jobname** Specifies the name of the job to retrieve. The name is required.
- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### Response Formats

- **xml** -- Specifies to return the data in XML format.
- **json** -- Specifies to return the data in JSON format.

#### Example: GET https://WSServer:9443/AEWS/job/sleep1

The command returns the following response based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/job/sleep1
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Wed, 01 Feb 2012 20:27:01 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><job><name>sleep1</name><jobType>CMD</jobType><machine>WSServer</machine><self rel="self" href="https://WSServer:9443/AEWS/job/sleep1"/><status>4</status></job>
```

- **JSON Format**

```
-- request --
```

```
GET https://WSServer:9443/AEWS/job/sleep1
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Wed, 01 Feb 2012 20:27:14 GMT
{"name":"sleep1","jobType":"CMD","machine":"WSServer","self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job/sleep1"},"status":"4"}
```

### **GET AEWS/job method-- Retrieve Information of a Single or Multiple Jobs**

The GET AEWS/job method returns information for one or more jobs based on the results of a specified query.

#### **HTTP Method**

GET

#### **URL Resource**

`https://webserver:port/AEWS/job?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **filter=FilterCriteria** (Optional) Specifies one or more expressions used to limit the data returned. Multiple expressions are represented by a job property, followed by an operator and a value. The job properties included in the response, such as name, boxName, machine, application, group, status, jobType, and description are allowed. The following operators are allowed:
  - == Match exactly
  - != Not an exact match
  - =@ Match the pattern. A wildcard can be used.
  - !@ Not match the pattern. A wildcard can be used.
  - < Less than
  - <= Less than or equal to
  - > Greater than
  - >= Greater than or equal to

Multiple expressions can be linked with either the and (;) and or (||) Boolean operators. For example, entering `application==abc;group==yyy|machine==localhost` returns all jobs that belong to the application abc and the group yyy or the jobs that are defined to run on the localhost machine. Multiple values can also be specified to create an or comparison. For example entering `group==abc,xyz` returns groups that match either abc or xyz.

**NOTE**

- When using `=@` or `!@` in the expression, you can use the asterisk (\*) as a wildcard for the entire name or for a partial name, where the \* represents zero or more characters. You can also use the question mark (?) to represent any single character.
- When you use `jobType` as the filter, specify the exact key word that you used in your job definition. The key word is case insensitive and wild cards are not allowed.
- From AutoSys Workload Automation Release 11.3.6 SP7 onwards, you can perform a case insensitive search for the description property.
- **count=*n***(Optional) Specifies the maximum number of records to return. If 0 is specified, previous and next references are not returned.
- **startIndex=*n*** (Optional) Specifies the starting logical page number, beginning with 1. A logical page contains the number of items specified by count. This option is ignored if count is set to 0.

**NOTE**

Although supported, we do not recommend using this parameter. All records must be read and processed before the requirement is fulfilled or cannot be met.

- **sortBy=*col1[,col2]...***  
(Optional) Specifies to order the results in the order of the specified elements. Multiple elements can be entered, separated by commas.  
**Default:** ascending order

**NOTE**

To specify descending order, prefix the element name with the hyphen (-).

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

**Response Formats**

- `xml` -- Specifies to return the data in XML format.
- `json` -- Specifies to return the data in JSON format.

**Example: GET https://WSServer:9443/AEWS/job?filter=machine==localhost**

The command returns the following response based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/job?filter=machine==localhost
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 03 Feb 2012 15:25:12 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?
><jobs><job><name>MAINT_CLEAN_AGENT_OUT</name><jobType>CMD</jobType><machine>localhost</
machine><self rel="self"href="https://WSServer:9443/AEWS/job/MAINT_CLEAN_AGENT_OUT"/
><status>4</status></job><job><name>MAINT_CLEAN_ARCHIVE</name><machine>localhost</
```

```
machine><self rel="self" href="https://WSServer:9443/AEWS/job/MAINT_CLEAN_ARCHIVE"/
><status>4</status></job><job><name>MAINT_CLEAN_SERVER_OUT</name><machine>localhost</
machine><self rel="self" href="https://WSServer:9443/AEWS/job/MAINT_CLEAN_SERVER_OUT"/
><status>4</status></job><self rel="self" href="https://WSServer:9443/AEWS/job?
filter=machine==localhost"/></jobs>
```

- **JSON Format**

```
-- request --
GET https://WSServer:9443/AEWS/job?filter=machine==localhost
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 03 Feb 2012 15:32:16 GMT
{"job":[{"name":"MAINT_CLEAN_AGENT_OUT","jobType":"CMD","machine":"localhost","self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job/
MAINT_CLEAN_AGENT_OUT"},"status":"4"},
{"name":"MAINT_CLEAN_ARCHIVE","machine":"localhost","self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job/
MAINT_CLEAN_ARCHIVE"},"status":"4"},
{"name":"MAINT_CLEAN_SERVER_OUT","machine":"localhost","self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job/
MAINT_CLEAN_SERVER_OUT"},"status":"4"}],"self":{"@rel":"self","@href":"https://
WSServer:9443/AEWS/job?filter=machine==localhost"}}
```

**Example: GET https://WSServer:9443/AEWS/job?filter=jobtype==b**

The command returns the following response based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/job?filter=jobtype==b
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 03 Feb 2012 15:25:12 GMT
```

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><jobs><job><name>box1</name><jobType>BOX</jobType><self href="https://WSServer:9443/AEWS/job/box1" rel="self"/><status>8</status></job><self href="https://WSServer:9443/AEWS/job?filter=jobtype==b" rel="self"/></jobs>
```

## • JSON Format

```
-- request --
GET https://WSServer:9443/AEWS/job?filter=jobtype==b
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 03 Feb 2012 15:32:16 GMT
{"job":[{"name":"box1","jobType":"BOX","self":{"@href":"https://WSServer:9443/AEWS/job/box1","@rel":"self"},"status":"8"}],"self":{"@href":"https://WSServer:9443/AEWS/job?filter=jobtype==b","@rel":"self"}}
```

## **GET AEWS/job-dependencies/jobname method -- Retrieve Job Dependency Information of a Specific Job**

The GET AEWS/job-dependencies/*jobname* method returns successor and predecessor information for a specific job.

### HTTP Method

GET

### URL Resource

`https://webserver:port/AEWS/job-dependencies/jobname?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### Parameters

- **jobname** Specifies the name of the job to retrieve. The name is required.
- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### Response Formats

- **xml** -- Specifies to return the data in XML format.
- **json** -- Specifies to return the data in JSON format.

### **Example: GET https://WSServer:9443/AEWS/job-dependencies/sleep1**

The command returns the following response based on the request type:

- **XML format**



```
-- request --
GET https://WSServer:9443/AEWS/job-dependencies/sleep1
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Content-Length: 597
Date: Thu, 12 Apr 2012 14:34:54 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?
><dependency><predecessors><predecessor><name>sleep1</name><type>job</type><self
rel="self" href="https://WSServer:9443/AEWS/job/sleep2"/></predecessor></
predecessors><self rel="self" href="https://WSServer:9443/AEWS/job-dependencies/sleep1"/
><successors><successor><self rel="self" href="https://WSServer:9443/AEWS/job/sleep3"/
><name>sleep3</name><type>job</type></successor></successors></dependency>
```

#### • JSON format

```
-- request --
GET https://WSServer:9443/AEWS/job-dependencies/sleep1
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Date: Thu, 12 Apr 2012 14:36:01 GMT
{"predecessors":{"predecessor":{"name":"sleep1","type":"job","self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job/sleep2"}}},"self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-dependencies/
sleep1"},"successors":{"successor":{"self":{"@rel":"self","@href":"https://
WSServer:9443/AEWS/job/sleep3"},"name":"sleep3","type":"job"}}}
```

### **GET AEWS/job-run-info/jobname method -- Retrieve Job Run Information of a Specific Job**

The GET AEWS/job-run-info/*jobname* method returns job run information for a specific job.

#### **HTTP Method**

GET

#### **URL Resource**

`https://webserver:port/AEWS/job-run-info/jobname?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **jobname**

Specifies the name of the job for which to retrieve the information. The name is required.

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **runnum=*n***  
(Optional) Specifies the run number for a job. The parameter cannot be used with iteration or runcount, and can only be used to retrieve information for an individual job. The jobname parameter is required when using this parameter.
- **runcount=*n***  
(Optional) Specifies the number of job runs to be retrieved for the input job. A value of 0 or 1 retrieves the last job run. This parameter cannot be used with iteration or runnum, and can only be used to retrieve information for an individual job. The jobname parameter is required when using this parameter  
**Default:** 0.
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US
- **verbose=0/1** (Optional) Specifies to return detailed information when verbose is set to 1 (true).  
**Default:** current server version
- **iteration=0[,*-1*][,*-2*]...**  
(Optional) Specifies the run iteration number from which to retrieve the information. The jobname parameter is required when using this parameter.

#### NOTE

0 equals the last iteration. Specify a negative number to get the results similar to the autorep -R command.

### Response Formats

- xml—Specifies to return the data in XML format.
- json—Specifies to return the data in JSON format.

#### Example: GET https://WSServer:9443/AEWS/job-run-info/test

The command returns the following response based on the request type:

#### • XML format

```
-- request --
GET https://WSServer:9443/AEWS/job/job-run-info/test
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Content-Length: 542
Date: Mon, 21 May 2012 16:23:09 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><jobRuns><jobRun><avgRunTime>0</avgRunTime><machine>localhost</machine><runEndTime>2012-05-21T15:36:27+0000</runEndTime><exitCode>0</exitCode><name>test</name><runMachine>WSServer</runMachine><runNum>64035</runNum><self rel="self" href="https://WSServer:9443/AEWS/job-run-info/test?runnum=64035"/><runStartTime>2012-05-21T15:36:27+0000</runStartTime><status>4</status></jobRun><self rel="self" href="https://WSServer:9443/AEWS/job-run-info/test"/></jobRuns>
```

#### • JSON format

```
-- request --
```

```
GET https://WSServer:9443/AEWS/job/job-run-info/test
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Date: Mon, 21 May 2012 16:17:12 GMT
{"jobRun":
{"avgRunTime":"0","machine":"localhost","runEndTime":"2012-05-21T15:36:27+0000","exitCode":"0",
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info/test?
runnum=64035"},"runStartTime":"2012-05-21T15:36:27+0000","status":"4"},"self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info/test"}}
```

### **GET AEWS/job-run-info method -- Retrieve Job Run Information of a Single or Multiple Jobs**

The GET AEWS/job-run-info method returns job run information for one or more jobs based on the results of a specified query.

#### **HTTP Method**

GET

#### **URL Resource**

`https://webserver:port/AEWS/job-run-info?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **filter=FilterCriteria**  
(Optional) Specifies one or more expressions used to limit the data returned. Multiple expressions are represented by a job property, followed by an operator and a value. The job properties included in the response, such as name, boxName, machine, application, group, status, and description are allowed. The following operators are allowed:

- == Match exactly
- != Not an exact match
- =@ Match the pattern. A wildcard can be used.
- !@ Not match the pattern. A wildcard can be used.
- < Less than
- <= Less than or equal to
- > Greater than
- >= Greater than or equal to

Multiple expressions can be linked with either the and (;) and or (|) Boolean operators. For example, entering `application==abc;group==yyy|machine==localhost` returns all jobs that belong to the application abc and the group yyy or the jobs that are defined to run on the localhost machine. Multiple values can also be specified to create an or comparison. For example entering `group==abc,xyz` returns groups that match either abc or xyz.

**NOTE**

- When using `=@` or `!@` in the expression, you can use the asterisk (\*) as a wildcard for the entire name or for a partial name, where the \* represents zero or more characters. You can also use the question mark (?) to represent any single character.
- The following elements are not supported for filtering:
  - evtNum
  - type
  - machine
  - ntry
  - status
  - gvar
  - gvarValue
  - text
  - queuedTime
  - processedTime
  - alarm
  - stepName
- From AutoSys Workload Automation Release 11.3.6 SP7 onwards, you can perform a case insensitive search for the description property.
- **count=*n***(Optional) Specifies the maximum number of records to return. If 0 is specified, previous and next references are not returned.
- **index=*n*** (Optional) Specifies a referenced starting position when returning a list (internal use only).
- **startIndex=*n*** (Optional) Specifies the starting logical page number, beginning with 1. A logical page contains the number of items specified by count. This option is ignored if count is set to 0.

**NOTE**

Although supported, we do not recommend using this parameter. All records must be read and processed before the requirement is fulfilled or cannot be met.

- **sortBy=*col1[,col2]...***  
(Optional) Specifies to order the results in the order of the specified elements. Multiple elements can be entered, separated by commas.  
**Default:** ascending order

**NOTE**

- To specify descending order, prefix the element name with the hyphen (-).
- The following elements are not supported for sorting:

- evtNum
- type
- machine
- ntry
- status
- gvar
- gvarValue
- text
- queuedTime
- processedTime
- alarm
- stepName

- **verbose=0/1**(Optional) Specifies to return detailed information when verbose is set to 1 (true).  
**Default:** 0
- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

### Response Formats

- **xml**—Specifies to return the data in XML format.
- **json**—Specifies to return the data in JSON format.

**Example: GET https://WSServer:9443/AEWS/job-run-info?filter=name=@test\***

The command returns the following response based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/job-run-info?filter=name=@test*
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Content-Length: 1336
Date: Mon, 21 May 2012 16:47:41 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><jobRuns><jobRun><avgRunTime>0</
avgRunTime><machine>localhost</machine><runEndTime>2012-05-21T15:36:27+0000</
runEndTime><exitCode>0</exitCode><name>test</name><runMachine>WSServer</
runMachine><runNum>64035</runNum><self rel="self" href="https://WSServer:9443/
AEWS/job-run-info/test?runnum=64035"/><runStartTime>2012-05-21T15:36:27+0000</
runStartTime><status>4</status></jobRun><jobRun><avgRunTime>0</
avgRunTime><machine>localhost</machine><runEndTime>2012-05-21T15:36:29+0000</
runEndTime><exitCode>0</exitCode><name>test1</name><runMachine>WSServer</
runMachine><runNum>64036</runNum><self rel="self" href="https://WSServer:9443/
AEWS/job-run-info/test1?runnum=64036"/><runStartTime>2012-05-21T15:36:29+0000</
runStartTime><status>4</status></jobRun><jobRun><avgRunTime>0</
avgRunTime><machine>localhost</machine><runEndTime>2012-05-21T15:36:31+0000</
runEndTime><exitCode>0</exitCode><name>test2</name><runMachine>WSServer</
```

```
runMachine><runNum>64037</runNum><self rel="self" href="https://WSServer:9443/AEWS/job-run-info/test2?runnum=64037"/><runStartTime>2012-05-21T15:36:31+0000</runStartTime><status>4</status></jobRun><self rel="self" href="https://WSServer:9443/AEWS/job-run-info?filter=name=@test*" /></jobRuns>
```

## • JSON Format

```
-- request --
GET https://WSServer:9443/AEWS/job-run-info?filter=name=@test*
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Date: Mon, 21 May 2012 16:52:29 GMT
{"jobRun":
[{"avgRunTime":"0","machine":"localhost","runEndTime":"2012-05-21T15:36:27+0000","exitCode":"0",
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info/test?
runnum=64035"},"runStartTime":"2012-05-21T15:36:27+0000","status":"4"},
{"avgRunTime":"0","machine":"localhost","runEndTime":"2012-05-21T15:36:29+0000","exitCode":"0",
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info/test1?
runnum=64036"},"runStartTime":"2012-05-21T15:36:29+0000","status":"4"},
{"avgRunTime":"0","machine":"localhost","runEndTime":"2012-05-21T15:36:31+0000","exitCode":"0",
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info/test2?
runnum=64037"},"runStartTime":"2012-05-21T15:36:31+0000","status":"4"}], "self":
{"@rel":"self","@href":"https://WSServer:9443/AEWS/job-run-info?filter=name=@test*"}}
```

## **POST AEWS/event/cancel-job -- Cancel a Job**

The POST AEWS/event/cancel-job method sends a KILLJOB event to a specific job, group, or application.

### **HTTP Method**

POST

### **URL Resource**

```
https://webserver:port/AEWS/event/cancel-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

## Content

To send the cancel-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <signal>2</signal>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","signal":"2","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **signal\_name/number**  
(Optional) Specifies the signal to send to the running job.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### Example: POST https://WSServer:9443/AEWS/event/cancel-job

The command sends a KILLJOB event to the job test.

#### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/cancel-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing cancel-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
```

```
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

#### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/cancel-job
Content-Type: application/json
{"jobName":"test","comment": "testing cancel-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

### **POST AEWS/event/force-start-job -- Start a Job by Force**

The POST AEWS/event/force-start-job method sends a FORCE\_STARTJOB event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

```
https://webserver:port/AEWS/event/force-start-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

#### **Content**

To send the force-start job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```



The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### Example: POST https://WSServer:9443/AEWS/event/force-start-job

The command sends the FORCE\_STARTJOB event to the job test.

##### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/force-start-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing force-start-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

##### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/force-start-job
Content-Type: application/json
{"jobName":"test","comment":"testing force-start-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
```

```
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

## **POST AEWS/event/change-status-job -- Change the Job Status**

The POST AEWS/event/change-status-job method changes the status of a job. You can change the job status to any one of the following statuses:

- FAILURE
- INACTIVE
- RUNNING
- STARTING
- SUCCESS
- TERMINATED

### **HTTP Method**

POST

### **URL Resource**

```
https://webserver:port/AEWS/event/change-status-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:**current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:**en-US

### **Request Formats**

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### **Content**

To send the change-status-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <jobStatus>4</jobStatus>
</command>
```

The following is an example of the content data in JSON format:

```
{ "jobName": "myJob", "jobStatus": "4" }
```

- **jobName**

Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.

- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **jobStatus**  
Specifies the status to which the job is changed to. In the example, the job status is changed to SUCCESS.

#### Example: POST https://WSServer:9443/AEWS/event/change-status-job

The command changes the status of the job.

#### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/change-status-job
Content-Type: application/xml
<command>
<jobName>myJob</jobName>
<jobStatus>4</jobStatus>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

#### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/change-status-job
Content-Type: application/json
{"jobName":"myJob","jobStatus":"4"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

## **POST AEWS/event/hold-job -- Put a Job in ON\_HOLD status**

The POST AEWS/event/hold-job method sends a JOB\_ON\_HOLD event to a specific job, group, or application.

### **HTTP Method**

POST

### **URL Resource**

```
https://webserver:port/AEWS/event/hold-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:**current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:**en-US

### **Request Formats**

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### **Content**

To send the hold-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

### **NOTE**

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.

- **comment**

(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

**Example: POST https://WSServer:9443/AEWS/event/hold-job**

The command sends the JOB\_ON\_HOLD event to the job test.

- **XML Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/hold-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing hold-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/hold-job
Content-Type: application/json
{"jobName":"test","comment":"testing hold-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

**POST AEWS/event/ice-job -- Put a Job in ON\_ICE status**

The POST AEWS/event/ice-job method sends a JOB\_ON\_ICE event to a specific job, group, or application.

**HTTP Method**

POST

**URL Resource**

https://webserver:port/AEWS/event/ice-job?parameters

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

**Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### Request Formats

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### Content

To send the ice-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
  - You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### Example: POST https://WSServer:9443/AEWS/event/ice-job

The command sends the JOB\_ON\_ICE event to the job test.

#### XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/ice-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
```

```
<comment>testing ice-job</comment>
</command>
 -- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

#### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/ice-job
Content-Type: application/json
{"jobName":"test","comment":"testing ice-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

### **POST AEWS/event/noexec-job -- Put a Job in ON\_NOEXEC status**

The POST AEWS/event/noexec-job method sends an ON\_NOEXEC event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

`https://webserver:port/AEWS/event/noexec-job?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

#### **Content**

To send the noexec-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### Example: POST https://WSServer:9443/AEWS/event/noexec-job

The command sends the ON\_NOEXEC event to the job test.

##### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/noexec-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing noexec-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 21 Feb 2012 18:15:54 GMT
```

##### • JSON Format

```
--request--
```



```

POST https://WSServer:9443/AEWS/event/noexec-job
Content-Type: application/json
{"jobName":"test","comment":"testing noexec-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 21 Feb 2012 18:23:02 GMT

```

### **POST AEWS/event/off-hold-job -- Take a Job Off Hold**

The POST AEWS/event/off-hold-job method sends a JOB\_OFF\_HOLD event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

```
https://webserver:port/AEWS/event/off-hold-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

#### **Content**

To send the off-hold-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```

<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>

```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**

Specifies the application with which the relevant jobs are associated.

- **group**

Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.

- **comment**

(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### Example: POST https://WSServer:9443/AEWS/event/off-hold-job

The command sends the JOB\_OFF\_HOLD event to the job test.

- **XML Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/off-hold-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing off-hold-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/off-hold-job
Content-Type: application/json
{"jobName":"test","comment":"testing off-hold-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

## **POST AEWS/event/off-ice-job -- Take a Job Off Ice**

The POST AEWS/event/off-ice-job method sends a JOB\_OFF\_ICE event to a specific job, group, or application.

### **HTTP Method**

POST

### **URL Resource**

```
https://webserver:port/AEWS/event/off-ice-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

### **Content**

To send the off-ice-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

### **NOTE**

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.

- **comment**

(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### **Example: POST https://WSServer:9443/AEWS/event/off-ice-job**

The command sends the JOB\_OFF\_ICE event to the job test.

- **XML Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/off-ice-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing off-ice-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/off-ice-job
Content-Type: application/json
{"jobName":"test","comment":"testing off-ice-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

#### **POST AEWS/event/off-noexec-job -- Take a Job Off the NOEXEC Status**

The POST AEWS/event/off-noexec-job method sends an OFF\_NOEXEC event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

```
https://webserver:port/AEWS/event/off-noexec-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### Request Formats

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### Content

To send the off-noexec-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### Example: POST https://WSServer:9443/AEWS/event/off-noexec-job

The command sends the OFF\_NOEXEC event to the job test.

#### XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/off-noexec-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
```

```
<comment>testing off-noexec-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 21 Feb 2012 18:26:57 GMT
```

#### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/off-noexec-job
Content-Type: application/json
{"jobName":"test","comment":"testing off-noexec-job"}
--response--
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 21 Feb 2012 18:26:57 GMT
```

### **POST AEWS/event/release-resources-job -- Release Resources Held by a Job**

The POST AEWS/event/release-resources-job method sends a RELEASE\_RESOURCE event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

```
https://webserver:port/AEWS/event/release-resources-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

#### **Content**

To send the release-resources-job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName** Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application** Specifies the application with which the relevant jobs are associated.
- **group** Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment** (Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### Example: POST https://WSServer:9443/AEWS/event/release-resources-job

The command sends the RELEASE\_RESOURCE event to the job test.

##### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/release-resources-job
Content-Type: application/xml
<command>
<jobName>myJob</jobName>
<comment>testing release-resources-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 24 Feb 2012 19:30:19 GMT
```

##### • JSON Format

```
-- request --
```

```

POST https://WSServer:9443/AEWS/event/release-resources-job
Content-Type: application/json
{"jobName":"test","comment":"testing release-resources-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Fri, 24 Feb 2012 19:48:14 GMT

```

### **POST AEWS/event/reply-job -- Reply to a Job**

The POST AEWS/event/reply-job method sends a REPLY\_RESPONSE event to a specific job.

#### **HTTP Method**

POST

#### **URL Resource**

```
https://webserver:port/AEWS/event/reply-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

#### **Content**

To send the reply-job event, you must include the command data specifying the job for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```

<command>
 <jobName>myJob</jobName>
 <response>YES</response>
</command>

```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","response":"YES"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **response**



Specifies the response that is sent to a running job.

**Example: POST https://WSServer:9443/AEWS/event/reply-job**

The command sends the REPLY\_RESPONSE event to the job test.

- **XML Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/reply-job
Content-Type: application/xml
<command>
<jobName>myJob</jobName>
<response>YES</response>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Wed, 29 Feb 2012 07:22:08 GMT
```

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/reply-job
Content-Type: application/json
{"jobName":"test","response":"YES"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Wed, 29 Feb 2012 07:15:25 GMT
```

**POST AEWS/event/restart-job -- Restart a Job**

The POST AEWS/event/restart-job method sends a RESTARTJOB event to a specific job. The command is applicable to a failed z/OS, Micro Focus, or Informatica job only.

**HTTP Method**

POST

**URL Resource**

```
https://webserver:port/AEWS/event/restart-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

**Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### Request Formats

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### Content

To send the restart job event, you must include the command data specifying the job and step name for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <stepName>myJobStepName</stepName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName": "myJob", "stepName": "myJobStepName", "comment": "reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **stepName**  
Specifies the step name or task name for the RESTARTJOB event. This parameter is mandatory for Microfocus jobs and optional for Informatica jobs. It is not supported for SAP BW Process Chain jobs.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### Example: POST https://WSServer:9443/AEWS/event/restart-job

The command sends the RESTART JOB event with step name to the job test.

#### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/restart-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<stepName>sn1</stepName>
<comment>testing restart-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
```

```
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/restart-job
Content-Type: application/json
{"jobName":"test","stepName":"sn1","comment":"testing restart-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

### **POST AEWS/event/start-job -- Start a Job**

The POST AEWS/event/start-job method sends the STARTJOB event to a specific job, group, or application.

#### **HTTP Method**

POST

#### **URL Resource**

`https://webserver:port/AEWS/event/start-job?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- **xml** -- Specifies to send the data in XML format.
- **json** -- Specifies to send the data in JSON format.

#### **Content**

To send the start job event, you must include the command data specifying the job, group, or application for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <jobName>myJob</jobName>
 <comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent. When specifying a job name, you cannot specify an application or group.
- **application**  
Specifies the application with which the relevant jobs are associated.
- **group**  
Specifies the group with which the relevant jobs are associated.

#### NOTE

- You can specify an application, a group, or both. However, you cannot specify a job name and an application or group. For example, if you specify an application with the name of Pay and a Group with the name of Pay, the event is sent to the jobs that match both parameters. If you send an event to a job with the name of Weekly and an application with the name of Pay, an error is returned.
- You can use an asterisk (\*) as a wildcard to represent multiple characters, or use a question mark (?) as a wildcard to represent any single character for the application or group name.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

#### Example: POST https://WSServer:9443/AEWS/event/start-job

The command sends the STARTJOB event to the job test.

#### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/start-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing start-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 31 Jan 2012 13:06:47 GMT
```

#### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/start-job
Content-Type: application/json
{"jobName":"test","comment":"testing start-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
```

```
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Mon, 06 Feb 2012 21:16:20 GMT
```

## **POST AEWS/event/suspend-job -- Suspend a Job**

The POST AEWS/event/suspend-job method sends a SUSPENDJOB event to a job. The command is applicable to a running Hadoop Oozie job only.

### **HTTP Method**

POST

### **URL Resource**

```
https://webserver:port/AEWS/event/suspend-job?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### **Request Formats**

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### **Content**

To send the SUSPENDJOB event, you must include the command data specifying the job for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
<jobName>myJob</jobName>
<comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- **jobName**  
Specifies the name of the job to which the event is sent.
- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### **Example: POST https://WSServer:9443/AEWS/event/suspend-job**

The command sends the SUSPENDJOB event to the job test.

- **XML Format**  
  
-- request --

```

POST https://WSServer:9443/AEWS/event/suspend-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing suspend-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 18 Aug 2015 09:46:14 GMT

```

#### • JSON Format

```

-- request --
POST https://WSServer:9443/AEWS/event/suspend-job
Content-Type: application/json
{"jobName":"test","comment":"testing suspend-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 18 Aug 2015 09:46:14 GMT

```

### **POST AEWS/event/resume-job -- Resume a Job**

The POST AEWS/event/resume-job method sends a RESUMEJOB event to a job. The command is applicable to a suspended Hadoop Oozie job only.

#### **HTTP Method**

POST

#### **URL Resource**

`https://webserver:port/AEWS/event/resume-job?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- `xml` -- Specifies to send the data in XML format.
- `json` -- Specifies to send the data in JSON format.

## Content

To send the RESUMEJOB event, you must include the command data specifying the job for which you are sending the event. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
<jobName>myJob</jobName>
<comment>reason for sending event</comment>
</command>
```

The following is an example of the content data in JSON format:

```
{"jobName":"myJob","comment":"reason for sending event"}
```

- ***jobName***  
Specifies the name of the job to which the event is sent.
- ***comment***  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

## Example: POST `https://WSServer:9443/AEWS/event/resume-job`

The command sends the RESUMEJOB event to the job test.

### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/resume-job
Content-Type: application/xml
<command>
<jobName>test</jobName>
<comment>testing resume-job</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 18 Aug 2015 09:46:14 GMT
```

### • JSON Format

```
-- request --
POST https://WSServer:9443/AEWS/event/resume-job
Content-Type: application/json
{"jobName":"test","comment":"testing resume-job"}
-- response --
201 Created
Server: Apache-Coyote/1.1
```

```
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Tue, 18 Aug 2015 09:46:14 GMT
```

## Manage Machines

Following are the procedures for managing machines:

### GET AEWS/machine/machinename method—Retrieve Information of a Specific Machine

The GET AEWS/machine/machinename method returns information for a specific machine.

#### HTTP Method

GET

#### URL Resource

```
https://webserver:port/AEWS/machine/machinename?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.
- **machinename** Specifies the name of the machine to retrieve. The name is required.

#### Parameters

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

#### NOTE

- If version =1, deleted machine is not returned.
- If version=2, the machine can be in deleted state.
  - Every machine has a field that is named "state". The value for the state field can be New, Updated, or Deleted.
  - The machineMethod is displayed.
  - The operating system type of the agent (detectedOS) is displayed. This information is displayed only after the agent communicates with the scheduler.
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### Response Formats

- **xml**—Returns the data in XML format.
- **json**—Returns the data in JSON format.

#### Example: GET https://WSServer:9443/AEWS/machine/mach1

The command returns the following response that is based on the request type:

#### • XML Format

```
-- request --
GET https://WSServer:9443/AEWS/machine/mach1
-- response --
```



```

200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Date: Wed, 28 May 2014 14:12:34 GMT
<?xml version="1.0" encoding="UTF-8"
standalone="yes"?><machine><advancedIntegrations>hadoop</
advancedIntegrations><agentBuild>795</agentBuild>
<agentMaintenanceLevel>2</agentMaintenanceLevel><agentName>WA_AGENT</
agentName><agentRelease>11.3</agentRelease>
<agentServicePack>4</agentServicePack><characterCode>ASCII</characterCode><factor>1.0</
factor>
<heartBeatAttempts>0</heartBeatAttempts><heartBeatFrequency>0</
heartBeatFrequency><name>mach1</name>
<status>o</status><machineMethod>0</machineMethod><detectedOS>Windows 64-bit</
detectedOS><maxLoad>0</maxLoad><nodeName>mach1</nodeName><platform>0</platform>
<pluginList>appservices,clearfiles,communicationmanager,dataservices,filebrowser,filemon,nop,oa
pluginList>
<port>7520</port><self href="https://WSServer:9443/AEWS/machine/mach1" rel="self"/
><type>a</type></machine>

```

## • JSON Format

```

-- request --
GET https://WSServer:9443/AEWS/machine/mach1
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-EncodingDate: Wed, 28 May 2014 14:12:34 GMT
{"machine":
[{"advancedIntegrations":"hadoop","agentBuild":"795","agentMaintenanceLevel":"2","agentName":"W
"agentRelease":"11.3","agentServicePack":"4","characterCode":"ASCII","factor":"1.0","heartBeatA
"heartBeatFrequency":"0","name":"mach1","status":"o","maxLoad":"0","nodeName":"mach1","platform
"pluginList":"appservices,clearfiles,communicationmanager,dataservices,filebrowser,filemon,nop,
"port":"7520","self":{"@href":"https://WSServer:9443/AEWS/machine/
mach1","@rel":"self"},"type":"a"}],
"self":{"@href":"https://WSServer:9443/AEWS/machine","@rel":"self"}}

```

## GET AEWS/machine method—Retrieve Information of a Single or Multiple Machines

The GET AEWS/machine method returns information for one or more machines that are based on the results of a specified query.

### HTTP Method

GET

### URL Resource

`https://webserver:port/AEWS/machine?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

### Parameters

- **filter=FilterCriteria** (Optional) Specifies one or more expressions that are used to limit the data returned. Multiple expressions are represented by a machine property, followed by an operator and a value. The machine properties included in the response, such as name, agentBuild, agentMaintenanceLevel, agentName, agentRelease, agentServicePack, characterCode, factor, heartBeatAttempts, heartBeatFrequency, description, status, machineMethod, detectedOS, maxLoad, nodeName, parentName, pluginList, advancedIntegrations, platform, port, createStamp, updateStamp, isActive, and type are allowed. The following operators are allowed:
  - == Match exactly
  - != Not an exact match
  - =@ Match the pattern. A wildcard can be used.
  - !@ Not match the pattern. A wildcard can be used.
  - < Less than
  - <= Less than or equal to
  - > Greater than
  - >= Greater than or equal to

Multiple expressions can be linked with either the and (;) and or (!) Boolean operators. For example, entering `port==7520;agentName==WA_AGENT|nodeName==mach1` returns all machines that have an agentName of WA\_AGENT and a port of 7520 or the machines that are on the physical machine named mach1. Multiple values can also be specified to create an or comparison. For example, entering `name==abc,xyz` returns names that match either abc or xyz.

### NOTE

- When using =@ or !@ in the expression, you can use the asterisk (\*) as a wildcard for the entire name or for a partial name, where the \* represents zero or more characters. You can also use the question mark (?) to represent any single character.
- If the max\_load attribute is not specified in the machine definition, the web service returns a value of 0.0 for the maxLoad parameter by default.
- If the factor attribute is not specified in the machine definition, the web service returns a value of 1.0 (for a real machine), 0.0 (for a virtual machine), or -1.0 (for a real machine associated with a virtual machine) for the factor parameter by default.
- The parentName parameter displays the machine pool name or the virtual machine name that is associated with a real machine.
- From AutoSys Workload Automation Release 11.3.6 SP7 onwards, you can perform a case insensitive search for the description property.

- **count=*n*** (Optional) Specifies the maximum number of records to return. If 0 is specified, previous and next references are not returned.
- **startIndex=*n*** (Optional) Specifies the starting logical page number, beginning with 1. A logical page contains the number of items specified by count. This option is ignored if count is set to 0.

**NOTE**

Although supported, we do not recommend using this parameter. All records must be read and processed before the requirement is fulfilled or cannot be met.

- **sortBy=*col1*[,*col2*]...**  
(Optional) Specifies to order the results in the order of the specified elements. Multiple elements can be entered, separated by commas.  
**Default:** ascending order

**NOTE**

To specify descending order, prefix the element name with the hyphen (-).

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version

**NOTE**

- If version =1, deleted machine is not returned.
- If version=2, the machine can be in deleted state.
  - Every machine has a field that is named "state". The value for the state field can be New, Updated, or Deleted.
  - The machineMethod is displayed.
  - The operating system type of the agent (detectedOS) is displayed. This information is displayed only after the agent communicates with the scheduler.

**Query Parameter**

The response header contains a new field that is called Query Parameter which specifies the database query execution time to retrieve the list of machines. This header value is used in subsequent API call as an updateStamp or createStamp filter value to get delta machine records from last API call.

For example, updateStamp>lastQueryTimeHeader|createStamp>lastQueryTimeHeader returns all machines that are created, updated, and deleted from the last query time (lastQueryTimeHeader).

updateStamp>lastQueryTimeheader|createStamp>lastQueryTimeHeader;isActive==1 returns all machines that are created and updated from the last query time (lastQueryTimeHeader).

**Response Formats**

- **xml**—Returns the data in XML format.
- **json**—Returns the data in JSON format.

**Example: GET https://WSServer:9443/AEWS/machine?filter=nodeName==mach1**

The command returns the following response that is based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/machine?filter=nodeName==mach1

-- response --
200 OK
Server: Apache-Coyote/1.1
```

```

Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Wed, 28 May 2014 15:45:07 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><machines><machine>
<advancedIntegrations>hadoop</advancedIntegrations><agentBuild>795</agentBuild>
<agentMaintenanceLevel>2</agentMaintenanceLevel><agentName>WA_AGENT</agentName>
<agentRelease>11.3</agentRelease><agentServicePack>4</agentServicePack>
<characterCode>ASCII</characterCode><factor>1.0</factor><heartBeatAttempts>0</
heartBeatAttempts>
<heartBeatFrequency>0</heartBeatFrequency><name>mach1</name><status>o</
status><machineMethod>0</machineMethod><detectedOS>Windows 64-bit</detectedOS>
<maxLoad>0</maxLoad><nodeName>mach1</nodeName><platform>0</platform>
<pluginList>appservices,clearfiles,communicationmanager,dataservices,filebrowser,filemon,nop,oa
pluginList>
<port>7520</port><self href="https://WSServer:9443/AEWS/machine/mach1" rel="self"/>
<type>a</type></machine><self href="https://WSServer:9443/AEWS/machine?
filter=nodeName==mach1" rel="self"/></machines>

```

## • JSON Format

```

-- request --
GET https://WSServer:9443/AEWS/machine?filter=nodeName==mach1

-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Content-Encoding: gzip
Vary: Accept-Encoding
Wed, 28 May 2014 15:45:07 GMT
{"machine":
[{"advancedIntegrations":"hadoop","agentBuild":"795","agentMaintenanceLevel":"2",
"agentName":"WA_AGENT","agentRelease":"11.3","agentServicePack":"4","characterCode":"ASCII",
"factor":"1.0","heartBeatAttempts":"0","heartBeatFrequency":"0","name":"mach1",
"status":"o","maxLoad":"0","nodeName":"mach1","platform":"0",
"pluginList":"appservices,clearfiles,communicationmanager,dataservices,filebrowser,filemon,nop,
"port":"7520","self":{"@href":"https://WSServer:9443/AEWS/machine/mach1","@rel":"self"},
"type":"a"}], "self":{"@href":"https://WSServer:9443/AEWS/machine?
filter=advancedIntegrations==hadoop","@rel":"self"}}

```

## Manage Global Variables

Following are the procedures for managing global variables:

### **GET AEWS/gvar/variableName method -- Retrieve Information of a Specific Global Variable**

The GET AEWS/gvar/variableName method returns information for a specific global variable.

#### **HTTP Method**

GET

#### **URL Resource**

```
https://webserver:port/AEWS/gvar/variableName?parameters
```

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **variableName**  
Specifies the name of the global variable for which to retrieve the value. The variable name is required.
- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Response Formats**

- **xml** -- Specifies to return the data in XML format.
- **json** -- Specifies to return the data in JSON format.

#### **Example: GET https://WSServer:9443/AEWS/gvar/TestVar**

The command returns the following response based on the request type:

- **XML format**

```
-- request --
GET https://WSServer:9443/AEWS/gvar/TestVar
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Content-Length: 243
Date: Fri, 11 May 2012 18:18:40 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><gvar><name>TestVar</name><value>TestVariables</value><self rel="self" href="https://WSServer:9443/AEWS/gvar/TestVar"/><updateTime>2012-05-11T17:35:22+0000</updateTime></gvar>
```

- **JSON format**

```
-- request --
```

```
GET https://WSServer:9443/AEWS/gvar/TestVar
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Date: Fri, 11 May 2012 18:25:29 GMT
{"name":"TestVar","value":"TestVariables","self":{"@rel":"self","@href":"https://
WSServer:9443/AEWS/gvar/TestVar"},"updateTime":"2012-05-11T17:35:22+0000"}
```

### **GET AEWS/gvar method -- Retrieve Information of a Single or Multiple Global Variables**

The GET AEWS/gvar method returns information for one or more global variables based on the results of a specified query.

#### **HTTP Method**

GET

#### **URL Resource**

`https://webserver:port/AEWS/gvar?parameters`

- **webserver** Specifies the name of the web server host computer.
- **port** Specifies the port number for the web server host computer.

#### **Parameters**

- **filter=FilterCriteria** (Optional) Specifies one or more expressions that are used to limit the data returned. Multiple expressions are represented by a global variable property, followed by an operator and a value. All global variable properties included in the response are valid, such as name and update time. The following operators are allowed:
  - == Match exactly
  - != Not an exact match
  - =@ Match the pattern. A wildcard can be used.
  - !@ Not match the pattern. A wildcard can be used.
  - < Less than
  - <= Less than or equal to
  - > Greater than
  - >= Greater than or equal to

Multiple expressions can be linked with either the and (;) and or (!) Boolean operators. For example, entering `name=@a*;updateTime>=2012-01-01T06:00:00-05:00|name==xyz` returns all global variables beginning with the letter a and updated on January 1, 2012 at 6:00 a.m. in the GMT-5 timezone, or all global variables with the name xyz. Multiple values can also be specified to create an or comparison. For example entering `name==abc,xyz` returns global variables that match either abc or xyz.

**NOTE**

When using `=@` or `!@` in the expression, you can use the asterisk (\*) as a wildcard for the entire name or for a partial name, where the \* represents zero or more characters. You can also use the question mark (?) to represent any single character.

- **count=*n*** (Optional) Specifies the maximum number of records to return. If 0 is specified, previous and next references are not returned.
- **index=*n*** (Optional) Specifies a referenced starting position when returning a list (internal use only).
- **startIndex=*n*** (Optional) Specifies the starting logical page number, beginning with 1. A logical page contains the number of items specified by count. This option is ignored if count is set to 0.

**NOTE**

Although supported, we do not recommend using this parameter. All records must be read and processed before the requirement is fulfilled or cannot be met.

- **sortBy=*col1[,col2]...*** (Optional) Specifies to order the results in the order of the specified elements. Multiple elements can be entered, separated by commas.

**Default:** ascending order

**NOTE**

To specify descending order, prefix the element name with the hyphen (-).

- **version=*n*** (Optional) Specifies the version for the request or response.

**Default:** current server version

**Response Formats**

- `xml` -- Specifies to return the data in XML format.
- `json` -- Specifies to return the data in JSON format.

**Example:** `GET https://WSServer:9443/AEWS/gvar?filter=name=@gvar*`

The command returns the following response based on the request type:

- **XML Format**

```
-- request --
GET https://WSServer:9443/AEWS/gvar?filter=name=@gvar*
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/xml
Content-Length: 518
Date: Mon, 21 May 2012 18:03:43 GMT
<?xml version="1.0" encoding="UTF-8" standalone="yes"?><gvars><gvar><name>gvar</
name><value>gvar gvar</value><self rel="self" href="https://WSServer:9443/AEWS/gvar/
gvar"/><updateTime>2012-05-21T17:43:05+0000</updateTime></gvar><gvar><name>gvar gvar</
name><value>gvar gvar</value><self rel="self" href="https://WSServer:9443/AEWS/gvar/
gvar%20gvar"/><updateTime>2012-05-21T17:42:55+0000</updateTime></gvar><self rel="self"
href="https://WSServer:9443/AEWS/gvar?filter=name=@gvar*" /></gvars>
```

- **JSON Format**

```
-- request --
```

```
GET https://WSServer:9443/AEWS/gvar?filter=name=@gvar*
-- response --
200 OK
Server: Apache-Coyote/1.1
Cache-Control: private
Expires: Wed, 31 Dec 1969 19:00:00 EST
Content-Type: application/json
Transfer-Encoding: chunked
Date: Mon, 21 May 2012 18:07:47 GMT
{"gvar":[{"name":"gvar","value":"gvar gvar","self":{"@rel":"self","@href":"https://WSServer/AEWS/gvar/gvar"},"updateTime":"2012-05-21T17:43:05+0000"}, {"name":"gvar gvar","value":"gvar gvar","self":{"@rel":"self","@href":"https://WSServer/AEWS/gvar/gvar%20gvar"},"updateTime":"2012-05-21T17:42:55+0000"}], "self":{"@rel":"self","@href":"https://WSServer:9443/AEWS/gvar?filter=name=@gvar*"} }
```

### **POST AEWS/event/delete-gvar/name -- Delete a Global Variable**

The POST AEWS/event/delete-gvar/*name* method sends a SET\_GLOBAL event to delete the specified global variable from the database.

#### **HTTP Method**

POST

#### **URL Resource**

`https://webserver:port/AEWS/event/delete-gvar/name?parameters`

- **webserver**  
Specifies the name of the web server host machine.
- **port**  
Specifies the port number for the web server host machine.
- **name**  
Specifies the name of the global variable.

#### **Parameters**

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

#### **Request Formats**

- `xml --` Specifies to send the data in XML format.
- `json --` Specifies to send the data in JSON format.

#### **Content**

To send the delete-gvar/*name* event, you must specify the name of the global variable. The global variable must be present in the database. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <comment>variable no longer used</comment>
```



```
</command>
```

The following is an example of the content data in JSON format:

```
{"comment":"variable no longer used"}
```

- **comment**

(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

**Example: POST https://WSServer:9443/AEWS/event/delete-gvar/name**

The command sends the SET\_GLOBAL event to delete the specified global variable.

- **XML Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/delete-gvar/DAY
Content-Type: application/xml
<command>
<comment>variable no longer used</comment>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Content-Type: application/xml
Content-Length: 0
Date: Wed, 14 Mar 2012 07:34:02 GMT
```

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/delete-gvar/DAY
Content-Type: application/json
{"comment":"variable no longer used"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Content-Type: application/json
Content-Length: 0
Date: Wed, 14 Mar 2012 09:31:12 GMT
```

**POST AEWS/event/set-gvar/name -- Create or Update a Global Variable**

The POST AEWS/event/set-gvar/name method sends a SET\_GLOBAL event to set a global variable in the database.

**HTTP Method**

POST

**URL Resource**

```
https://webserver:port/AEWS/event/set-gvar/name?parameters
```

- **webserver** Specifies the name of the web server host computer.

- **port** Specifies the port number for the web server host computer.
- **name**  
Specifies the name of the global variable.

### Parameters

- **version=*n*** (Optional) Specifies the version for the request or response.  
**Default:** current server version
- **locale=*locale\_code*** (Optional) Specifies the user locale. If the value entered is unsupported, en-US is used.  
**Default:** en-US

### Request Formats

- **xml --** Specifies to send the data in XML format.
- **json --** Specifies to send the data in JSON format.

### Content

To send the set-gvar/*name* event, you must define the name and value of the global variable. All the required parameters must be entered. JSON and XML formats are supported.

The following is an example of the content data in XML format:

```
<command>
 <gvarValue>Wednesday</gvarValue>
</command>
```

The following is an example of the content data in JSON format:

```
{"gvarValue": "Wednesday"}
```

- **gvarValue**  
Specifies the value of the global variable.

#### NOTE

You cannot specify the value as DELETE as it is a reserved word.

- **comment**  
(Optional) Specifies text to associate with the event you are sending. This field is for documentation purposes only. For example, you can provide an explanation of why an action was necessary.

### Example: POST https://WSServer:9443/AEWS/event/set-gvar/*name*

The command sends the SET\_GLOBAL event to set a global variable.

#### • XML Format

```
-- request --
POST https://WSServer:9443/AEWS/event/set-gvar/DAY
Content-Type: application/xml
<command>
<gvarValue>Wednesday</gvarValue>
</command>
-- response --
201 Created
Server: Apache-Coyote/1.1
Content-Type: application/xml
Content-Length: 0
```

Date: Wed, 14 Mar 2012 07:29:02 GMT

- **JSON Format**

```
-- request --
POST https://WSServer:9443/AEWS/event/set-gvar/DAY
Content-Type: application/json
{"gvarValue":"Wednesday"}
-- response --
201 Created
Server: Apache-Coyote/1.1
Content-Type: application/json
Content-Length: 0
Date: Wed, 14 Mar 2012 07:40:02 GMT
```

## Web Services Troubleshooting

The articles in this section can help you perform basic troubleshooting procedures.

### URI is Inaccessible

#### **Possible causes:**

- Web server is not running
- Incorrect host, port or resource name
- Application client not using SSL
- Firewall blocking access
- Database is inaccessible
- Oracle tnsnames.ora file unreadable
- NATIVEJDBCJARPATH is invalid

#### **Web server is not running**

##### **Action:**

Verify that the web server is running.

##### **UNIX:**

From a terminal session, issue the “unisrvctr status CA-WAAE” command to verify the status of the Web Server for the instance is running

```
unisrvctr status CA-WAAE
```

##### **Windows:**

Open the AutoSys Workload Automation Administrator GUI. Go the Services dialog and verify the status of the AutoSys Workload Automation Web Server for the instance is running.

##### **Resolution:**

Ensure that the Web Server has been started.

#### **Incorrect host, port or resource name**

##### **Action:**

- Verify that the hostname matches that where the web server is running.
- Open the `waae_webservices_wrapper.log` file that is created in the `AUTOUSER/out` directory and verify the port specified in the URI corresponds with that stated in the log file.
- Verify that the URI resource name is correct.

**Resolution:**

Ensure that the URI has been corrected as needed to correspond with the configured web server.

**Application client not using SSL****Action:**

Verify that the application is using HTTP over SSL (https). The web server is configured to run exclusively with SSL to avoid transferring data in clear text.

**Resolution:**

Ensure that the "https" is the protocol in the URI.

**Firewall blocking access****Action:**

Verify the `AUTOUSER/out/waae_webservices_access_log.YYYY_MM_DD.log` file for the current day to see if any requests are being logged from IP address of the application/source machine. If no entries are being logged, the requests are not being received by the web server.

**Resolution:**

Ensure that a firewall is not blocking the configured connection port on the web server.

**Database is inaccessible****Action:**

Verify the `AUTOUSER/out/waae_webservices_wrapper.log` file for messages indicating the database connection was refused.

```
[EL Severe]: 2012-08-09 11:46:41.033--ServerSession(1140137042)--
Exception [EclipseLink-4002] (Eclipse Persistence Services - 2.3.2.v20111125-
r10461): org.eclipse.persistence.exceptions.DatabaseException
```

```
INFO | jvm 1 | 2012/08/09 11:46:41 | Internal Exception: java.sql.SQLException: JZ006: Cau
```

```
INFO | jvm 1 | 2012/08/09 11:46:41 | Error Code: 0
```

**Resolution:**

Ensure that the Event Servers are accessible to the web server machine.

**Oracle tnsnames.ora file unreadable****Action:**

Verify the `AUTOUSER/out/waae_webservices_wrapper.log` file for messages indicating a database connection failed because it could not resolve a connection identifier.

```
INFO | jvm 1 | 2012/08/09 14:24:13 | [EL Severe]: 2012-08-09 14:24:13.593--
ServerSession(1632326726)--
Exception [EclipseLink-4002] (Eclipse Persistence Services - 2.3.2.v20111125-
r10461): org.eclipse.persistence.exceptions.DatabaseException
```

```
INFO | jvm 1 | 2012/08/09 14:24:13 | Internal Exception: java.sql.SQLException: Io excepti
```

```
INFO | jvm 1 | 2012/08/09 14:24:13 | Error Code: 17002
```

**Resolution:**

Ensure that the tnsnames.ora file is readable for the user ID running the web server.

**NATIVEJDBCJARPATH is incorrect****Action:**

Verify the *AUTOUSER*/out/waae\_webservices\_wrapper.log file for messages indicating a configuration error and com.sybase.jdbc4.jdbc.SybDriver is not found.

```
INFO | jvm 1 | 2012/09/13 13:04:24 | [EL Severe]: 2012-09-13 13:04:24.699--
ServerSession(1397300700)--
Exception [EclipseLink-4003] (Eclipse Persistence Services - 2.3.2.v20111125-
r10461): org.eclipse.persistence.exceptions.DatabaseException
```

```
INFO | jvm 1 | 2012/09/13 13:04:24 | Exception Description: Configuration error. Class [c
```

**Resolution:**

Ensure that the wrapper.conf file has NATIVEJDBCJARPATH is set correctly and is part of the classpath.

**Authentication Error (Status Code 403)****Possible causes:**

- User credentials are invalid
- User not authorized for web services
- DNS reverse lookup inconsistency
- CA EEM server is unavailable

**User credentials are invalid****Action:**

Verify the *AUTOUSER*/out/waae\_webservices\_wrapper.log file for warning messages indicating user authentication failures for the user such as the following:

```
WARNING: User authentication failure. User: autosys
```

**Resolution:**

Ensure that the CA EEM user id and password are correct. The CA EEM UI can be used to confirm the credentials for the WorkloadAutomationAE application. Have an EEM administrator define the CA EEM user id and password, or reset the password, if needed.

### User not authorized for web services

#### Action:

Verify the *AUTOUSER/out/waae\_webservices\_wrapper.log* file for warning messages indicating user authentication failures for the user such as the following:

```
WARNING: User authentication failure. User: autosys
```

#### Resolution:

Ensure that the CA EEM user id for the WorkloadAutomationAE application is a member of the WorkloadAutomationAEWebService group. Add if needed.

### DNS reverse lookup inconsistency

#### Action:

Verify that the DNS name and host name are exactly the same. CA EEM uses the host name as part of the encryption algorithm. It relies on the case being identical. A problem occurs if the hostname command and the nslookup command return the machine names differently. For example, if the hostname command returns the host names as host.example.com with an IP address of 192.168.255.255 and nslookup 192.168.255.255 returns HOST.example.com, you experience this problem.

#### Resolution:

Ensure that the host name known to the machine and the DNS lookup return the same values.

### CA EEM server is unavailable

#### Action:

Verify the *AUTOUSER/out/waae\_webservices\_wrapper.log* file for warning messages indicating login exception authenticating the user.

```
INFO | jvm 1 | 2012/08/09 14:20:53 | Aug 9, 2012 2:20:53 PM org.apache.catalina.realm.JAAS
INFO | jvm 1 | 2012/08/09 14:20:53 | WARNING: Login exception authenticating username "aut
INFO | jvm 1 | 2012/08/09 14:20:53 | javax.security.auth.login.LoginException: EE_SPONSORE
```

View the EEMTRACE.WS.SDK.log there is message:

```
ERROR 2012-08-09 14:20:53,665 [http-bio-9443-
exec-2] [Network] RunBatMethod - exception occurred calling Iclient runBatMethod

com.ca.itechnology.iclient.IclException: org.apache.http.conn.HttpHostConnectException: Connect
eemserver:5250 refused
```

**Resolution:**

1. Ensure that the CA EEM server is reachable from the web server machine.
2. Ensure that the CA EEM service is running.

Take the appropriate action to rectify the problem by correcting the configuration error, network problem, or restarting the CA EEM service.

**Web Server Will Not Start****Possible causes:**

- Port conflict

**Port conflict****Action:**

Verify the *AUTOUSER/out/waae\_webservices\_wrapper.log* file for severe initialization and bind exception messages.

```
INFO | jvm 1 | 2012/08/23 17:28:27 | INFO: Initializing ProtocolHandler ["http-
bio-9443"]

INFO | jvm 1 | 2012/08/23 17:28:29 | Aug 23, 2012 5:28:29 PM org.apache.coyote.AbstractPro

INFO | jvm 1 | 2012/08/23 17:28:29 | SEVERE: Failed to initialize end point associated wit
bio-9443"]

INFO | jvm 1 | 2012/08/23 17:28:29 | java.net.BindException: Address already in use: JVM_B
```

**Resolution:**

Ensure that the port is not shared, if shared change by opening the *server.xml* configuration file located in the *AUTOUSER/webserver/conf* directory. Assign available unused port numbers to the conflicting configured ports for the Server shutdown and the Connector. Save the configuration and restart the web server.

## Numeric and Character Codes

AutoSys Workload Automation web services use numeric codes to:

- Represent alarms that can be generated during job processing. For more information, see [Alarms](#).
- Track the current state or status of every job. For information about the statuses that a job can have, see [Status](#).
- Track the statuses that an event can have. For more information, see [Events](#).
- Represent the platform (operating system type) of a machine. For more information about the operating system type of a machine, see [opsys Attribute—Specify the Operating System Type of a Machine](#).

AutoSys Workload Automation web services use character codes to:

- Track the machine status. For more information about the statuses that a machine can have, see [Machine Status](#).
- Track the machine type. For more information about the type of a machine, see [type Attribute -- Specify the Machine Type](#).

# AutoSys Commands

AutoSys Workload Automation client utilities (or commands) let you manage your environment and workload. The product lets you run multiple commands in scripts (UNIX) or batch files (Windows) and create aliases for frequently used commands. Some commands support wildcards.

## Windows considerations:

- The AutoSys Workload Automation instance command prompt pre-sets the environment. Run commands from this prompt, not from the operating system prompt.
- You can run commands only if you have access to the Windows Registry. To gain access to the registry, take the following actions:
  - Open a command prompt and define the AUTOSERV, AUTOUSER, and AUTOSYS environment variables using the set command.
  - Set the PATH environment variable to %AUTOSYS%\bin.

## UNIX considerations:

- Run commands from the operating system command prompt.
- You can run commands only if the environment is set. To source the environment, run the shell that is sourced to use AutoSys Workload Automation.
- When you use wildcards, ensure that you enclose asterisks (\*) and question marks (?) in quotation marks.
- A reference page for each command is available when the MANPATH environment variable is set correctly. To reset this variable, edit the \$AUTOUSER/autosys.\* files.  
To access a command reference page, enter the following command:

```
man command
```

## Command Syntax Conventions

The conventional Windows instance command prompt commands have the following format:

```
name [option...] [command argument...]
```

For example:

```
x_stgd -h [-l [level]] [-L file] [-a port] [-d [level]] [-s server] [-m hostname] [-S status] [action -P|-R -p {-T -H -Y}]
```

The Command syntax conventions are:

- Options consist of one character and a hyphen proceeds the character.
- Options with no arguments can be grouped after a single hyphen.
- Brackets [ ] surround an option or command argument that is optional.
- Braces {} enclose options or arguments that are interdependent; everything that is enclosed must be treated as a unit.

## Send Events

AutoSys Workload Automation is event-driven. An *event* instructs the scheduler to initiate a specific action. The scheduler completes the action unless doing so violates internal rules or the event is not applicable.

AutoSys Workload Automation server components internally generate certain events in certain circumstances.

The sendevent command lets you manually generate events. Use manually generated events to complete certain system maintenance and workload management tasks.



## sendevent Command -- Send Events

The sendevent command lets you manually generate events.

### NOTE

This article provides an overview of the sendevent command. For information about individual events, see the [Maintain System](#) and [Control Workload](#) sections.

### Syntax

This command has the following format:

```
sendevent
 -E EVENT [OPTIONS]
 [-F file]
 [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-E *EVENT*** Sends the specified event.

### NOTE

- This parameter is not valid with the -F, -x, or -? option; otherwise, it is always required.
- Most events require a specific set of options. These options are described in the [Maintain System](#) and [Control Workload](#) sections.
- **-F *file*** (Optional) Processes the events in the specified file as a batch. Batch processing improves the client utility performance because the program initialization happens only once for all the sendevent commands included in the file.

### NOTE

To save multiple events in one file, enter the sendevent command for each event on a separate line of the text file.

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**

(Optional) Specifies external security credentials as follows:

**-usr *username* -pw *password***: Specifies a username and unencrypted password.

**-usr *username* -pwx *password***: Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact***: Specifies a Security Assertion Markup Language (SAML) artifact.

### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- -x(Optional) Returns the command version information to standard output.
- -?(Optional) Displays help for the command.

### Example: Process Events as a Batch

Suppose that you terminate several jobs simultaneously, but none of these jobs belong to the same group and none of them are associated with the same application. You do not want to issue a separate event for each job that you want to terminate, so you save the events as a batch in the file named jobs\_to\_kill. This example executes one command to process all the events in the jobs\_to\_kill file as a batch.

```
sendevent-F jobs_to_kill
```

## Maintain System

System maintenance utilities let you obtain system information, specify the time zone that the system uses, resolve system problems, or stop server processes.

### WARNING

Ensure that you stop all server processes before you run maintenance tasks on the server machines. We recommend that you stop the schedulers before you define or modify workload objects.

To resume workflow after you complete maintenance tasks or workload modifications, restart services. To restart services on UNIX, use the unisvrctr start command. To restart services in a Windows environment, use the Administrator utility.

### NOTE

- The unisvrctr start and unisvrctr stop commands are not AutoSys Workload Automation client utilities.
- The eventor command lets you start the scheduler on UNIX, but we recommend that you use the unisvrctr command to start all services simultaneously.
- For information about the unisvrctr commands or controlling the services using the Administrator utility, see [Control the Services](#).

The following are the system maintenance utilities:

## archive\_events Command -- Archive Data

The `archive_events` command is a scheduler component utility that archives data from the AutoSys Workload Automation database. It removes data that is older than the specified number of days from the AutoSys Workload Automation `ujo_proc_event`, `ujo_job_runs`, `ujo_audit_info`, `ujo_chkpnt_rstart`, `ujo_audit_msg`, `ujo_alarm`, `ujo_event2`, and `ujo_extended_jobrun_info` tables and archives it to a flat file. Use `archive_events` to prevent the database from becoming full.

In dual event server mode, the data is archived from both the event servers simultaneously. If information from these tables is not regularly purged from the database, the database can fill up, stopping all processing. We recommend that you run `archive_events` during the database maintenance cycle.

The DBMaint script, which runs daily by default, runs `archive_events` as a part of its process.

### Syntax

This command has the following format:

```
archive_events
 [-A] [-B batch_size] [-C delimiter] [-d directory_name] [-D
 {data_server:database|TNSname}] [-j {num_days|-l num_days|-n num_days|-r num_days}] [{-
 usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-A**  
(Optional) Deletes the information after archiving it to a file. If you omit the **-A** argument, the information is deleted without archiving.
- **-B *batch\_size***  
(Optional) Defines the number of events to archive at a time.  
**Limits:** 1-50000  
**Default:** 1000
- **NOTE**
  - Use the default value unless the database is full and the transaction log is likely to become full if 1000 events are archived at once. Because the database transaction file records each transaction (in this case, the deletion of a single event), and the transaction file shares file space with the actual data, only archive a few events at a time when the amount of available file space is low. If there is ample space in the transaction log file, a larger value can be specified to reduce the archival time.
  - If you do not maintain low or optimum job load, the transaction log fills up and the transactions are suspended. In dual event server mode, <UAJM> rolls over to single event server mode if the database transaction log in the primary event server is full.
- **-C *delimiter*** (Optional)  
Defines a character to use as a delimiter for columns in output archive files. **Default:** Comma (,)  
**Limits:** This value must be one character.
- **-d *directory\_name***  
(Optional) Defines the directory in which to store archived data. If you omit this parameter, the product archives data to the default directory (\$AUTOUSER/archive for UNIX, or %AUTOUSER%\archive for Windows).

### NOTE

This option is only valid when you also specify the **-A** option

- **-D *data\_server:database|TNSname***  
(Optional) Specifies the data source from which to archive events.  
– ***data\_server:database***

Specifies the Sybase data server and database from which to archive events.

– **TNSname**

Specifies the TNS alias of the Oracle data server from which to archive events.

- **-j num\_days**  
(Optional) Deletes job runs information that is older than the *num\_days* value from the *ujo\_job\_runs* table. The *num\_days* value indicates the number of days to leave information about job runs in the database. When a row in the table exceeds this value, the row is deleted.
- **-l num\_days**  
(Optional) Deletes audit information older than the *num\_days* value from the *ujo\_audit\_info* and *ujo\_audit\_msg* tables. The *num\_days* value indicates the number of days to leave audit information in the database. When a row in the table exceeds this value, the row is deleted.
- **-m num\_days**  
(Optional) Deletes machine information older than the *num\_days* value from the *ujo\_machine* table. The *num\_days* value indicates the number of days to leave machine information in the database. When a row in the table exceeds this value, the row is deleted.
- **-n num\_days**  
(Optional) Deletes events and alarms information older than the *num\_days* value from the *ujo\_proc\_event* and *ujo\_alarm* tables. The *num\_days* value indicates the number of days to leave events and their associated alarms in the database. When a row in the table exceeds this value, the row is deleted.
- **-r num\_days**  
(Optional) Deletes checkpoint information older than the *num\_days* value from the *ujo\_chkpnt\_rstart* table. The *num\_days* value indicates the number of days to leave information about checkpoints in the database. When a row in the table exceeds this value, the row is deleted.
- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**  
(Optional) Specifies external security credentials as follows:
  - usr username -pw password**: Specifies a username and unencrypted password.
  - usr username -pwx password**: Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (*as\_config* command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (*autosys\_secure* command) to encrypt passwords.

**-saml auth\_artifact**: Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is **STRICT**, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the **-usr** option and you do not specify the **-pw** or **-pwx** option, the utility prompts you to specify a password.
  - When you specify the **-pw** or **-pwx** option but do not specify the **-usr** option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is **EXTERNAL**, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy

identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.

- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**

(Optional) Returns the command version information to standard output.

- **-?**

(Optional) Displays help for the command.

### Example: Copy Events Older than Five Days

This example copies all events older than five days from the `ujo_proc_event` table to the default archive file and deletes them from the database.

```
archive_events -A -n 5
```

### Example: Copy All job\_runs Statistics Older than Five Days

This example copies all `job_runs` statistics older than five days to a specified archive directory and deletes them from the database.

```
archive_events -A -d \my_archive -j 5
```

## Archive Files

The `archive_events` command removes data from the following database tables and stores the data in the corresponding flat files:

- **ujo\_audit\_info**

The `archive_events` command archives data from the `ujo_audit_info` table to the following file:

- `archived_audit.$AUTOSERV.MM.dd.yyyy.hh.mm.ss` (UNIX)
- `archived_audit.%AUTOSERV%.MM.dd.yyyy.hh.mm.ss` (Windows)

**MM.dd.yyyy.hh.mm.ss**

Indicates the month, day, year, hour, minute, and second when the archive was created.

The format of each record in the flat file is as follows:

```
audit_info_num,entity,time,type
```

For example:

```
4324,user@domain,1104868885,J
```

Each audit event occupies one row in the archive. By default, a comma (,) delimits each column.

- **ujo\_audit\_msg**

The `archive_events` command archives data from the `ujo_audit_msg` table to the following file:

- `archived_audit_msg.$AUTOSERV.MM.dd.yyyy.hh.mm.ss` (UNIX)
- `archived_audit_msg.%AUTOSERV%.MM.dd.yyyy.hh.mm.ss` (Windows)

**MM.dd.yyyy.hh.mm.ss**

Indicates the month, day, year, hour, minute, and second when the archive was created.

The format of each record in the flat file is as follows:

```
audit_info_num,seq_no,attribute,value1+value2,is_edit
```

For example:

```
103,2,job_type,CMD,N
```

Each event occupies one row in the archive. By default, a comma (,) delimits each column.

If the command argument is more than 2000 characters, *value1* contains the first 2000 characters and *value2* contains the remaining characters.

If the command argument is less than 2000 characters, *value1* contains the command argument and *value2* is empty.

- **ujo\_job\_runs**

The archive\_events command archives data from the ujo\_job\_runs table to the following file:

- archived\_job\_runs.\$AUTOSERV.MM.dd.yyyy.hh.mm.ss (UNIX)
  - archived\_job\_runs.%AUTOSERV%.MM.dd.yyyy.hh.mm.ss (Windows)
- MM.dd.yyyy.hh.mm.ss**

Indicates the month, day, year, hour, minute, and second when the archive was created.

The format of each record in the flat file is as follows:

```
joid,run_num,ntry,job_ver,endtune,esp_lstatus,esp_status,evt_num,exit_code,over_num,run_mach
```

For example:

```
5,5,5,1,1287054429,,,3278,0,-1,localhost,1,1287054429,4,,,0,,1,WobId(1147.238_1/
WAAE_WF0.1/MAIN),1,1,0
```

Each job run occupies one row in the archive. By default, a comma (,) delimits each column.

- **ujo\_proc\_event**

The archive\_events command archives data from the ujo\_proc\_event table to the following file:

- archived\_events.\$AUTOSERV.MM.dd.yyyy.hh.mm.ss (UNIX)
  - archived\_events.%AUTOSERV%.MM.dd.yyyy.hh.mm.ss (Windows)
- MM.dd.yyyy.hh.mm.ss**

Indicates the month, day, year, hour, minute, and second when the archive was created.

The format of each record in the flat file is as follows:

```
eoid,alarm,AUTOSERV,event,event_time_gmt,evt_num,exit_code,global_name,
global_value,jc_pid,job_ver,joid,machine,ntry,orig_evt_num,over_num,pid,
priority,que_priority,que_status,que_status_stamp,run_num,stamp,status,
eventtext,wf_joid
```

For example:

```
Z5,0,ACE,106,1287054429,3888,0,,,53340618,1,1111,localhost,1,3288,-1,53340597,10,0,2,1999-12-
00:00:00.000,333,1999-12-31 00:00:00.000,3,"Some Text",1
```

Each event occupies one row in the archive. By default, a comma (,) delimits each column.

- **ujo\_chkpnt\_rstart**

The archive\_events command archives data from the ujo\_chkpnt\_rstart table to the following file:

- archived\_chkpnt\_rstart.\$AUTOSERV.MM.dd.yyyy.hh.mm.ss (UNIX)
  - archived\_chkpnt\_rstart.%AUTOSERV%.MM.dd.yyyy.hh.mm.ss (Windows)
- MM.dd.yyyy.hh.mm.ss**

Indicates the month, day, year, hour, minute, and second when the archive was created.

The format of each record in the flat file is as follows:

```
dest_machine,dest_app,as_evt_time,ubc_name,ubc_jobnumbr,ubc_sysid,ubc_date,ubc_time,ubc_proci
```

Each record occupies one row in the archive. By default, a comma (,) delimits each column.

## archive\_jobs Command—Delete Obsolete Job Versions

The archive\_jobs command deletes obsolete job versions from the database. Job versions are considered obsolete when all the following conditions apply:

- The job is inactive.
- The job is no longer referenced.
- All event data that is related to the job has been deleted.
- All job run data that is related to the job has been deleted.

AutoSys Workload Automation places jobs in the archive when you delete them using the `sendevent` command,

We recommend that you issue the `archive_events` command before issuing the `archive_jobs` command. We also recommend that you run `archive_jobs` as part of your usual database maintenance.

## Syntax

This command has the following format:

```
archive_jobs -j number_of_days
 [-d "directory_name"] [{-usr username {-pw|pwx} password|-saml auth_artifact}] [-A]

 [-x]
 [-?]
```

- **-j *number\_of\_days***  
Specifies that jobs in the archive for more than this number of days are permanently deleted.

- **-d "*directory\_name*"**  
(Optional) Specifies the name of the directory where the archive file is created.

- **-A**  
(Optional) Copies the job definitions to an archive file.

**Default:** \$AUTOUSER/archive

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**

(Optional) Specifies external security credentials as follows:

**-usr *username* -pw *password*:** Specifies a username and unencrypted password.

**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (`as_config` command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (`autosys_secure` command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

### NOTE

- When CA Workload Automation AE is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When CA Workload Automation AE is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode.
- **-x**  
(Optional) Displays the version number.
- **-?**  
(Optional) Displays help for the command.

**NOTE**

The name of the job being archived is written to the archive file.

**Example: Delete Obsolete Job Versions**

This example deletes obsolete job versions older than 7 days:

```
archive_jobs -j 7
```

This example deletes obsolete job versions older than 7 days and creates the archive flat file in the \$AUTOUSER/archive file (the default):

```
archive_jobs -j 7 -A
```

This example deletes obsolete job versions older than 7 days and creates the archive flat file in the /tmp/archive directory:

```
archive_jobs -j 7 -A -d "/tmp/archive"
```

**NOTE**

Obsolete job versions include all updated and deleted jobs.

**as\_info Command—Return Installation Information****Valid on UNIX**

The as\_info command is a client component utility that returns AutoSys Workload Automation installation information to standard output.



When you issue `as_info` with no parameters, the command returns the AutoSys Workload Automation components and version installed, installation log locations, configured instances, and status.

## Syntax

This command has the following format:

```
as_info [-a] [-c] [-C] [-d] [-f] [-i] [-l] [-L] [-p] [-s] [-t directory] [-v] [-x] [-z]
```

- **-a**  
(Optional) Returns all AutoSys Workload Automation installation information, including the current product status. It also collects the scheduler, agent, and event server log files.
- **-c**  
(Optional) Returns the list of all AutoSys Workload Automation components that are installed on the computer from which you issue the command.
- **-C**  
(Optional) Returns the configuration information (as displayed in the `$AUTOUSER/config.$AUTOSERV` file) for all AutoSys Workload Automation instances installed on a computer.
- **-d**  
(Optional) Returns information about the AutoSys Workload Automation database.
- **-f**  
(Optional) Returns the custom (or default) AutoSys Workload Automation component installation specifications.
- **-i**  
(Optional) Returns a list of all configured AutoSys Workload Automation instances.
- **-l**  
(Optional) Returns the names and locations of the AutoSys Workload Automation installation logs.
- **-L**  
(Optional) Returns only the latest event server, application server, and agent logs.

### NOTE

By default, all logs are collected. This option is used with the `-x` option and is ignored if the `-x` option is not specified.

- **-p**  
(Optional) Returns a list of the Product Interchange File (PIF) and the RPM packages included as part of this installation.

### NOTE

This parameter uses the RPM Package Manager and the CA lsm utility to retrieve the list. The process might take a few minutes.

- **-s**  
(Optional) Returns the current status of the AutoSys Workload Automation components.
- **--t *directory***  
Specifies the full path to the collection tar file.

### NOTE

By default, the collection tar file is created in the `/tmp/as_info_dir` directory. This option is used with the `-a` and `-x` options and is ignored if the `-x` option is not specified.

- **-v**  
(Optional) Returns the AutoSys Workload Automation version, installation date and time, and installation directory.
- **-x**

Returns the AutoSys Workload Automation log files and configuration files. If the `-z` option is not specified, it creates the tar file named `uajm_file_collection.tar`.

#### NOTE

- By default, the tar file is created in the `/tmp/as_info_dir` directory. You can use the `-t` directory option to specify another directory.
- The `as_info` command returns the <UAJM> components and version installed, installation log locations, configured instances, and status to the standard output. However, you can obtain additional information by creating and populating the following two files:
  - `$AUTOSYS/config/as_files`—Specifies the fully pathed files to be copied.
  - `$AUTOSYS/config/as_cmds`—Specifies the commands to run and collect information.

The result of the commands that are issued from the `as_cmds` file is stored in the `user_cmds.out` file within the `uajm_file_collection.tar` file.

- **-z**  
Returns the AutoSys Workload Automation log files and configuration files, but does not create the tar file. The results are stored in the `/tmp/as_info_dir` directory.

#### Example: Include /tmp/diag.file in the Collection of Files to be Copied to the Tar File

This example includes the `/tmp/diag.file` file in the collection of files that are copied to the tar file.

```
echo "/tmp/diag.file" >> $AUTOSYS/config/as_files
```

#### Example: Include the Output in the as\_cmds File

This example includes the output of the `ls -l /tmp/rslt` command in the `as_cmds` file.

```
echo "ls -l /tmp/rslt" >> $AUTOSYS/config/as_cmds
```

#### Example: Report Default Installation Information

This example returns the AutoSys Workload Automation components and version installed, the installation log locations, configured instances, and status to standard output.

```
as_info
```

#### Example: Report Specific Installation Information

This example returns the AutoSys Workload Automation version, installation date and time, installation directory, current status, and configured instances to standard output.

```
as_info -vsi
```

## as\_server Command—Start the Application Server

### Valid on UNIX

The `as_server` command is the application server component that runs the application server for a specific instance.

#### NOTE

The application server typically starts in the background using the `unisrvctr` command or boot scripts during startup. When CA support helps you resolve a problem, they might ask you to start the application server manually.

### Syntax

This command has the following format:

```
as_server -A instance_name [-?] [-x]
```

- **instance\_name**  
Specifies the instance that is associated with the application server you want to run.
- **-?**  
(Optional) Displays help for the command.
- **-x**  
(Optional) Displays the version information.

### Example: Start the Application Server

This example starts the application server for the instance ACE.

```
as_server -A ACE
```

## auto\_svcdesk Command—Open Service Desk Tickets

The `auto_svcdesk` command lets you open Service Desk helpdesk tickets that are related to jobs or other actions that might occur in the AutoSys Workload Automation environment. This command supplements the existing Service Desk integration provided by the AutoSys Workload Automation scheduler component.

### Syntax

This command has the following format:

```
auto_svcdesk {-G|-J JobName}
 [-a Attribute/Value pair] [-d Description] [-i Impact] [-p Priority] [-s Severity]
 [-y Summary] [-L Url] [-U Username -P Password] [-C Customer] [{-usr username {-pw|
 pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-G**  
Opens a generic Service Desk ticket request on behalf of AutoSys Workload Automation.
- **-J JobName**  
Opens a Service Desk ticket request on behalf of a specific job defined to AutoSys Workload Automation. The jobname that is specified in this option must exist in the database. The current attributes of the job do not need to employ its Service Desk attributes to send a request using this command. The various options of this command can be used to set description, impact, priority, and severity. If these options are not specified, they are set based on a pre-defined template that exists for AutoSys Workload Automation in Service Desk.

#### NOTE

Options -G and -J are mutually exclusive.

- **-a Attribute/Value pair**  
(Optional) Defines Service Desk request attributes and handles values to be set in the Service Desk request. This flag is the behavioral equivalent of the `svcdesk_attr` job attribute.

#### NOTE

If this flag is specified for a jobname Service Desk request, its value overrides the `svcdesk_attr` attribute value of the job in the database.

Use of this option requires advanced knowledge of Service Desk. Contact your Service Desk administrator for assistance.

- **-d Description**  
(Optional) Defines the message or string to be set as the description field in the Service Desk request. This flag is the behavioral equivalent of the `svcdesk_desc` job attribute.

**NOTE**

If this flag is specified for a jobname Service Desk request, its value overrides the `svcdesk_desc` attribute value of the job in the database.

- **-i Impact**

(Optional) Specifies the impact level to assign to the Service Desk request. This flag is the behavioral equivalent of the `svcdesk_imp` job attribute.

**NOTE**

If this flag is specified for a jobname Service Desk request, its value overrides the `svcdesk_imp` attribute value of the job in the database.

- **-p Priority**

(Optional) Specifies the priority level to assign to the Service Desk request. This flag is the behavioral equivalent of the `svcdesk_pri` job attribute.

**NOTE**

If this flag is specified for a jobname Service Desk request, its value overrides the `svcdesk_pri` attribute value of the job in the database.

- **-s Severity**

(Optional) Specifies the severity level to assign to the Service Desk request. This flag is the behavioral equivalent of the `svcdesk_sev` job attribute.

**NOTE**

If this flag is specified for a jobname Service Desk request, its value overrides the `svcdesk_sev` attribute value of the job in the database.

- **-y Summary**

(Optional) Defines the message or string to be set as the summary field in the Service Desk request. There is no job Service Desk attribute behavioral equivalent. The scheduler component when interfacing with Service Desk has a pre-defined summary, which it passes as part of the request.

- **-L Url**

(Optional) Specifies the URL of the Service Desk web server. If this option is not specified, the URL is obtained from the AutoSys Workload Automation environment (derived from the current AutoSys Workload Automation integration settings).

**NOTE**

For more information about AutoSys Workload Automation integration settings, see [Configure AutoSys Workload Automation to Work with Other Broadcom products](#).

- **-U username**

(Optional) Specifies a valid login identifier to access the Service Desk web server.

- **-p password** Specifies the user's password.

**NOTE**

- If you specify an incorrect credential or if you do not specify a password, the utility cannot access the Service Desk web server.
- If you do not specify this option, the utility uses the login credentials that are specified in the integration settings to access the Service Desk web server.
- Use this option to specify user credentials that grant access to the service desk. To authenticate the command with external security, use the `-usr` parameter.

- **-C Customer**

(Optional) Specifies a valid customer identifier to access the Service Desk web server. If this option is not specified, the customer identifier is obtained from the AutoSys Workload Automation environment (derived from the current AutoSys Workload Automation integration settings).

- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password:** Specifies a username and unencrypted password.

**-usr username -pwx password:** Specifies a username and encrypted password.

#### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

#### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**  
(Optional) Displays the version number.
- **-?**  
(Optional) Displays help for the command.

#### Example: Open a Service Desk Ticket

This example opens a Service Desk ticket for a job that is named 'svcdesk\_test'.

```
auto_svcdesk -J svcdesk_test
```

#### Example: Open a Service Desk Ticket and Set the Priority and Impact Levels

This example opens a Service Desk ticket for a job named 'svcdesk\_test' and sets the priority level to a value of 1 and the impact level to a value of 4:

```
auto_svcdesk -J svcdesk_test -p 1 -i 4
```

#### NOTE

This command overrides the jobs current priority and impact, if set in the database.

#### Example: Open a Service Desk Ticket and Set the Description

This example opens a Service Desk ticket for a job that is named 'svcdesk\_test' and sets the description to 'This is a test' and the urgency to a value of 5:

```
auto_svcdesk -J svcdesk_test -d "This is a test" -a "urgency/urg:1104"
```

#### NOTE

This command overrides the jobs current description and impact, if set in the database.

#### Example: Open a Service Desk Ticket and Direct it to a Specific Web Server

This example opens a Service Desk ticket for a job that is named 'svcdesk\_test' and directs it to a specific CA Service Web Server named 'unisvcdesk':

```
auto_svcdesk -J svcdesk_test -L http://unisvcdesk:8080/axis/services/USD_R11_WebService
```

## autoflags Command—Return System Information

The autoflags command is an agent, client, scheduler, and server component utility that returns information about the AutoSys Workload Automation release number, the database being used, and the operating system. You can also use autoflags to verify the proper host name and host ID for license key generation.

### Syntax

This command has the following format:

```
autoflags -a | -d | -f | -h | -i | -n | -o | -r | -v | -x | -c|-?
```

- **-a**  
(Optional) Writes all autoflags information to standard output in the following order:
  - Build number
  - Operating system
  - Database provider
  - Service pack number
  - Version number
  - Host ID
  - Host name
  - Fully qualified host name
  - AutoSys version format
- **-d**  
(Optional) Writes the database type (SYB for Sybase, MSQ for MSSQL, ORA for Oracle, or ANY for a non-server installation) to standard output.

#### NOTE

A non-server installation does not require a database. So, if you use the **-d** option on a non-server installation, it returns a value of ANY.

- **-f**  
(Optional) Writes the fully qualified host name to standard output.

#### NOTE

AutoSys Workload Automation performs system calls to get the host name of the machine the scheduler runs on when it starts up. The system calls return either the host name or the fully qualified host name. This host name is used to look up a machine definition for jobs that execute on the localhost. If the machine definition is not located or the [LocalMachineDefinition](#) (on UNIX) or Local Machine Definition (on Windows) parameter

is not set or does not specify a valid machine, the jobs do not start. The host name that is returned when you execute `autoflags -f` must be used as the machine name when defining the localhost machine definition.

- **-h**  
(Optional) Writes the host ID to standard output.

**NOTE**

The Windows operating system does not employ host IDs; therefore, on a Windows computer, the value that is displayed is 0.

- **-i**  
(Optional) Writes the build number to standard output.
- **-n**  
(Optional) Writes the host name to standard output.
- **-o**  
(Optional) Writes the operating system information to standard output.
- **-r**  
(Optional) Writes the service pack number to standard output.
- **-v**  
(Optional) Writes the version number to standard output.
- **-x**  
(Optional) Writes the product release number to standard output.
- **-c**  
(Optional) Writes the product version format to standard output.
- **-?**  
(Optional) Displays help for the command.

**NOTE**

You can specify multiple options at one time.

**Example: Return System Information from a Server Installation**

This example returns all AutoSys Workload Automation and system configuration information to standard output when executed on a server installation that uses Oracle as the database provider.

```
autoflags -a
```

On UNIX, the command displays the following information in standard output:

```
5 AIX ORA <RN> 0 c0a9e38d venice venice.europe.com
```

On Windows, the command displays the following information in standard output:

```
5 Windows_NT ORA <RN> 0 0 venice venice.europe.com
```

**Example: Return the Fully Qualified Host Name**

This example returns the fully qualified host name to standard output.

```
autoflags -f
```

The command displays the following information in standard output:

```
venice.europe.com
```

**Example: Return the Build Number**

This example returns the build number to standard output.

```
autoflags -i
```

The command displays the following information in standard output:

5

## autoping Command—Verify Server, Agent, and Client Communication

The autoping command is a client component utility that verifies that the server and agent computers are properly configured and are communicating successfully. It also verifies that the agent is able to communicate with the application server. If you are running multiple application servers, autoping verifies that the agent is able to communicate with all the application servers. When requested, the command generates an alarm if it detects problems.

Because the client/server communication facilities are critical, autoping provides valuable information for troubleshooting. We recommend that you use autoping early in the troubleshooting process.

You can issue autoping from any machine where the autoping executable resides, and the target can be any machine that has a valid machine definition defined.

### Syntax

This command has the following format:

```
autoping
 {-m machine | -m ALL}
 [-A] [-S] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-m *machine* | -m ALL** Specifies a valid machine to verify. The machine must be defined to the database and accessible over the network. Specify **-m ALL** to verify all machines.

#### NOTE

- If you specify **-m localhost**, the scheduler must have associated a valid machine definition to localhost on startup. If localhost is not associated with a valid machine, the autoping command fails.
- You cannot issue the autoping command to a machine of type 'v' (virtual machine).

- **-A**  
(Optional) Sends an alarm when errors occur.

- **-S**  
Tests the connectivity between the application server and the specified agent.

#### NOTE

- You cannot issue the autoping command with the -S option or cannot run jobs that contain globs or blobs as input or output, if you installed:
  - The agent using the Workload Automation Agent for UNIX, Linux, or Windows DVD where the native AutoSys Workload Automation client is not installed.
  - The 64-bit agent using earlier version than the Workload Automation Agent for UNIX, Linux, or Windows 11.4.00.0 (Build 1609).

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**  
(Optional) Specifies external security credentials as follows:
  - usr *username* -pw *password***: Specifies a username and unencrypted password.
  - usr *username* -pwx *password***: Specifies a username and encrypted password.



**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

**NOTE**

- When you issue the autoping command to a machine of type 'a', the client (the machine from which you issued autoping) sends a request to the application server and waits for the application server to respond. The application server contacts the scheduler and notifies it to ping the agent and waits for the scheduler to respond. The application server then pings the agent and prepares a response to autoping. If successful, the autoping command writes the following message to standard output on the server:

```
AutoPinging Machine [machine]
AutoPing WAS SUCCESSFUL!
```

- If there is a configuration problem, the autoping command writes a message indicating that the remote machine did not respond or that a more serious problem (such as a socket read error) exists. It also writes messages on behalf of the scheduler and the application server.

**Example: Verify Machine Configuration**

This example verifies that the machine named myhost is properly configured and that its agent can function properly.

```
autoping -m myhost
```

If successful, the following displays:

```
CAUAJM_I_50023 AutoPinging Machine [myhost]
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL.
```

### Example: Verify Agent and Application Server Connectivity

This example verifies agent and application server connectivity for all Workload Automation Agents defined in the event server.

```
autoping -m ALL -S
```

If successful, the following is displayed:

```
CAUAJM_I_50023 AutoPinging Machine [myhost1]
CAUAJM_I_50031 Checking the Agent's connectivity to the Application Server.
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL.

CAUAJM_I_50023 AutoPinging Machine [myhost2]
CAUAJM_I_50031 Checking the Agent's connectivity to the Application Server.
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL.
```

## autoprofm Command—Convert Job Profiles to the New Format (Windows Only)

### Valid on Windows

In Unicenter AutoSys JM 4.5.1 and r11, job profile information was stored in the Windows registry. To upgrade to r11.3, Release 11.3.5, or Release [assign the value for rn in your book], the job profile information in the registry must be converted to a file format that is compatible with the new Workload Automation Agent for UNIX, Linux, or Windows. The new Workload Automation Agent does not read job profile information from the Windows registry as past legacy agents did. Moving the job profile information to a flat file format makes the implementation consistent across both the UNIX and Windows platforms.

When you upgrade AutoSys Workload Automation, the upgrade process automatically converts the job profiles. You can also issue the autoprofm command to convert profiles manually after an upgrade. You can specify where these converted files are stored.

### NOTE

- The autoprofm command must be run from the environment of the instance which owns the job profiles.
- On UNIX, the job profiles are still defined in shell script files that contain the environment variables you want to source. You can specify the name of the script file (or the path to and name of the file) in the profile attribute.

### Syntax

This command has the following format:

```
autoprofm
 -P directory
 [-N agent_name] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-P *directory***  
Specifies the directory to store the converted profiles.
- **-N *agent\_name***  
(Optional) Specifies the name of an agent.  
**Default:** WA\_AGENT
- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**  
(Optional) Specifies external security credentials as follows:
  - usr *username* -pw *password*:** Specifies a username and unencrypted password.
  - usr *username* -pwx *password*:** Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

- saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

**NOTE**

- If you are converting profiles from different 4.5.1 instances and the profiles have the same name, the autoprofm command renames the files to *profilename\_instancename* to avoid overwriting the files.
- If you are converting profiles from different r11 instances and the profiles have the same name, the autoprofm command renames the files to *profilename\_11.0\_Base* to avoid overwriting the files.
- Similarly, if the r11.3, Release 11.3.5, or Release 11.3.6 instance already has job profiles with the same name, the autoprofm tool renames the migrated files to *profilename\_instancename* or *profilename\_11.0\_Base*. The files are renamed to avoid overwriting the existing job profiles.
- To convert the DEFAULT job profile, the autoprofm command collects the environment variables in all the DEFAULT profiles and writes them a single DEFAULT profile for AutoSys Workload Automation r11.3, Release 11.3.5, or Release [assign the value for rn in your book]. The agent sets the `oscomponent.environment.variables` parameter in its `agentparm.txt` file to this new DEFAULT profile. Jobs that do not have a profile that is associated with them uses the environment variables defined in the new DEFAULT profile.
- The migration activity is logged in the `%AUTOUSER%\out\autoprofm.instance_name` file.

**Example: Store the Current Job Profiles**

This example stores the current job profiles in the default profile location of the agent.

```
autoprofm - P "c:\program files\CA\Workload Automation AE\SystemAgent\WA_AGENT
\Profiles"
```

**autosyslog Command—Display Log Files**

The autosyslog command is a client component utility that displays the scheduler, application server, or agent log file, or the spool file for a job. The scheduler, application server, and agent write diagnostic messages to their respective logs as part of their normal operations and in response to detected error conditions. This utility is useful for diagnosing job failures.

The command displays information for supported job types only. The information that the command displays depends on the job type.

The command does not support the following job types:

- CPU Monitor (OMCPU)
- Disk Monitor (OMD)
- Entity Bean (ENTYBEAN)
- File Trigger (FT)
- IP Monitor (OMIP)
- JMS Publish (JMSPUB)
- JMX MBean Attribute Get (JMXMAG)
- JMX MBean Attribute Set (JMXMAS)
- JMX MBean Create Instance (JMXMC)
- JMX MBean Operation (JMXMOP)
- JMX MBean Remove Instance (JMXMREM)
- JMX MBean Subscribe (JMXSUB)
- Oracle E-Business Suite Copy Single Request (OACOPY)
- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)
- Process Monitoring (OMP)
- PeopleSoft (PS)
- RMI (RMI)
- Secure Copy (SCP)
- SNMP Value Get (SNMPGET)
- SNMP Value Set (SNMPSET)
- Text File Reading and Monitoring (OMTF)
- Windows Event Log Monitoring (OMEL)
- Windows Service Monitoring (OMS)
- Wake on LAN (WOL)

The command displays job log files for the following job types:

- File Watcher (FW)

**NOTE**

The command also displays the spool file log for these jobs.

- Informatica (INFORMATICA)

**NOTE**

The command displays the log of task name which is of type session for these jobs.

- SQL Server Agent (SQLAGENT)

**NOTE**

The command displays the step log file for these jobs.

- Web Service Documentation Literal (WSDOC)
- z/OS Regular (ZOS)

**NOTE**

The command displays associated job log files for these jobs depending on the data set name.

The command displays spool file logs for the following job types:

- Database Monitor (DBMON)
- Database Stored Procedure (DBPROC)
- Database Trigger (DBTRIG)
- File Watcher (FW)

**NOTE**

The command also displays job log files for these jobs.

- FTP (FTP)
- HTTP (HTTP)
- i5/OS (I5)
- Informatica (INFORMATICA)
- JMS Subscribe (JMSSUB)
- Micro Focus (MICROFOCUS)
- POJO (POJO)
- Process Automation Process Execution (PAPROC)
- Process Automation Start Request Form (PAREQ)
- Remote Execution (PROXY)
- SAP Batch Input Session (SAPBDC)
- SAP BW InfoPackage (SAPBWIP)
- SAP BW Process Chain (SAPBWPC)
- SAP Data Archiving (SAPDA)
- SAP Job Copy (SAPJC)
- SAP Process Monitor (SAPPM)
- SAP R/3 (SAP)
- Session Bean (SESSBEAN)
- SQL (SQL)
- Web Service RPC/Encoded (WBSVC)

The command displays standard output, error output, and a job log for command jobs and user defined jobs.

**NOTE**

On Windows, you can issue this command only when all the following conditions apply:

- You are a member of the administrative user group
- The user account control for the administrative user group is set to zero

AutoSys Workload Automation issues an error message if you issue this command when at least one of the required conditions does not apply.

**Syntax**

This command has the following format:

```
autosyslog [-e [-l num_lines|-p regex]]
 [-s]
 [-j job_name [-d dataset_name|FileName:Num | -r run_num | -n retry_num | -t
[A|P|O|E|S|L [-i StepId|-o StepName|TaskName]] | -z]]
 [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-e**

(Optional) Monitors the scheduler log and displays the log file. The scheduler continually updates the log file as processing occurs. To terminate the command, press Ctrl+C.

**Limits:** This parameter is only valid when you run the command on the machine that runs the scheduler or on a machine with access to the same \$AUTOUSER (UNIX) or %AUTOSERVER% (Windows) file system.

– **-l num\_lines**

(Optional) Specifies the maximum number of lines to display from the specified log file.

**Default:** If you do not specify this option, the command displays the last ten lines of the specified log file by default.

– **-p regex**

(Optional) Specifies a regular expression. The command displays the lines in the scheduler log file containing that expression. The interpretation of the regular expression depends on the operating system.

**NOTE**

- On UNIX, using autosyslog to view the scheduler log is the same as issuing the following command:  
`tail -f $AUTOUSER/out/event_demon.$AUTOSERV.`
- Only a power user or Administrator can view the scheduler log file on Windows 2003 and previous versions of Windows. On more recent versions of Windows, any user can issue this command.

• **-s**

(Optional) Monitors the application server log and displays the log file. To terminate the command, press Ctrl + C.

**Limits:** This parameter is only valid when you run the command on the machine that runs the application server or on a machine with access to the same \$AUTOUSER (UNIX) or %AUTOSERVER% (Windows) file system.

• **-j job\_name**

Displays the agent log for the specified job.

**NOTE**

- You do not need to precede the job name that contains a colon with a backslash.
- The -e, -s and the -j options are mutually exclusive.

• **-d dataset\_name|FileName:Num**

Displays the contents of a z/OS job log dataset or i5/OS job log file.

**NOTE**

To obtain a list of dataset names for a job, use the -j parameter with the -Z option.

• **-r run\_num**

(Optional) Displays the agent log for the specified run number of the job.

**Default:** 0. If you do not specify this option, the system displays the log for the most recent run.

**Limits:** This option is only valid when you also use the -j parameter.

• **-n retry\_num**

(Optional) Displays the agent log for the specified retry of the job run.

**Default:** 0. If you do not specify this option, the system displays the log for the last retry.

**Limits:** Specify a positive integer for this value. This option is only valid when you also use the -r option with the -j parameter.

• **-t**

(Optional) Specifies the type of log file to display.

**Limits:** When you use this parameter, you must also specify the **A, O, E, S, or L** option.

– **A**

(Default) Displays the agent log file for the job.

– **O**

Displays the standard output file for the job.

– **E**

Displays the standard error file for the job.

– **S**

Displays the spool file for the job.

– **L**

*Displays the step log file or task name log file for the job. When you use this option, you must also specify the -i StepId or -o StepName|TaskName option.*

- **-i StepId**

Displays the step log file for the specified SQL Server Agent job identification number. This option is only applicable for SQL Server Agent jobs.

- **-o *StepName|TaskName*** Displays the log file for the specified SQL Server Agent job step name or Informatica job task name. This option is only applicable for SQL Server Agent jobs and Informatica jobs.
- **-z**  
Displays a list of available z/OS job log types and dataset names or i5/OS job log file names.

#### NOTE

Use the -z option first to retrieve all the available log datasets or file names followed by the -d *DatasetName|FileName:Num* option to retrieve the contents of an individual log dataset or file where the i5/OS file is specified with both name and number.

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**  
(Optional) Specifies external security credentials as follows:  
**-usr *username* -pw *password***: Specifies a username and unencrypted password.  
**-usr *username* -pwx *password***: Specifies a username and encrypted password.

#### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact***: Specifies a Security Assertion Markup Language (SAML) artifact.

#### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.



**Example: View the Scheduler Log**

This example displays the scheduler log when entered on the command line of the system where the scheduler is running, or where it ran last:

```
autosyslog - e
```

**NOTE**

You can enter this command on any machine that can access the same file system (typically \$AUTOUSER on UNIX or %AUTOUSER% on Windows) as the scheduler.

**Example: View the Agent Log for a Job**

This example displays the agent log of the last run of the test\_install job.

```
autosyslog - J test_install
```

**Example: Append Profile Information to Log File on UNIX**

This example appends profile information to the log file for the test\_install job.

```
autosyslog - j test_install - tP
```

**Example: View the Agent Log for a Previous Run of a Job**

This example displays the agent log for a previous run of the test\_install job.

```
autosyslog - J test_install - r -1
```

**NOTE**

Specifying a value of -1 for the -r parameter displays the agent log for the run immediately prior to the current run. To view the agent log any other previous run, specify the actual run number using the -r parameter.

**Example: View the Agent Spool File for a Job**

This example displays the agent spool file for the last run of the test\_install job.

```
autosyslog - J test_install - T S
```

**autotimezone Command—Manage the ujo\_timezones Table**

The autotimezone command is a client component utility that lets you query, add, and delete entries in the ujo\_timezones table. The ujo\_timezones table maps cities and common aliases to valid POSIX TZ environment variables. The table contains entries for all common time zones recognized by most operating systems, and for many cities.

You can use the timezone attribute to specify entries from the ujo\_timezones table in a job definition.

The ujo\_timezones table has three entry types: Zone, Alias, and City, as shown in the following excerpt from the ujo\_timezones table:

Entry	Type	Zone
-----	-----	-----
US/Pacific	Zone	PST8PDT
US/Samoa	Alias	Pacific/Samoa
UTC	Alias	GMT+0
Vancouver	City	Canada/Pacific

All “Alias” and “City” types are eventually resolve to “Zone” types. The “Zone” types resolved to TZ variables (in the Zone column) that correspond to those recognized by the operating system for the machine on which the event server is running.

## Syntax

This command has the following format:

```
autotimezone
 [-a aliaszone] [-c cityzone] [-t timezone] [-d alias|city] [-
q alias|city|timezone|pattern] [-l][{-usr username {-pw|pwx} password|-
saml auth_artifact}]
 [-x]
 [-?]
```

- **-a *alias zone***

(Optional) Associates a name (*alias*) with a time zone (*zone*) and adds it to the ujo\_timezones table as an alias entry.

**Limits:** The *alias* value can be up to 50 alphanumeric characters and can include slash (/), hyphen (-), and underscore (\_) characters.

### NOTE

- The *zone* value must be a time zone recognized by the operating system.
- Separate the *alias* and *zone* values with a space.

- **-c *city zone***

(Optional) Associates a city (*city*) with a time zone (*zone*) and adds it to the ujo\_timezones table as a city entry. Entries added to the table with the -c parameter display as type "City" in a listing of the ujo\_timezones table.

**Limits:** The *city* value can be up to 50 alphanumeric characters and can include slash (/), hyphen (-), and underscore (\_) characters.

### NOTE

- The *zone* value must be a time zone recognized by the operating system.
- Separate the *city* and *zone* values with a space.

- **-t *timezone***

(Optional) Adds a time zone entry to the ujo\_timezones table.

**Limits:** A time zone entry must be formatted as a valid POSIX standard timezone (TZ) environment variable.

- **-d *alias|city***

(Optional) Deletes the entry associated with the specified alias or city from the ujo\_timezones table.

- **-q *alias|city|timezone|pattern***

(Optional) Queries the ujo\_timezones table for the settings of the specified alias, city, time zone, or pattern.

**Limits:** This value allows wildcard characters.

- **-l**

(Optional) Lists all entries in the ujo\_timezones table.

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**

(Optional) Specifies external security credentials as follows:

**-usr *username* -pw *password*:** Specifies a username and unencrypted password.

**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- -x  
(Optional) Returns the command version information to standard output.
- -?  
(Optional) Displays help for the command

**Example: Add a City to the ujo\_timezones Table**

This example adds a city San-Jose in the US/Pacific time zone to the ujo\_timezones table.

```
autotimezone - c San-Jose US/Pacific
```

**Example: Delete a City from the ujo\_timezones Table**

This example deletes the city San-Jose from the ujo\_timezones table.

```
autotimezone - d San-Jose
```

**Example: Query the ujo\_timezones Table**

This example queries the ujo\_timezones table for all entries beginning with the letter “d”.

```
autotimezone - q d%
```

The output from this command resembles the following:

```
Entry Type Zone

Dallas City US/Central
Denver City US/Mountain
Detroit City US/Eastern
```

## **TZ Variable Syntax**

### **WARNING**

The ujo\_timezones table is complete and accurate and does not need modification. If you change the ujo\_timezones table, you must not change or delete entries that are used by pre-existing STARTJOB and other events that were scheduled using the old ujo\_timezones table.

The TZ variable has the following format:

```
std offset [dst [offset] [,start [/time] ,end [/time]]]
```

- **std offset**

Defines standard time for the time zone, where *offset* indicates the amount of time to add to or subtract from the local time to arrive at GMT.

**Limits:** This value is required.

The *offset* value has the following format:

```
[-|+]hh[:mm[:ss]]
```

- **-|+**

(Optional) Defines whether the offset value represents a time zone west or east of GMT.

**Limits:** A + or no prefix indicates that the value represents a time zone west of GMT. A - prefix indicates that the value represents a time zone east of GMT.

- **hh**

Defines the number of hours offset from GMT.

**Limits:** This value is required. This value can be a number in the range 0 through 24.

- **mm**

(Optional) Defines the number of minutes offset from GMT.

**Limits:** This value can be a number in the range 0 through 59.

- **ss**

(Optional) Defines the number of seconds offset from GMT.

**Limits:** This value can be a number in the range 0 through 59.

- **dst offset**

(Optional) Defines daylight saving time for the time zone, where *offset* indicates the amount of time to add to the local time to arrive at GMT.

**Default:** When you do not specify dst, the product assumes that the time zone does not observe daylight saving time. When you specify dst without an offset value, the product assumes that daylight saving time is one hour ahead of standard time.

**Limits:** The *offset* value has the following format:

```
[-|+]hh[:mm[:ss]]
```

- **-|+**

(Optional) Defines whether the offset value represents a time zone west or east of GMT.

**Limits:** A + or no prefix indicates that the value represents a time zone west of GMT. A - prefix indicates that the value represents a time zone east of GMT.

- **hh**

Defines the number of hours offset from GMT.

**Limits:** This value is required. This value can be a number in the range 0 through 24.

- **mm**

(Optional) Defines the number of minutes offset from GMT.

**Limits:** This value can be a number in the range 0 through 59.

- **ss**

(Optional) Defines the number of seconds offset from GMT.

**Limits:** This value can be a number in the range 0 through 59.

- **start**

(Optional) Defines the day on which to change to daylight saving time from standard time.

**Limits:** This value can be one of the following:

– ***jn***

Defines the Julian day *n*, where *n* can be a number in the range 1 through 365. The product ignores leap days, meaning that December 31 is always day 365 and you cannot specify February 29.

– ***n***

Defines the Julian day *n*, where *n* can be a number in the range 0 through 365. In this case, the product counts leap days.

– ***mm.n. d***

Defines that the time change occurs on the *n*th occurrence of a particular day in a specific month, where:

- *mm* can be a number in the range 1 through 12. January is month 1.
- *n* can be a number in the range 0 through 5. Sunday is day 0. When *n* is 5, it means the last occurrence of the specified day in month *mm*.
- *d* can be a number in the range 0 through 6.

For example, to specify that the time change should occur on the first Sunday in April, you would specify 4.1.0.

**NOTE**

If you do not specify *start* and *end*, United States daylight saving time defaults are used. That is, *start* is the second Sunday in March and *end* is the first Sunday in November.

• ***end***

Defines the day on which to change to standard time from daylight saving time.

**Limits:** This value can be one of the following:

– ***jn***

Defines the Julian day *n*, where *n* can be a number in the range 1 through 365. The product ignores leap days, meaning that December 31 is always day 365 and you cannot specify February 29.

– ***n***

Defines the Julian day *n*, where *n* can be a number in the range 0 through 365. In this case, the product counts leap days.

– ***mm.n. d***

Defines that the time change occurs on the *n*th occurrence of a particular day in a specific month, where:

- *mm* can be a number in the range 1 through 12. January is month 1.
- *n* can be a number in the range 0 through 5. Sunday is day 0. When *n* is 5, it means the last occurrence of the specified day in month *mm*.
- *d* can be a number in the range 0 through 6.

For example, to specify that the time change should occur on the first Sunday in April, you would specify 4.1.0.

**NOTE**

If you do not specify *start* and *end*, United States daylight saving time defaults are used. That is, *start* is the second Sunday in March and *end* is the first Sunday in November.

• ***time***

Defines the time of day at which the change to daylight saving or standard time occurs. **Limits:** This value has the following format:

- hh[:mm[:ss]]

***hh***

Defines the hour at which the change occurs.

**Limits:** This value can be a number in the range 0 through 23.

This value is required.

• ***mm***

(Optional) Defines the minute at which the change occurs.

**Limits:** This value can be a number in the range 0 through 59.

- **ss**

Defines the seconds at which the change occurs.

**Default:** 02:00:00 (2:00 a.m.)

**Limits:** This value can be a number in the range 0 through 59.

**NOTE**

Do not enter spaces between any of the components of the TZ variable.

**UNIX Note:**

For more information about the format of the TZ variable, see your system time, timezone, or environ man page.

When processing a job definition that includes a time zone, AutoSys Workload Automation tries to resolve the specified time zone with the zones known to the operating system. If it is not found there, AutoSys Workload Automation looks up the zone in the ujo\_timezones table. If the time zone specification is not a TZ variable, the product reads the ujo\_timezones table repeatedly until the specification resolves to a TZ variable. For example, assume that a job definition included the following attribute:

```
timezone:Berlin.
```

The product resolves Berlin to Europe/Berlin the first time it reads the table. The second time the product reads the table that it resolves Europe/Berlin to METS-1METD, which is a TZ variable. If a time zone is not resolved after five tries, the product considers it invalid and the job specifying the time zone fails.

**Windows Note:**

TZ variable syntax is compatible with Windows NT TZ variables, but has been extended to allow full control over the day and time on which daylight saving time changes occur. If the time zone specification in a job definition is not a TZ variable, the product reads the ujo\_timezones table repeatedly until the specification resolves to a TZ variable. For example, assume that a job definition included the following attribute:

```
timezone:Berlin
```

The product resolves Berlin to Europe/Berlin the first time it reads the table. The second time the product reads the table, it resolves Europe/Berlin to MET, which is a TZ variable. If a time zone is not resolved after five tries, the product considers it invalid and the job specifying the time zone fails.

## chase Command—Verify Job STARTING or RUNNING Status

You can use the chase command to verify that jobs are running and to evaluate the health of the agent running a job. The chase command does the following:

1. Obtains a list of jobs from the database that have been in the STARTING state for more than 120 seconds
2. Obtains a list of jobs from the database that are in the RUNNING state
3. Passes the list of jobs that are in the RUNNING state to the associated agents

Each agent then verifies that the jobs are running and returns the process status to the chase command. If the chase command detects errors, it can be instructed to send an alarm or send an event to change the status of the job to FAILURE.

You can run the chase command at regular intervals to track network problems. For example, if a computer is unreachable while running a job, the chase command detects that the computer is down and can send an alarm to alert you about the problem.

**NOTE**

The database must be available and both the scheduler and the application server must be running.

## Syntax

This command has the following format:

```
chase
 [-A] [-E] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-A**(Optional) Sends an alarm when any job is stuck in the STARTING or RUNNING states.
- **-E**(Optional) Puts a job in FAILURE status when the job and its agent are not running on the client but the database indicates they should be. In this case, the job restarts if the job definition includes the `n_retrys` attribute. The scheduler must be running for chase to automatically restart jobs.

### NOTE

When chase cannot connect to an agent computer, it cannot determine the reason. To prevent jobs from being restarted when they might already have run, chase does not change the status of jobs in this case, even when you run it with the `-E` parameter.

- **{-usr username {-pw|-pwx} password |-saml auth\_artifact}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password:** Specifies a username and unencrypted password.

**-usr username -pwx password:** Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (`as_config` command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (`autosys_secure` command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the `-usr` option and you do not specify the `-pw` or `-pwx` option, the utility prompts you to specify a password.
  - When you specify the `-pw` or `-pwx` option but do not specify the `-usr` option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:

- The unauthenticated user mode setting is EXTERNAL.
- AutoSys Workload Automation is operating in native security mode
- -x  
(Optional) Returns the command version information to standard output.
- -?  
(Optional) Displays help for the command.

#### NOTE

- When chase detects errors, the client sends them to standard output. You can use the optional -A and -E parameters with chase to further define the actions to take for error conditions.
- When you run chase with no options, AutoSys Workload Automation performs all chase activities and writes the results to standard output, but sends no alarms or job restart events.
- Running the chase command does not result in the automatic restart of jobs that are stuck in the STARTING state. Instead, chase writes a message to standard output that manual intervention might be required. Jobs that are stuck in the STARTING state are not be automatically restarted -- it is possible (for example, due to network problems) that the job might already have run, and its state was not yet communicated to the database. Verify the actual status of these jobs before you manually restart them.
- You can reduce the alarms that are generated by the chase utility by setting the following environment variables on the computer where you are running the chase command:
  - CHASE\_SLEEP** Specifies the time (in seconds) that the chase utility pauses:
    - Between successive sends of chase data to an agent following a connection failure.  
**Default:** 5  
**Limits:** 1 or higher; increasing this value might delay recovery during a network failure. Decreasing this value might result in premature chase failures.
    - Before raising the CHASE\_ALARM alarm. During periods of network load or high workload activity, you can set this variable to let the chase utility extend the time that it waits for the scheduler to receive delayed job statuses from the agent; thus preventing the chase utility from raising false alarms.  
**Default:** 10
  - CHASE\_STARTING\_WAIT\_INTERVAL** Specifies the minimum time (in minutes) that the chase utility allows jobs to remain in the STARTING state before raising the CHASE\_ALARM alarm. During periods of network load or high workload activity, it might take more than the default two minutes for the agent to notify the scheduler that a job is executing. Setting this environment variable helps reduce the number of alarms that the chase utility generates during this time.  
**Default:** 2  
**Limits:** 1 to 35791394
- The chase command does not always work in an environment with agents installed in a load-balancing cluster. In that case, the chase command works only when the cluster name is configured to point to the same node where the job ran. For example, suppose that you have installed agents in a 3-node cluster that is named CLUSTERNODE that is configured to point to either NODE1, NODE2, or NODE3. Also, suppose that you create a machine definition that specifies CLUSTERNODE as the node\_name attribute value. If the cluster name, CLUSTERNODE, points to NODE1 when you start the job and you configure it to point to NODE2 after the job completes, then all chase requests are directed at NODE2. The chase command does not verify jobs running on NODE1 until you configure the cluster name to point back to NODE1.

#### Example: Verify that a Job is Running, Generate an Alarm, and Restart the Job

This example verifies whether the job is running, generates an alarm if there is an inconsistency, and restarts the job if necessary.

```
chase -A -E
```



## Running Chase Automatically

We recommend that you run chase automatically at regular intervals to track problems on the network. For example, if a computer becomes unreachable while running a job, chase detects that the computer is down and sends an alarm. If a user has a monitor running, they are also alerted to the problem.

On UNIX, you can run chase automatically by using AutoSys Workload Automation to run it as a job. The \$AUTOSYS/install/data/chase.jil file contains the job that instructs AutoSys Workload Automation to run chase every 10 minutes on the computer running the scheduler (“charley,” in the following example). You can change the specific parameters in this script to suit your own environment, and submit it to the jil command by running:

```
jil < $AUTOSYS/install/data/chase.jil
```

### Example: Run chase Automatically

This example defines a job for automatically running chase:

```
chase function
#
insert_job: chase
machine: charley
command: $AUTOSYS/bin/chase -A -E
date_conditions: yes
days_of_week: all
start_mins: 05,15,25,35,45,55
max_run_alarm: 5 /* Change if many jobs are running */
term_run_time: 10 /* Change if many jobs are running */
These output files can be changed
std_out_file: $AUTOUSER/out/chase.out
std_err_file: $AUTOUSER/out/chase.err
```

## chk\_auto\_up Command—Confirm Application Server, Scheduler, and Event Server Status

The chk\_auto\_up command is a client component that determines whether the event server (database), application server, and the scheduler are running. Because the event server and the scheduler must both be running for a job to start, the chk\_auto\_up command is the first utility that you should run when troubleshooting why events or jobs are not processing as scheduled.

**UNIX Note:** chk\_auto\_up uses the product environment variables to locate the configuration file \$AUTOUSER/config. \$AUTOSERV, and uses the information to determine the application server, data server, and database names. If chk\_auto\_up completes successfully, the product environment variables and the configuration file are set up correctly.

**Windows Note:** chk\_auto\_up uses the product environment variables to locate the instance configuration as defined in the AutoSys Workload Automation Administrator, and uses the information to determine the application server, data server, and database names. If chk\_auto\_up completes successfully, the product environment variables and the configuration are set up correctly.

When you issue the chk\_auto\_up command, the client (the machine from which you issued chk\_auto\_up) sends a request to the application server and waits for the application server to respond. The application server performs the following actions:

- look for schedulers who report their status as active in the database
- contact each active scheduler and wait for the schedulers to respond
- collect all results and prepare a response for chk\_auto\_up to report

If you are running the instance in High Availability mode with dual event servers, `chk_auto_up` also reports the state of the system including all schedulers and any occurrence of a database rollover or scheduler failover.

## Syntax

This command has the following format:

```
chk_auto_up
 [-Q] [-r ComponentFlags] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-Q**

(Optional) Returns the exit code without any descriptive message. This makes the command useful for inclusion in shell scripts. In this case, the return code is sufficient to indicate status. If you omit the `-Q` parameter, the command also returns a descriptive message.

- **-r *ComponentFlags***

(Optional) Checks one or more of the AutoSys Workload Automation components. You can use this option to verify the status of the event server and/or the application server and/or the scheduler. The value that is specified for the `-r` option determines the component to be checked.

**Limits:** This value can be one of the following:

- **001 or 1**  
Returns the status of the event server(s).
- **010 or 10**  
Returns the status of the scheduler(s).
- **011 or 11**  
Returns the status of the event server(s) and the scheduler(s).
- **100**  
Returns the status of the application server.
- **101**  
Returns the status of the application server and the event server(s).
- **110**  
Returns the status of the application server and the scheduler(s).
- **111**  
Returns the status of the application server, event server(s), and scheduler(s).

### NOTE

Use the `-r` option to verify the status of the application server. If you invoke the `chk_auto_up` command without any arguments, it just checks the status of the event server(s) and scheduler(s).

- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**

(Optional) Specifies external security credentials as follows:

**-usr *username* -pw *password*:** Specifies a username and unencrypted password.

**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (`as_config` command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (`autosys_secure` command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSysWorkload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

**Exit Codes**

The `chk_auto_up` command exits with two sets of exit codes, which include the following:

- Runs the application without any arguments.
- Runs the application with the - r option.

**Running chk\_auto\_up Without Any Arguments**

If you run `chk_auto_up` command without any arguments, the application exits with one of the following return codes:

- **0**  
Indicates that the scheduler or the event server is not running.
- **1**  
Indicates that the scheduler is not running, but the event server is running.
- **2**  
Indicates that the scheduler is not running, but the primary and dual event servers are running.
- **10**  
Indicates that the scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **11**

Indicates that the scheduler and the event server are running.

- **12**  
Indicates that the scheduler and the dual event servers are running.
- **20**  
Indicates that the shadow scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **21**  
Indicates that the shadow scheduler and the event server are running.
- **22**  
Indicates that the shadow scheduler and the dual event servers are running.
- **30**  
Indicates that the primary and shadow schedulers are running, but the event server is not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **31**  
Indicates that the primary and shadow schedulers and the event server are running.
- **32**  
Indicates that the primary and shadow schedulers and the dual event servers are running.
- **40**  
Indicates that the tie-breaker scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **41**  
Indicates that the tie-breaker scheduler and the event server are running.
- **42**  
Indicates that the tie-breaker scheduler and the dual event servers are running.
- **50**  
Indicates that the primary scheduler and the tie-breaker scheduler are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **51**  
Indicates that the primary scheduler, tie-breaker scheduler, and the event server are running.
- **52**  
Indicates that the primary scheduler, tie-breaker scheduler, and the dual event servers are running.
- **60**  
Indicates that the shadow scheduler and the tie-breaker scheduler are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **61**  
Indicates that the shadow scheduler, tie-breaker scheduler, and the event server are running.
- **62**  
Indicates that the shadow scheduler, tie-breaker scheduler, and the dual event servers are running.
- **70**  
Indicates that the primary, shadow, and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **71**  
Indicates that the primary scheduler, shadow scheduler, tie-breaker scheduler, and the event server are running.
- **72**

Indicates that the primary scheduler, shadow scheduler, tie-breaker scheduler, and the dual event servers are running.

- **81**  
Indicates that the event server is running, but the scheduler(s) are not responding.
- **82**  
Indicates that the dual event servers are running, but the scheduler(s) are not responding.
- **99**  
Indicates that one of the following has occurred:
  - One or more of the following environment variables is not set or is set incorrectly:
    - AUTOSYS
    - AUTOSERV
    - AUTOUSER.
  - The chk\_auto\_up command is issued with invalid arguments.
  - The application server is not running or not configured correctly.

### **Running chk\_auto\_up with the -r Argument**

If you run the chk\_auto\_up command with the - r argument, the application exits with any one of the following return codes.

If you run the chk\_auto\_up command with the - r 1 argument, it checks the event server and exits with one of the following return codes:

- **1**  
Indicates that the event server is running.
- **2**  
Indicates that the dual event servers are running.

If you run the chk\_auto\_up command with the - r 10 argument, it checks the scheduler and exits with one of the following return codes:

- **0**  
Indicates that the scheduler is not running.
- **10**  
Indicates that the primary scheduler is running.
- **20**  
Indicates that the shadow scheduler is running.
- **30**  
Indicates that the primary and shadow schedulers are running.
- **40**  
Indicates that the tie-breaker scheduler is running.
- **50**  
Indicates that the primary and tie-breaker schedulers are running.
- **60**  
Indicates that the shadow and tie-breaker schedulers are running.
- **70**  
Indicates that the primary, shadow, and tie-breaker schedulers are running.

If you run the chk\_auto\_up command with the - r 11 argument, it checks the scheduler and the event server, and exits with one of the following return codes:

- **0**  
Indicates that the scheduler or the event server is not running.
- **1**

- Indicates that the scheduler is not running, but the event server is running.
- **2**  
Indicates that the scheduler is not running, but the primary and dual event servers are running.
  - **10**  
Indicates that the scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **11**  
Indicates that the scheduler and the event server are running.
  - **12**  
Indicates that the scheduler and the dual event servers are running.
  - **20**  
Indicates that the shadow scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **21**  
Indicates that the shadow scheduler and the event server are running.
  - **22**  
Indicates that the shadow scheduler and the dual event servers are running.
  - **30**  
Indicates that the primary and shadow schedulers are running, but the event server is not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **31**  
Indicates that the primary and shadow schedulers and the event server are running.
  - **32**  
Indicates that the primary and shadow schedulers and the dual event servers are running.
  - **40**  
Indicates that the tie-breaker scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **41**  
Indicates that the tie-breaker scheduler and the event server are running.
  - **42**  
Indicates that the tie-breaker scheduler and the dual event servers are running.
  - **50**  
Indicates that the primary and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **51**  
Indicates that the primary scheduler, tie-breaker scheduler, and the event server are running.
  - **52**  
Indicates that the primary scheduler, tie-breaker scheduler, and the dual event servers are running.
  - **60**  
Indicates that the shadow and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
  - **61**  
Indicates that the shadow scheduler, tie-breaker scheduler, and the event server are running.
  - **62**

Indicates that the shadow scheduler, tie-breaker scheduler, and the dual event servers are running.

- **70**  
Indicates that the primary, shadow, and tie-breaker schedulers are running, but the event server is not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **71**  
Indicates that the primary scheduler, shadow scheduler, tie-breaker scheduler, and the event server are running.
- **72**  
Indicates that the primary scheduler, shadow scheduler, tie-breaker scheduler, and the dual event servers are running.
- **81**  
Indicates that the event server is running, but the scheduler(s) are not responding.
- **82**  
Indicates that the dual event servers are running, but the scheduler(s) are not responding.
- **99**  
Indicates that one of the following has occurred:
  - One or more of the following environment variables is not set or is set incorrectly:
    - AUTOSYS
    - AUTOSERV
    - AUTOUSER.
  - The `chk_auto_up` command is issued with invalid arguments.
  - The application server is not running or not configured correctly.

If you run the `chk_auto_up` command with the `- r 100` argument, it checks the application server, and exits with one of the following return codes:

- **0**  
Indicates that at least one or more application servers are not running.
- **100**  
Indicates that all the application servers are running.

If you run the `chk_auto_up` command with the `- r 101` argument, it checks the application server and the event server, and exits with one of the following return codes:

- **0**  
Indicates that the event server or at least one or more application servers are not running.
- **1**  
Indicates that the event server is running, but at least one or more application servers are not running.
- **2**  
Indicates that the dual event server is running, but at least one or more application servers are not running.
- **100**  
Indicates that all the application servers are running, but the event server is not running.
- **101**  
Indicates that the event server or all the application servers are running.
- **102**  
Indicates that the dual event server or all the application servers are running

If you run the `chk_auto_up` command with the `- r 110` argument, it checks the application server and the event server, and exits with one of the following return codes:

- **0**  
Indicates that the scheduler or at least one or more application servers are not running.
- **10**

- Indicates that the primary scheduler is running, but at least one or more application servers are not running.
- **20**  
Indicates that the shadow scheduler is running, but at least one or more application servers are not running.
- **30**  
Indicates that the primary and shadow schedulers are running, but at least one or more application servers are not running.
- **40**  
Indicates that the tie-breaker scheduler is running, but at least one or more application servers are not running.
- **50**  
Indicates that the primary and tie-breaker schedulers are running, but at least one or more application servers are not running.
- **60**  
Indicates that the shadow and tie-breaker schedulers are running, but at least one or more application servers are not running.
- **70**  
Indicates that the primary, shadow, and tie-breaker schedulers are running, but at least one or more application servers are not running.
- **100**  
Indicates that all the application servers are running, but the scheduler is not running.
- **110**  
Indicates that the primary scheduler or all the application servers are running.
- **120**  
Indicates that the shadow scheduler or all the application servers are running.
- **130**  
Indicates that the primary and shadow schedulers or all the application servers are running.
- **140**  
Indicates that the tie-breaker scheduler or all the application servers are running.
- **150**  
Indicates that the primary and tie-breaker schedulers or all the application servers are running.
- **160**  
Indicates that the shadow and tie-breaker schedulers or all the application servers are running.
- **170**  
Indicates that the primary, shadow, and tie-breaker schedulers or all the application servers are running.

If you run the `chk_auto_up` command with the `- r 111` argument, it checks the application server, scheduler, and the event server, and exits with one of the following return codes:

- **0**  
Indicates that the scheduler, event server, or at least one or more application servers are not running.
- **1**  
Indicates that the scheduler or at least one or more application servers are not running, but the event server is running.
- **2**  
Indicates that the scheduler or at least one or more application servers are not running, but the primary and dual event servers are running.
- **10**  
Indicates that the scheduler is running, but the event server or at least one or more application servers are not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **11**



Indicates that the scheduler and the event server are running, but at least one or more application servers are not running.

- **12**  
Indicates that the scheduler and the dual event servers are running, but at least one or more application servers are not running.
- **20**  
Indicates that the shadow scheduler is running, but the event server or at least one or more application servers are not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **21**  
Indicates that the shadow scheduler and the event server are running, but at least one or more application servers are not running.
- **22**  
Indicates that the shadow scheduler and the dual event servers are running, but at least one or more application servers are not running.
- **30**  
Indicates that the primary and shadow schedulers are running, but the event server or at least one or more application servers are not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **31**  
Indicates that the primary and shadow schedulers and the event server are running, but at least one or more application servers are not running.
- **32**  
Indicates that the primary and shadow schedulers and the dual event servers are running, but at least one or more application servers are not running.
- **40**  
Indicates that the tie-breaker scheduler is running, but the event server or at least one or more application servers are not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **41**  
Indicates that the tie-breaker scheduler and the event server are running, but at least one or more application servers are not running.
- **42**  
Indicates that the tie-breaker scheduler and the dual event servers are running, but at least one or more application servers are not running.
- **50**  
Indicates that the primary and tie-breaker schedulers are running, but the event server or at least one or more application servers are not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **51**  
Indicates that the primary and tie-breaker schedulers and the event server are running, but at least one or more application servers are not running.
- **52**  
Indicates that the primary and tie-breaker schedulers and the dual event servers are running, but at least one or more application servers are not running.
- **60**

Indicates that the shadow and tie-breaker schedulers are running, but the event server or at least one or more application servers are not running.

This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.

- **61**  
Indicates that the shadow and tie-breaker schedulers and the event server are running, but at least one or more application servers are not running.
- **62**  
Indicates that the shadow and tie-breaker schedulers and the dual event servers are running, but at least one or more application servers are not running.
- **70**  
Indicates that the primary, shadow, and the tie-breaker schedulers are running, but the event server or at least one or more application servers are not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **71**  
Indicates that the primary, shadow, and tie-breaker schedulers and the event server are running, but at least one or more application servers are not running.
- **72**  
Indicates that the primary, shadow, and tie-breaker schedulers and the dual event servers are running, but at least one or more application servers are not running.
- **81**  
Indicates that the event server is running, but the scheduler(s) are not responding and at least one or more application servers are not running.
- **82**  
Indicates that the dual event servers are running, but the scheduler(s) are not responding and at least one or more application servers are not running.
- **99** Indicates that one of the following has occurred:
  - One or more of the following environment variables is not set or is set incorrectly:
    - AUTOSYS
    - AUTOSERV
    - AUTOUSER.
  - The `chk_auto_up` command is issued with invalid arguments.
  - The application server is not running or not configured correctly.
- **100**  
Indicates that the scheduler or the event server is not running, but all the application servers are running.
- **101**  
Indicates that the scheduler is not running, but the event server or all the application servers are running.
- **102**  
Indicates that the scheduler is not running, but the primary and dual event servers or all the application servers are running.
- **110**  
Indicates that the scheduler is running, but the event server is not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **111**  
Indicates that the scheduler and the event server are running or all the application servers are running.
- **112**  
Indicates that the scheduler and the dual event servers are running or all the application servers are running.
- **120**

Indicates that the shadow scheduler is running, but the event server is not running.

This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.

- **121**  
Indicates that the shadow scheduler and the event server are running or all the application servers are running.
- **122**  
Indicates that the shadow scheduler and the dual event servers are running or all the application servers are running.
- **130**  
Indicates that the primary and shadow schedulers are running, but the event server is not running. This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **131**  
Indicates that the primary and shadow schedulers and the event server are running or all the application servers are running.
- **132**  
Indicates that the primary and shadow schedulers and the dual event servers are running or all the application servers are running.
- **140**  
Indicates that the tie-breaker scheduler is running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **141**  
Indicates that the tie-breaker scheduler and the event server are running or all the application servers are running.
- **142**  
Indicates that the tie-breaker scheduler and the dual event servers are running or all the application servers are running.
- **150**  
Indicates that the primary and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **151**  
Indicates that the primary and tie-breaker schedulers and the event server are running or all the application servers are running.
- **152**  
Indicates that the primary and tie-breaker schedulers and the dual event servers are running or all the application servers are running.
- **160**  
Indicates that the shadow and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **161**  
Indicates that the shadow and tie-breaker schedulers and the event server are running or all the application servers are running.
- **162**  
Indicates that the shadow and tie-breaker schedulers and the dual event servers are running or all the application servers are running.
- **170**  
Indicates that the primary, shadow, and tie-breaker schedulers are running, but the event server is not running.  
This could happen if the database password was changed in configuration file, but the application server was not restarted. The `chk_auto_up` command prints a detailed error message if it is not able to connect to the event server.
- **171**

Indicates that the primary, shadow, and tie-breaker schedulers and the event server are running or all the application servers are running.

- **172**  
Indicates that the primary, shadow, and tie-breaker schedulers and the dual event servers are running or all the application servers are running.
- **181**  
Indicates that the event server or all the application servers are running, but the scheduler(s) are not responding.
- **182**  
Indicates that the dual event servers or all the application servers are running, but the scheduler(s) are not responding.

### Example: Confirm Event Server and Scheduler Status

This example determines whether the event server and scheduler are up, and displays the results on your monitor:

```
chk_auto_up
```

### Example: Confirm Event Server, Application Server, and Scheduler Status

This example determines whether the event server, application server, and scheduler are up, and displays the results on your monitor:

```
chk_auto_up -r 111
```

## eventor Command—Start the Scheduler

### Valid on UNIX

The eventor command starts the scheduler (also called the event demon) on UNIX. By default, eventor runs in the background and performs the following tasks:

1. Confirms that another scheduler of the same instance is not running on the same machine (as determined by the \$AUTOSERV variable).
2. Writes scheduler log information from \$AUTOUSER/out/event\_demon.\$AUTOSERV to standard output using the astail command.

### Syntax

This command has the following format:

```
eventor [-f] [-q]
```

- **-f**  
(Optional) Runs the scheduler in the foreground and sends all its output to the display where the command was issued.

#### NOTE

We do not recommend the -f option for production use. The default behavior is to run the scheduler in the background. All output is written to the \$AUTOUSER/out/event\_demon.\$AUTOSERV file.

- **-q**  
(Optional) Runs eventor in quiet mode.

#### NOTE

- We recommend that you start the scheduler with no options set. In this situation, AutoSys Workload Automation performs all restart checks, sends alarms, and records output in the log file.
- Do not try to start the scheduler by invoking the event\_demon binary at the command line. Use the eventor command to properly verify and configure the scheduler environment.

### Example: Start the Scheduler

This example starts the scheduler in the background and runs the chase and astail commands.

```
eventor
```

## Log Files

### Valid on UNIX

The eventor command starts the scheduler, which writes log information to the file named \$AUTOUSER/out/event\_demon.\$AUTOSERV.

By default, eventor runs the astail -f command against the log file. This command is useful for monitoring the scheduler execution, particularly when there are problems with its startup. For example, if shared libraries that are required by the scheduler are missing or cannot be found, the scheduler does not start. In this case, the scheduler writes a message to its log file to indicate the problem. To exit the astail command, use Ctrl+C in the window displaying the scheduler log.

The shadow scheduler writes log information to the \$AUTOUSER/ out/event\_demon.shadow.\$AUTOSERV file.

The tie-breaker scheduler writes log information to the \$AUTOUSER/out/event\_demon.tie-breaker.\$AUTOSERV file.

## initautosys Command—Open the Instance Command Prompt Window

### Valid on Windows

The AutoSys Workload Automation instance command prompt lets you run client utilities, such as the jil and autorep commands. Usually, you open the instance Command Prompt window using the shortcut that is created when you install AutoSys Workload Automation.

The initautosys command lets you open the AutoSys Workload Automation instance Command Prompt window from the Windows Command Prompt and run a client utility. You can also specify the initautosys command in a BAT or CMD file to run a AutoSys Workload Automation command from a script.

### Syntax

This command has the following format:

```
initautosys -i instance -r "command_line"
```

- **-i *instance***  
Specifies the instance name.
- **-r "*command\_line*"**  
Specifies the full command line.

#### NOTE

- You cannot use a network drive letter. Specify a fully qualified network name (for example, \\taurus\C\$\tmp).
- The network drive (for example, \\taurus\C\$) must be shared so that a password is not required to access it.

#### NOTE

You do not have to set the environment. The AutoSys Workload Automation instance command prompt automatically sets the environment variables that are required to run AutoSys Workload Automation commands.

### Example: Open the Instance Command Prompt and Run a Command

This example opens a AutoSys Workload Automation instance Command Prompt window on the machine where the ace instance is installed and runs the autorep command. The autorep command generates a report that lists all machines defined on the data server. You can run the initautosys command from the Windows Command Prompt. You can also specify this command in a BAT or CMD file.

```
initautosys -i ace -r "autorep -M ALL"
```

## sendevent Command—Stop the Schedulers or Application Servers

Sometimes you must stop the server processes. For example, you must stop processes that are running on a machine so that you can run maintenance on it. You can use the sendevent command to stop server processes.

### Syntax

This command has the following format:

```
sendevent -E STOP_DEMON [-v arguments] [-S autoserv_instance] [-P priority] [-T "time_of_event"] [-C comment] [-v arguments]
```

- **-v arguments**

(Optional) Specifies the process to shut down. The event stops only the primary scheduler when you do not specify this option.

Specify *one* of the following arguments:

- **ALL**

Stops all server processes.

- **FAILOVER**

Forces a failover (from the primary scheduler to the shadow scheduler) or a fallback (from the shadow scheduler to the primary scheduler). The scheduler that is processing events shuts down and the other scheduler takes over processing events.

#### NOTE

- If AutoSys Workload Automation is running in High Availability mode, the active scheduler shuts down and the dormant scheduler takes over interpreting and processing events.
- If the primary scheduler is active and the shadow scheduler is dormant, use the option to force the shadow scheduler to take over processing events.
- If the shadow scheduler is active, the primary scheduler is dormant, and the primary fallback mode is set to 1 (Dormant), use this option to force the primary scheduler to resume processing events.

- **ROLE=values**

Stops the processes according to the combination of *values* specified. You can specify a single value or any combination of values as follows:

- **A**

Stops all the application servers.

- **P**

Stops the primary scheduler.

- **S**

Stops the shadow scheduler.

- **T**

Stops the tie-breaker scheduler.

#### NOTE

- Enter the combination of values with no spaces or commas. For example, entering ROLE=PST stops the primary, shadow, and tie-breaker schedulers.
- To process the STOP\_DEMON event and notify the processes to shutdown, either the primary scheduler must be active or the shadow scheduler must have failed over. If you send this event when there are no active schedulers, the system only processes the event when the primary scheduler is restored.
- The system writes warning messages to the scheduler log when you attempt to stop a process that does not exist.
- The active scheduler notifies all application servers to shut down when you specify -v ROLE=A or -v ALL. The active scheduler issues the APP\_SERVER\_SHUTDOWN alarm unless no active scheduler is available or the active scheduler must shut down.

- **-S *autoserv\_instance***  
(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
  - **-P *priority***  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.  
**Default:** 10  
**Limits:** 1-1000
  - **-T "*time\_of\_event*"**  
(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:
    - ***MM***  
Specifies the month.  
**Limits:** 01-12
    - ***dd***  
Specifies the day.  
**Limits:** 01-31
    - ***yyyy***  
Specifies the year.  
**Limits:** 1900-2100
    - ***hh***  
Specifies the hour.  
**Limits:** 00-23
    - ***mm***  
Specifies the minutes.  
**Limits:** 00-59
    - ***ss***  
(Optional) Specifies the seconds.  
**Limits:** 00-59
- NOTE**  
AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.
- **-C *comment***  
(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Stop the Primary Scheduler**

Suppose that you temporarily stop work so that you can modify the workflow. To stop processing events, shutdown the primary scheduler as follows:

```
sendevent -E STOP_DEMON
```

**Example: Stop Application Servers**

Suppose that you purchased new machines to replace your current application servers. This example shuts down all application servers so that you can replace them with the new machines.

```
sendevent -E STOP_DEMON -v ROLE=A
```

### Example: Initiate a Failover

Suppose that you run maintenance on the machine where the primary scheduler is running, but you do not want to interrupt your workflow. To ensure that the shadow scheduler takes over when you stop the primary scheduler for maintenance, initiate a failover as follows:

```
sendevent -E STOP_DEMON -v FAILOVER
```

#### **WARNING**

You cannot perform maintenance on a machine when server processes are running on it. After you initiate the failover, stop the all server processes.

### Example: Stop all Server Processes

This example stops all server processes.

```
sendevent -E STOP_DEMON -v ALL
```

## time0 Command—Calculate Internal AutoSys Workload Automation Time

The time0 command is a client component that calculates the internal AutoSys Workload Automation time and inserts it in MM/DD/YYYY hh:mm:ss format.

### Syntax

This command has the following format:

```
time0 [-A autotime | -T "datetime"] [-S] [-O "dateformat"] [-x] [-?]
```

- **-A autotime**  
(Optional) Converts the internal AutoSys Workload Automation date and time information to MM/DD/YYYY hh:mm:ss format.  
  
**Limits:** You cannot specify **-A autotime** if you specify **-T datetime**.
- **-T datetime**  
(Optional) Identifies a date and time to convert to the internal AutoSys Workload Automation format.  
  
**Limits:** You cannot specify **-T datetime** if you specify **-A autotime**.
- **-S**  
(Optional) Returns the internal AutoSys Workload Automation timeFormat.
- **-O dateformat**  
(Optional) Specifies a format with which to override the MM/DD/YYYY hh:mm:ss format.  
  
**Limits:** Enclose the dateformat value in quotation marks (").
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.



**NOTE**

When you use the time0 command with no arguments, the client returns the current AutoSys Workload Automation time and date information.

**Example: Return the Current AutoSys Workload Automation Date and Time Information**

This example returns the current AutoSys Workload Automation time and date information:

```
time0
```

The return from the command resembles the following:

```
CAUAJM I 50096 Current AutoTime(internal): 1121265931
```

**Example: Convert the AutoSys Workload Automation Date and Time to MM/DD/YYYY hh:mm:ss Format**

This example converts the AutoSys Workload Automation date and time to MM/DD/YYYY hh:mm:ss format:

```
time0 -A 1121265931
```

The return from the command resembles the following:

```
CAUAJM I 50097 External Time: 07/13/2005 07:45:31
```

**Maintain Databases**

Database maintenance utilities let you complete tasks that keep the system running smoothly.

The following are the database maintenance utilities:

**dbspace Command -- Calculate and Report Database Table Space**

The dbspace command calculates and displays a summary of the disk space used by the AutoSys Workload Automation database tables.

**Syntax**

This command has the following format:

```
dbspace [-x] [-?]
```

- **-x**  
(Optional) Returns the product version information.
- **-?**  
(Optional) Displays help for the command.

**NOTE**

If you use an Oracle database, consider the following points:

- To obtain the correct disk space that is used by the AutoSys Workload Automation database tables, run the dbstatistics command before running the dbspace command. The dbstatistics command updates the metadata maintained by Oracle in the all\_tables and dba\_tables tables.
- The dbspace command might erroneously report that the database is nearly full. This can occur because dbspace calculates how much space is not allocated for extents. The dbspace command reports entire extents as used space even when they are nearly empty.
- The dbspace command might incur privilege errors during execution. Privilege errors occur because the UJOADMIN account has not been granted the ANALYZE ANY privilege or the ujoadmin role does not have

SELECT privileges to DBA\_TABLES. If the ujoadmin role does not have the proper authority, the following messages are displayed

```
CAUAJM_E_18400 An error has occurred while interfacing with ORACLE.
CAUAJM_E_18401 Function <Oexec> invoked from <execute> failed <600>
CAUAJM_E_18402 ORA-00942: table or view does not exist
CAUAJM_I_18403 Processing OCI function not used(4)
dbspace: Failed to retrieve blocksize
```

To grant these privileges, run the following SQL commands:

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
grant ANALYZE ANY to UJOADMIN;
grant SELECT ON DBA_TABLES to UJOADMIN;
```

To revoke these privileges, run the following SQL commands:

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
revoke ANALYZE ANY from UJOADMIN;
revoke SELECT ON DBA_TABLES from UJOADMIN;
```

### Example: Calculate and Report Database Table Space

This example calculates and displays the disk space allocated for the AutoSys Workload Automation tables:

```
dbspace
```

## DBMaint Command -- Maintain the AutoSys Workload Automation Database

The DBMaint command runs the dbstatistics, archive\_events, and archive\_jobs commands to perform maintenance on the AutoSys Workload Automation database.

### Syntax

This command has the following format:

```
DBMaint [-x] [-?]
```

- **-x**  
(Optional) Returns version information.
- **-?**  
(Optional) Displays help for the command.

DBMaint runs the dbstatistics command to perform the following tasks:

- Updates the statistics for all the tables except the ujo\_meta\_enumerations, ujo\_meta\_properties, ujo\_meta\_rules, ujo\_meta\_types, and ujo\_real\_resource tables.
- Runs the dbspace command to determine the available space in the database. If the amount of free space is insufficient, the command issues warnings and generates a DB\_PROBLEM alarm.  
If you use an Oracle database, consider the following points:
  - The DBMaint command might erroneously report that the database is nearly full. This can occur because DBMaint calculates how much space is not allocated for extents. DBMaint reports entire extents as used space even when they are nearly empty.
  - The DBMaint command might incur privilege errors during the execution of either the dbstatistics or dbspace command. These privilege errors might also be incurred when running the programs standalone. Privilege errors occur because the UJOADMIN account has not been granted the ANALYZE ANY privilege or the ujoadmin role does not have SELECT privileges to DBA\_TABLES.

To grant these privileges, run the following SQL commands:

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
grant ANALYZE ANY to UJODADMIN;
grant SELECT ON DBA_TABLES to UJODADMIN;
```

To revoke these privileges, run the following SQL commands:

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
revoke ANALYZE ANY from UJODADMIN;
revoke SELECT ON DBA_TABLES from UJODADMIN;
```

- Calculates and updates the average job run statistics in the ujo\_avg\_job\_run table. The old data is overwritten with the new data. The command returns either a 0 or 1; 0 indicates success and 1 indicates failure.

DBMaint runs the archive\_events command to remove old information from the database tables. By default, the archive\_events command removes data that is more than seven days old. Specifically, archive\_events removes the following:

- Events and associated alarms from the ujo\_job\_runs table.
- Job run information from the ujo\_job\_runs table.
- autotrack log information from the ujo\_audit\_info and ujo\_audit\_msg tables.
- Machine information from the ujo\_machine table.

The output from DBMaint, \$AUTOUSER/out/DBMaint.out, reports the amount of space remaining in the database so you can monitor whether the event tables are getting full. By monitoring these values, you can calculate how many events you can safely maintain in a day before archiving.

DBMaint runs the archive\_jobs command to remove obsolete jobs from the database tables.

#### NOTE

- If you are archiving large event tables, your database connection might time out, causing DBMaint to fail. If this occurs, edit the DBMaint script to increase the value of the -t parameter in the archive\_events command. For information about editing the DBMaint script, see [General Database Maintenance](#).
- By default, the AutoSys Workload Automation OS user is "autosys". However, you can define a different ID for the OS user during installation.

## dbstatistics Command -- Generate Data Server Statistics

The dbstatistics command is a scheduler component that generates statistics in the data servers to maintain an optimal performance environment.

### Syntax

This command has the following format:

```
dbstatistics [-x] [-i] [-?]
```

- **-x**  
(Optional) Returns version information.
- **-i**  
(Optional) (Sybase only) Updates index statistics for all columns in the Sybase data servers.
- **-?**  
(Optional) Displays help for the command.

The dbstatistics command performs the following tasks:

- Updates the statistics for all the tables except the following tables:

- ujo\_intcodes
- ujo\_meta\_afm\_types
- ujo\_meta\_enumerations
- ujo\_meta\_parse\_exruntime
- ujo\_meta\_parse\_keywords
- ujo\_meta\_presentations
- ujo\_meta\_properties
- ujo\_meta\_restart\_afm\_types
- ujo\_meta\_root\_keywords
- ujo\_meta\_rules
- ujo\_meta\_types
- ujo\_meta\_ws\_elements
- ujo\_patch
- ujo\_real\_resource
- Updates the index statistics for all columns in the Sybase data servers. We recommend that you use the -i option once a week.
- Runs the dbspace command to determine the available space in the database. If the amount of free space is insufficient, the command issues warnings and generates a DB\_PROBLEM alarm.  
If you use an Oracle database, consider the following points:
  - The DBMaint command may erroneously report that the database is nearly full. This can occur because DBMaint calculates how much space is not allocated for extents. DBMaint reports entire extents as used space even when they are nearly empty.
  - The DBMaint command may incur privilege errors during the execution of either the dbstatistics or dbspace command. These privilege errors may also be incurred when running the programs standalone. Privilege errors occur because the UJOADMIN account has not been granted the ANALYZE ANY privilege or the ujoadmin role does not have SELECT privileges to DBA\_TABLES.  
To grant these privileges, run the following SQL commands:
 

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
grant ANALYZE ANY to UJOADMIN;
grant SELECT ON DBA_TABLES to UJOADMIN;
```

 To revoke these privileges, run the following SQL commands:
 

```
sqlplus sys/<passwd>@<ORACLE_SID> as sysdba
revoke ANALYZE ANY from UJOADMIN;
revoke SELECT ON DBA_TABLES from UJOADMIN;
```
- Calculates and updates the average job run statistics in the ujo\_avg\_job\_run table. The old data is overwritten with the new data. The command returns either a 0 or 1; 0 indicates success and 1 indicates failure.

#### NOTE

By default, the AutoSys Workload Automation OS user is "autosys". However, you can define a different ID for the OS user during installation.

## Manage Security

Security management utilities let you manage encryption keys and passwords, define security settings, manage the security policy repository, and test the security subsystem.

The following are the security management utilities:

## as\_config Command -- Manage Encryption Keys and Passwords

The `as_config` command lets you manage the encryption keys for the application server and for Workload Automation Agent for UNIX, Linux, or Windows. You can use `as_config` to encrypt a key and store it in `$AUTOUSER/cryptkey.txt`. If you want to use that encrypted key for the agent, you can copy the `cryptkey.txt` file to the `install_directory/SystemAgent/agent_name` directory.

### Syntax

This command has the following format:

```
as_config [-g key] [-a key] [-c file name [-e key string]] [-d] [-h] [-p | -p -
n pwvar] [-m message_key | -m message_key argument...] [-x] [-?]
```

- **-g key**

(Optional) Encrypts the specified key and stores it in `$AUTOUSER/cryptkey.txt`.

#### NOTE

To use the `-g` option, you must set the `AUTOUSER` environment variable.

- **-a key**

(Optional) Encrypts the specified key and stores it in `$AUTOUSER/cryptkey_alias.txt`. This option supports 32-character hexadecimal key only.

#### NOTE

To use the `-a` option, you must set the `AUTOUSER` environment variable.

- **-c file name**

(Optional) Compares the encrypted key in the file specified in this option with the key string specified in the `-e` option.

- **-e key string** (Optional) Defines a key string that is used to compare with the encrypted key in the file specified in the `-c` option.

#### NOTE

To use the `-e` option, you must also specify the `-c` option. If you do not use the `-e` option, the default encryption key is compared with file specified in the `-c` option.

- **-d**

Encrypts the default encryption key and stores it in `$AUTOUSER/cryptkey.txt`.

- **-h**

(Optional) Prints the usage statement.

- **-p**

(Optional) Encrypts the password specified by the `-n` option (environment variable) to standard output. If `-n` is not specified the password must be entered through stdin. This option invokes the AES encryption algorithm to encode the password, which is the equivalent of using the `autosys_secure` Get encrypted password function and Encrypt database access password (using AES).

- **-n pwvar**

(Optional) Defines an environment variable containing the password.

**Default:** If you do not define this parameter, the password is read from standard input.

#### NOTE

To use the `-n` option, you must also specify the `-a` or `-g` option.

- **-m message\_key**

(Optional) Displays the message that corresponds to the specified key.

**NOTE**

To use the -m option, you must set the AUTOSYS environment variable.

**Example:** `as_config -m CAUAJM_I_40005`

- **argument ...**  
Specifies one or more arguments to pass to the -m option.
- **-x**  
(Optional) Displays the version information.
- **-?**  
(Optional) Displays help for the command.

**Example: Encrypt the Default Encryption Key**

This example encrypts the default encryption key of the product and stores it in the \$AUTOUSER location as cryptkey.txt.

```
as_config - d
```

**Example: Encrypt a Custom Key for Instance Wide Encryption**

This example encrypts the custom key WAAE for instance-wide encryption and stores it in the \$AUTOUSER location as cryptkey\_alias.txt.

```
as_config - g WAAE
```

**Example: Encrypt an Agent's Encryption Key**

This example encrypts the 32-character hexadecimal key of the agent and stores it in the \$AUTOUSER location as cryptkey\_alias.txt.

```
as_config - a 0x0123456789ABCDEF01234567890ABCDE
```

**as\_safetool Command -- Manage the Repository of CA EEM Security Policies**

The `as_safetool` command manages the repository of CA EEM security policies that AutoSys Workload Automation uses.

AutoSys Workload Automation integrates with CA EEM, adding a comprehensive security layer. This integration results in consistent identity management, increased access security, comprehensive in-depth auditing, reduced application development costs, and improved responsiveness to business and regulatory demands. With `as_safetool`, you can install or remove security policies and grant administrative privileges to a CA EEM identity.

**Syntax**

This command has the following format:

```
as_safetool
[{-g | -r} user_name -b EEM_server_name [-a EEM_admin_name] [-q]]
[-i instance -b EEM_server_name [-a EEM_admin_name] [-f] [-q]]
[{-s | -S} -b EEM_server_name [-a EEM_admin_name]]
[-u instance -b EEM_server_name [-a EEM_admin_name] [-q]]
[-m -v version] [-b EEM_server_name] [-a EEM_admin_name] [-q]
```

[ -x ]

[ -? ]

- **-a *EEM\_admin\_name***  
(Optional) Specifies the CA EEM administrator.  
**Default:** EiamAdmin  
**Limits:** This option is only valid when you use the **-g**, **-i**, **-r**, **-s**, **-S**, or **-u** parameter. The value for the *EEM\_admin\_name* variable must match the name of an existing user with administrative privileges to the CA EEM repository.
- **-b *EEM\_server\_name***  
(Optional) Specifies the CA EEM server host.  
**Default:** localhost  
**Limits:** Specify this option when you define the **-i** or **-u** option.
- **-f**  
(Optional) Forces the reinstallation of security policies.  
**Limit:** This option is only valid when you also specify the **-i** option.
- **-g *user\_name***  
(Optional) Grants application instance administrative privileges to the specified user.
- **-i *instance***  
(Optional) Installs the CA EEM security policies on the specified instance.  
  
**NOTE**  
The repository on the CA EEM server stores these policies. For more information about setting, changing, and storing CA EEM security policies, see [Security Policy Customization](#) .
- **-q**  
(Optional) Issues the command in quiet mode.  
  
**NOTE**  
When you use this option, AutoSys Workload Automation does not display error messages.
- **-r *user\_name***  
(Optional) Revokes the application instance administrative privileges of the specified user.
- **-s**  
(Optional) Displays all instances with an installed security policy.
- **-S *instance***  
(Optional) Displays all Application Administrators for the specified instance.
- **-u *instance***  
(Optional) Uninstalls the CA EEM security policies on the specified instance.  
  
**NOTE**  
The repository on the CA EEM server stores these policies. For more information about setting, changing, and storing CA EEM security policies, see [Security Policy Customization](#) .
- **-m**  
(Optional) Updates the security policies after migration. This option updates the actions (that are added in a later release level than the release level that you specified in the **-v *version*** option) under each CA EEM policy to the default.  
  
**NOTE**  
Do not use this option on a machine that has customized policies as this option resets the actions to the default.
- **-v *version***

(Optional) Specifies the AutoSys Workload Automation release level from which the security policies are imported. Specify this value in the *major.minor.release.maintenance* format. For example, 11.3.6.SP6; specifies that the security policies are imported from Release 11.3.6 SP6.

- **-x**  
(Optional) Returns the as\_safetool version to standard output.
- **-?**  
Displays help for the command.

#### NOTE

- The **-g**, **-i**, **-s**, **-S**, **-u**, or **-m** option are only valid if the value of the ASSAFETOOLPW environment variable and the value of EiamAdmin user account password match.
- You cannot run the command in batch mode unless the value of the ASSAFETOOLPW environmental variable and the value of the CA EEM administrator password match.
- You can use the ASSAFETOOLFIPS environment variable to override the default FIPS compliance setting. If the default setting is off, you can enable FIPS compliance by setting the value of this variable to 1. If the default setting is on, you can disable FIPS compliance by setting the value of this variable to 0.

#### Example: Install the Default Security Policy for a AutoSys Workload Automation Instance

This example installs the default security policy for instance ACE into the local CA EEM server.

```
as_safetool - i ACE
```

#### Example: Show All AutoSys Workload Automation Instances with an Installed Security Policy

This example shows all instances with an installed security policy on host securityhost.

```
as_safetool - b securityhost - s
```

#### Examples Running as\_safetool in Interactive Mode

The following examples describe how to use as\_safetool in interactive mode:

#### Example: Deregister AutoSys Workload Automation from CA EEM

Suppose that you want to deregister a AutoSys Workload Automation instance with CA EEM in interactive mode. Do the following:

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
as_safetool
```

The following menu appears:

```
Please select from the following options:
```

```
[1] Manage CA EEM Backend Server Location.
[0] Exit CA WAAE IAM Toolkit Safetool Utility.
```

2. Enter 1 and press the Enter key.

The following menu appears:

```
Please select from the following options:
```

```
[1] Set and Connect CA EEM Backend Server Location as an Administrator.
[2] Set CA EEM Backend Server Location.
[3] Show current Backend Server Location.
[9] Exit from "Manage CA EEM Backend Server Location" menu.
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
```

3. Enter 1 and press the Enter key.



4. Enter the CA EEM backend server host name, the application instance name, and the CA EEM administrator user name and password when prompted.
5. Enter 9 and press the Enter key after connecting to the CA EEM backend server.  
The following menu appears:  
Please select from the following options:  
[1] Manage CA EEM Backend Server Location.  
[2] Generate Instance Authentication Certificate.  
[3] Manage Instance Security Policy.  
[4] Manage Application Instance Administrators.  
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
6. Enter 3 and press the Enter key.  
The following menu appears:  
Please select from the following options:  
[1] Install Default Security Policy.  
[2] Uninstall Default Security Policy.  
[3] Show Instances with Security Policy Installed.  
[4] Perform Permission Checks against Instance Security Policy.  
[9] Exit from "Manage Instance Security Policy" menu.  
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
7. Enter 2 and press the Enter key.
8. Enter the AutoSys Workload Automation instance name when prompted.  
The AutoSys Workload Automation instance is deregistered with CA EEM and the security policies are uninstalled.

### Example: Perform Permission Checks Against a Security Policy

Suppose that you want to verify a CA EEM security policy for a AutoSys Workload Automation instance in interactive mode. Do the following:

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:  
as\_safetool  
The following menu appears:  
Please select from the following options:  
[1] Manage CA EEM Backend Server Location.  
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
2. Enter 1 and press the Enter key.  
The following menu appears:  
Please select from the following options:  
[1] Set and Connect CA EEM Backend Server Location as an Administrator.  
[2] Set CA EEM Backend Server Location.  
[3] Show current Backend Server Location.  
[9] Exit from "Manage CA EEM Backend Server Location" menu.  
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
3. Enter 1 and press the Enter key.
4. Enter the CA EEM backend server host name, the application instance name, and the CA EEM administrator user name and password when prompted.
5. Enter 9 and press the Enter key after connecting to the CA EEM back-end server.  
The following menu appears:  
Please select from the following options:  
[1] Manage CA EEM Backend Server Location.

```
[2] Generate Instance Authentication Certificate.
[3] Manage Instance Security Policy.
[4] Manage Application Instance Administrators.
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
```

6. Enter 3 and press the Enter key.

The following menu appears:

Please select from the following options:

```
[1] Install Default Security Policy.
[2] Uninstall Default Security Policy.
[3] Show Instances with Security Policy Installed.
[4] Perform Permission Checks against Instance Security Policy.
[9] Exit from "Manage Instance Security Policy" menu.
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
```

7. Enter 4 and press the Enter key.

8. Enter the AutoSys Workload Automation instance name when prompted.

9. Enter the security policy that you want to verify when prompted. The syntax is as follows:

```
class resource access user
```

– **class**

Specifies the policy class. Options are the following:

- appl
- base\_jobtype
- calendar
- connectionprofile
- control
- cycle
- group
- gvar
- job
- joblog
- jobtype
- list
- machine
- owner
- resources
- sendevent

– **resource**

Specifies the name of the resource policy that you want to verify.

– **access**

Specifies the type of access. Options are the following:

- **x**  
Specifies execute access.
- **w**  
Specifies write access.
- **r**  
Specifies read access.
- **c**

Specifies create access.

- **d**

Specifies delete access.

For the sendevent policy class, the access levels are as follows:

- alarm
- change\_priority
- change\_status
- comment
- delete\_job
- force\_startjob
- job\_off\_hold
- job\_off\_ice
- job\_off\_noexec
- job\_on\_hold
- job\_on\_ice
- job\_on\_noexec
- killjob
- mach\_offline
- mach\_online
- release\_resource
- reply\_response
- restartjob
- resumejob
- send\_signal
- set\_global
- startjob
- suspendjob
- unsend\_event

- **user**

Specifies the user name.

The specified resource policy is verified.

### Example: Update Security Policies After Migration

Suppose that you want to update the security policies after migration in interactive mode. Do the following:

1. Enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
as_safetool
```

The following menu appears:

```
Please select from the following options:
```

```
[1] Manage CA EEM Backend Server Location.
```

```
[0] Exit CA WAAE IAM Toolkit Safetool Utility.
```

2. Enter 1 and press the Enter key.

The following menu appears:

```
Please select from the following options:
```

```
[1] Set and Connect CA EEM Backend Server Location as an Administrator.
```

```
[2] Set CA EEM Backend Server Location.
```

```
[3] Show current Backend Server Location.
```

```
[9] Exit from "Manage CA EEM Backend Server Location" menu.
```

[0] Exit CA WAAE EEM Toolkit Safetool Utility.

3. Enter 1 and press the Enter key.
4. Enter the CA EEM backend server host name, the application instance name, and the CA EEM administrator user name and password when prompted.
5. Enter 9 and press the Enter key after connecting to the CA EEM backend server.

The following menu appears:

Please select from the following options:

```
[1] Manage CA EEM Backend Server Location.
[2] Generate Instance Authentication Certificate.
[3] Manage Instance Security Policy.
[4] Manage Application Instance Administrators.
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
```

6. Enter 3 and press the Enter key.

The following menu appears:

Please select from the following options:

```
[1] Install Default Security Policy.
[2] Uninstall Default Security Policy.
[3] Show Instances with Security Policy Installed.
[4] Perform Permission Checks against Instance Security Policy.
[5] Update security policies post migration.
[9] Exit from "Manage Instance Security Policy" menu.
[0] Exit CA WAAE EEM Toolkit Safetool Utility.
```

7. Enter 5 and press the Enter key.
8. Enter the AutoSys Workload Automation release level from which you want to import the security policies when prompted.  
The security policies are updated. The actions (that are added in a later release level than the release level that you entered in this step) under each CA EEM policy are set to the default.

### **Exit Codes for the as\_safetool Command**

The following exit codes are associated with the as\_safetool command:

- **0**  
Indicates that the command was successful.
- **1**  
Indicates that the command contained a syntax error.
- **2**  
Indicates that the back-end server is not available or the EiamAdmin user account password is invalid.
- **3**  
Indicates that the AutoSys Workload Automation registration to CA EEM failed.
- **4**  
Indicates that installation of the AutoSys Workload Automation security policy failed.
- **5**  
Indicates that removal of the AutoSys Workload Automation security policy failed.
- **6**  
Indicates that the unregistration of AutoSys Workload Automation from CA EEM failed.
- **7**  
Indicates that the ASSAFETOOLPW environment variable is not defined.
- **8**

Indicates that the CA EEM client configuration file is missing or corrupted.

**NOTE**

The CA EEM client configuration file is located in the \$AUTOUSER/eiam.client.config (on UNIX) or %AUTOUSER%\eiam.client.config (on Windows) directory.

## autosec\_test Command -- Simulate a Call to the Security Subsystem

The autosec\_test command is a client component utility that is used to test the security subsystem by a AutoSys Workload Automation application for a given secured asset.

Use the autosec\_test command to test basic communication between a AutoSys Workload Automation secured application and the security subsystem and to verify enforcement of the security policies for AutoSys Workload Automation resources (jobs, calendars, machines, and so on).

When you customize external security policy, you specify an action (execute, write, read, create, or delete) to grant users a particular level of access to particular resources. If AutoSys Workload Automation is running in external security mode, the autosec\_test command returns a message indicating whether a security access check for a given resource to an action passes or fails as governed by the security policy. If no *username* value is specified, the application issues a security check as the currently logged-on user.

**NOTE**

For more information about customized security policy, see [Security Policy Customization](#).

### Syntax

This command has the following format:

```
autosec_test [class resource access [username]]
 [STAT]
 [-usr AuthUser [-pw AuthPassword | -pwx AuthEncryptedPassword] | -saml
AuthArtifact]
 [-x]
 [-?]
```

- **class** (Optional) Specifies the name of the resource class on which the security check is issued. The resource class name can be *one* of the following:

- APPL
- BASE\_JOBTYPE
- CALENDAR
- CONNECTIONPROFILE
- CONTROL
- CYCLE
- GLOBAL\_VAR
- GROUP
- JOB
- JOBLOG
- JOBTYPE
- LIST
- MACHINE
- OWNER
- RESOURCES
- SENDEVENT

#### NOTE

For more information about the resource classes, see [CA EEM Resource Classes for AutoSys Workload Automation](#)

- **resource**

(Optional) Specifies the name of the resource to which the user is granted a particular level of access to perform an action.

- **access**

(Optional) Specifies the access level to perform an action for a given resource class name. The access levels are as follows: C(create) | D(delete) | R(read) | W(write) | X(execute). The access levels for each resource class are as follows:

- APPL: R|W|X
- BASE\_JOBTYPE: X
- CALENDAR: C|D|R|W|X
- CONNECTIONPROFILE: C|D|R|W|X
- CONTROL: X
- CYCLE: C|D|R|W
- GLOBAL\_VAR: C|D|R|W|X
- GROUP: R|W|X
- JOB: C|D|R|W|X
- JOBLOG: R
- JOBTYPE: C|D|R|W|X
- LIST: R
- MACHINE: C|D|R|W|X
- OWNER: X|SENDEVENT\_JOBEXECUTE
- RESOURCES: C|D|R|W|X

The access levels for the SENDEVENT resource class are as follows:

- ALARM
  - CHANGE\_PRIORITY
  - CHANGE\_STATUS
  - COMMENT
  - DELETE\_JOB
  - FORCE\_STARTJOB
  - JOB\_OFF\_HOLD
  - JOB\_OFF\_ICE
  - JOB\_OFF\_NOEXEC
  - JOB\_ON\_HOLD
  - JOB\_ON\_ICE
  - JOB\_ON\_NOEXEC
  - KILLJOB
  - MACH\_OFFLINE
  - MACH\_ONLINE
  - RELEASE\_RESOURCE
  - REPLY\_RESPONSE
  - RESTARTJOB
  - RESUMEJOB
  - SEND\_SIGNAL
  - SET\_GLOBAL
  - STARTJOB
  - SUSPENDJOB
  - UNSEND\_EVENT
- **username** (Optional) Specifies the name of the user accessing the resource. The user is granted a particular level of access to particular resources.  
**Default:** If no user name is specified, access is checked on behalf of the currently logged-on user who is issuing the command.

#### NOTE

The issuing SERVER of the user attribute must be set in the CA EEM environment to allow issuance of a CA EEM security check on behalf of another user.

- **STAT**(Optional) Displays a message indicating the security mode that AutoSys Workload Automation is running under.
- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**  
 (Optional) Specifies external security credentials as follows:
  - usr username -pw password: Specifies a username and unencrypted password.
  - usr username -pwx password: Specifies a username and encrypted password.

#### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

-saml auth\_artifact: Specifies a Security Assertion Markup Language (SAML) artifact.

#### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

#### **Example: Determine Whether the Current User May Issue an Event for a Job**

This example simulates a request by the AutoSys Workload Automation secured application to the security subsystem to determine whether the current user has access to issue a STARTJOB event for the job good\_test:

```
autosec_test JOB good_test X
```

If the user has execute access to the job, the following message is displayed:

```
Security check passed!
```

If the user does not have execute access to the job, the following message is displayed:

```
Security check failed!
```

#### **Example: Determine Whether a Specified User May Issue an Event for a Job**

This example simulates a request by the AutoSys Workload Automation secured application to the security subsystem to determine whether the user admin@example.com can issue a STARTJOB event for the job good\_test:

```
autosec_test JOB good_test X admin@example.com
```

If admin@example.com has execute access to the job, the following message appears:

```
Security check passed!
```



If admin@example.com does not have execute access to the job, the following message appears:

```
Security check failed!
```

## autosys\_secure Command -- Define Security Settings

The autosys\_secure command is a scheduler and client component utility that defines the EDIT superuser, EXEC superuser, regular user, database password, remote authentication method, and Windows user IDs and passwords for AutoSys Workload Automation. You can also use autosys\_secure to enable external security (in interactive or batch mode) in AutoSys Workload Automation and perform basic CA EEM administration.

### Syntax

This command has the following format:

```
autosys_secure
 [-a {-editu | -execu} {user@host | user@domain} [-p password]]
 [-a -u {user@host | user@domain} -p password [-force]]
 [-a -u {user@host | user@domain} -sc SecurityCode {-kf KeyFile | -ks KeyStringFile}
 [-kp Passphrase]] [-a -u {user@host | user@domain} -al AliasName -vn VaultName [-force]]
 [-a -vn VaultName -vt VaultType]
 [-c {-editu | -execu} {user@host | user@domain} -h {NULL | host} | [-p password]]
 [-c -u {user@host | user@domain} -o {NULL | old_password} -p {NULL
 | new_password}]
 [-c -u {user@host | user@domain} -sc SecurityCode {-osc OldSecurityCode | {-
kf KeyFile | -ks KeyStringFile} [-kp Passphrase]]]
 [-c -u {user@host | user@domain} -oal OldAlias -al NewAlias [-d -u
{user@host | user@domain} [-p password]] [-d -u {user@host | user@domain} [-credtype
{PASSWORD|KEYCREDENTIAL|PASSWORDVAULT}]]]
 [-e -host host [-secadmu user -p password] [-regencert]]
 [-f set -u user -p password [-t {-1 | DDHH}]]
 [-f show]
 [-f delete]
 [-n -host host [-secadmu user -p password]]
 [-s [{-editu | -execu | -host | -key | -vault | -vn} ALL | -secadmu user -p
password]]]
 [-x]
 [-?]
```

### NOTE

You can define security settings using the AutoSys Security Utility interactive menu by entering the autosys\_secure command without specifying any optional parameters. We recommend this method, but you can also define security settings from the command line by specifying optional parameters when you run the command. This topic provides information about the command line options that you can use to define security settings.

- **-a {-editu|-execu} {user@host | user@domain} [-p password]**  
(Optional) Adds an EDIT superuser or EXEC superuser.
- **-p password**  
(Optional) Specifies the password of the user.
- **-a -u {user@host | user@domain} -p password**

- (Optional) Adds a regular user with password.
- **-p *password***  
(Optional) Specifies the password of the user.
  - **-a -u {*user@host* | *user@domain*} -sc *SecurityCode* {-kf *KeyFile* | -ks *KeyStringFile*} [-kp *Passphrase*]**  
(Optional) Adds a regular user with key credentials. Specify the security code and either the key file or the key string. You can also specify a passphrase for the key file or key string.
  - **-sc *SecurityCode***  
Specifies the security code that is used to authorize changes to the key credential user.
  - **-kf *KeyFile***  
Specifies the absolute path to the SSH private key file on the remote machine where the agent runs. A *key file* is the private key part of the Public key cryptography.  
You can generate private and public keys using the ssh-keygen OpenSSH program.  
**Limits:** Up to 255 characters  
**WARNING**  
KeyFile can be used for SCP (SFTP) or any of the CA WA Advanced Integration for Hadoop jobs.
  - **-ks *KeyStringFile***  
Specifies the path to the SSH private key file on the local machine to extract and store in the database.  
A *key string* is the content of the key file.  
**Limits:** Up to 255 characters  
**WARNING**  
KeyString is only applicable to the CA WA Advanced Integration for Hadoop job types.
  - **-kp *Passphrase***  
Specifies the passphrase that is associated with the SSH private key for a user with key credentials.  
**Limits:** Up to 255 characters  
**NOTE**  
Prior to using key credentials in jobs, you must configure the machine where the agent is installed and the remote machine that the agent tries to connect to so that they communicate using the Public key cryptography.
  - **[-a -u {*user@host* | *user@domain*} -al *AliasName* -vn *VaultName* [-force]]**  
(Optional) Adds a user with an alias.
  - **[-a -vn *VaultName* -vt *VaultType*]**  
(Optional) Adds a vault name for the vault type.
  - **-c {-editu|-execu} {*user@host* | *user@domain*} -h {NULL|*host*} | [-p *password*]**  
(Optional) Changes the host qualifier for the specified EDIT superuser or EXEC superuser. Specify the current host as part of the user ID with the -editu or -execu parameter. Specify the new host with the -h parameter. To remove the host qualifier, specify a value of NULL with the -h parameter.
  - **-p *password***  
(Optional) Specifies the password of the user.
  - **-c -u {*user@host* | *user@domain*} -o {NULL|*old\_password*} -p {NULL|*new\_password*}**  
(Optional) Changes the password for the specified user. To remove a password, specify a value of NULL for the -p parameter. To add a password, specify a value of NULL for the -o parameter.
  - **-c -u {*user@host* | *user@domain*} -sc *SecurityCode* {-osc *OldSecurityCode* | {-kf *KeyFile* | -ks *KeyStringFile*} [-kp *Passphrase*]}**  
(Optional) Changes the key credentials for the specified user. You can change either the security code or the key file or key string.  
To change the security code, specify the current security code with the -osc parameter and the new security code with the -sc parameter.  
To change the key file, specify the current security code with the -sc parameter and the new key file with the -kf parameter.

To change the key string, specify the current security code with the **-sc** parameter and the new key string with the **-ks** parameter.

To change the passphrase that you specified for the key file, specify the current security code with the **-sc** parameter, the current or new key file with the **-kf** parameter, and the new passphrase with the **-kp** parameter.

To change the passphrase that you specified for the key string, specify the current security code with the **-sc** parameter, the current or new key string with the **-ks** parameter, and the new passphrase with the **-kp** parameter.

- **[-c -u {user@host | user@domain} -oal OldAlias -al NewAlias]**  
(Optional) Changes the user with a new alias for an old alias.
- **-d -u {user@host | user@domain} [-p password]**  
Deletes the specified user.
- **-d -u {test@domain | user@domain} | -credtype {PASSWORD|KEYCREDENTIAL|PASSWORDVAULT}]**  
Deletes the user of specified credtype.
- **[-e -host host [-secadmu user -p password] [-regencert]]**  
(Optional) Activates external security on the specified host.

#### NOTE

- You can specify multiple CA EEM host names in a comma-separated format. However, each host name must be less than 80 characters and the entire list must be less than 255 characters.
- AutoSys Workload Automation requires a valid CA EEM authentication certificate to enable external security. If the certificate is missing or invalid, AutoSys Workload Automation requires the credentials of an external security user with administrative privileges to generate a new one.
- AutoSys Workload Automation stores the CA EEM authentication certificate in the database and makes it available to the AutoSys Workload Automation instance. If AutoSys Workload Automation cannot store the certificate, verify the database for problems.
- **-f set -u user -p password**  
(Optional) Specifies the user credentials to cache and the validity period for those credentials. The default validity period is 24 hours, but you can specify a different validity period using **-t** option.
- **-t {-1| DDHH}**  
(Optional) Specifies the validity period for the credentials.
- **-1**  
Specifies that the credentials never expire.
- **DDHH**  
Defines a specific validity period for the credentials in days and hours.

#### NOTE

- To specify that the credentials are valid for less than one day, enter a value of 00 for days and then specify a value from 01 through 24 for hours.
- If you specify that a non-zero value for days and a 00 value for hours, the credentials expire after the specified number of days simultaneously with the day that you generated them.
- If you specify 0000, the utility returns an error.
- The utility sets the validity period to 24 hours by default when no expiration date and time is specified.
- **-f show**  
(Optional) Displays cached credentials.
- **-f delete**  
(Optional) Deletes cached credentials. Your session remains active until cached credentials expire or you delete them. To protect the security of your user account, delete cached credentials.
- **-n -host host [-secadmu user -p password]**  
(Optional) Reverts to native security on the specified host.

**NOTE**

Run `autosys_secure` as an external security user with administrative privileges to revert to native security. You can perform this task when AutoSys Workload Automation is running in external security mode and you authenticate `autosys_secure` with external security as a privileged user.

- **[-s [{-editu | -execu | -host | -key | -vault | -vn} ALL | -secadmu user -p password]]**

(Optional) Displays security settings for all EDIT superusers, all EXEC superusers, all hosts, all key credential users, for all password vault names, for all password vault users, the specified administrative user, or all existing users. To display setting for all existing users, specify the parameter without extra options.

Use *one* of the following commands to verify the return code to determine if AutoSys Workload Automation is running under native security mode or external security mode:

- On UNIX:
 

```

csh echo $status
sh or ksh echo $?

```
- On Windows:
 

```

echo %ERRORLEVEL%

```

When you exit the program, `autosys_secure` returns *one* of the following values:

- **1**  
Indicates that a failure occurred.
- **2**  
Indicates that the system is running in native security mode.
- **3**  
Indicates that the system is running in external security mode.

**NOTE**

Run `autosys_secure` as an external security user with administrative privileges to view the CA EEM server list and unauthenticated user mode setting. You can only view this information when AutoSys Workload Automation is running in external security mode and you authenticate `autosys_secure` with external security as a privileged user.

- **-q**  
(Optional) Runs `autosys_secure` in quiet mode so it does not display messages.  
In quiet mode, you must use *one* of the following commands to verify the return code to see whether there were errors:
  - On UNIX:
 

On Bourne, Bash, or Korn shell, enter the following command:

```
echo $?
```

On C shell, enter the following command:

```
echo $status
```
  - On Windows:
 

```
echo %ERRORLEVEL%
```

When `autosys_secure` is successful, the return value is 0. When it is unsuccessful, the value is 1.
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

**Example: Add a Credentialed User from the Command Line**

This example adds *testuser* as the new credentialed user on the *machine1* host and assigns *testuserpass* as the password for the user.

```
autosys_secure -a -u testuser@machine1 -p testuserpass
```

**Example: List All Defined Users with Password from the Command Line**

This example lists all users with password.

```
autosys_secure -s -host ALL
```

**Example: List Defined Password Vault Names from the Command Line**

This example lists all password vault names.

```
autosys_secure -s -vn VaultName
```

**Example: Change Password Vault User using Alias from the Command Line**

This example changes a vault user replacing an existing alias with a new alias.

```
autosys_secure -s -oal OldAlias -al NewAlias
```

**Example: Enable External Security from the Command Line**

This example enables external security.

```
autosys_secure -e -host ext_sec_server_host -secadmu ExtSecAdmin -p ExtSecAdminpass
```

**Example : Generate cached credentials from the Command Line**

This example generates cached credentials that expire 47 hours after the time the command runs.

```
autosys_secure -f set -u ExtSecUser -p ExtSecUserPass -t 0123
```

**Example: Add a User and Password for a Hadoop Job from the Command Line**

This example adds *user1* as the new credentialed user for the edge node *HADOOP\_EDGE\_NODE* and assigns *hadoopuser1* as the password for the user.

```
autosys_secure -a -u user1@HADOOP_EDGE_NODE -p hadoopuser1
```

**Example: Add a User with Key File and Passphrase for a Hadoop Job from the Command Line**

This example adds *user2* as the new credentialed user for the edge node *HADOOP\_EDGE\_NODE* and assigns *seccode* as the security code, */tmp/id\_rsa* as the key file path and *passphrase* as the passphrase for the user.

```
autosys_secure -a -u user2@HADOOP_EDGE_NODE -sc seccode -kf /tmp/id_rsa -kp passphrase
```

**Example: Change the Key File and Passphrase for a User**

This example changes the key file and passphrase for *user2* user. The key file is changed to */tmp/new\_id\_rsa* and the passphrase is changed to *new\_passphrase*.

```
autosys_secure -c -u user2@HADOOP_EDGE_NODE -sc seccode -kf /tmp/new_id_rsa -kp
new_passphrase
```

**Native and Embedded Security**

AutoSys Workload Automation uses native or external security for protecting jobs, machines, and so on.

AutoSys Workload Automation native security includes the following:

- Job-level security
- Superuser privileges
- System-level security
- UNIX and Windows file permissions

Security is initiated when either a user or the scheduler sends events that affect a job run.

AutoSys Workload Automation external security is accomplished through integration with CA EEM. All GUI applications and command-line interfaces have callouts to security. User-defined classes in CA EEM are used to govern what types of resources can be controlled by which users.

**NOTE**

For more information about configuring AutoSys Workload Automation to work with CA EEM, see [Security Policy Customization](#).

**EDIT Superuser and EXEC Superuser**

Two users have administrator privileges: the EDIT superuser and the EXEC superuser.

The EDIT superuser is the only user with permission to do the following:

- Edit or delete any job, regardless of who owns it and what permissions are set for it.
- Change the owner of a job.
- Change the database password, remote authentication method, and Windows user passwords.

The EXEC superuser is the only user with permission to do the following:

- Start or kill any job, regardless of the execution permissions on the specified job. This user can affect how jobs run, typically by issuing the sendevent command.
- Shut down the scheduler (by sending the STOP\_DEMON event).

**Database Password**

The scheduler and command line utilities use the database password to securely access the database (event server). By default, this password is stored in the database as autosys, but the EDIT superuser can use autosys\_secure to change it.

Every event server in an instance must have the same database password. If you are running in dual event server mode, autosys\_secure changes the password on both event servers.

**NOTE**

If AutoSys Workload Automation has rolled over to single event server mode, do not change the password until you have re-established dual event server mode.

**Remote Authentication Method**

AutoSys Workload Automation uses remote authentication methods to verify the following:

- Whether a user has permission to start a job on a client.
- Whether a scheduler has permission to start a job on a client.

The remote authentication methods are stored in the database and are referenced when a client initializes.

**User and Host Management****Valid on Windows**

Windows user IDs and passwords must be entered into the database for jobs to run on Windows clients. When an agent runs a job on a computer, it logs in and impersonates the user who owns the job. To enable the agent to do this, the scheduler passes both the job information and an encrypted version of the job owner's password from the database to the agent. The EDIT superuser can use autosys\_secure interactively or from the command line to enter these passwords. After the EDIT superuser enters the IDs and passwords, any user who knows an existing user ID and password can change the password or can delete that user ID and password.

**NOTE**

Entering passwords using autosys\_secure is not related to Windows user Account Administration.

## Encrypted Passwords

The password is encrypted. It is associated with the autosys database user and is used for database connection. The encrypted password is stored in the configuration file.

## Define Workload Objects

You can use the following utilities to define workload objects :

### NOTE

For more information about the JIL scripting language and JIL object definitions, see [AE Job Information Language](#). For more information about calendar definitions and calendar management, see [Manage Your Calendars](#).

## autocal\_asc Command -- Manage Calendars

The autocal\_asc command provides a text-based command-line mechanism that lets you manage calendar definitions. Calendars let you customize job schedules.

A calendar specifies either fixed dates or criteria as follows:

- **Standard calendar**  
Specifies a list of fixed dates, such as holidays.
- **Cycle**  
Specifies a list date ranges (periods).
- **Extended calendar**  
Specifies a set of criteria that the utility uses to generate a list of fixed dates.

### NOTE

- An extended calendar might reference a cycle and apply rules to the periods in it. To customize a job schedule, use a standard or extended calendar (not a cycle).
- The interactive calendar utility lets you define, modify, regenerate, and delete calendars.
- Command line options do not support calendar regeneration; use the interactive utility to regenerate extended calendars after you modify criteria.
- Command line options support calendar creation, deletion, and modification. We recommend that you use the interactive utility to define calendars, delete calendars, and make modifications to a single calendar. To modify multiple calendars simultaneously, use command line options.
- Use the autocal\_asc command with the redirection operator to add or delete dates from a standard calendar. To add dates to a standard calendar, create a text file and include the calendar name, the A option, and the dates to be added to the calendar and then issue the autocal\_asc command with the redirection operator and specify the text file name. To delete dates from a standard calendar, create a text file and include the calendar name, the D option, and the dates to be deleted from the calendar and then issue the autocal\_asc command with the redirection operator and specify the text file name. All provided dates must be in MM/DD/YYYY format.

## Syntax

This command has the following format:

```
autocal_asc
 [{-D {-c|-e|-s} calendar_name|-E export_filename {-c|-e|-s [-f date_format]}
 {calendar_name|ALL}}|-l [{c|e|s}}|-I import_filename [-F] [-f date_format]]|[-
usr username {-pw|pwx} password|-saml auth_artifact}}]]
 [-x]
```

---

[ - ? ]

- **-D {-c|-e|-s} *calendar\_name***  
(Optional) Deletes the specified calendar or cycle from the database.
- **-E *export\_filename* {-c|-e|-s [-f *date\_format*] {*calendar\_name*|ALL}}**  
(Optional) Exports the specified calendars to the specified file.
- **-c|-e|-s [-f *date\_format*] *calendar\_name***  
Specifies a specific calendar of the relevant type.  
**Limits:** 1-64 of the following characters: Alphanumeric characters and the following symbols: period (.), underscore (\_), pound (#), and hyphen (-).
- **-c|-e|-s [-f *date\_format*] ALL**  
Specifies all calendars of the relevant type.
- **-f *date\_format***  
(Optional) Specifies the date format to use for a standard calendar in the file that is exported. If you do not specify this parameter, the default format MM/DD/YYYY is used.
- **-l[{-c|-e|-s}]**  
(Optional) lists all calendars of the specified type. If you do not specify a calendar type, the utility lists all calendars of all types.

**NOTE**

For each calendar, the calendar name, type, and description are displayed.

- **-I *import\_filename* [-F] [-f *date\_format*]**  
(Optional) Imports calendar definitions from the specified file.
- **[-F]**  
Force overwrites the calendar definitions from the specified file.
- **-f *date\_format***  
(Optional) Specifies the date format to use for a standard calendar in the file that is imported. If you do not specify this parameter, the default format MM/DD/YYYY is used.
- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**  
(Optional) Specifies external security credentials as follows:  
**-usr *username* -pw *password*:** Specifies a username and unencrypted password.  
**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:



- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

### Example: Export a Calendar

This example exports the standard calendar federal\_holidays to the file holidays\_06.txt.

```
autocal_asc -E holidays_06.txt -s federal_holidays
```

### Example: Define a Standard Calendar

This example defines a standard calendar named Holidays2007 in a text file.

```
calendar: Holiday2007
description: Holiday calendar for 2007
01/01/2007 00:00
01/02/2007 00:00
01/16/2007 00:00
02/20/2007 00:00
05/29/2007 00:00
07/02/2007 00:00
09/04/2007 00:00
11/23/2007 00:00
11/24/2007 00:00
12/25/2007 00:00
12/26/2007 00:00
```

### Example: Define a Cycle

This example defines a cycle with four 3-month periods named Quarter 2007 in a text file.

```
cycle: Quarter2007
description: Quarterly calendar for 2007
start_date: 01/01/2007
end_date: 03/31/2007
start_date: 04/01/2007
end_date: 06/30/2007
start_date: 07/01/2007
end_date: 09/30/2007
start_date:10/01/2007
end_date:12/31/2007
```

### Example: Define an Extended Calendar

This example defines an extended calendar named LastWeekCycleCal in a text file.

```
extended_calendar: LastWeekCycleCal
description:
workday: mo,tu,we,th,fr
non_workday: W
holiday:
holcal:
cyccal: Quarter2007
adjust: 0
condition: CWEEK#L
```

### Example: Import Calendars

#### WARNING

The definitions in the previous examples become calendar objects only when you import the text file that contains them. To ensure that updates are applied to calendar objects, import the file again after you modify it.

This example imports calendar definitions in the previous examples from the text file named Calendars2007.txt to the AutoSys Workload Automation database.

```
autocal_asc -I Calendars2007.txt
```

### Example: Add Dates to a Standard Calendar Using the Redirection Operator

This example creates a calendar named *holiday\_2016* and adds 01/04/2016, 02/04/2016, and 03/04/2016 dates to it. Create a text file named *dates.txt* and specify the calendar name, the A option, and the dates to be added to the calendar as follows:

```
holiday_2016
A
01/04/2016
02/04/2016
03/04/2016
```

Issue the `autocal_asc` command with the redirection operator and the text file name as follows:

```
autocal_asc < dates.txt
```

### Example: Delete Dates from a Standard Calendar Using the Redirection Operator

This example deletes 02/04/2016 and 03/04/2016 dates from the calendar named *holiday\_2016*. In the text file named *dates.txt*, specify the calendar name, the D option, and the dates to be deleted from the calendar as follows:

```
holiday_2016
D
02/04/2016
03/04/2016
```

Issue the `autocal_asc` command with the redirection operator and the text file name as follows:

```
autocal_asc < dates.txt
```

### NOTE

The interactive calendar utility lets you define and modify calendar objects directly. We recommend that you use this utility to create new calendars but that you manage definitions in a text file.

### Example: Delete an Extended Calendar

This example deletes the extended calendar `LastWeekCycleCal`.

```
autocal_asc -D -e LastWeekCycleCal
```

### Example: List all Standard Calendars

This example lists all standard calendars.

```
autocal_asc -ls
```

### Example: Modify the Description of a Standard Calendar Using the Calendar Utility Interactive Menu

This example modifies the description of a standard calendar named `Holidays2007`.

Issue the `autocal_asc` command with no additional options to open the Calendar Utility Interactive menu, and then follow these steps:

1. Type 1 and press Enter.
2. Type 7 and press Enter.
3. Enter `Holidays2007` when the utility prompts you to enter the calendar name and press Enter.
4. Enter "Holiday List for the Year 2007" when the utility prompts you to enter the calendar description and press Enter.
5. Enter 0 to exit the Calendar Utility interactive menu.

### Example: Specify the Date Format for a Standard Calendar in the File that is Imported

This example uses `yyyy/mm/dd` date format for the standard calendar named *caldates* in the file that is imported.

```
autocal_asc -I caldates -f yyyy/mm/dd
```

In the file that is imported, the dates are defined as follows:

```
calendar: caldates
description: date format for standard calendar
```

```
01/20/2016 00:00:00
01/29/2016 00:00:00
02/28/2016 00:00:00
```

## cron2jil Command -- Convert UNIX crontab Data to a JIL Script or Calendar File

### Valid on UNIX

The cron2jil command is a client component utility that converts each line in a UNIX crontab file to a corresponding JIL script (\*.jil file) and, if necessary, a run calendar file (\*.cal file).

Because the cron2jil command cannot comprehensively address all job processing requirements, you should use it as a first step in converting from cron to the environment. The second step requires editing the newly created JIL and calendar files to ensure the preferred job processing.

When cron2jil reads a crontab file, it assigns job names by combining the base name of the job's command and the line number of the file. For example, the following crontab entries would result in the job names "cp\_1" and "mail\_2" respectively.

```
>>0,59 0,23 * * 0,6 /bin/cp /etc/passwd /etc/passwd.bak

>>0,59 * * * 0,6 /usr/ucb/mail root@support1 < /tmp/errorLog
```

After conversion, cron commands involving pipes and I/O redirection perform merely as they did in the cron environment. If run calendars are created, cron2jil only generates calendars with one year duration. After conversion, pipe and I/O redirections might not take full advantage of the fault tolerance mechanisms of AutoSys Workload Automation. For example, the exit code of a failed command in a pipe might not result in the failure of the complete command expression. Therefore, you should edit converted JIL scripts and split pipes into separate jobs with the appropriate conditions and job control. With this approach, problems can be detected and reported at the point of failure.

cron2jil does not generate JIL files for jobs that are defined in crontab to start every minute; for example, with an asterisk (\*) specified in the first field of the cron listing. This special case should be remedied by defining successful job completion as a starting condition for the same job.

### NOTE

After the \*.jil or \*.cal files are generated, you must use the jil or autocal\_asc command to define the jobs and calendars to AutoSys Workload Automation.

### Syntax

This command has the following format:

```
cron2jil -f crontab_file [-d output_directory] [-i include_file] [-m machine] [-p prefix]
```

- **-f *crontab\_file***  
Defines the name of the crontab-formatted file to convert.
- **-d *output\_directory***  
(Optional) Defines the directory to which the product should write the \*.jil and \*.cal files.  
**Default:** The current working directory.
- **-i *include\_file***

(Optional) Defines a file containing JIL statements to include in every generated \*.jil file. You must create this file before the conversion; it can contain any default JIL statements.

- **-m *machine***

(Optional) Defines the name of the machine on which the conversion should occur.

**Default:** localhost

- **-p *prefix***

(Optional) Defines a prefix to insert before each job's name. For example, if you set *prefix* to AUTO, JobA becomes AUTOJobA.

### Example: Convert a crontab File to a JIL Script

This example converts the crontab file named daily:

```
cron2jil -f daily
```

## jil Command -- Run the Job Information Language Processor

The jil command is a client component utility that runs the Job Information Language (JIL) processor.

One advantage of using jil is that you can use available UNIX tools for file manipulation to create and control AutoSys Workload Automation job definitions. For example, when you run the same command on every workstation, you can create a “generic” JIL template (text file), and copy it for each machine, changing only the machine attribute of the job. You can also iteratively copy the template to a temporary file, replace the machine name, and redirect the temporary file into the jil definition.

You can use the jil command in two ways:

- Redirect a JIL script file to the jil command. This method lets you automate the maintenance of object definitions in the database. A JIL script file contains one or more subcommands (such as `insert_job`).
- Issue the jil command only and enter JIL subcommands at the prompts. This method lets you interactively maintain object definitions in the database. To exit interactive mode, enter “exit” at the prompt, or press Ctrl+D (UNIX only).

### Syntax

This command has the following format:

```
jil
 [-q] [-S autoserv_instance] [-V {none | job | session | batch | syntax}] [{-
usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-q**

(Optional) Runs the command in “quiet” mode so that it only outputs error messages. This is useful when entering many jobs, so that you can easily see errors.

**Default:** Output error messages and status messages that indicate the success or failure of the JIL subcommands. This information is useful and should typically be allowed to print.

- **-S *autoserv\_instance***

(Optional) Defines the three-character identifier of the AutoSys Workload Automation instance on which the job or global variable resides (for example, ACE).

**Default:** If you do not specify the instance name, the command uses the value of the AUTOSERV environment variable.

- **-V {none | job | session | batch | syntax}**

(Optional) Specifies which job dependency information that the JIL processor verifies.

**none**

Specifies that the JIL processor performs no job dependency verification.

**job**

Specifies that the JIL processor verifies job dependencies on a job-by-job basis when the jobs are submitted to the database.

**batch**

Specifies that the JIL processor verifies job dependencies in the database after the JIL file is entirely processed.

**session**

Specifies that the JIL processor verifies job dependencies for the jobs created in this session. This is the default.

**syntax**

Verifies the validity of the job definition without verifying job dependencies or writing the job definitions to the database.

**NOTE**

- When you set -V to job or batch, unsatisfied job dependencies (that is, jobs that are specified in the conditions that do not exist in the AutoSys Workload Automation database) are reported to standard output. The output resembles the following:

---

```
Insert/Updating Job: JobA
*** WARNING: The Following Jobs are referenced in the ***
Conditions for this Job, YET are not defined!
1) JobC
Database Change WAS Successful!
```

---

- The portion of a condition statement that includes job dependencies on undefined jobs evaluates to FALSE for all conditions except not running. For example, if JobA has the condition "**success(JobB) and success(JobC)**", and JobC is not defined in the database, the condition evaluates to FALSE. If JobA depends on "**not running(jobC)**", the condition evaluates to TRUE.  
**Default:** If you do not define the -V option, the JIL processor verifies job dependencies on a session basis when the jobs are submitted to the database

- {-usr username {-pw|-pwx} password [-saml auth\_artifact]}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password:** Specifies a username and unencrypted password.

**-usr username -pwx password:** Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- -x  
(Optional) Returns the command version information to standard output.
- -?  
(Optional) Displays help for the command.

#### Example: Redirect a File to jil

This example redirects a text file that is named job1 containing JIL statements into the jil definition:

```
jil < job1
```

#### Example: Redirect a File to jil Without Verifying Job Dependencies

This example redirects a text file that is named job1 containing JIL statements into jil and prohibits the JIL processor from verifying the existence of jobs specified in its job dependencies:

```
jil < job1 -V none
```

#### JIL Syntax Rules

You must follow these rules when using JIL subcommands in a script file or at the command line:

- Subcommands have the following format:  
***sub\_command:object\_name***  
*sub\_command*  
 Specifies one of the following JIL subcommands:

- delete\_blob
- delete\_box
- delete\_glob
- delete\_job
- delete\_job\_type
- delete\_machine
- delete\_monbro
- delete\_resource
- delete\_xinst
- insert\_blob
- insert\_glob
- insert\_job
- insert\_job\_type
- insert\_machine
- insert\_monbro
- insert\_resource
- insert\_xinst
- override\_job
- update\_job
- update\_job\_type
- update\_monbro
- update\_resource
- update\_xinst

#### *object\_name*

Indicates the user-specified name of the object to create, update, delete, or override.

- Follow each subcommand with its corresponding attribute statements, if applicable. Attribute statements can occur in any order, and are applied to the job specified in the preceding subcommand. A subsequent subcommand begins a new set of attributes for a different job.

Attribute statements have the following format:

**attribute\_keyword** : **value** *attribute\_keyword* Specifies a valid JIL attribute. *value* Defines the setting to apply to the attribute.

- You can enter multiple attribute statements on the same line.
- You must separate each attribute statement with at least one space.
- You must define a box job before you can put jobs in it.
- You must define a machine to the database before you can define a job to run on it.
- When defining a virtual machine or pool, you must first define all machines that it contains.
- Legal value settings can include uppercase and lowercase letters, numbers, colons (if the colon is escaped as explained in the next rule), underscores (\_), and @.
- Because JIL parses subcommands based on the keyword/colon combination, you must properly escape colons that are used in an attribute statement's value with quotation marks (") or a backslash (\). For example, to specify the time to start a job, specify "10:00" or 10\:00.

#### **NOTE**

**Windows Note:** When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C\:\tmp or "C:\tmp" is valid; C:\tmp is not.

- Characters within quoted strings are not interpreted. All characters within quotes are considered as part of the string. The string will not be scanned to interpret characters that might represent something else, such as a comment.

For example:

```
description: "This description /* includes special characters */"
```



```
command: grep "/* begin */" $HOME/*.log
```

- Comments in a JIL script file are interpreted using the following rules:
  - To comment an entire line, insert a pound sign (#) as the first non-blank character on a line.
  - A forward slash immediately followed by an asterisk (/\*) can be used to mark the start of a comment. This must be preceded by a blank or must be the first non-blank character on the line. The end of the comment is identified by an asterisk immediately followed by a forward slash (/). This must be immediately followed by a blank, except when it ends a line. There must be a corresponding end of the comment (\*/) delimiter for each start of the comment (/\*) delimiter.

This comment format syntax might span multiple lines. Following are a few examples:

```
/* this is a comment */
```

```
insert_job: job1
```

```
command: ls "/*" /* argument must be delimited in quotes so not to be interpreted
as a comment */
```

```
machine: localhost
```

```
insert_job: job2
```

```
command: sleep 5 /* sleep a while */
```

```
machine: localhost
```

```
insert_job: job3
```

```
/* application: accounting
```

```
box_name: master
```

```
*/
```

```
command: echo "application and box are commented out"
```

```
machine: localhost
```

```
insert_job: job4
```

```
command: rm $HOME/test/* /* delete files in my test directory */
```

```
machine: localhost
```

## Control Workload

The sendevent command lets you control workload.

### NOTE

The sendevent command also lets you stop server processes. For information about how to stop server processes, see [Maintain System](#). For basic information about the sendevent command, see [Send Events](#).

## sendevent Command -- Cancel Unprocessed Events

Sometimes you must cancel unprocessed events. For example, you issue a sendevent command that instructs the scheduler to process an event at some time in the future but later decide that the event should not be processed.

You can use the sendevent command to cancel (or "unsend") unprocessed events. You can cancel all unprocessed events of a certain type, or you can cancel a specific event.

### NOTE

This event does not terminate jobs or terminate events that are associated with jobs in progress. To terminate jobs in progress or events that are associated with those jobs, use the KILLJOB or SENDSIGNAL event.

## Syntax

This command has the following format:

```
sendevent -E EVENT -U [{-J job_name | -B group_name [-I application_name] | -I
application_name }][-S autoserv_instance] [-P priority] [-T "time_of_event"] [-C
comment]
```

- **-E *EVENT* -U**

Cancels unprocessed events that meet the specified parameters. Events of the type that are specified in the - E parameter are canceled when they meet all the following conditions:

- They have not been processed.
- They are associated with the job, group, or application that is specified in the - J, -B, or - I parameter and that job, group, or application is defined within the instance that is specified in the - S parameter.
- They reference a global variable that is defined within the instance that is specified in the - S parameter.

### NOTE

- AutoSys Workload Automation does not delete these events from the database, but the scheduler does not process them.
- To cancel events that are scheduled to run at a specific time and meet all other specified parameters, use option -T. The event cancels all unprocessed events that meet the specified parameters when you do not specify option -T.
- The event has no impact on events that the scheduler has already processed.

- **-J *job\_name* | -B *group\_name* [-I *application\_name*] | -I *application\_name***

(Optional) Specifies a job, or group of jobs as one of the parameters that are used to determine whether to cancel an event. Unprocessed events that are associated with the specified jobs and meet all other parameters are canceled when you specify this option. Events that are not associated with the specified jobs are not canceled even if they meet other parameters and have not been processed.

- **-J *job\_name***

Specifies a single job as a parameter that is used to determine whether to cancel an event. Use this option to cancel events that are associated with the specified job.

- **-B *group\_name***

Specifies a group of jobs as a parameter that is used to determine whether to cancel an event. Use this option to cancel events that are associated with all jobs in the group.

- **-I *application\_name***

Specifies an application as a parameter that is used to determine whether to cancel an event. Use this option to cancel events that are associated with all jobs that are associated with the specified application.

### NOTE

- You can use option **-B** and option **-I** together to specify that you want to cancel only events that are associated with jobs in the specified group that are also associated with the specified application.
- You can use option **-B** or option **-I** only if you have execute access as-control to SENDEVENT\_GRPAPP.
- **-S *autoserv\_instance*** (Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
- **-P *priority***  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.  
**Default:** 10  
**Limits:** 1-1000
- **-T "*time\_of\_event*"**

(Optional) Specifies the time of the event. Use the “MM/dd/yyyy hh:mm[:ss]” format, as follows:

- **MM**  
Specifies the month.  
**Limits:** 01-12
- **dd**  
Specifies the day.  
**Limits:** 01-31
- **yyyy**  
Specifies the year.  
**Limits:** 1900-2100
- **hh**  
Specifies the hour.  
**Limits:** 00-23
- **mm**  
Specifies the minutes.  
**Limits:** 00-59
- **ss**  
(Optional) Specifies the seconds.  
**Limits:** 00-59

#### NOTE

- AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.
- Use this option to specify a particular time as one of the parameters that is used to determine whether to cancel an event. Events that are scheduled to be processed at other times are not canceled even if they meet all other specified parameters.

- **-C comment**

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

#### NOTE

- Enclose values that contain spaces or special characters in quotation marks (“”).
- Avoid using single quotation marks as they are doubled.

### Example: Cancel an Unprocessed STARTJOB Events

Suppose that the job definition for a job that is named inventory does not specify date and time conditions, so it only runs when you issue a STARTJOB event against it. On January 2, 2012, you issue a STARTJOB event that instructs the scheduler to start inventory on January 5, 2012 at 14:00. On January 4, you determine that the scheduled January 5 run of inventory should be suspended. To prevent the scheduler from starting inventory, cancel the unprocessed STARTJOB event as follows:

```
sendevent -E STARTJOB -U -J inventory
```

### Example: Cancel Unprocessed SET\_GLOBAL Events

This example cancels unprocessed SET\_GLOBAL events that reference global variables that are associated with the application named JasperReports.

```
sendevent -E SET_GLOBAL -U -I JasperReports
```

## sendevent Command -- Communicate with a Job in Progress

Sometimes you want to communicate with an active job. For example, you want to send special instructions to processes that are associated with certain jobs.

You can use the SEND\_SIGNAL event to send any UNIX signal or Windows user-defined event to processes that are associated with command jobs. Specify the appropriate UNIX signal or Windows event name; otherwise, the SEND\_SIGNAL event is not valid.

You can use the SEND\_SIGNAL event to kill command jobs, but some applications do not process these instructions when you send them using this method. Use the KILLJOB event to stop non-command jobs or jobs running applications that do not process kill signals.

The KILLJOB event does not support the following job types:

- FTP
- Entity Bean
- i5/OS
- JMS Publish
- JMX-Mbean Attribute Get
- JMX-Mbean Attribute Set
- JMX-Mbean Create Instance
- JMX-Mbean Operation
- JMX-Mbean Remove Instance
- Process Automation Start Request
- RMI
- Secure Copy
- Session Bean
- SNMP Value Get
- SNMP Value Set
- Wake on LAN
- Web Service RPC/Encoded
- z/OS Data Set Trigger

### NOTE

- Issuing a KILLJOB event against a job that spawns other processes does not kill those processes.
- On UNIX, if the `oscomponent.wrapper.exec.force` parameter in the `agentparm.txt` file is set to false (the default), the agent does not send UNIX signals to jobs. To configure the agent to send UNIX signals to jobs, set the `oscomponent.wrapper.exec.force` parameter to true. When this parameter is set to true, the agent does not support chained commands (consisting of multiple commands that are separated by a semicolon). To support both the sending of UNIX signals and the processing of chained commands, do the following actions:
  - a. Set the `oscomponent.wrapper.exec.force` parameter to true.
  - b. Create a script file and add the chained commands to the script file.
  - c. Update the command attribute in the job definition to execute the script file instead of the chained command.

## Syntax

This command has the following format:

```
sendevent -E {SEND_SIGNAL -k {event_name | signal_num}| KILLJOB [-k {event_name |
signal_num}]} {-J job_name | -B group [-I application] | -I application} [-P priority]
[-T "time_of_event"] [-C comment]
```

- **-J job\_name**  
Specifies the job to send the event to.  
**Limit:** Enter a value of up to 64 alphanumeric characters.
- **-B group**  
Sends the specified event to the specified group of jobs.
- **-I application**  
Sends the event to the jobs that are associated with the specified application.

#### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J job\_name**Sends the event to the specified job. Use this argument alone.
  - **-B group**Sends the event to all the jobs that are in the specified group.
  - **-I application**Sends the event to all the jobs that are associated with the specified application.
  - **-B group -I application | -I application -B group**Sends the event to all jobs within the specified group that are also associated with the specified application.
- **-k event\_name | signal\_num**  
Specifies the UNIX signal number or the name of a Windows user-defined event.

#### NOTE

##### UNIX Notes:

- *On agents (r11.3, Release 11.3.5, or Release 11.3.6), you can send multiple signals by entering a comma-delimited list (specify only positive integers) as the value for the signal\_num variable. The agent sends the first signal, sleeps for three seconds, sends the next signal, and so on.*
- *On legacy agents (4.5 or r11), you can send multiple signals by entering a comma-delimited list as the value for the signal\_num variable. The agent sends the first signal, sleeps for a second, sends the next signal, and so on.*
- On agents or legacy agents, you can send non-kill signals by specifying this parameter with the KILLJOB event. AutoSys Workload Automation sends the kill signals that are specified in the configuration file when you send the KILLJOB event without specifying the **-k** option. AutoSys Workload Automation specifies the kill signals as 2 (interrupt job) and 9 (kill job) in the configuration file unless you specify other signals. To specify kill signals, open the configuration file on the machine where the scheduler is running and edit the following parameter:

```
KillSignals: {2, 9 | signal_num [, signal_num...]}
```

The scheduler instructs the agent to send the signals in the order that they appear in the configuration file. For example, the default behavior is to send the UNIX process interrupt signal (2) followed by the UNIX process kill signal (9).

- To send signals to an entire process group on agents, precede the appropriate signal\_numbers values with a minus sign (-).

##### Windows Notes:

- On agents, the KILLJOB event does not support the **-k** option and is ignored. However, you can issue the SEND\_SIGNAL event -k signals to agents or legacy agents.
  - To ensure that the scheduler interprets the value of the **-k** option as the name of a user-defined Windows event to signal, set the opsys attribute of the machine definitions where Windows jobs run to 'windows'.
  - The jobs that are associated with the event are programmed to create the Windows event that is specified in the **-k** option, to listen for this Windows event to be signaled, and to take a particular action upon receiving this signal. The scheduler instructs the Windows agent to signal the user-defined Windows event whose name matches the value that is specified in the **-k** option.
  - Process groups are not supported.
- **-P priority**  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.  
**Default:** 10  
**Limits:** 1-1000
  - **-T "time\_of\_event"**  
(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:
    - **MM**  
Specifies the month.  
**Limits:** 01-12
    - **dd**  
Specifies the day.  
**Limits:** 01-31
    - **yyyy**  
Specifies the year.  
**Limits:** 1900-2100
    - **hh**  
Specifies the hour.  
**Limits:** 00-23
    - **mm**  
Specifies the minutes.  
**Limits:** 00-59
    - **ss**  
(Optional) Specifies the seconds.  
**Limits:** 00-59
- NOTE**  
AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.
- **-C comment**  
(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.  
**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Kill a Job**

This example kills the job named quarterly\_report.

```
sendevent -E KILLJOB -J quarterly_report
```

### Example: Send a Signal to Pause a Job

This example pauses the job named `quarterly_report`.

```
sendevent -E SEND_SIGNAL -k {19 | pausejob} -J quarterly_report
```

#### NOTE

- On UNIX, the signal to pause a job is 19.
- On Windows, the example works only when you program the `quarterly_report` job to take the following actions:
  - a. Create a Windows event named `pausejob`.
  - b. Listen for the AutoSys Workload Automation agent to signal the `pausejob` event.
  - c. Stop running when the `pausejob` event is signaled.

### Example: Send a Signal to Resume a Paused Job

This example resumes the job named `quarterly_report`.

```
sendevent -E SEND_SIGNAL -k {18 | resumejob} -J quarterly_report
```

#### NOTE

- On UNIX, the signal to resume a job is 18.
- On Windows, the example works only when you program the `quarterly_report` job to take the following actions:
  - a. Create a Windows event named `resumejob`
  - b. Listen for the AutoSys Workload Automation agent to signal `resumejob` event
  - c. Resume when the `resumejob` event is signaled
- The job picks up where you paused it and continues running.

## sendevent Command -- Change the Online Status of a Machine

When a machine needs maintenance, take it offline. The jobs that are scheduled to run on the machine enter a queued state and do not run until you bring the machine online.

To change the online status of a machine, use the `sendevent` command to issue a `MACH_ONLINE` or `MACH_OFFLINE` event. Only users with execute access to a machine can change the online status of that machine.

The online status of a machine affects how the scheduler evaluates jobs with load balancing attributes or resource dependencies. When taking a machine offline, the scheduler places the job in the `PEND_MACH` status when all candidate machines are offline. While evaluating a job that is queued in the `QUE_WAIT` or `RESWAIT` status, the scheduler considers machine load units or machine-based resources as unavailable if the machine is offline. In that case, the job remains in the `QUE_WAIT` or `RESWAIT` status and does not transition into the `PEND_MACH` status. When bringing the machine back online, the scheduler evaluates jobs in the `QUE_WAIT` or `RESWAIT` status first followed by the jobs in the `PEND_MACH` status.

### Syntax

This command has the following format:

```
sendevent -E {MACH_ONLINE | MACH_OFFLINE} -N machine [-P priority] [-T "time_of_event"]
[-C comment]
```

- **-N *machine*** Specifies the machine to which the online status change should apply.  
**Limits:** Enter a value of up to 80 alphanumeric characters that represents the name of a real machine.

**WARNING**

Do not specify a host name, the name of a machine that represents a cluster, a node name, or the name of a virtual machine.

- **-P *priority*** (Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "*time\_of\_event*"**

(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:

- ***MM***

Specifies the month.

**Limits:** 01-12

- ***dd***

Specifies the day.

**Limits:** 01-31

- ***yyyy***

Specifies the year.

**Limits:** 1900-2100

- ***hh***

Specifies the hour.

**Limits:** 00-23

- ***mm***

Specifies the minutes.

**Limits:** 00-59

- ***ss***

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C *comment*** (Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Take a Machine Offline**

Suppose that you need to perform maintenance on the machine that is named retail, but you cannot perform this maintenance while retail is online. To perform maintenance, change the online status of retail as follows:

```
sendevent -E MACH_OFFLINE -N retail
```

**Example: Bring a Machine Online**

Suppose that you need to run the job that is named inventory on the machine that is named retail, but inventory is in the PEND\_MACH state because retail is offline for maintenance. To run inventory, release it from the PEND\_MACH state by changing the online status of retail as follows:



```
sendevent -E MACH_ONLINE -N retail
```

**NOTE**

The inventory job leaves the PEND\_MACH state when the retail comes online. The scheduler re-evaluates the starting conditions for inventory currently. The job starts only when it meets those starting conditions. The scheduler ignores unmet date and time conditions in this evaluation when those conditions were met while inventory is in the PEND\_MACH state.

**sendevent Command -- Change the Executable Status of a Job**

Before you solve an external problem with a job, place the job in a non-executable state. When the problem is solved, remove it from the non-executable state or force start it.

**WARNING**

Force start a job that is in a non-executable state only when you need the job to run immediately. To resume the regular run schedule, change the executable status of the job.

To change the executable status of a job, issue *one* of the following events:

- **JOB\_ON\_HOLD (112)**

Places the specified job in ON\_HOLD status. The job does not start, and downstream dependent jobs do not run. Boxes containing jobs in ON\_HOLD status cannot complete. The event has no effect on jobs with a status of STARTING or RUNNING. This event is manually generated.

- **JOB\_ON\_ICE (110)**

Suspends scheduled runs of the specified job by temporarily removing it from the job stream. Downstream dependent jobs run. The scheduler places the job in ON\_ICE status. If the job is in a box, the scheduler does not execute the job for the entire run of the box. The event has no effect on jobs with a status of STARTING or RUNNING. This event is manually generated.

**NOTE**

Use caution when you place jobs in this state. Downstream jobs with critical dependencies on jobs in the ON\_ICE state do not complete successfully. For example, a job that requires input that is contained in a file that is produced by another job cannot succeed when the job that produces the file is in the ON\_ICE state.

- **JOB\_ON\_NOEXEC (145)**

Places the specified job in the ON\_NOEXEC status and bypasses execution of commands that are associated with the job. The scheduler does not immediately schedule downstream dependent jobs nor does it evaluate their conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the INACTIVE status.

When the NOEXEC job meets its starting conditions, the scheduler evaluates this job as successfully completed with an exit code of zero even though the agent does not execute the commands. After the NOEXEC job is evaluated as successfully completed, the scheduler schedules downstream dependent jobs.

You can issue the JOB\_ON\_NOEXEC event to an ACTIVATED job that is within a running box. The job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job are met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.

The following conditions apply:

- The scheduler processes the JOB\_ON\_NOEXEC event similar to that of the CHANGE\_STATUS event to INACTIVE.
- The scheduler ignores the JOB\_ON\_NOEXEC event, if sent to:

- A non-box job that is in the STARTING, RUNNING, or ON\_ICE status
- A box job that is in the ON\_ICE or RUNNING status
- A box job with jobs (including the jobs contained in lower level boxes) in a status other than the following status: ON\_HOLD, ON\_NOEXEC, INACTIVE, SUCCESS, FAILURE, ACTIVATED, or TERMINATED.
- When a job that is in the ON\_NOEXEC status is in a box and the starting conditions of the box are met, the job enters the ACTIVATED status. The scheduler bypasses the execution of the job when the starting conditions of the job are met.
- When the starting conditions of a box that is in the ON\_NOEXEC status are met, the box enters the RUNNING status. The scheduler bypasses the execution of the box when the starting conditions of all jobs in the box (including all jobs contained in lower level boxes within the box) are met.
- If you send the JOB\_ON\_NOEXEC event to a job in a box, the effect is the same as sending the CHANGE\_STATUS event to INACTIVE for a job in a box. The scheduler sets the job to the ON\_NOEXEC status and also sets the internal secondary status to INACTIVE, so that the scheduler evaluates the overall box status as if the job entered the INACTIVE status. To complete a job that is in the ON\_NOEXEC status so that the scheduler evaluates the job as if the job entered the SUCCESS status, send the FORCE\_STARTJOB event.
- If you send the JOB\_ON\_NOEXEC event to a box, the effect is the same as sending the CHANGE\_STATUS event to INACTIVE for a box. The box enters the ON\_NOEXEC status and the scheduler sets the status of all jobs in the box (including all jobs contained in lower level boxes within the box) at all levels to ON\_NOEXEC.
- If you send the JOB\_ON\_NOEXEC event to change the status of an ACTIVATED job that is within a running job, the job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job are met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.
- The JOB\_ON\_NOEXEC event supersedes the JOB\_ON\_HOLD event effectively overwriting the ON\_HOLD status with the ON\_NOEXEC status.
- The JOB\_ON\_NOEXEC event does not supersede the JOB\_ON\_ICE event and does not overwrite the ON\_ICE status with the ON\_NOEXEC status.
- If the job is in the QUEWAIT or RESWAIT status, the scheduler removes the job from the load balancing and resource wait queues before placing it in the ON\_NOEXEC status.
- CHANGE\_STATUS events have no effect on the job status if the job is in the ON\_NOEXEC status. You must issue the JOB\_OFF\_NOEXEC event to take the job out of the ON\_NOEXEC status.
- If you send the CHANGE\_STATUS event to a box to set its status to INACTIVE, the scheduler does not change the status of ON\_NOEXEC jobs within the box to INACTIVE. Instead, the scheduler sets the internal secondary status to INACTIVE so that the scheduler can evaluate the jobs as if they were set to the INACTIVE status.
- The scheduler sends an internal JOB\_ON\_NOEXEC event to a job in the ON\_NOEXEC status instead of the CHANGE\_STATUS event to INACTIVE when:
  - The job is scheduled on a day that is excluded by the exclude\_calendar job attribute.
  - The next scheduled run of the job is outside the run window.
- When a box that is in the ON\_NOEXEC status completes, the scheduler resets the status of any remaining ACTIVATED jobs to ON\_NOEXEC behaving as if the jobs were reset to the INACTIVE status.
- **JOB\_OFF\_HOLD (113)**  
Removes the specified job from ON\_HOLD status. As soon as the starting conditions permit, the job starts. This event is manually generated.
- **JOB\_OFF\_ICE (111)**  
Returns the specified job to the job stream. The scheduler removes the job from ON\_ICE status. Jobs that meet their starting conditions while they are in the ON\_ICE state start when starting conditions recur. This event is manually generated.
- **JOB\_OFF\_NOEXEC (146)**  
Removes the specified job from the ON\_NOEXEC status. The agent executes the commands that are associated with the job the next time that the scheduler starts the job. This event is manually generated.

When the scheduler processes the JOB\_OFF\_NOEXEC event for a job, it places the job in the INACTIVE, ACTIVATED, or SUCCESS status based on the job's existing status while in non-execution mode. If the job is in a box, the scheduler does not evaluate the overall box status. If you send the JOB\_OFF\_NOEXEC to a box, all jobs in the box (including all jobs that are contained in lower level boxes within the box) are reset to the INACTIVE, ACTIVATED, or SUCCESS status.

#### NOTE

- If you issue the sendevent command to change the executable status of a job that is in the ON\_HOLD status to the ON\_ICE or ON\_NOEXEC status, AutoSys Workload Automation verifies that the security policy grants you access to send both the JOB\_OFF\_HOLD and JOB\_ON\_ICE or JOB\_ON\_NOEXEC events through the as-sendevent security resource class.
- The scheduler does not start jobs in the ON\_HOLD state which might affect whether downstream jobs run.
- When you send a JOB\_ON\_ICE event, the scheduler removes the job from the job stream, so downstream jobs still run. Downstream jobs with certain dependencies on jobs in the ON\_ICE state might not run to success.
- When a job is in the NOEXEC state, the scheduler sends the BYPASS event. This event simulates a job run but the scheduler does not execute job commands. These jobs evaluate as successfully completed, so downstream jobs run.

#### Syntax

This command has the following format:

```
sendevent -E EVENT {-J job_name | -B group [-I application] | -I application}{-S autoserv_instance} [-P priority] [-T "time_of_event"] [-C comment]
```

- **-E *EVENT*** Sends the specified event.
- **-J *job\_name*** Specifies the job to send the event to.  
**Limits:** Enter a value of up to 64 alphanumeric characters.
- **-B *group*** Sends the specified event to the specified group of jobs.
- **-I *application*** Sends the event to the jobs that are associated with the specified application.

#### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.
- **-S *autoserv\_instance***  
(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
- **-P *priority***  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "time\_of\_event"**

(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:

- **MM**  
Specifies the month.  
**Limits:** 01-12
- **dd**  
Specifies the day.  
**Limits:** 01-31
- **yyyy**  
Specifies the year.  
**Limits:** 1900-2100
- **hh**  
Specifies the hour.  
**Limits:** 00-23
- **mm**  
Specifies the minutes.  
**Limits:** 00-59
- **ss**  
(Optional) Specifies the seconds.  
**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C comment** (Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

### Example: Place a Job on Hold

Suppose the jobs that are named `summary_reports` is associated with the application named `JasperReports` and you need to upgrade `JasperReports`. Downstream jobs use information that is contained in the file that `summary_reports` contains. To suspend `summary_reports` and all downstream jobs while you upgrade `JasperReports`, place `summary_reports` on hold as follows:

```
sendevent -E JOB_ON_HOLD -J summary_reports
```

### Example: Return a Job to the Job Stream

Suppose that you placed the job named `charge_payments` is `ON_ICE` because you are sending gifts to your customers. You need to skip `charge_payments` but run the downstream job named `ship_orders`. After you ship the gifts, you need to start running `charge_payments` again. To resume the normal workflow, return `charge_payments` to the job stream as follows:

```
sendevent -E JOB_OFF_ICE -J charge_payments
```

### Example: Bypass Execution of Job Commands

Suppose that the job that is named `route_shipments` produces a route based on the information that is collected by the jobs named `warehouse1_data`, `warehouse2_data`, and `warehouse3_data` and starts only when those three jobs complete successfully. Warehouse #2 has a fire, so you need to skip `warehouse2_data` until the warehouse is repaired. To skip `warehouse2_data` and still run `route_shipments`, bypass executions of commands that are associated with `warehouse2_data` as follows:

```
sendevent -E JOB_ON_NOEXEC -J warehouse2_data
```

### Example: Change the Status of an Activated Job in a Box to ON\_NOEXEC

Suppose that the box that is named `BOX_ST` includes two child jobs; `JOB1` and `JOB2`. `JOB2` starts only when `JOB1` completes successfully. Another job named `JOB3` starts only when `JOB2` completes successfully. When `BOX_ST` starts running, `JOB1` starts running because it does not have any starting conditions and `JOB2` is placed in the `ACTIVATED` state. If for some reason you do not want to run `JOB2`, you can send the `JOB_ON_NOEXEC` event to `JOB2` to change its status from `ACTIVATED` to `ON_NOEXEC`, as follows:

```
sendevent -E JOB_ON_NOEXEC -J JOB2
```

When `JOB2` meets its starting conditions, the scheduler sends a `BYPASS` event and `JOB2` goes to the `ON_NOEXEC` state.

### sendevent Command -- Change the Queue Priority of a Job

Offline machines and machines with lack of available load units sometimes prevent jobs from running. The scheduler places these jobs in `QUE_WAIT` status. The scheduler starts the jobs when load units become available based on their queue priority.

When a job is ready to run on a designated machine but the current load on that machine is too large to accept the new job's load or the job has resource dependencies that are not satisfied, AutoSys Workload Automation queues the job for that machine so it runs when sufficient resources are available. For job queuing to take place, you must define the priority attribute in the job definition. The queue priority establishes the relative priority of all jobs queued for a given machine. A lower number attribute value indicates a higher priority. If you do not set the priority attribute or the priority is set to 0, the job is not queued behind other jobs and runs immediately on a machine if resource dependencies permit or if the job has no resource dependencies. The scheduler ignores any load unit values that are defined for the job or machine when the job has a priority value of zero.

#### NOTE

For more information about how AutoSys Workload Automation queues jobs, see [Specify Queuing Priority](#) and [How AutoSys Workload Automation Queues Jobs](#).

Sometimes you want to change the queue priority of a job. For example, you need the job to start as soon as possible. You can use the `sendevent` command to change the queue priority of a job manually.

#### NOTE

- This event applies to the current run of jobs in the `QUE_WAIT` or `RESWAIT` status. To change the priority of a job permanently, edit the job definition.
- The scheduler tries to evaluate the starting conditions of jobs that are associated with a group or application in order of their priority; however it does not ensure that jobs execute strictly in order of their priority.

### Syntax

This command has the following format:

```
sendevent -E CHANGE_PRIORITY -q queue_priority {-J job_name | -B group [-I application]
| -I application} [-S autoserv_instance] [-P priority] [-T "time_of_event"] [-C comment]
```

- **-q queue\_priority**

Specifies the new queue priority to assign to the job.

**Default:** 0

**Limits:** 0-99

- **-J *job\_name***

Specifies the job to send the event to.

**Limit:** Enter a value of up to 64 alphanumeric characters.

- **-B *group***

Sends the specified event to the specified group of jobs.

- **-I *application***

Sends the event to the jobs that are associated with the specified application.

#### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.

- **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

- **-P *priority***

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "*time\_of\_event*"**

(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:

- ***MM***

Specifies the month.

**Limits:** 01-12

- ***dd***

Specifies the day.

**Limits:** 01-31

- ***yyyy***

Specifies the year.

**Limits:** 1900-2100

- ***hh***

Specifies the hour.

**Limits:** 00-23

- ***mm***

Specifies the minutes.

**Limits:** 00-59

- ***ss***

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C comment**

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Change the Queue Priority of a Job**

Suppose that the job that is named `fin_report` is scheduled to run on the machine that is named `accounting`, but `fin_report` is in the `QUE_WAIT` state because `accounting` does not have enough load units available to run it. To ensure that `fin_report` runs as soon as resources become available, assign the highest queue priority to it as follows:

```
sendevent -E CHANGE_PRIORITY -q 1 -J fin_report
```

**sendevent Command -- Change the Status of a Job**

As a job progresses, the status of the job changes. Typically, the scheduler internally generates `CHANGE_STATUS` events for jobs in progress. Sometimes you need to change the status of a job manually. For example, when the terminal status of the job interferes with workflow.

You can use the `sendevent` command to change the status of a job manually. Only users with execute access for a job can change the status of that job.

**Syntax**

This command has the following format:

```
sendevent -E CHANGE_STATUS -s status {-J job_name | -B group [-I application] | -I application} [-S autoserv_instance] [-P priority] [-T "time_of_event"] [-C comment]
```

- **-s status**

Specifies the new status of the job. You can use this event to assign a job *one* of the following statuses:

- FAILURE
- INACTIVE
- RUNNING
- STARTING
- SUCCESS
- TERMINATED

**NOTE**

- If you issue the `sendevent` command to change the status of a job that is in the `ON_HOLD` status, AutoSys Workload Automation verifies that the security policy grants you access to send both the `JOB_OFF_HOLD` and `CHANGE_STATUS` events through the `as-sendevent` security resource class.
- If you issue the `sendevent` command to change the status of a box to the `RUNNING` status, AutoSys Workload Automation verifies that the security policy grants you access to send both the `CHANGE_STATUS`



and STARTJOB or FORCE\_STARTJOB events through the as-sendevent security resource class. To change the status of a box that specifies a condition attribute value in the box definition, you must have permissions to send the CHANGE\_STATUS and FORCE\_STARTJOB events. To change the status of a box that does not specify a condition attribute value in the box definition, you must have permissions to send the CHANGE\_STATUS and STARTJOB events.

- You can issue the sendevent command to change the status of a job that is in the ON\_NOEXEC status to either the INACTIVE or SUCCESS status while remaining in non-execution mode. When you change the status of a non-execution job to SUCCESS, the scheduler completes the job's run and schedules the downstream dependent jobs. Changing the status of a non-execution box job to INACTIVE also changes the status of all the jobs in that box job to INACTIVE in non-execution mode. Changing the status of a non-execution box job to the RUNNING status starts the box job.
- Changing the status of a non-box job to RUNNING changes the status of the job in the database but does not cause the job to execute. To execute the job, use the STARTJOB event or the FORCSTART event.
- The scheduler automatically reschedules a job in progress (with a status of RUNNING) to run at the next start time when you use this event to change the status that job to *one* of the following terminal states:
  - FAILURE
  - SUCCESS
  - TERMINATED
- The scheduler does not reschedule a job in progress (with a status of RUNNING) to run at the next start time when you use this event to change the status of that job to *one* of the following states:
  - INACTIVE
  - STARTING

- **-J *job\_name***

Specifies the job to send the event to.

**Limits:** Enter a value of up to 64 alphanumeric characters.

- **-B *group***

Sends the specified event to the specified group of jobs.

- **-I *application***

Sends the event to the jobs that are associated with the specified application.

#### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.

- **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

- **-P *priority***

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10



**Limits:** 1-1000

- **-T "time\_of\_event"**

(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:

- **MM**  
Specifies the month.  
**Limits:** 01-12
- **dd**  
Specifies the day.  
**Limits:** 01-31
- **yyyy**  
Specifies the year.  
**Limits:** 1900-2100
- **hh**  
Specifies the hour.  
**Limits:** 00-23
- **mm**  
Specifies the minutes.  
**Limits:** 00-59
- **ss**  
(Optional) Specifies the seconds.  
**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C comment** (Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

### Example: Change the Status of a Job

Suppose that your workflow includes the following jobs:

- process\_orders
- verify\_pay\_info
- charge\_payments
- prepare\_inventory
- confirm\_inventory
- prepare\_shipments
- ship\_orders

Each of these jobs depends on the success of the previous job. You need to ship orders to ship orders tonight, but verify\_pay\_info fails and prevents charge\_payments from starting. You cannot ship\_orders until charge\_payments starts and the entire workflow completes successfully. To complete the workflow, manually verify the information that is associated with verify\_pay\_info and change the status the job to SUCCESS as follows:

```
sendevent -E CHANGE_STATUS -J verify_pay_info -s SUCCESS
```

## sendevent Command -- Delete Jobs

Sometimes you need to delete jobs from the database. For example, when you have several versions of a job stored and some of the versions are obsolete.

You can use the sendevent command to delete jobs. Deleting obsolete jobs prevents the database from being overloaded. Only users with delete access can delete jobs.

### Syntax

This command has the following format:

```
sendevent -E DELETEJOB {-J job_name | -B group [-I application] | -I application}{-S autoserv_instance} [-P priority] [-T "time_of_event"] [-C comment]
```

- **-J *job\_name***  
Specifies the job to send the event to.  
**Limits:** Enter a value of up to 64 alphanumeric characters.
- **-B *group***  
Sends the specified event to the specified group of jobs.
- **-I *application***  
Sends the event to the jobs that are associated with the specified application.

### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.
- **-S *autoserv\_instance***  
(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
- **-P *priority***  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.  
**Default:** 10  
**Limits:** 1-1000
- **-T "*time\_of\_event*"**  
(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:
  - ***MM***  
Specifies the month.  
**Limits:** 01-12
  - ***dd***  
Specifies the day.  
**Limits:** 01-31
  - ***yyyy***  
Specifies the year.  
**Limits:** 1900-2100
  - ***hh***  
Specifies the hour.

**Limits:** 00-23

– ***mm***

Specifies the minutes.

**Limits:** 00-59

– ***ss***

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

• **-C *comment***

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Delete an Obsolete Job**

Suppose that the municipality stops requiring employers to report demographic information about their employees. This change renders the current version of the job named `employee_information` obsolete, so you create a job named `employee_information_v2`. To prevent the database from being overloaded, delete the obsolete version of the job as follows:

```
sendevent -E DELETEJOB -J employee_information
```

**Example: Delete a Box Job**

Suppose you have a job (`abc_box1`) of type `box` and there is job of type lets say "CMD" (`xyzjob1`) inside the box job (`abc_box1`) and user gives the delete `sendevent` command (`sendevent -e DELETEJOB -J abc_box1`) to delete the box job, then the box job and the job inside the box job (both) will be deleted.

```
/* ----- abc_box1 ----- */
```

```
insert_job: abc_box1
job_type: BOX
owner: Administrator@test004321
permission:
date_conditions: 1
days_of_week: all
start_mins: 00, 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55
alarm_if_fail: 1
alarm_if_terminated: 1
```

```
/* ----- xyzjob1 ----- */
```

```
insert_job: xyzjob1
job_type: CMD
box_name: abc_box1
command: sleep 10
machine: localhost
owner: Administrator@test004321
permission:
```

```
date_conditions: 0
alarm_if_fail: 1
alarm_if_terminated: 1
```

Generate a Report for the box job:

```
autorep -J abc_box1
```

The resulting report might resemble the following:

Job Name	Last Start	Last End	ST/Ex	Run/Ntry	Pri/Xit
abc_box1	-----	-----	IN	0/0	
xyzjob1	-----	-----	IN	0/0	

Delete the box job as follows:

```
sendevent -e DELETEJOB -J abc_box1
```

To verify after deleting the box job:

Generate a Report for the box job:

```
autorep -J abc_box1
```

The following error message is displayed:

**CAUAJM\_E\_50027 Invalid Job Name: abc\_box1**

## sendevent Command -- Post a Comment to the Event Log

Sometimes you want to post a comment to the event log. For example, you want to record the reason that no jobs ran today or you want to record the reason that you send the event.

You can use the sendevent command to post comments to the event log.

### Syntax

This command has the following format:

```
sendevent -E EVENT -C comment [{-J job_name | -B group_name [-I application_name]
| -I application_name }] [-S autoserv_instance] [-P priority] [-T "time_of_event"]
[OPTIONS]
```

- **-E *EVENT***

Sends the specified event. To post a general comment, send a COMMENT event. To post a comment that is associated with a particular event, specify the -C option when you issue the sendevent command for that event.

- **-C *comment***

Specifies the comment to post.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

#### NOTE

- Enclose values that contain spaces in quotation marks (").
- Avoid using single quotation marks as they are doubled.

- **-J *job\_name* | -B *group\_name* [-I *application\_name*] | -I *application\_name***

(Optional) Specifies a job, or group of jobs as one of the parameters that are used to associate with the posted comment.

- **-J *job\_name***

Specifies a single job as a parameter that is used to associate the job with the posted comment. Use this option to post a comment for the specified job.

- **-B *group\_name***

Specifies a group of jobs as a parameter that is used to associate the group of jobs with the posted comment. Use this option to post the same comment for each job in the group.

– **-I *application\_name***

Specifies an application as a parameter that is used to associate the application with the posted comment. Use this option to post the same comment for each job that is associated with the specified application.

**NOTE**

- You can use option **-B** and option **-I** together to specify that you want to post a comment that is associated with jobs in the specified group that are also associated with the specified application.
- You can use option **-B** or option **-I** only if you have execute access as-control to SENDEVENT\_GRPAPP.

• **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

• **-P *priority***

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

• **-T "*time\_of\_event*"**

(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:

– ***MM***

Specifies the month.

**Limits:** 01-12

– ***dd***

Specifies the day.

**Limits:** 01-31

– ***yyyy***

Specifies the year.

**Limits:** 1900-2100

– ***hh***

Specifies the hour.

**Limits:** 00-23

– ***mm***

Specifies the minutes.

**Limits:** 00-59

– ***ss***

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

• **OPTIONS**

(Optional) Specifies optional parameters. The options that are available depend on the event that you send.

**Example: Post a General Comment to the Event Log**

Suppose that today is a holiday and that all your job definitions exclude holidays from the run schedule. To record this information, post a comment to the event log as follows:

```
sendevent -E COMMENT -c "No jobs will run today because it is a holiday."
```

### Example: Attach a Comment to an Event and Post it to the Event Log

Suppose that you delete the job that is named inventory because it is obsolete. To record this information in the event log when you delete inventory, attach a comment to the DELETEJOB event as follows:

```
sendevent -E DELETEJOB -J inventory -c "Deleted this job because it is obsolete."
```

## sendevent Command -- Release a Held Resource

When a job does not release a resource to the pool, other jobs cannot use it. To avoid interruptions to the workflow, use the sendevent command to release these resources.

### WARNING

The scheduler does not release resources that are held by jobs that are still in progress. Issue the RELEASE\_RESOURCE event only when a job does not release resources after it completes.

### Syntax

This command has the following format:

```
sendevent -E RELEASE_RESOURCE {-J job_name | -B group [-I application] | -I application}
[-S autoserv_instance] [-P priority] [-C comment]
```

- **-J *job\_name***

Specifies the job to send the event to.

**Limits:** Enter a value of up to 64 alphanumeric characters.

- **-B *group***

Sends the specified event to the specified group of jobs.

- **-I *application***

Sends the event to the jobs that are associated with the specified application.

### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name*** - Sends the event to the specified job. Use this argument alone.
  - **-B *group*** - Sends the event to all the jobs that are in the specified group.
  - **-I *application*** - Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group*** - Sends the event to all jobs within the specified group that are also associated with the specified application.

- **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

- **-P *priority***

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "time\_of\_event"**

(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:

- **MM**  
Specifies the month.  
**Limits:** 01-12
- **dd**  
Specifies the day.  
**Limits:** 01-31
- **yyyy**  
Specifies the year.  
**Limits:** 1900-2100
- **hh**  
Specifies the hour.  
**Limits:** 00-23
- **mm**  
Specifies the minutes.  
**Limits:** 00-59
- **ss**  
(Optional) Specifies the seconds.  
**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C comment**

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

### Example: Release a Held Resource

Suppose that the job named profit\_loss uses the same resources as the job named revenue, but revenue did not release those resources after it completed. To run profit\_loss, release the held resources as follows:

```
sendevent -E RELEASE_RESOURCE -J revenue
```

### sendevent Command -- Resume a Hadoop Oozie Workflow Job

To resume a Hadoop Oozie workflow in the SUSPENDED state, use the sendevent command to issue the RESUMEJOB event. After you issue the RESUMEJOB event, the Hadoop Oozie workflow is placed in the RUNNING state.

#### Syntax

This command has the following format:

```
sendevent -E RESUMEJOB -j job_name
```

## sendevent Command -- Send an Alarm

*Alarms* are special events that notify operators of situations requiring their attention. The system issues alarms by internally generating an event, but these events are informational only. To resolve the issue that triggers an alarm, initiate the required action using the sendevent command.

Some problems do not trigger the scheduler to generate an alarm. For example, when a problem quickly resolves itself without manual intervention. Sometimes, you need to notify someone to follow up on such problems so that they do not recur. In these cases, use the sendevent command to generate an alarm manually.

The scheduler records alarms in the ujo\_alarm table. The ujo\_alarm table uses numeric codes to represent alarms. The numeric codes are resolved using the ujo\_intcodes table.

### Syntax

This command has the following format:

```
sendevent -E ALARM -A alarm [{-J job_name | -B group_name [-I application_name] |
-I application_name }] [-S autoserv_instance] [-P priority] [-T "time_of_event"] [-
C comment]
```

- **-A alarm**  
Specifies the alarm to send.
- **-J job\_name | -B group\_name [-I application\_name] | -I application\_name**  
(Optional) Specifies a job, or group of jobs as one of the parameters that are used to associate with the alarm.
  - **-J job\_name**  
Specifies a single job as a parameter that is used to associate the job with the alarm. Use this option to send an alarm for the specified job.
  - **-B group\_name**  
Specifies a group of jobs as a parameter that is used to associate the group of jobs with the alarm. Use this option to send the same alarm for each job in the group.
  - **-I application\_name**  
Specifies an application as a parameter that is used to associate the application with the alarm. Use this option to send the same alarm for each job that is associated with the specified application.

#### NOTE

- You can use option **-B** and option **-I** together to specify that you want to send an alarm that is associated with jobs in the specified group that are also associated with the specified application.
- You can use option **-B** or option **-I** only if you have execute access as-control to SENDEVENT\_GRPAPP.
- **-S autoserv\_instance**  
(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
- **-P priority**  
(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10



**Limits:** 1-1000

- **-T "time\_of\_event"**

(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:

- **MM**

Specifies the month.

**Limits:** 01-12

- **dd**

Specifies the day.

**Limits:** 01-31

- **yyyy**

Specifies the year.

**Limits:** 1900-2100

- **hh**

Specifies the hour.

**Limits:** 00-23

- **mm**

Specifies the minutes.

**Limits:** 00-59

- **ss**

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C comment**

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Send an Alarm**

Suppose that your database experiences a momentary problem but the scheduler recovers with no manual intervention. The scheduler does not generate an alarm, but you need the operator follow up on the problem. To notify the operator to follow up on the database problem, send an alarm as follows:

```
sendevent -E ALARM -A DB_PROBLEM -C "please contact the database administrator regarding
the problem on 01/01/2012 at 17:15:06"
```

**sendevent Command -- Send a Response to a Job that Requires Manual Intervention**

Sometimes you need to send information to a job before it can continue. For example, when a job is programmed to determine actions based on manual input. In these cases, the agent pauses the job and sends a message to the scheduler requesting the required information. The scheduler places the job in the WAIT\_REPLY state and forwards this message to you by sending an alarm. You can resume the job by using the sendevent command to provide the required response.

## Syntax

This command has the following format:

```
sendevent -E REPLY_RESPONSE {-J job_name | -B group [-I application] | -I application} -
r response [-S autoserv_instance] [-P priority] [-T "time_of_event"]
```

- **-J *job\_name***

Specifies the job to send the event to.

**Limits:** Enter a value of up to 64 alphanumeric characters.

- **-B *group***

Sends the specified event to the specified group of jobs.

- **-I *application***

Sends the event to the jobs that are associated with the specified application.

### NOTE

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.

- **-r *response***

Sends the specified response to the specified job.

- **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

- **-P *priority***

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "*time\_of\_event*"**(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:

- ***MM***  
Specifies the month.  
**Limits:** 01-12
- ***dd***  
Specifies the day.  
**Limits:** 01-31
- ***yyyy***  
Specifies the year.  
**Limits:** 1900-2100
- ***hh***  
Specifies the hour.

**Limits:** 00-23

– ***mm***

Specifies the minutes.

**Limits:** 00-59

– ***ss***

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

**Example: Manually Intervene to Resume a Job that is in the WAIT\_REPLY State**

Suppose that the job that is named `test_i5_tasks` must select between one of several actions at some point during the job execution. These actions are represented by the letters C, D, I, and R. To resume `test_i5_tasks` with the execution of action D, respond to the alarm as follows:

```
sendevent -E REPLY_RESPONSE -r D -j test_i5_tasks
```

**sendevent Command -- Set or Delete a Global Variable**

A *variable* is a symbolic name that represents a value. A *global variable* is a variable with indefinite scope.

Sometimes you need to modify the definition of a global variable to meet one of these conditions or to delete a global variable that is obsolete. You can use the `sendevent` command to modify global variable definitions or to delete global variables.

Only users with specific security permissions can issue this event:

- You can send this event when you have execute access to the specified global variable.
- You can use this event to set new global variables when you have create access to those variables.
- You can use this event to modify existing global variables when you have write access to those variables.
- You can use this event to delete global variables when you have delete access to those variables.

**NOTE**

Security policies can grant users execute and create access to global variables that do not yet exist. For example, you can set a new global variable that is named `TEST_1` when the security policy specifies that you have execute access and create access to all global variables with names that begin with `TEST`.

**Syntax**

This command has the following format:

```
sendevent -E SET_GLOBAL -G global_name=value [-S autoserv_instance] [-T "time_of_event"]
[-C comment]
```

• ***-G global\_name=value***

Specifies the name and value of the global variable you want to define or delete.

**NOTE**

- Certain job attributes support global variables. Global variables that you reference in the job definition resolve when the job executes. For more information about the job attributes that support global variables,

see [Global Variables vAE\\_12](#). Use the `$$ global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

- To delete a global variable from the database, set the value variable to DELETE:

```
sendevent -E SET_GLOBAL -G global_name=DELETE
```

#### Limits:

- For `global_name`, up to 64 of any of the following characters:
  - Alphanumeric characters
  - Period (.), underscore (\_), pound (#), hyphen (-), and space ( ).
- For `value`, up to 100 characters, including spaces.

#### NOTE

AutoSys Workload Automation ignores leading and trailing spaces.

- **-S *autoserv\_instance***

(Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.

- **-T "*time\_of\_event*"**

(Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:

- ***MM***  
Specifies the month.  
**Limits:** 01-12
- ***dd***  
Specifies the day.  
**Limits:** 01-31
- ***yyyy***  
Specifies the year.  
**Limits:** 1900-2100
- ***hh***  
Specifies the hour.  
**Limits:** 00-23
- ***mm***  
Specifies the minutes.  
**Limits:** 00-59
- ***ss***  
(Optional) Specifies the seconds.  
**Limits:** 00-59

#### NOTE

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C *comment***

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

**Example: Set a Global Variable**

Suppose that the job that is named `inventory_check` starts when the global variable named `NEXT_ACTION` evaluates to a value of `RUN_INVENTORY_CHECK` but `NEXT_ACTION` is not defined. To start `inventory_check`, define `NEXT_ACTION` as follows:

```
sendevent -E SET_GLOBAL -G NEXT_ACTION=RUN_INVENTORY_CHECK
```

**Example: Change the Value of a Global Variable**

Suppose that the job that is named `profit_analysis` is programmed to start when the global variable named `NEXT_ACTION` evaluates to a value of `RUN_PROFIT_ANALYSIS` but the value of `NEXT_ACTION` is set to `RUN_INVENTORY_CHECK`. To start `profit_analysis`, modify `NEXT_ACTION` as follows:

```
sendevent -E SET_GLOBAL -G NEXT_ACTION=RUN_PROFIT_ANALYSIS
```

**Example: Delete a Global Variable from the Database**

Suppose that you want the job that is named `profit_analysis` to run only at scheduled times, but it is programmed to also start whenever the global variable named `NEXT_ACTION` evaluates to a value of `RUN_PROFIT_ANALYSIS`. To prevent `profit_analysis` from being triggered by `NEXT_ACTION`, delete `NEXT_ACTION` as follows:

```
sendevent -E SET_GLOBAL -G NEXT_ACTION=DELETE
```

**sendevent Command -- Start Jobs**

Jobs with defined date and time conditions start on the schedule specified by those conditions. The `sendevent` command enables you to run a job with no defined starting conditions, override starting conditions and force a job to run, or restart a multi-step job at a specific step.

**WARNING**

You are authorized to start jobs manually only if the security policy grants you execute access to the jobs. For more information about security policy, see [Security Policy Customization](#).

**Syntax**

This command has the following format:

```
sendevent -E {STARTJOB|RESTARTJOB|FORCE_STARTJOB} {-J job_name [-o step_name] | -
B group [-I application] | -I application} [-S autoserv_instance] [-P priority] [-T
"time_of_event"] [-C comment]
```

- **-E EVENT**  
Sends the specified event.
- **{STARTJOB|RESTARTJOB|FORCE\_STARTJOB}**  
Specifies whether to start, restart, or force start the specified jobs.
- **STARTJOB**  
Starts a job with no defined date and time conditions.
- **RESTARTJOB**  
Resumes a failed Microfocus, Informatica, or SAP BW Process Chain job. The `RESTARTJOB` event is effective only on multistep jobs in the `FAILURE` state.

**NOTE**

- You can restart a failed Informatica workflow or its instance from the point of failure or from the beginning in recovery mode. If recovery mode is not set for a workflow or a session, the workflow starts normally. For example, if a workflow is terminated abnormally, you can restart the workflow.
- Due to a known issue in Informatica, workflow instances cannot be restarted in parallel.

- **FORCE\_STARTJOB**

Forces a job to start when the normal starting conditions for the job are not met.

**NOTE**

- When you force start a job that has resource dependencies and there are no available resources, the scheduler places the job in the RESWAIT state. You cannot force start a job that is in the RESWAIT state. The scheduler automatically resumes running the job after another job that is holding the shared resource releases it back into the resource pool.
- When a job is scheduled to run on an offline machine and you force start it, the scheduler repeatedly restarts the job until it makes contact with the agent or reaches the maximum number of retry attempts specified in the job definition.

**WARNING**

The product does not support concurrent runs of the same job. Errors might cause a job to stop progressing after the agent executes it. To avoid simultaneous executions, do not force start a job while it is in the STARTING or RUNNING state. In this case, change the status of the job. The RESTRICT\_FORCE\_STARTJOB environment variable enables you to force the scheduler to reject force start attempts on jobs that are executing. For information about setting environment variables, see [Environment Variables](#).

**WARNING**

When you force start a job that is in a non-executable state (ON\_HOLD, ON\_ICE), it returns to an executable state, runs, and does not revert to the previous (non-executable) state. Force start a job in a non-executable state only when you need to run the job immediately. To resume the regular run schedule, change the executable status of the job.

- **-J *job\_name*** Specifies the job to send the event to.  
**Limits:** Enter a value of up to 64 alphanumeric characters.
- **-o *step\_name*** (Optional) Restarts the job at the specified step or task.

**WARNING**

This option is required when restarting a Microfocus job, optional when restarting an Informatica job, and invalid when restarting an SAP BW Process Chain job.

- **-B *group*** Sends the specified event to the specified group of jobs.
- **-I *application*** Sends the event to the jobs that are associated with the specified application.

**NOTE**

- You can use the **-B** or **-I** option only when you have execute access to as-control SENDEVENT\_GRPAPP.
- The event is valid only when you specify one of the following combinations of arguments:
  - **-J *job\_name***Sends the event to the specified job. Use this argument alone.
  - **-B *group***Sends the event to all the jobs that are in the specified group.
  - **-I *application***Sends the event to all the jobs that are associated with the specified application.
  - **-B *group* -I *application* | -I *application* -B *group***Sends the event to all jobs within the specified group that are also associated with the specified application.

- **-S *autoserv\_instance*** (Optional) Specifies a AutoSys Workload Automation instance to send the event to. Use a three-character identifier for a defined AutoSys Workload Automation instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine.
- **-P *priority*** (Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "*time\_of\_event*"** (Optional) Specifies the time of the event. Use the "*MM/dd/yyyy hh:mm[:ss]*" format, as follows:
  - ***MM***  
Specifies the month.  
**Limits:** 01-12
  - ***dd***  
Specifies the day.  
**Limits:** 01-31
  - ***yyyy***  
Specifies the year.  
**Limits:** 1900-2100
  - ***hh***  
Specifies the hour.  
**Limits:** 00-23
  - ***mm***  
Specifies the minutes.  
**Limits:** 00-59
  - ***ss***  
(Optional) Specifies the seconds.  
**Limits:** 00-59

#### NOTE

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the [DateFormat](#) parameter in the configuration file. Use the date format that you specified in the configuration file.

- **-C *comment*** (Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.  
**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

#### NOTE

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

### Example: Restart a Job

Suppose that the Microfocus job that is named *income\_analysis* consists of the following steps:

1. revenue\_input
2. expenses\_input
3. net\_calculation

Suppose that *income\_analysis* fails at the second step (*expenses\_input*) and you do not want to restart it from the beginning. Solve the problem that caused the failure and restart the job at the *expenses\_input* step as follows:

```
sendevent -E RESTARTJOB -J net_income_analysis -o expenses_input
```

## sendevent Command -- Suspend a Hadoop Oozie Workflow

Sometimes, to take a corrective action, you might have to suspend a Hadoop Oozie workflow. To suspend a Hadoop Oozie workflow in the RUNNING state, use the sendevent command to issue the SUSPENDJOB event. After you issue the SUSPENDJOB event, the Hadoop Oozie workflow is placed in the SUSPENDED state.

### Syntax

This command has the following format:

```
sendevent -E SUSPENDJOB -j job_name
```

## Monitor and Report on Workload

Monitoring and reporting utilities help you analyze the workflow and detect issues that require attention.

The following are the monitoring and reporting utilities:

### as\_test Command -- Test Utility

The as\_test command is a utility that can run for a specified amount of time, write a message to either stdout or stderr or to both stdout and stderr, and exit with a specific exit code. When the scheduler is running in test mode to agents, command job commands are automatically replaced with the execution of this command. You can use as\_test to test job dependencies and error handling.

### Syntax

This command has the following format:

```
as_test [-t Seconds] [-e Exit Code] [-c Message] [-r Message] [-x] [-?]
```

- **-t *Seconds***  
(Optional) Specifies the number of seconds for the machine to sleep.
- **-e *Exit Code***  
(Optional) Specifies the return code for the job to complete.
- **-c *Message***  
(Optional) Specifies the message to write to the STDOUT file.
- **-r *Message***  
(Optional) Specifies the message to write to the STDERR file.
- **-x**  
(Optional) Displays version information.
- **-?**  
(Optional) Displays help for the command.

### Example: Run the as\_test Command on a Windows machine

This example runs the as\_test command on the Windows machine named winagent. The as\_test command waits for 5 seconds and exits with a return code of 0.

```
insert_job: win_cmd
job_type: CMD
machine: winagent
command: as_test -t 5 -e 0
```



## astail Command -- Display End of File

### Valid on UNIX

The astail command displays the last ten lines of the file specified in the command.

### Syntax

This command has the following format:

```
astail [-f filename] [-x]
```

- **-f *filename***  
(Optional) Defines the name of the file to display.
- **-x**  
(Optional) Returns the version number of the astail command to standard output.

### NOTE

To terminate the command, press Ctrl+C.

### Example: Display the Last ten Lines of the Scheduler Log File

This example displays the last ten lines of the scheduler log file where *instance\_name* is the unique 3-character identifier chosen for this installation.

```
astail -f $AUTOUSER/out/event_demon.instance_name
```

## autoaggr Command -- Report Aggregated Statistics

The autoaggr command is a client component utility that generates reports based on the aggregated job, alarm, and scheduler statistics retrieved from the database.

### Syntax

This command has the following format:

```
autoaggr
 [-d|-h|-m|-w] [-A] [-J] [-S] [-F from_date_time] [-T to_date_time] [-o filename] [-c]
 [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [?]
```

- **-d**  
(Optional) Generates a report on daily aggregated statistics.
- **-h**  
(Optional) Generates a report on hourly aggregated statistics.
- **-m**  
(Optional) Generates a report on monthly aggregated statistics.
- **-w**  
(Optional) Generates a report on weekly aggregated statistics.
- **-A**

- (Optional) Generates a report only on alarm statistics.
- **-J**  
(Optional) Generates a report only on job statistics.
- **-S**  
(Optional) Generates a report only on scheduler statistics.
- **-F "from\_date\_time"**  
(Optional) Specifies the aggregation start date and time in "MM/dd/yyyy hh:mm" format. The report displays the aggregated statistics starting from this specified date and time.

**NOTE**

- If you do not specify the **-F from\_date\_time** parameter, it defaults to the oldest processed event in the database.
- Enclose the start date and time in double quotation marks.
- **-T "to\_date\_time"**  
(Optional) Specifies the aggregation end date and time in "MM/dd/yyyy hh:mm" format. The report displays the aggregated statistics up to this specified date and time.

**NOTE**

- If you do not specify the **-T to\_date\_time** parameter, it defaults to the current time.
- Enclose the end date and time in double quotation marks.
- **-o "filename"**  
(Optional) Defines the path and file name where you want to direct all standard output.

**NOTE**

- You can specify only the file name in this option. If you specify only the file name, the file is written to the current directory.
- For readability purposes, we recommend that you direct the output to a file by using either the **-o** parameter or the command shell redirection command ("**>**" or "**>>**").
- Enclose the file name in double quotation marks.
- **-c**  
(Optional) Generates the output in a comma-separated format.
- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**  
(Optional) Specifies external security credentials as follows:
  - usr username -pw password**: Specifies a username and unencrypted password.
  - usr username -pwx password**: Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

- saml auth\_artifact**: Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the `-usr` option and you do not specify the `-pw` or `-pwx` option, the utility prompts you to specify a password.
- When you specify the `-pw` or `-pwx` option but do not specify the `-usr` option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is `EXTERNAL`, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is `EXTERNAL`.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

#### NOTE

- The **-d**, **-h**, **-m**, and **-w** options are mutually exclusive. You must specify *one* of these options.
- The **-A**, **-J**, and **-S** options can be specified in any combination. If you do not specify the **-A**, **-J**, or **-S** options, the report includes the job, alarm, and scheduler statistics.
- Verify that you issue the `autoaggr` command after the time range (specified using the **-F** and **-T** parameters) in which you want to generate a report has passed. For example, to generate a report on aggregated hourly statistics from 10/13/2011 16:00 to 10/13/2011 16:59:59, issue the `autoaggr` command after 17:00 on 10/13/2011.

#### Example: Generate a Report for Hourly Statistics on Windows

This example generates a report that displays the hourly job, alarm, and scheduler statistics from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the `c:\all.out` file.

```
autoaggr -h -o "c:\all.out" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

#### Example: Generate a Report for Hourly Statistics in Comma-separated Format on Windows

This example generates a report that displays the hourly job, alarm, and scheduler statistics in a comma-separated format from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the `c:\all.csv` file.

```
autoaggr -h -c -o "c:\all.csv" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

#### Example: Generate a Report for Hourly Job Statistics on UNIX

This example generates a report that displays the hourly job statistics from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the `/tmp/job.out` file.

```
autoaggr -h -J -o "/tmp/job.out" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

### Example: Generate a Report for Hourly Job Statistics in Comma-separated Format on UNIX

This example generates a report that displays the hourly job statistics in a comma-separated format from 09/26/2011 00:00 to 10/03/2011 00:00. The output is directed to the /tmp/job.csv file.

```
autoaggr -h -c -J -o "/tmp/job.csv" -F "09/26/2011 00:00" -T "10/03/2011 00:00"
```

### Example: Generate a Report for Daily Alarm Statistics on Windows

This example generates a report that displays the daily alarm statistics from 09/26/2011 00:00 to 09/29/2011 00:00. The output is directed to the c:\alarm.out file.

```
autoaggr -d -A -o "c:\alarm.out" -F "09/26/2011 00:00" -T "09/29/2011 00:00"
```

### Statistics Reported by the autoaggr Command

The autoaggr command generates reports on the aggregated job, alarm, and scheduler statistics.

#### NOTE

- The time column in the report represents the hourly time after the time hack. For example, if the reported time is 10:00:00, it represents the aggregated statistics generated from the activity that occurred from 10:00:00 to 10:59:59.
- You can also aggregate statistics using the Compliance Application, but this application collects only certain job run statistics.

The reports display job statistics that show the total number of jobs in STARTING, RUNNING, SUCCESS, BYPASS, FAILURE, TERMINATED, ON\_HOLD, OFF\_HOLD, ON\_ICE, OFF\_ICE, ON\_NOEXEC, OFF\_NOEXEC, INACTIVE, and RESTART status.

The reports display the following alarm statistics:

- **AVERAGE RESPONSE TIME**  
Displays the average time that is taken to respond to an alarm.
- **TOTAL ALARMS**  
Displays the total number of alarms irrespective of alarm status.

#### NOTE

The statistics that are displayed in the report includes the count of all the alarms that AutoSys Workload Automation generates.

- **UNANSWERED**  
Displays the total number of alarms that are open. This does not include alarms that are acknowledged or closed.
- **JOB FAILURE**  
Displays the total number of JOBFailure alarms due to jobs that are in the FAILURE state.
- **JOB TERMINATED**  
Displays the total number of JOBTerminated alarms due to jobs that are in the TERMINATED state.
- **START FAILURE**  
Displays the total number of STARTJOBFAIL alarms.
- **MAX RETRY**  
Displays the total number of MAX\_RETRY alarms.
- **MAX RUNTIME**  
Displays the total number of MAXRUNALARM alarms.
- **MIN RUNTIME**  
Displays the total number of MINRUNALARM alarms.
- **DATABASE ROLLOVER**

- Displays the total number of DB\_ROLLOVER alarms.
- **SCHEDULER FAILOVER**  
Displays the total number of EP\_ROLLOVER alarms.
- **SCHEDULER SHUTDOWN**  
Displays the total number of EP\_SHUTDOWN alarms.

The reports display the following scheduler statistics:

- **JOB RUNS**  
Displays the total number of job runs.
- **JOB FAILURES**  
Displays the total number of jobs that are in FAILURE or TERMINATED status.
- **JOB FORCE START**  
Displays the total number of jobs that were force started.
- **KILL JOB**  
Displays the total number of jobs that are killed using the KILLJOB event.
- **SERVICE DESK REQUESTS**  
Displays the total number of jobs for which service desk issues were opened.
- **TOTAL EVENTS**  
Displays the total number of events (including internal events) that AutoSys Workload Automation generates.
- **AVERAGE LATENCY**  
Displays the average latency (in seconds) in processing events, which is the average processing time that is required for all the events. The latency for an event is calculated by subtracting the time that the event was inserted into the database from the time it was picked up and processed by the scheduler. The average latency value is calculated by adding all the event latencies and then dividing the total latency by the number of events.
- **MAX LATENCY**  
Displays the highest latency (in seconds) for processed events within a given time range. The latency for an event is calculated by subtracting the time that the event was inserted into the database from the time it was picked up and processed by the scheduler.
- **AVERAGE LAG TIME**  
Displays the average lag time (in seconds) in processing events, which is the average processing time that is required for all the events. The lag time for an event is calculated by subtracting the time that the internal processing starts for the event from the time the scheduler completes processing it. Unlike average latency, the average lag time does not include the time that is required to fetch the event from the database. The average lag time value is calculated by adding all the event lag times and then dividing the total lag time by the number of events.
- **MAX LAG TIME**  
Displays the highest lag time (in seconds) for processed events within a given time range. The lag time for an event is calculated by subtracting the time that the internal processing starts for the event from the time the scheduler completes processing it. Unlike max latency, the max lag time does not include the time that is required to fetch the event from the database.

## autorep Command -- Report Job, Machine, and Variable Information

The autorep command is a client component utility that generates reports about objects such as jobs, machines, and global variables that are currently defined in the database. The autorep command retrieves data from the database to formulate the reports. You can use autorep to do the following:

- Display a summary of all currently defined jobs
- Display current machine load information
- Display connection profile information
- List the information of all job events that arrive during the span of the most recent run of a job or for a specified job run.
- Back up job definitions by extracting definitions and saving them to an output file
- List objects (for example, blobs, external instances, job, user-defined job types, global variables, machines, and resources) and their definitions.

**NOTE**

You can only view reports for the objects that you have read access to.

**Syntax**

This command has the following format:

```
autorep
 {-J job_name|-Y userdefined_jobtype|-M machine|-C connectionprofilename|-
 B group|-I application|-G global_name|-Q {ALL|job_name}|-X ext_instance|-
 V virtual_resource_name|-z {ALL|globalblob_name[*|%|?]} [-type blob_type]|-jb job_name
 [-type blob_type] [-name blob_name]}
 [{-d | -s | -q | -o overnum}] [-p] [-n] [-w] [-t] [-R run_num] [-L print_level] [-a] [-
 f blobfilepath]
 [{-usr username {-pw|pwd} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-J *job\_name***  
(Optional) Defines the job to report on. This value allows wildcard characters.

**NOTE**

- You do not need to precede the job name that contains a colon with a backslash.
- In the autorep report, the Time column reflects the client time zone and the ProcessTime column reflects the server time zone.

- **-M *machine***  
(Optional) Specifies the machine to report on. To generate a report about all machines, specify - M ALL.
- **-C *connectionprofilen***  
(Optional) Specifies the connection profile to report on. To generate a report for all connection profiles, specify -C ALL.
- **-G *global\_name***  
(Optional) Generates a global variable report, where *global\_name* is the name of a global variable set with the sendevent command. This value allows wildcard characters. To generate a report about all global variables, specify -G ALL.
- **-Q {ALL|*job\_name*}**  
(Optional) Generates a report on jobs that are queued due to insufficient load balancing units, machine resources, or global resources. To generate a report on all the jobs waiting in a queue, specify -Q ALL. To generate a report for all jobs waiting on the same queues as a specific queued job, specify -Q *job\_name*.
- **-s**  
(Optional) Generates a summary report. If you do not specify a report type to generate, AutoSys Workload Automation generates a summary report by default.  
This report contains the following information:

- For external instances, the summary report contains the name, type, machines, and port.
- For globs, the summary report contains the glob name and file associated with the glob.
- For jobs, the summary report contains the job name, start date and time, end date and time, current status, run number, and priority.  
When you specify **-r runnum**, the report contains information for the specified run; when you omit **-r runnum**, the report contains information for the most recent run.
- For machines, the summary report contains the machine name, maximum load, current load, factor, type, and status.
- For connection profiles, the summary report contains the connection profile name and type.
- For user-defined job types, the summary report contains the job type, command, and description.
- For virtual resources, the summary report contains the resource name, type, machine, and defined and available units.

**NOTE**

This parameter is ignored for global variable reports.

- **-n**  
(Optional) Reports truncated names.

**NOTE**

By default, the autorep command displays the full, non-truncated names of objects such as jobs, machines, resources, variables, and blobs. When you specify **-n**, the report displays narrow columns, which might truncate names.

- **-w**  
(Optional) Reports the long form of job conditions.  
By default, the autorep command displays the short form of conditions, for example, **condition: s(JobA)**. When you specify **-w**, the report displays the long form of conditions, for example, **condition: success(JobA)**.

**NOTE**

This option is valid only when the **-q** option is also specified.

- **-d**  
(Optional) Generates a detail report.  
This report contains the following information:
  - For jobs, the report covers all events from the most recent run of the specified jobs (the default). The report contains the following data for each event: status, the date and time of the event, try number, event state, process date and time, machine, whether the job ran with an override, and the override number.
  - For machines, the report contains the following data for each machine: machine name, maximum load, current load, factor, type, and status. For jobs currently running on each machine, the report also contains the following data: job name, machine, status, load, and priority.
  - For user-defined job types, the report contains the job type and the jobs that reference them.
  - For virtual resources, the report contains the resource name, type, machine, defined and available units, and the jobs currently using the resource.

**NOTE**

- This option does not apply to global variable reports, blob reports, or glob reports. The utility ignores this option if you specify it with the **-G**, **-f**, or **-z** parameter.
- The detailed report might also include events from another run of the job that arrive during the span of the requested job run.
- The **STATE\_CHANGE** events that are associated with the job entering and leaving the **RESWAIT** or **QUE\_WAIT** state are displayed in the detailed report of the previous run of the job.

- **-q**  
(Optional) Generates a query report, which contains the current JIL definition for the object.

**NOTE**

This parameter is ignored for global variable reports.

- **-o overnum**

(Optional) Generates an override report, which contains the overrides for the jobs that are specified in **-J job\_name**, **-B group\_name**, or **-I application\_name** parameters that match the override number specified by *overnum*. To report data for the most current override, set *overnum* to **0**.

**NOTE**

To use this parameter, you must also specify the **-J job\_name**, **-B group\_name**, or **-I application\_name** parameter.

- **-p**

(Optional) Generates a report that displays all the active agent plug-ins and advanced integrations on the machine and the agent release and build information. The report only shows the data for real machines.

**NOTE**

The agent release and build information is displayed only after the scheduler communicates with the agent.

- **-R run\_num**

(Optional) Generates the report for the job run specified in *run\_num*.

To specify a previous run relative to the most recent run, use the minus character (-). For example, when you specify **-2**, AutoSys Workload Automation generates a report for the job run two runs back.

If you omit this parameter or specify 0 for the *run\_num*, the report contains information about the most recent job run.

**NOTE**

- This parameter is only valid with the **-s** and **-d** parameters for job reporting.
- The detailed report might also include events from another run of the job that arrive during the span of the requested job run.

- **-L print\_level**

(Optional) Specifies the number of levels into the box job hierarchy about which to report. This value can be any number. For example, when you specify **-L 2**, the report contains data for the specified job (a box) and the top two levels of jobs in the box. To report on only the topmost level (that is, the box), specify **-L 0**.

**Default:** **-L** (report on all levels in the box)

**NOTE**

**-L** option is only applicable to boxes. However, if you specify the **-B**, **-I**, or both arguments with the **-L** option, autorep reports on all jobs matching the criteria, including jobs that are not contained in boxes. For the Box jobs reported, autorep applies the **-L** option to the jobs within each box.

- **-t**

(Optional) Includes the time zone (if one is specified in the job definition) in the report. The time zone appears in parentheses beneath the job name, and the Time column of the report is based on the time zone specified in the job.

- **-B group**

(Optional) Defines the group for which to generate a report.

**Limits:** When **-I application** is also defined, the command generates the report only for jobs that match both **-I application** and **-B group**. To search for jobs or boxes without group names, you can specify a blank group name by enclosing a blank with quotes, for example **-B ""**.

**NOTE**

- The **-J job\_name** and **-B group** or **-I application** are mutually exclusive. That is, you cannot specify the **-J job\_name** with **-B group** or **-I application** and conversely.
- This value allows wildcard characters.

- **-I application**



(Optional) Defines the application for which to generate a report.

**Limits:** When **-B group** is also defined, the command generates the report only for jobs that match both **-I application** and **-B group**. To search for jobs or boxes without application names, you can specify a blank application name by enclosing a blank with quotes, for example **-I ""**.

#### NOTE

- The **-J job\_name** and **-B group** or **-I application** are mutually exclusive. That is, you cannot specify the **-J job\_name** with **-B group** or **-I application** and conversely.
- This value allows wildcard characters.

- **-V virtual\_resource\_name**

(Optional) Specifies the virtual resource definition that you want to generate a report for.

#### NOTE

- You can use wildcard characters.
- If you specify **-s**, the command generates a summary report that displays the resource name, the machine it is defined for (if any), the total amount, and the amount currently available.
- If specify **-d [-M machine]**, the command generates a summary report and detail report. The detail report displays information about the machine that is using the resource or all machines that are currently using the resource if ALL is specified in the **-M** option.

- **-X ext\_instance**

(Optional) Specifies the external instance that you want to generate a report for.

- **-Y userdefined\_jobtype**

(Optional) Defines the user-defined job type that you want to generate a report for.

**Limits:** Specify **-Y ALL** to report on all user-defined job types.

#### NOTE

- When **-q** is also defined, the command generates a query report.
- When **-d** option is also defined, the command generates a detailed report.

- **-z {ALL|globalblob\_name[\*|%|?]} [-type blob\_type]**

(Optional) Specifies the global blobs on which the command reports.

- **ALL**

Specifies that the command generates a report on all global blobs that are defined within the instance.

- **globalblob\_name[\*|%|?]**

Specifies that the command generates a report on the specified global blobs only.

To generate a report on a single global blob, replace this variable with the name of that blob. Ensure that the value that you enter is the name of a global blob that is defined within the instance. You can use the **insert\_glob jil** subcommand to define a global blob.

To generate a report on all global blobs with names that contain a certain string of characters, replace the variable with a value that contains the string and *one* of the following wildcard characters:

- \*
- %
- ?

You can use any of the wildcard characters and can place the character anywhere within the value that you enter for this variable. **%** or **\*** match any number of characters and **?** matches one character only.

#### Examples:

- To generate a report on the global blob named MYBLOB1, enter the following command:  
`autorep -z MYBLOB1`
- To generate a report on all global blobs with names that contain MY, enter the following command:  
`autorep -z %MY%`
- To generate a report on all global blobs that have names beginning with MY, enter the following command:

```
autorep -z MY*
```

- To generate a report on global blobs that match the pattern M?BLOB1 (such as MYBLOB1, MRBLOB1, M1BLOB1), enter the following command:

```
autorep -z M?BLOB1
```

- **-type blob\_type**  
Specifies the type of glob that you want to generate a report for.
- **generic**  
Specifies that it is a generic glob.
- **emailtemplate**  
Specifies that it is an emailtemplate glob that includes the email template.

#### NOTE

To use the *-type* option, you must also specify *-z globalblob\_name* or *-jb job\_name -name blob\_name*.

- **-jb job\_name [-type blob\_type] [-name blob\_name]**  
(Optional) Defines the outputpayload or outputpayloadvar job blobs to report on.
- **-type blob\_type**  
Specifies the type of job blob that you want to generate a report for.

#### NOTE

- To use the *-type* option, you must also specify *-z globalblob\_name* or *-jb job\_name -name blob\_name*.
- A report for the outputpayload or outputpayloadvar job blobs is generated only if there was a successful job run of the current job definition.

- **outputpayloadvar**  
Specifies that the blob is an output payload var job blob.
- **outputpayload**  
Specifies that the blob is an output payload job blob.
- **-name blob\_name**  
Specifies the outputpayloadvar job blob that you want to generate the report for.

#### NOTE

To use the *-name blob\_name* option, you must also specify *-jb job\_name* and *-type outputpayloadvar*.

- **-a**  
(Optional) Downloads the appropriate globs in text file format. This option scans and converts all new line characters in the binary large object (blob) data to conform to the text file format of the operating system. If not specified, the glob is saved without modification to a file as binary data.

#### NOTE

To use the *-a* option, you must also specify *-z globalblob\_name*, *-J job\_name* with *-q*, or *-jb job\_name*.

- **-f blobfilepath**  
(Optional) Defines the directory to store files containing blob data. When you do not specify the directory to store blob files, AutoSys Workload Automation places the blob files into the designated temporary path for the operating system by default. On Windows, the default path is determined by the TEMP environment variable, whereas on UNIX the default path is /tmp.

#### NOTE

To use the *-f* option, you must also specify *-z globalblob\_name*, *-J job\_name* with *-q*, or *-jb job\_name*.

**{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password:** Specifies a username and unencrypted password.

**-usr username -pwx password:** Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

### **autorep Command Usage Notes**

Consider the following points when using the autorep command:

Read access to objects if CA EEM is enabled:

- When you invoke autorep, AutoSys Workload Automation verifies that you have read access to the objects (jobs, machines, global variables, and so on) you are requesting reports for. The list access for objects that are reported by autorep can be denied. You can only view reports for the objects in the list that you have read access to.
- There is no visual indication that a report list is incomplete. The autorep command does not display a security warning if it cannot obtain read access to one of the objects that match the input name qualifier. It also does not warn that access to view the entire list is denied.

Archived data:

Data archived with archive\_events does not appear in the reports.

Exit codes:

- The exit code and output of autorep vary depending on whether a report is generated for a single object or multiple objects.
- If the name of an explicit object is specified and the object does not exist or the read access to the object is denied, autorep returns an error message and a non-zero exit code value.
- If an object name with a wildcard is specified or the ALL qualifier is used and no reports for objects can be generated, autorep returns zero as the exit code value to indicate success.

Report format:

When autorep reports nested jobs, the subordinate jobs are indented to illustrate the hierarchy.

### **Types of autorep Reports**

You can use the autorep command to generate the following types of reports:

- **Job Summary**
  - When you use the -s parameter to request a summary, autorep returns the current status (if the job is still running) or the status for a previous run. The summary report reflects the last status posted to the database.
- **Job Detail**
  - When you use the -d parameter to request a detail report, autorep returns all events in the database ujo\_event table for the job. If the job is running, autorep returns events for the current run. If the job is not running, it returns events for the most recent run or for the specified run.
- **Machine Definition and Status in the Database**

When you use the -M parameter to request a machine report, autorep returns the current load, defined maximum load and factor, and current status for all machines or a specific machine defined in the database.

The valid machine statuses are:

  - **Online**  
Indicates that the machine is available and accepting jobs to run.
  - **Offline**  
Indicates that an operator used sendevent to take the machine offline.
  - **Missing**  
Indicates that, during proactive monitoring of machines, the scheduler determined that the machine is not available (for example, due to a crash).
  - **Unqualified**  
Indicates that the scheduler is attempting to qualify the status of an agent before switching the machine from an online to missing status.
  - **Empty**  
Indicates that a virtual machine or real machine pool does not contain any component machines. Jobs that are scheduled to machines in this status will not run.
  - **Blocked**  
Indicates that the agent blocks communication. All jobs that are scheduled to start on the agent are put in a PEND\_MACH status.
- **Job Overrides**

When you use the -o parameter to request an override report, autorep returns the overridden job definition fields and user, time, and run information for a specified override, based on job name and override number.

### **Columns in an autorep Report**

The columns in an autorep report vary with the type of report that is requested and might include the following:

- **Last Start**

Reports the date and time at which the specified job most recently started.

- **Last End**

Reports the date and time at which the specified job most recently ended.

- **ST/Ex**

Reports the completion status of the most recent run of the specified job or if the job is still running it reports the current status. If the job is in the ON\_NOEXEC status, autorep also reports the execution mode of the job as NE. Depending on the current status of the job, the ST/Ex column reports the status as IN/NE, AC/NE, RU/NE (for box jobs only), or SU/NE.

**NOTE**

When you use the -Q option, the report displays two ST columns. The first ST column appears next to the Job Name column and reports the status of the specified queued job. The second ST column appears next to the Mach Queue column and reports the current status of the machine.

The following table lists the abbreviations that are used in the ST column of the autorep report, and gives the status associated with each abbreviation:

Abbreviation	Status (Event)
AC	ACTIVATED
FA	FAILURE
IN	INACTIVE
NE	ON_NOEXEC
OH	ON_HOLD
OI	ON_ICE
PE	PEND_MACH
QU	QUE_WAIT
RE	RESTART
RU	RUNNING
RW	RESWAIT
ST	STARTING
SU	SUCCESS
SP	SUSPENDED
TE	TERMINATED

- **Run**

Reports the job run number and number of tries, separated by a slash (/).

- **Pri/Xit**

Reports the priority/exit code associated with the specified job. If the job is in QUE\_WAIT status, this column shows the priority in the queue; if the job completed, this column shows the exit code.

- **Status/[Event]**

Reports the status for the current run of the specified job.

**NOTE**

The STATE\_CHANGE events that are associated with the job entering and leaving the RESWAIT or QUE\_WAIT state are displayed in the detailed report of the previous run of the job.

- **Time**

Reports the date and time at which the scheduler or agent generated the CHANGE\_STATUS event for the specified job.

**NOTE**

The Time column reflects the client time zone. However, if you want the Time column to reflect the server time zone, set the TZ environment variable to the value that is set on the server.

- **Ntry**  
Reports the number of restart attempts for the event listed in the Status/[Event] column.
- **ES**  
Reports the processing state of the status (event).  
The following table lists the abbreviations that are used in the ES (event state) column of the autorep report, and gives the status associated with each abbreviation:

Abbreviation	Event State
ER	Error
PD	Processed
PG	Processing
UP	Unprocessed
US	Unsent

- **Process Time**  
Reports the date and time at which the scheduler processed the CHANGE\_STATUS event.

**NOTE**

The ProcessTime column reflects the server time zone.

- **Machine**  
Reports the name of the machine on which the specified job ran or is running.
- **Max Load**  
Reports the max\_load value defined in the specified machine's definition.
- **Current Load**  
Reports the current load on the specified machine.
- **Factor**  
Reports the factor value defined in the specified machine's definition. This is the value that is multiplied by a real machine's available CPU cycles to determine the relative available CPU cycles for the machine.
- **O/S**  
Reports the operating system in use on the specified machine.
- **Load**  
Reports the relative amount of processing power a job consumes on the specified machine.
- **Job Name** Reports the name of the job.
- **Priority**  
Reports the queue priority of a job on the specified machine.
- **Mach Queue**  
Reports the name of the load balancing queue on which the QUE\_WAIT job is queued.
- **Curr/Max**  
Reports the current load on the specified machine and the max\_load value defined in the specified machine's definition.
- **Resource**  
Reports the name of the resource queue on which the RESWAIT job is queued.
- **Req/Avl**  
Reports the required and available resources for the job to run.
- **Status (machine reports)**

Reports the current status of the machine. Options are the following:

- **Empty**  
Indicates that a virtual machine or real machine pool does not contain any component machines. Jobs that are scheduled to machines in this status will not run.
- **Missing**  
Indicates that the scheduler has verified that the machine is not responding and has automatically removed it from service. The machine will not accept jobs to run.
- **Offline**  
Indicates that the machine has been manually removed from service and will not accept jobs to run.
- **Online**  
Indicates that the machine is available and accepting jobs to run.
- **Unqualified**  
Indicates that the scheduler is attempting to qualify the status of an agent before switching the machine from an online to missing status. The machine will not accept jobs to run.
- **Blocked**  
Indicates that the agent blocks communication. All jobs that are scheduled to start on the agent are put in a PEND\_MACH status.
- **Blob Name**  
Reports the name of the outputpayloadvar job blob.
- **Blob Type**  
Reports the type of job blob or glob.
- **File Name**  
Reports the directory where the job blob is located.

### **Examples Generate Reports Using autorep**

The following examples generate reports using the autorep command:

#### **Example: Generate a Summary Report**

This example generates a summary report for a run of the Nightly\_Download box job:

```
autorep -J Nightly_Download
```

The resulting report might resemble the following:

Job Name Pri/Xit	Last Start	Last End	ST/Ex	Run/Ntry
Nightly_Download	05/21/2010 11:27:31	05/21/2010 11:27:33	SU	197/1 0
Watch_4_file	05/21/2010 11:27:32	05/21/2010 11:27:33	SU	197/1 0
filter_data	05/21/2010 11:27:32	05/21/2010 11:27:33	SU	197/1 0
update_DBMS	05/21/2010 11:27:32	05/21/2010 11:27:33	SU	197/1 0

#### **Example: Generate a Summary Report**

This example generates a summary report for a run of the nebox box job, where nebox\_b1 job is in the ON\_NOEXEC status. The report displays the current status (ST) and execution mode (Ex) of the jobs.

```
autorep -J nebox
```

The resulting report might resemble the following:

Job Name	Last Start	Last End	ST/Ex	Run/Ntry	Pri/Xit
nebox	09/07/2016 10:17:33	-----	RU	1620/1	
nebox_b1	09/07/2016 10:17:34	-----	RU/NE	1620/1	
nebox_b1_b1	-----	-----	AC/NE	1620/0	
nebox_b1_b1_j1	-----	-----	AC/NE	1620/0	
nebox_b1_j1	09/07/2016 10:17:35	09/07/2016 10:17:35	SU/NE	1620/1	0
nebox_j1	09/07/2016 10:17:34	09/07/2016 10:17:34	SU	1620/1	0

### Example: Generate a Job Detail Report

This example generates a detail report that shows each event and status for each job:

```
autorep -J Nightly_Download -d
```

The resulting report might resemble the following:

Job Name	Last Start	Last End	ST/Ex	Run/Ntry	Pri/Xit
Nightly_Download	05/21/2010 11:27:31	05/21/2010 11:27:33	SU	197/1	0

Status/[Event]	Time	Ntry	ES	ProcessTime	Machine
RUNNING	05/21/2010 11:27:31	1	PD	05/21/2010 11:27:31	
SUCCESS	05/21/2010 11:27:33	1	PD	05/21/2010 11:27:34	

Watch_4_file	05/21/2010 11:27:32	05/21/2010 11:27:33	SU		
197/1 0					

Status/[Event]	Time	Ntry	ES	ProcessTime	Machine
STARTING	05/21/2010 11:27:31	1	PD	05/21/2010 11:27:31	localhost
RUNNING	05/21/2010 11:27:32	1	PD	05/21/2010 11:27:32	localhost
<Executing at WA_AGENT>					
SUCCESS	05/21/2010 11:27:33	1	PD	05/21/2010 11:27:33	localhost

filter_data	05/21/2010 11:27:32	05/21/2010 11:27:33	SU		
197/1 0					

Status/[Event]	Time	Ntry	ES	ProcessTime	Machine
----------------	------	------	----	-------------	---------



```

STARTING 05/21/2010 11:27:31 1 PD 05/21/2010 11:27:31 localhost
RUNNING 05/21/2010 11:27:32 1 PD 05/21/2010 11:27:32 localhost
<Executing at WA_AGENT>
SUCCESS 05/21/2010 11:27:33 1 PD 05/21/2010 11:27:33 localhost

```

```

update_DBMS 05/21/2010 11:27:32 05/21/2010 11:27:33 SU
197/1 0

```

```

Status/[Event] Time Ntry ES ProcessTime Machine

STARTING 05/21/2010 11:27:31 1 PD 05/21/2010 11:27:31 localhost
RUNNING 05/21/2010 11:27:32 1 PD 05/21/2010 11:27:32 localhost
<Executing at WA_AGENT>
SUCCESS 05/21/2010 11:27:33 1 PD 05/21/2010 11:27:33 localhost

```

In this example, Nightly\_Download is a box job. Therefore, it does not enter the STARTING state, but goes directly to the RUNNING state. Box jobs do not list an associated machine, because the jobs in boxes might execute on different machines.

#### Example: Generate a Detailed Report for a Previous Job Run

This example generates a detailed report for the previous run of a job:

```
autorep -J RunData -d -r -1
```

The resulting report might resemble the following:

Job Name Xit	Last Start	Last End	ST/Ex	Run/Ntry	Pri/
RunData	05/21/2010 11:41:10	05/21/2010 11:41:10	FA	198/1	20007

```

Status/[Event] Time Ntry ES ProcessTime Machine

STARTING 05/21/2010 11:41:09 1 PD 05/21/2010 11:41:09 localhost
RUNNING 05/21/2010 11:41:10 1 PD 05/21/2010 11:41:10 localhost
FAILURE 05/21/2010 11:41:10 1 PD 05/21/2010 11:41:10
<Command file not found>
[*** ALARM ***]
JOBFAILURE 05/21/2010 11:41:10 1 PD 05/21/2010 11:41:11 localhost
[STARTJOB] 05/21/2010 11:41:41 0 PD 05/21/2010 11:41:41

```

#### Example: Generate a Machine Summary Report

This example generates a report that lists all machines defined on the data server:

```
autorep -M ALL
```

The resulting report might resemble the following:

Machine Name	Max Load	Current Load	Factor	O/S	Status
london	100	0	1.00	Unix	Online
berlin	90	0	0.90	NT	Online
v_italy.rome	0	0	0.00	Unix	Online
v_italy.venice	0	0	0.00	Unix	Online
v_italy.venice	100	0	1.00	NT	Online
v_france.cannes	75	0	1.00	NT	Online

#### Example: Generate an Override Report for the Current Override

This example generates an override report that shows the current one-time override in effect for the specified job:

```
autorep -J RunData -o 0
```

The resulting report might resemble the following:

```
/* ----- over ----- */
override_job: RunData
/* Over-Ride #2 Set by User: roger@venice on [07/28/1997 16:13:59] */
/* Over-Ride CURRENTLY IN EFFECT.*/
command: /bin/rundata2
```

#### Example: Generate an Override Report for a Previous Override

This example generates an override report that shows the first one-time override for the specified job:

```
autorep -J RunData -o 1
```

The resulting report might resemble the following:

```
/* ----- over ----- */
override_job: RunData
/* Over-Ride #1 Set by User: roger@venice on [07/25/1997 18:23:45] */
/* Was RUN on run_num=175, Started on: 07/25 18:24:01 */
command: /bin/rundata1
```

#### Example: Generate a Report for Jobs in a Group

This example generates a report of information for all jobs in the HumanResources group:

```
autorep -B HumanResources
```

The resulting report might resemble the following:

Job Name Xit	Last Start	Last End	ST/Ex	Run/Ntry	Pri/
hrbox	-----	-----	IN	0/0	
hrjob1	-----	-----	IN	0/0	
hrjob2	-----	-----	IN	0/0	

**Example: Generate a Report for Jobs in an Application**

This example generates a report of information for all jobs in the YearEndHR application:

```
autorep -I YearEndHR
```

The resulting report might resemble the following:

Job Name Xit	Last Start	Last End	ST/Ex	Run/Ntry	Pri/
yearendbox	-----	-----	IN	0/0	
yearendjob1	-----	-----	IN	0/0	

**Example: Generate a Report for a Specific Global Variable**

This example generates a report that lists the value of the global variable DAY:

```
autorep -G DAY
```

The resulting report might resemble the following:

```
Wednesday
```

**Example: Generate a Report for ALL Global Variables**

This example generates a report that lists the values of all global variables:

```
autorep -G ALL
```

The resulting report might resemble the following:

Global Name	Value	Last Changed
DAY	Wednesday	07/21/2005 08:15:14
AUDIT_DIR	/usr/audit	07/21/2005 08:15:12
DINNER_TIME	18:30	07/21/2005 08:15:13
MAX_VAL	2048	07/21/2005 08:15:13

**Example: Generate a Report for a Specific Glob**

This example generates a report that downloads the contents of the glob MYGLOB to location *download\_path* as binary data:

```
autorep -z MYGLOB -f download_path
```

The resulting report might resemble the following:

Glob Name	File Name
MYGLOB	<download_path>/MYGLOB

This example generates a report that downloads the contents of the glob MYGLOB to location *download\_path* as text data:

```
autorep -z MYGLOB -f download_path -a
```

The resulting report might resemble the following:

Glob Name	File Name
MYGLOB	<download_path>/MYGLOB.txt

### Example: Generate a Report for ALL Globbs

This example generates a report that downloads the contents of all globs to location *download\_path* as binary data:

```
autorep -z ALL -f download_path
```

The resulting report might resemble the following:

Glob Name	File Name
MYGLOB	<download_path>/MYGLOB
REPORT_CHART	<download_path>/REPORT_CHART
ARCHIVED_DATA	<download_path>/ARCHIVED_DATA
JOB_SNAPSHOT	<download_path>/JOB_SNAPSHOT

This example generates a report that downloads the contents of all globs to location *download\_path* as text data:

```
autorep -z ALL -f download_path -a
```

The resulting report might resemble the following:

Glob Name	File Name
MYGLOB	<download_path>/MYGLOB.txt
REPORT_CHART	<download_path>/REPORT_CHART.txt
ARCHIVED_DATA	<download_path>/ARCHIVED_DATA.txt
JOB_SNAPSHOT	<download_path>/JOB_SNAPSHOT.txt

### Example: Generate a Report for a Job Blob

This example generates a report for the job blob named testHTTPJob:

```
autorep -jb testHTTPJob
```

The resulting report might resemble the following:

Blob Name	Blob Type	File Name
	outputpayload	C:\Users\ADMINI~1\AppData\Local\Temp
\outputpayload		
var1	outputpayloadvar	C:\Users\ADMINI~1\AppData\Local\Temp
\var1		
var2	outputpayloadvar	C:\Users\ADMINI~1\AppData\Local\Temp
\var2		

**Note:** The blob name is displayed in the report only if the blob type is outputpayloadvar.

### Example: Generate a Report for a Specific Job Blob

This example generates a report for the job blob named outputpayloadvar\_jobblob that is located in the C:\Users\ADMINI~1\AppData\Local\Temp\2 directory and is of type outputpayloadvar:

```
autorep -jb jobblob -name outputpayloadvar_jobblob -type ouputpayloadvar
```

The resulting report might resemble the following:

Blob Name	Blob Type	File Name
outputpayloadvar_jobblob	outputpayloadvar	C:\Users\ADMINI~1\AppData\Local\Temp\2\outputpayloadvar_jobblob

### Example: Generate a Summary Report of Jobs in a Box

This example generates a summary report about the top two levels of jobs in the box job Box3:

```
autorep - J abox - L 2
```

The resulting report might resemble the following:

Job Name Xit	Last Start	Last End	ST/Ex	Run/Ntry	Pri/
abox	-----	-----	IN	0/0	
abox1	-----	-----	IN	0/0	
abox1job1	-----	-----	IN	0/0	
abox1job2	-----	-----	IN	0/0	
abox2	-----	-----	IN	0/0	

### Example: Generate a Detailed Report that Includes the Time Zone Specification

This example includes the time zone specification in a detailed report for the last run of the job MyJob, enter:

```
autorep -J MyJob -d -t
```

When you use the -t option, AutoSys Workload Automation translates the time in the Time column to the time zone specified in the job definition. The time reported in the ProcessTime column reflects the server time zone and is not affected by the -t option.

The resulting report might resemble the following:

Job Name Pri/Xit	Last Start	Last End	ST/Ex	Run/Ntry
MyJob	05/21/2010 11:27:32	05/21/2010 11:27:33	SU	197/1 0

(EST5EDT)

Status/[Event]	Time	Ntry	ES	ProcessTime	Machine
STARTING	05/21/2010 11:27:31	1	PD	05/21/2010 11:27:31	localhost
RUNNING	05/21/2010 11:27:32	1	PD	05/21/2010 11:27:32	localhost
<Executing at WA_AGENT>					

---

```
SUCCESS 05/21/2010 11:27:33 1 PD 05/21/2010 11:27:33 localhost
```

### Example: Extract Job Definitions in Script Format and Output to a File

This example uses the autorep command to extract job definitions and direct the output to a file:

```
autorep -J ALL -q > dump_file
```

- **dump\_file**

Identifies the file to which to write output.

The product formats the resulting output exactly as a job definition script like the following:

```
insert_job: test_job
job_type: c
command: sleep 60
machine: juno
#owner: jerry@ca
permission: gx,ge,wx
alarm_if_fail: 1
```

#### NOTE

The owner field of each job definition is commented out (#) unless the EDIT superuser ran the autorep command to generate this report. Only the EDIT superuser can change the owner field.

If the job has one or more input blobs that are tied to it, in addition to the job definition, the autorep command extracts each of the job blob definitions. In this case, the product formats the resulting output like the following:

```
insert_job: test_job_with_blob
job_type: c
command: sleep 60
machine: juno
owner: jerry@juno
permission:
std_in_file: $$blobt
alarm_if_fail: 1
/* -- test_job_with_blob:insert_blob #1 -- */
insert_blob: test_job_with_blob
blob_input: <auto_blobt>multi-lined text data for job blob 1
</auto_blobt>
/* -- test_job_with_blob:insert_blob #2 -- */
insert_blob: test_job_with_blob
blob_input: <auto_blobt> multi-lined text data for job blob 2
</auto_blobt>
/* -- test_job_with_blob:insert_blob #3 -- */
insert_blob: test_job_with_blob
blob_input: <auto_blobt> multi-lined text data for job blob 3
</auto_blobt>
```

Alternatively, you can specify a location to download the blobs using the -f parameter in the following way:

```
autorep -J ALL -q -f download_path > dump_file
```

- **download\_path**

Identifies the path to download any job input blobs.

- **dump\_file**

Identifies the file to which to write output.

In this case, the product formats the resulting output like the following:

```
insert_job: test_job_with_blob job_type: c
command: sleep 60
machine: juno
owner: jerry@juno
permission:
std_in_file: $$blobt
alarm_if_fail: 1
/* -- test_job_with_blob:insert_blob #1 -- */
insert_blob: test_job_with_blob
blob_file: <download_path>/test_job_with_blob_1.txt
/* -- test_job_with_blob:insert_blob #2 -- */
insert_blob: test_job_with_blob
blob_file: <download_path>/test_job_with_blob_2.txt
/* -- test_job_with_blob:insert_blob #3 -- */
insert_blob: test_job_with_blob
blob_file: <download_path>/test_job_with_blob_3.txt
```

You can save this file as a backup of job definitions, or you can use a text editor to quickly edit the job definitions. Use the following jil command to re-load the job definitions into the database:

```
jil < dump_file
```

### Example: Generate Summary and Detail Reports for a Global Virtual Resource

This example generates a summary report and a detail report for the ren1 virtual resource. The resource is global, so it is available to all machines.

```
autorep -V ren1 -d
```

The resulting report might resemble the following:

Resource Name	Type	Machine	Defined	Available
ren1	V	---	4	1

```
/**** Current Resource Usage ****/
```

ResName Use	JobName	Run/Ntry	Status	Machine	Amount in
ren1	job1	1/1	RUNNING	machine1	1
ren1	job2	1/1	RUNNING	machine2	2

The -M option is not specified in the command, so the report displays all the jobs that are currently using the resource.

### Example: Generate a Report for a Queued Jobs

This example generates a detailed report on all jobs that are in the queued status. Jobs j1 to j15 are standalone jobs that are defined on mach02-w7b machine and most of these jobs use the machine-based r1 resource. Job zeroprijob is a zero priority job with no load units that uses the g1 global resource. b1b1, b1b2, and b1b3 are box jobs and they also use the global g1 resource. B00001\_job20 and B00001\_job30 jobs are defined with load balancing attributes on the B00001\_mac machine and only B00001\_job20 uses the B00001\_res resource.

```
autorep -Q ALL -n
```

The resulting report might resemble the following:

Pri	Job Name Resource	ST Load Req/Avl	Mach Queue	ST	Curr/Max
0	zeroprijob g1	RW 0 10/1	----	----	----
1	j1 r1.mach02-w7b	RW 11 11/1	mach02-w7b	ONLN	0/11
1	j3 r1.mach02-w7b	RW 11 10/1	mach02-w7b	ONLN	0/11
1	j2 r1.mach02-w7b	RW 10 11/1	mach02-w7b	ONLN	0/11
1	j4 r1.mach02-w7b	RW 10 10/1	mach02-w7b	ONLN	0/11
1	j9 r1.mach02-w7b	RW 0 11/1	mach02-w7b	ONLN	0/11
1	j10 r1.mach02-w7b	RW 0 10/1	mach02-w7b	ONLN	0/11
1	b1b2 g1	RW 0 11/1	----	----	----
2	j15 ----	QU 11 ----	mach02-w7b	ONLN	0/11
2	j16 ----	QU 10 ----	mach02-w7b	ONLN	0/11



2	j5	QU 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
2	j7	QU 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
2	j6	QU 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
2	j8	QU 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
2	j13	RW 0	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
2	j14	RW 0	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
2	b1b1	RW 0	----	----	----
	g1	1/1			
3	b1b3	RW 0	----	----	----
	g1	10/1			
20	B00001_job20	QU 1	B00001_mac	OFLN	0/1
	B00001_res	5/5			
30	B00001_job30	QU 1	B00001_mac	OFLN	0/1
	----	----			

This example generates a report for the j1 job that is in the RESWAIT status with both load balancing and resource attributes:

```
autorep -Q j1 -n
```

The resulting report might resemble the following. The zeroprijob, b1b1, b1b2, b1b3, B00001\_job20 and B00001\_job30 jobs are filtered out because they do not share the same load balancing or resource queue criteria as the j1 job. Although the j1 job has a higher priority than all but the zero priority job, the report displays all the priority jobs that match the same queue criteria as they might be impacted by the j1 job.

Pri	Job Name	ST Load	Mach Queue	ST	Curr/Max
	Resource	Req/Avl			
1	j1	RW 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			

1	j3	RW 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
1	j2	RW 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
1	j4	RW 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
1	j9	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
1	j10	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j15	QU 11	mach02-w7b	ONLN 0/11
	----	----		
2	j16	QU 10	mach02-w7b	ONLN 0/11
	----	----		
2	j5	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j7	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j6	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j8	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j13	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j14	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		

This example generates a report for the b1b2 box job that is in the RESWAIT status with only resource attributes:

```
autorep -Q b1b2 -n
```

The resulting report might resemble the following. The B00001\_job20, B00001\_job30, and the j1 through j16 jobs are filtered out because they do not share the same resource queue criteria as the b1b2 box. All jobs that are queued on the r1 resource are displayed in the report.

Pri	Job Name	ST	Load	Mach	Queue	ST	Curr/Max
	Resource		Req/Avl				

0	zeroprijob	RW 0	----		----
	g1	10/1			
1	b1b2	RW 0	----		----
	g1	11/1			
2	b1b1	RW 0	----		----
	g1	1/1			
3	b1b3	RW 0	----		----
	g1	10/1			

This example generates a report for the j15 job that is in the QUE\_WAIT status with only load balancing attributes:

```
autorep -Q j15 -n
```

The resulting report might resemble the following. The B00001\_job20, B00001\_job30, zeroprijob, b1b1, b1b2, and b1b3 jobs are filtered out because they do not share the same load balancing queue criteria as the j15 job. Also, the jobs with zero priority and in the RESWAIT status are not displayed in the report. All jobs that are queued on the mach02-w7b machine are displayed in the report.

Pri	Job Name	ST Load	Mach Queue	ST	Curr/Max
Resource		Req/Avl			
1	j1	RW 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
1	j3	RW 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
1	j2	RW 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
1	j4	RW 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
1	j9	RW 0	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
1	j10	RW 0	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			

2	j15	QU 11	mach02-w7b	ONLN 0/11
----	----			
2	j16	QU 10	mach02-w7b	ONLN 0/11
----	----			
2	j5	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j7	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j6	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j8	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j13	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j14	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		

This example generates a report for the j8 job that is in the QUE\_WAIT status with both load balancing and resource attributes:

```
autorep -Q j8 -n
```

The resulting report might resemble the following. The zeroprijob, b1b1, b1b2, b1b3, B00001\_job20, and B00001\_job30 jobs are filtered out because they do not share the same load balancing or resource queue criteria as the j8 job. All jobs that are queued on the r1 resource or queued on the mach02-w7b machine are displayed in the report.

Pri	Job Name	ST Load	Mach Queue	ST	Curr/Max
	Resource	Req/Avl			
1	j1	RW 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
1	j3	RW 11	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			
1	j2	RW 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	11/1			
1	j4	RW 10	mach02-w7b	ONLN	0/11
	r1.mach02-w7b	10/1			

1	j9	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
1	j10	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j15	QU 11	mach02-w7b	ONLN 0/11
	----	----		
2	j16	QU 10	mach02-w7b	ONLN 0/11
	----	----		
2	j5	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j7	QU 11	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j6	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j8	QU 10	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		
2	j13	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	11/1		
2	j14	RW 0	mach02-w7b	ONLN 0/11
	r1.mach02-w7b	10/1		

### Example: Generate a Report for Agent Plug-ins, Advanced Integrations, Release, and Build Information

This example generates a report that displays all the active agent plug-ins and advanced integrations on the machine named *myhost* and the agent release and build information.

```
autorep -M myhost -p
```

The report displays the following information:

```
Machine Name: myhost
Type: a
Node Name: myagenthostname
Agent Name: WA_AGENT
Operating System: Windows 2003 for x86
Agent Release: R11.3
Agent Build: 738, Service Pack 1, Maintenance Level 1
Active Plug-ins:
 clearfiles
 communicationmanager
 filebrowser
```

```

filemon
nop
objmon
outbox
refresh
runner
setproperty
status
router

```

```

Adadvanced Integrations:
hadoop

```

## autostatus Command -- Report Job Status or Global Variable Value

The autostatus command is a client component utility that reports the current status of the specified job or the current value of a global variable to standard output. This command is especially useful in two circumstances:

- When one job needs to know the status of another.
- When complex starting conditions are required beyond those that can be easily specified in the job definition. For example, you may need to start a job when two out of a set of three jobs have completed successfully. You could handle these dependencies through the starting conditions, but the conditions would be very cumbersome to define. Instead, you can use autostatus to check the status of the three jobs and perform the appropriate action. For more information, see the examples later in this section.

When you invoke autostatus, AutoSys Workload Automation verifies that you have read access to the jobs and global variables for which you are requesting information.

### Syntax

This command has the following format:

```

autostatus
 {-J jobname | -G globalname}
 [-S instance] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]

```

- **-J *jobname***

Specifies the job to report on. The job's current status is returned to standard output.

**NOTE**

You do not need to precede the job name that contains a colon with a backslash. You can use wildcards, including the percent (%) character, in this value.

- **-G *globalname***

Specifies a global variable. The variable's value is returned to standard output.

**NOTE**

The specified global variable must have been set with the sendevent command.

- **-S *instance***

(Optional) Defines the three-character identifier of the AutoSys Workload Automation instance that the job or global variable resides on (for example, ACE).

**UNIX Default:** The value of \$AUTOSERV from the current environment.

**Windows Default:** The value of %AUTOSERV%.

**NOTE**

To use this option, you must also specify the -J or -G option.

- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password**: Specifies a username and unencrypted password.

**-usr username -pwx password**: Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact**: Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**

(Optional) Returns the command version information to standard output.

- **-?**

(Optional) Displays help for the command.

**Example: Check the Status of a Job on the Current Instance**

This example checks the current status of the test\_install job on the current instance.

```
autostatus -J test_install
```

The command writes the result to standard output. In the following example, SUCCESS is the current status and ON\_NOEXEC is the execution mode of the test\_install job.

```
SUCCESS/ON_NOEXEC
```

**Example: Check the Value of a Global Variable**

This example checks the value of a global variable named "Today".

```
autostatus -G Today
```

The command writes the result to standard output. For example:

```
12/20/2011
```

**Example: Replace a Condition Statement**

This example uses autostatus to replace a complex condition statement. The autostatus command runs the New\_Stats job when the following conditions are satisfied:

- The Account\_Run, Corp\_Run, and End\_Run jobs ran.
- At least two of the jobs completed successfully.
- It is not the 13th day of the month.

The autostatus command is used as follows:

1. Create a job named New\_Stats\_Starter with the following job definition:

```
#
JIL for New_Stats_Starter
#
insert_job: New_Stats_Starter
job_type: CMD
machine: mombo
command: new_stats_starter
condition: done(Account_Run) and
done(Corp_Run) and done(End_Run)
```

2. (UNIX only) Use the following code to create a Bourne shell script with the name new\_status\_starter to run for the job named New\_Stats\_Starter. This script determines whether to start the job New\_Stats. If the job exits with a 0 exit code (SUCCESS), New\_Stats can start; if the job exits with a non-zero exit code (FAILURE), New\_Stats cannot start.

```
#!/bin/sh
#
Program to determine when to start New_Stats
Check for 13th of month - if it is, exit
with 2
if [`date +%d` -eq 13]
then
exit 2
fi
Add up the Number of SUCCESS endings
SUCCESS=0
for JobName in Account_Run Corp_Run End_Run
do
if [`autostatus -J $JobName` = "SUCCESS"]
then
SUCCESS=`expr $SUCCESS + 1`
fi
done
```



```

if [$SUCCESS -gt 1]
then
exit 0
else
exit 1
fi

```

**NOTE**

To reference jobs running on other instances, you must set - S instance to the three-character identifier for that instance in the call to autostatus.

3. (Windows only) Use the following code to create a batch file with the name New\_Stats.bat to run for the job New\_Stats\_Starter. This batch file determines whether to start the job New\_Stats. If the job exits with a 0 exit code (SUCCESS), New\_Stats can start; if the job exits with a non-zero exit code (FAILURE), New\_Stats cannot start.

```

@REM ECHO OFF
@REM Program to determine when to start New_Stats
@REM
@REM Check for 13th of month, if it is, exit with 2
@echo | more | date > testdate.out
@findstr /c:"/13/" testdate.out
@if errorlevel 1 goto not13
@if errorlevel 0 echo 13th of the month
@del testdate.out
@autostatus -J %1 > test1.dat
@findstr SUCCESS test1.dat
@if errorlevel 1 goto nosuccess
@if errorlevel 0 goto sayok
:nosuccess
@del test1.dat
@REM false is supplied in %AUTOSYS%\bin
@false 1
@goto exit:
:sayok
@del test1.dat
@goto exit
:not13
@del testdate.out
@REM false is supplied in %AUTOSYS%\bin
@false 2
:exit

```

**NOTE**

To reference jobs running on other instances, you must set - S instance to the three-character identifier for that instance in the call to autostatus.

4. Create the job New\_Stats and set the following starting condition in the job definition to specify that it should start when the previous job (New\_Stats\_Starter) completes successfully:
 

```
condition: success(New_Stats_Starter)
```

## autostatad Command -- Report the Status of a CA AutoSys Workload Automation Adapter Job

### Valid on UNIX

The autostatad command reports the current status of the specified AutoSys Workload Automation Adapter job to standard output. For example, you might use the autostatad command to report the status of a AutoSys Workload Automation for SAP job.

When AutoSys Workload Automation runs an adapter job, the adapter stores the job's current status in the AutoSys Workload Automation database as a global variable. The global variable name is the job name and its value is the job status and the date and time the status was written to the database.

### Syntax

This command has the following format:

```
autostatad -J job_name [-h] [-s] [-t] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
[-x]
[-?]
```

- **-J job\_name**

Specifies the job to report on. The job's current status is returned to standard output.

#### NOTE

- You can use wildcards, including the percent (%) character, in this value.
- Adapter jobs do not support colons in the job name. If you specify a colon in the adapter job name, the job fails during execution.

- **-h**

(Optional) Displays help for the command.

- **-s**

(Optional) Displays only the status for the specified job, without the job name or any labeling. This parameter is useful for creating stripped output to include in scripts.

- **-t**

(Optional) Displays the date and time the job's status was written to the AutoSys Workload Automation database.

- **{-usr username {-pw|-pwx} password |-saml auth\_artifact}**

(Optional) Specifies external security credentials as follows:

**-usr username -pw password:** Specifies a username and unencrypted password.

**-usr username -pwx password:** Specifies a username and encrypted password.

#### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

#### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Writes the version number to standard output.

#### NOTE

- When you invoke autostatad, AutoSys Workload Automation verifies that you have read access to the jobs that you are requesting information for.
- The exit code and output of autostatad vary depending on whether the status is requested for a single job or multiple jobs. If the name of an explicit job is given, and either the job does not exist or the read access to the job is denied, autostatad returns an error message and a non-zero exit code value. However, if a job name with a wildcard is specified and no status for jobs can be returned, autostatad returns zero as the exit code value to indicate success.
- If the list access for jobs that are reported by autostatad is denied, you can only view the status for jobs in the list to which you are granted read access. There is no visual indication that the status list is incomplete. autostatad does not display a security warning if it cannot obtain read access to one of the jobs that match the input name qualifier. It does not warn that access to view the entire list is denied.

#### Example: Report the Status of Similarly Named Adapter Jobs

This example returns the job name, status, and timestamp of all Adapter jobs whose names begin with autoMB.

```
autostatad -J autoMB% -t
```

The output resembles the following:

```
Job: autoMB1 Status:completed Time: 10/13/05 09:09:00
Job: autoMB2 Status:aborted Time: 10/13/05 09:15:31
Job: autoMB3 Status:failed Time: 10/13/04 09:09:21
```

#### Example: Report the Status of an Adapter Job with No Labeling

This example returns only the status of the job autoMB1 with no labeling.

```
autostatad -J autoMB1 -s
```

If appropriate, you can redirect the stripped output from this command to a file for use in a script.

## autotrack Command -- Tracks Changes to the Database

The autotrack command is a client component utility that tracks changes to the database (for example, job definition changes, sendevent calls, and job overrides) and writes this information to the database.

When you query for this information, autotrack prints a report to the screen, or you can redirect the output to a file. autotrack can track changes that are made to job definitions from jil, Web UI, or the SDK. It cannot track changes that are made directly to the database through SQL commands.

This command is especially useful in sites that require careful monitoring of the job definition environment or where multiple users have permission to edit job definitions or send AutoSys Workload Automation events.

AutoSys Workload Automation stores autotrack output in two tables: ujo\_audit\_info and ujo\_audit\_msg. By default, these tables reside in the same database (or tablespace) as all other AutoSys Workload Automation tables. If you frequently unload all your jobs from the database and reload, turn off autotrack while you unload and reload your jobs to prevent the database from growing large. After you have unloaded and reloaded your jobs, turn on autotrack.

Running archive\_events helps to prevent the database from filling up with autotrack output. The archive\_events -l num\_of\_days command archives information older than the specified number of days from the ujo\_audit\_info and ujo\_audit\_msg tables.

### Syntax

This command has the following format:

```
autotrack
 [{-u level|-l}] [-v] [-F "from_time"] [-T "to_time"] [-U username] [-m machine] [-J jobname] [-B group] [-I application] [-R resource] [-t type] [{-usr username {-pw| pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

- **-u level**

(Optional) Defines the level of detail that autotrack writes to the database.

**Default:** 0

**Limits:** This value can be one of the following:

- **0**  
Writes no tracking detail.
- **1**  
Tracks the following and condenses each tracked event to a one-line summary:
  - Job, calendar, monitor, browser, and machine definition changes
  - Job overrides
  - autosys\_secure, autotrack, and sendevent calls
- **2**  
Tracks the same information as level 1, but also writes the job definition for overrides and job definition changes. This level is very database intensive and significantly impairs JIL performance.

### NOTE

- Only the EXEC or EDIT superuser can change the tracking level.

- The autotrack command writes the entire job definition for overrides and job definition changes only when you use the level 2 option with the -v option.
- Only specified mandatory attributes and specified optional common attributes at the time of job creation will be displayed.
- **-l**  
(Optional) Displays the current tracking level (0, 1, or 2).
- **-v**  
(Optional) Activates verbose autotrack reporting.
- **-F "from\_time"**  
(Optional) Defines the date and time that starts the interval about which to report.  
**Limits:** This value can be in "MM/dd/yyyy hh:mm" format, as follows:
  - **MM**  
This value can be a number in the range 01 through 12 specifying the month in which to begin the reporting interval.
  - **dd**  
This value can be a number in the range 01 through 31 specifying the day on which to begin the reporting interval.
  - **yyyy**  
This value can be a number in the range 00 through 99 or 1900 through 2100 specifying the year in which to begin the reporting interval. When you enter a two-digit year, AutoSys Workload Automation saves the setting to the database as a four-digit year. If you enter 79 or less, the product precedes your entry with 20; if you enter 80 or greater, the product precedes your entry with 19.
  - **hh**  
This value can be a number in the range 00 through 23 specifying the hour in which to begin the reporting interval.
  - **mm**  
This value can be a number in the range 00 through 59 specifying the minute in which to begin the reporting interval.

**NOTE**

You must enclose this value in quotation marks (").

- **-T "to\_time"**  
(Optional) Defines the date and time that ends the interval about which to report.  
**Limits:** This value can be in "MM/dd/yyyy hh:mm" format, as follows:
  - **MM**  
This value can be a number in the range 01 through 12 specifying the month in which to end the reporting interval.
  - **dd**  
This value can be a number in the range 01 through 31 specifying the day on which to end the reporting interval.
  - **yyyy**  
This value can be a number in the range 00 through 99 or 1900 through 2100 specifying the year in which to end the reporting interval. When you enter a two-digit year, AutoSys Workload Automation saves the setting to the database as a four-digit year. If you enter 79 or less, the product precedes your entry with 20; if you enter 80 or greater, the product precedes your entry with 19.
  - **hh**  
This value can be a number in the range 00 through 23 specifying the hour in which to end the reporting interval.
  - **mm**  
This value can be a number in the range 00 through 59 specifying the minute in which to end the reporting interval.

**NOTE**

You must enclose this value in quotation marks (").

- **-U username**  
(Optional) Reports changes or events initiated by the specified user.

- **-m machine**  
(Optional) Reports changes or events initiated from the specified machine.
- **-J jobname**  
(Optional) Defines the job about which to report.

**NOTE**

- You can use the percent (%) and underscore (\_) characters as wildcards in this value.
- You do not need to precede the job name that contains a colon with a backslash.

- **-B group**  
(Optional) Defines the group about which to report.

**Limits:** When **-I application** is also defined, the command generates the report only for jobs that match both **-I application** and **-B group**. To search for jobs or boxes without group names, you can specify a blank group name by enclosing a blank with quotes, for example **-B " "**.

**NOTE**

- The **-J jobname** and **-B group** or **-I application** are mutually exclusive. That is, you cannot specify the **-J jobname** with **-B group** or **-I application** and conversely.
- This value allows wildcard characters.
- You can use the percent (%) and underscore (\_) characters as wildcards in this value.
- You do not need to precede the box name that contains a colon with a backslash.

- **-I application**  
(Optional) Defines the application about which to report.

**Limits:** When **-B group** is also defined, the command generates the report only for jobs that match both **-I application** and **-B group**. To search for jobs or boxes without application names, you can specify a blank application name by enclosing a blank with quotes, for example **-I " "**.

**NOTE**

- The **-J jobname** and **-B group** or **-I application** are mutually exclusive. That is, you cannot specify the **-J jobname** with **-B group** or **-I application** and conversely.
- This value allows wildcard characters.
- You can use the percent (%) and underscore (\_) characters as wildcards in this value.
- You do not need to precede the box name that contains a colon with a backslash.

- **-R resource**  
(Optional) Defines the virtual resource about which to report.

**NOTE**

- This value allows wildcard characters.
- You can use the percent (%) and underscore (\_) characters as wildcards in this value.

- **-t type**  
(Optional) Reports a specific event. This value can be one of the following:

- **A**  
Reports the calls that are generated by the autosys\_secure command.
- **B**  
Reports Monitor/Browser definition changes generated by jil.
- **C**  
Reports calendar definition changes generated by the autocal\_asc command.
- **J**  
Reports job definition changes, sendevent -J, or overrides to a specific job generated by jil.
- **M**

- Reports machine definition changes generated by jil.
- **N**  
Reports connection profile definition changes generated by jil.
- **O**  
Reports override definition changes generated by jil.
- **S**  
Reports the calls that are generated by the sendevent command evoked through client utilities.
- **T**  
Reports the calls that are generated by the autotrack command (for example, changes to the tracking level).
- **{-usr username {-pw|-pwx} password [-saml auth\_artifact]}**  
(Optional) Specifies external security credentials as follows:
  - usr username -pw password**: Specifies a username and unencrypted password.
  - usr username -pwx password**: Specifies a username and encrypted password.

**NOTE**  
Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact**: Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

  - When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
    - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
    - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
    - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
    - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
    - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
  - When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
  - Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
    - The unauthenticated user mode setting is EXTERNAL.
    - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

#### Example: Display Level 1 Verbose Reporting About a Job

This example displays verbose reporting about the job NightlyReport.

```
autotrack -J NightlyReport -v
```

The output resembles the following:

```
jane@taurus
11/21 10:04:54
job definition change
::
jane@taurus
11/21 10:05:49
job definition change
command: date
::
jane@taurus
11/21 10:06:29
sendevent issued
```

### Example: Display Level 2 Verbose Reporting About a Job

This example displays a report that includes the entire job definition with changes indicated by asterisks.

```
autotrack -J NightlyReport -v
```

The output resembles the following:

```
jane@taurus
05/26/2010 13:13:51
CAUAJM_I_50191 Job definition change.
 insert_job: NightlyReport
 job_type: CMD
 box_name: holdBox
 alarm_if_fail: 1
 auto_delete: -1
 auto_hold: 0
 box_terminator: 0
 command: rpt_util nightly
 date_conditions: 0
 interactive: 0
 job_load: 0
 job_terminator: 0
 machine: taurus
 max_exit_success: 0
 max_run_alarm: 0
 min_run_alarm: 0
 n_retrys: 0
 owner: "jane@taurus"
 priority: 0
 service_desk: 0
 term_run_time: 0
```



```

::
jane@taurus
05/26/2010 13:14:04
CAUAJM_I_50152 Sendevent issued.

eoid: WCCz10000628
job_name: env
command: sendevent "-E" "STARTJOB" "-J" "NightlyReport"
::
jane@taurus
05/26/2010 13:23:31
CAUAJM_I_50191 Job definition change.

insert_job: NightlyReport
job_type: CMD
* command: rpt_util NIGHTLY

```

### Example: Report Changes After a Specific Date and Time

This example requests a report of all the changes that occurred to the job NightlyReport after 1 a.m. on November 12, 2009.

```
autotrack -F "11/12/2009 01:00" -J NightlyReport
```

### Example: Report Changes Made by a Specific User

This example requests a report of all changes made by the user Sue over the weekend of May 8 and 9, 2010.

```
autotrack -U sue -F "5/8/2010 01:00" -T "5/9/2010 23:59"
```

### Example: Report autosys\_secure Changes from a Specific Machine

This example requests a report of all autosys\_secure changes that occurred from the machine gemini.

```
autotrack -t A -m gemini
```

The output from this command resembles the following:

```

jane@gemini
11/05 19:08:12
autosys_secure change
EDIT Super-User: jane
EXEC Super-User: jane
password: *****

```

### Example: Report Job Deletions from the Database

This example requests a report of all jobs that have been deleted from the database from 1.00 a.m. on May 8, 2010 to 8.00 p.m. on May 20, 2010.

#### UNIX:

```
autotrack -t J -F "5/8/2010 01:00" -T "5/20/2010 20:00" -v | grep delete_job
```

#### Windows:

```
autotrack -t J -F "5/8/2010 01:00" -T "5/20/2010 20:00" -v | findstr delete_job
```

**NOTE**

You can also use the GetDeletedJobsReq SDK API to get the list of jobs that have been deleted from the database. For more information about how to view the documentation about GetDeletedJobsReq SDK API, see [AE Architecture](#).

**Example: autotrack for rename\_job Level 1**

```
autotrack -J RenamedTestJob -v
::
root@exatest0000
06/16/2020 07:58:18
CAUAJM_I_50191 Job definition change.

rename_job: TestJob
new_name: RenamedTestJob
```

**Example: autotrack for rename\_job Level 2**

```
autotrack -J RenamedTestJob -v
::
root@exatest0000
06/16/2020 08:07:56
CAUAJM_I_50191 Job definition change.

insert_job: RenamedTestJob
job_type: CMD
command: sleep 3
machine: localhost
* new_name: "TestJob"
owner: "root@exatest0000"
old_name: TestJob
```

**forecast Command -- Report Job Flow**

The forecast command reports job flows based on a date or date range. Forecast reports can help you predict the outcome for a set of predefined conditions. You can see what happens when values are changed for each forecast period and can use this information to plan workflow. Based on the current job definitions, the reported job flow displays a list of jobs predicted to run on the dates you specify.

The forecast report displays day-based jobs and event-based jobs that are expected to occur during a specified time frame of any length. The memory the system uses to generate a forecast report is directly correlated to the length of the time frame, the number of jobs that are scheduled to run during that time frame, and the types of those jobs.

For example, a database containing 45,375 jobs (including 12,107 jobs with date-time conditions) has the following consumption (actual values might differ on your system):

- A one day (24 hours) forecast consumes 50 MB of memory and 80 percent CPU, takes 14 seconds, and generates a 72,307 line report.
- A one-week (168 hours) forecast consumes 130 MB of memory and 80 percent CPU, takes 2 minutes and 22 seconds, and generates a 834,578 line report.

**Syntax**

This command has the following formats:

```
forecast { -M machine_name | -J job_name | -M machine_name -J job_name] -F {"mm/dd/
yyyy HH:MM | mm/dd/yyyy HH:MM-mm/dd/yyyy HH:MM"} [-T "mm/dd/yyyy HH:MM"] [OPTIONS] [{-
usr username {-pw|pwx} password|-saml auth_artifact}]
[-x]
[-?]
```

- **-M machine\_name**  
Specifies the name of the machine whose workflow the report forecasts.
- **-J job\_name**  
Specifies the name of the job whose workflow the report forecasts.

#### NOTE

- To forecast workflow for all jobs that run on a particular machine, use the -M parameter alone. To forecast workflow all runs of a particular job regardless of which machine executes the run, use the -J parameter alone. To forecast workflow only the runs of a job that a specific machine executes, use the -M and -J parameters together.
- You do not need to precede the job name that contains a colon with a backslash.
- **-F "mm/dd/yyyy HH:MM[:ss]" | "mm/dd/yyyy HH:MM[:ss]-mm/dd/yyyy HH:MM[:ss]"**  
Specifies the period for which the report forecasts workflow.
- **-T "mm/dd/yyyy HH:MM[:ss]"**  
(Optional) Specifies when to generate the forecast report.

#### NOTE

Note: Use the date and time format that is specified in the configuration file when you specify the -F parameter or the -T option. The default format is mm/dd/yyyy HH:MM[:ss]. For example, to forecast workflow information for the period between January 2, 2013 at 5 p.m. and January 3, 2013 at 5 p.m. using the default format, specify the -F parameter as follows:

```
-F 01/02/2013 17:00:00-01/03/2013 17:00:00
```

- **OPTIONS**  
(Optional) Specifies optional attributes that you can use to control what the information the forecast report displays.
- **{-usr username {-pw|-pwx} password |-saml auth\_artifact}**  
(Optional) Specifies external security credentials as follows:
  - usr username -pw password:** Specifies a username and unencrypted password.
  - usr username -pwx password:** Specifies a username and encrypted password.

#### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml auth\_artifact:** Specifies a Security Assertion Markup Language (SAML) artifact.

#### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:

- When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
- When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
- When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
- When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
- When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode
- **-x**  
(Optional) Returns the command version information to standard output.
- **-?**  
(Optional) Displays help for the command.

#### NOTE

- If the -l option is specified, the command considers all the jobs that start a day before the specified time frame. Those jobs' successors are evaluated to identify any jobs that occur within that time frame.
- To determine which day-based jobs occur in the specified time frame, the forecast command considers the following job attributes:
  - start\_mins
  - start\_times
  - run\_window
  - run\_calendar
  - exclude\_calendar
  - days\_of\_week

To determine which event-based jobs occur, the command then traces the event-based jobs based on the box and child relationships, dependency relationships, and average run-time information of the predecessor job.
- The forecast command interprets the logical operators specified in a condition (job dependencies) as OR. For example, consider jobs A, B, and C. Suppose that jobs A and B have start times of 10:00 and 11:00, and job C depends on the success of jobs A and B. In an AND condition, the start time of job C in its first run would be evaluated to whichever time occurs later (11:00 in this example). After both jobs A and B run to SUCCESS, job C would start at both 10:00 and 11:00 in subsequent runs. However, the forecast command interprets the condition as OR, so the generated report shows the start time for job C as both 10:00 and 11:00 for all runs.
- To project jobs during dependency traces, the forecast command uses the average run time for the job. If no average run time information is available, the command uses 1 minute.
- The forecast command ignores global variables, job overrides, cross-instance dependencies, DST changes, and the timezone attribute associated with a job. The forecast report is based on the local time zone.

### Example: Create a Forecast Report for Jobs Running on a Specific Machine

This example generates forecast report for all jobs on the testm machine. The report forecasts when each job runs between 00:00 on January 16, 2012 and 23:59 on January 16, 2012.

```
forecast -m testm -F "01/16/2012"
```

### Example: Create a Forecast Report for a Specific Job

This example generates a report that forecasts when test3 runs on the testm machine between 00:00 on January 16, 2012 and 23:59 on January 16, 2012.

```
forecast -m testm -j test3 -F "01/16/2012"
```

### Example: Create a Forecast Report for a Job for Multiple Days

This example generates report that forecasts when test2 runs between 00:00 on January 1, 2012 and 23:59 on January 3, 2012.

```
forecast -F "01/01/2012" -T "01/03/2012" -J test2 -n
```

## job\_depends Command -- Generate Detailed Reports of Job Dependencies and Conditions

The `job_depends` command is a client component that provides detailed reports about a job's dependencies and conditions. You can use this command to determine the current state of a job, its job dependencies. For a box job, you can use this command to determine the nested hierarchies specified in the definition. This command can generate a report of what jobs run during a given interval.

When you invoke `job_depends`, AutoSys Workload Automation verifies that you have read access to the jobs that you are requesting information for.

### Syntax

This command has the following format:

```
job_depends
 -c | -d | -r | -t
 [-B group] [-F from_datetime] [-I application] [-J job_name] [-L print_level] [-e] [-s]
 [-T to_datetime] [-w][{-usr username {-pw|pwx} password|-saml auth_artifact}]
 [-x]
 [-?]
```

#### NOTE

The `-c`, `-d`, `-r`, and `-t` parameters are mutually exclusive. You must specify one of these parameters.

- **-B group**

(Optional) Defines the group to generate a report for.

**Limits:** When `-I application` is also defined, the command generates the report only for jobs that match both `application` and `-B group`. A null group value might be specified as `-B ""`.

#### NOTE

- The `-J job_name` and `-B group` are mutually exclusive. You cannot specify the `-J job_name` with `-B group`.
- This value allows wildcard characters.

- **-c**

(Optional) Returns the current state of the job and the names of any jobs that depend on it.

- **-d**  
(Optional) Returns the starting conditions for the job but includes no indication of its current status. This option displays jobs in a box hierarchically (use the **-L** parameter to set how many levels of nesting to display).
- **-r**  
(Optional) Returns a job's virtual resource dependencies, the amount of resource required, and whether the requirements are met. The job is specified using **-J *job\_name*** option.
- **-t**  
(Optional) Returns the starting conditions for a job; however, the top level of jobs (or boxes) reported is limited to those that start during the interval specified by the job or box's date conditions. When a box satisfies those date conditions, all the jobs in the box are also returned (use the **-L** parameter to set how many levels of nesting to display).  
When you set this parameter, **job\_depends** does not calculate all complex job dependencies when reporting on the jobs and boxes that are scheduled to run. For example, JobB might have a date condition and depend on JobA. The date conditions for JobB might be met for a given day, while those for JobA are not. As a result, JobA does not run, nor does JobB. However, JobB appears on the **job\_depends** report. Therefore, the starting conditions are also printed. For example, the following condition is reported for JobB:

```
success (JobA)
```

- **-F *from\_datetime***  
(Optional) Defines the date and time that starts the interval about which to report.  
**Limits:** This value can be in "**MM/dd/[yy]yy hh:mm**" format, as follows:
  - **MM**  
This value can be a number in the range 01 through 12 specifying the month in which to begin the reporting interval.
  - **dd**  
This value can be a number in the range 01 through 31 specifying the day on which to begin the reporting interval.
  - **yyyy**  
This value can be a number in the range 00 through 99 or 1900 through 2100 specifying the year in which to begin the reporting interval. When you enter a two-digit year, AutoSys Workload Automation saves the setting to the database as a four-digit year. If you enter 79 or less, the product precedes your entry with 20; if you enter 80 or greater, the product precedes your entry with 19.
  - **hh**  
This value can be a number in the range 00 through 23 specifying the hour in which to begin the reporting interval.
  - **mm**  
This value can be a number in the range 00 through 59 specifying the minute in which to begin the reporting interval.
- **-I *application***  
(Optional) Defines the application for which to generate a report.  
**Limits:** When **-B *group*** is also defined, the command generates the report only for jobs that match both **-I *application*** and **-B *group***. A null application value might be specified as **-I ""**.

#### NOTE

- **-J *job\_name*** and **-I *application*** are mutually exclusive. You cannot specify **-J *job\_name*** with **-I *application***.
- This value allows wildcard characters.
- **-J *job\_name***  
(Optional) Identifies the job about which to report.  
**Limits:** To report on all jobs, enter **-J ALL**.

**NOTE**

- **-J *job\_name*** is mutually exclusive with **-B *group*** and **-I *application***. You cannot specify **-J *job\_name*** with either **-B *group*** or **-I *application***.
- This value allows wildcard characters.
- You do not need to precede the job name that contains a colon with a backslash.

- **-L *print\_level***

(Optional) Specifies the number of levels into the box job hierarchy about which to report. For example, when you specify -L 2, the report contains data for the specified job (a box) and the top two levels of jobs in the box. This value can be any number.

**Default:** -L (report about all levels in the box)

**Limits:** To report about only the topmost level (that is, the box); set this value to -L 0.

To report about all levels in the box, omit the *print\_level* value.

This parameter is valid only when you define the -d or -t parameters.

- **-e**

Generates all start times between the dates and times specified in the -F and -T options.

**Limits:** This parameter is only valid with the -t parameter.

- **-s**

(Optional) Sorts the jobs in an order based on their next start time.

**Limits:** This parameter is only valid with the -t parameter.

- **-T *to\_datetime***

(Optional) Defines the date and time that ends the interval about which to report.

**Limits:** This value can be in “*MM/dd/[yy]yy hh:mm*” format, as follows:

- ***MM***

This value can be a number in the range 01 through 12 specifying the month in which to end the reporting interval.

- ***dd***

This value can be a number in the range 01 through 31 specifying the day on which to end the reporting interval.

- ***yyyy***

This value can be a number in the range 00 through 99 or 1900 through 2100 specifying the year in which to end the reporting interval. When you enter a two-digit year, AutoSys Workload Automation saves the setting to the database as a four-digit year. If you enter 79 or less, the product precedes your entry with 20; if you enter 80 or greater, the product precedes your entry with 19.

- ***hh***

This value can be a number in the range 00 through 23 specifying the hour in which to end the reporting interval.

- ***mm***

This value can be a number in the range 00 through 59 specifying the minute in which to end the reporting interval.

- **-w**

(Optional) Displays all output columns in widened format.

- **{-usr *username* {-pw|-pwx} *password* [-saml *auth\_artifact*}**

(Optional) Specifies external security credentials as follows:

**-usr *username* -pw *password*:** Specifies a username and unencrypted password.

**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

**NOTE**

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

**NOTE**

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.
- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- **-x**

(Optional) Returns the command version information to standard output.

- **-?**

(Optional) Displays help for the command.

**Example: Report About Jobs Scheduled to Run on a Specific Day**

This example lists the jobs that are scheduled to run on Christmas day, 2007.

```
job_depends -t -J ALL -F "12/25/2007 00:00" -T "12/26/2007 00:00"
```

**Example: Report the Current Condition Status of a Specific Job**

This example generates a report about the current condition status of the JobX job.

```
job_depends -c -J JobX
```

**monbro Command -- Run a Monitor or Report**

The monbro command is a client component that runs a pre-defined monitor or report (browser) and returns the results to standard output. The command can connect to any database that the instance is configured to use. To run a monitor or report, connect to the database where the definition of the monitor or report resides.

You can configure a monitor to play sound clips when it runs by turning on the sound before you start the configuration process. The monitor uses the sound capabilities of the machine on which runs.

**Notes:**

- On UNIX, the command does not play sound clips unless all the following conditions apply:



- The sound clips are pre-recorded.
- The machine that runs the command has access to the \$AUTOUSER/sounds directory.
- You can use this command to run a monitor only when you have read access to the monitored jobs.

## Syntax

This command has the following format:

```
monbro -N name [-P poll_frequency] [-q] [{-usr username {-pw|pwx} password|-saml auth_artifact}]
[-x]
[-?]
```

- **-N *name***  
Defines the monitor or report to run.  
**Limits:** To run all monitors or reports, specify **-N ALL**.  
You can use the percent (%) character as a wildcard in this value.
- **-P *poll\_frequency***  
(Optional) For monitors only, *poll\_frequency* defines the interval (in seconds) at which to poll the database.  
**Default:** 10
- **-q**  
(Optional) Displays monbro definition in JIL format.
- **{-usr *username* {-pw|-pwx} *password* |-saml *auth\_artifact*}**  
(Optional) Specifies external security credentials as follows:  
**-usr *username* -pw *password*:** Specifies a username and unencrypted password.  
**-usr *username* -pwx *password*:** Specifies a username and encrypted password.

### NOTE

Automated password encryption is useful if you regularly change your password or run commands embedded in scripts. The configuration utility (as\_config command) lets you automate password encryption. Automated encryption may present security risks. To mitigate these risks, use the security utility (autosys\_secure command) to encrypt passwords.

**-saml *auth\_artifact*:** Specifies a Security Assertion Markup Language (SAML) artifact.

### NOTE

- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is STRICT, only utilities that successfully authenticate can access objects that are protected by security policy. The following conditions apply:
  - When you do not specify one of the external security command line options, the utility attempts to authenticate with cached credentials.
  - When the utility attempts to authenticate with cached credentials and no credentials are cached, the utility prompts you to specify credentials.
  - When the utility attempts to authenticate with cached credentials and the credentials are expired, the utility does not run and exits with an error.
  - When you specify the -usr option and you do not specify the -pw or -pwx option, the utility prompts you to specify a password.
  - When you specify the -pw or -pwx option but do not specify the -usr option, the utility does not run and exits with an error.
- When AutoSys Workload Automation is operating in external security mode and the unauthenticated user mode setting is EXTERNAL, access to objects is governed by security policy but authentication is not handled by the external security server. In this case, the security subsystem assigns security policy identities to all users who validate their identities using the specified external authentication system. If you

are not authenticated by the external system, you have no access to objects protected by security policy and may be unable to successfully execute utilities that require access to those objects.

- Utilities ignore command line options that specify external security authentication information and cached credentials when any of the following conditions apply:
  - The unauthenticated user mode setting is EXTERNAL.
  - AutoSys Workload Automation is operating in native security mode

- -x  
(Optional) Returns the command version information to standard output.
- -?  
(Optional) Displays help for the command.

### Example: Run a Monitor

This example runs the monitor Mon1 (which is defined in the default database):

```
monbro -N mon1
```

### Example: Run a Monitor and Display Monitor Definitions

This example runs the monitor Mon1 (which is defined in the default database) and displays the monitor definitions:

```
monbro -N mon1 -q
```

The monitor definitions returned resemble the following:

```
insert_monbro: xxx
mode: m
all_events: Y
job_filter: a
sound: N
alarm_verif: N
insert_monbro: xxx2
mode: b
all_events: N
alarm: Y
all_status: N
running: N
success: Y
failure: Y
terminated: N
suspended: Y
starting: N
restart: N
job_filter: b
job_name: box
currun: N
after_time: "11/11/2014 12:12"
```

## sendevent Command -- Aggregate Statistics

The scheduler aggregates job, alarm, and scheduler statistics at regular intervals. Sometimes you want to force the aggregation of these statistics before the next aggregation is scheduled to occur. For example, you need to produce a report based on the current statistics.

You can use the sendevent command to instruct the scheduler to aggregate these statistics before the next scheduled aggregation interval. The scheduler processes the event, and the system generates the statistics. The statistics are stored in the following tables:

- ujo\_rep\_hourly
- ujo\_rep\_daily
- ujo\_rep\_monthly
- ujo\_rep\_weekly

AutoSys Workload Automation records messages that are related to this event in the following locations:

- The scheduler log file  
Records starting and completed messages.
- The aggregation log file  
Records detailed messages

### NOTE

- You can configure the scheduler to aggregate job statistics by setting the AggregateStatistics configuration variable to 1. For more information about specifying configuration parameters, see [Retrieve Aggregated Job, Alarm, and Scheduler Statistics](#).
- You can also delete previously aggregated statistics by issuing this event.
- The aggregation log file (aggregation instance) is located in the \$AUTOUSER/out (on UNIX) or %AUTOUSER%\out (on Windows) directory. The aggregation log file and the scheduler log file have the same properties. The aggregation log file rolls over based on the LOGROLLOVER parameter value (on UNIX) or LOGROLLOVER environment variable setting (on Windows).
- The aggregation process terminates when the scheduler shuts down while the aggregation process is active. AutoSys Workload Automation displays a message to indicate termination of the process.

## Syntax

This command has the following format:

```
sendevent -E AGGREGATE {[-d] -l {HOURLY | DAILY | WEEKLY | MONTHLY}} [-d] [-S autoserv_instance] [-P priority] [-T "time_of_event"] [-C comment]
```

- **-l {HOURLY | DAILY | WEEKLY | MONTHLY}**  
Specifies the type of statistics to aggregate.

### NOTE

- The scheduler aggregates statistics in smaller intervals before starting the specified aggregation. For example, if you schedule a monthly aggregation process, the scheduler aggregates the hourly, daily, and weekly statistics before starting the monthly aggregation.
- The scheduler does not aggregate statistics for the *current* month, week, day, or hour.
- **-d**  
Deletes all aggregated statistics from the database. Optionally, you can use this parameter to modify the -l parameter. Using -d and -l together deletes previously stored statistics from the database before aggregating and storing current statistics.

**NOTE**

- To delete previously stored statistics before you aggregate and store current statistics, use option -d and the -l parameter together as follows:

```
sendevent -E AGGREGATE -d -l {HOURLY | DAILY | WEEKLY | MONTHLY}
```

- For information about deleting aggregated statistics from the database, see [Delete Aggregated Statistics](#).

- **-S autoserv\_instance**

(Optional) Specifies a <UAJM> instance to send the event to. Use a three-character identifier for a defined <UAJM> instance, such as ACE. Ensure that the instance you specify and the instance from which you send the event are installed on the same machine. Use this option to aggregate statistics that are associated with jobs that reside on the specified instance. Specify a three-character identifier for a defined <UAJM> instance, such as ACE.

- **-P priority**

(Optional) Assigns a priority to the event. To ensure that the scheduler processes an event immediately, assign the highest priority (1) to the event.

**Default:** 10

**Limits:** 1-1000

- **-T "time\_of\_event"**

(Optional) Specifies the time of the event. Use the "MM/dd/yyyy hh:mm[:ss]" format, as follows:

- **MM**

Specifies the month.

**Limits:** 01-12

- **dd**

Specifies the day.

**Limits:** 01-31

- **yyyy**

Specifies the year.

**Limits:** 1900-2100

- **hh**

Specifies the hour.

**Limits:** 00-23

- **mm**

Specifies the minutes.

**Limits:** 00-59

- **ss**

(Optional) Specifies the seconds.

**Limits:** 00-59

**NOTE**

AutoSys Workload Automation sets this format by default when you install the product unless you specify another format using the DateFormat parameter in the configuration file. Use the date format that you specified in the configuration file. For more information about the DateFormat parameter, see [DateFormat](#).

- **-C comment**

(Optional) Attaches the specified comment to the event. Use this parameter to post a comment about the event to the event log.

**Limits:** Up to 255 alphanumeric or special characters in a single-line, including spaces

**NOTE**

- Enclose values that contain spaces or special characters in quotation marks (").
- Avoid using single quotation marks as they are doubled.

### Example: Delete Stored Statistics and Aggregate Current Statistics

This example deletes the stored statistics and aggregates the following statistics:

- Hourly statistics for the period between 00:00 of the day of the previous aggregation and the last completed hour
- Daily statistics for each complete day between the previous aggregation and yesterday
- Weekly statistics for each full week between the previous aggregation and the last completed week ending on Saturday
- Monthly statistics for each full month between the previous aggregation and last month

```
sendevent -E AGGREGATE -d -l MONTHLY
```

#### NOTE

Suppose that you issue this command at 16:30 on April 16, 2012, and the statistics that are stored in the database were aggregated on January 15, 2012. The scheduler deletes the stored statistics and aggregates the following statistics:

- Hourly statistics for the period between 00:00 on January 15, 2012 and 16:00 on April 16, 2012
- Daily statistics for January 15 - April 15, 2012
- Weekly statistics for January 15 - April 14, 2012
- Monthly statistics for January, February, and March, 2012

## AutoSys Job Information Language

Job information language (JIL) is a scripting language that lets you define jobs, machines, resources, external instances, monitors, reports, blobs, and job types (or *JIL objects*).

This article provides information about the JIL subcommands and attribute statements.

#### NOTE

The `jil` command lets you manage object definitions interactively or import script files to JIL. For more information, see [Run the Job Information Language Processor](#) in the [Commands](#) article.

## JIL Job Definitions

This section describes the JIL subcommands and attributes that you can use to define and manage jobs. To define or manage a job, you specify the appropriate JIL subcommand and attribute statements to the `jil` command. JIL attributes define a job's properties. For example, the `job_type` attribute defines the type of job that you are creating or updating, and the `machine` attribute defines the name of a machine definition that describes where the job runs.

#### NOTE

When issuing commands or defining jobs that run on a different operating system (for example, Windows to UNIX or UNIX to Windows), use the syntax appropriate to the operating system of the agent machine where the job runs.

### delete\_box Subcommand -- Delete a Box Job from the Database

The `delete_box` subcommand deletes the specified box job and all the jobs in that box from the database. If the box job or any jobs in it are already scheduled, they are deleted without running.

#### NOTE

To delete a box job but not the jobs in it, use the `delete_job` subcommand instead.

#### Syntax

This subcommand has the following format:

```
delete_box: box_name
```

- ***box\_name***  
Defines the name of a box job that is currently defined in the database.

### Example: Delete a Box Job from the Database

This example deletes the box job Box1 and all jobs in it:

```
delete_box: Box1
```

## delete\_job Subcommand -- Delete a Job from the Database

The delete\_job subcommand deletes the specified job from the database. If the job is already scheduled to run, it will not run.

If running jil in job verification mode, the delete\_job subcommand checks the ujo\_job\_cond table and notifies you if the deleted job had dependent conditions. If the specified job is a box job, the box job is deleted and the jobs in the box become stand-alone jobs.

### NOTE

To delete a box job and the jobs in it, use the delete\_box subcommand instead.

### Syntax

This subcommand has the following format:

```
delete_job: job_name
```

- ***job\_name***  
Defines the name of a job or box job that is currently defined in the database.  
**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), colon (:), and pound (#); do not include embedded spaces or tabs.

### NOTE

Enclose the job name that contains a colon with quotation marks (" ") or precede it with a backslash (\).

### Example: Delete a Job from the Database

This example deletes the job Job1:

```
delete_job: Job1
```

## insert\_job Subcommand -- Add a Job to the Database

The insert\_job subcommand adds a job definition to the database. Depending on the type of job you are adding, the insert\_job subcommand requires one or more additional attributes.

### NOTE

You must define the machine using the insert\_machine command before you can insert a job that uses that machine. The job will not insert if the machine definition does not exist.

### Syntax

This subcommand has the following format:

```
insert_job: job_name
```

- ***job\_name***

Defines the name of the job that you want to schedule.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), colon (:), and pound (#); do not include embedded spaces or tabs.

#### NOTE

- Enclose the job name that contains a colon with quotation marks (" ") or precede it with a backslash (\).
- Adapter jobs do not support colons in the job name. If you specify a colon in the adapter job name, the job fails during execution.

#### Example: Add a Command Job to the Database

This example creates the command job `time_stamp` to run on the real machine `prod` that runs the `time_stamp.bat` batch file:

```
insert_job: time_stamp
machine: prod
command: time_stamp.bat
```

The `job_type` attribute is optional when defining a command job. To specify a command job, enter:

```
job_type: CMD
```

#### Example: Add a File Watcher Job to the Database

This example creates the file watcher job `EOD_batch_watch` to run on the real machine `prod` that watches for a file named `C:\myapps\EOD_batch.bat`:

```
insert_job: EOD_batch_watch
job_type: fw
machine: prod
watch_file: "C:\myapps\EOD_batch.bat"
```

#### Example: Add a Box Job to the Database

This example creates the box job `end_of_day`:

```
insert_job: end_of_day
job_type: box
```

### override\_job Subcommand -- Define a Job Override

The `override_job` subcommand defines a one-time override of the specified attributes. The override applies to the next run of the job you are defining.

If AutoSys Workload Automation generates a RESTART event because of system problems, the job override is reissued until the job runs once or until the maximum number of retries set in the `n_retrys` attribute is reached. The override is then discarded.

The `jil` command does not accept overrides that result in an invalid job definition. For example, if a job definition has one starting condition, `start_times`, you cannot set the `start_times` attribute to NULL because removing the start condition makes the job definition invalid. In this situation, no start time can be calculated.

AutoSys Workload Automation assigns an `over_num` value to the override when you save the job definition. To determine which overrides were applied after a job run, you can reference the `over_num` value. For example, when applying a job override, the scheduler specifies the override it is using as follows:

```
Job: JOB_NAME is using Over-Ride #14
```

**NOTE**

You cannot edit a job override specified using jil. If you specify an override for a job, but an override already exists, the new override replaces the original one. However, the original override (the over\_num value) is maintained in the database.

**Syntax**

This subcommand has the following formats:

- To override one or more attributes:  

```
override_job: job_name
attribute_keyword: value | NULL
attribute_keyword: value | NULL
```
- To delete an override:  

```
override_job: job_name delete
```
- **job\_name** Specifies a job or Box job that is defined in the database.

**NOTE**

Enclose the job name that contains a colon with quotation marks ( " ") or precede it with a backslash (\).

- **delete**  
 Cancels the previously specified job override for job\_name.
- **attribute\_keyword**  
 Defines the job attribute to override. You can specify one of the following:
  - auto\_hold
  - command
  - condition
  - date\_conditions
  - days\_of\_week
  - exclude\_calendar
  - machine
  - max\_run\_alarm
  - min\_run\_alarm
  - n\_retrys
  - profile
  - run\_calendar
  - run\_window
  - start\_mins
  - start\_times
  - std\_err\_file
  - std\_in\_file
  - std\_out\_file
  - term\_run\_time
  - watch\_file
  - watch\_file\_min\_size
  - watch\_interval
- **value**  
 Defines the value to override the specified attribute with.
- **NULL**  
 Deletes or negates the current value of the specified attribute.



### Example: Define a One-time Job Override

This example defines a one-time job override to change the standard output file for the job, JobA.

```
override_job: JobA
std_out_file: "C:\OUTPUT\SpecialRun.out"
```

### Example: Cancel a Job Override

This example cancels the job override for JobA.

```
override_job: JobA delete
```

## rename\_job Subcommand -- Rename a Job

The `rename_job` subcommand renames an existing job, and all dependencies reflect the changed name for that specific job.

### NOTE

A job cannot be renamed when it is in either STARTING or RUNNING state.

To rename a job, the user must have write permissions on the existing job which is to be renamed.

### Supported Job Types

All job types support the `rename_job` subcommand.

The `rename_job` subcommand is also supported across cross-instances.

### NOTE

If one instance of a job is renamed and this job is being referred to from other instances, then those conditions are also renamed accordingly if the instance is at Version 12.0 or higher. For all other instances, you must manually change the condition.

### Syntax

This subcommand has the following format:

```
rename_job: old_job_name
new_name: new_job_name
```

- **old\_job\_name**  
Specifies the existing name for the job.
- **new\_job\_name**  
Specifies the new name for the job.

**Limits:** Up to 64 alphanumeric characters; can include the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), colon (:), and hyphen (-); cannot include embedded spaces or tabs.

When a job which has multiple references from the other jobs is renamed, then all the jobs which have references to the renamed job will also be modified accordingly.

The user is notified with renamed jobs that have dependent conditions.

### Example:

```
insert_job:jobExample
```

```
command:echo
machine: machine_name
condition: s(old_job_name)
```

If the `old_job_name` is renamed to `new_job_name` then the `jobExample` condition should also be updated with the new name for the job.

## update\_job Subcommand -- Update an Existing Box or Job Definition in the Database

The `update_job` subcommand updates an existing box or job definition in the database. The `update_job` argument is followed by a list of arguments to update.

### NOTE

- When many attributes must be updated for a job, it may be more efficient to delete and re-insert the job definitions.
- When you update a job, you cannot update the resources attribute in the existing job definition if the job has a resource dependency and has held the resource.

### Syntax

This subcommand has the following format:

```
update_job: job_name
```

- **job\_name**

Specifies the name of the job whose definition you want to update.

#### Limits:

Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), colon (:), and pound (#); do not include embedded spaces or tabs.

### NOTE

Enclose the job name that contains a colon with quotation marks (" ") or precede it with a backslash (\).

### Example: Update a Command

This example changes the preexisting command job `time_stamp` to run on the machine `paris` instead of on the originally specified machine:

```
update_job: time_stamp

machine: paris
```

## Common Job Attributes

All valid job definitions specify the job name and the machine that the job runs as follows:

```
insert_job: job_name
```

machine: *machine\_name*

Optionally, a job definition may include statements that use any of the following attributes:

alarm_if_fail	
application	notification_msg
auto_delete	owner <b>Note:</b> This attribute is not valid for file trigger jobs.
auto_hold	permission
avg_runtime	priority
box_name	resources
box_terminator	run_calendar
condition	run_window
date_conditions	send_notification
days_of_week	service_desk
description	start_mins
exclude_calendar	start_times
group	svcdesk_attr
job_class <b>Note:</b> This attribute is not valid for z/OS job types.	svcdesk_desc
job_load	svcdesk_imp
job_type	svcdesk_pri
max_run_alarm	svcdesk_sev
min_run_alarm	term_run_time
must_complete_times	timezone
must_start_times	
n_retrys	

## alarm\_if\_fail Attribute -- Specify Whether to Post an Alarm for FAILURE Status

The alarm\_if\_fail attribute specifies whether to post an alarm to the scheduler to alert the operator to take corrective action when the job completes with a FAILURE status.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
alarm_if_fail: y | n
```

- **y**  
Posts an alarm to the scheduler when the job fails. This is the default.

**NOTE**

You can specify 1 instead of y.

- **n**  
Does not post an alarm to the scheduler when the job fails.

**NOTE**

You can specify 0 instead of n.

**Example: Do not Generate an Alarm when a Job Fails**

This example does not post an alarm to the scheduler when the job fails:

```
alarm_if_fail: n
```

**alarm\_if\_terminated Attribute -- Specify Whether to Post the JOBTERMINATED for TERMINATED Status**

The alarm\_if\_terminated attribute specifies whether to post the JOBTERMINATED alarm to the scheduler to alert the operator to take corrective action when the job completes with a TERMINATED status.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

```
alarm_if_terminated: y | n
```

- **y**  
Posts the JOBTERMINATED alarm to the scheduler when the job is terminated. This is the default.

**NOTE**

You can specify 1 instead of y.

- **n**  
Does not post the JOBTERMINATED alarm to the scheduler when the job is terminated.

**NOTE**

You can specify 0 instead of n.

**NOTE**

- When you insert a job, the alarm\_if\_terminated attribute is set to y.
- For existing jobs, when you upgrade, if the alarm\_if\_fail attribute is set to n, then the alarm\_if\_terminated attribute is also set to n. However, if the alarm\_if\_fail attribute is set to y, then the alarm\_if\_terminated attribute is also set to y. So, any user actions that you might have set for the JOBFailure alarm might have to be revisited to accommodate the JOBTERMINATED alarm.

**Example: Do not Generate the JOBTERMINATED Alarm when a Job is Terminated**

This example does not post the JOBTERMINATED alarm to the scheduler when the job is terminated:

```
alarm_if_terminated: n
```

## application Attribute -- Associate a Job with an Application

The application attribute associates a job with a specific application so users can classify, sort, and filter jobs by application name.

You can use the application attribute with the sendevent, job\_depends, and autorep commands. For example, you could use a single sendevent command to start all of the jobs associated with a specific application.

Consider the following when you use the application attribute:

- Users with execute permission can add new application attributes to a job definition.
- You can only associate a job with one application at a time.
- When a job is associated with a group (using the group attribute) and an application, the product assumes that the application attribute is a subset of the group attribute.

### NOTE

If both the box and child job belongs to the same application, you have to specify the application attribute only for a box job.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
application: application_name
```

- ***application\_name***

Defines the name of the application with which to associate the job. You can only associate a job with one application, but an application may have many jobs associated with it.

#### Limits:

Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

### Example: Associate a Job with an Application

This example associates a job with the application summary\_reports:

```
application: summary_reports
```

## arc\_obj\_name Attribute -- Identify Name of SAP Archiving Object

The arc\_obj\_name attribute identifies the name of an archiving object defined on the SAP system.

### Supported Job Type

This attribute is required for the [SAP Data Archiving](#) (SAPDA) job type.

### Syntax

This attribute has the following format:

```
arc_obj_name: object_name
```

- ***object\_name*** Specifies the name of the archiving object.

**Limits:** Up to 256 characters; case-sensitive

### Example: Specify an Archiving Object

This example defines a job that stores information described in the BC\_ARCHIVE Archiving Object into an SAP data archive. The archiving object variant is BC\_ARCVARIANT.

```
insert_job: SAPDA_job
job_type: SAPDA
machine: sapagent
owner: WAAESAP@sapagent
arc_obj_name: BC_ARCHIVE
arc_obj_variant: BC_ARCVARIANT
sap_print_parms: dest=LP01,prt_arc_mode=PRINT
```

## arc\_obj\_variant Attribute -- Identify Name of SAP Archiving Object Variant

The arc\_obj\_variant attribute identifies the name of an archiving object variant defined on the SAP system.

### Supported Job Type

This attribute is required for the [SAP Data Archiving \(SAPDA\) job type](#).

### Syntax

This attribute has the following format:

```
arc_obj_variant: object_variant
```

- **object\_variant**

Specifies the name of the archiving object variant.

Limits: Up to 256 characters; case-sensitive

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

### Example: Specify an Archiving Object Variant

This example defines a job that stores information described in the BC\_ARCHIVE Archiving Object into an SAP data archive. The archiving object variant is BC\_ARCVARIANT.

```
insert_job: SAPDA_job
job_type: SAPDA
machine: sapagent
owner: WAAESAP@sapagent
arc_obj_name: BC_ARCHIVE
arc_obj_variant: BC_ARCVARIANT
sap_print_parms: dest=LP01,prt_arc_mode=PRINT
```

## arc\_parms Attribute -- Specify Archive Parameters

The arc\_parms attribute specifies the archive parameters for an SAP Data Archiving object.

#### NOTE

To use this attribute, you must specify prt\_arc\_mode=ARCHIVE or prt\_arc\_mode=BOTH in the sap\_print\_parms attribute.

### Supported Job Type

This attribute is optional for the SAP Data Archiving (SAPDA) job type.

## Syntax

This attribute has the following format:

```
arc_parms: arc_doc_type=value,arc_info=value,arc_obj_type=value
[,keyword=value...]
```

- **arc\_doc\_type=value**

Specifies the SAP ArchiveLink type. This keyword corresponds to the Doc type field on the Define Background Archive Parameters dialog.

**Limits:** Up to 10 valid SAP characters; case-sensitive

**Example:** archive

**NOTE**

This keyword is required if prt\_arc\_mode is set to BOTH or ARCHIVE.

- **arc\_info=value**

Specifies the SAP ArchiveLink information.

**Limits:** Up to three valid SAP characters; case-sensitive

**Example:** inf

**NOTE**

This keyword is required if prt\_arc\_mode is set to BOTH or ARCHIVE.

- **arc\_obj\_type=value**

Specifies the type of external system archive object. This keyword corresponds to the Obj. type field on the Define Background Archive Parameters dialog.

**Limits:** Up to 10 valid SAP characters; case-sensitive

**Example:** archive

**NOTE**

This keyword is required if prt\_arc\_mode is set to BOTH or ARCHIVE.

- **keyword=value**

(Optional) Specifies an archive parameter. You can specify multiple parameters. Separate each keyword=value pair with a comma. Options include the following:

- **arc\_client=value**

Specifies the archive link client.

**Limits:** 0-999

**Example:** 800

- **arc\_connect=value**

Specifies the archive link communication connection.

**Limits:** Up to 14 valid SAP characters; case-sensitive

**Example:** arcconnect

**NOTE**

Escape the double quotation marks with a backslash.

- **arc\_date=value**

Specifies the archive link archiving date. The format is *yyyymmdd*.

**Limits:** Up to eight numeric digits

**Example:** 20080702

- **arc\_doc\_class=value**

Specifies the SAP ArchiveLink document class.

**Limits:** Up to 20 valid SAP characters; case-sensitive

**Example:** arcdocclass

- **arc\_format=value**

Specifies the SAP ArchiveLink output format.

**Limits:** Up to 16 valid SAP characters; case-sensitive

**Example:** X\_65\_132

- **arc\_host\_link=value**  
Specifies the SAP ArchiveLink RPC host.  
**Limits:** Up to 32 valid SAP characters; case-sensitive  
**Example:** rpchost
- **arc\_path=value**  
Specifies the standard archive path.  
**Limits:** Up to 70 valid SAP characters; case-sensitive  
**Example:** /export/home/archive
- **arc\_printer=value**  
Specifies the SAP ArchiveLink target printer.  
**Limits:** Up to four valid SAP characters; case-sensitive  
**Example:** LP01
- **arc\_protocol=value**  
Specifies the archive storage connection protocol.  
**Limits:** Up to eight valid SAP characters; case-sensitive  
**Example:** prtcl12
- **arc\_report=value**  
Specifies the report name.  
**Limits:** Up to 30 valid SAP characters; case-sensitive

**NOTE**

Enclose the values that contain spaces in double quotation marks (").

**Example:** "Archive Report"

- **arc\_service=value**  
Specifies the RPC service/RFC destination.  
**Limits:** Up to 32 valid SAP characters; case-sensitive  
**Example:** rpcdest
- **arc\_storage=value**  
Specifies the SAP ArchiveLink target storage system.  
**Limits:** Up to two valid SAP characters; case-sensitive  
**Example:** st
- **arc\_text=value**  
Specifies the archive link Text Information field.  
**Limits:** Up to 40 valid SAP characters; case-sensitive

**NOTE**

Enclose the values that contain spaces in double quotation marks (").

**Example:** "Archive text"

- **arc\_userid**  
Specifies the archive data element for user.  
**Limits:** Up to 12 valid SAP characters; case-sensitive  
**Example:** arcuser
- **arc\_version=value**  
Specifies the archive version.  
**Limits:** Up to four valid SAP characters  
**Example:** 4.1

**NOTE**

- The entire value can be up to 4096 characters.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.



**Example: Specify Archive Parameters**

This example specifies the archive parameters in an SAP Data Archiving job.

```
insert_job: test_SAPDA
job_class: SAP
job_type: SAPDA
machine: localhost
owner: WAAESAP@waae-jobstest
arc_obj_name: BC_ARCHIVE
arc_obj_variant: PRODUKTIVMODUS
sap_print_parms: prt_arc_mode=BOTH,dest=LP01,title="testing"
arc_parms: arc_doc_type=ar01, arc_info=ar1, arc_obj_type=obj01, arc_client=202
```

**auth\_string Attribute -- Define an Authorization String**

The `auth_string` attribute defines the authorization string the scheduling manager uses to post the correct job. The scheduling manager posts the manual job complete when the manually submitted job runs under the corresponding user ID.

**Supported Job Type**

This attribute is optional for the [z/OS Manual \(ZOSM\) job type](#).

**Syntax**

This attribute has the following format:

```
auth_string: string
```

- ***string*** Defines the authorization string the scheduling manager uses to post and track the correct job.

**Limits:** Up to 80 characters

**NOTE**

Contact your agent administrator to confirm whether the agent checks the authorization string. Depending on the scenario, the action to take differs as follows:

- If authorization checking (AUTHSTR) is not set up for this agent, do not specify the authorization string.
- If authorization checking (AUTHSTR) is set to RACUSER, specify the user ID that owns the job.
- If authorization checking (AUTHSTR) is set to ACCOUNT1, ACCOUNT2, ACCOUNT3, or ACCOUNT4, specify the value found in the corresponding position of the job's Job Control Language (JCL). For example, if AUTHSTR is set to check the ACCOUNT2 field, specify the value found in the second accounting position in the job's JCL.

**Example: Post a z/OS Manual Job as Complete Based on the User ID**

This example posts a z/OS Manual job as complete when the manually-submitted job ABC runs under user CYBER. The ZOS1 agent monitors job ABC.

```
insert_job: ABC_job
job_type: ZOSM
machine: ZOS1
zos_jobname: ABC
owner: zosuser
auth_string: CYBER
```

## auto\_delete Attribute -- Automatically Delete a Job on Completion

AutoSys Workload Automation can automatically delete a job after it completes. The `auto_delete` attribute specifies the time to wait to delete a job after it goes to the SUCCESS state for the first time. The job is deleted at the designated time even if it is started manually for the second time and is in the RUNNING state. If the job is a box, the box and all the jobs it contains are deleted.

The `auto_delete` attribute is useful for scheduling and running a one-time batch job.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
auto_delete: hours | 0 | -1
```

- **hours**  
Defines the number of hours to wait after the job completes. At that time, the job is automatically deleted.  
**Limits:** 1-17520
- **0**  
Deletes the job immediately after successful completion. If the job does not complete successfully, the job definition is kept for seven days before it is automatically deleted.
- **-1**  
Does not automatically delete the job. This is the default.

### Example: Delete a Job on Successful Completion

This example automatically deletes the job five hours after the job completes successfully.

```
insert_job: unix_script
job_type: CMD
machine: unixagent
command: /usr/common/backup
auto_delete: 5
```

## auto\_hold Attribute -- Put a Job on Hold when Its Container's Status Changes to RUNNING

When a job is in a box, it inherits the starting conditions of the box. When the box's status changes to RUNNING, all the jobs in the box also start unless other conditions are not satisfied.

The `auto_hold` attribute specifies whether to change the state of a job that is in a box to ON\_HOLD when the status of the containing box changes to RUNNING.

### NOTE

This attribute statement is valid only when you also specify the `box_name` attribute statement in the job definition.

### Supported Job Types

This attribute is optional for all job types.

This attribute only applies to jobs that are in a box.

### Syntax

This attribute has the following format:

`auto_hold: y | n`

- **y**  
Automatically changes the state of the job to ON\_HOLD when its containing box starts running.  
**Note:** You can specify 1 instead of y.
- **n**  
Does not put the job on hold when its containing box starts running. This is the default.

**NOTE**

You can specify 0 instead of n.

**NOTE**

To take a job off hold, use the sendevent command to send the JOB\_OFF\_HOLD event for the job.

**Example: Put a Job on Hold when its Container's Status Changes to RUNNING**

This example defines a box that contains two jobs. When the status of the box changes to RUNNING, the autoHold job runs, but the autohold job does not run because it is on hold.

```
insert_job: holdBox
job_type: BOX
owner: root@localhost
insert_job: autohold
job_type: CMD
box_name: holdBox
command: ls
machine: localhost
owner: root@localhost
auto_hold: y
insert_job: autoHold
job_type: CMD
box_name: holdBox
command: ls
machine: localhost
owner: root@localhost
```

**avg\_runtime Attribute -- Define Average Run Time for a New Job**

The avg\_runtime attribute defines an average run time (in minutes) for a job that is newly submitted to the database. The attribute is used when the job has not run.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

`avg_runtime: value`

- **value**  
Defines the average run time (in minutes) to use for a job that has not run.  
Default: 0  
Limits: Up to 527040

**NOTE**

When the DBMaint script runs, it recalculates the average run time for each job in the database.

**NOTE**

**Windows Note:** AutoSys Workload Automation only uses the avg\_runtime attribute to establish an average run time for a new job in the ujo\_avg\_job\_runs table.

**Example: Define Average Run Time for a New Job**

This example sets the average run time for a new job to five minutes.

```
avg_runtime: 5
```

**bdc\_err\_rate Attribute -- Specify the Maximum Acceptable Error Rate**

The bdc\_err\_rate attribute specifies the maximum acceptable error rate in an SAP Batch Input Session job.

**Supported Job Type**

This attribute is optional for the [SAP Batch Input Session \(SAPBDC\) job type](#).

**Syntax**

This attribute has the following format:

```
bdc_err_rate: percent
```

- ***percent***

Specifies the maximum acceptable error rate as a percentage of the total transactions. When set to 100, all errors are acceptable. When set to zero (0), the job fails if any transaction contains an error. When set to 5, for example, the job fails if more than five percent of the transactions contain errors.

Limits: 0-100

**Example: Specify a Maximum Acceptable Error Rate**

In this example, the maximum acceptable error rate is five percent. If more than five percent of transactions contain errors, the job fails.

```
insert_job: test_SAPBDC
job_type: SAPBDC
machine: sapagent
owner: WAAESAP@waae-jobstest
sap_job_name: ZBDCTEST
sap_release_option: A
sap_step_parms: abap_name=ZBDCTEST
bdc_err_rate: 5
```

**bdc\_ext\_log Attribute -- Generate Advanced Logging of the Batch Input Session**

The bdc\_ext\_log attribute specifies whether to generate advanced logging of the Batch Input Session (BDC) running on the SAP system.

**NOTE**

If you do not specify the bdc\_ext\_log attribute in your job definition, the job does not generate advanced logging (the default).

**Supported Job Type**

This attribute is optional for the [SAP Batch Input Session \(SAPBDC\) job type](#).

## Syntax

This attribute has the following format:

```
bdc_ext_log: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Generates advanced logging.
- **N | F | NO | FALSE**  
Does not generate advanced logging. This is the default.

### Example: Generate Advanced Logging of the Batch Input Session

This example generates advanced logging of the BDC running on the SAP system.

```
insert_job: test_SAPBDC
job_type: SAPBDC
machine: sapagent
owner: WAAESAP@waae-jobstest
sap_job_name: ZBDCTEST
sap_release_option: A
sap_step_parms: abap_name=ZBDCTEST
bdc_ext_log: Y
```

## bdc\_proc\_rate Attribute -- Specify Minimum Acceptable Process Rate

The `bdc_proc_rate` attribute specifies the minimum acceptable process rate within an SAP Batch Input Session job.

### Supported Job Type

This attribute is optional for the [SAP Batch Input Session \(SAPBDC\) job type](#).

## Syntax

This attribute has the following format:

```
bdc_proc_rate: percent
```

- **percent**  
Specifies the minimum acceptable process rate as a percentage of the total transactions. When set to 100, the job fails if all transactions are not processed. When set to zero (0), no minimum processed transactions are required. When set to 90, for example, the job fails if less than 90 percent of the transactions are processed.  
Limits: 0-100

### Example: Specify a Minimum Acceptable Process Rate

In this example, the minimum acceptable process rate is 90 percent. If less than 90 percent of transactions are processed, the job fails.

```
insert_job: test_SAPBDC11_SIMPLE
job_type: SAPBDC
machine: sapagent
owner: WAAESAP@waae-jobstest
permission:
date_conditions: n
alarm_if_fail: y
sap_job_name: ZBDCTEST
```

```

sap_office: N
sap_release_option: I
bdc_ext_log: N
bdc_proc_rate: 90
sap_step_parms:
 abap_name="ZBDCTEST",banner_page=N,release=N,print_imm=N,new_spool=N,footer=N

```

## bdc\_system Attribute -- Specify the Name of a Background Server

The `bdc_system` attribute specifies the name of the background server that processes the Batch Input Session (BDC) running on the SAP system.

### Supported Job Type

This attribute is optional for the [SAP Batch Input Session \(SAPBDC\) job type](#).

### Syntax

This attribute has the following format:

```
bdc_system: destination
```

- ***destination***  
Specifies the name of the background server.  
**Limits:** Up to 256 valid SAP characters; case-sensitive

### Example: Specify the Background Processing Server Name

This example defines a job that uses a background server named `dest` to process the Batch Input Session running on the SAP system.

```

insert_job: test_SAPBDC11_SIMPLE2
job_type: SAPBDC
machine: sap_agent
owner: WAAESAP@waae-jobstest
sap_job_name: ZBDCTEST
sap_office: N
sap_release_option: I
bdc_ext_log: N
sap_step_parms:
 abap_name="ZBDCTEST",banner_page=N,release=N,print_imm=N,new_spool=N,footer=N
bdc_system:dest

```

## bean\_name Attribute -- Specify the JNDI Name of the Bean

The `bean_name` attribute specifies the JNDI name of the bean in an Entity Bean and Session Bean job.

### Supported Job Types

This attribute is required for the following job types:

- [Entity Bean \(ENTYBEAN\)](#)
- [Session Bean \(SESSBEAN\)](#)

### Syntax

This attribute has the following format:

```
bean_name: bean
```

- **bean** Specifies the JNDI name of the session or entity bean.

**Limits:** Up to 256 characters; case-sensitive

#### Example: Invoke a Method on a Stateless Session Bean

Suppose that you want to invoke the reverse method on the CybEJBTestBean stateless session bean. The reverse method has one parameter, with type java.lang.String and value a23. The output from the reverse method is saved in the C:\Makapt15 file. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is corbaloc:iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere application server and 2809 is the ORB port. When the job runs, the output of the reverse method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"
```

#### blob\_file Attribute -- Specify File for the Blob

The blob\_file attribute specifies the file containing the data that is used to create an input job blob. When you specify this attribute in a job definition, the blob is uploaded to the database when the job is defined. This input job blob is associated with the job. The job itself can then use the blob as input at run time. Other jobs that run on different computers can also access this blob by specifying the appropriate keyword in the std\_in\_file attribute.

##### NOTE

Alternatively, you can create an input job blob after a job is defined by using the insert\_blob subcommand.

#### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#) and [HTTP job type](#).

#### Syntax

This attribute has the following format:

```
blob_file: file
```

- **file**  
Specifies the name of the file that contains the input data.

**Limits:** Up to 255 characters

##### Examples:

- /tmp/blob\_input
- "c:\temp\blob\_input"

#### Example: Create an Input Job Blob Using a File

This example creates a job blob from the data contained in the file named `blob_input_file.txt` and uses the blob data as input for the command job named `test_blob`.

```
insert_job: test_blob
job_type: CMD
machine: unixagent
command: cat
blob_file: /blob_input_file.txt
std_in_file: $$blobt
owner: produser@unixagent
```

## blob\_input Attribute -- Specify Input for the Blob

The `blob_input` attribute specifies one or more lines of text to contain in a blob. Unlike the `blob_file` attribute that lets you specify a file that contains the data to store in the blob, the `blob_input` attribute lets you type the blob's data directly in the job definition.

When you specify the `blob_input` attribute in a job definition, the blob is uploaded to the database when the job is defined. This input job blob is associated with the job. The job itself can then use the blob as input at run time. Other jobs that run on different computers can also access this blob by specifying the appropriate keyword in the `std_in_file` attribute.

For the HTTP and WSDOC job types, the blobs can now have global variables and substitution variables defined. The scheduler will resolve those variables when the job is run.

### NOTE

Alternatively, you can create an input job blob after a job is defined by using the `insert_blob` subcommand.

## Supported Job Type

This attribute is optional for the following job types:

- [Command \(CMD\)](#)
- [HTTP \(HTTP\)](#)
- [Web Service Document/Literal Job \(WSDOC\)](#)

## Syntax

This attribute has the following format:

```
blob_input: <auto_blobt>input</auto_blobt>
```

- ***input***

Specifies the text to insert for the blob.

### NOTE

You can specify multiple lines of text. All text enclosed in the `auto_blobt` XML-style metatags is stored literally in the database.

## Example: Create an Input Job Blob

This example creates a job blob from the text that is specified in the `blob_input` attribute statement and uses the blob as input data for the command job named `test_blob`.

```
insert_job: test_blob
job_type: CMD
command: cat
machine: unixagent
blob_input: <auto_blobt>multi-lined text data for job blob
</auto_blobt>
```



```
std_in_file: $$blobt
owner: produser@unixagent
```

## box\_failure Attribute -- Define Criteria for Box Job Failure

The `box_failure` attribute defines dependencies that indicate when a box job has not completed successfully. The `box_failure` condition is in addition to the status of each job in the box.

By default, AutoSys Workload Automation considers a box job to have failed when any job in it completes with a failure condition. However, a box job can contain complex logic that can take a number of different paths, any of which might constitute failure or one of which can include recovery from a failed job. You can use the `box_failure` attribute to define specific conditions for box job failure in situations where there might be multiple failure or recovery scenarios. Jobs within the box that fail still result in the box being marked as failed. The `box_failure` condition allows jobs outside the box to be included in determining the box status in addition to other conditions such as global variables.

You can base box job failure on one or more of the following:

- Job status; for example, `failure(DB_BACKUP)`
- Job status across instances; for example, `failure(jobB^PRD)`
- Jobs status with look-back; for example, `success (job1,01.00)`
- Job exit code; for example, `exitcode (my_job)=4`
- Global variables; for example, `VALUE(TODAY)=Friday`

You can also define conditions that meet *any* of the following criteria:

- **Specify that a job that is contained in the box job enters a certain state**  
AutoSys Workload Automation evaluates the overall failure of the box job when the specified job enters the specified state, regardless of the states of other jobs that are contained in the box job. The condition is not met if the specified job is not scheduled. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the `RUNNING` state.
- **Specify that a job that is not contained in the box job enters a certain state**  
*AutoSys Workload Automation evaluates the overall failure of the box job when one of the jobs that is contained in the box job completes after the specified job (which is not contained in the box) enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the `RUNNING` state.*
- **Specify that a job that is defined on an external instance enters a certain state**  
AutoSys Workload Automation evaluates the overall failure of the box job when one of the jobs that is contained in the box completes after the specified external job enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified external job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the box job and it remains in the `RUNNING` state.
- **Specify a global variable**  
AutoSys Workload Automation evaluates the overall failure of the box job when *one* of the jobs that is contained in the box job completes *after* the global variable is defined and set to the value that is specified in the condition. The condition is not met if all of the jobs that are contained in the box job complete before the global variable is set to the specified value. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the `RUNNING` state.

### NOTE

To define a global variable and store it in the database, issue a [SET\\_GLOBAL event](#) using the `sendevent` command.

### NOTE

For more information about how AutoSys Workload Automation evaluates a box job completion state, see [Control How AutoSys Workload Automation Evaluates a Box Job Completion State](#).

## Support Job Types

This attribute applies only to the BOX job type and is optional.

### Syntax

This attribute has the following format:

```
box_failure: conditions
```

- **conditions** Specifies the job failure conditions using a logical expression. The box job is evaluated against these conditions in addition to all job statuses of jobs within the box. If no job failure condition is specified, the box job fails if at least one job in the box fails.

**Limits:** Up to 4096 characters

Keep the following in mind when defining conditions for box job failure:

- You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- Escape the job name that contains a colon with a backslash (\).
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following attribute definition indicates that the box job is considered to have failed when either JobA and JobB return FAILURE or JobD returns FAILURE (for example, if you set up JobD to attempt to recover a failed job):

```
box_failure: (failure(JobA) AND failure(JobB)) OR failure(JobD)
```

### Example: Fail a Box Job when JobA or JobB but Not JobC Fails

This example sets the status of the box job to FAILURE when JobA or JobB fails, but not when JobC fails:

```
box_failure: failure(JobA) OR failure(JobB)
```

### Example: Fail a Box Job when JobA, JobB, and JobC Fail

This example sets the status of the box job to FAILURE only when all three jobs (JobA, JobB, and JobC) fail:

```
box_failure: failure(JobA) AND failure(JobB) AND failure(JobC)
```

## box\_name Attribute -- Identify a Box as Container for a Job

The box\_name attribute identifies an existing box job to put a job in. Boxes let you process a set of jobs as a group. This is useful for setting starting conditions at the box level to “gate” the jobs in the box. You can specify each job’s starting conditions relative to the other jobs in the box as appropriate.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
box_name: name
```

- **name**  
Specifies an existing box job to put the job in.  
**Limits:** Up to 64 characters

**NOTE**

Enclose the box job name that contains a colon with quotation marks (") or precede it with a backslash (\).

**Example: Identify a Box as Container for a Job**

This example puts the job you are defining in an existing box job named Box\_1:

```
box_name: Box_1
```

**box\_success Attribute -- Define Criteria for Box Job Success**

The box\_success attribute defines which of the dependencies specified in the condition attribute should indicate when a box job has completed successfully.

By default, AutoSys Workload Automation considers a box job successful when every job in it completes successfully. However, a box job can contain complex logic that can take a number of different paths, any of which might constitute success. In such cases, some jobs in the box might never need to run, but the default box behavior would be to interpret this as a box failure. You can use the box\_success attribute to define specific conditions for box job success in situations where there might be multiple success scenarios.

You can base box job success on one or more of the following:

- Job status; for example, success (DB\_BACKUP)
- Job status across instances; for example, success (jobB^PRD)
- Jobs status with look-back; for example, success (job1,01.00)
- Job exit code; for example, exitcode (my\_job)=4
- Global variables; for example, VALUE(TODAY)=Friday

You can also define conditions that meet *any* of the following criteria:

- **Specify that a job that is contained in the box job enters a certain state**  
AutoSys Workload Automation evaluates the overall success of the box job when the specified job enters the specified state, regardless of the states of other jobs that are contained in the box job. The condition is not met if the specified job is not scheduled. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.
- **Specify that a job that is not contained in the box job enters a certain state**  
*AutoSys Workload Automation evaluates the overall success of the box job when one of the jobs that is contained in the box job completes after the specified job (which is not contained in the box) enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.*
- **Specify that a job that is defined on an external instance enters a certain state**  
AutoSys Workload Automation evaluates the overall success of the box job when one of the jobs that is contained in the box completes after the specified external job enters the specified state. The condition is not met if all of the jobs that are contained in the box job complete before the specified external job enters the specified state. In this case, AutoSys Workload Automation does not evaluate the box job and it remains in the RUNNING state.
- **Specify a global variable**  
AutoSys Workload Automation evaluates the overall success of the box job when *one* of the jobs that is contained in the box job completes *after* the global variable is defined and set to the value that is specified in the condition. The condition is not met if all of the jobs that are contained in the box job complete before the global variable is set to the specified value. In this case, AutoSys Workload Automation does not evaluate the completion state of the box job and it remains in the RUNNING state.

**NOTE**

To define a global variable and store it in the database, issue a [SET\\_GLOBAL event](#) using the sendevent command.

**NOTE**

For more information about how AutoSys Workload Automation evaluates a box job completion state, see [Control How AutoSys Workload Automation Evaluates a Box Job Completion State](#).

**Supported Job Types**

This attribute applies only to the BOX job type and is optional.

**Syntax**

This attribute has the following format:

```
box_success: conditions
```

- **conditions** Specifies the job success conditions using a logical expression. The box job is evaluated against these conditions. If no job success conditions are specified, the box job succeeds only if all the jobs in the box complete successfully.

**Limits:** Up to 4096 characters

Keep the following in mind when defining conditions for box job success:

- You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- Escape the job name that contains a colon with a backslash (\).
- You can specify even more complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following attribute definition indicates that the box job is considered successful when either JobA and JobB return SUCCESS or JobD and JobE return SUCCESS, FAILURE, or TERMINATED:

```
box_success: (success(JobA) AND success(JobB)) OR (done(JobD) AND done(Job E))
```

**NOTE**

In multiple conditions (that is, if JobC is dependent on SUCCESS of JobA and SUCCESS of JobB), JobC runs when both JobA and JobB succeed, no matter when the first SUCCESS event occurs. This may not be the desired result, for example, for a daily processing cycle where JobA finished yesterday but did not run today, and JobB succeeded today. To avoid this result, you should group the jobs in a box.

**Example: Set a Box Job to SUCCESS when JobA or JobB Succeeds**

This example sets the status of the box job to SUCCESS when JobA succeeds or JobB succeeds, but ignores the status of JobC:

```
box_success: success(JobA) OR success(JobB)
```

**Example: Set a Box Job to SUCCESS when JobA and JobB Complete Successfully, and JobC Completes**

This example sets the status of the box job to SUCCESS when jobs JobA and JobB succeed and JobC completes (regardless of its status):

```
box_success: success(JobA) AND success(JobB) AND done(JobC)
```

**box\_terminator -- Terminate Box on Job Failure**

The box\_terminator attribute statement controls whether the scheduler terminates the parent box when the job fails.

To terminate the parent box, set the value portion of the statement to y or 1 as follows:

```
box_terminator: {y|1}
```

### WARNING

This attribute statement is valid only when you also specify the `box_name` attribute statement in the job definition.

### Supported Job Types

This attribute applies to all job types and is optional.

### Syntax

This attribute has the following format:

```
box_terminator: {y|1}|{n|0}}
```

- **y**  
Instructs the scheduler to terminate the parent box when the job ends in FAILURE.

### NOTE

The scheduler sends a KILLJOB event to terminate the parent box.

- **n**  
(Default) Does not terminate the parent box when the job ends in FAILURE.

### Example: Terminate Box on Job Failure

This example terminates the containing Box job if the job you are defining fails or is terminated.

```
box_terminator: y
box_name: terminatebox
```

## broadcast\_address Attribute -- Specify the Broadcast Address

The `broadcast_address` attribute specifies the IP address of the LAN or subnet of the computer that receives the Wake on LAN (WOL) signal.

### Supported Job Type

This attribute is required for the [Wake on LAN \(WOL\) job type](#).

### Syntax

This attribute has the following format:

```
broadcast_address: ip_address
```

- **ip\_address**  
Specifies the IP address of the LAN or subnet of the computer that receives the Wake on LAN (signal). On UNIX, you can use `ifconfig` to obtain the broadcast address. On Windows, you must calculate the broadcast address by performing a bitwise OR operation on the IP address and the bit complement of the subnet mask. You can obtain the IP address and subnet mask using `ipconfig`. The special broadcast address of 255.255.255.255 will broadcast to the subnet that is local to the agent machine.

**Limits:** Up to 45 characters

### NOTE

The agent must be on the same subnet as the computer it is trying to wake.

### Example: Broadcast the WOL Signal to a Broadcast Address

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server with the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent

pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
```

## chk\_files Attribute -- Verify File Space Required to Start a Job

The `chk_files` attribute defines the minimum amount of file space that must be available on the specified UNIX file system or Windows drive before the job can start.

At run time, the agent checks whether the required space is available on the machine where the job runs. If the requirements are met, the job starts. If the requirements are not met, the agent generates an alarm and the job does not start. The system tries to verify the file space again and start the job. The number of tries is determined using the `n_retrys` attribute. If the `n_retrys` attribute is not specified in the job definition, the number of tries is determined by the `MaxRestartTrys` parameter in the configuration file (UNIX) or the `Max Restart Trys` field in the Administrator utility (Windows). If the required space is still not available after all the restart attempts, the job fails.

By default, the available file space is not checked.

### Supported Job Type

This attribute is optional for the [Command](#) (CMD) job type.

### Syntax

This attribute has the following format:

```
chk_files: locationsize [locationsize...]
```

- **location** Specifies a Windows drive or the full path name to a UNIX file system where the file space is required.

#### NOTE

You can specify multiple location and size pairs. Specify each pair with a space.

UNIX: You can specify the full path to a directory in a file system.

Windows: Only drives are checked. If directories are specified, they are ignored.

- **size** Specifies the required amount of file space. This value must be less than the amount of available space on the file system or drive. Otherwise, the job fails. Optionally, you can specify the size qualifier. Options are the following:
  - **B**  
Specifies that the value is in bytes.
  - **KB**  
Specifies that the value is in kilobytes. This is the default.
  - **M**  
Specifies that the value is in megabytes.
  - **G**  
Specifies that the value is in gigabytes.

**Limits:** Any integer  
**Examples:** 200, 20G

**NOTE**

The entire value can be up to 255 characters.

**Example: Verify the Available File Space on UNIX**

This example checks whether the file system named roots has 100 KB of available space. This example also checks whether the file system named auxfs1 has 120 KB of available space. The specified file space must be available before the job can start.

```
insert_job: unix_chk
job_type: CMD
machine: unixagent
command: /ul/procrun.sh
chk_files: /roots 100 /auxfs1 120
```

**Example: Verify the Available File Space on Windows**

This example checks whether the C: drive has 100 KB of available space and the D: drive has 120 KB of available space. The specified file space must be available before the job can start.

```
insert_job: win_chk
job_type: CMD
machine: winagent
command: "C:\Programs\Payroll\pay.exe"
chk_files: "C: 100 D: 120"
```

**class\_name Attribute -- Specify a Class Name**

The class\_name attribute specifies the name of the Java class to instantiate in a POJO job or the fully qualified Java class of the MBean object in a JMX-MBean Create Instance job.

**Supported Job Types**

This attribute is required for the following job types:

- [JMX-MBean Create Instance \(JMXMC\)](#)
- [POJO](#)

**Syntax**

This attribute has the following format:

```
class_name: class
```

- **class**  
Specifies the Java class to instantiate in a POJO job or the fully qualified Java class of the MBean object in a JMX-MBean Create Instance job.  
Limits: Up to 1024 characters; case-sensitive

**Example: Specify Java Class to Instantiate in a POJO Job**

Suppose that you want to define a POJO job that creates a Java String with value "5" and calls the parseInt method with the created Java String object as an argument. The parseInt method returns a Java Integer object.

```
insert_job: ignore
job_type: POJO
machine: appagent
class_name: java.lang.Integer
method_name: parseInt
```

```
j2ee_parameter: java.lang.String=5
```

### Example: Specify Java Class of the MBean Object

Suppose that you want to create an MBean instance on a JMX server. The job uses the `cdc.jmx.SimpleDynamic` class. The constructor of the class takes a single string parameter with the value "Hello".

```
insert_job: create
job_type: JMXMC
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"
mbean_name: "DefaultDomain:index=CreateInsl,type=cdc.jmx.SimpleDynamic"
class_name: cdc.jmx.SimpleDynamic
jmx_parameter: java.lang.String="Hello"
```

## command Attribute -- Specify a Command or Script to Run

### Contents

The command attribute specifies a command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met. The attribute can also specify arguments to pass to the command.

### Supported Job Type

This attribute is required for the Command (CMD) job type.

### Syntax

This attribute has the following format:

```
command: commandargument...
```

- **command**

Specifies the command, executable, UNIX shell script, application, or batch file to run when all the starting conditions are met.

**NOTE**

Enclose values that contain embedded blanks in quotation mark

**Example:** "C:\Program Files\Application\program.exe"

- **argument...**

Specifies one or more arguments to pass to the command or script at run time.

**NOTE**

Separate each argument with a space. You must specify each argument in the order it is expected in the program.

Single arguments that contain embedded blanks must be delimited by quotes.

**Example:** "a value"



**NOTE**

- The entire value can be up to 1000 characters. You cannot insert a job if the value specified in the command exceeds 1000 characters. To avoid this limitation, embed the entire command in a batch or shell script and specify the full path to the script as the command.
- When issuing commands or defining jobs that run on a different operating system (for example, Windows to UNIX or UNIX to Windows), you must use the syntax appropriate to the operating system where the job will run.
- The specified command, batch file, UNIX shell script, or executable does not have to exist at job definition time. However, it must exist at run time.
- You can use global variables to specify the command name or the command's run-time arguments. Use the `$global_name`, `$$global_name`, or `$$global_name` format to reference global variables.
- Global variable names can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#). Global variable names can also contain spaces.

**UNIX Considerations for the command Attribute**

When you define a CMD job to run a UNIX script or command, consider the following points when specifying the command attribute:

Permission level of the job owner:

The job owner must have execute permission for the command on the client.

Syntax restrictions:

- UNIX systems are case-sensitive. The specified path and script name must match the path and script name on the UNIX system or the job will fail.
- Piping or redirection of standard input, output, and error files is not allowed. Use shell scripts to execute piped commands and attributes (such as `std_in_file` for standard input) to provide the necessary functionality.

Running a C script:

If you are running a C-Shell (csh) script, the system attempts to source a `.cshrc` file when it begins interpreting the file. Although this may be preferred, the system also overwrites variables defined in the profile script. If you do not want to have the `.cshrc` file sourced, you must invoke the csh script with the `-f` option. In this case, the first line of the script resembles the following:

```
#!/bin/csh -f
```

Using environment variables that are defined in the job profile:

- The command specified in the command attribute runs in the environment defined in the `/etc/auto.profile` file, the job profile, and the `envvars` attribute. The job profile is a shell script and can be specified in the job definition using the profile attribute. For more information about the job profile, see the documentation about profile attribute and [Job Profiles](#).
- If `$PATH` is assigned in the job profile, AutoSys Workload Automation searches that path to find the command. When you specify the full path name, you can use variables exported from the profile script in the path name specification. If you use variable substitution, enclose the variable in braces (`{ }`). For example, you can specify the following:  
`${PATH}`
- You can define the entire environment needed for the command in the profile that will be sourced. Alternatively, you can define environment variables at the job level using the `envvars` attribute.
- If a command works properly when issued at a shell prompt, but fails to run properly when specified as a command attribute, the shell and AutoSys Workload Automation environments are probably different. In this situation, check that all required command variables are specified in the job profile or `envvars` attribute.

Specifying the command or script name without the full path:

The command attribute usually requires the full path to the command or script name you want to run. However, you can specify the name without the full path if all of the following conditions are true:

- The agent is running under the root account.
- The agent is configured to resolve environment variables.
- Using the owner attribute, you specify a user ID that has the authority to run the job on the agent computer. The agent uses the user default shell.
- The path to the command name is set in the PATH system environment variable for the specified user ID.
- The `oscomponent.lookupcommand=true` parameter is set in the `agentparm.txt` file of the agent.

Specifying a script that calls a second script:

When you define a job that runs a script that calls a second script, the fully qualified path of the called script must be provided.

Specifying the command script name using an environment variable:

You can specify the command or script name using an environment variable, for example `$MY_PATH/myscript.sh`, if all of the following conditions are true:

- The agent is running under the root account.
- The agent is configured to resolve environment variables.
- Using the owner attribute, you specify a user ID that has the authority to run the job on the agent computer. The agent uses the user default shell.
- The environment variable used, for example `$MY_PATH`, is set in the specified user ID's profile file.

Determining which shell is used to run the UNIX script:

The shell can be specified in different places. To run a UNIX script, the agent uses, in the following order, the shell specified in the following places:

1. The shell attribute (if specified in the job definition)
2. The first line of the script (if the shell attribute is not specified)
3. The `oscomponent.defaultshell` parameter in the `agentparm.txt` file (if not specified in the shell attribute or in the script)
4. The user default shell defined in the user profile (if not specified in one of the previous three locations)

#### NOTE

- If the parameter that instructs the system to automatically verify shell validity (`oscomponent.checkvalidshell`) is set to true in the agent parameter file (`agentparm.txt`), take the following actions:
  - Specify all shells in the valid shell parameter (`oscomponent.validshell`).
  - Ensure that the path defined in the first line of the script or in the job definition matches the corresponding path in the valid shell parameter.
- Use only shells that are defined on the agent, otherwise the job fails.

Supporting chained commands:

To support chained commands (commands separated by a semicolon) on UNIX, ensure that the `oscomponent.wrapper.exec.force` parameter in the `agentparm.txt` file is set to false (the default). When this parameter is set to false, the `SEND_SIGNAL` event is not supported. If you set the `oscomponent.wrapper.exec.force` parameter to true, the `exec` function is supported and chained commands are not supported. To support chained commands and the `SEND_SIGNAL` event, take the following actions:

- Set the `oscomponent.wrapper.exec.force` parameter to true and move the chained commands to a script file.
- Update the command attribute in the job definition to execute the script file instead of the chained command.

**NOTE**

For more information about supporting chained commands, see [Upgrade Unicenter AutoSys JM 4.5.1 or r11 to the Current Release](#).

**Examples Specifying the command Attribute on UNIX**

The following examples are Command Jobs on UNIX:

**Example: Run a Command on UNIX**

This example runs the UNIX date command on the UNIX machine named unixagent. If the /bin directory is included in the search path, either in the /etc/ auto.profile or in the user-defined profile, you do not have to specify the full path to the command. Instead, you can specify the command name only (for example, command: date).

```
insert_job: unix_cmd
job_type: CMD
machine: unixagent
command: /bin/date
```

**Example: Run a Script on UNIX**

This example runs the Backup script that is located in the /usr/common directory. The job runs on the UNIX machine named unixagent. If the /usr/common directory is included in the job's runtime environment path, you do not have to specify the full path to the script. Instead, you can specify the script name only (for example, command: backup)

```
insert_job: unix_script
job_type: CMD
machine: unixagent
command: /usr/common/backup
```

**Example: Remove All Files from a Subdirectory on UNIX**

This example removes all files from the tmp subdirectory on the unixagent machine. The tmp subdirectory is located in the directory specified in the MY\_BACKUPS global variable.

```
insert_job: unix_remove
job_type: CMD
machine: unixagent
command: rm ${MY_BACKUPS}/tmp/*
```

**Example: Redirect I/O and Pipe Output on UNIX**

This example redirects I/O and pipes output from a command. The agent runs the UNIX command ps -ef|grep someValue and pipes output to the /tmp/log file. The shell interpreter is invoked in the command attribute. The command values are passed as positional parameters to the interpreter.

```
insert_job: unix_job
job_type: CMD
machine: unixagent
command: /usr/bin/ksh -c "ps -ef|grep someValue >>/tmp/log"
owner: user1
```

**Windows Considerations for the command Attribute**

When you define a CMD job to run a Windows command, consider the following points when specifying the command attribute:

### Permission level of the job owner:

When the command defined in the attribute affects a file system that supports Windows security mechanisms (for example, NTFS), the job's owner must have read and execute permission for that command. The job's owner must also have access to all resources and files referenced in the command.

### Syntax restrictions:

- You cannot redirect standard input, output, and error files in the command attribute. Use the `std_in_file`, `std_out_file`, and `std_err_file` attributes instead.
- When you specify a drive letter in a job definition, escape the colon with quotation marks or backslashes as in the following examples:  
`C:\tmp\`  
`"C:\tmp"`

### Redirecting the standard error file:

If the Windows command specified in the job definition does not exist, the job does not run. The standard error file is not created and the job log indicates that the error file is not found.

### Starting a job interactively:

By default, all Command jobs start in batch mode. You can define a CMD job to start interactively on Windows. Interactive mode lets users view and interact with jobs that invoke Windows Terminal Services or user interface processes. To start a job interactively, specify the **interactive: y** attribute in your job definition.

### Using the environment in the job profile:

- Environment variables for the command are defined by a default profile or the profile specified in the job definition.
- Although system environment variables are automatically set in the command's environment, user environment variables are not. You must define all other required environment variables in the job's profile.
- If a command works properly when issued at the command prompt, but fails to run properly when specified as a command attribute, the necessary user-defined environment variables and the variables defined in the job profile are probably different. If this is the case, use the Profiles Manager to verify that all required user environment variables are defined in the job profile.

### Running Windows executable files:

- You can run the following executable file types using the command attribute:
  - Binary files (.exe)
  - Batch files (.bat, .cmd)
  - Command files (.com)
- To run Windows applications, such as `ipconfig.exe`, specify the command file in the command attribute, as shown in the following example:  
`command: "c:\winnt\system32\ipconfig.exe"`
- To run Windows operating system commands, such as `DIR` or `TYPE`, ask your agent administrator to set the following parameters in the `agentparm.txt` file to true:  
`oscomponent.lookupcommand=true`  
`oscomponent.cmdprefix.force=true`

When these parameters are set, the path to the command interpreter is automatically prefixed to the command. Therefore, you can specify the command and arguments in the command attribute (for example, `command: "dir c:\temp\"`). If these agent parameters are not set, you must explicitly invoke the command interpreter in the command attribute, as shown in the following example:

```
command: "c:\winnt\system32\cmd.exe /c dir c:\temp\"
```

- To allow commands that have spaces in their path run without error, ask your agent administrator to set the following parameter in the `agentparm.txt` file to true:

```
oscomponent.cmdprefix.force.quotes.full=true
```

When you set this parameter to true, the agent wraps the entire command in double quotes before the Windows command interpreter runs the command.

- You can run other types of executable files from the Windows command line than those that are directly supported by the command attribute. (The executable files you can run are defined in the Windows system variable PATHEXT.) There are two ways to run those executable files using the agent:
  - Run the executable from a batch file (.bat) or command file (.cmd)
  - Invoke the program's interpreter and pass the command as a parameter using the command attribute.
- For some GUI applications, if there is a command line interface, you can run the application by specifying the batch file name using the command attribute. For example, you can transfer files using the FTP DOS commands or send e-mail in a batch file using software such as BatMail.

Specifying a directory other than the agent's working directory:

By default, the agent's working directory is the directory where you install the agent. If your executable file inputs or outputs to a different directory, or runs another executable file in a different directory, specify that directory. For example, if your batch file runs an executable file, you can do the following:

- Specify the full path of the executable within the batch file:
 

```
C:\Program files\prog\espcmd1.bat
C:\Program files\prog\espcmd2.bat
C:\payroll\data\pay.exe
```
- Use the Change Directory (cd) command to switch to the directory:
 

```
cd C:\Program files\prog
espcmd1.bat
espcmd2.bat
cd C:\payroll\data
pay.exe
```

If your batch file takes input and output information, use the cd command to switch to the directory where the input and output files reside. In this example, file ftptest.bat reads the ftp command from the file ftpscript.src. The batch file outputs to file ftp.out in the current working directory.

```
cd d:\temp\script
ftp -n -i -s:ftpscript.src 131.50.30.28 >ftp.out
user guest
verbose
lcd d:\temp\data
cd /u1/guest
get data1
quit
```

If you run ftptest.bat from the command line, manually switch to the d:\temp\script directory before running the file. The Windows operating system knows the current working directory, so you do not need to include the cd d:\temp\script command in the batch file.

If you run ftptest.bat using the agent, however, the agent does not know the working directory. Specify the cd d:\temp\script command in the batch file, or the ftp command will fail.

Specifying the command or script name without the full path:

The command attribute usually requires the full path to the command or script name you want to run. However, you can specify the command or script name without the full path if all of the following conditions are true:

- The agent is configured to search for paths to command or script files.
- The script or command file is located in one of the following directories:

- The directory the agent is installed in
- WINDOWS\system32 directory on the agent computer
- WINDOWS\system directory on the agent computer
- WINDOWS directory on the agent computer
- Any other directory whose path is set in the system path or user path on the agent computer

### **Examples Specifying the command Attribute on Windows**

The following examples are Command Jobs on Windows:

#### **Example: Run a Command on Windows**

This example runs the `as_test` command on the Windows machine named `winagent`. The `as_test` command waits for 5 seconds and exits with a return code of 0.

```
insert_job: win_cmd
job_type: CMD
machine: winagent
command: as_test -t 5 -e 0
```

#### **Example: Run a Batch File on Windows**

This example runs the Backup batch file that is located in the `C:\COMMON` directory. The job runs on the Windows machine named `winagent`. If `C:\COMMON` is included in the run-time environment, either through the system environment variables or a job profile, you do not have to specify the full path to the file. Instead, you can specify the file name only (for example, `command: Backup`).

```
insert_job: win_batch
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
```

#### **Example: Remove All Files from a Subdirectory on Windows**

This example removes all files from the `tmp` subdirectory on the `winagent` machine. The `tmp` subdirectory is located in the directory specified in the `MY_BACKUPS` global variable.

```
insert_job: win_remove
job_type: CMD
machine: winagent
command: c:\:\WINDOWS\system32\cmd.exe "/c del /s /q ${MY_BACKUPS}\tmp*.*"
```

#### **Example: Run a Visual Basic Script on Windows**

In this example, the agent invokes the Visual Basic interpreter `wscript.exe` and passes the Visual Basic script `notify.vbs` as a parameter using the command attribute.

```
insert_job: win_vbjob
job_type: CMD
machine: winagent
command: "c:\winnt\system32\wscript.exe" "c:\programs\vbs\notify.vbs"
```

#### **Example: Run the Windows Notepad Application in Interactive Mode**

This example opens the configuration text file (`config.txt`) in the Windows Desktop application named Notepad.

```
insert_job: edit_file
```

```

job_type: CMD
machine: winagent
description: "Edit/review a configuration file"
command: notepad.exe "c:\run_info\config.txt"
interactive: y

```

## compatibility Attribute -- Specify the Agent Compatibility Version

Specifies the version of the HTTP Job.

**Default:** 11

Options are as follows:

- **11**  
Specifies releases before Version 12.0
- **12**  
Specifies Version 12.0 and higher.

If the specified value is 11, the job definition allows only keywords from before Version 12.0. If the specified value is 12, all new keywords are allowed.

### NOTE

#### Version 12 - Compatibility Considerations

- If **preemptive\_authentication** is set to true, it is mandatory to specify **j2ee\_conn\_user**. Also, you must specify BASIC as one of the authentication protocols. If not, the HTTP job is not inserted.
- If the **blob\_input** attribute is specified, **compatibility** must be specified as 12.
- **provider\_url** can contain path and query parameters.  
**Example:** provider\_url: "https://webservicemachine:9443/AEWS/jil?verify=true"

## condition Attribute -- Define Starting Conditions for a Job

### Contents

The condition attribute specifies job dependencies to use as starting conditions for the job you are defining. The job cannot run until all specified dependencies evaluate to true. You can base dependencies on one or more of the following:

- Job status; for example, success(DB\_BACKUP)
- Job status across instances; for example, success(jobB^PRD)
- Job exit code; for example, exitcode(my\_job)=4
- Jobs status with look-back; for example, success(job1,01.00)
- Jobs status across instances with look-back; for example, success(jobA^PRD,24)
- Jobs exit code with look-back; for example, exitcode(my\_job2,12)=4
- Jobs exit code across instances with look-back; for example, exitcode(my\_job3^PRD,02.30)=1
- Global variables; for example, VALUE(TODAY)=Friday

These conditions are described in the sections that follow.

### Supported Job types

This attribute is optional for all job types.

### Syntax

This attribute has the following formats:

```
condition: [(] condition[]] [(AND|OR) [(] condition[]]]
condition: [(] condition,look_back[]] [(AND|OR) [(] condition,look_back[]]]
```

- **condition**

Defines any combination of dependencies based on job status, job exit codes, or global variables.

**NOTE**

- To configure complex conditions, combine a series of conditions with the AND or the OR logical operators. You can use the pipe symbol (|) instead of the word OR, and the ampersand symbol (&) instead of the word AND.
- Spaces between conditions and delimiters are optional.
- Escape the job name that contains a colon with a backslash (\).
- You can specify complex conditions by grouping the expressions in parentheses. The parentheses force precedence, and the equation is evaluated from left to right. For example, the following condition definition lets the dependent job start when either JobA and JobB return SUCCESS or JobD and JobE return SUCCESS, FAILURE, or TERMINATED:  

```
condition: (success(JobA) AND success(JobB)) OR (done(JobD) AND done(JobE))
```
- In multiple conditions (that is, if JobC is dependent on SUCCESS of JobA and SUCCESS of JobB), JobC runs when both JobA and JobB succeed, no matter when the first SUCCESS event occurs. This may not be the appropriate result, for example, for a daily processing cycle where JobA finished yesterday but did not run today, and JobB succeeded today. To avoid this result, group the jobs in a box.

- **look\_back**

Defines the last run of the condition on the predecessor job. To support look-back in AutoSys Workload Automation, you must add an additional operand to the condition statement.

**NOTE**

- The look-back feature is applied to job status, cross instance or external dependency job status, and exit code condition types only. Look-back is not applied to the global variable condition type.
- For more information about the look-back syntax and how to use it, see [Look-Back Dependencies](#).
- If you specify a condition for an undefined job, the condition evaluates as false and any jobs dependent on the condition do not run. Use job\_depends to check for this type of invalid condition statement.
- You can use the autorep -J command to generate a report that lists the job definitions stored in the database. The autorep command inserts spaces between the job's conditions so that the report is easy to read. For example, if you specify **condition: s(a)&s(b)&s(c)** in your job definition, autorep outputs **condition: s(a) & s(b) & s(c)** in the report.

## **Job Status Dependencies**

You can define a starting condition for a job so that it starts when another job returns a specific status. For example, you might specify that JobB starts when JobA returns a SUCCESS status and JobC starts when JobB returns a SUCCESS status.

To define a starting condition based on a job's status on AutoSys Workload Automation, use the following format in the condition attribute:

```
status(job_name)
```

- **status**

Specifies one of the following:

- **Done**

Indicates that the job you are defining may run when *job\_name*'s status is SUCCESS, FAILURE, or TERMINATED.



**NOTE**

You can specify d instead of Done.

**Example:** d(jobA)

- **Failure**

Indicates that the job you are defining may run when *job\_name*'s status is FAILURE.

**NOTE**

You can specify f instead of Failure.

**Example:** f(jobA)

- **Notrunning**

Indicates that the job you are defining might run when *job\_name*'s status is any value except STARTING, RUNNING, WAIT\_REPLY, RESTART, or SUSPENDED. Set the status to notrunning to prevent the dependent job from running at the same time as *job\_name*.

**NOTE**

You can specify n instead of Notrunning.

**Example:** n(jobA)

- **Success**

Indicates that the job you are defining may run when *job\_name*'s status is SUCCESS.

**NOTE**

You can specify s instead of Success.

**Example:** s(jobA)

- **Terminated**

Indicates that the job you are defining may run when *job\_name*'s status is TERMINATED. A status of TERMINATED means the job was killed.

**NOTE**

You can specify t instead of Terminated.

**Example:** t(jobA)

- ***job\_name***

Specifies the name of a job. The job you are defining depends on this job.

**NOTE**

- You can use the max\_exit\_success attribute to control the value of the SUCCESS status for a specific job. When you define the max\_exit\_success attribute, a job that exits with an exit code less than or equal to the specified value is treated as a success. FAILURE means the job exited with an exit code higher than the max\_exit\_success value. All other status settings are internally defined and you cannot control them.
- You can use uppercase or lowercase characters to define conditions. You cannot mix case.

### **Example: Define Starting Conditions Based on Job Success**

This example (specified in the job definition for JobA) runs JobA if JobB succeeds.

```
condition: success(JobB)
```

### **Example: Define Starting Conditions Based on the Status of Multiple Jobs**

This example (specified in the job definition for JobC) starts JobC only when both JobA and JobB complete successfully or when both JobD and JobE complete (regardless of whether they failed, succeeded, or terminated).

```
condition: (success(JobA) AND success(JobB)) OR (done(JobD) AND done(JobE))
```

### **Example: Define Starting Conditions Based on Job Failure**

This example (specified in the job definition for JobA) runs JobA if JobB fails.

```
condition: failure(JobB)
```

#### Example: Define Starting Conditions Based on Job Success in a Given Duration

This example (specified in the job definition for JobA) runs JobA if JobB has completed successfully in the last 12 hours.

```
condition: success(JobB,12)
```

#### Example: Define Starting Conditions Based on the Status of Multiple Jobs in a Given Duration

This example (specified in the job definition of JobA) runs JobA if JobB has failed in the last 30 minutes and JobC has completed successfully on external instance XYZ in the last 90 minutes.

```
condition: failure(JobA,00.30) AND success(JobC^XYZ,01.30)
```

#### Example: Define Starting Conditions with the notrunning Operator

One use of the notrunning operator could be to avoid a database dump (DB\_DUMP) and a file backup (BACKUP) at the same time, which would cause frequent hard disk access. This example (specified in the job definition for JobA) runs JobA only when both of these resource-intensive jobs are **not** running.

```
condition: notrunning(DB_DUMP) AND notrunning(BACKUP)
```

### Cross-Instance Job Dependencies

To define a cross-instance job dependency, add the following attribute to your job definition:

```
condition: status(JOB_NAME^INS) [AND|OR status(JOB_NAME^INS) ...]
```

- **status**

Specifies the status that the external job must have before your job starts. When the external job completes with the specified status, the job dependency is satisfied. Options are the following:

- done
- failure
- notrunning
- success
- terminated

- **JOB\_NAME**

Specifies the name of the external job that the local job depends on.

**Example:** JOBA

**CA AutoSys WA Connect Option Notes:** Due to naming limitations in the mainframe environment, the names of jobs specified as job dependencies between AutoSys Workload Automation and CA AutoSys WA Connect Option must follow these guidelines:

- The first character of a job name must be an uppercase letter (A-Z), a pound sign (#), an at sign (@), or a dollar sign (\$).
- The remaining characters in the job name can be any combination of uppercase letters (A-Z), numbers (0-9), pound signs (#), at signs (@), and dollar signs (\$).
- All letters (A-Z) must be in uppercase.
- Job names can be up to 8 alphanumeric characters.

#### **NOTE**

**CA UJMA Notes:** If the external job runs on a distributed system, the names of jobs specified as job dependencies between AutoSys Workload Automation and CA UJMA must follow these guidelines: If the external job runs on the mainframe, the names of jobs specified as job dependencies between AutoSys

Workload Automation and CA UJMA must follow the same guidelines as previously described for CA AutoSys WA Connect Option.

- **^**  
Indicates that the job resides on an external instance.
- **INS**  
Specifies the ID of the external instance that *JOB\_NAME* runs on.  
**Limits:** The value must be three uppercase alphanumeric characters.  
**Example:** PRD

#### NOTE

You can specify multiple job dependencies in a definition.

#### Example: Specify a Cross-Instance Job Dependency

This example defines a job that runs only when the following starting conditions are met:

- jobA on the same instance returns SUCCESS
- jobB on the external instance PRD returns SUCCESS

```
insert_job: dep_job1
machine: localhost
job_type: CMD
command: sleep 100
condition: success(jobA) AND success(jobB^PRD)
```

#### Example: Define Starting Conditions Based on a Cross-Instance Job Status

This example (specified in the job definition for JobA) runs JobA only when the job DB\_BACKUP (which resides on another AutoSys Workload Automation instance named PRD) succeeds.

```
condition: success(DB_BACKUP^PRD)
```

#### Exit Code Dependencies

You can base job dependencies on exit codes that indicate completed tasks. For example, you can specify that the system runs JobB if JobA fails with an exit code of 4.

To define a starting condition based on a job's exit code, use the following format in the condition attribute:

```
exitcode (job_name) operatorvalue
```

- **job\_name**  
Specifies the name of a job. The job you are defining depends on this job.  
**Limits:** Up to 64 characters
- **operator**  
Specifies one of the following comparison operators:  
= Equals  
!= Does not equal  
< Less than  
> Greater than  
<= Less than or equal to  
>= Greater than or equal to
- **value**  
Defines the value to compare against.  
**Limits:** Up to 14 characters

#### Example: Define Starting Conditions Based on Job Exit Code

This example (specified in the job definition for JobB) runs JobB when JobA completes with an exit code of 4.

```
condition: exitcode (JobA) = 4
```

### **Global Variable Dependencies**

You can base job dependencies on global variables set using the sendevent command. When using global variables, the value of the variable must evaluate to TRUE to satisfy the job dependency.

To define a starting condition based on a global variable, use the following format in the condition attribute:

```
VALUE(global_name) operatorvalue
```

- ***global\_name***  
Identifies a global variable that is already defined.  
Limits: Up to 64 alphanumeric characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-)
- ***operator***  
Specifies one of the following comparison operators:  
= Equals  
!= Does not equal  
< Less than  
> Greater than  
<= Less than or equal to  
>= Greater than or equal to
- ***value***  
Defines the value to compare against.  
**Limits:** Up to 100 alphanumeric characters. Values with embedded blanks must be enclosed with quotation marks.  
**Example:** value(VAR1)="SOME DATA"

### **Example: Define Starting Conditions Based on a Global Variable Value**

This example (specified in the job definition for JobA) runs JobA only if the value of the global variable OK\_TO\_RUN is greater than 2.

```
condition: VALUE(OK_TO_RUN)>2
```

### **Example: Define Starting Conditions Based on Multiple Condition Types**

This example (specified in the job definition for JobA) runs JobA only if the job BACKUP completes with a SUCCESS status and the global variable TODAY has a value of Friday.

```
condition: success(BACKUP) AND value(TODAY)=Friday
```

### **Look-Back Dependencies**

The look-back feature is defined as the last run of the condition or predecessor job. The last run is defined as the ending time of the last successful run of the job. If the condition or predecessor job has run successfully after the job which has the condition or predecessor job defined, then the condition or predecessor is satisfied and the job starts. If not, then the condition or predecessor job is not satisfied and the job for which the predecessor job condition is defined is not started.

To support the look-back feature, you must add an additional operand to the condition statement. You can apply the look-back feature to job status, cross-instance or external dependency job status, and exit code condition types only. You cannot apply it to the global variable condition type.

To define a starting condition based on a look-back value, use the following format in the condition attribute:

```
status(job_name, hhhh.mm)
status(job_name^INS, hhhh.mm)
exitcode(job_name, hhhh.mm) operator value
exitcode(job_name^INS, hhhh.mm) operator value
```

- **hhhh**

Indicates the hours taken for the last run of the condition or predecessor job. You can look back approximately 416.58 days.

**Limits:** 0-9999

- **mm**

Indicates the minutes taken for the last run of the condition or predecessor job.

**Limits:** 0-59 when specifying minutes; 0-9998 for *hhhh* when specifying hours and minutes (for example, 9998.59)

**Examples:** 0, 00.15, 06.30, 23.30, 24, 98.30, 720, 9998.59, 9999

#### NOTE

- When the elapsed time to be specified is less than one hour, you must specify the hourly value (hhhh) as "00". For example, "00.30" represents 30 minutes. A value of "30" indicates an elapsed time of 30 hours. Specifying an elapsed time of ".30" is not valid.
- When you specify 0, AutoSys Workload Automation examines the last end time of the job first. It then examines the last end time of the condition job. If the condition job has run since the last run of the job for which the condition is coded for, the job is allowed to start. If the condition job has not run since the last run of the job for which the condition is coded for, the job is not allowed to start.
- When you specify 9999, AutoSys Workload Automation looks back indefinitely (the default behavior of Unicenter AutoSys JM 4.5.1 and previous releases). If the condition job has run, the job is allowed to start. This is the same as not specifying a look-back value.
- Instead of using a period (.) to separate hhhh and mm, you can use a colon that must be escaped (\:) as shown in the following example:  

```
condition: success(JobA,01\:00) and failure(JobB,02\:15)
```

You must use an escape character (backslash) with any colons used in an attribute's value because JIL parses on the combination of keywords followed by a colon.
- If the predecessor job being evaluated for the look-back condition is currently in an ON\_ICE status, it always evaluates to true. That is, any look-back evaluation is ignored.

#### Example: Specify a Look-back Dependency

This example starts the dependent job if JobA is successful during the last hour and JobB failed during the last 2 hours and 15 minutes.

```
condition: success(JobA,01.00) and failure(JobB,02.15)
```

#### Example: Specify Look-Back Conditions

This example shows a job definition with look-back conditions.

In the following job definition, the command job test\_sample\_04 can only start if all of the following conditions are met:

- The last run of test\_sample\_01 completed successfully during the last 12 hours.
- The last run of test\_sample\_02 completed with a FAILURE status during the last 24 hours.
- The last run of test\_sample\_03 completed successfully at any time.

```
insert_job: test_sample_04
machine: localhost
command: sleep 10
```

```
condition: success(test_sample_01,12.00) AND failure(test_sample_02,24.00) AND
success(test_sample_03)
```

## condition\_code Attribute -- Specify a Code to Indicate Success or Failure

The `condition_code` attribute specifies a condition code that indicates success or failure. You can define condition codes for jobs, steps, procedure steps, or programs.

For each condition code or range of condition codes, you can indicate whether the scheduling manager interprets the code or range as a success or a failure. You can also indicate whether the job continues or stops running. Each specification becomes a separate item. Together these specifications form a list of condition codes.

### Supported Job Type

This attribute is optional for the [z/OS Regular](#) (ZOS) job type.

### Syntax

This attribute has the following format:

```
condition_code: rc=return_code
 [,result= SUCCESS|FAILURE]
 [,action= CONTINUE|STOP]
 [,proc_step= procedure_name]
 [,program= program_name]
 [,step_name= step_name]
```

- **rc=return\_code**

Specifies the return code using one of the following values:

- A condition code for one number or a range of numbers between 1 and 4095 (ranges are separated by colons, such as 1:4095).
- A system abend code (*Sccc*), such as S0C1 or SB37.
- A user abend code (*Unnnn*), such as U0001 or U0462. The *nnnn* must be four decimal digits and cannot exceed 4095.

**Limits:** Up to 9 characters

- **result=SUCCESS|FAILURE**

(Optional)

Specifies whether to consider the job as failed based on the condition code. Options are the following:

- **SUCCESS**  
Considers the job not failed if the condition code is matched. This is the default.
- **FAILURE**  
Considers the job failed if the condition code is matched.

- **action=CONTINUE|STOP**

(Optional) Specifies whether the job continues with the next step regardless of whether the job is considered as failed. Options are the following:

- **CONTINUE**  
Continues with the next step if the condition code is matched. This is the default.
- **STOP**  
Stops the job if the condition code is matched.

- **proc\_step=procedure\_name**

(Optional)

Specifies a valid z/OS procedure name.

**Limits:** Up to eight characters; first character must not be numeric

- **program=program\_name**

(Optional)

Specifies a valid z/OS program name.

**Limits:** Up to eight characters; first character must not be numeric

- **step\_name=step\_name**

(Optional)

Specifies a valid z/OS step name.

**Limits:** Up to eight characters; first character must not be numeric

#### NOTE

- You can specify multiple condition codes. Use a separate condition\_code attribute for each code. For example, to specify the 111 and 112 condition codes for your job, define a condition\_code attribute for 111 and a condition\_code attribute for 112.
- The entire value can be up to 4096 characters.

### Example: Specify Multiple Condition Codes

This example runs a z/OS Regular job. The job is considered failed if the return code is 114 or 115. The job is not considered failed if the return code is 111, 112, or 113. The job continues with the next step only if the return code is 111, 112, or 114. Otherwise, the job stops running.

```
insert_job: zos_job
job_type: ZOS
machine: zagent
jcl_library: CYBJAK1.JCL
jcl_member: CYBJAK11
condition_code: rc=111,result=SUCCESS,action=CONTINUE,step_name=xyz
condition_code: rc=112,result=SUCCESS,action=CONTINUE
condition_code: rc=113,result=SUCCESS,action=STOP
condition_code: rc=114,result=FAILURE,action=CONTINUE
condition_code: rc=115,result=FAILURE,action=STOP
```

## connect\_string Attribute -- Specify Database Resource Location

The SQL attribute specifies the database resource location for a database job.

### Supported Job Types

This attribute is optional for the following job types:

- [Database Monitor](#) (DBMON)
- [Database Stored Procedure](#) (DBPROC)
- [Database Trigger](#) (DBTRIG)
- [SQL](#)

### Syntax

This attribute has the following format:

```
connect_string: "dburl"
```

- **"dburl"**

Specifies a JDBC database resource location for a job. Use the appropriate format for your database type:

- For an Oracle database:

```
connect_string: "jdbc:oracle:thin:@host:port:dbname"
```

- For a Microsoft SQL Server database:

```
connect_string: "jdbc:sqlserver://host:port;DatabaseName=dbname"
```

- For an IBM DB2 database:

```
connect_string: "jdbc:db2://host:port/dbname"
```

- For a Sybase database:

```
connect_string: "jdbc:sybase:Tds:host:port/dbname"
```

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- If you do not specify this attribute in the job definition, a default database resource location must be defined in the db.default.url parameter in the agent's agentparm.txt file. Otherwise, the job fails.
- The connect\_string attribute overrides the default resource location specified in the agent's agentparm.txt file.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- You can use global variables to specify the database resource location. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables. AutoSys Workload Automation does not perform any validations if the value of the connect\_string attribute starts with a global variable. However, if the global variable is included in the middle or at the end of the value, AutoSys Workload Automation validates the value specified before the global variable.

#### Example: Specify an Oracle Database

This example uses the Oracle database named ORDERS that is reachable on port 1433 on host 172.31.255.255.

```
insert_job: QRY1

job_type: SQL

machine: DB_agent

owner: sys@orcl

sql_command: SELECT * from NEWWORDS

destination_file: "c:\log\qry1.txt"

connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

#### Example: Specify a Microsoft SQL Server Database



This example uses the Microsoft SQL Server database named ORDERS that is reachable on port 1433 on a host named myhost.

```
insert_job: QRY2

job_type: SQL

machine: DB_agent

owner: sa@myhost

sql_command: SELECT * from NEWWORDS

destination_file: "c:\log\qry2.txt"

connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=ORDERS"
```

### Example: Specify an IBM DB2 Database

This example updates a record in the STAFF table under the user entadm. The job changes the years to 3 for the employee with the name Jonson.

```
insert_job: UPDATE

job_type: SQL

machine: DB_AGENT

owner: entadm@myhost

sql_command: UPDATE ENTADM.STAFF SET YEARS=3 where NAME='Jonson'

destination_file: "c:\log\updatejob.txt"

connect_string: "jdbc:db2://172.31.255.255:50000/SAMPLE"
```

### Example: Specify a Sybase Database

This example monitors the ap\_invoices table for changes. The job connects to the Sybase database named APDB that is reachable on port 5001 on a host named myhost. The job completes when the table has changed.

```
insert_job: db_sqltrig_upd

job_type: DBTRIG

machine: localhost

dbtype: Sybase
```

```
connect_string: "jdbc:sybase:Tds:myhost:5001/APDB"

trigger_type: UPDATE

tablename: ap_invoices

owner: admin@myhost
```

## connection\_factory Attribute -- Specify the JNDI Name of the Connection Factory

The connection\_factory attribute specifies the connection factory JNDI name in a JMS job.

### Supported Job Types

This attribute is required for the following job types:

- [JMS Publish \(JMSPUB\)](#)
- [JMS Subscribe \(JMSSUB\)](#)

### Syntax

This attribute has the following format:

```
connection_factory: factory
```

- ***factory***

Specifies the connection factory JNDI name. The connection factory contains all the bindings needed to look up the referenced topic or queue. JMS jobs use the connection factory to create a connection with the JMS provider.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

You can use global variables to specify the database resource location. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables. AutoSys Workload Automation does not perform any validations if the value of the connection\_factory attribute starts with a global variable. However, if the global variable is included in the middle or at the end of the value, AutoSys Workload Automation validates the value specified before the global variable.

### Example: Specify Connection Factory in a JMS Publish Job

This example publishes the message "this is my message" to the queue named Queue. The Java class of the message is String. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the cyberuser@host JNDI user name to gain access to the connection factory named ConnectionFactory.

```
insert_job: publish
job_type: JMSPUB
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue
use_topic: FALSE
```

```
message_class: String
j2ee_user: cyberuser@host
j2ee_parameter: java.lang.String="this is my message"
```

**NOTE**

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

**connection\_retry Attribute -- Specify the Number of Times to Retry a Connection Failure**

The `connection_retry` attribute specifies the number of times to retry when there is a connection failure.

**Supported Job Types**

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

This attribute has the following format:

```
connection_retry: number_of_retries
```

- ***number\_of\_retries***  
Specifies the number of times to retry when there is a connection failure.  
**Default: 3**

**connection\_timeout Attribute -- Defines When the Connection Times Out**

The `connection_timeout` attribute specifies the duration, in seconds, after which the request times out.

**Supported Job Types**

This attribute is optional for the [HTTP job type](#) and [Define a Web Service Document/Literal Job](#) type.

**Syntax**

This attribute has the following format:

```
connection_timeout: time_in_seconds
```

- ***time\_in\_seconds*** Specifies the duration after which the request times out.  
**Default: 0**

**continuous Attribute -- Specify Whether the Job Monitors Continuously**

The `continuous` attribute specifies whether a job continuously monitors for a condition.

**NOTE**

If you do not specify the `continuous` attribute in your job definition, the job completes the first time the condition is met (the default).

**Supported Job Types**

This attribute is optional for the following job types:

- Database Monitor (DBMON)
- Database Trigger (DBTRIG)
- File Trigger (FT)
- JMS Subscribe (JMSSUB)
- JMX-MBean Subscribe (JMXSUB)
- SAP Event Monitor (SAPEVT)
- SAP Process Monitor (SAPPM)

**NOTE**

This attribute does not apply to File Trigger jobs if EXIST or NOTEXIST is specified for the watch\_file\_type attribute.

**Syntax**

This attribute has the following format:

```
continuous: Y | N
```

- **Y**  
Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the scheduler log file (event\_demon.\$AUTOSERV on UNIX and event\_demon.%AUTOSERV% on Windows).

**NOTE**

- To end continuous monitoring, you must complete the job manually by issuing the following command:  
sendevent -E KILLJOB -J job\_name
- To specify continuous: Y for SAP Event Monitor jobs, you must specify N in the sap\_is\_trigger attribute.

- **N**  
Specifies that the job completes the first time the condition is met. This is the default.

**File Trigger Notes:**

- If the file trigger type is CREATE, EXPAND, SHRINK, or UPDATE, the File Trigger job will not complete unless it is forced to complete.
- If the file trigger type is DELETE, the File Trigger job completes when all the monitored files are deleted or it is forced to complete.
- The following applies to using wildcards in the file name value with the continuous attribute:
  - If the file trigger type is CREATE or DELETE, the file name value must contain a wildcard (\* or ?), for example, /usr/data/pay\*.
  - If the file trigger type is EXPAND or UPDATE and the file name value contains a wildcard, the agent monitors all files that match the file name criteria when the file trigger is set. When the criteria is satisfied for any of the selected files, the job triggers and the agent continues to monitor the files.

**NOTE**

If the continuous attribute is specified with the watch\_no\_change attribute and a wildcard is used in the file name value, all of the matching files must remain unchanged for the duration of the watch\_no\_change interval. If another file satisfying the trigger criteria is updated during the watch\_no\_change interval, the trigger waits until the trigger interval lapses on the first file before monitoring the second file for the duration of the watch\_no\_change interval.

**Example: Continuously Monitor the Same File Using Multiple File Trigger Jobs**

In this example, two File Trigger jobs monitor the same file for a change in size by 10 KB. One job monitors for an increased change in size and the other job monitors for a decreased change in size. The jobs are independent and do not relate to each other in any way.

```

insert_job: ftjob1
job_type: FT
machine: ftagent
watch_file: "c:\data\totals"
watch_file_type: EXPAND
continuous: Y
watch_file_change_type: DELTA
watch_file_change_value: 10240 /* The value must be entered in bytes (10 x 1024 bytes = 10240) */
insert_job: ftjob2
job_type: FT
machine: ftagent
watch_file: "c:\data\totals"
watch_file_type: SHRINK
continuous: Y
watch_file_change_type: DELTA
watch_file_change_value: 10240 /* The value must be entered in bytes (10 x 1024 bytes = 10240) */

```

Suppose that the initial file size of totals (c:\data\totals) is 100 KB and it changes as follows:

100 KB (initial size), 80 KB, 90 KB, 110 KB, 50 KB, 60 KB

The following triggers occur:

- Because the file trigger type is EXPAND, the first job (ftjob1) writes an alert to the scheduler log file once when the file size changes from 90 KB to 110 KB.
- Because the file trigger type is SHRINK, the second job (ftjob2) writes an alert to the scheduler log file twice. The triggers occur when the file shrinks from 100 KB to 80 KB and then again when the file shrinks from 110 KB to 50 KB.

The jobs end when they are forced to complete.

## copy\_jcl Attribute -- Specify a Data Set Name that Receives the Copy JCL

The copy\_jcl attribute specifies the data set name that receives the working copy of the JCL you submitted.

### Supported Job Type

This attribute is optional for the [z/OS Regular \(ZOS\) job type](#).

### Syntax

This attribute has the following format:

```
copy_jcl: data_set_name
```

- **data\_set\_name** Specifies the data set name that receives the working copy of the JCL you submitted. Data set names typically have up to four positions; positions must each be eight characters or less and separated by periods.

**Limits:** Up to 44 characters

**Example:** prod.copyjcl.weekly

### Example: Store a Working Copy of the JCL that You Submitted

Suppose that the agent ZOS1 submits the JCL in member CYBDL01A in the CYBDL01.JCLLIB library. If the job fails, you can modify a working copy of the JCL in the CYBDL01.COPY.JCLLIB data set, and resubmit the job without affecting the JCL source.

```

insert_job: CYBDL01A
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB

```

```
jcl_member: CYBDL01A
owner: CYBDL01
copy_jcl: CYBDL01.COPY.JCLLIB
```

## correlation\_id Attribute -- Specify a Message Correlation ID

The `correlation_id` attribute specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue.

### NOTE

If the message is published to a topic or consumed from a topic, the correlation ID is ignored.

### Supported Job Types

This attribute is optional for the following job types:

- [JMS Publish \(JMSPUB\)](#)
- [JMS Subscribe \(JMSSUB\)](#)

### Syntax

This attribute has the following format:

```
correlation_id: correlation_id
```

- ***correlation\_id*** Specifies the correlation ID that is associated with a message that is sent to a queue or consumed from a queue. **Limits:** Up to 64 characters

### NOTE

- If you do not specify the `correlation_id` attribute in your job definition, the system generates a unique correlation id. The job that wants to consume the message must refer to the predecessor job so that the `correlation_id` value of the message that was last posted by the job is fetched and the same is consumed by the successor job. In this case, the predecessor job must run first and then the successor.
- If you specify the `correlation_id` attribute in your job definition and it does not refer to a job, it is treated as a literal string that the consumer consumes if the `correlation_id` attribute value matches any of the messages in the queue. In this case, the order of the predecessor and successor does not matter as we are referring to a literal value and not a system generated id that can change with each run of the job.

### Example: Specify the Message Correlation ID in a JMS Job

This example defines a JMS Publish job that is named `jmssub3` and a JMS Subscribe job that is named `jmssub3`.

The `jmssub3` job publishes the message "Hello World" to the `weblogic.examples.jms.exampleQueue` queue. The Java class of the message is `String`. The correlation ID of the message is `jms_corr_id`. The initial context factory that is supplied by the JNDI service provider is `weblogic.jndi.WLInitialContextFactory`. The service provider's URL is `t3://jms_server:7001`, where `jms_server:7001` is the IP address of the Weblogic Application Server and 7001 is the port. The job uses the `weblogic@mach` user name to gain access to the connection factory that is named `weblogic.examples.jms.QueueConnectionFactory`.

```
insert_job: jmssub3
job_type: JMSPUB
machine: jms
owner: weblogic@mach
connection_factory: weblogic.examples.jms.QueueConnectionFactory
initial_context_factory: weblogic.jndi.WLInitialContextFactory
use_topic: false
```

```

provider_url: "t3://jms_server:7001"
message_class: String
destination_name: weblogic.examples.jms.exampleQueue
j2ee_parameter: String="Hello World"
correlation_id: jms_corr_id

```

The jmssub3 job consumes the message with the jms\_corr\_id correlation ID from the weblogic.examples.jms.exampleQueue queue.

```

insert_job: jmssub3
job_type: JMSSUB
machine: jms
owner: weblogic@mach
connection_factory: weblogic.examples.jms.QueueConnectionFactory
initial_context_factory: weblogic.jndi.WLInitialContextFactory
use_topic: false
provider_url: "t3://jms_server:7001"
destination_name: weblogic.examples.jms.exampleQueue
correlation_id: jms_corr_id

```

### Example: Specify the Job Name as the Correlation ID

This example defines a JMS Publish job that is named jmssub4 (the correlation\_id is not specified in this job definition) and a JMS Subscribe job that is named jmssub4.

The jmssub4 job publishes the message "Hello World" to the weblogic.examples.jms.exampleQueue queue. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is weblogic.jndi.WLInitialContextFactory. The service provider's URL is t3://jms\_server:7001, where jms\_server:7001 is the IP address of the WebLogic Application Server and 7001 is the port. The job uses the weblogic@mach user name to gain access to the connection factory that is named weblogic.examples.jms.QueueConnectionFactory.

```

insert_job: jmssub4
job_type: JMSPUB
machine: jms
owner: weblogic@mach
connection_factory: weblogic.examples.jms.QueueConnectionFactory
initial_context_factory: weblogic.jndi.WLInitialContextFactory
use_topic: false
provider_url: "t3://jms_server:7001"
message_class: String
destination_name: weblogic.examples.jms.exampleQueue
j2ee_parameter: String="Hello World"

```

The jmssub4 job uses the jmssub4 job name as its correlation id to consume the message from the weblogic.examples.jms.exampleQueue queue.

```

insert_job: jmssub4
job_type: JMSSUB
machine: jms
owner: weblogic@mach
connection_factory: weblogic.examples.jms.QueueConnectionFactory
initial_context_factory: weblogic.jndi.WLInitialContextFactory

```

```

use_topic: false
provider_url: "t3://jms_server:7001"
destination_name: weblogic.examples.jms.exampleQueue
correlation_id: jmspub4

```

## create\_method Attribute -- Specify a Create Method

The `create_method` attribute specifies the name of the create method in a Session Bean job.

### Supported Job Type

When accessing a stateful session bean, this attribute is required for the [Session Bean \(SESSBEAN\)](#) job type.

### Syntax

This attribute has the following format:

```
create_method: create_method
```

- ***create\_method***

Specifies the name of the create method.

**Default:** create

**Limits:** Up to 256 characters; case-sensitive; it must always begin with create

**Example:** createaccount

### Example: Create a Stateful Session Bean for an Online Shopping Cart

Suppose that you want to access a stateful session bean for an online shopping cart. The `createaddbook` method creates the `Shoppingcart` stateful bean for the duration of the job. The `addbook` method adds books to the shopping cart using the book's ISBN number. In this example, the Session Bean job adds two books to the shopping cart with ISBN numbers 1551929120 and 1582701709. When the job runs, two books are added to the shopping cart.

```

insert_job: addbook
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
bean_name: Shoppingcart
create_method: createaddbook
method_name: addbook
create_parameter: String="ISBN"
j2ee_parameter: Integer[2]="1551929120,1582701709"

```

## create\_name Attribute -- Specify a Create Method

The `create_name` attribute specifies the name of the create method in an Entity Bean job.

### NOTE

If you do not specify the `create_name` attribute in your job definition, the job uses `create` for the create method (the default).

### Supported Job Type

When the operation type is CREATE, this attribute is optional for the [Entity Bean \(ENTYBEAN\)](#) job type.

### Syntax

This attribute has the following format:



```
create_name: create_method
```

- **create\_method**

Specifies the name of the create method.

**Default:** create

**Limits:** Up to 256 characters; case-sensitive; it must always begin with create

**Example:** createaccount

### Example: Create an Entity Bean

Suppose that you want to create an entity bean that stores information about a customer such as the customer ID and phone number. The initial context factory supplied by the JNDI service provider is weblogic.jndi.WLInitialContextFactory. The service provider's URL is t3://localhost:7001, where localhost is the domain name of the WebLogic application server and 7001 is the ORB port. When the job runs, the entity bean instance is created.

```
insert_job: create
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
create_name: createcustomer
operation_type: CREATE
create_parameter: String="customerid", String="800-555-0100"
```

## create\_parameter Attribute -- Specify the Create Parameters

The create\_parameter attribute specifies the create parameters in an Entity Bean and Session Bean job. You can specify create parameters to create an entity bean in a relational database on your application server or to access a stateful session bean.

### NOTE

To access stateful session beans, you must specify create parameters.

### Supported Job Type

This attribute is optional for the following job types:

- [Entity Bean](#) (ENTYBEAN)
- [Session Bean](#) (SESSBEAN)

### Syntax

This attribute has the following format:

```
create_parameter: type=value | payload_job=job_name[, type=value |
payload_job=job_name...]
```

- **type=value**

Specifies the Java class and String value for the parameter.

**Limits:** Up to 1024 characters; case-sensitive

**Examples:** String="5:00PM", String[2]="winnt1,winnt2", double=2000

### NOTE

If the package of the Java class is not specified, java.lang is assumed.

- **payload\_job=job\_name** Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

**NOTE**

- You can define up to 64 create parameters using one `create_parameter` attribute or multiple entries of `create_parameter`.
- In each `create_parameter` entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4096 characters.
- Enclose the job name that contains a colon with quotation marks (") or precede it with a backslash (\).
- Order is important. Specify the parameters in the same order as they appear in the create method. The job fails if the parameters are listed in the wrong order.
- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

**Example: Specify Create Parameter to Access a Stateful Session Bean**

Suppose that you want to access a stateful session bean for an online shopping cart. The `createaddbook` method creates the `Shoppingcart` stateful bean for the duration of the job. The `addbook` method adds books to the shopping cart using the book's ISBN number. In this example, the `Session Bean` job adds two books to the shopping cart with ISBN numbers 1551929120 and 1582701709. When the job runs, two books are added to the shopping cart.

```
insert_job: addbook
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
bean_name: Shoppingcart
create_method: createaddbook
method_name: addbook
create_parameter: String="ISBN"
j2ee_parameter: Integer[2]="1551929120,1582701709"
```

**cpu\_usage Attribute -- Specify Whether to Monitor Available or Used CPU**

The `cpu_usage` attribute specifies whether the job monitors for available or used CPU.

**NOTE**

If you do not specify the `cpu_usage` attribute in your job definition, the job monitors for available CPU (the default).

**Supported Job Type**

This attribute is optional for the [CPU Monitoring \(OMCPU\) job type](#).

To use this attribute, you must specify `WAIT` or `CONTINUOUS` for the `monitor_mode` attribute. This attribute does not apply when the monitor mode is `NOW`.

**Syntax**

This attribute has the following format:

```
cpu_usage: FREE | USED
```

- **FREE**  
Specifies that the job monitors the available CPU processing capacity.  
This is the default.
- **USED**  
Specifies that the job monitors the used CPU processing capacity.

**NOTE**

- The CPU usage reported as greater than 100 percent occurs when a partitioned or virtualized machine has overcommitted its CPU allocation and was able to use unused CPU from the rest of the machine.
- On UNIX, the CPU calculations are based on load averaging as displayed by the command `top`. You can scale the results the agent returns with the `objmon.cpu.scalefactor` parameter. For more information about the `objmon.cpu.scalefactor` parameter, see [Workload Automation Agents Parameters for Troubleshooting](#).
- You must specify the `lower_boundary` attribute, the `upper_boundary` attribute, or both when using the `monitor_mode` attribute set to the `WAIT` value.
- The `cpu_usage` attribute does not apply when using the `monitor_mode` attribute set to the `NOW` value.

**Example: Monitor Used CPU**

This example monitors the used CPU on the unixagent computer. The job completes when the used CPU is less than 20 percent or greater than 80 percent.

```
insert_job: omcpu_job
job_type: OMCPU
machine: unixagent
lower_boundary: 20
upper_boundary: 80
inside_range: FALSE
cpu_usage: USED
```

**date\_conditions Attribute -- Base Job Start on Date and Time Attribute Values**

The `date_conditions` attribute specifies whether to use the date or time conditions defined in the following attributes to determine when to run the job:

- `autocal_asc`
- [days\\_of\\_week](#)
- [exclude\\_calendar](#)
- `must_complete_times`
- `must_start_times`
- `run_calendar`
- `run_window`
- `start_mins`
- `start_times`
- `timezone`

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

```
date_conditions: y | n
```

- **y**  
Uses the date and time conditions defined in other attributes to determine when to run the job.

**NOTE**

You can specify 1 instead of y.

- **n**  
Ignores the date and time conditions defined in other attributes to determine when to run the job. This is the default.

**NOTE**

You can specify 0 instead of n.

**Example: Base Job Start on Date and Time Attribute Values**

This example uses starting date and time conditions specified in the `autocal_asc`, `days_of_week`, `exclude_calendar`, `run_calendar`, `run_window`, `start_mins`, `start_times`, `must_start_times`, `must_complete_times`, and `timezone` attributes in the job definition.

```
date_conditions: y
```

**days\_of\_week Attribute -- Specify which Days of the Week to Run a Job**

The `days_of_week` attribute specifies one or more days of the week on which to run the job you are defining.

When you define the `days_of_week` attribute, you must also define either the `start_times` or `start_mins` attribute. AutoSys Workload Automation schedules the job to run at the times specified in the `start_times` or `start_mins` attribute on the specified days of the week.

**NOTE**

The `days_of_week` attribute has no effect when you also define the `run_calendar` attribute.

The `date_conditions` attribute controls the usage of this attribute.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following formats:

```
days_of_week: day [,day...]
days_of_week: all
```

- **day**  
Specifies the days of the week when the job runs. Options are the following:
  - `mo` -- Specifies Monday.
  - `tue` -- Specifies Tuesday.
  - `we` -- Specifies Wednesday.
  - `th` -- Specifies Thursday.
  - `fr` -- Specifies Friday.
  - `sa` -- Specifies Saturday.
  - `su` -- Specifies Sunday.

**Default:** No days are selected.

**NOTE**

- You can specify multiple days. Separate each day with a comma (,).
- The entire value can be up to 80 characters.

- **all**  
Specifies that the job runs every day of the week.

**Example: Specify to Run a Job Only on Weekdays**

This example specifies that the job runs only on weekdays.

```
date_conditions: y
days_of_week: mo, tu, we, th, fr
```

**Example: Specify to Run a Job on Every Day of the Week**

This example specifies that the job runs every day of the week.

```
date_conditions: y
days_of_week: all
```

**dbtype Attribute -- Specify the Type of Database**

The dbtype attribute specifies the type of the database that the job monitors. Depending on the database type, AutoSys Workload Automation generates the proper syntax for the database trigger if it monitors multiple actions. For example, Microsoft SQL Server requires the actions to be separated by commas, and Oracle requires that the actions be separated by the "or" operand. AutoSys Workload Automation automatically changes the actions to the proper syntax according to the dbtype value, if needed.

**NOTE**

The dbtype attribute is not used for validating the connection URL type.

**Supported Job Type**

This attribute is required for the [Database Trigger](#) (DBTRIG) job type.

**Syntax**

This attribute has the following format:

```
dbtype: Oracle | MSSQL | DB2 | SYBASE | POSTGRESQL | TERADATA
```

- **Oracle**  
Specifies that the job monitors an Oracle database.
- **MSSQL**  
Specifies that the job monitors a Microsoft SQL Server database.
- **DB2**  
Specifies that the job monitors an IBM DB2 database.
- **SYBASE**  
Specifies that the job monitors a Sybase database.
- **POSTGRESQL**  
Specifies that the job monitors a PostgreSQL database.
- **TERADATA** Specifies that the job monitors a TERADATA database.

**NOTE**

You can specify only one [trigger\\_type](#) for the DB2 and TERADATA databases.

**Example: Specify the Oracle Database Type**

This example monitors the Inventory\_List table in the Oracle database named ORDERS. Because the dbtype is Oracle, AutoSys Workload Automation changes the "INSERT,DELETE" syntax to "INSERT OR DELETE". When a row is inserted or deleted, the job completes.

```

insert_job: dbtrig_oracle
job_type: DBTRIG
machine: dbagent
tablename: Inventory_List
dbtype: Oracle
trigger_type: INSERT,DELETE
owner: sys@orcl
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"

```

### Example: Specify the Sybase Database Type

This example monitors the `ap_invoices` table for changes. The job connects to the Sybase database named `APDB` that is reachable on port 5001 on a host named `myhost`. The job completes when the table has changed.

```

insert_job: db_sqltrig_upd
job_type: DBTRIG
machine: localhost
dbtype: Sybase
connect_string: "jdbc:sybase:Tds:myhost:5001/APDB"
trigger_type: UPDATE
tablename: ap_invoices
owner: admin@myhost

```

## description Attribute -- Define a Text Description for a Job

The `description` attribute defines a text description for the job you are defining.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
description: "text"
```

- **"text "**  
Defines an optional free-form text description of the job.  
**Limits:** Up to 500 alphanumeric characters (including spaces)

### Example: Define a Text Description for a Job

This example specifies that the job you are defining is an incremental daily backup of the database.

```
description: "incremental daily backup of the database"
```

## destination\_file Attribute -- Specify an Output Destination File

The `destination_file` attribute specifies the output destination file in a job. In a payload producing job, the file stores the Java serialized object produced by the job. In an SQL job, the file stores the SQL query results. In a HTTP job, the file stores the output of the invocation, which if not specified would have been stored in a spool file.

**NOTE**

- If you omit this attribute in a payload producing job, the Java serialized object is stored in the spool directory on the agent computer using the job name and a numeric suffix as the file name. If you omit this attribute in an SQL or HTTP job, the output is stored in the job's spool file.
- For Web Service Document/Literal payload producing jobs, only the portion of the returned data that satisfies the `job_criteria` specification will be stored in the output file for use as payload.

**Supported Job Types**

This attribute is optional for the following job types:

- [HTTP](#) (HTTP)
- [JMS Subscribe](#) (JMSSUB)
- [POJO](#)
- [RMI](#) (JAVARMI)
- [Session Bean](#) (SESSBEAN)
- [SNMP Value Get](#) (SNMPGET)
- [SNMP Value Set](#) (SNMPSET)
- [SQL](#)
- [Web Service Document/Literal](#) (WSDOC)
- [Web Service RPC/Encoded](#) (WBSVC)

**Syntax**

This attribute has the following format:

```
destination_file: destination
```

- ***destination*** Specifies the output destination file. In a payload producing job, the file stores the Java serialized object produced by a job. In an SQL job, the file stores the SQL query results. In a HTTP job, the file stores the output of the invocation.

**Limits:** Up to 256 characters; case-sensitive

**Examples:** /tmp/myoutput, "D:\Stock output\STOCK3"

**NOTE**

- Specify the values that contain embedded blanks with double quotation marks.
- Your agent administrator can specify a default destination file for all jobs by setting the `spooldir` parameter in the agent's `agentparm.txt` file.
- The `destination_file` attribute overrides the default file specified in the agent's `agentparm.txt`.
- For HTTP jobs, the `destination_file` attribute is not applicable if you have set the `invocation_type` attribute value to `delete`.
- You can use global variables to specify the output destination file. Use the `$$global_name`, `$(global_name)`, or `$$${global_name}` format to reference global variables.

**Example: Save Output from a Session Bean Job in a File**

Suppose that you want to invoke the `reverse` method on the `CybEJBTestBean` stateless session bean. The `reverse` method has one parameter, with type `java.lang.String` and value `a23`. The output from the `reverse` method is saved in the `C:\Makapt15` file. The initial context factory supplied by the JNDI service provider is `com.ibm.websphere.naming.WsnInitialContextFactory`. The service provider's URL is `corbaloc:iiop://172.24.0.0:2809`, where `172.24.0.0` is the IP address of the WebSphere application server and `2809` is the ORB port. When the job runs, the output of the `reverse` method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
```

```

machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"

```

### Example: Return Data from a Database Table that Match a Condition

This example queries a database table named emp for an emp\_name value that starts with S, followed by at least two characters. If the result starts with S, followed by one other character, the job completes. Otherwise, the job fails. The job's output is written to c:\temp\sql\_job1\_out.log.

```

insert_job: sql_job1
job_type: SQL
machine: SQL_server1
sql_command: select emp_name from emp where emp_name like 'S%'
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
success_criteria: emp_name=S.*
destination_file: "c:\temp\sql_job1_out.log"

```

### destination\_name Attribute -- Specify the JNDI Name of the Topic or Queue

The destination\_name attribute specifies the JNDI name of the topic or queue in a JMS job.

#### NOTE

Topics are not supported on IBM WebSphere or IBM WebSphere MQ.

### Supported Job Types

This attribute is required for the following job types:

- [JMS Publish](#) (JMSPUB)
- [JMS Subscribe](#) (JMSSUB)

### Syntax

This attribute has the following format:

```
destination_name: destination
```

- **destination** Specifies the JNDI name of the topic or queue. The job uses the JNDI name to indicate the destination where messages are received.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- You can use global variables to specify the JNDI name of the topic or queue. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.
- In a JMS Publish job, you can also specify the reply\_to attribute as a value for the destination\_name attribute, as follows:

```
destination_name: reply_to=subscriber
```



The `destination_name` is populated with the queue name that is specified in the `reply_to` attribute of the publisher job, which is consumed and stored by subscriber job that is named subscriber.

### Example: Monitor a Queue on a WebLogic Application Server

This example continuously monitors the queue that is named Queue1 (residing on WebLogic) for a message matching the filter criteria. The consumed messages from the queue are stored in the file `/export/home/user1/outputfile1`. The service provider's URL is `t3://172.24.0.0:7001`, where 172.24.0.0 is the IP address of the WebLogic Application server and 7001 is the ORB port.

```
insert_job: monitor
job_type: JMSSUB
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://172.24.0.0:7001"
connection_factory: ConnectionFactory
destination_name: Queue1
continuous: Y
filter: abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
use_topic: FALSE
destination_file: /export/home/user1/outputfile1
j2ee_user: cyberuser@host
```

In this example, the regular expression used as the filter criteria can be defined as follows:

```
abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
```

- **abc\s**  
Specifies the text abc, followed by white space.
- **...\s**  
Specifies any three characters, followed by white space.
- **[a-zA-Z]+\s**  
Specifies at least one letter, followed by white space.
- **Filter![\sa-z0-9]+**  
Specifies the text Filter!, followed by at least one of the following: white space or digit or lower case letter.

**Example:** abc vvv B Filter! 95

### destination\_user Attribute -- Specify the User Name to Connect to the Topic or Queue

The `destination_user` attribute specifies the user name that is required to connect to the topic or queue in a JMS job.

#### Supported Job Types

This attribute is optional for the following job types:

- [JMS Publish \(JMSPUB\)](#)
- [JMS Subscribe \(JMSSUB\)](#)

#### Syntax

This attribute has the following format:

```
destination_user: user@queue
```

- **user@queue** Specifies the user name that is required to connect to the topic or queue in a JMS job. **Limits:** Up to 145 characters

**NOTE**

If you specify the `destination_user` attribute in your job definition, you must set the password that is associated with the destination user using the `autosys_secure` command.

**Example: Specify the User Name to Connect to the Queue**

This example publishes the message "this is my message" to the queue that is named Queue1 whose owner is queueuser@host. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the cyberuser@host JNDI user name to gain access to the connection factory that is named ConnectionFactory.

```
insert_job: publish
job_type: JMSPUB
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue1
destination_user: queueuser@host
use_topic: FALSE
message_class: String
j2ee_user: cyberuser@host
j2ee_parameter: java.lang.String="this is my message"
```

**NOTE**

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

**disk\_drive Attribute -- Specify the Disk Space to Monitor**

The `disk_drive` attribute specifies the disk space to monitor.

**Supported Job Type**

This attribute is required for the [Disk Monitoring \(OMD\) job type](#).

**Syntax**

This attribute has the following format:

```
disk_drive: drive
```

- **drive**

Specifies the path to the disk, logical partition, or auxiliary storage pool to be monitored.

**Limits:** Up to 256 characters; case-sensitive **Example:** /dev/QASP02

**UNIX/Windows:** Specify the path to the disk or logical partition to monitor.

**i5/OS:**

- Specify the path to a file system mounted on the i5/OS operating system.
- To specify the system ASP, type the forward slash (/).
- To specify any other ASP, use the following format, where digit1 and digit2 indicate the ASP number:

```
/dev/QASPdigit1digit2
```

### Example: Monitor Disk Space on UNIX

This example monitors /export/home for available space. The job completes when the available space is between 8 and 10 gigabytes (GB), and the available space changes by more than 15 percent. If the change in value is less than or equal to 15, the job does not register a change.

```
insert_job: omd_unix
job_type: OMD
machine: unixagt
disk_drive: /export/home
disk_space: FREE
disk_format: GB
lower_boundary: 8
upper_boundary: 10
inside_range: TRUE
no_change: 15
```

### Example: Monitor Disk Space on Windows

This example monitors the C drive on a Windows computer for used space. When the value of disk space used falls below 16 percent or exceeds 95 percent, the job completes.

```
insert_job: omd_win
job_type: OMD
machine: winagent
disk_drive: C
disk_format: PERCENT
lower_boundary: 16
upper_boundary: 95
inside_range: FALSE
disk_space: USED
```

### Example: Monitor Disk Space on i5/OS

This example monitors the system auxiliary storage pool on an i5/OS computer for used space. When 90 to 100 percent of the storage space is used, the job completes.

```
insert_job: omd_i5
job_type: OMD
machine:i5agent
disk_drive: /
disk_format: PERCENT
lower_boundary: 90
upper_boundary: 100
inside_range: TRUE
disk_space: USED
```

### disk\_format Attribute -- Specify the Unit of Measurement Used to Monitor Disk Space

The disk\_format attribute specifies the unit of measurement used to monitor available or used disk space.

**NOTE**

If you do not specify the `disk_format` attribute in your job definition, the job monitors for disk space in percent (the default).

**Supported Job Type**

This attribute is optional for the [Disk Monitoring \(OMD\) job type](#).

**Syntax**

This attribute has the following format:

```
disk_format: PERCENT | GB | MB | KB | B
```

- **PERCENT**

Monitors disk usage by percentage. This is the default.

**NOTE**

If you specify `PERCENT`, the values specified by the `lower_boundary`, `upper_boundary`, and `no_change` attributes can be greater than 100.

- **GB**

Monitors disk usage in gigabytes.

- **MB**

Monitors disk usage in megabytes.

- **KB**

Monitors disk usage in kilobytes.

- **B**

Monitors disk usage in bytes.

**Example: Monitor Disk Space on UNIX**

This example monitors `/export/home` for available space. The job completes when the available space is between 8 and 10 gigabytes (GB), and the available space changes by more than 15 percent. If the change in value is less than or equal to 15, the job does not register a change.

```
insert_job: omd_unix
job_type: OMD
machine: unixagt
disk_drive: /export/home
disk_space: FREE
disk_format: GB
lower_boundary: 8
upper_boundary: 10
inside_range: TRUE
no_change: 15
```

**disk\_space Attribute -- Specify Whether to Monitor Available or Used Disk Space**

The `disk_space` attribute specifies whether the job monitors for available or used disk space.

**NOTE**

If you do not specify the `disk_space` attribute in your job definition, the job monitors for available disk space (the default).

**Supported Job Type**

This attribute is optional for the [Disk Monitoring \(OMD\) job type](#).

## Syntax

This attribute has the following format:

```
disk_space: FREE | USED
```

- **FREE**  
Specifies that the job reads the available disk space for monitoring.
- **USED**  
Specifies that the job reads the used disk space for monitoring.

### NOTE

For a newly created file system on any platform, the operating system usually reports the file system overhead as used bytes.

### Example: Monitor Available Disk Space on UNIX

This example monitors /export/home for available space. The job completes when the available space is between 8 and 10 gigabytes (GB), and the available space changes by more than 15 percent. If the change in value is less than or equal to 15, the job does not register a change.

```
insert_job: omd_unix
job_type: OMD
machine: unixagt
disk_drive: /export/home
disk_space: FREE
disk_format: GB
lower_boundary: 8
upper_boundary: 10
inside_range: TRUE
no_change: 15
```

### Example: Monitor Used Disk Space on Windows

This example monitors the C drive on a Windows computer for used space. When the value of disk space used falls below 16 percent or exceeds 95 percent, the job completes.

```
insert_job: omd_win
job_type: OMD
machine: winagent
disk_drive: C
disk_format: PERCENT
lower_boundary: 16
upper_boundary: 95
inside_range: FALSE
disk_space: USED
```

### Example: Monitor Used Disk Space on i5/OS

This example monitors the system auxiliary storage pool on an i5/OS computer for used space. When 90 to 100 percent of the storage space is used, the job completes.

```
insert_job: omd_i5
job_type: OMD
machine: i5agent
```

```
disk_drive: /
disk_format: PERCENT
lower_boundary: 90
upper_boundary: 100
inside_range: TRUE
disk_space: USED
```

## elevated Attribute -- Specify Whether to Run a Command Job in Elevated Mode on Windows

The elevated attribute specifies whether to run Command jobs (that are run by standard users and require administration privileges) in elevated mode on Microsoft Windows operating systems. In elevated mode, the User Account Control (UAC) feature is enabled. UAC improves the security of Microsoft Windows by limiting application software to standard user privileges until an administrator authorizes an elevation.

If you specify to use elevated mode in a Command job definition, the agent runs the job in elevated mode on Windows. You do not need to respond to the UAC prompt.

### NOTE

For more information about UAC, see the Microsoft documentation.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#) on Windows. This attribute is ignored for Command jobs that run on UNIX.

### Syntax

This attribute has the following format:

```
elevated: y | n
```

- **y**  
Runs the Command job in elevated mode on Windows.
- **n**  
Does not run the Command job in elevated mode on Windows. This is the default.

### NOTE

- The agent works according to Microsoft restrictions, as follows:
  - To schedule elevated jobs, the user specified to run the job must be logged in with an active session.
  - To schedule elevated jobs, the user can have only one active session. If more than one session exists for the specified user, the agent cannot determine which session to use and therefore fails the job.
- If the agent administrator sets the `oscomponent.logon.elevated` parameter to true in the `agentparm.txt` file, the agent runs all Command jobs in elevated mode on Windows, regardless of the elevated attribute. To use the elevated attribute in a job definition, the `oscomponent.logon.elevated` parameter must be set to false.
- By default, the agent uses the default Windows shell `explorer.exe`. The agent administrator can specify an alternative Windows shell for elevated jobs by setting the `oscomponent.shell` parameter in the `agentparm.txt` file.
- If you want to run a Command job that requires administrative privileges, the owner of the job must be part of the administrators group. The job fails if the job owner is not part of the administrators group.
- In general, the behavior of the UAC prompt for administrators and standard users is as follows:
  - When a user from the administrator group (other than the administrator) runs a Command job, the prompt for consent appears if the consent option is enabled. However, if you run the Command job in elevated

mode and the job owner is a member of the administrator group, the job runs without prompting for consent.

- When a standard user runs a Command job, the prompt for consent does not appear. Instead, the prompt to enter the administrator credentials appears. The standard user must provide the administrator user credentials to execute administration tasks or deny the request depending on whether the consent option is enabled or not. However, when the standard user runs a Command job, the job fails as there is no option to provide the administrator credentials from AutoSys Workload Automation.

### Example: Run a Command Job in Elevated Mode on Windows

This example runs a Command job in elevated mode on Windows. The job opens the Computer Management window on the Windows desktop. You do not need to respond to the UAC prompt.

```
insert_job: open_window
job_type: CMD
machine: winagent
description: "Open the Computer Management window"
command: compmgmt.msc
interactive: y
elevated: y
```

### encoding Attribute -- Specify the Character Encoding of a File

The encoding attribute specifies the character encoding of the text file to be monitored.

#### NOTE

If you do not specify the encoding attribute in your job definition, the job monitors the file as US-ASCII (the default).

### Supported Job Type

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\)](#) job type.

### Syntax

This attribute has the following format:

```
encoding: char_set_name
```

- ***char\_set\_name***

Specifies the name of the character set used to encode the data in the file.

**Default:** US-ASCII

**Limits:** Up to 42 characters

**Examples:** ISO-8859-1, UTF-8, UTF-16BE, UTF-16LE, UTF-16

#### NOTE

The supported character sets depends on your operating system and JVM.

### Example: Monitor a Text File Encoded in ISO-8859-1

This example monitors a text file that contains data encoded in the ISO Latin Alphabet No. 1 (also named ISO-LATIN-1). The job checks the text file immediately and completes successfully if the specified string is found. If the string is not found, the job fails.

```
insert_job: textfile_job
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transactions.log
```

```

text_file_filter: ERROR MESSAGE
text_file_mode: LINE
lower_boundary: 1
upper_boundary: 50
encoding: ISO-8859-1
monitor_mode: NOW

```

## endpoint\_URL Attribute -- Specify a Target Endpoint URL

The endpoint\_URL attribute specifies the target endpoint address URL in a Web Service or Process Automation job.

### Supported Job Types

This attribute is required for the following job types:

- [Process Automation Process Execution](#) (PAPROC)
- [Process Automation Start Request Form](#) (PAREQ)
- [Web Service Document/Literal](#) (WSDOC) job type

This attribute is optional for the Web Service RPC/Encoded (WBSVC) job type.

### Syntax

This attribute has the following format:

```
endpoint_URL: endpoint_url
```

- **endpoint\_url** Specifies the target endpoint address URL. In a published WSDL file, the URL defining the target endpoint address is found in the location attribute of the port's soap:address element. In a Process Automation job, it specifies the target endpoint address URL on the CA Process Automation server.  
**Limits:** Up to 256 characters (Web Service RPC/Encoded ) and up to 2048 characters (Process Automation and Web Service Document/Literal); case-sensitive  
**Example:** "http://machinename:8080/itpam/soap"

### NOTE

- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, you must specify both the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL\_URL and endpoint\_URL attributes, then the service\_name and port\_name attributes are optional.
- You can use global variables to specify the target endpoint address URL. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Specify the Target Endpoint Address URL for Getting Stock Quotes

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is http://host:8080/axis/services/StockQuote?wsdl. The WSDL port name within the target namespace http://ws.ca.com is StockQuote. The target endpoint address URL is http://host:8080/axis/services/StockQuote. The job calls the operation GetQuote within the StockQuoteService web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.



```

insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false

```

### Example: Specify the Target Endpoint Address URL for a Process Automation Job

This example executes the /CAWAAE/Test Process process on the Process Automation server at the specified target endpoint address URL.

```

insert_job: execproc
job_type: PAPROC
machine: localhost
endpoint_URL: "http://capamach:8080/itpam/soap"
pa_name: "/CAWAAE/Test Process"
owner: pamadmin@capamach
pa_monitor_progress: Y
pa_parameter: Name="strTestName", Value="PA Test"

```

## envvars Attribute -- Specify a Job's Environment Variables

The envvars attribute specifies environment variables to define the local environment where the script, command, or batch file runs. You can modify existing environment variables or create your own.

### Supported Job Types

This attribute is optional for the following job types:

- [Command \(CMD\)](#)
- [Micro Focus \(MICROFOCUS\)](#)
- [PeopleSoft \(PS\)](#)
- [Remote Execution \(PROXY\)](#)
- [z/OS Regular \(ZOS\)](#)

### Syntax

This attribute has the following format:

```
envvars: parm_name=parm_value[, parm_name=parm_value...]
```

- **parm\_name**  
Defines the name of a new environment variable or specifies the name of an existing environment variable.

**NOTE**

Enclose names that contain spaces in quotation marks.

**Example:** "CMDS 1"

- **parm\_value**

Specifies the value of the environment variable.

**NOTE**

Enclose values that contain spaces in quotation marks.

**Example:** "cmd1 cmd2 cmd3"

**NOTE**

- You can specify multiple parm\_name=parm\_value combinations. Separate each combination with a comma.
- You can use multiple entries of envvars to define different sets of environment variables. For example, you can specify two sets of combinations using two envvars attributes as follows:  

```
envvars: path1="c:\test", "Data Dir"="c:\prod"
```
- Each parm\_name=parm\_value combination can be up to 1024 characters.
- Programs, commands, batch files, and scripts can access environment variables using the getenv() function. Specify variable names for your environment as follows:
  - For UNIX scripts or commands to access environment variables, prefix the variable name with a dollar sign (\$).
  - For Windows batch files or programs to access environment variables, specify the variable name with a leading and trailing percent sign (%).
- The envvars attribute does not override Windows system variables. To reset Windows system-defined variables, set these variables in a batch file. For example, code the following to set variables TMP and TEMP in a batch file:  

```
set TMP=d:\spool
set TEMP=d:\agent\temp
```
- The environment variables defined using the envvars attribute are processed before the job profile specified in the profile attribute. For more information about environment variables and how the environment for a job is sourced, see [Job Profiles](#).

**Example: Pass UNIX Environment Variables to a Script**

This example includes two envvars attributes that pass environment variables to a script and a third envvars attribute that defines the Present Working Directory (PWD). The parameter "user 1" is enclosed with quotation marks because it contains a space.

```
insert_job: unix_job
job_type: CMD
machine: unixprod
command: /home/scripts/pay
envvars: NAME="user 1"
envvars: JOB=PAYROLL
envvars: PWD=/usr/scripts/dailyrun
```

In this example, the pay script can reference these variables:

Environment Variable	Value Passed
NAME	user 1

JOB	PAYROLL
PWD	/usr/scripts/dailyrun

**NOTE**

If the parameter `oscomponent.loginshell` is set to true in the agent's `agentparm.txt` file, the agent invokes the user environment while running the script. To override the value of a shell variable that is already defined in the user login file, reassign a value to this variable in the script.

**Example: Define Two Environment Variables on Windows**

This example runs the `processrun.exe` command file on a Windows computer named `winprod`. The job uses two local environment variables named `PATH` and `TEMP`.

```
insert_job: cmd_job1
job_type: CMD
machine: winprod
command: "c:\cmds\processrun.exe"
envvars: PATH="c:\windows\system32"
envvars: TEMP="c:\temp"
```

**exclude\_calendar Attribute -- Define Days to Exclude a Job from Running**

The `exclude_calendar` attribute identifies a custom calendar to use to determine the days of the week on which to exclude the job you are defining from running. You can use a standard calendar or an extended calendar as the `exclude_calendar`.

Custom calendars are useful for complex date specification, such as when you need to exclude a job from running on the last business day of the month. Custom calendars can specify any number of dates on which to exclude associated jobs from running. Each calendar is stored as a separate uniquely named object in the database and can be associated with one or more jobs. The specified calendar must have previously been created using the `autocal_asc` command.

When you specify an `exclude_calendar` as the only date condition and the job has other implicit or explicit start conditions, the scheduler inspects the calendar before starting the job. If the current date is on the calendar, the job does not start and its status changes to `INACTIVE`. If the job's status changes to `INACTIVE` and it is in a box, the box job completes when all other conditions are satisfied. Also, if the job is a box job and its status changes to `INACTIVE`, all the jobs in the box change to `INACTIVE`.

**NOTE**

Times set in the `start_times` and `start_mins` attributes override times set in a custom calendar. The `date_conditions` attribute controls the usage of this attribute.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

```
exclude_calendar: calendar_name
```

- ***calendar\_name***

Defines the name of a custom calendar that was created with the `autocal_asc` command.

**Default:** No calendar is used.

**Limits:** Up to 64 alphanumeric characters

**Example: Specify the Exclusion Calendar for a Job**

This example specifies the custom calendar `HOLIDAY` that defines days when the job cannot run.

```
date_conditions: y
exclude_calendar: HOLIDAY
```

## external\_os\_user Attribute -- Specify an External User

The `external_OS_user` attribute specifies the user ID of the user with the authority to download the file from the remote FTP server or upload the file to the remote FTP server.

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) and FTP job types.

### Syntax

This attribute has the following format:

```
external_os_user: OS_user
```

- **OS\_user** Specifies the user ID of the user with the authority to download the file from the remote FTP server or upload the file to the remote FTP server. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command.

**Limits:** 80 characters

### NOTE

- You can specify the `external_os_user` attribute only if you set the `run_external` attribute to T.
- You can use this user ID to access remote resources that the agent user does not have access to. You are restricted to how you can access data on remote computers. To access restricted remote resources, you can run the job under a user ID that has access to those resources.
- This user ID must have access to the file location that you specified using the `scp_local_name` attribute.

## fail\_codes Attribute -- Define Exit Codes to Indicate Job Failure

The `fail_codes` attribute defines which exit codes indicate job failure. For example, you can define the `fail_codes` attribute to be 40-50. If a job returns an exit code in the range 40 to 50, the job is considered to have failed.

### NOTE

If you do not specify the `fail_codes` attribute in your job definition, the scheduling manager interprets any exit code other than (0) zero as job failure (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [Command](#) (CMD)
- [HTTP](#) (HTTP)
- [i5/OS](#) (I5)
- [Micro Focus](#) (MICROFOCUS)
- [Remote Execution](#) (PROXY)
- [Web Service Document/Literal](#) (WSDOC)

### Syntax

This attribute has the following format:

```
fail_codes: exit_codes
```

- **exit\_codes** Defines which exit codes indicate job failure. Specify a single exit code (for example, 4), a range of exit codes (for example, 0-9999), a list of multiple exit codes separated by commas, or a list of ranges separated by commas.  
**Limits:** Up to 256 characters. The exit code must be an integer in the range -2147483647 to 2147483647.  
**Default:** Any exit code other than 0 (zero) indicates job failure  
**Examples:** 1,150,20-30

#### NOTE

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

#### Example: Define a Range of Exit Codes to Indicate Job Failure

Suppose that you want a job named CMDJOB to run the procjob.exe file. The job is considered to have failed if the exit code is in the 40-50 range.

```
insert_job: CMDJOB
job_type: CMD
machine: winagt
command: "c:\temp\procjob.exe"
fail_codes: 40-50
```

## filter Attribute -- Specify a Filter Using Regular Expression Logic

The filter attribute specifies a regular expression to use as a filter in a JMS Subscribe, JMX-MBean Subscribe, and HTTP job.

#### Supported Job Types

This attribute is required for the [JMX-MBean Subscribe](#) (JMXSUB) job type if you specify Types in the filter\_type attribute.

This attribute is optional for the following job types:

- [HTTP](#)
- [JMS Subscribe](#) (JMSSUB)

#### Syntax

This attribute has the following format:

- For the JMSSUB and HTTP job types:  
`filter: filter`
- For the JMXSUB job type:  
`filter: attribute | type[,attribute | type...]`
- **filter**  
Specifies a regular expression to use as a filter. In a JMS Subscribe job, this attribute filters messages from the topic or queue. In an HTTP job, this attribute filters the output of the invoked servlet. In a JMX-MBean Subscribe job, this attribute filters the notifications by attribute or by notification type.  
**Example:** .+(CA)+.
- **attribute**  
Identifies an attribute to monitor for change.  
**Limits:** Case-sensitive
- **type**  
Identifies a notification type to monitor for.  
**Limits:** Case-sensitive  
**Example:** jmx.attribute.change (sends a notification every time an attribute changes)

**NOTE**

- Specify the values that contain embedded blanks with double quotation marks.
- The entire value can be up to 256 characters.
- To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**Example: Monitor a Queue on a WebLogic Application Server for a Message Matching Filter Criteria**

This example continuously monitors the queue named Queue (residing on WebLogic) for a message matching the filter criteria. The consumed messages from the queue are stored in the file `/export/home/user1/outputfile1`. The service provider's URL is `t3://172.24.0.0:7001`, where 172.24.0.0 is the IP address of the WebLogic Application server and 7001 is the ORB port.

```
insert_job: monitor
job_type: JMSSUB
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://172.24.0.0:7001"
connection_factory: ConnectionFactory
destination_name: Queue
continuous: Y
filter: abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
use_topic: FALSE
destination_file: /export/home/user1/outputfile1
j2ee_user: cyberuser@host
```

In this example, the regular expression used as the filter criteria can be defined as follows:

```
abc\s...\s[a-zA-Z]+\sFilter![\sa-z0-9]+
```

- **abc\s**  
Specifies the text abc, followed by white space.
- **...\s**  
Specifies any three characters, followed by white space.
- **[a-zA-Z]+\s**  
Specifies at least one letter, followed by white space.
- **Filter![\sa-z0-9]+**  
Specifies the text Filter!, followed by at least one of the following: white space or digit or lower case letter.

**Example:** abc vvv B Filter! 95

**Example: Monitor for a Change to a Specific MBean Attribute Using a Filter**

Suppose that you want to monitor for a change to the `cachesize` attribute of the MBean named Config. The job filters the notifications the MBean sends by attribute. When the `cachesize` attribute changes, the job completes.

```
insert_job: change
job_type: JMSSUB
machine: agent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
filter_type: Attributes
filter: cachesize
```

## filter\_type Attribute -- Specify the Filter Type

The filter\_type attribute specifies whether to filter notifications by attribute or by notification type in a JMX-MBean Subscribe job.

### NOTE

If you do not specify the filter\_type attribute in your job definition, the job filters notifications by attribute (the default).

### Supported Job Type

This attribute is optional for the [JMX-MBean Subscribe \(JMXSUB\) job type](#).

### Syntax

This attribute has the following format:

```
filter_type: Attributes | Types
```

- **Attributes**  
Filters notifications by attribute. This is the default.
- **Types**  
Filters notifications by notification type.

### NOTE

If you specify filter\_type: Types, you must also specify the filter attribute.

### Example: Monitor for a Change to a Specific MBean Attribute Using a Filter

Suppose that you want to monitor for a change to the cachesize attribute of the MBean named Config. The job filters the notifications the MBean sends by attribute. When the cachesize attribute changes, the job completes.

```
insert_job: change
job_type: JMXSUB
machine: agent
URL: "service:jmx:rmi://docops.ca.com/jndi/rmi://localhost:9999/server" scope="local">rmi:///jndi/rmi://
localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
filter_type: Attributes
filter: cachesize
```

### Example: Monitor for Changes to Any MBean Attribute

Suppose that you want to set up continuous monitoring for changes to any attribute of the MBean named Config. Each time an attribute changes, an alert is written to the scheduler log file.

```
insert_job: change
job_type: JMXSUB
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
continuous: Y
filter_type: Types
filter: jmx.attribute.change
```

## finder\_name Attribute -- Specify a Finder Method

The `finder_name` attribute specifies the name of the finder method in an Entity Bean job. You require the finder method to update the property values of an existing entity bean or to remove an entity bean from the database. To find the entity bean, the agent uses the bean's Java Naming and Directory Interface (JNDI) name along with its finder method.

### Supported Job Type

When the operation type is UPDATE or REMOVE, this attribute is required for the [Entity Bean \(ENTYBEAN\)](#) job type.

### Syntax

This attribute has the following format:

```
finder_name: finder_method
```

- ***finder\_method***  
Specifies the name of the finder method.  
**Limits:** Up to 256 characters; case-sensitive

### Example: Update the Phone Number for the Acme Company in a Database

Suppose that you want to update the phone number for the Acme company to 800-555-0199. The customer entity bean stores the customer ID and phone number. The primary key for the customer is the customer ID. To find the entity bean, the job uses the Acme's customer ID. When the job runs, the Acme company's phone number is changed.

```
insert_job: update
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: UPDATE
method_name: changephone
finder_name: findByPrimaryKey
finder_parameter: String="customerid"
modify_parameter: String="800-555-0199"
```

## finder\_parameter Attribute -- Specify Finder Parameters

The `finder_parameter` attribute specifies the finder parameters in an Entity Bean job. You specify finder parameters to update the property values of an existing entity bean or to remove an entity bean from the database.

### Supported Job Type

When the operation type is UPDATE or REMOVE, this attribute is required for the [Entity Bean \(ENTYBEAN\)](#) job type.

### Syntax

This attribute has the following format:

```
finder_parameter: type=value | payload_job=job_name[, type=value |
payload_job=job_name...]
```

- ***type=value***  
Specifies the Java class and String value for the parameter.  
**Limits:** Up to 1024 characters; case-sensitive  
**Examples:** String="5:00PM", String[2]="winnt1,winnt2", double=2000



**NOTE**

If the package of the Java class is not specified, java.lang is assumed.

- **payload\_job=job\_name**

Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

**NOTE**

- Enclose the job name that contains a colon with quotation marks ( " ") or precede it with a backslash (\).
- You can define up to 64 finder parameters using one finder\_parameter attribute or multiple entries of finder\_parameter.
- In each finder\_parameter entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4096 characters.
- Order is important. Specify the parameters in the same order as they appear in the finder method. The job fails if the parameters are listed in the wrong order.

**Example: Update the Phone Number for the Acme Company in a Database**

Suppose that you want to update the phone number for the Acme company to 800-555-0199. The customer entity bean stores the customer ID and phone number. The primary key for the customer is the customer ID. To find the entity bean, the job uses the Acme's customer ID. When the job runs, the Acme company's phone number is changed.

```
insert_job: update
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: UPDATE
method_name: changephone
finder_name: findByPrimaryKey
finder_parameter: String="customerid"
modify_parameter: String="800-555-0199"
```

**form\_parameter Attribute -- Specify Input Parameters**

The form\_parameter attribute specifies the parameters to be passed to a PUT or POST request.

Supported Job Type

HTTP

**Syntax**

This attribute has the following format:

```
name=value
```

Specifies the parameter name and value; comma separated multiple values can be specified.

Limits: Upto 4000 characters; case sensitive

**Example:**

```
jobName="testV2HTTPPostJobWithFormParms_25075534365",comment="start job"
```

**NOTE**

- This attribute is not applicable for GET and DELETE invocation types.
- `media_type` must be specified as "application/x-www-form-urlencoded" when `form_parameter` is specified.
- This attribute is mutually exclusive with `blob_input` and `payload_uri`.

**ftp\_command -- Specify Site-specific FTP Commands**

The `ftp_command` attribute specifies the commands to execute prior to file transfer. You can use this attribute to send site-specific FTP commands to FTP servers.

**Supported Job Type**

This attribute is optional for the [FTP job type](#).

**Syntax**

This attribute has the following format:

```
ftp_command: command[, command...]
```

- **command**

Defines a command that is to be executed prior to file transfer. **Limits:** Up to 4096 characters

**NOTE**

- You can specify multiple commands. Separate each command with a comma.
- You can use multiple entries of `ftp_command` to define different sets of commands. For example, you can specify two sets of commands using two `ftp_command` attributes as follows:
 

```
ftp_command: command1,command2
ftp_command: command3,command4
```
- Each entry of `ftp_command` can be up to 4096 characters.
- This attribute does not support all FTP commands. Use this attribute only to set up the FTP connection so that the data transfers correctly. For example, you can specify this attribute to run the `LOCSITE` command.

**Example: Send FTP Commands to an FTP Server**

This example sends two FTP commands to the FTP server prior to transferring a file.

```
insert_job: CYBJK.FTP
job_type: FTP
machine: ftpagent
ftp_server_name: ftp.ca.com
ftp_server_port: 21
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: B
ftp_compression: 8
ftp_remote_name: /pub/cazip.exe
ftp_local_name: /tmp/bla
ftp_command: locsite blksize=FB
ftp_command: locsite automount
owner: user1@ftp.ca.com
```

**ftp\_compression Attribute -- Specify the Data Compression Level**

The `ftp_compression` attribute specifies the compression level for FTP data transfer.

**NOTE**

If you do not specify the `ftp_compression` attribute in your job definition, the data is compressed using the level set in the `ftp.data.compression` parameter on the agent FTP client. This parameter is automatically set to 0 by default (no data compression).

**Supported Job Type**

This attribute is optional for the [FTP job type](#).

**Syntax**

This attribute has the following format:

```
ftp_compression: level
```

- ***level***

Specifies the data compression level.

**Limits:** 0-9 (0 is no data compression and 9 is the highest data compression)

**Default:** 0

**NOTE**

- To use the compression feature, both the FTP client and the FTP server must run on the agent software. If this value is specified and the FTP server or the FTP client does not run on the agent, the data will be transferred without compression.
- On the agent FTP client, your agent administrator can specify the default compression level for all FTP jobs by setting the `ftp.data.compression` parameter in the agent's `agentparm.txt` file.
- The `ftp_compression` attribute overrides the default compression level specified in the agent's `agentparm.txt`.

**Example: Compress a File**

The local computer in this example has an agent running as an FTP client. The remote server has an agent running as an FTP server. Both computers operate on a low bandwidth network.

The following FTP job downloads a large file named `largefile.txt` from the remote server to the FTP client. The computers are on a low bandwidth network, so the data is compressed at compression level 3.

```
insert_job: FTPJOB
job_type: FTP
machine: ftpagent
ftp_server_name: aixunix
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_compression: 3
ftp_remote_name: /files_to_download/largefile.txt
ftp_local_name: "c:\downloaded_files\largefile.txt"
owner: ftpuser@aixunix
```

**ftp\_local\_name Attribute -- Specify Local Filenames within an FTP Job**

The `ftp_local_name` attribute specifies the name of one or more files on the agent computer involved in an FTP transfer.

**Supported Job Type**

This attribute is required for the [FTP job type](#).

**Syntax**

This attribute has the following format:

```
ftp_local_name: file_name[; file_name...]
```

- **file\_name** Specifies the destination of the file (if downloading) or the source location of the file (if uploading).

**Limits:** Case-sensitive

**UNIX/Windows:**

- If you are downloading a file, you must specify the full path and file name.
- If you are downloading multiples files using wildcards in the ftp\_remote\_name, you must specify the full path to a directory name.
- If you are uploading files, you can use wildcards in the file name. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.
- You cannot use wildcards in the path.

**i5/OS:**

- To specify a file in the root file system, use UNIX path and file formats.
- To specify a \*FILE object in QSYS, use the following format:

```
/QSYS.LIB/libraryname.LIB/objectname.FILE/membername.MBR
```

#### NOTE

- The entire value can be up to 256 characters.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- If a wildcard is used in a local file name for upload, the remote file name (the target) must refer to a directory. A wildcard transfer is equivalent to an mget transfer using an FTP client.
- You can specify multiple files. Separate each file name with a semi-colon (the default delimiter to separate multiple files). The number of files specified in the ftp\_local\_name and ftp\_remote\_name attributes must match.
- Your agent administrator can change the default delimiter that separates file names by setting the ftp.client.separator parameter in the agent's agentparm.txt file.
- You can use global variables to specify the local file name involved in an FTP transfer. Use the \$global\_name, \$\$ (global\_name), or \${global\_name} format to reference global variables.

#### Example: Download a Single File

The FTP job in this example downloads an ASCII file named textfile from the remote UNIX server to the /export/home/ftpfiles/ftpdata/Folder directory on the agent computer.

```
insert_job: FTPDL_SINGLE
job_type: FTP
machine: ftpagent
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: u1/files/ftpdata/textfile
ftp_local_name: /export/home/ftpfiles/ftpdata/Folder/textfile
owner: ftpuser@hprsupp
```

#### Example: Download Multiple Local Files

The FTP job in this example downloads multiple ASCII files from a remote UNIX server. Note that the number of files listed in the ftp\_remote\_name and ftp\_local\_name attributes must match.

```
insert_job: DOWNLOAD_MULTIPLE
```

```

job_type: FTP
machine: RAGENT
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/files/scripts/echo;/u1/files/scripts/echo1;/u1/files/scripts/echo2
ftp_local_name: /export/home/ftpfiles/ftpdata/echo;/export/home/ftpfiles/ftpdata/
echo_1;/export/home/ftpfiles/ftpdata/echo_2
owner: ftpuser@hprsupp

```

### Example: Upload Files to a Directory Using a Wildcard

If a wildcard is used in a local file name for upload, the remote file name (the target) must refer to a directory. In this example, the remote file name is specified as the directory /tmp.

```

insert_job: FTPUPLOAD
job_type: FTP
machine: ftpagent
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: UPLOAD
ftp_transfer_type: A
ftp_remote_name: /tmp
ftp_local_name: /export/home/ftpfiles/ftpdata/text*
owner: ftpuser@hprsupp

```

#### NOTE

For Windows examples, substitute a Windows path in the ftp\_local\_name attribute and enclose the path in quotation marks.

### Example: Download a QSYS.LIB EBCDIC-encoded File

This example downloads an EBCDIC-encoded file in the QSYS file system from an i5/OS server to another i5/OS system. The file names are specified in the path format.

```

insert_job: FTP_EBCDIC
job_type: FTP
machine: I5AGENT
ftp_server_name: i5agent
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: E
ftp_local_name: /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
ftp_remote_name: /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
owner: ftpuser@i5agent

```

### ftp\_local\_user Attribute -- Specify the Local User ID on the Agent

The ftp\_local\_user attribute specifies the local user ID on agent machine. The job uses this user ID to determine the level of access for the file on the local system. For example, if ftp\_transfer\_direction attribute is set to DOWNLOAD and ftp\_local\_user is set to JDOE, the downloaded file is created with the JDOE as the owner.

**NOTE**

- If you do not specify the `ftp_local_user` attribute in your job definition, the job uses the user who inserted the job (the default).
- The `ftp_local_user` attribute differs from the owner attribute. The owner attribute specifies the FTP user ID that is used to log in to the FTP server to run the job. The FTP local user is used to check whether the specified file can be accessed.

**Supported Job Type**

This attribute is optional for the [FTP job type](#).

**Syntax**

This attribute has the following format:

```
ftp_local_user: user
```

- **user**  
Specifies the local user ID on the agent machine.  
**Limits:** Up to 80 characters  
**Default:** The user who inserted the job

**NOTE**

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

**ftp\_remote\_name Attribute -- Specify Remote Filename within an FTP Job**

The `ftp_remote_name` attribute specifies the name of one or more files on a remote server involved in an FTP transfer.

**Supported Job Type**

This attribute is required for the [FTP job type](#).

**Syntax**

This attribute has the following format:

```
ftp_remote_name: file_name[; file_name...]
```

- **file\_name** Specifies the source location of the file (if downloading) or the destination of the file (if uploading).  
**Limits:** Case-sensitive  
**UNIX/Windows:**
  - If you are uploading a file, you must specify the full path and file name.
  - If you are uploading multiple files by specifying a file name containing wildcard characters for the `ftp_local_name` attribute, you must specify the full path of a directory where the files will be stored.
  - If you are downloading files, you can use wildcards in the file name. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.
  - You cannot use wildcards in the path.
  - On UNIX, if you want to use a Windows file as a remote file, you must use a forward slash at the beginning of the path statement and between the directories and file name, as shown in the following example:  

```
ftp_remote_name: "/C:/Temp/textfile"
```
- **i5/OS:**
  - To specify a file in the root file system, use UNIX path and file formats.
  - To specify a \*FILE object in QSYS, use the following format:  

```
/QSYS.LIB/libraryname.LIB/objectname.FILE/membername.MBR
```

**NOTE**

- The entire value can be up to 256 characters.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- If a wildcard is used in a remote file name for download, the local file name (the target) must refer to a directory. A wildcard transfer is equivalent to an mget transfer using an FTP client.
- You can specify multiple files. Separate each file name with a semi-colon (the default delimiter to separate multiple files). The number of files specified in the ftp\_local\_name and ftp\_remote\_name attributes must match.
- Your agent administrator can change the default delimiter that separates file names by setting the ftp.client.separator parameter in the agent's agentparm.txt file.
- You can use global variables to specify the remote file name involved in an FTP transfer. Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

**Example: Download a Single File**

The FTP job in this example downloads an ASCII file named textfile from the remote UNIX server to the /export/home/ftpfiles/ftpdata/textfile\_dn directory on the agent computer.

```
insert_job: DOWNLOAD_SINGLE
job_type: FTP
machine: RAGENT
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/files/ftpdata/textfile
ftp_local_name: /export/home/ftpfiles/ftpdata/textfile_dn/textfile
owner: testuser@hprsupp
```

**Example: Download Multiple Local Files**

The FTP job in this example downloads multiple ASCII files from a remote UNIX server. Note that the number of files listed in the ftp\_remote\_name and ftp\_local\_name attributes must match.

```
insert_job: DOWNLOAD_MULTIPLE
job_type: FTP
machine: RAGENT
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/files/scripts/echo;/u1/files/scripts/echo1;/u1/files/scripts/echo2
ftp_local_name: /export/home/ftpfiles/ftpdata/echo;/export/home/ftpfiles/ftpdata/
echo_1;/export/home/ftpfiles/ftpdata/echo_2
owner: ftpuser@hprsupp
```

**Example: Download Files to a Directory Using a Wildcard**

If a wildcard is used for the remote file name, the corresponding local file name (the target) must refer to a directory. In this example, the directory is WC.

```
insert_job: FTP_WILDCARD
```

```

job_type: FTP
machine: ftpagent
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/files/scripts/r*
ftp_local_name: /export/home/ftpfiles/ftpdata/WC
owner: test@hprsupp

```

### Example: Upload a File to a Remote Windows Computer

In this example, a file named textfile is uploaded from a local UNIX computer and copied to a remote Windows computer. Both locations include a complete path in the ftp\_remote\_name and ftp\_local\_name attributes. The remote Windows computer uses an agent as the FTP server. Its IP address is 172.16.0.0.

```

insert_job: FTP_REMOTE
job_type: FTP
machine: unixagent
ftp_server_name: 172.16.0.0
ftp_server_port: 123
ftp_transfer_direction: UPLOAD
ftp_transfer_type: A
ftp_remote_name: "/C:/Temp/textfile"
ftp_local_name: /export/home/ftpfiles/ftpdata/textfile
owner: test@172.16.0.0

```

#### NOTE

To transfer a file from a Windows computer, substitute a Windows path in the ftp\_local\_name attribute and enclose the entire path in quotation marks.

### Example: Download a QSYS.LIB EBCDIC-encoded File

This example downloads an EBCDIC-encoded file in the QSYS file system from an i5/OS server to another i5/OS system. The file names are specified in the path format.

```

insert_job: FTP_EBCDIC
job_type: FTP
machine: I5AGENT
ftp_server_name: i5agent
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: E
ftp_local_name: /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
ftp_remote_name: /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
owner: ftpuser@i5agent

```

### ftp\_server\_name Attribute -- Specify FTP Server Name within an FTP Job

The ftp\_server\_name attribute specifies the name of the FTP server in an FTP job.

#### Supported Job Type



This attribute is required for the [FTP job type](#).

### Syntax

This attribute has the following format:

```
ftp_server_name: server_address
```

- **server\_address** Specifies the DNS name or IP address of a remote server. **Limits:** Up to 256 characters; case-sensitive

**Example:** 172.24.36.107 (IPv4), 0:0:0:0:FFFF:192.168.00.00 (IPv6), or machine1 (DNS name)

### NOTE

- Do not use embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the DNS name or IP address of a remote server. Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

### Example: Download a File from a UNIX Computer

This example downloads a UNIX file from the hpsupport server with port 5222.

```
insert_job: FTPT1A
job_type: FTP
machine: ftpagent
ftp_server_name: hpsupport
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/qatest/ftpdata/textfile
ftp_local_name: /export/home/ftpfiles/test1
owner: ftpuser@hpsupport
```

### Example: Download a File from a UNIX Computer to a Windows Computer

The FTP job in this example downloads a file from a UNIX server with IP address 172.16.0.0 and port 21 to a Windows computer.

```
insert_job: FTPT1A
job_type: FTP
machine: ftpagent
ftp_server_name: 172.16.0.0
ftp_server_port: 21
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/ftpdata/textfile
ftp_local_name: "c:\ftpfiles"
owner: ftpuser@172.16.0.0
```

### Example: Download a QSYS.LIB EBCDIC-encoded File

This example downloads an EBCDIC-encoded file in the QSYS file system from an i5/OS server to another i5/OS system. The file names are specified in the path format.

```
insert_job: FTP_EBCDIC
job_type: FTP
machine: I5AGENT
ftp_server_name: i5agent
```

```
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: E
ftp_local_name: /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
ftp_remote_name: /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
owner: ftpuser@i5agent
```

## ftp\_server\_port Attribute -- Specify the Server Port of a Remote FTP Server

The `ftp_server_port` attribute specifies the port number of the remote FTP server involved in an FTP file transfer.

### Supported Job Type

This attribute is optional for the [FTP job type](#).

### Syntax

This attribute has the following format:

```
ftp_server_port: port
```

- ***port***  
Specifies the port number of the remote server.  
**Limits:** 0-65535  
**Example:** 5222

### Example: Specify a Server Port Number

The FTP job in this example downloads a file from a UNIX server with IP address 172.16.0.0 and port 21 to a Windows computer.

```
insert_job: FTPT1A
job_type: FTP
machine: ftpagent
ftp_server_name: 172.16.0.0
ftp_server_port: 21
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/ftpdata/textfile
ftp_local_name: "c:\ftpfiles"
owner: ftpuser@172.16.0.0
```

## ftp\_transfer\_direction Attribute -- Specify Transfer Direction within an FTP Job

The `ftp_transfer_direction` attribute specifies the file transfer direction in an FTP job.

### NOTE

If you do not specify the `ftp_transfer_direction` attribute in your job definition, the job downloads the data (the default).

### Supported Job Type

This attribute is optional for the [FTP job type](#).

### Syntax

This attribute has the following format:

```
ftp_transfer_direction: UPLOAD | DOWNLOAD
```

- **UPLOAD**

Transfers files from the agent computer to the remote server.

- **DOWNLOAD**

Transfers files from the remote server to the agent computer. This is the default.

### Example: Upload Files

If a wildcard is used in a local file name for upload, the remote file name (the target) must refer to a directory. In this example, the remote file name is specified as the directory /tmp.

```
insert_job: FTPUPLOAD
job_type: FTP
machine: ftpagent
ftp_server_name: hprsupp
ftp_server_port: 5222
ftp_transfer_direction: UPLOAD
ftp_transfer_type: A
ftp_remote_name: /tmp
ftp_local_name: /export/home/ftpfiles/ftpdata/text*
owner: ftpuser@hprsupp
```

#### NOTE

For Windows examples, substitute a Windows path in the ftp\_local\_name attribute and enclose the path in quotation marks.

### ftp\_transfer\_type Attribute -- Specify Data Code Type within an FTP Job

The ftp\_transfer\_type attribute specifies the type of data involved in an FTP transfer (binary, ASCII, or EBCDIC).

#### NOTE

If you do not specify the ftp\_transfer\_type attribute in your job definition, the job performs a binary data transfer (the default).

### Supported Job Type

This attribute is optional for the [FTP job type](#).

### Syntax

This attribute has the following format:

```
ftp_transfer_type: A | B | E
```

- **A**  
Indicates an ASCII transfer.  
**i5/OS:**
  - If the ASCII file to be transferred already exists on the target computer, the file is written using the encoding of the existing file.
  - If the file does not exist, the file is written using the ASCII CCSID (Coded Character Set Identifier) defined on the agent. The default is 819.
- **B** Indicates a binary transfer. This is the default.
- **E** Indicates an EBCDIC transfer.

**NOTE**

E applies to Workload Automation Agent for i5/OS only. If an EBCDIC file to be transferred already exists on the target computer, the file is written using the encoding of the existing file. If the file does not exist, the file is written using the EBCDIC CCSID (Coded Character Set Identifier) defined on the agent. The default is 37.

**Example: Download a Binary File on UNIX**

The FTP job in this example uses a binary transfer.

```
insert_job: FTPBINARY
job_type: FTP
machine: unixagent
ftp_server_name: hpunix
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: B
ftp_remote_name: /u1/qatest/ftpdata/binaryfile
ftp_local_name: /export/home/qatest/ftpdata/transf.bin
owner: test@hpunix
```

**Example: Upload an ASCII-encoded File in the Root File System from an i5/OS System to a UNIX System**

This example uploads the textdoc file in the root file system from an i5/OS system and copies it to a UNIX system.

```
insert_job: UPLOAD_i5
job_type: FTP
machine: I5AGENT
ftp_server_name: hpunix
ftp_server_port: 5222
ftp_transfer_direction: UPLOAD
ftp_transfer_type: A
ftp_remote_name: /u1/ftpfiles/ftpdata/textdoc
ftp_local_name: /home/ftp/textdoc
owner: test@hpunix
```

**Example: Download a QSYS EBCDIC-encoded File**

This example downloads an EBCDIC-encoded file named DATAFILE in the QSYS file system from an i5/OS system to another i5/OS system. The file names are specified in the path format.

```
insert_job: EBCDIC_FILE
job_type: FTP
machine: I5AGENT
ftp_server_name: i5agent
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_transfer_type: E
ftp_remote_name: /QSYS.LIB/DATALIB.LIB/DATAFILE.FILE/DATA.MBR
ftp_local_name: /QSYS.LIB/ESPLIB.LIB/DOWNLOAD.FILE/DATA.MBR
owner: test@i5agent
```

## ftp\_use\_SSL Attribute -- Specify SSL FTP or Regular FTP

The ftp\_use\_SSL attribute specifies whether to transfer the data with Secure Sockets Layer (SSL) communication or regular communication.

### NOTE

If you do not specify the ftp\_use\_SSL attribute in your job definition, the data is transferred using the communication type set in the ftp.client.ssl parameter on the agent FTP client. This parameter is automatically set to false by default (regular communication is used).

### Supported Job Type

This attribute is optional for the [FTP](#) job type.

### Syntax

This attribute has the following format:

```
ftp_use_SSL: TRUE | FALSE
```

- **TRUE**  
Transfers data using SSL FTP.
- **FALSE**  
Transfers data using regular FTP. This is the default.

### NOTE

- To transfer data using SSL, check for the following:
  - The FTP server must have SSL FTP enabled.
  - The FTP client must have SSL FTP configured (SSL FTP can be enabled or disabled).

The agent administrator can enable or disable SSL on the agent FTP client using the ftp.client.ssl parameter in the agent parameter file (agentparm.txt) as follows:

  - If the agent FTP client has SSL FTP enabled, all FTP jobs on that agent automatically use SSL FTP.
  - If the agent FTP client does not have SSL FTP enabled, all FTP jobs on that agent automatically use regular FTP.
- The ftp\_use\_ssl attribute overrides the ftp.client.ssl parameter specified in the agent's agentparm.txt.
- If the FTP client has SSL FTP enabled, but the FTP server does not, you must set the ftp\_use\_SSL attribute to FALSE. Otherwise, the job will fail.

### Example: Upload a File from a Local Computer to a Remote Windows Server Using SSL FTP

This example defines a job named FTP\_UPLOAD to upload the filename.txt file from the agent FTP client to the c:\uploaded\_files directory on the remote Windows server named winserver using SSL FTP.

This example uploads a file from a local computer to a remote Windows server using SSL FTP.

```
insert_job:FTP_UPLOAD
job_type: FTP
machine: winagent
owner: user1@winserver
ftp_server_name: winserver
ftp_server_port: 21
ftp_transfer_direction: UPLOAD
ftp_use_SSL: TRUE
ftp_remote_name: "c:\uploaded_files\filename.txt"
ftp_local_name: "d:\files_to_upload\filename.txt"
```

### Example: Download a File from a Remote UNIX Server That Does Not Support SSL FTP to a Local Computer That Supports SSL FTP

In this example, the agent runs on a local computer as an FTP client and has SSL FTP enabled (all FTP jobs on the agent computer run using SSL FTP by default). The remote UNIX server does not support SSL FTP.

The following FTP job downloads the filename.txt file from the remote UNIX server to the d:\downloaded\_files directory on the local computer. Because the FTP server does not support SSL FTP, the ftp\_use\_ssl attribute is set to FALSE so the job does not fail.

```
insert_job: FTP_DOWNLOAD
job_type: FTP
machine: winagent
owner: user1@hpunix
ftp_server_name: hpunix
ftp_server_port: 5222
ftp_transfer_direction: DOWNLOAD
ftp_use_ssl: FALSE
ftp_remote_name: /files_to_download/filename.txt
ftp_local_name: "d:\downloaded_files\filename.txt"
```

### ftp\_user\_type Attribute -- Specify How the Owner Attribute Value is Passed to the FTP Server for Authentication

The ftp\_user\_type attribute specifies how the value of the owner attribute is passed to the FTP server for authentication.

#### NOTE

If you do not specify the ftp\_user\_type attribute in your job definition, only the user part of the owner attribute value is passed to the FTP server for authentication (the default).

#### Supported Job Type

This attribute is optional for the [FTP job type](#).

#### Syntax

This attribute has the following format:

```
ftp_user_type: Simple | Windows | Qualified
```

- **Simple**

Indicates that only the user part of the owner attribute value is passed to the FTP server for authentication. This is the default.

**Example:** If you set the owner attribute to userxyz@domain, userxyz is passed to the FTP server for authentication.

- **Windows**

Indicates that the value of the owner attribute is passed in the form of domain\user to the FTP server for authentication.

**Example:** If you set the owner attribute to userxyz@domain, domain\userxyz is passed to the FTP server for authentication.

- **Qualified**

Indicates that the value of the owner attribute is passed in the form of an email address or user@domain (fully qualified domain) to the FTP server for authentication.

**Example:** If you set the owner attribute to userxyz@domain, userxyz@domain is passed to the FTP server for authentication.

## group Attribute -- Associate a Job with a Group

The group attribute associates a job with a specific group so users can classify, sort, and filter jobs by group name.

You can use the group attribute with the sendevent, job\_depends, and autorep commands. For example, you could use a single sendevent command to start all of the jobs associated with a specific group.

Consider the following when you use the group attribute:

- Users with execute permission can add new group attributes to a job definition.
- You can only associate a job with one group at a time.
- When a job is associated with a group and an application (using the application attribute), AutoSys Workload Automation assumes that the application attribute is a subset of the group attribute.

### NOTE

If both the box and child job belongs to the same group, you have to specify the group attribute only for a box job.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
group: group_name
```

- **group\_name**

Defines the name of the group with which to associate the job. You can only associate a job with one group, but a group may have many jobs associated with it.

**Limits:** Up to 64 characters; valid characters are a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); do not include embedded spaces or tabs

### Example: Associate a Job with a Group

This example associates a job with the group year\_end\_reports.

```
group: year_end_reports
```

## hadoop\_pollinterval Attribute - Specify the Number of Seconds to Wait Between Job Status Checks

The hadoop\_pollinterval attribute specifies the number of seconds to wait between job status checks.

### Supported Job Type

This attribute is optional for HDFS, Oozie, Hive, Sqoop, and Pig job types.

### Syntax

This attribute has the following format:

```
hadoop_pollinterval: time
```

- **time**

Specifies the number of seconds to wait between job status checks.

**Default:** 60

**Limits:** 1-86400

---

## hadoop\_hdfs\_create\_parentdir Attribute - Specify Whether to Create a Parent Directory

The `hadoop_hdfs_create_parentdir` attribute specifies whether you want to create a parent directory.

### Supported Job Type

When the `hadoop_hdfs_type` is `MKDIR`, this attribute is optional for the HDFS job type.

### Syntax

This attribute has the following format:

```
hadoop_hdfs_create_parentdir: true | false
```

- **true**  
Creates a parent directory.
- **false**  
Does not create a parent directory. This is the default.

## hadoop\_hdfs\_destination Attribute - Specify the Destination Path of the File or Directory

The `hadoop_hdfs_destination` attribute specifies the path of a file or directory that is copied or moved to the destination location.

### Supported Job Type

When the `hadoop_hdfs_type` is `COPY`, `COPYFROMLOCAL`, `COPYTOLOCAL`, `MOVE`, or `MOVEFROMLOCAL`, this attribute is required for the HDFS job type.

### Syntax

This attribute has the following format:

```
hadoop_hdfs_destination: destinationpath
```

- ***destinationpath***  
Specifies the path of a file or directory to which the file or directory is copied or moved.  
**Limits:** Up to 255 characters

#### NOTE

- If the path includes a colon, enclose it in double quotation marks.
- The destination path can be absolute or relative to the HDFS home directory.
- You can use global variables to specify the destination path of a file or directory. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## hadoop\_hdfs\_overwrite Attribute - Specify Whether to Overwrite a File or Directory

The `hadoop_hdfs_overwrite` attribute specifies whether you want to overwrite an existing file or directory in the destination location when a file or directory with the same name is copied from the source location.

### Supported Job Type

When the `hadoop_hdfs_type` is `COPY` or `COPYFROMLOCAL`, this attribute is optional for the HDFS job type.

### Syntax

This attribute has the following format:

```
hadoop_hdfs_overwrite: true | false
```



- **true**  
Overwrites an existing file or directory in the destination location when a file or directory with the same name is copied from the source location.
- **false**  
Does not overwrite the file or directory. This is the default.

## **hadoop\_hdfs\_path Attribute - Specify the Path to the HDFS Home Directory**

The `hadoop_hdfs_path` attribute specifies the path of the file or directory where the HDFS operation is performed. The path can be absolute or relative to the HDFS home directory. The HDFS home directory is set when a user is created on the Hadoop edge node.

### **Supported Job Type**

This attribute is required for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_path: path
```

- ***path***  
Specifies the absolute or relative path to the HDFS home directory.  
**Limits:** Up to 1024 characters.

### **NOTE**

- If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.
- To specify multiple paths for files or directories, separate each path with a comma. Each path can be up to 1024 characters.
- You can specify multiple `hadoop_hdfs_path` attributes with different values in your job definition.
- You can use global variables to specify the path of the file or directory where the HDFS operation is performed. Use the `$$global_name`, `$(global_name)`, or `$$ {global_name}` format to reference global variables.

## **hadoop\_hdfs\_preserve\_status - Specify Whether to Retain the Timestamps, Ownership, and Permission Attributes for a File or Directory**

The `hadoop_hdfs_preserve_status` attributes specifies whether you want to retain the timestamps, ownership, and permission attributes for the file or directory after it is copied from the source location to the destination location.

### **Supported Job Type**

When the `hadoop_hdfs_type` is `COPY`, `COPYFROMLOCAL`, or `COPYTOLOCAL`, this attribute is optional for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_preserve_status: true | false
```

- **true**  
Retains the timestamps, ownership, and permission attributes for the file or directory after it is copied from the source location to the destination location.
- **false**  
Does not retain the timestamps, ownership, and permission attributes for the file or directory. This is the default.

## **hadoop\_hdfs\_recursive Attribute - Specify Whether to Invoke the File Operation Recursively**

The `hadoop_hdfs_recursive` attribute specifies whether you want to invoke the file operation on the file or directory and all of its sub-directories.

### **Supported Job Type**

When the `hadoop_hdfs_type` is REMOVE, this attribute is optional for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_recursive: true | false
```

- **true**  
Invokes the file operation on the file or directory and all of its subdirectories. Based on the value you set for `hadoop_hdfs_type`, the file operations are invoked recursively as follows:
  - REMOVE  
Deletes the specified file or directory and all of its sub-directories.
- **false**  
Does not invoke the file operation on the file or directory and all of its sub-directories. This is the default.

## **hadoop\_hdfs\_skiptrash Attribute - Specify Whether to Skip Trash**

The `hadoop_hdfs_skiptrash` attribute specifies whether you want to bypass the trash and delete the file or directory permanently.

### **Supported Job Type**

When the `hadoop_hdfs_type` is REMOVE, this attribute is optional for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_skiptrash: true | false
```

- **true**  
Bypasses the trash and deletes the file or directory permanently.
- **false**  
Does not bypass the trash. This is the default.

## **hadoop\_hdfs\_source Attribute - Specify the Source Path of the File or Directory**

The `hadoop_hdfs_source` attribute specifies the path of one or more files or directories that are to be copied or moved from the source location.

### **Supported Job Type**

When the `hadoop_hdfs_type` is COPY, COPYFROMLOCAL, COPYTOLOCAL, MOVE, or MOVEFROMLOCAL, this attribute is required for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_source: sourcepath
```

- ***sourcepath***

Specifies the path of one or more files or directories that are to be copied or moved from the source location.

**Limits:** Up to 1024 characters

**NOTE**

- If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.
- The source path can be absolute or relative to the HDFS home directory.
- To specify multiple paths for files or directories, separate each path with a comma. Each path can be up to 1024 characters.
- You can specify multiple `hadoop_hdfs_source` attributes with different values in your job definition.
- You can use global variables to specify the source path of a file or directory. Use the `$$global_name`, `$(global_name)`, or `$$${global_name}` format to reference global variables.

## **hadoop\_hdfs\_type Attribute - Specify the File Operation Performed on HDFS**

The `hadoop_hdfs_type` attribute specifies the file operation that the job performs on HDFS.

### **Supported Job Type**

This attribute is required for the HDFS job type.

### **Syntax**

This attribute has the following format:

```
hadoop_hdfs_type: TOUCH | MKDIR | REMOVE | COPY | COPYFROMLOCAL | COPYTOLOCAL | MOVE |
MOVEFROMLOCAL
```

- **TOUCH**  
Creates one or more zero-byte files.
- **MKDIR**  
Creates one or more directories.
- **REMOVE**  
Deletes one or more files or directories.
- **COPY**  
Copies one or more files or directories within HDFS.
- **COPYFROMLOCAL**  
Copies one or more files or directories from the local file system to HDFS.
- **COPYTOLOCAL**  
Copies one or more files or directories from HDFS to the local file system.
- **MOVE**  
Moves one or more files or directories within HDFS.
- **MOVEFROMLOCAL**  
Moves one or more files or directories from the local file system to HDFS.

## **hadoop\_hive\_additional\_params - Specify an Additional Command Line Argument**

The `hadoop_hive_additional_params` attribute specifies an additional Hadoop Beeline command line argument and its value.

### **Supported Job Type**

This attribute is optional for the Hive job type.

## Syntax

This attribute has the following format:

```
hadoop_hive_additional_params: parameter
```

- ***parameter***

Specifies an additional Hadoop Beeline command line argument and its value.

**Example:** --color=true --showHeader=true**Limits:** Up to 2048 characters

### NOTE

If you specify multiple additional parameters, separate each additional parameter with a blank space.

## hadoop\_hive\_configparams - Specify a Hive Configuration Property

The `hadoop_hive_configparams` attribute specifies a Hive configuration property and its value in the `property=value` format.

### Supported Job Type

This attribute is optional for the Hive job type.

## Syntax

This attribute has the following format:

```
hadoop_hive_configparams: config_param
```

- ***config\_param***

Specifies a Hive configuration property and its value in the `property=value` format.

### NOTE

- If you specify multiple configuration properties, separate each configuration property with a comma. Each configuration property can be up to 1024 characters.
- You can specify multiple `hadoop_hive_configparams` attributes with different values in your job definition.

## hadoop\_hive\_initfile - Specify the Name of the Initialization Script

The `hadoop_hive_initfile` attribute specifies the name of the initialization SQL script file that you provide to the Hadoop Beeline client. This initialization SQL script file must be located on the Hadoop edge node.

### Supported Job Type

This attribute is optional for the Hive job type.

## Syntax

This attribute has the following format:

```
hadoop_hive_initfile: init_file
```

- ***init\_file***

Specifies the name of the initialization SQL script file that you provide to the Hadoop Beeline client.

**Limits:** Up to 255 characters

## hadoop\_hive\_scriptfile - Specify the Name of the SQL Script

The `hadoop_hive_scriptfile` attribute specifies the name of the SQL script file that you provide to the Hadoop Beeline client. This script file must be located on the Hadoop edge node.

### Supported Job Type

This attribute is required for the Hive job type.

### Syntax

This attribute has the following format:

```
hadoop_hive_scriptfile: script_file
```

- ***script\_file***  
Specifies the name of the SQL script file that you provide to the Hadoop Beeline client.  
**Limits:** Up to 255 characters

## hadoop\_hive\_user - Specify a Hive User Name

The `hadoop_hive_user` attribute defines the user name to connect to the database.

### Supported Job Type

This attribute is optional for the Hive job type.

### Syntax

This attribute has the following format:

```
hadoop_hive_user: user_name
```

- ***user\_name***  
Defines the user name to connect to the database. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command.  
**Default:** anonymous (the Hive default user)  
**Limits:** Up to 145 characters; up to 64 characters for user and up to 80 characters for token.

### NOTE

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

## hadoop\_hive\_variables - Specify a Hive Variable

The `hadoop_hive_variables` attribute specifies a Hive variable and its value in the `variable=value` format. These variables are set at session level and used in Hive commands or queries.

### Supported Job Type

This attribute is optional for the Hive job type.

### Syntax

This attribute has the following format:

```
hadoop_hive_variables: variables
```

- ***variables***

Specifies a Hive variable and its value in the variable=value format.

**NOTE**

- If you specify multiple variables, separate each variable with a comma. Each variable can be up to 1024 characters.
- You can specify multiple `hadoop_hive_variables` attributes with different values in your job definition.

## **hadoop\_oozie\_cfgfile Attribute - Specify the Path to the Job Configuration File**

The `hadoop_oozie_cfgfile` attribute specifies the path to the job configuration file (.xml or .properties) on the edge node. This can be a fully qualified path name or a path that is relative to the home directory of the user on the edge node.

**Supported Job Type**

This attribute is required for the Oozie job type.

**Syntax**

This attribute has the following format:

```
hadoop_oozie_cfgfile: path
```

- ***path***

Specifies the path to the job configuration file (.xml or .properties) on the edge node.

**Limits:** Up to 255 characters.

**NOTE**

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

## **hadoop\_pig\_additional\_params - Specify an Additional Command Line Argument**

The `hadoop_pig_additional_params` attribute specifies an additional Hadoop Pig command line argument and its value.

**Supported Job Type**

This attribute is optional for the Pig job type.

**Syntax**

This attribute has the following format:

```
hadoop_pig_additional_params: parameter
```

- ***parameter***

Specifies an additional Hadoop Pig command line argument and its value.

**Limits:** Up to 2048 characters

**NOTE**

- You can specify multiple additional parameters. Separate each additional parameter with a blank space. For example, `-verbose -warning -stop_on_failure -printCmdDebug`.
- When you specify multiple additional parameters, use the same format that you use while specifying Hadoop Pig command line arguments.

## **hadoop\_pig\_parameterfile - Specify the Path to the Parameter File**

The `hadoop_pig_parameterfile` attribute specifies the path to the Hadoop Pig parameter file.

---

### Supported Job Type

This attribute is optional for the Pig job type.

#### Syntax

This attribute has the following format:

```
hadoop_pig_parameterfile: parameter_file
```

- ***parameter\_file***  
Specifies the path to the Hadoop Pig parameter file.  
**Limits:** Up to 255 characters

#### NOTE

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

### hadoop\_pig\_propertyfile - Specify the Path to the Property File

The `hadoop_pig_propertyfile` attribute specifies the path to the Hadoop Pig property file.

#### Supported Job Type

This attribute is optional for the Pig job type.

#### Syntax

This attribute has the following format:

```
hadoop_pig_propertyfile: property_file
```

- ***property\_file***  
Specifies the path to the Hadoop Pig property file.  
**Limits:** Up to 255 characters

#### NOTE

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

### hadoop\_pig\_scriptfile - Specify the Path to the Script File

The `hadoop_pig_scriptfile` attribute specifies the path to the Hadoop Pig script file.

#### Supported Job Type

This attribute is required for the Pig job type.

#### Syntax

This attribute has the following format:

```
hadoop_pig_scriptfile: script_file
```

- ***script\_file***

Specifies the path to the Hadoop Pig script file.

**Limits:** Up to 255 characters

#### NOTE

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

## hadoop\_sqoop\_additional\_params - Specify an Additional Command Line Argument

The `hadoop_sqoop_additional_params` attribute specifies an additional Hadoop Sqoop command line argument and its value.

### Supported Job Type

This attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_additional_params: parameter
```

- ***parameter***

Specifies an additional Hadoop Sqoop command line argument and its value.

**Limits:** Up to 2048 characters

#### NOTE

- You can specify multiple additional parameters. Separate each additional parameter with a blank space. For example, `-m 1 --outdir directorypath --validate`.
- When you specify multiple additional parameters, use the same format that you use while specifying Hadoop Sqoop command line arguments.

## hadoop\_sqoop\_boundary\_query - Specify the Boundary Query

The `hadoop_sqoop_boundary_query` attribute specifies the boundary query to use for creating splits.

### Supported Job Type

When the `hadoop_sqoop_type` is `IMPORT`, this attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_boundary_query: boundary_query
```

- ***boundary\_query***

Specifies the boundary query to use for creating splits. This value is passed to the `--boundary-query` Hadoop Sqoop import control argument.

**Limits:** Up to 1024 characters



---

## hadoop\_sqoop\_clearstagingtable Attribute - Specify Whether to Delete Data in the Staging Table

The `hadoop_sqoop_clearstagingtable` attribute specifies whether you want to delete the data in the staging table.

### NOTE

To use this attribute, you must also specify the `hadoop_sqoop_stagingtable` attribute in your job definition.

### Supported Job Type

When the `hadoop_sqoop_type` is `EXPORT`, this attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_clearstagingtable: true | false
```

- **true**  
Deletes the data in the staging table.
- **false**  
Does not delete the data in the staging table. This is the default.

## hadoop\_sqoop\_columns - Specify the Columns to Import

The `hadoop_sqoop_columns` attribute specifies the columns to import from the database table.

### Supported Job Type

When the `hadoop_sqoop_type` is `IMPORT`, this attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_columns: columns
```

- ***columns***  
Specifies the columns to import from the database table. This value is passed to the `--columns` Hadoop Sqoop import control argument.  
**Limits:** Up to 1024 characters

## hadoop\_sqoop\_destination - Specify the HDFS Destination Directory

The `hadoop_sqoop_destination` attribute specifies the HDFS destination directory to which the data is imported.

### Supported Job Type

When the `hadoop_sqoop_type` is `IMPORT`, this attribute is required for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_destination: destination
```

- ***destination***  
Specifies the HDFS destination directory to which the data is imported.  
**Limits:** Up to 255 characters

**NOTE**

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

**hadoop\_sqoop\_jdbctable - Specify the Database Table**

The `hadoop_sqoop_jdbctable` attribute specifies the database table from which the data is imported or to which the data is exported.

**Supported Job Type**

This attribute is required for the sqoop job type.

**Syntax**

This attribute has the following format:

```
hadoop_sqoop_jdbctable: jdbctable
```

- ***jdbctable***

Specifies the database table from which the data is imported or to which the data is exported.

**Limits:** Up to 128 characters

**hadoop\_sqoop\_jdbcuser - Specify a JDBC User Name**

The `hadoop_sqoop_jdbcuser` attribute defines the JDBC user name to connect to the database.

**Supported Job Type**

This attribute is required for the Sqoop job type.

**Syntax**

This attribute has the following format:

```
hadoop_sqoop_jdbcuser: user_name
```

- ***user\_name***

Defines the JDBC user name to connect to the database. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command.

**Limits:** Up to 145 characters; up to 64 characters for user and up to 80 characters for token.

**NOTE**

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

**hadoop\_sqoop\_nullnonstring Attribute - Specify a Null String for Non-string Columns**

The `hadoop_sqoop_nullnonstring` attribute specifies the string that indicates a null value for non-string columns.

**Supported Job Type**

This attribute is optional for the Sqoop job type.

**Syntax**

This attribute has the following format:

---

```
hadoop_sqoop_nullnonstring: null_nonstring
```

- ***null\_nonstring***  
Specifies the string that indicates a null value for non-string columns.  
**Limits:** Up to 20 characters

## hadoop\_sqoop\_nullstring Attribute - Specify a Null String for String Columns

The `hadoop_sqoop_nullstring` attribute specifies the string that indicates a null value for string columns.

### Supported Job Type

This attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_nullstring: null_string
```

- ***null\_string***  
Specifies the string that indicates a null value for string columns.  
**Limits:** Up to 20 characters

## hadoop\_sqoop\_source - Specify the HDFS Source Directory

The `hadoop_sqoop_source` attribute specifies the HDFS source directory from where the data is exported.

### Supported Job Type

When the `hadoop_sqoop_source` is EXPORT, this attribute is required for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_source: source_dir
```

- ***source\_dir***  
Specifies the HDFS source directory from where the data is exported.  
**Limits:** Up to 255 characters

### NOTE

If the path includes a colon, escape it with a backslash or enclose it in double quotation marks.

## hadoop\_sqoop\_stagingtable Attribute - Specify the Table that Stages Data

The `hadoop_sqoop_stagingtable` attribute specifies the table in which the data is staged before being inserted into the destination table.

### Supported Job Type

When the `hadoop_sqoop_type` is EXPORT, this attribute is optional for the Sqoop job type.

### Syntax

This attribute has the following format:

```
hadoop_sqoop_stagingtable: staging_table
```

- ***staging\_table***

Specifies the table in which the data is staged before being inserted into the destination table.

**Limits:** Up to 128 characters

## **hadoop\_sqoop\_type - Specify the Sqoop Job Operation**

The `hadoop_sqoop_type` attribute specifies the operation that the Sqoop job performs.

### **Supported Job Type**

This attribute is required for the Sqoop job type.

### **Syntax**

This attribute has the following format:

```
hadoop_sqoop_type: IMPORT | EXPORT
```

- **IMPORT**

Imports data from a relational database (such as MySQL and Oracle) to Hadoop HDFS.

- **EXPORT**

Exports data from Hadoop HDFS to a relational database.

## **hadoop\_sqoop\_whereclause - Specify the WHERE clause to Use During Import**

The `hadoop_sqoop_whereclause` attribute specifies the WHERE clause to use during import.

### **Supported Job Type**

When the `hadoop_sqoop_type` is `IMPORT`, this attribute is optional for the Sqoop job type.

### **Syntax**

This attribute has the following format:

```
hadoop_sqoop_whereclause: whereclause
```

- ***whereclause***

Specifies the WHERE clause to use during import. This value is passed to the `--where` Hadoop Sqoop import control argument.

**Limits:** Up to 1024 characters

## **header\_filter Attribute -- Specify a Filter on the Header**

The `header_filter` attribute specifies a filter that is applied on the header of a response in a HTTP job.

### **Supported Job Type**

This attribute is optional for the [HTTP job type](#).

### **Syntax**

This attribute has the following format:

```
header_filter: filter
```

- ***filter***

Specifies a filter (a regular expression) that is applied on the header of a response in a HTTP job. If any line from the server response matches the filter, the job completes. Else, the job fails.

**Limits:** Up to 256 characters

**Example:** gr[ae]y

#### NOTE

You can use global variables to specify a filter for the header. Use the \$\$global\_name, \$\$ (global\_name), or \${global\_name} format to reference global variables.

## headers Attribute -- Specify a Header

The headers attribute specifies the header to use for the request in an HTTP job.

#### NOTE

The following characters are not supported in the HTTP header: @ ( ) [ ] \ : space , ; > < ? / "

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
headers: name1="value1", name2="value2"
```

- **name1="value1"**

Specifies the header to use for the request in an HTTP job. Specify the header in a comma separated name="value" format. You must specify the value within double quotation marks. However, if the value itself is in double quotation marks, include the double quotation marks again and escape the quotation marks.

**Limits:** Up to 4000 characters for each name="value" format

#### NOTE

You can use global variables to specify a header. Use the \$\$global\_name, \$\$ (global\_name), or \${global\_name} format to reference global variables.

If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

## heartbeat\_interval Attribute -- Set the Monitoring Frequency for a Job

The heartbeat\_interval attribute defines the time interval (in minutes) between consecutive heartbeats from a command job. The scheduler checks this time interval before raising an alarm for late or missing heartbeats from jobs.

Heartbeats are events that the scheduler uses to monitor the progress of the application that the agent executes. You can program the application to send a HEARTBEAT event to the event servers at a regular interval using the AutoSys Workload Automation SDK.

The scheduler checks for an occurrence of the HEARTBEAT event within the time interval specified by the heartbeat\_interval job attribute. If the job does not send a heartbeat during the specified time interval, the scheduler generates the MISSING\_HEARTBEAT alarm.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#).

### Syntax

This attribute has the following format:

```
heartbeat_interval: mins
```

- **mins**

Defines the time interval (in minutes) between consecutive heartbeats from a command job. The scheduler, while monitoring heartbeats, checks whether the HEARTBEAT event occurs during this time interval.

**Limits:** 0-2147483647

**Default:** 0; the scheduler does not monitor heartbeats from the job

#### UNIX Notes:

- You must configure how often the scheduler checks for heartbeats. For more information about configuring the scheduler on UNIX to check for heartbeats, see [Configure a Scheduler](#).
- To send a heartbeat from a C++ program, call the `autoheartbeat` routine defined in the following C++ header file and link your program with the `libautosys_api` UNIX shared object in the `$AUTOSYS/lib` directory:  
`$AUTOSYS/code/autosys_api_auto.h`
- You must install the AutoSys Workload Automation client for each instance on all machines that run a C++ program linked with the `libautosys_api` UNIX shared object. You must also modify the configuration file to connect to an existing application server for the instance.

#### Windows Notes:

- You must configure how often the scheduler checks for heartbeats. For more information about configuring the scheduler on Windows to check for heartbeats, see [Configure a Scheduler](#).
- To send a heartbeat from a C++ program, call the `autoheartbeat` routine in the following C++ header file and link your program with the `autosys_api` Windows dynamic-link library in the `%AUTOSYS%\bin` directory:  
`%AUTOSYS%\code\autosys_api_auto.h`
- You must install the AutoSys Workload Automation client for each instance on all machines that run a C++ program linked with the `autosys_api` Windows dynamic-link library. You must also modify the configuration file to connect to an existing application server for the instance.

#### Example: Specify the Interval for a Job that Sends a Heartbeat Every Two Minutes

This example defines a Command job that runs a C++ program named `backup` that uses the AutoSys Workload Automation SDK. The job sends a heartbeat every two minutes.

```
insert_job: cplusplus_program
job_type: CMD
machine: unixagent
command: /usr/common/backup
heartbeat_interval: 2
```

### i5\_action Attribute -- Specify Whether to Run a Program or Issue a Command

The `i5_action` attribute specifies whether to run a program or issue a command on an i5/OS system.

#### NOTE

The default value is `COMMAND`. If you do not specify the `i5_action` attribute in your job definition, the job interprets the `i5_name` value to be a command.

#### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

#### Syntax

This attribute has the following format:

```
i5_action: RUN_PROGRAM | RUN_PROG_IN_FILE | COMMAND
```

- **RUN\_PROGRAM**

Runs a program on an i5/OS system.

- **RUN\_PROG\_IN\_FILE**

Runs a program from a database file member on an i5/OS system. This is similar to running a shell script on a UNIX system or a batch file on a Windows system.

- **COMMAND**

Issues a command on an i5/OS system when the i5/OS job runs. This is the default.

### Example: Run a Program on an i5/OS System

This example runs the program named PARAM under the ASYSOPR user ID. The program takes three parameters in the following order: ABC, C1, and P.

```
insert_job: i5job_runprog
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PARAM
i5_params: "ABC C1 P"
owner: ASYSOPR
```

### Example: Run a Command on an i5/OS System

This example issues the DISPLIBL command to display the job's library list. When the job completes, you can view the library list in the job's spool file.

```
insert_job: i5job_displib
job_type: I5
machine: i5agent
i5_action: COMMAND
i5_name: DSPLIBL
i5_lib: *LIBL
i5_curr_lib: AGTLIB
i5_job_name: PAYJOBA
i5_job_desc: PRDLIB/PRDOPR
i5_job_queue: ABCSYS/PAYROLL
```

## i5\_cc\_exit Attribute -- Specify the Type of Exit Code to Return

The i5\_cc\_exit attribute specifies the type of exit code returned by an i5/OS job. The exit code can be the job's ending severity code, the return code of an ILE program or module, or the return code of an OPM program.

### NOTE

The default type is \*SEVERITY. If you do not specify the i5\_cc\_exit attribute in your job definition, the job sends the ending severity code as the exit code.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

### Syntax

This attribute has the following format:

```
i5_cc_exit: *SEVERITY | *USER | *PROGRAM
```

- **\*SEVERITY**

Sends the job's ending severity code. This is the default.

- **\*USER**

Sends the return code of an ILE program or module. For example, if your job runs an ILE C or ILE RPG program that contains an exit or return statement, that return code is sent as the exit code.

**NOTE**

If the program does not set a return code, the i5 system returns a 0 (zero), which is sent as the exit code. By default, an exit code of zero (0) is interpreted as job success.

- **\*PROGRAM**

Sends the return code of an OPM program. For example, if your job runs an OPM RPG or OPM COBOL program that contains an exit or return statement, that return code is sent as the exit code.

**NOTE**

- If the program does not set a return code, the i5 system returns a 0 (zero), which is sent as the exit code. By default, an exit code of zero (0) is interpreted as job success.
- Ensure that you specify the appropriate exit code type for the program you are running. If you specify the wrong exit code type (for example, you specify \*PROGRAM for an ILE program), the agent sends 0 (zero) as the exit code. By default, AutoSys Workload Automation interprets exit codes of 0 (zero) as job success.
- If you specify \*PROGRAM or \*USER, the agent adds a custom message before the final completion message in the job's spool file (job log). This message is required to track the program's return code.

### Example: Send a C Program's Return Code as the Job's Exit Code

This example runs a C language program named SALARY. The agent sends the SALARY program's return code to AutoSys Workload Automation.

```
insert_job: i5job_returnC
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: SALARY
i5_cc_exit: *USER
```

### Example: Send an OPM COBOL Program's Return Code as the Job's Exit Code

This example runs an OPM COBOL program named PAYROLL. The agent sends the PAYROLL program's return code to AutoSys Workload Automation.

```
insert_job: i5job_returnOPM
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYROLL
i5_cc_exit: *PROGRAM
```

## i5\_curr\_lib Attribute -- Specify the Name of the Current Library

The i5\_curr\_lib attribute specifies the name of the current library to use when running the job.

**NOTE**

If you do not specify a current library, the agent uses the current library of the user profile the agent runs under.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).



## Syntax

This attribute has the following formats:

```
i5_curr_lib: current_library
i5_curr_lib: /QSYS.LIB/current_library.LIB
```

- ***current\_library***

Specifies the current library for the job.

### NOTE

The entire value can be up to 46 characters; it cannot contain delimiters (such as spaces).

## Example: Specify the Current Library for a Job

This example runs a program named MFGDATA on an i5/OS system. The current library for the job is PAY1, and the job description is DFTJOB in library CYB1.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: MFGDATA
i5_curr_lib: PAY1
i5_job_desc: CYB1/DFTJOB
```

## i5\_job\_desc Attribute -- Specify the Job Description

The i5\_job\_desc attribute specifies the name of the job description for the Control Language (CL) program that the job submits.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

## Syntax

This attribute has the following formats:

```
i5_job_desc: description
i5_job_desc: library/description
i5_job_desc: /QSYS.LIB/library.LIB/description.JOB
```

- ***library***

Specifies the name of the library that contains the job description. The value must be a valid i5/OS library name.

- ***description***

Specifies the name of the job description for the program submitted. The value must be a valid i5/OS job description name.

**Limits:** Up to 10 characters; the first character must be one of the following: A-Z, \$, #, or @; the second to tenth characters can be A-Z, 0-9, \$, #, \_, ., or @

### NOTE

- The entire value can be up to 46 characters; it cannot contain delimiters (such as spaces).
- If you do not specify the library name, the i5/OS system will search for it using the library list for the job.
- Your agent administrator can specify a default job description for all i5/OS jobs by setting the os400.default.jobname parameter in the agent's agentparm.txt file.
- The i5\_job\_desc attribute overrides the default job description specified in the agent's agentparm.txt.

### Example: Specify the Library Name and Description for a Job

This example runs a program named MFGDATA on an i5/OS system. The current library for the job is PAY1, and the job description is DFTJOB in library CYB1.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: MFGDATA
i5_curr_lib: PAY1
i5_job_desc: CYB1/DFTJOB
```

### i5\_job\_name Attribute -- Specify the Job Name

The i5\_job\_name attribute specifies the name for an i5/OS job. This is the name displayed in the job queue.

#### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

#### Syntax

This attribute has the following format:

```
i5_job_name: job_name
```

- **job\_name**

Specifies the name of the i5/OS job.

**Limits:** Up to 10 characters; the first character must be one of the following: A-Z, \$, #, or @; the second to tenth characters can be A-Z, 0-9, \$, #, \_, ., or @

### Example: Specify the Name of an i5/OS Job

This example defines an i5/OS job that runs a program named PROCprog. The name of this job that is displayed in the job queue of the i5/OS system is PROG1.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PROCprog
i5_job_name: PROG1
```

### i5\_job\_queue Attribute -- Specify the Job Queue for a Program

The i5\_job\_queue attribute specifies the i5/OS job queue for the submitted program.

#### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

#### Syntax

This attribute has the following formats:

```
i5_job_queue: jobqueue
i5_job_queue: library/jobqueue
i5_job_queue: /QSYS.LIB/library.LIB/jobqueue.JOBQ
```

- **library**

Specifies the name of the library that contains the job queue. The value must be a valid i5/OS library name.

- **job\_queue**

Specifies the name of the job queue for the submitted program. The value must be a valid i5/OS job queue name.

**NOTE**

- The entire value can be up to 46 characters; it cannot contain delimiters (such as spaces).
- If you do not specify the library name, the i5/OS system will search for it using the library list for the job.
- Your agent administrator can specify a default job queue for all i5/OS jobs by setting the `os400.default.jobqname` parameter in the agent's `agentparm.txt` file.
- The `i5_job_desc` attribute overrides the default job queue specified in the agent's `agentparm.txt`.

**Example: Specify the Library Name and Job Queue for an i5/OS Job**

This example runs a program named PAYLOAD on an i5/OS system. The current library for the job is PAY1, and the job queue is JQUEUE in library QBASE.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYLOAD
i5_curr_lib: PAY1
i5_job_queue: /QYS.LIB/QBASE.LIB/JQUEUE.JOBQ
```

## i5\_Ida Attribute -- Specify the Data for the Local Data Area

The `i5_Ida` attribute specifies data for the local data area in an i5/OS job. The local data area is a temporary 1024-byte storage area that exists for the duration of the job. You can use the local data area to pass data to the job and to other programs that run as part of the job.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

### Syntax

This attribute has the following format:

```
i5_lda: "data" | X'hh...'
```

- **"data"**

Specifies the data in character format.

**Limits:** Up to 1024 characters; case-sensitive.

**NOTE**

Specify the values that contain embedded blanks with double quotation marks.

**Example:** "My data"

- **X'hh...'**

Specifies the data in hexadecimal format, with each pair of hexadecimal digits representing one byte of data.

**Limits:** Up to 2051 characters

**Example:** X'C1C2C340C1C2C3'

**Example: Specify Data for the Local Data Area in Hexadecimal Format**

This example defines an i5/OS job with data for the local data area. When the job is submitted, the agent initializes the local data area with the hexadecimal data abcd. When the job completes, the local data area is destroyed automatically by the operating system.

```
insert_job: i5job_lda
job_type: I5
machine: i5agent
i5_action: COMMAND
i5_name: IVP
i5_lda: x'abcd'
```

## i5\_lib Attribute -- Specify the Name of a Library

The i5\_lib attribute specifies the name of the library that contains the Control Language (CL) program, the source file for the CL program, or the command that you want to run.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

### Syntax

This attribute has the following formats:

```
i5_lib: library
i5_lib: /QSYS.LIB/library.LIB
```

- **library** Specifies the name of the library that contains the program, the source file for the program, or the command that you want to run. The value must be a valid i5/OS library name.

### NOTE

- The entire value can be up to 46 characters; it cannot contain delimiters (such as spaces).
- Instead of using the i5\_lib attribute, you can specify the library name in the i5\_name attribute, or you can specify the library name using the i5\_curr\_lib or i5\_library\_list attribute.
- Do not use the i5\_lib attribute if the library name is already specified in the i5\_name attribute. If you specify the library name in both attributes, you will get a submission error. The job status will be JOB FAILURE.
- If you do not specify the library name in the i5\_name attribute or the i5\_lib attribute, the agent uses the job's library list.

### Example: Specify the Library that Contains the CL Program to Run

This example runs a program named MFGDATA on an i5/OS system. The MFGDATA program is contained in the library named PRODJOB.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: MFGDATA
i5_lib: PRODJOB
```

## i5\_library\_list Attribute -- Specify the Libraries for an i5/OS Job

The i5\_library\_list attribute specifies the names of the libraries that the job uses.

### Supported Job Type

This attribute is optional for the [i5/OS](#) job type.

### Syntax

This attribute has the following format:

```
i5_library_list: "library..."
```

- **library**

Specifies the name of a library that you want to add to the library list.

**Limits:** Up to 10 characters; it cannot contain delimiters (such as spaces)

### NOTE

- You can specify up to 25 libraries separated by spaces. The libraries are searched in the same order as they are listed.
- The entire value can be up to 274 characters.
- Enclose the entire list in quotation marks.
- The i5\_library\_list attribute specifies the initial user part of the library list that is used to search for operating system object names that do not have a library qualifier.
- The i5\_library\_list attribute overrides the library list specified in the job description for the job.
- You can use global variables to specify the library name for an i5/OS job. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Add a Library to the Library List of an i5/OS job

This example runs a program named PAYJOB on an i5/OS system. The library named PAYROLL is added to the library list.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYJOB
i5_library_list: "PAYROLL"
```

### Example: Add More Than One Library to the Library List of an i5/OS Job

This example runs a program named ACCTJOB on an i5/OS system. The libraries named PAYROLL and FINANCE are added to the library list.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: ACCTJOB
i5_library_list: "PAYROLL FINANCE"
```

## i5\_name Attribute -- Specify a CL Program, Source File, or Command

The i5\_name attribute specifies the program, the source file for the program, or the command that you want to run on an i5/OS system.

### NOTE

Use this attribute with the i5\_action attribute, which specifies whether the job runs a program or issues a command. The format of this attribute value depends on the action type specified in the i5\_action attribute.

### Supported Job Type

This attribute is required for the [i5/OS job type](#).

## Syntax

This attribute has the following formats:

- To specify a program:
  - `i5_name: program`
  - `i5_name: library/program`
  - `i5_name: /QSYS.LIB/library.LIB/program.PGM`

### NOTE

- The entire value can be up to 60 characters; it cannot contain delimiters (such as spaces).
  - To specify a program, the value of the `i5_action` attribute must be `RUN_PROGRAM`.
- To specify the source file for a program:
  - `i5_name: file`
  - `i5_name: file(member)`
  - `i5_name: library/file(member)`
  - `i5_name: /QSYS.LIB/library.LIB/file.FILE/member.MBR`

### NOTE

- The entire value can be up to 60 characters; it cannot contain delimiters (such as spaces).
  - To specify a source file, the value of the `i5_action` attribute must be `RUN_PROG_IN_FILE`.
- To specify a command:
  - `i5_name: command`
  - `i5_name: library/command`
  - `i5_name: /QSYS.LIB/library.LIB/command.CMD`

### NOTE

- The entire value can be up to 120 characters; enclose values that contain delimiters (such as spaces) in single quotation marks.
  - To specify a command, the value of the `i5_action` attribute must be `COMMAND`.
- program***  
Specifies the name of the i5/OS program to run. The value must be a valid i5/OS program name.
- library***  
Specifies the name of the library that contains the program, the source file for the program, or the command that you want to run. The value must be a valid i5/OS library name.
- file***  
Specifies the file containing the CL source for the program. The value must be a valid i5/OS file name.

### NOTE

If you specify the file name without a member name, `*FIRST` is used by default.

- member***  
Specifies the member in the file that contains the CL source for the program. The value must be a valid i5/OS member name.
- command***  
Specifies the name of the i5/OS command to run. The value must be a valid i5/OS command name.

**NOTE**

- If the library name is already specified in the `i5_name` attribute, do not use the `i5_lib` attribute. If you specify the library name in both attributes, you will get a submission error. The job status will be JOB FAILURE.
- If you do not specify the library name in the `i5_name` attribute or the `i5_lib` attribute, the agent uses the job's library list.
- You can use global variables to specify the program name, the source file name for the program, or the command to run on an i5/OS system. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Specify a Program to Run**

This example runs a program named MFGDATA on an i5/OS system. The MFGDATA program is contained in the library named PRODJOB.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: MFGDATA
i5_lib: PRODJOB
```

**Example: Specify a File that Contains the Program to Run**

This example runs a program that is contained in the BACKUP member of the CLSRC file. The library name is not specified in the `i5_name` attribute or the `i5_lib` attribute, so the agent uses the job's library list.

```
insert_job: i5job_file
job_type: I5
machine: i5agent
i5_action: RUN_PROG_IN_FILE
i5_name: CLSRC(BACKUP)
```

**Example: Specify a Command to Run**

This example issues the DISPLIBL command to display the job's library list. When the job completes, you can view the library list in the job's spool file.

```
insert_job: i5job_displib
job_type: I5
machine: i5agent
i5_action: COMMAND
i5_name: DSPLIBL
i5_lib: *LIBL
i5_curr_lib: AGTLIB
i5_job_name: PAYJOBA
i5_job_desc: PRDLIB/PRDOPR
i5_job_queue: ABCSYS/PAYROLL
```

**i5\_others Attribute -- Pass Keyword Parameters to SBMJOB**

When AutoSys Workload Automation submits a job to the i5/OS system, the job must pass through the SBMJOB command to execute. Keyword parameters are additional parameters that AutoSys Workload Automation passes to the

i5/OS SBMJOB command. The `i5_others` attribute lets you specify any valid SBMJOB command keyword and value combination. You can also specify multiple combinations.

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

### Syntax

This attribute has the following format:

```
i5_others: keyword=value[,keyword=value...]
```

- **keyword=value**

Specifies any valid SBMJOB command keyword and value combination. The value corresponds to the SBMJOB command keyword (the value that would appear in brackets after the keyword in the SBMJOB command).

#### NOTE

Enclose values that contain spaces in single quotation marks.

#### NOTE

- You can specify up to 64 keyword=value combinations. Separate each combination with a comma.
- Each *keyword=value* combination can be up to 1024 characters.
- You can use multiple entries of `i5_others` to define different sets of keyword=value combinations. For example, you can specify two `i5_others` attributes as follows:

```
i5_others: PRTDEV=*JOB
```

```
i5_others: OUTQ=*JOB
```

The entire value of each `i5_others` entry can be up to 4096 characters.

- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- AutoSys Workload Automation does not validate the keyword and value combinations.
- The agent uses the user's default output queue if the owner attribute is specified in the job definition and the `i5_others` attribute does not include the SBMJOB OUTQ() command keyword and value that defines the output queue for the job.

### Example: Specify the Printer and Output Queue for an i5/OS Job

This example runs a program named PAYJOB on an i5/OS system. The printer and output queue information is taken from the job definition.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYJOB
i5_others: PRTDEV=*JOB,OUTQ=*JOB
```

### i5\_params Attribute -- Pass Positional Parameters

The `i5_params` attribute defines the parameter values that you want to pass to the program at the time the program is invoked.

#### NOTE

To use this attribute, you must specify RUN\_PROGRAM or COMMAND for the `i5_action` attribute.

### Supported Job Type



This attribute is optional for the [i5/OS](#) job type.

### Syntax

This attribute has the following format:

```
i5_params: value[value...]
```

- **value[value...]**

Specifies one or more parameter values to pass to the program. Separate each parameter with a space.

**Limits:** Up to 64 parameters; each parameter can be up to 1024 characters.

#### NOTE

To pass a parameter containing spaces, enclose it in double quotation marks.

#### NOTE

- You must specify each parameter in the order it is expected in the program.
- You can specify multiple i5\_params attributes in a job definition, as shown in the following example:
 

```
i5_params: parameter1 parameter2 parameter3
i5_params: parameter4 parameter5 parameter6
i5_params: "parameter 7" "parameter 8" "parameter 9"
```
- You can use global variables to specify the parameter values to pass to the program. Use the \$global\_name, \$\$ (global\_name), or \${global\_name} format to reference global variables.
- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### Example: Pass Multiple Parameters to an i5/OS Job

This example passes six parameters to an i5/OS program named PAYJOB. The parameter VALUE C is enclosed with double quotation marks because it contains a space.

```
insert_job: i5job_lib
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYJOB
i5_params: ABC 1 P "VALUE C" X r
```

### i5\_process\_priority Attribute -- Specify the Process Priority of an i5/OS Job

The i5\_process\_priority attribute specifies the process priority of an i5/OS job. Process priority determines the order in which processes are scheduled on the processor. Depending on the priority level, process priority can speed up or slow down a process.

#### NOTE

If you do not specify the i5\_process\_priority attribute in your job definition, the job uses the normal priority (the default).

### Supported Job Type

This attribute is optional for the [i5/OS job type](#).

### Syntax

This attribute has the following format:

```
i5_process_priority: HIGH | ABOVE_NORMAL | NORMAL | BELOW_NORMAL | IDLE
```

- **HIGH**  
Indicates processes that must be executed immediately. These processes can use nearly all available CPU time.
- **ABOVE\_NORMAL**  
Indicates processes that have priority above the normal level but below the high level.
- **NORMAL**  
Indicates processes without special scheduling needs. This is the default.
- **BELOW\_NORMAL**  
Indicates processes that have priority above the idle level but below the normal level.
- **IDLE**  
Indicates processes that will run only when the system is idle.

**NOTE**

For UNIX jobs running on an i5/OS system, you can increase the process priority above normal only if the agent is running under a profile that has \*JOBCTL authority. If the agent is not running under a profile that has \*JOBCTL authority and you set the process priority to a level above normal, the job runs with the normal process priority.

**Example: Send a C Program's Return Code as the Job's Exit Code**

This example sets the process priority for an i5/OS job to HIGH:

```
insert_job: i5_priority
job_type: I5
machine: i5agent
i5_action: RUN_PROGRAM
i5_name: PAYROLL
i5_process_priority: HIGH
```

**informatica\_folder\_name Attribute -- Specify the Informatica PowerCenter Repository Folder Name**

The `informatica_folder_name` attribute specifies the Informatica PowerCenter repository folder name where the specified workflow resides.

**Supported Job Type**

This attribute is required for the Informatica job type.

**Syntax**

This attribute has the following format:

```
informatica_folder_name: foldername
```

- **foldername** Specifies the Informatica PowerCenter repository folder name where the specified workflow resides. You can obtain this value from the Informatica server. **Limits:** Up to 79 characters

**NOTE**

You can use global variables to specify the Informatica PowerCenter repository folder name. Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

**informatica\_param\_file Attribute -- Specify the Path and Name of the Parameter File**

The `informatica_param_file` attribute specifies the path and name of a parameter file on the Informatica server. Informatica PowerCenter processes the parameter file when running a workflow. The parameter file contains values for symbolic

variables and must adhere to the format that Informatica PowerCenter expects. A property value that is set in a parameter file overrides a value for this property that is set at the session level or elsewhere.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_param_file: paramfile
```

- ***paramfile*** Specifies the path and name of a parameter file on the Informatica server. **Limits:** Up to 256 characters

#### NOTE

You can use global variables to specify the path and name of a parameter file. Use the \$\$global\_name, \$(global\_name), or \$\$ {global\_name} format to reference global variables.

### Example: Process a Parameter File in a Workflow

This example processes the wf2.prm parameter file on the Informatica server in the WF\_2 workflow. Based on the status of tasks, the workflow status is reported.

The wf2.prm parameter file includes symbolic variables that override the default log directory and file name:

```
;Workflow Parameters [INFAPLUGIN.WF:wf_3] ;Session parameters $PMSessionLogFile=wf3.log
$PMSessionLogDir=C:\Infa\Log ;mapping parameters
insert_job: informatica
job_type: INFORMATICA
machine: 160_infor
owner: root@informatica
informatica_folder_name: FSTL
informatica_workflow_name: WF_2
informatica_pass_on_success: 1
informatica_param_file: C:\Infa\wf2.prm
```

### informatica\_pass\_on\_success Attribute -- Specify Whether the Status of the Workflow is Retrieved Directly From Informatica

The informatica\_pass\_on\_success attribute specifies whether the status of the workflow is retrieved directly from Informatica or inferred depending on the successful execution of all the tasks.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_pass_on_success: 0 | 1
```

- **0**  
Returns the workflow status as reported by Informatica. In Informatica, a workflow can succeed even if one or more tasks in the workflow fail.
- **1**

Returns the workflow status based on the status of tasks in the Informatica workflow. If all tasks in the workflow complete successfully, the agent reports the status of the workflow as successful. If any task in the workflow fails, the agent reports the status of the workflow as failed.

#### NOTE

If you do not specify any value, the agent uses the default value that is specified in the `informatica.passonsuccessonly.default` parameter in the `agentparm.txt` file.

## informatica\_repository\_name Attribute -- Specify the Name of the Repository

The `informatica_repository_name` attribute specifies the name of the repository that is created on the Informatica server. The repository can be obtained from the Informatica Administrative console. The corresponding property file (`repositoryName.properties`) that is stored on the agent contains the database connection properties for a specified Informatica target (`infaTarget`).

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_repository_name: repositoryname
```

- ***repositoryname*** Specifies the name of the repository that is created on the Informatica server

**Limits:** Up to 128 characters

#### NOTE

- If you do not specify any value, the agent uses the default value specified in the `informatica.repository.name.default` parameter in the `informatica.properties` file stored on the agent.
- You can use global variables to specify the name of the repository that is created on the Informatica server. Use the `$$global_name`, `$(global_name)`, or `$$${global_name}` format to reference global variables.

## informatica\_security\_domain Attribute -- Specify the Name of the LDAP Security Domain

The `informatica_security_domain` attribute specifies the name of the LDAP security domain that is configured with the Informatica Server. If you specify this attribute in your job definition, the user is authenticated against the specified LDAP security domain. Else, the user is authenticated with the Informatica Server using the value you specified in the [informatica\\_user](#) attribute.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_security_domain: ldap_security_domain
```

- **ldap\_security\_domain** Specifies the name of the LDAP security domain that is configured with the Informatica Server. **Limits:** Up to 80 characters

#### NOTE

You can use global variables to specify the name of the LDAP security domain. Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

## informatica\_target\_name Attribute -- Specify the Name of the Informatica Target

The informatica\_target\_name attribute specifies the name of the Informatica target (infaTarget). The corresponding directory (install\_dir/config/informatica/infaTarget) stored on the agent contains the properties for connecting to the Informatica server and one or more database repositories.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_target_name: targetname
```

- **targetname** Specifies the name of the Informatica target (infaTarget). **Limits:** Up to 128 characters

#### NOTE

- If you do not specify any value, the agent uses the default value specified in the informatica.server.target.default parameter in the agentparm.txt file.
- You can use global variables to specify the name of the Informatica target (infaTarget). Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

## informatica\_task\_name Attribute -- Specify the Name of the Task

The informatica\_task\_name attribute specifies the name of the task to start the workflow from. If the task name is not specified, the workflow starts from the beginning.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_task_name: taskname
```

- **taskname** Specifies the name of the task to start the workflow from. **Limits:** Up to 79 characters

#### NOTE

You can use global variables to specify the name of the task. Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

## informatica\_user Attribute -- Specify the Informatica Server User Name

The `informatica_user` attribute specifies the Informatica server user name.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_user: informatica_user@token
```

- ***informatica\_user@token*** Specifies the Informatica server user name. Specify this value in the `user@token` format and it must match the `user@host` value that you specify using the `autosys_secure` command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against `user@host` using the `autosys_secure` command. **Limits:** Up to 145 characters; up to 64 characters for user and up to 80 characters for token.

### NOTE

- If you specify the `informatica_user` attribute in your job definition, you must set the password that is associated with the Informatica user using the `autosys_secure` command.
- If you do not specify any value, the agent uses the default value that is specified in the `informatica.user.id.default` parameter in the `informatica.properties` file that is stored on the agent.

## informatica\_workflow\_inst\_name Attribute -- Specify the Name of the Workflow Instance

The `informatica_workflow_inst_name` attribute specifies the name of the workflow instance to be started. When the task name and the workflow instance name are specified, the workflow instance is started from the specified task. When the task name is not specified, the workflow instance starts from the beginning. If the instance name is not specified, the workflow is started.

### Supported Job Type

This attribute is optional for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_workflow_inst_name: workflow_instname
```

- ***workflow\_instname*** Specifies the name of the workflow instance to be started. **Limits:** Up to 79 characters

### NOTE

- Before you specify the workflow instance name, you must specify the workflow using the `informatica_workflow_name` attribute.
- You can use global variables to specify the name of the workflow instance to be started. Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

## informatica\_workflow\_name Attribute -- Specify the Name of the Informatica PowerCenter Workflow

The `informatica_workflow_name` attribute specifies the name of the Informatica PowerCenter workflow to be started.

### Supported Job Type

This attribute is required for the Informatica job type.

### Syntax

This attribute has the following format:

```
informatica_workflow_name: workflowname
```

- **workflowname** Specifies the name of the Informatica PowerCenter workflow to be started. **Limits:** Up to 79 characters

### NOTE

You can use global variables to specify the name of the Informatica PowerCenter workflow. Use the `$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## initial\_context\_factory Attribute -- Specify an Initial Context Factory

The `initial_context_factory` attribute specifies the initial context factory to use when creating the initial context in a Entity Bean, Session Bean, JMS Publish, and JMS Subscribe job.

### Supported Job Types

This attribute is required for the following job types:

- [Entity Bean \(ENTYBEAN\)](#)
- [JMS Publish \(JMSPUB\)](#)
- [JMS Subscribe \(JMSSUB\)](#)
- [Session Bean \(SESSBEAN\)](#)

### Syntax

This attribute has the following format:

```
initial_context_factory: factory
```

- **factory** Specifies the initial context factory to use when creating the initial context. The initial context is required within the Java Naming and Directory Interface (JNDI) framework. The initial context factory is supplied by a specific provider of the naming and directory service. The factory acquires an arbitrary initial context that the application can use to connect to the application server. **Limits:** Up to 256 characters; case-sensitive

**Example:** `weblogic.jndi.WLInitialContextFactory`

### NOTE

You can use global variables to specify the initial context factory to use when creating the initial context. Use the `$$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

### Example: Specify Initial Context Factory for a WebSphere Application Server

Suppose that you want to invoke the `reverse` method on the `CybEJBTestBean` stateless session bean. The `reverse` method has one parameter, with type `java.lang.String` and value `a23`. The output from the `reverse` method is saved in the `C:\Makapt15` file. The initial context factory supplied by the JNDI service provider is `com.ibm.websphere.naming.WsnInitialContextFactory`. The service provider's URL is `corbaloc:iiop://172.24.0.0:2809`, where `172.24.0.0` is the IP address of the WebSphere application server and `2809` is the ORB port. When the job runs, the output of the `reverse` method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
```

```

provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"

```

## input\_cookies Attribute -- Input the Cookies in the Database

The `input_cookies` attribute specifies the name of an existing job, the cookies of which will be consumed by the current job.

This specified existing job should have the `persist_output_cookies` attribute set to 1.

### NOTE

If the specified job does not exist or does not have the `persist_output_cookies` value set to 1, the job insertion still succeeds, but with a warning message.

### Supported Job Types

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
input_cookies: jobname
```

### Examples:

Predecessor job definition:

```

insert_job: testae_getjob job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "https://github.gwd.broadcom.net:443/ESD/ae"
results_file_overwrite: 0
persist_output_cookies: 1

```

Successor job definition:

```

insert_job: testae_getjob1 job_type: HTTP
machine: localhost
invocation_type: GET
provider_url: "https://github.gwd.broadcom.net:443/ESD/ae"
input_cookies: testae_getjob
results_file_overwrite: 0

```

## inside\_range Attribute -- Specify Whether to Monitor for CPU or Disk Usage Within or Outside a Range

The `inside_range` attribute specifies whether the job completes (or triggers if monitoring continuously) when the CPU or disk usage value is inside or outside the specified boundaries.



**NOTE**

If you do not specify the `inside_range` attribute in your job definition, the job completes or triggers if the usage value is inside the lower and upper boundaries (the default).

**Supported Job Types**

This attribute is optional for the following job types:

[CPU Monitoring \(OMCPU\) job type](#)

[Disk Monitoring \(OMD\) job type](#)

To use this attribute, you must specify `WAIT` or `CONTINUOUS` for the `monitor_mode` attribute. This attribute does not apply when the monitor mode is `NOW`.

**Syntax**

This attribute has the following format:

```
inside_range: TRUE | FALSE
```

- **TRUE**

Specifies that the job completes (or triggers if monitoring continuously) when the value of CPU usage or disk usage is inside the range specified by the `lower_boundary` and `upper_boundary` attributes. This is the default.

- **FALSE**

Specifies that the job completes (or triggers if monitoring continuously) when the value of CPU usage or disk usage is outside the range specified by the `lower_boundary` and `upper_boundary` attributes.

**Example: Monitor Available CPU**

This example continuously monitors the CPU available on the `winagt` computer. An alert is written to the scheduler log file each time the available CPU is less than 25 percent or greater than 75 percent, and the CPU usage changes by more than 10 percent. If the change in value is less than or equal to 10, the job does not register a change.

```
insert_job: omcpu_job
job_type: OMCPU
machine: winagt
cpu_usage: FREE
lower_boundary: 25
upper_boundary: 75
inside_range: FALSE
no_change: 10
monitor_mode: CONTINUOUS
```

**Example: Monitor Disk Space Usage Outside of a Range**

This example monitors the C drive on a Windows computer for used space. When the value of disk space used falls below 16 percent or exceeds 95 percent, the job completes.

```
insert_job: omd_win
job_type: OMD
machine: winagent
disk_drive: C
disk_format: PERCENT
lower_boundary: 16
upper_boundary: 95
inside_range: FALSE
disk_space: USED
```

## interactive Attribute -- Specify Whether to Run a Command Job in Interactive Mode on Windows

The interactive attribute specifies whether to submit a Command job in interactive mode or in batch mode on Windows. Interactive mode lets users view and interact with jobs that invoke Windows Terminal Services or user interface processes. For example, you can define a job to open a GUI application, such as Notepad, on the Windows desktop.

### NOTE

If you do not specify the interactive attribute in your job definition, the job runs in batch mode (the default).

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#) on Windows. This attribute is ignored for Command jobs that run on UNIX.

### Syntax

This attribute has the following format:

```
interactive: y | n
```

- **y**  
Runs the Command job in interactive mode on Windows.
- **n**  
Runs the Command job in batch (background) mode on Windows. This is the default.

### NOTE

- The agent works according to Microsoft restrictions, as follows:
  - To schedule interactive jobs, the user specified to run the job must be logged in with an active session.
  - To schedule interactive jobs, the user can have only one active session. If more than one session exists for the specified user, the agent cannot determine which session to use and therefore fails the job.
- If the agent administrator sets the `oscomponent.interactive` parameter to true in the `agentparm.txt` file, the agent submits all Windows jobs in interactive mode, regardless of the interactive attribute. To use the interactive attribute in a job definition, the `oscomponent.interactive` parameter must be set to false.
- If the agent administrator sets the `oscomponent.interactive.sessionzero`, `oscomponent.su.newconsole`, `oscomponent.cmdprefix.force.redir.inline`, and `oscomponent.cmdprefix.force.redir.ifstdon` parameters to true in the `agentparm.txt` file, the agent runs interactive jobs in session zero. To run interactive jobs using terminal services, set the `oscomponent.interactive.sessionzero` parameter to false.
- By default, the agent uses the default Windows shell `explorer.exe`. The agent administrator can specify an alternative Windows shell for interactive jobs by setting the `oscomponent.shell` parameter in the `agentparm.txt` file.

### Example: Run a Command Job in Interactive Mode on Windows

This example opens the configuration text file (`config.txt`) in the Windows Desktop application named Notepad.

```
insert_job: edit_file
job_type: CMD
machine: winagent
description: "Edit/review a configuration file"
command: notepad.exe "c:\run_info\config.txt"
interactive: y
```

## invocation\_type Attribute -- Specify the HTTP Method Type

The `invocation_type` attribute indicates the HTTP method type, which can be GET, POST, PUT or DELETE in an HTTP job.

### NOTE

If you do not specify the `invocation_type` attribute in your job definition, the job uses the POST method type (the default).

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
invocation_type: GET | POST | PUT | DELETE
```

- **GET**  
Sends the URL over HTTP using the GET method. The GET method requests data and sends the data as part of the URL.
- **POST**  
Sends the URL over HTTP using the POST method. The POST method submits data and is the preferred method for sending lengthy form data. This is the default.
- **PUT**  
Sends the URL over HTTP using the PUT method. The PUT method submits data and is the preferred method for updating or creating resource identified by the URL as the operation is idempotent.
- **DELETE**  
Sends the URL over HTTP using the DELETE method. The DELETE method removes the data.

### Example: Use the GET Method to Perform a Google Search

Suppose that you want to define a job to perform a Google search and have the results returned to the job's spool file. In this example, the job uses the HTTP GET method to perform the Google search on "ca workload automation". When the job runs, the job's spool file includes all the results of the search.

```
insert_job: google
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://google.com/search"
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_parameter: q="ca workload automation"
```

## ip\_host Attribute -- Specify the IP Address to Monitor

The `ip_host` attribute specifies an IP address to monitor.

### Supported Job Type

This attribute is required for the [IP Monitoring \(OMIP\) job type](#).

### Syntax

This attribute has the following format:

```
ip_host: ip_address
```

- ***ip\_address***

Specifies the DNS name or IP address.

**Limits:** Up to 100 characters; case-sensitive

**Example:** 172.24.36.107 (IPv4), 0:0:0:0:FFFF:192.168.00.00 (IPv6), or machine1 (DNS name)

**NOTE**

Do not use embedded blanks, single quotation marks, or double quotation marks.

**Example: Monitor an IP Address Specified in Dotted Decimal Format**

This example monitors a device at IP address 172.31.255.255 and port 7510. When the job runs, it immediately checks if the device is running. If the device is running, the job completes successfully.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: 172.31.255.255
ip_port: 7510
monitor_mode: NOW
ip_status: RUNNING
```

## **ip\_port Attribute -- Specify the Port Number at the IP Address to Monitor**

The `ip_port` attribute specifies the port number at the IP address to monitor.

### **Supported Job Type**

This attribute is optional for the [IP Monitoring \(OMIP\) job type](#).

### **Syntax**

This attribute has the following format:

```
ip_port: ip_port
```

- ***ip\_port***

Specifies the port number for the IP address being monitored. The agent attempts to connect to this port.

**Limits:** 0-65535

**Example:** 5800

**NOTE**

If you specify 0 or no value for the port, the agent uses ping for monitoring.

**Example: Monitor an IP Address and Port**

This example monitors a device at IP address 172.31.255.255 and port 7510. When the job runs, it immediately checks if the device is running. If the device is running, the job completes successfully.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: 172.31.255.255
ip_port: 7510
monitor_mode: NOW
ip_status: RUNNING
```

## ip\_status Attribute -- Specify the Status of the IP Address to Monitor

The ip\_status attribute specifies the status of the IP address to monitor.

### NOTE

If you do not specify the ip\_status attribute in your job definition, the job monitors the IP address for a stopped status (the default).

### Supported Job Type

This attribute is optional for the [IP Monitoring \(OMIP\) job type](#).

### Syntax

This attribute has the following format:

```
ip_status: RUNNING | STOPPED
```

- **RUNNING**

Specifies that the job monitors the IP address for a running status.

- **STOPPED**

Specifies that the job monitors the IP address for a stopped status. This is the default.

### Example: Monitor an IP Address for a Running Status

This example monitors a device at IP address 172.31.255.255 and port 7510. When the job runs, it immediately checks if the device is running. If the device is running, the job completes successfully.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: 172.31.255.255
ip_port: 7510
monitor_mode: NOW
ip_status: RUNNING
```

### Example: Monitor an IP Address for a Stopped Status

This example monitors a device with DNS name myhost. When the device stops running, the job completes.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: myhost
ip_status: STOPPED
monitor_mode: WAIT
```

## j2ee\_authentication\_order Attribute -- Specify Authentication Protocols

The j2ee\_authentication\_order attribute specifies a list of protocols to be used by a web server for authentication in an HTTP job.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following formats:

```
j2ee_authentication_order: BASIC | DIGEST | NTLM
j2ee_authentication_order: { [BASIC] }
 { , [DIGEST] }
 { , [NTLM] }
```

- **BASIC**  
Indicates the BASIC protocol.
- **DIGEST**  
Indicates the DIGEST protocol.
- **NTLM**  
Indicates the NTLM protocol.

#### NOTE

- You can specify any of the following protocols in any order: BASIC, DIGEST, and NTLM. To specify more than one protocol, separate the protocols by commas.
- If you are connecting to an old web server that cannot negotiate authentication protocols, enter the following list in the specified order: BASIC,DIGEST,NTLM.

#### Example: Connect to a Web Server that Cannot Negotiate Authentication Protocols

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

insert\_job: HTTP.CON\_USER

```
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

#### j2ee\_conn\_domain Attribute -- Specify a Domain for NTLM Connection Authentication

The j2ee\_conn\_domain attribute specifies the domain for connection authentication in an HTTP job. This attribute is required for NTLM authentication.

#### Supported Job Type

This attribute is optional for the [HTTP job type](#).

#### Syntax

This attribute has the following format:

```
j2ee_conn_domain: origin_domain
```

- **origin\_domain**

Specifies the domain for NTLM connection authentication.

**Limits:** Up to 80 characters; case-sensitive

**NOTE**

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the domain for NTLM connection authentication. Use the `$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Specify Connection Domain and Origin for NTLM Authentication**

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## **j2ee\_conn\_origin Attribute -- Specify an Origin Host Name for NTLM Connection Authentication**

The `j2ee_conn_origin` attribute specifies the origin host name for NTLM connection authentication in an HTTP job.

**NOTE**

If you do not specify the `j2ee_conn_origin` attribute in your job definition, the job uses the computer name where the agent is running (the default).

### **Supported Job Type**

This attribute is optional for the [HTTP job type](#).

### **Syntax**

This attribute has the following format:

```
j2ee_conn_origin: origin_host
```

- **origin\_host** Specifies the origin host name for NTLM connection authentication.  
**Default:** The computer name where the agent is running  
**Limits:** Up to 80 characters; case-sensitive

**NOTE**

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the origin host name for NTLM connection authentication. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Specify Connection Domain and Origin for NTLM Authentication**

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

**j2ee\_conn\_user Attribute -- Specify a User Name for Connection Authentication**

The `j2ee_conn_user` attribute specifies the user name in an HTTP job.

**Supported Job Types**

This attribute is optional for the [HTTP job type](#).

**Syntax**

This attribute has the following format:

```
j2ee_conn_user: authenticated_user@connhost
```

- ***authenticated\_user@connhost***  
Specifies the user name for connection authentication.  
**Limits:** Up to 145 characters

**NOTE**

- If you specify the `j2ee_conn_user` attribute in your job definition, you must set the password that is associated with the connection user using the `autosys_secure` command.
- You can use global variables to specify the user name for connection authentication. Use the `$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Define an HTTP Job to Perform an HTTP Query**

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the



global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## j2ee\_no\_global\_proxy\_defaults Attribute -- Specify Whether to Use Global Proxy Defaults

The j2ee\_no\_global\_proxy\_defaults attribute specifies whether the agent uses the global proxy defaults specified in the agentparm.txt file.

### NOTE

If you do not specify the j2ee\_no\_global\_proxy\_defaults attribute in your job definition, the job does not use the global proxy configuration specified by the proxy parameters in the agentparm.txt file (the default).

### Supported Job Type

This attribute is optional for the HTTP job type.

### Syntax

This attribute has the following format:

```
j2ee_no_global_proxy_defaults: Y | N
```

- **Y**  
Ignores the global proxy configuration specified by the proxy parameters in the agentparm.txt file. Specify this option if you want to ignore the proxy defaults for this job only. For example, you might ignore the proxy defaults if the request is going to a server on the LAN that does not require the proxy. This is the default.
- **N**  
Uses the global proxy configuration specified by the proxy parameters in the agentparm.txt file.

### Example: Override Global Proxy Defaults in Agent Parameter File

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
```

```
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## j2ee\_parameter Attribute -- Specify Input Parameters

The j2ee\_parameter attribute specifies input parameters in a job.

### Supported Job Types

This attribute is required for the JMS Publish (JMSPUB) job type.

This attribute is optional for the following job types:

- [HTTP](#)
- [POJO](#)
- [RMI](#) (JAVARMI)
- [Session Bean](#) (SESSBEAN)

### Syntax

This attribute has the following format:

```
j2ee_parameter: type=value | payload_job=job_name[, type=value |
payload_job=job_name...]
```

- **type=value**  
Specifies the parameter type and value. In most job types, it specifies the Java class and String value of the parameter. In an HTTP job, it specifies the variable name and value that the agent passes to the program the job invokes.

**Limits:** Up to 4000 characters; case-sensitive

**Examples:** String="5:00PM", studentName=SMITH

String=\$\$TIME, studentName=\${NAME}

#### NOTE

If the package of the Java class is not specified, java.lang is assumed.

- **payload\_job=job\_name** Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

#### NOTE

- This parameter does not apply to HTTP jobs.
- Enclose the job name that contains a colon with quotation marks (" ") or precede it with a backslash (\).

**NOTE**

- You can define up to 64 parameters using one `j2ee_parameter` attribute or multiple entries of `j2ee_parameter`.
- In each `j2ee_parameter` entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4000 characters.
- Order is important. Specify the parameters in the same order as they appear in the method. The job fails if the parameters are listed in the wrong order.
- For HTTP jobs, the `j2ee_parameter` attribute is not applicable if you have set the `invocation_type` attribute value to `delete`.
- You can use global variables to specify the input parameters in a job. You can specify both the key and value pairs as global variables provided that either `$(global_name)` or `${global_name}` format is used to reference the key and value.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.
- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

**Example:**

```
j2ee_parameter: parm1=filter\=name\=datawith\,value
j2ee_parameter: parm1="filter=name=datawith,value"
```

**NOTE****JMS Publishing:**

- To construct a text message, you require a single parameter (String). To construct an object message, the number of parameters that you require depends on the object's constructor. For example, an Integer object requires a single constructor (String). Some objects require no parameters or multiple parameters for construction.
- Leading and trailing spaces are not supported in JMS Publish messages.

**Example: Define an RMI Job to Start a Remote Server Immediately**

Suppose that you want to invoke a method that starts a remote server using remote object activation. You want the server to start immediately.

```
insert_job: start
job_type: JAVARMI
machine: appagent
remote_name: "rmi://remotehost/Test"
method_name: startserver
j2ee_parameter: String="now"
```

**Example: Define an HTTP Job to Perform a Google Search**

Suppose that you want to define a job to perform a Google search and have the results returned to the job's spool file. In this example, the job uses the HTTP GET method to perform the Google search on "ca workload automation". When the job runs, the job's spool file includes all the results of the search.

```
insert_job: google
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://google.com/search"
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_parameter: q="ca workload automation"
```

## j2ee\_proxy\_domain Attribute -- Specify a Domain for Proxy Authentication

The `j2ee_proxy_domain` attribute specifies the domain for proxy authentication in an HTTP job. This attribute is required for NTLM authentication.

### NOTE

If you do not specify the `j2ee_proxy_domain` attribute in your job definition, the job uses the proxy domain set in the `http.proxyDomain` parameter on the agent.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
j2ee_proxy_domain: proxy_domain
```

- **proxy\_domain** Specifies the domain for proxy authentication. **Limits:** Up to 80 characters; case-sensitive

### NOTE

- Your agent administrator can specify a default proxy domain for all HTTP jobs by setting the `http.proxyDomain` parameter in the agent's `agentparm.txt` file.
- The `j2ee_proxy_domain` attribute overrides the default proxy domain specified in the agent's `agentparm.txt`.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the domain for proxy authentication. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

### Example: Override Global Proxy Defaults in Agent Parameter File

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## j2ee\_proxy\_host Attribute -- Specify a Proxy Host

The `j2ee_proxy_host` attribute specifies the proxy host name to use for the request in an HTTP job.

**NOTE**

If you do not specify the `j2ee_proxy_host` attribute in your job definition, the job uses the proxy host set in the `http.proxyHost` parameter on the agent.

**Supported Job Type**

This attribute is optional for the [HTTP](#) job type.

**Syntax**

This attribute has the following format:

```
j2ee_proxy_host: proxy_host
```

- **proxy\_host** Specifies the proxy host name to use for the request.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

- Your agent administrator can specify a default proxy host name for all HTTP jobs by setting the `http.proxyHost` parameter in the agent's `agentparm.txt` file.
- The `j2ee_proxy_host` attribute overrides the default proxy host name specified in the agent's `agentparm.txt`.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the proxy host name to use for the request. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Override Global Proxy Defaults in Agent Parameter File**

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

**j2ee\_proxy\_origin\_host Attribute -- Specify an Origin Host Name for Proxy Authentication**

The `j2ee_proxy_origin_host` attribute specifies the origin host name for proxy authentication in an HTTP job. This attribute is used for NTLM authentication.

**NOTE**

If you do not specify the `j2ee_proxy_origin_host` attribute in your job definition, the job uses the computer name where the agent is running (the default).

## Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
j2ee_proxy_origin_host: proxy_origin
```

- **proxy\_origin** Specifies the origin host name for proxy authentication.  
**Limits:** Up to 256 characters; case-sensitive  
**Default:** The computer name where the agent is running

### NOTE

You can use global variables to specify the origin host name for proxy authentication. Use the `$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

### Example: Override Global Proxy Defaults in Agent Parameter File

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## j2ee\_proxy\_port Attribute -- Specify a Proxy Port

The `j2ee_proxy_port` attribute specifies the proxy port to use for the request in an HTTP job.

### NOTE

If you do not specify the `j2ee_proxy_port` attribute in your job definition, the job uses port 80 (the default).

## Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
j2ee_proxy_port: proxy_port
```

- **proxy\_port**  
Specifies the proxy port to use for the request.

**Limits:** 0-65535

### Example: Override Global Proxy Defaults in Agent Parameter File

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

### j2ee\_proxy\_user Attribute -- Specify a User for Proxy Authentication

The j2ee\_proxy\_user attribute specifies the user name required for proxy authentication.

#### NOTE

If you do not specify the j2ee\_proxy\_user attribute in your job definition, the job uses the proxy user set in the http.proxyUser parameter on the agent.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
j2ee_proxy_user: proxy_user
```

- **proxy\_user** Specifies the user name required for proxy authentication.

**Limits:** Up to 80 characters; case-sensitive; it cannot contain delimiters

#### NOTE

- Your agent administrator can specify a default proxy user name for all HTTP jobs by setting the http.proxyUser parameter in the agent's agentparm.txt file.
- The j2ee\_proxy\_user attribute overrides the default proxy user name specified in the agent's agentparm.txt.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the user name required for proxy authentication. Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

### Example: Override Global Proxy Defaults in Agent Parameter File

This example performs an HTTP query using the HTTP GET method. The output of the invocation is returned in the job's spool file. In this example, the job specifies the connection domain and origin for NTLM authentication, overrides the

global proxy defaults specified in the agentparm.txt file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: HTTP.CON_USER
job_type: HTTP
machine: appagent
invocation_type: GET
provider_url: "http://host.example.com/protected"
j2ee_conn_origin: host.example.com
j2ee_conn_domain: windows_domain
j2ee_conn_user: myuser@windows_domain
j2ee_no_global_proxy_defaults: Y
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_proxy_domain: "http://host.domain.proxy"
j2ee_proxy_host: proxy.example.com
j2ee_proxy_origin_host: "http://host.origin.proxy"
j2ee_proxy_port: 90
j2ee_proxy_user: user01
```

## j2ee\_user Attribute -- Specify a JNDI User Name

The `j2ee_user` attribute specifies the JNDI user name in an Entity Bean, Session Bean, JMS Publish, and JMS Subscribe job. This user name refers to the application server within the JNDI framework. If specified, this authentication information is supplied when creating the initial context.

### Supported Job Types

This attribute is optional for the following job types:

- [Entity Bean \(ENTYBEAN\)](#)
- [JMS Publish \(JMSPUB\)](#)
- [JMS Subscribe \(JMSSUB\)](#)
- [Session Bean \(SESSBEAN\)](#)

### Syntax

This attribute has the following format:

```
j2ee_user: user_name
```

- ***user\_name***  
Specifies the JNDI user ID to be used to gain access to the connection factory.  
**Limits:** Up to 145 characters; case-sensitive; it cannot contain delimiters

### NOTE

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

### Example: Specify a JNDI User ID in a JMS Publish Job

This example publishes the message "this is my message" to the queue named Queue. The Java class of the message is String. The initial context factory supplied by the JNDI service provider is `com.ibm.websphere.naming.WsnInitialContextFactory`. The service provider's URL is `iiop://172.24.0.0:2809`, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the `cyberuser@host` JNDI user name to gain access to the connection factory named `ConnectionFactory`.

```
insert_job: publish
job_type: JMSPUB
```



```

machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue
use_topic: FALSE
message_class: String
j2ee_user: cyberuser@host
j2ee_parameter: java.lang.String="this is my message"

```

**NOTE**

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

**jcl\_library Attribute -- Specify a JCL Library**

The `jcl_library` attribute specifies the JCL library that contains the JCL for your job.

**Supported Job Type**

This attribute is required for the [z/OS Regular \(ZOS\) job type](#).

**Syntax**

This attribute has the following format:

```
jcl_library: library
```

- ***library***

Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job. The JCLLIB is a z/OS data set name. Data set names typically have up to four positions. Each position can be up to eight characters. Separate the positions using periods. The JCL library can be any one of the following:

- A valid z/OS partitioned data set (PDS) name.
- A valid z/OS data set name, prefixed with `lib-` to indicate Librarian input.
- A valid z/OS data set name, prefixed with `pan-` to indicate Panvalet input.
- A valid z/OS data set name, prefixed with `seq-` to indicate sequential input.

**Limits:** Up to 44 characters

**NOTE**

- For partitioned, Librarian, and Panvalet data sets, you can specify a data set name with or without a member name. If the member name is not included in the data set name, specify the `jcl_member` attribute in your job definition.
- For sequential data sets, you must specify the `jcl_member` attribute in your job definition. The `jcl_member` attribute value is used to set the z/OS job name. This job name must match with the job name in the mainframe job control or JCL.

**Example:Specify a JCL Library Using a Partitioned Data Set Name**

This example submits a zOS job named CYBDL01A from the JCL member named CYBDL01A in the CYBDL01.JCLLIB library.

```

insert_job: ZOS_JOB
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB(CYBDL01A)

```

**NOTE**

If the `jcl_member` attribute is not specified, the member name that is specified as a part of the `jcl_library` attribute is used to set the zOS job name.

**Example: Specify a JCL Library Using a Fully Qualified Partitioned Data Set Name**

This example submits a zOS job named CYBTS000 from the JCL member named TAPEJCL in the CYBDL01.JCLLIB library.

```
insert_job: ZOS_TAPE_JOB
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB(TAPEJCL)
jcl_member: CYBTS000
```

**NOTE**

- If the data set name is fully qualified, the `jcl_member` attribute is used to set the zOS job name.
- If the `jcl_member` attribute is not specified, the member name that is specified as a part of the `jcl_library` attribute is used to set the zOS job name.
- A fully qualified data set name includes the member name.

**Example: Specify a JCL Library Using a Sequential Data Set Name**

This example submits a zOS job named CYBTS002 from the CYBSL02.SEQJCL data set.

```
insert_job: ZOS_SEQ_JOB
job_type: ZOS
machine: ZOS1
jcl_library: seq-CYBSL02.SEQJCL
jcl_member: CYBTS002
```

**NOTE**

The `jcl_member` attribute is required when the `jcl_library` attribute specifies a sequential data set.

**jcl\_member Attribute -- Specify a JCL Member**

The `jcl_member` attribute specifies either the JCL member that contains the JCL for your job or the zOS job name when the `jcl_library` attribute specifies a fully qualified data set name.

**NOTE**

A fully qualified data set name includes the member name.

**Supported Job Type**

This attribute is required for the [z/OS Regular \(ZOS\) job type](#) when the `jcl_library` attribute does not specify a fully qualified data set name, or it is specified using a sequential data set name.

**Syntax**

This attribute has the following format:

```
jcl_member: member
```

- ***member***

Specifies the JCL member that contains the JCL for your job. If the `jcl_library` attribute specifies a fully qualified data set name, the `jcl_member` attribute specifies the zOS job name.

**Limits:** Up to 8 characters

**Example: Specify a JCL Member of a Partitioned Data Set**

This example submits a zOS job named CYBDL01A from the JCL member named CYBDL01A in the CYBDL01.JCLLIB library.

```
insert_job: CYBDL01A
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB
jcl_member: CYBDL01A
```

#### Example: Specify a JCL Member from a Sequential Data set

This example submits a zOS job named CYBTS002 from the CYBSL02.SEQJCL data set.

```
insert_job: ZOS_SEQ_JOB
job_type: ZOS
machine: ZOS1
jcl_library: seq-CYBSL02.SEQJCL
jcl_member: CYBTS002
```

#### NOTE

The jcl\_member attribute is required when the jcl\_library attribute specifies a sequential data set.

#### Example: Specify a JCL Member Using a Fully Qualified Partitioned Data Set Name

This example submits a zOS job named CYBTS004 from the JCL member named CYBBR14 in the CYBDL01.JCLLIB library.

```
insert_job: ZOS_TAPE_JOB
job_type: ZOS
machine: ZOS1
jcl_library: CYBDL01.JCLLIB (CYBBR14)
jcl_member: CYBTS004
```

#### NOTE

- If the data set name is fully qualified, the jcl\_member attribute is used to set the zOS job name.
- If the jcl\_member attribute is not specified, the member name that is specified as a part of the jcl\_library attribute is used to set the zOS job name.

## jmx\_parameter Attribute -- Specify JMX Parameters

The jmx\_parameter attribute specifies the new value that the attribute should be set to in a JMX-MBean Attribute Set job, the CREATE parameter list in a JMX-MBean Create Instance job, and the parameter list in a JMX-MBean Operation job.

### Supported Job Types

This attribute is optional for the following job types:

- [JMX-MBean Attribute Set \(JMXMAS\)](#)
- [JMX-MBean Create Instance \(JMXMC\)](#)
- [JMX-MBean Operation \(JXMOP\)](#)

### Syntax

This attribute has the following format:

---

```
jmx_parameter: type=value | payload_job=job_name[, type=value | payload_job=job_name...]
```

- **type=value**

Specifies the Java class and String value for the parameter.

**Limits:** Up to 1024 characters; case-sensitive

**Examples:** String="5:00PM", String[2]="winnt1,winnt2", double=2000

**NOTE**

If the package of the Java class is not specified, java.lang is assumed.

- **payload\_job=job\_name**

Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

**NOTE**

- You can define only a single parameter in a JMX-MBean Attribute Set (JMXMAS) job.
- You can define up to 64 parameters in a JMX-MBean Create Instance (JMXMC) job or a JMX-MBean Operation (JMXMOP) job. Specify the parameters using one jmx\_parameter attribute or multiple entries of jmx\_parameter.
- Enclose the job name that contains a colon with quotation marks (") or precede it with a backslash (\).
- In each jmx\_parameter entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4096 characters.
- Order is important. Specify the parameters in the attribute in the same order as they appear in the parameter list. The job fails if the parameters are listed in the wrong order.
- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

### Example: Change the Value of an MBean Attribute Using a Serialized Java Object

Suppose that you want to set the value of the State attribute of the cdc.jmx.SimpleDynamic MBean to the serialized Java object returned by a JMX-MBean Attribute Set job named size.

```
insert_job: change
job_type: JMXMAS
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"
mbean_name: "DefaultDomain:index=1,type=cdc.jmx.SimpleDynamic"
mbean_attr: State
jmx_parameter: payload_job=size
condition: success(size)
```

### Example: Invoke an Operation on an MBean to Reset the Value of a Parameter

Suppose that you want to invoke the resetmem operation on the config MBean to reset the value of the memory parameter to 50.

```
insert_job: reset
job_type: JMXMOP
machine: agent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_operation: resetmem
jmx_parameter: Integer=50
```

## jmx\_user Attribute -- Specify a JMX User Name

The `jmx_user` attribute specifies the user name in a JMX job.

### Supported Job Types

This attribute is optional for the following job types:

- [JMX-MBean Attribute Get \(JMXMAG\)](#)
- [JMX-MBean Attribute Set \(JMXMAS\)](#)
- [JMX-MBean Create Instance \(JMXMC\)](#)
- [JMX-MBean Operation \(JMXMOP\)](#)
- [JMX-MBean Remove Instance \(JMXMREM\)](#)
- [JMX-MBean Subscribe \(JMXSUB\)](#)

### Syntax

This attribute has the following format:

```
jmx_user: user_name
```

- **user\_name**  
Specifies the JMX user name formatted as follows: `user@token`. Specify the user name in your job definition only when the JMX job requires an ID. The password is resolved by the value that is stored using `autosys_secure`.  
**Limits:** Up to 145 characters

#### NOTE

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

## job\_class Attribute -- Assign a Job Class to a Job

The `job_class` attribute assigns a job class value to a job. When this attribute is specified, the agent interprets the job class against the job classes defined in the `agentparm.txt` file. Each defined job class has a specified number of initiators, which allows for manual load balancing against system resources.

### Supported Job Types

This attribute is optional and applies to all job types *except* the z/OS job types (ZOS, ZOSM, and ZOSDST).

### Syntax

This attribute has the following format:

```
job_class: class
```

- **class**  
Specifies the job class that this job runs under. The agent maintains a list of job classes and the number of initiators assigned to each job class. A job class with more initiators can process more jobs more quickly. For higher-priority jobs, assign a job class that contains more initiators.  
**Limits:** Up to 32 characters; it cannot contain delimiters (such as spaces)

#### NOTE

- The specified class must be defined in the `agentparm.txt` file. If the class name is unavailable on the target machine, the job submission will fail.
- If all the initiators in a specified class are being used, the job does not run. The job waits until the class has an initiator available.
- The agent administrator must specify the number of initiators for job classes in the `agentparm.txt` file. The following example specifies three job classes and a default job class:

```

initiators.class_1=Default,1000
initiators.class_2=JOBCLASS1,20
initiators.class_3=JOBCLASS2,500
initiators.class_4=FILEM,1000

```

- The agent administrator can set the `initiators.afmjobclassmap_N` parameter to automatically assign a job class based on the combination of verbs and subverbs that appear in a message. For more information, see [Workload Automation Agent Parameters](#).
- The `job_class` attribute overrides the default job class specified in the agent's `agentparm.txt`.

### Example: Specify a Job Class in a Command Job on UNIX

This example runs a UNIX Command job under job class JOBCLASS2.

```

insert_job: unix_cmd
job_type: CMD
machine: unixagent
command: /export/home/payroll/data
job_class: JOBCLASS2

```

## job\_criteria Attribute -- Defines Optional Success Criteria

The `job_criteria` attribute defines optional success criteria in a Web Service Document/Literal job.

### Supported Job Type

This attribute is optional for the [Web Services Document/Literal \(WSDOC\)](#) job type.

### Syntax

This attribute has the following format:

```
job_criteria: [criteria_xpath="xpath expression"]criteria_filter="filter expression"
```

- **criteria\_xpath="*xpath expression*"**  
(Optional) Checks the xpath expression that is used to evaluate a return string.

#### NOTE

You can use global variables to check the xpath expression that is used to evaluate a return string. Use the `$global_name`, `$(global_name)`, or `$$ {global_name}` format to reference global variables.

- **criteria\_filter="*filter expression*"**  
Defines a regular expression that is used to evaluate a return string. If the return string matches the regular expression, the job completes successfully. Otherwise, the job fails.

#### NOTE

You can use global variables to define a regular expression that is used to evaluate a return string. Use the `$global_name`, `$(global_name)`, or `$$ {global_name}` format to reference global variables.

#### NOTE

- To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".
- Each return string includes the field name from the SELECT statement and its value, separated by an equal sign (=). For example, consider the query `SELECT ORD_NUM FROM SALES`. To match order number A2976, specify the regular expression `ORD_NUM=A2976`. Specifying the regular expression A2976 does not

match any return string causing the job to fail. You can also specify the regular expression `.*A2976.*`, which matches any return string that ends with A2976.

- If the `criteria_xpath` operand is not present, the `criteria_filter` operand applies to the entire response returned by the call. The response is transformed into a formatted XML document before being checked against the filter. In defining the filter, it may be useful to know how the returned XML document is formatted by the agent. The returned XML is formatted to be easier to read with a four-space indent for nested tags. Specifically, the following keys are used for transformation:  
`"indent" : "yes"`  
`"standalone" : "yes"`  
`"{http://xml.apache.org/xslt}indent-amount" : "4"`
- For Web Service Document/Literal payload producing jobs, only the portion of the returned data that satisfies the `job_criteria` specification will be stored in the output file for use as payload.
- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

### Example: Invoke a Web Service to Get a Stock Quote

The following job calls the `getPrice` operation in the `StockQuoteService` service on the server at `http://wsservice:8080/axis2/services/StockQuoteService`, using the default proxy settings from the `agentparm.txt` file.

The job passes in the following XML:

```
<getPrice><symbol>ABC</symbol></getPrice>
```

The job checks the returned XML for the value in an element that matches the `criteria_xpath` `"//ns:return"`. If the return values contain 42, the job completes successfully. Otherwise, the job fails.

```
insert_job: sql
job_type: WSDOC
machine: wsagent
wsdl_operation: getPrice
port_name: StockQuoteServiceHttpSoap11Endpoint
service_name: StockQuoteService
endpoint_URL: "http://wsservice:8080/axis2/services/StockQuoteService"
WSDL_URL: "http://wsservice:8080/axis2/services/StockQuoteService?wsdl"
ws_parameter: param_name="/getPrice",param_value=""
ws_parameter: param_name="/getPrice/symbol", param_value="ABC"
ws_global_proxy_defaults: Y
job_criteria: criteria_xpath="//ns:return", criteria_filter=".*42.*"
```

### job\_load Attribute -- Define a Job's Relative Processing Load

The `job_load` attribute defines the relative amount of processing power the job you are defining consumes. The range of possible values is arbitrary and user-defined. You can use the `max_load` attribute to assign a machine a "maximum job load" (that is, a measure of CPU load that is desirable to place on a machine at any given time). Then, you can use the `job_load` attribute to assign the job a load value that indicates the relative amount of the machine's load that the job should consume. This scheme allows you to control machine loading and to prevent a machine from being overloaded.

When a job is ready to run on a designated machine but the current load on that machine is too large to accept the new job's load, AutoSys Workload Automation queues the job for that machine so that it runs when sufficient resources are available. However, for this scheme to function properly, you must specify a job load for each job to run on a controlled machine; otherwise, a job could start on a machine without the machine showing the additional load.

When you force a job to start, for example by using the sendevent command to issue a FORCE\_STARTJOB event, the job runs even if its load exceeds the machine's max\_load value.

The default job priority is 0, which means that the job should run immediately and ignore any available load units. Therefore, whenever you set job\_load for a job, you should also set the priority attribute for the job to 1 or higher to ensure that the job\_load setting is used.

**NOTE**

You cannot use the priority or job\_load attribute if you specify a user-defined load balancing script in the machine attribute.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

```
job_load: load_units
```

- **load\_units**

Defines the relative load of the job. This number can be any value in the user-defined range of possible values, which are arbitrary. Define a value that has some relationship to the value defined in the max\_load attribute.

**Default:** 0 (unrestricted)

**Limits:** Up to 10 digits

**Example: Define the Relative Processing Load for a Job**

This example sets the load for a job that typically uses 10% of the CPU, with a range of possible load values from 1-100.

```
job_load: 10
```

**job\_terminator Attribute -- Kill a Job if Its Box Fails**

The job\_terminator attribute specifies whether to use a KILLJOB event to terminate the job if its containing box job completes with a FAILURE or TERMINATED status.

Use this attribute with the box\_terminator attribute to control how nested jobs behave when a job fails.

**Supported Job Types**

This attribute only applies to jobs that are in a box. This attribute is optional for the following job types:



- [Box](#) (BOX)
- [Command](#) (CMD)
- [CPU Monitoring](#) (OMCPU)
- [Database Monitor](#) (DBMON)
- [Database Stored Procedure](#) (DBPROC)
- [Database Trigger](#) (DBTRIG)
- [Disk Monitoring](#) (OMD)
- [File Trigger](#) (FT)
- [File Watcher](#) (FW)
- [IP Monitoring](#) (OMIP)
- [JMS Subscribe](#) (JMSSUB)
- [JMX-MBean Subscribe](#) (JMXSUB)
- [Micro Focus](#) (MICROFOCUS)
- [Oracle Applications Copy Single Request](#) (OACOPY)
- [Oracle Applications Request Set](#) (OASET)
- [Oracle Applications Single Request](#) (OASG)
- [PeopleSoft](#) (PS)
- [Process Automation Process Execution](#) (PAPROC)
- [Process Monitoring](#) (OMP)
- [SAP Batch Input Session](#) (SAPBDC)
- [SAP BW InfoPackage](#) (SAPBWIP)
- [SAP BW Process Chain](#) (SAPBWPC)
- [SAP Data Archiving](#) (SAPDA)
- [SAP Event Monitor](#) (SAPEVT)
- [SAP Job Copy](#) (SAPJC)
- [SAP Process Monitor](#) (SAPPM)
- [SAP R/3](#) (SAP)
- [SQL](#)
- [Text File Reading and Monitoring](#) (OMTF)
- [Windows Event Log Monitoring](#) (OMEL)
- [Windows Service Monitoring](#) (OMS)
- [z/OS Manual](#) (ZOSM)
- [z/OS Regular](#) (ZOS)

## Syntax

This attribute has the following format:

```
job_terminator: Y | N
```

- **Y**  
Terminates the job if the containing box fails or terminates.  
**Note:** You can specify 1 instead of Y.
- **N**  
Does not terminate the job if the containing box fails or terminates. This is the default.

### NOTE

You can specify 0 instead of N.

## Example: Terminate a Job on Box Failure

This example terminates the jobterm1 job if its containing box, unix\_box, fails or terminates.

```
insert_job: jobterm1
job_type: C
box_name: unix_box
command: sleep 120
machine: unixagent
job_terminator: y
```

## **job\_type Attribute -- Specify Job Type**

The `job_type` attribute specifies a job's type.

### **NOTE**

If you do not specify the `job_type` attribute in your job definition, the job type is set to CMD (the default).

### **Supported Job Types**

This attribute is optional for all job types.

### **Syntax**

This attribute has the following format:

```
job_type: type
```

### ***type***

Specifies the type of the job that you are defining. You can specify *one* of the following values:

- 
- BOX -- Box
  - CMD -- Command
  - DBMON -- Database Monitor
  - DBPROC -- Database Stored Procedure
  - DBTRIG -- Database Trigger
  - ENTYBEAN -- Entity Bean
  - FT -- File Trigger
  - FTP -- File Transfer Protocol
  - FW -- File Watcher
  - HTTP -- HTTP
  - HDFS -- Hadoop HDFS
  - HIVE -- Hadoop Hive
  - I5 -- i5/OS
  - INFORMATICA -- Informatica
  - JAVARMI -- Java Remote Method Invocation
  - JMSPUB -- JMS Publish
  - JMSSUB -- JMS Subscribe
  - JMXMAG -- JMX-MBean Attribute Get
  - JMXMAS -- JMX-MBean Attribute Set
  - JMXMC -- JMX-MBean Create Instance
  - JMXMOP -- JMX-MBean Operation
  - JMXMREM -- JMX-MBean Remove Instance
  - JMXSUB -- JMX-MBean Subscribe
  - MICROFOCUS -- Micro Focus
  - OACOPY -- Oracle E-Business Suite Copy Single Request
  - OASET -- Oracle E-Business Suite Request Set
  - OASG -- Oracle E-Business Suite Single Request
  - OMCPU -- CPU Monitoring
  - OMD -- Disk Monitoring
  - OMEL -- Windows Event Log Monitoring
  - OMIP -- IP Monitoring
  - OMP -- Process Monitoring
  - OMS -- Windows Service Monitoring
  - OMTF -- Text File Reading and Monitoring
  - OOZIE -- Hadoop Oozie Workflow
  - PAPROC -- Process Automation Process Execution
  - PAREQ -- Process Automation Start Request Form
  - PIG -- Hadoop Pig
  - POJO -- Plain Old Java Object
  - PROXY -- Remote Execution
  - PS -- PeopleSoft
  - SAPBDC -- SAP Batch Input Session
  - SAPBWIP -- SAP BW InfoPackage
  - SAPBWPC -- SAP BW Process Chain
  - SAPDA -- SAP Data Archiving
  - SAPEVT -- SAP Event Monitor
  - SAPJC -- SAP Job Copy
  - SAPPM -- SAP Process Monitor
  - SAP -- SAP R/3
  - SCP -- Secure Copy
  - SESSBEAN -- Session Bean
  - SNMPGET -- Simple Network Management Protocol Value Get
  - SNMPSET -- Simple Network Management Protocol Value Set
  - SQL -- Structured Query Language Command

**Default:** CMD

#### NOTE

- For backward compatibility, the following values are supported:
  - c (in place of CMD)
  - f (in place of FW)
  - b (in place of BOX)
- In Unicenter AutoSys JM, it was acceptable to enter, for example, "file" or "foo" for the file watcher job type, because only the first character of the input string was checked. However, in Unicenter AutoSys JM r11 and later versions, the syntax checking no longer allows this. You must enter "b" or "box" for box jobs, "c" or "cmd" for command jobs, or "f" or "fw" for file watcher jobs.

#### Example: Specify a CMD Job

This example defines a Command job to run a script on a UNIX computer named unixprod.

```
insert_job: cmd_job1
job_type: CMD
machine: unixprod
command: "/usr/scripts/calcpoc.sh"
```

### lower\_boundary Attribute -- Define the Start of the Range to be Searched (OMTF Jobs)

The lower\_boundary attribute defines the start of the range to be searched in a text file.

This attribute is used with the text\_file\_mode attribute, which specifies the search mode (line, regular expression, or date and time).

#### Supported Job Type

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\) job type](#).

#### Syntax

This attribute has the following format:

```
lower_boundary: start_range
```

- **start\_range**

Defines the start of the range to be searched. The format of this value depends on the text file search mode. The formatting options are as follows:

- Numeric -- Specify a numeric value if the search mode is LINE.
- Regular expression -- Specify a regular expression if the search mode is REGEX.
- Date and time -- Specify date and time values if the search mode is DATETIME. Define the date and time using the format specified by the time format.

**Limits:** Up to 256 characters; case-sensitive. When the search mode is LINE, the upper limit is 9223372036854775807.

**NOTE**

- If you do not specify the `text_file_mode` attribute in the job definition, the search mode is `LINE` by default.
- If you do not specify the `lower_boundary` attribute value in the job definition, the job starts searching from the first line of the text file to the upper boundary value (if specified).
- If you specify the `lower_boundary` attribute value in the job definition and the text file does not include this value, the job starts searching from the last line of the text file, that is, it takes the last line as the starting point for the search. This is applicable only when the monitor mode is `WAIT` or `CONTINUOUS`.
- When the search mode is `REGEX`, you can specify a regular expression in the `lower_boundary` attribute. To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules using a Google search for `java pattern`.
- You can use escape sequences in the regular expression. The following characters have a special meaning in regular expressions. To use these characters literally, precede the characters with one backward slash (`\`).
  - `\` -- back slash
  - `|` -- vertical bar
  - `(` -- opening parenthesis
  - `)` -- closing parenthesis
  - `[` -- opening bracket
  - `{` -- opening brace
  - `^` -- caret (circumflex)
  - `$` -- dollar sign
  - `*` -- asterisk
  - `+` -- plus sign
  - `?` -- question mark
  - `.` -- period
 For example, to match the characters `.*` literally, specify `\*\.*` in your regular expression. The backward slashes escape the characters' special meanings.

**Example: Define the Start of a Range When the Search Mode is LINE**

This example searches the `c:\ca\log` file in line mode. The job starts searching the content from line 143 of the file. The upper boundary is not defined, so the job searches to the last line of the file. The job completes successfully if the `ERROR MESSAGE` string is found.

```
insert_job: omtf_line
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ERROR MESSAGE
text_file_mode: LINE
lower_boundary: 143
monitor_mode: NOW
```

**Example: Define the Start of a Range When the Search Mode is REGEX**

This example searches the `c:\ca\log` file in regular expression mode. The lower boundary is not defined, so the job searches the content from the first line of the file to the upper boundary (a line that contains the word `service`). The job completes successfully if the `ARCHIVE` string is found.

```
insert_job: omtf_regex
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ARCHIVE
text_file_mode: REGEX
upper_boundary: service
```

```
monitor_mode: NOW
```

### Example: Define the Start of a Range When the Search Mode is DATETIME

This example searches the /export/home/logs/transmitter.log file in date and time mode. The job searches the content between May 20, 2010 at midnight and May 27, 2010 at 11:59 p.m. The date and time values are defined using the format specified in the time\_format attribute. The job completes successfully if the transmitted string is found.

```
insert_job: omtf_timedate
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transmitter.log
text_file_filter: transmitted
text_file_mode: DATETIME
lower_boundary: "Thu May 20 00:00:00.000 EDT 2010"
upper_boundary: "Thu May 27 23:59:59.999 EDT 2010"
time_format: "EEE MMM dd HH:mm:ss.SSS zzz yyyy"
time_position: 12
monitor_mode: NOW
```

### lower\_boundary Attribute -- Define the Minimum CPU or Disk Usage to Monitor (OMCPU and OMD Jobs)

The lower\_boundary attribute defines the minimum CPU or disk usage to monitor for.

#### Supported Job Types

This attribute is required for the following job types if WAIT or CONTINUOUS is specified for the monitor\_mode attribute:

- [CPU Monitoring \(OMCPU\)](#)
- [Disk Monitoring \(OMD\)](#)

This attribute does not apply when the monitor mode is NOW.

#### Syntax

This attribute has the following format:

- To specify the lower boundary for CPU usage:

```
lower_boundary: min_percent
```

#### – **min\_percent**

Defines the minimum amount of CPU usage to monitor for in percent.

**Default:** 0

**Limits:** 0-100

#### **NOTE**

If you specify the lower\_boundary attribute without the upper\_boundary attribute, the monitoring takes place between the lower boundary and 100 percent.

- To specify the lower boundary for disk usage:

```
lower_boundary: min_value
```

#### – **min\_value**

Defines the minimum amount of disk usage to monitor for. The disk\_format attribute specifies the unit for this value.

**Limits:**

- 0-100 (This limit applies when the disk format is PERCENT)
- Up to 20 characters (This limit applies for all other disk formats.)

**Example:** 35000000

**NOTE**

If the disk format is PERCENT, the lower boundary must be less than the upper boundary.

**Example: Specify a Lower Boundary of 25 Percent for Monitoring Available CPU**

This example continuously monitors the CPU available on the winagt computer. An alert is written to the scheduler log file each time the available CPU is less than 25 percent or greater than 75 percent, and the CPU usage changes by more than 10 percent. If the change in value is less than or equal to 10, the job does not register a change.

```
insert_job: omcpu_job
job_type: OMCPU
machine: winagt
cpu_usage: FREE
lower_boundary: 25
upper_boundary: 75
inside_range: FALSE
no_change: 10
monitor_mode: CONTINUOUS
```

**Example: Specify a Lower Boundary of 8 GB for Monitoring Available Disk Space**

This example monitors /export/home for available space. The job completes when the available space is between 8 and 10 gigabytes (GB), and the available space changes by more than 15 percent. If the change in value is less than or equal to 15, the job does not register a change.

```
insert_job: omd_unix
job_type: OMD
machine: unixagt
disk_drive: /export/home
disk_space: FREE
disk_format: GB
lower_boundary: 8
upper_boundary: 10
inside_range: TRUE
no_change: 15
```

**mac\_address Attribute -- Specify the Media Access Control (MAC) Address**

The mac\_address attribute specifies the Media Access Control (MAC) address of the computer that receives the Wake on LAN (WOL) signal. The MAC address is an integral part of the Ethernet card (NIC) of the computer's motherboard.

**Supported Job Type**

This attribute is required for the [Wake on LAN \(WOL\) job type](#).

**Syntax**

This attribute has the following format:

```
mac_address: xx-xx-xx-xx-xx-xx
```

- **xx-xx-xx-xx-xx-xx**

Specifies the Media Access Control (MAC) address of the computer that receives the Wake on LAN (WOL) signal. The MAC address consists of six 2-digit hexadecimal values separated by dashes (-). On some UNIX versions, you can obtain the MAC address using ifconfig (listed under HWaddr). On Windows, you can obtain the MAC address using ipconfig /all (listed under Physical Address).

**Limits:** Up to 17 characters

**NOTE**

You can also separate the values using colons. If you use colons, enclose the values in quotes.

**Example: Broadcast the WOL Signal to a MAC Address**

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server with the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
```

**machine Attribute -- Define the Client Where a Job Runs**

The machine attribute defines how agent selection for the job you are defining will be determined. The owner of the job must have permission to access this machine and to execute the specified command on this machine. The machine must already be defined to AutoSys Workload Automation (using the insert\_machine subcommand). If the machine was not previously defined, the product will not create the job.

You cannot use the priority or job\_load attribute if you specify a script for making the selection in the machine attribute.

**Supported Job Types**

This attribute is required for all job types. This attribute does not apply to Box jobs.

**Syntax**

This attribute has the following format:

```
machine: {machine_name [, machine_name]...| `machine chooser`}
```

- **machine\_name** Defines the name of a pre-defined machine object.

**Limits:** Up to 80 alphanumeric characters

**NOTE**

If you specify machine: localhost, the scheduler tries to resolve the localhost value when it runs the job. By default, the localhost value is resolved to the name of the machine where the scheduler was started. If the local machine definition setting is defined to a value other than "localhost", the localhost value is resolved to the name of the specified real machine. For more information about the localmachinedefinition parameter, see [Configure a Scheduler](#). For more information about how the localhost value is resolved, see [The localhost Definition](#).

- **machine chooser**

Defines the name of a program or user-written batch file for the scheduler to execute at run time to determine which machine to use. The scheduler runs this program, writes the name of the machine to its log file, and substitutes this output as the machine name. You must enclose the fully qualified program or batch file name in grave accent marks (`).

**NOTE**

When specifying drive letters for Windows in job definitions, you must escape the colon with backslashes. For example, 'C:\tmp' is valid; 'C:\tmp' is not.

**Example: Run a Job on Either of Two Machines**



This example specifies that the job should run on either the machine named prod or the machine named test:

```
machine: prod, test
```

#### Example: Run a Batch File to Determine the Client on Which to Run a Job (Windows)

This example runs the batch file C:\MYSTUFF\MYCHOOSER.BAT at job run time to determine which machine to use:

```
machine: `C\:\MYSTUFF\MYCHOOSER.BAT`
```

#### Example: Run a Script to Determine the Client on Which to Run a Job (UNIX)

This example runs the script /usr/local/bin/my-machine-chooser at job run time to determine which machine to use:

```
machine: `/usr/local/bin/my-machine-chooser`
```

## machine\_method Attribute -- Specify the Method to Achieve Load Balancing

The machine\_method attribute specifies the method that is used to achieve load balancing. This attribute only applies to virtual machines of type "n" and "v".

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [delete\\_machine Subcommand -- Delete a Real or Virtual Machine](#)
- [insert\\_machine Subcommand -- Add a Machine Definition](#)
- [update\\_machine Subcommand -- Update a Machine Definition](#)

### Syntax

This attribute has the following format:

```
machine_method: cpu_mon|rstatd|job_load|roundrobin|roundrobin_jobload
```

- **cpu\_mon**  
Specifies that the cpu\_mon method is used to determine the percentage of available CPU cycles on the agent. The scheduler runs the CPU Monitoring (OMCPU) job on the target computer to get the available CPU cycles. This is the default.
- NOTE**
- The cpu\_mon method does not apply to z/OS machines (Workload Automation Agent on z/OS) because the OMCPU job is not supported on z/OS.
- **rstatd**  
Specifies that the rstatd method is used by the UNIX schedulers to determine the percentage of available CPU cycles on UNIX computers. The scheduler makes a UNIX remote procedure call to the remote kernel statistics daemon (rstatd) on the target computer to get the available CPU cycles.

**NOTE**

- The rstatd method does not apply to type "n" machines.
- You cannot configure the rstatd method for the Windows scheduler.
- Ensure that the rstatd daemon is running on the UNIX scheduler and on all target UNIX computers.
- *Set the value of the opsys attribute for type 'a' machines to one of the following values that support the rstatd method: aix, hpux, linux, or openvms.*
- If the load balancing request is sent to an agent with an opsys attribute value that does not support the rstatd method, AutoSys Workload Automation uses the cpu\_mon method to obtain the available CPU cycles.

- **job\_load**

Specifies that the job\_load method is used to determine the machine where the job runs. The scheduler uses the job\_load and max\_load attributes to calculate the available load for a computer.

- **roundrobin**

Specifies that the roundrobin method is used to determine the real machine where the job runs from the list of available real machines belonging to a virtual machine. The jobs are run on the available real machines in a round robin manner.

- **roundrobin\_jobload**

Specifies that the job\_load method is used first and then the roundrobin method is used on the resulting real machines to determine the real machine where the job runs. However, if none of the machines have sufficient load units, AutoSys Workload Automation puts the job in QUE\_WAIT status. The job stays in QUE\_WAIT status until one of the machines has sufficient load units available. When sufficient load units are available, the job runs on that machine. In this situation, the roundrobin method is not considered.

**NOTE**

For information about how load balancing works, see [How Load Balancing Works](#). For information about how AutoSys Workload Automation queues jobs, see [Queuing Jobs](#) and [How AutoSys Workload Automation Queues Jobs](#).

**NOTE**

- The cpu\_mon or rstatd settings verify the CPU usage statistics of candidate machines, whereas the job\_load setting checks only the job load and no real-time usage of machine is required.
- If you set the [MachineMethod](#) parameter value and also specify the machine\_method attribute in your job definition, the machine\_method attribute value overrides the value you specified for the MachineMethod parameter.

**Example: Use roundrobin Method to Determine the Real Machine to Run the Job**

This example defines a virtual machine VM to include five real machines (mach1, mach2, mach3, mach4, and mach5). The roundrobin method is used to determine the real machine where the job runs from the list of available real machines belonging to a virtual machine.

```
insert_machine: VM
machine: mach1
machine: mach2
machine: mach3
machine: mach4
machine: mach5
machine_method: roundrobin
```

If 10 jobs named Job1, Job2, Job3, Job4, Job5, Job6, Job7, Job8, Job9, and Job10 are scheduled to run on the virtual machine VM, the jobs are scheduled to run in a round robin manner as follows:

Job1 is run on mach1

Job2 is run on mach2  
 Job3 is run on mach3  
 Job4 is run on mach4  
 Job5 is run on mach5  
 Job6 is run on mach1  
 Job7 is run on mach2  
 Job8 is run on mach3  
 Job9 is run on mach4  
 Job10 is run on mach5

### Example: Use job\_load and roundrobin Methods to Determine the Real Machine to Run the Job

This example defines a virtual machine VM\_JL1 to include four real machines (M1, M2, M3, and M4) with varying max\_load values. The roundrobin\_jobload method is used to determine the real machine where the job runs.

```

insert_machine: VM_JL1
type: v
machine_method: roundrobin_jobload
machine: M1
max_load: 30

machine: M2
max_load: 10

machine: M3
max_load: 10

machine: M4
max_load: 10

```

Suppose that you define four jobs (JobL1, JobL2, JobL3, and JobL4) with job\_load and priority as follows and they are scheduled to run on the virtual machine VM\_JL1:

```

insert_job: JobL1
command: sleep 5
machine: VM_JL1
job_load: 10
priority: 1

insert_job: JobL2
command: sleep 5
machine: VM_JL1
job_load: 10
priority: 2

insert_job: JobL3
command: sleep 5
machine: VM_JL1
job_load: 10
priority: 3

```

```

insert_job: JobL4
command: sleep 5
machine: VM_JL1
job_load: 30
priority: 4

```

The jobs are scheduled to run as follows based on the `max_load` value for each real machine and the job's priority and `job_load` attributes:

```

JobL1 is scheduled on M1
JobL2 is scheduled on M2
JobL3 is scheduled on M3
JobL4 is scheduled on M1

```

## max\_exit\_success Attribute -- Specify Maximum Exit Code for Success

The `max_exit_success` attribute specifies the maximum exit code the job can finish with and still be considered successful. An exit code equal to or less than the specified value is considered a success. AutoSys Workload Automation uses this attribute when a command can exit with more than one exit code, indicating either “degrees of success” or other conditions that cannot indicate a failure. This attribute is useful when defining complex branching logic based on real-time processing.

### NOTE

If you do not specify the `max_exit_success` attribute in your job definition, the scheduling manager interprets only an exit code of zero (0) as job success (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [Command \(CMD\)](#)
- [i5/OS \(I5\)](#)
- [Micro Focus \(MICROFOCUS\)](#)

### Syntax

This attribute has the following format:

```
max_exit_success: exit_code
```

- **exit\_code**  
Defines the maximum exit code that the job can exit with and be considered a success.  
**Default:** 0  
**Limits:** 0-2147483647 (CMD and i5); 0-999999999 (Micro Focus)

### NOTE

- The agent assumes that zero (0) is the maximum exit code for success unless you use this attribute statement to specify a different maximum exit code.
- The agent evaluates job success based on exit code when the job completes, and then assigns the appropriate terminal status (SUCCESS or FAILURE).
- When you manually terminate (kill) a job (send a KILLJOB event), the agent assigns a non-zero exit code to the job and ignores this attribute. In this case, the job enters the TERMINATED status.

### Example: Specify the Maximum Exit Code for Success

This example runs the backup script on a UNIX computer. The job is considered successful if it completes with an exit code of 2 or less.

```
insert_job: unix_script
job_type: CMD
machine: unixagent
command: /usr/common/backup
max_exit_success: 2
```

## max\_run\_alarm Attribute -- Define the Maximum Run Time for a Job

The `max_run_alarm` attribute specifies the maximum run time (in minutes) that a job requires to finish normally. This “reasonability” test can catch errors, such as when a job is stuck in a loop or waiting on a system event that never occurs. If the job runs longer than the specified time, AutoSys Workload Automation generates an alarm to alert the operator to take corrective action.

### NOTE

You can use the `term_run_time` attribute to automatically terminate a job that has run for too long. If you do not define the `term_run_time` attribute for a job, it continues running until it completes or you manually interrupt it.

### Supported Job Types

This attribute is optional for all job types.

We recommend that you define the `max_run_alarm` attribute for file watcher jobs to keep them from running indefinitely. For example, defining the `max_run_alarm` attribute for a file watcher job would address situations in which a communication link is down and preventing the arrival of a file.

### Syntax

This attribute has the following format:

```
max_run_alarm: mins
```

- **mins**

Defines the maximum number of minutes the job requires to finish normally.

**Default:** No maximum set.

**Limits:** 0-2147483647; must be an integer.

### Example: Define the Maximum Run Time for a Job

This example specifies that the product should generate an alarm if the job runs for more than two hours:

```
max_run_alarm: 120
```

## mbean\_attr Attribute -- Specify the MBean Attribute to Query or Set

The `mbean_attr` attribute specifies the name of the MBean attribute you want to query or set in a JMX-MBean Attribute job. You specify the `mbean_attr` attribute to query a JMX server for the value of an MBean attribute or to change the value of an MBean attribute on a JMX server. You can change the value of an MBean attribute using a set value for the attribute or using a serialized Java object passed by another job.

### Supported Job Types

This attribute is required for the following job types:

- [JMX-MBean Attribute Get \(JMXMAG\)](#)
- [JMX-MBean Attribute Set \(JMXMAS\)](#)

## Syntax

This attribute has the following format:

```
mbean_attr: attribute
```

- ***attribute***  
Specifies the name of the MBean attribute that you want to query or set.  
**Limits:** Up to 256 characters; case-sensitive

### Example: Query a JMX Server for the Value of an MBean Attribute

Suppose that you want to know the value of the `cacheSize` attribute for the `Config` MBean. The URL for the JMX server is `service:jmx:rmi:///jndi/rmi://localhost:9999/server`, where `localhost` is the host name and `9999` is the port number.

```
insert_job: query
job_type: JMXMAG
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_attr: cacheSize
```

### Example: Change the Value of an MBean Attribute

Suppose that you want to set the value of the `State` attribute of the `cdc.jmx.SimpleDynamic` MBean to the serialized Java object returned by a JMX-MBean Attribute Set job named `size`.

```
insert_job: change
job_type: JMXMAS
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://agenttest:5099/jmxserver"
mbean_name: "DefaultDomain:index=1,type=cdc.jmx.SimpleDynamic"
mbean_attr: State
jmx_parameter: payload_job=size
condition: success(size)
```

## mbean\_name Attribute -- Specify the Name of the MBean

The `mbean_name` attribute specifies the name of an MBean in a JMX job. An MBean is a managed bean (Java object) that represents an application, a device, or any resource that you want to manage.

### Supported Job Types

This attribute is required for the following job types:

- [JMX-MBean Attribute Get \(JMXMAG\)](#)
- [JMX-MBean Attribute Set \(JMXMAS\)](#)
- [JMX-MBean Create Instance \(JMXMC\)](#)
- [JMX-MBean Operation \(JMXMOP\)](#)
- [JMX-MBean Remove Instance \(JMXMREM\)](#)
- [JMX-MBean Subscribe \(JMXSUB\)](#)

## Syntax

This attribute has the following format:

```
mbean_name: "domain_name:key=value[,key=value...]"
```

- ***"domain\_name:key=value[,key=value...]"***

Specifies the full object name of an MBean.

- **domain\_name**

Specifies the default domain name.

- **key=value[,key=value...]**

Specifies one or more key=value pairs.

**Limits:** Up to 715 characters; case-sensitive

**Example:** "DefaultDomain:type=SimpleDynamic,index=3"

### Example: Query a JMX Server for the Value of an MBean Attribute

Suppose that you want to know the value of the cachesize attribute for the Config MBean. The URL for the JMX server is service:jmx:rmi:///jndi/rmi://localhost:9999/server, where localhost is the host name and 9999 is the port number.

```
insert_job: query
job_type: JMXMAG
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_attr: cachesize
```

## mbean\_operation Attribute -- Specify the Operation to be Invoked

The mbean\_operation attribute specifies the MBean operation to be invoked in a JMX-MBean Operation job.

### Supported Job Type

This attribute is required for the [JMX-MBean Operation \(JMXMOP\) job type](#).

### Syntax

This attribute has the following format:

```
mbean_operation: operation
```

- ***operation***

Specifies the operation to be invoked.

**Limits:** Up to 256 characters; case-sensitive

### Example: Invoke an Operation on an MBean

Suppose that you want to invoke the resetmem operation on the config MBean to reset the value of the memory parameter to 50.

```
insert_job: reset
job_type: JMXMOP
machine: agent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_operation: resetmem
jmx_parameter: Integer=50
```

## media\_type Attribute -- Specify the Type of Payload

The media\_type attribute specifies the type of payload data for a POST or PUT request in an HTTP job.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

## Syntax

This attribute has the following format:

```
media_type: type
```

- **type**

Specifies the type of payload data for a POST or PUT request in an HTTP job.

**Limits:** Up to 128 characters

**Examples:** text/plain, text/html, image/jpeg, application/xml, application/atom+xml, application/javascript, application/json and so on.

### NOTE

- The media\_type attribute is not applicable if you have set the invocation\_type attribute value to delete.
- You can use global variables to specify the type of payload. Use the \$\$global\_name, \$\$ (global\_name), or \$ {global\_name} format to reference global variables.

## message\_class Attribute -- Specify the Java Class of the JMS Message

The message\_class attribute specifies the Java class of the JMS message in a JMS Publish job.

### Supported Job Type

This attribute is required for the [JMS Publish \(JMSPUB\) job type](#).

## Syntax

This attribute has the following format:

```
message_class: class
```

- **class**

Specifies the Java class of the JMS message.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

If the package is not specified, java.lang is assumed.

### Example: Specify Java Class of the JMS Message

This example publishes the message "this is my message" to the queue named Queue. The Java class of the message is String. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the cyberuser@host JNDI user name to gain access to the connection factory named ConnectionFactory.

```
insert_job: publish
job_type: JMSPUB
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue
use_topic: FALSE
message_class: String
j2ee_user: cyberuser@host
j2ee_parameter: java.lang.String="this is my message"
```



**NOTE**

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

## method\_name Attribute -- Specify the Method or Operation to be Invoked

The method\_name attribute specifies the method to be invoked in a job.

**NOTE**

- From AutoSys Workload Automation Release 12.0 onwards, the method\_name attribute is not applicable to HTTP jobs. You can specify the path to the servlet to be invoked using the [provider\\_url](#) attribute.
- When you upgrade from a previous release to AutoSys Workload Automation Release 12.0, the method\_name attribute value is appended to the [provider\\_url](#) attribute value.

### Supported Job Types

This attribute is required for the following job types:

- [Entity Bean \(ENTYBEAN\)](#) when the operation type is UPDATE
- [POJO](#)
- [RMI \(JAVARMI\)](#)
- [Session Bean \(SESSBEAN\)](#)
- [HTTP job type](#)

### Syntax

This attribute has the following format:

```
method_name: method
```

- ***method***

Specifies the method to be invoked. In a Session Bean or Entity Bean job, it specifies the method to be invoked on the application server. In a POJO job, it specifies the Java method to call on the instance of the Java object. In an HTTP job, it specifies the path to the servlet to be invoked. In a RMI job, it specifies the method of the remote Java class to invoke.

**Limits:** Up to 256 characters; case-sensitive

**Example:** GetQuote

### Example: Invoke a Method on a Stateless Session Bean

Suppose that you want to invoke the reverse method on the CybEJBTestBean stateless session bean. The reverse method has one parameter, with type java.lang.String and value a23. The output from the reverse method is saved in the C:\Makapt15 file. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is corbaloc:iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere application server and 2809 is the ORB port. When the job runs, the output of the reverse method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
```

```
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"
```

## mf\_jcl\_name Attribute -- Specify the Path to and Name of a JCL File

The `mf_jcl_name` attribute specifies the path to and name of a JCL file to run. Depending on the value specified in the `mf_jcl_type` attribute, the contents or the name of the JCL file is passed to the Micro Focus Enterprise Server.

### Supported Job Type

This attribute is required for the [Micro Focus \(MICROFOCUS\) job type](#).

### Syntax

This attribute has the following format:

```
mf_jcl_name: jcl_file
```

- ***jcl\_file***

Specifies the path to and name of the JCL file to pass to the Micro Focus Enterprise Server.

**Limits:** Up to 256 characters

Note: If the path contains a colon or an embedded blank, enclose the path with double quotation marks. Embedded double quotation marks must be prefixed with a backslash as an escape character.

**Example:** "C:\Program Files\micro focus\es-jcldemo\if.jcl"

### Example: Specify the Path and Name of a JCL File

This example passes the contents of the JCL file using a Micro Focus Enterprise Server named JCLDEMO. The contents of the `sleepsmp2.jcl` file is read and passed to the server.

```
insert_job: mfjcl_job
job_type: MICROFOCUS
machine: mfagent
mf_jcl_type: CONTENT
mf_server_address_type: NAME
mf_server: JCLDEMO
mf_jcl_name: "C:\Program Files\Micro Focus\es-jcldemo\sleepsmp2.jcl"
```

## mf\_jcl\_type Attribute -- Specify Whether to Pass the JCL by Content or by Reference

The `mf_jcl_type` attribute specifies whether to pass the contents or name of the JCL file to the Micro Focus Enterprise Server.

### NOTE

If you do not specify the `mf_jcl_type` attribute in your job definition, the job passes the contents of the JCL file to the Micro Focus Enterprise Server (the default).

### Supported Job Type

This attribute is required for the [Micro Focus \(MICROFOCUS\) job type](#).

### Syntax

This attribute has the following format:

```
mf_jcl_type: CONTENT | REFERENCE
```

- **CONTENT**

Passes the contents of the JCL file to the Micro Focus Enterprise Server. The submitted JCL is read from the file specified in the `mf_jcl_name` attribute, and the contents of the JCL file are physically sent to the server. This is the default.

- **REFERENCE**

Passes the name of the JCL file to the Micro Focus Enterprise Server as specified in the `mf_jcl_name` attribute. The server then reads and submits the named file.

**NOTE**

To edit the JCL before submitting the job, or to restart the job from a particular step, you must specify `CONTENT`.

**Example: Pass the Contents of the JCL File**

This example passes the contents of the JCL file to the Micro Focus Enterprise Server with IP address 172.31.255.255 and port 2222. The contents of the `if.jcl` file is read and passed to the server.

```
insert_job: mf_job_jcl_by_content
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: ADDRESS
mf_server: "tcp:172.31.255.255:2222"
mf_jcl_type: CONTENT
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
```

**mf\_server Attribute -- Specify the Server Name or IP Address of a Micro Focus Server**

The `mf_server` attribute specifies the server name or the IP address and port of the Micro Focus Enterprise Server. This attribute must be used with the `mf_server_address_type` attribute.

**Supported Job Type**

This attribute is required for the [Micro Focus \(MICROFOCUS\) job type](#).

**Syntax**

This attribute has the following format:

```
mf_server: server_name | "tcp:ip_address:port"
```

- **server\_name**

Specifies the server name of a Micro Focus Enterprise Server.

**Limits:** Up to 256 characters

**NOTE**

The value of the `mf_server_address_type` attribute must be `NAME`.

(This is the default.)

- **"tcp:ip\_address:port"**

Specifies the network protocol, IP address and port of a Micro Focus Enterprise Server. These values must be enclosed in quotation marks.

This parameter includes the following:

- **ip\_address**

Specifies IP address of the Micro Focus Enterprise Server.

- **port**

Specifies the port of the Micro Focus Enterprise Server.

**Example:** "tcp:172.31.255.255:2222"

**NOTE**

The value of the `mf_server_address_type` attribute must be ADDRESS.

**Example: Specify the IP Address and Port of a Micro Focus Server**

This example passes the contents of the JCL file to the Micro Focus Enterprise Server with IP address 172.31.255.255 and port 2222. The contents of the `if.jcl` file is read and passed to the server.

```
insert_job: mf_job_with_server_by_address
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: ADDRESS
mf_server: "tcp:172.31.255.255:2222"
mf_jcl_type: CONTENT
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
```

**mf\_server\_address\_type Attribute -- Specify Whether the Micro Focus Server Value is a Server Name or an IP Address**

The `mf_server_address_type` attribute specifies whether the `mf_server` attribute value is the server name or the IP address and port of the Micro Focus Enterprise Server.

**NOTE**

If you do not specify the `mf_server_address_type` attribute in your job definition, the job interprets the `mf_server` attribute value as a server name (the default).

**Supported Job Type**

This attribute is required for the Micro Focus (MICROFOCUS) job type.

**Syntax**

This attribute has the following format:

```
mf_server_address_type: NAME | ADDRESS
```

- **NAME**

Specifies that the `mf_server` attribute value is the server name of the Micro Focus Enterprise Server. This is the default.

- **ADDRESS**

Specifies that the `mf_server` attribute value is the IP address and port of the Micro Focus server.

**Example: Specify the IP Address and Port of a Micro Focus Server**

This example passes the contents of the JCL file to the Micro Focus Enterprise Server with IP address 172.31.255.255 and port 2222. The contents of the `if.jcl` file is read and passed to the server.

```
insert_job: mf_job_test_with_server_by_address
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: ADDRESS
mf_server: "tcp:172.31.255.255:2222"
mf_jcl_type: CONTENT
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
```

**Example: Specify the Name of a Micro Focus Server**

This example passes the contents of the JCL file using a Micro Focus Enterprise Server named JCLDEMO. The contents of the sleepsmp2.jcl file is read and passed to the server.

```
insert_job: mf_job_with_server_by_name
job_type: MICROFOCUS
machine: mfagent
mf_jcl_type: CONTENT
mf_server_address_type: NAME
mf_server: JCLDEMO
mf_jcl_name: "C:\Program Files\Micro Focus\es-jcldemo\sleepsmp2.jcl"
```

## mf\_user Attribute -- Specify a Micro Focus User ID

The mf\_user attribute specifies a Micro Focus user ID that the job runs under.

### Supported Job Type

This attribute is optional for the [Micro Focus \(MICROFOCUS\) job type](#).

### Syntax

This attribute has the following format:

```
mf_user: user_id@identifier
```

- ***user\_id@identifier***

Specifies the Micro Focus user ID that the job runs under. The user\_id must be defined on the Micro Focus Enterprise server where identifier specifies the name of the server.

**Limits:** Up to 80 characters

#### NOTE

The *user\_id* and *identifier* combination obtains the credentials of the user. For the job to run successfully, the credentials of the user must be defined on AutoSys Workload Automation using the autosys\_secure command.

### Example: Run a Micro Focus Job Under a Micro Focus User ID

This example runs a Micro Focus job under the authority of the Micro Focus user ID named MFUSER01.

```
insert_job: mf_job_with_mf_user
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: NAME
mf_server: mfprod
mf_jcl_type: REFERENCE
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
mf_user: MFUSER01@mfprod
```

## mf\_version Attribute -- Specify the Mainframe Environment

The mf\_version attribute specifies the environment on the mainframe that manages the job you want to run.

#### NOTE

If you do not specify the mf\_version attribute in your job definition, the job assumes that the environment is JES2 (the default).

## Supported Job Type

This attribute is optional for the Micro Focus (MICROFOCUS) job type.

### Syntax

This attribute has the following format:

```
mf_version: VSE/POWER | JES2 | JES3
```

- **VSE/POWER**  
Specifies the VSE/POWER environment.
- **JES2**  
Specifies the JES2 version of the Job Entry Subsystem. This is the default.
- **JES3**  
Specifies the JES3 version of the Job Entry Subsystem.

### Example: Run a Job in the JES3 Environment

This example runs a Micro Focus job in the JES3 environment.

```
insert_job: mf_job_with_version_jes3
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: NAME
mf_server: mfprod
mf_jcl_type: REFERENCE
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
mf_version: JES3
```

## min\_run\_alarm Attribute -- Define the Minimum Run Time for a Job

The min\_run\_alarm attribute specifies the minimum run time (in minutes) that a job should require to finish normally. This “reasonability” test can catch errors, such as when an input file is truncated due to an error. If the job runs in less than the specified time, AutoSys Workload Automation generates an alarm to alert the operator to take corrective action.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
min_run_alarm: mins
```

- ***mins***  
Defines the minimum number of minutes the job should ever require to finish normally.  
**Default:** No minimum set.  
**Limits:** This value can be any integer.

### Example: Define the Minimum Run Time for a Job

This example specifies that the product should generate an alarm if the job finishes in less than two hours:

```
min_run_alarm: 120
```

## modify\_parameter Attribute -- Specify Modify Parameters

The `modify_parameter` attribute specifies the modify parameters in an Entity Bean job. You specify modify parameters to update the property values of an existing entity bean. The modify parameters are used by the setter method specified in the `method_name` attribute.

### Supported Job Type

When the operation type is UPDATE, this attribute is optional for the [Entity Bean \(ENTYBEAN\)](#) job type.

### Syntax

This attribute has the following format:

```
modify_parameter: type=value | payload_job=job_name[, type=value |
payload_job=job_name...]
```

- **type=value**

Specifies the Java class and String value for the parameter.

**Limits:** Up to 1024 characters; case-sensitive

**Examples:** String="5:00PM", String[2]="winnt1,winnt2", double=2000

#### NOTE

If the package of the Java class is not specified, `java.lang` is assumed.

- **payload\_job=job\_name**

Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

#### NOTE

- You can define up to 64 modify parameters using one `modify_parameter` attribute or multiple entries of `finder_parameter`.
- Enclose the job name that contains a colon with quotation marks (" ") or precede it with a backslash (\).
- In each `modify_parameter` entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4096 characters.
- Order is important. Specify the parameters in the same order as they appear in the setter method. The job fails if the parameters are listed in the wrong order.

### Example: Update the Phone Number for the Acme Company in a Database

Suppose that you want to update the phone number for the Acme company to 800-555-0199. The customer entity bean stores the customer ID and phone number. The primary key for the customer is the customer ID. To find the entity bean, the job uses the Acme's customer ID. When the job runs, the Acme company's phone number is changed.

```
insert_job: update
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: UPDATE
method_name: changephone
finder_name: findByPrimaryKey
finder_parameter: String="customerid"
modify_parameter: String="800-555-0199"
```

## monitor\_cond Attribute -- Specify a Condition to Monitor For

The monitor\_cond attribute specifies a condition to monitor the database for. The job completes when the condition is met.

### Supported Job Type

This attribute is optional for the [Database Monitor \(DBMON\) job type](#).

### Syntax

This attribute has the following format:

```
monitor_cond: condition
```

#### condition:

Specifies the condition to monitor the database. This condition is equivalent to an SQL where clause.

#### Limits:

Up to 500 characters; case-sensitive.

#### NOTE

- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.
- You can use global variables to specify the condition to monitor in the database. Use the \$\$ global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Monitor for an increase in the Number of Rows in a Table Using a Condition

This example monitors the Inventory\_List table for an increase in the number of rows. If the number of rows increases and the number of units of ProductA has fallen below 1000, the job completes. The database user ID is set to the user who invokes jil to define the job (the default owner).

```
insert_job: dbmon_job
job_type: DBMON
machine: dbagt
tablename: Inventory_List
monitor_type: INCREASE
monitor_cond: "ProductA < 1000"
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

## monitor\_mode Attribute -- Specify Whether to Monitor Conditions Immediately or Continuously

The monitor\_mode attribute specifies whether the job waits until the monitor conditions are met or tries to verify them immediately.

#### NOTE

- If you do not specify the monitor\_mode attribute in your OMIP, OMP, or OMS job definition, the job checks for the conditions immediately and completes (the default).
- If you do not specify the monitor\_mode attribute in your OMCPU, OMD, OMTF, or OMEL job definition, the job waits until the specified conditions are met before completing (the default).

### Supported Job Types



This attribute is optional for the following job types:

- [CPU Monitoring](#) (OMCPU)
- [Disk Monitoring](#) (OMD)
- [IP Monitoring](#) (OMIP)
- [Process Monitoring](#) (OMP)
- [Text File Reading and Monitoring](#) (OMTF)
- [Windows Event Log Monitoring](#) (OMEL)
- [Windows Service Monitoring](#) (OMS)

## Syntax

This attribute has the following format:

- To specify a mode for OMIP, OMP, and OMS jobs:

```
monitor_mode: WAIT | NOW
```

- To specify a mode for OMCPU, OMD, OMTF, and OMEL jobs:

```
monitor_mode: WAIT | NOW | CONTINUOUS
```

- **WAIT**

Waits for the conditions to occur. When the conditions are met, the job completes. This is the default for OMCPU, OMD, OMTF, and OMEL jobs.

### NOTE

- When using the WAIT value in Disk Monitoring jobs and the disk\_format attribute is set to any value other than PERCENT, you must also specify the lower\_boundary attribute, the upper\_boundary attribute, or both.
- When using the WAIT value in Text File Reading and Monitoring jobs, do not specify the upper\_boundary attribute.

- **NOW**

Checks for the conditions immediately. If the conditions are met, the job completes successfully. If the conditions are not met, the job fails. This is the default for OMIP, OMP, and OMS jobs.

- **CONTINUOUS**

Monitors for the conditions continuously. Each time the specified conditions occur, an alert is written to the scheduler log file (event\_demon.\$AUTOSERV on UNIX and event\_demon.%AUTOSERV% on Windows).

### NOTE

- When using the CONTINUOUS value in Disk Monitoring jobs and the disk\_format attribute is set to any value other than PERCENT, you must also specify the lower\_boundary attribute, the upper\_boundary attribute, or both.
- When using the CONTINUOUS value in Text File Reading and Monitoring jobs, consider the following points:
  - If the job finds one or more occurrences of the searched text, the first trigger takes place at the first occurrence only. Subsequently, the job only searches for new lines that are added to the text file. In other words, the job resumes the search from the first line following the last line of the file from the previous search.
  - You cannot specify the upper\_boundary attribute.
  - When using the text\_file\_filter\_exists attribute, it must be set to TRUE (the default). The job continues to monitor the file for the existence of the specified text string until the job is completed manually. Each time the text string is found, an alert is triggered.
- This value does not apply to IP Monitoring (OMIP), Process Monitoring (OMP), and Windows Service Monitoring (OMS) jobs.
- To end continuous monitoring, you must complete the job manually by issuing the following command:

```
sendevent -E KILLJOB -J job_name
```

### Example: Monitor a Text File Continuously

This example searches the transmitter.log file for the text string "Warning".

```
insert_job: textfile_job
job_type: OMTF
machine: monagt
text_file_name: /export/home/log/transmitter.log
text_file_filter: Warning
text_file_mode: LINE
lower_boundary: 25
text_file_filter_exists: TRUE
monitor_mode: CONTINUOUS
```

When the job first runs, it searches the content between line 25 and the end of the file. An alert is written to the scheduler log file the first time that the string is found. In other words, suppose that the file contains multiple occurrences of "Warning" between lines 25 to the end of the file, as follows:

```
.
.
.
25
26 Warning
27
28 Warning
29
30
31 Warning
.
.
.
EOF
```

When the job first runs, the trigger only occurs at the first occurrence of the text (line 26). Subsequently, the job continues monitoring only the new data that is appended to the file. An alert is triggered each time the string is found in the appended data.

#### NOTE

Alerts are not triggered for new occurrences of the "Warning" string in the data that has already been searched. For example, suppose that the job has already searched lines 25 to 100 of the file. The file is then modified to include "Warning" on line 30. During continuous monitoring, an alert is not triggered for that occurrence.

This job runs until it is completed manually.

### Example: Check a Process Status Immediately

The job in this example checks the process status immediately and completes successfully if the process is running. If the process is not running, the job fails.

```
insert_job: omp_unix
job_type: OMP
machine: unixagt
process_name: 2568
process_status: RUNNING
```

```
monitor_mode: NOW
```

### Example: Monitor an IP Address Until It Stops

This example monitors a device with DNS name myhost. When the device stops running, the job completes.

```
insert_job: omip_job
job_type: OMIP
machine: monagt
ip_host: myhost
ip_status: STOPPED
monitor_mode: WAIT
```

## monitor\_type Attribute -- Specify the Type of Database Change to Monitor For

The monitor\_type attribute specifies the type of database change to monitor for.

### NOTE

If you do not specify the monitor\_type attribute in your job definition, the job monitors for an increase or a decrease in the number of rows in the database table (the default).

### Supported Job Type

This attribute is optional for the [Database Monitor \(DBMON\) job type](#).

### Syntax

This attribute has the following format:

```
mon_type: INCREASE | DECREASE | VARIANCE
```

- **INCREASE**  
Monitors for an increase in the number of rows in the database table.
- **DECREASE**  
Monitors for a decrease in the number of rows in the database table.
- **VARIANCE**  
Monitors for an increase or a decrease in the number of rows in the database table. This is the default.

### NOTE

By default, the specified table is polled every 10 seconds for changes to the number of rows.

### Example: Monitor for an Increase in the Number of Rows in a Table

This example monitors the Inventory\_List table for an increase in the number of rows. By default, the database is checked for changes every 10 seconds. When the number of rows increases, the job completes. This job runs on the dbagt agent computer and the database user ID is set to the user who invokes jil to define the job (the default owner).

```
insert_job: dbmon_job1
job_type: DBMON
machine: dbagt
tablename: Inventory_List
monitor_type: INCREASE
connect_string: "jdbc:db2://172.31.255.255:50000/SAMPLE"
```

## must\_complete\_times Attribute -- Specify the Time a Job Must Complete By

The `must_complete_times` attribute defines the time or a list of times that a job must complete by. If a job run does not complete by the specified time, an alarm (MUST\_COMPLETE\_ALARM) is issued.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following formats:

```
must_complete_times: "hh:mm[, hh:mm...]"
must_complete_times: +minutes
```

- **"hh:mm[, hh:mm...]"**

Defines one or more absolute times that a job must complete by before an alarm is issued. Specify the time in 24-hour format. Separate each time with a comma.

**Limits:** 00:00-71:59 (2 calendar days ahead of the current calendar day)

**Notes:**

- To define absolute times, you must also specify the following attributes in your job definition:
  - `date_conditions`
  - `start_times`
- If you specify the `start_mins` attribute in the job definition, you can only define relative times. You will get an error if you define absolute times with `start_mins`.
- If a job has multiple start times, you must specify the same number of must complete times in the `must_complete_times` attribute. For example, if the job runs at three different times during the day, you must specify three must complete times, corresponding to each run of the job. If the number of must complete times does not match the number of start times, the job terminates.
- The must complete time for a run must be earlier than the start time for the next run. For example, the following values are invalid because the must complete time for the job's second run (11:10) occurs later than the start time of the job's third run (11:00):

```
start_times: "10:00, 10:30, 11:00"
must_complete_times: "10:10, 11:10, 11:20"
```

- Absolute times can cross up to 2 calendar days. For example, suppose that you define a job that starts at 11:00 a.m. and you want to specify a must complete time of 10:00 a.m. the next day. This must complete time crosses the day boundary. Because the value of the must complete time is less than the value of the start time, you cannot specify 10:00 a.m. as the must complete time. If 10:00 a.m. is specified, the job issues an error message and terminates the client request. Instead, you must specify the must complete time as 34:00. This time is calculated as follows:

```
must complete time + 24 hours
= 10:00 + 24 hours
= 34:00
```

- **+minutes**

Defines a relative time in minutes that a job must complete by before an alarm is issued.

**Limits:** +0 to +4319 (2 calendar days from the current calendar day)

**Notes:**

- To define a relative time, you must also specify the following attributes in your job definition:
  - `date_conditions`
  - `start_mins` or `start_times`

The must complete times are calculated relative to the `start_mins` or `start_times` attributes.
- A relative time can cross up to 2 calendar days. For example, suppose that you define a job that starts at 11:57 p.m. and you want the job to complete within 30 minutes of the start time. The must start time is calculated as follows:

```

start time + relative must complete time
= 11:57 + 30
= 12:27 a.m.

```

This must complete time crosses the day boundary.

#### NOTE

- The entire value can be up to 255 characters.
- You cannot use both absolute times and relative times in the same `must_complete_times` attribute. You must use one format for all specified times.
- You *can* specify both `must_start_times` and `must_complete_times` attributes in the same job definition.

#### Example: Specify Absolute Must Complete Times

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. The job must complete by 10:08 a.m., 11:08 a.m., and 12:08 p.m. respectively. Otherwise, an alarm is issued for each missed complete time.

```

insert_job: test_must_start_complete
command: /opt/StartTransactions.sh
machine: localhost
date_conditions: y
days_of_week: all
start_times: "10:00, 11:00, 12:00"
must_start_times: "10:02, 11:02, 12:02"
must_complete_times: "10:08, 11:08, 12:08"

```

#### NOTE

The number of must complete times must match the number of start times. Otherwise, the job terminates. For example, the job terminates if it has one start time and two must complete times, as follows:

```

start_times: "10:00"
must_complete_times: "10:08, 12:08"

```

#### Example: Specify an Absolute Must Complete Time on the Next Day

This example defines a job to run every day at 12:00 a.m. Suppose that you want each job run to end by 10:12 a.m. the next day. You must specify 34:12 in the `must_complete_times` attribute.

```

insert_job: job3
command: echo "hello"
machine: localhost
date_conditions: y
days_of_week: all
start_times: "12:00"
must_start_times: "34:10"
must_complete_times: "34:12"

```

The must complete time is calculated as follows:

```

must complete time + 24 hours
= 10:12 + 24
= 34:12

```

#### Example: Specify an Absolute Must Complete Time That Crosses Two Days

This example defines a job to run every day at 16:00 (4:00 p.m.). Suppose that you want each job run to end by 6:12 a.m. two days later. You must specify 54:12 in the `must_complete_times` attribute.

```
insert_job: job3
job_type: CMD
machine: localhost
command: echo "hello"
date_conditions: y
days_of_week: all
start_times: "16:00"
must_start_times: "54:10"
must_complete_times: "54:12"
```

The must complete time is calculated as follows:

```
must complete time + 48 hours
= 6:12 + 48 hours
= 54:12
```

### Example: Specify Relative Must Complete Times

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. Each job run must complete within 8 minutes after each start time (10:08 a.m., 11:08 a.m., and 12:08 p.m.). Otherwise, an alarm is issued for each missed complete time.

```
insert_job: test_must_start_complete
job_type: CMD
machine: localhost
command: /opt/StartTransactions.sh
date_conditions: y
days_of_week: all
start_times: "10:00, 11:00, 12:00"
must_start_times: +3
must_complete_times: +8
```

### Example: Specify Relative Must Complete Times With start\_mins

This example defines a job to run every day at 10 minute intervals in each hour (for example, 2:00 p.m., 2:10 p.m., 2:20 p.m., and so on). Each job run must complete within 7 minutes after the specified start times. Otherwise, an alarm is issued for each missed complete time. For instance, the 2:10 p.m. job run must complete by 2:17 p.m.

```
insert_job: test_must_start_complete
job_type: CMD
machine: localhost
command: /opt/StartTransactions.sh
date_conditions: y
days_of_week: all
start_mins: 10, 20, 30, 40, 50, 00
must_start_times: +2
must_complete_times: +7
```

### Example: Specify an Unmatched Number of Must Start and Must Complete Times

Suppose that you want to define a job to run twice. You specify two start times and only one must start time and one must complete time. The job is not inserted into the database because the number of must start times and must complete times must match the number of start times.

```
insert_job: job2
job_type: CMD
machine: localhost
command: echo "Hello"
date_conditions: y
days_of_week: all
start_times: "12:00, 13:00"
must_start_times: "12:01"
must_complete_times: "12:02"
```

To successfully insert the job definition, you must specify two must start times and two must complete times, as follows:

```
insert_job: job2
job_type: CMD
machine: localhost
command: echo "Hello"
date_conditions: y
days_of_week: all
start_times: "12:00, 13:00"
must_start_times: "12:01, 13:01"
must_complete_times: "12:02, 13:02"
```

CHK\_START and CHK\_COMPLETE events for the must start times and must complete times are inserted to database. The job's first run occurs at 12:00 p.m. If the job runs for three minutes, the MUST\_START\_ALARM is not generated because the job started by the specified must start time (12:01). However, the MUST\_COMPLETE\_ALARM is issued because the job ran past the specified must complete time (12:02).

## must\_start\_times Attribute -- Specify the Time a Job Must Start By

The must\_start\_times attribute defines the time or a list of times that a job must start by. If the job does not start by the specified time, an alarm (MUST\_START\_ALARM) is issued.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following formats:

```
must_start_times: "hh:mm[, hh:mm...]"
```

```
must_start_times: +minutes
```

- **"hh:mm[, hh:mm...]"**  
Defines one or more absolute times that a job must start by before an alarm is issued. Specify the time in 24-hour format. Separate each time with a comma.

**Limits:** 00:00-71:59 (2 calendar days ahead of the current calendar day)

**Notes:**

- To define absolute times, you must also specify the following attributes in your job definition:
  - `date_conditions`
  - `start_times`
- If you specify the `start_mins` attribute in the job definition, you can only define relative times. You will get an error if you define absolute times with `start_mins`.
- If a job has multiple start times, you must specify the same number of must start times in the `must_start_times` attribute. For example, if the job runs at three different times during the day, you must specify three must start times, corresponding to each run of the job. If the number of must start times does not match the number of start times, the job terminates.
- The must start time for a run must be earlier than the start times for the next run. For example, the following values are invalid because the must start time for the job's second run (11:10) occurs later than the start time of the job's third run (11:00):

```
start_times: "10:00, 10:30, 11:00"
```

```
must_start_times: "10:10, 11:10, 11:20"
```

- Absolute times can cross up to 2 calendar days. For example, suppose that you define a job that starts at 11:00 a.m. and you want to specify a must start time of 10:00 a.m. the next day. This must start time crosses the day boundary. Because the value of the must start time is less than the value of the start time, you cannot specify 10:00 a.m. as the must start time. If 10:00 a.m. is specified, the job issues an error message and terminates the client request. Instead, you must specify the must start time as 34:00. This time is calculated as follows:

```
must start time + 24 hours
```

```
= 10:00 + 24 hours
```

```
= 34:00
```

- **+minutes**

Defines a relative time in minutes that a job must start by before an alarm is issued.

**Limits:** +0 to +4319 (2 calendar days from the current calendar day)

**Notes:**

- To define a relative time, you must also specify the following attributes in your job definition:
  - `date_conditions`
  - `start_mins` or `start_times`
 The must start times are calculated relative to the `start_mins` or `start_times` attributes.
- A relative time can cross up to 2 calendar days. For example, suppose that you define a job that starts at 11:57 p.m. and you want the job to start within 30 minutes of the start time. The must start time is calculated as follows:

```
start time + relative must start time
```

```
= 11:57 + 30
```

```
= 12:27 a.m.
```

This must start time crosses the day boundary.



**NOTE**

- The entire value can be up to 255 characters.
- You cannot use both absolute times and relative times in the same `must_start_times` attribute. You must use one format for all specified times.
- You *can* specify both `must_start_times` and `must_complete_times` attributes in the same job definition.

**Example: Specify Absolute Must Start Times**

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. The job must start by 10:02 a.m., 11:02 a.m., and 12:02 p.m. respectively. Otherwise, an alarm is issued for each missed start time.

```
insert_job: test_must_start_complete

command: /opt/StartTransactions.sh

machine: localhost

date_conditions: y

days_of_week: all

start_times: "10:00, 11:00, 12:00"

must_start_times: "10:02, 11:02, 12:02"

must_complete_times: "10:08, 11:08, 12:08"
```

**NOTE**

The number of must start times must match the number of start times. Otherwise, the job terminates. For example, the job terminates if it has one start time and two must start times, as follows:

```
start_times: "10:00"

must_start_times: "10:02, 12:02"
```

**Example: Specify an Absolute Must Start Time on the Next Day**

This example defines a job to run every day at 12:00 a.m. Suppose that you want each job run to start by 10:10 a.m. the next day. You must specify 34:10 in the `must_start_times` attribute.

```
insert_job: job3

command: echo "hello"

machine: localhost

date_conditions: y

days_of_week: all
```

```
start_times: "12:00"

must_start_times: "34:10"

must_complete_times: "34:12"
```

The must start time is calculated as follows:

```
must start time + 24 hours

= 10:10 + 24 hours

= 34:10
```

### **Example: Specify an Absolute Must Start Time That Crosses Two Days**

This example defines a job to run every day at 16:00 (4:00 p.m.). Suppose that you want each job run to start by 6:10 a.m. two days later. You must specify 54:10 in the `must_start_times` attribute.

```
insert_job: job3

job_type: CMD

machine: localhost

command: echo "hello"

date_conditions: y

days_of_week: all

start_times: "16:00"

must_start_times: "54:10"

must_complete_times: "54:12"
```

The must start time is calculated as follows:

```
must start time + 48 hours

= 6:10 + 48 hours

= 54:10
```

### **Example: Specify Relative Must Start Times**

This example defines a job to run every day at 10:00 a.m., 11:00 a.m., and 12:00 p.m. Each job run must start within 3 minutes after each start time (10:03 a.m., 11:03 a.m., and 12:03 p.m.). Otherwise, an alarm is issued for each missed start time.

```
insert_job: test_must_start_complete

job_type: CMD

machine: localhost

command: /opt/StartTransactions.sh

date_conditions: y

days_of_week: all

start_times: "10:00, 11:00, 12:00"

must_start_times: +3

must_complete_times: +8
```

#### **Example: Specify Relative Must Start Times With start\_mins**

This example defines a job to run every day at 10 minute intervals in each hour (for example, 2:00 p.m., 2:10 p.m., 2:20 p.m., and so on). Each job run must start within 2 minutes after the specified start times. Otherwise, an alarm is issued for each missed start time. For instance, the 2:10 p.m. job must start by 2:12 p.m.

```
insert_job: test_must_start_complete

job_type: CMD

machine: localhost

command: /opt/StartTransactions.sh

date_conditions: y

days_of_week: all

start_mins: 10, 20, 30, 40, 50, 00

must_start_times: +2

must_complete_times: +7
```

#### **Example: Specify a Relative Must Start Time on the Next Day**

This example specifies a relative must start time of 12:01 a.m. the next day. The must start time is calculated by adding three minutes to the start time. The calculated time crosses into the next day.

```
insert_job: job2

job_type: CMD

machine: localhost

command: echo "Hello"

date_conditions: y

days_of_week: all

start_times: "23:58"

must_start_times: +3
```

#### Example: Specify an Unmatched Number of Must Start and Must Complete Times

Suppose that you want to define a job to run twice. You specify two start times and only one must start time and one must complete time. The job is not inserted into the database because the number of must start times and must complete times must match the number of start times.

```
insert_job: job2
job_type: CMD
machine: localhost
command: echo "Hello"
date_conditions: y
days_of_week: all
start_times: "12:00, 13:00"
must_start_times: "12:01"
must_complete_times: "12:02"
```

To successfully insert the job definition, you must specify two must start times and two must complete times, as follows:

```
insert_job: job2
job_type: CMD
machine: localhost
command: echo "Hello"
date_conditions: y
days_of_week: all
start_times: "12:00, 13:00"
must_start_times: "12:01, 13:01"
must_complete_times: "12:02, 13:02"
```

CHK\_START and CHK\_COMPLETE events for the must start times and must complete times are inserted to database. The job's first run occurs at 12:00 p.m. If the job runs for three minutes, the MUST\_START\_ALARM is not generated because the job started by the specified must start time (12:01). However, the MUST\_COMPLETE\_ALARM is issued because the job ran past the specified must complete time (12:02).

## **n\_retrys Attribute -- Define the Number of Times to Restart a Job After a Failure**

The `n_retrys` attribute specifies how many times to attempt to restart the job after it exits with a FAILURE status. If a job exits with a TERMINATED status, it does not restart.

This attribute applies to application failures (for example, AutoSys Workload Automation cannot find a file or a command, permissions are not properly set, and so on). It does not apply to system or network failures (for example, when a computer is unavailable, socket connection time-outs, failure of a fork in the agent, failure of the file system space resource check, and so on).

### **UNIX Notes:**

- Job restarts after system or network failures are controlled by the `MaxRestartTrys` parameter in the configuration file.
- The delay between restarts is determined by the `RestartConstant` and `RestartFactor` parameters in the configuration file, up to the maximum specified by the `MaxRestartWait` parameter.

### **Windows Notes:**

- Job restarts after system or network failures are controlled by the setting in the Max Restart Trys field on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator.
- The delay between restarts is determined by the Restart Constant and Restart Factor settings on the Scheduler - AutoSys Workload Automation Administrator window of AutoSys Workload Automation Administrator, up to the maximum specified by the Max Restart Wait setting.

### **Supported Job Types**

This attribute is optional for all job types.

### **Syntax**

This attribute has the following format:

```
n_retrys: attempts
```

- **attempts** Defines the number of times to attempt to restart the job after it exits with a FAILURE status.  
**Limits:** This value can be any integer in the range 0 to 20.  
**Default:** 0

### **NOTE**

The `n_retrys` value applies when the command associated with the job fails. When a job fails for other reasons, such as invalid credentials or unavailable space, the `MaxRestartTrys` value in the configuration file defines how many times the job should be restarted. When the `MaxRestartTrys` value is zero, the job is not restarted.

### **Example: Set the Job to Automatically Restart Up to Five Times**

This example sets the job to automatically restart up to five times after an application failure:

```
n_retrys: 5
```

This means that the job would start as scheduled, and if it fails, it would restart up to five times for a total of six attempts.

## no\_change Attribute -- Define the Minimum Change Required in CPU or Disk Usage

The no\_change attribute defines the value that the CPU or disk usage must change by to trigger the job. If the change in CPU or disk usage is within this specified value, the job does not trigger.

### Supported Job Types

This attribute is optional for the following job types:

- CPU Monitoring (OMCPU)
- Disk Monitoring (OMD)

#### NOTE

To use this attribute with OMD jobs, you must specify WAIT or CONTINUOUS for the monitor\_mode attribute. This attribute does not apply to OMD jobs when the monitor mode is NOW.

### Syntax

This attribute has the following format:

- To specify a no\_change value for CPU usage:  
`no_change: delta_percent`
  - **delta\_percent**  
 Defines the percentage value that the CPU usage must change to trigger the job.  
**Limits:** 1-100
- To specify a no\_change value for disk usage:  
`no_change: delta_value`
  - **delta\_value**  
 Defines the value that the CPU usage must change by to trigger the job. The unit for this value is specified by the disk\_format attribute.  
**Limits:** Up to 10 characters  
**Example:** 100

#### NOTE

The job will trigger only if the following conditions are met:

- The change value is within the range specified by the lower\_boundary and upper\_boundary attributes.
- The change value is greater than the delta specified by the last scanned amount. That is, the first time CPU or disk usage matches the job criteria, an alert is written to the scheduler log file. Subsequently, an alert is triggered only if the change in CPU or disk usage is greater than the delta calculated using the last scanned amount that was registered in a trigger.

### Example: Monitor for a Change in CPU Usage Greater Than 10 Percent

This example continuously monitors the CPU available on the winagt computer. An alert is written to the scheduler log file each time the available CPU is less than 25 percent or greater than 75 percent, and the CPU usage changes by more than 10 percent. If the change in value is less than or equal to 10, the job does not register a change.

```
insert_job: omcpu_job
job_type: OMCPU
machine: winagt
cpu_usage: FREE
lower_boundary: 25
upper_boundary: 75
inside_range: FALSE
no_change: 10
```

```
monitor_mode: CONTINUOUS
```

### Example: Monitor for a Change in Disk Usage Greater than 100 KB

This example continuously monitors the available disk space in kilobytes (KB) on the local Windows C drive. When the available space is in the 35000000 to 36000000 KB range, the first alert is written to the scheduler log file.

Subsequently, an alert is triggered each time the available disk space is within the specified boundaries and the disk usage changes by more than 100 KB. If the amount of change is less than or equal to 100 KB, the job does not register a change.

```
insert_job: omd_job
job_type: OMD
machine: winagent
disk_drive: C
disk_space: FREE
disk_format: KB
lower_boundary: 35000000
upper_boundary: 36000000
inside_range: TRUE
no_change: 100
monitor_mode: CONTINUOUS
```

The following table shows four sequential scans:

Scan	Scanned Amount (Kilobytes)	Does the Trigger Occur?
1	35018896	Yes
2	35018900	No. Comparing scan 2 to scan 1, the delta value is only 4 KB. This scanned amount will not be included in the next calculation.
3	35018795	Yes. Comparing scan 3 to scan 1, the delta value is greater than 100 KB. The delta value of the next scan will be calculated using the scan 3 value of 35018795.
4	36000001	No. The scanned amount is outside the lower and upper boundary range.

### notification\_alarm\_types Attribute -- Specify the Alarm For Which to Send the Email Notification

The notification\_alarm\_types attribute specifies the alarm for which to send the email notification.

#### Supported Job Types

This attribute is optional for all job types.

#### Syntax

This attribute has the following format:

```
notification_alarm_types: alarm_type
```

- **alarm\_type**  
Specifies the alarm for which to send the email notification. You can specify any of the following alarms:

- MINRUNALARM
- JOBFailure
- MAX\_RETRIES
- STARTJOBFAIL
- EVENT\_HDLR\_ERROR
- EVENT\_QUE\_ERROR
- JOBNOT\_ONICEHOLD
- MAXRUNALARM
- RESOURCE
- MISSING\_HEARTBEAT
- CHASE
- DATABASE\_COMM
- VERSION\_MISMATCH
- DUPLICATE\_EVENT
- INSTANCE\_UNAVAILABLE
- AUTO\_PING
- EXTERN\_DEPS\_ERROR
- SERVICEDESK\_FAILURE
- UNINOTIFY\_FAILURE
- CPI\_JOBNAME\_INVALID
- CPI\_UNAVAILABLE
- MUST\_START\_ALARM
- MUST\_COMPLETE\_ALARM
- WAIT\_REPLY\_ALARM
- KILLJOBFAIL
- SENDSIGFAIL
- REPLY\_RESPONSE\_FAIL
- RETURN\_RESOURCE\_FAIL
- RESTARTJOBFAIL
- JOBNOT\_ONNOEXEC
- QUEUEDJOB\_STARTFAIL
- SUSPENDJOBFAIL
- RESUMEJOBFAIL

#### NOTE

- You can specify multiple alarms, separated by semi-colons or commas. If you set the `notification_alarm_types` attribute value to ALL or all, an email notification is sent when any of the above listed alarms are raised.
- To send an email notification using the email notification template when an alarm is raised, set the `send_notification` attribute value to **y** or **f** and specify the `notification_template`, `notification_alarm_types`, and the `notification_emailaddress_on_alarm` or `notification_emailaddress` attributes in your job definition. In a job run, for each alarm type, you can use the [RepeatAlarmNotification](#) parameter to configure whether you want to send an email notification for all alarms, just the first alarm, or only the last alarm.

#### Example: Specify Alarms for Which to Send an Email Notification

This example specifies that an email notification is sent to `admin@example.com` using the `emailtemplate` email notification template when the `JOBFailure` or `MAXRUNALARM` alarm is raised.

```
insert_job: alarm_notification
```



```

machine: localhost
job_type: c
command: as_test -t 2
send_notification:y
notification_alarm_types: JOBFAILURE, MAXRUNALARM
notification_template: emailtemplate
notification_emailaddress_on_alarm :admin@example.com

```

## notification\_emailaddress Attribute -- Identify the Recipient of the Email Notification by Email Address

The notification\_emailaddress attribute specifies the email address of the recipient of the email notification. The email notification is generated when you set the send\_notification attribute to **y** or **f** and the job completes successfully, fails, or terminates or an alarm is raised.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
notification_emailaddress: address
```

- **address**

Specifies the email address of the user who receives the email notification.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ , ; \_ -) for each line.

### NOTE

- You can specify multiple email addresses, separated by semi-colons or commas.
- Alternatively, you can use multiple entries of notification\_emailaddress to specify the email addresses of the users to receive the email notification. For example, you can specify multiple email addresses using two notification\_emailaddress attributes as follows:

```

notification_emailaddress:
 user1@example.com;user2@example.com;user3@example.com
notification_emailaddress:
 user4@example.com;user5@example.com;user6@example.com

```

When the SMTP server rejects an email address, AutoSys Workload Automation treats it as an invalid email address. If you specify one or more such invalid email addresses, AutoSys Workload Automation ignores these invalid email addresses and sends the email notification to only the valid email addresses. For each invalid email address, a message is displayed to indicate that the email notification is ignored. When you add these invalid email addresses to the SMTP server, they become valid. However, you must pause and resume the AutoSys Workload Automation services so that the changes take effect and AutoSys Workload Automation sends the email notifications to these valid email addresses next time.

### Example: Send Email Notification to Multiple Recipients When the Job Completes Successfully

This example specifies that an email notification is sent to

admin@example.com, admin1@example.com, admin2@example.com, and success@example.com when email\_job\_1 completes successfully.

```

insert_job: email_job_1
machine: localhost

```

```

job_type: c
command: as_test -t 2
send_notification:1
notification_emailaddress: admin@example.com, admin1@example.com, admin2@example.com
notification_emailaddress_on_success: success@example.com
notification_emailaddress_on_terminated: terminate@example.com
notification_msg:"email_job_1 has completed"

```

## notification\_emailaddress\_on\_alarm Attribute -- Specify the Email Address of the Recipient of the Email Notification When an Alarm is Generated

The notification\_emailaddress\_on\_alarm attribute specifies the email address of the recipient of the email notification when an alarm is raised.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
notification_emailaddress_on_alarm: email_alarm
```

- **email\_alarm**

Specifies the email address of the recipient of the email notification when an alarm is raised. You can specify the alarm for which to send the email notification using the [notification\\_alarm\\_types](#) attribute.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ , ; \_ -) for each line.

### NOTE

- To send an email notification using the email notification template when an alarm is raised, set the send\_notification attribute value to **y** or **f** and specify the notification\_template, notification\_alarm\_types, and the notification\_emailaddress\_on\_alarm or notification\_emailaddress attributes in your job definition. In a job run, for each alarm type, you can use the [RepeatAlarmNotification](#) parameter to configure whether you want to send an email notification for all alarms, just the first alarm, or only the last alarm.
- You can specify multiple email addresses, separated by semi-colons or commas.
- Alternatively, you can use multiple entries of notification\_emailaddress\_on\_alarm to specify the email addresses of the recipients to receive the email notification when an alarm is raised. For example, you can specify multiple email addresses using two notification\_emailaddress\_on\_alarm attributes as follows:

```

notification_emailaddress_on_alarm:
 user1@example.com;user2@example.com;user3@example.com
notification_emailaddress_on_alarm:
 user4@example.com;user5@example.com;user6@example.com

```

When the SMTP server rejects an email address, AutoSys Workload Automation treats it as an invalid email address. If you specify one or more such invalid email addresses, AutoSys Workload Automation ignores these invalid email addresses and sends the email notification to only the valid email addresses. For each invalid email address, a message is displayed to indicate that the email notification is ignored. When you add these invalid email addresses to the SMTP server, they become valid. However, you must pause and resume the AutoSys Workload Automation services so that the changes take effect and AutoSys Workload Automation sends the email notifications to these valid email addresses the next time you run the job.

### Example: Specify Email Addresses of the Recipient of the Email Notification When an Alarm is Raised

This example specifies that an email notification is sent to email@example.com and admin@example.com using the emailtemplate email notification template when the JOBFAILURE or MAXRUNALARM alarm is raised.

```
insert_job: alarm_notification
machine: localhost
job_type: c
command: as_test -t 2
send_notification:y
notification_alarm_types: JOBFAILURE, MAXRUNALARM
notification_template: emailtemplate
notification_emailaddress_on_alarm :admin@example.com
notification_emailaddress: email@example.com
```

## notification\_emailaddress\_on\_failure Attribute -- Specify the Email Address of the Recipient of the Email Notification When the Job Fails

The notification\_emailaddress\_on\_failure attribute specifies the email address of the recipient of the email notification when the job fails.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

```
notification_emailaddress_on_failure: email_failure
```

- **email\_failure**

Specifies the email address of the recipient of the email notification when the job fails.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ , ; \_ -) for each line.

### NOTE

- You can specify multiple email addresses, separated by semi-colons or commas.
- If you specify both the notification\_emailaddress and notification\_emailaddress\_on\_failure attributes in your job definition, the email notification is sent to the email addresses that you specified for these two attributes. The email notification that is sent to the email addresses that you specified in the notification\_emailaddress\_on\_failure use the email notification template.
- Alternatively, you can use multiple entries of notification\_emailaddress\_on\_failure to specify the email addresses of the recipients to receive the email notification when the job fails. For example, you can specify multiple email addresses using two notification\_emailaddress\_on\_failure attributes as follows:

```
notification_emailaddress_on_failure:
 user1@example.com;user2@example.com;user3@example.com
notification_emailaddress_on_failure:
 user4@example.com;user5@example.com;user6@example.com
```

When the SMTP server rejects an email address, AutoSys Workload Automation treats it as an invalid email address. If you specify one or more such invalid email addresses, AutoSys Workload Automation ignores these invalid email addresses and sends the email notification to only the valid email addresses. For each invalid email address, a message is displayed to indicate that the email notification is ignored. When you add these invalid email addresses to the SMTP server, they become valid. However, you must pause and resume the AutoSys Workload Automation services so that the changes take effect and AutoSys Workload Automation sends the email notifications to these valid email addresses the next time you run the job.

### Example: Send an Email Notification when a Job Fails

This example specifies that an email notification is sent to admin2@example.com and admin@example.com when the email\_job\_2 fails. The email notification that is sent to admin@example.com uses the emailtemplate2 email notification template.

```
insert_job: email_job_2
machine: localhost
job_type: c
command: as_test -t 2
send_notification:f
notification_template: emailtemplate2
notification_emailaddress: admin2@example.com
notification_emailaddress_on_failure:admin@example.com
```

## notification\_emailaddress\_on\_success Attribute -- Specify the Email Address of the Recipient of the Email Notification When the Job Completes Successfully

The notification\_emailaddress\_on\_success attribute specifies the email address of the recipient of the email notification when the job completes successfully.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

```
notification_emailaddress_on_success: email_success
```

- **email\_success**

Specifies the email address of the recipient of the email notification when the job completes successfully.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -)

### NOTE

:

- You cannot specify the notification\_template attribute in your job definition.
- You can specify multiple email addresses, separated by semi-colons or commas.
- Alternatively, you can use multiple entries of notification\_emailaddress\_on\_success to specify the email addresses of the users to receive the email notification on when the job completes successfully. For example, you can specify multiple email addresses using two notification\_emailaddress\_on\_success attributes as follows:

```
notification_emailaddress_on_success:
 user1@example.com;user2@example.com;user3@example.com
notification_emailaddress_on_success:
 user4@example.com;user5@example.com;user6@example.com
```

When the SMTP server rejects an email address, AutoSys Workload Automation treats it as an invalid email address. If you specify one or more such invalid email addresses, AutoSys Workload Automation ignores these invalid email addresses and sends the email notification to only the valid email addresses. For each invalid email address, a message is displayed to indicate that the email notification is ignored. When you add these invalid email addresses to the SMTP server, they become valid. However, you must pause and resume the AutoSys Workload Automation services so that the changes take effect and AutoSys Workload Automation sends the email notifications to these valid email addresses the next time you run the job.

### Example: Send an Email Notification When a Job Completes Successfully

This example specifies that an email notification is sent to admin@example.com using the emailtemplate2 email notification template when the email\_job\_2 completes successfully.

```
insert_job: email_job_2
machine: localhost
job_type: c
command: as_test -t 2
send_notification:y
notification_template: emailtemplate2
notification_emailaddress_on_success:admin@example.com
```

## notification\_emailaddress\_on\_terminated Attribute -- Specify the Email Address of the Recipient of the Email Notification When the Job Terminates

The notification\_emailaddress\_on\_terminated attribute specifies the email address of the recipient of the email notification when the job terminates.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
notification_emailaddress_on_terminated: email_terminate
```

- **email\_terminate**

Specifies the email address of the user who receives the email notification when the job terminates.

**Limits:** Up to 1024 characters including alphanumeric and special characters (@ . , ; \_ -)

### NOTE

- You can specify multiple email addresses, separated by semi-colons or commas.
- Alternatively, you can use multiple entries of notification\_emailaddress\_on\_terminated to specify the email addresses of the users to receive the email notification when the job terminates. For example, you can specify multiple email addresses using two notification\_emailaddress\_on\_terminated attributes as follows:

```
notification_emailaddress_on_terminated:
 user1@example.com;user2@example.com;user3@example.com
notification_emailaddress_on_terminated:
 user4@example.com;user5@example.com;user6@example.com
```

When the SMTP server rejects an email address, AutoSys Workload Automation treats it as an invalid email address. If you specify one or more such invalid email addresses, AutoSys Workload Automation ignores these invalid email addresses and sends the email notification to only the valid email addresses. For each invalid email address, a message is displayed to indicate that the email notification is ignored. When you add these invalid email addresses to the SMTP server, they become valid. However, you must pause and resume the AutoSys Workload Automation services so that the changes take effect and AutoSys Workload Automation sends the email notifications to these valid email addresses the next time you run the job.

### Example: Send an Email Notification when a Job Terminates

This example specifies that an email notification is sent to admin@example.com using the emailtemplate2 email notification template when email\_job\_2 terminates.

```
insert_job: email_job_2
machine: localhost
```

```

job_type: c
command: as_test -t 2
send_notification: y
notification_template: emailtemplate2
notification_emailaddress_on_terminated: admin@example.com

```

## notification\_msg Attribute -- Define the Message to Include in the Notification

The notification\_msg attribute defines the message to include in the notification generated when you have set the send\_notification attribute to **y**, **1**, or **f** and the job you are defining completes with the appropriate status.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
notification_msg: value
```

- **value** Defines a message to include in the notification.  
**Limits:** Up to 255 alphanumeric characters  
**Example:** "The server appears to be down."

### NOTE

- The notification\_msg and notification\_template attributes are mutually exclusive.
- You must specify the notification\_msg and the [notification\\_emailaddress Attribute -- Identify the Recipient of the Email Notification by Email Address](#) in your job definition.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.

### Example: Define a Message to Include in a Notification

This example defines a message to include in a notification:

```
notification_msg: "The server appears to be down."
```

## notification\_template -- Specify the Template for the Email Notification

The notification\_template attribute specifies the email notification template that is used to send the email notification.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
notification_template: template_name
```

- **template\_name**  
Specifies the name of the email notification template that is used to send the email notification.  
**Limits:** 256 characters

**NOTE**

- The [notification\\_msg Attribute -- Define the Message to Include in the Notification](#) and `notification_template` attributes are mutually exclusive.
- To send an email notification using the email notification template when an alarm is raised, set the `send_notification` attribute value to **y** or **f** and specify the `notification_template`, `notification_alarm_types`, and the `notification_emailaddress_on_alarm` or `notification_emailaddress` attributes in your job definition.
- To send an email notification using the email notification template when a job completes successfully or terminates, set the `send_notification` attribute value to **y** and specify the `notification_template` and the `notification_emailaddress`, `notification_emailaddress_on_success`, or `notification_emailaddress_on_terminated` attributes in your job definition.
- To send an email notification using the email notification template when a job fails, set the `send_notification` attribute value to **f** and specify the `notification_template` and the `notification_emailaddress` or `notification_emailaddress_on_failure` attributes in your job definition.

A sample email notification template is available in the `$AUTOSYS/install/email_templates` (UNIX) or `%AUTOSYS%\install\email_templates` (Windows) directory. You can edit the sample template to create an email template glob using the `insert_glob` subcommand and this glob can be used in the email notification template. If you update the glob that is used in the email notification template, you must also update it using the [update\\_glob](#) subcommand for the changes to get reflected in the database.

The sample email notification template includes a section that is sent when the job completes successfully, a section that is sent when the job terminates, a section that is sent when the job fails, and a section when an alarm is raised. Based on the job status or the alarm raised, the appropriate section is used to send the email notification.

The sample notification template includes the following fields:

- **[SUCCESS] | [FAILURE] | [TERMINATED] | [ALARM]**  
Indicates the alarm or the status of the job for which you want to send a notification.
- **SUBJECT**  
Defines an email subject to include in the email notification.
- **MESSAGE**  
Defines the message to include in the email notification.
- **STDOUT**  
Specifies whether to attach the STDOUT file to the email notification. The valid values are yes and no.  
**Default:** no; does not attach the STDOUT file to the email notification.
- **STDERR**  
Specifies whether to attach the STDERR file to the email notification. The valid values are yes and no.  
**Default:** no; does not attach the STDERR file to the email notification.
- **LOG\_LINES**  
Specifies the maximum number of lines to include in the STDOUT or STDERR file that is attached to the email notification. If you set this value to 0 (zero), the complete STDOUT or STDERR file is attached to the email notification.  
**Default:** 1000

You can include the following standard tags and the jil attributes and global variables in the email notification template:

**Standard tags:**

- **Instance: <<instance>>**  
Specifies the name of the AutoSys Workload Automation instance.
- **Scheduler host: <<scheduler\_host>>**  
Specifies the host name of the scheduler.
- **Job name: <<job\_name>>**  
Specifies the name of the job.
- **Status: <<job\_status>>**

Specifies the status of the job.

- **Job run machine: <<run\_machine>>**  
Specifies the name of the machine where the job runs.
- **Job run identifier: <<run\_identifier>>**  
Specifies the job run identifier.
- **Job start time: <<start\_time>>**  
Specifies the time that the job starts.
- **Job completion time: <<completion\_time>>**  
Specifies the time the job completes by.
- **Alarm name: <<alarm\_name>>**  
Specifies the name of the alarm.
- **Alarm message: <<alarm\_message>>**  
Specifies the message to include in the alarm.
- **Exit code: <<exit\_code>>**  
Specifies the exit code of the job.

#### JIL attributes:

- **Command: <<command>>**  
Specifies the command to run the job.
- **Job Description: <<description>>**  
Specifies the description of the job.
- **Application: <<application>>**  
Specifies the name of the application that is associated with the job.
- **Group: <<group>>**  
Specifies the name of the group that is associated with the job.
- **Profile: <<profile>>**  
Specifies the name of the profile that is associated with the job.
- **Machine: <<machine>>**  
Specifies the name of the machine where the job runs.

#### NOTE

You can include any JIL attribute in the notification template in the *jil\_attribute*: <<jil\_attribute>> format. In the notification, <<jil\_attribute>> is replaced with the attribute value.

#### Global variables:

- **Global Variable: <<\$\$GV1>>**  
Specifies the global variable upon which the job depends.

#### NOTE

When the global variable contains special characters, ensure that the global variable is enclosed in curly braces.

The following example contains a period (.) and the name is added inside curly braces.

Print GLOB: <<\${GLOBAL.VARIABLE}>>

#### Example: Send an Email Notification Using a Template When the Completes Successfully

This example specifies that an email notification is sent to admin@example.com and user@example.com using the emailtemplate email notification template when email\_job\_2 completes successfully.

```
insert_job: email_job_2
machine: localhost
job_type: c
command: as_test -t 2
```



```

send_notification: y
notification_emailaddress: admin@example.com
notification_emailaddress_on_success: user@example.com
notification_template: emailtemplate

```

## one\_way Attribute -- Specify Whether to Invoke the Service One Way

The `one_way` attribute specifies whether the job completes without waiting for a response after the agent invokes the operation.

### NOTE

If you do not specify the `one_way` attribute in your job definition, the job waits for a response after the agent invokes the operation before completing (the default).

### Supported Job Type

This attribute is optional for the Web Service (WBSVC) job type.

### Syntax

This attribute has the following format:

```
one_way: TRUE | FALSE
```

- **TRUE**  
Invokes the service one way. After the agent invokes the operation, the Web Service job completes without waiting for a response.
- **FALSE**  
Does not invoke the service one way. This is the default.

### Example: Invoke a Web Service Operation

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```

insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote

```

```

endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false

```

## operation\_type Attribute -- Specify the Operation Type

The `operation_type` attribute indicates the operation to perform on the entity bean in an Entity Bean job. The Entity Bean job lets you create an entity bean, update the property values of an existing entity bean, or remove an entity bean from the database.

### Supported Job Type

This attribute is required for the [Entity Bean \(ENTYBEAN\) job type](#).

### Syntax

This attribute has the following format:

```
operation_type: CREATE | UPDATE | REMOVE
```

- **CREATE**  
Creates an entity bean that is stored in a relational database on your application server.
- **UPDATE**  
Updates the property values of an entity bean instance in a relational database on your application server.
- **REMOVE**  
Removes an instance of an entity bean stored in a relational database on your application server.

### NOTE

- To update an entity bean, you require the `finder_name`, `finder_parameter`, `method_name`, and `modify_parameter` attributes.
- To remove an entity bean, you require the `finder_name` and `finder_method` attributes.

### Example: Remove an Entity Bean

Suppose that you want to remove the customer record for the Acme customer. The record is stored in the database by the customer ID. When the job runs, the row in the customer table that corresponds to the Acme customer ID is removed.

```

insert_job: remove
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
operation_type: REMOVE
finder_name: findByPrimaryKey
finder_parameter: String="customerid"

```

## oracle\_appl\_name\_type Attribute -- Specify the Type of Oracle Applications Name

The `oracle_appl_name_type` attribute specifies whether the name of the Oracle E-Business Suite application is the short name or the display name.

**NOTE**

- This attribute must be used with the `oracle_appl_name` attribute.
- The default type is `SHORT`. If you do not specify the `oracle_appl_name_type` attribute in your job definition, the job interprets the `oracle_appl_name` value to be a short name.

**Supported Job Types**

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)

**Syntax**

This attribute has the following format:

```
oracle_appl_name_type: SHORT | DISPLAY
```

- **SHORT**

Indicates that the Oracle E-Business Suite application is specified by its short name. The short name is defined by the Oracle Applications Concurrent Manager. This is the default.

- **DISPLAY**

Indicates that the Oracle E-Business Suite application is specified by its display name.

**Example: Specify the Display Name of an Oracle E-Business Suite Application**

This example runs a single request program named `FNDSCARU`. The single request belongs to the application with the display name `Application Object Library` in Oracle E-Business Suite. The job runs under the `SYSADMIN` user with System Administrator responsibility. The history of the program is included in the description.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
```

**Example: Specify the Short Name of an Oracle E-Business Suite Application**

This example runs a single request program named `FNDSCARU`. The single request belongs to the application with the short name `ACCOUNTS` in Oracle E-Business Suite. The job runs under the `SYSADMIN` user with System Administrator responsibility.

```
insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

## oracle\_appl\_name Attribute -- Specify the Name of an Oracle Applications Application

The `oracle_appl_name` attribute specifies the name of the Oracle Applications application that the single request or request set belongs to.

This attribute must be used with the `oracle_appl_name_type` attribute.

### Supported Job Type

This attribute is required for the following job types:

- Oracle Applications Request Set (OASET)
- Oracle Applications Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_appl_name: short_name | display_name
```

#### ***short\_name***

Specifies the short name of the Oracle Applications application. The short name is defined by the Oracle Applications Concurrent Manager.

**Limits:** Up to 50 characters; case-sensitive.

#### **NOTE**

The value of the `oracle_appl_name_type` attribute must be SHORT.

#### ***display\_name***

Specifies the display name of the Oracle Applications application. In Oracle Applications, the display name is part of the request definition and is found in the Application field.

**Limits:** Up to 256 characters; case-sensitive.

#### **NOTE**

The value of the `oracle_appl_name_type` attribute must be DISPLAY.

### Example: Specify the Display Name of an Oracle Applications Application

This example runs a single request program named FNDSCARU. The single request belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The history of the program is included in the description.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
```

### Example: Specify the Short Name of an Oracle Applications Application

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator

```

## oracle\_args Attribute -- Define Argument Values to Pass to a Single Request

The `oracle_args` attribute defines the argument values to pass to an Oracle E-Business Suite single request. The argument values override the default values that are defined by the registered Oracle Applications Concurrent Manager program.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_args: argument[,argument...]
```

- **argument**

Defines each argument value to pass to an Oracle Applications single request. In Oracle Applications, arguments are part of the program definition and are found in the Concurrent Program Parameters dialog.

#### NOTE

- You can specify up to 100 arguments. Separate each argument with a comma.
- The entire value can be up to 500 characters.
- Do not add a space after a comma unless the argument value starts with a space. If a value starts with a space, enclose the value in single quotation marks.
- To use a default value that is defined in Oracle E-Business Suite, enter one comma in the argument position and specify Y in the `oracle_use_arg_def` attribute.
- You can use global variables to specify the argument. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example:** X22F,,X56R143,, 'X23T'

### Example: Specify Argument Values

Suppose that you want to pass the argument values, T,DefArg2,X23,DefArg5, to a single request program. The job uses the argument values that are specified in the job definition and the default values defined in Oracle E-Business Suite as follows:

- The first argument, T, and the third argument, X23, are specified in the job definition.
- The second argument, DefArg2, and the fifth argument, DefArg5, are defined as defaults in Oracle E-Business Suite.
- The fourth argument is not specified in the job definition or defined as a default on Oracle E-Business Suite, so the agent passes an empty string for that argument.

```

insert_job: oasg_args
job_type: oasg

```

```

machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_program: FNDSCARU
oracle_use_arg_def: Y
oracle_args: T,,X23,,

```

## oracle\_custom\_property Attribute -- Specify Value Set Expressions for Default Values

The `oracle_custom_property` attribute specifies value set expressions for default values. For example, if the default value of an argument is an SQL statement that contains profile (`:$PROFILES$`) or flexfield (`:$FLEX$`) expressions, you can specify how the agent resolves the expressions using this attribute.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_custom_property: prop_key=expression, prop_value=value
```

- **prop\_key=expression**  
Specifies the expression to resolve.  
**Limits:** Up to 256 characters; case-sensitive  
**Examples:**
  - `":$FLEX$.EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT"`
  - `":$PROFILES$.GL_ACCESS_SET_ID"`
- **prop\_value=value**  
Specifies the default value for the expression. The value can contain a constant or an SQL SELECT query to run.  
**Limits:** Up to 256 characters; case-sensitive  
**Example:** `SELECT EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT from SOME_TABLE where SOME_KEY = 'Default'`

#### NOTE

- If you want the agent to quote the resolved value, put the default value in single quotes, for example, `'SELECT '1' FROM DUAL'`.
- The agent does not resolve Field or Segment validation value sets.

#### NOTE

- To specify multiple value set expressions, use multiple entries of `oracle_custom_property`.
- Enclose values that contain special characters in double quotation marks.
- Within quoted strings, double quotation marks (") and backslashes (\) must be prefixed with a backslash as an escape character.

### Example: Specify a Default Value Set Expression

Suppose that the default value of an argument in a single request program is the following SQL statement:

```
SELECT inventory_item_id from mtl_system_items where SEGMENT1 = :$FLEX
$.EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT
```

To resolve the flexfield expression, :\$FLEX\$.EAM\_SRS\_MAINTAINED\_ITEM\_NAMES\_W\_ACT, you define the following default value set expression in the job:

```
:$FLEX$.EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT='SELECT EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT
from SOME_TABLE where SOME_KEY = 'Default''
```

The default value for the expression is enclosed in single quotes so that the agent quotes the resolved value.

Assuming that the SQL SELECT statement returns Acct123, the following SQL query determines the default argument value:

```
SELECT inventory_item_id from mtl_system_items where SEGMENT1 = 'Acct123'
```

The following job definition specifies the default value set expression:

```
insert_job: oaset_resp
job_type: oaset
machine: cyboa
oracle_appl_name_type: DISPLAY
oracle_appl_name: Process Manufacturing Financials
oracle_resp: System Administrator
oracle_user: SYSADMIN
oracle_use_arg_def: Y
oracle_custom_property: key=":$FLEX$.EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT",
Value="'SELECT EAM_SRS_MAINTAINED_ITEM_NAMES_W_ACT from SOME_TABLE where SOME_KEY =
'Default''"
```

## oracle\_desc Attribute -- Describe a Single Request Job

The oracle\_desc attribute defines a description for an Oracle E-Business Suite Single Request job.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_desc: description
```

- **description** Defines a description for an Oracle E-Business Suite Single Request job. The description displays in the Name column of the Requests dialog.

**Limits:** Up to 30 characters; case-sensitive

### NOTE

- Your agent administrator can specify a default description for all Oracle E-Business Suite Single Request jobs by setting the oa.default.desc parameter in the agent's agentparm.txt file.
- The oracle\_desc attribute overrides the default description specified in the agent's agentparm.txt.

### Example: Specify a Description for an Oracle E-Business Suite Single Request Job

This example runs a single request program named FNDSCARU. The single request belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The history of the program is included in the description.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
```

## oracle\_mon\_children Attribute -- Specify Whether to Monitor the Children Programs

The oracle\_mon\_children attribute specifies whether the children of the Oracle E-Business Suite programs are monitored. Program children are programs that are released by the parent program.

**Note:** To use this attribute, you must also specify the oracle\_mon\_children\_delay attribute in your job definition.

### Supported Job Types

- [Oracle E-Business Suite Copy Single Request \(OACOPY\)](#)
- [Oracle E-Business Suite Request Set \(OASET\)](#)
- [Oracle E-Business Suite Single Request \(OASG\)](#)

### Syntax

This attribute has the following format:

```
oracle_mon_children: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Specifies that the children of the Oracle E-Business Suite programs are monitored.
- **N | F | NO | FALSE**  
Specifies that the children of the Oracle E-Business Suite programs are not monitored. This is the default.

### NOTE

Up to five levels of children can be monitored for each job.

### Example: Specify that the Children of a Single Request Program are Monitored

This example runs a request set named FNDRSSUB873. The request set belongs to the application with the short name FND in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The children of the single request program are monitored.

```
insert_job: OA7
job_type: OASET
machine: localhost
oracle_appl_name: FND
oracle_appl_name_type: SHORT
oracle_req_set: FNDRSSUB873
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_mon_children: Y
```



```
oracle_mon_children_delay: 10
oracle_save_output: Y
oracle_use_arg_def: Y
```

## oracle\_mon\_children\_delay Attribute -- Define the Time to Wait Before Monitoring Children Programs

The `oracle_mon_children_delay` attribute defines the number of seconds to wait after an Oracle E-Business Suite program completes before monitoring its children.

### NOTE

To use this attribute, you must also specify the `oracle_mon_children` attribute in your job definition.

### Supported Job Types

This attribute is optional for the following job types:

- [Oracle E-Business Suite Copy Single Request \(OACOPY\)](#)
- [Oracle E-Business Suite Request Set \(OASET\)](#)
- [Oracle E-Business Suite Single Request \(OASG\)](#)

### Syntax

This attribute has the following format:

```
oracle_mon_children_delay: seconds
```

- **seconds**

Defines the number of seconds to wait after an Oracle E-Business Suite program completes before monitoring its children.

**Limits:** 0-99999

**Example:** 60

### NOTE

When the `oracle_mon_children_delay` attribute is defined in an Oracle E-Business Suite Request Set job, the setting is applied to all the programs in the request set. You cannot specify a different setting for each program.

### Example: Delay Monitoring Children by 60 Seconds

This example runs a single request program named FNDSCARU. The single request belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The children of the single request program are monitored 60 seconds after the program completes.

```
insert_job: oasg_nomon
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_mon_children: Y
oracle_mon_children_delay: 60
```

## oracle\_notify\_display\_users Attribute -- Specify Users to Notify Using Display Names

The oracle\_notify\_display\_users attribute specifies a list of users to notify when the single request completes using display names.

### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_notify_display_users: 'display_user' [, 'display_user'...]
```

- **display\_user** Specifies a list of users to notify when a single request completes using display names.

**Limits:** case-sensitive

### NOTE

- The entire value can be up to 1024 characters.
- Enclose the user name in single quotes.
- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- You can specify multiple users; separated by comma.

### Example: Specify Display User Names

In this example, the Applications Manager and System Administrator users are notified when the single request completes.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
oracle_save_output: N
oracle_notify_display_users: 'Applications Manager', 'System Administrator'
oracle_output_format: PDF
```

## oracle\_notify\_users Attribute -- Specify Users to Notify Using Short Names

The oracle\_notify\_users attribute specifies a list of users to notify when the single request completes using short names.

### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_notify_users: 'short_user'[, 'short_user'...]
```

- **short\_user** Specifies a list of users to notify when a single request completes using short names.

**Limits:** case-sensitive

#### NOTE

- The entire value can be up to 1024 characters.
- Enclose the user name in single quotes.
- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- You can specify multiple users; separated by comma.

#### Example: Specify Short User Names

In this example, the APPSMGR and SYSADMIN users are notified when the single request completes.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
oracle_save_output: N
oracle_notify_users: 'APPSMGR', 'SYSADMIN'
oracle_output_format: PDF
```

#### oracle\_oprunit Attribute -- Specify Operating Unit Name for a Single Request

The oracle\_oprunit specifies the operating unit name for a single request.

#### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

#### Syntax

This attribute has the following format:

```
oracle_oprunit: oprunit_name
```

- ***oprunit\_name*** Specifies the operating unit name for a single request. In Oracle Applications, the operating unit name is specified in the Operating Unit field on the OA Single Request Submission dialog.  
**Limits:** Up to 240 bytes  
**Example:** MyOrgName

#### Example: Specify the Operating Unit Name

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The output format is PDF. The template name is OraTemplate, the template language is English, and the template territory is US. The operating unit name is "MyOrgName".

```
insert_job: oasg_short
job_type: oasg
machine: oaagent
```

```

oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_output_format: PDF
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_template_name: OraTemplate
oracle_template_language: en
oracle_template_territory: US
oracle_oprunit: "MyOrgName"

```

## oracle\_output\_format Attribute -- Specify Output Format for a Single Request or Request Set

The `oracle_output_format` specifies the output format for a single request or request set.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)

#### NOTE

When you define an OASET job, you can specify the output format only for an individual program in the request set.

- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_output_format: output_format
```

- ***output\_format*** Specifies the output format for a single request or request set. In Oracle Applications, the output format is specified as a request definition option and is found in the Format column of the Upon Completion dialog.

**Limits:** Up to 10 characters; case-sensitive

**Example:** PDF

#### NOTE

- Enclose values that contain delimiters (such as spaces) in quotation marks.
- Your agent administrator can specify a default description for all Oracle E-Business Suite jobs by setting the `oa.default.outputFormat` parameter in the agent's `agentparm.txt` file.
- The `oracle_output_format` attribute overrides the default output format specified in the agent's `agentparm.txt`.

### Example: Specify the Output Format

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The output format is PDF. The template language is English and the template territory is US.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS

```

```

oracle_output_format: PDF
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_template_language: en
oracle_template_territory: US

```

## oracle\_print\_copies Attribute -- Specify the Number of Copies to Print

The `oracle_print_copies` attribute specifies the number of copies to print for an Oracle E-Business Suite single request or request set.

**Supported Job Types** This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)

### NOTE

You can also specify the `oracle_print_copies` attribute for an individual program in an Oracle E-Business Suite request set.

- Oracle E-Business Suite Single Request (OASG)

## Syntax

This attribute has the following format:

```
oracle_print_copies: num_copies
```

- ***num\_copies***

Specifies the number of copies to print. In Oracle E-Business Suite, the number of copies is specified as a request definition option and is found in the Copies column of the Upon Completion dialog.

**Limits:** Up to six numeric digits.

**Example:** 100

### NOTE

- In an Oracle E-Business Suite Request Set job, you can specify the number of copies to print for an individual program in the request set by using the `oracle_programdata` attribute. This attribute overrides the `oracle_print_copies` attribute for that particular program.
- Your agent administrator can specify a default number of copies for all Oracle Applications jobs by setting the `oa.default.printCopies` parameter in the agent's `agentparm.txt` file. The `oracle_print_copies` attribute overrides the default number of copies specified in the agent's `agentparm.txt`.

## Example: Print 15 Copies of an Oracle E-Business Suite Single Request

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The job specified 15 copies to be printed to the Q\_DEVELOPMENT printer in PORTRAIT style.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN

```

```
oracle_resp: System Administrator
oracle_printer:Q_DEVELOPMENT
oracle_print_style: PORTRAIT
oracle_print_copies: 15
```

## oracle\_print\_style Attribute -- Specify a Print Style

The oracle\_print\_style attribute specifies an Oracle E-Business Suite print style.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)

#### NOTE

You can also specify the oracle\_print\_style attribute for an individual program in an Oracle E-Business Suite request set.

- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_print_style: print_style
```

- ***print\_style***

Specifies a print style. The value must match the name of an Oracle Applications print style. In Oracle Applications, the print style is specified as a request definition option and is found in the Style field of the Upon Completion dialog.

**Limits:** Up to 30 characters; case-sensitive

**Example:** PORTRAIT

#### NOTE

- Do not use embedded double quotation marks. Enclose values that contain delimiters (such as spaces) in double quotation marks.
- Your agent administrator can specify a default print style for all Oracle E-Business Suite Request Set and Single Request jobs by setting the oa.default.printStyle parameter in the agent's agentparm.txt file.
- The oracle\_print\_style attribute overrides the default print style specified in the agent's agentparm.txt.
- In an Oracle E-Business Suite Request Set job, you can specify the print style for an individual program in the request set by using the oracle\_programdata attribute. This attribute overrides the oracle\_print\_style attribute and the oa.default.printStyle default parameter for that particular program.

### Example: Specify an Oracle E-Business Suite Print Style

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-business Suite. The job runs under the YADMIN user with System Administrator responsibility. The job specifies 15 copies to be printed on the Q\_DEVELOPMENT printer in portrait style.

```
insert_job: oast_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: SYSADMIN
oracle_resp: System Administrator
```

```
oracle_printer:Q_DEVELOPMENT
oracle_print_style: PORTRAIT
oracle_print_copies: 15
```

## oracle\_printer Attribute -- Specify a Printer

The `oracle_printer` attribute specifies the name of a printer to be used by Oracle E-Business Suite.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)

#### NOTE

You can also specify the `oracle_printer` attribute for an individual program in an Oracle E-Business Suite request set.

- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_printer: printer_name
```

- ***printer\_name***

Specifies the name of a printer that Oracle Applications can print to. This printer must be defined in Oracle Applications, the printer name is specified as a request definition option and is found in the Printer column of the Upon Completion dialog.

**Limits:** Up to 30 characters; case-sensitive

**Example:** "\\printer path\K6700"

#### NOTE

Do not use embedded double quotation marks. Enclose values that contain delimiters (such as spaces) in double quotation marks.

#### NOTE

- Your agent administrator can specify a default printer for all Oracle E-Business Suite Request Set and Single Request jobs by setting the `oa.default.printer` parameter in the agent's `agentparm.txt` file.
- The `oracle_printer` attribute overrides the default printer specified in the agent's `agentparm.txt`.
- In an Oracle E-Business Suite Request Set job, you can specify the printer for an individual program in the request set by using the `oracle_programdata` attribute. This attribute overrides the `oracle_printer` attribute and the `oa.default.printer` default parameter for that particular program.

### Example: Specify a Printer for Oracle E-Business Suite

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The job specifies 15 copies to be printed to the Q\_DEVELOPMENT printer in PORTRAIT STYLE.

```
insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
```

```

oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_printer:Q_DEVELOPMENT
oracle_print_style: PORTRAIT
oracle_print_copies: 15

```

## oracle\_program Attribute -- Specify the Name of a Single Request Program

The oracle\_program attribute specifies the name of an Oracle E-Business Suite single request program.

This attribute must be used with the oracle\_program\_name\_type attribute.

### Supported Job Type

This attribute is required for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_program: program_short_name | program_display_name
```

- **program\_short\_name**

Specifies the short name of the Oracle Applications single request program. The program short name must be registered in the Oracle Applications Concurrent Manager. In Oracle Applications, the program short name is part of the program definition and is found in the Short Name field of the Concurrent Programs dialog.

**Limits:** Up to 30 characters; case-sensitive

#### NOTE

The value of the oracle\_program\_name\_type attribute must be SHORT.

- **program\_display\_name**

Specifies the display name of the Oracle Applications single request program. In Oracle Applications, the display name is part of the program definition and is found in the Display Name field of the Concurrent Programs dialog.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

The value of the oracle\_program\_name\_type attribute must be DISPLAY.

### Example: Specify the Name of an Oracle E-Business Suite Single Request Program

In this example, the single request program is identified by its display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The history of the program is included in the description.

```

insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name: ACCOUNTS
oracle_program_name_type: DISPLAY
oracle_program: Application Object Library
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212

```



## oracle\_program\_name\_type Attribute -- Specify the Type of Oracle Applications Program Name

The `oracle_program_name_type` attribute specifies whether the name of the Oracle E-Business Suite single request program is the short name or the display name.

### NOTE

- This attribute must be used with the `oracle_program` attribute.
- The default type is `SHORT`. If you do not specify the `oracle_program_name_type` attribute in your job definition, the job interprets the `oracle_program` value to be a short name.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_program_name_type: SHORT | DISPLAY
```

- **SHORT**

Indicates that the Oracle E-Business Suite single request program is specified by its short name. The short name is defined by the Oracle Applications Concurrent Manager. This is the default.

- **DISPLAY**

Indicates that the Oracle E-Business Suite single request program is specified by its display name.

### Example: Specify the Display Name of an Oracle E-Business Suite Program

In this example, the single request program is identified by its display name Application Object Library in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The history of the program is included in the description.

```
insert_job: oasg_disp
job_type: oasg
machine: oaagent
oracle_appl_name: ACCOUNTS
oracle_program_name_type: DISPLAY
oracle_program: Application Object Library
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_desc: CONFIRMED REQ 2010808 by TYB212
```

### Example: Specify the Short Name of an Oracle E-Business Suite Program

In this example, the single request program is identified by its short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrators responsibility.

```
insert_job: oast_short
job_type: oasg
machine: oaagent
oracle_appl_name: FNDSCARU
oracle_program_name_type: SHORT
oracle_program: ACCOUNTS
oracle_user: YSADMIN
```

---

```
oracle_resp: System Administrator
```

## oracle\_programdata Attribute -- Specify Data for a Program in a Request Set

The `oracle_programdata` attribute specifies data for an individual program in an Oracle E-Business Suite request set. This data overrides the print parameters specified for the entire request set.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Request Set (OASET) job type.

### Syntax

This attribute has the following format:

```
oracle_programdata: index=number,
args="argument[,argument...]",
printer=printer_name,
print_style=print_style,
print_copies=num_copies,
saveop=Y | N,
quote_in_default=Y | N,
notify_users='short_user'[, 'short_user'...],
notify_display_users='display_user'[, 'display_user'...],
oprunit=oprunit_name,
output_format=output_format,
template_name=template_name,
template_language=template_language,
template_territory=template_territory
```

#### NOTE

To specify data for multiple programs in a request set, define a separate `oracle_programdata` attribute for each program.

- **`index=program_number`**

Specifies the order number of a program in an Oracle Applications request set.

**Limits:** Integer

**Example:** 1 (specifies the first program in the request set)

- **`args="argument[,argument...]"`**

Defines the argument values to override the default values that are defined by the registered Oracle Applications Concurrent Manager program.

**Limits:** Up to 100 argument values, for a total of 500 characters

#### NOTE

- You can specify multiple arguments, enclose the arguments in double quotation marks. Separate each argument with a comma.
- Do not add a space after a comma unless the argument value starts with a space. If a value starts with a space, enclose the value in single quotation marks.
- To use a default value defined in Oracle Applications, enter one comma in the argument position and specify Y in the `oracle_use_arg_def` attribute.
- The entire value can be up to 4096 characters.

**Example:** `args="X22F,,X56R143,,X23T"`

- **printer=printer\_name**

Specifies the name of a printer that Oracle Applications can print to. This printer must be defined in Oracle Applications. In Oracle Applications, the printer name is specified as a request definition option and is found in the Printer column of the Upon Completion dialog.

**Limits:** Up to 30 characters; case-sensitive

**NOTE**

Enclose paths that contain delimiters (such as spaces) in quotation marks.

**Example:** "\\printer path\K6700"

- **print\_style=print\_style**

Specifies a print style. The value must match the name of an Oracle Applications print style. In Oracle Applications, the print style is specified as a request definition option and is found in the Style field of the Upon Completion dialog.

**Limits:** Up to 30 characters; case-sensitive

**Example:** PORTRAIT

- **print\_copies=num\_copies**

Specifies the number of copies to print. In Oracle E-Business Suite, the number of copies is specified as a request definition option and is found in the Copies column of the Upon Completion dialog.

**Limits:** Up to six numeric digits

**Example:** 100

**Default:** 0

- **saveop=Y | N | T | F | YES | NO | TRUE | FALSE**

- **Y | T | YES | TRUE**

Saves output from an Oracle Applications program. In Oracle Applications, this option is specified as a request definition option and corresponds to selecting the Save all Output Files check box of the Upon Completion dialog.

- **N | F | NO | FALSE**

Does not save output from an Oracle Applications program. In Oracle Applications, this option is specified as a request definition option and corresponds to clearing the Save all Output Files check box of the Upon Completion dialog.

- **quote\_in\_default=Y | N | T | F | YES | NO | TRUE | FALSE**

- **Y | T | YES | TRUE**

Specifies that resolved expressions in default values are quoted.

- **N | F | NO | FALSE**

Specifies that resolved expressions in default values are not quoted.

- **notify\_users='short\_user['short\_user'...]**

Specifies a list of users to notify when a single request completes using short names.

**Limits:** case-sensitive

**NOTE**

- The entire value can be up to 1024 characters.
- Enclose the user name in single quotes.
- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- You can specify multiple users; separated by comma.

- **notify\_display\_users='display\_user['display\_user'...]**

Specifies a list of users to notify when a single request completes using display names.

**Limits:** case-sensitive

**NOTE**

- The entire value can be up to 1024 characters.
  - Enclose the user name in single quotes.
  - Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
  - You can specify multiple users; separated by comma.
- **oprunit=oprunit\_name**  
Specifies the operating unit name for a single request or request set.  
**Limits:** Up to 240 bytes
  - **output\_format=output\_format**  
Specifies the output format for a single request or request set. In Oracle Applications, the output format is specified as a request definition option and is found in the Format column of the Upon Completion dialog.  
**Limits:** Up to 10 characters; case-sensitive  
**Example:** PDF

**NOTE**

- Enclose values that contain delimiters (such as spaces) in quotation marks.
  - Your agent administrator can specify a default description for all Oracle E-Business Suite jobs by setting the oa.default.outputFormat parameter in the agent's agentparm.txt file.
  - The oracle\_output\_format attribute overrides the default output format specified in the agent's agentparm.txt.
- **template\_name=template\_name**  
Specifies the template name for a program in a request set.  
**Limits:** Up to 120 characters; case-sensitive
  - **template\_language=template\_language**  
Specifies the template language for a program in a request set. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.  
**Limits:** Up to 10 characters; case-sensitive  
**Example:** en

**NOTE**

- Enclose values that contain delimiters (such as spaces) in quotation marks.
  - The oracle\_output\_format operand must be specified with the oracle\_template\_language operand.
- **template\_territory=template\_territory**  
Specifies the template territory for a program in a request set. In Oracle Applications, the template territory is specified as a request definition option and is found in the For Territory column of the Upon Completion dialog.  
**Limits:** Up to 10 characters; case-sensitive  
**Example:** US

**NOTE**

- Enclose values that contain delimiters (such as spaces) in double quotation marks.
- The oracle\_output\_format operand must be specified with the oracle\_template\_territory operand.

**Examples: Specify Data for Two Programs in an Oracle E-Business Suite Request Set**

Suppose that you want to run an Oracle E-Business Suite request set named EXTRACTS on the oaagent agent. The request set belongs to the application with the display name Application Object Library in Oracle E-Business Suite. The job definition specifies the following default settings for all programs in the request set:

- The number of copies to be printed is 1.
- The print style is LANDSCAPE.

- The printer is \\printer path\Q8.

The first oracle\_programdata attribute overrides the default values for the first program in the request set. The first program uses the arguments T23 and R1 and prints two copies using the \\printer path\Q1 printer in PORTRAIT style.

The second oracle\_programdata attribute overrides the default values for the fifth program in the request set. The fifth program uses arguments R and R1 and prints three copies using the \\printer path\Q2 printer in PORTRAIT style.

The other programs in the request set use the default values.

```
insert_job: oaset_prog
job_type: oaset
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_req_set: EXTRACTS
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_printer: "\\printer path\Q8"
oracle_print_style: LANDSCAPE
oracle_print_copies: 1
oracle_programdata: index=1, args="T23,,R1", printer="//printer\Q1",
 print_style=PORTRAIT, print_copies=2, saveop=Y
oracle_programdata: index=5, args="R,R1,", printer="//printer\Q2", print_style=PORTRAIT,
 print_copies=3, saveop=N
```

Suppose that you want to run an Oracle E-Business Suite request set named FNRSSUB1310 on the local host. The request set belongs to the application with the short name BIS in Oracle E-Business Suite. The SYSADMIN and ASGUEST users are notified when the single request completes on QATEST1.

The first oracle\_programdata attribute overrides the default values for the first program in the request set. The first program uses the output format as PDF, the template name as OraTemplate, the template language is English, and template territory is US.

The second oracle\_programdata attribute overrides the default values for the fifth program in the request set. The second program uses the output format as EXCEL, operating unit name as MyOrgName, the template language is English, and template territory is US.

The other programs in the request set use the default values.

```
delete_job: test_OASET
insert_job: oaset_fmt
job_type: OASET
machine: localhost
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_name_type: DISPLAY
oracle_req_set: FNRSSUB1310
oracle_use_arg_def: Y
```

```

oracle_programdata: index=1, output_format=PDF, template_name=OraTemplate,
 template_language=en, template_territory=US,
 notify_users="'SYSADMIN','ASGUEST','QATEST1'"
oracle_programdata: index=2, output_format=EXCEL, oprunit=MyOrgName,
 template_language=en, template_territory=US,
 notify_users="'SYSADMIN','ASGUEST','QATEST1'"
group: OA

```

## oracle\_quote\_in\_default Attribute -- Specify Whether to Quote Resolved Expressions in Default Values

The `oracle_quote_in_default` attribute specifies whether to quote resolved expressions in default values.

### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_quote_in_default: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Specifies that resolved expressions in default values are quoted.
- **N | F | NO | FALSE**  
Specifies that resolved expressions in default values are not quoted. This is the default.

### Example: Quote Resolved Expressions in Default Values

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. Resolved expressions in default values are quoted. The job runs under the SYSADMIN user with System Administrator responsibility.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_quote_in_default: Y

```

## oracle\_req\_set Attribute -- Specify the Name of a Request Set

The `oracle_req_set` attribute specifies the name of an Oracle E-Business Suite request set.

This attribute must be used with the `oracle_req_set_type` attribute.

### Supported Job Type

This attribute is required for the Oracle E-Business Suite Request Set (OASET) job type.

### Syntax

This attribute has the following format:

```
oracle_req_set: request_set_short_name | request_set_display_name
```

- **request\_set\_short\_name** Specifies the short name of the Oracle Applications request set. In Oracle Applications, the request set short name is part of the Request Set definition and is found in the Set Code field of the Request Set dialog. **Limits:** Up to 30 characters; case-sensitive **Note:** The value of the oracle\_req\_set\_type attribute must be SHORT.
- **request\_set\_display\_name** Specifies the display name of the Oracle Applications request set. In Oracle Applications, the request set display name is part of the Request Set definition and is found in the Set Code field of the Request Set dialog. **Limits:** Up to 30 characters; case-sensitive

#### NOTE

The value of the oracle\_req\_set\_type attribute must be DISPLAY.

### Example: Specify the Name of an Oracle E-Business Suite Request Set

This example runs a request set named FNDRSSUB1310 on the oaagent agent. The request set belongs to the application with the short name BIS in Oracle E-Business Suite. The job runs under the SYSADMIN user with Business Intelligence Super User, Progress S&L responsibility. The job's output is saved.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_user: SYSADMIN
oracle_resp: "Business Intelligence Super User, Progress S&L"
oracle_save_output: Y
```

### oracle\_req\_set\_type Attribute -- Specify the Type of Request Set Name

The oracle\_req\_set\_type attribute specifies whether the name of the Oracle E-Business Suite request set is the short name or the display name.

#### NOTE

- This attribute must be used with the oracle\_req\_set attribute.
- The default type is SHORT. If you do not specify the oracle\_req\_set\_type attribute in your job definition, the job interprets the oracle\_req\_set value to be a short name.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Request Set (OASET) job type.

### Syntax

This attribute has the following format:

```
oracle_req_set_type: SHORT | DISPLAY
```

- **SHORT**

Indicates that the Oracle E-Business Suite request is specified by its short name. The short name is defined by the Oracle Applications Concurrent Manager. This is the default.

- **DISPLAY**

Indicates that the Oracle E-Business Suite request is specified by its display name.

**Limits:** Up to 30 characters; case-sensitive

### Example: Specify the Name of an Oracle E-Business Suite Request Set

This example runs a request set named FNDRSSUB1310 on the oaagent agent. The request set belongs to the application with the short name BIS in Oracle E-Business Suite. The job runs under the SYSADMIN user with Business Intelligence Super User, Progress S&L responsibility. The job's output is saved.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_user: SYSADMIN
oracle_resp: "Business Intelligence Super User, Progress S&L"
oracle_save_output: Y
```

### oracle\_resp Attribute -- Specify a Responsibility Name

The oracle\_resp attribute specifies an Oracle E-Business Suite responsibility name.

#### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Copy Single Request (OACOPY)
- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)

#### Syntax

This attribute has the following format:

```
oracle_resp: responsibility_name
```

- ***responsibility\_name*** Specifies an Oracle Applications responsibility name.

**Limits:** Up to 100 characters; case-sensitive

#### NOTE

Do not use embedded double quotation marks. Enclose values that contain delimiters (such as spaces) in double quotation marks.

#### Single Request or Request Set Job Notes:

- If you do not specify the oracle\_resp attribute in the job definition, a default responsibility name must be defined in the oa.default.responsibility parameter in the agent's agentparm.txt file. Otherwise, the job fails.
- The oracle\_resp attribute overrides the default responsibility name specified in the agent's agentparm.txt file.

#### Copy Single Request Job Notes:

- To override the responsibility name specified in the existing single request, both the user name and responsibility name are required:



- If a default user name is not specified in the `oa.default.user` parameter in the agent's `agentparm.txt` file, you must specify the `oracle_user` attribute in the job definition.
- If a default responsibility name is not specified in the `oa.default.responsibility` parameter in the agent's `agentparm.txt` file, you must specify the `oracle_resp` attribute in the job definition.
- If the user name or responsibility name is not specified in either the job definition or in the `agentparm.txt` file, the job copies the user name and responsibility name from the original single request.

### Example: Specify an Oracle E-Business Suite Responsibility Name

This example runs a request set named `FNDRSSUB1310` on the `oaagent` agent. The request set belongs to the application with the short name `BIS` in Oracle E-Business Suite. The job runs under the `SYSADMIN` user with Business Intelligence Super User, Progress S&L responsibility. The job's output is saved.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_user: SYSADMIN
oracle_resp: "Business Intelligence Super User, Progress S&L"
oracle_save_output: Y
```

### Example: Copy a Single Request Job and Override the Responsibility Name

Suppose that you want to copy an existing single request defined on Oracle E-Business Suite. In this example, the job copies the single request job with request ID `2255470` and overrides the Oracle E-Business Suite user name and responsibility name defined in the `agentparm.txt` file.

```
insert_job: oacopy_single
job_type: oacopy
machine: oaagent
request_id: 2255470
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

## oracle\_save\_output Attribute -- Specify Whether to Save Output

The `oracle_save_output` attribute specifies whether to save the output from an Oracle E-Business Suite Single Request or Request Set job.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)

#### NOTE

When you define an OASET job, you can specify whether to save the output only for an individual program in the request set.

- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_save_output: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**

Saves output from an Oracle Applications program. In Oracle Applications, this option is specified as a request definition option and corresponds to selecting the Save all Output Files check box of the Upon Completion dialog.

- **N | F | NO | FALSE**

Does not save output from an Oracle Applications program. In Oracle Applications, this option is specified as a request definition option and corresponds to clearing the Save all Output Files check box of the Upon Completion dialog. This is the default.

**NOTE**

In an Oracle E-Business Suite Request Set job, you can specify whether to save the output for an individual program in the request set by using the oracle\_programdata attribute. This attribute overrides the oracle\_save\_output attribute for that particular program.

**Example: Save the Output of an Oracle E-Business Suite Request Set Job**

This example runs a request set named FNDRSSUB1310 on the oaagent agent. The request set belongs to the application with the short name BIS in Oracle E-Business Suite. The job runs under the SYSADMIN user with Business Intelligence Super User, Progress S&L responsibility. The job's output is saved.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_user: SYSADMIN
oracle_resp: "Business Intelligence Super User, Progress S&L"
oracle_save_output: Y
```

## oracle\_template\_language Attribute -- Specify the Template Language for a Single Request

The oracle\_template\_language attribute specifies the template language for a single request. The oracle\_output\_format attribute must be specified with the oracle\_template\_language attribute.

### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_template_language: template_language
```

- ***template\_language***

Specifies the template language for a single request. In Oracle Applications, the template language is specified as a request definition option and is found in the Template Language column of the Upon Completion dialog.

**Limits:** Up to ten characters; case-sensitive

**Example:** en

**NOTE**

- Do not use colons and double quotation marks.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the `oa.default.templateLanguage` parameter in the `agentparm.txt` file of the agent.
- The `oracle_template_language` attribute overrides the default setting specified in the `oa.default.templateLanguage` parameter in the `agentparm.txt` file of the agent.

**Example: Specify the Template Language**

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The output format is PDF. The template language is English and the template territory is US.

```
insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_output_format: PDF
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_template_language: en
oracle_template_territory: US
```

**oracle\_template\_name Attribute -- Specify the Template Name for a Single Request**

The `oracle_template_name` attribute specifies the template name for a single request. The `oracle_output_format` attribute must be specified with the `oracle_template_name` attribute.

**Supported Job Types**

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

**Syntax**

This attribute has the following format:

```
oracle_template_name: template_name
```

- ***template\_name***

Specifies the template name for a single request. In Oracle Applications, the template name is specified as a request definition option and is found in the Template Name column of the Upon Completion dialog.

**Limits:** Up to 120 characters; case-sensitive **Example:** OraTemplate

**Example: Specify the Template Name**

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The output format is PDF. The template template name is OraTemplate, the template language is English and the template territory is US.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_output_format: PDF
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator

oracle_template_name: OraTemplate
oracle_template_language: en
oracle_template_territory: US

```

## oracle\_template\_territory Attribute -- Specify the Template Territory for a Single Request

The `oracle_template_territory` attribute specifies the template territory for a single request. The `oracle_output_format` attribute must be specified with the `oracle_template_territory` attribute.

### Supported Job Types

This attribute is optional for the Oracle E-Business Suite Single Request (OASG) job type.

### Syntax

This attribute has the following format:

```
oracle_template_territory: template_territory
```

- **template\_territory**

Specifies the template territory for a single request. In Oracle Applications, the template territory is specified as a request definition option and is found in the For Territory column of the Upon Completion dialog.

**Limits:** Up to ten characters; case-sensitive

**Example:** US

#### NOTE

- Do not use colons and double quotation marks.
- Your agent administrator can specify a default setting for all Oracle E-Business Suite jobs by setting the `oa.default.templateTerritory` parameter in the agent's `agentparm.txt` file.
- The `oracle_template_territory` attribute overrides the default setting specified in the `oa.default.templateTerritory` parameter in the agent's `agentparm.txt`.

### Example: Specify the Template Territory

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility. The output format is PDF. The template language is English and the template territory is US.

```

insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT

```

```

oracle_appl_name: ACCOUNTS
oracle_output_format: PDF
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_template_language: en
oracle_template_territory: US

```

## oracle\_use\_arg\_def Attribute -- Specify Whether to Use Argument Defaults

The `oracle_use_arg_def` attribute specifies whether to use default values for arguments that not defined using the `oracle_args` attribute or the `oracle_programdata` attribute. The default arguments are defined in Oracle E-Business Suite.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_use_arg_def: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**

Specifies that the job uses the default values defined in Oracle Applications for arguments that are not specified in the job definition.

- **N | F | NO | FALSE**

Specifies that the job does not use the default values defined in Oracle Applications for arguments that are not specified in the job definition. This is the default.

### Example: Use Oracle E-Business Suite Argument Defaults

Suppose that you want to pass the argument values, T,DefArg2,X23,,DefArg5, to a single request program. The job uses the argument values that are specified in the job definition and the default values defined in Oracle E-Business Suite as follows:

- The first argument, T, and the third argument, X23, are specified in the job definition.
- The second argument, DefArg2, and the fifth argument, DefArg5, are defined as defaults in Oracle E-Business Suite.
- The fourth argument is not specified in the job definition or defined as a default on Oracle E-Business Suite, so the agent passes an empty string for that argument.

```

insert_job: oasg_args
job_type: oasg
machine: oaagent
oracle_appl_name_type: DISPLAY
oracle_appl_name: Application Object Library
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_program: FNDSCARU
oracle_use_arg_def: Y
oracle_args: T,,X23,,

```

## oracle\_use\_set\_defaults\_first Attribute -- Specify Whether Request Set Defaults Take Precedence over Program Defaults

The oracle\_use\_set\_defaults\_first attribute specifies whether request set defaults take precedence over concurrent program defaults in Oracle Applications.

### Supported Job Type

This attribute is optional for the Oracle E-Business Suite Request Set (OASET) job type.

### Syntax

This attribute has the following format:

```
oracle_use_set_defaults_first: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Specifies that request set defaults take precedence over concurrent program defaults.
- **N | F | NO | FALSE**  
Specifies that concurrent program defaults take precedence over request set defaults. This is the default.

### NOTE

- Your agent administrator can specify a default setting for all Oracle E-Business Suite Request Set jobs by setting the oa.default.useSetDefaultsFirst attribute in the agent's agentparm.txt file.
- The oracle\_use\_set\_defaults\_first attribute overrides the default setting specified in the oa.default.useSetDefaultsFirst attribute in the agent's agentparm.txt.

### Example: Use Request Set Defaults

In this example, request set defaults take precedence over concurrent program defaults.

```
insert_job: oaset_resp
job_type: oaset
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: BIS
oracle_req_set_type: SHORT
oracle_req_set: FNDRSSUB1310
oracle_output_format: PDF
oracle_user: SYSADMIN
oracle_resp: System Administrator
oracle_use_set_defaults_first: Y
```

## oracle\_user Attribute -- Specify a User Name

The oracle\_user attribute specifies an Oracle E-Business Suite user name that the job runs under.

### Supported Job Types

This attribute is optional for the following job types:

- Oracle E-Business Suite Copy Single Request (OACOPY)
- Oracle E-Business Suite Request Set (OASET)
- Oracle E-Business Suite Single Request (OASG)

### Syntax

This attribute has the following format:

```
oracle_user: user_name
```

- **user\_name** Specifies the Oracle Applications user name that the job runs under.  
**Limits:** Up to 100 characters; case-sensitive; it cannot contain delimiters (such as spaces)

#### NOTE

Do not use embedded double quotation marks. Enclose values that contain delimiters (such as spaces) in double quotation marks.

#### Single Request or Request Set Job Notes:

- If you do not specify the `oracle_user` attribute in the job definition, a default user name must be defined in the `oa.default.user` parameter in the agent's `agentparm.txt` file. Otherwise, the job fails.
- The `oracle_user` attribute overrides the default user name specified in the agent's `agentparm.txt`.

#### Copy Single Request Job Notes:

- To override the user name specified in the existing single request, both the user name and responsibility name are required:
  - If a default user name is not specified in the `oa.default.user` parameter in the agent's `agentparm.txt` file, you must specify the `oracle_user` attribute in the job definition.
  - If a default responsibility name is not specified in the `oa.default.responsibility` parameter in the agent's `agentparm.txt` file, you must specify the `oracle_resp` attribute in the job definition.
- If the user name or responsibility name is *not* specified in either the job definition or in the `agentparm.txt` file, the job copies the user name and responsibility name from the original single request.

#### Example: Specify an Oracle E-Business Suite User Name

This example runs a single request program named FNDSCARU. The single request belongs to the application with the short name ACCOUNTS in Oracle E-Business Suite. The job runs under the SYSADMIN user with System Administrator responsibility.

```
insert_job: oasg_short
job_type: oasg
machine: oaagent
oracle_appl_name_type: SHORT
oracle_appl_name: ACCOUNTS
oracle_program: FNDSCARU
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

#### Example: Copy a Single Request Job and Override the User Name

Suppose that you want to copy an existing single request defined on Oracle E-Business Suite. In this example, the job copies the single request job with request ID 2255470 and overrides the Oracle E-Business Suite user name and responsibility name defined in the `agentparm.txt` file.

```
insert_job: oacopy_single
job_type: oacopy
machine: oaagent
request_id: 2255470
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

## os\_user Attribute -- Specify an Operating System User ID

The `os_user` attribute specifies an operating system user ID that the job runs under.

### Supported Job Type

This attribute is optional for the Micro Focus (MICROFOCUS) job type.

### yntax

This attribute has the following format:

```
os_user: os_userid@identifier
```

- **os\_userid@identifier** Specifies the Operating System user ID that the job runs under. The `os_userid` must meet the operating system requirements and should be defined on the Micro Focus Enterprise server where `identifier` specifies the name of the server. **Limits:** Up to 80 characters

### NOTE

- If the agent runs as a service under the system account, the Micro Focus job is executed under the system account. If the Micro Focus Enterprise Server runs under a particular user ID, and you have specified a Micro Focus Enterprise Server name for the job, you must specify the same operating system user ID for the job to find the server.
- The `os_userid` and `identifier` combination obtains the credentials of the user. For the job to run successfully, the credentials of the user must be defined on AutoSys Workload Automation using the `autosys_secure` command.
- If the credentials of the user are invalid, or do not meet the requirements of the `os_user` attribute, the following message is displayed:

```
Owner attribute error. Logon failure: unknown user name or bad password.
```

To resolve this, you must change the password of the user specified in the `os_user` attribute using `autosys_secure`. This is not related to the owner attribute.

### Example: Run a Micro Focus Job Under an Operating System User ID

This example runs a Micro Focus job under the Micro Focus user ID named MFUSER01, and the operating system user ID named OSUSER.

```
insert_job: mf_job_with_os_user
job_type: MICROFOCUS
machine: mfagent
mf_server_address_type: NAME
mf_server: mfprod
mf_jcl_type: REFERENCE
mf_jcl_name: "C:\Program Files\micro focus\es-jcldemo\if.jcl"
mf_user: MFUSER01@mfprod
os_user: OSUSER@osserver
```

## output\_var Attribute -- Specify the Output Variables

The `output_var` attribute specifies the output variables (`var_name`, `var_value`, `var_format`, `var_type`, `var_group`) that can be consumed by the successor jobs for their execution.

### Supported Job Types

This attribute is optional for the [HTTP job type](#) and [Web Service Document/Literal Job](#).



## Syntax

This attribute has the following format:

```
output_var: var_name="var", var_value="value", var_format="regex|xpath|jsonpath",
var_type="payload|header", var_group="group"
```

- **var**  
Specifies the name of the output variable.  
**Limits:** Up to 256 characters
- **value**  
Specifies the value of the output variable. You can specify the xpath for XML, jsonpath for JSON, or an ordinary regex expression for Plain Text.  
**Limits:** Up to 3950 characters
- **regex|xpath|jsonpath|key\_index**  
Specifies the format of the output variable. Valid values are xpath (for XML), jsonpath (for JSON), or regex (for plain text), key\_index(for var\_type header).
- **payload|header**  
Specifies the type of the output variable. Valid values are payload and header. The header var\_types do not apply to the xpath format.

### NOTE

An HTTP job on its execution produces payload that can be consumed by the successor jobs for their execution. The payload is in the form of XML, JSON, or Plain Text. When a job with output variables of type payload completes successfully, the scheduler parses the incoming payload for the elements whose path is specified in the var\_value variable and the resultant is stored as a job blob with the name that is specified in the var\_name variable.

- **group**  
Specifies the numerical group ID in any order. This group ID enables the extraction of the groups that are matched while applying the regex. The var\_group variable applies to the regex and key\_index formats.**Limits:** Up to 3950 characters

### NOTE

If you do not specify the var\_group variable, the first result of the pattern that matches the regex that is specified in the var\_value variable is stored as the output\_var value.

**Example:** If the payload contains a string in the form “This is a sample string” and you want “This string” as a variable, specify two group IDs <<\$1>> <<\$5>> to pick the value from group 1 and 5.

### NOTE

While updating a job definition, if you update the output\_var attribute value, the already defined ones in the previous version are deleted from the database and the newer ones are populated. If you update the output\_var attribute to include an empty value, all the previously defined ones are deleted.

## Examples:

- Specify a regex based output variable, as follows:  

```
output_var:var_name="var1", var_value="[a-zA-Z]+", var_format="regex",
var_type=payload
output_var:var_name="var1", var_value="[a-zA-Z]+", var_format="regex",
var_type=payload, var_group="<<$1>>"
```
- Specify a xml based output variable when the HTTP response payload is XML based, as follows:

```
output_var: var_name="var1", var_value="/store/book[1]/title", var_type=payload,
var_format="xpath"
```

- Specify a jsonpath based output variable when the HTTP response payload is JSON based, as follows:

```
output_var: var_name="var1", var_value="x.store.book[0].title" var_type=payload,
var_format="jsonpath"
```

- Specify a header based output variable, as follows:

```
output_var: var_name="var1", var_value="Transfer-Encoding" var_type=header,
var_format="key_index", var_group="<<$1>>"
```

- You can specify var\_group only when the header contains multiple values.

You cannot specify more than one value for var\_group.

In the following example, if the first value against header3 needs to be retrieved, then the group format would be <<\$0>>. If the second one is required then the group format would be <<\$1>>.

```
[
 {
 header3:value3
 },
 {
 header3:value4
 }
];
```

## owner Attribute -- Define the Owner of the Job

### Contents

The owner attribute defines the owner of the job. By default, the operating system user who invokes jil to define a job is the owner of that job. You can change the owner a job by specifying a different user ID in the job definition.

To change the owner of a job, specify the user ID of a AutoSys Workload Automation user in the owner attribute job definition. AutoSys Workload Automation accesses the associated password or key credentials from the database. Prior to using key credentials in jobs, you must configure the machine where the agent is installed and the remote machine that the agent tries to connect to so that they communicate using the Public key cryptography.

### WARNING

- KeyFile can be used for SCP (SFTP) or any of the CA WA Advanced Integration for Hadoop jobs. KeyString is only applicable to the CA WA Advanced Integration for Hadoop job types. For more information about KeyFile or KeyString, see [autosys\\_secure Command -- Define Security Settings](#).
- Jobs that run on other software, such as PeopleSoft and databases, can run only when the owner specified in the job definition is defined as a user on AutoSys Workload Automation and on the authenticated software. For example, to run an SAP job, specify an SAP user that is also defined as a AutoSys Workload Automation user.

You can use autosys\_secure command to define users on AutoSys Workload Automation.

### NOTE

- If AutoSys Workload Automation is running in native security mode, you can change the owner value only if you have EDIT superuser permissions.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you can specify the owner value only if as-owner security policies grant you access to the owner value. You can update the job

to specify a different owner value only if as-owner security policies grant you access to both the existing and new owner values.

- AutoSys Workload Automation uses the owner value for all job types except for File Trigger. AutoSys Workload Automation does not use the `oscomponent.default.user` parameter located in the agent's `agentparm.txt` file.
- Other application specific uses of the owner attribute apply to certain job types. For more information about these job type specific uses, see the documentation on individual job types.

## Supported Job Types

This attribute is optional for all job types except for File Trigger (FT) jobs.

### NOTE

File Trigger jobs run under the user that runs the agent (when the jobs run as threads), or they run under the `oscomponent.default.user` value (when the jobs run as external processes).

## UNIX Syntax

This attribute has the following format:

```
owner: user@host
```

- ***user@host***

Specifies a valid user that is used as the owner of the job. For a command job, the user must be a valid user with an account on the specified real machine. The user must have an account on all machines where a command job can run. For other job types, the credentials can be verified in different ways either through the operating system or a specific application.

**Default:** *user@host*, where *user* is the user who invokes jil to define the job and *host* is the real machine that user is logged on to.

The specified owner owns and has edit permission for all jobs defined during the session.

For command jobs, the command that is specified in the job definition runs under the user ID that is specified in the owner attribute of the job definition. When a command starts on the agent, the uid of the process is changed to the owner of the job. Only the specific user on the specific machine can edit the job. The job only runs when the user has execute permission for the command specified in the job, on the client for the job.

For command jobs, if you used the `autosys_secure` command to activate remote authentication, AutoSys Workload Automation verifies the permissions for that user on the agent at job run time. If the user does not have the correct permissions on the agent, the log in is not authenticated and the job does not run. For example, if you activate remote authentication for a job owned by the user named `admin@accounting`. The agent verifies the permissions for `admin@agent_name` and then runs the job. If the agent cannot log in to the client as `admin@agent_name` for `admin@accounting`, verify the following information:

- The user named `admin@agent_name` has the permissions that are required to run the job that is owned by the user named `admin@accounting`.
- The `rshd` and `rlogind` processes are enabled on the client, and the `/etc/hosts.equiv` and the `.rhosts` files are configured correctly.

### NOTE

When the `rshd` and `rlogind` files are disabled on a client, users can still run jobs on that client as long as the `.rhosts` files are configured correct. However, users cannot log in to the client when the `rshd` and `rlogind` files are disabled. When agent cannot identify the login for a job that uses remote authentication, the job does not run.

## Windows Syntax

This attribute has the following format:

```
owner: user@host| user@domain
```

**user**

Specifies the name of the user.

**host**

Specifies the name of the local host machine on which the user is defined.

**domain**

Specifies the name of the domain on which the user is defined.

**WARNING**

The agent cannot log in to the client as the owner unless the owner attribute specifies the name of a user that is defined within the instance and has the required permissions.

**NOTE**

- When defining the job, the owner can grant other users edit or execute permission for that job. Execute permission controls which users can issue sendevent commands, such as STARTJOB or KILLJOB on the job. However, this does not affect whose permissions the job's command runs under.
- To determine the job owner, issue the following command at an instance command prompt:  

```
autorep -J job_name -q
```
- The EDIT superuser can use the update\_job subcommand to change the owner of an individual job. To change a large number of jobs, the EDIT superuser can use the autorep command to dump multiple job definitions to an output file, change the owner, and reload the changed job definitions using the jil command.
- Jobs cannot run on the Primary Domain Controller when the owner attribute specifies a local user (user@host). To ensure that these jobs run, specify a network user (user@domain) as the owner.
- You cannot log in to the domain controller machine. To use the domain controller machine, log in to the network machine on that domain.
- For command jobs, the agent logs in to the client as the current user on the machine where the job is defined to run. The job runs as long as that user has all the necessary Windows permissions to access the command that is associated with the job and any resources needed to execute that command.
- If you change your computer name on the Identification tab of the Windows Control Panel Network dialog, the job will not run unless you also change the Host Name on the DNS tab of the TCP/IP Protocol dialog to the same name.

**Example: Define the Owner of a Job**

This example changes the owner of the job to chris. Only the EDIT superuser can change the value. The user chris on any machine in the network can edit the job. The job's command runs with the permissions of chris.

```
owner: chris
```

**Changing the Owner in Multiple Jobs**

To change a large number of jobs, the EDIT superuser can invoke the autorep command to dump multiple job definitions to an output file, change the owner, and reload the changed job definitions using the jil command. The following example shows how to save all job definitions to a file:

```
autorep -J ALL -q > dump_file
```

The output of this command is formatted exactly as a job definition script as shown in the following example:

```
insert_job: test_job
job_type: c
```

```
command: sleep 60
machine: juno
owner: jerry@jupiter
permission: wx
alarm_if_fail: 1
```

The owner field of each job definition is typically commented out, unless the EDIT superuser runs the autorep command to generate the report. This is because only the EDIT superuser can change the owner field. After generating this report, the EDIT superuser can use a text editor to change the owner field and reload the job definitions into the database using the jil command, as follows:

```
jil < dump_file
```

### **Job Owners and Security**

Workload Automation Agent for UNIX, Linux, or Windows provides a local security feature that controls which users are allowed to submit jobs on behalf of other users. However, this security rule does not apply to AutoSys Workload Automation. On AutoSys Workload Automation, jobs are always submitted to run under the user specified in the owner attribute. If local security is enabled on the agent, the agent checks the permissions of the job owner only. The agent does not check the AutoSys Workload Automation user who submits the job.

#### **NOTE**

For more information about how AutoSys Workload Automation works with local security on the agent, see [Configure AutoSys Workload Automation to Work with Workload Automation Agent for UNIX, Linux, Windows, or i5/OS](#).

### **pa\_monitor\_progress Attribute -- Indicate Whether to Track the Process to Completion**

The pa\_monitor\_progress attribute indicates whether the job tracks the process to completion.

#### **NOTE**

If you specify the pa\_monitor\_progress attribute in your job definition, the job track the process to completion (the default).

### **Supported Job Types**

This attribute is optional for the following job types:

- [Process Automation Process Execution \(PAPROC\)](#)
- [Process Automation Start Request Form \(PAREQ\)](#)

### **Syntax**

This attribute has the following format:

```
pa_monitor_progress: y | n
```

- **y**  
Tracks the process to completion. This is the default.
- **n**  
Does not track the process to completion.

### **Example: Track a Process Automation Process to Completion**

The pa\_monitor\_progress attribute in the following job definition overrides the default value.

```
insert_job: pam1
job_type: PAPROC
```

```

endpoint_url: "http://canor01-vm2k3c:8080/itpam/soap"
machine: paagent
owner: pamadmin@capamach
pa_name: "/ITPAM Tutorials/S01 Getting started/L01 Creating a Simple Process/L01-
2 Running the Process/Running the Process"
pa_monitor_progress: Y
pa_trace: N

```

## pa\_name Attribute -- Specify the Name of the Process or Start Request Form

The `pa_name` attribute specifies the fully qualified name of the process or the name of the Start Request Form in a Process Automation job.

### Supported Job Types

This attribute is required for the following job types:

- [Process Automation Process Execution \(PAPROC\)](#)
- [Process Automation Start Request Form \(PAREQ\)](#)

### Syntax

This attribute has the following format:

```
pa_name: name
```

- ***name***  
Specifies the name of the process or Start Request form.  
**Limits:** Up to 256 characters; case-sensitive

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### Example: Execute a Process Automation Process

This example executes a CA Process Automation process named `/CAWAAE/Test Process` using the agent on the localhost machine and connecting to a CA Process Automation installation on `capamach`. It uses the credentials from `pamadmin@capamach` to authenticate on the Process Automation installation and passes values for the process variable named `strTestName`. It will monitor the process until the process completes and report success or failure at that time.

```

insert_job: execproc
job_type: PAPROC
machine: localhost
endpoint_URL: "http://capamach:8080/itpam/soap"
pa_name: "/CAWAAE/Test Process"
owner: pamadmin@capamach
pa_monitor_progress: Y
pa_parameter: param_name="strTestName", param_value="PA Test"

```

## pa\_parameter Attribute -- Specify Process Parameters

The `pa_parameter` attribute specifies the parameters in a Process Automation job. When the job runs, the parameters are passed to the executing process.

### Supported Job Types

This attribute is optional for the following job types:

- [Process Automation Process Execution \(PAPROC\)](#)
- [Process Automation Start Request Form \(PAREQ\)](#)

### Syntax

This attribute has the following format:

```
pa_parameter: param_name=name, param_value=value
```

- **param\_name=name, param\_value=value**  
Specifies the parameter name and scalar value.
  - **param\_name=name**  
Specifies the name or path of the element.
  - **param\_value=value**  
Specifies the scalar string value.**Limits:** Up to 2048 characters; case-sensitive

### NOTE

- To specify multiple parameters, use multiple entries of pa\_parameter.
- In each pa\_parameter entry, separate the name/value pair with a comma. The entire value can be up to 2048 characters.
- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

### Example: Specify Multiple Parameters in the Start Request Form

This example specifies the Process Automation server name, port, and alarm parameters in the Start Request Form.

```
insert_job: srfl
job_type: PAREQ
endpoint_URL: "http://paserver:8080/itpam/soap"
machine: paagent
owner: pamadmin@paserver
pa_path: "/ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/"
pa_name: "Start Request Form"
pa_parameter: param_name="AlarmProcessingProduct", param_value="Test Alarm Product"
pa_parameter: param_name="Server", param_value="paserver"
pa_parameter: param_name="Port", param_value=80
pa_monitor_progress: Y
pa_trace: Y
```

## pa\_path Attribute -- Specify the Path of the Start Request Form

The pa\_path attribute specifies the path of the Start Request Form.

### Supported Job Type

This attribute is required for the [Process Automation Start Request Form job type \(PAREQ\)](#).

### Syntax

This attribute has the following format:

```
pa_path: path
```

- **path**

Specifies the path of the Start Request Form.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

- The path specified must end in a / character.
- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.

**Example:** /CAWAAE/

**Example: Specify the Path to the Start Request Form**

This example specifies the Start Request Form located at /ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/. It is being passed values for three parameters that are defined in that form, and it will continue to monitor the progress of the job until the job completes.

```
insert_job: srf1
job_type: PAREQ
endpoint_URL: "http://paserver:8080/itpam/soap"
machine: paagent
owner: pamadmin@paserver
pa_path: "/ITPAM Tutorials/S05 External Interactions/L06 Start Req Form for Input/"
pa_name: "Start Request Form"
pa_parameter: Name="AlarmProcessingProduct", Value="Test Alarm Product"
pa_parameter: Name="Server", Value="paserver"
pa_parameter: Name="Port", Value=80
pa_monitor_progress: Y
pa_trace: N
```

**pa\_trace Attribute -- Indicate Whether to Write SOAP Messages to the Spool File**

The pa\_trace attribute indicates whether the job writes a trace of SOAP messages to the spool file.

**Supported Job Types**

This attribute is optional for the following job types:

- [Process Automation Process Execution \(PAPROC\)](#)
- [Process Automation Start Request Form \(PAREQ\)](#)

**Syntax**

This attribute has the following format:

```
pa_trace: y | n
```

- **y**  
Writes a trace of SOAP messages to the spool file.
- **n**  
Does not write a trace of SOAP messages to the spool file. This is the default.

**Example: Write a Trace of SOAP Messages to the Spool File**

This example writes a trace of SOAP messages to the spool file. The process named Test Process is tracked to completion.



```

insert_job: execproc
job_type: PAPROC
machine: localhost
endpoint_URL: "http://capamach:8080/itpam/soap"
pa_name: "Test Process"
owner: pamadmin@capamach
pa_monitor_progress: Y
pa_trace: Y
pa_parameter: Name="strTestName", Value="PA Test"

```

## payload\_uri Attribute -- Specify the Payload Data

The `payload_uri` attribute specifies the payload data for the POST or PUT request in an HTTP job.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
payload_uri: data
```

- ***data***  
Specifies the payload data for the POST or PUT request in an HTTP job.  
**Limits:** Up to 256 characters  
**Example:** "file://host/path"

### NOTE

You can use global variables to specify the payload data. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## permission Attribute -- Specify the Users with Edit and Execute Permissions

### Contents

The permission attribute specifies which users have edit and execute permissions for the job you are defining.

**UNIX Note:** The permission attribute is based on the same permissions used in native UNIX. It uses the user ID (uid), and group ID (gid) from the UNIX environment to control who can edit job definitions and who can execute the actual command specified in the job.

**Windows Note:** The permission attribute provides users (by type) with edit and execute permissions for a specific job. By default, only a job's owner has edit and execute permissions on a job.

### NOTE

When issuing commands or defining jobs that run on a different operating system (for example, Windows to UNIX or UNIX to Windows), you must use the syntax appropriate to the operating system where the job will run.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
permission: permission [, permission]
```

- **permission**

Specifies the comma-delimited permission levels to associate with the job. The order in which permission values are specified is not important.

**Limits:**

This value can be up to 30 alphanumeric characters in length.

Valid values are:

- **gx**  
(UNIX only) Assigns group execute permissions to the job.
- **ge**  
(UNIX only) Assigns group edit permissions to the job.
- **me**  
Indicates that any authorized user may edit the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).
- **mx**  
Indicates that any authorized user may execute the job, regardless of the machine they are on. Otherwise, the user must be logged on to the machine specified in the owner field (for example, user@host\_or\_domain).
- **we**  
Assigns world edit permissions to the job.
- **wx**  
Assigns world execute permissions to the job.  
The job's owner always has full edit and execute permissions.  
**Default:** Machine permissions are turned off.

**Example: Set Permissions to Execute and Edit a Job on UNIX**

This example sets permissions on UNIX so that any user can execute the job, but only members of your group can edit it:

```
permission: ge, wx
```

**Example: Set Permissions to Execute a Job on Windows**

This example sets permissions on Windows so that any authorized user can execute the job regardless of the machine they are logged on to:

```
permission: mx
```

## User Types

AutoSys Workload Automation supports the following types of users for any job:

- **Owner**  
Indicates the user who created the job.
- **World**  
Indicates any user.
- **Group**  
(UNIX only) Indicates any user who is in the same group as the owner.

**NOTE**

Windows does not support the UNIX group permission attribute.

The default owner is the user who initiated JIL to define the job. The owner always has both edit and execute permissions for the job on the machine on which it was defined. Permission attributes can extend edit and execute capability to other users and other machines.

### **Permission Types on UNIX**

On UNIX, AutoSys Workload Automation supports multiple levels of permissions associated with a job. Every job has the following levels of permissions:

- **Edit**  
Indicates that the user can edit, override, or delete the job definition.
- **Execute**  
Indicates that the user can affect how the job runs, typically by issuing a sendevent command. Events that affect how a job runs are:
  - CHANGE\_STATUS
  - FORCE\_STARTJOB
  - JOB\_OFF\_HOLD
  - JOB\_OFF\_ICE
  - JOB\_ON\_HOLD
  - JOB\_ON\_ICE
  - JOB\_ON\_NOEXEC
  - KILLJOB
  - SEND\_SIGNAL
  - STARTJOB
- **Machine**  
Indicates that, by default, all edit and execute permissions are valid only on the machine on which the job was defined. Permission attributes can extend this permission to other machines.

The job owner has edit permission on the job, and the UNIX command specified in the job runs under the owner's user ID.

When a command runs on the machine specified in the job definition, the `setuid(uid)` system call changes the uid of the process to that of the job's owner.

### **Permission Types on Windows**

On Windows, AutoSys Workload Automation supports the following permission levels:

- **Edit**  
Indicates that the user can edit, override, or delete the job definition.
- **Exec**  
Indicates that the user can affect how the job runs, typically by issuing a sendevent command. Events that affect how a job runs are:
  - CHANGE\_STATUS
  - FORCE\_STARTJOB
  - JOB\_OFF\_HOLD
  - JOB\_OFF\_ICE
  - JOB\_ON\_HOLD
  - JOB\_ON\_ICE
  - JOB\_ON\_NOEXEC
  - KILLJOB
  - STARTJOB
- **Machine**

Indicates that, by default, all edit and execute permissions are valid only on the machine on which the job was defined. Permission attributes can extend this permission to other machines.

## **User and Permission Types on UNIX**

### **Valid on UNIX**

When a job is created, AutoSys Workload Automation retrieves the user ID from the environment and associates it with the job. Then, the product uses the owner's current umask value to supply default permissions to the job. The product uses the umask "write" permission as the default "edit" permission for the job, and it uses the umask "execute" permission as the default "execute" permission for the job.

### **Job Permissions and Windows**

If you are defining jobs and running them on different operating systems, keep the following in mind:

- When you define a job to run on a Windows client, you can set group permissions, but they are ignored. Group permissions are only used when a job is edited or executed on a UNIX computer.
- When you edit a job from a Windows computer, the group edit permission is ignored. In this case, the user editing the job must be the job's owner or World Edit permissions must be specified for the job.
- When you run a job from a Windows computer, the group execute permission is ignored. In this case, the user executing the job must be the job's owner or World Execute permissions must be specified for the job.

## **persist\_output\_cookies Attribute -- Persist the Cookies Output in the Database**

The `persist_output_cookies` attribute specifies whether you want to persist the cookies output in the database as a blob so that it can be consumed by the successor jobs for their execution.

Any successor job can refer to persisted cookies with the attribute `input_cookies:<predecessor jobName>`

### **Supported Job Types**

This attribute is optional for the [HTTP job type](#).

### **Syntax**

This attribute has the following format:

```
persist_output_cookies: true | false
```

- **true**  
Specifies to persist the cookies output in the database as a blob so that it can be consumed by the successor jobs for their execution.
- **false**  
Does not persist the cookies output in the database.

**Default:** false

### **NOTE**

Only after a successful job run with the current job definition, the outputcookies job blobs are available for consumption by the successor jobs.

## **persist\_output\_header Attribute -- Persist the Header Output in the Database**

The `persist_output_header` attribute specifies whether you want to persist the header output in the database as a blob so that it can be consumed by the successor jobs for their execution.

### **Supported Job Types**

This attribute is optional for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
persist_output_header: true | false
```

- **true**  
Specifies to persist the header output in the database as a blob so that it can be consumed by the successor jobs for their execution.
- **false**  
Does not persist the header output in the database.

**Default:** false

### NOTE

Only after a successful job run with the current job definition, the outputheaders or outputpayloadvar job blobs are available for consumption by the successor jobs.

## persist\_output\_payload Attribute -- Persist the Payload Output in the Database

The persist\_output\_payload attribute specifies whether you want to persist the payload output in the database as a blob so that it can be consumed by the successor jobs for their execution.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

### Syntax

This attribute has the following format:

```
persist_output_payload: true | false
```

- **true**  
Specifies to persist the payload output in the database as a blob so that it can be consumed by the successor jobs for their execution.
- **false**  
Does not persist the payload output in the database.

**Default:** false

### NOTE

Only after a successful job run with the current job definition, the outputpayload or outputpayloadvar job blobs are available for consumption by the successor jobs.

## ping\_host--Verify that the Target Host is up

The ping\_host attribute specifies the host name or IP address of the target host that receives the Wake on LAN (WOL) signal. You must specify a value for this field to enable the ping\_timeout and ping\_ports fields.

### NOTE

If you do not specify the ping\_host, the job completes immediately after the WOL signal is sent without verifying whether the signal worked.

### Supported Job Type

This attribute is optional for the [Wake on LAN \(WOL\) job type](#).

## Syntax

This attribute has the following format:

```
ping_host: target_hostname
```

- ***target\_hostname***  
Specifies the name of the host computer.  
**Limits:** Up to 80 characters

### Example: Broadcast the WOL Signal and Ping a Port

This example broadcasts the WOL signal to the server subnet identified by the 172.16.255.255 broadcast IP address and the server identified by the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the WOL signal, it pings port 7 of the host computer to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails. The job uses the default timeout for the ping.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
```

## ping\_ports Attribute -- Specify the Ports to Contact

After the agent sends the Wake on LAN (WOL) signal, it attempts to connect to the ports specified in the ping\_ports attribute. The agent connects to the ports to confirm that the computer is awake by verifying that an application is listening on the port. The agent creates a TCP connection; it does not send and receive ICMP ping packets. If at least one of the specified ports is available, the job completes successfully; otherwise, it fails.

### NOTE

If a firewall blocks the specified port or the machine is not addressable on the network, the connection attempt fails.

### Supported Job Type

This attribute is optional for the [Wake on LAN \(WOL\)](#) job type.

## Syntax

This attribute has the following format:

```
ping_ports: port | port,port,...
```

- ***(optional) port***  
Specifies the ports to contact after the agent sends the Wake on LAN (WOL) signal. If you do not enter a value, the agent contacts the defaults ports.  
**Defaults:** 21 (ftp), 22 (ssh), 23 (telnet), 80 (http), 111 (sunrpc), 135 (epmap), 139 (netbios-ssn), 445 (microsoft-ds)

### NOTE

- To specify multiple ports, separate each port with a comma.
- The entire value can be up to 256 characters.

### Example: Broadcast the WOL Signal to a Broadcast IP address and Ping a Port

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server with the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
```

### Example: Broadcast the WOL Signal to Ping the Default Port

This example broadcasts the WOL signal to the subnet with the 172.16.255.255 IP address and the 00-11-43-73-38-DC MAC address. After the agent sends the WOL signal, the agent pings the default ports at the 172.16.1.101 IP address to verify if these ports are available. If at least one of these ports is available, the job completes successfully; otherwise, it fails.

```
insert_job: wol_default_job
job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-11-43-73-38-DC
ping_host: 172.16.1.101
```

## ping\_timeout Attribute--Specify the Timeout for the Ping

The `ping_timeout` attribute specifies the maximum amount of time in seconds that the agent attempts to connect to the target host before a timeout occurs.

### NOTE

You must specify the `ping_host` attribute when you use this attribute.

### Supported Job Type

This attribute is optional for the [Wake on LAN \(WOL\) job type](#).

### Syntax

This attribute has the following format:

```
ping_timeout: timeout
```

- ***timeout***  
Specifies the timeout for the ping in seconds.  
**Limits:** Up to 5 digits  
**Default:** 120 seconds

### Example: Specify a Timeout Limit When you Broadcast the WOL Signal to a MAC address

This example broadcasts the WOL signal to the subnet identified by the 172.16.255.255 broadcast IP address and the server identified by the 00-1E-4F-C1-0F-FE MAC address. After the agent sends the Wake on LAN (WOL) signal, the agent pings port 7 at the 172.16.1.101 IP address to ensure it is available. If port 7 becomes available within 5 minutes (300 seconds), the job completes successfully; otherwise, it exceeds the timeout limit and fails.

```
insert_job: wol_job
```

```

job_type: WOL
machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 00-1E-4F-C1-0F-FE
ping_host: 172.16.1.101
ping_ports: 7
ping_timeout: 300

```

## poll\_interval Attribute -- Define the Frequency to Poll CPU or Disk Usage

The poll\_interval attribute defines how often the CPU or disk usage is polled.

### NOTE

If you are monitoring continuously and you do not specify the poll\_interval attribute in your job definition, the CPU or disk usage is polled every 10 seconds (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [CPU Monitoring \(OMCPU\)](#)
- [Disk Monitoring \(OMD\)](#)

### NOTE

To use this attribute, you must specify WAIT or CONTINUOUS for the monitor\_mode attribute. This attribute does not apply when the monitor mode is NOW.

### Syntax

This attribute has the following format:

```
poll_interval: seconds
```

- **seconds**

Defines the interval (in seconds) between successive scans of the CPU or disk usage.

**Default:** 10

**Limits:** 1-2147483647

### NOTE

- Your agent administrator can specify a default poll interval for all CPU and Disk Monitoring jobs by setting the objmon.scaninterval parameter in the agent's agentparm.txt file. That agent parameter is specified in milliseconds.
- The poll\_interval attribute overrides the default poll interval specified in the agent's agentparm.txt.
- When you update a job, if you do not specify a value for the poll\_interval attribute, the CPU or disk usage is polled every 10 seconds (the default).

### Example: Poll CPU Usage Every 60 Seconds

This example continuously monitors the CPU available on the unixagent computer. The job polls the CPU usage every 60 seconds. Each time the available CPU is within 75 and 95 percent, an alert is written to the scheduler log file. The job continues monitoring the CPU usage until it is ended manually.

```

insert_job: omcpu_job
job_type: OMCPU
machine: unixagent
cpu_usage: FREE

```



```
inside_range: TRUE
poll_interval: 60
monitor_mode: CONTINUOUS
lower_boundary: 75
upper_boundary: 95
```

## port Attribute -- Specify an SNMP Port

The port attribute specifies the port used to communicate with the network device.

### NOTE

If you do not specify the port attribute in your job definition, the job uses port 161 (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [SNMP Value Get](#) (SNMPGET)
- [SNMP Value Set](#) (SNMPSET)

### Syntax

This attribute has the following format:

```
port: port
```

- ***port***  
Specifies the port of the network device.  
**Limits:** Up to 10 digits  
**Default:** 161

### Example: Specify the Port in an SNMP Value Get Job

Suppose that you want to know the value of the agentVersion variable hosted by a network device. In this example, the host name of the network device is host.example.com and its port is 161. The SNMP version is v2 and the read community string is public. The name of the MIB file is cybermation.mib, which is located in the agent installation directory.

```
insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public
```

## port\_name Attribute -- Specify the Port Name Within the Namespace

The port\_name attribute specifies the WSDL port name within the target namespace in a Web Service job.

### Supported Job Types

This attribute is required for the [Web Service Document/Literal \(WSDOC\)](#) job type.

This attribute is optional for the [Web Service RPC/Encoded \(WBSVC\)](#) job type.

### Syntax

This attribute has the following format:

```
port_name: portname
```

- **portname**

Specifies the WSDL port name within the target namespace. A WSDL port describes the operations exposed by a web service and defines the connection point to the web service.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

- Only valid XML characters are allowed.
- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, you must specify both the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL\_URL and endpoint\_URL attributes, then the service\_name and port\_name attributes are optional.
- You can use global variables to specify the WSDL port name within the target namespace. Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

**Example: Specify the WSDL Port Name for Getting Stock Quotes**

**NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

## preemptive\_authentication Attribute -- Authenticate Application Credentials

The preemptive-authentication attribute specifies whether or not to authenticate the application credentials using preemptive authentication. In preemptive authentication, credentials are submitted along with the initial request. No authentication process occurs between the target server and the client.

Before the addition of this attribute, all authentication was done in non-preemptive mode.

There are two modes of authentication. The basic mode supports pre-emptive and non-preemptive authentication. The digest mode supports only non-pre-emptive authentication.

### Supported Job Types

This attribute is optional for the [HTTP job type](#) and [Define a Web Service Document/Literal Job](#) type.

### Syntax

This attribute has the following format:

```
preemptive_authentication: false
```

**Default:** false

## priority Attribute -- Define the Queue Priority of the Job

The priority attribute defines the queue priority of the job you are defining.

When a job is ready to run on a designated machine but the current load on that machine is too large to accept the new job's load or the job has resource dependencies that are not satisfied, AutoSys Workload Automation queues the job for that machine so it runs when sufficient resources are available.

The queue priority establishes the relative priority of all jobs queued for a given machine. A lower number attribute value indicates a higher priority. If you do not set the priority attribute or the priority is set to 0, the job is not queued behind other jobs and runs immediately on a machine if resource dependencies permit or if the job has no resource dependencies. The scheduler ignores any load unit values defined for the job or machine when the job has a priority value of zero.

### NOTE

For more information about how AutoSys Workload Automation queues jobs, see [Specify Queuing Priority](#) and [How AutoSys Workload Automation Queues Jobs](#)

You can assign priorities to box jobs. Box jobs can be queued in the RESWAIT state if they have resource dependencies that are not met. Assigning priorities to jobs in boxes allows the scheduler to evaluate the starting conditions of the jobs in order of their priority, instead of in the order in which they were defined, which is the default.

### NOTE

- You cannot use the priority or job\_load attribute if you specify a user-defined load balancing script in the machine attribute.
- The scheduler tries to evaluate the starting conditions of jobs that are associated with a group or application in order of their priority; however it does not guarantee that jobs execute strictly in order of their priority.

### Supported Job Types

This attribute is optional for all job types. This attribute does not apply to box jobs.

### Syntax

This attribute has the following format:

```
priority: priority_level
```

- **priority\_level**

Defines the queue priority of the job. The lower the value, the higher the priority; therefore, 1 is the highest possible queued priority. 0 signifies to run the job immediately, regardless of the current machine load

**Limits:** Any integer 0 or greater

**Default:** 0

**Example: Set the Job to Run Immediately**

This example sets the job to always run immediately, regardless of the current load on the client:

```
priority: 0
```

**Example: Set the Job to Run with Highest Priority**

This example sets the job to run with the highest priority without overriding the machine load control mechanism:

```
priority: 1
```

**Example: Set the Job to Run with Low Priority**

This example sets the job to run with low priority:

```
priority: 100
```

**process\_name Attribute -- Specify the Name of the Process to be Monitored**

The process\_name attribute specifies the name of the process to be monitored.

**Supported Job Type**

This attribute is required for the [Process Monitoring \(OMP\) job type](#).

**UNIX and Windows Syntax**

This attribute has the following format:

```
process_name: process_name
```

- **process\_name**  
Specifies the name of the process to be monitored.  
**UNIX:** Specify a PID or process name.  
**Windows:** Specify a full path or process name.  
**Limits:** Up to 256 characters; case-sensitive

**i5/OS Syntax**

This attribute has the following format:

```
process_name: jobnumber/username/jobname
```

- **jobnumber**  
Specifies the six-digit job number or \*ALL.
- **username**  
Specifies the user ID the job runs under. The value can be a generic name or \*ALL.  
**Limits:** Up to 10 characters
- **jobname**  
Specifies the job name on the i5/OS system. The value can be a generic name or \*ALL.  
**Limits:** Up to 10 characters

**NOTE**

- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- If you specify only the PID or process name and the job runs on an i5/OS computer, the agent searches the active UNIX workload for a matching PID or process name.
- On i5/OS, you can use generic names to specify the process name. A generic name starts with characters that are part of a valid name and ends with an asterisk (\*). The asterisk denotes any number of characters and can only be placed at the end of a generic name. A name cannot contain a single asterisk and no other characters. To specify all possible names, use the special value \*ALL.
- On all operating systems, if you have multiple processes on a system with the same process name, the job completes successfully as follows:

Status and Monitor Mode	Job Completion Requirements
process_status: RUNNING monitor_mode: NOW	The job completes successfully if at least one of the processes with the specified name is running.
process_status: RUNNING monitor_mode: WAIT	The job completes successfully when at least one of the processes with the specified name starts running.
process_status: STOPPED monitor_mode: NOW	The job completes successfully if all the processes with the specified name are stopped.
process_status: STOPPED monitor_mode: WAIT	The job completes successfully when all the processes with the specified name stops running.

**Example: Monitor Multiple Instances**

This example monitors the Microsoft SQL Server processes of two instances using the full path name. When the server process stops, the job monitoring that instance completes successfully. The first job monitors the sqlserver.exe process in the ...\\MSSQL.1\\MSSQL\\Binn directory. The second job monitors the sqlserver.exe process in the ...\\MSSQL.2\\MSSQL\\Binn directory.

```
insert_job: mon_sql_server_instance1
job_type: OMP
machine: mssqlserver
process_name: "C:\\Program Files\\Microsoft SQL Server\\MSSQL.1\\MSSQL\\Binn\\sqlservr.exe"
process_status: stopped
monitor_mode: wait
insert_job: mon_sql_server_instance2
job_type: OMP
machine: mssqlserver
process_name: "C:\\Program Files\\Microsoft SQL Server\\MSSQL.2\\MSSQL\\Binn\\sqlservr.exe"
process_status: stopped
monitor_mode: wait
```

**Example: Monitor the Agent Process on i5/OS**

This example monitors the APPLPGM process on an i5/OS computer. The job checks the process status immediately and completes successfully if the process is running.

```
insert_job: omp_i5_onejob
job_type: OMP
machine: i5agt
process_name: 123456/PROD/APPLPGM
process_status: RUNNING
```

```
monitor_mode: NOW
```

### Example: Monitor i5/OS Processes That Have Similar Names

This example monitors all processes running on an i5/OS computer under the JDOE user profile and whose names start with CALC. When all of these processes stop running, the job completes successfully.

```
insert_job: omp_i5
job_type: OMP
machine: i5agt
process_name: *ALL/JDOE/CALC*
process_status: STOPPED
monitor_mode: WAIT
```

### Example: Monitor an i5/OS Process That Has Any Job Number and Is Running Under Any User Name

This example monitors for any process named CYBAGENT, regardless of the first part (job number) and the second part (user name) of the process ID. The job completes successfully if at least one process named CYBAGENT is running. The entire process ID is enclosed in quotation marks so that the text following /\* is not interpreted as a comment.

```
insert_job: omp_i5_all
job_type: OMP
machine: i5agt
process_name: "*ALL/*ALL/CYBAGENT"
process_status: RUNNING
monitor_mode: NOW
```

### Example: Monitor a UNIX Process on an i5/OS Computer

This example monitors for the BMPROC process in the PASE environment on an i5/OS computer. If the BMPROC process is running, the job completes.

```
insert_job: omp_i5_unix
job_type: OMP
machine: i5agt
process_name: BMPROC
process_status: RUNNING
monitor_mode: NOW
```

## process\_status Attribute -- Specify the Status of the Process to be Monitored (11.3.6 SP6)

The process\_status attribute specifies the status of the process to be monitored. The job checks whether the status of the process matches the status specified by this attribute.

### NOTE

If you do not specify the process\_status attribute in your job definition, the job checks whether the process is stopped (the default).

### Supported Job Type

This attribute is optional for the [Process Monitoring \(OMP\) job type](#).

### Syntax

This attribute has the following format:

```
process_status: RUNNING | STOPPED
```

- **RUNNING**

Specifies that the job monitors the process for a running status.

- **STOPPED**

Specifies that the job monitors the process for a stopped status. This is the default.

### Example: Monitor a Running Process

This example monitors the process with ID 2568. If the process is running, the job completes successfully. If the process is not running, the job continues monitoring it until the process starts running.

```
insert_job: omp_unix
job_type: OMP
machine: unixagt
process_name: 2568
process_status: RUNNING
monitor_mode: WAIT
```

### Example: Monitor a Stopped Process

This example monitors the cybAgent.exe process. The job checks the process status immediately and completes successfully if the process is stopped. If the process is running, the job fails.

```
insert_job: omp_win
job_type: OMP
machine: winagt
process_name: "c:\Program files\Agent\cybAgent.exe"
process_status: STOPPED
monitor_mode: NOW
```

## profile Attribute -- Specify a Job Profile

The profile attribute specifies a profile that defines the non-system environment variables that a job uses. The variables in the profile are sourced before the job runs.

System environment variables are the only other variables defined for the command's execution. Therefore, you must define non-system environment variables in a user-defined job profile. System environment variables are set before profile variables. Therefore, you can use references to system environment variables in job profiles.

### NOTE

If a command that typically runs when entered at the command line fails when run as a job, it is usually due to the incomplete specification of the required environment for the command in the job's profile.

### Supported Job Types

This attribute is optional for the following job types:

- [Command \(CMD\)](#)
- [File Watcher \(FW\)](#)

### UNIX Syntax

This attribute has the following format:

```
profile: path_name
```

- ***path\_name***

Defines the full path to and name of the job profile. The job profile must be a shell script.

**Limits:** Up to 255 characters

**NOTE**

You cannot use variable substitution in this value.

**NOTE**

- File Watcher jobs with the profile are also supported for the System Agent.
- AutoSys Workload Automation searches for the specified profile on the machine where the command is to run. The agent always spawns a process and starts the shell in that process, passing it the name of the profile to source. This profile typically includes the definitions and exports of environment variables, which can be referenced in the job's command. This is helpful if the command is a shell script.
- Redirection of the standard input, standard output, and standard error files will probably fail.
- You must set the \$PATH variable in the profile, because AutoSys Workload Automation uses it to locate the command specified in the job.
- For Korn shell users, we recommend that you explicitly set any other required environment variables in the shell script that is specified as the command to run. Alternatively, you can source additional shell scripts from your main shell script.
- If you want to set permissions for std\_out and std\_err to -rw-r--r--, you must set umask 022 set it in the specified profile. If you do not set umask 022, the standard output and standard error files will have world write permissions.

## Windows Syntax

This attribute has the following format:

```
profile: profile_name
```

- ***profile\_name***

Specifies the name of a job profile. You can specify a profile that is located on the computer where the job runs. Alternatively, you can specify the path to a profile file on a remote machine, as follows:

```
\\machine_name\share_name\profile_name.txt
```

**Limits:** Up to 255 characters

**NOTE**

- The owner of the job must have read access to the specified profile.  
For more information about creating, viewing, and deleting job profiles on Windows, see [\(Windows\) Managing Job Profiles Using the Administrator Utility](#).

## Example: Assign a Job Profile on UNIX

This example assigns the my\_profile profile to a Command job. The profile is located in the user's home directory, /usr/home.

```
insert_job: test_run
job_type: CMD
machine: unixagent
command: /bin/touch /tmp/test_run.out
profile: /usr/home/my_profile
```

## Example: Assign a Job Profile on Windows

This example assigns the eng profile to a Command job. The eng profile was defined in the Job Profiles window in the AutoSys Workload Automation Administrator utility.

```
insert_job: test_run
job_type: CMD
```



```
machine: winagent
command: "c:\bin\test.bat"
profile: eng
```

### Example: Assign a Job Profile that is Located on Another Windows Machine

This example assigns the cmdprof.txt profile to a Command job. The cmdprof.txt profile is defined on the dev machine, which is different from the machine where the job runs.

```
insert_job: test_run
job_type: CMD
machine: winagent
command: "c:\bin\test.bat"
profile: \\dev\share\cmdprof.txt
```

## provider\_url Attribute -- Specify a Service Provider URL (EJB and JMS Jobs)

The provider\_url attribute specifies the URL of the service provider using dotted decimal notation or DNS name in an Entity Bean, Session Bean, JMS Publish, and JMS Subscribe job. The service provider implements a context or initial context. This context can be plugged in dynamically to the JNDI architecture used by the JNDI client.

### Supported Job Types

This attribute is required for the following job types:

- [Entity Bean](#) (ENTYBEAN)
- [JMS Publish](#) (JMSPUB)
- [JMS Subscribe](#) (JMSSUB)
- [Session Bean](#) (SESSBEAN)

### Syntax

This attribute has the following format:

```
provider_url: "location"
```

#### • "*location*"

Specifies the JNDI service provider URL.

- For WebLogic servers, the format is the following:

```
"t3://WLIPAddress[:ORBPort]"
```

#### **WLIPAddress**

Specifies the IP address or domain name of the WebLogic Application Server.

#### **ORBPort**

(Optional) Specifies the ORB port.

#### **Default:**

7001

#### **Examples:**

- "t3://localhost:7001"
- "t3://172.24.36.107:7001" (IPv4)
- "t3://0:0:0:0:FFFF:7001" (IPv6)
- For WebSphere servers, the format is the following:

```
"iiop://WSIPAddress[:ORBPort]"
```

#### **WSIPAddress**

Specifies the IP address or domain name of the WebSphere Application Server.

**ORBPort**

(Optional) Specifies the ORB port.

**Default:**

2809

**Examples:**

- "iiop://172.24.0.0:2809"
- "iiop://172.24.36.107:2809" (IPv4)
- "iiop://0:0:0:0:FFFF:192.168.00.00:2809" (IPv6)

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

You can use global variables to specify the JNDI service provider URL. Use the `$$global_name`, `$(global_name)`, or `$$${global_name}` format to reference global variables.

**Example: Specify the Service Provider URL for a WebLogic Application Server**

Suppose that you want to create an entity bean that stores information about a customer such as the customer ID and phone number. The initial context factory supplied by the JNDI service provider is `weblogic.jndi.WLInitialContextFactory`. The service provider's URL is `t3://localhost:7001`, where `localhost` is the domain name of the WebLogic application server and `7001` is the ORB port. When the job runs, the entity bean instance is created.

```
insert_job: create
job_type: ENTYBEAN
machine: appagent
initial_context_factory: weblogic.jndi.WLInitialContextFactory
provider_url: "t3://localhost:7001"
bean_name: customer
create_name: createcustomer
operation_type: CREATE
create_parameter: String="customerid", String="800-555-0100"
```

**Example: Specify the Service Provider URL for a WebSphere Application Server**

Suppose that you want to invoke the `reverse` method on the `CybEJBTestBean` stateless session bean. The `reverse` method has one parameter, with type `java.lang.String` and value `a23`. The output from the `reverse` method is saved in the `C:\Makapt15` file. The initial context factory supplied by the JNDI service provider is `com.ibm.websphere.naming.WsnInitialContextFactory`. The service provider's URL is `corbaloc:iiop://172.24.0.0:2809`, where `172.24.0.0` is the IP address of the WebSphere application server and `2809` is the ORB port. When the job runs, the output of the `reverse` method is stored in the output destination file.

```
insert_job: reverse
job_type: SESSBEAN
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "corbaloc:iiop://172.24.0.0:2809"
bean_name: CybEJBTestBean
method_name: reverse
destination_file: "C:\Makapt15"
j2ee_user: cyberuser
j2ee_parameter: java.lang.String="a23"
```

## provider\_url Attribute -- Specify a Host URL (HTTP Jobs)

The provider\_url attribute specifies the host where the program you want to invoke resides in a HTTP job.

### Supported Job Type

This attribute is required for the [HTTP job type](#).

### Syntax

This attribute has the following format:

```
provider_url: "servlet_url"
```

- **"servlet\_url"**

Specifies the host where the program or servlet you want to invoke resides. The URL has the following format:

```
"http://host[:port][/action]"
```

- **host**

Specifies the name of the computer running the application server.

- **port**

(Optional) Specifies the port the host uses to listen for HTTP requests.

**Default:** 80

- **action**

(Optional) Specifies the path to the servlet to be invoked.

**Limits:** Up to 4000 characters; case-sensitive

**Example:** "http://localhost/cgi-bin/test.sh"

### NOTE

- HTTP and HTTPS are supported.
- From AutoSys Workload Automation Release 12.0 onwards, the [method\\_name](#) attribute is not applicable to HTTP jobs and so you can specify the path to the servlet to be invoked using the provider\_url attribute.
- When you upgrade from a previous release to AutoSys Workload Automation Release 12.0, the [method\\_name](#) attribute value is appended to the provider\_url attribute value.

### NOTE

- If you omit the servlet path (action) in this attribute, you must specify the method\_name attribute in the job definition. If the method\_name attribute is omitted from the job definition, the agent assumes that the provider\_url attribute specifies the full path to the servlet.
- If you specify the servlet path (action) in this attribute, do not include the method\_name attribute in the job definition. If the method\_name attribute is included in the job definition, the agent assumes that the provider\_url attribute specifies only the server host name.
- You can use global variables to specify the host where the program or servlet you want to invoke resides. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Specify the Host URL in an HTTP Job

Suppose that you want to define a job to perform a Google search and have the results returned to the job's spool file. In this example, the job uses the HTTP GET method to perform the Google search on "ca workload automation". When the job runs, the job's spool file includes all the results of the search.

```
insert_job: google
job_type: HTTP
machine: appagent
invocation_type: GET
```

```
provider_url: "http://google.com/search"
j2ee_authentication_order: BASIC,DIGEST,NTLM
j2ee_parameter: q="ca workload automation"
```

## ps\_args Attribute -- Pass Additional Parameters for the PeopleSoft Report

The ps\_args attribute specifies additional parameters to pass to a PeopleSoft report.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_args: argument ...
```

- **argument**  
Specifies additional argument values to pass to the PeopleSoft report.  
**Limits:** Case-sensitive

#### NOTE

You can pass multiple strings as a single argument by enclosing them in quotation marks: "parm1 parm2"

#### NOTE

- The entire value can be up to 254 characters.
- Specify multiple arguments by separating them with blank spaces.
- To pass an argument with delimiters (such as spaces and semicolons), enclose the argument in quotation marks.
- If you specify more than one ps\_args attribute, the second ps\_args attribute within the job definition overrides the first, with the exception of a blank character string, which is ignored.

### Example: Ignore Parameters in the PS\_PRCSDDEFN Table and Pass Additional Parameters to the nVision Report

In this example, the job parameters are not updated with data in the PS\_PRCSDDEFN table. The ps\_args attribute passes additional parameters to the nVision report.

```
insert_job: pass_args
job_type: PS
machine: psagt
owner:VP1@server01
ps_process_name: NVSRUN
ps_process_type: nVision-Report
ps_dest_type: FILE
ps_dest_format: XLS
ps_output_dest: "c:\test\testnv.xls"
ps_skip_parm_updates: Yes
ps_args: "-NRNVARIABLE -NBUAUS01 -NHLhttp://10.1.1.40/psp/ps/EMPLOYEE/ERP/c/
REPORT_BOOKS.IC_RUN_DRILLDOWN.GBL?Action=A"
```

## ps\_dest\_format Attribute -- Specify the Output Format for a PeopleSoft Report

The ps\_dest\_format attribute specifies the output format for the PeopleSoft report.

## Supported Job Type

This attribute is optional for the PeopleSoft (PS) job type.

### Syntax

This attribute has the following format:

```
ps_dest_format: output_format
```

- **output\_format**

Specifies the field name of the output destination format. PeopleSoft stores the list of output destination formats in the PSXLATITEM table. This value corresponds to the Format field in PeopleSoft.

**Limits:** Up to 8 characters; case-sensitive; it cannot contain delimiters (such as spaces)

#### NOTE

Do not use the following special characters: left parenthesis ( ( ), right parenthesis ( ) ), and apostrophe (').  
Use caution when using other special characters, such as backslash (\) and at (@).

Format options are the following:

Field Number	Field Name	Description
1	ANY	Any
2	NONE	(None)
3	PDF	Acrobat (*.pdf)
4	CSV	Comma delimited (*.csv)
5	HP	HP Format (*.lis)
6	HTM	HTML Documents (*.htm)
7	LP	Line printer format (*.lis)
8	WKS	Lotus 1-2-3 files
9	XLS	Microsoft Excel Files (*.xls)
10	DOC	Microsoft Word (*.doc)
11	PS	Postscript (*.lis)
12	RPT	Crystal Report (*.rpt)
13	RTF	Rich Text File (*.rtf)
14	SPF	SQR Portable Format (*.spf)
15	TXT	Text Files (*.txt)
16	OTHER	Other
17	Default	Default
18	XML	XML Format (*.xml)
19	DAT	Data Mover Data File (*.dat)

#### NOTE

- Your agent administrator can specify a default output format for all PeopleSoft jobs by setting the ps.default.outDestFormat parameter in the agent's agentparm.txt file.
- The ps\_dest\_format attribute overrides the default output format specified in the agent's agentparm.txt.

### Example: Format the Report Output as a Text File

This example runs the PAYROLL process that has the Application Engine process type and PS\_ALL run control ID. The PAYROLL process output is formatted as a text file.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
owner: VF1@server01
ps_output_dest:"c:\temp\payroll.txt"
ps_process_name: PAYROLL
ps_process_type: Application Engine
ps_dest_type: FILE
ps_dest_format: TXT
ps_run_cntrl_id: PS_ALL
```

## ps\_dest\_type Attribute -- Specify the Output Type for a PeopleSoft Report

The ps\_dest\_type attribute specifies the output destination type for a PeopleSoft report.

### NOTE

If you do not specify the ps\_dest\_type attribute in your job definition, the job uses the default output type specified in the agent's agentparm.txt file if it is defined.

### Supported Job Type

This attribute is optional for the [PeopleSoft](#) (PS) job type.

### Syntax

This attribute has the following format:

```
ps_dest_type: NONE | FILE | PRINTER | EMAIL | WEB
```

- **NONE**  
Specifies no output destination type.
- **FILE**  
Sends the output of the PeopleSoft report to a file.
- **PRINTER**  
Sends the output of the PeopleSoft report to a printer.
- **EMAIL**  
Sends the output of the PeopleSoft report as an email message.
- **WEB**  
Posts the output of the PeopleSoft report on a website.

### NOTE

- PeopleSoft stores the list of output destination types in the PSXLATITEM table.
- Your agent administrator can specify a default output type for all PeopleSoft jobs by setting the ps.default.outDestType parameter in the agent's agentparm.txt file.
- The ps\_dest\_type attribute overrides the default output type specified in the agent's agentparm.txt.
- If you specify EMAIL or WEB as the output destination type, you can distribute the PeopleSoft report electronically to operators, groups of people, or individuals.

This attribute corresponds to the Type field in PeopleSoft.

### Example: Format the Report Output as a Text File

This example runs the PAYROLL process that has the Application Engine process type and PS\_ALL run control ID. The PAYROLL process output is formatted as a text file.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
owner: VP1@server01
ps_output_dest:"c:\\temp\\payroll.txt"
ps_process_name: PAYROLL
ps_process_type: Application Engine
ps_dest_type: FILE
ps_dest_format: TXT
ps_run_cntrl_id: PS_ALL
```

## ps\_detail\_folder Attribute -- Specify the Name of a PeopleSoft Distribution Detail Folder

The ps\_detail\_folder attribute specifies the name of a report folder for the contents of a distribution list.

### NOTE

To use this attribute, you must specify EMAIL or WEB for the ps\_dest\_type attribute.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_detail_folder: folder_name
```

- **folder\_name**

Specifies the name of a distribution detail folder. This value corresponds to the Folder Name field in PeopleSoft.

**Limits:** Up to 18 characters; it cannot contain delimiters (such as spaces)

### Example: Specify a Distribution Detail Folder

This example runs the AEMINITEST process that has the process type, Application Engine. The output is sent to a website. The distribution details are stored in the GENERAL folder.

```
insert_job: ps_dsfolder
job_type: ps
machine: psagt
ps_process_name: AEMINITEST
ps_process_type: Application Engine
ps_dest_type: WEB
ps_dest_format: PDF
ps_run_cntrl_id: PS_ALL
owner: VP1@server01
ps_detail_folder: GENERAL
ps_server_name: PSPROD
```

## ps\_dlist\_roles Attribute -- Specify a Distribution List of Roles

The ps\_dlist\_roles attribute specifies a distribution list of the roles that represent the individuals who are receiving the PeopleSoft report. For example, you can have a group of users defined as operators. Rather than specifying all the individual users to send the report, you can use this attribute to reference the role and let PeopleSoft handle it.

**NOTE**

To use this attribute, you must specify EMAIL or WEB for the ps\_dest\_type attribute.

**Supported Job Type**

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

**Syntax**

This attribute has the following format:

```
ps_dlist_roles: role[,role...]
```

- **role**

Specifies a role or a list of roles to send the report to. This value corresponds to the ID Type field (with Role selected) and the Distribution ID field in PeopleSoft.

**Limits:** Case-sensitive

**NOTE**

- You can specify multiple roles. Separate each role with a comma.
- The entire value can be up to 256 characters.
- An alternative to using this attribute is to use operator IDs as specified by the ps\_dlist\_users attribute.

**Example: Specify Multiple Roles**

This example runs a Crystal report under the VP1 operator ID. The report output type is WEB and the output format is PDF. The report is distributed to the CLERK, BANK MANAGER, and Employee roles.

```
insert_job: ps_roles
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: WEB
ps_dest_format: PDF
ps_dlist_roles: CLERK,BANK MANAGER,Employee
owner: VP1@server01
```

**Example: Specify One Role**

This example runs a Crystal report under the VP1 operator ID. The report output type is WEB and the output format is PDF. The report is distributed to the BANK MANAGER role.

```
insert_job: ps_roles
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: WEB
ps_dest_format: PDF
ps_dlist_roles: BANK MANAGER
owner: VP1@server01
```

**ps\_dlist\_users Attribute -- Specify a Distribution List of Operator IDs**

The ps\_dlist\_users attribute specifies a distribution list of operator IDs to send a PeopleSoft report to.



**NOTE**

To use this attribute, you must specify EMAIL or WEB for the `ps_dest_type` attribute.

**Supported Job Type**

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

**Syntax**

This attribute has the following format:

```
ps_dlist_users: operator_id[,operator_id...]
```

- **operator\_id**

Specifies an operator ID to send the report to. This value corresponds to the ID Type field (with User selected) and the Distribution ID field in PeopleSoft.

**Limits:** Case-sensitive

**Default:** The operator ID specified in the owner attribute

**NOTE**

- You can specify multiple operator IDs. Separate each operator ID with a comma.
- The entire value can be up to 256 characters.
- By default, the report output is sent to the operator ID specified in the owner attribute. If you do not want to distribute the output to any operator IDs, you must include a `ps_dlist_users` attribute in the job definition and set it to two single quotes or provide no value.
- Instead of specifying a list of operator IDs to distribute the report output to, you can distribute the report output to a list of roles using the `ps_dlist_roles` attribute.
- If the output destination type is EMAIL or WEB, you can also distribute the report output to a list of email addresses using the `ps_email_address` attribute.

**Example: Specify Multiple Operator IDs**

This example runs a Crystal report under the VP3 operator ID. The report is formatted as PDF and distributed in an email to the VP1, VP2, and VP3 operator IDs.

```
insert_job: ps_users
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_dlist_users: VP1,VP2,VP3
ps_operator_id: VP3@ps1
ps_run_cntrl_id: test
```

**Example: Distribute a Report to Email Addresses Instead of a Distribution List**

This example runs a Crystal report under the VP2 operator ID. The output is stored as a PDF web report. The `ps_dlist_users` attribute has no value, so the output is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title and body text. The web report and the job logs are included with the email.

```
insert_job: ps_nousers
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
```

```

ps_dest_type: WEB
ps_dest_format: PDF
ps_dlist_users:
owner: VP2@server01
ps_email_web_report: YES
ps_email_address: user1@example.com;user2@example.com
ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.
ps_email_log: YES

```

## ps\_email\_address Attribute -- Specify the Email Addresses on a Distribution List

The `ps_email_address` attribute specifies the email addresses of the recipients on a distribution list. The PeopleSoft report is emailed to the recipients.

### NOTE

To use this attribute, you must specify EMAIL or WEB for the `ps_dest_type` attribute.

### Supported Job Type

This attribute is optional for the [PeopleSoft](#) (PS) job type.

### Syntax

This attribute has the following format:

```
ps_email_address: address[;address...]
```

- **address**

Specifies an email address on a distribution list to send the report to. This value corresponds to the Email Address List field in PeopleSoft.

**Limits:** Case-sensitive

### NOTE

- You can specify multiple email addresses. Separate each address with a semi-colon.
- The entire value can be up to 256 characters.

### Example: Send a PeopleSoft Report to Two Email Addresses

This example runs a Crystal report and emails the output to recipients. The Crystal report runs under the VP2 operator ID. The output is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title.

```

insert_job: ps_email
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_operator_id: VP2@ps1
ps_email_address: user1@example.com;user2@example.com
ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.

```

## ps\_email\_address\_expanded Attribute -- Specify Additional Email Addresses on a Distribution List

The `ps_email_address_expanded` attribute lets you specify additional email addresses for the recipients on a distribution list. When the email addresses you specify in the `ps_email_address` attribute exceeds the limit, you can use this attribute to extend the email address list.

### NOTE

To use this attribute, you must specify the `ps_email_address` attribute.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_email_address_expanded: address[;address...]
```

- **address**

Specifies an email address on a distribution list to send the report to. This value corresponds to the Email Address List field in PeopleSoft.

**Limits:** Case-sensitive

### NOTE

- You can specify multiple email addresses. Separate each address with a semi-colon.
- The entire value can be up to 256 characters.

## ps\_email\_log Attribute -- Specify Whether to Email PeopleSoft Job Logs

The `ps_email_log` attribute specifies whether to email job logs along with the PeopleSoft report to recipients on a distribution list.

### NOTE

- To use this attribute, you must specify EMAIL or WEB for the `ps_dest_type` attribute. You must also specify the `ps_email_address` attribute in the job definition.
- If you do not specify the `ps_email_log` attribute in your job definition, job logs are not emailed to recipients (the default).

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_email_log: YES | NO
```

- **YES**

Emails job logs along with the report to the recipients on the distribution list.

- **NO**

Does not email job logs to the recipients on the distribution list. This is the default.

### NOTE

This attribute corresponds to the Email With Log field in PeopleSoft.

### Example: Email PeopleSoft Job Logs

This example emails the report in PDF format and the PeopleSoft job logs to the VP1 user. The email includes a subject title and body text.

```
insert_job: ps_logs
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_dlist_users: VP1
owner: VP2@server01
ps_email_subject: PeopleSoft Report Logs
ps_email_text: This report is available for distribution.
ps_email_log: YES
```

## ps\_email\_subject Attribute -- Define an Email Subject

The `ps_email_subject` attribute defines an email subject to include in the email to distribute to recipients of a PeopleSoft report.

### NOTE

To use this attribute, you must specify EMAIL or WEB for the `ps_dest_type` attribute.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_email_subject: subject_text
```

- ***subject\_text***

Defines an email subject to include in the email. This value corresponds to the Email Subject field in PeopleSoft.

**Limits:** Up to 256 characters; case-sensitive

### Example: Include an Email Subject in an Email

This example runs a Crystal report and emails the output to recipients. The Crystal report runs under the VP2 operator ID. The output is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title.

```
insert_job: ps_email
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_operator_id: VP2@ps1
ps_email_address: user1@example.com;user2@example.com
ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.
```

## ps\_email\_text Attribute -- Define the Body Text of an Email

The ps\_email\_text attribute defines the body text of the email to distribute to recipients of a PeopleSoft report.

### NOTE

To use this attribute, you must specify EMAIL or WEB for the ps\_dest\_type attribute.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_email_text: email_text
```

- **email\_text**

Defines the body text of the email. This value corresponds to the Message Text field in PeopleSoft.

**Limits:** Up to 1000 characters; case-sensitive

### Example: Include Body Text in an Email

This example runs a Crystal report and emails the output to recipients. The Crystal report runs under the VP2 operator ID. The output is sent to the email addresses specified in the ps\_email\_address attribute. The email includes a subject title.

```
insert_job: ps_email
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: EMAIL
ps_dest_format: PDF
ps_operator_id: VP2@ps1
ps_email_address: user1@example.com;user2@example.com
ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.
```

## ps\_email\_web\_report Attribute -- Specify Whether to Email a PeopleSoft Web Report

The ps\_email\_web\_report attribute specifies whether to email a web report to recipients on a distribution list.

### NOTE

- To use this attribute, you must specify WEB for the ps\_dest\_type attribute. You must also specify one or more addresses using the ps\_email\_address attribute.
- If you do not specify the ps\_email\_web\_report attribute in your job definition, a web report is not emailed to recipients (the default).

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_email_web_report: YES | NO
```

- **YES**

Emails a web report to the recipients on the distribution list.

- **NO**

Does not email a web report to the recipients on the distribution list. This is the default.

**NOTE**

This attribute corresponds to the Email Web Report field in PeopleSoft.

**Example: Email a PeopleSoft Web Report**

This example stores the output as a PDF web report. The web report is sent to the email addresses specified in the `ps_email_address` attribute. The email includes a subject title and body text.

```
insert_job: ps_web
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Crystal
ps_dest_type: WEB
ps_dest_format: PDF
ps_operator_id: VP2@ps1
ps_email_web_report: YES
ps_email_address: user1@example.com;user2@example.com
ps_email_subject: PeopleSoft Report Status
ps_email_text: This report is available for distribution.
```

**ps\_operator\_id Attribute -- Specify a PeopleSoft Operator ID**

The `ps_operator_id` attribute specifies the operator ID under whose authority the PeopleSoft reports run. The value you specify for this attribute overrides the default defined on the agent by the `ps.default.oprId` parameter in the `agentparm.txt` file.

**Supported Job Type**

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

**NOTE**

A PeopleSoft job must have a valid operator ID to run successfully. Check with your agent administrator to determine whether a default operator ID is set on the agent. If a default is not set, you must specify this attribute.

**Syntax**

This attribute has the following format:

```
ps_operator_id: psopr@tag
```

- ***psopr@tag***

Specifies a valid operator ID under whose authority the PeopleSoft reports run, where *psopr* is the PeopleSoft operator ID and the entire string is used to look up the password that is stored using `autosys_secure`.

**Limits:**

- For r11.3, up to 30 characters total; *psopr* is restricted to five characters.
- For Release 11.3.5 and Release 11.3.6, up to 111 characters total; *psopr* is restricted to 30 characters.

**Example:** `ptdmo@peoplesoft`

**NOTE**

- The `ps_operator_id` must be defined on AutoSys Workload Automation using the `autosys_secure` command unless the `ps.skipOprPswdValidation=true` parameter is configured in the `agentparm.txt` file.
- You can omit the `ps_operator_id` if the default operator ID, specified by the `ps.default.oprld` parameter in the `agentparm.txt`, applies for the job.

**ps\_output\_dest Attribute -- Specify an Output Path for a PeopleSoft Job**

The `ps_output_dest` attribute specifies the output destination for the PeopleSoft request. The destination can be a file directory or a printer. The value you specify for this attribute overrides the default destination defined on the PeopleSoft system.

**NOTE**

To use this attribute, you must specify `FILE` or `PRINTER` for the `ps_dest_type` attribute.

**Supported Job Type**

This attribute is required for the [PeopleSoft \(PS\) job type](#) if `FILE` or `PRINTER` is specified for the `ps_dest_type` attribute.

**Syntax**

This attribute has the following format:

```
ps_output_dest: file_path | printer_path
```

- **file\_path** Specifies the path to the output directory and the output file name.

**Limits:** Up to 127 characters; case-sensitive

**NOTE**

- Specify this path when the `ps_dest_type` is `FILE`.
- You can use global variables to specify the `file_path`. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

- **printer\_path** Specifies the network location of the printer including the printer server and shared printer name.

**Limits:** Up to 127 characters; case-sensitive

**NOTE**

- Specify this path when the `ps_dest_type` is `PRINTER`.
- The specified printer must be set up on the PeopleSoft system.
- You can use global variables to specify the `printer_path`. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**NOTE**

- This value corresponds to the Output Destination field in PeopleSoft.
- Enclose the values that contain embedded blanks in double quotation marks.

**Example: Direct the Output of a PeopleSoft Job to a File**

This example runs an SQR Report. The report is formatted as PDF and outputted to a file.

```
insert_job: ps_file
job_type: ps
machine: psagent
ps_process_name: XRFWIN
ps_process_type: SQR Report
```

```
ps_dest_type: FILE
ps_dest_format: PDF
ps_output_dest: /export/home/PSoutput/report1.pdf
ps_run_cntrl_id: test
owner: VP1@ps1
```

### Example: Direct the Output of a PeopleSoft Job to a Printer

This example runs an SQR Report. The report is formatted as PS and outputted to a printer.

```
insert_job: ps_printer
job_type: ps
machine: psagent
ps_process_name: XRFWIN
ps_process_type: SQR Report
ps_dest_type: PRINTER
ps_dest_format: PS
ps_output_dest: \\printers\PRINTER1
ps_run_cntrl_id: test
ps_operator_id: VP1@ps1
```

## ps\_process\_name Attribute -- Specify a PeopleSoft Process to Run

The `ps_process_name` attribute specifies the name of the PeopleSoft process that the job runs.

### Supported Job Type

This attribute is required for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_process_name: process_name
```

- ***process\_name***  
Specifies the name of the PeopleSoft report to run. This value corresponds to the Process Name field in PeopleSoft. PeopleSoft stores the list of process names in the PS\_PRCSEDEFN table.  
**Limits:** Up to 12 characters; case-sensitive

#### NOTE

- Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- If the `ps_process_type` attribute is set to PSJob, the value of the `ps_process_name` attribute is the PeopleSoft job name.

### Example: Run a PeopleSoft Process

This example runs the process named AEMINTEST. The process has the Application Engine type and PS\_ALL run control ID.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
ps_process_name: AEMINTEST
ps_process_type: Application Engine
```



```
ps_run_cntrl_id: PS_ALL
```

## ps\_process\_type Attribute -- Specify the Type of PeopleSoft Process to Run

The ps\_process\_type attribute specifies the type of the PeopleSoft process that the job runs.

### Supported Job Type

This attribute is required for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_process_type: process_type
```

- **process\_type**

Specifies the PeopleSoft process type corresponding to the process that the job runs. PeopleSoft stores the list of process types in the PS\_PRCSTYPEDEFN table. This value corresponds to the Process Type field in PeopleSoft.

**Limits:** Up to 30 characters; case-sensitive

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.

Examples:

- Application Engine
- COBOL SQL
- Crw Online
- Crystal
- Crystal Check
- Cube Builder
- Database Agent
- Demand Planning Upload
- Essbase
- HyperCube Builder
- Message Agent API
- nVision
- nVision-Report
- Optimization Engine
- PSJob
- SQR PO-Special Process
- SQR Process
- SQR Report
- SQR Report For WF Delivery
- Winword

### Example: Run an Application Engine PeopleSoft Process

This example runs the process named AEMINTEST. The process has the Application Engine type and PS\_ALL run control ID.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
```

```
ps_process_name: AEMINTEST
ps_process_type: Application Engine
ps_run_cntrl_id: PS_ALL
```

## ps\_restarts Attribute -- Specify Whether to Disable the Restart of Failed PeopleSoft Jobs

The `ps_restarts` attribute specifies whether to disable a restart feature for previously failed jobs from the point where the job failed.

### NOTE

- If you do not specify the `ps_restarts` attribute in your job definition, the restart feature is enabled (the default). When a previously failed job is resubmitted, it restarts from where it stopped. Set the `ps_restarts` attribute value to `NO` to explicitly disable the restart feature for the job.
- AutoSys Workload Automation does not support the use of `RESTARTJOB` events to resubmit PeopleSoft (PS) jobs. Peoplesoft application developers, system administrators, or application users who want to resubmit the batch processes that are associated with Peoplesoft jobs that enable the restart feature must use the PeopleSoft Process Scheduler directly.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\)](#) job type.

### Syntax

This attribute has the following format:

```
ps_restarts: YES | NO
```

- **NO**  
Disables the restart feature. Previously-failed jobs restart from the beginning.
- **YES**  
Does not disable the restart feature. When a failed job is resubmitted, it restarts from where it stopped. This is the default.

### NOTE

The restart feature only applies to certain PeopleSoft Process Types such as Application Engine.

### Example: Disable the Restart of Failed PeopleSoft Jobs

This example disables the restart feature. Previously failed jobs restart from the beginning.

```
insert_job: ps_norestart
job_type: PS
machine: psagt
ps_process_name: XRFWIN
ps_process_type: Application Engine
ps_restarts: Yes
```

## ps\_run\_cntrl\_args Attribute -- Specify the Arguments for a PeopleSoft Run Control Table

The `ps_run_cntrl_args` attribute specifies the run control arguments for a run control table on the PeopleSoft system. You can use this attribute with the `ps_run_control_table` attribute to add parameters to the run control table.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\)](#) job type.

### Syntax

This attribute has the following format:

```
ps_run_cntrl_args: argument[, argument...]
```

- **argument**

**Limits:** Case-sensitive; each argument can be up to 1024 characters.

**NOTE**

Enclose the values that contain delimiters (such as spaces) or special characters in quotation marks.

**Examples:** BI\_WF\_0001, 0, "third arg"

**NOTE**

- You can specify multiple arguments. Separate each argument with a comma.
- You can specify multiple entries of ps\_run\_cntrl\_args to define different sets of arguments. For example, you can specify two sets of arguments using two ps\_run\_cntrl\_args attributes as follows:

```
ps_run_cntrl_args : BI_WF_0001, 0, 0, N, 0, N, , "b\", sysdate
```

```
ps_run_cntrl_args : BI_WF_0002, 0, 0, N, 0, N, , "b\", sysdate
```

- Each entry of ps\_run\_cntrl\_args can be up to 4096 characters.
- If the number of values specified in the ps\_run\_cntrl\_args attribute is less than the number of columns in the corresponding run control table, the agent populates the remaining columns with default values. The default value for strings is a space. The default value for all other data types is null.
- If an argument cannot be updated with a default value, the job fails with an exception. For example, the job can fail if the database does not allow a null value within the table.
- For Oracle, if you specify date values other than SYSDATE in the job definition, you must use the following format:

```
"DD-MM-YYYY HH24:MI:SS"
```

Oracle date values are processed using the to\_date() function. For example, for the argument value 26-07-2009, the agent invokes to\_date('26-07-2009','DD-MM-YYYY HH24:MI:SS') and "2009-07-26 00:00:00.0" is inserted into the run control table.

**Example: Add a Row of Process Parameters to a Run Control Table**

In this example, the ps\_run\_cntrl\_table attribute specifies the name of the run control table that contains the run control parameters for the XYZ process. The job inserts a new row in the PS\_AEREQUESTTBL run control table with data from the run control arguments (ps\_run\_cntrl\_args).

```
insert_job: ps_params
job_type: ps
machine: psagent
ps_process_name: XYZ
ps_process_type: Application Engine
owner: VP1@server01
ps_run_cntrl_id: TEST
ps_run_control_table: PS_AEREQUESTTBL
ps_run_cntrl_args: BI_WF_0001, 0, 0, N, 0, N, , "b\", sysdate
```

The columns in the PS\_AEREQUESTTBL run control table correspond to the run control arguments as follows:

Column	Data Type	Value
AE_APPLID	VARCHAR2(12)	BI_WF_0001
CURTEMPINSTANCE	NUMBER(38)	0
PROCESS_FREQUENCY	VARCHAR2(1)	0

AE_PROCESS_STATUS	VARCHAR2(1)	N
PROCESS_INSTANCE	NUMBER(10)	0
PROCESS_ORIG	VARCHAR2(1)	N
LAST_RUN_DTM	DATE	null
MARKET	VARCHAR2(3)	b\
ASOF_DT	DATE	2010-07-10 14:30:00

## ps\_run\_cntrl\_id Attribute -- Specify PeopleSoft Run Parameters

The ps\_run\_cntrl\_id attribute specifies a set of PeopleSoft run parameters for a PeopleSoft process.

### Supported Job Type

This attribute is optional for the [PeopleSoft](#) (PS) job type.

### Syntax

This attribute has the following format:

```
ps_run_cntrl_id: control_id
```

- **control\_id**

Specifies the value assigned to the run control identifier. This value corresponds to the Run Control ID field in PeopleSoft.

**Limits:** Up to 30 characters; case-sensitive

**Example:** FLOOR8\_COLOR

#### NOTE

- In PeopleSoft, some processes spawn children using the CreateProcessRequest PeopleCode function. PeopleSoft may mark the parent process as a successful completion even if one or more of its children fail. If you want the agent to check the status of children when determining the completion status of the parent, add ".\*" to the end of the Run Control ID, for example, PS\_ALL.\*.
- You can use global variables to specify the control\_id. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

#### NOTE

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- If you do not specify this attribute in the job definition, a default run control ID must be defined in the ps.default.runCntlId parameter in the agent's agentparm.txt file. Otherwise, the job fails.
- The ps\_run\_cntrl\_id attribute overrides the default run control ID specified in the agent's agentparm.txt file.

### Example: Specify a Run Control ID

This example runs the process named AEMINTEST. The process has the Application Engine type and PS\_ALL run control ID.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
ps_process_name: AEMINTEST
ps_process_type: Application Engine
ps_run_cntrl_id: PS_ALL
```

### Example: Check Children Status

This example runs the AEMINITEST process on the agent named psagt. The job completes successfully if the AEMINITEST process and all of its children complete successfully.

```
insert_job: ps_children
job_type: ps
machine: psagt
ps_process_name: AEMINITEST
ps_process_type: Application Engine
owner: VP1@server01
ps_run_cntrl_id: HDL8010.*
```

## ps\_run\_control\_table Attribute -- Specify the Table that Contains the PeopleSoft Run Parameters

The `ps_run_control_table` attribute specifies the name of the table that contains the run control parameters for a PeopleSoft process. You can use this statement with the `ps_run_cntrl_args` attribute to add parameters to the run control table.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_run_control_table: control_table
```

- ***control\_table***

Specifies the name of the table that contains the run parameters for a given PeopleSoft process.

**Limits:** Up to 50 characters; case-sensitive

**Example:** PS\_AEREQUESTTBL

#### NOTE

Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

### Example: Add a Row of Process Parameters to a Run Control Table

In this example, the `ps_run_cntrl_table` attribute specifies the name of the run control table that contains the run control parameters for the XYZ process. The job inserts a new row in the PS\_AEREQUESTTBL run control table with data from the run control arguments (`ps_run_cntrl_args`).

```
insert_job: ps_params
job_type: ps
machine: psagent
ps_process_name: XYZ
ps_process_type: Application Engine
owner: VP1@server01
ps_run_cntrl_id: TEST
ps_run_control_table: PS_AEREQUESTTBL
ps_run_cntrl_args: BI_WF_0001, 0, 0, N, 0, N, , "b\", sysdate
```

The columns in the PS\_AEREQUESTTBL run control table correspond to the run control arguments as follows:

Column	Data Type	Value
AE_APPLID	VARCHAR2(12)	BI_WF_0001
CURTEMPINSTANCE	NUMBER(38)	0

PROCESS_FREQUENCY	VARCHAR2(1)	O
AE_PROCESS_STATUS	VARCHAR2(1)	N
PROCESS_INSTANCE	NUMBER(10)	0
PROCESS_ORIG	VARCHAR2(1)	N
LAST_RUN_DTM	DATE	null
MARKET	VARCHAR2(3)	b\
ASOF_DT	DATE	2010-07-10 14:30:00

## ps\_server\_name Attribute -- Specify a Server to Run the PeopleSoft Job

The `ps_server_name` attribute specifies the name of the target server that runs the PeopleSoft job.

### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

### Syntax

This attribute has the following format:

```
ps_server_name: server_name
```

- **server\_name**

Specifies the name of the target server that runs the PeopleSoft job. PeopleSoft stores the list of server names in the `PS_SERVERDEFN` table. This value corresponds to the Server Name field in PeopleSoft.

**Limits:** Up to 8 characters; cannot contain delimiters (such as spaces)

**Example:** PSNT

### NOTE

- Your agent administrator can specify a default run server name for all PeopleSoft jobs by setting the `ps.default.serverName` parameter in the agent's `agentparm.txt` file.
- The `ps_server_name` attribute overrides the default run server name specified in the agent's `agentparm.txt`.
- If the server name is not defined on the agent or using the `ps_server_name` attribute, the PeopleSoft Scheduler determines the server that will run the job.

### Example: Specify a Server to Run a PeopleSoft Process

This example runs a process named `DDDAUDIT`. The server named `PSPR` runs the job. The job runs in Central time in the Continental United States.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
ps_process_name: DDDAUDIT
ps_process_type: SQR Report
ps_server_name: PSPR
ps_time_zone: CST
```

## ps\_skip\_parm\_updates Attribute -- Specify Whether to Update Job Parameters

The `ps_skip_parm_updates` attribute specifies whether the agent updates job parameters with data in the `PS_PRCSEFN` table.

**NOTE**

- If you do not specify the `ps_skip_parm_updates` attribute in your job definition, the agent updates job parameters with data in the `PS_PRCSEFN` table (the default).
- We recommend that you skip parameter updates when some bind variables in the `PS_PRCSEFN` table may not be suitably defined.

**Supported Job Type**

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

**Syntax**

This attribute has the following format:

```
ps_skip_parm_updates: YES | NO
```

- **YES**

Specifies that the agent does not update job parameters with data in the `PS_PRCSEFN` table. If you specify YES, you can use the `ps_args` attribute to pass additional argument values.

- **NO**

Specifies that the agent updates job parameters with data in the `PS_PRCSEFN` table. This is the default.

**Example: Run a PeopleSoft Process Without Updating the PS\_PRCSEFN Table**

This example runs the OMC3220- process on the agent named psagt. The job parameters are not updated with data in the `PS_PRCSEFN` table. The `ps_run_cntrl_args` attribute passes additional parameters for the report.

```
insert_job: ps_skipparms
job_type: ps
machine: psagt
ps_process_name: OMC3220-
ps_process_type: Crystal
ps_dest_type: WEB
ps_dest_format: PDF
ps_run_cntrl_id: test
owner: VP1@server01
ps_skip_parm_updates: YES
ps_args: -TEST, SHARE
```

**Example: Ignore Parameters in the PS\_PRCSEFN Table and Pass Additional Parameters to the nVision Report**

In this example, the job parameters are not updated with data in the `PS_PRCSEFN` table. The `ps_run_cntrl_args` attribute passes additional parameters for the nVision report.

```
insert_job: ps_skipparms2
job_type: ps
machine: psagt
ps_process_name: NVSRUN
ps_process_type: nVision-Report
ps_dest_type: FILE
ps_dest_format: XLS
ps_output_dest: "c:\test\testnv.xls"
ps_skip_parm_updates: YES
ps_args: "-NRNARIABLE -NBUAUS01 -NHLhttp://10.1.1.40/psp/ps /EMPLOYEE/ERP/c/
REPORT_BOOKS.IC_RUN_DRILLDOWN.GBL?Action=A"
```

### Example: Update Job Parameters with Data in the PS\_PRCSEDEFN Table

This example updates job parameters with data in the PS\_PRCSEDEFN table.

```
insert_job: ps_updateparms
job_type: ps
machine: psagt
ps_process_name: OMC3220-
ps_process_type: Crystal
ps_run_cntrl_id: sample
ps_skip_parm_updates: NO
owner: VP2@server01
```

### ps\_time\_zone Attribute -- Specify a Time Zone

The ps\_time\_zone attribute specifies a different time zone for the PeopleSoft report being run.

#### Supported Job Type

This attribute is optional for the [PeopleSoft \(PS\) job type](#).

#### Syntax

This attribute has the following format:

```
ps_time_zone: time_zone
```

- **time\_zone**

Specifies a different time zone for the report being run. This value corresponds to the Time Zone field in PeopleSoft.

**Limits:** Up to 9 characters; cannot contain delimiters (such as spaces)

#### NOTE

You can view the time zone settings in PeopleSoft.

### Example: Specify a Time Zone

This example runs a process named DDDAUDIT. The server named PSPR runs the job. The job runs in Central time in the Continental United States.

```
insert_job: ps_txtfile
job_type: ps
machine: psagt
ps_process_name: DDDAUDIT
ps_process_type: SQR Report
ps_server_name: PSPR
ps_time_zone: CST
```

### remote\_command Attribute -- Specify a Remote Command or Script

The remote\_command attribute specifies a command or script to run on a remote computer.

#### Supported Job Type

This attribute is required for the [Remote Execution \(PROXY\) job type](#).

#### Syntax

This attribute has the following format:



```
remote_command: command
```

- **command**

Specifies a command or script to run on a remote computer.

**Limits:** Up to 1024 characters; case-sensitive

**NOTE**

- You cannot use redirection operators in your commands, but you can use pipes. Piping is supported between commands.
- The remote\_command attribute commands are captured in the job log and in the agent logs. Only the result of the last command to execute is captured in the spool file.
- On UNIX, the working directory is changed to the spool directory before executing remote\_command attribute commands.
- You can specify multiple remote\_command attributes on HP Integrity NonStop or OpenVMS, but UNIX does not support multiple remote\_command attributes.
- If the command operand contains delimiters, the command operand must be enclosed in double quotation marks to denote the start and end of the operand.
- When using the fail\_codes and success\_codes attributes with multiple remote\_command attributes, the agent tests only the return code of the last executed remote\_command attribute.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Run Multiple Commands on a Remote System**

Suppose that you want to specify command attributes command1 and command2 for a Remote Execution job. The job runs on the remote machine Garfunkel.

```
insert_job: proxy_job
job_type: PROXY
machine: proxyagent
remote_target: garfunkel
remote_command: command1
remote_command: command2
```

## remote\_name Attribute -- Specify a Remote Class Name

The remote\_name attribute specifies the naming class of the Java object you want to invoke a method on in an RMI job.

### Supported Job Type

This attribute is required for the [RMI \(JAVARMI\) job type](#).

### Syntax

This attribute has the following format:

```
remote_name: "remote_class"
```

- **"remote\_class"**

Specifies the reference location of the object you want to invoke a method on.

The format is the following:

```
"rmi://[host:port/]name"
```

- **host**

(Optional) Specifies the name of the local or remote computer where the RMI registry that hosts the remote object runs.

**Default:** local host

– **port**

(Optional) Specifies the RMI registry port number the host uses to listen for requests.

**Default:** 1099, the default port of the host's RMI registry

– **name**

Specifies the name of a remote object.

**Limits:** Up to 256 characters; case-sensitive

**Example:** "rmi://smith:5000/test"

### Example: Invoke a Method to Start a Remote Server

Suppose that you want to invoke a method that starts a remote server using remote object activation. You want the server to start immediately.

```
insert_job: start
job_type: JAVARMI
machine: appagent
remote_name: "rmi://remotehost/Test"
method_name: startserver
j2ee_parameter: String="now"
```

## remote\_target Attribute -- Specify the Name of the Custom Properties File for the Remote System

The `remote_target` attribute specifies the name of the custom properties file that is created on the agent for the remote system.

### NOTE

The custom properties file includes the IP address or host name of the remote computer where the job is scheduled to run.

### Supported Job Type

This attribute is required for the [Remote Execution \(PROXY\) job type](#).

### Syntax

This attribute has the following format:

```
remote_target: name
```

- **name**

Specifies the name of the custom properties file that is created on the agent for the remote system. The name should not include the `.properties` extension. For example, if the name of the custom properties file is `hpserver.properties`, you would specify `hpserver` for this attribute.

**Limits:** Up to 128 alphanumeric characters; case-sensitive; must start with an alpha character; cannot contain pound or hash (#), underscore (\_), hyphen (-), or period(.).

### Example: Specify a Target System

This example identifies the `hpserver` as the remote target where the `ls_cmd` job should run.

```
insert_job: ls_cmd
job_type: PROXY
machine: proxyagent
remote_target: hpserver
```

```
remote_command: ls
spool_file: >>/opt/rootdev/CA_USER/test/ls.out
owner: user1@proxytest
```

## reply\_to Attribute -- Specify the JNDI Name of the Reply to Queue

The `reply_to` attribute specifies the JNDI name of the reply to a queue.

### NOTE

The `reply_to` attribute only applies to a queue.

### Supported Job Types

This attribute is optional for the [JMS Publish \(JMSPUB\)](#) job type.

### Syntax

This attribute has the following format:

```
reply_to: replyname
```

- ***replyname*** Specifies the JNDI name of the reply to a queue. This is applicable only to the message ID pattern. **Limits:** 256 characters; case-sensitive

### NOTE

In a JMS Publish job, you can also specify the `reply_to` attribute as value for the `destination_name` attribute, as follows:

```
destination_name: reply_to=subscriber
```

The `destination_name` is populated with the queue name that is specified in the `reply_to` attribute of the publisher job, which is consumed and stored by the subscriber job that is named subscriber.

### Example: Specify the JNDI Name of the Reply to Queue

This example defines two JMS Publish jobs (named publisher1 and publisher2) and two JMS Subscribe jobs (named subscriber1 and subscriber2).

The publisher1 job publishes the message "Hello World" to the queue that is named Queue1. The Java class of the message is String. The initial context factory that is supplied by the JNDI service provider is `com.ibm.websphere.naming.WsnInitialContextFactory`. The service provider's URL is `iiop://172.24.0.0:2809`, where 172.24.0.0:2809 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the `cyberuser@host` user name to gain access to the connection factory that is named `ConnectionFactory`.

The queue to which the reply must be sent is Queue2.

```
insert_job: publisher1
job_type: JMSPUB
machine: jms
connection_factory: ConnectionFactory
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
use_topic: false
provider_url: "iiop://172.24.0.0:2809"
message_class: String
destination_name: Queue1
```

```
j2ee_parameter: String="Hello World"
j2ee_user: cyberuser@host
reply_to: Queue2
```

The subscriber1 job uses the publisher1 job name as its correlation id to consume the publisher1 message from the queue that is named Queue1.

```
insert_job: subscriber1
job_type: JMSSUB
machine: jms
connection_factory: ConnectionFactory
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
use_topic: false
provider_url: "iiop://172.24.0.0:2809"
destination_name: Queue1
j2ee_user: cyberuser@host
correlation_id: publisher1
```

The publisher2 job acts on the message that is received by subscriber1 and sends its response to the reply\_to queue that is specified by publisher1, which is Queue2.

```
insert_job: publisher2
job_type: JMSPUB
machine: jms
connection_factory: ConnectionFactory
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
use_topic: false
provider_url: "iiop://172.24.0.0:2809"
message_class: String
destination_name: reply_to=subscriber1
j2ee_parameter: String="Hello World"
j2ee_user: cyberuser@host
correlation_id: subscriber1
```

The subscriber2 job uses the publisher2 job name as its correlation id to consume the publisher2 message from the queue that is named Queue2.

```
insert_job: subscriber2
job_type: JMSSUB
machine: jms
connection_factory: ConnectionFactory
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
use_topic: false
provider_url: "iiop://172.24.0.0:2809"
destination_name: Queue2
j2ee_user: cyberuser@host
correlation_id: publisher2
```

## request\_id Attribute -- Specify Request ID of the Oracle E-Business Suite Request to Copy

The `request_id` attribute specifies the request ID of the Oracle E-Business Suite request you want to copy. You can copy an existing single request defined on Oracle E-Business Suite and run it under the agent.

### Supported Job Type

This attribute is required for the [Oracle E-Business Suite Copy Single Request \(OACOPY\)](#) job type.

### Syntax

This attribute has the following format:

```
request_id: request_id
```

- **request\_id**

Specifies the request ID of the single request you want to copy. In Oracle Applications, the request ID is found in the Request ID column of the Requests dialog.

**Limits:** Up to 20 numeric digits

**Example:** 493895459709656750

### Example: Copy a Single Request

Suppose that you want to copy an existing single request defined on Oracle E-Business Suite. In this example, the job copies the single request job with request ID 2255470 and overrides the Oracle E-Business Suite user name and responsibility name defined in the `agentparm.txt` file.

```
insert_job: oacopy_single
job_type: oacopy
machine: oaagent
request_id: 2255470
oracle_user: SYSADMIN
oracle_resp: System Administrator
```

## resources Attribute -- Define or Update Virtual Resource Dependencies in a Job

The `resources` attribute defines or updates virtual resource dependencies in a job.

### NOTE

- To define a virtual resource dependency, the virtual resource must already be defined on the database. Otherwise, you get an error.
- You can specify only global resources on a box job. However, the resource that is specified for a box applies only to the box and not to the jobs within the box.
- When you update a job, you cannot update the `resources` attribute in the existing job definition if the job has a resource dependency and has held the resource.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_job](#)
- [update\\_job](#)

### Syntax

This attribute has the following formats:

- To specify one or more virtual resource dependencies:

```
resources: (virtual_resource_name, quantity=amount|ALL[, free=Y|N|A]) [AND ...]
```

- **virtual\_resource\_name**

Specifies the name of a virtual resource.

**NOTE**

The virtual resource must already be defined in the database.

- **quantity=amount | ALL**

Specifies the amount of virtual resource required. The options are as follows:

- *amount* -- Specifies the number of units that the job requires. The number must be a positive integer.
- *ALL* -- Specifies that the job requires all the units of the resource.

- **free=Y | N | A**

(Optional for renewable virtual resources only) Specifies whether the units of the virtual resource are freed from the job. The options are as follows:

- *Y* -- Frees the units only if the job completes successfully.
- *N* -- The units are not freed. To free the resources, issue the following command:

```
sendevent -E RELEASE_RESOURCE -J job_name
```

- *A* -- Frees the units unconditionally.

**Default:** Y

### Example: Define Virtual Resource Dependencies

This example defines a Command job that has virtual resource dependencies. Before the job can start running, it needs a machine that satisfies all the following dependencies:

- 1 unit of the depletable resource named D1
- 3 units of the threshold resource named T1
- 4 units of the renewable resource named R1
- SYSTEM\_OS\_TYPE is AIX 5.3
- SYSTEM\_PHYSICAL\_MEMORY is greater than 2 GB

```
insert_job: res_dep_job
job_type: CMD
machine: unixagent
command: /u1/procrun.sh
resources: (D1, QUANTITY=1) AND (T1, QUANTITY=3) AND
(R1, QUANTITY=4, FREE=Y) AND
(SYSTEM_OS_TYPE, VALUEOP=EQ, VALUE=AIX, VERSION=5.3) AND
(SYSTEM_PHYSICAL_MEMORY, VALUEOP=GT, VALUE=2048)
```

### Example: Define a Global Resource on a Box Job

This example defines a global resource that is named *glb\_resource1* on the box that is named *box1\_resource\_Y*. The global resource applies only to the box and not to the jobs within the box.

```
insert_job: box1_resource_Y
job_type: box
resources: (glb_resource1, QUANTITY=1, FREE=Y)
```

Issue the following command to generate a detailed report for the *glb\_resource1* virtual resource. The resource is global, so it is available to all machines.

```
autorep -V glb_resource1 -d
```

The resulting report might resemble the following:

Resource Name	Type	Machine
Defined Available		

glb_resource1	R	---
10 9		

/\*\*\*\* Current Resource Usage \*\*\*\*/

ResName	JobName
Run/Ntry	Amount in Use
Status	Machine

glb_resource1	box1_resource_Y
1608/1	1
RUNNING	**NOMACHINE**

## result\_type Attribute -- Specify the Data Type to be Returned from the Stored Procedure

The `result_type` attribute specifies the data type to be returned from the stored procedure. If the type returned from the stored procedure and the value specified for the `result_type` attribute do not match, the job fails.

### Supported Job Type

This attribute is optional for the [Database Stored Procedure \(DBPROC\) job type](#).

### Syntax

This attribute has the following format:

```
result_type: data_type
```

- **data\_type**

Specifies the data type to be returned from the stored procedure.

The following table lists the supported data types for different databases. The data types listed in the database columns are defined in the procedure definition within the database. In the job definition, use the corresponding JDBC data type from the first column.

JDBC Data Type	Valid Input Format	Oracle	Microsoft SQL Server	DB2
CHAR	Plain text	CHAR	CHAR	CHAR
VARCHAR	Plain text	VARCHAR2(x) VARCHAR(x)	VARCHAR(x)	VARCHAR(x)
LONGVARCHAR		Not supported	Not supported	Not supported
NUMERIC	Number	NUMBER, NUMERIC	NUMERIC	NUMERIC
DECIMAL	Number	DECIMAL	DECIMAL	DECIMAL
BIT		Not supported	BIT	Not supported
BOOLEAN		Not supported	Not supported	Not supported
TINYINT		Not supported	TINYINT	Not supported

SMALLINT	Number	SMALLINT	SMALLINT	SMALLINT
INTEGER	Number	INTEGER	INTEGER	INTEGER
BIGINT	Number	Not supported	BIGINT	BIGINT
REAL	Number [dot Number]	REAL	REAL	REAL
FLOAT	Number [dot Number]	FLOAT	FLOAT	FLOAT
DOUBLE				DOUBLE
DATE	yyyy-mm-dd	DATE		DATE
DATE	String (for example, SYSDATE)			
TIME	hh:mm:ss			TIME
TIMESTAMP	yyyy-mm-dd hh:mm:ss.ffffff	TIMESTAMP		TIMESTAMP

## results\_file Attribute -- Specify the Path to the Results File

The results\_file attribute specifies the path to the results file. The results file stores the output (in text format) of the invoked web service. It contains HTTPResponse only.

### NOTE

If you omit the results\_file attribute in a HTTP job, the output is stored in the job's spool file.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

### Syntax

This attribute has the following format:

```
results_file: path
```

- **path** Specifies the path to the results file. The results file stores the output (in text format) of the invoked web service. It contains HTTPResponse only. By default, this file is always appended.

**Limits:** Up to 256 characters; case-sensitive

**Examples:** /tmp/myoutput, "D:\Stock output\STOCK3"

### NOTE

- Specify the values that contain embedded blanks with double quotation marks.
- The results\_file attribute is not applicable if you have set the invocation\_type attribute value to delete.
- You can use global variables to specify the results file. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

## results\_file\_overwrite Attribute -- Specify Whether to Overwrite the Results File

The results\_file\_overwrite attribute specifies whether you want to overwrite the results file when a file with the same name exists.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.



## Syntax

This attribute has the following format:

```
results_file_overwrite: true | false
```

- **true**  
Overwrites the results file. This is the default.
- **false**  
Does not overwrite the results file.

### NOTE

Your agent administrator can set a default value by setting the `http.results.file.overwrite` parameter in the agent's `agentparm.txt` file. The `results_file_overwrite` attribute value overrides the default value that is specified in the agent's `agentparm.txt` file.

## Example: Perform a HTTP Query Using the HTTP POST Method

This example performs a HTTP query using the HTTP POST method. In this example, the job does not overwrite the results file, it overrides the global proxy defaults specified in the `agentparm.txt` file, and specifies the BASIC, DIGEST, and NTLM protocols for web server authentication.

```
insert_job: httppost
job_type: HTTP
machine: localhost
invocation_type: POST
provider_url: "http://google.com/search"
j2ee_no_global_proxy_defaults: 1
j2ee_authentication_order: BASIC, DIGEST, NTLM
j2ee_proxy_port: 80
j2ee_parameter: q="pangea"
media_type: text/plain
headers: Pragma="no-cache"
headers: X-Forwarded-For="129.78.138.66, \"129.78.64.103\""
header_filter: gr[ae]y
retry_on_failure_count: 2
retry_on_failure_delay: 10
retry_on_failure_details: yes
results_file: /tmp/results_file.txt
results_file_overwrite: false
fail_codes: 201
success_codes: 400-401
payload_uri: "file://host/path"
```

## retry\_on\_failure\_count Attribute -- Define the Number of Times an HTTP Request is Retried

The `retry_on_failure_count` attribute specifies the number of times an HTTP request is retried when the HTTP exit code is a failure code or the specified filter criteria does not match the response.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

## Syntax

This attribute has the following format:

```
retry_on_failure_count: attempts
```

- **attempts**  
Specifies the number of times an HTTP request is retried when the HTTP exit code is a failure code or the specified filter criteria does not match the response.  
**Default:** 0 (no retries)

## retry\_on\_failure\_delay Attribute -- Specify the Number of Seconds to Wait Between Retrying HTTP Requests

The `retry_on_failure_delay` attribute specifies the number of seconds to wait between retrying HTTP requests if the `retry_on_failure_count` attribute is set to a value that is greater than zero.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

## Syntax

This attribute has the following format:

```
retry_on_failure_delay: delay
```

- **delay**  
Specifies the number of seconds to wait between retrying HTTP requests if the `retry_on_failure_count` attribute is set to a value that is greater than zero..  
**Default:** 10

### NOTE

Your agent administrator can set a default value by setting the `RetryOnFailureDelay` parameter in the agent's `agentparm.txt` file. The `retry_on_failure_delay` attribute value overrides the default value that is specified in the agent's `agentparm.txt` file.

## retry\_on\_failure\_details Attribute -- Specify Whether to Send the Failure Details

The `retry_on_failure_details` specifies whether to send the failure details (indicating the failure and retry count) to the manager when the `retry_on_failure_count` attribute is set to a value that is greater than zero.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

## Syntax

This attribute has the following format:

```
retry_on_failure_details: TRUE | FALSE
```

- **TRUE**  
Sends the failure details to the manager when the `retry_on_failure_count` attribute is set to a value that is greater than zero.
- **FALSE**  
Does not send the failure details to the manager when the `retry_on_failure_count` attribute is set to a value that is greater than zero.

**Default:** FALSE

## **return\_class\_name Attribute -- Specify a Return Class Name**

The `return_class_name` attribute specifies the Java class name of the return value in a Web Service job.

### **Supported Job Type**

This attribute is optional for the [Web Service \(WBSVC\) job type](#).

### **Syntax**

This attribute has the following format:

```
return_class_name: return_class
```

- ***return\_class***

Specifies the Java class name of the return value.

**Limits:** Up to 1024 characters; case-sensitive

### **Example: Specify Java Class Name of Stock Quote**

#### **NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. The `GetQuote` operation returns a `java.lang.String` object, which maps to the XML type `string` in the return namespace `http://ws.ca.com`. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
return_class_name: java.lang.String
return_xml_name: string
return_namespace: "http://ws.ca.com"
```

## return\_namespace Attribute -- Specify an XML Namespace

The `return_namespace` attribute specifies the XML namespace for the `return_xml_name` attribute in a Web Service job. If you omit this attribute, the return namespace defaults to the target namespace specified by the `target_namespace` attribute.

### Supported Job Type

This attribute is optional for the [Web Service \(WBSVC\)](#) job type.

### Syntax

This attribute has the following format:

```
return_namespace: return_namespace
```

- **return\_namespace**

Specifies the XML namespace for the XML type that maps to the Java class name of the return value.

**Limits:** Up to 256 characters; case-sensitive

### Example: Specify Return Namespace for the XML Type of Stock Quote

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. The `GetQuote` operation returns a `java.lang.String` object, which maps to the XML type `string` in the return namespace `http://ws.ca.com`. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
return_class_name: java.lang.String
return_xml_name: string
return_namespace: "http://ws.ca.com"
```

## return\_xml\_name Attribute -- Specify an XML Type

The return\_xml\_name attribute specifies the XML type that maps to the return\_class\_name attribute in a Web Service job.

### Supported Job Type

This attribute is optional for the [Web Service \(WBSVC\) job type](#).

#### NOTE

This attribute is required if the return\_class\_name attribute is specified in the job definition.

### Syntax

This attribute has the following format:

```
return_xml_name: XML_type
```

- **XML\_type**

Specifies the XML type that maps to the Java class name of the return value.

**Limits:** Up to 256 characters; case-sensitive

### Example: Specify XML Type of Stock Quote

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. The `GetQuote` operation returns a `java.lang.String` object, which maps to the XML type `string` in the return namespace `http://ws.ca.com`. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
return_class_name: java.lang.String
return_xml_name: string
return_namespace: "http://ws.ca.com"
```

## run\_calendar Attribute -- Identify a Custom Calendar

The `run_calendar` attribute identifies a custom calendar to use to determine the days of the week on which to run the job you are defining. You can use a standard calendar or an extended calendar as the run calendar.

Custom calendars are useful for complex date specification, such as when you need to run a job on the last business day of the month. Custom calendars can specify any number of dates on which to run associated jobs. Each calendar is stored as a separate uniquely named object in the database and can be associated with one or more jobs. The specified calendar must have previously been created using the `autocal_asc` command.

AutoSys Workload Automation schedules the job to run on every day defined in the specified calendar, at the times defined in the calendar or in the `start_times` or `start_mins` attributes.

### NOTE

Times set in the `start_times` and `start_mins` attributes override times set in a custom calendar. The `date_conditions` attribute controls the usage of this attribute.

You cannot specify both the `days_of_week` and `run_calendar` attributes in the same job definition. If you do, AutoSys Workload Automation ignores both attributes and generates an error.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
run_calendar: calendar_name
```

- **calendar\_name**

Defines the name of a custom calendar that was created with the `autocal_asc` command.

**Default:** No calendar is used.

**Limits:** Up to 64 alphanumeric characters

### Example: Specify a Job to Run on Days Specified in the Previously Created Custom Calendar

This example specifies that the job should run on the days specified in the previously created custom calendar `last_business`:

```
date_conditions: y
run_calendar: last_business
```

## run\_external Attribute -- Specify Whether to Define a User ID

The `run_external` attribute specifies whether to define a user ID that runs the job on behalf of the agent user.

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) and FTP job types.

### Syntax

This attribute has the following format:

```
run_external: T|F
```

- **T**  
Specifies that you can define a user ID that runs the job on behalf of the agent user. You can define the user ID using the `external_os_user` attribute.
- **F**

Specifies that you cannot define a user ID that runs the job on behalf of the agent user.

**Default:** F

## run\_window Attribute -- Define an Interval for a Job to Start

The run\_window attribute defines an interval during which the job you are defining may start. This attribute controls only when the job starts -- not when it stops.

When a job definition contains the run\_window attribute and the job becomes eligible to run (based on its starting conditions), AutoSys Workload Automation verifies whether the specified run window includes the current time. If it does, the job starts. If it does not, the product determines whether to run the job based on the end of the previous run window and the beginning of the next run window.

- When the current time is closer to the beginning of the next run window, the product schedules the job to start when the next run window starts.
- When the current time is closer to the end of the previous run window, the product does not start the job and changes its status to INACTIVE.

The run\_window attribute is not in itself a starting condition -- it is an additional control over when a job may start after its starting conditions are satisfied. A run window cannot span more than 24 hours.

This attribute is especially useful, for example, when you do not know when a monitored file may arrive and there are specific times when a job dependent on the monitored file should not run.

### NOTE

Jobs must have starting conditions in addition to the run\_window attribute for the job to start automatically.

The date\_conditions attribute controls the usage of this attribute.

If the job is in a box, its status changes to ACTIVATED when the box starts running. However, if the current time is not in the specified run window for the job, its status changes to INACTIVE.

- When the current time is closer to the end of the previous run\_window, the job's status changes to INACTIVE. The box job can still run to completion.
- When the current time is closer to the beginning of the next run\_window, the product issues a future STARTJOB event for the job for the next run\_window.

These actions enable a box job to run to completion when the run\_window for a job in the box has just closed.

For example, assume JobA, JobB, and JobC are in Box1, and JobA has a run\_window of 02:00 to 04:00 and another starting condition such as start\_mins or start\_times. If Box1 starts at 04:05, JobB and JobC can run and JobA becomes INACTIVE so that the box can complete that day. If Box1 instead starts at 16:05, JobA will have a STARTJOB event set for 02:00 the next day, and the box continues running until the job starts the next day.

You can use the sendevent command to define times of the day when you do not want the job to start by putting the job into JOB\_ON\_HOLD status and then changing its status to JOB\_OFF\_HOLD later.

## Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
run_window: "time-time"
```

- *time - time*

Defines the interval during which a job may start in the format "hh:mm-hh:mm", where hh denotes hours (in 24-hour format) and mm denotes minutes. You must enclose the interval in quotation marks (") and separate the beginning and ending times with a hyphen (-). The specified interval can overlap midnight, but cannot encompass more than 24 hours.

**Limits:** Up to 20 alphanumeric characters

**Default:** No run window is used.

#### Example: Specify an Interval for the Job to Start

This example specifies that the job may start only between 11:00 p.m. and 2:00 a.m., regardless of other conditions.

```
date_conditions: y
run_window: "23:00-02:00"
start_mins: "10,20"
```

## sap\_abap\_name Attribute -- Run ABAP Steps Within an SAP Job

The sap\_abap\_name attribute identifies the ABAP step to be run.

#### Supported Job Type

This attribute is optional for the [SAP Process Monitor \(SAPPM\) job type](#) when sap\_process\_status is set to RUNNING or STOPPED.

#### NOTE

You cannot use this attribute with sap\_process\_status set to WAITING.

#### Syntax

This attribute has the following format:

```
sap_abap_name: abap
```

- **abap**

Specifies the valid SAP system ABAP name. The ABAP name corresponds to the SAPGUI ABAP program Name field on the Create Step dialog.

**Limits:** Up to 40 valid SAP characters

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

#### Example: Run an ABAP Step

This example defines a job that runs the RSPARAM ABAP.

```
insert_job: SAPPM_JOB
job_type: SAPPM
machine: SAPAGENT
sap_process_status: RUNNING
sap_abap_name: RSPARAM
```

## sap\_chain\_id Attribute -- Identify Name of Business Warehouse Process Chain on SAP

The sap\_chain\_id attribute identifies the name of a Business Warehouse Process Chain on the SAP system.

#### Supported Job Type



This attribute is required for the [SAP BW Process Chain \(SAPBWPC\)](#) job type.

### Syntax

This attribute has the following format:

```
sap_chain_id: process_chain
process_chain
```

Specifies the name of the Business Warehouse Process Chain.

**Limits:** Up to 30 valid SAP characters; case-sensitive

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. Do not use embedded double quotation marks.

**Example:** ZPAK\_6C7ZW2JAXS90AK71WD1CYIN

### Example: Run a Process Chain

This example runs the PC\_7A3929SH Process Chain on the SAP system.

```
insert_job: BWPCTEST
job_type: SAPBWPC
machine: SAPHTAGENT
sap_chain_id: PC_7A3929SH
```

## sap\_client Attribute -- Specify SAP Client Number

The sap\_client attribute specifies the SAP client within the SAP system.

#### NOTE

Your agent administrator can define a default SAP client number using the jco.client.client property in the connection properties file. The sap\_client attribute overrides the default value.

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Batch Input Session](#) (SAPBDC)
- [SAP BW InfoPackage](#) (SAPBWIP)
- [SAP BW Process Chain](#) (SAPBWPC)
- [SAP Data Archiving](#) (SAPDA)
- [SAP Event Monitor](#) (SAPEVT)
- [SAP Job Copy](#) (SAPJC)
- [SAP Process Monitor](#) (SAPPM)
- [SAP R/3](#) (SAP)

### Syntax

This attribute has the following format:

```
sap_client: client_number
```

- ***client\_number***

Specifies the client within the SAP system. The value corresponds to the General data, Target server field on the Define Background Job dialog.

**Limits:** Up to three digits

**NOTE**

You can specify a single digit (for example, `sap_client: 3`), but JIL automatically prefixes zeroes to the value. The value will be stored as `sap_client: 003` in the database.

**Example: Specify the Numeric Client within the SAP System**

This example runs the BTCTEST ABAP. The SAP client in the SAP system is 850.

```
insert_job: SAPTEST
job_type: SAPBWPC
machine: SAPHTAGENT
sap_client: 850
sap_chain_id: DEMO_CHAIN
```

**sap\_event\_id**

The `sap_event_id` attribute specifies the SAP event to monitor or trigger.

**Supported Job Types**

This attribute is required for the [SAP Event Monitor \(SAPEVT\)](#) job type.

**Syntax**

This attribute has the following format:

```
sap_event_id: event_name
```

- **event\_name**

Specifies the name of the SAP event to monitor or trigger.

**Limits:** Up to 32 valid SAP characters; case-sensitive

**NOTE**

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Trigger an SAP event**

This example triggers the SAP event named `SAP_Event`.

```
insert_job: SAPMON
job_type: SAPEVT
machine: SAPAGENT
sap_event_id: SAP_Event
```

**sap\_event\_parm**

The `sap_event_parm` attribute specifies the name of an SAP event parameter.

**Supported Job Types**

This attribute is optional for the [SAP Event Monitor \(SAPEVT\)](#) job type.

**Syntax**

This attribute has the following format:

```
sap_event_parm: parameters
```

- **parameters**

Specifies the name of the SAP event parameter.

**Limits:** Up to 64 valid SAP characters

**NOTE**

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Specify an SAP Event Parameter**

This example triggers the SAP event named SAP\_Event. The event parameter is 2.

```
insert_job: SAPTRIG
job_type: SAPEVT
machine: SAPAGENT
sap_event_id: SAP_Event
sap_is_trigger: Y
sap_event_parm: 2
```

**sap\_ext\_table Attribute -- Specify Data Selection Criteria**

The sap\_ext\_table attribute specifies the SAP Business Warehouse InfoPackage data selection criteria.

**Support Job Type**

This attribute is optional for the [SAP BW InfoPackage \(SAPBWIP\) job type](#).

**Syntax**

This attribute has the following format:

```
sap_ext_table: field_name=field, object_name=object, sign=I|E, operation=BT|EQ,
low=low_number, high=high_number
```

- **field\_name=*field***  
Specifies the Business Warehouse InfoPackage Technical Name.  
**Limits:** Up to 100 characters
- **object\_name=*object***  
Specifies the Business Warehouse InfoPackage Technical InfoObject.  
**Limits:** Up to 100 characters
- **sign=I|E**  
Specifies the Business Warehouse InfoPackage Technical Sign. Options are the following:
  - I -- Specifies include.
  - E -- Specifies exclude.
- **operation=BT|EQ**  
Specifies the Business Warehouse InfoPackage Technical Operation. Options are the following:
  - BT -- Specifies between.
  - EQ -- Specifies equal.
- **low=*low\_number***  
Specifies the Business Warehouse InfoPackage Technical From Value.  
**Limits:** Up to 256 characters  
**Default:** Uses the SAP default.

**NOTE**

You must specify this value if you have specified a value for the high keyword.

- **high=*high\_number***  
Specifies the Business Warehouse InfoPackage Technical To Value.

**Limits:** Up to 256 characters

**Default:** Uses the SAP default.

**NOTE**

You must specify this value if you have specified a value for the low keyword and set operation=BT.

**NOTE**

- To specify multiple data selection criteria, define a separate sap\_ext\_table attribute for each set of criteria. The combination of all the sap\_ext\_table attributes form a single row in the SELECTION table.
- If a keyword has no value, you can omit the keyword from the attribute.

### Example: Specify Multiple Data Selection Criteria

This example specifies three sets of criteria in the SAP table.

```
insert_job: SAPJOB1
job_type: SAPBWIP
machine: sapagent
sap_info_pack: InfoBw2
sap_ext_table: field_name=SPTAG, object_name=0CALDAY, sign=I, operation=BT,
 low=20010101,high=20060606
sap_ext_table: field_name=VBELN, object_name=0CALDAY, sign=I, operation=BT,
 low=20010101,high=20060606
sap_ext_table: field_name=PKUNRE,object_name=0CALDAY, sign=I, operation=BT,
 low=20010101,high=20060606
```

### Example: Specify a Row of Criteria With Select Keywords

This example specifies one set of criteria in the SAP table. The object\_name, sign, and operation fields have no values, so they are omitted from the job definition.

```
insert_job: SAPJOB2
job_type: SAPBWIP
machine: sapagent
sap_ext_table: field_name=SPTAG,low=20010101,high=20060606
```

## sap\_fail\_msg Attribute -- Specify Failure Message for SAP Job

The sap\_fail\_msg attribute specifies a string that indicates the failure of a job. If the string is found in the job's spool file, the job is considered failed.

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Job Copy \(SAPJC\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_fail_msg: message
```

- ***message***

Specifies a string that indicates the job failed. If the string is found in the job's spool file, the job is considered failed even if the job succeeds on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive.

**NOTE**

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Use a Failure Message to Determine a Job's Failure**

This example specifies the failure message, "Internal problem". If the string is found in the SAP job log or the SAP spool output, AutoSys Workload Automation treats this job as failed.

```
insert_job: SAPTEST
job_type: SAP
machine: SAPAGENT
sap_fail_msg: Internal problem
sap_step_parms: abap_name=BTCTEST, variant=TEST
sap_job_name: SAP_JOB
```

**sap\_info\_pack Attribute -- Identify Name of Business Warehouse InfoPackage on SAP System**

The sap\_info\_pack attribute identifies the name of a Business Warehouse InfoPackage on the SAP system.

**Supported Job Type**

This attribute is required for the [SAP BW InfoPackage \(SAPBWIP\) job type](#).

**Syntax**

This attribute has the following format:

```
sap_info_pack: infopackage
```

- ***infopackage***  
Specifies the name of the Business Warehouse InfoPackage.  
**Limits:** Up to 30 valid SAP characters; case sensitive

**Example: Define a Business Warehouse InfoPackage Job**

This example runs the ZPAK\_6C7ZW2JAXS90AK71WD1CYIN Business Warehouse InfoPackage on the SAP system.

```
insert_job: BWIP
job_type: SAPBWIP
machine: sapagent
sap_info_pack: ZPAK_6C7ZW2JAXS90AK71WD1CYIN
```

**sap\_is\_trigger Attribute**

The sap\_is\_trigger attribute specifies whether to trigger or monitor an SAP event.

**NOTE**

If you do not specify the sap\_is\_trigger attribute in your job definition, the job monitors the SAP event (the default).

**Supported Job Type**

This attribute is optional for the [SAP Event Monitor \(SAPEVT\) job](#).

**Syntax**

This attribute has the following format:

```
sap_is_trigger: Y | N | 0 | 1
```

- **Y | 1**

Triggers an SAP event.

**NOTE**

You must specify N in the continuous attribute.

- **N | 0**

Monitors an SAP event. This is the default.

### Example: Trigger an SAP Event

This example triggers the SAP event named SAP\_Event. The event parameter is 2.

```
insert_job: SAPTRIG
job_type: SAPEVT
machine: SAPAGENT
sap_event_id: SAP_Event
sap_is_trigger: Y
sap_event_parm: 2
```

## sap\_job\_class Attribute -- Specify SAP Job Class

The sap\_job\_class attribute specifies the SAP system job class the job will run under.

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_job_class: class
```

- **class**

Specifies a valid SAP system job class.

**Limits:** One character

### Example: Specify an SAP System Job Class

This example runs the SAP R/3 job under the SAP system job class C.

```
insert_job: SAPTEST
job_type: SAP
machine: SAPAGENT
sap_job_class: C
sap_job_name: SAP_JOB
sap_step_parms: abap_name=BTCTEST
```

## sap\_job\_count Attribute -- Specify the Job ID of the SAP Job to Copy

The sap\_job\_count attribute specifies the ID of the job to be copied.

## Supported Job Type

This attribute is optional for the [SAP Job Copy \(SAPJC\)](#) job type.

### Syntax

This attribute has the following format:

```
sap_job_count: count
```

- **count**  
Specifies the ID of the job to be copied.  
**Limits:** Up to 8 alphanumeric characters

### Example: Specify the ID of the Job to be Copied

This example defines an SAP Job Copy job that copies the AM job with job count 113315AM and runs the new copy.

```
insert_job: SAPJC_job
job_type: SAPJC
owner: WAAESAP@sapagent
machine: sapagent
sap_job_name: AM
sap_job_count: 113315AM
```

## sap\_job\_name Attribute -- Specify SAP Job Name

The sap\_job\_name attribute specifies the SAP R/3 job name.

### Supported Job Types

This attribute is required for the following job types:

- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP Job Copy \(SAPJC\)](#)
- [SAP R/3 \(SAP\)](#)

This attribute is optional for the SAP BW InfoPackage (SAPBWIP) job type.

### Syntax

This attribute has the following format:

```
sap_job_name: job_name
```

- **job\_name**  
Specifies the following:
  - For SAP R/3 jobs, the SAP job name that identifies the workload in the SAP system.
  - For SAP Batch Input Session jobs, the name of the ABAP job that creates the BDC session on the SAP system.
  - For SAP Job Copy jobs, the name of the SAP R/3 job to be copied.
  - For SAP BW InfoPackage jobs, specifies the name of the Business Warehouse InfoPackage job on the SAP system.**Limits:** Up to 32 characters

**NOTE**

- This attribute corresponds to the General data, Job name field on the Define Background Job dialog in SAP.
- You can use special characters, blanks, and spaces. Enclose the values that contain delimiters (such as spaces) in double quotation marks. Embedded double quotation marks must be prefixed with a backslash as an escape character.

**Example: Specify an SAP Job Name**

This example specifies a job name that does not contain blanks or delimiters, so the job name does not need to be enclosed in single quotation marks.

```
insert_job: SAPTEST
job_type: SAP
machine: sapagent
sap_job_name: prodrun1
sap_step_parms: abap_name=BTCTEST, step_user=J01PROD, variant=TEST
```

**Example: Specify an SAP Job Name That Contains Blanks and Other Characters**

This example specifies a job name that contains a space and a single quotation mark, so the job name must be enclosed in quotation marks. The quotation mark in the job name must also be duplicated.

```
insert_job: SAPTEST
job_type: SAP
machine: sapagent
sap_job_name: "Today''s Work"
sap_step_parms: abap_name=BTCTEST, step_user=J01PROD, variant=TEST
```

**Example: Specify a Job to be Copied**

This example copies the proctest job.

```
insert_job: SAPCOPY
job_type: SAPJC
machine: sapagent
sap_job_name: proctest
sap_job_count: 00458131
```

**sap\_lang Attribute -- Specify SAP Language for Login**

The sap\_lang attribute specifies the language to use to log in to the SAP system.

**NOTE**

Your agent administrator can define a default language using the jco.client.lang property in the connection properties file. The sap\_lang attribute overrides the default value.

**Supported Job Types**

This attribute is optional for the following job types:



- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP BW InfoPackage \(SAPBWIP\)](#)
- [SAP BW Process Chain \(SAPBWPC\)](#)
- [SAP Data Archiving \(SAPDA\)](#)
- [SAP Event Monitor \(SAPEVT\)](#)
- [SAP Process Monitor \(SAPPM\)](#)
- [SAP Job Copy \(SAPJC\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_lang: language
```

- **language**  
Specifies a character code representing a valid language for SAP.  
**Limits:** Up to two characters; case-sensitive  
**Default:** The SAP system language  
**Example:** EN

#### NOTE

This attribute corresponds to the SAPGUI Language field on the login dialog.

### Example: Specify an SAP Language

This example defines the SAPBWPC3 job that runs the DEMO\_CHAIN1 Process Chain on the SAP system. The language used to log in to the SAP system is English.

```
insert_job: SAPBWPC3
job_type: SAPBWPC
machine: sapagent
owner: WAAESAP@sapagent
sap_chain_id: DEMO_CHAIN1
sap_lang: EN
sap_rfc_dest: SM1
```

### sap\_mon\_child Attribute -- Enable Monitoring of Children Jobs

The sap\_mon\_child attribute specifies whether to monitor children jobs. Children jobs are jobs spawned by a parent job.

#### NOTE

If you do not specify the sap\_mon\_child attribute in your job definition, the job does not monitor children jobs (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Job Copy \(SAPCP\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_mon_child: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Monitors the job's children.
- **N | F | NO | FALSE**  
Does not monitor the job's children. This is the default.

**NOTE**

- Your agent administrator can specify a default setting for all SAPCP and SAP jobs by setting the `sap.job.children.monitor` parameter in the agent's `agentparm.txt` file.
- The `sap_mon_child` attribute overrides the default setting specified in the agent's `agentparm.txt`.

**Example: Enable Monitoring of Children**

In this example, children spawned by SAPTEST are monitored.

```
insert_job: SAPTEST
job_type: SAP
machine: SAPHTAGENT
sap_mon_child: Y
sap_job_name: SAP_JOB
sap_step_parms: abap_name=ZCHILDJOB1
```

**sap\_office Attribute -- Save Outgoing Documents to SAPoffice Outbox**

The `sap_office` attribute specifies whether to save outgoing documents to the SAPoffice outbox of the SAP user associated with the job.

**NOTE**

If you do not specify the `sap_office` attribute in your job definition, the job does not save outgoing documents (the default).

**Supported Job Types**

This attribute is optional for the following job types:

- [SAP Batch Input Session](#) (SAPBDC)
- [SAP R/3](#) (SAP)

**Syntax**

This attribute has the following format:

```
sap_office: Y | N | T | F | YES | NO | TRUE | FALSE
```

- **Y | T | YES | TRUE**  
Saves outgoing documents to the SAPoffice outbox.
- **N | F | NO | FALSE**  
Does not save outgoing documents to the SAPoffice outbox. This is the default.

**Example: Save Outgoing Mail to the SAPoffice Outbox**

In this example, outgoing mail is sent to the SAPoffice outbox of the SAP user J01Pro.

```
insert_job: SAPTEST
job_type: SAP
machine: SAPAGENT
sap_job_name: SAP_JOB
sap_step_parms: abap_name=RSPARAM, step_user=J01PRO
```

```
sap_recipients: recipient=SU,rcp_type=SU
sap_office: Y
```

## sap\_print\_parms Attribute -- Specify Print Parameters

The sap\_print\_parms attribute specifies print parameters.

### Supported Job Types

This attribute is optional for the [SAP Data Archiving \(SAPDA\) job type](#).

### Syntax

This attribute has the following format:

```
sap_print_parms: dest=value[,keyword=value...]
```

- **dest=value**  
Specifies the output device name. This keyword corresponds to the SAPGUI Output device field on the Background Print Parameters dialog.  
**Limits:** Up to four characters; case-sensitive  
**Example:** dev1
- **keyword=value**  
(Optional) Specifies a print parameter. You can specify multiple parameters. Separate each keyword=value pair with a comma. Options include the following:
  - **authorization=value**  
Specifies the SAP authorization user for viewing the print spool list. This keyword corresponds to the SAPGUI Authorization field on the Background Print Parameters dialog.  
**Limits:** Up to 12 valid SAP characters; case-sensitive
  - **banner\_page=Y | N | T | F | YES | NO | TRUE | FALSE**  
Indicates whether to include an SAP cover page with the report output.
    - Y | T | YES | TRUE  
Prints the cover page.
    - N | F | NO | FALSE  
Does not print the cover page.  
**Default:** N
  - **copies=value**  
Specifies the number of print copies.  
**Limits:** 1-255
  - **dataset=value**  
Specifies the print spool data set name.  
**Limits:** Up to six characters; case-sensitive  
**Example:** LIST1S
  - **dept=value**  
Specifies the department name to print on the cover page.  
**Limits:** Up to 12 characters; case-sensitive
  - **expiration=value**  
Specifies the number of days before the spool request is deleted.  
**Limits:** 0-9
  - **footer=Y | N | T | F | YES | NO | TRUE | FALSE**  
Indicates whether to print a footer on an SAP report.
    - Y | T | YES | TRUE

- Prints the footer.
  - N | F | NO | FALSE  
Does not print the footer.
  - **format=value**  
Specifies an output format for an SAP report.  
**Limits:** Up to 16 valid SAP characters; case-sensitive  
**Example:** X\_65\_80
  - **host\_page=Y | N | 0 | 1 | D**  
Indicates whether to print an operate system page.
    - Y | 1  
Prints the operating system page.
    - N | 0  
Does not print the operating system page.
    - D  
Uses the SAP default.
  - **new\_spool=Y | N | T | F | YES | NO | TRUE | FALSE**  
Indicates whether to create a new spool request or append the spool request to an existing spool request with similar attributes (if any).
    - Y | T | YES | TRUE  
Creates a new spool request.
    - N | F | NO | FALSE  
Appends spool request to an existing spool request.  
**Default:** Y
  - **num\_columns=value**  
Specifies the line width of the list.  
**Limits:** 1-255  
**Example:** 251
- NOTE**  
The maximum line width of a list for viewing on the screen is 255.
- **num\_lines**  
Specifies the number of lines per list page. The length of the list is determined by its content. You can use 0 or blank only when viewing the list online. You cannot use 0 or blank to format a list to be printed.  
**Limits:** 0-486  
**Example:** 15
  - **print\_imm=Y | N | T | F | YES | NO | TRUE | FALSE**  
Indicates whether to print the job immediately after completion. This keyword corresponds to the SAPGUI Spool options, Print immediately field on the Background Print Parameters dialog.
    - Y | T | YES | TRUE  
Job prints immediately after completion
    - N | F | NO | FALSE  
The job output remains on spool.  
**Default:** N
  - **print\_priority**  
Specifies the priority of the SAP spool request. The priority is a number between 1 (highest priority) and 9 (lowest priority).  
**Default:** 5  
**Example:** print\_priority=2
  - **prt\_arc\_mode=PRINT | ARCHIVE | BOTH**  
Specifies whether the spool file is printed, archived, or both.

- PRINT
- ARCHIVE
- BOTH
 

**Example:** prt\_arc\_mode=BOTH

**Default:** PRINT
- **recipient\_name**

Specifies the spool request recipient's name to appear on the SAP cover page. This keyword corresponds to the SAPGUI Cover sheets options, Recipient field on the Background Print Parameters dialog.

**Limits:** Up to 12 valid SAP characters; case-sensitive
- **release=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether the spool request associated with the ABAP is deleted after printing. This keyword corresponds to the SAPGUI Spool options, Delete after output field on the Background Print Parameters dialog.

  - **Y | T | YES | TRUE**  
The spool request is deleted after printing.
  - **N | F | NO | FALSE**  
The spool request is not deleted after printing.
- **req\_type=value**

Specifies the print request type.

**Limits:** Up to 12 valid SAP characters; case-sensitive

**Example:** req\_type=prtreq12
- **sap\_banner=Y | N | 0 | 1 | D**

Indicates whether to include an SAP cover page. The SAP cover page contains information such as recipient name, department name, and format used.

  - **Y | 1**  
Prints the SAP cover page.
  - **N | 0**  
Does not print the SAP cover page.
  - **D**  
Uses the SAP default.

**Example:** sap\_banner=Y
- **spool\_name**

Specifies the name of the print spool.

**Limits:** Up to 12 valid SAP characters; case-sensitive

**Example:** prtspool12
- **title**

Specifies the spool request description.

**Limits:** Up to 68 characters

#### NOTE

- The entire value can be up to 4096 characters.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

#### Example: Specify Print Parameters

This example specifies the print parameters for an SAP Data Archiving job.

```
insert_job: test_SAPDA
job_class: SAP
job_type: SAPDA
machine: sapagent
```

```
owner: WAAESAP@sapserver
arc_obj_name: BC_ARCHIVE
arc_obj_variant: PRODUKTIVMODUS
sap_print_parms: dest=INHYP502,prt_arc_mode=PRINT,title="testing", num_lines=65,
num_columns=132, format=X_65_132, copies=1
```

## sap\_process\_client Attribute -- Specify the Client to Monitor for a Specified Process

The sap\_process\_client attribute specifies the client to be monitored for the specified process.

### Supported Job Type

This attribute is optional for the [SAP Process Monitor \(SAPPM\)](#) job type when sap\_process\_status is set to RUNNING or STOPPED.

### Syntax

This attribute has the following format:

```
sap_process_client: client
```

- **client**  
Specifies the client monitored for the specified process.  
**Limits:** Up to three digits  
**Example:** 800

## sap\_process\_status Attribute -- Specify the SAP Process Status to Monitor

The sap\_process\_status attribute specifies the SAP process status to monitor.

### Supported Job Types

This attribute is required for the [SAP Process Monitor \(SAPPM\)](#) job type.

### Syntax

This attribute has the following format:

```
sap_process_status: RUNNING | STOPPED | WAITING
```

- **RUNNING**  
Monitors the running status.
- **STOPPED**  
Monitors the stopped status.
- **WAITING**  
Monitors the waiting status.

### Example: Monitor for Jobs Running in the Background

This example monitors for running background jobs and checks for the SAP user named user14.

```
insert_job: PMTEST
job_type: SAPPM
machine: SAPHTAGENT
sap_process_status: RUNNING
sap_proc_type: BGD
sap_proc_user: user14
```

## sap\_proc\_type Attribute -- Specify the SAP Process Type to Monitor

The sap\_proc\_type attribute specifies the SAP process type to monitor.

### Supported Job Type

This attribute is optional for the [SAP Process Monitor \(SAPPM\)](#) job type.

### Syntax

This attribute has the following format:

```
sap_proc_type: BGD | DIA | ENQ | SPO | UPD | UP2
```

- **BGD**  
Monitors the background process type.
- **DIA**  
Monitors the dialog process type.
- **ENQ**  
Monitors the lock process type.
- **SPO**  
Monitors the spool process type.
- **UPD**  
Monitors the update process type.
- **UP2**  
Monitors the update2 process type.

### Example: Monitor for Jobs Running in the Background

This example monitors for running background jobs and checks for the SAP user named user14.

```
insert_job: PMTEST
job_type: SAPPM
machine: SAPHTAGENT
sap_process_status: RUNNING
sap_proc_type: BGD
sap_proc_user: user14
```

## sap\_proc\_user -- Specify a SAP Process User

The sap\_proc\_user attribute specifies an SAP process user.

### Supported Job Type

This attribute is optional for the [SAP Process Monitor \(SAPPM\)](#) job type when sap\_process\_status is set to RUNNING or STOPPED.

### Syntax

This attribute has the following format:

```
sap_proc_user: user
```

- **user**  
Specifies an SAP user name. When a process matches the specified process status, the job the process runs is checked for a match to this user.  
**Limits:** Up to 12 characters; case-sensitive

**NOTE**

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Specify the User for an SAP Process**

This example monitors for running background jobs and checks for the SAP user named user14.

```
insert_job: PMTEST
job_type: SAPPM
machine: SAPHTAGENT
sap_process_status: RUNNING
sap_proc_type: BGD
sap_proc_user: user14
```

**sap\_recipients Attribute -- Specify SAP Spool Recipient**

The sap\_recipients attribute specifies a recipient for the SAP spool. The recipient receives the spool upon job completion or failure.

**Supported Job Types**

This attribute is optional for the following job types:

- [SAP Batch Input Session](#) (SAPBDC)
- [SAP R/3](#) (SAP)

**Syntax**

This attribute has the following format:

```
sap_recipients: rcp_type=type[,keyword=value...]
```

- **rcp\_type=type** Specifies the recipient type. Options include the following:
  - **SU**  
Specifies that the type is SAP user.
  - **DLI**  
Specifies that the type is SAPoffice distribution list.
  - **INT**  
Specifies that the type is email address.
- **keyword=value**  
(Optional) Specifies a parameter for the recipient. You can specify multiple parameters. Separate each keyword=value pair with a comma. Options include the following:
  - **blind\_copy=Y | N | T | F | YES | NO | TRUE | FALSE**  
Specifies whether to send the email as a blind carbon copy (bcc).
    - **Y | T | YES | TRUE**  
Sends the email as a blind carbon copy.
    - **N | F | NO | FALSE**  
Does not send the email as a blind carbon copy.
  - **copy=Y | N | T | F | YES | NO | TRUE | FALSE**  
Specifies whether to send the email as a carbon copy (cc).
    - **Y | T | YES | TRUE**  
Sends the email as a carbon copy.
    - **N | F | NO | FALSE**



Does not send the email as a carbon copy.

– **express=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether to send instant messages to recipients if they are logged into SAP.

- **Y | T | YES | TRUE**

Sends instant messages.

- **N | F | NO | FALSE**

Does not send instant messages.

**Example:** express=Y

**NOTE**

This parameter only applies when rcp\_type is set to SU or DLI.

– **no\_forward=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether the SAPoffice users can forward the document.

- **Y | T | YES | TRUE**

SAPoffice users cannot forward the document.

- **N | F | NO | FALSE**

SAPoffice users can forward the document.

– **no\_print=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether the SAPoffice users can print the document.

- **Y | T | YES | TRUE**

SAPoffice users cannot print the document.

- **N | F | NO | FALSE**

SAPoffice users can print the document.

– **recipient=value**

Specifies the target recipient's distribution address. The value corresponds to the rcp\_type value.

**Limits:** Up to 256 characters

**Examples:**

- rcp\_type=SU,recipient=SAPUSER
- rcp\_type=DLI,recipient=SAPDLIST
- rcp\_type=DLI,recipient=\$\$RECIPIENT
- rcp\_type=INT,user@example.com

**NOTE**

- This parameter is required when rcp\_type is set to SU, DLI, or INT.
- You can use global variables to specify the target recipient's distribution address. Use the \$global\_name, \$\$global\_name, or \$\$global\_name format to reference global variables.

**NOTE**

- You can specify multiple spool recipients. Use one entry of sap\_recipients to define each recipient and its properties. For example, to specify two recipients user1@company.com and user2@company.com, define an sap\_recipients attribute for user1 and an sap\_recipients attribute for user2.
- The entire value can be up to 4096 characters.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Specify Multiple Spool Recipients**

This example sends the SAP spool file to five recipients when the job completes or fails.

```
insert_job: sap_job
machine: sapagent
job_type: SAP
```

```

sap_job_name: BTCTEST
sap_step_parms: abap_name=BTCABAP
sap_recipients: rcp_type=INT,recipient=J01PROD@example.com
sap_recipients: rcp_type=INT,recipient=J01TEST@example.com,blind_copy=Y
sap_recipients: rcp_type=INT,recipient=J02PROD@example.com,Copy=Y
sap_recipients: rcp_type=DLI,recipient="SAPoffice Dist List",
 Express=Y,Copy=Y,no_forward=Y,no_print=Y
sap_recipients: rcp_type=SU,recipient=WAAESAP,blind_copy=Y

```

## sap\_release\_option Attribute -- Specify Release Option for a Job

The sap\_release\_option attribute specifies the action to take with a job after it is defined.

### NOTE

If you do not specify the sap\_release\_option attribute in your job definition, the job releases the job as soon as possible (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP Job Copy \(SAPJC\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_release_option: I | A
```

- **I**  
Releases the job immediately. If no free background processing is available, the job is not released and stays in the scheduled SAP job state.
- **A**  
Releases the job as soon as possible. This is the default.

### Example: Release a Job as Soon as Possible

This example runs the ZBDCTEST ABAP that creates a Batch Input Session (BDC ABAP) on the default SAP system defined to the agent. The job runs as soon as possible after the job is defined. After this job runs, the BDC job starts the data transfer.

```

insert_job: bdcjob
job_type: SAPBDC
machine: sapagent
owner: WAAESAP@sapagent
sap_job_name: ZBDCTEST
sap_release_option: A
sap_step_parms: abap_name="ZBDCTEST"

```

## sap\_rfc\_dest Attribute -- Specify SAP Destination

The sap\_rfc\_dest attribute specifies the destination value for the Remote Function Call (RFC) connection and gateway information.

## Supported Job Types

This attribute is optional for the following job types:

- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP BW InfoPackage \(SAPBWIP\)](#)
- [SAP BW Process Chain \(SAPBWPC\)](#)
- [SAP Data Archiving \(SAPDA\)](#)
- [SAP Event Monitor \(SAPEVT\)](#)
- [SAP Job Copy \(SAPJC\)](#)
- [SAP Process Monitor \(SAPPM\)](#)
- [SAP R/3 \(SAP\)](#)

## Syntax

This attribute has the following format:

```
sap_rfc_dest: destination
```

- **destination**  
Specifies the destination for the RFC connection and gateway information for an SAP R/3 system. This value corresponds to the destination@properties agent configuration file name containing the SAP connection information.  
**Limits:** Up to 256 characters; case-sensitive

## Example: Specify the SAP Destination

This example runs the Business Warehouse InfoPackage 0PAK\_D2XZMZ1HD5WFVFL3EN1NVIT4V at the SAP destination, SM1.

```
insert_job: InfoBW2
job_type: SAPBWIP
machine: sapagent
owner: WAAESAP@sapagent
sap_job_name: InfoBW2
sap_rfc_dest: SM1
sap_info_pack: 0PAK_D2XZMZ1HD5WFVFL3EN1NVIT4V
```

## sap\_step\_num Attribute -- Specify the Number of the First Step to Copy

The sap\_step\_num attribute specifies the number of the first step to start copying job data from.

## Supported Job Type

This attribute is optional for the [SAP Job Copy \(SAPJC\) job type](#).

## Syntax

This attribute has the following format:

```
sap_step_num: step
```

- **step**  
Specifies the number of the first step to start copying job data from.  
**Limits:** 0-highest step number

### NOTE

If you specify 0 or 1, all steps are copied.

## Example: Specify an SAP Step Number

This example defines an SAP Job Copy job that starts copying job data from step 1.

```
insert_job: test_SAPJC_2
job_class: SAP
job_type: SAPJC
machine: sapagent
owner: WAAESAP@sapserver
sap_job_name: AM
sap_job_count: 11331500
sap_release_option: I
sap_target_jobname: "SAPJC_DUP"
sap_step_num:1
```

## sap\_step\_parms Attribute -- Specify SAP R/3 Step Specifications

The sap\_step\_parms attribute defines the SAP R/3 step specifications.

### Supported Job Types

This attribute is required for the following job types:

- [SAP R/3 \(SAP\)](#)
- [SAP Batch Input Session \(SAPBDC\)](#)

### Syntax

This attribute has the following format:

```
sap_step_parms: abap_name=abap[, keyword=value...]
```

- **abap\_name=abap**  
Specifies the valid SAP system ABAP name. This keyword corresponds to the SAPGUI ABAP program Name field on the Create Step dialog.  
**Limits:** Up to 40 valid SAP characters  
**Example:** abap\_name=BTCTEST
  - **keyword=value**  
(Optional) Specifies a parameter for the step. You can specify multiple parameters. Separate each keyword=value pair with a comma. Options include the following:
    - **abap\_lang=value**  
Specifies a valid language for the ABAP.  
**Default:** The SAP system default log in language  
**Limits:** Up to two characters  
**Examples:** abap\_lang=E
    - **arc\_client=value**  
Specifies the archive link client.  
**Limits:** 0-999
- NOTE**  
To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.
- Example:** arc\_client=800
- **arc\_connect=value**  
Specifies the archive link communication connection.  
**Limits:** Up to 14 valid SAP characters; case-sensitive  
**Example:** arc\_connect=arcconnect

**NOTE**

- To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.
- Escape the double quotation marks with a backslash.

- **arc\_date= value** Specifies the archive link archiving date. The format is *yyyymmdd*. **Limits:** Up to eight digits  
**Example:** arc\_date=20080630

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_doc\_class=value**  
Specifies the archive link document class.  
**Limits:** Up to 20 valid SAP characters; case-sensitive  
**Example:** arc\_doc\_class=arcdocclass

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_doc\_type=value**  
Specifies the name of the document type. This keyword corresponds to the Doc. type field on the Define Background Archive Parameters dialog.  
**Limits:** Up to 10 valid SAP characters; case-sensitive  
**Example:** arc\_doc\_type=archive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_format=value**  
Specifies the archive output format.  
**Limits:** Up to 16 valid SAP characters; case-sensitive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_host\_link=value**  
Specifies the SAP archive link RPC host.  
**Limits:** Up to 32 valid SAP characters; case-sensitive  
**Example:** arc\_host\_link=rpchost

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_info=value**  
Specifies the archive link information.  
**Limits:** Up to three valid SAP characters; case-sensitive  
**Example:** arc\_info=inf

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_obj\_type=value**  
Specifies the type of external system archive object. This keyword corresponds to the Obj. type field on the Define Background Archive Parameters dialog.  
**Limits:** Up to 10 valid SAP characters; case-sensitive  
**Example:** arc\_obj\_type=archive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_path=value**  
Specifies the standard archive path.  
**Limits:** Up to 70 valid SAP characters; case-sensitive

**Example:** arc\_path=/export/home/archive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_printer=value**

Specifies the archive target printer.

**Limits:** Up to four valid SAP characters; case-sensitive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_protocol=value**

Specifies the archive storage connection protocol.

**Limits:** Up to eight valid SAP characters; case-sensitive

**Example:** arc\_protocol=prtcl12

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_report=value**

Specifies the archive link report name.

**Limits:** Up to 30 valid SAP characters; case-sensitive

**Example:** arc\_report=Archive Report

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_service=value**

Specifies the SAP archive link RPC service/destination.

**Limits:** Up to 32 valid SAP characters; case-sensitive

**Example:** arc\_service=rpcdest

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_storage=value**

Specifies the archive link target storage system.

**Limits:** Up to two valid SAP characters; case-sensitive

**Example:** arc\_storage=st

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_text=value**

Specifies the archive link Text Information field.

**Limits:** Up to 40 valid SAP characters; case-sensitive

**Example:** arc\_text='Archive text'

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_userid=value**

Specifies the user ID for archiving the report.

**Limits:** Up to 12 valid SAP characters; case-sensitive

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **arc\_version=value**

Specifies the archive version.

**Limits:** Up to four valid SAP characters

**Example:** arc\_version=4.1

**NOTE**

To use this keyword, prt\_arc\_mode must be set to ARCHIVE or BOTH.

- **authorization=value**

Specifies the SAP authorization user for viewing the print spool list. This keyword corresponds to the SAPGUI Authorization field on the Background Print Parameters dialog.

**Limits:** Up to 12 valid SAP characters; case-sensitive

- **banner\_page=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether to include an SAP cover page with the report output.

- **Y | T | YES | TRUE**

Prints the cover page.

- **N | F | NO | FALSE**

Does not print the cover page.

**Default:** N

- **copies=value**

Specifies the number of print copies.

**Limits:** 1-255

- **dataset=value**

Specifies the print spool data set name.

**Limits:** Up to six characters; case-sensitive

**Example:** LIST1S

- **dept=value**

Specifies the department name to print on the cover page.

**Limits:** Up to 12 characters; case-sensitive

- **dest=value**

Specifies the output device name. This keyword corresponds to the SAPGUI Output device field on the Background Print Parameters dialog.

**Limits:** Up to four characters; case-sensitive

**Example:** dev1

- **expiration=value**

Specifies the number of days before the spool request is deleted.

**Limits:** 0-9

- **footer=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether to print a footer on an SAP report.

- **Y | T | YES | TRUE**

Prints the footer.

- **N | F | NO | FALSE**

Does not print the footer.

- **format=value**

Specifies an output format for an SAP report.

**Limits:** Up to 16 valid SAP characters; case-sensitive

**Example:** X\_65\_80

- **host\_page=Y | N | 0 | 1 | D**

Indicates whether to print an operate system page.

- **Y | 1**  
Prints the operating system page.
- **N | 0**  
Does not print the operating system page.
- **D**  
Uses the SAP default.

- **new\_spool=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether to create a new spool request or append the spool request to an existing spool request with similar attributes (if any).

- **Y | T | YES | TRUE**  
Creates a new spool request.
- **N | F | NO | FALSE**  
Appends spool request to an existing spool request.

**Default:** Y

- **num\_columns=value**

Specifies the line width of the list.

**Limits:** 1-255

**Example:** 251

**NOTE**

The maximum line width of a list for viewing on the screen is 255.

- **num\_lines=value**

Specifies the number of lines per list page. The length of the list is determined by its content. You can use 0 or blank only when viewing the list online. You cannot use 0 or blank to format a list to be printed.

**Limits:** 0-486

**Example:** 15

- **print\_imm=Y | N | T | F | YES | NO | TRUE | FALSE**

Indicates whether to print the job immediately after completion. This keyword corresponds to the SAPGUI Spool options, Print immediately field on the Background Print Parameters dialog.

- **Y | T | YES | TRUE**  
Job prints immediately after completion.
- **N | F | NO | FALSE**  
The job output remains on spool.

**Default:** N

- **print\_priority=value**

Specifies the priority of the SAP spool request. The priority is a number between 1 (highest priority) and 9 (lowest priority).

**Default:** 5

**Example:** print\_priority=2

- **prt\_arc\_mode=value**

Specifies whether the spool file is printed, archived, or both.



- PRINT
- ARCHIVE
- BOTH

**Example:** prt\_arc\_mode=BOTH

- **recipient\_name=value**  
Specifies the spool request recipient's name to appear on the SAP cover page. This keyword corresponds to the SAPGUI Cover sheets options, Recipient field on the Background Print Parameters dialog.  
**Limits:** Up to 12 valid SAP characters; case-sensitive  
**Default:** Current user name
  - **release=Y | N | T | F | YES | NO | TRUE | FALSE**  
Indicates whether the spool request associated with the ABAP is deleted after printing. This keyword corresponds to the SAPGUI Spool options, Delete after output field on the Background Print Parameters dialog.
    - **Y | T | YES | TRUE**  
The spool request is deleted after printing.
    - **N | F | NO | FALSE**  
The spool request is not deleted after printing.
  - **req\_type=value**  
Specifies the print request type.  
**Limits:** Up to 12 valid SAP characters; case-sensitive  
**Example:** req\_type=prtreq12
  - **sap\_banner=Y | N | 0 | 1 | D**  
Indicates whether to include an SAP cover page. The SAP cover page contains information such as recipient name, department name, and format used.
    - **Y | 1**  
Prints the SAP cover page.
    - **N | 0**  
Does not print the SAP cover page.
    - **D**  
Uses the SAP default.
- Example:** sap\_banner=Y
- **sap\_fail\_msg=value**  
Specifies a string that indicates the failure of the step. If the string matches the SAP ABAP output for the step, the step is considered failed even if the step succeeds on the SAP system. **Limits:** Up to 128 valid SAP characters; case-sensitive

#### NOTE

- This keyword does not apply to SAPBDC jobs.
- Enclose the values that contain spaces in double quotation marks ("").

**Example:** sap\_fail\_msg="Internal problem"

- **sap\_maillist="address[,address...]"**  
Specifies one or more email addresses to send the spool list results to.  
**Limits:** Up to 2048 characters; each email address can be up to 1024 characters  
**Example:** sap\_maillist="user1@example.com,user2@example.com"
- **sap\_success\_msg=value**  
Specifies a string that indicates the success of the step. If the string matches the SAP ABAP output for the step, the step is considered successfully completed even if the step fails on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive

**NOTE**

- This keyword does not apply to SAPBDC jobs.
- Enclose the values that contain spaces in double quotation marks (").

**Example:** sap\_success\_msg="Program Selections"

- **spool\_name=value**

Specifies the name of the print spool.

**Limits:** Up to 12 valid SAP characters; case-sensitive

**Example:** spool\_name=prtspool12

- **step\_user=value**

Specifies the SAP R/3 system user the ABAP program runs under.

**Limits:** Up to 12 valid alphanumeric SAP characters

**Example:** step\_user=PROD01

**NOTE**

- If the value exceeds the limits, the job is not inserted.
- You can use global variables to specify the SAP R/3 system user the ABAP program runs under. Use the \$\$global\_name, \$\$ (global\_name), or \$\$ {global\_name} format to reference global variables.
- If the value exceeds the limits after the global variable expansion, the job goes into a RESTART state. The job remains in the RESTART state until you specify a value within the limits.

- **title=value**

Specifies the spool request description.

**Limits:** Up to 68 characters

- **variant=value**

Specifies the variant to pass. This keyword corresponds to the SAPGUI ABAP program Variant field on the Create Step dialog.

**Limits:** Up to 14 valid alphanumeric SAP characters

**Example:** variant=test

**NOTE**

- If the value exceeds the limits, the job is not inserted.
- You can use global variables to specify the variant to pass. Use the \$\$global\_name, \$\$ (global\_name), or \$\$ {global\_name} format to reference global variables.
- If the value exceeds the limits after the global variable expansion, the job goes into a RESTART state. The job remains in the RESTART state until you specify a value within the limits.

**NOTE**

- You can specify multiple steps. Use one entry of sap\_step\_parms to define each step and its properties. For example, to specify two steps, define a sap\_step\_parms attribute for the first step and a sap\_step\_parms attribute for the second step.
- The entire value can be up to 4096 characters.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Define an SAP R/3 Job With Multiple Steps**

This example runs an SAP R/3 job named SAP 3 STEPS\_@#\$. The job runs three steps. Each step runs the same ABAP but with different parameters.

```
insert_job: TEST_1
job_type: SAP
machine: localhost
```

```

job_class: c
sap_job_name: "SAP 3 STEPS_@# $"
sap_release_option: I
sap_step_parms: abap_name=BTCTEST, variant=test, arc_printer=LP01, copies=2, prt_arc_mode=BOTH, arc_o
sap_step_parms: abap_name=BTCTEST, variant=test2, arc_printer=LP01, copies=3, prt_arc_mode=BOTH, prin
sap_step_parms: abap_name=BTCTEST, variant=test, arc_printer=LP01, copies=1, prt_arc_mode=BOTH, sap_b

```

## sap\_success\_msg Attribute -- Specify Success Message for SAP Job

The sap\_success\_msg attribute specifies a string that indicates the success of the job.

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Job Copy \(SAPJC\)](#)
- [SAP R/3 \(SAP\)](#)

### Syntax

This attribute has the following format:

```
sap_success_msg: message
```

- **message**

Specifies a string that indicates the job completed successfully. If the string is found in the job's spool file, the job is considered successfully completed even if the job fails on the SAP system.

**Limits:** Up to 128 valid SAP characters; case-sensitive.

#### NOTE

Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

### Example: Use a Success Message to Determine a Job's Success

This example defines the success message, "Program Selections". If the string is found in the SAP job log, AutoSys Workload Automation considers the job as successful.

```

insert_job: SAPTEST
job_type: SAP
machine: SAPHTAGENT
sap_job_name: SAP_JOB
sap_success_msg: Program Selections
sap_step_parms: abap_name=BTCTEST, step_user=J01PROD, variant=TEST

```

## sap\_target\_jobname Attribute -- Specify Job Name to Copy

The sap\_target\_jobname attribute specifies the name of the target job to copy.

#### NOTE

If you do not specify the sap\_target\_jobname attribute in your job definition, the target job will have the same name as the source job (the default).

### Supported Job Type

This attribute is optional for the [SAP Job Copy \(SAPJC\)](#) job type.

## Syntax

This attribute has the following format:

```
sap_target_jobname: name
```

- ***name***  
Specifies the name of the target job to copy.  
**Default:** The name of the source job  
**Limits:** Up to 32 valid SAP characters

### NOTE

Enclose the values that contain delimiters (such as spaces) in double quotation marks.

## Example: Specify the Name of a Job to Copy

This example defines an SAP Job Copy job that copies the job named SAPJC\_DUP.

```
insert_job: test_SAPJC_2
job_class: SAP
job_type: SAPJC
machine: sapagent
owner: WAAESAP@sapserver
sap_job_name: AM
sap_job_count: 11331500
sap_release_option: I
sap_target_jobname: "SAPJC_DUP"
group: SAPJC
```

## sap\_target\_sys Attribute -- Specify SAP Application Server

The sap\_target\_sys attribute specifies the name of the SAP system application server where the SAP job is to run.

### Supported Job Types

This attribute is optional for the following job types:

- [SAP Batch Input Session \(SAPBDC\)](#)
- [SAP Data Archiving \(SAPDA\)](#)
- [SAP Job Copy \(SAPJC\)](#)
- [SAP Process Monitor \(SAPPM\)](#)
- [SAP R/3 \(SAP\)](#)

## Syntax

This attribute has the following format:

```
sap_target_sys: application_server
```

- ***application\_server*** Specifies the host computer within the SAP system architecture that is capable of running SAP system workload.  
**Limits:** Up to 256 characters; case-sensitive

**NOTE**

- If the `sap_target_sys` attribute is not defined, the SAP system will select an application server based on available resources.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the name of the SAP system application server where the SAP job is to run. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Specify the SAP System Application Server**

This example defines a job that runs the BTCTEST ABAP at the tst005 SAP system regardless of the entries specified in the destination properties file.

```
insert_job: SAPTEST
job_type: SAP
machine: SAPHTAGENT
sap_job_name: SAP_JOB
sap_step_parms: abap_name=BTCTEST, step_user=J01PROD, variant=TEST
sap_target_sys: tst005
```

**scp\_create\_targetdir Attribute -- Create Target Directory**

The `scp_create_targetdir` attribute specifies whether to create a target directory before you upload or download a file. The target directory is created if the path to the file that you upload or download includes nonexistent target directories.

**NOTE**

Specify this attribute only if you are using SFTP protocol.

**Supported Job Type**

This attribute is optional for the Secure Copy (SCP) job type.

**Syntax**

This attribute has the following format:

```
scp_create_targetdir: T | F
```

- **T**  
Creates the target directory.
- **F**  
Does not create the target directory.

**Default:** F

**NOTE**

- You can specify the `scp_create_targetdir` attribute even when multiple source files are specified using wildcards (\* or ?) in the source file names.
- If you specify the local user using the `scp_local_user` attribute and the `scp_transfer_direction` attribute is set to `DOWNLOAD`, the agent must ensure that the directories created are owned by the user that you specified using the `scp_local_user` attribute.

## scp\_delete\_sourcedir Attribute -- Specify Whether to Delete the Directory That Contains Source Files

The scp\_delete\_sourcedir attribute specifies whether to delete the directory that contains the local source files (if uploading) or the remote source files (if downloading) after the files are uploaded or downloaded.

### NOTE

Specify this attribute only if you are using SFTP protocol.

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) job type.

### Syntax

This attribute has the following format:

```
scp_delete_sourcedir: T | F
```

- **T**  
Deletes the directory that contains the local source files (if uploading) or the remote source files (if downloading) after the files are uploaded or downloaded.
- **F**  
Does not delete the directory that contains the local source files or the remote source files.

**Default:** F

### NOTE

- You must have EXECUTE permissions to delete the source directory.
- If the directory is not empty or if it contains files other than the source files you are uploading or downloading, the source directory cannot be deleted.
- You can specify the scp\_delete\_sourcedir attribute only if you specified the scp\_delete\_sourcefile attribute in your job definition.
- You can specify the scp\_delete\_sourcedir attribute even when multiple source files are specified using wildcards (\* or ?) in the source file names.
- You can specify the scp\_delete\_sourcedir attribute along with the scp\_delete\_sourcefile and scp\_rename\_sourcefile attributes provided that the file being renamed to refers to a different directory location than the one specified in the scp\_delete\_sourcedir attribute.

### Example: Delete the Source File Directory After the Source File is Downloaded From the Remote FTP Server

Suppose that you want to download the abc.txt file from the /ftproot/download directory on the SFTP server to the C:\temp directory on the agent computer and delete the /ftproot/download directory after the file is downloaded.

The file is downloaded using user authentication. The /ftproot/download directory contains only the abc.txt file.

```
insert_job: sftp_download
machine: AGENT_SFTP
scp_transfer_direction: DOWNLOAD
scp_server_name: abcde01-i12345
scp_remote_dir: /ftproot/download
scp_remote_name: abc.txt
scp_local_name: "C:\temp"
scp_protocol: SFTP
scp_delete_sourcefile: T
scp_delete_sourcedir: T
```

```
owner: causer@AGENT_SFTP
```

## scp\_delete\_sourcefile Attribute -- Specify Whether to Delete Source Files

The `scp_delete_sourcefile` attribute specifies whether you want to delete the local source files (if uploading) after they are uploaded from the agent computer or the remote source files (if downloading) after they are downloaded from the remote SFTP server.

### NOTE

Specify this attribute only if you are using SFTP protocol.

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) job type.

### Syntax

This attribute has the following format:

```
scp_delete_sourcefile: T | F
```

- **T**  
Deletes the local source files (if uploading) after they are uploaded from the agent computer or the remote source files (if downloading) after they are downloaded from the remote SFTP server.
- **F**  
Does not delete the local source files or the remote source files.

**Default:** F

### NOTE

- To delete the source file, you must have EXECUTE permissions to the directory that contains the source file.
- To delete multiple source files, use wildcards. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.

### Example: Delete the Source File After Uploaded From the Agent Computer

This example uploads the files in the `c:\temp\upload` directory to the `/u1/build/uploaded` directory on the aixunix server and deletes the files in the `c:\temp\upload` directory after it is uploaded. The files are uploaded using user authentication. The job uses SFTP.

```
insert_job: sftp_up_mult
job_type: SCP
machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: aixunix
scp_remote_dir: /u1/build/uploaded
scp_remote_name: "*"
scp_local_name: "c:\temp\upload*"
scp_protocol: SFTP
scp_delete_sourcefile: T
owner: causer@WINAGENT
```

## scp\_local\_name Attribute -- Specify the Local File to Transfer

The `scp_local_name` attribute specifies a file on the agent computer to be downloaded or uploaded.

## Supported Job Type

This attribute is required for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_local_name: file_name
```

- **file\_name**

Specifies the file's destination (if downloading) or the file's source location (if uploading). The local file or files are located on the computer where the agent is installed.

**Limits:** Up to 256 characters; case-sensitive

### NOTE

- For downloads, if the scp\_remote\_name does not contain wildcard characters, you must specify the full path and file name without wildcards. If the scp\_remote\_name does contain wildcard characters, the scp\_local\_name must specify the explicit name of the existing target directory.
- For uploads, you can use wildcards for the file name if the scp\_protocol attribute is set to SFTP. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character. If a wildcard is used in a local file name for upload, the scp\_remote\_name attribute is ignored. A wildcard transfer is equivalent to an mget transfer using an FTP client.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- You cannot rename files if wildcards are used.
- You cannot use wildcards in the path.
- You cannot use wildcards if the scp\_protocol attribute is set to SCP.
- You cannot use a semi-colon (;) to separate file names. To transfer multiple files, you can use wildcards.
- You can use global variables to specify the local file name to transfer. Use the \$\$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

### Example: Download a File from a Remote Server to the Agent Computer

This example downloads the install.log1 file from the root directory on the chi-linux server using the Secure Copy Protocol (SCP).

```
insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
owner: causer@WINAGENT
```

### Example: Upload Multiple Files from the Agent Computer to a Remote Server

This example uploads the files in the c:\temp\upload directory to the /u1/build/uploaded directory on the aixunix server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_up_mult
job_type: SCP
```



```

machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: aixunix
scp_remote_dir: /u1/build/uploaded
scp_remote_name: "*"
scp_local_name: "c:\temp\upload*"
scp_protocol: SFTP
owner: causer@WINAGENT

```

## scp\_local\_user Attribute -- Specify a User ID on the Agent Computer

The `scp_local_user` attribute specifies a user ID on the computer where the agent is installed. This user ID determines the access permissions on the agent computer.

### NOTE

- To use this attribute, you must specify `DOWNLOAD` for the `scp_transfer_direction` attribute.
- If you do not specify the `scp_local_user` attribute in your job definition, the job uses the user ID that defined the job (the default).

### Supported Job Type

This attribute is optional for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_local_user: user_id
```

- ***user\_id***  
Specifies a user ID on the computer where the agent is installed.  
**Default:** The user ID that defined the SCP job  
**Limits:** Up to 80 characters; case-sensitive; cannot contain delimiters

### NOTE

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- If the `scp_transfer_direction` attribute is set to `DOWNLOAD`, the downloaded file is created with this user as the file owner.
- The local user ID does not require a password to be stored on the scheduling manager.
- The `scp_local_user` attribute overrides the default setting specified in the `ftp.download.owner` parameter in the agent's `agentparm.txt`.

### Example: Download a File from a Remote Server to the Agent Computer

This example downloads the `install.log1` file from the root directory on the `chi-linux` server using the Secure Copy Protocol (SCP).

```

insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log

```

```
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
owner: causer@WINAGENT
scp_local_user: causer@WIAGENT
```

## scp\_protocol Attribute -- Specify the SCP Transfer Protocol to Use

The `scp_protocol` indicates whether the SCP data transfer uses Secure File Transfer Protocol (SFTP) or regular Secure Copy (SCP).

### NOTE

If you do not specify the `scp_protocol` attribute in your job definition, the job uses SFTP (the default).

### Supported Job Type

This attribute is optional for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_protocol: SCP | SFTP
```

- **SCP**  
Transfers files using the Secure Copy Protocol (SCP).
- **SFTP**  
Transfers files using Secure File Transfer Protocol (SFTP). This is the default.

### Example: Download a File Using SCP

This example downloads the `install.log1` file from the root directory on the `chi-linux` server using the Secure Copy Protocol (SCP).

```
insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
owner: causer@WINAGENT
```

### Example: Upload a File Using SFTP

This example uploads the `logs.tar` file to the `/u/tmp` directory on the `hpsupport` server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_upload
job_type: SCP
machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: hpsupport
scp_remote_dir: /u/tmp
scp_remote_name: logs.tar
scp_local_name: "D:\temp\logs.tar"
scp_protocol: SFTP
```

```
owner: causer@WINAGENT
```

## scp\_remote\_dir Attribute -- Specify a Remote Directory

The `scp_remote_dir` attribute specifies the file's remote source directory (if downloading) or the file's remote destination directory (if uploading).

### Supported Job Type

This attribute is required for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_remote_dir: directory
```

- **directory**

Specifies the file's remote source directory (if downloading) or the file's remote destination directory (if uploading).

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- Specify the values that contain embedded blanks with double quotation marks.
- You can use global variables to specify the file's remote directory. Use the `$$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

### Example: Upload Multiple Files from the Agent Computer to a Directory on a Remote Server

This example uploads the files in the `c:\temp\upload` directory to the `/u1/build/uploaded` directory on the `aixunix` server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_up_mult
job_type: SCP
machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: aixunix
scp_remote_dir: /u1/build/uploaded
scp_remote_name: "*"
scp_local_name: "c:\temp\upload*"
scp_protocol: SFTP
owner: causer@WINAGENT
```

## scp\_remote\_name Attribute -- Specify a Remote File Name

The `scp_remote_name` attribute specifies the file's source location (if downloading) or the file's destination (if uploading).

### Supported Job Type

This attribute is required for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_remote_name: file_name
```

- **file\_name**

Specifies the file's source location (if downloading) or the file's destination (if uploading). The remote file or files are located on the computer designated by the `scp_server_name` attribute value.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- For downloads, you can use wildcards for the file name if the `scp_protocol` attribute is set to SFTP. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character. If a wildcard is used, the `scp_local_name` (the target) must refer to a directory. A wildcard transfer is equivalent to an `mget` transfer using an FTP client.
- Specify the values that contain embedded blanks with double quotation marks.
- You cannot use wildcards if the `scp_protocol` attribute is set to SCP.
- You cannot rename files if wildcards are used.
- You cannot use a semi-colon (;) to separate file names.
- You can use global variables to specify the remote file name. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

#### Example: Download a File from a Remote Server to the Agent Computer

This example downloads the `install.log1` file from the root directory on the `chi-linux` server using the Secure Copy Protocol (SCP).

```
insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
owner: causer@WINAGENT
```

#### Example: Upload a File from the Agent Computer to a Remote Server

This example uploads the `logs.tar` file to the `/u/tmp` directory on the `hpsupport` server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_upload
job_type: SCP
machine: WINAGENT
scp_transfer_direction: UPLOAD
scp_server_name: hpsupport
scp_remote_dir: /u/tmp
scp_remote_name: logs.tar
scp_local_name: "D:\temp\logs.tar"
scp_protocol: SFTP
owner: causer@WINAGENT
```

#### **scp\_rename\_sourcefile Attribute -- Specify the Name That the Source File Must be Renamed to**

The `scp_rename_sourcefile` attribute specifies the name that the source file must be renamed to after the file is uploaded or downloaded.

#### NOTE

Specify this attribute only if you are using SFTP protocol.

#### Supported Job Type

This attribute is optional for the Secure Copy (SCP) job type.

### Syntax

This attribute has the following format:

```
scp_rename_sourcefile: sourcefilename
```

- **sourcefilename**

Specifies the name that the source file must be renamed to after the file is uploaded or downloaded. When renaming a local source file or a remote source file, you cannot use wildcards (\* or ?).

**Limits:** 256 characters

### NOTE

If you do not have WRITE permissions, the source file is downloaded but it is not renamed.

## scp\_server\_name Attribute -- Specify a Remote Server Name

The scp\_server\_name attribute specifies a remote server name.

### Supported Job Type

This attribute is required for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_server_name: server
```

- **server**

Specifies the DNS name or IP address of a remote server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** 172.24.36.107 (IPv4), 0:0:0:0:FFFF:192.168.00.00 (IPv6), or machine1 (DNS name)

### NOTE

Do not use embedded blanks, single quotation marks, or double quotation marks.

### Example: Specify a Remote Server Name

This example downloads the install.log1 file from the root directory on the chi-linux server using the Secure Copy Protocol (SCP).

```
insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
owner: causer@WINAGENT
```

## scp\_server\_port Attribute -- Specify a Remote Server Port

The `scp_server_port` specifies the port number of the remote server.

### NOTE

If you do not specify the `scp_server_port` attribute in your job definition, the job uses port 22 (the default).

### Supported Job Type

This attribute is optional for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_server_port: port
```

- **port**  
Specifies the port number of the remote server.  
**Default:** 22  
**Limits:** 0-65535

### Example: Specify the Port Number of the Remote Server

This example defines a Secure Copy Protocol job that downloads a file from a remote server using port 22.

```
insert_job: scp_download
job_type: SCP
machine: WINAGENT
scp_transfer_direction: DOWNLOAD
scp_server_name: chi-linux
scp_remote_dir: /root
scp_remote_name: install.log
scp_local_name: "C:\temp\install.log1"
scp_protocol: SCP
scp_server_port: 22
owner: causer@WINAGENT
```

## scp\_target\_os Attribute -- Specify the Remote Operating System Type

The `scp_target_os` specifies the remote operating system type, which is used to determine the path separator on the remote system.

### NOTE

If you do not specify the `scp_target_os` attribute in your job definition, the job uses the UNIX type (the default).

### Supported Job Type

This attribute is optional for the [Secure Copy \(SCP\) job type](#).

### Syntax

This attribute has the following format:

```
scp_target_os: UNIX | Windows | OpenVMS | Tandem | I5OS | MVS
```

- **UNIX**  
Specifies that the remote operating system is UNIX. This is the default.
- **Windows**

- Specifies that the remote operating system is Windows.
- **OpenVMS**  
Specifies that the remote operating system is OpenVMS.
- **Tandem**  
Specifies that the remote operating system is Tandem.
- **i5OS**  
Specifies that the remote operating system is i5/OS.
- **MVS**  
Specifies that the remote operating system is MVS.

### Example: Specify the Remote Operating System

This example defines a Secure Copy job that downloads a file from a remote UNIX server.

```
insert_job: scp_download
job_type: scp
machine: localhost
owner: ftpuser@ftpserver
scp_local_name: "c:\temp\scp.out"
scp_remote_dir: /etc
scp_remote_name: profile.CA
scp_server_name: ftpserver
scp_target_os: unix
scp_transfer_direction: download
```

### scp\_transfer\_direction Attribute -- Specify the File Transfer Direction

The `scp_transfer_direction` specifies the file transfer direction between the agent computer and the remote server.

#### NOTE

If you do not specify the `scp_transfer_direction` attribute in your job definition, the job downloads the specified file (the default).

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) job type.

### Syntax

This attribute has the following format:

```
scp_transfer_direction: UPLOAD | DOWNLOAD
```

- **UPLOAD**  
Transfers files from the agent computer to the remote server.
- **DOWNLOAD**  
Transfers files from the remote server to the agent computer. This is the default.

### Example: Upload Files Using SFTP

This example uploads the files in the `c:\temp\upload` directory to the `/u1/build/uploaded` directory on the aixunix server. The job uses the Secure File Transfer Protocol (SFTP).

```
insert_job: sftp_up_mult
job_type: SCP
machine: WINAGENT
```

```
scp_transfer_direction: UPLOAD
scp_server_name: aixunix
scp_remote_dir: /ul/build/uploaded
scp_remote_name: "*"
scp_local_name: "c:\temp\upload*"
scp_protocol: SFTP
owner: causer@WINAGENT
```

## scp\_transfer\_mode Attribute -- Specify the Type of Data to Transfer

The scp\_transfer\_mode specifies the type of data you are transferring.

### Supported Job Type

This attribute is optional for the Secure Copy (SCP) job type.

### Syntax

This attribute has the following syntax:

```
scp_transfer_mode: A | B
```

- **A**  
Specifies an ASCII transfer.
- **B**  
Specifies a binary transfer.

**Default:** B

### NOTE

- (i5/OS only) If the ASCII file to be transferred already exists on the target computer, the file is written using the encoding of the existing file. If the file does not exist, the file is written using the ASCII CCSID (Coded Character Set Identifier) defined on the agent. The default is 819.
- To transfer ASCII files, we recommend that the SFTP server that your agent computer communicates with is compliant with protocol level 4 or higher. To transfer ASCII files to or from an SFTP server that is compliant with protocol level 3 or lower, specify the operating system type of the SFTP server using the scp\_target\_os attribute.

## search\_bw Attribute -- Define the Time Used to Search Backwards for a z/OS Manual Job

The search\_bw attribute defines the number of hours that the agent does a backward search for a manually-submitted job.

### Supported Job Type

This attribute is optional for the [z/OS Manual \(ZOSM\) job type](#).

### Syntax

This attribute has the following format:

```
search_btw: hours
```

- **hours**  
Defines the number of hours that the agent does a backward search for a manually-submitted job.  
**Limits:** Up to 5 numeric digits

**Example: Search One Hour Backwards**



This example searches one hour backwards for a manually-submitted job.

```
insert_job: test_ZOSM
job_type: ZOSM
machine: zosagent
owner: user1@call
search_bw: 01
zos_jobname: CYBJAK11
```

## send\_notification Attribute -- Specify Whether to Send a Notification

The send\_notification attribute specifies whether to send a notification when the job completes successfully, fails, or terminates or when an alarm is raised.

### NOTE

If you do not specify the send\_notification attribute in your job definition, the job does not send a notification when it completes (the default).

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
send_notification: y | n | f
```

- **y**  
Sends a notification when the job completes successfully or terminates or when an alarm is raised.  
  

### NOTE

  - You can specify 1 instead of y.
  - The notification\_msg and notification\_template attributes are mutually exclusive.
  - You must specify the notification\_msg and the notification\_emailaddress attributes in your job definition.
  - To send an email notification using the email notification template when a job completes successfully or terminates, set the send\_notification attribute value to **y** and specify the notification\_template and the notification\_emailaddress, notification\_emailaddress\_on\_success, or notification\_emailaddress\_on\_terminated attributes in your job definition.
  - To send an email notification using the email notification template when an alarm is raised, set the send\_notification attribute value to **y or f** and specify the notification\_template, notification\_alarm\_types, and the notification\_emailaddress\_on\_alarm or notification\_emailaddress attributes in your job definition.
- **n**  
Does not send a notification. This is the default.  
  

### NOTE

You can specify 0 instead of n.
- **f**  
Sends a notification if the job completes with a FAILURE status or when an alarm is raised.

**NOTE**

- You can specify 2 or F instead of f.
- To send an email notification using the email notification template when a job fails, set the `send_notification` attribute value to **f** and specify the `notification_template` and the `notification_emailaddress` or `notification_emailaddress_on_failure` attributes in your job definition.

**Example: Send a Notification by Notification ID When the Job Completes**

This example specifies that a notification is sent to the user `ccsuser1` when the job completes, regardless of its status.

```
insert_job: email_job_1
machine: localhost job_type: c
command: as_test -t 2
send_notification: 1
notification_msg: The server appears to be down.
```

**Example: Send a Notification by Notification ID Only When the Job Completes with a FAILURE Status**

This example specifies that a notification is sent to the user `ccsuser1` when the job completes with a FAILURE status.

```
insert_job: email_job_1
machine: localhost job_type: c
command: as_test -t 2
send_notification: f
notification_msg: "notify_job_1 has failed."
```

**Example: Send an Email Notification When the Job Completes**

This example specifies that an email notification is sent to `admin@example.com` when the job completes, regardless of its status.

```
insert_job: email_job_1
machine: localhost job_type: c
command: as_test -t 2
send_notification: 1
notification_emailaddress: admin@example.com
notification_msg: The server appears to be down.
```

**Example: Send an Email Notification Using the Template When the Job Completes Successfully**

This example specifies that an email notification is sent using the emailtemplate email notification template to `admin@example.com` and `operator@example.com` when `email_job_1` job completes successfully.

```
insert_job: email_job_1
machine: localhost job_type: c
command: as_test -t 2
send_notification: y
notification_template: emailtemplate
notification_emailaddress: admin@example.com
notification_emailaddress_on_success: operator@example.com
notification_emailaddress_on_terminated: terminate@example.com
```

## service\_desk Attribute -- Specify Whether to Open a CA Service Desk Ticket When a Job Fails

The `service_desk` attribute specifies whether to open a CA Service Desk ticket (request or incident) when the job you are defining completes with a FAILURE status. When a job is defined to open a service desk ticket, the AutoSys Workload Automation scheduler initiates the opening of the ticket during terminal status processing. The scheduler prepares and sends the ticket. Messages are written to the scheduler log indicating whether the ticket was sent and processed successfully.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
service_desk: y | n
```

- **y**  
Opens a CA Service Desk request when the job fails.  
**Note:** You can specify 1 instead of y.
- **n**  
Does not open a CA Service Desk request when the job fails. This is the default.  
**Note:** You can specify 0 instead of n.

#### NOTE

Only the `service_desk` attribute is required to open a service desk ticket for a job. You can also specify the following optional service desk-related attributes in your job definition:

- `svcdesk_desc`
- `svcdesk_pri`
- `svcdesk_imp`
- `svcdesk_sev`
- `svcdesk_attr`

If the optional attributes are not set, the job uses the AutoSys Workload Automation service desk template values set in CA Service Desk. This template is included in the CA Service Desk installation. Before initiating a service desk ticket, you must ensure that the web services for CA Service Desk are active.

### Example: Initiate a Service Desk Incident

This example initiates a service desk incident with a priority of 1 for a job named `service_desk_on_failure_1`.

```
insert_job: service_desk_on_failure_1
machine: localhost
command: false
owner: user@localhost
service_desk: y
svcdesk_pri: 1
svcdesk_desc: "service_desk_on_failure_1 has failed."
```

### Example: Initiate a Service Desk Request

This example initiates a service desk request with an impact of 3 and a severity of 4 for a job named `service_desk_on_failure_2`.

```
insert_job: service_desk_on_failure_2
machine: localhost
```

```

command: false
owner: user@localhost
service_desk: y
svcdesk_imp: 3
svcdesk_sev: 4
svcdesk_desc: "service_desk_on_failure_2 has failed."

```

## service\_name Attribute -- Specify the Web Service Name Within the Target Namespace

The service\_name attribute specifies the web service name within the target namespace in a Web Service job.

### Supported Job Types

This attribute is required for the [Web Service Document/Literal \(WSDOC\) job type](#).

This attribute is optional for the [Web Service RPC/Encoded \(WBSVC\) job type](#).

### Syntax

This attribute has the following format:

```
service_name: service
```

- **service** Specifies the web service name within the target namespace.

**Limits:** Up to 255 characters; case-sensitive

### NOTE

- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, you must specify both the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL\_URL and endpoint\_URL attributes, then the service\_name and port\_name attributes are optional.
- You can use global variables to specify the web service name within the target namespace. Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.

### Example: Invoke Web Service to Get Stock Quote

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```

insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"

```

```
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

## shell Attribute -- Specify a UNIX Shell to Run a Script or Command

The shell attribute specifies the shell used to run a script or command in a CMD job on a UNIX computer.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#) on UNIX.

### Syntax

This attribute has the following format:

```
shell: shell_name
```

- ***shell\_name***

Specifies the name of the shell used to execute the script or command file.

You can specify any of the following UNIX shells:

- /bin/ksh (Korn shell)
- /bin/sh (Bourne shell)
- /bin/bash (Bourne again shell)
- /bin/csh (C shell)
- /usr/local/bin/perl (Perl shell)

**Limits:** Up to 20 characters

### NOTE

- The shell attribute applies to CMD jobs on UNIX only. This attribute is ignored for CMD jobs on Windows.
- The shell can be specified in different places. To run a UNIX script, the agent uses, in the following order, the shell specified in:
  - a. The shell attribute (if specified in the job definition)
  - b. The first line of the script (if the shell attribute is not specified)
  - c. The `oscomponent.defaultshell` parameter in the `agentparm.txt` file (if not specified in the shell attribute or in the script)
  - d. The user default shell defined in the user profile (if not specified in one of the previous three locations)
- The shell attribute only applies to the command attribute. The script is invoked using the specified shell.
- The shell attribute does not apply to the profile attribute. The job profile is always sourced using the job owner's default shell, which is set for the user in the `etc/passwd` file.
- If the `oscomponent.checkvalidshell` parameter in the agent's `agentparm.txt` file is set to true (the default), all shells that are used must be specified using the `oscomponent.validshell` parameter. If the shell you want to use in your job definition or script is not defined on the agent, the job fails. For more information about specifying valid shells in the `agentparm.txt` file, see [Workload Automation Agent Parameters](#).
- Different shells provide different facilities and have different characteristics. Some functions are specific to certain shells and may be incompatible with other shells.

### Example: Specify a Shell to Run a Script

This example uses the C shell to run the sort script.

```
insert_job: MFGSORT
job_type: CMD
machine: unixagt
command: /mfg/test/sort
shell: /bin/csh
```

## snmp\_auth\_protocol Attribute -- Specify the SNMP v3 Authentication Protocol

The `snmp_auth_protocol` attribute specifies the SNMP v3 authentication protocol to use when connecting with the user specified in the `snmp_privacy_user` attribute.

### NOTE

If you do not specify the `snmp_auth_protocol` attribute in your job definition, the job uses the Message Digest 5 (MD5) protocol (the default).

### Supported Job Types

If your SNMP version is v3, this attribute is required for the following job types:

- [SNMP Value Get](#) (SNMPGET)
- [SNMP Value Set](#) (SNMPSET)

### NOTE

If you use this attribute with SNMP v1 or v2, an error is displayed and the job is not inserted.

### Syntax

This attribute has the following format:

```
snmp_auth_protocol: MD5 | SHA
```

- **MD5**  
Specifies the Message Digest 5 (MD5) authentication protocol. This is the default.
- **SHA**  
Specifies the Secure Hash (SHA) authentication protocol.

### Example: Specify the Authentication Protocol in an SNMP Value Set Job

Suppose that you want to set the value of the `out456789` variable using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The credentials of user `user1` are used for authorization.

```
insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8
```

## snmp\_comm\_string Attribute -- Specify the SNMP v1 or v2 Read or Write Community String

The `snmp_comm_string` attribute specifies the SNMP v1 or v2 read community string for `SNMPGET` and the SNMP v1 or v2 write community string for `SNMPSET`.

### NOTE

If you do not specify the `snmp_comm_string` attribute in your job definition, the job uses `public` (the default).

### Supported Job Types

If your SNMP version is v1 or v2, this attribute is required for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

### NOTE

If you use the `snmp_comm_string` attribute with SNMP v3, the attribute is ignored by the agent.

### Syntax

This attribute has the following format:

```
snmp_comm_string: community
```

- ***community***  
Specifies the community string that is used to authenticate against the network device.  
**Limits:** Up to 256 characters; case-sensitive  
**Default:** `public`

### Example: Specify the Read Community String in an SNMP Value Get Job

Suppose that you want to know the value of the `agentVersion` variable hosted by a network device. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the read community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```
insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public
```

## snmp\_context\_engine\_id Attribute -- Specify the SNMP v3 Context Engine ID

The `snmp_context_engine_id` attribute specifies the context engine ID, which uniquely identifies an SNMP entity within an administrative domain. An SNMP entity can access management information in many contexts. Since an item of management information can exist in more than one context, you can use the `snmp_context_engine_id` attribute, together with the `snmp_context_name` attribute, to identify a unique variable within an administrative domain.

### Supported Job Types

If your SNMP version is v3, you can specify this attribute for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

**NOTE**

If you use this attribute with SNMP v1 or v2, an error is displayed and the job is not inserted.

**Syntax**

This attribute has the following format:

```
snmp_context_engine_id: context_engine_id'
```

- ***context\_engine\_id***

Specifies the context engine ID in hexadecimal format.

**Limits:** Up to 12 hexadecimal digits (24 characters)

**Example: Specify a Context Engine ID in an SNMP Value Set Job**

Suppose that you want to set the value of a context engine ID that belongs to a context using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The out456789 variable belongs to the context named contxtname within the SNMP entity identified by the hexadecimal value CAB0. The credentials of user user1 are used for authorization.

```
insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_context_name: contxtname
snmp_context_engine_id: 8000000006
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8
```

**snmp\_context\_name Attribute -- Specify a SNMP v3 Context Name**

The snmp\_context\_name attribute specifies the name of the context that the variable belongs to. An SNMP context is a collection of management information accessible by an SNMP entity. Since an item of management information can exist in more than one context, you can use the snmp\_context\_name attribute, together with the snmp\_context\_engine\_id attribute, to identify a unique variable within an administrative domain.

**Supported Job Types**

If your SNMP version is v3, you can specify this attribute for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

**NOTE**

If you use this attribute with SNMP v1 or v2, an error is displayed and the job is not inserted.

**Syntax**

This attribute has the following format:

```
snmp_context_name: context_name
```



- **context\_name**

Specifies the name of the context that the variable belongs to.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

Enclose values that contain spaces in quotation marks.

**Example:** "contxt name"

**Example: Set the Value of a Variable that Belongs to a Context**

Suppose that you want to set the value of a variable that belongs to a context using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The out456789 variable belongs to the context named contxtname within the SNMP entity identified by the hexadecimal value CAB0. The credentials of user user1 are used for authorization.

```
insert_job: setvarv3

job_type: SNMPSET

machine: snmpagent

snmp_mib: cybermation.mib

snmp_host: host.example.com

snmp_oid: out456789

snmp_version: 3

snmp_context_name: contxtname

snmp_context_engine_id: X'cab0'

snmp_privacy: AES

snmp_privacy_user: user1

snmp_auth_protocol: MD5

snmp_value: int=8
```

**snmp\_host Attribute -- Specify an SNMP Host Name or IP Address**

The snmp\_host attribute specifies the host name or IP address of the network device that hosts the MIB variable.

**Supported Job Types**

This attribute is required for the following job types:

- [SNMP Value Get](#) (SNMPGET)
- [SNMP Value Set](#) (SNMPSET)

### Syntax

This attribute has the following format:

```
snmp_host: host
```

### *host*

Specifies the host name or IP address of the network device.

**Limits:** Up to 256 characters; case-sensitive

### Example: Specify the Host Name in an SNMP Value Get Job

Suppose that you want to know the value of the agentVersion variable hosted by a network device. In this example, the host name of the network device is host.example.com and its port is 161. The SNMP version is v2 and the read community string is public. The name of the MIB file is cybermation.mib, which is located in the agent installation directory.

```
insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public
```

## snmp\_mib Attribute -- Specify the Name of the MIB File

The snmp\_mib attribute specifies the name of the MIB (management information base) file in an SNMP job.

### Supported Job Types

This attribute is optional for the following job types:

- [SNMP Value Get](#) (SNMPGET)
- [SNMP Value Set](#) (SNMPSET)

### Syntax

This attribute has the following format:

```
snmp_mib: file_name
```

- ***file\_name***  
Specifies the relative or absolute path to and name of the MIB **file**.  
**Limits:** Up to 256 characters; case-sensitive

#### NOTE

If the MIB file is not loaded successfully, the job fails.

#### NOTE

Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

### Example: Specify the MIB File Name Using a Relative Path

Suppose that you want to know the value of the `agentVersion` variable hosted by a network device. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the read community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```
insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public
```

## snmp\_oid Attribute -- Specify an SNMP OID

The `snmp_oid` attribute specifies the SNMP OID (object identifier).

### Supported Job Types

This attribute is required for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

### Syntax

This attribute has the following format:

```
snmp_oid: oid
```

- **oid**

Specifies the SNMP OID (object identifier) in numeric or string format. In an SNMP Value Get job, it identifies the variable whose value you want to retrieve. In an SNMP Value Set job, it identifies the variable whose value you want to change.

**Limits:** Up to 80 characters; case-sensitive

**Examples:** `agentLogLevel`, `.1.3.6.1.2.1.1.6.0`

#### NOTE

If you specify the `snmp_oid` attribute in string format, you must also specify the `snmp_mib` attribute for the job to complete successfully.

### Example: Specify the OID in String Format

Suppose that you want to know the value of the `agentVersion` variable hosted by a network device. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the read community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```
insert_job: getvarv2
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
```

```
snmp_version: 2
snmp_comm_string: public
```

### Example: Specify the OID in Numeric Format

Suppose that you want to set the value of a variable to an IP address. In this example, the job specifies the OID in numeric format, the DES privacy protocol, and the MD5 authentication protocol. The credentials of user user1 are used for authorization.

```
insert_job: setvarip
job_type: SNMPSET
machine: localhost
owner: Administrator@localhost
snmp_oid: .1.3.1.1.2.1.9.3
snmp_host: snmp
snmp_value: addr=172.24.2.20
snmp_version: 3
snmp_auth_protocol: MD5
snmp_privacy: DES
snmp_privacy_user: user1
```

## snmp\_privacy Attribute -- Specify the SNMP v3 Privacy Protocol

The snmp\_privacy attribute specifies the SNMP v3 privacy protocol to use.

### NOTE

If you do not specify the snmp\_privacy attribute in your job definition, the job uses the Data Encryption Standard (DES) protocol (the default).

### Supported Job Types

If your SNMP version is v3, this attribute is required for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

### NOTE

If you use this attribute with SNMP v1 or v2, an error is displayed and the job is not inserted.

### Syntax

This attribute has the following format:

```
snmp_privacy: DES | AES
```

- **DES**  
Specifies the Data Encryption Standard (DES) privacy protocol. DES uses a 16-character encryption key. This is the default.
- **AES**  
Specifies the Advanced Encryption Standard (AES) privacy protocol. AES uses a 32-character encryption key. AES is the algorithm required by U.S. Government organizations to protect sensitive (unclassified) information (FIPS-140-2 compliance).

### Example: Specify the Privacy Protocol in an SNMP Value Set Job

Suppose that you want to set the value of the out456789 variable using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The credentials of user user1 are used for authorization.

```

insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8

```

## snmp\_privacy\_user Attribute -- -- Specify the SNMP v3 User Name

The `snmp_privacy_user` attribute specifies the user name whose credentials are used for authorization in SNMP v3.

### Supported Job Types

If your SNMP version is v3, this attribute is required for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

#### NOTE

- If you use this attribute with SNMP v1 or v2, an error is displayed and the job is not inserted.
- For the job to run successfully, you must define the `snmp_privacy_user` attribute using `autosys_secure`. The password specified using `autosys_secure` and the password in the `management.snmp.agent.user.priv.password` agent parameter in the `agentparm.txt` file must match.

### Syntax

This attribute has the following format:

```
snmp_privacy_user: user_name
```

- ***user\_name***  
Specifies the user name whose credentials are used for authentication.  
**Limits:** Up to 145 characters; case-sensitive; it cannot contain delimiters (such as spaces)

### Example: Specify the SNMP User in an SNMP Value Set Job

Suppose that you want to set the value of the `out456789` variable using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The credentials of user `user1` are used for authorization.

```

insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8

```

## snmp\_subtree Attribute -- Specify the SNMP Subtree Value

The `snmp_subtree` attribute specifies whether to walk the SNMP subtree starting with the `snmp_oid` attribute.

### NOTE

If you do not specify the `snmp_subtree` attribute in your job definition, the job does not walk the SNMP subtree (the default).

### Supported Job Type

This attribute is optional for the [SNMP Value Get \(SNMPGET\)](#) job type.

### Syntax

This attribute has the following format:

```
snmp_subtree: y | n
```

- **y**  
Specifies that the job walks the SNMP subtree. You can specify this value in either uppercase or lowercase.  
  
**NOTE**  
You can specify 1 instead of y.
- **n**  
Specifies that the job does not walk the SNMP subtree. You can specify this value in either uppercase or lowercase. This is the default.

### NOTE

You can specify 0 instead of n.

### Example: Query a Network Device for the Values of an SNMP Subtree in an SNMP Value Get Job

Suppose that you want to retrieve the values of an SNMP subtree starting with the `agentVersion` OID.

```
insert_job: getsubtree
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentVersion
port: 161
snmp_version: 2
snmp_comm_string: public
snmp_subtree: y
```

## snmp\_table\_view Attribute -- Specify the Format for MIB Data

The `snmp_table_view` attribute specifies whether the job retrieves the MIB data in a table format. The table is stored as job output variables, which you can use as input to other jobs in the application.

### NOTE

- To use the `snmp_table_view` attribute, you must specify y or 1 for the `snmp_subtree` attribute.
- If you do not specify the `snmp_table_view` attribute in your job definition, the job retrieves the MIB data in the `oid=value` format (the default).

### Supported Job Type

This attribute is optional for the [SNMP Value Get \(SNMPGET\)](#) job type.

## Syntax

This attribute has the following format:

```
snmp_table_view: y | n
```

- **y**  
Specifies that the job retrieves the MIB data in table format. You can specify this value in either uppercase or lowercase.

### NOTE

You can specify 1 instead of y.

- **n**  
Specifies that the job retrieves the MIB data in the oid=value format. You can specify this value in either uppercase or lowercase. This is the default.

### NOTE

You can specify 0 instead of n.

## Example: Specify a table format to retrieve MIB data in an SNMP Value Get Job

Suppose that you want to specify the format to retrieve the MIB data in a table format.

```
insert_job: getsubtree
job_type: SNMPGET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: tcpipChannelsTable
port: 161
snmp_version: 3
snmp_privacy_user: user@host
snmp_comm_string: public
snmp_subtree: y
snmp_table_view: y
```

## snmp\_value Attribute -- Specify the SNMP Parameter Type and Value

The snmp\_value attribute specifies the type and value of the variable that you are changing in an SNMP Value Set job.

### Supported Job Type

This attribute is required for the [SNMP Value Set \(SNMPSET\)](#) job type.

## Syntax

This attribute has the following format:

```
snmp_value: type=value[, type=value...]
```

- **type=value**  
Specifies the SNMP parameter type and value. The value is converted to the corresponding type. Options for the type include the following:
  - **int**  
Specifies an integer value.
  - **uint**

Specifies an unsigned integer value.

– **ticks**

Specifies a time ticks (measured in hundredths of a second) value.

– **addr**

Specifies an IP address value.

**Note:** SNMP only supports IPv4 addresses. IPv6 addresses are not supported.

– **oid**

Specifies an OID (object identifier) value.

– **string**

Specifies an octet string value.

– **string.x**

Specifies an octet string value represented in hexadecimal format.

– **counter32**

Specifies a counter32 value. A counter32 value is a non-negative integer that increases until it reaches a maximum value of 232 - 1 (4294967295). It then wraps around and starts increasing again from zero.

– **gauge32**

Specifies a gauge32 value. A gauge32 value is a non-negative integer that can increase or decrease, but cannot exceed a maximum value. The maximum value cannot exceed 232 - 1 (4294967295). The value of a Gauge has its maximum value whenever the information being modeled is greater or equal to that maximum value; if the information being modeled then decreases below the maximum value, the Gauge also decreases.

**Limits:** Up to 1024 characters; case-sensitive

**NOTE**

- You can define multiple parameters using one `snmp_value` attribute or multiple entries of `snmp_value`.
- In each `snmp_value` entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4096 characters.
- Enclose values that contain spaces in quotation marks.

**Example: Change the Value of an SNMP Variable using an Integer Value**

Suppose that you want to set the value of the `agentLogLevel` variable to its highest level to diagnose a problem. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the write community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```
insert_job: setvarint
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentLogLevel
port: 161
snmp_version: 2
snmp_comm_string: public
snmp_value: int=8
```

**Example: Change the Value of an SNMP Variable using an Address Value**

Suppose that you want to set the value of a variable to an IP address. In this example, the job specifies the OID in numeric format, the DES privacy protocol, and the MD5 authentication protocol. The credentials of user `user1` are used for authorization.

```
insert_job: setvarip
```



```

job_type: SNMPSET
machine: localhost
owner: Administrator@localhost
snmp_oid: .1.3.1.1.2.1.9.3
snmp_host: snmp
snmp_value: addr=172.24.2.20
snmp_version: 3
snmp_auth_protocol: MD5
snmp_privacy: DES
snmp_privacy_user: user1

```

## snmp\_version Attribute -- Specify the SNMP Version

The `snmp_version` attribute specifies the SNMP version used when connecting to the network device.

### NOTE

If you do not specify the `snmp_version` attribute in your job definition, the job uses SNMP Version 1 (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [SNMP Value Get \(SNMPGET\)](#)
- [SNMP Value Set \(SNMPSET\)](#)

### Syntax

This attribute has the following format:

```
snmp_version: 1 | 2 | 3
```

- **1**  
Specifies SNMP Version 1 (v1). This is the default.
- **2**  
Specifies SNMP Version 2 (v2).
- **3**  
Specifies SNMP Version 3 (v3).

### Example: Change the Value of an SNMP Variable using SNMP v2

Suppose that you want to set the value of the `agentLogLevel` variable to its highest level to diagnose a problem. In this example, the host name of the network device is `host.example.com` and its port is 161. The SNMP version is v2 and the write community string is `public`. The name of the MIB file is `cybermation.mib`, which is located in the agent installation directory.

```

insert_job: setvarint
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: agentLogLevel
port: 161
snmp_version: 2
snmp_comm_string: public

```

```
snmp_value: int=8
```

### Example: Change the Value of an SNMP Variable using SNMP v3

Suppose that you want to set the value of the out456789 variable using SNMP v3. In this example, the job uses the AES privacy protocol and the MD5 authentication protocol. The credentials of user user1 are used for authorization.

```
insert_job: setvarv3
job_type: SNMPSET
machine: snmpagent
snmp_mib: cybermation.mib
snmp_host: host.example.com
snmp_oid: out456789
snmp_version: 3
snmp_privacy: AES
snmp_privacy_user: user1
snmp_auth_protocol: MD5
snmp_value: int=8
```

## soap\_action Attribute -- Defines the SOAP HTTP Request

The soap\_action attribute specifies the HTTP request header field that indicates the intent of the SOAP HTTP request.

### Supported Job Types

This attribute is optional for the [Define a Web Service Document/Literal Job](#) type.

### Syntax

This attribute has the following format:

```
soap_action: webmethod/soapaction
```

### Example:

```
http://quickstart.samples/intGetMax
```

## soap\_version Attribute -- Displays the SOAP Version

The soap\_version attribute displays the version of the SOAP message.

### Supported Job Types

This attribute is optional for the [Define a Web Service Document/Literal Job](#) type.

### Syntax

This attribute has the following format:

```
soap_version: 1.2
```

**Default:** 1.1

## sp\_arg Attribute -- Specify a Parameter to Pass to a Stored Procedure

The sp\_arg attribute specifies a parameter to pass to a database stored procedure.

### Supported Job Type

This attribute is optional for the [Database Stored Procedure \(DBPROC\) job type](#).

### Syntax

This attribute has the following format:

```
sp_arg: [ignore=TRUE|FALSE,]name=arg_name, argtype=IN|OUT|INOUT, datatype=type,
value=arg_value
```

- **ignore=TRUE | FALSE** (Optional) Specifies whether the value is returned in the spool file. Options are the following:
  - TRUE -- Specifies that the value is not returned.
  - FALSE -- Specifies that the value is returned. This is the default.

#### NOTE

This argument only applies to output parameters.

- **name=arg\_name** Specifies the name of the parameter.

**Limits:** Up to 64 characters

#### NOTE

You can use global variables to specify the name of the parameter. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

- **argtype=IN | OUT | INOUT** Specifies the parameter type. Options are the following:
  - IN -- Specifies that the parameter is an input parameter.
  - OUT -- Specifies that the parameter is an output parameter.
  - INOUT -- Specifies that the parameter is an input-output parameter.
- **datatype=type** Specifies the database [data type](#) of the parameter. The following data types are supported:
  - CHAR
  - VARCHAR
  - LONGVARCHAR
  - NUMERIC
  - DECIMAL
  - BIT
  - BOOLEAN
  - TINYINT
  - SMALLINT
  - INTEGER
  - BIGINT
  - REAL
  - FLOAT
  - DOUBLE
  - DATE
  - TIME
  - TIMESTAMP

- **value=arg\_value** Specifies the value of the parameter.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

- This value applies to input parameters only.
- Enclose values that contain spaces in quotation marks.
- You can use global variables to specify the value of the parameter. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

**NOTE**

To specify multiple parameters to pass to a stored procedure, define a separate `sp_arg` attribute for each parameter. Specify the parameters in the order that they need be passed to the stored procedure.

**Example: Pass Two Parameters to a Stored Procedure**

This example passes two parameters to the `UPDATE_INV` stored procedure. `QUANTITY` is an input parameter of type integer with a value of 3000, and `NEW_INV` is an output parameter of type integer.

```
insert_job: SPJOB
job_type: DBPROC
machine: DB_agent
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
sp_name: UPDATE_INV
sp_arg: name=QUANTITY, argtype=IN, datatype=INTEGER, value=3000
sp_arg: name=NEW_INV, argtype=OUT, datatype=INTEGER
```

**Supported Data Types**

The following table lists the supported data types for different databases:

**NOTE**

The data types listed in the database columns are defined in the procedure definition within the database. In the job definition, use the corresponding JDBC data type from the first column.

BC Data Type	Valid Input Format	Oracle	Microsoft SQL Server	DB2
CHAR	Plain text	CHAR	CHAR	CHAR
VARCHAR	Plain text	VARCHAR2(x) VARCHAR(x)	VARCHAR(x)	VARCHAR(x)
LONGVARCHAR		Not supported	Not supported	Not supported
NUMERIC	Number	NUMBER, NUMERIC	NUMERIC	NUMERIC
DECIMAL	Number	DECIMAL	DECIMAL	
BIT		Not supported	BIT	Not supported
BOOLEAN		Not supported	Not supported	Not supported
TINYINT		Not supported	TINYINT	Not supported
SMALLINT	Number	SMALLINT	SMALLINT	SMALLINT
INTEGER	Number	INTEGER	INT	INTEGER
BIGINT	Number	Not supported	BIGINT	BIGINT
REAL	Number [dot Number]	REAL	REAL	REAL
FLOAT	Number [dot Number]	FLOAT	FLOAT	FLOAT
DOUBLE			FLOAT	DOUBLE, DECIMAL
DATE	yyyy-mm-dd	DATE	Not supported	DATE
TIME	hh:mm:ss		Not supported	TIME
TIMESTAMP	yyyy-mm-dd hh:mm:ss.ffffff	TIMESTAMP	datetime smalldatetime	TIMESTAMP

BINARY	Hex (such as 010203FF)		BINARY	
--------	------------------------	--	--------	--

### Handling Unsupported Data Types

Although JDBC provides functionality for Booleans, the Oracle database driver does not. To handle unsupported data types such as Booleans, you can call a wrapper procedure in the job definition. The wrapper procedure converts the inputted values to values supported by the database driver and calls the desired stored procedure from inside the database.

#### Example: Run a Wrapper Procedure Named boolwrap

In the following example, the agent runs a wrapper procedure named boolwrap, which converts the inputted integer value to a Boolean and calls the boolproc stored procedure.

```
CREATE OR REPLACE PROCEDURE boolproc(x boolean) AS BEGIN
[...]
```

```
END;
```

```
CREATE OR REPLACE PROCEDURE boolwrap(x int) AS BEGIN
IF (x=1) THEN
 boolproc(TRUE);
ELSE
 boolproc(FALSE);
END IF;
END;
```

### sp\_name Attribute -- Specify Stored Procedure or Stored Function to Run

The sp\_name attribute specifies the database stored procedure to run. If you use an Oracle or SQL Server database, you can also use the sp\_name attribute to specify a stored function.

#### Supported Job Type

This attribute is required for the [Database Stored Procedure \(DBPROC\) job type](#).

#### Syntax

This attribute has the following format:

```
sp_name: stored_procedure
```

- **stored\_procedure**

Specifies the stored procedure to run. You can specify a stored function if you use an Oracle or SQL Server database.

**Limits:** Up to 300 characters; case-sensitive

#### NOTE

- The agent does not support stored functions (also called user-defined functions) on DB2.
- You can use global variables to specify the stored procedure or stored function to run. Use the \$global\_name, \$\$global\_name, or \$\$global\_name format to reference global variables.

#### Example: Run a Stored Procedure

This example invokes the PAYROLL procedure and checks whether the employee name (ename) starts with John. The procedure is stored in a SQL Server database named ORDERS. The database user ID is set to the user who invokes jil

to define the job (the default owner). When the procedure runs, the parameter `ename`, which is both an input and output parameter of type `VARCHAR`, is passed with the value John Evans.

```
insert_job: SPJOB
job_type: DBPROC
machine: DB_agent
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=ORDERS"
sp_name: PAYROLL
success_criteria: ename=John.*
sp_arg: name=ename, argtype=INOUT, datatype=VARCHAR, value=John Evans
```

## spool\_file Attribute -- Specify the Path and File Name of the Spool File

The `spool_file` attribute specifies the path to the spool file.

### NOTE

If you do not specify the `spool_file` attribute in your job definition, the job uses the default spool location specified in the agent's custom properties file.

### Supported Job Type

This attribute is optional for the [Remote Execution \(PROXY\) job type](#).

### Syntax

This attribute has the following format:

`spool_file: path`

- ***path***

Specifies the path to the spool file when you do not want to use the default location. If a spool file is specified, it overwrites a previous spool file with the same name unless the name starts with `>>`.

**Limits:** Up to 256 characters; can contain any characters and delimiters; case-sensitive

### NOTE

- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- The `spool_file` attribute overrides the default spool location specified in the custom properties file for the remote system.
- The user who executes the job must have read and write access to the spool file directory.
- The job log resides in the spool directory on the agent computer.

### NOTE

#### OpenVMS

- The spool file has to be a complete file name ending with `.LOG`. For example, `TEST.LOG`.
- If you do not provide a device and directory, your default device and directory are used to store the spool file. For example, if the spool file definition is `TEST.LOG`, `TEST.LOG` is stored under the current default directory, which can be found using the `SHOW DEFAULT` command after login.
- If you provide an incomplete spool file definition, the agent reports an invocation error. The following examples are incomplete spool file definitions because they do not end with `.LOG`: `TEST`, `TEST.OUT`, `DEV $USER:[CAUSR05]TEST`. The following example has a valid directory name but is incomplete because it does not have a valid file name: `DEV$USER:[CAUSR05]`.
- Appending to the spool file is not supported.
- Using a logical name in the `spool_file` attribute is not supported.

### Example: Specify a Path to a Spool File

This example specifies a spool file location other than the agent default without overwriting previous spool files with the same name.

```
insert_job: ls_cmd
job_type: PROXY
machine: proxyagent
remote_target: hpserver
remote_command: ls
spool_file: >>/opt/rootdev/CA_USER/test/ls.out
owner: user1@proxytest
```

### sql\_command Attribute -- Specify a SQL Statement to Run

The sql\_command attribute specifies an SQL statement to run against a database table.

#### Supported Job Type

This attribute is required for the [SQL job type](#).

#### Syntax

This attribute has the following format:

```
sql_command: sql_statement
```

- **sql\_statement**

Specifies the SQL statement to run against a database table.

**Limits:** Up to 2048 characters; case-sensitive

**Example:** "select emp\_name from emp where emp\_name like 'S%'"

#### NOTE

The value can contain any SQL statement including SELECT, DELETE, UPDATE, or INSERT.

#### NOTE

- To use the sql\_command attribute, you must know how to construct an SQL query.
- Specify the values that contain embedded blanks with double quotation marks.
- You can use global variables to specify the SQL statement to run. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Return Data from a Table that Match a Condition

This example queries a database table named emp for an emp\_name value that starts with S, followed by at least two characters. If the result starts with S, followed by one other character, the job completes. Otherwise, the job fails. The job's output is written to c:\temp\sql\_job1\_out.log.

```
insert_job: sql_job1
job_type: SQL
machine: SQL_server1
sql_command: select emp_name from emp where emp_name like 'S%'
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
success_criteria: emp_name=S.*
destination_file: "c:\temp\sql_job1_out.log"
```

## sqlagent\_domain\_name Attribute -- Specify the Windows Domain Name

The sqlagent\_domain\_name attribute specifies the Windows domain name when Windows authentication is used to connect to the database server.

### Supported Job Type

This attribute is optional for the SQL Server Agent job type.

### Syntax

This attribute has the following format:

```
sqlagent_domain_name: domain_name
```

- **domain\_name** Specifies the Windows domain name when Windows authentication is used to connect to the database server. **Limits:** Up to 24 alphanumeric characters; you can also specify hyphen (-) and period (.).

### NOTE

- If you specify both the sqlagent\_user\_name and sqlagent\_domain\_name attributes in your job definition, the agent assumes that the sqlagent\_user\_name attribute value contains the Windows domain user. In this case, Windows domain authentication is used.
- If you specify the sqlagent\_user\_name attribute and do not specify the sqlagent\_domain\_name attribute in your job definition, the agent assumes that the sqlagent\_user\_name attribute value contains the Microsoft SQL Server user. In this case, Microsoft SQL Server authentication is used.
- If you do not specify both the sqlagent\_user\_name and sqlagent\_domain\_name attributes in your job definition, the agent uses the default user settings specified in the *target\_database.properties* file.

## sqlagent\_jobid Attribute -- Specify the Job Identification Number

The sqlagent\_jobid attribute specifies the SQL Server Agent job identification number.

### Supported Job Type

This attribute is required for the SQL Server Agent job type.

### Syntax

This attribute has the following format:

```
sqlagent_jobid: jobid
```

- **jobid** Specifies the SQL Server Agent job identification number. **Limits:** Up to 36 characters

### NOTE

The sqlagent\_jobname and sqlagent\_jobid attributes are mutually exclusive. You can specify any one of these in your job definition.

## sqlagent\_jobname Attribute -- Specify the Job Name

The sqlagent\_jobname attribute specifies the name for an SQL Server Agent job.

### Supported Job Type

This attribute is required for the SQL Server Agent job type.

### Syntax

This attribute has the following format:



`sqlagent_jobname: jobname`

- ***jobname***

Specifies the name for an SQL Server Agent job. **Limits:** Up to 128 characters

**NOTE**

- The `sqlagent_jobname` and `sqlagent_jobid` attributes are mutually exclusive. You can specify any one of these in your job definition.
- The agent does not support double quotes in the job name. If the job name includes double quotes, remove the double quotes from the job definition using Microsoft SQL Server Management Studio.
- You can use global variables to specify the name for an SQL Server Agent job. Use the `$$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

## **sqlagent\_server\_name Attribute -- Specify the MSSQL Server**

The `sqlagent_server_name` attribute specifies the MSSQL server to start the job on.

### **Supported Job Type**

This attribute is optional for the SQL Server Agent job type.

### **Syntax**

This attribute has the following format:

`sqlagent_server_name: mssql_server`

- ***mssql\_server***

Specifies the MSSQL server to start the job on.

**Limits:** Up to 128 characters

**NOTE**

You can use global variables to specify the MSSQL server. Use the `$$global_name`, `$(global_name)`, or `$(global_name)` format to reference global variables.

## **sqlagent\_step\_name Attribute -- Specify the Name of the Job Step**

The `sqlagent_step_name` attribute specifies the name of the job step to start the job from.

### **Supported Job Type**

This attribute is optional for the SQL Server Agent job type.

### **Syntax**

This attribute has the following format:

`sqlagent_step_name: step_name`

- ***step\_name***

Specifies the name of the job step to start the job from. The default is to start the job from the first step. To start the job from another step, specify the `sqlagent_step_name` attribute in your job definition. **Limits:** Up to 128 characters

**NOTE**

You can use global variables to specify the name of the job step. Use the `$$global_name`, `$(global_name)`, or `$(global_name)` format to reference global variables.

## sqlagent\_target\_db Attribute -- Specify the Target Server Database

The sqlagent\_target\_db attribute specifies the target server database.

### Supported Job Type

This attribute is optional for the SQL Server Agent job type.

### Syntax

This attribute has the following format:

```
sqlagent_target_db : target_db
```

- **target\_db**

Specifies the target server database. The agent retrieves the database connection information from the corresponding property file (target\_database.properties) stored on the agent. **Limits:** Up to 128 alphanumeric or underscore characters. The first character cannot be a number.

#### NOTE

- If you do not specify the target database in your job definition, the agent uses the default value specified in the mssql.default.TargetDB parameter in the agentparm.txt file.
- If the target server database is not specified in either the job definition or the mssql.default.TargetDB parameter, the agent reports an error.
- You can use global variables to specify the target server database. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

## sqlagent\_user\_name Attribute -- Specify the sqlagent User Name

The sqlagent\_user\_name attribute specifies the user that the agent uses to connect to the database. The user can be a Windows domain user for Windows domain authentication or a Microsoft SQL Server user for Microsoft SQL Server authentication.

### Supported Job Type

This attribute is optional for the SQL Server Agent job type.

### Syntax

This attribute has the following format:

```
sqlagent_user_name: sqlagent_user@token
```

- **sqlagent\_user@token**

Specifies the user that the agent uses to connect to the database. Specify this value in the user@token format and it must match the user@host value that you specify using the autosys\_secure command. The token portion of the value can be any arbitrary string that enables the retrieval of credentials entered against user@host using the autosys\_secure command. **Limits:** Up to 145 characters; up to 64 characters for user and up to 80 characters for token. Do not use colons, embedded blanks, single quotation marks, or double quotation marks.

#### NOTE

If you specify the sqlagent\_user\_name attribute in your job definition, you must set the password that is associated with the sqlagent user using the autosys\_secure command.

## start\_mins Attribute -- Define the Minutes to Run a Job

The `start_mins` attribute defines the number of minutes after each hour at which the job should start on the days specified in the `days_of_week` attribute or on the dates specified in the calendar identified in the `run_calendar` attribute.

### NOTE

Times set in the `start_times` and `start_mins` attributes override times set in a custom calendar. The `date_conditions` attribute controls the usage of this attribute.

You cannot specify both the `start_times` and `start_mins` attributes in the same job definition. If you do, AutoSys Workload Automation ignores both attributes and generates an error.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
start_mins: mins [, mins]...
```

- **mins**

Defines the number of minutes after each hour to run the job.

**Limits:** This value can be a number in the range 0 to 59. This value can contain up to 255 characters, and you can use multiple lines without specifying a continuation character. Separate multiple values with commas.

**Default:** No start time is set.

### Example: Specify a Job to Run at a Quarter Past and Half Past Each Hour

This example specifies that the job should run at a quarter past and half past each hour:

```
date_conditions: y
days_of_week: all
start_mins: 15, 30
```

## start\_times Attribute -- Define the Time of the Day to Run a Job

The `start_times` attribute defines the times of day, in 24-hour format, at which the job should start on the days specified in the `days_of_week` attribute or on the dates specified in the calendar identified in the `run_calendar` attribute.

### NOTE

Times set in the `start_times` and `start_mins` attributes override times set in a custom calendar. The `date_conditions` attribute controls the usage of this attribute.

You cannot specify both the `start_times` and `start_mins` attributes in the same job definition. If you do, AutoSys Workload Automation ignores both attributes and generates an error.

When AutoSys Workload Automation writes a job definition containing the `start_times` attribute to the database within one minute of the specified start time, the job definition takes effect at the specified start time on the following day. However, when the product writes a job definition containing the `start_times` attribute to the database two or more minutes before the specified start time, the job definition takes effect at the specified start time on the same day.

For example, if you define a job to run daily at 12:00 and you edit its definition at 11:59, the job does not run with your changes until the following day at 12:00. However, if you edit the job's definition at 11:58 or before, the job runs with your changes at 12:00 on the same day.

### Supported Job Types

This attribute is optional for all job types.

## Syntax

This attribute has the following format:

```
start_times: "time [, time]..."
```

- ***time***

### Example: Specify the Time of the Day to Run a Job

This example specifies that the job should run at 10:00 a.m. and 2:00 p.m. on every specified day or date:

```
date_conditions: y
days_of_week: all
start_times: "10:00, 14:00"
```

or

```
date_conditions: y
days_of_week: all
start_times: 10\:00, 14\:00
```

## status Attribute—Set an Initial Status for a Job During Insertion

The status attribute sets an initial status for a job during insertion. You cannot use this attribute while updating or overriding a job definition.

### NOTE

When you set the initial status of a job using the status attribute, downstream dependent jobs do not run. However, if you set the initial status to FAILURE, SUCCESS, or TERMINATED, the downstream jobs can start when other jobs complete and utilize this job as a dependency.

### Supported Job Types

This attribute is optional for all job types.

## Syntax

This attribute has the following format:

```
status: initial_status
```

- ***initial\_status***

Sets an initial status for a job during insertion. The valid values are FAILURE, INACTIVE, ON\_HOLD, ON\_ICE, ON\_NOEXEC, SUCCESS, or TERMINATED. **Default:** INACTIVE

### Example: Set the Initial Status of a Job to ON\_ICE

This example sets the initial status of a job to ON\_ICE:

```
insert_job: test_js2
command:dir
machine:localhost
status:on_ice
```

## std\_err\_file Attribute -- Redirect the Standard Error File

The `std_err_file` attribute identifies the location to redirect the standard error file for the job you are defining. The error output can be redirected to a file or a blob.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#).

### Syntax

This attribute has the following format:

```
std_err_file: file_name [environment_variable] | blob_name
```

- **file\_name**

Defines the path and file name where you want to redirect all standard error output.

If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter `>>` as the first characters in the value. To overwrite the file, enter `>` as the first character in the value.

**Limits:** Up to 255 characters; cannot contain spaces between the `>` or `>>` characters and the full path or file name

#### NOTE

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location. Enclose variables referenced in the job profile in braces (for example, `"${PATH}"`). Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.
- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

#### UNIX:

- This value overrides the instance-wide setting for the `AutoInstWideAppend` parameter in the configuration file.
- This value overrides the machine-specific setting for the `AutoMachWideAppend` parameter in the `/etc/auto.profile` file.

#### Windows:

- This value overrides the instance-wide setting for the `Append stdout/stderr` field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, `C:\tmp` or `"C:\tmp"` is valid; `C:\tmp` is not.
- If the Windows command specified in the job definition does not exist, the job does not run. The standard error file is not created and the job log indicates that the error file is not found.

**Workload Automation Agent Default:** The agent directory (*installation\_directory/SystemAgent/agent\_name/*)

- **environment\_variable**

(Optional) Specifies an environment variable to add to the file name, so you can generate flexible filenames. When the job executes, the variable resolves as part of the filename. You can set environment variables and values for those variables at the system level or as part of the job definition.

#### NOTE

If you want to create a unique identifier, you must specify an environment variable which generates a unique value upon each job execution.

For more information about environment variables, see [Environment Variables](#).

- **blob\_name**

Specifies the format, type, and name associated with the blob. Blobs may be associated with the job or created as globally accessible. They can be created in either binary or text formats. Unlike file output, blobs may not be appended to.

Blob names and types are identified by the following keywords:

- **\$\$blob**  
Captures the output and stores it as a binary formatted error blob associated with the current job.
- **\$\$blobt**  
Captures the output and stores it as a textual formatted error blob associated with the current job.
- **\$\$glob.<global\_blob\_name>**  
Captures the output and stores it as a binary formatted blob under the *global\_blob\_name* specified.
- **\$\$globt.<global\_blob\_name>**  
Captures the output and stores it as a textual formatted blob under the *global\_blob\_name* specified

#### Example: Specify the File to Receive Standard Error File Output for a Job on UNIX

This example redirects the job's standard error file output to the /tmp/test.err file.

```
insert_job: unix_err
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_err_file: /tmp/test.err
```

#### Example: Append New Information to the Standard Error File on UNIX

This example appends new information to the standard error file.

```
insert_job: unix_appenderr
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_err_file: >>/tmp/test.err
```

#### Example: Specify the Standard Error File for a Job on Windows

This example redirects the standard error file to the C:\tmp\TEST.ERR file.

```
insert_job: win_err
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_err_file: "C:\tmp\TEST.ERR"
```

#### Example: Append New Information to the Standard Error File on Windows

This example appends new information to the standard error file.

```
insert_job: win_appenderr
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_err_file: ">>C:\tmp\TEST.ERR"
```

### Example: Append a Unique Identifier on UNIX

This example redirects the job's standard error file to a file with a unique name. The \$\$ characters append the process ID to the file name.

```
insert_job:
 unix_unique_namejob_type: CMDmachine: unixagentcommand: /usr/common/
 backupstd_err_file: /tmp/myfile.$$
```

### Example: Embed a Unique Identifier on UNIX

This example redirects the job's standard error file to a file with a unique name. The \$\$ characters embed the process ID in the middle of the file name. The following example embeds the process ID in the middle of the file name:

```
insert_job:
 unix_unique_namejob_type: CMDmachine: unixagentcommand: /usr/common/backup
 std_err_file: /tmp/my_file.$${}err
```

#### NOTE

A separator in the form of a dot, slash, space or braces ({} ) must follow the \$\$ characters. If you omit the required separator character, AutoSys Workload Automation will interpret the additional text as a global variable. If AutoSys Workload Automation cannot find that global variable, it erases the \$\$ characters and the associated text from the file name.

### Example: Store Standard Error File Output as a Job Blob in Textual Format

This example captures the standard error output and stores it as a textual format blob associated with the job.

```
insert_job: report_job1
job_type: CMD
machine: localhost
command: myapplication
std_err_file: $$blobt
```

### Example: Store Standard Error File Output as a Global Blob in Textual Format

This example captures the standard error output and stores it as a textual format global blob named "some\_stderr\_data".

```
insert_job: report_job2
job_type: CMD
machine: localhost
command: myapplication
std_err_file: $$globt.some_stderr_data
```

## std\_in\_file Attribute -- Redirect the Standard Input File

The std\_in\_file attribute specifies where you want to redirect the standard input file from. The standard input file can be a file or a blob.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#).

### Syntax

This attribute has the following format:

```
std_in_file: file_name [environment_variable]|blob_name
```

- **file\_name** Specifies the path and file name where you want to redirect standard input from.

**Limits:** Up to 255 characters

**NOTE**

- The job owner must have read permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location. Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$*global\_name*, \$\$(*global\_name*), or \$\$({*global\_name*}) format to reference global variables.
- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

**Windows:**

- If you do not specify a full path name, the path defaults to the job owner's home directory.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Default:** Do not redirect standard input.

- **environment\_variable** (Optional) Specifies an environment variable to add to the file name, so you can generate flexible filenames. When the job executes, the variable resolves as part of the filename. You can set environment variables and values for those variables at the system level or as part of the job definition.

**NOTE**

- If you want to create a unique identifier, you must specify an environment variable which generates a unique value upon each job execution.
- For more information about environment variables, see [Environment Variables](#).

- **blob\_name**

Specifies the format, type, and name associated with the blob. Blobs may be associated with the job or created as globally accessible. They may be created in either binary or text formats.

Blob names and types are identified by the following keywords:

- **\$\$blobt**  
Uses the highest version of textual formatted input job blob associated with the current job. These blobs are manually added through jil, using the insert\_blob subcommand.
- **\$\$blob.<jobname>**  
Uses an existing binary formatted output blob associated with the specified job.
- **\$\$blobt.<jobname>**  
Uses an existing textual formatted output blob associated with the specified job.
- **\$\$glob.<global\_blob\_name>**  
Uses an existing binary formatted blob under the *global\_blob\_name* specified.
- **\$\$globt.<global\_blob\_name>**  
Uses an existing textual formatted blob under the *global\_blob\_name* specified.

**NOTE**

- You cannot use \$\$blob keyword to specify the use of the current job's input blob.
- You can include colons in the job name that is associated with the blob. Enclose the job name that contains a colon with quotation marks (") or precede it with a backslash (\).
- You can define a job to use the output blob of its previous run as input. To accomplish this, you must define the job so that the name of the job is in the std\_in\_file attribute, using either \$\$blob.<job name> or \$\$blobt.<job name> keywords. After you define the job, apply a one-time override of the std\_in\_file attribute, so that the job reads from a local file on the computer on its first run.

**Example: Specify the Standard Input File on UNIX**



This example specifies the /tmp/test.in file as the standard input file.

```
insert_job: unix_stdin
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_in_file: /tmp/test.in
```

#### Example: Specify a File Name that Contains a Global Variable on UNIX

This example specifies a standard input file whose path contains the global variable named Today. You can use the sendevent command to set the value of a global variable to today's date.

```
insert_job: unix_globalvar
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_in_file: /tmp/${Today}.in
```

#### Example: Specify a Standard Input File on Windows

This example specifies the C:\tmp\TEST.IN file as the standard input file.

```
insert_job: win_stdin
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_in_file: "C:\tmp\TEST.IN"
```

#### Example: Specify a File Name that Contains a Global Variable on Windows

This example specifies a standard input file whose path contains the global variable named Today. You can use the sendevent command to set the value of a global variable to today's date.

```
insert_job: win_globalvar
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_in_file: "C:\tmp\$$Today.in"
```

#### Example: Retrieve Standard Input File from a Job Blob in Textual Format

This example defines a CMD job that creates and uses an input job blob. When the job is defined, a blob is created using the textual data contained in the my\_args.txt file. The blob is associated with the job. When the job runs, it uses the created blob as input and treats the blob data as textual data.

```
insert_job: report_job1
job_type: CMD
command: myapplication
machine: localhost
std_in_file: $$blobt
insert_blob: report_job1
blob_file: my_args.txt
```

#### Example: Retrieve Standard Input File from Another Job in Textual Format

This example defines two Command jobs. The report\_job1 job creates an output job blob in textual format. This output job blob is associated with the job. The report\_post\_job1 job retrieves the output job blob created from report\_job1 and uses it as input.

```
insert_job: report_job1
job_type: CMD
command: myapplication
std_out_file: $$blobt
machine: localhost
insert_job: report_post_job1
job_type: CMD
command: my_other_application
machine: WinProd1
std_in_file: $$blobt.report_job1
```

### Example: Append a Unique Identifier on UNIX

This example redirects the job's standard input file to a file with a unique name. The \$\$ characters append the process ID to the file name.

```
insert_job: unix_unique_name
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_in_file: /tmp/myfile.$$
```

### Example: Embed a Unique Identifier on UNIX

This example redirects the job's standard input file to a file with a unique name. The \$\$ characters embed the process ID in the middle of the file name. The following example embeds the process ID in the middle of the file name:

```
insert_job: unix_unique_name
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_in_file: /tmp/my_file.$${}in
```

#### NOTE

A separator in the form of a dot, slash, space or braces ({} ) must follow the \$\$ characters. If you omit the required separator character, AutoSys Workload Automation will interpret the additional text as a global variable. If AutoSys Workload Automation cannot find that global variable, it erases the \$\$ characters and the associated text from the file name.

## std\_out\_file Attribute -- Redirect the Standard Output File

The std\_out\_file attribute specifies where you want to redirect the standard output file. The standard output can be redirected to a file or a blob.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type](#).

### Syntax

This attribute has the following format:

`std_out_file: file_name [environment_variable]|blob_name`

- **file\_name**

Defines the path and file name where you want to redirect all standard output. If you are running jobs across platforms, the scheduler of the issuing instance controls the default behavior. The default is to append this file. To append the file, enter >> as the first characters in the value. To overwrite the file, enter > as the first character in the value.

**Limits:** Up to 255 characters; cannot contain spaces between the > or >> characters and the full path or file name

**NOTE**

- The job owner must have write permission to the specified file on the client computer.
- When specifying standard input, output, and error file locations, we recommend that you enter absolute paths to the files instead of relative paths or file names alone.
- You can use variables exported from the job profile or global variables to specify the file name and location. Enclose variables referenced in the job profile in braces (for example, "\${PATH}"). Use the \$global\_name, \${global\_name}, or \${global\_name} format to reference global variables.
- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.

**UNIX:**

- This value overrides the instance-wide setting for the AutoInstWideAppend parameter in the configuration file.
- This value overrides the machine-specific setting for the AutoMachWideAppend parameter in the /etc/auto.profile file.

**Windows:**

- This value overrides the instance-wide setting for the Append stdout/stderr field in the AutoSys Workload Automation Administrator utility.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.
- If the Windows command specified in the job definition does not exist, the job does not run. The standard error file is not created and the job log indicates that the error file is not found.

**Default:** /dev/null (UNIX) or NULL (Windows)

- **environment\_variable**

(Optional) Specifies an environment variable to add to the file name, so you can generate flexible filenames. When the job executes, the variable resolves as part of the filename. You can set environment variables and values for those variables at the system level or as part of the job definition.

**NOTE**

If you want to create a unique identifier, you must specify an environment variable which generates a unique value upon each job execution. For more information about environment variables, see [Environment Variables](#).

- **blob\_name**

Specifies the format, type, and name associated with the blob. Blobs may be associated with the job or created as globally accessible. They can be created in either binary or text formats. Unlike file output, blobs may not be appended to.

Blob names and types are identified by the following keywords:

- **\$\$blob**  
Captures the output and stores it as a binary formatted output blob associated with the current job.
- **\$\$blobt**  
Captures the output and stores it as a textual formatted output blob associated with the current job.
- **\$\$glob.<global\_blob\_name>**  
Captures the output and stores it as a binary formatted blob under the *global\_blob\_name* specified.
- **\$\$globt.<global\_blob\_name>**  
Captures the output and stores it as a textual formatted blob under the *global\_blob\_name* specified.

**Example: Specify the File to Receive Standard Output for a Job on UNIX**

This example specifies the /tmp/test.out file to receive standard output for the job.

```
insert_job: unix_stdout
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_out_file: /tmp/test.out
```

**Example: Append New Information to the Output File on UNIX**

This example appends new information to the /tmp/test.out output file.

```
insert_job: unix_appendout
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_out_file: >>/tmp/test.out
```

**Example: Specify a File Name with a Global Variable on UNIX**

This example redirects the job's standard output to a file whose path contains the global variable named Today. You can use the sendevent command to set the value of a global variable to today's date.

```
insert_job: unix_globalvar
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_out_file: /tmp/${Today}.out
```

**Example: Append a Unique Identifier on UNIX**

This example redirects the job's standard output file to a file with a unique name. The \$\$ characters append the process ID to the file name.

```
insert_job:
unix_unique_name
job_type: CMD
machine: unixagent
command: /usr/common/
backup
std_out_file: /tmp/myfile.$$
```

**Example: Embed a Unique Identifier on UNIX**

This example redirects the job's standard output file to a file with a unique name. The \$\$ characters embed the process ID in the middle of the file name. The following example embeds the process ID in the middle of the file name:

```
insert_job:
unix_unique_name
job_type: CMD
machine: unixagent
command: /usr/common/backup
std_out_file: /tmp/my_file.${$}out
```

**NOTE**

A separator in the form of a dot, slash, space or braces ({} ) must follow the \$\$ characters. If you omit the required separator character, AutoSys Workload Automation will interpret the additional text as a global variable. If AutoSys Workload Automation cannot find that global variable, it erases the \$\$ characters and the associated text from the file name.

**Example: Specify the File to Receive Standard Output for a Job on Windows**

This example specifies the C:\tmp\TEST.OUT file to receive standard output for the job.

```
insert_job: win_stdout
```

```

job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_out_file: "C:\tmp\TEST.OUT"

```

### Example: Append New Information to the Output File on Windows

This example appends new information to the C:\tmp\TEST.OUT output file.

```

insert_job: win_appendout
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_out_file: ">>C:\tmp\TEST.OUT"

```

### Example: Specify a File Name with a Global Variable on Windows

This example redirects the job's standard error file output to a file whose path contains the global variable named Today. You can use the sendevent command to set the value of a global variable to today's date.

```

insert_job: win_globalvar
job_type: CMD
machine: winagent
command: "C:\COMMON\Backup"
std_out_file: "C:\tmp\${Today}.out"

```

### Example: Store Standard Output File Output as a Job Blob in Textual Format

This example captures the standard output and stores it as a textual format blob associated with the job.

```

insert_job: report_job1
job_type: CMD
machine: localhost
command: myapplication
std_out_file: $$blobt

```

### Example: Store Standard Output File Output as a Global Blob in Textual Format

This example captures the standard output and stores it as a textual format global blob named "some\_stdout\_data".

```

insert_job: report_job1
job_type: CMD
machine: localhost
command: myapplication
std_out_file: $$globt.some_stdout_data

```

## submit\_modifier Attribute -- Specify Variables and Flags for Remote Execution Jobs

The submit\_modifier attribute specifies setup commands or options on which the job submission commands depend.

### Supported Job Type

This attribute is optional for the [Remote Execution \(PROXY\) job type](#).

### Syntax

This attribute has the following format:

```
submit_modifier: Option1="value1"[,Option2="value2"... , OptionN="valueN"]
```

- **OptionN="valueN"**

Specifies the submit\_modifier commands you want to run or the options to set prior to executing a job on a remote system. Running these setup commands provides additional flexibility to pass additional information to commands executed on the target system. These commands are captured in the job log and in the agent logs.

**Limits:** Up to 600 characters for each value; case-sensitive; can contain alphanumeric characters and delimiters.

**NOTE**

- If the option value contains delimiters, the value must be enclosed in single quotes to denote the start and end of the value. If the value contains a single quote, double up that single quote so it does not denote the end of the value. Do not use double quotes; it must be two single quotes.
- You can use multiple Option keywords to run multiple submit\_modifier commands; for example, submit\_modifier: Option1="command1", Option2="command2" requires that commands command1 and command2 run before AutoSys Workload Automation executes the job.

**NOTE**

**UNIX:** The submit\_modifier attribute specifies commands to run prior to remote\_command. The commands specified must be valid platform-specific commands.

**NOTE**

**OpenVMS:** The submit\_modifier attribute specifies options for the submit command. The commands specified must be valid platform-specific commands.

**NOTE**

**HP Integrity NonStop:** Option1 is used to specify the TACL run-options. Option2 is used to specify the backup CPU number. Option3 is used to specify the program parameters.

### Example: Specify NonStop Options Using the submit\_modifier Attribute

This example shows how to specify a HP NonStop job to run with the following run-options: HIGHPIN ON, INSPECT ON with a backup CPU id of 2, and the ABENDONABEND parameter.

```
insert_job: tan_job
job_type: PROXY
machine: proxagent
owner: user1@NonStop1400
remote_target: NonStop1400
remote_command: who
submit_modifier: Option1="HIGHPIN ON,INSPECT ON",option2="2",option3="ABENDONABEND"
```

### Example: Specify UNIX Option Using the submit\_modifier Attribute

This example shows how to specify the UNIX option using the submit\_modifier attribute. The UNIX job generates the setTest.out file using the option specified in the submit\_modifier attribute.

```
insert_job: ux_job
job_type: PROXY
machine: proxagent
owner: user1@uxmach
remote_target: uxmach
remote_command: "ls -l /u1/user1/setTest.out"
submit_modifier: Option1="pwd;set>/u1/user1/setTest.out"
```

## success\_codes Attribute -- Define Exit Codes to Indicate Job Success

The `success_codes` attribute defines which exit codes indicate job success. For example, you can define the `success_codes` attribute to be 20-30. If a job returns an exit code between 20 and 30, the job is considered to have completed successfully.

### NOTE

- This attribute applies only to jobs that run on the Workload Automation Agent for UNIX, Linux, or Windows.
- If you do not specify the `success_codes` attribute in your job definition, the scheduling manager interprets an exit code of zero (0) as job success (the default). Any exit code other than zero is interpreted as job failure.

### Supported Job Types

This attribute is optional for the following job types:

- [Command](#) (CMD)
- [HTTP](#) (HTTP)
- [i5/OS](#) (I5)
- [Micro Focus](#) (MICROFOCUS)
- [Remote Execution](#) (PROXY)
- [Web Service Document/Literal](#) (WSDOC)

### Syntax

This attribute has the following format:

```
success_codes: exitcodes
```

- **exitcodes**

Defines which exit codes indicate job success. Specify a single exit code (for example, 4), a range of exit codes (for example, 0-9999), or a list of multiple exit codes or ranges separated by commas.

**Example:** 1,150,20-30

**Limits:** Up to 256 characters. The exit code must be an integer in the range -2147483647 to 2147483647.

**Default:** 0

### NOTE

If you specify multiple exit codes, enter the most specific codes first followed by the ranges.

### Example: Define Multiple success\_codes

Suppose that you want a job named HTTPJOB to run multiple exit codes and indicate job success. The job is considered to have completed successfully if the exit code is in the 20-30 or 100-150 ranges.

```
insert_job: HTTPJOB
job_type: HTTP
machine: localhost
```

```
success_codes: 20-30,100-150
```

## success\_criteria Attribute -- Define the Evaluation Criteria for a Return String

The `success_criteria` attribute defines a regular expression that is used to evaluate a string returned by an SQL statement or a stored procedure.

### Supported Job Types

This attribute is optional for the following job types:

- [Database Stored Procedure \(DBPROC\)](#)
- [SQL](#)

## Syntax

This attribute has the following format:

```
success_criteria: regexp
```

- ***regexp***  
Defines a regular expression that is used to evaluate a return string. If the return string matches the regular expression, the job completes successfully. Otherwise, the job fails.

**Limits:** Up to 500 characters; case-sensitive

### NOTE

Each return string includes the field name from the SELECT statement and its value, separated by an equal sign (=). For example, consider the query SELECT ORD\_NUM FROM SALES. To match order number A2976, specify the regular expression ORD\_NUM=A2976. Specifying the regular expression A2976 does not match any return string causing the job to fail. You can also specify the regular expression .\*A2976, which matches any return string that ends with A2976.

### NOTE

- The success\_criteria attribute only applies to SQL queries that are SELECT statements.
- To compose a regular expression, follow the rules for Java class java.util.regex.Pattern. You can find these rules using a Google search for java pattern.
- You can use escape sequences in the regular expression. The following characters have a special meaning in regular expressions. To use these characters literally, precede the characters with one backward slash (\).
  - \ -- backward slash
  - [ -- opening bracket
  - \* -- asterisk
  - | -- vertical bar
  - { -- opening brace
  - + -- plus sign
  - ( -- opening parenthesis
  - ^ -- caret (circumflex)
  - ? -- question mark
  - ) -- closing parenthesis
  - \$ -- dollar sign
  - . -- period

For example, to match the characters `.*` literally, specify `\*\.` in your regular expression. The backward slashes escape the characters' special meanings.
- Embedded double quotation marks must be prefixed with a backslash as an escape character. Enclose the values that contain embedded blanks with double quotation marks.
- When you are trying to check the return value from a stored procedure, then you must use both result\_type and success\_criteria together to make sure the job goes to SUCCESS state based on the return value from the stored procedure.
- You can use global variables to define the regular expression that is used to evaluate a return string. Use the \$\$global\_name, \$\$ (global\_name), or \$\$ {global\_name} format to reference global variables.

## Example: Define Basic Success Criteria

This example queries the Inv\_List table. If the SQL query returns a PartNo that begins with IDG, the job completes successfully.



```

insert_job: QRY1
job_type: SQL
machine: DB_agent
sql_command: SELECT PartNo FROM Inv_List WHERE Stock > 30
success_criteria: PartNo=IDG.*
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=ORDERS"
destination_file: "c:\log\qry1.txt"

```

### Example: Define a More Complex Success Criteria

This example queries the Proc\_List table. If the SQL query returns an ID that matches the specified regular expression, the job completes successfully.

```

insert_job: QRY2
job_type: SQL
machine: DB_agent
sql_command: SELECT ID FROM Proc_List WHERE Orders > 30
success_criteria: ID=\w{2,4}
connect_string: "jdbc:db2://172.31.255.255:50000/SAMPLE"
destination_file: /export/home/user1/log/qry2.txt

```

The regular expression is interpreted as follows:

- \w -- A word character [a-zA-Z0-9]
- {2,4} -- Match at least 2 times, but not more than 4 times

To illustrate the last item (2, 4), consider the syntax:

```
ID=b1{1,3}c
```

Evaluating this expression yields the following conditions:

- The line contains the text b1.
- Numeric 1 should exist at least once, but not more than three times.
- The specified string must be followed by the letter c.

### Example: Define a Job to Check Stored Procedure's Return Code as Success Criteria

This example checks the return value from stored procedure to be an integer and value to be 10, the following job will go to SUCCESS

```

insert_job: test_db_proc_success_return_10_to_10_check
job_type: DBPROC
machine: localhost
result_type: INTEGER
sp_name: ujo_sample_procedure
success_criteria: "return=10"

```

### success\_criteria\_json Attribute -- Specify the JSON Path

The success\_criteria\_json attribute specifies the JSON path whose evaluated value is validated against the value that you specified in the success\_criteria\_value attribute. If the value matches, the job is successful. Else, the job fails.

### Supported Job Type

This attribute is optional for the [HTTP job type](#).

## Syntax

This attribute has the following format:

```
success_criteria_json: json_path
```

- ***json\_path***

Specifies the JSON path whose evaluated value is validated against the value that you specified in the `success_criteria_value` attribute.

**Limits:** Up to 256 characters

### NOTE

- If you specify the `success_criteria_value` attribute, you must specify either the `success_criteria_json` or `success_criteria_xpath` attribute in your job definition. The `success_criteria_json`, `success_criteria_xpath`, and `success_criteria_regex` attributes are mutually exclusive.
- You can use global variables to specify the JSON path. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## success\_criteria\_regex Attribute -- Specify the Regex Expression

The `success_criteria_regex` attribute specifies the Regex expression which is validated against the job's output payload. If the Regex expression is evaluated to true, the job is successful. Else, the job fails.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

This attribute has the following format:

```
success_criteria_regex: regex_expression
```

- ***regex\_expression***

Specifies the Regex expression which is validated against the job's output payload.

**Limits:** Up to 256 characters, case-sensitive.

### NOTE

- The `success_criteria_regex`, `success_criteria_json`, and `success_criteria_xpath` attributes are mutually exclusive.
- You can use global variables to specify the Regex expression. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

The value specified against `success_criteria_regex`, `success_criteria_json` and `success_criteria_xpath` is checked for compliance against regex, jsonpath, and xpath format rules respectively.

### NOTE

Validation is skipped if global variables are referenced as part of `success_criteria_regex`, `success_criteria_json`, and `success_criteria_xpath`.

## success\_criteria\_value Attribute -- Specify the Success Criteria

The `success_criteria_value` attribute specifies the success criteria to determine the success or failure of an HTTP job.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal](#) job types.

This attribute has the following format:

`success_criteria_value: regexp`

- ***regexp***

Defines a regular expression that is used to evaluate the xpath expression (that you specify using the `success_criteria_xpath`) or the JSON path ((that you specify using the `success_criteria_json`). If the evaluated xpath or JSON expression value matches the regular expression, the job completes successfully. Otherwise, the job fails.

**Limits:** Up to 256 characters

**NOTE**

- If you specify the `success_criteria_value` attribute, you must specify either the `success_criteria_json` or `success_criteria_xpath` attribute in your job definition. The `success_criteria_json` and `success_criteria_xpath` attributes are mutually exclusive.
- You can use global variables to specify the success criteria. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## success\_criteria\_xpath Attribute -- Specify the xpath Expression

The `success_criteria_xpath` attribute specifies the xpath expression whose evaluated value is validated against the value that you specified in the `success_criteria_value` attribute.

### Supported Job Types

This attribute is optional for the [HTTP](#) and [Web Service Document/Literal Type](#) job types.

This attribute has the following format:

`success_criteria_xpath: xpath_expression`

- ***xpath\_expression***

Specifies the xpath expression whose evaluated value is validated against the value that you specified in the `success_criteria_value` attribute.

**Limits:** Up to 256 characters

**NOTE**

- If you specify the `success_criteria_value` attribute, you must specify either the `success_criteria_json` or `success_criteria_xpath` attribute in your job definition. The `success_criteria_json` and `success_criteria_xpath` attributes are mutually exclusive.
- You can use global variables to specify the xpath expression. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

## success\_pattern Attribute -- Specify a Success Pattern Using Regular Expression Logic

The `success_pattern` attribute specifies a regular expression that is used to check for an expected value in a JMX and Web Service job. If the output matches the success pattern, the job completes; otherwise, it fails.

### Supported Job Types

This attribute is optional for the following job types:

- [JMX-MBean Attribute Get \(JMXMAG\)](#)
- [JMX-MBean Attribute Set \(JMXMAS\)](#)
- [JMX-MBean Operation \(JMXMOP\)](#)
- [Web Service \(WBSVC\)](#)

### Syntax

This attribute has the following format:

```
success_pattern: pattern
```

- ***pattern***

Specifies a regular expression to use as a filter. To match any character zero or more times, add `.*` to the regular expression. To match an entire line of text using part of the line, add `.*` before and after the part. For example, you can use `.*web.*` to match "this is a web service job".

**Limits:** Up to 256 characters; case-sensitive

Example: `<StockQuotes><Stock><Symbol>CA.*`

**NOTE**

To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules on the Internet by searching for "java pattern".

**Example: Validate an Email Address in a Web Service Job**

Suppose that you want to invoke a web service that validates an email address. The URL for the WSDL that describes the web service and its location is `http://www.webserviceX.net/ValidateEmail.asmx?wsdl`. The job calls the `IsValidEmail` operation within the `ValidateEmail` web service. When the job invokes the web service, the email address is passed to the operation. If the email address is valid, the operation returns `true` and the job completes successfully. If the email address is invalid, the operation returns `false` and the job fails.

```
insert_job: subscribe
job_type: WBSVC
machine: wsagent
target_namespace: "http://www.webserviceX.NET/"
service_name: ValidateEmail
port_name: ValidateEmailSoap
wsdl_operation: IsValidEmail
WSDL_URL: "http://www.webserviceX.net/ValidateEmail.asmx?wsdl"
endpoint_URL: "http://www.webserviceX.net/ValidateEmail.asmx"
web_parameter: xsd:string="john.smith@example.com"
return_class_name: java.lang.Boolean
return_xml_name: boolean
return_namespace: "http://www.webserviceX.net"
success_pattern: true
```

## svcdesk\_attr Attribute -- Define Service Desk Request Attributes

The `svcdesk_attr` attribute defines Service Desk request attributes and values to set in the Service Desk request generated when you have set the `service_desk` attribute to `y` or `1` and the job you are defining completes with a `FAILURE` status.

**NOTE**

For information about valid attributes and values, see the *Service Desk Modification Guide*.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
svcdesk_attr: attribute/value[,attribute/value,attribute/value...]
```

- **attribute**  
Identifies a Service Desk attribute to override with the specified value in the generated request.
- **value**  
Defines the value to which to set the associated Service Desk request attribute.

**Limits:** The svcdesk\_attr value can be up to 255 alphanumeric characters in length. You must use a slash to separate *attribute* from *value*, and you must separate each *attribute/value* pair with a comma.

#### Example: Set the Service Desk Priority Attribute

This example sets the Service Desk priority attribute to 3, as an alternative to using the svcdesk\_pri attribute:

```
svcdesk_attr: priority/3
```

#### Example: Set Four Attributes at a Time

This example sets four attributes at once, such as priority=4, severity=1, impact=2, and description=This is a test:

```
svcdesk_attr: priority/4,severity/1,impact/2,description/"This is a test"
```

#### Example: Set the Service Desk Urgency Attribute

This example sets the Service Desk urgency attribute equal to 1:

```
svcdesk_attr: urgency/urg:1100
```

Only users with advanced knowledge of Service Desk should attempt to use an attribute, such as urgency with svcdesk\_attr, as it requires specific predefined values, such as urg:1100. (urg:1104 indicates an urgency of 5).

#### NOTE

For more information, see the Service Desk documentation.

### svcdesk\_desc Attribute -- Define the Message to Include in the CA Service Desk Request

The svcdesk\_desc attribute defines the message to include in the CA Service Desk request generated when you have set the service\_desk attribute to **y** or **1** and the job you are defining completes with a FAILURE status.

#### Supported Job Types

This attribute is optional for all job types.

#### Syntax

This attribute has the following format:

```
svcdesk_desc: value
```

- **value**  
Specifies a description to include in the CA Service Desk request generated when the job fails.

#### Limits:

Up to 255 alphanumeric characters

#### Example: Define a Message to Include in the CA Service Desk Request

This example defines the description of the problem that caused a job to fail:

```
svcdesk_desc: The server appears to be down.
```

---

## svcdesk\_imp Attribute -- Specify the Impact Level of the Service Desk Request

The svcdesk\_imp attribute specifies the impact level to assign the Service Desk request generated when you have set the service\_desk attribute to **y** or **1** and the job you are defining completes with a FAILURE status. The impact level indicates how much you expect the request to affect work being performed.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
svcdesk_imp: level
```

- **level**

Specifies the impact level of the request.

**Limits:** This value is one of the following:

- **0**  
None. The request has no impact.
- **1**  
High
- **2**  
Medium-High
- **3**  
Medium
- **4**  
Medium-Low
- **5**  
Low

### Example: Set the Impact Level of the Service Desk Request

This example sets the impact level of a request generated when the job fails to medium:

```
svcdesk_imp: 3
```

## svcdesk\_pri Attribute -- Specify the Priority Level of the Service Desk Request

The svcdesk\_pri attribute specifies the priority level of the Service Desk request generated when you set the service\_desk attribute to **y** or **1**, and the job you are defining completes with a FAILURE status.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
svcdesk_pri: level
```

- **level**

Specifies the priority level of the request.

**Limits:** Set this value to one of the following:

- **0**

- Indicates an unassigned priority.
- **1**  
Indicates that the priority level is high.
- **2**  
Indicates that the priority level is medium high.
- **3**  
Indicates that the priority level is medium.
- **4**  
Indicates that the priority level is medium low.
- **5**  
Indicates that the priority level is low.

### Example: Set the Priority Level of the Service Desk Request

This example sets the priority level of a request generated when the job fails to medium:

```
svcdesk_pri: 3
```

### svcdesk\_sev Attribute -- Specify the Severity Level of the Service Desk Request

The `svcdesk_sev` attribute specifies the severity level to assign the Service Desk request generated when you have set the `service_desk` attribute to **y** or **1** and the job you are defining completes with a FAILURE status. The severity level indicates how much you expect the request to affect other users.

#### Supported Job Types

This attribute is optional for all job types.

#### Syntax

This attribute has the following format:

```
svcdesk_sev: level
```

- ***level***  
Specifies the severity level of the request.  
**Limits:** This value is one of the following:
  - **0**  
Low. This is the same as setting the value to 5.
  - **1**  
High
  - **2**  
Medium-High
  - **3**  
Medium
  - **4**  
Medium-Low
  - **5**  
Low

### Example: Set the Severity Level of the Service Desk Request

This example sets the severity of a request generated when the job fails to medium:

```
svcdesk_sev: 3
```

## tablename Attribute -- Specify the Database Table to Monitor

The tablename attribute specifies the name of the database table to monitor for the changes specified in the Database Monitor or Database Trigger job.

### Supported Job Type

This attribute is required for the following job types:

- [Database Monitor \(DBMON\)](#)
- [Database Trigger \(DBTRIG\)](#)

### Syntax

This attribute has the following format:

```
tablename: table
```

- ***table***  
Specifies the name of the database table to monitor for changes.  
**Limits:** Up to 300 characters; case-sensitive

#### NOTE

- Do not use colons and double quotation marks.
- You can use global variables to specify the database table name to monitor. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

### Example: Monitor for a Decrease in the Number of Rows in a Table

This example monitors the Inventory\_List table for a decrease in the number of rows. If the number of rows decreases, the job completes. The database user ID is set to the user who invokes jil to define the job (the default owner).

```
insert_job: dbmon_job
job_type: DBMON
machine: DB_agent
tablename: Inventory_List
monitor_type: DECREASE
connect_string: "jdbc:db2://172.31.255.255:50000/SAMPLE"
```

## target\_namespace Attribute -- Specify a Target Namespace

The target\_namespace attribute specifies the target namespace in a Web Service job.

### Supported Job Type

This attribute is required for the [Web Service \(WBSVC\)](#) job type.

### Syntax

This attribute has the following format:

```
target_namespace: target_namespace
```

- ***target\_namespace***  
Specifies the target namespace used for the names of messages, port type, binding, and services defined in the WSDL for the web service. Complex data types such as arrays require the target namespace.  
**Limits:** Up to 256 characters; case-sensitive

### Example: Specify Target Namespace for Getting Stock Quote



**NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

**term\_run\_time Attribute -- Specify the Maximum Runtime**

The `term_run_time` attribute specifies the maximum run time (in minutes) that the job you are defining should require to finish normally. If the job runs longer than the specified time, AutoSys Workload Automation terminates it.

**Supported Job Types**

This attribute is optional for all job types.

**Syntax**

This attribute has the following format:

```
term_run_time: mins
```

- **mins**

Defines the maximum number of minutes the job should ever require to finish normally.

**Limits:** 0-527040

**Default:** 0 (the job is allowed to run forever)

**Example: Terminate the Job if it Runs Beyond the Maximum Run Time**

This example specifies that the product should terminate the job if it runs for more than two hours:

```
term_run_time: 120
```

## text\_file\_filter Attribute -- Specify a Text String to Search For

The text\_file\_filter attribute specifies the text to search for.

### Supported Job Type

This attribute is required for the [Text File Reading and Monitoring \(OMTF\) job type](#).

### Syntax

This attribute has the following format:

```
text_file_filter: textstring
```

- **textstring**

Defines the text string to search for. You can specify the text string as a regular expression.

**Limits:** Up to 1024 characters; case-sensitive

### NOTE

- The OMTF job searches the entire file specified by the text\_file\_name attribute. When the job finds the first match to the text string, the job completes.
- If monitor\_mode: CONTINUOUS is specified in the job definition, and the job finds one or more occurrences of the searched text, the first trigger takes place at the first occurrence only. Subsequently, the job only searches for new lines that are added to the text file. In other words, the job resumes the search from the first line following the last line of the file from the previous search.
- To compose a regular expression, follow the rules for Java class java.util.regex.Pattern. You can find these rules using a Google search for java pattern.
- You can use escape sequences in the regular expression. The following characters have a special meaning in regular expressions. To use these characters literally, precede the characters with one backward slash (\).
  - \ -- back slash
  - | -- vertical bar
  - ( -- opening parenthesis
  - ) -- closing parenthesis
  - [ -- opening bracket
  - { -- opening brace
  - ^ -- caret (circumflex)
  - \$ -- dollar sign
  - \* -- asterisk
  - + -- plus sign
  - ? -- question mark
  - . -- period

For example, to match the characters \*.\* literally, specify \\*\. in your regular expression. The backward slashes escape the characters' special meanings.

### Example: Monitor a File for a String

This example shows a search of the text file DATA SOURCE NAME for the text string EVENT ""COMPUTER() ". If this string is found, the job completes.

```
insert_job: textfile_job1
job_type: OMTF
machine: monagt
text_file_name: DATA SOURCE NAME
text_file_filter: "EVENT ""COMPUTER() "
text_file_mode: LINE
```

```
lower_boundary: 123
upper_boundary: 876
```

### Example: Monitor a File for a String Using Special Characters

In this example, because the text string contains escape sequences, back slashes must precede the special characters asterisk (\*), period (.), and zed (Z). The required text string is a wild card search for the text string "=jars\*.\*" that must appear at the end of the line (\Z).

```
insert_job: textfile_job2
job_type: OMTF
machine: monagt
text_file_name: /export/home/agentdir/agentparm.txt
text_file_filter: "=jars*.*\Z"
text_file_mode: LINE
lower_boundary: 1
upper_boundary: 110
monitor_mode: now
```

### Example: Monitor a File for a String using a Regular Expression

In this example, the text string contains regular expression pattern matching syntax. The search range is also a regular expression as indicated by the text\_file\_mode attribute.

```
insert_job: omtf2
job_type: OMTF
machine: monagt
text_file_name: /export/home/agentdir/agentparm.txt
text_file_filter: ^\w{4,10}\.
text_file_mode: REGEX
lower_boundary: "log*.*"
monitor_mode: now
```

The regular expression can be interpreted as follows:

- ^ or \A -- match only at the beginning of string (line)
- \Z or \$ -- match only at the end of string
- \w -- a word character [a-zA-Z0-9]
- \W -- a non-word character
- \s -- a whitespace character
- {4,10} -- match at least 4 times but not more than 10 times

To illustrate the last item (4, 10), consider the syntax:

```
text_file_filter: b1{1,3}c
```

Evaluating this expression yields the following conditions:

- The line contains the text b1.
- Numeric 1 should exist at least once, but not more than three times.
- The specified text string must be followed by the letter c.

### Example: Monitor a Windows Text File for a Windows Path

Because back slashes are interpreted as an escape sequence, the two back slashes in the text string, C:\\Program Files, must each be preceded by a back slash, giving four back slashes in the text\_file\_filter attribute below.

```

insert_job: textfile_job4
job_type: OMTF
machine: monagt
text_file_name: /export/home/agentdir/agentparm.txt
text_file_filter: "C:\\\\Program Files"
text_file_mode: LINE
lower_boundary: 1
upper_boundary: 200
monitor_mode: now

```

## text\_file\_filter\_exists Attribute -- Specify Whether to Monitor for the Existence of Text

The `text_file_filter_exists` attribute specifies whether the job monitors for the existence of a text string in a text file.

### NOTE

If you do not specify the `text_file_filter_exists` attribute in your job definition, the job checks whether the text string exists (the default).

### Supported Job Type

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\) job type](#).

### Syntax

This attribute has the following format:

```
text_file_filter_exists: TRUE | FALSE
```

- **TRUE**

Monitors whether a text string exists in a specified text file. If the text string exists, the job completes successfully or an alert is triggered (if monitoring continuously). This is the default.

- **FALSE**

Monitors whether a text string does not exist in a specified text file. If the text string does not exist, the job completes successfully. If the text string exists, the job fails.

### NOTE

To use the `FALSE` setting, you must specify `NOW` in the `monitor_mode` attribute.

### Example: Check Whether a String Does Not Exist in a File

This example searches lines 1 to 200 of the `transmitter.log` file for the text string "Warning". If the string is not found, the job completes successfully. If the string is found, the job fails.

```

insert_job: textfile_job
job_type: OMTF
machine: monagt
text_file_name: /export/home/log/transmitter.log
text_file_filter: Warning
text_file_mode: LINE
lower_boundary: 1
upper_boundary: 200
text_file_filter_exists: FALSE
monitor_mode: NOW

```

## text\_file\_mode Attribute -- Specify the Search Mode

The text\_file\_mode attribute specifies the search mode when monitoring a text file.

This attribute is used with the lower\_boundary and upper\_boundary attributes, which define the search boundaries in the text file.

### NOTE

If you do not specify the text\_file\_mode attribute in your job definition, the job searches for the text in the specified line boundaries (the default).

### Supported Job Type

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\) job type](#).

### Syntax

This attribute has the following format:

```
text_file_mode: LINE | REGEX | DATETIME
```

- **LINE**

Searches for a text string between line numbers that define the upper and lower boundaries to search in the text file. This is the default. The values for the lower\_boundary and upper\_boundary attributes must be numeric.

- **REGEX**

Searches for a text string between regular expressions that define the lower and upper boundaries to search in the text file. The values for the lower\_boundary and upper\_boundary attributes must be regular expressions.

- **DATETIME**

Searches for a text string between a date range specified by the upper and lower boundaries. You must specify the time\_format attribute when using this value.

### NOTE

- You must specify the lower\_boundary attribute, upper\_boundary attribute, or both. Most searches require you to specify both attributes.
- For all search modes, the upper and lower boundaries are determined as follows:
  - If you only specify the lower\_boundary attribute, the agent searches from the lower boundary to the last line of the text file.
  - If you only specify the upper\_boundary attribute, the agent starts searching from the first line of the text file to the upper boundary.
- For regular expressions, the upper and lower boundaries are determined as follows:
  - If you specify the upper\_boundary attribute as a regular expression and there are no strings in the file that match it, the agent searches for the text string to the last line of the file.
  - If you specify the lower\_boundary attribute as a regular expression and there are no strings in the file that match it, the agent does not search for the text string.

### Example: Define the Search Mode as LINE

This example searches the c:\callog file in line mode. The job starts searching the content from line 143 of the file. The upper boundary is not defined, so the job searches to the last line of the file. The job completes successfully if the ERROR MESSAGE string is found.

```
insert_job: omtf_line
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ERROR MESSAGE
text_file_mode: LINE
```

```
lower_boundary: 143
monitor_mode: NOW
```

### Example: Define Search Mode as REGEX

This example searches the c:\ca\log file in regular expression mode. The lower boundary is not defined, so the job searches the content from the first line of the file to the upper boundary (a line that contains the word service). The job completes successfully if the ARCHIVE string is found.

```
insert_job: omtf_regex
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ARCHIVE
text_file_mode: REGEX
upper_boundary: service
monitor_mode: NOW
```

### Example: Define the Search Mode as DATETIME

This example searches the /export/home/logs/transmitter.log file in date and time mode. The job searches the content between May 20, 2010 at midnight and May 27, 2010 at 11:59 p.m. The date and time values are defined using the format specified in the time\_format attribute. The job completes successfully if the transmitted string is found.

```
insert_job: omtf_timedate
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transmitter.log
text_file_filter: transmitted
text_file_mode: DATETIME
lower_boundary: "Thu May 20 00:00:00.000 EDT 2010"
upper_boundary: "Thu May 27 23:59:59.999 EDT 2010"
time_format: "EEE MMM dd HH:mm:ss.SSS zzz yyyy"
time_position: 12
monitor_mode: NOW
```

### Example: Monitor a UNIX Text File for a String at the Beginning of a Line

This example searches the /export/home/systemagent/agentparm.txt file. The search starts at the first line that contains the word "agent" at the beginning of the line (as specified by \A in the regular expression specified for lower\_boundary) and until it finds the string "level=2" at the end of a line (as specified by \Z in the regular expression specified for upper\_boundary).

```
insert_job: omtf_unix_line
job_type: OMTF
machine: monagt
text_file_name: /export/home/systemagent/agentparm.txt
text_file_filter: \.0/MAIN$
text_file_mode: REGEX
lower_boundary: \Aagent
upper_boundary: level=2\Z
monitor_mode: NOW
```

### Example: Monitor a UNIX Text File for a String Between Regular Expressions

This example searches for the string "# Runner plugin parameters", which is evaluated as a regular expression. The job searches from the first line of the file to the line that contains the word service.

```

insert_job: omtf_unix_regex
job_type: OMTF
machine: monagt
text_file_name: /export/home/systemagent/agentparm.txt
text_file_filter: \A\W\sRunner
text_file_mode: REGEX
upper_boundary: service
monitor_mode: now

```

The criteria \A, \W, and the word \sRunner are evaluated as follows:

- \A means match only at the beginning of the line.
- \W means match a non-word character.
- \sRunner means look for a white space before the word Runner.

### Example: Monitor an i5/OS Text File for a String in the First 20 Lines of the File

The following job searches for a text string "Create file failed". This job searches the DATA member for this text string between lines 1 (the lower boundary) and 20 (the upper boundary). The job completes successfully when the text string is found.

```

insert_job: omtf_i5os_line
job_type: OMTF
machine: i5agt
text_file_name: /QSYS.LIB/LIBRARY.LIB/RESULTS.FILE/DATA.MBR
text_file_filter: Create file failed
text_file_mode: LINE
lower_boundary: 1
upper_boundary: 20
monitor_mode: now

```

## text\_file\_name Attribute -- Specify a Text File Name and Location

The text\_file\_name attribute specifies the name and location (path) of the text file to search.

### Supported Job Type

This attribute is required for the [Text File Reading and Monitoring \(OMTF\) job type](#).

### UNIX and Windows Syntax

This attribute has the following format:

```
text_file_name: file_name
```

- **file\_name**  
Specifies the path to and name of the text file to search.  
**Limits:** Up to 256 characters; case-sensitive

#### NOTE

**Windows Note:** When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**Example:** "C:\Program Files\agent\prodagt"

**NOTE**

- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- You can use global variables to specify the text file name and location to search. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

**Example: Specify a Text File Name on Windows**

This example searches for a text string in the log text file on a Windows computer. The path to the text file is enclosed in quotation marks because the path contains a colon. The job searches for the string "ERROR MESSAGE" between lines 1234 and 1876. The job completes successfully if the string is found.

```
insert_job: textfile_job1
job_type: OMTF
machine: monagt
text_file_name: "c:\program files\agent\log"
text_file_filter: ERROR MESSAGE
text_file_mode: LINE
lower_boundary: 1234
upper_boundary: 1876
monitor_mode: NOW
```

**Example: Specify a Text File Name on UNIX**

This example searches the `/export/home/logs/transmitter.log` file in date and time mode. The job searches the content between May 20, 2010 at midnight and May 27, 2010 at 11:59 p.m. The date and time values are defined using the format specified in the `time_format` attribute. The job completes successfully if the transmitted string is found.

```
insert_job: omtf_timedate
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transmitter.log
text_file_filter: transmitted
text_file_mode: DATETIME
lower_boundary: "Thu May 20 00:00:00.000 EDT 2010"
upper_boundary: "Thu May 27 23:59:59.999 EDT 2010"
time_format: "EEE MMM dd HH:mm:ss.SSS zzz yyyy"
time_position: 12
monitor_mode: NOW
```

**time\_format Attribute -- Define a Time Format**

The `time_format` attribute defines the date and time pattern to use when searching a text file with a time stamp.

**Supported Job Type**

If `DATETIME` is specified for the `text_file_mode` attribute, this attribute is required for the [Text File Reading and Monitoring \(OMTF\) job type](#).

**Syntax**

This attribute has the following format:

```
time_format: time_format
```



- ***time\_format***

Defines the date and time pattern to use when the upper and lower boundaries are specified as date and time. The upper and lower boundaries are used to search inside a log file.

**Limits:** Up to 256 characters; case-sensitive.

**NOTE**

To specify the time format, construct a time pattern string that is used as a mask for searching the particular time stamp you want to find. In this pattern, all ASCII letters are reserved as pattern letters.

The following table displays the symbols you can use to build a time format pattern:

Time Format Pattern Characters			
Character	Type	Definition	Example
G	Era designator	Text	AD
y	Year	Number	2010
M	Month in year	Text and Number	July; 07
d	Day in month	Number	10
h	Hour in am/pm (1 to 12)	Number	12
H	Hour in day (0 to 23)	Number	0
m	Minute in hour	Number	30
s	Second in minute	Number	55
S	Millisecond	Number	978
E	Day in week	Text	Tuesday; Tues
D	Day in year	Number	189
F	Day of week in month	Number	2 (2nd Wednesday in July)
w	Week in year	Number	27
W	Week in month	Number	2
a	AM/PM marker	Text	PM
k	Hour in day (1 to 24)	Number	24
K	Hour in AM/PM (0 to 11)	Number	0
z	Time zone	Text	EST; Eastern Standard Time
Z	RFC 822 time zone	Sign and Number	-0500
'	Escape for test	Delimiter	
"	Single quote	Literal	'

The following table displays sample time format patterns for July 10, 2010 at 12:08 p.m. Eastern Standard Time:

Format Pattern	Result
"EEE, MMM d, "yy"	Wed, July 10, '10
"h:mm a"	12:08 PM
"hh 'o'clock' a, zzzz"	12 o'clock PM, EST
"K:mm a, z"	12:08 PM, EST
"yyyy.MM.dd G 'at' hh:mm:ss z"	2010.07.10 AD at 12:08:56 EST
"yyyyy.MMMMM.dd GGG hh:mm aaa"	2010.July.10 AD 12:08 PM

### Example: Define a Time Format

In this example, the `time_format` attribute specifies the time pattern format that is applied to the values specified by the `lower_boundary` and `upper_boundary` attributes. The time values are enclosed in quotation marks because they contain colons.

The values for the `time_format` attribute are as follows:

- `MM` -- month in the year
- `dd` -- day in the month
- `yyyy` -- year
- `HH:mm:ss.SSS` -- hours, minutes, seconds, and milliseconds.
- `zZ` -- time zone and RFC 822 time zone

The job completes successfully if the string is found.

```
insert_job: timeformat_job
job_type: OMTF
machine: monagt
text_file_name: /export/home/agentdir/log/transmitter.log
text_file_filter: UNAGR
text_file_mode: DATETIME
time_format: "MM/dd/yyyy HH:mm:ss.SSS zZ"
lower_boundary: "08/17/2010 00:00:00.000 EST-0500"
upper_boundary: "08/25/2010 23:59:59.999 EST-0500"
monitor_mode: now
```

### time\_position Attribute -- Specify the First Column of a Time Stamp

The `time_position` attribute defines the first column of the time stamp in a text file.

#### NOTE

To use this attribute, you must specify `DATETIME` for the `text_file_mode` attribute.

#### Supported Job Type

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\) job type](#).

#### Syntax

This attribute has the following format:

```
time_position: timepos
```

- ***timepos***  
Specifies the first column of the time stamp in the log file.  
**Limits:** Up to 5 digits  
**Examples:** 12, 12345

### Example: Define the First Column of a Time Stamp

In this example, the `time_format` attribute provides the time pattern format that is applied to the values specified by the `lower_boundary` and `upper_boundary` attributes. The first column of the time stamp is 12. The time values are enclosed in quotation marks because they contain colons.

The values for the `time_format` attribute are as follows:

- h:mm -- hour in a.m./p.m., minute in hour
- a -- a.m./p.m. marker

The job completes successfully if the string is found.

```
insert_job: timeformat_job
job_type: OMTF
machine: monagt
monitor_mode: now
text_file_name: /export/home/logs/transaction.log
text_file_filter: UNAGR
text_file_mode: DATETIME
time_format: "h:mm a"
time_position: 12
lower_boundary: "10:08 PM"
upper_boundary: "12:30 PM"
monitor_mode: now
```

## timezone Attribute -- Define the Time Zone

The timezone attribute defines the time zone for the job you are defining. For example, if you define a start time of 01:00 for a job running on a machine in Denver, and set timezone to San Francisco (which is in the Pacific time zone, one hour earlier than Denver), the job starts at 2:00 a.m. Denver time.

The start event for jobs with time-based starting conditions that do not specify a time zone is scheduled based on the time zone under which the scheduler is running.

### Supported Job Types

This attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
timezone: zone
```

#### • **zone**

Specifies the time zone for the job. The job's time settings are based on this time zone, regardless of the server location. Specify a string that corresponds to an entry in the ujo\_timezones table or is recognized by the operating system or is any valid POSIX value.

**Limits:** Up to 50 characters; valid characters are a-z, A-Z, decimal digits, slash (/), hyphen (-), and underscore (\_); not case-sensitive; do not include spaces or periods.

#### **NOTE**

Enclose values that contain colons in quotation marks. If you do not enclose a value that contains a colon in quotation marks, the colon is interpreted as a delimiter.

**Examples:** "IST-5:30", MountainTime (instead of Mountain Time), SolomonIs (instead of Solomon Is)

#### **NOTE**

- Some regions might change to daylight saving time (DST) on different days and times than the time zone that the server is running in. Jobs scheduled to start in a DST time zone run one hour earlier (or later) until

the time zone in the job definition and the time zone where the server is running are both in the same time scheme (either DST or Standard Time).

- To display all the entries in the `ujo_timezones` table, enter the **autotimezone -l** command at the instance command prompt.

#### NOTE

##### UNIX:

- The `ujo_timezones` table contains entries for all the common time zones maintained by the operating system and for many cities in the United States.
- The scheduling manager interprets the string and matches it to a time zone value on your operating system. If the string is not found, the `ujo_timezones` table is read up to five times to resolve a zone value. If the zone value is not resolved after five attempts, the job fails. We recommend that you use a time zone value available from your operating system to help ensure that the scheduling manager is using the same values as other applications running on that computer.

#### NOTE

##### Windows:

- The `ujo_timezones` table contains entries for all the cities and time zones supported by Windows and for many cities in the world.
- Windows time zones are located in the Date/Time applet in the Control Panel. Most of the values in the Time Zone menu of the Date/Time applet are cities. You can specify any of those values as the time zone in your job definition. For example, if you want a job to run in the “Bogota, Lima, Quito,” time zone as shown on Windows, you can specify “Bogota,” “Lima,” or “Quito.”

#### Example: Set the Time Zone to Chicago Time

This example sets the time zone for a job definition to Chicago time:

```
timezone: Chicago
```

#### Example: Set the Time Zone to Pacific Time

This example sets the time zone for a job definition to Pacific time:

```
timezone: US/Pacific
```

## trigger\_cond Attribute -- Specify a Condition to Monitor For

The `trigger_cond` attribute specifies a condition to monitor the database for. The job completes when the condition is met.

### Supported Job Type

This attribute is optional for the [Database Trigger \(DBTRIG\) job type](#).

### Syntax

This attribute has the following format:

```
trigger_cond: condition
```

- **condition**  
Specifies the condition to monitor in the database.
  - For Oracle and DB2, this condition is the WHEN clause.
  - For SQL Server, this condition is the IF clause.

**Limits:** Up to 500 characters; case-sensitive

**NOTE**

For the specific database syntax, refer to your database vendor's documentation.

**NOTE**

- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.
- You can use global variables to specify the condition to monitor in the database. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

**Example: Specify a Trigger Condition for Oracle**

This example monitors the Inventory\_List table for updates. The job connects to the Oracle database named ORDERS. When the INVENTORY\_LIST is updated, if the number of units of productA has fallen below 1000, the job completes.

```
insert_job: dbtrig_oracle
job_type: DBTRIG
machine: dbagent
tablename: Inventory_List
trigger_type: UPDATE
trigger_cond: "new.ProductA < 1000"
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

**Example: Specify a Trigger Condition for SQL Server**

This example monitors the sales table for changes to the ord\_date and qty columns. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job completes only when both columns have changed.

```
insert_job: dbtrig_sqlsvr
job_type: DBTRIG
machine: dbagent
dbtype: MSSQL
owner: sa@myhost
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
tablename: sales
trigger_type: UPDATE
trigger_cond: UPDATE(ord_date) and UPDATE(qty)
```

**Example: Monitor a SQL Server Database Table for Added Rows with a Trigger Condition**

This example monitors the sales table for added rows. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job connects to the default database resource location defined on the agent. When the qty for inserted title ID TC7777 is greater than or equal to 20, the job completes.

```
insert_job: dbtrig3
job_type: DBTRIG
machine: DB_agent
dbtype: MSSQL
trigger_type: INSERT
trigger_cond: (select QTY from INSERTED where TITLE_ID='TC7777')>=20
tablename: sales
```

```
owner: sa@myhost
```

### Example: Specify a Trigger Condition for IBM DB2

This example monitors the STAFF table for changes. The job connects to the SAMPLE database and runs under the entadm user, who is authorized to create triggers on the database or schema the table belongs to. The job completes when the table has changed.

```
insert_job: dbtrig_db2
job_type: DBTRIG
machine: dbagent
dbtype: DB2
owner: entadm@myhost
connect_string: "jdbc:db2://172.31.255.255:50000/SAMPLE"
tablename: STAFF
dbtype: DB2
trigger_type: UPDATE
```

### trigger\_type Attribute -- Specify the Type of Database Change to Monitor For

The trigger\_type attribute specifies the type of database change to monitor for.

#### NOTE

If you do not specify the trigger\_type attribute in your job definition, the job monitors for the insertion of a row in the database table (the default).

### Supported Job Type

This attribute is optional for the Database Trigger (DBTRIG) job type.

### Syntax

This attribute has the following formats:

```
trigger_type: INSERT
trigger_type: DELETE
trigger_type: UPDATE
trigger_type: INSERT,DELETE
trigger_type: INSERT,UPDATE
trigger_type: DELETE,UPDATE
trigger_type: INSERT,DELETE,UPDATE
```

- **INSERT**  
Monitors for the insertion of a row in the database table. This is the default.
- **DELETE**  
Monitors for the deletion of a row in the database table.
- **UPDATE**  
Monitors for an update to any of the rows in the database table.
- **INSERT,DELETE**  
Monitors for the insertion or deletion of a row in the database table.

#### NOTE

This argument applies to Oracle and SQL Server only.

- **INSERT,UPDATE**

Monitors for the insertion of a row or an update to any of the rows in the database table.

**NOTE**

This argument applies to Oracle and SQL Server only.

- **DELETE,UPDATE**

Monitors for the deletion of a row or an update to any of the rows in the database table.

**NOTE**

This argument applies to Oracle and SQL Server only.

- **INSERT,DELETE,UPDATE** Monitors for the insertion of a row, deletion of a row, or an update to any of the rows in the database table.

**NOTE**

This argument applies to Oracle and SQL Server only.

**NOTE**

- Depending on the dbtype attribute, AutoSys Workload Automation generates the proper syntax for that database type using the values specified in the trigger\_type attribute.
- Each Database Trigger job creates a trigger on the database. We recommend that you speak to your database administrator before creating a Database Trigger job.

### Example: Specify a Trigger Type for Oracle

This example monitors the Inventory\_List table for an added row. The job runs on the dbagent agent computer and connects to the ORDERS database. When a row is inserted, the job completes.

```
insert_job: dbtrig_oracle
job_type: DBTRIG
machine: dbagent
tablename: Inventory_List
dbtype: Oracle
trigger_type: INSERT
owner: sys@orcl
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

### Example: Specify a Trigger Type for SQL Server

This example monitors the stores table for an added row or a deleted row. The job runs under the sa user, who owns the table and is authorized to create triggers on the database or schema the table belongs to. The job remains in a RUNNING state waiting for an added or deleted row. When a row is either added or deleted, the job completes.

```
insert_job: dbtrig1
job_type: DBTRIG
machine: DB_agent
trigger_type: DELETE,INSERT
tablename: stores
dbtype: MSSQL
owner: sa@myhost
connect_string: "jdbc:sqlserver://myhost:1433;DatabaseName=pubs"
```

### Example: Specify a Trigger Type for IBM DB2

This example monitors the STAFF table for changes. The job connects to the SAMPLE database and runs under the entadm user, who is authorized to create triggers on the database or schema the table belongs to. The job completes when the table has changed.

```
insert_job: dbtrig_db2
job_type: DBTRIG
machine: dbagent
dbtype: DB2
owner: entadm@myhost
connect_string:"jdbc:db2://172.31.255.255:50000/SAMPLE"
tablename: STAFF
dbtype: DB2
trigger_type: UPDATE
```

## ulimit Attribute -- Specify UNIX Resource Limits

The ulimit attribute modifies the shell's resource usage limit on a UNIX computer.

### Supported Job Type

This attribute is optional for the [Command \(CMD\) job type on UNIX](#).

### Syntax

This attribute has the following format:

```
ulimit: resource_type="soft_value,hard_value"
[, resource_type="soft_value,hard_value"...]
```

- **resource\_type**  
Specifies the resource type. The following options are available:
  - c -- Specifies the core file size in blocks.
  - d -- Specifies the data segment size in kilobytes (KB).
  - f -- Specifies the maximum file size in blocks.
  - m -- Specifies the process virtual size in kilobytes (KB).
  - n -- Specifies the number of files.
  - s -- Specifies the stack size in kilobytes (KB).
  - t -- Specifies the CPU time in seconds.
- **soft\_value**  
Defines the usage (soft) limit for the specified resource type. The soft limit can be increased up to the specified hard limit and can be changed without root authority.  
**Limits:** Any numeric digits or you can specify unlimited; the value must be less than or equal to the hard limit
- **hard\_value**  
Defines the maximum usage (hard) limit for the specified resource type. The hard limit can only be increased by the root user. Any user ID can decrease a hard limit.  
**Limits:** Any numeric digits or you can specify unlimited; the value must be greater than or equal to the soft limit.

### NOTE

If the job runs under a non-root user ID and the value specified in the job definition is greater than the current hard limit on the UNIX system, the hard limit will not be increased.



**NOTE**

- To find out the lower and upper limits of your operating system, contact your system administrator.
- You can specify multiple *resource\_type="soft\_value,hard\_value"* combinations. Separate each combination with a comma.
- You can use multiple entries of ulimit to define different sets of resource combinations. For example, you can specify two sets of combinations using two ulimit attributes as follows:
 

```
ulimit: c="100,200", s="250,300"
ulimit: t="4000,unlimited", m="3332,unlimited"
```
- Each *resource\_type="soft\_value,hard\_value"* combination can be up to 1024 characters.
- Each entry of ulimit can be up to 4096 characters.
- The ulimit values are applied before the user profile is sourced. Therefore, the values can be overridden by the */etc/profile* script, the *.profile* script, or any other user login scripts.
- This attribute is ignored for CMD jobs on Windows.

**Example: Specify Multiple ulimit Values**

This example runs the *procrun.sh* script on the *unixagent* computer. The job modifies the following resource limits on the UNIX computer:

- The core file size limit is 100 KB (soft limit). The size can be increased to 200 KB (hard limit).
- The stack size limit is 250 KB (soft limit). The size can be increased to 300 KB (hard limit).
- The CPU time can be up to 4000 seconds.
- The process virtual size limit is 3332 KB (soft limit). The size can be increased to an unlimited value.

```
insert_job: cmd_job
job_type: CMD
machine: unixagent
command: /ul/procrun.sh
ulimit: c="100,200", s="250,300", t="4000,unlimited", m="3332,unlimited"
```

**NOTE**

The resource limits are modified for the current job definition only. When you run another job, the default values will be used.

**upper\_boundary Attribute -- Define the End of the Range to be Searched**

The *upper\_boundary* attribute defines the end of the range to be searched in a text file.

This attribute is used with the *text\_file\_mode* attribute, which specifies the search mode (line, regular expression, or date and time).

**Supported Job Type**

This attribute is optional for the [Text File Reading and Monitoring \(OMTF\) job type](#).

To use this attribute, you must specify *NOW* for the *monitor\_mode* attribute. This attribute does not apply when the monitor mode is set to *WAIT* or *CONTINUOUS*.

**Syntax**

This attribute has the following format:

```
upper_boundary: end_range
```

- ***end\_range***

Defines the end of the range to be searched. The format of this value depends on the text file search mode. The formatting options are as follows:

- Numeric -- Specify a numeric value if the search mode is LINE.
- Regular expression -- Specify a regular expression if the search mode is REGEX.
- Date and time -- Specify date and time values if the search mode is DATETIME. Define the date and time using the format specified by the time format.

**Limits:** Up to 256 characters; case-sensitive

#### NOTE

You cannot specify the `upper_boundary` attribute when using the `monitor_mode` attribute set to the `WAIT` value.

#### NOTE

- If you do not specify the `text_file_mode` attribute in the job definition, the search mode is `LINE` by default.
- If you do not specify the `upper_boundary` attribute value in the job definition, the job starts searching from the `lower_boundary` value (if specified) to the end of the file.
- When the search mode is `REGEX`, you can specify a regular expression in the `upper_boundary` attribute. To compose a regular expression, follow the rules for Java class `java.util.regex.Pattern`. You can find these rules using a Google search for `java pattern`.
- You can use escape sequences in the regular expression. The following characters have a special meaning in regular expressions. To use these characters literally, precede the characters with one backward slash (`\`).
  - `\` -- back slash
  - `|` -- vertical bar
  - `(` -- opening parenthesis
  - `)` -- closing parenthesis
  - `[` -- opening bracket
  - `{` -- opening brace
  - `^` -- caret (circumflex)
  - `$` -- dollar sign
  - `*` -- asterisk
  - `+` -- plus sign
  - `?` -- question mark
  - `.` -- period

For example, to match the characters `.*` literally, specify `\.*\.` in your regular expression. The backward slashes escape the characters' special meanings.

#### Example: Define the End of a Range When the Search Mode is Line

This example searches the `c:\callog` file in line mode. The job starts searching the content from line 143 of the file. The upper boundary is not defined, so the job searches to the last line of the file. The job completes successfully if the `ERROR MESSAGE` string is found.

```
insert_job: omtf_line
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ERROR MESSAGE
text_file_mode: LINE
lower_boundary: 143
monitor_mode: NOW
```

#### Example: Define the End of a Range When the Search Mode is RegEx

This example searches the c:\ca\log file in regular expression mode. The lower boundary is not defined, so the job searches the content from the first line of the file to the upper boundary (a line that contains the word service). The job completes successfully if the ARCHIVE string is found.

```
insert_job: omtf_regex
job_type: OMTF
machine: monagt
text_file_name: "c:\ca\log"
text_file_filter: ARCHIVE
text_file_mode: REGEX
upper_boundary: service
monitor_mode: NOW
```

### Example: Define the End of a Range When the Search Mode is DateTime

This example searches the /export/home/logs/transmitter.log file in date and time mode. The job searches the content between May 20, 2010 at midnight and May 27, 2010 at 11:59 p.m. The date and time values are defined using the format specified in the time\_format attribute. The job completes successfully if the transmitted string is found.

```
insert_job: omtf_timedate
job_type: OMTF
machine: monagt
text_file_name: /export/home/logs/transmitter.log
text_file_filter: transmitted
text_file_mode: DATETIME
lower_boundary: "Thu May 20 00:00:00.000 EDT 2010"
upper_boundary: "Thu May 27 23:59:59.999 EDT 2010"
time_format: "EEE MMM dd HH:mm:ss.SSS zzz yyyy"
time_position: 12
monitor_mode: NOW
```

## upper\_boundary Attribute -- Define the Maximum CPU or Disk Usage to Monitor (OMCPU and OMD Jobs)

The upper\_boundary attribute defines the maximum CPU or disk usage to monitor for.

### Supported Job Types

This attribute is required for the following job types if WAIT or CONTINUOUS is specified for the monitor\_mode attribute:

- [CPU Monitoring \(OMCPU\)](#)
- [Disk Monitoring \(OMD\)](#)

The lower\_boundary attribute does not apply when the monitor mode is NOW.

### Syntax

This attribute has the following format:

- To specify the upper boundary for CPU usage:
 

```
upper_boundary: max_percent
```

  - **max\_percent**  
Defines the maximum amount of CPU usage to monitor for in percent.  
**Default:** 100  
**Limits:** 0-100
- To specify the upper boundary for disk usage:

```
upper_boundary: max_value
```

– **max\_value**

Defines the maximum amount of disk usage to monitor for. The `disk_format` attribute specifies the unit for this value.

**Limits:**

- 0-100 (This limit applies when the disk format is PERCENT)
- Up to 20 characters (This limit applies for all other disk formats.)

**Example:** 35000000

**NOTE**

- If you specify the `upper_boundary` attribute without the `lower_boundary` attribute, the range is between zero and the upper boundary.
- The `upper_boundary` attribute value must be greater than the `lower_boundary` attribute value.

**Example: Specify an Upper Boundary of 75 Percent for Monitoring Available CPU**

This example continuously monitors the CPU available on the winagt computer. An alert is written to the scheduler log file each time the available CPU is less than 25 percent or greater than 75 percent, and the CPU usage changes by more than 10 percent. If the change in value is less than or equal to 10, the job does not register a change.

```
insert_job: omcpu_job
job_type: OMCPU
machine: winagt
cpu_usage: FREE
lower_boundary: 25
upper_boundary: 75
inside_range: FALSE
no_change: 10
monitor_mode: CONTINUOUS
```

**Example: Specify an Upper Boundary of 10 GB for Monitoring Available Disk Space**

This example monitors /export/home for available space. The job completes when the available space is between 8 and 10 gigabytes (GB), and the available space changes by more than 15 percent. If the change in value is less than or equal to 15, the job does not register a change.

```
insert_job: omd_unix
job_type: OMD
machine: unixagt
disk_drive: /export/home
disk_space: FREE
disk_format: GB
lower_boundary: 8
upper_boundary: 10
inside_range: TRUE
no_change: 15
```

## URL Attribute -- Specify the URL of the JMX Server

The URL attribute specifies the URL to connect to the JMX server in a JMX job.

### Supported Job Types

This attribute is required for the following job types:

- [JMX-MBean Attribute Get](#) (JMXMAG)
- [JMX-MBean Attribute Set](#) (JMXMAS)
- [JMX-MBean Create Instance](#) (JMXMC)
- [JMX-MBean Operation](#) (JMXMOP)
- [JMX-MBean Remove Instance](#) (JMXMREM)
- [JMX-MBean Subscribe](#) (JMXSUB)

### Syntax

This attribute has the following format:

URL: "url"

- **"url"**

Specifies the URL to connect to the JMX server using an RMI connector.

The format is the following:

```
"service:jmx:rmi:///jndi/rmi://hostName:portNum/jmxServerName"
```

- **hostName**

Specifies the host name or IP address of the JMX server.

- **portNum**

Specifies the port number of the JMX server.

- **jmxServerName**

Specifies the name of the JMX server.

**Limits:** Up to 256 characters; case-sensitive

**Example:** "service:jmx:rmi:///jndi/rmi://localhost:9999/server"

### Example: Query a JMX Server for the Value of an MBean Attribute

Suppose that you want to know the value of the cachesize attribute for the Config MBean. The URL for the JMX server is service:jmx:rmi:///jndi/rmi://localhost:9999/server, where localhost is the host name and 9999 is the port number.

```
insert_job: query
job_type: JMXMAG
machine: appagent
URL: "service:jmx:rmi:///jndi/rmi://localhost:9999/server"
mbean_name: "DefaultDomain:index=1,type=Config"
mbean_attr: cachesize
```

### use\_topic Attribute -- Publish or Subscribe to a Topic or Queue

The use\_topic attribute specifies whether the job sends messages to a topic or queue in a JMS Subscribe job or publishes messages to a topic or queue in a JMS Publish job.

#### NOTE

If you do not specify the use\_topic attribute in your job definition, the job sends or publishes messages to a topic (the default).

### Supported Job Types

This attribute is optional for the following job types:

- [JMS Publish](#) (JMSPUB)
- [JMS Subscribe](#) (JMSSUB)

### Syntax

This attribute has the following format:

```
use_topic: TRUE | FALSE
```

- **TRUE**  
Sends or publishes messages to a topic.
- **FALSE**  
Sends or publishes messages to a queue. This is the default.

### Example: Publish a Message to a Queue

This example publishes the message "this is my message" to the queue named Queue. The Java class of the message is String. The initial context factory supplied by the JNDI service provider is com.ibm.websphere.naming.WsnInitialContextFactory. The service provider's URL is iiop://172.24.0.0:2809, where 172.24.0.0 is the IP address of the WebSphere MQ server and 2809 is the ORB port. The job uses the cyberuser@host JNDI user name to gain access to the connection factory named ConnectionFactory.

#### NOTE

```
insert_job: publish
job_type: JMSPUB
machine: appagent
initial_context_factory: com.ibm.websphere.naming.WsnInitialContextFactory
provider_url: "iiop://172.24.0.0:2809"
connection_factory: ConnectionFactory
destination_name: Queue
use_topic: FALSE
message_class: String
j2ee_user: cyberuser@host
j2ee_parameter: java.lang.String="this is my message"
```

#### NOTE

The agent does not support JMS messaging on IBM WebSphere. If you have IBM WebSphere MQ, your agent administrator can set up the agent plug-in to run JMS Publish and JMS Subscribe for JMS queues. JMS topics are not supported on IBM WebSphere MQ.

## user\_role Attribute -- Specify Database Resource Location

The user\_role attribute specifies the Oracle database user type for a database job.

### Supported Job Types

This attribute is optional for the following job types:

- Database Monitor (DBMON)
- Database Stored Procedure (DBPROC)
- Database Trigger (DBTRIG)
- SQL

### Syntax

This attribute has the following format:

```
user_role: user_type
```

- **user\_type**  
Specifies the role of the Oracle user to log in as. The user role must be defined in the Oracle database.  
**Limits:** Up to 30 characters; case-sensitive

**NOTE**

- If you omit this attribute from the job definition, the job runs with normal privileges.
- In Oracle, a user with sufficient authority can log in with different system privileges. For example, if a job requires sysdba privileges, enter **as sysdba** in this field.
- Your agent administrator can specify a default user type for all database jobs by setting the db.default.userType parameter in the agent's agentparm.txt file.
- The user\_role attribute overrides the default user type specified in the agent's agentparm.txt.

**Example: Specify a Database User Type**

This example specifies a database user named dbuser1, who is logged in with sysdba privileges. The job connects to the Oracle database named ORDERS, and the output is stored in the job's spool file.

```
insert_job: QRY1
job_type: SQL
machine: dbagent
owner: dbuser1@orcl
user_role: as sysdba
sql_command: SELECT * from NEWORDS
connect_string: "jdbc:oracle:thin:@172.31.255.255:1433:ORDERS"
```

**wake\_password Attribute -- Specify the Wake on LAN Password**

The wake\_password attribute specifies the Wake on LAN (WOL) password for network cards that support the Secure On security feature. After the agent broadcasts the WOL signal, the network card wakes up the computer only if the specified password is correct.

**NOTE**

The password is not encrypted on the scheduling manager.

**Supported Job Type**

This attribute is optional for the [Wake on LAN \(WOL\) job type](#).

**Syntax**

This attribute has the following formats:

```
wake_password: xx-xx-xx-xx
wake_password: xx-xx-xx-xx-xx-xx
```

- **xx-xx-xx-xx**  
Specifies the WOL password as a set of four 2-digit hexadecimal values separated by dashes.  
**Limits:** Up to 17 characters

**NOTE**

You can also separate the values using colons or periods. If you use colons, enclose the values in quotes.

**Example: Broadcast the WOL Signal Including a Password**

This example broadcasts the WOL signal including a password to the subnet with the 172.16.255.255 broadcast IP address and the server with the 11-22-33-44-55-66 MAC address. If the specified password matches the password stored on the server's network card, the server wakes up. The job completes successfully after the scheduler sends the WOL signal without verifying that the machine starts.

```
insert_job: wol_pwd_job
job_type: WOL
```

```

machine: agentnme
broadcast_address: 172.16.255.255
mac_address: 11-22-33-44-55-66
wake_password: AA-BB-CC-DD-EE-FF

```

## watch\_file Attribute -- Specify a File to Monitor

### Contents

The watch\_file attribute specifies the path to and name of a file to monitor.

### Supported Job Types

This attribute is required for the following job types:

- [File Trigger \(FT\)](#)
- [File Watcher \(FW\)](#)

### UNIX and Windows Syntax

This attribute has the following format:

```
watch_file: file
```

- **file**

Specifies the path to and name of one or more files to monitor.

To specify multiple files, use the asterisk (\*) and question mark (?) wildcard characters. Use \* for any number of characters and ? for any single character. Do not use the \* and ? in the path.

**Limits:** Up to 255 characters; case-sensitive

**UNIX Notes:**

- You can specify variables exported from the profile script or global variables in the name.
- Wildcard characters are expanded according to the wildcard expansion rules of the Bourne Shell.
- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, \${PATH}). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.

**Windows Notes:**

- You can specify system environment variables, job profile environment variables, or global variables in the name. If the variable is referenced in the job profile, we recommend that you enclose the variable in braces (for example, %{PATH}%). Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.
- When specifying drive letters in job definitions, you must escape the colon with quotation marks or backslashes. For example, C:\tmp or "C:\tmp" is valid; C:\tmp is not.

**NOTE**

An application that generates the monitored file may crash, or a communication link may be interrupted after the minimum size file is written. In these situations, AutoSys Workload Automation may incorrectly determine that the file is complete when it is not. If you have the permissions to control the application that generates the monitored file, you can do the following to avoid an incorrect result:

- Instruct the application to create a separate, zero-length file when it finishes writing the monitored file.
- Set the job to monitor for the completion of the zero-length file.

If the job detects the zero-length file, the job is complete.

### File Trigger Notes

When you define a File Trigger (FT) job, consider the following points:



## Scan intervals:

File Trigger jobs monitor file activity using a polling interval, which is every 30 seconds by default. File Trigger jobs do not detect multiple updates during the polling interval. They also do not detect changes that cancel each other out.

For example, if the job monitors for updates to a file, and the file is updated twice during the polling interval, the trigger occurs only once for the two updates. If the job monitors for the creation of a file, and the file is created and deleted during the polling interval, the trigger does not occur. Because the file did not exist when the directory was polled, the job does not detect the file creation and deletion.

### NOTE

To change the number of seconds between scans, specify the `watch_no_change` attribute in your job definition.

## Specifying wildcards in the file name:

- You can use wildcards in the file name value. The asterisk (\*) is a wildcard for zero or more characters and the question mark (?) is a wildcard for a single character.

In the following example, the agent monitors for any file in the `/usr/data` directory:

```
watch_file: "/usr/data/*"
```

In the following example, the agent monitors for any file in the `/usr/data` directory that has a file name with four characters and any extension:

```
watch_file: /usr/data/????.*
```

- You cannot use a wildcard as a prefix or within the directory names in the file path.
- When using a wildcard immediately after a forward slash (/), enclose the path in quotation marks. The quotation marks prevent part of the path from being recognized as a comment line. For example, this statement might not produce the desired results:

```
watch_file: /usr/data/*
```

To include the wildcard as part of the path, enclose the path in quotation marks, as follows:

```
watch_file: "/usr/data/*"
```

## Specifying a file on a remote Windows computer:

You can specify UNC (Universal Naming Convention) names. A UNC name is the name of a file or other resource that begins with two backslashes (\\), indicating that it exists on a remote computer.

## Specifying an i5/OS object:

You can use generic names to specify library names and object names on the i5/OS system. A generic name starts with characters that are part of a valid name and ends with an asterisk (\*). The asterisk denotes any number of characters and can only be placed at the end of a generic name.

The following `watch_file` attribute prompts the agent to monitor for the creation of any file object in the QSYS file system with a file name that matches the `PAY*` generic name:

```
watch_file: "LIB/PAY*/*FILE"
```

### NOTE

A name cannot contain a single asterisk and no other characters. To specify all possible names, use the special value `*ALL`.

## Examples Specifying UNIX files using the `watch_file` Attribute

The following examples specify UNIX files to monitor:

### NOTE

The following examples are File Trigger jobs.

- This example monitors the `/tmp/batch.input` file.

```
insert_job: ft_unix1
job_type: FT
machine: unixagt
watch_file: /tmp/batch.input
```

- This example monitors a file whose name has been assigned to the global variable file\_1.

```
insert_job: ft_unix2
job_type: FT
machine: unixagt
watch_file: ${file_1}
```

- This example monitors files with the format, /usr/reports/Augustxx.out (such as /usr/reports/August01.out and /usr/reports/August02.out).

```
insert_job: ft_unix3
job_type: FT
machine: unixagt
watch_file: /usr/reports/August??out
```

- This example monitors the /tmp directory for files with names that begin with "pay".

```
insert_job: ft_unix4
job_type: FT
machine: unixagt
watch_file: /tmp/pay*
```

### **Examples Specifying Windows files using the watch\_file Attribute**

The following examples specify Windows files to monitor:

#### **NOTE**

The following examples are File Trigger jobs. If you are using a legacy agent, you can define these examples as File Watcher jobs by modifying the job\_type attribute to **job\_type: FW**.

- This example monitors the C:\tmp\BATCH.IN file.

```
insert_job: ft_win1
job_type: FT
machine: winagt
watch_file: "C:\tmp\BATCH.IN"
```

- This example monitors a file whose name has been assigned to the global variable file\_1.

```
insert_job: ft_win2
job_type: FT
machine: winagt
watch_file: ${file_1}
```

- This example monitors files with the format, c:\reports\Augustxx.out (such as C:\reports\August01.out and C:\reports\August02.out).

```
insert_job: ft_win3
job_type: FT
machine: winagt
watch_file: "c:\reports\August??out"
```

- This example monitors the temp directory for files with names that begin with "pay".

```
insert_job: ft_win4
job_type: FT
```

```
machine: winagt
watch_file: "c:\temp\pay*"
```

### **Examples Specifying i5/OS files using the watch\_file Attribute**

The following examples specify i5/OS files to monitor:

- This example monitors a daily update to a payroll file object on an i5/OS computer. When the file is updated, the job completes.

```
insert_job: ft_i5os1
job_type: FT
machine: i5agt
watch_file: /QSYS.LIB/LIBRARY.LIB/DEPT.FILE/PAYROLL.MB
watch_file_type: UPDATE
```

- This example monitors for files that are created in the /home/cybesp/ directory in the root file system. The job completes if a file is created with a file name that matches the following criteria:

- Starts with PID
- Ends with four characters
- Has any extension.

```
insert_job: ft_i5os2
job_type: FT
machine: i5agt
watch_file: /home/cybesp/PID????.*
watch_file_type: CREATE
```

- This example uses a generic name to specify a QSYS file object in an i5/OS computer. The job completes when any file object with a file name that starts with PAY and ends with any characters is created in the QSYS file system. Note that the file name is enclosed in quotation marks so that the text following /\* is not interpreted as a comment.

```
insert_job: ft_i5os3
job_type: FT
machine: i5agt
watch_file: "LIB/PAY*/*FILE"
watch_file_type: CREATE
```

- This example monitors the size of a QSYS file object in an i5/OS computer. The job completes when the EXPOBJ file object in the QSYS file system increases by 10 bytes or more. Since no member is specified in the file name, the job monitors the entire object size. The entire object size includes the size of the file object itself and the total size of the file object's members.

```
insert_job: ft_i5os3
job_type: FT
machine: i5agt
watch_file: "LIB/EXPOBJ/*FILE"
watch_file_type: EXPAND
watch_file_type: SIZE
watch_file_change_value: 10
```

#### **NOTE**

- The file name is enclosed in single quotation marks so that the text following /\* is not interpreted as a comment.
- To monitor only the total size of the file object's members, specify \*ALL as the member name.

## watch\_file\_change\_type Attribute -- Specify the Type of Change in File Size

The watch\_file\_change\_type attribute specifies the type of change to detect when monitoring a file for an increase or decrease in size.

### NOTE

- If you do not specify the watch\_file\_change\_type attribute in your job definition, the job monitors the file to reach a specified size (the default).
- To use this attribute, you must include the watch\_file\_change\_value attribute in your job definition. These attributes correspond to each other.

### Supported Job Type

This attribute is optional for the [File Trigger \(FT\) job type](#) as follows:

- If EXPAND or SHRINK is specified for the watch\_file\_type attribute, you can specify DELTA, PERCENT, or SIZE.
- If CREATE is specified for the watch\_file\_type attribute, you can specify SIZE.

### Syntax

This attribute has the following format:

```
watch_file_change_type: DELTA | PERCENT | SIZE
```

- **DELTA**  
Specifies that the job monitors for a change in the file size. The unit is bytes.
- **PERCENT**  
Specifies that the job monitors for a change in the file size by percentage.
- **SIZE**  
Specifies that the job monitors the file to reach a specified size. The unit is bytes. This is the default.

### NOTE

Some file systems will create files with extra space in anticipation of further file expansion requests. Therefore, specifying exact byte counts will not always work.

### Example: Monitor for an Increase in File Size by a Certain Amount

This example defines a job named ft\_unix\_delta to monitor the file named record in the credit directory and issue a trigger when the file increases in size by at least 200000 bytes and then remains unchanged for 30 minutes.

```
insert_job: ft_unix_delta
job_type: FT
machine: unixagt
watch_file: /credit/record
watch_file_type: EXPAND
watch_file_change_type: DELTA
watch_file_change_value: 200000
watch_no_change: 30
```

### Example: Monitor for a Decrease in File Size to a Certain Amount

This example defines a job named ft\_unix\_shrink to issue a trigger when the file named distribute that is located in the /cash/items directory is 1000 bytes or smaller.

```
insert_job: ft_unix_shrink
job_type: FT
machine: unixagt
watch_file: /cash/items/distribute
watch_file_type: SHRINK
```

```
watch_file_change_type: SIZE
watch_file_change_value: 1000
```

### Example: Monitor for a Decrease in File Size by a Certain Percentage

This example defines a job named `ft_win_percent` to issue a trigger when the file named `test` in the `c:\amount` directory shrinks by at least 65% and then remains unchanged for 30 minutes.

```
insert_job: ft_win_percent
job_type: FT
machine: winagt
watch_file: "c:\amount\test"
watch_file_type: SHRINK
watch_file_change_type: PERCENT
watch_file_change_value: 65
watch_no_change: 30
```

### watch\_file\_change\_value Attribute -- Specify the Change in File Size

The `watch_file_change_value` attribute specifies the amount of change or the limit in size to detect when monitoring a file for an increase or decrease in size.

#### NOTE

To use this attribute, you must include the `watch_file_change_type` attribute in your job definition. The `watch_file_change_type` specifies the type of unit associated with this value.

### Supported Job Type

This attribute is required for the [File Trigger \(FT\) job type](#) if `EXPAND` or `SHRINK` is specified for the `watch_file_type` attribute.

This attribute is optional for the File Trigger job type if `watch_file_type: CREATE` and `watch_file_change_type: SIZE` are specified.

### Syntax

This attribute has the following format:

```
watch_file_change_value: value
```

#### • **value**

Defines one of the following:

- When the file change type is `DELTA`, this attribute defines the number of bytes the file size must change for the file trigger to occur.
- When the file change type is `PERCENT`, this attribute defines the percentage the file size must change for the file trigger to occur.
- When the file change type is `SIZE`, this attribute defines a limit on the file size. If the type of activity is `CREATE` or `EXPAND`, the file trigger occurs when the file size is equal to or greater than the specified size. If the type of activity is `SHRINK`, the file trigger occurs when the file size is equal to or less than the specified size.

**Limits:** Up to 19 digits

**i5/OS:** When monitoring a \*FILE object in QSYS with no member specified in the file name, the job monitors the entire object size. The entire object size includes the size of the file object itself and the total size of the file object's members. To only monitor the total size of the file object's members, specify \*ALL as the member name.

**NOTE**

- The maximum value is 9,223,372,036,854,775,807.
- If `watch_file_type` is set to EXPAND or SHRINK, the value must be greater than or equal to 1.
- If `watch_file_type` is set to SHRINK and `watch_file_change_value` is set to PERCENT, this value must be between 1 and 100.
- If `watch_file_type` is set to EXPAND and `watch_file_change_value` is set to PERCENT, this value must be between 1 and 10000.

**Example: Monitor for an Increase in File Size by a Certain Percentage**

This example monitors file test in directory amount. When the file expands in size by 90% or more, the job completes. For instance, suppose that the current file size is 1000 bytes. The file trigger occurs when the file size expands to 1900 bytes or more.

```
insert_job: ft_win_percent
job_type: FT
machine: unixagt
watch_file: /amount/test
watch_file_type: EXPAND
watch_file_change_type: PERCENT
watch_file_change_value: 90
```

**watch\_file\_groupname Attribute -- Specify the Group that Owns the File**

The `watch_file_groupname` attribute specifies the name of the group that owns the file to be monitored.

**Supported Job Type**

This attribute is optional for the [File Trigger \(FT\) job type](#) on UNIX.

**Syntax**

This attribute has the following format:

```
watch_file_groupname: group
```

- ***group***  
Specifies the name of the group that owns the file to be monitored.  
**Limits:** Up to 80 characters; case-sensitive; it cannot contain delimiters (such as spaces)

**NOTE**

- Applies only to UNIX jobs and i5/OS jobs (if the file is not a QSYS object).
- If the file is not owned by the specified group, the following occurs:
  - The job continues monitoring if the type of file activity is CREATE.
  - The job completes if the type of file activity is DELETE or NOTEXIST.
  - The job fails if the type of file activity is EXPAND, EXIST, SHRINK, or UPDATE.

**Example: Monitor for the Existence of a File that Is Owned by a Specified UNIX Group**

Suppose that you want a job to monitor for the existence of the `/data/payroll.dat` file that is owned by the UNIX group ACCTS:

- If the file exists and the file is owned by the group ACCTS when the job is readied, the job completes immediately.
- If the file exists and the file is not owned by the group ACCTS when the job is readied, the job fails.

```
insert_job: ft_unixgroup
job_type: FT
machine: unixagent
watch_file: /data/payroll.dat
watch_file_type: EXIST
watch_file_groupname: ACCTS
```

## watch\_file\_min\_size Attribute -- Specify the Minimum Size that a File Must Reach

The `watch_file_min_size` attribute specifies the minimum size that the watched file must reach for the job to complete. The watched file must maintain a steady state during the interval specified in the `watch_file` attribute.

### NOTE

If you do not specify the `watch_file_min_size` attribute in your job definition, the job completes if the file exists (the default).

### Supported Job Type

This attribute is optional for the [File Watcher \(FW\) job type](#).

### Syntax

This attribute has the following format:

```
watch_file_min_size: bytes
```

- **bytes**

Specifies the minimum file size (in bytes) that must be reached to cause this job to complete.

**Limits:** Up to 19 digits

**Default:** 0 (The job completes if the file exists.)

### NOTE

The maximum value is 9,223,372,036,854,775,807.

### Example: Specify the Minimum File Size

This example monitors the `watch_file.log` file on the unixagent computer. The unixagent is a legacy agent. The job completes when the file reaches 10000 bytes and maintains a steady state for 60 seconds.

```
insert_job: fw_job
job_type: FW
machine: unixagent
watch_file: /tmp/watch_file.log
watch_file_min_size: 10000
watch_interval: 60
```

## watch\_file\_owner Attribute -- Specify the Owner of the File to be Monitored

The `watch_file_owner` attribute specifies the user ID that owns the file to be monitored.

### Supported Job Type

This attribute is optional for the [File Trigger \(FT\) job type](#) on UNIX.

### Syntax

This attribute has the following format:

```
watch_file_owner: owner
```

- **owner**

Specifies the user ID that owns the file to be monitored.

**Limits:** Up to 80 characters; case-sensitive; it cannot contain delimiters (such as spaces)

**NOTE**

- Applies only to UNIX jobs and i5/OS jobs (if the file is not a QSYS object).
- If the file is not owned by this group:
  - The job continues monitoring if the type of file activity is CREATE.
  - The job completes if the type of file activity is DELETE or NOTEXIST.
  - The job fails if the type of file activity is EXPAND, EXIST, SHRINK, or UPDATE.

**Example: Monitor a File Owned by a Specific User**

This example defines a job named `ft_job` to issue a trigger in the following situations:

- When a file named payroll with attributes that specify JDOE as the file owner is created and is not modified within the first 30 minutes of existence.
- When a file named payroll with attributes that specify JDOE as the file owner exists before the job starts, is located during the first scan, and is not modified within the following 30 minutes.
- When the owner of the file named payroll changes to JDOE and the file is not modified within the next 30 seconds.

```
insert_job: ft_job
job_type: FT
machine: ftagt
watch_file: /usr/data/payroll
watch_file_type: CREATE
watch_file_owner: JDOE
watch_no_change: 30
```

**watch\_file\_recursive Attribute -- Specify Whether to Monitor Subdirectories for File Activity**

The `watch_file_recursive` attribute specifies whether the job monitors for file activity in the specified directory only or in the specified directory and all of its subdirectories.

**NOTE**

If you do not specify the `watch_file_recursive` attribute in your job definition, the job monitors for file activity in the specified directory only (the default).

**Supported Job Type**

This attribute is optional for the [File Trigger \(FT\) job type](#).

**Syntax**

This attribute has the following format:

```
watch_file_recursive: Y | N
```

- **Y**  
Specifies that the job monitors for file activity in the specified directory and all of its subdirectories.
- **N**  
Specifies that the job monitors for file activity in the specified directory only. This is the default.

**Example: Monitor for an Update to a File in a Directory and its Subdirectories**



This example defines a job named `ft_unix_update` to monitor the `/usr/data/` directory and its subdirectories for the file named `payroll` and to issue a trigger when the file is updated.

```
insert_job: ft_unix_update
job_type: FT
machine: unixagt
watch_file: /usr/data/payroll
watch_file_type: UPDATE
watch_file_recursive: Y
```

### Example: Monitor for Updates to All Files in a Directory and its Subdirectories

This example defines a job named `ft_unix_update` to monitor the `/usr/data/` directory and all of its subdirectories and to issue a trigger when any file that is located in those directories is updated and then is not modified again during the next 30 minutes.

```
insert_job: ft_unix_update
job_type: FT
machine: unixagt
watch_file: "/usr/data/*"
watch_file_type: UPDATE
watch_file_recursive: Y
watch_no_change: 30
```

### Example: Monitor a Directory and Its Subdirectories for File Expansion

This example monitors a file named `inventory.txt` and all other jobs in that directory and its subdirectories. When the size of the `inventory.txt` file or any other file in that same directory increases by 10 bytes, the job completes and the output is recorded in the `ftjob1_out.log` file.

```
insert_job: ft_job
job_type: FT
machine: ftagt
watch_file: "c:\fwjobs\inventory.txt"
watch_file_type: Expand
watch_file_change_type: DELTA
watch_file_change_value: 10
watch_file_recursive: Y
```

## watch\_file\_type Attribute -- Specify the Type of File Activity to Monitor For

### Contents

The `watch_file_type` attribute specifies the type of file activity that a File Trigger job monitors for.

#### NOTE

If you do not specify the `watch_file_type` attribute in your job definition, the job monitors for the creation of the file (the default).

### Supported Job Type

This attribute is optional for the [File Trigger \(FT\) job type](#).

### Syntax

This attribute has the following format:

watch\_file\_type: CREATE | DELETE | EXIST | EXPAND |  
NOTEXIST | SHRINK | UPDATE | GENERATE

- **CREATE**

(Default) Indicates that the file trigger occurs when the file is located and remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is located. A file that exists when the job starts is located during the first scan, which occurs 30 seconds after the job starts. Subsequent scans occur every 30 seconds, so a file that does not exist when the job starts is located a maximum of 30 seconds after it is created.

- **DELETE**

Indicates that the file trigger occurs when the file is deleted.

- **EXIST**

Indicates that the file trigger occurs if the file exists. If the file does not exist, the job fails.

- **EXPAND**

Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirement is met when the file is equal to or larger than the specified size or expands by at least the specified amount, depending on the type of size change that you specify in the job definition.

**NOTE**

To monitor for this type of file activity, you must specify the watch\_file\_change\_type attribute in your job definition.

- **NOTEXIST**

Indicates that the file trigger occurs if the file does not exist. If the file exists, the job fails.

- **SHRINK**

Indicates that the file trigger occurs when the file meets the specified size requirements and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the size requirements are met. When the file meets the size requirements before the job starts, the job detects this during the first scan. In this case, the trigger occurs 30 seconds after the job starts unless the watch\_no\_change attribute is specified. The size requirements are met when the file is equal to or smaller than the specified size or shrinks by at least the specified amount.

**NOTE**

To monitor for this type of file activity, you must specify the watch\_file\_change\_type attribute in your job definition.

- **UPDATE**

Indicates that the file trigger occurs when the file is updated and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is updated.

- **GENERATE**

Indicates that the trigger occurs when the file is created or updated and then remains unchanged for the amount of time specified in the watch\_no\_change attribute. When the watch\_no\_change attribute is not specified, the trigger occurs as soon as the file is created or updated.

**NOTE**

If the file exists when the job starts and is not updated after the job starts, no trigger occurs. In this case, the job does not complete and remains in the RUNNING state indefinitely.

### **CREATE File Trigger Type Notes**

When specifying the CREATE value in the watch\_file\_type attribute, consider the following points:

- If the job definition specifies a file name that contains wildcards, the job monitors the first file that matches the name.
- If the job definition specifies a directory that does not exist or the directory is deleted during monitoring, the job fails.
- The file trigger occurs only when all conditions specified in the job definition are satisfied, including the owner or group criteria if they are specified. If the criteria do not match, the job continues to run and the trigger occurs when file attributes are updated to meet the criteria.

#### **Example: Monitor the Creation of a File Owned by a Specific User**

This example defines a job named ft\_job to issue a trigger in the following situations:

- When a file named payroll with attributes that specify JDOE as the file owner is created and is not modified within the first 30 minutes of existence.
- When a file named payroll with attributes that specify JDOE as the file owner exists before the job starts, is located during the first scan, and is not modified within the following 30 minutes.
- When the owner of the file named payroll changes to JDOE and the file is not modified within the next 30 minutes.

```
insert_job: ft_job
job_type: FT
machine: ftagt
watch_file: /usr/data/payroll
watch_file_type: CREATE
watch_file_owner: JDOE
watch_no_change: 30
```

### **DELETE File Trigger Type Notes**

When specifying the DELETE value in the watch\_file\_type attribute, consider the following points:

- If the file does not exist when the job starts, the trigger occurs immediately.
- If the job definition specifies a directory that does not exist or the directory is deleted during monitoring, the job fails.
- You can specify the continuous attribute only when you are using a wildcard (\* or ?) in the file name value.

#### **Example: Monitor for the Deletion of Files That Have Names Beginning with Pay**

This example continuously monitors the /usr/data/ directory for files that have names beginning with pay. When all files that have a name beginning with pay are deleted, the job completes successfully.

```
insert_job: ftjob
job_type: FT
machine: ftagt
watch_file: /usr/data/pay*
watch_file_type: DELETE
continuous: Y
```

### **EXIST File Trigger Type Notes**

When specifying the EXIST value in the watch\_file\_type attribute, consider the following points:

- The job fails if the file does not exist in the specified location at the time that the job starts.
- When the job definition specifies a file name that contains wildcards and more than one matching file exists, the file job monitors the file that was most recently modified. To determine which file is being monitored, view the debug output.

**Example: Verify that a File Exists in a Directory**

This example defines a job named `ft_win_exist` to search the `c:\bank\account` directory for a file named `money` and to issue a trigger if the file exists.

```
insert_job: ft_win_exist
job_type: FT
machine: winagt
watch_file: "c:\bank\account\money"
watch_file_type: EXIST
```

**Example: Verify that a File Exists in a Directory or Its Subdirectories**

This example defines a job named `ft_win_recursive` to search the `c:\bank\account` directory and all of its subdirectories for a file named `money` and to issue a trigger if the file exists.

```
insert_job: ft_win_recursive
job_type: FT
machine: winagt
watch_file: "c:\bank\account\money"
watch_file_type: EXIST
watch_file_recursive: Y
```

**EXPAND File Trigger Type Notes**

When specifying the `EXPAND` value in the `watch_file_type` attribute, consider the following points:

- The job fails when the monitored file is deleted before the trigger occurs or when the file does not exist at the time that the job starts.
- When the job is defined to monitor continuously for a specific size, the job continues to run after the first trigger and subsequent triggers occur as long as the file is at least as big as the specified size. If the size of the file decreases, triggers continue to occur until the file is smaller than the specified size.
- When the job monitors for a percent expansion, the trigger occurs when the size of the file increases by the specified percentage. For example, when a job monitors for 50 percent and the file is 2MB at the time that the job starts, the trigger occurs when the file is 3MB ( $2\text{MB} + (0.5 \times 2\text{M})$ ).
- When the job is defined to monitor continuously for a percent expansion, the job continues to run after the first trigger and subsequent triggers occur when the file expands exponentially by the specified percentage. For example, when the file is 3MB at the time of the first trigger and the job monitors for 50 percent, subsequent triggers occur when the file is 4.5MB, 6.75MB, and so on.
- When the job monitors for a delta expansion, the trigger occurs when the size of the file expands by the specified number of bytes. For example, when a job monitors for a delta expansion of 400 and the file is 200 bytes when the job starts, the trigger occurs when the file is 600 bytes.
- When the job is defined to monitor continuously for a delta expansion, the job continues to run after the first trigger occurs and subsequent triggers occur each time the file expands by the specified number of bytes. For example, when the file is 600 bytes at the time that the first trigger occurs and the job monitors for a delta expansion of 400, subsequent triggers occur when the file is 1000 bytes, 1400 bytes, and so on.
- When the job definition specifies a file name that contains wildcards and more than one matching file exists, the file job monitors the file that was most recently modified. To determine which file is being monitored, view the debug output.

**Example: Monitor for an Increase in File Size to a Certain Amount**

This example monitors the test file in the data directory. When that file size reaches 100 bytes or more, the job completes.

```
insert_job: ft_unix_size
job_type: FT
```

```

machine: unixagt
watch_file: /data/test
watch_file_type: EXPAND
watch_file_change_type: SIZE
watch_file_change_value: 100

```

### Example: Monitor for an Increase in File Size by a Certain Amount

This example defines a job named `ft_unix_delta` to monitor the file named `record` in the `credit` directory and issue a trigger when the file increases in size by at least 200000 bytes and then remains unchanged for 30 minutes.

```

insert_job: ft_unix_delta
job_type: FT
machine: unixagt
watch_file: /credit/record
watch_file_type: EXPAND
watch_file_change_type: DELTA
watch_file_change_value: 200000
watch_no_change: 30

```

### Example: Monitor a File for Size Increase

This example monitors a file named `inventory.txt` in the `c:\fwjobs` directory. When the size of the file increases to 1000 bytes, the job completes.

```

insert_job: ftjob
job_type: FW
machine: ftagt
watch_file: "c:\fwjobs\inventory.txt"
watch_file_type: EXPAND
watch_file_change_type: SIZE
watch_file_change_value: 1000

```

### NOTEXIST File Trigger Type Notes

When specifying the `NOTEXIST` value in the `watch_file_type` attribute, consider the following points:

- If the file does not exist when the job starts, the trigger occurs immediately.
- If the job definition specifies a directory that does not exist or the directory is deleted during monitoring, the job fails.
- If the file name specified in the `watch_file` attribute contains wildcards, the file with the most recent modification time that matches the criteria is monitored.

### Example: Verify that a File Does Not Exist in a Directory

This example defines a job named `ft_unix_notexist` to search the `/start/term/` directory for a file named `vacation` and to issue a trigger if the file does not exist.

```

insert_job: ft_unix_notexist
job_type: FT
machine: unixagt
watch_file: /start/term/vacation
watch_file_type: NOTEXIST

```

## **SHRINK File Trigger Type Notes**

When specifying the SHRINK value in the watch\_file\_type attribute, consider the following points:

- The job fails when the monitored file is deleted before the trigger occurs or when the file does not exist at the time that the job starts.
- When the job is defined to monitor continuously for a specific size, the job continues to run after the first trigger and subsequent triggers occur as long as the file is smaller than the specified size. If the size of the file shrinks, triggers continue to occur until the file reaches the specified size.
- When the job monitors for a percent shrinkage, the trigger occurs when the size of the file shrinks by the specified percentage. For example, when a job monitors for 50 percent and the file is 200MB at the time that the job starts, the trigger occurs when the file is 100MB ( $200\text{MB} + (0.5 \times 200\text{MB})$ ).
- When the job is defined to monitor continuously for a percent shrinkage, the job continues to run after the first trigger and subsequent triggers occur when the file shrinks exponentially by the specified percentage. For example, when the file is 100MB at the time of the first trigger and the job monitors for 50 percent, subsequent triggers occur when the file is 50MB, 25MB, and so on.
- When the job monitors for a delta shrinkage, the trigger occurs when the size of the file shrinks by the specified number of bytes.
- When the job is defined to monitor continuously for a delta shrinkage, the job continues to run after the first trigger occurs and subsequent triggers occur each time the file shrinks by the specified number of bytes. For example, when the file is 25 bytes at the time that the first trigger occurs and the job monitors for a delta shrinkage of 4, a subsequent trigger occurs when the job scans the file and finds that it is 21 bytes or smaller.
- When the job definition specifies a file name that contains wildcards and more than one matching file exists, the file job monitors the file that was most recently modified. To determine which file is being monitored, view the debug output.
- If the job definition specifies a directory that does not exist or the directory is deleted during monitoring, the job fails.
- The file trigger occurs only when all conditions specified in the job definition are satisfied, including the owner or group criteria if they are specified. If the criteria do not match, the job continues to run and the trigger occurs when file attributes are updated to meet the criteria.

### **Example: Monitor for a Specific Size**

This example defines a job named ft\_unix\_shrink to issue a trigger when the file named distribute that is located in the /cash/items directory is 1000 bytes or smaller.

```
insert_job: ft_unix_shrink
job_type: FT
machine: unixagt
watch_file: /cash/items/distribute
watch_file_type: SHRINK
watch_file_change_type: SIZE
watch_file_change_value: 1000
```

### **Example: Monitor for a Specific Amount of Shrinkage**

This example defines a job named ft\_win\_shrink to issue a trigger when the file named cash in the c:\cost directory shrinks by 10 bytes.

```
insert_job: ft_win_shrink
job_type: FT
machine: winagt
watch_file: "c:\cost\cash"
watch_file_type: SHRINK
watch_file_change_type: DELTA
```

```
watch_file_change_value: 10
```

### Example: Monitor for Percent Shrinkage

This example defines a job named `ft_win_percent` to issue a trigger when the file named `test` in the `c:\amount` directory shrinks by at least 65% and then remains unchanged for 30 minutes.

```
insert_job: ft_win_percent
job_type: FT
machine: winagt
watch_file: "c:\amount\test"
watch_file_type: SHRINK
watch_file_change_type: PERCENT
watch_file_change_value: 65
watch_no_change: 30
```

### UPDATE File Trigger Type Notes

When specifying the UPDATE value in the `watch_file_type` attribute, consider the following points:

- The job fails when the file does not exist at the time the job starts or when it is deleted without being modified.
- The file trigger occurs only when all conditions specified in the job definition are satisfied, including the owner or group criteria if they are specified. If the criteria do not match, the job continues to run and the trigger occurs when file attributes are updated to meet the criteria.
- When the job definition specifies a file name that uses wildcards, the job monitors the file that was most recently updated.

### Example: Monitor for an Update to a File in a Directory and its Subdirectories

This example defines a job named `ft_unix_update` to monitor the `/usr/data/` directory and its subdirectories for the file named `payroll` and to issue a trigger when the file is updated.

```
insert_job: ft_unix_update
job_type: FT
machine: unixagt
watch_file: /usr/data/payroll
watch_file_type: UPDATE
watch_file_recursive: Y
```

### Example: Monitor for Updates to All Files in a Directory and its Subdirectories

This example defines a job named `ft_unix_update` to monitor the `/usr/data/` directory and all of its subdirectories and to issue a trigger when any file that is located in those directories is updated and then is not modified again during the next 30 minutes.

```
insert_job: ft_unix_update
job_type: FT
machine: unixagt
watch_file: "/usr/data/*"
watch_file_type: UPDATE
watch_file_recursive: Y
watch_no_change: 30
```

## GENERATE File Trigger Type Notes

When specifying the GENERATE value in the watch\_file\_type attribute, consider the following points:

- If the job definition specifies a directory that does not exist or the directory is deleted during monitoring, the job fails.
- The file trigger occurs only when all conditions specified in the job definition are satisfied, including the owner or group criteria if they are specified. If the criteria do not match, the job continues to run and the trigger occurs when file attributes are updated to meet the criteria.
- If the job definition specifies continuous monitoring and a file name that contains wildcards, the job monitors all files that match the name. The job issues an alert for each monitored file that remains unchanged for the amount of time specified in the watch\_no\_change attribute.
- If the job definition specifies that monitoring is not continuous and a file name that contains wildcards, the job monitors the first file that matches the name.

### Example: Monitor a File for Inactivity

This example defines a job named FTP\_input\_companyXYZ to issue a trigger when the file named /in\_coming/FTP/companyXYZ.dat is created or updated and remains unchanged for 3 minutes.

```
insert_job: FTP_input_companyXYZ
job_type: FT
machine: localhost
watch_file: /in_coming/FTP/companyXYZ.dat
watch_file_type: GENERATE
watch_no_change: 3
```

## watch\_file\_win\_user Attribute -- Specify a Windows User for Monitoring UNC Files

The watch\_file\_win\_user attribute specifies the Windows user ID that monitors a network resource. For example, you can specify UNC names and share names in your job definition. The Windows user ID is used to monitor the specified file.

Usually, when running a Windows program as a service, you are restricted to how you can access data on remote computers. For example, to access data on a remote computer as a specified user ID, you must run the Windows service with that user ID. With the agent, however, those restrictions do not apply. Instead of running the agent service with a specific user ID, you can specify the user ID in the job definition.

### NOTE

To use the watch\_file\_win\_user attribute, you *cannot* specify the watch\_file\_owner and watch\_file\_groupname attributes in the same job definition.

### Supported Job Type

This attribute is optional for the [File Trigger \(FT\) job type](#) on Windows.

### Syntax

This attribute has the following format:

```
watch_file_win_user: user
```

- **user**  
Specifies the Windows user ID used to monitor the remote file.user ID and the domain the user ID belongs to.  
**Limits:** Up to 80 characters; case-sensitive; it cannot contain delimiters (such as spaces)

### Example: Monitor a Remote Windows File

In this example, the path c:\WINNT\Profiles\Visitor\Desktop\ has the share name MyDesktop. The command notify.txt is in that path on the CYBNT server. JDOE is a user ID in the CYBDOM domain and has access to the notify.txt command.



JDOE's password is defined on the scheduling manager. The job uses JDOE to monitor for the existence of the notify.txt file. If the file exists in the directory, the job completes. If the file does not exist in that directory, the job fails.

```
insert_job: ft_job
job_type: FT
machine: winagent
watch_file: \\CYBNT\MyDesktop\notify.txt
watch_file_type: EXIST
watch_file_win_user: CYBDOM\JDOE
```

## watch\_interval Attribute -- Specify the Frequency to Monitor a File

The watch\_interval attribute specifies how often the job checks the existence and size of the watched file.

### Supported Job Type

This attribute is optional for the [File Watcher \(FW\) job type](#).

### Syntax

This attribute has the following format:

```
watch_interval: seconds
```

- **seconds**

Specifies the frequency (in seconds) the File Watcher job checks for the existence and size of the watched file.

**Default:** 60

**Limits:** Up to 2147483647

### NOTE

- If you are monitoring for the existence of a file (not the size) and the file already exists when the job runs, the job completes immediately. The watch\_interval attribute is ignored.
- The job is considered complete when the watched file reaches the minimum size and remains in a steady state (has not grown) during the specified interval. We recommend that you specify a reasonable interval to help ensure that the file does not appear to be in a steady state when it is still being modified over a longer period of time.
- If the FW job runs on the machine where Workload Automation Agent for UNIX, Linux, Windows, or iOS is installed, the agent uses the watch\_interval as a second polling interval. The first polling interval is set by default on the agent for 30 seconds. If the job does not complete after the first polling interval, the agent waits for the second polling interval specified by the watch\_interval attribute to ensure the file remains steady. If the file has not changed after the second polling interval elapses, the agent returns the status. If the file does change, the agent goes back to sleep for the duration of the second polling interval until the file eventually stabilizes.

### Example: Monitor a File Every 120 Seconds

This example monitors the watch\_file.log file on the winagent computer. The job completes when the file reaches 10,000 bytes and maintains a steady state for at least 120 seconds (30 seconds for the agent's global poll interval plus 90 seconds for the watch\_interval).

```
insert_job: fw_job
job_type: FW
machine: winagent
watch_file: "c:\tmp\watch_file.log"
watch_file_min_size: 10000
watch_interval: 90
```

## watch\_no\_change Attribute -- Specify the Time the File Must Remain Unchanged

The watch\_no\_change attribute defines the time the file must remain unchanged to satisfy the monitor condition.

### Supported Job Type

This attribute is optional for the [File Trigger \(FT\) job type](#).

#### NOTE

This attribute is valid only when CREATE, EXPAND, SHRINK, UPDATE, or GENERATE is specified for the watch\_file\_type attribute.

### Syntax

This attribute has the following format:

```
watch_no_change: nS|M|H|D
```

- **nS|M|H|D**

Specifies the time in seconds (S), minutes (M), hours (H), or days (D) that the file must remain unchanged to satisfy the monitor condition. Specify the time in the following format: numeric value (n) followed by the character (S,M,H, or D).

**Limits:** 0 to 9223372036854775 for S, 0 to 153722867280912 for M, 0 to 2562047788015 for H, 0 to 106751991167 for D

**Default:** 1 (1 minute)

#### NOTE

- You cannot specify combinations of time periods. For example, 12D3H is not a valid value.
- You cannot specify a character in between, but the specified value can end with only S, M, H, or D character. For example, 12D4 is not a valid value. 4056S is a valid value.
- If you specify a number only, AutoSys Workload Automation assumes minutes by default.
- If the FT job runs on the machine where Workload Automation Agent for UNIX, Linux, Windows, or i5/OS is installed, the agent uses the watch\_no\_change value as a second polling interval. The first polling interval is the filemonplugin.sleepperiod parameter value which is set by default for 30 seconds. If the job does not complete after the first polling interval, the agent waits for the second polling interval specified by the watch\_no\_change value to ensure the file remains steady. If the file has not changed after the second polling interval elapses, the agent returns the status. If the file does change, the agent goes back to sleep for the duration of the second polling interval until the file eventually stabilizes. For more information about the filemonplugin.sleepperiod parameter, see the [Workload Automation Agents](#) documentation.
- The filemonplugin.sleepperiod parameter controls how often the agent checks the files. Recursive file triggers use more CPU and disk since it has to look at all of the files in a directory tree. So, specifying a lower filemonplugin.sleepperiod parameter value might increase the CPU and disk consumption.

### Example: Monitor for an Increase in File Size to a Certain Amount with an Unchanged Condition

In this example, the unixagt agent monitors the analysis file in the /research directory. If the file size expands to 1 byte or more and remains unchanged for 120 minutes or more, the job completes.

```
insert_job: ft_unix_nochange
job_type: FT
machine: unixagt
watch_file: /research/analysis
watch_file_type: EXPAND
watch_file_change_type: SIZE
```

```
watch_file_change_value: 1
watch_no_change: 120
```

## web\_parameter Attribute -- Specify Operation Parameters

The `web_parameter` attribute specifies the operation parameters in a Web Service RPC/Encoded job. When the job invokes the web service, the parameters are passed to the operation.

### Supported Job Type

This attribute is optional for the [Web Service RPC/Encoded \(WBSVC\) job type](#).

### Syntax

This attribute has the following format:

```
web_parameter: xsd\:type=value | payload_job=job_name[, xsd\:type=value |
payload_job=job_name...]
```

- **xsd\:type=value**  
Specifies the XML schema type and value for the operation parameter. The agent supports the following types:

xsd\:string	xsd\:anySimpleType
xsd\:integer	xsd\:duration
xsd\:int	xsd\:gYearMonth
xsd\:long	xsd\:gYear
xsd\:short	xsd\:gMonthDay
xsd\:decimal	xsd\:gDay
xsd\:float	xsd\:gMonth
xsd\:double	xsd\:normalizedString
xsd\:boolean	xsd\:token
xsd\:byte	xsd\:language
xsd\:unsignedInt	xsd\:Name
xsd\:unsignedShort	xsd\:NCName
xsd\:unsignedByte	xsd\:ID
xsd\:QName	xsd\:NMToken
xsd\:dateTime	xsd\:NMTOKENS
xsd\:date	xsd\:nonPositiveInteger
xsd\:time	xsd\:negativeInteger
xsd\:anyURI	xsd\:nonNegativeInteger
xsd\:base64Binary	xsd\:unsignedLong
xsd\:hexBinary	xsd\:positiveInteger

**Limits:** Up to 4000 characters; case-sensitive

- **payload\_job=job\_name** Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

**NOTE**

- You can define up to 64 parameters using one `web_parameter` attribute or multiple entries of `web_parameter`.
- Enclose the job name that contains a colon with quotation marks ( " ") or precede it with a backslash (\).
- In each `web_parameter` entry, separate each set of parameter values with a comma. The entire value of each entry can be up to 4000 characters.
- Order is important. The agent passes the parameters in the order that you specify them.
- You can use global variables to specify the parameters in a Define a Web Service RPC/Encoded Job. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.
- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.

**Example: Pass Parameter to Web Service for Getting Stock Quotes****NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

**web\_user Attribute -- Specify a Web Service User Name**

The `web_user` attribute specifies the user name in a Web Service job.

**Supported Job Type**

This attribute is optional for the [Web Service \(WBSVC\) job type](#).

**Syntax**

This attribute has the following format:

```
web_user: user_name
```

- ***user\_name***  
Specifies the user name.  
**Limits:** Up to 145 characters

## **win\_event\_category Attribute -- Specify a Windows Event Category**

The win\_event\_category attribute specifies the event category as displayed in the Windows Event Viewer.

### **Supported Job Type**

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\) job type](#).

### **Syntax**

This attribute has the following format:

```
win_event_category: event_category
```

- ***event\_category***  
Specifies the event category as displayed in the Windows Event Viewer.  
**Limits:** Up to 256 characters; case-sensitive

### **Example: Monitor for a Windows Application Event Log**

This example monitors for a Windows application event log. When the job finds an application log in the DISK category and has an event ID equal to 14, the job completes.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_category: DISK
win_event_op: EQ
win_event_id: 14
```

## **win\_event\_computer Attribute -- Specify a Local Computer for Windows Event Log**

The win\_event\_computer attribute specifies the name of the local machine that the Windows event log applies to.

### **Supported Job Type**

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\) job type](#).

### **Syntax**

This attribute has the following format:

```
win_event_computer: computer_name
```

- ***computer\_name***  
Specifies the local computer name where the event log is monitored.  
**Limits:** Up to 80 characters; case-sensitive

### **Example: Monitor a Windows Event Log in a Specific Computer**

This example monitors the D9GBJG11 computer for an event source of Security.

```
insert_job: eventlog_job
job_type: OMEL
```

```

machine: monagt
win_log_name: Security
win_event_type: AUDITS
win_event_source: Security
win_event_computer: D9GBJG11
win_event_category: System Event
win_event_op: GT
win_event_id: 0

```

## win\_event\_datetime Attribute -- Specify the Date and Time of a Windows Event Log

The win\_event\_datetime attribute specifies the date and time of a Windows event log. The job monitors for an event log that occurs on or after the specified date and time.

### Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring](#) (OMEL) job type.

### Syntax

This attribute has the following format:

```
win_event_datetime: "yyyymmdd hh:mm:ss"
```

- **"yyyymmdd hh:mm:ss"**

Specifies the date and time of the Windows event log. This parameter includes the following:

- **yyyy**  
Specifies the four-digit year.
  - **mm**  
Specifies the two-digit month.
  - **dd**  
Specifies the two-digit day.
  - **hh**  
Specifies the two-digit hour.
  - **mm**  
Specifies the two-digit minute.
  - **ss**  
Specifies the two-digit second.
- Example:** "20090501 06:30:00"

### Example: Monitor an Application Log that Occurs on or After a Specified Date

This example monitors an application log that occurs any time on or after July 1, 2010. When this event occurs in the application log, the job completes successfully.

```

insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_datetime: "20100701 00:00:00"

```

## win\_event\_description Attribute -- Specify a Windows Event Description

The win\_event\_description attribute specifies the description of the Windows event as shown in the Event Viewer.

## Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\)](#) job type.

### Syntax

This attribute has the following format:

```
win_event_description: event_description
```

- ***event\_description***

Specifies the event description as displayed in the Windows Event Viewer.

**Limits:** Up to 256 characters; case-sensitive

### Example: Monitor for Words Excluded in the Event Description

In this example, the event description must include the words conflict and state as indicated by the plus signs but must exclude the words deny, master, or browser as indicated by the minus sign. The plus sign is the default and is optional.

```
insert_job: eventlog_job1
job_type: OMEL
machine: monagt
win_log_name: System
win_event_type: ERROR
win_event_description: "+conflict +state -deny -master -browser"
```

### Example: Monitor for Words Included in the Event Description

In this example, the event description must include the words Normal shutdown. This example does not use the plus sign, and by default, the specified words are included in the search.

```
insert_job: eventlog_job2
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_type: INFO
win_event_description: "Normal shutdown"
```

## win\_event\_id Attribute -- Specify a Windows Event ID

The win\_event\_id attribute specifies the Windows event ID.

### Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\)](#) job type.

### Syntax

This attribute has the following format:

```
win_event_id: id
```

- ***id***

Specifies the Windows event ID to monitor.

**Limits:** Up to 10 digits

**Example:** 4000

### Example: Monitor for an Event ID Less Than a Specified Value

This example checks for an event ID number less than 1. The job returns the first application event from the application log that has an event ID equal to 0.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_op: LT
win_event_id: 1
```

## win\_event\_op Attribute -- Specify a Comparison Operator for a Windows Event ID

The win\_event\_op attribute specifies a comparison operator against the value of a Windows Event ID.

### Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\) job type](#).

### Syntax

This attribute has the following format:

```
win_event_op: GT | LT | LE | EQ | GE
```

- **EQ**  
Specifies the equal to (=) operator against the value of a Windows Event ID. This is the default.
- **LE**  
Specifies the less than or equal to (<=) operator against the value of a Windows Event ID.
- **LT**  
Specifies the less than (<) operator against the value of a Windows Event ID.
- **GE**  
Specifies the greater than or equal to (>=) operator against the value of a Windows Event ID.
- **GT**  
Specifies the greater than (>) operator against the value of a Windows Event ID.

### Example: Monitor for an Event ID Less Than a Specified Value

This example checks for an event ID number less than 1. The job returns the first application event from the application log that has an event ID equal to 0.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_op: LT
win_event_id: 1
```

## win\_event\_source Attribute -- Specify a Windows Event Source

The win\_event\_source attribute specifies the Windows event source.

### Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\) job type](#).

### Syntax

This attribute has the following format:

```
win_event_source: source_name
```



- **source\_name**  
Specifies the event source as displayed in the Windows Event Viewer.  
**Limits:** Up to 256 characters; case-sensitive

### Example: Monitor a Windows Event Source

This example monitors a Windows event source named Service Control Manager.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: System
win_event_type: ERROR
win_event_source: Service Control Manager
```

### win\_event\_type Attribute -- Specify a Windows Event Type

The win\_event\_type attribute specifies the event type to monitor in the Windows event log.

#### NOTE

If you do not specify the win\_event\_type attribute in your job definition, the job monitors for the Error event type (the default).

### Supported Job Type

This attribute is optional for the [Windows Event Log Monitoring \(OMEL\) job type](#).

### Syntax

This attribute has the following format:

```
win_event_type: ERROR | WARN | INFO | AUDITS | AUDITF
```

- **ERROR**  
Specifies the Error event type. This is the default.
- **WARN**  
Specifies the Warning event type.
- **INFO**  
Specifies the Information event type.
- **AUDITS**  
Specifies the Success Audit event type.
- **AUDITF**  
Specifies the Failure Audit event type.

### Example: Monitor an Application Event Log for Info Events

This example monitors the Application log for INFO events.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Application
win_event_type: INFO
win_event_category: None
win_event_source: Userenv
```

```
win_event_computer: LLUSER
win_event_id: 1000
win_event_op: EQ
```

### Example: Monitor for a Successful Audit

In this example, the security log is monitored for a successful audit of a security access attempt. The event category is System Event, the term succeeded is excluded, but the words Audit and log are included in the event description.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: Security
win_event_type: AUDITS
win_event_category: System Event
win_event_source: Service Control Manager
win_event_description: "-succeeded +Audit log"
```

## win\_log\_name Attribute -- Specify the Name of a Windows Event Log

The win\_log\_name attribute specifies the name of a Windows event log to monitor.

### Supported Job Type

This attribute is required for the [Windows Event Log Monitoring \(OMEL\)](#) job type.

### Syntax

This attribute has the following format:

```
win_log_name: log_name
```

- ***log\_name***  
Specifies the name of the event log.  
**Limits:** Up to 256 characters; case-sensitive

### Example: Monitor a Windows Event Log named System

This example monitors a system event log for an event type of WARN, event source of MrxSmb, and event category of None.

```
insert_job: eventlog_job
job_type: OMEL
machine: monagt
win_log_name: System
win_event_type: WARN
win_event_source: MrxSmb
win_event_category: None
```

## win\_service\_name Attribute -- Specify the Name of the Windows Service to be Monitored

The win\_service\_name attribute specifies the name of the Windows service to be monitored.

### Supported Job Type

This attribute is required for the [Windows Service Monitoring \(OMS\)](#) job type.

### Syntax

This attribute has the following format:

```
win_service_name: service_name
```

- **service\_name**

Specifies the name of the local Windows service to be monitored.

**Limits:** Up to 256 characters; case-sensitive

**NOTE**

You can only specify the service name. The service display name is shown in the Windows Services Manager.

### Example: Specify a Windows Service Name

This example monitors a Windows service named Proc Server. The job checks the status immediately and completes successfully if the service is paused. If the service is not paused, the job fails.

```
insert_job: oms_job1
job_type: OMS
machine: winagt
win_service_name: Proc Server
win_service_status: PAUSED
monitor_mode: NOW
```

### Example: Specify a Path to a Windows Service Executable

This example monitors a Windows service named Log App. The job waits until the service status is CONTINUE\_PENDING before it completes.

```
insert_job: oms_job2
job_type: OMS
machine: winagt
win_service_name: "C:\Program Files\Log App\apptask.exe"
win_service_status: CONTINUE_PENDING
monitor_mode: WAIT
```

## win\_service\_status Attribute -- Specify the Status of the Windows Service to be Monitored

The win\_service\_status attribute specifies the status of the Windows Service to be monitored. The job checks if the status of the service matches the status specified by this attribute.

**NOTE**

If you do not specify the win\_service\_status attribute in your job definition, the job checks if the Windows service is running (the default).

### Supported Job Type

This attribute is optional for the [Windows Service Monitoring \(OMS\) job type](#).

### Syntax

This attribute has the following format:

```
win_service_status: RUNNING | STOPPED | CONTINUE_PENDING |
 PAUSE_PENDING | PAUSED | START_PENDING |
 STOP_PENDING | EXISTS | NOTEXISTS
```

- **RUNNING**  
Specifies that the job monitors the Windows service for a running status. This is the default.
- **STOPPED**  
Specifies that the job monitors the Windows service for a stopped status.
- **CONTINUE\_PENDING**  
Specifies that the job monitors the Windows service for a continue pending status.
- **PAUSE\_PENDING**  
Specifies that the job monitors the Windows service for a pause pending status.
- **PAUSED**  
Specifies that the job monitors the Windows service for a paused status.
- **START\_PENDING**  
Specifies that the job monitors the Windows service for a start pending status.
- **STOP\_PENDING**  
Specifies that the job monitors the Windows service for a stop pending status.
- **EXISTS**  
Specifies that the job checks whether the Windows service exists.
- **NOTEXISTS**  
Specifies that the job checks whether the Windows service does not exist.

#### **Example: Monitor for the Existence of a Windows Service**

This example monitors a Windows service named Proc Server. The job completes successfully if the service exists. By default, the job checks for the condition immediately and completes, so the `monitor_mode` attribute is not required in the job definition.

```
insert_job: oms_job1
job_type: OMS
machine: winagt
win_service_name: Proc Server
win_service_status: EXISTS
```

#### **Example: Monitor for a Running Status**

This example monitors a Windows service named App Server. The `win_service_status` attribute is not specified in the job definition, so the job monitors for a **RUNNING** status by default. The job completes when the service is running.

```
insert_job: oms_job2

job_type: OMS

machine: winagt

win_service_name: App Server

monitor_mode: WAIT
```

## ws\_authentication\_order Attribute -- Specify Authentication Protocols

The ws\_authentication\_order attribute specifies a list of protocols used by a web server for authentication in a Web Service Document/Literal job.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

### Syntax

This attribute has the following formats:

```
ws_authentication_order: BASIC | DIGEST | NTLM
ws_authentication_order: {[BASIC]}
 {, [DIGEST]}
 {, [NTLM]}
```

- **BASIC**  
Indicates the BASIC protocol.
- **DIGEST**  
Indicates the DIGEST protocol.
- **NTLM**  
Indicates the NTLM protocol.

### NOTE

- You can specify any of the following protocols in any order: BASIC, DIGEST, and NTLM. To specify more than one protocol, separate the protocols by commas.
- If you are connecting to an old web server that cannot negotiate authentication protocols, enter the following list in the specified order: BASIC, DIGEST, NTLM.

### Example: Authenticate Credentials Using NTLM

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2 - and is expected to return 4. The job will attempt to authenticate the myuser@mydomain credentials using NTLM authentication.

```
insert_job: getmax1
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://wsmach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://wsmach:8080/axis2/services/TestWS?wsdl"
ws_conn_user: myuser@mydomain
ws_conn_domain: mydomain
ws_conn_origin: myhost.example.com
ws_authentication_order: NTLM
ws_parameter: param_name="/intGetMax", param_value=""
ws_parameter: param_name="/intGetMax/i[3]", param_value="1,4,2"
job_criteria: criteria_xpath="//ns:return", criteria_filter="4"
```

## ws\_conn\_domain Attribute -- Specify a Domain for NTLM Connection Authentication

The `ws_conn_domain` attribute specifies the domain for connection authentication in a Web Service Document/Literal job. This attribute is required for NTLM authentication.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal](#) (WSDOC) job type.

### Syntax

This attribute has the following format:

```
ws_conn_domain: origin_domain
```

- ***origin\_domain*** Specifies the domain for NTLM connection authentication.

**Limits:** Up to 128 characters; case-sensitive

### NOTE

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the domain for NTLM connection authentication. Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

### Example: Specify Domain for NTLM Connection Authentication

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2 - and is expected to return 4. The job will attempt to authenticate the `myuser@mydomain` credentials using NTLM authentication.

```
insert_job: getmax1
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://wsmach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://wsmach:8080/axis2/services/TestWS?wsdl"
ws_conn_user: myuser@mydomain
ws_conn_domain: mydomain
ws_conn_origin: myhost.example.com
ws_authentication_order: NTLM
ws_parameter: param_name="/intGetMax", param_value=""
ws_parameter: param_name="/intGetMax/i[3]", param_value="1,4,2"
job_criteria: criteria_xpath="//ns:return", criteria_filter="4"
```

## ws\_conn\_origin Attribute -- Specify an Origin Host Name for NTLM Connection Authentication

The `ws_conn_origin` attribute specifies the origin host name for NTLM connection authentication in a Web Service Document/Literal job.

### NOTE

If you do not specify the `ws_conn_origin` attribute in your job definition, the job uses the computer name where the agent is running (the default).

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal](#) (WSDOC) job type.

### Syntax

This attribute has the following format:

```
ws_conn_origin: origin_host
```

- ***origin\_host*** Specifies the origin host name for NTLM connection authentication.  
**Default:** The computer name where the agent is running  
**Limits:** Up to 128 characters; case-sensitive

#### NOTE

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the origin host name for NTLM connection authentication. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

### Example: Specify the Origin Host Name for NTLM Connection Authentication

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2 - and is expected to return 4. The job will attempt to authenticate the myuser@mydomain credentials using NTLM authentication.

```
insert_job: getmax1
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://wsmach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://wsmach:8080/axis2/services/TestWS?wsdl"
ws_conn_user: myuser@mydomain
ws_conn_domain: mydomain
ws_conn_origin: myhost.example.com
ws_authentication_order: NTLM
ws_parameter: param_name="/intGetMax", param_value=""
ws_parameter: param_name="/intGetMax/i[3]", param_value="1,4,2"
job_criteria: criteria_xpath="//ns:return", criteria_filter="4"
```

### ws\_conn\_user Attribute -- Specify a User Name for Connection Authentication

The `ws_conn_user` attribute specifies the user name in a Web Service job.

#### Supported Job Types

This attribute is optional for the following job types:

[Web Service Document/Literal \(WSDOC\) job type](#)

[Web Service RPC/Encoded \(WBSVC\) job type](#)

#### Syntax

This attribute has the following format:

```
ws_conn_user: authenticated_user@connhost
```

- ***authenticated\_user@connhost***  
Specifies the user name for connection authentication.  
**Limits:** Up to 145 characters

**NOTE**

- If you specify the `ws_conn_user` attribute in your job definition, you must set the password that is associated with the connection user using the `autosys_secure` command.
- You can use global variables to specify the user name for connection authentication. Use the `$global_name`, `$(global_name)`, or `$$global_name` format to reference global variables.

**Example: Specify User Name for NTLM Connection Authentication**

This job calls a web service that returns the maximum of an arbitrary number of integers that are passed to it. In this case, the web service is passed three integers - 1, 4, and 2 - and is expected to return 4. The job will attempt to authenticate the `myuser@mydomain` credentials using NTLM authentication.

```
insert_job: getmax1
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://wsmach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://wsmach:8080/axis2/services/TestWS?wsdl"
ws_conn_user: myuser@mydomain
ws_conn_domain: mydomain
ws_conn_origin: myhost.example.com
ws_authentication_order: NTLM
ws_parameter: param_name="/intGetMax", param_value=""
ws_parameter: param_name="/intGetMax/i[3]", param_value="1,4,2"
job_criteria: criteria_xpath="//ns:return", criteria_filter="4"
```

**ws\_global\_proxy\_defaults Attribute -- Specify Whether to Use Global Proxy Defaults**

The `ws_global_proxy_defaults` attribute specifies whether the agent uses the global proxy defaults specified in the `agentparm.txt` file.

**NOTE**

If you do not specify the `ws_global_proxy_defaults` attribute in your job definition, the job uses the global proxy configuration specified by the proxy parameters in the `agentparm.txt` file (the default).

**Supported Job Type**

This attribute is optional for the [Web Service Document/Literal \(WSDOC\)](#) job type.

**Syntax**

This attribute has the following format:

```
ws_global_proxy_defaults: Y | N
```

- **Y**  
Uses the global proxy configuration specified by the proxy parameters in the `agentparm.txt` file. This is the default.
- **N**  
Ignores the global proxy configuration specified by the proxy parameters in the `agentparm.txt` file. Specify this option if you want to ignore the proxy defaults for this job only. For example, you might ignore the proxy defaults if the request is going to a server on the LAN that does not require the proxy

**Example: Use Global Proxy Defaults in Agent Parameter File**



This job calls the ListAllMyBuckets operation at the Amazon S3 web service. It passes its parameters in the form of an XML document:

```
<ListAllMyBuckets>
 <AWSAccessKeyId>0x0102030405060708</AWSAccessKeyId>
 <Timestamp>2011-11-10T21:24:21</Timestamp>
 <Signature>0x0102030405060708</Signature>
</ListAllMyBuckets>
```

It will use all the global proxy default values from the agentparm.txt file.

```
insert_job: execws
job_type: WSDOC
machine: wsmach
endpoint_URL: "https://s3.amazonaws.com/soap"
WSDL_URL: "http://s3.amazonaws.com/doc/2006-03-01/AmazonS3.wsdl"
service_name: AmazonS3
port_name: AmazonS3
wsdl_operation: ListAllMyBuckets
ws_parameter: param_name="/ListAllMyBuckets", param_value=""
ws_parameter: param_name="/ListAllMyBuckets/AWSAccessKeyId",
 param_value="0x0102030405060708"
ws_parameter: param_name="/ListAllMyBuckets/Timestamp",
 param_value="2011-11-10T21:24:21"
ws_parameter: param_name="/ListAllMyBuckets/Signature", param_value="0x0102030405060708"
ws_global_proxy_defaults: Y
```

## ws\_parameter Attribute -- Specify Web Service Parameters

The ws\_parameter attribute specifies the parameters in a Web Service Document/ Literal job in the form of flattened XML. When the job runs, the parameters are passed to the executing process as a literal XML document.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

### Syntax

This attribute has the following formats:

```
ws_parameter: param_name=name, param_value="value"
ws_parameter: param_name=name[i], param_value="value,value,..."
ws_parameter: param_name=name, payload_job=jobname
```

- **param\_name=name, param\_value="value"**  
Specifies the parameter name and value.
  - **param\_name=name**  
Specifies the name or path of the element.
  - **param\_value="value"**  
Specifies the scalar string value.  
**Limits:** Up to 2048 characters; case-sensitive
- **param\_name=name[i], param\_value="value,value,..."**  
Specifies the parameter name and array of values.

- **param\_name=name[*i*]**  
Specifies the name or path of the element. When the parameter name contains [*i*], it indicates an array of values.
- **param\_value="value,value,..."**  
Specifies an array of scalar string values. The individual array values are separated by commas and the entire quoted value is enclosed in double quotes.  
**Limits:** Up to 2048 characters; case-sensitive
- **param\_name=name, payload\_job=jobname**  
Specifies the parameter name and the name of a payload producing job.
  - **param\_name=name**  
Specifies the value is the name or path of the element.
  - **payload\_job=jobname**  
Specifies the name of the payload producing job that produced the binary output to be used as an input parameter. A payload producing job is a job that produces binary output that is persisted as a serialized Java object. The serialized Java object can be passed as input to a payload consuming job.

## NOTE

- Enclose the values that contain embedded blanks in double quotation marks. All embedded double quotation marks must be escaped with a backslash.
- Enclose the job name that contains a colon with quotation marks (") or precede it with a backslash (\).
- You can use global variables to specify the parameters in a Web Service Document/ Literal job. Use the \$global\_name, \$\$ (global\_name), or \${global\_name} format to reference global variables. **Examples:**

```
ws_parameter: param_name="/ListAllMyBuckets/$$WSDOCGVAR/$
testHTTPJobWithXPathVar@<v>XPathVar</v>", param_value="$$WSDOCGVAR/$
testHTTPJobWithXPathVar@<v>XPathVar</v>"
ws_parameter: param_name="/$$testHTTPJobWithXPathVar@<v>XPathVar</v>/$
WSDOCGVAR/", payload_job="$$WSDOCGVAR/$testHTTPJobWithXPathVar@<v>XPathVar</
v>"
```
- To specify an XML hierarchy of the document, use multiple entries of ws\_parameter.
- In each ws\_parameter entry, separate the name/value pair with a comma. The entire value can be up to 2048 characters.
- In the XML document, these parameters are represented in a flattened view. Consider the following example:
 

```
<tag1>
<tag2>value</tag2>
</tag1>
```

In this example, the full path to the value is </tag1/tag2>.
- For tags that do not hold any value, the full path is required because the no-value tags could represent arrays of structures and appear in the multiple levels. Consider the following example:
 

```
ws_parameter: param_name="/tag1", param_value=""
ws_parameter: param_name="/tag1/tag2", param_value="value"
```
- For parameters that contain attributes, the attributes are added to the full path of the parameter separating them with the '@' symbol. Consider the following example:
 

```
<tag1>
<tag2 key="thetag">value</tag2>
</tag1>
```
- For Web Service Document/Literal payload producing jobs, only the portion of the returned data that satisfies the job\_criteria specification will be stored in the output file for use as payload.  
In this example, the list of parameters would be as follows:
 

```
ws_parameter: param_name="/tag1", param_value=""
ws_parameter: param_name="/tag1/tag2", param_value="value"
```

ws\_parameter: param\_name="/tag1/tag2@key", param\_value="thetag"

- If the value is not enclosed in double quotes, then the characters equal to (=) and comma (,), if they are part of the data, must be escaped with a backslash (\).

### Example: Add Two Numbers Passing Two Parameters

This job calls a simple calculator web service at <http://ws1.parasoft.com/glue/calculator>. The web service requests the "add" operation to add two numbers, passing in two parameters: "x", which is the result from the last execution of the getmax1 job and "y", which is the literal value 1.00. The job uses specific values for the proxy host, port, user, and domain.

```
insert_job: incpayload
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://ws1.parasoft.com/glue/calculator"
WSDL_URL: "http://soaptest.parasoft.com/calculator.wsdl"
service_name: Calculator
port_name: ICalculator
wsdl_operation: add
ws_parameter: param_name="/add", param_value=""
ws_parameter: param_name="/add/x", payload_job="getmax1"
ws_parameter: param_name="/add/y", param_value="1.00"
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain
```

### ws\_proxy\_domain Attribute -- Specify a Domain for Proxy Authentication

The `ws_proxy_domain` attribute specifies the domain for proxy authentication in a Web Service Document/Literal job. This attribute is required for NTLM authentication.

#### NOTE

If you do not specify the `ws_proxy_domain` attribute in your job definition, the job uses the proxy domain set in the `http.proxyDomain` parameter on the agent unless the `ws_global_proxy_defaults` attribute is set to N.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

### Syntax

This attribute has the following format:

```
ws_proxy_domain: proxy_domain
```

- **proxy\_domain** Specifies the domain for proxy authentication.

**Limits:** Up to 128 characters; case-sensitive

#### NOTE

- Your agent administrator can specify a default proxy domain for all Web Service Document/Literal jobs by setting the `http.proxyDomain` parameter in the agent's `agentparm.txt` file.
- The `ws_proxy_domain` attribute overrides the default proxy domain specified in the agent's `agentparm.txt`.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the domain for proxy authentication. Use the `$$global_name`, `$(global_name)`, or `$$ {global_name}` format to reference global variables.

### Example: Specify Domain for Proxy Authentication

This job calls a simple calculator web service at `http://ws1.parasoft.com/glue/calculator`. The web service requests the "add" operation to add two numbers, passing in two parameters: "x", which is the result from the last execution of the `getmax1` job and "y", which is the literal value 1.00. The job uses specific values for the proxy host, port, user, and domain.

```
insert_job: incpayload
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://ws1.parasoft.com/glue/calculator"
WSDL_URL: "http://soaptest.parasoft.com/calculator.wsdl"
service_name: Calculator
port_name: ICalculator
wsdl_operation: add
ws_parameter: param_name="/add", param_value=""
ws_parameter: param_name="/add/x", payload_job="getmax1"
ws_parameter: param_name="/add/y", param_value="1.00"
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain
```

### ws\_proxy\_host Attribute -- Specify a Proxy Host

The `ws_proxy_host` attribute specifies the proxy host name to use for the request in a Web Service Document/Literal job.

#### NOTE

If you do not specify the `ws_proxy_host` attribute in your job definition, the job uses the proxy host set in the `http.proxyHost` parameter on the agent unless the `ws_global_proxy_defaults` attribute is set to N.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

### Syntax

This attribute has the following format:

```
ws_proxy_host: proxy_host
```

- ***proxy\_host***

Specifies the proxy host name to use for the request.

**Limits:** Up to 128 characters; case-sensitive

#### NOTE

- Your agent administrator can specify a default proxy host name for all Web Service Document/Literal jobs by setting the `http.proxyHost` parameter in the agent's `agentparm.txt` file.
- The `ws_proxy_host` attribute overrides the default proxy host name specified in the agent's `agentparm.txt`.
- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- You can use global variables to specify the proxy host name to use for the request. Use the `$$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

### Example: Specify a Proxy Host

This job calls a simple calculator web service at <http://ws1.parasoft.com/glue/calculator>. The web service requests the "add" operation to add two numbers, passing in two parameters: "x", which is the result from the last execution of the getmax1 job and "y", which is the literal value 1.00. The job uses specific values for the proxy host, port, user, and domain.

```
insert_job: incpayload
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://ws1.parasoft.com/glue/calculator"
WSDL_URL: "http://soaptest.parasoft.com/calculator.wsdl"
service_name: Calculator
port_name: ICalculator
wsdl_operation: add
ws_parameter: param_name="/add", param_value=""
ws_parameter: param_name="/add/x", payload_job="getmax1"
ws_parameter: param_name="/add/y", param_value="1.00"
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain
```

## ws\_proxy\_origin Attribute -- Specify an Origin Name for Proxy Authentication

The `ws_proxy_origin` attribute specifies the origin machine name for proxy authentication in a Web Service Document/Literal job. This attribute is used for NTLM authentication.

### NOTE

If you do not specify the `ws_proxy_origin` attribute in your job definition, the job uses the machine set in the `http.proxyOrigin` parameter on the agent unless the `ws_global_proxy_defaults` attribute is set to N.

## Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

## Syntax

This attribute has the following format:

```
ws_proxy_origin: proxy_origin
```

- ***proxy\_origin***  
Specifies the origin name for proxy authentication.  
**Limits:** Up to 128 characters; case-sensitive

### NOTE

- Your agent administrator can specify a default proxy origin for all Web Service Document/Literal jobs by setting the `http.proxyOrigin` parameter in the agent's `agentparm.txt` file.
- The `ws_proxy_origin` attribute overrides the default proxy origin specified in the agent's `agentparm.txt`.
- You can use global variables to specify the origin name for proxy authentication. Use the `$$global_name`, `$(global_name)`, or `$${global_name}` format to reference global variables.

## Example: Invoke a Web Service to Get Movie Information

This job calls the `GetTheatersAndMovies` operation in the `MovieInformation` service hosted at <http://www.ignyte.com/webservices/ignyte.whatsshowing.webservice/moviefunctions.asmx>, requesting information on movies showing around the 12590 zip code. It passes the request through the `proxyhost.example.com` proxy on port 80, and for proxy authentication specifies the user `myuser@mydomain`, the domain `mydomain`, and the origin of `myhost.example.com`.

```

insert_job: showing
job_type: WSDOC
machine: wsagent
endpoint_URL: "http://www.ignyte.com/webservices/ignyte.whatsshowing.webservice/
moviefunctions.asmx"
WSDL_URL: "http://www.ignyte.com/webservices/ignyte.whatsshowing.webservice/
moviefunctions.asmx?wsdl"
service_name: MovieInformation
port_name: MovieInformationSoap
wsdl_operation: GetTheatersAndMovies
ws_parameter: param_name="/GetTheatersAndMovies", param_value=""
ws_parameter: param_name="/GetTheatersAndMovies/zipCode", param_value="12590"
ws_parameter: param_name="/GetTheatersAndMovies/radius", param_value="20"
ws_proxy_host: proxyhost.example.com
ws_proxy_port: 80
ws_poxy_origin: myhost.example.com
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain

```

## ws\_proxy\_port Attribute -- Specify a Proxy Port

The `ws_proxy_port` attribute specifies the proxy port to use for the request in a Web Service Document/Literal job.

### NOTE

If you do not specify the `ws_proxy_port` attribute in your job definition, the job uses the proxy host set in the `http.proxyPort` parameter on the agent unless the `ws_global_proxy_defaults` attribute is set to N.

## Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

## Syntax

This attribute has the following format:

```
ws_proxy_port: proxy_port
```

- ***proxy\_port***  
Specifies the port on the proxy host to use for the request.

### NOTE

- Your agent administrator can specify a default proxy port for all Web Service Document/Literal jobs by setting the `http.proxyPort` parameter in the agent's `agentparm.txt` file.
- The `ws_proxy_port` attribute overrides the default proxy port specified in the agent's `agentparm.txt`.

## Example: Invoke a Web Service to Add Two Numbers With a Default Proxy Port

This job calls a simple calculator web service at `http://ws1.parasoft.com/glue/calculator`. The web service requests the "add" operation to add two numbers, passing in two parameters: "x", which is the result from the last execution of the `getmax1` job and "y", which is the literal value 1.00. The job uses specific values for the proxy host, port, user, and domain.

```

insert_job: incpayload
job_type: WSDOC
machine: wsmach
endpoint_URL: "http://ws1.parasoft.com/glue/calculator"
WSDL_URL: "http://soaptest.parasoft.com/calculator.wsdl"
service_name: Calculator
port_name: ICalculator
wsdl_operation: add
ws_parameter: param_name="/add", param_value=""
ws_parameter: param_name="/add/x", payload_job="getmax1"
ws_parameter: param_name="/add/y", param_value="1.00"
ws_proxy_host: 172.31.255.255
ws_proxy_port: 80
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain

```

## ws\_proxy\_user Attribute -- Specify a User for Proxy Authentication

The `ws_proxy_user` attribute specifies the user name required for proxy authentication.

### NOTE

If you do not specify the `ws_proxy_user` attribute in your job definition, the job uses the proxy user set in the `http.proxyUser` parameter on the agent unless the `ws_global_proxy_defaults` attribute is set to N.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\) job type](#).

### Syntax

This attribute has the following format:

```
ws_proxy_user: authenticated_user@proxydomain
```

- ***authenticated\_user@proxydomain***

Specifies the user name required for proxy authentication.

**Limits:** Up to 128 characters; case-sensitive; it cannot contain delimiters

### NOTE

- Do not use colons, embedded blanks, single quotation marks, or double quotation marks.
- If you specify the `ws_proxy_user` attribute in your job definition, you must set the password that is associated with the proxy user using the `autosys_secure` command.
- Your agent administrator can specify a default proxy user name and password for all Web Service Document/Literal jobs by setting the `http.proxyUser` and `http.proxyPassword.encr` parameters in the agent's `agentparm.txt` file. The password must be encrypted using the password utility located in the agent directory.
- The `ws_proxy_user` attribute overrides the default proxy user name specified in the agent's `agentparm.txt`.
- The password you specify using the `autosys_secure` command overrides the password specified in the agent's `agentparm.txt` file.
- You can use global variables to specify the user name required for proxy authentication. Use the `$global_name`, `$$global_name`, or `$$global_name` format to reference global variables.

### Example: Invoke a Web Service to Get Global Weather Information for a Weather Station

This job calls the `GetWeather` operation in the `GlobalWeather` service hosted at `http://www.webservicex.net/globalweather.asmx`, requesting weather information for a specific weather station. It passes the request through the

proxyhost.example.com proxy on port 80, and for proxy authentication specifies the user myuser@mydomain, the domain mydomain, and the origin of myhost.example.com.

```
insert_job: weather
job_type: WSDOC
machine: wsagent
endpoint_URL: "http://www.webservices.net/globalweather.asmx"
WSDL_URL: "http://www.webservices.net/globalweather.asmx?wsdl"
service_name: GlobalWeather
port_name: GlobalWeatherSoap
wsdl_operation: GetWeather
ws_parameter: param_name="/GetWeather", param_value=""
ws_parameter: param_name="/GetWeather/CityName", param_value="Raleigh / Durham, Raleigh-Durham International Airport"
ws_parameter: param_name="/GetWeather/CountryName", param_value="United States"
ws_proxy_host: proxyhost.example.com
ws_proxy_port: 80
ws_proxy_user: myuser@mydomain
ws_proxy_domain: mydomain
ws_proxy_origin: myhost.example.com
```

## ws\_security Attribute -- Specify Additional Web Services Security Options

The `ws_security` attribute specifies SOAP-level WS-Security (WSS) options that the agent uses to communicate with the web service.

### Supported Job Type

This attribute is optional for the [Web Service Document/Literal \(WSDOC\)](#) job type.

### Syntax

This attribute has the following formats:

```
ws_security: wss_type=USERTOKEN, wss_user=user_name, wss_digest=TRUE | FALSE
ws_security: wss_type=TIMESTAMP, wss_ttl=time_to_live
ws_security: wss_type=SIGNATURE
ws_security: wss_type=ENCRYPTION, wss_alias=alias, wss_keytype=ISSUER_SERIAL
ws_security: wss_type=ENCRYPTION, wss_alias=alias, wss_keytype=EMBEDDED_KEYNAME,
wss_keyname=key_name, wss_keyvalue=key_value
```

- **wss\_type=USERTOKEN**  
Adds a user name and password to the SOAP message that is validated in the server's SOAP layer.
- **wss\_user=user\_name@identifier**  
Specifies the user name that is added to the SOAP message. The `autosys_secure` command stores the password encrypted in the database for the given user name.

### NOTE

You can use global variables to specify the user name that is added to the SOAP message. Use the `$global_name`, `$(global_name)`, or `${global_name}` format to reference global variables.

- **wss\_digest=TRUE | FALSE**  
Specifies whether you want to pass the password in digest form. Options are as follows:



- **TRUE**  
Sends the password in the encrypted form to the SOAP request.
- **FALSE**  
Send the password in the plain form to the SOAP request.
- **wss\_type=TIMESTAMP**  
Adds a timestamp to the SOAP message with an optional time-to-live option to prevent reuse of old messages.
- **wss\_ttl=time\_to\_live**  
Specifies the time-to-live in seconds.
- **wss\_type=SIGNATURE**  
Adds a digital signature to the SOAP message, using the certificate specified in the agentparm.txt file.  
  
**NOTE**  
Your agent administrator must specify the ws.wss.keystore, ws.wss.keystore.alias, and ws.wss.keystore.password parameters in the agentparm.txt file.
- **wss\_type=ENCRYPTION**  
Encrypts the message. The server will decrypt the message in the SOAP layer.  
  
**NOTE**  
Your agent administrator must specify the ws.wss.keystore, ws.wss.keystore.alias, and ws.wss.keystore.password parameters in the agentparm.txt file if wss\_keytype is ISSUER\_SERIAL.
- **wss\_alias=alias**  
Specifies the alias in the keystore defined by properties for the Web Service certificate.  
**Limits:** Up to 512 characters; case-sensitive  
  
**NOTE**  
You can use global variables to specify the alias in the keystore defined by properties for the Web Service certificate. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.
- **wss\_keytype=ISSUER\_SERIAL**  
Uses X509 certificate for encryption. It is obtained from the keystore defined by the properties specified in the agentparm.txt file.
- **wss\_keytype=EMBEDDED\_KEYNAME, wss\_keyname=key\_name, wss\_keyvalue=key\_value**  
Uses the following key specified in the job definition for encryption.
  - **wss\_keyname=key\_name**  
Specifies the key name as defined by the web service to be invoked.  
  
**NOTE**  
You can use global variables to specify the key name as defined by the web service to be invoked. Use the \$\$global\_name, \$(global\_name), or \${global\_name} format to reference global variables.
  - **wss\_keyvalue=key\_value**  
Specifies the key in hexadecimal format as defined by the web service to be invoked. The agent sends the key encrypted.  
  
**NOTE**
    - The entire value can be up to 2048 characters.
    - The WS-Security options provide the following additional SOAP-level security for the web services messages:
      - Transport-level authentication

- USERTOKEN -- a user name token and password are added to each message and validated in the web service's SOAP layer.
- TIMESTAMP -- a timestamp, with a time-to-live parameter, is added to each message to prevent message hijacking and reuse.
- Message-level security
  - SIGNATURE -- messages are signed with a certificate.
  - ENCRYPTION -- messages are encrypted at the message level, increasing security when going through proxies that may require decrypting or re-encrypting of the transport stream.
- For more information about controlling the agent's use of certificates for the SIGNATURE and ENCRYPTION WS-Security options, see [Workload Automation Agents](#).

### Example: Get Maximum Number

This job calls the get maximum number web service at <http://thismach:8080/axis2/services/TestWS>. The web service requests the "intGetMax" operation to get the maximum number, passing in three parameters. The USERTOKEN and TIMESTAMP WS-Security options specify transport level authentication. The SIGNATURE and ENCRYPTION WS-Security options specify message level security.

```
insert_job: getmax2
job_type: WSDOC
machine: thismach
endpoint_URL: "http://thismach:8080/axis2/services/TestWS"
wsdl_operation: intGetMax
port_name: TestWSHttpSoap11Endpoint
service_name: TestWS
WSDL_URL: "http://thismach:8080/axis2/services/TestWS?wsdl"
ws_parameter: param_name="/intGetMax",param_value=""
ws_parameter: param_name="/intGetMax/ii[3]", param_value="1,4,2"
ws_authentication_order: DIGEST,BASIC,NTLM
ws_global_proxy_defaults: Y
ws_conn_user: causer@thismach
job_criteria: criteria_xpath="//ns:return", criteria_filter=".*4.*"
ws_security: wss_type=USERTOKEN, wss_user=myuser@hostname, wss_digest=True
ws_security: wss_type=TIMESTAMP, wss_ttl=120
ws_security: wss_type=SIGNATURE
ws_security: wss_type=ENCRYPTION, wss_alias=securestore, wss_keytype=EMBEDDED_KEYNAME,
wss_keyname=SessionKey, wss_keyvalue=31FDCBDAFBCD6BA8E619A7BF51F7C73E
```

### wsdl\_operation Attribute -- Specify the Operation to be Invoked

The `wsdl_operation` attribute specifies the web service operation to be invoked in a Web Service job.

#### Supported Job Types

This attribute is required for the [Web Service Document/Literal \(WSDOC\) job type](#).

This attribute is optional for the [Web Service RPC/Encoded \(WBSVC\) job type](#).

#### Syntax

This attribute has the following format:

```
wsdl_operation: operation
```

- **operation** Specifies the operation to be invoked.

**Limits:** Up to 256 characters; case-sensitive

### Example: Invoke a Web Service Operation

#### NOTE

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

## WSDL\_URL Attribute -- Specify a WSDL URL

The `WSDL_URL` attribute specifies the URL to the Web Service Description Language (WSDL) of the web service to invoke in a Web Service job.

### Supported Job Types

This attribute is required for the [Web Service Document/Literal](#) (WSDOC) job type.

This attribute is optional for the [Web Service RPC/Encoded](#) (WBSVC) job type.

### Syntax

This attribute has the following format:

```
WSDL_URL: wsdl_url
```

- **wsdl\_url** Specifies the URL to the Web Service Description Language (WSDL) of the web service to invoke.  
**Limits:** Up to 256 characters (Web Service RPC/Encoded) and up to 2048 characters (Web Service Document/Literal); case-sensitive

**NOTE**

- Enclose the values that contain a colon or embedded blanks in double quotation marks. Do not use embedded double quotation marks.
- In a Web Service RPC/Encoded job, if you specify the WSDL\_URL attribute but not the endpoint\_URL attribute, you must specify both the service\_name and port\_name attributes. For the job to run successfully without the endpoint\_URL attribute, the agent must be running on the same computer as the application server such as WebLogic or JBoss. If you specify both the WSDL\_URL and endpoint\_URL attributes, then the service\_name and port\_name attributes are optional.
- You can use global variables to specify the URL to the WSDL of the web service to invoke. Use the \$global\_name, \${global\_name}, or \${\${global\_name}} format to reference global variables.

**Example: Specify WSDL URL for Getting Stock Quotes****NOTE**

This is just a sample and not a working example.

Suppose that you want to invoke a web service that returns a company stock quote. The URL for the WSDL that describes the web service and its location is `http://host:8080/axis/services/StockQuote?wsdl`. The WSDL port name within the target namespace `http://ws.ca.com` is `StockQuote`. The target endpoint address URL is `http://host:8080/axis/services/StockQuote`. The job calls the operation `GetQuote` within the `StockQuoteService` web service. When the job invokes the web service, the company's stock symbol is passed to the operation. When the job completes, the stock quote for CA is stored as a serialized Java object in the job's spool directory.

```
insert_job: quote
job_type: WBSVC
machine: localhost
owner: ejmcommander
max_run_alarm: 0
alarm_if_fail: y
send_notification: n
wsdl_url: "http://host:8080/axis/services/StockQuote?wsdl"
target_namespace: "http://ws.ca.com"
service_name: StockQuoteService
port_name: StockQuote
wsdl_operation: GetQuote
endpoint_url: "http://host:8080/axis/services/StockQuote"
web_parameter: "xsd:string=CA"
one_way: false
```

**zos\_dataset Attribute -- Specify a JCL Library for a z/OS Data Set Trigger Job**

The `zos_dataset` attribute specifies the JCL library that contains the JCL for your job.

**Supported Job Type**

This attribute is required for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

**Syntax**

This attribute has the following format:

```
zos_dataset: jcl_library
```

- **jcl\_library**

Specifies the Job Control Language (JCL) library name. The JCL library or JCLLIB contains the JCL for the z/OS job. The JCLLIB is a z/OS data set name.

Data set names typically have up to four positions; positions must each be eight characters or less and separated by periods.

**Limits:** Up to 44 characters

**Example:** prod.jcllib

**NOTE**

Instead of specifying a single data set, you can also use the hyphen wildcard to specify a group of data sets that match the selection criteria.

For example, the following table shows how the hyphen wildcard matches a group of data sets:

Specified data set name	Matching data sets
CYB1.PAYROLL.A	CYB1.PAYROLL.A
CYB1.PAYROLL.G-	CYB1.PAYROLL.G0145V00 CYB1.PAYROLL.G0146V00 CYB1.PAYROLL.G0147V00 CYB1.PAYROLL.GRANTED.MORE
CYB1.P-	CYB1.PAYROLL CYB1.POSTJOBS CYB1.PROD.INPUT

**Example: Specify a JCL Library**

Suppose that you want a z/OS Data Set Trigger job named PROD.NIGHTLY to release its successors when the data set PROD.CICS.FILE1602 is closed (created or updated). The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.NIGHTLY
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: PROD.CICS.FILE1602
```

## zos\_dsn\_renamed Attribute -- Specify Whether to Monitor When a Data Set is Renamed

The zos\_dsn\_renamed attribute specifies whether the job monitors when a data set is renamed. When the data set is renamed, the job triggers.

**NOTE**

If you specify the zos\_dsn\_renamed attribute in your job definition, you cannot specify the zos\_dsn\_updated attribute.

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

### Syntax

This attribute has the following format:

```
zos_dsn_renamed: Y | N
```

- **Y**

Monitors when the specified data set is renamed.

- **N**  
Does not monitor when the specified data set is renamed. This is the default.

### Example: Monitor When a Data Set is Renamed

This example monitors a data set named CYB2.QA.DSTRIG1. When the data set is renamed, the job triggers.

```
insert_job: test_ZOSDST
job_type: ZOSDST
machine: zosagent
owner: user1@ca11
zos_dsn_updated: N
zos_trigger_on: 0
zos_dataset: CYB2.DA.DSTRIG1
zos_dsn_renamed: y
```

### zos\_dsn\_updated Attribute -- Specify Whether to Monitor for Updates to a Data Set

The `zos_dsn_updated` attribute specifies whether the job monitors for updates to a data set. When the data set is updated, the job triggers.

#### NOTE

If you specify the `zos_dsn_updated` attribute in your job definition, you cannot specify the `zos_dsn_renamed` attribute.

### Supported Job Type

This attribute is optional for the z/OS Data Set Trigger (ZOSDST) job type.

### Syntax

This attribute has the following format:

```
zos_dsn_updated: Y | N
```

- **Y**  
Monitors for updates to the specified data set.
- **N**  
Does not monitor for updates to the specified data set. This is the default.

### Example: Monitor When a Data Set is Updated

This example monitors a data set named CYB2.QA.DSTRIG1. When the data set is updated, the job triggers.

```
insert_job: test_ZOSDST_Upd
job_type: ZOSDST
machine: zosagent
owner: user1@ca11
zos_dsn_updated: Y
zos_dataset: CYB2.DA.DSTRIG1
```

## zos\_explicit\_dsn Attribute -- Specify Whether to Monitor for an Explicit Data Set Notification

The `zos_explicit_dsn` attribute specifies whether the job monitors for an explicit data set notification (used when the data set activity does not generate an SMF record). The job triggers when it receives notification that the specified data set is being updated.

### NOTE

If you do not specify the `zos_explicit_dsn` attribute in your job definition, the job does not monitor for an explicit data set notification (the default).

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

### Syntax

This attribute has the following format:

```
zos_explicit_dsn: TRUE | FALSE
```

- **TRUE**

Monitors for an explicit data set notification.

- **FALSE**

Does not monitor for an explicit data set notification. This is the default.

### Example: Monitor for an Explicit Data Set Notification

This example monitors a data set named CYB2.DA.DSTRIG1. When the job receives notification that the data set is being updated, the job completes.

```
insert_job: test_ZOSDST
job_type: ZOSDST
machine: zosagent
owner: user1@call
zos_dsn_updated: N
zos_trigger_on: 0
zos_dataset: CYB2.DA.DSTRIG1
zos_explicit_dsn: y
```

## zos\_ftp\_direction Attribute -- Specify Whether to Monitor for an FTP Transfer To or From a Remote Computer

The `zos_ftp_direction` attribute specifies whether the job monitors for an FTP transfer to a remote computer or from a remote computer.

### NOTE

To use this attribute, you must specify FALSE for the `zos_explicit_dsn` attribute.

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

### Syntax

This attribute has the following format:

```
zos_ftp_direction: RECEIVE | SEND
```

- **RECEIVE**

Monitors for an FTP transfer from a remote computer to the local mainframe computer.

- **SEND**

Monitors for an FTP transfer from the local mainframe computer to a remote computer.

### Example: Monitor for a Data Set Received from a Remote FTP Partner

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release its successors when a file is successfully received from a remote FTP partner, creating generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.G-
zos_ftp_direction: RECEIVE
```

### zos\_ftp\_host Attribute -- Specify a Remote Computer for an FTP Transfer

The `zos_ftp_host` attribute specifies the remote computer used in an FTP transfer. The job only triggers when the FTP transfer occurs on the specified computer. The job does not trigger for FTP transfers on other computers.

#### NOTE

To use this attribute, you must specify FALSE for the `zos_explicit_dsn` attribute.

#### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger](#) (ZOSDST) job type.

#### Syntax

This attribute has the following format:

```
zos_ftp_host: host_name
```

- **host\_name**

Specifies the name of the remote computer involved in the FTP transfer. The data is transferred to or from the local mainframe computer.

**Limits:** Up to 128 characters

**Example:** 172.24.36.107 (IPv4), 0:0:0:0:FFFF:192.168.00.00 (IPv6), or machine1 (DNS name)

#### NOTE

Do not use embedded blanks, single quotation marks, or double quotation marks.

### Example: Restrict Triggering to a Specific Host

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a remote FTP partner with IP address 172.16.0.0 successfully transfers a file creating the data set CYBER.XFER.001. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.XFER.001
zos_ftp_direction: RECEIVE
zos_ftp_host: 172.16.0.0
zos_ftp_userid: CYB1
```



## zos\_ftp\_userid Attribute -- Specify a User ID for an FTP Connection

The `zos_ftp_userid` attribute specifies an FTP user ID used to connect to a remote computer. The job only triggers for FTP transfers by the specified log in user ID. The job does not trigger for FTP transfers by other user IDs.

### NOTE

To use this attribute, you must specify FALSE for the `zos_explicit_dsn` attribute.

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\)](#) job type.

### Syntax

This attribute has the following format:

```
zos_ftp_userid: user
```

- **user**

Specifies the FTP user ID used to connect to a remote computer.

**Limits:** Up to 8 characters

### Example: Restrict Triggering to a Specific Log In ID

Suppose that you want the z/OS Data Set Trigger job CYBER.XFER to release its successors when a remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming that the remote FTP partner logged on to the FTP server with the CYBER005 user ID. The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.XFER.001
zos_ftp_direction: RECEIVE
zos_ftp_host: 172.16.0.0
zos_ftp_userid: CYBER005
```

## zos\_jobname Attribute -- Specify the z/OS Job Name

The `zos_jobname` attribute specifies the name of the z/OS job that is being monitored. The job is submitted outside of AutoSys Workload Automation, such as a job that is submitted manually by a user.

### Supported Job Type

This attribute is required for the [z/OS Manual \(ZOSM\)](#) job type.

### Syntax

This attribute has the following format:

```
zos_jobname: name
```

- **name**

Specifies the name of the z/OS job that is being monitored.

**Limits:** Up to 8 alphanumeric or national (\$, #, @) characters. The first character must be alphabetic or national. If your z/OS system uses ANSI tapes, the name must contain only alphanumeric characters.

### Example: Post a z/OS Manual Job as Complete Based on the User ID

This example posts a z/OS Manual job as complete when the manually-submitted job ABC runs under user CYBER. The ZOS1 agent monitors job ABC.

```

insert_job: ABC_job
job_type: ZOSM
machine: ZOS1
zos_jobname: ABC
owner: zosuser
auth_string: CYBER

```

## zos\_trigger\_by Attribute -- Specify a Job Name or a User ID that Triggers the Job

The `zos_trigger_by` attribute specifies the name of the job or the user who performs the data set activity that triggers the job. The job trigger is restricted to data set activity performed by the specified job or user.

### NOTE

To use this attribute, you must also specify the `zos_trigger_type` attribute in your job definition.

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

### Syntax

This attribute has the following format:

```
zos_trigger_by: name
```

- ***name***

Specifies the name of the job or user who performs the data set activity that triggers the job.

**Limits:** Up to 8 characters

### Example: Restrict the Trigger to Specific Data Sets Created by a Particular Job

Suppose that you want a z/OS Data Set Trigger job named `PROD.PAY_DATA` to release its successors when job `ABC` creates generation data set `USER1.PAYROLL (USER1.PAYROLL.G-)`. The agent `ZOS1` monitors the data set under user `CYBDL01`.

```

insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.G-
zos_trigger_type: zos_job_name
zos_trigger_by: ABC

```

### Example: Restrict the Trigger to Specific Data Sets Created by a Particular User

Suppose that you want the z/OS Data Set Trigger job `PROD.PAY_DATA` to release its successors when the user `CYB1` creates generation data set `USER1.PAYROLL (USER1.PAYROLL.G-)`. The agent `ZOS1` monitors the data set under user `CYBDL01`.

```

insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.G-
zos_trigger_type: zos_user_id
zos_trigger_by: CYB1

```

### Example: Restrict the Trigger to an FTP Transfer from a Specific User ID

This example releases the job's successors when a remote FTP partner successfully transfers a file creating the data set CYBER.XFER.001, assuming that the user ID prefix of the local FTP partner is CYB (CYB-). The agent ZOS1 monitors the FTP transfer under user CYBDL01.

```
insert_job: CYBER.XFER
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.XFER.001
zos_trigger_type: zos_user_id
zos_trigger_by: CYB-
zos_ftp_direction: RECEIVE
zos_ftp_userid: CYB-
```

## zos\_trigger\_on Attribute -- Specify the Number of Actions that Must Occur

The `zos_trigger_on` attribute specifies the number of actions that must occur before the job triggers.

### Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger \(ZOSDST\) job type](#).

### Syntax

This attribute has the following format:

```
zos_trigger_on: number
```

- ***number***  
Specifies the number of actions that must occur before the job triggers.  
**Limits:** 0-100

### Example: Run a Compress Job After 100 Closures of a Data Set

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release a compress job named COPYJCL.COMPRESS after every 100 closures of the data set CYBER.COPY.JCL. The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: CYBER.COPY.JCL
zos_trigger_on: 100
```

#### NOTE

Define the compress job, COPYJCL.COMPRESS, as a successor to the z/OS Data Set Trigger job.

## zos\_trigger\_type Attribute -- Specify Whether to Monitor for Data Set Activity by a Job or a User ID

The `zos_trigger_type` attribute specifies whether the job monitors data set activity by a job or a user ID.

When a data set is updated or renamed by the specified job or user ID, the job triggers.

#### NOTE

To use this attribute, you must also specify the `zos_trigger_by` attribute in your job definition.

## Supported Job Type

This attribute is optional for the [z/OS Data Set Trigger](#) (ZOSDST) job type.

### Syntax

This attribute has the following format:

```
zos_trigger_type: zos_job_name | zos_user_id
```

- **zos\_job\_name**  
Monitors data set activity by a job.
- **zos\_user\_id**  
Monitors data set activity by a user ID.

### Example: Restrict the Trigger to Specific Data Sets Created by a Particular Job

Suppose that you want a z/OS Data Set Trigger job named PROD.PAY\_DATA to release its successors when job ABC creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.G-
zos_trigger_type: zos_job_name
zos_trigger_by: ABC
```

### Example: Restrict the Trigger to Specific Data Sets Created by a Particular User

Suppose that you want the z/OS Data Set Trigger job PROD.PAY\_DATA to release its successors when the user CYB1 creates generation data set USER1.PAYROLL (USER1.PAYROLL.G-). The agent ZOS1 monitors the data set under user CYBDL01.

```
insert_job: PROD.PAY_DATA
job_type: ZOSDST
machine: ZOS1
owner: CYBDL01
zos_dataset: USER1.PAYROLL.G-
zos_trigger_type: zos_user_id
zos_trigger_by: CYB1
```

## JIL Job Type Definitions

This section describes the JIL subcommands and attributes that you can use to define, update, and delete AutoSys Workload Automation job types.

### delete\_job\_type Subcommand -- Delete a Job Type from the Database

The delete\_job\_type subcommand deletes a job type from the database.

**Note:** AutoSys Workload Automation issues an error and rejects attempts to delete a job type when the database contains job definitions for that job type.

### Syntax

This subcommand has the following format:

```
delete_job_type: value
```

- ***value***

Defines the value between 0-9, of the job type that is currently defined in the database.

### Example: Delete a Job Type from the Database

This example deletes the job type 1:

```
delete_job_type: 1
```

## insert\_job\_type Subcommand -- Add a Job Type to the Database

You can use the `insert_job_type` subcommand to add user-defined job types to the database. This is useful when you want to run multiple jobs that specify the same command. Define the command that is associated with this job type by specifying the `command` attribute when you issue the `insert_job_type` subcommand. Optionally, you can also specify the `description` attribute. You can reference a user-defined job type in the definition of any job that executes the command specified in the `command` attribute.

### NOTE

You cannot reference a user-defined job type in a job definition until you define the command that is associated with that job type.

### Syntax

This subcommand has the following format:

```
insert_job_type: value
command: file
[description: text]
```

### ***value***

Specifies a job type to add.

**Limits:** 0-9

### ***file***

Specifies the binary file to use for the command in a job.

**Limits:** Up to 255 characters specifying the full pathname of an executable file.

**Windows Note:** Enclose the colon with backslashes when specifying drive letters. For example, C:\tmp.

### ***text***

(Optional) Describes the job type.

**Limits:** Up to 255 characters

### Example: Add a Job Type with a Command to the Database

This example adds a definition for job type 1 to the database. The definition specifies that job type 1 executes the binary file `wv`.

```
insert_job_type: 1
command: /tmp/wv
description: executes tmp/wv
```

## update\_job\_type Subcommand -- Update a Job Type in the Database

You can use the `update_job_type` command to change the definition of a user-defined job type. Update user-defined job definitions when you make changes to the binary file that is specified in the command attribute changes. For example, when you change the location, name, or parameters of the file.

### Syntax

This subcommand has the following format:

```
update_job_type: value
command: file
[description: text]
```

#### **value**

Specifies a job type to add.

**Limits:** 0-9

#### **file**

Specifies the binary file to use for the command in a job.

**Limits:** Up to 255 characters specifying the full pathname of an executable file.

**Windows Note:** Enclose the colon with backslashes when specifying drive letters. For example, C:\tmp.

#### **text**

(Optional) Describes the job type.

**Limits:** Up to 255 characters

### Example: Update a Job Type with a New Command of ftp to the Database

This example updates job type 1 to specify that the command for the job executes the binary file called ftp.

```
update_job_type: 1
command: /bin/ftp
description: executes bin/ftp
```

## command Attribute -- Specify Command for the Job Type

The command attribute specifies the binary file to use for the job type.

### Syntax

This attribute has the following format:

```
command: file
```

- **file**

Specifies the binary file to use for the command in a job.

**Limits:** Up to 255 characters

**Windows Note:** When specifying drive letters in job definitions, you must escape the colon with backslashes. For example, C:\tmp is valid; C:\tmp is not.

## description Attribute -- Specify a Description for the Job Type

The description attribute specifies a description to use for the job type.

This attribute has the following format:

```
description: text
```

### ***text***

Specifies the text description for the job type.

**Limits:** Up to 255 characters

## JIL Machine Definitions

This section describes the JIL subcommands and attributes that you can use to define and delete the machines where jobs run. You can define real and virtual machines in the Windows or UNIX operating environments. To define or delete a machine, you specify the appropriate JIL subcommand and attribute statements to the jil command.

## delete\_machine Subcommand -- Delete a Real or Virtual Machine

### Contents

The delete\_machine subcommand deletes the following from the AutoSys Workload Automation database:

- A real machine
- A virtual machine
- Virtual machine and real machine pool references to a real machine

### Syntax

This subcommand has the following format:

```
delete_machine: machine_name
```

- ***machine\_name***  
Identifies a real or virtual machine that currently exists in the AutoSys Workload Automation database.

### Optional Attributes

You can specify the following optional attributes when deleting a virtual machine:

- [machine](#)

You can specify the following optional attributes when deleting a real machine:

- [force](#)
- [remove\\_references](#)

### NOTE

If AutoSys Workload Automation is running in external security mode using CA EEM, you must have the following authority:

- DELETE authority for the corresponding as-machine resource policy to delete the real or virtual machine.
- WRITE authority for the corresponding as-machine resource policies to remove references to real machines or to update the virtual machines that reference it.

### Example: Delete a Machine

This example deletes the machine mach1, which could be a virtual or a real machine.

```
delete_machine: mach1
```

### Example: Delete a Real Machine From the Virtual Machine

This example deletes the real machine socrates from the virtual machine cheetah without deleting cheetah.

```
delete_machine: cheetah
machine: socrates
```

### Delete a Real Machine

To delete a real machine that is not referenced by any virtual machines, use the following syntax:

```
delete_machine: real_machine_name
```

To delete a real machine that is referenced by virtual machines, you must do *one* of the following:

- Individually delete the references from all virtual machines that reference it.
- Delete all references at one time and the real machine itself.

### Example: Delete a Real Machine

This example deletes the real machine named realmach1. This machine is not referenced by any virtual machines.

```
delete_machine: realmach1
```

### Delete a Virtual Machine

To delete a virtual machine, use the following syntax:

```
delete_machine: virtual_machine_name
```

The virtual machine is deleted, but the real machines it references are not deleted.

### Delete a Real Machine Pool

To delete a real machine pool, use the following syntax:

```
delete_machine: real_machine_pool_name
```

The real machine pool is deleted, but the real machines it references are not deleted.

### Delete a Reference to a Real Machine

To delete a virtual machine or real machine pool reference to a real machine, use the following syntax:

```
delete_machine: virtual_machine_name
machine: real_machine_name_referenced
```

Or

```
delete_machine: real_machine_pool_name
machine: real_machine_name_referenced
```

The reference to the real machine is deleted. The real machine definition itself is *not* deleted.



**NOTE**

You can individually delete all real machine references in a virtual machine or a real machine pool until there is only one reference remaining. You cannot delete the last reference. If you try to delete the virtual machine's or real machine pool's last machine reference, the command will fail because a virtual machine or real machine pool definition with no references is invalid. To delete the last reference in a virtual machine or real machine pool, you must delete the virtual machine or real machine pool itself.

The real machine can be deleted after the previous command is issued for the last virtual machine or real machine pool that references it.

**Example: Delete a Real Machine that is the One of Several References of a Virtual Machine**

Suppose that a virtual machine named `virtmach1` has references to multiple real machines. One of the references is to the real machine named `realmach1`. This example deletes `realmach1`.

```
delete_machine: virtmach1
machine: realmach1
```

**Delete All References to a Real Machine and the Machine Itself**

Instead of individually deleting references to a real machine from virtual machines or real machine pools, you can delete all references at one time. The real machine itself is also deleted.

To delete the real machine and all references to it, use the following syntax:

```
delete_machine: real_machine_name
remove_references: y
```

**NOTE**

This command deletes all references to the specified real machine from all virtual machines or real machine pools even if a virtual machine or real machine pool only has that one reference. After that one reference is deleted, the virtual machine or real machine pool is considered empty. You cannot re-define that virtual machine or real machine pool because it already exists. However, you can re-insert real machine references to it. To display all empty virtual machines or real machine pools that you can use, issue the following command:

```
autorep - M ALL
```

**Example: Delete a Real Machine and All References to It**

Suppose that the real machine named `realmach1` is referenced by virtual machines. This example deletes `realmach1` and all references to it.

```
delete_machine: realmach1
remove_references: y
```

**insert\_machine Subcommand -- Add a Machine Definition****Contents**

The `insert_machine` subcommand adds a new machine definition to the AutoSys Workload Automation database for a real machine or virtual machine.

You can also add a machine definition for the following agents:

- Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS
- CA NSM Job Management Option
- CA Universal Job Management Agent (UJMA)
- CA AutoSys Workload Automation Connect Option

After you define a machine, you can schedule jobs to run on those machines. You can identify any machine accessible through the TCP/IP protocol in the machine attribute of a job.

#### NOTE

- You must define the machine using the `insert_machine` subcommand before you can insert a job that uses that machine. The job will not insert if the machine definition does not exist.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you must have CREATE authority for the corresponding as-machine resource policy to define a machine.
- When you insert a machine, the scheduler calibrates to the actual machine status and sets the machine status accordingly.

### Syntax

This subcommand has the following format:

```
insert_machine: machine_name
```

- ***machine\_name***  
Defines a unique name for the machine definition. When defining jobs, specify this name in the machine attribute.  
**Limits:** Up to 80 alphanumeric characters or the following special characters: hyphen (-), underscore (\_), period (.), and hash (#). The machine name cannot contain spaces or tab.

#### NOTE

- If you do not specify the `node_name` attribute in your machine definition, the node name is set to this *machine\_name* by default. In this situation, the *machine\_name* must be the DNS name of the agent machine. Otherwise, AutoSys Workload Automation cannot connect to the agent.
- Do not specify ALL or localhost as the machine name.

### Optional Attributes

You can specify the following optional attributes when defining a machine:

- `description`
- `factor`
- `machine`
- `max_load`
- `node_name`
- `type`

If you are defining a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS machine, the following attributes are also optional:

- `agent_name`
- `character_code`
- `encryption_type`
- `key_to_agent`
- `opsys`
- `port`

### Example: Define an r11 Real Machine

This example defines a legacy (r11) real Windows machine named plato:

```
insert_machine: plato
type: n
```

#### Example: Define an r11.3, Release 11.3.5, or Release 11.3.6 Real Machine

This example defines an r11.3, Release 11.3.5, or Release [assign the value for rn in your book] real Windows machine named cheetah:

```
insert_machine: cheetah
type: a
opsys: windows
```

#### Example: Define a Virtual Machine to Include Two Real Machines

This example defines a virtual machine virtual\_b to include two real machines (france and italy) without specifying factors or loads:

```
insert_machine: virtual_b
type: v
machine: france
machine: italy
```

#### Example: Define Two Real Machines and a Virtual Machine with Load Balancing Criteria

This example defines two real Windows machines (elm and maple) and a virtual machine (trees) that includes both the real machines:

```
insert_machine: elm
type: n max_load: 100 factor: 1.0
insert_machine: maple
type: n max_load: 200 factor: 1.0
insert_machine: trees
type: v
machine: elm
max_load: 50 factor: .1
machine: maple
max_load: 100 factor: 1.0
```

### **Real Machines**

*Real machines* are the definitions of actual machines where an agent has been installed.

You can use the max\_load attribute in a real machine specification to define how many load units are allowed on that machine simultaneously. You can also use the factor attribute to specify the relative processing capacity of the machine. Both of these attributes are assigned from an arbitrary, user-defined range of values. max\_load is used with the job\_load attribute to limit the load placed on a machine at any one time. To avoid overloading a machine based on these attribute values, the scheduler queues jobs to run when load units become available again.

When you specify multiple real machines for a virtual machine, the scheduler determines which of the real machines have sufficient load units to run the job. If more than one real machine has sufficient units to run the job, the scheduler queries each machine's available CPU cycles, multiplies that number by the machine's factor, and chooses the machine with the largest value.

## Virtual Machines

*Virtual machines* are the machine definitions that reference one or more existing real machine definitions. Virtual machines are typically used for load balancing, but they can be used for other things, such as creating an alias for a machine. For example, you can define a virtual machine named "produnix", which references the real name of a production UNIX machine. If job definitions reference the virtual machine name, the production UNIX machine can be changed in the virtual machine definition without having to change all jobs that referenced it.

If AutoSys Workload Automation is running in external security mode using CA EEM, you must have CREATE authority for the corresponding as-machine resource policy to define the virtual machine.

To define a virtual machine, follow the `insert_machine` subcommand with one or more machine attributes to specify the real machines it contains, as follows:

```
insert_machine: virtual_machine_name
machine: real_machine_name
machine: real_machine_name
.
```

The following attributes can be added to the definition:

- [description](#)
- [factor](#)
- [max\\_load](#)
- [type](#)

## update\_machine Subcommand -- Update a Machine Definition

### Contents

The `update_machine` subcommand updates a machine definition in the AutoSys Workload Automation database.

#### NOTE

- If AutoSys Workload Automation is running in external security mode using CA EEM, you must have WRITE authority for the corresponding as-machine resource policy to update a machine.
- When you update the `port`, `node_name`, `encryption_type`, `agent_name`, or `key_to_agent` machine attribute, the scheduler calibrates to the actual machine status and sets the machine status accordingly.

### Syntax

This subcommand has the following format:

```
update_machine: machine_name
```

- **machine\_name**  
Specifies the name of the machine to update.  
**Limits:** Up to 80 alphanumeric characters or the following special characters: hyphen (-), underscore (\_), period (.), and hash (#). The machine name cannot contain spaces or tab.

### Optional Attributes

You can specify the following optional attributes when updating a machine:

- [description](#)
- [factor](#)
- [machine](#)
- [max\\_load](#)
- [node\\_name](#)

If you are updating a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS machine, the following attributes are also optional:

- [agent\\_name](#)
- [character\\_code](#)
- [encryption\\_type](#)
- [key\\_to\\_agent](#)
- [opsys](#)
- [port](#)

### Updating Virtual Machines

If AutoSys Workload Automation is running in external security mode using CA EEM, you must have WRITE authority for the corresponding as-machine resource policy to update the virtual machine.

If you are updating a virtual machine that contains real machines, only the following attributes can be updated:

- [description](#)
- [factor](#)
- [max\\_load](#)

## agent\_name Attribute -- Specify the Name of an Agent

The agent\_name attribute specifies the name of an agent.

### NOTE

- If you do not specify the agent\_name attribute in your machine definition, the agent name is set to WA\_AGENT (the default).

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
agent_name: agent
```

- **agent**

Specifies the name of an agent. This name must match *one* of the following values:

- The agentname parameter in the agent's agentparm.txt file (on Workload Automation Agent for UNIX, Linux, Windows, or i5/OS)
- The ZOSAGENT(*name*) parameter in the AGENTDEF data set (on Workload Automation Agent for z/OS)

**Default:** WA\_AGENT

**Limits:** Up to 16 alphanumeric characters or the following special characters: hyphen (-), underscore (\_), and period (.).

**NOTE**

For more information, see [Workload Automation Agent Parameters](#).

**character\_code Attribute -- Specify the Character Code for the Machine**

The `character_code` attribute specifies the character code for an agent.

**NOTE**

- If you do not specify the `character_code` attribute in your job definition, the machine is defined with the ASCII character code (the default).

**Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

**Syntax**

This attribute has the following format:

```
character_code: ASCII | EBCDIC
```

- **ASCII**  
Specifies the ASCII character code.  
Specify this value if you are defining a Workload Automation Agent for UNIX, Linux, Windows, or i5/OS instance. This is the default.
- **EBCDIC**  
Specifies the EBCDIC character code.  
Specify this value if you are defining a Workload Automation Agent for z/OS instance.

**description Attribute -- Specify a Description for an Agent**

The `description` attribute specifies a description for the agent you are defining.

**Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

**Syntax**

This attribute has the following format:

```
description: text
```

- **text**  
Specifies a description for the agent.  
**Limits:** Up to 256 characters

**NOTE**

If the description contains a colon (:), precede the colon with a backslash (\) or enclose the description with double quotation marks ("").

## encryption\_type Attribute -- Specify the Type of Encryption

The encryption\_type attribute specifies the type of encryption between AutoSys Workload Automation and the agent machine you are defining.

### NOTE

- If you do not specify the encryption\_type attribute in your machine definition, the machine uses DEFAULT encryption (the default).

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
encryption_type: NONE | DEFAULT | AES
```

- **NONE**  
Specifies that the machine uses no encryption.
- **DEFAULT**  
Specifies that the machine uses the default encryption key and type. This is the default.
- **AES**  
Specifies that the machine uses AES 128-bit encryption.

### NOTE

You must specify a key using the key\_to\_agent attribute.

### NOTE

(Workload Automation Agent for z/OS) To use encryption, both the agent on z/OS and AutoSys Workload Automation must support AES encryption. You must specify the same AutoSys Workload Automation encryption setting in the AGENTDEF data set of the agent. For more information about the encryption types that the agent supports, see the Workload Automation Agent for z/OS documentation.

## factor Attribute -- Define a Factor for an Agent Machine

You can use the factor attribute to determine the relative processing power for a machine. To calculate the relative processing power for each machine, the scheduler multiplies the available CPU cycles by the factor attribute value:

(Available CPU Cycles) x (Factor Attribute Value) = Relative Processing Power

The scheduler determines which of the agent machines specified in the job definition have the best calculated usage (highest relative processing power). The scheduler starts the job on the agent machine with the best calculated usage (highest relative processing power).

Setting the factor value to zero (0) results in a calculated usage of zero (0) but does not disqualify the machine. The scheduler selects an agent machine with a factor value of zero (0) only when all other available machines specified in the job definition also have a factor value of zero (0).

**NOTE**

- You can specify the factor attribute only in a real machine definition or in component machines that are defined in a virtual machine definition.
- The factor attribute does not support type 'c' or type 'u' machines.
- Sometimes the scheduler identifies multiple machines as having the best calculated usage (highest relative processing power). In such cases, the scheduler randomly selects one of those machines and starts the job on it. To allow the scheduler to start the job on any machine specified in the job definition, set the factor attribute value for all of those machines to zero (0).

**Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

**Syntax**

This attribute has the following format:

```
factor: real_number
```

- ***real\_number***

Defines a real number from a user selected range of values.

**Limits:**

- 0-1024; up to 7 characters; you can specify a decimal.
- The number of decimal places cannot exceed hundredths (for example, 1023.99 is valid, but 1.999 is not valid).
- The examples in the following section use a range of 0.0 to 1.0; however, any reasonable convention is valid. For example, 100.00 to 999.99 is a valid range, but -1.0 and 1024.01 are not valid.

**Default:** 1.0

**Example: Set the Factor for a Very High-Performance Real Machine**

This example sets the factor for a very high-performance real machine on a scale of 0.0 to 1.0:

```
factor: 1.0
```

**NOTE**

Real machines and virtual machines can have different `max_load` and `factor` values. The scheduler handles the load balancing according to the machine specified in the job definition. If the virtual machine is specified, the values associated with the virtual machine definition are used. If a real machine is specified, the values associated with the corresponding real machine definition are used.

**Example: Set the Factor for a Relatively Low-Performance Real Machine**

This example sets the factor for a relatively low-performance real machine on a scale of 0.0 to 1.0:

```
factor: 0.4
```

Assume the following virtual machines are defined:

```
insert_machine: marmot
machine: cheetah
factor: 1
machine: hippogriff
factor: .8
machine: goat
```



```
factor: .3
```

If the machine attribute for a job that is ready to start is set to the virtual machine marmot, the scheduler performs the necessary calculations to determine on which machine to run the job and reports these calculations in its output log. For example:

```
EVENT: STARTJOB JOB: test_mach
Machine usages obtained for job <test_mach>
Machine <cheetah>: Method <CPU monitor>. Available usage <78*[1.00]=78>
Machine <hippogriff>: Method <CPU monitor>. Available usage <80*[0.80]=64>
Machine <goat>: Method <CPU monitor>. Available usage <2*[0.30]=0>
EVENT: CHANGE_STATUS STATUS: STARTING JOB: test_mach
[cheetah connected for test_mach]
```

The factors weigh each machine to account for variations in processing power. Note that the product of the CPU cycles and the factor is rounded off to the integer value as decimals are not supported. In this example, even though the raw available CPU for cheetah was less than that of hippogriff, the product chose to run the job on cheetah based on its factor calculation:

```
(78 * 1.0 > 80 * 0.8)
```

That is, cheetah had more relative CPU cycles available.

When max\_load attributes are specified for the real machines in a virtual machine, the product determines which of the real machines have sufficient load units to run the job. If more than one real machine has sufficient units to run the job, the scheduler queries each machine's available CPU cycles, multiplies that number by the machine's factor, and chooses the machine with the largest value.

## force Attribute -- Force a Machine or Connection Profile to be Deleted

The force attribute deletes the machine or connection profile whether it is currently being used or not.

### Supported JIL Subcommand

This attribute is optional for the [delete\\_machine](#) and [delete\\_connectionprofile](#) subcommands.

### Syntax

This attribute has the following format:

```
force: y | n
```

- **y**  
Deletes the machine or connection profile whether it is currently being used or not.
- **n**  
Does not delete the machine or connection profile if it is currently being used. This is the default.

### NOTE

After you delete the machine, its jobs, containers (such as virtual machines), and resources are no longer associated with any machine. You can re-insert the machine to associate the objects with it.

### Example: Force a Machine to be Deleted

This example deletes the real machine named realmach1, which is currently being used.

```
delete_machine: realmach1
force: y
```

---

## heartbeat\_attempts Attribute -- Specify the Number of Heartbeat Signals

The heartbeat\_attempts attribute specifies the number of heartbeat signals the scheduler tries to detect before it sends an SNMP message indicating inactivity.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
heartbeat_attempts: number_of_signals
```

- ***number\_of\_signals***  
Specifies the number of heartbeat signals the scheduler tries to detect before it sends an SNMP message indicating inactivity.  
**Default:** 1

## heartbeat\_freq Attribute -- Specify How Frequently the Scheduler Sends Heartbeat Signals

The heartbeat\_freq attribute specifies how frequently the scheduler sends the heartbeat signal (in minutes).

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
heartbeat_freq: minutes
```

- ***minutes***  
Specifies how frequently the scheduler sends the heartbeat signal (in minutes)  
**Default:** 5

## key\_to\_agent Attribute -- Specify the Agent Encryption Key

The key\_to\_agent attribute specifies the key used to encrypt data from AutoSys Workload Automation to the agent.

### NOTE

This attribute only applies to the Workload Automation Agents.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
key_to_agent: key
```

- **key**  
Specifies the key used to encrypt data from AutoSys Workload Automation to the agent. This value must match the security.cryptkey parameter in the agent's agentparm.txt file, without the prefix 0x. If the values do not match, AutoSys Workload Automation cannot communicate with the agent. You must specify one of the following:
  - A 32-digit hexadecimal key
  - A passphrase with up to 16 characters

## machine Attribute -- Define or Update a Real Machine As a Component of the Virtual Machine or Real Machine Pool

The machine attribute defines a real machine as a component of the virtual machine or real machine pool that is being defined or updated by the insert\_machine subcommand. If the specified virtual machine or real machine pool does not exist, it is created.

### NOTE

This attribute differs from the machine attribute used in the job definition subcommands. In a job definition, the machine attribute assigns the job to one or more real or virtual machines.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [delete\\_machine](#)
- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
machine: machine_name
```

- **machine\_name**  
Specifies a real machine as a component of the virtual machine or real machine pool. The specified machine must have been defined to AutoSys Workload Automation as a real machine.  
**Limits:** Up to 80 alphanumeric characters or the following special characters: hyphen (-), underscore (\_), period (.), and hash (#). The machine name cannot contain spaces or tab.

### NOTE

- The type of the specified machine cannot be v, w, or p. You can define virtual resources for any real machine defined on AutoSys Workload Automation.
- You can specify multiple real machines. Separate each machine with a comma.
- When you specify multiple real machines, AutoSys Workload Automation uses one of various load balancing methods to select the real machine that is best suited to run the job. For more information, see [How Load Balancing Works](#).

### Example: Define a Virtual Machine to Include Two Real Machines

This example defines a virtual machine virtual\_a to include two real machines (test and prod). If the virtual machine already exists, the component machines are inserted.

```
insert_machine: virtual_a
machine: test
```

```
machine: prod
```

## max\_load Attribute -- Define the Maximum Load an Agent Machine Can Handle

The `max_load` attribute defines the maximum load (in load units) that a machine can reasonably handle. Load units are arbitrary values. The range of values is user-defined.

You can assign a `max_load` value to each machine to indicate its relative processing capacity. For example, you might define that a powerful machine can handle up to 100 load units while a less powerful machine can handle only 10 load units. Similarly, you can use the `job_load` attribute to assign values to your jobs that indicate how much relative processing power the jobs require. When a job is ready to run, the product compares the available `max_load` value for each machine with the `job_load` value for the job to determine which machine is the best fit for the job.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
max_load: load_units
```

- **load\_units**

Defines how many load units are allowed on the machine simultaneously. This number can be any value in the user-defined range of possible values. The range is also arbitrary.

**Limits:** 0-9999999999

#### NOTE

- Set the value to 0 for unlimited load units.
- If you do not set a `max_load` value, AutoSys Workload Automation does not limit the load on the machine.

#### NOTE

- If `job_load` is not set for a job, the job runs without checking for load units. If a priority is not set for a job, the priority defaults to 0 and the `job_load` value is ignored.
- When you specify multiple machines in the job's machine attribute, each machine is checked for available load units when the job is ready to run. If none of the machines currently has the necessary load units available, the job is queued on all of the specified machines. The job will be submitted to the first machine where the necessary load units are available.
- The `job_load` attribute is not related to the priority or ordering of jobs in a queue. The `job_load` attribute only controls the machine where the job runs when it exceeds the `max_load` of a machine, thus eliminating that machine.
- You can use the `max_load` attribute in real and virtual machine definitions. Real machines and virtual machines can have different `max_load` and factor values. The scheduler handles the load balancing according to the machine specified in the job definition. If the virtual machine is specified, the values associated with the virtual machine definition are used. If a real machine is specified, the values associated with the corresponding real machine definition are used.

### Example: Set the Maximum Load for a Very High-Performance Real Machine

This example sets the `max_load` for a very high-performance real machine on a scale of 1 to 100.

```
max_load: 100
```

### Example: Set the Maximum Load for a Relatively Low-Performance Real Machine

This example sets the `max_load` for a relatively low-performance real machine on a scale of 1 to 100.

```
max_load: 20
```

### node\_name Attribute -- Specify the IP Address of a Machine

The `node_name` attribute specifies the IP address or DNS name of the machine you are defining.

#### NOTE

If you do not specify the `node_name` attribute in your machine definition, the value specified in the `insert_machine: machine_name` command is used (the default).

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
node_name: address
```

- **address**  
Specifies the IP address or DNS name of the machine.  
**Default:** The value specified in the `insert_machine: machine_name` command. In this situation, the `machine_name` must be the DNS name of the agent machine. Otherwise, AutoSys Workload Automation cannot connect to the agent.  
**Limits:** Up to 80 characters

#### NOTE

If you specify the DNS name, it must be a valid name or IPv4 or IPv6 address.

**Example:** 172.24.36.107 (IPv4) or 0:0:0:0:FFFF:192.168.00.00 (IPv6)

### opsys Attribute -- Specify the Operating System Type of a Machine

The `opsys` attribute specifies the operating system type of the machine you are defining.

#### NOTE

This attribute only applies to Workload Automation Agent machines. To use this attribute, you must specify the `type: a` attribute.

### Supported JIL Subcommands

This attribute is optional for the following subcommands if the `type` attribute is set to `a`:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
opsys: type
```

- **type**

Specifies the operating system type of the machine. Options are the following:

- aix
- hpux
- linux
- openvms
- i5os
- tandem
- windows
- zos

#### NOTE

- If the machine is running on Windows and the jobs run under users that belong to a Windows network domain, you must add the opsys: windows attribute to the machine definition.
- If you specify opsys: zos in your machine attribute, you must also specify encryption\_type: NONE or encryption\_type: AES.
- AutoSys Workload Automation web services use the following numeric codes to represent the operating system of a machine (6—solaris is not supported):
  - 1—aix
  - 2—hpux
  - 3—linux
  - 4—openvms
  - 5—i5os
  - 7—tandem
  - 8—windows
  - 9—zos

## port Attribute -- Specify the Port Used to Communicate with the Machine

The port attribute specifies the port that the agent uses to listen for traffic.

#### NOTE

- If you do not specify the port attribute in your machine definition, the machine is defined with port 7520 (the default).

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_machine](#)
- [update\\_machine](#)

### Syntax

This attribute has the following format:

```
port: port_number
```

- ***port\_number***

Specifies the port that the agent uses to listen for traffic.

This name must match *one* of the following values:

- The communication.input.port parameter in the agent's agentparm.txt file (on Workload Automation Agent for UNIX, Linux, Windows, or i5/OS)
- The COMMCHAN PORT(port) parameter in the AGENTDEF data set (on Workload Automation Agent for z/OS)

**Default:** 7520

**Limits:** 1-65535

#### NOTE

For more information about the agent parameters, see [Workload Automation Agent Parameters](#).

## remove\_references Attribute -- Remove All Machine References from Virtual Machines or Real Machine Pools

The remove\_references attribute removes all references to a real machine and deletes the machine. This attribute only applies to real machines.

#### NOTE

The remove\_references attribute does not delete the real machine if it is being referenced by virtual resources or active jobs.

### Supported JIL Subcommand

This attribute is optional for the [delete\\_machine subcommand](#).

### Syntax

This attribute has the following format:

```
remove_references: y | n
```

- **y**  
Removes all references from virtual machines or real machine pools and deletes the real machine definition.
- **n**  
Does not remove all references from virtual machines or real machine pools and does not delete the real machine definition. This is the default.

### Example: Delete a Real Machine and Remove All References to It

This example deletes the real machine named realmach1 and removes all references to it from virtual machines or real machine pools.

```
delete_machine: realmach1
remove_references: y
```

## type Attribute -- Specify the Machine Type

The type attribute specifies the type of machine you are defining. The type attribute is optional in a machine definition.

#### NOTE

If you do not specify the type attribute in your machine definition, the machine is defined as type a (the default).

### Supported JIL Subcommand

This attribute is optional for the [insert\\_machine subcommand](#).

### Syntax

This attribute has the following format:

type: a | c | l | L | n | r | u | v

- **a**  
Specifies a Workload Automation Agent for UNIX, Linux, Windows, i5/OS, or z/OS machine. This is the default.
- **c**  
Specifies a CA AutoSys Workload Automation Connect Option machine.
- **l**  
Specifies a 4.5.1 real UNIX machine. You must specify a lowercase l.
- **L**  
Specifies a 4.5.1 real Windows machine. You must specify a capital L.
- **n**  
Specifies an r11 real Windows machine or a virtual machine that consists only of r11 real Windows machines (type n).
- **r**  
Specifies an r11 real UNIX machine.
- **u**  
Specifies a CA NSM or a CA Universal Job Management Agent (CA UJMA) machine.
- **v**  
Specifies a virtual machine. The virtual machine can consist of Workload Automation Agent machines (type a), r11 real UNIX machines (type r), and r11 real Windows machines (type n).

#### NOTE

- In Release 11.3.6, the scheduler component has the capability to run jobs on the r11.3, Release 11.3.5, and Release 11.3.6 agent machines, r11 agent machines, and 4.x remote agent machines. Due to the differences in the communication protocol used by the different versions of the agents, the scheduler component must know which communication protocol to invoke before contacting the agent machine. This is done by examining the agent's machine definition type attribute.
- The machine types "t" and "z" available in the 4.x releases of the product are now replaced with "u" and "c" respectively. Using JIL, if you input the machine type as "t" it is converted to "u" and if you input the machine type as "z" it is converted to "c", and a warning message appears.
- The machine type "w" in the 4.x releases of the product is now replaced with "n" respectively. Using JIL, if you input the machine type as "w" it is converted to "n" and a warning message appears.
- Suppose that you define an r11 virtual machine with type "v" and it only contains r11 real Windows machines (type "n"), as shown in the following example:

```
insert_machine: realmach1
type: n
insert_machine: realmach2
type: n
insert_machine: virtmach
type: v
machine: realmach1
machine: realmach2
```



## JIL Connection Profile Definitions

This section describes the JIL subcommands and attributes that you can use to define, update, and delete connection profiles.

### delete\_connectionprofile Subcommand -- Delete a Connection Profile

The delete\_connectionprofile subcommand deletes a connection profile from the database.

#### NOTE

If AutoSys Workload Automation is running in external security mode using CA EEM, you must have the DELETE authority for the corresponding as-connectionprofile resource class to delete a connection profile.

#### Syntax

This subcommand has the following format:

```
delete_connectionprofile: conn_name
```

- **conn\_name**  
Specifies the name of the connection profile that you want to delete.

#### Optional Attributes

You can specify the following optional attribute when deleting a connection profile:

- **force**

#### Example: Delete a Connection Profile

This example deletes the connection profile conn1.

```
delete_connectionprofile: conn1
```

### insert\_connectionprofile Subcommand -- Add a Connection Profile

The insert\_connectionprofile subcommand adds a new connection profile to the database. After you create a connection profile, you can assign it to a Hadoop (HDFS, Hive, Oozie, Sqoop, or Pig) job to connect to the Hadoop cluster, Oozie server, Hive database, or Scoop database.

#### NOTE

- You must define the connection profile using the insert\_connectionprofile subcommand before you assign it to a Hadoop job.
- If AutoSys Workload Automation is running in external security mode using CA EEM, you must have CREATE authority for the corresponding as-connectionprofile resource class to define a connection profile.

#### Syntax

This subcommand has the following format:

```
insert_connectionprofile: conn_name
```

- **conn\_name**  
Defines a unique name for the connection profile.  
**Limits:** Up to 64 alphanumeric characters; the value cannot contain embedded blanks or tabs.

#### Required Attributes

To define a connection profile, you must specify the following attributes following the insert\_connectionprofile subcommand:

- connectionprofile\_type
- hadoop\_edgenode

To define a connection profile to connect to the Oozie server, you must also specify the following attribute following the insert\_connectionprofile subcommand:

- hadoop\_oozie\_server

To define a connection profile to connect to the Hive database, you must also specify the following attribute following the insert\_connectionprofile subcommand:

- hadoop\_hive\_jdbcurl

To define a connection profile to connect to the Sqoop database, you must also specify the following attribute following the insert\_connectionprofile subcommand:

- hadoop\_sqoop\_jdbcurl

### Optional Attributes

You can specify the following optional attributes when defining a connection profile:

- description
- hadoop\_edgenode\_port
- hadoop\_binpath
- hadoop\_hdfs\_namenode
- hadoop\_hive\_driverclass
- hadoop\_sqoop\_driverclass

### Example: Add a Connection Profile

This example creates a connection profile *conn1* that includes the parameters required to connect to the Hadoop cluster, Oozie server, Hive database, or the Sqoop database. You can specify this connection profile in your job definition when you define a HDFS, Hive, Sqoop, Pig, or Oozie job.

```
insert_connectionprofile: conn1
connectionprofile_type: HADOOP
hadoop_edgenode: HADOOP_EDGE_NODE
hadoop_edgenode_port: 22
hadoop_binpath: /usr/bin
hadoop_hdfs_namenode: "hdfs://HADOOP_EDGE_NODE:8020"
hadoop_oozie_server_url: "http://HADOOP_EDGE_NODE:11000/oozie"
hadoop_hive_jdbcurl: "jdbc:hive2://HADOOP_EDGE_NODE:10000"
hadoop_hive_driverclass: org.apache.hive.jdbc.HiveDriver
hadoop_sqoop_jdbcurl: "jdbc:sqlserver://
DATABASE_SERVER_MACHINE:1433;databaseName=SAMPLE"
hadoop_sqoop_driverclass: com.microsoft.sqlserver.jdbc.SQLServerDriver
```

### update\_connectionprofile Subcommand -- Update a Connection Profile

The update\_connectionprofile subcommand updates an existing connection profile in the database. The update\_connectionprofile subcommand is followed by a list of attribute:value statements. Attributes in the existing connection profile that are not explicitly replaced in the update\_connectionprofile input retain their original settings.

**NOTE**

- If AutoSys Workload Automation is running in external security mode using CA EEM, you must have WRITE authority for the corresponding as-connectionprofile resource class to update a connection profile.
- When you update a connection profile, you cannot update the connectionprofile\_type attribute. If you want to update the connectionprofile\_type, delete the connection profile and redefine it.

**Syntax**

This subcommand has the following format:

```
update_connectionprofile: conn_name
```

- ***conn\_name***

Specifies the name of the connection profile to update.

**Limits:** Up to 64 alphanumeric characters; the value cannot contain embedded blanks or tabs.

**Optional Attributes**

You can specify the following optional attributes when updating a connection profile:

- description
- hadoop\_edgenode
- hadoop\_oozie\_server\_url
- hadoop\_hive\_jdbcurl
- hadoop\_sqoop\_jdbcurl
- hadoop\_edgenode\_port
- hadoop\_binpath
- hadoop\_hdfs\_namenode
- hadoop\_hive\_driverclass
- hadoop\_sqoop\_driverclass

**Example: Update a Connection Profile**

This example updates the path to the Hadoop client binaries on the Hadoop edge node in the *conn1* connection profile.

```
update_connectionprofile: conn1
hadoop_binpath: /user/root/bin
```

**connectionprofile\_type Attribute - Specify the Connection Profile Type**

The connectionprofile\_type attribute specifies the type of connection profile you are defining.

**Supported JIL Subcommand**

This attribute is required for the insert\_connectionprofile sub command.

**Syntax**

This attribute has the following format:

```
connectionprofile_type: HADOOP
```

- **HADOOP**

Specifies a Hadoop connection profile. You can use this connection profile to connect to the Hadoop cluster, Oozie server, Hive database, or Scoop database.

**Example: Add a Connection Profile**

This example creates a connection profile *conn2* that includes the parameters required to connect to the Hadoop cluster. You can specify this connection profile in your job definition when you define a HDFS job.

```
insert_connectionprofile: conn2
connectionprofile_type: HADOOP
hadoop_edgenode: HADOOP
```

#### NOTE

AutoSys Workload Automation web services uses 501 numeric code to represent the "HADOOP" connection profile type.

## connectionprofile Attribute - Specify the Name of the Connection Profile

The connectionprofile attribute specifies the name of a connection profile.

### Supported Job Type

This attribute is required for HDFS, Oozie, Hive, Sqoop, and Pig job types.

### Syntax

This attribute has the following format:

```
connectionprofile: conn_name
```

- ***conn\_name*** Specifies a unique name for the connection profile.  
**Limits:** Up to 64 alphanumeric characters; the value cannot contain embedded blanks or tabs.

### Example: Create a File

This example creates a zero-byte file named *file1*. *The attributes that are required for connecting to the Hadoop environment are specified in the conn1 connection profile*

```
insert_job: HDFS_TOUCH_JOB
job_type: HDFS
machine: HADOOPAI
hadoop_hdfs_type: TOUCH
owner: user1@HADOOP_EDGE_NODE
hadoop_hdfs_path: /tmp/file1
connectionprofile: conn1
```

## description Attribute -- Specify a Description for the Connection Profile

The description attribute specifies a description for the connection profile you are defining.

### Supported JIL Subcommands

This attribute is optional for the insert\_connectionprofile and update\_connectionprofile subcommands.

### Syntax

This attribute has the following format:

```
description: text
```

- ***text***  
Specifies a description for the connection profile.  
**Limits:** Up to 256 characters

**NOTE**

If the description contains a colon (:), precede the colon with a backslash (\) or enclose the description with double quotation marks ("").

## JIL Resource Definitions

This section describes the JIL subcommands and attributes that you can use to define, update, and delete resources. To define, update, or delete a resource, you specify the appropriate JIL subcommand and attribute statements to the jil command.

### delete\_resource Subcommand -- Delete a Virtual Resource

The delete\_resource subcommand deletes a virtual resource definition from the database.

**NOTE**

You cannot delete a virtual resource if it is referenced as a dependency in a job. To delete the resource, you must delete all the related job dependencies first. If a resource is deleted while a job that references it as a dependency is active or running, the job continues to run and is not affected.

**Syntax**

This subcommand has the following format:

```
delete_resource: resource_name
```

- **resource\_name**  
Specifies the name of the virtual resource that you want to delete.

**Optional Attribute**

To delete a virtual resource, you can specify the following optional attribute following the delete\_resource subcommand:

- machine

**Example: Delete a Global Virtual Resource**

This example deletes the global virtual resource named glob\_resource.

```
delete_resource: glob_resource
```

**Example: Delete a Machine-Level Virtual Resource**

This example deletes the virtual resource named mach\_resource that is associated with the winagent machine.

```
delete_resource: mach_resource
machine: winagent
```

### insert\_resource Subcommand -- Define a Virtual Resource

The insert\_resource subcommand adds a virtual resource definition to the database. A virtual resource is representational only and is not directly tied to a physical system. Using virtual resources helps you control job execution and to improve your environment's performance.

For example, you can define a virtual resource to represent the maximum number of floating product licenses available in your enterprise. Each time a qualified job runs, a unit of that resource is used. When all the units are used, no more jobs can run.

**NOTE**

You can define resources on distributed machines only. Before you can define a resource on a machine, the machine must already be defined on the database.

**Syntax**

This subcommand has the following format:

```
insert_resource: resource_name
```

- **resource\_name**

Defines the name of the virtual resource.

**Limits:** Up to 128 characters; can include a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); cannot include embedded spaces or tabs

**NOTE**

The virtual resource name must be unique across all resource types.

**NOTE**

- You cannot define the same virtual resource at the machine level and global level.
- You can define the same virtual resource on multiple machines.

**Required Attributes**

To define a virtual resource, you must specify the following attributes following the insert\_resource subcommand:

- [amount](#)
- [res\\_type](#)

**Optional Attributes**

You can specify the following optional attributes when defining a virtual resource:

- [description](#)
- [machine](#)

**Example: Define a Virtual Resource at the Machine Level**

This example defines a renewable resource named renew\_res1 for the unixagent1 machine. The resource is assigned 10 units.

```
insert_resource: renew_res1
res_type: R
machine: unixagent1
amount: 10
```

Suppose that you issue the following subcommand after adding the previous resource. The machine attribute is not specified, so the subcommand tries to add a renewable resource also named renew\_res1 at the global level. Machine-level and global-level resources cannot have the same name, so the subcommand fails.

```
insert_resource: renew_res1
res_type: R
amount: 25
```

**Example: Define the Same Virtual Resource on Two Machines**

This example defines renewable virtual resources on machine1 and machine2 with the same name. The resources are successfully created because you can define the same resource on multiple machines if the res\_type value is also the same.

```
insert_resource: renew_count
machine: machine1
res_type: R
amount: 10
```

```
insert_resource: renew_count
machine: machine2
res_type: R
amount: 20
```

Suppose that the following job is defined to run on a virtual machine named `virtmach`, which contains `machine1` and `machine2`. The `renew_count` resource is defined as a dependency, as follows:

```
insert_job: res_dep_job
job_type: CMD
machine: virtmach
command: /u1/procrun.sh
resources: (renew_count, QUANTITY=5, free=Y)
```

AutoSys Workload Automation searches both `machine1` and `machine2`, chooses the "best" machine that satisfies the job's resource dependency, and runs the job on that machine. If both machines satisfy the virtual resource requirements, the job runs on the machine with the most available CPU resources.

For example, if both `machine1` and `machine2` satisfy the virtual resource requirements and have equal amount of CPU resources available, the scheduler randomly selects a machine and runs the job on the selected machine. If both `machine1` and `machine2` satisfy the virtual resource requirements and `machine1` has less amount of CPU resource available, the scheduler runs the job on `machine2`.

#### NOTE

A job can have both virtual and real resource dependencies. In this scenario, the resource dependencies are evaluated as follows:

- AutoSys Workload Automation gathers machines that satisfy the virtual resource dependencies.
- The job runs on that machine.
- If AutoSys Workload Automation is not integrated with CA Automation Suite for Data Centers, the real resource dependencies are ignored and the job is submitted on the machine that satisfies the virtual resource requirements and has the most available CPU resources.

## update\_resource Subcommand -- Update a Virtual Resource

The `update_resource` subcommand updates the amount and description properties of a virtual resource definition in the database.

#### NOTE

You cannot update resource types or machine names using the `update_resource` subcommand. To update resource types or machine names, you must delete the resource and add it to database again with the new properties.

### Syntax

This subcommand has the following format:

```
update_resource: resource_name
```

- ***resource\_name***

Defines the name of the virtual resource that you want to update. This resource must be defined in the database.

### Optional Attributes

To update a virtual resource, you can specify the following optional attributes following the `update_resource` subcommand:

- [amount](#)
- [description](#)

To update a virtual resource that is associated with a machine, you must specify the following attribute:

- `machine`

### Example: Update a Virtual Resource that is Associated with a Machine

This example updates a virtual resource that is associated with the `unixagent` machine. The number of units is changed to 35.

```
update_resource: mach_res
machine: unixagent
amount: 35
```

Suppose that the `amount` attribute is defined as follows:

```
amount: +20
```

In this situation, the command adds 20 units to the available resource count. For example, if the resource already has 35 units, the command adds 20 units and the total would be 55.

Similarly, suppose that the `amount` attribute is defined as follows:

```
amount: -20
```

If the resource has 35 units, the command removes 20 units and the total would be 15.

### Example: Update a Global Virtual Resource

This example defines a global virtual resource and assigns 10 units to it.

```
insert_resource: glob_res
res_type: D
amount: 10
```

This following command updates the same resource and assigns 25 units to it:

```
update_resource: glob_res
amount: 25
```

## amount Attribute -- Specify the Amount of Virtual Resource

The `amount` attribute specifies the number of units to assign to a virtual resource.

### Supported JIL Subcommands

This attribute is required for the [insert\\_resource](#) subcommand.

This attribute is optional for the [update\\_resource](#) subcommand.

### Syntax

This attribute has the following format:



amount: *value*

- **value**  
Defines the number of units to assign to the virtual resource.  
**Limits:** Integer; up to 2000000000

#### NOTE

If you are updating an existing resource, you can specify an absolute number or a relative number (for example, +3 or -5). Specifying a relative number lets you increase or decrease the amount without assigning a specific number. The amount cannot be decremented to a negative value.

### Example: Update a Global Virtual Resource Using an Absolute Amount

This example assigns 30 units to the count\_res resource. The machine attribute is not specified, so the count\_res resource must be a global resource.

```
update_resource: count_res
amount: 30
```

### Example: Update a Machine-Level Virtual Resource Using a Relative Amount

This example updates the win\_res resource that is associated with the winagent machine. The amount is increased by 5 units.

```
update_resource: win_res
machine: winagent
amount: +5
```

## description Attribute -- Specify a Description for a Virtual Resource

The amount attribute specifies a description for a virtual resource.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_resource](#)
- [update\\_resource](#)

### Syntax

This attribute has the following format:

```
description: "text"
```

- **"text"**  
Defines a free-form text description for the virtual resource.  
**Limits:** Up to 256 alphanumeric characters (including spaces)

### Example: Specify a Description for a Virtual Resource

This example defines a new resource that includes a description.

```
insert_resource: renew_res
machine: winagent
res_type: R
amount: 10
description: "This renewable resource is only available for the winagent machine."
```

You can issue the following `update_resource` subcommand to change the amount and description:

```
update_resource: renew_res
machine: winagent
amount: 25
description: "This renewable resource is only available for the winagent machine. The
number of units is 25."
```

## machine Attribute -- Specify the Machine for a Virtual Resource

The machine attribute specifies the machine name that a virtual resource is associated with. A resource defined for a specific machine is only available to jobs submitted to that machine.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- `delete_resource`
- `insert_resource`
- `update_resource`

#### NOTE

- If you omit the machine attribute from a new virtual resource definition, the resource is added as a global resource that is available to all machines.
- You cannot update the machine name by specifying the machine attribute with the `update_resource` subcommand. To update a machine name, you must delete the resource and add it to the database again with the same name.
- You can specify the machine attribute with the `update_resource` or `delete_resource` subcommands to identify the virtual resource you want to update or delete.

### Syntax

This attribute has the following format:

```
machine: machine_name
```

- ***machine\_name***

Specifies the name of the machine that the virtual resource is defined for.

**Limits:** Up to 80 characters; can include a-z, A-Z, 0-9, period (.), underscore (\_), hyphen (-), and pound (#); cannot include embedded spaces or tabs

#### NOTE

- The machine must already be defined to AutoSys Workload Automation.
- The type of the specified machine cannot be v, w, or p. You can define virtual resources for any machine defined on AutoSys Workload Automation.
- The specified machine cannot be a virtual machine or real machine pool.

### Example: Define a Global Virtual Resource

This example defines a virtual depletable resource named `glob_res`. The machine attribute is not specified, so the resource is available to all machines.

```
insert_resource: glob_res
res_type: D
amount: 50
```

```
description: "This resource is permanently consumed."
```

### Example: Update a Resource Amount

Suppose that `threshold_res` is an existing resource on the `prodserv` machine and you issue the following `update_resource` subcommand:

```
update_resource: threshold_res
amount: 25
```

The machine attribute is not specified, so the subcommand tries to update a global resource named `threshold_res`, which does not exist. The subcommand fails.

To update the `threshold_res` resource, you must include the machine attribute in the `update_resource` subcommand as follows:

```
update_resource: threshold_res
machine: prodserv
amount: 25
```

## res\_type Attribute -- Specify the Type of Virtual Resource

The `res_type` attribute specifies the type of virtual resource.

### Supported JIL Subcommand

This attribute is required for the [insert\\_resource subcommand](#).

### Syntax

This attribute has the following format:

```
res_type: D | R | T
```

- **D**  
Defines a depletable resource. A depletable resource is a consumed resource. When a job that uses this resource is submitted, the used resource units are permanently removed from the available resource pool. If you want to use the depletable resource again in a job, you must use the `update_resource` subcommand to replenish the resource units back to the resource pool.
- **R**  
Defines a renewable resource. A renewable resource is a borrowed resource. When a job that uses this resource is submitted, the used resource units are temporarily removed from the available resource pool. When the job completes, the resource units are returned to the pool, or the units are held until they are manually released back to the pool. You specify whether the units are returned or not when you define the resource dependency in a job.
- **T**  
Defines a threshold resource. A threshold resource is a sizing resource. Only the jobs that require the specified resource amount or fewer are submitted to run. For example, if the threshold resource is set to 2, AutoSys Workload Automation submits the jobs that require 2 or fewer units. The used resource units are not removed from the resource pool.

### NOTE

You cannot update a resource's type using the `update_resource` subcommand. To update a resource's type, you must delete the resource and add it to database again with the new type.

### Example: Define a Global Renewable Resource

This example defines a depletable resource that is available to all machines. The resource is assigned 10 units.

```
insert_resource: renewable_res
res_type: R
amount: 10
```

Suppose that the `renewable_res` resource is defined as a dependency to the following job:

```
insert_job: res_dep_job
job_type: CMD
machine: unixagent
command: /u1/procrun.sh
resources: (renewable_res, QUANTITY=5, FREE=Y)
```

When this job is submitted, five resource units are temporarily removed from the resource pool. If the job completes successfully, the resource units are returned to the pool. If the job does not complete successfully, the units are held until they are manually released back to the resource pool.

### Example: Define a Machine-Level Depletable Resource

This example defines a depletable resource for the `winagent` machine. The resource is assigned 25 units. Units that are used by jobs are permanently removed from the available resource pool.

```
insert_resource: depletable_res
res_type: D
machine: winagent
amount: 25
```

### Example: Define a Machine-Level Threshold Resource

This example defines a threshold resource for the `unixagent` machine. The resource is assigned 10 units, so only jobs that require 10 or fewer units of this resource are submitted to the `unixagent` machine.

```
insert_resource: threshold_res
res_type: T
machine: unixagent
amount: 10
```

## JIL Cross-Instance Definitions

This section describes the JIL subcommands and attributes that you can use to define, update, and delete external instances that AutoSys Workload Automation can communicate with. An external instance can be another AutoSys Workload Automation instance or it can be a CA Workload Automation product running on another platform, including mainframe. To define, update, or delete an external instance, you specify the appropriate JIL subcommand and attribute statements to the `jil` command.

### delete\_xinst Subcommand -- Delete an External Instance

The `delete_xinst` subcommand deletes the specified external instance from the AutoSys Workload Automation database.

#### Syntax

This subcommand has the following format:

```
delete_xinst: external_instance
```

- ***external\_instance***  
Identifies the external instance to delete from the AutoSys Workload Automation database.

**Limits:**

Must be three alphanumeric characters  
; uppercase

**Example: Delete an External Instance**

This example deletes the external instance ACE.

```
delete_xinst: ACE
```

**insert\_xinst Subcommand -- Define an External Instance**

The insert\_xinst subcommand defines a new external instance in the AutoSys Workload Automation database. The external instance can be one of the following Broadcom scheduling managers:

Scheduling Manager	Required Integration Software	Environment
CA Job Management Option	CA UJMA	Distributed
CA Jobtrac Job Management	AutoSys WA Connect Option or CA UJMA	Mainframe
CA Scheduler Job Management	AutoSys WA Connect Option or CA UJMA	Mainframe
AutoSys Workload Automation	None	Distributed
CA Workload Automation EE	None	Mainframe
CA Workload Automation SE	AutoSys WA Connect Option or CA UJMA	Mainframe

**Syntax**

This subcommand has the following format:

```
insert_xinst: external_instance
```

- ***external\_instance***

Defines the unique name of the external instance to add.

**Limits:**

Must be three alphanumeric characters

**NOTE**

Values specified in lowercase characters are converted to uppercase.

**Required Attributes**

To define an external instance, you must specify the following attributes following the insert\_xinst subcommand:

- [xtype](#)
- [xmachine](#)

If you are defining an external AutoSys Workload Automation instance, the following attribute is also required:

- [xport](#)

If you are defining an external CA Workload Automation EE instance, the following attributes are also required:

- [xmanager](#)
- [xport](#)

**Optional Attributes**

You can specify the following optional attributes when defining an external AutoSys Workload Automation or CA Workload Automation EE instance:

- [xcrypt\\_type](#)
- [xkey\\_to\\_manager](#)

**NOTE**

If you are defining an external AutoSys Workload Automation instance and it uses a shadow scheduler, you must add the `xmachine` and `xport` attributes for that shadow scheduler. You can also optionally add the `xcrypt_type` and `xkey_to_manager` attributes.

**Example: Define an External AutoSys Workload Automation Instance**

This example defines an external AutoSys Workload Automation application server instance named ACE to run on machineA and port 9000 using the default encryption.

```
insert_xinst: ACE
xtype: a
xmachine: machineA
xport: 9000
```

**Example: Define Multiple External AutoSys Workload Automation Instances**

This example defines multiple external AutoSys Workload Automation application server external instances named ACE to run on machineA and machineB at port 9000.

```
insert_xinst: ACE
xtype: a
xmachine: machineA,machineB
xport: 9000
```

**Example: Define an External CA AutoSys WA Connect Option Instance**

This example defines an external CA AutoSys WA Connect Option instance named CA7 to run on the machine called MAINFR1.

```
insert_xinst: CA7
xtype: c
xmachine: MAINFR1
```

**Example: Define a CA Job Management Option External Instance**

This example defines an external CA Job Management Option instance named JMO to run on the machine called unimachA.

```
insert_xinst: JMO
xtype: u
xmachine: unimachA
```

**update\_xinst Subcommand -- Update an External Instance Definition**

The `update_xinst` subcommand updates an existing external instance definition in the AutoSys Workload Automation database. You can update the external instance definition when the connection information changes. The connection information must be accurate so that the scheduling managers can communicate with each other.

**Syntax** This subcommand has the following format:

```
update_xinst: external_instance
```

- ***external\_instance***

Identifies the external instance definition to update.

**Limits:** Must be three alphanumeric characters; uppercase

### Optional Attributes

To update an external instance, you can specify the following attribute following the `update_xinst_subcommand`:

- `xmachine`

To update an external AutoSys Workload Automation instance, you can specify the following attributes:

- `xport`
- `xcrypt_type`
- `xkey_to_manager`

To update an external CA Workload Automation EE instance, you can specify the following attributes:

- `xmanager`
- `xport`
- `xcrypt_type`
- `xkey_to_manager`

### `xcrypt_type` Attribute -- Specify the Encryption Type

The `xcrypt_type` attribute specifies the encryption type to use.

#### NOTE

- This attribute only applies to AutoSys Workload Automation (xtype: a) and CA Workload Automation EE (xtype: e) instances.
- If you do not specify this attribute, the instance is defined with the default encryption type for AutoSys Workload Automation or no encryption for CA Workload Automation EE.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- `insert_xinst`
- `update_xinst`

### Syntax

This attribute has the following formats:

- To specify the encryption type for AutoSys Workload Automation:  
`xcrypt_type: NONE | DEFAULT | AES`
- To specify the encryption type for CA Workload Automation EE:  
`xcrypt_type: NONE | AES`
- **DEFAULT**  
Specifies that the default encryption key and type are used. This is the default for AutoSys Workload Automation instances (xtype: a).
- **NONE**  
Specifies that no encryption is used. This is the default for CA Workload Automation EE instances (xtype: e).
- **AES**  
Indicates that the data is encrypted using AES 128-bit encryption. If you are using AES 128-bit encryption to encrypt the data, you must generate the instance-wide communication alias encryption file (`cryptkey_alias.txt`) using the `as-config` command.

**NOTE**

When `xcrypt_type` is set to AES, you must also specify the `xkey_to_manager` attribute.

**NOTE**

On CA Workload Automation EE, you must specify the same AutoSys Workload Automation encryption setting in the AGENTDEF data set. Therefore, to use encryption, AutoSys Workload Automation and the external instance must support the same encryption type.

## **xkey\_to\_manager Attribute -- Specify an Encryption Key for Manager Communication**

The `xkey_to_manager` attribute specifies the encryption key used to encrypt data sent to the scheduling manager.

**NOTE**

- This attribute only applies to AutoSys Workload Automation and CA Workload Automation EE instances. Therefore, you must specify the `xtype: a` or `xtype: e` attribute.
- To use this attribute, you must specify AES for the `xcrypt_type` attribute.

### **Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_xinst](#)
- [update\\_xinst](#)

### **Syntax**

This attribute has the following format:

```
xkey_to_manager: key
```

- **key**

Specifies the key used to encrypt data between AutoSys Workload Automation and the external instance. You must specify one of the following:

- A 32-digit hexadecimal key
- A passphrase with up to 16 characters (valid only for `xtype: a`)

**NOTE**

For CA Workload Automation EE (`xtype: e`), specify the encryption key defined by ENCRYPT KEYNAME (keyname) in the CA Workload Automation EE AGENTDEF data set. You must prefix the hexadecimal identifier 0x to this value, as shown in the following example:

```
xkey_to_manager: 0x0123456789ABCDEF0123456789ABCDEF
```

## **xmachine Attribute -- Define Connection Information for an External Instance**

The `xmachine` attribute defines connection information for an external instance.

### **Supported JIL Subcommands**

This attribute is required for the [insert\\_xinst](#) subcommand.

This attribute is optional for the [update\\_xinst](#) subcommand.

### **Syntax**

This attribute has the following format:

```
xmachine: ext_instance_node_names
```



- ***ext\_instance\_node\_names***

Defines the external instance's node names to be used for communication as follows:

- If the xtype attribute is set to **a** (an external AutoSys Workload Automation instance), specify the application server machine name. To specify multiple application servers, separate each machine name with a comma.
- If the xtype attribute is set to **c** (a scheduling manager with CA AutoSys WA Connect Option) or **u** (a scheduling manager with CA UJMA), specify the machine name.
- If the xtype attribute value is **e** (CA Workload Automation EE), specify the host name of CA Workload Automation EE.

**Limits:** Up to 512 characters

### Example: Specify a Machine Name for an External Instance

This example defines an external AutoSys Workload Automation instance named ACE. The instance's machine name is machineB. The instance communicates using port 9100 and default encryption.

```
insert_xinst: ACE
xtype: a
xmachine: machineB
xport: 9100
xcrypt_type: DEFAULT
```

## xmanager Attribute -- Specify the Alias Name of a CA Workload Automation EE Instance

The xmanager attribute specifies the alias name of the CA Workload Automation EE manager.

### NOTE

This attribute only applies to CA Workload Automation EE instances. To use this attribute, you must specify **e** for the xtype attribute.

### Supported JIL Subcommands

This attribute is required for the [insert\\_xinst](#) subcommand if the xtype attribute is set to **e**.

This attribute is optional for the [update\\_xinst](#) subcommand.

### Syntax

This attribute has the following format:

```
xmanager: manager_name
```

- ***manager\_name***

Specifies the alias name of the CA Workload Automation EE manager.

**Limits:** Up to 16 characters

### NOTE

You cannot use the same alias name for multiple CA Workload Automation EE managers. If AutoSys Workload Automation detects another instance with the same alias name, you will get an error.

## xport Attribute -- Specify the Port Number of the External Instance

The xport attribute specifies the port number of the external instance.

### NOTE

This attribute only applies to AutoSys Workload Automation and CA Workload Automation EE instances. To use this attribute, you must specify **a** or **e** for the xtype attribute.

## Supported JIL Subcommands

This attribute is required for the [insert\\_xinst](#) subcommand if the xtype attribute is set to a or e.

This attribute is optional for [update\\_xinst](#) subcommand.

### Syntax

This attribute has the following format:

```
xport: port
```

- **port**  
Specifies the port number of the external instance.  
**Limits:** 1-65535

## xtype Attribute -- Specify the External Instance Type

The xtype attribute specifies the type of external instance.

### Supported JIL Subcommands

This attribute is required for the [insert\\_xinst](#) subcommand.

### Syntax

This attribute has the following format:

```
xtype: ext_instance_type
```

- **ext\_instance\_type**  
Specifies the external instance type. Options in the following:
  - a -- Identifies a remote AutoSys Workload Automation application server instance.
  - c -- Identifies a CA AutoSys Workload Automation Connect Option instance.
  - u -- Identifies a CA Universal Job Management Agent or CA NSM instance.
  - e -- Identifies a CA Workload Automation EE instance.

### Example: Define an External AutoSys Workload Automation Instance

This example defines an external AutoSys Workload Automation application server instance named ACE to run on machineA and port 9000 using default encryption:

```
insert_xinst: ACE
xtype: a
xmachine: machineA
xport: 9000
```

## JIL Monitor and Report Definitions

This section describes the JIL subcommands and attributes that you can use to monitor and generate reports about AutoSys Workload Automation jobs. To define, update, or delete a monitor or report, you specify the appropriate JIL subcommand and attribute statements to the jil command.

### delete\_monbro Subcommand -- Delete a Monitor or Report

The delete\_monbro subcommand deletes the specified monitor or report from the AutoSys Workload Automation database.

## Syntax

This subcommand has the following format:

```
delete_monbro: monbro_name
```

### ***monbro\_name***

Identifies an existing monitor or report to delete from the AutoSys Workload Automation database.

**Limits:** Up to 30 alphanumeric characters

### **Example: Delete a Monitor**

This example deletes the monitor track\_alarm:

```
delete_monbro: track_alarm
```

## insert\_monbro Subcommand -- Define a Monitor or Report

The insert\_monbro subcommand defines a new monitor or report. Monitors are used to monitor AutoSys Workload Automation events and to watch for specific occurrences, such as alarms. Reports are used to filter and report AutoSys Workload Automation events.

To use a monitor or report, you must define it and run it. Defining it alone has no effect.

## Syntax

This subcommand has the following format:

```
insert_monbro: monbro_name
```

### ***monbro\_name***

Defines a unique name for the monitor or report to define.

**Limits:** Up to 30 alphanumeric characters; the value can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); the value cannot contain embedded blanks or tabs

### **Required Attributes**

To define a monitor or report, you must specify the following attribute following the insert\_monbro subcommand:

- [mode](#)

### **Optional Attributes**

You can specify the following optional attributes when defining a monitor or report:

- [alarm](#)
- [all\\_events](#)
- [all\\_status](#)
- [failure](#)
- [job\\_filter](#)
- [job\\_name](#) (required to monitor or report a single box or job)
- [restart](#)
- [running](#)
- [starting](#)
- [success](#)
- [terminated](#)

If you are defining a monitor, you can specify the following additional optional attribute:

- [alarm\\_verif](#)

If you are defining a report, you can specify the following additional optional attributes:

- [after\\_time](#)
- [currn](#)

### Example: Define a Report

This example defines the report `success_report` that browses all jobs for success in the current or most recent run of the job:

```
insert_monbro: success_report
mode: b /* "browser" can also be specified */
success: y
job_filter: a /* the default */
currn: y /* the default */
```

### Example: Define a Monitor

This example defines the monitor `alarm_monitor` to watch for alarms on all jobs.

```
insert_monbro: alarm_monitor
mode: m /* "monitor" can also be specified */
alarm: y
job_filter: a /* the default */
```

## update\_monbro Subcommand -- Update a Monitor or Report

The `update_monbro` subcommand updates an existing monitor or report.

Attributes in the existing monitor or report definition that are not explicitly replaced by attributes in the `update_monbro` input retain their original settings after the update. When you need to reset many attributes, it may be more efficient to delete and redefine the monitor or report.

### Syntax

This subcommand has the following format:

```
update_monbro: monbro_name
```

#### ***monbro\_name***

Identifies an existing monitor or report to delete from the AutoSys Workload Automation database.

**Limits:** Up to 30 alphanumeric characters

### Optional Attributes

You can specify the following optional attributes when updating a monitor or report:

- [alarm](#)
- [all\\_events](#)
- [all\\_status](#)
- [failure](#)
- [job\\_filter](#)
- [job\\_name](#)
- [mode](#)
- [restart](#)
- [running](#)
- [starting](#)
- [success](#)
- [terminated](#)

If you are updating a monitor, you can specify the following additional optional attribute:

- [alarm\\_verif](#)

If you are updating a report, you can specify the following additional optional attributes:

- [after\\_time](#)
- [currun](#)

### Example: Update a Monitor

This example changes the monitor `alarm_monitor`, so that it tracks all occurrences of jobs changing to `TERMINATED` status in addition to the alarms already specified in the monitor definition:

```
update_monbro: alarm_monitor
terminated: y
```

## after\_time Attribute -- Specify the Report Start Date and Time

The `after_time` report attribute specifies the start date and time for the report you are defining. Only events that occur after the specified date and time are reported.

### Supported JIL Subcommands

This attribute is optional for the following subcommands (report definitions only):

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
after_time: "date_time"
```

### date\_time

Defines the reporting start date in the format "`MM/DD/[YY]YY hh:mm`", where *MM* is the month, *DD* is the day, *[YY]YY* is the year, *hh* is the hour (in 24-hour format), and *mm* is the minutes. You must enclose the `date_time` value in quotation marks ("").

**Default:** When the `currun` attribute is set to **no**, the `after_time` attribute defaults to 12:00 a.m. on the specified day. When the `currun` attribute is set to **yes**, the `after_time` attribute is ignored.

When you omit the date, the `after_time` attribute defaults to the current day.

**NOTE**

When you enter a two-digit year, AutoSys Workload Automation saves the setting to the database as a four-digit year. If you enter 79 or less, the product precedes your entry with 20; if you enter 80 or greater, the product precedes your entry with 19.

**Example: Specify the Start Data and Time for a Report**

This example report all events that occur after 2:00 p.m. on October 1, 2005:

```
after_time: "10/01/2005 14:00"
```

**alarm Attribute -- Specify Whether to Track Alarms**

The alarm monitor/report attribute specifies whether to track alarms for the specified jobs in the monitor or report you are defining. You can track alarms and other events in the same monitor or report.

**NOTE**

When you define the all\_events attribute, all alarms are tracked, regardless of the alarm attribute setting.

**Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

**Syntax**

This attribute has the following format:

```
alarm: y | n
```

- **y**  
Tracks alarms in the monitor or report.  
**Note:** You can specify 1 instead of y.
- **n**  
Does not track alarms in the monitor or report. This is the default.  
**Note:** You can specify 0 instead of n.

**Example: Set the Monitor or Report to Track Alarms**

This example sets the monitor or report to track alarms:

```
alarm: y
```

**alarm\_verif Attribute -- Specify Whether the Monitor Requires an Operator Response to Alarms**

The alarm\_verif monitor attribute specifies whether the monitor you are defining requires an operator response to alarm events. When the monitor detects an alarm event, it prompts the operator for a response and repeats the prompt every 20 seconds until the operator responds. When the operator responds (typically by entering a comment at the prompt), the monitor time stamps the response and records it in the database along with the alarm event.

**Supported JIL Subcommands**

This attribute is optional for the following subcommands (monitor definitions only):

- [insert\\_monbro](#)
- [update\\_monbro](#)

## Syntax

This attribute has the following format:

```
alarm_verif: y | n
```

### y

Issues alarms until an operator responds.

#### NOTE

You can specify 1 instead of y.

### n

Specifies that the monitor does not require operator verification of alarms. This is the default.

#### NOTE

You can specify 0 instead of n.

## Example: Set the Monitor to Require Operator Verification of Alarms

This example sets the monitor to require operator verification of alarms:

```
alarm_verif: y
```

## all\_events Attribute -- Specify Whether to Track all Events for a Job

The all\_events monitor/report attribute specifies whether to track all events for the specified jobs in the monitor or report you are defining. This attribute specifies whether any event filtering is in effect.

When you set this attribute to **y** or **1**, the product ignores other event filtering attributes reports all events (including job status events, alarms, and manually generated events such as when you start a job), regardless of source, for the specified jobs.

When you set this attribute to **n** or **0**, the product uses other event filtering attributes (such as alarm, currun, failure, and so on) to select the events to track for the selected jobs.

#### NOTE

To monitor all events for all jobs, use the following command to display the scheduler log in real time instead of running a monitor:

```
autosyslog -e
```

#### NOTE

Running a monitor adds another connection to the database and establishes a process that continually polls the database. This can significantly impact system performance. Using the scheduler log to monitor all jobs is more efficient and provides more diagnostic information than a monitor.

## Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

## Syntax

This attribute has the following format:

```
all_events: y | n
```

**y**

Tracks all events for the specified jobs.

**Note:** You can specify 1 instead of y.

**n**

Does not track all events for the specified jobs. This is the default.

**Note:** You can specify 0 instead of n.

**Example: Set the Monitor or Report to Track all Events**

This example sets the monitor or report to track all events for the specified jobs:

```
all_events: y
```

**all\_status Attribute -- Specify Whether to Track all Job Status Events**

The all\_status monitor/report attribute specifies whether to track all job status events for the specified jobs in the monitor or report you are defining. Job status events occur whenever a job's status changes.

When the all\_events attribute is set to **y** or **1**, the product ignores the all\_status attribute.

**Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

**Syntax**

This attribute has the following format:

```
all_status: y | n
```

**y**

Tracks all job status events for the specified jobs.

**Note:** You can specify 1 instead of y.

**n**

Does not track all job status events for the specified jobs. This is the default.

**Note:** You can specify 0 instead of n.

**Example: Set the Monitor or Report to Track all Job Status Events**

This example sets the monitor or report to track all job status events:

```
all_status: y
```

**currn Attribute -- Specify Whether to Report Events in the Current Run**

The currn report attribute specifies whether to report only the events in the current or most recent run of the specified jobs in the report you are defining.

When you set the currn attribute to **n** or **0**, you must define the after\_time attribute.

**Supported JIL Subcommands**

This attribute is optional for the following subcommands (report definitions only):



- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
currun: y | n
```

#### **y**

Reports events only for the current or most recent job run. This is the default.

**Note:** You can specify 1 instead of y.

#### **n**

Does not restrict reporting to events for the current or most recent job run.

**Note:** You can specify 0 instead of n.

### Example: Report Only Events in the Current Run

This example reports only events in the current or most recent run of the specified jobs:

```
currun: y
```

## failure Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to FAILURE

The failure monitor/report attribute specifies whether to track the job status event generated when a job changes to the FAILURE status.

The product ignores this attribute when either the all\_events or the all\_status attribute is set to **y** or **1**.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
failure: y | n
```

#### **y**

Tracks the job status event generated when a job enters the FAILURE status.

#### **NOTE**

You can specify 1 instead of y.

#### **n**

Does not track the job status event generated when a job enters the FAILURE status. This is the default.

#### **NOTE**

You can specify 0 instead of n.

- Example: Set Monitor or Report to Track Occurrences of Jobs Changing to FAILURE Status
- This example sets the monitor or report to track occurrences of jobs changing to FAILURE status:

```
failure: y
```

## job\_filter Attribute -- Specify the Jobs to Track

The `job_filter` monitor/report attribute specifies which jobs to track in the monitor or report you are defining. The combination of the event filters and the job filter determines which events to track.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
job_filter: a | b | j
```

#### **a**

Tracks all jobs (no job filtering). This is the default.

**Note:** You can specify **all** instead of **a**.

#### **b**

Tracks a single box and the jobs it contains. You must also define the `job_name` attribute to specify the name of the job to track.

**Note:** You can specify **box** instead of **b**.

#### **j**

Tracks a single job. You must also define the `job_name` attribute to specify the name of the job to track.

**Note:** You can specify **job** instead of **j**.

### Example: Set the Monitor or Report to Only Track Events Specified by job\_name Attribute

This example sets the monitor or report to only track events in the box specified by the `job_name` attribute:

```
job_filter: b
```

## job\_name Attribute -- Identify the Job for Which to Track Events

The `job_name` monitor/report attribute identifies the box or job for which the monitor or report you are defining should track events. The combined event filters, job filter, and job name determines which events to track. The `job_name` attribute is required when you set the `job_filter` attribute to **j** (a single job) or **b** (a box and its jobs); otherwise, it is ignored.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
job_name: name
```

**name**

Identifies the existing job or box job for which to track events.

**Limits:** Up to 64 alphanumeric characters; can include the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-); cannot include embedded spaces or tabs

#### Example: Track Events in a Box Job

This example only tracks events in the box job EOD\_Box (and the jobs it contains), set the job\_filter attribute to **b**:

```
job_name: EOD_Box
```

## mode Attribute -- Specify Whether to Define a Monitor or Report

The mode monitor/report attribute specifies whether to define a monitor or report.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
mode: monitor | browser
```

#### monitor

Defines a monitor.

**Note:** You can specify **m** instead of monitor.

#### browser

Defines a report.

**Note:** You can specify **b** instead of browser.

#### Example: Define a Report

This example defines a report:

```
mode: b
```

## restart Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to RESTART

The restart monitor/report attribute specifies whether to track the job status event generated when a job changes to the RESTART status.

The product ignores this attribute when either the all\_events or the all\_status attribute is set to **y** or **1**.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
restart: y | n
```

### **y**

Tracks the job status event generated when a job enters the RESTART status.

**Note:** You can specify 1 instead of y.

### **n**

Does not track the job status event generated when a job enters the RESTART status. This is the default.

**Note:** You can specify 0 instead of n.

### **Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to RESTART Status**

This example sets the monitor or report to track occurrences of jobs changing to RESTART status:

```
restart: y
```

## **running Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to RUNNING**

The running monitor/report attribute specifies whether to track the job status event generated when a job changes to the RUNNING status.

The product ignores this attribute when either the all\_events or the all\_status attribute is set to **y** or **1**.

### **Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### **Syntax**

This attribute has the following format:

```
running: y | n
```

### **y**

Tracks the job status event generated when a job enters the RUNNING status.

**Note:** You can specify 1 instead of y.

### **n**

Does not track the job status event generated when a job enters the RUNNING status. This is the default.

**Note:** You can specify 0 instead of n.

### **Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to RUNNING Status**

This example sets the monitor or report to track occurrences of jobs changing to RUNNING status:

```
running: y
```

## **starting Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to STARTING**

The starting monitor/report attribute specifies whether to track the job status event generated when a job changes to the STARTING status.

The product ignores this attribute when either the `all_events` or the `all_status` attribute is set to **y** or **1**.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
starting: y | n
```

#### **y**

Tracks the job status event generated when a job enters the STARTING status.

**Note:** You can specify 1 instead of y.

#### **n**

Does not track the job status event generated when a job enters the STARTING status. This is the default.

**Note:** You can specify 0 instead of n.

### Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to STARTING Status

This example sets the monitor or report to track occurrences of jobs changing to STARTING status:

```
starting: y
```

## success Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to SUCCESS

The success monitor/report attribute specifies whether to track the job status event generated when a job changes to the SUCCESS status.

The product ignores this attribute when either the `all_events` or the `all_status` attribute is set to **y** or **1**.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### Syntax

This attribute has the following format:

```
success: y | n
```

#### **y**

Tracks the job status event generated when a job enters the SUCCESS status.

**Note:** You can specify 1 instead of y.

#### **n**

Does not track the job status event generated when a job enters the SUCCESS status. This is the default.

**Note:** You can specify 0 instead of n.

### Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to SUCCESS Status

This example sets the monitor or report to track occurrences of jobs changing to SUCCESS status:

```
success: y
```

## **suspended Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to SUSPENDED**

The suspended monitor/report attribute specifies whether to track the job status event generated when a job changes to the SUSPENDED status.

The product ignores this attribute when either the all\_events or the all\_status attribute is set to **y** or **1**.

### **Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### **Syntax**

This attribute has the following format:

```
suspended: y | n
```

#### **y**

Tracks the job status event generated when a job enters the SUSPENDED status.

**Note:** You can specify 1 instead of y.

#### **n**

Does not track the job status event generated when a job enters the SUSPENDED status. This is the default.

**Note:** You can specify 0 instead of n.

### **Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to SUSPENDED Status**

This example sets the monitor or report to track occurrences of jobs changing to SUSPENDED status:

```
suspended: y
```

## **terminated Attribute -- Specify Whether to Track Job Events When a Jobs Status Changes to TERMINATED**

The terminated monitor/report attribute specifies whether to track the job status event generated when a job changes to the TERMINATED status.

The product ignores this attribute when either the all\_events or the all\_status attribute is set to **y** or **1**.

### **Supported JIL Subcommands**

This attribute is optional for the following subcommands:

- [insert\\_monbro](#)
- [update\\_monbro](#)

### **Syntax**

This attribute has the following format:

```
terminated: y | n
```

**y**

Tracks the job status event generated when a job enters the TERMINATED status.

**Note:** You can specify 1 instead of y.

**n**

Does not track the job status event generated when a job enters the TERMINATED status. This is the default.

**Note:** You can specify 0 instead of n.

**Example: Set the Monitor or Report to Track Occurrences of Jobs Changing to TERMINATED Status**

This example sets the monitor or report to track occurrences of jobs changing to TERMINATED status:

```
terminated: y
```

## JIL Blob and Glob Definitions

This section describes the JIL subcommands and attributes that you can use to insert and delete AutoSys Workload Automation binary large objects tied to jobs (blobs) and global binary large objects (globs). To insert or delete a blob or glob, you specify the appropriate JIL subcommand and attribute statements to the jil command.

### delete\_blob Subcommand -- Delete a Blob from the Database

The delete\_blob subcommand decouples the highest version of a blob from an existing job and deletes it from the database.

**Syntax**

This subcommand has the following format:

```
delete_blob: blob_name
```

- **blob\_name**  
Specifies the name of an existing job defined in the database to which the blob belongs.  
**Limits:** Up to 64 characters

**Required Attribute**

To delete a blob, specify the following attribute:

- **blob\_type**; only in, out, and inputpayload blob types are supported.

**NOTE**

If you delete a blob, all the jobs that are using this blob might get effected.

**Example: Delete a Blob from the Database**

This example deletes the highest version of input blob Blob1:

```
delete_blob: Blob1
blob_type: in
```

### delete\_glob Subcommand -- Delete a Glob from the Database

The delete\_glob subcommand deletes a glob from the database.

**Syntax**

This subcommand has the following format:

```
delete_glob: glob_name
```

- **glob\_name**  
Specifies the name of a glob that is currently defined in the database.  
**Limits:** Up to 64 characters

### Example: Delete a Glob from the Database

This example deletes the glob Glob1:

```
delete_glob: Glob1
```

## insert\_blob Subcommand -- Add a Blob to the Database

The insert\_blob subcommand adds a new input blob definition tied to an existing job to the database.

### Syntax

This subcommand has the following format:

```
insert_blob: blob_name
```

- **blob\_name**  
Specifies the name of an existing job defined in the database to which the blob belongs.  
**Limits:** Up to 64 characters; the name can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-)

### Required Attributes

To add a blob, specify one of the following attributes:

- [blob\\_file](#)
- [blob\\_input](#)

### Optional Attribute

You can specify the following optional attribute:

- [blob\\_type](#); only in and inputpayload blob types are supported.

### Example: Add a Blob with Input to the Database

This example creates an input blob with textual input:

```
insert_blob: blob_input1
blob_input: <auto_blobt>Testing this blob</auto_blobt>
```

### Example: Add a Blob with a File to the Database

This example creates an input blob with a file:

```
insert_blob: blob_input1
blob_file: /tmp/test.pl
```

### Example: Add an inputpayload Blob to the Database

This example creates an inputpayload blob with textual input:

```
insert_blob: httppayload
blob_type: inputpayload
```



```
blob_input: <auto_blobt>{"firstName": "John","lastName": "Doe","phoneNumber":
 "555-555-5555"}</auto_blobt>
```

## insert\_glob Subcommand -- Add a Glob to the Database

The insert\_glob subcommand adds a new glob definition to the database.

### Syntax

This subcommand has the following format:

```
insert_glob: glob_name
```

- **glob\_name**  
Defines a glob name.  
**Limits:** Up to 64 characters; the name can only contain the following characters: a-z, A-Z, 0-9, period (.), underscore (\_), pound (#), and hyphen (-)

### Required Attributes

To add a glob, you must specify the following attributes:

- [blob\\_mode](#)
- [blob\\_file](#) or [blob\\_input](#)

### Optional Attribute:

You can specify the following optional attribute:

- [blob\\_type](#)

### Example: Add a Glob with Text Input to the Database

This example creates a glob with text input:

```
insert_glob: glob_input1
blob_mode: text
blob_input: <auto_blobt>Testing this glob</auto_blobt>
```

### Example: Add a Glob with a Binary File to the Database

This example creates the glob with a binary file:

```
insert_glob: blob_input1
blob_mode: binary
blob_file: /tmp/test
```

## update\_blob Subcommand -- Update a Blob

The update\_blob subcommand updates only the content of an existing blob.

### Syntax

This subcommand has the following format:

```
update_blob: blob_name
```

- **blob\_name**  
Specifies the name of an existing blob (tied to an existing job) that you want to update.

## Required Attribute

To update a blob, specify the following attributes:

- `blob_file` or `blob_input`
- `blob_type`; only in and inputpayload blob types are supported.

### NOTE

If you update a blob, all jobs that are using this blob might get effected.

## Example: Update a Blob

This example updates the content of the Blob1 blob:

```
update_blob: Blob1
blob_input: <auto_blobt>Updating this blob</auto_blobt>
```

## update\_glob Subcommand -- Update a Glob

The `update_glob` subcommand updates only the content of an existing glob.

### Syntax

This subcommand has the following format:

```
update_glob: glob_name
```

- ***glob\_name***  
Specifies the name of an existing glob you want to update.

### Required Attributes

To update a glob, specify the following attributes:

- `blob_mode`
- `blob_file`

## Example: Update a Glob

This example updates the content of the Glob1 glob:

```
update_blob: Glob1
blob_mode:text
blob_file:"updated file path"
```

## blob\_file Attribute -- Specify the File for the Blob

The `blob_file` attribute specifies the file containing the data to be stored in the database.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- `insert_blob`
- `insert_glob`
- `update_blob`
- `update_glob`

### NOTE

If you do not specify this attribute, you must specify the `blob_input` attribute.

## Syntax

This attribute has the following format:

```
blob_file: file
```

- **file**  
Specifies the file to insert for the blob.  
**Limits:** Up to 255 characters

## blob\_input Attribute -- Specify the Input for the Blob

The blob\_input attribute specifies one or more lines of text to be stored in the database. All text in the auto\_blob XML-style metatags is stored literally and may span multiple lines.

### Supported JIL Subcommands

This attribute is optional for the following subcommands:

- [insert\\_blob](#)
- [insert\\_glob](#)
- [update\\_blob](#)

#### NOTE

If you do not specify this attribute, you must specify the blob\_file attribute.

## Syntax

This attribute has the following format:

```
blob_input: <auto_blobt>input</auto_blobt>
```

- **input**  
Specifies the text to insert for the blob.

## blob\_mode Attribute -- Specify File Format for the Glob or Blob

The blob\_mode attribute specifies whether the file is a binary or a text file. The file is stored in the database as the specified format.

### Supported JIL Subcommands

This attribute is required for the [insert\\_glob](#) subcommand.

This attribute is optional for the following subcommands:

- [insert\\_blob](#)
- [insert\\_job](#)
- [update\\_glob](#)

## Syntax

This attribute has the following format:

```
blob_mode: b | t
```

- **b**  
Specifies that the file is a binary file.  
**Note:** You can specify **binary** instead of b.
- **t**

Specifies that the file is a text file.

**Note:** You can specify **text** instead of **t**.

## blob\_type Attribute -- Specify the Type of the Blob or Glob

The `blob_type` attribute specifies the type of a blob or glob.

### Supported JIL Subcommands

When the `blob_type` is `in`, this attribute is optional for `insert_blob` and `update_blob` subcommands.

When the `blob_type` is `in` or `out`, this attribute is required for the [delete\\_blob](#) subcommand.

### Supported Job Types

When the `blob_type` is `generic` or `emailtemplate`, this attribute is optional for all job types.

### Syntax

This attribute has the following format:

```
blob_type: in | out | generic | emailtemplate | outputpayload | outputpayloadvar
```

- **in**  
Specifies that the latest version of the job input blob is deleted. This is the default for blobs.  
**Note:** You can specify **i** instead of **in**.
- **out**  
Specifies that the output or error blobs for the job are deleted.  
**Note:** You can specify **o** instead of **out**.
- **generic**  
Specifies that the glob is a generic blob. This is the default for globs.
- **emailtemplate**  
Specifies that the glob is an emailtemplate blob that includes the email template. You can use this emailtemplate blob in the [notification template](#) that is used to send the email notification.

### NOTE

If you set the `blob_type` attribute value to `emailtemplate`, the [blob\\_mode](#) attribute value must be set to `text`.

- **outputpayload**  
Specifies that the blob is an output payload job blob.
- **outputpayloadvar**  
Specifies that the blob is an output payload var job blob.

### Example: Specify a Generic Blob

This example creates a glob named `glob_input` with text input that includes the generic content that is specified in the `blob_input` attribute.

```
insert_glob: glob_input
blob_mode: text
blob_type: generic
blob_input: <auto_blobt>Testing this glob</auto_blobt>
```

### Example: Specify an emailtemplate Blob

This example creates a glob named `email_input` with text input that includes the email template that is included in the `sample_template` file.

```
insert_glob: email_input
```

```
blob_mode: text
blob_type: emailtemplate
blob_file: $AUTOSYS/install/email_templates/sample_template
```

## AutoSys System States

This article describes the system events, alarms, statuses, and exit codes that AutoSys Workload Automation processes.

### Events

AutoSys Workload Automation is event-driven. An event instructs the scheduler to initiate a specific action. The scheduler completes the action unless doing so violates internal rules or the event is not applicable. AutoSys Workload Automation server components internally generate certain events in certain circumstances, but you can also manually generate events using the `sendevent` command.

AutoSys Workload Automation records events in the following tables:

- `ujo_event`  
Records events that the scheduler has not yet processed.
- `ujo_proc_event`  
Records processed events other than alarms.
- `ujo_alarm` tables  
Records processed alarms.

These tables use numeric codes to represent events and event status. The numeric codes are resolved using the `ujo_intcodes` table.

An event may have *one* of the following statuses:

- **UNPROCESSED (<0|0)**  
Indicates that the scheduler has not yet processed the event.
- **PROCESSED (>0)**  
Indicates that the scheduler has processed the event and completed the specified action.

The scheduler continually polls the database, processes events, and updates their status. The scheduler processes events on receipt unless you specify a time when you issue the `sendevent` command.

The scheduler processes the following events:

#### **AGGREGATE (144)**

Aggregates job, alarm, and scheduler statistics. Records these statistics in the `ujo_rep_hourly`, `ujo_rep_daily`, `ujo_rep_monthly`, and `ujo_rep_weekly` tables. You can also use this event to delete statistics. This event is manually generated.

#### **ALARM (106)**

Indicates the occurrence of one of the specific [alarms](#). The scheduler internally generates this event to send alarms. You can also generate this event manually.

#### **ALERT (139)**

Reports the occurrence of a monitored condition for continuously monitored jobs. The scheduler internally generates this event.

**BYPASS (147)**

Bypasses execution of a job that is in the NO\_EXEC state. The scheduler internally generates this event for any job that is in the NO\_EXEC state. Bypassed jobs complete successfully but the scheduler does not execute job commands.

**CHANGE\_PRIORITY (120)**

Modifies the job queue priority of the specified job to the priority specified in the -q parameter of the sendevent command. This event is manually generated.

**NOTE**

The priority change applies to the next run job run only, unless the job is in QUE\_WAIT status. The change applies to the current run only for jobs in QUE\_WAIT status. To change the priority of a job permanently, edit the job definition.

**CHANGE\_STATUS (101)**

Changes the status of the specified job to the status defined in the -s parameter of the sendevent command. The scheduler generates this event internally to change the status of jobs in progress. You can also manually generate this event. The scheduler initiates any action that depends on the status of the job.

You can issue the CHANGE\_STATUS event to change the status of a job that is in the ON\_NOEXEC status to either the INACTIVE or SUCCESS status while remaining in non-execution mode. When you change the status of a non-execution job to SUCCESS, the scheduler completes the job's run and schedules the downstream dependent jobs. Changing the status of a non-execution box job to INACTIVE also changes the status of all the jobs in that box job to INACTIVE in non-execution mode. Changing the status of a non-execution box job to the RUNNING status starts the box job.

**CHECK\_HEARTBEAT (116)**

Verifies that all jobs with a heartbeat\_interval generate heartbeats as scheduled. You can configure the scheduler to generate CHECK\_HEARTBEAT events automatically.

**CHK\_BOX\_TERM (118)**

Verifies that a box job completes within the time limit specified in the term\_run\_time attribute. The scheduler internally generates this event for all box jobs with a run-time term specified in the job definition.

**CHK\_COMPLETE (136)**

Verifies that the job completes by the time specified in the job definition. The scheduler internally generates this event for all jobs that have times specified in the must\_complete\_time attribute of the job definition.

**CHK\_MAX\_ALARM (114)**

Verifies that a job completes within the time limit specified in the job definition. The scheduler internally generates this event for all jobs with maximum running time specified in the max\_run\_time attribute.

**CHK\_RUN\_WINDOW (122)**

Verifies that a job runs within the time frame specified in the run\_window attribute. The scheduler internally generates this event for all jobs with a runtime window specified in the job description.

**CHK\_START (135)**

Verifies that the job starts by the time specified in the job definition. The scheduler internally generates this event for all jobs with times specified in the must\_start\_times attribute of the job definition.

**CHK\_TERM\_RUNTIME (138)**

Verifies that a job runs within the time limit specified in the `term_run_time` attribute of the job definition. If the run time for a job exceeds the specified time frame, the job terminates. The scheduler internally generates this event for all jobs with an interval specified in the `term_run_time` attribute.

**COMMENT (117)**

Generates a comment in the scheduler log. This event is manually generated. Use with the `-J` parameter of the `sendevent` command to associate the comment with a specific job.

**DELETEJOB (119)**

Deletes the specified job from the database. If the job is a box job, the event deletes the box job and all the jobs in it. This event is manually generated.

**EXTERNAL\_DEPENDENCY (127)**

Adds or removes a dependency that is associated with an external job.

The local scheduler generates this event internally when any of the following circumstances apply:

- A job that is scheduled to run on the remote application server of a type 'a' external instance has dependencies on a job that is scheduled to run on another instance.
- A job that is scheduled to run on the remote CA Workload Automation ESP Edition manager of a type 'e' external instance has dependencies on a job that is scheduled to run on a AutoSys Workload Automation instance.
- You delete an external job that has cross-instance or cross-platform dependencies.
- The manager of an external instance other than type-'a' sends a request to the scheduler. For example, a job on the external instance depends on a job on the local instance and the external manager requests that the scheduler remove this dependency.

The local application server generates this event internally when any of the following circumstances apply:

- You create a local job with a condition attribute that establishes a dependency on an external job.
- You update or override the condition attribute in the definition of a local job to add or remove a dependency on an external job.
- You delete a local job that has dependencies on an external job.

The scheduler that is running on a type-'a' external instance sends a request to the application server. For example, a job on the external instance depends on a job on the local instance and the external scheduler requests that the local application server remove this dependency.

**FORCE\_STARTJOB (108)**

Overrides starting conditions and starts the specified job. Use this event to start a job that failed, was killed, did not execute properly, has unmet starting conditions, remains indefinitely in the `STARTING` or `RUNNING` state, is scheduled to run on a machine that is not online, or did not release resources when it completed.

The following conditions apply:

- The event has no effect on jobs in the `RESWAIT` state. Other jobs with unmet resource conditions enter the `RESWAIT` state when you send the event.
- The event starts a job even if it is in a non-executable state (`ON_HOLD`, `ON_ICE`). Use this option with a job in a non-executable state only when you need to start the job immediately. To resume a job's regular run schedule, change the

executable status of the job. When you force start a job that is in a non-executable state, the job does not revert to that state after it completes.

- Jobs scheduled to run on machines that are not online start only when the scheduler successfully connects to the agent before exhausting the maximum number of retry attempts specified in the job definition.

The following conditions apply:

- The scheduler changes the status of a missing or unqualified machine to online when the job starts.
- The scheduler starts the job on an offline machine without changing the status of the machine.

### **WARNING**

The product does not support concurrent runs of the same job. Errors may cause a job to stop progressing after the agent executes it. To avoid simultaneous executions, do not force start a job while it is in the STARTING or RUNNING state. In this case, change the status of the job. The RESTRICT\_FORCE\_STARTJOB environment variable enables you to force the scheduler to reject force start attempts on jobs that are executing. For information about setting environment variables, see [Configuring AutoSys Workload Automation](#).

Use caution when you force start jobs in the ON\_HOLD, ON\_ICE, or QUE\_WAIT state, jobs that require held resources, or jobs scheduled to run on offline machines. When possible, start these jobs by sending one of the following events:

- **CHANGE\_PRIORITY**  
Jobs in the QUE\_WAIT state start when you update their priority to 1.
- **MACH\_ONLINE**  
Jobs in the PEND\_MACH state on the specified machine start.
- **JOB\_OFF\_HOLD** or **JOB\_OFF\_ICE**  
Jobs that skipped scheduled runs while in the ON\_HOLD state start. Jobs that skipped runs while in the ON\_ICE state are rescheduled.

### **HEARTBEAT (115)**

Posts a heartbeat for a specific job. The scheduler generates this event internally for jobs that are defined to post heartbeats.

### **INVALIDATE\_MACH (133)**

Notifies the scheduler when you update or delete the machine definition.

### **JOB\_OFF\_HOLD (113)**

Removes the specified job from ON\_HOLD status. As soon as the starting conditions permit, the job starts. This event is manually generated.

### **JOB\_OFF\_ICE (111)**

Returns the specified job to the job stream. The scheduler removes the job from ON\_ICE status. Jobs that meet their starting conditions while they are in the ON\_ICE state start when starting conditions recur. This event is manually generated.

### **JOB\_OFF\_NOEXEC (146)**

Removes the specified job from the ON\_NOEXEC status. The agent executes the commands that are associated with the job the next time that the scheduler starts the job. This event is manually generated.

When the scheduler processes the JOB\_OFF\_NOEXEC event for a job, it places the job in the INACTIVE, ACTIVATED, or SUCCESS status based on the job's existing status while in non-execution mode. If the job is in a box, the scheduler



does not evaluate the overall box status. If you send the `JOB_OFF_NOEXEC` to a box, all jobs in the box (including all jobs contained in lower level boxes within the box) are reset to the `INACTIVE`, `ACTIVATED`, or `SUCCESS` status.

### **JOB\_ON\_HOLD (112)**

Places the specified job in `ON_HOLD` status. The job does not start, and downstream dependent jobs do not run. Boxes containing jobs in `ON_HOLD` status cannot complete. The event has no effect on jobs with a status of `STARTING` or `RUNNING`. This event is manually generated.

### **JOB\_ON\_ICE (110)**

Suspends scheduled runs of the specified job by temporarily removing it from the job stream. Downstream dependent jobs run. The scheduler places the job in `ON_ICE` status. If the job is in a box, the scheduler does not execute the job for the entire run of the box. The event has no effect on jobs with a status of `STARTING` or `RUNNING`. This event is manually generated.

#### **NOTE**

Use caution when you place jobs in this state. Downstream jobs with critical dependencies on jobs in the `ON_ICE` state do not complete successfully. For example, a job that requires input that is contained in a file that is produced by another job cannot succeed when the job that produces the file is in the `ON_ICE` state.

### **JOB\_ON\_NOEXEC (145)**

Places the specified job in the `ON_NOEXEC` status and bypasses execution of commands that are associated with the job. The scheduler does not immediately schedule downstream dependent jobs nor does it evaluate their conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the `INACTIVE` status.

When the `NOEXEC` job meets its starting conditions, the scheduler evaluates this job as successfully completed with an exit code of zero even though the agent does not execute the commands. After the `NOEXEC` job is evaluated as successfully completed, the scheduler schedules downstream dependent jobs.

You can issue the `JOB_ON_NOEXEC` event to an `ACTIVATED` job that is within a running box. The job remains in the `ACTIVATED` state in non-execution mode until its starting conditions are met. After the starting conditions of the job is met, the scheduler sends a `BYPASS` event and the job goes to the `SUCCESS` state in non-execution mode.

The following conditions apply:

- The scheduler processes the `JOB_ON_NOEXEC` event similar to that of the `CHANGE_STATUS` event to `INACTIVE`.
- The scheduler ignores the `JOB_ON_NOEXEC` event, if sent to:
  - A non-box job that is in the `STARTING`, `RUNNING`, or `ON_ICE` status
  - A box job that is in the `ON_ICE` or `RUNNING` status
  - A box job with jobs (including the jobs contained in lower level boxes) in a status other than the following status: `ON_HOLD`, `ON_NOEXEC`, `INACTIVE`, `SUCCESS`, `FAILURE`, `ACTIVATED`, or `TERMINATED`
- When a job that is in the `ON_NOEXEC` status is in a box and the starting conditions of the box are met, the job enters the `ACTIVATED` status. The scheduler bypasses the execution of the job when the starting conditions of the job are met.
- When the starting conditions of a box that is in the `ON_NOEXEC` status are met, the box enters the `RUNNING` status. The scheduler bypasses the execution of the box when the starting conditions of all jobs in the box (including all jobs contained in lower level boxes within the box) are met.
- If you send the `JOB_ON_NOEXEC` event to a job in a box, the effect is the same as sending the `CHANGE_STATUS` event to `INACTIVE` for a job in a box. The scheduler sets the job to the `ON_NOEXEC` status and also sets the internal secondary status to `INACTIVE`, so that the scheduler evaluates the overall box status as if the job entered the

INACTIVE status. To complete a job that is in the ON\_NOEXEC status so that the scheduler evaluates the job as if the job entered the SUCCESS status, send the FORCE\_STARTJOB event.

- If you send the JOB\_ON\_NOEXEC event to a box, the effect is the same as sending the CHANGE\_STATUS event to INACTIVE for a box. The box enters the ON\_NOEXEC status and the scheduler sets the status of all jobs in the box (including all jobs contained in lower level boxes within the box) at all levels to ON\_NOEXEC.
- If you send the JOB\_ON\_NOEXEC event to change the status of an ACTIVATED job that is within a running job, the job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job is met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.
- The JOB\_ON\_NOEXEC event supersedes the JOB\_ON\_HOLD event effectively overwriting the ON\_HOLD status with the ON\_NOEXEC status.
- The JOB\_ON\_NOEXEC event does not supersede the JOB\_ON\_ICE event and does not overwrite the ON\_ICE status with the ON\_NOEXEC status.
- If the job is in the QUEWAIT or RESWAIT status, the scheduler removes the job from the load balancing and resource wait queues before placing it in the ON\_NOEXEC status.
- CHANGE\_STATUS events have no effect on the job status if the job is in the ON\_NOEXEC status. You must issue the JOB\_OFF\_NOEXEC event to take the job out of the ON\_NOEXEC status.
- If you send the CHANGE\_STATUS event to a box to set its status to INACTIVE, the scheduler does not change the status of ON\_NOEXEC jobs within the box to INACTIVE. Instead, the scheduler sets the internal secondary status to INACTIVE so that the scheduler can evaluate the jobs as if they were set to the INACTIVE status.
- The scheduler sends an internal JOB\_ON\_NOEXEC event to a job in the ON\_NOEXEC status instead of the CHANGE\_STATUS event to INACTIVE when:
  - The job is scheduled on a day that is excluded by the exclude\_calendar job attribute
  - The next scheduled run of the job is outside the run window
  - When a box that is in the ON\_NOEXEC status completes, the scheduler resets the status of any remaining ACTIVATED jobs to ON\_NOEXEC behaving as if the jobs were reset to the INACTIVE status.

#### • **KILLJOB (105)**

Notifies the agent to stop execution of the specified job and place it in TERMINATED status.

You get an error when you issue this event to the following unsupported job types:

- FTP
- SCP
- JAVARMI
- ENTYBEAN
- SESSBEAN
- POJO
- JMSPUB
- JMXMAG
- JMXMAS
- JMXMC
- PAREQ
- JMXMREM
- JMXMOP
- WBSVC
- I5OS
- ZOSDST
- SNMPGET
- SNMPSET
- WOL
- WSDOC

The agent stops execution of the specified job and kills all processes that are associated with the job when you issue the event against most job types. Exceptions apply to the following job types:

- Command jobs on UNIX  
The event kills the currently running process and the associated process group but does not kill orphan processes.
- Box jobs  
The event changes the status of the specified job to TERMINATED. Jobs that are contained in the box job but are not yet in STARTING or RUNNING status do not start. The scheduler also sends KILLJOB events to jobs that are contained in the box job and meet the following conditions:
  - Are in progress
  - Have a job\_terminator value of Y
 Other jobs that are in the box job and are in progress complete.
- File Watcher jobs  
The event kills the file watcher job and changes its status to TERMINATED.

- **MACH\_OFFLINE (130)**

Takes the machine that the -N parameter specifies offline. The scheduler automatically changes the status of jobs that are scheduled to run on the machine to PEND\_MACH. The scheduler generates this event when *one* of the following situations occurs:

- The scheduler cannot connect to the specified machine.
- The agent machine shuts down.

You can also generate this event manually.

- **MACH\_ONLINE (131)**

Places the machine specified in the -N parameter online after the scheduler calibrates that the machine is actually available and accepts jobs to run. If the machine is not available, the machine status is set to missing or blocked. The scheduler automatically removes jobs that are scheduled to run on the machine from QUE\_WAIT, RESWAIT, or PEND\_MACH status and starts the jobs. The scheduler internally generates this event when one of the following situations occurs:

- The scheduler successfully contacts a previously missing machine.
- The agent starts.

You can also manually generate this event.

- **REFRESH\_EXTINST (129)**

Indicates that the scheduler detects changes in the external instance definitions. The scheduler internally generates this event.

- **RELEASE\_RESOURCE (137)**

Releases a held resource to the back to the pool of usable resources.

When a job is in process, you cannot use this event to release resources that the job consumes. In this case, wait until the job reaches completion status and then issue the event.

This event is manually generated.

- **REPLY\_RESPONSE (140)**

Issues a response to a WAIT\_REPLY\_ALARM. Jobs automatically trigger this alarm when they enter the WAIT\_REPLY state. These jobs cannot continue running until you manually generate this event.

- **RESEND\_EXTERNAL\_STATUS (128)**

Resends a CHANGE\_STATUS event to an external instance. The scheduler generates this event after the INSTANCE\_UNAVAILABLE alarm indicates that the original CHANGE\_STATUS event failed because the external instance is not available.

- **RESTARTJOB (143)**

Restarts a job that ends in failure. This event applies to Micro Focus, Informatica, and SAP BW Process Chain jobs only. The scheduler records occurrences of this event in the logfile. Reports that are generated using the autorep -

J -d command display occurrences of this event that were generated during the most recent job runs. This event is manually generated.

- **RESUMEJOB (149)**

Resumes a Hadoop Oozie workflow and places it in the RUNNING state. This event is manually generated.

- **SEND\_SIGNAL (126)**

Sends a signal to a running job. This event is manually generated.

**NOTE**

- You can send this event only when you specify the -k and -J parameters in the sendevent command.
- On Windows, you can use this event to signal a semaphore with a specific name. Signaling a specific semaphore enables you to program your application to watch for the semaphore and react to it.
- On UNIX, if the oscomponent.wrapper.exec.force parameter in the agentparm.txt file is set to false (the default), the SEND\_SIGNAL event is not supported. To send UNIX signals, set the oscomponent.wrapper.exec.force parameter to true. When this parameter is set to true, chained commands are not supported. To support chained commands and the SEND\_SIGNAL event, take the following actions:
  - Set the oscomponent.wrapper.exec.force parameter to true and move the chained commands to a script file.
  - Update the command attribute in the job definition to execute the script file instead of the chained command.

- **SET\_GLOBAL (125)**

Sets a global variable. AutoSys Workload Automation automatically assigns a high priority to this event, so that the scheduler processes the variable before processing jobs that reference the variable. This event is manually generated.

**NOTE**

You can send this event only when you specify the -G parameter.

- **STARTJOB (107)**

Starts the specified job when its starting conditions are satisfied. This event is manually generated.

**NOTE**

- Using the sendevent command to generate this event is the recommended method for manually starting a job.
- This event considers all starting conditions, such as time and date conditions and dependencies on other jobs.
- Jobs that are contained in box jobs inherit the starting conditions of the box job. These jobs start only when the box job starts. To start a job that is contained in a box job, issue this event against the box job.
- You cannot issue the STARTJOB event to start a job that has a virtual resource dependency with free=Y or free=N and has already held the resource.

- **STATE\_CHANGE (141)**

Indicates a change in the state of a job. The scheduler internally generates this event and records the state change in its log file.

**NOTE**

You can change the status of a job manually only by generating a CHANGE\_STATUS event. You cannot generate the STATE\_CHANGE event manually.

- **STOP\_DEMON (109)**

Shuts down the scheduler or the application server, depending on the options that you specify. Optionally, you can instruct the shadow scheduler to take over processing events when you use this event to shut down the primary scheduler. This event is manually generated.

- **SUSPENDJOB (148)**

Suspends the Hadoop Oozie workflow and places it in the SUSPENDED state. This event is manually generated.

**NOTE**

To resume the Hadoop Oozie workflow, issue the RESUMEJOB event manually.

**Alarms**

*Alarms* are special events that notify operators of situations requiring their attention. The system issues alarms by internally generating an event, but these events are informational only. To resolve the issue that triggers an alarm, initiate the required action using the sendevent command.

Some problems do not trigger the scheduler to generate an alarm. For example, when a problem quickly resolves itself without manual intervention. Sometimes, you need to notify someone to follow up on such problems so that they do not recur. In these cases, use the sendevent command to generate an alarm manually.

The scheduler records alarms in the ujo\_alarm table. The ujo\_alarm table uses numeric codes to represent alarms. The numeric codes are resolved using the ujo\_intcodes table.

The scheduler records the following alarms:

**APP\_SERVER\_SHUTDOWN (545)**

Indicates that all the application servers are shutting down, typically when you send a STOP\_DEMON event with one of the following options:

- -v ALL
- -v ROLE=A

This alarm can trigger a user-specified notification procedure.

**AUTO\_PING (526)**

Indicates a problem communicating with the agent. This alarm is raised in the following situations:

- The autoping -M -A command raises the AUTO\_PING alarm when it does not complete successfully.
- The scheduler raises the AUTO\_PING alarm when the agent does not immediately acknowledge its request to perform a job operation (start a job, kill a job, send a signal and so on).

The alarm text includes the name of the machine that is associated with the unresponsive agent.

**CHASE (514)**

Indicates that chase command identified a problem with a job in progress. The alarm includes the job name and the problem encountered.

**CPI\_JOBNAME\_INVALID (535)**

Indicates that the cross-platform interface has received a job name that is not valid for the agent it is submitting the job to.

**CPI\_UNAVAILABLE (536)**

Indicates that the cross-platform Interface is not running.

**DATABASE\_COMM (516)**

Indicates that the agent cannot send an event to the database. Does not necessarily indicate job failure. Determine the cause of the alarm by inspecting the logfile.

**DB\_PROBLEM (523)**

Indicates that there is a problem with one of the databases; for example, lack of free space. This alarm can trigger a user-specified notification procedure.

**DB\_ROLLOVER (519)**

Indicates a change from dual event server mode to single event server mode. This alarm can trigger a user-specified notification procedure.

**DBMAINT\_FAILURE (548)**

Indicates that the DBMaint command failed during automated database maintenance.

**DUPLICATE\_EVENT (524)**

Indicates that the event server received duplicate events. Typically, this error occurs when two schedulers are running. Sometimes event server configuration errors trigger this alarm.

**EP\_HIGH\_AVAIL (522)**

Indicates *one* of the following situations:

- The machine that resolves contentions between two schedulers is not reachable.
- The scheduler is shutting down.
- Other problems interfere with the scheduler takeover process.

This alarm can trigger a user-specified notification procedure.

**EP\_ROLLOVER (520)**

Indicates that the shadow scheduler takes over processing. This alarm can trigger a user-specified notification procedure.

**EP\_SHUTDOWN (521)**

Indicates that the active scheduler (either the primary scheduler or the shadow scheduler after it takes over processing events while running in high availability mode) is shutting down because of a normal shutdown process or an error condition. This alarm can trigger a user-specified notification procedure.

The EP\_SHUTDOWN alarm is raised with text containing the role designator, the host, and the time of the scheduler shutdown. If AutoSys Workload Automation is configured to send SNMP traps, the scheduler sends two traps with an identifier of 109 (STOP\_DEMON) and 521 (EP\_SHUTDOWN).

**EVENT\_HDLR\_ERROR (507)**

Indicates that the scheduler cannot process an event due to an error. Inspect the job that is associated with the event to determine if manual intervention is required.

**EVENT\_QUEUE\_ERROR (508)**

Indicates that an event cannot be marked as processed, typically due to a problem with the event server. For assistance, contact CA Support.

**EXTERN\_DEPS\_ERROR (529)**

Indicates that the cross-platform interface cannot send external dependencies to the remote node. The alarm identifies the failing remote node.

**FORKFAIL (501)**

Indicates that the agent cannot start the user command because it cannot get a process slot on the UNIX machine. AutoSys Workload Automation automatically attempts a RESTART. This alarm only occurs when a job is running on a UNIX machine.

**INSTANCE\_UNAVAILABLE (525)**

Indicates that an attempt to communicate with an event server on the receiving instance fails, typically when that event server is down.

**JOBFAILURE (503)**

Indicates that a job fails.

**JOBTERMINATED (552)**

Indicates that a job is terminated.

**JOBNOT\_ONICEHOLD (509)**

Indicates that the JOB\_ON\_HOLD or JOB\_ON\_ICE event fails, typically when you send the event after the job starts.

**JOBNOT\_ONNOEXEC (546)**

Indicates that the JOB\_ON\_NOEXEC event fails, typically because the job (or the box job containing it) starts before you send the event or if the job's box is in the ON\_NOEXEC state.

**KILLJOBFAIL (540)**

Indicates that the KILLJOB event fails.

**MACHINE\_UNAVAILABLE (532)**

Indicates that a machine is not responding to ping requests from the scheduler. The scheduler takes the machine offline after issuing this event.

**MACHINE\_DISKTHRESHOLD (549)**

Indicates that a disk threshold has been breached on the agent. This alarm is raised when the notice, severe, or critical threshold is breached.

**MAX\_RETRYS (505)**

Indicates that AutoSys Workload Automation cannot restart the job after the maximum number of restart attempts specified in the n\_retrys attribute. After you resolve the problem, start the job manually.

**MAXRUNALARM (510)**

Indicates that a job does not complete within the time limit specified in the max\_run\_alarm attribute of the job definition. This alarm does not stop the job.

**MINRUNALARM (502)**

Indicates that a job finishes before the end of the interval specified in the min\_run\_alarm attribute of the job definition.

**MISSING\_HEARTBEAT (513)**

Indicates that the CHECK\_HEARTBEAT event cannot detect a heartbeat for a job. This alarm only occurs when you specify the heartbeat\_interval attribute in the job definition and the scheduler fails to generate a HEARTBEAT event within that interval.

**MUST\_COMPLETE\_ALARM (538)**

Indicates that a job is not completed at the time specified in the must\_complete\_times attribute of the job definition.

**MUST\_START\_ALARM (537)**

Indicates that a job does not start by the time specified in the must\_start\_times attribute of the job definition.

**QUEUEDJOB\_STARTFAIL (547)**

Indicates that a job leaving the queue fails to meet the starting conditions specified in the job definition. The job cannot start without manual intervention.

**NOTE**

This failure only occurs when AutoSys Workload Automation is configured to re-evaluate the starting conditions of queued jobs before running those jobs. For more information about this configuration option, see [Configure a Scheduler](#).

**REPLY\_RESPONSE\_FAIL (542)**

Indicates that the reply\_response event fails.

**NOTE**

Jobs in WAIT\_REPLY status cannot continue running until they receive a user response. Use the following command to send a response:

```
sendevent - E REPLY_RESPONSE - J job_name - r response
```

**RESOURCE (512)**

Indicates a resource problem, typically that a job cannot run because a resource it requires is not available. The alarm text includes specific information about the problem. AutoSys Workload Automation attempts to restart the job after a suitable delay.

**RESTARTJOBFAIL (544)**

Indicates that the manually generated sendevent -E RESTARTJOB command fails, typically due to *one* of the following reasons:

- The job is not in a FAILURE state.
- The job is not an SAP BW Process Chain, Microfocus, or Informatica job.

AutoSys Workload Automation displays RESTARTJOBFAIL alarms when you create a report using the autorep -J -d command.



**RESUMEJOBFAIL(551)**

Indicates that the RESUMEJOB event fails; typically when you send the event when the Hadoop Oozie workflow is not in the SUSPENDED state.

**RETURN\_RESOURCE\_FAIL (543)**

Indicates that the resource manager cannot return a resource.

**SENDSIGFAIL (541)**

Indicates that the SEND\_SIGNAL event fails.

**SERVICEDESK\_FAILURE (533)**

Indicates that the scheduler cannot open a CA Service Desk ticket for the failing job.

**STARTJOBFAIL (506)**

Indicates that AutoSys Workload Automation cannot start a job, typically due to communication problems with the remote machine. AutoSys Workload Automation attempts to restart the job.

**SUSPENDJOBFAIL (550)**

Indicates that the SUSPENDJOB event fails; typically when you send the event when the Hadoop Oozie workflow is not in the RUNNING state.

**TELEMETRY\_FAILURE (553)**

Indicates that Telemetry failed to collect or send data.

**UNINOTIFY\_FAILURE (534)**

Indicates that the scheduler cannot send a notification for the requesting job to the Notification Services component of CA NSM.

**VERSION\_MISMATCH (518)**

Indicates that the version number that the calling routine generates and the version number of the agent do not match. Inspect the agent log file, and install the proper agent version.

**WAIT\_REPLY\_ALARM (539)**

Indicates that a job cannot continue running until it receives a user reply.

## Status

AutoSys Workload Automation tracks the current state, or status, of every job. AutoSys Workload Automation displays job status in the job report that the autorep command generates and in the Web UI.

The scheduler records job run information, including job status. The ujo\_job\_status table stores job status for the current run, and the ujo\_job\_runs table stores job status for previous runs. These tables use numeric codes to represent job status. The numeric codes are resolved using the ujo\_intcodes table. A job can have one of the following statuses:

**ACTIVATED (9)**

Indicates that job is contained in a box job with a status of RUNNING but that the job itself is waiting to start.

**FAILURE (5)**

Indicates that a job fails to complete successfully. The scheduler issues this alarm when the alarm\_if\_fail attribute in a job definition is set to Y and that job fails.

**INACTIVE (8)**

Indicates that a newly created job has not yet run for the first time.

**PEND\_MACH (14)**

Indicates that the offline status of the machine to which a job is assigned prevents the job from starting. These jobs can otherwise logically start, so the scheduler attempts to start them when the offline machine returns to service.

**NOTE**

You can use the GlobalPendMach configuration options to control how jobs in PEND\_MACH status start and what happens to new jobs scheduled to currently offline machines. For more information about the GlobalPendMach configuration options, see [Configure a Scheduler](#).

**ON\_HOLD (11)**

Indicates that the job is on hold and cannot run until you take it off hold.

**NOTE**

- You can place a job in this status only by sending the JOB\_ON\_HOLD event.
- To take a job off hold, send the JOB\_OFF\_HOLD event.

**ON\_ICE (7)**

Indicates that the job is removed from the job stream but is still defined.

**NOTE**

- You can place a job in this status only by sending the JOB\_ON\_ICE event.
- You cannot manually change the status of a job from ON\_ICE to INACTIVE. To return a job that is on ice to the job stream and resume running it, send the JOB\_OFF\_ICE event.

**ON\_NOEXEC (16)**

Indicates that the scheduler bypasses execution of the job.

**NOTE**

- You can place a job in this status only by sending the JOB\_ON\_NOEXEC event.
- When you instruct the scheduler to bypass the execution of a job, the scheduler places the job in the ON\_NOEXEC status and the effect is the same as sending the CHANGE\_STATUS event to INACTIVE for the job. The scheduler does not immediately schedule downstream jobs that have dependency on the

NOEXEC job nor does it evaluate their success conditions to success. Instead, the scheduler evaluates the conditions of downstream dependent jobs as if the predecessor job is set to the INACTIVE status.

- When you instruct the scheduler to bypass the execution of a box job, the scheduler automatically places all jobs in the box job (including all jobs contained in lower level boxes within the box job) in ON\_NOEXEC status. The effect is the same sending a CHANGE\_STATUS event to INACTIVE for a job in a box.
- When you instruct the scheduler to change the status of an ACTIVATED job that is within a running box. The job remains in the ACTIVATED state in non-execution mode until its starting conditions are met. After the starting conditions of the job is met, the scheduler sends a BYPASS event and the job goes to the SUCCESS state in non-execution mode.
- When the starting conditions of a NOEXEC job are met, the scheduler bypasses the execution of the job and places the job in the ON\_NOEXEC status. The effect is the same as sending a CHANGE\_STATUS event to SUCCESS with an exit code of 0. The scheduler schedules downstream jobs that have a dependency on the NOEXEC job and evaluates their success conditions to success.
- To resume executing bypassed jobs, send the JOB\_OFF\_NOEXEC event.

### **QUE\_WAIT (12)**

Indicates that a lack of available load units on the machines to which a job is assigned prevents the job from starting. It also indicates that the load balancing job is blocked behind one or more higher priority jobs that are queued in the QUE\_WAIT or RESWAIT state. When all higher priority queued jobs exit their queued state and the required load units become available, the job starts.

### **RESTART (10)**

Indicates that the scheduler is attempting to restart a job that failed to start as scheduled. The scheduler attempts to restart the job at periodic intervals until the job starts or the maximum number of restart attempts is exceeded.

#### **NOTE**

You can set the maximum number of restart attempts to zero (0). If the scheduler reaches the maximum number of restart attempts without successfully starting the job, the job fails. The maximum number of restart attempts depends on the following global settings:

- Max Restart Trys (on the Windows Administrator) or MaxRestartTrys (in the UNIX configuration file)
- Max Restart Wait (on the Windows Administrator) or MaxRestartWait (in the UNIX configuration file)

### **RESWAIT (15)**

Indicates that a lack of available resources to which a job is assigned prevents the job from starting. It also indicates that the resource job is blocked behind one or more higher priority jobs that are queued in the QUE\_WAIT or RESWAIT state. When all higher priority queued jobs exit their queued state and the resource is available, the job starts.

### **RUNNING (1)**

Indicates *one* of the following situations:

- The agent is executing the job.
- The scheduler has instructed the agent to start jobs that are contained in the box job.

#### **NOTE**

A box job acts as a container for other jobs but performs no action itself. The agent does not execute box jobs. The scheduler places box jobs in RUNNING status to send a message to the agent that it can start jobs that are contained in the box job. These jobs start as soon as they meet the starting conditions that are specified in the job definitions.

**STARTING (3)**

Indicates that the scheduler initiated the start job procedure with the agent. This status does not apply to box jobs.

**SUCCESS (4)**

Indicates that the job exits with a code equal to or less than the maximum exit code for success specified in the job definition. A box job enters this status in the following situations:

- All jobs that are contained in the box job succeed.
- The exit condition for the box job evaluates to TRUE.

**SUSPENDED (17)**

Indicates that the Hadoop Oozie workflow is suspended.

**TERMINATED (6)**

Indicates that the job ends while it is still in the RUNNING state. The scheduler issues an alarm when a job is terminated. A job is terminated when *one* of the following situations occurs:

- You send a KILLJOB event.
- You issue the kill command (UNIX).
- The job definition specifies that the scheduler should terminate the job under specific circumstances and those circumstances occur. For example, you can specify that the scheduler should terminate a job when the box job containing it fails.
- The job does not complete within the maximum run time specified in the job definition.

**WAIT\_REPLY (13)**

Indicates that the job cannot continue running without manual intervention. The scheduler issues an alarm when a job requires a user response to continue running.

**NOTE**

The job continues running only when you reply to this alarm by sending the REPLY\_RESPONSE event.

**Exit Codes**

AutoSys Workload Automation reports the exit codes of jobs as they complete. Exit codes typically indicate reasons for job failures or command failures for command jobs. A command exit code of zero typically indicates successful execution of the command. If a job completes successfully, AutoSys Workload Automation reports an exit code of zero. When a job encounters an error and the job does not complete successfully, the log files for that job display an exit code that is returned by the operating system or the application where the job ran.

AutoSys Workload Automation job reports may display the following internal exit codes for jobs that the agent cannot execute or for jobs that the agent lost track of after executing:

- **-655**  
Indicates that the job fails because the scheduler has exceeded the number of restart attempts.
- **-656**  
Indicates that the job execution has not yet started or is in progress.
- **-657**  
Indicates that you verify the RUNNING or STARTING status of a job using the chase command and receive the following results:

- The job is in RUNNING status
- The command cannot find some processes that are associated with the job, typically because those processes are not running.

**NOTE**

To verify the RUNNING or STARTING status a job and the processes that are associated with that job, issue the following command:

```
chase -E -A
```

# Messages

This section includes a listing of all messages, including workarounds.

- **Error Messages**

Messages that begin with an *E* indicate an error condition. The function cannot be completed until this condition is corrected.

- **Informational Messages**

Messages that begin with an *I* are informational. They are commonly used to indicate successful completion of a function, or give you additional information about it.

- **Warning Messages**

Messages that begin with a *W* warn that caution is required. They are commonly used to caution that proceeding with an action may have unexpected results.

Use this box to search the content in the Messages section:

## NOTE

You can use wildcards, such as "?" for single characters and "\*" for undefined multiple characters, and matched phrase or excluded term searches.

Messages

## AutoSys Workload Automation Messages

While working with AutoSys Workload Automation, you see the following three types of messages:

- Information messages that display information and call for no action.
- Warning messages that warn you of a potential error and may require you to take some action.
- Error messages that tell you what went wrong and what you need to do to resolve it.

Each message is explained giving the exact message string, the reason, and the suggested solution.

## NOTE

If AutoSys Workload Automation cannot locate the catalog files containing the messages, it issues a non-cataloged, generic error.

## Generic Error Resource Bundle Is Not Available

**Resource bundle is not available while looking for key="*catalog\_value*"**

### Reason:

All messages are compiled into the AutoSys Workload Automation message catalog files. The message catalog files are located in the \$AUTOSYS/messages (on UNIX) or %AUTOSYS%\messages (on Windows) directory. If you run the AutoSys Workload Automation product or SDK application without sourcing the environment with the required variables, AutoSys Workload Automation cannot locate the message catalog files.

The "Resource bundle is not available" message is a generic error that can occur for the following reasons:

- You have not set the AUTOSYS environment variable before running AutoSys Workload Automation or the AUTOSYS environment variable is set to an incorrect value.
- The message catalog files are missing from the \$AUTOSYS/messages (on UNIX) or the %AUTOSYS%\messages (on Windows) directory. Missing catalog files indicate an issue with the installation.
- You applied new patched binaries containing new cataloged messages, but you have not patched the message catalog files. Therefore, the new cataloged messages are not in the catalog files.

**Action:**

Perform the appropriate action:

- Set the AUTOSYS environment variable using the shell script (on UNIX) or by executing commands from the instance command prompt (on Windows).

**NOTE**

On Windows, if the AUTOSYS environment variable is set at the System level (by clicking Computer, System properties, Advanced system settings, Environment Variables, System variables), then the AutoSys Workload Automation instance command prompt uses the AUTOSYS environment variable that is set at the System level.

- Reinstall the product or SDK runtime environment.
- Patch the message catalog files in addition to the binaries.

## Information Messages

You may encounter information messages while working with AutoSys Workload Automation. Information messages display information and usually do not require you to take any action.

### CAUAJM\_I\_00004

**New listener started on port [{0,number,1}]. Proceeding..."**

**Reason:**

New listener started on the specified port. Proceeding...

**Action:**

Required action unknown.

### CAUAJM\_I\_00044

**The UNIX function getaddrinfo successfully resolved hostname [{0}] after retrying.**

**Reason:**

Informational only.

**Action:**

No Action required.

### CAUAJM\_I\_00098

**Resending email notifications for job <{0}>.**

**Reason:**

The scheduler is attempting to resend email notifications after ignoring invalid email addresses.

**Action:**

No action required.

### CAUAJM\_I\_00117

**Service Desk request pending JOB: {0}**

**Reason:**

The product is currently in the process of creating a Service Desk request.

**Action:**

No action required.

**CAUAJM\_I\_00119**

**Service Desk request successful REQUEST: {0} REQUEST HANDLE: {1} JOB: {2}**

**Reason:**

The product has successfully opened the Service Desk request.

**Action:**

No action required.

**CAUAJM\_I\_00120**

**Notification Services request pending JOB: {0}**

**Reason:**

The product is currently in the process of sending a Notification Services request.

**Action:**

No action required.

**CAUAJM\_I\_00122**

**Notification Services request successful JOB: {0}**

**Reason:**

The product has successfully sent the Notification Services request.

**Action:**

No action required.

**CAUAJM\_I\_00128**

**Notification Services TRACKING ID: {0} JOB: {1}**

**Reason:**

Unicenter Notification Services tracking identifier.

**Action:**

No action is necessary.

**CAUAJM\_I\_00140**

**High availability system status has returned to normal. Continuing as shadow in high availability mode.**

**Reason:**

Informational only.



**Action:**

No action is required.

**CAUAJM\_I\_00145**

**System is in the failover state due to manual {0} event.**

**Reason:** Informational only.

**Action:** No action required.

**CAUAJM\_I\_00146**

**The {0} application servers were notified to shut down due to manual {1} event.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_00149**

**The shadow has been shutdown. The system is no longer in high availability mode.**

**Reason:**

Information only.

**Action:**

No action necessary.

**CAUAJM\_I\_00150**

**The tie-breaker has been shutdown. The system is no longer in high availability mode.**

**Reason:**

Informational only.

**Action:**

No action necessary.

**CAUAJM\_I\_00151**

**The system is running in high availability mode.**

**Reason:**

Information only.

**Action:**

No action necessary.

## CAUAJM\_I\_00152

The primary has been shutdown. The system is no longer in high availability mode.

**Reason:**

Information only.

**Action:**

No action is required.

## CAUAJM\_I\_00155

The {0} primary scheduler has resumed updating its status.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_00156

System is in the failback state due to manual {0} event.

**Reason:** Informational only.

**Action:** No action required.

## CAUAJM\_I\_00157

The {0} shadow scheduler has resumed updating its status.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_00158

The {0} tie-breaker scheduler has resumed updating its status.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_00162

Service Desk request pending.

**Reason:**

---

The product is currently in the process of creating a Service Desk request.

**Action:**

No action required.

**CAUAJM\_I\_00164**

**Service Desk request successful REQUEST: {0} REQUEST HANDLE: {1}**

**Reason:**

The product has successfully opened the Service Desk request.

**Action:**

No action required.

**CAUAJM\_I\_00183**

**Database indicates remote notification to shut down. The shadow must shut down.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_00184**

**Database indicates remote notification to shutdown. The tie-breaker must shutdown.**

**Reason:**

Informational only

**Action:**

No action required.

**CAUAJM\_I\_00197**

**Database indicates remote notification to fail back. The shadow scheduler must restart.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_00201**

**High availability status <Event Server {0}>**

**Reason:**

Informational only.

**Action:**

---

No action required.

## **CAUAJM\_I\_00202**

**State**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_00203**

**Heartbeat({0})**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_00209**

**High availability system status has returned to failover. Continuing as dormant primary in high availability mode.**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_00210**

**Database indicates remote notification to shut down. The primary scheduler must shut down.**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_00216**

**The shadow scheduler has shut down. The primary is taking over.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_00400

**Native socket libraries successfully loaded.**

**Reason:**

SSA socket libraries are not loaded.

**Action:**

No action required.

## CAUAJM\_I\_10021

**Could not find {0} variable, trying {1}...\n**

**Reason:**

Could not find HOMEPATH variable.

**Action:**

Required action unknown.

## CAUAJM\_I\_10022

**Unable to obtain resources for {0} variable!\n**

**Reason:**

Unable to obtain resources for HOMEPATH variable.

**Action:**

Required action unknown.

## CAUAJM\_I\_10023

**Could not get drive letter for {0} variable!\n**

**Reason:**

Could not get drive letter for HOMEPATH variable.

**Action:**

Required action unknown.

## CAUAJM\_I\_10028

**Unable to obtain resources for {0} and drive variable!\n**

**Reason:**

Unable to obtain resources for HOMEPATH and drive variable.

**Action:**

Required action unknown.

## CAUAJM\_I\_10031

Could not find {0} variable, querying system...\n

**Reason:**

Could not find HOME variable, querying system.

**Action:**

Required action unknown.

## CAUAJM\_I\_10064

The incoming year was not four digits.

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10065

\*\*\*Daylight Savings Change. Time moved up to: {0}

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10066

Event <{0}> successfully sent to external instance <{1}>

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10073

AutoSys --> Cross Platform Interface:\tmachine={0}\t{1}={2}

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10074

{0} {1} for {2} is not supported in this version of CA Workload Automation AE.

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10082

[{0} connected for {1}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10103

start\_rem:read

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10116

Localhost machine definition has been successfully set to {0}.

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10122

Job "{0}" scheduled: {1}

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10149

Job was to start at : {0}

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10153

Setting environment variable: <{0}={1}>

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10154

to run at : {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10165

Previously failed query below succeeded after {0} attempt(s):

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10167

Previously failed scheduler operation [{0}] succeeded after {1,number,1} attempt(s).

**Reason:**

Information only.

**Action:**

No action required.



**CAUAJM\_I\_10168**

Received PAUSE request from the Windows Service Control Manager.

**Reason:**

Informational only.

**Action:**

No action is required.

**CAUAJM\_I\_10169**

Received RESUME request from the Windows Service Control Manager.

**Reason:**

Informational only.

**Action:**

No action is required.

**CAUAJM\_I\_10170**

Received STOP request from the Windows Service Control Manager.

**Reason:**

Informational only.

**Action:**

No action is required.

**CAUAJM\_I\_10173**

**SERVICE STOPPED BY USER:**

**Reason:**

Informational only.

**Action:**

No action is required.

**CAUAJM\_I\_10177**

Service has detected a change in the {0} environment!

**Reason:**

Informational only.

**Action:**

No action is required.

---

## CAUAJM\_I\_10186

No jobs to archive.

**Reason:**

No Jobs found.

**Action:**

Required action unknown.

## CAUAJM\_I\_10188

{0} Scheduler Midnight Log Rollover has occurred

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10189

{0} Scheduler Size Log Rollover has occurred.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10190

{0} Application Server Midnight Log Rollover has occurred.

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_10191

{0} Application Server Size Log Rollover has occurred.

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_10196

**Job:** {0} is using Over-Ride #{1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10197

**{0} Aggregate Statistics Midnight Log Rollover has occurred**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10198

**{0} Aggregate Statistics Size Log Rollover has occurred**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10203

**Machine chooser script for JOB: {0} returned MACHINE: {1}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10208

**Machine usages obtained for job <{0}>:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10209

Machine <{0}> selected for job <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10211

Machine <{0}>: Method <job load>. Current load <{1}>. Available usage <{2}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10214

Machine <{0}>: Method <vmstat>. Available usage <{1}\*[{2}]=<{3}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10215

Machine <{0}>: Method <Windows performance counters>. Available usage <{1}\*[{2}]=<{3}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10218

No machines found for virtual machine: {0}

**Reason:**

No machines for Virtual Machine.

**Action:**

Required action unknown.

## CAUAJM\_I\_10234

Machine <{0}>: Method <rstatd>. Available usage <{1}\*[{2}]=<3}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10239

Jobs which are dependent on the \nDELETED Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10258

Machine <{0}>: Method <CPU monitor>. Available usage <{1}\*[{2}]=<3}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10268

Jobs which are dependent on the deleted job:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10298

Over-Rides are not in effect for Job: {0}

**Reason:**

Over-Rides are not in effect for Job.

**Action:**

Required action unknown.

---

## CAUAJM\_I\_10326

Machine <{0}>: Usage ignored for force starts. Factor is {1}. Availability <unknown>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10327

Machine <{0}>: Usage ignored for force starts. Factor is {1}. Availability <online>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10348

{0} Job: {1} has no Over-Ride in Effect.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10349

{0} Job: {1} has no OverRide# {2}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10350

{0} Over-Ride #{1} Set by User: {2} on [{3}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10351

**{0} Over-Ride CURRENTLY IN EFFECT.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10352

**{0} Was RUN on run\_num={1}, Started on: {2}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10353

**{0} Over-Ride WAS NEVER Used!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10358

**Server: {0} is already connected.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10375

**Job <{0}> is in the {1} status. Respond with a {2} event or end the job with a {3} event.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10378

Job <{0}> is in the {1} status. {2} has no effect.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10379

Job <{0}> is not in the {1} status. {2} has no effect.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10468

Only the {0} {1} superuser can use the {2} utility.

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_I\_10474

Resource manager initialization complete.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10475

Resources Execute Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.



---

## CAUAJM\_I\_10478

**{0} Manager Msg: <{1}>, code: <{2}>**

**Reason:**

Message from DCA Manager

**Action:**

Check the reason for the message and take the appropriate action.

## CAUAJM\_I\_10485

**{0} security session initialized with server <{1}>.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10490

**No Resource dependencies found for input job <{0}>**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10492

**Machine chooser script <{0}> specified for job <{1}>.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10493

**{0} security session terminated.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10534

Attempting to connect to the Agent Service on port={0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10535

Attempting to connect to "{0}" on port={1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10546

Job <{0}> is in the {1} status. Changing the status to {2} has no effect. Send the {3} event instead.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10549

Machine <{0}> selected by Resource Manager.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10553

Machine <{0}>: Method <vmstat>. Usage ignored because this is the only available machine.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10554

**Machine <{0}>: Method <Windows performance counters>. Usage ignored because this is the only available machine.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10555

**Machine <{0}>: Method <rstatd>. Usage ignored because this is the only available machine.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10556

**Machine <{0}>: Method <CPU monitor>. Usage ignored because this is the only available machine.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10573

**Contacting active agents to send updated scheduler communication attributes.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10574

**Completed the sending of updated scheduler communication attributes to agents.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10582

[initiated evaluation for box {0}]

Reason:

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10583

Job <{0}> is in the {1} status. Changing the status to {2} has no effect. Change the status to {3} or {4} or send the {5} event instead.

Reason:

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10633

Reconnection attempt for database <{0}> succeeded.

Reason:

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10647

The {0} Primary Scheduler has rolled over to single server mode.

Reason:

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10654

System is running in dual server mode. Event server 1: {0}. Event server 2: {1}.

Reason:

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10655

System is running in single server mode. Event server: {0}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10659

The {0} Shadow Scheduler has rolled over to single server mode.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10667

Notification completed successfully.

**Reason:**

Information only

**Action:**

Required action unknown.

## CAUAJM\_I\_10686

Switching to a different Application Server. Try contacting the Server again.

**Reason:**

Informational only.

**Action:**

Informational only.

## CAUAJM\_I\_10719

**AUTOEXTVJ Information:**

**Reason:**

Job failed custom validation.

**Action:**

Required action unknown.

## CAUAJM\_I\_10782

[bypassed execution for {0}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10901

The database monitoring system is beginning validation of database connections.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10902

The database monitoring system has completed validation of database connections.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10903

The database monitoring system will resume monitoring for database problems.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10940

Attempting to purge {0} log files older than {1} day(s).

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_10942

**{0} log files purged successfully.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10943

**No {0} log files found matching the purge criteria.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10944

**Successfully purged <{0}>.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10949

**Connection profile Execute Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_I\_10960

**Resource dependency evaluations for job <{0}>:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10961

The resource manager successfully satisfied all resource dependencies for job <{0}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10963

Global virtual resource <{0}>: Amount in use <{1,number,1}>. Available units <{2,number,1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10964

Global virtual resource <{0}> is already reserved from a previous run on machine <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10969

Virtual resource <{0}> on machine <{1}>: Amount in use <{2,number,1}>. Available units <{3,number,1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_10970

Virtual resource <{0}> on machine <{1}> is already reserved from a previous run.

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_10978

Global virtual resource <{0}> is already reserved from a previous run.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18030

Event Server: <{0}> Completed Stored Procedure: <{1}>

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18031

Event Server: <{0}> Completed Query: <{1}>

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18035

Database has initiated the connection failover process.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18036

Database has successfully completed the connection failover process.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18403

Processing OCI function {0}({1})

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18404

SQL block being parsed: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_18417

{0} for {1}

**Reason:**

The user's account is about to expire and the password needs to be changed.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_I\_18608

{0} - Executed Stored Procedure: <{1}>

**Reason:**

Informational only.

**Action:**

No action necessary.

## CAUAJM\_I\_18609

{0} - Executed Query: <{1}>

**Reason:**

Informational only.

**Action:**

No action necessary.

## CAUAJM\_I\_18800

**Server:** {0} is already connected.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_19002

**Job:** <{0}> current run is not in a completion status. Cannot release resource consumption.

**Reason:**

Unable to release resources because the current job run is in STARTING/RUNNING status.

**Action:**

No action is required.

## CAUAJM\_I\_20092

**string** { "CAUAJM\_I\_20092 Connected to the {0} Agent: [{1}:{2,number,1}]" }

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20095

**string** { "CAUAJM\_I\_20095 Connected to the {0} Application Server: [{1}:{2,number,1}]" }

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20197

**{0}** operational on port {1,number,0}.

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_20209

Email notification sent successfully for job <{0}> for alarm <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20210

Email notification sent successfully for job <{0}> for status <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20215

Email notification sent successfully for job <{0}> for {1}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20216

Email notification pending for job <{0}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20218

Executing Command:{0}, DBMaint {1}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20219

**Maintenance Done SUCCESSFULLY!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20228

**Telemetry data sent SUCCESSFULLY!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20352

**{0} Operational on Port {1}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20366

**{0} operational on agent listener port {1,number,0}.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_20367

**{0} operational on auxiliary agent listener port {1,number,0}.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_30000

**{0} Application Server exiting.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30001

**{0} Application Server startup complete.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30004

**{0} Application Server shutdown complete.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30024

**Waiting for Scheduler to notify Rollover.**

**Reason:**

AppServer is waiting for rollover notification from the Scheduler.

**Action:**

No action required.

## CAUAJM\_I\_30030

**Client [{0}][{1}][{2}] [{3}] session established.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30031

Client [{0}][{1}][{2}][{3}] API ID [{4}] execution started.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30032

Client [{0}][{1}][{2}][{3}] API ID [{3}] execution completed. Total time: {4} seconds.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30033

Client [{0}][{1}][{2}] session ended.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30038

Client [{0}] [{1}][{2}] authentication request type [{3}] started.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_30039

Client [{0}] [{1}][{2}] authentication request type [{3}] completed. Total time: {4} seconds.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40000

**JOB:** [{0}] Already running or had run - Will not be restarted.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40002

**Scheduler running in TEST mode level {0}!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40005

**Cross Platform Interface initialization in progress...**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40008

**Comment ignored for REPLY\_RESPONSE event.**

**Reason:**

Comment will not be used for REPLY\_RESPONSE.

**Action:**

Do not specify comment for REPLY\_RESPONSE.

## CAUAJM\_I\_40013

**Received {0} request from the {1} {2} program - {3}**

**Reason:**

Informational only.

**Action:**

No action required.



---

## CAUAJM\_I\_40014

**Unimplemented Event: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40015

**Cross Platform Interface is now active**

**Reason:**

The Cross Platform Interface has been initialized successfully.

**Action:**

No action required.

## CAUAJM\_I\_40024

**No statistics found for the specified time range.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40026

**Unrecognized sender = {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40027

**Request will be ignored....**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40029

**Blank Trigger/External Dependency jobname sent from node {0} -- request ignored!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40030

**Blank Trigger/External Dependency jobset name sent from node {0} -- request ignored!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40031

**Blank Trigger/External Dependency job number sent from node {0} -- request ignored!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40039

**Job Name: {0} is currently in the {1} status.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40043

**Job Name: {0} currently has dependent jobs**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40052

Connection profile definition change.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40055

Skipped record {0} Dependency = {1} Length of name invalid = {2}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40056

Skipped record{0} Length of name invalid = {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40058

{0} criteria has been satisfied for JOB: {1}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40060

Job run for {0} ending at "{1}" meets the scheduled criteria ["{2}", "{3}"] of dependency request from external instance <{4}>. No new dependency established.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40061

[Sending event <{0}>: external dependency acknowledgement for Job: {1} to external instance <{2}>]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40077

SendToConnect: Skipped dependency - dependency name is NULL

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40090

No new {0} statistics available to aggregate. Aggregation skipped.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40091

Aggregated {0} statistics from [{1}] to [{2}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40092

{0} aggregation skipped [{1}] to [{2}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40095

**Deleting all statistics.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40096

**All statistics have been successfully deleted.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40097

**Skipping {0} aggregation of statistics for [{1}] due to DST fallback.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40100

**Evaluating queued job <{0}> for load balancing queue name <{1}>.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40101

**Completed {0} job start(s) for load balancing queue name <{1}>.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_40109

Scheduled [{0}] due to {1} event.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40113

Job starts are commencing for online machine <{0}>. Added {1} job(s).

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40119

Bringing job <{0}> out of the pending state due to online machine <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40120

Completed {0} job start(s) for online machine <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40121

Job starts are in progress for online machine <{0}>. Added {1} job(s).

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40122**

Job <{0}> is already running or completed and cannot be started.

**CAUAJM\_I\_40126**

joid == 0! exit!

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40130**

Sending event <{0}> due to online external instance <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40131**

Completed {0} event send(s) for online external instance <{1}>.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40134**

Job <{0}> remains in the {1} state as it did not meet its load requirements.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40136**

The current status of job <{0}> is {1}. Cannot start job.

**CAUAJM\_I\_40137**

Job <{0}> remains in the {1} state as it did not meet its resource requirements.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40138**

Job <{0}> remains in the {1} state as higher priority jobs are queued.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40139**

The parent box for job <{0}> is not in a {1} state. Cannot start job.

**CAUAJM\_I\_40142**

Provision Successful for Machine {0}.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40148**

Job <{0}> has missed its run window. Sending the {1} event to reschedule the job on its next available start time.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40149**

Cancelling restart for job <{0}> and setting to {1}.

**Reason:**

Informational only.

**Action:**

No action required.



**CAUAJM\_I\_40151**

Job <{0}> cannot start today due to exclusion calendar <{1}>. Setting status to {2}.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40154**

Already scheduled job <{0}>. Cannot start job.

**CAUAJM\_I\_40156**

Conditions do not allow Job to start. Cancelled job start

**CAUAJM\_I\_40159**

The date/time conditions for job <{0}> have not been met. Cannot start job.

**CAUAJM\_I\_40162**

The starting conditions for job <{0}> have not been met. Cannot be started.

**CAUAJM\_I\_40163**

Job <{0}> is outside the run window. Sending the {1} event to reschedule the job on its next available start time.

**CAUAJM\_I\_40164**

Job <{0}> is outside the run window. Sending the {1} event to reschedule the job when the next run window begins.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40167**

Job <{0}> put in an {1} status because global auto hold is active.

**CAUAJM\_I\_40180**

Job <{0}> is not in {1} status. {2} request skipped.

**CAUAJM\_I\_40181**

Cannot terminate job <{0}> because it is not in a {1} state.

**CAUAJM\_I\_40182**

Cannot signal job <{0}> because it is not in a {1} status. Request skipped.

**CAUAJM\_I\_40183**

Cannot signal job <{0}> because the process identifier (PID)for the job is not available. Request skipped.

**CAUAJM\_I\_40188**

Pending job {0} due to offline machine(s).

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40191**

Dequeing job <{0}> and setting to {1}.

**CAUAJM\_I\_40195**

\*\*\*\*\* Global {0} IS ON ! \*\*\*\*\*

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40206**

Job <{0}> has no running process to kill because it bypasses execution. {1} request skipped.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40209**

Job <{0}> has no running process to signal because it bypasses execution. {1} request skipped.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40211**

Using TZ = {0}

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40218**

\*\*\*\* Time offset between the {0} and Event Server {1} is {2} seconds. \*\*\*\*

**Reason:**

Informational only.

**Action:**

Time should be reset on these two machines.

**CAUAJM\_I\_40237**

Machine <{0}> is responsive and so it is automatically placed in the online state.

**Reason:**

The agent on this machine is responsive.

**Action:**

No action required.

**CAUAJM\_I\_40242**

This instance has been configured to use only FIPS-approved security functions. Access to the non-FIPS compliant legacy database password has been disabled.

**Reason:**

The value of the EnableFIPSMODE configuration parameter has been set to 1, and access to the legacy database password has been disabled.

**Action:**

Informational only.

**CAUAJM\_I\_40243**

This instance has been configured to no longer use only FIPS-approved security functions. Access to the non-FIPS compliant legacy database password has been restored.

**Reason:**

The value of the EnableFIPSMODE configuration parameter has been set to 0, and access to the legacy database password has been restored.

**Action:**

Informational only.

**CAUAJM\_I\_40244**

{0} value set to {1}

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40245****EVENT:****Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40246**

{0} scheduler startup complete.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40247**

{0} scheduler processing of events complete.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40248**

{0} scheduler shutdown complete. Exiting.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40249**

Debug level set to <{0}>.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40251**

Service has detected the termination of the {0} Scheduler.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40252**

{0} Service has stopped

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40255**

{0} Operational

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_40256**

[Sending event <{0}>: {1} to {2} for Job: {3} to external instance <{4}>]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40260

{0} scheduler shutdown complete. Waiting at most one poll interval for scheduler acknowledgments...

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40261

Machine <{0}> is reachable. Its status remains manually set to offline.

**Reason:**

The agent on this machine is reachable while its status is manually set to offline.

**Action:**

No action required.

## CAUAJM\_I\_40262

{0} signal caught!

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40263

**EVENTU:**

**Reason:**

Identifies an event received from a Unicenter based agent.

**Action:**

Informational only.

## CAUAJM\_I\_40264

\*\*\*\* Internal Database Maintenance <Date:{0}> \*\*\*\*

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40265

\*\*\*\*<Date:{0}>\*\*\*\*

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40266

**DBMaint is ended with: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40268

**Instance <{0}> automatically brought online due to detected status change.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40275

**Log Rollover level set to {0}.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40276

**STATUS:**

**Reason:**

Identifies a status sent by the Scheduler component which has been converted to an event and will be dispatched to a remote manager.

**Action:**

Informational only.

---

## CAUAJM\_I\_40277

**EVENT:****Reason:**

Identifies an event to be sent to the database.

**Action:**

Informational only.

## CAUAJM\_I\_40285

**{0} scheduler restart aborted.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40286

**{0} scheduler restarting...**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40289

**Machine <{0}> is already in an online status. Nothing to do.**

**Reason:**

The machine is currently in an online status. The scheduler ignores the request to place it online.

**Action:**

Informational only.

## CAUAJM\_I\_40295

**{0} set to <{1}>**

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_40300

Starting {0} {1} aggregation of product statistics.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40301

Completed {0} {1} aggregation of product statistics.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40314

Request to cancel status for job {0} on external instance {1} cannot be sent as other dependencies exist.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40315

Cross Platform Interface processing terminated.

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_40318

{0} shadow scheduler active.

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_40319

**{0} tie-breaker scheduler active.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40323

**Reestablished connection with remote application ({0},{1}).**

**Reason:**

Reestablished connection with the identified remote application.

**Action:**

Informational only.

## CAUAJM\_I\_40334

**{0} primary scheduler active in dormant failback mode.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40335

**{0} primary scheduler active in immediate failback mode.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40340

**Completed aggregation of product statistics.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_40356

**SNMP trap generation : {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40362

**Request for status for job {0} on external instance {1} has already been sent.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40378

**One Automation Event Management messaging interface initialized successfully.**

**Reason:**

Informational only.

**Action:**

No action is required.

## CAUAJM\_I\_40447

**Placing job <{0}> into {1} state as it did not meet its resource requirements.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40448

**Placing job <{0}> into {1} state as it did not meet its load requirements.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_40449

Placing job <{0}> into {1} state as higher priority jobs are queued.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50010

The Primary Scheduler is not {0}\n

**Reason:**

The primary scheduler is not RUNNING.

**Action:**

Required action unknown.

## CAUAJM\_I\_50011

The Shadow Scheduler is not {0}\n

**Reason:**

The Shadow Scheduler is not RUNNING.

**Action:**

Required action unknown.

## CAUAJM\_I\_50012

The Tie-Breaker Scheduler is not {0}\n

**Reason:**

The Tie-Breaker Scheduler is not RUNNING.

**Action:**

Required action unknown.

## CAUAJM\_I\_50019

\7 \*\*\* Response required \*\*\*\n\n

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50023**

**AutoPinging Machine [{0}]**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50025**

**AutoPing WAS SUCCESSFUL.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50028**

**Checking the Agent's connectivity to the Event Server.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50030**

**Successfully added timezone {0}**

**Reason:**

Successfully added timezone.

**Action:**

Required action unknown.

**CAUAJM\_I\_50031**

**Checking the Agent's connectivity to the Application Server.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50038

Checking {0} on Machine: {1}\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50039

No Scheduler is {0} on machine: {1}\n

**Reason:**

The Scheduler was not running on the remote machine.

**Action:**

Required action unknown.

## CAUAJM\_I\_50040

Could Not inspect machine: {0}\n

**Reason:**

Error connecting to remote machine.

**Action:**

Required action unknown.

## CAUAJM\_I\_50042

Cannot check the status of the Scheduler because the Event Server(s) is not available\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50044

Primary Scheduler is {0} on machine: {1}\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50045

Shadow Scheduler is {0} on machine: {1}\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50046

\*\*\* The Shadow Scheduler on {0} appears to have TAKEN OVER \*\*\*\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50047

Tie-Breaker Scheduler is {0} on machine: {1}\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50049

No Scheduler is {0}\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50050

Timezone {0} already in the database, request denied.

**Reason:**

Timezone is already in the database.

**Action:**

Required action unknown.

**CAUAJM\_I\_50051**

Unable to obtain Scheduler information\n

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50053**

Timezone {0} does not exist in the database, request denied.

**Reason:**

Timezone does not exist in the database.

**Action:**

Required action unknown.

**CAUAJM\_I\_50054**

Attempting ({0}) to Connect with Database: {1}\n

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50055**

\*\*\* Have Connected successfully with Database: {0}. \*\*\*\n

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50057**

\*\*\* There was an Automatic DB Rollover. \*\*\*\n\n

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_50065

Cleaning log files on machine: {0}...\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50067

Removed {0} of {1} total log files on machine {2}.\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50068

(Could not remove {0} log file{1} because the corresponding job{2} still running.)\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50069

Successfully deleted timezone {0}

**Reason:**

Successfully deleted timezone.

**Action:**

Required action unknown.

## CAUAJM\_I\_50071

Cannot delete timezone {0}, used in one or more jobs!

**Reason:**

Cannot delete timezone.

**Action:**

Required action unknown.

## CAUAJM\_I\_50072

Cannot delete timezone {0}, used by one or more Alias/City!

**Reason:**

Cannot delete the timezone.

**Action:**

Required action unknown.

## CAUAJM\_I\_50086

Interrupted... Exiting.\n

**Reason:**

The monbro program has caught a signal to come down.

**Action:**

No action required.

## CAUAJM\_I\_50089

**Timezones:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50092

No time zones were found in the database that matched the request.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50096

Current {0}(internal): {1,number,0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50097

**External Time:** {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50099

**{0}(internal):** {1,number,0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50121

**Successfully added alias** {0}

**Reason:**

Successfully added timezone.

**Action:**

Required action unknown.

## CAUAJM\_I\_50122

**Successfully added city** {0}

**Reason:**

Successfully added timezone.

**Action:**

Required action unknown.

## CAUAJM\_I\_50128

**Connected with Event Server:** {0}\n\n

**Reason:**

Successfully connected to the event server.

**Action:**

No action required.

## CAUAJM\_I\_50129

**Monitoring {0} :**\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50133

**Monitoring {0} output file: {1}\n**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50134

**\*\*\* To break out type control-c (^c) \*\*\*\n\n**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50150

**{0} Level={1}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50151

**Override definition change.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50152

{0} issued.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50154

**Running the Chase Process.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50155

**Chase is done!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50156

**There are no Jobs in a {0} or {1} state!\n**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50157

**Chase is successful.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50158

Checking on Jobs in the {0} state...\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50159

There are NO Jobs that have been in the {0} state

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50160

more than {0} Seconds.\n\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50161

Examining Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50162

Job has been in the {0} state more than {1} Seconds. Manual intervention may be required.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50164

There are NO Jobs in the {0} state.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50166

Examining Job: {0} On Machine: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50167

Machine definition change.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50171

{0} change.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50172

OK! Both the {0} daemon and the job are still running.\n

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50173

The {0} job is still running.\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50174

{0} daemon MAY be in the process of sending the final event back. Check again in a few seconds.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50175

An event for this Job has come in during Examination... Everything is fine.\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50176

Can not determine the status of {0}. Manual investigation is necessary.\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50179

\*\*\* ALARM Sent! \*\*\*\n

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_50181

\*\*\* {0} to {1} sent! \*\*\*\n

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50189

**Monitor/browser definition change.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50190

**Calendar definition change.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50191

**Job definition change.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50192

**Invalid audit type "{0}" detected.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50200

**Attempting to Delete Job: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50202

**Delete was successful!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50204

**Inserting/Updating Job: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50205

**Database Change WAS Successful!**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50207

**Over-Ride turned OFF!**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_50208

Installing Over-Rides for Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50209

[Over-Ride# {0} IS IN EFFECT!]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50210

Checking Syntax for Insert/Update Job Type: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50211

Inserting/Updating Job Type: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50212

Checking Syntax for Delete Job Type: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50213

**Deleting Job Type: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50218

**Deleting {0}: {1}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50219

**\*\*\*{0} {1} does not exist! Nothing to Delete!\*\*\***

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50222

**Inserting/Updating {0}: {1}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50227

**Deleting machine: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50229

Component Machine: {0} deleted...

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50232

Inserting/Updating Machine: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50239

{0} interrupted... Exiting.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50240

{0} interrupted... Exiting after work in progress is finished.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50243

{0} is {1} on machine: {2}\n

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50244**

{0} is NOT {1} on machine: {2}\n

**Reason:**

CA Workload Automation AE Application Server is not running on the remote machine.

**Action:**

Required action unknown.

**CAUAJM\_I\_50245**

Checking Syntax for Insert/Update Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50247**

JIL syntax check PASSED

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50248**

Checking Syntax for Delete Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_50249**

Checking Syntax for Delete Machine: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50250

Checking Syntax for Insert/Update Machine: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50251

Checking Syntax for Insert/Update {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50253

Checking Syntax for Delete {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50255

JIL syntax check PASSED WITH WARNINGS

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50259

Cancelling override for Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50260

Checking Syntax for Cancel Over-Ride for Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50261

Checking Syntax for Install Over-Ride for Job: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50262

Inserting {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50263

Checking Syntax for Insert {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50264

Total versions of input blobs for job {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_50273

Total versions of output blobs for job {0}: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50280

AutoPing from the Scheduler WAS SUCCESSFUL.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50282

AutoPing from the Application Server WAS SUCCESSFUL.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50284

AutoPing from the Application Server SKIPPED.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50286

Job in {0} state, but the user process is not running on machine: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50287

**ERROR:** File watcher in {0} state, but the file is not being watched by the agent on machine: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50288

The {0} job is in {1} state, but the job is not running on machine: {2}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50291

**Possible chase anomaly:** The agent is running, but the job is not running on machine: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50297

**Job has been in the {0} state for more than {1} minutes. Manual intervention may be required.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50298

**Job and agent are not {0}. chase is sending a {1} event.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50299

Job is not {0}. chase is sending a {1} event.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50322

Checking Syntax for {0} {1}: {2}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50323

{0} {1}: {2}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50326

No dates entered. Calendar {0} was not created.

**Reason:**

The calendar was not inserted into the database.

**Action:**

No action required.

## CAUAJM\_I\_50337

Standard calendar <{0}> is used as holiday calendar for extended calendar <{1}>, jobs depending on <{1}> will be rescheduled.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50339

Cycle <{0}> is used in extended calendar <{1}>, jobs depending on <{1}> will be rescheduled.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50340

{0} Job {1} referred in the external instances job. External Instance and Job {2} referred are listed below:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50400

Insert Successful. Date added to Calendar: {0}

**Reason:**

The date was added to the calendar definition in the database.

**Action:**

No action required.

## CAUAJM\_I\_50402

Date already exists in calendar.

**Reason:**

The input date already exists in calendar.

**Action:**

No action required.

## CAUAJM\_I\_50403

Select Failed.\n\n.\n.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50404

No jobs are currently running on this machine.

**Reason:**

Informational only.

**Action:**

No action necessary.

## CAUAJM\_I\_50407

Reading external instance information

**Reason:**

Reading external instance information.

**Action:**

No action required.

## CAUAJM\_I\_50408

Instance=[{0}]: Type=[{1}] Server=[{2}] Port=[{3,number,0}] Manager Alias=[{4}]

**Reason:**

All valid external instance definitions are displayed when the scheduler is started.

**Action:**

No action required.

## CAUAJM\_I\_50410

Instance=[{0}]: Type=[{1}] Server=[{2}] Port=[{3,number,0}]

**Reason:**

All valid external instance definitions are displayed when the scheduler is started.

**Action:**

No action required.

## CAUAJM\_I\_50423

Insert Successful for Extended Calendar: {0}

**Reason:**

The calendar was successfully inserted into the database.

**Action:**

No action required.

## CAUAJM\_I\_50424

**Insert Failed for Extended Calendar: {0}**

**Reason:**

The calendar was not inserted into the database.

**Action:**

Check the status of the database. Check that the extended calendar is valid.

## CAUAJM\_I\_50425

**Update Successful for Extended Calendar: {0}**

**Reason:**

The calendar definition was successfully updated in the database.

**Action:**

No action required.

## CAUAJM\_I\_50426

**Update Failed for Extended Calendar: {0}**

**Reason:**

The calendar definition was not updated in the database.

**Action:**

Check that the calendar definition entered was valid.

## CAUAJM\_I\_50427

**Delete Successful for Extended Calendar: {0}**

**Reason:**

The calendar was successfully deleted from the database.

**Action:**

No action required.

## CAUAJM\_I\_50428

**Delete Failed for Extended Calendar: {0}**

**Reason:**

The calendar was not successfully deleted from the database.

**Action:**

Check that the calendar name refers to an existing calendar.

## CAUAJM\_I\_50429

**Date(s) deleted from Calendar: {0}**

**Reason:**

The calendar date was successfully deleted from the database.

**Action:**

No action required.

## CAUAJM\_I\_50430

**Insert Successful for Calendar: {0}**

**Reason:**

The calendar was successfully inserted into the database.

**Action:**

No action required.

## CAUAJM\_I\_50431

**Insert Failed for Calendar: {0}**

**Reason:**

The calendar was not inserted into the database.

**Action:**

Check that the calendar definition has a valid name and date set.

## CAUAJM\_I\_50432

**Insert Failed. Date was not added to Calendar: {0}**

**Reason:**

The date was not added to the calendar definition in the database.

**Action:**

Check that the date is valid.

## CAUAJM\_I\_50434

**Insert Failed for Cycle: {0}**

**Reason:**

The cycle period was not added to the calendar definition.

**Action:**

Refer to the UAJM Reference Guide for further information on cycles.

## CAUAJM\_I\_50435

**Regenerate dates Successful for Extended Calendar: {0}**

**Reason:**

The calendar was successfully regenerated and added to the database.

**Action:**

No action required.

## CAUAJM\_I\_50436

**Regenerate dates Failed for Extended Calendar: {0}**

**Reason:**

The calendar dates failed to regenerate. No changes were made to the database.

**Action:**

No action required.

## CAUAJM\_I\_50437

**Insert Successful for Cycle: {0}**

**Reason:**

The cycle period was added to the cycle calendar definition.

**Action:**

No action required.

## CAUAJM\_I\_50438

**Delete Successful for Calendar: {0}**

**Reason:**

The calendar was successfully deleted from the database.

**Action:**

No action required.

## CAUAJM\_I\_50439

**Insert Successful for Cycle: {0}**

**Reason:**

The cycle calendar was successfully inserted into the database.

**Action:**

No action required.



## CAUAJM\_I\_50440

**Insert Failed for Cycle: {0}**

**Reason:**

The cycle calendar was not inserted into the database.

**Action:**

Check that the calendar definition has a valid name and valid dates.

## CAUAJM\_I\_50441

**Delete Successful for Cycle: {0}**

**Reason:**

The cycle calendar was successfully deleted from the database.

**Action:**

No action required.

## CAUAJM\_I\_50503

**The agent may be in the process of sending an event back. Check the status of the Scheduler.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_50517

**The description is modified successfully for calendar: {0}**

**Reason:**

The calendar description was modified successfully.

**Action:**

No action required.

## CAUAJM\_I\_50518

**The description modification failed for calendar: {0}**

**Reason:**

The calendar description did not get updated. No changes were made to the database.

**Action:**

Required action unknown.

---

## CAUAJM\_I\_52301

**Exit Code = {0}**

**Reason:**

Information only.

**Action:**

No action required

## CAUAJM\_I\_60000

**Job ({0}) is already running. {1} has no effect.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60001

**Job ({0}) has already started. {1} has no effect.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60006

**Archiving Checkpoints...**

**Reason:**

Information only, archival beginning

**Action:**

No action required.

## CAUAJM\_I\_60007

**Archiving Events...**

**Reason:**

Information only, archival beginning.

**Action:**

No action required.

---

## CAUAJM\_I\_60013

**Number of Events Deleted:** {0,number,1}

**Reason:**

Indicates the number of events that were deleted.

**Action:**

No action required.

## CAUAJM\_I\_60014

**Archiving Job Runs...**

**Reason:**

Information only. Job run archival is beginning.

**Action:**

No action required.

## CAUAJM\_I\_60017

**Number of job runs deleted:** {0,number,1}

**Reason:**

Indicates the number of job runs that were deleted.

**Action:**

No action required.

## CAUAJM\_I\_60018

**Archiving Audit Info...**

**Reason:**

Informational only, audit info archival is beginning.

**Action:**

No action required.

## CAUAJM\_I\_60019

**dbstatistics: Running dbstatistics at:** {0}

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60020

**dbspace:** Running dbspace at : {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60021

**Number of {0} Deleted:** {1,number,1}

**Reason:**

Indicates the number of audit infos that were deleted.

**Action:**

No action required.

## CAUAJM\_I\_60022

**The {0} syntax is:**\n\n\t{1}, or\n\t {2} to Delete the Global.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60031

{0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60032

**Un-Send EVENT eoid:** {0} **Event Time:** {1}

**Reason:**

Un-Send EVENT eoid.

**Action:**

Required action unknown.

---

## CAUAJM\_I\_60051

**Running Scheduler(s) have been detected.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60052

**Shutdown the Scheduler(s) before changing the remote authentication method.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60056

**Shutdown the Scheduler(s) before changing the database password.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60059

**Password change successful.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60062

**Password= {0}**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60063

Initializing {0} {1} and {2} {3}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60064

Initializing {0} {1} {2}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60068

Request ignored.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60069

User successfully added.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60074

Please try again.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60081**

**Try a different password.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60084**

**Restoring current password.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60087**

**Current remote authentication method:**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60088**

**No remote authentication.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60089**

**Agent user authentication only.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60090**

**Scheduler authentication only.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60091**

**Both Agent and Scheduler authentication.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60092**

**Unix ruserok authentication only.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60093**

**Remote Authentication is not supported for type "a" machines or agents.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60094**

**Both Unix ruserok and Scheduler authentication.**

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_60098

Remote authentication method change successful.

**Reason:**

Informational only.

**Action:**

Required action unknown.

## CAUAJM\_I\_60104

Use {0} option to create user@host password.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60105

This user and the attribute already exist.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60106

Use option {0} to change the user.

**Reason:**

Informational only.

**Action:**

Required action unknown.

## CAUAJM\_I\_60107

Use {0} option to change password.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60112**

**No user name entered. Canceling request.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60114**

**Domain successfully added.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60116**

**No host or domain name entered. Canceling request.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60119**

**User successfully deleted.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60120**

**Enter the new host or domain for user {0} (or hit enter if no host):**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60123

**User successfully modified.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60125

**Domain successfully deleted.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60126

**Use option {0} to create the user.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60135

**User create successful.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60137

**User password change successful.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60139

**User delete successful.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60140

**Listing all password users:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60141

**No users found for this instance.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60148

**Valid values are {0}, {1}, {2} or {3}.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60150

**Attempting to regenerate the CA EEM certificate and reconnect to the CA EEM server.**

**Reason:**

Informational only

**Action:**

No action required.

## CAUAJM\_I\_60151

Instance security is set to {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60153

Listing {0} cached credentials:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60154

No {0} cached credentials exist.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60155

Deleting {0} cached credentials: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60162

{0} cached credentials successfully deleted.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60164

Format must be \"{0}\" where the timeout is specified in days ({1}) and hours ({2}), or \"{3}\".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60170

Valid values are between {0} and {1} when hour values other than 00 are specified. Otherwise, specify a value between {2} and {3}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60171

Valid values are between {0} and {1} when day values other than 00 are specified. Otherwise, specify a value between {2} and {3}.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60175

No password vault name entered. Canceling request.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60176

Password vault add successful.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60177

**Listing all password vaults:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60178

**No password vault found for this instance.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60179

**Password vault delete successful.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60181

**Use the Manage users with password vault menu.**

**Reason:**

User credential is of password vault type.

**Action:**

Use the Manage users with password vault menu.

## CAUAJM\_I\_60182

**Modify a user with alias: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60183

**Listing all password vault users:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60185

**User password will be managed through password.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60423

**Please select password vault type from the following options:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60429

**Create a user with password vault: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60451

**Try a different alias.**

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_I\_60191

The {0} server location was changed successfully.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60195

This instance remains under NATIVE security control due to end-user response.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60200

{0} certificate generated successfully.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60201

{0} instance security successfully set.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60202

NATIVE instance security successfully set.

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60205

**Listing all {0} {1} superusers:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60206

**No {0} {1} superusers are defined.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60207

**Create an {0} {1}: {2}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60208

**Modify a {0} {1}: {2}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60209

**Delete an {0} {1}: {2}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60210

**Security setting:** {0} **Policy User:** {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60216

**Security check FAILED.**

**Reason:**

The resulting security check with the supplied parameters failed.

**Action:**

Informational only.

## CAUAJM\_I\_60217

**Security check PASSED.**

**Reason:**

The resulting security check with the supplied parameters passed.

**Action:**

Informational only.

## CAUAJM\_I\_60226

**Enter the CA Workload Automation AE release level from which you imported the security policies [enter to cancel]:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60228

**CA EEM server:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60229

No application instance security policy is installed.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60230

Enter CA EEM server host name [{0}]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60231

Connection successful: {0} User: {1} Application: {2}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60232

Enter the instance name [enter to cancel]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60233

Usage: <class> <resource> <access> <user>

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60235

**Input:** [enter to cancel] >

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60241

Successfully upgraded CA EEM application instance \"{0}\" to version {1}

## CAUAJM\_I\_60242

Policy granting access: [{0}] delegator: [{1}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60243

Class: {0} Resource: {1} User: {2} Access: {3}

**Reason:**

Information regarding the access check.

**Action:**

No action required.

## CAUAJM\_I\_60244

Policy denying access: [{0}] delegator: [{1}]

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60245

Enter the \"{0}\" account password [enter to cancel]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60248

Successfully registered CA EEM application instance \"{0}\".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60252

Successfully created security policies for instance {0}!

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60253

{0} instance(s) currently registered.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60257

Successfully deleted security policies for instance {0}!

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60259

Successfully unregistered CA EEM application instance \"{0}\".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60262

Enter the name of an administrative user [{0}]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60264

Add privileges for user:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60265

Remove privileges for user:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60266

Administrative privileges successfully added for user \"{0}\

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60268

Administrative privileges successfully removed from user \"{0}\

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60270

User \"{0}\" is the only user with administrative privileges to the CA EEM application instance \"{1}\".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60273

Successfully generated an authentication certificate for instance {0}!

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60275

authentication artifact:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60276

Enter the artifact [enter to cancel]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60278

Artifact Authentication passed! User account: \"{0}\"

**Reason:**

Informational only

**Action:**

No action required.



---

## CAUAJM\_I\_60279

Upgrading common security policies for all instances

## CAUAJM\_I\_60280

Upgrading security policies for instance \"{0}\"

## CAUAJM\_I\_60281

Administrative policy '{0}' exists

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60282

Folder '{0}' exists

## CAUAJM\_I\_60288

CA EEM server host name is already set to \"{0}\". No further action is necessary.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60290

{0} credentials generated and cached successfully.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60292

Successfully updated the security policies.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60296

Updating the actions for security policies that are migrated from the CA EEM application instance \"{0}\".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60298

The provided CA Workload Automation AE release level \"{0}\" is at the current level. Hence, an update after migration is not required.

**Reason:**

The provided release level is already at an updated level.

**Action:**

No action required.

## CAUAJM\_I\_60301

Use option \"{0}\" on the main menu to set and connect the CA EEM server.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60302

Enter directory location to store certificate file [{0}]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60303

Upgrading version of CA EEM application instance \"{0}\"

## CAUAJM\_I\_60304

Certificate location: {0}

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60306**

Obtaining new certificate from the CA EEM application instance \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60307**

Creating UserGroup \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60308**

Creating folder \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60309**

Creating default policy \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60310**

Creating administrative policy \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60311

Removing folder "{0}"

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60312

Starting registration of the CA EEM application instance \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60313

Removing registration of the CA EEM application instance \"{0}\"

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60314

Adding \"{0}\" resource class

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60318

Attempting connection to CA EEM server: {0}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60320

Default policy '{0}' exists

## CAUAJM\_I\_60326

Currently connected to the CA EEM server as user "{0}".

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60327

Enter application instance [1({0})/0({1})]:

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60338

Restricting access to all unauthenticated users is the most secure option but client utilities may become unusable. Also, the scheduler will not execute the database maintenance script. You must define a job to execute the script and generate cached credentials for the job owner on the machine where the job runs.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60339

Qualifying unauthenticated remote users with a computer name assists in distinguishing those users from authenticated users.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60342

Unauthenticated user mode: {0}.

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60350**

**Attempting {0} against {1}**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60353**

**Job profile migration started...**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60355**

**Phase 1 initiated (locate release 11.0 job profiles)**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60357**

**Phase 2 initiated (locate any pre-release 11.0 job profiles)**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60358**

**Phase 3 initiated (migrate release 11.0 to flat files)**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60359**

### **Migrating job profiles**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60361**

### **Phase 4 initiated (migrate pre-release 11.0 to flat files)**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60362**

### **Migrating job profiles for instance <{0}>**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60363**

### **Phase 5 initiated (summary)**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_I\_60364**

### **Job profiles found (release 11.0) :{0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60365

**Job profiles found (pre-release 11.0) :{0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60366

**Total job profiles found :{0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60367

**Job profiles migrated (release 11.0) :{0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60368

**Job profiles migrated (pre-release 11.0) :{0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60369

**Total job profiles migrated :{0}**

**Reason:**

Informational only.

**Action:**

No action required.



**CAUAJM\_I\_60373**

Job profile <{0}> written to file < {1} >

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60375**

Job profile <{0}> is found.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60376**

Attempting to locate job profiles for instance <{0}>

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60378**

Successfully created shared security policies.

**CAUAJM\_I\_60380**

Job profile migration complete.

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60382**

No Job profiles found.

**Reason:**

Informational only.

**Action:**

---

No action required.

### **CAUAJM\_I\_60383**

**Merge successfully completed.**

**Reason:**

Informational only.

**Action:**

No action required.

### **CAUAJM\_I\_60384**

**Deploying DEFAULT profile environment variables as agent wide variables in agentparm.txt...**

**Reason:**

Informational only.

**Action:**

No action required.

### **CAUAJM\_I\_60385**

**Deployment completed successfully.**

**Reason:**

Informational only.

**Action:**

No action required.

### **CAUAJM\_I\_60389**

**The file {0} does not contain any Sendevent commands.**

**Reason:**

Incorrect file specified.

**Action:**

Specify correct file.

### **CAUAJM\_I\_60390**

**No domain name entered. Canceling request.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60391

**Listing all trusted domains.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60392

**No trusted domains are defined.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60393

**Add a trusted domain: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60395

**Delete a trusted domain: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60398

**Successfully changed unauthenticated user mode to {0}.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60399

**You must build server and client authentication libraries and install them on computers with server and client installations.**

**Reason:**

When the unauthenticated user mode setting is EXTERNAL, the product invokes routines in custom server and client libraries to authenticate users.

**Action:**

Ensure that custom libraries are installed on client and server machines, or change the unauthenticated user mode to a setting that does not require them.

## CAUAJM\_I\_60402

**Invalid response.**

**Reason:**

The selection was not in the range of valid selections.

**Action:**

Select a value in the specified range.

## CAUAJM\_I\_60405

**Use the Manage users with key credentials menu.**

**Reason:**

User is of key credentials type.

**Action:**

Use the Manage users with key credentials menu.

## CAUAJM\_I\_60406

**Use the Manage users with password menu.**

**Reason:**

User is of password type credentials.

**Action:**

Use the Manage users with password menu.

## CAUAJM\_I\_60415

**Try a different security code.**

**Reason:**

Informational only.

**Action:**

No action required.

---

## CAUAJM\_I\_60416

**Create a user with password: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60417

**Modify key credential user security code:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60418

**Modify a user with key credentials: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60420

**Listing all key credential users:**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_60425

**Modify a user with password: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

---

**CAUAJM\_I\_60426**

**Delete a user: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60427**

**No key credential file path entered. Canceling request.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_60428**

**Create a user with key credentials: {0}**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_I\_80007**

**Cross Platform Interface proceeding to shutdown.**

**Reason:**

The Cross Platform Interface is proceeding to shutdown due errors.

**Action:**

Informational only.

**CAUAJM\_I\_80014**

**Attempting resend status to PRIMARYCCISYSID {0}**

**Reason:**

Send failure occurred to primary node. Attempting resent to PRIMARYCCISYSID node.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_I\_80017

Invalid input: {0} found. Ignoring!

**Reason:**

Information only.

**Action:**

No action required

## CAUAJM\_I\_80018

Invalid value: {0} specified for {1}.

**Reason:**

Invalid value specified.

**Action:**

Valid values are yes, YES, y, Y, no, NO, n and N

## CAUAJM\_I\_80021

The time taken for updating the status of {0} agents is {1} seconds and {2} microseconds.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_80022

The agent status monitor has set the status of the agent on host {0}:{1} to online.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_I\_80023

The agent status monitor has set the status of the agent on host {0}:{1} to offline.

**Reason:**

Informational only.

**Action:**

No action required.

## Warning Messages

You may encounter warning messages while working with AutoSys Workload Automation. Warning messages warn you of a potential error situation and may require you to take some action.

### CAUAJM\_W\_00002

**Listener could not be started on port [{0,number,1}]. Proceeding...**

**Reason:**

Listener could not be started on the specified port. Proceeding...

**Action:**

Required action is not known.

### CAUAJM\_W\_00018

**Error retrieving network data from socket [{0,number,1}]. Discarding socket...**

**Reason:**

Error retrieving network data from the specified socket. Discarding socket...

**Action:**

Required action unknown.

### CAUAJM\_W\_00019

**Error retrieving acknowledgement from socket [{0,number,1}]. Discarding socket..."**

**Reason:**

Error retrieving acknowledgement from the specified socket. Discarding socket...

**Action:**

Required action unknown.

### CAUAJM\_W\_00043

**The UNIX function getaddrinfo did not respond in a timely fashion for hostname [{0}]. Retrying...**

**Reason:**

The UNIX function getaddrinfo is experiencing long delays in resolving the hostname into a network IP address or the system clock was artificially advanced by a large increment.

**Action:**

Check the network or name server settings on the computer to troubleshoot delays in resolving the hostname. If you are utilizing any system clock synchronization algorithms, configure them so that the system clock is not advanced by a large increment.

### CAUAJM\_W\_00060

**Timed out waiting for incoming data at socket [{0,number,1}] after {1,number,1} seconds. Discarding socket...**

**Reason:**



---

Timed out waiting for incoming data at the specified socket after the specified number of seconds. Discarding socket...

**Action:**

Required action unknown.

## CAUAJM\_W\_00068

Timed out waiting for acknowledgement at socket [{0,number,1}] after {1,number,1} seconds. Discarding socket...

**Reason:**

Timed out waiting for acknowledgement at the specified socket after the specified number of seconds. Discarding socket...

**Action:**

Required action unknown.

## CAUAJM\_W\_00093

No email notifications were sent for job <{0}>.

**Reason:**

The email server was unable to forward mail to any of the email addresses associated with the specified job.

**Action:**

Verify all the email addresses and update the job definition.

## CAUAJM\_W\_00097

Ignoring email address <{0}> on behalf of job <{1}> due to error: <{2}>

**Reason:**

The email server was unable to forward mail to this email address on behalf of the specified job.

**Action:**

Verify the email address and update the job definition.

## CAUAJM\_W\_00127

Unable to obtain Notification Services tracking ID JOB: {1}

**Reason:**

Unable to obtain Notification Services tracking ID.

**Action:**

Check extended diagnostic messages for further indication of the problem.

## CAUAJM\_W\_00129

Unable to obtain host name. Notification header will be incomplete.\n

**Reason:**

Unable to obtain host name. Will not be able to set detailed machine information in Notification Services header.

**Action:**

No action required.

## CAUAJM\_W\_00130

**No job\_runs entry available to store Service Desk handle REQUEST HANDLE: {0} JOB: {1}**

**Reason:**

There is no job\_runs entry available for this job. A job that has a bad owner/password or bad machine will not generate a job\_runs entry.

**Action:**

No action required.

## CAUAJM\_W\_00134

**The {0} shadow scheduler has failed to update its status. Please investigate.**

**Reason:**

The shadow scheduler has failed to update its status. Please investigate.

**Action:**

Determine why the shadow did not update its status in the database and restart it if necessary.

## CAUAJM\_W\_00135

**The {0} tie-breaker scheduler has failed to update its status. Please investigate.**

**Reason:**

The tie-breaker scheduler has failed to update its status. Please investigate.

**Action:**

Determine why the tie-breaker did not update its status in the database and restart it if necessary.

## CAUAJM\_W\_00136

**Database indicates failover status. The shadow is taking over.**

**Reason:**

Database indicates failover status. The shadow is taking over.

**Action:**

Determine problem that caused failover and restart system.

## CAUAJM\_W\_00137

**The primary appears to be missing. Changing high availability system status to pending.**

**Reason:**

The primary appears to be missing. Changing high availability system status to pending.

**Action:**

Determine why the primary did not update its status in the database and restart it if necessary.

## CAUAJM\_W\_00138

**Changing high availability system status to failover.**

**Reason:**

The primary is not updating database.

**Action:**

Investigate the state of the primary.

## CAUAJM\_W\_00139

**Unable to mark high availability system as failover. Continuing as shadow.**

**Reason:**

Unable to mark high availability system as failover. Continuing as shadow.

**Action:**

Attempt to mark database as failover returned a non-failover status.

## CAUAJM\_W\_00141

**Unable to mark high availability system as pending. Continuing as shadow.**

**Reason:**

Unable to mark high availability system as pending. Continuing as shadow.

**Action:**

Attempt to mark database as pending returned a non-pending status.

## CAUAJM\_W\_00142

**Successfully changed high availability system status. The shadow is taking over.**

**Reason:**

Successfully changed high availability system status. The shadow is taking over.

**Action:**

Determine cause of failover and restart or failback system.

## CAUAJM\_W\_00143

**Database indicates failback status. Restarting in shadow mode.**

**Reason:**

Database indicates failback status. Restarting in shadow mode.

**Action:**

Determine problem that caused failback.

## CAUAJM\_W\_00144

**The {0} shadow scheduler appears to have failed over prematurely. Consider a higher value for the HAPollInterval setting. Proceeding with shutdown...**

**Reason:**

The shadow scheduler appears to have failed over prematurely. Consider a higher value for the HAPollInterval setting. Proceeding with shutdown...

**Action:**

After recovery from failover, set a higher value for the HAPollInterval across all schedulers.

## CAUAJM\_W\_00147

**The primary and tie-breaker are both missing. Database indicates shadow can fail over. Changing high availability system status to pending.**

**Reason:**

The primary and tie-breaker are both missing. Database indicates shadow can fail over. Changing high availability system status to pending.

**Action:**

Determine why the primary and tie-breaker did not update their statuses in the database and restart it if necessary.

## CAUAJM\_W\_00153

**The database indicates that a failover to the shadow has occurred. The system is no longer in high availability mode.**

**Reason:**

The database indicates that a failover to the shadow has occurred. The system is no longer in high availability mode.

**Action:**

No action necessary.

## CAUAJM\_W\_00154

**The {0} primary scheduler has failed to update its status. Please investigate.**

**Reason:**

The primary scheduler has failed to update its status. Please investigate.

**Action:**

Determine why the primary did not update its status in the database and restart it if necessary.

## CAUAJM\_W\_00163

**System does not appear to be in high availability mode. Skipping failover and shutting down.**

**Reason:**

System does not appear to be in high availability mode. Skipping failover and shutting down.

**Action:**

---

Wait until the system enters high availability mode before sending the STOP\_DEMON with the value specified as FAILOVER.

## **CAUAJM\_W\_00167**

**The primary's status is in question. The system is no longer in high availability mode.**

**Reason:**

The primary's status is in question. The system is no longer in high availability mode.

**Action:**

Determine the problem with the primary scheduler and restart it if necessary.

## **CAUAJM\_W\_00168**

**The shadow's status is in question. The system is no longer in high availability mode.**

**Reason:**

The shadow's status is in question. The system is no longer in high availability mode.

**Action:**

Determine the problem with the shadow scheduler and restart it if necessary.

## **CAUAJM\_W\_00169**

**The tie-breaker's status is in question. The system is no longer in high availability mode.**

**Reason:**

The tie-breaker's status is in question. The system is no longer in high availability mode.

**Action:**

Determine the problem with the tie-breaker scheduler and restart it if necessary.

## **CAUAJM\_W\_00173**

**System appears to have a running primary scheduler. Checking for primary activity.**

**Reason:**

System appears to have a running primary scheduler. Checking for primary activity.

**Action:**

No action required.

## **CAUAJM\_W\_00174**

**System appears to have a running shadow scheduler. Checking for shadow activity.**

**Reason:**

System appears to have a running shadow scheduler. Checking for shadow activity.

**Action:**

No action required.

## CAUAJM\_W\_00175

**System appears to have a running tie-breaker scheduler. Checking for tie-breaker activity.**

**Reason:**

System appears to have a running tie-breaker scheduler. Checking for tie-breaker activity.

**Action:**

No action required.

## CAUAJM\_W\_00176

**The primary is not active. Startup proceeding.**

**Reason:**

The primary is not active. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_00177

**The shadow is not active. Startup proceeding.**

**Reason:**

The shadow is not active. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_00178

**The tie-breaker is not active. Startup proceeding.**

**Reason:**

The tie-breaker is not active. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_00180

**System does not appear to have a running shadow scheduler. Skipping the shutdown of the shadow scheduler.**

**Reason:**

System does not appear to have a running shadow scheduler. Skipping the shutdown of the shadow scheduler.

**Action:**

Run the shadow scheduler before sending the STOP\_DEMON event with the value specified as ROLE and the shadow scheduler specified in the role.

## CAUAJM\_W\_00181

**System does not appear to have a running primary scheduler. Skipping the shutdown of the primary scheduler.**

**Reason:**

System does not appear to have a running primary scheduler. Skipping the shutdown of the primary scheduler.

**Action:**

Run the primary scheduler before sending the STOP\_DEMON event with the value specified as ROLE and the primary scheduler specified in the role.

## CAUAJM\_W\_00182

**System does not appear to have a running tie-breaker scheduler. Skipping the shutdown of the tie-breaker scheduler.**

**Reason:**

System does not appear to have a running tie-breaker scheduler. Skipping the shutdown of the tie-breaker scheduler.

**Action:**

Run the tie-breaker scheduler before sending the STOP\_DEMON event with the value specified as ROLE and the tie-breaker scheduler specified in the role.

## CAUAJM\_W\_00185

**System does not appear to have any running application servers. Skipping the shutdown of the application server.**

**Reason:**

System does not appear to have any running application servers. Skipping the shutdown of the application server.

**Action:**

Run the application server before sending the STOP\_DEMON event with the value specified as ROLE and the application server specified in the role.

## CAUAJM\_W\_00187

**Shutdown notification for the primary scheduler was written into one of the databases.**

**Reason:**

Shutdown notification for the primary scheduler was written into one of the databases.

**Action:**

Investigate the state of both databases.

## CAUAJM\_W\_00188

**Shutdown notification for the shadow scheduler was written into one of the databases.**

**Reason:**

Shutdown notification for the shadow scheduler was written into one of the databases.

**Action:**

---

Investigate the state of both databases.

## **CAUAJM\_W\_00189**

**Shutdown notification for the tie-breaker scheduler was written into one of the databases.**

**Reason:**

Shutdown notification for the tie-breaker scheduler was written into one of the databases.

**Action:**

Investigate the state of both databases.

## **CAUAJM\_W\_00190**

**Shutdown notifications for all application servers were written into one of the databases.**

**Reason:**

Shutdown notifications for all application servers were written into one of the databases.

**Action:**

Investigate the state of both databases.

## **CAUAJM\_W\_00195**

**Failover notification was written into one of the databases.**

**Reason:**

Failover notification was written into one of the databases.

**Action:**

Investigate the state of both databases.

## **CAUAJM\_W\_00204**

**Restart notification for the shadow scheduler was written into one of the databases.**

**Reason:**

Restart notification for the shadow scheduler was written into one of the databases.

**Action:**

Investigate the state of both databases.

## **CAUAJM\_W\_00206**

**Changing high availability system status to failback.**

**Reason:**

The shadow is not updating the database.

**Action:**

Investigate the state of the shadow scheduler, otherwise contact technical support.



## CAUAJM\_W\_00207

**Successfully changed high availability system status. The primary is taking over.**

**Reason:**

Successfully changed high availability system status. The primary is taking over.

**Action:**

Determine cause of failback and restart shadow if necessary.

## CAUAJM\_W\_00208

**The shadow appears to be missing. Changing high availability system status to pending.**

**Reason:**

The shadow appears to be missing. Changing high availability system status to pending.

**Action:**

Determine why the shadow did not update its status in the database and restart it if necessary.

## CAUAJM\_W\_00211

**Unable to mark high availability system to failback. Continuing as primary in dormant failback mode.**

**Reason:**

Unable to mark high availability system to failback. Continuing as primary in dormant failback mode.

**Action:**

Attempt to mark database in failback mode returned a non-failback status.

## CAUAJM\_W\_00212

**The {0} primary scheduler appears to have failed back prematurely. Consider a higher value for the HAPollInterval setting. Proceeding with restart...**

**Reason:**

The primary scheduler appears to have failed back prematurely. Consider a higher value for the HAPollInterval setting. Proceeding with restart...

**Action:**

After shutting down all schedulers, set a higher value for the HAPollInterval across all schedulers.

## CAUAJM\_W\_00213

**The shadow and tie-breaker are both missing. Database indicates primary can fail back. Changing high availability system status to pending.**

**Reason:**

The shadow and tie-breaker are both missing. Database indicates primary can fail back. Changing high availability system status to pending.

**Action:**

---

Determine why the shadow and tie-breaker are missing and restart it if necessary.

## **CAUAJM\_W\_00214**

**Unable to mark high availability system as pending. Continuing as primary in dormant failback mode.**

**Reason:**

Unable to mark high availability system as pending. Continuing as primary in dormant failback mode.

**Action:**

Attempt to mark database as pending returned a non-pending status.

## **CAUAJM\_W\_00215**

**Database indicates failover status. Restarting primary scheduler in dormant failback mode.**

**Reason:**

Database indicates failover status. Restarting primary scheduler in dormant failback mode.

**Action:**

Determine problem that caused failover.

## **CAUAJM\_W\_00217**

**System does not appear to be in high availability mode. Skipping failback.**

**Reason:**

System does not appear to be in high availability mode. Skipping failback.

**Action:**

Wait until the system enters high availability mode before sending the STOP\_DEMON with the value specified as FAILOVER.

## **CAUAJM\_W\_00218**

**The database indicates that a failback to the primary has occurred. The system is no longer in high availability mode.**

**Reason:**

The database indicates that a failback to the primary has occurred. The system is no longer in high availability mode.

**Action:**

No action necessary.

## **CAUAJM\_W\_00220**

**The shadow's status is in question. The primary is taking over.**

**Reason:**

Informational only.

**Action:**

---

No action required.

## CAUAJM\_W\_00221

**Unable to mark high availability system as pending. Continuing as primary in immediate failback mode.**

**Reason:**

Unable to mark high availability system as pending. Continuing as primary in immediate failback mode.

**Action:**

Attempt to mark database as pending returned a non-pending status.

## CAUAJM\_W\_00222

**Unable to mark high availability system to failback. Continuing as primary in immediate failback mode.**

**Reason:**

Unable to mark high availability system to failback. Continuing as primary in immediate failback mode.

**Action:**

Attempt to mark database in failback mode returned a non-failback status.

## CAUAJM\_W\_00223

**Database indicates failover status. Restarting primary scheduler in immediate failback mode.**

**Reason:**

Database indicates failover status. Restarting primary scheduler in immediate failback mode.

**Action:**

Determine problem that caused failover.

## CAUAJM\_W\_00231

**The query parameters specified in the 'provider\_url' will be ignored by the Agent.**

**Reason:**

The job has an invalid combination of attributes specified.

**Action:**

Define the job with a valid combination of attributes

## CAUAJM\_W\_00232

**The value specified against 'method\_name' will overwrite the method name and query parameters specified against 'provider\_url'.**

**Reason:**

The job has an invalid combination of attributes specified.

**Action:**

Define the job with a valid combination of attributes.

## CAUAJM\_W\_10015

**Response for [{0}] has expired after {1} seconds. Discarding...**

**Reason:**

An incoming response for an expired request arrived from the application server after the specified number of seconds.

**Action:**

Review the application server log file for any delays in processing requests from the client host. Verify that traffic over the network is not delayed.

## CAUAJM\_W\_10039

**Discarding expired message...**

**Reason:**

Discarding expired message.

**Action:**

No action is required.

## CAUAJM\_W\_10047

**Config file contains invalid input: {0}**

**Reason:**

Bad input from config file.

**Action:**

Required action is not known.

## CAUAJM\_W\_10048

**{0} is deprecated for {1}. Defaulting to {2}**

**Reason:**

vmstat is deprecated for MachineMethod.

**Action:**

No action required.

## CAUAJM\_W\_10049

**{0} is deprecated**

**Reason:**

It is a deprecated configuration option.

**Action:**

Remove the deprecated option from the config file.

## CAUAJM\_W\_10063

The configured value for FileSystemThreshold ({0}) is out of range. Resetting threshold to 20480KB (20MB).

**Reason:**

The value specified for FileSystemThreshold is not within the allowable range.

**Action:**

Specify a FileSystemThreshold value between 8192KB (8MB) and 102400KB (100MB) and recycle the Scheduler component.

## CAUAJM\_W\_10108

No valid machine definition found for "{0}". The Scheduler requires a machine definition in order to start jobs on localhost.

**Reason:**

No valid machine definition found for the specified machine. The Scheduler requires a machine definition in order to start jobs on localhost.

**Action:**

Localhost cannot be resolved to any machine in the database. Define a valid machine.

## CAUAJM\_W\_10109

Please define a machine with the name "{0}" or specify an existing machine as the LocalMachineDefinition configuration variable.

**Reason:**

Please define a single machine with the Scheduler hostname or specify an existing machine as the LocalMachineDefinition configuration variable.

**Action:**

Localhost cannot be resolved to any machine in the database. Define a valid machine.

## CAUAJM\_W\_10119

Calendar "{0}" has no more valid dates.

**Reason:**

The specified calendar has no more valid dates.

**Action:**

Verify the specified calendar definition.

## CAUAJM\_W\_10120

Job "{0}" can no longer be scheduled because it uses a calendar without any available dates.

**Reason:**

The specified job can no longer be scheduled because it uses a calendar without any available dates.

**Action:**

---

Verify the specified calendar definition.

## **CAUAJM\_W\_10128**

**No valid machine definition found for the LocalMachineDefinition configuration variable.**

**Reason:**

No valid machine definition found for the LocalMachineDefinition configuration variable.

**Action:**

LocalMachineDefinition configuration variable cannot be resolved to any machine in the database. Define a valid machine.

## **CAUAJM\_W\_10134**

**{0} failed to establish a listener at host {1}. Acquiring a new agent listener port...**

**Reason:**

The specified product failed to establish a listener at the specified host.

**Action:**

Required action is not known.

## **CAUAJM\_W\_10138**

**Duplicate machines found for "{0}". The Scheduler must resolve to one machine definition in order to start jobs on localhost.**

**Reason:**

Duplicate machines found for the specified machine. The Scheduler must resolve to one machine definition in order to start jobs on localhost.

**Action:**

Localhost cannot be resolved to a single machine in the database.

## **CAUAJM\_W\_10150**

**Function {0} failed to get environment from the registry. Details={1}**

**Reason:**

Specified registry function failed to get Environments.

**Action:**

Required action unknown.

## **CAUAJM\_W\_10158**

**Ignored invalid MaxRestartTrys file entry at line {0}.**

**Reason:**

Invalid MaxRestartTrys file entry at specified line.

**Action:**

---

Examine the \$AUTOUSER/MaxRestartTrys.\$AUTOSERV file and make the necessary corrections.

## **CAUAJM\_W\_10175**

**Error has occurred reading the System Environment registry key.**

**Reason:**

Error opening the registry key associated with the system environments.

**Action:**

Required action unknown.

## **CAUAJM\_W\_10176**

**Service will not be able to detect changes in the {0} environment**

**Reason:**

Due to a registry error, new PATH environment changes will not be detected.

**Action:**

Required action unknown.

## **CAUAJM\_W\_10193**

**Job "{0}" will be scheduled when it is taken {1}.**

**Reason:**

The specified job will be scheduled when it is taken OFF ICE.

**Action:**

Send the JOB\_OFF\_ICE event for the Scheduler to recalculate the next start time.

## **CAUAJM\_W\_10194**

**Job "{0}" is restarting. The next run will be scheduled after the current run completes.**

**Reason:**

The specified job is restarting. The next run will be scheduled after the current run completes.

**Action:**

Allow the job to complete its run.

## **CAUAJM\_W\_10210**

**Machine <{0}>: Usage is not supported for UUJMA based machines.**

**Reason:**

Usage is not supported for the specified UUJMA based machine.

**Action:**

Required action unknown.

## CAUAJM\_W\_10225

Only the {0} {1} superuser can set the owner.

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_W\_10226

Only the {0} {1} superuser can send the {2} event!

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action is required.

## CAUAJM\_W\_10227

Only the {0} {1} superuser can modify a machine.

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_W\_10235

Service has detected a change in the {0} environment but an error occurred: {1}

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_10264

**WARNING - Job: {0} DOES NOT EXIST!**

**Reason:**

Job does not exist.

**Action:**

Required action unknown.



## CAUAJM\_W\_10265

**WARNING - No {0} Information about Job: {1} Check that Instance.**

**Reason:**

No status information about job.

**Action:**

Required action unknown.

## CAUAJM\_W\_10269

**Access rights could not be obtained to view all jobs for Instance {0}!**

**Reason:**

Access rights could not be obtained to view all jobs for Instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10276

**Access rights could not be obtained to view all job types for Instance {0}!**

**Reason:**

Access rights could not be obtained to view all jobs for Instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10282

**WARNING for Job: {0}. {1}**

**Reason:**

Warning for job.

**Action:**

Required action unknown.

## CAUAJM\_W\_10284

**Access rights could not be obtained to view all calendars for Instance {0}!**

**Reason:**

Access rights could not be obtained to view all calendars for Instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10291

**WARNING! Requested Over-Ride fields match defined fields. No Over-Rides defined!**

**Reason:**

Requested Over-Ride fields match defined fields. Nothing to override.

**Action:**

Required action unknown.

## CAUAJM\_W\_10296

**Job "{0}" was modified by custom job validation.**

**Reason:**

Job was modified by custom job validation.

**Action:**

Required action unknown.

## CAUAJM\_W\_10300

**Attempt to send the {0} event has failed. Please issue the event manually.**

**Reason:**

Attempt to send the REFRESH\_EXTINST event has failed. Please issue the event manually.

**Action:**

Send the REFRESH\_EXTINST event manually.

## CAUAJM\_W\_10330

**Access rights could not be obtained to view all machines for Instance {0}!**

**Reason:**

Access rights could not be obtained to view all machines for Instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10335

**No machine selected for job <{0}>.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_10345

**Warning! The machine <{0}> has jobs in the {1} or {2} state. Proceeding with sending {3} event.**

**Reason:**

The machine has jobs in the RUNNING or STARTING state. Proceeding with sending MACH\_OFFLINE event.

**Action:**

Required action unknown.

## CAUAJM\_W\_10360

**Warning: truncating {0} from "{1}" to "{2}".**

**Reason:**

Truncation warning.

**Action:**

Required action unknown.

## CAUAJM\_W\_10367

**WARNING! Requested update fields match already defined fields.**

**Reason:**

Requested fields match already defined fields.

**Action:**

No action required.

## CAUAJM\_W\_10371

**This job's type does not support the KILLJOB operation. Event not sent for job <{0}>.**

**Reason:**

This job's type does not support the KILLJOB operation. Event not sent for the specified job.

**Action:**

Do not issue a KILLJOB event for this job.

## CAUAJM\_W\_10374

**Access rights could not be obtained to view all jobs or global variables for Instance {0}!**

**Reason:**

Access rights could not be obtained to view all jobs for Instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10380

**Only the job timestamp will be updated!**

**Reason:**

Only the job time stamp will be updated.

**Action:**

No action required.

## CAUAJM\_W\_10382

**WARNING! Unable to get environments for instance {0}!**

**Reason:**

The specified registry function failed to get Environments.

**Action:**

Required action unknown.

## CAUAJM\_W\_10389

**WARNING for Machine: < {0} >**

**Reason:**

Warning for job Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_W\_10391

**WARNING for Monbro: < {0} >**

**Reason:**

Warning for machine Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_W\_10392

**WARNING for Job: < {0} >**

**Reason:**

Warning for job.

**Action:**

Required action unknown.

## CAUAJM\_W\_10394

**WARNING for Blob/Glob: < {0} >**

**Reason:**

Warning for blob/glob Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_W\_10396

**WARNING for External Instance: < {0} >**

**Reason:**

Warning for External Instance Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_W\_10400

**Application Execute Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact your security policy administrator.

## CAUAJM\_W\_10401

**Calendar Read Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10402

**Calendar Write Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10403**

### **Calendar Execute Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10404**

### **Calendar Create Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10405**

### **Calendar Delete Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10406**

### **Control Execute Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10407**

### **Cycle Read Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10408

### Cycle Write Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10409

### Cycle Create Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10410

### Cycle Delete Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10411

### Global Variable Read Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10412

### Global Variable Write Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10413

### Global Variable Execute Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10414

### Global Variable Create Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10415

### Global Variable Delete Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10416

### Group Execute Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10417

### Job Read Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.



## **CAUAJM\_W\_10418**

### **Job Write Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10419**

### **Job Execute Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10420**

### **Job Create Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10421**

### **Job Delete Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10422**

### **List Read Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10423

### Machine Read Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10424

### Machine Write Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10425

### Machine Execute Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10426

### Machine Create Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10427

### Machine Delete Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10428

### Owner Execute Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10429

### Resources Read Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10430

### Resources Write Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10431

### Resources Create Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10432

### Resources Delete Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10438

**Explicit Deny Policy:** \"{0}\"

**Reason:**

Information regarding last failed access check.

**Action:**

Required action unknown.

## CAUAJM\_W\_10439

**No Policies Granting Access To Resource**

**Reason:**

Information regarding last failed access check.

**Action:**

Required action unknown.

## CAUAJM\_W\_10440

**Class: {0} Resource: {1} User: {2} Access: {3}**

**Reason:**

Information regarding last failed access check.

**Action:**

Required action unknown.

## CAUAJM\_W\_10441

**Time: {0,number,0} Delegator: {1}**

**Reason:**

Information regarding last failed access check.

**Action:**

Required action unknown.

## CAUAJM\_W\_10442

**Time: {0,number,0} Delegator: None**

**Reason:**

Information regarding last failed access check.

**Action:**

Required action unknown.

## CAUAJM\_W\_10446

### Security Access to Resource Denied!

**Reason:**

Access to resource denied.

**Action:**

No action required.

## CAUAJM\_W\_10447

### New install detected! Please run the {0} menu utility to initiate the security engine!

**Reason:**

The product is making access level checks with an uninitialized security engine.

**Action:**

Run autosys\_secure first before using the product.

## CAUAJM\_W\_10450

### Only the {0} {1} superuser can activate external security.

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_W\_10452

### The Windows {0} User Account does not have sufficient authority to access {1} assets.

**Reason:**

An access level check was made by the default Windows Service Local System account.

**Action:**

No action required.

## CAUAJM\_W\_10453

### Job Log Read Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10454**

### **Job Type Read Access Denied!**

#### **Reason:**

Security policy denied access to the resource.

#### **Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10455**

### **Job Type Write Access Denied!**

#### **Reason:**

Security policy denied access to the resource.

#### **Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10456**

### **Job Type Execute Access Denied!**

#### **Reason:**

Security policy denied access to the resource.

#### **Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10457**

### **Job Type Create Access Denied!**

#### **Reason:**

Security policy denied access to the resource.

#### **Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10458**

### **Job Type Delete Access Denied!**

#### **Reason:**

Security policy denied access to the resource.

#### **Action:**

Contact security policy administrator.

## CAUAJM\_W\_10463

**Input job does not have any resource dependencies [{0} {1,number,1}.{2,number,1}.{3,number,1}]**

**Reason:**

The input job does not exist or has\_resource flag indicates no resource dependencies.

**Action:**

Check the job table for the input job definition.

## CAUAJM\_W\_10470

**Application Read Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10471

**No orphaned renewable usage detected for job <{0}>.**

**Reason:**

The input job does not exist have entries in the resource status table.

**Action:**

No action required.

## CAUAJM\_W\_10472

**Resource manager encountered the following during initialization: <{0}>.**

**Reason:**

Resource manager encountered the specified issue during initialization.

**Action:**

Required action unknown.

## CAUAJM\_W\_10473

**Resource manager will ignore non-virtual resource constraints for all jobs.**

**Reason:**

Resource manager will ignore non-virtual resource constraints for all jobs.

**Action:**

Required action unknown.

## CAUAJM\_W\_10489

**Resource manager is not available. Any defined real resource constraints will be ignored.**

**Reason:**

Resource manager is not available. Any defined real resource constraints will be ignored.

**Action:**

Resource manager is not loaded during startup. Check the log file for errors during startup. Fix the problem and restart.

## CAUAJM\_W\_10491

**Found invalid machine <{0}> for job <{1}>.**

**Reason:**

Problem detected with job definition.

**Action:**

Fix the job definition with the correct machine name.

## CAUAJM\_W\_10496

**Agent on [{0}] has not responded in a timely fashion. Try again later.**

**Reason:**

Agent on [{0}] has not responded in a timely fashion. Try again later.

**Action:**

Could indicate that the Agent is too busy or cannot reach the Manager port.

## CAUAJM\_W\_10499

**Scheduler shutdown in progress. Communications interrupted.**

**Reason:**

Scheduler shutdown in progress. Communications interrupted.

**Action:**

No action required.

## CAUAJM\_W\_10508

**Application Write Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.



## CAUAJM\_W\_10509

### Group Read Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10511

### Group Write Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10544

**Machine <{0}> is unavailable due to insufficient load units.**

**Reason:**

Machine is unavailable due to insufficient load units.

**Action:**

Required action unknown.

## CAUAJM\_W\_10545

**Only the {0} or {1} superuser can use the {2} utility!**

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_W\_10548

**Machine <{0}> is unavailable due to lack of resources.**

**Reason:**

Machine is unavailable due to lack of resources.

**Action:**

Required action unknown.

## CAUAJM\_W\_10550

**Machine <{0}> is unavailable as queued {1} jobs have a higher priority.**

**Reason:**

Machine is unavailable as queued QUE\_WAIT or RESWAIT jobs have a higher priority.

**Action:**

Required action unknown.

## CAUAJM\_W\_10559

**Unable to set groups: {0}**

**Reason:**

The user name was not found or the group list could not be set.

**Action:**

Correct the problem and try again.

## CAUAJM\_W\_10575

**Network performance warning: Detected an out of sequence, repeated message for job [{0}] from agent [{1}].**

**Reason:**

The product has detected an out of sequence, repeated network message for the specified job from the specified agent.

**Action:**

Please contact CA support.

## CAUAJM\_W\_10576

**Received message sequence {0} after sequence {1}.**

**Reason:**

The product has detected an out of sequence, repeated network message.

**Action:**

Please contact CA support.

## CAUAJM\_W\_10577

**Discarding message:**

**Reason:**

The product is discarding the specified network message.

**Action:**

No action required.

## CAUAJM\_W\_10578

**Postponing the sending of updated scheduler communication attributes to agents due to shutdown.**

**Reason:**

The scheduler received notification to shut down and must postpone sending updated communication attributes to the agents.

**Action:**

Restart the scheduler to resume contacting the agents.

## CAUAJM\_W\_10613

**Notification attributes ignored for job <{0}>. The notification method is currently set to <{1}>.**

**Reason:**

The notification attributes were ignored for this job as there is no active notification method.

**Action:**

Select an active notification method or remove the notification attributes from the job definition.

## CAUAJM\_W\_10614

**Email notification is not sent for job <{0}>.**

**Reason:**

Email address is not specified in the job definition.

**Action:**

Specify the email address in the job definition.

## CAUAJM\_W\_10631

**Error with database <{0}>. Checking connection.**

**Reason:**

The database client libraries have indicated that an error condition occurred.

**Action:**

No action required.

## CAUAJM\_W\_10632

**Attempting to reconnect to database <{0}>. Attempt number [{1}].**

**Reason:**

Attempting to reconnect to the database.

**Action:**

No action required.

## CAUAJM\_W\_10634

Exceeded maximum number of database reconnection attempts.

**Reason:**

Exceeded maximum number of retries.

**Action:**

No action is required.

## CAUAJM\_W\_10635

Rolling over to single server mode. Using database <{0}>.

**Reason:**

Rolling to single server mode.

**Action:**

No action required.

## CAUAJM\_W\_10648

The database: {0} is unavailable. Attempting to rollover to single server mode.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_10688

Database <{0}> indicates that a rollover has occurred. Updating connections accordingly.

**Reason:**

The other scheduler has rolled over.

**Action:**

No action required.

## CAUAJM\_W\_10697

**\*\*\* WARNING: Syntax mode cannot verify against all attributes for updates or overrides. Only requested attributes will be verified.**

**Reason:**

Syntax mode cannot verify against all attributes for updates or overrides. Only requested attributes will be verified.

**Action:**

No action required.

## CAUAJM\_W\_10698

**WARNING For Machine: {0} < Circular definition: Cannot add self to virtual machine list >**

**Reason:**

Circular definition: Cannot add self to virtual machine list.

**Action:**

No action required.

## CAUAJM\_W\_10699

**WARNING For Machine: {0} < machine "{1}" not defined >**

**Reason:**

The Machine is not defined.

**Action:**

No action required.

## CAUAJM\_W\_10700

**CONDITION WARNING: Job name <{0}> may be invalid for external instance <{1}>. Please review the details for this external instance.**

**Reason:**

The job name in the condition may be invalid for this external instance.

**Action:**

Please refer to the UAJM documentation for more information on valid job name lengths for external instances.

## CAUAJM\_W\_10701

**Machine "{0}" is defined.**

**Reason:**

Machine is defined.

**Action:**

No action required.

## CAUAJM\_W\_10702

**Machine "{0}" is not defined.**

**Reason:**

Machine is not defined.

**Action:**

No action required.

## CAUAJM\_W\_10703

**CONDITION WARNING: External instance name <{0}> is not defined**

**Reason:**

The external instance name is currently not defined to the product.

**Action:**

No action required.

## CAUAJM\_W\_10704

**{0} {1} is defined.**

**Reason:**

The object is defined.

**Action:**

No action required.

## CAUAJM\_W\_10705

**\*\*\* WARNING: Syntax mode cannot verify virtual machine type compatibility for machine {0}. Please check the type of this machine before inserting.**

**Reason:**

Syntax mode cannot verify machine type compatibility against virtual machines.

**Action:**

Please check the type of this machine before inserting.

## CAUAJM\_W\_10706

**GLOBBLOB WARNING: Job {0} is not defined.**

**Reason:**

The job is not defined.

**Action:**

No action required.

## CAUAJM\_W\_10707

**{0} {1} is not defined.**

**Reason:**

Object is not defined.

**Action:**

No action required.

## CAUAJM\_W\_10708

**External Instance does not exist. Nothing to Delete.**

**Reason:**

No External Instance is defined to delete.

**Action:**

No action required.

## CAUAJM\_W\_10709

**External Instance {0} is defined.**

**Reason:**

The external Instance is defined.

**Action:**

No action required.

## CAUAJM\_W\_10710

**External Instance {0} is not defined.**

**Reason:**

The external Instance is not defined.

**Action:**

No action required.

## CAUAJM\_W\_10711

**Job Type: {0} is defined**

**Reason:**

The job type is defined.

**Action:**

Required action unknown.

## CAUAJM\_W\_10712

**Job Type: {0} is not defined**

**Reason:**

The job type is not defined.

**Action:**

Required action unknown.

## CAUAJM\_W\_10713

There are job(s) defined with type "{0}".

**Reason:**

Cannot delete the job type.

**Action:**

Required action unknown.

## CAUAJM\_W\_10714

Database change ignored.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_10720

**AUTOEXTVJ Warning:**

**Reason:**

Job failed custom validation.

**Action:**

Required action unknown.

## CAUAJM\_W\_10721

Ignored attempt by the AUTOEXTVJ\_ValidateJob function to change name of existing job "{0}".

**Reason:**

Ignored attempt by the AUTOEXTVJ\_ValidateJob function to change an existing job name.

**Action:**

Custom criteria library cannot change the job name of an ex.

## CAUAJM\_W\_10723

Virtual machine "{0}" is in the machine chooser list and will not be selected at runtime.

**Reason:**

Virtual machine "{0}" is in the machine chooser list and will not be selected at runtime.

**Action:**

Required action unknown.



## CAUAJM\_W\_10724

**All machines in the machine chooser list are virtual. No machine will be selected at runtime.**

**Reason:**

All machines in the machine chooser list are virtual. No machine will be selected at runtime.

**Action:**

Required action unknown.

## CAUAJM\_W\_10726

**Failed to process an incoming AFM string because the AFM header is invalid: {0}**

**Reason:**

Failed to process an incoming AFM string because the AFM header is invalid.

**Action:**

Required action unknown.

## CAUAJM\_W\_10727

**Failed to process an incoming AFM string because it is of an unknown format.**

**Reason:**

Failed to parse an AFM response string because it is of an unknown format.

**Action:**

Check to see that the Agent encryption key is properly set. Unknown data could also be indication of a network attack.

## CAUAJM\_W\_10729

**Ignored attempt by the AUTOEXTVJ\_ValidateJob function to add unknown keyword/value "{0}/{1}" to job "{2}"**

**Reason:**

Ignored attempt by the AUTOEXTVJ\_ValidateJob function to add an unknown keyword/value pair to job.

**Action:**

Custom criteria library cannot add, update or delete a keyword that is not recognized by WAAE.

## CAUAJM\_W\_10734

**Send Start Job Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10735

### Send Force Start Job Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10736

### Send Job On Hold Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10737

### Send Job Off Hold Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10738

### Send Job On Ice Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10739

### Send Job Off Ice Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10740

### Send Job On Noexec Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10741

### Send Job Off Noexec Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10742

### Send Kill Job Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10743

### Send Release Resource Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10744

### Send Send Signal Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10745**

### **Send Change Priority Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10746**

### **Send Change Status Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10747**

### **Send Comment Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10748**

### **Send Delete Job Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10749**

### **Send Restart Job Event Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10750

### Send Reply Response Event Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10751

### Send Resume Job Event Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10752

### Send Suspend Job Event Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10753

### Send Alarm Event Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10754

### Send Machine Online Event Access Denied!

#### Reason:

Security policy denied access to resource.

#### Action:

Contact security policy administrator.

## CAUAJM\_W\_10755

### Send Machine Offline Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10756

### Un-send Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10757

### Send Set Global Event Access Denied!

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## CAUAJM\_W\_10788

### Attribute {0} unsupported for machine type "{1}".

**Reason:**

An attribute was specified that is unsupported for the machine type.

**Action:**

Attribute is not saved.

## CAUAJM\_W\_10886

### Access rights could not be obtained to view all connection profiles for instance {0}.

**Reason:**

Access rights could not be obtained to view all connection profiles for instance.

**Action:**

Required action unknown.

## CAUAJM\_W\_10900

**The database monitoring system has detected a potential problem with the database.**

**Reason:**

The database monitoring system has detected a potential problem with the database.

**Action:**

No action required.

## CAUAJM\_W\_10908

**System appears to have another running application server with the same {0} on machine: {1}. Checking for application server activity.**

**Reason:**

System appears to have another running application server with the same AutoServerId or AutoServerAliasId. Checking for application server activity.

**Action:**

No action required.

## CAUAJM\_W\_10909

**The {0} application server with the same {1} is in an unknown state on machine: {2}.**

**Reason:**

The application server with the same AutoServerId or AutoServerAliasId is in an unknown state on the specified machine.

**Action:**

Investigate the state of the application server and the AutoServerId or AutoServerAliasId configuration on the specified machine.

## CAUAJM\_W\_10910

**No application server is active with the same {0} or {1}. Startup proceeding.**

**Reason:**

No application server is active with the same AutoServerId or AutoServerAliasId. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_10921

**The credentials that are cached are outdated and cannot be used to authenticate.**

**Reason:**

The expiration date and time for these credentials is in the past.

**Action:**

---

Specify a new expiration date when you regenerate the cached credentials and then reissue the command, or contact your administrator for assistance.

## **CAUAJM\_W\_10945**

**Connection profile Read Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10946**

**Connection profile Write Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10947**

**Connection profile Create Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10948**

**Connection profile Delete Access Denied!**

**Reason:**

Security policy denied access to resource.

**Action:**

Contact security policy administrator.

## **CAUAJM\_W\_10955**

The job's time zone <{0}> is not recognized on this host. The reported times will be based on this host's time zone.

**Reason:**

The application server did not calculate the offset between this host and the job time zone and so autorep must do the calculation. However, autorep cannot because the job time zone is in a format that is not recognized on this host.

**Action:**



---

Apply the available maintenance on the application server to resolve Problem 4434.

## **CAUAJM\_W\_10962**

**The resource manager is unable to satisfy all resource dependencies for job <{0}>.**

**Reason:**

Review previous log messages to see which resources could not be reserved.

**Action:**

No action required. The scheduler automatically executes the job as soon as the resource dependencies are met.

## **CAUAJM\_W\_10965**

**Global virtual resource <{0}> is unavailable due to insufficient units.**

**Reason:**

The specified global virtual resource is unavailable due to insufficient units.

**Action:**

No action required.

## **CAUAJM\_W\_10966**

**Global virtual resource <{0}> is not evaluated due to database query failures.**

**Reason:**

The specified global virtual resource is not evaluated due to database query failures.

**Action:**

Review previous log messages to determine why the database queries failed to execute.

## **CAUAJM\_W\_10967**

**Global virtual resource <{0}> is unavailable as queued {1} jobs have a higher priority.**

**Reason:**

The specified global virtual resource is unavailable as queued RESWAIT jobs have a higher priority.

**Action:**

No action required.

## **CAUAJM\_W\_10968**

**Global virtual resources cannot be reserved for machine <{1}> because the machine is unavailable.**

**Reason:**

Global virtual resources cannot be reserved for the specified machine because the machine is unavailable.

**Action:**

No action required.

## CAUAJM\_W\_10971

**Virtual resource <{0}> on machine <{1}> is unavailable due to insufficient units.**

**Reason:**

The specified machine-based virtual resource is unavailable due to insufficient units.

**Action:**

No action required.

## CAUAJM\_W\_10972

**Virtual resource <{0}> on machine <{1}> is not evaluated due to database query failures.**

**Reason:**

The specified machine-based virtual resource is not evaluated due to database query failures.

**Action:**

Review previous log messages to determine why the database queries failed to execute.

## CAUAJM\_W\_10973

**Virtual resource <{0}> on machine <{1}> is unavailable as queued {2} jobs have a higher priority.**

**Reason:**

The specified machine-based virtual resource is unavailable as queued RESWAIT jobs have a higher priority.

**Action:**

No action required.

## CAUAJM\_W\_10974

**Virtual resource <{0}> is not defined for machine <{1}>.**

**Reason:**

The specified virtual resource is not defined on the specified machine.

**Action:**

No action required.

## CAUAJM\_W\_10975

**Virtual resource <{0}> on machine <{1}> is not evaluated because the machine is unavailable.**

**Reason:**

The specified machine-based virtual resource is not evaluated because the machine is unavailable.

**Action:**

No action required.

## CAUAJM\_W\_10976

**Real resource dependencies for node name <{0}> is not evaluated due to database query failures.**

**Reason:**

Real resource dependencies for the specified node name is not evaluated due to database query failures.

**Action:**

Review previous log messages to determine why the database queries failed to execute.

## CAUAJM\_W\_10977

**Real resource dependencies for node name <{0}> is not evaluated as queued {1} jobs have a higher priority.**

**Reason:**

Real resource dependencies for the specified node name is not evaluated as queued RESWAIT jobs have a higher priority.

**Action:**

No action required.

## CAUAJM\_W\_10991

**The attribute \"{0}\" value has been re-set to \"{1}\".**

**Reason:**

Warning message for the job.

**Action:**

Required action unknown.

## CAUAJM\_W\_18910

**No rules are found for rule: {0}. Please add rules for this rule name.**

**Reason:**

Please add rules for this rule name.

**Action:**

The job definition is saved.

## CAUAJM\_W\_18950

**DCA Manager is not loaded/available. Machine provisioning has been skipped.**

**Reason:**

DCA Manager is not loaded/available. Machine provisioning has been skipped.

**Action:**

Ensure that SDK client and/or the DCA libraries are installed. Also, check that credentials for the DCA Manager URL are correct.

## CAUAJM\_W\_18952

**Preparing machine "{0}" failed.**

**Reason:**

Unable to prepare machine.

**Action:**

Contact the DCA Machine administrator.

## CAUAJM\_W\_18969

**WARNING for {0}: {1}. {2}**

**Reason:**

Warning for job.

**Action:**

Required action unknown.

## CAUAJM\_W\_18985

**Unable to find a machine resource <{0}> that matches job machine definition. Job will get stuck in RESWAIT state.**

**Reason:**

Machine Resources matching job machine definition not found. Job will be stuck in RESWAIT state.

**Action:**

Machine Resources matching the job machine must defined to be able to run the job.

## CAUAJM\_W\_19009

**Payload producing job "{0}" does not exists for JIL keyword "{1}".**

**Reason:**

Payload producing Job does not exists.

**Action:**

Object definition is saved.

## CAUAJM\_W\_19018

**External Instance "{0}" is being used and will not be deleted.**

**Reason:**

External Instance is being used.

**Action:**

Delete the condition that uses this external instance.

## CAUAJM\_W\_19022

**WARNING For Machine: {0} < Cannot add localhost to a virtual machine list >**

**Reason:**

Machine is not defined.

**Action:**

No is action required.

## CAUAJM\_W\_19024

**The value of {0} attribute {1} failed to be encrypted.**

**Reason:**

Invalid encryption key.

**Action:**

Specify a different encryption key.

## CAUAJM\_W\_19054

**Connection profile "{0}" is already defined.**

**Reason:**

A connection profile with this name is already defined.

**Action:**

Specify another name for the connection profile.

## CAUAJM\_W\_19057

**Connection profile \"{0}\" is still being referenced by active jobs.**

**Reason:**

Connection profile is referenced in active jobs.

**Action:**

Redefine connection profile.

## CAUAJM\_W\_19059

**Connection profile name: {0} does not exist.**

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value.

## CAUAJM\_W\_19065

Cannot remove \"{0}\" in connection profile \"{1}\" as it is still being referenced by active jobs.

**Reason:**

Connection profile is referenced in active jobs.

**Action:**

Remove reference in jobs and then redefine connection profile.

## CAUAJM\_W\_19078

New job name and the old job name cannot be the same

**Reason:**

The specified job name is the same as the old job name.

**Action:**

Specify a new job name.

## CAUAJM\_W\_19102

Added job {0} into an already running box. Sent the {1} event to schedule the job immediately, if necessary, or on its appointed time.

**Reason:**

When you add a job to a running box, the STARTJOB event is sent to ensure that the scheduler starts the job at its earliest available time.

**Action:**

No action required.

## CAUAJM\_W\_19103

Added job {0} into an already running box. Job's run number ({1,number,1}) equals or exceeds the containing box's run number ({2,number,1}). No {3} event sent.

**Reason:**

When you add a job to a running box, the STARTJOB event is sent unless the job's run number equals or exceeds the containing box run number.

**Action:**

No action required.

## CAUAJM\_W\_19104

Added job {0} into an already running box. Job's status is {1}. No {2} event sent.

**Reason:**

When you add a job to a running box, the STARTJOB event is sent unless the job status is STARTING, RUNNING, or ON\_ICE.

**Action:**

---

No action required.

## CAUAJM\_W\_20206

**Port {0,number,0} has not been configured through the CA Secure Sockets Adapter**

**Reason:**

The specified port has not been configured through the CA Secure Sockets Adapter.

**Action:**

Examine the CA Secure Sockets Adapter configuration settings for the port.

## CAUAJM\_W\_20223

**Email address is blank. Email notification request skipped for job <{0}>.**

**Reason:**

The specified notification\_emailaddress is invalid. Attribute value cannot be blank.

**Action:**

Specify a valid notification\_emailaddress.

## CAUAJM\_W\_20226

**Notification Message is truncated to 255 characters limit after global variable expansion for job {0}.**

**Reason:**

The expanded Notification message is greater than 255 characters in length.

**Action:**

Modify global variable to less than 255 characters.

## CAUAJM\_W\_30005

**Version mismatch detected [Product: {0}; Database: {1}]. Operating in limited functionality mode.**

**Reason:**

Product version is different from database version.

**Action:**

Required action unknown.

## CAUAJM\_W\_30010

**The Application Server for instance {0} is already running.**

**Reason:**

Only one Application Server may be running for each instance on any given machine.

**Action:**

Avoid starting the Application Server when it is already running.

## CAUAJM\_W\_30023

**Received notification to rollover.**

**Reason:**

Scheduler has indicated that a database rollover is necessary.

**Action:**

Figure out the problem with the database and follow recovery procedures for recovering from a rollover.

## CAUAJM\_W\_30025

**Have not received rollover notification from the Scheduler. Attempting reconnection 1 time to database <{0}>**

**Reason:**

No Notification from Scheduler.

**Action:**

No action required.

## CAUAJM\_W\_30036

**Client [{0}][{1}][{2}] API ID [{3}] request discarded.**

**Reason:**

The application server could not execute the request before its expiration time.

**Action:**

Review previous messages for an indication of delays that could cause requests to be discarded. If you cannot determine the cause of the problem, contact CA Support.

## CAUAJM\_W\_30100

**Request has expired after {0} seconds. Discarding...**

**Reason:**

Request has expired after the specified number of seconds. Discarding...

**Action:**

Application Server is too busy to process the request.

## CAUAJM\_W\_30101

**Request queue is too full. Discarding...**

**Reason:**

Request queue is too full. Discarding...

**Action:**

Application Server is too busy to process the request.



## CAUAJM\_W\_30102

**Flushing request queue due to shutdown. Discarding...**

**Reason:**

Flushing request queue due to shutdown. Discarding...

**Action:**

Application Server is shutting down. Unprocessed requests will be discarded.

## CAUAJM\_W\_30103

**The system has insufficient resources to process this request in a timely fashion. Discarding...**

**Reason:**

The system has insufficient resources to allow the application server to create the threads it needs to process the request in a timely fashion.

**Action:**

Collect information about the processes that are currently executing including the amount of memory and CPU that they consume. Stop any applications that consume too many system resources.

## CAUAJM\_W\_40023

**Unable to aggregate statistics. There is already an active aggregate statistics process in progress. Request ignored.**

**Reason:**

Received a sendevent request to aggregate statistics, but there was already an active aggregate statistics process in progress. There can only be one active aggregate statistics process at a time.

**Action:**

No action required.

## CAUAJM\_W\_40028

**{0} does not support this trigger type {1} {2}**

**Reason:**

Trigger type is not supported. Request ignored.

**Action:**

No action required.

## CAUAJM\_W\_40032

**A {0} or {1} operation is already in progress for job <{2}>. Request ignored.**

**Reason:**

There is already an active KILLJOB or SEND\_SIGNAL operation in progress for the specified job.

**Action:**

No action required.

## **CAUAJM\_W\_40037**

**A {0} operation is already in progress for job <{1}>. Request ignored.**

**Reason:**

There is already an active operation in progress for the specified job.

**Action:**

No action required.

## **CAUAJM\_W\_40102**

**Cancelled the execution of job <{0}> for load balancing queue name <{1}>.**

**Reason:**

The scheduler was unable to start the job for the specified load balancing queue name.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_W\_40104**

**Skipping the evaluation of the remaining jobs for load balancing queue name <{0}> due to database error.**

**Reason:**

Skipping the evaluation of the remaining jobs for the specified load balancing queue name due to database error.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_W\_40105**

**Skipping the evaluation of the remaining jobs for load balancing queue name <{0}> due to insufficient load.**

**Reason:**

Skipping the evaluation of the remaining jobs for the specified load balancing queue name due to insufficient load.

**Action:**

No action required.

## **CAUAJM\_W\_40106**

**Skipping the evaluation of the remaining jobs for load balancing queue name <{0}> due to higher priority queued jobs.**

**Reason:**

Skipping the evaluation of the remaining jobs for the specified load balancing queue name due to higher priority queued jobs.

**Action:**

No action required.

**CAUAJM\_W\_40107**

**Skipping the evaluation of the remaining jobs for load balancing queue name <{0}> due to offline machine.**

**Reason:**

Skipping the evaluation of the remaining jobs for the specified load balancing queue name due to offline machine.

**Action:**

No action required.

**CAUAJM\_W\_40114**

**Failed to retrieve jobs in a {0} status. {0} Delay processing skipped.**

**Reason:**

Informational only.

**Action:**

No action required.

**CAUAJM\_W\_40115**

**Pending remaining jobs for machine <{0}> due to offline machine.**

**Reason:**

Pending remaining jobs for the specified machine due to offline machine.

**Action:**

No action required.

**CAUAJM\_W\_40118**

**Postponing job starts for online machine <{0}> due to shutdown.**

**Reason:**

Postponing job starts for the specified online machine due to shutdown.

**Action:**

Scheduler is shutting down. MACH\_ONLINE event will be reprocessed on startup.

**CAUAJM\_W\_40135**

**Skipped starting job <{0}> due to manual status change while bringing machine <{1}> online.**

**Reason:**

The job's status was manually changed after it came out of PEND\_MACH status. Starting of the job has been skipped.

**Action:**

No action required.

## CAUAJM\_W\_40176

**The current job run has no associated agent information. {0} canceled.**

**Reason:**

The job in question more than likely was placed in a RUNNING state through executing a manual sendevent.

**Action:**

If the job was placed in a RUNNING state manually use sendevent -E CHANGE\_STATUS to mark the job as INACTIVE. Otherwise contact technical support for further assistance.

## CAUAJM\_W\_40179

**Global variable <{0}> is not defined for job <{1}>**

**Reason:**

Global variable is not defined.

**Action:**

Create a global variable.

## CAUAJM\_W\_40207

**An unexpected problem occurred while fetching events. Continuing....**

**Reason:**

etBatchEvent failed.

**Action:**

Required action unknown.

## CAUAJM\_W\_40253

**Machine <{0}> is not responding. Taking offline.**

**Reason:**

The machine is not responding.

**Action:**

Determine if the Agent is running on the machine or if there is a connectivity problem.

## CAUAJM\_W\_40254

**Machine <{0}> is in question. Its status remains manually set to offline.**

**Reason:**

The specified machine is in question while its status is manually set to offline.

**Action:**

Determine if the Agent is running on the machine or if there is a connectivity problem.

## CAUAJM\_W\_40258

**Machine <{0}> has explicitly rejected communication. Placing in a blocked state.**

**Reason:**

The specified machine has explicitly rejected communication.

**Action:**

Check the agent configuration and verify that the scheduler information is configured properly. If a severe disk threshold breach has occurred clear disk space so the agent will resume communication.

## CAUAJM\_W\_40259

**External instance <{0}> has explicitly rejected communications. Taking offline.**

**Reason:**

The specified external instance has explicitly rejected communications. Taking offline.

**Action:**

Check the remote manager configuration and verify that the scheduler information is configured properly.

## CAUAJM\_W\_40272

**Machine <{0}> cannot be placed in an automatic offline status because the machine auto offline feature is disabled.**

**Reason:**

The scheduler cannot automatically place the specified machine in a blocked or offline status because the machine auto offline feature is disabled.

**Action:**

Enable the machine auto offline feature by clearing the PE\_AUTO\_DISABLE environment variable and restarting the scheduler.

## CAUAJM\_W\_40273

**Machine <{0}> cannot be automatically brought online because status was manually set offline.**

**Reason:**

Machine cannot be automatically brought online because machine status was manually set offline.

**Action:**

Issue a MACH\_ONLINE event to manually bring the machine online.

## CAUAJM\_W\_40278

**Unable to set machine <{0}> to an automatic offline status because the status was manually set to offline.**

**Reason:**

Machine was manually set to an offline status. Cannot move it to an automatic offline status.

**Action:**

---

Manually set the machine status to online.

## CAUAJM\_W\_40280

The following events were processing during the last run of the Scheduler and will be re-queued.

**Reason:**

The following events were processing during the last run of the Scheduler and will be re-queued:.

**Action:**

<No action mentioned>

## CAUAJM\_W\_40281

RE-QUEUEING: {0}

**Reason:**

Display list of requeued events.

**Action:**

<No action mentioned>

## CAUAJM\_W\_40282

External instance <{0}> is not responding. Taking offline.

**Reason:**

The external instance is not responding.

**Action:**

Determine if the manager is running on the machine or if there is a connectivity problem.

## CAUAJM\_W\_40283

Error from Instance <{0}>: \"{1}\".

**Reason:**

Error from specified external instance.

**Action:**

The SEND\_EVENT\_INTERNAL API returned a failure.

## CAUAJM\_W\_40284

Postponing send of event <{0}> due to offline external instance <{1}>.

**Reason:**

Postponing send of event due to the specified offline external instance.

**Action:**

Determine if the manager is running on the machine or if there is a connectivity problem.

## CAUAJM\_W\_40287

Unable to set machine <{0}> to a blocked status because the status was manually set to offline.

**Reason:**

Machine was manually set to an offline status. Cannot move it to a blocked status.

**Action:**

Manually set the machine status to online.

## CAUAJM\_W\_40288

Skipping the execution of internal database maintenance because the unauthenticated user mode setting requires user authentication.

**Reason:**

The scheduler skips the execution of the database maintenance script when the unauthenticated user mode is set to STRICT or EXTERNAL.

**Action:**

Define a job to execute the script and generate cached credentials for the job owner on the machine where the job runs.

## CAUAJM\_W\_40290

Machine <{0}> is in question. Placing machine in the unqualified state.

**Reason:**

The specified machine is in question. Placing machine in the unqualified state.

**Action:**

Determine if the Agent is running on the machine or if there is a connectivity problem.

## CAUAJM\_W\_40291

Machine <{0}> has been updated. Placing machine in the unqualified state.

**Reason:**

The specified machine has been updated. Placing machine in the unqualified state.

**Action:**

No action required.

## CAUAJM\_W\_40292

Machine <{0}> cannot be placed in a blocked status because the machine auto offline feature is disabled.

**Reason:**

The scheduler cannot automatically place the specified machine in a blocked status because the machine auto offline feature is disabled.

**Action:**

---

Enable the machine auto offline feature by clearing the PE\_AUTO\_DISABLE environment variable and restarting the scheduler.

## CAUAJM\_W\_40293

**Machine <{0}> is not responding or has shut down. Automatically placing offline.**

**Reason:**

The specified machine is not responding or has shut down.

**Action:**

Check prior messages to determine why the agent is not responding or has shut down and take action as required.

## CAUAJM\_W\_40299

**The primary scheduler is restarting in dormant failback mode as the shadow scheduler has taken over.**

**Reason:**

The primary scheduler is restarting in dormant failback mode as the shadow scheduler has taken over.

**Action:**

No action required.

## CAUAJM\_W\_40304

**Statistical aggregation has been aborted due to a shutdown request. Statistics have been only partially aggregated.**

**Reason:**

Statistical aggregation was aborted due to a shutdown request.

**Action:**

Statistics will be aggregated on the next automatic aggregation or can be manually aggregated through sendevent once the scheduler has been restarted.

## CAUAJM\_W\_40305

**The shadow scheduler is restarting as the primary scheduler has taken over.**

**Reason:**

The shadow scheduler is restarting as the primary scheduler has taken over.

**Action:**

No action required.

## CAUAJM\_W\_40306

**The primary scheduler has taken over. The system is no longer in high availability mode.**

**Reason:**

The primary scheduler has taken over. The system is no longer in high availability mode.

**Action:**



---

Investigate the cause of high availability failover.

## **CAUAJM\_W\_40307**

**System appears to be in failback state. Skipping system-wide shutdown or failback.**

**Reason:**

System appears to be in failback state. Skipping system-wide shutdown or failback.

**Action:**

No action required.

## **CAUAJM\_W\_40308**

**The {0} shadow scheduler is restarting due to {1} event.**

**Reason:**

The shadow scheduler is restarting due to STOP\_DEMON event.

**Action:**

No action required.

## **CAUAJM\_W\_40309**

**The primary scheduler is shutting down as the shadow scheduler has taken over.**

**Reason:**

The primary scheduler is shutting down as the shadow scheduler has taken over.

**Action:**

No action required.

## **CAUAJM\_W\_40310**

**The shadow scheduler has taken over. The system is no longer in high availability mode.**

**Reason:**

The shadow scheduler has taken over. The system is no longer in high availability mode.

**Action:**

Investigate the cause of high availability failover.

## **CAUAJM\_W\_40311**

**Cross Platform Scheduling has been disabled.**

**Reason:**

The cross platform scheduling feature of CA Workload Automation AE has been disabled.

**Action:**

See prior error messages for additional information.

## CAUAJM\_W\_40326

**Invalid option "{0}" specified for cross platform scheduling. Cross platform scheduling disabled.**

**Reason:**

Invalid option specified for cross platform scheduling. Cross platform scheduling disabled.

**Action:**

Valid options are '0','1', and '2'. Correct the option and recycle the Scheduler.

## CAUAJM\_W\_40328

**System appears to be in failover state. Checking for shadow activity.**

**Reason:**

System appears to be in failover state. Checking for shadow activity.

**Action:**

No action required.

## CAUAJM\_W\_40329

**The shadow is still active. The primary cannot continue.**

**Reason:**

The shadow is still active. The primary cannot continue.

**Action:**

No action required.

## CAUAJM\_W\_40330

**The shadow is not active. Startup proceeding.**

**Reason:**

The shadow is not active. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_40331

**System check interrupted due to shutdown.**

**Reason:**

System check interrupted due to shutdown.

**Action:**

No action required.

## CAUAJM\_W\_40332

**System appears to be in failover state. Skipping system-wide shutdown or failover.**

**Reason:**

System appears to be in failover state. Skipping system-wide shutdown or failover.

**Action:**

No action required.

## CAUAJM\_W\_40333

**System appears to be in failover state. Startup proceeding in dormant failback mode.**

**Reason:**

System appears to be in failover state. Startup proceeding in dormant failback mode.

**Action:**

No action required.

## CAUAJM\_W\_40336

**The primary scheduler is restarting in immediate failback mode as the shadow scheduler has taken over.**

**Reason:**

The primary scheduler is restarting in immediate failback mode as the shadow scheduler has taken over.

**Action:**

No action required.

## CAUAJM\_W\_40337

**System appears to be in failover state. Startup proceeding in immediate failback mode.**

**Reason:**

System appears to be in failover state. Startup proceeding in immediate failback mode.

**Action:**

No action required.

## CAUAJM\_W\_40338

**System no longer appears to be in failover state. Startup proceeding.**

**Reason:**

System no longer appears to be in failover state. Startup proceeding.

**Action:**

No action required.

## CAUAJM\_W\_40357

The {0} tie-breaker scheduler is only required to run in dual server mode. Shutting down.

**Reason:**

The tie-breaker scheduler is only required to run in dual server mode. Shutting down.

**Action:**

No action required.

## CAUAJM\_W\_40358

The disk partition containing the {0} Scheduler log file is full.

**Reason:**

Disk partition containing the Scheduler log file is full!

**Action:**

Free up space on the disk partition containing the Scheduler log file.

## CAUAJM\_W\_40359

The {0} Scheduler will shutdown if the disk partition has less than {1} bytes available.

**Reason:**

Disk partition containing the Scheduler log file is full.

**Action:**

Free up space on the disk partition containing the Scheduler log file.

## CAUAJM\_W\_50018

Machine <{0}> is of a type that does not support the autoping operation.

**Reason:**

The specified machine is of a type that does not support the autoping operation.

**Action:**

Execute autoping against a non virtual/pool machine.

## CAUAJM\_W\_50064

Jobs did not run on Legacy machines. No files to clean.\n

**Reason:**

Jobs did not run on Legacy machines.

**Action:**

Required action unknown.

## CAUAJM\_W\_50076

**Warning - Levels not supported for "c" (current state) Option.\n\n**

**Reason:**

Levels are not supported with the -c option.

**Action:**

Specify valid parameters.

## CAUAJM\_W\_50094

**A partial list is displayed.\n**

**Reason:**

A partial list is being displayed because of a security violation.

**Action:**

No action is required.

## CAUAJM\_W\_50112

**No machines have been returned.\n**

**Reason:**

No machines have been returned.

**Action:**

Required action unknown.

## CAUAJM\_W\_50131

**Line number specification <{0}> is not valid. Returning the default number of lines.**

**Reason:**

Invalid line number.

**Action:**

Specify a line number greater than 0.

## CAUAJM\_W\_50183

**No Global Variables match masked specification: {0}\n**

**Reason:**

No global variables match the masked specification in CA Workload Automation AE.

**Action:**

Required action unknown.

## CAUAJM\_W\_50185

No Job Names were found matching Group Name: {0}\n

**Reason:**

The group name is not currently defined in CA Workload Automation AE.

**Action:**

No required action.

## CAUAJM\_W\_50186

No Job Names were found matching Application Name: {0}\n

**Reason:**

The application name is currently not defined in CA Workload Automation AE.

**Action:**

No required action.

## CAUAJM\_W\_50187

No Job Names were found matching Group Name: {0} and Application Name: {1}\n

**Reason:**

No jobs were found matching the group and application specification in AutoSys. If a fully qualified name was specified this indicates it is currently not defined within AutoSys.

**Action:**

No required action.

## CAUAJM\_W\_50196

No user defined job types have been returned.\n

**Reason:**

No user defined job types have been returned.

**Action:**

No required action.

## CAUAJM\_W\_50197

**\*\*\* WARNING: failed to install interrupt handler.**

**Reason:**

Failed to install interrupt handler.

**Action:**

Required action unknown.

## CAUAJM\_W\_50203

No jobs have been returned that utilize a user defined job type.\n

**Reason:**

No jobs have been returned that utilize a user defined job type.

**Action:**

No action is required.

## CAUAJM\_W\_50215

**\*\*\* WARNING: {0} is NO LONGER SUPPORTED.**

**Reason:**

The field is NO LONGER SUPPORTED.

**Action:**

Required action unknown.

## CAUAJM\_W\_50224

No jobs have been returned for job type {0}.\n

**Reason:**

No jobs have been returned for the identified user defined job type.

**Action:**

No action required.

## CAUAJM\_W\_50230

Machine type "{0}" is no longer supported for {1} based machines. Type changed to "{2}".

**Reason:**

Machine type is no longer supported and has been changed accordingly.

**Action:**

No action required.

## CAUAJM\_W\_50241

**\*\*\* WARNING: The Following Jobs are referenced in the Conditions for this Job, yet are not defined:**

**Reason:**

Conditions not defined.

**Action:**

Required action unknown.

## CAUAJM\_W\_50252

**\*\*\* WARNING: This update has resulted in the loss of the following machine definitions:**

**Reason:**

The Machine type has changed to a non-virtual type or the machine was explicitly removed from the virtual machine list.

**Action:**

Required action unknown.

## CAUAJM\_W\_50257

**Attribute ignored for <{0}>: {1}**

**Reason:**

The specified attribute is ignored in this context.

**Action:**

No action required.

## CAUAJM\_W\_50258

**\*\*\* WARNING: The following Jobs have starting conditions that depend on nonexistent Jobs or Global Variables:**

\n\n\n

**Reason:**

Jobs with starting conditions that depend on nonexistent Jobs detected.

**Action:**

No action required.

## CAUAJM\_W\_50268

**{0} could not be uploaded:**

**Reason:**

An error occurred while uploading the blob.

**Action:**

Required action unknown.

## CAUAJM\_W\_50269

**Job or global input blob will not be available to job**

**Reason:**

Job inserted/updated without the input blob.

**Action:**

Required action unknown.



## CAUAJM\_W\_50270

JIL Keyword {0} is discarded because its value is NULL, null or ALL.

**Reason:**

Invalid values specified.

**Action:**

Required action unknown.

## CAUAJM\_W\_50274

JIL Keyword {0} is discarded because it is not valid for {1} definition.

**Reason:**

Invalid values specified.

**Action:**

Examine the errors reported and re-enter the JIL attributes.

## CAUAJM\_W\_50279

Cannot download the blob with blob name: {0} blob type: {1} for job: {2}

**Reason:**

An error occurred while downloading the blob.

**Action:**

Required action unknown.

## CAUAJM\_W\_50285

No job blobs exist.

**Reason:**

The blobs did not match the specification.

**Action:**

Required action unknown.

## CAUAJM\_W\_50289

Cannot download the blob of blob type: {1} for job: {2}

**Reason:**

An error occurred while downloading the blob.

**Action:**

Required action unknown.

## CAUAJM\_W\_50317

**Could not download blob**

**Reason:**

An error occurred while downloading blob.

**Action:**

Required action unknown.

## CAUAJM\_W\_50319

**No Global Blob Names match masked specification: {0}\n**

**Reason:**

No jobs match masked specification.

**Action:**

Required action unknown.

## CAUAJM\_W\_50320

**No input blobs have been defined for this job.**

**Reason:**

No input blobs have been defined for this job.

**Action:**

No action required.

## CAUAJM\_W\_50331

**Machine \"{0}\" is still being referenced by machine containers.**

**Reason:**

Machine is referenced in machine containers.

**Action:**

Redefine the machine.

## CAUAJM\_W\_50332

**Machine \"{0}\" is still being referenced by active jobs.**

**Reason:**

Machine is referenced in active jobs.

**Action:**

Redefine the machine.

## CAUAJM\_W\_50333

Machine \"{0}\" is still being referenced by resources.

**Reason:**

Machine is referenced in active jobs.

**Action:**

Redefine the machine.

## CAUAJM\_W\_50335

Ignoring invalid input in Workday string \"{0}\".

**Reason:**

The Workdays string is not in the correct format.

**Action:**

Check the value of the input string.

## CAUAJM\_W\_50340

The renamed job {0} is being referred from the following external jobs(s)

**Reason:**

Please check the action.

**Action:**

Please update the following external jobs accordingly.

## CAUAJM\_W\_50401

The machine: {0} does not exist.

**Reason:**

The machine name does not exist in the database.

**Action:**

No action required.

## CAUAJM\_W\_50412

No External Instances match masked specification: {0}\n

**Reason:**

No external instances match masked specification.

**Action:**

Required action unknown.

## CAUAJM\_W\_50413

**\*\*\* WARNING - The Holiday calendar: {0} does not exist.**

**Reason:**

Informational only.

**Action:**

Define the Holiday calendar before referring it in the extended calendar.

## CAUAJM\_W\_50414

**\*\*\* WARNING - The Cycle calendar: {0} does not exist.**

**Reason:**

Informational only.

**Action:**

Cycle calendar can be created later.

## CAUAJM\_W\_50433

**Warnings = {0}**

**Reason:**

Total Warnings.

**Action:**

Review JIL output for warnings.

## CAUAJM\_W\_50505

**The user process is not running on machine: {0}. The Scheduler is inactive.**

**Reason:**

The user process is not running on the specified machine. The Scheduler is not running thus the status of the job may not be accurately reflected.

**Action:**

If the user process has truly completed, start the Scheduler and monitor the status of the job.

## CAUAJM\_W\_50506

**The {0} job is not running on machine: {1}, The Scheduler is inactive.**

**Reason:**

The job is not running on the specified machine. It has been determined that the Scheduler is not running.

**Action:**

Start the Scheduler.

## CAUAJM\_W\_50507

**The user process is not running on machine: {0}, an attempt to communicate with the Scheduler failed.**

**Reason:**

The user process is not running on the specified machine, an attempt to communicate with the Scheduler failed.

**Action:**

Check the status of the Scheduler.

## CAUAJM\_W\_50508

**The {0} job is not running on machine: {1}, an attempt to communicate with the Scheduler failed.**

**Reason:**

The job is not running on the specified machine, an attempt to communicate with the Scheduler failed.

**Action:**

Check the status of the Scheduler.

## CAUAJM\_W\_50509

**The user process is not running on machine: {0}, it seems Scheduler is unresponsive.**

**Reason:**

The user process is not running on the specified machine, it seems the Scheduler is unresponsive.

**Action:**

Check the status of the Scheduler.

## CAUAJM\_W\_50510

**The {0} job is not running on machine: {1}, it seems Scheduler is unresponsive.**

**Reason:**

The job is not running on the specified machine, it seems the Scheduler is unresponsive.

**Action:**

Check the status of the Scheduler.

## CAUAJM\_W\_50511

**The user process is not running on machine: {0}, as the Scheduler status is pending high availability failover.**

**Reason:**

The user process is not running on the specified machine, as the Scheduler status is pending high availability failover.

**Action:**

Wait for the Scheduler's High Availability status to be resolved.

## CAUAJM\_W\_50512

**The {0} job is not running on machine: {1}, as the Scheduler status is pending high availability failover.**

**Reason:**

The job is not running on the specified machine, as the Scheduler status is pending high availability failover.

**Action:**

Wait for the Scheduler's High Availability status to be resolved.

## CAUAJM\_W\_50513

**The user process is not running on machine: {0}. The Scheduler status is unknown.**

**Reason:**

The user process is not running on the specified machine, but the Scheduler status is unknown.

**Action:**

Check the status of the database.

## CAUAJM\_W\_50514

**The {0} is not running on machine: {1}. The Scheduler status is unknown.**

**Reason:**

The job is not running on the specified machine. The Scheduler status is unknown.

**Action:**

Wait for the Scheduler's High Availability status to be resolved.

## CAUAJM\_W\_50515

**The user process is not running on machine: {0}, but the Scheduler has not yet reestablished communications with the Agent.**

**Reason:**

The user process is not running on the specified machine, but the Scheduler has not yet reestablished communications with the Agent.

**Action:**

Check the status of the database.

## CAUAJM\_W\_50516

**The {0} job is not running on machine: {1}, the Scheduler has not yet reestablished communication with the Agent.**

**Reason:**

The job is not running on the specified machine, the Scheduler has not yet reestablished communication with the agent.

**Action:**

Wait for the Scheduler's High Availability status to be resolved.

## CAUAJM\_W\_50519

Invalid connection profile name: {0}\n.

**Reason:**

No connection profile is found matching the specification.

**Action:**

No required action.

## CAUAJM\_W\_60002

Batch\_size exceeds {0}, changing batch\_size to {1}

**Reason:**

Specifying a Batch\_size more than MAX\_UPDATE may impact performance.

**Action:**

No action required.

## CAUAJM\_W\_60045

Only an {0} {1} {2} has access to all options!

**Reason:**

Only an EDIT/EXEC superuser has access to all options!

**Action:**

Run autosys\_secure as an EDIT/EXEC superuser to have access to all options.

## CAUAJM\_W\_60050

Insufficient privileges to execute options {0,number,0} through {1,number,0} } and {2,number,0}. .

**Reason:**

Insufficient privileges to execute options 1 through 4 (native) or 3 through 4 (external).

**Action:**

Required action unknown.

## CAUAJM\_W\_60144

Instance security is already set to {0}. Request skipped.

**Reason:**

The system is already running in the requested toggle mode.

**Action:**

No action required.

## CAUAJM\_W\_60190

**The CA EEM server location remains unchanged.**

**Reason:**

The CA EEM server location remains unchanged.

**Action:**

Review any previous error messages for further details.

## CAUAJM\_W\_60192

**Failed to authenticate as user {0}. Proceeding as the current unauthenticated user.**

**Reason:**

The external security server did not authenticate user credentials. The program proceeds as the current unauthenticated user when login fails.

**Action:**

Verify user credentials and attempt to log in again.

## CAUAJM\_W\_60194

**This instance remains under {0} security control due to end-user response.**

**Reason:**

This instance remains under CA EEM security control due to end-user response.

**Action:**

No action necessary.

## CAUAJM\_W\_60213

**Ignoring additional program arguments.**

**Reason:**

The program has been passed additional program arguments based on asset type. The program will ignore these arguments.

**Action:**

No action required.

## CAUAJM\_W\_60218

**Instance configuration is unavailable.\n.**

**Reason:**

The instance configuration is needed to determine the database type.

**Action:**

Set the product environment or avoid requesting the database type.



## CAUAJM\_W\_60221

**Not connected to the CA EEM server.**

**Reason:**

Not connected the CA EEM server.

**Action:**

Use option "1" of the CA Workload Automation AE Safetool to set and connect to the CA EEM server.

## CAUAJM\_W\_60222

**Not connected to the CA EEM global application instance.**

**Reason:**

Not connected the CA EEM global application instance.

**Action:**

This utility must be connected to a CA EEM global application instance to perform all the options in this submenu.

## CAUAJM\_W\_60223

**Some of the options in this menu will not be available!**

**Reason:**

Some of the options in this menu will not be available.

**Action:**

This utility must be connected to the appropriate application instance to perform all the options in this submenu.

## CAUAJM\_W\_60224

**Failed to change the user to owner of {0}. <{1}> call failed with <{2}>.**

**Reason:**

User may not have permissions to login.

**Action:**

User may not have permissions to login, check the \$AUTOUSER owner permissions.

## CAUAJM\_W\_60225

**Failed to set Write and Execute permissions on Certificate file to owner of {0}. <{1}> call failed with <{2}>.**

**Reason:**

User may not have permissions.

**Action:**

User may not have permissions, check the \$AUTOUSER owner permissions.

## CAUAJM\_W\_60247

CA EEM application instance \"{0}\" already registered.

**Reason:**

CA EEM application instance for the product already registered.

**Action:**

No action required.

## CAUAJM\_W\_60250

Security policies for instance {0} already exist!

**Reason:**

Security policies for the product instance already exist.

**Action:**

No action required.

## CAUAJM\_W\_60254

CA EEM application instance \"{0}\" not registered

**Reason:**

CA EEM application instance for the product is not registered.

**Action:**

No action required.

## CAUAJM\_W\_60255

No security policies for instance {0} exist.

**Reason:**

No instance security policies exist.

**Action:**

No action required.

## CAUAJM\_W\_60305

**\*\*WARNING\*\*** The generated certificate file must be placed in the {0} directory of instance {1} in order for security to properly function!

**Reason:**

The generated certificate file must be placed in the instance specific AUTOUSER directory for security to function properly.

**Action:**

If the generated certificate file was not placed in the instance specific AUTOUSER directory during the generate operation it should be copied to this location.

## CAUAJM\_W\_60322

Unable to verify user's existence.\nAre you sure you want to continue?

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_60323

Unable to verify user's existence.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_60337

Disabling the unauthenticated user mode may have security implications.

**Reason:**

Disabling the unauthenticated user mode may have security implications.

**Action:**

No action required.

## CAUAJM\_W\_60340

Trusting all unauthenticated remote users as authenticated users may have security implications.

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_60341

An error occurred attempting to set {0} as the unauthenticated user mode. Instance remains under mode {1}.

**Reason:**

An error occurred attempting to set the specified mode as the unauthenticated user mode.

**Action:**

See any previous messages which may have been outputted for indication for error cause. Otherwise, call technical support.

## CAUAJM\_W\_60345

**Trusting all unauthenticated users as authenticated users may have security implications.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_60374

**A file for profile <{0}> already exists**

**Reason:**

Duplicate file name.

**Action:**

Required action unknown.

## CAUAJM\_W\_60377

**Default security policies shared by all instances are missing.**

**Reason:**

At least one default security policy shared by all instances is missing.

**Action:**

No action required.

## CAUAJM\_W\_60381

**Merging pre-release 11.0 DEFAULT profiles...**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_W\_60433

**The {0} application server is running at a version that does not support option {1}.**

**Reason:**

The application server is running at a version that does not support the selected option.

**Action:**

Upgrade the application server to the latest supported version.

## CAUAJM\_W\_60452

Only an {0} {1} {2} can delete the password vault.

**Reason:**

The user running the security utility does not have the required privileges to perform the requested operation.

**Action:**

The request must be executed by an account that has sufficient privileges.

## CAUAJM\_W\_60453

The {0} application server is running at a version that does not support managing password vault users.

**Reason:**

The application server is running at a version that does not support this option.

**Action:**

Upgrade the application server to the latest supported version.

## CAUAJM\_W\_90016

**Reason:**

License warning.

**Action:**

Required action unknown.

## CAUAJM\_W\_101001

Unable to deregister the WAAE dependency for CAPKI.

**Reason:**

Unable to deregister the CAPKI WAAE dependency.

**Action:**

For more information, refer to the log file for the reason of failure and try to manually remove the dependency.

## CAUAJM\_W\_101101

The SSA uninstallation has failed. For more information, refer to the SSAInstall.log file under the TEMP directory.

**Reason:**

The SSA uninstallation has failed.

**Action:**

For more information, refer to the log file for the reason of failure and try to rerun the uninstallation manually.

## CAUAJM\_W\_101102

Unable to configure the SSA dependency or port. Execute the "{0}" command to manually configure the SSA dependency or port.

**Reason:**

Unable to configure the SSA dependency or port.

**Action:**

Run the command mentioned in the error message manually.

## CAUAJM\_W\_101103

The SSA executable {0} not found.

**Reason:**

The SSA executable was not found.

**Action:**

Refer to the path mentioned in the error message, and ensure that the SSA executable is present at the location specified above.

## CAUAJM\_W\_101200

Unable to configure the root user for csutils.

**Reason:**

Failed to configure the root user for the csutils.

**Action:**

Run the command manually to configure the root user.

**Examples:**

On Linux:

```
$AUTOSYS/csutils/bin/assrvPath useradd --name=root all
```

On Windows:

```
%AUTOSYS%\csutils\bin\assrvPath useradd --name=root all
```

## CAUAJM\_W\_101201

Unable to configure the "{0}" owner for csutils. Execute the "{1}" command manually to configure.

**Reason:**

Failed to configure the owner for csutils.

**Action:**

Run the command manually to configure the owner.

**Examples:**

On Linux:

---

```
$AUTOSYS/csutils/bin/ casrvc useradd --name=<owner> all
```

On Windows:

```
%AUTOSYS%\csutils\bin\ casrvc useradd --name=<owner> all
```

## CAUAJM\_W\_104701

**Unable to update the "{0}" configuration file. {1}**

**Reason:**

An error occurred while updating the configuration file.

**Action:**

Refer to the error message and try to update the configuration file manually.

## CAUAJM\_W\_104702

**Unable to merge the user comments in the updated "config.{0}" file.**

**Reason:**

Failed to read the content of the file.

**Action:**

Manually update the user comments section in the config.<INSTANCE> file present in the backup folder.

## CAUAJM\_W\_104703

**Unable to update the vars file for the "{0}" instance.**

**Reason:**

An error occurred updating configuration files.

**Action:**

The file vars<INSTANCE> contains a list of installed components and their names for each instance. Based on the installed components, update the file.

For example, if the scheduler is installed, agent and client are dependent components.

```
IA_AE_INSTALLED_SCHEDULER=1
```

```
IA_AE_INSTALLED_AGENT=1
```

```
IA_AE_INSTALLED_CLIENT=1
```

## CAUAJM\_W\_104900

**Unable to assign an owner and group.**

**Reason:**

Failed to assign the owner and group.

**Action:**

Manually assign the owner and group for the AutoSys installer directory.

For example, on Linux, you can run the following command:

```
chown -hR <user name>:<group> <install path>
```

## CAUAJM\_W\_105000

**Unable to create a service for the "{0}" component.**

**Reason:**

Failed to create a service for the component.

**Action:**

Could not register service with either chkconfig or with the Windows service.

## CAUAJM\_W\_105001

**Unable to start the "{0}" service.**

**Reason:**

Failed to start the service.

**Action:**

Manually start the service.

**Examples:**

On Linux:

```
unisrvctrstart <service_name>;
```

Windows: Start the service from Service Manager.

## CAUAJM\_W\_105003

**Unable to deregister the service for the "{0}" component.**

**Reason:**

Failed to deregister the service for the component.

**Action:**

Manually deregister the service.

**Examples:**

On Linux:

```
Uniregister -D <service_name>
```

On Windows:

```
sc delete <service_name>
```

## CAUAJM\_W\_109000

**Unable to delete the shortcut for AutoSys utilities from the Start menu.**

**Reason:**



---

Failed to delete the AutoSys utilities from the start menu.

**Action:**

Remove the shortcut from the Start menu manually.

## **CAUAJM\_W\_109999**

**Skipping the installation as per the user's choice.**

**Reason:**

The user selected to skip the installation.

**Action:**

IA\_AE\_SKIP\_INSTALL is set to 1, so the install phase is skipped. To install the product, unset the variable.

## **CAUAJM\_W\_110014**

**The "{0}" port is already configured for a different application. Specify a different value for "{1}".**

**Reason:**

The specified port is already in use.

**Action:**

Specify a port that is not in use.

## **CAUAJM\_W\_110026**

**This operating system version is not yet certified. For more information, refer to AutoSys Support Matrix.**

**Reason:**

The operating system is not certified.

**Action:**

Refer to the [support matrix](#) for supported operating systems.

## **CAUAJM\_W\_110032**

**A 32-bit CA EEM Server is present on this system. An upgrade of the 32-bit CA EEM Server is not supported on this platform. The other AutoSys components will be upgraded.**

**Reason:**

A 32-bit CA EEM Server is present on this system. An upgrade of the 32-bit CA EEM Server is not supported on this platform.

**Action:**

Upgrade other AutoSys components that can be upgraded, or install the remote EEM server, migrate the policies and uninstall the 32-bit EEM server and proceed with the installation. For more information, see [AutoSys Common Components Upgrade](#).

## CAUAJM\_W\_110033

**The Web UI collector service should be running for the Web UI service to run. The Web UI collector "{0}" service will be started post-installation.\n**

**Reason:**

The WebUI collector service needs to be running for the Web UI service to run.

**Action:**

Check the start service *or* select YES to continue to start the service after installation.

## CAUAJM\_W\_110034

**The Web UI collector service should be running for the Web UI service to run. The Web UI collector "{0}" service will be started on system startup.\n**

**Reason:**

The Web UI collector services configuration should match the Web UI configuration.

**Action:**

Change the configuration *or* select YES to continue to start service on system startup.

## CAUAJM\_W\_110036

**The AutoSys installation configurations are corrupted. Unable to find any installed components.**

**Reason:**

Configurations are corrupted.

**Action:**

Installed components could not be determined. If you proceed with the uninstallation, some of the configurations (such as the service deregistration or profile file in /etc directory, etc.) will not be deleted.

Verify that the AUTOSYS\_INSTALL\_LOCATION variable is available, and check if globalVars in the AUTOSYS directory is valid.

## CAUAJM\_W\_110042

**An upgrade of the CA EEM server is not supported on this platform. The other AutoSys components will be upgraded.**

**Reason:**

CA EEM upgrade is not supported on this platform.

**Action:**

Upgrade other AutoSys components that can be upgraded, or install the remote EEM server, migrate the policies and uninstall the 32-bit EEM server and proceed with the installation. For more information, see [AutoSys Common Components Upgrade](#).

## CAUAJM\_W\_110043

**Unable to determine the operating system version. However, all the required AutoSys components will be enabled for installation. For more information, refer to the AutoSys Support Matrix.**

**Reason:**

The OS version was not detected.

**Action:**

Refer to the [support matrix](#) and if the operating system and components are supported continue with the installation.

**CAUAJM\_W\_110044**

**Unable to find the 64-bit "libnsl.\*" package on this system. It is required for the successful working of AutoSys components.**

**Reason:**

Libnsl.\* package was not found.

**Action:**

Install libnsl.\* package.

**CAUAJM\_W\_112105**

**Unable to find the "{0}" CA EEM application.**

**Reason:**

The CA EEM application was not found.

**Action:**

Verify whether the specified application exists on the CA EEM server.

**CAUAJM\_W\_112107**

**An EEM application instance "{0}" created using AutoSys versions prior to this release have been detected.**

**Reason:**

A CA EEM application created using an older version of AutoSys has been detected.

**Action:**

Select YES to continue to upgrade the CA EEM application instance.

**CAUAJM\_W\_112110**

**Unable to find any instances in the CA EEM application.**

**Reason:**

Failed to retrieve the AutoSys instance list for the CA EEM application.

**Action:**

Verify whether the connection to the CA EEM server is successful and the instance list exists.

**CAUAJM\_W\_112112**

**Unable to retrieve the version of the CA EEM server.**

**Reason:**

Failed to retrieve the version of the CA EEM server.

**Action:**

Verify whether the connection to the CA EEM server is successful and version can be retrieved successfully. Refer to the [support matrix](#), and if EEM is supported continue with the installation.

## CAUAJM\_W\_112114

**The CA EEM application "{0}" is of a higher version.**

**Reason:**

The CA EEM application is of a higher version.

**Action:**

Use the same version of the DVD as the application.

**Example:**

If the CA EEM application is of version R12, then use the R12 DVD.

## CAUAJM\_W\_112131

**A 32-bit CA EEM server installation is present on this system which cannot be upgraded. Therefore, skipping the CA EEM server upgrade.**

**Reason:**

A 32-bit CA EEM server is installed.

**Action:**

Select YES to skip the upgrade of the CA EEM server, or install a remote CA EEM server, migrate policies and uninstall the 32-bit CA EEM server and proceed with the installation. For more information, see [AutoSys Common Components Upgrade](#).

## CAUAJM\_W\_112133

**Unable to install or upgrade the CA EEM server.**

**Reason:**

Failed to install or upgrade the CA EEM server.

**Action:**

Refer to the logs for errors, and try to rerun the CA EEM server installation manually.

Check the eiam\_install.log in the AUTOSYS\_INSTALL\_LOCATION Logs directory.

If you are trying to upgrade, try standalone EEM.

If you are performing a fresh install, and other components are installed, select Add Component to install EEM.

If the installation continues to fail, contact Broadcom Support.

## CAUAJM\_W\_112502

**Unable to find "sqlcmd" in the PATH environment variable.**

**Reason:**

Failed to locate the sqlcmd in the PATH environment variable.

**Action:**

Set the sqlcmd path in the PATH environment variable.

## CAUAJM\_W\_112505

The "{0}" database is already configured with the latest schema version.

**Reason:**

The database is at the current schema version.

**Action:**

Select YES to skip creating the database, *or* specify the details of an empty database.

## CAUAJM\_W\_112506

The "{0}" database did not have "TRANSACTION ISOLATION LEVEL" with the "READ\_COMMITTED\_SNAPSHOT" option set to "ON".  
To avoid database deadlocks, stop all applications connected to the database and set the option to "ON".  
Perform this task before or after the AutoSys installation.

**Reason:**

The READ\_COMMITTED\_SNAPSHOT is not on.

**Action:**

Connect to the MSSQL server and run the following command to turn READ\_COMMITTED\_SNAPSHOT on.

```
ALTER DATABASE <DATABASE_NAME> SET READ_COMMITTED_SNAPSHOT ON
GO
```

## CAUAJM\_W\_112518

The current database server "{0}" version is "{1}" and is not yet supported. Therefore, a few functionalities may not work as expected. For more information, refer the AutoSys Support Matrix.

**Reason:**

The database server version is not supported yet.

**Action:**

Refer to the [support matrix](#) and specify the supported database version.

## CAUAJM\_W\_112525

The specified database admin user does not have the 'dbcreator' role. Ensure that the user has at least 'create any database' and 'alter any login' permissions. For example, execute the below commands to grant the required permissions.  
GRANT CREATE ANY DATABASE TO <LOGIN\_NAME>  
GRANT ALTER ANY LOGIN TO <LOGIN\_NAME>.

**Reason:**

The specified database admin user does not have the required permissions.

**Action:**

Run the following commands to grant the required permissions.

```
GRANT CREATE ANY DATABASE TO <LOGIN_NAME>
GO
GRANT ALTER ANY LOGIN TO <LOGIN_NAME>.
GO
```

## CAUAJM\_W\_112527

The specified database admin user does not have the 'securityadmin' role. Ensure that the user has at least the 'alter any user' or 'alter any login' permission.

**Reason:**

The specified database admin user does not have the required permissions.

**Action:**

Run the following commands to grant the required permissions.

```
GRANT ALTER ANY LOGIN TO <LOGIN_NAME>.
```

## CAUAJM\_W\_112529

The "{0}" database schema is at a lower version. It will be upgraded to the latest version.

**Reason:**

The database schema is at a lower version.

**Action:**

Select YES to upgrade the database schema version.

## CAUAJM\_W\_112540

Unable to find the TNS alias name: "{0}" in the file: "{1}". The database connection may fail.

**Reason:**

Failed to find the TNS alias name in the tnsnames.ora file.

**Action:**

Make sure that the TNS alias name is present in the tnsnames.ora file.

## CAUAJM\_W\_112541

Unable to find the "{0}" server name in the "{1}" interfaces file.

**Reason:**

Failed to find the server name in the interfaces file.

**Action:**

Make sure that the server name is present in the file.

## CAUAJM\_W\_112556

The number of available Sybase connections "{0}" is less than the recommended number "{1}". AutoSys will attempt to increase the Sybase connections.

**Reason:**

There are fewer available Sybase connections than recommended.

**Action:**

Select YES to increase the number of Sybase connections, or manually increase the number of Sybase connections.

## CAUAJM\_W\_112562

The Oracle service name and TNS alias name should be the same.

**Reason:**

The Oracle service name and TNS alias name need to be the same.

**Action:**

Ensure that the Oracle service name and the TNS alias are the same. If they are different ensure that the connection is successful using the TNS alias.

## CAUAJM\_W\_112568

An error occurred while connecting to the database server using the "{0}" service name.\n{1}

**Reason:**

An error occurred while connecting to the database server using the service name.

**Action:**

Check the database server details to resolve the error.

## CAUAJM\_W\_112585

The specified data tablespace "{0}" already exists.

**Reason:**

The specified data tablespace already exists.

**Action:**

Check if the provided data tablespace is indeed the intended device.

## CAUAJM\_W\_112586

The specified index tablespace "{0}" already exists.

**Reason:**

The specified index tablespace already exists.

**Action:**

Check if the provided index tablespace is indeed the intended device.

## CAUAJM\_W\_112588

The specified data device "{0}" already exists.

**Reason:**

The specified data device already exists.

**Action:**

Check if the provided data device is indeed the intended device.

## CAUAJM\_W\_112589

The specified log device "{0}" already exists.

**Reason:**

The specified log device already exists.

**Action:**

Check if the provided log device is indeed the intended device.

## CAUAJM\_W\_112594

The database schema is at a lower version for the following instances: "{0}". If configured to start post-installation, the AutoSys services will fail to start.

**Reason:**

The present schema in the database is at a lower version. This path will not upgrade the database, and the installer will not start the services post-installation (as the services will fail to start due to schema incompatibilities).

**Action:**

Start the services after the database schema is upgraded to the installer level.

## CAUAJM\_W\_112597

Unable to find sqlplus in the PATH "{0}".

**Reason:**

Unable to find the Oracle client binary in the PATH.

**Action:**

Make sure that the client binary is available in the PATH and retry the installer.

## CAUAJM\_W\_112598

Unable to find "{0}" in the PATH: "{1}".

**Reason:**

Unable to find the SYBASE client binary in the PATH.

**Action:**

Make sure that the client binary is available in the PATH and retry the installer.



## CAUAJM\_W\_113004

The dependent component(s) identified and selected by the AutoSys installer are:

"{0}"

**Reason:**

The required components for some components are not selected.

**Action:**

The dependent component(s) will be selected for installation by the AutoSys installer. No action is required.

## CAUAJM\_W\_113007

A 32-bit CA EEM Server is present on this system. An upgrade of the 32-bit CA EEM Server is not supported on this platform. However, the other AutoSys components can be upgraded. Note: AutoSys version 12.0 Server works fine with a 32-bit CA EEM server.

**Reason:**

A 32-bit CA EEM Server is present on this system. An upgrade of the 32-bit CA EEM Server is not supported on this platform.

**Action:**

Upgrade the other AutoSys components. Perform the CA EEM migration later. For more information, see [AutoSys Common Components Upgrade](#).

## CAUAJM\_W\_113011

The CA EEM server uses iGateway. The installer has detected products that are using the same iGateway. If you continue with AutoSys installation or upgrade, other products might get impacted.

**Reason:**

The CA EEM server uses iGateway. The installer has detected products that are using the same iGateway. As part of AutoSys installation or upgrade process, iGateway may be upgraded. This can cause the other products using iGateway to be impacted if they support the latest iGateway.

**Action:**

Check the compatibility matrix of the other products to see if the iGateway laid with CA EEM server is supported by them.

## CAUAJM\_W\_113013

A few warnings occurred while validating the AutoSys component(s). For more information, refer to the log file.

**Reason:**

Multiple warnings occurred while validating the AutoSys component(s).

**Action:**

Refer to the log file to resolve the issues.

## CAUAJM\_W\_114005

The following host(s) are unreachable:\n"{0}"

**Reason:**

The host(s) are not reachable from the machine.

**Action:**

Make sure that the host names provided are valid and that they will be available for the AutoSys product to work properly. If they will be provisioned later, proceed with the installation.

**CAUAJM\_W\_114012**

The "{0}" port is in use. Specify a different value for "{1}".

**Reason:**

Another process is running on the machine which is listening on the port specified.

**Action:**

The port is required for the installation of the CA directory. If it is already installed on the system as part of other CA Broadcom software, you can continue with the installation, or change the port to an unused port.

**CAUAJM\_W\_116009**

The Web UI connection.properties configuration file is corrupted. {0}

**Reason:**

The Web UI configuration file <CA\_WCC\_INSTALL\_LOCATION>\data\config\application\config\resources\connection.properties is corrupted.

**Action:**

Check the properties files to address the problem mentioned in the message, and rerun the uninstaller or select the **Ignore** option and proceed with the uninstallation.

**CAUAJM\_W\_116203**

The specified account(s) "{0}" do not exist. \n AutoSys installer will create "{1}".

**Reason:**

The accounts specified are not yet defined on the machine, so the installer will create them.

**Action:**

No action is required.

**CAUAJM\_W\_116405**

An error occurred while performing the post-install cleanup. {0}

**Reason:**

An error occurred while performing the post-install cleanup of the Windows registry used in the earlier release of AutoSys.

**Action:**

Manually delete the following Windows registry entries if they exist.

**NOTE**

If these Windows registry entries are not present, it will not impact the functionality of the product, but may interfere with future upgrades.

- HKEY\_CLASSES\_ROOT\Installer\Features\D79FA0A0F351D8644BC9E030C2D3B00D
- HKEY\_CLASSES\_ROOT\Installer\Products\D79FA0A0F351D8644BC9E030C2D3B00D
- HKEY\_LOCAL\_MACHINE\SOFTWARE\Classes\Installer\Features\D79FA0A0F351D8644BC9E030C2D3B00D
- HKEY\_LOCAL\_MACHINE\SOFTWARE\Classes\Installer\Products\D79FA0A0F351D8644BC9E030C2D3B00D
- HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Installer\UserData\S-1-5-18\Products\D79FA0A0F351D8644BC9E030C2D3B00D
- HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\InstallShield\_{0A0AF97D-153F-468D-B49C-0E032C3D0BD0}
- HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\{0A0AF97D-153F-468D-B49C-0E032C3D0BD0}
- HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\ComputerAssociates\License
- HKEY\_LOCAL\_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Uninstall\CA Workload Control Center
- HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Uninstall\CA Workload Control Center Command Sponsor

**CAUAJM\_W\_116406**

**An error occurred while setting owner, groups, or permission. {0}**

**Reason:**

An IOException occurred while setting the owner, groups, or permission.

**Action:**

Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

**CAUAJM\_W\_116410**

**An error occurred while creating the "environment.{0}" file. {1}**

**Reason:**

An error occurred while creating the environment file.

**Action:**

Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

**CAUAJM\_W\_116411**

**An error occurred while creating the "notify.{0}" file. {1}**

**Reason:**

An error occurred while creating the file.

**Action:**

Manually create the file in the AUTOUSER folder, with the following content:

```
EPHighAvail =
AppSrvShutdown =
DBProblem =
```

```
DBRollover =
EPShutdown =
EPRollover =
```

## CAUAJM\_W\_116412

**An error occurred while assigning the owner and group to files. {0}**

**Reason:**

An error occurred while assigning the owner and group to files.

**Action:**

Manually update the permission of the files by running the following command.

**Example:**

Linux:

```
chown -hR <user name>:<group> <install path>
```

## CAUAJM\_W\_116414

**Unable to install or upgrade the "{0}" agent. The agent installer return code is "{1}".**

**Reason:**

Unable to install or upgrade the agent.

**Action:**

Check the agent installation log in the folder <InstallPath>\SystemAgent\<Agent name> for more information to resolve the issue.

## CAUAJM\_W\_116415

**Uninstallation of the "{0}" agent has failed. {1}**

**Reason:**

Uninstallation of the agent has failed.

**Action:**

Try to manually uninstall the agent by executing the binary <InstallPath>\SystemAgent\<Agent name>\UninstallData\Uninstall\_WA\_Agent.

If the file is not present, you may need to manually delete some configuration such as the services and profiles.

## CAUAJM\_W\_116416

**The database has been updated. However, a reconciliation of communication tables was unsuccessful.**

**Reason:**

The database has been updated. However, a reconciliation of communication tables was unsuccessful.

**Action:**

The communication tables need to be reconciled. Contact Broadcom Support.

## CAUAJM\_W\_116417

Unable to install or upgrade the "{0}" agent. Because the installer is unable to find the "{1}" template response file.

**Reason:**

Unable to install or upgrade the agent, because the installer is unable to find the "{1}" template response file.

**Action:**

Run the agent setup directly from ISO\modules\Agent\setup, or contact Broadcom Support.

## CAUAJM\_W\_116418

Unable to install or upgrade the "{0}" agent. {1}

**Reason:**

Failed to create a silent response file to install or upgrade the agent due to an IOException.

**Action:**

Run the agent setup directly from ISO\modules\Agent\setup, or contact Broadcom Support.

## CAUAJM\_W\_116420

Unable to register or upgrade the Web UI EEM application.

**Reason:**

Unable to register or upgrade the Web UI EEM application.

**Action:**

Refer to the log file for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

## CAUAJM\_W\_116602

The connection has failed for the specified proxy URL.

**Reason:**

The connection has failed for the specified proxy URL.

**Action:**

Check if the firewall is open and the URL is accessible from the machine.

## CAUAJM\_W\_119007

The specified key is not matching with the key configured for the selected agent.

**Reason:**

The specified key is not the key with which the agent was installed. Communication cannot be established using the specified key.

**Action:**

Specify the key of the selected agent.

## CAUAJM\_W\_119011

**An error occurred while verifying the specified key with the key used during the selected agent installation. {0}**

**Reason:**

Failed to validate if the specified key matches the selected agent key.

**Action:**

Make sure that the agent key matches the specified key, and proceed with the installation.

## CAUAJM\_W\_119012

**Unable to configure the CA EEM security for the "{0}" instance. Execute the "{1}" command to configure the instance security.**

**Reason:**

Failed to configure the CA EEM security.

**Action:**

Manually execute the command mentioned in the message to configure the CA EEM security.

## CAUAJM\_W\_119015

**Unable to install or upgrade the CA EEM server. Because the installer is unable to find the "{0}" template response file.**

**Reason:**

Failed to create a response file to install the CA EEM server.

**Action:**

Run the CA EEM setup directly from ISO\modules\ EEM\setup, or contact Broadcom Support.

## CAUAJM\_W\_119016

**Unable to install or upgrade the CA EEM server.**

**Reason:**

Failed to install or upgrade the CA EEM server.

**Action:**

Refer to the log file for more information and rerun the installer, or run the CA EEM setup directly from ISO\modules\ EEM \setup.

## CAUAJM\_W\_119017

**Unable to update the telemetry configuration for the "{0}" instance. {1}**

**Reason:**

Failed to update the "telemetry.properties" telemetry configuration.

**Action:**

Manually restore the file from the backup folder.

## CAUAJM\_W\_119018

**Uninstallation of the CA EEM server has failed. {0}**

**Reason:**

Failed to perform the uninstallation of the CA EEM server.

**Action:**

Refer to the eiam-uninstall.log in the TEMP directory for information. Try to rerun the CA EEM uninstaller by executing the <Shared Component Path>\EmbeddedEntitlementsManager\uninstall\eamuninstall binary, or contact Broadcom Support.

## CAUAJM\_W\_119019

**Unable to update the "{0}" configuration file. {1}**

**Reason:**

Failed to update the mentioned configuration file.

**Action:**

Manually update the attribute represented by "{0}" in the configuration file represented by {1}. Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

## CAUAJM\_W\_119020

**Unable to modify the "Enable Reporting" status in the Web UI database. {0}**

**Reason:**

Failed to update the reporting status in the Web UI database.

**Action:**

Manually update the reporting status in the database by running the following query.

```
INSERT INTO CFG_GLOBAL_PREF(PREFKEY, PREFVALUE) values ('internal.reporting.database.enabled','<true/false>')
INSERT INTO CFG_GLOBAL_PREF(PREFKEY, PREFVALUE) values (internal.eem.safeApplication.version,'12.0')
```

## CAUAJM\_W\_119022

**Unable to create the machine definition for the local agent. After the installation, create a machine definition manually using the AutoSys jil command.**

**Reason:**

Failed to create a machine definition for the local agent.

**Action:**

Manually create a machine definition using the AutoSys jil command.

## CAUAJM\_W\_119023

**Unable to get the agent port for "{0}". Therefore, using the default port 7520.**

**Reason:**

Failed to retrieve the communication port of the agent, so a machine definition for the local agent with the default agent port was created.

**Action:**

If the agent port is incorrect, manually update the machine definition for the local agent with proper agent port details.

## CAUAJM\_W\_119024

Unable to create an instance profile for the "{0}" instance.

**Reason:**

Failed to create the auto.<Instance name> instance profile.

**Action:**

Manually update the profile as follows, or contact Broadcom Support.

Sample content of file. Create the file if it doesn't exist and add instance section if file exists.

```
DISPLAY=":0.0"
```

```
export DISPLAY
```

```
Set a PATH so executables can be found
```

```
PATH=".: $AUTOSYS/bin: $AUTOSYS/test/bin: /bin: /usr/bin: /usr/local/bin: /usr/openwin/bin: /usr/bin/X11: /bin: /usr/ucb: /usr/etc: $PATH"
```

```
export PATH
```

```
Set the library path
```

```
case `uname` in
```

```
 AIX) LIBPATH=$AUTOSYS/lib:$LIBPATH
```

```
 export LIBPATH;;
```

```
 HP-UX) SHLIB_PATH=$AUTOSYS/lib:$SHLIB_PATH
```

```
 export SHLIB_PATH;;
```

```
 *) LD_LIBRARY_PATH=$AUTOSYS/lib:$LD_LIBRARY_PATH
```

```
 export LD_LIBRARY_PATH;;
```

```
esac
```

```
The AUTOSYS and AUTOUSER environment variables are needed if the job's
```

```
command uses AutoSys Workload Automation programs.
```

```
#
```

```
AUTOSYS is already set in the environment for cybAgent, the Agent.
```

```
AUTOUSER can be different for each instance in the case statement below.
```

```
case $AUTOSERV in
```

```
ACE)
```

```
 # Generic R11 setup, replaced during installation.
```

```
 hostname=`$AUTOSYS/bin/autoflags -n`
```

```
 AUTOUSER=/opt/CA/WorkloadAutomationAE/autouser.ACE
```

```
 test -f $AUTOUSER/autosys.sh.$hostname &&
```

```
 . $AUTOUSER/autosys.sh.$hostname
```

```
 ;;
```

```
esac
```



## CAUAJM\_W\_119025

Unable to update the "/etc/auto.

profile" file for the "{0}" instance.

### Reason:

Failed to update the profile file with instance details.

### Action:

Manually update the profile as shown below, or contact Broadcom Support.

Sample auto.profile file content:

```
#* ----- *
"@(#) [file] [version] [modtime]"
#* ----- *
#
Set AutoSys Workload Automation Environmental Variables:
#
This must be a Bourne shell script, and the variables must be exported
for your command to have access to them.

Windowing system environment variable
DISPLAY=":0.0"
export DISPLAY

Set a PATH so executables can be found
PATH=".:$AUTOSYS/bin:$AUTOSYS/test/bin:/bin:/usr/bin:/usr/local/bin:/usr/openwin/bin:/usr/bin/X11:/bin:/usr/
ucb:/usr/etc:$PATH"
export PATH

Set the library path
case `uname` in
 AIX) LIBPATH=$AUTOSYS/lib:$LIBPATH
 export LIBPATH;;
 HP-UX) SHLIB_PATH=$AUTOSYS/lib:$SHLIB_PATH
 export SHLIB_PATH;;
 *) LD_LIBRARY_PATH=$AUTOSYS/lib:$LD_LIBRARY_PATH
 export LD_LIBRARY_PATH;;
esac

The AUTOSYS and AUTOUSER environment variables are needed if the job's
command uses AutoSys Workload Automation programs.
#
AUTOSYS is already set in the environment for cybAgent, the Agent.
AUTOUSER can be different for each instance in the case statement below.

case $AUTOSERV in
 ORA)
 # Generic R11 setup, replaced during installation.
 hostname=`$AUTOSYS/bin/autoflags -n`
 AUTOUSER=/opt/CA/WorkloadAutomationAE/autouser.ORA
 test -f $AUTOUSER/autosys.sh.$hostname &&
```

```

 . $AUTOUSER/autosys.sh.$hostname
 ;;
esac

```

## CAUAJM\_W\_119027

**Unable to create the "{0}" AutoSys super user.**

**Reason:**

Failed to create the mentioned super user.

**Action:**

Create the user manually by running the autosys\_secure utility.

## CAUAJM\_W\_119028

**Unable to create a symbolic link "{0}" for the "{1}" file.**

**Reason:**

Failed to create a symbolic link for the mentioned file.

**Action:**

Manually create the symbolic link for the file mentioned in the message.

## CAUAJM\_W\_119029

**Unable to create the following AutoSys environment variable in the "{0}" file.**

**Reason:**

Failed to update the mentioned file with the environment variables.

**Action:**

Manually update the mentioned file with the variables.

```

AUTOSYS_INSTALL_LOCATION=/opt/CA/WorkloadAutomationAE; export AUTOSYS_INSTALL_LOCATION

AUTOSYS_CSUTILS=/opt/CA/WorkloadAutomationAE/autosys/csutils; export AUTOSYS_CSUTILS

PATH=$AUTOSYS_CSUTILS/bin:$PATH; export PATH
Example:
Sample content of the /etc/profile.CA
CA_SECTION=${CA_SECTION:-${1:-}}; export CA_SECTION #Set CA_SECTION to the first argument
CA_ETC_CAPROFILE() {
CASHCOMP=/opt/CA/SharedComponents; export CASHCOMP
["$CA_SECTION" = "minimum" -a CASHCOMP] && return 0
CALIB=/opt/CA/SharedComponents/lib; export CALIB
CABIN=/opt/CA/SharedComponents/bin; export CABIN
CSAM_SOCKADAPTER=/opt/CA/SharedComponents/Csam/SockAdapter; export CSAM_SOCKADAPTER
if [-f "$CSAM_SOCKADAPTER/scripts/csamsetlibpath.sh"] ; then
. "$CSAM_SOCKADAPTER/scripts/csamsetlibpath.sh"
fi
CSAM_LOGGER_CONF=/opt/CA/SharedComponents/Csam/SockAdapter/cfg/logger.cfg; export CSAM_LOGGER_CONF
}; CA_ETC_CAPROFILE $CA_SECTION

```

```
AUTOSYS_INSTALL_LOCATION=/opt/CA/WorkloadAutomationAE; export AUTOSYS_INSTALL_LOCATION
AUTOSYS_CSUTILS=/opt/CA/WorkloadAutomationAE/autosys/csutils; export AUTOSYS_CSUTILS
PATH=$AUTOSYS_CSUTILS/bin:$PATH; export PATH
WD_COREDIR="core.d"; export WD_COREDIR
WD_SAVECORE="false"; export WD_SAVECORE
IGW_LOC="/opt/CA/SharedComponents/iTechnology"; export IGW_LOC
["${CANOLOG_flag:-${CANOLOG_flag:-}} " = "off"] && set +o nolog; CANOLOG_flag="" # return to system setting
["${CACSUTILS_setu:-${CACSUTILS_setu:-}} " = "1"] && set -u; CACSUTILS_setu="" # return to system setting
```

## CAUAJM\_W\_119030

**Unable to generate the keystore. Use the following command to manually generate the keystore.\n{0}**

### Reason:

Failed to generate the keystore file to be consumed by Tomcat.

### Action:

Run the command mentioned in the message string to manually generate the keystore file.

## CAUAJM\_W\_119031

**Unable to delete the "{0}" Web UI collector record from the database.**

### Reason:

Failed to delete the collector node information from the database during uninstallation/deletion of the Web UI collector.

### Action:

Manually delete the collector node information from the database by running the following query.

```
DELETE FROM CFG_COLLECTOR_TASKS WHERE NODE_ID=(SELECT NODE_ID FROM CFG_COLLECTOR_NODES WHERE
COLLECTOR_NAME=<collector name> AND HOST=<localhost>)
DELETE_WCC_COLLECTOR=DELETE FROM CFG_COLLECTOR_NODES WHERE COLLECTOR_NAME=<collector name> AND
HOST=<localhost>
```

## CAUAJM\_W\_119032

**Unable to delete the "{0}" Web UI collector directory.\n{1}**

### Reason:

Failed to delete the Web UI collector folder as part of the collector deletion.

### Action:

Examine the exception mentioned in the message and take action accordingly.

## CAUAJM\_W\_119033

**Unable to delete the Web UI node record from the database.**

### Reason:

During uninstallation of the Web UI, the uninstaller failed to delete the current node information from the database.

### Action:

Manually delete the current node information from the database by running the following query.

```
DELETE FROM CFG_HA_INSTANCE WHERE NODENAME=<localhost>
```

## CAUAJM\_W\_119034

Unable to configure the CA EEEM certificate with a key length of 2048. For more information, refer to the online documentation.

**Reason:**

Failed to configure the CA EEM certificate with a key length of 2048.

**Action:**

Manually configure the CA EEM certificate with a key length of 2048. For more information, see [Configure CA EEM for Root Certificates Generated with Different Key Lengths](#).

## Error Messages

You may encounter error messages while working with AutoSys Workload Automation. Error messages tell you that an error has occurred and the action you need to take to resolve the situation.

### CAUAJM\_E\_00003

Could not allocate size[{}]. Exiting.

**Reason:**

Allocation failure.

**Action:**

Required action unknown.

### CAUAJM\_E\_00006

Error to access api.

**Reason:**

Error to access api.

**Action:**

Required action unknown.

### CAUAJM\_E\_00017

Unable to determine class. Cannot continue.

**Reason:**

JNIEnv::IsInstanceOf() did not return a known class.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_00020

**Local error in attempt to send request. RC={0}**

**Reason:**

Local error in attempt to send request.

**Action:**

Required action unknown.

## CAUAJM\_E\_00023

**Unable to copy file <{0}> to <{1}>**

**Reason:**

A copy file error has occurred.

**Action:**

Review messages that follow and take action as required.

## CAUAJM\_E\_00025

**Unable to read file <{0}>**

**Reason:**

A read file error has occurred.

**Action:**

Review messages that follow and take action as required.

## CAUAJM\_E\_00035

**Unable to open file <{0}>**

**Reason:**

An open file error has occurred.

**Action:**

Review messages that follow and take action as required.

## CAUAJM\_E\_00036

**Function failed: <{0}> Errno: <{1,number,1}> Error: <{2}>**

**Reason:**

An error has occurred performing the indicated operation.

**Action:**

Review the error information and take action as required.

## CAUAJM\_E\_00039

**Local error in send {0} request to the server.**

**Reason:**

Local error in send AuthUser request to the server.

**Action:**

Required action unknown.

## CAUAJM\_E\_00040

**Unable to create object: Invalid class.**

**Reason:**

Unable to create object: Invalid class.

**Action:**

Required action unknown.

## CAUAJM\_E\_00042

**Unable to create Response: Invalid class.**

**Reason:**

Unable to create Response: Invalid class.

**Action:**

Required action unknown.

## CAUAJM\_E\_00045

**Exceeded wait time for the UNIX function getaddrinfo. Hostname [{0}] was not resolved.**

**Reason:**

All attempts to resolve the hostname have been exhausted.

**Action:**

Verify that the hostname can be resolved into a network IP address and is reachable over the network.

## CAUAJM\_E\_00046

**The UNIX function getaddrinfo failed to resolve hostname [{0}] after retrying. System message: {1}.**

**Reason:**

The name server returned an error while resolving the hostname to a network IP address

**Action:**

Review the system message for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_00047

**Unable to create Filter: can not find java classes.**

**Reason:**

Unable to create Filter: cannot find java classes.

**Action:**

Required action unknown.

## CAUAJM\_E\_00048

**Unable to create Filter: Invalid class.**

**Reason:**

Unable to create Filter: Invalid class.

**Action:**

Required action unknown.

## CAUAJM\_E\_00049

**Could not setdetachstate UNDETACHED for [{0}][{1}].**

**Reason:**

Could not setdetachstate.

**Action:**

Required action unknown.

## CAUAJM\_E\_00050

**Could not setdetachstate DETACHED for [{0}][{1}].**

**Reason:**

Could not setdetachstate.

**Action:**

Required action unknown.

## CAUAJM\_E\_00051

**Unable to create thread! [{0}][{1}]**

**Reason:**

Unable to create thread.

**Action:**

Required action unknown.

## CAUAJM\_E\_00053

**[[{0}]] not joined: {1}**

**Reason:**

Join error.

**Action:**

Required action unknown.

## CAUAJM\_E\_00054

**The string provided for field <{0}> is too long: {1}**

**Reason:**

The string is too long.

**Action:**

Choose a shorter string.

## CAUAJM\_E\_00056

**Unable to add Filter: cannot locate JFilter object.**

**Reason:**

Unable to add Filter: cannot locate JFilter Object.

**Action:**

Required action unknown.

## CAUAJM\_E\_00061

**Machine <{0}> is not defined in the database.**

**Reason:**

Machine is unavailable.

**Action:**

Define the machine to the database.

## CAUAJM\_E\_00062

**Unsupported machine type for authentication.**

**Reason:**

The machine type defined is unsuitable for authentication.

**Action:**

Specify a different machine.



## CAUAJM\_E\_00063

**Database error during authentication.**

**Reason:**

A database error occurred retrieving a machine definition used for authentication.

**Action:**

Check the application server log for errors.

## CAUAJM\_E\_00064

**Return code {0} received processing request.**

**Reason:**

An unexpected error occurred identified by the return code.

**Action:**

Contact CA Support for further assistance.

## CAUAJM\_E\_00065

**Validation Internal Error: User failed authentication**

**Reason:**

The product instance has not been configured with internal user.

**Action:**

Define the specified user according to the security implemented.

## CAUAJM\_E\_00066

**Validation Failure: Artifact \"{0}\" failed authentication.**

**Reason:**

The artifact provided is invalid.

**Action:**

Acquire a new artifact and retry.

## CAUAJM\_E\_00067

**Validation Internal Error: Artifact \"{0}\" failed authentication.**

**Reason:**

The product instance has not been configured to run under CA EEM security.

**Action:**

Reconfigure the product or change the authentication method.

## CAUAJM\_E\_00076

{0} {1} failed. Error: {2}\n" }

**Reason:**

WSAStartup failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_00088

Could not get host information for [{0}]. Not started.

**Reason:**

Could not get host information.

**Action:**

Required action unknown.

## CAUAJM\_E\_00092

Unable to load the jvm library <{0}>.

**Reason:**

Could not load the required library.

**Action:**

Ensure that the required library is available on the system.

## CAUAJM\_E\_00094

Unable to obtain routine address for <{0}> in library <{1}>.

**Reason:**

The system cannot locate the address for the specified function in the library, typically when the specified library is unavailable.

**Action:**

Verify that the required library is available on the system.

## CAUAJM\_E\_00095

Unable to load the email notification library <{0}>.

**Reason:**

Unable to load required library.

**Action:**

Make sure the required library is available on the system.

## CAUAJM\_E\_00096

**Unable to determine current user credentials.**

**Reason:**

Unknown.

**Action:**

Unknown.

## CAUAJM\_E\_00102

**Notification request aborted.**

**Reason:**

Unable to load required library. The notification request has been aborted.

**Action:**

Make sure the Notification component is installed and the required library is available on the system.

## CAUAJM\_E\_00103

**Unable to obtain routine address for {0} in library {1}; Notification request aborted.**

**Reason:**

Unable to locate function address in library. Notification request aborted.

**Action:**

Make sure the Notification component is installed and the required library is available on the system.

## CAUAJM\_E\_00104

**Failed to open the Notification interface rc = {0}**

**Reason:**

Unable to open the Notification interface.

**Action:**

Make sure the Notification component is installed and is active.

## CAUAJM\_E\_00105

**Notification provider was not supplied. Request aborted.**

**Reason:**

The notification provider was not supplied.

**Action:**

Review the attributes of the job for which the notification was being sent.

## CAUAJM\_E\_00106

Failed to set {0} rc = {1}

**Reason:**

Unable to set parameter.

**Action:**

Review the return code from Unicenter Notification and take action as required.

## CAUAJM\_E\_00107

Failed to issue Notification Services request rc = {0} JOB: {1}

**Reason:**

Unable to issue notification request.

**Action:**

Review the return code from Unicenter Notification Services and take action as required.

## CAUAJM\_E\_00108

Service Desk login failure rc = {0}

**Reason:**

Service Desk login failure.

**Action:**

Make sure Service Desk Web Service is properly installed and accessible through the network.

## CAUAJM\_E\_00109

{0}

**Reason:**

Service Desk SOAP FAULT message.

**Action:**

Review message and take action as required.

## CAUAJM\_E\_00110

{0}

**Reason:**

Service Desk SOAP FAULT DETAIL message.

**Action:**

Review message and take action as required.

## CAUAJM\_E\_00111

{0}

**Reason:**

Service Desk SOAP FAULT LOCATION message.

**Action:**

Review message and take action as required.

## CAUAJM\_E\_00112

**Failure obtaining Service Desk user handle for user {0} rc = {1}**

**Reason:**

Failure obtaining Service Desk user handle for user.

**Action:**

Make sure the user has been defined to the Service Desk Web Service.

## CAUAJM\_E\_00113

**Failure obtaining Service Desk customer handle for customer {0} rc = {1}**

**Reason:**

Failure obtaining ServiceDesk customer handle for customer.

**Action:**

Make sure the customer has been defined to the ServiceDesk Web Service.

## CAUAJM\_E\_00114

**Service Desk request failure rc = {0}**

**Reason:**

Failure creating ServiceDesk request.

**Action:**

Review the return code and take action as needed.

## CAUAJM\_E\_00115

**Unable to obtain Service Desk login user and password from config file/registry.**

**Reason:**

Unable to obtain Service Desk login user and password from config file/registry.

**Action:**

Ensure the user/encrypted password combination is set for configuration variable ServiceDeskUser. Try re-generating and setting encrypted password.

## CAUAJM\_E\_00116

**Service Desk user is not available. Request aborted.**

**Reason:**

Service Desk user is not available. Request aborted.

**Action:**

Ensure the user/encrypted password combination is set for configuration variable ServiceDeskUser. Try re-generating, setting encrypted password and restart the Scheduler.

## CAUAJM\_E\_00118

**Service Desk request failed JOB: {0}**

**Reason:**

Unable to create a Service Desk request.

**Action:**

Refer to prior messages for cause and take action as required.

## CAUAJM\_E\_00121

**Notification Services request failure JOB: {0}**

**Reason:**

Unable to send Notification Services request.

**Action:**

Refer to prior messages for cause and take action as required.

## CAUAJM\_E\_00123

**Error parsing svcdesk\_attr keyword/keyword value pair. Request aborted.**

**Reason:**

Error parsing svcdesk\_attr keyword/keyword value pair.

**Action:**

Review the JIL svcdesk\_attr attribute to ensure the input string has been coded in keyword/keyword value pairs. If multiple attributes are specified they must be comma separated.

## CAUAJM\_E\_00124

**Duplicate Service Desk keyword detected "{0}". Requested aborted.**

**Reason:**

A duplicate keyword has been detected within the Service Desk JIL input.

**Action:**

Review the JIL job definition for the specified duplicate keyword and remove accordingly.

## CAUAJM\_E\_00125

**Notify request was rejected.**

**Reason:**

Unicenter Notification rejected the request.

**Action:**

Using the Unicenter Notification Status UI check the request for indications of the problem. Take action as required.

## CAUAJM\_E\_00126

**UNS Diagnostic: {0}**

**Reason:**

Unicenter Notification diagnostic message for the failing request.

**Action:**

Take action as required by the message(s).

## CAUAJM\_E\_00132

**Database indicates failover status. Shutting down.**

**Reason:**

Database indicates failover status. Shutting down.

**Action:**

Determine problem that caused failover and restart system.

## CAUAJM\_E\_00133

**The shadow and tie-breaker are both missing. The primary must shutdown.**

**Reason:**

The shadow and tie-breaker are both missing. The primary must shutdown.

**Action:**

Determine why shadow and tie-breaker are missing and restart system.

## CAUAJM\_E\_00148

**The primary and tie-breaker are both missing. The shadow must shutdown.**

**Reason:**

The primary and tie-breaker are both missing. The shadow must shutdown.

**Action:**

Determine why shadow and tie-breaker are missing and restart system.

## CAUAJM\_E\_00159

Unexpected error in NTthread.cpp. Function: {0}; File: {1}; Line: {2}

**Reason:**

Unknown.

**Action:**

Unknown.

## CAUAJM\_E\_00160

WaitForSingleObject failed. Return code [{0}].

**Reason:**

Unknown.

**Action:**

Unknown.

## CAUAJM\_E\_00161

GetLastError(): [{0}]

**Reason:**

Unknown.

**Action:**

Unknown.

## CAUAJM\_E\_00165

The {0} primary scheduler is already running on machine: {1}. Startup aborted.

**Reason:**

The primary scheduler is already running on the specified machine. Startup aborted.

**Action:**

Shutdown the active primary scheduler before starting a new scheduler on this machine.

## CAUAJM\_E\_00166

The {0} shadow scheduler is already running on machine: {1}. Startup aborted.

**Reason:**

The shadow scheduler is already running on the specified machine. Startup aborted.

**Action:**

Shutdown the active shadow scheduler before starting a new scheduler on this machine.



## CAUAJM\_E\_00170

**The {0} tie-breaker scheduler is already running on machine: {1}. Startup aborted.**

**Reason:**

The tie-breaker scheduler is already running on the specified machine. Startup aborted.

**Action:**

Shutdown the active tie-breaker scheduler before starting a new scheduler on this machine.

## CAUAJM\_E\_00171

**Database error encountered while trying to determine if scheduler is already running, startup aborted.**

**Reason:**

Database error encountered while trying to determine if scheduler is already running, startup aborted.

**Action:**

Check the status of the database.

## CAUAJM\_E\_00172

**Database error encountered while initializing the high availability system. Startup aborted.**

**Reason:**

Database error encountered while initializing the high availability system. Startup aborted.

**Action:**

Check the status of the database.

## CAUAJM\_E\_00179

**Unable to load the CA Secure Sockets Adapter shared library dynamically. Process terminating.**

**Reason:**

The CA Secure Sockets Adapter shared library could not be loaded. The library could have been moved, renamed, or deleted.

**Action:**

Check if the CA Secure Sockets Adapter files have been moved, renamed, or deleted.

## CAUAJM\_E\_00191

**Shutdown notification for the primary scheduler was not written into any database.**

**Reason:**

Shutdown notification for the primary scheduler was not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00192

**Shutdown notification for the shadow scheduler was not written into any database.**

**Reason:**

Shutdown notification for the shadow scheduler was not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00193

**Shutdown notification for the tie-breaker scheduler was not written into any database.**

**Reason:**

Shutdown notification for the tie-breaker scheduler was not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00194

**Shutdown notifications for all application servers were not written into any database.**

**Reason:**

Shutdown notifications for all application servers were not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00196

**Failover notification was not written into any database.**

**Reason:**

Failover notification was not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00198

**Unable to allocate internal Service Desk work objects. Request aborted.**

**Reason:**

The scheduler failed to allocate Service Desk work objects required to prepare the request. This is more than likely due to system resource constraints.

**Action:**

Ensure the system running the scheduler process has ample memory otherwise contact CA Technical Support for further assistance.

## CAUAJM\_E\_00200

**Encryption/Decryption Failed. Module: {0}, Error: {1}, Possible Cause: {2}\n**

**Reason:**

Encryption/Decryption Failed

**Action:**

Required action unknown.

## CAUAJM\_E\_00205

**Restart notification for the shadow scheduler was not written into any database.**

**Reason:**

Restart notification for the shadow scheduler was not written into any database.

**Action:**

Investigate the state of both databases.

## CAUAJM\_E\_00224

**The {0} library experienced an internal error and may not have initialized properly.**

**Reason:**

The CAPKI library experienced an internal error during initialization.

**Action:**

If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_00225

**The {0} library experienced an internal error. Error message: {1}**

**Reason:**

The OPENSSL library experienced an internal error during initialization.

**Action:**

If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_00226

**The data cannot be decrypted as the provided encrypted data is not in the correct format.**

**Reason:**

The encrypted data is not in the expected format.

**Action:**

If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_00227

Unsupported algorithm found in the header.

Unsupported AES algorithm mode found in the header.

Unsupported padding found in the header.

Unsupported key size found in the header.

Unsupported value found in the unused section of the header.

**Reason:**

The encrypted data does not have the correct format.

**Action:**

If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_00228

Error when retrieving agent version from the database.

**Reason:**

Error when retrieving agent version from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_00233

The path specified against 'payload\_uri' should follow file URI conventions and begin with 'file:/'.

**Reason:**

The path specified against payload\_uri is in incorrect format.

**Action:**

Specify the proper file prefix to the path against payload uri.

## CAUAJM\_E\_00234

The attribute "j2ee\_authentication\_order" should contain "BASIC" when "preemptive\_authentication" is specified.

**Reason:**

The job has an invalid combination of attributes specified.

**Action:**

Define the job with a valid combination of attributes.

## CAUAJM\_E\_00235

ERROR for job: {0} < "{1}" cannot be specified for "{2}" operation.

**Reason:**

---

Invalid attribute has been specified.

**Action:**

Specify valid attributes.

## **CAUAJM\_E\_00236**

**The attribute "{0}" needs to be specified when "{1}" is specified with value "{2}".**

**Reason:**

The job has an invalid combination of attributes specified.

**Action:**

Define the job with a valid combination of attributes."

## **CAUAJM\_E\_00237**

**The blob data defined against the job contains missing "{0}" tag(s).**

**Reason:**

The blob data defined against the job contains either missing beginning or ending tags.

**Action:**

Redefine the job with matching beginning and ending tags.

## **CAUAJM\_E\_00337**

**Service Desk request failed due to error.**

**Reason:**

Unable to create a Service Desk request.

**Action:**

Refer to prior messages for cause and take action as required.

## **CAUAJM\_E\_10000**

**Invalid value "{0}" specified for keyword {1}.**

**Reason:**

Invalid value specified for job keyword.

**Action:**

Re-submit the request with a proper value for the keyword.

## **CAUAJM\_E\_10001**

**Unable to execute a request originating from a source that is not authenticated.**

**Reason:**

The unauthenticated user mode is set to STRICT or EXTERNAL, and the request executed by the CA Workload Automation AE SDK program is missing required user authentication information.

**Action:**

Troubleshoot the CA Workload Automation AE SDK program code to ensure that it authenticates the user and sets the user in the request and then recompile and execute the program, or contact your security administrator for assistance.

**CAUAJM\_E\_10002**

**Unable to retrieve connection information for instance <{0}>.**

**Reason:**

The specified instance does not exist in the external configuration information, or it is not of AppServer type.

**Action:****CAUAJM\_E\_10003**

**User <{0}> on machine <{1}> is not authenticated.**

**Reason:**

The request contains information for a user that is not authenticated.

**Action:**

Troubleshoot the CA Workload Automation AE SDK program code to ensure that it authenticates the user and sets the user in the request and then recompile and execute the program, or contact your security administrator for assistance.

**CAUAJM\_E\_10004**

**Error retrieving current user and machine.**

**Reason:**

Error retrieving current user and machine.

**Action:**

Required action unknown.

**CAUAJM\_E\_10005**

**Attempt to create a local API connection when one has already been created.**

**Reason:**

API connection has already been created.

**Action:**

Required action unknown.

**CAUAJM\_E\_10006**

**Unable to load libtx\_local.**

**Reason:**

Unable to load libtx\_local.

**Action:**

---

Required action unknown.

## **CAUAJM\_E\_10007**

**Unable to get address for createTx.**

**Reason:**

Unable to get address for createTx.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10008**

**Unable to get address for deleteTx.**

**Reason:**

Unable to get address for deleteTx.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10009**

**Unable to load libtx\_remote.**

**Reason:**

Unable to load libtx\_remote.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10010**

**Fatal error. Process terminating. [{0}][{1}]**

**Reason:**

Fatal Error.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10011**

**The specified job field "{0}" can not be saved as type {1}.**

**Reason:**

The set method was called with a job field that is not of the correct type.

**Action:**

Call the set method with the correct datatype for the specified job field.

## CAUAJM\_E\_10012

Unable to get address for createTxRemote.

**Reason:**

Unable to get address for createTxRemote.

**Action:**

Required action unknown.

## CAUAJM\_E\_10013

Failed to load {0} shared library.

**Reason:**

Failed to load DRM shared library.

**Action:**

Required action unknown.

## CAUAJM\_E\_10014

Error establishing a database connection.

**Reason:**

Error establishing database connection.

**Action:**

Required action unknown.

## CAUAJM\_E\_10016

This request [{0}] came from a previous instance and cannot be processed.

**Reason:**

Request cannot be processed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10018

CONDITION ERROR: Unbalanced parentheses in logical expression.

**Reason:**

Unbalanced parentheses in logical expression.

**Action:**

Required action unknown.



## CAUAJM\_E\_10019

**CONDITION ERROR:** Invalid external instance {0}: exceeds {1} characters.

**Reason:**

Error with external instance.

**Action:**

Required action unknown.

## CAUAJM\_E\_10020

**CONDITION ERROR:** Invalid external instance {0}: illegal use of characters.

**Reason:**

Invalid characters specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_10024

**CONDITION ERROR:** Invalid job name <{0}> exceeds {1} characters

**Reason:**

Error with jobname in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10025

**CONDITION ERROR:** Invalid {0}: {1} <{2}>

**Reason:**

Error with jobname in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10026

**CONDITION ERROR:** Invalid lookback expression {0}: exceeds {1} characters

**Reason:**

Error with lookback in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10027

**CONDITION ERROR: Invalid lookback expression {0}: must have a non-decimal numeric value**

**Reason:**

Error with lookback in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10029

**Communication attempt with the {0} Application Server has failed! [{1}:{2,number,1}]**

**Reason:**

Could not send.

**Action:**

Make sure the application server is running and the node it is on is available.

## CAUAJM\_E\_10030

**Error encountered while reading API results!**

**Reason:**

Could not DQ.

**Action:**

Required action unknown.

## CAUAJM\_E\_10032

**Unable to create server object for host [{0}]**

**Reason:**

The host is invalid.

**Action:**

Change the host.

## CAUAJM\_E\_10033

**CONDITION ERROR: Invalid lookback expression {0}: hours not within range (1-9998)**

**Reason:**

Error with lookback in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10034

**CONDITION ERROR: Invalid lookback expression {0}: minutes not within range (0-59)**

**Reason:**

Error with lookback in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10035

**{0} failed! Error: {1}\n**

**Reason:**

WSAStartup failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10036

**CONDITION ERROR: Logical operator must have 2 operands**

**Reason:**

Logical operator needs to have 2 operands.

**Action:**

Required action unknown.

## CAUAJM\_E\_10037

**CONDITION ERROR: Illegal Key Word: {0}.**

**Reason:**

Illegal Key Word.

**Action:**

Required action unknown.

## CAUAJM\_E\_10038

**Unsupported encryption type specified.**

**Reason:**

An unsupported encryption type was requested to be used.

**Action:**

Specify a supported encryption type.

## CAUAJM\_E\_10041

**CONDITION ERROR: Using both ANDs and ORs requires the use of parenthesis.**

**Reason:**

Mixing ANDs and ORs. Please use parenthesis.

**Action:**

Required action is not known.

## CAUAJM\_E\_10042

**CONDITION ERROR: Invalid global variable name <{0}> exceeds {1} characters.**

**Reason:**

Error with global variable name in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10045

**An invalid encryption key was specified.**

**Reason:**

The encryption key specified is invalid.

**Action:**

Correct the encryption key.

## CAUAJM\_E\_10046

**Could not open file: {0}**

**Reason:**

The file could not be opened.

**Action:**

Required action unknown.

## CAUAJM\_E\_10048

**template: {0} file contains invalid input: {1}**

**Reason:**

Bad input from template file.

**Action:**

Specify the input as per the template format.

## CAUAJM\_E\_10049

**Unable to retrieve instance configuration settings.**

**Reason:**

The configuration settings could not be found.

**Action:**

Run this application from the product's command prompt.

## CAUAJM\_E\_10050

**Invalid {0} in config file: {1}**

**Reason:**

Invalid MachineMethod in config file.

**Action:**

Required action unknown.

## CAUAJM\_E\_10051

**The environmental variable {0} is not set.**

**Reason:**

The given environment variable is not set within the product's environment.

**Action:**

This indicates that the product's environment was not set or the product is not correctly installed.

## CAUAJM\_E\_10053

**CONDITION ERROR: Missing job or global variable name.**

**Reason:**

Errors encountered parsing the name value.

**Action:**

Required action unknown.

## CAUAJM\_E\_10054

**The environmental variable {0} is larger then the acceptable length.**

**Reason:**

The given environment variable is too long.

**Action:**

Specify a valid value for the indicated environmental variable.

## CAUAJM\_E\_10056

**CONDITION ERROR: Invalid lookback expression {0}: value exceeds allowable maximum of 9999 hours.**

**Reason:**

Error with lookback in condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10057

**ERROR logging audit data://{0} may have unpredictable results\n**

**Reason:**

ERROR logging audit data//autotrack may have unpredictable results.

**Action:**

Required action unknown.

## CAUAJM\_E\_10058

**Could not load intcodes. Exiting!**

**Reason:**

Could not load intcodes.

**Action:**

Required action unknown.

## CAUAJM\_E\_10059

**Unable to encode password.**

**Reason:**

Unable to encode password.

**Action:**

Required action unknown.

## CAUAJM\_E\_10060

**Unable to decode password.**

**Reason:**

Unable to decode password.

**Action:**

Required action unknown.

## CAUAJM\_E\_10061

Cannot locate AUTOUSER for instance: {0}

**Reason:**

Unable to locate AUTOUSER for this instance

**Action:**

Required action unknown.

## CAUAJM\_E\_10062

Failed to get initial configuration from {0} Application Server(s)

**Reason:**

Failed to get initial configuration the product's Application Server(s).

**Action:**

Required action unknown.

## CAUAJM\_E\_10067

Failed to get an event identifier. Scheduler cannot recover and must shutdown.

**Reason:**

Get EventID failed.

**Action:**

Required action is not known.

## CAUAJM\_E\_10068

The specified attribute "{0}" was not found.

**Reason:**

The attribute was not found.

**Action:**

Call the get method with an attribute that does exist.

## CAUAJM\_E\_10069

sigaction failed\n

**Reason:**

sigaction failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10072

**The Cross Platform Interface is currently not active.**

**Reason:**

The Cross Platform Interface is currently not active..

**Action:**

Set the CrossPlatformScheduling option to a value of '1' or '2' depending on your installation requirements.

## CAUAJM\_E\_10075

**Error initiating communications**

**Reason:**

opening stream socket.

**Action:**

No action is required.

## CAUAJM\_E\_10076

**The connection to machine: {0} TIMED OUT.**

**Reason:**

Connection timed out.

**Action:**

No action is required.

## CAUAJM\_E\_10077

**Either that machine, or the network to it, is having problems.**

**Reason:**

See previous CAUAJM\_E\_10076 message.

**Action:**

No action is required.

## CAUAJM\_E\_10078

**Could NOT connect to machine: {0}**

**Reason:**

The connection to the machine failed.

**Action:**

Check if the agent is installed properly.



## **CAUAJM\_E\_10079**

**ERROR for Job: {0} < {1} >**

**Reason:**

Error for job.

**Action:**

Define a resource.

## **CAUAJM\_E\_10080**

**The Agent Service may not be installed properly.**

**Reason:**

See previous CAUAJM\_E\_10078 message.

**Action:**

No action required.

## **CAUAJM\_E\_10081**

**The internet daemon or the Agent Service may not be configured properly.**

**Reason:**

See previous CAUAJM\_E\_10078 message.

**Action:**

No action required.

## **CAUAJM\_E\_10083**

**read failed Win32 socket error={0}\n**

**Reason:**

Win socket error.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10084**

**read failed stream socket.**

**Reason:**

Read failed stream socket.

**Action:**

Required action unknown.

## CAUAJM\_E\_10085

\*\*\* @@ Agent Process not started. \*\*\*

**Reason:**

The agent was not started.

**Action:**

Required action unknown.

## CAUAJM\_E\_10086

\*\*\* @@ Agent Process returned ERROR. \*\*\*

**Reason:**

Error occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_10087

\t## socket read <{0}>, rc={1}

**Reason:**

Socket read error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10088

write failed stream socket.

**Reason:**

Write failed stream socket.

**Action:**

Required action unknown.

## CAUAJM\_E\_10089

socket write length mismatch rc={0} wlen={1}

**Reason:**

Socket write length mis-match.

**Action:**

Required action unknown.

## CAUAJM\_E\_10090

\*\*\*\tThe {0} IS NOT Running, yet the Job is.

**Reason:**

The auto\_remote IS NOT Running, yet the Job is.

**Action:**

Required action unknown.

## CAUAJM\_E\_10091

\*\*\*\tThis must be investigated Manually.

**Reason:**

This must be investigated manually.

**Action:**

Required action unknown.

## CAUAJM\_E\_10092

Could not {0}/{1} job: {2} The {3} IS NOT Running, yet the Job is!

**Reason:**

Failure sending KILLJOB.

**Action:**

Required action unknown.

## CAUAJM\_E\_10093

Could not {0}/{1} job <{2}>: error code <{3}>.

**Reason:**

Could not SEND\_SIGNAL/KILLJOB.

**Action:**

Required action unknown.

## CAUAJM\_E\_10094

For further information, see {0} log file.

**Reason:**

See previous CAUAJM\_E\_10093 message.

**Action:**

No action required.

## CAUAJM\_E\_10095

**Shadow Processor NOT started successfully - Exit Code={0}**

**Reason:**

The shadow processor was not started successfully.

**Action:**

Required action unknown.

## CAUAJM\_E\_10096

**Could not Open the Shadow Processor Output File:\n\t {0}\n\n**

**Reason:**

Could not open the Shadow Processor output file.

**Action:**

Required action unknown.

## CAUAJM\_E\_10097

**Make sure the Shadow Machine is properly setup, and that**

**Reason:**

See previous CAUAJM\_E\_10096 message.

**Action:**

No action required.

## CAUAJM\_E\_10098

**if {0} is different on that machine that the correct value is**

**Reason:**

See previous CAUAJM\_E\_10096 message.

**Action:**

No action required.

## CAUAJM\_E\_10099

**defined and exported in {0}.**

**Reason:**

See previous CAUAJM\_E\_10096 message.

**Action:**

No action required.

---

## CAUAJM\_E\_10100

Could not open Agent File in directory: {0}

**Reason:**

Could not Open the Agent File.

**Action:**

Required action unknown.

## CAUAJM\_E\_10101

DATABASE setup ERROR: {0}

**Reason:**

Database setup error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10102

\*\*\* ERROR :Bad msg\_type {0} \*\*\*

**Reason:**

Bad msg\_type.

**Action:**

Required action unknown.

## CAUAJM\_E\_10104

socket close failed

**Reason:**

Socket close failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10105

Could not get {0}. Exiting!

**Reason:**

Could not get EvtNum.

**Action:**

Required action unknown.

## CAUAJM\_E\_10106

Failed to determine currently logged in user.\n

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10107

We Cannot Proceed!!! Might have to switch this server off.

**Reason:**

We cannot proceed, might have to switch this server off.

**Action:**

Required action unknown.

## CAUAJM\_E\_10110

Database error while fetching row for machine {0}.

**Reason:**

Database error while fetching row for machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_10111

Trouble getting the next {0} for server: {1}

**Reason:**

Trouble getting the next EvtNum for server.

**Action:**

Required action unknown.

## CAUAJM\_E\_10112

[\*\* Unable to get text for intcode {0} \*\*]

**Reason:**

Unable to get text for intcode.

**Action:**

Required action unknown.

## CAUAJM\_E\_10113

[\*\*\* Unable to get intcode for text {0} \*\*\*]

**Reason:**

Unable to get intcode for text.

**Action:**

Required action unknown.

## CAUAJM\_E\_10114

Call to gen\_key failed.

**Reason:**

Call to gen\_key failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10115

The agent encountered an error and cannot retrieve log information for job {0}.

**Reason:**

The agent encountered an unidentified problem when attempting to retrieve the requested job log information.

**Action:**

Review additional messages from the agent to determine the problem, and take required action.

## CAUAJM\_E\_10117

check\_heartbeat: Unable to get run\_machine for joid {0}

**Reason:**

Unable to get run\_machine for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10118

{0} {1} for Job: {2}. Trying to send {3} ALARM

**Reason:**

send\_event failed for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10121

Failed to calculate "{0}" for Job: {1}

**Reason:**

Failed to calculate calculate next\_start for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10123

Trouble updating "{0}" for job: {1}

**Reason:**

Trouble updating next\_start for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10124

{0} failed to validate hostname {1}. Exiting.

**Reason:**

The specified product failed to validate its hostname. Exiting.

**Action:**

Make sure that the operating system is able to resolve the specified hostname.

## CAUAJM\_E\_10125

{0} failed to establish a listener at host {1}. Exiting.

**Reason:**

The specified product failed to establish a listener at the specified host. Exiting.

**Action:**

Check the network settings and port availability for the specified hostname.

## CAUAJM\_E\_10126

{0} cannot use auxiliary agent listener port [{1,number,0}] because it is not in a valid range. Valid values are {2}. Exiting.

**Reason:**

The Scheduler/App Server cannot use the specified auxiliary agent listener port because it is not in a valid range. Exiting.

**Action:**

Make sure that the [Scheduler/AppServer]AuxiliaryPort configuration variable specifies a port that is in the valid range.



## CAUAJM\_E\_10127

**{0} cannot use communications port [{1,number,0}] as an auxiliary agent listener port because it has been configured through the CA Secure Sockets Adapter. Exiting.**

**Reason:**

Cannot use the specified port as an auxiliary agent listener because it has been configured through the CA Secure Sockets Adapter. Exiting.

**Action:**

Make sure that the [Scheduler/AppServer]AuxiliaryPort configuration variable specifies a port that has not been configured through the CA Secure Sockets Adapter.

## CAUAJM\_E\_10129

**{0} cannot transmit because the Secure Sockets Adapter port configuration is not correct.**

**Reason:**

The specified product cannot transmit because the Secure Sockets Adapter port configuration is not correct.

**Action:**

Make sure that the specified product is trying to communicate with a Manager or EWA Agent that has been configured through the CA Secure Sockets Adapter or that the [Scheduler/AppServer]AuxiliaryPort is a non-SSA enabled port.

## CAUAJM\_E\_10130

**Unable to contact the {0} Agent at [{1}:{2,number,0}]**

**Reason:**

Unable to contact the agent.

**Action:**

Make sure that the agent is running and the node it is on is available.

## CAUAJM\_E\_10131

**Error creating system registry key {0}. EXITING {1}**

**Reason:**

RegCreateKey failed for the given key.

**Action:**

No action required, program exits.

## CAUAJM\_E\_10132

**Unable to contact the Manager at [{0}:{1,number,0}].**

**Reason:**

Unable to contact the manager.

**Action:**

---

Make sure the manager is running and the node it is on is available.

## **CAUAJM\_E\_10133**

**Database error occurred while accessing manager information. Exiting.**

**Reason:**

Database error occurred while accessing manager information. Exiting.

**Action:**

Check the status of the database server.

## **CAUAJM\_E\_10137**

**The specified job field "{0}" can not be set to null.**

**Reason:**

The setNull method was called with a job field that can not be set to null.

**Action:**

The setNull request is ignored.

## **CAUAJM\_E\_10139**

**Localhost cannot be resolved to a machine definition.**

**Reason:**

Localhost cannot be resolved to a machine definition.

**Action:**

Define a single machine with the Scheduler hostname or specify an existing machine as the LocalMachineDefinition configuration variable.

## **CAUAJM\_E\_10140**

**Unable to create {0} <{1}>. Request aborted.**

**Reason:**

Unable to create file for the run request.

**Action:**

Review the error code and take action as needed.

## **CAUAJM\_E\_10143**

**The {0} application server cannot start because the {1} configuration variable has not been set properly. Exiting.**

**Reason:**

The application server cannot start because the AutoServerId/AutoServerAliasId configuration variable has not been set properly. Exiting.

**Action:**

Ensure that the AutoServerId/AutoServerAliasId configuration variables are not empty, not equal and the AutoServerAliasId length is at most 16 characters long.

**CAUAJM\_E\_10145**

**{0}: Failed to delete previous {1} events.**

**Reason:**

Failed to delete previous STARTJOB events.

**Action:**

Required action unknown.

**CAUAJM\_E\_10146**

**Trouble sending the event {0} to schedule job: {1}**

**Reason:**

Trouble sending event.

**Action:**

Required action unknown.

**CAUAJM\_E\_10147**

**Unable to create process <{0}> <{1}>.**

**Reason:**

A failure occurred when attempting to run the specified process.

**Action:**

Review the error code and take action as needed.

**CAUAJM\_E\_10148**

**Failed to start Alarm timer. {0}d\n**

**Reason:**

Failed to start Alarm timer.

**Action:**

Required action unknown.

**CAUAJM\_E\_10151**

**Trouble sending the event {0} for Job: {1}**

**Reason:**

Trouble sending the event.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10152**

**Expansion of variable {0} too large. Not expanding {0}**

**Reason:**

Expansion of variable too large.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10155**

**Minutes (MM) cannot be greater than 59.**

**Reason:**

Minutes (MM) cannot be greater than 59.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10157**

**Failed to retrieve pending jobs for machine <{0}>.**

**Reason:**

Problem with database.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10159**

**Hours (HH) cannot be greater than 23.**

**Reason:**

Hours (HH) cannot be greater than 23.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10160**

**The execution of the query experienced an unrecoverable error and will not be retried:**

**Reason:**

The product has detected unrecoverable errors while executing the specified query and will not attempt to execute the query again.

**Action:**

Examine the database errors in the log file or the output and check the state of the database.

## CAUAJM\_E\_10161

**Failed to determine hostname. Service desk request aborted.**

**Reason:**

The system call to retrieve the machine's hostname failed.

**Action:**

Contact technical support for further assistance.

## CAUAJM\_E\_10162

**Failed to get next sequence number.**

**Reason:**

Database query failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10163

**Exhausted maximum number of query retries.**

**Reason:**

Retried the deadlocked query the maximum number of times.

**Action:**

Required action unknown.

## CAUAJM\_E\_10164

**Communication table entries for qualifier <{0}> could not be modified properly due to a database failure.**

**Reason:**

Communication table entries for the specified qualifier could not be modified properly due to a database failure.

**Action:**

Check the availability of the database and examine the database logs for error.

## CAUAJM\_E\_10166

**Failed to retrieve eoid from database.**

**Reason:**

Database query failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10171

**Open Event: {0} Failed.**

**Reason:**

Opening of event failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10172

**ERROR SERVICE STOPPED:**

**Reason:**

External error prompted service stop.

**Action:**

Required action unknown.

## CAUAJM\_E\_10174

**WIN32 Function {0} has failed**

**Reason:**

External error prompted service stop.

**Action:**

Required action unknown.

## CAUAJM\_E\_10178

**Start time is outside run window for job: {0}**

**Reason:**

Start time is outside run window for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10179

**SNMP trap send failed to <{0}> in <{1}> for <{2}> with error code <{3}>**

**Reason:**

SNMP specific failure.

**Action:**

Contact Technical Support for further assistance.

## CAUAJM\_E\_10180

**Internal error - failed to find next start date in "{0}"**

**Reason:**

Internal error - failed to find next start date in get\_next\_day.

**Action:**

Required action unknown.

## CAUAJM\_E\_10181

**Internal ERROR 3451. Call Tech Support!**

**Reason:**

Internal ERROR 3451. Call Tech Support!

**Action:**

Call Technical Support.

## CAUAJM\_E\_10182

**Invalid time specifications...**

**Reason:**

Invalid time specifications.

**Action:**

Required action unknown.

## CAUAJM\_E\_10183

**Query failed: <{0}>.**

**Reason:**

A database query failed.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_10184

**SNMP trap send failed to <{0}> in <{1}> with error code <{2}>**

**Reason:**

SNMP specific failure.

**Action:**

Contact Technical Support for further assistance.

## CAUAJM\_E\_10185

**Trouble sending the event {0} for job : {1}**

**Reason:**

Trouble sending event.

**Action:**

Required action unknown.

## CAUAJM\_E\_10187

**Failed to delete job row.**

**Reason:**

Failed to delete job row

**Action:**

Required action unknown.

## CAUAJM\_E\_10192

**The {0} application server with the same {1} is already running on machine: {2}. Shutting down.**

**Reason:**

The application server with the same AutoServerId/AutoServerAliasId is already running on the specified machine.  
Shutting down.

**Action:**

Shutdown the active application server or configure a different AutoServerId/AutoServerAliasId before starting a new one.

## CAUAJM\_E\_10195

**SNMP trap send failed to <{0}> due to incomplete argument list for {1}.**

**Reason:**

SNMP specific failure.

**Action:**

Contact Technical Support for further assistance.

## CAUAJM\_E\_10199

**Machine <{0}> is unavailable because it is not online.**

**Reason:**

Machine is unavailable because it is not online.

**Action:**

Required action unknown.



## CAUAJM\_E\_10200

<The command to get the machine name for job {0} has timed out.>

**Reason:**

Timeout occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_10201

<Could not execute command to get machine name for job {0}.>

**Reason:**

Failure executing command.

**Action:**

Check the value of the job definition machine attribute.

## CAUAJM\_E\_10202

Did not get a machine name from the executed command for job {0}.

**Reason:**

Did not get a machine name from the executed command.

**Action:**

Check the value of the job definition machine attribute.

## CAUAJM\_E\_10204

Communication attempt with the Manager has failed! [{0}:{1,number,1}]

**Reason:**

Communication attempt with the Manager has failed.

**Action:**

Make sure the manager is running and the node it is on is reachable on the network.

## CAUAJM\_E\_10205

The redirected STDERR file path exceeds the maximum allowable path length for job {0}. Unable to proceed.

**Reason:**

The combination of the owner's temporary path plus the string "MACHCMD\_<jobname>" exceeds the maximum allowable path length.

**Action:**

Shorten the owner's temporary path or the jobname itself and then restart the job.

## CAUAJM\_E\_10206

**Scheduler is not active. -l or -p option cannot be used.**

**Reason:**

Scheduler is not active.

**Action:**

Start the scheduler before using -l or -p option.

## CAUAJM\_E\_10207

**Machine <{0}> is unavailable to obtain usage.**

**Reason:**

Machine is unavailable to obtain usage.

**Action:**

Required action unknown.

## CAUAJM\_E\_10212

**Machine <{0}> is not defined in the database.**

**Reason:**

Machine is unavailable.

**Action:**

Required action unknown.

## CAUAJM\_E\_10213

**The internet daemon may not be configured properly for rstatd. {0}**

**Reason:**

The internet daemon may not be configured properly for rstatd.

**Action:**

Required action unknown.

## CAUAJM\_E\_10216

**Unsupported machine method.**

**Reason:**

Unsupported machine method.

**Action:**

Required action unknown.

## CAUAJM\_E\_10217

Failed to obtain the best machine to run job <{0}>."

## CAUAJM\_E\_10219

Scheduler has not responded in a timely fashion. Try again later.

**Reason:**

Scheduler has not responded in a timely fashion. Try again later.

**Action:**

Could indicate that the Scheduler is too busy to process this request.

## CAUAJM\_E\_10220

Machine <{0}> could not find method vmstat in the {0} environment variable. Execute permission denied.

**Reason:**

Machine could not find method vmstat in the {0} environment variable. Execute permission denied.

**Action:**

Required action unknown.

## CAUAJM\_E\_10221

Exhausted list of available Application Server(s). Failing request.

**Reason:**

Exhausted list of Application Server(s). Failing request.

**Action:**

Verify the status of the Application Server(s).

## CAUAJM\_E\_10222

{0} (HH:MM) cannot be greater than 23:59

**Reason:**

The run\_window (HH:MM) cannot be greater than 23:59.

**Action:**

Required action unknown.

## CAUAJM\_E\_10223

Duplicate entry in Comma list.

**Reason:**

Duplicate entry in comma list.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10224**

**Internal Date/Time ERROR #3449. Call Tech Support.**

**Reason:**

Internal Date/Time ERROR #3449. Call Tech Support.

**Action:**

Call Technical Support.

## **CAUAJM\_E\_10228**

**Cannot check permissions: Job \"{0}\" does not exist!**

**Reason:**

Cannot check permissions.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10229**

**Communication attempt with the {0} Agent has failed! [{1}:{2,number,1}]**

**Reason:**

Communication attempt with the agent has failed!

**Action:**

Make sure the agent is running and the node it is on is reachable on the network.

## **CAUAJM\_E\_10230**

**User:{0} does not have {1} permission on job:{2}**

**Reason:**

User does not have the required permission.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10231**

**Internal error checking super user access**

**Reason:**

Internal error in the super user security key generation algorithm.

**Action:**

Required action unknown.

## CAUAJM\_E\_10232

**FATAL SECURITY ERROR: ONLY one of the super user fields has been initialized.**

**Reason:**

The product has detected corrupt or missing security data in the database.

**Action:**

Contact technical support.

## CAUAJM\_E\_10233

**Owner UserId/Password error! Could not find User [{0}] on Host [{1}]!**

**Reason:**

Owner UserId/Password error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10236

**Communication attempt with the {0} Agent has failed: [{1}:{2,number,1}]**

**Reason:**

Could not send.

**Action:**

Make sure Agent is running and the node it is on is available.

## CAUAJM\_E\_10237

**The hostname for the {0} Agent at [{1}:{2,number,0}] is either invalid or unreachable over the network.**

**Reason:**

The hostname for the agent is invalid or unreachable over the network.

**Action:**

Make sure the hostname is pingable and can be resolved by the Domain Name Server or hosts files.

## CAUAJM\_E\_10238

**Query FAILED. Assuming {0} = 0.**

**Reason:**

Query FAILED. Assuming AutoTrack = 0.

**Action:**

Required action unknown.

## CAUAJM\_E\_10240

**ERROR:** Unable to insert {0} table for Job:{1}{2}{3}

**Reason:**

Unable to insert ext\_job table for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10241

**Error** interrogating {0} database table for rows where {1}.

## CAUAJM\_E\_10242

**ERROR:** Could not post change to {0} table!!!

**Reason:**

Could not post change to job\_status table.

**Action:**

Required action unknown.

## CAUAJM\_E\_10243

**ERROR:** Could not INSERT information for running job [{0}]!!!

**Reason:**

Could not INSERT into ujo\_job\_runs table.

**Action:**

Required action unknown.

## CAUAJM\_E\_10244

**ERROR:** Could not UPDATE information for running job [{0}]!!!

**Reason:**

Could not UPDATE ujo\_job\_runs table.

**Action:**

Required action unknown.

## CAUAJM\_E\_10245

**Trouble with the stored procedure:** chase\_state.

**Reason:**

Trouble with the stored procedure: chase\_state..

**Action:**

Required action unknown.

## **CAUAJM\_E\_10246**

**Cannot change database password because the server is configured to run in the external authentication mode.**

**Reason:**

Changing the database password is not applicable.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10247**

**DB Server error during {0} SP.**

**Reason:**

DB Server error during send\_event SP.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10248**

**{0} SP detected a duplicate event.**

**Reason:**

send\_event SP detected a duplicate event.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10249**

**{0} SP insertion failed.**

**Reason:**

send\_event SP insertion failed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10250**

**We cannot send {0} event, Because {1} job(s) depend on job:{2}**

**Reason:**

We cannot send EXTERNAL\_DEPENDENCY event.

**Action:**

Required action unknown.

## CAUAJM\_E\_10251

Eoid during {0} is empty.

**Reason:**

The event is not sent out.

**Action:**

Required action unknown.

## CAUAJM\_E\_10252

**MONBRO ERROR: Monitor/Browser {0} < Monitor/Browser name contains one or more invalid characters or symbols >**

**Reason:**

Error with monitor/browser name.

**Action:**

Required action unknown.

## CAUAJM\_E\_10253

**Trouble sending info to the database(s).**

**Reason:**

Trouble sending info to the database(s).

**Action:**

Please check connectivity to database.

## CAUAJM\_E\_10254

**MONBRO ERROR: < Monitor/Browser must have a name >**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_10255

**MONBRO ERROR: < Mode missing - choose \*Monitor\* or \*Browser\* >**

**Reason:**

Informational only.

**Action:**

No action required.



## CAUAJM\_E\_10256

**MONBRO ERROR: < Job Name or "ALL Jobs" assignment missing >**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_10257

**Potential corruption of job data for joid <{0}>**

**Reason:**

Potential corruption of job data for joid.

**Action:**

Check the ujo\_job database table to verify the job definition.

## CAUAJM\_E\_10259

**Job Name must be specified.**

**Reason:**

The job name must be specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_10260

**Error getting number of global variables.**

**Reason:**

Error in getting number of global variables.

**Action:**

Required action unknown.

## CAUAJM\_E\_10261

**The hostname for the Manager at [{0}:{1,number,0}] is either invalid or unreachable over the network.**

**Reason:**

The hostname for the manager is invalid or unreachable over the network.

**Action:**

Make sure the hostname is pingable and can be resolved by the Domain Name Server or hosts files.

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## CAUAJM\_E\_10262

**get\_jobs({0}) FAILED.**

**Reason:**

get\_jobs FAILED.

**Action:**

Required action unknown.

## CAUAJM\_E\_10263

**{0}() {1} for Job: {2}. Trying to send {3} Event to kid joid: {4}**

**Reason:**

send\_event() FAILED.

**Action:**

Required action unknown.

## CAUAJM\_E\_10266

**Cannot Evaluate Conditions on Job: {0} THIS JOB WILL NOT START!**

**Reason:**

Cannot evaluate conditions on job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10267

**Could not get status and {0} for Job: {1} in {2}()**

**Reason:**

Could not get status and exit\_code for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10270

**Failed to load {0} shared library for <{1}>.**

**Reason:**

Failed to load DRM shared library for DB.

**Action:**

Required action unknown.

## CAUAJM\_E\_10271

**ERROR - Database error for job <{0}>. The override will not be CANCELLED.**

**Reason:**

Could not access the specified table for the given job. The override will not be CANCELLED.

**Action:**

Required action unknown.

## CAUAJM\_E\_10272

**ERROR - Could not update the {0} table for job <{1}>. The override will not be CANCELLED.**

**Reason:**

Could not update the specified table for the given job. The override will not be CANCELLED.

**Action:**

Required action unknown.

## CAUAJM\_E\_10273

**CONDITION ERROR: Missing operand.**

**Reason:**

Missing operand.

**Action:**

Required action unknown.

## CAUAJM\_E\_10274

**CONDITION ERROR: Empty expression detected.**

**Reason:**

Empty expression.

**Action:**

Required action unknown.

## CAUAJM\_E\_10275

**CONDITION ERROR: Illegal Comparison Expression: {0}.**

**Reason:**

Illegal Comparison Expression: {0}.

**Action:**

Required action unknown.

## CAUAJM\_E\_10277

**ERROR for job: {0} < "{1}" is a reserved keyword and cannot be specified for jobname >**

**Reason:**

Invalid job name has been specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_10278

**MONBRO ERROR: Monitor/Browser {0} < "{1}" is a reserved keyword and cannot be specified for Monitor/Browser name >**

**Reason:**

Invalid job name has been specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_10279

**Invalid Job Name: {0} {1} not performed.**

**Reason:**

Invalid job name.

**Action:**

Required action unknown.

## CAUAJM\_E\_10280

**Failed to delete {0}.**

**Reason:**

Failed to delete job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10281

**ERROR for Job: {0} < {1} >**

**Reason:**

Error for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10283

**Exhausted maximum number of retries for scheduler operation [{0}]**

**Reason:**

Exhausted maximum number of retries for the specified scheduler operation.

**Action:**

Check the state of the database tables and verify access to the database.

## CAUAJM\_E\_10285

**Failed to reconstruct job's condition from its atomic conditions for {0}.**

**Reason:**

Failed to reconstruct job's condition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10286

**Failed to generate job\_key.**

**Reason:**

Failed to generate the job\_key.

**Action:**

Required action unknown.

## CAUAJM\_E\_10287

**\*\*\* CANNOT INSERT {0}: {1}, because it ALREADY EXISTS \*\*\***

**Reason:**

Cannot insert monbro.

**Action:**

Required action unknown.

## CAUAJM\_E\_10288

**\*\*\* CANNOT INSERT Job: {0}, because it ALREADY EXISTS \*\*\***

**Reason:**

Cannot insert the job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10289

**There was some trouble with setting the Condition for job: {0}.**

**Reason:**

There was some trouble with Condition processing for the job. The job minus the condition attribute has been added to the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10290

**There was some trouble with setting the Box {0} Condition for job: {1}**

**Reason:**

There was some trouble with setting the Box Condition for the job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10292

**Unable to check conditions for job: {0}**

**Reason:**

Unable to check conditions for the job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10293

**{0} FAILED. Trying to {1} for job {2}**

**Reason:**

send\_event FAILED. Trying to DELETEJOB for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10294

**Error retrieving job <{0}>.**

**Reason:**

Error retrieving the job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10295

**ERROR! Job "{0}" failed custom job validation.**

**Reason:**

Job failed custom validation.

**Action:**

Required action unknown.

## CAUAJM\_E\_10297

**Invalid job name: {0} OVER-RIDE not performed**

**Reason:**

Invalid job name.

**Action:**

Required action unknown.

## CAUAJM\_E\_10299

**{0} FAILED. Trying to {1} for machine {2}.**

**Reason:**

send\_event FAILED. Trying to INVALIDATE\_MACH for machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_10301

**Trouble with Setting Starting Condition for joid: {0}**

**Reason:**

Trouble with setting the starting condition for the joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10302

**Database Change WAS NOT successful.**

**Reason:**

The database change was not successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_10303

**ERROR for Machine: {0} < machine "{1}" does not exist >**

**Reason:**

The machine does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10304

**Error validating Machine {0}**

**Reason:**

Unable to determine the machine's existence.

**Action:**

Required action unknown.

## CAUAJM\_E\_10305

**\*\*\* CANNOT INSERT Machine: {0}, because it ALREADY EXISTS \*\*\***

**Reason:**

Cannot insert the machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_10306

**Unable to Insert Machine {0}**

**Reason:**

Unable to Insert Machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_10307

**Unable to Delete Machine {0}**

**Reason:**

Unable to perform Delete for name.

**Action:**

Required action unknown.



## CAUAJM\_E\_10308

**Invalid Machine Name: {0} {1} not performed.**

**Reason:**

Invalid Machine Name.

**Action:**

Refer to the UAJM Reference Guide for information on valid characters for machine names.

## CAUAJM\_E\_10309

**Failed to retrieve atomic conditions for {0}.**

**Reason:**

Failed to retrieve the atomic conditions for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10310

**Failed to get job for joid <{0}>.**

**Reason:**

Failed to get the job for the joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10311

**Failed to retrieve dependents for {0}.**

**Reason:**

Failed to retrieve dependents for the job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10312

**Failed to retrieve joids from the database.**

**Reason:**

Failed to retrieve joids from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10313

Unable to retrieve job for joid <{0}>.

**Reason:**

Unable to retrieve the job for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10314

Memory boundary failure while checking list of job identifiers.

**Reason:**

Memory boundary failure occurred while checking list of job identifiers.

**Action:**

Required action unknown.

## CAUAJM\_E\_10315

Error retrieving children for joid <{0}>.

**Reason:**

Error retrieving children for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10316

Error getting joids.

**Reason:**

Error getting joids.

**Action:**

Required action unknown.

## CAUAJM\_E\_10317

Cannot add non "{0}"-type compatible Machine {1} to "{0}"-type virtual machine list

**Reason:**

Cannot add non-compatible Machine to virtual machine list.

**Action:**

Required action unknown.

## CAUAJM\_E\_10318

**ERROR Circular definition: Cannot add self to virtual machine list**

**Reason:**

ERROR Circular definition: Cannot add self to virtual machine list.

**Action:**

Required action unknown.

## CAUAJM\_E\_10319

**Server timezone initialization error. You need to execute the Scheduler component at least once before running any client application.**

**Reason:**

Server timezone initialization error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10321

**ERROR for Box: {0}. The specified resource {1} is not a global virtual resource.**

**Reason:**

Box jobs can only accept virtual global resources.

**Action:**

Specify a global virtual resource.

## CAUAJM\_E\_10320

**Failed to retrieve jobs from database.**

**Reason:**

Failed to retrieve jobs from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10322

**Failed to retrieve details for joid {0} with run\_num {1}.**

**Reason:**

Failed to retrieve details for the joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10323

Failed to retrieve details for job <{0}>. The run number specified is invalid.

**Reason:**

Failed to retrieve the details for the job. The run number specified is invalid.

**Action:**

Specify a valid run number.

## CAUAJM\_E\_10325

Failed to retrieve details information for joid <{0}> with run\_num <{1}>.

**Reason:**

Failed to retrieve details information for the joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10328

Error retrieving global variable information from the database.

**Reason:**

Error retrieving global variable information from the database occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_10329

Access rights could not be obtained to view all global variables for Instance {0}!

**Reason:**

Access rights could not be obtained to view all global variables for Instance.

**Action:**

Required action unknown.

## CAUAJM\_E\_10331

Error loading machine details from the database.

**Reason:**

An error occurred loading the machine details from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10332

**Error loading machine jobs from the database.**

**Reason:**

Error loading machine jobs from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10333

**Error retrieving active override for joid <{0}>. Possible job definition corruption.**

**Reason:**

The base job definition indicates the job has an active override but no active override definition found.

**Action:**

Redefine the job's active override and restart the job.

## CAUAJM\_E\_10334

**Error retrieving machine jobs from the database.**

**Reason:**

An error occurred retrieving machine jobs from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10336

**There are No EVENTS to Un-Send.**

**Reason:**

There are no events to un-send.

**Action:**

Required action unknown.

## CAUAJM\_E\_10337

**Could not mark event <{0}> as unsent.**

**Reason:**

Could not mark event unsent.

**Action:**

Required action unknown.

**CAUAJM\_E\_10338**

**Cannot change to requested status manually.**

**Reason:**

Cannot change to requested status manually.

**Action:**

Required action unknown.

**CAUAJM\_E\_10339**

**Invalid status.**

**Reason:**

Invalid status.

**Action:**

Required action unknown.

**CAUAJM\_E\_10340**

**Invalid job name.**

**Reason:**

Invalid job name.

**Action:**

Required action unknown.

**CAUAJM\_E\_10341**

**Unable to get job/box/application/group name.**

**Reason:**

Unable to get job/box name.

**Action:**

Required action unknown.

**CAUAJM\_E\_10342**

**Invalid job queue priority specified.**

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value and resubmit the request.

## CAUAJM\_E\_10343

Real machine <{0}> does not exist.

**Reason:**

Real machine does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10344

Failed to read job statuses from the database.

**Reason:**

Failed to read job statuses from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10346

Signals can only be sent to {0} jobs.

**Reason:**

Signals can only be sent to command jobs.

**Action:**

Required action unknown.

## CAUAJM\_E\_10347

Subroutine {0}() failed, event not sent!

**Reason:**

Subroutine send\_event failed. The event was not sent.

**Action:**

Required action unknown.

## CAUAJM\_E\_10354

Machine: {0} defined as virtual. Virtual machines may not belong to other virtual machines.

**Reason:**

The machine is defined as virtual. Virtual machines may not belong to other virtual machines.

**Action:**

Required action unknown.

## CAUAJM\_E\_10355

**An invalid security class has been specified.**

**Reason:**

The API has been passed as an invalid security class.

**Action:**

Correct the class name and resubmit the request.

## CAUAJM\_E\_10356

**An invalid access level has been specified.**

**Reason:**

The API has been passed an invalid access level.

**Action:**

Correct the access level and resubmit the request.

## CAUAJM\_E\_10357

**ERROR for Machine: < cannot be a keyword > {0} not performed.**

**Reason:**

Invalid Machine Name. Name cannot be a keyword.

**Action:**

Machine names cannot be keywords.

## CAUAJM\_E\_10359

**Server: {0} Not Defined!!! ERROR!!!.**

**Reason:**

Server not defined.

**Action:**

Required action unknown.

## CAUAJM\_E\_10361

**DBSETLHOST() failed with hostname of "{0}".**

**Reason:**

DBSETLHOST() failed.

**Action:**

Required action unknown.



## CAUAJM\_E\_10362

ct\_con\_props(CS\_USERNAME) failed with user of "{0}".

**Reason:**

ct\_con\_props(CS\_USERNAME) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10363

ct\_con\_props(CS\_PASSWORD) failed with password of "{0}".

**Reason:**

ct\_con\_props(CS\_PASSWORD) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10364

ct\_con\_props(CS\_APPNAME) failed with appname of "{0}".

**Reason:**

ct\_con\_props(CS\_APPNAME) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10366

**Error turning on Sybase password encryption.**

**Reason:**

An error occurred turning on the Sybase password encryption.

**Action:**

Required action unknown.

## CAUAJM\_E\_10368

**Database error occurred while retrieving the database password.**

**Reason:**

Database error occurred while retrieving the database password.

**Action:**

Verify the state of the database.

## CAUAJM\_E\_10369

**Database error updating the extended runtime information for job ID {0}**

**Reason:**

Database error updating the extended runtime information for job ID.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10370

**Database error retrieving the extended runtime information for job ID {0}**

**Reason:**

Database error retrieving the extended runtime information for job ID.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10372

**This job's type does not support the {0} operation. Event not sent for job <{1}>**

**Reason:**

This job's type does not support this operation. Event has not been sent for the specified job.

**Action:**

No action required.

## CAUAJM\_E\_10373

**Job <{0}> has no extended runtime information.**

**Reason:**

The specified job has no extended runtime information.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10376

**Timed out waiting for the rstatd internet daemon on [{0}].**

**Reason:**

The job timed out waiting for the rstatd internet daemon on the specified machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_10377

Expanded value for attribute <{0}> exceeds {1,number,0} characters. Job "{2}" cannot start.

**Reason:**

Expanded value for the specified attribute exceeds the specified number of characters. Job cannot start.

**Action:**

Ensure that the global variable substitution in the attribute value fits within the specified number of characters.

## CAUAJM\_E\_10381

\*\*\* CANNOT INSERT Job Type: {0}, because it ALREADY EXISTS \*\*\*

**Reason:**

Job type already exists.

**Action:**

Required action unknown.

## CAUAJM\_E\_10383

This job's type does not support the {0} operation. Event not sent for job <{1}>.

**Reason:**

This job's type does not support the SUSPENDJOB operation. Event not sent for the specified job.

**Action:**

Do not issue a SUSPENDJOB event for this job.

## CAUAJM\_E\_10384

Failed to obtain status for global variable {0} from the database.

**Reason:**

Failed to obtain the status for the global variable from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10385

Error setting CS\_DIAG\_TIMEOUT timeout to true.

**Reason:**

ct\_con\_props(CS\_DIAG\_TIMEOUT) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10386

Receive timed out. Agent on [{0}] is in question.

**Reason:**

Receive from Agent timed out.

**Action:**

Could indicate loss of network connection.

## CAUAJM\_E\_10387

10387 Invalid data received from the Agent on [{0}]

**Reason:**

Invalid data received from the agent component.

**Action:**

Action required unknown.

## CAUAJM\_E\_10388

ERROR for Machine: {0} < {1} >

**Reason:**

Error for job Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10390

ERROR for Monbro: {0} < {1} >

**Reason:**

Error for machine Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10393

ERROR for Blob/Glob: {0} < {1} >

**Reason:**

Error for blob/glob Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10395

**ERROR for External Instance: {0} < {1} >**

**Reason:**

Error for External Instance Jil definition.

**Action:**

Required action unknown.

## CAUAJM\_E\_10397

**Unknown job type: {0} for job: {1}**

**Reason:**

Job has unknown job type.

**Action:**

Required action unknown.

## CAUAJM\_E\_10398

**\*\*\* Cannot Insert/Update/Query/Delete Job Type: {0}, because it DOES NOT EXIST \*\*\***

**Reason:**

Job type does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10399

**Insert/Update/Query/Delete of reserved job type: {0} is NOT allowed**

**Reason:**

Cannot Insert/Update/Query/Delete reserved job type.

**Action:**

Required action unknown.

## CAUAJM\_E\_10433

**Could not obtain security server from the database!**

**Reason:**

Error occurred while accessing the database for the security server name.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10434**

### **Error initiating security session!**

**Reason:**

Error occurred while initiating the security session.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10435**

### **Memory allocation error!**

**Reason:**

Error allocating memory.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10436**

### **Security server unreachable or invalid authentication certificate file!**

**Reason:**

Security server information may be inaccurate or certificate file may be corrupt.

**Action:**

Try running autosys\_secure to reset the security server or regenerate the certificate file.

## **CAUAJM\_E\_10437**

### **Detailed Error Information:**

**Reason:**

Information regarding last failure.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10443**

### **Security server unreachable or invalid credentials!**

**Reason:**

Security server information or credentials may be inaccurate.

**Action:**

Try running autosys\_secure to reset the security server or enter valid credentials.

## CAUAJM\_E\_10444

**Error generating certificate file!**

**Reason:**

Security server failed to generate certificate file.

**Action:**

Required action unknown.

## CAUAJM\_E\_10445

**Error accessing the security information in the database!**

**Reason:**

A Database error has occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_10448

**FATAL SECURITY ERROR: Tampering or corruption of the security-related database keys detected!**

**Reason:**

The product has detected corrupt or missing security data in the database.

**Action:**

Contact technical support.

## CAUAJM\_E\_10449

**Please contact {0} technical support!**

**Reason:**

A serious error has occurred that requires the attention of technical support.

**Action:**

Contact CA WAAE technical support.

## CAUAJM\_E\_10451

**Someone may have been tampering with the Database and attempted to change the {0} Super User: {1} by hand!.**

**Reason:**

The product has detected corrupt or missing security data in the database.

**Action:**

Contact Technical Support.

## CAUAJM\_E\_10459

**Security server information unavailable! Run autosys\_secure option 1 to enable eTrust IAM Toolkit security for this instance!**

**Reason:**

Security server information is unavailable.

**Action:**

Run autosys\_secure option 1 to enable eTrust IAM Toolkit security for this instance.

## CAUAJM\_E\_10460

**The MACH\_OFFLINE and MACH\_ONLINE events cannot be used against VIRTUAL machine <{0}>.**

**Reason:**

The machine offline and online event cannot be used against a virtual machine.

**Action:**

Specify a real machine name and resubmit the request.

## CAUAJM\_E\_10461

**{0} ERROR: {1} < {2} >**

**Reason:**

Error for job or global blob.

**Action:**

Required action unknown.

## CAUAJM\_E\_10462

**Failed to retrieve time offset from database**

**Reason:**

Database Error on query.

**Action:**

Required action unknown.

## CAUAJM\_E\_10464

**Could not find valid machines for job [{0} {1,number,1}.{2,number,1}.{3,number,1}]**

**Reason:**

The input machine list does not contain any valid machines.

**Action:**

Please check the input machine list to make sure it contains valid machines.



## CAUAJM\_E\_10465

**{0} shadow scheduler encountered a database error while notifying agents.**

**Reason:**

The shadow scheduler encountered a database error while notifying agents.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10466

**Unable to query virtual resource dependency details for job [{0} {1,number,1}.{2,number,1}.{3,number,1}]**

**Reason:**

Database Error on query.

**Action:**

Review the log file for possible reasons of query failure and take the appropriate action.

## CAUAJM\_E\_10467

**Unable to query real resource dependency details for job [{0} {1,number,1}.{2,number,1}.{3,number,1}]**

**Reason:**

Database Error on query.

**Action:**

Review the log file for possible reasons of query failure and take the appropriate action.

## CAUAJM\_E\_10469

**Unable to delete resource usage for job [{0} {1,number,1}.{2,number,1}.{3,number,1}]**

**Reason:**

Database Error on delete.

**Action:**

Review the log file for possible reasons of query failure and take the appropriate action.

## CAUAJM\_E\_10476

**Error loading Resource Manager.**

**Reason:**

Error loading Resource Manager.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10477**

### **Error connecting to the {0} Manager**

#### **Reason:**

Error connecting to the DCA Manager with the specified url

#### **Action:**

Check the credentials of the DCA Manager.

## **CAUAJM\_E\_10479**

### **Error - Input machine list empty**

#### **Reason:**

The input machine list is empty.

#### **Action:**

Need a valid CSV input machine list.

## **CAUAJM\_E\_10480**

### **Error - No target machine specified**

#### **Reason:**

Target machine not specified.

#### **Action:**

Need an input machine.

## **CAUAJM\_E\_10481**

### **Error - No target machine user specified**

#### **Reason:**

Target machine user not specified.

#### **Action:**

Need target machine user for deployment.

## **CAUAJM\_E\_10482**

### **Error - No password of the target machine user specified**

#### **Reason:**

Target machine password not specified.

#### **Action:**

Need target machine user password for deployment.

## CAUAJM\_E\_10483

**Error - No JobId specified**

**Reason:**

Prep JobId not specified.

**Action:**

Need a valid prep JobId.

## CAUAJM\_E\_10484

**Machine {0}: Provision Error - {1}**

**Reason:**

PrepStatus returned error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10486

**{0} primary scheduler encountered a database error while notifying agents.**

**Reason:**

The primary scheduler encountered a database error while notifying agents.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10487

**An error occurred uploading the certificate file to the database.**

**Reason:**

A program or database error prevents the storage of the authentication certificate in the database.

**Action:**

Check previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_10488

**Resource Manager not available. Unable to process request.**

**Reason:**

Resource Manager not loaded during startup.

**Action:**

Check the server log file for errors during startup. Fix the problem and restart server.

## CAUAJM\_E\_10494

**Missing value for \"{0}\"**

**Reason:**

Required value is missing.

**Action:**

Specify the required value.

## CAUAJM\_E\_10495

**Database error encountered while accessing virtual resource data.**

**Reason:**

Database error encountered while accessing virtual resource data

**Action:**

Check the log file for possible reasons of database query failure and act accordingly.

## CAUAJM\_E\_10497

**{0} scheduler encountered a database error while resolving the localhost machine.**

**Reason:**

Scheduler encountered a database error while resolving the localhost machine.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10498

**{0} scheduler failed to establish the required listeners.**

**Reason:**

Scheduler failed to establish the required listeners.

**Action:**

Check the Manager port and the Secure Sockets Adapter configurations.

## CAUAJM\_E\_10500

**Invalid job specification for retrieving successors.**

**Reason:**

Must specify non-wildcard job name.

**Action:**

Re-execute the query with a non-wildcard job name.

## CAUAJM\_E\_10501

**Invalid state for alarm.**

**Reason:**

API called with an incorrect state parameter.

**Action:**

Recall API with valid state parameter.

## CAUAJM\_E\_10502

**Alarm does not exist.**

**Reason:**

Alarm eoid specified in update does not exist.

**Action:**

Recall API with valid alarm eoid.

## CAUAJM\_E\_10503

**{0} primary scheduler encountered a problem starting up and must shut down.**

**Reason:**

The primary scheduler encountered a problem starting up and must shut down.

**Action:**

Check the scheduler log for details.

## CAUAJM\_E\_10504

**Alarm response is too long.**

**Reason:**

Alarm response specified in API call is too long.

**Action:**

Recall API with shorter response string.

## CAUAJM\_E\_10505

**Invalid object type specified in call to API.**

**Reason:**

Invalid object type specified in call to API.

**Action:**

Recall API with valid object type.

## CAUAJM\_E\_10506

Failed to retrieve job with joid <{0}>.

**Reason:**

Job is not in the database.

**Action:**

Call API with existing job.

## CAUAJM\_E\_10507

Failed to retrieve job <{0}>.

**Reason:**

Job is not in the database.

**Action:**

Call API with existing job.

## CAUAJM\_E\_10510

Error modifying jobs affected by calendar change.

**Reason:**

Could not modify job start times based on change of calendar.

**Action:**

Required action unknown.

## CAUAJM\_E\_10512

Failed to check existence of {0} extended calendar.

**Reason:**

Query failed. Could not check the calendar.

**Action:**

Required action unknown.

## CAUAJM\_E\_10513

Failed to retrieve cal\_id of holiday calendar.

**Reason:**

QUERY failed. Could to retrieve a database field.

**Action:**

Required action unknown.

## CAUAJM\_E\_10514

**Holiday calendar may not be an extended calendar**

**Reason:**

Holiday calendar must be a standard calendar.

**Action:**

Required action unknown.

## CAUAJM\_E\_10515

**Request was not fulfilled within its expiration time.**

**Reason:**

Request was not fulfilled within its expiration time.

**Action:**

Set a higher value for the MaxExpirationSeconds attribute of the request.

## CAUAJM\_E\_10516

**Request was not fulfilled within its expiration time by the {0} Application Server:**

**Reason:**

Request was not fulfilled within its expiration time by the Application Server.

**Action:**

Set a higher value for the MaxExpirationSeconds attribute of the request.

## CAUAJM\_E\_10517

**Failed to retrieve id of extended calendars.**

**Reason:**

QUERY failed. Could not create a new calendar ID.

**Action:**

Required action unknown.

## CAUAJM\_E\_10518

**Failed to modify extended calendars, calendar does not exist.**

**Reason:**

Attempted to modify an extended calendar that does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10520

**Syntax error in extended calendar {0}.**

**Reason:**

The condition field has a syntax error. The extended calendar will generate no dates.

**Action:**

Correct the condition field in the extended calendar.

## CAUAJM\_E\_10521

**Invalid keyword in extended calendar {0}.**

**Reason:**

The condition field has an invalid keyword. The extended calendar might generate incorrect dates.

**Action:**

Correct the condition field in the extended calendar.

## CAUAJM\_E\_10523

**Can not delete a regular calendar.**

**Reason:**

Attempted to delete a regular calendar when trying to delete extended calendar.

**Action:**

Required action unknown.

## CAUAJM\_E\_10524

**Failed to check existence of cycle.**

**Reason:**

QUERY failed. Can not check existence of cycle.

**Action:**

Required action unknown.

## CAUAJM\_E\_10525

**Cannot Insert Cycle, periods overlap.**

**Reason:**

Cycle has overlapping periods.

**Action:**

Enter periods for the cycle that do not overlap.



## CAUAJM\_E\_10526

**Failed to modify cycle , cycle does not exist.**

**Reason:**

Attempting to modify a cycle that does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10527

**Timed out waiting for response from the {0} Application Server.**

**Reason:**

Application server may not be reachable.

**Action:**

Make sure the application server is running and the node it is on is available.

## CAUAJM\_E\_10528

**Timed out waiting for response from the {0} Agent: [{1}:{2,number,1}]**

**Reason:**

The agent may not be reachable.

**Action:**

Make sure the agent is running and the node it is on is available.

## CAUAJM\_E\_10530

**The Agent communication port has not been configured. Returning {0}**

**Reason:**

The communication port for the Agent component has not been configured. Returning FAILURE.

**Action:**

Required action unknown.

## CAUAJM\_E\_10532

**Unable to communicate with machine <{0}> because the agent does not support data encryption.**

**Reason:**

Unable to communicate with the specified machine because the agent does not support data encryption.

**Action:**

Communications with a legacy agent cannot be encrypted. Turn off encryption in the instance.

## CAUAJM\_E\_10536

**Agent remote authentication! ERROR: {0}**

**Reason:**

Agent remote authentication error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10537

**Agent owner not Job owner! ERROR: {0}**

**Reason:**

Agent owner is not the Job owner.

**Action:**

Required action unknown.

## CAUAJM\_E\_10538

**Agent could not locate service owner! ERROR: {0}**

**Reason:**

The agent could not locate the service owner.

**Action:**

Required action unknown.

## CAUAJM\_E\_10539

**Agent unable to log in owner! ERROR: {0}**

**Reason:**

Agent is unable to log in owner.

**Action:**

Required action unknown.

## CAUAJM\_E\_10540

**Owner User/Password error due to: {0}**

**Reason:**

Owner User/Password error.

**Action:**

User credentials are not found in the CA Workload Automation AE database. Use the autosys\_secure command to enter the user credentials and try again.

## CAUAJM\_E\_10541

**Permission ERROR: {0}**

**Reason:**

Permission error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10542

**Remote authentication ERROR: {0}**

**Reason:**

Remote authentication error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10543

**VERSION Control Error: {0}**

**Reason:**

Version control error.

**Action:**

Required action unknown.

## CAUAJM\_E\_10551

**Database error encountered while attempting to reserve virtual resources for job [{0} {1,number,1}.{2,number,1}. {3,number,1}].**

**Reason:**

Database error encountered while attempting to reserve virtual resources for the specified job.

**Action:**

Review the log file for possible reasons of database query failure and take the appropriate action.

## CAUAJM\_E\_10552

**Database error encountered while querying for the virtual resources held by job [{0} {1,number,1}.{2,number,1}. {3,number,1}].**

**Reason:**

Database error encountered while querying for the virtual resources held by the specified job.

**Action:**

Review the log file for possible reasons of database query failure and take the appropriate action.

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## CAUAJM\_E\_10560

Unable to chdir to {0}: {1}

**Reason:**

The environment is not set or the directory is unavailable.

**Action:**

Correct the problem and try again.

## CAUAJM\_E\_10561

Unable to set uid: {0}

**Reason:**

The out directory is not available or you may not set uid.

**Action:**

Correct the problem and try again.

## CAUAJM\_E\_10562

Unable to write in {0}: {1}

**Reason:**

The directory does not exist or you do not have the appropriate permission.

**Action:**

Correct the problem and try again.

## CAUAJM\_E\_10563

Unable to fork:

**Reason:**

Possible memory shortage.

**Action:**

Required action unknown.

## CAUAJM\_E\_10564

Unable to set sid:

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10565

Unable to query {0}:

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10569

Global Blob: {0} does not exist

**Reason:**

Specified Global Blob does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10570

**CONDITION ERROR: Unbalanced double-quotes in logical expression \"{0}\".**

**Reason:**

Unbalanced double-quotes in logical expression.

**Action:**

Required action unknown.

## CAUAJM\_E\_10571

**CONDITION ERROR: Invalid backslash at the end of logical expression \"{0}\".**

**Reason:**

Invalid backslash at the end of logical expression logical expression.

**Action:**

Required action unknown.

## CAUAJM\_E\_10579

**CONDITION ERROR: Cross-Instance global variable dependency invalid.**

**Reason:**

Cross-Instance global variable dependency invalid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10600

**Could not check existence of calendar.**

**Reason:**

Query failed. Count not check existence of calendar.

**Action:**

Required action unknown.

## CAUAJM\_E\_10601

**Cannot insert calendar {0}, calendar name already exists.**

**Reason:**

Query failed. Calendar name is already in use.

**Action:**

Required action unknown.

## CAUAJM\_E\_10602

**Failed to insert {0} into {1} calendar.**

**Reason:**

Query FAILED. An error occurred inserting a new calendar date.

**Action:**

Enter a valid date.

## CAUAJM\_E\_10603

**Failed to modify calendars, calendar does not exist.**

**Reason:**

Modifying a calendar that does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_10604

**Failed to check existence of standard calendars.**

**Reason:**

QUERY failed. Could not check existence of standard calendars.

**Action:**

Required action unknown.

## CAUAJM\_E\_10605

**A database error occurred while fetching information from the communication tables. Exiting.**

**Reason:**

A database error occurred while fetching information from the communication tables. Exiting.

**Action:**

Check the status of the database server.

## CAUAJM\_E\_10607

**Could not check existence of password vault.**

**Reason:**

Query failed. Count not check existence of password vault.

**Action:**

Required action unknown.

## CAUAJM\_E\_10612

**Error retrieving uninotify row for joid {0}.**

**Reason:**

Error retrieving uninotify row for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_10618

**Failed to uncompress incoming response buffer. Error code [{0}].**

**Reason:**

Internal zlib decompression algorithm failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10619

**Return values from dual database writes disagree. Databases may no longer be synchronized.**

**Reason:**

**Action:**

Shut down and resynchronize the databases.

## CAUAJM\_E\_10620

**Return values from dual database writes disagree. Databases may no longer be synchronized.**

**Reason:**

Unknown.

**Action:**

Shutdown and resynchronize databases.

**CAUAJM\_E\_10622**

**Cannot Insert Cycle {0}, cycle name already exists**

**Reason:**

Cycle name is already in use.

**Action:**

Enter a new cycle.

**CAUAJM\_E\_10629**

**Failed to get next run\_num.**

**Reason:**

Database query failed.

**Action:**

Required action unknown.

**CAUAJM\_E\_10637**

**Could not create the notify call: "{0}". Error code: {1}**

**Reason:**

Failed to create child.

**Action:**

Required action unknown.

**CAUAJM\_E\_10638**

**Could not execute the notify call: '{0}'**

**Reason:**

Call to system() failed.

**Action:**

Required action unknown.

**CAUAJM\_E\_10639**

**Could not find SystemRoot variable.**

**Reason:**

SystemRoot environment variable is not defined.



**Action:**

Required action unknown.

**CAUAJM\_E\_10642**

**Notify could not open Window Station '{0}'**

**Reason:**

Could not open Window Station.

**Action:**

Required action unknown.

**CAUAJM\_E\_10644**

**Could not execute the notify call = {0}**

**Reason:**

Failed on WaitForSingleObject().

**Action:**

Required action unknown.

**CAUAJM\_E\_10645**

**Could not mark the database: {0} in single server mode. Cannot complete rollover.**

**Reason:**

Attempt to update DB as single server mode failed.

**Action:**

Required action unknown.

**CAUAJM\_E\_10646**

**Failed to update configuration information regarding rollover.**

**Reason:**

Failed to update the configuration information regarding rollover.

**Action:**

Required action unknown.

**CAUAJM\_E\_10649**

**Server {0} was not available during connection operation.**

**Reason:**

Unable to connect to the server.

**Action:**

Check database name and connectivity and try again.

## CAUAJM\_E\_10650

**Unable to mark the database mode.**

**Reason:**

The update to the ujo\_alamode table failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10651

**Failed to determine what mode the database is in.**

**Reason:**

Query to retrieve the DB mode from ujo\_alamode failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10652

**Configuration indicates that the system is in dual server mode but the database indicates otherwise.**

**Reason:**

Configuration and ujo\_alamode disagree as to whether the system is in dual server mode.

**Action:**

Inspect configuration. If it is set correctly, restart the primary scheduler to reset the database mode.

## CAUAJM\_E\_10653

**Configuration indicates that the system is in single server mode but the database indicates otherwise.**

**Reason:**

Configuration and ujo\_alamode disagree as to whether the system is in dual server mode.

**Action:**

Inspect configuration. If it is set correctly, restart the primary scheduler to reset the database mode.

## CAUAJM\_E\_10656

**The database <{0}> has encountered a critical error. Please contact the database administrator.**

**Reason:**

The database has encountered a critical error. Please contact the database administrator.

**Action:**

Recover the database and restart the system.

## CAUAJM\_E\_10657

**Failed to update DBAccess information.**

**Reason:**

Failed to update DBAccess information.

**Action:**

Required action unknown.

## CAUAJM\_E\_10658

**Cannot insert External Instance: {0} because it already exists.**

**Reason:**

Cannot insert External Instance as it is already defined to the database.

**Action:**

Change the External Instance name to INSERT or use the UPDATE attribute to update the External Instance.

## CAUAJM\_E\_10660

**ERROR: Could not INSERT to ujo\_asext\_config for External Instance : {0}!!!**

**Reason:**

Could not INSERT to ujo\_asext\_config.

**Action:**

Required action unknown.

## CAUAJM\_E\_10661

**Only "{0}" and "{1}" agents support this operation.**

**Reason:**

Only z/OS and i5/OS agents support this operation type.

**Action:**

Correct the input parameters and resubmit the request.

## CAUAJM\_E\_10662

**Cannot delete External Instance: {0} because it does not exist.**

**Reason:**

Cannot delete External Instance as it does not exist in the database.

**Action:**

Change the External Instance name to update or use the INSERT attribute to insert the External Instance.

## CAUAJM\_E\_10663

**Cannot update External Instance: {0} because it does not exist.**

**Reason:**

Cannot update External Instance as it does not exist in the database.

**Action:**

Change the External Instance name to update or use the INSERT attribute to insert the External Instance.

## CAUAJM\_E\_10664

**Error retrieving AFM information from the database.**

**Reason:**

Error retrieving AFM information from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10665

**Agents of machine type {0} do not support this operation.**

**Reason:**

Agents of the specified machine type do not support this operation.

**Action:**

The Agent does not support the requested operation.

## CAUAJM\_E\_10666

**Error retrieving machine information from the database.**

**Reason:**

Error retrieving machine information from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_10668

**Unable to determine the file name of the requested log.**

**Reason:**

The database query failed.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_10669

Unable to determine the node that has the requested log.

**Reason:**

The database query failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10670

Error retrieving remote agent directory.

**Reason:**

The database query failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10671

Error retrieving job output {0}:

**Reason:**

Error retrieving job output blob/glob.

**Action:**

Required action unknown.

## CAUAJM\_E\_10672

Input field contains illegal apostrophe.

**Reason:**

Client input error.

**Action:**

Remove the apostrophe and retry.

## CAUAJM\_E\_10673

The requested server operation failed. Operation ID = {0}.

**Reason:**

The application server operation for the specified ID value has failed. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

## CAUAJM\_E\_10674

**The requested server operation failed because the requester has insufficient privileges. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because the requester has insufficient privileges. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

## CAUAJM\_E\_10675

**The requested server operation failed because it did not pass the external job validation criteria. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because it did not pass the external job validation criteria. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

## CAUAJM\_E\_10676

**The requested server operation failed because one of the specified fields was invalid. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because one of the specified fields was invalid. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

## CAUAJM\_E\_10677

**The requested server operation failed because required attributes were not specified. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because required attributes were not specified. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

## CAUAJM\_E\_10678

**The requested server operation failed because specified attributes were invalid. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because specified attributes were invalid. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

**CAUAJM\_E\_10679**

**The requested server operation failed because specified filter fields were invalid. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because specified filter fields were invalid. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

**CAUAJM\_E\_10680**

**The requested server operation failed because the specified object does not exist. Operation ID = {0}.**

**Reason:**

The application server operation for the specified ID value has failed because the object does not exist. This message typically follows other error messages that specify the actual causes for the server operation failure.

**Action:**

Review the preceding error messages for the actual cause for the server operation failure.

**CAUAJM\_E\_10681**

**The requested calendar <{0}> does not exist.**

**Reason:**

Calendar has not previously been defined.

**Action:**

Create the calendar, or change the name of the requested calendar.

**CAUAJM\_E\_10682**

**Requested log file for the specified job could not be found.**

**Reason:**

Requested log file for the specified job could not be found.

**Action:**

Pick a job that creates job logs.

**CAUAJM\_E\_10683**

**CONDITION ERROR: External instance name <{0}> is not defined**

**Reason:**

The external instance name is currently not defined to the product.

**Action:**

Define the external instance through JIL.

**CAUAJM\_E\_10684**

**Unable to establish a network connection to [{0}:{1}].**

**Reason:**

Network failure or specified host is bad.

**Action:**

Correct the host name or network failure.

**CAUAJM\_E\_10685**

**Specified filter is invalid.**

**Reason:**

The generated where clause is empty.

**Action:**

Change the filter.

**CAUAJM\_E\_10687**

**The specified binary object is too large.**

**Reason:**

The binary object is too large.

**Action:**

Make the binary object smaller or adjust the MaxBinaryTransferSize configuration parameter.

**CAUAJM\_E\_10689**

**The time interval type is invalid.**

**Reason:**

Specified type is invalid.

**Action:**

Change the type and re-run the API.

**CAUAJM\_E\_10690**

**No statistics fields were specified.**

**Reason:**

No statistics were included in request.

**Action:**

Add statistics to request and re-run the API.



## CAUAJM\_E\_10691

**At least one of the statistics fields supplied was invalid.**

**Reason:**

The specified statistic is invalid.

**Action:**

Change and re-run the API.

## CAUAJM\_E\_10692

**The requested server operation failed to complete within the expiration time of {0} seconds. Operation ID = {1}.**

**Reason:**

The requested server operation failed to complete within the expiration time.

**Action:**

Set a higher value for the MaxExpirationSeconds attribute of the request.

## CAUAJM\_E\_10693

**This job's type does not support the {0} operation. Event not sent for job <{1}>.**

**Reason:**

This job's type does not support the RESUMEJOB operation. Event not sent for the specified job.

**Action:**

Do not issue a RESUMEJOB event for this job.

## CAUAJM\_E\_10694

**Required parameters missing for {0} log retrieval.**

**Reason:**

The input command requires additional parameters to retrieve log files from a z/OS or i5/OS platform.

**Action:**

Input the missing parameters and resubmit the request.

## CAUAJM\_E\_10695

**Parameter must specify file name and number as FileName:Number.**

**Reason:**

The -d parameter must be specified with both file name and number.

**Action:**

Add the file number to the parameter and resubmit the request.

## CAUAJM\_E\_10696

**Step or Task Logs cannot be retrieved for this job type.**

**Reason:**

Step or Task Logs cannot be retrieved for this job type.

**Action:**

Required action unknown.

## CAUAJM\_E\_10715

**Library loaded does not contain the AUTOEXTVJ\_ValidateJob function.**

**Reason:**

Invalid custom validation library.

**Action:**

Customer must rebuild the custom validation library so it includes AUTOEXTVJ\_ValidateJob function.

## CAUAJM\_E\_10716

**All custom job validations will fail.**

**Reason:**

The job failed custom validation.

**Action:**

Required action unknown.

## CAUAJM\_E\_10718

**AUTOEXTVJ Error:**

**Reason:**

Job failed custom validation.

**Action:**

Required action unknown.

## CAUAJM\_E\_10725

**Failed to construct an Automated Framework Message to start job <{0}>.**

**Reason:**

Failed to construct an Automated Framework Message to start the specified job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10728

**Failed to process an incoming AFM string: {0.**

**Reason:**

Failed to process an incoming AFM string.

**Action:**

Required action unknown.

## CAUAJM\_E\_10730

**The product has detected corrupt meta string.**

**Reason:**

Meta string is corrupted.

**Action:**

Contact technical support.

## CAUAJM\_E\_10731

**Failed to retrieve job information from the database.**

**Reason:**

Database query failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_10732

**Failed to parse AFM attribute \"{0}\". The remaining unparsed data is <<{1}>>.**

**Reason:**

The agent is sending data that causes the parser logic to fail.

**Action:**

Contact technical support.

## CAUAJM\_E\_10733

**Database error retrieving the predecessor and successor information.**

**Reason:**

Database error retrieving the predecessor and successor information.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10780

The {0} value for {1} must be {2} bytes in length.

**Reason:**

The value's length is incorrect.

**Action:**

Machine definition is not saved.

## CAUAJM\_E\_10781

The {0} value for {1} contains an invalid {2} value.

**Reason:**

The value is not of correct format.

**Action:**

Machine definition is not saved.

## CAUAJM\_E\_10785

Attribute {0} must be between {1} and {2}.

**Reason:**

The value is outside the acceptable range.

**Action:**

Machine definition is not saved.

## CAUAJM\_E\_10786

Attribute {0} must be greater than or equal to {1}.

**Reason:**

The value is outside the acceptable range.

**Action:**

Machine definition is not saved.

## CAUAJM\_E\_10787

Attribute {0} unsupported for machine type "{1}".

**Reason:**

An attribute was specified that is unsupported for the machine type.

**Action:**

Machine definition is not saved.

## CAUAJM\_E\_10789

Error retrieving connection profile information from the database.

**Reason:**

Error retrieving connection profile information from the database.

**Action:**

Check the status of the database.

## CAUAJM\_E\_10790

StepId cannot be specified for this job

**Reason:**

StepId cannot be specified for this job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10802

must\_start/must\_complete Hours (HH) cannot be greater than 71.

**Reason:**

Hours (HH) cannot be greater than 71.

**Action:**

Required action unknown.

## CAUAJM\_E\_10873

Unable to update resource <{0}> at machine <{1}>. Reason: {2}.

**Reason:**

Unable to Insert/Update/Delete Resource.

**Action:**

Required action unknown.

## CAUAJM\_E\_10874

Unable to {0} global resource <{1}>. Reason: {2}.

**Reason:**

Unable to Insert/Update/Delete Resource.

**Action:**

Required action unknown.

## CAUAJM\_E\_10875

Unable to {0} resource <{1}> at machine <{2}>. Reason: {3}.

**Reason:**

Unable to Insert/Update/Delete Resource.

**Action:**

Required action unknown.

## CAUAJM\_E\_10876

Unable to update resource <{0}>. Reason: {1}.

**Reason:**

Unable to Insert/Update/Delete Resource.

**Action:**

Required action unknown.

## CAUAJM\_E\_10877

Unable to update resource {0}. Reason: {1}.

**Reason:**

Unable to Insert/Update/Delete Resource.

**Action:**

Required action unknown.

## CAUAJM\_E\_10878

Unable to {0} resource dependency {1}. Reason: {2}.

**Reason:**

Unable to Insert/Update/Delete Resource dependency.

**Action:**

Required action unknown.

## CAUAJM\_E\_10879

Unable to read resource id {0}. Reason: Resource does not exist, inconsistent resource data..

**Reason:**

Unable to read Resource.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_10880

**Error encountered while reading timezones. Reverting to gmt timezone.**

**Reason:**

Potential corruption of cfg file.

**Action:**

Check the cfg file to verify timezone definitions.

## CAUAJM\_E\_10881

**Failed to enable core file : {0}.**

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10882

**Maximum number of {0} exceeded**

**Reason:**

Too many subtype parameters have been specified.

**Action:**

Specify the correct number of parameters.

## CAUAJM\_E\_10883

**Unable to redirect {0} to /dev/null [{1}]: errno={2}**

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10884

**Username not found or uid not set properly: {0}**

**Reason:**

Reason unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_10885

**Job {0} cannot be updated while consuming resources.**

**Reason:**

Job cannot be updated while consuming resources.

**Action:**

Wait for the job to complete and manually free the resources if necessary to update the job.

## CAUAJM\_E\_10887

**The expression \"{0}\" against keyword \"{1}\" is in an invalid \"{2}\" format.**

**Reason:**

The format of the specified expression is invalid.

**Action:**

Specify the expression in a valid format.

## CAUAJM\_E\_10888

**The value against JIL keyword {0} is in an invalid {1} format.**

**Reason:**

The specified JIL keyword value is in an invalid format.

**Action:**

Specify the JIL keyword value in a valid format.

## CAUAJM\_E\_10889

**The JIL keywords {0} and {1} cannot be specified together, when the expression {2} is valid.**

**Reason:**

The JIL keyword does not meet the rule requirement.

**Action:**

Specify correct keywords to save the job definition.

## CAUAJM\_E\_10890

**The JIL keywords {0} and {1} cannot be specified together.**

**Reason:**

The JIL keywords does not meet the rule requirement.

**Action:**

Specify correct keywords to save the job definition.



## CAUAJM\_E\_10891

The JIL keywords \{0}\ or \{1}\ must be specified.

**Reason:**

The JIL keywords does not meet the rule requirement.

**Action:**

Specify correct keywords to save the job definition.

## CAUAJM\_E\_10892

10892 The specified combination {0} : {1} is not valid for the JIL keyword \{2}\.

**Reason:**

An invalid combination of values is specified.

**Action:**

Specify a valid combination of values.

## CAUAJM\_E\_10904

**ERROR: Unable to delete {0} table for Job:{1}{2}{3}**

**Reason:**

Unable to delete ext\_job table for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_10905

**Please run the {0} {1} database scripts to synchronize data from event server {2} to {3}.**

**Reason:**

Please run the CA WAAE BCP database scripts to synchronize data from one event server to the other.

**Action:**

Execute the BCP database scripts and restart the Scheduler.

## CAUAJM\_E\_10906

**Detected critical dual event synchronization deadlock. Detailed information below. Shutting down.**

**Reason:**

A critical dual event synchronization deadlock was detected. Detailed information below. Shutting down.

**Action:**

The Scheduler cannot reconcile events from both databases. A development fix is required.

## CAUAJM\_E\_10907

**The start date/time must be less than the end date/time.**

**Reason:**

An invalid from date/time has been specified. The start date/time must be less than the end date/time.

**Action:**

Please correct the start or end date/time and resubmit the request.

## CAUAJM\_E\_10911

**The Local Settings Application Data directory is not accessible.**

**Reason:**

The user credentials that you provided are either invalid or identify you as a user who is not authorized to access the directory.

**Action:**

Verify user permissions, log in as a user who is permitted to access the directory, and reissue the command.

## CAUAJM\_E\_10912

**The home directory does not exist or is not accessible.**

**Reason:**

The user credentials that you provided are either invalid or identify you as a user who is not authorized to access the directory or whose HOME environment variable is not properly set.

**Action:**

Verify home directory assignments or HOME environment variable settings, log in as a user with the necessary permissions or environment variable settings, and reissue the command.

## CAUAJM\_E\_10920

**Authentication of user <{0}> has failed. Command aborted..**

**Reason:**

Unable to authenticate using the supplied credentials.

**Action:**

Re-run the utility with valid credentials or contact your security administrator for assistance.

## CAUAJM\_E\_10922

**Unable to authenticate using cached credentials. Command aborted.**

**Reason:**

Cached credentials are invalid.

**Action:**

Regenerate cached credentials using a valid user name and password combination and then re-issue the command, or contact your security administrator for assistance.

**NOTE**

For more information on cached credentials, see the Generate Cached Credentials section in the [Manage the Unauthenticated User Mode](#) page.

## CAUAJM\_E\_10923

**Unable to retrieve credentials from the external client authentication library. Command aborted.**

**Reason:**

The external client authentication library is not installed, is invalid, or does not recognize you as a valid user.

**Action:**

Verify that a valid external client authentication library is installed and that your user is valid, and then re-issue the command, or contact your security administrator for assistance.

## CAUAJM\_E\_10924

**Unable to authenticate using the credentials obtained from the external client authentication library. Command aborted.**

**Reason:**

The credentials obtained from the external client authentication library are invalid.

**Action:**

Verify that a valid external client authentication library is installed and that your user is valid, and then re-issue the command, or contact your security administrator for assistance.

## CAUAJM\_E\_10925

**The external authentication library is missing symbol <{0}>.**

**Reason:**

The definition for the specified function is missing from the external authentication library code.

**Action:**

Add a definition for the specified function to the external authentication library code and then recompile, test, and re-install the library, or contact your security administrator for assistance.

## CAUAJM\_E\_10926

**Caught an exception in external authentication library function {0}: <{1}>**

**Reason:**

The operating system raised an exception while executing the external authentication library code due to a programming error.

**Action:**

Troubleshoot the programming error, recompile, and test the external authentication library code and then re-install the library, or contact your security administrator for assistance.

## CAUAJM\_E\_10927

**Authentication using the artifact has failed. Command aborted..**

**Reason:**

The specified artifact is invalid.

**Action:**

Generate a new artifact and then re-issue the command, or contact your security administrator for assistance.

## CAUAJM\_E\_10928

**Function failed: <{0}> Errno: <{1,number,1}> Error:**

**Reason:**

An error has occurred performing the indicated operation.

**Action:**

Review the error information and take action as required.

## CAUAJM\_E\_10930

**Encryption error. File <{0}> cannot be written.**

**Reason:**

Cached credential file data is corrupt.

**Action:**

Re-issue the command. If encryption failures continue, contact technical support.

## CAUAJM\_E\_10931

**Decryption error. Unable to read file <{0}>.**

**Reason:**

Cached credential file data is corrupt.

**Action:**

Regenerate the cached credentials and then re-issue the command. If decryption failures continue, contact technical support.

## CAUAJM\_E\_10932

**Time out error. Cannot unlock file <{0}>.**

**Reason:**

The file is locked because another active client utility is using it.

**Action:**

Manually terminate other client utility processes that are in a running, hung, or deadlocked state and then re-issue the command.

## CAUAJM\_E\_10933

**File system error. Unable to acquire a lock for file <{0}>.**

**Reason:**

The file does not exist, the permissions for the file are not set correctly, or some other unspecified file system or operating system error prevents the utility from acquiring a lock.

**Action:**

Verify that the file exists and that the file permissions are set correctly, review any other error messages and take appropriate necessary corrective action, then re-issue the command.

## CAUAJM\_E\_10934

**File system error. Unable to open file <{0}>.**

**Reason:**

The permissions for the file are not set properly, or some other unspecified file system or operating system error prevents the utility from opening the file.

**Action:**

Verify that the file permissions are set correctly, review any other error messages and take necessary corrective action, then re-issue the command.

## CAUAJM\_E\_10935

**File system error. Unable to write file <{0}>.**

**Reason:**

You do not have write permissions, or some other unspecified file system or operating system error is preventing the utility from creating the file.

**Action:**

Verify your permissions, review any other error messages and take necessary corrective action, then re-issue the command.

## CAUAJM\_E\_10936

**File system error. Unable to read file <{0}>.**

**Reason:**

The file does not exist, you do not have read access to it, or some other unspecified file system or operating system error is making the file unreadable.

**Action:**

Verify that the file exists and that you have read access to it, review other error messages and take necessary corrective action, then re-issue the command.

## CAUAJM\_E\_10937

**File system error. Unable to delete file <{0}>.**

**Reason:**

You do not have permission to delete the file, or some other unspecified file system or operating system error is preventing the utility from deleting the file.

**Action:**

Verify your permissions, review any other error messages and take necessary corrective action, then re-issue the command.

## **CAUAJM\_E\_10938**

**Invalid file permissions. Only the owner may read or write file <{0}>.**

**Reason:**

Users other than the owner have read or write access to cached credentials file.

**Action:**

Regenerate the cached credentials and then re-issue the command. If the problem persists, contact CA Support.

## **CAUAJM\_E\_10941**

**Failed to purge {0} log file <{1}>.**

**Reason:**

An error has occurred performing the purge operation.

**Action:**

Verify if the log file is open, review other error messages, and take corrective action.

## **CAUAJM\_E\_10950**

**Unable to encode passphrase.**

**Reason:**

Unable to encode passphrase.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10951**

**Unable to encode key string.**

**Reason:**

Unable to encode key string.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10952**

**Unable to decode passphrase.**

**Reason:**

Unable to decode passphrase.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10953**

Unable to decode keystring.

**Reason:**

Unable to decode keystring.

**Action:**

Required action unknown.

## **CAUAJM\_E\_10979**

Found more than one type of credential. Current operation is not supported.

**Reason:**

An older client is trying to access the credentials that are created by newer client.

**Action:**

Use the latest client to access the credentials.

## **CAUAJM\_E\_10980**

Blob name is not specified.

**Reason:**

Client input error.

**Action:**

Specify the blob name and retry.

## **CAUAJM\_E\_10981**

Blob name must be specified only when the blob type is outputpayloadvar.

**Reason:**

Client input error.

**Action:**

Modify the blob type and retry.

## **CAUAJM\_E\_10954**

An error occurred uploading the key string file to the database.

**Reason:**

A program or database error prevents the storage of the key string in the database.

**Action:**

Check previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact CA Technologies Technical Support.

## **CAUAJM\_E\_18000**

**Database function call <{0}> failed in <{1}> with return code: {2}**

**Reason:**

Sybase Database specific error from ct\_library.

**Action:**

Contact CA Technical Support for further assistance.

## **CAUAJM\_E\_18001**

**Rollback for Insert/Delete of blob failed with return code: {0}**

**Reason:**

Failed to rollback transaction.

**Action:**

Contact CA Support to get information about rolling back blob insert manually.

## **CAUAJM\_E\_18002**

**Stored Procedure execution returned a failure: {0}**

**Reason:**

Stored procedure returned non-zero value referring to an error.

**Action:**

Contact CA Technical Support for further assistance.

## **CAUAJM\_E\_18003**

**Failed to insert Blob/Glob**

**Reason:**

Failed to insert the blob insert query with null's for Blob field.

**Action:**

Previous errors may have some information about the failure, otherwise, Contact CA Technical Support for further assistance with complete list of errors displayed in the log.

## **CAUAJM\_E\_18004**

**Failed to retrieve the global variables to update the blob fields.**

**Reason:**

Query selecting the global variables failed.

**Action:**



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Earlier messages (in the log) will have more information about the problem.

## **CAUAJM\_E\_18005**

**Bind failed for <{0}> of <{1}>.**

**Reason:**

Bind failed for a column of IODescriptor. Could be a result of some other process trying to access the global variables.

**Action:**

Re-executing query may resolve the problem.

## **CAUAJM\_E\_18006**

**next\_row call failed for IO Descriptor in Blob implementation.**

**Reason:**

Failed to fetch the next row from the database.

**Action:**

Next messages in the log should give additional information about the problem.

## **CAUAJM\_E\_18007**

**Failed to update the Blob Global Variable.**

**Reason:**

Failed to update the Blob Global Variable.

**Action:**

Check if global variables are updated by another process. Contact CA Technical Support for further assistance.

## **CAUAJM\_E\_18008**

**Unknown type passed for binding.**

**Reason:**

ct\_bind of ct\_library failed.

**Action:**

Contact CA Technical Support for further assistance.

## **CAUAJM\_E\_18009**

**nextRow failed to retrieve next record from database.**

**Reason:**

Failed to fetch the next row from the database. Something unexpected happened.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18010

**Database function call ct\_diag client msg initialization failed**

**Reason:**

ct\_diag of ct\_library failed.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18011

**Database function ct\_diag server msg initialization failed**

**Reason:**

ct\_diag of ct\_library failed.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18012

**Event Server: {0} Failed Stored Procedure: {1}**

**Reason:**

Error from Sybase DMI occurred while executing the stored procedure.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18013

**Event Server: {0} Failed Query: {1}**

**Reason:**

Error from Sybase DMI occurred while executing the query.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18014

**OS error: {0}**

**Reason:**

Error was received from Sybase DMI.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18015

**Client message:** Message number: layer ({0}), Origin ({1}), text ({2}), SEVERITY = ({3}), NUMBER = ({4})

**Reason:**

Error from Sybase DMI.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18016

**Server:** {0}

**Reason:**

Error from Sybase DMI.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18017

**Procedure:** {0}

**Reason:**

Error from Sybase DMI.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18018

**Server message:** Message number({0}), Severity({1}), State({2}), Line({3}), Message String: {4}

**Reason:**

Error from Sybase DMI.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18019

**Failed to retrieve size of input Blob/Glob.**

**Reason:**

Failed to retrieve size of input Blob/Glob.

**Action:**

Check if Blob/Glob has data. The Blob file might be empty.

## CAUAJM\_E\_18020

**Failed to retrieve bits of input blob.**

**Reason:**

Failed to retrieve bits of input blob.

**Action:**

Check if the input blob has valid data.

## CAUAJM\_E\_18021

**ProcessTimestamp failed in updateBlobData.**

**Reason:**

Failed to process the timestamp Global Variable while updating Blob.

**Action:**

Re-try blob insert again, if problem persists, Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18022

**Encountered an error while processing results of ct\_send.**

**Reason:**

Process failed to process the results from ct\_send database call.

**Action:**

Next messages in the log should give additional information about the problem.

## CAUAJM\_E\_18023

**Database function call ct\_results returned unexpected result type from updateBlobData.**

**Reason:**

ct\_results of ct\_library failed.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18024

**Database returned unexpected data format while processing Timestamp.**

**Reason:**

timestamp data format is different from CS\_TIMESTAMP.

**Action:**

Execute Query again, Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18025

**Initializing ct\_diag failed.**

**Reason:**

Initializing ct\_diag failed..

**Action:**

Required action unknown.

## CAUAJM\_E\_18026

**Setting CS\_EXTRA\_INF failed.**

**Reason:**

Error turning on Sybase password encryption.

**Action:**

Required action unknown.

## CAUAJM\_E\_18027

**Failed to change default database.**

**Reason:**

Could not change the user from default database.

**Action:**

Make user's default database as automdb.

## CAUAJM\_E\_18028

**The database connection is dead. Notified caller on problematic connection.**

**Reason:**

Connection is dead

**Action:**

Check if connection is valid and make sure if isql can connect to database

## CAUAJM\_E\_18029

**Failed to parse values from internal SQL to retrieve blob fields.**

**Reason:**

Internal problem. Bad query.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18032

ct\_con\_props(CS\_SEC\_NETWORKAUTH) failed with appname of "{0}".

**Reason:**

ct\_con\_props(CS\_SEC\_NETWORKAUTH) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_18033

ct\_con\_props(CS\_SEC\_MUTUALAUTH) failed with appname of "{0}".

**Reason:**

ct\_con\_props(CS\_SEC\_MUTUALAUTH) failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_18034

ct\_con\_props(CS\_SEC\_SERVERPRINCIPAL) failed with appname of "{0}".

**Reason:**

ct\_con\_props(CS\_SEC\_SERVERPRINCIPAL) failed

**Action:**

Required action unknown.

## CAUAJM\_E\_18400

**An error has occurred while interfacing with ORACLE.**

**Reason:**

Failure encountered

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18401

**Function <{0}> invoked from <{1}> failed <{2}>**

**Reason:**

Failure encountered.

**Action:**

Contact CA Technical Support for further assistance.

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## CAUAJM\_E\_18402

{0}

**Reason:**

Failure encountered.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18405

bad argument to {0}

**Reason:**

bad argument to bind

**Action:**

Required action unknown.

## CAUAJM\_E\_18406

Unable to find the class <{0}>.

**Reason:**

Could not find the class.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_18407

Failed to connect to Oracle server: {0}

**Reason:**

Failed to connect to Oracle server.

**Action:**

Required action unknown.

## CAUAJM\_E\_18408

Invalid connect string specified.

**Reason:**

Invalid connect string specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_18409

Unable to load Oracle client libraries <{0}>

**Reason:**

Could not locate Oracle client libraries.

**Action:**

Check if ORACLE\_HOME is set, Check permissions on folder, Try setting PATH variables.

## CAUAJM\_E\_18410

Error loading symbols from Oracle client libraries.

**Reason:**

Error loading symbols from Oracle client libraries.

**Action:**

Required action unknown.

## CAUAJM\_E\_18411

OCIEnvCreate() failed.

**Reason:**

OCIEnvCreate() failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_18412

The database client has been interrupted while query execution is in progress.

**Reason:**

The database client has been interrupted while query execution is in progress.

**Action:**

No action required.

## CAUAJM\_E\_18413

Unknown error code.

**Reason:**

Unknown error code.

**Action:**

Required action unknown.



## CAUAJM\_E\_18414

OCIHandleAlloc() failed.

**Reason:**

OCIHandleAlloc() failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_18415

{0} failed with <{1}>.

**Reason:**

JNI API call failed.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_18416

Event Server: {0} Failed Query: {1}

**Reason:**

Database error.

**Action:**

Contact your database administrator.

## CAUAJM\_E\_18418

Oracle Function "{0}" returned [{1}]: {2}

**Reason:**

Oracle Function returned an error.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18419

A failure occurred while a transaction was attempting to commit.

**Reason:**

Failover could not automatically determine instance status.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18420

Unable to find the method <{0}> using signature <{1}>.

**Reason:**

Could not find the method.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_18426

Unable to retrieve the value of the JSONPATH expression <{0}> from the JSON document, error: <{1}>.

**Reason:**

Could not retrieve the value of the JSONPATH expression from the JSON document.

**Action:**

Check the error and take the required action.

## CAUAJM\_E\_18427

Cannot validate the JSON document, error: <{0}>.

**Reason:**

Could not validate the JSON document.

**Action:**

Provide a valid JSON document.

## CAUAJM\_E\_18428

Cannot validate the JSONPATH expression <{0}>, error: <{1}>.

**Reason:**

Could not validate the JSONPATH expression.

**Action:**

Provide a valid JSONPATH expression.

## CAUAJM\_E\_18429

The provided payload data does not comply with \"{0}\" media type.

**Reason:**

The provided payload data is not of the correct media type.

**Action:**

Provide the payload data in a format that is appropriate for the media type.

## CAUAJM\_E\_18431

**{0} failed at {1}: {2}.**

**Reason:**

JNI API call failed.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_18600

**Database function call <{0}> failed in <{1}> with return code: {2}**

**Reason:**

SQL Database specific error from odbc library.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18601

**SQLSTATE: {0}, Native error: {1}, Message: {2}**

**Reason:**

Native error.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18604

**bind failed: Unknown type for bind variable.**

**Reason:**

Type of bind variable unknown.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18605

**Failed to retrieve size of input LOB.**

**Reason:**

Getsize failed for blob.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18606

Failed to retrieve bits of input blob.

**Reason:**

copyBits failed for blob.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18607

Event Server: {0} Failed Stored Procedure: {1}

**Reason:**

Error from odbc DMI while executing stored procedure

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18611

Event Server: {0} Failed Query: {1}

**Reason:**

Error from odbc DMI while executing Query.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_18801

Server: {0} Not Defined!!! ERROR!!!.

**Reason:**

Server not defined.

**Action:**

Required action unknown.

## CAUAJM\_E\_18802

Error from {0}.

**Reason:**

Error encountered.

**Action:**

Required action unknown.

## CAUAJM\_E\_18803

Database operation to modify machine <{0}> could not complete due to an internal error.

**Reason:**

The ujo\_put\_machine stored procedure experienced an internal error.

**Action:**

Please contact CA Support.

## CAUAJM\_E\_18900

Error for {0} \"{1}\". <The JIL keyword \"{2}\"OR JIL keyword \"{3}\" should be specified.

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18901

Invalid value \"{0}\" for JIL Keyword: {1}.

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value.

## CAUAJM\_E\_18902

Invalid value \"{0}\" for JIL Keyword: \"{1}\". Accepted values are \"{2}\".

**Reason:**

Invalid value specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18903

Invalid subtype keyword {0} for JIL keyword {1}.

**Reason:**

Invalid subtype keyword specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18904

Invalid value \"{0}\" for JIL Keyword: \"{1}\". Accepted values are any combination of \"BASIC\", \"DIGEST\", or \"NTLM\" separated by commas.

**Reason:**

Invalid Job Type specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18905

The hour portion of the datetime format for JIL keyword \"{0}\" must be less than {1}.

**Reason:**

Invalid absolute time format.

**Action:**

Specify a valid time format for an absolute time. For eg: must\_start\_times: 11:30, 52:35

## CAUAJM\_E\_18906

Unsigned short specified \"{0}\" for JIL keyword \"{1}\" is out of range.

**Reason:**

The short integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18907

Invalid integer specified \"{0}\" for JIL keyword \"{1}\".

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18908

The value of JIL keyword \"{0}\" must be less than the value of JIL keyword \"{1}: {2}\".

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18909

Invalid subtype keyword \"{0}\" for JIL Keyword: \"{1}\". Accepted subtype keywords are \"{2}\".

**Reason:**

Invalid subtype keyword specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18911

Database query execution could not complete due to an internal error.

**Reason:**

The database experienced an internal error while processing a query.

**Action:**

Please contact CA Support.

## CAUAJM\_E\_18912

Database operation to fetch query results could not complete due to an internal error.

**Reason:**

The database experienced an internal error while fetching query results.

**Action:**

Please contact CA Support.

## CAUAJM\_E\_18913

The value \"{0}\" is a reserved word. Please use another name.

**Reason:**

Invalid value specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18914

Invalid syntax \"{0}\" for subtype keyword \"{1}\". Syntax of subtype: {2}

**Reason:**

An invalid syntax was specified for subtype keyword.

**Action:**

Object definition is not saved.

## CAUAJM\_E\_18915

Invalid double value for JIL keyword \"{0}\". {1}.

**Reason:**

Invalid double value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18916

Invalid object\_type specified: {0}

**Reason:**

Invalid object Type specified.

**Action:**

object definition is not saved.

## CAUAJM\_E\_18917

Invalid keyword specified: {0}

**Reason:**

Invalid keyword specified.

**Action:**

Object definition is not saved.

## CAUAJM\_E\_18918

Required JIL subtype keyword \"{0}\" is missing from JIL Keyword \"{1}\".

**Reason:**

Required keyword is missing.

**Action:**

Specify required jil keyword in job definition.

## CAUAJM\_E\_18919

The 'from' date is past the 'to' date

**Reason:**

The to date precedes the from date

**Action:**

The from date should precede the to date



## CAUAJM\_E\_18920

**Value of keyword {0} contains only whitespace**

**Reason:**

The keyword value contains only whitespaces.

**Action:**

Refer to the product documentation for more information.

## CAUAJM\_E\_18921

**Error for {0} \"{1}\". <The JIL keyword \"{2}\" cannot be specified together with JIL keyword \"{3}\".>**

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Modify job definition.

## CAUAJM\_E\_18922

**The value of the JIL keyword \"{0}\" does not have a valid operator. The valid operator is \"{1}\".**

**Reason:**

Error for value of jil keyword.

**Action:**

Value is in incorrect format.

## CAUAJM\_E\_18923

**A relative time must be specified for JIL keyword \"must\_start\_times\" and \"must\_complete\_times\" because JIL keyword \"start\_mins\" is specified.**

**Reason:**

JIL keyword \"start\_mins\" is specified.

**Action:**

Specify a relative time for JIL keyword 'must\_start\_times' and 'must\_complete\_times'. For eg: must\_complete\_times: +6

## CAUAJM\_E\_18924

**The number of elements for JIL keyword \"{0}\" must be the same as the number of elements for JIL keyword \"{1}\".**

**Reason:**

Number of elements are different for the 2 specified JIL keywords.

**Action:**

Specify the same number of elements for the 2 specified JIL keywords.

## CAUAJM\_E\_18925

The value \"{0}\" for element[{1}] of JIL keyword \"{2}\" must be less than the value \"{3}\" for the same element of JIL keyword \"{4}\".

**Reason:**

Invalid time combination entered.

**Action:**

Specify a valid time combination.

## CAUAJM\_E\_18926

Invalid datetime format JIL keyword \"{0}\". The hh(hours portion) of the (hh:mm) format is incorrect.

**Reason:**

Invalid absolute time format.

**Action:**

Specify a valid time format for an absolute time. For eg: must\_start\_times: 11:30, 11:35

## CAUAJM\_E\_18927

Invalid datetime (hh:mm) format specified for JIL keyword \"{0}\". The mm(minutes portion) of the (hh:mm) format is incorrect.

**Reason:**

Invalid absolute time format.

**Action:**

Specify a valid time format for an absolute time. For eg: must\_start\_times: 11:30, 11:35

## CAUAJM\_E\_18928

Invalid datetime (hh:mm) format specified for JIL keyword \"{0}\". The mm(minutes portion) of the (hh:mm) format is out of range.

**Reason:**

Invalid absolute time format.

**Action:**

Specify a valid time format for an absolute time. For eg: must\_start\_times: 11:30, 11:35

## CAUAJM\_E\_18929

Only integers (between 0 and 23) are allowed for the hour portion of the JIL keyword \"{0}\". Please specify value in the format of: hh:mm, hh:mm, etc.

**Reason:**

The datetime format is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18930

Only integers (between 0 and 59) that are separated by commas are allowed for the JIL keyword `\{0\}`. Please specify value in the format of: mm, mm, etc

### Reason:

The datetime format is invalid.

### Action:

Job definition is not saved.

## CAUAJM\_E\_18931

Only integers (between 0 and 59) are allowed for the minutes portion of the JIL keyword `\{0\}`. Please specify value in the format of: hh:mm, hh:mm, etc

### Reason:

The datetime format is invalid.

### Action:

Job definition is not saved.

## CAUAJM\_E\_18932

Invalid datetime format for JIL keyword `\{0\}`. Please specify value in the format of: hh:mm, hh:mm, etc.

### Reason:

The datetime format is invalid.

### Action:

Job definition is not saved.

## CAUAJM\_E\_18933

The value `\{0\}` for the JIL keyword `\must_complete_times\` must be greater than the value `\{1\}` for JIL keyword `\must_start_times\`.

### Reason:

The `\must_start_times\` is greater than the `\must_complete_times\`

### Action:

Correct the 'must\_start\_times' and 'must\_complete\_times' values.

## CAUAJM\_E\_18934

A mixture of relative time and absolute time cannot be specified for the JIL keyword `\must_start_times\` and `\must_complete_times\`. You must specify either absolute or relative time.

### Reason:

Inconsistent time formats specified for `\must_start_times\` and `\must_complete_times\` attributes.

**Action:**

Specify either absolute or relative times for both attributes.

**CAUAJM\_E\_18935**

**Problem with metadata for jil subtype {0}. Please contact technical support.**

**Reason:**

Invalid data in metadata.

**Action:**

Job definition is not saved.

**CAUAJM\_E\_18936**

**Required JIL keyword \"{0}\" is missing.**

**Reason:**

Required keyword is missing

**Action:**

Specify required jil keyword in job definition

**CAUAJM\_E\_18937**

**Invalid integer specified \"{0}\" for JIL keyword \"{1}\" of type EXIT\_CODES.**

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

**CAUAJM\_E\_18938**

**Unable to retrieve credentials to prepare the machine \"{0}\" for DCA. Please use autosys\_secure option 5 to enter a correct userid and password.**

**Reason:**

Unable to get password to prepare machine for DCA.

**Action:**

Check to see if administrator is entered correctly.

**CAUAJM\_E\_18939**

**JIL keywords start\_times and start\_mins are mutually exclusive.**

**Reason:**

Invalid keywords combination.

**Action:**

Specify either start\_times or start\_mins, not both. Job definition is not saved.

## CAUAJM\_E\_18940

JIL keywords must\_start\_times and/or must\_complete\_times requires JIL keyword start\_times or start\_mins

### Reason:

Invalid keywords combination.

### Action:

Job definition is not saved.

## CAUAJM\_E\_18941

It is invalid for JIL keyword \"{0}\" to have a relative time format that have multiple values.

### Reason:

Too many values specified for a relative time format.

### Action:

Specify only 1 value for a relative time. For eg: must\_start\_times: +5

## CAUAJM\_E\_18942

Invalid object\_name specified: {0}

### Reason:

Invalid object\_name specified.

### Action:

object definition is not saved.

## CAUAJM\_E\_18943

job {0} does not exist.

### Reason:

Job does not exist

### Action:

object definition is not saved.

## CAUAJM\_E\_18944

Binding of database columns failed.

### Reason:

Binding of database columns failed.

### Action:

Please contact technical support.

## CAUAJM\_E\_18945

Failed to get next oid for JIL Keyword {0}.

**Reason:**

Problem with retrieving next oid for JIL Keyword from the database.

**Action:**

Please contact technical support.

## CAUAJM\_E\_18946

Database operation to modify job <{0}> could not complete due to an internal error.

**Reason:**

The ujo\_put\_job stored procedure experienced an internal error.

**Action:**

Please contact CA Support.

## CAUAJM\_E\_18947

Invalid box name {0} specified, it does not exist.

**Reason:**

Box name specified does not exist.

**Action:**

Please specify a valid box\_name.

## CAUAJM\_E\_18948

\*\*\* CANNOT DELETE Job: {0}, because it DOES NOT EXISTS \*\*\*

**Reason:**

Cannot delete job.

**Action:**

Job does not exist.

## CAUAJM\_E\_18949

The relative time \"{0}\" specified for JIL Keyword \"start\_mins\" must be greater than the previous time specified \"{1}\" plus the relative value \"{2}\" specified for JIL Keyword \"{3}\"

**Reason:**

The datetime format is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18951

The value \"{0}\" of JIL keyword \"{1}\" must be less than the value \"{2}\" of JIL keyword \"{3}\".

**Reason:**

Invalid time combination entered.

**Action:**

Specify a valid time combination

## CAUAJM\_E\_18953

{0} has not been defined by the user.

**Reason:**

User-Defined Job Type has not been defined by the user.

**Action:**

Please define this user-defined job type.

## CAUAJM\_E\_18954

Non-matching brackets or quotes for JIL keyword \"{0}\".

**Reason:**

Non-matching brackets or quotes.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18955

Matching brackets not found \"{0}\" for JIL keyword \"{1}\"

**Reason:**

Non-matching brackets

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18956

Value of keyword {0} contains Unprintable characters.

**Reason:**

The keyword value contains unprintable characters

**Action:**

Refer to the product documentation for more information

## CAUAJM\_E\_18957

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". It contains one or more invalid characters or symbols.

**Reason:**

The keyword value contains illegal characters

**Action:**

Refer to the product documentation for more information on legal characters for the keyword.

## CAUAJM\_E\_18958

Error for {0} \"{1}\". <The JIL keyword \"{2}\" is only valid for jil keyword \"{3}\" of value {4}.>

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved

## CAUAJM\_E\_18959

The JIL keyword \"{0}\" must be specified with JIL keyword(s) \"{1}\".

**Reason:**

The value for this JIL keyword does not meet the rule requirement. Job definition is not saved

**Action:**

Enter the appropriate JIL keywords and resubmit the request.

## CAUAJM\_E\_18960

Integer specified \"{0}\" for JIL keyword \"{1}\" is out of range.

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18961

Error for {0}: {1} < {2} exceeds {3} bytes.>

**Reason:**

The value specified is too long, hence it is rejected.

**Action:**

Job definition is NOT saved.



## CAUAJM\_E\_18962

Numeric value specified \"{0}\" for JIL keyword \"{1}\" is out of allowable range.

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18963

The {0} value for the JIL keyword {1} must be {2} bytes in length.

**Reason:**

The value's length is incorrect.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18964

The value \"{0}\" for JIL keyword \"{1}\" must begin with \"{2}\"

**Reason:**

The keyword value must begin with the specified value.

**Action:**

Modify the keyword value as specified and define the job.

## CAUAJM\_E\_18965

Error for {0} \"{1}\". <The JIL keyword \"{2}\" cannot be specified together with JIL keyword \"{3}\".>

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18966

Updates to job\_type is not allowed for job \"{0}\". Please delete this job definition and re-define it again.

**Reason:**

Cannot update job type.

**Action:**

Required action unknown.

## CAUAJM\_E\_18967

Invalid datetime value specified \"{0}\" for JIL keyword \"{1}\". Valid format: yyyyMMdd HH:mm:ss

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18968

Error validating Machine {0}: \"{1}\" value out of allowable range for attribute {2}

**Reason:**

Unable to determine machine existence

**Action:**

Required action unknown.

## CAUAJM\_E\_18970

The {0} attribute {1} must be specified if {2} is {3}.

**Reason:**

Invalid keyword combination. Name cannot be a keyword

**Action:**

Machine names cannot be keywords.

## CAUAJM\_E\_18971

The {0} attribute {1} cannot be specified if {2} is {3}. It can only be specified if {4} is {5}.

**Reason:**

Invalid keyword combination. Name cannot be a keyword

**Action:**

Machine names cannot be keywords.

## CAUAJM\_E\_18972

The {0} attributes \"{1} and {2}\" must be the same for {3} with the same agent name {4}.

**Reason:**

Invalid keyword combination. Name cannot be a keyword

**Action:**

Machine names cannot be keywords.

## CAUAJM\_E\_18973

The {0} value for the JIL keyword {1} contains an invalid {2} value.

**Reason:**

The value is not of correct format.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18974

Invalid \"{0}\" specified in the metadata for JIL keyword \"{1}\". Please contact ca.

**Reason:**

The integer value is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18975

Invalid subtype keyword/value combination for JIL keyword \"{0}\". {1}

**Reason:**

Invalid subtype keyword or value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18976

Invalid value for JIL keyword \"{0}\". {1}

**Reason:**

Invalid value for JIL keyword.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18977

Box \"{0}\" lineage does not exist. Database might be corrupted.

**Reason:**

Unable to verify box lineage.

**Action:**

Required action unknown.

## CAUAJM\_E\_18978

**ERROR Circular definition: Cannot add job to a box that belongs to self.**

**Reason:**

Circular Definition of boxes or jobs.

**Action:**

Required action unknown.

## CAUAJM\_E\_18979

**Error for JIL Keyword "{0}". The soft\_limit cannot be greater than the hard\_limit for ulimit\_id of "{1}".**

**Reason:**

Invalid value for ulimit keyword.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18980

**The value of JIL keyword \"{0}\" must be greater than the value of JIL keyword \"{1}\".**

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18981

**Error for {0} \"{1}\": {2}**

**Reason:**

The job definition does not satisfy this rule.

**Action:**

Job definition is NOT saved.

## CAUAJM\_E\_18982

**There are duplicate entries in the value for the JIL keyword \"{0}\".**

**Reason:**

The datetime format is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18983

Invalid absolute or relative time specified for JIL keyword \"{0}\". Absolute time must be in the format of hh:mm. Relative time must be in the format of +mm.

**Reason:**

Invalid time format specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18984

This virtual machine definition does not contain any real machines.

**Reason:**

Virtual machine definition is empty.

**Action:**

Add one or more real machines to this virtual machine definition.

## CAUAJM\_E\_18986

ERROR for {0}: {1} < Resource cannot be defined to localhost >

**Reason:**

Machine Resources cannot be defined to localhost.

**Action:**

Machine Resources cannot be defined to localhost.

## CAUAJM\_E\_18987

ERROR for {0}: {1} < machine \"{2}\" does not exist >

**Reason:**

Machine does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_18988

Error For {0}: {1} < machine \"{2}\" not defined >

**Reason:**

Machine is not defined.

**Action:**

No action required.

## CAUAJM\_E\_18989

The value for JIL keyword \"{0}\" must be less than or equal to \"{1}\" when jil keyword \"{2}\" value is {3}

**Reason:**

Replace the value for the keyword to be less than the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18990

Invalid User-Defined Job Type: \"{0}\".

**Reason:**

Invalid User-Defined Job Type.

**Action:**

Required action unknown.

## CAUAJM\_E\_18991

Comma-delimited list of \"{0}\" is only valid for JIL keyword \"{1}\" of value \"{2}\" and \"{3}\".

**Reason:**

Invalid use of comma delimited list.

**Action:**

Required action unknown.

## CAUAJM\_E\_18992

Invalid value \"{0}\" for JIL keyword \"{1}\". The correct format is: keyword(value),keyword(value), ... etc.

**Reason:**

Invalid name(value) format.

**Action:**

Required action unknown.

## CAUAJM\_E\_18993

The JIL keyword \"{0}\" must be numeric if JIL keyword \"{1}\" has the value \"{2}\".

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18994

The JIL keyword \"{0}\" cannot be zero or blank.

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18995

Invalid number specified \"{0}\" for JIL keyword \"{1}\". Reason: {2}.

**Reason:**

The integer is invalid.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18996

The value for JIL keyword \"{0}\" must be greater than or equal to \"{1}\" for jil keyword \"{2}\" of value {3}.

**Reason:**

Replace the value for the keyword to be greater than specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_18997

Decrypting value of JIL keyword \"{0}\" failed for \"{1}\" \"{2}\".

**Reason:**

Decrypting key for update of external instance failed

**Action:**

External instance definition is NOT saved.

## CAUAJM\_E\_18998

Invalid value for JIL keyword \"chk\_files\": missing filesystem or size parameter. The correct format is: filesystem size [filesystem size .....].

**Reason:**

Missing size parameter

**Action:**

Please specify in the format of: filesystem size filesystem size.

## CAUAJM\_E\_18999

The value for JIL keyword \"{0}: {1}\" must match the date time format {2}

**Reason:**

Replace the value for the keyword with a value that matches the datetime mask.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19000

The date time format specified in the value for JIL keyword \"{0}: {1}\" is invalid.

**Reason:**

Replace the value for the keyword with a value that matches the datetime mask.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19001

Job \"{0}\" is still in STARTING status. Please try again later.

**Reason:**

Unable to update job because it is still in STARTING status.

**Action:**

Object definition is not saved.

## CAUAJM\_E\_19003

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". It contains a semi-colon.

**Reason:**

The keyword value contains illegal characters.

**Action:**

Refer to the product documentation for more information.

## CAUAJM\_E\_19004

The value of JIL keyword \"{0}\" does not allow any wildcard characters in JIL keyword \"{1}\".

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.



## CAUAJM\_E\_19005

**ERROR for job: {0} < "{1}" is a reserved keyword and cannot be specified for "{2}".>**

**Reason:**

Invalid job name has been specified.

**Action:**

Required action unknown.

## CAUAJM\_E\_19006

**The value of JIL keyword \"{0}\" contains only blanks or spaces or is empty.**

**Reason:**

Error for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_19007

**The JIL subtype keyword \"{0}\" can only be specified once for each JIL keyword \"{1}\".**

**Reason:**

This JIL subtype keyword can only be specified once for this JIL keyword.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19008

**Job \"{0}\" specified for JIL keyword \"{1}\" is not a payload producing job.**

**Reason:**

Job specified is not a payload producing job.

**Action:**

Object definition is NOT saved.

## CAUAJM\_E\_19010

**The JIL subtype keyword \"payload\_job\" is not valid for JIL keyword \"{0}\" because this job \"{1}\" is not a payload consuming job.**

**Reason:**

Job specified is not a payload producing job.

**Action:**

Object definition is NOT saved.

## CAUAJM\_E\_19011

The number of files specified in the values of keywords \"{0}\" and \"{1}\" must be the same.

**Reason:**

The number of files are not the same for the 2 keywords.

**Action:**

Job definition is NOT saved.

## CAUAJM\_E\_19012

The exit codes specified in the values of keywords \"{0}\" and \"{1}\" overlap each other.

**Reason:**

The exit codes specified in the values of the keywords overlap each other.

**Action:**

Job definition is NOT saved.

## CAUAJM\_E\_19013

Invalid exit\_code range specified for JIL keyword \"{0}\" because \"{1}\" is greater than or equal to \"{2}\".

**Reason:**

Invalid exit\_code values specified.

**Action:**

Object definition is NOT saved.

## CAUAJM\_E\_19014

Invalid range of values \"{0}\" specified for JIL keyword \"{1}\".

**Reason:**

Invalid exit\_code values specified.

**Action:**

Object definition is NOT saved.

## CAUAJM\_E\_19015

Duplicate exit\_code value \"{0}\" specified for JIL keyword \"{1}\".

**Reason:**

Invalid exit\_code values specified.

**Action:**

Object definition is NOT saved.

## CAUAJM\_E\_19016

Updates to type is not allowed for machine "{0}". Please delete this machine definition and re-define it again.

**Reason:**

Updates to machine type is not allowed.

**Action:**

Machine definition is NOT saved.

## CAUAJM\_E\_19017

External Instance "{0}" is being used and will not be deleted.

**Reason:**

External Instance is being used.

**Action:**

Delete the condition that uses this external instance.

## CAUAJM\_E\_19019

Duplicate value "{0}" for subtype keyword "{1}" for JIL keyword "{2}".

**Reason:**

Invalid subtype keyword value specified.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19020

The component machine "{0}" that you have defined does not exist.

**Reason:**

Component machine does not exist.

**Action:**

Please define the component machine first.

## CAUAJM\_E\_19021

ERROR for Machine: {0} < Cannot add localhost to a virtual machine list >

**Reason:**

Machine does not exist.

**Action:**

Required action unknown.

## CAUAJM\_E\_19023

The {0} attribute {1}: {2} is not valid for {3}: {4}.

**Reason:**

Invalid keyword value combination.

**Action:**

Specify a different encryption type.

## CAUAJM\_E\_19025

Updates to xtype is not allowed for external instance "{0}". Please delete this definition and re-define it again.

**Reason:**

Cannot update external instance type.

**Action:**

Please delete this definition and define it again.

## CAUAJM\_E\_19026

Invalid value for JIL keyword "\"chk\_files\"": Invalid filesystem <{0}> specified. {1}

**Reason:**

Invalid filesystem specified.

**Action:**

Please specify only drive letters. Eg: c: or d:

## CAUAJM\_E\_19027

The maximum number of values for keyword "\"{0}\"" that can specified for "\"{1}\"" of "\"{2}\"" is {3}.

**Reason:**

The number of values that can be specified exceeds the maximum.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19028

Job <{0}> could not be modified properly due to a database failure. The job definition is in an unknown state.

**Reason:**

The specified job could not be modified properly due to a database failure.

**Action:**

Check the connectivity to the database and inspect the job definition. You may have to delete and redefine the job.

## CAUAJM\_E\_19029

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be a one or more valid email addresses separated by \";\" or \",\".

**Reason:**

Replace the value for the keyword to match correct email address.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19030

Only the {0} {1} superuser can set the {2}.

**Reason:**

The user does not have sufficient privileges to perform this operation.

**Action:**

No action required.

## CAUAJM\_E\_19031

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The first character must be alphabetic.

**Reason:**

The keyword value contains illegal characters

**Action:**

Refer to the product documentation for more information.

## CAUAJM\_E\_19032

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be a valid IP address.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19033

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be a valid MAC address.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19034

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be 4 or 6 hexadecimal numbers separated by \"-\", \".\" or \":\".

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19035

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be a valid hostname or IP address.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19036

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be one or more single-quoted values separated by commas.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19037

The value for JIL keyword \"{0}\" must end with \"{1}\".

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19038

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must be in the format of user@host.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19039

The JIL subtype keyword \"{0}\" must be specified with JIL keyword(s) \"{1}\".

**Reason:**

The value for this JIL subtype keyword does not meet the rule requirement. Job definition is not saved.

**Action:**

Enter the appropriate JIL keywords and resubmit the request.

## CAUAJM\_E\_19040

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must contain only hexadecimal digits.

**Reason:**

Replace the value for the keyword to begin with the specified value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19041

Error for {0} \"{1}\". <The JIL subtype keyword \"{2}\" is only valid for jil keyword \"{3}\" of value {4}.>

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19042

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". An email address can be up to 254 characters.

**Reason:**

One of the email address is greater than 254 characters.

**Action:**

Re-enter the correct email address.

## CAUAJM\_E\_19043

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". User part of the email address cannot be greater than 64 chars.

**Reason:**

Local-part cannot be more than 64 characters.

**Action:**

Re-enter the correct email address.

## CAUAJM\_E\_19044

**Duplicate subtype keyword "{0}". There is already an instance of "{2}: {0}={1}" in the job definition.**

**Reason:**

Replace the value for the keyword or remove the keyword and value.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19045

**"seq-" cannot be specified with a jcl\_library value that also contains a member name.**

**Reason:**

The keyword value contains an illegal member name specification. A sequential dataset does not contain members.

**Action:**

Remove the member name and resubmit the request.

## CAUAJM\_E\_19046

**Error for {0}: {1} < {2} subtype keyword {3} exceeds {4} bytes.>**

**Reason:**

The specified subtype keyword value is too long.

**Action:**

Modify the subtype keyword value as specified and define the job.

## CAUAJM\_E\_19047

**Invalid JIL keyword value "{0}" for keyword "{1}". It contains one or more unpaired delimiters.**

**Reason:**

The attribute value is missing or contains an extraneous delimiter.

**Action:**

Specify the corresponding delimiter. Prefix any embedded delimiters with an escape character.

## CAUAJM\_E\_19048

**Updates to job attribute "{0}" is not allowed. Usage valid for the insert operation only.**

**Reason:**

The specified attribute can be used only during job insertion to set the initial status of job. The attribute cannot be updated.

**Action:**

Remove the specified attribute and resubmit the JIL. If the specified attribute is the only attribute specification no action is required.



## CAUAJM\_E\_19049

Invalid JIL keyword value \"{0}\" for keyword \"{1}\". The value must contain only octal digits.

**Reason:**

The specified keyword value is not octal.

**Action:**

Specify an octal value.

## CAUAJM\_E\_19050

JIL Keyword: {0} cannot be empty.

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value.

## CAUAJM\_E\_19051

Connection profile <{0}> could not be modified properly due to a database failure. The connection profile definition is in an unknown state.

**Reason:**

The specified connection profile could not be modified properly due to a database failure.

**Action:**

Check the connectivity to the database and inspect the connection profile definition. You may need to delete and redefine the connection profile.

## CAUAJM\_E\_19052

Error validating connection profile {0}.

**Reason:**

Unable to determine connection profile existence.

**Action:**

Required action unknown.

## CAUAJM\_E\_19053

Cannot insert connection profile: {0} as it already exists.

**Reason:**

A connection profile with this name already exists.

**Action:**

Specify another name for the connection profile.

## CAUAJM\_E\_19055

**Invalid connection profile name: {0} {1} not performed.**

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value.

## CAUAJM\_E\_19056

**Cannot delete connection profile \"{0}\" as it is still being referenced by active jobs.**

**Reason:**

Could not delete connection profile as it is referenced in active jobs.

**Action:**

To delete references in active jobs, change the connection profile that those jobs are using or delete those active jobs.

## CAUAJM\_E\_19058

**Connection profile <{0}> could not be deleted properly due to a database failure. The connection profile definition is in an unknown state.**

**Reason:**

The specified connection profile could not be deleted properly due to a database failure.

**Action:**

Check the connectivity to the database and inspect the connection profile definition. You may need to delete the connection profile again.

## CAUAJM\_E\_19060

**Updates to connectionprofile\_type is not allowed for connection profile \"{0}\".**

**Reason:**

Cannot update connection profile type.

**Action:**

Delete this connection profile definition and redefine it.

## CAUAJM\_E\_19061

**Invalid connection profile type: {0} {1} not performed.**

**Reason:**

Invalid value specified.

**Action:**

Specify a valid type of connection profile.

## CAUAJM\_E\_19062

**Connection profile name: {0} is inaccurate {1} not performed.**

**Reason:**

The connection profile is not complete.

**Action:**

Redefine the connection profile.

## CAUAJM\_E\_19063

**Connection profile \"{0}\" cannot be specified for this job because it is missing JIL keyword(s) \"{1}\".**

**Reason:**

The connection profile is not complete for this job.

**Action:**

Create a new connection profile or update the existing profile for this job.

## CAUAJM\_E\_19064

**Cannot remove \"{0}\" in connection profile \"{1}\" as it is still being referenced by active jobs.**

**Reason:**

Connection profile is referenced in active jobs.

**Action:**

To update references in active jobs, please change the connection profile that those jobs are using or delete those active jobs.

## CAUAJM\_E\_19066

**The value for JIL keyword \"{0}\" cannot contain alphabets in between but can end with only \"{1}\" alphabets.**

**Reason:**

The value cannot contain alphabets in between but can end with one of the specified alphabets.

**Action:**

Modify the keyword value.

## CAUAJM\_E\_19067

**The value for JIL keyword \"{0}\" is not in the range \"{1} - {2}\" .**

**Reason:**

The keyword value is not in the range.

**Action:**

Specify a value for the keyword that is within the range.

## CAUAJM\_E\_19068

The email notification template \"{0}\" does not exist.

**Reason:**

The email notification template does not exist.

**Action:**Specify the email notification template.

## CAUAJM\_E\_19071

Error for {0} \"{1}\". <The JIL keyword \"{2}\" is only applicable for virtual machines.>

**Reason:**

The value for this JIL keyword does not meet the rule requirement.

**Action:**

Job definition is not saved.

## CAUAJM\_E\_19073

An invalid combination of values is specified for the attribute {0} : {1}. The supported combination of values is \"{2}\".

**Reason:**

An invalid combination of values is specified.

**Action:**

Specify a valid combination of values.

## CAUAJM\_E\_19077

Job name cannot be empty while renaming the job.

**Reason:**

The specified job name is empty.

**Action:**

Specify a proper job name.

## CAUAJM\_E\_19080

Invalid combination specified. The keyword "j2ee\_parameter" cannot be specified when the keyword "media\_type" has empty value.

## CAUAJM\_E\_19100

You cannot move job \"{0}\" into box \"{1}\" while it is in the {2} status and the box is in the {3} status.

**Reason:**

Unable to update the job because a job that is in the ON\_ICE status cannot be placed into a box that is in the ON\_NOEXEC status.

**Action:**

Object definition is not saved. Please send the JOB\_OFF\_ICE event on behalf of the job or the JOB\_OFF\_NOEXEC event on behalf of the box and try again.

**CAUAJM\_E\_19101**

Please send the {0} event on behalf of the job or the {1} event on behalf of the box and try again.

**Reason:**

Please send the JOB\_OFF\_ICE event on behalf of the job or the JOB\_OFF\_NOEXEC event on behalf of the box and try again.

**Action:**

Object definition is not saved. Please send the JOB\_OFF\_ICE event on behalf of the job or the JOB\_OFF\_NOEXEC event on behalf of the box and try again.

**CAUAJM\_E\_19969**

{0} is not allowed for {1}:{2} as {3} is deprecated.

**Reason:**

Deprecated attribute cannot be specified for Update or Override.

**Action:**

Specify valid attributes.

**CAUAJM\_E\_20048**

Error uploading {0}: {1}:

**Reason:**

An error occurred while uploading the blob.

**Action:**

Required action unknown.

**CAUAJM\_E\_20049**

Unable to initialize the curl library interface. Output payload and output payload vars creation skipped for job <{0}>.

**Reason:**

System error.

**Action:**

Required action unknown.

**CAUAJM\_E\_20050**

Unable to initialize the curl library interface. Skipped URL encoding.

**Reason:**

System error.

**Action:**

Required action unknown.

## **CAUAJM\_E\_20052**

**AutoPing cannot run as FAT Client.**

**Reason:**

AutoPing cannot run as FAT Client.

**Action:**

Run AutoPing as a thin client.

## **CAUAJM\_E\_20054**

**Chase cannot run as FAT Client.**

**Reason:**

Chase cannot run as FAT Client.

**Action:**

Run Chase as a thin client.

## **CAUAJM\_E\_20064**

**Error opening log file: {0}**

**Reason:**

Error opening log file.

**Action:**

Required action unknown.

## **CAUAJM\_E\_20065**

**Exhausted available memory**

**Reason:**

Exhausted available memory.

**Action:**

Required action unknown.

## **CAUAJM\_E\_20069**

**Disk error while accessing log file: {0}.**

**Reason:**

Disk error while accessing log file.

**Action:**

---

Required action unknown.

## CAUAJM\_E\_20132

**Function "{0}" returned error {1}.**

**Reason:**

glob() or stat() failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_20211

**Exceeded the maximum numbers of errors when attempting to perform email notification**

**Reason:**

Maximum number of errors have been exceeded when attempting to perform email notification.

**Action:**

Check for any prior error messages and correct the situation.

## CAUAJM\_E\_20212

**Email notification has been disabled. The scheduler's notification method has been reset to <{0}>.**

**Reason:**

The email notification method has been disabled due to a non-recoverable error.

**Action:**

Check for any prior error messages and correct the situation.

## CAUAJM\_E\_20213

**Email notification library <{0}> is not loaded. Email notification skipped for job <{1}>.**

**Reason:**

The email notification library is not present. Unable to initiate email notification.

**Action:**

Make sure the required library is available on the system.

## CAUAJM\_E\_20214

**Email notification failed for job <{0}>. Detail information <{1}>.**

**Reason:**

An error occurred during the email notification process.

**Action:**

Please review the error message for possible causes. Otherwise contact CA technical support.

## CAUAJM\_E\_20217

Unable to initialize the email notification library interface. Email notification skipped for job <{0}>.

**Reason:**

System error.

**Action:**

Required action unknown.

## CAUAJM\_E\_20220

DB Maintenance returned an exit code of {0}.

**Reason:**

DB Maintenance returned an error exit code.

**Action:**

Required action unknown.

## CAUAJM\_E\_20221

SMTP host is blank. Email notification request skipped for job <{0}>.

**Reason:**

The specified SMTP host name is invalid. Host name cannot be blank.

**Action:**

Specify a valid SMTP host name.

## CAUAJM\_E\_20222

Could not create the DB Maintenance process: {0}.

**Reason:**

System error.

**Action:**

Required action unknown.

## CAUAJM\_E\_20225

The DBMaintCmd parameter is not set to a valid path.

**Reason:**

The DBMaintCmd parameter value is empty.

**Action:**

Set the DBMaintCmd parameter to a valid path.



## CAUAJM\_E\_20227

Could not start the Telemetry process: {0}.

**Reason:**

System error.

**Action:**

Required action unknown.

## CAUAJM\_E\_20235

Wait on object failed <{0}>.

**Reason:**

Wait on single object call failed. This indicates the program that was started never completed.

**Action:**

Review the scheduler logfile for error messages or execute the program outside the scheduler. Review errors and take action as needed.

## CAUAJM\_E\_20291

**CANNOT RENAME Job : {0}, because it ALREADY EXISTS?**

Rename the specified job. If the specified job is referred in other jobs, the RenameJob verifies that the caller has update permissions for the specified job, and then renames it. Execution of this request can result in the following exceptions:

- AsGeneralErrorException: A general error has occurred.
- AsSecurityException: The set user does not have permission to modify this job.

**Reason:**

Job does not exist.

**Action:**

Object definition is not saved.

## CAUAJM\_E\_20322

WIN32 Function {0} failed for command: [{1}] with rc=({2}).

**Reason:**

WIN32 CreateProcess() for command failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_30002

Port number not in required interval [1024,...,65535]. Exiting.

**Reason:**

Port number not in required interval.

**Action:**

Select a new port number for the Windows Service or UNIX Daemon to run on.

**CAUAJM\_E\_30003**

**The argument list is incorrect. Exiting.**

**Reason:**

The argument list is incorrect.

**Action:**

Correct the argument list and restart.

**CAUAJM\_E\_30021**

**Requested API index [{0}] is out of bounds.**

**Reason:**

The received request contained an out of bounds index.

**Action:**

Verify that all client binaries are compatible with the Application Server version.

**CAUAJM\_E\_30034**

**Client [{0}][{1}][{2}][{3}] is not reachable. API ID [{3}] execution terminated. Total time: {4} seconds.**

**Reason:**

The application server was unable to send all responses back to the SDK client.

**Action:**

Review previous messages for an indication of errors. Also, verify that the SDK client remains active until all responses are received.

**CAUAJM\_E\_30035**

**Client [{0}][{1}][{2}] is not reachable. Session aborting...**

**Reason:**

The application server was unable to complete the session information exchange with the SDK client.

**Action:**

Review previous messages for an indication of errors. Also, verify that the SDK client remains active until all responses are received.

**CAUAJM\_E\_30037**

**Client [{0}][{1}] is not reachable. API ID [{2}] execution cancelled.**

**Reason:**

Hostname of the SDK client is invalid or unreachable over the network.

**Action:**

Verify that the hostname can be resolved into an IP address and is pingable.

**CAUAJM\_E\_30040**

Client [{0}] [{1}][{2}] is not reachable. Authentication request type [{3}] terminated. Total time: {4} seconds.

**Reason:**

The application server was unable to send all responses back to the SDK client.

**Action:**

Review previous messages for an indication of errors. Also, verify that the SDK client remains active until you receive all responses.

**CAUAJM\_E\_40001**

Could not display event <{0}>.

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

**CAUAJM\_E\_40003**

The specified parameters are not valid.

**Reason:**

Invalid parameters were passed to the program.

**Action:**

Correct the input parameters and resubmit the request.

**CAUAJM\_E\_40004**

Invalid num\_days specified.

**Reason:**

Invalid value specified.

**Action:**

Specify valid value.

**CAUAJM\_E\_40006**

Job Name required for REPLY\_RESPONSE.

**Reason:**

Invalid parameters were passed to program.

Specify Job name for REPLY\_RESPONSE.

## CAUAJM\_E\_40007

**Application and Group are invalid arguments for REPLY\_RESPONSE.**

## Reason:

Invalid parameters were passed to program.

**Action:**

Specify Job name for REPLY\_RESPONSE.

**CAUAJM\_E\_40009**

The {0} syntax is:\n\n\tsevent {1} {2} {3}.

## Reason:

Incorrect syntax while using `REPLY_RESPONSE`.

**Action:**

Check the syntax.

**CAUAJM\_E\_40010**

**Invalid override # specified.**

## Reason:

Invalid value specified.

**Action:**

Specify value of 0 or greater.

**CAUAJM\_E\_40011**

**The {0} step name argument is only valid when using the {1} event.**

## Reason:

The RESTARTJOB operation only supports the step name argument.

**Action:**

Specify RESTARTJOB as input for -E argument or remove the -o step name argument from the current command and resubmit the command.

**CAUAJM\_E\_40012**

**REPLY\_RESPONSE** requires a non-blank response.

## Reason:

A non-blank response is required.

**Action:**

---

Specify a response that contains non-blank characters.

## **CAUAJM\_E\_40016**

**Create Event: {0} Failed.**

**Reason:**

Creation of event failed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40017**

**Parameters {0}, {1}, {2}, and {3} are mutually exclusive. Only one of these parameters can be specified.**

**Reason:**

Invalid parameters were passed to the program.

**Action:**

Specify only one of the aforementioned parameters and resubmit the request.

## **CAUAJM\_E\_40018**

**An invalid date/time has been specified: {0}**

**Reason:**

An invalid date/time combination has been specified or the date/time is not valid on this operating system.

**Action:**

Please correct the date/time format and resubmit the request.

## **CAUAJM\_E\_40019**

**Application(-I) and Group(-B) are invalid arguments for {0}.**

**Reason:**

Invalid parameters were passed to program.

**Action:**

Specify Job name.

## **CAUAJM\_E\_40020**

**Job Name(-J) required for {0}.**

**Reason:**

Invalid parameters were passed to program.

**Action:**

Specify Job name.

## CAUAJM\_E\_40021

Unable to create file {0}

**Reason:**

The freopen function was able to create and open the specified file.

**Action:**

Specify a valid filename or contact CA Technical Support for further assistance.

## CAUAJM\_E\_40022

Ensure the date format is specified as: {0} {1}

**Reason:**

A date value with an invalid format was specified as input.

**Action:**

Specify a date value using a valid format.

## CAUAJM\_E\_40025

Report option missing. One of the following options must be specified: {0}, {1}, {2}, or {3}.

**Reason:**

Missing report option.

**Action:**

Specify a report option and resubmit the request.

## CAUAJM\_E\_40033

The RESTARTJOB event requires a non-blank step name based on this job's type.

**Reason:**

A non-blank step name is required as part of issuing the RESTARTJOB event for this job's type.

**Action:**

Specify a step name and resubmit the request.

## CAUAJM\_E\_40034

Failed to locate an external instance definition for node <{0}>. Trigger request from node {1} ignored.

**Reason:**

The external instance definition not found in database.

**Action:**

Insert an external instance definition matching machine name.

## CAUAJM\_E\_40035

The <{0}> From date/time must be less than the <{1}> To date/time or the current date/time if <{2}> was not specified.

**Reason:**

An invalid from date/time has been specified. The from date/time must be less than the to or current date/time.

**Action:**

Please correct the from date/time and resubmit the request.

## CAUAJM\_E\_40036

Could not get list of children for Box: {0}

**Reason:**

Could not get list of children for box.

**Action:**

Required action unknown.

## CAUAJM\_E\_40038

The <{0}> To date/time must be less than the current date/time.

**Reason:**

An invalid To date/time has been specified. The To date/time must be less than the current date/time.

**Action:**

Correct the To date/time and resubmit the request.

## CAUAJM\_E\_40040

Checkpoint/Restart event retrieval aborted <{0}>

**Reason:**

Checkpoint/Restart processing with remote manager has been aborted.

**Action:**

Contact Technical Support for further assistance.

## CAUAJM\_E\_40041

This job's box is in the {0} state. Job <{1}> cannot be taken off {2}.

**Reason:**

Cannot take a job off NOEXEC when its box is in a NOEXEC status.

**Action:**

Take the box off NOEXEC status.

## CAUAJM\_E\_40042

Database error getting dependencies for Job Name: {0}

**Reason:**

Database error getting dependencies for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_40044

Could not get row for joid [{0}].

**Reason:**

Could not get row for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_40045

Cannot take job <{0}> off {1} because it is not {2}.

**Reason:**

Cannot take job off the ON\_HOLD, ON\_ICE, or ON\_NOEXEC status. The job is currently not in the ON\_HOLD, ON\_ICE, or ON\_NOEXEC status.

**Action:**

No action is required.

## CAUAJM\_E\_40046

Could not take job off {0}. {1}

**Reason:**

Could not take job HOLD, ICE or NOEXEC status.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_E\_40047

Could NOT set Global: {0} To: {1}

**Reason:**

Could not set global.

**Action:**

Required action unknown.



## CAUAJM\_E\_40048

Could not place job in the {0} status. {1}

**Reason:**

Could not place job on HOLD, ICE or NOEXEC status.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_E\_40049

Failed starting job <{0}> while bringing machine <{1}> online.

**Reason:**

chkNstart failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_40050

Could not setup the {0} Table.

**Reason:**

Could not setup the ext\_job Table.

**Action:**

Required action unknown.

## CAUAJM\_E\_40051

Cannot put box job <{0}> in {1} status. Active job descendants found.

**Reason:**

Active child jobs prevent the scheduler from putting the box job in ON\_NOEXEC status.

**Action:**

Wait for all active descendant jobs to complete or manually change the status of those jobs to INACTIVE or to a valid terminal status (FAILURE, SUCCESS or TERMINATED).

## CAUAJM\_E\_40059

Invalid maximum retry attempts specified.

**Reason:**

Invalid value specified.

**Action:**

Specify a valid value and resubmit the request.

## CAUAJM\_E\_40064

**Trouble starting Job: {0}**

**Reason:**

Trouble starting job.

**Action:**

Required action unknown.

## CAUAJM\_E\_40068

**{0} FAILED for Job: {1}. Trying to send {2} to Servers**

**Reason:**

send\_event FAILED for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_40071

**Invalid record type passed from the Scheduler component: {1}**

**Reason:**

Invalid record type.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_E\_40072

**Consecutive single quotes encountered, error in the command field of job: {0}**

**Reason:**

Invalid parameters specified.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_40073

**Remove this parameter if it is meant to be empty.**

**Reason:**

See previous CAUAJM\_E\_40072 message.

**Action:**

No action required.

---

## CAUAJM\_E\_40074

**Consecutive double quotes encountered, error in command field of job: {0}**

**Reason:**

Invalid parameters specified.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_40076

**Invalid parameter encountered <{0}>\n**

**Reason:**

Invalid parameters specified.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_40078

**SendToConnect: Skipped dependency = {0} length of name invalid = {1}**

**Reason:**

Name was invalid.

**Action:**

Required action unknown.

## CAUAJM\_E\_40079

**SendToConnect: Connect type dependencies cannot exceed a maximum of 8 characters.**

**Reason:**

See previous CAUAJM\_E\_40078 message.

**Action:**

No action required.

## CAUAJM\_E\_40080

**Database error retrieving the runtime information for box {0}.**

**Reason:**

Database error retrieving the runtime information for box

**Action:**

Required action unknown.

## CAUAJM\_E\_40081

**Database error retrieving the runtime information for job {0}.**

**Reason:**

Database error retrieving the runtime information for job

**Action:**

Required action unknown.

## CAUAJM\_E\_40103

**Failed the evaluation of job <{0}> for load balancing queue name <{1}>.**

**Reason:**

The job start evaluation failed due to a database error.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_40111

**Unable to fetch job details using internal job identifier <{0}>. Event processing for {1} aborted.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Verify that the job exists in the database. If you cannot find the job in the database, contact technical support.

## CAUAJM\_E\_40112

**Timezone "{0}" cannot be found.**

**Reason:**

Timezone cannot be found.

**Action:**

Required action unknown.

## CAUAJM\_E\_40116

**Invalid record type from EP: {0}**

**Reason:**

Invalid record type.

**Action:**

Required action unknown.

## CAUAJM\_E\_40117

Invalid record type [{0}]

**Reason:**

Invalid record type.

**Action:**

Required action unknown.

## CAUAJM\_E\_40123

Failed to obtain jobs in box {0}.

**Reason:**

Failed to obtain jobs in box.

**Action:**

Check the status of the database.

## CAUAJM\_E\_40124

{0} {1} for Job: {2}. Trying to send {3} status to itself.

**Reason:**

send\_event FAILED for Job.

**Action:**

Required action unknown.

## CAUAJM\_E\_40125

Cannot change the status of job <{0}> to {1} because it is in the {2} status. Change the status to {3} or {4} or send the {5} event instead.

**Reason:**

If the status of a job is ON\_NOEXEC, you can only send the CHANGE\_STATUS event to place the job in the INACTIVE or SUCCESS status.

**Action:**

Send the CHANGE\_STATUS event to place the job in the INACTIVE or SUCCESS status or send the JOB\_OFF\_NOEXEC event.

## CAUAJM\_E\_40127

Cannot change the status of job <{0}> to {1} because it is in the {2} status. Please send the {3} event to take the job off the {4} status.

**Reason:**

Cannot change the status of the job because it is on ice or on noexec.

**Action:**

---

Send the JOB\_OFF\_ICE event to place the job off ice or the JOB\_OFF\_NOEXEC event to place the job off noexec.

## **CAUAJM\_E\_40140**

**ERROR: Failed to retrieve machines from PROVISION\_QUEUE**

**Reason:**

Failed to query on ujo\_machine table.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40141**

**ERROR: Failed to remove machine {0} from PROVISION\_QUEUE**

**Reason:**

Failed to update machine from ujo\_machine table.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40150**

**Error obtaining a run number for job <{0}>. Job will not be placed in <{1}> status.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. Place the job in GlobalPendMachStatus.

## **CAUAJM\_E\_40152**

**Unable to send the {0} event for job <{1}>. The sendevent command failed.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40153**

**Unable to check the run window ({0}) for job <{1}>. The sendevent command failed.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40155**

**Unable to remove job <{0}> from the resource wait queue.**

**Reason:**

A database error prevents the system from removing the specified job from the queue.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40157**

**System Restart - Job [{0}] was unable to start.**

**Reason:**

Job was unable to start.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40158**

**Could not update to status <{0}> for {1} <{2}>.**

**Reason:**

Could not update status.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40160**

**Cannot generate change status event to set job <{0}> to {1} status. The sendevent command failed.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40161**

**Failed to update status for job <{0}>. Failed sql <{1}>.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40165**

**Error setting the run number in table <{0}> for job <{1}>.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take the required action. If you cannot determine the cause of the problem, contact CA Technologies Technical Support.

## **CAUAJM\_E\_40171**

**Machines referenced by job <{0}> are not defined. Cannot start job.**

**Reason:**

Missing machines definitions prevent the specified job from starting.

**Action:**

Verify that definitions for machines referenced by the job are in the database.

## **CAUAJM\_E\_40172**

**Cannot find a machine definition to use in order to authenticate the user.**

**Reason:**

Cannot find a machine definition to use in order to authenticate the user.

**Action:**

Insert a machine definition in order to authenticate the user.

## **CAUAJM\_E\_40173**

**Job <{0}> not {1} when {2} event occurred. No {3} process to be killed.**

**Reason:**

Informational only.

**Action:**

No action required.

## **CAUAJM\_E\_40174**

**Error updating the ujo\_job\_status table. Query: {0}**

**Reason:**

Error updating the ujo\_job\_status table.

**Action:**



Required action unknown.

## **CAUAJM\_E\_40175**

**Unable to send the {0} alarm to indicate that resources could not be released for job <{1}>.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40177**

**Error obtaining a valid machine on which to start job <{0}>.**

**Reason:**

The job definition specifies an invalid or missing machine.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40178**

**Error obtaining an event identifier for {0} event.**

**Reason:**

Error obtaining an event identifier for event.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40184**

**Problem with status change for {0}[{1}] status[{2}] Something must be out of sequence.**

**Reason:**

Problem with status change for box\_joid.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40185**

**Unable to send the {0} alarm to indicate that job <{1}> failed its starting condition checks.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40186**

**Unable to send event. Failed to put job {0} in its GlobalPendMachStatus.**

**Reason:**

Failed to put the job in its GlobalPendMachStatus. The status could not be sent through sendevent.

**Action:**

Review any prior error messages. Otherwise, contact CA Technical Support for further assistance.

## **CAUAJM\_E\_40187**

**Failed to retrieve job from RESWAIT\_QUE**

**Reason:**

Failed to query on ujo\_reswait\_que table.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40189**

**Failed to put job <{0}> in pending status.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40190**

**Invalid GlobalPendMachStatus specified: {0}**

**Reason:**

Invalid global pending machine status has been specified.

**Action:**

Specify a valid status for the GlobalPendMachStatus configuration variable. Valid statuses are FAILURE, INACTIVE, ON\_ICE, PEND\_MACH, SUCCESS and TERMINATED.

## **CAUAJM\_E\_40194**

**Failed to update job <{0}^{1}> in the external instance job table. Failed sql <{2}>.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40196**

**Error retrieving event\_time\_gmt row for joid {0}.**

**Reason:**

Error retrieving event\_time\_gmt row for joid.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40197**

**Error evaluating next start time for job <{0}>. Cannot reschedule the job.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40198**

**{0} FAILS for joid [{1}].**

**Reason:**

cond\_depends FAILS for joid.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40199**

**{0} failed trying to {1} for job {2}**

**Reason:**

send\_event failed trying to DELETEJOB.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40200**

**{0} failed.**

**Reason:**

Failure encountered.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40201**

**Thread creation failed.**

**Reason:**

Thread creation failed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40202**

**{0} failed attempting to Terminate box joid = {1}**

**Reason:**

send\_event failed attempting to Terminate box.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40203**

**This box joid [{0}] does not exist!**

**Reason:**

This box does not exist.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40204**

**Unable to initialize the global process space for execution. Program terminating.**

**Reason:**

An unrecoverable program error has occurred.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40205**

**The {0} scheduler has failed to initialize startup information and must shut down. Exiting.**

**Reason:**

The scheduler experienced an unrecoverable error while starting up and must shut down.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40208**

**Unable to initialize the scheduler setting for "{0}" due to database errors.**

**Reason:**

An unrecoverable database error prevents the scheduler from updating the specified row of the ujo\_alamode table.

**Action:**

Review previous messages for an indication of what caused the problem and take required action. If you cannot determine the cause of the problem, contact technical support.

## **CAUAJM\_E\_40210**

**Unable to retrieve the "{0}" from the ujo\_alamode table due to database errors.**

**Reason:**

An unrecoverable database error prevents the scheduler from starting as the BINARY\_ID is missing from the ujo\_alamode table.

**Action:**

Check if the database has been upgraded to the current level. If the problem still persists, contact Technical Support.

## **CAUAJM\_E\_40212**

**Scheduler does not start as the AEDB version does not match with the product version.**

**Reason:**

BINARY\_ID of the product is not matching with the database contents in the ujo\_alamode table.

**Action:**

Check if the database has been upgraded to the current level. If the problem still persists, contact Technical Support.

## **CAUAJM\_E\_40213**

**Application server does not start as the AEDB version does not match with the product version.**

**Reason:**

BINARY\_ID of the product is not matching with the database contents in the ujo\_alamode table.

**Action:**

Check if the database has been upgraded to the current level. If the problem still persists, contact Technical Support.

## **CAUAJM\_E\_40214**

**The TZ environment variable is not defined.**

**Reason:**

The TZ environment variable is not set in the environment of the scheduler process.

**Action:**

---

Before starting the scheduler, set the TZ environment variable to the actual time zone of the scheduler.

## **CAUAJM\_E\_40216**

**Erm.jobst.joid == 0! exiting !**

**Reason:**

Unknown Error Type.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40221**

**The event: {0} Could NOT be marked as processed. Sending Alarm!**

**Reason:**

Event could NOT be marked as processed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40222**

**{0}: Could not send event [{1}]!**

**Reason:**

Could not send event.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40224**

**DBErr 20003! No ALARM sent!**

**Reason:**

DBErr 20003! No ALARM sent.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40225**

**Trouble processing Event [{0}]!**

**Reason:**

Trouble processing event.

**Action:**

Required action unknown.

## CAUAJM\_E\_40228

Execution of the following SQL failed! {0}

**Reason:**

SQL did not succeed.

**Action:**

Required action unknown.

## CAUAJM\_E\_40229

Error retrieving owner from the database JOB: {0}

**Reason:**

Job has no owner specification.

**Action:**

Required action unknown.

## CAUAJM\_E\_40230

Error retrieving job ID from the database JOB: {0}

**Reason:**

Error retrieving job ID from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_40231

Could not find job in the database JOB: {0}

**Reason:**

Could not find the job in the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_40232

Credentials error: Username, password and domain total length has exceeded the allowable maximum of 70 characters!

**Reason:**

Username, password and domain total length has exceeded the allowable maximum of 70 characters.

**Action:**

Specify valid credentials.

## CAUAJM\_E\_40233

**Please specify another user/password/domain combination that will not exceed this limit.**

**Reason:**

See previous CAUAJM\_E\_40232 message.

**Action:**

No action required.

## CAUAJM\_E\_40234

**Credentials error JOB: {0}/{1} has no owner specification!**

**Reason:**

Credentials error: Job has no owner specification.

**Action:**

Required action unknown.

## CAUAJM\_E\_40235

**Could not locate job in database using cross platform entryno {0}**

**Reason:**

Could not find the job in the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_40236

**Could not find job in the database using joid {0}.**

**Reason:**

Could not find the job in the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_40240

**Cross Platform Interface unable to obtain new eoid. Retry request in {0} seconds.**

**Reason:**

Unable to obtain new eoid from database.

**Action:**

If problem persists contact CA Technical Support for further assistance.



## CAUAJM\_E\_40241

**This instance is configured to use FIPS-approved security functions but it does not have either instance-wide or zOS data encryption enabled.**

**Reason:**

The value of the EnableFIPSMODE configuration parameter is 1, but the value of either the UseEncryption or the UseCommAliasEncryption configuration parameter is 0.

**Action:**

To operate CA Workload Automation AE in FIPS mode, verify that the value of the EnableFIPSMODE configuration parameter is set to 1, set both the UseEncryption and the UseCommAliasEncryption configuration parameter to a value greater than 0, and restart CA Workload Automation AE. If you do not want to operate in FIPS mode, set the value of the EnableFIPSMODE configuration parameter to 0.

## CAUAJM\_E\_40250

**Function {0} failed to create Scheduler binary watch thread. Error=({1})!**

**Reason:**

WIN32 CreateThread() failed to create scheduler binary watch thread.

**Action:**

Required action unknown.

## CAUAJM\_E\_40257

**External instance <{0}> is invalid or unreachable over the network.**

**Reason:**

An error occurred while attempting to communicate with the specified external instance.

**Action:**

Verify that the hostname associated with the external instance can be resolved into an IP address and is pingable

## CAUAJM\_E\_40267

**Cross Platform Interface aborting startup.**

**Reason:**

Indicates a fatal error has occurred and startup processing will be aborted.

**Action:**

See prior error messages for startup failure.

## CAUAJM\_E\_40269

**Cross Platform Interface sendevent failure [{0,number,0}..{1,number,0}..{2,number,0}..{3,number,0}]. Retry request in {4} seconds.**

**Reason:**

Sendevent has failed. Unable to place event into the database. See message CAUAJM\_E\_40238 or CAUAJM\_E\_40239 to identify event, job and status. Also review return codes [sendevent rc.job.event.status] for additional information.

**Action:**

Make sure the database is accessible. If accessible, contact CA Technical Support for further assistance.

## **CAUAJM\_E\_40270**

**Configuration file config.{0} is corrupt. Scheduler shutting down.**

**Reason:**

Configuration file is corrupt.

**Action:**

Check for a proper configuration file in the \$AUTOUSER directory.

## **CAUAJM\_E\_40271**

**Invalid message type [{0}] has been specified. Request ignored.**

**Reason:**

An invalid event type has been passed to the Cross Platform Interface. See message CAUAJM\_E\_40266 to identify failing job.

**Action:**

Contact CA Technical Support for further assistance.

## **CAUAJM\_E\_40274**

**A database error occurred while trying to bring machine <{0}> online.**

**Reason:**

A database error occurred while trying to bring machine <{0}> online.

**Action:**

Issue a MACH\_ONLINE event to manually bring the machine online."

## **CAUAJM\_E\_40279**

**Unable to process external dependency**

**Reason:**

SQL did not succeed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40302**

**Unable to create the aggregation object.**

**Reason:**

---

An internal error has occurred attempting to create the aggregation object needed to aggregate statistics.

**Action:**

Contact CA Technical Support for further assistance.

**CAUAJM\_E\_40303**

**Failed to complete aggregation of product statistics.**

**Reason:**

Unable to complete the aggregation of product statistics.

**Action:**

Check the aggregation log file for further details. Contact CA Technical Support for further assistance if necessary.

**CAUAJM\_E\_40313**

{0}

**Reason:**

An error has occurred during CCI initialization.

**Action:**

Review error messages and take action as required.

**CAUAJM\_E\_40316**

{0}

**Reason:**

An error has occurred during CCI initialization.

**Action:**

Review error messages and take action as required.

**CAUAJM\_E\_40317**

**Unable to communicate with any database. Shutting down.**

**Reason:**

All databases are either unavailable or have indicated they have had unrecoverable errors.

**Action:**

Investigate database problems and restart system.

**CAUAJM\_E\_40320**

**User authentication failure for {0}. User does not exist or bad password has been specified.**

**Reason:**

User authentication failure. User does not exist or bad password has been specified.

**Action:**

---

Correct username in remote job definition if it does not exist. Otherwise make sure correct password has been specified for the username.

## **CAUAJM\_E\_40321**

**Unable to process checkpoint restart event.**

**Reason:**

SQL did not succeed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40322**

**Unable to contact remote application ({0},{1}).**

**Reason:**

The identified remote application cannot be reached. If the remote application was previously active it is now marked inactive.

**Action:**

Make sure the remote application the Cross Platform Interface is attempting to communicate with is active.

## **CAUAJM\_E\_40324**

**User does not have sufficient privileges to {0} this job.**

**Reason:**

User does not have sufficient privileges to start this job.

**Action:**

Grant privileges or change the user in the remote job definition under which the job will be executed.

## **CAUAJM\_E\_40325**

**Unable to communicate with local agent to perform authentication. Request aborted.**

**Reason:**

Unable to communicate with local agent to perform authentication.

**Action:**

Make sure the local agent is started.

## **CAUAJM\_E\_40327**

**Unable to determine external instance type for Server=[{0}]**

**Reason:**

Unable to determine external instance type for identified server.

**Action:**

---

Make sure there is an external instance entry defined for the identified server.

### **CAUAJM\_E\_40360**

**No disk space left to write the {0} Scheduler log file.**

**Reason:**

No disk space left to write product logfile.

**Action:**

Delete old/unnecessary files.

### **CAUAJM\_E\_40361**

**ERROR: Unable to determine available disk space for the {0} Scheduler log file.**

**Reason:**

Error determining available space for disk partition containing the Scheduler log file.

**Action:**

Required action unknown.

### **CAUAJM\_E\_40372**

**FIPS mode is enabled, but external instance <{0}> does not use encryption. Taking offline.**

**Reason:**

The value of the EnableFIPSMODE configuration parameter has been set to 1, and non-FIPS-compliant communication has been blocked.

**Action:**

If you do not want to operate in FIPS mode, set the value of the EnableFIPSMODE configuration parameter to 0.

### **CAUAJM\_E\_40373**

**FIPS mode is enabled, but target machine [{0}] does not use encryption.**

**Reason:**

The value of the EnableFIPSMODE configuration parameter has been set to 1, and non-FIPS-compliant communication has been blocked.

**Action:**

If you do not want to operate in FIPS mode, set the value of the EnableFIPSMODE configuration parameter to 0.

### **CAUAJM\_E\_40374**

**The scheduler log is unavailable because the scheduler is not running.**

**Reason:**

Communication to the scheduler failed.

**Action:**

---

Start the scheduler and repeat the request.

## **CAUAJM\_E\_40379**

**An unrecoverable error has occurred or the maximum number of allowable errors has been exceeded. One Automation Event Management messaging has been disabled.**

**Reason:**

An unrecoverable error has occurred or the maximum number of allowable errors has been exceeded. One Automation Event Management messaging is disabled.

**Action:**

Review any previous error messages pertaining to One Automation Event Management. Take action as required and recycle the Scheduler to reinstate One Automation Event Management messaging.

## **CAUAJM\_E\_40380**

**Get userid failed; error code={0}.**

**Reason:**

Unable to get current userid. The message is not forwarded to One Automation Event Management.

**Action:**

Contact CA Technical Support.

## **CAUAJM\_E\_40381**

**Get machine name failed; error code={0}.**

**Reason:**

Unable to get current machine name. The message is not forwarded to One Automation Event Management.

**Action:**

Contact CA Technical Support.

## **CAUAJM\_E\_40382**

**Unable to initialize the curl library interface. One Automation notification skipped.**

**Reason:**

System error.

**Action:**

Required action unknown.

## **CAUAJM\_E\_40383**

**Unable to decrypt the authorization key. CA Automatic One Automation notification disabled.**

**Reason:**

Unable to decrypt the authorization key.

**Action:**

---

Update the authorization key with a proper encrypted value and restart the scheduler.

### **CAUAJM\_E\_40384**

**Authorization key is empty. CA Automic One Automation notification disabled.**

**Reason:**

Authorization key is empty.

**Action:**

Update the authorization key with a proper encrypted value and restart the scheduler.

### **CAUAJM\_E\_40385**

**Event Type is empty. CA Automic One Automation notification disabled.**

**Reason:**

Event Type is empty

**Action:**

Update the Event Type with a proper value and restart the scheduler.

### **CAUAJM\_E\_40386**

**Failed to publish One Automation event. Detail information <{1}>.**

**Reason:**

An error occurred during the One Automation event publishing.

**Action:**

Please review the error message for possible causes. Otherwise contact CA technical support.

### **CAUAJM\_E\_40387**

**Rest URL is empty. CA Automic One Automation notification disabled.**

**Reason:**

Rest URL is empty.

**Action:**

Update the Rest URL with a proper value and restart the scheduler.

### **CAUAJM\_E\_40388**

**An error has occurred calling One Automation Event Management - API Response Code rc={0}.**

**Reason:**

An error has occurred calling One Automation Event Management.

**Action:**

Review overall return code. Take action as needed. Otherwise contact CA Technical Support.

## CAUAJM\_E\_40389

**Telemetry data collection failed.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_40390

**Telemetry data transmission failed.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_40446

**A database error occurred while trying to place machine <{0}> in unqualified state.**

**Reason:**

A database error occurred while trying to place machine <{0}> in unqualified state.

**Action:**

No action required.

## CAUAJM\_E\_50001

**Error in command line for {0}.**

**Reason:**

Invalid arguments were passed to the program.

**Action:**

Specify the proper command arguments.

## CAUAJM\_E\_50002

**Cannot specify both {0} and {1} options.**

**Reason:**

Invalid arguments were passed to the program.

**Action:**

Specify the proper command arguments.



---

## CAUAJM\_E\_50003

**ERROR: Could not load intcodes table...**

**Reason:**

There was a failure trying to load the intcodes table.

**Action:**

Required action unknown.

## CAUAJM\_E\_50004

**Invalid job/box name: {0}**

**Reason:**

An invalid job/box name was specified.

**Action:**

Specify a valid job/box name.

## CAUAJM\_E\_50005

**Invalid Global Variable Name: {0}\n**

**Reason:**

An invalid global variable name was specified.

**Action:**

Specify a valid global variable name.

## CAUAJM\_E\_50006

**Internal API error.**

**Reason:**

There was a failure in the internal API.

**Action:**

Required action unknown.

## CAUAJM\_E\_50007

**Hostname [{0}] is invalid or unreachable over the network.**

**Reason:**

Hostname is invalid or unreachable over the network.

**Action:**

Verify that the hostname can be resolved into an IP address and is pingable.

## CAUAJM\_E\_50008

**The Scheduler is not available to test agent communication.**

**Reason:**

The Scheduler is not available to test agent communication.

**Action:**

Verify that the Scheduler is up and running.

## CAUAJM\_E\_50009

**The Scheduler is not available to resolve localhost.**

**Reason:**

The Scheduler is not available to resolve localhost.

**Action:**

Verify that the Scheduler is up and running.

## CAUAJM\_E\_50013

**Bad box name: {0}.**

**Reason:**

Bad box name.

**Action:**

Required action unknown.

## CAUAJM\_E\_50014

**Could not get jobs contained by box {0}.**

**Reason:**

Could not get jobs contained by box.

**Action:**

Required action unknown.

## CAUAJM\_E\_50016

**{0}: bad {1}.**

**Reason:**

Bad job filter.

**Action:**

Required action unknown.

## CAUAJM\_E\_50017

**Interactive mode does not support the {0} option.\n\n**

**Reason:**

Interactive mode does not support the {0} option.

**Action:**

Remove the -h option or specify a valid run command.

## CAUAJM\_E\_50020

**Failed to start Beeper timer. {0}**

**Reason:**

Failed to start Beeper timer.

**Action:**

Required action unknown.

## CAUAJM\_E\_50021

**Could not get list of machines from the database.**

**Reason:**

Could not get list of machines from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_50022

**No machines in the database to check.**

**Reason:**

No machines in the database to check.

**Action:**

Required action unknown.

## CAUAJM\_E\_50024

**The current operating system is not supported. Program execution is aborted.\n\n**

**Reason:**

The current operating system is not supported. Program execution is aborted.

**Action:**

Refer to the product documentation for supported operating systems that this program will execute on.

---

## CAUAJM\_E\_50026

**ERROR: AutoPing WAS NOT SUCCESSFUL.**

**Reason:**

ERROR: AutoPing WAS NOT SUCCESSFUL.

**Action:**

Required action unknown.

## CAUAJM\_E\_50027

**Invalid Job Name: {0}\n**

**Reason:**

An invalid job name was specified.

**Action:**

Specify a valid job name.

## CAUAJM\_E\_50029

**Internal error 5001 in "{0}".**

**Reason:**

Internal error 5001.

**Action:**

Required action unknown.

## CAUAJM\_E\_50032

**Environment/Configuration setup failed!\n**

**Reason:**

Initialization of the environment failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_50033

**Error initializing tx subsystem:**

**Reason:**

Error encountered during tx subsystem initialization.

**Action:**

Required action unknown.

## CAUAJM\_E\_50034

Unable to get event server information\n

**Reason:**

Unable to retrieve event server information.

**Action:**

Required action unknown.

## CAUAJM\_E\_50035

Could not connect with Server: {0}\n\n

**Reason:**

Connection to the server could not be established.

**Action:**

Required action unknown.

## CAUAJM\_E\_50036

**ERROR: Agent failed to access the Event Server data.**

**Reason:**

ERROR: Agent failed to access the Event Server data.

**Action:**

Check AAgent's connectivity to the Application Server or legacy Agent's access to the database.

## CAUAJM\_E\_50037

**ERROR: Agent failed to communicate with Application Server.**

**Reason:**

ERROR: Agent failed to communicate with Application Server.

**Action:**

Check AAgent's connectivity to the Application Server.

## CAUAJM\_E\_50041

**ERROR: Agent returned <{0}>.**

**Reason:**

ERROR: Agent returned the displayed message.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_50056

Unable to Connect with Database: {0} after {1} attempts.\n\n

**Reason:**

Unable to connect to the event server.

**Action:**

Required action unknown.

## CAUAJM\_E\_50059

Database error trying to add timezone {0}

**Reason:**

Database error trying to add timezone.

**Action:**

Required action unknown.

## CAUAJM\_E\_50061

{0}() failed.

**Reason:**

Failure encountered.

**Action:**

Required action unknown.

## CAUAJM\_E\_50062

Internal error 5011/{0} in "{1}".

**Reason:**

Internal Error 5011.

**Action:**

Required action unknown.

## CAUAJM\_E\_50063

Trouble getting machines from Database...\n\n

**Reason:**

A failure occurred while getting machine names from the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_50066

Could not clean files on machine: {0}!\n

**Reason:**

A failure occurred while cleaning files.

**Action:**

Required action unknown.

## CAUAJM\_E\_50070

Database error trying to delete timezone {0}

**Reason:**

Database error trying to delete timezone.

**Action:**

Required action unknown.

## CAUAJM\_E\_50073

JobName is required.\n\n

**Reason:**

Job name was not specified.

**Action:**

Specify the job name.

## CAUAJM\_E\_50074

Must specify Mode:\nCAUAJM\_E\_50074\|tc Current Condition Status\nCAUAJM\_E\_50074\|td Dependencies Only (from Job Definition)\nCAUAJM\_E\_50074\|tt Time Dependencies (Forecasts)\n\n

**Reason:**

Mode was not specified.

**Action:**

Specify the mode.

## CAUAJM\_E\_50075

FROM Date/Time required for "t" (time dependency) Option.\n\n

**Reason:**

From date/time was not specified.

**Action:**

Specify FROM date/time.

## CAUAJM\_E\_50077

Timezone {0} not found in the database, delete failed.

**Reason:**

Timezone not found in the database.

**Action:**

Required action unknown.

## CAUAJM\_E\_50078

Invalid {0} date specified. ({1} HH:MM)\n\n

**Reason:**

An invalid date was specified.

**Action:**

Specify a valid date.

## CAUAJM\_E\_50080

The TO time is BEFORE the FROM time!\n\n

**Reason:**

The TO time is before the FROM time.

**Action:**

Specify a proper time range.

## CAUAJM\_E\_50081

Job {0} not found in database.

**Reason:**

The specified job was not found in the database.

**Action:**

No action required.

## CAUAJM\_E\_50082

Internal error 5012/{0} in "{1}".

**Reason:**

Internal error 5012.

**Action:**

Required action unknown.



## CAUAJM\_E\_50084

**\*\*\* Error in command line for {0}.\*\*\*\n**

**Reason:**

There was an error in the command line for job\_depends.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_50088

**Invalid {0} Name: {1}\n**

**Reason:**

Invalid monbro name specified.

**Action:**

Specify a valid monbro name.

## CAUAJM\_E\_50090

**Database problems with {0}: {1}\n**

**Reason:**

There was a database failure in monbro.

**Action:**

Required action unknown.

## CAUAJM\_E\_50095

**Internal error 5013/{0} in "{1}".**

**Reason:**

Internal error 5013.

**Action:**

Required action unknown.

## CAUAJM\_E\_50098

**The input date format does not match \"{0}\"**

**Reason:**

An invalid format was used for the date format.

**Action:**

Required action unknown.

## CAUAJM\_E\_50100

**This date cannot be represented in a tm struct!**

**Reason:**

The data cannot be displayed in a tm struct.

**Action:**

Required action unknown.

## CAUAJM\_E\_50101

**An incorrect argument was specified for the {0} (add alias) option.**

**Reason:**

An incorrect argument was specified for the option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50102

**The glob <{1}> input is not in a valid email template format.**

**Reason:**

The glob input is not as per the email template format.

**Action:**

Redefine the glob input as per the valid email template format.

## CAUAJM\_E\_50103

**Machine name specification cannot be masked.\n**

**Reason:**

Machine name specification cannot be masked.

**Action:**

Specify the actual machine name and resubmit the request.

## CAUAJM\_E\_50104

**Bad report mode detected.\n**

**Reason:**

A bad report mode has been specified.

**Action:**

Specify a valid report mode and resubmit the request.

## CAUAJM\_E\_50105

**Job name specification cannot be masked.\n**

**Reason:**

Job name specification cannot be masked.

**Action:**

Specify the actual Job name and resubmit the request.

## CAUAJM\_E\_50106

**An incorrect argument was specified for the {0} (add city) option.**

**Reason:**

An incorrect argument was specified for the option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50109

**An incorrect argument was specified for the {0} (delete entry) option.**

**Reason:**

An incorrect argument was specified for the option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50111

**Invalid Machine Name: {0}\n**

**Reason:**

Machine is not defined.

**Action:**

Required action unknown.

## CAUAJM\_E\_50113

**Invalid Resource Name: {0}\n**

**Reason:**

Resource is not defined.

**Action:**

Required action unknown.

## CAUAJM\_E\_50114

This option is applicable only for real machines.\n

**Reason:**

This option is applicable only for real machines.

**Action:**

Provide a real machine.

## CAUAJM\_E\_50120

An incorrect argument was specified for the {0} (add time zone) option.

**Reason:**

An incorrect argument was specified for the option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50130

Argument missing for option -{0}

**Reason:**

Argument missing for option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50132

The log file \"{0}\" cannot be opened. (Error {1}: {2})

**Reason:**

Logfile is missing.

**Action:**

Required action unknown.

## CAUAJM\_E\_50135

Only Administrators or Power Users can execute autosyslog.

**Reason:**

The user account control settings for the administrator user group are not configured correctly, or you are neither an administrator nor a power user.

**Action:**

Verify that the user account control settings for the administrator user group are set to zero, ensure that you are logged in as an administrator or power user, and attempt to execute the command again.

## CAUAJM\_E\_50137

**The run number specified for job {0} is not valid.**

**Reason:**

The run number specified for the job does not exist.

**Action:**

Correct the run number and resubmit the request.

## CAUAJM\_E\_50138

**No agent files found for job {0}.**

**Reason:**

The job either does not exist or there are no agent job log files present for the job.

**Action:**

No action is required if the job or job log files do not exist. Otherwise, correct the input parameters and resubmit the request.

## CAUAJM\_E\_50139

**Internal error 5002 in "{0}".**

**Reason:**

Internal error 5002.

**Action:**

Required action unknown.

## CAUAJM\_E\_50141

**Instance {0} is not present on this system\n**

**Reason:**

Instance was not present.

**Action:**

Required action unknown.

## CAUAJM\_E\_50142

**Command \"{0}\" was not found\n**

**Reason:**

Command was not found.

**Action:**

Required action unknown.

## CAUAJM\_E\_50143

The {0} for Command \"{1}\" was not found\n

**Reason:**

PATH for the command was not found.

**Action:**

Required action unknown.

## CAUAJM\_E\_50144

Access Denied for Command \"{0}\" \n

**Reason:**

Access denied for the command.

**Action:**

Required action unknown.

## CAUAJM\_E\_50145

Unable to start command \"{0}\" Error = {1}\n

**Reason:**

Unable to start command.

**Action:**

Required action unknown.

## CAUAJM\_E\_50146

An invalid value was specified for a {0} or {1} request.

**Reason:**

An invalid value was specified for the request.

**Action:**

Required action unknown.

## CAUAJM\_E\_50147

Please specify one of the follow options: {0}, {1}, {2}, {3}, {4}, or {5}

**Reason:**

Please specify a proper option.

**Action:**

Required action unknown.

## CAUAJM\_E\_50148

Invalid {0} time argument.

**Reason:**

Invalid time argument

**Action:**

Required action unknown.

## CAUAJM\_E\_50149

Invalid {0} time argument.

**Reason:**

Invalid time argument.

**Action:**

Required action unknown.

## CAUAJM\_E\_50153

**ERROR: Receiving information from the agent.**

**Reason:**

An error was encountered while receiving information from the agent.

**Action:**

Required action unknown.

## CAUAJM\_E\_50165

Unable to inspect any jobs on machine: {0}

**Reason:**

Unable to inspect any jobs on machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50168

**ERROR: Job in {0} state, but run\_machine is empty. Job may have been manually changed to {1} state.\n**

**Reason:**

Chase Error: Job in RUNNING state, but run\_machine is empty. Job may have been manually changed to RUNNING state.

**Action:**

Required action unknown.

## CAUAJM\_E\_50169

The index pointed to a zero "joid": chase[{0}].joid = 0. chase\_string={1}

**Reason:**

An error was encountered while receiving information from the agent.

**Action:**

Required action unknown.

## CAUAJM\_E\_50170

The index for {0} pointed to a zero "joid": chase[{1}].joid = 0. chase\_string={2}

**Reason:**

An error was encountered while receiving information from the agent.

**Action:**

Required action unknown.

## CAUAJM\_E\_50178

Could not get row for joid = {0}

**Reason:**

Could not get row for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_50180

Chase failed to send the {0} {1} event.\n

**Reason:**

Chase failed to send the CHANGE\_STATUS FAILURE event.

**Action:**

Required action unknown.

## CAUAJM\_E\_50182

Invalid Global Variable Name: {0}\n

**Reason:**

The global variable is not defined.

**Action:**

Required action unknown.



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## CAUAJM\_E\_50188

Could not get row for machine {0}

**Reason:**

Could not get row for machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50193

Could not get row for {0} {1}

**Reason:**

Could not get row for monbro.

**Action:**

Required action unknown.

## CAUAJM\_E\_50195

Update FAILED !!

**Reason:**

Update FAILED.

**Action:**

Required action unknown.

## CAUAJM\_E\_50198

Exit Code = {0}

**Reason:**

Error occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_50199

Invalid Job Type: {0} <{1}>

**Reason:**

The job type is not valid.

**Action:**

Required action unknown.

## CAUAJM\_E\_50201

**Delete was not performed!**

**Reason:**

Delete was not performed.

**Action:**

Required action unknown.

## CAUAJM\_E\_50206

**Could not turn off Over-Ride for Job: {0}**

**Reason:**

Could not turn off Over-Ride for job.

**Action:**

Required action unknown.

## CAUAJM\_E\_50216

**Invalid attribute name for <{0}>: {1}**

**Reason:**

Invalid attribute name

**Action:**

Required action unknown

## CAUAJM\_E\_50217

**Cannot use Field "{0}" for an Over-ride value.**

**Reason:**

Cannot use field for an Over-ride value.

**Action:**

Required action unknown.

## CAUAJM\_E\_50220

**Deletion of {0}: {1} was NOT successful!**

**Reason:**

Deletion of monbro was NOT successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_50221

Invalid {0} Name: {1} Update NOT performed.

**Reason:**

Invalid monbro name.

**Action:**

Required action unknown.

## CAUAJM\_E\_50223

There are job(s) defined with type "{0}", job type "{0}" cannot be deleted

**Reason:**

Cannot delete job type

**Action:**

Required action unknown.

## CAUAJM\_E\_50225

Deletion of {0}: {1} was NOT successful!

**Reason:**

Deletion of External Instance was NOT successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_50226

Invalid blank machine name. Delete NOT performed.

**Reason:**

Invalid blank machine name.

**Action:**

Required action unknown.

## CAUAJM\_E\_50228

Deletion of machine: {0} was NOT successful!

**Reason:**

Deletion of machine was NOT successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_50231

**Invalid blank machine name. Insert/Update NOT performed.**

**Reason:**

Invalid blank machine name. Insert/Update NOT performed.

**Action:**

Required action unknown.

## CAUAJM\_E\_50233

**Database change for machine: {0} WAS NOT successful.**

**Reason:**

Database change for machine WAS NOT successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_50234

**Cannot specify "{0}" for a {1} or {2} machine.**

**Reason:**

Cannot specify identified keyword for a Unicenter or Unicenter AutoSys Connect machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50235

**Incorrect {0} agent machine name: {1}**

**Reason:**

Incorrect TNG agent machine name.

**Action:**

Required action unknown.

## CAUAJM\_E\_50236

**Illegal machine type: {0}**

**Reason:**

Illegal machine type.

**Action:**

Required action unknown.

## CAUAJM\_E\_50237

Cannot specify a "{0}" for a real or legacy (non-virtual) machine.

**Reason:**

Cannot specify a 'machine' for a real or legacy (non-virtual) machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50238

Deletion of {0}: {1} was NOT successful!

**Reason:**

Deletion of Blob/Glob was NOT successful.

**Action:**

Required action unknown.

## CAUAJM\_E\_50242

Failed to establish communication with the {0} Application Server ({1}).

**Reason:**

Connection timed out.

**Action:**

Make sure the Application Server is running and its node is available.

## CAUAJM\_E\_50246

JIL syntax check FAILED

**Reason:**

Syntax error detected in JIL

**Action:**

Examine the syntax errors reported and re-enter the JIL attributes

## CAUAJM\_E\_50254

Unexpected line for <{0}>: {1}

**Reason:**

Unexpected line.

**Action:**

Enter a valid attribute name.

## CAUAJM\_E\_50256

**Invalid attribute value for <{0}>: Missing attribute name or unquoted \"\:"**

**Reason:**

Invalid attribute value. Missing attribute name

**Action:**

Enter a valid attribute name.

## CAUAJM\_E\_50265

**Error deleting {0}:**

**Reason:**

An error occurred while deleting the blob.

**Action:**

Required action unknown.

## CAUAJM\_E\_50266

**Cannot define both {0} and {1} attributes for the job input blob.**

**Reason:**

Cannot define both blob\_input and blob\_file attributes for the job input blob.

**Action:**

Required action unknown.

## CAUAJM\_E\_50267

**Cannot specify {0} or {1} attributes without defining {2} as a job input blob.**

**Reason:**

Cannot specify blob\_input or blob\_file attributes without defining std\_in\_file as a job input blob.

**Action:**

Required action unknown.

## CAUAJM\_E\_50271

**Error in JIL. Name \"{0}\" is too long.**

**Reason:**

Syntax error detected in JIL

**Action:**

Enter a shorter name.

## CAUAJM\_E\_50272

**Valid job types are: "0" through "9"**

**Reason:**

Valid job types are 0 through 9

**Action:**

Required action unknown.

## CAUAJM\_E\_50278

**Invalid blob type: {0}\n**

**Reason:**

The specified blob type cannot be retrieved.

**Action:**

Specify a valid blob type.

## CAUAJM\_E\_50281

**AutoPing from the Scheduler WAS NOT SUCCESSFUL.**

**Reason:**

The scheduler fails to contact the agent when you issue the autoping command.

**Action:**

Verify that the agent is running and that there is network connectivity between the scheduler and the agent, and then reissue the command.

## CAUAJM\_E\_50283

**AutoPing from the Application Server WAS NOT SUCCESSFUL.**

**Reason:**

The Application Server fails to contact the agent when you issue the autoping command.

**Action:**

Verify that the agent is running and that there is network connectivity between the Application Server and the agent, and then reissue the command.

## CAUAJM\_E\_50290

**Manual intervention is required.**

**Reason:**

Manual intervention is required to recover from the previous error.

**Action:**

See the message displayed before this one for details.

## CAUAJM\_E\_50292

The agent is not running, but the job is running on machine = {0}

**Reason:**

The agent is not running, but the job is running on the machine.

**Action:**

Manual intervention is required.

## CAUAJM\_E\_50293

File watcher is in {0} state, but the agent is not available on machine: {1}

**Reason:**

File watcher is in RUNNING state, but the agent is not available on the machine.

**Action:**

Manual intervention is required.

## CAUAJM\_E\_50294

Job is in {0} state, but both the agent and the user process are not running on machine: {1}

**Reason:**

File watcher job is in RUNNING state, but both the agent and the user process are not running on the machine.

**Action:**

Manual intervention is required.

## CAUAJM\_E\_50295

Could not connect to machine: {0}. The machine or the network must be down.

**Reason:**

Could not connect to the machine. The machine or the network must be down.

**Action:**

Manual intervention is required.

## CAUAJM\_E\_50296

Error in finding the actual state of the job: {0} on machine = {1}

**Reason:**

Could not get information about the actual state of the specified job on the specified machine.

**Action:**

Inspect the agent log files to determine the state of the job.



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## CAUAJM\_E\_50300

API Call GET\_CALENDAR\_DAYS failed

**Reason:**

API Call GET\_CALENDAR\_DAYS failed

**Action:**

n.

## CAUAJM\_E\_50301

Call to GET\_EXT\_CALENDAR\_DAYS failed.\n

**Reason:**

The request failed.

**Action:**

No action required.

## CAUAJM\_E\_50302

Could not retrieve extended calendar definition.\n

**Reason:**

The API could not complete successfully.

**Action:**

Check the status of the application server, check that the calendar exists and that the user has permissions to access it.

## CAUAJM\_E\_50303

Call to GET\_GENERATED\_EXT\_CALENDAR\_DAYS failed.

**Reason:**

Call to API failed.

**Action:**

No action required.

## CAUAJM\_E\_50304

Delete failed.

**Reason:**

The request failed.

**Action:**

No action required.

---

## CAUAJM\_E\_50305

Invalid date {0} entered.

**Reason:**

The input date did not match the correct format.

**Action:**

Check the value of the input date.

## CAUAJM\_E\_50306

Call to GET\_CALENDAR\_NAMES failed.\n.

**Reason:**

Call to API failed.

**Action:**

No action required.

## CAUAJM\_E\_50307

Unknown type of calendar being imported

**Reason:**

Not a recognized calendar type

**Action:**

Verify import file

## CAUAJM\_E\_50308

Standard calendar {0} does not exist.

**Reason:**

The standard calendar does not exist.

**Action:**

Specify an existing standard calendar.

## CAUAJM\_E\_50309

Extended calendar {0} does not exist.

**Reason:**

The extended calendar does not exist.

**Action:**

Specify an existing extended Calendar.

## CAUAJM\_E\_50310

**Cycle {0} does not exist.**

**Reason:**

The cycle does not exist.

**Action:**

Specify an existing cycle.

## CAUAJM\_E\_50311

**Invalid Calendar Name: {0}**

**Reason:**

Invalid calendar name specified.

**Action:**

Enter a valid calendar name.

## CAUAJM\_E\_50312

**Call to GET\_CYCLE\_NAMES failed.**

**Reason:**

Call to API failed.

**Action:**

No action required.

## CAUAJM\_E\_50313

**Invalid Cycle Period. Periods must be between 1 and 366 days in length.**

**Reason:**

Please enter a period that is greater than 1 day and less than 366.

**Action:**

Refer to the UAJM Reference Guide for further information on cycles.

## CAUAJM\_E\_50314

**Invalid Calendar Name: {0} has exceeded the maximum number ({1}) of characters.**

**Reason:**

The calendar name has exceeded the maximum number of characters allowed.

**Action:**

Refer to UAJM Documentation for more information on maximum length of calendar names.

## CAUAJM\_E\_50315

**Invalid Calendar Name: {0} contains illegal characters**

**Reason:**

The calendar name contains illegal characters

**Action:**

Refer to the UAJM Documentation for more information on legal characters for calendar names.

## CAUAJM\_E\_50316

**Invalid Cycle Period. Valid dates are 1980 or later or wild card year.**

**Reason:**

Please enter a period with a valid year.

**Action:**

Refer to the UAJM Reference Guide for further information on cycle periods.

## CAUAJM\_E\_50318

**{0} Condition field has a maximum length of 255 characters.**

**Reason:**

Check the length of the condition field.

**Action:**

Please refer to the UAJM documentation for more information on [extended calendar field definitions](#).

## CAUAJM\_E\_50321

**Database error while fetching row for job {0}.**

**Reason:**

Database error while fetching row for job {0}.

**Action:**

Required action unknown.

## CAUAJM\_E\_50324

**Either the scheduler is down or it is located on a remote machine.**

**Reason:**

Either the scheduler is down or it is located on a remote machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50325

Add -l and/or -p to access the Scheduler log on a remote machine.

**Reason:**

Add -l and/or -p to access the Scheduler log on a remote machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50327

Cannot delete machine \"{0}\" as it is still being referenced by machine containers. To delete references in machine containers, please specify \"remove\_references: y\".

**Reason:**

Could not delete machine as it is referenced in machine containers.

**Action:**

To delete references in machine containers, please specify: remove\_references: y.

## CAUAJM\_E\_50328

This machine \"{0}\" is the last component machine in this virtual machine \"{1}\". Please delete the virtual machine instead.

**Reason:**

Last machine in this virtual machine.

**Action:**

Please delete the virtual machine instead.

## CAUAJM\_E\_50329

Cannot delete machine \"{0}\" as it is still being referenced by active jobs.

**Reason:**

Could not delete machine as it is referenced in active jobs.

**Action:**

To delete references in active jobs, please change the machine that those jobs at or delete those active jobs.

## CAUAJM\_E\_50330

Cannot delete machine \"{0}\" as it is still being referenced by resources.

**Reason:**

Could not delete machine as it is referenced in resources.

**Action:**

To delete references in resources, please change the machine that those resources use or delete those resources.

## CAUAJM\_E\_50334

**Either the application server is down or it is located on a remote machine.**

**Reason:**

Either the application server is down or it is located on a remote machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50336

**Failed to retrieve dependencies on standard calendar {0}.**

**Reason:**

QUERY failed. Could not get existence of dependencies on standard calendar

**Action:**

Required action unknown.

## CAUAJM\_E\_50338

**Failed to retrieve dependencies on cycle {0}.**

**Reason:**

QUERY failed. Could not get existence of dependencies on cycle

**Action:**

Required action unknown.

## CAUAJM\_E\_50405

**Delete failed. The following dates are invalid:**

**Reason:**

Specified dates did not exist for the specified calendar.

**Action:**

Specify valid dates.

## CAUAJM\_E\_50406

**The machine: {0} does not exist.**

**Reason:**

The machine name does not exist in the database.

**Action:**

Specify a valid machine.

## CAUAJM\_E\_50411

**Invalid External Instance Name: {0}\n**

**Reason:**

The external instance name is not defined.

**Action:**

Required action unknown.

## CAUAJM\_E\_50415

**The Holiday calendar: {0} is already defined as Extended Calendar.**

**Reason:**

Holiday Calendar cannot be an Extended Calendar.

**Action:**

Re-enter the Holiday Calendar.

## CAUAJM\_E\_50416

**The Extended Calendar: {0} is already defined as a Holiday Calendar for another Extended Calendar.**

**Reason:**

Extended Calendar cannot be a Holiday Calendar.

**Action:**

Re-enter the Extended Calendar.

## CAUAJM\_E\_50417

**The Holiday calendar name should be different from {0}.**

**Reason:**

Holiday Calendar cannot be the same as the calendar name.

**Action:**

Re-enter the Holiday Calendar.

## CAUAJM\_E\_50418

**Please enter valid Non-Workday Action.**

**Reason:**

Valid Action required for Non-Workday.

**Action:**

Re-enter the Non-Working Action.

## CAUAJM\_E\_50419

**Please enter valid Holiday Action.**

**Reason:**

Valid Action required for Holiday.

**Action:**

Re-enter the Holiday Action.

## CAUAJM\_E\_50420

**Please enter valid Workdays.**

**Reason:**

Valid Workdays are required.

**Action:**

Re-enter the Workdays.

## CAUAJM\_E\_50421

**Make three more attempts to inspect machine: {0}, before giving up.**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_50422

**Three attempts failed to inspect jobs on machine : {0}.**

**Reason:**

Unable to inspect any jobs on machine.

**Action:**

Required action unknown.

## CAUAJM\_E\_50442

**Invalid attribute value for <{0}>: Unexpected end of character string: {1}**

**Reason:**

Invalid attribute value. Unexpected end of character string.

**Action:**

Add missing quote to attribute value.



## CAUAJM\_E\_50443

**Attributes cannot be accepted without a valid JIL subcommand.**

**Reason:**

Attributes cannot be accepted without a valid JIL subcommand.

**Action:**

Required action unknown.

## CAUAJM\_E\_50444

**Database error occurred while retrieving the database version.**

**Reason:**

Database error occurred while retrieving the database version.

**Action:**

Verify the state of the database.

## CAUAJM\_E\_50445

**Invalid attribute value for <{0}>: Missing attribute name or unquoted \"\:\" after \"{1}\"**

**Reason:**

Invalid attribute value. Missing attribute name

**Action:**

Enter a valid attribute name.

## CAUAJM\_E\_50500

**There was a failure in attempting to process the request.**

**Reason:**

A heap allocation error has occurred.

**Action:**

Increase memory on the machine.

## CAUAJM\_E\_50501

**The "jobname" argument was not specified.**

**Reason:**

The 'jobname' argument was not specified.

**Action:**

Specify the jobname argument.

## CAUAJM\_E\_50502

**"{0}" is not an adapter job or it's status is currently unavailable.**

**Reason:**

Job is not an adapter job or it's status is currently unavailable.

**Action:**

Specify a valid job or try again when status is available.

## CAUAJM\_E\_52302

**The calendar description has exceeded the maximum number ({0}) of characters.**

**Reason:**

The calendar description has exceeded the maximum number of characters allowed.

**Action:**

Refer to the product documentation for information on the maximum length of the calendar description.

## CAUAJM\_E\_60003

**Invalid Date/Time: {0}**

**Reason:**

Invalid Date/Time.

**Action:**

Required action unknown.

## CAUAJM\_E\_60004

**Date must be specified as: {0} {1}**

**Reason:**

Date specified incorrectly.

**Action:**

Required action unknown.

## CAUAJM\_E\_60005

**or date is invalid on this Operating System**

**Reason:**

Date is invalid on this Operating System.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_60008

**Could not open the archive file: {0}**

**Reason:**

Problem opening the archive file.

**Action:**

Required action unknown.

## CAUAJM\_E\_60009

**Shutting Down. {0}**

**Reason:**

Program is exiting because of failure to open archive file.

**Action:**

Required action unknown.

## CAUAJM\_E\_60010

**{0}() failed. Exiting...**

**Reason:**

Failure encountered.

**Action:**

No action required, program exits.

## CAUAJM\_E\_60011

**Thread creation failed. Exiting...**

**Reason:**

There was a failure while creating a new thread.

**Action:**

No action required, program exits.

## CAUAJM\_E\_60012

**Keyword {0} can not be a Global Name.**

**Reason:**

Invalid global name.

**Action:**

Required action unknown.

## CAUAJM\_E\_60015

**Failed to retrieve space usage of {0}. Verify the ujoadmin account has select privileges on ALL\_TABLES.**

**Reason:**

Missing select privilege on ALL\_TABLES.

**Action:**

Grant select privilege to ujoadmin account on ALL\_TABLES.

## CAUAJM\_E\_60016

**Failed to retrieve database information.**

**Reason:**

Database information is wrong.

**Action:**

Specify correct information for database.

## CAUAJM\_E\_60023

**An archive directory was specified, BUT the [{0}] flag to Save the events to a file was NOT. Nothing will be saved.**

**Reason:**

The -d option was specified without a corresponding -A option.

**Action:**

Specify the proper command arguments.

## CAUAJM\_E\_60024

**Job Name must not be set for {0}.**

**Reason:**

Job Name must not be set for the specified event.

**Action:**

Required action unknown.

## CAUAJM\_E\_60025

**Global Name Missing.**

**Reason:**

Global Name Missing.

**Action:**

Required action unknown.

## CAUAJM\_E\_60026

Global Name can only be {0} characters.

**Reason:**

Invalid global name.

**Action:**

Required action unknown.

## CAUAJM\_E\_60027

Value can only be {0} characters.

**Reason:**

Invalid value name.

**Action:**

Required action unknown.

## CAUAJM\_E\_60028

Global Name contains one or more invalid characters.

**Reason:**

Global name contains one or more invalid characters.

**Action:**

Required action unknown.

## CAUAJM\_E\_60029

You must specify a node for this event.

**Reason:**

You must specify a node for this event.

**Action:**

Required action unknown.

## CAUAJM\_E\_60030

The [{0} Signals(s)] Option is required for {1}.\nCAUAJM\_E\_60030 This is an ordered comma list of the signals to be sent to the Job.

**Reason:**

The [-K Signals(s)] Option is required for SEND\_SIGNAL

**Action:**

Required action unknown.

---

## CAUAJM\_E\_60033

{0} API failed.

**Reason:**

API failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_60034

dbspace: Unable to send alarm.

**Reason:**

Unknown.

**Action:**

Required action unknown.

## CAUAJM\_E\_60035

INVALID Event: {0}

**Reason:**

INVALID Event.

**Action:**

Required action unknown.

## CAUAJM\_E\_60036

The Value of the ALARM is missing.

**Reason:**

The Value of the ALARM is missing.

**Action:**

Required action unknown.

## CAUAJM\_E\_60037

INVALID Alarm: {0}

**Reason:**

INVALID Alarm.

**Action:**

Required action unknown.

---

## CAUAJM\_E\_60038

The Value of the {0} is missing.

**Reason:**

The Value of the STATUS is missing.

**Action:**

Required action unknown.

## CAUAJM\_E\_60039

The [{0}] option is valid only when using {1}.

**Reason:**

The [-v StopArg] option is valid only when using -E STOP\_DEMON.

**Action:**

Correct the input and resubmit the request.

## CAUAJM\_E\_60040

INVALID Status: {0}

**Reason:**

INVALID Status.

**Action:**

Required action unknown.

## CAUAJM\_E\_60041

You cannot manually change the status to {0}.

**Reason:**

You cannot change Status manually.

**Action:**

Required action unknown.

## CAUAJM\_E\_60042

Usage error encountered: \n\n

**Reason:**

A usage error was encountered.

**Action:**

Please refer to the applications help information for further assistance.

## CAUAJM\_E\_60043

**Invalid parameters specified.**

**Reason:**

Invalid parameters were passed to program.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_60044

**Could not open file {0}**

**Reason:**

File does not exist.

**Action:**

Specify valid file.

## CAUAJM\_E\_60046

**Invalid specification {0} for the [{1}] option. Valid values are {2}. {3} values should not be comma separated.**

**Reason:**

An invalid value was specified for the [-v StopArg] option. Valid values are ALL, FAILOVER, OR ROLE=APST.

**Action:**

Correct the input and resubmit the request.

## CAUAJM\_E\_60047

**Unable to determine if Schedulers are running.**

**Reason:**

Unable to determine if Schedulers are running.

**Action:**

Required action unknown.

## CAUAJM\_E\_60048

**Only an {0} {1} {2} can add new users or set the security mode.**

**Reason:**

The user running the security utility does not have the required privileges to perform the requested operation.

## CAUAJM\_E\_60049

**Initialization error. Cannot execute options 1 through 5 and 7.**



**Reason:**

Initialization error. Cannot execute options 1 through 5 and 7.

**Action:**

This occurred as a result of not being able to connect to the Application Server.

**CAUAJM\_E\_60053**

**The arguments {0} specified for the [{1}] option are invalid. Valid values are {2}. These values must not be duplicated or comma separated.**

**Reason:**

The arguments specified for the ROLE option are invalid. Valid values are A, P, S, or T. These values must not be duplicated or comma separated.

**Action:**

Correct the input and resubmit the request.

**CAUAJM\_E\_60054**

**Duplicate value {0} specified for the [{1}] argument. Role setting flags must not be duplicated.**

**Reason:**

Duplicate role setting flag detected. Only one occurrence of a flag can be specified within the ROLE= value.

**Action:**

Correct the input and resubmit the request.

**CAUAJM\_E\_60055**

**The [{0}] and [{1}] options are valid only when using {2}.**

**Reason:**

The [-d] option is valid only when using -E AGGREGATE.

**Action:**

Correct the input and resubmit the request.

**CAUAJM\_E\_60057**

**Invalid specification {0} for the [{1}] option.**

**Reason:**

An invalid value was specified for the [-I AggrArg] option.

**Action:**

Correct the input and resubmit the request.

**CAUAJM\_E\_60058**

**Could not change the database password.**

**Reason:**

Could not change the database password.

**Action:**

Required action unknown.

**CAUAJM\_E\_60060**

**The new password was not correctly confirmed. Be sure that the confirmation password exactly matches the password.**

**Reason:**

Password confirmation failed.

**Action:**

Re-type the password and confirmation.

**CAUAJM\_E\_60061**

**Error encoding this password.**

**Reason:**

Error encoding this password.

**Action:**

Required action unknown.

**CAUAJM\_E\_60065**

**Machine Name can only be {0} characters.**

**Reason:**

Machine name exceeded defined maximum length.

**Action:**

Make sure machine name should not exceed maximum length.

**CAUAJM\_E\_60066**

**An internal program abort has occurred.**

**Reason:**

An internal program abort has occurred.

**Action:**

Contact technical support.

**CAUAJM\_E\_60067**

**User {0} is already a {1}. Request aborted.**

**Reason:**

The user specified as input is already a super user. The request will be aborted.

**Action:**

No action required.

## **CAUAJM\_E\_60070**

**Unable to decode current password.**

**Reason:**

Unable to decode current password.

**Action:**

Required action unknown.

## **CAUAJM\_E\_60071**

**Job name, application, or group name required.**

**Reason:**

Missing parameters necessary to identify applicable jobs.

**Action:**

Specify job name, application, or group parameters.

## **CAUAJM\_E\_60072**

**The passphrase exceeds {0} characters in length.**

**Reason:**

The passphrase exceeds the specified number of characters in length.

**Action:**

Choose a new passphrase with fewer characters.

## **CAUAJM\_E\_60073**

**The provided password is not valid.**

**Reason:**

The provided password is not valid.

**Action:**

Please enter the valid password.

## **CAUAJM\_E\_60075**

**Password must be at least {0} characters in length.**

**Reason:**

Password must be at least nn characters in length.

**Action:**

Required action unknown.

## **CAUAJM\_E\_60076**

**Password exceeds {0} characters in length.**

**Reason:**

Password exceeds maximum character length limit.

**Action:**

Specify a password with in the character limit.

## **CAUAJM\_E\_60077**

**Password contains an illegal character.**

**Reason:**

Password contains an illegal character.

**Action:**

Required action unknown.

## **CAUAJM\_E\_60078**

**Confirmation of password failed.**

**Reason:**

Confirmation of password failed.

**Action:**

Required action unknown.

## **CAUAJM\_E\_60079**

**Failed to enter a valid database password after three attempts. Aborting request!**

**Reason:**

Failed to enter a valid database password after three attempts. Aborting request.

**Action:**

Required action unknown.

## **CAUAJM\_E\_60080**

**Unable to encrypt password.**

**Reason:**

Unable to encrypt password.

**Action:**

Required action unknown.

## CAUAJM\_E\_60082

**Unable to update encoded password in database.**

**Reason:**

Unable to update encoded password in database.

**Action:**

Required action unknown.

## CAUAJM\_E\_60083

**Unable to verify database update of new password.**

**Reason:**

Unable to verify database update of new password.

**Action:**

Required action unknown.

## CAUAJM\_E\_60085

**Unable to restore password in database.**

**Reason:**

Unable to restore password in database.

**Action:**

Required action unknown.

## CAUAJM\_E\_60086

**Unable to retrieve current remote authentication method from database.**

**Reason:**

Unable to retrieve current remote authentication method from database.

**Action:**

Required action unknown.

## CAUAJM\_E\_60095

**Invalid remote authentication method: {0}.**

**Reason:**

Invalid remote authentication method.

**Action:**

Required action unknown.

**CAUAJM\_E\_60096**

**Unable to set remote authentication method into the database.**

**Reason:**

Unable to set remote authentication method into the database.

**Action:**

Required action unknown.

**CAUAJM\_E\_60097**

**Could not change the AutoSys remote authentication method.**

**Reason:**

Could not change the AutoSys remote authentication method.

**Action:**

Required action unknown.

**CAUAJM\_E\_60099**

**An unexpected error has occurred. A null argument was passed to {0}.**

**Reason:**

An internal error has occurred.

**Action:**

Please contact CA technical support for further assistance.

**CAUAJM\_E\_60100**

**Failed to retrieve space usage from {0}.**

**Reason:**

Failure encountered.

**Action:**

Required action unknown.

**CAUAJM\_E\_60101**

**Encoded password validation failed.**

**Reason:**

Encoded password validation failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_60102

The length of user@host or user@domain has exceeded {0} characters.

**Reason:**

The length of user@host or user@domain has exceeded the maximum allowable length.

**Action:**

Specify a user@host or user@domain value that does not exceed the maximum allowable lengthy.

## CAUAJM\_E\_60103

User {0} does not exist. Request aborted.

**Reason:**

This user does not exist. The request will be aborted.

**Action:**

Required action unknown.

## CAUAJM\_E\_60108

Unable to store {0} in database.

**Reason:**

Unable to store to the database

**Action:**

Required action unknown.

## CAUAJM\_E\_60109

Unable to delete {0} from database.

**Reason:**

Unable to delete from database.

**Action:**

Required action unknown.

## CAUAJM\_E\_60110

Unable to set {0}.

**Reason:**

Unable to set superuser.

**Action:**

Required action unknown.

## CAUAJM\_E\_60111

**Domain {0} is already a trusted domain. Request aborted.**

**Reason:**

The domain specified is already a trusted domain. The request will be aborted.

**Action:**

Required action unknown.

## CAUAJM\_E\_60113

**The username is not valid.**

**Reason:**

The username is not valid.

**Action:**

Required action unknown.

## CAUAJM\_E\_60115

**One or more CA EEM server names are empty.**

**Reason:**

One or more CA EEM server names are empty.

**Action:**

Enter a non-empty CA EEM server name and resubmit the request.

## CAUAJM\_E\_60117

**The host or domain is not valid.**

**Reason:**

The host or domain is not valid.

**Action:**

Enter a valid host or domain.

## CAUAJM\_E\_60118

**User {0} is not a super user. Request aborted.**

**Reason:**

User is not a superuser. The request will be aborted.

**Action:**

Required action unknown.



## CAUAJM\_E\_60121

**The new host or domain is not valid.**

**Reason:**

The new Host or Domain is not valid.

**Action:**

Required action unknown.

## CAUAJM\_E\_60122

**The CA EEM server name must be a valid, unique host name that is less than {0} characters.**

**Action:**

Enter a valid, unique host name of the proper length in the CA EEM server field and resubmit the request.

## CAUAJM\_E\_60124

**Domain {0} is not a trusted domain. Request aborted.**

**Reason:**

Domain is not a trusted domain. The request will be aborted.

**Action:**

Required action unknown.

## CAUAJM\_E\_60127

**Error encountered while creating {0} {1}.**

**Reason:**

Error encountered while creating superuser.

**Action:**

Required action unknown.

## CAUAJM\_E\_60128

**{0} superuser functions are not available when instance security is set to {1}.**

**Reason:**

The EDIT/EXEC superuser functions are only permitted when running in a native security mode.

**Action:**

No action required.

## CAUAJM\_E\_60129

**Error encountered while modifying {0} {1}.**

**Reason:**

Error encountered while modifying superuser.

**Action:**

Required action unknown.

**CAUAJM\_E\_60130**

**Cannot delete the last {0} {1}. Request aborted.**

**Reason:**

Cannot delete last superuser. The request will be aborted.

**Action:**

Required action unknown.

**CAUAJM\_E\_60131**

**CA EEM server name <{0}> exceeds {1} characters.**

**Reason:**

The specified CA EEM server name exceeds the specified number of characters.

**Action:**

Enter a properly sized CA EEM server name and resubmit the request.

**CAUAJM\_E\_60132**

**Error encountered while deleting {0} {1}.**

**Reason:**

Error encountered while deleting superuser.

**Action:**

Required action unknown.

**CAUAJM\_E\_60133**

**Insufficient privileges to execute option 1 or option 4.**

**Reason:**

Insufficient privileges to execute option 1 or option 4.

**Action:**

Required action unknown.

**CAUAJM\_E\_60134**

**User create aborted.**

**Reason:**

User create aborted.

**Action:**

Required action unknown.

**CAUAJM\_E\_60136**

**User change aborted.**

**Reason:**

User change aborted.

**Action:**

Required action unknown.

**CAUAJM\_E\_60138**

**User delete aborted.**

**Reason:**

User delete aborted.

**Action:**

Required action unknown.

**CAUAJM\_E\_60142**

**CA EEM server name <{0}> is not unique.**

**Reason:**

The specified CA EEM server name is not unique.

**Action:**

Enter CA EEM server names that are not duplicated in the server list and resubmit the request.

**CAUAJM\_E\_60143**

**The CA EEM server list cannot exceed {0} characters.**

**Reason:**

The CA EEM server list cannot exceed the specified number of characters.

**Action:**

Enter a properly sized list of CA EEM servers and resubmit the request.

**CAUAJM\_E\_60145**

**Specifying an archive directory requires the [{0}] flag to save events to a file.**

**Reason:**

The -d option was specified without a corresponding -A option.

**Action:**

Specify the proper command arguments.

---

## CAUAJM\_E\_60146

**Number of Days for {0} should be a zero or a positive number.**

**Reason:**

Invalid parameters were passed to program.

**Action:**

Specify valid parameters.

## CAUAJM\_E\_60147

**Directory {0} does not exist or file permissions are insufficient**

**Reason:**

Directory does not exist or read and write permissions are not present.

**Action:**

Specify valid directory or check read and write permissions.

## CAUAJM\_E\_60149

**Missing arguments. The {0} event requires either the [{1}] or [{2}] argument.**

**Reason:**

Missing arguments. The -E AGGREGATE event requires the -l or -d argument.

**Action:**

Correct the input and resubmit the request.

## CAUAJM\_E\_60152

**You must specify the CA EEM server names previously used to enable external security.**

**Reason:**

The CA EEM server names entered and the names of servers previously used to enable external security do not match.

**Action:**

Enter CA EEM server names that match the names of servers previously used to enable external security.

## CAUAJM\_E\_60156

**{0} cached credentials are unreadable.**

**Reason:**

The home directory or Local Settings Application Data directory permissions do not authorize you to read cached credentials.

**Action:**

Check file and directory permissions in the user's home directory or Local Settings Application Data directory.

## CAUAJM\_E\_60157

The {0} option requires the {1} and the {2} option.

**Reason:**

The cached credential operation is missing required parameters.

**Action:**

Specify the required options and resubmit the request.

## CAUAJM\_E\_60158

The {0} option requires a valid argument. Specify \"{1}\", \"{2}\" or \"{3}\".

**Reason:**

An invalid argument was specified as input to the cached credential operation.

**Action:**

Specify a valid argument and resubmit the request.

## CAUAJM\_E\_60159

The cached credential time argument cannot exceed {1} characters.

**Reason:**

The time value has exceeded the maximum allowable input.

**Action:**

Change the input and resubmit the request.

## CAUAJM\_E\_60160

The user portion of the \"{0}\" argument cannot exceed {1} characters.

**Reason:**

The user value has exceeded the maximum allowable input.

**Action:**

Change the input and resubmit the request.

## CAUAJM\_E\_60161

The host or domain portion of the \"{0}\" argument cannot exceed {1} characters.

**Reason:**

The host or domain value of the input argument has exceeded the maximum allowable input.

**Action:**

Change the input and resubmit the request.

## CAUAJM\_E\_60163

**Invalid format specified for the credential timeout. Missing day or hour specification.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60165

**Invalid format specified for the credential timeout. Day specification is not numeric.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60166

**Invalid format specified for the credential timeout. Hour specification is not numeric.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60167

**Invalid format specified for the credential timeout. Day and hour specification cannot be zero.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60168

**Invalid format specified for the credential timeout. Day specification is invalid.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60169

**Invalid format specified for the credential timeout. Hour specification is invalid.**

**Reason:**

The credential timeout value has been incorrectly formatted.

**Action:**

Enter a valid credential timeout and resubmit the request.

## CAUAJM\_E\_60172

**{0} cached credentials are not deleted.**

**Reason:**

The home directory or local application directory permissions do not authorize you to delete cached credentials.

**Action:**

Verify the home directory or local application data directory permissions, log in as a user who is authorized to delete cached credentials, and re-issue the command.

## CAUAJM\_E\_60173

**The passphrase contains an illegal character.**

**Reason:**

The passphrase contains an illegal character.

**Action:**

Enter the passphrase without illegal characters.

## CAUAJM\_E\_60174

**Confirmation of the passphrase failed.**

**Reason:**

Confirmation of the passphrase failed.

**Action:**

Please retry.

## CAUAJM\_E\_60187

**Alias exceeds {0} characters in length.**

**Reason:**

Alias exceeds character length limit.

**Action:**

Specify an alias with in the limit.

---

## CAUAJM\_E\_60188

**Alias contains an illegal character.**

**Reason:**

Alias contains an illegal character.

**Action:**

Required action unknown.

## CAUAJM\_E\_60189

**Confirmation of alias failed.**

**Reason:**

Confirmation of alias failed.

**Action:**

Required action unknown.

## CAUAJM\_E\_60193

**You are not authorized to perform this action.**

**Reason:**

You do not have sufficient privileges to perform this action.

**Action:**

Sufficient privileges are required to perform this action. Request aborted.

## CAUAJM\_E\_60196

**An error occurred attempting to set {0} instance security. Instance remains under {1} security control.**

**Reason:**

This instance remains under NATIVE security control due to error attempting to reset internal security flag.

**Action:**

Review prior messages to determine the cause of the error. Otherwise, contact technical support for further assistance.

## CAUAJM\_E\_60197

**An error occurred attempting to set {0} instance security. Instance remains under {1} security control.**

**Reason:**

The instance remains under CA EEM security control due to an error attempting to reset the internal security flag.

**Action:**

Review prior messages to determine the cause of the error. Otherwise, contact technical support for further assistance.



## CAUAJM\_E\_60198

**Unable to regenerate {0} database keys.**

**Reason:**

An internal error has occurred while attempting to regenerate the superuser database keys.

**Action:**

Contact CA Technical Support.

## CAUAJM\_E\_60199

**Unable to generate the {0} certificate. See previous error messages for details.**

**Reason:**

The external security server cannot generate a CA EEM certificate.

**Action:**

Review previous error messages for details and take required action.

## CAUAJM\_E\_60203

**Program aborting due to an invalid security environment.**

**Reason:**

The state of the security environment prevents the program from running.

**Action:**

If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_60204

**The instance is running under CA EEM security control but CA EEM is not available.**

**Reason:**

CA WAAE cannot establish a session with at least one of the specified CA EEM servers.

**Action:**

Check previous messages for an indication of what caused the problem, take required action, and try to reconnect. If you cannot determine the cause of the problem, contact technical support.

## CAUAJM\_E\_60211

**An invalid asset type has been specified.**

**Reason:**

An invalid asset type has been specified. Program aborts processing.

**Action:**

Review usage and correct asset.

## CAUAJM\_E\_60212

The value for the "{0}" argument cannot exceed {1} characters.

**Reason:**

The argument value has exceeded the maximum allowable input.

**Action:**

Change the input and resubmit the request.

## CAUAJM\_E\_60214

Missing program arguments.

**Reason:**

The program is missing one or more valid arguments and will abort.

**Action:**

Review usage and correct the input program arguments.

## CAUAJM\_E\_60215

Username name length exceeds maximum allowable length.

**Reason:**

Username name length exceeds maximum allowable length. Username cannot exceed 64 characters.

**Action:**

Correct username and resubmit the request.

## CAUAJM\_E\_60219

Reserved word "{0}" cannot be specified as target.

**Reason:**

Reserved word cannot be specified as target.

**Action:**

Change input and resubmit the request.

## CAUAJM\_E\_60220

Reserved word "{0}" must be specified as target for {1} command.

**Reason:**

Reserved word must be specified as target for command.

**Action:**

Change target and resubmit the request.

---

## CAUAJM\_E\_60227

**{0}: environment variable is not set.\n**

**Reason:**

The program requires this variable to be set in the environment.

**Action:**

Set the variable and try again.

## CAUAJM\_E\_60234

**{0}: Not a valid key.\n**

**Reason:**

The key specified is not valid.

**Action:**

Specify a valid key and try again.

## CAUAJM\_E\_60236

**Invalid command: {0}**

**Reason:**

The command is invalid.

**Action:**

Enter a command with the proper usage.

## CAUAJM\_E\_60237

**Invalid asset class name <{0}>.**

**Reason:**

The asset class name is invalid

**Action:**

Enter a valid asset class name.

## CAUAJM\_E\_60238

**Invalid asset access level <{0}>.**

**Reason:**

The asset access level is invalid

**Action:**

Enter a valid asset access level.

## CAUAJM\_E\_60239

### **Certificate authentication error:**

#### **Reason:**

Certificate authentication error.

#### **Action:**

Authentication certificate for the instance must be regenerated.

## CAUAJM\_E\_60240

### **Error establishing a session with the CA EEM Backend:**

#### **Reason:**

An error occurred establishing a session with the CA EEM Backend.

#### **Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60246

### **Error authenticating the administrative user account \"{0}\":**

#### **Reason:**

An error occurred authenticating the administrative user account

#### **Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60251

### **Error creating security policies for product instance {0}:**

#### **Reason:**

An error occurred creating security policies for the product instance.

#### **Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60249

### **Error registering CA EEM application instance \"{0}\":**

#### **Reason:**

An error occurred registering the CA EEM application instance for the product.

#### **Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60256

**Error deleting security policies for product instance {0}:**

**Reason:**

An error occurred deleting security policies for the product instance.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60258

**Error unregistering CA EEM application instance \"{0}\":**

**Reason:**

An error occurred unregistering the CA EEM application instance for the product.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60260

**Error upgrading CA EEM application instance \"{0}\":**

**Reason:**

Informational only.

**Action:**

No action required.

## CAUAJM\_E\_60261

**Error generating an authentication certificate for instance {0}:**

**Reason:**

An error occurred generating an authentication certificate for a product instance.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60267

**Error adding administrative privileges to user \"{0}\".**

**Reason:**

An error occurred adding administrative privileges to user.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60269

**Error removing administrative privileges from user \"{0}\".**

**Reason:**

An error occurred removing administrative privileges from user.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60271

**Error obtaining the list of administrative users.**

**Reason:**

An error occurred obtaining the list of administrative users.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60272

**File <{0}> could not be read or does not have a valid key file format.\n**

**Reason:**

The specified file name does not have read permissions or does not have a valid key file format.

**Action:**

Check the read permissions of the specified file or specify the name of a file that was generated with a valid key format.

## CAUAJM\_E\_60274

**Error generating the session artifact:**

**Reason:**

An error occurred generating the session artifact

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60277

**Artifact authentication failed.**

**Reason:**

Artifact could not be authenticated.

**Action:**

Artifact may have expired or is invalid. Generate a new one.

## CAUAJM\_E\_60283

**Error checking for the registration of CA EEM application instance \"{0}\":**

**Reason:**

An error occurred checking for the registration of CA EEM application instance for the product.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60284

**Error checking for the registration of instance {0}:**

**Reason:**

An error occurred checking for the registration of the specified instance.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60285

**Error checking for default policies shared by all instances:**

**Reason:**

An error occurred checking for default policies shared by all instances.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60286

**Error loading CA EEM configuration files from directory: {0}.**

**Reason:**

An error occurred loading CA EEM configuration files from the specified directory.

**Action:**

Verify that the necessary EEM configuration files are present in the specified directory.

## CAUAJM\_E\_60287

**Error initializing the CA EEM library.**

**Reason:**

An error occurred initializing the CA EEM library.

**Action:**

Please review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60289

Unable to generate the {0} credentials. See previous error messages for details.

**Reason:**

The external security server cannot generate a CA EEM credentials.

**Action:**

Review previous error messages for details and take required action.

## CAUAJM\_E\_60291

{0} credentials are not cached.

**Reason:**

The home directory or local application directory permissions do not authorize you to generate cached credentials.

**Action:**

Verify the home directory or local application data directory permissions, log in as a user who is authorized to delete cached credentials, and re-issue the command.

## CAUAJM\_E\_60315

Invalid instance {0}: must be {1} characters in length. Operation aborted.

**Reason:**

An instance was entered with an invalid number of characters. Operation aborted.

**Action:**

Enter an instance name with the specified number of characters.

## CAUAJM\_E\_60316

60316 Invalid instance {0}: illegal use of characters. Operation aborted.

**Reason:**

An instance was entered with illegal use of characters. Operation aborted.

**Action:**

Enter an instance name with valid characters.

## CAUAJM\_E\_60317

CA EEM server host name not specified. Operation aborted.

**Reason:**

The CA EEM server host name is a required operand.

**Action:**

Specify the CA EEM server host name through use of the -b operand.



## CAUAJM\_E\_60319

**Connection to CA EEM server failed.**

**Reason:**

Unable to connect to CA EEM server.

**Action:**

Reviewing preceding error messages for indications on why failure occurred.

## CAUAJM\_E\_60321

**User object \"{0}\" is not defined to CA EEM or object does not currently have administrative privileges to the application instance.**

**Reason:**

User is not defined to CA EEM or does not currently have administrative privileges to application instance.

**Action:**

Specify an existing user with administrative privileges.

## CAUAJM\_E\_60324

**Error obtaining registered product instance count.**

**Reason:**

An error has occurred obtaining the registered product instance count.

**Action:**

Please review any preceding messages for an indication of the problem.

## CAUAJM\_E\_60325

**Operation requires connection to CA EEM be made as \"{0}\".**

**Reason:**

This operation requires connection to CA EEM be made as EiamAdmin

**Action:**

You will be prompted to connect with the EiamAdmin account

## CAUAJM\_E\_60328

**Option -{0} out of range. Valid values are 0-5.**

**Reason:**

The identified option is out of range. Valid values are 0-5.

**Action:**

Specify a valid value.

## CAUAJM\_E\_60329

**60329 Option {0} was not specified OR was not available through the product's environment.**

**Reason:**

Option was not specified and was available through the product's environment.

**Action:**

Specify the option or update the specific configuration option within the the product's environment.

## CAUAJM\_E\_60330

**Option {0} requires option {1}.**

**Reason:**

Specification of the user option requires the password option.

**Action:**

Specify the password option.

## CAUAJM\_E\_60331

**Unable to extract password from user string as set in the product's environment.**

**Reason:**

The Service Desk user and password combination has not been properly set within the the product's environment.

**Action:**

Verify the Service Desk user and password combination is properly set.

## CAUAJM\_E\_60332

**Unable to locate job {0} in the database.**

**Reason:**

Job does not exist in database. The job must exist in the database in order to use this test tool.

**Action:**

Specify a jobname that exists in the product's database.

## CAUAJM\_E\_60333

**Error retrieving service desk attributes for job {0} joid {1}.**

**Reason:**

Error retrieving uninotify row for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_60334

**Option {0} was not specified OR was not available through the product's environment.**

**Reason:**

Option was not specified and was available through the product's environment.

**Action:**

Specify the option or update the specific configuration option within the the product's environment.

## CAUAJM\_E\_60335

**Missing required option {0} or {1}.**

**Reason:**

Missing required option.

**Action:**

Specify the required option.

## CAUAJM\_E\_60336

**Missing required option {0} and {1} for generic request.**

**Reason:**

Missing required option.

**Action:**

Specify the required option.

## CAUAJM\_E\_60343

**Invalid run number specified for job {0} joid {1} with run number {2}.**

**Reason:**

No run number row for joid was found.

**Action:**

Required action unknown.

## CAUAJM\_E\_60344

**Error retrieving job run data for job {0} joid {1} with run number {2}.**

**Reason:**

Error retrieving run number row for joid.

**Action:**

Required action unknown.

## CAUAJM\_E\_60293

The CA EEM application instance is at \"{0}\" which is not the current release level.

**Reason:**

The CA EEM application instance is not at a compatible level to perform the update.

**Action:**

Upgrade the CA EEM application instance to the compatible level.

## CAUAJM\_E\_60294

The CA Workload Automation AE release level \"{0}\" is either incorrect or not in a proper format.

**Reason:**

The release level is either incorrect or not in a proper format.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_60295

The provided CA Workload Automation AE release level \"{0}\" is either incorrect or not in a proper format.

**Reason:**

The provided release level is either incorrect or not in a proper format.

**Action:**

Provide the release level in the \"Major.Minor.Revision.Maintenance\" format."

## CAUAJM\_E\_60297

Error updating the security policies for the product.

**Reason:**

An error occurred while updating the security policies for the product.

**Action:**

Review any subsequent messages for an indication of the problem.

## CAUAJM\_E\_60299

The provided CA Workload Automation AE release level \"{0}\" is at an unsupported level.

**Reason:**

The provided release level is at an unsupported level.

**Action:**

Provide a supported level.

## CAUAJM\_E\_60300

The provided CA Workload Automation AE release level \"{0}\" is at a higher level.

**Reason:**

The provided release level is at a higher level. Hence, an update is not relevant.

**Action:**

No action required.

## CAUAJM\_E\_60346

An invalid option combination has been specified. Options {0} and {1} are mutually exclusive.

**Reason:**

The specified options are mutually exclusive.

**Action:**

Remove one of the options from the command and resubmit the request.

## CAUAJM\_E\_60347

Required parameter {0} missing. Command aborted.

**Reason:**

The combination of user authentication parameters is incomplete or incorrect.

**Action:**

Specify the missing parameter when you re-issue the command. Ensure that you specify the -pw parameter or the -pwx parameter but not both parameters.

## CAUAJM\_E\_60348

Unable to authenticate when an empty value is specified for the {0} parameter. Command aborted.

**Reason:**

Authentication completes only when a valid user name and password combination is specified.

**Action:**

Specify a valid user name when you re-issue the command.

## CAUAJM\_E\_60349

Unable to complete authentication when an empty value is specified for {0} parameter or {1} parameter. Command aborted.

**Reason:**

Authentication completes only when a valid user name and password combination is specified.

**Action:**

Specify a valid password when you re-issue the command. Ensure that you specify either the -pw parameter or the -pwx parameter, but not both parameters. Use -pwx for encrypted passwords and -pw for non-encrypted passwords.

---

## CAUAJM\_E\_60351

**Cannot {0} {1} event manually**

**Reason:**

INVALID Event.

**Action:**

Required action unknown.

## CAUAJM\_E\_60352

**The directory does not exist or you do not have the appropriate permission**

**Reason:**

Invalid path specified

**Action:**

Correct the problem and try again.

## CAUAJM\_E\_60354

**Failed to read key <{0}>**

**Reason:**

The user may not have the registry access permissions

**Action:**

Required action unknown.

## CAUAJM\_E\_60356

**Phase {0} failed**

**Reason:**

The user may not have the registry access permissions

**Action:**

Required action unknown.

## CAUAJM\_E\_60360

**Unable to authenticate when an empty value is specified for the user name. Command aborted.**

**Reason:**

Authentication completes only when a valid user name and password combination is specified.

**Action:**

Specify a valid user name when you re-issue the command.

---

**CAUAJM\_E\_60370**

**Failed to read job profile <{0}>**

**Reason:**

The user may not have the registry access permissions

**Action:**

Required action unknown.

**CAUAJM\_E\_60371**

**Failed to create file <{0}>**

**Reason:**

WriteFile API failed

**Action:**

Required action unknown.

**CAUAJM\_E\_60372**

**Failed to write job profile <{0}> to file <{1}>**

**Reason:**

WriteFile API failed

**Action:**

Required action unknown.

**CAUAJM\_E\_60379**

**Error creating shared security policies:**

**Reason:**

An error occurred creating shared security policies.

**Action:**

Please review any subsequent messages for an indication of the problem.

**CAUAJM\_E\_60386**

**Merge failed**

**Reason:**

The user may not have the registry access permissions.

**Action:**

Required action unknown.

**CAUAJM\_E\_60387:****Deployment failed****Reason:**

The user may not have write permissions on the file.

**Action:**

Required action unknown.

**CAUAJM\_E\_60388**

**One or more sendevent commands have Invalid parameters specified.**

**Reason:**

Invalid parameters were passed to program.

**Action:**

Specify valid parameters.

**CAUAJM\_E\_60394**

**Error encountered while adding trusted domain.**

**Reason:**

Error encountered while adding trusted domain.

**Action:**

Required action unknown.

**CAUAJM\_E\_60396**

**Error encountered while removing trusted domain.**

**Reason:**

Error encountered while removing trusted domain.

**Action:**

Required action unknown.

**CAUAJM\_E\_60397**

**The unauthenticated mode is already set to {0}. Request aborted.**

**Reason:**

The unauthenticated mode is already set. The request will be aborted.

**Action:**

Pick another unauthenticated user mode that has not been set.



## CAUAJM\_E\_60400

**Required parameter {0} or {1} missing. Command aborted.**

**Reason:**

The combination of user authentication parameters is incomplete or incorrect.

**Action:**

Specify the missing parameter when you re-issue the command. Ensure that you specify the -pw parameter or the -pwx parameter but not both parameters.

## CAUAJM\_E\_60401

**Unable to complete authentication when parameters {0} and {1} are specified. Command aborted.**

**Reason:**

Authentication completes only when a valid user name and password combination is specified.

**Action:**

Specify a valid password when you re-issue the command. Ensure that you specify either the -pw parameter or the -pwx parameter, but not both parameters. Use -pwx for encrypted passwords and -pw for non-encrypted passwords.

## CAUAJM\_E\_60403

**You do not have administrative rights to delete user <{0}>. A password or security code is required.**

**Reason:**

You do not have administrative rights to delete user. A password or security code is required.

**Action:**

Required action unknown.

## CAUAJM\_E\_60407

**The provided security code is not valid.**

**Reason:**

The provided security code is not valid.

**Action:**

Enter a valid security code.

## CAUAJM\_E\_60408

**The provided alias is not valid.**

**Reason:**

The provided alias is not valid.

**Action:**

Please enter the valid alias.

---

## CAUAJM\_E\_60411

**The security code exceeds {0} characters in length.**

**Reason:**

The security code exceeds the maximum character length limit.

**Action:**

Specify a security code within the character limit.

## CAUAJM\_E\_60412

**The security code contains an illegal character.**

**Reason:**

The security code contains an illegal character.

**Action:**

Specify a valid value.

## CAUAJM\_E\_60413

**Confirmation of security code failed.**

**Reason:**

Confirmation of security code failed.

**Action:**

Re-enter the security code.

## CAUAJM\_E\_60414

**Unable to encrypt the security code.**

**Reason:**

Unable to encrypt the security code.

**Action:**

Provide another security code.

## CAUAJM\_E\_60419

**User change key credentials aborted.**

**Reason:**

User change key credentials aborted.

**Action:**

Retry.

---

## CAUAJM\_E\_60421

The option {0} or {1} is not applicable for user {2}. Change aborted.

**Reason:**

The option is not applicable for this type of user.

**Action:**

Use the appropriate option.

## CAUAJM\_E\_60422

The option {0} is not applicable for user {2}. Delete aborted.

**Reason:**

The option is not applicable for this type of user.

**Action:**

Use an appropriate option.

## CAUAJM\_E\_60424

Invalid value specified.

**Reason:**

Not a valid credential type.

**Action:**

Specify a valid credential type.

## CAUAJM\_E\_60430

File system error. The path <{0}> is not a file.

**Reason:**

The specified path is not a file.

**Action:**

Specify the path with the file name.

## CAUAJM\_E\_60431

File system error. The file <{0}> is empty.

**Reason:**

The specified file doesnot contain any data.

**Action:**

Specify the file holding the keyfile content.

## CAUAJM\_E\_60432

The {0} application server is running at a version that does not support this operation.

**Reason:**

The application server is running at a version that does not support this operation.

**Action:**

Upgrade the application server to the latest supported version.

## CAUAJM\_E\_60434

User {0} credential information is corrupted. Request aborted.

**Reason:**

The request will be aborted.

**Action:**

Contact CA Technologies Technical Support.

## CAUAJM\_E\_60435

User {0} or the attribute {1} for the user do not exist. Request aborted.

**Reason:**

This user does not exist. The request will be aborted.

**Action:**

Required action unknown.

## CAUAJM\_E\_60443

Could not add password vault name {0}.

**Reason:**

Could not perform the operation.

**Action:**

Required action unknown.

## CAUAJM\_E\_60444

Could not find {0} password vault

**Reason:**

Could not perform the operation.

**Action:**

Enter valid password vault.

---

## CAUAJM\_E\_60445

Could not add password vault name {0}. Invalid password vault type.

**Reason:**

Could not perform the operation.

**Action:**

Enter valid password vault type.

## CAUAJM\_E\_60446

Could not delete password vault name {0}.

**Reason:**

Could not perform the operation.

**Action:**

Required action unknown.

## CAUAJM\_E\_60447

Could not add any more password vaults.

**Reason:**

Could not perform the operation.

**Action:**

Required action unknown.

## CAUAJM\_E\_60448

Could not add password vault name {0}. It is already exists.

**Reason:**

Could not perform the operation.

**Action:**

Required action unknown.

## CAUAJM\_E\_60452

The password vault alias provided for the user <{0}> is incorrect.

**Reason:**

An incorrect password vault alias has been specified.

**Action:**

Specify the correct alias for the user.

## CAUAJM\_E\_60454

An error has occurred while resolving password vault alias to password for user: {0}. Return code [{1}].

**Reason:**

Unknown.

**Action:**

Check the Symantec Privileged Access Manager documentation for information on return code.

## CAUAJM\_E\_60455

An error has occurred while resolving password vault alias to password for user: {0}. Error message [{1}].

**Reason:**

Unknown.

**Action:**

Check the Symantec Privileged Access Manager documentation for information on the error message.

## CAUAJM\_E\_60456

Password vault alias contains one or more invalid characters.

**Reason:**

Password vault alias contains one or more invalid characters.

**Action:**

Specify the alias name with valid characters.

## CAUAJM\_E\_60459

The provided password vault name is not valid.

**Reason:**

The provided password vault name is not valid.

**Action:**

Enter a valid password vault name.

## CAUAJM\_E\_60461

Error initializing CA Privileged Access Manager A2AClient environment. Error message [{1}].

**Reason:**

The CA Privileged Access Manager A2AClient is either not installed or corrupted.

**Action:**

Install the CA Privileged Access Manager A2AClient correctly.

## CAUAJM\_E\_60675

Create a user aborted as password vault name <{0}> does not exist.

**Reason:**

Create a user aborted as password vault name does not exist.

**Action:**

Specify the correct password vault name.

## CAUAJM\_E\_70295

You cannot change the status manually to <{0}>.

**Reason:**

You cannot change the status manually.

**Action:**

Review the documentation for the supported status and correct accordingly.

## CAUAJM\_E\_70296

You cannot change the status to INACTIVE when job is in the ON\_ICE status.

**Reason:**

You cannot change the status manually.

**Action:**

Review the documentation for the supported status and correct accordingly.

## CAUAJM\_E\_71254

The {0} specified to run the command should have {1}.

**Reason:**

Error occurred while parsing the request.

**Action:**

Specify the required parameters.

## CAUAJM\_E\_71255

The specified command [ {0} ] is not present in the list of allowable commands.

**Reason:**

Error occurred while parsing the request.

**Action:**

Command should be specified in the filter file. Contact your administrator.

## CAUAJM\_E\_71256

The command [ {0} ] requires user and password to be specified in the request.

**Reason:**

Error occurred while parsing the request.

**Action:**

Specify the required parameters.

## CAUAJM\_E\_71257

Execution of the command [ {0} ] encountered an exception {1}.

**Reason:**

Error occurred while executing the command.

**Action:**

Check the response for the error message.

## CAUAJM\_E\_71258

The execution of the command [ {0} ] exceeded the specified timeout value of {0}ms.

**Reason:**

Execution of the command was interrupted.

**Action:**

Increase the timeout value.

## CAUAJM\_E\_71259

The specified timeout value should be a positive integer.

**Reason:**

Error occurred while parsing the request.

**Action:**

Specify a valid timeout value.

## CAUAJM\_E\_71260

Incorrect type specified. Valid values are job, machine, resource, connection, and xinst.

## CAUAJM\_E\_80001

CCI send error Oid({0},{1}) to Did({2},{3})

**Reason:**

Unable to send data from origin identifier to destination identifier.

**Action:**



---

Review extended error code information in messages CAUAJM\_E\_80004 and CAUAJM\_E\_80005.

## CAUAJM\_E\_80002

**CCI receive error has occurred for Oid({0})**

**Reason:**

Unable to set CCI receiver.

**Action:**

Review extended error code information in messages CAUAJM\_E\_80004 and CAUAJM\_E\_80005.

## CAUAJM\_E\_80003

**Bad environment: {0} not set. Exiting.**

**Reason:**

Bad environment.

**Action:**

Required action unknown.

## CAUAJM\_E\_80004

**Error Detail: [{0}.{1}.{2}]**

**Reason:**

Extended error detail provided as part of message CAUAJM\_E\_00001 or CAUAJM\_E\_00002.

**Action:**

No action required.

## CAUAJM\_E\_80012

**Unable to extract CCI message data. Request ignored.**

**Reason:**

Unable to extract CCI message data.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_E\_80005

**Error Text: {0}**

**Reason:**

Extended error text provided as part of message CAUAJM\_E\_00001 or CAUAJM\_E\_00002.

**Action:**

No action required.

## CAUAJM\_E\_80006

**Maximum number of CCI receive errors exceeded.**

**Reason:**

Maximum number of CCI receive errors exceeded.

**Action:**

Review error code information in messages CAUAJM\_E\_80002, CAUAJM\_E\_80004 and CAUAJM\_E\_80005.

## CAUAJM\_E\_80008

**Unable to obtain CCI Receive load point addresses for receive processing. Aborting.**

**Reason:**

Unable to obtain CCI Receive load point addresses for receive processing.

**Action:**

Make sure the CCI shared libraries are installed and accessible on the system.

## CAUAJM\_E\_80009

**Unable to obtain CCI Send load point address for send processing. Aborting.**

**Reason:**

Unable to obtain CCI Send load point address for send processing.

**Action:**

Make sure the CCI shared libraries are installed and accessible on the system.

## CAUAJM\_E\_80010

**CCI init error.**

**Reason:**

Unable to successfully initialize CCI environment.

**Action:**

Review extended error code information in messages CAUAJM\_E\_80004 and CAUAJM\_E\_80005.

## CAUAJM\_E\_80011

**An internal communication error has occurred. Unable to {0} internal request.**

**Reason:**

An internal communication error has errored.

**Action:**

Contact CA Technical support for further assistance.

## CAUAJM\_E\_80013

**CCI is inactive or has experienced an unrecoverable error <{0}><{1}>. Receiver Oid<{2}> terminating.**

**Reason:**

CCI is inactive or has experienced some unrecoverable error.

**Action:**

Restart the CCI component if inactive. Otherwise, contact CA technical support for further assistance.

## CAUAJM\_E\_80015

**Unable to send status. Remote application <{0}.<{1}> still pending setup.**

**Reason:**

Unable to send status because the remote application is pending setup.

**Action:**

The remote node initiated a setup request with the Cross Platform Interface. This setup has not successfully completed. Contact CA Technical support for further assistance.

## CAUAJM\_E\_80016

**Unable to locate procedure {0} within the {1} shared library.**

**Reason:**

The CCI shared library that was loaded does not contain the noted procedure.

**Action:**

Contact CA Technical Support for further assistance.

## CAUAJM\_E\_80019

**Could not find the {0} tag for the MESSAGE.**

**Reason:**

{0} tag not found.

**Action:**

{0} tag needs to be provided.

## CAUAJM\_E\_80020

**Unable to read the template file {0}.**

**Reason:**

An error occurred while reading the template file.

**Action:**

Required action unknown.

## CAUAJM\_E\_80022

The attachment could not be sent due to an internal error.

**Reason:**

An internal error occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_80024

The attachment could not be sent due to an internal error.

**Reason:**

An internal error occurred.

**Action:**

Required action unknown.

## CAUAJM\_E\_80025

Buffer overflow occurred in file {0} at line number {1} with return value {2}.

**Reason:**

An internal error has occurred.

**Action:**

Please contact CA technical support for further assistance.

## CAUAJM\_E\_90010

Unable to load {0} <{1}>.

**Reason:**

Unable to load Library

**Action:**

Check file permissions and file existence

## CAUAJM\_E\_90011

Unable to load {1} for Sybase 15.0 or {2} for Sybase 12.5.x <{3}>

**Reason:**

Unable to load loadlibrary

**Action:**

Check file permissions and file existence

## CAUAJM\_E\_90012

Unable to get the address of {0} < {1} >

**Reason:**

Unable to get the address of procedure

**Action:**

Required action unknown

## CAUAJM\_E\_90015

**Reason:**

License Error

**Action:**

Required action unknown.

## CAUAJM\_E\_101000

The CAPKI installation has failed. {0}

**Reason:**

The uninstallation of the CAPKI failed.

**Action:**

For more information, refer to the log file for the reason of failure and try to manually uninstall the CAPKI.

## CAUAJM\_E\_101100

The SSA installation has failed. For more information, refer to the SSAInstall.log file under the TEMP directory.

**The SSA installation has failed. For more information, refer to the SSAInstall.log file under the TEMP directory.**

**Reason:**

Unable to install the SSA.

**Action:**

For more information, refer to the log file for the reason of failure and rerun the SSA install.

## CAUAJM\_E\_105002

Unable to stop the "{0}" service. Stop the service manually and run the AutoSys installer again.

**Reason:**

Failed to stop the service.

**Action:**

Manually stop the service.

**Examples:**

On Linux:

---

```
unisrvcntr stop <service_name>
```

On Windows: Stop the service from Service Manager.

## CAUAJM\_E\_110000

**A few errors occurred. For more information, refer to the log file.**

**Reason:**

Errors occurred during the installation.

**Action:**

Refer to the log file for detailed information.

## CAUAJM\_E\_110004

**Unable to determine the installation path.**

**Reason:**

The AutoSys installation path could not be determined.

**Action:**

Refer to the log for details. Verify that the values for the following settings are valid and not empty:

- USER\_INSTALL\_DIR in the response file
- AUTOSYS\_INSTALL\_LOCATION path

## CAUAJM\_E\_110005

**A few manual AutoSys instances have been detected. Convert these manually copied instance(s) environment into an environment that can be upgraded successfully. For more information, refer to the AutoSys documentation.\n"{0}"**

**Reason:**

Manually copied instance(s) were detected.

**Action:**

Refer to the AutoSys documentation to convert to the environment that can be upgraded.

## CAUAJM\_E\_110007

**The AutoSys instance name is already in use. Specify a different instance name.**

**Reason:**

The instance name is already in use.

**Action:**

Specify an instance name which is not in use.

## CAUAJM\_E\_110008

**An error occurred while loading the configuration file(s). For more information, refer to the log file.**

**Reason:**

Error loading the configuration file.

**Action:**

Verify whether the configuration file exists in the appropriate path.

**CAUAJM\_E\_110009**

**An error occurred while loading the "{0}" configuration file.\n{1}**

**Reason:**

Error loading the configuration file.

**Action:**

Verify that the configuration file exists in the appropriate location and is in the proper format.

**CAUAJM\_E\_110010**

**The following configurations are corrupted:\n"{0}"\nCorrect the configurations and retry the AutoSys installation.**

**Reason:**

The configuration is corrupted.

**Action:**

Verify that the values in the configuration file are valid and in the proper format.

**CAUAJM\_E\_110011**

**The specified keys are in an invalid format.**

**Reason:**

The specified keys are in an invalid format.

**Action:**

Specify a valid key.

Hexadecimal:

- Exactly 32 characters
- Allowed characters: 0-9 and A-F

Passphrase: 1-16 characters

**CAUAJM\_E\_110012**

**The "Verify key" value does not match with the specified key.**

**Reason:**

The specified key and verify key do not match.

**Action:**

Specify the same value for both keys.

## CAUAJM\_E\_110015

**Unable to find instance-specific configuration file(s) (config.<INS>), despite having instance components.**

**Reason:**

Instance-specific components were detected, but the configurations are missing in the installation.

**Action:**

Verify that the environment is in the proper state and that config.<INSTANCE> is present in the autouser.<INSTANCE> directory.

## CAUAJM\_E\_110016

**The start port configuration is corrupted for "{0}". Ensure that the configuration in the server.xml is correct.**

**Reason:**

The start port was not found in the configuration file.

**Action:**

Verify that the start port is available in the configuration file.

## CAUAJM\_E\_110017

**The stop port configuration is corrupted for "{0}". Ensure that the configuration in the server.xml is correct.**

**Reason:**

The stop port was not found in the configuration file.

**Action:**

Verify that the stop port is available in the configuration file.

## CAUAJM\_E\_110020

**"{0}" should not be empty.**

**Reason:**

The field is empty.

**Action:**

Specify the proper value.

## CAUAJM\_E\_110021

**The required configuration file {0} is missing for "{1}"**

**Reason:**

An install configuration corruption was found, missing <value>.

**Action:**

Verify that the value is set correctly in the configuration file.



## CAUAJM\_E\_110022

**The Sybase database user password cannot exceed 30 characters.**

**Reason:**

The Sybase database user password exceeds 30 characters.

**Action:**

Specify a value which is less than or equal to 30 characters.

## CAUAJM\_E\_110023

**The Verify password and Password fields must match.**

**Reason:**

The passwords do not match.

**Action:**

Specify the same value for the password and verify password fields.

## CAUAJM\_E\_110024

**Select at least one option to proceed.**

**Reason:**

No option is selected.

**Action:**

Select at least one option to proceed.

## CAUAJM\_E\_110025

**This operating system is not supported for AutoSys installation. For a list of supported operating systems, refer to AutoSys Support Matrix.**

**Reason:**

The operating system is not supported.

**Action:**

Refer to the [support matrix](#) for supported operating systems.

## CAUAJM\_E\_110027

**The "{0}" Web UI collector name cannot exceed 255 characters.**

**Reason:**

The Web UI collector name exceeds 255 characters.

**Action:**

Specify a value that has 255 characters or fewer.

## CAUAJM\_E\_110028

The "{0}" Web UI collector name has invalid characters. Provide a valid name.

**Reason:**

The Web UI collector name contains invalid characters.

**Action:**

Specify a valid value.

## CAUAJM\_E\_110029

The "{0}" Web UI collector name is already configured. Provide a different name.

**Reason:**

The specified Web UI collector name is already in use.

**Action:**

Specify a different collector name which is not in use.

## CAUAJM\_E\_110030

Unable to delete the specified Web UI collector "{0}" as it is absent.

**Reason:**

The specified collector name does not exist.

**Action:**

Specify a collector name that exists.

## CAUAJM\_E\_110031

Server components are not supported on this platform.

**Reason:**

Server components are not supported.

**Action:**

Do not select the server components.

## CAUAJM\_E\_110035

The installation path variable AUTOSYS\_INSTALL\_LOCATION is empty. The environment is corrupted. Point the variable to the AutoSys installation directory.

**Reason:**

AUTOSYS\_INSTALL\_LOCATION is empty.

**Action:**

Specify a valid installation location for AUTOSYS\_INSTALL\_LOCATION.

## CAUAJM\_E\_110037

The "Create database" option is not supported on this platform.

**Reason:**

The Create DB option is not supported on this platform.

**Action:**

Refer to the [support matrix](#) and run on the supported platform.

## CAUAJM\_E\_110038

Unable to find the JRE directory in the AutoSys installation directory.

**Reason:**

The JRE folder was not found in the installation directory.

**Action:**

Verify whether the JRE folder is available in the installation directory.

## CAUAJM\_E\_110039

An error occurred while detecting the installed JRE.\n{0}

**Reason:**

An error occurred while detecting the installed JRE.

**Action:**

Verify whether the JRE is properly installed and compatible.

## CAUAJM\_E\_110040

AutoSys installer cannot be executed from the Universal Naming Convention (UNC) path ("{0}"). Instead, use "Map network drive" to install AutoSys.

**Reason:**

The installer was executed from the UNC path.

**Action:**

Map a network share to a drive and run the installer.

## CAUAJM\_E\_110041

A 32-bit CA EEM Server is present on this system. An upgrade of the 32-bit CA EEM Server is not supported on this platform.

**Reason:**

A 32-bit EEM server update is not supported on this platform.

**Action:**

Upgrade other AutoSys components that can be upgraded, or install the remote EEM server, migrate the policies and uninstall the 32-bit EEM server, and proceed with the installation. For more information, see [AutoSys Common Components Upgrade](#).

## CAUAJM\_E\_111001

**Invalid value is specified for attribute "{0}": {1}.**

**Reason:**

An invalid value is specified for the attribute.

**Action:**

Specify a valid value.

## CAUAJM\_E\_112001

**Unable to identify the modification mode:"{0}".**

**Reason:**

An invalid value is specified for the modification mode.

**Action:**

Specify a valid value for the modification mode.

## CAUAJM\_E\_112002

**The "Add Component" option cannot be used when all the AutoSys components are already installed.**

**Reason:**

All components are installed.

**Action:**

Add component is not applicable when all AutoSys components are already installed.

## CAUAJM\_E\_112003

**The "Add Instance" option can be used only when at least one AutoSys instance is already installed.**

**Reason:**

Add instance is applicable only when at least one AutoSys instance is already installed.

**Action:**

Install any AutoSys instance component like client, scheduler, application server or webserver, to enable add instance.

## CAUAJM\_E\_112004

**The "Modify Instance" option can be used only when at least one AutoSys instance is already installed.**

**Reason:**

Modify instance is only applicable when at least one AutoSys instance is already installed.

**Action:**

---

Install any AutoSys instance component like client, scheduler, application server or webserver, to enable modify instance.

## **CAUAJM\_E\_112005**

The "Delete Instance" option can be used only when more than one AutoSys instance is already installed.

**Reason:**

Delete instance is applicable only when there are more than one AutoSys instance.

**Action:**

Install more than one instance to enable delete instance.

## **CAUAJM\_E\_112006**

The "Add Agent" option can be used only when at least one agent is already installed.

**Reason:**

The Add Agent option can be used only when at least one agent is already installed.

**Action:**

Install at least one agent to enable add agent.

## **CAUAJM\_E\_112007**

The "Add Web UI Collector" option can be used only when the Web UI collector is already installed.

**Reason:**

Web UI Collector is not installed.

**Action:**

Install Web UI collector to enable the Add Web UI Collector option.

## **CAUAJM\_E\_112008**

The "Delete Web UI Collector" option can be used only when at least one Web UI Collector is already installed.

**Reason:**

There is only one Web UI Collector.

**Action:**

Install another Web UI collector which is not the default one to enable the Delete Web UI Collector option.

## **CAUAJM\_E\_112100**

Unable to connect to the CA EEM server due to incorrect password.\n{0}

**Reason:**

The CA EEM server password is incorrect.

**Action:**

Specify the correct password for CA EEM server.

## CAUAJM\_E\_112101

Unable to connect to the CA EEM server.\n{0}

**Reason:**

Failed to establish a connection to CA EEM server.

**Action:**

Ensure that the connection to the CA EEM server is successful.

## CAUAJM\_E\_112102

Unable to retrieve the version of the CA EEM server.

**Reason:**

Failed to retrieve version of the CA EEM server.

**Action:**

Verify whether the connection to CA EEM server is successful.

## CAUAJM\_E\_112103

The version of the CA EEM server is not supported.

**Reason:**

The CA EEM server version is not supported.

**Action:**

BBBB

Refer to the [support matrix](#) and specify the supported CA EEM server version

## CAUAJM\_E\_112104

Unable to find the "{0}" CA EEM application.

**Reason:**

he CA EEM application was not found.

**Action:**

Verify whether the specified application exists on the CA EEM server.

## CAUAJM\_E\_112108

Unable to retrieve the AutoSys instance list for the CA EEM application.

**Reason:**

Failed to retrieve the AutoSys instance list for the CA EEM application.

**Action:**

Verify whether the connection to the CA EEM server is successful and instance list exists.

## CAUAJM\_E\_112109

Unable to find any instances in the CA EEM application.

**Reason:**

No instances were found in the CA EEM application.

**Action:**

Verify whether the connection to the CA EEM server is successful and the instance list exists.

## CAUAJM\_E\_112111

An internal error occurred while interacting with the CA EEM server.\n{0}

**Reason:**

Failed to connect to the CA EEM server.

**Action:**

Make sure the connection to the CA EEM server is successful, and inspect the reason for any connectivity issues.

## CAUAJM\_E\_112113

The CA EEM application "{0}" is of a higher version.

**Reason:**

The CA EEM application is of a higher version.

**Action:**

Use the same version of the DVD as the application.

**Example:**

If the CA EEM application is of version R12, then use the R12 DVD.

## CAUAJM\_E\_112116

The available disk space is: "{0}" MB. The required disk space must be: "{1}" MB. Free up some disk space and try again.

**Reason:**

The available disk space is less than required.

**Action:**

Free up some disk space and verify there is sufficient space available.

## CAUAJM\_E\_112117

The database size "{0}" is not a number. Specify a valid value.

**Reason:**

The number is not specified.

**Action:**

The database size should be a valid number.

## CAUAJM\_E\_112118

The database size is "{0}" MB. It is less than the minimum required database size of 256 MB. Specify a valid value.

**Reason:**

Database size is less than 256MB.

**Action:**

Specify a value greater than or equal to 256MB.

## CAUAJM\_E\_112120

Unable to connect to the CA EEM server with the available certificate and key (cert:" + "{0}" + " key:" + {1} + ")")

**Reason:**

Connection to the EEM server with the certificate failed.

**Action:**

Verify whether the EEM connection is successful with the certificate. Regenerate the certificate if required and retry.

## CAUAJM\_E\_112121

An error occurred while authenticating the CA EEM server with the certificate-based authentication.\n{0}.

**Reason:**

Connection to the EEM server with the certificate failed.

**Action:**

Verify whether the EEM connection is successful with the certificate. Regenerate the certificate if required and retry.

## CAUAJM\_E\_112122

An error occurred while generating the certificate for the CA EEM server authentication.\n{0}.

**Reason:**

Failed while generating the certificate for the CA EEM server.

**Action:**

Verify whether the EEM connection is successful with the certificate. Regenerate the certificate if required and retry.

## CAUAJM\_E\_112124

The instance security state is "{0}". The REST web service can be configured only in the CA EEM security mode. Therefore, activate the CA EEM security for this instance.

**Reason:**

The instance security state is not in EEM security mode.

**Action:**



---

Change the security to EEM mode to install the REST web service.

## **CAUAJM\_E\_112125**

**CA EEM server(s) "{0}" configured for AutoSys are unreachable.**

**Reason:**

CA server(s) are unreachable.

**Action:**

Verify that the CA EEM server(s) are reachable and connecting successfully.

## **CAUAJM\_E\_112126**

**The CA EEM server(s) field value length should be less than 256 characters. Enter a valid value.**

**Reason:**

CA EEM server(s) field length exceeds 256 characters.

**Action:**

Specify a value which is less than or equal to 256 characters.

## **CAUAJM\_E\_112127**

**Unable to retrieve the version of the "{0}" CA EEM application.{1}**

**Reason:**

Failed to retrieve the version of CAA EEM application.

**Action:**

Verify that the CA EEM connection is successful and that the version is retrieved properly.

## **CAUAJM\_E\_112128**

**An error occurred while retrieving the CA EEM server information. For more information, refer to the log file.**

**Reason:**

Failed to retrieve the CA EEM server information.

**Action:**

Verify that the CA EEM connection is successful.

## **CAUAJM\_E\_112129**

**Unable to determine the security state of instance "{0}".\n {1}**

**Reason:**

Failed to determine the security state of the instance. An error occurred while retrieving the security from the database.

**Action:**

Run autosys\_secure to check the security state of the instance.

## CAUAJM\_E\_112130

The password specified for "{0}" did not match with the CA EEM server password.

**Reason:**

The specified password is wrong.

**Action:**

Specify the correct password for the CA EEM server.

## CAUAJM\_E\_112132

An error occurred during the decryption of the "{0}" value. Provide a properly encrypted value.\n{1}

**Reason:**

Failed to decrypt the value.

**Action:**

Verify that the value is properly decrypted. Use as\_config to check that the value has the proper decrypted value.

## CAUAJM\_E\_112134

The CA EEM password must contain at least 5 characters.

**Reason:**

The specified password value is less than 5 characters.

**Action:**

Specify a value which has at least 5 characters.

## CAUAJM\_E\_112135

Unable to resolve the local host name.

**Reason:**

Failed to resolve the local host name.

**Action:**

Verify that the FQDN is properly resolved on the operating system.

## CAUAJM\_E\_112136

The "{0}" local hostname does not match the DNS host name "{1}".

**Reason:**

The local hostname does not match the DNS hostname.

**Action:**

Make sure that local hostname and the DNS hostname match.

## CAUAJM\_E\_112137

The "{0}" host name is unreachable.

**Reason:**

The hostname is unreachable.

**Action:**

Verify that hostname is reachable.

## CAUAJM\_E\_112138

The "{0}" host should not resolve to the 'localhost'.

**Reason:**

The hostname should not resolve to the localhost.

**Action:**

Ensure that the hostname does not resolve to localhost.

## CAUAJM\_E\_112139

An error occurred while resolving the host name. "{0}"

**Reason:**

Failed to resolve the hostname.

**Action:**

Verify that the hostname is resolved to proper value.

Check the host file.

Linux:

```
/etc/hosts
```

Windows:

```
C:\Windows\System32\drivers\etc
```

## CAUAJM\_E\_112140

A few errors occurred while processing the CA EEM information.\n"{0}".\n For more information, refer to the log file.

**Reason:**

Multiple errors occurred while processing the CA EEM information.

**Action:**

Refer to the log file for more information.

## CAUAJM\_E\_112501

The database type "{0}" is not applicable for "{1}".

**Reason:**

The database type is not applicable.

**Action:**

Refer to the [support matrix](#) and specify the applicable database.

## CAUAJM\_E\_112503

**The Secondary Event Server name must be different than the Primary Event Server name "{0}".**

**Reason:**

The secondary and primary event server names must not be the same.

**Action:**

Specify different hostnames for the secondary and primary event server names.

## CAUAJM\_E\_112504

**The "{0}" database is already configured for instance "{1}". Specify details of another database.**

**Reason:**

The database is already configured for the other instance.

**Action:**

Specify an empty database which is not used by any other instance.

## CAUAJM\_E\_112507

**The "{0}" database schema is at a higher version.**

**Reason:**

The database schema is at a higher version.

**Action:**

Specify a database schema which is not at a higher version.

## CAUAJM\_E\_112508

**An error occurred while connecting to the "{0}" database server. \n {1}**

**Reason:**

Failed to connect to the database server.

**Action:**

Verify that the database is running and the connection is successful.

## CAUAJM\_E\_112509

**The specified database "{0}" is absent.**

**Reason:**

The specified database is not present.

**Action:**

Specify an existing database.

**CAUAJM\_E\_112510**

**An error occurred while retrieving the instance name from the "{0}" database.\n {1}**

**Reason:**

Failed to retrieve the instance name from the database.

**Action:**

Verify that the database is running and the connection is successful.

**CAUAJM\_E\_112511**

**The "{0}" database is already configured with the latest schema version.**

**Reason:**

The database is already configured with the latest schema version.

**Action:**

Specify an empty database.

**CAUAJM\_E\_112512**

**An error occurred while retrieving the AEDB\_VERSION from the "{0}" database.\n {1}**

**Reason:**

Failed to retrieve the AEDB\_VERSION from the database.

**Action:**

Verify whether the database is up and connection is successful and AEDB\_VERSION can be retrieved successfully.

**CAUAJM\_E\_112513**

**An error occurred while checking the "READ\_COMMITTED\_SNAPSHOT" option of "TRANSACTION ISOLATION LEVEL" for the database: "{0}"\n. {1}\n If turned "OFF", to avoid database deadlocks, stop all applications connected to the database(s), and set it to "ON".\n Perform this task before or after the AutoSys installation.**

**Reason:**

An error occurred while checking the READ\_COMMITTED\_SNAPSHOT.

**Action:**

Verify that the database is running and the connection is successful, and that READ\_COMMITTED\_SNAPSHOT is set to ON.

**CAUAJM\_E\_112514**

**An error occurred while checking if another scheduler is configured for the "{0}" database.\n{1}**

**Reason:**

An error occurred while checking if another scheduler is configured.

**Action:**

Verify that the database is running and the connection is successful.

**CAUAJM\_E\_112515**

Unable to connect to the "{0}" database server using the database client utility. For more information, refer to the log file.

**Reason:**

Failed to connect to the database using the client utilities.

**Action:**

Ensure that the database connection is successful using the client utilities.

**CAUAJM\_E\_112516**

An error occurred while retrieving the database server version from the "{0}" database.\n{1}

**Reason:**

Failed to retrieve the database version from database.

**Action:**

Ensure that the database connection is successful.

**CAUAJM\_E\_112517**

The current database server "{0}" version is "{1}". The minimum database version required is not met. For more information, refer the AutoSys Support Matrix.

**Reason:**

The database server is not supported.

**Action:**

Refer to the [support matrix](#) and specify the supported database version.

**CAUAJM\_E\_112519**

An error occurred while checking if the "{0}" database server is enabled for the Mixed Mode Authentication.\n{1}.

**Reason:**

Failed to check the database server mixed mode authentication.

**Action:**

Verify that the database is running and that mixed mode authentication is enabled.

**CAUAJM\_E\_112520**

The "{0}" database server is not configured for the Mixed Mode Authentication.

**Reason:**

The database server is not configured for the Mixed Mode Authentication.

**Action:**

Enable Mixed Mode Authentication for the database server.

**CAUAJM\_E\_112521**

**An error occurred while connecting to the "{0}" database server using the database client utility.\n{1}**

**Reason:**

Failed to connect to the database using the client utility.

**Action:**

Ensure the database is running and that the connection from the client utility is successful.

**CAUAJM\_E\_112523**

**An error occurred while checking if another Web UI is configured for the "{0}" database.\n{1}**

**Reason:**

Checking if another Web UI is configured for the database has failed.

**Action:**

Verify that the database is running and can retrieve the information about the Web UI configuration.

**CAUAJM\_E\_112524**

**An error occurred while retrieving the Web UI VERSION from the "{0}" database.\n{1}**

**Reason:**

Checking the Web UI version for the database has failed.

**Action:**

Verify that the database is running and can retrieve the information about the Web UI version.

**CAUAJM\_E\_112526**

**An error occurred while checking if the specified database user has the 'dbcreator' role.\n"{0}"**

**Reason:**

Checking if the specified database user has the dbcreator role has failed.

**Action:**

Verify that the database is running and can retrieve the information about the dbcreator role.

**CAUAJM\_E\_112528**

**An error occurred while checking if the specified database user has the 'securityadmin' role.\n"{0}"**

**Reason:**

Checking if the specified database user has the 'securityadmin' role has failed.

**Action:**

Verify that the database is running and can retrieve the information about the 'securityadmin' role.

## CAUAJM\_E\_112530

**An error occurred while retrieving the MSSQL server instance name of the "{0}" database server.\n{1}**

**Reason:**

Failed to retrieve the MSSQL server instance name for the specified database server.

**Action:**

Verify that the database is running and can retrieve the information about the MSSQL server instance name.

## CAUAJM\_E\_112531

**The MSSQL server instance name "{0}" detected from the database does not match with the specified value. Correct the instance name and retry.**

**Reason:**

The specified name and the MSSQL instance name do not match.

**Action:**

Specify the correct value for the MSSQL instance name.

## CAUAJM\_E\_112532

**An error occurred while checking the existence of the "{0}" user in the "{1}" database server.\n {2}**

**Reason:**

Checking for the existence of the user in the database has failed.

**Action:**

Verify that the database is running and can retrieve the information about the user in the database.

## CAUAJM\_E\_112533

**The specified database "{0}" does not exist.**

**Reason:**

The database does not exist.

**Action:**

Specify a database that exists.

## CAUAJM\_E\_112534

**The "{0}" database is at a lower schema version. First upgrade the database schema.**

**Reason:**

The database is at a lower schema version.

**Action:**

First, upgrade the scheduler machine, and then any other database-dependent components like the application server or REST web server.



## CAUAJM\_E\_112535

The specified database user "{0}" is present. However, AutoSys failed to connect to the "{1}" database server.  
\n{2}

**Reason:**

Connecting with the specified user failed.

**Action:**

Specify a valid value for the user password to ensure that the connection is successful.

## CAUAJM\_E\_112536

An error occurred while checking if the specified database user has the 'db\_owner' role.\n{1}

**Reason:**

Checking if the specified database user has the db\_owner role has failed.

**Action:**

Verify that the database is running and can retrieve the information about db\_owner role.

## CAUAJM\_E\_112537

The specified database user does not have the 'UJOADMIN' role.

**Reason:**

The database user does not have the UJOADMIN role.

**Action:**

Ensure that the database user has the UJOADMIN role.

## CAUAJM\_E\_112538

An error occurred while checking if the specified database user has the 'UJOADMIN' role.\n{1}

**Reason:**

Checking if the specified database user has UJOADMIN role has failed.

**Action:**

Verify that the database is running and can retrieve the information about the UJOADMIN role.

## CAUAJM\_E\_112539

A few warnings occurred while processing the database information of the below instances. For more information, refer to the log file.\n"{0}"

**Reason:**

Multiple warnings occurred while processing the database information of the instances.

**Action:**

Refer to the log for details.

## CAUAJM\_E\_112542

**The Oracle home directory does not exist.**

**Reason:**

The Oracle home directory does not exist.

**Action:**

Ensure that the Oracle home directory exists and that you are pointing to a valid directory.

## CAUAJM\_E\_112543

**Unable to find the "{0}" file.**

**Reason:**

The file was not found.

**Action:**

Ensure that the file is available.

## CAUAJM\_E\_112544

**The Oracle wallet jar path does not exist at "{0}".**

**Reason:**

The Oracle wallet jar path does not exist.

**Action:**

Ensure that Oracle wallet jar path exists in the specified location.

## CAUAJM\_E\_112545

**The Oracle wallet JARs: oraclepki.jar, osdt\_cert.jar, and osdt\_core.jar do not exist at "{0}".**

**Reason:**

The Oracle wallet JARs are not available.

**Action:**

Ensure that the following Oracle wallet JARs are present in the specified directory: oraclepki.jar, osdt\_cert.jar, and osdt\_core.jar.

## CAUAJM\_E\_112546

**Sybase home does not exist at "{0}".**

**Reason:**

Sybase home does not exist.

**Action:**

Ensure that the specified directory is pointing to a valid Sybase home directory.

## CAUAJM\_E\_112547

Unable to find the SYBASE\_OCS variable.

**Reason:**

Failed to retrieve the SYBASE\_OCS variable.

**Action:**

Ensure that the SYBASE\_OCS variable is available in the environment.

## CAUAJM\_E\_112548

The Sybase interface file does not exist at "{0}".

**Reason:**

Unable to find the Sybase interface file.

**Action:**

Ensure that the Sybase interface file exists in the specified path.

## CAUAJM\_E\_112549

The "JDBC JAR path" field must contain the absolute path to the JConn JAR file including the file name. Or, the JAR file does not exist at "{0}".

**Reason:**

The JDBC JAR path does not exist.

**Action:**

Specify the Jconn absolute path to the jconn jar file, including the file name.

## CAUAJM\_E\_112550

Unable to find the oci.dll file under the Oracle home directory.

**Reason:**

The oci.dll file was not found.

**Action:**

Ensure that the oci.dll file is available in the %ORACLE\_HOME% or %ORACLE\_HOME%\bin directory.

## CAUAJM\_E\_112551

Unable to find the libclntsh.so file under the Oracle home directory.

**Reason:**

The libclntsh.so file was not found.

**Action:**

Ensure that the libclntsh.so file is available in the \$ORACLE\_HOME or \$ORACLE\_HOME/lib or \$ORACLE\_HOME/bin directory.

## CAUAJM\_E\_112552

**Sybase libraries: libsybintl64, libsybtcl64, libsybcomn64, libsybcs64, and libsybct64 do not exist.**

**Reason:**

The Sybase libraries were not found.

**Action:**

Ensure that the Sybase libraries are available in the %SYBASE\_OCS%\dll or %SYBASE\_OCS%\bin directory.

## CAUAJM\_E\_112553

**Sybase libraries: libsybintl\_r64, libsybtcl\_r64, libsybcomn\_r64, libsybcs\_r64, and libsybct\_r64 do not exist.**

**Reason:**

The Sybase libraries were not found.

**Action:**

Ensure that the Sybase libraries are available in the \$SYBASE\_OCS/lib directory.

## CAUAJM\_E\_112554

**The Sybase page size is too small to run the selected components.**

**Reason:**

The page size is too small.

**Action:**

Change the page size to greater than or equal to 4k.

## CAUAJM\_E\_112557

**The "{0}" database is not a pluggable database.**

**Reason:**

The specified database is not a pluggable database.

**Action:**

Specify a database that pluggable.

## CAUAJM\_E\_112558

**The Oracle user does not have a create role.**

**Reason:**

The Oracle user does not have the create role.

**Action:**

Specify an Oracle user who has the create role.

## CAUAJM\_E\_112559

**Oracle Managed Files (OMF) is disabled**

**Reason:**

Oracle Managed Files (OMF) is disabled.

**Action:**

Configure Oracle Managed Files (OMF).

## CAUAJM\_E\_112560

**Automation Storage Management (ASM) is disabled.**

**Reason:**

Automation Storage Management (ASM) is disabled.

**Action:**

Specify a database setup that supports Automation Storage Management (ASM).

## CAUAJM\_E\_112563

**The 'sso\_role' role has not been assigned to the admin user.**

**Reason:**

The admin user does not have the sso\_role.

**Action:**

Provide a user that has been assigned the sso\_role.

## CAUAJM\_E\_112564

**The 'sa\_role' role has not been assigned to the admin user.**

**Reason:**

The admin user does not have the sa\_role.

**Action:**

Provide a user that has been assigned the sa\_role.

## CAUAJM\_E\_112565

**The schema user is not having the following privileges: 'CREATE SESSION', 'CREATE TABLE', 'CREATE VIEW', 'CREATE SEQUENCE', and 'CREATE PROCEDURE'.**

**Reason:**

The schema user does not have the following privileges:

- CREATE SESSION
- CREATE TABLE
- CREATE VIEW
- CREATE SEQUENCE
- CREATE PROCEDURE

**Action:**

Provide a schema user that has the listed privileges, or grant the required privileges to the schema user.

## CAUAJM\_E\_112566

The schema user is not having the following privileges: 'CREATE SESSION', 'CREATE TABLE', and 'CREATE SEQUENCE'.

**Reason:**

The schema user does not have the following privileges:

- CREATE SESSION
- CREATE TABLE
- CREATE SEQUENCE

**Action:**

Provide a schema user that has the listed privileges, or grant the required privileges to the schema user.

## CAUAJM\_E\_112567

An error occurred while connecting to the database server using the "{0}" service name.\n{1}

**Reason:**

An error occurred while connecting to the database server using the service name.

**Action:**

Check the database server details to resolve the error.

## CAUAJM\_E\_112569

An attempt to update the number of user connections in the Sybase database has failed. Manually increase the number of Sybase user connections and then rerun the install. For example, to increase the number of connections to 150, run the following query: sp\_configure "number of user connections", 150.

**Reason:**

An attempt to update the number of user connections in the Sybase database has failed.

**Action:**

Manually increase the number of Sybase user connections, and rerun the installation.

**Example:** To increase the number of connections to 150, run the following query:

```
sp_configure "number of user connections", 150.
```

## CAUAJM\_E\_112570

The specified credentials do not correspond to the "{0}" user.

**Reason:**

The user name provided does not match the user name mapped to the credentials in the database.

**Action:**

Provide the correct user credentials.

**CAUAJM\_E\_112571**

**The specified TNS admin directory does not exist at "{0}" or is not a valid directory.**

**Reason:**

The specified TNS admin directory does not exist or is not a valid directory.

**Action:**

Provide a valid directory.

**CAUAJM\_E\_112572**

**The specified wallet location does not exist at "{0}" or is not a valid file.**

**Reason:**

The specified wallet location does not exist or is not a valid file.

**Action:**

Provide a valid wallet file.

**CAUAJM\_E\_112573**

**Specify an Oracle user and its password. Or, specify a TNS\_ADMIN that has been granted the DBA ROLE, DBA\_TABLESPACES SELECT, and CREATE SESSION authorities.**

**Reason:**

The Oracle admin user must have the following authorities:

- DBA ROLE
- DBA\_TABLESPACES SELECT
- CREATE SESSION

**Action:**

Provide a user with the required authorities.

**CAUAJM\_E\_112574**

**The specified "{0}" data device does not exist.**

**Reason:**

Unable to find the data device name in the database.

**Action:**

Specify another data device name *or* select the **Create the database devices** option to continue with the installation.

## CAUAJM\_E\_112575

The specified "{0}" log device does not exist.

**Reason:**

Unable to find the log device name in the database.

**Action:**

Specify another log device name *or* select the **Create the database devices** option to continue with the installation.

## CAUAJM\_E\_112576

The data device size must be at least 800 MB.

**Reason:**

The data device to be created must be at least 800 MB in size. If you are using an existing data device, it must have at least 800 MB free disk space.

**Action:**

Provide a value greater than 800 MB.

## CAUAJM\_E\_112577

The log device size must be at least 100 MB.

**Reason:**

The log device to be created must be at least 100 MB in size. If you are using an existing device, it must have at least 100 MB free disk space.

**Action:**

Provide a value greater than 100 MB.

## CAUAJM\_E\_112578

Unable to find the data tablespace name in the Oracle database. Specify another data tablespace name. Or, select the "Create tablespace, database user, and roles" option to continue with the installation.

**Reason:**

Unable to find the data tablespace name in the Oracle database.

**Action:**

Specify another data tablespace name *or* select the **Create tablespace, database user, and roles** option to continue with the installation.

## CAUAJM\_E\_112579

Unable to find the index tablespace name in the Oracle database. Specify another index tablespace name. Or, select the "Create tablespace, database user, and roles" option to continue with the installation.

**Reason:**

Unable to find the index tablespace name in the Oracle database.

**Action:**



Specify another index tablespace name *or* select the **Create tablespace, database user, and roles** option to continue with the installation.

## CAUAJM\_E\_112580

**The data tablespace size must be at least 800 MB.**

### Reason:

The data tablespace to be created must be at least 800 MB in size.

### Action:

Provide a value greater than or equal to 800 MB.

## CAUAJM\_E\_112581

**The index tablespace size must be at least 80 MB.**

### Reason:

The index tablespace to be created must be at least 80 MB in size.

### Action:

Provide a value greater than or equal to 80 MB.

## CAUAJM\_E\_112582

**Unable to find the "{0}" data disk group. Ensure that the name is written correctly.**

### Reason:

Unable to find the data disk group

### Action:

Provide an existing disk group. Check if the disk group is present by running the following query.

```
SELECT * from v$parameter where name = 'db_create_file_dest' and upper(value) = upper(<disk group name>)
```

## CAUAJM\_E\_112583

**Unable to find the "{0}" index disk group. Ensure that the name is written correctly.**

### Reason:

Unable to find the index disk group

### Action:

Provide an existing disk group. Check if the disk group is present by running the following query.

```
SELECT * from v$parameter where name = 'db_create_file_dest' and upper(value) = upper(<disk group name>)
```

## CAUAJM\_E\_112584

**The Data and Index tablespace's name and path should be different. Specify a different name or directory.**

### Reason:

The name and the location in which the tablespaces have to be created must be different for Data and Index.

**Action:**

Provide a different name or location.

**CAUAJM\_E\_112587**

**The name and group of Data and Index tablespaces should be different. Specify a different name or group.**

**Reason:**

The name and the group in which the tablespace have to be created must be different for Data and Index.

**Action:**

Provide a different name or group.

**CAUAJM\_E\_112590**

**An error occurred while altering the database session to "{0}" for the "{1}" database server.\n{2}**

**Reason:**

An error occurred while altering the database session.

**Action:**

Make sure that the user who is the schema admin user is the owner of the schema, and retry.

**CAUAJM\_W\_112593**

**The "{0}" database schema is at a lower version. If configured to start post-installation, the AutoSys services will fail to start.**

**Reason:**

The present schema in the database is at a lower version. This path will not upgrade the database, and the installer will not start the services post-installation (as the services would fail to start due to schema incompatibilities).

**Action:**

Start the services after the database schema is upgraded to the installer level.

**CAUAJM\_E\_112595**

**The specified database "{0}" does not have the schema defined.**

**Reason:**

The database does not have the AutoSys schema with tables and other objects, but the **Create the database schema** is not selected.

**Action:**

Provide a database with the schema already created or select the **Create the database schema** option and proceed with the installation.

**CAUAJM\_E\_112596**

**The Microsoft SQL Server database is not supported on this platform. Specify a different database type.**

**Reason:**

The Microsoft SQL Server database is not supported on this platform.

**Action:**

Refer to the [support matrix](#) and specify a supported database type.

## CAUAJM\_E\_112599

**The database schema exists. However, the specified database user "{0}" is absent.**

**Reason:**

The AutoSys schema with tables and other objects is already present in the database, but the application user provided is not present.

**Action:**

Provide an application user which is already defined to be used with the database schema.

## CAUAJM\_E\_112600

**The "Use reporting module for Web UI" option did not match the reporting module value present in the database, which is set to "{0}". Select the "Use reporting module for Web UI" option if the value is already "true". Otherwise, remove the option.**

**Reason:**

The **Use reporting module for Web UI** option did not match the reporting module value present in the database.

**Action:**

Select the **Use reporting module for Web UI** option if the value is already TRUE.

Otherwise, remove the option.

## CAUAJM\_E\_112601

**An error occurred while retrieving the reporting configuration from the "{0}" database server.\n{1}**

**Reason:**

An error occurred while retrieving the reporting configuration from the database server.

**Action:**

Address the database connection error and retry the installation.

## CAUAJM\_E\_112602

**The schema user is not having the SELECT\_CATALOG\_ROLE.**

**Reason:**

The schema admin user must have the SELECT\_CATALOG\_ROLE.

**Action:**

Assign the SELECT\_CATALOG\_ROLE to the schema admin user.

## CAUAJM\_E\_112603

An error occurred while connecting to the "{0}" database server. This is due to the corrupted JAAS.conf file "{1}".  
\\n {2}

**Reason:**

An error occurred while connecting to the database server, because the JAAS.conf file is corrupted.

**Action:**

Modify the JAAS.conf in order for the database server connection to succeed. The file is located in the following path:

- R11.3.X : <Installation Path>\autosys\config\cmplapp\JAAS.conf
- R12: <Installation Path>\autosys\config\JAAS.conf

## CAUAJM\_E\_112604

The specified database user "{0}" is absent.

**Reason:**

The **Create the tablespace, database user, and roles** option was selected, but the database user is absent.

**Action:**

Provide a user which is already defined or select the **Create the tablespace, database user, and roles** option and proceed.

## CAUAJM\_E\_112605

An error occurred while fetching the "{0}" database device size.\\n{1}

**Reason:**

An error occurred while fetching the free database device size.

**Action:**

Refer to the log file and resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

## CAUAJM\_E\_112606

The specified login user "{0}" is already present. However, AutoSys is unable to connect to the database server "{1}".\\n{2}

**Reason:**

The specified login user is already present with different credentials.

**Action:**

Provide the correct credentials.

## CAUAJM\_E\_112607

The specified database user doesn't have the 'db\_owner' role.

**Reason:**

The admin user must have the db\_owner role.

**Action:**

Provide a user which has the db\_owner role.

**CAUAJM\_E\_113001**

**Select at least one component to install.**

**Reason:**

No components were specified to be installed.

**Action:**

Select at least one component to install.

**CAUAJM\_E\_113002**

**Unable to detect the installation mode: "{0}".**

**Reason:**

Unable to detect whether the machine is in FRESH, UPGRADE or MAINTENANCE mode.

**Action:**

Refer to log file and resolve the issue. Depending on your installation process, check the following possible causes:

- If you are performing a silent install, verify that the silent install check value is correct.
- If you are upgrading to R12, check the registry or PIF install records.
- If you are in maintenance mode, check that the environment variables are set correctly and that the global variables file exists.

If you cannot resolve the issue, contact Broadcom Support.

**CAUAJM\_E\_113003**

**A few errors occurred while validating the AutoSys component(s). For more information, refer to the log file.**

**Reason:**

Multiple errors occurred while validating the AutoSys component(s).

**Action:**

Refer to the log file and resolve the issues.

**CAUAJM\_E\_113005**

**The AutoSys installer has detected a CA EEM server that is already installed. To upgrade the CA EEM server, execute the setup file available at <ISO>\modules\EEM.**

**Reason:**

A CA EEM server is already installed on the machine (possibly from directly executing the CA EEM server installer).

**Action:**

To upgrade the CA EEM server, execute the setup file available at <ISO>\modules\EEM.

## CAUAJM\_E\_113006

**A discrepancy is detected in the installed components. For more information, refer to the log file.**

**Reason:**

The installer detected that CA EEM and Command Sponsor and the configuration and bitness of iGateway differ, indicating that the product is in a corrupted status.

**Action:**

Refer to the log file to resolve the issue.

## CAUAJM\_E\_113010

**The iGateway configuration is corrupted. For more information, refer to the log file.**

**Reason:**

The iGateway configuration is corrupted.

**Action:**

Refer to the log file.

## CAUAJM\_E\_113012

**An internal error occurred. For more information, refer to the log file.**

**Reason:**

An internal error has occurred.

**Action:**

Refer to the log file to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

## CAUAJM\_E\_113014

**The following component(s) are not supported on this platform: "{0}". For more information, refer to the AutoSys Support Matrix.**

**Reason:**

The components are not supported on the platform.

**Action:**

Refer to the [support matrix](#) and modify the selection.

## CAUAJM\_E\_113015

**"{0}" is not installed. Use the "Add Component" option and install the component.**

**Reason:**

You can only enable a component in the Add Instance and Modify Instance modes if it is already installed on the machine.

**Action:**

Use the **Add Component** option and install the component.

## CAUAJM\_E\_113016

**"{0}" is already enabled for this instance.**

**Reason:**

The instance-specific component is already enabled for this instance.

**Action:**

No action is required.

## CAUAJM\_E\_113017

**"{0}" is not an instance-specific component.**

**Reason:**

The component is not an instance-specific component (Scheduler, Application Server, Client and REST web service). Only they can be enabled in the Add Instance and Modify Instance modes.

**Action:**

Use the **Add component** mode to install non instance-specific components:

- Agent
- Software Development Kit (SDK)
- Secure Socket Adapter (SSA)
- SOAP web service
- Web UI (WCC)
- Web UI Collector
- CA Embedded Entitlements Manager (CA EEM)

## CAUAJM\_E\_113018

**Perl is not set in the operating system environment variable PATH.**

**Reason:**

Either Perl is not installed on the machine, or the PATH is not configured to point to the Perl binary.

**Action:**

Make sure Perl is installed on the machine and that the PATH is configured to point to a Perl binary.

## CAUAJM\_E\_113019

**Keytool is absent at the specified path "{0}". Select a different JVM.**

**Reason:**

The keytool utility is not present in the JRE selected for the AutoSys installation.

**Action:**

The keytool utility is used by Autosys, so provide a JRE that has the keytool.

## CAUAJM\_E\_113020

**The dependent component(s) of the installed AutoSys components is missing: "{0}"**

**Reason:**

The dependent component(s) of the installed AutoSys components is missing.

**Action:**

If the product is functioning, check the following configuration to see if the dependent components are also properly registered.

- **Version R11.3.X**
  - (Windows) Check the following Windows registry path:  
HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\ComputerAssociates\Unicenter AutoSys JM\11.3\<instance name>\CurrentVersion
    - Scheduler: EventProcessorInstalled
    - Application Server: AppServerInstalled
    - REST Webserver: WebServerInstalled
    - Client: ConsoleInstalled
  - (UNIX) The services are present in the box:
    - Scheduler: waee\_sched.<instance name>
    - Application Server: waee\_server.<instance name>
    - REST Webserver: waee\_webserver.<instance name>
    - Client: AutoSys client binary
- **Version R12:** (Windows and UNIX) Check <AUTOUSER>\vars<InstanceName> for the following entries:
  - IA\_AE\_INSTALLED\_SCHEDULER = <1/0>
  - IA\_AE\_INSTALLED\_APPLICATIONSERVER = <1/0>
  - IA\_AE\_INSTALLED\_CLIENT = <1/0>
  - IA\_AE\_INSTALLED\_RESTWS = <1/0>

## CAUAJM\_E\_113021

**A 32bit iGateway being used by other products found. Cannot proceed with CA EEM installation or upgrade.**

**Reason:**

An iGateway 32-bit installation that is being used by other products was found, so iGateway could not be upgraded to 64-bit installation.

**Action:**

Check the other products that are using iGateway to see if it can be freed up so that it can be used for AutoSys installation.

## CAUAJM\_E\_113022

**iGateway 32-bit installation is detected. Cannot install/upgrade the CA EEM.**

**Reason:**

iGateway 32-bit installation was detected. CA EEM server uses a 64-bit iGateway for communication. A 32-bit and 64-bit iGateway cannot co-exist.

**Action:**

If no other process is consuming iGateway, uninstall the 32-bit iGateway and rerun the installer.



## CAUAJM\_E\_114001

The "{0}" port is outside the range of 1024-65535. Specify a different value for "{1}".

**Reason:**

The acceptable values for the port are in the range of 1024-65535. The value provided is outside this range.

**Action:**

Provide a number that is within the correct range.

## CAUAJM\_E\_114002

The "{0}" port is already in use. Specify a different value for "{1}".

**Reason:**

Another process is running on the machine which is listening on the port specified.

**Action:**

Provide a port that is not being used.

## CAUAJM\_E\_114003

The "{0}" port for "{1}" must be outside the virtual port range (49152-50176). Specify a different port.

**Reason:**

The port should not be in the virtual port range (49152-50176).

**Action:**

Provide a value which is in the range 1024 – 49151 or 50177 – 65535.

## CAUAJM\_E\_114004

The "{0}" port is reserved for multiplexing. Specify a different value for "{1}".

**Reason:**

The specified port number is reserved for multiplexing use for the product.

**Action:**

Provide another valid port.

## CAUAJM\_E\_114007

The following host(s) are unreachable:\n"{0}"

**Reason:**

The host(s) are not reachable from the machine.

**Action:**

Make sure that the host names provided are valid and reachable for the AutoSys product to work properly.

## CAUAJM\_E\_114008

The "{0}" port is already specified for another attribute during the AutoSys installation. Specify a different value for "{1}".

**Reason:**

The port is already specified for another attribute during the AutoSys installation. All the ports used should be unique.

**Action:**

Provide a unique port value.

## CAUAJM\_E\_114009

The event server port "{0}" is invalid.

**Reason:**

The port value specified in the event server field (EventServer\_1/ EventServer\_2) of the instance configuration file is not a number.

**Action:**

Check the field and correct the configuration file.

## CAUAJM\_E\_114010

The "{0}" port for "{1}" is invalid.

**Reason:**

The value provided is not a number.

**Action:**

Provide a port number which is in the range of 1024-65535.

## CAUAJM\_E\_114011

The "{0}" port is not in the virtual port range (49152 to 50176). Specify a different port.

**Reason:**

The port should be in the virtual port range (49152-50176).

**Action:**

Provide a value which is in the range (49152-50176).

## CAUAJM\_E\_114501

The "{0}" AutoSys instance contains Command Sponsor without the Application Server.

**Reason:**

Found an AutoSys instance that has Command Sponsor and no Application Server component. This is not a valid component combination.

**Action:**

Check that the instance is properly configured.

## CAUAJM\_E\_114502

A few errors occurred while processing the command sponsor information for the below instances. For more information, refer to the log file. {0}

**Reason:**

Multiple errors occurred while processing the Command Sponsor information for the listed instances.

**Action:**

Refer to the log file to resolve the issues. If you cannot resolve the issues, contact Broadcom Support.

## CAUAJM\_E\_114503

The Port field for instance(s) "{0}" should contain a valid port number.

**Reason:**

No port value was provided for installing the REST web server for the instance(s).

**Action:**

Provide a unique port value for each REST web server.

## CAUAJM\_E\_115001

The specified installation path is already in use: "{0}". Specify a different path.

**Reason:**

The installation path is not empty.

**Action:**

Make sure the installation path is empty, or use the Custom mode to provide a different path for installation.

## CAUAJM\_E\_115002

The installation path is invalid: "{0}". Specify a valid path.

**Reason:**

The installation path provided is an invalid path.

**Action:**

Provide a valid path.

## CAUAJM\_E\_115003

The default installation path is already in use: "{0}". Use the custom mode for AutoSys installation.

**Reason:**

In Express mode, the default installation path (C:\Program Files\CA\WorkloadAutomationAE, /opt/CA/WorkloadAutomationAE) must be empty.

**Action:**

Make sure the default installation path is empty, or use the Custom mode to provide a different path for installation.

## CAUAJM\_E\_115004

**Cannot specify the root directory as the installation path. Specify a valid path.**

**Reason:**

The root directory cannot be specified as the installation path.

**Action:**

The AutoSys product cannot be installed on the root path. Provide a subdirectory for installation.

## CAUAJM\_E\_115005

**The available disk space is: "{0}" MB. The minimum required disk space is: "{1}" MB. Free some space and try again.**

**Reason:**

The available disk space is not sufficient for the installation.

**Action:**

Provide a path that has sufficient free disk space.

## CAUAJM\_E\_115006

**The available disk space is: "{0}" MB. The minimum required disk space is: "{1}" MB. Free some space in the TEMP directory and try again.**

**Reason:**

The available disk space in the TEMP directory is not sufficient for the installer.

**Action:**

Make sure that the TEMP directory has sufficient empty disk space. In the case of UNIX, set IATEMPDIR to a directory that has sufficient space to be used as the TEMP directory.

## CAUAJM\_E\_115007

**Insufficient user privileges. Must have the Write access to the path: "{0}".**

**Reason:**

There is no write access permission to the path specified.

**Action:**

Provide a path that the user has write access to.

## CAUAJM\_E\_115008

**The USER\_INSTALL\_DIR "{0}" value must match with the installation path "{1}" value.**

**Reason:**

The USER\_INSTALL\_DIR variable in the silent response file does not match the detected AutoSys installation path.

**Action:**

---

Check if the AUTOSYS\_INSTALL\_LOCATION environment variable and USER\_INSTALL\_DIR variable are set to the AutoSys installation location and retry.

## CAUAJM\_E\_115009

**An internal error occurred. For more information, refer to the log file. Or, contact Broadcom Support.**

**Reason:**

Failed to detect the AutoSys Installation path.

**Action:**

Check if the AUTOSYS\_INSTALL\_LOCATION environment variable is set to the AutoSys installation location and retry.

## CAUAJM\_E\_115010

**The installation path "{0}" contains a space. Specify a valid path without any space.**

**Reason:**

The installation path contains a space character. Spaces are not allowed in the installation path.

**Action:**

Provide an installation path without spaces.

## CAUAJM\_E\_116001

**An error occurred while retrieving the CA EEM server information.\n"{0}" Verify the database connection and try again.\n**

**Reason:**

Unable to communicate with the database to retrieve the CA EEM server information.

**Action:**

Check the database connectivity, and check if the instance is properly configured with a CA EEM server.

## CAUAJM\_E\_116003

**The configuration file is corrupted. "{0}"\n Unable to proceed with the selected AutoSys instance or Web UI. Check the configuration file.**

**Reason:**

The configuration file is corrupted.

**Action:**

Fix the corruption and retry the installer.

## CAUAJM\_E\_116005

**The specified AutoSys instance does not exist. Specify another instance name.**

**Reason:**

The specified AutoSys instance (IA\_AE\_INSTANCE\_NAME) does not exist.

**Action:**

The instance name is not installed yet. Use the **Add Instance** option to create the instance.

## CAUAJM\_E\_116006

**Unable to retrieve the components enabled for instance "{0}".**

### Reason:

Unable to retrieve the components enabled for the instance.

### Action:

If the instance components are running properly, check the following location to ensure that the installer properly detects the components enabled for the instance.

- **Version R11.3.X**
  - (Windows) Check the following Windows registry path:  
HKEY\_LOCAL\_MACHINE\SOFTWARE\WOW6432Node\ComputerAssociates\Unicenter AutoSys JM  
\11.3\<instance name>\CurrentVersion
    - Scheduler: EventProcessorInstalled
    - Application Server: AppServerInstalled
    - REST Webserver: WebServerInstalled
    - Client: ConsoleInstalled
  - (UNIX) The services are present in the box:
    - Scheduler: waae\_sched.<instance name>
    - Application Server: waae\_server.<instance name>
    - REST Webserver: waae\_webserver.<instance name>
    - Client: AutoSys client binary
- **Version R12:** (Windows and UNIX) Check <AUTOUSER>\vars<InstanceName> for the following entries:
  - IA\_AE\_INSTALLED\_SCHEDULER = <1/0>
  - IA\_AE\_INSTALLED\_APPLICATIONSERVER = <1/0>
  - IA\_AE\_INSTALLED\_CLIENT = <1/0>
  - IA\_AE\_INSTALLED\_RESTWS = <1/0>

## CAUAJM\_E\_116007

**All components are already enabled for instance "{0}".**

### Reason:

All the installed instance-specific components are already enabled for the mentioned instance.

### Action:

If you intend to install a new component on the machine, use the **Add Component** option.

## CAUAJM\_E\_116008

**The configuration file is corrupted. "{0}"\n Unable to proceed with the selected Web UI collector. Check the configuration file in the Web UI collector directory at <install\_dir>\WCC\collectors.**

### Reason:

The Web UI collector configuration file <CA\_WCC\_INSTALL\_LOCATION>\collectors\<collector name>\data\config\application\config\resources\connection.properties is corrupted.

**Action:**

Check the properties files to address the problem mentioned in the message, and rerun the installer.

**CAUAJM\_E\_116100**

The "{0}" port(s) are in an invalid format.

**Reason:**

The application server ports provided are not in a valid format.

**Action:**

If you provide multiple ports, list them as comma-separated numbers.

**CAUAJM\_E\_116101**

The "{0}" port is not a number. Specify a valid value for "{1}".

**Reason:**

The port must be a number.

**Action:**

Provide a number in the range of 1024 to 65535 for port.

**CAUAJM\_E\_116102**

Host name(s) are in an invalid format.

**Reason:**

Host name(s) are in an invalid format.

**Action:**

If you provide multiple host names, specify them separated by commas.

**CAUAJM\_E\_116201**

The "{0}" owner name is invalid. Allowed characters are [a-z][A-Z]\_[0-9].

**Reason:**

The owner name is invalid. The owner name can have only the following characters: [a-z][A-Z]\_[0-9].

**Action:**

Make sure that the owner name only contains allowed characters.

**CAUAJM\_E\_116202**

The "{0}" group name is invalid. Allowed characters are [a-z][A-Z]\_[0-9].

**Reason:**

The group name is invalid. The group name can have only the following characters: [a-z][A-Z]\_[0-9].

**Action:**

Make sure that the group name only contains allowed characters.

## CAUAJM\_E\_116401

The "{0}" agent name cannot exceed 16 characters.

**Reason:**

The agent name cannot exceed 16 characters.

**Action:**

Provide an agent name which has 16 characters or less.

## CAUAJM\_E\_116402

The "{0}" agent name has invalid characters. Provide a valid name.

**Reason:**

The agent name has invalid characters.

**Action:**

Make sure that the agent name adheres to the following format.

- Windows: ^[^\|\\\*!?!:;\"'<>\\|]+\$
- UNIX: ^[^\|\\`~]+\$

## CAUAJM\_E\_116403

The "{0}" agent name is already configured. Provide a different name.

**Reason:**

Agent names must be unique. Another agent with the same name is already installed on the machine.

**Action:**

Provide a unique agent name.

## CAUAJM\_E\_116404

An error occurred while taking a backup of AutoSys. {0}

**Reason:**

An error occurred while backing AutoSys up.

**Action:**

Try to resolve the problem by looking at the message, and retry the installation.

## CAUAJM\_E\_116408

An error occurred while creating or upgrading the {0} database. For more information, refer to the log file.

**Reason:**

An error occurred while creating or upgrading the database.

**Action:**

Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.



## CAUAJM\_E\_116413

An error occurred while creating the "{0}" owner. {1}

**Reason:**

Unable to create the owner.

**Action:**

Create the owner manually by running the following command, and retry the installation.

**Example:**

Linux:

```
useradd -g <group name> <user name>
```

## CAUAJM\_E\_116419

An error occurred while creating the "{0}" group. {1}

**Reason:**

Unable to create the group.

**Action:**

Create the group manually by running the following command, and add the user to this group.

- AIX:

```
mkgroup <groupname>
```

- Others:

```
groupadd <groupname>
```

## CAUAJM\_E\_116421

Failed granting database privileges to WebUI database user. {0}

**Reason:**

Failed to grant database privileges to WebUI database user.

**Action:**

Manually grant the following permissions to the Web UI database user for the WebUI schema tables and sequences:

- SELECT
- INSERT
- UPDATE
- DELETE
- ALTER
- SEQ\_GEN\_SEQUENCE

## CAUAJM\_E\_116601

An invalid URL format is specified.

**Reason:**

An invalid URL format is specified.

**Action:**

Provide a properly formatted URL.

## CAUAJM\_E\_116603

An invalid value is specified for the "Enterprise Site ID" field.

**Reason:**

An invalid value is specified for the **Enterprise Site ID** field.

**Action:**

Provide the Enterprise Site ID registered with Broadcom Support.

## CAUAJM\_E\_116604

An invalid value is specified for the "Company Domain" field.

**Reason:**

An invalid value is specified for the "Company Domain" field.

**Action:**

Provide the company domain, which is in the format `^[a-zA-Z0-9]{1,}([-]{0,}[a-zA-Z0-9]{1,}){0,}[.][a-zA-Z]{1,}$`

## CAUAJM\_E\_116606

The selected cluster group name "{0}" is absent in the available cluster group names "{1}".

**Reason:**

The cluster group name specified to register the AutoSys services is not available on the machine.

**Action:**

Provide a cluster group which is already present on the machine.

## CAUAJM\_E\_116607

The cluster resource groups are absent on the system.

**Reason:**

There are no cluster resource groups available on the system.

**Action:**

Provide a resource group which has resources (IP address, network, physical disk, etc.) that are only used for AutoSys services. If such a resource group is not already available, create one to configure services for the cluster.

## CAUAJM\_E\_116701

The AutoSys installer does not bundle a JVM for this operating system. Specify a pre-installed JRE.

**Reason:**

The AutoSys installer does not bundle a JVM for this operating system.

**Action:**

Refer to the [support matrix](#) and provide a valid installed JRE.

**CAUAJM\_E\_116702**

**The specified JRE "{0}" is incompatible. Specify a compatible pre-installed JRE.**

**Reason:**

The specified JRE is incompatible.

**Action:**

Refer to the [support matrix](#) and provide a valid installed JRE or use the bundled JRE option.

**CAUAJM\_E\_116703**

**The specified JRE path "{0}" is inside the AutoSys installation path "{1}". Specify another pre-installed JRE.**

**Reason:**

The specified JRE path is inside the AutoSys installation.

**Action:**

Provide a JRE which is installed outside of the AutoSys installation path.

**CAUAJM\_E\_116704**

**An error occurred while validating the specified JRE "{0}".\n{1}**

**Reason:**

An IOException occurred while validating the specified JRE.

**Action:**

Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

**CAUAJM\_E\_116705**

**The executable JRE specified at "{0}" is absent in the JRE home specified at "{1}".**

**Reason:**

The executable JRE (ex: java.exe) is not in the sub folder of the JRE home.

**Action:**

Provide an executable which is present in the sub folder JRE home.

**CAUAJM\_E\_119001**

**Unable to resolve the specified path "{0}". Specify a valid path.**

**Reason:**

Unable to resolve the specified path as file does not exist or an I/O error occurs.

**Action:**

Make sure that the file exists and retry.

## CAUAJM\_E\_119002

**The specified directory or file "{0}" is absent. Specify a valid directory or file.**

**Reason:**

The specified directory or file does not exist.

**Action:**

Make sure that the file exists and retry.

## CAUAJM\_E\_119003

**Unable to update the properties in the "{0}" file. {1}\n**

**Reason:**

Unable to update the properties in the file due to an IOException.

**Action:**

Refer to the log for more information to resolve the issue. If you cannot resolve the issue, contact Broadcom Support.

## CAUAJM\_E\_119005

**Unable to encrypt the specified key.**

**Reason:**

Unable to encrypt the specified key.

**Action:**

Contact Broadcom Support.

## CAUAJM\_E\_119006

**The selected agent name is absent on this machine.**

**Reason:**

The agent name (IA\_AE\_LOCAL\_MAC\_DEF\_AGENT\_NAME) specified to create a local machine definition is absent on this machine.

**Action:**

Provide an agent name which is already installed on the machine.

## CAUAJM\_E\_119008

**Unable to determine the bitness of the library. Or, "oci.dll" is not a 64-bit client library.**

**Reason:**

Unable to determine the bitness of the library *or* oci.dll is not a 64-bit client library.

**Action:**

Make sure the Oracle client is a 64-bit client. If so, run the binary in the <ISO>\modules\installUtils\checkbit.exe oci.dll and verify that it shows the result as 64-bit. If an error occurs while executing checkbit.exe, see if the required Microsoft VC redist 2019 64bit is properly installed.

## CAUAJM\_E\_119009

Unable to determine the bitness of the library. Or, "libcIntsh.so" is not a 64-bit client shared library.

**Reason:**

Unable to determine the bitness of the library or libcIntsh.so is not a 64-bit client shared library.

**Action:**

Ensure that the Oracle client is a 64-bit client. If the libcIntsh.so file name contains the version number, create a soft link with name libcIntsh.so and point the link to the original file. For example, if the file name with version number is "libcIntsh.so.19.1" then create a soft link with the "libcIntsh.so" name as "ln -s libcIntsh.so.19.1 libcIntsh.so".

## CAUAJM\_E\_119013

Unable to revoke the read-only flag for the "{0}" directory.

**Reason:**

Failed to revoke the read-only flag for the directory.

**Action:**

Manually run the following command to revoke the read-only flag.

**Example:**

```
attrib -r "<directory>*" /s
```

## CAUAJM\_E\_119014

Unable to rename the existing Web UI directory: "{0}". Stop the Web UI services, close open files or directories and try again.

**Reason:**

Failed to rename the existing Web UI directory as a backup folder.

**Action:**

The files might be in use. Stop the Web UI services, close open files or directories, and try to manually rename the folder to avoid unnecessary references to the old Web UI installation.

## CAUAJM\_E\_119021

Unable to register the Derby database service for the Web UI.

**Reason:**

Failed to register the Derby database service.

**Action:**

Contact Broadcom Support.

## CAUAJM\_E\_119035

The "{0}" agent is already defined with a different key. The agent name can be mapped only to a single key. Modify the key.

**Reason:**

Another agent with the same name using a different key is already configured in the database. AutoSys expects the agent name to be unique.

**Action:**

Provide the key that is defined for the agent in the database, or use a different agent name.

# Troubleshooting

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This section includes troubleshooting procedures for AutoSys Workload Automation and Web UI.

## Troubleshooting AutoSys

Problems with AutoSys Workload Automation usually involve interactions between the four primary components (that is, the application server, the scheduler, the agent, and the event server) instead of the individual components themselves.

This section describes several common problems, their symptoms, and possible solutions. It provides useful information about troubleshooting the primary AutoSys Workload Automation components.

To troubleshoot AutoSys Workload Automation more effectively, you must understand the stages in the life of a job, the order in which they occur, and the roles played by the four primary components.

When you define a job, AutoSys Workload Automation saves its starting conditions to the event server (database), and the following steps occur:

- When the starting conditions of the job are met, the scheduler submits the job to an agent.
- The agent runs the job and returns the exit status of the job to the application server.
- The application server updates the event server.
- After the job completes, it does not run again until its starting conditions are met.

### NOTE

On UNIX, Sybase and Oracle are supported. On Windows and Linux, Microsoft SQL Server, Oracle, and Sybase are supported. Database specific tools like SQLPLUS (Oracle), ISQL (Sybase) and SQLCMD (Microsoft SQL Server) are recommended for any database-specific tasks.

## Agent Troubleshooting

This article describes scenarios for troubleshooting the agent.

You can use the autoping command to verify the agent and the agent's database connection from the application server are functioning correctly. The autoping command also verifies whether the server and client computers are properly configured and are communicating successfully.

To verify whether the agent is functioning correctly, issue the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autoping -m machine_name
```

- ***machine\_name***

Identifies the machine to verify. The machine must be defined in the database and accessible over the network.

Specify **-m ALL** to verify all machines.

The IP address or DNS name of the machine must be listed in the /etc/hosts file (on UNIX) or accessible through TCP/IP (on Windows) on the machine from which you issue the autoping command.

### NOTE

- When you issue the autoping command to a machine of type 'a', the client (the machine from which you issued autoping) sends a request to the application server and waits for the application server to respond. The application server contacts the scheduler and notifies it to ping the agent and waits for the scheduler to respond. The application server then pings the agent and prepares a response to autoping. If successful, the autoping command writes the following message to standard output on the server.

```
AutoPinging Machine [machine]
```

AutoPing WAS SUCCESSFUL!

- If there is a configuration problem, the autoping command writes a message indicating that the remote machine did not respond or that a more serious problem (such as a socket read error) exists. It also writes messages on behalf of the scheduler and the application server.

### Example: Verify Application Server Access

This example verifies that the machine "venice" is properly configured and that the agent's application server connection in executing an SDK call for blob operations is functioning properly.

```
autoping -m venice -S
CAUAJM_I_50023 AutoPinging Machine [venice]
CAUAJM_I_50031 Checking the Agent's connectivity to the Application Server.
CAUAJM_I_50025 AutoPing WAS SUCCESSFUL.
```

### Agent Not Responding

#### Symptom:

The autosyslog -e command displays a message similar to the following:

```
COMM_ERR_5 Communication attempt with Agent on machine [machine_name] has failed.
CAUAJM_E_40157 System Restart Job [Jobxxx] was unable to start
CAUAJM_W_40290 Machine <machine_name> is in question. Placing machine in the
unqualified state.
```

#### Solution:

##### **(UNIX) To verify the status of the agent**

1. Run the shell that is sourced to use AutoSys Workload Automation.
2. Enter the following command at the operating system prompt:

```
unisrvcntr status waae_agent-WA_AGENT
```

##### – **WA\_AGENT**

Defines the name of the agent for which you are verifying the status.

The agent's current status is displayed.

3. If the agent is not running, enter the following command at the operating system prompt:

```
unisrvcntr start waae_agent-WA_AGENT
```

##### – **WA\_AGENT**

Defines the name of the agent to start.

The agent starts.

##### **(Windows) To verify the status of the agent**

1. Click Start, Programs, AutoSys, Administrator.  
The Instance - AutoSys Workload Automation Administrator window opens.
2. Click the Services icon on the toolbar.  
The Services - AutoSys Workload Automation Administrator window appears, displaying a list of services installed on the selected instance. The Status column indicates the status of the agent.



- If the agent is not running, right-click the agent service, and click Start.  
The agent starts. The Status column indicates the status.

**NOTE**

You must verify the machine definition to ensure that the parameters you specify when you define an agent on AutoSys Workload Automation match the corresponding parameters in the agentparm.txt file. If the agent\_name, node\_name, port, and key\_to\_agent attribute values do not match, it can result in communication problems.

**Agent Starts, Command Runs No RUNNING Event Is Sent****Symptom:**

- Job does not advance from STARTING state.
- The scheduler log or the output of the autorep command on the job contains the following event with nothing after it, but the job runs to completion on the client computer:

```
CHANGE_STATUS Status: STARTING Job: test_install
```

**Solution:**

This is a common problem and occurs when the agent is unable to contact the scheduler. You must verify the following:

- Ensure that network problems are not preventing the communication between the agent and the scheduler computers.
- Verify the encryption settings between the scheduler and the agent in the receiver.log file. The receiver.log file is located in the log subdirectory in the SystemAgent directory. If the agent detects a problem with encryption, the receiver.log file will include messages related to the encryption problem.

**NOTE**

For the communication between the scheduler and the agent to be successful, you must ensure that the agent encryption settings on AutoSys Workload Automation match the encryption settings defined in the agentparm.txt file on the agent. If you detect any encryption problems, you must modify the encryption type and encryption key on AutoSys Workload Automation to match the encryption settings defined in the agentparm.txt file on the agent. For more information about modifying the encryption type and encryption key on AutoSys Workload Automation, see [Modify the Encryption Type and Encryption Key on AutoSys Workload Automation](#).

The agent must be able to contact the scheduler, and the scheduler must be able to connect to the database to send the RUNNING, SUCCESS, FAILURE, or TERMINATED status events.

To verify the problem, issue the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosyslog -J job_name
```

- job\_name***  
Defines the name of the job.

The agent log for the job is displayed.

**Using different database types for Web UI**

To use different database types for Web UI, follow these steps:

- Disable reporting for the Web UI.
- Install Web UI collector using the AutoSys installer.
- Enable reporting for the Web UI.
- Add reporting details for the collector.

## Web UI Services Fail to Start

**Symptom:** When the connectivity of the Web UI Collector service and/or CA-wcc service with the database is lost, for example, due to either the database taking a longer time to start or network connectivity issues, then the Web UI Collector and/or CA-wcc service fail to start.

**Note:** This problem is evident when the database is installed on the same machine where Web UI is being installed. At the end of the Web UI installation, the installer requests to restart the machine. After restart, if the database is taking a longer time to start and in the meantime if the Web UI services checks for database connectivity, the services eventually fail to start.

**Solution:** Restart WCC services after the database is up.

## FATAL SanityCheckAction

**Symptom:** If the DNS name and hostname of the machine where EEM is to be installed do not match, the EEM installation fails with the below error.

**FATAL SanityCheckAction - Local hostname name is <hostname> and DNS hostname is <FQDN of host>. Hostnames don't match.**

**Solution:** The DNS name and hostname of the machine should match for EEM installation.

## EEM Hangs on Linux

**Symptom:** On Linux machines, the EEM connectivity hangs at "Creating safecontext for <eem-server> ". This occurs on certain machines where the entropy to generate the random output is less. To confirm the insufficient entropy, on your Linux machine, run `cat /dev/random`. You may notice it hangs.

**Solution:**

To prevent the hang, on your Linux machine, use the following commands to install the rng-tools package and then start the rngd daemon.

```
yum install rng-tools // installs the rng-tools package.
service rngd start // starts the rngd daemon.
```

For more information, refer to [Using the Random Number Generator](#).

## Application Server Troubleshooting

This article describes scenarios for troubleshooting the application server.

The output from the application server is redirected to the following log file:

- On UNIX -- \$AUTOUSER/out/as\_server.\$AUTOSERV
- On Windows -- %AUTOUSER%\out\as\_server.%AUTOSERV%

To view the application server log file, enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
autosyslog -s
```

The application server log file displays error messages as they occur.

**NOTE**

- You can enable run-time traces to view incoming client requests in the order they were received by the application server and use them for troubleshooting communications with the AutoSys Workload Automation client or an application using the SDK.
- To terminate autosyslog, press Ctrl+C.

**Application Server Is Down****Symptom:**

- AutoSys Workload Automation client utilities on the local machine time out.
- When I issue the `chk_auto_up` command, a message similar to the following is displayed:  

```
CAUAJM_E_50033 Error initializing tx subsystem: CAUAJM_E_10527 Timed out waiting for
response from the Unicenter AutoSys JM Application Server.
CAUAJM_E_10062 Failed to get initial configuration from Unicenter AutoSys JM
Application Server: [<application server machine>:9,000]
```
- The application server log has not registered an error message since it was started.

**Solution:**

Do one of the following to verify whether the application server is down:

- Issue the `chk_auto_up` command. This command verifies if the application server is running.
- Issue the `autosyslog -s` command. This command displays the application server log file. Check for date and time stamps of the last run and any other error messages.
- (UNIX) Issue the following command to verify the status of the application server:  

```
unisrvcntr status waae_server.$AUTOSERV
```
- To test that communication from the application server to the event server is set up properly, log in to the event server from the computer where the application server is available by using the following:
  - For Microsoft SQL Server, use the ISQL/w graphical query interface.
  - For Oracle, use the SQL\*Plus command language interface.
  - For Sybase, use the ISQL utility.

**NOTE**

Use the AutoSys Workload Automation user name and password to log on to the event server.

- (UNIX) Check for running `as_server` processes for the given `$AUTOSERV` using the `ps` command. If the application server is down, enter the following command at the operating system prompt:

```
unisrvcntr start waae_server.$AUTOSERV
```

The application server starts.

- (Windows) Check the status of the application server using the Services - AutoSys Workload Automation Administrator window of the Administrator utility. You can also check the status of the application server using the Windows Control Panel Services dialog. If the application server is down, you must start it using the Services - AutoSys Workload Automation Administrator window of the Administrator utility.

**NOTE**

For more information about how to start the application server or verify the application server status using the Administrator utility, see [Control the Services](#).

## **Application Server Will Not Start**

### **Symptom:**

The autosyslog -s command displays messages indicating that it cannot connect to the database.

### **(UNIX) Solution:**

This problem occurs if the database is down or there are problems with the database installation. To test that communication from the application server to the event server is set up properly, log in to the event server from the application server computer by using the following:

- For Oracle, use the SQL\*Plus command language interface.
- For Sybase, use the ISQL utility.

#### **NOTE**

Use the AutoSys Workload Automation user name and password to log in to the event server.

### **(Windows) Solution:**

To verify whether the application server is down, log on to the event server computer and issue the chk\_auto\_up command. If the database is running, there is a possibility that AutoSys Workload Automation is configured to the wrong application server or communication between AutoSys Workload Automation and the application server is not successful.

To test that communication from the application server to the event server is set up properly, log in to the event server from the application server computer by using the following:

- For Microsoft SQL Server, use the ISQL/w graphical query interface.
- For Oracle, use the SQL\*Plus command language interface.
- For Sybase, use the ISQL utility.

#### **NOTE**

Use the AutoSys Workload Automation user name and password to log in to the event server.

### **Symptom:**

The application server is not running and does not write log output to the \$AUTOUSER/out/as\_server.\$AUTOSERV (UNIX) or %AUTOUSER%\out\as\_server.%AUTOSERV% (Windows) file.

### **(UNIX) Solution:**

This problem could have various causes and the solution depends on which of the following message is displayed:

- **The environment variable AUTOSYS is not set.**  
This message is displayed if the \$AUTOSYS system environment variable is not available to the application server. You must ensure that the AutoSys Workload Automation source file has been sourced in your session.
- **The environment variable AUTOSYS is too long.**  
This message is displayed if the \$AUTOSYS system environment variable value is set to a path that is more than 80 characters in length. You must uninstall AutoSys Workload Automation, and reinstall it to a directory path that is fewer than 80 characters in length.
- **Application server cannot open its log file as\_server.\$AUTOSERV. Some directory in the path is not accessible to the SYSTEM.**  
This message is displayed when the application server cannot create the as\_server.\$AUTOSERV log file. You must ensure that the log file has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.

### **(Windows) Solution:**

This problem could have various causes; and the solution depends on which of the following message is displayed on the Event Log Viewer dialog. You can access the Event Log Viewer dialog using the Windows Control Panel Administrative Tools dialog.

- **The environment variable AUTOSYS is not set.**

This message is displayed if the %AUTOSYS% system environment variable is not available to the application server. You must ensure that the %AUTOSYS% system environment variable is set properly by using the Windows Control Panel or the System - AutoSys Workload Automation Administrator window of the Administrator utility.

**NOTE**

For more information about how to add, modify, or delete a system environment variable using the Administrator utility, see [Environment Variables](#).

- **The environment variable AUTOSYS is too long.**

This message is displayed if the %AUTOSYS% system environment variable value is set to a path that is more than 80 characters in length. You must uninstall AutoSys Workload Automation, and reinstall it to a directory path that is fewer than 80 characters in length.

- **Application server cannot open its log file as\_server.%AUTOSERV%. Some directory in the path is not accessible to the SYSTEM.**

This message is displayed when the application server cannot create the as\_server.%AUTOSERV% log file. You must ensure that the log file has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.

### **Application Server Starts, Client on Remote Machine Times out**

#### **Symptom:**

When I issue the chk\_auto\_up command from a remote machine, a message similar to the following is displayed:

```
CAUAJM_E_50033 Error initializing tx subsystem: CAUAJM_E_10527 Timed out waiting for
response from the Unicenter AutoSys JM Application Server.
CAUAJM_E_10062 Failed to get initial configuration from Unicenter AutoSys JM
Application Server: [<application server machine>:9,000]
```

#### **Solution:**

You must ensure that network problems are not preventing communication between the client and the application server computers through the Operating System ping command.

#### **Follow these steps:**

1. On the client computer, change to the \$CSAM\_SOCKADAPTER/bin (UNIX) or %CSAM\_SOCKADAPTER/bin (Windows) directory and enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
csamconfigedit Port=value display
```

- **value**

Defines the port number to display.

**Default:** 9000

2. On the application server computer, change to the \$CSAM\_SOCKADAPTER/bin (UNIX) or %CSAM\_SOCKADAPTER/bin (Windows) directory and enter the following command at the UNIX operating system prompt or the Windows instance command prompt:

```
csamconfigedit Port=value display
```

- **value**

Defines the port number to display.

**Default:** 9000

3. Compare the outputs in Step 1 and Step 2 and ensure that both the EnablePmux and EnableSSL settings are identical.
4. On both the client and application server computers, enter the following command at the UNIX operating system prompt or the Windows instance command prompt::

```
csamconfigedit PortRange=49152-50176 display
```

The port settings are compared. If the settings do not match, change the settings such that the settings match on both the computers, and restart the AutoSys Workload Automation services.

If the settings match, verify that physical port 7163 is not being blocked by a firewall software on either of the computers.

## **Not Enough Procedure Cache**

### **Symptom:**

On Sybase, the application server fails to execute any queries and shuts down. The following error message is displayed:

```
CAUAJM_E_18018 Server message: Message number(701), Severity(17), State(3), Line(1),
Message String: There is not enough procedure cache to run this procedure, trigger, or
SQL batch. Retry later, or ask your SA to reconfigure ASE with more procedure cache.
```

### **Solution:**

Verify if the current procedure cache size is sufficient. If it is insufficient, increase the procedure cache size. For information about increasing the procedure cache size, contact your database administrator or see the database documentation.

For example, to increase the procedure cache size to 28000, follow these steps:

1. Connect to the Sybase database and run the following command:

```
sp_monitorconfig "procedure cache"
go
```

2. Add the Num\_free and Num\_active parameters from the output of the above command. This gives the current procedure cache size.
3. Increase the max memory allocated based on the current procedure\_cache size, as follows:

```
sp_configure "max memory", 114455
go
```

4. Increase the procedure cache size, as follows:

```
sp_configure "procedure cache size", 28000
go
```

5. Restart the Sybase server.
6. Start the application server.

## **Reintroduce the ujoadmin Role Privileges**

If you install AutoSys Workload Automation on an Oracle database and then install Unicenter AutoSys JM r11 on the same database, the MDB is created where the AEDB already existed. This removes the ujoadmin role privileges on the AEDB and results in errors when AutoSys Workload Automation tries to access the AEDB. So, you must reintroduce the ujoadmin role privileges to use the database for AutoSys Workload Automation.

**NOTE**

The schema owner example is 'aedbadmin' and the database user example is 'autosys'.

**Follow these steps:**

1. Log in as the SYS user and issue the following commands at the instance command prompt (Windows) or the operating system prompt (UNIX):

```
sqlplus SYS/password@Instance as sysdba
GRANT ANALYZE ANY TO ujoadmin;
GRANT SELECT ON DBA_TABLESPACES TO ujoadmin;
GRANT ujoadmin TO aedbadmin;
GRANT ujoadmin TO autosys;
```

The ujoadmin role is granted ANALYZE ANY and SELECT privileges and these privileges are passed on to the schema owner (aedbadmin) and database user (autosys).

2. Log in as the schema owner and issue the following commands at the instance command prompt (Windows) or the operating system prompt (UNIX):

```
sqlplus aedbadmin/password@Instance
alter session set current_schema = aedbadmin;
GRANT SELECT on ujo_afm to ujoadmin;
GRANT DELETE on ujo_afm to ujoadmin;
GRANT UPDATE on ujo_afm to ujoadmin;
GRANT INSERT on ujo_afm to ujoadmin;
GRANT SELECT on ujo_afm_strings to ujoadmin;
GRANT DELETE on ujo_afm_strings to ujoadmin;
GRANT UPDATE on ujo_afm_strings to ujoadmin;
GRANT INSERT on ujo_afm_strings to ujoadmin;
GRANT SELECT on ujo_agent_alias to ujoadmin;
GRANT DELETE on ujo_agent_alias to ujoadmin;
GRANT UPDATE on ujo_agent_alias to ujoadmin;
GRANT INSERT on ujo_agent_alias to ujoadmin;
GRANT SELECT on ujo_alamode to ujoadmin;
GRANT DELETE on ujo_alamode to ujoadmin;
GRANT UPDATE on ujo_alamode to ujoadmin;
GRANT INSERT on ujo_alamode to ujoadmin;
GRANT SELECT on ujo_alarm to ujoadmin;
GRANT DELETE on ujo_alarm to ujoadmin;
GRANT UPDATE on ujo_alarm to ujoadmin;
GRANT INSERT on ujo_alarm to ujoadmin;
GRANT SELECT on ujo_asbnode to ujoadmin;
GRANT DELETE on ujo_asbnode to ujoadmin;
GRANT UPDATE on ujo_asbnode to ujoadmin;
GRANT INSERT on ujo_asbnode to ujoadmin;
GRANT SELECT on ujo_asext_config to ujoadmin;
GRANT DELETE on ujo_asext_config to ujoadmin;
GRANT UPDATE on ujo_asext_config to ujoadmin;
GRANT INSERT on ujo_asext_config to ujoadmin;
GRANT SELECT on ujo_audit_alamode to ujoadmin;
```

---

```
GRANT DELETE on ujo_audit_alamode to ujoadmin;
GRANT UPDATE on ujo_audit_alamode to ujoadmin;
GRANT INSERT on ujo_audit_alamode to ujoadmin;
GRANT SELECT on ujo_audit_info to ujoadmin;
GRANT DELETE on ujo_audit_info to ujoadmin;
GRANT UPDATE on ujo_audit_info to ujoadmin;
GRANT INSERT on ujo_audit_info to ujoadmin;
GRANT SELECT on ujo_audit_msg to ujoadmin;
GRANT DELETE on ujo_audit_msg to ujoadmin;
GRANT UPDATE on ujo_audit_msg to ujoadmin;
GRANT INSERT on ujo_audit_msg to ujoadmin;
GRANT SELECT on ujo_avg_job_runs to ujoadmin;
GRANT DELETE on ujo_avg_job_runs to ujoadmin;
GRANT UPDATE on ujo_avg_job_runs to ujoadmin;
GRANT INSERT on ujo_avg_job_runs to ujoadmin;
GRANT SELECT on ujo_blobs to ujoadmin;
GRANT DELETE on ujo_blobs to ujoadmin;
GRANT UPDATE on ujo_blobs to ujoadmin;
GRANT INSERT on ujo_blobs to ujoadmin;
GRANT SELECT on ujo_calendar to ujoadmin;
GRANT DELETE on ujo_calendar to ujoadmin;
GRANT UPDATE on ujo_calendar to ujoadmin;
GRANT INSERT on ujo_calendar to ujoadmin;
GRANT SELECT on ujo_calendar_desc to ujoadmin;
GRANT DELETE on ujo_calendar_desc to ujoadmin;
GRANT UPDATE on ujo_calendar_desc to ujoadmin;
GRANT INSERT on ujo_calendar_desc to ujoadmin;
GRANT SELECT on ujo_chase to ujoadmin;
GRANT DELETE on ujo_chase to ujoadmin;
GRANT UPDATE on ujo_chase to ujoadmin;
GRANT INSERT on ujo_chase to ujoadmin;
GRANT SELECT on ujo_chkpnt_rstart to ujoadmin;
GRANT DELETE on ujo_chkpnt_rstart to ujoadmin;
GRANT UPDATE on ujo_chkpnt_rstart to ujoadmin;
GRANT INSERT on ujo_chkpnt_rstart to ujoadmin;
GRANT SELECT on ujo_comm_rcv_seq to ujoadmin;
GRANT DELETE on ujo_comm_rcv_seq to ujoadmin;
GRANT UPDATE on ujo_comm_rcv_seq to ujoadmin;
GRANT INSERT on ujo_comm_rcv_seq to ujoadmin;
GRANT SELECT on ujo_comm_send_alias to ujoadmin;
GRANT DELETE on ujo_comm_send_alias to ujoadmin;
GRANT UPDATE on ujo_comm_send_alias to ujoadmin;
GRANT INSERT on ujo_comm_send_alias to ujoadmin;
GRANT SELECT on ujo_comm_send_seq_lock to ujoadmin;
GRANT DELETE on ujo_comm_send_seq_lock to ujoadmin;
GRANT UPDATE on ujo_comm_send_seq_lock to ujoadmin;
```



---

```
GRANT INSERT on ujo_comm_send_seq_lock to ujoadmin;
GRANT SELECT on ujo_comm_send_nodes to ujoadmin;
GRANT DELETE on ujo_comm_send_nodes to ujoadmin;
GRANT UPDATE on ujo_comm_send_nodes to ujoadmin;
GRANT INSERT on ujo_comm_send_nodes to ujoadmin;
GRANT SELECT on ujo_comm_send_seq to ujoadmin;
GRANT DELETE on ujo_comm_send_seq to ujoadmin;
GRANT UPDATE on ujo_comm_send_seq to ujoadmin;
GRANT INSERT on ujo_comm_send_seq to ujoadmin;
GRANT SELECT on ujo_command_job to ujoadmin;
GRANT DELETE on ujo_command_job to ujoadmin;
GRANT UPDATE on ujo_command_job to ujoadmin;
GRANT INSERT on ujo_command_job to ujoadmin;
GRANT SELECT on ujo_config to ujoadmin;
GRANT DELETE on ujo_config to ujoadmin;
GRANT UPDATE on ujo_config to ujoadmin;
GRANT INSERT on ujo_config to ujoadmin;
GRANT SELECT on ujo_connectionprofile to ujoadmin;
GRANT DELETE on ujo_connectionprofile to ujoadmin;
GRANT UPDATE on ujo_connectionprofile to ujoadmin;
GRANT INSERT on ujo_connectionprofile to ujoadmin;
GRANT SELECT on ujo_cred to ujoadmin;
GRANT DELETE on ujo_cred to ujoadmin;
GRANT UPDATE on ujo_cred to ujoadmin;
GRANT INSERT on ujo_cred to ujoadmin;
GRANT SELECT on ujo_cycle to ujoadmin;
GRANT DELETE on ujo_cycle to ujoadmin;
GRANT UPDATE on ujo_cycle to ujoadmin;
GRANT INSERT on ujo_cycle to ujoadmin;
GRANT SELECT on ujo_event to ujoadmin;
GRANT DELETE on ujo_event to ujoadmin;
GRANT UPDATE on ujo_event to ujoadmin;
GRANT INSERT on ujo_event to ujoadmin;
GRANT SELECT on ujo_event2 to ujoadmin;
GRANT DELETE on ujo_event2 to ujoadmin;
GRANT UPDATE on ujo_event2 to ujoadmin;
GRANT INSERT on ujo_event2 to ujoadmin;
GRANT SELECT on ujo_ext_calendar to ujoadmin;
GRANT DELETE on ujo_ext_calendar to ujoadmin;
GRANT UPDATE on ujo_ext_calendar to ujoadmin;
GRANT INSERT on ujo_ext_calendar to ujoadmin;
GRANT SELECT on ujo_ext_event to ujoadmin;
GRANT DELETE on ujo_ext_event to ujoadmin;
GRANT UPDATE on ujo_ext_event to ujoadmin;
GRANT INSERT on ujo_ext_event to ujoadmin;
GRANT SELECT on ujo_ext_job to ujoadmin;
```

---

```
GRANT DELETE on ujo_ext_job to ujoadmin;
GRANT UPDATE on ujo_ext_job to ujoadmin;
GRANT INSERT on ujo_ext_job to ujoadmin;
GRANT SELECT on ujo_extended_jobrun_info to ujoadmin;
GRANT DELETE on ujo_extended_jobrun_info to ujoadmin;
GRANT UPDATE on ujo_extended_jobrun_info to ujoadmin;
GRANT INSERT on ujo_extended_jobrun_info to ujoadmin;
GRANT SELECT on ujo_file_watch_job to ujoadmin;
GRANT DELETE on ujo_file_watch_job to ujoadmin;
GRANT UPDATE on ujo_file_watch_job to ujoadmin;
GRANT INSERT on ujo_file_watch_job to ujoadmin;
GRANT SELECT on ujo_ftp_job to ujoadmin;
GRANT DELETE on ujo_ftp_job to ujoadmin;
GRANT UPDATE on ujo_ftp_job to ujoadmin;
GRANT INSERT on ujo_ftp_job to ujoadmin;
GRANT SELECT on ujo_generic_job to ujoadmin;
GRANT DELETE on ujo_generic_job to ujoadmin;
GRANT UPDATE on ujo_generic_job to ujoadmin;
GRANT INSERT on ujo_generic_job to ujoadmin;
GRANT SELECT on ujo_glob to ujoadmin;
GRANT DELETE on ujo_glob to ujoadmin;
GRANT UPDATE on ujo_glob to ujoadmin;
GRANT INSERT on ujo_glob to ujoadmin;
GRANT SELECT on ujo_globblob to ujoadmin;
GRANT DELETE on ujo_globblob to ujoadmin;
GRANT UPDATE on ujo_globblob to ujoadmin;
GRANT INSERT on ujo_globblob to ujoadmin;
GRANT SELECT on ujo_ha_process to ujoadmin;
GRANT DELETE on ujo_ha_process to ujoadmin;
GRANT UPDATE on ujo_ha_process to ujoadmin;
GRANT INSERT on ujo_ha_process to ujoadmin;
GRANT SELECT on ujo_hdfs_job to ujoadmin;
GRANT DELETE on ujo_hdfs_job to ujoadmin;
GRANT UPDATE on ujo_hdfs_job to ujoadmin;
GRANT INSERT on ujo_hdfs_job to ujoadmin;
GRANT SELECT on ujo_hive_job to ujoadmin;
GRANT DELETE on ujo_hive_job to ujoadmin;
GRANT UPDATE on ujo_hive_job to ujoadmin;
GRANT INSERT on ujo_hive_job to ujoadmin;
GRANT SELECT on ujo_spark_job to ujoadmin;
GRANT DELETE on ujo_spark_job to ujoadmin;
GRANT UPDATE on ujo_spark_job to ujoadmin;
GRANT INSERT on ujo_spark_job to ujoadmin;
GRANT SELECT on ujo_informatica_job to ujoadmin;
GRANT DELETE on ujo_informatica_job to ujoadmin;
GRANT UPDATE on ujo_informatica_job to ujoadmin;
```

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```
GRANT INSERT on ujo_informativa_job to ujoadmin;
GRANT SELECT on ujo_sqlagent_job to ujoadmin;
GRANT DELETE on ujo_sqlagent_job to ujoadmin;
GRANT UPDATE on ujo_sqlagent_job to ujoadmin;
GRANT INSERT on ujo_sqlagent_job to ujoadmin;
GRANT SELECT on ujo_i5_job to ujoadmin;
GRANT DELETE on ujo_i5_job to ujoadmin;
GRANT UPDATE on ujo_i5_job to ujoadmin;
GRANT INSERT on ujo_i5_job to ujoadmin;
GRANT SELECT on ujo_intcodes to ujoadmin;
GRANT SELECT on ujo_job to ujoadmin;
GRANT DELETE on ujo_job to ujoadmin;
GRANT UPDATE on ujo_job to ujoadmin;
GRANT INSERT on ujo_job to ujoadmin;
GRANT SELECT on ujo_job_cond to ujoadmin;
GRANT DELETE on ujo_job_cond to ujoadmin;
GRANT UPDATE on ujo_job_cond to ujoadmin;
GRANT INSERT on ujo_job_cond to ujoadmin;
GRANT SELECT on ujo_job_resource_dep to ujoadmin;
GRANT DELETE on ujo_job_resource_dep to ujoadmin;
GRANT UPDATE on ujo_job_resource_dep to ujoadmin;
GRANT INSERT on ujo_job_resource_dep to ujoadmin;
GRANT SELECT on ujo_job_runs to ujoadmin;
GRANT DELETE on ujo_job_runs to ujoadmin;
GRANT UPDATE on ujo_job_runs to ujoadmin;
GRANT INSERT on ujo_job_runs to ujoadmin;
GRANT SELECT on ujo_job_status to ujoadmin;
GRANT DELETE on ujo_job_status to ujoadmin;
GRANT UPDATE on ujo_job_status to ujoadmin;
GRANT INSERT on ujo_job_status to ujoadmin;
GRANT SELECT on ujo_job_tree to ujoadmin;
GRANT DELETE on ujo_job_tree to ujoadmin;
GRANT UPDATE on ujo_job_tree to ujoadmin;
GRANT INSERT on ujo_job_tree to ujoadmin;
GRANT SELECT on ujo_jobblob to ujoadmin;
GRANT DELETE on ujo_jobblob to ujoadmin;
GRANT UPDATE on ujo_jobblob to ujoadmin;
GRANT INSERT on ujo_jobblob to ujoadmin;
GRANT SELECT on ujo_jobtype to ujoadmin;
GRANT DELETE on ujo_jobtype to ujoadmin;
GRANT UPDATE on ujo_jobtype to ujoadmin;
GRANT INSERT on ujo_jobtype to ujoadmin;
GRANT SELECT on ujo_keymaster to ujoadmin;
GRANT DELETE on ujo_keymaster to ujoadmin;
GRANT UPDATE on ujo_keymaster to ujoadmin;
GRANT INSERT on ujo_keymaster to ujoadmin;
```

---

```
GRANT SELECT on ujo_last_eoid_counter to ujoadmin;
GRANT DELETE on ujo_last_eoid_counter to ujoadmin;
GRANT UPDATE on ujo_last_eoid_counter to ujoadmin;
GRANT INSERT on ujo_last_eoid_counter to ujoadmin;
GRANT SELECT on ujo_ma_process to ujoadmin;
GRANT DELETE on ujo_ma_process to ujoadmin;
GRANT UPDATE on ujo_ma_process to ujoadmin;
GRANT INSERT on ujo_ma_process to ujoadmin;
GRANT SELECT on ujo_machine to ujoadmin;
GRANT DELETE on ujo_machine to ujoadmin;
GRANT UPDATE on ujo_machine to ujoadmin;
GRANT INSERT on ujo_machine to ujoadmin;
GRANT SELECT on ujo_meta_afm_types to ujoadmin;
GRANT SELECT on ujo_meta_restart_afm_types to ujoadmin;
GRANT SELECT on ujo_meta_enumerations to ujoadmin;
GRANT SELECT on ujo_meta_parse_keywords to ujoadmin;
GRANT SELECT on ujo_meta_parse_extruntime to ujoadmin;
GRANT SELECT on ujo_meta_presentation to ujoadmin;
GRANT SELECT on ujo_meta_properties to ujoadmin;
GRANT SELECT on ujo_meta_root_keywords to ujoadmin;
GRANT SELECT on ujo_meta_rules to ujoadmin;
GRANT SELECT on ujo_meta_types to ujoadmin;
GRANT SELECT on ujo_meta_ws_elements to ujoadmin;
GRANT SELECT on ujo_micro_focus_job to ujoadmin;
GRANT DELETE on ujo_micro_focus_job to ujoadmin;
GRANT UPDATE on ujo_micro_focus_job to ujoadmin;
GRANT INSERT on ujo_micro_focus_job to ujoadmin;
GRANT SELECT on ujo_monbro to ujoadmin;
GRANT DELETE on ujo_monbro to ujoadmin;
GRANT UPDATE on ujo_monbro to ujoadmin;
GRANT INSERT on ujo_monbro to ujoadmin;
GRANT SELECT on ujo_monitor_object_job to ujoadmin;
GRANT DELETE on ujo_monitor_object_job to ujoadmin;
GRANT UPDATE on ujo_monitor_object_job to ujoadmin;
GRANT INSERT on ujo_monitor_object_job to ujoadmin;
GRANT SELECT on ujo_monitor_winevent_log to ujoadmin;
GRANT DELETE on ujo_monitor_winevent_log to ujoadmin;
GRANT UPDATE on ujo_monitor_winevent_log to ujoadmin;
GRANT INSERT on ujo_monitor_winevent_log to ujoadmin;
GRANT SELECT on ujo_msg_ack to ujoadmin;
GRANT DELETE on ujo_msg_ack to ujoadmin;
GRANT UPDATE on ujo_msg_ack to ujoadmin;
GRANT INSERT on ujo_msg_ack to ujoadmin;
GRANT SELECT on ujo_next_oid to ujoadmin;
GRANT DELETE on ujo_next_oid to ujoadmin;
GRANT UPDATE on ujo_next_oid to ujoadmin;
```

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```
GRANT INSERT on ujo_next_oid to ujoadmin;
GRANT SELECT on ujo_oozie_job to ujoadmin;
GRANT DELETE on ujo_oozie_job to ujoadmin;
GRANT UPDATE on ujo_oozie_job to ujoadmin;
GRANT INSERT on ujo_oozie_job to ujoadmin;
GRANT SELECT on ujo_oraapps to ujoadmin;
GRANT DELETE on ujo_oraapps to ujoadmin;
GRANT UPDATE on ujo_oraapps to ujoadmin;
GRANT INSERT on ujo_oraapps to ujoadmin;
GRANT SELECT on ujo_oraapps_steps to ujoadmin;
GRANT DELETE on ujo_oraapps_steps to ujoadmin;
GRANT UPDATE on ujo_oraapps_steps to ujoadmin;
GRANT INSERT on ujo_oraapps_steps to ujoadmin;
GRANT SELECT on ujo_oraapps_props to ujoadmin;
GRANT DELETE on ujo_oraapps_props to ujoadmin;
GRANT UPDATE on ujo_oraapps_props to ujoadmin;
GRANT INSERT on ujo_oraapps_props to ujoadmin;
GRANT SELECT on ujo_patch to ujoadmin;
GRANT DELETE on ujo_patch to ujoadmin;
GRANT UPDATE on ujo_patch to ujoadmin;
GRANT INSERT on ujo_patch to ujoadmin;
GRANT SELECT on ujo_peoplesoft_job to ujoadmin;
GRANT DELETE on ujo_peoplesoft_job to ujoadmin;
GRANT UPDATE on ujo_peoplesoft_job to ujoadmin;
GRANT INSERT on ujo_peoplesoft_job to ujoadmin;
GRANT SELECT on ujo_pig_job to ujoadmin;
GRANT DELETE on ujo_pig_job to ujoadmin;
GRANT UPDATE on ujo_pig_job to ujoadmin;
GRANT INSERT on ujo_pig_job to ujoadmin;
GRANT SELECT on ujo_proc_event to ujoadmin;
GRANT DELETE on ujo_proc_event to ujoadmin;
GRANT UPDATE on ujo_proc_event to ujoadmin;
GRANT INSERT on ujo_proc_event to ujoadmin;
GRANT SELECT on ujo_real_resource to ujoadmin;
GRANT SELECT on ujo_rep_daily to ujoadmin;
GRANT DELETE on ujo_rep_daily to ujoadmin;
GRANT UPDATE on ujo_rep_daily to ujoadmin;
GRANT INSERT on ujo_rep_daily to ujoadmin;
GRANT SELECT on ujo_rep_hourly to ujoadmin;
GRANT DELETE on ujo_rep_hourly to ujoadmin;
GRANT UPDATE on ujo_rep_hourly to ujoadmin;
GRANT INSERT on ujo_rep_hourly to ujoadmin;
GRANT SELECT on ujo_rep_monthly to ujoadmin;
GRANT DELETE on ujo_rep_monthly to ujoadmin;
GRANT UPDATE on ujo_rep_monthly to ujoadmin;
GRANT INSERT on ujo_rep_monthly to ujoadmin;
```

---

```
GRANT SELECT on ujo_rep_weekly to ujoadmin;
GRANT DELETE on ujo_rep_weekly to ujoadmin;
GRANT UPDATE on ujo_rep_weekly to ujoadmin;
GRANT INSERT on ujo_rep_weekly to ujoadmin;
GRANT SELECT on ujo_req_job to ujoadmin;
GRANT DELETE on ujo_req_job to ujoadmin;
GRANT UPDATE on ujo_req_job to ujoadmin;
GRANT INSERT on ujo_req_job to ujoadmin;
GRANT SELECT on ujo_restart to ujoadmin;
GRANT DELETE on ujo_restart to ujoadmin;
GRANT UPDATE on ujo_restart to ujoadmin;
GRANT INSERT on ujo_restart to ujoadmin;
GRANT SELECT on ujo_reswait_que to ujoadmin;
GRANT DELETE on ujo_reswait_que to ujoadmin;
GRANT UPDATE on ujo_reswait_que to ujoadmin;
GRANT INSERT on ujo_reswait_que to ujoadmin;
GRANT SELECT on ujo_sap_arcspec to ujoadmin;
GRANT DELETE on ujo_sap_arcspec to ujoadmin;
GRANT UPDATE on ujo_sap_arcspec to ujoadmin;
GRANT INSERT on ujo_sap_arcspec to ujoadmin;
GRANT SELECT on ujo_sap_infodetail to ujoadmin;
GRANT DELETE on ujo_sap_infodetail to ujoadmin;
GRANT UPDATE on ujo_sap_infodetail to ujoadmin;
GRANT INSERT on ujo_sap_infodetail to ujoadmin;
GRANT SELECT on ujo_sap_job to ujoadmin;
GRANT DELETE on ujo_sap_job to ujoadmin;
GRANT UPDATE on ujo_sap_job to ujoadmin;
GRANT INSERT on ujo_sap_job to ujoadmin;
GRANT SELECT on ujo_sap_jobstep to ujoadmin;
GRANT DELETE on ujo_sap_jobstep to ujoadmin;
GRANT UPDATE on ujo_sap_jobstep to ujoadmin;
GRANT INSERT on ujo_sap_jobstep to ujoadmin;
GRANT SELECT on ujo_sap_prtspec to ujoadmin;
GRANT DELETE on ujo_sap_prtspec to ujoadmin;
GRANT UPDATE on ujo_sap_prtspec to ujoadmin;
GRANT INSERT on ujo_sap_prtspec to ujoadmin;
GRANT SELECT on ujo_sap_recipient to ujoadmin;
GRANT DELETE on ujo_sap_recipient to ujoadmin;
GRANT UPDATE on ujo_sap_recipient to ujoadmin;
GRANT INSERT on ujo_sap_recipient to ujoadmin;
GRANT SELECT on ujo_sched_info to ujoadmin;
GRANT DELETE on ujo_sched_info to ujoadmin;
GRANT UPDATE on ujo_sched_info to ujoadmin;
GRANT INSERT on ujo_sched_info to ujoadmin;
GRANT SELECT on ujo_service_desk to ujoadmin;
GRANT DELETE on ujo_service_desk to ujoadmin;
```

---

```
GRANT UPDATE on ujo_service_desk to ujoadmin;
GRANT INSERT on ujo_service_desk to ujoadmin;
GRANT SELECT on ujo_snmp_job to ujoadmin;
GRANT DELETE on ujo_snmp_job to ujoadmin;
GRANT UPDATE on ujo_snmp_job to ujoadmin;
GRANT INSERT on ujo_snmp_job to ujoadmin;
GRANT SELECT on ujo_sql_job to ujoadmin;
GRANT DELETE on ujo_sql_job to ujoadmin;
GRANT UPDATE on ujo_sql_job to ujoadmin;
GRANT INSERT on ujo_sql_job to ujoadmin;
GRANT SELECT on ujo_sql_proc_parms to ujoadmin;
GRANT DELETE on ujo_sql_proc_parms to ujoadmin;
GRANT UPDATE on ujo_sql_proc_parms to ujoadmin;
GRANT INSERT on ujo_sql_proc_parms to ujoadmin;
GRANT SELECT on ujo_sqoop_job to ujoadmin;
GRANT DELETE on ujo_sqoop_job to ujoadmin;
GRANT UPDATE on ujo_sqoop_job to ujoadmin;
GRANT INSERT on ujo_sqoop_job to ujoadmin;
GRANT SELECT on ujo_strings to ujoadmin;
GRANT DELETE on ujo_strings to ujoadmin;
GRANT UPDATE on ujo_strings to ujoadmin;
GRANT INSERT on ujo_strings to ujoadmin;
GRANT SELECT on ujo_sys_ha_state to ujoadmin;
GRANT DELETE on ujo_sys_ha_state to ujoadmin;
GRANT UPDATE on ujo_sys_ha_state to ujoadmin;
GRANT INSERT on ujo_sys_ha_state to ujoadmin;
GRANT SELECT on ujo_temp_jobruns to ujoadmin;
GRANT DELETE on ujo_temp_jobruns to ujoadmin;
GRANT UPDATE on ujo_temp_jobruns to ujoadmin;
GRANT INSERT on ujo_temp_jobruns to ujoadmin;
GRANT SELECT on ujo_timezones to ujoadmin;
GRANT DELETE on ujo_timezones to ujoadmin;
GRANT UPDATE on ujo_timezones to ujoadmin;
GRANT INSERT on ujo_timezones to ujoadmin;
GRANT SELECT on ujo_uninotify to ujoadmin;
GRANT DELETE on ujo_uninotify to ujoadmin;
GRANT UPDATE on ujo_uninotify to ujoadmin;
GRANT INSERT on ujo_uninotify to ujoadmin;
GRANT SELECT on ujo_virt_resource to ujoadmin;
GRANT DELETE on ujo_virt_resource to ujoadmin;
GRANT UPDATE on ujo_virt_resource to ujoadmin;
GRANT INSERT on ujo_virt_resource to ujoadmin;
GRANT SELECT on ujo_virt_resource_lookup to ujoadmin;
GRANT DELETE on ujo_virt_resource_lookup to ujoadmin;
GRANT UPDATE on ujo_virt_resource_lookup to ujoadmin;
GRANT INSERT on ujo_virt_resource_lookup to ujoadmin;
```

---

```
GRANT SELECT on ujo_virt_resource_status to ujoadmin;
GRANT DELETE on ujo_virt_resource_status to ujoadmin;
GRANT UPDATE on ujo_virt_resource_status to ujoadmin;
GRANT INSERT on ujo_virt_resource_status to ujoadmin;
GRANT SELECT on ujo_wait_que to ujoadmin;
GRANT DELETE on ujo_wait_que to ujoadmin;
GRANT UPDATE on ujo_wait_que to ujoadmin;
GRANT INSERT on ujo_wait_que to ujoadmin;
GRANT SELECT on ujo_web_services to ujoadmin;
GRANT DELETE on ujo_web_services to ujoadmin;
GRANT UPDATE on ujo_web_services to ujoadmin;
GRANT INSERT on ujo_web_services to ujoadmin;
GRANT SELECT on ujo_web_services2 to ujoadmin;
GRANT DELETE on ujo_web_services2 to ujoadmin;
GRANT UPDATE on ujo_web_services2 to ujoadmin;
GRANT INSERT on ujo_web_services2 to ujoadmin;
GRANT SELECT on ujo_wol_job to ujoadmin;
GRANT DELETE on ujo_wol_job to ujoadmin;
GRANT UPDATE on ujo_wol_job to ujoadmin;
GRANT INSERT on ujo_wol_job to ujoadmin;
GRANT SELECT on ujo_workflow to ujoadmin;
GRANT DELETE on ujo_workflow to ujoadmin;
GRANT UPDATE on ujo_workflow to ujoadmin;
GRANT INSERT on ujo_workflow to ujoadmin;
GRANT SELECT on ujo_ws_criteria to ujoadmin;
GRANT DELETE on ujo_ws_criteria to ujoadmin;
GRANT UPDATE on ujo_ws_criteria to ujoadmin;
GRANT INSERT on ujo_ws_criteria to ujoadmin;
GRANT SELECT on ujo_ws_parm to ujoadmin;
GRANT DELETE on ujo_ws_parm to ujoadmin;
GRANT UPDATE on ujo_ws_parm to ujoadmin;
GRANT INSERT on ujo_ws_parm to ujoadmin;
GRANT SELECT on ujo_ws_security to ujoadmin;
GRANT DELETE on ujo_ws_security to ujoadmin;
GRANT UPDATE on ujo_ws_security to ujoadmin;
GRANT INSERT on ujo_ws_security to ujoadmin;
GRANT SELECT on ujo_zos_conddcodes to ujoadmin;
GRANT DELETE on ujo_zos_conddcodes to ujoadmin;
GRANT UPDATE on ujo_zos_conddcodes to ujoadmin;
GRANT INSERT on ujo_zos_conddcodes to ujoadmin;
GRANT SELECT on ujo_zos_dsn_trigger to ujoadmin;
GRANT DELETE on ujo_zos_dsn_trigger to ujoadmin;
GRANT UPDATE on ujo_zos_dsn_trigger to ujoadmin;
GRANT INSERT on ujo_zos_dsn_trigger to ujoadmin;
GRANT SELECT on ujo_zos_job to ujoadmin;
GRANT DELETE on ujo_zos_job to ujoadmin;
```



---

```
GRANT UPDATE on ujo_zos_job to ujoadmin;
GRANT INSERT on ujo_zos_job to ujoadmin;
GRANT SELECT on ujo_ws_info to ujoadmin;
GRANT DELETE on ujo_ws_info to ujoadmin;
GRANT UPDATE on ujo_ws_info to ujoadmin;
GRANT INSERT on ujo_ws_info to ujoadmin;
GRANT SELECT on ujo_eventvu to ujoadmin;
GRANT DELETE on ujo_eventvu to ujoadmin;
GRANT UPDATE on ujo_eventvu to ujoadmin;
GRANT INSERT on ujo_eventvu to ujoadmin;
GRANT SELECT on ujo_eventvul to ujoadmin;
GRANT DELETE on ujo_eventvul to ujoadmin;
GRANT UPDATE on ujo_eventvul to ujoadmin;
GRANT INSERT on ujo_eventvul to ujoadmin;
GRANT SELECT on ujo_jobrow to ujoadmin;
GRANT DELETE on ujo_jobrow to ujoadmin;
GRANT UPDATE on ujo_jobrow to ujoadmin;
GRANT INSERT on ujo_jobrow to ujoadmin;
GRANT SELECT on ujo_jobrowl to ujoadmin;
GRANT DELETE on ujo_jobrowl to ujoadmin;
GRANT UPDATE on ujo_jobrowl to ujoadmin;
GRANT INSERT on ujo_jobrowl to ujoadmin;
GRANT SELECT on ujo_job_runstatus to ujoadmin;
GRANT DELETE on ujo_job_runstatus to ujoadmin;
GRANT UPDATE on ujo_job_runstatus to ujoadmin;
GRANT INSERT on ujo_job_runstatus to ujoadmin;
GRANT SELECT on ujo_jobst to ujoadmin;
GRANT DELETE on ujo_jobst to ujoadmin;
GRANT UPDATE on ujo_jobst to ujoadmin;
GRANT INSERT on ujo_jobst to ujoadmin;
GRANT SELECT on ujo_proc_eventvu to ujoadmin;
GRANT DELETE on ujo_proc_eventvu to ujoadmin;
GRANT UPDATE on ujo_proc_eventvu to ujoadmin;
GRANT INSERT on ujo_proc_eventvu to ujoadmin;
GRANT SELECT on ujo_proc_eventvul to ujoadmin;
GRANT DELETE on ujo_proc_eventvul to ujoadmin;
GRANT UPDATE on ujo_proc_eventvul to ujoadmin;
GRANT INSERT on ujo_proc_eventvul to ujoadmin;
GRANT SELECT on alamode to ujoadmin;
GRANT DELETE on alamode to ujoadmin;
GRANT UPDATE on alamode to ujoadmin;
GRANT INSERT on alamode to ujoadmin;
GRANT SELECT on event to ujoadmin;
GRANT DELETE on event to ujoadmin;
GRANT UPDATE on event to ujoadmin;
GRANT INSERT on event to ujoadmin;
```

---

```
GRANT SELECT on intcodes to ujoadmin;
GRANT DELETE on intcodes to ujoadmin;
GRANT UPDATE on intcodes to ujoadmin;
GRANT INSERT on intcodes to ujoadmin;
GRANT SELECT on proc_event to ujoadmin;
GRANT DELETE on proc_event to ujoadmin;
GRANT UPDATE on proc_event to ujoadmin;
GRANT INSERT on proc_event to ujoadmin;
GRANT SELECT on ujo_calc_machload_helper_vu to ujoadmin;
GRANT DELETE on ujo_calc_machload_helper_vu to ujoadmin;
GRANT UPDATE on ujo_calc_machload_helper_vu to ujoadmin;
GRANT INSERT on ujo_calc_machload_helper_vu to ujoadmin;
GRANT SELECT on ujo_queuedjobs_machres_vu to ujoadmin;
GRANT DELETE on ujo_queuedjobs_machres_vu to ujoadmin;
GRANT UPDATE on ujo_queuedjobs_machres_vu to ujoadmin;
GRANT INSERT on ujo_queuedjobs_machres_vu to ujoadmin;
GRANT SELECT on timezones to ujoadmin;
GRANT DELETE on timezones to ujoadmin;
GRANT UPDATE on timezones to ujoadmin;
GRANT INSERT on timezones to ujoadmin;
grant execute on ujo_audit_info_log to ujoadmin;
grant execute on ujo_audit_msg_log to ujoadmin;
grant execute on ujo_batch_pkg to ujoadmin;
grant execute on ujo_batch_get_event to ujoadmin;
grant execute on ujo_batch_move_event to ujoadmin;
grant execute on ujo_ext_batch_pkg to ujoadmin;
grant execute on ujo_batch_get_ext_event to ujoadmin;
grant execute on ujo_bump_oid to ujoadmin;
grant execute on ujo_chase_state to ujoadmin;
grant execute on ujo_chk_cond to ujoadmin;
grant execute on ujo_chk_cond_pkg to ujoadmin;
grant execute on ujo_cleanup_jobruns to ujoadmin;
grant execute on ujo_comm_update to ujoadmin;
grant execute on ujo_cred_delete to ujoadmin;
grant execute on ujo_cred_get to ujoadmin;
grant execute on ujo_cred_set to ujoadmin;
grant execute on ujo_delete_dependent_strings to ujoadmin;
grant execute on ujo_delete_job_dependants to ujoadmin;
grant execute on ujo_delete_job to ujoadmin;
grant execute on ujo_det_rep to ujoadmin;
grant execute on ujo_det_rep_pkg to ujoadmin;
grant execute on ujo_eval_has_notrunning to ujoadmin;
grant execute on ujo_event_state to ujoadmin;
grant execute on ujo_get_eoid_pkg to ujoadmin;
grant execute on ujo_get_id to ujoadmin;
grant execute on ujo_get_job_status to ujoadmin;
```

```
grant execute on ujo_get_job to ujoadmin;
grant execute on ujo_get_job_pkg to ujoadmin;
grant execute on ujo_get_jobstart_depends to ujoadmin;
grant execute on ujo_get_jobstart_depends_pkg to ujoadmin;
grant execute on ujo_get_job_for_jil to ujoadmin;
grant execute on ujo_get_job_for_jil_pkg to ujoadmin;
grant execute on ujo_get_legacy_job to ujoadmin;
grant execute on ujo_get_legacy_job_pkg to ujoadmin;
grant execute on ujo_get_mach_jobs to ujoadmin;
grant execute on ujo_get_mach_jobs_pkg to ujoadmin;
grant execute on ujo_get_machine to ujoadmin;
grant execute on ujo_get_machine_pkg to ujoadmin;
grant execute on ujo_get_primary_info to ujoadmin;
grant execute on ujo_get_run_num to ujoadmin;
grant execute on ujo_get_sched_info to ujoadmin;
grant execute on ujo_get_sched_info_pkg to ujoadmin;
grant execute on ujo_glob_set to ujoadmin;
grant execute on ujo_job_lookup to ujoadmin;
grant execute on ujo_post_runtime_info to ujoadmin;
grant execute on ujo_post_virt_resources to ujoadmin;
grant execute on ujo_put_cmd to ujoadmin;
grant execute on ujo_put_fw to ujoadmin;
grant execute on ujo_put_keyed_blob to ujoadmin;
grant execute on ujo_get_resource_ref to ujoadmin;
grant execute on ujo_put_job_resource_dep to ujoadmin;
grant execute on ujo_put_job_resource_dep_pkg to ujoadmin;
grant execute on ujo_put_machine to ujoadmin;
grant execute on ujo_put_monbro to ujoadmin;
grant execute on ujo_put_notification to ujoadmin;
grant execute on ujo_put_svcdesk to ujoadmin;
grant execute on ujo_put_virt_resource to ujoadmin;
grant execute on ujo_put_workflow to ujoadmin;
grant execute on ujo_release_virt_resources to ujoadmin;
grant execute on ujo_sendafm to ujoadmin;
grant execute on ujo_send_event to ujoadmin;
grant execute on ujo_afm_sendevent to ujoadmin;
grant execute on send_event to ujoadmin;
grant execute on ujo_sendevent to ujoadmin;
grant execute on ujo_send_ext_event to ujoadmin;
grant execute on ujo_setup_next to ujoadmin;
grant execute on ujo_post_event to ujoadmin;
grant execute on ujo_set2act to ujoadmin;
grant execute on ujo_set_eoid_counter to ujoadmin;
grant execute on ujo_set_job_status to ujoadmin;
grant execute on setnget_status to ujoadmin;
grant execute on ujo_setnget_pkg to ujoadmin;
```

```

grant execute on ujo_sum_rep to ujoadmin;
grant execute on ujo_time_offset to ujoadmin;
grant execute on ujo_update_alamode to ujoadmin;
grant execute on ujo_update_batches to ujoadmin;
grant execute on ujo_update_sap_ver_dependants to ujoadmin;
grant execute on ujo_update_job_ver_dependants to ujoadmin;
grant execute on ujo_update_job_ver to ujoadmin;
grant execute on ujo_put_job to ujoadmin;
grant execute on ujo_unset_override to ujoadmin;
grant execute on ujo_calculate_avg_jobruns to ujoadmin;
grant execute on ujo_get_preds_and_succs to ujoadmin;
grant execute on ujo_get_preds_and_succs_pkg to ujoadmin;
grant execute on ujo_get_queued_jobs to ujoadmin;
grant execute on ujo_get_queued_jobs_pkg to ujoadmin;

```

All the required privileges on the AutoSys Workload Automation database objects are granted to the ujoadmin role for AutoSys Workload Automation to work properly.

## Back Out Upgrade Troubleshooting

After backing out from an upgraded Version 12.0 installation, when the Version 12.0 upgrade is initiated again on the computer where the Version 12.0 instance was backed out, the installer fails with errors. The reason is that after the Version 12.0 installation is complete, the RPM database on the upgraded computer is wiped off.

Pre-Version 12.0 installations used RPM databases, but Version 12.0 uses InstallAnywhere technology for installation and upgrade. The Version 12.0 install logic depends on the RPM database to identify the components that were installed during the previous installation.

### Symptom:

The following steps confirm that the upgrade fails:

1. Running 'ls -l | grep 11.3' displays CAWorkloadAutomationAE as the folder that you replaced for the \$CASHCOMP folder on your computer:

```

[root@ibntest005089 ~]# ls -l | grep 11.3
PIF CAWorkloadAutomationAE 11.3.6.1643 Linux

```

2. Running 'ls -l -O rpm|grep ca-waae' command displays empty output as the RPM database is lost.

```

[root@ibntest005089 ~]# ls -l -O rpm|grep ca-waae
[root@ibntest005089 ~]#
[root@ibntest005089 ~]#

```

3. Running Version 12.0 installation results in the CAUAJM\_E\_115001 or CAUAJM\_E\_113020 error:

- Scenario #1: When AutoSys is installed without the WebServices components, it results in the error CAUAJM\_E\_115001:

```

[root@ibntest005089 Linux]# ./install.bin
Preparing to install
Extracting the JRE from the installer archive...
Unpacking the JRE...
Extracting the installation resources from the installer archive...
Configuring the installer for this system's environment...

- - - - -
- - - - -
- - - - -

```

```

- - - - -
=====
Upgrade Options

Select an option.
 ->1- Upgrade the product [Upgrades AutoSys components]
 2- Create or Upgrade the database schema [Creates or Upgrades the AutoSys database schema]
ENTER THE NUMBER FOR YOUR CHOICE, OR PRESS <ENTER> TO ACCEPT THE DEFAULT:
=====
Choose Java Virtual Machine

Please choose a 64-bit Java VM for use by the installed application
 ->1- Use the Java VM installed with this application
 2- /usr/bin/java
 3- Choose a Java VM already installed on this system
ENTER THE NUMBER FOR THE JAVA VM, OR PRESS <ENTER> TO ACCEPT THE
 CURRENT SELECTION:
=====
Installation Path

Specify the installation path.Installation path (Default: /opt/CA/WorkloadAutomationAE):
=====
Installation Path Validation

Error:
[CAUAJM_E_115001] The specified installation path is already in use:"/opt/CA/WorkloadAutomationAE".
Specify a different path.
PRESS <ENTER> TO ACCEPT THE FOLLOWING (OK):

```

- **Scenario #2: When AutoSys is installed with the WebServices components, it results in the error CAUAJM\_E\_113020:**

```

[root@ibntest005089 Linux]# ./install.bin
Preparing to install
Extracting the JRE from the installer archive...
Unpacking the JRE...
Extracting the installation resources from the installer archive...
Configuring the installer for this system's environment...
Launching installer...=====
AutoSys Workload Automation (created with InstallAnywhere)

Preparing CONSOLE Mode Installation...
=====
OS Validation

Error:[CAUAJM_E_113020] The dependent component(s) of the installed AutoSyscomponents is missing:
"REST web service : Application server
SOAP web service : Application server"

```

### Solution:

1. Identify the list of RPMs installed during the earlier install or upgrade.  
Change your present working directory to \$CASHCOMP/installer/administration/admi/  
CAWorkloadAutomationAE/11.3.\* and run the following command to get the list of RPMs installed during installation.

```
[root@ibntest005089 log]# ls
```

```
$CASHCOMP/installer/administration/admi/CAWorkloadAutomationAE/11.3.*
CAWorkloadAutomationAE.@prm
[root@ibntest005089 log]# cd
$CASHCOMP/installer/administration/admi/CAWorkloadAutomationAE/11.3.*
```

In the above output because CAWorkloadAutomationAE.@prm is the most recent file, the next command should also contain the same file; if this is a different file in your environment, then you must use that file name from your environment in the next command we execute.

```
[root@ibntest005089 11.3.6.1643]# grep installedproducts CAWorkloadAutomationAE.@prm | awk ' BEGIN { FS="
" } { print $2 } ' | sed 's/,/\n/g' | grep ca-waae
ca-waae-jre-32
ca-waae-sockadapter
ca-waae-jre-64
ca-waae-safex
ca-waae-base
ca-waae-webserver
ca-waae-common
ca-waae-common-cs
ca-waae-common-cs2
ca-waae-sdk
ca-waae-agent-helper
ca-waae-common-sac
ca-waae-client
ca-waae-asapi
ca-waae-db
ca-waae-scheduler
ca-waae-server
```

2. Load the CA WAAE r11.3.6 SP8 DVD or change to the folder where you have the Release 11.3.6 SP8 DVD available.

```
[root@ibntest005089 Linux]# cd /share1/UAJM/11.3.6/SP8/waae1136_unix
[root@ibntest005089 waae1136_unix]# ls -l
acknowledgements.txt
Documentation
Linux
VendorEULA.txt
version_check.sh
Wa_setup.sh
```

3. Change the directory to Linux/WorkloadAutomationAE/RPM/ in the DVD location.

```
[root@ibntest005089 waae1136_unix]# cd Linux/WorkloadAutomationAE/RPM/
[root@ibntest005089 RPM]# ls -l
ca-waae-agent-helper-11.3.6-1643.i386.rpm
ca-waae-common-11.3.6-1643.i386.rpm
ca-waae-db-11.3.6-1643.i386.rpm
ca-waae-asapi-11.3.6-1643.i386.rpm
ca-waae-common-cs-11.3.6-1643.i386.rpm
ca-waae-scheduler-11.3.6-1643.i386.rpm
ca-waae-base-11.3.6-1643.i386.rpm
ca-waae-common-cs2-11.3.6-1643.i386.rpm
ca-waae-sdk-11.3.6-1643.i386.rpm
ca-waae-client-11.3.6-1643.i386.rpm
ca-waae-common-sac-11.3.6-1643.i386.rpm
ca-waae-server-11.3.6-1643.i386.rpm
```

4. Run the following commands in the same order by selecting the RPM list from the #1 above; if any of the item is missing in your earlier investigation, then you need not run the RPM command for that specific RPM package:

```

rpm -Uvh --justdb ca-waae-base-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-common-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-agent-helper-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-asapi-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-common-cs2-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-common-sac-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-common-cs-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-client-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-db-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-server-11.3.6-1643.i386.rpm
rpm -Uvh --justdb ca-waae-scheduler-11.3.6-1643.i386.rpm

```

5. Go back to the AutoSys Workload Automation Version 12.0 DVD location and start the installation; you will notice that the installation is successful.

```

[root@ibntest005089 Linux]# ./install.bin
Preparing to install
Extracting the JRE from the installer archive...
Unpacking the JRE...
Extracting the installation resources from the installer archive...
Configuring the installer for this system's environment...

- - - - -
- - - - -
- - - - -
- - - - -

=====
Installation Complete

AutoSys Workload Automation has been successfully installed to:
/opt/CA/WorkloadAutomationAE
Post-installation logs will be written to:
/opt/CA/WorkloadAutomationAE/Logs/AutoSys_Workload_Automation_<Install_time_stamp>.log
/opt/CA/WorkloadAutomationAE/Logs/cawa_installer_06_08_2020_02_18_34.log
PRESS <ENTER> TO EXIT THE INSTALLER:

```

6. After the installation, the product is up and running as expected.

```

[root@ibntest005089 Linux]# su
[root@ibntest005089 Linux]# cd /opt/CA/WorkloadAutomationAE/autouser.ACE/
[root@ibntest005089 autouser.ACE]# unisrvcntr start waae_agent-WA_AGENT waae_sched.ACE waae_server.ACE
Starting waae_agent-WA_AGENT (via systemctl): [OK]
Executed waae_agent-WA_AGENT start.....OK
Starting waae_sched.ACE (via systemctl): [OK]
Executed waae_sched.ACE start.....OK
Starting waae_server.ACE (via systemctl): [OK]
Executed waae_server.ACE start.....OK
[root@ibntest005089 autouser.ACE]#
[root@ibntest005089 autouser.ACE]#
[root@ibntest005089 autouser.ACE]# chk_auto_up

CAUAJM_I_50054 Attempting (1) to Connect with Database: AEORA01
CAUAJM_I_50055 *** Have Connected successfully with Database: AEORA01. ***

CAUAJM_I_50128 Connected with Event Server: AEORA01

```

```
CAUAJM_I_50038 Checking AutoSys Primary Scheduler on Machine:
ibntest005089.abc.example.net
CAUAJM_I_50044 Primary Scheduler is RUNNING on machine: ibntest005089.abc.example.net
```

---

**NOTE**

- This procedure may not be required if you do not backout.
- Use this procedure only when you do not have the option for a fresh installation or when your system is unusable.

## Event Server Troubleshooting

This article describes scenarios for troubleshooting the event server.

### Event Server Is Down

**Symptom:**

When I issue the `chk_auto_up` command, a message similar to the following is displayed:

```
Couldn't connect with Server: AUTOSYS:autosys
```

**Solution:**

Either the database server is down or the process in question cannot access the database server.

To verify whether the database server is down, log in to the event server and check if the database processes are active.

If the database is running, the problem could be that AutoSys Workload Automation is configured to the wrong event server or communication between AutoSys Workload Automation and the event server is not configured correctly.

### Deadlocks

**Symptom:**

The database server error log or the scheduler log (the output of the `autosyslog -e` command) displays a message similar to the following:

```
Your server command (process id #11) was deadlocked with another process and has been
chosen as deadlock victim. Re-run your command.
```

**Solution:**

A deadlock is a condition that occurs when two users have a lock on separate objects, and they each want to acquire an additional lock on the other user's object. The first user is waiting for the second user to release the lock, but the second user will not release it until the lock on the first user's object is released.

The database server detects the situation and selects the user whose process has accumulated the least amount of CPU time. The database server rolls back that user's transaction, notifies the application with the indicated error message, and lets the other user's processes continue.

AutoSys Workload Automation tries to rerun the command until it is successful or until it has exceeded the maximum number of retries.



**NOTE**

AutoSys Workload Automation defines database table indices to optimize the performance of database queries. A database deadlock is an indication that the database table index definitions have been manually altered or removed.

**Not Enough User Connections****Symptom:**

AutoSys Workload Automation processes cannot make connections to the database; they cannot start the Web UI or send events.

**Solution:**

Verify the maximum number of user connections your system can support. If the current number of connections does not exceed the capacity of your environment, you can increase the number of user connections.

**NOTE**

For more information about increasing the maximum number of user connections, contact your database administrator or see the database documentation.

**Job Troubleshooting**

This article describes scenarios for troubleshooting job failures and problems.

**Agent Will Start Command Job Will Not Run**

Each time the agent starts on a computer, it creates a log file for command jobs in the spool subdirectory in the SystemAgent directory. This log file contains all the instructions passed to the agent by the scheduler, the results of any resource checks, and a record of all actions taken. Any problems encountered by the agent are recorded in this log file.

To retrieve the most recent instance of the agent log for a given job, enter the following command on the computer where the job last ran:

```
autosyslog -J job_name
```

- ***job\_name***  
Defines the name of the job.

To retrieve a particular instance of the agent log for a given job run, enter the following command on the computer where the job last ran:

```
autosyslog -J job_name -r run_num -n ntry
```

- ***run\_num***  
Defines the job's run number.
- ***ntry***  
Defines the number of tries or restarts.

**Symptom:**

When I issue the autosyslog -e command, the scheduler log displays a message similar to *one* of the following:

```
Owner UserId/Password error! ERROR: The password specified for USER@HOSR_OR_DOMAIN is
invalid! Run "autosys_secure" to enter the correct password.
```

or

Owner UserId/Password error! ERROR: No valid password was found for USER@HOST or USER@DOMAIN. Cannot run job for user USER! Run "autosys\_secure" to enter the user password.

When I issue the autorep -J *job\_name* command, the agent log might also display a message similar to *one* of the following:

The password specified for USER@DOMAIN is invalid! Run "autosys\_secure" to enter the correct password.

or

No valid password was found for USER@HOST or USER@DOMAIN. Cannot run job for user USER! Run "autosys\_secure" to enter the user password.

#### **Solution:**

The password for *user@host\_or\_domain* does not exist or is invalid. To fix this problem, issue the autosys\_secure command to enter or change the user ID and password.

#### **Symptom:**

When I issue the autosyslog -e command, the scheduler log indicates that the job immediately returned a FAILURE status.

#### **Solution:**

To verify this problem, issue the autosyslog -e command on the scheduler computer and the autorep -J *job\_name* command on the computer where the job should have run, and review the resultant error messages.

For example, if the job's standard output file was read-only, a message indicating this is included in the scheduler log.

You should also verify the following:

- Ensure that the default profile or the job's specified user-defined profile defines the appropriate job environment. In particular, ensure that the path variable, if defined in a job profile, is correct. You should always include the following in any job profile that defines a path variable to help ensure that all system path directories are accessible:
  - On UNIX -- \$PATH
  - On Windows -- %PATH%
- Ensure that the file system where the job command resides is accessible from the computer where the job should have run.
- Ensure that the system permissions are correct for the command job to run.
- Ensure that the permissions are correct on any standard input and output files specified for redirection.

#### **NOTE**

A valuable debugging technique is to specify a file to use for standard output and standard error for a job that is having run problems. If there are any command problems, most of the error messages are recorded in that file.

#### **Symptom:**

When I issue the autosyslog -e command, the scheduler log displays a message similar to the following when a job starts:

COMM\_ERR\_5 Communication attempt with Agent on machine [*machine\_name*:49154] has failed.

#### **Solution:**

This message is displayed in the following situations:

- Performance problems with the network or machine
- Network problems
- Incompatible encryption settings between the scheduler and agent
- Machine definition for the agent is not correct
- Port number that is used by the agent is not correct

Occasionally, performance problems on the network or computer might cause communication failures. The network might be down or slow due to high traffic volume. The computer might be underpowered, or you are trying to do too much on it at one time.

#### NOTE

For the communication between the scheduler and the agent to be successful, you must ensure that the agent encryption settings on AutoSys Workload Automation match the encryption settings defined in the agentparm.txt file on the agent. If you detect any encryption problems, you must modify the encryption type and encryption key on AutoSys Workload Automation to match the encryption settings defined in the agentparm.txt file on the agent. For more information about modifying the encryption type and encryption key on AutoSys Workload Automation, see [Modify the Encryption Type and Encryption Key on AutoSys Workload Automation](#).

### **(UNIX) Agent Not Found**

#### **Symptom:**

When I issue the autosyslog -e command, the scheduler log displays the following message when I try to start a job or try to start the scheduler with a shadow scheduler:

```
COMM_ERR_2 Hostname for machine [machine_name] is invalid or unreachable over the
network.
```

#### **Solution:**

This message is displayed in the following situations:

1. There is a network problem and the scheduler cannot connect to the agent computer.
2. The agent computer is not in /etc/hosts or DNS.
3. The AutoSys Workload Automation configuration file lists the computer; however, there is a space after the computer name. Check /etc/hosts or DNS for the computer name, and correct it if necessary.

#### **Follow these steps:**

1. Check the node\_name attribute of the AutoSys Workload Automation machine definition corresponding to the machine in question.
2. Enter the following command at the operating system prompt:

```
autorep -m machine -q
```

The machine definition is exported to a text file.

3. Edit the configuration file (remove anything after the name of the computer and before the \$ that marks the end of the line) using an editor, such as vi (with the :set list option).
4. If a problem is found with the node\_name attribute of the machine, re-create the machine definition using the jil utility.

### **(Windows) Job Fails Multiple Interactive Log in Sessions**

#### **Symptom:**

When a command job with the interactive attribute set to true is scheduled, the job fails when the userid defined to run the job is currently logged on to Windows with more than one session. The autorep -d command displays the following error message:

More than one interactive logon session

The job status appears as a failure.

#### **Solution:**

The agent works according to Microsoft restrictions. To schedule interactive jobs, follow these two rules:

1. The user specified to run the job **MUST** be logged on with an active session.
2. The user can have **ONLY** one active session.

If more than one session exists for the specified user, the agent cannot determine which session to use and therefore fails the job.

### **(UNIX) Jobs Run Only From the Command Line**

#### **Symptom:**

Jobs run from the command line, but they fail when run.

#### **Solution:**

This problem is nearly always in the shell environment where the job runs. The following are the possible reasons:

- The profile in the job definition is not a Bourne shell (sh) type profile. If this is the case, the profile fails.
- The default profile does not produce the proper environment for the job to run. The default profile for all jobs is /etc/auto.profile, not the job owner's log in profile \$HOME/.profile. If the job owner's profile is not specified in the job definition, it is never sourced.
- The oscomponent.defaultshell.force parameter is set to true in the agentparm.txt file for the agent. This loads the /etc/auto.profile in the shell specified in the oscomponent.defaultshell parameter rather than the job owner's login shell.

#### **To verify the difference between the job definition and the user environment**

1. Log in as the owner of the job on the computer where the job runs, and enter the following command at the operating system prompt:

```
env >user.env
```

The current owner's environment is written to a file.

2. Enter the following commands at the operating system prompt:

```
insert_job: auto_env
machine: client_hostname
owner: owner
command: env
std_out_file: /tmp/auto.env
std_err_file: /tmp/auto.err
```

– **client\_hostname**

Defines the host name of the computer where the problem job runs.

– **owner**

Defines the owner of the job that will not run.

The agent environment is written to a file.

3. Enter the following command at the operating system prompt:

```
sendevent -E STARTJOB -J auto_env
```

The problem job runs.

4. Enter the following command at the operating system prompt:

```
diff /tmp/auto.env user.env
```

The differences in the two files (std\_out\_file and std\_err\_file) are verified. The diff command displays where the environment and the user environment files differ. Make the necessary changes in the job definition and the user profile.

It is also useful to define the std\_err\_file file for the job that fails, because you can check the errors from the shell to get an indication about what is missing.

#### **NOTE**

No spaces are allowed between the >> characters and the full path or file name in the std\_out\_file or std\_err\_file fields in a job definition.

### **Job Remains in STARTING or RUNNING State**

#### **Symptom:**

The state of a job remains in STARTING or RUNNING.

#### **Solution:**

A job remaining in the STARTING or RUNNING state is indicative of the following issues:

- Agent is unable to communicate with the scheduler on the agent listener port of the scheduler  
Verify that the Secure Sockets Adapter port configurations are identical on both the agent machine and on the scheduler machine for the 49152-50176 port range.  
Run chk\_auto\_up to exercise the agent listener port of the scheduler. If chk\_auto\_up reports that the scheduler is running, then the scheduler is capable of accepting communications on its agent listener port.  
Run autoping -m <machine> to exercise the communication path between the agent and the scheduler. If autoping reports success then the agent is capable of communicating with the scheduler on its agent listener port. Autoping returns the following to indicate that the agent is experiencing an internal delay:  

```
CAUAJM_W_10496 Agent on [<machine>] has not responded in a timely fashion. Try again later.
```
- Agent has encountered an internal limitation preventing it from executing the job or sending status  
The agent logs in both the SystemAgent/<agent\_name> and SystemAgent/<agent\_name>/logs subdirectories of the install path need to be consulted to determine the nature of the limitation:
  - 1) The nohup.stderr file may contain miscellaneous operating system error messages.
  - 2) The default\_agent.log file may contain agent-specific errors not applicable to any one agent component.
  - 3) The various agent component logs contain errors specific to a function of the agent.
 The agent may have temporarily exhausted all of its internal initiators. This means that the agent is trying to start too many jobs at once. For information on how to increase the number of initiators, refer to [Workload Automation Agents](#). Check that the SystemAgent/<agent\_name> subdirectory of the install path on the agent machine has sufficient disk space.

### **(UNIX) Unable to Run Jobs Using Cross Platform Scheduling**

#### **Symptom:**

When running a job from a cross platform scheduling manager to AutoSys Workload Automation, the job does not start. The scheduler log displays the following error message:

```
CAUAJM_E_40320 User authentication failure for {0}. User does not exist or bad password has been specified.
```

#### **Solution:**

The problem is with the agent associated with the job. To correct the problem, add the following line to the agentparm.txt file for the agent.

```
oscomponent.auth.pam.svc=sshd
```

### NOTE

- By default, the agent verifies the user password using the default service, for example login. The sshd service changes the way the agent verifies the password.
- For more information about the oscomponent.auth.pam.svc parameter, see [Workload Automation Agent Parameters](#).

## **A Job with Date or Time Condition does not Start OR Next STARTJOB Event Missing for Job**

### Symptom:

A job with date/time conditions does not start on time even when conditions like start\_mins, start\_times, and run\_window attributes have valid values and the calendar attached to the job contains valid future dates.

### Cause:

This can happen when the job's date/time conditions and its parent job's date/time conditions are different and the box\_job does not stay in RUNNING state until the date/time conditions of the job are valid.

### Example

Consider a scenario, where a job is part of a box job and both have date/time conditions. When the box job starts, all jobs without any conditions and future STARTJOB events will be started and then based on the conditions of the jobs, other jobs in the box will start. When an unprocessed STARTJOB event is present, the scheduler will not attempt to start the child job even if the conditions of the job are good enough to start the job. The box job will stay in RUNNING status until the child job with date/time conditions are met.

If for any external reasons a STARTJOB event is triggered, then there is a possibility of the parent job going into its terminal status before the actual start time of the child job; when the child job's STARTJOB event is processed, the box\_job will be in terminal status and will lead to child job's STARTJOBFAIL error.

### Solution:

Option #1: Check if the child job has a future STARTJOB event matching the date/time conditions; if the STARTJOB event is missing, you can issue a CHANGE\_STATUS SUCCESS/FAILURE to make sure the job gets a future STARTJOB event.

Option #2: Issue a JOB\_ON\_HOLD to the child job and make sure the child job will not start immediately when the box\_job goes into RUNNING status. Place a future JOB\_OFF\_HOLD event after child job's date/time condition time. This will make sure the box\_job will stay in RUNNING status and the job will start due to JOB\_OFF\_HOLD event and generate a next future STARTJOB event that matches the job definition.

## **Scheduler Troubleshooting**

This article describes scenarios for troubleshooting the scheduler.

Output from the scheduler is redirected to the following log file:

- On UNIX -- \$AUTOUSER/out/event\_demon.\$AUTOSERV
- On Windows -- %AUTOUSER%\out\event\_demon.%AUTOSERV%

You must issue the autosyslog -e command to view the scheduler log file. This log file contains a record of all the actions taken by the scheduler (in the order performed). Network problems are usually reflected in this log file. This log file is very useful for reconstructing what happened when a problem occurs.

### NOTE

To terminate autosyslog, press Ctrl+C.

You can view the following short video, to see how to capture a dump file from a running process.

**NOTE**

Click the YouTube logo to open this video in a new tab with all normal YouTube functionality.

**Scheduler Is Down****Symptom:**

- Jobs do not start.
- When I issue the `chk_auto_up` command, a message similar to the following is displayed:  

```
No Scheduler is RUNNING.
```
- The scheduler log has not registered a date and time stamp for a while. The scheduler log should register date and time stamps each minute.

**Solution:**

Do one of the following to verify whether the scheduler is down:

- Issue the `chk_auto_up` command. This command verifies if the scheduler is running.
- Issue the `autosyslog -e` command. This command displays the scheduler log file. Check for date and time stamps.

On UNIX, check for running AutoSys Workload Automation scheduler processes. If the scheduler is down, you must issue the `eventor` command to start it.

On Windows, verify the status of the scheduler using the Services - AutoSys Workload Automation Administrator window of the Administrator utility. If the scheduler is down, you must start it using the Services - AutoSys Workload Automation Administrator window of the Administrator utility.

**NOTE**

For more information about how to verify the scheduler status or start the scheduler using the Administrator utility, see [Control the Services](#).

**Scheduler Will Not Start****Symptom:**

The `autosyslog -e` command displays messages indicating that it cannot connect to the database.

**Solution:**

This problem occurs if the database is down or there are database problems. To correct this problem, verify that the database is running and that you can connect to it by issuing the `autoping` command. After the database is accessible, the scheduler should be able to connect to the database.

**Symptom:**

- The `autosyslog -e` command displays messages indicating that the scheduler log file does not exist, or that no entries were made when the scheduler service was started.
- The scheduler service does not remain running or never starts.

**(UNIX) Solution:****To correct this problem:**

1. Check for a file named `event_demon.$AUTOSERV` in the `$AUTOUSER/out` directory.
2. If the file exists, enter the following command at the operating system prompt:

```
cat $AUTOUSER/out/event_demon.$AUTOSERV | more
```

You can view the `event_demon.$AUTOSERV` file.

3. Identify the problems at the end of the file, correct them, and restart the scheduler.  
The problem is corrected.

**NOTE**

The scheduler appends the event\_demon.\$AUTOSERV file each time it starts.

**(Windows) Solution:****To correct this problem:**

1. Check for a file named event\_demon.%AUTOSERV% in the %AUTOUSER%\out directory.
2. If the file exists, enter the following command at the instance command prompt:

```
cat %AUTOUSER%\out\EVENT_DEMON.%AUTOSERV% | more
```

You can view the event\_demon.%AUTOSERV% file.

3. Identify the problems at the end of the file, correct them, and restart the scheduler.  
The problem is corrected.

**NOTE**

The scheduler appends the event\_demon.%AUTOSERV% file each time it starts.

**Symptom:**

The scheduler does not remain running and does not write log output to the \$AUTOUSER/out/event\_demon.\$AUTOSERV (UNIX) or %AUTOUSER%\out\event\_demon.%AUTOSERV% (Windows) file.

**(UNIX) Solution:**

This problem could have various causes and the solution depends on which of the following message is displayed:

- **The log file \$AUTOUSER/out/event\_demon.\$AUTOSERV is missing!**  
The scheduler must have been started on the computer at least once or this message is displayed. If the scheduler has been started, ensure that permissions are set on the log file that enables a system program to read and write to it.
- **The environment variable AUTOSYS is not set.**  
This message is displayed if the \$AUTOSYS system environment variable is not available to the scheduler. You must ensure that the AutoSys Workload Automation source file has been sourced in your session.
- **The AutoSys Workload Automation environment has not been installed correctly.**  
This message is displayed when the scheduler runs the chk\_auto\_up command on initialization, and it reports that the setup is incorrect. You must ensure that the AutoSys Workload Automation source file has been sourced in your session.
- **The primary, shadow, or the tie-breaker scheduler is already running. Startup aborted.**  
This message is displayed when the scheduler starts, and it detects another scheduler running with the same instance ID. Only one scheduler can run in an instance. Either stop the other scheduler, or do not try to start this scheduler.
- **Scheduler cannot open its log file event\_demon.\$AUTOSERV. Some directory in the path is not accessible to the SYSTEM.**  
This message is displayed when the scheduler cannot create the event\_demon.\$AUTOSERV log file. You must ensure that the log file has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.
- **Could not rename the LARGE scheduler file: event\_demon.\$AUTOSERV to backup archive file: event\_demon.\$AUTOSERV.date. Fix file and directory permissions so accessible by SYSTEM, or remove the files.**  
This message is displayed when the scheduler starts and checks the size of the event\_demon.\$AUTOSERV log file. If this file is larger than 100 MB, the scheduler tries to rename it to event\_demon.\$AUTOSERV.date and creates a new event\_demon.\$AUTOSERV log file. If the scheduler cannot do this, verify that event\_demon.\$AUTOSERV has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.



**NOTE**

You can use the LOGROLLOVER environment variable to specify when the scheduler or the application server log rolls over.

**(Windows) Solution:**

This problem could have various causes; and the solution depends on which of the following message is displayed on the Event Log Viewer dialog. You can access the Event Log Viewer dialog using the Windows Control Panel Administrative Tools dialog.

- **The log file %AUTOUSER%\out\event\_demon.%AUTOSERV% is missing!**

The scheduler must have been started on the computer at least once or this message is displayed. If the scheduler has been started, ensure that permissions are set on the log file that enables a system program to read and write to it.

- **The environment variable AUTOSYS is not set.**

This message is displayed if the %AUTOSYS% system environment variable is not available to the scheduler. You must ensure that the %AUTOSYS% system environment variable is set properly by using the Windows Control Panel or the System - AutoSys Workload Automation Administrator window of the Administrator utility.

**NOTE**

For more information about how to add, modify, or delete a system environment variable using the Administrator utility, see [Environment Variables](#).

- **The environment variable AUTOSYS is too long.**

This message is displayed if the %AUTOSYS% system environment variable value is set to a path that is more than 80 characters in length. You must uninstall AutoSys Workload Automation, and reinstall it to a directory path that is fewer than 80 characters in length.

- **chk\_auto\_up process is missing. Scheduler not operational. Call Tech support.**

This message is displayed when the scheduler runs the chk\_auto\_up command on initialization, and that process is terminated without properly notifying the scheduler. This indicates a serious problem with your local system account. You must try setting the scheduler to log in as the administrator. If this is successful, you can run the scheduler. However, we recommend that you uninstall and reinstall AutoSys Workload Automation.

- **chk\_auto\_up times out while waiting for response from application server**

This message is displayed when the application server does not respond. You must verify whether the application server is running by using the Services - AutoSys Workload Automation Administrator window of the Administrator utility.

**NOTE**

For more information about how to verify the status of the application server using the Administrator utility, see [Control the Services](#).

- **chk\_auto\_up is taking a while to complete...**

This message is displayed when the scheduler runs the chk\_auto\_up command on initialization, and it takes more than five minutes to complete. This might occur on large or slow networks where the chk\_auto\_up command must query every machine listed in the Authorized Manager List pane on the Agent - AutoSys Workload Automation Administrator window of the Administrator utility. To test this problem, issue the chk\_auto\_up command at the instance command prompt, and check how long it takes to complete. This message is only a warning, and the scheduler waits for the command to complete before starting.

- **Wait for chk\_auto\_up process failed. Windows Error Code**

This message is displayed when the scheduler runs the chk\_auto\_up command on initialization, and it terminates prematurely with a Windows error code. You must verify that chk\_auto\_up.exe is located in the %AUTOSYS%\bin directory and has the proper permissions for system programs to execute.

- **The AutoSys Workload Automation environment has not been installed correctly.**

This message is displayed when the scheduler runs the chk\_auto\_up command on initialization, and it reports that the setup is incorrect. You must uninstall and reinstall AutoSys Workload Automation.

- **The primary, shadow, or tie-breaker scheduler is already running. Startup aborted.**

This message is displayed when the scheduler starts, and it detects another scheduler running with the same instance ID. Only one scheduler can run in an instance. Either stop the other scheduler, or do not try to start this scheduler.

- **Scheduler cannot open its log file event\_demon.%AUTOSERV%. Some directory in the path is not accessible to the SYSTEM.**

This message is displayed when the scheduler cannot create the event\_demon.%AUTOSERV% log file. You must ensure that the log file has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.

- **Could not rename the LARGE scheduler file: event\_demon.%AUTOSERV% to backup archive file: event\_demon.%AUTOSERV%.date. Fix file and directory permissions so accessible by SYSTEM, or remove the files.**

This message is displayed when the scheduler starts and checks the size of the event\_demon.%AUTOSERV% log file. If this file is larger than 256 KB, the scheduler tries to rename it to event\_demon.%AUTOSERV%.date and creates a new event\_demon.%AUTOSERV% log file. If the scheduler cannot do this, verify that event\_demon.%AUTOSERV% has permissions that enable a system program to read and write it. Also, verify that the disk drive has not run out of space.

#### NOTE

You can use the LOGROLLOVER environment variable to specify when the scheduler or the application server log rolls over.

## (Windows) Services Troubleshooting

You can start the application server, scheduler, and agents using the Services - AutoSys Workload Automation Administrator window of the Administrator utility. You can start the event server (the Microsoft SQL Server, Oracle, or Sybase service) using the Windows Control Panel Services dialog. You can find details as to why a service did not start using the Event Viewer option in the Windows Control Panel Administrative Tools dialog.

Typically, problems with starting services using the Administrator utility indicate that the software is not installed successfully. In such cases, the best approach is to remove the existing AutoSys Workload Automation installation and reinstall it.

#### NOTE

For more information about how to remove AutoSys Workload Automation, see [\(Windows\) Uninstall AutoSys Workload Automation](#). For more information about starting AutoSys Workload Automation services using the Administrator utility, see [Control the Services](#).

To verify whether the event server service (the database service) is started, look at the Windows Control Panel Services dialog. You can verify the following details:

- If you are running Microsoft SQL Server, verify the status of the MSSQLServer service.
- If you are running Oracle, verify the status of the following services (substitute your Oracle SID for the asterisk): OracleService\*, OracleStart\*, and OracleTNSListener.
- If you are running Sybase, verify that a service with a name that starts with SYBSQL is started. It is possible that a different name was selected for the service when Sybase was installed.

## Capture Diagnostic Information

We recommend that you capture diagnostic information and provide it to the Broadcom Support team. This information helps in root cause analysis and troubleshooting of an issue.

#### NOTE

This article details how to capture diagnostic information for issues where the AutoSys Workload Automation application server or the scheduler component deadlocks or terminates ungracefully.

You can capture diagnostic information using any of these procedures:

- Collect runtime debugging trace using the [ISDBGACTIV](#) parameter in the configuration file.
- Collect the process dump or pstack or procstack or gdb.

### **Collect the process dump or pstack or procstack or gdb**

You can collect the process dump for:

- Issues where the application server or the scheduler processes appear to stall and do not shutdown.
- Issues where the application server or scheduler terminate ungracefully.

#### **NOTE**

Collecting the process dump only once shows the state of threads at that point of time alone and might not be very useful. Repeat this process at least four to five times with a gap of one minute between each collection to see if the process really stalled.

**This video describes how to capture diagnostic information when the scheduler deadlocks:**

#### **NOTE**

Click the YouTube logo to open this video in a new tab with all normal YouTube functionality.

**To collect the process dump for issues where the application server or the scheduler processes appear to stall and do not shut down, follow these steps:**

- **Windows:**  
Collect the process dump using Microsoft's User Mode Process Dumper. For more information about how to use the Userdump.exe tool to create a dump file, see <http://support.microsoft.com/kb/241215>.
- ***process\_ID***  
Defines the process ID of the scheduler or application server.
- ***aehang.txt***  
Specifies the name of the file that includes the process stack trace information.
- **AIX:** Open the operating system command prompt and execute the following command:  

```
procstack process_ID > /tmp/aehang.txt
```
- **HP-UX**
  - a. Install HP Wildebeest Debugger – WDB.
  - b. Open the operating system command prompt and execute the following commands:  

```
script /tmp/aehang.txt
gdb -p process_ID
```
  - c. At the gdb prompt, enter the following gdb commands:  

```
set pagination off [enter]
info threads [enter]
thread apply all bt [enter]
quit [enter]
```
  - d. Press Ctrl+D to complete the recording of your console session into the /tmp/aehang.txt file.
  - e. Upload the /tmp/aehang.txt file to the STAR issue.

**This video describes how to extract stack information from the core file:**

**To collect the process dump for issues where the scheduler or the application server terminate ungracefully, follow these steps:**

- **Windows:**

Collect the process dump using Microsoft's User Mode Process Dumper. For more information about how to use the Userdump.exe tool to create a dump file, see <http://support.microsoft.com/kb/241215>.

- **UNIX or Linux:**

- a. Obtain the core file.

- Usually, when a process experiences a fatal process error, the UNIX operating system generates a core file containing the memory image of the process at the time of the fatal process error. Your UNIX operating system might provide utilities that control whether the core files are generated, and if they are, where the generated core files are placed and what names to assign them. Consult your operating system documentation to configure your UNIX machine to generate core files at a known location with a known name such that it is easy to find the core file.
    - If your UNIX machine is configured with the default values to generate core files, you might find the core file by performing a recursive search for files that are named core within the AutoSys Workload Automation installation directory (/opt/CA/WorkloadAutomationAE).
    - If a core file is not generated, consult your operating system documentation to configure your UNIX machine to generate a core file at a known location with a known name and try to reproduce the fatal process error. For example, some UNIX systems use the ulimit utility to configure the operating system to generate core files. To use the ulimit utility, log in to the UNIX machine as the autosys user and execute the ulimit -a UNIX command. Verify that the output for the coredump row reads as unlimited. If the coredump row does not read as unlimited, set it by running the ulimit -c unlimited command. However, to ensure that the core file is properly generated on your UNIX machine, we recommend that you consult your operating system documentation.

- b. Execute the following command to verify that the core file got generated by running the AutoSys Workload Automation binary:

```
file complete_path_of_core_file
```

- c. Extract the process stack information from the core file that is generated.

- **Linux or HP-UX:**

- a. Install HP Wildebeest Debugger – WDB.

- b. Run the shell that is sourced to use AutoSys Workload Automation and execute the following commands from the operating system prompt:

```
script /tmp/aestack.txt
```

- c. Execute the following command if the core file resulted from running the scheduler:

```
gdb $AUTOSYS/bin/event_demon complete_path_to_core_file
```

- d. Execute the following command if the core file resulted from running the application server:

```
gdb $AUTOSYS/bin/as_server complete_path_to_core_file
```

- e. Enter the following commands at the gdb prompt:

```
(gdb) set pagination off [enter]
(gdb) info threads [enter]
(gdb) thread apply all bt [enter]
(gdb) quit [enter]
```

- f. Press Ctrl+D to complete the recording of your console session into the /tmp/aestack.txt file.

- g. Upload the /tmp/aestack.txt file to the STAR issue.

- **AIX:**

- a. Run the shell that is sourced to use AutoSys Workload Automation and execute the following commands from the operating system prompt:

```
script /tmp/aestack.txt
```

- b. Execute the following command if the core file resulted from running the scheduler:

```
dbx $AUTOSYS/bin/event_demon complete_path_to_core_file
```

- c. Execute the following command if the core file resulted from running the application server:

```
dbx $AUTOSYS/bin/as_server complete_path_to_core_file
```

- d. Enter the following commands at the dbx prompt:

```
(dbx) thread [enter]
```

A list of all thread numbers in the form of \$tN are displayed; where N is the thread number.

- e. Cut and paste this output into a text editor.

- f. Edit the text file to create the following list of commands for all thread numbers:

```
thread current 1
where
thread current 2
where
...
thread current N
where <newline>
```

- g. Copy these commands and paste them into the dbx command prompt.

- h. Enter the following command to quit the dbx command prompt:

```
(dbx) quit [enter]
```

- i. Press Ctrl+D to complete the recording of your console session into the /tmp/aestack.txt file.

- j. Upload the /tmp/aestack.txt file to the STAR issue.

#### NOTE

- If you are not able to get the process stack trace after following the instructions in this article, upload the core file to the STAR issue.
- If you upload the core file, provide detailed information about your UNIX machine including the version of your UNIX operating system and all installed patches. Also, provide detailed information about the database client libraries, including the provider (Sybase or Oracle) and the complete database product version and all patch information.
- Depending on the version or patch level of your UNIX machine or database client, it might not be possible for Broadcom Support to find an equivalent machine to extract the process stack from the uploaded core file. To ensure that Broadcom Support has the most accurate information about the state of the scheduler or application server process following an issue, extract the process stack information from the core file and upload the process stack file to the STAR issue.

## Enabling Core File Generation

On UNIX, core file creation is enabled for the server (scheduler and application server) processes to enhance supportability.

Core file creation is enabled only when the server processes are started using the `unisvrcntr` command or by a root user. During the post-installation process, the AutoSys Workload Automation source files in the `$AUTOUSER` directory

are configured to set the limit of the core file size to unlimited; this enables core file generation if there is an abnormal termination. Later, you can modify this value, but we do not recommend you to do so.

**NOTE**

- To enable core file creation on Oracle, set the following variable in the sqlnet.ora file:  
`DIAG_SIGHANDLER_ENABLED=FALSE`
- For information about configuring your operating system settings to enable creating core files, contact your system administrator.

## (Windows) AutoSys Workload Automation Error Code 20007

**Problem:**

When you submit a job to a Windows machine, the following error message is displayed in autosyslog:

```
FAILURE dd/mm/yyyy hh:mm:ss <Command file not found>
```

**Environment:**

Workload Automation Agent for Windows Release 11.3.x

**Cause:**

The Workload Automation Agent for Windows is unable to locate the file that must be executed.

**Resolution:**

Set the `oscomponent.lookupcommand` parameter in the `agentparm.txt` file as follows:

```
oscomponent.lookupcommand=true
```

**NOTE**

- Specify the `oscomponent.lookupcommand` parameter value (true or false) in lower case.
- The `agentparm.txt` file is located in the agent installation directory.
- For more information about the `oscomponent.lookupcommand` parameter, see the [Workload Automation Agent Parameters](#).

## autoping Failed From CA Workload Automation AE Server to a Particular Agent

**Problem:**

When you issue the `autoping -m` command from the CA Workload Automation AE server to an agent (for example, `xxxxxxxxSB1`), the communication attempt failed.

```
autoping -m xxxxxxxxSB1
```

The following error messages are displayed:

```
CAUAJM_I_50023 AutoPinging Machine [xxxxxxxSB1]
CAUAJM_E_10229 Communication attempt with the CA WAAE Agent has failed!
[xxxxxxxSB1:7530]
CAUAJM_E_50281 AutoPing from the Scheduler WAS NOT SUCCESSFUL.
```

```
CAUAJM_E_10229 Communication attempt with the CA WAAE Agent has failed!
[xxxxxxxSB1:7530]
CAUAJM_E_50283 AutoPing from the Application Server WAS NOT SUCCESSFUL.
```

```
CAUAJM_E_50026 ERROR: AutoPing WAS NOT SUCCESSFUL.
```

### **Environment:**

CA Workload Automation AE Release 11.3.5 INCR3 HADS on Solaris with Oracle database

### **Cause:**

This issue is with network. The agent waits only for 10 seconds between a message being sent and an acknowledgement being received. If this time exceeds, the connection is aborted.

### **Workaround:**

For slower networks, increase the value of the communication.timeout parameter in the agentparm.txt. By default, this parameter is set to 10000 (10 seconds).

Although you resolve this problem, some jobs might fail. Work with your network administrator to fix the network issue.

### **NOTE**

For more information about the communication.timeout parameter, see the [Workload Automation Agent for UNIX, Linux, or Windows documentation](#).

## **CAUAJM\_E\_10436 - CA Workload Automation AE CLIs does not work**

### **Problem:**

CA Workload Automation AE commands fail with the following error messages soon after the application server restart:

```
[autosys@waaeserver1 ~]$ autosyslog -e
CAUAJM_E_10436 Security server unreachable or invalid authentication certificate file.
CAUAJM_E_10434 Error initiating security session.
[EE_BADOBJECT Bad Object]
[ISP_ERROR_NOGATEWAY igateway not running]
[Authenticate Error: Authentication Failed]
[Identity Attempted:]
[CertificateReader::loadPEM - cannot read certificate]

CAUAJM_W_10406 Control Execute Access Denied!
```

```
CAUAJM_E_10434 Error initiating security session.
CAUAJM_W_10440 Class: as-control Resource: ACE.EPLOG User: autosys Access: execute
CAUAJM_W_10442 Time: 1455405684 Delegator: None
```

```
[autosys@waaeserver1 ~]$ autorep -J ALL
/*CAUAJM_E_10436 Security server unreachable or invalid authentication certificate file.
CAUAJM_E_10434 Error initiating security session.
[EE_BADOBJECT Bad Object]
[ISP_ERROR_NOGATEWAY igateway not running]
[Authenticate Error: Authentication Failed]
[Identity Attempted:]
[CertificateReader::loadPEM - cannot read certificate]
```

The application server log (\$AUTOUSER/out/as\_server.\$AUTOUSER) repeatedly logs the below messages soon after its restart:

```
[02/14/2016 04:49:06] CAUAJM_I_40275 Log Rollover level set to
<MIDNIGHT,SIZE(100),PURGE(7)>.
[02/14/2016 04:49:06] CAUAJM_I_40244 EnableIPCaching value set to <0>.
[02/14/2016 04:49:06] CAUAJM_I_40244 LogMaxEndLines value set to <0>.
[02/14/2016 04:49:06] CAUAJM_I_40211 Using TZ = Asia/Kolkata.
[02/14/2016 04:49:06] CAUAJM_I_10655 System is running in single server mode.
Event server: autosysdb.
[02/14/2016 04:49:11] CAUAJM_I_20197 CA WAAE Application Server operational on port
9000.
[02/14/2016 04:49:11] CAUAJM_I_40244 CA EEM unauthenticated user mode value set to
<**UNKNOWN**>.
[02/14/2016 04:49:11] CAUAJM_I_20366 CA WAAE Application Server operational on
agent listener port 49156.
[02/14/2016 04:49:11] CAUAJM_I_20367 CA WAAE Application Server operational on
auxiliary agent listener port 7500.
ERROR: [0x036ccb70] ispUtil::TcpConnect: Error in delayed connection() 111 - Connection
refused
ERROR: [0x036ccb70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x036ccb70] ispUtil.HttpPostRequest: ObtainPC Failed
log4cxx: No appender could be found for logger (PozFactory).
log4cxx: Please initialize the log4cxx system properly.
ERROR: [0x036ccb70] ispUtil::TcpConnect: Error in delayed connection() 111 - Connection
refused
ERROR: [0x036ccb70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x036ccb70] ispUtil.HttpPostRequest: ObtainPC Failed
....
...
..
```



```

ERROR: [0x036ccb70] ispUtil::TcpConnect: Error in delayed connection() 111 - Connection
refused
ERROR: [0x036ccb70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x036ccb70] ispUtil.HttpPostRequest: ObtainPC Failed
ERROR: [0x036ccb70] CertificateReader::loadPEM : etpki_pem_file_to_cert failed
[errorcode : -1]
[02/14/2016 04:49:12] CAUAJM_E_10436 Security server unreachable or invalid
authentication certificate file.
[02/14/2016 04:49:12] CAUAJM_E_10434 Error initiating security session.
[02/14/2016 04:49:12] CAUAJM_E_10437 Detailed Error Information:
[02/14/2016 04:49:12] [EE_BADOBJECT Bad Object]
[02/14/2016 04:49:12] [ISP_ERROR_NOGATEWAY igateway not running]
[02/14/2016 04:49:12] [Authenticate Error: Authentication Failed]
[02/14/2016 04:49:12] [Identity Attempted:]
[02/14/2016 04:49:12] [CertificateReader::loadPEM - cannot read certificate]
ERROR: [0x03158b70] ispUtil::TcpConnect: Error in delayed connection() 111 - Connection
refused
ERROR: [0x03158b70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x03158b70] ispUtil.HttpPostRequest: ObtainPC Failed
....
...
..

```

**Environment:**

All supported and CA EEM multi-write configured environments.

**Cause:**

- CA EEM is configured in High Availability environment (multi-write)
- CA Workload Automation AE external security is not configured with CA EEM server-list as per CA EEM High Availability setup
- CA Workload Automation AE application server is restarted when the Primary EEM is down

**Resolution:**

Configure the CA Workload Automation AE security to work with all the nodes of the CA EEM cluster.

- Ensure that all the CA EEM clustered servers are up and running.
- Use the autosys\_secure utility to set the CA EEM server location and regenerate the certificate by adding all the nodes of the CA EEM cluster with comma separated values.

```
[autosys@waaeserver1 ~]$ autosys_secure
```

CA Workload Automation AE Security Utility

---

Please select from the following options:

- [1] Revert to NATIVE instance security.
- [2] Manage CA EEM security settings.
- [3] Change database password.
- [4] Change remote authentication method.
- [5] Manage user@host or user@domain users.
- [6] Get encrypted password.
- [0] Exit CA WAAE Security Utility.

> 2

Manage CA EEM security settings

Please select from the following options:

- [1] Manage CA EEM server settings.
- [2] Manage cached credentials.
- [9] Exit from "Manage CA EEM security settings" menu.
- [0] Exit CA WAAE Security Utility.

> 1

Manage CA EEM server settings

Please select from the following options:

- [1] Show current CA EEM server settings.
- [2] Set CA EEM server location and regenerate certificate.
- [3] Set unauthenticated user mode.
- [9] Exit from "Manage CA EEM server settings" menu.
- [0] Exit CA WAAE Security Utility.

> 1

CAUAJM\_I\_60228 CA EEM server: eemserver1  
CAUAJM\_I\_60342 Unauthenticated user mode: OFF

Please select from the following options:

- [1] Show current CA EEM server settings.
- [2] Set CA EEM server location and regenerate certificate.
- [3] Set unauthenticated user mode.

```

[9] Exit from "Manage CA EEM server settings" menu.
[0] Exit CA WAAE Security Utility.
> 2
Input the CA EEM server name(s) (or hit enter to
cancel): eemserver1,eemserver2
Input the CA EEM administrator name (or hit enter to cancel): EiamAdmin
Input the CA EEM administrator password:

Confirm the CA EEM administrator password:

```

```

CAUAJM_I_60200 CA EEM certificate generated successfully.
CAUAJM_I_60191 The CA EEM server location was changed successfully.

```

```

Please select from the following options:
[1] Show current CA EEM server settings.
[2] Set CA EEM server location and regenerate certificate.
[3] Set unauthenticated user mode.
[9] Exit from "Manage CA EEM server settings" menu.
[0] Exit CA WAAE Security Utility.
> 0

```

#### A quick test after security changes:

- CA Workload Automation AE CLIs no longer fail.
- While the EEM primary is down, restart the application server and monitor the log (\$AUTOUSER/out/as\_server.\$AUTOSERV) for the following messages:

```

[02/14/2016 06:59:13] CAUAJM_I_20367 CA WAAE Application Server operational on
auxiliary agent listener port 7500.
ERROR: [0x07371b70] ispUtil::TcpConnect: Error in delayed connection() 111 -
Connection refused
ERROR: [0x07371b70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x07371b70] ispUtil::TcpConnect: Error in delayed connection() 111 -
Connection refused
ERROR: [0x07371b70] ispUtil::TcpConnect Failed to connect to host [eemserver1]
ERROR: [0x07371b70] ispUtil.HttpPostRequest: ObtainPC Failed
log4cxx: No appender could be found for logger (PozFactory).
log4cxx: Please initialize the log4cxx system properly.
CAUAJM_I_10485 CA EEM security session initialized with server
<eemserver1,eemserver2>.

```

```
[02/14/2016 06:59:14] CAUAJM_W_10472 Resource manager encountered the following
during initialization: <Library "libDCAMClient" has not been loaded>.
[02/14/2016 06:59:14] CAUAJM_W_10473 Resource manager will ignore non-virtual
resource constraints for all jobs.
[02/14/2016 06:59:14] CAUAJM_I_10474 Resource manager initialization complete.
[02/14/2016 06:59:15] CAUAJM_I_30001 CA WAAE Application Server startup
complete.
```

## NOTE

- For more information about failover configuration, see [EEM Failover Configuration](#).
- For more information about configuring a new CA EEM instance with existing Web UI security policies, see [Configure a New CA EEM Instance with Existing Web UI Security Policies](#).

## Jobs are Stuck in STARTING Status With "Waiting for initiator" in the Job Detail Report

### Problem:

Jobs on an agent host are stuck in STARTING status. The job detail report displays a STATE\_CHANGE event with comment <Waiting for initiator>. For example, when you issue the autorep command to get the detailed report for test\_perl job:

```
autorep -J test_perl -d
```

The resulting report might resemble the following:

Job Name	Last End	ST Run/Ntry Pri/Xit	Last Start
test_perl	00:43:13	----- <b>ST</b> 39164/1	02/02/2016

Status/[Event]	Time	Ntry	ES	ProcessTime	Machine
<b>STARTING</b>	02/02/2016 00:43:13	1	PD	02/02/2016 00:43:13	unix-agent-113
<b>[STATE_CHANGE]</b>	02/02/2016 00:43:13	1	PD	02/02/2016 00:43:14	unix-agent-113
<b>&lt;Waiting for initiator&gt;</b>					

Further analysis of the initiator logs on the Workload Automation Agent for UNIX, Linux, or Windows reveals the following:

### **initiatormanager.log:**

```
02/02/2016 00:29:22.811 EST-0500 5 Core
Mailbox.outbound.CybInitiatorManager.releaseInitiator[:918] - 176.39162_1/WAAE_WF0.1/
MAIN released initiator of class DEFAULT. 1 available
```

```

02/02/2016 00:43:02.525 EST-0500 5 Core
Mailbox.inbound.CybInitiatorManager.executeWOB[:715] - 177.39163_1/WAAE_WF0.1/MAIN
acquired initiator of class DEFAULT. 0 left
02/02/2016 00:43:13.845 EST-0500 5 Core
Mailbox.inbound.CybInitiatorManager.executeWOB[:785] - |20160202 00431381+0500 WA_AGENT
tcpip@CVC_SCH 176.39164_1/WAAE_WF0.1/MAIN RUN .
Data(Shell=/usr/bin/perl,Args="\\"hello world!\\";"",Script="print")
ComponentType(AUTOSYS.APPSRV) Attribute1(Type=AsServers,Value=waae-lnx-prd)
Attribute2(Type=AsPort,Value=9000)
Attribute3(Type=AsToken,Value=D64E6C75A88A8A12393F67A8C310B71E61210331160F5AF02086D01175E01692B
Env(AUTOSERV=CVC,AUTO_JOB_NAME="test_perl",AUTORUN="39164-
1",STDOUT=>>/tmp/$AUTO_JOB_NAME.log,STDERR=>>/tmp/$AUTO_JOB_NAME.err) User(autosys)
Password() MFUser(autosys)
WOBRequestID(78570E47BB8D601B49354E5EDBB294E02E8865ED14543917938360) | has been put to
waiting queue

```

#### **initiators\_wait\_default.log:**

```

Log Started Tue Feb 02 00:43:13 EST 2016 with level 5

02/02/2016 00:43:13.844 EST-0500 3 Core Mailbox.inbound.CybObjectList.put[:1150] -
put 20160202 00431381+0500 WA_AGENT tcpip@CVC_SCH 176.39164_1/WAAE_WF0.1/MAIN RUN .
Data(Shell=/usr/bin/perl,Args="\\"hello
world!\\";"",Script="print") ComponentType(AUTOSYS.APPSRV)
Attribute1(Type=AsServers,Value=waae-lnx-prd) Attribute2(Type=AsPort,Value=9000)
Attribute3(Type=AsToken,Value=D64E6C75A88A8A12393F67A8C310B71E61210331160F5AF02086D01175E01692B
Env(AUTOSERV=CVC,AUTO_JOB_NAME="test_perl",AUTORUN="39164-
1",STDOUT=>>/tmp/$AUTO_JOB_NAME.log,STDERR=>>/tmp/$AUTO_JOB_NAME.err) User(autosys)
Password() MFUser(autosys)
WOBRequestID(78570E47BB8D601B49354E5EDBB294E02E8865ED14543917938360)

```

#### **Environment:**

Workload Automation Agent for UNIX, Linux, or Windows Release 11.3.x on all supported platforms.

#### **Cause:**

If the Workload Automation Agent for UNIX, Linux, or Windows Release 11.3 has a set of job-class initiators defined and the number of jobs for that job-class submitted to that agent has exceeded the number specified for that job-class, jobs enter "Waiting for initiator." status. By default, the agent allows up to 1000 jobs of the Default job-class at a given time.

#### **Resolution:**

Edit the `initiators.class_n` parameter in the `agentparm.txt` file to increase the maximum concurrent jobs and restart the Workload Automation Agent for UNIX, Linux, or Windows.

For example, set the `initiators.class_n` parameter as follows to allow up to 2,000 concurrent jobs of the Default job class.

```
initiators.class_1=Default,2000
```

**NOTE**

- The agentparm.txt file is located at /opt/CA/WorkloadAutomationAE/SystemAgent/WA\_AGENT (UNIX) or C:\Program Files\CA\WorkloadAutomationAE\SystemAgent\WA\_AGENT (Windows).
- For more information about the initiators.class\_n parameter, see the [Workload Automation Agent for UNIX, Linux, or Windows documentation](#).

## Add a Time Zone to CA Workload Automation AE

**Problem:**

How to add a time zone using the autotimezone command in AutoSys Workload Automation?

**Environment:**

CA Workload Automation AE Release 11.3.X and Release 11.4, and AutoSys Workload Automation 12.X.

**Resolution:****Follows these steps:**

1. Define the zone using the -t option as follows:

```
autotimezone -t 'EST3EDT,M10.3.0/00:00:00,M3.3.0/00:00:00'
```

2. Define an alias for that time zone as follows:

```
autotimezone -a ART 'EST3EDT,M10.3.0/00:00:00,M3.3.0/00:00:00'
```

## autoping to a Specific Workload Automation Agent Fails Although the Agent is Active

**Problem:**

The autoping -m *agent\_name* command to a specific agent fails with the following error messages:

```
CAUAJM_W_10496 Agent on [agent_name] has not responded in a timely fashion. Try again
later. [CA WAAE Autoping]
CAUAJM_E_50281 AutoPing from the Scheduler WAS NOT SUCCESSFUL.
```

**Environment:**

CA Workload Automation AE Release 11.3.x and AutoSys Workload Automation 12.x..

**Cause:**

The problem is caused by the name resolution. Workload Automation Agent is not able to resolve the AutoSys Workload Automation manager host name. The following error messages are seen in the Workload Automation Agent transmitter.log file:

```
04/08/2016 14:52:44.126 SGT+0800 1 TCP/IP Controller Plugin.Transmitter pool thread
<Slow:1>.CybTargetHandlerChannel.constructConversation[:1045] - Error connecting to
ACE_SCH:
cybermation.library.communications.CybConversationConnectException: Unknown host:
manager_name.com.xxx
```

### **Resolution:**

To ensure that the Workload Automation Agent can successfully resolve the AutoSys Workload Automation manager's host name (either the short name or the fully qualified name depending on how it is defined in the agentparm.txt file), either add the AutoSys Workload Automation manager hostname to the hosts file on the Workload Automation Agent machine or update the DNS so that the AutoSys Workload Automation manager name is resolved successfully.

## **How AutoSys Workload Automation Configures the Oracle Database Connection Properties**

### **Introduction:**

This document explains how the database connection is determined and established between AutoSys Workload Automation and Oracle.

### **Background:**

There is no clear understanding how AutoSys Workload Automation is configured to connect to the database as there is no setting or configuration which controls it. The connection is determined based on the presence of the TNS\_ADMIN environment variable and based on this, the connection type and details are determined.

### **Environment:**

This document applies to CA Workload Automation AE Release 11.3.X and AutoSys Workload Automation 12.x

### **Instructions:**

To establish a connection with Oracle, AutoSys Workload Automation and its components use the EventServer definition in the config file (config.HAL). If \$TNS\_ADMIN is set, it uses the values of EventServer for the parameters in the connection string.

### **Autosys Configuration File**

In this example, the AutoSys Workload Automation instance is HAL, Oracle Service Name is HA1136, port is 1521, and host is win-db2. EventServer holds the parameters that must build the database connection string.

### **\$AUTOUSER/config.HAL**

```
EventServer_1=HA1136:aedbadmin,1521,win-db2
```

```
EventServer_1=SERVICE_NAME:SCHEMA_OWNER,PORT,HOST
```

**Note:** If you want to make use of LDAP for tnsnames instead of tnsnames.ora, follow the instructions mentioned in [Configure the Java Database Connectivity Driver to Use Oracle Call Interface](#).

### **Oracle \$TNS\_ADMIN Environment Variable**

If \$TNS\_ADMIN is set, the appropriate database parameters are passed on the JDBC driver.

The \$TNS\_ADMIN variable must be set in the AutoSys Workload Automation profile that is sourced in when the services are started.

If \$TNS\_ADMIN is set, AutoSys Workload Automation:

- Uses the SERVICENAME as the ALIAS to select the matching entry from tnsnames.ora
- Passes in all the values from the tnsnames.ora entry to the JDBC driver
- JDBC Driver uses - oracle.tns.connection.string

If \$TNS\_ADMIN is not set, AutoSys Workload Automation:

- Uses the SERVICENAME, PORT & HOST from the AutoSys Workload Automation config file to establish a connect
- JDBC Driver uses - oracle.connection.string

#### **Additional Information:**

Oracle tnsnames.ora File

The formatting of the file is important.

- Use Oracle Net Manager to make any updates to tnsnames.ora (Recommended)
- Each sub-line requires a two-space indentation
- The Alias should NOT have any periods
- If an “Internal Exception: IO Error: could not resolve” message is seen, there is a likely formatting issue and the JDBC driver is not able to connect to the database.

#### **Example: tnsnames.ora**

##### **\$TNS\_ADMIN/tnsnames.ora**

```
HA1136 =
 (DESCRIPTION =
 (ADDRESS = (PROTOCOL = TCP) (HOST = win-db2) (PORT = 1521))
 (CONNECT_DATA =
 (SERVER = DEDICATED)
 (SERVICE_NAME = HA1136)
)
)
```

#### **JDBC Driver Connection Properties File**

The connection.properties file must remain as default as the JDBC driver handles the different scenarios automatically. Making any updates or hard-coding values into this file is not recommended.

#### **NOTE**

SSL\_SERVER\_DN\_MATCH will be read from the connection.properties first. If not found, then it would be read from TNS\_ADMIN/sqlnet.ora

oracle.net.ssl\_server\_dn\_match=false is added to connection.properties during installation.

If oracle wallet is chosen, then the installer will update the values as entered by the user during the interview phase.

#### **Example: connection.properties**

##### **\$AUTOSYS/config/connection.properties**

```

```



```
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this Java property file contains the definition of different database

connection properties

#

connection configuration for Microsoft SQL Server

msql.driver.class.name=com.microsoft.sqlserver.jdbc.SQLServerDriver

msql.connection.string=jdbc:sqlserver://${servername}:${port};databasename=${dbname}

msql_trusted.driver.class.name=com.microsoft.sqlserver.jdbc.SQLServerDriver

msql_trusted.connection.string=jdbc:sqlserver://${servername}:${port};databasename=
${dbname};integratedSecurity=true;

connection configuration for Oracle Thin driver
oracle.driver.class.name=oracle.jdbc.OracleDriveroracle.connection.string=jdbc:oracle:thin:@//
${servername}:${port}/${dbname}

oracle.tns.connection.string=jdbc:oracle:thin:@${dbname}
oracle.net.ssl_server_dn_match=false

connection configuration for SyBase

sybase.driver.class.name=com.sybase.jdbc4.jdbc.SybDriver

sybase.connection.string=jdbc:sybase:Tds:${servername}:${port}/${dbname}
```

## Error CAUAJM\_E\_18402 ORA-24347: Warning of a NULL column

### **Problem:**

When the scheduler tries to generate the aggregator statistics, the following errors are displayed in the scheduler log:

---

CAUAJM\_E\_18402 ORA-24347: Warning of a NULL column in an aggregate function

CAUAJM\_E\_40303 Failed to complete aggregation of product statistics.

**Environment:**

The scheduler failed to generate aggregator statistics after upgrading CA Workload Automation AE to Release 11.3.6 SP3.

This problem is fixed in CA Workload Automation AE Release 11.3.6 SP4.

**Cause:**

In CA Workload Automation AE Release 11.3.6 SP3, new columns are added to the aggregator tables (ujo\_rep\_hourly, ujo\_rep\_daily, ujo\_rep\_weekly, ujo\_monthly) to include the job states: SUSPENDED and RESUME. The existing records would have NULL for these newly added columns and so the database aggregator function like SUM or AVERAGE fail on NULL values.

**Workaround:**

Log in as the aedbadmin (for oracle) or sa (for Sybase or MS-SQL) user and run the following queries:

```
update ujo_rep_daily set ac_resume_fail_n=0 where ac_resume_fail_n is NULL;

update ujo_rep_daily set ac_suspend_fail_n=0 where ac_suspend_fail_n is NULL;

update ujo_rep_daily set js_suspend_n=0 where js_suspend_n is NULL;

update ujo_rep_daily set js_suspend_p=0 where js_suspend_p is NULL;

update ujo_rep_hourly set ac_resume_fail_n=0 where ac_resume_fail_n is NULL;

update ujo_rep_hourly set ac_suspend_fail_n=0 where ac_suspend_fail_n is NULL;

update ujo_rep_hourly set js_suspend_n=0 where js_suspend_n is NULL;

update ujo_rep_hourly set js_suspend_p=0 where js_suspend_p is NULL;

update ujo_rep_weekly set ac_resume_fail_n=0 where ac_resume_fail_n is NULL;

update ujo_rep_weekly set ac_suspend_fail_n=0 where ac_suspend_fail_n is NULL;

update ujo_rep_weekly set js_suspend_n=0 where js_suspend_n is NULL;

update ujo_rep_weekly set js_suspend_p=0 where js_suspend_p is NULL;
```

```
update ujo_rep_monthly set ac_resume_fail_n=0 where ac_resume_fail_n is NULL;

update ujo_rep_monthly set ac_suspend_fail_n=0 where ac_suspend_fail_n is NULL;

update ujo_rep_monthly set js_suspend_n=0 where js_suspend_n is NULL;

update ujo_rep_monthly set js_suspend_p=0 where js_suspend_p is NULL;

commit;
```

## Why autoping is not Working from a Client or Agent Machine?

### **Introduction:**

When you try to autoping an agent from a client or agent machine, the following errors occur:

```
D:\Program Files (x86)\CA\Workload Automation AE>autoping -m <machine_name>

CAUAJM_E_10527 Timed out waiting for response from the CA WAAE Application Server:
[<AutoServer>:<AutoServerPort>]

CAUAJM_E_50033 Error initializing tx subsystem: CAUAJM_E_10062 Failed to get initial
configuration from CA WAAE Application Server(s).
```

### **Problem:**

Why autoping does not work on the client or agent machine, but works on the scheduler?

The cryptkey.txt file is the same on both the agent machine and the client or agent machine.

### **Resolution:**

If the client or agent port is open bidirectionally, the connection is successful using telnet and the cryptkey.txt files included in the agent directories are the same.

### **Follows these steps:**

1. Locate the config.\$AUTOSERV files (or config.%AUTOSERV% files) from client or agent machine and scheduler.
2. Within the files, search for the 'UseEncryption' parameter and ensure they match.
3. Go to the scheduler and get the cryptkey.txt located in the \$AUTOUSER path (on Linux/UNIX) or %AUTOUSER% path (on Windows) and copy this file over to the problematic client or agent machine under the same path.
4. Restart the services on the problem machine and try to autoping the agent again.

## CAUAJM\_E\_10527 Timed Out Waiting for Response Returned by autorep on Linux RedHat 7.2

### **Problem:**

When you try to run autorep on a newly installed client node on Red Hat 7 client, autorep times out with the following error messages:

```
CAUAJM_E_10527 Timed out waiting for response from the CA WAAE Application Server:
[server-name:9460]
```

```
CAUAJM_E_50033 Error initializing tx subsystem: CAUAJM_E_10062 Failed to get initial
configuration from CA WAAE Application Server(s).
```

### **Environment:**

CA Workload Automation AE Release 11.3.6 SP4 on Linux Red Hat 7.2.

### **Resolution:**

The encountered problem is related to the CA Workload Automation AE Release 11.3.6 SP4 server and client utilities not being compatible with Linux Red Hat 7.

#### **NOTE**

For more information, see [AutoSys Compatibility Matrix](#).

The recommended solution is to use Release 11.3.6 SP5 for compatibility with Linux Red Hat 7.

## Job Fails with the Exit Code - 656

### **Problem:**

Client gets an error message -656 and wants information about this error code and ask for a solution.

### **Environment:**

CA Workload Automation AE Release 11.3.6 on UNIX or Linux.

### **Cause:**

The job fails because the scheduler has exceeded the number of restart attempts.

### **Resolution:**

-655 exit code indicates that the job fails because the scheduler has exceeded the number of restart attempts.

You can use the "n\_retrys" attribute in the job definition:

### **Syntax**

This attribute has the following format:

```
n_retrys: attempts
```

- **attempts**

Defines the number of times to attempt to restart the job after it exits with a FAILURE status.

**Limits:** This value can be any integer in the range 0 through 20.

**Default:** 0

**NOTE**

: The `n_retrys` value applies when the command that is associated with the job fails. When a job fails for other reasons, such as invalid credentials or unavailable space, the `MaxRestartTrys` value in the configuration file defines how many times the job should be restarted. When the `MaxRestartTrys` value is zero, the job is not restarted.

## autoping Fails with CAUAJM\_E\_18416 and CAUAJM\_E\_18402 ORA-01400 Messages

**Problem:**

The autoping command fails with the following messages:

```
CAUAJM_I_30033 Client [CA WAAE AutoPing:x][x][xx] session ended
CAUAJM_E_18416 Event Server:<x> Failed Query: <BEGIN :RetVal := ujo_put_job ..
CAUAJM_E_18402 ORA-01400: cannot insert NULL in
 ("AEDBADMIN"."UJO_SCHED_INFO"."JOB_VER")
ORA-06512: a "AEDBADMIN.UJO_PUT_JOB", ligne 590
ORA-06512: a ligne 1
```

....

```
CAUAJM_I_10165 The previously failed query below succeeded after 1 attempt(s)
```

**Cause:**

Most likely there is an intermittent database OCI API failure and so the `as_server` command failed to complete the SQL operation at that time.

**Resolution:**

No action required. This is just a temporarily error at the base level. The SQL error is generally successful after 1 re-attempt (1 second after the problem occurs).

## File Watcher Processes are Still Running After I Run CHANGE\_STATUS to SU

**Introduction:**

The file watcher processes are running even after you run the sendevent `CHANGE_STATUS` command to change the status of the job to SUCCESS (SU).

**Environment:**

CA Workload Automation AE Release 11.3.x on all platforms

**Resolution:**

The file watcher processes continue to run until they are complete. If you delete these processes, the job status changes to FAILURE (FA). The sendevent CHANGE\_STATUS command does not delete the job status. We recommend that you use the KILLJOB event to terminate the job.

## autorep Command Very Slow

**Problem:**

You experience intermittent slow autorep -J *job\_name* responses. The problem persists even after you rebuild the indexes and bounce the application servers.

Also, the responses from the agent on the shadow scheduler to the application server on the shadow server gets delayed intermittently. It runs fine, but suddenly throws a 16 seconds delay.

**Environment:**

CA Workload Automation AE Release 11.3.6 SP1

**Resolution:****Follow these steps:**

1. Open \$AUTOUSER/autosys.sh.*hostname* file and add or update the AS\_IAM\_CACHE\_UPDATE environment variable, as follows:

```
AS_IAM_CACHE_UPDATE=300;
```

2. Use the export command to export it in the \$AUTOUSER/autosys.sh.*hostname* file, as follows:

```
export AS_IAM_CACHE_UPDATE
```

**NOTE**

For more information about the AS\_IAM\_CACHE\_UPDATE parameter, see [Environment Variables -- Application Server](#).

## log4cxx Error in the Scheduler Log

**Problem:**

The following messages are displayed in the scheduler log file:

```
log4cxx: No appender could be found for logger (PozFactory).
log4cxx: Please initialize the log4cxx system properly.
```

**Resolution:**

You might see these messages in the logs when the application server and/or the scheduler lose connectivity with the gateway process. Check the network connectivity between AutoSys Workload Automation and CA EEM hosts. Additionally, ensure that the process gateway is running without errors on the CA EEM host.

**Access Denied to autosys\_secure****Problem:**

You are unable to access some of the autosys\_secure options and the following errors are displayed:

```
$ autosys_secure
CA WAAE Security Utility
CAUAJM_W_10450 Only the CA WAAE EDIT superuser can activate external security.

CAUAJM_W_60050 Insufficient privileges to execute options 1 through 4.
CAUAJM_W_60045 Only an CA WAAE EDIT superuser has access to all options!

Please select from the following options:
[5] Manage user@host or user@domain users.
[6] Get encrypted password.
[0] Exit CA WAAE Security Utility.
> 0
$
```

**Environment:**

CA Workload Automation AE Release 11.3.5 and later.

**Cause:**

AutoSys Workload Automation was running in CA EEM mode and is changed to the native security mode. So, you cannot launch options from 1 to 4 as the autosys or root user. To revert back to the CA EEM mode, you must manually reset the security settings for autosys\_secure.

**Resolution:**

To manually reset the security setting for autosys\_secure, access the AEDB database and run the following SQL queries:

- ORACLE

```
delete from aedbadmin.ujo_keymaster where hostid = 'SECURITY';
update aedbadmin.ujo_alamode set int_val=0 where type = 'JOB';
update aedbadmin.ujo_alamode set int_val=0 where type = 'EVT';
delete from aedbadmin.ujo_alamode where type = 'SEC';
commit;
```

- SQL SERVER

```
BEGIN TRANSACTION;
delete from ujo_keymaster where hostid = 'SECURITY';
```

```
update ujo_alamode set int_val=0 where type = 'JOB';
update ujo_alamode set int_val=0 where type = 'EVT';
delete from ujo_alamode where type = 'SEC';
COMMIT TRANSACTION;
```

### **WARNING**

**DISCLAIMER:** The above procedure bypasses the AutoSys Workload Automation external security (CA EEM security mode) and hence all security authorization policies that are setup in CA EEM for the AutoSys Workload Automation instance in question are not available until the external security is reactivated. Ensure that you stop the AutoSys Workload Automation application server(s) and scheduler(s) before you perform the above procedure so that no unauthorized actions (autorep, sendevent, and so on) are processed while the AutoSys Workload Automation instance security is being reset.

## **Find Jobs With the Same Owner in AutoSys Workload Automation**

This video describes how to find the job names with one specific owner "XXXXX" in AutoSys Workload Automation:

### **NOTE**

Click the YouTube logo to open this video in a new tab with all normal YouTube functionality.

### **Problem:**

How to get the list of jobs with the same owner (the owner that you specify in your job definition) in AutoSys Workload Automation.

### **Environment:**

Applicable only on the Linux and Unix environment.

### **Resolution:**

Currently, you cannot get the list of jobs with the same owner in AutoSys Workload Automation. However, you can create a script and get this information using the autorep command. For example, issue the following command:

```
autorep -J ALL -q|egrep owner -b3|egrep "insert_job"|awk '{print $2}'
```

## **What Causes the CAUAJM\_W\_00043 Warning Message Regarding getaddrinfo?**

### **Problem:**

The scheduler log contains many messages that read "CAUAJM\_W\_00043 The UNIX function getaddrinfo did not respond in a timely fashion for hostname [XXXXXXXXXX]. Retrying...". What is the root cause of this message and what can be done to address it?

### **Environment:**

CA Workload Automation AE and AutoSys Workload Automation: All Releases

Operating Systems: All



**Resolution:**

When the scheduler or application server needs to resolve a hostname, they do so by making a call to the `getaddrinfo()` UNIX function. When the system call is initiated, there is a default timeout of 15 seconds set for it to return a result. If the result is not returned within the timeout period, the CAUAJM\_W\_00043 message is posted to the log of the component making the call. At that point, a second call to `getaddrinfo()` is made. If the second attempt also times out, the machine is placed offline. The machine is automatically placed online once communication with it is restored.

When there are many occurrences of this warning message in the scheduler log, it is a good indication that there are intermittent communication issues between the scheduler machine and a DNS server and should be investigated by your system administrator or network team.

Environments that have IP caching disabled are more likely to see these warnings because the call is made every time a hostname needs resolution.

The timeout interval is configurable using an environment variable that is read by the schedule or application Server on startup. This is accomplished by setting the variable in the `$AUTOUSER/autosys.sh.<hostname>` file, which is read by the startup scripts for both components.

For Release 11.3.x and 12.x, the variable is `AS_RESOLVEHOST_TIMEOUT`

The value set for this variable can be from 1 to 120 and is added to the default 15-second timeout. For example, if the value is set to 20, the timeout period is increased from 15 to 35 seconds.

Increasing the timeout reduces the frequency of the warning messages. However, it does not address the underlying issue with `getaddrinfo` performance, that should still be investigated by the system admins or network admins. Slow performance of this function within an environment that does not have IP caching enabled causes a significant performance hit to the instance.

## **When delete\_job is run via jil the job is still in the ujo\_job table in the database. Is this expected?**

**Problem:**

When you delete a job from AutoSys Workload Automation, the job is not deleted from `ujo_job` table in the database. You can no longer see it from Web UI or autorep, but it is still in the database. Is this expected?

**Example:**

```
SQL> select job_name from aedbadmin.ujo_job where job_name like '%ZZ694CC_test_R113Agent_parm_value_7%';
```

```
JOB_NAME
```

```

```

```
ZZ694CC_test_R113Agent_parm_value_7
```

```
SQL>
```

```
[root]# autorep -j ZZ694CC_test_R113Agent_parm_value_7
```

```
CAUAJM_E_50027 Invalid Job Name: ZZ694CC_test_R113Agent_parm_value_7
```

**Environment:**

CA Workload Automation AE Release 11.3.6

Linux

Oracle

**Resolution:**

In CA Workload Automation AE Release 11.3.x versions when a job is deleted, the `is_active` flag in the `ujo_job` table is changed from 1 to 0. The job is later deleted from the database using the `archive_jobs` command, which by default runs as part of DBMaint each day.

Regarding the Reference Guide:

The `archive_jobs` command deletes obsolete job versions from the database.

Job versions are considered obsolete when all the following conditions apply:

- The job is inactive.
- The job is no longer referenced.
- All event data that is related to the job has been deleted.
- All job run data that is related to the job has been deleted.

The job must not be referenced by any other job, it must not have any events, alarms, or `job_runs` in the database and the job def must be flagged for deletion for N number of days.

N, is similar to `archive_events` usage. When the data is older than N days then the job is deleted.

**Example:**

A job is created, never run and then immediately deleted using `jil`. The job remains in the database for seven days before being deleted (by default).

**Example:**

A job is created, run once and then deleted using `jil`. It remains in the database for seven days before trying to be deleted (by default). If one had adjusted the `archive_events` to keep event data for 30 days, then the `archive_jobs` would not delete the job until first all the events are purged. So 30 days must past, then the events are deleted, then the job could be deleted.

Additional Information:

[Delete Obsolete Job Versions](#)

## Trying to Use the AutoSys Workload Automation API

**Problem:**

We are trying to get the AutoSys Workload Automation API working on a server which has the CA Workload Automation AE Release 11.3.5 client installed on it. The AutoSys Workload Automation client on Server2 is working fine and is able to communicate with the application server on Server1.

For example, you can give `chk_auto_up` and `sendevent` commands from the AutoSys Workload Automation client on Server2 that are sent to Server1 and the output is returned.

Now that we have the client on Server2 communicating with the application server on Server1, the applications development team must be able to interface the AutoSys Workload Automation API on Server2 so that they can issue `sendevent` commands from their application scripts. However, when they run tests, they get timeout errors.

**Environment:**

CA Workload Automation AE Release 11.3.5.

**Resolution:**

The problem was related to the encryption type in the config file where the default encryption was being used.

Resolved by setting AsConstants.ENCRYPTION\_TYPE\_DEFAULT in the API as in the following example:

```
AsApi api = new AsApi(args[0], Integer.parseInt(args[1]),
AsConstants.ENCRYPTION_TYPE_DEFAULT, "");
```

## Obtain a List of Job Names from the AutoSys Workload Automation Database

**Introduction:**

How to gather a list of job names within the AutoSys Workload Automation database for a specific status, between a specific date or time range?

**Environment:**

CA Workload Automation AE Release 11.3 or higher.

**Instructions:**

1. You must translate the dates you want into "UNIX time" (which is in seconds). To do this, you use the "time0" utility that is available with AutoSys Workload Automation, located in \$AUTOSYS/bin. You need both the start and ending date or time from the time0 command.

**Example:**

Beginning date/time:

```
time0 -T "04/01/2016 00:00"
```

```
CAUAJM_I_50099 Autotime(internal): 1459483200
```

Ending date/time:

```
time0 -T "04/01/2016 23:59"
```

```
CAUAJM_I_50099 Autotime(internal): 1459569540
```

2. Log in to the database as aedbadmin.

3. Now you can query the database for specific information:

```
select j.job_name, p.status from ujo_job j inner join ujo_proc_event p on j.joid=p.joid where event_time_gmt from 1459483200 through 1459569540;
```

The select statement above, retrieves the job name and status of the job. You can also query the same table for specific statuses.

**Example:**

```
select j.job_name, p.status from ujo_job j inner join ujo_proc_event p on j.joid=p.joid where p.status = 7 and event_time_gmt from 1459483200 through 1459569540;
```

This query obtains jobs that were placed ON\_ICE during a particular date/time frame.

**NOTE**

Status Codes:

1 RUNNING

3 STARTING  
4 SUCCESS  
5 FAILURE  
6 TERMINATED  
7 ON\_ICE  
8 INACTIVE  
9 ACTIVATED  
10 RESTART  
11 ON\_HOLD  
12 QUE\_WAIT

## Remote Collector Error when Changing the Collector DB or Collector Port

### Problem:

After executing the wcc\_config.bat/wcc\_config.sh, the cmd/shell sometimes hangs.

### NOTE

The following error message is included in the log file:

```
org.springframework.remoting.RemoteConnectFailureException: Could not connect to HTTP invoker remote service at [https://localhost:8443/wcc-configuration/cfmremote/remoteGlobalPrefService]; nested exception is java.net.ConnectException: Connection refused (Connection refused)
```

### Cause:

The remote collector is not able to communicate with the Web UI host.

### Resolution:

Resolve the connectivity issue of the Web UI machine and the Collector machine.

## Installation/Upgrade Fails With java/lang/OutOfMemoryError Exception

### Problem:

During the CA EEM validation steps, the AutoSys installation/upgrade fails with a "java/lang/OutOfMemoryError" exception.

### Root-Cause:

When the CA EEM server has a large number of policies, the InstallAnywhere installer runs out of Memory Heap during validation. By default, it runs with 134m of Max Memory Heap.

### Resolution:

Set the Java Max Memory provided by the InstallAnywhere on the command line when executing the installer. Set jvmxmx memory to a higher value to avoid the error.

### Example:

```
./install.bin -jvmxmx 250m
```

## Installing Second AutoSys Instance Fails With Manually Installed Agent

### **Problem:**

The following error message is displayed:

**The dependent component(s) of the installed AutoSys components is missing: "Scheduler : Agent"**

### **Root Cause:**

When trying to install a second AutoSys instance, after manually installing the Agent, the instance installation fails.

### **Resolution:**

Add `IA_AE_INSTALLED_AGENT = 1` to the file: `%AUTOSYS%\globalVars`, because the Agent was manually installed.

## MSSQL Named Instance Server on Linux Fails

### **Problem:**

When installing AutoSys on Linux using MSSQL Server with a named instance, the following error message is displayed:

**Error: [CAUAJM\_E\_112515] Unable to connect to the "HOST\INSTANCE" database server using the database client utility. For more information, refer to the log file.**

### **Root Cause:**

The Linux installer tests the connection to the database using both a bundled JDBC and the 'sqlcmd' command (from the Linux package `mssql-tools`).

The `sqlcmd` command on Linux requires two backslashes as follows:

```
sqlcmd -U user -P password -S HOST\\INSTANCE
```

If using a non default port for the database (default is 1433), the command must be as follows:

```
sqlcmd -U user -P password -S HOST\\INSTANCE,PORT
```

The installation asks for the `HOST\INSTANCE`, so the `sqlcmd` command ends up badly formatted with a single backslash, causing it to fail. If you try using double backslashes during the install, it also fails as this affects the formatting of the JDBC connect string used by the installer.

### **Resolution:**

Use the following workaround to install AutoSys on Linux using a MS-SQL Named Instance:

1. Skip the Database Check.

You can export the following variable to instruct the installer to skip the DB checks:

```
export IA_AE_SKIP_DB_CHECK=1
```

2. Run the AutoSys installation.

During the install, enter the database information using a single backslash : `HOST\INSTANCE`

This is important, as it will be used to set the `EventServer` parameter in the `$AUTOUSER/config.$AUTOSERV` file correctly.

3. Run the `CreateAEDB.pl` script.

After the installation is complete, you will need to create the database schema for AutoSys using the `$AUTOSYS/dbobj/MSQ/CreateAEDB.pl` script.

Consider the following before you run the script:

- Use the HOST\\INSTANCE format with 2 double backslashes.
  - Add the port number if using a non-default port: HOST\\INSTANCE,PORT.
4. Start the AutoSys Scheduler and Application Server, after the script had successfully created the AutoSys DB.

## Troubleshooting Web UI

The troubleshooting section describes common problems, their symptoms, and possible solutions. It provides useful information about troubleshooting the various applications of Web UI.

### Agent Inventory - Troubleshooting

The following list contains some common error messages that can be found in the Agent Inventory GUI and Logs:

- `SunCertPathBuilderException`: unable to find valid certification path to requested target
- `java.lang.RuntimeException`: Unexpected error: `java.security.InvalidAlgorithmParameterException`: the trustAnchors parameter must be non-empty
- `javax.ws.rs.ProcessingException`: `javax.net.ssl.SSLHandshakeException`: `sun.security.validator.ValidatorException`: PKIX path validation failed: `java.security.cert.CertPathValidatorException`: signature check failed
- `E190098 <server_alias_name>` (web services validation) - A connection exception was encountered: Certificate for `<app_server_name>` doesn't match any of the subject alternative names: [].

To resolve the above errors, follow these steps:

#### NOTE

After completing each step, you must validate the functionality. If the problem persists, move to the next step.

1. Check your Web Services URL in the Server definition in the Configuration tab. If the URL has a fully qualified hostname, remove the domain name and then validate server.  
**Example:** If the Web Services URL is `https://hostname.broadcom.com:9443/AEWS`, use `https://hostname:9443/AEWS`.

2. Check the following parameters in the 'connection.properties':

- `main.app.sslEnabled`
- `wcc.host`
- `main.app.port`

All the above params must have same value as in the Web UI URL.

**Example:** If the Web UI URL is `https://hostname.broadcom.net:8443/wcc/ui/Login.html`, then the params in 'connection.properties' must be:

- `main.app.sslEnabled = true`
- `wcc.host = hostname.broadcom.net`
- `main.app.port = 8443`

#### NOTE

When using load balancer,

- If load balancer is using https, ensure that Web UI also uses https.
- If Web UI is configured with load balancer, then the hostname in 'connection.properties' must be the actual hostname of system and not the one which is used by load balancer.

3. If the above steps did not resolve the issue, then you must import a new SSL certificate (from a trusted source) into the Web UI keystore.
4. Follow these steps to export a certificate from AE Web Services and import it into the WCC keystore:
  - a. Go to the Web Services directory, generally found at the following location (for UNIX) in the computer where AE is installed:

```
/opt/CA/WorkloadAutomationAE/autouser.ACE/webserver/conf
```

- b. Run the following command to list all alias in the keystore:

```
$JAVA_HOME/bin/keytool -provider
org.bouncycastle.jcajce.provider.BouncyCastleFipsProvider -providerpath
$AUTOSYS/lib/bc-fips.jar -storepass <keystorepass> -storetype BCFKS -list -v -
keystore .keystore
```

- c. Run the following command to export the certificate:

```
$JAVA_HOME/bin/keytool -export -alias tomcat -file cert.cer -storepass
<keystorepass> -storetype BCFKS -keystore .keystore
```

#### NOTE

- “cert.cer” is the name of the exported certificate file.
- Replace the keystorepass with the correct password; default is *changeit*.

- d. To import the certificate into Web UI, go to the following location in the computer where Web UI is installed:

```
$CA_WCC_INSTALL_LOCATION/data/config
```

#### NOTE

Make sure you have copied the new certificate to the Web UI computer.

- e. Run the following command to import the certificate:

```
keytool -import -alias <alias_name> -file <new_cert_file_path> -keystore .keystore
```

#### NOTE

If asked for a password, enter <keystorepass>.

The default value for <keystorepass> is “changeit”, which can also be found in  
CA\_WCC\_INSTALL\_LOCATION/tomcat/conf/server.xml

The <alias\_name> must be unique for the import. Ensure that it does not overwrite the existing  
<alias\_name> in the Web UI keystore.

#### NOTE

After each step, validate the server. If the validation is successful, then wait for configurable collection time  
(Refresh Time, defined in Agent Inventory configuration) to allow the next collection to start.

## Agent Parameters File Update Fails

### Problem:

When you update the agentparm.txt file the following error message displays:

**Failed to update agent parameters. Try again.**

### Solution:

Check if the agentparm.txt file is locked, enable the file for updates, and try again.

## (Windows) Web UI Error Codes E140178 and E140179

### Problem:

When you submit a job to a machine, the following error messages are displayed:

**ERROR: E140179 Unexpected error occurred while performing the agent action. File is not available**

**ERROR: E140178 Unexpected error occurred while executing agent command. Communication attempt with the CA WAAE agent has failed!**  
**ERROR: E140179 Unexpected error occurred while downloading agent file. Refused by agent security**

**Environment:**

Workload Automation Agent for Windows Release 11.3.x

**Cause:**

The Workload Automation Agent is unable to locate the file that must be executed.

**Resolution:**

Set the `oscomponent.lookupcommand` parameter in the `agentparm.txt` file as follows:

```
oscomponent.lookupcommand=true
```

**NOTE**

- Specify the `oscomponent.lookupcommand` parameter value (true or false) in lower case.
- The `agentparm.txt` file is located in the agent installation directory.
- For more information about the `oscomponent.lookupcommand` parameter, see the [Workload Automation Agent Parameters](#).

## Action Status Update Fails for Agents with Java Runtime Environment (JRE) Version 1.6

**Problem:**

Action Status is not updated for agents with Java Runtime Environment (JRE) version 1.6, because the JRE uses the TLSv1 protocol for sending status to the Agent Inventory.

**Solution:**

Update the “CA\Workload Control Center\tomcat\conf\wrapper.CONF” file with “wrapper.java.additional.14=Djdk.tls.ephemeralDHKeySize=1024”.

**NOTE**

If the value is 2048, the status update for the agents fails in the Agent List table.

## Download Logs Fail Due to File Size

**Problem:**

Downloads fail when the log directory is greater than 500 MB (uncompressed) or 250 MB (compressed).

**Solution:**

Remove the logs that are not required.

## EEM Cluster Primary, Secondary Synchronization Problem SafeException: EE\_SPONSORERROR

**Problem:**



After adding the secondary EEM server to the primary EEM server using the eiam-clustersetup utility, the sync command from the secondary EEM server fails with the following error:

```
C:\Program Files (x86)\CA\SC\EmbeddedEntitlementsManager\bin>java -jar eiam-
clustersetup.jar -p eemserver1.cluster.com
Feb 2, 2016 1:45:16 PM IclUtil itechLibInit
INFO: iTechSDK initialized successfully
INFO - EIAM_HOME [C:\Program Files (x86)\CA\SC\EmbeddedEntitlementsManager\]
INFO - IGW_LOC [C:\Program Files (x86)\CA\SC\iTechnology\]
INFO - DXHOME [C:\Program Files (x86)\CA\Directory\dxserver/]
INFO - Hostname identified as [eemserver2.cluster.com]
EiamAdmin password :
INFO - Checking server status
INFO - igateway status [started]
INFO - dxserver status [started]
Are you sure you want to continue? [Y/N]:Y
[eemserver2.cluster.com]>sync
=====
INFO - Select current machine hostname
=====
[Authenticate Error: Authentication Failed, Identity Attempted: EiamAdmin]
com.ca.eiam.SafeException: EE_SPONSORERROR iSponsor Error
 at com.ca.eiam.SafeContext.authenticateWithPassword(SafeContext.java:1683)
 at
 com.ca.eiam.clustersetup.FailoverUtility.fetchServerConfiguration(FailoverUtility.java:31)
 at
 com.ca.eiam.clustersetup.FailoverConfigurator.startSecondaryFailoverConfiguration(FailoverConf
 at
 com.ca.eiam.clustersetup.FailoverConfigurator.performAction(FailoverConfigurator.java:380)
 at
 com.ca.eiam.clustersetup.FailoverConfigurator.commandPrompt(FailoverConfigurator.java:356)
 at com.ca.eiam.clustersetup.FailoverConfigurator.main(FailoverConfigurator.java:106)
ERROR - EE_SPONSORERROR iSponsor Error
[eemserver2.cluster.com]>
```

### Environment:

CA EEM on all supported platforms. [Click here](#) for the Web UI compatibility matrix.

### Cause:

- Communication problems between the Primary and Secondary EEM servers:
  - Network firewall blocking ports.
  - Name lookup (DNS) issues.
- Local Firewall (Windows firewall, Linux IPTABLES) on EEM hosts.
- EiamAdmin password incorrect or difference between EEM servers.

### Resolution:

Complete the following steps to resolve the issue:

**Follow these steps:**

1. Ensure the following:
  - a. The port for CA iTechnology iGateway [TCP/5250] is not blocked on the Network, Windows, or Linux firewall. Enable the port for both inbound and outbound communication.
  - b. The CA Directory port [TCP/509] is also enabled for bidirectional communication.
  - c. Name resolution between EEM server hosts is working.
  - d. The EiamAdmin password is the same on all the EEM server hosts.
2. Run a connection verification:
  - a. Attempt to connect the primary EEM server URL through browser on the Secondary host.
  - b. Log in to the 'Global' application as 'EiamAdmin' user from the Secondary host and do the reverse.

**Example:**

The following output is from a successful sync activity through eiam-clustersetup.jar after enabling port TCP/5250 on the Windows firewall between the primary and secondary EEM servers:

```
C:\Program Files (x86)\CA\SC\EmbeddedEntitlementsManager\bin>java -jar eiam-
clustersetup.jar -p eemserver1.cluster.com
Feb 2, 2016 2:24:44 PM IclUtil itechLibInit
INFO: iTechSDK initialized successfully
INFO - EIAM_HOME [C:\Program Files (x86)\CA\SC\EmbeddedEntitlementsManager\]
INFO - IGW_LOC [C:\Program Files (x86)\CA\SC\iTechnology\]
INFO - DXHOME [C:\Program Files (x86)\CA\Directory\dxserver/]
INFO - Hostname identified as [eemserver2.cluster.com]
EiamAdmin password :
INFO - Checking server status
INFO - igateway status [started]
INFO - dxserver status [started]
Are you sure you want to continue? [Y/N]:y
[eemserver2.cluster.com]>sync
=====
INFO - Select current machine hostname
=====
INFO - [1] eemserver1.cluster.com:509
INFO - [2] eemserver2.cluster.com:509
Select Hostname from [1 - 2] : 2
INFO - Synchronization level
INFO - [1] [NEW] secondary node is being added first time
INFO - [2] [DELTA] secondary node is being synced to update configurations
Select Synchronization mode from [1 - 2] : 1
=====
INFO - Syncing with primary server

Primary server Secondary server

eemserver1.cluster.com eemserver2.cluster.com

```

---

```
INFO - Re-configuring of server may lead to loss of data, it is advised to perform a
 backup of configuration and data store.
Are you sure you want to continue? [Y/N]:y
INFO - Stopping dxserver service
INFO - Stopping igateway service
INFO - Configuring the knowledge group file
INFO - -----
INFO - The knowledge group file iTechPoz.dxc resides in the directory: C:\Program Files
 (x86)\CA\Directory\dxserver/\config\
knowledge\itechpoz.dxc
INFO - Node was removed successfully
INFO - Fetching configuration from eemserver1.cluster.com
INFO - Adding self node [eemserver2.cluster.com]
INFO - Generating: : C:\Program Files (x86)\CA\SC\iTechnology\iAuthority.conf
INFO - Generating: : C:\Program Files (x86)\CA\SC\iTechnology\iControl.conf
INFO - Generating: : C:\Program Files (x86)\CA\SC\iTechnology\rootcert.cer
INFO - Generating: : C:\Program Files (x86)\CA\SC\iTechnology\rootcert.key
INFO - Generating certificate for host : eemserver1.cluster.com
INFO - Generating file : C:\Program Files (x86)\CA\Directory\dxserver/config\ssld
\itechpoz-trusted.pem
INFO - Adding failover node [eemserver1.cluster.com]
INFO - Generating: : C:\Program Files (x86)\CA\SC\iTechnology\iControl.conf
INFO - Configuring DSA itechpoz for host eemserver2.cluster.com
INFO - Configuring the itechpoz knowledge file
INFO - -----
INFO - The itechpoz knowledge file C:\Program Files (x86)\CA\Directory\dxserver/\config
\knowledge\itechpoz.dxc.dxc resides
in the directory: {2}
INFO - Writing the itechpoz knowledge file
INFO - Configuring DSA itechpoz-eemserver1.cluster.com for host eemserver1.cluster.com
INFO - Configuring the itechpoz-eemserver1.cluster.com knowledge file
INFO - -----
INFO - The itechpoz-eemserver1.cluster.com knowledge file C:\Program Files (x86)\CA
\Directory\dxserver/\config\knowledg
e\itechpoz-eemserver1.cluster.com.dxc.dxc resides in the directory: {2}
INFO - Writing the itechpoz-eemserver1.cluster.com knowledge file
INFO - Configuring the knowledge group file
INFO - -----
INFO - The knowledge group file iTechPoz.dxc resides in the directory: C:\Program Files
 (x86)\CA\Directory\dxserver/\config\
knowledge\itechpoz.dxc
INFO - Configuring the settings file
INFO - -----
INFO - C:\Program Files (x86)\CA\Directory\dxserver/\config/settings\itechpoz.dxc
INFO - Clearing secondary server dsa-db
INFO - Starting dxserver service
```

---

```
INFO - Starting igateway service
INFO - Run [status] to get server details.
[eemserver2.cluster.com]>exit
C:\Program Files (x86)\CA\SC\EmbeddedEntitlementsManager\bin>
```

## Error: "C:\Windows\System32\<command>" (Web UI)

The C:\Windows\System32\<command> error can occur when you try to run a job on 64-bit versions of Windows using a command that is located in the %WinDir%\System32 folder.

The 64-bit versions of Windows supports the *File System Redirection* feature. On 64-bit Windows, the %WinDir%\System32 folder is reserved for 64-bit applications. When a 32-bit application tries to access the System32 folder, access is redirected to %WinDir%\SysWOW64.

### NOTE

- By default, file system redirection is enabled.
- 32-bit applications can access the native system directory by substituting %WinDir%\Sysnative for %WinDir%\System32. WOW64 recognizes Sysnative as a special alias that indicates that the file system must not redirect the access.

## Error: CAUAJM\_E\_10682 Requested log file for the specified job could not be found.

When attempting to view or retrieve Standard Output or Standard Error logs for a job through Web UI QuickView, the following errors may occur:

- **CAUAJM\_E\_10682 Requested log file for the specified job could not be found.**
- **E125001 There was an error trying to load the log: CAUAJM\_E\_10682 Requested log file for the specified job could not be found.**

### NOTE

This error is seen when a job uses a UNC path in the std\_out\_file / std\_err\_file attributes.

To resolve these errors, follow these steps:

1. Grant the agent machine account access to the share.
2. Identify the agent machine.
3. Locate the job definition below.

The agent machine is named: Pasteur

```
insert_job: unc_std_out_path job_type: CMD
command: dir
machine: Pasteur
owner: louis.pasteur@glitch.com
std_out_file: "\\PASCAL\Temp\%AUTO_JOB_NAME%.out"
std_err_file: "\\PASCAL\Temp\%AUTO_JOB_NAME%.err"
alarm_if_fail: 1
```

4. Find the std\_out/std\_err files on machine "PASCAL":
5. Set the agent service on machine: "Pasteur" to "Log on as" in the default Local System.
6. Grant the agent machine "Pasteur" permissions to the "Temp" share.
7. Open Properties for Temp.
8. Select the Permission tab.

9. Click the Add button.
10. Click the Object Types button.
11. Select/Check Computers from the list of Object Types.
12. Add the agent machine in the object names to select box.  
The machine PASTEUR has now been granted permissions to the TEMP share.
13. Retrieve the log details.  
Log details are displayed.

## Unable to login into Web UI. Results in Error "E150005: Login timed out."

### Problem:

When you log in to the Web UI User Interface, it results in the following error:

E150005: Login timed out.

### Environment:

WCC Releases 11.3.5, 11.3.6, and 11.4 on Linux and UNIX.

### Cause:

- You can see the following error messages in the CA-wcc.log file:  
Caused by: org.springframework.remoting.RemoteAccessException: Could not access HTTP invoker remote service at [http://<host>:10132/wcc-configuration/cfmremote/remoteGlobalPrefService]; nested exception is com.ca.wcc.config.api.exceptions.WCCException: Did not receive successful HTTP response: status code = 500, status message = [Internal Server Error]
- You can also see the following error messages in the CA-wcc-services.log file:
  - [] ERROR #BaseLoopImpl # Hazelcast Instance is not active!
  - java.lang.IllegalStateException: Hazelcast Instance is not active!
  - INFO | jvm 1 | 2017/03/04 20:11:22 | SEVERE: Servlet.service() for servlet [remoteservices] in context with path [/RemoteServices] threw exception [Handler processing failed; nested exception is java.lang.OutOfMemoryError: Java heap space] with root cause
  - INFO | jvm 1 | 2017/03/04 20:11:22 | **java.lang.OutOfMemoryError: Java heap space**
  - INFO | jvm 1 | 2017/03/04 20:11:22 |
  - INFO | jvm 1 | 2017/03/04 20:11:22 | Mar 04, 2017 8:11:22 PM org.apache.catalina.core.StandardWrapperValve invoke
  - INFO | jvm 1 | 2017/03/04 20:11:22 | SEVERE: Servlet.service() for servlet [remoteservices] in context with path [/RemoteServices] threw exception [Handler processing failed; nested exception is java.lang.OutOfMemoryError: Java heap space] with root cause
  - INFO | jvm 1 | 2017/03/04 20:11:22 | **java.lang.OutOfMemoryError: Java heap space**
  - INFO | jvm 1 | 2017/03/04 20:11:22 |

### Resolution:

You must address the OutOfMemory: Java heap space problem with the Web UI services.

### Follow these steps:

1. Stop the CA-wcc, CA-wcc-services services.
2. Take a backup copy of and edit the %CA\_WCC\_INSTALL\_LOCATION%\tomcat\_32\conf\wrapper.conf file (\$CA\_WCC\_INSTALL\_LOCATION\tomcat\_32\conf\wrapper.conf, on Linux and UNIX) and change the following:  
From:  
wrapper.java.maxmemory=812  
To:  
wrapper.java.maxmemory=1600

**NOTE**

This value is associated with the Tomcat server for the CA WCC 32-bit service. The maximum and recommended value is 1600 on Windows and 1900 on Linux and UNIX (2GB).

3. Take a backup copy of and edit the %CA\_WCC\_INSTALL\_LOCATION%\tomcat\conf\wrapper.conf file (\$CA\_WCC\_INSTALL\_LOCATION\tomcat\conf\wrapper.conf, on Linux and UNIX) and change the following:  
From:  
wrapper.java.maxmemory=2048  
To:  
wrapper.java.maxmemory=4096
4. Start the CA-wcc, CA-wcc-services services back up.

## Web UI ECLI message : Command processing exception encountered

### Symptom:

When I run any operating system command similar to the following in WCC 11.3 Linux, Enterprise Command Line Interface,

```
- Command: "echo `hostname;date;uptime` > /tmp/uptime.out"
```

the command execution fails with the following message :

- **Output:**  
The command could not be executed. Command processing exception encountered.
- **Errors:**  
Command not allowed due to filter file.

### Solution:

Add the required commands to run through Web UI ECLI in the "AutoSysCommandISponsorFilters.txt" filter file, located at \$AUTOSYS/config.

You can use the regular expression syntax for adding the required commands.

### Example :

```
^autorep(\s.+)?
^jil(\s.+)?
^sendevent(\s.+)?
^ls(\s.+)?
^echo(\s.+)?
^hostname(\s.+)?
^date(\s.+)?
^uptime(\s.+)?
```

## Web Services URL Validation Fails

When the Web Services URL is specified while adding a server in the Web UI configuration tab, validation fails with the following error:

**Error: E190098 PRD\_SP7 (web services validation) - A connection exception was encountered: Certificate for <user01-l10279.xyz.com> doesn't match any of the subject alternative names.**

**Problem:**

This error is due to the SSL certificate exception for the hostname mismatch.

**Solution:**

Verify the system hostname and specify the value in the Web Services URL. If it does not work, find the hostname using the Fully Qualified Domain Name (FQDN).

## Web Services URL Validation Fails with Status Code 403

When the Web Services URL is specified while adding a server in the WCC configuration tab, validation fails with the following error:

**Error: E190098 <APPLICATION\_SERVER\_HOST> (web services validation) - A connection exception was encountered: https://<APPLICATION\_SERVER\_HOST>:9443/AEWS returned status 403.**

Also the "Datasources Status" dialog in the Agent Inventory tab displays the following information against the respective AutoSys server:

**Status Details : WlaInternalServerErrorException: Unexpected exception connecting to Scheduler <AutoSysServerName>**

**Problem:** The error is due to the "Monitor user" of the respective AutoSys server does not have the required EEM permissions on the AutoSys Web Service.

**Solution:** Ensure that the "Monitor user" belongs to the WorkloadAutomationAEWebService application group.

## Web Services URL Validation Fails with Status Code 404

When the Web Services URL is specified while adding a server in the WCC configuration tab, validation fails with the following error:

**Error: E190098 <APPLICATION\_SERVER\_HOST> (web services validation) - A connection exception was encountered: https://<APPLICATION\_SERVER\_HOST>:9443/AEW returned status 404.**

**Problem:** The error is due to a wrong Web Services URL.

**Solution:** Check for any typos in the Web Services URL.

**Example:** https://<APPLICATION\_SERVER\_HOST>:9443/AEWS

## ECLI command execution fails with Status 403 Forbidden

When a command is executed from ECLI, it fails with the following error:

```
<!doctype html><html lang="en"><head><title>HTTP Status 403 - Forbidden</title><style type="text/css">body
```

Also the "Datasources Status" dialog in the Agent Inventory tab displays the following information against the respective AutoSys server Status Details :

WlaInternalServerErrorException: Unexpected exception connecting to Scheduler <AutoSysServerName>

**Problem:** The error is due to the "Monitor user" of the respective AutoSys server does not have the required EEM permissions on the AutoSys Web service.

**Solution:** Enusure the "Monitor user" belongs to the WorkloadAutomationAEWebService application group.

## ECLI Command Execution Fails - Wrong Web Services URL

When a command is executed against an AutoSys server, you see following error in the "Errors" section of ECLI:

**Not found URL while connecting to Server StatusCode: 404:**

**https://<APPLICATION\_SERVER\_HOST>:9443/AEW/command,**  
**URL: https://<APPLICATION\_SERVER\_HOST>:9443/AEW**

**Problem:** The error is due to a wrong Web Services URL.

**Solution:**

**Follow these steps:**

1. Go to the Configuration tab.
2. Edit the respective AutoSys server definition.
3. Check for any typos in the Web Services URL.

**Example:** https://<APPLICATION\_SERVER-HOST>:9443/AEWS.

## ECLI Command Execution Fails - Missing Web Services URL

When a command is executed against an AutoSys server, you may see the following error in the "Errors" section of ECLI:  
**Missing AutoSys Restful Web Services URL in server definition**

You must configure the AutoSys RESTful Web Services URL in the Configuration Tab.

**Problem:** The respective AutoSys server definition in the Configuration tab is not configured for "Web Services URL".

**Solution:**

**Follow these steps:**

1. Go to the Configuration tab.
2. Edit the respective AutoSys server definition.
3. Add the RESTful webservice URL of this AutoSys server in the "Web Services URL" field.

**Example:** https://<APPLICATION\_SERVER\_HOST>:9443/AEWS

## ECLI Command Execution Fails - Missing suid

On Linux and AIX, the underlying file system must be mounted with suid.

**Problem:** On Linux and AIX, the set-user-ID set on AutoSysCommandWrapper is not honored if the underlying file system is mounted with nosuid.

**Solution:** Remount the file system with suid using the following command:

```
mount -o remount,suid <filesystem>
```

## External Applications Not Working After Upgrade

**Symptom:**

External applications do not work after an upgrade.

**Solution:**

You must modify the \$CA\_WCC\_INSTALL\_LOCATION/tomcat/conf/server.xml file to include the external application.

### NOTE

For more information on how to configure external applications, see the [Manage External Applications](#) section.

## ECLI JIL Import Fails

ECLI JIL import fails with following error, though the /tmp has full permissions for the webservice user:



sh: /tmp/133792.in: Permission denied

Status Code: 200

**Problem:** AutoSys RESTful service creates a temporary file that is imported from ECLI. So it should be mandatory to have umask 0022 for the user who owns the RESTful web server service. If you do not set umask 022, the RESTful webservice user will not have the permission to read the temporary file.

**Solution:** Set the umask to 022 for that user who owns the RESTful web server service in their profile and restart the webserver. By default, this service is owned by AutoSys, unless changed by the user during installation.

## Troubleshoot SSA

This article explains how to troubleshoot SSA issues.

### SSA Installation Fails

Valid on UNIX

#### Symptom:

The SSA installation fails on UNIX.

#### Solution:

Before you install SSA:

- Ensure that the SSA installation is started by a root user on UNIX and as Administrator on Windows.
- Verify that the csampmxf daemon is not active during the SSA installation or upgrade.

To verify that the csampmxf daemon is active, issue the following command:

```
ps -ef | grep csam
```

If the csampmxf daemon is active, and the \$CSAM\_SOCKADAPTER/bin/csampmux binary is installed, issue the following command to stop the csampmxf daemon:

```
$CSAM_SOCKADAPTER/bin/csampmxf stop
```

If the csampmux binary is not installed, issue the following command to kill csampmxf daemon:

```
kill csampmxf
```

#### NOTE

If the csampmxf daemon starts again even after you killed it, then another process is starting it. Verify which process is starting csampmxf and shut it down before you install SSA.

### csampmux status Command Reports that Service Controller is Not Running

Valid on UNIX

#### Symptom:

When I issue the csampmux status command, the following message is displayed:

```
Service Controller is not running/responding
```

When I look at the process list, the csampmux process is running.

**Solution:**

This issue may occur for the following two reasons:

- You moved the csam/etpki libraries from one directory to another while the csam server was running.  
To fix this issue, do the following:
  - a. Issue the following command:

```
ps -ef | grep csampmuxf
```

The process ID of the csam server is obtained.

- b. Issue the following command:

```
kill csampmuxf
```

The csampmuxf daemon (csam server) is killed.

- c. Ensure that you set the library path to the csam libraries correctly.
  - d. Restart csam.

**NOTE**

If the csampmuxf daemon starts again even after you killed it, then another process is starting it. Verify which process is starting csampmuxf and shut it down before you install SSA.

- You installed SSA in the following order:
  - a. Install SSA using the CA Common Components DVD with certain path settings.
  - b. Uninstall SSA.
  - c. Reinstall SSA manually without running the profile.ca file.

The problem occurs when the Broadcom product that uses SSA is looking for the server in one location and csam looks for it in another location. To fix this issue, ensure that the profile.ca file has the proper symbolic link path for CSAM\_SOCKADAPTER and CSAM\_LOGGER\_CONF (based on the CCS path).

---

## Web UI Help Systems

---

AutoSys Web UI (Web UI) has a unified Help system with an Index for better search performance. The Help categories provide information that is related to the Help tabs on the Web UI Graphical User Interface and are described in this section. The Help categories are as follows: Agent Inventory, Application Editor, Configuration, Credentials, Dashboard, Enterprise Command Line, Forecast, Monitoring, Reporting, and Resources.

### WARNING

This section contains confidential subsections that are not viewable to anonymous users. To view all subsections, you must first log in. Click Log-in in the upper-right corner and enter your CA Support Online credentials. If you are already logged in, ignore this message.

## Web UI Agent Inventory

The Agent Inventory display lets you see and export information about each separately installable component (system agent, plug-ins, and advanced integrations).

The Agent Inventory tab is available to Web UI users with permission for Agent Inventory functions. Agents are shown only for AutoSys Workload Automation instances that have the AutoSys Workload Automation Web Services component enabled.

### NOTE

- Agent Inventory requires the CA Workload Automation AE servers to be at version 11.3.6 SP2 or later.
- Agent Inventory supports only the agents that are in AutoSys compatible mode.

### Agent List

The Agent Inventory tab contains the Agent List providing a list of machines for a AutoSys Workload Automation server. The list contains the following columns:

- **Machine Name**  
Displays the AutoSys Workload Automation machine name (as defined to AutoSys Workload Automation server) where the agent is installed.  
Hover the cursor over the required **Machine Name** row, click the button next to the machine name to [retrieve the following agent information](#):

- Plugin Details
- Stop Agent
- Restart Agent
- Update AgentParams
- Quiesce Agent
- Unquiesce Agent
- Status Details
- Reset Action Status
- Stage Agent
- Upgrade Agent
- Upgrade Status
- Upgrade Timestamp
- Stage/Upgrade Status Details
- Cancel Scheduled Upgrade
- Rollback Upgrade

- **Machine Status**

Indicates the connection status of the machine as reported by the AutoSys Workload Automation Scheduler. A machine can have one of following statuses:

- **Online**  
Indicates that the machine is available and accepting jobs to run.
- **Offline**  
Indicates that the machine has been manually removed from service and does not accept jobs to run. In this status, the agent is accessible.

**NOTE**

This status is applicable for machines in schedulers from CA Workload Automation AE Release 11.3.6 SP6 or earlier releases. From CA Workload Automation AE Release 11.3.6 SP7, this status is displayed as **Offline** and **OfflineDown**.

- **Offline Down**  
Indicates that the machine has been manually removed from service and does not accept jobs to run. In this status, the agent is not accessible.
- **Missing**  
Indicates that the scheduler has verified that the machine is not responding and has automatically removed it from service. The machine does not accept jobs to run.
- **Unqualified**  
Indicates that the scheduler is attempting to qualify the status of an agent before switching the machine from an online to missing status. The machine does not accept jobs to run.
- **Blocked**  
Indicates that the agent blocks communication. All jobs that are scheduled to start on the agent are put in a PEND\_MACH status.

**NOTE**

For more information about machine status, see the [Machine Status](#) section.

Status changes can be delayed by the specified refresh duration and Agent Status Poll Interval. The check interval can be set on the AutoSys Workload Automation Scheduler using [AgentStatusPollInterval](#).

- **Agent Type**  
Indicates the agent type.
- **Agent Name**Indicates the name of the agent.
- **Host**

Indicates the hostname of the agent.

- **Port**  
Indicates the port on which the agent is listening.
- **Version**  
Indicates the version of the agent. The format is <version\_number>-<build\_number>. If the version is not available, -- is displayed (that is, two dashes).
- **Platform**  
Indicates the platform where agent is being run. If the platform is not available, -- is displayed (that is, two dashes).

#### NOTE

For CA Workload Automation AE Release 11.3.6 SP7, the value of 'opsys' as entered under the Machine Name configuration or the agent detected value is displayed. If 'opsys' value is not entered, -- is displayed for 11.3.6 SP6 and previous releases.

- **Server Name**  
Displays the AutoSys Workload Automation server name (as defined to Web UI) where the agent is installed.
- **Action Status**  
Displays the agent status such as stopping, stopped successfully, failed, restarted, quiesced, or unquiesced.
- **Action Timestamp**  
Displays the time dynamically corresponding to the agent status.

#### NOTE

The Action Timestamp column data cannot be located using the Search.

- **Plugins**  
Displays all plugins (internal and external), that are configurable by the user.

#### NOTE

The internal plugins “clearfiles, communicationmanager, config, filebrowser, nop, outbox, refresh, router, setproperty, and status” are not displayed in the Agent Inventory table. When you export data, it contains all the plugin details except the mentioned internal plugins.

- **Description**  
Displays the description property value of the machine.

#### NOTE

##### More information:

- [Administer the Agent Inventory](#)

## Security

The following security requirements control how Agent Inventory displays information and what functions a user can perform.

- Users must have configuration access to use Agent Inventory. Only users with Agent Inventory through EEM access see the Agent Inventory tab.
- The Monitor ID and Monitor Password must be specified in the configuration tab of Web UI per server description. The [Monitor User](#) must have AutoSys Workload Automation web service access enabled.
- If the web services URL is empty in Web UI, Agent Inventory ignores the server.
- Download logs, download params, view agent details, plugin details, stop agent, restart agent, status details, and reset status are not supported when local job security is turned on (security.level=on setting in agentparm.txt). In this case, you might receive the following error message:

**Error: E140179 Unexpected error occurred while performing the action. Refused by agent security**

## Search Bar

The search bar lets you specify multiple search terms (separated by spaces). Rows are displayed if any of the search terms are found anywhere within a row of data. A case-insensitive search is performed. Searches also support the asterisk (\*) wildcard character.

For example, you enter the following items in the search bar:

```
11.3.0 solaris
```

In this case, rows are displayed where 11.3.0 or solaris were found in the agent data. For example, agents with versions like 11.3.06.01-0478, or platforms and host names containing solaris are displayed.

## Export Table

You can export the Agent List to CSV (Comma Separated Values) so that you can filter, sort, edit, and create custom reports.

To export the Agent List, click the **EXPORT TABLE** button on the top of the Agent List table.

The CSV file is filtered with the same criteria, and displays the same columns with one column for each plugin, as the Agent List table, described earlier in this article.

## Example:

```
Machine Name,Status,Agent Type,Agent Name,Host,Port,Version,Platform,Server Name,Action
Status,Action Timestamp,Application Services Agent,BigData Agent,DataServices
Agent,Database Agent,FTP Agent,File Monitoring Agent,Hadoop Agent,I5OS Agent,ITPAM
Agent,Informatica Agent,MSSQL Agent,Management Agent,Microfocus Agent,Object Monitoring
Agent,Oracle Applications Agent,PeopleSoft Agent,Remote Agent,Runner Agent,SAP
Agent,SNMP Agent,SQL Agent,WebServices Agent,Description
P13res,Online,System Agent,WA_AGENT,user01-I152613,7520,11.4.00.00-1607,Linux for
amd64,P13,,,No,No,No,No,Yes,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,test
user01-I152613,Online,System Agent,WA_AGENT,user01-I152613,7520,11.4.00.00-1607,Linux
for
amd64,P13,,,No,No,No,No,Yes,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,Created
by the installer.
i5agentnw,Online,System Agent,DEMO832,USILIA09,7832,11.3.04.03-0833,OS/400 for
PowerPC,P13,,,No,No,No,No,No,No,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,
user01-I152612,Online,System Agent,WA_AGENT,user01-I152612,7520,11.3.07.00-0979,Linux
for
i386,P13,,,Yes,No,Yes,Yes,Yes,Yes,No,No,Yes,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,Yes,
user01-I8901,Online,System Agent,WA_AGENT,user01-I8901,7520,11.3.07.00-0979,Windows
Server 2012 R2 for
x86,P13,,,No,No,No,No,Yes,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,
user01-I8901,Online,System Agent,WA_AGENT1a,user01-I8901,7521,11.3.07.00-0979,Windows
Server 2012 R2 for
x86,P13,,,No,No,No,No,Yes,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,
user01-I8901,Online,System Agent,WA_AGENT2,user01-I8901,7522,11.3.07.00-0979,Windows
Server 2012 R2 for
x86,P13,,,No,No,No,No,Yes,Yes,No,No,No,No,No,No,No,No,Yes,No,No,No,No,Yes,No,No,No,No,No,
```

## **Datasources**

The **Datasources** icon informs the user of the overall state of all AutoSys Workload Automation Servers (Datasources) connected to Agent Inventory.

The possible conditions are:

- **Green (Ok)**  
Indicates that all enabled AutoSys Workload Automation schedulers are connected and providing agent data.
- **Red (Error)**  
Indicates one of the following conditions:
  - When no AutoSys Workload Automation servers are defined, the overall Datasources status is Error.
  - When one of the AutoSys Workload Automation servers is not connected or providing agent data as expected, the overall Datasources status is Error.

Clicking the **Datasources** icon takes you to the Datasources page.

### **NOTE**

#### **More information:**

- [Datasources](#)

## **Supported Browsers**

Some browsers do not support Agent Inventory inside the Web UI tab, including IE10 and IE11. In this case, you see the following message:

**The application could not be loaded inside Web UI tab.**

**Please click here to open in a new window.**

If you click on click here, Agent Inventory opens as a new browser tab where it is fully functional.

## **Retrieve Information from Agents**

Agent Inventory lets authorized users retrieve information from Agents. This information includes agent details, logs, and parameters.

### **NOTE**

The selected action is performed only on one of the agents in the cluster configuration.

## **Download Logs**

Agent Inventory provides the capability to download the agent log files. The files are collected on the agent, compressed, and downloaded to your browser. All the files in the agent log directory are downloaded.

To download the agent log files, click to select a row in the Agent Inventory table. Select the **Download Logs** option in the **Machine Name** button drop-down list. When downloading logs, a warning dialog appears to notify you that the download can take some time. If you click Continue, it downloads the log files; otherwise it does not. After the logs are downloaded, you can use a standard "unzip" utility of your choice to recover the original log files as they existed on the agent host.

Other considerations:

- Supported versions are described in the Retrieve Information from Agents topic.
- The machine (agent) must be accessible by AutoSys Workload Automation for the download to be successful. You cannot download logs from an agent that is not running.
- While not required, we recommend that you take the machines associated with the agent offline. Next, ensure that the agent is not processing any critical jobs that the log download could interfere with.
- Retrieving the agent logs can be quite time consuming (several minutes). So be patient until the files are available on the browser or an error message is displayed.
- If the logs cannot be downloaded for some reason, the UI presents an error message. As noted previously, it might be some time before the error message appears.
- Downloads fail when the log directory is greater than 500 MB (uncompressed) or 250 MB (compressed). The error message reflects this size problem.
- As part of the process, files are staged in a temporary directory on the agent. For this reason, sufficient agent disk space must exist for the download to succeed. As a rule of thumb, you need about twice the amount of space that the agent log directory consumes.
- When reviewing agent logs, remember that the log retrieval process itself might appear in the logs. Repeated attempts to download the same logs can cause the logs themselves to grow.
- Only files in the agent log directory are downloaded. Log files that are archived to other storage areas are not downloaded.
- Assume a standard agent configuration where agent log folder location is not customized.

### **Download AgentParams**

Agent Inventory provides the capability to download the Agent parameters file (agentparm.txt). The agentparm.txt file is collected from the agent, and a text file is downloaded to your browser.

To download the agent parameters, click a row in the Agent Inventory table. Select **Download AgentParams** option in the **Machine Name** button drop-down list. Once downloaded, you can use a standard editor of your choosing to open the Agent details.

Other considerations:

- Supported versions are described in the Retrieve Information from Agents topic.
- The machine (agent) must be accessible by AutoSys Workload Automation for the download to be successful. You cannot download parameters from an agent that is not running.
- If the parameters cannot be downloaded for some reason, the UI displays an error message.
- Downloads fail when the agentparm.txt file is greater than 10- MB. The error message reflects this size problem.

### **View Agent Details**

Agent Inventory lets you view various details about an agent like enabled functionality and other integrations and plug-ins. The displayed details vary based on the version of the agent.

To view the Agent details, click a row in the Agent Inventory table. Select the **View Agent Details** option in the **Machine Name** button drop-down list.

Other considerations:

- Supported versions are described in the Retrieve Information from Agents topic.
- The machine (agent) must be accessible by AutoSys Workload Automation for the View Agent Details to succeed. You cannot View Agent Details from an agent that is not running.
- The displayed details are the same as the details that are displayed with the “cybAgent -v[v]” command of the agent.
- A modal window in the browser displays the details.



To download the agent details, click a row in the Agent Inventory table. Select the **View Agent Details** option in the **Machine Name** button drop-down list. Click the **DOWNLOAD** button. Once downloaded, you can use a standard editor of your choice to open the agent parameters.

### Update AgentParams

Agent Inventory provides the capability to update the parameters in the agentparm.txt file.

#### **Follow these steps:**

1. Click ... against the machine (agent) name for which you want to update the parameters in the **Machine Name** column of the **Agent List** table.
2. Select **Update AgentParams**.  
The agentparm.txt file is collected from the agent and the **Update Agent Parameters** dialog displays the agent parameters.

#### **NOTE**

A message displays if no content is present in the agentparm.txt file.

3. Add new content or modify the existing content as appropriate and Click **Update**.  
The updates are applied and the agent is restarted.

#### **NOTE**

- To update the agentparm.txt file, you must have Configure access under the AgentAccess policy.
- If the updated agentparm.txt file contains any errors, the agent restart fails.

### Stop Agent

Agent Inventory lets you stop agents that are running. Stopping the agent stops all the plug-ins of that agent.

#### **Follow these steps:**

1. To stop the agent, select the agent that you want to stop, in the Inventory table. Select the **Stop Agent** option in the **Machine Name** button drop-down list.
2. You can stop more than one agent, by selecting the checkboxes against the agents; to stop more than one agent, select the **STOP AGENTS** option that is displayed from the **ACTIONS** button drop-down list.  
After you select Stop Agent, a message displays.
3. Click CONTINUE to stop the agents.  
The Action Status and Action Timestamp column in the Inventory table are updated.

#### **NOTE**

Legacy agents are not supported. If you select a legacy agent and use the **Stop Agent** option, a message displays.

### Restart Agent

Agent Inventory lets you Stop Agents that are running and restart those agents.

#### **Follow these steps:**

1. To restart the agent, select the agent that you want to restart, in the Inventory table. Select the **Restart Agent** option in the **Machine Name** button drop-down list.
2. You can restart more than one agent, by selecting the checkboxes against the agents; to restart more than one agent, select the **RESTART AGENTS** option that is displayed from the **ACTIONS** button drop-down list.  
After you select Restart Agent, a message displays.
3. Click CONTINUE to restart the agents.  
The Action Status and Action Timestamp column in the Inventory table are updated.

**NOTE**

Legacy agents are not supported. If you select a legacy agent and use the **Restart Agent** option, a message displays.

**Reset Action Status**

Agent Inventory lets you reset the status an agent, when the agent failed to stop or restart and the issue was manually resolved.

**Follow these steps:**

1. To reset the status of an agent, select the agent which status you want to reset, in the Inventory table. Select the **Reset Action Status** option in the **Machine Name** button drop-down list.
2. You can reset the status of more than one agent, by selecting the checkboxes against the agents; to reset more than one agent, select the **RESET ACTION STATUS** option that is displayed from the **ACTIONS** button drop-down list. The Action Status field is cleared.

**Quiesce Agents**

Agent Inventory lets you quiesce agents. Jobs already running continue to execute, but no newly submitted jobs are executed. Already running jobs cannot be canceled.

**NOTE**

- Legacy agents are not supported.
- You must have control and upgrade policy access to perform these actions. For more information, see the *Manage Access Policies in CA Embedded Entitlements Manager* section in [Administer the Agent Inventory](#).
- Only unquiesce can be performed on the quiesced agent.

You can select the agents that are to be quiesced. Running jobs are either completed or timeout is reached. Upon completion of jobs or timeout, whichever is earlier, except the current upgrade job, all other jobs are quiesced.

**Follow these steps:**

1. To quiesce the agent, select the agent that you want to quiesce, in the Agents List table. Select the **Quiesce Agent** option in the **Machine Name** button drop-down list.
2. You can quiesce more than one agent, by selecting the checkboxes against the agents; to quiesce more than one agent, select the **QUIESCE AGENTS** option that is displayed from the **ACTIONS** button drop-down list. The Action Status column in the Agents List table is updated.

**Unquiesce Agents**

Agent Inventory lets you unquiesce the agents to resume the execution of the jobs. You can select the agents that are to be unquiesced. Only a quiesced agent can be unquiesced.

**Follow these steps:**

1. To unquiesce the agent, select the agent that you want to unquiesce, in the Agents List table. Select the **Unquiesce Agent** option in the **Machine Name** button drop-down list.
2. You can unquiesce more than one agent, by selecting the checkboxes against the agents; to unquiesce more than one agent, select the **UNQUIESCE AGENTS** option that is displayed from the **ACTIONS** button drop-down list. The Action Status column in the Agents List table is updated.

**Status Details**

You can view the status of the agent that was stopped, restarted, or the agent parameter was updated.

To view the status details of an agent, select the agent which status you want to reset, in the Inventory table. Select the **Status Details** option in the **Machine Name** button drop-down list.

A message displays the status details of the agent.

### **Plugin Details**

Agent Inventory lets you view various details about an agent like enabled functionality and plug-ins.

To view the Agent and Plugin details, click a row in the Agent Inventory table. Select the **Plugin Details** option in the **Machine Name** button drop-down list.

#### **NOTE**

When you select the Plugin Details option, columns not displayed in the Agent List table are also listed under Agent Details.

### **Stage/Upgrade Status Details**

Agent Inventory lets you view various details about the staging or upgrading status of an agent.

To view the Staging or Upgrading Status Details details, click a row in the Agent Inventory table. Select the **Stage/Upgrade Status Details** option in the **Machine Name** button drop-down list.

## **Centralized Agents Upgrade**

You can upload and stage agent installers from the Web UI Server and upgrade the Agent.

#### **NOTE**

You must have control and upgrade policy access to perform these actions. For more information, see the *Manage Access Policies in CA Embedded Entitlements Manager* section in [Administer Agent Inventory](#).

### **Upload Agent Installer to the Web UI Server**

You can upload the agent installer to the Web UI server so that agents can be seamlessly upgraded.

#### **Follow these steps:**

1. Click Agent Installer.
2. Select AutoSys or Agent against Upload Installer.  
The Upload Agent Installer window opens with the Upload Details options.

#### **NOTE**

If AutoSys is selected, only the following operating systems are displayed in the drop-down for selection:

AIX PPC, HP-UX IA, Linux x86, and Windows.

3. Enter Version Details.  
Options are Major Version, Minor Version, Service Pack, and Maintenance Levels of the Agent.

#### **NOTE**

- Upgrade of Workload Automation Agent Release 11.3 and higher is supported.
- When a row is selected, the Version Details are disabled.

4. (Optional) Enter Description.

**Limit:** 1000 characters

Displays the reason or purpose of the upload. For example, why the ISO or Version is uploaded.

5. Select the Operating systems from the drop-down list.
6. Click Choose File against Browse Location to choose the agent or autosys iso file from your computer.
7. Click SUBMIT.

The agent installer is uploaded to the Web UI server.

The Agent Installer Inventory page displays the Agent Version, Description, Operating System, and Upload Status of agents that are already uploaded.

You can also click Delete to remove the selected Agent Installer version from agent installer inventory and delete the installer files from the Web UI server.

### **Edit the Agent Installer for Upload**

You can edit the description and add or remove operating systems of a single agent installer.

#### **Follow these steps:**

1. Select the checkbox from the Agent Installers List for the agent installer that you want to edit.
2. Click Upload.

The Upload Agent Installer window opens with the Version Details disabled.

#### **NOTE**

If more than one row is selected, Upload is disabled.

3. Update Description as required.
4. Add or remove operating systems as required.

#### **NOTE**

If you add a new operating system, you must choose the relevant agent or autosys iso file.

5. Click SUBMIT.

The agent installer is uploaded to the Web UI server.

### **Stage Agent Installer for Upload from the Web UI Server to the Agent**

You can stage the agent installer for upload from the Web UI server to the agent so that agents can be seamlessly upgraded.

#### **Follow these steps:**

1. Select the machine names from the Agents List for the agents that you want to upgrade.
2. Click Upgrade Actions, Stage Agents.  
The Stage Agents window opens.
3. Select an installer version and click Continue.

#### **NOTE**

- If the selected agent version is higher than the installer version, an error message displays.
- Legacy agents are not supported.
- Agents where stopping and starting are in progress are not supported.

You can ignore these and proceed with staging the agents for upload.

The Description field displays the reason or purpose of the upload. For example, why the ISO, Version, or Fix is uploaded.

The staged agents are displayed in the Agent List.

You can now upload the agent installers from the Web UI server to the Agents.

### **Upgrade Agent Installer**

You can upgrade the agent/s where the installers are already staged.

#### **NOTE**

- You can select the Upgrade using Java VM bundle from installer checkbox to use the JVM bundled with the product.

Or, you can also specify the JVM path in your computer against the operating system.

**Follow these steps:**

1. Select the agent that you want to upgrade.

**NOTE**

Agent installer must be staged before you begin the upgrade.

2. Select the machine names from the Agents List for the agents that you want to upgrade.
3. Click Upgrade Actions, Upgrade Agents.  
The Upgrade Agents window opens.

**NOTE**

- If the selected agent version is higher than the installer version, an error message displays.
- Legacy agents are not supported.
- Agents where stopping and starting are in progress are not supported.

You can ignore these and proceed with upgrading the agents.

During the upgrade process the agent is,

- quiesced
- stopped
- upgraded
- restarted
- unquiesced

The upgraded agents are displayed in the Agent List.

**NOTE**

If the agent upgrade fails on z/Linux or AIX, see the [\(z/Linux and AIX\) Unable to Restart the Agent](#) section in the Workload Automation System Agent documentation.

## **Schedule the Upgrade**

You can select the date and time at which you want to schedule the upgrade.

**Follow these steps:**

1. Select the agent that you want to upgrade.

**NOTE**

Agent installer must be staged before you begin the upgrade.

- a. For a single agent, select Upgrade Agent from the drop-down next to the Machine Name.
  - b. For multiple agents, select the Machine Name in the Agents List and click UPGRADE ACTIONS, Upgrade Agents.  
The Upgrade Agents window opens.
2. Select one of the following:
    - a. Upgrade Now to proceed with the upgrade.
    - b. Schedule Upgrade to schedule the upgrade for a later time.
 If you selected Schedule Upgrade, the Upgrade Scheduling Details pane opens.
  3. Enter the following information:
 

**Schedule Date**  
Displays the selected date for the upgrade.

**Schedule Time**  
Displays the selected time for the upgrade.
  4. (Optional) Select the Upgrade using Java VM bundle from the installer checkbox to use the JVM bundled with the product (or) Specify the JVM path in your computer against the operating system.

**NOTE**

If the checkbox is clear, you must specify the custom JVM path.

The JVM bundled with the product is used for the upgrade.

5. Click Continue to proceed with the upgrade.

If upgrade cannot be performed for any of your selected Machine Names, the reasons are listed.

**Cancel the Scheduled Upgrade**

You can cancel the scheduled upgrade.

**Follow these steps:**

Select the agent that you had scheduled to upgrade.

- For a single agent, select Cancel Scheduled Upgrade from the drop-down next to the Machine Name.
- For multiple agents, select the Machine Name in the Agents List and click UPGRADE ACTIONS, Cancel Scheduled Upgrade.

The upgrade scheduled by you is canceled.

If no upgrade is scheduled for the agent, a failed to cancel the upgrade message displays.

**Upgrade Status**

You can view the following Upgrade status:

**Time outs** - Stage Timeout, Quiesce Timeout, Upgrade Timeout, and Rollback Timeout

**Successful upgrades** - Upgrading, Quiescing, Stopping, Upgrading, and Upgraded

**Cancelled upgrades** - Upgrade Cancelled

**Failures** - Stage failed, Upgrade failed, Rollback failed

To view the detailed Stage or Upgrade Status details, click a row in the Agent Inventory table. Select the Stage/ Upgrade Status Details option in the Machine Name button drop-down list.

**Rollback Agent Upgrade**

You can rollback only an agent that was upgraded using Agent Inventory. You can use rollback and revert only to the previously replaced version.

You can rollback an upgraded agent in the following conditions:

- Agent was successfully upgraded.
- Rollback failed.

You can perform any other action such as Stop or Restart agent on the upgraded agent and then roll back to the previously replaced version only once.

You cannot perform rollback if an upgrade is scheduled.

**Follow these steps:**

Select the machine names from the Agents List for the agents that you want to rollback.

- For a single agent, select Rollback Upgrade from the drop-down next to the Machine Name.
- For multiple agents, select the Machine Names in the Agents List and click UPGRADE ACTIONS, Rollback Upgrade.

The selected agents are rolled back to the version from which they were upgraded.

## Web UI Application Editor

Application Editor provides a graphical representation of all the relationships between AutoSys Workload Automation jobs and their dependencies. It displays the conditions of the relationships that must be met before a job can run, and the successor jobs triggered when a job finishes running.

### NOTE

The new job that is dragged from the palette and dropped on to the graph is red in color. After entering the mandatory and optional fields as required and applied, the color changes to teal.

You can create a job flow or modify an existing job flow by adding or deleting objects graphically using drag-and-drop. If your site is implementing this feature, your administrator can create job templates with predefined properties for use. Changes made to the job flow or object properties can be committed to the selected AutoSys Workload Automation server.

### NOTE

If committed to a AE r11.3 server, the following job types are ignored and a warning message is displayed:

- Micro Focus
- Process Automation Process Execution
- Process Automation Start Request Form
- Remote Execution
- SNMP Value Get
- SNMP Value Set
- Wake on LAN
- Web Service Document/Literal

### Application Editor Navigation

Application Editor is accessed by clicking the Application Editor tab in the main Web UI page. The Application Editor page contains two sections: Flow and Properties. Initially, only the Flow section is displayed. To edit a job, a page navigation link lets you move from the Flow to the Properties section.

### NOTE

- Access to Application Editor is controlled through the ApplicationAccess resource in CA EEM.
- The Properties section appears when you right-click a job object in the Graph and select Edit.  
If the job has an error condition, an error message opens and the display does not automatically scroll down to the Properties section. You can manually scroll down to the Properties section and correct the error.

The Flow section consists of the following panes:

- The upper left pane is the Palette, which lists object types that you can add to the job flow.
- The lower left pane is the Overview, which displays a thumbnail view of all the jobs and dependencies imported into Application Editor.
- The left pane also contains the Delete List section, displaying a list of the jobs that are selected for deletion.
- The center pane is the Graph, which graphically displays an editable view of the jobs and dependencies that you imported into Application Editor or manually added when you started a new flow, or that were displayed previously in the Application Editor.

- You can define the conditions required to run a job by dragging objects from the Palette and dropping them onto the Graph. Job properties and dependencies can be edited, and multiple changes can be committed to the server you select in a single operation.
- If you save object property and job flow changes before you leave Application Editor and close Web UI, you can redisplay the job flow that was displayed when you left by selecting Load Flow from the drop-down list the next time you start and log in to Web UI, and access Application Editor.
- If some objects in the imported job flows are not visible in the Graph, the boundary box in the Overview pane can be dragged to display them in the Graph.
- The right pane consists of the SAP filters sections. SAP drag-and-drop functionality let you use an SAP job that exists on an SAP server as a template to create a new SAP job, and is implemented by the SAP Filters and SAP Filter Results panes. The SAP Filters and SAP Filter Results panes are displayed by default when you select any one or all the following job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. The SAP Filters pane contains filters with SAP job properties which you can use to search for jobs on an SAP server. The SAP Filter Results pane contains the search results. You can drag a job from the search results into the job flow to create a new SAP job, modify the job properties, and then commit the job flow to create the new job on the specified AutoSys Workload Automation server.
- The Properties section lets you edit the properties of the job that has been imported or added to the job flow, and appears when you right-click the job and select Edit. When you add a job to the Graph, it is displayed with a red border and an asterisk (\*) next to its name, indicating that a required property is missing.
- The Delete List pane contains a list of jobs that you deleted from the job flow displayed in the Graph. The jobs are deleted from the specified server when you commit the job flow. If needed, you can remove jobs from the list before you commit the job flow.

#### **NOTE**

If the icon of a dependent job shown in the Graph contains a question mark (?), the job cannot be edited in Application Editor.

### **Workload Objects**

For information about the jobs and workload objects that are supported in Web UI, see Manage the Job Flow and Manage Workload Objects sections.

### **Logical Operators**

A logical operator uses the logically combined result of two or more objects in the job flow to control subsequent processing. Objects are connected to logical operators using dependency links. You must define the property of the dependency link, such as the result of a job run (success, failure, and so on), from objects that precede the logical operator. The dependency link from a logical operator to the next object in the job flow has no properties to define; the operator itself controls subsequent processing.

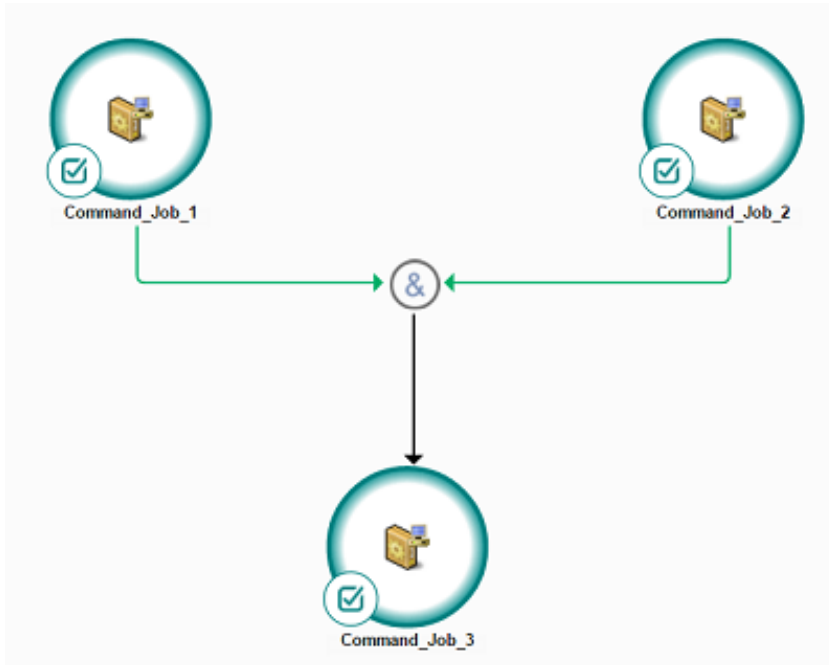
The available logical operators are:

- **AND**

Subsequent processing occurs as the result of the logical AND of two or more preceding objects. For example, if you want Command\_Job\_3 to run when Command\_Job\_1 and Command\_Job\_2 complete successfully, the dependency statement for Command\_Job\_3 is: s(Command\_Job\_1) & s(Command\_Job\_2).

The following diagram shows how this dependency statement appears in Application Editor:

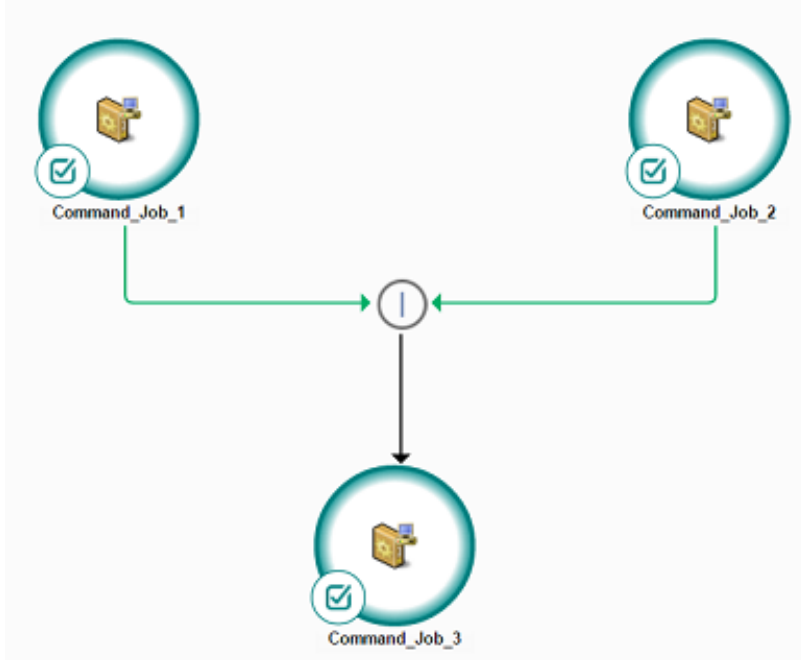




- **OR**

Subsequent processing occurs as the result of the logical OR of two or more preceding objects. For example, if you want Command\_Job\_3 to run when Command\_Job\_1 or Command\_Job\_2 completes successfully, the dependency statement for Command\_Job\_3 is: `s(Command_Job_1) | s(Command_Job_2)`.

The following diagram shows how this dependency statement appears in Application Editor:



## **Dependencies**

A dependency is created by linking two objects in the Graph. The link appears as an arrow pointing from the dependency object to the object that depends on it. The dependency link property defines the condition of the object that must be met

before the next step in the job flow is performed. You can select the properties for a dependency link originating from a job or a global variable by right-clicking the link and selecting Edit.

For the dependency link from a job, you can select one of the following conditions:

- **SUCCESS** -- If the job completes successfully, processing continues with the next step in the flow. The link color is green.
- **FAILURE** -- If the job fails, processing continues with the next step in the flow. The link color is red.
- **DONE** -- If the job indicates that it is done, processing continues with the next step in the flow. The link color is blue.
- **TERMINATED** -- If the job is terminated, processing continues with the next step in the flow. The link color is magenta.
- **NOTRUNNING** -- If the job is not running, processing continues with the next step in the flow. The link color is gray.
- **EXITCODE** -- If the job returns the exit code you select, processing continues with the next step in the flow. The link color is black.

For the dependency link from a global variable or from a job when the selected dependency condition is EXITCODE, you can select one of the following operators and a value:

- **= (Equal)** -- If the global variable or job exit code is equal to the value specified, processing continues with the next step in the flow.
- **!= (Not equal)** -- If the global variable or job exit code is not equal to the value specified, processing continues with the next step in the flow.
- **< (Less than)** -- If the global variable or job exit code is less than the value specified, processing continues with the next step in the flow.
- **> (Greater than)** -- If the global variable or job exit code is greater than the value specified, processing continues with the next step in the flow.
- **<= (Less than or equal to)** -- If the global variable or job exit code is less than or equal to the value specified, processing continues with the next step in the flow.
- **>= (Greater than or equal to)** -- If the global variable or job exit code is greater than or equal to the value specified, processing continues with the next step in the flow.

### **Customize Dialog**

You can use the Customize dialog to set default values in Application Editor.

This dialog contains the following tabs:

- **Filter Job Types** -- You can use the Filter Job Types tab to select one or more preferred job types from the list of job types created by your administrator in the Filter Job Types Batch Interface. Only the selected job types are available in the Palette.
- **General** -- You can use the General tab in the Customize dialog to hide the SAP drag-and-drop panes, specify the number of rows per page to display in the SAP drag-and-drop panes, specify default values to use for display parameters, and to customize other settings.

### **Set Display Preferences**

You can set your display preferences on the General tab of the Customize dialog.

#### **Follow these steps:**

1. Click Customize.  
The Customize dialog opens.
2. Click the General tab, make the appropriate changes, and click Save.  
Your preferences are saved, the dialog closes, and the display is updated.

---

## Set Job Type Preferences

You can use the Filter Job Types tab of the Customize dialog to select the job types to be displayed in the Palette.

### Follow these steps:

1. Click Customize on the title bar.

#### NOTE

The first time you open the Customize dialog, all available job types are selected on the Filter Job Types tab.

2. Clear the Select check box for the job types you do not need.

#### NOTE

When a group name is selected, all job types within the group are selected. Clearing the group name check box clears all job types within the group. If you clear certain job types within a selected group, the group name check box is cleared also. Similarly, if you select all job types within an unselected group, the group name check box is selected also.

3. Click Save.

Your job type preferences are saved.

#### NOTE

- Only the job types (and corresponding job templates, if any) displayed in the Palette can be added to the job flow in the Graph.
- Job templates, if any, are displayed in the Palette automatically when the corresponding job type is selected in the Customize dialog.

## Add a Job with Conditions to a Job Flow in the Application Editor

This article explains how to add a Job with conditions to the Job Flow in the Application Editor. As a scheduler, you want to insert one or more new jobs into an existing job flow as needed. Application Editor lets you accomplish this task using a graphical workspace with drag-and-drop functionality.

#### NOTE

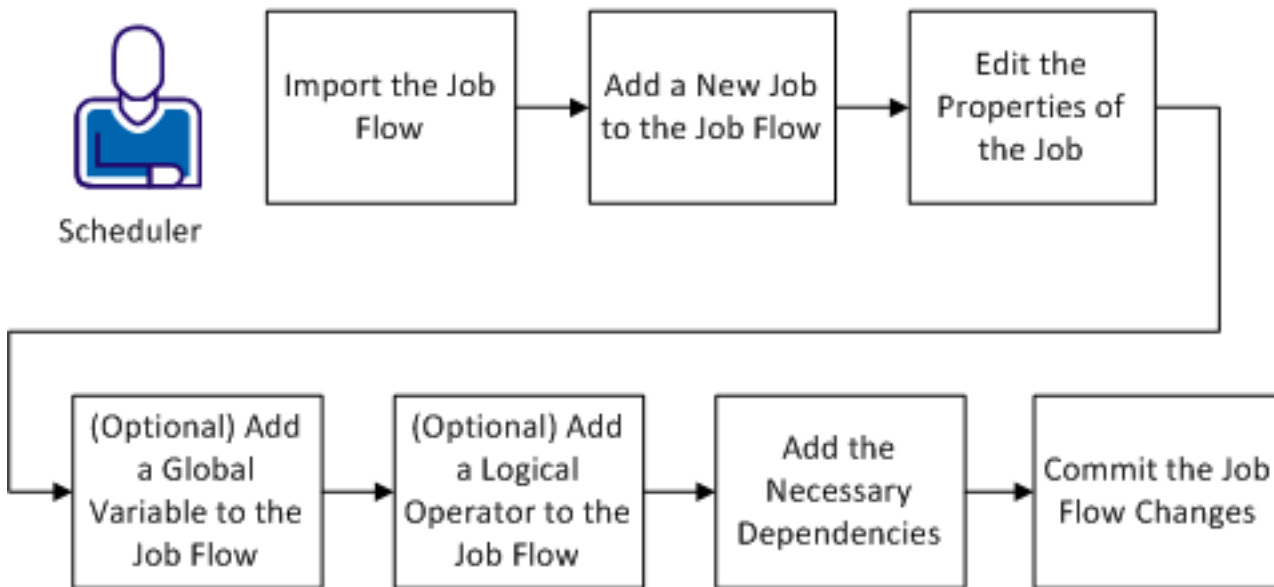
- For HTTP and WSDOC job types, the server field is displayed with the first server selected by default. The displayed job properties are based on the selected server version.
- When selecting a different server, a warning dynamically displays error messages if the specified resources such as calendars, machine, and templates are not available in the selected server.

### Server

Specifies the server from the drop-down list.

The following scenario walks you through the process of importing an existing job flow into the workspace, creating a job, editing the job properties, adding a global variable and a logical operator, adding dependencies between objects, and committing the modified job flow to the scheduling manager.

## How to Add a Job with Conditions to a Job Flow



To add a job with conditions to a job flow, follow these steps:

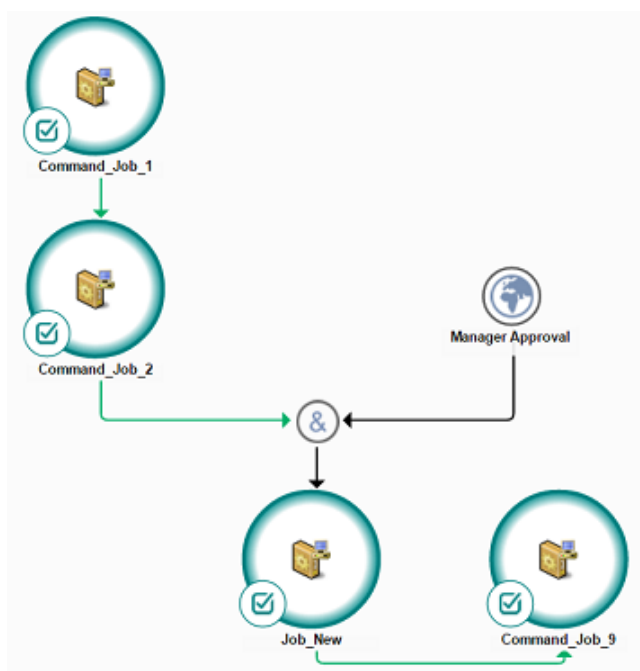
1. [Import the job flow](#)
2. [Add a new job to the job flow](#)
3. [Edit the properties of the job](#)
4. (Optional) [Add a global variable to the job flow](#)
5. (Optional) [Add a logical operator to the job flow](#)
6. [Add the necessary dependencies](#)
7. [Commit the job flow changes](#)

### Example: Add a New Job with a Manager Approval Condition to a Job Flow

Suppose you have a job flow that contains three jobs named Command\_Job\_1, Command\_Job\_2, and Command\_Job\_3. When Command\_Job\_1 runs and ends successfully, Command\_Job\_2 starts. When Command\_Job\_2 runs and ends successfully, Command\_Job\_3 starts and runs to completion. The job flow imported into Application Editor appears as follows:



You must add a new job to the flow that runs between Command\_Job\_2 and Command\_Job\_9. Also, add a manager approval condition (as a global variable) that lets the new job run. If manager approval is granted and Command\_Job\_2 ends successfully, the new job starts. When the new job ends successfully, Command\_Job\_9 starts. You can modify the job flow as follows:



The final step is to commit the modified job flow so that the changes are applied to the scheduling manager. The modified job flow is then ready to run.

## **Import Jobs**

You can import one or more jobs into Application Editor to view the flow, add jobs, edit job properties, add dependencies, or modify existing dependencies.

### **Follow these steps:**

1. Select Import from Scheduler from the drop-down list and click Go in the Flow section.  
The Import Jobs dialog opens.
2. Select a server from the Server drop-down list.
3. Specify the job name, optionally specify one or both of the following options, and then click Go:
  - Group
  - Application

#### **NOTE**

The following options are permitted:

- Valid job name, group, or application—Finds the specified jobs on the selected server that match the specified criteria.
- Asterisk (\*), percent (%), or question mark (?), used as a wildcard with a partial name, where the \* or % represents zero or more characters, or ? represents a single character—Finds all jobs on the selected server that contain the partial job, group, or application name. For example, entering n\* in the Name field only would find all jobs on the server that begin with the letter n.
- Asterisk (\*) or percent (%)—Finds all jobs on the selected server when the wildcard is specified for the Name field and the other two fields are blank.

A list of jobs appears in the Search Results section.

#### **NOTE**

The Search Results section can include jobs and box jobs.

4. Do one of the following actions:
  - Select the Select check box for the jobs you want to import and click Import Selected Jobs.  
The Import Jobs dialog closes, and the selected jobs, box jobs, and their dependencies appear in the Graph.
  - Click Import All.  
The Import Jobs dialog closes, and all jobs and box jobs in the Search Results section and their dependencies appear in the Graph.

#### **NOTE**

If the number of jobs to be imported is greater than the value specified for Maximum Number of Imported Jobs in the Customize dialog, an error message is displayed and the import is not performed.

## **Add a Job**

You can add a job to a job flow shown in the Graph.

#### **NOTE**

The contents of a box job are displayed or hidden by clicking the expand control on the box job icon.

### **Follow these steps:**

1. Drag a job object or a job template object, if applicable, from the Palette to the Graph.

#### **NOTE**

- The job types and corresponding job templates displayed in the Palette depend on the job types selected in the Customize dialog.
- You can add a job to a box job by dragging it inside the box job.

A job icon appears in the graph.

### NOTE

The icon appears with a red border if a required job property has not been specified.

2. Right-click the job object and select Edit or select the job object and press Enter.  
The Properties section appears.

### NOTE

If the job has an error condition, an error message opens and the display does not automatically scroll down to the Properties section. You can manually scroll down to the Properties section and correct the error.

3. Edit the job properties and click Apply in the Properties section.  
The job properties are saved, the Properties section and navigation link are no longer displayed, and the display scrolls up to the Flow section. If all required property fields have been populated, the red border is removed from the job icon.
4. Add the necessary dependency links to connect the new job to other objects.
5. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.  
For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

### NOTE

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

## Edit a Job

You can edit the properties of a job displayed in the Graph. Job properties appear in the Properties section.

### NOTE

- If the icon of a dependent job shown in the flow contains a question mark (?), the job cannot be edited in Application Editor.
- The contents of a box job can be alternately displayed or hidden by clicking the expand control on the box job icon.

### Follow these steps:

1. Right-click the job object in the Graph and select Edit or select the job object and press Enter.  
The Properties section appears.

### NOTE

If the job has an error condition, an error message opens and the display does not automatically scroll down to the Properties section. You can manually scroll down to the Properties section and correct the error.

2. Edit the job properties as necessary and click Apply in the Properties section.

### NOTE

If the Apply button is not visible, you must scroll up to the beginning of the Properties section.

The changes are saved, the Properties section and navigation link are no longer displayed, and the display scrolls up to the Flow section.

3. Select Save Flow from the drop-down list in the Flow section and click Go.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Add a Global Variable**

You can add a global variable to a job flow shown in the Graph.

**Follow these steps:**

1. Drag the Global Variable object from the Palette to the Graph.  
The global variable in the Graph is highlighted.
2. Right-click the global variable and select Edit or select the global variable and press Enter.  
The Edit Global Variable dialog opens.
3. Enter a name and click OK.  
The dialog closes and the name appears with the global variable.
4. Add the necessary dependency links to connect the new global variable to other objects.
5. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Add a Logical Operator**

You can add a logical object to a job flow shown in the Graph.

**Follow these steps:**

1. Drag a logical operator object from the Palette to the Graph.
2. Add the necessary dependency links to connect the new logical operator to other objects.
3. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Add a Dependency**

You can add a dependency to an object by creating a link between two objects in the Graph. The dependency appears as an object with a line pointing to the object that depends on it. The dependency link property defines the condition that must be met before the next step in the job flow is performed.

**Follow these steps:**

1. Click



on the toolbar.

**Default mode:** Select

2. Click the object serving as the dependency and drag a line to the object that depends on it without releasing the mouse button.



**NOTE**

You must wait for the object to which you are dragging the line to be highlighted before you release the mouse button.

A dependency link appears between the two objects.

3. Right-click the dependency link and select Edit or select the dependency link and press Enter.  
One of the following occurs:
  - If the dependency link originates from a job object, the Edit Job Dependency dialog opens.
  - If the dependency link originates from a global variable object, the Edit Global Variable Dependency dialog opens.

**NOTE**

A dependency link from a logical operator to another object has no editable properties.

4. Select the appropriate properties, enter values if necessary, and click OK.  
The dependency link between the two objects is complete. You can also view the dependency in the Condition field of the properties for the second object if it is a job.
5. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Commit Changes**

You must commit job property and job flow changes to the appropriate AutoSys Workload Automation server to make the changes permanent on that server.

**NOTE**

- You must apply your job property changes before you perform a commit; however, it is not necessary to save the Application Editor workspace before you perform a commit.
- Jobs are committed to the server that was selected during the commit process and not to the server version selected for the job.

**WARNING**

If you have added a new job to the Graph and a job with the same name and job type exists on the server to which you are committing the flow, the new job replaces the existing job; however, if both job names are the same but the job types are different, an error message is displayed in the Job Commit Progress dialog, and the existing job remains unchanged.

**Follow these steps:**

1. Select Commit to Scheduler from the drop-down list and click Go in the Flow section.  
The Commit Jobs dialog opens.
2. Select the server you want to commit the changes to, select the appropriate options, and click OK.

**NOTE**

You must select Permanently Delete Listed Jobs to delete the jobs and box jobs in the Delete List on the specified server.

The Job Commit Progress dialog appears and the changes are committed to the server.

3. Click Close when the commit process completes.  
The Job Commit Progress dialog closes.

## Create a SAP Job from an Existing SAP Job in the Application Editor

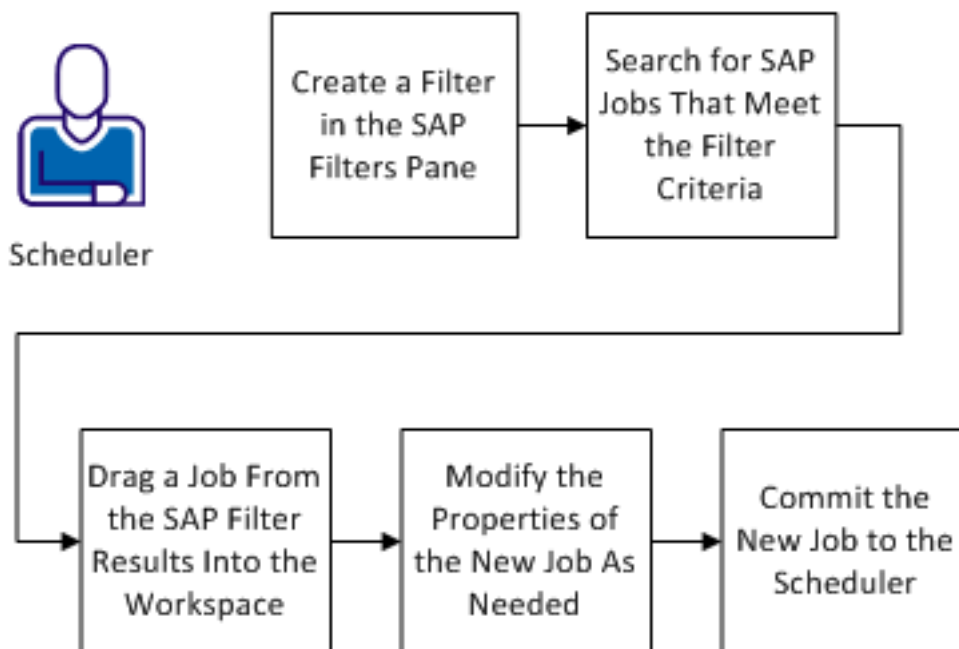
This article explains how to create an SAP Job from an existing SAP Job in the Application Editor. As a scheduler, you want to quickly create a AutoSys Workload Automation job that has many of the same properties as an existing SAP job on your SAP server. You can use the SAP drag-and-drop feature to create an SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage job. The new job can be scheduled on the AutoSys Workload Automation server you commit it to.

### NOTE

This process uses the SAP Filters and SAP Filter Results panes. If they are not visible, you can display them by selecting any one or more of the following job types using the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Panes in the Customize dialog is not checked.

The following scenario walks you through the process of creating a filter in the SAP Filters pane, using the filter to find the appropriate SAP job, creating a job by dragging a job from the Filter Results into the Application Editor workspace, modifying the properties of the new job, and committing the job.

### How to Create a New SAP Job from an Existing SAP Job



To create a AutoSys Workload Automation job from an existing SAP job, follow these steps:

1. [Create a filter in the SAP Filters pane](#)
2. [Search for SAP jobs that meet the filter criteria](#)
3. [Drag a job from the SAP Filter Results into the workspace](#)
4. [Modify the properties of the new job as needed](#)
5. [Commit the new job to the scheduler](#)

### Example: Create a AutoSys Workload Automation Job Based on an Existing SAP BW Process Chain Job

Suppose you have an SAP BW Process Chain job that runs a sequence of background processes on your SAP system. You want to create a AutoSys Workload Automation job definition that has many of the same properties of your existing

SAP job; however, you want it to run a different BW process chain at a different time. For this, you first create an SAP filter to find your existing job and drag that job into the workspace. Then you can change the job name, the chain identifier, and the date and time dependencies. Finally, you must commit the job to the scheduler. The new job is ready to run.

### **Add an SAP Filter**

You can add a filter containing the SAP properties of jobs that you want to retrieve from an SAP server. This filter is displayed in the list in the SAP Filters pane and, when selected, returns those jobs in the SAP Filter Results pane so you can use them as templates for new SAP jobs. Unique filters can be added to assist you in creating the following SAP job types: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, and SAP BW InfoPackage.

#### **NOTE**

- To display the SAP Filters and SAP Filter Results panes, select one or more of the following SAP job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Pane is not checked in the Customize dialog.
- The filters that you add are not available to other users.

#### **Follow these steps:**

1. Click Add in the SAP Filters pane.  
The Create New SAP Filter dialog opens.
2. Select the Filter type that corresponds to the job type for which you intend to use it.  
The Filter criteria fields for the specified SAP filter type are displayed.
3. Specify a name in the Filter name field.
4. Specify the following required Logon parameters:
  - **Server**  
Specifies the server on which to perform the search.
  - **Machine**  
Specifies the name of the computer on which the SAP agent runs.
5. Specify the following required filter criteria:
  - **SAP job name**  
Specifies the SAP job name.  
You can use an asterisk (\*) as a wildcard character to represent multiple characters.
  - **SAP user name**  
Specifies the SAP user name.  
You can use an asterisk (\*) as a wildcard character to represent multiple characters.

#### **NOTE**

These fields apply to SAP R/3 and SAP Batch Input Sessions filter types only.

6. (Optional) Specify other filter criteria to narrow the job search.

#### **NOTE**

The Filter criteria fields that are displayed in the dialog depend on the filter type selected.

7. Click Save.  
The filter is saved and its name is added to the SAP Filters list.

### **Find Jobs on an SAP Server**

You can find SAP jobs on an SAP server that match the filter criteria you have defined.

#### **NOTE**

To display the SAP Filters and SAP Filter Results panes, select one or more of the following SAP job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Pane is not checked in the Customize dialog.

To find SAP jobs on an SAP server, click the filter in the SAP Filters pane that contains the SAP properties for which you want to search.

**NOTE**

If many jobs that match the search criteria are found, the FTP Credentials dialog opens. You must enter your credentials to continue. However, if the No such plugin error appears, you must install or enable the FTP plug-in on the agent.

The SAP jobs found on the SAP server are displayed in the Results for SAP Filter Criteria pane.

### **Add an SAP Job Based on an Existing SAP Server Job**

You can add a new SAP job containing the properties of a job on an SAP server to your job flow.

**NOTE**

To display the SAP Filters and SAP Filter Results panes, select one or more of the following SAP job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Pane is not checked in the Customize dialog.

**Follow these steps:**

1. Click the appropriate filter in the SAP Filters pane.  
SAP jobs that match the filter properties are displayed in the Results for SAP Filter Criteria pane.
2. Drag the job you want from the Results for SAP Filter Criteria pane to the Graph.  
An SAP job icon appears in the graph.

**NOTE**

The icon appears with a red border if a required job property has not been specified.

3. Right-click the job object and select Edit.  
The Properties section appears.

**NOTE**

If the job has an error condition, an error message opens and the display does not automatically scroll down to the Properties section. You can manually scroll down to the Properties section and correct the error.

4. Edit the job properties and click Apply in the Properties section.  
The job properties are saved, the Properties section and navigation link are no longer displayed and, if all mandatory property fields have been populated, the red border is removed from the job icon.
5. Add the necessary dependency links to connect the new job to other objects.
6. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

### **Edit a Job**

You can edit the properties of a job displayed in the Graph. Job properties appear in the Properties section.

**NOTE**

- If the icon of a dependent job shown in the flow contains a question mark (?), the job cannot be edited in Application Editor.
- The contents of a box job can be alternately displayed or hidden by clicking the expand control on the box job icon.

**Follow these steps:**

1. Right-click the job object in the Graph and select Edit.  
The Properties section appears.

**NOTE**

If the job has an error condition, an error message opens and the display does not automatically scroll down to the Properties section. You can manually scroll down to the Properties section and correct the error.

2. Edit the job properties as necessary and click Apply in the Properties section.

**NOTE**

If the Apply button is not visible, you must scroll up to the beginning of the Properties section.

The changes are saved, the Properties section and navigation link are no longer displayed, and the display scrolls up to the Flow section.

3. Select Save Flow from the drop-down list in the Flow section and click Go.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Commit Changes**

You must commit job property and job flow changes to the appropriate AutoSys Workload Automation server to make the changes permanent on that server.

**NOTE**

You must apply your job property changes before you perform a commit; however, it is not necessary to save the Application Editor workspace before you perform a commit.

**WARNING**

If you have added a new job to the Graph and a job with the same name and job type exists on the server to which you are committing the flow, the new job replaces the existing job; however, if both job names are the same but the job types are different, an error message is displayed in the Job Commit Progress dialog, and the existing job remains unchanged.

**Follow these steps:**

1. Select Commit to Scheduler from the drop-down list and click Go in the Flow section.  
The Commit Jobs dialog opens.
2. Select the server you want to commit the changes to, select the appropriate options, and click OK.

**NOTE**

You must select Permanently Delete Listed Jobs to delete the jobs and box jobs in the Delete List on the specified server.

The Job Commit Progress dialog appears and the changes are committed to the server.

3. Click Close when the commit process completes.  
The Job Commit Progress dialog closes.

**Manage the Job Flow in the Application Editor**

This article explains how to manage the Job flow in the Application Editor.

**NOTE**

For CA WCC releases upto r11.4 SP5 to display jobnames with clarity in the jobflow, we recommend one of the following options:

- (Windows) Install Arial font on your computers.
- (Linux and Solaris) Install equivalent font, similar to Arial for Windows, on your computers.

**Start a New Job Flow**

If there are objects in the Graph, you can use this procedure to create a job flow. If the Graph is empty, you can start a new job flow by dragging objects from the Palette to the Graph or by importing jobs.

**Follow these steps:**

1. Select Clear from the drop-down list and click Go in the Flow section.  
A confirmation prompt appears.
2. Click OK.  
The Graph is cleared.

**Save Job Properties and Job Flow Changes**

You should save job property and job flow changes before you log out and close Web UI. When you return to Application Editor at another time, you can redisplay the flow as it last appeared.

**Follow these steps:**

1. Click Apply in the Properties section.  
The job properties are saved to the Application Editor workspace and the Properties section closes.
2. Select Save Flow from the drop-down list and click Go in the Flow section.  
The Application Editor workspace, which contains your job properties and job flow changes, is saved.

**NOTE**

- If you do not save your job property changes before you save the Application Editor workspace, your job flow changes are saved; however, your job property changes are lost when you close Web UI.
- If you apply your job property changes but do not save the Application Editor workspace, your job property and job flow changes are lost when you close Web UI.
- For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Display the Previous Job Flow**

When you start Web UI, you can display the job flow that was last saved in Application Editor during the previous Web UI session.

**NOTE**

If you leave Application Editor but do not close Web UI, when you return to Application Editor, the job flow in the Graph appears as it did when you left.

**Follow these steps:**

1. Start and log in to Web UI.  
The Web UI main page appears.
2. Click the Application Editor tab.  
The Application Editor page appears.
3. Select Load Flow from the drop-down list and click Go in the Flow section.  
The last saved job flow is displayed in the Graph.

### **Display the Previous Job Flow After the Session Expires**

If you try to perform an action in the Graph after the Web UI session expires, the following message appears: "The session has expired. Please reload the page." As an alternative to reloading the page, you can display the job flow as it was last saved.

#### **Follow these steps:**

1. Select Clear from the drop-down list and click Go in the Flow section.  
The login page appears with the Application Editor tab selected.
2. Enter the appropriate credentials and click Log In.  
The Application Editor page appears with no job flow in the Graph section.
3. Select Load Flow from the drop-down list and click Go in the Flow section.  
The last saved job flow is displayed in the Graph.

### **Revert to the Original Flow**

If you have deleted jobs from a job flow you have imported and then decide that you want to add the jobs back into the flow, you can display the original job flow by re-importing the jobs.

#### **NOTE**

If you have deleted a job from the job flow and have committed the workspace to the server from which the job was originally imported with the Permanently Delete Listed Jobs option selected, the job no longer exists on that server and cannot be imported.

#### **Follow these steps:**

1. Select Clear from the drop-down list and click Go in the Flow section.  
A confirmation prompt appears.
2. Click OK.  
The Graph and Delete List are cleared.
3. Select Import from scheduler from the drop-down list and click Go in the Flow section.  
The Import Jobs dialog opens.
4. Select the server on which the jobs reside from the Server drop-down list.
5. Do one of the following actions and click Go:
  - Specify the job name.
  - Specify the search criteria you used when you imported the jobs previously.
 The jobs appear in the Search Results section.
6. Do one of the following actions:
  - Select the Select check box for the jobs you want to import and click Import Selected Jobs.  
The Import Jobs dialog closes, and the selected jobs, box jobs, and their dependencies appear in the Graph.
  - Click Import All.  
The Import Jobs dialog closes, and all jobs and box jobs in the Search Results section and their dependencies appear in the Graph.

### **Download and Save the Job Flow**

You can download the job flow as a PDF document or PNG image and save it to a file.

#### **NOTE**

Post migration, job flows saved in releases before r11.4 SP5 are not displayed.

Click



on the toolbar, select Download as PDF or Download as PNG.

The file is saved in the Downloads folder, by default.

**Limits:** 200 nodes maximum for PNG image.

### **Print the Job Flow Image**

The Flow toolbar lets you zoom in on and print portions of the flow. This feature is helpful with complex flows because it lets you examine each portion of the flow in detail.

#### **Follow these steps:**

1. Click



on the toolbar.

2. Select File, Print from the menu options.
3. Select a printer, modify its settings if necessary, and click Print.

#### **NOTE**

- You can select the Print background colors and images option in your internet browser for the page to print correctly.
- If your browser is blocking popup windows, the print dialog does not appear.

The job flow image is printed.

## **Manage Workload Objects in the Application Editor**

This article explains how to manage Workload objects in the Application Editor.

### **Search for Jobs in Job Flow**

The Application Editor flow toolbar lets you search the job flow by job name. You can quickly locate the object in the Graph and highlight it in the viewing area.

#### **Follow these steps:**

1. Enter the value to match for the job or box job name in the Search field.

#### **NOTE**

This value is case-sensitive and can contain wildcard characters. You can use an asterisk ( \* ) or a percentage symbol ( % ) as a wildcard to represent multiple characters, or use a question mark ( ? ) as a wildcard to represent any single character. For example, you can enter \*JOB which returns all jobs that end with JOB. If no job matches these criteria, then nothing is returned. Alternatively, you can enter JOB in the Search field, and all jobs that contain JOB in the name are returned.

2. Select the appropriate job.  
The job is selected in the flow.

#### **NOTE**

- Alternatively, if you do not know the name of the job, you can search for the job using the Select by status drop-down to find a job by status.
- The Search Results section contains a maximum of 100 jobs. If more results are available, you need to refine the search to limit the number of results.



**Expand/Collapse All**

Expand/Collapse All functionality, in a particular view, is supported through the flow toolbar buttons and is synced when these flow actions are triggered.

**NOTE**

Double-click on the box job in the graph to expand and collapse.

**View Tooltip**

Hover the cursor on the job object or the link object to view the corresponding tooltip.

**Edit a Job or Box Job**

You can modify date and time conditions, change dependencies or triggers, manage permissions, or change notification settings for a specific job.

**Follow these steps:**

1. Find the appropriate job or box job.

**NOTE**

Box jobs are displayed in a hierarchical tree view. You can expand a top-level box job to view, edit, copy, or delete any nested box jobs and jobs.

2. Click the name in the Search Results section.
3. Edit the job definition as necessary.

**NOTE**

The Name field is read-only.

4. Click Commit.  
The changes are saved.

**Edit a Global Variable Name**

You can change the name of a global variable displayed in the Graph.

**Follow these steps:**

1. Right-click the global variable and select Edit or select the global variable and press Enter.  
The Edit Global Variable dialog opens.
2. Modify the name.
3. Click OK.  
The global variable name is changed.
4. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

**Edit a Dependency**

You can edit the dependency link between objects to modify the condition that must be met before the next step in the job flow is performed.

**Follow these steps:**

1. Right-click the dependency link and select Edit or select the dependency link and press Enter.

One of the following results occur:

- If the dependency link originates from a job object, the Edit Job Dependency dialog opens.
- If the dependency link originates from a global variable object, the Edit Global Variable Dependency dialog opens.

**NOTE**

A dependency link from a logical operator to another object has no editable properties.

2. Edit the dependency properties as necessary and click OK.  
The dependency link property is modified.
3. Select Save Flow from the drop-down list and click Go in the Flow section.  
The object properties and flow changes are saved.

**NOTE**

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

### **Edit Job Dependency Dialog**

The Edit Job Dependency dialog opens when you select a dependency link from a job in the Graph, right-click, and select Edit. This lets you specify a job result that determines whether the next step in the job flow is performed. For example, if JOB\_A runs successfully, processing continues.

This dialog contains the following fields:

- **[Dependency condition]**  
Specifies the dependency condition (SUCCESS, FAILURE, DONE, TERMINATED, NOTRUNNING, EXITCODE).
- **[Operator]**  
Specifies the operator to use when comparing the specified value with the exit code (=, !=, <, >, <=, >=).

**NOTE**

This field is only active when EXITCODE is selected for the dependency condition.

- **[Value]**  
Specifies the value to which the exit code is compared.

**NOTE**

This field is only active when EXITCODE is selected for the dependency condition.

### **Look-back**

**NOTE**

Specifies the time taken for the last run of the condition or predecessor job. The format is as follows:

- **hhhh**  
Indicates the hours taken for the last run of the condition or predecessor job.  
**Limits:** One to four numeric characters from 0-9999, which makes it possible to look back approximately 416.58 days

**NOTE**

When specifying minutes also, the valid values range from 0-9998 instead.

- **mm**  
Indicates the minutes taken for the last run of the condition or predecessor job.  
**Limits:** One to two numeric characters from 0-59

**Examples:** 0, 00.15, 06.30, 23.30, 24, 98.30, 720, 9998.59, and 9999

**NOTE**

- A period (.) or colon (:) can be used to separate hours and minutes.

- When the elapsed time to be specified is less than one hour, you must specify the hourly value as 00. For example, 00.30 represents 30 minutes. If hhhh.mm is specified as 30, it implies an elapsed time of 30 hours. Specifying an elapsed time of .30 is not valid.
- When you specify 0, <UAJM> examines the last end time of the job first. It then examines the last end time of the condition or predecessor job. If the condition or predecessor job has run after the last run of the job for which it is a dependency, that job is allowed to start. If the condition or predecessor job has not run after the last run of the job for which it is a dependency, that job is not allowed to start.
- For more information about look-back, see the *Workload Scheduling documentation*.

## Delete an Object

You can delete one or more jobs, logical operators, or dependencies from the job flow in the Graph.

### Follow these steps:

1. Select the object to be deleted.

#### NOTE

You can select more than one object by holding down the Ctrl key while clicking objects.

2. Do one of the following actions:

- Click  the toolbar.
- Right-click and select Delete from the pop-up menu.

The object is deleted from the Graph. When you delete a job or a box job, its name is added to the Delete List.

#### NOTE

A confirmation prompt is not displayed.

#### NOTE

For the object property, dependency, and job flow changes to be saved on a AutoSys Workload Automation server, you must commit the flow to the appropriate server.

When you delete a job from the job flow, the job is added to the Delete List. When you commit the modified flow, the jobs in the Delete List are deleted on the specified server if you select the option to do so in the Commit Jobs dialog.

## Remove a Job from the Delete List

When you delete a job from the Graph, its name is added to the Delete List. If you decide that you do not want to permanently delete the job on a server, you can remove it from the Delete List. When you commit the flow, only jobs appearing in the Delete List is deleted on the server.

### Follow these steps:

1. Select the Select check box for the job you want to remove from the Delete List.
2. Click the Remove From List link.  
The job is removed from the list.

#### NOTE

The job does not reappear in the Graph.

## Manage SAP Filters in the Application Editor

This article explains how to manage SAP filters in the Application Editor.

## SAP Filters

The SAP Filters section displays a list of user-defined filters for the SAP drag-and-drop feature. When a filter name is clicked, a search is performed on the SAP server that is specified by the filter. Jobs that match the filter properties are displayed in the SAP Filter Results pane. You can add, edit, and delete filters as needed.

### Edit an SAP Filter

The available SAP filters are listed in the SAP Filters pane. You can edit the properties of a filter.

#### NOTE

To display the SAP Filters and SAP Filter Results panes, select one or more of the following SAP job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Pane is not checked in the Customize dialog.

#### Follow these steps:

1. Click  the Actions column of the filter you want to edit. The Edit SAP Filter dialog opens.
2. Edit the properties as necessary and click Save. The dialog closes and your changes are saved.

#### NOTE

If you have edited the filter for which SAP job results are currently displayed, you must click the filter name or click Refresh to update the search results.

### Delete an SAP Filter

The available SAP filters are listed in the SAP Filters pane. You can delete an SAP filter when it is no longer needed.

#### NOTE

To display the SAP Filters and SAP Filter Results panes, select one or more of the following SAP job types in the Customize dialog: SAP R/3, SAP Batch Input Session, SAP BW Process Chain, or SAP BW InfoPackage. Also, verify that Hide SAP drag-and-drop Pane is not checked in the Customize dialog.

#### Follow these steps:

1. Select the check box next to the filter name in the SAP Filters pane and click Delete. A confirmation prompt appears.
2. Click OK. The filter is removed from the list.

## Web UI Configuration Application

Web UI provides full system configuration capabilities through the Configuration application. This application lets you do all system setup for Web UI. You can configure Quick Edit and Application Editors, Monitoring, Quick View, Forecast, Reporting, Resources, and Enterprise Command Line run-time parameters, and connectivity to back-end servers.

The Configuration page comprises two sections:

- You can use the Servers section to search for a specific server and view the search results. You can also create a new server or edit an existing server.
- You can use the Preferences section to view and modify environment properties for the instance of Web UI in use. The environment properties are displayed by category.

**NOTE**

For more information on the Servers and Preferences, see the sub-sections of this page and also the [Server Properties and Application Preferences](#) section.

## Server Management Console (SMC)

You can use the Server Management Console dialog to create, modify, copy, or delete server definitions. This dialog is opened by selecting the actions available in the Servers section or clicking the SMC button.

The Server Management Console dialog comprises three panes: Command, Commands, and Command Attributes. Depending on how you open the Server Management Console determines the content displayed in the Command pane. For example, clicking the SMC button opens the Server Management Console dialog with a blank Command pane in which to create your servers. You can use the links in the Commands and Command Attributes panes to select valid commands and attributes to define your servers. Clicking any command link in the Commands and Command Attributes panes populates the Command pane with the correct command syntax.

**NOTE**

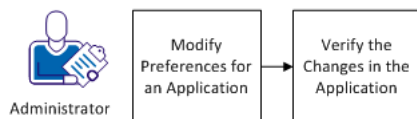
- You can use cut and paste in the left pane (Command pane) to reuse commands and attributes. To do so, select the commands that you want to duplicate, and then copy and paste them as needed. Remember to modify the server name and attributes as appropriate before executing.
- Each command is executed sequentially. If an error occurs in a command, the command execution is terminated and the next command starts executing.

## Customize Web UI Application Preferences

This article explains how to customize Web UI application preferences. As an administrator, you want to customize the Web UI application preferences for your users. You can accomplish this using the Configuration application. For example, you can modify the following application preferences:

- Number of table rows to display per page in Forecast
- Height of the Flow section in Quick View
- Collector run interval in Reporting
- Severity ranking for the various job statuses in Monitoring

### Customize Web UI Application Preferences



To customize Web UI application preferences, follow these steps:

1. [Modify preferences for an application.](#)
2. [Verify the changes in the application.](#)

### Example: Improve Monitoring Performance

Suppose that you want to improve the performance of the Monitoring application in an environment where both AutoSys Workload Automation and Web UI use CA Embedded Entitlements Manager (CA EEM) for security. Security for jobs is checked against AutoSys Workload Automation policies by default, resulting in slower Monitoring updates. However, you have decided a security check for each job is not necessary in your enterprise. To improve Monitoring performance, you open the application preferences for Monitoring, and uncheck the Use AutoSys Workload Automation policies in EEM option. When you open the Monitoring application, you see that performance has improved.

## **Application Preferences**

Environment and user interface properties for Web UI applications are set to logical defaults. Use the Preferences section to modify the properties of the following applications as required:

- Application Editor
- Configuration
- Enterprise Command Line
- Forecast
- Monitoring
- Quick Edit
- Quick View
- Reporting
- Resources

### **NOTE**

Some application preferences take effect immediately upon saving. Some applications require that you log out and then log back in to Web UI before taking effect.

## **Edit Application Preferences**

You can use the Preferences section to change the general properties of the applications for this Web UI instance.

### **Follow these steps:**

1. Select the appropriate application from the View/Modify Preferences drop-down list in the Preferences section, and click Go.
2. Edit the fields as appropriate, and click Save.  
The properties are saved and are now in effect.

## **Verify Preference Changes**

You can verify Configuration preference changes in the appropriate application. For example, change the Maximum Flow Height in Application Editor preferences. You can verify that the Flow section size has changed in Application Editor.

## **Manage Servers in the Configuration Application**

This article explains how to manage servers in the Configuration application.

## **Create a Server Using the Property Fields**

If you want information that is collected from a specific back-end server to be displayed in Web UI, you must define that server to Web UI.

### **Follow these steps:**

1. Click Create.
2. Complete the fields in the Create Server section as appropriate, and click Save.

### **NOTE**

- Every server must have a unique name.
- The Cluster Name and Instance Name combination must be unique for each AutoSys Workload Automation server.

The server configuration is saved and is available for information retrieval.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Create One or More Servers Using the SMC**

As an alternative to completing property fields to define a server, you can use the Server Management Console (SMC) to define servers with preset commands.

**Follow these steps:**

1. Click SMC in the Servers section.
2. Click the insert\_ae\_server link in the Commands pane to add a AutoSys Workload Automation server definition. The command and its required attributes are displayed in the Command pane.
3. Edit the server definition as appropriate.

**NOTE**

- Every server must have a unique name.
- The Cluster Name and Instance Name combination must be unique for each AutoSys Workload Automation server.

4. Repeat Steps 2 and 3 for each server definition you want to create.

**NOTE**

To add additional server definitions, click the insert\_ae\_server link in the Commands pane for the next server you want to create, use the browser copy and paste commands to create additional server definitions, or do both.

5. Click Execute.
6. Web UI verifies the validity of the server definitions. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed. The servers are created and are available for information retrieval.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Verify a Server Status**

To update the current status of a server, or to view the error message for a server when a problem is encountered, we recommend that you validate the status of the server definition in Configuration.

To verify a server status, select the check box to the left of the server in the Server Results section. Next, select Validate from the Select and drop-down list, and click Go.

A status message is displayed and the Server Results section refreshes to display the status of the selected server.

**NOTE**

Validation automatically updates the AutoSys Workload Automation version number.

**Find a Server**

You can use the Server Search section to find one or more servers that are currently defined on the Web UI server.

To find a server, enter a server name in the Name field, and click Go.

**NOTE**

You can enter any of the following values in the Name field:

- Valid name -- Finds the specified server.
- Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all servers in the enterprise that contain the partial name. For example, entering n\* in the Name field finds all servers that begin with the letter n. Similarly, entering \*n finds all servers that end with the letter n.
- Asterisk (\*), percent (%), or blank -- Finds all servers.

The Server Results section lists the servers that meet your search criteria.

**Copy a Server**

After you set up a server definition for a certain server type, you can use that definition as the basis for creating new definitions of that type to reduce data entry.

**Follow these steps:**

1. Select the Select check boxes to the left of the servers that you want to copy. Next, select Edit in SMC from the Select and drop-down list, and click Go.
2. Change the modify\_ae\_server command to insert\_ae\_server for each server and edit the server definitions as appropriate.

**NOTE**

- Every server must have a unique name.
- The Cluster Name and Instance Name combination must be unique for each AutoSys Workload Automation server.
- The instance\_name parameter must be added to each CA Workload Automation server definition you modify.

3. Click Execute.

Web UI verifies the validity of the server definition. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed. The server is created and is available for information retrieval.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Edit a Server Using the Property Fields**

If the configuration of a specific back-end server has changed, you can modify its properties.

**Follow these steps:**

1. Click the name of the server that you want to edit.
2. Modify the fields in the Edit Server section as appropriate, and click Save.

**NOTE**

- If you selected a server that was deleted by another user during your session, a message appears informing you that the selected server no longer exists.
- If you try to add or update a AutoSys Workload Automation server that has the same instance name and cluster name as another AutoSys Workload Automation server, an error message is displayed.



The server configuration is saved and the modified parameters are now in effect for the server definition.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Edit One or More Servers Using the SMC**

As an alternative to completing property fields to edit a server, you can use the Server Management Console (SMC) to edit server definitions.

**Follow these steps:**

1. Select the Select check boxes for the servers to be modified. Next, select Edit in SMC from the Select and drop-down list, and click Go.
2. Modify the server definitions as appropriate.
3. Click Execute.

Web UI verifies the validity of the server definitions. If the validity check encounters problems, a message is displayed describing the error. If the validity check succeeds, the commands are executed and the servers are modified. The modified parameters are now in effect for the server definitions.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Delete a Server**

If a specific back-end server definition is no longer needed, you can remove it.

**NOTE**

Deleting a server does not immediately remove the server from the available servers list in Monitoring.

**Follow these steps:**

1. Select the check boxes to the left of the servers that you want to delete.
2. Select Delete from the Select and drop-down list, and click Go. Alternatively, you can select Delete in SMC.

**NOTE**

If another user has deleted the server during your session, a message appears informing you that the selected server no longer exists.

3. Click OK to confirm.

**NOTE**

For changes to server configurations to take effect in Resources, you must log out and then log back in to Web UI.

**Preferences Section in the Configuration Application**

This article contains detailed information about the Preferences sections in Configuration.

## Agent Inventory Preferences

The Agent Inventory properties display when you select Agent Inventory from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Agent Inventory Preferences subsection to specify properties for the Agent Inventory application for this Web UI instance.

The properties in the Agent Inventory Preferences subsection are displayed by category.

The Agent Table category contains the following fields:

- **Table Columns**  
Specifies the comma-separated values for the columns to be displayed in the Agent List table. **Valid Values:** actionStatus, actionTimestamp, agentName, description, host, name, platform, plugins, port, serverName, status, type, upgradeStatus, upgradeTimestamp, and version

### NOTE

'name' is a mandatory column value.

- **Refresh Time(Delta)** Specifies the maximum duration by which Agent Inventory must fetch the updated or modified data from the AE server. Collects data from AutoSys Workload Automation Release 11.3.6 SP7 or higher.  
**Limits** (in seconds): 5-86400  
**Default:** 60 seconds
- **Refresh Time(Full)**  
Specifies the maximum duration by which Agent Inventory must fetch the entire data and must refresh the Agent Inventory page. Collects data from CA Workload Automation AE Release 11.3.6 SP6 or earlier.  
**Limits** (in seconds): 60-86400  
**Default:** 300 seconds

The Agent Action Timeout category contains the following fields:

- **Stop Timeout**  
Specifies the maximum duration by which the agent is stopped.  
**Default:** 300 seconds
- **Restart Timeout**  
Specifies the maximum duration by which the agent is restarted.  
**Default:** 300 seconds
- **Update Agent Parameters Timeout**  
Specifies the maximum duration by which the agent parameters file is updated and the Agent Inventory page is refreshed.  
**Default:** 300 seconds

The Agent Installer Directory category contains the following field:

- **Installer Directory on Server**  
Specifies the location where the Agent installer is available on the server.

The Agent Upgrade Parameters category contains the following fields:

- **Staging Directory (Windows)**  
Specifies the relevant Agent installer file location for the Windows operating system.
- **Staging Directory (UNIX)**  
Specifies the relevant Agent installer file location for the UNIX operating system.
- **Post-Quiesce Timeout Action for Upgrade/Rollback**  
Select the required action after the post-quiesce timeout. Options are:
  - **Cancel Action**

- **Continue Action**  
Continues with the upgrade.
- **Delete staged installer after successful upgrade**  
**Default:** Selected checkbox
- **Delete backup directory after successful upgrade**  
**Default:** Blank checkbox

The Agent Upgrade using Java VM category contains the following field:

- **Agent Upgrade using Java VM**  
Specifies the Java VM path against the operating system and while upgrading, the same path is replicated in the Agent inventory tab.

The Agent Upgrade Timeout Preferences category contains the following fields:

- **Stage Timeout**  
Specifies the duration after which the staging times out.  
**Default:** 300
- **Quiesce Timeout**  
Specifies the duration to complete running jobs during upgrade or rollback.  
**Default:** 600
- **Upgrade/Rollback Timeout**  
Specifies the duration after which the upgrade or rollback times out.  
**Default:** 600

### **Application Editor Preferences**

The Application Editor properties display when you select Application Editor from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Application Editor Preferences subsection to specify properties for the Application Editor application for this Web UI instance.

The properties in the Application Editor Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for SAP Filters**  
Specifies the maximum number of rows to display per page in the SAP Filters table.  
**Default:** 20
- **Rows per Page for SAP Filter Results**  
Specifies the maximum number of rows to display per page in the SAP Filter Results table.  
**Default:** 50
- **Rows per Page for Import Jobs Dialog**  
Specifies the maximum number of rows to display per page in the Search Results table in the Import Jobs dialog.  
**Default:** 50
- **Rows per Page for Search Results in Dialog**  
Specifies the maximum number of rows to display per page in the Search Results table in search dialogs.  
**Default:** 20
- **Hide SAP Drag and Drop Panes**  
Indicates whether to hide the SAP drag and drop panes.  
**Default:** Not selected
- **Description Column Length**  
Specifies the maximum number of characters to display in the Description column of the search results in the Import Jobs dialog.

**Default:** 500

**Limits:** 1-500, or 0 to not display a description

- **Maximum Number of Imported Jobs**

Specifies the maximum number of jobs to import into Application Editor at any given time.

**Default:** 1000

**Limits:** 1-5000

- **New Job Name Default**

Specifies the default naming convention to use for new jobs dropped on to the Graph.

**Default:** {1} Job {0}

#### NOTE

- In the default expression, {0} is required; {1} is optional.
- When the job is dropped on to the Graph:
  - {0} is replaced by a unique number.
  - {1} is replaced by the SAP job name if a job is dragged from the SAP Filter Results pane.
  - {1} is replaced by the job type if a non-template SAP job is dragged from the Palette.
  - {1} is replaced by the template name if a job template is dragged from the Palette.

- **Maximum Flow Height**

Specifies the maximum height of the Graph in pixels.

**Default:** 570

**Limits:** 507-9999

- **Left Panes Width**

Specifies the width of the Palette, Delete List, and Overview panes in pixels.

**Default:** 250

**Limits:** 250-500

The System category contains the following fields:

- **Owner Field Mandatory**

Indicates whether the owner field is mandatory when creating a job. Selecting this check box requires the user to enter a value in the owner field; clearing this check box specifies that the AutoSys Workload Automation default should be used.

**Default:** Not selected

- **Graph Tooltip Length**

Specifies the maximum number of characters to be displayed for each value in the Show tooltips feature.

**Default:** 64

**Limits:** 0-256

- **Import All Button**

Indicates whether the Import buttons should be displayed in Application Editor's job import dialog. Selecting this check box displays the Import All and Import Selected buttons; clearing this check box only displays the OK and Cancel buttons.

**Default:** Selected

- **Commit Threads**

Specifies the number of threads that should be used to commit jobs to the back-end server.

**Default:** 20

**Limits:** 1-50

- **Commit Dialog Refresh Interval**

Specifies the interval (in milliseconds) at which the page refreshes after a job is committed.

**Default:** 2000

**Limits:** 2000-10000

- **Submit Owner on Commit**

Indicates whether the owner field should be submitted by default when the flow is committed.

**Default:** Selected

- **Proxy Server Name**(Optional) Specifies the name of the proxy server to use for Web Service jobs.  
**Limits:** 128 or fewer characters
- **Proxy Port**(Optional) Specifies the port assigned to the proxy server for Web Service jobs.  
**Limits:** 80-65535
- **Proxy User Name**  
(Optional) Specifies the user name to use to connect to the proxy server for Web Service jobs.  
**Limits:** 128 or fewer characters
- **Proxy Password**(Optional) Specifies the password associated with the proxy user name entered.  
**Limits:** 128 or fewer characters

## **Configuration Preferences**

The Configuration properties display when you select Configuration from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Configuration Preferences subsection to specify additional properties for the Configuration application for this Web UI instance.

The properties in the Configuration Preferences subsection are displayed by category.

The EEM category contains the following fields:

- **EEM Connection Retry Interval**Specifies the interval (in seconds) at which Web UI attempts to re-connect to the CA EEM server.  
**Default:** 1  
**Limits:** 1-100
- **Maximum EEM Connection Retries**Specifies the maximum number of times that Web UI attempts to connect to the CA EEM server.  
**Default:** 1  
**Limits:** 1-100

The AE category contains the following fields:

- **Enable SSL**Indicates whether Secure Socket Layer (SSL) is enabled for communications with the AutoSys Workload Automation application server.  
**Default:** Not selected

### **NOTE**

The value displayed here must match the SSL setting for the AutoSys Workload Automation agents that communicate with the target AutoSys Workload Automation application server.

- **Enable PMUX**Indicates whether port multiplexing (PMUX) is enabled for communications with the AutoSys Workload Automation application server.  
**Default:** Selected

### **NOTE**

The value displayed here must match the PMUX setting for the AutoSys Workload Automation agents that communicate with the target AutoSys Workload Automation application server.

The Credentials category contains the following fields:

- **Validation Thread Pool Minimum Size**  
Specifies the minimum size to use for the Credentials application validation thread pool.  
**Default:** 2  
**Limits:** 2-10
- **Validation Thread Pool Maximum Size**

Specifies the maximum size to use for the Credentials application validation thread pool.

**Default:** 4

**Limits:** 2-10

- **Validation Thread Pool Timeout**

Specifies the interval (in seconds) before the Credentials application validation thread pool times out.

**Default:** 600

**Limits:** 600-5000

- **Persist Default User**

Indicates whether to keep the default user across sessions.

**Default:** Not selected

- **Validation Refresh Interval**

Specifies the interval (in milliseconds) at which to refresh the Credentials application when a validation is in progress.

**Default:** 5000

**Limits:** 1000-60000

The Single Sign-On category contains the following fields:

- **Single Sign-On**

Indicates whether single sign-on using NTLM authentication protocol or integration with CA Siteminder is enabled.

Web UI uses NTLM protocol or CA Siteminder to authenticate your session when single sign-on is active. To disable single sign-on, select None. If you disable single sign-on, Web UI requires you to authenticate your session by entering login credentials.

**Default:** Not selected

#### NOTE

SSO parameters do not apply to ECLI and Forecast tabs

- **Force Kerberos Compatibility**

Indicates whether to force NTLM authentication when the browser is Kerberos capable. When this check box is selected and single sign-on is active, Web UI overrides Kerberos protocol and uses NTLM protocol to authenticate your session. If you disable this option, Web UI requires you to authenticate sessions that you open on Kerberos compatible browsers by entering your login credentials.

**Default:** Selected

The Security category contains the following fields:

- **Session Timeout**

Specifies the time (in minutes) that a user session can remain idle before timing out.

**Default:** 30

**Limits:** 1-1440

- **Auto Refresh Prevents Session Timeout** Indicates that Auto refresh is no longer a user action when you are idle in the application and remain logged in. **Default:** Not Selected

- **Single Session per User** Indicates whether to allow only one active session per user. The change applies only to the new session and does not affect a user that is already logged in to Web UI.

**Default:** Not selected

- **URL Parameters Encryption**

Specifies if the URL parameters are encrypted to allow protection of potentially sensitive information. **Default:** Not selected.

#### NOTE

The affected URL parameters are jobName, serverName, server, and objectName.

The Environment Category contains the following fields:

- **Default Time Zone** Indicates which time zone is used as a default for Web UI users.

**Default:** Empty Selection

The Integrations Category contains the following fields:

- **AM base URL** Specifies that the base URL where AM is running. AM base URL must include HTTP or HTTPS protocol prefix and port.

**NOTE**

For example, the AM running on `http://camhostname:8080/wla_am/ui`, the preference value should be `http://camhostname:8080/`.

### **Enterprise Command Line Preferences**

The Enterprise Command Line properties display when you select Enterprise Command Line from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Enterprise Command Line Preferences subsection to view and edit properties for the Enterprise Command Line application for this Web UI instance.

The properties in the Enterprise Command Line Preferences subsection are displayed by category.

The My Commands Settings category contains the following fields:

- **Maximum My Commands Saved**  
Specifies the maximum number of commands that can be saved in the My Commands table.  
**Default:** 100
- **Rows per Page for My Commands**  
Specifies the maximum number of rows to display per page in the My Commands table.  
**Default:** 10

The Global Commands Settings category contains the following fields:

- **Maximum Global Commands Saved**  
Specifies the maximum number of commands that can be saved in the Global Commands table.  
**Default:** 100
- **Rows per Page for Global Commands**  
Specifies the maximum number of rows to display per page in the Global Commands table.  
**Default:** 10

The General Settings category contains the following fields:

- **Maximum Number of Servers**  
Specifies the maximum number of target servers available for a command.  
**Default:** 30
- **Command Timeout**  
Specifies the maximum time (in milliseconds) allowed for a command to complete processing before it times out.  
**Default:** 86400000
- **Maximum Auto Complete Items**  
Specifies the maximum number of items to save in the auto-complete history table.  
**Default:** 25
- **Default Export File Name**  
Specifies the default name to use for the exported output file.  
**Default:** export.txt

### **Forecast Preferences**

The Forecast properties display when you select Forecast from the View/Modify Preference drop-down list in the Preference section, and click Go.

Use the Forecast Preferences subsection to view and edit properties for the Forecast application for this Web UI instance.

The properties in the Forecast Preferences subsection are displayed by category.

The User Interface category contains the following fields:

- **Rows per Page for Forecast**  
Specifies the maximum number of rows to display per page in the Forecast table.  
**Default:** 50
- **Rows per Page for Table View**  
Specifies the maximum number of rows to display per page in the Forecast Output Table View.  
**Default:** 50
- **Rows per Page for Timeline View**  
Specifies the maximum number of rows to display per page in the Forecast Output Timeline view.  
**Default:** 50

The Command Parameters category contains the following field:

- **Execution Timeout**  
Specifies the interval (in milliseconds) allowed for the forecast command to run and the report being generated.  
**Default:** 300000  
**Limits:** 60000-2147483647

## **Monitoring Preferences**

The Monitoring properties are displayed when you select Monitoring from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Monitoring Preferences subsection to view and edit properties for the Monitoring application for this Web UI instance.

The environment properties in the Monitoring Preferences subsection are displayed by category.

The UI category contains the following fields:

- **Auto Refresh Interval**  
Specifies the interval (in seconds) at which the display on the page refreshes.  
**Default:** 15  
**Limits:** 15-2147483647
- **Views Table Columns**  
Specifies the names of the columns to be displayed in the Views table. Multiple names can be entered, separated by commas. Valid values are name, job\_status, server\_status, alert\_severity\_status, jobs, collections, filters, inactive\_filters, unbound\_filters, status\_time, last\_edit\_time, last\_edit\_user, description, and servers.  
**Default:** name,job\_status,server\_status,alert\_severity\_status,jobs,collections,status\_time
- **Servers Table Columns**  
Specifies the names of the columns to be displayed in the Servers table. Multiple names can be entered, separated by commas. Valid values are name, status, status\_time, status\_detail, filters, version, and type.  
**Default:** name,status,status\_time,type,filters
- **Jobs Table Columns**  
Specifies the names of the columns to be displayed in the Jobs table. Multiple names can be entered, separated by commas. Valid values are agent\_status, agent\_lstatus, application, box\_name, description, duration, end\_time, exit\_code, group, instance, job\_name, job\_type, joid, machine, next\_start, owner, run\_machine, server, start\_time, status, status\_time, and timezone.  
**Default:** job\_name,job\_type,status,server,run\_machine,status\_time,duration
- **Alerts Table Columns**  
Specifies the names of the columns to be displayed in the Alerts table. Multiple names can be entered, separated by commas. Valid values are alert, severity, status, job, server, create\_time, status\_time, response, comment, text, eioid, modified\_user, alert\_policy, url, and alert\_type.



**Default:** alert,severity,status,job,server,create\_time,status\_time

- **Scheduler Log Default Lines**

Specifies the number of lines to retrieve for the Scheduler Log.

**Default:** 50

**Limits:** 1-99999

- **Scheduler Log Auto Refresh Interval**

Specifies the interval (in seconds) at which the display on the page refreshes.

**Default:** 15

**Limits:** 15-2147483647

The Job Level Security category contains the following field:

- **Use AutoSys Workload Automation Security**

Indicates to use the AutoSys Workload Automation as-job, as-appl, and as-group policies in CA EEM to control access to jobs.

**Default:** Selected

**NOTE**

If changes are made in the as-job, as-appl, and as-group policies after the Monitoring collector has collected the data from the AutoSys Workload Automation server, changes to existing jobs contained in the views and monitoring database are not reflected.

The Back-end Services category contains the following fields:

- **Collector Mode**

Specifies whether to collect data for all defined servers or only the servers that are part of the active view filters.

**Default:** all

**Limits:** all or filter

- **Job Collector Run Interval**

Specifies the interval (in seconds) at which to run the collectors (get data from servers) and synchronize server changes from configuration.

**Default:** 5

**Limits:** 5-2147483647

- **Alarm Collector Run Interval**

Specifies the interval (in seconds) at which to run the alarm collector.

**Default:** 5

**Limits:** 5-2147483647

- **Alerts Deletion Threshold**

Specifies the number of days to store alerts in the database before deleting them.

**Default:** 7

**Limits:** 1-365

- **Alert Cleaner Run Interval**

Specifies the interval (in hours) in which to run database cleanup activity.

**Default:** 24

**Limits:** 1-2147483647

- **Security Cache Timeout**

Specifies the time (in minutes) after which the security cache is invalidated.

**Default:** 60

**Limits:** 0,1-2147483647

**NOTE**

When any CA EEM policy that affects Monitoring is changed, by default, it takes 60 minutes to be reflected in Web UI. The default timeout value reduces the overhead of checking CA EEM for every request. However,

you can set the timeout to a lower value to reflect CA EEM security changes in Monitoring more quickly. Setting the value to 0 allows security changes to be reflected immediately.

- **DB Statistics Update Threshold**

(Oracle only) Specifies the number of records to be updated in the Oracle database before the DB statistics updater starts.

**Default:** 5000

**Limits:** 1-1000000, or 0 to disable the updater.

The Flow category contains the following fields:

- **Flow Threshold Per Flow**

Specifies the maximum number of nodes that can be displayed in a single flow. If the number of nodes in the flow is greater than this value, the flow is not rendered.

**Default:** 5000

**Limits:** 1-20000

- **Flow Threshold Collapsed**

Specifies the maximum number of nodes that can be displayed by default in an expanded flow. If the number of nodes is greater than this value, the flow is initially collapsed.

**Default:** 2000

**Limits:** 1-10000

- **Flow Expand Box Jobs**

Indicates whether all box jobs in the flow are expanded. Selecting this check box specifies that all box jobs are expanded when the flow is initially rendered. Clearing this check box specifies that all box jobs are collapsed when the flow is initially rendered.

**Default:** Selected

- **Maximum Flow Height**

Specifies the maximum height of the flow in pixels.

**Default:** 500

**Limits:** 500-3000

The Server Status Order category contains the fields that specify the severity value that should be assigned to this server status. This value is used to prioritize the overall server status for the view. The lowest number assigned indicates the highest severity. The following fields are available:

- **Failure**

Specifies the Failure severity value

**Default:** 0

**Limits:** 0-99

- **Disconnected**

Specifies the Disconnected severity value.

**Default:** 1

**Limits:** 0-99

- **Unknown**

Specifies the Unknown severity value.

**Default:** 2

**Limits:** 0-99

- **Not Configured**

Specifies the Not Configured severity value.

**Default:** 3

**Limits:** 0-99

- **Connected**

Specifies the Connected severity value.

**Default:** 50

**Limits:** 0-99

The Job Status Order category contains the fields that define the severity value that should be assigned to this job status. This value is used to prioritize the overall job status for the view. The lowest number assigned indicates the highest severity. If the priority is changed, the view status is updated on the next job run. The following fields are available:

- **Failure**  
Defines the Failure severity value.  
**Default:** 111  
**Limits:** 80-127
- **Waiting on Reply**  
Defines the Waiting on Reply severity value.  
**Default:** 112  
**Limits:** 80-127
- **Pending Machine**  
Defines the Pending Machine severity value.  
**Default:** 113  
**Limits:** 80-127
- **Waiting on Resources**  
Defines the Waiting on Resources severity value.  
**Default:** 114  
**Limits:** 80-127
- **Queue Wait**  
Defines the Queue Wait severity value.  
**Default:** 115  
**Limits:** 80-127
- **Restart**  
Defines the Restart severity value.  
**Default:** 116  
**Limits:** 80-127
- **Terminated**  
Defines the Terminated severity value.  
**Default:** 117  
**Limits:** 80-127
- **On Ice**  
Defines the On Ice severity value.  
**Default:** 118  
**Limits:** 80-127
- **On Hold**  
Defines the On Hold severity value.  
**Default:** 119  
**Limits:** 80-127
- **Unknown**  
Defines the Unknown severity value.  
**Default:** 120  
**Limits:** 80-127
- **Success**  
Defines the Success severity value.  
**Default:** 121  
**Limits:** 80-127
- **Running**  
Defines the Running severity value.  
**Default:** 122

**Limits:** 80-127

- **Starting**

Defines the Starting severity value.

**Default:** 123

**Limits:** 80-127

- **Activated**

Defines the Activated severity value.

**Default:** 124

**Limits:** 80-127

- **Inactive**

Defines the Inactive severity value.

**Default:** 125

**Limits:** 80-127

- **On No Execution**

Defines the On No Execution severity value.

**Default:** 126

**Limits:** 80-127

### **Quick Edit Preferences**

The Quick Edit properties display when you select Quick Edit from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Quick Edit Preferences subsection to specify properties for the Quick Edit application for this Web UI instance.

The properties in the Quick Edit Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for Search Results**

Specifies the maximum number of results per page to display in the Search Results section.

**Default:** 50

- **Rows per Page for Search Results in Dialog**

Specifies the maximum number of rows per page to display in the Search Results table in search dialogs.

**Default:** 20

- **Search History Items**

Specifies the maximum number of entries to display in the auto-complete history for the Name field in the Search section.

**Default:** 30

**Limits:** 1-100

- **Description Column Length**

Specifies the maximum number of characters to display for the Description field in the Search Results section.

**Default:** 500

**Limits:** 0-500

- **Single Result Redirect**

Indicates whether to automatically display the search result in the Properties section if the search returns a single item. Selecting this check box specifies that a single result displays directly in the Properties section instead of in the Search Results section; clearing this check box specifies that the result is displayed in the Search Results section.

**Default:** Selected

- **Display Calendar with 24 Months**

Indicates whether to display a 24-month calendar view instead of a 12-month calendar view. Selecting this check box specifies that the calendar view displays 24 months; clearing this check box specifies that the calendar view displays 12 months.

**Default:** Not selected

The System category contains the following fields:

- **Owner Field Mandatory** Indicates whether the owner field is mandatory when creating a job. Selecting this check box requires the user to enter a value in the owner field; clearing this check box specifies that the AutoSys Workload Automation default should be used.  
**Default:** Not selected
- **Copy Job Owner**  
Indicates whether to copy the owner field when a job is copied. Selecting this check box retains the same owner name as the original job. If this field is not selected and the Owner Field Mandatory option is selected, the user must enter a value in the owner field; if this field is not selected and the Owner Field Mandatory option is cleared, the default user specified by the AutoSys Workload Automation instance is used.  
**Default:** Selected
- **User Defined Job Description Maximum Length**  
Specifies the maximum number of characters to display in the Description field for a User Defined job.  
**Default:** 50  
**Limits:** 1-256
- **User Defined Job Command Maximum Length**  
Specifies the maximum number of characters to display in the Command field for a User Defined job.  
**Default:** 50  
**Limits:** 1-510
- **Proxy Server Name**  
(Optional) Specifies the name of the proxy server to use for Web Service jobs.  
**Limits:** 128 or fewer characters
- **Proxy Port**(Optional) Specifies the port assigned to the proxy server for Web Service jobs.  
**Limits:** 80-65535
- **Proxy User Name**  
(Optional) Specifies the user name to use to connect to the proxy server for Web Service jobs.  
**Limits:** 128 or fewer characters
- **Proxy Password**(Optional) Specifies the password associated with the proxy user name entered.  
**Limits:** 128 or fewer characters

### **Quick View Preferences**

The Quick View properties display when you select Quick View from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Quick View Preferences subsection to view and edit properties for the Quick View application for this Web UI instance.

The properties in the Quick View Preferences subsection are displayed by category.

The User Defaults category contains the following fields:

- **Rows per Page for Search Results**  
Specifies the number of rows to display per page in the Search Results table.  
**Default:** 50
- **Rows per Page for Reports**  
Specifies the number of rows to display per page in a report table.  
**Default:** 50
- **Search History Items**  
Specifies the maximum number of entries to display in the auto-complete history for the Name field in the Search section.  
**Default:** 30

**Limits:** 1-100

- **Job Details View**

Specifies whether the job details should be displayed as a tree or as JIL in the Job Details section.

**Default:** Tree

- **Maximum Flow Height**

Specifies the maximum height of the Job Flow section in pixels.

**Default:** 400

**Limits:** 150-1500

- **Show Dependent Jobs**

Indicates whether to display the Dependent Jobs report section.

**Default:** Selected

- **Show Starting Conditions**

Indicates whether to display the Starting Conditions report section, which includes the Job Dependencies and Job Resource Dependencies subsections.

**Default:** Selected

- **Show Job Runs**

Indicates whether to display the Job Runs report section, which includes job run events and logs.

**Default:** Selected

- **Show Jobs from AE instead of Web UI collector DB** Displays jobs from AE even when they are not in the Web UI collector database.

**Default:** Clear

#### **NOTE**

Restart the application to apply any changes to the checkbox.

## **Reporting Preferences**

The Reporting properties are displayed when you select Reporting from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Reporting Preferences subsection to view and edit properties for the Reporting application for this Web UI instance.

The properties in the Reporting Preferences subsection are displayed by category.

The Collectors category contains the following fields:

- **Enable Collector**

Indicates to automatically run the collectors for all servers.

**Default:** Selected

- **Collector Run Interval**

Specifies the time interval between collector runs, in seconds.

**Default:** 60

The Database Cleanup category contains the following fields:

- **Enable Cleanup**

Indicates to automatically perform a periodic database cleanup.

**Default:** Selected

- **Cleanup Interval**

Specifies the number of days between database cleanup activity.

**Default:** 2

- **Purge Data**

Specifies the number of days that the data can exist in the database before being removed during database cleanup.

**Default:** 30

## Resources Preferences

The Resources properties display when you select Resources from the View/Modify Preferences drop-down list in the Preferences section, and click Go.

Use the Resources Preferences subsection to view and edit properties for the Resources application for this Web UI instance.

The properties in the Resources Preferences subsection are displayed by category.

The Table Controls category contains the following fields:

- **Rows per Page for All Tables**  
Specifies the maximum number of rows to display per page in the section tables.  
**Default:** 50
- **Rows per Page for All Dialog Tables**  
Specifies the number of rows to display per page in the dialog tables.  
**Default:** 20

The Search History Controls category contains the following field:

- **Search History Items**  
Specifies the maximum number of entries to display in the auto-complete history for the Name, Amount, and Description fields in the Search section, the Name and Amount fields in the Properties section, and the Name field in Resource Usage Search section.  
**Default:** 30  
**Limits:** 1-100

## Web UI Credentials Application

The Credentials application, accessed by clicking the Credentials tab from the Web UI Dashboard, lets you manage login credentials. Users can add, modify, and delete their credentials for back-end servers in real time by selecting the appropriate server and operation. They can also validate the status of the credentials displayed.

CA EEM policies control the ability of each user to add or modify credentials. These policies can be used to allow or deny users from creating a specific user ID, credentials for a specific server, and so on. The Credentials application lists all the servers the logged-in user has access to, based on the active ServerAccess policies.

You can configure the following types of credentials in the Credentials application, depending on your user rights:

- **User credentials**  
Valid for a specific server and only the user who configures these user credentials
- **Global credentials**  
Valid for a specific server and all users; set by administrators
- **Default credentials**  
Valid for all servers and only for the user who configures these user credentials

Web UI captures the credentials of the user during login to use as the default credentials. These default credentials are passed directly to the back-end servers, unless alternative credentials are supplied in the Credentials application. To supply alternative credentials, the user can either modify the default credentials for the current session or supply server-specific credentials. If users add server-specific credentials, these credentials override the default credential settings. Users can add multiple user names and passwords for each back-end server, although only one user can be active for each server. That active user for each server is also set with this application.

If you have configured more than one credential type for a server, the credentials with the highest priority are used. The credential type priority is as follows:

1. User credentials (if set active)

2. Global credentials
3. Default credentials
4. Web UI user name and password

### **Credentials Application**

Consider the following list before using the Credentials application:

- The credentials defined here are used to connect to AutoSys Workload Automation servers that do not share a common CA EEM instance with Web UI.
- The credentials specified here are used only for Enterprise Command Line and Forecast if the following two criteria are met:
  - The AutoSys Workload Automation server uses the same CA EEM instance as Web UI
  - EEM Enabled is selected when the server is defined in Web UI Configuration

#### **NOTE**

(CA AE 11.3.6 and later) The credentials specified are used only to access CA AE server, however the commands ran from Enterprise Command Line and Forecast are performed under the context of the user logged in Web UI. Thus, you need not define all Web UI user accounts on CA AE server, only one user account could be defined on CA AE server and added as a global credential; CA AE commands are still performed under Web UI logged in user.

- The level of access for a credential set is determined by the user rights configured for the corresponding user ID in the AutoSys Workload Automation server operating system.
- The CA EEM ServerAccess policy determines which AutoSys Workload Automation servers are visible in Web UI.
- Your administrator must enable Persist Default User in Web UI Configuration, to save the default credentials between sessions.

## **Manage Credentials**

This article explains how to manage Credentials.

### **Add New Credentials for a Server**

To set up credentials for a new user on a back-end server, you can enter the credentials directly into the Credentials application.

#### **NOTE**

You can add multiple credentials to a server.

#### **Follow these steps:**

1. Select the option button next to the appropriate server.
2. Enter a new user ID in the User Name field.
3. Enter the password for that user ID in the Password and Confirm Password fields.
4. Click Add.

The new credentials for the back-end server are saved.

### **Edit Credentials for a Server**

To change the user password on a back-end server, you can edit the user credentials directly in the Credentials application.



**NOTE**

- You can change only the password for a user ID you have added; you cannot change the user ID.
- To change a user ID, you must add a new credential user ID. You can then either delete the existing ID, or set the new user ID as the active credential.

**Follow these steps:**

1. Select the option button next to the appropriate server.
2. Select the appropriate user ID from the User Name drop-down list.
3. Modify the user password in both the Password and Confirm Password fields, as necessary.
4. Click Update.  
The updated credentials are passed to the back-end server.

**Delete Credentials for a Server**

You can use the Credentials application to delete the user credentials from a back-end server when they are no longer required.

**NOTE**

The Default credentials cannot be deleted.

**Follow these steps:**

1. Select the option button next to the appropriate server.
2. Select the appropriate user ID from the User Name drop-down list.
3. Click Delete.  
The credentials are deleted from the server.

**Add Global Credentials for a Server**

To set up global credentials for a back-end server, you can enter the credentials directly into the Credentials application.

**Follow these steps:**

1. Select the option button next to the appropriate server.
2. Enter the following statement in the User Name field:  
`_GLOBAL_:CA_WA_AE_instance_user_id`
3. Enter the password for that user ID in the Password and Confirm Password fields.
4. Click Add.  
The global credentials for the back-end server are saved. All Web UI users can now automatically access this particular server at whatever level of access is set up for these credentials on that server.

**Example: Add Global Credentials for a AutoSys Workload Automation Server**

Suppose you are the administrator of a small environment with a few AutoSys Workload Automation servers that do not use CA EEM security. For ease of administration, you want to have a user ID and password that works for all users accessing these servers. You can define a generic back-end user ID and password, such as AEuser/AEpw, that enables basic access to any of the AutoSys Workload Automation servers in use at your site. You must add these same global credentials for each server.

**Add Default User Credentials**

To set up default user credentials for all back-end servers, you can enter the credentials directly into the Credentials application.

**NOTE**

- As a prerequisite, the designated user ID and password must exist on every AutoSys Workload Automation server you want to use the default user credentials.
- Your administrator must have enabled Persist Default User in the Configuration preferences section of Configuration.
- If user-specific or global credentials are already configured for one or more back-end servers, they take precedence and those credentials are used for those servers.

**Follow these steps:**

1. Select the option button next to Default.
2. Enter the appropriate user name in the User Name field.
3. Enter the appropriate password in the Password and Confirm Password fields.
4. Click Update.

The default user credentials are saved. You can now access all back-end servers to which you have server access rights with the indicated user name and password.

**Example: Simplify Login to AutoSys Workload Automation Servers in Your Environment**

Suppose you use Web UI in an environment with many AutoSys Workload Automation servers, all of which are configured to use the same credentials. You currently must enter credentials for each server separately, because the required credentials are different from your Web UI login credentials. To log in to any AutoSys Workload Automation server in your environment automatically, you can add default user credentials to your Web UI user account. You can then use these credentials to log in to any AutoSys Workload Automation server you connect to, including servers added later. Also, when you have to edit the credentials, for example, to change the password, you do not have to edit the settings for each server.

**Modify the Default User Credentials**

You can modify the user name, password, or both for the default user.

**NOTE**

This option is not recommended as a best practice.

**Follow these steps:**

1. Select the option button next to Default.
2. Select the default user ID from the User Name drop-down list.
3. Modify the user name, as necessary.
4. Modify the user password in both the Password and Confirm Password fields, as necessary.
5. Click Update.

The updated credentials are passed to the back-end server.

**Set the Active User for a Server**

You can specify the active user for a back-end server, which indicates the credentials currently in use for that back-end server, in the Credentials application.

**NOTE**

Only one set of credentials can be active at a time. If no credentials have been configured, your Web UI user name and password are used for authentication.

**Follow these steps:**

1. Select the option button next to the appropriate server.
2. Select the appropriate user ID from the User Name drop-down list.

**NOTE**

If the User Name drop-down list contains more than one name, the current active user name is preceded by a tilde (~).

3. Click the Set Active User link.  
The active user is set.

**Example: Change the Credentials You Use to Connect to a AutoSys Workload Automation Server**

Suppose that, as a scheduler, you must be able to create, edit, and delete all job types, including Oracle E-Business Suite application jobs on the AutoSys Workload Automation server AS1. However, for security reasons, the credentials you typically use are not configured on the operating system of the Oracle database server. In this case, you must use two user IDs with server AS1:

- **AS1\_Scheduler**  
AS1\_Scheduler is the user ID that lets you create, edit, and delete most job types on server AS1. However, if you create an Oracle E-Business Suite application job using this user ID, the job owner ID is not recognized by the Oracle database server. This causes the job to fail when run.
- **AS1\_Ora\_Scheduler**  
AS1\_Ora\_Scheduler is the user ID configured on the Oracle database server operating system. Oracle E-Business Suite application jobs that you create with this user ID can be run.

**Validate Credentials**

You can view the status of the credentials that you have set up for your back-end servers, for active users only, using the Credentials application.

To validate the credentials against the back-end server, click Validate.

The Status column in the Credentials application displays the status of the credentials as follows:

- **Valid**  
Indicates that the credentials for active users have been validated against the back-end server and are valid for that instance.
- **Invalid**  
Indicates that the credentials for active users have been validated against the back-end server, and no matching credentials are available on that instance.
- **Not Validated**  
Indicates that the credentials for active users have not been validated on the back-end server.
- **Connection Error**  
Indicates that there is a connection problem with the back-end server.

**NOTE**

You must refresh the view to see the updated statuses, if any.

**Verify Validation of a Credential User**

If the validation of a Credential User fails under the Credentials Tab, you can see the following message in the \$CA\_WCC\_INSTALL\_LOCATION/log/application/config/configui.log file.

```
ERROR #VerificationManager # Exception caught trying to login to AutoSys server (hostname): E180011 Login failed
using: User=username Password=<hidden>
```

You can verify that the provided credentials are valid on the WA Agent server.

**Follow these steps:**

On your WA Agent server:

1. Navigate to the `/opt/CA/WorkloadAutomationAE/SystemAgent/WA_AGENT` directory.
2. Run `./password <cred user password>`.
3. Run `./chkusr <cred username> <encrypted password>`.  
"User and Password okay" is displayed.

#### NOTE

If the verification fails, then the issue could be that the username does not exist on AE or the password is invalid.

### **Disable Credentials for a Server**

You can use the Credentials application to disable all credentials for a back-end server. When you disable the credentials for a server, the default credentials (your login ID) are used instead. You can disable the credentials for a server if you have set up multiple credentials and instead want to use your default login ID.

#### NOTE

Follow this same procedure to re-enable credentials you have previously disabled.

#### **Follow these steps:**

1. Select the option button next to the appropriate server.
2. Select the appropriate user ID from the User Name drop-down list.
3. Click Enable/Disable Users.  
The credentials are disabled and the status for the server changes to Users Disabled.

## **Web UI Dashboard**

This article describes the widgets, links, and tabs available from the main Web UI page, Dashboard.

### **Dashboard Widgets**

The Dashboard contains one or more of the following widgets that may be useful to schedulers, console operators, and executives in your enterprise:

- **Alerts**  
Displays all server-related alerts and alarms. From this widget, you can initiate an action on one or more alerts.
- **Connection Status**  
Displays the status of the connections to Web UI databases and the Cluster Collector Node.
- **Servers**  
Displays all the servers configured in Web UI to which you have access. From this widget, you can monitor the status of the servers. You can also issue send event commands to AutoSys Workload Automation servers.
- **Views**  
Displays information about all the views defined in Web UI to which you have access. By default, all available views are displayed. When you double-click the link for a view name, the Monitoring page opens, displaying the jobs and alerts associated with the selected view.
- **Jobs**  
Displays information about all the jobs, contained in the specified view or server, to which you have access. You have the option of selecting a view-based or server-based list of jobs.

#### NOTE

Multiple jobs widgets can be added to a single dashboard.

- **Active Users**

Displays a list of the users and active user sessions that are currently connected to the Web UI server.

- **Machines**

Displays a list of the machines that are currently defined on the CA AE instance connected to Web UI server.

The availability of each widget is role-based and is determined by your administrator using the CA EEM WidgetAccess security policy. Clicking



on the Dashboard tab opens a menu from which you can select the widgets that you want to display on the Dashboard (from those available).

**NOTE**

- The auto-refresh check box in the monitoring views widget remains persistent between user sessions.
- The visibility of the widget filter is persistent between user sessions.

**WARNING**

By default, the Jobs and Active User widgets are not displayed on the Dashboard, so you must select Jobs or Active User from this menu to add it.

## **Dashboard Links**

The Dashboard contains the following links:

- **My Profile**

Opens the My Profile dialog, which lets you select the time zone and the format in which Web UI displays date and time. In this dialog, you can also change your password.

**NOTE**

The selections made in the My Profile dialog affect only the date and time displayed in Web UI. They do not affect the formatting requirements of fields in Web UI where you enter date and time. When you enter date and time values in Web UI, you should use the short format of the selected format locale. Only SAP jobs require a different date and time format.

- **EEM**

Opens the CA EEM application, accessible to authorized users only. Security administrators can use the EEM component to access and configure the CA EEM security subsystem. For more information, see the CA Embedded Entitlements Manager documentation.

**NOTE**

When the CA EEM login dialog opens, select the appropriate Web UI application from the Application drop-down list and enter valid credentials for a security administrator.

- **Help**

Opens an application-specific help system, depending on which tab is selected.

- **About**

Opens the About dialog, which displays Web UI release and build information. The About dialog also provides access to environmental information about the system. The system information available is primarily of use for troubleshooting with CA Support.

## **Application Tabs**

Web UI applications are accessed using tabs in the user interface. When you move from one application to another, the displayed information is retained and redisplayed when you return.

In addition to the Dashboard, the following applications are available:

Application	Description
Agent Inventory	Lets you see and export information about each separately installable component (system agent, plug-ins, and advanced integrations).
Application Editor	Provides a graphical representation of all the relationships between AutoSys Workload Automation jobs and their dependencies. It displays the conditions of the relationships that must be met before a job can run, and the successor jobs triggered when a job finishes running.
Configuration	Lets you perform all system setup for Web UI. For example, you can configure Quick Edit, Application Editor, and Monitoring run-time parameters, and connectivity to back end servers.
Credentials	Provides a summary of the servers to which the logged-in user has access. Credential also lets you add or delete credentials for a selected server, update the password for a selected user and server, and validate credentials for the active users against a back-end server.
Enterprise Command Line	Lets you run a command-line interface command such as jil or autorep on the selected AutoSys Workload Automation server. Enterprise Command Line also lets you save commands, modify and run saved commands, and import and export AutoSys Workload Automation jobs and calendars.
Forecast	Lets you create, modify, and delete report definitions that specify the time period of the forecast window. Forecast report output includes jobs that are forecasted to start during the forecast window period. You can view forecast report output in table and Gantt chart formats.
Monitoring	Lets you add, edit, delete, export, and import views, monitor jobs and box jobs within the views, and view any associated alerts. You can view the jobs and box jobs by server (workload automation scheduling manager instances) or by user-defined views.
Quick Edit	Lets you manage all the properties of your AutoSys Workload Automation workload objects, such as jobs and calendars, from a single page view.
Quick View	Provides detailed AutoSys Workload Automation job information in a single page view. Quick View lets you view a job definition, access Quick Edit, view a job flow, send an event to one or more jobs, view a prior job run report, and view logs. Quick View can be used as an alternative to searching through a command line report for the job status information that you want.
Reporting	Lets you run predefined reports for multiple servers. These reports display server statistical information and job run information in a table and graphical view.
Resources	Lets you add, edit, or delete virtual resources, monitor resources, and view resource dependencies for a job or collection of jobs in a graphical view. You can specify box job or job predecessors that must complete successfully before a box job or job can run. Similarly, you can specify virtual and real resource requirements that must be met before a box job or job can run.
+	Lets you add, edit, or delete external applications in additional tabs.

**NOTE**

- Application availability is role-based and is determined by the CA EEM ApplicationAccess security policy.
- External application availability is role-based and is determined by the CA EEM ExternalApplicationAccess security policy.

## Manage Your Dashboard

This article explains how to complete general tasks to manage your dashboard.

### Change Your Password

After you have logged in to Web UI, you can change your current password.

**NOTE**

If the CA EEM Server is configured to use an external authentication directory, the option to change your password is not available.

**Follow these steps:**

1. Click My Profile in the main user interface.  
The My Profile dialog opens.
2. Click Change Password.  
The Change Password dialog opens.
3. Enter your current password in the Current Password field, your new password in the New Password and Confirm New Password fields, and click OK.  
A confirmation message that your password has been changed appears.

### Change the Date and Time Format

You can change the time zone and the format in which the date and time are displayed in Web UI.

**NOTE**

The initial time zone and date and time format are the same as those used in your browser.

**Follow these steps:**

1. Click My Profile in the main user interface.  
The My Profile dialog opens.
2. Specify the locale, time zone, date and time format, and whether to use 24-hour format, and click OK.  
The dialog closes and your settings are saved.

**NOTE**

For your changes to take effect throughout Web UI, you must log out and then log back in to Web UI.

### Print the System Information Dialog

The System Information dialog that you can access by clicking the System Information button in the About dialog can be printed. You can print the information in the dialog to get a hard copy record of the current configuration.

To print the System Information dialog, click Print.

The Print dialog opens so you can select a printer, modify its settings, and print the information.

**NOTE**

You may need to select the Print background colors and images option in your internet browser for the page to print correctly.

## Add a Widget

You can add a widget to the Dashboard page. A widget displays an overview of information from an application.

### NOTE

- The option to add and remove widgets is determined by the CA EEM ConfigurationControl policy.
- Widget availability is role-based and is determined by the CA EEM WidgetAccess policy. The default setting for this policy is that all configured widgets are displayed on the Dashboard tab for all users, and that all users can remove and add widgets.

To add a widget,

click 

the Dashboard tab and select the widget you want from the list of available widgets.

### NOTE

If you have permission to view the Jobs widget, but do not have permission to add a widget, the Jobs widget is not displayed unless it was previously added to the Dashboard.

The widget is added to the Dashboard page.

## Detach a Widget from the Dashboard

You can detach a widget from the Dashboard page and open it in a stand-alone browser to monitor the widget while continuing to use other applications in Web UI.

### Follow these steps:

1. Click  the title bar of the widget you want to detach. The open widget options are displayed.
2. Click  The selected widget is opened in a stand-alone browser.

## Remove a Widget

You can remove a widget from the Dashboard tab that you no longer need.

### NOTE

- The option to add and remove widgets is determined by the CA EEM ConfigurationControl policy.
- Widget availability is role-based and is determined by the CA EEM WidgetAccess policy. The default setting for this policy is that all configured widgets are displayed on the Dashboard tab for all users, and that all users can remove and add widgets.

To remove a widget,

click 

the title bar of the widget.

The widget is removed from the Dashboard tab.

## Add a Dashboard Tab

You can add a customizable Dashboard tab to the tab panel.

### Follow these steps:

1. Click the "+" tab on the tab panel to add a new application and create a Dashboard tab.



The new Dashboard tab is created.

2. Click  the Dashboard tab to customize your dashboard by selecting Rename, Remove, Set as Primary, or Add a Widget. The Dashboard Tab is customized.

### **Change a Dashboard Tab Title**

You can change the displayed Dashboard tab title in the tab panel.

Click  the Dashboard tab to select Rename to title your dashboard.

### **Remove a Dashboard Tab**

You can remove Dashboard tab that you no longer need. To remove a Dashboard tab, click  the Dashboard tab and select Remove Dashboard.

A confirmation window appears to complete or reject the Dashboard tab removal.

### **Set a Dashboard Tab as Primary**

You can select a particular dashboard as primary over other various created dashboards. You can also designate the newly created dashboard as your default dashboard.

Click  the Dashboard tab to customize your dashboard and select Set as Primary. The newly created Dashboard tab takes priority in tab order along the tab panel.

## **Manage External Applications**

This article explains how to manage external applications. In addition to the Web UI applications, you can display external applications in the Web UI user interface. Those applications are accessible by tabs that you create.

### **NOTE**

Your security administrator controls the availability of the + tab and the option to add, edit and delete external applications.

- External application availability is role-based and is determined by the CA EEM ExternalApplicationAccess policy. The default setting for this policy is that all users have access to all configured external applications.
- If Web UI is running in SSL mode, a warning dialog appears when users try to access a non-secure external application. Clicking the appropriate button in the dialog allows the non-secure content to be displayed.

### **Follow these steps:**

1. Click the + tab and select Add External Application.  
The Add External Application dialog opens.
2. Specify the following:
  - Tab Name
  - URL
  - EEM Resource ID
3. (Optional) Specify the following:
  - Help Page URL
  - Tab Tooltip

4. (Optional) Do one or both of the following, as appropriate:
  - Select the Propagate Username as checkbox and specify the external application parameter.
  - Select the Propagate Artifact as checkbox and specify the external application parameter.
 Single sign-on is enabled for external applications.
5. Click OK.
  - A tab containing the external application is created.

### **Configuration Changes in Web UI for External Applications**

To configure external applications for use in Web UI you must modify the

\$CA\_WCC\_INSTALL\_LOCATION/tomcat/conf/server.xml file as follows:

#### **From**

```
<Valve className="com.ca.wcc.catalina.ContentSecurityPolicyValve" enable="true" value="default-src 'self';
img-src 'self' data:; script-src 'self' 'unsafe-inline' 'unsafe-eval'; style-src 'self' 'unsafe-inline'
https://fonts.googleapis.com; font-src 'self' https://fonts.gstatic.com; frame-src 'self' blob;"/>
```

#### **To**

```
<Valve className="com.ca.wcc.catalina.ContentSecurityPolicyValve" enable="true" value="default-src 'self';
img-src 'self' data:; script-src 'self' 'unsafe-inline' 'unsafe-eval'; style-src 'self' 'unsafe-inline'
https://fonts.googleapis.com; font-src 'self' https://fonts.gstatic.com; frame-src 'self' url1 url2 blob;"/>
```

#### **Example:**

The following example shows how you can configure an external application, using <https://tomcat.apache.org>

```
<Valve className="com.ca.wcc.catalina.ContentSecurityPolicyValve" enable="true" value="default-src 'self';
img-src 'self' data:; script-src 'self' 'unsafe-inline' 'unsafe-eval'; style-src 'self' 'unsafe-inline'
https://fonts.googleapis.com; font-src 'self' https://fonts.gstatic.com; frame-src 'self' https://
tomcat.apache.org blob;"/>
```

### **Configuration Changes in External Applications**

Web UI uses the <iframe> tag to display an external-application content in a tab. For an external web-application to be displayed in a Web UI tab, the external-application must be iframe enabled or aware.

An external-application may not be displayed when it is enabled for "X-Frame-Options" HTTP response header, to avoid the clickjacking security risk. For more information about X-Frame-Options" HTTP response header, see <https://developer.mozilla.org>

You can try the following options to ensure that external-applications are displayed in Web UI tabs:

- Some browsers have plugins available which can ignore the "X-Frame-Options" so that the external-application can be displayed in Web UI.

#### **NOTE**

Ignoring the X-Frame-Options header can be a security risk, so use it with caution.

- For Chrome browser, download the "Ignore X-Frame headers" plugin, and verify Web UI for the external-application.
- For Firefox browser, download the "Ignore X-Frame-Options" plugin, and verify Web UI for the external-application.
- When Web UI and the external-applications are deployed on the same machine and the "X-Frame-Options: SAMEORIGIN" is used in the external-application web.xml, then this issue may not arise. For more information, see the 'Browser Compatibility' section at <https://developer.mozilla.org>.
- When Web UI and the external-applications are deployed on different machines (general scenario), you can update the external-application web.xml to use "X-Frame-Options: ALLOW-FROM [https://wcchost\[:<port>\]](https://wcchost[:<port>])". This enables the external-application to be displayed from "wcchost". For more information, see the 'Browser Compatibility' section at <https://developer.mozilla.org>.

**NOTE**

To display CA iDash application as an external-application in Web UI, see the iDash documentation on how to enable iDash for iframes.

## Manage the Active User Widget

This article explains how to manage the Active User widget.

### Customize the Active Users Table

You can customize the columns that display in the Active Users table by selecting from the following options:

- User Name
- HA Node
- On Since
- Duration

#### **Follow these steps:**

1. Click  any column in the Active Users table.
2. Select  from the drop-down list.
3. Either select the checkboxes for the columns you want to display, clear the checkboxes for the columns you do not want to display, or do both.
4. Drag the columns to the left or right to order the columns in the table.
5. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### Find an Active User

You can use the Active User widget to monitor the Web UI users that are currently connected to the Web UI nodes in the HA cluster, and to view a list of active user sessions.

#### **Follow these steps:**

1. Click  to display the filter.
2. Either enter the user name in the User Name field, enter the name of the computer in the HA Node field, or do both.
3. Click Go.

**NOTE**

The users that match your criteria are listed in the Active Users table.

## Manage the Alerts Widget

This article explains how to manage the Alerts widget.

## **Customize the Alerts Table**

You can customize the columns that display in the Alerts table by selecting from the following options:

- Alert
- Type
- Job
- Status
- Alarm
- Severity
- Server
- Alert Policy
- Response
- Create Time
- Status Time
- Comment
- Text
- Eoid
- User
- URL

### **NOTE**

- You can customize the Alerts table in the Dashboard and Monitoring tabs.
- If you open the Alerts table from the Views table, only the columns specified in the view definition are available for selection.
- If you open the Alerts table from the Servers table, only the columns specified in the Alerts Table Columns property field in Configuration are available for selection.
- Also, you can select the columns displayed in the Alerts table and order them using the Alert Columns selection pane in the View Editor.

### **Follow these steps:**

1. Open the Alerts table in the right pane.
2. Click  any column in the Alerts table.
3. Select  the drop-down list.
4. Either select the checkboxes for the columns you want to display, clear the checkboxes for the columns you do not want to display, or do both.
5. Drag the columns to the left or right to order the columns in the table.
6. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

## **Find an Alert**

You can use the filter section in the Alerts widget to find one or more specific alerts.

### **Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Perform one or more of the following actions:

- Select an Alarm from the Alarm drop-down list.
- Enter the name of the server in the Servers field or select the server from the Servers drop-down list.
- Enter the name of one or more jobs in the Jobs field.

**NOTE**

You can enter any of the following options in the Servers and Jobs fields:

- Select one or more severity from the Severity drop-down list.
- Select one or more alarms from the Alarms drop-down list.
- Select one or more statuses from the Status drop-down list.
- Select one or more job statuses from the Job Status drop-down list.
- Enter a start time in the Start Time field or select a time from the Start Time drop-down list.
- Enter an end time in the End Time field or select a time from the End Time drop-down list.
- Enter a number of days from the current date back in the Past N Days in which an alert occurred.

## 3. Click Go.

The alerts that meet your search criteria are listed in the Alerts table.

**NOTE**

- The title changes to indicate that a filter has been applied to the table.
- The filter is persisted through user sessions until you change or clear the filter criteria.

## 4. (Optional) Click the Display Filter icon to hide the search filter.

**Display Alert Details**

You can display a table of server-related alerts or job-related alerts, depending on whether the Alerts table is opened from the Servers table or from the Views table.

To display the details for an alert, find the appropriate alert and click .

The information is displayed directly below the selected alert.

**NOTE**

When job level security is enabled in Configuration, only the alerts associated with the jobs to which you have access are displayed in the Alerts table.

## Manage the Servers Widget

This article explains how to manage the Servers widget.

**Customize the Servers Table**

You can customize the columns that display in the Servers table by selecting from the following options:

- Name
- Status
- Type
- Version
- Views
- Status Time
- Status Details

**NOTE**

- You can customize the Servers table in the Dashboard and Monitoring tabs.
- Only the columns specified in the Servers Table Columns property field in Configuration are available for selection.

**Follow these steps:**

1. Click All Servers in the Servers section of the Navigation pane.  
The Servers table is displayed in the right pane.
2. Click  any column in the Servers table.
3. Select  from the drop-down list.
4. Either select the checkboxes for the columns you want to display, clear the checkboxes for the columns you do not want to display, or do both.
5. Drag the columns to the left or right to order the columns in the table.
6. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

**Find a Server**

You can use the filter in the Servers widget to find one or more specific servers.

**Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Either enter the server name in the Server Name field, select the version of the server from the Version drop-down list, or do both.

**NOTE**

You can enter any of the following characters in the Server Name field:

- Valid name—Finds the specified server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all servers in the enterprise that contain the partial name. For example, entering n\* in the Server Name field would find all servers that begin with the letter n, and entering \*n would find all servers that end with the letter n.
  - Asterisk (\*), percent (%), or blank—Finds all servers.
  - Multiple names, separated by a comma.
  - Select the appropriate instance name for the server from the Instance drop-down list.
3. Click Go.  
The servers that meet your search criteria are listed in the Server table.

**Display Server Details**

You can display server-related jobs or box jobs and the associated alerts on a server from the Dashboard tab or the Monitoring page.

**Follow these steps:**

1. Find the appropriate server.
2. Perform one of the following actions:

- Click the name of the server in the Servers table on the Monitoring tab.
- Right-click a server name in the Servers table, select Open from the pop-up menu, and select the Jobs and Alerts tab.
- Double-click the server name in the Servers table, and select the Jobs and Alerts tab.

The Jobs and Alerts tables open in Monitoring, displaying the jobs, box jobs, and all alerts that are associated with the selected server.

#### NOTE

- When you display server details, the Open Servers folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open servers.
- When job level security is enabled in Configuration, only the jobs to which you have access and the alerts that are associated with those jobs are displayed in the tables.

## Manage the Views Widget

This article explains how to manage the Views widget.

### Customize the Views Table

You can customize the columns that display in the Views table by selecting from the following options:

- Name
- Alert Severity
- Job Status
- Server Status
- Jobs
- Box Jobs
- Modified by
- Modified Time
- Status Time
- Active Filters
- Inactive Filters
- Description
- Servers

#### NOTE

- You can customize the Views table in the Dashboard tab and the Monitoring tab.
- Only the columns specified in the Views Table Columns property field in Configuration are available for selection.

### Follow these steps:

1. Click  any column in the Views table.
2. Select  the drop-down list.
3. Either select the checkboxes for the columns you want to display, clear the checkboxes for the columns you do not want to display, or do both.
4. Drag the columns to the left or right to order the columns in the table.

5. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### **Find a View**

You can use the filter section in the Views widget to find one or more specific views defined in Web UI.

#### **Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Perform one or more of the following actions:
  - Enter the view name in the Name field.
  - Enter the name of the server in the Servers field or select the server from the Servers drop-down list.

#### **NOTE**

You can enter any of the following options in the Name and Servers fields:

- Valid name—Finds the specified view.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character—Finds all views or servers in the enterprise that contain the partial name. For example, entering n\* in the Name field would find all views that begin with the letter n, and entering \*n would find all views that end with the letter n.
  - Asterisk (\*), percent (%), or blank—Finds all views or servers.
  - Multiple names, separated by a comma.
- Select one or more job-based statuses from the Job Status drop-down list.
  - Select one or more of the following options from the Filter drop-down list:
    - Active -- Finds all views that have an active filter.
    - Inactive -- Finds all views that have an inactive filter.
    - Unbound -- Finds all views that have a filter specifying a server that is no longer configured.
    - None -- Finds all views that do not have a filter defined.
3. Click Go.  
The views that meet your search criteria are listed in the Views table.

#### **NOTE**

- The title changes to indicate that a filter has been applied to the table.
  - The filter is persisted through user sessions until you change or clear the filter criteria.
4. (Optional) Click the Display Filter icon to hide the search filter.

### **Display View Details**

You can display a table of jobs or box jobs represented in the view from the Dashboard tab or the Monitoring page.

#### **Follow these steps:**

1. Right-click a view name and select Open from the pop-up menu or double-click a view name in the Views table.
2. Select the Jobs and Alerts tab.  
The Jobs and Alerts tables open in Monitoring, displaying the jobs and box jobs contained in the view and all associated alerts.

#### **NOTE**

- When you display view details, the Open Views folder in the Navigation pane on the Monitoring tab expands, displaying a list of all open views.
- When job level security is enabled in Configuration, only the jobs to which you have access and the alerts associated with those jobs are displayed in the tables.



## Manage the Machines Widget

This article explains how to manage the Machines widget.

### Configure the Machines Widget

You can add the Machines widget by clicking



and selecting Machines from the menu. The Machines widget contains information about all the machines available.

### Customize the Machines Table

You can customize the columns that display in the Machines widget table by selecting from the following options:

- Name
- Server
- Status
- Status Time
- Type

#### **Follow these steps:**

1. Click



in any column in the Machines table.

2. Select



Columns from the drop-down list.

3. Do one or both of the following steps:

- Select the checkboxes for the columns you want to display.
- Clear the checkboxes for the columns you do not want to display.

4. Drag the columns to the left or right to order the columns in the table.

5. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### Find a Machine

You can use the filter section in the Machines widget to find one or more specific machines.

#### **Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Do one or more of the following steps:
  - Enter the machine name in the Name field.
  - Select a machine status from the Status drop-down list.
  - Select a machine type from the Type drop-down list.
  - Select one or more servers from the Servers drop-down list.

#### **NOTE**

- You can enter any of the following characters in the Name field:

- Valid name -- Finds the specified machine.
- Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all machines in the enterprise that contain the partial name. For example, entering n\* in the Name field would find all machines that begin with the letter n, and entering \*n would find all machines that end with the letter n.
- Asterisk (\*), percent (%), or blank -- Finds all machines.
- Multiple names, separated by a comma.

3. Click Go.

The machines that meet your search criteria are listed in the machines table.

**NOTE**

The title changes to indicate that a filter has been applied to the table.

4. (Optional) Click the Display Filter icon to hide the search filter.

### **Machine context menu actions**

You can do the following actions in Machines Widget using right-click for opening context menu:

- Every machine can use "Edit..." if you select only one instance.
- "a - Sys Agent Type" machines have in additional "Send Machine Online Event" and "Send Machine Offline Event" for selection single instance and also select of multiple instance of same type.

### **WLA Agent Monitor integration**

To integrate with WLA Agent monitor, define base url of WLA Agent monitor in Configuration, under Configuration section.

**NOTE**

To create a "Launch WLA Agent Monitor" button to open the Workload Automation Agent Monitor UI, you must define the URL to WLA Agent Monitor in Web UI Configuration.

## **Manage the Jobs Widget**

This article explains how to manage the Jobs widget.

### **Configure the Jobs Widget**

You can add and configure the Jobs widget by clicking



and selecting Jobs from the menu. The Jobs widget contains information about all the jobs to which you have access that are contained in a selected view or server. You can configure this Jobs widget to specify a view or server for which to view the jobs.

If a Jobs widget already exists, click



and select Configure from the menu.

### **Follow these steps:**

1. Enter a name in the Widget Name field.
2. Do one of the following options:
  - Select a Server from the Server drop-down list.
  - Select a View from the View drop-down list.
3. Click Save.

The Jobs widget is configured.

### **Customize the Jobs Table**

You can customize the columns that display in the Jobs widget table by selecting from the following options:

- Agent Status
- Agent Status Long
- Application
- Box Name
- Description
- Duration
- End Time
- Exit Code
- Group
- Instance
- Job Type
- Joid
- Machine
- Name
- Next Start
- Nth Run Try
- Owner
- Run Machine
- Run Number
- Server
- Start Time
- Status
- Status Time
- TimeZone

#### **NOTE**

- If you specified a view when you configured the Jobs widget, the columns specified in the view definition are available for selection.
- If you specified a server when you configured the Jobs widget, the columns specified in the Jobs Table Columns property field in the Monitoring Preferences section in Configuration are available for selection.
- Also, you can select the columns displayed in the view and order them using the Job Columns selection pane in the View Editor.

### **Follow these steps:**

1. Click



in any column in the Jobs table.

2. Select



Columns from the drop-down list.

3. Do one or both of the following steps:

- Select the checkboxes for the columns you want to display.
- Clear the checkboxes for the columns you do not want to display.

4. Drag the columns to the left or right to order the columns in the table.
5. Select the number of rows to display in the table from the Show drop-down list.  
The changes are saved.

### **Find a Job**

You can use the filter section in the Jobs widget to find one or more specific jobs in the selected view or server.

#### **Follow these steps:**

1. Click the Display Filter icon to display the filter.
2. Do one or more of the following steps:
  - Enter the job name in the Name field.
  - Select a job status from the Job Status drop-down list.
  - Select a job type from the Job Type drop-down list.
  - Select one or more servers from the Servers drop-down list.
  - Enter the application name in the Application field.
  - Enter the group name in the Group field.

#### **NOTE**

- If the Jobs widget is configured for a server, the server criteria is disabled.
- You can enter any of the following characters in the Name, Servers, Application, or Group field:
  - Valid name -- Finds the specified job, server, application, or group.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all jobs in the enterprise that contain the partial name. For example, entering n\* in the Job Name field would find all jobs that begin with the letter n, and entering \*n would find all jobs that end with the letter n.
  - Asterisk (\*), percent (%), or blank -- Finds all jobs.
  - Multiple names, separated by a comma.

3. Click Go.  
The jobs that meet your search criteria are listed in the Jobs table.

#### **NOTE**

The title changes to indicate that a filter has been applied to the table.

4. (Optional) Click the Display Filter icon to hide the search filter.

## **Web UI Enterprise Command Line Application**

The Enterprise Command Line application lets you run a AutoSys Workload Automation command, such as jil or autorep, on the selected AutoSys Workload Automation server from within Web UI. This application, which is accessed by clicking the Enterprise Command Line tab from the Web UI Dashboard, displays a command-line interface.

#### **NOTE**

For detailed information about the AutoSys Workload Automation commands you can run using this application, see the AutoSys Workload Automation *Reference section*.

Also, the following capabilities are available from Enterprise Command Line:

- Once a command has been run, you can view, and if needed, save the command output (stdout). If they occur, errors (stderr) can also be viewed.
- You can save commands, modify saved commands, and run saved commands. Frequently used commands can be saved to a list accessible only to you (My Commands) or to a list accessible to all users (Global Commands).
- You can import and export AutoSys Workload Automation jobs, calendars, machines, resources, connection profiles, and review the import/export data. You may want to search for a specific job, calendar, machine, resource, or

connection profile, and export it to a text file. Then, you can modify its properties using a text editor and can import the modified job back to the AutoSys Workload Automation server. This same process can be used to modify multiple jobs, machines, resources, and connection profiles and calendars in batch.

#### NOTE

A file on the AutoSys Workload Automation server controls the commands that are allowed to be run. This ensures that system commands cannot be run with the Enterprise Command Line application. An administrator can modify this file to control the allowable commands. For more information, contact CA Support.

### Enterprise Command Line Considerations

The following list displays important considerations for using the Enterprise Command Line application:

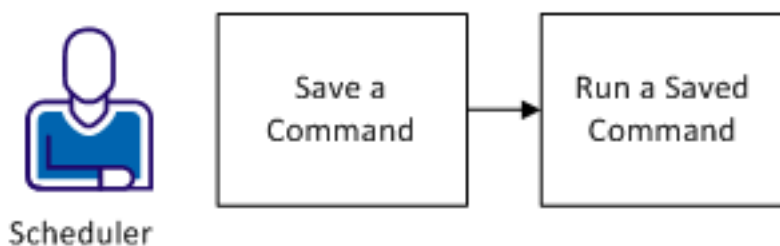
- Users must be authorized to edit, remove, or run commands.
- Specify the Web Services URL in the Configuration tab for the configured AutoSys server where you want to execute the commands.
- You cannot run interactive commands such as the `autosys_secure` command. The response from the server is blocked, so you would not be able to make the selection that is required to complete the request.
- By default, you can select up to seven servers for command execution; however, you can select only one server for an import or export.
- The maximum number of commands that can be saved is determined by settings specified by your administrator in Web UI Configuration.
- The Use Fixed-Width Font option controls the appearance of the command output and error text that is displayed. If this option is checked, all characters have the same width, resulting in detail lines with columns that are vertically aligned. If this option is not checked, all characters have proportional widths, resulting in detail lines with columns that are not vertically aligned.
- You do not need to specify a job, calendars, machines, resources, or connection profile name when you import; you simply select a text file.
- You can use wildcard symbols to export jobs but not calendars. The keyword ALL can be used to include all objects of the selected type in the export file.

### Save and Run a Command

This article explains how to save and run a command. As a scheduler, you want to run a command regularly on a AutoSys Workload Automation server without having to type the command attributes every time. The Enterprise Command Line application lets you save and then execute a AutoSys Workload Automation command at any time.

The following scenario walks you through the process of saving a custom command and executing the command.

#### How to Save and Run a Command



To save and run a command, follow these steps:

1. [Save a command](#)

## 2. Run a saved command

### Example: Saving and Running a Command for Creating a Daily Report

Suppose at the beginning of each workday you want to view a detailed report of all the payroll jobs on a particular AutoSys Workload Automation server, ServerPAY. To do this task, you can specify ServerPAY and enter the following command:

```
autorep -J payroll* -d
```

Next, you can name and save the command. You can then run the saved command daily by selecting it in the appropriate commands list and clicking the Execute link.

### Save a Command

You can enter a command to run on specific servers, and save it to the My Commands or Global Commands list.

#### **NOTE**

The maximum number of commands that can be saved is determined by the settings that are specified in the preferences section in Configuration.

#### **Follow these steps:**

1. Enter the name of the server on which the command should be run in the Servers field in the Input section. If you enter multiple servers, separate each with a comma and a space. Alternatively, click  select the servers on which the command should be run.

#### **NOTE**

You can populate the Servers and Command fields by clicking an existing command's name in the My Commands or Global Commands list.

2. Enter the command in the Command field in the Input section.
3. Click . The Command Editor dialog opens, populated with the information you have entered.
4. Enter a name and description for the command.
5. Select the list to which you want to save the command from the Save To drop-down list (My Commands or Global Commands).
6. Click Save To. The command and server information is saved.
7. Click OK. The dialog closes and the command appears in the list you selected.

### Run a Saved Command

You can use the Enterprise Command Line application to run a saved command.

#### **Follow these steps:**

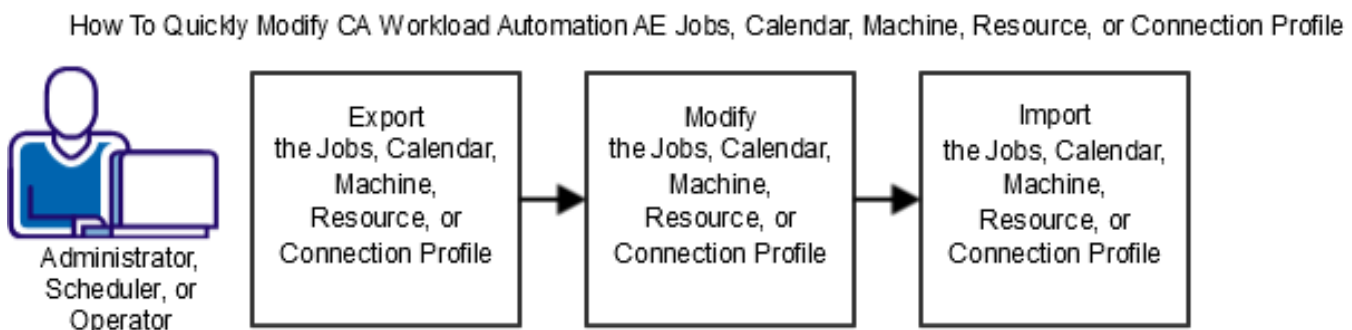
1. Click the Select option for the command in the My Commands or Global Commands list and click Execute. A confirmation dialog opens.
2. Click OK. The command runs and the results are displayed in the Output section. Errors, if any, are displayed in the Errors section.

## Modify Jobs, Calendar, Machine, Resource, or Connection Profile

This article explains how to modify Jobs, Calendars, Machines, Resources, and Connection Profiles. As an administrator, scheduler, or operator, you want to modify one or more jobs, a single calendar, machine, resource, or connection profile definitions without accessing Application Editor or Quick Edit. Additionally, you sometimes want to copy the modified jobs, calendars, machines, resources, or connection profiles to a different AutoSys Workload Automation server in your Web UI environment. The Enterprise Command Line application lets you export one or more jobs, a single or all calendars, machines, resources, or connection profiles residing on a AutoSys Workload Automation server to a file. You can modify the calendar, job, machine, resource, or connection profile definitions using a text editor, and then import the modified file to any AutoSys Workload Automation server you specify.

The following scenario walks you through the process of exporting, modifying, and importing jobs, a single calendar, machine, resource, or connection profile.

**Figure 83: UWCC--How To Quickly Modify AutoSys Workload Automation Jobs, Calendar, Machine, Resource, or Connection Profile--11.4 SP2**



To modify jobs in a batch or modify a single calendar, follow these steps:

1. [Export a AutoSys Workload Automation Job, Calendar, Machine, Resource, or Connection Profile](#)
2. [Import a AutoSys Workload Automation Job, Calendar, Machine, Resource, or Connection Profile](#)

### Example: Modifying Jobs on a AutoSys Workload Automation Server and Copying Them to a Different Server

Suppose that you have several jobs on ServerA with the prefix payroll, and you want to both modify them and use them on ServerB. To do this task in Enterprise Command Line, you can enter payroll\*, specify ServerA, and click Export. Next, you can open the exported file in a text editor and can make changes. For example, use search and replace to change the calendar that is associated with the jobs so they run on a modified schedule. Once saved, you can return to Enterprise Command Line, enter the name of the exported file, specify ServerB, and click Import. The modified payroll jobs are copied to ServerB.

### Export a AutoSys Workload Automation Job, Calendar, Machine, Resource, or Connection Profile

If you want to edit the properties of a AutoSys Workload Automation job or calendar, you can use the Enterprise Command Line application to export the job or calendar to a text file, and then make the changes using a text editor.

#### Follow these steps:

1. Select the appropriate object type in the AutoSys Workload Automation Import/Export section. If you select Calendar, also select the calendar type (Standard, Extended, or Cycle).
2. Enter the name of the job or calendar in the Name field.

**NOTE**

- A job name can contain an asterisk (\*), percent (%), or question mark (?) as a wildcard character, which may cause more than one job to be exported. A calendar name cannot contain a wildcard character.
- You can specify ALL to include all objects based on your selection.

3. Enter the name of the AutoSys Workload Automation server that hosts the job or calendar you want to export in the Server field. Alternatively, click  select the server
4. Click Export.  
The File Download dialog opens.

**NOTE**

The default file name is specified in Configuration Manager.

5. Click Save.  
The specified job or calendar is exported as a text file.

**Import a AutoSys Workload Automation Job, Calendar, Machine, Resource, or Connection Profile**

After you have modified the text file for an exported job or calendar, you must import the modified file back to the appropriate AutoSys Workload Automation server for the changes to go into production. You can use the Enterprise Command Line application to do this.

**NOTE**

You can select the Overwrite option when importing a calendar from a text file. This option helps you to overwrite an existing calendar when you import a calendar.

**Follow these steps:**

1. Select the appropriate object type in the AutoSys Workload Automation Import/Export section.
2. Click Browse next to the File Name field and select the file you want to import.
3. Enter the name of the AutoSys Workload Automation server into which you want to import the file in the Server field. Alternatively, click  select the server.
4. Click Import.  
The file containing the job or calendar changes is imported back to the specified server.

**Manage Commands Using the Enterprise Command Line Application**

This article explains how to manage commands using the Enterprise Command Line application.

**Run a Command**

You can use the Enterprise Command Line application to run a command. A single command can be sent to run on one or more servers.

**Follow these steps:**

1. Enter the name of the server or servers on which the command should be run in the Servers field in the Input section. If you enter multiple servers, separate each with a comma and a space. Alternatively, click  and select servers.



**NOTE**

The maximum number of servers that can be selected is determined by the settings specified in Configuration.

2. Enter the command you want to run in the Command field. Alternatively, click



and select a previously used command.

3. Click Execute.

**NOTE**

When executing a command against multiple servers, you can use the directional arrows or the server drop-down list in the Output section to view all results. Only the servers on which the command has been run are listed.

The command runs and the results are displayed in the Output section. Errors, if any, are displayed in the Errors section.

**Run Multiple Commands**

By default, within Web UI you can send Autosys commands to your Autosys server to be executed using the Enterprise Command Line (ECLI) tab. Enterprise Command Line passes the command through the AE Web Server which can only accept one command at a time.

However, we can make use of the AutoSysCommandFilters.txt file to configure which commands are allowed. This file can contain multiple Autosys commands, by default.

**Follow these steps:**

1. Create a file 'ecli\_commands.sh' (UNIX) or 'ecli\_commands.cmd' (Windows) with the following commands:
  - autoflags -a &
  - chk\_auto\_up -r 111 &
  - autorep -M ALL

**NOTE**

Place an '&' (ampersand) after each command, except the last.

Verify that the file is accessible and executable (UNIX: check the owner and group permissions).

2. Place this file in a directory that is listed in the PATH.  
For example, you can place the file in \$AUTOSYS/bin (UNIX) or %AUTOSYS%\bin (Windows).
3. Configure the AutoSys Web Server to allow the execution of 'ecli\_commands.sh' or 'ecli\_commands.cmd' as a command.

You can complete the configuration through updates to the file AutoSysCommandFilters.txt that is located under the \$AUTOSYS/config (unix) or %AUTOSYS%\config (windows) directory.

Add following line at bottom of the file and save:

**UNIX**

^ecli\_commands\.sh(\s.+)?

**Windows**

^ecli\_commands\.cmd(\s.+)?

4. Log in to the Web UI application and select the ECLI tab.  
In the right pane, under 'Input' complete the following actions:
  - a. Select the Server.
  - b. In the 'Command field' enter as follows:

**UNIX**

ecli\_commands.sh

**Windows**

ecli\_commands.cmd

- c. Click 'Execute'.

### **Modify a Saved Command**

You can modify a command saved in the My Commands or Global Commands list using the Command Editor dialog.

#### **NOTE**

When modifying a command, clicking Reset does not clear any fields. It instead repopulates the Command field with the saved version of the selected command, allowing you to begin again.

#### **Follow these steps:**

1. Click the Select option for the command in the My Commands or Global Commands list and click the Edit link.  
The Command Editor dialog opens.
2. Make the necessary changes.
3. Select the list to which you want to save the command from the Save To drop-down list (My Commands or Global Commands).
4. Click Save To.  
The command and server information is saved.

### **Remove a Saved Command**

You can remove a saved command from the My Commands or Global Commands list.

#### **Follow these steps:**

1. Click the Select option for the command in the My Commands or Global Commands list and click Remove.  
A confirmation dialog opens.
2. Click OK.  
The command is removed from the list.

## **Web UI Forecast Application**

The Forecast application lets you create, edit, and delete forecast report definitions, generate and view report data, and search within generated reports for specific information. This application is accessed by clicking the Forecast tab on the main Web UI page.

### **Forecast**

Forecast lets you create and generate forecast reports that show jobs that started or that are scheduled to start during a specific time period. Report output is available in table and Timeline chart formats. You can use a forecast report to plan your future workload.

#### **NOTE**

- Only authorized users have the rights to create and generate new reports.
- Specify the Web Services URL in the Configuration tab for the configured AutoSys server where you want to generate the reports.
- Generating a report is a resource-intensive activity.
- You can edit the report definition only. You cannot edit report output.

The following are important points about the report:

- The forecast window (the time period defined for the report) is based on forecasted job start time, not the start time in the job definition. For example, consider that job\_2 is defined to start at 9:00 a.m. If job\_2 has a predecessor

job (job\_1) and job\_1 completes at 9:15 a.m., Forecast will use 9:15 a.m. as the start time of job\_2, not 9:00 a.m. Therefore, if you defined the forecast window as 5:00 a.m. to 9:00 a.m., job\_2 will not appear in the report.

- Jobs that start but do not end during the forecast window are included in the report.
- If the time zone set as your user preference in the My Profile dialog differs from the time zone of the AutoSys Workload Automation server on which the report runs, the times displayed in report output are converted to the user-preferred time zone.

#### **NOTE**

In some situations, the conversion from the AutoSys Workload Automation server time to the time that appears may not be accurate. For example, if the user-preferred time zone adheres to Daylight Savings Time but the AutoSys Workload Automation server time zone does not, the conversion may not compensate for the 1 hour time difference.

- Dependencies (predecessors, for example) are assumed to be met.
- Global variables are ignored; they are treated as though the conditions are met.
- On hold jobs are treated as though they will be taken off hold and, therefore, are included in the report.
- On ice jobs are not forecast to run and, therefore, are not included in the report.

### **Forecast Navigation**

Forecast is accessed by clicking the Forecast tab on the main Web UI page. When you open the Forecast application, the Forecast page appears with links to the Reports, Table View, and Timeline View sections.

#### **NOTE**

Forecast access is controlled by the ApplicationAccess policy in CA EEM.

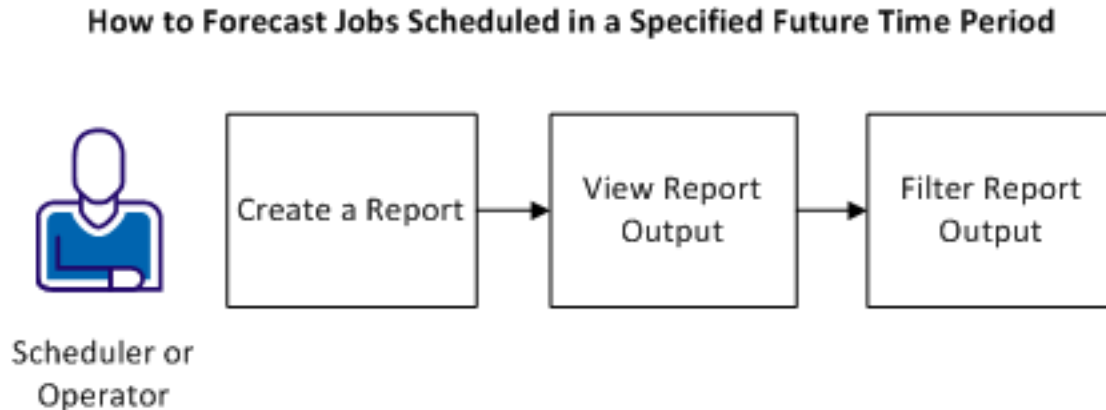
This page contains the following sections:

- Reports Search -- Lets you search for one or more reports on specific servers, or for all reports across servers.
- Reports -- Lists the report definitions that match your search criteria, and lets you perform actions on those reports.
- Search in Table -- Lets you filter the report data in the Table View section by searching for specific jobs, boxes, machines, or start time period. You can also display output for a different report by specifying the report name.
- Table View -- Displays the report data in table format for the report you select in the Reports section or for the report you specified in the Search in Table section if its status is Ready.
- Export -- The report data exports to a CSV format.
- Search in Timeline -- Lets you filter the report data in the Timeline View section by searching for specific jobs, boxes, machines, or start time period. You can also display output for a different report by specifying the report name.
- Timeline View -- Displays the report data in Timeline chart format for the report you select in the Reports section or for the report you specified in the Search in Timeline section if its status is Ready.

### **Forecast Jobs Scheduled in a Specified Future Time Period**

This article explains how to forecast jobs schedule in a future time period. As a scheduler or console operator, you want to know the jobs that are scheduled to start during a specific time period. A forecast report provides that information. The time period (forecast window) of the report is typically in the future; however, you can specify a time period in the past.

The following scenario walks you through the process of creating, viewing, and filtering a report.

**Figure 84: How to Forecast Jobs Scheduled in a Specified Future Time Period**

To forecast the jobs scheduled in a specified future time period, follow these steps:

1. [Create a report](#)
2. [View report output](#)
3. Filter report output in [Table View](#) or [Timeline View](#)

#### **Example: Plan for a Server Outage**

Suppose you know that a server will be down for maintenance on Saturday between 11:00 am and 11:00 pm. You want to know the payroll jobs that are scheduled to run on that server during that time period. With this information, you can force start the jobs before the scheduled outage to help ensure that payroll checks still go out on time. You can create a report to show all jobs scheduled to run on the affected server in that 12 hour period. You can then filter the report to display payroll jobs only.

#### **Create a Report**

You can create a forecast report definition to show the jobs that started or are scheduled to start during a specific time period.

#### **Follow these steps:**

1. Click Create in the Reports section.  
The Create Report dialog opens.
2. Enter a Report Name.
3. Select the target AutoSys Workload Automation instance from the Server drop-down list.
4. Enter a name in one or both of the following fields:
  - Job
  - Machine

#### **NOTE**

A tooltip appears when you hover the mouse on a field. The tooltip provides the definition and limits of the field.

5. (Optional) Enter text in the Description field.
6. In the Forecast Window section, do the following:
  - a. Select Start on Date from the drop-down list.

- b. Select the beginning date for the forecast window using the date picker. Only jobs that start on this date are included in the report.
- c. Specify the beginning time of the forecast window. Only jobs that start at this time or at a time within the forecast window are included in the report.
- d. Enter values in the Report Duration fields.

The forecast window is defined.

7. Click OK.

The Create a Report dialog closes, and the report is saved to the server specified and the report is generated.

**NOTE**

It will take several minutes to generate the report. If the new report name and server match the current report search criteria, it is listed in the Reports section; otherwise, you can perform a search for the new report. You can view the report data when the report status is Ready.

### **View Report Output**

You can view the output for a forecast report that has run successfully.

**Follow these steps:**

1. Search for the report you want to view using the Reports Search section.  
The Reports section contains the search results.
2. Click the link in the Name column of the Reports section for the report you want to view.

**NOTE**

You can only view a report that has a status of Ready.

The Table View and Timeline View sections are populated.

3. Do one or both of the following:
  - Scroll down to the Table View section to view the report details in table format.
  - Scroll down and, if collapsed, click  the Timeline View section to view the duration and start time of jobs in Timeline chart format.

### **Filter Report Output in Table View**

You can filter the jobs returned in a report that has run successfully by searching the report using specific criteria. By default, the report for which results are currently displayed is selected. For example, if your initial report results show a large number of jobs, you can specify a job name in the Search in Table section and perform a search. The Table View section then displays runs of the specified job only.

**Follow these steps:**

1. (Optional) Specify a different report name in the Search in Table section.

**NOTE**

If you change the name in the Report Name field, the report you search for must have a status of Ready for the report output to be displayed; otherwise, an error message is displayed.

2. Specify values for one or more of the search criteria fields and click Go.  
The Table View section displays the filtered report results.

### **Filter Report Output in Timeline View**

You can filter the jobs returned in a report that has run successfully by searching the report using specific criteria. By default, the report for which results are currently displayed is selected. For example, if your initial report results show a

large number of jobs, you can specify a job name in the Search in Timeline section and perform a search. The Timeline View section then displays runs of the specified job only.

**Follow these steps:**

1. Click  expand the Search in Timeline section, if necessary.
2. (Optional) Specify a different report name in the Search in Timeline section.

**NOTE**

If you change the name in the Report Name field, the report you search for must have a status of Ready for the report output to be displayed; otherwise, an error message is displayed.

3. Specify values for one or more of the search criteria fields and click Go.  
The Timeline View section displays the filtered report results.

## Manage Forecast Reports

This article explains how to manage Forecast reports.

### Find a Report

You can find all existing forecast reports or specify report and server names to limit the search results. The Reports section displays either the available reports for your user ID or the results of the most recent search.

**Follow these steps:**

1. Enter a report name and a server name in the Report Search section. Alternatively, click



and select the servers on which you want to search.

**NOTE**

- In the Report Name and Server fields, you can enter one or more names, separated by commas. You can use an asterisk (\*) or a percentage sign (%) as a wildcard character to represent multiple characters, or use a question mark (?) as a wildcard character to represent any single character.
- If you leave the Report Name and Server fields blank, all forecast reports found on the available servers are returned in the search results.

2. Click Go.  
The reports that you are authorized to view and that meet the search criteria are listed in the Reports section.

### Edit a Report

You can edit the properties of a report listed in the Reports section. The changes are reflected in the report output when the report is generated.

**NOTE**

- You can edit a report definition only if the report has a status of Ready or Error.
- The forecast window that you specify in the report definition can begin at any time. For <UAJM> r11.3, the duration is limited to 24 hours.

**Follow these steps:**

1. Click



in the Actions column of the Reports section for the report that you want to edit.

The Edit Report dialog opens.

2. Edit the fields as needed and click OK.

The report properties are saved, the report is generated, and a confirmation message is displayed.

**NOTE**

It will take several minutes to generate the report. You can view the report data when the report status is Ready.

### **Regenerate a Report**

You can regenerate a report listed in the Reports section. For example, suppose your report definition specifies a time period that begins in three days. You can regenerate the report at any time to show the jobs that are scheduled to run during the specified time period beginning in three days.

**NOTE**

You can regenerate only those reports with a status of Ready or Error.

To regenerate a report, click



in the Reports section for the report you want to regenerate.

The report status changes to Pending, then to Ready when the report regeneration is complete. You can view the report output when its status is Ready.

### **Delete a Report**

You can delete one or more reports listed in the Reports section when they are no longer needed.

To delete a report, select the check box for the report that you want to delete, click the Delete link, and click OK in the dialog that appears to confirm your selection.

The report is removed from the Reports section and is deleted from the Reports database.

## **Web UI Quick Edit Application**

With Quick Edit, you can schedule AutoSys Workload Automation jobs, and manage jobs on distributed networks. You can also view detailed information about all currently defined workload objects and define new ones, as needed.

To access Quick Edit, click the Quick Edit tab in the main Web UI page.

**NOTE**

In external security mode, use the ApplicationAccess resource to control access to Quick Edit. For more information about access policies, see [Securing](#).

### **Quick Edit Navigation**

Quick Edit consists of the following sections:

- **Title bar**  
Lets you refresh page content, customize search and display parameters, set preferred job types, and create workload objects.
- **Search section**  
Displays key search fields that help you find specific workload objects. You can search for specific objects, character strings, wildcard characters, or object groups. All searches are limited to the selected server and object type.
- **Search Results**

Displays the results of your search criteria in a table. You can use the Search Results section to select one or more objects for deletion, perform an action such as viewing or creating a job override, create a copy of any object, or select an object to view its detailed properties. The fields that are displayed are dependent on the type of object selected.

- **Properties section**

Lets you view or edit object properties that can be displayed when you select or create an object from the Search Results section.

### **Use the Customize Dialog**

The Customize dialog opens when you click Customize in the Quick Edit title bar. You can use the Customize dialog to set default values in Quick Edit.

This dialog contains the following tabs:

- **Filter Job Types**

Specifies the preferred job types to display in the Create Object, Create Job, and Select Job Types dialogs.

- **General**

Specifies default values to use for Quick Edit search and display parameters.

### **Set Display Preferences**

You can set your display preferences on the General tab of the Customize dialog.

#### **Follow these steps:**

1. Click Customize on the title bar.
2. Select the General tab.
3. Make the appropriate changes and click Save.  
Your preferences are saved, the dialog closes, and the display is updated.

### **Set Job Type Preferences**

You can use the Filter Job Types tab in the Customize dialog to select the job types to be displayed in the Create Object, Create Job, and Select Job Types dialogs.

#### **Follow these steps:**

1. Click Customize on the title bar.

#### **NOTE**

The first time you open the Customize dialog, all available job types are selected on the Filter Job Types tab.

2. Clear the Select check box for the job types you do not need.

#### **NOTE**

When a group name is selected, all job types within the group are selected. Clearing the group name check box clears all job types within the group. If you clear certain job types within a selected group, the group name check box is cleared also. Similarly, if you select all job types within an unselected group, the group name check box is selected also.

3. Click Save.  
Your job type preferences are saved.

### **Use the Create Object Dialog**

The Create Object dialog opens when you click Create in the Quick Edit title bar. You can use the Create Object dialog to create a job, box job, calendar, cycle, global variable, or machine type.



**NOTE**

If the selected server has no credentials set, the object tree does not appear. An error message is displayed instead.

**Use the Create Job Dialog**

The Create Job dialog opens when you click the Create Job in this Box Job icon from the Search Results table, or select the Create Job in this Box action from a box job's properties panel while in edit mode.

**NOTE**

When you select a job type, the following limitations occur:

- The User Defined job type appears only if that job type is defined on the back-end server.
- Only the job types defined on the Filter Job Types tab in the Customize dialog, and those to which you have been authorized in CA EEM, are displayed.
- Job templates for a specific job type, if available, appear in the list below the job type with which they are associated. Selecting a job template creates the job with specific properties predefined.

**View and Manage Jobs in Quick Edit**

This article explains how to manage jobs using Quick Edit.

**Find a Job or Box Job**

You can use Quick Edit to find one or more AutoSys Workload Automation jobs or box jobs residing on a server.

**Follow these steps:**

1. Select the appropriate server from the Server drop-down list.
2. Do one of the following
  - a. Select Job from the Type drop-down list.
  - b. Select Top-level Box Job from the Type drop-down list.
3. Enter one of the following in the Name field:
  - Valid name -- Finds the specified job on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all jobs on the selected server that contain the partial name. For example, entering n\* would find all jobs on the server that begin with the letter n, and entering \*n would find all jobs on the server that end with the letter n.
  - Asterisk (\*) or percent (%) -- Finds all jobs on the selected server.
4. (Optional) Enter search criteria in any of the optional fields to further refine your search.
5. Click Go.

The Search Results section lists the jobs or box jobs that meet your search criteria.

**NOTE**

- Box jobs are displayed in a hierarchical tree view. You can expand a top-level box job to view, edit, copy, or delete any nested box jobs and jobs.
- When searching by Job Types, you can specify multiple job types using a comma-delimited list.

## **View a Job or Box Job**

You can view the date and time conditions, dependencies or triggers, permissions, and notification settings for a specific job.

### **Follow these steps:**

1. Find the appropriate job or box job.

#### **NOTE**

Box jobs are displayed in a hierarchical tree view. You can expand a top-level box job to view, edit, copy, or delete any nested box jobs and jobs.

2. Click the name in the Search Results section.  
The detailed properties are displayed.

## **Copy a Job or Box Job**

You can copy an existing job to create a new job with the same properties. You must specify a different name for the new job; otherwise, the job you copied is simply updated.

### **Follow these steps:**

1. Find the appropriate job.

#### **NOTE**

Box jobs are displayed in a hierarchical tree view. You can expand a top-level box job to view, edit, copy, or delete any nested box jobs and jobs.

2. Do one of the following:

-  Click the Actions column to the right of that job or box job.
- Click the name in the Search Results section. Next, select Copy from the Select and drop-down list, and click Go. The Properties section for the specific job type is displayed, with the relevant categories. For example, if you select a command job, the Properties section displays the Primary, Command Job, Environment, Termination, Schedule, Notification, and Native Security Permissions categories.

3. Define your job by doing the following:
  - a. Enter a name for the new job in the Name field.

#### **NOTE**

The name can contain up to 64 characters.

- b. Edit the job definition as necessary.
4. Click Commit.  
The new job is saved.

#### **NOTE**

Copying a job or box job also copies all its dependencies.

## **Edit a Job or Box Job**

You can modify date and time conditions, change dependencies or triggers, manage permissions, or change notification settings for a specific job.

### **Follow these steps:**

1. Find the appropriate job or box job.

**NOTE**

Box jobs are displayed in a hierarchical tree view. You can expand a top-level box job to view, edit, copy, or delete any nested box jobs and jobs.

2. Click the name in the Search Results section.
3. Edit the job definition as necessary.

**NOTE**

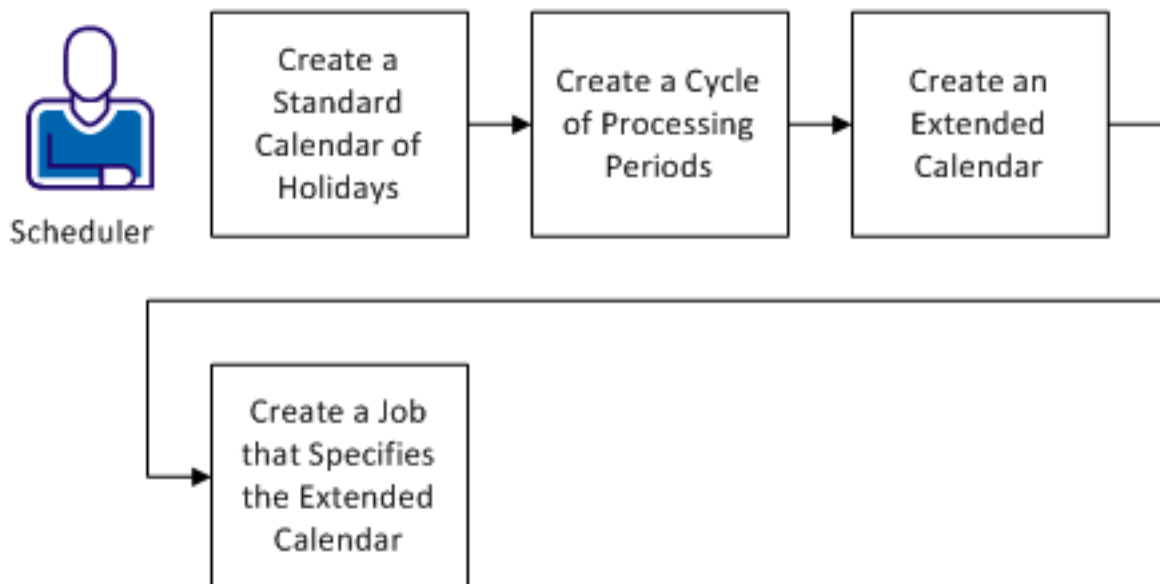
The Name field is read-only.

4. Click Commit.  
The changes are saved.

**How to Create a Job that Runs on a Complex Schedule**

As a scheduler, you want to create a job that runs on a complex schedule such as one that includes specific days in specific months but does not run on holidays. You can create a complex run schedule with an extended calendar. For example, you can specify one day of the week as a run day, create a calendar of holidays on which the job should not run, and specify that the job runs on the weekday before a holiday if these conflict. You can also create a cycle so that the job runs during specific time periods only.

The following scenario walks you through the process of creating a standard calendar, a cycle, an extended calendar, and a job that specifies the extended calendar as a property.

**How to Create a Job that Runs on a Complex Schedule**

To create a job that runs on a complex schedule, follow these steps:

1. [Create a standard calendar of holidays](#)
2. [Create a cycle of processing periods](#)
3. [Create an extended calendar](#)
4. [Create a job that specifies the extended calendar](#)

**Example: Create a Job that Runs on a Complex Schedule**

Suppose that you want to create a job named ContractorPayroll to run every Friday during the months of January, February, November, and December. Those months are the busiest for your business so you hire part-time contractors to help. If a holiday occurs on a Friday, you want the job to run on the nearest non-holiday weekday before Friday instead. To do this task, you can create a standard calendar of holidays for the year. Next, you can create a cycle named StretchMonths that contains the periods from January 1st to February 28th and November 1st to December 31st. Finally, you can create an extended calendar named ContractorPaydays with the following properties:

- Conditions -- cycle and fri
- Cycle name -- StretchMonths (the cycle you created)
- Adjustment -- 0
- Reschedule rule -- Schedule on the previous workday
- Calendar -- Holidays (the holiday calendar you created)
- Non-workday action -- Schedule only on workdays
- Workdays -- Select Mo, Tu, We, Th, Fr

When you create the ContractorPayroll job, you simply specify the extended calendar as the run calendar.

### **Create a Standard Calendar**

You can create a standard calendar to define specified dates on which jobs should run.

#### **Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Calendar link.  
The Properties section is displayed.
3. Define your calendar by doing the following:
  - a. Enter a name for the calendar in the Name field.

#### **NOTE**

The name can contain up to 30 characters. If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- b. Change the month or year, if necessary, by selecting from the associated drop-down lists and clicking Go to navigate the calendar and select days.
- c. Do **one** of the following:
  - Click individual days in the calendar to toggle them between set and unset.
  - Define and apply rules in the Generate Days section for determining processing and blocked days in the calendar and for rescheduling processing when schedule conflicts arise.
- d. Provide optional description (up to 1024 characters)

#### **NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

4. Click Commit.  
The calendar definition is saved.

### **Create a Cycle**

You can create a cycle to define date ranges to be used as input to an extended calendar.

#### **Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Cycle link.  
The Properties section is displayed.

3. Define your cycle by doing the following:
  - a. Enter a name for the cycle in the Name field.

**NOTE**

The name can contain up to 30 characters. If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- b. (Optional) Select the Repeat every year check box to repeat the cycle every year.
- c. Select Add from the Select an action drop-down list and click Go to add periods to the cycle.
  - One cycle has a maximum of 30 periods.
  - You can optionally
    - click  select the period dates. This is available only when the Repeat every year check box is not selected.
    - At least one period must be added to the definition.
- d. (Optional) Select a period, select Add Before from the Select an action drop-down list, and click Go to add a new period before the selected period.
- e. (Optional) Select a period, select Move Up from the Select an action drop-down list, and click Go to move the selected period up one level.
- f. (Optional) Select a period, select Move Down from the Select an action drop-down list, and click Go to move the selected period down one level.
- g. (Optional) Select a period, select Delete from the Select an action drop-down list, and click Go to delete the selected period from the cycle.
- h. (Optional) Provide description (up to 1024 characters)

**NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

4. Click Commit.  
The cycle definition is saved.

**Example: Create a Cycle and Associate it with an Extended Calendar**

To associate a cycle with an extended calendar, you first create the cycle with the periods defined. Then, when you create the extended calendar with which you want to associate it, enter CYCLE in the Conditions field, enter the name of the cycle calendar in the Cycle name field, and click Generate Preview to view the processing dates selected. You can also

click 

the right of the Cycle name field to select a cycle from the Search Cycle dialog.

**Create an Extended Calendar**

You can create an extended calendar to define processing days using any combination of holiday calendars, cycles, and conditions for scheduled jobs.

**Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Extended Calendar link.  
The Properties section is displayed.
3. Define your extended calendar by doing the following:
  - a. Enter a name for the extended calendar in the Name field.

**NOTE**

The name can contain up to 30 characters. If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- b. (Optional) Enter date conditions for the extended calendar in the Conditions field.

**Examples:**

- DAILY - Selects all dates after the current date
  - MON - Selects all Mondays after the current date
  - FOMWEEK - Selects the first weekday of each month after the current date
  - EOMWEEK - Selects the last weekday of each month after the current date
  - XFOM - Selects all days except the first day of each month after the current date
  - CYCLE - Selects any day within the date ranges, or periods defined in the cycle.
- c. (Optional) Enter the name of an existing cycle in the Cycle Name field or select a cycle from the Search Cycle dialog. Also, enter CYCLE in the Conditions field.
  - d. Enter any integer in the Adjustment field to adjust the days generated by the other criteria to either a previous day or a specified number of days after the actual generated day.
  - e. (Optional) Select a rule for holidays from the Reschedule rule list, and then enter the name of a standard calendar in the Calendar field or select a calendar from the Search Calendar dialog.
  - f. (Optional) Select an action for non-workdays from the Non Workday Action list, and then select at least one workday from the Workdays panel.
  - g. (Optional) Click Generate preview to open the Extended Calendar Preview dialog so you can preview the dates generated by the rules in the extended calendar.
  - h. (Optional) Provide description (up to 1024 characters)

**NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

## 4. Click Commit.

The extended calendar definition is saved.

**NOTE**

If you create an extended calendar by entering a value in the Name field only and saving it, all days are selected from the current day forward, for a total of 366 days.

**Example: Create an Extended Calendar Based on Field Adjustment**

To create an extended calendar that processes associated jobs one day after each holiday, you would first create a calendar defining the holidays for the year, then create an extended calendar with the following options selected:

- Your holidays calendar in the Calendar field
- Schedule only on a holiday in the Holiday Action list
- 1 in the Adjustment field

**Create a Job or Box Job**

You can create jobs to manage and schedule workload in your enterprise. Additionally, You can create a box job to be a container for all jobs that have similar starting conditions or complex logic flows. A box job can be created to group jobs with the same scheduling parameters.

**Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list.
3. Click the link for the type of job you want to create or click the box job link.

**NOTE**

You can select a specific job type or, if available, a job template containing predefined properties to create a new job.

4. Enter a name in the Name field.

**NOTE**

The name can contain up to 64 characters.

5. Complete the fields in each category as appropriate.
6. Click Commit.

The job or box job is saved.

**Create a Box Job**

You can create a box job to be a container for all jobs that have similar starting conditions or complex logic flows. A box job can be created to group jobs with the same scheduling parameters.

**Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Box Job link.
3. Perform the following actions to define your box job:
  - a. Enter a name for the new box job in the Name field.

**NOTE**

The name can contain up to 64 characters.

- b. Complete the fields in each category as appropriate.
  4. Click Commit.
- The box job is saved.

**Create a Job inside a Box Job**

You can use a box job to group jobs with like starting conditions (such as date/time conditions or job dependency conditions). For example, if you create a new job that will have the same starting conditions as the jobs in a previously-created box job, you can create the job inside that box job.

**Follow these steps:**

1. Find the appropriate box job.
2. Click the box job name in the Search Results section.
3. Select Create a Job in this Box from the Select an action drop-down list, and then click Go.
4. Click the link for the type of job you want to create.  
The Properties section for the specific job type you selected is displayed. The Box field contains the name of the selected Box Job.
5. Define your job by performing the following actions:
  - a. Enter a name for the new job in the Name field.

**NOTE**

The name can contain up to 64 characters.

- b. Complete the fields in each category as appropriate.
  6. Click Commit.
- The job is saved.

**Add a Job to a Box Job**

You can use a box job to group jobs with like starting conditions (such as date/time conditions or job dependency conditions). For example, if you have several jobs that run daily at 1:00 a.m., you can put all these jobs in a box and assign a daily start condition to the box.

**Follow these steps:**

1. Find the appropriate job.
2. Click the job name in the Search Results section.

**NOTE**

You can select a box job if you want to nest it in another box job.

3. Enter the name of the target box job in the Box field.

Clicking 

the Search Box Job dialog, from which you can select a box job.

4. Click Commit.  
The job is added to the box job.

**Remove a Job from a Box Job**

If you no longer need to group a specific job in a box of other jobs because, for example, its date/time conditions have changed, you can remove it from a box job.

**Follow these steps:**

1. Find the appropriate job.
2. Click the job name in the Search Results section.

**NOTE**

This job can be a nested box job.

3. Remove the name of the box job from the Box field.
4. Click Commit.  
The job is removed from the box job.

**Delete a Job or Box Job**

If you no longer need to manage and schedule a job in your enterprise, you can delete it. If you no longer need a box job, along with all jobs contained within it, you can delete it.

**WARNING**

Deleting a box job also deletes all jobs contained within it.

**Follow these steps:**

1. Find the appropriate job.
2. Do one of the following:
  - Click the name in the Search Results section, and then click Delete.
  - Select the Select check box in front of the appropriate job and click the Delete link.

**NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

The Delete Object dialog opens and displays the dependencies of the selected job object with the Job Name and Dependents.

A confirmation prompt is displayed.

3. Click OK.  
The job or box job definition is deleted.



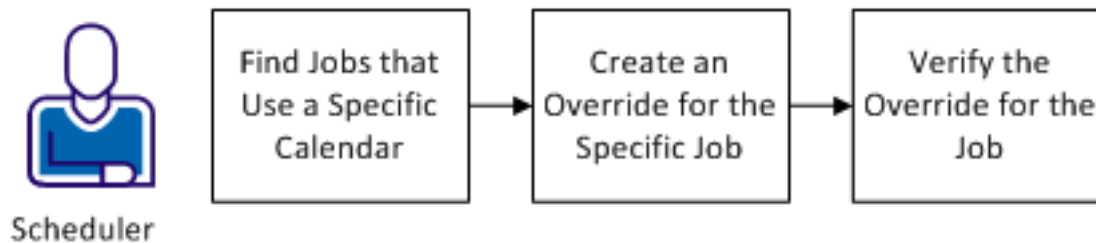
## How to Create an Override for a Scheduled Job

As a scheduler, for one run only, you want to modify the properties of a job that runs on a regular basis. An override lets you modify the properties of a job for the next run only.

For example, you have several jobs that are scheduled to run on a specific day of the week, but you want one of those jobs not to run this week. You can create an override in which you clear the run calendar option for the job. The job does not run the next day that it is normally scheduled to run; however it runs on its scheduled day in the weeks that follow.

The following scenario walks you through the process of finding the job, and scheduling and verifying an override.

### How to Find a Job, and Create and Verify an Override



1. Find jobs that use a specific calendar
2. Create an override for the specific job
3. Verify the override for the job

#### Example: Create an Override for a Job

Suppose that you have some jobs that specify a calendar named RunFridays in their properties and run every Friday. You plan to be out of the office on Friday of this week and you do not want the print jobs to run that day. You want them to run on the following Monday instead. To do this task, you can find the jobs that specify the RunFridays calendar. You can then create an override for just the print jobs, in which you remove the specified calendar and specify Monday as the run day instead. Once you commit the override, you can search for the jobs and view the override details to verify that they are correct. The jobs do not run on the Friday after the day the override is created but run on Fridays as usual thereafter.

#### Find a Job Using a Specific Calendar

You can use Quick Edit to find one or more AutoSys Workload Automation jobs based on when they are scheduled to run.

##### Follow these steps:

1. Select the appropriate server from the Server drop-down list.
2. Select Job from the Type drop-down list.
3. Enter \* or % in the Name field.
4. Type the name of the calendar in the Run calendar field, or

click 

the right of the field and select the appropriate calendar.

#### NOTE

You can also search for jobs based on a specific exclude calendar, using the Exclude calendar field.

5. Click Go.  
The Search Results section lists the job or jobs that meet your search criteria.

## Create a Job Override

You can modify the values of specific properties for a job or box job to create a one-time override. The values you enter are effective only for the next job run.

### Follow these steps:

1. Find the appropriate job.
2. Do one of the following:
  - Click  the Actions column to the right of that job.
  - Click the job name in the Search Results section, select Create Override from the Select and drop-down list in the Properties section, and then click Go.

The Properties section displays with only certain fields that can be modified in each category.
3. Edit the properties as necessary.
4. Click Commit.  
The job override for the next run is saved.

## View a Job Override

If a one-time override has been defined for a job or box job, you can view the properties and values for the latest override set for the next run.

### Follow these steps:

1. Find the appropriate job.
2. Perform one of the following actions:
  - Click  the Actions column to the right of that job.
  - Click the job name in the Search Results section. Next, select View Override from the Select an action drop-down list in the Properties section, and click Go.

The Properties section for the specific job displays the latest properties and values set for the next run.

## Delete a Job Override

If a one-time override has been defined for a job or box job and you no longer need it, you can delete it.

### Follow these steps:

1. Find the appropriate job.
2. Perform one of the following actions:
  - Click  the Actions column to the right of that job.
  - Click the job name in the Search Results section. Next, select View Override from the Select and drop-down list, and click Go.

The Properties section for the specific job displays the properties and values set for the next run.+
3. Click Delete.  
A confirmation prompt is displayed.
4. Click OK.  
The job override is deleted.

## Supported Workload Objects in Quick Edit

This article explains the Workload Objects that you can manage using Quick Edit.

### Jobs

Job definitions control all aspects of job processing, from submission through execution and completion. This release of Web UI supports numerous job types using agent technology.

Agents let you automate, monitor, and manage workload on all major platforms, applications, and databases. To run workload on a particular system, your agent administrator needs to install the appropriate agent on that system. For example, if your workload must run on a UNIX computer, your administrator must install and configure the Workload Automation Agent for UNIX, which lets you create jobs to run UNIX scripts, execute UNIX commands, transfer files using FTP, monitor file activity on the agent computer, and perform many other tasks.

The Workload Automation Agent for UNIX, Linux or Windows is provided with your scheduling manager, giving you a broad set of default job types. Additional agents are available to augment that set.

By extending the functionality of your scheduling manager with agents, you can use Web UI to define and schedule the following types of jobs:

- [Box jobs](#)
- [Command jobs](#)
- [File Watcher jobs](#)
- [File Trigger jobs](#)
- [FTP jobs](#)
- [i5/OS jobs](#)
- [Secure Copy jobs](#)
- [User Defined jobs](#)
- [z/OS jobs](#)
- [Application and Web Services jobs](#)
- [Database jobs](#)
- [Object Monitoring Jobs](#)
- [Oracle E-Business Suite jobs](#)
- [PeopleSoft jobs](#)
- [Process Automation jobs](#)
- [SAP jobs](#)
- [Micro Focus jobs](#)
- [Remote Execution jobs](#)
- [SNMP jobs](#)
- [Wake on LAN jobs](#)
- [Informatica jobs](#)
- [SQL Server Agent jobs](#)

#### NOTE

For more information about each job type, including how to create them and their job properties, see *Workload Scheduling*.

## **Box Jobs**

A box job is a AutoSys Workload Automation-specific jobset, a collection of jobs or other box jobs that have similar starting conditions or complex logic flows. Box jobs do not group jobs organizationally, but instead group jobs with the same scheduling parameters. For example, if several jobs run daily at 1:00 a.m., you can bundle them in a box job and assign that one job a daily start condition. However, you should not group jobs with diverse starting conditions in the same box job.

You can use a box job to organize and control process flow. The box performs no actions, although it can trigger the execution of other jobs. An important feature of this type of job is that you can nest boxes inside of other boxes. When you nest box jobs, you can group jobs with similar starting conditions and maintain and execute them logically.

## **Command Jobs**

The Command job is the most common type of job you can schedule. The command can be a shell script, an executable program, and so on. When you run a Command job, the result is the execution of a specific command on a client or target computer that is specified in the job definition.

## **File Watcher Jobs**

A File Watcher job starts a process that looks for the existence and size of a specific operating system file. When that file reaches a specified minimum size (and is no longer increasing in size), the job completes successfully.

## **File Trigger Jobs**

You can define a File Trigger job to monitor file activity for UNIX, Windows, or i5/OS systems. The File Trigger job can monitor when a file is created, updated, deleted, expanded, or shrunk, and when a file exists or does not exist.

## **FTP Jobs**

You can automate FTP transfers with an FTP job. The FTP job can upload data to or download data from an existing FTP server or another agent as an FTP server. The FTP job always acts as an FTP client.

### **NOTE**

To run these jobs on i5/OS systems, your system requires Workload Automation Agent for i5/OS.

## **i5/OS Jobs**

You can define an i5/OS job that lets you run a program or issue a command on an i5/OS system. You can run i5/OS jobs in the root file system, open systems file system (QOpenSys), or library file system (QSYS).

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for i5/OS.

## **Secure Copy Jobs**

The Secure Copy job utilizes technology similar to an FTP job. The Secure Copy job uses the Secure Copy protocol to automate file transfers, offering more security than FTP protocol.

## **User Defined Jobs**

User Defined jobs let you associate a job driver and description with a job type. User Defined jobs can only have a job type of 0 through 9. Therefore, at most, there can only be 10 user-defined job types. User Defined job types must be defined on the AutoSys Workload Automation scheduler.

## **z/OS Jobs**

z/OS jobs let you run mainframe workload. You can schedule z/OS jobs, create dependencies on z/OS jobs submitted outside of the scheduling manager, and create dependencies on data set activities.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for z/OS.

You can define and run the following types of z/OS jobs:

- z/OS Data Set Trigger
- z/OS Manual
- z/OS Regular

## **Application and Web Services Jobs**

Application Services jobs let you perform tasks such as making calls to J2EE session and entity beans, publishing and monitoring messages to J2EE JMS queues and topics, invoking methods on Java classes in the agent class path, making Java Remote Method Invocation (RMI) calls, and performing HTTP/HTTPS queries on a servlet.

Web Services jobs let you call an operation within a web service and pass parameters to the operation. The parameters can be actual values, literal values, or a serialized Java object passed by another job.

### **NOTE**

To run these jobs, your system requires either Workload Automation Agent for Application Services or Workload Automation Agent for Web Services, depending on the job type.

You can define and run the following types of application and web services jobs:

- Entity Bean
- HTTP
- JMS Publish
- JMS Subscribe
- JMX-MBean Attribute Get
- JMX-MBean Attribute Set
- JMX-MBean Create Instance
- JMX-MBean Operation
- JMX-MBean Remove Instance
- JMX-MBean Subscribe
- POJO
- RMI
- Session Bean
- Web Service RPC/Encoded
- Document/Literal Style Web Service

## **Database Jobs**

Database jobs let you automate common database tasks on Oracle, Microsoft SQL Server, and IBM DB2 databases. You can perform tasks such as executing SQL queries, running stored procedures contained in relational databases, and monitoring relational databases for changes to a table.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for Databases.

You can define and run the following types of database jobs:

- Database Monitor
- Database Stored Procedure
- Database Trigger
- SQL

### **Object Monitoring Jobs**

Object Monitoring jobs let you monitor different aspects of your system. A set of object monitoring jobs comes with your scheduling manager.

#### **NOTE**

To run these jobs on i5/OS systems, your system requires Workload Automation Agent for i5/OS.

You can define the following types of object monitoring jobs:

- CPU Monitoring
- Disk Monitoring
- File Trigger
- IP Monitoring
- Process Monitoring
- Text File Reading and Monitoring
- Windows Event Log Monitoring
- Windows Service Monitoring

### **Oracle E-Business Suite Jobs**

Oracle E-Business Suite jobs let you execute Oracle E-Business Suite (OA) single requests and OA request sets, monitor the status of OA programs and child processes, and report on OA program execution status.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for Oracle E-Business Suite.

You can define and run the following types of Oracle E-Business Suite jobs:

- Oracle E-Business Suite Copy Single Request
- Oracle E-Business Suite Request Set
- Oracle E-Business Suite Single Request

### **PeopleSoft Jobs**

PeopleSoft jobs let you perform tasks such as running PeopleSoft programs, monitoring and reporting on the status of those programs, deleting PeopleSoft jobs, and removing PeopleSoft spool and trace files.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for PeopleSoft.

### **SAP Jobs**

SAP jobs let you perform tasks such as running SAP R/3 workload, running SAP programs, monitoring ABAP status, and deleting SAP jobs.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for SAP.

You can define and run the following types of SAP jobs:

- SAP Batch Input Session
- SAP BW InfoPackage
- SAP BW Process Chain
- SAP Data Archiving
- SAP Event Monitor
- SAP Job Copy
- SAP Process Monitor
- SAP R/3

### **Micro Focus Jobs**

Micro Focus jobs let you run your mainframe JCL and COBOL programs in a Windows environment.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Micro Focus.

### **Process Automation Jobs**

Process Automation jobs let you launch CA Process Automation processes and monitor them to completion.

You can define the following Process Automation jobs:

- **Process Automation Process Execution**  
Lets you directly execute a CA Process Automation process.
- **Process Automation Start Request Form**  
Lets you submit a Start Request Form to execute a CA Process Automation process. The Start Request Form sets the values for the current execution of the process.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Web Services.

### **Remote Execution Jobs**

Remote Execution jobs let you run commands on a remote UNIX, HP Integrity NonStop, or OpenVMS computer through Secure Shell (SSH2) or Telnet.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows and Workload Automation Agent for Remote Execution. The Workload Automation Agent for Remote Execution supports only remote target environments of UNIX, HP Integrity NonStop, and OpenVMS.

### **Wake on LAN Jobs**

Wake on LAN jobs let you use the Wake on LAN (WOL) feature of the agent to automate the startup and shutdown of your computers. You can define and schedule a Wake on LAN job to send a signal to a server to power it up. When the server is no longer needed, you can schedule a different command job to power it down.

#### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

## **Informatica Jobs**

You can define an Informatica job to start a predefined workflow on Informatica PowerCenter. You can start the workflow from the beginning of the workflow or from a specified task. You can also start a workflow instance.

### **NOTE**

- To run these jobs, your system requires Workload Automation Agent for Linux or Windows and Workload Automation Agent for Informatica.
- Due to a known issue in Informatica, you cannot run workflow instances in parallel.

## **SQL Server Agent Jobs**

You can define a SQL Server Agent job to execute predefined jobs in the SQL Server Agent to perform administrative tasks.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for Windows and Workload Automation Agent for Microsoft SQL Server.

## **SNMP Jobs**

You can define the following types of SNMP jobs:

- Simple Network Management Protocol Value Get (SNMPGET)
- Simple Network Management Protocol Value Set (SNMPSET)

The SNMP Value Get job queries a network device for the value of a variable that is assigned to a Management Information Base (MIB) address. You can use the SNMP Value Get job to retrieve information about a network device to determine whether an administrator is required to be notified.

The SNMP Value Set job modifies a variable on a network device. The variable is assigned to the MIB address that you specify. You can use the SNMP Value Set job to update a variable that reports on the failure or success of a mission-critical policy.

### **NOTE**

To run these jobs, your system requires Workload Automation Agent for UNIX, Linux, or Windows.

## **HADOOP Jobs**

CA WA Advanced Integration for Hadoop lets you integrate with various distributions of Hadoop including Cloudera, Hortonworks, and Apache. You can define the following job types using CA WA Advanced Integration for Hadoop:

- HDFS (Hadoop Distributed File System)
- Hive
- Oozie
- Pig
- Sqoop

### **NOTE**

To run these jobs, your system requires CA WA Advanced Integration for Hadoop.



## Job Type Icons

The following job type icons are used in Web UI:

 (BOX)	 Copy (SCP)	Secure
 (CMD)	 Command Defined (specify 0-9)	User
 Watcher (FW)	 OS Regular (ZOS)	z/
 Trigger (FT)	 OS Data Set Trigger (ZOSDST)	z/
 (FTP)	 OS Manual (ZOSM)	z/
 OS (I5)	i5/	
<b>Application and Web Services Jobs</b>		
 Bean (ENTYBEAN)	 MBean Operation (JMXMOP)	JMX-
 (HTTP)	 MBean Remove Instance (JMXMREM)	JMX-
 Publish (JMSPUB)	 MBean Subscribe (JMXSUB)	JMX-
 Subscribe (JMSSUB)	 (POJO)	POJO
 MBean Attribute Get (JMXMAG)	 (JAVARMI)	RMI
 MBean Attribute Set (JMXMAS)	 Bean (SESSBEAN)	Session
 MBean Create Instance (JMXMC)	 Service RPC/Encoded (WBSVC)	Web
 Literal Style Web Service (WSDOC)	Document/	
<b>Database Jobs</b>		
 Monitor (DBMON)	 Database Trigger (DBTRIG)	Databas
 Stored Procedure (DBPROC)	 base (SQL)	SQL

Object Monitoring Jobs		
 Monitoring (OMCPU)	 File Reading and Monitoring (OMTF)	Text
 Monitoring (OMD)	 Event Log Monitoring (OMEL)	Windows
 Monitoring (OMIP)	 Service Monitoring (OMS)	Windows
 Monitoring (OMP)	Process	
ERP Jobs		
 E-Business Suite Copy Single Request (OACOPY)	 Oracle BW Process Chain (SAPBWPC)	SAP
 E-Business Suite Request Set (OASET)	 Data Archiving (SAPDA)	SAP
 E-Business Suite Single Request (OASG)	 Event Monitor (SAPEVT)	SAP
 (PS)	 PeopleSoft Job Copy (SAPJC)	SAP
 Batch Input Session (SAPBDC)	 Process Monitor (SAPPM)	SAP
 BW InfoPackage (SAPBWIP)	 R/3 (SAP)	SAP
HADOOP Jobs		
 Hadoop Distributed File System (HDFS)	 Oozie	
 Pig	 Pig	Pig
 Sqoop	Sqoop	
SNMP Jobs		
 Value Get (SNMPGET)	 Value Set (SNMPSET)	SNMP
Other Jobs		
 Focus (MICROFOCUS)	 Execution (PROXY)	Remote
 Automation Process Execution (PAPROC)	 Wake on LAN (WOL)	Wake

 Automation Start Request Form (PAREQ)	 fess
 MSSQL Server	MSSQL

Informat

## Calendars

Calendars are used for scheduling jobs to identify the days when jobs should or should not process. You can define a custom calendar to include any number of dates. You can then associate any calendar with one or more jobs. A custom calendar can be created using a standard calendar or an extended calendar.

A *standard calendar* is a set of manually selected, user-specified dates. There are two types of standard calendars: run and exclude. A *run calendar* specifies the days on which a job should run, such as the first weekday of every month. Conversely, an *exclude calendar* specifies the days on which a job should not run. For example, to help ensure that a job never runs on any corporate holiday, you could create a calendar named Holidays that has each corporate holiday selected.

An *extended calendar* is a dynamic calendar that is generated automatically based on a combination of holiday calendars, cycles, and conditions defined.

Each calendar is stored in the database as a separate object with a unique name.

## Cycles

Cycles are used as processing periods in an extended calendar definition. These user-defined date ranges make it easy to define rules for non-standard processing periods, such as business cycles, quarters, and accounting periods. Cycles can be set to repeat every year.

## Global Variables

When you use a global variable as a job dependency, the job dependency is satisfied when the global variable dependency expression evaluates to TRUE.

Assume, for example, that you have a set of jobs in a box that must only run on a manager's approval. You could set the global variable named "manager-ok" to "OK," and make the top-level box job dependent on this global variable.

## Machines

A machine can refer to either a real machine or a virtual machine. Each machine definition is stored in a database as a separate object with a unique name.

### Real Machines

A *real machine* is any single physical machine that meets the following criteria:

- It has been identified in the appropriate network so that AutoSys Workload Automation and Web UI can access it.
- An agent, CA AutoSys Workload Automation Connect Option, CA Network and Systems Management, or AutoSys Workload Automation is installed, which lets AutoSys Workload Automation run jobs on it.
- It is defined to AutoSys Workload Automation as a real machine using JIL or Web UI.

A real machine must meet these criteria to run jobs.

## Virtual Machines

A *virtual machine* is a machine definition that references one or more existing real machine definitions. By defining virtual machines and submitting jobs to run on those machines, you can specify run-time resource policies (or constraints) at a high level and ensure that AutoSys Workload Automation automatically runs those policies in a multi-machine environment.

## Manage Calendars and Cycles in Quick Edit

This article explains how to manage calendars and cycles using Quick Edit.

### Create a Standard Calendar

You can create a standard calendar to define specified dates on which jobs should run.

#### Follow these steps:

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Calendar link.
3. Define your calendar by performing the following actions:
  - a. Enter a name for the calendar in the Name field.

#### NOTE

- The name can contain up to 30 characters.
  - If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.
- b. Select the month or year from the drop-down lists and click Go to navigate the calendar and select days.
  - c. Perform one of the following actions:
    - Click individual days in the calendar to toggle between set and unset.
    - Define and apply rules in the Generate Days section. The rules are used to determine the processing and blocked days and to resolve schedule conflicts.
4. (Optional) Provide a description.
    - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

#### NOTE

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

5. Click Commit.  
The calendar definition is saved.

### Find a Calendar

You can use Quick Edit to find one or more AutoSys Workload Automation calendars residing on a specific server.

#### Follow these steps:

1. Select the appropriate server from the Server drop-down list.
2. Select Calendar from the Type drop-down list.
3. Enter one of the following in the Name field and click Go:
  - Valid calendar name -- Finds the specified calendar on the selected server.

Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all calendars on the selected server that contain the partial name. For example, entering n\* would find all calendars on the server that begin with the letter n, and entering \*n would find all calendars on the server that end with the letter n.

Asterisk (\*) or percent (%) -- Finds all calendars on the selected server.

4. (Optional) Enter search criteria in any of the optional fields to further refine your search.  
The Search Results section lists the calendar or calendars that meet your search criteria.

### **View a Calendar**

You can review the specific definitions for a calendar.

#### **Follow these steps:**

1. Find the appropriate calendar.
2. Click the calendar name in the Search Results section.
3. Click Go.  
The Search Results section is displayed.

### **Edit a Calendar**

You can edit a calendar to modify its properties. For a standard calendar, you can change the specific days on which a job may process by manually selecting them or setting the rules that control processing days. For an extended calendar, you can modify specific days on which a job may process.

#### **Follow these steps:**

1. Find the appropriate calendar.
2. Click the calendar name in the Search Results section.  
The detailed properties for the calendar are displayed.
3. Edit the calendar definition as necessary.

#### **NOTE**

The Name field is read-only when you edit an existing calendar.

4. Click Commit.  
The changes are saved.

### **Delete a Calendar**

If you no longer have any jobs associated with a calendar, you can delete it.

#### **Follow these steps:**

1. Find the appropriate calendar.
2. Do one of the following:
  - Click the calendar name in the Search Results section and click Delete.
  - Select the Select check box in front of the appropriate calendar, and click the Delete link.

#### **NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

A confirmation prompt is displayed.

- Click OK.  
The calendar definition is deleted.

## Create an Extended Calendar

You can create an extended calendar to define processing days using any combination of holiday calendars, cycles, and conditions for scheduled jobs.

### Follow these steps:

1. Click Create.  
The Create Object dialog opens.
2. Select the appropriate server from the Server drop-down list and click the Extended Calendar link.
3. Define your extended calendar by performing the following actions:
  - a. Enter a name for the extended calendar in the Name field.

#### NOTE

- The name can contain up to 30 characters.
  - If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.
- b. (Optional) Enter date conditions for the extended calendar in the Conditions field.  
Examples:
    - DAILY - Selects all dates after the current date
    - MON - Selects all Mondays after the current date
    - FOMWEEK - Selects the first weekday of each month after the current date
    - EOMWEEK - Selects the last weekday of each month after the current date
    - XFOM - Selects all days except the first day of each month after the current date
    - CYCLE - Selects any day within the date ranges, or periods defined in the cycle.
  - c. (Optional) Enter the name of an existing cycle in the Cycle Name field or select a cycle from the Search Cycle dialog. Also, enter CYCLE in the Conditions field.
  - d. Enter any integer in the Adjustment field to adjust the days generated by the other criteria to either a previous day or a specified number of days after the actual generated day.
  - e. (Optional) Select a rule for holidays from the Reschedule rule list, and then enter the name of a standard calendar in the Calendar field or select a calendar from the Search Calendar dialog.
  - f. (Optional) Select an action for non-workdays from the Non Workday Action list, and then select at least one workday from the Workdays panel.
  - g. (Optional) Click Generate preview to open the Extended Calendar Preview dialog so you can preview the dates generated by the rules in the extended calendar.
4. (Optional) Provide a description
    - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

#### NOTE

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

5. Click Commit.  
The extended calendar definition is saved.

#### NOTE

If you create an extended calendar by entering a value in the Name field only and saving it, all days are selected from the current day forward, for a total of 366 days.

### Example: Create an Extended Calendar Based on Field Adjustment

To create an extended calendar that processes associated jobs one day after each holiday, you would first create a calendar defining the holidays for the year, then create an extended calendar with the following options selected:

- Your holidays calendar in the Calendar field
- Schedule only on a holiday in the Holiday Action list
- 1 in the Adjustment field

### **Copy an Existing Calendar**

You can copy an existing calendar to create a new calendar with the same properties. You must specify a different name for the new calendar; otherwise, the calendar you copied is simply updated.

#### **Follow these steps:**

1. Find the calendar to copy.
2. Do one of the following:
  - Click  in the Actions column to the right of that calendar.
  - Click the calendar name in the Search Results section. Next, select Copy from the Select and drop-down list, and click Go.

The Calendar or Extended Calendar Properties section is displayed.
3. Define your calendar by doing the following:
  - a. Enter a name for the calendar in the Name field.

#### **NOTE**

- The name can contain up to 30 characters.
- If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- b. Edit the calendar definition as necessary.
4. Click Commit.  
The new calendar definition is saved.

### **Create a Cycle**

You can create a cycle to define date ranges to be used as input to an extended calendar.

#### **Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Cycle link.
3. Define your cycle by performing the following actions:
  - a. Enter a name for the cycle in the Name field.

#### **NOTE**

- The name can contain up to 30 characters.
  - If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.
- b. (Optional) Select the Repeat every year check box to repeat the cycle every year.
  - c. Select Add from the Select an action drop-down list and click Go to add periods to the cycle.
    - One cycle has a maximum of 30 periods.
    - You can optionally click  select the period dates. This is available only when the Repeat every year check box is not selected.
    - At least one period must be added to the definition.

- d. (Optional) Select a period and select Add Before from the Select an action drop-down list. Click Go to add a new period before the selected period.
  - e. (Optional) Select a period, select Move Up from the Select an action drop-down list, and click Go to move the selected period up one level.
  - f. (Optional) Select a period, select Move Down from the Select an action drop-down list, and click Go to move the selected period down one level.
  - g. (Optional) Select a period, select Delete from the Select an action drop-down list, and click Go to delete the selected period from the cycle.
4. (Optional) Provide a description
    - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

#### NOTE

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

5. Click Commit.  
The cycle definition is saved.

#### Example: Create a Cycle and Associate it with an Extended Calendar

To associate a cycle with an extended calendar, you first create the cycle with the periods defined. When you create the extended calendar, enter CYCLE in the Conditions field. Next, enter the name of the cycle calendar in the Cycle name field, and click Generate Preview. You can now view the processing dates that are selected. You can also

click 

the right of the Cycle name field to select a cycle from the Search Cycle dialog.

#### Find a Cycle

You can use Quick Edit to find one or more AutoSys Workload Automation cycles residing on a specific server.

##### Follow these steps:

1. Select the appropriate server from the Server drop-down list.
2. Select Cycle from the Type drop-down list.
3. Enter one of the following in the Name field and click Go:
  - Valid cycle name -- Finds the specified cycle on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all cycles on the selected server that contain the partial name. For example, entering n\* would find all cycles on the server that begin with the letter n, and entering \*n would find all cycles on the server that end with the letter n.
  - Asterisk (\*) or percent (%) -- Finds all cycles on the selected server.
4. (Optional) Enter search criteria in any of the optional fields to further refine your search.

The Search Results section lists the cycle or cycles that meet your search criteria.

#### View a Cycle

You can review the date ranges to be used as input to an extended calendar.

##### Follow these steps:

1. Find the appropriate cycle.
2. Click the cycle name in the Search Results section.



3. Click Go.  
The Search Results section is displayed.

### **Copy an Existing Cycle**

You can copy an existing cycle to create a new cycle with the same properties. You must specify a different name for the new cycle; otherwise, the cycle you copied is simply updated.

#### **Follow these steps:**

1. Find the cycle to copy.
2. Do one of the following:
  -  Click in the Actions column to the right of that cycle.
  - Click the cycle name in the Search Results section. Next, select Copy from the Select and drop-down list, and click Go.  
The Cycle Properties section is displayed.
3. Define your cycle by doing the following:
  - a. Enter a name for the cycle in the Name field.

#### **NOTE**

- The name can contain up to 30 characters.
- If using only CA AE 11.3.6 SP2 (and any later versions), the limit of up to 64 characters applies.

- b. Edit the cycle definition as necessary.
4. (Optional) Provide a description
  - **Description**  
A non-mandatory description that contains any text.  
**Limits:** Up to 1024 characters.

#### **NOTE**

The description is available only if AE 11.3.6 SP2 (and any later versions) is used.

5. Click Commit.  
The new cycle definition is saved.

### **Edit a Cycle**

You can edit a cycle to modify existing periods and add or delete new periods.

#### **Follow these steps:**

1. Find the appropriate cycle.
2. Click the cycle name in the Search Results section.
3. Edit the cycle definition as necessary.

#### **NOTE**

The Name field is read-only when you edit an existing cycle.

4. Click Commit.  
The changes are saved.

## **Delete a Cycle**

If you no longer need specific date ranges defined for use in an extended calendar, you can delete the cycle that defines these date ranges.

### **Follow these steps:**

1. Find the appropriate cycle.
2. Do one of the following:
  - Click the cycle name in the Search Results section, and then click Delete.
  - Select the Select check box in front of the appropriate cycle, and click the Delete link.
 A confirmation prompt is displayed.

#### **NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

3. Click OK.  
The cycle is deleted.

## **Manage Global Variables in Quick Edit**

This article explains how to manage global variables using Quick Edit.

### **Find a Global Variable**

You can use Quick Edit to find one or more AutoSys Workload Automation global variables residing on a specific server.

#### **Follow these steps:**

1. Select the appropriate server from the Server drop-down list.
2. Select Global Variable from the Type drop-down list.
3. Enter one of the following in the Name field and click Go:
  - Valid global variable name -- Finds the specified cycle on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all global variables on the selected server that contain the partial name. For example, entering n\* would find all global variables on the server that begin with the letter n, and entering \*n would find all global variables on the server that end with the letter n.
  - Asterisk (\*) or percent (%) -- Finds all global variables on the selected server.

#### **NOTE**

The Search Results section lists the global variable or global variables that meet your search criteria.

### **View a Global Variable**

You can review the criteria specified for jobs on a particular server.

#### **Follow these steps:**

1. Find the appropriate global variable.
2. Click the global variable name in the Search Results section.  
The detailed properties for the global variable are displayed.
3. Click Go.  
The Search Results section is displayed.

## **Create a Global Variable**

You can create a global variable if you have a set of jobs you want to run based on specific criteria.

### **Follow these steps:**

1. Click Create.  
The Create Object dialog opens.
2. Select the appropriate server from the Server drop-down list and click the Global Variable link.  
The Properties section is displayed with the properties relevant to global variables.
3. Define your global variable by doing the following:
  - a. Enter a name for the new global variable in the Name field.

#### **NOTE**

The name can contain up to 64 characters.

- b. Enter a value for the new global variable in the Value field.

#### **NOTE**

The value can contain up to 100 characters.

4. Click Commit.  
The global variable is saved.

## **Copy a Global Variable**

You can copy an existing global variable to create a new global variable with the same properties. You must specify a different name for the new global variable; otherwise, the global variable you copied is simply updated.

### **Follow these steps:**

1. Find the global variable to copy.
2. Do one of the following:
  -  Click  in the Actions column to the right of that global variable.
  - Click the global variable name in the Search Results section. Next, select Copy from the Select and drop-down list, and click Go.  
The global variable Properties section is displayed.
3. Define your global variable by doing the following:
  - a. Enter a name for the global variable in the Name field.

#### **NOTE**

The name can contain up to 64 characters.

- b. Edit the value as necessary.

#### **NOTE**

The value can contain up to 100 characters.

4. Click Commit.  
The new global variable is saved.

## **Edit a Global Variable**

You can edit a global variable to modify its value.

### **Follow these steps:**

1. Find the appropriate global variable.
2. Click the global variable name in the Search Results section.

The global variable Properties section is displayed.

3. Edit the value as necessary.

**NOTE**

The Name field is read-only when you edit an existing global variable.

4. Click Commit.

The changes are saved.

### **Delete a Global Variable**

If you no longer need a global variable defined for a specific server, you can delete it.

**Follow these steps:**

1. Find the appropriate global variable.
2. Do one of the following:
  - Click the global variable name in the Search Results section, and click Delete.
  - Select the Select check box in front of the appropriate global variable and click the Delete link.

**NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

A confirmation prompt is displayed.

3. Click OK.

The global variable is deleted.

## **Manage Machines in Quick Edit**

This article explains how to manage your machines in Quick Edit.

### **Find a Machine**

You can use Quick Edit to find one or more machines in your environment.

**Follow these steps:**

1. Select the appropriate server from the Server drop-down list.
2. Select Machine from the Type drop-down list.
3. Enter one of the following in the Name field and click Go:
  - Valid machine name -- Finds the specified machine on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all machines on the selected server that contain the partial name. For example, entering n\* would find all machines on the server that begin with the letter n, and entering \*n would find all machines on the server that end with the letter n.
  - Asterisk (\*) or percent (%) -- Finds all machines on the selected server.

The Search Results section lists the machine or machines that meet your search criteria.

### **Create a Real Machine**

Although a real machine is a physical machine, it must be defined in AutoSys Workload Automation to run jobs or be added to a virtual machine. You can define a real machine using Quick Edit.

**Follow these steps:**

1. Click Create.
2. Select the appropriate server from the Server drop-down list and click the Real Machine link.  
The Properties section is displayed with the properties relevant to real machines.
3. Define your machine by doing the following:
  - a. Enter a name for the new machine in the Name field.

**NOTE**

The name can contain up to 80 characters.

- b. Complete the fields in each category as appropriate.
4. Click Commit.  
The real machine is saved.

**Create a Virtual Machine**

You can define virtual machines on which to run jobs you create.

**Follow these steps:**

1. Click Create.  
The Create Object dialog opens.
2. Select the appropriate server from the Server drop-down list and click the Virtual Machine link.  
The Properties section is displayed with the properties relevant to virtual machines.
3. Define your virtual machine by doing the following:
  - a. Enter a name for the new machine in the Name field.

**NOTE**

The name can contain up to 80 characters.

- b. Complete the fields as appropriate.
4. Click Commit.  
The virtual machine is saved.

**Add a Real Machine to an Existing Virtual Machine Definition**

You can add one or more real machines to a virtual machine definition.

**Follow these steps:**

1. Find the appropriate virtual machine.
2. Click the machine name in the Search Results section.  
The detailed properties for the machine are displayed.
3. Enter the name of the real machine to associate with this virtual machine in the Machines field. Alternatively, you can click



to open the Search Machine dialog, from which you can select one or more machines to add to the virtual machine definition.

**NOTE**

- Only real machines can be associated with a virtual machine. If a virtual machine name is entered, an error is displayed when your changes are committed.
  - The Search Machine dialog contains both virtual and real machines.
4. (Optional) Enter the maximum load for the machine in the Max load field. This field is available for v (virtual) and n (Windows) type virtual machines only.

5. (Optional) Enter the multiplication factor in the Factor field. This field is available for v (virtual) and n (Windows) type virtual machines only.
6. Click Commit.  
The real machine is added to the virtual machine definition.

### **Remove a Real Machine from an Existing Virtual Machine Definition**

If you no longer need a specific machine associated with a virtual machine, you can remove the real machine from the virtual machine definition.

#### **Follow these steps:**

1. Find the appropriate virtual machine.
2. Click the machine name in the Search Results section.  
The detailed properties for the virtual machine are displayed.
3. Select the check box next to the appropriate machine in the Machines section and click the Delete link.

#### **NOTE**

- You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.
- At least one real machine must be associated with a virtual machine definition.

4. Click Commit.  
The real machine is removed from the virtual machine definition.

### **Edit a Machine**

You can use Quick Edit to edit the properties of a machine.

You can modify all properties of a machine, except the machine name and type.

#### **Follow these steps:**

1. Find the appropriate machine.
2. Click the machine name in the Search Results section.  
The detailed properties for the machine are displayed.
3. Edit the machine definition as necessary.
4. Click Commit.  
The changes are saved.

### **Delete a Machine**

If you no longer need a machine, you can delete it.

#### **Follow these steps:**

1. Find the appropriate machine.
2. Do one of the following:
  - Click the machine name in the Search Results section, and click Delete.
  - Select the check box next to the appropriate machine and click the Delete link.
 A confirmation prompt is displayed.

#### **NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

3. Click OK.

The machine definition is deleted.

## Manage Connection Profiles in Quick Edit

This article explains how to manage connection profiles.

### Find a Connection Profile

You can use Quick Edit to find a AutoSys Workload Automation connection profile residing on a specific server.

#### Follow these steps:

1. Select the appropriate server from the Server drop-down list.
2. Select Connection Profile from the Type drop-down list.
3. Enter one of the following in the Name field and click Go:
  - Valid connection profile name -- Finds the specified cycle on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all connection profiles on the selected server that contain the partial name. For example, entering n\* would find all connection profiles on the server that begin with the letter n, and entering \*n would find all connection profiles on the server that end with the letter n.
  - Asterisk (\*) or percent (%) -- Finds all connection profiles on the selected server.

### View a Connection Profile

You can review the criteria specified for jobs on a particular server.

#### Follow these steps:

1. Find the appropriate connection profile.
2. Click the connection profile name in the Search Results section.  
The detailed properties for the connection profile is displayed.
3. Click Go.  
The Search Results section is displayed.

### Create a Connection Profile

Before you define a Hadoop job, create a connection profile. You can create one or more connection profiles specific to the Hadoop environment.

#### Follow these steps:

1. Create a Connection Profile.
2. Specify the following properties: value combinations (as appropriate) based on whether you want to connect to the Hadoop cluster, Oozie server, Hive database or Scoop database:
  - Connection name** Defines the name of the connection profile.
  - Description** (Optional) Specifies a description for the connection profile.
  - Host** Defines the name of the Hadoop edge node where the specified operation is performed. This can be a simple name, fully qualified domain name, or IP address.
  - Port** (Optional) Specifies the Secure Shell (SSH) port on the Hadoop edge node.
  - Default:** 22
  - Hadoop bin directory path** (Optional) Specifies the path to the Hadoop client binaries on the Hadoop edge node.
  - Notes:**

- If the path includes a colon, enclose it in double quotation marks.
- Specify this attribute only if the Hadoop client binaries are not in the user's path. The hadoop bin directory path value is prepended to the client name at execution time.

#### **Hadoop name node**

(Optional) Specifies the namenode of the Hadoop HDFS cluster.

#### **Oozie server URL**

(Optional) Specifies the Oozie server URL.

3. Specify the following attributes for Hive database connection:

**JDBC URL** Specifies the JDBC URL that you provide to the Hadoop Beeline client to connect to the database.

**Note:** If you are creating a connection profile to connect to the Hive database, you must specify the JDBC URL attribute in the connection profile.

#### **Drive class**

(Optional) Specifies the name of the JDBC driver class that you provide to the Hadoop Beeline client.

4. Specify the following attributes for Sqoop database connection:

**JDBC URL** Specifies the JDBC URL that you provide to the Hadoop Sqoop client to connect to the database.

**Note:** If you are creating a connection profile to connect to the Sqoop database, you must specify the JDBC URL attribute in the connection profile.

#### **Drive class**

(Optional) Specifies the name of the JDBC driver class to execute on the database.

The connection profile is created and added in the database.

**Note:** The availability to create, modify, and delete a connection profile depends on the user's access rights defined in CA EEM.

### **Copy a Connection Profile**

You can copy an existing global variable to create a new connection profile with the same properties. You must specify a different name for the new connection profile; otherwise, the connection profile you copied is simply updated.

#### **Follow these steps:**

1. Find the connection profile to copy.
2. Do one of the following:

- Click



in the Actions column to the right of that connection profile.

located

- Click the connection profile name in the Search Results section. Next, select Copy from the Select and drop-down list, and click Go.

The connection profile Properties section is displayed.

3. Define your connection profile by doing the following:
  - a. Enter a name for the connection profile in the Name field.

**Note:** The name can contain up to 64 characters.

- b. Edit the value as necessary.

**Note:** The value can contain up to 100 characters.

4. Click Commit.

The new connection profile is saved.

### **Edit a Connection Profile**

You can edit a connection profile to modify its value.



**Follow these steps:**

1. Find the appropriate connection profile.
2. Click the connection profile name in the Search Results section.  
The connection profile Properties section is displayed.
3. Edit the value as necessary.  
**Note:** The Name field is read-only when you edit an existing connection profile.
4. Click Commit.  
The changes are saved.

**Delete a Connection Profile**

If you no longer need a connection profile defined for a specific server, you can delete it.

**Follow these steps:**

1. Find the appropriate connection profile.
2. Do one of the following:
  - Click the connection profile name in the Search Results section, and click Delete.
  - Select the Select check box in front of the appropriate connection profile and click the Delete link.

**NOTE**

You can delete multiple objects at one time by selecting the Select check boxes for all of the objects to be deleted and clicking the Delete link.

A confirmation prompt is displayed.

3. Click OK.  
The connection profile is deleted.

## Web UI Quick View Application

Quick View can be used as a diagnostic tool to monitor critical aspects of jobs that you run from AutoSys Workload Automation, as an alternative to searching through a command line report for the job status information you want. If needed, you can print the visible portion of the job flow or save an image of the entire job flow. You can access Quick View by clicking the Quick View tab on the main Web UI page.

For added convenience, from within Quick View, you can access Quick Edit to edit the properties of the selected job, or access the Send Event dialog to send an event to one or more selected jobs.

**Quick View**

Quick View lets you access consolidated job detail information for AutoSys Workload Automation servers. Whenever you search for a job, Quick View accesses the AutoSys Workload Automation API, then parses the resulting information for display.

For the selected AutoSys Workload Automation job, Quick View displays the job details, job execution flow, dependencies, starting conditions, prior job run logs, and prior job run events. Job details include specific job properties such as job type, start and end time, and priority. Only run time properties (Start time, End time, Status, Run, Priority, Exit Code) and the properties for which values have been entered in the job definition are displayed.

**NOTE**

You cannot edit job properties directly in Quick View; however, if you are authorized to use Quick Edit, clicking Quick Edit in the Job Details section opens a new window that contains a subset of the functionality of the Quick Edit application. If you need the full functionality of Quick Edit, you can access the application by clicking the QEDIT tab. For more information, see *Quick Edit Help*.

All upstream and downstream levels are displayed by default in the graphical Job Flow. Job dependencies are additional jobs that can run only if a condition placed on the selected job is satisfied, similar to successors. For example, if the selected job runs successfully, one dependent job runs; if it fails, a different dependent job runs. Starting conditions are conditions that must be satisfied for the selected job to start. Each starting condition listed in the table represents one clause in a string, separated by AND or OR logic. The selected job can start only if the entire string evaluates to true. Resource dependencies are resources that the selected job needs in order to run. Details of prior job runs can be displayed, and you can also view the Standard Out log, Standard Error log, or Agent Log for a prior run of the selected AutoSys Workload Automation command or user defined job. Additionally, logs can be displayed for the job types that support agent logs. Events that occurred during prior job runs can be displayed with their associated event details, and resulting alarms, if any.

#### NOTE

- Because the data displayed in Quick View is retrieved from three different sources (AutoSys Workload Automation server requests, the Web UI DB, and the Flow Model), job details, dependencies, and job status information can display different results. For example, a job created in Quick Edit and retrieved in Quick View can be displayed in the Search Results section but its details may not be displayed if the job collector run is not completed. Additionally, job status in the Job Details section can indicate running while the latest job status in the Job Run section can indicate success, due to timing differences resulting from the way the requests are handled by the AutoSys Workload Automation server.
- Performing a Refresh updates the information displayed on the page. The Job Collector Run Interval specifies the interval at which data is obtained from the AutoSys Workload Automation servers and server changes are synchronized. This value is set in the Monitoring Preferences section in Configuration Manager. For more information, see *Configuration Manager Help*.

#### Quick View Navigation

When you start a Quick View session, the initial page displays the Search section. You can then specify a server and search for a specific job, list of multiple jobs separated by a comma, spaces, new line separators, or find jobs with names that contain the characters you specify. All of these job combinations support wildcards.

#### NOTE

New line separators in job name are supported by Chrome, Firefox, and Safari.

#### AutoSys Workload Automation Job Information

Quick View displays the following information for the AutoSys Workload Automation job you specify:

- Job Details -- Displays job properties such as job name, job type, start and end times, status, and machine.

#### NOTE

Only the run time properties (Start time, End time, Status, Run, Priority, Exit Code) and properties for which values have been specified are displayed in Quick View.

- Job Flow -- Graphically displays the flow of the selected job or box job, including its upstream (predecessor) and downstream (successor) jobs or box jobs. If applicable, job resource and global variable dependencies are also displayed. Additionally, jobs on other Web UI servers that are dependencies of the selected job (cross-reference job dependencies) are displayed with the status and job name. However, if the data for the cross-referenced job is not collected in Web UI (for example, a CA Workload Automation CA 7 Edition job) and the job is a predecessor to a job

in the view, the dependency is displayed with an unknown job icon, and the job and instance names. Cross-reference dependency links are displayed with a dashed line.

- **Dependent Jobs** -- Lists the additional jobs that can run only if a condition placed on the selected job is satisfied, similar to successors. For example, if the selected job runs successfully, one dependent job runs; if it fails, a different dependent job runs.
- **Starting Conditions** -- Displays the conditions that must be satisfied for the selected job to start, similar to predecessors.
  - **Job Dependencies** -- Lists dependencies and their details, including atomic condition, expected value, current status, and whether the condition was satisfied.

#### **NOTE**

Quick View displays each job or global variable in the condition string, enabling you to pinpoint which specific starting conditions were met.

- **Job Resource Dependencies** -- Lists the resources that the selected job is dependent on. Resource details are also displayed.
- **Job Runs** -- Lists the details of prior job runs such as start and end times, status, run, exit code, and machine.
  - **Events** -- Lists each event for a prior job run, including event status, event text, the time the event was sent, number of retries, event state, the time the event was processed, the name of the machine that the job was run on, and any related alarms.
  - **Logs** -- Displays the log type you select for a prior job run. For a command or user defined job, one of the following can be selected: Standard Out, Standard Error, or Agent Log. Additionally, logs can be displayed for the job types that support agent logs.

#### **NOTE**

For some job types, no log is available.

## **Display Preferences**

You can customize your Quick View display by selecting the following options:

- The number of rows per page to display in the Search Results table.
- The number of rows per page to display in a report table.
- The number of search history items to display.
- The job details default format (tree or JIL).
- The maximum height of the Job Flow section.
- The report types to display.

Three report types are available for the selected job: Dependent Jobs, Starting Conditions, which includes job resource dependencies, and Job Runs, which includes job run events. Initially, reports of all types are displayed, but you can specify the report types that you want to be displayed for your jobs using the Customize dialog. For example, if you plan to use Quick View to determine why a job has not started or whether a job will run, you can select the Dependent Jobs and Starting Conditions report types. If you plan to use Quick View to determine why a job has failed or to view past performance, you can select the Job Runs report type.

## **Security**

Depending on the CA EEM security rights assigned to you by your administrator, you may not have access to all functionality in Quick View. For example, if you do not have send event rights, the send event functionality is not available in the Search Results and Job Flow sections.

## Prior Job Run Logs

You can display a prior job run log in the Job Runs section. The following table contains the job types that support logging and the logs that you can display:

Job Type	Available Logs
Command (CMD)	Standard Output, Error Output, Agent Log
Database Monitor (DBMON)	Spool File
Database Stored Procedure (DBPROC)	Spool File
File Watcher (FW)	Agent Log
FTP (FTP)	Spool File
HDFS (HDFS)	Spool File
Hive (HIVE)	Spool File
HTTP	Spool File
i5/OS (I5)	Spool File
Informatica	Spool File, Workflow Log
Micro Focus (MICROFOCUS)	Spool file, JES file
MS SQL Server	Step Log
Oracle E-Business Suite Copy Single Request (OACOPY)	View Log, View Output, View Completion Details
Oracle E-Business Suite Request Set (OASET)	View Log, View Output, View Completion Details
Oracle E-Business Suite Single Request (OASG)	View Log, View Output, View Completion Details
Oozie (OOZIE)	Spool File
Pig (PIG)	Spool File
POJO	Spool File
Process Automation Process Execution (PAPROC)	Spool File
Process Automation Start Request Form (PAREQ)	Spool File
PeopleSoft (PS)	Trace File, Log File
Remote Execution (PROXY)	Spool file, Remote Spool File, Remote Spool File Reference
SAP Batch Input Session (SAPBDC)	Spool File
SAP BW InfoPackage (SAPBWIP)	Spool File
SAP BW Process Chain (SAPBWPC)	Spool File, Get Chain Process Log, Get Chain Log
SAP Data Archiving (SAPDA)	Spool File
SAP Job Copy (SAPJC)	Spool File, Job Log Read, Job Get Spool List
SAP R/3 (SAP)	Spool File, Job Log Read, Job Get Spool List
Secure Copy (SCP)	Spool File
SQL (SQL)	Spool File
Sqoop (SQOOP)	Spool File
User Defined	Standard Output, Error Output, Agent Log
Web Service Document/Literal (WSDOC)	Spool File
z/OS Regular (ZOS)	Spool File

### NOTE

Prior job run logs are not available for the following job types:

- Box (BOX)
- CPU Monitoring (OMCPU)
- Database Trigger (DBTRIG)
- Disk Monitoring (OMD)
- Entity Bean (ENTYBEAN)
- File Trigger (FT)
- IP Monitoring (OMIP)
- JMS Publish (JMSPUB)
- JMS Subscribe (JMSSUB)
- JMX-MBean Attribute Get (JMXMAG)
- JMX-MBean Attribute Set (JMXMAS)
- JMX-MBean Create Instance (JMXMC)
- JMX-MBean Operation (JMXMOP)
- JMX-MBean Remove Instance (JMXMREM)
- JMX-MBean Subscribe (JMXSUB)
- Process Monitoring (OMP)
- RMI (JAVARMI)
- SAP Event Monitor (SAPEVT)
- SAP Process Monitor (SAPPM)
- Session Bean (SESSBEAN)
- SNMP Value Get (SNMPGet)
- SNMP Value Set (SNMPSet)
- Text File Reading and Monitoring (OMTF)
- Web Service (WBSVC)
- Windows Event Log Monitoring (OMEL)
- Windows Service Monitoring (OMS)
- Wake on LAN (WOL)
- z/OS Data Set Trigger (ZOSDST)
- z/OS Manual (ZOSM)

## View Consolidated Job Details in Quick View

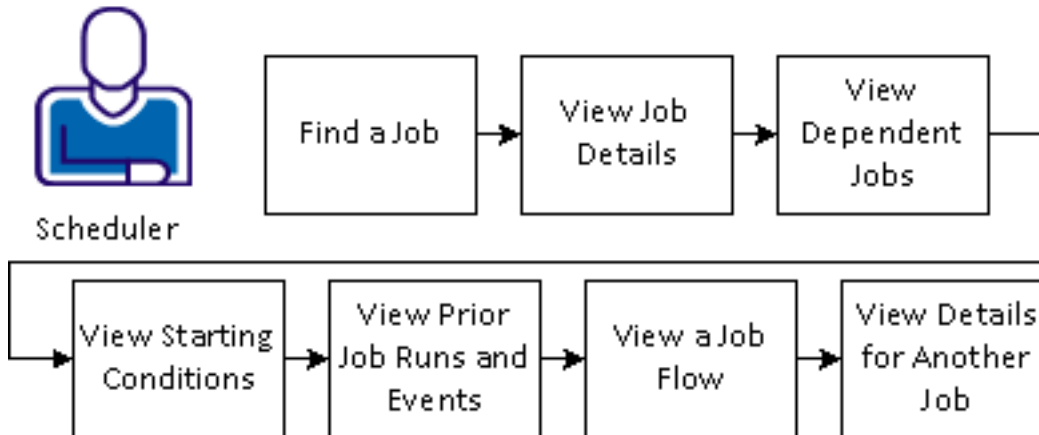
This article explains how to view consolidate Job details in Quick View. As a scheduler, you can use Quick View to search for jobs, monitor critical aspects of jobs, execute commands on selected jobs, and access job definitions. You can access Quick View by clicking the Quick View tab, or it can be opened for a selected job in Monitoring.

You can use Quick View to do any of the following tasks:

- Search for a job and view its details
- View box hierarchy for a job or job stream
- View dependency information for the selected job in text or graphical format
- View the dependencies in a job stream
- View starting conditions and dependencies, including resource dependencies for the selected job
- View job properties for the selected job
- View the JIL for the selected job
- View job runs with job run events
- View agent logs, and the standard and error outputs for the selected job run

The following scenario walks you through the process of finding a specific job, and viewing job details.

## How to View Consolidated Job Details



To find a job and view the job details, you can follow these steps:

1. [Find a job](#)
2. [View job details](#)
3. [View dependent jobs](#)
4. [View starting conditions](#)
5. [View prior job runs and events](#)
6. [View a job flow](#)
7. [View details for another job](#)

### Example: View Detailed Properties for the WORLD01\_001 Job

Perhaps you are interested in seeing a consolidated view of the WORLD01\_001 job properties, dependencies, and past run history. You can find the job in Quick View and then view the job properties, dependencies, and run history on a single page. After viewing the details of that job, you can view the details for dependent job WORLD01\_002B.

#### Find a Job

You can use the Search section to find one or more jobs with common characters in their names that reside on a specific server. Details such as start and end time, status, run, and priority/exit code are displayed for the jobs that are found.

#### Follow these steps:

1. Select a server from the Server drop-down list.
2. Enter one of the following in the Job Name field and click Go:
  - Valid job name -- Finds the specified job on the selected server.
  - Asterisk (\*), percent (%), or question mark (?), used as a wildcard with a partial name, where the \* or % represents zero or more characters, or ? represents a single character -- Finds all jobs on the selected server that contain the partial name. For example, entering n\* would find all jobs on the server that begin with the letter n.
  - Asterisk (\*), percent (%), or ALL -- Finds all jobs on the selected server.

One of the following occurs:

- If only one job is found, detailed information about the job and the selected reports are displayed.
- If two or more jobs are found, the jobs are listed in the Search Results section.

#### NOTE

- Multiple names are not permitted.
- The Search Results section includes jobs and box jobs.

## **View Job Details**

You can view the details for a job listed in the Search Results table to quickly evaluate its current status. In Quick View, all job details can be displayed on one page, either in a table or as JIL. In the table view, you can expand or collapse the property categories to show or hide job definition details as needed.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.

### **NOTE**

- Only the properties for which values have been specified are displayed.
- Job details are displayed in the Job Details section using the format selected in the Customize dialog (Tree or JIL). You can toggle the display to show job details as JIL by clicking Show JIL, or in table format by clicking Show Tree.

## **View Dependent Jobs**

A job flow is comprised of the selected job and its dependencies. Dependencies are additional jobs that can run only if a condition placed on the selected job is satisfied, similar to successors. You can view the dependencies for the selected job in the Dependent Jobs section.

### **NOTE**

To display this section, Show Dependent Jobs must be selected in the Customize dialog.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Dependent Jobs navigation link.  
The Dependent Jobs section appears.

## **View Starting Conditions**

Starting conditions are job dependencies and resource dependencies that must be satisfied for the selected job to start. The Starting Conditions section displays this information in two sub-sections: Job Dependencies and Job Resource Dependencies.

### **NOTE**

To display this section, Show Starting Conditions must be selected in the Customize dialog.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.

2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Starting Conditions navigation link.  
The Starting Conditions section appears, and the job dependencies and resource dependencies for the last job run are displayed.

### **View Job Dependencies**

Job dependencies are conditions that must be satisfied for the selected job to start. Each condition listed in the table represents one clause in a string, separated by AND or OR logic. The selected job can start only if the entire string evaluates to true. You can view the job dependency conditions for the selected job in the Job Dependencies sub-section of the Starting Conditions section.

#### **NOTE**

To display this section, Show Starting Conditions must be selected in the Customize dialog.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Starting Conditions navigation link.  
The Starting Conditions section appears and job dependencies are displayed.

### **View Job Resource Dependencies**

A job requires resources to run. You can view the resources that the selected job is dependent on in the Job Resource Dependencies sub-section of the Starting Conditions section.

#### **NOTE**

To display this section, Show Starting Conditions must be selected in the Customize dialog.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Starting Conditions navigation link.  
The Starting Conditions section appears and job resource dependencies for the last job run are displayed.

### **View Prior Job Runs and Events**

Prior job runs are listed in the table in the Job Runs section. Event details for each job run can be displayed including resulting alarms, if any.

#### **NOTE**

To display this section, Show Job Runs must be selected in the Customize dialog.

#### **Follow these steps:**

1. Find a job.



One of the following occurs:

- If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
  3. Click the Job Runs navigation link.  
The Job Runs section appears with a list of prior job runs.
  4. Click  the Details column for the appropriate job run.  
Event details for the job run are displayed.

### **View a Job Flow**

A job flow is comprised of the selected job and its upstream and downstream jobs. You can view the job flow for the selected job graphically in the Job Flow section. You can also initiate an action to be performed on a job shown in the flow.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Job Flow navigation link.  
The Job Flow section appears
4. Click  the Job Flow section.  
The Job Flow section expands, showing the selected job, and its upstream and downstream jobs.

#### **NOTE**

You can select 0-10 upstream and downstream levels using the controls in the Job Flow section.

### **View Details for Another Job**

Job details are displayed for the job selected from the search results (the current job). You can view the details for another job displayed in the Job Flow.

#### **NOTE**

Job details can be displayed for only one job at a time.

To display the details for another job, right-click the job you want in the Job Flow section and select Jump to.

Details are displayed for that job.

## **Manage Jobs in Quick View**

This article explains how to manage jobs in Quick View.

### **Edit Job Properties**

You may want to modify the properties of a job. From Quick View, you can access Quick Edit to modify job properties if you have the required user permissions.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click Quick Edit.  
Quick Edit opens in a separate browser window.
4. Modify the job properties as needed and save the changes.

**NOTE**

For information about editing job definitions in Quick Edit, see the *Quick Edit Help*.

5. Click Go.  
Quick View refreshes and reflects the changes.

**View Box Details**

If a job listed in the initial Search Results table is contained in a box, you can view the box details to quickly evaluate its current status.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. Click the link in the Box Name column of the Search Results table for which you want to view box details.  
A tree of the jobs contained in the box is displayed in the Search Results section, and the box details are displayed in the Job Details section. Initially, the tree is collapsed.

**NOTE**

- You can display the details for a specific job within the box by clicking a job name.
- Only the properties for which values have been specified are displayed.
- If the name in the Box Name field is the same name as the box for which details are displayed, clicking this link causes the box details tree to collapse.

**View a Prior Job Run Log**

You can view the following logs for the selected Command or User Defined job run: Standard Out, Standard Error, and Agent Log. Additionally, you can view the Agent Log for a File Watcher job.

**NOTE**

To display this section, Show Job Runs must be selected in the Customize dialog.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.

3. Click the Job Runs navigation link or scroll down to the Job Runs section, and select the job run for which you want to display the log.
4. Select one of the following logs from the Action drop-down list and click Go: Standard Out, Standard Error, or Agent Log.  
The selected log is displayed.

**NOTE**

You may receive an error message at the top of the page if log files are not defined for the job or if the log file locations defined for the job, using either the AutoSys Workload Automation command line or Quick Edit, were not entered as absolute paths. Log names should be entered with absolute paths. Additionally, to retrieve previous standard output and standard error logs, the file name must include the AutoSys Workload Automation variable AUTORUN. For more information, see the *AutoSys Workload Automation Reference* section.

**View an Agent Log**

For job types that support agent logs, you can view the agent log for the selected job run.

**NOTE**

To display this section, Show Job Runs must be selected in the Customize dialog.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Click the Job Runs navigation link or scroll down to the Job Runs section, and select the job run for which you want to display the log.

**NOTE**

If agent logs are not supported by the selected job, you cannot select a job run.

4. Select one of the following logs from the Action drop-down list and click Go:
  - Spool File -- Displays the job run output messages for the selected Database Monitor, Database Stored Procedure, FTP, HTTP, Informatica, SAP Batch Input Session, Micro Focus, POJO, Process Automation Process Execution, Process Automation Start Request Form, Remote Execution, SAP BW InfoPackage, SAP BW Process Chain, SAP Data Archiving, SAP Job Copy, SAP R/3, Secure Copy, SQL, Web Service Document/Literal, or User Defined job run in text format.
  - Spool File (z/OS regular) -- Displays the job run output messages for the z/OS regular job run.
  - Spool File (i5/OS) -- Displays the job run output messages for the i5/OS job run.
  - Remote Spool File -- Displays the remote job run output messages for the Remote Execution job run in text format.
  - Remote Spool File Reference -- Displays the reference for the spool file Remote Execution job.
  - JES File -- Displays the spool file for the Micro Focus job.
  - Job Get Spool List -- Displays the output that the selected ABAP produced according to the print parameters for the selected SAP R/3 or SAP Job Copy job run in text format.
  - Job Log Read -- Displays the SAP system job log for the selected SAP R/3 or SAP Job Copy job in table format.
  - Get Chain Log -- Displays the log for the selected SAP BW Process Chain job run in table format.
  - Get Chain Process Log -- Displays the process log of a SAP BW Process Chain job run in table format.
  - Step Log -- Displays the log file for the MS SQL Server job
  - View Log -- Displays the log for the selected Oracle job run in text format.
  - View Output -- Displays the output for the selected Oracle job run in text format.

**NOTE**

Some Oracle jobs generate empty output files or no output files at all.

- View Completion Details -- Displays the completion details for the selected Oracle job run in text format.
  - Workflow Log - Displays the log file for the Informatica job
  - Log File -- Displays the log file for a PeopleSoft job in text format.
  - Trace File -- Displays the trace file for a PeopleSoft job in text format.
5. (Optional) Depending on your selection one of the following may occur:
- If you selected Spool File, the Spool File Attributes dialog opens so that you can specify selection criteria or specify the number of lines from the end of the spool file to display.
  - For an i5/OS job, you must select one of the available spool files. For a z/OS Regular job, you can only select one of the available spool files. Enter the required information and click OK.
  - If you selected Job Get Spool List, the Job Get Spool List Attributes dialog opens. Enter the step number that you want to display and click OK.

**NOTE**

Depending on the amount of data retrieved for SAP log types, the FTP Credentials dialog may open. If this occurs, enter the required information and click OK.

- If you selected Get Chain Process Log, the Process list table is displayed. Select a process from the table and click OK.

The agent log is displayed.

**Set Display Preferences**

You can customize your Quick View display by selecting the following options:

- The number of rows per page to display in the Search Results table.
- The number of rows per page to display in a report table.
- The number of search history items to display.
- The job details default format (tree or JIL).
- The maximum height of the Job Flow section.
- The report types to display.

**Follow these steps:**

1. Click Customize.  
The Customize dialog opens.
2. Select the display options and report types that you want to display and click Save.  
The dialog closes and your changes are saved.

**NOTE**

Reports for the report types you have selected are populated and displayed only after you click Go again either with the existing search criteria or new search criteria. If multiple jobs are found, select the job that you want from the search results list.

**Print the Job Flow Image**

The Flow toolbar lets you zoom in on and print portions of the flow. This is helpful with complex flows because it lets you examine each portion of the flow in detail.

**Follow these steps:**

1. Click



on the toolbar.

2. Select File, Print from the menu options.
3. Select a printer, modify its settings if necessary, and click Print.

**NOTE**

- You may need to select the Print background colors and images option in your internet browser for the page to print correctly.
- If your browser is blocking popup windows, the print dialog may not appear.

The job flow image is printed.

**Save the Job Flow Image**

You can download the job flow as a .png image or .pdf document and save it to a file.

**Follow these steps:**

1. Click



on the toolbar and click Save.

2. Select a folder, enter a file name in the Save As dialog, and click Save.  
The image is saved.

**Start a Job**

You can use the Start command to send a STARTJOB event to start the selected job or top-level box job when its starting conditions are satisfied. The STARTJOB event ignores date and time conditions, but it enforces all other starting conditions such as job dependencies.

**NOTE**

You cannot send a STARTJOB event to a job that is contained in a box job. Jobs in box jobs inherit the starting conditions of these box jobs; as a result, they start when the box job starts. If the job is contained in a box job, the STARTJOB event has no effect.

**Follow these steps:**

1. Find a job.

One of the following occurs:

- If only one job is found, detailed information about the job is displayed.
- If two or more jobs are found, the jobs are listed in the Search Results section.

2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.

3. Do one of the following:

- Select Start from the drop-down list in the Search Results section and click Go.
- Click



in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Start from the pop-up menu.

A confirmation prompt appears.

4. Click Yes.

The selected job is scheduled to start when its starting conditions, other than date and time, are satisfied.

## **Force Start a Job**

You can use the Force Start command to send a FORCE\_STARTJOB event to the selected job or box job. The manually-generated FORCE\_STARTJOB event starts the specified job, regardless of whether its starting conditions are satisfied. Because you can use this command to start multiple instances of the same job, we recommend that you use the FORCE\_STARTJOB event only in extreme situations.

### **NOTE**

If you assign a resource as a dependency to a job, and that resource is not available, a FORCE\_STARTJOB event will not run.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Force Start from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Force Start from the pop-up menu.

A confirmation prompt appears.
4. Click Yes.  
The selected job is started.

## **Put a Job on Hold**

You can use the On Hold command to send a JOB\_ON\_HOLD event to the selected job or box job. Jobs that are on hold do not start, temporarily preventing the remainder of the job flow from executing. If the job's status is STARTING or RUNNING, the On Hold command has no effect.

### **NOTE**

A box job cannot successfully complete if a job in it is on hold.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select On Hold from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select On Hold from the pop-up menu.

A confirmation prompt appears.
4. Click Yes.

The job is put on hold.

### **Take a Job off Hold**

You can use the Off Hold command to send a JOB\_OFF\_HOLD event to the selected job or box job. The JOB\_OFF\_HOLD event instructs the scheduling manager to remove the specified job from ON\_HOLD status. If the job's starting conditions are met, the job starts.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Off Hold from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Off Hold from the pop-up menu.  
A confirmation prompt appears.
4. Click Yes.  
The selected job is taken off hold.

### **Put a Job on Ice**

You can use the On Ice command to send a JOB\_ON\_ICE event to the selected job or box job, changing the job's status to ON\_ICE. Putting a job on ice is the functional equivalent of deactivating the job definition. If the job's status is STARTING or RUNNING, the On Ice command has no effect.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select On Ice from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select On Ice from the pop-up menu.  
A confirmation prompt appears.
4. Click Yes.  
The selected job is put on ice.

### **Take a Job off Ice**

You can use the Off Ice command to send a JOB\_OFF\_ICE event to the selected job or box job. The JOB\_OFF\_ICE event instructs the scheduling manager to remove the specified job from ON\_ICE status. Jobs that are taken off ice do not start until the next time their starting conditions are met. If the specified job is in a box job with a RUNNING status, the scheduling manager attempts to start it, conditions permitting.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Off Ice from the drop-down list in the Search Results section and click Go
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Off Ice from the pop-up menu.
 A confirmation prompt appears.
4. Click Yes.  
The selected job is taken off ice.

### **Put a Job On No Execution**

You can use the On No Execution command to send an ON\_NOEXEC event to the selected job or box job, changing the job status to ON\_NOEXEC. Putting a job on no execution is the functional equivalent of bypassing the job during the scheduled run.

#### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select On No Execution from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select On No Execution from the pop-up menu.
 A confirmation prompt appears.
4. Click Yes.  
The selected job is put on no execution.

### **Kill a Job**

You can use the Kill command to send a KILLJOB event to the selected job or box job. The manually-generated KILLJOB event instructs the scheduling manager to kill the specified job if it is RUNNING and change its status to TERMINATED. If the job's status is not RUNNING, the Kill command has no effect.



**NOTE**

If the selected job is a box job, the scheduling manager changes the box job's status to TERMINATED and (if configured to do so) also kills the jobs it contains. The result of this command depends on the type of job selected and the type of server (UNIX or Windows) on which the job is running.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Kill from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Kill from the pop-up menu.  
A confirmation prompt appears.
4. Click Yes.  
The selected job is terminated.

**Take a Job Off No Execution**

You can use the Off No Execution command to send an OFF\_NOEXEC event to the selected job or box job. The OFF\_NOEXEC event instructs the scheduling manager to remove the specified job from ON\_NOEXEC status. Jobs that are taken off no execution status start the next time their starting conditions are met.

**Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Off No Execution from the drop-down list in the Search Results section and click Go
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Off No Execution from the pop-up menu.  
A confirmation prompt appears.
4. Click Yes.  
The selected job is taken off no execution.

**Release a Resource**

Renewable resources can be defined using the Free attribute with one of the following options: Y (free resource on success only), A (free resource unconditionally always), or N (do not free resources).

If you assign a resource as a dependency to a job with the attribute Free=N, and that job is terminated, the resource continues to be held by that job. For this resource to be available to other dependent jobs waiting to run, you must use the RELEASE\_RESOURCE sendevent command to free the resource.

#### NOTE

The Free attribute is only valid for renewable resources.

#### Follow these steps:

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Release Resources from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, select the appropriate job or box job in the flow, right-click, and select Release Resources from the pop-up menu.  
A confirmation prompt appears.
4. Click Yes.  
The resources for the selected job are released.

#### Send a Reply to an i5/OS Job

To send a reply to an i5/OS agent notification requesting information required for an i5/OS job run to continue, you can use the Reply command.

#### Follow these steps:

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the i5/OS job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Reply from the drop-down list in the Search Results section and click Go.
  - Click  in the Actions column for the job in the Search Results section.
  - Click  in the Job Flow section, select the appropriate job in the flow, right-click, and select Reply from the pop-up menu.  
Reply and  appear only if the selected i5/OS job is waiting on a reply.  
The Send Reply dialog opens.
4. Enter the appropriate reply in the Reply field and click OK.

The Send Reply dialog closes, a confirmation message is displayed, and the reply is sent to the i5/OS agent.

### **Resume an Event to a Hadoop Oozie Job**

To resume a reply to an Oozie agent notification requesting information required for an Oozie job run to continue, you can use the Resume command.

#### **Follow these steps:**

1. Find an Oozie job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the Oozie job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Resume from the drop-down list in the Send event dialog section and click Go.
  - Click



in the Actions column for the job in the Search Results section.

- Select the appropriate job in the flow, right-click, and select Resume from the pop-up menu.

Resume and



appear only if the selected Oozie job is suspended.

### **Suspend an Event to a Hadoop Oozie Job**

To suspend a reply to an Oozie agent notification that requests information required for an Oozie job run to continue, you can use the Suspend command.

#### **Follow these steps:**

1. Find an Oozie job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the Oozie job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Suspend from the drop-down list in the Send event dialog section and click Go.
  - Click



in the Actions column for the job in the Search Results section.

- Select the appropriate job in the flow, right-click, and Suspend from the pop-up menu.

Suspend and



appear only if the selected Oozie job is running.

## **Send an Event**

You can send an event of the appropriate type to change the state, status, or priority for one or more jobs or box jobs. You can also send an event to have a UNIX signal sent for one or more jobs.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:
  - Select Send Event from the drop-down list in the Search Results section and click Go.
  - Click  in the Job Flow section, right-click the jobs you want to send the event to, and select Send Event.  
The Send Event dialog opens with the selected job or jobs listed.
4. Select an event to send from the Event drop-down list.
5. Do one of the following, if applicable:
  - If you selected the Reply option in Step 4, enter the response in the Reply field.
  - If you selected the Send Signal option in Step 4, enter the UNIX signal to send in the Signal field.
  - If you selected the Change Priority option in Step 4, enter the new priority for the job in the Priority field.
  - If you selected the Change Status option in Step 4, select the new status for the job from the Status drop-down list.
  - If you selected the Set Global option in Step 4, enter a name for the global variable to set in the Global Name field and a value for the variable in the Global Value field.
6. (Optional) Enter a comment describing the event and your reasons for sending it in the Comment field.
7. (Optional) Enter a value in the Send Priority field.
8. (Optional) Specify a date and time at which to send the selected event.  
The event will not be processed until the specified date and time.
9. Click OK.  
The Send Event dialog closes, a confirmation message is displayed, and when the specified criteria are met, if any, the event is sent.

## **Cancel an Event**

You can cancel an event associated with one or more jobs or box jobs, for example, to prevent a job from being taken off hold.

### **Follow these steps:**

1. Find a job.  
One of the following occurs:
  - If only one job is found, detailed information about the job is displayed.
  - If two or more jobs are found, the jobs are listed in the Search Results section.
2. (Optional) Click the name of the job in the Search Results section for which you want to display details.  
Detailed information about the job is displayed.
3. Do one of the following:

- Select Cancel Event from the drop-down list in the Search Results section and click Go.
- Click



in the Job Flow section, right click the jobs for which you want to cancel the event, and select Cancel Event.

The Cancel Event dialog opens with the selected job or jobs listed.

4. Select the event to cancel from the Event Type drop-down list.
5. (Optional) Specify the date and time that the scheduled event you want to cancel was originally sent.
6. Click OK.

The Cancel Event dialog closes, a confirmation message is displayed, and when the specified criteria are met, if any, the event is cancelled.

## Web UI Resources Application

The Resources application lets you manage, create, update or delete virtual resources, run resource reports, and view job resource usage, from a single page. This application is accessed by clicking the Resources tab on the main Web UI page.

### AutoSys Workload Automation Resources

In the AutoSys Workload Automation environment, a resource is a system component that can be quantified. Resources govern job execution and play a vital role in scheduling. Before starting a job, AutoSys Workload Automation compares the resources required with the resources available. For example, you may have specified resource requirements that must be met before a job can run. If the required resources are not available, AutoSys Workload Automation defers the job run until resources are available. The only type of resource is virtual.

*Virtual resources* are only representative and are not tied to a physical system. The number of licenses available to all machines controlled by AutoSys Workload Automation is an example of a virtual resource. As each job that requires a license is submitted to the system, the number of licenses in the pool is decremented. When the number of licenses reaches zero, no more jobs can be submitted until the license pool is replenished.

### Types of Virtual Resources

Based on their properties and behavior, virtual resources are of the following types:

- **Depletable**  
Defines a resource where the amount allocated to the job decreases as it is used, for example, the disk space. When AutoSys Workload Automation starts a job, the depletable resources needed for the job are permanently removed from the resource pool.
- **Renewable**  
Defines a resource that is returned to the resource pool after use, for example, write access to a database. When AutoSys Workload Automation starts a job, the job borrows the renewable resource from the resource pool, and then returns it after use.
- **Threshold**  
Defines a resource that is used as a limit. For example, if the threshold resource quantity is set to two, AutoSys Workload Automation submits only those jobs that require two or fewer units of that resource.

In summary, depletable resources limit the number of usages, renewable resources control concurrent usage, and threshold resources set a limit to job submission.

### Global and Machine-Level Virtual Resources

You can define virtual resources by the name and machine with which they are associated (machine-level resources) or as global resources.

**NOTE**

A single resource cannot be defined as more than one type. For example, if a virtual resource is defined as a global resource, then it cannot also be defined as a machine-level resource.

The following is an example of a machine-level resource:

**Example: Resource R1 defined as a renewable resource with a quantity of 1 at machine M1**

This definition indicates that resource R1 is available to any job submitted to machine M1; however you can submit only one job at a time because only one resource is available.

The following are examples of global resources:

**Example: DISK defined as a renewable global resource with a quantity of 1**

This definition indicates that one DISK is available for all AutoSys Workload Automation machines. For example, if you have two jobs that require the DISK resource, with JobA requiring the DISK resource at M1 and JobB requiring the DISK resource at M2, you can submit only one of them because only one DISK resource is available globally.

**Example: CHECKS defined as a depletable global resource with a quantity of 10**

This definition indicates that a total of 10 checks are available for all AutoSys Workload Automation machines. As jobs requiring checks are submitted at various machines, the check pool is decremented. When the check pool number reaches zero, no more jobs can be submitted until the check pool is replenished.

**Resources Navigation**

Resources consists of the following sections:

- **Search**  
Displays key fields you can use to quickly find specific virtual resources that can be managed in your enterprise.
- **Search Results**  
Displays the results of your search criteria in a table with the complete virtual resource definition. You can use the Search Results section to select one or more virtual resources for deletion, to update virtual resource properties, or to display and run reports against existing resources.
- **Properties**  
Displays detailed information for the selected virtual resource and lets you create, update, or copy virtual resources to any AutoSys Workload Automation instance.
- **Resource Usage Report**  
Displays a search section from which you can search for a specific virtual resource and display the amount available for that resource or view the jobs that use the resource.
- **Job Resource Dependencies**  
Displays detailed information about the resource dependencies for the selected job and, optionally, shows them in a flow diagram. You can specify whether the flow should display as job-centric, display as resource-centric, or not display at all when the report is run.
- **Job Resource Dependencies Flow**  
Displays a graphical view of the selected job's resource dependencies in either a resource-centric or job-centric flow, determined by the selection you make for the Flow field in the Job Resource Dependencies Search. You can use this section to quickly check whether those resource requirements have been satisfied. For example, a solid red line indicates that the resource dependency is unsatisfied; a solid green line indicates that the resource has been satisfied. Right-clicking a job opens a pop-up menu, letting you edit the job in Quick Edit, see a consolidated view of the job in Quick View, or release resources. This section also includes a toolbar from which you can control the viewing area.

**NOTE**

To simplify the page presentation, you can click the Hide Instructions button, located in the upper right corner of the page, to remove the panel instructions from each section; to display instructions, you can click the Show Instructions button. By default, the page instructions are displayed.

## Security

### Contents

Depending on the CA EEM security rights assigned by your administrator, you may not have access to all functionality in Resources. For example, if you are not authorized to send events, you cannot release resources from the Job Resource Dependencies Report.

Resources access is controlled by the ApplicationAccess security policy in CA EEM.

The server list is controlled by CA EEM policies. Only the servers to which you have access are listed.

The server list includes only those AutoSys Workload Automation instances defined in Configuration Manager that, unless the AutoSys Workload Automation and the Web UI server are sharing the same CA EEM server, have valid credentials defined in the Credentials application.

### Quick Edit and Quick View Access

If you are authorized to use Quick Edit,

click 

the Job Resource Dependencies Report section to open a new window that contains a subset of the functionality of the Quick Edit application. The Search and Search Results sections and the Create and Customize buttons are not available. If you need the full functionality of Quick Edit, you can access the application by clicking the QEDIT tab. For information about using Quick Edit, see *Quick Edit Help*.

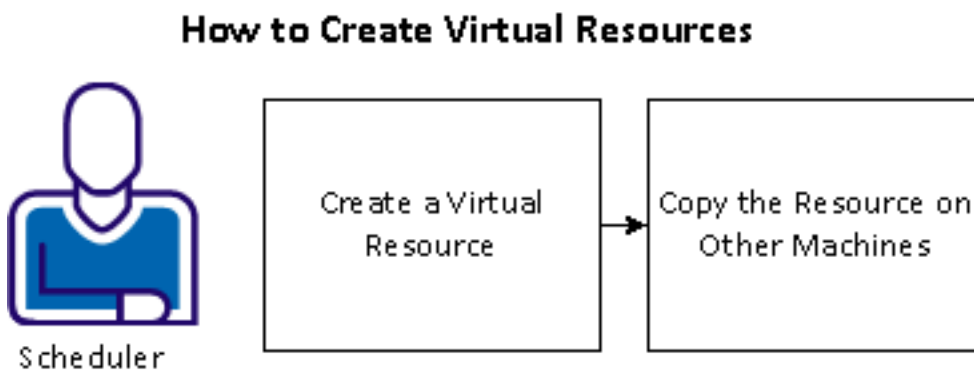
If you are authorized to use Quick View, click  in the Job Resource Dependencies Report section to open a new window that contains a subset of functionality of the Quick View application. The Search section and the Customize button are not available. If you need the full functionality of Quick View, you can access the application by clicking the QVIEW tab. For information about using Quick View, see *Quick View Help*.

## Create Virtual Resources

This article explains how to create virtual resources. As a scheduler, you use virtual resources to place certain restrictions on job execution.

This scenario walks you through the process of creating a virtual resource and copying it to other machines.

**Figure 85: How to Create Virtual Resources**



To create a virtual resource in a machine pool, follow these steps:

1. [Create a virtual resource.](#)
2. [Copy the resource on other machines.](#)

### Example: Create a Virtual Resource to Control Tape Drive Access

Suppose you have several jobs that are dependent on the availability of a tape drive. However, you have limited tape drives available. You can create a renewable virtual resource named Tape on each machine in a machine pool and assign those virtual resources to jobs as dependencies. When a job completes, the resource is returned to the machine in the pool so that other jobs with similar requirements can use it. Because you want to have four machines with a tape drive as a virtual resource, you create a virtual resource with the amount of 1 on MachineA. After you create the virtual resource on MachineA, you can copy that virtual resource definition to MachineB, MachineC, and MachineD.

#### NOTE

Not all machines in a machine pool need to have resources defined. Jobs requiring the resource are only sent to machines with sufficient resources. Therefore, if MachineE is in the machine pool but does not have a resource assigned, the job is never sent to that machine.

### Create a Virtual Resource

You can create a virtual resource in Resources, and use that resource as a dependency for a job's execution. Virtual resources are created in the Properties section of the Resources main page.

#### Follow these steps:

1. (Optional) Click Reset to clear all the fields in the Properties section.
2. Perform the following actions:
  - a. Select the server on which you want the virtual resource defined from the Server drop-down list.
  - b. Enter a name for the new resource in the Name field.

#### NOTE

Multiple names can be entered in the Name field in a comma-separated list.

- c. Select the type of resource (Renewable, Depletable, or Threshold) in the Type field.
- d. Enter the name of the machine on which the resource will reside in the Machine field.

Clicking 

the Select Machines dialog, from which you can select one or more machines.

#### NOTE

- Multiple names can be entered in the Machine field in a comma-separated list.
  - The machine name must be the name of a machine that is defined on the server to which the resource is being saved. If a machine name is not entered, then the resource will be saved as a global virtual resource.
  - You can create multiple machine-level resources with the same names on multiple machines on the same server.
  - You can create only one global resource with a given name on the same server; however that global resource cannot have the same name as any machine-level resource.
- e. Specify the number of available resources in the Amount field. Valid operators are =, !=, <, >, <=, and >=.

#### NOTE

When you create a virtual resource on multiple machines, the number entered in the Amount field is the same for each machine.

- f. (Optional) Enter a description in the Description field.
3. Select Create from the drop-down list, and click Commit.  
The resource is saved.



**NOTE**

If the resource is already defined on the selected server, an error message is displayed stating that the resource could not be created.

**Copy a Virtual Resource**

You can copy an existing virtual resource from one server or machine to another server or machine. The new resource has the same properties as the copied resource; however, if the new resource already exists on the selected server or machine, the create action fails.

**Follow these steps:**

1. Select the Select check box in front of the resource or resources you want to copy in the Search Results section.
  2. Select the server to which you want to copy the resource from the Server drop-down list.
  3. Click  to open the Select Machines dialog and select one or more machines to which you want to copy the resource. Alternatively, multiple names can be entered in the Machine field in a comma-separated list.
  4. Perform one of the following actions:
    - Select Create from the Select and drop-down list to copy an existing resource to the specified server and machine.
    - Select Create Global from the Select and drop-down list to copy the existing resource to the specified server as a global resource. The content in the Machine field is ignored.
  5. Click Go.
  6. Click OK to confirm.
- The virtual resource is saved to the new server.

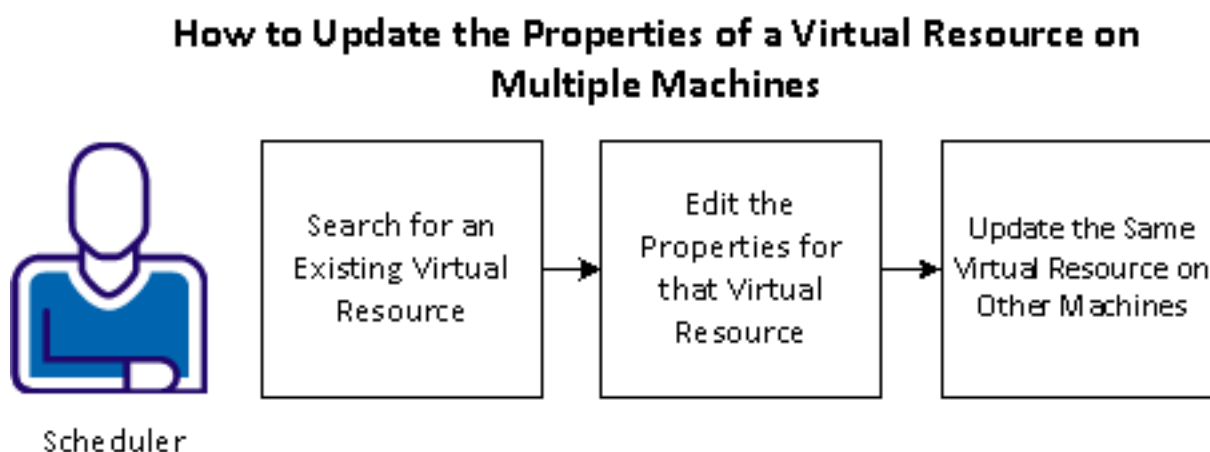
**NOTE**

- You must enter at least one machine name when using the Create command.
- To copy a resource as a global resource, you must select the Create Global command from the Select and drop-down list.
- In the Properties section, you can copy one or more virtual resources from one server or machine to another server or machine. To do this, select the server on which to copy the resources, and enter the resource names and machines in a comma-separated list in the appropriate fields. Next, select Create from the drop-down list, and click Commit.
- The resources must not exist on the selected target server in order to be copied. Resources are copied until a resource with the same name as one in the specified list is found on the target server. If this occurs, the action stops and an error message is displayed.

**Update the Properties of a Virtual Resource**

This article explains how to update the properties of a virtual resource. As a scheduler, sometimes you want to update the properties of a virtual resource on all of the machines on which it is defined. For example, increasing the amount of a renewable resource enables jobs requiring the higher resource amount to run.

This scenario walks you through the process of updating the properties of a virtual resource across machines.

**Figure 86: How to Update the Properties for a Virtual Resource on Multiple Machines**

To update the properties for a virtual resource on multiple machines, follow these steps:

1. [Search for an existing virtual resource.](#)
2. [Edit the properties for that virtual resource.](#)
3. [Update the same virtual resource on other machines.](#)

#### **Example: Update the Virtual Resources for Tape Drives**

To handle the heavy dependencies on your tape drives, your company has now added an additional tape drive to each of the four computers in the machine pool. Therefore, you want to increase the amount for the renewable resource on each of the machines. To increase that amount, search for the virtual resource. On one of the machines found, increase the resource amount in the properties and commit the changes on that machine. After you modify the properties of the resource on one machine, you can update the properties of the same resource on the remaining machines.

#### **Find a Virtual Resource**

You can use the Search section in Resources to find one or more AutoSys Workload Automation virtual resources residing on a server.

#### **Follow these steps:**

1. Select the appropriate server from the Server drop-down list.

#### **NOTE**

- The server list is controlled by CA EEM policies. Only the servers to which you have access are listed.
  - The server list includes only those AutoSys Workload Automation instances defined in Configuration Manager that, unless the AutoSys Workload Automation and the Web UI server are sharing the same CA EEM server, have valid credentials defined in the Credentials application.
2. Enter one of the following in the Name field:
    - Valid name -- Finds the specified resource on the selected server.
    - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all resources on the selected server that contain the partial name. For example, entering res\* finds all resources on the server that begin with the letters res, and entering \*res finds all resources on the server that end with the letters res.
    - Asterisk (\*) or percent (%) -- Finds all resources on the selected server.

3. (Optional) Select the machine on which to search for the machine-dependent virtual resource. Leaving the field blank, only returns global virtual resources.

**NOTE**

If a machine name is specified in the Machine field, and the Global check box is selected in the Type field, then machine-specific and global virtual resources are returned for the selected type in the Search Results section. For example, if \* is entered in the Machine field, and the Threshold and Global check boxes are selected in the Type field, all global and machine-specific threshold virtual resources are listed in the Search Results section.

4. (Optional) Enter search criteria in any of the optional fields to further refine your search.

**NOTE**

You can select multiple check boxes in the Type field to return multiple resource types matching the other search criteria. When you select multiple check boxes, Web UI uses OR logic to evaluate the criteria. At least one resource type must be selected. If no resource type is selected, an error message will be displayed. All types are selected by default.

5. Click Go.

The Search Results section lists the resources that meet your search criteria.

**NOTE**

- Only the resources that the user has been authorized to view appear in the search results list.
- To sort by a specific column, click the column heading. Clicking it again toggles between ascending and descending order.

**Edit a Virtual Resource**

You can edit a resource definition, for example, to change the amount defined for a selected virtual resource.

**NOTE**

The resources must exist on the selected server in order to be modified. Existing resources are updated until a resource in the specified list is not found. If this occurs, the action stops and an error message is displayed.

**Follow these steps:**

1. Click the name of the resource(s) you want to edit. You can edit multiple virtual resources, on one or more machines, at a single time. To do this, enter the resource names and machines in comma-separated lists in the appropriate fields.
2. Edit the resource(s) by changing the values in the Amount or Description fields.
3. Select Update from the drop-down list, and click Commit.  
The changes are saved.

**Update a Virtual Resource on a Different Server or Machine**

After you modify the properties of a resource on one server or machine, you can update the properties of the resource of the same name on another server or machine. For example, if you have made changes to the Amount field for a resource on Server1 and a resource with the same name exists on other machines, you can update the properties of the resource on all machines at the same time instead of editing each one individually.

**NOTE**

The resources must exist on the selected server in order to be modified. Existing resources are updated until a resource in the specified list is not found. If this occurs, the action stops and an error message is displayed.

**Follow these steps:**

1. (Optional) Select the Select check box for the resource(s) in the Search Results section that you want to update on other servers or machines. You can update multiple virtual resources, on one or more machines, at a single time. To do so, enter the resource names and (if applicable) machines in comma-separated lists in the appropriate fields.
2. Select another server on which the resource(s) should be updated from the Server drop-down list.

3. (Optional) Select the name of the machine, if you are updating a machine-level resource, on which the resource to be updated resides. Next, click  to open the Select Machines dialog, and select one or more machines to which you want to copy the resource. Alternatively, multiple names can be entered in the Machine field in a comma-separated list.
4. Select Update or Update Global from the Select and drop-down list:

**NOTE**

Selecting Update Global ignores the Machine field and updates the properties of the existing resource on the specified server as a global resource.

5. Click OK to confirm.
6. Repeat Steps 1 through 5 on any additional servers or machines as necessary.  
The virtual resources are updated on the selected servers.

**NOTE**

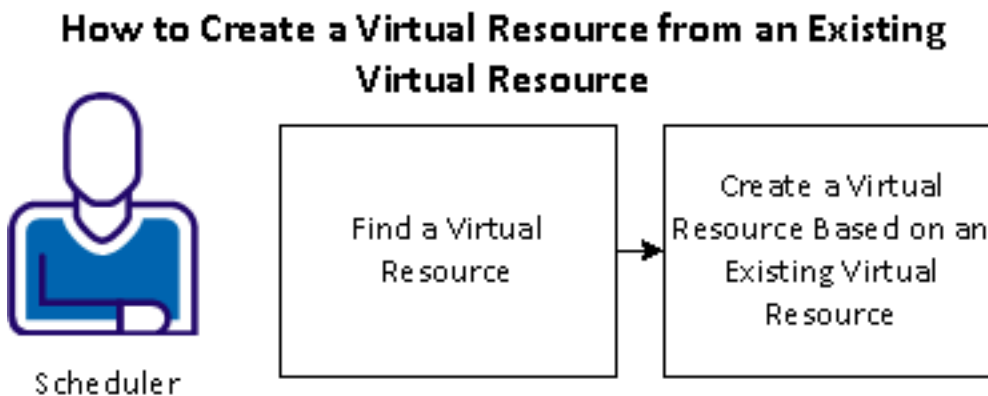
You must enter at least one machine name when using the Update command.

## Create a Virtual Resource from an Existing Virtual Resource

This article explains how to create a Virtual Resource from an Existing Resource. As a scheduler, you use virtual resources to place certain restrictions on job execution. You can use existing virtual resources to create additional virtual resources.

This scenario walks you through the process of creating a global virtual resource from an existing global virtual resource.

**Figure 87: How to Create a Virtual Resource from an Existing Virtual Resource**



To create a resource based on an existing resource, follow these steps:

1. [Find a virtual resource.](#)
2. [Create a virtual resource based on an existing virtual resource.](#)

### Example: Use the HR\_Report Resource to Create a New Resource

Suppose your organization restricts the number of HR reports running at the same time to 15. That restriction is implemented using a global resource named HR\_Report that has a resource amount of 15. You want to also restrict the number of sales reports running at the same time to 15. Because you already have a global resource for your HR reports, you can copy it to create a global resource to use for sales reports.

## Find a Virtual Resource

You can use the Search section in Resources to find one or more AutoSys Workload Automation virtual resources residing on a server.

### Follow these steps:

1. Select the appropriate server from the Server drop-down list.

#### NOTE

- The server list is controlled by CA EEM policies. Only the servers to which you have access are listed.
- The server list includes only those AutoSys Workload Automation instances defined in Configuration Manager that, unless the AutoSys Workload Automation and the Web UI server are sharing the same CA EEM server, have valid credentials defined in the Credentials application.

2. Enter one of the following in the Name field:
  - Valid name -- Finds the specified resource on the selected server.
  - Asterisk (\*), percent (%), or question mark (?) with a partial name, where the \* or % represents zero or more characters and ? represents a single character -- Finds all resources on the selected server that contain the partial name. For example, entering res\* finds all resources on the server that begin with the letters res, and entering \*res finds all resources on the server that end with the letters res.
  - Asterisk (\*) or percent (%) -- Finds all resources on the selected server.
3. (Optional) Select the machine on which to search for the machine-dependent virtual resource. Leaving the field blank, only returns global virtual resources.

#### NOTE

If a machine name is specified in the Machine field, and the Global check box is selected in the Type field, then machine-specific and global virtual resources are returned for the selected type in the Search Results section. For example, if \* is entered in the Machine field, and the Threshold and Global check boxes are selected in the Type field, all global and machine-specific threshold virtual resources are listed in the Search Results section.

4. (Optional) Enter search criteria in any of the optional fields to further refine your search.

#### NOTE

You can select multiple check boxes in the Type field to return multiple resource types matching the other search criteria. When you select multiple check boxes, Web UI uses OR logic to evaluate the criteria. At least one resource type must be selected. If no resource type is selected, an error message will be displayed. All types are selected by default.

5. Click Go.  
The Search Results section lists the resources that meet your search criteria.

#### NOTE

- Only the resources that the user has been authorized to view appear in the search results list.
- To sort by a specific column, click the column heading. Clicking it again toggles between ascending and descending order.

## Create a Resource Based on an Existing Resource

You can create a virtual resource that is based on an existing resource.

### Follow these steps:

1. Click the name of the resource you want to copy.
2. Modify the name of the resource in the Properties pane.
3. Make any other needed parameter changes.
4. Select Create from the drop-down list, and click Commit.

The changes are saved and the new resource is displayed in the Search Results pane.

## Monitor Resource Usage

This article explains how to monitor resource usage. The reporting section of the Resources page consists of the Resource Usage Report and Job Resource Dependencies Report panes. Use this section to manage resource usage and monitor the resources for specified jobs.

As a scheduler, you can view a report showing the resource usage for a specified resource, or resource usage by jobs.

This process walks you through the steps for monitoring job resources.

### Figure 88: How to Monitor Resource Usage

To monitor the resources of a job, you can follow these steps:

1. [View the Resource Usage report.](#)
2. [View the Job Dependencies report.](#)
3. [Monitor the resources of a job.](#)
4. [Release a resource.](#)

### Example: Monitor Resource Usage by Resources and Jobs

As a scheduler, you are responsible for monitoring resource usage by resources and jobs. Today, you want to view a report showing the tape drive requirements on MachineD. You know that MachineD, which contains two of the available resources, is currently unavailable. That leaves some jobs suspended while they are waiting for resources. You can run a Resource Usage report to display all of the jobs using those resources.

Additionally, you are interested in a specific set of Human Resources jobs that must complete on time. You can use the Job Resource Dependencies report to display those jobs and the resources they are using. In the report, you notice that a specific job does not have enough resources available to complete. You can edit the resource so that its amount is sufficient for successful completion of the job. You also notice that a renewable resource is being held by a job. You can release this resource so that it is available for other jobs to use.

### View a Resource Report

In the Resource Report section, you can view a report to see the amount remaining for specific virtual resources on selected machines and the jobs that are using those resources.

To run a resource report, perform one of the following actions:

Click 

the Actions column of the Search Results section for a single resource.

- Select one or more resources in the Search Results section, and select Report from the Select and drop-down list. Next, click Go, and click OK to confirm.
- Enter the appropriate search criteria for one or more resources in the Resource Usage Search subsection and click Go.

The detailed information for the specified resource or resources is displayed in the Resource Usage Summary Report and Resource Usage Detail Report subsections.

#### NOTE

When a single resource is selected from the Search Results section, the Resource Usage Search subsection is populated with that resource name; however, when multiple resources are selected from the Search Results section, the Resource Usage Search panel remains blank.

## **View a Job Resource Dependencies Report**

You can use the Job Resource Dependencies Report section to view the status of a job and its resource dependencies.

### **Follow these steps:**

1. Perform one of the following actions:
  - View a Resource Report. Next, click a job name in the Resource Usage Detail Report section to populate the Job Resource Dependencies Search section.
  - Enter the appropriate search criteria in the Job Resource Dependencies Search section.
2. Click Go.  
The detailed information for the job is displayed in the Job Resource Dependencies Report.

### **NOTE**

When you click Go, a confirmation dialog opens, displaying the progress of the search. You can press the Cancel button at any time to stop the search and display a partial listing of the results.

## **Monitor the Resources of a Job**

You can use the Job Resource Dependencies Report section to monitor the status of a job and view its resource dependencies. For example, if a job's status is waiting on resources, you can view the resources available for the job and edit them as needed to successfully run the job.

### **Follow these steps:**

1. View the Job Resource Dependencies Report.
2. Click the name of that resource where the number of resources in the Available column do not meet or exceed the number of resources needed to run the job.
3. Edit the resource so that its amount is sufficient for successful completion of the job.
4. Select Update from the drop-down list, and click Commit.  
The changes are saved.
5. Repeat Steps 2 through 4 for each resource as needed.
6. Click Go in the Job Resource Dependencies Search pane to view the status of the job in the Job Resource Dependencies Report.  
The job status displays the updated status for the job and the Satisfied column indicates that the resources have been satisfied.

## **Release a Resource**

Renewable resource dependencies can be defined using the Free attribute with one of the following options:

- Y -- Free resource on success only
- A -- Free resource unconditionally always
- N -- Do not free resources

If you assign a resource as a dependency to a job with the attribute Free=N, and that job is terminated, the resource continues to be held by that job. For this resource to be available to other dependent jobs waiting to run, you must use the RELEASE\_RESOURCE Send Event command to free the resource.

### **NOTE**

The Free attribute is only valid for renewable resource dependencies.

To release a resource, find the appropriate job in the Job Resource Dependencies Report,

click 

the Actions column, and click OK to confirm.

in

## Manage Your Resources

This article explains how to manage your resources.

### View the Properties of a Virtual Resource

You can view the detailed properties for a selected resource in the Properties section of the Resources main page.

To view the properties of a virtual resource, find the appropriate resource and click the resource name.

The detailed properties for the resource are displayed in the Properties section.

### Delete a Virtual Resource

If you no longer need to manage a virtual resource in your environment, you can delete it.

#### **Follow these steps:**

1. Select the Select check boxes in front of the resource(s) you want to delete, and select Delete from the Select and drop-down list.
2. Click Go, and click OK to confirm.  
The virtual resources are deleted.

#### **NOTE**

If the resource is in use by a job, the resource cannot be deleted. In that case, an error message is displayed explaining the reason for the failure.

### Edit the Properties of a Job

You can access Quick Edit from Resources to modify the properties for a job.

#### **Follow these steps:**

1. Find the appropriate job in the Job Resource Dependencies Report, and  
click  in the Actions column.
2. Edit the job definition as necessary, and click Commit.  
The changes to the job definition are saved.
3. Click Close when you are finished.

#### **NOTE**

For more information about editing job definitions in Quick Edit, see the *Quick Edit Help*.

### Display Job Details

You can access Quick View from Resources to view the job details, dependencies, starting conditions, and associated events for a job.

#### **Follow these steps:**

1. Find the appropriate job in the Job Resource Dependencies Report, and  
click  in the Actions column.
2. Scroll down as necessary to view all of the available job information.
3. Click Close when you are finished viewing the details.



**NOTE**

For more information about Quick View, see the *Quick View Help*.

## Additional Resources

Additional resources for AutoSys Workload Automation:

- [Workload Automation AE r11.3.5 / r11.3.6 Support Cookbook](#)
- [Workload Automation AE r11.3.5 / r11.3.6 Troubleshooting Cookbook](#)
- [AutoSys Workload Automation Solutions and Patches](#)
- [AutoSys Workload Automation Compatibility Matrix](#)
- [AutoSys Workload Automation Release and Support Lifecycle Dates](#)
- [Workload Automation Agents](#)
- [CA Embedded Entitlements Manager](#)
- [Broadcom Enterprise Software](#)
- [Broadcom Automation](#)
- [Broadcom Support](#)
- [Broadcom Community](#)
- [Broadcom Education](#)
- [Broadcom Software Tech Blogs](#)
- [Broadcom Twitter](#)

## Enable Ports on AWS or Microsoft Azure Security Group

AutoSys Workload Automation requires access to the following default ports. Ensure that the ports are enabled in the Amazon Web Services (AWS) Security Group for the respective EC2 instances.

Server	Inbound Ports	Outbound Ports
<b>AutoSys Workload Automation Server with Agent</b>	7163 (SSA)	7163 (SSA)
	7520 (Agent port)	5250 (CA EEM port)
	9443 (Web server port)	1521 (Oracle port) 1433 (MSSQL port)
	7080 (Soap Web server Port)	
<b>Web UI Server</b>	8080, 8443 (SSL), 8005 (64-bit Tomcat ports)	1521 (Oracle port) 1433 (MSSQL port)
		1527 (Derby)
	7163 (SSA)	7163 (SSA)
	7004 (Event bus port)	5250 (CA EEM port)
	1527 (Derby)	
<b>CA Workload Automation iXp Server</b>	8080 (Tomcat port)	1521 (Oracle port) 1433 (MSSQL port)
	7163 (SSA)	7163 (SSA)
<b>CA EEM Server</b>	5250 (CA EEM port)	
<b>AutoSys Workload Automation Server with Remote Agent</b>	7163 (SSA)	7163 (SSA)
	7520 (Agent port)	5250 (CA EEM port)

	9433 (Web server port)	1521 (Oracle port) 1433 (MSSQL port)
	7080 (Soap Web server port)	
	7500 (Application server port)	
	7507 (Scheduler port)	
<b>Remote Agent</b>	7520 (Agent port)	7500 (Application server port)
	7163 (SSA)	7507 (Scheduler port)
		7163 (SSA)
<b>Only AutoSys Workload Automation Client</b>	7163 (SSA)	7163 (SSA)

#### NOTE

- Enable the ports that are related to AutoSys Workload Automation as well as traffic in the respective network.
- For Agents, enable 443 (HTTPS port), for HTTPS communication.
- In Amazon Web Services (AWS) environment, ensure that the IP address is resolvable across the respective domains to enable proper communication between the Broadcom products. For example, you can resolve the IP address by updating the host file, providing the DNS server details, or by navigating to EC2 configService settings > EC2 Service properties and selecting the Set Computer Name option.
- You can connect to CA Workload Automation iXp with the AutoSys Workload Automation database using the Amazon RDS database. To connect, set the `ixp.server.db.test.host.reachable` operating system environment variable to false.

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## Product Names Abbreviations

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This space references the following products and abbreviations:

- CA Workload Control Center (CA WCC)
- CA Workload Automation AE (CAUAJM)
- Automatic Storage Management (ASM)
- Application Programming Interfaces (API)
- Domain Name Server (DNS)
- Extended Binary Coded Decimal Interchange Code (EBCDIC)
- File Transfer Protocol (FTP)
- Java Architecture for XML Binding (JAXB)
- JavaServer Pages Standard Tag Library (JSTL)
- Job Control Language (JCL)
- Job Information Language (JIL)
- Java Message Service (JMS)
- Java Management Extensions (JMX)
- Open Research Online (ORO)
- Perl Compatible Regular Expressions (PCRE)
- Plain Old Java Object (POJO)
- Portable Operating System Interface for UNIX (POSIX)
- Remote Method Invocation (RMI)
- Remote Procedure Call (RPC)
- SAP Business Warehouse (SAP BW)
- Software Development Kit (SDK)
- Simple Network Management Protocol (SNMP)
- Software Security Assurance (SSA)
- Secure Sockets Layer (SSL)
- Structured Query Language (SQL)
- Transmission Control Protocol/Internet Protocol (TCP/IP)

## Documentation for Previous Versions

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To see the documentation for other versions of AutoSys Workload Automation Workload Control Center, select **Versions** in the upper-right corner.

If you cannot locate the version that you need in this space, go to the [techdocs.broadcom.com landing page](https://techdocs.broadcom.com) and click the Legacy Bookshelves and PDFs link.

- To view the documentation from release 4.0 to 11.3 SP1, go to the [root page](#), click the Bookshelves and PDFs link, and enter "CA Workload Automation AE (AutoSys Edition)" in the "Search for your product" field.
- To view the bookshelves from release 11.3 to 11.3.6, go to the [root page](#), click the Bookshelves and PDFs link, and enter "CA Workload Automation AE" in the "Search for your product" field.

## Usage Data (Telemetry)

Telemetry is a capability that is integrated into AutoSys Workload Automation to collect and send the product usage and system configuration information of the AutoSys Workload Automation environment to Broadcom. This data helps Broadcom gain insights into customers' product usage, understand their software needs, and focus on the features and platforms that are most commonly used.

By default, the server stores the collected data in the database. The collected data is sent to Broadcom only after you enable the **Do you want to send telemetry data?** option in the **Activate Your Product** panel.

### WARNING

Telemetry does not collect any personally identifiable information (PII) or sensitive information.

### What Data Do We Collect?

The following table describes the data that is provided by customers:

### NOTE

For more information on data collection, see [Telemetry](#).

Data	Description
<b>Is the product installation or upgrade a new or additional planned usage under the portfolio license agreement?</b>	Details of whether the AutoSys Workload Automation installation or upgrade is a new or additional planned usage under the portfolio license agreement (PLA) with Broadcom.
<b>Send telemetry data to Broadcom</b>	Details of whether you want to send the telemetry data to Broadcom
<b>Domain name</b>	Your company's domain name (the last part of your company's email address)
<b>Enterprise site ID</b>	Your company's site ID that is listed on your company's license agreement and the Broadcom Support portal
<b>Internal identifier</b>	(Optional) Your cost-center or department's non-personally identifiable information (non-PII) identifier that you use for your company's tracking needs
<b>Proxy server and credentials</b>	(Optional) Details of the proxy server and user credentials to access the proxy server. <b>Note:</b> Provide these details only when the telemetry service cannot be reached through the outgoing port 443 from the AutoSys Workload Automation server.

The following table describes the data that is generated by AutoSys Workload Automation:

Data	Description
<b>Instance ID</b>	The instance ID of AutoSys Workload Automation
<b>Product usage</b>	Details of jobs, agents, and installed plug-ins that are defined in the AutoSys Workload Automation environment
<b>System configuration</b>	Version and component details of the AutoSys Workload Automation server. Version details of the operating system that hosts the server, and the database that is used with the server.
<b>Date when the data is collected</b>	The date when the product usage data is captured

Usage data includes actual usage of the product based on how it was licensed. Our privacy policy is located [here](#).

Parameter	Description	Example
Enterprise Site ID	Your company's unique login site ID that is listed on your PLA and the CA Support portal.	12345
Domain	The domain name of your company. It is the last part of your company's email address.	broadcom.com
Internal Identifier	Optional business unit, environment, cost center, department to enable you to charge back to your respective area.	IT-Sales-2367
PLA Flag	Indicates whether the installation or upgrade is related to a new or additional planned usage under a Portfolio License Agreement (PLA). true indicates yes and false indicates no.	True or False
Instance ID	Internally generated id for each instance of the product.	xxx
Date	Date on which data was collected.	yyyy-mm-dd
Usage	Maximum actual usage for the day: Examples: For Autosys Agents metric, it will be the number of agents and installed plugins as on that day. For Autosys Scheduler metric, the usage is always sent as "1" for each scheduler instance.	200
Product SKU	Product Identifier	ATASPV990
Product Name.	Product Name.	AutoSys Workload Automation Server
Product Version	Product Version #.	Version 12.0
FlushWaitTime= <i>wait_time</i>	Defines the maximum amount of time (in seconds) to wait before sending the telemetry data to Broadcom. <b>Default: 60</b>	
SKU=ATASPV990	Specifies the Stock Keeping Unit (SKU) of AutoSys Workload Automation. The AutoSys Workload Automation SKU is ATASPV990. <b>Note:</b> Do not change any of the SKU values until notified by Broadcom.	
SKUDescription=AutoSys Workload Automation Server	Specifies the description of the SKU. The AutoSys Workload Automation SKU description is AutoSys Workload Automation Server.	
AgentSKU=WAAAEV990	Specifies the SKU of the agent. The agent SKU is WAAAEV990.	

Parameter	Description	Example
AgentSKUDescription= AutoSys Workload Automation Agent	Specifies the description of the agent SKU. The agent SKU description is AutoSys Workload Automation Agent.	
EnableProxy=0 1	Specifies whether you want to use a proxy to send the telemetry data to CA. 1 indicates yes and 0 indicates no. <b>Note:</b> If you enable proxy, you must specify the URI in the ProxyURI parameter.	
ProxyURI= <i>web_address</i>	Specifies the URI to access the proxy server.	
ProxyURIUsername= <i>user_name</i>	Specifies the user name that is required for proxy authentication.	
ProxyURIPassword= <i>password</i>	Specifies the AES encrypted password that is associated with the proxy URI user name. <b>Note:</b> If you are updating the password in the configuration file, use the <code>autosys_secure</code> utility to generate the password in AES encrypted format.	
Operating System	The name of the Operating System of the Server.	
Operating System Version	The version number of the Operating System.	
Database	The name of the Database of the Server.	
Database Version	The Version number of the database.	
Web UI Installed	Specifies if the Web UI is installed or not.	
WebServer Installed	Specifies if the web server is installed or not.	
Security Type	Specifies if the instance is running in NATIVE or CA EEM mode.	
HA Mode	Specifies whether the scheduler is running in HA mode or not.	
Num_of_Jobs_Defined	The number of jobs that are defined.	
Num_of_JobTypes_Defined	The number of job types that are defined.	
Num_of_JobRuns	The number of jobs that ran.	
Num_of_Jobs_by_JobType_defined	The number of jobs that are defined for a specific job type.	
System Agents	Specifies the number of system agents pointing to the current scheduler instance.	



Parameter	Description	Example
Agent Plug-ins	Specifies the number of agent plug-ins (Application Services, Databases , Micro Focus, Oracle Applications, PeopleSoft , SAP , Remote Execution Agent, Informatica , Microsoft SQL Server ) installed on the agents pointing to the current scheduler instance.	
Unique HTTP EndPoint URLs Count	The number of Unique HTTP endpoint urls used in the jobs that are executed on a given day within a year. <b>Note:</b> The endpoint url details are collected until the first path parameter in the url. Example: https://TestMachine:9443/AEWS	
Unique SOAP EndPoint URLs Count	The number of Unique SOAP endpoint urls used in the jobs that are executed on a given day within a year. <b>Note:</b> The endpoint url details are collected until the first path parameter in the url. Example: https://TestMachine:9443/AESOA	
lic_identifier	Original URL as specified in the job definition.	
masked_lic_identifier	Masked URL hiding the machine, port, and application names.	
date_collected	Date on which data was collected.	yyyy-mm-dd

## How To Report Usage Data

As an administrator, during installation or upgrade, you can configure AutoSys Workload Automation to send telemetry data to Broadcom by filling in the fields in the *Activate Your Product* panel. **Follow these steps:**

1. Select from the following options:
  - a. Is this install or upgrade related to new or additional planned usage under a Portfolio License Agreement.  
**Default:** Yes.
  - b. Send telemetry data.  
You can send the telemetry data to Broadcom either automatically or manually.
    - Send the telemetry data to Broadcom:  
**Default:** The check box is selected.  
If the check box is selected, the telemetry data is **automatically** sent.  
If the check box is clear, the telemetry data is stored in the telemetrydata.txt file inside the telemetry folder in the following directory and must be **manually** sent by running the [telemetry script](#):  
install\_dir/server/c2o/

You can specify proxy server details to send the usage data to Broadcom.

Field name	Silent Install Variable	Description
Do you want to send telemetry data?	IA_AE_SEND_TEL_DATA	Select Yes if you want to send telemetry data to Broadcom. Otherwise, select No.

Use a proxy to send usage data	IA_AE_USE_PROXY	Select this checkbox if you are using a proxy server to send the usage data. Otherwise, select No. Provide these details only when the telemetry service cannot be reached through the outgoing port 443 from the AutoSys scheduler.
Proxy URI	IA_AE_PROXY_URI	The URI of your proxy server.
Username	IA_AE_PROXY_USER_NAME	The username to access your proxy server.
Password	IA_AE_PROXY_PWD	The password to access your proxy server.

### **How Frequently Do We Collect the Telemetry Data?**

By default, telemetry collects and stores the data daily at midnight.

### **How to Configure Telemetry?**

As an administrator, you can configure telemetry as follows:

- During the server installation or [upgrade](#)—by specifying the details in the **Activate Your Product** panel.

#### **NOTE**

You are required to configure telemetry during the server installation or upgrade. If the configuration is not successful during the installation or upgrade, we recommend that you configure telemetry by using the AutoSys Workload Automation.

- After the server installation or upgrade—by specifying the details in the **Activate Your Product** panel on the product topology.

#### **NOTE**

You can use AutoSys to [Configure AutoSys Workload Automation to Collect and Send Telemetry Data to Broadcom](#), only when the telemetry configuration is not successful during the installation or upgrade. Also, review the [Parameters in the telemetry.properties Configuration File](#).

### **How to Export the Telemetry Data that is Being Collected?**

At any time, you can view the data that is being collected.

To export the telemetry data into a file, see [Telemetry Script](#).

### **How to Verify that the Collected Data is Sent to Broadcom?**

You can verify whether the telemetry data that is collected from the AutoSys environment is sent to Broadcom. For more information, see [Extract Telemetry Data Offline](#).

By default, the server stores the collected data in the database. The collected data is sent to Broadcom only after you enable the **Do you want to send telemetry data?** option in the **Activate Your Product** panel.

### **Where do I see Usage Data?**

- As a PLA customer, you can see the usage data that you submit on the Digital Hub. <https://digitalhub.broadcom.com>
- Your Account Team / Customer Success Team can provide you with the Usage Report.

# Configure AutoSys Workload Automation to Collect and Send Telemetry Data to Broadcom

## WARNING

Before you configure AutoSys Workload Automation to send telemetry data to Broadcom, ensure that the outbound traffic from the scheduler computer to the port 443 is allowed. The scheduler uses this port to send the telemetry data of your AutoSys Workload Automation environment to Broadcom. You can optionally secure this port by using a proxy server to allow only authenticated access to this port.

As an administrator, you can configure AutoSys Workload Automation to collect and send telemetry data to Broadcom using any of the following ways:

- During installation or upgrade—By filling the fields in the Activate Your Product panel. For information about the fields on the Activate Your Product panel, see (UNIX) Installation Checklist or (Windows) Installation Checklist.
- Post installation—By editing the [parameters in the telemetry.properties configuration file](#). This configuration file is located in the \$AUTOUSER/telemetry (UNIX) or %AUTOUSER%\telemetry (Windows) directory.

## NOTE

- If you have completed the fields in the Activate Your Product panel during the installation, then those values are auto populated in the telemetry.properties configuration file. However, post installation, if you edit the parameters in the telemetry.properties configuration file and save it, the updated values take effect once telemetry runs.
- You can also extract the telemetry data offline by using the telemetry.bat (Windows) or telemetry.sh (UNIX) script. For more information about the telemetry script, see [telemetry Script—Extracts Telemetry Data Offline](#).
- The telemetry.log file is located in the \$AUTOUSER/out (UNIX) or %AUTOUSER%\out (Windows) directory. You can configure the log file properties by editing the logging.properties file in the \$AUTOUSER/telemetry (UNIX) or %AUTOUSER%\telemetry (Windows) directory. By default, the logging level is set to severe.

## Parameters in the telemetry.properties Configuration File

The telemetry.properties configuration file includes the following parameters:

- **PLAEnabled=0|1** Specifies whether the installation or upgrade is related to a new or additional planned usage under a Portfolio License Agreement (PLA). 1 indicates yes and 0 indicates no.
- **SendTelemetryData=0|1** Specifies whether you want to send the telemetry data to CA or not. 1 indicates yes and 0 indicates no.  
**Default:** 1
- **CompanyDomain=domain\_name**  
Specifies the domain name of your company. It is the last part of your company's email address.  
**Example:** xyz.com
- **EnterpriseSiteID=unique\_site\_ID**  
Specifies your company's 4 to 9 digit unique login site ID that is listed on the Broadcom Support portal. To find the site ID, go to Broadcom Support and select **Profile** from the **My Account** menu.  
**Example:** 10986
- **InternalIdentifier=unique\_ID**  
(Optional) Specifies your department's non-personally identifiable information (non-PII) identifier that you use for your company's tracking needs.  
**Example:** IT-Sales-1234
- **EnableProxy=0|1** Specifies whether you want to use a proxy to send the telemetry data to Broadcom. 1 indicates yes and 0 indicates no.

**NOTE**

If you enable proxy, you must specify the URI in the ProxyURI parameter.

- **ProxyURI=web\_address**  
Specifies the URI to access the proxy server.
- **ProxyURIUsername=user\_name**  
(Optional) Specifies the user name that is required for proxy authentication.
- **ProxyURIPassword=password**  
(Optional) Specifies the AES encrypted password that is associated with the proxy URI user name.

**NOTE**

If you are updating the password in the configuration file, use the autosys\_secure utility to generate the password in AES encrypted format.

- **FlushWaitTime=wait\_time**  
Defines the maximum amount of time (in seconds) to wait before sending the telemetry data to Broadcom.  
**Default:** 60
- **SKU=ATASPV990** Specifies the Stock Keeping Unit (SKU) of AutoSys Workload Automation. The AutoSys Workload Automation SKU is ATASPV990.

**NOTE**

Do not change the SKU value until notified by Broadcom.

- **SKUDescription=AutoSys Workload Automation Server** Specifies the description of the SKU.  
The AutoSys Workload Automation SKU description is AutoSys Workload Automation Server.

**NOTE**

Do not change the SKU description until notified by Broadcom.

- **AgentSKU=WAAAEV990** Specifies the SKU of the agent. The agent SKU is WAAAEV990.

**NOTE**

Do not change the agent SKU value until notified by Broadcom.

- **AgentSKUDescription=AutoSys Workload Automation Agent**  
Specifies the description of the agent SKU. The agent SKU description is AutoSys Workload Automation Agent.

**NOTE**

Do not change the agent SKU description until notified by Broadcom.

## telemetry Script—Extracts Telemetry Data Offline

The telemetry.bat (Windows) or telemetry.sh (UNIX) script extracts the telemetry data offline from your AutoSys Workload Automation environment.

The telemetry script is available in the %AUTOSYS%\bin (Windows) or \$AUTOSYS%/bin (UNIX) directory. When you run this script, the scheduler and agent data is extracted as follows:

- The scheduler data is extracted to the %AUTOUSER%\telemetry\scheduler\_telemetry\_data.%AUTOSERV%.*timestamp.csv* (Windows) or \$AUTOUSER/telemetry/scheduler\_telemetry\_data.\$AUTOSERV.*timestamp.csv* (UNIX) file.
- The agent data is extracted to the %AUTOUSER%\telemetry\agent\_telemetry\_data.%AUTOSERV%.*timestamp.csv* (Windows) or \$AUTOUSER/telemetry/agent\_telemetry\_data.\$AUTOSERV.*timestamp.csv* (UNIX) file.

The telemetry script has the following format:

(UNIX) telemetry.sh -export [-F *From Date*] [-T *To Date*]

(Windows) telemetry.bat -export [-F *From Date*] [-T *To Date*]

- **-export** Specifies that the telemetry script is used to extract the telemetry data.
- **-F *From Date***  
(Optional) Specifies the start date from which to extract the telemetry data to a .csv file in the MM/dd/yyyy format. The .csv file includes the telemetry data from this specified date.
- **-T *To Date***(Optional) Specifies the end date until which to extract the telemetry data to a .csv file in the MM/dd/yyyy format. The .csv file includes the telemetry data up to this specified date.

**NOTE**

- If you do not specify both the start date and end date, it defaults to the current month.
- If you do not specify the end date, it defaults to the current date.

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